

Exotic External SUGRA Bases — Quiver Diagrams (v2)

gravity_blocks.v2 catalog ($c_{\text{ext}} \leq 16$ for $T > 1$), T_{max} via f_{multi}

Contents

1	$\hat{1}$ External (87 combos, 16930 bases)	33
1.1	$\mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	33
1.2	$\mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	34
1.3	$\mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	38
1.4	$\mathfrak{e}_8 + \hat{1} \rightarrow (-1)$	39
1.5	$\mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	46
1.6	$\hat{1} \rightarrow (-1)$	46
1.7	$\hat{1} \rightarrow (-2)$	49
1.8	$\mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1330
1.9	$\mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1331
1.10	$\mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1331
1.11	$\mathfrak{su}_2 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1332
1.12	$\mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1334
1.13	$\mathfrak{su}_2 + \mathfrak{e}_8 + \hat{1} \rightarrow (-1)$	1335
1.14	$\mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1336
1.15	$\mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1337
1.16	$\mathfrak{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$	1338
1.17	$\mathfrak{su}_2 + \hat{1} \rightarrow (-2)$	1338
1.18	$\mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1392
1.19	$\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1393
1.20	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1393
1.21	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1394
1.22	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1394
1.23	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1394
1.24	$\mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1395
1.25	$\mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$	1397
1.26	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1397
1.27	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1398
1.28	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1399
1.29	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1399
1.30	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1402
1.31	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1404
1.32	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1413
1.33	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1414
1.34	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1424
1.35	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1424
1.36	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1424
1.37	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1424
1.38	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1424
1.39	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1430
1.40	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1430
1.41	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1437
1.42	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1437
1.43	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1439
1.44	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1439
1.45	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1443
1.46	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1444
1.47	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1446
1.48	$\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$	1446

1.49	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1446
1.50	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1446
1.51	$\mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1447
1.52	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1448
1.53	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1448
1.54	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1449
1.55	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1449
1.56	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1449
1.57	$\mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1450
1.58	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1450
1.59	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1450
1.60	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1450
1.61	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1451
1.62	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1451
1.63	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1451
1.64	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1451
1.65	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow (-1)$	1452
1.66	$\mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$	1452
1.67	$\mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-2)$	1453
1.68	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$	1454
1.69	$\mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1455
1.70	$\mathfrak{su}_3 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1455
1.71	$\mathfrak{su}_3 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1455
1.72	$\mathfrak{su}_3 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1455
1.73	$\mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1456
1.74	$\mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1456
1.75	$\mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1457
1.76	$\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1458
1.77	$\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1458
1.78	$\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1458
1.79	$\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1458
1.80	$\mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1459
1.81	$\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1459
1.82	$(-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1459
1.83	$(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1459
1.84	$(-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1460
1.85	$(-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1460
1.86	$(-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1460
1.87	$(-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1461
2	$\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$ External (5 combos, 6 bases)	1462
2.1	$\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1462
2.2	$\mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1462
2.3	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1462
2.4	$(-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1462
2.5	$\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2)$	1462
3	$\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$ External (67 combos, 581 bases)	1463
3.1	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	1463
3.2	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	1463
3.3	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1463
3.4	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1464
3.5	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1464
3.6	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1464
3.7	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$	1464
3.8	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	1464

3.9	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1465
3.10	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	1465
3.11	$\text{NHC}(-2)(-3) + \mathfrak{e}_7 + (-3) \rightarrow (-2)$	1465
3.12	$\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + (-3) \rightarrow (-2)$	1465
3.13	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1466
3.14	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$	1466
3.15	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	1466
3.16	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1467
3.17	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	1467
3.18	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1467
3.19	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$	1469
3.20	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + (-3) \rightarrow (-2)$	1469
3.21	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1469
3.22	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1470
3.23	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1471
3.24	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	1473
3.25	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \text{su2n3mix}$	1473
3.26	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	1473
3.27	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	1474
3.28	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	1474
3.29	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1474
3.30	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1476
3.31	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1478
3.32	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6$	1504
3.33	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4$	1504
3.34	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8$	1505
3.35	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1505
3.36	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1506
3.37	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	1507
3.38	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1508
3.39	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \text{su2n3mix}$	1509
3.40	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	1509
3.41	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1511
3.42	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1511
3.43	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1513
3.44	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1514
3.45	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8$	1518
3.46	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1519
3.47	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	1521
3.48	$\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1522
3.49	$\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1522
3.50	$\mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1522
3.51	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1522
3.52	$\mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1522
3.53	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1522
3.54	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1523
3.55	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1523
3.56	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1523
3.57	$\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1524
3.58	$\mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1524
3.59	$\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1524
3.60	$(-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1524
3.61	$(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1525
3.62	$(-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1525
3.63	$(-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1525

3.64	$(-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1526
3.65	$(-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1526
3.66	$(-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1526
3.67	$\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2)$	1527
4	NHC $(-2)(-3)$ External (1227 combos, 49200 bases)	1528
4.1	NHC $(-2)(-3)$	1528
4.2	NHC $(-2)(-3) + \mathfrak{e}_7$	1568
4.3	NHC $(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2$	1568
4.4	NHC $(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	1569
4.5	NHC $(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	1569
4.6	NHC $(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1569
4.7	NHC $(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1570
4.8	NHC $(-2)(-3) + \mathfrak{so}_8$	1571
4.9	NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2$	1571
4.10	NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1571
4.11	NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1571
4.12	NHC $(-2)(-3) + \mathfrak{su}_2$	1572
4.13	NHC $(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	1576
4.14	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	1577
4.15	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1580
4.16	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1581
4.17	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1584
4.18	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	1584
4.19	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	1585
4.20	NHC $(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$	1586
4.21	NHC $(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	1586
4.22	NHC $(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \text{su2n3mix}$	1587
4.23	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	1587
4.24	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1587
4.25	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1587
4.26	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1587
4.27	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1588
4.28	NHC $(-2)(-3) + \mathfrak{su}_3$	1588
4.29	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	1589
4.30	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1592
4.31	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1592
4.32	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1592
4.33	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1592
4.34	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1592
4.35	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1592
4.36	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	1593
4.37	NHC $(-2)(-3) + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	1593
4.38	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1593
4.39	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1593
4.40	NHC $(-2)(-3)$	1593
4.41	NHC $(-2)(-3) + \mathfrak{e}_6$	1771
4.42	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_6$	1780
4.43	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_6 + \mathfrak{su}_2$	1781
4.44	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_7$	1781
4.45	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}'_7$	1781
4.46	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{f}_4$	1782
4.47	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{f}_4 + \mathfrak{su}_2$	1783
4.48	NHC $(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$	1784
4.49	NHC $(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \text{NHC}(-2)(-3)$	1785

4.50	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$	1786
4.51	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	1786
4.52	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	1788
4.53	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	1788
4.54	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	1789
4.55	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	1789
4.56	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	1790
4.57	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3$	1790
4.58	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8$	1790
4.59	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$	1793
4.60	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$	1793
4.61	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	1800
4.62	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1801
4.63	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$	1801
4.64	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	1802
4.65	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$	1803
4.66	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$	1804
4.67	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2$	1804
4.68	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{e}_6$	1861
4.69	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$	1861
4.70	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	1862
4.71	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	1864
4.72	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	1865
4.73	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$	1865
4.74	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	1872
4.75	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	1873
4.76	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$	1873
4.77	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	1874
4.78	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	1919
4.79	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	1920
4.80	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	1921
4.81	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1921
4.82	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	1938
4.83	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	1939
4.84	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1941
4.85	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	1942
4.86	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	1954
4.87	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	1957
4.88	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1958
4.89	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1967
4.90	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	1967
4.91	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	1968
4.92	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	1969
4.93	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix}$	1971
4.94	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	1985
4.95	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_3$	1985
4.96	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	1986
4.97	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$	2013
4.98	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$	2013
4.99	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	2013
4.100	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	2026
4.101	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	2028
4.102	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2029
4.103	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{su2n3mix}$	2031
4.104	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	2032

4.105NHC(-2)(-3) + e ₆ + su ₂ + su ₃ + su ₃ + su ₂	2036
4.106NHC(-2)(-3) + e ₆ + su ₂ + su ₃ + su ₃ + su ₃	2037
4.107NHC(-2)(-3) + e ₆ + su ₂ + (-3)→(-2)	2037
4.108NHC(-2)(-3) + e ₆ + su ₂ →(-3)	2038
4.109NHC(-2)(-3) + e ₆ + su ₂ →(-3) + NHC(-2)(-3)	2040
4.110NHC(-2)(-3) + e ₆ + su ₂ →(-3) + so ₈	2040
4.111NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₂	2041
4.112NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₂ + su ₂	2050
4.113NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₂ + su ₂ →(-3)	2051
4.114NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₂ + su2n3mix	2052
4.115NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₂ + su ₃	2052
4.116NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₂ →(-3)	2054
4.117NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₂ →(-3) + su ₂	2055
4.118NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su2n3mix	2056
4.119NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su2n3mix + su ₂	2057
4.120NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₃	2057
4.121NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₃ + su ₂	2059
4.122NHC(-2)(-3) + e ₆ + su ₂ →(-3) + su ₃ + su ₃	2061
4.123NHC(-2)(-3) + e ₆ + su ₂ →(-3) + (-3)→(-2)	2061
4.124NHC(-2)(-3) + e ₆ + su2n3mix	2061
4.125NHC(-2)(-3) + e ₆ + su2n3mix + su ₂	2076
4.126NHC(-2)(-3) + e ₆ + su2n3mix + su ₂ + su ₂ →(-3)	2090
4.127NHC(-2)(-3) + e ₆ + su2n3mix + su ₂ + su ₃	2090
4.128NHC(-2)(-3) + e ₆ + su2n3mix + su ₂ →(-3)	2090
4.129NHC(-2)(-3) + e ₆ + su2n3mix + su ₂ →(-3) + su ₂	2092
4.130NHC(-2)(-3) + e ₆ + su2n3mix + su2n3mix	2092
4.131NHC(-2)(-3) + e ₆ + su2n3mix + su ₃	2092
4.132NHC(-2)(-3) + e ₆ + su2n3mix + su ₃ + su ₂	2095
4.133NHC(-2)(-3) + e ₆ + su ₃	2096
4.134NHC(-2)(-3) + e ₆ + su ₃ + NHC(-2)(-3)	2107
4.135NHC(-2)(-3) + e ₆ + su ₃ + NHC(-2)(-3) + su ₂	2108
4.136NHC(-2)(-3) + e ₆ + su ₃ + NHC(-2)(-3) + su ₃	2108
4.137NHC(-2)(-3) + e ₆ + su ₃ + so ₈	2108
4.138NHC(-2)(-3) + e ₆ + su ₃ + so ₈ + su ₂	2109
4.139NHC(-2)(-3) + e ₆ + su ₃ + su ₂	2110
4.140NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + NHC(-2)(-3)	2137
4.141NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + so ₈	2137
4.142NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + su ₂	2137
4.143NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + su ₂ + su ₂	2150
4.144NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + su ₂ + su ₃	2152
4.145NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + su ₂ →(-3)	2153
4.146NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + su2n3mix	2155
4.147NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + su ₃	2156
4.148NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + su ₃ + su ₂	2160
4.149NHC(-2)(-3) + e ₆ + su ₃ + su ₂ + su ₃ + su ₃	2161
4.150NHC(-2)(-3) + e ₆ + su ₃ + su ₂ →(-3)	2161
4.151NHC(-2)(-3) + e ₆ + su ₃ + su ₂ →(-3) + su ₂	2163
4.152NHC(-2)(-3) + e ₆ + su ₃ + su ₂ →(-3) + su ₃	2165
4.153NHC(-2)(-3) + e ₆ + su ₃ + su2n3mix	2166
4.154NHC(-2)(-3) + e ₆ + su ₃ + su2n3mix + su ₂	2169
4.155NHC(-2)(-3) + e ₆ + su ₃ + su ₃	2169
4.156NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + NHC(-2)(-3)	2173
4.157NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + su ₂	2173
4.158NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + su ₂ + su ₂	2177
4.159NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + su ₂ + su ₃	2178

4.160NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2179
4.161NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	2179
4.162NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	2180
4.163NHC(-2)(-3) + $\mathfrak{e}_6 + (-3) \rightarrow (-2)$	2180
4.164NHC(-2)(-3) + $\mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	2181
4.165NHC(-2)(-3) + $\mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	2181
4.166NHC(-2)(-3) + $\mathfrak{e}_7$	2181
4.167NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{e}_6$	2183
4.168NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{f}_4$	2183
4.169NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-3)$	2183
4.170NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	2184
4.171NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	2186
4.172NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	2186
4.173NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{so}_8$	2187
4.174NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2$	2187
4.175NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	2189
4.176NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	2190
4.177NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3$	2190
4.178NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2$	2191
4.179NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	2205
4.180NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8$	2206
4.181NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	2208
4.182NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$	2209
4.183NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	2240
4.184NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	2241
4.185NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	2256
4.186NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2257
4.187NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	2258
4.188NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2259
4.189NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	2262
4.190NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	2263
4.191NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \text{su2n3mix}$	2263
4.192NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	2265
4.193NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	2270
4.194NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2271
4.195NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	2271
4.196NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2271
4.197NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	2276
4.198NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	2279
4.199NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	2279
4.200NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	2280
4.201NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	2280
4.202NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	2280
4.203NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	2282
4.204NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \text{su2n3mix}$	2282
4.205NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$	2290
4.206NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	2291
4.207NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_3$	2291
4.208NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3$	2292
4.209NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	2307
4.210NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	2312
4.211NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2313
4.212NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	2313
4.213NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2314
4.214NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	2315

4.215NHC(-2)(-3) + $e_7 + su_2 + su_3 + su_{2n3mix}$	2316
4.216NHC(-2)(-3) + $e_7 + su_2 + su_3 + su_3$	2316
4.217NHC(-2)(-3) + $e_7 + su_2 + su_3 + su_3 + su_2$	2318
4.218NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3)$	2318
4.219NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + NHC(-2)(-3)$	2319
4.220NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + so_8$	2320
4.221NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2$	2320
4.222NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_2$	2324
4.223NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_2 + su_2$	2327
4.224NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_2 + su_3$	2328
4.225NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_2 \rightarrow (-3)$	2328
4.226NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_{2n3mix}$	2328
4.227NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_3$	2328
4.228NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_3 + su_2$	2330
4.229NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2330
4.230NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2$	2331
4.231NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2331
4.232NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_{2n3mix}$	2331
4.233NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_3$	2331
4.234NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_{2n3mix}$	2332
4.235NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_{2n3mix} + su_2$	2332
4.236NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_{2n3mix} + su_2 \rightarrow (-3)$	2332
4.237NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3$	2333
4.238NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3 + su_2$	2333
4.239NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3 + su_2 + su_2$	2335
4.240NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3 + su_2 \rightarrow (-3)$	2335
4.241NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3 + su_3$	2335
4.242NHC(-2)(-3) + $e_7 + su_{2n3mix}$	2335
4.243NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2$	2339
4.244NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 + su_2$	2347
4.245NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 + su_2 \rightarrow (-3)$	2349
4.246NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 + su_3$	2349
4.247NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 \rightarrow (-3)$	2349
4.248NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 \rightarrow (-3) + su_2$	2350
4.249NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2350
4.250NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_{2n3mix}$	2350
4.251NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_3$	2351
4.252NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_3 + su_2$	2352
4.253NHC(-2)(-3) + $e_7 + su_3$	2352
4.254NHC(-2)(-3) + $e_7 + su_3 + NHC(-2)(-3)$	2355
4.255NHC(-2)(-3) + $e_7 + su_3 + so_8$	2356
4.256NHC(-2)(-3) + $e_7 + su_3 + su_2$	2356
4.257NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2$	2372
4.258NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 + su_2$	2377
4.259NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 + su_2 \rightarrow (-3)$	2378
4.260NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 + su_3$	2378
4.261NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 \rightarrow (-3)$	2379
4.262NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 \rightarrow (-3) + su_2$	2380
4.263NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_{2n3mix}$	2380
4.264NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_3$	2380
4.265NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_3 + su_2$	2383
4.266NHC(-2)(-3) + $e_7 + su_3 + su_2 \rightarrow (-3)$	2383
4.267NHC(-2)(-3) + $e_7 + su_3 + su_2 \rightarrow (-3) + su_2$	2384
4.268NHC(-2)(-3) + $e_7 + su_3 + su_2 \rightarrow (-3) + su_2 + su_2$	2385
4.269NHC(-2)(-3) + $e_7 + su_3 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2386

4.270NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	2386
4.271NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_{2n3mix}$	2386
4.272NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2$	2387
4.273NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$	2387
4.274NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	2389
4.275NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	2392
4.276NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2392
4.277NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	2392
4.278NHC(-2)(-3) + $\mathfrak{e}_7 + (-3) \rightarrow (-2)$	2393
4.279NHC(-2)(-3) + $\mathfrak{e}'_7$	2393
4.280NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{e}_6$	2395
4.281NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{f}_4$	2395
4.282NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3)$	2395
4.283NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	2396
4.284NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	2397
4.285NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	2397
4.286NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8$	2397
4.287NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_2$	2398
4.288NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	2400
4.289NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	2400
4.290NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_3$	2401
4.291NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2$	2401
4.292NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	2415
4.293NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8$	2416
4.294NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	2418
4.295NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$	2418
4.296NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	2449
4.297NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	2450
4.298NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	2454
4.299NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2455
4.300NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	2457
4.301NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix}$	2457
4.302NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	2459
4.303NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	2463
4.304NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2463
4.305NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	2464
4.306NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2464
4.307NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	2468
4.308NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	2470
4.309NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	2470
4.310NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_{2n3mix}$	2470
4.311NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	2471
4.312NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	2472
4.313NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix}$	2472
4.314NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2$	2480
4.315NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	2481
4.316NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_3$	2482
4.317NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$	2482
4.318NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	2497
4.319NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	2501
4.320NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2501
4.321NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	2501
4.322NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2501
4.323NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	2503
4.324NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_{2n3mix}$	2503

4.325NHC(-2)(-3) + $e'_7 + su_2 + su_3 + su_3$	2503
4.326NHC(-2)(-3) + $e'_7 + su_2 + su_3 + su_3 + su_2$	2505
4.327NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3)$	2505
4.328NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + NHC(-2)(-3)$	2506
4.329NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + so_8$	2507
4.330NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2$	2507
4.331NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_2$	2511
4.332NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_2 + su_3$	2513
4.333NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_2 \rightarrow (-3)$	2513
4.334NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_{2n3mix}$	2514
4.335NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_3$	2514
4.336NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_3 + su_2$	2515
4.337NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2515
4.338NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2$	2516
4.339NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2516
4.340NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_{2n3mix}$	2516
4.341NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_3$	2517
4.342NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_{2n3mix}$	2517
4.343NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_{2n3mix} + su_2$	2517
4.344NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_{2n3mix} + su_2 \rightarrow (-3)$	2518
4.345NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3$	2518
4.346NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3 + su_2$	2518
4.347NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3 + su_2 + su_2$	2520
4.348NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3 + su_2 \rightarrow (-3)$	2520
4.349NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3 + su_3$	2520
4.350NHC(-2)(-3) + $e'_7 + su_{2n3mix}$	2521
4.351NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_2$	2525
4.352NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_2 + su_2$	2532
4.353NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_2 + su_2 \rightarrow (-3)$	2534
4.354NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_2 + su_3$	2534
4.355NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_2 \rightarrow (-3)$	2534
4.356NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_2 \rightarrow (-3) + su_2$	2535
4.357NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2535
4.358NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_{2n3mix}$	2535
4.359NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_3$	2535
4.360NHC(-2)(-3) + $e'_7 + su_{2n3mix} + su_3 + su_2$	2537
4.361NHC(-2)(-3) + $e'_7 + su_3$	2537
4.362NHC(-2)(-3) + $e'_7 + su_3 + NHC(-2)(-3)$	2540
4.363NHC(-2)(-3) + $e'_7 + su_3 + so_8$	2540
4.364NHC(-2)(-3) + $e'_7 + su_3 + su_2$	2541
4.365NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2$	2556
4.366NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 + su_2$	2560
4.367NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 + su_2 \rightarrow (-3)$	2560
4.368NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 + su_3$	2560
4.369NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 \rightarrow (-3)$	2560
4.370NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 \rightarrow (-3) + su_2$	2562
4.371NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_{2n3mix}$	2562
4.372NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_3$	2562
4.373NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_3 + su_2$	2564
4.374NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3)$	2565
4.375NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3) + su_2$	2565
4.376NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3) + su_2 + su_2$	2567
4.377NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2567
4.378NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3) + su_3$	2567
4.379NHC(-2)(-3) + $e'_7 + su_3 + su_{2n3mix}$	2567

4.380NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{su}_2$	2569
4.381NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2569
4.382NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2571
4.383NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2573
4.384NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$	2573
4.385NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2573
4.386NHC(-2)(-3) + $\mathfrak{e}'_7$ + $(-3) \rightarrow (-2)$	2574
4.387NHC(-2)(-3) + $\mathfrak{e}_8$	2574
4.388NHC(-2)(-3) + $\mathfrak{e}_8$ + NHC(-2)(-3)	2574
4.389NHC(-2)(-3) + $\mathfrak{e}_8$ + NHC(-2)(-3) + $\mathfrak{su}_2$	2574
4.390NHC(-2)(-3) + $\mathfrak{e}_8$ + NHC(-2)(-3) + $\mathfrak{su}_3$	2575
4.391NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{so}_8$	2575
4.392NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	2575
4.393NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$	2575
4.394NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + NHC(-2)(-3)	2578
4.395NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	2578
4.396NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2578
4.397NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2586
4.398NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2590
4.399NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2590
4.400NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2591
4.401NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2592
4.402NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2593
4.403NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2596
4.404NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2597
4.405NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2598
4.406NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$	2598
4.407NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + NHC(-2)(-3)	2599
4.408NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2599
4.409NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2602
4.410NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2603
4.411NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2604
4.412NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2604
4.413NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2604
4.414NHC(-2)(-3) + $\mathfrak{f}_4$	2605
4.415NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}_6$	2614
4.416NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$	2615
4.417NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}_7$	2616
4.418NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}'_7$	2617
4.419NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$	2617
4.420NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$	2618
4.421NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2620
4.422NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	2621
4.423NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$	2621
4.424NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$	2621
4.425NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + su2n3mix	2621
4.426NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$	2622
4.427NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3)	2622
4.428NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{so}_8$	2623
4.429NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_2$	2623
4.430NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	2625
4.431NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$	2625
4.432NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$	2625
4.433NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + su2n3mix	2626
4.434NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_3$	2626

4.435	NHC(-2)(-3) + f ₄ + s ₀₈	2627
4.436	NHC(-2)(-3) + f ₄ + s ₀₈ + NHC(-2)(-3)	2630
4.437	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u2}	2630
4.438	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u2} + s _{u2}	2639
4.439	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u2} + s _{u2} →(-3)	2640
4.440	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u2} + su2n3mix	2641
4.441	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u2} + s _{u3}	2641
4.442	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u2} →(-3)	2642
4.443	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u2} →(-3) + s _{u2}	2643
4.444	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u2} →(-3) + s _{u3}	2643
4.445	NHC(-2)(-3) + f ₄ + s ₀₈ + su2n3mix	2643
4.446	NHC(-2)(-3) + f ₄ + s ₀₈ + su2n3mix + s _{u2}	2644
4.447	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u3}	2644
4.448	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u3} + s _{u2}	2646
4.449	NHC(-2)(-3) + f ₄ + s ₀₈ + s _{u3} + s _{u2} →(-3)	2647
4.450	NHC(-2)(-3) + f ₄ + s _{u2}	2647
4.451	NHC(-2)(-3) + f ₄ + s _{u2} + e ₆	2705
4.452	NHC(-2)(-3) + f ₄ + s _{u2} + f ₄	2706
4.453	NHC(-2)(-3) + f ₄ + s _{u2} + f ₄ + s _{u2}	2707
4.454	NHC(-2)(-3) + f ₄ + s _{u2} + f ₄ + s _{u2} →(-3)	2708
4.455	NHC(-2)(-3) + f ₄ + s _{u2} + NHC(-2)(-3)	2708
4.456	NHC(-2)(-3) + f ₄ + s _{u2} + NHC(-2)(-3) + s _{u2} →(-3)	2709
4.457	NHC(-2)(-3) + f ₄ + s _{u2} + s ₀₈	2710
4.458	NHC(-2)(-3) + f ₄ + s _{u2} + s ₀₈ + s _{u2}	2718
4.459	NHC(-2)(-3) + f ₄ + s _{u2} + s ₀₈ + s _{u2} →(-3)	2719
4.460	NHC(-2)(-3) + f ₄ + s _{u2} + s ₀₈ + su2n3mix	2719
4.461	NHC(-2)(-3) + f ₄ + s _{u2} + s ₀₈ + s _{u3}	2719
4.462	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2}	2720
4.463	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} + f ₄	2762
4.464	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} + s ₀₈	2762
4.465	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} + s _{u2} →(-3)	2763
4.466	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} + s _{u2} →(-3) + s _{u3}	2769
4.467	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} + su2n3mix	2769
4.468	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} + s _{u3}	2777
4.469	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} + s _{u3} + s _{u2} →(-3)	2786
4.470	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3)	2787
4.471	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3) + f ₄	2800
4.472	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3) + NHC(-2)(-3)	2800
4.473	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3) + s ₀₈	2801
4.474	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3) + s _{u2}	2801
4.475	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3) + s _{u2} + s _{u3}	2806
4.476	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3) + su2n3mix	2807
4.477	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3) + s _{u3}	2808
4.478	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u2} →(-3) + s _{u3} + s _{u2}	2811
4.479	NHC(-2)(-3) + f ₄ + s _{u2} + su2n3mix	2812
4.480	NHC(-2)(-3) + f ₄ + s _{u2} + su2n3mix + s ₀₈	2842
4.481	NHC(-2)(-3) + f ₄ + s _{u2} + su2n3mix + s _{u2}	2842
4.482	NHC(-2)(-3) + f ₄ + s _{u2} + su2n3mix + s _{u2} →(-3)	2850
4.483	NHC(-2)(-3) + f ₄ + s _{u2} + su2n3mix + s _{u3}	2850
4.484	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u3}	2852
4.485	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u3} + s ₀₈	2879
4.486	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u3} + s _{u2}	2879
4.487	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u3} + s _{u2} + s _{u2} →(-3)	2888
4.488	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u3} + s _{u2} →(-3)	2889
4.489	NHC(-2)(-3) + f ₄ + s _{u2} + s _{u3} + s _{u2} →(-3) + s _{u2}	2893

4.490NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + su2n3mix	2894
4.491NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2895
4.492NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + (-3)→(-2)	2898
4.493NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3)	2898
4.494NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{f}_4$	2903
4.495NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$	2903
4.496NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + NHC(-2)(-3)	2903
4.497NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + NHC(-2)(-3) + $\mathfrak{su}_2$	2904
4.498NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{so}_8$	2904
4.499NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	2905
4.500NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$	2906
4.501NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$	2906
4.502NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$	2919
4.503NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ + NHC(-2)(-3)	2919
4.504NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	2920
4.505NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2920
4.506NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2925
4.507NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ + su2n3mix	2926
4.508NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2927
4.509NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2930
4.510NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$ →(-3)	2931
4.511NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + su2n3mix	2932
4.512NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + su2n3mix + $\mathfrak{su}_2$	2934
4.513NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_3$	2934
4.514NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_3$ + $\mathfrak{so}_8$	2937
4.515NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2937
4.516NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2941
4.517NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2941
4.518NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ →(-3) + (-3)→(-2)	2942
4.519NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix	2942
4.520NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{f}_4$	2969
4.521NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + NHC(-2)(-3)	2969
4.522NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{so}_8$	2969
4.523NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	2970
4.524NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_2$	2970
4.525NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	3000
4.526NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	3001
4.527NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ →(-3)	3008
4.528NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	3008
4.529NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_2$ →(-3)	3010
4.530NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_2$ →(-3) + $\mathfrak{su}_2$	3012
4.531NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + su2n3mix	3012
4.532NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_3$	3013
4.533NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	3021
4.534NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	3023
4.535NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$	3023
4.536NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{f}_4$	3036
4.537NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + NHC(-2)(-3)	3037
4.538NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$	3037
4.539NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	3039
4.540NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ →(-3)	3040
4.541NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	3040
4.542NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	3067
4.543NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	3067
4.544NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ →(-3)	3076

4.545NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	3077
4.546NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3081
4.547NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{su2n3mix}$	3082
4.548NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	3083
4.549NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	3086
4.550NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	3089
4.551NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3089
4.552NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	3093
4.553NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	3094
4.554NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix}$	3094
4.555NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	3102
4.556NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_3$	3104
4.557NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$	3105
4.558NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	3108
4.559NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	3111
4.560NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \text{su2n3mix}$	3111
4.561NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	3112
4.562NHC(-2)(-3) + $\mathfrak{f}_4 + (-3) \rightarrow (-2)$	3112
4.563NHC(-2)(-3) + $\mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	3113
4.564NHC(-2)(-3) + $\mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	3113
4.565NHC(-2)(-3) + NHC(-2)(-3)	3113
4.566NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6$	3117
4.567NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)$	3117
4.568NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8$	3118
4.569NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2$	3118
4.570NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	3119
4.571NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	3120
4.572NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3$	3120
4.573NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	3121
4.574NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	3121
4.575NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{f}_4$	3121
4.576NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8$	3121
4.577NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2$	3122
4.578NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	3123
4.579NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	3123
4.580NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3124
4.581NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{f}_4 + \text{su2n3mix}$	3124
4.582NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3$	3124
4.583NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{so}_8$	3125
4.584NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6$	3126
4.585NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4$	3126
4.586NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8$	3126
4.587NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2$	3127
4.588NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2$	3130
4.589NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	3130
4.590NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$	3131
4.591NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3131
4.592NHC(-2)(-3) + NHC(-2)(-3) + su2n3mix	3131
4.593NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_3$	3132
4.594NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2$	3132
4.595NHC(-2)(-3) + $\mathfrak{so}_8$	3133
4.596NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6$	3153
4.597NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$	3155
4.598NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$	3155
4.599NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	3159

4.600NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	3160
4.601NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	3160
4.602NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$	3161
4.603NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$	3161
4.604NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	3162
4.605NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7$	3163
4.606NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$	3163
4.607NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$	3164
4.608NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$	3164
4.609NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_3$	3164
4.610NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7$	3164
4.611NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$	3165
4.612NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$	3165
4.613NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3)$	3165
4.614NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_3$	3166
4.615NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4$	3166
4.616NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$	3168
4.617NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$	3168
4.618NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$	3174
4.619NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	3175
4.620NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	3176
4.621NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$	3176
4.622NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	3176
4.623NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3178
4.624NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	3178
4.625NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix}$	3178
4.626NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	3179
4.627NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$	3179
4.628NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$	3181
4.629NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	3181
4.630NHC(-2)(-3) + $\mathfrak{so}_8 + \text{NHC}(-2)(-3)$	3181
4.631NHC(-2)(-3) + $\mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{e}_6$	3183
4.632NHC(-2)(-3) + $\mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{f}_4$	3183
4.633NHC(-2)(-3) + $\mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$	3183
4.634NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8$	3183
4.635NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$	3187
4.636NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{so}_8$	3187
4.637NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$	3188
4.638NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	3200
4.639NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	3204
4.640NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \text{su2n3mix}$	3205
4.641NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$	3205
4.642NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	3208
4.643NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3209
4.644NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	3210
4.645NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	3210
4.646NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix}$	3210
4.647NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2$	3213
4.648NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_3$	3213
4.649NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$	3214
4.650NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$	3216
4.651NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	3220
4.652NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \text{su2n3mix}$	3220
4.653NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$	3220
4.654NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2$	3221

4.710NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{f}_4$	3600
4.711NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{f}_4$ + $\mathfrak{su}_2$	3600
4.712NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{so}_8$	3600
4.713NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	3602
4.714NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{so}_8$ + $\mathfrak{su}_3$	3603
4.715NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$	3603
4.716NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	3641
4.717NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	3641
4.718NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$ + su2n3mix	3645
4.719NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$	3647
4.720NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$	3651
4.721NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + su2n3mix	3655
4.722NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + su2n3mix + $\mathfrak{su}_2$	3659
4.723NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_3$	3660
4.724NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_3$ + $\mathfrak{so}_8$	3668
4.725NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $(-3) \rightarrow (-2)$	3668
4.726NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$	3668
4.727NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{e}_6$	3689
4.728NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$	3690
4.729NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{e}_7$	3691
4.730NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{e}'_7$	3691
4.731NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{f}_4$	3691
4.732NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$	3692
4.733NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$	3693
4.734NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$	3693
4.735NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	3695
4.736NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$	3699
4.737NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + su2n3mix	3699
4.738NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$	3699
4.739NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	3700
4.740NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{e}_6$	3740
4.741NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$	3741
4.742NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	3741
4.743NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	3744
4.744NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$	3744
4.745NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{f}_4$	3746
4.746NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{so}_8$	3746
4.747NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$	3747
4.748NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + su2n3mix	3747
4.749NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{so}_8$	3755
4.750NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	3755
4.751NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$	3761
4.752NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	3761
4.753NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $(-3) \rightarrow (-2)$	3763
4.754NHC(-2)(-3) + $\mathfrak{so}_8$ + $(-3) \rightarrow (-2)$	3763
4.755NHC(-2)(-3) + $\mathfrak{so}_8$ + $(-3) \rightarrow (-2)$ + $\mathfrak{su}_2$	3764
4.756NHC(-2)(-3) + $\mathfrak{so}_8$ + $(-3) \rightarrow (-2)$ + $\mathfrak{su}_2 \rightarrow (-3)$	3766
4.757NHC(-2)(-3) + $\mathfrak{so}_8$ + $(-3) \rightarrow (-2)$ + su2n3mix	3767
4.758NHC(-2)(-3) + $\mathfrak{so}_8$ + $(-3) \rightarrow (-2)$ + $\mathfrak{su}_3$	3767
4.759NHC(-2)(-3) + $\mathfrak{so}_8$ + su3n2mix	3767
4.760NHC(-2)(-3) + $\mathfrak{su}_2$	3768
4.761NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{e}_6$	4143
4.762NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{e}_6$ + $\mathfrak{e}_6$	4152
4.763NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{e}_6$ + $\mathfrak{f}_4$	4152
4.764NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{e}_6$ + NHC(-2)(-3)	4153

4.765NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	4155
4.766NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	4155
4.767NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$	4155
4.768NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$	4158
4.769NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	4159
4.770NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$	4159
4.771NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$	4160
4.772NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	4182
4.773NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$	4183
4.774NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	4184
4.775NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	4184
4.776NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	4201
4.777NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	4201
4.778NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	4204
4.779NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	4204
4.780NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	4215
4.781NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	4218
4.782NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$	4219
4.783NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	4222
4.784NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	4222
4.785NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	4223
4.786NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \text{su2n3mix}$	4224
4.787NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_3$	4225
4.788NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$	4225
4.789NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$	4236
4.790NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$	4236
4.791NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	4237
4.792NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	4248
4.793NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	4250
4.794NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	4251
4.795NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{su2n3mix}$	4252
4.796NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	4252
4.797NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	4255
4.798NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	4256
4.799NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	4257
4.800NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4$	4257
4.801NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{e}_6$	4266
4.802NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4$	4267
4.803NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2$	4267
4.804NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	4268
4.805NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$	4268
4.806NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	4269
4.807NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$	4269
4.808NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$	4272
4.809NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	4273
4.810NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su2n3mix}$	4274
4.811NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$	4274
4.812NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$	4275
4.813NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4$	4298
4.814NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$	4298
4.815NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	4299
4.816NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	4304
4.817NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	4305
4.818NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$	4313
4.819NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	4321

4.820NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	4322
4.821NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$	4330
4.822NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$	4330
4.823NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	4330
4.824NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	4330
4.825NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	4336
4.826NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	4337
4.827NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	4337
4.828NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	4340
4.829NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix}$	4341
4.830NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{so}_8$	4356
4.831NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	4357
4.832NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	4364
4.833NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3$	4364
4.834NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$	4366
4.835NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$	4378
4.836NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$	4379
4.837NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	4387
4.838NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	4388
4.839NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	4391
4.840NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix}$	4392
4.841NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$	4394
4.842NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	4396
4.843NHC(-2)(-3) + $\mathfrak{su}_2 + \text{NHC}(-2)(-3)$	4396
4.844NHC(-2)(-3) + $\mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	4399
4.845NHC(-2)(-3) + $\mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	4400
4.846NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8$	4400
4.847NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6$	4455
4.848NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$	4457
4.849NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	4458
4.850NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$	4459
4.851NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4$	4459
4.852NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$	4462
4.853NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	4463
4.854NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix}$	4463
4.855NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$	4463
4.856NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$	4464
4.857NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$	4472
4.858NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	4476
4.859NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix}$	4477
4.860NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$	4477
4.861NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	4480
4.862NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$	4523
4.863NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$	4525
4.864NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	4525
4.865NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	4531
4.866NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	4531
4.867NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	4533
4.868NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix}$	4536
4.869NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{so}_8$	4557
4.870NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	4557
4.871NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix} + \text{su2n3mix}$	4560
4.872NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$	4562
4.873NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$	4603
4.874NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$	4603

4.930NHC(-2)(-3) + su ₂ + su ₂ →(-3) + su ₂ →(-3)	4907
4.931NHC(-2)(-3) + su ₂ + su ₂ →(-3) + su ₂ →(-3) + so ₈	4915
4.932NHC(-2)(-3) + su ₂ + su ₂ →(-3) + su ₂ →(-3) + su ₂	4916
4.933NHC(-2)(-3) + su ₂ + su ₂ →(-3) + su2n3mix	4940
4.934NHC(-2)(-3) + su ₂ + su ₂ →(-3) + su2n3mix + su ₂	4978
4.935NHC(-2)(-3) + su ₂ + su ₂ →(-3) + su ₃	5025
4.936NHC(-2)(-3) + su ₂ + su ₂ →(-3) + su ₃ + su ₂	5026
4.937NHC(-2)(-3) + su ₂ + su ₂ →(-3) + su ₃ + su ₂ + su ₂	5026
4.938NHC(-2)(-3) + su ₂ + su2n3mix	5027
4.939NHC(-2)(-3) + su ₂ + su2n3mix + f ₄	5201
4.940NHC(-2)(-3) + su ₂ + su2n3mix + f ₄ + su ₂	5201
4.941NHC(-2)(-3) + su ₂ + su2n3mix + f ₄ + su ₃	5202
4.942NHC(-2)(-3) + su ₂ + su2n3mix + so ₈	5202
4.943NHC(-2)(-3) + su ₂ + su2n3mix + so ₈ + su ₂ →(-3)	5203
4.944NHC(-2)(-3) + su ₂ + su2n3mix + su ₂	5203
4.945NHC(-2)(-3) + su ₂ + su2n3mix + su ₂ + su ₂ →(-3)	5307
4.946NHC(-2)(-3) + su ₂ + su2n3mix + su ₂ + su2n3mix	5308
4.947NHC(-2)(-3) + su ₂ + su2n3mix + su ₂ →(-3)	5308
4.948NHC(-2)(-3) + su ₂ + su2n3mix + su ₂ →(-3) + su ₂	5346
4.949NHC(-2)(-3) + su ₂ + su2n3mix + su2n3mix	5393
4.950NHC(-2)(-3) + su ₂ + su2n3mix + su2n3mix + su ₂	5427
4.951NHC(-2)(-3) + su ₂ + su ₃	5436
4.952NHC(-2)(-3) + su ₂ + su ₃ + e ₆	5588
4.953NHC(-2)(-3) + su ₂ + su ₃ + e ₆ + NHC(-2)(-3)	5595
4.954NHC(-2)(-3) + su ₂ + su ₃ + e ₆ + so ₈	5595
4.955NHC(-2)(-3) + su ₂ + su ₃ + e ₆ + su ₂	5595
4.956NHC(-2)(-3) + su ₂ + su ₃ + e ₆ + su ₂ + su ₂	5605
4.957NHC(-2)(-3) + su ₂ + su ₃ + e ₆ + su ₂ + su ₃	5608
4.958NHC(-2)(-3) + su ₂ + su ₃ + e ₆ + su ₃	5609
4.959NHC(-2)(-3) + su ₂ + su ₃ + e ₆ + su ₃ + su ₂	5612
4.960NHC(-2)(-3) + su ₂ + su ₃ + e ₆ + su ₃ + su ₃	5613
4.961NHC(-2)(-3) + su ₂ + su ₃ + f ₄	5613
4.962NHC(-2)(-3) + su ₂ + su ₃ + f ₄ + so ₈	5619
4.963NHC(-2)(-3) + su ₂ + su ₃ + f ₄ + su ₂	5620
4.964NHC(-2)(-3) + su ₂ + su ₃ + f ₄ + su ₂ + su ₂ →(-3)	5626
4.965NHC(-2)(-3) + su ₂ + su ₃ + f ₄ + su ₂ →(-3)	5627
4.966NHC(-2)(-3) + su ₂ + su ₃ + f ₄ + su ₂ →(-3) + su ₂	5631
4.967NHC(-2)(-3) + su ₂ + su ₃ + f ₄ + su2n3mix	5631
4.968NHC(-2)(-3) + su ₂ + su ₃ + f ₄ + su ₃	5633
4.969NHC(-2)(-3) + su ₂ + su ₃ + NHC(-2)(-3)	5636
4.970NHC(-2)(-3) + su ₂ + su ₃ + so ₈	5637
4.971NHC(-2)(-3) + su ₂ + su ₃ + so ₈ + e ₆	5673
4.972NHC(-2)(-3) + su ₂ + su ₃ + so ₈ + f ₄	5673
4.973NHC(-2)(-3) + su ₂ + su ₃ + so ₈ + so ₈	5674
4.974NHC(-2)(-3) + su ₂ + su ₃ + so ₈ + su ₂	5676
4.975NHC(-2)(-3) + su ₂ + su ₃ + su ₂	5677
4.976NHC(-2)(-3) + su ₂ + su ₃ + su ₂ + su ₂	5689
4.977NHC(-2)(-3) + su ₂ + su ₃ + su ₂ + su ₂ + su ₂	5695
4.978NHC(-2)(-3) + su ₂ + su ₃ + su ₂ + su ₂ + su ₂ →(-3)	5696
4.979NHC(-2)(-3) + su ₂ + su ₃ + su ₂ + su ₂ →(-3)	5696
4.980NHC(-2)(-3) + su ₂ + su ₃ + su ₂ + su ₂ →(-3) + su ₂	5698
4.981NHC(-2)(-3) + su ₂ + su ₃ + su ₂ →(-3)	5698
4.982NHC(-2)(-3) + su ₂ + su ₃ + su ₂ →(-3) + su ₂	5707
4.983NHC(-2)(-3) + su ₂ + su ₃ + su ₂ →(-3) + su ₂ + su ₂	5712
4.984NHC(-2)(-3) + su ₂ + su ₃ + su2n3mix	5713

4.985	NHC(-2)(-3) + su ₂ + su ₃ + su _{2n3mix} + su ₂	5722
4.986	NHC(-2)(-3) + su ₂ + su ₃ + su ₃	5723
4.987	NHC(-2)(-3) + su ₂ + su ₃ + su ₃ + su ₂	5725
4.988	NHC(-2)(-3) + su ₂ + su ₃ + su ₃ + su ₂ + su ₂	5727
4.989	NHC(-2)(-3) + su ₂ + su ₃ + su ₃ + su ₂ + su ₂ + su ₂	5727
4.990	NHC(-2)(-3) + su ₂ + su ₃ + (-3)→(-2)	5727
4.991	NHC(-2)(-3) + su ₂ + (-3)→(-2)	5729
4.992	NHC(-2)(-3) + su ₂ + (-3)→(-2) + e ₆	5756
4.993	NHC(-2)(-3) + su ₂ + (-3)→(-2) + f ₄	5756
4.994	NHC(-2)(-3) + su ₂ + (-3)→(-2) + so ₈	5756
4.995	NHC(-2)(-3) + su ₂ + (-3)→(-2) + su ₂	5757
4.996	NHC(-2)(-3) + su ₂ + (-3)→(-2) + su ₂ + su ₂ →(-3)	5758
4.997	NHC(-2)(-3) + su ₂ + (-3)→(-2) + su ₂ →(-3)	5758
4.998	NHC(-2)(-3) + su ₂ + (-3)→(-2) + su ₂ →(-3) + su ₂	5760
4.999	NHC(-2)(-3) + su ₂ + (-3)→(-2) + su _{2n3mix}	5760
4.100	NHC(-2)(-3) + su ₂ + (-3)→(-2) + su ₃	5760
4.100	NHC(-2)(-3) + su ₂ + su _{3n2mix}	5762
4.100	NHC(-2)(-3) + su ₃	5766
4.100	NHC(-2)(-3) + su ₃ + e ₆	5851
4.100	NHC(-2)(-3) + su ₃ + e ₆ + NHC(-2)(-3)	5856
4.100	NHC(-2)(-3) + su ₃ + e ₆ + NHC(-2)(-3) + su ₂	5857
4.100	NHC(-2)(-3) + su ₃ + e ₆ + NHC(-2)(-3) + su ₃	5857
4.100	NHC(-2)(-3) + su ₃ + e ₆ + so ₈	5857
4.100	NHC(-2)(-3) + su ₃ + e ₆ + so ₈ + su ₂	5858
4.100	NHC(-2)(-3) + su ₃ + e ₆ + su ₂	5859
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + NHC(-2)(-3)	5877
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + so ₈	5877
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + su ₂	5878
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + su ₂ + su ₂	5890
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + su ₂ + su ₃	5893
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + su ₂ →(-3)	5894
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + su _{2n3mix}	5896
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + su ₃	5896
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + su ₃ + su ₂	5900
4.101	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ + su ₃ + su ₃	5901
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ →(-3)	5902
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ →(-3) + su ₂	5903
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su ₂ →(-3) + su ₃	5905
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su _{2n3mix}	5905
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su _{2n3mix} + su ₂	5907
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su ₃	5907
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su ₃ + NHC(-2)(-3)	5909
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su ₃ + su ₂	5910
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su ₃ + su ₂ + su ₂	5914
4.102	NHC(-2)(-3) + su ₃ + e ₆ + su ₃ + su ₂ + su ₃	5915
4.103	NHC(-2)(-3) + su ₃ + e ₆ + su ₃ + su ₂ →(-3)	5915
4.103	NHC(-2)(-3) + su ₃ + e ₆ + su ₃ + su ₃	5916
4.103	NHC(-2)(-3) + su ₃ + e ₆ + su ₃ + su ₃ + su ₂	5916
4.103	NHC(-2)(-3) + su ₃ + e ₇	5917
4.103	NHC(-2)(-3) + su ₃ + e ₇ + NHC(-2)(-3)	5917
4.103	NHC(-2)(-3) + su ₃ + e ₇ + so ₈	5918
4.103	NHC(-2)(-3) + su ₃ + e ₇ + su ₂	5918
4.103	NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂	5921
4.103	NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₂	5922
4.103	NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₂ →(-3)	5923

4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₃	5923
4.104	NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ → (-3)	5923
4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ → (-3) + su ₂	5924
4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ n3mix	5924
4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₃	5924
4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₃ + su ₂	5925
4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3)	5925
4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₂	5925
4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₂ + su ₂	5926
4.104	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₃	5926
4.105	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ n3mix	5926
4.105	NHC(-2)(-3) + su ₃ + e ₇ + su ₂ n3mix + su ₂	5927
4.105	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃	5927
4.105	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂	5928
4.105	NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂ + su ₂	5928
4.105	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂ → (-3)	5929
4.105	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₃	5929
4.105	N NHC(-2)(-3) + su ₃ + e ₇	5929
4.105	N NHC(-2)(-3) + su ₃ + e ₇ + NHC(-2)(-3)	5930
4.105	N NHC(-2)(-3) + su ₃ + e ₇ + so ₈	5930
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂	5930
4.106	NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂	5933
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₂	5934
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₂ → (-3)	5934
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₃	5935
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ → (-3)	5935
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ → (-3) + su ₂	5936
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ n3mix	5936
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₃	5936
4.106	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₃ + su ₂	5937
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3)	5937
4.107	NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₂	5937
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₂ + su ₂	5938
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₃	5938
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ n3mix	5938
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ n3mix + su ₂	5939
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃	5939
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂	5940
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂ + su ₂	5940
4.107	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂ → (-3)	5940
4.108	N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₃	5941
4.108	NHC(-2)(-3) + su ₃ + f ₄	5941
4.108	N NHC(-2)(-3) + su ₃ + f ₄ + f ₄	5947
4.108	N NHC(-2)(-3) + su ₃ + f ₄ + NHC(-2)(-3)	5947
4.108	N NHC(-2)(-3) + su ₃ + f ₄ + so ₈	5948
4.108	N NHC(-2)(-3) + su ₃ + f ₄ + so ₈ + su ₂	5949
4.108	N NHC(-2)(-3) + su ₃ + f ₄ + so ₈ + su ₂ → (-3)	5950
4.108	N NHC(-2)(-3) + su ₃ + f ₄ + su ₂	5950
4.108	N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + so ₈	5975
4.108	N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + su ₂	59

4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6577
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4$	6577
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$	6578
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6581
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix}$	6581
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$	6582
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$	6582
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$	6587
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$	6588
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	6589
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6593
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	6594
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	6595
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	6595
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$	6600
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	6600
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \text{su3n2mix}$	6601
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2)$	6601
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8$	6606
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	6606
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	6608
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + \text{su3n2mix}$	6609
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + \text{su3n2mix} + \mathfrak{su}_3$	6610

5 NHC (-2)(-3)(-2) External (163 combos, 2345 bases)

6611

5.1 NHC(-2)(-3)(-2)	6611
5.2 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6612
5.3 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6612
5.4 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$	6612
5.5 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6613
5.6 NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$	6614
5.7 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$	6614
5.8 NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$	6614
5.9 NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2$	6614
5.10 NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3$	6615
5.11 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)(-2)$	6615
5.12 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)(-2)$	6615
5.13 NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-3)(-2)$	6615
5.14 NHC(-2)(-3) + $\mathfrak{e}_7' + \text{NHC}(-2)(-3)(-2)$	6616
5.15 NHC(-2)(-3) + $\mathfrak{e}_8 + \text{NHC}(-2)(-3)(-2)$	6616
5.16 NHC(-2)(-3) + $\mathfrak{f}_4 + \text{NHC}(-2)(-3)(-2)$	6616
5.17 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$	6617
5.18 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$	6617
5.19 NHC(-2)(-3)(-2)	6617
5.20 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$	6634
5.21 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{e}_6$	6636
5.22 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{f}_4$	6636
5.23 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	6636
5.24 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)$	6636
5.25 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	6637
5.26 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	6637
5.27 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{so}_8$	6637
5.28 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$	6638
5.29 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$	6638
5.30 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{su}_2$	6638

5.31	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	6644
5.32	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$	6644
5.33	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	6645
5.34	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	6653
5.35	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6654
5.36	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6654
5.37	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	6658
5.38	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	6658
5.39	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$	6658
5.40	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6659
5.41	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	6659
5.42	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \text{su2n3mix}$	6659
5.43	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3$	6659
5.44	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$	6661
5.45	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$	6661
5.46	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6661
5.47	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6665
5.48	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	6665
5.49	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	6666
5.50	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	6666
5.51	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	6667
5.52	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4$	6667
5.53	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{e}_6$	6668
5.54	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{f}_4$	6669
5.55	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$	6669
5.56	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8$	6669
5.57	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$	6670
5.58	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	6671
5.59	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$	6671
5.60	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2$	6671
5.61	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$	6678
5.62	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$	6679
5.63	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	6685
5.64	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6686
5.65	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	6687
5.66	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$	6690
5.67	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	6693
5.68	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6693
5.69	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	6694
5.70	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6694
5.71	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	6696
5.72	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{su2n3mix}$	6696
5.73	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	6698
5.74	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3$	6701
5.75	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3$	6702
5.76	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$	6704
5.77	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$	6704
5.78	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6707
5.79	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	6708
5.80	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix}$	6708
5.81	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$	6708
5.82	$\text{NHC}(-2)(-3)(-2) + \text{NHC}(-2)(-3)(-2)$	6709
5.83	$\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8$	6709
5.84	$\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{e}_6$	6714
5.85	$\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$	6714

5.141NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6855
5.142NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$	6855
5.143NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6856
5.144NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	6856
5.145NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4$	6856
5.146NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$	6857
5.147NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$	6857
5.148NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$	6860
5.149NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6860
5.150NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix}$	6861
5.151NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$	6861
5.152NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{so}_8$	6862
5.153NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6$	6864
5.154NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$	6864
5.155NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$	6865
5.156NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2$	6865
5.157NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6867
5.158NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6867
5.159NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6868
5.160NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	6868
5.161NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6868
5.162NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	6869
5.163NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \text{su2n3mix}$	6869

6 NHC (-2)(-2)(-3) External (186 combos, 2947 bases)

6870

6.1 NHC(-2)(-2)(-3)	6870
6.2 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2$	6871
6.3 NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6871
6.4 NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	6872
6.5 NHC(-2)(-2)(-3)	6872
6.6 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$	6916
6.7 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{e}_6$	6918
6.8 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{f}_4$	6918
6.9 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	6919
6.10 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)$	6919
6.11 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	6920
6.12 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	6920
6.13 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8$	6921
6.14 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$	6922
6.15 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	6923
6.16 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	6924
6.17 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$	6924
6.18 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2$	6925
6.19 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	6934
6.20 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$	6934
6.21 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	6936
6.22 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	6936
6.23 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	6950
6.24 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	6950
6.25 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6953
6.26 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix}$	6954
6.27 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6954
6.28 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	6961
6.29 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	6964
6.30 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$	6965

6.31	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	6966
6.32	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6966
6.33	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	6966
6.34	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	6967
6.35	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix}$	6967
6.36	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_2$	6968
6.37	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_3$	6968
6.38	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3$	6968
6.39	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$	6972
6.40	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$	6972
6.41	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6972
6.42	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6980
6.43	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	6983
6.44	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	6984
6.45	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{su2n3mix}$	6984
6.46	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	6984
6.47	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	6985
6.48	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	6986
6.49	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	6987
6.50	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_7$	6987
6.51	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}'_7$	6987
6.52	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4$	6987
6.53	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{e}_6$	6989
6.54	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4$	6990
6.55	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2$	6990
6.56	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6990
6.57	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$	6990
6.58	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8$	6990
6.59	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$	6992
6.60	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	6993
6.61	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su2n3mix}$	6994
6.62	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$	6994
6.63	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2$	6995
6.64	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4$	7004
6.65	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$	7004
6.66	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	7006
6.67	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	7009
6.68	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	7009
6.69	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_3$	7014
6.70	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$	7015
6.71	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	7018
6.72	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{su2n3mix}$	7018
6.73	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	7019
6.74	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$	7020
6.75	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	7020
6.76	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	7020
6.77	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	7023
6.78	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	7024
6.79	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	7025
6.80	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix}$	7026
6.81	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{so}_8$	7031
6.82	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	7031
6.83	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_3$	7036
6.84	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3$	7036
6.85	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3 + \mathfrak{su}_2$	7038

6.86	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3$	7038
6.87	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$	7043
6.88	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$	7043
6.89	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	7046
6.90	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{su2n3mix}$	7047
6.91	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	7047
6.92	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	7048
6.93	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix}$	7049
6.94	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	7050
6.95	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$	7051
6.96	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	7052
6.97	$\text{NHC}(-2)(-2)(-3) + \text{NHC}(-2)(-3)$	7052
6.98	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8$	7052
6.99	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6$	7059
6.100	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$	7059
6.101	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	7061
6.102	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$	7061
6.103	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4$	7062
6.104	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$	7062
6.105	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	7064
6.106	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix}$	7064
6.107	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$	7064
6.108	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8$	7065
6.109	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$	7066
6.110	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	7069
6.111	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix}$	7070
6.112	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$	7070
6.113	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2$	7072
6.114	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$	7085
6.115	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	7086
6.116	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	7088
6.117	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	7088
6.118	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	7088
6.119	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix}$	7089
6.120	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{so}_8$	7095
6.121	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	7095
6.122	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \text{su2n3mix}$	7096
6.123	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3$	7097
6.124	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$	7104
6.125	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$	7104
6.126	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$	7104
6.127	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	7105
6.128	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7106
6.129	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8$	7123
6.130	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$	7125
6.131	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	7126
6.132	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix}$	7126
6.133	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	7127
6.134	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	7127
6.135	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3$	7127
6.136	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6$	7144
6.137	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$	7145
6.138	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$	7146
6.139	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$	7146
6.140	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	7153

6.141NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	7155
6.142NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$	7156
6.143NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	7157
6.144NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	7158
6.145NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4$	7158
6.146NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$	7160
6.147NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$	7160
6.148NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	7163
6.149NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	7164
6.150NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	7164
6.151NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	7165
6.152NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix}$	7166
6.153NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	7167
6.154NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$	7167
6.155NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \text{NHC}(-2)(-3)$	7168
6.156NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8$	7169
6.157NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6$	7174
6.158NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$	7174
6.159NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$	7175
6.160NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2$	7176
6.161NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	7178
6.162NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	7178
6.163NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	7178
6.164NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \text{su2n3mix}$	7182
6.165NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3$	7184
6.166NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2)$	7184
6.167NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	7184
6.168NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	7185
6.169NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7186
6.170NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3$	7187
6.171NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$	7187
6.172NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-2)(-3)$	7187
6.173NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-2)(-3)$	7188
6.174NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7189
6.175NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$	7189
6.176NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-2)(-3)$	7189
6.177NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7190
6.178NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$	7190
6.179NHC(-2)(-3) + $\mathfrak{e}_8 + \text{NHC}(-2)(-2)(-3)$	7190
6.180NHC(-2)(-3) + $\mathfrak{e}_8 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7191
6.181NHC(-2)(-3) + $\mathfrak{e}_8 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$	7191
6.182NHC(-2)(-3) + $\mathfrak{f}_4 + \text{NHC}(-2)(-2)(-3)$	7191
6.183NHC(-2)(-3) + $\text{NHC}(-2)(-2)(-3)$	7191
6.184NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	7191
6.185NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	7192
6.186NHC(-2)(-3) + $\mathfrak{su}_3 + \text{NHC}(-2)(-2)(-3)$	7192

7 NHC (-2)(-4) External (1 combos, 1 bases)

7193

7.1 nhc ₂₄	7193
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1 $\hat{1}$ External (87 combos, 16930 bases)

1.1 $\mathbf{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^1 1^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 184, \Delta = -291, \det = -6, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 184, T \in [13, 184]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^1 21^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 173, \Delta = -262, \det = 5, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 173, T \in [13, 173]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^1 221^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 101, \Delta = -233, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 101, T \in [13, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 1^1 2221^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -204, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \quad 1^1 22221^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -175, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^1 23131^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 101, \Delta = -192, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 3 & 3 \\ 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 3 \quad 21^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 101, \Delta = -186, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 2 & 2 \\ 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [8] \quad 1^1 231321^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 223131^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 101, \Delta = -163, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 4 & 4 \\ 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \quad 1^1 3141321^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [11] \quad 1^1 23215131^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 17, T_{\max} = 102, T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 4 & 4 \\ 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1 \overset{1}{5} \overset{1}{131} \overset{1}{+ \mathbf{e}_6 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 102, \Delta = -157, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 17, T_{\max} = 102, T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [12] \ 1 \overset{1}{6} \overset{1}{131} \overset{1}{+ \mathbf{e}_6 + \hat{1} \rightarrow 1} \quad T = 11, T_{\max} = 102, \Delta = -156, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18, T_{\max} = 102, T \in [18, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

1.2 $\mathbf{e}_7 + \hat{1} \rightarrow (-1)$

$$T = 3 \quad [0] \ 1 \overset{1}{1} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 3, T_{\max} = 190, \Delta = -346, \det = -8, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 190, T \in [15, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & -1 & -1 \\ 2 & -5 & -1 & -5 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \ 1 \overset{1}{21} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 4, T_{\max} = 188, \Delta = -317, \det = 7, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 188, T \in [15, 188]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 2 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \ 1 \overset{1}{221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 5, T_{\max} = 186, \Delta = -288, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 186, T \in [15, 186]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \ 1 \overset{1}{2221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 6, T_{\max} = 170, \Delta = -259, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 170, T \in [15, 170]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \ 1 \overset{1}{22221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 7, T_{\max} = 102, \Delta = -230, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1 \overset{1}{3} \overset{1}{21} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 7, T_{\max} = 114, \Delta = -241, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 114, T \in [16, 114]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 1 & -3 \\ 1 & 1 & 2 & 2 \\ 2 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \ 1 \overset{1}{222221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 8, T_{\max} = 102, \Delta = -201, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1 \overset{1}{231321} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 8, T_{\max} = 103, \Delta = -219, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1 \overset{1}{2222221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^1 2231321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 9, T_{\max} = 103, \Delta = -190, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 103, T \in [17, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[10] $1^1 2231321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 9, T_{\max} = 103, \Delta = -190, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 103, T \in [17, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [11] $1^1 22313221^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 10, T_{\max} = 102, \Delta = -161, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 17, T_{\max} = 102, T \in [17, 102]$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[12] $1^1 23141321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 10, T_{\max} = 104, \Delta = -189, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 17.33 \rightarrow 18, T_{\max} = 104, T \in [18, 104]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [13] $1^1 231413221^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 103, \Delta = -160, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 18, T_{\max} = 103, T \in [18, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^1 231413221^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 103, \Delta = -160, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 18, T_{\max} = 103, T \in [18, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^1 231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 105, \Delta = -184, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 18.33 \rightarrow 19, T_{\max} = 105, T \in [19, 105]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 1 & -2 \\ 1 & 1 & 3 & 3 \\ 2 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

[17] $1^1 231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 105, \Delta = -186, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 18.33 \rightarrow 19, T_{\max} = 105, T \in [19, 105]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

[16] $1^1 231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 105, \Delta = -186, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 18.33 \rightarrow 19, T_{\max} = 105, T \in [19, 105]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 12$ [22] $1^1 2231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 12, T_{\max} = 104, \Delta = -157, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 19, T_{\max} = 104, T \in [19, 104]$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^1 2314141321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 12, T_{\max} = 104, \Delta = -159, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 19, T_{\max} = 104, T \in [19, 104]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

[20] $1^1 2231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 12, T_{\max} = 104, \Delta = -157, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 19, T_{\max} = 104, T \in [19, 104]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\hat{1}2231} \ 1^{\hat{1}} \ 5 \ 1321^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 12, \ T_{\max} = 103, \ \Delta = -155, \ \det = 2, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 19, \ T_{\max} = 103, \ T \in [19, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 1 & -2 \\ 1 & 1 & 4 & 4 \\ 2 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\hat{1}2231} \ 1^{\hat{1}} \ 5 \ 1321^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 12, \ T_{\max} = 103, \ \Delta = -155, \ \det = 2, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 19, \ T_{\max} = 103, \ T \in [19, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\hat{1}2321612321} \ 1^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 12, \ T_{\max} = 106, \ \Delta = -185, \ \det = 3, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 19.33 \rightarrow 20, \ T_{\max} = 106, \ T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \ 1^{\hat{1}2321} \ 1^{\hat{1}} \ 7 \ 12321^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 13, \ T_{\max} = 107, \ \Delta = -183, \ \det = -3, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 20.33 \rightarrow 21, \ T_{\max} = 107, \ T \in [21, 107]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [1] \ 1^{\hat{1}231513151321} \ 1^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 14, \ T_{\max} = 106, \ \Delta = -157, \ \det = 2, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 21, \ T_{\max} = 106, \ T \in [21, 106]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\hat{1}2321} \ 1^{\hat{1}} \ 8 \ 12321^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 14, \ T_{\max} = 107, \ \Delta = -182, \ \det = 3, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 21.33 \rightarrow 22, \ T_{\max} = 107, \ T \in [22, 107]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 1 & -2 \\ 1 & 1 & 3 & 3 \\ 2 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 21^{\hat{1}2321} \ 1^{\hat{1}} \ 8 \ 12321^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 14, \ T_{\max} = 107, \ \Delta = -182, \ \det = 3, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 21.33 \rightarrow 22, \ T_{\max} = 107, \ T \in [22, 107]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [5] \ 1^{\hat{1}2321613151321} \ 1^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 15, \ T_{\max} = 106, \ \Delta = -156, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 22, \ T_{\max} = 106, \ T \in [22, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\hat{1}2321613151321} \ 1^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 15, \ T_{\max} = 106, \ \Delta = -156, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 22, \ T_{\max} = 106, \ T \in [22, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [7] \ 1^{\hat{1}23216131612321} \ 1^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 16, \ T_{\max} = 107, \ \Delta = -155, \ \det = 2, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 23, \ T_{\max} = 107, \ T \in [23, 107]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\hat{1}23217123151321} \ 1^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 16, \ T_{\max} = 107, \ \Delta = -155, \ \det = 2, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 23, \ T_{\max} = 107, \ T \in [23, 107]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\hat{1}23217123151321} \ 1^{\hat{1}} + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 16, \ T_{\max} = 107, \ \Delta = -155, \ \det = 2, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 23, \ T_{\max} = 107, \ T \in [23, 107]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [9] \quad 1^{1232181223151321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 17, \quad T_{\max} = 108, \quad \Delta = -154, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 24, \quad T_{\max} = 108, \quad T \in [24, 108]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1^{123151321}} 8^{12321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 17, \quad T_{\max} = 108, \quad \Delta = -154, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 24, \quad T_{\max} = 108, \quad T \in [24, 108]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1232181223151321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 17, \quad T_{\max} = 108, \quad \Delta = -154, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 24, \quad T_{\max} = 108, \quad T \in [24, 108]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1^{123151321}} 8^{12321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 17, \quad T_{\max} = 108, \quad \Delta = -154, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 24, \quad T_{\max} = 108, \quad T \in [24, 108]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [13] \quad 1^{12321712321712321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 18, \quad T_{\max} = 109, \quad \Delta = -154, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 25, \quad T_{\max} = 109, \quad T \in [25, 109]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [14] \quad 1^{1^{12321}} 8^{12321712321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 19, \quad T_{\max} = 110, \quad \Delta = -153, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 26, \quad T_{\max} = 110, \quad T \in [26, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 4 & 4 \\ 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1^{12321}} 8^{12321712321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 19, \quad T_{\max} = 110, \quad \Delta = -153, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 26, \quad T_{\max} = 110, \quad T \in [26, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [16] \quad 1^{1^{12321}} 8^{12321812321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 20, \quad T_{\max} = 111, \quad \Delta = -152, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 27, \quad T_{\max} = 111, \quad T \in [27, 111]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 22 \quad [17] \quad 1^{123218123151321812321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 22, \quad T_{\max} = 113, \quad \Delta = -153, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 22, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 29, \quad T_{\max} = 113, \quad T \in [29, 113]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [0] \quad 1^{12321812321712321812321^1} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 24, \quad T_{\max} = 115, \quad \Delta = -152, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 31, \quad T_{\max} = 115, \quad T \in [31, 115]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 25 \quad [1] \quad 1^{1^{2321812321}} \quad 12321812321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 25, \quad T_{\max} = 116, \quad \Delta = -151, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 32, \quad T_{\max} = 116, \quad T \in [32, 116]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

1.3 $\mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^{1^1} + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 3, \quad T_{\max} = 187, \quad \Delta = -318, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 3, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.57 \rightarrow 14, \quad T_{\max} = 187, \quad T \in [14, 187]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 2 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^{1^1} 21^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 4, \quad T_{\max} = 185, \quad \Delta = -289, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 4, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.67 \rightarrow 14, \quad T_{\max} = 185, \quad T \in [14, 185]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{1^1} 221^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 5, \quad T_{\max} = 171, \quad \Delta = -260, \quad \det = -5, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.80 \rightarrow 14, \quad T_{\max} = 171, \quad T \in [14, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 1^{1^1} 2221^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 6, \quad T_{\max} = 101, \quad \Delta = -231, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \quad 1^{1^1} 22221^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 7, \quad T_{\max} = 101, \quad \Delta = -202, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 3^1 21^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 7, \quad T_{\max} = 102, \quad \Delta = -213, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 1 & -2 \\ 1 & 1 & 2 & 2 \\ 2 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{1^1} 222221^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -173, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1^1} 231321^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 8, \quad T_{\max} = 102, \quad \Delta = -191, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.33 \rightarrow 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 1^{1^1} 2231321^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -162, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & 5 & 5 \\ 2 & -5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1^1} 2231321^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -162, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [10] \quad 1^{123141321^1} + \mathbf{c}'_7 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 10, T_{\max} = 102, \Delta = -161, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 17, T_{\max} = 102, T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \quad 1^{1231} \quad 1^5 \quad 1321^1 + \mathbf{c}'_7 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 102, \Delta = -156, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 102, T \in [18, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1231512321^1} + \mathbf{c}'_7 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 103, \Delta = -158, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 103, T \in [18, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & 5 & 5 \\ 2 & -5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1231512321^1} + \mathbf{c}'_7 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 103, \Delta = -158, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 103, T \in [18, 103]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [14] \quad 1^{12321612321^1} + \mathbf{c}'_7 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 12, T_{\max} = 104, \Delta = -157, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 19, T_{\max} = 104, T \in [19, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \quad 1^{12321} \quad 1^7 \quad 12321^1 + \mathbf{c}'_7 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 13, T_{\max} = 104, \Delta = -155, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 20, T_{\max} = 104, T \in [20, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [2] \quad 1^{12321} \quad 1^8 \quad 12321^1 + \mathbf{c}'_7 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 14, T_{\max} = 105, \Delta = -154, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 21, T_{\max} = 105, T \in [21, 105]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^{12321} \quad 1^8 \quad 12321^1 + \mathbf{c}'_7 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 14, T_{\max} = 105, \Delta = -154, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 21, T_{\max} = 105, T \in [21, 105]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

1.4 $\mathbf{e}_8 + \hat{\mathbf{1}} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^{11^1} + \mathbf{e}_8 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 3, T_{\max} = 193, \Delta = -461, \det = -12, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.33 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -11 & 0 & -9 \\ 1 & 0 & -1 & -1 \\ 2 & -9 & -1 & -9 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^{121^1} + \mathbf{e}_8 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 4, T_{\max} = 193, \Delta = -432, \det = 11, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.36 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -10 & 0 & -8 \\ 1 & 0 & -1 & -1 \\ 2 & -8 & -1 & -8 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 5 \quad [2] \quad 1^1 221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 5, T_{\max} = 193, \Delta = -403, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.40 \rightarrow 19, T_{\max} = 193, T \in [19, 193] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & -1 & -1 \\ 2 & -7 & -1 & -7 \end{pmatrix}_{b_0} \\
\\
& T = 6 \quad [3] \quad 1^1 2221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 6, T_{\max} = 193, \Delta = -374, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.44 \rightarrow 19, T_{\max} = 193, T \in [19, 193] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & -1 & -1 \\ 2 & -6 & -1 & -6 \end{pmatrix}_{b_0} \\
\\
& T = 7 \quad [4] \quad 1^1 22221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 7, T_{\max} = 193, \Delta = -345, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.50 \rightarrow 19, T_{\max} = 193, T \in [19, 193] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & -1 & -1 \\ 2 & -5 & -1 & -5 \end{pmatrix}_{b_0} \\
\\
& T = 8 \quad [5] \quad 1^1 222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 8, T_{\max} = 192, \Delta = -316, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 192, T \in [19, 192] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 2 & -4 & -1 & -4 \end{pmatrix}_{b_0} \\
\\
& T = 9 \quad [6] \quad 1^1 2222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 9, T_{\max} = 190, \Delta = -287, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.67 \rightarrow 19, T_{\max} = 190, T \in [19, 190] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0} \\
\\
& [7] \quad 1^1 2231321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 9, T_{\max} = 192, \Delta = -305, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.57 \rightarrow 20, T_{\max} = 192, T \in [20, 192] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 2 & -4 & -1 & -4 \end{pmatrix}_{b_0} \\
\\
& T = 10 \quad [8] \quad 1^1 22222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 10, T_{\max} = 174, \Delta = -258, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.80 \rightarrow 19, T_{\max} = 174, T \in [19, 174] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0} \\
\\
& [9] \quad 1^1 22313221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 10, T_{\max} = 190, \Delta = -276, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.67 \rightarrow 20, T_{\max} = 190, T \in [20, 190] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0} \\
\\
& T = 11 \quad [10] \quad 1^1 222222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 11, T_{\max} = 106, \Delta = -229, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19, T_{\max} = 106, T \in [19, 106] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0} \\
\\
& [11] \quad 1^1 231413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 11, T_{\max} = 191, \Delta = -275, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.67 \rightarrow 21, T_{\max} = 191, T \in [21, 191] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 3 & 3 \\ 2 & -7 & 3 & -3 \end{pmatrix}_{b_0} \\
\\
& T = 12 \quad [12] \quad 1^1 2222222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 12, T_{\max} = 106, \Delta = -200, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.33 \rightarrow 20, T_{\max} = 106, T \in [20, 106] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0} \\
\\
& [13] \quad 1^1 2231413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 12, T_{\max} = 152, \Delta = -246, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 20.80 \rightarrow 21, T_{\max} = 152, T \in [21, 152]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & 2 & 2 \\ 2 & -5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1^1 2231} 1321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 12, T_{\max} = 187, \Delta = -270, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21.67 \rightarrow 22, T_{\max} = 187, T \in [22, 187]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 1 & -5 \\ 1 & 1 & 3 & 3 \\ 2 & -5 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1^1 2231512321} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 12, T_{\max} = 192, \Delta = -272, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21.67 \rightarrow 22, T_{\max} = 192, T \in [22, 192]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \ 1^{1^1 22222222221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 13, T_{\max} = 106, \Delta = -171, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1^1 2231} 13221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 13, T_{\max} = 120, \Delta = -241, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21.80 \rightarrow 22, T_{\max} = 120, T \in [22, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 1 & -5 \\ 1 & 1 & 4 & 4 \\ 2 & -5 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1^1 23141413221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 13, T_{\max} = 147, \Delta = -245, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21.80 \rightarrow 22, T_{\max} = 147, T \in [22, 147]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 4 & 4 \\ 2 & -7 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [3] \ 1^{1^1 223141413221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 14, T_{\max} = 109, \Delta = -216, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22, T_{\max} = 109, T \in [22, 109]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 2 & 2 \\ 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [4] \ 1^{1^1 2314141413221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 15, T_{\max} = 110, \Delta = -215, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23, T_{\max} = 110, T \in [23, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 5 & 5 \\ 2 & -7 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1^1 2231513151321} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 15, T_{\max} = 138, \Delta = -243, \det = -5, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.80 \rightarrow 24, T_{\max} = 138, T \in [24, 138]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [6] \ 1^{1^1 22314141413221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 16, T_{\max} = 110, \Delta = -186, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.33 \rightarrow 24, T_{\max} = 110, T \in [24, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 2 & 2 \\ 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1^1 22315131513221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 16, T_{\max} = 111, \Delta = -214, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24, T_{\max} = 111, T \in [24, 111]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1^1 23216131513221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 16, T_{\max} = 134, \Delta = -242, \det = 5, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.80 \rightarrow 25, T_{\max} = 134, T \in [25, 134]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 3 & 3 \\ 2 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [9] \quad 1^1 23141414141413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 17, \quad T_{\max} = 111, \quad \Delta = -185, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.33 \rightarrow 25, \quad T_{\max} = 111, \quad T \in [25, 111]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 6 & 6 \\ 2 & -7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 232171231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 17, \quad T_{\max} = 124, \quad \Delta = -241, \quad \det = -5, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.80 \rightarrow 26, \quad T_{\max} = 124, \quad T \in [26, 124]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 3 & 3 \\ 2 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [11] \quad 1^1 223141414141413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 18, \quad T_{\max} = 109, \quad \Delta = -156, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 25, \quad T_{\max} = 109, \quad T \in [25, 109]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^1 2321812231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 18, \quad T_{\max} = 120, \quad \Delta = -240, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.80 \rightarrow 27, \quad T_{\max} = 120, \quad T \in [27, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 3 & 3 \\ 2 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^1 223151321 \quad 12321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 18, \quad T_{\max} = 120, \quad \Delta = -240, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.80 \rightarrow 27, \quad T_{\max} = 120, \quad T \in [27, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 3 & 3 \\ 2 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [14] \quad 1^1 2314141414141413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 19, \quad T_{\max} = 110, \quad \Delta = -155, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 26, \quad T_{\max} = 110, \quad T \in [26, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 7 & 7 \\ 2 & -7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 23151316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 19, \quad T_{\max} = 114, \quad \Delta = -213, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 27, \quad T_{\max} = 114, \quad T \in [27, 114]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [16] \quad 1^1 223151316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 20, \quad T_{\max} = 114, \quad \Delta = -184, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.33 \rightarrow 28, \quad T_{\max} = 114, \quad T \in [28, 114]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 3 & 3 \\ 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^1 232161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 20, \quad T_{\max} = 115, \quad \Delta = -212, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, \quad T_{\max} = 115, \quad T \in [28, 115]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [18] \quad 1^1 2315132181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 21, \quad T_{\max} = 116, \quad \Delta = -212, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 29, \quad T_{\max} = 116, \quad T \in [29, 116]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 22 \quad [19] \quad 1^1 22315132181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 22, T_{\max} = 116, \Delta = -183, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 29.33 \rightarrow 30, T_{\max} = 116, T \in [30, 116]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 3 & 3 \\ 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 23 \quad [0] \quad 1^1 231513161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 23, T_{\max} = 117, \Delta = -183, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.33 \rightarrow 31, T_{\max} = 117, T \in [31, 117]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 232171232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 23, T_{\max} = 118, \Delta = -211, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31, T_{\max} = 118, T \in [31, 118]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [2] \quad 1^1 2231513161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 24, T_{\max} = 115, \Delta = -154, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31, T_{\max} = 115, T \in [31, 115]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 2321613161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 24, T_{\max} = 117, \Delta = -182, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.33 \rightarrow 32, T_{\max} = 117, T \in [32, 117]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 2321 \quad 8 \quad 1232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 24, T_{\max} = 118, \Delta = -210, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32, T_{\max} = 118, T \in [32, 118]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 25 \quad [5] \quad 21 \quad \begin{matrix} 1^1 2231513221 \\ (12) \end{matrix} \quad 1223151321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 25, T_{\max} = 120, \Delta = -211, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 120, T \in [33, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1 \quad \begin{matrix} 1 \\ (12) \end{matrix} \quad 1223151321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 25, T_{\max} = 120, \Delta = -211, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 120, T \in [33, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 1 & -4 \\ 1 & 1 & 4 & 4 \\ 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 26 \quad [7] \quad 1 \quad \begin{matrix} 1 \\ (12) \end{matrix} \quad 12231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 26, T_{\max} = 119, \Delta = -182, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.33 \rightarrow 34, T_{\max} = 119, T \in [34, 119]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 1 & -4 \\ 1 & 1 & 5 & 5 \\ 2 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 21 \quad \begin{matrix} 1^1 2231513221 \\ (12) \end{matrix} \quad 12231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 26, T_{\max} = 119, \Delta = -182, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.33 \rightarrow 34, T_{\max} = 119, T \in [34, 119]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 3 & 3 \\ 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 27 \quad [9] \quad 1^1 2315131613161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 27, T_{\max} = 118, \Delta = -153, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 34, T_{\max} = 118, T \in [34, 118]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 2315132181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 27, T_{\max} = 120, \Delta = -182, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 34.33 \rightarrow 35, T_{\max} = 120, T \in [35, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 28 \quad [12] \quad 1^1 22315132181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 28, T_{\max} = 119, \Delta = -153, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35, T_{\max} = 119, T \in [35, 119]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^1 23216131613161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 28, T_{\max} = 119, \Delta = -152, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35, T_{\max} = 119, T \in [35, 119]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 29 \quad [13] \quad 1^1 232171232181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 29, T_{\max} = 122, \Delta = -181, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.33 \rightarrow 37, T_{\max} = 122, T \in [37, 122]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 30 \quad [14] \quad 1^1 2321 \quad 8 \quad 1232181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 30, T_{\max} = 123, \Delta = -180, \det = 3, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.33 \rightarrow 38, T_{\max} = 123, T \in [38, 123]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 2231513221 \quad (12) \quad 1223151321812321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 30, T_{\max} = 124, \Delta = -210, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38, T_{\max} = 124, T \in [38, 124]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 1 & 1 \\ 2 & -3 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 33 \quad [0] \quad 1^1 2315132181232181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 33, T_{\max} = 124, \Delta = -152, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40, T_{\max} = 124, T \in [40, 124]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 35 \quad [1] \quad 1^1 232171232181232181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 35, T_{\max} = 126, \Delta = -151, \det = -2, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42, T_{\max} = 126, T \in [42, 126]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 232181232181231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 35, T_{\max} = 129, \Delta = -209, \det = -4, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43, T_{\max} = 129, T \in [43, 129]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 36 \quad [3] \quad 1 \quad \overset{1}{8} \quad 1232181232181232181231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 36, T_{\max} = 127, \Delta = -150, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43, T_{\max} = 127, T \in [43, 127]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 37 \quad [4] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 37, T_{\max} = 130, \Delta = -181, \det = -3, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.33 \rightarrow 45, T_{\max} = 130, T \in [45, 130]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 1 & -4 \\ 1 & 1 & 5 & 5 \\ 2 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 38 \quad [5] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 38, T_{\max} = 129, \Delta = -152, \det = 2, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45, T_{\max} = 129, T \in [45, 129]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 1 & -3 \\ 1 & 1 & 5 & 5 \\ 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$T = 42 \quad [6] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)1223151321812321^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 42, T_{\max} = 135, \Delta = -180, \det = 3, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.33 \rightarrow 50, T_{\max} = 135, T \in [50, 135]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 48 \quad [0] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)1223151321^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 48, T_{\max} = 141, \Delta = -180, \det = 3, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.33 \rightarrow 56, T_{\max} = 141, T \in [56, 141]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & 4 & 4 \\ 2 & -5 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 49 \quad [1] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 49, T_{\max} = 140, \Delta = -151, \det = -2, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 140, T \in [56, 140]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 1 & -4 \\ 1 & 1 & 6 & 6 \\ 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 49, T_{\max} = 140, \Delta = -151, \det = -2, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 140, T \in [56, 140]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 4 & 4 \\ 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 54 \quad [0] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221(12)1223151321812321^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 54, T_{\max} = 145, \Delta = -150, \det = 2, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 61, T_{\max} = 145, T \in [61, 145]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

1.5 $\mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^1 1^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 3, T_{\max} = 174, \Delta = -265, \det = -5, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 174, T \in [12, 174]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^1 21^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 4, T_{\max} = 106, \Delta = -236, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 106, T \in [12, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^1 221^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -207, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 1^1 2221^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -178, \det = 2, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 1321^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -187, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 2 & 2 \\ 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 2231^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -187, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 1 & -1 \\ 2 & 1 & 2 & 2 \\ 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [6] \quad 1^1 1^2 321^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 7, T_{\max} = 99, \Delta = -160, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 2 \\ 2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 13221^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 7, T_{\max} = 99, \Delta = -158, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^1 23131^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

1.6 $\hat{\mathbf{1}} \rightarrow (-1)$

$$T = 2 \quad [0] \quad 1^1 1^1 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 2, T_{\max} = 120, \Delta = -242, \det = 4, (n_+, n_-, n_0) = (1, 2, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 120, T \in [10, 120]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 3 \quad [2] \quad 1^1 21^1 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -213, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 21^1 2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -213, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [4] \ 1^1 221^1 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -184, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 222^1 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -184, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [6] \ 1^1 2221^1 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -155, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 2221^2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -155, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [9] \ 2^3 21^1 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -145, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 22221^2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 85, \Delta = -126, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 85, T \in [10, 85]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 22221^1 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 85, \Delta = -126, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 85, T \in [10, 85]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [11] \ 1^1 222221^1 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 13, T \in [10, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[10] \ 222221^2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 13, T \in [10, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[12] \ 2^3 1321^1 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 30, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 30, T \in [11, 30]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [14] \ 1^1 2222221^1 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[15] \ 2^3 13221^1 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[13] \quad 2^{2222221^1 + \hat{1} \rightarrow 1} \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [16] \quad 2^{22222221^1 + \hat{1} \rightarrow 1} \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{122222221^1 + \hat{1} \rightarrow 1} \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[18] \quad 31^{1^{231} 13 + \hat{1} \rightarrow 1} \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 2^{1^{23141} 3 + \hat{1} \rightarrow 1} \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [22] \quad 31^{1^{2231} 13 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[23] \quad 2^{1^{23141} 3 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1222222221^1 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

$$[20] \quad 2^{222222221^1 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \quad 321^{321} 8 12321^1 + \hat{1} \rightarrow 1 \quad T = 12, T_{\max} = 28, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{1^{2321} 7 12321^1 + \hat{1} \rightarrow 1} \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [2] \quad 31^{1^{2315131} 6 13 + \hat{1} \rightarrow 1} \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [3] \quad 316^{31} 131612321^1 + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 17, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [4] \quad 3221 \overset{1}{2} 231513221 (12) \quad 1223 + \hat{1} \rightarrow 1 \quad T = 19, T_{\max} = 22, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

1.7 $\hat{1} \rightarrow (-2)$

$$T = 12 \quad [11] \quad 22 \overset{1}{1} 181223151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 321 \overset{512321}{8} 12 \overset{1}{1} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18, T_{\max} = 71, T \in [18, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 321 \overset{151321}{8} 12 \overset{1}{1} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18, T_{\max} = 49, T \in [18, 49]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 18123215132 \overset{1}{1} 2 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 71, T \in [19, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1812315132 \overset{1}{1} 2 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 49, T \in [19, 49]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 22 \overset{1}{1} 1812315123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 49, T \in [19, 49]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 22 \overset{1}{1} 1812321513 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 60, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 60, T \in [19, 60]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 22 \overset{1}{1} 1812231512 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1 \overset{151321}{8} 12 \overset{1}{1} 2 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1 \overset{1512321}{8} 12 \overset{1}{1} 2 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 26, T \in [19, 26]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 22 \overset{1}{1} 2181231512 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 22 \overset{1}{1} 2181232151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 26, T \in [19, 26]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

- [14] $22^1 218123151 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [15] $8122315132^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [16] $31^1 6^{21} 131 6^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 18, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [17] $31^1 6^{21} 1316131 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 18, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [18] $31^1 6^{21} 1316122^1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 18, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [19] $1612321712^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [20] $3141321812^1 31 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [21] $21612321712^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [22] $14141321812^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 28, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [23] $81231414132^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 82, T \in [19, 82]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [0] $1^{2151321} 8^{12} 12^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [10] $22^1 181231513 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 39, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 39, T \in [19, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [64] $1612^1 3217^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [65] $161232^1 17^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [28] $1 \begin{smallmatrix} 1 \\ 8 \\ 151321 \end{smallmatrix} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [29] $1 \begin{smallmatrix} 21 \\ 215132^1 1 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [30] $1 \begin{smallmatrix} 21 \\ 1512^1 321 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [31] $1 \begin{smallmatrix} 21 \\ 151232^1 1 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [32] $1 \begin{smallmatrix} 21 \\ 15132^1 1 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [33] $1 \begin{smallmatrix} 32^1 1513221 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [34] $1 \begin{smallmatrix} 512321 \\ 8 \end{smallmatrix} 122^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 71, T \in [19, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [35] $1 \begin{smallmatrix} 151321 \\ 8 \end{smallmatrix} 122^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [36] $1 \begin{smallmatrix} 1513221 \\ 8 \end{smallmatrix} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [37] $1 \begin{smallmatrix} 32^1 151321 \\ 8 \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [38] $1 \begin{smallmatrix} 215132^1 21 \\ 8 \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [39] $1 \begin{smallmatrix} 15132^1 21 \\ 8 \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [40] $1 \begin{smallmatrix} 215132^1 1 \\ 8 \end{smallmatrix} 122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [41] $1 \overset{151232^1}{8} 122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [42] $1 \overset{1512^1 321}{8} 122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [43] $1 \overset{15132^1}{8} 122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [44] $21 \overset{512321}{8} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 71, T \in [19, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [45] $21 \overset{151321}{8} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [46] $2321 \overset{51321}{8} 12^1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.33 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [47] $21 \overset{215132^1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [48] $21 \overset{1512^1 321}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [49] $21 \overset{151232^1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [50] $21 \overset{15132^1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [51] $32^1 1613151315 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 82, T \in [19, 82]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [52] $32^1 2181223151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [53] $21812231512^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [54] $22181231512^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$

- [55] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$ $2218122^1 31512 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [56] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ $2218122^1 3151 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [57] $2221812^1 31512 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [58] $222^1 181231512 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [59] $222181232^1 151 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [60] $222^1 181232151 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [61] $2221812^1 32151 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [62] $222^1 18123151 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [63] $2221812^1 3151 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [26] $1 \begin{matrix} 1 \\ 15132^1 21 \end{matrix} 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [25] $1 \begin{matrix} 1 \\ 215132^1 21 \end{matrix} 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [66] $1712321612^1 3 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [67] $1 \begin{matrix} 32^1 1 \\ 7 \end{matrix} 1232161 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [68] $1612^1 3217122 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$

- [24] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $1 \xrightarrow{32^1 151321} 8 \xrightarrow{1} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [69] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $161232^1 17122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [70] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $17 \xrightarrow{21} 12^1 321612 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [71] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $17 \xrightarrow{21} 1232^1 1612 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [72] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $14132^1 1812313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [73] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 $141321812^1 313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [74] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 $16123217122^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
- [75] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $32^1 1612321712 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
- [76] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $2161232^1 17122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [77] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $21612^1 3217122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [78] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $41232^1 1812313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [79] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $412^1 321812313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [80] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 $412321812^1 313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [81] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 $412321812^1 32 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 20, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

- [82] $412^1 132181232 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 20, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [83] $41232^1 181232 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 20, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [84] $4141321^{32^1 1} 8^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 82, T \in [19, 82]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [85] $41413218122^1 3 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 82, T \in [19, 82]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [86] $41413221812^1 3 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 82, T \in [19, 82]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [27] $1_{512321}^1 8^1 12^1 3 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 71, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 71, T \in [19, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [88] $1_{3151232^1 1}^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 60, \Delta = -120, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 60, T \in [18, 60]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [89] $1_{2151322^1 1}^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [90] $1_{315132^1 1}^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 39, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 39, T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [91] $1_{151322^1 1}^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [92] $1_{512^1 1321}^1 8^1 123 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 71, \Delta = -121, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [93] $1_{512^1 315131}^1 6^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 29, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 29, T \in [18, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [94] $1_{3215132^1 21}^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

- [95] $1 \overset{1}{8} \overset{315132^1 21}{1} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 39, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 39, T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [96] $1 \overset{512^1 321}{8} 1223 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [97] $1 \overset{3215132^1 1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [98] $1 \overset{3151232^1 1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 60, \Delta = -120, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 60, T \in [18, 60]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [99] $1 \overset{2151322^1 1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [100] $1 \overset{315132^1 1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 39, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 39, T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [101] $1 \overset{151322^1 1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [102] $21 \overset{512^1 321}{8} 123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [103] $1612315132^1 15 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 32, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 32, T \in [18, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [104] $1612^1 31513215 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 32, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 32, T \in [18, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [105] $1612^1 31513151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 29, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 29, T \in [18, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [106] $2161315132^1 15 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 29, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 29, T \in [18, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [107] $512^1 321613151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 35, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 35, T \in [18, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [108] $218122^1 315123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

- [109] $218122^1 31513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [110] $221812^1 315123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [111] $221812^1 321513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [112] $221812^1 231512 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 24$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [113] $221812^1 31513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [114] $221812^1 23151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 24$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [115] $6122315132^1 15 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 32$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 32$, $T \in [18, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [116] $6122^1 31513215 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 32$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 32$, $T \in [18, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [117] $6122^1 31513151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 29$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 29$, $T \in [18, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [118] $161232^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [119] $316131612^1 2 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 18$, $\Delta = -96$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 16.25 \rightarrow 17$, $T_{\max} = 18$, $T \in [17, 18]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0}$
- [120] $1712^1 3216123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [121] $171232^1 161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [122] $161232^1 1712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [123] $41315132^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [124] $314132^1 181231 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 24$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [125] $1612^1 32171223 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [126] $32161232^1 1712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [127] $4132171232^1 15 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [128] $41315132^1 1712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [129] $41315132^1 2171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [130] $18 \overset{21}{12} 1^1 314141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [131] $18 \overset{1}{12} 2^1 314141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [132] $1414132^1 18122 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [133] $1414132^1 21812 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [134] $413161232^1 161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [87] $1 \overset{1}{\underset{3215132^1 1}{8}} 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [188] $413151322^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [137] $1 \overset{1}{\underset{15132^1 1}{8}} 123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

- [151] $1 \overset{31512^1 321}{8} 12 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [152] $21 \overset{51232^1 1}{8} 123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [153] $21 \overset{15132^1 1}{8} 123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [154] $21 \overset{51321}{8} 12^1 32 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [155] $51232^1 1613151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 35$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 35$, $T \in [18, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [156] $15132^1 1613151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 29$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 29$, $T \in [18, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [157] $18122^1 3151232 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [158] $1812231512^1 32 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [159] $18122^1 315132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [160] $32218122^1 3151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [161] $322^1 181223151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [162] $21812^1 2315123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [163] $2181231512^1 32 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [164] $21812^1 3151232 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [165] $21812^1 3215132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[166] $21812^1 231513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[167] $21812^1 315132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[168] $22181232^1 1513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[169] $612^1 231513215 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 32$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 32$, $T \in [18, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[170] $612^1 231513151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 29$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 29$, $T \in [18, 29]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[171] $612^1 315131513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[172] $1612^1 32171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[173] $171232^1 16123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[174] $1712^1 32161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[175] $1612^1 321712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[176] $612321^{232^1} 71 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[177] $612^1 32171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[178] $1316131612^1 23 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 18$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 18$, $T \in [18, 18]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[179] $161232^1 171223 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [180] $1612321712^1 23 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [181] $232161232^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [182] $232^1 161232171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [183] $321612^1 321712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [184] $41321712^1 3215 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [185] $4132^1 17123215 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [186] $4132^1 17123151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [187] $41321712^1 3151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [135] $1 \begin{smallmatrix} 8 \\ 31512^1 321 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [189] $1812^1 3141413 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [190] $1812^1 2314141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [191] $4132^1 1613161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 35$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 35$, $T \in [18, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [192] $31414132^1 1812 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [193] $14141322^1 1812 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [194] $31414132^1 2181 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$

- [195] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $4141321812^1 23 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
- [196] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $4132^1 16131612 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 35, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 35, T \in [18, 35]$
- [197] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ $4131612^1 32161 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
- [198] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ $4132^1 1812314 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
- [199] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ $4132^1 18122314 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
- [200] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $413218122^1 314 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
- [201] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $141414132^1 171 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 28, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 28, T \in [18, 28]$
- [202] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ $7122^1 31414141 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 28, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 28, T \in [18, 28]$
- [136] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $1 \xrightarrow{1} \begin{matrix} 8 \\ 5123211 \end{matrix} 123 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 71, \Delta = -121, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
- [204] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $1812^1 32151322 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 71, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
- [205] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $1812^1 2315132 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
- [206] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $1812^1 3151322 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
- [207] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $8122^1 3151322 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
- [208] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ $231414132^1 181 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$

- [209] $314141322^1 181 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [210] $8122^1 31414132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [211] $712^1 314141413 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [212] $712^1 231414141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [203] $1812^1 23151232 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [218] $31612^1 3151315 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [215] $1^{151322^1} 8^{123} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [216] $1^{5132^1 21} 8^{1232} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [217] $21^{5132^1 1} 8^{1232} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [214] $1^{231512^1 321} 8^{1232} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [213] $1^{5132^1 1} 8^{1232} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [220] $3221812^1 23151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [221] $2181232^1 15132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [222] $812^1 23151322 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [223] $61232^1 1^{2321} 7^1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [224] $2321612^1 32171 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [225] $414132^1 1^1 8^1 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [226] $414132^1 181223 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [227] $812^1 314141322 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [228] $414132^1 218123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [229] $812^1 231414132 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [230] $41321812^1 2314 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [219] $5132^1 16131513 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [231] $315132^1 161315 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.94 \rightarrow 17, T_{\max} = 82, T \in [17, 82]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [233] $4141322^1 18123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.84 \rightarrow 17, T_{\max} = 82, T \in [17, 82]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [232] $181232^1 151322 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.84 \rightarrow 17, T_{\max} = 71, T \in [17, 71]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [234] $1^{51322^1 1} 8^1 1232 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.76 \rightarrow 17, T_{\max} = 71, T \in [17, 71]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $T = 13$ [267] $21^{22^1 22221} (12) 122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [258] $1^{22^1 2222221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 99, T \in [20, 99]$

66

- [290] $(12)1222222222^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 99$, $T \in [20, 99]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [291] $1612321 \overset{1}{8} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [292] $16123218122^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [293] $321 \overset{2}{8} 1232161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [294] $1 \overset{22}{8} 12321612 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [295] $1612321812^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 36$, $T \in [20, 36]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [296] $1712321712^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [297] $321612321812^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 40$, $T \in [20, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [298] $216123218122^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [299] $22^1 17123217122 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [300] $413151321812^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 36$, $T \in [20, 36]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [237] $1 \overset{22}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 99$, $T \in [20, 99]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [343] $21(12)1222222^1 22 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -9$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 99$, $T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [301] $1 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -9$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 99$, $T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [329] $222^1 221 (12) 122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [330] $222^1 21 (12) 1222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [331] $222^{21} 221 (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [332] $222^{21} 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [333] $32^1 17123151315 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 77, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 77, T \in [20, 77]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [334] $132181231512^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [335] $31321812^1 31512 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [336] $3132^1 181231512 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [337] $3132^1 181232151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [338] $31321812^1 32151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [339] $3132181232^1 151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [340] $31321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [341] $3132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [342] $1(12)12222222^1 22 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [302] $1 (12) 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $222^1 22221$

$$T_{\min} = 19.22 \rightarrow 20, \quad T_{\max} = 99, \quad T \in [20, 99]$$

[344] 221(12)1222222¹22 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -9$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [358] $1 \overset{32^1 21}{8} 1232161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 40, T \in [20, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{16} 12^1 32181222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{16} 12321812^1 22 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{16} 1232^1 181222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{18} 12321612^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 40, T \in [20, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{21} 8 1232161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 40, T \in [20, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{221}{1} 8 12^1 321612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{221}{1} 8 1232^1 1612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{21} 8 1232^1 1612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{21} 8 12^1 321612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{17} 123217122^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 36, T \in [20, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{1} 7 12321712 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 36, T \in [20, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{17} 12^1 3217122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{17} 1232^1 17122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[372] \quad 612321^{2321} 8 \ 12^1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.33 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[373] \quad 32^1 16131612315 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 72, \Delta = -121, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 72, T \in [20, 72]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[374] \quad 161315131612^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 61, T \in [20, 61]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[375] \quad 32^1 16123218122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 40, T \in [20, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[376] \quad 21612321812^1 22 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[377] \quad 21612^1 32181222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[378] \quad 2161232^1 181222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[379] \quad 217123217122^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 36, T \in [20, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[380] \quad 221712^1 3217122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[381] \quad 612315131612^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 72, \Delta = -121, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 72, T \in [20, 72]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[382] \quad 412^1 3218123214 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[383] \quad 41232^1 18123214 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[384] \quad 414141321812^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 77, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 77, T \in [20, 77]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[303] \quad 1 \begin{pmatrix} 12 \\ 222^1 2221 \end{pmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[illegible]

- [412] $17122^1 131513215 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [413] $17122^1 131513151 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [414] $21712315132^1 15 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [415] $21712^1 131513215 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [416] $21712^1 131513151 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [417] $512^1 3217123215 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [418] $15132171232^1 15 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 33, \Delta = -116, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 33, T \in [19, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [419] $1321812^1 1315123 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [420] $1321812^1 321513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 25, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 25, T \in [19, 25]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [421] $1321812^1 131513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 20, \Delta = -98, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [422] $1(12)1222222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [423] $21(12)122222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [424] $221(12)12222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [425] $2221(12)1222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [386] $1 \begin{pmatrix} 12 \\ 2222^1 2221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$

- [427] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $22221(12)1222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
- [428] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $2222^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
- [429] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $(12)12222222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
- [430] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $41315132^1 181 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [431] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $161232^1 181 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [432] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $1712^1 3217123 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [433] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$ $1613161232^1 15 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
- [434] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $161315132^1 161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 27, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
- [435] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $1^3 8^3 1232^1 161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
- [436] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $1812^1 3216123 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
- [437] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $21^3 8^3 1232^1 161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
- [438] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $161232^1 18122 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [439] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $1712^1 32171223 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [440] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $1^3 7^3 1232^1 1712 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$

- $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [456] $6122315132^1 161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 28$, $\Delta = -113$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [457] $6122^1 315132216 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 28$, $\Delta = -113$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [458] $413161232^1 171 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [459] $413161232^1 1712 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [460] $4132171232^1 161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [387] $1 \begin{pmatrix} 12 \\ 2222^1 221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [468] $1 \begin{pmatrix} 12 \\ 22222^1 22221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [537] $612^1 2315132161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 28$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [464] $1 \begin{pmatrix} 12 \\ 22222^1 1 \end{pmatrix} 1222 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [465] $1 \begin{pmatrix} 12 \\ 22222^1 21 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [466] $1 \begin{pmatrix} 12 \\ 22222^1 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [467] $21 \begin{pmatrix} 12 \\ 22222^1 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [462] $1 \begin{pmatrix} 12 \\ 22222^1 221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [483] $15132^1 17123151 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [484] $132181232^1 1513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 25, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 25, T \in [19, 25]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [485] $1(12)122222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [486] $21(12)12222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [487] $221(12)1222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [488] $2221(12)1222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [489] $22221(12)122^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [490] $22222^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [491] $7122^1 315131513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [492] $(12)1222222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [493] $16^1 12^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [494] $171232^1 17123 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [495] $16131612^1 3215 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [496] $16131612^1 3151 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 27, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [497] $1^1 8^1 12^1 32161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[498] \quad 1 \quad \overset{322^1}{8} \quad 1232161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[499] \quad 1 \quad 1812321612^1 32 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[500] \quad 1 \quad 1812^1 32161232 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[501] \quad 1 \quad 181232^1 16123 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[502] \quad 21 \quad \overset{321}{8} \quad 12^1 32161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[503] \quad 31 \quad 1612^1 3218122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[504] \quad 1 \quad 171232^1 171223 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[505] \quad 1 \quad 1712321712^1 23 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[506] \quad 1 \quad \overset{2321}{7} \quad 1232^1 171 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 50, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 50, T \in [19, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[507] \quad 1 \quad \overset{232^1}{7} \quad 1232171 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 50, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 50, T \in [19, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[508] \quad 1 \quad \overset{321}{7} \quad 12^1 321712 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[509] \quad 512321 \quad 4132^1 1 \quad \overset{1}{8} \quad 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[510] \quad 51232^1 \quad 41321 \quad \overset{1}{8} \quad 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [511] $41321 \overset{15132^1}{8} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [512] $41321 \overset{15132^1}{1} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [513] $413151322 \overset{1}{181} + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [514] $612 \overset{31}{1} 31513161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [515] $612 \overset{1}{1} 231513161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [516] $612321 \overset{232^1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [517] $612 \overset{232^1}{1} 321 \overset{1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [518] $216131612 \overset{1}{1} 3215 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 29, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [461] $1 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $\overset{22222^1}{22221}$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [536] $612 \overset{1}{1} 3151321613 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 72, \Delta = -121, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [519] $216131612 \overset{1}{1} 3151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [520] $161232 \overset{1}{1} 1613151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 29, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [521] $321612 \overset{1}{1} 3218122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [522] $232161232 \overset{1}{1} 1812 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [523] $232^1 1612321812 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [524] $21712321712^1 23 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [525] $2171232^1 171223 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [526] $4132^1 181223151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [527] $413218122^1 3151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [528] $4132^1 218123215 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [529] $413221812^1 3215 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [530] $413221812^1 3151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [531] $4132^1 218123151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [532] $413151322^1 1812 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [533] $612^1 3216131513 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 77$, $\Delta = -122$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 77$, $T \in [19, 77]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [534] $612^1 3151316123 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 72$, $\Delta = -121$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 72$, $T \in [19, 72]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [535] $612^1 2315131612 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 27$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 27$, $T \in [19, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [548] $4141321812^1 314 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 77$, $\Delta = -122$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 77$, $T \in [19, 77]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [547] $141414132^1 2181 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 28$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [546] $141414132^1 1812 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [545] $1812^1 31414141 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [544] $4132^1 171232161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [543] $41321712^1 32161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [542] $4131612^1 321712 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [541] $4132^1 161316131 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 19, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [540] $4131612^1 32171 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [539] $612^1 2315132216 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [538] $61315131612^1 32 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [463] $1 \begin{pmatrix} 221 \\ 22222^1 21 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [567] $3141414132^1 181 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [566] $(12)122222^1 22222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [550] $1 \begin{pmatrix} 21 \\ 22222^1 21 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [551] $1 \begin{pmatrix} 221 \\ 22222^1 1 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [568] $1414141322^1 181 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [569] $8122^1 314141413 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [570] $712^1 3141414141 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [549] $1 \begin{pmatrix} 12 \\ 222222^1 221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [572] $1 \begin{pmatrix} 12 \\ 222222^1 1 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [573] $1 \begin{pmatrix} 12 \\ 222222^1 221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [574] $1 \begin{pmatrix} 12 \\ 222222^1 21 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [575] $1 \begin{pmatrix} 12 \\ 222222^1 1 \end{pmatrix} 122 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [576] $21 \begin{pmatrix} 12 \\ 222222^1 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [577] $321712^1 3151315 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [578] $5132^1 171231513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [579] $1(12)12222^1 222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [580] $21(12)1222^1 222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [581] $221(12)122^1 222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [582] $2221(12)12^1 222222 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [583] $(12)122222^1 222222 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [584] $16131612^1 315 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [571] $1 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $222222^1 21$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [586] $1 \overset{2321}{7} 12^1 32171 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 50$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 50$, $T \in [19, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [587] $61232^1 1 \overset{2321}{8} 12 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [588] $3216131612^1 315 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 72$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 72$, $T \in [19, 72]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [589] $31612^1 32161315 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 77$, $\Delta = -122$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 77$, $T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [590] $31612^1 31513161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [591] $2321612^1 321812 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [592] $41321812^1 23151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [593] $41322^1 18123215 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [594] $41322^1 18123151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [595] $61232^1 16131513 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 77$, $\Delta = -122$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 77$, $T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [596] $612315132^1 1613 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 72$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 72$, $T \in [19, 72]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [597] $812^1 3141414132 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [598] $812^1 2314141413 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [599] $414132^1 1812314 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [585] $181232^1 161232 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 61, \Delta = -120, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [600] $1 \begin{pmatrix} 12 \\ 22222222^1 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [601] $1 \begin{pmatrix} 12 \\ 22222222^1 \end{pmatrix} 21 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [602] $1 \begin{pmatrix} 12 \\ 22222222^1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [603] $51321712^1 31513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 77, T \in [18, 77]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [604] $1(12)1222^1 2222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [605] $21(12)122^1 2222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [606] $221(12)12^1 2222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [607] $(12)1222^1 2222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [608] $16131612^1 31513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 61, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 61, T \in [18, 61]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [609] $3161232^1 161315 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 77, T \in [18, 77]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$

- [610] $1 \overset{22222222^1}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [611] $1(12)122^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [612] $21(12)12^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [613] $(12)1222^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [614] $41414132^1 18123 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 77, T \in [18, 77]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [615] $1(12)12^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -100, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.76 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [616] $(12)122^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -100, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.76 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [617] $(12)12^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -121, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.69 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [618] $6131612^1 315132 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -121, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.69 \rightarrow 18, T_{\max} = 72, T \in [18, 72]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $T = 14$ [630] $3221 \overset{1}{(12)1222^1} + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [632] $1 \overset{21}{(12)} 12 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [622] $1 \overset{221}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [623] $1 \overset{22^1}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [624] $1 \overset{1}{(12)} 1222^1 2 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$

- [639] $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 ${}^{221}_{3221} 3221(12)12^1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [640] ${}^{2^1 21}_{3221} 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [641] ${}^{22^1 1}_{313221} 21(12)12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [642] ${}^{13221}_{13221} 1(12)122222^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [643] ${}^{132^1 21}_{13221} 1(12)122222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [644] ${}^{2313221}_{2313221} 1(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [645] ${}^{22313221}_{22313221} 1(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [646] ${}^{22^1 313221}_{22^1 313221} 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [647] ${}^{313221}_{313221} 1(12)12222^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [648] ${}^{132^1 21}_{132^1 21} 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [649] ${}^{13221}_{13221} 21(12)12222^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [650] ${}^{13221}_{13221} 221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [651] ${}^{13221}_{13221} 22^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$

- $$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [652] $221 \overset{132^1 21}{(12)} 12222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$
- [653] $22^1 21 \overset{13221}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [654] $2221 \overset{132^1 21}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$
- [655] $3221 \overset{3221}{(12)} 12222^1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 87, T \in [20, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [656] $21 \overset{2313221}{(12)} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [657] $22^1 1 \overset{2313221}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [658] $21 \overset{22^1 313221}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$
- [659] $21 \overset{22313221}{(12)} 12^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [660] $21 \overset{313221}{(12)} 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [661] $221 \overset{313221}{(12)} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [662] $22^1 1 \overset{313221}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [663] $22^1 181231513215 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 29, T \in [21, 29]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [664] $22^1 181231513151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 28, T \in [21, 28]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [665] $22222221(12)122^1 31 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [666] $22^1 222221(12)12231 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [667] $22^1 2221(12)1223132 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [668] $22^1 221(12)12231322 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [669] $22221(12)1223132^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [670] $22^1 22221(12)122313 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [671] $8123151315132^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 73, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 73, T \in [21, 73]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [672] $1712321 \overset{1}{8} \overset{21}{123} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [673] $1712321 \overset{21}{8} \overset{221}{1} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [674] $1 \overset{321}{7} 12321812^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 34, T \in [21, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [675] $17123218122^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [676] $1 \overset{221}{8} 12321712^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [677] $1 \overset{221}{8} 123217122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [678] $321 \overset{21}{8} 12321712 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [679] $21 \overset{21}{8} 12321712^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$

- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [680] $321612321812^1 31 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 21$, $\Delta = -98$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [681] $3221712321812^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 34$, $T \in [21, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [682] $2217123218122^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [683] $22^1 171232181222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [684] $4131612321812^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 34$, $T \in [21, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 1 \\ 2313221 \end{pmatrix}$
 [621] $1 (12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 87$, $T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 1 \\ 132^1 21 \end{pmatrix}$
 [620] $1 (12) 122222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 1 \\ 13221 \end{pmatrix}$
 [619] $1 (12) 12222^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 32^1 1 \\ 1 \end{pmatrix}$
 [739] $1 7 123218122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 34$, $T \in [21, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 1 \\ 3132^1 21 \end{pmatrix}$
 [689] $1 (12) 12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 84$, $T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 21 \\ 1322^1 1 \end{pmatrix}$
 [690] $1 (12) 12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 21 \\ 13221 \end{pmatrix}$
 [691] $1 (12) 122^1 22 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 221 \\ 13221 \end{pmatrix}$
 [692] $1 (12) 12^1 22 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [693] $1 \begin{smallmatrix} 221 \\ 1322^1 1 \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [694] $2221 \begin{smallmatrix} 1 \\ 3221 \end{smallmatrix} (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [695] $1 \begin{smallmatrix} 21 \\ 313221 \end{smallmatrix} (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [696] $1 \begin{smallmatrix} 21 \\ 3132^1 21 \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [697] $1 \begin{smallmatrix} 221 \\ 3132^1 21 \end{smallmatrix} 122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [698] $21 \begin{smallmatrix} 21 \\ 1322^1 1 \end{smallmatrix} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [699] $21 \begin{smallmatrix} 21 \\ 13221 \end{smallmatrix} (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [700] $21 \begin{smallmatrix} 221 \\ 1322^1 1 \end{smallmatrix} (12) 122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [701] $21 \begin{smallmatrix} 221 \\ 3221 \end{smallmatrix} (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [702] $21 \begin{smallmatrix} 221 \\ 3132^1 21 \end{smallmatrix} (12) 12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [703] $1 \begin{smallmatrix} 513151321 \end{smallmatrix} 8 \begin{smallmatrix} 221 \\ 1322^1 1 \end{smallmatrix} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 73, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 73, T \in [21, 73]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [704] $1 \begin{smallmatrix} 1322^1 1 \end{smallmatrix} (12) 1222222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [705] $1 \begin{smallmatrix} 13221 \end{smallmatrix} (12) 12222^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

- [733] $1712^1 1^1 321^{221} 81 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [734] $171232^1 1^1 81 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [735] $171232^1 1^1 812 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [736] $1712^1 1^1 321^{221} 812 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [737] $18123217122^1 3 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 34$, $T \in [21, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [687] $1^{12} 12^{122} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 87$, $T \in [21, 87]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [686] $1^{12} 1222^1 22 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [685] $1^{12} 12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [741] $1712321812^1 22 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [742] $1712^1 1^1 32181222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [743] $171232^1 1^1 81222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [744] $1812^1 1^1 3217122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [745] $181232^1 1^1 7122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [746] $21^1 812321712 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 34, T \in [21, 34] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[747] & \begin{matrix} 21 \\ 21812^1 3217122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[748] & \begin{matrix} 21 \\ 2181232^1 17122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[749] & 5131612321712^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 73, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 73, T \in [21, 73] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[750] & 1613151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 51, T \in [21, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[751] & 32^1 161315132171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 51, T \in [21, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[752] & 32^1 161232181231 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[753] & 32^1 217123218122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 34, T \in [21, 34] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[754] & 221712321812^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[755] & 221712^1 32181222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[756] & 22171232^1 181222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[757] & 7122315131612^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 51, T \in [21, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[758] & 6123151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 62, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 62, T \in [21, 62] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[759] & 6123216131612^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 62, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 62, T \in [21, 62] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[760] & \begin{matrix} 1 \\ 161316131612^1 3 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 14, T_{\max} = 37, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 37, T \in [21, 37] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [738] $1 \overset{32^1 21}{8} 12321712 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 34, T \in [21, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [688] $1 \overset{1}{(12)} 122^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [740] $2321 \overset{2^1 1}{8} 1232171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.33 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [761] $1 \overset{1}{512315132^1 1} 8 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [762] $1 \overset{1}{512^1 3151321} 8 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [763] $1 \overset{1}{151315132^1 1} 8 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [764] $1 \overset{1}{(12)} 122^1 222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [765] $1 \overset{1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [766] $1 \overset{1}{(12)} 12^1 222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [767] $1 \overset{1}{(12)} 12222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [768] $1 \overset{21}{(12)} 12^1 222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [769] $22^1 21 \overset{1}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.25 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [770] $1 \overset{221}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [774] $21 \begin{pmatrix} 12 \\ 23132^{121} \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [775] $21 \begin{pmatrix} 12 \\ 31322^{11} \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 84$, $T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [776] $1 \begin{pmatrix} 8 \\ 512315132^{121} \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [777] $1 \begin{pmatrix} 8 \\ 512^{131513221} \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [778] $1 \begin{pmatrix} 8 \\ 151315132^{121} \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [779] $1 \begin{pmatrix} 8 \\ 512^{13151321} \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [780] $1 \begin{pmatrix} 8 \\ 512315132^{11} \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [781] $1 \begin{pmatrix} 8 \\ 151315132^{11} \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [782] $1 \begin{pmatrix} 8 \\ 512321 \end{pmatrix} 1232^{115} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [783] $1 \begin{pmatrix} 8 \\ 512^{1321} \end{pmatrix} 123151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [784] $1 \begin{pmatrix} 12 \\ 13221 \end{pmatrix} 1222^{1222} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [785] $1 \begin{pmatrix} 12 \\ 23132^{121} \end{pmatrix} 12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [786] $1 \begin{pmatrix} 12 \\ 2313221 \end{pmatrix} 12^{1222} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$

- $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [801] $221812^1 31513215 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [802] $221812^1 31513151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [803] $151322181232^1 15 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [804] $2222^1 2221(12)12231 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [805] $2222^1 21(12)1223132 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [806] $222221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [807] $2222^1 1(12)12231322 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [808] $2222221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 84$, $T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [809] $2222^1 221(12)122313 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 84$, $T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [810] $161315132^1 171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 26$, $T \in [20, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [811] $1712^1 32181223 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [812] $1712^1 3218^{21} 123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [813] $1^1 7^{321} 1232^1 181 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [814] $1712^1 321613151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

- [815] $161315132^1 1712 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 26, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [816] $1712^1 315131612 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 26, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [817] $161315132^1 2171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 26, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [818] $1712^1 315132161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [819] $1712315132^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [820] $16^3 1232^1 181231 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 20, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [821] $18^2 12^1 32171223 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [822] $1^3 8^2 1232^1 1712 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [823] $1^3 7^2 1232^1 18122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [824] $21^3 8^2 1232^1 1712 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [825] $6122^1 315132171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [826] $6122315132^1 171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [827] $161232171232^1 15 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [828] $1612^1 3217123215 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[illegible]

- [844] $1613161232^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [845] $413161232^{21} 181 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [846] $4132171232^1 171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [847] $21613161232^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [848] $414132181232^1 15 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 51, T \in [20, 51]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [849] $413161232^1 18122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [850] $41322181232^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 37, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 37, T \in [20, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [851] $4132171232^1 1712 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [852] $4132181232^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 37, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 37, T \in [20, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [773] $21^{22} (12) 1223 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.25 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [771] $1^{21} (12) 1222 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [772] $1^{221} (12) 122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [920] $21613151322^1 171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 26, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [884] $15132^1 218123151 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [858] $1 \begin{smallmatrix} 21 \\ 223132^1 21 \end{smallmatrix} (12) \ 12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [859] $21 \begin{smallmatrix} 221 \\ 3221 \end{smallmatrix} (12) 12^1 23 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [860] $21 \begin{smallmatrix} 21 \\ 231322^1 1 \end{smallmatrix} (12) \ 12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [861] $1 \begin{smallmatrix} 5123151322^1 1 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [862] $1 \begin{smallmatrix} 3151315132^1 1 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 62, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 62, T \in [20, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [863] $1 \begin{smallmatrix} 1513151322^1 1 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [864] $1 \begin{smallmatrix} 512321 \\ 8 \end{smallmatrix} 12^1 3215 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 37, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 37, T \in [20, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [865] $1 \begin{smallmatrix} 51232^1 1 \\ 8 \end{smallmatrix} 123151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [866] $1 \begin{smallmatrix} 512321 \\ 8 \end{smallmatrix} 12^1 3151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [867] $1 \begin{smallmatrix} 151321 \\ 8 \end{smallmatrix} 12^1 3151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [868] $1 \begin{smallmatrix} 13221 \\ (12) 122^1 2222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [869] $1 \begin{smallmatrix} 231322^1 1 \\ (12) 12222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [870] $1 \begin{smallmatrix} 223132^1 21 \\ (12) 1222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

- [871] $1^{313221}(12)12^12222 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [872] $21^{13221}(12)12^12222 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [873] $22221^{3221}(12)12^123 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [874] $21^{231322^11}(12)1222 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [875] $221^{231322^11}(12)122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [876] $21^{223132^121}(12)122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [877] $18122^{1315131513} + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 62, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 62, T \in [20, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [878] $322^{118123151315} + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [879] $21812^{1231513215} + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [880] $21812^{1315131513} + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 62, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 62, T \in [20, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [881] $21812^{1231513151} + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 28, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [882] $15132^{1218123215} + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [883] $1513221812^{13215} + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [856] $2221^{13221}(12)12^123 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [855] $1 \overset{221}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [854] $1 \overset{1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [887] $22222 \overset{1}{1}(12)1223132 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [888] $222221(12)12^1 23132 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [889] $22221(12)122^1 31322 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [890] $22222 \overset{1}{21}(12)122313 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [891] $1712321812 \overset{1}{23} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [892] $171232 \overset{1}{18}1223 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [893] $1 \overset{2321}{7} 1232 \overset{1}{18}1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [894] $1 \overset{2321}{7} 1232181 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [895] $171232 \overset{1}{18}123 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [896] $1 \overset{321}{7} 12^1 32181 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [897] $171232 \overset{1}{16}13151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [898] $161315132 \overset{31}{17}1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$

- [899] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 1 \\ 1613151322 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
- [900] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 31 \\ 1612 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 20, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
- [901] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 21 \\ 1812321712 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
- [902] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 21 \\ 181232 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
- [903] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 3221 \\ 1812 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
- [904] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 3221 \\ 1812321712 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
- [905] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 321 \\ 1712 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
- [906] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 2321 \\ 1712321812 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
- [907] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 2321 \\ 1712321812 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
- [908] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 2321 \\ 2181232171 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
- [909] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 2321 \\ 2181232 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
- [910] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 321 \\ 21812 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
- [911] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 31 \\ 7122 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
- [912] $\begin{smallmatrix} 1 \\ 612 \end{smallmatrix} \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 27, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 27, T \in [20, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[913] & 1612321712^1 3215 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[914] & 161232^1 17123215 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[915] & 31513161232^1 171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 62, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 62, T \in [20, 62] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[916] & 161232^1 17123151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[917] & 1612321712^1 3151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[918] & 15131612^1 321712 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[919] & 32161315132^1 171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 51, T \in [20, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[853] & 1 \begin{pmatrix} 12 \\ 13221 \end{pmatrix} 12^1 2222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[921] & 31612315132^1 171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 51, T \in [20, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[922] & 16123151322^1 171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 27, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 27, T \in [20, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[923] & 321612^1 32181231 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[924] & 322^1 17123218122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[925] & 3221712^1 3218122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[926] & 612^1 32171231513 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 73, \Delta = -121, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 73, T \in [20, 73] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [927] $712315131612^1 32 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [928] $613151321712^1 32 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [929] $7122^1 3151316123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [930] $612^1 31513217123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [931] $612^1 23151321712 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [932] $7122^1 3151321613 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [933] $612^1 23151322171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [934] $61223151322^1 171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [935] $16131612^1 32161 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [936] $4131612^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [937] $41321712^1 32171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [938] $4132^1 171232171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [939] $612^1 3216131^1 61 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [940] $216131612^1 32161 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [941] $4141321812^1 3215 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$

- [942] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 4141321812¹3151 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [943] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 4131612¹3218122 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
- [944] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 413221812¹32161 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [945] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 4132¹2181232161 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [946] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 41321712¹321712 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
- [947] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 4132¹1712321712 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
- [948] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 612¹32161316123 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
- [949] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 41321812¹32161 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [950] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 4132¹181232161 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [951] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 61316131612¹32 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
- [952] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 413161315132¹16 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
- [886] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$ 22222¹221(12)12231 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
- [857] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 1 (12)²²¹_{231322¹1} + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
- [885] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 1513221812¹3151 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$

- [954] $222^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.89 \rightarrow 19, T_{\max} = 87, T \in [19, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [957] $1^{2231322^1} (12) 1222 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [958] $222^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.89 \rightarrow 19, T_{\max} = 87, T \in [19, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [959] $21^{2231322^1} (12) 122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [960] $1812^1 2315131513 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 62, \Delta = -120, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 62, T \in [20, 62]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [961] $1812^1 3151315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [962] $222222^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [963] $22221(12)12^1 231322 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [964] $222222^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [965] $8122^1 3151315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [966] $712^1 2315131^1 61 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 41, \Delta = -118, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [967] $712^1 32161315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [968] $712^1 31513161232 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [969] $712^1 23151316123 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 51, T \in [20, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [970] $712^1 23151321613 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [971] $14141414132^1 181 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [972] $8122^1 3141414141 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [955] $1 \begin{pmatrix} 12 \\ 2231322^1 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [956] $1 \begin{pmatrix} 12 \\ 13221 \end{pmatrix} 12^1 22222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [953] $1 \begin{pmatrix} 12 \\ 2231322^1 1 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [973] $1 \begin{pmatrix} 8 \\ 51315132^1 1 \end{pmatrix} 123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [974] $1 \begin{pmatrix} 8 \\ 3151321 \end{pmatrix} 12^1 315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [975] $3221812^1 3151315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [976] $3218122^1 3151315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [977] $5132^1 2181231513 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [978] $31513221812^1 315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [979] $151322^1 18123215 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [980] $151322^1 18123151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [981] $2222222^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [982] $812^1 23151315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [983] $812^1 31513151322 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [984] $1^{\frac{2321}{7}} 12^1 32181 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 41, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [985] $1^{\frac{2321}{7}} 12^1 321812 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 41, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [986] $21^{\frac{2321}{8}} 12^1 32171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 41, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [987] $315131612^1 32171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 62, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 62, T \in [20, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [989] $31612321712^1 315 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [990] $31612^1 321712315 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [991] $161315132^1 17123 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 51, T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [992] $31612^1 315132171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 51, T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [993] $61232^1 171231513 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [994] $612315132^1 17123 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 62, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 62, T \in [20, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [995] $7122315132^1 1613 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 51, T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [996] $61232^1 1613161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20, T_{\max} = 51, T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [997] $1613161232^1 1613 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [998] $414132^1 18123215 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [999] $414132^1 18123151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1000] $41316131612^1 315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1001] $41322^1 181232161 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1002] $61232^1 161316123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1003] $812^1 31414141413 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1004] $812^1 23141414141 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [988] $513161232^1 17123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1005] $1^{315132^1 1} 8^{12315} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1006] $2222^1 1^{3221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.69 \rightarrow 19$, $T_{\max} = 87$, $T \in [19, 87]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1007] $513221812^1 31513 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1008] $315132^1 21812315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1009] $5131612^1 3217123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1010] $3161232^1 1712315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1011] $612321712^1 31513 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 64, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1012] $16131612^1 321613 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19, T_{\max} = 51, T \in [19, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1013] $321812^1 23151315 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 81, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.84 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1014] $71232^1 161315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 81, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.84 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1015] $51322^1 181231513 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 100, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.76 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1016] $61315132^1 171232 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 100, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.76 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1017] $3151322^1 1812315 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 121, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.69 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

$T = 15$ [1037] $21(12)^{21} 122314132^1 2 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 85, \Delta = -123, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1018] $1(12)^1 122^1 31 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1023] $22^1 1(12)^{313221} 12231 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1024] $221(12)^{313221} 122^1 31 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1025] $22^1 313221(12)122313 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 29, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1026] $1^1 8^{22} 12321^1 8^1 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1027] $18^{21} 12321^1 8^1 123 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 32, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 32, T \in [21, 32]$

- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
 [1028] $1 \overset{22^1 1}{8} \overset{21}{12321} 8 \overset{1}{12} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 2221}{1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 221}{21} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{2221}{22^1 1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 21}{221} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 1}{1} (12) 1223141322 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{221}{1} (12) 122314132^1 2 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 21}{1} (12) 122314132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{22313221}{1} (12) 122^1 31 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 313221}{1} (12) 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{1} (12) 122^1 31 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 21}{21} (12) 12231413 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 79, T \in [22, 79]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{221}{22^1 1} (12) 12231413 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 79, T \in [22, 79]$

- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1055] $21613151321812^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1056] $16123151321812^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1057] $22^1 2181232181222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -106$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [1058] $81232161315132^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 63$, $T \in [22, 63]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1059] $61223151321812^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1060] $41321712321812^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 32$, $T \in [22, 32]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1022] $21 \begin{pmatrix} 2313221 \\ 12 \end{pmatrix} 122^1 31 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1038] $21 \begin{pmatrix} 22^1 1 \\ 12 \end{pmatrix} 122314132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 85$, $T \in [22, 85]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1036] $1 \begin{pmatrix} 22^1 221 \\ 12 \end{pmatrix} 12231413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 79$, $T \in [22, 79]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1120] $71223151321712^1 3 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 42$, $T \in [22, 42]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [1086] $221 \begin{pmatrix} 222^1 1 \\ 12 \end{pmatrix} 1223141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
 [1066] $1 \begin{pmatrix} 22313221 \\ 12 \end{pmatrix} 12^1 231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [1067] $1 \begin{pmatrix} 3132^1 21 \\ 12 \end{pmatrix} 1223132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1068] $21 \begin{pmatrix} 2313221 \\ 12 \end{pmatrix} 12^1 231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 27$, $T \in [22, 27]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [1069] $^{313221} 221 (12) 12^1 231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $^{3132^{121}} 221 (12) 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $^{313221} 32^1 21 (12) 1223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $^{313221} 21 (12) 122^1 313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $^{313221} 22313221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $^{21} 18 123218 122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $^{1} 18 123218 1232 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.33 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- $^{221} 18 12^1 3218 1 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $^{21} 18 123218 12^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $^{221} 18 1232^1 18 12 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $^{221} 18 12^1 3218 12 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $^{21} 218 12^1 3218 12 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $^{222221} 1 (12) 122^1 3141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $^{222^1 221} 1 (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 22221 \\ [1083] \end{matrix} 21(12)122^1 3141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 222^1 21 \\ [1084] \end{matrix} 21(12)1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 2221 \\ [1085] \end{matrix} 221(12)122^1 3141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ [1064] \end{matrix} 1(12)12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1063] \end{matrix} 13221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1062] \end{matrix} 1(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 222^1 1 \\ [1089] \end{matrix} 21(12)12231413 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 79, T \in [22, 79] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 32^1 1 \\ [1090] \end{matrix} 181232161315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 63, T \in [22, 63] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1091] \end{matrix} 1812315131612^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 32^1 1 \\ [1092] \end{matrix} 181231513161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 32^1 21 \\ [1093] \end{matrix} 18123218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 32, T \in [22, 32] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1094] \end{matrix} 181232^1 181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1095] \end{matrix} 1812^1 32181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1110] & 16131513218122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1111] & 32^1 1613151321812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1112] & 16131513221812^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1113] & 32^1 1613151322181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1114] & 32^1 1712321712315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 63, T \in [22, 63] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1115] & 17123151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1116] & 222^1 181232181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
[1117] & 2221812^1 32181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1118] & 61231513218122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 52, T \in [22, 52] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1119] & 61231513221812^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 52, T \in [22, 52] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1061] & \begin{matrix} 1 \\ 12 \end{matrix} 12^1 231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& \begin{matrix} 2313221 \\ 2313221 \end{matrix} \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1121] & 16131612321712^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1122] & 32^1 1613161232171 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1088] & \begin{matrix} 222^1 21 \\ 12 \end{matrix} 12231413 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 79, T \in [22, 79] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1065] $1 \begin{smallmatrix} 21 \\ 313221 \end{smallmatrix} 12^1 231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1087] $1 \begin{smallmatrix} 222^1 1 \\ 313221 \end{smallmatrix} 122314132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1126] $1 \begin{smallmatrix} 21 \\ 31322^1 1 \end{smallmatrix} 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1127] $1 \begin{smallmatrix} 313221 \\ 313221 \end{smallmatrix} 122^1 3132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1128] $1 \begin{smallmatrix} 31322^1 1 \\ 313221 \end{smallmatrix} 1223132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1129] $21 \begin{smallmatrix} 23132^1 21 \\ 31322^1 1 \end{smallmatrix} 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1130] $221 \begin{smallmatrix} 31322^1 1 \\ 313221 \end{smallmatrix} 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1131] $21 \begin{smallmatrix} 313221 \\ 313221 \end{smallmatrix} 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1132] $23132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1133] $22313221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1134] $6122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1135] $6122^1 3151321 181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1136] $161315132^1 181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1137] $161315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1138] $1712^1 1315132171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1139] $1812^{21} 132181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1125] $132^{11} 21(12)1223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.25 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1124] $1(12)12^{11} 2313 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1123] $1(12)12231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1143] $21(12)1223141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1144] $21(12)12^{11} 23141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1145] $221(12)12^{11} 23141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1146] $1(12)122^{11} 31413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1147] $1(12)12231413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1148] $21(12)122^{11} 31413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1149] $221(12)122^{11} 31413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1150] $1^1 8^{512321} 1232^{11} 161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1151] $1 \overset{512^1 321}{8} 1232161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1152] $1 \overset{151321}{8} 1232^1 161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1153] $1 \overset{21}{8} 12^1 321613151 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [1154] $1 \overset{1}{16} 1315132^1 18122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1155] $1 \overset{21}{8} 12^1 315131612 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1156] $1 \overset{1}{16} 1315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1157] $1 \overset{1}{18} 122^1 315131612 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1158] $1 \overset{21}{8} 12^1 315132161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1159] $1 \overset{21}{8} 12315132^1 161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1160] $1 \overset{1}{18} 122^1 315132161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1161] $1 \overset{1}{18} 122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1162] $1 \overset{1}{17} 1232171232^1 15 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [1163] $1 \overset{1}{17} 12^1 3217123215 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [1164] $1 \overset{1}{17} 12^1 3217123151 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1165] $1712315132^1 1712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1166] $1712^1 3151321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1167] $1712^1 3151322171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1168] $1712315132^1 2171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1169] $1^{\frac{321}{7}} 1232^1 181231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 21$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1170] $1^{\frac{3221}{8}} 1232^1 18122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1171] $21^{\frac{321}{8}} 1232^1 18122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1172] $413221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.25 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1173] $6122^1 315132181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1174] $6122315132^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1175] $6122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1176] $6122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1177] $1413221(12)1222^1 222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1178] $141322^1 1(12)1222222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1179] \quad 231413221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 85, T \in [21, 85]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1180] \quad 314132^1 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 79, T \in [21, 79]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1181] \quad 31413221(12)122^1 222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 79, T \in [21, 79]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1182] \quad 1612^1 32181223151 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1183] \quad 1513161232^1 18122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1184] \quad 2161315132^1 18122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1185] \quad 2161315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1186] \quad 1612315132^1 18122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1187] \quad 1612^1 31513218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1188] \quad 1612315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1189] \quad 1612^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1190] \quad 21712^1 3217123215 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1191] \quad 2171232171232^1 15 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1192] \quad 21712^1 3217123151 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1193] \quad 21712^1 3151321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1208] $21613161232^1 1712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1209] $1612^1 32171232161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1210] $413161232^1 181231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 21$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1211] $4132171232^1 18122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1212] $41322181232^1 1712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1213] $4132181232^1 1712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1214] $414132181232^1 161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 35$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 35$, $T \in [21, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1142] $1^{2222} (12) 1223141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1140] $1^{21} 8 12^{21} 3218123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1141] $1^{222221} (12) 12^1 23141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1260] $21^{232} 8^1 12321812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1217] $1^{223132} (12) 12231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1326] $4132^1 21812321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1328] $7123216131612^1 32 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [1219] $21 \ (12) \ 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1220] $322^1 1 \ (12) \ 1223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1221] $231322^1 1 (12) 1223132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1222] $223132^1 21 (12) 122313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1223] $612^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1224] $161315132^1 181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1225] $1613151322^1 181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1226] $181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1258] $21 \ 8 \ 12^1 3218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1227] $1812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1228] $1812^1 32181232 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1229] $1812321812^1 32 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1230] $181232^1 18123 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1231] $1 \ (12) \ 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{aligned}
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1246] & \overset{1}{1812^1 2315131612} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1247] & \overset{31}{161315132^1 2181} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1248] & \overset{1}{1812^1 3151321613} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 42, T \in [21, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1249] & \overset{1}{1812^1 2315132161} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1250] & \overset{1}{171232^1 17123215} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 30, T \in [21, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1251] & \overset{1}{1712321712^1 3215} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 30, T \in [21, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1252] & \overset{1}{171232^1 17123151} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 28, T \in [21, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1253] & \overset{1}{1712321712^1 3151} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 28, T \in [21, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1254] & \overset{1}{17123151322^1 171} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1255] & \overset{321}{1712^1 32181231} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 21, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1256] & \overset{3221}{1712^1 3218122} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1257] & \overset{322^1}{1712^1 123218122} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1259] & \overset{2321}{21712^1 1812} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1261] $413221(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1262] $4132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1263] $612^1 321 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1264] $613151321 \quad 232^1 1 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1265] $612^1 3151321 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1266] $612^1 23151321 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1267] $612^1 231513221 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1268] $61223151322^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1269] $1413221(12)122^1 2222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1270] $2314132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1271] $3141322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1272] $31413221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1273] $161232^1 181223151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1274] $16123218122^1 3151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1275] $5131612321812^1 23 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1276] $31513161232^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1277] $15131612^1 3218122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1278] $1613151321812^1 23 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1279] $232^1 161315132181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1280] $32161315132^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1281] $32161315132^1 2181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1282] $21613151322^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1283] $31612315132^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1284] $16123151322^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1285] $31612315132^1 2181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1286] $1712^1 32171231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1287] $2171232^1 17123215 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 30$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 30$, $T \in [21, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1288] $21712321712^1 3215 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 30$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 30$, $T \in [21, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1289] $21712321712^1 3151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1290] $2171232^1 17123151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1291] $1712^1 31513217123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1292] $1712^1 23151321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1293] $1712^1 23151322171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1294] $322^1 171232181231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 21$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1295] $3221712^1 32181231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 21$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1296] $4132^1 18123151315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1297] $612^1 315131613151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1298] $612^1 321812231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1299] $8122315131612^1 32 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1300] $6131513221812^1 32 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1301] $612^1 315132181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1302] $6123151321812^1 23 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1303] $612^1 315132218123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1304] $612^1 231513218122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1305] $61223151322^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1306] $612^1 231513221812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1307] $7123151321712^1 32 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1308] $7122^1 31513217123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1309] $16131612^1 321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1310] $171232^1 16131612 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1311] $16^1 13161232^1 171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1312] $41321712^1 32181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1313] $4132^1 171232181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1314] $413221812^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1315] $4132^1 2181232171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1316] $4132^1 181232171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1317] $41321812^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1318] $16131612^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1319] $321613161232^1 171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 42, T \in [21, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1320] & 216131612^1 321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1321] & 161232^1 171232161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1322] & 413161315132^1 171 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1323] & 4131612^1 32181231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 21, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1324] & 41321712^1 3218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1325] & 4132^1 17123218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1327] & 413221812^1 321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1329] & 6131612321712^1 32 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 63, T \in [21, 63] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1330] & 612^1 321712321613 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 63, T \in [21, 63] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1331] & 4132^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1332] & 41321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1333] & 413161315132^1 217 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1334] & 4141321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 35, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 35, T \in [21, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1335] & 4131613161232^1 16 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 42, T \in [21, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1218] $1 \overset{313221}{(12)} 12^1 23132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1215] $1 \overset{1}{(12)} 12231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1216] $13221 \overset{1}{(12)} 12^1 23 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1356] $712^1 231513217123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1358] $4131613151322^1 17 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1357] $712^1 321613161232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1336] $1322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.89 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [1337] $1 \overset{2231322^1 1}{(12)} 12231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1338] $3221 \overset{3132^1 21}{(12)} 1223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.89 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [1339] $2231322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1340] $1(12)122^1 3141322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1341] $1(12)12^1 2314132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1342] $21(12)122^1 3141322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1343] $21(12)12^1 2314132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1344] $1613151322^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1345] $413221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.89 \rightarrow 20$, $T_{\max} = 85$, $T \in [20, 85]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[1346] $1413221(12)12^1 22222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1347] $22314132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1348] $23141322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1349] $231513161232^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1350] $232161315132^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1351] $321613151322^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1352] $316123151322^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1353] $8122^1 31513161232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1354] $712^1 321712315132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1355] $712^1 315132171232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1401] $414132^1 181232161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 35$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 35$, $T \in [21, 35]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1400] $61232171232^1 1613 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1403] $812^1 314141414141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
 [1359] $1^{18}1232^{121}181232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 221
 [1360] $1(12)12^{121}23141322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [1361] $21(12)12^{121}23141322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 1
 [1362] $181232^{121}16131513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 321
 [1363] $1^8 12^{1321}2161315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- $^{5132^1 1}$
 [1364] $1^8 12321613 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 51321
 [1365] $1^8 1232^{121}1613 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 321
 [1366] $1^8 12^{131513161} + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- 1
 [1367] $1812315132^{121}1613 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 2321
 [1368] $21^8 12^{1321}21812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- 3221
 [1369] $41322^{121}(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 3151321
 [1370] $61232^{121}1^8 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 321
 [1371] $612315132^{121}1^8 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1372] $223141322^{121}(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1373] $5132^1 16131613151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1374] $1612321812^1 23151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1375] $513161232^1 181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1376] $315131612^1 321812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1377] $316123218122^1 315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1378] $31612^1 3218122315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1379] $161315132^1 181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1380] $161315132^1 218123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1381] $31612^1 3151321812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1382] $31612^1 3151322181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1383] $171232^1 171231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1384] $321712^1 321712315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1385] $321712321712^1 315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1386] $1712315132^1 17123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1387] $41321812^1 3151315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1388] $61232^1 1812231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1389] $812^1 321613151322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1390] $812^1 231513161232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1391] $612315132^1 181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1392] $612315132^1 218123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1393] $7122315132^1 17123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1394] $16^1 131612^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1395] $41322^1 181232171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1396] $1613161232^1 17123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1397] $3216131612^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1398] $41322^1 1812321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1399] $61232^1 1712321613 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1402] $41316131612^1 3216 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1412] $6123218122^1 31513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -64$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.94 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1413] $16131612^1 3217123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 42, T \in [20, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1414] & 612321712^1 321613 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1404] & 3221 \begin{smallmatrix} 31322^1 1 \\ (12) \end{smallmatrix} 1223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.69 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1405] & 1 \begin{smallmatrix} 321 \\ 8 \end{smallmatrix} 1232^1 161315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1406] & 1 \begin{smallmatrix} 51321 \\ 8 \end{smallmatrix} 12^1 321613 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1407] & 612321 \begin{smallmatrix} 315132^1 1 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1408] & 5131612^1 32181223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1409] & 3161232^1 18122315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1410] & 1712321712^1 31513 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 52, T \in [20, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1411] & 32171232^1 1712315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1418] & 71232^1 1712315132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1419] & 71232^1 1613161232 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1417] & 6123151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 52, T \in [20, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1415] & 2315131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1416] & 1613151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 42, T \in [20, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1422] $61315132^1 2181232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1424] $613161232^1 171232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1423] $712315132^1 171232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1420] $61315132^1 1^{2321} 8 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1421] $31612321812^1 2315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1427] $712321712^1 315132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.69 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1428] $6131612^1 32171232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.69 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1425] $612321812^1 231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.69 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1426] $81232^1 1613151322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.69 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1429] $613151322^1 181232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -169$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.59 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$T = 16$ [1503] $413221812321812^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1502] $216131612321812^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1461] $21^{21513221} (12) 1222^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1430] $1^{21} (12) 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $22^1 31513221$
 $T_{\min} = 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1431] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 2321513221
 $T_{\min} = 23, T_{\max} = 64, T \in [23, 64]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1432] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 231513221
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1433] $1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 321513221
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1434] $1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 31513221
 $T_{\min} = 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1435] $1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 21513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1436] $1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1437] $1 \begin{smallmatrix} 2^1 21 \\ (12) \end{smallmatrix} 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1438] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 321513221
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1439] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 31513221
 $T_{\min} = 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1440] $1 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 21513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1441] $1 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1442] $1 \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} 12^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 21513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1443] & 1 \quad (12) \quad 12^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1444] & 3221 \quad (12) \quad 12^1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1445] & 21 \quad (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1446] & 21 \quad (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1447] & 1 \quad (12) \quad 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 43, T \in [23, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1448] & 1 \quad (12) \quad 12^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 43, T \in [23, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1449] & 1 \quad (12) \quad 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 64, T \in [23, 64] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1450] & 1 \quad (12) \quad 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 43, T \in [23, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1451] & 1 \quad (12) \quad 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 43, T \in [23, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1452] & 1 \quad (12) \quad 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 39, T \in [23, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1453] & 1 \quad (12) \quad 12222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1454] & 1 \quad (12) \quad 12222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1455] 1513221 3221 (12) $1222^1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1456] $^{22^1 31513221}$ 21 (12) $12 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1457] 2321513221 21 (12) $12^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 64$, $T \in [23, 64]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1458] 231513221 21 (12) $12^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1459] 321513221 21 (12) $122^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1460] 31513221 21 (12) $122^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1505] $^{2^1 21}$ 413221(12)122314 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1462] 1513221 21 (12) $1222^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1504] 1 41321812321812 $^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1464] 31513221 22 $^1 1$ (12) $122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1465] 21513221 22 $^1 1$ (12) $1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1466] 21513221 221 (12) $122^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1467] 1513221 22 $^1 1$ (12) $1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1468] $221 \overset{1513221}{(12)} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1469] $22^1 21(12)1223151322 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1470] $2221(12)122315132^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1471] $22^1 221(12)1223151232 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 64, T \in [23, 64]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1472] $22^1 221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1473] $22^1 2221(12)122315123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1474] $22^1 2221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1475] $22^1 22221(12)12231512 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1476] $22^1 22221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1477] $1 \overset{132^1 21}{(12)} 122314132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1478] $21 \overset{132^1 21}{(12)} 12231413 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1479] $17123151321812^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1480] $1413221(12)1223132^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1481] $22^1 31413221(12)12231 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1482] $2231413221(12)122^131 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1483] $512321712321812^12 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 30, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 30, T \in [23, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1484] $151321712321812^12 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1485] $217123151321812^12 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1486] $171223151321812^12 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1487] $812321712315132^12 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1488] $1^{22^1221}(12)122314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1489] $21^{22^121}(12)122314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1490] $1^{21}(12)12231414132^12 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1491] $22^11^{221}(12)122314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1492] $1^{22^11}(12)12231414132 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1493] $1^{22^121}(12)1223141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 75, T \in [23, 75]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1494] $21^{22^11}(12)1223141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 75, T \in [23, 75]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1495] $1^{16131612321812^12} + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

- [1496] $41413221^{3221}(12)122^1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 80, T \in [22, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1497] $141413221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1498] $22^1 3141413221(12)122 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1499] $223141413221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1500] $23141413221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1501] $3141413221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 75, T \in [23, 75]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1463] $22^1 1^{321513221}(12) 122 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1591] $32^1 16131612321812 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1590] $161316123218122^1 3 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1593] $612321712321712^1 3 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1506] $1^{321513221}(12) 12^1 22 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1507] $1^{32^1 1513221}(12) 1222 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1508] $1^{31513221}(12) 12^1 22 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1509] $1^{21513221}(12) 122^1 22 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_b$$

$$T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_b$$

15132¹21

$$\min_{(0 \ 3 \ 2)} = 22.2$$

$$T = \langle 12, 12, 12, 12 \rangle, T = \langle 12, 12, 12, 12 \rangle, T = \langle 12, 12, 12, 12 \rangle$$

$$\min = 22.2$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 10 \end{pmatrix}$$

$$T : \quad = 22 \ 22 \rightarrow 23 \quad T \quad = 13 \quad T \in [23 \ 13]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}.$$

$$T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.22 \rightarrow 23, \quad T_{\max} = 43, \quad T \in [23, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_b$$

$$T_{\min} = 22.22 \rightarrow 23, \quad T_{\max} = 39, \quad T \in [23, 39]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_b$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix},$$

$$\begin{array}{r} 1 \\ 15132 \end{array}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_L$$

$$\begin{array}{r} 15132 \\ \times 2317 \\ \hline \end{array}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_1}$$

$$\sqrt{2} \approx 1.414213562$$

- [1534] $^{1513221} 2221 \ (12) \ 122^1 3 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1535] $^{32^1 1513221} 21 \ (12) \ 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1536] $^{321513221} 21 \ (12) \ 12^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1537] $^{31\bar{5}13221} 21 \ (12) \ 12^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1538] $^{215132^1 21} 21 \ (12) \ 12222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1539] $^{21513221} 21 \ (12) \ 122^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1540] $^{15132^1 21} 21 \ (12) \ 12222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1541] $^{1513221} 21 \ (12) \ 122^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1542] $^{32^1 1513221} 221 \ (12) \ 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1543] $^{215132^1 21} 221 \ (12) \ 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1544] $^{21513221} 221 \ (12) \ 12^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1545] $^{1513221} 221 \ (12) \ 12^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1546] $^{15132^1 21} 221 \ (12) \ 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1547] $222^1 1(12)1223151322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1548] $222^1 21(12)1223151232 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 64$, $T \in [23, 64]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1549] $222^1 21(12)122315132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1550] $222221(12)12231512^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1551] $222^1 221(12)122315123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1552] $222^1 221(12)12231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1553] $2222221(12)122^1 31512 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1554] $222^1 2221(12)12231512 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1555] $2222221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1556] $222^1 2221(12)1223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1557] $1^{2313221} (12) 122^1 3141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1558] $1^{1322^1 1} (12) 122314132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1559] $32^{3221} 21(12)1223141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1560] $21^{313221} (12) 122^1 3141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -102$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

- [1561] $^{3132^1 21} 21 (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [1562] $^{3132^1 21} 1 (12) 12231413 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1563] $^{1322^1 1} 21 (12) 12231413 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1564] $^{32^1 1} 1 \ 8 \ 12321712315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1565] $^{51321} 1 \ 8 \ 12321712^1 3 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1566] $^1 18123151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1567] $^{32^1 1} 1 \ 8 \ 12315132171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1568] $^{32^1 21} 413221(12)12231 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1569] $^{32^1 1} 71223151321 \ 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1570] $^{14132^1 21} 14132^1 21(12)12231322 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1571] $^{2231413221} 2231413221(12)12^1 231 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1572] $^{231413221} 231413221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 29, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1573] $^{513217123218122^1 3} 513217123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1574] $^{32^1 1} 17123218122315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1575] $171231513218122^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1576] $32^1 17123151321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1577] $171231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1578] $32^1 17123151322181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1579] $712231513218122^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1580] $712231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1581] $1^{222^1 21} (12) 122314141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [1582] $21^{222^1 1} (12) 122314141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [1583] $1^{222^1 1} (12) 1223141413 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 75$, $T \in [23, 75]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1584] $1^{18123216131612^1 3} + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1585] $1^{32^1 1} 8 12321613161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1586] $41413221^{32^1 21} (12) 122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 80$, $T \in [23, 80]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1587] $141413221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1588] $23141413221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 80$, $T \in [23, 80]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1589] $3141413221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 75$, $T \in [23, 75]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1592] $613151316131612^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 53$, $T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $^{2151322^1 1}$
 [1623] $21 \quad (12) \quad 12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 1
 [1711] $4132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 3221
 [1674] $413221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.25 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 1
 [1594] $1 \quad (12) \quad 1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{3215132^1 21}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1595] $1 \quad (12) \quad 1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{315132^1 21}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1596] $1 \quad (12) \quad 12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{2151322^1 1}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1597] $1 \quad (12) \quad 12^1 222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 21513221
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1598] $1 \quad (12) \quad 12^1 222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1599] $1 \quad (12) \quad 12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{151322^1 1}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $^{22^1 1}$
 [1600] $1 \quad (12) \quad 1223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 21.25 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 221
 [1601] $1 \quad (12) \quad 12 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{3215132^1 21}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1627] $221 \overset{315132^1}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1628] $221 \overset{2151322^1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1629] $221 \overset{151322^1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1630] $512^1 3218123151315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1631] $2222^1 1(12)1223151232 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1632] $2222^1 1(12)122315132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1633] $222221(12)122^1 315123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1634] $2222^1 21(12)122315123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1635] $2222^1 21(12)12231513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1636] $222221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1637] $2222^1 221(12)12231512 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1638] $2222221(12)12^1 231512 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1639] $2222^1 221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1640] $2222221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

- [1654] $21^{13221}(12)122^1 31413 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -102$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1655] $1712^1 32181223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1656] $18^{21}12^1 3217123215 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1657] $18^{21}1232171232^1 15 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [1658] $1^{512321}8^{1232^1}1712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1659] $1^{512^1}8^{321}12321712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1660] $18^{21}12^1 3217123151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1661] $1^{151321}8^{1232^1}1712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1662] $17^{12315132^1}18122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1663] $17^{12^1}131513218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1664] $18^{21}12^1 3151321712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1665] $18^{21}12315132^1 1712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1666] $17^{12315132^1}21812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1667] $17^{12^1}131513221812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1668] $\overset{1}{18122315132^1 1712} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{18122^1 3151321712} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{21}{1812315132^1 2171} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{21}{1812^1 3151322171} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{18122^1 3151322171} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{18122315132^1 2171} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1714] 414132181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 32, \Delta = -114, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1675] 141322^1 1(12)12231322 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1713] 414132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 32, \Delta = -114, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1677] 314132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 29, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1678] 231413221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 29, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $[1679] 21712^1 32181223151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $[1680] 512^1 3217123218122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 30, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $[1681] 51232171232^1 18122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 30, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

- $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1697] $1712^1 32171232161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1698] $171232171232^1 161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1699] $41413221^{3221}(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.25 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1700] $41322181232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1701] $4132181232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1702] $1613161232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1703] $1414132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1704] $141413221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1705] $3141413221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 75$, $T \in [22, 75]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1706] $21613161232^1 18122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1707] $1612^1 321712321712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1708] $161232171232^1 1712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1709] $41322181232^1 18122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1710] $1612^1 32181232161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1712] & 413221^{22^1 1}(12)122314 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 80, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.25 \rightarrow 22, T_{\max} = 80, T \in [22, 80] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1752] & 22222^{1^1}(12)122315123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1867] & 41321812315132^{1^1}16 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1811] & 1712^{1^1}231513218122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1715] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 23215132^{1^1}21 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1716] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 232^{1^1}1513221 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1717] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 2315132^{1^1}21 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1718] & 1 \quad \begin{pmatrix} 1 \\ 12 \end{pmatrix} \quad 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 32151322^{1^1}1 \\ 12 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1719] & 1 \quad \begin{pmatrix} 1 \\ 12 \end{pmatrix} \quad 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 3151322^{1^1}1 \\ 12 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1720] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 12^{1^1}23 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 1513221 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1721] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 15132^{1^1}21 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1722] & 1 \quad \begin{pmatrix} 21 \\ 12 \end{pmatrix} \quad 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 23215132^{1^1}21 \\ 21 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1723] $1 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $232^1 1513221$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1724] $1 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $231^1 15132^1 21$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1725] $1 \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $32151322^1 1$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1726] $1 \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $3151322^1 1$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1727] $21 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12^1 23 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1728] $21 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $15132^1 21$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1729] $21 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $32151322^1 1$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1730] $21 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $3151322^1 1$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1731] $1 \begin{smallmatrix} 232^1 1513221 \\ (12) \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1732] $1 \begin{smallmatrix} 23215132^1 21 \\ (12) \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1733] $1 \begin{smallmatrix} 2315132^1 21 \\ (12) \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1734] $1 \begin{smallmatrix} 32151322^1 1 \\ (12) \end{smallmatrix} 12222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

- [1735] $1 \overset{3151322^1 1}{(12)} 12222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1736] $1 \overset{21513221}{(12)} 12^1 2222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1737] $1 \overset{1513221}{(12)} 12^1 2222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1738] $2221 \overset{1513221}{(12)} 12^1 23 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1739] $2221 \overset{15132^1 21}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1740] $21 \overset{23215132^1 21}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1741] $21 \overset{232^1 1513221}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1742] $21 \overset{2315132^1 21}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1743] $21 \overset{32151322^1 1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1744] $21 \overset{3151322^1 1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1745] $221 \overset{32151322^1 1}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1746] $221 \overset{3151322^1 1}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1747] $51232^1 18123151315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [1748] $15132^1 18123151315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 33$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 33$, $T \in [22, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1749] $22221(12)12231512^1 32 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1750] $22221(12)122^1 3151232 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1751] $22221(12)122^1 315132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1870] $4131613161232^1 171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1753] $222221(12)12^1 2315123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1754] $22222^1 1(12)12231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1755] $222221(12)12^1 231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1756] $22222^1 21(12)12231512 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1757] $22222^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1758] $171232^1 18123215 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1759] $1712321812^1 3215 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1760] $1712321812^1 3151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1761] $171232^1 18123151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1762] $17123151322^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 231322^1 \\ [1763] \end{matrix} 1 (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 13221 \\ [1764] \end{matrix} 1 (12) 122^1 314132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 3221 \\ [1765] \end{matrix} 322^1 1 (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 313221 \\ [1766] \end{matrix} 1 (12) 12^1 231413 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 13221 \\ [1767] \end{matrix} 21 (12) 12^1 231413 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1768] \end{matrix} 1812^1 32171231513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 3151321 \\ [1769] \end{matrix} 1^1 8^1 1232^1 171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1770] \end{matrix} 17123218122^1 3151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1771] \end{matrix} 171232^1 181223151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ [1772] \end{matrix} 181232^1 17123215 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 30, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ [1773] \end{matrix} 1812321712^1 3215 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 30, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 512321 \\ [1774] \end{matrix} 1^1 8^1 12^1 321712 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 30, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 51232^1 1 \\ [1775] \end{matrix} 1^1 8^1 12321712 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 30, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1776] $18^{21}12321712^13151 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1777] $18^{21}1232^117123151 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1778] $1^{15132^11}8^{12321712} + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1779] $1^{151321}8^{12^1321712} + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1780] $1812^131513217123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1781] $1^{321}8^{12315132^1171} + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1782] $17123151322^11812 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1783] $1812^123151321712 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1784] $18^{21}123151322^1171 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1785] $1812^123151322171 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1786] $181223151322^1171 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1787] $413221^{322^11}(12)12231 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1788] $4132^{3221}21(12)12231 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1789] $7123151321^{232^11}8^1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1868] $4141321812^1 32171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1791] $1413221(12)122^1 31322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1792] $31413221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1793] $3141322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1794] $2314132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1795] $1513161315132^1 171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1796] $1712^1 321812231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1797] $51321712321812^1 23 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1798] $315132171232^1 1812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1799] $217123218122^1 3151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1800] $2171232^1 181223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1801] $51232^1 17123218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1802] $512321712^1 3218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1803] $15132^1 17123218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1804] $151321712^1 3218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

- [1805] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 1712¹315132181223 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1806] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 17123151321812¹23 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1807] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 232¹1712315132181 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- [1808] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 321712315132¹1812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1809] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 1712¹315132218123 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1810] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 321712315132¹2181 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1813] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 171223151322¹1812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1812] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 217123151322¹1812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1869] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 4141321812¹321712 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
- [1814] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ 1712¹231513221812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1815] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 7122¹315131613151 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
- [1816] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 71223151321812¹23 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1817] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 7122¹315132181223 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1818] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 81223151321712¹32 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- [1819] 71231513221812¹32 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [1820] $7122^1 315132218123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 22221
 [1821] $1(12)12^1 2314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- 2221
 [1822] $21(12)12^1 2314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- 221
 [1823] $221(12)12^1 2314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- 2221
 [1824] $1(12)122^1 3141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 75, T \in [22, 75]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 221
 [1825] $21(12)122^1 3141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 75, T \in [22, 75]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1826] $1812^1 32161316123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1827] $16131612^1 3218122 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 21
 [1828] $181232^1 16131612 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 31
 [1829] $1613161232^1 1812 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 36, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 36, T \in [22, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [1830] $1712321712^1 32161 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 1
 [1831] $171232^1 171232161 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $^{322^1 1}$
 [1832] $41413221(12)122 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 80, T \in [22, 80]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 1
 [1833] $413161315132^1 181 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 36, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1834] $4132^1 21812321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1835] $413221812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1836] $612^1 315131613161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1837] $6131612321^{232^1 1} 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1838] $41321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1839] $4132^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1840] $16131612^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1841] $1613161232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1842] $141413221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1843] $14141322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1844] $31414132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 75$, $T \in [22, 75]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1845] $16131612321812^1 23 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1846] $232^1 1613161232181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1847] $321613161232^1 1812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$

- [1848] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 216131612¹3218122 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1849] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 1612321712¹321712 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1850] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 161232¹1712321712 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1851] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 3161232171232¹171 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
- [1852] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 413161315132¹1812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
- [1853] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 413161315132¹2181 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
- [1854] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 4132¹181232161315 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- [1855] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 4132¹181231513161 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1856] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 413221812¹3218122 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
- [1857] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 4132¹218123218122 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
- [1858] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 612¹3216131613151 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
- [1859] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ 612¹3151316131612 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
- [1860] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 612¹3217123217123 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- [1861] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 161232¹181232161 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
- [1862] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 4132¹18123218122 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [1863] $41321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1864] $612^1 321812321613 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [1865] $4132^1 21^{221} 122314 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [1866] $4132^1 181231513216 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1790] $7122^1 3151321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1914] $4131613151322^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1911] $231414132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1912] $314141322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 75$, $T \in [22, 75]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1871] $1^{21} (12)^{21} 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1872] $1^{221} (12)^{221} 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1873] $1^{221} (12)^{221} 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1874] $1^{21} (12)^{21} 12 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1875] $1^{21} (12)^{21} 12 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$

- [1889] $3221(12)122^1 3141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.89 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [1890] $413221(12)12^1 231 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.89 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [1891] $712^1 321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1892] $712^1 3151321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1893] $712^1 23151321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1894] $1413221(12)12^1 231322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1895] $22314132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1896] $23141322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1897] $2315132171232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1898] $2321712315132^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1899] $3217123151322^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1900] $712^1 3151316131513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1901] $712^1 2315131613151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1902] $712^1 3218122315132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1903] $712^1 2315132181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$

- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [1904] $8122^1 315132171232 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [1905] $712^1 3151322181232 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [1906] $712^1 2315132218123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 36, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [1907] $1(12)^{221} 122^1 31414132 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 36, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 80, T \in [22, 80]$
- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [1908] $1(12)^{2221} 12^1 23141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 75, T \in [22, 75]$
- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [1909] $21(12)^{21} 122^1 31414132 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 36, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 80, T \in [22, 80]$
- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [1910] $21(12)^{221} 12^1 23141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 75, T \in [22, 75]$
- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [1913] $2321613161232^1 181 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
- $$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [1967] $31612321712^1 32171 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
- $$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [1979] $414132^1 1812321712 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 32, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
- $$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [1915] $1(12)^{21} 1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
- $$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [1916] $1(12)^{221} 1223 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
- $$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$
- [1917] $21(12)^{21} 1223 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
- $$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

- [1918] $1 \overset{223151322^1}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1919] $2221 \overset{151322^1}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1920] $21 \overset{223151322^1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1921] $512321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1922] $151321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 33, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 33, T \in [22, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1923] $2221(12)12^1 23151322 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1980] $41316131612^1 32171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 36, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 36, T \in [22, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1924] $181232^1 171231513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1925] $1 \overset{3151321}{8} 12^1 32171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1926] $1712321812^1 23151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1927] $1 \overset{321}{8} 12321712^1 315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1928] $1 \overset{321}{8} 12^1 321712315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1929] $1 \overset{51321}{8} 1232^1 17123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1930] $1 \overset{5132^1}{8} 123217123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [1931] $1812315132^1 17123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 321
 [1932] $1\ 8\ 12^1 315132171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- 3221
 [1933] $41322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 321
 [1934] $7122315132^1 1\ 8\ 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1935] $223141322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1936] $171232^1 1812231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1937] $5132^1 171232181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1938] $5132171232^1 181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1939] $321712^1 3218122315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1940] $3217123218122^1 315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1941] $3151321712^1 321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1942] $21712321812^1 23151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1943] $1712315132^1 181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1944] $321712^1 3151321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1945] $1712315132^1 218123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1946] $321712^1 3151322181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1947] $812^1 3217123151322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1948] $7122315132^1 181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1949] $812^1 2315132171232 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1950] $7122315132^1 218123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 21
 [1951] $1(12)122^1 314141322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 221
 [1952] $1(12)12^1 231414132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [1953] $21(12)12^1 231414132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 1
 [1954] $161316131612^1 315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1955] $181232^1 161316123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 321
 [1956] $1812^1 321613161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- 31
 [1957] $16131612^1 321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- 3221
 [1958] $414132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 21
 [1959] $41322^1 181232181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1960] $1\overset{31}{6}131612^132181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1961] $2231414132^121(12)122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1962] $2314141322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1963] $2161316131612^1315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1964] $1613161232^1181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1965] $3216131612^1321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1966] $31612^132171232171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1968] $41321812^132161315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1970] $41322^118123218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1971] $61232^116131613151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1972] $61232171232^117123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1973] $61232^117123217123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1974] $61232^11812321613 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1975] $61232181232^11613 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [1976] $41322^1 1(12)122314 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1977] $41321812^1 31513216 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1978] $414132^1 181232171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1969] $41321812^1 31513161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1981] $3221(12)12^1 23141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1997] $612321812^1 321613 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1983] $1^1 8^1 1232171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 43$, $T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1984] $1^1 8^1 1232^1 1712315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1985] $1^1 8^1 12^1 3217123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1986] $17123218122^1 31513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 43$, $T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1987] $51321712^1 32181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1988] $32171232^1 18122315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1989] $315132^1 1712321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 43$, $T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1990] $1(12)12^1 2314141322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 80$, $T \in [21, 80]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

- [1991] $1\ 8\ 1232^1 1613161 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 39, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 39, T \in [21, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1992] $22314141322^1 1(12)122 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 80, \Delta = -122, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 80, T \in [21, 80]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1993] $16131612^1 32181223 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 39, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 39, T \in [21, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1994] $3161232^1 171232171 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 43, \Delta = -118, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 43, T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1995] $4132181232^1 161315 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 53, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1996] $612321712^1 3217123 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 53, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1982] $1812321712^1 31513 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 43, \Delta = -118, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 43, T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1998] $71232^1 1\ 8\ 1 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 53, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1999] $23151321712^1 32181 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 53, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2000] $17123151322^1 18123 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 39, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 39, T \in [21, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2001] $71232^1 18122315132 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 53, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2002] $71223151322^1 18123 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 39, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 39, T \in [21, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2003] $4141322^1 1(12)122 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 80, \Delta = -122, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 80, T \in [21, 80]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2004] $23216131612^1 32181 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 53, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2010] $613161232^1 1\ 8\ 1 + \hat{1} \rightarrow 2\ T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2005] $712315132^{\overset{2321}{1}} 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2006] $321712321812^{\overset{1}{2315}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2007] $2321712^{\overset{1}{315132181}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2008] $8122315132^{\overset{1}{171232}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2009] $712315132^{\overset{1}{2181232}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2016] $6131612^{\overset{2321}{321}} 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2011] $712321^{\overset{2315132^{\overset{1}{1}}}{8}} 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2012] $1712321812^{\overset{1}{231513}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 43, T \in [21, 43]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2013] $2315132^{\overset{1}{171232181}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2014] $7123218122^{\overset{1}{315132}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2015] $81232^{\overset{1}{17123151322}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2017] $7123151322^{\overset{1}{181232}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 169, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.59 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2018] $812321712^{\overset{1}{3151322}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 196, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.55 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2019] $712321812^{\overset{1}{2315132}} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 225, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.52 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

- $T = 17$ [2028] $22^1 313221(12)1223151 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2050] $4141413221(12)12^1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 76, T \in [23, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2047] $1(12)1223141414132^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 76, T \in [24, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2049] $1(12)122314141413 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 65, \Delta = -120, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 65, T \in [24, 65]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2051] $14141413221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 31, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2046] $1(12)12231414141 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 31, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2045] $1612321712321812^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2044] $23141413221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2043] $1(12)1223141413 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2042] $8123218122315132^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2041] $4131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2040] $2181223151321812^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2039] $22^1 181231513218122 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2038] $22^1 181232181223151 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [2037] $812321^{22^1 3151321} 8^1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2036] $4131513221^{221} (12) 12^1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2035] $4131513221^{22^1 21} (12) 12 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2034] $4131513221^{22^1 221} (12) 1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2033] $413221^{1513221} (12) 122^1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 40, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 40, T \in [23, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2032] $181223151321812^{12} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2031] $18^{21} 123151321812^{12} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2030] $1^{151321} 8^1 12321812^{12} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2026] $13221^{12} 122315132^{12} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2025] $132^{121} 21^{12} 1223151322 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2024] $3221^{2^1 1513221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 24, T \in [22, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [2023] $21^{321513221} (12) 122^1 31 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2022] $1^{231513221} (12) 122^1 31 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2021] $1^{2321513221} (12) 122^1 31 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2020] $1 \overset{1}{(12)} 122^1 31 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 321513221
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2029] $1 \overset{512321}{8} 12321812^1 2 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2027] $22^1 313221(12)12231512 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2052] $22^1 314141413221(12)12 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 76, T \in [24, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2053] $2314141413221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 76, T \in [24, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2054] $314141413221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 65, \Delta = -120, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 65, T \in [24, 65]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2055] $4141321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 32, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2056] $413221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 76, T \in [23, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2048] $21(12)12231414141 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 31, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2076] $3132^1 21(12)122315132 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2075] $3132^1 21(12)1223151232 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2074] $22313221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2073] $22313221(12)122^1 31512 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2072] $2313221(12)12231512^1 3 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2071] $1322^1 1(12)1223151322 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2070] $32^1 21^{1513221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2069] $32^1 21^{21513221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2077] $1^3 8^{321} 12321812315 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 44, T \in [24, 44]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2067] $21^{321} 1513221 (12) 12231 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2066] $13221^{1513221} (12) 122^1 3 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2065] $1^{31513221} (12) 122^1 313 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [2064] $1^{321513221} (12) 122^1 313 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [2063] $1^{321} 1513221 (12) 122313 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2062] $1^{151321} (12) 1223132 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2061] $1^{2151321} (12) 1223132 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2060] $1^{231513221} (12) 12^1 231 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [2059] $1^{2321513221} (12) 12^1 231 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [2058] $\overset{1}{32^1 1513221} (12) 12231 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{321513221} (12) 12^1 231 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{18123151321812^1 3} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{51321}{1} 8^1 123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{1} 8^1 123218122315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{181231513218122^1 3} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{1} 8^1 123151321812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{181231513221812^1 3} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{1} 8^1 123151322181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{222^1 21}{4131513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{222^1 1}{4131513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{5131613151321812^1 3} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 65, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 65, T \in [24, 65]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{32^1 218123218122315} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [2089] $2181231513218122^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2090] $1812231513218122^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2091] $32^1 181223151321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [2092] $1812231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2093] $4131513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2094] $1 \begin{pmatrix} 3132^1 21 \\ 12 \end{pmatrix} 122314141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [2095] $1 \begin{pmatrix} 1322^1 1 \\ 12 \end{pmatrix} 1223141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2096] $612321712321 \begin{pmatrix} 32^1 1 \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 44$, $T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2097] $23141413221(12)12^1 231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2098] $3141413221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2099] $14132^1 21(12)122314132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2100] $1712321712321712^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 40$, $T \in [24, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2101] $7123151316131612^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 44$, $T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2102] $6123217123218122^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 44$, $T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2103] $14132^1 21 \begin{pmatrix} 1 \\ 12 \end{pmatrix} 12231413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$

- [2104] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}^1$ 612321812321712¹3 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 44$, $T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2105] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}^{222^1 1}$ 1(12)12231414141 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [2106] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}^{32^1 21}$ 4141413221(12)12 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 76$, $T \in [24, 76]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2107] 14141413221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2108] 314141413221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 65$, $T \in [24, 65]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2109] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}^1$ 613161316131612¹3 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2068] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}^{321513221}$ 21 (12) 12¹231 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2137] 31322¹1(12)1223151232 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2136] 22313221(12)12¹23151 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2197] 414132181232¹18122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2138] 31322¹1(12)122315132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2139] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{21}$ 1812¹3218123215 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2140] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}^{21}$ 181232181232¹15 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2141] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{21}$ 1812¹3218123151 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [2155] $^{21}1812^131513221812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2156] $^{11}18122315132^121812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2157] $^{11}18122^131513221812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2158] $^{1513221}413221(12)12^12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.25 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2159] $^{22221}41315132^121(12)1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2160] $^{22221}4131513221(12)1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2161] $^{2221}41315132^121(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2162] $^{221}41315132^121(12)122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2163] $^{1612}1612^13218123151315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2164] $^{513161232181232}115 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2165] $^{161315132181232}115 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2166] $^{221812}132181223151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2167] $^{221812}131513218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2168] $^{218122315132}118122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2169] $^{218122}131513218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2170] & 218122^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2171] & 4131513221(12)122^1 222 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2172] & 41315132^1 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2173] & 612315132181232^1 15 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 34, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 34, T \in [23, 34] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 313221 \\ 1 \end{matrix} \\
[2174] & 1 (12) 122^1 314141 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 31322^1 1 \\ 1 \end{matrix} \\
[2175] & 1 (12) 122314141 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ 1 \end{matrix} \\
[2176] & 181232171232^1 161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ 1 \end{matrix} \\
[2177] & 1812^1 32171232161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2178] & 171232171232^1 1712 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2179] & 1712^1 321712321712 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2180] & 171232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2181] & 1712^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2182] & 1712^1 32171232171 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2183] 141413221(12)122¹3132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2184] 1414132¹21(12)1223132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2185] 3141413221(12)12¹2313 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2186] 141322¹1(12)122314132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2187] 314132¹21(12)12231413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2188] 161232171232¹18122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2189] 1612¹3217123218122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2190] 21712¹321712321712 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2191] 1413221¹(12)122¹31413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2192] 141322¹1(12)12231413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2193] 161232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2194] 1612¹321812321712 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2195] 414132181232¹181 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2110] 1¹(12) 12231 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
3215132¹21

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 $\begin{smallmatrix} 31513221 \\ 1 \end{smallmatrix} (12) 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{smallmatrix} 1513221 \\ 1 \end{smallmatrix} (12) 1223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.25 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{smallmatrix} 3215132^1 21 \\ 1 \end{smallmatrix} (12) 12231 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{smallmatrix} 315132^1 21 \\ 1 \end{smallmatrix} (12) 12231 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{smallmatrix} 21513221 \\ 1 \end{smallmatrix} (12) 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{smallmatrix} 2151322^1 1 \\ 1 \end{smallmatrix} (12) 122313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{smallmatrix} 151322^1 1 \\ 1 \end{smallmatrix} (12) 122313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{smallmatrix} 1513221 \\ 1 \end{smallmatrix} (12) 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[2131] 2313221(12)122^1 315123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $[2132] 23132^1 21(12)122315123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[2133] 2313221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $[2134] 23132^1 21(12)12231513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[2135] 22313221(12)12^1 231512 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2196] 14141413221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2232] 1⁵¹³²¹8^{12321812¹23} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2231] 1^{1812¹321812231513} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2292] 612¹321712321³²¹8¹1 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2291] 7122¹315131613161¹ + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[2290] 1^{1712321712¹321712} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2289] 1^{171232¹1712321712} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2288] 1^{1812¹321712321613} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2287] 1^{181232¹171232161} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2286] 1^{1812321712¹32161} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2285] 1^{1613161315132¹171} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2284] 1¹³²²¹(12)122¹3141413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2283] 1³¹³²²¹(12)12¹2314141 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[2282] 812231513221812¹32 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 32, T \in [23, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 32, T \in [23, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 34, T \in [23, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 34, T \in [23, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [2267] $218123151321812^1 23 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2266] $2218123218122^1 3151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2265] $22181232^1 181223151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2264] $21812^1 321812231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2263] $322^1 18123218122315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2262] $1513161315132^1 2181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2261] $1513161315132^1 1812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2260] $1613151321812^1 3151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2259] $1613151321812^1 3215 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2258] $5131612321812^1 3151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2257] $5131612321812^1 3215 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2256] $161232^1 18123151315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2255] $81223151321^{232^1 1} 8^1 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2254] $413151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2253] $413151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2252] $413151322^1 (12) 1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [2251] $4132^1 21 (12) 122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [2250] $413221^{15132^1 21} (12) 122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [2249] $1812^1 231513221812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2248] $181223151322^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2247] $18123151322^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [2246] $1812^1 231513218122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2245] $1812315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2244] $1812^1 315132218123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2243] $1812315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2242] $1812315132181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2241] $18123151321812^1 23 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2240] $1812^1 315132181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2239] $1812315132^1 18122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2238] $1 \quad 8 \quad 12^{1321} 3218122 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2237] $1 \quad 8 \quad 12^{1321} 3218122 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2236] $1 \quad 8 \quad 12^{512321} 23218122 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2235] $1 \quad 8 \quad 12^{21} 3218122^{1321} 181223151 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2234] $1 \quad 8 \quad 12^{21} 3218122^{1321} 3151 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2198] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2199] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 23$, $\Delta = -97$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2200] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2201] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2202] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2203] $1 \quad (12) \quad 12^{21513221} 23132 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2204] $1 \quad (12) \quad 12^{1513221} 23132 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2205] $1 \quad (12) \quad 12^{32151322} 2313 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹₃₁₅₁₃₂₂
 [2206] $1 \quad (12) \quad 122313 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹₁₅₁₃₂²¹
 [2207] $13221 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- ¹₁₅₁₃₂₂₁
 [2208] $13221 \quad (12) \quad 12^1 23 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- ¹₃₂₁₅₁₃₂₂
 [2209] $21 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹₃₁₅₁₃₂₂
 [2210] $21 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 23$, $\Delta = -97$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹₂₁₅₁₃₂₂₁
 [2211] $322^1 1 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- ¹₁₅₁₃₂₂₁
 [2212] $322^1 1 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- ¹₂₃₁₃₂₂¹
 [2213] $1(12)122315123 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹₂₃₁₃₂₂₁
 [2214] $1(12)12^1 2315123 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹₂₃₁₃₂₂₁
 [2215] $1(12)12^1 231513 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹₂₃₁₃₂₂¹
 [2216] $1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹₂₂₃₁₃₂¹
 [2217] $21(12)12231512 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹₂₂₃₁₃₂¹
 [2218] $21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 24, T \in [23, 24] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2219] & 313221(12)12231512^1 32 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2220] & 313221(12)122^1 3151232 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2221] & 313221(12)122^1 315132 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2222] & 1812^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2223] & 1812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 30, T \in [23, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2224] & 181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 30, T \in [23, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2225] & 181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2226] & 1812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2227] & 1812^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2228] & 1812^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2229] & 1812^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2294] & 171232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2295] & 171232^1 171232171 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2296] 14141322¹1(12)1223132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2297] 141413221(12)12¹23132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2298] 31414132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2299] 1413221(12)122¹314132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2300] 3141322¹1(12)12231413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2301] 15131613161232¹171 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 30, T \in [23, 30]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2302] 21613161315132¹171 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2303] 161232¹17123218122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2304] 1612321712¹3218122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2305] 3161232171232¹1812 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2306] 1712¹3217123217123 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2307] 2171232¹1712321712 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2308] 4132¹1812321712315 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2309] 41321812315132¹171 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [2310] $4132^1 1812315132171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2311] $7122^1 3151316131612 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2312] $612^1 32171232181223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2313] $612321712321812^1 23 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2314] $712321712321712^1 32 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2315] $1413221(12)12^1 231413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2316] $161232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2317] $1612321812^1 321712 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2318] $3161232181232^1 171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2319] $612^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2320] $41321812315132^1 217 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2321] $4132^1 1812315132217 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2322] $1(12)122^1 31414141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2323] $21(12)122^1 31414141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2324] $4141413221(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 76, T \in [23, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2325] & 4141321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 32, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 32, T \in [23, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2326] & 4131613161232^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 34, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 34, T \in [23, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2327] & 612^1 3216131613161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2328] & 141414132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 31, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 31, T \in [23, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2329] & 4141321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 32, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 32, T \in [23, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2330] & 4131613161232^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 34, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 34, T \in [23, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2331] & 4132^1 1812321613161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2332] & 612^1 32161316131612 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2333] & 4132^1 21(12)12231414 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 76, \Delta = -121, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 76, T \in [23, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2293] & 1712321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2230] & 1812^1 315131613151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 30, T \in [23, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2233] & 1^{3151321} 8^{1232^1} 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2337] & 3221^{215132^1} (12)^{21} 1223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.89 \rightarrow 22, T_{\max} = 24, T \in [22, 24] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2338] $3221 \overset{15132^1 21}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.89 \rightarrow 22, T_{\max} = 24, T \in [22, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [2339] $13221(12)122 \overset{1}{1} 3151322 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2340] $2231322^1(12)12231512 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2341] $2231322^1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2342] $313221(12)12^1 23151232 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2343] $313221(12)12^1 2315132 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2344] $1 \overset{23151321}{8} 1232^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2345] $1 \overset{2321}{8} 12315132^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2346] $1 \overset{321}{8} 123151322^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2347] $8122^1 3151321 \overset{2321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2348] $31513161315132^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 54, \Delta = -119, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 54, T \in [23, 54]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2349] $15131613151322^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 30, T \in [23, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2350] $1812^1 3218122315132 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2351] $1812^1 2315132181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [2352] $23218122315132^1 181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2353] $1812^1 2315132218123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2354] $8122^1 3151316131513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2355] $8122^1 3151322181232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2356] $1^{13221} (12) 12^1 23141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2357] $712^1 3151316131^{31} 61 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2358] $712^1 2315131613161^1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2359] $231414132^1 21(12) 12231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2360] $314141322^1 1(12) 122313 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2361] $1413221(12) 12^1 2314132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2362] $712^1 32161316131513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2363] $712^1 31513161316123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2364] $712^1 23151316131612 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2365] $712^1 32171232171232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [2366] $413218123151322^1 17 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2367] $1(12)12^1 231414141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2368] $21(12)12^1 231414141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2369] $1(12)12^1 314141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2370] $21(12)12^1 314141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2371] $1414141322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2372] $3141414132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2336] $1322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.89 \rightarrow 22$, $T_{\max} = 24$, $T \in [22, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [2335] $1(12)12^1 12231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2334] $1(12)12^1 12231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -100$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2441] $414132^1 18123218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2373] $1322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2378] $1812315132^1 18123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2379] $181232^1 1812231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2380] & \begin{matrix} 51321 \\ 1 & 8 & 1232^1 181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2381] & \begin{matrix} 5132^1 1 \\ 1 & 8 & 1232181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2382] & \begin{matrix} 321 \\ 1 & 8 & 123218122^1 315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2383] & \begin{matrix} 321 \\ 1 & 8 & 12^1 3218122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2384] & \begin{matrix} 3151321 \\ 1 & 8 & 12^1 321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[2385] & \begin{matrix} 21 \\ 1 & 8 & 12321812^1 23151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2386] & \begin{matrix} 1 \\ 1 & 8 & 12315132^1 181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2387] & \begin{matrix} 321 \\ 1 & 8 & 12^1 3151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2388] & \begin{matrix} 1 \\ 1 & 8 & 12315132^1 218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2389] & \begin{matrix} 321 \\ 1 & 8 & 12^1 3151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2390] & \begin{matrix} 1513221 \\ 41322^1 1 & (12) & 122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2391] & \begin{matrix} 151322^1 1 \\ 413221 & (12) & 122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2392] & \begin{matrix} 223151321 \\ 812^1 321 & 8 & 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2393] $812^1 23151321^{\overline{2321}} 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2394] $1612321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 33, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 33, T \in [23, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2395] $513161232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2396] $513161232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 33, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 33, T \in [23, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2397] $161315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 33, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 33, T \in [23, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2398] $161315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 31, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2399] $31613151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 65, \Delta = -120, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 65, T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2400] $3221812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2401] $32218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2402] $2181232^1 1812231513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2403] $221812321812^1 23151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2404] $21812315132^1 181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2405] $18122315132^1 181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2406] $3218122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2407] $18122315132^1 218123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [2408] $812^1 31513161315132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2409] $812^1 23151316131513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2410] $612315132^1 18123215 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 34$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 34$, $T \in [23, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2411] $612315132^1 18123151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2412] $812^1 32181223151322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2413] $812^1 23151322181232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2414] $181232171232^1 1613 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2415] $181232^1 1712321613 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2416] $61232^1 1712321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2417] $61232171232^1 1 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2418] $2314141322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2419] $151316131612^1 32171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2420] $31612321712^1 321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2421] $31612^1 321712321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2422] $171232^1 17123217123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2423] $41321812^1 321712315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [2424] $41321812321712^1 315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2425] $41321812^1 315132171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [2426] $61232171232^1 181223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2427] $61232^1 171232181223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2428] $31612^1 32181232171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2429] $31612321812^1 32171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2430] $61232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2431] $61232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2432] $41321812^1 315132217 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [2433] $1(12)122^1 3141414132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 76$, $T \in [23, 76]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [2434] $1(12)12^1 2314141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [2435] $21(12)12^1 2314141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [2436] $414132^1 181232181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

- [2437] $41316131612^1 32181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 34$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 34$, $T \in [23, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2438] $61232^1 16131613161 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [2439] $23141414132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 76$, $T \in [23, 76]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2440] $31414141322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2376] $1 \ 8 \ 12^1 321812315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2442] $41316131612^1 321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 34$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 34$, $T \in [23, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2443] $41321812^1 321613161 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2444] $61232^1 161316131612 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [2445] $41322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 76$, $T \in [23, 76]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2446] $413221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 76$, $T \in [23, 76]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2375] $181232^1 181231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2377] $1 \ 8 \ 12321812^1 315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2374] $13221(12)12^1 23151322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2461] $612321712^1 321 \ 8 \ 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$

- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2451] $1 \overset{1}{18} 123218122^1 31513 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{51321}{1} \overset{1}{8} 12^1 32181223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{321}{1} \overset{1}{8} 1232^1 18122315 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- $\overset{315132^1}{1} \overset{1}{8} 12321812 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $[2455] \overset{1}{51} 31612^1 3218123215 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 37, T \in [22, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2456] \overset{1}{51} 31612^1 3218123151 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 33, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 33, T \in [22, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2457] \overset{1}{32} 2181232^1 18122315 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- $[2458] \overset{1}{21} 8123218122^1 31513 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2459] \overset{1}{61} 3151321812^1 31513 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 65, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 65, T \in [22, 65]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{1}{18} 12321812^1 31513 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{1}{32} 21 \overset{1}{15} 1322^1 1223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22, T_{\max} = 24, T \in [22, 24]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{2151322^1}{32} 21 \overset{1}{15} 1322^1 1223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22, T_{\max} = 24, T \in [22, 24]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2463] \overset{1}{17} 12321712^1 3217123 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2464] \overset{1}{41} 32181232^1 1712315 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$

- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [2465] $612321712^1 32181223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2466] $3161232^1 181232171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2467] $612321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2468] $1(12)122^1 31414141322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [2469] $1(12)12^{21} 23141414132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [2470] $223141414132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2471] $231414141322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2472] $4132181232^1 1613161 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 37$, $T \in [22, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [2450] $1 \ 8 \ 1232^1 1812315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [2462] $3161232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [2460] $1812321712^1 321613 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2473] $1 \ 8 \ 12^{23151321} 32181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
 [2474] $18123151322^1 18123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 37$, $T \in [22, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
 [2475] $513161315132^1 18123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 65$, $T \in [22, 65]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2476] $181232^1 18122315132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2477] $321812^1 23151321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 37$, $T \in [22, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2478] $181223151322^1 18123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 37$, $T \in [22, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2479] $71232^1 161316131513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 54$, $T \in [22, 54]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2480] $71232^1 171232171232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 54$, $T \in [22, 54]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2481] $1(12)12^{11} 231414141322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2482] $41414132^{3221} 21(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2483] $2231414141322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2484] $413221(12)12^{21} 231414 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2485] $1\ 8\ 12321812^1 2315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2486] $1\ 8\ 12^{2321} 315132181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2487] $8122315132^1 1\ 8\ 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2488] $3221812321812^1 2315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2489] $23218122^1 315132181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$

- $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [2490] $61315132^1 181231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 65$, $T \in [22, 65]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2491] $8122315132^1 2181232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2492] $71232171232^1 171232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 54$, $T \in [22, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2493] $1812321812^1 231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [2494] $1^{2315132^1} 8^{1232181} + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
 [2495] $81232^1 1^{223151321} 8^{1232181} + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
 [2496] $3161315132^1 1812315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 65$, $T \in [22, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [2497] $18123218122^1 315132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [2499] $81232^1 181223151322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
 [2500] $712321712^1 32171232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 54$, $T \in [22, 54]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [2501] $414141322^1 1^{3221} (12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2498] $21812321812^1 231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [2502] $2321812^1 2315132181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.59 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [2503] $81223151322^1 181232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.59 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [2504] $812321^{22315132^1 1} 1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.55 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2505] $8123218122^{13151322} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.55 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2506] $1812321812^{12315132} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -225, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.52 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2507] $812321812^{123151322} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -361, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.41 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- $T = 18$ [2508] $1^{22^1 315131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2509] $1^{315131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 55, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2510] $1^{51231513221} (12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2511] $1^{15131513221} (12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2512] $1^{5131513221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 66, T \in [24, 66]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2513] $1^{51231513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2514] $1^{15131513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2515] $1^{1513221} (12) 1223151 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2516] $1^{22^1 315131513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 66, T \in [25, 66]$

- $T_{\min} = 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2531] $22^1 221(12)122315131513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2532] $22^1 2221(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2533] $22^1 2221(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2534] $1^1 12321712321812^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2535] $41413221 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12^1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2536] $21712321712321812^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2537] $1612321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2538] $314141413221(12)122^1 31 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2539] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 1223141414141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2540] $1414141413221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2541] $22^1 31414141413221(12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2542] $31414141413221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2577] $1414141413221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2576] $414141413221 \begin{smallmatrix} 32^1 21 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 66, T \in [25, 66]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2548] $1^{51231513221} (12) 122^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{15131513221} (12) 122^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{5131513221} 221 (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $1^{51231513221} 21 (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{15131513221} 21 (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{31513221} 513221 (12) 122^1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.33 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- $1^{222^1 1(12)1223151315132} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{222^1 21(12)122315131513} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{222^1 221(12)12231513215} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{222^1 221(12)12231513151} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{3132^1 21} 4131513221 (12) 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $1^{14132^1 21(12)1223151232} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 26, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 26, T \in [25, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{31413221(12)12231512^1 3} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $1^{1812321712321712^1 3} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 38, T \in [25, 38]$

- [2546] $1 \begin{smallmatrix} 2^1 1 \\ 31513221 \end{smallmatrix} (12) \quad 122315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.33 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2545] $1 \begin{smallmatrix} 21 \\ 5131513221 \end{smallmatrix} (12) \quad 122^1 3 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2544] $1 \begin{smallmatrix} 1 \\ 15131513221 \end{smallmatrix} (12) \quad 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2547] $1 \begin{smallmatrix} 315131513221 \\ 1 \end{smallmatrix} (12) \quad 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2543] $1 \begin{smallmatrix} 1 \\ 51231513221 \end{smallmatrix} (12) \quad 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2581] $1 \begin{smallmatrix} 221 \\ 512315132^1 21 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2582] $1 \begin{smallmatrix} 221 \\ 512^1 31513221 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2583] $1 \begin{smallmatrix} 221 \\ 151315132^1 21 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2584] $21 \begin{smallmatrix} 21 \\ 512^1 31513221 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2585] $21 \begin{smallmatrix} 21 \\ 512315132^1 21 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2586] $21 \begin{smallmatrix} 21 \\ 151315132^1 21 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2587] $1 \begin{smallmatrix} 21 \\ 512315132^1 21 \end{smallmatrix} (12) \quad 12^1 222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2588] $1 \begin{smallmatrix} 512315132^1 21 \\ 1 \end{smallmatrix} (12) \quad 12222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2589] \quad 1 \quad (12) \quad 12222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2590] \quad 1 \quad (12) \quad 12^1 222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2591] \quad 1 \quad (12) \quad 12222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2592] \quad 22^1 1 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.25 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2593] \quad 21 \quad (12) \quad 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2594] \quad 21 \quad (12) \quad 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2595] \quad 21 \quad (12) \quad 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2596] \quad 221 \quad (12) \quad 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2597] \quad 221 \quad (12) \quad 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2598] \quad 221 \quad (12) \quad 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2599] \quad 22^1 1 \quad (12) \quad 1223151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.25 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2600] \quad 2222^1 1 (12) 122315131513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2601] \quad 2222^1 21 (12) 12231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

- [2615] $41315132^{121} (12) 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2616] $4131513221^{11} (12) 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2617] $41315132^{121} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 24, \Delta = -96, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2618] $141322^1 (12) 1223151232 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 26, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 26, T \in [24, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2619] $141322^1 (12) 122315132 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 24, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2620] $31413221 (12) 122^1 315123 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2621] $314132^1 21 (12) 122315123 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2622] $314132^1 21 (12) 12231513 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2623] $31413221 (12) 122^1 31513 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2624] $231413221 (12) 12^1 231512 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 24, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2625] $231413221 (12) 12^1 23151 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 24, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2626] $21712^1 3218123151315 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2627] $5132171232181232^1 15 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 35, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2628] $1712315132181232^1 15 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 33, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2629] $4131513221 (12) 122^1 3132 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [26444] $1612^1 32181232161315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [26445] $1612^1 32181231513161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [26446] $2171232171232^1 18122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [26447] $21712^1 3217123218122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [26448] $612315132181232^1 161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [26449] $161232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2650] $1612^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2651] $21712^1 321812321712 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2652] $14141413221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2653] $141413221(12)122^1 31413 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2654] $1414132^1 21(12)12231413 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2655] $3141413221(12)12^1 23141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2579] $1 \quad \begin{matrix} 1 \\ 512315132^1 21 \end{matrix} \quad 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2578] $1 \quad \begin{matrix} 1 \\ 512^1 31513221 \end{matrix} \quad 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2603] $222221(12)122315132^1 15 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[2666] $1 \begin{pmatrix} 3151315132^1 21 \\ (12) \end{pmatrix} 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2660] $1 \begin{pmatrix} 21 \\ (12) \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2661] $1 \begin{pmatrix} 221 \\ (12) \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2662] $1 \begin{pmatrix} 221 \\ (12) \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2663] $1 \begin{pmatrix} 21 \\ (12) \end{pmatrix} 122^1 3151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2664] $21 \begin{pmatrix} 21 \\ (12) \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2659] $1 \begin{pmatrix} 21 \\ (12) \end{pmatrix} 12^1 23 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2658] $1 \begin{pmatrix} 1 \\ (12) \end{pmatrix} 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2657] $1 \begin{pmatrix} 1 \\ (12) \end{pmatrix} 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2668] $1 \begin{pmatrix} 1513151322^1 1 \\ (12) \end{pmatrix} 12222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2669] $221 \begin{pmatrix} 5131513221 \\ (12) \end{pmatrix} 12^1 23 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [2670] $21^{3151315132^1 21} (12) 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2671] $21^{5123151322^1 1} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2672] $21^{1513151322^1 1} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2673] $221^{5123151322^1 1} (12) 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2674] $221^{1513151322^1 1} (12) 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2675] $221^{1513221} (12) 122^1 3151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2676] $22221(12)122^1 315131513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2677] $22222^1(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2678] $222221(12)12^1 231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2679] $22222^1(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2680] $222221(12)12^1 231513151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2681] $1^{32151322^1 1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2682] $1^{3151322^1 1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2683] $1^{21513221} (12) 12^1 231413 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
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 [2684] $1(12)12^1 231413 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2685] $171232^1 18123151315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [2686] $413151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹³²²¹
 [2687] $413151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2688] $1413221(12)12231512^1 32 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 26, \Delta = -99, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 26, T \in [24, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2689] $1413221(12)122^1 3151232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 26, \Delta = -99, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 26, T \in [24, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2690] $1413221(12)122^1 315132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2691] $31413221(12)12^1 2315123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2692] $3141322^1 1(12)122315123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2693] $31413221(12)12^1 231513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2694] $3141322^1 1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2695] $2314132^1 21(12)12231512 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2696] $2314132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2697] $2171232^1 18123151315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$

$$\begin{aligned}
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2713] & \begin{matrix} 1 \\ 1613161315132^1 2181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2714] & \begin{matrix} 1 \\ 1812^1 3217123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2715] & \begin{matrix} 321 \\ 1 \ 8 \ 1232171232^1 171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2716] & \begin{matrix} 1 \\ 1712321712^1 3218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2717] & \begin{matrix} 1 \\ 171232^1 17123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2718] & \begin{matrix} 21 \\ 1 \ 8 \ 1232^1 1712321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2719] & \begin{matrix} 21 \\ 1 \ 8 \ 12321712^1 321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2720] & \begin{matrix} 15132^1 21 \\ 41413221 \ (12) \ 12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2721] & \begin{matrix} 51321 \\ 4132^1 1812321 \ 8 \ 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2722] & \begin{matrix} 1 \\ 41321812315132^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2723] & \begin{matrix} 1 \\ 4132^1 1812315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2724] & \begin{matrix} 232^1 1 \\ 712321712321 \ 8 \ 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 45, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 45, T \in [24, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2725] & \begin{matrix} 1 \\ 171232^1 181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2726] $18^{21}123218^{12}132161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2727] $18^{21}123218^{12}1232161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2728] $18^{12}13218^{12}12321613 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2729] $17^{12}123218^{12}1321712 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2730] $17^{12}123218^{12}12321712 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2731] $612^{12}13218^{12}123218^11 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2732] $16^{12}1316131513218^11 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2733] $17^{12}12321712^{12}13218^11 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2734] $17^{12}12321712^{12}123218^11 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2735] $16^{12}1316123218^{12}13215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2736] $16^{12}1316123218^{12}13151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2737] $15^{12}13161316123218^{12}12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2665] $21^{21} \begin{pmatrix} 12 \\ 1513151322^11 \end{pmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2739] $216^{12}1316131513212181 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [2740] $161232^1 181232161315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2741] $161232^1 181231513161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2742] $1712^1 32171232181223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2743] $1712321712321812^1 23 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2744] $232^1 171232171232181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2745] $32171232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2746] $2171232^1 17123218122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2747] $21712321712^1 3218122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2748] $4132^1 18123218122315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2749] $41321812315132^1 1812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2750] $4132^1 18123151321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2751] $41321812315132^1 2181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2752] $4132^1 18123151322181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2753] $8123151316131612^1 32 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2754] $6131613151321812^1 32 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [2755] $612^1 321812315131613 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 55, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2756] $6123151321812^1 32161 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2757] $612^1 315132181232161 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2758] $161232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2759] $1612321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2760] $3161232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2761] $1712^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2762] $2171232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2763] $612^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2764] $612321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2765] $712321812321712^1 32 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 45, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 45, T \in [24, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2766] $16131613161232^1 171 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2767] $141414132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2768] $141413221(12)12^1 231413 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2769] $14141322^1 1(12)12231413 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2770] & 31414132^1 21(12)1223141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[2771] & 216131613161232^1 171 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2772] & 4132181232171232^1 16 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2773] & 4132^1 18123217123216 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2774] & 414141413221^{322^1 1} (12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2775] & 4132^1 21^{1} (12)1223141414 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2656] & 1^{221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2667] & 1^{5123151322^1 1} (12) 12222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2738] & 21613161315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2779] & 1^{231513151322^1 1} (12) 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 36, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.22 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2780] & 1^{31513151322^1 1} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.22 \rightarrow 24, T_{\max} = 55, T \in [24, 55] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2781] & 21^{231513151322^1 1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 36, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.22 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2782] & 21^{31513151322^1 1} (12) 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.22 \rightarrow 24, T_{\max} = 55, T \in [24, 55] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2783] $22^1 1^{31513221} (12) 122315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.78 \rightarrow 23$, $T_{\max} = 66$, $T \in [23, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 25 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2784] $2221(12)122^1 3151315132 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2785] $22221(12)12^1 2315131513 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2786] $1413221(12)12^1 23151232 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 26$, $\Delta = -99$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 26$, $T \in [24, 26]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2787] $1413221(12)12^1 2315132 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 24$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2788] $23141322^1 1(12)12231512 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 24$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2789] $23141322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 24$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2790] $712^1 315132181231513 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2791] $712^1 231513218123215 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2792] $712^1 231513218123151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2793] $1613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2794] $16131613151322^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2795] $8122^1 31513161316^1 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2796] $712^1 321712321^1 81 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2797] $315131613161232^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2798] $321613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2799] $21613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2800] $232171232171232^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2801] $41321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2802] $8122^1 31513161316123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2803] $712^1 32181232171232 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2804] $712^1 3216131613161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2805] $1414141322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2806] $3141414132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2807] $314141322^1 1(12)1223141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 77 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [2808] $712^1 321613161316123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2809] $4131613161315132^1 17 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2810] $1(12)122^1 3141414141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2811] $21(12)122^1 3141414141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

- [2812] $14141414132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2778] $1 \begin{pmatrix} 21 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $31513151322^1 1$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2777] $1 \begin{pmatrix} 221 \\ 12 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $31513151322^1 1$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2776] $1 \begin{pmatrix} 21 \\ 12 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $23151315132^1 21$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2839] $1 \begin{pmatrix} 321 \\ 8 \end{pmatrix} 12^1 32171232171 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2813] $1 \begin{pmatrix} 1 \\ 12 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $223151315132^1 21$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2818] $1 \begin{pmatrix} 223151315132^1 21 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2819] $1 \begin{pmatrix} 231513151322^1 1 \\ 12 \end{pmatrix} 122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2820] $221 \begin{pmatrix} 51315132^1 21 \\ 12 \end{pmatrix} 1223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2821] $21 \begin{pmatrix} 231513151322^1 1 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2822] $221 \begin{pmatrix} 31513221 \\ 12 \end{pmatrix} 122^1 315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2823] $221 \begin{pmatrix} 1513221 \\ 12 \end{pmatrix} 12^1 23151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2824] $221(12)122^1 31513151322 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2825] $2221(12)12^1 23151315132 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2826] & \overset{1}{1712321812^1 3151315} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 32, T \in [24, 32] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2827] & 21712321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 32, T \in [24, 32] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2828] & 5132^1 17123218123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2829] & 5132171232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2830] & 5132171232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2831] & 5132^1 17123218123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2832] & 1712315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2833] & 1712315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2834] & 7122315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2835] & 7122315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2836] & \overset{1}{181232^1 16131613151} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2837] & \overset{1}{181232^1 17123217123} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2816] & \overset{21}{1 \quad (12) \quad 122^1 315} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& \quad \quad \quad 31513221 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2815] & \overset{21}{1 \quad (12) \quad 1223} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& \quad \quad \quad 51315132^1 21 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66]
\end{aligned}$$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [2814] $1 \quad \begin{matrix} 21 \\ (12) \end{matrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $231513151322^1 1$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 1513221 \\ (12) \end{matrix} \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 151322^1 1 \\ (12) \end{matrix} \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 5132^1 1 \\ 8 \end{matrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 51321 \\ 8 \end{matrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 1 \\ 41321812^1 315132181 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 31 \\ 812^1 2315131613161 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 1 \\ 181232181232^1 1613 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 1 \\ 181232^1 1812321613 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 1 \\ 61232181232^1 18 \end{matrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 1 \\ 61232^1 18123218 \end{matrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- $1613161232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $1613161232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $151316131612^1 321812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[2854] \quad 316131612321812^1 315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2855] \quad 1612321812^1 32161315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2856] \quad 1612321812^1 31513161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2857] \quad 171232^1 171232181223 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[2858] \quad 171232171232^1 181223 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2859] \quad 321712321712^1 321812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[2860] \quad 321712^1 321712321812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2861] \quad 413218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2862] \quad 41321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2863] \quad 41321812^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2864] \quad 41321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2865] \quad 812^1 321613161315132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2866] \quad 812^1 315131613161232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2867] \quad 812^1 231513161316123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2868] \quad 61315132181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2869] $61232^1 1812315131613 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2870] $612315132^1 181232161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2871] $31612321812^1 321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2872] $31612^1 321812321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2873] $171232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2874] $171232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2875] $61232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2876] $61232181232^1 181223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2877] $161316131612^1 32171 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2878] $31414141322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2879] $2161316131612^1 32171 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2880] $41321812321712^1 3216 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2881] $41321812^1 3217123216 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2882] $1(12)12^{221} 23141414141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
 [2883] $21^{21}(12)12^1 23141414141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [2884] $1^{21}(12)122^1 31414141413 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [2885] $141414141322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [2886] $314141414132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [2887] $41322^1 1(12)1223141414 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2888] $414132^1 21(12)12231414 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2817] $1^{21}(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2840] $1^{321} 8 12321712^1 32171 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [2838] $1^{18} 1232171232^1 17123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2895] $51321712^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 33, T \in [23, 33]$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2894] $51321712^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 35, T \in [23, 35]$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2892] $221^{315132^1 21} (12) 122315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2891] $1^{2231513151322^1 1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [2890] $1^{21} (12) 122315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
- $315132^1 21$

- $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 66$, $T \in [23, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2897] $1812321712^1 3217123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2898] $181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2899] $4132181232^1 18^1 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 41$, $T \in [23, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2900] $1812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2901] $612321812^1 3218^1 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 41$, $T \in [23, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2902] $16131612^1 3218123215 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2903] $16131612^1 3218123151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2904] $161232181232^1 161315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2905] $1712321712^1 32181223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2906] $32171232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2907] $4132181232^1 18122315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 41$, $T \in [23, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2908] $6131612321812^1 31513 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 55$, $T \in [23, 55]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2909] $613151321812^1 321613 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 55$, $T \in [23, 55]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2910] $3161232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[29111] & 1712321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 38, T \in [23, 38] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2912] & 612321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 41, T \in [23, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2913] & 4132181232^1 17123216 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 41, T \in [23, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2914] & 1(12)122^1 314141414132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2915] & 1(12)12^{21} 231414141413 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2916] & 2314141414132^{21} 21(12)12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2917] & 3141414141322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2896] & 7123151321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2893] & 221(12)12^1 231513151322 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2889] & 1 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \quad 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& 2231513151322^{11} \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2927] & 71232^1 16131613161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.84 \rightarrow 23, T_{\max} = 38, T \in [23, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2922] & 3151316131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 45, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.84 \rightarrow 23, T_{\max} = 45, T \in [23, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2923] & 5131613161232^1 18123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.84 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2924] & 1613161315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.84 \rightarrow 23, T_{\max} = 41, T \in [23, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2925] $2321712321712^1 32181 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 45, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 45, T \in [23, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2926] $71232^1 181232171232 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 45, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 45, T \in [23, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2920] $4141322^1 1^{1513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 38, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 38, T \in [23, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2928] $71232^1 1613161316123 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 41, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 41, T \in [23, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2929] $1^{12} (12) 12^1 2314141414132 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2930] $22314141414132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2931] $23141414141322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2932] $413221^{12} 122^1 3141414 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2933] $4141322^1 1^{12} 12231414 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2921] $71232^1 1712321^{2321} 8 \ 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 45, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 45, T \in [23, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2919] $221^{513151322^1 1} (12) 1223 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2918] $1^{21} (12) 1223 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2934] $1^{21} (12) 12^1 2315 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 100, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.76 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2935] $221^{31513221} (12) 12^1 2315 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 100, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.76 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2950] & 81232^1 1613161315132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2951] & 612321812^1 315131613 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2952] & 712321812^1 32171232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 45, \Delta = -118, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 45, T \in [23, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2953] & 4141414132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2954] & 413221(12)12^1 23141414 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2955] & 51316131612^1 3218123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 144, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.64 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2956] & 316131612^1 321812315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 169, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.59 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2957] & 613161315132^1 181232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 169, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.59 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2958] & 41414141322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 169, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.59 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
T = 19 & [2974] 22^1 1(12)122315131612 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 31, T \in [26, 31] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[2959] & 1^{315131513221} (12) 122^1 31 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2964] & 16^{31} 131513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 39, T \in [26, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2965] & 16^{31} 131513221(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 39, T \in [26, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2966] $16131513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2967] $3221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2968] $1(12)1223151316123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2969] $1(12)12231513161232 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 56, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 56, T \in [26, 56]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2970] $1(12)122315131612 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2971] $16131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 39, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 39, T \in [26, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2972] $21(12)1223151316123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2962] $16131513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2961] $16131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2960] $132^1 21(12)1223151315132 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2976] $21(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2977] $22^1 1(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2978] $1(12)1223151321613 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [2979] $21^{22^1 1}(12)1223151321613 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2980] $413221^{5131513221}(12)12^1 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 56, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2981] $61231513221^{3221}(12)122^1 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 46, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2982] $612231513221^{22^1 221}(12)1 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2983] $612231513221^{22^1 21}(12)12 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2984] $612231513221^{221}(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2985] $3216131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2986] $23216131513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 56, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 56, T \in [26, 56]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2987] $216131513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2988] $161231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2989] $3161231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2990] $22^1 18123218123151315 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 32, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2991] $612231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

- [2992] $18^{21}12321712321812^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2993] $1712321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2994] $22^1 18123217123218122 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2995] $21712321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2996] $141414141413221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2997] $(12)12231414141414132^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 56, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 56, T \in [26, 56]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2963] $16131513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2975] $1^{22^1 221}(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2973] $21^{22^1 21}(12)122315131612 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3012] $1^{222^1 21}(12)122315131612 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [2998] $1^{315131513221}(12)12^1 231 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 27, \Delta = -99, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3003] $32^1 1513221(12)122315123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3004] $31^{32^1 1513221(12)12231} 1 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3005] $16131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ¹
 [3006] $16131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ²¹
 [3007] $16131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 39$, $T \in [26, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ¹
 [3008] $16131513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ^{32^1 21}
 [3009] $221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 1}
 [3010] $1(12)1223151316123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ¹
 [3001] $1322^1 1(12)1223151315132 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -100$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ¹⁵¹³¹⁵¹³²²¹
 [3000] $1(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 27$, $\Delta = -99$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ⁵¹²³¹⁵¹³²²¹
 [2999] $1(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -100$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ²²¹
 [3014] $21(12)122315131612^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 1}
 [3015] $21(12)122315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 21}
 [3016] $1(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 1}
 [3017] $21(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 1}
 [3018] $1(12)1223151321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ^{32¹21}
 [3019] 61231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 46$, $T \in [26, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- ^{222¹21}
 [3020] 612231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ^{222¹1}
 [3021] 612231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ^{32¹16131513221(12)12222} + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3023] 3216131513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3024] 23216131513221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 56$, $T \in [26, 56]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3025] 216131513221(12)1222¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3026] 161231513221(12)1222¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3027] 3161231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3028] 32¹18123151321812315 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 56$, $T \in [26, 56]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3029] 612231513221(12)1222¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ¹
 [3030] 18123217123218122¹3 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

- [3031] $1 \overset{32^1}{8} 12321712321812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3032] $1812321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3033] $1 \overset{32^1}{8} 1232181232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3034] $1812321712321812^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3035] $218123217123218122^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3036] $613151321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 46$, $T \in [26, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3037] $17123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3038] $32^1 1712321812321812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3039] $161316131612321812^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 39$, $T \in [26, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3040] $32^1 16131613161232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 39$, $T \in [26, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3041] $41414141413221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 56$, $T \in [26, 56]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3002] $3132^1 21(12)122315131513 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3013] $16131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 39$, $T \in [26, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3011] $1(12)122315131612^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[3066] & \begin{matrix} 31 \\ 16 \end{matrix} 131513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 39, T \in [25, 39] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3047] & 2313221(12)122^1 31513215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3044] & \begin{matrix} 51231513221 \\ 1 \end{matrix} (12) 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3043] & \begin{matrix} 512^1 31513221 \\ 1 \end{matrix} (12) 122313 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3049] & 2313221(12)122315132^1 15 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3050] & 2313221(12)122^1 31513151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3051] & 23132^1 21(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3052] & 31322^1 1(12)122315131513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3053] & 3215132^1 21(12)122315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3054] & \begin{matrix} 321513221(12)122^1 31 \\ 31 \end{matrix} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3055] & \begin{matrix} 3215132^1 21(12)12231 \\ 31 \end{matrix} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3056] & \begin{matrix} 31513221(12)122^1 31 \\ 31 \end{matrix} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 27, T \in [25, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3057] $\overset{1}{161315132^1}21(12)12222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3058] $\overset{1}{16131513221}(12)12^1222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3059] $\overset{1}{161315132^1}21(12)^{21}1222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3060] $\overset{1}{161315132^1}21(12)^{221}122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3061] $\overset{1}{16131513221}(12)122^1222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3062] $\overset{1}{161315132^1}21(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3063] $\overset{3221}{22^1}1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.25 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3064] $\overset{2222^1}{1}(12)122315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3065] $\overset{22221}{1}(12)122^1315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3042] $\overset{512315132^1}{1}(12)^{2221}122313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -100$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3067] $\overset{2221}{21}(12)122^1315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3068] $\overset{221}{221}(12)122^1315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3069] $\overset{22221}{1}(12)122^1315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3070] $1^{22221}(12)122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $1^{2222^1 1}(12)122315132161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $1^{2221}(21)(12)122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $1^{2221}(21)(12)122^1 315132161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $1^{221}(221)(12)122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $1^{221}(221)(12)122^1 315132161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $1^{21}(18)12^1 3218123151315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $1^{51321}(1)8^{12}1232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $1^{12}(18)12315132181232^1 15 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $1^{3221}(61)231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.25 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $1^{22221}(61)22315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $1^{2222^1 1}(61)2231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $1^{22221}(61)22^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [3083] $6122315132^{\overline{1}21}(12)12 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3084] $6122^{\overline{1}31513221}(12)12 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3085] $6122^{\overline{1}31513221}(12)122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3086] $6122315132^{\overline{1}21}(12)122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3087] $3216131513221(12)12^{\overline{1}222} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3088] $216131513221(12)122^{\overline{1}222} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3089] $2161315132^{\overline{1}21}(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3090] $1612^{\overline{1}31513221}(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3091] $161231513221(12)122^{\overline{1}222} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3092] $1612315132^{\overline{1}21}(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3093] $3161231513221(12)12^{\overline{1}222} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3094] $221812^{\overline{1}3218123151315} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3095] $51322181232181232^{\overline{1}15} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$

- [3096] $21812315132181232^1 15 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3097] $18122315132181232^1 15 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3098] $612231513221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3099] $6122315132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3100] $6122^1 31513221(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3101] $1712^1 32181232161315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3102] $1712^1 32181231513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3103] $181232171232^1 18122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3104] $1812^1 3217123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3105] $612315132181232^1 171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3106] $1712^1 32181232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3107] $171232181232^1 181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3108] $1712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3109] $171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3110] & 181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3111] & 1812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3112] & 1812^1 32171232181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3113] & 1414132^1 21(12)122315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 25, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3114] & 141413221(12)122^1 315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 25, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3115] & 1612^1 321812321712315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 33, T \in [25, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3116] & 513161232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 32, T \in [25, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3117] & 161315132181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3118] & 1612^1 321812315132171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3119] & 221812^1 3217123218122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3120] & 4131513221(12)122^1 31413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3121] & 41315132^1 21(12)12231413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3122] & 61232171232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 35, T \in [25, 35] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3123] & 612315132181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]
\end{aligned}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3124] $7122315132181232^1 161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3125] $2171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[3126] $21712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3127] $1613161232181232^1 161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3048] $23132^1 21(12)12231513215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.75 \rightarrow 25$$
, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3046] $1^{15131513221}(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 27$, $\Delta = -99$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 27$, $T \in [25, 27]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3045] $1^{151315132^1 21}(12)122313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 27$, $\Delta = -99$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 27$, $T \in [25, 27]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3177] $32161315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25$$
, $T_{\max} = 42$, $T \in [25, 42]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3218] $181232^1 1712321^1 81 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3220] $141413221(12)12^1 2315123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 25$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25$$
, $T_{\max} = 25$, $T \in [25, 25]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3221] $161232^1 1812321712315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3222] $5131612321812^1 321712 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3128] $1^{3151315132^1 21}(12)12231 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 27$, $\Delta = -99$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 27$, $T \in [25, 27]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [3129] $1^{5123151322^1 1}(12)122313 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3130] $1^{1513151322^1 1}(12)122313 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 27, T \in [25, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3131] $231322^1 1(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3132] $2313221(12)12^1 231513215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3133] $231322^1 1(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3134] $2313221(12)12^1 231513151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3135] $313221(12)122^1 315131513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3136] $32151322^1 1(12)122315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3137] $31^{321513221(12)12^1 231} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3138] $31^{32151322^1 1(12)12231} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3139] $31^{31513221(12)12^1 231} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 27, T \in [25, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3140] $1^{1613151322^1 1(12)12222} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3141] $1^{1613151322^1 1(12)1222} 21 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3142] & \begin{matrix} 1 & & 221 \\ 1613151322^1 1(12)122 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3143] & \begin{matrix} 31 & & 1 \\ 161315132^1 21(12)1222 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 39, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 39, T \in [25, 39] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3144] & \begin{matrix} 31 & & 221 \\ 161315132^1 21(12)12 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 39, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 39, T \in [25, 39] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3145] & \begin{matrix} 1 \\ 1613151322^1 1(12)122222 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3146] & \begin{matrix} 1 \\ 16131513221(12)12^1 2222 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3147] & \begin{matrix} 322^1 1 \\ 221(12)12231513161 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 42, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3148] & \begin{matrix} 2221 \\ 1(12)122^1 3151316123 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 42, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3149] & \begin{matrix} 221 \\ 1(12)122315131612^1 32 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 56, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3150] & \begin{matrix} 22221 \\ 1(12)12^1 2315131612 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3151] & \begin{matrix} 31 \\ 161315132^1 21(12)12222 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 39, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 39, T \in [25, 39] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3152] & \begin{matrix} 221 \\ 21(12)122^1 3151316123 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 42, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3153] & \begin{matrix} 21 \\ 21(12)122315131612^1 32 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 56, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \end{matrix}
\end{aligned}$$

- [3154] $21(12)12^{2221}2315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3155] $221(12)12^{221}2315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3156] $1(12)12^{22221}2315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3157] $21(12)12^{2221}2315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3158] $221(12)12^{221}2315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3159] $1(12)122^{2221}3151321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3160] $21(12)122^{221}3151321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3161] $181232^{21}18123151315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3162] $18^{51321}12321812^13215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3163] $18^{51321}12321812^13151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3164] $1812^1131513218123215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3165] $18123151321812^13215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3166] $1812^1131513218123151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3167] $18123151321812^13151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3168] $4132^{\hat{1}}21^{5131513221}(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3169] $61231513221^{322^{\hat{1}}1}(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [3170] $612^{\hat{1}}31513221^{3221}(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3171] $61223151322^{\hat{1}}1^{22221}(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [3172] $612^{\hat{1}}231513221^{22221}(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3173] $61223151322^{\hat{1}}1^{2221}(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [3174] $612^{\hat{1}}231513221^{2221}(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [3176] $612^{\hat{1}}231513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [3178] $232^{\hat{1}}16131513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [3179] $216131513221(12)12^{\hat{1}}2222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3180] $21613151322^{\hat{1}}1(12)122222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3181] $16123151322^{\hat{1}}1(12)122222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3182] $161231513221(12)12^{\hat{1}}2222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3183] 31612315132¹21(12)12222 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3184] 22181232¹18123151315 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3185] 513221812321812¹3215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3186] 15132¹18123218122315 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3187] 218123151321812¹3215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3188] 21812¹31513218123215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3189] 218123151321812¹3151 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3190] 21812¹31513218123151 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3191] 181223151321812¹3215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3192] 18122¹31513218123215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3193] 181223151321812¹3151 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3194] 18122¹31513218123151 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3195] 61223151322¹1(12)122222 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3196] 612¹231513221(12)122222 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3197] $612231513221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3198] $171232^1 181232161315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3199] $171232^1 181231513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3200] $1812321712321812^1 23 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3201] $1812^1 32171232181223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3202] $1^1 8^1 1232171232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3203] $1^1 8^1 1232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3204] $1^1 8^1 12321712^1 3218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3205] $1^1 8^1 1232^1 17123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3206] $612^1 315132181232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3216] $712321812321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3208] $1712321812^1 3218^1 1 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3209] $171232^1 18123218^1 1 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3210] $1812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [3211] $1 \ 8 \ 1232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [3212] $171232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3213] $1712321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [3214] $181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3215] $1812321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [3223] $1613151321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3224] $3161315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3225] $1612321812315132^1 171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3226] $161232^1 1812315132171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [3227] $21812^1 32171232181223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3228] $21812321712321812^1 23 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3229] $22181232^1 17123218122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3230] $413151322^1 1(12)12231413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3231] $4131513221(12)12^1 231413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$

- [3232] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 612321712321812¹3215+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
- [3233] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 612¹3217123218123215+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
- [3234] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 612¹3217123218123151+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
- [3235] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 612321712321812¹3151+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
- [3236] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ 612¹3151321812321712+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3237] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 6123151321812¹321712+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3238] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ 7122¹315132181232161+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3239] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 71223151321812¹32161+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3240] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 171232181¹2321812¹23+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
- [3241] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 1712¹32181¹232181223+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
- [3242] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 232¹171232181¹232181+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
- [3243] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 32171232181¹232¹1812+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
- [3244] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 2171232¹181¹23218122+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3245] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 21712321812¹3218122+ $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

- [3246] $16131613161232^1 181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3247] $16131613161232^1 1812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3248] $1812^1 32161316131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3249] $4141413221^{15132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3250] $216131613161232^1 1812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3251] $16131612321812^1 32161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3252] $4132181232171232^1 171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3253] $4132^1 181232171232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3254] $81232161316131612^1 32 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 46$, $T \in [25, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3255] $61316131612321812^1 32 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 46$, $T \in [25, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3256] $612^1 3218123216131613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 46$, $T \in [25, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3257] $612321812321812^1 314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 39$, $T \in [25, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3258] $612^1 321812321812314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 39$, $T \in [25, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3259] $414141413221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 56$, $T \in [25, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3260] $414141413221(12)122^1 314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3217] \quad 1812^1 13217123218123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3175] \quad 61223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3219] \quad 14141322^1 1(12)122315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 25, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3207] \quad 6123151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3265] \quad 1613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3266] \quad 1(12)12^1 23151316123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3267] \quad 1(12)122^1 31513161232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3268] \quad 1613151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3269] \quad 21(12)12^1 23151316123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3270] \quad 21(12)122^1 31513161232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3271] \quad 1(12)12^1 23151321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3272] \quad 21(12)12^1 23151321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3273] \quad 321613151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

15.0

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3274] \quad 232161315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3275] \quad 316123151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3276] \quad 1812^1 315132181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3277] \quad 1812^1 231513218123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3278] \quad 1812^1 231513218123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3279] \quad 8122^1 315132181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3280] \quad 1 \quad 8 \quad 1232171232^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3281] \quad 712^1 321812321 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3282] \quad 712^1 3151316131613151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3283] \quad 712^1 3151321812321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3284] \quad 712^1 2315132181232161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3285] \quad 232171232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3286] \quad 1 \quad 6 \quad 131613161232^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3287] \quad 3216131613161232^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3288] $4131613161315132^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3289] $4131613161315132^1 218 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3290] $41316131613161232^1 17 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3264] $16^{31} 13151322^1 1(12) 1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3263] $313221(12) 12^1 2315131513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3262] $13221(12) 122^1 3151315132 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3261] $1^{31513151322^1} (12) 12231 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 27, T \in [25, 27]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3295] $1^{2221} (12) 122315132^1 1613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3296] $21^{221} (12) 122315132^1 1613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3297] $18^{21} 12321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[3298] $1^{51321} 8^{1232^1} 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3299] $1^{5132^1} 8^{123218123215} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3300] $1^{51321} 8^{1232^1} 18123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [3301] $1 \overset{5132^1}{8} 123218123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3302] $1812315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3303] $1812315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3304] $41322^1 \overset{5131513221}{1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3305] $41322^1 \overset{51315132^1 21}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3306] $612315132^1 \overset{3221}{21} (12) 122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3307] $2321613151322^1 1(12) 1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [3308] $31612^1 31513221 (12) 12222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3309] $221812321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3310] $51322181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3371] $4141413221 (12) 122^1 31414 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3294] $21 \overset{21}{(12)} 12^1 231513161232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3293] $1 \overset{221}{(12)} 12^1 231513161232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3311] $5132^1 218123218123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3312] \quad 151321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3313] \quad 1513218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3314] \quad 3218123151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3315] \quad 21812315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3316] \quad 21812315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3317] \quad 18122315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3318] \quad 18122315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3319] \quad 812^1 3151321812315132 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3320] \quad 812^1 2315132181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3321] \quad 1712321812^1 32161315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3322] \quad 1712321812^1 31513161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3323] \quad 181232^1 171232181223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3324] \quad 181232171232^1 181223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3325] \quad 1^1 812^1 321712321812 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

- [3326] $1\ 8\ 12321712^1 321812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3327] $612315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3328] $181232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3329] $181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3330] $1\ 8\ 12^1 32181232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3331] $1\ 8\ 12321812^1 32171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3332] $181232^1 17123218123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3333] $181232171232^1 18123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3334] $1612321812321712^1 315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3335] $1612321812^1 321712315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3336] $513161232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3337] $161315132^1 1812321712 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3338] $31613151321812^1 32171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3339] $1612321812^1 315132171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3340] $2181232171232^1 181223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

3350] 171232181232+181223 $\rightarrow$ 1 $\rightarrow$ 2 $T = 1$
 $T = 24.06 \rightarrow 25$ $T = 26$ $T \in [25, 26]$

$$T_{\min} = 24.06$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \end{pmatrix}$$

$$3351] \ 171232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 1$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25 \quad T_{\max} = 36 \quad T \in [25, 36]$$

$$I_{\min} = 24.00 \rightarrow 25, \quad I_{\max} = 30, \quad I \in [25, 30]$$

$$\begin{pmatrix} 0 & 7 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix} b_0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

- [3355] $\overset{1}{161316131612^1 321812} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3356] $\overset{1}{181232^1 161316131612} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [3357] $\overset{151322^1}{4141413221} (12) \overset{1}{1} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3358] $\overset{1}{2161316131612^1 321812} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3359] $\overset{1}{1613161232^1 181232161} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3360] $\overset{1}{41321812^1 32171232171} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3361] $\overset{1}{41321812321712^1 32171} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3362] $\overset{1}{812^1 3216131613161232} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3363] $\overset{1}{61232^1 18123216131613} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3364] $\overset{1}{613161232181232^1 1613} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3365] $\overset{1}{61232^1 1812321812314} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3366] $\overset{1}{61232181232^1 1812314} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3367] $\overset{1}{41316131613151322^1 18} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3368] $\overset{21}{1(12)122^1 314141414141} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$

- [3292] $^{3221}221(12)122^131513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3291] $13221(12)12^123151315132 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -100$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3369] $1414141414132^121(12)122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [3370] $414141413221(12)12^12314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 56$, $T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3378] $^{1171232181232^1161315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3372] $^{51321}1^8 12^13218123215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3373] $^{51321}1^8 12^13218123151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3374] $513221812^13218123215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3375] $15132181232^118122315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3376] $18123151321812^131513 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3379] $^{11812321712^132181223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3380] $^{321}1^8 1232^11712321812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3381] $^{11812321812^13217123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3382] $^{321}1^8 1232^1181232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3383] $1812321712^1 3218123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 36, T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3384] $161232181232^1 1712315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3385] $5131612^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3386] $21812321712^1 32181223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 36, T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3387] $612321712^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3388] $612321712^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3389] $613151321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3390] $7123151321812^1 321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3391] $1712321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 36, T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3392] $32171232^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 36, T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3393] $16131612^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3394] $4132181232^1 171232171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 36, T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3395] $6131612321812^1 321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3396] $612321812^1 321812314 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24, T_{\max} = 39, T \in [24, 39]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3397] $1(12)^{21} 12^1 2314141414141 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[3398] \quad 1(12)122^1 3141414141413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[3399] \quad 1414141414141322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3400] \quad 3141414141414132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3377] \quad 81223151321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3416] \quad 4141413221(12)12^1 231414 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3411] \quad 32161316131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 39, T \in [24, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[3401] \quad 221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3402] \quad 413221^{513151322^1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3403] \quad 6123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3404] \quad 321812^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3405] \quad 1^1 8 12321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[3406] \quad 71232^1 1812321^1 8 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[3407] \quad 2321712321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 42, T \in [24, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[3408] & \overset{31}{161316131612^1 32181} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 36, T \in [24, 36] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3409] & \overset{1513221}{41414132^1 21(12)} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 36, T \in [24, 36] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3410] & 16131613161232^1 18123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 39, T \in [24, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3415] & 314141414141322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 46, T \in [24, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3414] & 231414141414132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3413] & \overset{1}{1(12)12^1 23141414141413} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 46, T \in [24, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3412] & 413161316131612^1 3217 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 36, T \in [24, 36] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[3417] & 41414132^1 21(12)12231414 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3426] & \overset{1}{2321712^1 32181232181} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.76 \rightarrow 24, T_{\max} = 42, T \in [24, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3427] & 613161232^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3418] & 51322^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.76 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3419] & 151321812321812^1 2315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.76 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3420] & 1812315132^1 181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} =
\end{aligned}$$

15.0

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3421] \ 8122315132^1 181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3422] \ 1 \ 8 \ 12^{2321} 32171232181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[3423] \ 71232181232^1 1 \ 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3424] \ 61315132^1 18123217123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3425] \ 712315132^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3429] \ (12)122^1 314141414141322 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3428] \ 2314141414141322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3436] \ 414141322^1 1^{1513221} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 36, T \in [24, 36]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3431] \ 8123151321812^1 315132 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3432] \ 1 \ 8 \ 1232^1 171232181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3433] \ 712321812^1 321 \ 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3434] \ 3161315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

- [3430] $321812315132^1 1812315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3437] $81232^1 16131613161232 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [3438] $612321812^1 3216131613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3439] $6131612^1 321812321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3440] $(12)12^1 2314141414141322 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3441] $4141414132^1 21(12)122314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3442] $414141322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3435] $232171232^1 181232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 42$, $T \in [24, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3443] $161316131612^1 3218123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -144$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.64 \rightarrow 24$, $T_{\max} = 39$, $T \in [24, 39]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3444] $812315132^1 1812315132 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -169$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.59 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3445] $6131613161232^1 181232 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.59 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3446] $61232181232^1 16131613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.59 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3447] $414141414132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -169$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 23.59 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3448] & 41414141322^1 1(12)122314 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -169, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.59 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3449] & 4141414141322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -225, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.52 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3450] & 61316131612^1 32181232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -289, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.46 \rightarrow 24, T_{\max} = 46, T \in [24, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& T = 20 \quad [3460] \quad 21(12)12231513217123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 40, T \in [27, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3459] & 1(12)1223151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3452] & 171231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3453] & 171231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3454] & 171231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3455] & 171231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[3456] & 3221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 40, T \in [26, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[3457] & 1(12)12231513217123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 40, T \in [27, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3458] & 1(12)122315132171232 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 47, T \in [27, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3461] $21^{22^1 21}(12)1223151321712 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3462] $22^1 1^{221}(12)1223151321712 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3463] $1^{22^1 221}(12)1223151322171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3464] $21^{22^1 21}(12)1223151322171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3465] $22^1 1^{221}(12)1223151322171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3466] $712231513221^{3221}(12)122^1 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3467] $23216131513221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 28, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 28, T \in [27, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3468] $32171231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 40, T \in [27, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3469] $232171231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 47, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 47, T \in [27, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3470] $2171231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3471] $1712231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3472] $41316131513221^{22^1 21}(12)1 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 37, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 37, T \in [27, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3473] $41316131513221^{22^1 1}(12)12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 37, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 37, T \in [27, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

- [3474] $18^{21}123218^112321812^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3475] $5131612321812321812^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3476] $1613151321812321812^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3477] $41316131513221(12)1222^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 37, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 37, T \in [27, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3478] $6123151321812321812^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3479] $22^{21}18123218123218122 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3480] $413221(12)^{21}12231513161 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3481] $413221(12)^{21}12231513216 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 43, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 43, T \in [26, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3451] $1513221^{5131513221}(12)12^1 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 36, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3483] $171231513221(12)^{21}12^122 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3484] $16^{31}131513221(12)122^1313 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3482] $171231513221(12)^{21}12^122 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3486] $221^{321}122315132171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3487] $1^{2221}(12)1223151321712^13 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3501] $32171231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3502] $232171231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 47, T \in [27, 47]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3503] $2171231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3504] $1712231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3505] $4131513221(12)12231512^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3506] $41316131513221(12)1 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 37, T \in [27, 37]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3507] $613151321812321 \overset{222^1}{8} 1 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 43, T \in [27, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3508] $1812321812321812^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3509] $18123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3510] $1 \overset{32^1}{8} 12321812321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3511] $41316131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 37, T \in [27, 37]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3512] $6131513218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 43, T \in [27, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3513] $6123218123151321812^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 47, T \in [27, 47]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [3514] $7123151321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 43$, $T \in [27, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3515] $218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 36$, $T \in [27, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3516] $6131612321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 43$, $T \in [27, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3485] $171231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 32$, $T \in [27, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3534] $171231513221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3535] $1712315132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3523] $1712315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3524] $1712315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3525] $1712^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3526] $16131513221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3527] $161315132^1 21(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3528] $1(12)122315131612 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -96$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3529] $1(12)122^1 315131612 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -96$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3530] $16131513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -104$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3531] \quad 1 \quad (12) \quad 122315132161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3532] \quad 1 \quad (12) \quad 122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3533] \quad 1 \quad (12) \quad 122^1 315132161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3518] \quad 161315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -96, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3517] \quad 16131513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -96, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3536] \quad 1712^1 31513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3537] \quad 22^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.25 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3538] \quad 1(12)122^1 3151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3539] \quad 1(12)122315132^1 1712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3540] \quad 1(12)1223151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3541] \quad 21(12)122^1 3151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3542] \quad 21(12)122315132^1 1712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3543] \quad 221(12)122^1 3151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ²²¹
 [3544] $221(12)122315132^1 1712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ²²²²¹
 [3545] $1(12)122^1 3151322171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ²²²²¹
 [3546] $1(12)122315132^1 2171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ^{2222^1 1}
 [3547] $1(12)1223151322171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- ²²²¹
 [3548] $21(12)122^1 3151322171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ²²²¹
 [3549] $21(12)122315132^1 2171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ²²¹
 [3550] $221(12)122^1 3151322171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ²²¹
 [3551] $221(12)122315132^1 2171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ^{31322^1 1}
 [3552] $612231513221(12)1 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [3553] $6122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [3554] $6122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ³²²¹
 [3555] $712231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.25 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3556] $141322^1 1(12)122315131513 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3557] & 314132^1 21(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3558] & 31413221(12)122315132^1 15 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3559] & 31413221(12)122^1 31513215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3560] & 314132^1 21(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3561] & 31413221(12)122^1 31513151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3562] & 216131513221(12)122^1 3132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3563] & 2161315132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3564] & 3216131513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3565] & 1612^1 31513221(12)1223132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3566] & 161231513221(12)122^1 3132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3567] & 1612315132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3568] & 3161231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[3569] 32171231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3570] 21712315132¹21(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3571] 2171231513221(12)122¹222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3572] 21712¹31513221(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3573] 17122¹31513221(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3574] 1712231513221(12)122¹222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3575] 17122315132¹21(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3576] 4131513221(12)122¹315123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3577] 41315132¹21(12)122315123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3578] 4131513221(12)122¹31513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3579] 41315132¹21(12)12231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3580] 6122315132¹21(12)1223132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3581] 6122¹31513221(12)1223132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3582] & 612231513221(12)122^1 3132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3583] & \begin{matrix} 51321 \\ 1 \end{matrix} 8^{12} 1232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3584] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1^1 32181232161315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3585] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1^1 32181231513161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3586] & \begin{matrix} 1 \\ 1 \end{matrix} 8^{12} 15132181232^1 161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3587] & \begin{matrix} 1 \\ 1 \end{matrix} 7^{12} 1^1 321812321712315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3588] & \begin{matrix} 1 \\ 1 \end{matrix} 7^{12} 1^1 321812315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3589] & 612315132181232^1 \begin{matrix} 21 \\ 1 \end{matrix} 8^1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3590] & 7122315132181232^1 \begin{matrix} 1 \\ 1 \end{matrix} 7^1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3591] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1^1 321812321 \begin{matrix} 1 \\ 1 \end{matrix} 8^1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3592] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3593] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3594] & 1612^1 3218123218122315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3595] $513161232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3596] $161315132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3597] $1612^1 3218123151321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3598] $1612^1 3218123151322181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3599] $171232171232181232^1 15 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 34$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 34$, $T \in [26, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3600] $21712^1 321812321712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3601] $21712^1 321812315132171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3602] $41316131513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 37$, $T \in [26, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3603] $612315132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3604] $7122315132181232^1 1712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3605] $221812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3606] $61232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 35$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 35$, $T \in [26, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3607] $1712^1 321812321613161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3608] $1613161232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3609] $61232171232181232^1 161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$

- $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹
 [3617] $16131513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [3618] $1613151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹³²²¹
 [3619] $1(12)122^1 3151316123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [3620] $1(12)12^1 2315131612 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -96$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- ³¹
 [3621] $161315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [3622] $1(12)12^1 2315132161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- ¹³²²¹
 [3623] $1(12)122^1 3151321613 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [3624] $171231513221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [3625] $17123151322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ³²²¹
 [3626] $221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ^{322^1 1}
 [3627] $221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- ²²²¹
 [3628] $1(12)122^1 31513217123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ²²¹
 [3629] $1(12)1223151321712^1 32 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

- [3643] $7122^1 31513221^{3221}(12)122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3644] $712231513221^{322^1 1}(12)122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3645] $1413221(12)122^1 315131513 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3646] $3141322^1 1(12)12231513215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3647] $31413221(12)12^1 231513215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3702] $1712321712321812^1 3215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 34$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 34$, $T \in [26, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3648] $3141322^1 1(12)12231513151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3703] $1712^1 3217123218123215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 34$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 34$, $T \in [26, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3649] $31413221(12)12^1 231513151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3650] $232^1 16131513221(12)12231 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3651] $21613151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3652] $216131513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3653] $32161315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3654] 16123151322¹1(12)1223132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3655] 161231513221(12)12¹23132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3656] 31612315132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3657] 321712315132¹21(12)12222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3658] 232¹171231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3659] 217123151322¹1(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3660] 2171231513221(12)12¹2222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3661] 1712231513221(12)12¹2222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3662] 171223151322¹1(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3663] 1712¹231513221(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3664] 413151322¹1(12)122315123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3665] 4131513221(12)12¹2315123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [3666] $413151322^1 1(12)12231513 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3667] $4131513221(12)12^1 231513 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3668] $61223151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3669] $612^1 231513221(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3670] $612231513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3671] $1^1 8^1 12321812^1 32161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3672] $18^1 1232^1 181232161315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3673] $18^1 1232^1 181231513161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3674] $1812^1 321812315131613 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3675] $18123151321812^1 32161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3676] $1812^1 315132181232161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3677] $171232^1 1812321712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3678] $1712321812315132^1 171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3679] $171232^1 1812315132171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3680] \quad 413161315132^{2221}21(12)1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 37, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 37, T \in [26, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3681] \quad 413161315132^{221}21(12)12 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 37, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 37, T \in [26, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3682] \quad 612^13151321812321^2181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[3683] \quad 6123151321812^1321^2181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3684] \quad 7122^13151321812321^171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[3685] \quad 71223151321812^1321^171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3686] \quad 1812^1321^1812321^18123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3687] \quad 181232^{21}1812321^181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3688] \quad 1812321812321812^123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3689] \quad 1812^1321^181232181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3690] \quad 1^232^11^1232181232181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3691] \quad 1^321^181232^11812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3692] \quad 1812321812^13218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [3693] $181232^{\overline{21}}18123218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3694] $161232^{\overline{1}}18123218122315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3695] $5131612321812^{\overline{1}}3218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3704] $1712^{\overline{1}}3217123218123151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3705] $1712321712321812^{\overline{1}}3151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3706] $2171232^{\overline{1}}1812321712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3707] $21712321812315132^{\overline{1}}171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3708] $2171232^{\overline{1}}1812315132171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3709] $413161315132^{\overline{1}}21(12)12222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 37$, $T \in [26, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3710] $613151321812321812^{\overline{1}}23 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3711] $612^{\overline{1}}32181231513218123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 47$, $T \in [26, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3712] $6123151321812^{\overline{1}}3218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3713] $612^{\overline{1}}31513218123218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3714] $71223151321812^{\overline{1}}321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3715] $7122^{\overline{1}}3151321812321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3716] $21812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3717] $21812^1 181232181223 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3718] $22181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3719] $4132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3720] $612^1 3218123218123215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 35$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 35$, $T \in [26, 35]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3721] $612321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 35$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 35$, $T \in [26, 35]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3722] $612^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3723] $612321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3724] $413221(12)122315132^1 16 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3725] $4132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3726] $171232^1 1812321613161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3727] $4132^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3728] $4132181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3729] $16131612321812^1 321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3730] $31613161232181232^1 171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3731] $4132^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3732] $4132181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3733] $612^1 32171232181232161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3698] $161232^1 18123151321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3735] $4132^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3736] $4132181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3737] $414141413221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3696] $1613151321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3697] $3161315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3734] $612321712321812^1 32161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3738] $16^{31} 13151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -96$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3742] $1(12)12^{221} 231513217123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3743] $1(12)122^{221} 315132171232 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 47$, $T \in [26, 47]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3757] 15131613161315132¹181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3758] 31612321812315132¹181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3759] 16123218123151322¹181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3760] 4131613151322¹1(12)12222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 37$, $T \in [26, 37]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3761] 8122¹3151316131613151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3762] 8122¹3151321812321613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3763] 712¹32171232181231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 47$, $T \in [26, 47]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3764] 712¹31513218123217123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3765] 712¹23151321812321712 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3766] 712¹3151316131613161¹ + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3767] 712¹32161316131613151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3768] 712¹31513161316131612 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3769] 41316131613161232¹181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3741] 1¹³²²¹(12)12¹23151321613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [3740] $1613151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -104, \det = 36, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3739] $1(12)12^1 23151316123 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 26, \Delta = -97, \det = 36, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3770] $1(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3771] $1(12)1223151 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3772] $21(12)1223151 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3773] $21(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3774] $1(12)122315132^1 1613 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 26, \Delta = -97, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3775] $221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3776] $1(12)122315132^1 17123 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3777] $1(12)12^1 2315132171232 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3778] $21(12)122315132^1 17123 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3779] $21(12)12^1 2315132171232 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3780] $7122315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3781] 2321613151322¹1(12)12231 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3782] 31612¹31513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3783] 321712¹31513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3784] 23217123151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3785] 1⁵¹³²8¹1232181232161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3786] 1⁵¹³²¹8¹1232¹181232161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3787] 1²¹8¹12321812¹32161315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3788] 1²¹8¹12321812¹31513161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3789] 1¹181232¹1812315131613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3790] 1¹1812315132¹181232161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3791] 1¹1712321812¹321712315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3792] 1¹1712321812321712¹315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3793] 1¹1712321812¹315132171 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

- [3794] $414132^{121} \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3795] $41413221^{51315132^1 21} \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3796] $61315132181232^{321} 1 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3798] $7122315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3799] $181232^1 181232^1 18123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3800] $181232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3801] $181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3802] $1 \quad 8 \quad 12^{321} 321812321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3803] $1 \quad 8 \quad 12321812^{321} 321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3804] $16123218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3805] $1612321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3806] $513161232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3807] $161315132^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3808] $31613151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3809] $1612321812^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3810] \quad 31612^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3811] \quad 1612321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3812] \quad 171232^1 17123218123215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 34, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 34, T \in [26, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3813] \quad 171232171232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 34, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 34, T \in [26, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3814] \quad 171232^1 17123218123151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3815] \quad 171232171232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3816] \quad 21712321812^1 321712315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3817] \quad 21712321812321712^1 315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3818] \quad 21712321812^1 315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3819] \quad 812^1 31513161316131513 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3820] \quad 812^1 23151316131613151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3821] \quad 61315132181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3822] \quad 61232^1 181231513218123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3823] \quad 612315132^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3824] \quad 812^1 23151321812321613 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3825] \quad 8122315132181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3826] \quad 712315132181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3827] \quad 7122315132^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3828] \quad 2181232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3829] \quad 2181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3830] \quad 413221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3831] \quad 41322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3832] \quad 61232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 35, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 35, T \in [26, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3833] \quad 61232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 35, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 35, T \in [26, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3834] \quad 61232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3835] \quad 61232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3836] \quad 413221(12)122^1 31513216 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3837] \quad 41322^1 1(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3838] \quad 1712321812^1 321613161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3839] \quad 41321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[3840] \quad 41321812^1 32171232181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3841] \quad 1613161232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3842] \quad 316131612321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[3843] \quad 41321812^1 321712321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3844] \quad 41321812321712^1 321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[3845] \quad 613161232181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3846] \quad 61232^1 171232181232161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3847] \quad 61232171232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3848] \quad 41321812^1 32181232171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3849] \quad 41321812321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3850] \quad 414141413221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3797] \quad 612315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3865] \quad 2171232181232^1 1712315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 64, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.94 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3866] 613151321812¹32181223 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3856] 613151321812¹321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3857] 1812321812¹321¹8123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3858] 1812321812¹32181223 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3859] 1³²¹8 1232¹1812321812 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3860] 161232181232¹18122315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3861] 5131612¹3218123218122 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3862] 3161232¹1812315132181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3863] 1712321712¹3218123215 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 34$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 34$, $T \in [25, 34]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3854] 1²¹8 1232181232¹161315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3853] 1⁵¹³²¹8¹²¹32181232161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3852] 141414141414132¹21(12)12 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3867] 81223151321812¹321613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3868] 712321712321812¹31513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 47$, $T \in [25, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [3869] $7123151321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3870] $21812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3871] $612321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 35$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3872] $612321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3873] $171232181232^1 1613161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3874] $4132181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3875] $16131612^1 321812321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3876] $4132181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3877] $6131612321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3878] $612321712^1 32181232161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3879] $4132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3855] $171232181232^1 1712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3864] $1712321712^1 3218123151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3851] $1(12)122^1 31414141414141 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3881] 1414141414141322¹1(12)12 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3884] 21^{513151322¹1}(12) 1223151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3885] 221³²²¹(12)12¹2315132171 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3886] 71223151322¹1³²²¹(12)122 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3887] 4141322¹1⁵¹³¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3888] 41413221^{513151322¹1}(12) 1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3889] 1²³²¹8¹12321812¹32181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3890] 612321812315132¹18123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3891] 71232¹171232181231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3892] 413221²¹(12)12¹231513161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3893] 413221²¹(12)12¹231513216 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3894] 71232¹161316131613151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3895] 413161316131612¹32181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3882] 3141414141414132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3883] & 1 \quad \begin{matrix} 1 \\ 513151322^1 1 \end{matrix} \quad 1223151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.84 \rightarrow 25, T_{\max} = 36, T \in [25, 36] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3880] & 1 \quad \begin{matrix} 1 \\ 313141414141414141 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.84 \rightarrow 25, T_{\max} = 32, T \in [25, 32] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3896] & 3141414141414141322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3897] & (12)122^1 314141414141414132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3898] & \begin{matrix} 221 \\ 1(12)122315132^1 171232 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3899] & \begin{matrix} 21 \\ 21(12)122315132^1 171232 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3900] & 2321712^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3901] & 61315132^1 1812321 \begin{matrix} 321 \\ 8 \ 1 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3902] & 1 \quad \begin{matrix} 2321 \\ 8 \ 12^1 32181232181 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 40, T \in [25, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3904] & 31612321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3905] & 61315132^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3906] & 8122315132^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3907] 71232171232¹181231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 100$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3908] 712315132¹18123217123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3909] 613161232¹18123217123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3903] 1612321812321812¹2315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.76 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3910] (12)12¹23141414141414132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3911] 1812321812¹315131613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3912] 1²³²¹8 1232¹181232181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3913] 3161315132¹1812321812 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3914] 612321812¹31513218123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3915] 712321712¹32181231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3916] 31613161232¹181232171 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3917] 6131612¹3218123217123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3918] 4141414132¹21(12)1223151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3919] \ 316131612^1 32181232171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.59 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[3920] \ 41414141322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.59 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [3956] \ 22^1 1(12)12231513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3925] \ 1^1 8^1 1231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3926] \ 1^1 8^1 1231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3927] \ 1^1 8^1 1231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3928] \ 1^1 8^1 12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3929] \ 1^1 8^1 12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3930] \ 1^1 8^1 12231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3931] \ 1^1(12)12231513161315132^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 58, T \in [28, 58]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3932] \ 1513221^1(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3933] \ 1^1(12)1223151316131513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 28, T_{\max} = 48, T \in [28, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3934] $1(12)122315131613151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3935] $3221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 58, T \in [27, 58]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3936] $21(12)122315131613151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3937] $1(12)122315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3938] $1 \ 8 \ 1231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3939] $1 \ 8 \ 1231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 44, T \in [28, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3940] $1 \ 8 \ 1231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3924] $1 \ 8 \ 1231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3923] $1 \ 8 \ 1231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 44, T \in [28, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3922] $1 \ 8 \ 1231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3944] $3221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 38, T \in [27, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [3945] $21(12)1223151321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[3946] & \begin{matrix} 22^1 21 \\ 21(12)12231513218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3947] & \begin{matrix} 221 \\ 221(12)1223151321812^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[3948] & \begin{matrix} 221 \\ 22^1 1(12)12231513218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3949] & \begin{matrix} 1 \\ 1812231513221(12)12222^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[3950] & \begin{matrix} 2^1 21 \\ 3221(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[3951] & \begin{matrix} 22^1 21 \\ 1(12)122315132218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 38, T \in [28, 38] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3952] & \begin{matrix} 22^1 1 \\ 1(12)1223151322181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 44, T \in [28, 44] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3953] & \begin{matrix} 22^1 221 \\ 1(12)12231513221812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3954] & \begin{matrix} 22^1 1 \\ 21(12)122315132218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 38, T \in [28, 38] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3955] & \begin{matrix} 22^1 21 \\ 21(12)12231513221812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3921] & \begin{matrix} 1 \\ 181231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[3957] & \begin{matrix} 1513221 \\ 61231513221(12)12^1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 35, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 35, T \in [27, 35] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[3958] $22^1 3151316131513221(12)12 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 58, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 58, T \in [28, 58]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3959] $23151316131513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 58, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 58, T \in [28, 58]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3960] $3151316131513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 48, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 48, T \in [28, 48]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3961] $151316131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3962] $232171231513221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 28, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3963] $322181231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3964] $22181231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3965] $22^1 181231513221(12)122222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3966] $321812231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3967] $2321812231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 44, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 44, T \in [28, 44]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3968] $21812231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3969] $51321712321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3970] $17123151321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3971] \quad 71223151321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3972] \quad 413221(12)^{21} 122315132171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3973] \quad 413221(12)^{21} 122315132217 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3974] \quad 16131612321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3941] \quad 1(12)^{22221} 1223151321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3943] \quad 21(12)^{2211} 122315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3942] \quad 1(12)^{221221} 12231513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3978] \quad 181231513221(12)^1 122^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3979] \quad 181231513221(12)^{3211} 12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3980] \quad 181231513221(12)^{21} 12^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3981] \quad 1812231513221(12)^1 122^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3982] \quad 1812231513221(12)^{21} 12^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$$

15.0

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3983] \quad 1(12)122315131613151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -110, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3984] \quad 21(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 58, T \in [28, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3985] \quad 513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.33 \rightarrow 27, T_{\max} = 58, T \in [27, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3986] \quad 1(12)122315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3987] \quad 1(12)12231513218122^1 3 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3988] \quad 1 \ 8 \ 1231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3989] \quad 1 \ 8 \ 1231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3990] \quad 1 \ 8 \ 1231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 44, T \in [28, 44]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3991] \quad 1 \ 8 \ 1231513221(12)1222^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3992] \quad 1(12)12231513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[3993] \quad 21(12)12231513218122^1 3 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [3994] $221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3995] $21(12)12231513218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [3996] $1812231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3997] $221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3998] $1(12)12231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3999] $1(12)122315132218123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [4000] $1(12)12231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [4001] $21(12)12231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [4002] $21(12)12231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [4003] $6131513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.33 \rightarrow 27$, $T_{\max} = 58$, $T \in [27, 58]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [4004] $3151316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 48$, $T \in [28, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [4005] $151316131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [4006] $232171231513221(12)12^1 231 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 28$, $\Delta = -98$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 28$, $T \in [28, 28]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4007] $32^1 171231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4008] $32171231513221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4009] $322181231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4010] $32^1 2181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4011] $22181231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4012] $321812231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4013] $32^1 1812231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4014] $2321812231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 44, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 44, T \in [28, 44]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4015] $21812231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4016] $7123151321812321 \overset{32^1 1}{8} 1 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 41, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4017] $32^1 1812321712321812315 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 48, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 48, T \in [28, 48]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4018] $17123218123151321812^1 3 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 41, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4019] $32^1 1712321812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4020] $41316131513221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -103$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4021] $71231513218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4022] $81223151321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4023] $6131612321812321 \overset{32^1 1}{8} 1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4024] $81231513161316131612^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4025] $61316123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3976] $1 \overset{32^1}{8} 1231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3975] $181231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3977] $1 \overset{32^1}{8} 1231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4106] $6122315132^1 21(12)12231413 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4026] $181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.25 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4031] $1812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4032] $1812315132^1 21^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4033] $1812^1 131513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4034] $18122315132^1 21^{12}(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4035] $18122^1 131513221^1(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4036] $1812231513221^1(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4037] $18122315132^1 21^{21}(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4038] $18122^1 131513221^{21}(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4039] $18122315132^1 21^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4040] $18122^1 131513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4041] $1712^1 131513221(12)1223132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4042] $1712315132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4043] $171231513221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4044] $181231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4045] $181231513221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4046] $1812^1 31513221(12)122222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4047] $1812315132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4048] $1(12)122315132^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4049] $1(12)122^1 31513218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4050] $1(12)12231513218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4051] $22^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.25 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4052] $21(12)122^1 31513218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4053] $21(12)122315132^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4054] $221(12)122315132^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [4055] \quad 221^{221}(12)122^1 31513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4056] \quad 1812231513221(12)122^1 222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [4057] \quad 18122315132^1 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [4058] \quad 18122^1 31513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4059] \quad 22^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.25 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4060] \quad 1^{22221}(12)122315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [4061] \quad 1^{22221}(12)122^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4062] \quad 1^{2222^1 1}(12)12231513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [4063] \quad 21^{2221}(12)122^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4064] \quad 21^{2221}(12)122315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [4065] \quad 221^{221}(12)122^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [4066] $221(12)122315132^1 21812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4067] $151316131513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4068] $2171231513221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4069] $21712^1 31513221(12)1223132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4070] $21712315132^1 21(12)1223132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4071] $32171231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4072] $17122^1 31513221(12)1223132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4073] $17122315132^1 21(12)1223132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4074] $1712231513221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4075] $322181231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4076] $221812315132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4077] $221812^1 31513221(12)122222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4078] $22181231513221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4079] 321812231513221(12)12¹222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4080] 218122¹31513221(12)12222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4081] 21812231513221(12)122¹222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4082] 218122315132¹21(12)12222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4083] ¹1712¹3218123218122315 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4084] ¹181232171232181232¹15 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[4085] ²¹1812¹321812321712315 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4086] ⁵¹³²¹181232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4087] ¹1712¹3218123151321812 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4088] ²¹1812¹321812315132171 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4089] ¹1812315132181232¹1712 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [4090] $1712^1 3218123151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4091] $7122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4092] $1712^1 321812321812315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4093] $1712^1 321812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4094] $141322^1 1(12)1223151316123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4095] $141322^1 1(12)1223151321613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4096] $31413221(12)122315132^1 161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4097] $31413221(12)122^1 315132161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4098] $314132^1 21(12)122315132161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4099] $21712^1 3218123218122315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4100] $2181232171232181232^1 15 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [4101] $5132171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4102] $1712315132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4103] $21712^1 3218123151321812 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4104] $21712^1 3218123151322181 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4029] $1812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4028] $1812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4027] $181231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4108] $612231513221(12)122^1 31413 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4109] $7122315132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4110] $171232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[4111] $41315132^1 21(12)1223151314 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4112] $1812^1 321812321613161 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4113] $61232171232181232^1 171 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [4114] $1613161232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4115] $1612^1 32181232171232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4116] $61232171232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4117] $61232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4030] $1^1 8 12315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4107] $6122^1 31513221(12)12231413 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4105] $41316131513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4227] $612231513221(12)12^1 231413 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4122] $1^1 8 12315132^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4123] $1^1 8 12315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4124] $1^1 8 1231513221(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 44$, $T \in [27, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [4125] $1^1 8 12315132^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [4126] $18123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [4127] $181223151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [4128] $1812^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [4129] $181223151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [4130] $1812^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [4131] $1812^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [4132] $181223151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [4133] $21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [4134] $1(12)122^1 315131613151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [4135] $21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 58$, $T \in [27, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [4136] $21(12)122^1 315131613151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [4137] $17123151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4138] $1^{13221} 171231513221(12)12^1 23132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4139] $1^{13221} 1(12)122^1 31513217123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4140] $1^{2221} 1(12)1223151321812^1 23 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4141] $1^{2221} 1(12)122^1 315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4142] $1^{321} 1812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4143] $1^{232^1 1} 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4144] $1^{21} 181231513221(12)12^1 2222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4145] $1^{21} 18123151322^1 1(12)122222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4146] $1^{2221} 1(12)12^1 231513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4147] $1^{221} 21(12)1223151321812^1 23 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4148] $1^{221} 21(12)122^1 315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

- [4160] $1(12)12^{2221}12^1 231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4161] $1(12)12231513221^{2221}1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4162] $21(12)122^{221}315132218123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4163] $21(12)12231513221812^{21}32 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 44$, $T \in [27, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [4164] $21(12)12^{2221}231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4165] $21(12)12231513221^{2221}1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4166] $221(12)12^{221}231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4167] $221(12)12231513221^{221}1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4168] $612^{1513221}131513221(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 35$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 35$, $T \in [27, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4169] $61231513221^{15132^1 21}(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 35$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 35$, $T \in [27, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4170] $1513161315132^{21}21(12)12222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4171] $232^{171}231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 28$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 28$, $T \in [27, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[4172] $217123151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4173] $2171231513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4174] $321712315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4175] $1712^1 231513221(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4176] $171223151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4177] $1712231513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4178] $3221812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4179] $322^1 181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4180] $2218123151322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4181] $22181231513221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4182] $3218122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4183] $232^1 1812231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 44, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4184] $2181223151322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4185] 21812231513221(12)12¹2222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4186] 21812¹231513221(12)122222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4187] 171232¹18123218122315 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[4188] 1812¹3217123218123215 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4189] 1812321712321812¹3215 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4190] 1812¹3217123218123151 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4191] 1812321712321812¹3151 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4192] 181232²¹1812321712315 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[4193] 1⁵¹³²¹8¹²³²¹12321812¹321712 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4194] 1712321812315132¹1812 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4195] 171232¹18123151321812 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

- [4196] $1812321812315132^1 171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4197] $181232^1 1812315132171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4198] $18123151321812^1 321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [4199] $1812^1 3151321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4200] $171232^1 18123151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4201] $1712321812315132^1 2181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4202] $7122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4203] $71223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [4204] $171232^1 1812321812315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [4205] $171232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4206] $1712321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4207] $1413221(12)122^1 3151316123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[4208] 1413221(12)122¹3151321613 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4209] 3141322¹1(12)122315132161 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[4210] 31413221(12)12¹2315132161 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4211] 2171232¹18123218122315 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[4212] 21812¹3217123218123215 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4213] 21812321712321812¹3215 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4214] 21812321712321812¹3151 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4215] 21812¹3217123218123151 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4216] 51321712321812¹3218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4226] 61223151322¹1(12)12231413 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4218] 1712¹31513218123218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4219] 17123151321812¹3218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4220] 2171232¹18123151321812 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[4221] \quad 21712321812315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4222] \quad 21712321812315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4223] \quad 2171232^1 18123151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[4224] \quad 413161315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4225] \quad 612^1 231513221(12)12231413 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[4121] \quad 1 \quad 8 \quad 232^1 1 \quad 221 \quad 1231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[4120] \quad 1 \quad 8 \quad 321 \quad 1 \quad 12315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4119] \quad 1 \quad 181231513221(12)12^1 23 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4229] \quad 8123151321812321712^1 32 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4230] \quad 7123218123151321812^1 32 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4231] \quad 71223151321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[4232] \quad 7122^1 31513218123218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$$

15.0

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[4233] \quad 1712^1 13218123218123215 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[4234] \quad 1712321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4235] \quad 1712^1 13218123218123151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[4236] \quad 1712321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4237] \quad 4132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4238] \quad 413221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[4239] \quad 413221(12)122315132^1 217 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4240] \quad 4132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4241] \quad 413151322^1 1(12)1223151314 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -97, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4242] \quad 181232^1 1812321613161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[4243] \quad 1812^1 13218123216131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4244] $612^1 32171232181232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4245] $612321712321812^1 32171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4246] $4132181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4247] $4132^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4248] $16131612321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4249] $31613161232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4250] $161232^1 181232171232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4251] $161232181232171232^1 171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4252] $6131612321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4253] $612^1 321712321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4254] $612321712321812^1 321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4255] $4132181232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [4256] $4132^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 36$, $T \in [27, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4257] $612321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [4258] $612^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [4259] $612^1 31513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 41$, $T \in [27, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4260] $4132^1 18123218123151316 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 41$, $T \in [27, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [4118] $1812^1 31513221(12)1223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4228] $7123151321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 41$, $T \in [27, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4217] $1712^1 32181231513218123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 41$, $T \in [27, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4264] $1^1 8^{2321} 12315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 44$, $T \in [27, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4265] $1(12)122^1 3151316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 48$, $T \in [27, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4266] $1(12)12^1 2315131613151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4267] $21(12)122^1 3151316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 48, T \in [27, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4268] & \begin{smallmatrix} 221 \\ 1(12)12^1 2315131613151 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4269] & \begin{smallmatrix} 13221 \\ 1(12)12^1 231513217123 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4270] & \begin{smallmatrix} 2221 \\ 1(12)12^1 2315132181223 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4271] & \begin{smallmatrix} 321 \\ 1 \ 8 \ 123151322^1 1(12)12222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4272] & \begin{smallmatrix} 2321 \\ 1 \ 8 \ 12315132^1 21(12)1222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4273] & \begin{smallmatrix} 221 \\ 21(12)12^1 2315132181223 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4274] & \begin{smallmatrix} 3221 \\ 221(12)1223151322^1 181 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4275] & \begin{smallmatrix} 2221 \\ 1(12)12^1 2315132218123 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4276] & \begin{smallmatrix} 221 \\ 1(12)122^1 3151322181232 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4277] & \begin{smallmatrix} 221 \\ 21(12)12^1 2315132218123 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4278] & \begin{smallmatrix} 21 \\ 21(12)122^1 3151322181232 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]
\end{aligned}$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4279] 31513161315132¹21(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4280] 15131613151322¹1(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4281] 2321712315132¹21(12)12231+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 28$, $\Delta = -98$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4282] 3217123151322¹1(12)122313+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4283] 32218123151322¹1(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4284] 32181223151322¹1(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4285] 23218122315132¹21(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4286] 17123218123151322¹181+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4287] 712¹3151321812321³²¹81+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4288] 712¹23151321812321²¹81+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4289] 1413221(12)12¹23151316123+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4290] 1413221(12)12¹23151321613+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4291] $1812^1 32171232181231513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4292] $321712321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4293] $217123218123151322^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4294] $4131613151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -103$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4295] $712^1 315132181232181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4296] $712^1 321812315132181232 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4297] $8122^1 31513218123217123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4298] $712^1 231513218123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4299] $712^1 32181232181231513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4300] $413221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4301] $16131613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4302] $8122^1 3151316131613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4303] $151316131613161232^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4304] $216131613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4305] $8122^1 31513161316131612 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4306] $712^1 321712321812321613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4307] $712^1 32161316131613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4308] $712^1 321613161316131612 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4263] $1^{\frac{321}{8}} 123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4262] $1^{\frac{2321}{8}} 12315132^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4261] $1^{\frac{321}{8}} 123151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4360] $321812321712321812^1 315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4313] $1^{\frac{2321}{8}} 123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[4311] $1^{\frac{2321}{8}} 123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

- [4310] $1 \overset{321}{8} 12^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [4316] $21 \overset{1513221}{(12)} 122^1 31513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [4317] $1 \overset{221}{(12)} 12^1 23151316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 48$, $T \in [27, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4318] $21 \overset{21}{(12)} 12^1 23151316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 48$, $T \in [27, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4319] $21 \overset{5132^1 21}{(12)} 122315131613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 58$, $T \in [27, 58]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [4320] $1 \overset{13221}{(12)} 122315132^1 17123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4321] $1 \overset{2221}{(12)} 122315132^1 181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4322] $1 \overset{321}{8} 12^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [4323] $1 \overset{2321}{8} 123151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 44$, $T \in [27, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [4324] $21 \overset{221}{(12)} 122315132^1 181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4325] $221 \overset{3221}{(12)} 122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [4326] $221 \overset{3221}{(12)} 122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4327] & \begin{matrix} 2221 \\ 1(12)122315132^1 218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4328] & \begin{matrix} 221 \\ 1(12)12^1 23151322181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4329] & \begin{matrix} 221 \\ 21(12)122315132^1 218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4330] & \begin{matrix} 21 \\ 21(12)12^1 23151322181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4331] & \begin{matrix} 151322^1 1 \\ 61231513221 (12) 12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 35, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 35, T \in [27, 35] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[4332] & \begin{matrix} 1513221 \\ 612315132^1 21 (12) 12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 35, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 35, T \in [27, 35] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4333] & 231513161315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 58, T \in [27, 58] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[4334] & 315131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 48, T \in [27, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[4335] & 23217123151322^1 1(12)12231 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 28, \Delta = -98, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 28, T \in [27, 28] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[4336] & 321712^1 31513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4337] & 3221812^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4338] & 3218122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4339] $232181223151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[4340] $1712321812^1 3218122315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4341] $17123218123218122^1 315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4342] $181232^1 17123218123215 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4343] $181232171232^1 18123215 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4344] $181232^1 17123218123151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4345] $181232171232^1 18123151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4346] $1812321812321712^1 315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4347] $1812321812^1 321712315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4348] $1^1 8^1 12321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[4349] $1^1 8^1 1232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4350] $1712321812^1 3151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4351] & \overset{21}{1} 8 12321812^1 315132171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4352] & \overset{1}{1} 812315132^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[4353] & \overset{1}{1} 712321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4354] & 712315132181232^1 \overset{321}{1} 8 1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4355] & 7122315132^1 1812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[4356] & \overset{1}{1} 712321812321812^1 315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[4357] & \overset{1}{1} 712321812^1 321812315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[4358] & \overset{1}{1} 712321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4359] & 1413221(12)122315132^1 1613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4309] & \overset{1}{1} 812315132^1 21 \overset{221}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4361] & 217123218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[4362] & 21712321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4363] 2181232¹17123218123215 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4364] 2181232171232¹18123215 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4365] 2181232171232¹18123151 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4366] 2181232¹17123218123151 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4367] 5132¹17123218123218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[4368] 5132171232¹18123218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4369] 171232¹181231513218123 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4370] 321712¹321812315132181 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4371] 1712315132¹18123218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4372] 21712321812¹3151321812 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4373] 21712321812¹3151322181 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4374] 812¹321712321812315132 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4375] $712315132181232^1 181223 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 41, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4376] $812^1 315132181232171232 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 48, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4377] $812^1 231513218123217123 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 41, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4378] $8122315132181232^1 17123 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 41, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4379] $7122315132^1 18123218122 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4380] $171232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4381] $171232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4382] $171232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4383] $171232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4384] $413221(12)^{21} 122^1 315132171 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4385] $41322^1 1(12)^{21} 122315132171 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4386] $413221(12)^{21} 122^1 315132217 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4387] $41322^1 1(12)122315132217 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4388] $1812321812^1 321613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4389] $181232^1 18123216131613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4390] $812^1 3151316131613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4391] $812^1 23151316131613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[4392] $613161232181232^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4393] $61232171232^1 181232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4394] $61232^1 171232181232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[4395] $41321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4396] $41321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4397] $1613161232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4398] $316131612321812^1 321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[4399] $1612321812^1 32171232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4400] $1612321812321712^1 32171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[4401] $4141413221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4402] $812^1 321613161316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4403] $812^1 315131613161316123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4404] $812^1 231513161316131612 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[4405] $613161232181232^1 181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4406] $61232171232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4407] $61232^1 1712321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[4408] $71232171232181232^1 1613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4409] $414132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4410] $41413221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4411] $41321812^1 321812321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4412] $41321812321812^1 321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4413] $61232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4414] $61232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4415] $61231513221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4416] $612315132^1 21(12)12231414 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4417] $41321812^1 3218123151316 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4315] $21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4312] $1^1 8^1 12^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4314] $1(12)122^1 31513161315132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 58, T \in [27, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4419] $1(12)12^1 231513161315132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[4422] $2315131613151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[4423] & \begin{matrix} 1 \\ 171232181232^1 18122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[4424] & \begin{matrix} 1 \\ 1812321712^1 3218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 34, T \in [26, 34] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4425] & \begin{matrix} 1 \\ 1812321712^1 3218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4426] & \begin{matrix} 21 \\ 181232181232^1 1712315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[4427] & \begin{matrix} 51321 \\ 1 \ 8 \ 12^1 321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[4428] & \begin{matrix} 321 \\ 7123151321812^1 321 \ 8 \ 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4429] & \begin{matrix} 1 \\ 171232181232^1 1812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[4430] & \begin{matrix} 1 \\ 1812321712321812^1 31513 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4431] & \begin{matrix} 1 \\ 2171232181232^1 18122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[4432] & \begin{matrix} 1 \\ 21812321712^1 3218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 34, T \in [26, 34] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4433] & \begin{matrix} 1 \\ 21812321712^1 3218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4434] & \begin{matrix} 1 \\ 51321712^1 3218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]
\end{aligned}$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[4435] $32171232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[4436] $7123151321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4437] $81223151321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4438] $1712321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 34, T \in [26, 34]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4439] $1712321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4440] $712321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4441] $181232181232^1 1613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[4442] $6131612321812^1 321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4443] $612321712^1 32181232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[4444] $4132181232^1 1812321812^1 321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4445] $16131612^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[4446] $161232181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[4447] 6131612321812¹32181223 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4448] 612321712¹321812321712 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[4449] 712321712321812¹321613 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4450] 4132181232¹1812321812 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4451] 612321812¹32181232161 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[4452] 4132181232¹18123151316 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[4421] 2231513161315132¹21(12)12 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4418] 1(12)122¹315131613151322 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[4420] 613151322¹315132¹21(12)12 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4456] 21(12)12¹231513161 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.84 \rightarrow 26, T_{\max} = 34, T \in [26, 34]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[4454] 18123151322¹1(12)1223 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[4458] 221(12)12¹23151321812 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$$

- $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
 [4459] $221^{3221}(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- 2221
 [4460] $1(12)1223151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- 221
 [4461] $21(12)1223151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- 1513221
 [4462] $6123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 35, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 35, T \in [26, 35]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [4463] $22315131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [4464] $321812^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [4465] $181232^1 171232181231513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [4466] $321812^1 321712321812315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 48, T \in [26, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [4467] $1712321812315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [4468] $71232^1 1812315132181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 48, T \in [26, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- 21
 [4469] $413221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- 1
 [4470] $71232^1 181232181231513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[4471] & 413221(12)^{21}12^1 2315132217 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4472] & 1513161316131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[4473] & 41414132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[4474] & 4141413221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4475] & 71232^1 1712321812321613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[4476] & 41413221(12)^1 12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[4477] & 4141322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4478] & 6123151322^1 1(12)12231414 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[4479] & 61231513221(12)^1 12^1 231414 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4480] & 71232^1 161316131613161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[4481] & 71232^1 1613161316131612 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [4457] $21(12)122^{3221}31513161315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [4455] $1(12)12^{12315131613151322} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4453] $14141414141414132^{121(12)1} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -108$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [4485] $1^1 8^{2321} 12^{1231513221(12)12} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [4486] $21^{51322^{11}} (12) 122315131613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4487] $1^1 8^{2321} 12^{1231513221(12)1222} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [4488] $1(12)122315132^{12181232} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4489] $21(12)122315132^{12181232} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4490] $61315132^{121(12)12} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4491] $2321712^{131513221(12)12231} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 28$, $\Delta = -98$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [4492] $23218122^{131513221(12)1222} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [4493] $1712321812321812^{12315} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4494] 712315132¹1812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4495] 181232171232¹181231513 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4496] 21712321812321812¹2315 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4497] 321712321812¹315132181 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[4498] 712315132¹181232181223 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4499] 812315132181232¹171232 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 48, T \in [26, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4500] 8122315132¹18123217123 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4501] 71232181232¹181231513 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4502] 613161232¹1812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4503] 613161232¹181232181223 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4504] 71232171232¹1812321613 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4505] 41321812321812¹3151316 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4484] & 1 \ 8 \ 12^{2321} 131513221^{221} (12)1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.76 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[4482] & 14141414141414141322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.76 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[4483] & (12)122^1 3141414141414141413 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4510] & 1812321712^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4506] & (12)12^1 23141414141414141413 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[4507] & 21(12)12^{3221} 231513161315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[4512] & 1712321812^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[4513] & 81232^1 1712321812315132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[4514] & 812321712321812^1 315132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4515] & 8123151321812^1 32171232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4516] & 712321812^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4517] & 1812321812^1 3216131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

- $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 38$, $T \in [26, 38]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [4518] $6131612^1 321812321^{321} 8\ 1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 41$, $T \in [26, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [4519] $31613161232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 38$, $T \in [26, 38]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [4520] $414141322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4521] $81232^1 1613161316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [4522] $6131612^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 41$, $T \in [26, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [4523] $712321712^1 321812321613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [4508] $21(12)122^1 315131613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [4511] $32181232171232^1 1812315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 48$, $T \in [26, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [4509] $6131513221^{3151322^1 1} (12) 12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.69 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [4524] $32181232^1 1712321812315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -144$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.64 \rightarrow 26$, $T_{\max} = 48$, $T \in [26, 48]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 59 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4528] $2321812^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.59 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [4525] $1(12)1223151322^{221} 181232 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[4530] & 81232171232^1 1812315132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[4531] & 812315132^1 181232171232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[4532] & 712321812315132^1 181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[4533] & 181232181232^1 16131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[4534] & 316131612^1 321812321812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[4527] & 613151322^1 1^{31513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[4529] & 321812321712^1 321812315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[4526] & 21^{21} (12) 1223151322^1 181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[4535] & 21^{513221} (12) 12^1 2315131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -196, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.55 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[4536] & 812321712^1 321812315132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.55 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[4537] & 712321812^1 315132181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -225, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.52 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- $T = 22$ [26] $612321712321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [25] $413221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3] $161316131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [4] $3221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 42, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 42, T \in [28, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [5] $1(12)12231513161316123 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 42, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 42, T \in [29, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [6] $1(12)1223151316131612 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [7] $161316131513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 39, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 39, T \in [29, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [8] $21(12)1223151316131612 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [9] $1^1 8^1 12321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [10] $18123151321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [11] $4132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 38, T \in [29, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [12] $4132181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 38, T \in [29, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [13] $413221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1] $712231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [14] $\overset{1}{161316131513221}(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [15] $\overset{1}{161316131513221}(12)12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [16] $\overset{31}{161316131513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 39, T \in [29, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [17] $\overset{22^1}{32161316131513221}(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 42, T \in [29, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [18] $\overset{22^1}{232161316131513221}(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 49, T \in [29, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [19] $\overset{22^1}{2161316131513221}(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [20] $\overset{22^1}{22^1 18123218123218122315} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [21] $\overset{22^1}{22^1 18123218123151321812} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [22] $\overset{22^1}{22^1 18123218123151322181} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [0] $\overset{2^1}{1513221}(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [23] $\overset{2^1}{4132181231513221}(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 38, T \in [29, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [24] $\overset{2^1}{413221}(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2] $\overset{2^1}{3151316131513221}(12)122^1 31 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$

- [31] $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$ 151316131513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 29$, $T \in [29, 29]$
- [41] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 21(12)1223151316131612¹3 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
- [28] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 1 8 1231513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [29] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$ 71231513221 (12) 12¹ + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.33 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
- [30] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 3151316131513221(12)12¹231 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 29$, $T \in [29, 29]$
- [32] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 322181231513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [33] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$ 32¹2181231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [34] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 321812231513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [35] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$ 32¹1812231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [36] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 161316131513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [37] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$ 21(12)122315131613161 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
- [38] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$ 1(12)1223151316131612¹3 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
- [39] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$ 1(12)1223151316131612 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$

- [40] $161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [42] $18123218123151321812^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [43] $1812321812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [44] $4132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [45] $8122315132181232181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [46] $161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [47] $32161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [48] $32^1 161316131513221(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [49] $2161316131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [50] $32^1 21812321812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [51] $218123218123151321812^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [52] $4132181231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [53] $812231513218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [54] $32^1 1812321812321812315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 45$, $T \in [29, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [55] $612321812321712321812^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 45$, $T \in [29, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [56] $712321712321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [57] $812321613161316131612^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [27] $1^{\frac{32}{1}} 8^1 1231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [79] $1712^1 32181232171232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [59] $151316131513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 29$, $T \in [28, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [78] $181232171232181232^1 161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [77] $2181232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [60] $322181231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [61] $321812231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [62] $161316131513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [63] $1812^1 3218123218122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [64] $1^{\frac{51321}{1}} 8^1 1232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [65] $1812315132181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [66] $1812^{21} 3218123151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [74] $221812^1 3218123151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [75] $4132181231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [76] $61315132181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 37, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [67] $1812^{21} 3218123151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [68] $181232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [69] $1812^{21} 321812321812315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [70] $1812^{21} 321812315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [71] $2161316131513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [72] $221812^1 3218123218122315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [73] $221812^1 3218123151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [80] $61232171232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [81] $61232181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [82] $1612^1 321812321712321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [83] $171232171232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [84] $61232171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [85] $171232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [86] $61232181232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [58] $1^1 8^1 1231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [148] $413221(12)^{21} 122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [158] $612321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [134] $22181232^1 18123151321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [132] $18123218123151321812^1 32 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [146] $21812^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [87] $21^1 (12)^{1513221} 122315132^1 171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [88] $21^1 (12)^{15132^1 21} 122315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [89] $1^1 8^1 1231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 28$, $T \in [28, 28]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [90] $1 \overset{321}{8} 12315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{1513221}{7122^1 31513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{15132^1 21}{712231513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [93] $1513161315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 29, T \in [28, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [94] $3221812315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [95] $322^1 181231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [96] $232^1 1812231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [97] $3218122315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{1613161315132^1 21(12)12222} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{322^1 1}{21(12)122315131613161} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $\overset{21}{1(12)1223151316131612^1 32} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 49, T \in [28, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $\overset{2221}{1(12)122^1 3151316131612} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

- [102] $21^{221}(12)122^1 3151316131612 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [103] $18^1 1232^1 18123218122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [104] $1^{51321} 8^1 12321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [105] $1812^1 32181231513218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [106] $1812^1 31513218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [107] $18123151321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [108] $18^1 1232^1 18123151321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [109] $18^1 12321812315132^1 1812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [110] $18^1 1232^1 18123151322181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [111] $18^1 12321812315132^1 2181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [112] $4132^1 181231513221^{2221}(12)1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [119] $1812^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [120] $1812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [121] $413221^{21}(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [122] $4132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [123] $1613161315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [124] $1613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [125] $181232^1 1812321812315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [126] $181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [127] $1812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [128] $232^1 161316131513221(12)122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [129] $21613161315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [130] $22181232^1 18123218122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [131] $322^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [116] $8123151321812321^{232^1 1} 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [133] $21812^1 32181231513218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [147] $4132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [135] 221812321812315132¹1812+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [136] 22181232¹18123151322181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [137] 221812321812315132¹2181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [138] 4132¹181231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [139] 41321812315132¹21(12)12222 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [140] 613151321812321812¹3215+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 37$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 37$, $T \in [28, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [141] 613151321812321812¹3151+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 35$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 35$, $T \in [28, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [142] 81223151321812321812¹23+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [143] 21812¹3218123218123215 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [144] 21812321812321812¹3215 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [145] 21812321812321812¹3151 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [115] 4132¹181231513221²²¹(12)12+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [114] 41321812315132¹21²²¹(12)12+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [113] 41321812315132¹21²²²¹(12)1+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [149] $4132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [150] $413221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [151] $4132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [152] $1812321712321812^1 32161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [117] $1812^1 3218123218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [153] $1812^1 32171232181232161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [154] $171232^1 181232171232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [155] $171232181232171232^1 171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [156] $612^1 321712321812321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [157] $612321712321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [175] $4132^1 181232181232161316 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [174] $4141413221(12)122315132^1 16 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [173] $4132^1 181232181231513217 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [172] $7122^1 31513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [171] $612321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [170] $612^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [169] $41413221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [168] $1712321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [167] $1712^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [166] $612^1 3217123218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [165] $612321712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [164] $612^1 3218123217123218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [163] $1712^1 321712321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [162] $1712321712321812^1 321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [161] $161232^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [160] $161232181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [159] $612^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

- [118] $1812321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [176] $1^{\frac{2321}{8}} 12315132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 28$, $T \in [28, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [177] $1^{\frac{321}{8}} 123151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [178] $712^{\frac{31513221}{12}} 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [179] $712^{\frac{1513221}{12}} 231513221 (12) 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [180] $31513161315132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 29$, $T \in [28, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [182] $32218123151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [183] $23218122315132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 28$, $T \in [28, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [184] $32181223151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [185] $16131613151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [186] $1^{\frac{221}{12}} (12)122^1 31513161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [187] $1^{\frac{2221}{12}} (12)12^1 23151316131612 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [188] $16^{\frac{31}{12}} 13161315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$

- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [189] $21^{21}(12)122^1 31513161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 221
 [190] $21(12)12^1 23151316131612 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- 321
 [191] $1\ 8\ 12321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [192] $1\ 8\ 123218123151322^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 2221
 [193] $413218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 221
 [194] $413218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 321
 [195] $8122^1 3151321812321\ 8\ 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [196] $16131613151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 221
 [197] $16131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 31
 [198] $1\ 6\ 13161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [199] $1\ 6\ 13161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [200] $321613161315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [201] $216131613151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$

- [202] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 3221812321812315132¹181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
- [203] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 1812¹321812315132181232+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$
- [204] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 2218123218123151322¹181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- [205] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 413218123151322¹1(12)12222 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
- [206] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 8122¹315132181232181223+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
- [207] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 1812¹32181232181231513 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
- [208] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 413221²¹(12)1223151322¹181 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
- [209] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 3161232181232171232¹181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
- [210] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 712¹3217123218123217123+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
- [211] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 712¹321812321812321613 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
- [212] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 712¹231513221¹(12)12231414 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
- [213] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 41321812321812315132¹17+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
- [214] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 161316131613161232¹181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- [215] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 2161316131613161232¹181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[181] $15131613151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 29, T \in [28, 29]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[244] $812^1 23151321812321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[245] $181232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[272] $2181232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[247] $181232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[246] $181232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{5131513221} (12) 122^1 3151315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[217] $21^{1513221} (12) 122^1 315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[218] $21^{151322^1 1} (12) 122315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{2321} 8 123151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{321} 8 12^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[221] $7122315132^1 21^{1513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[222] $712231513221^{151322^1 1} (12) 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

- [223] $315131613151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 29$, $T \in [28, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [224] $3221812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [225] $232181223151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 28$, $T \in [28, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [226] $3218122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [227] $1(12)12^{221} 231513161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [228] $1(12)12^{21} 315131613161232 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [229] $16^{31} 131613151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [230] $21^{21} (12)12^1 231513161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [231] $18^{21} 12321812^1 3218122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [232] $18^{21} 123218123218122^1 315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [233] $1^{5132} 8^1 123218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [234] $1^{51321} 8^1 1232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [235] $1^{1} 181232^1 181231513218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [236] $1\ 8\ 12^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
 [237] $1812315132^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [238] $1\ 8\ 12321812^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [239] $1\ 8\ 12321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [240] $41321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [241] $41321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [248] $181232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [249] $41322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [250] $413221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [251] $16131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [252] $16131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [253] $1\ 8\ 12321812^1 321812315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [254] $1\ 8\ 12321812321812^1 315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [255] $1\ 8\ 12321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[256] \ 3216131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[257] \ 2321613161315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 49, T \in [28, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[258] \ 2218123218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[259] \ 221812321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[260] \ 3221812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[261] \ 2181232^1 181231513218123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[262] \ 221812321812^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[263] \ 221812321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[264] \ 41321812^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[265] \ 61315132181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 37, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[266] \ 61315132181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 35, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[267] \ 812^1 2315132181232181223 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[268] \ 8122315132181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[269] \ 321812321812321812^1 315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 45, T \in [28, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[270] $2181232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[271] $2181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[273] $2181232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[274] $41322^1 1(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 49, T \in [28, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[275] $41322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[276] $413221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[277] $41322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[278] $413221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[279] $812^1 321812321812315132 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 45, T \in [28, 45]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[280] $181232^1 171232181232161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[281] $181232171232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[282] $1712321812321712^1 32171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

- [283] $1712321812^1 32171232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [242] $812^1 3151321812321^{2321} 8^1 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [312] $41321812^1 32181232161316 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [311] $41316131613161315132^1 18 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [310] $4141413221(12)122^1 31513216 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [309] $812^1 3216131613161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [308] $4141413221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [307] $812^1 32161316131613161^1 61 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [306] $41321812^1 32181231513217 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [305] $7122315132^1 21(12)12231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [304] $712231513221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [303] $71232181232181232^1 1613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [302] $61232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [301] $61232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[300] \quad 41413221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[299] \quad 414132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[298] \quad 171232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[297] \quad 171232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[296] \quad 71232171232181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[295] \quad 61232171232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[294] \quad 61232^1 17123218123218122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[293] \quad 61232^1 18123217123218123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 45, T \in [28, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[292] \quad 171232^1 1712321812321712 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[291] \quad 171232171232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[290] \quad 31612^1 32181232171232181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[289] \quad 1612321812^1 321712321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[288] \quad 1612321812321712^1 321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[287] \quad 61232^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [286] $612321^1 81232^1 1812321^1 71 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [285] $61232171232^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 16 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [284] $61232^1 1712321812321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [243] $8122315132181232^1 1^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [321] $1^1 81232181232^1 1812315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [330] $21812321^1 812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [313] $71231513221^1 (12)^1 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 49$, $T \in [27, 49]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [314] $1^1 (12)12^1 2315131613161232 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 49$, $T \in [27, 49]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [315] $1^1 81232181232^1 18122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [316] $1^1 8^1 12^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [317] $1^1 8^1 1232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [318] $81223151321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [319] $1^1 1812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$

- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [320] $1812321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [322] $23216131613151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 49$, $T \in [27, 49]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [323] $22181232181232^1 18122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [324] $322181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [325] $613151321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 37$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 37$, $T \in [27, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [326] $613151321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 35$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 35$, $T \in [27, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [327] $81223151321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [328] $1812321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [344] $4132181232^1 181232161316 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [343] $4132181232^1 181231513217 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [342] $712321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [341] $612321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [340] $1712321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [339] $712321712321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

- [338] $612321712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [337] $1712321712^1 321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [336] $3161232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [335] $161232181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [334] $612321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [333] $612321712^1 32181232181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [332] $171232181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [331] $1812321712^1 32181232161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [329] $21812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [360] $31612321812321712^1 32181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [361] $61232181232171232^1 18123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [346] $21 \begin{pmatrix} 15 & 13 & 22 & 1 \end{pmatrix} 12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [347] $1 \begin{pmatrix} 13 & 22 & 1 \end{pmatrix} 1223151322^1 18123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 27$, $\Delta = -95$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [348] $71223151322^1 1 \begin{pmatrix} 15 & 13 & 22 & 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [349] $321812^1 231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[350] \quad 21(12)122^1 315131613161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[351] \quad 1812321812315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[362] \quad 71232^1 17123218123217123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[352] \quad 413221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[353] \quad 181232^1 1812315132181232 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[354] \quad 21812321812315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[355] \quad 181232^1 181232181231513 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[356] \quad 321812^1 321812321812315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[357] \quad 413221(12)122^1 31513161315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 49, T \in [27, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[358] \quad 413221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[359] \quad 413221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[363] \quad 4141322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [364] $41413221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [365] $71232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [366] $71223151322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [367] $712231513221(12)12^1 231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [368] $1613161316131612^1 32181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [369] $21613161316131612^1 32181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [370] $4141413221(12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [371] $41414132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [372] $4141413221(12)12^1 231513216 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [373] $41414132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [345] $1^{5131513221}(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [380] $812315132181232^1 1^8 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [382] $221812321812321812^1 2315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[381] $8122315132^1 1812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[383] $3221812321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[384] $61315132^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 37, \Delta = -113, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 37, T \in [27, 37]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[378] $1812321812321812^1 2315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[377] $23218122^1 31513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[376] $712315132^1 21 \overset{31513221}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 49, T \in [27, 49]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[375] $1 \overset{2321}{8} 12^1 31513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[374] $(12)122^1 314141414141414141 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[385] $61315132^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 35, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 35, T \in [27, 35]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[386] $8122315132^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[387] $181232181232^1 181231513 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[388] $31612321812^1 32171232181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[389] $71232171232^1 18123217123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[390] \quad 1 \quad 71232181232^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[391] \quad 41321812321812^1 31513217 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[392] \quad 41321812321812^1 32161316 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[379] \quad 1 \quad 8 \quad 12321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[412] \quad 4132181232181232^1 161316 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[395] \quad 21(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[396] \quad 1 \quad 1812321812^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[397] \quad 8123151321812^1 321 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[398] \quad 21812321812^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[399] \quad 1 \quad 1812321812^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[400] \quad 32181232181232^1 1812315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[401] \quad 1 \quad 413221(12)12^1 231513161315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 49, T \in [27, 49]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

- [402] $81232^1 1812321812315132 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [403] $812321812321812^1 315132 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [405] $612321812^1 3217123218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [407] $712321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [408] $81232^1 16131613161316^1 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [409] $414141322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [410] $81232^1 16131613161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [411] $414141322^1 1(12)12231513216 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [393] $(12)12^1 2314141414141414141 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [394] $71231513221^1 (12) 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 49$, $T \in [27, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [406] $712321712^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [404] $3161232181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [413] $32181232^1 1812321812315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 144$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.64 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 59 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[414] 612321812321712¹3218123 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 144$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.64 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[415] 7123151322¹1³¹⁵¹³²²¹(12) 12 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 49, T \in [27, 49]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[416] 2321812¹231513221(12)12231 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[417] 812315132¹1812321²³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[418] 1812321812315132¹181232 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[419] 321812321812¹321812315 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[420] 81232181232¹1812315132 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[421] 61232181232¹17123218123 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[422] 812321812¹321812315132 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.55 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[423] 1812321812¹315132181232 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 225$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.52 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$T = 23$ [424] 181231513221¹(12)1223151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[425] 1(12)1223151321812315132¹2 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[426] 1(12)122315132181231513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 30, T_{\max} = 46, T \in [30, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[427] \quad 1(12)12231513218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[428] \quad 1(12)12231513218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[429] \quad 3221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 29, T_{\max} = 50, T \in [29, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[430] \quad 1513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[431] \quad 21(12)12231513218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[432] \quad 21(12)12231513218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[433] \quad 1513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[434] \quad 22^1 315132181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[435] \quad 2315132181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[436] \quad 315132181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 46, T \in [30, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[437] \quad 51232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[438] 15132181231513221(12)1222¹2+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[439] 413161316131513221²²1(12)1+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[440] 1712321712321812321812¹2+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[441] 413161316131513221(12)122¹2+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[442] 612321812321812321812¹2+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[449] 812231513221³¹⁵¹³²²¹(12)12¹+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.33 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[447] 513221²1(12)12231513218123+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.33 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[460] 712321812321812321712¹3+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[459] 612321812321812321812¹3+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 43, T \in [30, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[458] 7123217123218123218122¹3+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[457] 413161316131513221(12)12¹22+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[456] 32¹171232181232171232181+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[455] 1712321812321712321812¹3 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[454] 712321712321812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[453] 4132181231513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[452] 15132181231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[451] 51232181231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[450] 315132181231513221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 46, T \in [30, 46]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[448] 21⁵¹³²²¹(12) 1223151321812¹3 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[446] 21³²¹21(12) 1223151321812315 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[445] 1²²²¹(12) 12231513218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[444] 1²²²¹(12) 12231513218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[443] 31513221²¹(12) 122315132181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.33 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[465] 512¹32181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[466] 15132181231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[464] 51232181231513221(12)12¹222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 36, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[463] 81231513221²³¹⁵¹³²²¹(12) 12¹+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 50, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 6 \\ 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[462] 21²²¹(12)122315132181232¹15 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 36, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[461] 1²²²¹(12)122315132181232¹15 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 36, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[469] 1712¹32181232171232181 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[471] 181232171232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[467] 4132181231513221(12)12¹2313+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[468] 712315132181232181232¹15 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 37, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[470] 1712¹321812321712321812 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[472] 1812¹32181232171232171 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[473] 181232181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

- [474] $1712^1 132181232181232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [475] $61232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [476] $171232171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [477] $21712^1 321812321712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [478] $613161232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 37$, $T \in [29, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [479] $61315132181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [480] $2181232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [481] $171232181232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [482] $61232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [532] $712321712321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [499] $612^1 31513221^{5131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [509] $171232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[533] $812321712321812321712^1 32 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[508] $171232181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[507] $7123151321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[506] $7123151321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[505] $41321812315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[504] $4132^1 181231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[503] $151321812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[502] $15132^1 181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[501] $512321812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[500] $51232^1 181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[511] $171232181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[498] $21 \overset{1513221}{(12)} 122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [497] $^{15132^1 21}$ 21 (12) 1223151322181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [496] 221 21(12)1223151321812¹3151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [495] 221 21(12)122¹31513218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [494] 221 21(12)122¹31513218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [493] 221 21(12)1223151321812¹3215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [492] $^{15132^1 21}$ 21 (12) 1223151321812 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [491] 1513221 21 (12) 122315132¹1812 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [490] $^{322^1 1}$ 21(12)1223151321812315 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 50$, $T \in [29, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [489] 221 1(12)122¹31513218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [488] 221 1(12)1223151321812¹3151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [487] 221 1(12)122¹31513218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [486] 221 1(12)1223151321812¹3215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 36, T \in [29, 36] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[485] & 1 \quad (12) \quad 12231513161315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[484] & 1 \quad 1812^1 31513221^{21} (12) 1223151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[483] & 1 \quad 181231513221^{21} (12) 122^1 3151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[512] & 1 \quad 1812^1 321712321812321712 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[513] & 1 \quad 1812321712321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[514] & 1 \quad 181232^1 181232171232171 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[515] & 1 \quad 181232181232171232^1 171 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[516] & 1 \quad 1812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[517] & 1 \quad 1812321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[518] & 1 \quad 41413221^{21} (12) 122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[519] & 1 \quad 171232181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [520] $171232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [521] $612^1 32181232181232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [522] $612321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [523] $1712^1 3218123217123218123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [534] $712321812321712321812^1 32 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 46$, $T \in [29, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [535] $21812^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [536] $21812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [537] $1712321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [524] $1712321712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [538] $1712^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [525] $1712^1 3217123218123218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [539] $41413221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[540] $41413221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[541] $612^1 3218123218123218123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 43, T \in [29, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[542] $4132^1 21(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[543] $612^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[544] $612321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[545] $4132^1 1812321812315132218 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[546] $4141413221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[547] $4141413221(12)122315132^1 217 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[526] $2171232181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[527] $2171232^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[528] $4132^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[529] $6131612321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 37$, $T \in [29, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [530] $6131612321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [531] $613151321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 1
 [510] $171232^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [575] $41321812321812315132^1 218 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 221
 [549] $1(12)122^1 315132181231513 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 46$, $T \in [29, 46]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [574] $712^1 3218123218123217123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [576] $4141413221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 1
 [573] $41413221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [572] $1812^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [571] $712^1 32181232171232181232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 46$, $T \in [29, 46]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [570] $712^1 32171232181232181223 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [569] $41321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[568] 4131613161315132¹21(12)1222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 39, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[567] 32171232181232171232¹181+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 40, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[566] 712¹321712321812321³²¹8 1+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 40, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[565] 4131613161315132²¹12(12)12+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 39, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[564] 4131613161315132²²¹21(12)1+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 39, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[563] 712¹31513218123218123151+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[562] 712¹31513218123218123215+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 37, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[561] 413218123151322¹1(12)122313+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[560] 321613161315132¹21(12)12231+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 29, \Delta = -96, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[559] 1513218123151322¹1(12)12222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[558] 5123218123151322¹1(12)12222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 36, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[557] 3151321812315132¹21(12)1222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 46, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[556] \quad 8122^{\overset{31513221}{1}}1513221 \quad (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[555] \quad 21 \quad (12) \quad 1223151322^{\overset{1513221}{1}}181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[554] \quad 21 \quad (12) \quad 12^{\overset{221}{1}}231513218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[553] \quad 21 \quad (12) \quad 12^{\overset{221}{1}}231513218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[552] \quad 21 \quad (12) \quad 122^{\overset{21}{1}}315132181231513 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[551] \quad 1 \quad (12) \quad 12^{\overset{2221}{1}}231513218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[550] \quad 1 \quad (12) \quad 12^{\overset{2221}{1}}231513218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[548] \quad 21 \quad (12) \quad 122315132^{\overset{31513221}{1}}181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[579] \quad 1 \quad (12) \quad 12231513161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[577] \quad 181231513221 \quad (12) \quad 12^{\overset{21}{1}}23151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[633] \quad 41316131613151322^{\overset{1}{1}}1(12)1222 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 39, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[632] 414132¹21(12)12231513161315 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[631] 21712321812321712¹321812 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[630] 21712321812¹321712321812 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[629] 171232171232¹18123218122 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[628] 171232¹17123218123218122 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[627] 321712¹32181232171232181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[626] 171232¹18123217123218123 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[625] 61232181232¹1812321²¹81 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[624] 61232¹1812321812321²¹81 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[578] 1812315132¹21²¹(12)1223151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[623] 1712321812321812¹32171 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[622] 1712321812¹32181¹232171 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[621] $414132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[620] $41413221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[619] $181232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[618] $181232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[617] $71232171232181232^1 1^3 81 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[616] $41316131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[615] $41316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[614] $1812321812^1 32171232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[613] $1812321812321712^1 32171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[612] $181232^1 1712321812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[611] $181232171232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[610] $1712321812^1 321712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[609] $1712321812321712^1 321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[608] $1712321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[607] $1712321812321712^1 32181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[606] $712315132181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[605] $712315132181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 37$, $T \in [29, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[604] $41321812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[603] $3216131613151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 29$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 29$, $T \in [29, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[602] $151321812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[601] $512321812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[600] $31513218123151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 46$, $T \in [29, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[599] $23151321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 50$, $T \in [29, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

- [598] $812^{1231513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [597] $812^{131513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [596] $612315132^{121} (12) 1 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 39, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [595] $61231513221^{51315132^{121}} (12) 1 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 39, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [594] $21^{151322^{121}} (12) 1223151322181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [593] $21^{1513221} (12) 122^{13151322181} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [592] $21^{221} (12) 122315132^{181} 18123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [591] $21^{221} (12) 122315132^{181} 18123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [590] $21^{151322^{121}} (12) 1223151321812 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [589] $21^{1513221} (12) 122^{13151321812} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [588] $21^{21} (12) 12^{12315132181} 1231513 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [587] $21^{5132^{121}} (12) 12231513218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

- [586] $^{3221}21(12)1223151321812^1315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 50$, $T \in [29, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [585] $^{2221}1(12)122315132^118123151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [584] $^{2221}1(12)122315132^118123215 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [635] $812^131513161316131613151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [636] $613161232181232^118123215 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 37$, $T \in [29, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [637] $613161232181232^118123151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [638] $61315132181232^1181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [639] $71232171232181232^1181223 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [640] $812^132171232181232171232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 46$, $T \in [29, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [641] $^{1}2181232181232^1181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [642] $^{1}2181232^1181232181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [643] $^{1}181232181232181232^11613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[644] $1712321^{181232^1}1812321712 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[645] $1712321^{1812321}1812321812321712 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[646] $414132^{121}(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[647] $41413221^{121}(12)122^13151321812 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[648] $41413221^{121}(12)122^13151322181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[649] $414132^{121}(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[650] $61232^{1812321}18123218123218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 43, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 43, T \in [29, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[651] $41322^{11}(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[652] $61232181232^{1812321}18123218122 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[653] $61232^{1812321}18123218123218122 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[654] $71232181232181232^{1812321}17123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[655] 413218123218123151322¹18 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[656] 41321812¹321812315132218 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[657] 4141413221(12)122¹315132171 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[658] 4141413221(12)122¹315132217 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[581] 1^{151322¹1}(12) 12231513161315 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[659] 413161316131613161232¹18 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[580] 1⁵¹³¹⁵¹³²²¹(12) 122¹31513161 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[583] 1²²¹(12)12¹2315132181231513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[582] 1²¹(12)122¹3151321812315132 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[634] 41321812¹321812315132181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[662] 1²¹(12)12¹23151321812315132 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[664] 21²¹(12)1223151321812¹31513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [661] $21 \quad (12) \quad 122315132181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [663] $1(12)1223151321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [692] $4132181232^1 1812315132218 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 39, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [691] $71232181^1 2321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [690] $612321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [689] $1712321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [688] $1812321812321812^1 321613 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [687] $21812321812^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [686] $712321712321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [685] $613151321812^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [684] $6131612321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [683] $6131612321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 37, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[682] $4132181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[681] $2171232181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[680] $1712321712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[679] $32171232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[678] $612321812^1 321812321^1 81 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[677] $171232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[676] $1812321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[675] $712321712321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[674] $181232181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[673] $1812321712^1 321812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[672] $171232181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[671] $171232181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[670] 7123151321812¹3218123151 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[669] 7123151321812¹3218123215 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 37, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[668] 315132¹181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 46, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[667] 231513218123151322¹1(12)122 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 50, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[666] 223151321812315132¹21(12)12 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 50, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[665] 812231513221^{315132¹21}(12) 12 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 46, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[660] 1¹(12)122¹31513218123151322 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 50, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[713] 71232¹171232181232181223 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 40, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[701] 21¹⁵¹³²²¹(12) 12¹23151322181 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[718] 4141322¹1(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[719] 41413221¹(12)12¹23151322181 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[720] $61232181232181232^1 18123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 43, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 43, T \in [28, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[721] $413221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[722] $71232^1 18123218123217123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[723] $41414132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[724] $4141413221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[725] $4141413221(12)12^1 2315132217 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[726] $41414132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[716] $41413221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[693] $18123151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[694] $1^{5131513221} (12) 12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[695] $1^{513151322^1 1} (12) 12231513161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[696] $1^{1513221} (12) 122^1 31513161315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

- [697] $1(12)12^1 231513218123151322 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 50$, $T \in [28, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [698] $21(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 50$, $T \in [28, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [699] $21(12)122315132^1 18123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 50$, $T \in [28, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [712] $41413221(12)122^1 31513161315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [711] $4141322^1 1(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [710] $321712321812321712^1 32181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 40$, $T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [709] $171232181232171232^1 18123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 40$, $T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [708] $4141322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [707] $41413221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [714] $71232^1 181232171232181232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [706] $71232^1 1712321812321 \ 8 \ 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 40$, $T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [717] $4141322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[715] $181232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[705] $1413221(12)1223151322^1 18123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 28$, $\Delta = -94$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[704] $2231513218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[703] $61231513221^{513151322^1 1} (12) 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[702] $6123151322^1 1^{5131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[700] $21^{1513221} (12) 12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[735] $71232171232^1 1812321^{321} 8 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[744] $71232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[745] $41321812321812^1 315132218 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[743] $181232181232^1 1812321613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[741] $71232171232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[740] $61315132^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

- $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [739] 613161232¹18123218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 35$, $T \in [28, 35]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [738] 613161232¹18123218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 37$, $T \in [28, 37]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [737] 41321812321812¹315132181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [736] 321712321812¹32171232181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 40$, $T \in [28, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [734] 712315132¹18123218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 35$, $T \in [28, 35]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [733] 712315132¹18123218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 37$, $T \in [28, 37]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [732] 3151321812¹31513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [731] 8122315132¹21³¹⁵¹³²²¹(12) 12 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [730] 21²¹(12)122315132¹181231513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [729] 21^{51322¹1}(12) 12231513218123 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 50$, $T \in [28, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [728] 1²²¹(12)122315132¹181231513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [727] 21³¹⁵¹³²²¹(12) 122¹315132181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 27.76 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[742] & 81232171232181232^1 171232 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.76 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[757] & 32171232181232^1 171232181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[770] & 4131613161316131612^1 3218 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 38, T \in [28, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[748] & 1(12)1223151321812^1 315132 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[749] & 21(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[750] & 21(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[751] & 21(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[752] & 81231513221^{2315132^1 21} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[753] & 812231513221^{3151322^1 1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[754] & 2315132^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[755] & 712321712^1 321812321^1 8 1 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[756] $1712321812^1 3217123218123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[747] $21^1 (12) 122315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[746] $1^1 (12) 12^1 231513161315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[760] $6131612^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[761] $712321712^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[762] $812321712321812^1 32171232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[763] $81232^1 171232181232171232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[764] $1812321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[758] $41413221(12)12^1 231513161315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[765] $612321812^1 3218123218123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 43, T \in [28, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[766] $413221(12)12^1 2315131613161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[767] $712321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[768] 414141322¹1(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[769] 414141322¹1(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[759] 6131612¹3218123218123215+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[771] 1712321812321712¹3218123+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -144$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.64 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[772] 612321812321812¹3218123+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -144$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.64 \rightarrow 28, T_{\max} = 43, T \in [28, 43]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[773] 21³¹⁵¹³²²¹(12)¹12¹2315132181+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[774] 1²¹(12)122315132¹1812315132+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[775] 812315132¹21²³¹⁵¹³²²¹(12)¹12+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[777] 23151321812¹31513221(12)122+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[778] 171232181232¹17123218123+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[779] 81232171232¹181232171232+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[780] 71232181232171232¹181232+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[781] \quad 61232181232^1 18123218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 43, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 43, T \in [28, 43]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[776] \quad 81223151322^1 1^{31513221} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[784] \quad 22315132^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.55 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[785] \quad 812321712^1 32181232171232 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -196, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.55 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[782] \quad 1^{12} (12) 1223151321812^1 3151322 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.55 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[783] \quad 21^{513221} (12) 12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.55 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[786] \quad 81231513221^{23151322^1} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -225, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.52 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[787] \quad 712321812^1 32171232181232 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -225, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.52 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[788] \quad 1^{12} (12) 122315132^1 18123151322 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -256, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.48 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[789] \quad 223151321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -256, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.48 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[790] \quad 8123151322^1 1^{231513221} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -289, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.46 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[791] 712321812321712¹32181232 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -289$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.46 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[792] 71232181232¹171232181232 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -324$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.43 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$T = 24$ [797] 61232181231513221³²²¹(12)¹12 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[804] 21812321712321812321812¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[793] 1²²¹(12)1223151316131613151 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[794] 1²²¹²¹(12)122315132181232161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[795] 21²²¹¹(12)122315132181232161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[796] 1²²¹¹(12)1223151321812321613 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 44, T \in [31, 44]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[798] 22¹31513161316131513221(12)1 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 51, T \in [31, 51]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[799] 31513161316131513221(12)12¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 47, T \in [31, 47]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[800] 1513161316131513221(12)122¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[801] 161232181231513221(12)122¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[802] 3161232181231513221(12)122¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31, T_{\max} = 44, T \in [31, 44]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[803] \overset{1}{1812321712321812321812} \overset{32^1}{12} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[805] \overset{1}{1712321812321812321812} \overset{32^1}{12} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[815] \overset{1}{712321812321812321} \overset{32^1}{8 \ 1} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[806] \overset{1}{51232181231513221(12)122^1} 313 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 32, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 32, T \in [31, 32]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[807] \overset{32^1}{1(12)122315131613161315} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 51, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 51, T \in [31, 51]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[808] \overset{222^1}{1(12)122315132181232161} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[809] \overset{32^1}{61232181231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 47, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 47, T \in [31, 47]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[810] \overset{1}{1513161316131513221(12)12^1} 22 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[811] \overset{1}{161232181231513221(12)122^1} 22 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[817] \overset{32^1}{32^1 1812321712321812321812} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[812] \overset{1}{3161232181231513221(12)12^1} 22 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 44, T \in [31, 44]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[818] \overset{1}{1812321812321812321712^1} 3 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [819] $32^1 181232181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [820] $7123218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [816] $18123217123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [813] $1812321812321712321812^1 3 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [814] $1^3 8^1 1232181232171232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [824] $1^{2221} (12) 122315132181232^1 161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [843] $613161232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [825] $21^{221} (12) 122315132181232^1 161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [821] $512^1 32181231513221 (12) 122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 32$, $T \in [30, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [822] $51232181231513221 (12) 12^1 2313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 32$, $T \in [30, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [823] $15132181231513221 (12) 12^1 2313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 30$, $T \in [30, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [826] $161232181231513221 (12) 12^1 222 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [827] $1612^1 32181231513221 (12) 12222 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[828] & 512^1 321812321812315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 37, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[829] & 8122315132181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 37, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[830] & 181232171232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[831] & 181232171232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[832] & 1812^1 321812321712321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[833] & 61315132181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[834] & 181232181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[835] & 181232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[836] & 1712^1 321812321812321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[837] & 1812^1 32181232181232171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[838] & 2181232171232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[839] & 61315132181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[840] 712315132181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[841] 171232181232181232¹18122 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[842] 2181232181232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[853] 1²²²¹(12)122¹315132181232161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[880] 171232181232181232¹1812 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[845] 7122¹31513221⁵¹³¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[846] 51232¹181231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 32, T \in [30, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[847] 512321812315132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 32, T \in [30, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[848] 15132¹181231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[849] 151321812315132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[850] 1^{15132¹21}(12) 122315131613161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[851] 1^{322¹1}(12)122315131613161315 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 51, T \in [30, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [854] $21(12)122^{221}315132181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [852] $1(12)1223151321812^132161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [855] $21(12)1223151321812^132161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [903] $4132^118123218123217123217 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [902] $6131612321812321812^132161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [901] $4141413221(12)122315132^12181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [900] $4141413221(12)122315132^11812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [899] $4141413221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [856] $61232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 47$, $T \in [30, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [857] $612^132181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 47$, $T \in [30, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [858] $1612321812315132^121(12)12222 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [859] $161232^1181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[860] $51232^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[861] $15132^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[862] $81223151321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[863] $81223151321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[864] $4132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[865] $612^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[866] $612^1 31513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[867] $1812321712321812^1 32181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[868] $1812^1 32171232181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[869] $1812^1 3218123217123218123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[870] $1812321712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[871] $1812^1 3217123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [872] $181232181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [873] $181232^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [874] $613151321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [875] $812321712321812321^{232^1} 8^1 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [876] $1812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [877] $1812^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [878] $1812^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [879] $1812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [898] $812321812321812321712^1 32 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [897] $712321812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [896] $712321812321812321812^1 32 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [895] $21812^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[894] $21812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[893] $1712^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[892] $1712321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[891] $32181232181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[890] $7123151321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[889] $613151321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[888] $21812^1 3217123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[887] $21812321712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[886] $32181232171232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[885] $232^1 181232171232181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[884] $1812321712321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[883] $1^1 81232181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
 [882] $181232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [881] $171232^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [844] $1^{5131513221} (12) 122315132^1 171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [904] $712^1 231513221^{5131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [905] $3151321812315132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 30$, $T \in [30, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [906] $5123218123151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 32$, $T \in [30, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [907] $1513218123151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 30$, $T \in [30, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [908] $1^{221} (12)122^1 3151316131613151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [909] $21^{21} (12)122^1 3151316131613151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [910] $1^{2221} (12)12^1 2315132181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [911] $21^{221} (12)12^1 2315132181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [912] $1^{221} (12)122^1 3151321812321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$

- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [913] $21^{21}(12)122^1 3151321812321613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [914] $15131613161315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [915] $1612321812315132^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [916] $31612321812315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [917] $512321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [918] $151321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [919] $8122^1 31513218123218123215 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [920] $8122^1 31513218123218123151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [921] $1^1 8^{321} 1232181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [922] $712^1 3218^1 12321812321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [923] $1812^1 32171232181232181223 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [924] $232181232171232181232^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$

- [928] $712^1 32181232181232181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [929] $4141413221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [930] $4132181232181232171232^1 17 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [925] $712^1 315132181232181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [927] $712^1 32181232181232181232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [926] $1812^1 3218123218123217123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1000] $812^1 32181232181232171232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [998] $2181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [932] $1^{5131513221} (12) 122^1 315132171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [933] $7122315132^1 21^{5131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [934] $712231513221^{51315132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [935] $31513218123151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 30$, $T \in [30, 30]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[936] $512321812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 32, T \in [30, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[937] $151321812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[938] $1^{151322^1} (12) 122315131613161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[939] $1^{21}(12)122^1 31513161316131513 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1001] $4141413221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1002] $812^1 31513161316131613161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[940] $1^{221}(12)12^1 23151316131613151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[941] $1^{5132^1 21} (12) 1223151316131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 51, T \in [30, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[942] $21^{21}(12)12^1 23151316131613151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[943] $1^{2221}(12)122315132^1 181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[944] $21^{221}(12)122315132^1 181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[945] $1^{221}(12)12^1 23151321812321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[946] $1(12)122315132181232^1 1613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[947] $21(12)122315132181232^1 1613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[948] $21(12)12^1 23151321812321613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[949] $61232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[950] $315131613161315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[951] $151316131613151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[952] $1612321812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[953] $31612^1 32181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[954] $316123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[955] $512321812^1 321812315132181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[956] $151321812^1 321812315132181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1006] $812^1 321613161316131613151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1007] $812^1 315131613161316131612 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1008] $613161232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1009] $41321812^1 3218123217123217 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[957] $812^1 315132181232181231513 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[958] $812^1 231513218123218123215 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[959] $8122315132181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[960] $8122315132181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[961] $812^1 231513218123218123151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[962] $161315132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[963] $413221(12)1223151321812^1 315 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[964] $41322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [965] $61231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [966] $612315132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [967] $612315132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [968] $181232^1 1712321812321^2 81 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [969] $181232171232^1 1812321^2 81 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [970] $181232^1 18123217123218123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [971] $1^3 812^1 32181232171232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [972] $181232^1 17123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [973] $181232171232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [974] $1^3 812321812^1 321712321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [975] $1^3 812321812321712^1 321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [976] $61315132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[977] $812^1 321712321812321^{2321} 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[999] $71232181232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[978] $181232181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[979] $181232^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[980] $181232181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[981] $181232^1 1812321812321712 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[982] $1712321812^1 321812321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[983] $1712321812321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[984] $18^{21} 12321812^1 32181232171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[985] $18^{21} 12321812321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[986] $71232181232181232^1 1 \ 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[987] $181232171232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[988] $321812321712321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[989] $2181232^1 17123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[990] $2181232171232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[991] $61315132181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[992] $712315132181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[993] $181232181232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[994] $321812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1004] $4141413221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[931] $1^{51315132^1 21} (12) 122315132171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1005] $414132^1 21(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[995] $171232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[996] $171232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[997] $2181232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1003] $4141413221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1041] $712321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1027] $181232181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1028] $613151321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1029] $1812321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1030] $1812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1010] $315132^1 181231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 30, T \in [29, 30]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1011] $1(12)122^1 315131613161315132 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1012] $1(12)12^1 231513161316131513 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1013] $1(12)1223151321812^1 321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1014] $21(12)1223151321812^1 321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [1015] $61316131513221^{315132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1016] $2315131613161315132^{1 21} (12) 12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1017] $3151316131613151322^1 1 (12) 122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1018] $3161232^1 181231513221 (12) 1222 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1019] $51232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 37$, $T \in [29, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1020] $15132181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1021] $8123151321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1022] $81223151321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 37$, $T \in [29, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1023] $81223151321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1024] $1812321712^1 32181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1031] $171232181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1032] $181232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$

- [1025] $1 \overset{321}{8} 1232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1026] $1812321712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1033] $712321812321812^1 321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1034] $1812321712321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1035] $21812321712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1036] $613151321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1037] $7123151321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1038] $1812321812321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1039] $1712321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1040] $21812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1043] $4132181232^1 18123217123217 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1042] $6131612321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1076] $41321812321812321712^1 3217 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1071] $4141413221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1045] $1^{5131513221}(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1046] $71223151322^1 1^{5131513221}(12)1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1047] $71223151322^1 1^{5131513221}(12)1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1048] $1^1(12)12^1 2315131613161315132 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1049] $1^{1513221}(12)122^1 315131613161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1050] $612321812315132^1 21^{3221}(12)12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1051] $22315131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1052] $2315131613161315132^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1053] $1613151322^1 1^1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1054] $413221^1(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1055] $6123151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1056] $6123151322^1 1(12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1057] $6123151322^1 1(12)12231513216 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1058] $181232181232171232^1 18123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1059] $1^3 8 12321812321712^1 32181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1060] $71232^1 1812321812321^3 8 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1061] $181232^1 171232181232181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1062] $2321812321712321812^1 32181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1063] $321812^1 321712321812321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1064] $181232^1 18123218123217123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1065] $321812^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1066] $71232^1 181232181232181232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1067] $71232^1 181232181232181223 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1068] $41414132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1069] $4141413221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1073] $41414132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1074] $41413221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1075] $4141322^1 1(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1072] $4141413221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1070] $41414132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1044] $1^{513151322^1 1} (12) 122315132171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1099] $613161232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.76 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1090] $61315132^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.76 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

- [1097] $71232181232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1096] $181232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1095] $712315132^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1100] $41321812321812^1 3217123217 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1098] $81232181232181232^1 171232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1094] $61315132^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1093] $181232171232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1092] $71232181232^1 1812321 \ 8 \ 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1091] $81232171232181232^1 1 \ 8 \ 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1077] $3151321812^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 30$, $T \in [29, 30]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1078] $1 \ (12) \ 1223151316131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1079] $1(12)122315132^1 1812321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

- $$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [1080] $21(12)122315132^1 1812321613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [1081] $223151316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$
- [1082] $31612321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [1083] $512321812321812^1 315132181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [1084] $151321812321812^1 315132181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [1085] $812315132181232^1 181231513 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [1086] $8122315132^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [1087] $8122315132^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [1088] $61315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [1089] $1 \ 8 \ 12321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [1118] $32181232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$
- [1117] $1812321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1121] & 812321812321812^1 32171232 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 44, T \in [29, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1122] & 414141322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1123] & 414141322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1124] & 414141322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1119] & 712321812^1 32181232181223 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1125] & 41413221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1126] & 81232^1 1613161316131613151 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 36, T \in [29, 36] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1127] & 6131612^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 36, T \in [29, 36] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1128] & 4132181232181232^1 17123217 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1101] & 1^{1513221} (12) 12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 36, T \in [29, 36] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1102] & 1^{3221} (12) 122^1 315131613161315 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 51, \Delta = -118, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 51, T \in [29, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1103] $61316131513221^{3151322^1 1} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1104] $61232181231513221^{3221} (12) 12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1105] $612321812^{3221} 31513221 (12) 12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1106] $8123151321812^{321} 32181231513 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1107] $413221^{1} (12) 122315132^{321} 1812315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1108] $413221^{1} (12) 12^{321} 23151321812315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1109] $6131513221^{1} (12) 122^{321} 315131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1110] $1812321812^{321} 3217123218123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1111] $1^3 8 1232181232^{321} 171232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1112] $81232^{321} 1712321812321^{321} 8 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1113] $812321712321812^{321} 321^{321} 8 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1114] $712321812^{321} 321812321^{321} 8 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1115] 1812321712¹32181232181223+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1116] 32181232171232¹1812321812+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1120] 81232¹181232181232171232+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.69 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1131] 32181232¹181232181232171+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 144, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.64 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1129] 1812321812321712¹3218123+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 144, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.64 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1130] 32181232¹1712321812321812+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 144, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.64 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1133] 613161315132¹21³¹⁵¹³²²¹(12) 1+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1134] 812315132¹181232181231513+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1135] 613151322¹1(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1136] 181232181232¹17123218123+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1137] 81232171232¹1812321²³²¹8 1+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

- $$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
 [1138] $2321812^1 32171232181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$
 [1139] $321812321712^1 321812321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$
 [1140] $321812321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$
 [1141] $71232181232181232^1 181232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$
 [1142] $81232181232^1 181232171232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
 [1132] $1(12)12^1 2315131613161315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
 [1145] $812321712^1 321812321^{2321} 81 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.55 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$
 [1144] $6131513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 196$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.55 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
 [1146] $812321812^1 32181232171232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.55 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$
 [1143] $1(12)122^1 3151316131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 196$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.55 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
 [1148] $712321812^1 32181232181232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 225$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.52 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$
 [1147] $232181232171232^1 181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 225$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.52 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[1149] $61316131513221^{31513221}1(12)1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 256, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.48 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1152] $712321812321812^{132181232}1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 289, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.46 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1151] $232181232^{171232181232181}1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 289, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.46 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 373 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[1150] $1(12)12^{513221}23151316131613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 289, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.46 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1154] $71232181232^{181232181232}1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 324, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.43 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[1153] $2321812321712^{132181232181}1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 324, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.43 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 413 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$T = 25$ [1174] $221^{221}(12)12231513221(12)12222^12 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1178] $3221^{211}(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1195] $218123218^{12321812321812}2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1194] $613151321812321812321812^12 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1193] $21613161316131513221(12)122^12 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1192] & \begin{smallmatrix} 1 \\ 321613161316131513221 \end{smallmatrix} (12) 12^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1191] & \begin{smallmatrix} 1 \\ 1812321812321812321812 \end{smallmatrix} 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1190] & \begin{smallmatrix} 31 \\ 1613161316131513221 \end{smallmatrix} (12) 12^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1189] & \begin{smallmatrix} 22^1 1 \\ 1(12)12231513161316131612 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1155] & \begin{smallmatrix} 22^1 221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 1 \\ (12)12222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1156] & \begin{smallmatrix} 22^1 21 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 221 \\ (12)1223 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1157] & \begin{smallmatrix} 2221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 2^1 21 \\ (12)1223 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1158] & \begin{smallmatrix} 22221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 21 \\ (12)122^1 2 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1159] & \begin{smallmatrix} 22^1 221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 21 \\ (12)1222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1160] & \begin{smallmatrix} 22^1 221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 221 \\ (12)122 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1161] & \begin{smallmatrix} 22221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 221 \\ (12)12^1 2 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1162] & 21^{221}(12)12231513221^{2^1 21}(12)1223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1163] & 21^{22^1 1}(12)12231513221^{221}(12)1223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1164] & 21^{22^1 21}(12)12231513221^{21}(12)1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1165] & 21^{22^1 21}(12)12231513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1166] & 21^{2221}(12)12231513221^{221}(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1167] & 22^1 1^{221}(12)12231513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1168] & 1^{22^1 221}(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1169] & 1^{22221}(12)12231513221(12)12222^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1173] & 21^{22^1 21}(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1188] & 1^{1613161316131513221}(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1175] & 22^1 1^{221}(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1170] & \begin{matrix} 3221 \\ 221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1171] & \begin{matrix} 2^1 21 \\ 3221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1172] & \begin{matrix} 2221 \\ 21(12)12231513221(12)12222^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1176] & \begin{matrix} 22^1 2221(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1187] & \begin{matrix} 1 \\ 1613161316131513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1186] & \begin{matrix} 22^1 1 \\ 2171232181231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1185] & \begin{matrix} 22^1 1 \\ 232171232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 48, T \in [32, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1184] & \begin{matrix} 22^1 1 \\ 32171232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1183] & \begin{matrix} 1 \\ 171232181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1182] & \begin{matrix} 1 \\ 171232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1181] & \begin{matrix} 22^1 1 \\ 21(12)1223151321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[1180] $1(12)1223151321812321712 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1179] $1(12)12231513218123217123 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1177] $171232181231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1197] $1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1196] $1(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1209] $21(12)12231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1208] $221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1207] $221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1206] $1(12)12231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1205] $1(12)12231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1204] $21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1203] $21^{222^1 1} (12) 12231513221^{21} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1202] $21^{221} (12) 12231513221^{221} (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1201] $1^{222^1 21} (12) 12231513221^{21} (12) 122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1200] $1^{222^1 21} (12) 12231513221^{21} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1199] $1^{22221} (12) 12231513221^{21} (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1198] $1^{2221} (12) 12231513221^{221} (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1211] $221^{221} (12) 12231513221 (12) 1222^1 22 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1212] $222^1 221 (12) 12231513221 (12) 122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1213] $1^{171232181231513221} (12) 122^1 22 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1214] $21^{32^1 21} (12) 122315132181232171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1215] $1^{221} (12) 1223151321812321712^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1216] $1(12)1223151321812321712 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1217] $21(12)1223151321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1218] $171232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1219] $32171232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1223] $1613161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1220] $32^1 171232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1221] $2171232181231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1222] $32^1 21(12)1223151316131613161 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1230] $32^1 1812321812321812321812 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 40, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1229] $18123218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 40, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1228] $21613161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1227] $32^1 1613161316131513221(12)122 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1226] & 1 \overset{32^1}{8} 1232181232181232181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1225] & 1812321812321812321812^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1224] & 1 \overset{21}{(12)} 12231513161316131612^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1210] & 21 \overset{2221}{(12)} 12231513221(12)1222^1 22 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1262] & 21 \overset{221}{(12)} 122315132181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1247] & 1 \overset{22221}{(12)} 122315132^1 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1276] & 8122315132181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1277] & 71232171232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1278] & 712315132181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1279] & 2181232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1280] & 613161232181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1281] & 613161232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} =
\end{aligned}$$

15.0

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1274] $1612^1 321812321812315132181 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1273] $1812^{21} 321812321812321812 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1272] $181232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1271] $181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1270] $712315132181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1269] $61315132181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1268] $21712^1 32181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1267] $2171232181231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1266] $1612^1 32181231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 31, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1265] $161232181231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 31, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1264] $1712^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1263] $\overset{1}{1712^1 32181231513221(12)1222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1261] $\overset{2221}{1(12)122315132181232^1 1712 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1260] $\overset{1}{171232181231513221(12)12^1 222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1259] $\overset{1}{1712^1 32181231513221(12)12222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1258] $\overset{2221}{2222^1 21(12)12231513221(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1257] $\overset{2221}{222221(12)122^1 31513221(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1256] $\overset{221}{221(12)122^1 31513221(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1255] $\overset{221}{221(12)12231513221(12)122^1 222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1254] $\overset{221}{221(12)122315132^1 21(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1253] $\overset{2221}{21(12)122^1 31513221(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1252] $\overset{2221}{21(12)122315132^1 21(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1251] $\overset{2221}{21(12)12231513221(12)122^1 222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{3221} \\
[1250] & 221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{3221} \\
[1249] & 22^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.25 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{22221} \\
[1245] & 1(12)12231513221(12)122^1 222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{221} \\
[1244] & 221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{2221} \\
[1243] & 21(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}^{2221} \\
[1242] & 21(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{2221} \\
[1241] & 21(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{221} \\
[1240] & 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.25 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{2222^1 1} \\
[1239] & 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}^{2221} \\
[1238] & 1(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}^{22221} \\
[1237] & 1(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1236] & \begin{matrix} 22221 & 21 \\ 1(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1235] & \begin{matrix} 22221 & 21 \\ 1(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1234] & \begin{matrix} 2222^1 1 & 21 \\ 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1233] & \begin{matrix} 2221 & 22^1 1 \\ 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.25 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1232] & \begin{matrix} 22221 & 1 \\ 1(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1231] & \begin{matrix} 2222^1 1 & 1 \\ 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1248] & \begin{matrix} 22221 \\ 1(12)122^1 31513221(12)122222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1246] & \begin{matrix} 2222^1 1 \\ 1(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1275] & \begin{matrix} 61315132181232181232^1 18122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1362] & \begin{matrix} 1 \\ 4132^1 21(12)12231513218123216 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 45, T \in [31, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1320] & \begin{matrix} 221 \\ 21(12)1223151321812^1 321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1327] \quad & 232^1 171232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 48, T \in [31, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1328] \quad & 21712321812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1329] \quad & 2171232^1 181231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1330] \quad & 16131513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1331] \quad & 712^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1332] \quad & 61231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1333] \quad & 612^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1334] \quad & 61231513221(12)122315132^1 217 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1335] \quad & 612^1 31513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1336] \quad & 1(12)1223151316131613161 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1337] \quad & 1(12)12231513161316131612^1 32 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 48, T \in [31, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1338] $613151321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1339] $7123151321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1340] $1812321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1341] $1812^1 32181232181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1342] $1812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1343] $1812321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1344] $1812^1 3218123218123218123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1345] $181232181232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [1346] $181232^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1347] $812321812321812321^{232^1} 81 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1348] $232^1 1613161316131513221(12)12 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 48$, $T \in [31, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1349] $161232^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1350] & 4132^1 21(12)122315131613161315 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 48, T \in [31, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1351] & 613151321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1352] & 81223151321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1353] & 712321712321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1354] & 712321712321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1355] & 7123151321812321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1356] & 1812321812321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1357] & 232^1 181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1358] & 32181232181232181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1359] & 21812^1 3218123218123218122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1360] & 21812321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1325] & 161232^1 181231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 31, \Delta = -96, \det = -25, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 25, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 31$, $T \in [31, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1324] $1^{12}171232^{11}181231513221(12)12+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1323] $1^{12}1712321812315132^{121}21(12)12+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1364] $4132^{11}181232181232171232181+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1365] $6131612321812321812^{13}21712+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1366] $712321812321812321812^{13}14+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1282] $1^{12}1812^{13}1513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1283] $1^{22221}(12)12^{12}231513221(12)12222+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1284] $1^{2221}(12)12231513221(12)12^{12}23+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1285] $1^{2221}(12)122^{13}1513221(12)1223+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1286] $1^{22221}(12)12^{12}231513221(12)12222+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1287] $1^{22221}(12)1223151322^{121}1(12)1222+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1288] & \begin{matrix} 22221 & & 221 \\ 1(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1289] & \begin{matrix} 22221 & & 221 \\ 1(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1290] & \begin{matrix} 221 & & 221 \\ 21(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1291] & \begin{matrix} 221 & & 221 \\ 21(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1292] & \begin{matrix} 2221 & & 21 \\ 21(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1293] & \begin{matrix} 2221 & & 221 \\ 21(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1361] & \begin{matrix} 1 \\ 413221(12)122315132181232^1 16 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 45, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 45, T \in [31, 45] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1322] & \begin{matrix} 1 & & 1 \\ 171232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1363] & \begin{matrix} 1 \\ 6131612321812321812^1 32171 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1294] & \begin{matrix} 2221 & & 221 \\ 21(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1295] & \begin{matrix} 221 & & 221 \\ 221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1296] $1^{5131513221}(12)122315132^11812 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 38, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1326] $1612321812315132^121(12)122313 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 31, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1297] $1^{15132^121}(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 39, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 39, T \in [31, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1298] $1^{5131513221}(12)122315132^12181 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 38, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1299] $1^{22221}(12)12231513221(12)12^12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1300] $1^{22221}(12)1223151322^11(12)122222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1301] $1^{22221}(12)12^1231513221(12)122222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1302] $221^{322^11}(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1303] $221^{3221}(12)122315132^121(12)12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1304] $21^{2221}(12)12231513221(12)12^12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1305] $21^{2221}(12)1223151322^11(12)122222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1306] $21^{2221}(12)12^1231513221(12)122222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

15.0

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1307] \quad {}^{221}_{12}(12)12^1 231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1308] \quad {}^{221}_{12}(12)12231513221(12)12^1 2222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1309] \quad {}^{221}_{12}(12)1223151322^1 1(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1310] \quad 22222^1 1(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1311] \quad 222221(12)12^1 231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1312] \quad {}^1_{17}1232^1 181231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1313] \quad {}^1_{17}12321812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1314] \quad {}^{322^1}_{21}(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1315] \quad {}^{3221}_{21}(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1316] \quad {}^{21}_{12}(12)1223151321812321712^1 32 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 48, T \in [31, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1317] \quad {}^{2221}_{12}(12)122^1 3151321812321712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1318] & \begin{matrix} 2221 \\ 1(12)1223151321812^1 321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1319] & \begin{matrix} 221 \\ 21(12)122^1 3151321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1321] & \begin{matrix} 1 \\ 1712321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1380] & \begin{matrix} 1 \\ 321712321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1400] & \begin{matrix} 1 \\ 4132181232181232171232^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1377] & \begin{matrix} 1 \\ 17123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1376] & \begin{matrix} 1 \\ 17123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1375] & \begin{matrix} 221 \\ 21(12)12^1 23151321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1374] & \begin{matrix} 21 \\ 21(12)122^1 31513218123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1373] & \begin{matrix} 2221 \\ 1(12)12^1 23151321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1372] & \begin{matrix} 221 \\ 1(12)122^1 31513218123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1371] $17123218123151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1370] $221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1369] $1^{5131513221}(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1378] $16123218123151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 31$, $\Delta = -96$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 31$, $T \in [31, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1382] $712^1 31513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 48$, $T \in [31, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1383] $712^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1384] $61231513221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1368] $21(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1367] $1(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1381] $217123218123151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1385] $16131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1379] $31612321812315132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 31$, $\Delta = -96$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1386] & \begin{matrix} 1 \\ 16131613161315132^1 21(12)12+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1387] & \begin{matrix} 1 \\ 16131613161315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1388] & \begin{matrix} 221 \\ 1(12)122^1 31513161316131612+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1389] & \begin{matrix} 21 \\ 21(12)122^1 31513161316131612+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1390] & \begin{matrix} 1 \\ 712^1 315132181232181232171+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1391] & \begin{matrix} 321 \\ 1 \ 8 \ 1232181232181232^1 181+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1392] & \begin{matrix} 1 \\ 216131613161315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1393] & \begin{matrix} 1 \\ 1612321812321812315132^1 181+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1394] & \begin{matrix} 1 \\ 8122^1 315132181232181232161+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1395] & \begin{matrix} 1 \\ 712^1 3217123218123218123215+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1396] & \begin{matrix} 1 \\ 712^1 3217123218123218123151+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1397] $712^1 3151321812321812321712 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1398] $1812^1 32181232181232181223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1399] $232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1401] $712^1 321812321812321812314 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1413] $51316131513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1425] $1612321812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 31$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1447] $21(12)12^1 231513161316131612 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1448] $61315132181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1449] $712315132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1450] $181232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1451] $181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1452] $181232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1453] & \begin{matrix} 1 \\ 181232^1 18123218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1454] & \begin{matrix} 1 \\ 181232^1 18123218123218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1455] & \begin{matrix} 321 \\ 1 \ 8 \ 12^1 32181232181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1456] & \begin{matrix} 21 \\ 1 \ 8 \ 12321812^1 321812321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1457] & \begin{matrix} 21 \\ 1 \ 8 \ 12321812321812^1 321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1458] & \begin{matrix} 1 \\ 812^1 321812321812321 \ 8 \ 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1459] & \begin{matrix} 2321 \\ 3216131613161315132^1 21(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1460] & \begin{matrix} 2321 \\ 2161316131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1461] & \begin{matrix} 2321 \\ 1612321812^1 321812315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1462] & \begin{matrix} 21 \\ 414132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 45, T \in [31, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1463] & \begin{matrix} 21 \\ 41413221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 45, T \in [31, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1464] & \begin{matrix} 21 \\ 41322^1 1(12)122315131613161315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

(1, 25, 0), $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 48, T \in [31, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1465] 61315132181232¹18123218122+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1466] 812315132181232181232¹1613+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 45, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 45, T \in [31, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1467] 812¹3151321812321812321613+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 45, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 45, T \in [31, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1468] 812¹2315132181232181232161+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1469] 8122315132181232¹181232161+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1470] 71232171232181232¹18123215+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 38, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1471] 71232171232181232¹18123151+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1472] 712315132181232¹1812321712+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1473] 181232181232181232¹181223+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 40, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1474] 321812321812321812¹321812+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 40, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1475] 2181232181232¹18123218122+ $\hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [1476] $2181232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1477] $413221(12)1223151321812^1 3216 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 45$, $T \in [31, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1478] $41322^1 1(12)12231513218123216 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 45$, $T \in [31, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1479] $613161232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1480] $41321812^1 32181232171232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1481] $613161232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1482] $71232181232181232^1 1812314 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1483] $812^1 321613161316131613161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [1484] $812^1 3216131613161316131612 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [1408] $1^{151322^1} (12) 1223151321812315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 39$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 39$, $T \in [31, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1407] $1^{5131513221} (12) 122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1406] $1^{51315132^1 21} (12) 1223151321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1405] & \begin{matrix} 221 & & 221 \\ 21(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1404] & \begin{matrix} 2221 & & 221 \\ 1(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1403] & \begin{matrix} 1 & & 1 \\ 181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1402] & \begin{matrix} 1 & & 1 \\ 1812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1445] & \begin{matrix} 221 \\ 1(12)12^1 231513161316131612 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[1444] & \begin{matrix} 21 \\ 1(12)122^1 315131613161316123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1443] & \begin{matrix} 1 \\ 161316131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1446] & \begin{matrix} 31 \\ 16131613161315132^1 21(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1442] & \begin{matrix} 31 & & 21 \\ 16131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1441] & \begin{matrix} 1 & & 21 \\ 161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1440] & \begin{matrix} 1 & & 221 \\ 161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1439] $61231513221(12)122^1 315132217 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1438] $612315132^1 21(12)122315132217 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1437] $61231513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1410] $1^{5131513221}(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1436] $612315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1435] $7122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1434] $712231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1433] $16131513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1432] $161315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1431] $21712321812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1430] $2321712321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 48, T \in [31, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1429] $3217123218123151322^1(12)1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[1428] $321712^132181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1427] $31612^132181231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 31, \Delta = -96, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1426] $316123218123151322^11(12)12231 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 31, \Delta = -96, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[1411] $1^{51315132^121}(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[1412] $221^{3221}(12)122^131513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1414] $1712321812^131513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1415] $21^{3221}(12)1223151321812^132171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[1416] $1^{221}(12)12^1231513218123217123 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1417] $1^{221}(12)122315132181232^117123 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1418] $1^{21}(12)122^1315132181232171232 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 48, T \in [31, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1419] $1^{2221}(12)122315132^11812321712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1420] & 21(12)12^{21}231513218123217123+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1421] & 21(12)122315132181232^1 17123+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1422] & 21(12)122315132^1 1812321712+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1423] & 1712321812^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1424] & 1712321812^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1409] & 1^{1513221} (12) 1223151321812^1 315+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 39, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 39, T \in [31, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1515] & 712321812321812^1 321812314+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1514] & 6131612321812^1 321812321712+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1513] & 4132181232^1 181232171232181+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1512] & 6131612321812^1 32181232171+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1511] & 21812321812^1 3218123218122+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]
\end{aligned}$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1510] $1812321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1509] $7123151321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1508] $712321712321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1507] $712321712321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1506] $81223151321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1505] $8123151321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1504] $613151321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1503] $161232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1501] $32161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1500] $181232181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1499] $181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1498] $1812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1497] & \begin{matrix} 1 & 1 & 21 \\ 1812321812^1 32181232181 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1496] & \begin{matrix} 1 \\ 7123151321812^1 32181232171 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1495] & \begin{matrix} 21 \\ 613151321812^1 32181232181 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1494] & \begin{matrix} 31 \\ 161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 41, T \in [30, 41] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1493] & \begin{matrix} 21 \\ 1(12)12^1 2315131613161316123 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 42, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1492] & \begin{matrix} 1 \\ 1(12)122^1 3151316131613161232 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 48, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1491] & \begin{matrix} 31 \\ 161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 41, T \in [30, 41] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1490] & \begin{matrix} 1 \\ 23217123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 48, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1489] & \begin{matrix} 1 \\ 32171232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 42, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1488] & \begin{matrix} 1 \\ 3161232^1 181231513221(12)12231 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 31, \Delta = -96, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 31, T \in [30, 31] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1487] & \begin{matrix} 21 \\ 21(12)1223151321812^1 3217123 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 42, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \end{matrix}
\end{aligned}$$

- [1486] $1(12)^{21}12^1 2315132181232171232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.94 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1485] $1(12)^{221}1223151321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.94 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1502] $23216131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.94 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1547] $321812^1 321812321812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1528] $21(12)^{3221}122^1 315132181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1531] $712231513221(12)^1 12^1 231513161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1529] $16131513221(12)^1 12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1527] $513161315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1526] $51316131513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1525] $221(12)^{3221}12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 41$, $T \in [30, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1524] $1^{513151322^1} (12) 1223151322181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1523] $1^{5131513221} (12) 12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ [1522] \end{matrix} 1 \quad (12) \quad 122^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 39, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 39, T \in [30, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 513151322^1 1 \\ [1521] \end{matrix} 1 \quad (12) \quad 1223151321812 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 5131513221 \\ [1520] \end{matrix} 1 \quad (12) \quad 12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1519] \end{matrix} 21(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 41, T \in [30, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1550] \end{matrix} 71232^1 1812321812321812314 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& [1549] 41321812321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1548] \end{matrix} 413221(12)122^1 31513218123216 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 45, T \in [30, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1546] \end{matrix} 2321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1545] \end{matrix} 181232^1 181232181232181223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& [1544] 71232^1 17123218123218123151 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& [1543] 71232^1 17123218123218123215 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1542] $4141322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 45$, $T \in [30, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1541] $41413221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 45$, $T \in [30, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1540] $232161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1539] $1^{321} 8 12321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1538] $1^{12} 181232181232181232^1 18123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1537] $1^{12} 1(12)12^1 23151316131613161232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1536] $6123151322^1 1(12)122315132217 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1535] $61231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1534] $6123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1533] $61231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1518] $1^{2221} (12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 41$, $T \in [30, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1517] $1^{12} 18123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1516] & \begin{matrix} 1 \\ 181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1530] & \begin{matrix} 1 \\ 1613151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1532] & \begin{matrix} 1 \\ 71223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1569] & \begin{matrix} 1 \\ 181232181232^1 181232181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1570] & \begin{matrix} 1 \\ 613161232^1 181232181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1571] & \begin{matrix} 1 \\ 41321812321812^1 32171232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1572] & \begin{matrix} 1 \\ 613161232^1 1812321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1573] & \begin{matrix} 1 \\ 71232181232^1 1812321812314 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1568] & \begin{matrix} 1 \\ 712315132^1 1812321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1567] & \begin{matrix} 1 \\ 71232171232^1 18123218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1566] & \begin{matrix} 1 \\ 71232171232^1 18123218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1565] $8122315132^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1564] $812315132181232^1 1812321613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1560] $1 \overset{321}{8} 12321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1551] $1 \overset{221}{(12)} 122315132^1 18123217123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1552] $1 \overset{21}{(12)} 122315132181232^1 171232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1553] $21 \overset{21}{(12)} 122315132^1 18123217123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1554] $31612321812^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 31$, $\Delta = -96$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 31, T \in [30, 31]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1555] $321712321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1556] $2321712^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1557] $712315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1558] $61315132^1 1812321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1559] $712315132^1 1812321812321 \overset{1}{7} 1 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1562] $1612321812321812^1 315132181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1561] $81232181232181232^1 1^8 1 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1563] $61315132^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1579] $1(12)1223151321812^1 32171232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1575] $1^{1513221}(12)122315132^1 1812315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 39$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1594] $32181232181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1595] $413221(12)122315132^1 18123216 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1596] $413221(12)12^1 231513218123216 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1597] $6131612^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1598] $4132181232181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1599] $6131612^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1600] $712321812^1 321812321812314 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} =$

- [1584] $1 \overset{321}{8} 1232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1583] $1812321812^1 3218123218123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1582] $1(12)122^1 3151316131613161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1581] $71231513221(12)122^1 315131613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1580] $232171232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1578] $21(12)122315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1577] $21(12)12^1 2315132181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1576] $5131613151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1593] $1812321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1603] $1812321812321812^1 3218123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -144$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.64 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1604] $32181232^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -144$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.64 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1605] $1(12)122315132^1 181232171232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1607] & \begin{matrix} 1 \\ 7123151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -169, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1608] & \begin{matrix} 3221 \\ 1(12)12^1 23151316131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[1609] & \begin{matrix} 1 \\ 181232181232^1 18123218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1610] & \begin{matrix} 1 \\ 81232181232^1 1812321 \end{matrix} \begin{matrix} 2321 \\ 8 \end{matrix} 1 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1611] & 413221(12)12^1 2315131613161315 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -169, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1612] & 812315132^1 1812321812321613 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 45, T \in [30, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1613] & \begin{matrix} 1 \\ 2321812^1 32181232181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1614] & \begin{matrix} 1 \\ 321812321812^1 321812321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1606] & 2321712321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -169, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1615] & \begin{matrix} 1 \\ 71231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -196, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.55 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1616] & \begin{matrix} 1 \\ 812321812^1 321812321 \end{matrix} \begin{matrix} 2321 \\ 8 \end{matrix} 1 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.55 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1617] $232181232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -225$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.52 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1618] $232181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.46 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1619] $2321812321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -324$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.43 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$T = 26$ [1643] $7123151321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1630] $413221(12)^{2^1 1} 12231513221(12)^{2221} 1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1636] $181232181231513221(12)^{21} 122^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1637] $181232181231513221(12)^{22^1 1} 12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1638] $322181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1639] $22181232181231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1640] $22^1 181232181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1641] $413221(12)^{21} 12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1634] $181232181231513221(12)^{2^1 1} 1223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1633] $413221(12)^{2^1 1} 12231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1632] $413221(12)^{21} 12231513221(12)^{22^1 1} 12 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1645] $6131612321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1620] $3221(12)^{2^1 21} 12231513221(12) 122313 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1642] $413221(12)^{2^1 1} 12231513221(12) 12222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1629] $21(12)^{221} 1223151321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1628] $21(12)^{22^1 1} 12231513218123218122 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1627] $3221(12)^{2^1 1} 1223151321812321812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 42, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1626] $1(12)^{2221} 1223151321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1625] $1(12)^{22^1 21} 12231513218123218122 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1624] $1^1 81232181231513221(12) 1222^1 2 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1623] \overset{2321}{1 \ 8 \ 1232181231513221(12)12^1 2 + \hat{1} \rightarrow 2} \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 46, T \in [33, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1622] \overset{321}{1 \ 8 \ 1232181231513221(12)122^1 2 + \hat{1} \rightarrow 2} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1621] \overset{22^1}{1(12)122315132181232181223 + \hat{1} \rightarrow 2} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1644] 4131613161316131513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1635] \overset{321}{1 \ 8 \ 1232181231513221(12)1 + \hat{1} \rightarrow 2} \overset{22^1}{1} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1631] 413221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \overset{21}{22^1 21} \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1657] \overset{32^1}{1 \ 8 \ 1232181231513221(12)1 + \hat{1} \rightarrow 2} \overset{221}{221} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1661] 32^1 2181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1662] 22181232181231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1663] 413221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \overset{21}{21} \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1664] 6131513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \overset{1}{1} \quad T = 26, T_{\max} = 49, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 49, T \in [33, 49]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1665] & 8123151321812321812321712^1 3 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1660] & 322181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1646] & 221(12)^{3221} 12231513221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1647] & 221(12)^{32^1 21} 12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1648] & 1(12)^{221} 12231513218123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1649] & 1 \ 8 \ 1232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1650] & 1 \ 8 \ 1232181231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1651] & 1 \ 8 \ 1232181231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1652] & 1(12)^{222^1 1} 12231513218123218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
[1658] & 1 \ 8 \ 1232181231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1659] & 1 \ 8 \ 1232181231513221(12)^{21} 12 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1656] $181232181231513221(12)122^13 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1655] $413221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 9$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 41$, $T \in [33, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1654] $21(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1653] $21(12)12231513218123218122^13 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1673] $1812^132181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1672] $1812^132181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1674] $22181232181231513221(12)12^1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1675] $221812^132181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1676] $413221(12)12231513221(12)12^1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1677] $1712^1321812321812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1678] $712315132181232181232^1181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1679] $8122315132181232181232^1171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1680] 181232171232181232181232¹15+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1682] 712315132181232181232¹18122+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1683] 8122315132181232181232¹1712+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1684] 712321¹81232181232181232¹15+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1685] 613161232181232181232¹1²¹81+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1686] 613161232181232181232¹18122+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1687] 71232171232181232181232¹161+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1671] 21²²¹(12)122315132181232¹18122+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1670] 1²²²¹(12)122315132181232¹18122+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1669] 1²¹81232181231513221(12)12¹222+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1668] 1²¹812¹32181231513221(12)12222+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [1667] $^{3221}221(12)12231513221(12)12^12313+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -102$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1666] $^{3221}22^11(12)12231513221(12)122313+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -102$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.25 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1681] $21712^1321812321812315132181+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1713] $^{21}1812321812315132^121(12)1222+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1700] $^{221}21(12)122^131513218123218122+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1704] $4132^121(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1705] $413221(12)122315132^121(12)1+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1706] $4132^121(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1707] $413221(12)122315132^121(12)12+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1708] $61231513221(12)122315132^1181+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1709] $612^131513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [1710] $^{21}181232181231513221(12)12^123+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1711] $1812^1 32181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1712] $1^{232} 8^{11} 1232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 46, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 46, T \in [32, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1702] $612^1 32181231513221^{1513221}(12)1 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 39, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1714] $1^{21} 8^{11} 1232^1 181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1715] $1^{21} 8^{11} 1232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1716] $1^{21} 8^{11} 12321812315132^1 21^{221}(12)12 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1717] $513161316131513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 40, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1718] $322^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1719] $221812321812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1720] $22181232^1 181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1721] $612^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1722] $16131513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1723] $16131513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1724] $1712^1 31513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1725] $4132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1726] $413221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1727] $61231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1728] $612^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1729] $61231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1730] $612^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1731] $712231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1732] $7122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1733] $7122^1 31513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1734] $171232^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1735] $7123151321812321812^{21} 32181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1736] $81223151321812321812^{1} 32171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1737] $1812321712321812321812^{1} 3215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1738] $1812321712321812321812^{1} 3151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1739] $2171232^{1} 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1740] $7123151321812321812^{1} 3218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1741] $81223151321812321812^{1} 321712 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1742] $413221(12)122315132181232^{1} 171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1743] $4132^{1} 21(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1744] $712321812321812321812^{1} 3215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1745] \quad 712321\overset{1}{8}12321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1746] \quad 6131612321812321812^1 321\overset{21}{8}1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1747] \quad 4132^1 21(12)1223151316131613161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1748] \quad 6131612321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1749] \quad 712321712321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1750] \quad 4132^1 181232181232181\overset{1}{2}32181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1751] \quad 41413221(12)122315132181232^1 16 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1698] \quad 21(12)122315132181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1697] \quad 21(12)1223151321812321812 + \overset{322^1}{\hat{1}} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1696] \quad 21(12)1223151321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1695] \quad 1(12)1223151321812^1 3218122 + \overset{2221}{\hat{1}} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1694] & \begin{matrix} 2221 \\ 1(12)122^1 31513218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1693] & \begin{matrix} 21 \\ 181232^1 181231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1692] & \begin{matrix} 21 \\ 1812321812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1691] & \begin{matrix} 232^1 1 \\ 181232181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 46, T \in [32, 46] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1690] & \begin{matrix} 221 \\ 1(12)1223151321812321812^1 23 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1689] & \begin{matrix} 322^1 1 \\ 221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1688] & \begin{matrix} 3221 \\ 221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1701] & \begin{matrix} 15132^1 21 \\ 61232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 39, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1699] & \begin{matrix} 221 \\ 21(12)1223151321812^1 3218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1703] & \begin{matrix} 1 \\ 1812^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1770] & \begin{matrix} 1 \\ 8122^1 31513221(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 46, T \in [32, 46]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1761] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1768] & \begin{matrix} 1 \\ 16131513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1769] & \begin{matrix} 21 \\ 413221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1766] & \begin{matrix} 21 \\ 2218123218123151322^1 1(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1771] & \begin{matrix} 1 \\ 61231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1772] & \begin{matrix} 1 \\ 712^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1765] & \begin{matrix} 21 \\ 3221812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1764] & \begin{matrix} 21 \\ 18123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1760] & \begin{matrix} 21 \\ 413221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1759] & \begin{matrix} 21 \\ 413221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1773] & \begin{matrix} 1 \\ 712^1 231513221(12)122315132217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1774] $1712321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1775] $712^1 3151321812321812321^1 81 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1776] $8122^1 3151321812321812321^1 71 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1777] $1812^1 3217123218123218123215 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -110, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1778] $1812^1 3217123218123218123151 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1779] $21712321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1780] $712^1 31513218123218123218122 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1781] $8122^1 3151321812321812321712 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1782] $712^1 3218123218123218123215 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -110, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1783] $712^1 3218123218123218123151 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1784] $712^1 32171232181232181232161 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1785] $4132181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 40, \Delta = -112, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1752] & \begin{matrix} 3221 \\ 221(12)1223151322^1 1(12)122313+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1753] & \begin{matrix} 221 \\ 1(12)122^1 315132181232181223+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1754] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812315132^1 21(12)1222+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1755] & \begin{matrix} 21 \\ 1 \ 8 \ 123218123151322^1 1(12)12222+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1756] & \begin{matrix} 2221 \\ 1(12)12^1 231513218123218122+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1757] & \begin{matrix} 21 \\ 21(12)122^1 315132181232181223+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1758] & \begin{matrix} 221 \\ 21(12)12^1 231513218123218122+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1763] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812315132^1 21(12)12+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1767] & \begin{matrix} 1 \\ 316131513221(12)122315132^1 181+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 46, T \in [32, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1762] & \begin{matrix} 21 \\ 1 \ 8 \ 123218123151322^1 1(12)1222+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1796] & \begin{matrix} 3221 \\ 21(12)1223151321812^1 321812+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1831] $61231513221(12)122^1 3151321812+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1798] $61232181231513221^{1513221} (12) 1+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 39, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1799] $61232^1 181231513221^{1513221} (12) 1+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 39, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1800] $1812315132^1 21(12)12231513161+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1801] $181231513221(12)122^1 31513161+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1802] $41322^1 1(12)12231513221^{21} (12)1+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1803] $413221^{21} (12)122^1 31513221^{2221} (12)1+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1804] $41322^1 1(12)12231513221^{21} (12)12+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1805] $413221^{21} (12)122^1 31513221^{221} (12)12+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1806] $61231513221(12)122^1 315132181+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1807] $612315132^1 21(12)122315132181+\hat{1}\rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

- [1808] $181232^1 181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1809] $181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1810] $181232181231513221^1(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1811] $1812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 46$, $T \in [32, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1812] $1812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1813] $181232181231513221^1(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1814] $181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1815] $1812321812^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1816] $161316131513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1817] $513161316131513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1818] $161315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1819] $3221812^1 32181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1820] \quad & 32218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1821] \quad & 221812321812^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1822] \quad & 612315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1823] \quad & 16131513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1824] \quad & 161315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1825] \quad & 161315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1826] \quad & 16131513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1827] \quad & 1712315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1828] \quad & 41322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1829] \quad & 413221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1830] \quad & 812^1 231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 46, T \in [32, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1795] $21(12)^{21}12^1 2315132181232181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1794] $21(12)^{21}122315132181232^1 181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1793] $1(12)^{2221}122315132^1 18123218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1834] $61231513221(12)^1 122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1835] $7122315132^1 21(12)^1 122315132171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1836] $712231513221(12)^1 122^1 315132171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1837] $7122315132^1 21(12)^1 122315132217 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1838] $1712321812^1 321812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1839] $712315132181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1840] $812^1 2315132181232181232171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [1841] $8122315132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1842] $181232171232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1843] \ 181232171232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1844] \ 21712321812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1845] \ 812^1 32171232181232181231513 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 46, T \in [32, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1846] \ 812315132181232181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1847] \ 812^1 31513218123218123217123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1848] \ 712315132181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1849] \ 8122315132181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1850] \ 812^1 23151321812321812321712 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[1851] \ 413221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1852] \ 41322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1853] \ 71232181232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [1854] $71232181232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1855] $41316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1856] $613161232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1857] $41316131613161315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1858] $41322^1 1(12)1223151316131613161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1859] $613161232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1860] $71232171232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1861] $41321812^1 32181232181232181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1862] $414132^1 21(12)12231513218123216 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1863] $41413221(12)1223151321812^1 3216 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1792] $1812321812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1791] $1^1 812321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 46$, $T \in [32, 46]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1790] $1 \ 8 \ 12^{321} 12^1 32181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1789] $1 \ 8 \ 123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1788] $1(12)122315132181232^1 181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1787] $1(12)12^{221} 12^1 2315132181232181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1786] $221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -102$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1833] $612315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1832] $612315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1797] $21(12)122315132^1 18123218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1866] $1 \ 8 \ 123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1867] $21(12)1223151321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1868] $1 \ 8 \ 1232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

- [1869] $1 \ 8 \ 123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1870] $1 \ 8 \ 1232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1871] $32171232^1 181231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 31$, $\Delta = -95$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 31$, $T \in [31, 31]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1872] $322181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1873] $171232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1874] $7123151321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1875] $81223151321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1876] $1812321712321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1877] $1812321712321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1878] $2171232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1879] $812321712321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1880] $8123151321812321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 43$, $T \in [31, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1881] 7123151321812¹3218123218122+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1882] 81223151321812¹321812321712+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1883] 712321812321812¹3218123215+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1884] 712321812321812¹3218123151+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1885] 413161316131613151322²¹1(12)1+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1886] 6131612321812¹32181232181+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1887] 413161316131613151322¹1(12)122 + $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1888] 6131612321812¹3218123218122+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1889] 712321712321812¹32181232161+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1890] 4132181232¹181232181232181+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1865] 1³²¹81232¹181231513221(12)1222+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1864] 1²²¹(12)1223151321812¹32181223+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1902] \quad 161316131513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1927] \quad 71232^1 171232181232181232161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[1900] \quad 61231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1930] \quad 41413221(12)122^1 31513218123216 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 43, T \in [31, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1929] \quad 4141322^1 1(12)12231513218123216 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 43, T \in [31, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1891] \quad 1^{13221}(12)1223151322^1 1^{221}(12)1223 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 31, \Delta = -94, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1892] \quad 221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1893] \quad 21^{3221}(12)122^1 3151321812321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1894] \quad 612321812315132^1 21^{1513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 39, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 39, T \in [31, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1895] \quad 1^{181231513221}(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1896] \quad 1^{18123151322^1 1}(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

15.0

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1911] \quad 16131513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1912] \quad 17123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1913] \quad 413221(12)^{21} 12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1914] \quad 6131513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 49, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 49, T \in [31, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1915] \quad 6123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1916] \quad 61231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1917] \quad 6123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1918] \quad 61231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1919] \quad 71223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1920] \quad 712231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1921] \quad 71223151322^1 1(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 30.84 \rightarrow 31, \quad T_{\max} = 38, \quad T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1922] & 181232^1 17123218123218123215 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 38, \quad \Delta = -110, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, \quad T_{\max} = 38, \quad T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1923] & 181232^1 17123218123218123151 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 38, \quad \Delta = -108, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, \quad T_{\max} = 38, \quad T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1924] & 413221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 42, \quad \Delta = -114, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, \quad T_{\max} = 42, \quad T \in [31, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1925] & 71232^1 18123218123218123215 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 38, \quad \Delta = -110, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, \quad T_{\max} = 38, \quad T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1926] & 71232^1 18123218123218123151 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 38, \quad \Delta = -108, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, \quad T_{\max} = 38, \quad T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1928] & 41321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 40, \quad \Delta = -112, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, \quad T_{\max} = 40, \quad T \in [31, 40] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1945] & 8122315132^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 38, \quad \Delta = -106, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, \quad T_{\max} = 38, \quad T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1950] & 812315132181232^1 18123217123 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 43, \quad \Delta = -115, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, \quad T_{\max} = 43, \quad T \in [31, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1954] & 71232181232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 38, \quad \Delta = -108, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, \quad T_{\max} = 38, \quad T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1931] & 1(12)122315132^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 42, \quad \Delta = -114, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, \quad T_{\max} = 42, \quad T \in [31, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1932] & 1^321 8 12321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, \quad T_{\max} = 42, \quad \Delta = -114, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, \quad T_{\max} = 42, \quad T \in [31, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1933] $1 \ 8 \ 12^{2321} 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1934] $21(12)122315132^1 181232181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1935] $1 \ 8 \ 12321812^1 31513221^{221}(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1936] $1 \ 8 \ 12^{2321} 32181231513221^{21}(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1937] $1 \ 8 \ 12321812^1 31513221^{21}(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1938] $321712321812^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 31$, $\Delta = -95$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 31$, $T \in [31, 31]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1939] $3221812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1940] $316131513221^1(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1941] $61315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 49$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 49$, $T \in [31, 49]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1942] $8122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1956] $613161232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1957] $71232171232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1958] & 41321812321812^1 32181232181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1951] & 712315132^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1943] & 1712321812321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1944] & 712315132^1 1812321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1953] & 71232181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1955] & 613161232^1 1812321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1946] & 181232171232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1947] & 181232171232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1948] & 21712321812321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1949] & 81232171232181232^1 181231513 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1952] & 8122315132^1 1812321812321712 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1969] & 16131513221(12)12^1 231513161315 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 121, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 26, 0), $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1970] 61231513221(12)12¹231513161315 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1971] 3161315132¹21(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 46, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1968] 5131613161315132¹21(12)1223151 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 40, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1967] 16131613151322¹1(12)1223151315 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 40, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1966] 1²³²¹8 1232¹181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 46, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1965] 18123218123151322¹1(12)1223 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 42, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1964] 1812321812¹31513221(12)1223 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 42, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1963] 612321812¹⁵¹³²²¹31513221 (12) 1 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 39, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 39, T \in [31, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1973] 812231513221(12)122¹315131613 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 46, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1974] 1812321712¹3218123218123215 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1975] 1812321712¹3218123218123151 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1976] 81232¹171232181232181231513+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1977] 812321712321812¹32181231513+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1978] 8123151321812¹3218123217123+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 43, T \in [31, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1979] 413221(12)¹122315132¹181232171 + $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1980] 413221(12)¹12¹2315132181232171 + $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1981] 712321812¹3218123218123215+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1982] 712321812¹3218123218123151+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1983] 6131612¹321812321812321²¹81+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1984] 413221(12)¹122¹3151316131613161 + $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1985] 6131612¹3218123218123218122+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1986] 712321712¹32181232181232161+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1987] 4132181232181232¹181232181+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1988] & 41413221(12)12^1 231513218123216 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 43, T \in [31, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1989] & 41413221(12)122315132^1 18123216 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 43, T \in [31, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1962] & 6123218123151322^1 1^{1513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 39, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 39, T \in [31, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1961] & 21(12)12^1 23151321812321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1960] & 21(12)122315132^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1959] & 1^1 8^{2321} 1232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1972] & 6131513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 49, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 49, T \in [31, 49] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1990] & 1^1 8^{2321} 12321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.59 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1991] & 1^1 8^{2321} 12321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.59 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1992] & 51316131613151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.59 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1993] & 316131513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.59 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1994] $613151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 49$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 49$, $T \in [31, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1995] $81223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1996] $81232171232^1 181232181231513 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1997] $812315132^1 18123218123217123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 43$, $T \in [31, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1998] $413221(12)12^1 23151316131613161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1999] $31613151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.55 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2000] $6131513221^1 (12)12^1 231513218123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 49$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.55 \rightarrow 31$, $T_{\max} = 49$, $T \in [31, 49]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2001] $812231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.55 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2002] $812321712^1 32181232181231513 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.55 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- $T = 27$ [2005] $1^{22} 1^{21} (12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34$, $T_{\max} = 39$, $T \in [34, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2009] $21^{22} 1^1 (12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34$, $T_{\max} = 39$, $T \in [34, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2010] $21^{22} 1^{31} 513221 (12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -4$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 34, T_{\max} = 50, T \in [34, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2011] 21 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 231513221 \\ (12) \end{smallmatrix}$ 12231513221(12)12¹2+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 50, T \in [34, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2012] 21 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix}$ 12231513221(12)122¹2+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 47, T \in [34, 47]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2013] 1513221²¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$ 12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2014] 21 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix}$ 12231513221(12)1222¹2+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 39, T \in [34, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2015] 413221²¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$ 12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[2016] 4132181232181231513221²²¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$ (12)1+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2017] 41413221¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 22^1 21 \\ (12) \end{smallmatrix}$ 12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2018] 41413221¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix}$ 12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2019] 15131613161316131513221(12)12¹2 + $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 39, T \in [34, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2020] 22¹1812321812321812315132181+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 39, T \in [34, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2021] 4132181232181231513221(12)122¹2 + $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 34, T_{\max} = 42, T \in [34, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2022] $(12)1223151316131613161315132^1 2 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34, T_{\max} = 50, T \in [34, 50]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2023] $81223151321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34, T_{\max} = 39, T \in [34, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 1 \\ & & \uparrow \end{matrix}$
 [2024] $41413221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34, T_{\max} = 41, T \in [34, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 2221 \\ & & \uparrow \end{matrix}$
 [2006] $1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 22^1 1 \\ & & \uparrow \end{matrix}$
 [2004] $1(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34, T_{\max} = 47, T \in [34, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 1 \\ & & \uparrow \end{matrix}$
 [2003] $1(12)12231513221(12)122315132^1 2 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34, T_{\max} = 50, T \in [34, 50]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 221 \\ & & \uparrow \end{matrix}$
 [2008] $21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 2^1 1 \\ & & \uparrow \end{matrix}$
 [2007] $3221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33, T_{\max} = 50, T \in [33, 50]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 1 \\ & & \uparrow \end{matrix}$
 [2034] $71231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34, T_{\max} = 47, T \in [34, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 31513221 \\ & & \uparrow \end{matrix}$
 [2031] $21(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34, T_{\max} = 47, T \in [34, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{matrix} & & 1513221 \\ & & \uparrow \end{matrix}$
 [2032] $21(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34, T_{\max} = 39, T \in [34, 39]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2033] & 413221(12)^{21}12231513221(12)122^1313 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 39, T \in [34, 39] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[2029] & 21(12)^{21}12231513221(12)^{2^1 1}12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.33 \rightarrow 33, T_{\max} = 47, T \in [33, 47] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2035] & 8123151321812321812321^{32^1 1}8^1 1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2036] & 41413221(12)^112231513221(12)^{222^1 1}1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[2037] & 5131613161316131513221(12)122^13 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 50, T \in [34, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2038] & 4132181232181231513221(12)12^122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2039] & 81231513218123218123218122^13 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2040] & 41413221(12)^112231513221(12)122^122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[2028] & 21(12)^{32^1 21}12231513221(12)^{21}122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 50, T \in [34, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2030] & 31513221(12)^{2^1 1}12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.33 \rightarrow 33, T_{\max} = 47, T \in [33, 47] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2027] & 21(12)^{3221}12231513221(12)^{2^1 1}122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.33 \rightarrow 33, T_{\max} = 50, T \in [33, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2026] $1(12)12231513221(12)1223151+\hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 39$, $T \in [34, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2025] $1(12)12231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.33 \rightarrow 33$, $T_{\max} = 47$, $T \in [33, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2041] $1(12)12231513221(12)122315132 + \hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 6 \\ 2 & 6 & 3 \end{pmatrix}_{b_0}$
- [2042] $231513221(12)12231513221(12)122 + \hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [2043] $21(12)12231513221(12)12^1222 + \hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2045] $1812^1321812321812315132181+\hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2046] $8122315132181232181232^1181+\hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2047] $221812^1321812321812315132181 + \hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2048] $8122315132181232181232^118122 + \hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2049] $181232181232181232181232^115+\hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2050] $41413221(12)12231513221(12)12^1222 + \hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33$, $T_{\max} = 41$, $T \in [33, 41]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2051] $71232171232181232181232^1171+\hat{1}\rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2052] \quad 1612^1 32181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2053] \quad 71232171232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2054] \quad 71232181232181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2044] \quad 413221^{21}(12)12231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2059] \quad 21^{221}(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2084] \quad 4132^1 181232181231513221^{21}(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2061] \quad 2231513221^{2^1 1}(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.80 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 30 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2062] \quad 21^{15132^1 21}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2063] \quad 21^{1513221}(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2064] \quad 1^{1513221}(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2065] \quad 1^{15132^1 21}(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2066] & \begin{matrix} 1 \\ 1812^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2067] & \begin{matrix} 1 \\ 181231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2068] & \begin{matrix} 1 \\ 712231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2069] & \begin{matrix} 1 \\ 7122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2070] & \begin{matrix} 1 \\ 51316131513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 40, \Delta = -111, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2058] & \begin{matrix} 322^1 1 \\ 21(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 50, T \in [33, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2057] & \begin{matrix} 2221 \\ 1(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2071] & \begin{matrix} 1 \\ 7122^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 40, \Delta = -111, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2072] & \begin{matrix} 1 \\ 171231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2073] & \begin{matrix} 1 \\ 1712^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2074] & \begin{matrix} 1 \\ 171231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2075] $1712^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2076] $413221^{21}(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2077] $4132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2078] $7122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2079] $712231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2080] $712231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2081] $7122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2082] $181232^{21} 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2083] $4132^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2056] $1(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2055] $1(12)12231513221(12)1223151322 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.80 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2106] $712321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2105] $712321712321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2104] $41413221(12)122315132181232^1 171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2103] $161232^1 181232181232171232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2102] $712321712321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2101] $4132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2100] $413221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2099] $41413221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2098] $1812321812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2097] $1812321812321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2096] $81223151321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2095] $8123151321812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) =$

(1, 27, 0), $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2094] $612^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[2093] $4132^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2092] $22181232^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2091] $5131613161316131513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 50, T \in [33, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2090] $1613161316131513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2089] $4132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2088] $413221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2087] $41413221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2086] $41413221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2085] $81223151321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2060] $21(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [2109] $21^{21}(12)122^1 31513221^{21}(12)12231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 47$, $T \in [33, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2123] $41321812321812315132^1 21^{221}(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2111] $21^{31513221}(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 47$, $T \in [33, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2112] $21^{1513221}(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2113] $712^{31513221} 32181231513221^{31513221}(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 50$, $T \in [33, 50]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2114] $712^1 231513221^1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2115] $712^1 231513221(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2116] $171231513221^1(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2117] $413221^{21}(12)1223151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -101$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2118] $712^1 31513221^1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 47$, $T \in [33, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2119] $712^1 231513221^1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2120] $712231513221^1(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) =$

$$(1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2107] \quad 1(12)122^{\overset{21}{1}}31513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2124] \quad 41321812321812315132^{\overset{21}{1}}21(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2125] \quad 8122^{\overset{21}{1}}315132181232181232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2126] \quad 41413221(12)1223151322^{\overset{2221}{1}}1(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2127] \quad 41413221(12)1223151322^{\overset{221}{1}}1(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2128] \quad 221812321812321812315132^{\overset{1}{1}}181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2129] \quad 41321812321812315132^{\overset{1}{1}}21(12)1222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2130] \quad 8122^{\overset{1}{1}}31513218123218123218122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2131] \quad 1812^{\overset{1}{1}}3218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2132] \quad 1812^{\overset{1}{1}}3218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2133] \quad 41413221(12)1223151322^{\overset{1}{1}}1(12)12222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2134] $712^1 32171232181232181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2135] $161232181232181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2136] $712^1 321712321812321812321712 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2137] $712^1 32181232181232181232161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2108] $1(12)^{2221} 12^1 231513221(12)^{21} 1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2121] $712^1 231513221(12)^1 1223151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2110] $21(12)^{221} 12^1 231513221(12)^{21} 1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2122] $18^1 12321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2142] $21(12)^{3221} 12231513221(12)^{21} 122^1 315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 50, T \in [33, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2143] $21(12)^{21} 12^1 231513221(12)^{21} 12231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2144] $21(12)^{221} 12231513221(12)^{21} 12^1 23151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2145] $21^{221}(12)122315132^1 21^{21}(12)1223151 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2146] $21^{231513221}(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 50, T \in [33, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[2147] $21^{31513221}(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[2148] $21^{1513221}(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[2149] $21^{151322^1 1}(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[2150] $5132^1 181231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 43, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2151] $5132181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 43, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2152] $1^{1513221}(12)1223151321812^1 32171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[2153] $1^{151322^1 1}(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[2154] $1^{1812315132^1 21}(12)122315132171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2155] $1^{181231513221}(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2156] $712231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) =$

(1, 27, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2157] $7122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2158] $51316131513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2159] $7122315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2160] $171231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2161] $1712315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2162] $171231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2163] $1712315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2164] $413221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -101$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2165] $41322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -101$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2166] $7122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2167] $712231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2168] & \quad 7122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2169] & \quad 712231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2170] & \quad 1(12)122^{21} 31513161316131613151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2171] & \quad 1^1 812321812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2172] & \quad 413218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2173] & \quad 41321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2174] & \quad 41321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2175] & \quad 413218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2176] & \quad 812315132181232181232^1 1^1 81 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2177] & \quad 812^1 3151321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2178] & \quad 812^1 23151321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2141] $1(12)122315132^{\overline{2221}}21(12)1223151+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2140] $1(12)12231513221(12)12^{\overline{21}}23151+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2139] $1(12)12^{\overline{221}}231513221(12)12231513 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2182] $414132^{\overline{1}}21(12)12231513221(12)12 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2183] $41413221(12)122^{\overline{1}}31513221(12)12 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2184] $413221(12)1223151321812^{\overline{1}}32181 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2185] $41322^{\overline{1}}1(12)122315132181232181 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2186] $1613161316131513221(12)12^{\overline{1}}23151 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2187] $151316131613161315132^{\overline{1}}21(12)122 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[2188] $31613161316131513221(12)122^{\overline{1}}315 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 50, T \in [33, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2189] $161316131513221(12)122^{\overline{1}}31513161 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[2190] $221812321812^{\overline{1}}321812315132181 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2191] 413218123218123151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2192] 41321812¹32181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2193] 612315132¹21(12)122315131613161 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[2194] 812¹315132181232181232181223 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2195] 812315132181232181232¹181223 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2196] 812¹231513218123218123218122 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[2197] 8122315132181232¹18123218122 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[2198] 181232181232181232¹18123215 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2199] 181232181232181232¹18123151 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2200] 414132¹21(12)12231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2201] 41413221(12)122¹31513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2202] 41322¹1(12)1223151321812321812 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
& [2203] \quad 413221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
& [2204] \quad 812^1 32181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2205] \quad 71232171232181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [2206] \quad 1612321812^1 32181232171232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2207] \quad 41413221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
& [2208] \quad 414132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [2209] \quad 81232171232181232181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [2210] \quad 812^1 321712321812321812321613 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2211] \quad 71232171232181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [2212] \quad 71232181232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
& [2138] \quad 1(12)122^1 31513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 50, T \in [33, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2181] $414132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2179] $8122315132181232^1 1812321 81 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[2180] $41413221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2213] $1(12)122^1 31513221(12)1223151322 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2214] $1(12)12^1 231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2215] $1(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32, T_{\max} = 47, T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2216] $21(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32, T_{\max} = 47, T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2217] $21(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2218] $21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2219] $21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32, T_{\max} = 47, T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2220] $1 8 1232^1 181231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 32, \Delta = -94, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32, T_{\max} = 32, T \in [32, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

- [2221] $71232181231513221^{315132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2222] $322181232^1 181231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 32$, $\Delta = -94$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 32$, $T \in [32, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2223] $1(12)122^1 315131613161316131513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [2224] $1(12)12^1 231513161316131613151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [2225] $181232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2226] $4132181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2227] $4132181232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2228] $8123151321812321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2229] $81223151321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2230] $3151316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2231] $1513161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2232] $22181232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2233] & 4132181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2234] & 613161316131513221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2235] & 8123151321812321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 43, T \in [32, 43] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2236] & 81223151321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2237] & 1812321812321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2238] & 1812321812321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2239] & 812321812321812321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 44, T \in [32, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2240] & 712321712321812^1 32181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2241] & 161232181232^1 181232171232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2242] & 812321712321812321812^1 321613 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 44, T \in [32, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2243] & 712321712321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2244] & 712321812321812^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} =
\end{aligned}$$

15.0

$$T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2296] $71232^1 1712321812321812321712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2287] $181232^1 18123218123218123215 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2297] $71232^1 181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2295] $41413221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2294] $4141322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2293] $1612321812321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2292] $71232^1 171232181232181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2291] $413221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2290] $4141322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2245] $1(12)12^1 231513221(12)1223151322 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2246] $1(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2247] & \begin{matrix} 3221 & & 21 \\ 21(12)122^1 31513221(12)122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2248] & \begin{matrix} 221 & & 21 \\ 21(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2249] & \begin{matrix} 2231513221 \\ 21(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2250] & \begin{matrix} 1513221 \\ 21(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2251] & \begin{matrix} 5132181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 43, T \in [32, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2252] & \begin{matrix} 51321812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 43, T \in [32, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2253] & \begin{matrix} 3221 \\ 221(12)12^1 231513221(12)1223141 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 32, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2254] & \begin{matrix} 1513221 \\ 1(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2255] & \begin{matrix} 31513221 \\ 71232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2256] & \begin{matrix} 1 \\ 18123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2257] & \begin{matrix} 1 \\ 181231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2258] $413221^{21}(12)12^1 231513221^{13221}(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 32$, $\Delta = -93$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 32$, $T \in [32, 32]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2259] $71223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2260] $71223151322^1 (12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2289] $41413221^1 (12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2261] $513161315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2262] $51316131513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2263] $712231513221(12)122^1 31513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2264] $71223151322^1 1(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2265] $17123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2266] $17123151322^1 (12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2267] $17123151322^1 (12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2268] $17123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2269] & 413221(12)12^{21}231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -101, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2270] & 71231513221(12)122315132^118123 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2271] & 71223151322^11(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2272] & 712231513221(12)12^123151321812 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2273] & 71223151322^11(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2274] & 712231513221(12)12^123151322181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2275] & 1(12)12^12315131613161316131513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2276] & 41413221(12)12^1231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2277] & 4141322^11(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2278] & 41413221(12)12^1231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2279] & 4141322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41]
\end{aligned}$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2280] $413221(12)122^1 315132181232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2281] $23151316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2282] $31513161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 47, T \in [32, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2283] $1613161315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2284] $161316131513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2288] $181232^1 18123218123218123151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2285] $61231513221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2286] $6123151322^1 1(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2298] $1(12)122315132^1 21(12)12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 47, T \in [32, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2306] $41321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2308] $812315132181232^1 1812321^1 81 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2307] $41321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2299] & \begin{matrix} 3221 & & 21 \\ 21(12)12231513221(12)12^1 2315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2300] & \begin{matrix} 21 & & 21 \\ 21(12)122315132^1 21(12)12231513 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2301] & \begin{matrix} 31513221 \\ 21(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2302] & \begin{matrix} 321 \\ 1812321812^1 31513221(12)12231 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 32, \Delta = -94, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2303] & \begin{matrix} 321 \\ 3221812321812^1 31513221(12)12231 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 32, \Delta = -94, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2310] & \begin{matrix} 321 \\ 231513161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2311] & \begin{matrix} 321 \\ 31613161316131513221(12)12^1 2315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2312] & \begin{matrix} 321 \\ 221812321812321812^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2313] & \begin{matrix} 321 \\ 41321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2314] & \begin{matrix} 321 \\ (12)122^1 31513161316131613151322 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2315] & \begin{matrix} 321 \\ 61315132^1 21(12)1223151316131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2316] & \begin{matrix} 321 \\ 812315132181232^1 181232181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -100, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 27, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2317] $8122315132^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2318] $181232181232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2319] $181232181232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2320] $81232181232^1 181231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2321] $71232171232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2322] $1612321812321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2323] $81232171232181232^1 1812321613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2324] $71232171232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2325] $71232181232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2309] $8122315132^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2304] $712315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 31.76 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2305] & \begin{matrix} 21 \\ 1812321812321812^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 39, \quad \Delta = -105, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, \quad T_{\max} = 39, \quad T \in [32, 39] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2328] & \begin{matrix} 3221 \\ 21(12)122315132^1 21(12)122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2330] & \begin{matrix} 21 \\ 21(12)12231513221(12)12^1 231513 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 47, \quad \Delta = -116, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 47, \quad T \in [32, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2331] & \begin{matrix} 2315132^1 21 \\ 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2332] & \begin{matrix} 3151322^1 1 \\ 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 47, \quad \Delta = -116, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 47, \quad T \in [32, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2333] & \begin{matrix} 51321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 43, \quad \Delta = -114, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 43, \quad T \in [32, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2334] & \begin{matrix} 513218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 43, \quad \Delta = -114, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 43, \quad T \in [32, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2335] & \begin{matrix} 1513221 \\ 1(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 39, \quad \Delta = -109, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 39, \quad T \in [32, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2336] & \begin{matrix} 1513221 \\ 1(12)122315132^1 181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 39, \quad \Delta = -109, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 39, \quad T \in [32, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2337] & \begin{matrix} 3151322^1 1 \\ 71232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2338] & \begin{matrix} 5131613151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 40, \quad \Delta = -111, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 40, \quad T \in [32, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2339] $712231513221(12)12^1 231513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2340] $71231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2341] $8123151321812^1 321812321^1 81 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2342] $413221(12)12^1 23151321812321^1 81 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2343] $413221(12)122315132^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2344] $16131613161315132^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2345] $16131613151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2346] $613161316131513221(12)12^1 231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2347] $(12)12^1 231513161316131613151322 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2348] $61316131513221(12)122^1 315131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2349] $61231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2350] $8123151321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

- [2351] $1812321812^1 3218123218123215 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2352] $1812321812^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2353] $413221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2354] $413221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2355] $81232^1 181232181232181231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2356] $812321812321812^1 32181231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2357] $712321712^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [2358] $161232181232181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2359] $41413221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [2360] $41413221(12)122315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2361] $81232^1 1712321812321812321613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2362] $812321712321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2363] & 712321712^1 321812321812321712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2364] & 712321812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2329] & 21(12)^{3221} 12^1 231513221(12)^{21} 122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2326] & 1(12)^{21} 12231513221(12)^{21} 122^1 315132 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2327] & 1(12)^{221} 12231513221(12)^{21} 12^1 231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2370] & 712321812315132^1 21(12)^{31513221} 1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2365] & 1(12)^{21} 122315132^1 21(12)^{21} 122315132 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2367] & 21(12)^{21} 1223151322^1 1(12)^{21} 12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2366] & 1(12)^{221} 1223151322^1 1(12)^{21} 12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2372] & 812315132^1 1812321812321^{321} 8 1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 43, T \in [32, 43] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2373] & 161316131613151322^1 1(12)^{21} 1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2374] $51316131613161315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2375] $613151322^1 1(12)1223151316131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2376] $613161315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2377] $812315132^1 181232181232181223 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2378] $81232181232^1 181232181231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2379] $81232171232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2369] $21^1 (12) 12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[2371] $7123151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2368] $21^1 (12) 122^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2383] $71231513221^1 (12)12^1 231513218123 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -196$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.55 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2384] $6131513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -196$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.55 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2385] $61316131513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -196$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.55 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2386] & \begin{matrix} 1 \\ 812321812^1 32181232181231513 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 44, \quad \Delta = -115, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 44, \quad T \in [32, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2387] & \begin{matrix} 1 \\ 812321712^1 321812321812321613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 44, \quad \Delta = -115, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 44, \quad T \in [32, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1(12)12231513221(12)122^1 3151322 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 3221 \\ 21(12)1223151322^1 1(12)122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2382] & \begin{matrix} 22315132^1 21 \\ 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2388] & \begin{matrix} 21 \\ 1(12)12231513221(12)12^1 2315132 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -225, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.52 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2389] & \begin{matrix} 23151322^1 1 \\ 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -225, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.52 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[2390] & \begin{matrix} 31513221 \\ 712321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -225, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.52 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2391] & \begin{matrix} 513161316131613151322^1 1(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -225, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.52 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2392] & \begin{matrix} 1 \\ 1(12)122315132^1 21(12)1223151322 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.48 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2393] & \begin{matrix} 2231513221 \\ 21(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.48 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2394] $7123218123151322^{\overline{1}1} (12)^{\overline{1}1} 1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -256, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.48 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2395] $6131613161315132^{\overline{1}21}(12)12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -256, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.48 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2396] $6131613151322^{\overline{1}1}(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -256, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.48 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2397] $1(12)1223151322^{\overline{1}1}(12)122315132 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -289, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.46 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2398] $21^{\overline{1}1}(12)^{\overline{1}1} 231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -289, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.46 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[2399] $316131613161315132^{\overline{1}21}(12)122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -289, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.46 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2400] $6131513221(12)12^{\overline{1}1} 23151316131613 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -289, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.46 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2403] $61316131613151322^{\overline{1}1}(12)12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -361, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.41 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[2401] $1(12)12231513221(12)12^{\overline{1}1} 23151322 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -361, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.41 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[2402] $21^{\overline{1}1}(12)^{\overline{1}1} 2231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -361, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.41 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[2404] $3161316131613151322^{\overline{1}1}(12)122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -400, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.39 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 139 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2406] $21^{\overline{1}1}(12)^{\overline{1}1} 12^{\overline{1}1} 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -441, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.37 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[2405] & \begin{matrix} 1 \\ 1(12)1223151322^1 1(12)1223151322 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -441, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.37 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
T = 28 & \quad [2411] \quad 315132181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 48, \quad T \in [35, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2410] & \begin{matrix} 22^1 315132181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 51, \quad T \in [35, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2412] & \begin{matrix} 51232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2413] & \begin{matrix} 15132181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2414] & \begin{matrix} 16131613161316131513221(12)12^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2415] & \begin{matrix} 4141413221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 41, \quad \Delta = -111, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 41, \quad T \in [35, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2416] & \begin{matrix} 4141413221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 41, \quad \Delta = -111, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 41, \quad T \in [35, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2417] & \begin{matrix} 216131613161316131513221(12)12^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2418] & \begin{matrix} 4141413221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 41, \quad \Delta = -111, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 41, \quad T \in [35, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2419] & \begin{matrix} 712321712321812321812321812^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2407] & \begin{matrix} 1513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 34, \quad \Delta = -96, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0
\end{aligned}$$

$(1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34$, $T_{\max} = 34$, $T \in [34, 34]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2409] $1(12)12231513218123218123151 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2408] $1(12)12231513218123218123215 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 4$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2424] $51232181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2423] $413221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.33 \rightarrow 34$, $T_{\max} = 48$, $T \in [34, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[2425] $15132181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2426] $181231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 45$, $T \in [35, 45]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2427] $812231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 45$, $T \in [35, 45]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2428] $41413221(12)12231513221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2429] $1(12)122315131613161316131612^1 3 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2430] $4141413221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2431] $16131613161316131513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2432] $32^1 16131613161316131513221(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [2433] $4141413221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2434] $812321712321812321812321712^1 3 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- 513221
 [2422] $1(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35, T_{\max} = 51, T \in [35, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $^{2^1 1}$
 [2420] $31513221(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.33 \rightarrow 34, T_{\max} = 34, T \in [34, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $^{32^1 21}$
 [2421] $1(12)1223151321812321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35, T_{\max} = 51, T \in [35, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- 21
 [2441] $71232171232181232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [2440] $1712^1 32181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2438] $512^1 32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 21
 [2437] $21(12)122315132181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- 221
 [2436] $1(12)122315132181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2443] $21712^1 32181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 16, (n_+, n_-, n_0) =$

(1, 28, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2444] 4141413221(12)12231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2445] 71232171232181232181232¹18122 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2446] 1812321¹81232181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[2447] 71232181232181232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2439] 41413221¹(12)12231513221(12)12¹2313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2442] 71232181232181232181232¹171 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2435] 21¹⁵¹³²²¹(12)12231513221(12)12¹2313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 34$, $T \in [34, 34]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2467] 51316131513221(12)122315132¹2181 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2452] 1¹⁵¹³²²¹(12)122315132181232¹1812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2453] 1^{15132¹21}(12)1223151321812321812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2454] 1²²¹(12)1223151321812321812¹3215 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2455] & \begin{matrix} 221 \\ 1(12)1223151321812321812^1 3151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2456] & \begin{matrix} 21 \\ 21(12)1223151321812321812^1 3215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2457] & \begin{matrix} 21 \\ 21(12)1223151321812321812^1 3151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2458] & \begin{matrix} 1 \\ 1812^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2459] & \begin{matrix} 1 \\ 181231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2460] & \begin{matrix} 1 \\ 1812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2461] & \begin{matrix} 1 \\ 1812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2462] & \begin{matrix} 1 \\ 181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2463] & \begin{matrix} 1 \\ 4132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 48, T \in [34, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2464] & \begin{matrix} 1 \\ 181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2465] & \begin{matrix} 1 \\ 1812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2451] $1(12)1223151321812321812315 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 51, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 51, T \in [34, 51]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2450] $1812^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2449] $21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 34, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 34, T \in [34, 34]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2469] $15132^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2470] $612^1 32181231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 43, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 43, T \in [34, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2471] $612^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 42, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2472] $21812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2473] $21812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2474] $2181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2475] $18122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2476] $81231513221(12)1223151321812^1 32 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 51, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34, T_{\max} = 51, T \in [34, 51]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [2477] $161316131513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2478] $7122^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2479] $41413221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2480] $171232^1 181232181232171232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2481] $4141413221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2482] $4141413221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2483] $41413221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2484] $712321712321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2485] $712321812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2486] $16131613161316131513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2487] $232^1 16131613161316131513221(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 48$, $T \in [34, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2488] $2171232^1 181232181232171232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2489] 4141413221(12)122315132¹21(12)12222 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2490] 41413221(12)122315132181232¹1812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2491] 712321712321812321812¹3218122 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2492] 1812321812321812321812¹32161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2493] 712321812321812321812¹321712 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2494] 4132¹181232181232181231513218 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 43, T \in [34, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2448] 21¹⁵¹³²²¹(12)122315132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2468] 51232¹181232181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2466] 51316131513221(12)122315132¹1812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2498] 1²²¹(12)122¹31513218123218123215 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2499] 1²²¹(12)122¹31513218123218123151 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2500] 21²¹(12)122¹31513218123218123215 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2501] & \begin{matrix} 21 \\ 12 \end{matrix} 122^1 31513218123218123151 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2502] & \begin{matrix} 1 \\ 18 \end{matrix} 1231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2503] & 51316131513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2504] & 512321812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2505] & 151321812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2506] & 1812^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 45, T \in [34, 45] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2507] & 2181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2508] & 1812^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2509] & 8122^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 45, T \in [34, 45] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2510] & 712^1 31513221(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 48, T \in [34, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2511] & 712^1 231513221(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2512] $41413221(12)1223151322^11(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2513] $171232181232181232171232^1181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2514] $4141413221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2515] $4141413221(12)1223151322^11(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2516] $712^132171232181232181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2517] $712^132181232181232181232171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2518] $2171232181232181232171232^1181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2519] $4141413221(12)1223151322^11(12)12222 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2520] $712^13217123218123218123218122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2521] $1812^132181232181232181232161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2522] $712^1321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2497] $1(12)122315132181232^1181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 48, T \in [34, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2496] \quad 21 \quad (12) \quad 1223151322^1 1(12)122313 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2495] \quad 21 \quad (12) \quad 122315132^1 21(12)12231 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2542] \quad 1812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [2537] \quad 21(12)12^1 231513218123218123215 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2528] \quad 1 \quad (12) \quad 12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 51, T \in [34, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
& [2529] \quad 1(12)122^1 315132181232181231513 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 48, T \in [34, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2530] \quad 1 \quad (12) \quad 1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
& [2531] \quad 1 \quad (12) \quad 1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [2532] \quad 1(12)122315132181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [2533] \quad 1(12)12^1 231513218123218123215 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2534] \quad 1(12)12^1 231513218123218123151 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2526] & \begin{matrix} 1 \\ 1812315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2525] & \begin{matrix} 151322^1 1 \\ 21 \quad (12) \quad 12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 34, \quad \Delta = -96, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2524] & \begin{matrix} 1513221 \\ 21 \quad (12) \quad 122^1 31513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 34, \quad \Delta = -96, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2538] & \begin{matrix} 21 \\ 21(12)12^1 231513218123218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2539] & \begin{matrix} 21 \\ 21(12)122315132181232^1 18123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2540] & \begin{matrix} 231513221 \\ 812^1 32181231513221 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 51, \quad T \in [34, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2541] & \begin{matrix} 1 \\ 1812315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -106, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2523] & \begin{matrix} 31513221 \\ 21 \quad (12) \quad 1223151322^1 1(12)12231 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 34, \quad \Delta = -96, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2543] & \begin{matrix} 1 \\ 181231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -106, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2544] & \begin{matrix} 1 \\ 1812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -106, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2545] & \begin{matrix} 1 \\ 181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -106, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2546] $413221^1(12)12231513221^1(12)12+\hat{1}\rightarrow 2$ $T = 28, T_{\max} = 48, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 48, T \in [34, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2547] $41322^11(12)12231513221^1(12)12+\hat{1}\rightarrow 2$ $T = 28, T_{\max} = 48, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 48, T \in [34, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2548] $181232^1181231513221^1(12)1223151 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2549] $181232181231513221^1(12)12^123151 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2550] $161315132^121(12)1223151321812315 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 41, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2551] $16131513221(12)1223151321812^1315 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 41, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2552] $51316131513221(12)122^13151321812 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2553] $51316131513221(12)122^13151322181 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2554] $3151321812321812315132^121(12)122 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 48, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 48, T \in [34, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2555] $512321812^132181231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2556] $5123218123218123151322^11(12)1222 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2557] $151321812^132181231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2558] & 1513218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2559] & 61232^1 181231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2560] & 61232181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2561] & 61231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2562] & 612315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2563] & 21812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2564] & 21812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2565] & 2181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2566] & 18122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2567] & 812^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 51, T \in [34, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2568] & 812^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 45, T \in [34, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2569] $161316131513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2570] $7122315132^1 21(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2571] $41413221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2572] $414132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2573] $161316131613161315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2574] $1(12)122^1 315131613161316131612 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [2575] $1712321812^1 32181232171232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2576] $4141413221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2577] $4141413221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2578] $41413221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2579] $414132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2580] $71232171232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2581] & 71232181232181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2582] & 161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2583] & 2161316131613161315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2584] & 21712321812^1 32181232171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2585] & 4141413221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2586] & 41413221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[2587] & 414132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2588] & 81232171232181232181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2589] & 812^1 3217123218123218123217123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2590] & 71232171232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2591] & 181232181232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2592] & 81232181232181232181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

15.0

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 44, T \in [34, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2593] $812^1 321812321812321812321613 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 44, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 44, T \in [34, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2594] $71232181232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2595] $41321812^1 32181232181231513218 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 43, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 43, T \in [34, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2596] $41321812321812321812315132^1 18 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 43, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 43, T \in [34, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2536] $21^{21}(12)122315132181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2535] $1^{221}(12)122315132181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2527] $1^{3221}(12)1223151321812321812^1 315 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 51, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 51, T \in [34, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2600] $1^{21}(12)12^1 2315132181232181231513 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 48, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2601] $1^{21}(12)1223151321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 48, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2602] $1^{221}(12)1223151321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2603] $1^{221}(12)1223151321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2604] & \begin{matrix} 21 \\ 21(12)1223151321812^1 3218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2605] & \begin{matrix} 21 \\ 21(12)1223151321812^1 3218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2606] & 23151321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 51, \quad T \in [33, 51] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2607] & 315132^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2608] & 31513218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2609] & 51232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2610] & 15132181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2611] & 4132181232^1 181231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 33, \quad \Delta = -93, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2612] & \begin{matrix} 1 \\ 1(12)122^1 3151316131613161316123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 44, \quad \Delta = -114, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 44, \quad T \in [33, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2613] & \begin{matrix} 1 \\ 1613161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2614] & \begin{matrix} 21 \\ 1(12)12^1 2315131613161316131612 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2615] & \begin{matrix} 31 \\ 161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 43, \quad \Delta = -113, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2616] & \overset{1}{171232181232^1 181232171232181} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2617] & 712321712321812^1 32181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2618] & \overset{1}{712321812321812^1 32181232171} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2619] & \overset{1}{1613161316131613151322^1 1(12)1} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2620] & \overset{31}{161316131613161315132^1 21(12)1} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2621] & 32161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2622] & 21613161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2623] & 2171232181232^1 181232171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2624] & 812321712321812321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2625] & 712321712321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2626] & \overset{1}{1812321812321812^1 32181232161} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2627] $812321812321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2628] $712321812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2629] $4132181232^1 181232181231513218 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2599] $1(12)122^1 3151321812321812315132 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2598] $1^{315132^1 21} (12) 122315132181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2597] $21^{315132^1 21} (12) 12231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 34, T \in [33, 34]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2657] $218123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2658] $218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2635] $1^{513221} (12) 122315132181232^1 18123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2636] $1^{1513221} (12) 122^1 3151321812321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2637] $18123151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2638] $181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) =$

(1, 28, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2639] $18123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2640] $18123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2641] $181231513221^1 (12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2642] $413221^1 (12)122^1 31513221^1 (12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2643] $1812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2644] $16131513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 41$, $T \in [33, 41]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2645] $1613151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 41$, $T \in [33, 41]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2646] $51316131513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2647] $513161315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2648] $513161315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2649] $51316131513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [2650] $223151321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2651] $231513218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2652] $612321812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 43$, $T \in [33, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2653] $61232181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 43$, $T \in [33, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2654] $6123151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2655] $61231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2633] $1^{31513221} (12) 1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2632] $1^{181231513221} (12)122^1 31513161315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2631] $1^{18123151322^1} 1(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2659] $1^{2181231513221} (12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2660] $1^{181223151322^1} 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2661] $1^{812231513221} (12)122315132^1 18123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2662] 1613161315132¹21(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2663] 161316131513221(12)12¹2315132171 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2664] 71223151322¹1(12)122315131613161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2665] 712231513221(12)122¹315131613161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2666] 41413221¹(12)12¹231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2667] 4141322¹1(12)12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2668] 1¹(12)12¹23151316131613161316123 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2669] 16³¹13161316131613151322¹1(12)12 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2670] 1712321812321812321712¹32181 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2671] 4141413221(12)12¹231513221²²²¹(12)1 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2672] 41414132¹21(12)12231513221²²²¹(12)1 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2673] 4141413221(12)12¹231513221²²¹(12)12 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$(1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2674] \quad 41414132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2675] \quad 41413221(12)122^1 315132181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2676] \quad 4141322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2677] \quad 71232^1 171232181232181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2678] \quad 71232^1 181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2679] \quad 16^1 13161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2680] \quad 232161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2681] \quad 321613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2682] \quad 21712321812321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2683] \quad 41414132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2684] \quad 4141413221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [2685] $4141322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2686] $41413221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2687] $71232^1 17123218123218123218122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2688] $181232^1 181232181232181232161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2689] $71232^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2634] $1(12)12^1 23151321812321812315132 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2630] $21(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 34$, $T \in [33, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2656] $181231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2692] $1(12)122315132181232^1 181231513 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.76 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2694] $1(12)122315132^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.76 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2695] $21(12)122315132^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.76 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2696] $21(12)122315132^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2697] \quad 413221(12)12231513221 \overset{1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& [2698] \quad 22315132181232181231513221 \overset{51322}{1} (12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
& [2699] \quad 3151321812 \overset{1}{32181231513221} (12)122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2700] \quad 512321812321812 \overset{1}{31513221} (12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2701] \quad 151321812321812 \overset{1}{31513221} (12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2702] \quad 1812315132 \overset{1}{21} (12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2703] \quad 8122315132 \overset{1}{21} (12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2704] \quad 41321812321812 \overset{1}{31513221} (12)12231 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 33, \Delta = -93, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2705] \quad 712315132 \overset{1}{21} (12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2706] \quad 1712321812321812 \overset{1}{32171232181} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2707] \quad 71232171232 \overset{21}{1812321812321} 81 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2708] $71232181232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2709] $2321613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2710] $21712321812321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2711] $81232171232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2712] $71232171232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2713] $181232181232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2714] $81232181232181232^1 1812321613 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2715] $71232181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2716] $41321812321812^1 32181231513218 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2693] $1(12)122315132^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2690] $21 \begin{pmatrix} 31513221 \\ 12 \end{pmatrix} 122^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 34, T \in [33, 34]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2691] $1 \begin{pmatrix} 51322^1 \\ 12 \end{pmatrix} 12231513218123218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2763] & \overset{1}{712321812^1 321812321812321712} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2721] & \overset{3221}{1(12)122^1 3151321812321812315} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 51, \quad T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2722] & \overset{3221}{1(12)122315132181232^1 1812315} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 51, \quad T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2723] & \overset{21}{1(12)1223151321812^1 32181231513} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2724] & \overset{1513221}{1(12)122315132^1 1812321812} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2725] & \overset{1513221}{1(12)12^1 23151321812321812} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2726] & \overset{2315132^1 21}{81232181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 51, \quad T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2727] & \overset{231513221}{81232^1 181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 51, \quad T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2728] & \overset{1}{413221(12)12^1 231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2729] & \overset{1}{413221(12)122315132^1 21(12)12} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2730] & \overset{1}{18123218123151322^1 1(12)1223151} + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2731] & \overset{1}{1812321812^1 31513221(12)1223151} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2732] & 16131513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2720] & \overset{1}{1(12)1223151321812321812^1 315132} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2719] & \overset{3151322^1 1}{1(12)122315132181232181} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2718] & \overset{1}{181231513221(12)12^1 231513161315} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2736] & 2315132^1 181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2737] & 315132181232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2738] & 612321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2739] & 6123218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2740] & 61231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2741] & 61231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2742] & \overset{1}{181231513221(12)122^1 31513218123} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2743] & \begin{matrix} 1 \\ 812231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2744] & \begin{matrix} 1 \\ 16131613151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2745] & \begin{matrix} 1 \\ 712231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2746] & \begin{matrix} 1 \\ 171232181232181232^1 171232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2747] & \begin{matrix} 1 \\ 414141322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2748] & \begin{matrix} 221 \\ 414141322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2749] & \begin{matrix} 1 \\ 41413221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2750] & \begin{matrix} 1 \\ 41413221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2751] & \begin{matrix} 21 \\ 712321712^1 32181232181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2752] & \begin{matrix} 1 \\ 712321812^1 32181232181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2753] & \begin{matrix} 1 \\ 2171232181232181232^1 171232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2754] 414141322¹1(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2755] 41413221(12)12¹23151321812321812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2756] 41413221(12)122315132¹1812321812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2757] 812321712321812¹3218123217123 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2758] 81232¹17123218123218123217123 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2759] 712321712¹3218123218123218122 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2760] 1812321812¹32181232181232161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2761] 81232¹1812321812321812321613 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2762] 812321812321812¹321812321613 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2717] 21^{3151322¹1}(12) 12231513221(12)12231 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 34, T \in [33, 34]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2764] 4132181232181232¹181231513218 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2733] 16131513221(12)12¹23151321812315 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

- [2735] $5131613151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.69 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [2734] $5131613151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.69 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [2765] 513221
 $1(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 144$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.64 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2777] $81223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2766] 31513221
 $21(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 34$, $T \in [33, 34]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [2771] 21
 $1(12)122315132^1 181232181231513 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2772] $23151321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2773] $3151321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2774] $18123151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2775] $81231513221(12)122315132^1 181232 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2776] $812315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2767] 31513221
 $1(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2778] & 7123151322^1 1(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2779] & 161316131613161315132^1 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2780] & 81232171232^1 18123218123217123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2781] & 81232181232^1 1812321812321613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2782] & 41321812321812321812^1 31513218 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2770] & 1(12)12^1 23151321812321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 169, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2768] & 1(12)122315132181232^1 1812315132 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2769] & 1(12)1223151321812^1 321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 169, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2786] & 22315132^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 196, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.55 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2785] & 413221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 196, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.55 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2788] & 181231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 196, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.55 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2789] $812231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2790] $71231513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2791] $812321712^1 3218123218123217123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 44$, $T \in [33, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2792] $812321812^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 44$, $T \in [33, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2784] $1(12)122^1 31513218123218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2783] $1(12)1223151321812^1 321812315132 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2787] $2315132181232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [2793] $1(12)122315132^1 181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 225$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.52 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [2794] $81232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 225$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.52 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2795] $81231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 225$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.52 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2796] $1613161316131613151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 225$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.52 \rightarrow 33$, $T_{\max} = 44$, $T \in [33, 44]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2797] $1(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 256$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 32.48 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& [2798] \overset{1}{1}(12)122315132^1 1812321812315132 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 256, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.48 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& [2799] 223151321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 256, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.48 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
& [2800] 23151321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 256, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.48 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
& [2802] \overset{513221}{1}(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& [2805] \overset{1}{8}123151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
& [2806] 71231513221(12)12^1 23151316131613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& [2803] 812321812315132^1 21 \overset{231513221}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
& [2804] 22315132181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& [2801] \overset{3221}{1}(12)122315132^1 1812321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
& [2807] \overset{513221}{1}(12)122315132^1 18123218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 324, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.43 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& [2809] 223151321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 361, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.41 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2808] $812321812^1 31513221^{231513221} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 361, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.41 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[2810] $81231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 361, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.41 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[2811] $81232181231513221^1 (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 441, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.37 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

$T = 29$ [2819] $712321812321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[2816] $161232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2817] $3161232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2818] $1812321712321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2815] $1^{22^1 1}(12)122315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[2814] $1^{5131513221} (12) \ 12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2813] $1^{5131513221} (12) \ 12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2812] $1^{22^1 21} (12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [2820] $1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 41$, $T \in [36, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2821] $1(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.33 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2823] $41413221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.33 \rightarrow 35$, $T_{\max} = 46$, $T \in [35, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2824] $61232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 49$, $T \in [36, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2825] $161232181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 41$, $T \in [36, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2826] $61316131513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 49$, $T \in [36, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2827] $812321712321812321812321 8 1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2828] $4141413221(12)12231513221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 37$, $\Delta = -99$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 37$, $T \in [36, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2829] $8123217123218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2830] $812321812321812321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2822] $1(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 41$, $T \in [36, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2834] $1612^1 32181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2835] & 812315132181232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[2836] & 1812^{21} 32181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2837] & 1712^1 32181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2838] & 71232181232181232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2839] & 181232171232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2840] & 4141413221(12)12231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 37, T \in [35, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2841] & 181232181232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[2842] & 71232181232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2832] & 1(12)122315132181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2833] & 21(12)122315132181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2831] & 1 \quad \begin{matrix} 5131513221 \\ (12) \end{matrix} 12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2845] $1^{15132^1 21} (12) 12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 42, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[2847] $1712^1 32181231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 42, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2848] $1712^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2849] $7122^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2850] $1812^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[2851] $1^{221} (12)1223151321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2852] $21^{21} (12)1223151321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2853] $612^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 49, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 49, T \in [35, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2854] $61232181232181231513221^{322^1 1} (12)1 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 49, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 49, T \in [35, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[2855] $161316131513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2856] $161316131513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2857] $161232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2858] & 4132^1 21(12)1223151321812321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 49, T \in [35, 49] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2859] & 612^1 32181231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2860] & 8123151321812321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2861] & 8123151321812321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -108, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2862] & 61232181231513221(12)122315132^1 16 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2863] & 612^1 32181231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2864] & 181232^1 181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2865] & 171232^1 181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2866] & 712321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2867] & 1812321712321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2868] & 4141413221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 37, T \in [35, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2869] & 812321712321812321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 44, T \in [35, 44]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2870] $1812321812321812321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2871] $712321812321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2846] $1^{5131513221} (12) 122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2843] $1^{2221} (12) 122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2844] $1^{5131513221} (12) 122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2872] $1^{2221} (12) 12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2873] $1^{5131513221} (12) 1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2874] $1^{5131513221} (12) 1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2875] $712^1 231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2876] $1^{221} (12) 122^1 315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2877] $21^{21} (12) 122^1 315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

- [2878] 3161316131513221(12)122315132¹181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 46$, $T \in [35, 46]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2879] 161316131513221(12)122315132¹181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2880] 1612321812321812315132¹21(12)1222 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2881] 8122¹31513221(12)1223151316131613 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 46$, $T \in [35, 46]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2882] ²¹181232181232181232171232¹181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2883] ¹1712321812321812321¹81232¹181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2884] ¹712¹321812321812321812321²¹81 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2885] 1812¹3217123218123218123218122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2886] 4141413221(12)1223151322¹1(12)122313 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 37$, $\Delta = -99$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 37$, $T \in [35, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2887] 1812¹321812321812321812321712 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2888] ¹712¹3218123218123218123218122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2934] ¹71232181232181232¹1812321²¹81 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2927] & 812315132181232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -108, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 151322^1 1 \quad 21 \\
[2894] & 1 (12) 12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 5131513221 \\
[2895] & 1 (12) 122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& 51315132^1 21 \\
[2896] & 1 (12) 12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2897] & 171232^1 181231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2898] & 171232181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2899] & 1712315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2900] & 171231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2901] & 712231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2902] & 7122315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 1 \\
[2903] & 1812315132^1 21(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[2904] & 1(12)122315132181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2905] & 1(12)12^{21}2315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[2906] & 21(12)122315132181232^{18}181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2907] & 21(12)12^{21}2315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[2908] & 1(12)122315132181232181232^{16}1613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2909] & 1(12)122^{21}3151321812321812321613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2910] & 41413221(12)12231513221^{5132^{121}}(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2911] & 414132^{513221}21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2912] & 61232^{3221}181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 49, T \in [35, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2913] & 161316131513221(12)122^{13}151321812 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2914] & 161316131513221(12)122^{13}151322181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2915] & 16123218123218123151322^{11}(12)1222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2916] 1612321812¹32181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2917] 31612¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2918] 31612321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2919] 41322¹1(12)1223151321812321812315 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 49, T \in [35, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2920] 413221(12)1223151321812321812¹315 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 49, T \in [35, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2921] 812¹231513221(12)1223151316131613 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2922] 61232181231513221(12)122¹31513161 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2923] 61232¹181231513221(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[2924] 812¹31513218123218123218123215 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2925] 812315132181232181232¹18123215 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2926] 812¹31513218123218123218123151 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2892] 1^{51315132¹21}(12)²²¹12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2891] & 1^{5131513221} (12) 122^1 31513221^{221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2890] & 1^{2221} (12) 12231513221^1 (12) 122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2930] & 1^{21} 1812321812^1 32181232171232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2931] & 812^1 321712321812321812321^{321} 81 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2932] & 81232171232181232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2933] & 1^{1712321812^1 32181232181232181} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2889] & 1^{2221} (12) 122315132^1 21^{12} 1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2935] & 181232171232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2936] & 4141413221(12) 122^1 31513221(12) 122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 37, T \in [35, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2937] & 812^1 32171232181232181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2938] & 81232171232181232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2939] & 1^{181232181232181232^1 1812321712} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) =
\end{aligned}$$

$(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

$[2940] \quad 81232181232181232181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

$[2941] \quad 812^1 3218123218123218123217123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[2942] \quad 71232181232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[2929] \quad 61232^1 181231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

$[2893] \quad 1 \quad (12) \quad 12231513221(12)122^1 315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

$[2928] \quad 61232181231513221(12)122^1 31513216 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[2957] \quad 812321712321812321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -64, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34, T_{\max} = 44, T \in [34, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

$[2947] \quad 161232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -64, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

$[2948] \quad 316123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -64, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34, T_{\max} = 46, T \in [34, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

$[2949] \quad 3161232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -64, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34, T_{\max} = 46, T \in [34, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

$[2950] \quad 8123151321812321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -64, (n_+, n_-, n_0) =$
 $(1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

- [2951] $8123151321812321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2952] $181232181232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2953] $812321712321812321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2954] $171232181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2955] $712321812321812^1 321812321^1 81 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2945] $1(12)1223151321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2944] $21(12)1223151321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2943] $1(12)1223151321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2959] $812321812321812321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2960] $712321812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2961] $413161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 43$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 43$, $T \in [34, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2962] $413161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 43$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.94 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2946] & 1(12)12^{\overset{21}{1}}23151321812321812321613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -64, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.94 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2958] & 1812321812321812^{\overset{1}{1}}321812321712 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -64, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.94 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2956] & 1812321712321812^{\overset{1}{1}}3218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -64, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.94 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2963] & 1(12)12231513221(12)12^{\overset{1}{1}}23151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2964] & 1(12)12231513221^{\overset{2221}{1}}1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2965] & 1^{\overset{5131513221}{1}}(12)12^{\overset{221}{1}}231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2966] & 1^{\overset{5131513221}{1}}(12)12231513221^{\overset{221}{1}}(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2967] & 1^{\overset{1513221}{1}}(12)122^{\overset{21}{1}}31513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2968] & 1^{\overset{5131513221}{1}}(12)12^{\overset{1}{1}}231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2969] & 1^{\overset{5131513221}{1}}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2970] & 1712321812315132^{\overset{1}{1}}21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2971] \quad & 171232181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2972] \quad & 17123151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2973] \quad & 171231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2974] \quad & 712231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2975] \quad & 71223151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2976] \quad & \begin{matrix} 1 \\ 181231513221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2977] \quad & \begin{matrix} 1 \\ 18123151322^1 1(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2978] \quad & \begin{matrix} 513221 \\ 41413221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2979] \quad & \begin{matrix} 513221 \\ 4141322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2980] \quad & 161316131513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2981] \quad & 1613161315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2982] \quad & 161316131513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2983] & 1613161315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2985] & 61232181231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2986] & 612321812315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2987] & 612321812315132^1 21(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2988] & 61232181231513221(12)12^1 231513216 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2989] & 18^{21} 12321812321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2990] & 1712321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2991] & 71232^1 18123218123218123218^1 1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2992] & 181232^1 17123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2993] & 4141413221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 37, T \in [34, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2994] & 41414132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 37, T \in [34, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2995] & 181232^1 1812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2996] & 71232^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2997] & 4131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 43, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2998] & 4131613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 43, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2984] & 61316131513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2999] & 1^{1513221} (12) 12231513221^{21} (12)12^1 2315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3000] & 1^{221} (12)122315132^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3001] & 21^{21} (12)122315132^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3002] & 1^{21} (12)122315132181232^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3003] & 41413221(12)12231513221^{51322^1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3004] & 3161316131513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3005] & 1612321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3006] $31612321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3007] $8122315132^1 21(12)1223151316131613 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3008] $812315132181232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3009] $812315132181232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3010] $1812321812321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3011] $81232171232181232^1 1812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3012] $1712321812321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3013] $71232181232^1 1812321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3014] $181232171232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [3015] $81232171232181232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3016] $181232181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3017] $81232181232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3018] $71232181232^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.76 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3027] $181231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[3023] $171231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3024] $171231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3025] $712231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3022] $1712321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3021] $17123218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3020] $1^{1513221} (12) 12^1 231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3029] $41413221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 46, T \in [34, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3030] $41413221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 46, T \in [34, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3031] $612321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 49, T \in [34, 49]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3032] $612321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -121, (n_+, n_-, n_0) =$

(1, 29, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3033] 16131613151322¹1(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3034] 16131613151322¹1(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3035] 3161232181232¹181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[3036] 413221(12)122¹3151321812321812315 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3037] 413221(12)122315132181232¹1812315 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3038] 61316131513221(12)122¹31513218123 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3039] 612321812¹31513221(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3040] 6123218123151322¹1(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3041] 8123151321812¹3218123218123215 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3042] 8123151321812¹3218123218123151 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -108$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3043] 612321812¹31513221(12)12231513216 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3044] 6123218123151322¹1(12)12231513216 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3045] & \overset{21}{181232181232181232^1 171232181} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3046] & 812321712321812^1 321812321 \overset{321}{81} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3047] & 81232^1 1712321812321812321 \overset{321}{81} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3048] & \overset{1}{171232181232181232^1} \overset{1}{181232181} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3049] & \overset{1}{712321812^1} \overset{21}{321812321812321} \overset{21}{81} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3050] & 1812321712^1 3218123218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3051] & 414141322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 37, T \in [34, 37] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3052] & 812321712321812^1 32181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3053] & 81232^1 171232181232181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3054] & \overset{1}{1812321812^1} \overset{1}{321812321812321712} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3055] & \overset{1}{812321812321812^1} \overset{1}{3218123217123} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3056] $81232^1 18123218123218123217123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [3057] $712321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3019] $1^{1513221} (12)^{21} 122315132^1 21 (12)^{21} 122315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3026] $712231513221 (12)^{21} 23151321812315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [3028] $1^{21} (12)^{21} 1223151321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3060] $6123218123218123151322^1 1 (12)^1 1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [3062] $31612321812321812^1 31513221 (12)^1 122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [3063] $413221 (12)^1 12^1 23151321812321812315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3064] $413221 (12)^1 1223151321812^1 321812315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3065] $613161315132^1 21 (12)^1 12231513218123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3066] $81223151322^1 1 (12)^1 1223151316131613 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3067] $812315132^1 18123218123218123215 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3068] & 812315132^1 18123218123218123151 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -108, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3069] & 81232171232^1 1812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3070] & 81232171232^1 181232181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3071] & 81232181232^1 18123218123217123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3058] & 1(12)122315132^1 1812321812321613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3059] & 61232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3061] & 3161316131513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3076] & 812231513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3074] & 31613161315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[3077] & 812321712^1 321812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3078] & 812321712^1 32181232181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3079] & 812321812^1 3218123218123217123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [3072] \quad 1 \quad (12) \quad 1223151322^1 (12) 122315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [3073] \quad 41413221(12)1223151322^1 (12) 12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [3075] \quad 61316131513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [3080] \quad 6131613151322^1 (12) 12231513218123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -256, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.48 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& [3081] \quad 612321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -289, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.46 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& [3082] \quad 316131613151322^1 (12) 122315132181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -289, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.46 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
& [3083] \quad 413221(12)122315132^1 1812321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -289, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.46 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
& [3084] \quad 812231513221(12)12^1 23151316131613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -289, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.46 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& T = 30 \quad [3095] \quad 171232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 4, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37, T_{\max} = 42, T \in [37, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
& [3089] \quad 61231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 4, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37, T_{\max} = 42, T \in [37, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& [3090] \quad 61231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 4, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37, T_{\max} = 42, T \in [37, 42]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3091] $16131513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3092] $316131513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 47, T \in [37, 47]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3093] $61231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3094] $171232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3088] $6131513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36, T_{\max} = 50, T \in [36, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[3096] $1(12)1223151321812321812321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3097] $32171232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3098] $2171232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3099] $1812321812321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3085] $1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3086] $21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

- [3087] $1(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37, T_{\max} = 47, T \in [37, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3109] $1(12)1223151321812321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3104] $16131513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3105] $316131513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 47, T \in [37, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3106] $61231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3107] $1(12)122315132181232181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3108] $171232181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3103] $61231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3110] $32^1 171232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3111] $2171232181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3112] $812321812321812321812321 \ 8 \ 1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3113] $8123218123218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 9, (n_+, n_-, n_0) =$

$$(1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3100] \quad 1 \quad \overset{5131513221}{(12)} \quad 12231513221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3101] \quad 1 \overset{222^1 1}{(12)} 12231513221 \overset{1}{(12)} 12231513161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3102] \quad 6131513221 \overset{1}{(12)} 12231513221 \overset{32^1 21}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 50, T \in [37, 50]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3117] \quad 1712^1 32181232181231513221 \overset{221}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3118] \quad 1712^1 32181232181231513221 \overset{21}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3119] \quad 1712^1 32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3120] \quad 1 \overset{221}{(12)} 122315132181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3121] \quad 21 \overset{21}{(12)} 122315132181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3122] \quad 21712^1 32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3123] \quad 1812^1 32181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3124] \quad 812315132181232181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3125] $181232181232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[3115] $16131513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[3114] $1^{5131513221}(12)12231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 37, T \in [36, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3116] $61231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3127] $1812^1 32181231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3130] $21^{221}(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3131] $6131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 50, T \in [36, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3132] $612^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3133] $61231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3134] $612^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3135] $61231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3136] $21812^1 32181231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3137] $5132181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 43, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3138] $21812^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3139] $5132181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3140] $16131513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3141] $61231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3142] $612^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3143] $171232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3144] $171232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3145] $1(12)122315132181232181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3146] $1(12)122315132181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3147] $1(12)1223151321812321812321712^1 32 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 50, T \in [36, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[3148] $171232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3149] $1(12)1223151321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3150] $21(12)1223151321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3151] $16131513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3152] $232^1 171232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 50, T \in [36, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3153] $2171232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3154] $612^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3155] $61231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3156] $612^1 32181231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3157] $61232181231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3158] $612^1 32181231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

- [3159] $61232181231513221(12)122315132^1 217 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [3160] $181232^1 181232181232181232181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3161] $8123151321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3162] $1812321812321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3163] $812321812321812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3164] $4132^1 21(12)12231513218123218123216 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 47$, $T \in [36, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3165] $413221(12)122315132181232181232^1 16 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 47$, $T \in [36, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [3166] $4132^1 18123218123218123217123218 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3129] $1(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3128] $1812^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3126] $1(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 37$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 37$, $T \in [36, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3170] $1(12)122^1 31513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 47$, $T \in [36, 47]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3171] & 21(12)122^1 31513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 47, T \in [36, 47] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3172] & 61231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3173] & 61231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3174] & 5132181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3175] & 16131513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3176] & 316131513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 47, T \in [36, 47] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3177] & 61231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3178] & 1712321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3179] & 1712321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3180] & 1712321812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3181] & 1(12)122^1 3151321812321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3182] $21(12)122^1 3151321812321812321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3183] $21712321812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3184] $712^1 32181231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 50, T \in [36, 50]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3185] $61232181231513221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3186] $181232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3187] $1812^1 3218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3169] $21(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3168] $1(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3167] $1^1 (12) 1223151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 37, \Delta = -98, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 37, T \in [36, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3221] $1(12)1223151321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3220] $17123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [3193] $1812315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

657

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3205] 2181232181231513221(12)122¹3151315 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3206] 5132¹181232181231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 43$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3207] 5132181232181231513221(12)12¹23151 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 43$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3208] 2181231513221(12)1223151321812¹315 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3209] 21812315132¹21(12)1223151321812315 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3210] 5132¹181231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3211] 5132181231513221(12)122¹3151322181 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3212] 161315132¹21(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3213] 16131513221(12)122¹31513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3214] 316131513221(12)1223151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 47, T \in [36, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3215] 61231513221(12)122¹31513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3216] 612315132¹21(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3217] & \begin{matrix} 1 \\ 1712321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3218] & \begin{matrix} 1 \\ 17123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[3219] & \begin{matrix} 1 \\ 1712321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3191] & \begin{matrix} 1 \\ 181232181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3190] & \begin{matrix} 1 \\ 181232^1 181231513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[3189] & \begin{matrix} 51315132^1 21 \\ 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 37, T \in [36, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[3223] & \begin{matrix} 1 \\ 17123218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[3224] & \begin{matrix} 21 \\ 1(12)122^1 31513218123218123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3225] & \begin{matrix} 21 \\ 1(12)122315132181232181232^1 17123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3226] & \begin{matrix} 221 \\ 1(12)122315132181232^1 1812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[3227] & \begin{matrix} 221 \\ 1(12)12^1 23151321812321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[3228] $21(12)122315132181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3229] $21(12)12^1 23151321812321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[3230] $161315132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3231] $16131513221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[3232] $321712321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3233] $321712^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3234] $21712321812^1 32181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3235] $217123218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3236] $612315132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3237] $61231513221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[3238] $61232^1 181231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3239] $61232181231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

- [3240] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 $61232181231513221(12)122^1 315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
- [3241] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 $61232^1 181231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
- [3242] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 $18^{21} 12321812^1 3218123218^1 1232181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
- [3243] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 $812^1 321812321812321812321^1 8^1 1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
- [3244] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 $812321812321812321812321^1 8^1 1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
- [3245] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $812^1 315132181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
- [3246] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
 $812315132181232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
- [3247] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
 $181232181232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
- [3248] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $812^1 32181232181232181232181223 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
- [3249] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 $81232181232181232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
- [3250] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $413221(12)1223151321812321812^1 3216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 47, T \in [36, 47]$
- [3251] $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 $41322^1 1(12)12231513218123218123216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 47, T \in [36, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3252] & 4132181232181232181232171232^1 18 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[3253] & 41321812^1 3218123218123217123218 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3222] & 1712321812^1 32181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3188] & 1^{5131513221}(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 37, T \in [36, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3192] & 1^{181231513221(12)1223151321812^1 315} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3257] & 1^{171232181232^1 181231513221(12)1222} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3258] & 1^{21}(12)12^1 231513218123218123217123 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[3259] & 1^{21}(12)1223151321812321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3260] & 1^{221}(12)1223151321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3261] & 1^{21}(12)1223151321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3262] & 3217123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3263] $32171232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 46$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 46$, $T \in [35, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3264] $2321712321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 50$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 50$, $T \in [35, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3265] $2171232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3266] $1(12)122^1 315131613161316131613151 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3267] $1^1 81232181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3268] $812321812321812321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3269] $1513161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3270] $8123151321812321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3271] $1812321812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3272] $812321812321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3273] $4132181232^1 18123218123217123218 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3256] $1(12)122^1 315132181232181232171232 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 50$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 50$, $T \in [35, 50]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[3255] & \begin{matrix} 1 \\ 171232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -105, \quad \det = 64, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3254] & \begin{matrix} 1 \\ 171232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -105, \quad \det = 64, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3306] & \begin{matrix} 1 \\ 61232181231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -109, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3274] & \begin{matrix} 513151322^1 1 \\ 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 37, \quad \Delta = -98, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 37, \quad T \in [35, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3279] & \begin{matrix} 1 \\ 18123151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -109, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3280] & \begin{matrix} 2221 \\ 1(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3281] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3282] & \begin{matrix} 221 \\ 21(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3283] & \begin{matrix} 221 \\ 21(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3284] & \begin{matrix} 1 \\ 6131513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 50, \quad \Delta = -116, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 50, \quad T \in [35, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3285] & \begin{matrix} 1 \\ 61231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -108, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3286] $6123151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3287] $6123151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3288] $61231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3289] $21812321812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3290] $2181232181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3291] $2181231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3292] $218123151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3293] $5132181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3294] $51321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3295] $1613151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3296] $16131513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3297] $6123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3298] $1(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3299] $1(12)12^1 2315132181232181232171232 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3300] $16131513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3301] $1613151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3302] $23217123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3303] $71232^1 181231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3304] $61231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3305] $6123151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3277] $1812321812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3307] $612321812315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3308] $61232181231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3309] $612321812315132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3310] & \overset{1}{1}(12)12^1 2315131613161316131613151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3311] & \overset{21}{1} 812321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[3312] & 31513161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3313] & 15131613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3314] & \overset{1}{1} 81232^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[3276] & \overset{1}{1} 81232181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3278] & \overset{1}{1} 81231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3275] & \overset{5131513221}{1} (12) 12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, T_{\max} = 37, T \in [35, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3317] & \overset{1}{61315132^1} 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3319] & \overset{1}{1712321812321812^1} 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3320] & \overset{1}{1712321812321812^1} 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3321] $1(12)122315132181232181232^1 171232 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3322] $1712321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3323] $1(12)122315132181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3324] $1(12)122315132^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3325] $21(12)122315132^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3326] $321712321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3327] $2321712^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3328] $21712321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3329] $1812321812321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3330] $81232181232181232^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3331] $31513161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[3332] $(12)122^1 31513161316131613161315132 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3333] & 812315132181232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3334] & 181232181232^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3335] & 81232181232181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3336] & 41321812321812^1 3218123217123218 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3318] & 316131513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3315] & 1(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3316] & 21(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3369] & 181232181232181232^1 181232181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3350] & 2181231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[3342] & 21(12)12231513221(12)122^1 315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3343] & 6131513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3344] $6131513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3345] $21812321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3346] $218123218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3347] $51321812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 43, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 43, T \in [35, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3340] $181231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[3339] $181231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[3338] $18123218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3351] $51321812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3352] $513218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[3353] $3161315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3354] $1(12)122^1 315132181232181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3355] $1(12)122315132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3356] & 1(12)1223151321812321812^1 32171232 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3357] & 1(12)1223151321812^1 3218123217123 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3358] & 16131513221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3359] & 16131513221(12)122315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3360] & 32171232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3361] & 232171232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3362] & 71232181231513221(12)122^1 315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3363] & 61231513221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3364] & 61231513221(12)122315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3365] & 612321812^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3366] & 6123218123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3367] & 6123218123151322^1 1(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]
\end{aligned}$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3368] $612321812^1 31513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3337] $1812321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3370] $812321812321812^1 321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3371] $81232^1 1812321812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3372] $(12)12^1 231513161316131613161315132 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 50$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3373] $8123151321812^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3374] $1812321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3375] $812321812321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3376] $81232^1 181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3377] $413221(12)122315132181232^1 18123216 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3378] $413221(12)122^1 31513218123218123216 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3379] $41321812321812321812321712^1 3218 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3380] & 4132181232181232^1 18123217123218 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3349] & 2181231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[3348] & 51321812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 43, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 43, T \in [35, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3341] & \begin{matrix} 221 & & 1 \\ 1(12)12231513221(12)122^1 315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3381] & \begin{matrix} 221 & & 1 \\ 1(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3385] & 5132181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 43, \Delta = -111, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 43, T \in [35, 43] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3386] & \begin{matrix} & & 1 \\ 316131513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3387] & \begin{matrix} 3221 \\ 1(12)1223151321812^1 32181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[3388] & \begin{matrix} 3221 \\ 1(12)12^1 2315132181232181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3389] & \begin{matrix} & & 1 \\ 1(12)122315132181232^1 181232171232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3390] & \begin{matrix} & & 21 \\ 1(12)122315132^1 18123218123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3391] $321712321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3392] $2321712321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3393] $712321812315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3394] $81232181232^1 1812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3395] $812315132^1 181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[3396] $81232181232^1 181232181232181223 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3397] $413221(12)1223151321812^1 3218123216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3398] $413221(12)12^1 231513218123218123216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[3399] $41321812321812321812^1 3217123218 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[3384] $513218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 43, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 43, T \in [35, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3383] $613151322^1 1(12)12231513221 \overset{3221}{(12)12} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3382] $21 \overset{21}{(12)1223151322^1 1(12)122315131613} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [3400] $1(12)12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3401] $21(12)12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3403] $31613151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3404] $1(12)1223151321812^1 32181232171232 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3405] $232171232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [3406] $71232181231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3407] $812321812^1 321812321812321^1 81 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3408] $812321812^1 32181232181232181223 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3409] $4132181232181232181232^1 17123218 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3402] $6131513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3410] $712321812^1 31513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 225, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.52 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3411] $1(12)122315132^1 181232181232171232 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 256, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 34.48 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3412] & 2321712321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 256, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.48 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3413] & 7123218123151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 256, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.48 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3414] & 51321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 43, \Delta = -111, \det = 289, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.46 \rightarrow 35, T_{\max} = 43, T \in [35, 43] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3415] & 1(12)122315132^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 289, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.46 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[3416] & 413221(12)122315132^1 18123218123216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.46 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
\\
T = 31 & [3419] 71231513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[3432] & 22^1 181232181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3417] & 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3418] & 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3420] & 712231513221(12)12231513221(12)1^1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3421] & 712231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3422] $171231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3423] $712231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3424] $181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3425] $181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3426] $181232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3427] $1(12)1223151321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3428] $1(12)12231513218123218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3429] $21(12)1223151321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3430] $322181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3431] $22181232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3448] $32^1 16131613161316131513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3433] $513221(12)^{2^1 1} 12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.33 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3434] $1(12)^{222^1 1} 12231513221(12)^1 122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 43, T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[3435] $71231513221(12)^1 12231513221(12)^{32^1 21} 12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[3436] $712231513221(12)^1 12231513221(12)^{222^1 1} 1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 43, T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[3437] $171231513221(12)^1 12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 43, T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[3438] $712231513221(12)^1 12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 43, T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[3439] $181232181232181231513221(12)^1 122^1 3 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[3440] $1^1 8^{32^1 1} 1232181232181231513221(12)^{21} 1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[3441] $1(12)^{21} 12231513218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[3442] $1(12)^{32^1 21} 1223151321812321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[3443] $1^1 8^{32^1 1} 1232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

- [3444] $18^{21}1232181232181231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 43$, $T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3445] $32^12181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 46$, $T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3446] $22181232181232181231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 43$, $T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3447] $6131513221(12)1223151321812321812^13 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 51$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 51$, $T \in [38, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3458] $8123151321812321812321^1171 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3449] $171231513221(12)12231513221(12)12^1222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3450] $61231513221(12)12231513221(12)12^12313 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 37$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 37$, $T \in [37, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3459] $812321712321812321812321^115 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3460] $812315132181232181232181232^11712 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3451] $712231513221(12)12231513221(12)12^1222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3452] $18^{21}12^132181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3453] $18^{21}12^132181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3454] $1812^{21}32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.75 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3455] $1(12)122315132181232181232^{221}18122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.75 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3456] $21(12)122315132181232181232^{21}18122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.75 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3457] $221812^{21}32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.75 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3502] $61232181231513221(12)122315132^{21}1812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3516] $(12)122315131613161316131612^{21}32 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3503] $612^{21}32181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3461] $1(12)12231513221(12)122315132^{2221}171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3462] $1(12)122^{2221}31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3463] $21(12)122^{221}31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3464] $21(12)12231513221(12)122315132^{221}171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3465] $71231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3466] $7122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3467] $712231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3468] $7122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3469] $712231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3470] $171231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3471] $1712^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3472] $61231513221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 37, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3473] $612^1 31513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 37, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3474] $7122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3475] $712231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3476] $181232181232181231513221(12)12^123 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3477] $1812^132181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3478] $1^{232}8^{11}1232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3479] $18^{21}1232^1181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3480] $18^{21}1232^1181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3481] $1812^132181231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3482] $1(12)1223151321812321812321812^123 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3483] $1(12)1223151321812321812321812 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3484] $1(12)122315132181232181232^11812 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3485] $1^{232}8^{11}1232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3486] $18^{21}1232^1181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3487] $1(12)1223151321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3488] $21(12)1223151321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3489] $61231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3490] $612^1 31513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3491] $61232181231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3492] $612^1 32181231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3493] $16131513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3494] $171232181231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3495] $1712^1 32181231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3496] $322^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3497] $22181232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3498] $612^1 32181232181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3499] & 61232181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3500] & 612^1 31513221(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[3504] & 61232181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3505] & 612^1 32181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[3506] & 712231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3507] & 7122^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[3508] & 4132^1 21(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[3509] & 8123151321812321812321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[3510] & 4132^1 21(12)122315132181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3511] & 413221(12)122315132181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3512] & 812321712321812321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3513] & 812321712321812321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3514] $8123151321812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3515] $812321812321812321812321812^1 314 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 44, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3501] $61231513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3517] $1(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3518] $21(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3519] $712^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3520] $712^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3521] $712231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3522] $712^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3523] $712231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3524] $171231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [3525] $61231513221(12)1223151322^11(12)122313 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 37, \Delta = -96, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3526] $712^1231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3527] $712231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3528] $18^112321812321812315132^121(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3529] $18^112321812321812315132^121(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3530] $18^112321812321812315132^121(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3531] $1(12)122^1131513218123218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3532] $21(12)122^1131513218123218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3533] $316131513221(12)122315132181232^1181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3534] $221812321812321812315132^121(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3535] $61232181231513221(12)1223151322^1181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3536] $712^1231513221(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3591] & 61232^1 181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[3544] & 7122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3537] & 21(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3538] & 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[3539] & 1(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3540] & 1(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3546] & 712231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3547] & 171231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3548] & 1712315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3549] & 61231513221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 37, \Delta = -96, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3550] & 612315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 37, \Delta = -96, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 37, T \in [37, 37]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3551] $7122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3552] $712231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3553] $181232^1 181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3554] $1^3 8^{321} 12321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3555] $1^3 8^{321} 12^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3556] $18^{21} 123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3557] $18^{21} 12321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3558] $18^{21} 123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3559] $18^{21} 12321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3560] $181232^1 181231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3561] $181232181231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[3562] \quad & \overset{21}{1(12)122^1 315132181232181232181223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3563] \quad & \overset{21}{1(12)122315132181232181232^1 181223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3564] \quad & \overset{3221}{1(12)1223151321812321812^1 321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[3565] \quad & \overset{321}{1 \ 8 \ 12^1 32181232181231513221(12)122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3566] \quad & \overset{321}{1 \ 8 \ 12321812321812315132^1 21(12)122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[3567] \quad & \overset{21}{1 \ 8 \ 123218123218123151322^1 1(12)1222} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[3568] \quad & \overset{21}{1 \ 8 \ 12321812^1 32181231513221(12)1222} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3569] \quad & \overset{21}{1(12)122315132181232^1 18123218122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[3570] \quad & \overset{221}{1(12)12^1 231513218123218123218122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[3571] \quad & \overset{21}{21(12)122315132181232^1 18123218122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[3572] \quad & \overset{21}{21(12)12^1 231513218123218123218122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[3573] \quad & \overset{1}{612315132^1 21(12)122315132181232181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) =
\end{aligned}$$

$(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

$[3574] \ 61231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

$[3575] \ 61232^1 181231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$

$[3576] \ 61232181231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

$[3577] \ 16131513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

$[3578] \ 161315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

$[3579] \ 171232181231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

$[3580] \ 171232^1 181231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$

$[3581] \ 3221812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

$[3582] \ 3221812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

$[3583] \ 221812321812^1 32181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

$[3584] \ 2218123218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) =$
 $(1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

$[3585] \ 61232181232181231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -110, \det = -49, (n_+, n_-, n_0) =$

$$(1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3586] \ 61232^1 181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3587] \ 612315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3588] \ 61231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[3589] \ 61232^1 181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3590] \ 61232181231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3541] \ 21(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3592] \ 61232181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3593] \ 712231513221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[3594] \ 7122315132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3595] \ 41322^1 1(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3596] \ 812^1 315132181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[3597] $812315132181232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[3598] $413221(12)1223151321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[3599] $41322^1 1(12)122315132181232181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[3600] $81232171232181232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3601] $812^1 3217123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3602] $81232171232181232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3603] $812^1 3217123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3604] $812315132181232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[3605] $812^1 3151321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$

[3606] $812^1 321812321812321812321812314 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 44, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3607] $81232181232181232181232^1 1812314 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 44, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3545] $7122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

- [3542] $21(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3543] $712231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3614] $1(12)1223151321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3617] $1^8 12321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3618] $181232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3612] $181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3611] $181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3610] $1^8 1232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3609] $1^8 123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3608] $1^8 12321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3619] $1(12)1223151321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3620] $21(12)1223151321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) =$

$$(1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3621] \quad 32218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3622] \quad 322181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3623] \quad 22181232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3624] \quad 8123151321812321812^1 32181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3625] \quad 812321712321812321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3626] \quad 812321712321812321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3627] \quad 8123151321812321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3628] \quad 812321812321812321812^1 321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3629] \quad 1613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3630] \quad 1613161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3631] \quad 1(12)122^1 3151316131613161316131612 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3632] $21613161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3616] $1^3 8 1232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[3613] $1^3 1(12)12^1 2315132181232181232181223 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[3615] $1^3 8 123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3633] $513221 21(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3634] $513221 21(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3635] $2221 1(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3636] $2221 1(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3637] $221 21(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3638] $221 21(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3639] $71231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3640] $71223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) =$

$$(1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3641] \quad 712231513221^1(12)12^1231513221^{221}(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3642] \quad 71223151322^11(12)12231513221^{221}(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3643] \quad 712231513221^1(12)12^1231513221^{221}(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3644] \quad 1613151322^11(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 36, \Delta = -95, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3645] \quad 16131513221^1(12)12^1231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 36, \Delta = -95, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3646] \quad 17123151322^11(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3648] \quad 61231513221^1(12)12^1231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 37, \Delta = -96, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 37, T \in [36, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3649] \quad 6123151322^11(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 37, \Delta = -96, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 37, T \in [36, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3650] \quad 71223151322^11(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3651] \quad 712231513221^1(12)12^1231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3652] \quad 1^2321123218123218123151322^11(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3653] & \overset{1}{1812321812315132^1 21(12)12231513161} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3654] & \overset{1}{181232181231513221(12)12^1 231513161} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3655] & \overset{2321}{1 \ 8 \ 123218123218123151322^1 1(12)12} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3656] & \overset{1}{61231513221(12)122^1 315132181232181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3657] & \overset{1}{6123151322^1 1(12)122315132181232181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3658] & \overset{1}{612321812315132^1 21(12)122315132181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3659] & \overset{1}{61232181231513221(12)12^1 2315132181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3660] & \overset{1}{16131513221(12)122^1 3151321812321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3661] & \overset{1}{1613151322^1 1(12)1223151321812321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3662] & \overset{1}{316131513221(12)1223151321812^1 32181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3663] & \overset{1}{171232181231513221(12)12^1 2315132171} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3664] $1712321812315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3665] $6131513221(12)122315132181232^1 18123 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 51, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 51, T \in [36, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3666] $6123151322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3667] $61231513221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3668] $61232181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3669] $612321812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3670] $61232181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3671] $612321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3672] $712231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3673] $71223151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3674] $413221(12)122^1 31513221(12)1223151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3675] $16131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

- [3676] $16131613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3677] $1(12)12^1 23151316131613161316131612 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3678] $1613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 45$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 45$, $T \in [36, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3679] $321613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 46$, $T \in [36, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3680] $216131613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3647] $171231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3687] $1812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3700] $81232171232181232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3681] $21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 46$, $T \in [36, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3682] $712315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 48$, $T \in [36, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3683] $316131513221(12)122^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 36$, $\Delta = -95$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 36$, $T \in [36, 36]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3684] $1812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3702] & \overset{1}{8}1232181232181232^1 1812321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3703] & \overset{31}{1}613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 45, \Delta = -112, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 45, T \in [36, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3704] & 3216131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3705] & (12)122^1 315131613161316131613161232 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3686] & \overset{21}{1}812321812321812^1 31513221 \overset{221}{(12)}1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3685] & \overset{321}{1}812321812^1 32181231513221 \overset{21}{(12)}1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3688] & \overset{21}{1}(12)122315132181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3689] & \overset{321}{1}812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3690] & \overset{2321}{1}812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3691] & \overset{21}{1}812321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3692] & \overset{221}{1}(12)122315132^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3693] $21^{21}(12)122315132^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3694] $3221812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3695] $221812321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3696] $61315132^1 21(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 51, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 51, T \in [36, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3697] $413221(12)12231513221(12)122^1 3151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3698] $812315132181232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3699] $81232171232181232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3701] $812315132181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3739] $612321812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3750] $81232^1 17123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3728] $3161315132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[3725] $6123218123151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[3724] $612321812^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3723] $61231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3722] $61231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[3721] $1^2321 1^8 1232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[3720] $1^321 1^8 1232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[3719] $1(12)122^1 3151321812321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3718] $1(12)122315132181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[3706] $21(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3717] $1(12)1223151321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[3716] $1812321812^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3715] $18123218123151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[3714] $1^321 1^8 1232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) =$

$$(1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3713] \quad 1 \quad 8 \quad 1232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3712] \quad 1812321812^1 32181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3711] \quad 1812321812321812315132^1 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3710] \quad 3161315132^1 21(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 36, \Delta = -95, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3727] \quad 16131513221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[3709] \quad 71231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3729] \quad 1712321812^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3741] \quad 712231513221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3730] \quad 17123218123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3731] \quad 322181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3732] \quad 612321812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

- [3733] $612321812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3734] $61231513221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [3735] $61231513221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3736] $6123218123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [3737] $612321812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [3738] $6123218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [3740] $712231513221(12)122315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3708] $71231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3707] $21(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3742] $413221(12)12^1 231513221(12)1223151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [3743] $413221(12)122315132^1 21(12)1223151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3744] $8123151321812^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3745] & 413221(12)122^1 315132181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3746] & 413221(12)122315132181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3747] & 812321712321812^1 3218123218123215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3748] & 81232^1 17123218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3749] & 812321712321812^1 3218123218123151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3754] & (12)12^1 231513161316131613161232 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3753] & 812321812321812^1 321812321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3752] & 81232^1 1812321812321812321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3751] & 8123151321812^1 321812321812321712 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3726] & 16131513221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3755] & 6131513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 51, \Delta = -116, \det = -144, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.64 \rightarrow 36, T_{\max} = 51, T \in [36, 51] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3775] & 413221(12)12^1 2315132181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3779] & \overset{1}{81232181232^1 1812321812321812314} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3758] & \overset{1}{181232181232^1 181231513221(12)1223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3759] & \overset{1}{18123218123218123151322^1 1(12)1223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3760] & \overset{2321}{1 \ 8 \ 12321812^1 32181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3761] & \overset{321}{1 \ 8 \ 12321812321812^1 31513221(12)1} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3762] & \overset{21}{1(12)122315132^1 181232181232181223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3763] & \overset{3221}{1(12)1223151321812^1 321812321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[3764] & \overset{3221}{1(12)12^1 23151321812321812321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3765] & \overset{321}{1 \ 8 \ 12321812321812^1 31513221(12)122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3766] & \overset{2321}{1 \ 8 \ 12321812^1 32181231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3767] & \overset{2321}{316131513221(12)122^1 315132181232181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3768] $3221812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3769] $61232181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -110, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[3770] $6123218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -110, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3771] $613151322^1 1(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 51, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 51, T \in [36, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3772] $413221(12)^1 12231513221(12)^1 12^1 23151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3773] $812315132^1 181232181232181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[3774] $413221(12)1223151321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

[3756] $712315132^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3776] $81232171232^1 18123218123218123215 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3777] $81232171232^1 18123218123218123151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3778] $812315132^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[3757] $316131513221(12)^1 12^1 231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 36, \Delta = -95, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [3780] $^{513221} 21 (12) 12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3781] $^{3221} 71231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3782] $^1 31613151322^1 1(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 36, \Delta = -95, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3783] $^{2321} 1 \ 8 \ 1232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [3784] $^{2321} 1 \ 8 \ 1232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [3785] $^1 31613151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [3786] $6131513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 51, \Delta = -116, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 51, T \in [36, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3787] $^1 413221(12)1223151322^1 1(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3788] $812321712^1 3218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3789] $812321712^1 3218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3790] $^1 812321812^1 321812321812321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3791] $316131513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -225, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.52 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[3792] & \begin{matrix} 2321 \\ 1 \end{matrix} 8 \ 12321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 48, \ \Delta = -115, \ \det = -256, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.48 \rightarrow 36, \ T_{\max} = 48, \ T \in [36, 48] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3793] & \begin{matrix} 2321 \\ 1 \end{matrix} 8 \ 12321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 48, \ \Delta = -115, \ \det = -256, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.48 \rightarrow 36, \ T_{\max} = 48, \ T \in [36, 48] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3794] & 316131513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 48, \ \Delta = -115, \ \det = -256, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.48 \rightarrow 36, \ T_{\max} = 48, \ T \in [36, 48] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[3798] & 6131513221(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 51, \ \Delta = -116, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 51, \ T \in [36, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3795] & \begin{matrix} 1 \\ 18 \end{matrix} 12321812321812^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 46, \ \Delta = -113, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 46, \ T \in [36, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3797] & 612321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 43, \ \Delta = -110, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 43, \ T \in [36, 43] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3796] & \begin{matrix} 3221 \\ 1 \end{matrix} (12)122315132^1 1812321812321812 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 46, \ \Delta = -113, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 46, \ T \in [36, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[3799] & 413221(12)122315132^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 46, \ \Delta = -113, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 46, \ T \in [36, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[3800] & 6131513221(12)122315132^1 18123218123 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 51, \ \Delta = -116, \ \det = -324, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.43 \rightarrow 36, \ T_{\max} = 51, \ T \in [36, 51] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
\\
T = 32 & [4] \begin{matrix} 1 \\ 18 \end{matrix} 1231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 32, \ T_{\max} = 44, \ \Delta = -105, \ \det = 4, \ (n_+, n_-, n_0) = \\
& (1, 32, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 39, \ T_{\max} = 44, \ T \in [39, 44] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[0] & \begin{matrix} 22^1 21 \\ 1 \end{matrix} (12)12231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, \ T_{\max} = 44, \ \Delta = -109, \ \det = 4, \ (n_+, n_-, n_0) = \\
& (1, 32, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 39, \ T_{\max} = 44, \ T \in [39, 44] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1] $^{22^1 1} 21(12)12231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2] $^1 181231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3] $^1 181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [5] $^{2^1 1} 3221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [6] $^{22^1 1} 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [7] $^{22^1 21} 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [8] $^{22^1 1} 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [9] $^{22^1 21} 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [10] $^{22^1 1} 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [11] $^1 812231513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [12] $^1 51316131513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

- [13] $32181231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [14] $232181231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 52, T \in [39, 52]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [15] $2181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [16] $1812231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [17] $4132181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [18] $8123151321812321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [19] $513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.33 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [31] $32181231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [21] $61231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 45, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.33 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [22] $181231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [23] $181231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [24] $21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 9, (n_+, n_-, n_0) =$

$(1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[27] $21(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[28] $1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[29] $812231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $51316131513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[36] $71232181231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 49$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 49$, $T \in [39, 49]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $71231513221(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 49$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 49$, $T \in [39, 49]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[34] $1812231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[33] $2181231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[32] $32^1 181231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [20] $1^{222^1} (12) 12231513221(12) 12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [37] $1^{181} 1231513221(12) 12231513221(12) 12^1 222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [38] $51316131513221(12) 12231513221(12) 12^1 222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [39] $2181231513221(12) 12231513221(12) 12^1 222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [40] $1812231513221(12) 12231513221(12) 12^1 222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [41] $812315132181232181232181232^{21} 181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [42] $812315132181232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [43] $81232181232181232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [44] $81232171232181232181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [49] $1^{181} 12^1 31513221(12) 12231513221(12) 1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [76] $7122^1 31513221(12) 122315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[51] $1812^1 31513221(12)^1 12231513221(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[50] $181231513221(12)^1 122315132^1 21(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[53] $181231513221(12)^1 122315132^1 21(12) 12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[54] $21(12)^{322^1} 12231513221(12)^1 122315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[55] $1(12)^{21} 12231513221(12)^1 1223151321812^1 32 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 52, T \in [38, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[56] $1(12)^{2221} 12231513221(12)^1 122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[57] $1(12)^{2221} 122^1 31513221(12)^1 1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[58] $21(12)^{221} 122^1 31513221(12)^1 1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[59] $21(12)^{221} 12231513221(12)^1 122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[60] $1(12)^{2221} 122^1 31513221(12)^1 1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[61] $1(12)^{2221} 12231513221(12)^1 122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

- [62] $21(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [63] $21(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [64] $6131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [65] $812231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [66] $51316131513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [48] $181231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [47] $612^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 45$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [46] $21(12)122^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [45] $1(12)122^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [67] $232^1 181231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 52$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 52$, $T \in [38, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [68] $2181231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [69] $21812^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$(1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[70] \quad 18122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[71] \quad 1812231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[72] \quad 181232181231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[73] \quad 1812^1 32181231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[74] \quad 181231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[75] \quad 1812^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[99] \quad 812321712321812321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[98] \quad 812321812321812321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[97] \quad 812321812321812321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[96] \quad 8123151321812321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[95] \quad 413221(12)122315132181232181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[94] $4132^1 21(12)1223151321812321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[93] $4132^1 1812321812321812321513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[92] $8123151321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[91] $4132^1 21(12)122315132181232181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[90] $413221(12)122315132181232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[89] $4132^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[88] $4132^1 21(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[87] $712231513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[86] $7122^1 31513221(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[85] $171232181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[84] $1712^1 32181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[83] $171231513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[82] & 1712^1 31513221(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[81] & 1712^1 32181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[80] & 171232181231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[79] & 171232181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[78] & 1712^1 32181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[77] & 712231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[52] & 1812^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[100] & 1(12)12^1 231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 36, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[101] & 21(12)12^1 231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 36, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[102] & 181231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[103] & 181231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [104] $181231513221(12)1223151322^11(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [105] $21(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [106] $1(12)122^131513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [107] $1(12)12^1231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [108] $21(12)122^131513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [109] $21(12)12^1231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [110] $1(12)12^1231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [111] $1(12)12231513221(12)1223151322^1181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [112] $21(12)12231513221(12)1223151322^1181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [113] $21(12)12^1231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [114] $8122^131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [115] $51316131513221(12)1223151322^11(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) =$

$$(1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[116] \quad 32181231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[117] \quad 2181231513221(12)122315132^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[118] \quad 1812^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[119] \quad 1812231513221(12)122315132^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[120] \quad 712^1 231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[121] \quad 171232181231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[122] \quad 712^1 32181232181231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 52, T \in [38, 52]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[123] \quad 712^1 31513221(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[124] \quad 712^1 32181231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[125] \quad 712^1 231513221(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[126] \quad 413221(12)12231513221(12)122315132^1 17 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [170] $1712315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [140] $1(12)122^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 52$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 52$, $T \in [38, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [145] $21(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [147] $1(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [128] $21(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [129] $1(12)122315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [130] $21(12)122315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [131] $61231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 45$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [132] $612315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 45$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [133] $1812315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [134] $181231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [135] $1812315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) =$

(1, 32, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[136] $181231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[137] $1812315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[138] $181231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[139] $1(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[194] $81232171232181232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[193] $812^1 32171232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[192] $6131513221(12)12231513221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[191] $812^1 3218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[190] $81232181232181232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[189] $812^1 3218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[188] $81232181232181232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[187] 812315132181232181232¹18123218122 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[186] 812¹31513218123218123218123218122 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[185] 41322¹1(12)1223151321812321812321812 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[184] 413221(12)1223151321812321812¹321812 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[183] 41321812321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[182] 41321812¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[181] 812315132181232181232¹18123218²¹1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[180] 812¹31513218123218123218123218²¹1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[179] 41322¹1(12)12231513218123218123218¹1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[178] 413221(12)1223151321812321812¹3218¹1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[177] 41321812321812321812315132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[176] 41321812¹32181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[175] & 41322^1 1(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[174] & 7122315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[173] & 712231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[172] & 171232181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[171] & 171232^1 181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[169] & 171231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[168] & 171232181231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[167] & 171232^1 181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[166] & 171232181232181231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[165] & 171232^1 181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[164] & 7122315132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[163] & 712231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[162] & \overset{1}{181231513221(12)1223151321812^1 32171} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[161] & \overset{1}{1812315132^1 21(12)122315132181232171} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[160] & \overset{1}{181232181231513221(12)122^1 315132171} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[159] & \overset{1}{181232^1 181231513221(12)122315132171} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[158] & \overset{1}{18122315132^1 21(12)12231513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[157] & \overset{1}{1812231513221(12)122^1 31513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[156] & \overset{1}{21812315132^1 21(12)12231513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[155] & \overset{1}{2181231513221(12)122^1 31513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[154] & \overset{1}{232181231513221(12)122315132^1 21(12)122} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 52, T \in [38, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[153] & \overset{1}{32181231513221(12)1223151322^1 1(12)1222} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[152] & \overset{1}{51316131513221(12)122^1 31513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[151] & 812^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[150] & 6131513221(12)12231513221(12)151322^1 1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[149] & 21^{221}(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[148] & 21^{221}(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[146] & 1^{2221}(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[144] & 21^{221}(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[143] & 21^{21}(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[142] & 1^{2221}(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[141] & 1^{2221}(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[127] & 21^{5132^1 21}(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[195] & 1^{21}(12)12^1 231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.94 \rightarrow 37, T_{\max} = 52, T \in [37, 52]
\end{aligned}$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[196] $232181231513221 \overset{1}{(12)}1223151322^11(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[197] $71232181232181231513221(12)122^131513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[198] $413218123218123218123151322^11(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[199] $4132181232^1181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[200] $8123151321812321812^132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[201] $4132181232^1181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[202] $413218123218123218123151322^11(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[203] $8123151321812321812^13218123218122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[204] $812321812321812321812^13218123215 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[205] $812321812321812321812^13218123151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[206] $812321712321812321812^132181232161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[212] $21(12)1223151322^11(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 81, (n_+, n_-, n_0) =$

$$(1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[228] \quad {}^{221}_1 21(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[208] \quad {}^{513221}_1 21(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[209] \quad {}^{513221}_1 21(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[211] \quad {}^{2221}_1 1(12)1223151322^1 1(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[213] \quad {}^{221}_1 21(12)12231513221(12)122^1 31513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[214] \quad {}^{513221}_1 61231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[215] \quad {}^{513221}_1 6123151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[216] \quad {}^1_{18} 1231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[217] \quad {}^1_{18} 123151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[218] \quad {}^1_{18} 1231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[219] \quad {}^1_{18} 123151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[220] & \begin{matrix} 1 & 1 \\ 18123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[221] & \begin{matrix} 1 & 1 \\ 181231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[222] & \begin{matrix} 3221 & 1 \\ 21(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[210] & \begin{matrix} 2221 & 1 \\ 1(12)12231513221(12)122^1 31513161315 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[223] & \begin{matrix} 221 & 1 \\ 1(12)12231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[224] & \begin{matrix} 2221 & 1 \\ 1(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[225] & \begin{matrix} 2221 & 1 \\ 1(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[226] & \begin{matrix} 21 & 1 \\ 21(12)12231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[227] & \begin{matrix} 221 & 1 \\ 21(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[266] & \begin{matrix} 1 \\ 6131513221(12)122315132^1 21(12)12231414 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[265] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)12^1 231414 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[264] & 413221(12)122^1 31513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[263] & 71223151322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[262] & 712231513221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[261] & 71232181231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[260] & 71232^1 181231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[259] & 71231513221(12)122315132181232^1 18123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[258] & 71232^1 181232181231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[257] & 171232181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[256] & 1712321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[255] & 17123151322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[254] & 171231513221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[253] & 171232181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[252] & 1712321812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[251] & 712231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[250] & 71223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[249] & 18123151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[248] & 181231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[247] & 181232181231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[246] & 1812321812315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[245] & 71223151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 37, \Delta = -94, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[244] & 712231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 37, \Delta = -94, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[243] & 1812231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[242] & 181223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [241] $218123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [240] $2181231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [239] $321812^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [238] $171231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 37$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 37$, $T \in [37, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [237] $17123151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 37$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 37$, $T \in [37, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [236] $51316131513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [235] $513161315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [234] $812231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [233] $6131513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [232] $21(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [231] $21(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [230] $1(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) =$

$(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 2221 & 1 \\ 1(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 $[207] \quad 4131613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 81, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 $[267] \quad 41316131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 51322^1 1 \\ 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 51322^1 1 \\ 61231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 3221 & 1 \\ 21(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 221 & 1 \\ 1(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 21 & 1 \\ 21(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[273] \quad 61315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[274] \quad 8122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[275] \quad 32181231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[276] & 712315132^1 21(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[277] & 413221(12)12231513221(12)122^1 31513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[278] & 41321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[279] & 812315132181232^1 181232181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[280] & 41321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[281] & 812315132181232^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[282] & 81232181232181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[283] & 81232181232181232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[284] & 61315132^1 21(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[285] & 81232171232181232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[291] & 61231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 45, T \in [37, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[294] $1(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[336] $812321712321812^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[337] $81232^1 171232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

[288] $1(12)12231513221(12)12^1 231513161315 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[287] $21(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[286] $21(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[332] $81232^1 18123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[331] $81232^1 18123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[330] $812321812321812^1 3218123218123215 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[329] $8123151321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[328] $413221(12)122315132181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[327] $413221(12)122^1 3151321812321812321812 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[326] & 4132181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[325] & 8123151321812^1 321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[324] & 413221(12)122^1 3151321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[323] & 413221(12)122315132181232^1 181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[322] & 4132181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[321] & 413221(12)12^1 231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[320] & 413221(12)122315132^1 21(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[319] & 712231513221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[318] & 712231513221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[317] & 71232181231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[316] & 71232181232181231513221(12)12^1 231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[315] & 1712321812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[314] \quad 17123218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[290] \quad 61231513221(12)12^{513221} 231513221 (12) 12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[333] \quad 812321812321812^1 3218123218123151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[289] \quad 21^{221}(12)12231513221(12)12^1 231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[313] \quad 171231513221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[312] \quad 171231513221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[311] \quad 17123218123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[310] \quad 1712321812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[309] \quad 1712321812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[308] \quad 1712321812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[307] \quad 712231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[306] \quad 712231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) =$$

$(1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[305] $\overset{1}{181231513221}(12)122315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[304] $\overset{1}{181231513221}(12)12^1 2315132181232171 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[303] $\overset{1}{18123218123151322^1 1}(12)122315132171 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[302] $\overset{1}{1812321812^1 31513221}(12)122315132171 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[301] $\overset{1}{321812315132^1 21}(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[300] $\overset{1}{5131613151322^1 1}(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[299] $\overset{1}{812231513221}(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[298] $\overset{1}{812231513221}(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[297] $\overset{1}{6131513221}(12)1223151322^1 1 \overset{1513221}{(12) 1} + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[296] $\overset{1}{6131513221}(12)122^1 31513221 \overset{1513221}{(12) 1} + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.69 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[295] $\overset{21}{21}(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[334] & 6131513221(12)122^1 31513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[292] & 21(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[293] & 21(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[335] & 6131513221(12)1223151322^1 1(12)12231414 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[338] & 71231513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 144, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.64 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[361] & 812315132^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[349] & 17123218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[359] & 413221(12)1223151321812^1 321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[358] & 41321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[357] & 812315132^1 1812321812321812321 81 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[356] & 413221(12)12^1 2315132181232181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [355] $413221(12)1223151321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [354] $41321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [353] $413221(12)12231513221(12)12^1 231513217 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [363] $81232181232^1 18123218123218123151 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [352] $712321812315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 49$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 49$, $T \in [37, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [351] $7123151322^1 1(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 49$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 49$, $T \in [37, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [350] $171232181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [348] $2321812^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 52$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 52$, $T \in [37, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [365] $81232171232^1 181232181232181232161 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [364] $613151322^1 1(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [346] $32181231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [345] $81223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[344] & 613151322^1 1(12)12231513221^{1513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[343] & 21^{21} (12)1223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[342] & 1(12)12231513221^{21} (12)122315132^1 181232 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[341] & 1(12)122315132^{21} 21(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[340] & 1(12)1223151322^{221} 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[339] & 21^{3221} (12)12231513221^{12} 12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[362] & 81232181232^1 18123218123218123215 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[347] & 232181231513221^{12} 122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[360] & 413221(12)12^1 23151321812321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[366] & 21^{513221} (12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 196, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[367] & 61231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 196, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.55 \rightarrow 37, T_{\max} = 45, T \in [37, 45]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[368] & \begin{matrix} 3221 & 1 \\ 21(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[369] & \begin{matrix} 221 & 1 \\ 1(12)12231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[370] & \begin{matrix} 21 & 1 \\ 21(12)12231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[371] & \begin{matrix} 1 & 1513221 \\ 6131513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -110, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[372] & \begin{matrix} 1 & 3221 \\ 812231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[373] & \begin{matrix} 1 \\ 3218123151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[374] & \begin{matrix} 1 \\ 71231513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 49, \Delta = -115, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 49, T \in [37, 49] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[375] & \begin{matrix} 1 \\ 71232181231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 49, \Delta = -115, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 49, T \in [37, 49] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[376] & \begin{matrix} 1 & 1 \\ 413221(12)1223151322^1 1(12)12231513217 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[377] & \begin{matrix} 1 \\ 812321812^1 3218123218123218123215 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -108, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[378] & \begin{matrix} 1 \\ 812321812^1 3218123218123218123151 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -106, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [379] $6131513221(12)12^1 231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [380] $812321712^1 32181232181232181232161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [383] $712321812^1 32181231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 225, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.52 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [381] $1(12)12231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 225, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.52 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [384] $712321812^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 49, \Delta = -115, \det = 225, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.52 \rightarrow 37, T_{\max} = 49, T \in [37, 49]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [382] $2321812315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 225, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.52 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [386] $7123218123151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 49, \Delta = -115, \det = 256, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.48 \rightarrow 37, T_{\max} = 49, T \in [37, 49]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [385] $712321812321812315132^1 21(12)12231513 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 256, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.48 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [389] $1712321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -108, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.46 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [391] $413221(12)122315132^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.46 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [392] $413221(12)122315132^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.46 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [388] $232181231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.46 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[387] & \begin{matrix} 21 & 1 \\ 1(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.46 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[390] & 71231513221(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.46 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[394] & 71231513221(12)122315132^1 18123218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 324, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.43 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[393] & 71232181232^1 181231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 324, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.43 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[397] & 7123218123218123151322^1 1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 361, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.41 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[395] & \begin{matrix} 21 & 1 \\ 1(12)12231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 361, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.41 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[396] & \begin{matrix} 1 \\ 23218123151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 361, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.41 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[398] & 712321812321812^1 31513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.33 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
T = 33 & [399] \begin{matrix} 22^1 21 \\ 1(12)12231513221(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40, T_{\max} = 45, T \in [40, 45] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[400] & \begin{matrix} 22^1 1 \\ 21(12)12231513221(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40, T_{\max} = 45, T \in [40, 45] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[401] & \begin{matrix} 22^1 1 \\ 1(12)12231513221(12)1223151316131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40, T_{\max} = 49, T \in [40, 49] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [402] $61316131513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39$, $T_{\max} = 50$, $T \in [39, 50]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [404] $3161316131513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [405] $51232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [406] $15132181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [407] $(12)1223151321812321812321812315132^1 2 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 53$, $T \in [40, 53]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [403] $161316131513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [410] $1(12)12231513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [418] $812231513221(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [417] $32^1 181232181231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [416] $181232181231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [415] $32^1 181232181232181231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 53$, $T \in [40, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [414] $5132181232181232181231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 53$, $T \in [40, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

- [413] 3161316131513221(12)12231513221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [412] 161316131513221(12)12231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [408] 513221²¹(12)12231513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.33 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [409] 712231513221(12)12231513221⁵¹³²²¹(12)12¹ + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.33 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [411] 61316131513221(12)12231513221³²¹(12)12 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [420] 1²¹(12)122315132181232181232181232¹15 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [421] 161316131513221(12)12231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [422] 512¹32181232181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [423] 81232181232181232181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [424] 81232171232181232181232181232¹171 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [425] 81232171232181232181232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [419] 51316131513221(12)12231513221(12)12¹2313 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 39$, $\Delta = -97$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 39$, $T \in [39, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[427] $7122^1 31513221(12)12231513221^{513221}(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[428] $71231513221^1(12)12231513221^{15132^1 21}(12)^1 1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[429] $51316131513221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 39, \Delta = -97, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[430] $181232181231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[431] $1812^1 32181231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[432] $1(12)122^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[433] $21(12)122^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[434] $181232181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[435] $1812^1 32181232181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[436] $1(12)1223151321812321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[437] $1(12)1223151321812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [438] $1812^1 132181231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [439] $181232181231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [440] $181231513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [441] $1812^1 131513221(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [442] $1812^1 32181231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [443] $181232181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [444] $61316131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [445] $161316131513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [446] $5132181232181232181231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [447] $2181232181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [448] $21812^1 32181232181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [449] $51232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[450] & 15132^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& 513221 \\
[426] & 21(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[452] & 21812^1 32181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[453] & 21812^1 32181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[454] & 2181232181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[455] & 81231513221(12)1223151321812321812^1 32 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 53, T \in [39, 53] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[456] & 81232181231513221(12)1223151321812^1 32 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 53, T \in [39, 53] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[457] & 4132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[458] & 4132^1 21(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[459] & 812321812321812321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[460] & 812321712321812321812321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[461] & 812321712321812321812321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[451] \quad 2181232181231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[462] \quad 712^1 231513221(12)12231513221 \overset{513221}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[463] \quad 712^1 31513221(12)12231513221 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[464] \quad 51316131513221(12)1223151322^1 1(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[465] \quad 1(12)12^1 231513221(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[466] \quad 21(12)12^1 231513221(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[467] \quad 1(12)122^1 31513221(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[468] \quad 21(12)122^1 31513221(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[469] \quad 181232181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[470] \quad 161316131513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[471] \quad 3161316131513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

- [472] $1812^1 32181232181231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [473] $1812^1 32181231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [474] $32181232181231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [475] $2181232181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [476] $8122^1 31513221(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [477] $413221^1(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [478] $413221^1(12)12231513221(12)122315132^1 218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [479] $712^1 31513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [485] $51316131513221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 39, \Delta = -97, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [516] $2181232^1 181231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [483] $7122315132^1 21(12)12231513221^1(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [484] $71231513221^1(12)12231513221^1(12)1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[481] & \begin{matrix} 513221 \\ 21(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[486] & \begin{matrix} 1 \\ 181232^1 181231513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[487] & \begin{matrix} 1 \\ 181232181231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[488] & \begin{matrix} 2221 \\ 1(12)122315132^1 21(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[489] & \begin{matrix} 221 \\ 21(12)122315132^1 21(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[490] & \begin{matrix} 221 \\ 1(12)12^1 231513221(12)1223151316131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 49, T \in [39, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[491] & \begin{matrix} 21 \\ 21(12)12^1 231513221(12)1223151316131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 49, T \in [39, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[492] & \begin{matrix} 1 \\ 181232181232181231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[493] & \begin{matrix} 1 \\ 181232^1 181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[494] & \begin{matrix} 21 \\ 1(12)122315132181232181232^1 18123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[495] & \begin{matrix} 21 \\ 1(12)122^1 31513218123218123218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[496] $1(12)122^1 31513218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[497] $1(12)122315132181232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[498] $181232^1 181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$

[499] $181232181231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[500] $1812315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[502] $181232181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[503] $181232^1 181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$

[504] $161316131513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[505] $3161316131513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[506] $5132^1 181232181232181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[507] $32181232181232181231513221(12)122^1 315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[508] $2181232181232181231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[509] 2181232¹181232181231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[480] 21^{5132¹21}(12)12231513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[510] 512321812¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[511] 512321812321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[512] 151321812321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[513] 151321812¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[514] 2181232181231513221(12)122¹3151321812 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[482] 712231513221(12)12231513221^{5132¹21}(12)12 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[515] 2181232¹181231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[531] 812¹321712321812321812321812321712 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[530] 81232171232181232181232¹1812321712 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[529] 81232171232181232181232¹181232171¹ + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[528] $812^1 32171232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[527] $81232181232181232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[526] $812^1 32181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[525] $71231513221(12)12231513221(12)122^1 31414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[524] $41322^1 1(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[523] $413221(12)12231513221(12)1223151322^1 18 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[522] $41322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[521] $812^1 231513221(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[520] $812^1 32181231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[519] $812^1 31513221(12)122315132181232181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[518] $812^1 32181232181231513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[517] $2181232181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) =$

$$(1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[501] \quad 1^{12}181231513221(12)1223151321812^1321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[532] \quad 1^{12}(12)122^1315132181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[534] \quad 1^{21}(12)1223151321812321812^13218123215 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[535] \quad 1^{21}(12)12^1231513218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[536] \quad 1^{21}(12)1223151321812321812^13218123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[537] \quad 1^{21}(12)12^1231513218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[538] \quad 181232181232181231513221(12)122^131513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[539] \quad 3151321812321812321812315132^121(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[540] \quad 315132^1181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[541] \quad 5123218123218123218123151322^11(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[542] \quad 51232181232^1181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

- [543] $1513218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [544] $15132181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [545] $812321812321812321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [546] $812321712321812321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [547] $812321712321812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [533] $1(12)1223151321812321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [584] $321812^1 32181231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [582] $181232^1 181231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [581] $31513218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [585] $2181232181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [586] $21812321812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [587] $2181232181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [588] $21812321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [589] $812231513221(12)122315132181232^1 18123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [590] $413221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [591] $413221(12)122^1 31513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [592] $71231513221(12)12231513221(12)12^1 231414 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [593] $71231513221(12)122315132^1 21(12)12231414 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [594] $15131613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [575] $61316131513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [576] $1613161315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [577] $161316131513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [583] $181232181231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [578] $181232^1 181232181231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[579] $23151321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[580] $321812^1 32181232181231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[573] $181232181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[572] $181231513221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[571] $18123151322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[570] $181232181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[569] $1812321812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[568] $1(12)12^1 2315132181232181232181231513 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[567] $21(12)1223151322^1 1(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[566] $21(12)12231513221(12)122^1 315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[565] $1(12)1223151322^1 1(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[564] $1(12)12231513221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[563] $181232181231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[562] $1812321812315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[561] $181223151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[560] $1812231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[559] $218123151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[558] $2181231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[557] $321812^1 31513221(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -94, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[556] $51316131513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 39, T \in [38, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[555] $513161315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 39, T \in [38, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[554] $71231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[574] $1812321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[552] & \overset{1}{181231513221}(12)12^{\overset{1}{1}231513221(12)122313} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[551] & 712231513221(12)122^{\overset{513221}{1}31513221}(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[550] & 71223151322^{\overset{513221}{1}1}(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[549] & \overset{513221}{21}(12)122315132^{\overset{1}{1}21}(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[548] & \overset{513221}{21}(12)12231513221(12)12^{\overset{1}{1}2315132171} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[553] & \overset{1}{18123151322^{\overset{1}{1}1}}(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[620] & 81232171232181232^{\overset{1}{1}1812321812321712} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[596] & 712231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[597] & 712315132^{\overset{1}{1}21}(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[598] & \overset{1}{32181231513221}(12)122^{\overset{1}{1}31513221}(12)12231 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -94, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[599] & \overset{221}{1}(12)122315132^{\overset{1}{1}21}(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[600] & 21^{21}(12)122315132^1 21(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[601] & 1^{12}(12)122315132181232181232^1 181231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[595] & 51322^1 1^{21} 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[602] & 21^{21} 1(12)122315132181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[603] & 21^{21} 1(12)122315132181232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[604] & 3161316131513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[605] & 231513218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[606] & 32181232181232181231513221(12)12^1 2315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[607] & 3151321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[608] & 512321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[609] & 151321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[610] & 32181232181231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[611] & (12)122^1 31513218123218123218123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[612] & 8122315132^1 21(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[613] & 413221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[614] & 413221(12)12231513221(12)122^1 315132218 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[615] & 712315132^1 21(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[616] & 81232181232181232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[617] & 81232171232181232^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[618] & 151316131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[619] & (12)122^1 31513161316131613161316131513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[629] & 18123218123151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[632] & 21(12)12231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [657] $21812321812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [658] $21812321812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [659] $218123218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [660] $81232^1 181232181231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [661] $81232181232181231513221(12)122^1 315132 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [628] $321812315132^1 21(12)12231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 38, \Delta = -94, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [662] $(12)12^1 231513218123218123218123151322 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [663] $81232^1 181231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [664] $413221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [665] $413221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [666] $413221(12)12^1 231513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [667] $413221(12)122315132^1 21(12)122315132218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[668] & 71231513221(12)1223151322^1 1(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[669] & 71231513221(12)122^1 31513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[670] & 81232^1 181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[671] & 812321812321812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[672] & 812321712321812^1 32181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[673] & 81232^1 171232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[674] & (12)12^1 23151316131613161316131513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[675] & 812321712321812^1 321812321812321712 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[676] & 81232^1 1712321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[634] & 1812321812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[631] & 1(12)12231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[636] \quad & \overset{21}{1(12)1223151321812^1 3218123218123215} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[630] \quad & \overset{1}{1812321812^1 31513221(12)122315132181} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[633] \quad & \overset{1}{1(12)1223151321812321812^1 32181231513} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[637] \quad & \overset{21}{1(12)1223151321812^1 3218123218123151} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[638] \quad & \overset{1}{1812321812^1 31513221(12)1223151321812} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[639] \quad & \overset{1}{18123218123151322^1 1(12)1223151321812} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[640] \quad & \overset{1}{181231513221(12)12^1 23151321812321812} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[641] \quad & \overset{1}{181231513221(12)122315132^1 1812321812} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[642] \quad & \overset{1}{1812321812^1 31513221(12)1223151322181} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[643] \quad & \overset{1}{18123218123151322^1 1(12)1223151322181} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[644] \quad & \overset{3221}{61316131513221(12)1223151322^1 1(12)12} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[645] \quad & \overset{3221}{61316131513221(12)122^1 31513221(12)12} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -121, (n_+, n_-, n_0) =
\end{aligned}$$

- (1, 33, 0), $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [646] 16131613151322¹1(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [647] 181232181232181231513221(12)12¹231513 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [648] 2315132¹181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [649] 51321812¹32181232181231513221(12)1223 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [650] 315132181232¹181232181231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 23 & 3 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [651] 21812321812321812315132¹21(12)1223151 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [652] 21812321812¹32181231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [653] 51232181232181232¹181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [656] 218123218123151322¹1(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [654] 15132181232181232¹181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [655] 181232181231513221(12)122¹31513218123 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [621] 21 (12) 122¹31513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) =$

$$\begin{aligned}
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& \text{[622]} \quad 21 \text{ (12) } 1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& \text{[623]} \quad 712231513221(12)122315132^1 21 \text{ (12) } 12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
& \text{[624]} \quad 712231513221(12)12^1 231513221 \text{ (12) } 12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& \text{[625]} \quad 71231513221(12)122^1 31513221 \text{ (12) } 1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& \text{[626]} \quad 71231513221(12)1223151322^1 1 \text{ (12) } 1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& \text{[627]} \quad 5131613151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 39, T \in [38, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& \text{[635]} \quad 1812321812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& \text{[679]} \quad 812231513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -144, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.64 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \text{[677]} \quad 32181232^1 181232181231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -144, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.64 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 59 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& \text{[678]} \quad 32181232^1 181231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -144, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.64 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& \text{[710]} \quad 81232171232^1 1812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[699] $1812321812315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[705] $413221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[704] $81223151322^1 1(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[703] $81232181231513221(12)122315132^1 181232 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[702] $812315132^1 21(12)122315132181232181232 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[701] $81231513221(12)122315132181232^1 181232 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[700] $32181232181231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[698] $151321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[697] $512321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[696] $2181232181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[695] $218123218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[694] $3151321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

- [693] $51321812321812321812315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [692] $5132181232^1 181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [691] $23151321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [690] $3161316131513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [689] $613161315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [688] $1(12)122315132^1 18123218123218123151 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [687] $1(12)122315132^1 18123218123218123215 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [686] $181232181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [685] $18123218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [684] $1(12)122315132181232^1 181232181231513 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [683] $21(12)1223151322^1 1(12)1223151316131613 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [682] $1(12)1223151322^1 1(12)1223151316131613 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[681] & 32181231513221(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -94, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[680] & 7123151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[707] & 7123151322^1 1(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[708] & 81232181232^1 181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[709] & 81232171232^1 181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[706] & 413221(12)12231513221(12)12^1 2315132218 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[728] & 71231513221(12)12^1 231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[719] & 31613161315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[720] & 2315132181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[721] & 315132181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[722] & 181232181231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [723] $321812321812315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [724] $(12)1223151321812321812321812^1 3151322 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [725] $812231513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [726] $413221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [727] $413221(12)1223151322^1 1(12)122315132218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [718] $61316131513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [729] $812321812^1 32181232181232181232161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [730] $812321712^1 32181232181232181232171 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [731] $812321712^1 321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [713] $71231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [715] $1(12)12231513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [711] $21(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[712] \quad 712231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[716] \quad 21(12)12231513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[717] \quad 1(12)1223151321812^1 32181232181231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[714] \quad 3218123151322^1 1(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -94, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[732] \quad 1812321812^1 32181231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[736] \quad 81232181231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[735] \quad 81232181232181231513221(12)12^1 2315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[734] \quad 1812321812^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[733] \quad 513218123218123218123151322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[737] \quad 1(12)122315132^1 181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.48 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[740] \quad 23151321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.48 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[739] \quad 1812321812321812315132^1 21(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -256, (n_+, n_-, n_0) =$$

$(1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[741] \ 3151321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

$[742] \ 18123218123151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[743] \ (12)122315132181232181232^1 18123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[738] \ 6131613151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[744] \ 1(12)12231513221(12)12^1 23151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

$[746] \ 1812321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[747] \ 316131613151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

$[748] \ 2315132181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

$[749] \ 51321812321812^1 32181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[750] \ 321812321812321812315132^1 21(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[751] \ 21812321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[752] \ 3218123218123151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -289, (n_+, n_-, n_0) =$

$(1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

[753] $(12)1223151321812321812^1 3218123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

[754] $8123151322^1 (12)122315132181232181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[755] $81231513221(12)1223151321812^1 32181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[756] $812321812315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[757] $812231513221(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

21
[745] $21(12)12231513221(12)12^1 23151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[759] $321812321812^1 32181231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -324, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.43 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[761] $812231513221(12)122315132^1 18123218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -324, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.43 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[760] $321812321812^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -324, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.43 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[758] $181232181232^1 181231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -324, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.43 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

[766] $(12)122315132181232^1 18123218123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[767] $81231513221(12)122^1 315132181232181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) =$

$$(1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[768] \ 81232181231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[769] \ 812321812^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[765] \ 812321812^1 32181231513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[764] \ 5132181232181232^1 181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[763] \ 23151321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[762] \ 18123218123218123151322^1 1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[770] \ 3218123218123218123151322^1 1(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -400, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.39 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 139 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[771] \ (12)1223151321812^1 3218123218123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -400, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.39 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[773] \ 812321812321812315132^1 21(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -441, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.37 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[772] \ 32181232181232^1 181231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -441, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.37 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 167 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[774] \ 8123218123151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -441, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.37 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[775] \ (12)122315132^1 18123218123218123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -484, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.36 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[776] \ 51321812321812321812^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -529, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.34 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[777] \ 81232181232^1 181231513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -529, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.34 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 413 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[778] \ 81231513221(12)12^1 2315132181232181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -529, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.34 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$$

$$[779] \ 1812321812321812^1 31513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -576, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.33 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[780] \ 8123218123218123151322^1 1(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -625, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.31 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 474 & 42 \\ 2 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[781] \ 81231513221(12)122315132^1 181232181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -625, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.31 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[782] \ 321812321812321812^1 31513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -729, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.29 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 269 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[783] \ 812321812321812^1 31513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -961, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.25 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 743 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$$

$$T = 34 \quad [784] \ 1^{22^1 21}(12)12231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41, T_{\max} = 46, T \in [41, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[785] \ 21^{22^1 1}(12)12231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41, T_{\max} = 46, T \in [41, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[786] \ 5132181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41, T_{\max} = 46, T \in [41, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[787] \ 161232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 4, (n_+, n_-, n_0) =$$

(1, 34, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 41$, $T_{\max} = 46$, $T \in [41, 46]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[788] 812321712321812321812321812321812¹2 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 4$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41$, $T_{\max} = 46$, $T \in [41, 46]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[789] (12)12231513161316131613161316131612¹3 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 40.22 \rightarrow 41$, $T_{\max} = 48$, $T \in [41, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[792] 513221²1¹(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.33 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[793] 181231513221(12)12231513221²1¹(12)122315 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.33 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[794] 5132181231513221(12)12231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 40.22 \rightarrow 41$, $T_{\max} = 46$, $T \in [41, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[795] 61232181232181232181231513221(12)122¹3 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 51$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 40.22 \rightarrow 41$, $T_{\max} = 51$, $T \in [41, 51]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[790] 1²²²1¹(12)12231513221(12)1223151321812315 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 40.22 \rightarrow 41$, $T_{\max} = 46$, $T \in [41, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[791] 513221²1¹(12)12231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.33 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[796] 5132181231513221(12)12231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.75 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[797] 1²¹(12)122315132181232181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.75 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[798] 1612¹32181232181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.75 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[799] $81232181232181232181232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.75 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[800] $81232181232181232181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.75 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[801] $81232171232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.75 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[802] $81232171232181232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.75 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[809] $1812^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[811] $1(12)1223151321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[812] $161232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[813] $61232181232181232181231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 51$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 51, T \in [40, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[814] $612^1 32181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 51$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 51, T \in [40, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[815] $4132^1 21(12)12231513221(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 50$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[816] $812321812321812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[817] $812321812321812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) =$

$$(1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[818] \quad 812321712321812321812321812^1 321 81 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[819] \quad 812321712321812321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[810] \quad 5132181231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[808] \quad 812231513221(12)12231513221 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[807] \quad 1 \quad (12) \quad 12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[806] \quad 21 \quad (12) \quad 12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[805] \quad 21(12)122^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[804] \quad 21 \quad (12) \quad 12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[803] \quad 1(12)122^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[824] \quad 8122^1 31513221(12)12231513221 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[820] \quad 1(12)12^1 231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [821] \quad 21(12)12^1 231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [827] \quad 8122^1 31513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 48, T \in [40, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [823] \quad 1(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [825] \quad 5132181231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [826] \quad 41413221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 48, T \in [40, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [822] \quad 21(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [835] \quad 21(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [836] \quad 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
& [837] \quad 812^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 54, \Delta = -116, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 54, T \in [40, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [838] \quad 812231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
& [839] \quad 812^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[840] & \begin{matrix} 1 & & 21 \\ 181231513221(12)12231513221(12)122^1 315 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[841] & \begin{matrix} 1 & & 21 \\ 1812315132^1 21(12)12231513221(12)122315 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[842] & \begin{matrix} 5132181231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[843] & \begin{matrix} 5132^1 181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[844] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 48, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 48, T \in [40, 48] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[845] & \begin{matrix} 21 \\ 1(12)122315132181232181232^1 181232161 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[846] & \begin{matrix} 21 \\ 1(12)122^1 315132181232181232181232161 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[847] & \begin{matrix} 1 \\ 1(12)122315132181232181232^1 1613 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 50, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[848] & \begin{matrix} 1612321812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[834] & \begin{matrix} 5132^1 21 \\ 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[850] & \begin{matrix} 31612^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 50, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 39.06 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\ \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[851] & \begin{matrix} 61232^1 181232181232181231513221(12)1223 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 51, \Delta = -115, \det = 49, (n_+, n_-, n_0) = \end{matrix}
\end{aligned}$$

$$(1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 51, T \in [40, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[852] \quad 414132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[853] \quad 812^1 231513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[854] \quad 812231513221(12)12231513221(12)122^1 31414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[855] \quad 41322^1 1(12)12231513221(12)12231513161316 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[856] \quad 81232181232181232181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[857] \quad 812^1 321812321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[858] \quad 81232181232181232181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[859] \quad 812^1 32181232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[860] \quad 812^1 32171232181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[861] \quad 81232171232181232181232^1 181232181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[862] \quad 812^1 3217123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[863] & 81232171232181232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& 513221 \\
[830] & 21(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& 2221 \\
[828] & 1(12)12231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[833] & 21(12)122315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[832] & 21(12)12231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 5132^1 21 \\
[831] & 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
& 2221 \\
[829] & 1(12)122315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[849] & 1612321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[870] & 16123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 64, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[869] & 161232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 64, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& 1 \\
[868] & 1(12)122^1 3151321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 64, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.94 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [867] $1(12)1223151321812321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [866] $1(12)12^1 2315132181232181232181232161 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [865] $1(12)1223151321812321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [864] $81231513221(12)12231513221 \begin{smallmatrix} 1 \\ 315132^1 21 \end{smallmatrix} (12) 1 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 54, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 54, T \in [39, 54]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [876] $812321712321812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [875] $812321712321812321812^1 321812321 \begin{smallmatrix} 21 \\ 81 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [874] $812321812321812321812^1 321812321 \begin{smallmatrix} 1 \\ 171 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [873] $812321812321812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [872] $3161232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [871] $31612321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [885] $21 \begin{smallmatrix} 513221 \\ (12) \end{smallmatrix} 12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [878] $21613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[891] & 51321812315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[893] & 6131513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[903] & 413221(12)122^1 31513221(12)12231513161316 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[902] & 812231513221(12)12231513221(12)12^1 231414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[901] & 812231513221(12)122315132^1 21(12)12231414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[900] & 4141322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[899] & 41413221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[898] & 316123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[897] & 1(12)12^1 23151321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[896] & 1613161315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 39, \Delta = -94, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[895] & 161316131513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 39, \Delta = -94, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [894] $6131513221(12)^1 122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [892] $5132181231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [890] $181231513221(12)122^1 31513221(12)122315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [889] $18123151322^1 1(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [888] $812231513221(12)122315132^1 21(12)^1 1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [887] $1^{1513221}(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [886] $21^{513221}(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [884] $21^{221}(12)1223151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [883] $21^{221}(12)12231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [882] $21^{513221}(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [881] $21^{513221}(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [880] $1^{2221}(12)12231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) =$

$(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 2221 \\ [879] \end{matrix} 1(12)1223151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 81, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 1 \\ [877] \end{matrix} 16131613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 81, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 $[907] (12)122^1 315131613161316131613161316123 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 1 \\ [904] \end{matrix} 161316131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 $[906] 2161316131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 51322^1 1 \\ [908] \end{matrix} 21 (12) 12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 51322^1 1 \\ [909] \end{matrix} 21 (12) 12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 25 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 1513221 \\ [910] \end{matrix} 1 (12) 12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[911] 8122315132^1 21(12)12231513221 \begin{matrix} 1513221 \\ (12) \end{matrix} 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[912] 181231513221(12)12231513221(12)12^1 2315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 1 \\ [913] \end{matrix} 61315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$

$$(1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 47, T \in [39, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[946] \ 16131613151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 39, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[945] \ 6131513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[944] \ 6131513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[943] \ 51321812^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[942] \ 513218123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[941] \ 181231513221(12)12^1 231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[940] \ 181231513221(12)122315132^1 21(12)122315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[939] \ 812231513221(12)122^1 31513221^{1513221} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[938] \ 812231513221(12)1223151322^1 1(12) \ 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[937] \ 81231513221(12)12231513221^{3151322^1} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 54, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 54, T \in [39, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[947] \ 1(12)1223151321812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[935] & 1(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[934] & 21(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[933] & 21(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& 221 \\
[932] & 21(12)12231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
& 221 \\
[931] & 21(12)12231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[930] & 21(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[929] & 21(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& 2221 \\
[928] & 1(12)12231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& 2221 \\
[927] & 1(12)12231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[926] & (12)12^1 231513161316131613161316123 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
& 31 \\
[925] & (12)12^1 23151316131613161316131613161 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 47, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 47, T \in [39, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[948] $1(12)1223151321812321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[949] $161232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[950] $3161232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[951] $612321812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 51, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 51, T \in [39, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[952] $41413221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[953] $41413221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[954] $812231513221(12)1223151322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[955] $812231513221(12)122^1 31513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[956] $413221(12)122315132^1 21(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[957] $413221(12)12^1 231513221(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[958] $812321812321812^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[959] $81232^1 1812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[960] & 81232^1 181232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[961] & 812321812321812^1 32181232181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[962] & 81232^1 171232181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[963] & 812321712321812^1 32181232181232181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[964] & 812321712321812^1 3218123218123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[965] & 81232^1 17123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[936] & 1^{1513221} (12) 12^1 231513221 (12) 122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[980] & 413221^1 (12) 12231513221 (12) 122^1 31513161316 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[984] & 81232171232^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 169, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[977] & 612321812321812321812315132^1 21 (12) 1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 169, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[976] & 61232181232^1 181232181231513221 (12) 1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 169, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[975] & 31612321812321812^1 32181231513221 (12) 12 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 169, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 34, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 50$, $T \in [39, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[974] $1612321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[973] $1(12)122315132181232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 50$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 50$, $T \in [39, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[972] $1(12)122315132^1 181232181232181232161 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[971] $3161316131513221(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 39$, $\Delta = -94$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 39$, $T \in [39, 39]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[970] $613151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[969] $81223151322^1 1(12)12231513221^{1513221}(12)1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[981] $81232181232^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[968] $81231513221(12)122315132^1 21^{31513221}(12)1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 54$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 54$, $T \in [39, 54]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[979] $81223151322^1 1(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[967] $812315132^1 21(12)12231513221^{31513221}(12)1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 54$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 54$, $T \in [39, 54]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[978] $41413221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[966] & \overset{1513221}{1} (12) 12231513221 \overset{1}{(12)12^1 2315132181} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[983] & 81232171232^1 1812321812321812321 \overset{21}{81} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[982] & \overset{1}{81232181232^1 181232181232181232171} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[993] & 3161232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[985] & \overset{513221}{21} (12) 12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[986] & \overset{513221}{21} (12) 12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[987] & \overset{1513221}{1} (12) 1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[988] & \overset{1}{812231513221(12)12^1 231513221} \overset{1513221}{(12)1} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[989] & \overset{1}{181231513221(12)1223151322^1 1(12)122315} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[990] & \overset{1}{6131513221(12)12^1 231513221(12)1223151315} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[996] & \overset{1}{413221(12)1223151322^1 1(12)12231513161316} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [991] $31613161315132^1 21(12)12231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 39$, $\Delta = -94$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 39$, $T \in [39, 39]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [992] $1(12)1223151321812^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 50$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 50$, $T \in [39, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [997] $812321812^1 321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [998] $812321812^1 32181232181232181232171 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [999] $812321712^1 32181232181232181232181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [1000] $812321712^1 3218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [995] $812231513221(12)12^1 231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [994] $41413221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1002] $6123218123218123218123151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 51$, $\Delta = -115$, $\det = 225$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.52 \rightarrow 39$, $T_{\max} = 51$, $T \in [39, 51]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1001] $81231513221(12)122^1 31513221^{31513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 54$, $\Delta = -116$, $\det = 225$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.52 \rightarrow 39$, $T_{\max} = 54$, $T \in [39, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1003] $81231513221(12)1223151322^1 1^{31513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 54$, $\Delta = -116$, $\det = 256$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.48 \rightarrow 39$, $T_{\max} = 54$, $T \in [39, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1004] $1(12)122315132^1 1812321812321812321613 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 50$, $\Delta = -114$, $\det = 256$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 38.48 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1005] & 31612321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 256, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.48 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1006] & 413221(12)12231513221(12)12^1 231513161316 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 256, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.48 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1007] & 8123151322^1 1(12)12231513221^{31513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 54, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.46 \rightarrow 39, T_{\max} = 54, T \in [39, 54] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1008] & 316131613151322^1 1(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 39, \Delta = -94, \det = 289, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.46 \rightarrow 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[1009] & 612321812321812^1 32181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.46 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1010] & 81231513221(12)12^1 231513221^{31513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 54, \Delta = -116, \det = 361, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.41 \rightarrow 39, T_{\max} = 54, T \in [39, 54] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1011] & 61232181232181232^1 181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 361, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.41 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1012] & 612321812321812321812^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 529, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.34 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
T = 35 & [1015] 61232181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -107, \det = \\
& -4, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 47, T \in [42, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1018] & 812321812321812321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -102, \det = -4, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 47, T \in [42, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1017] & 2171232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 47, T \in [42, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1016] $171232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -4$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1013] $61232181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -4$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1014] $61232181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -4$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1023] $32^1 171232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1022] $171232181232181232181231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1021] $1(12)1223151321812321812321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1020] $61232181231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1019] $61232181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1031] $81232181232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.75 \rightarrow 41$, $T_{\max} = 47$, $T \in [41, 47]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1024] $5132181231513221(12)12231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.75 \rightarrow 41$, $T_{\max} = 41$, $T \in [41, 41]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1030] $81232181232181232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.75 \rightarrow 41$, $T_{\max} = 47$, $T \in [41, 47]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1029] $21712^1 32181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.75 \rightarrow 41$, $T_{\max} = 47$, $T \in [41, 47]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1028] & \begin{matrix} 1 \\ 1712^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.75 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1027] & \begin{matrix} 21 \\ 1(12)122315132181232181232181232^1 1712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.75 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1026] & \begin{matrix} 1 \\ 1712^1 32181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.75 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1025] & \begin{matrix} 1 \\ 61232181231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.75 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1040] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122315132^1 16 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1033] & \begin{matrix} 15132^1 21 \\ 61316131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1034] & \begin{matrix} 2221 \\ 61232181231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1035] & \begin{matrix} 2221 \\ 612^1 32181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1036] & \begin{matrix} 221 \\ 61232181231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1037] & \begin{matrix} 221 \\ 612^1 32181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1041] & \begin{matrix} 1 \\ 171232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1042] $1(12)1223151321812321812321812^1321712 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1043] $171232^1181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1044] $171232181232181232181231513221(12)12^123 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1045] $1712^132181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1049] $812321812321812321812321812^132181 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1048] $812321812321812321812321812^13218122 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1047] $2171232^1181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1046] $232^1171232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 52, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1038] $61232181231513221(12)122315132^121(12)12222 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1039] $612^132181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[1032] $5132181231513221(12)122315132^121(12)122313 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 41, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1054] $61232181231513221(12)1223151322^11(12)12222 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [1053] $61232181231513221(12)1223151322^11(12)12 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1052] $61232181231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1051] $712^131513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 48, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1050] $5132181231513221(12)1223151322^11(12)122313 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 41, \Delta = -96, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1058] $61316131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1075] $321712^132181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1057] $71231513221(12)12231513221(12)122^13151315 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 48, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1073] $1712321812321812321812315132^121(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1072] $1712321812^132181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1071] $1(12)122315132181232181232^11812321712 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1070] $1(12)122^13151321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [1074] $171232^1181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 50, \quad T \in [41, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1069] & \begin{matrix} 1 \\ 1712321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1068] & \begin{matrix} 1 \\ 1712321812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1067] & \begin{matrix} 1 \\ 1(12)122315132181232181232181232^1 17123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 50, \quad \Delta = -113, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 50, \quad T \in [41, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1066] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122^1 31513216 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -110, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1065] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -109, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1064] & \begin{matrix} 1 \\ 61232^1 181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[1063] & \begin{matrix} 1 \\ 61232181231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1062] & \begin{matrix} 221 \\ 61232^1 181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[1061] & \begin{matrix} 221 \\ 61232181231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1060] & \begin{matrix} 2221 \\ 61232181231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1059] & \begin{matrix} 2221 \\ 61232^1 181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1055] $5132^1 181231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1076] $21712321812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1077] $21712321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1078] $812^1 3218123218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1079] $81232181232181232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1080] $81232181232181232181232^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1081] $812^1 321812321812321812321812321^1 81 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1082] $414132^1 21(12)12231513221(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1056] $5132181231513221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1087] $1(12)12^1 23151321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.94 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1096] $812321812321812321812^1 321812321^1 81 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.94 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1084] $1(12)1223151321812321812321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) =$

(1, 35, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1086] $17123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1088] $1(12)1223151321812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1089] $17123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1090] $171232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1091] $32171232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1092] $321712321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1093] $217123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1094] $2171232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1095] $812321812321812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1085] $171232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1083] $1(12)122^1 31513218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1097] 5132181231513221(12)12¹231513221(12)122313+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 41, T \in [40, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1098] 51321812315132¹21(12)12231513221(12)122313+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 41, T \in [40, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1099] 71231513221(12)12231513221(12)12¹23151315+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 48$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1100] 71231513221(12)122315132¹21(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 48$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1101] 61316131513221(12)122315132¹⁵¹³²²¹21 (12) 1 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1102] 61232181231513221(12)12¹231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1103] 612321812315132²²²¹21(12)12231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1104] 612321812315132²²¹21(12)12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1105] 61232181231513221(12)12²²¹231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1107] 61232181231513221(12)12¹231513221(12)12222+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1108] 6131513221(12)12231513221(12)12¹231513161+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1110] $6131513221(12)122315132^1 21(12)12231513216+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1111] $6131513221(12)12231513221(12)12^1 231513216+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1112] $1(12)12^1 231513218123218123218123217123 + \hat{1}\rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1113] $2321712321812321812321812315132^1 21(12)1+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 52, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1114] $3217123218123218123218123151322^1 1(12)12+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1115] $41413221(12)122^1 31513221(12)12231513161316+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1116] $4141322^1 1(12)12231513221(12)12231513161316+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1106] $612321812315132^1 21(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1109] $6131513221(12)122315132^1 21(12)12231513161+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1117] $4131613161316131613161315132^1 21(12)+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -110, \det = -100, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1118] $712315132^1 21(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 48, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.76 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1119] $61315132^1 21(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1120] $61315132^1 21(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1121] $1(12)122315132181232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1122] $1712321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1123] $1(12)122315132181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1124] $1712321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1125] $23217123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1126] $2321712^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1127] $321712321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1128] $21712321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1129] $81232181232181232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1130] $81232181232181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1147] $6131513221(12)122^1 31513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) =$

$$(1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1161] 41413221(12)122315132^1 21(12)12231513161316+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1131] 41316131613161316131613151322^1 1(12)+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 10 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1132] 51321812^1 31513221(12)12231513221(12)122313+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 41, \Delta = -96, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 41, T \in [40, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1133] 513218123151322^1 1(12)12231513221(12)122313+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 41, \Delta = -96, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 41, T \in [40, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1134] 71231513221(12)1223151322^1 1(12)1223151315+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 48, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1135] 71231513221(12)122^1 31513221(12)1223151315+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 48, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1136] 61316131513221(12)122^1 31513221 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1137] 61316131513221(12)1223151322^1 1 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1138] 612321812^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1139] 6123218123151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1143] 6123218123151322^1 1(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1144] $6131513221(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1145] $6131513221(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1146] $6131513221(12)1223151322^1 1(12)12231513216 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1140] $6123218123151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1148] $1(12)1223151321812321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1149] $171232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1150] $1(12)1223151321812^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1151] $171232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1152] $1712321812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1153] $232171232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 52, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1154] $32171232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1155] $2171232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -121, (n_+, n_-, n_0) =$

(1, 35, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1156] $812321812321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1157] $81232^1 18123218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1158] $812321812321812^1 32181232181232181 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1159] $81232^1 181232181232181232181232181 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1160] $41413221(12)12^1 231513221(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1141] $612321812^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1142] $612321812^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1168] $1(12)122315132^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -169$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.59 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[1167] $1712321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -169$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.59 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1166] $1(12)122315132181232^1 18123218123217123 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.59 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1165] $613151322^1 1(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.59 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{aligned}
& T_{\min} = 39.59 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
& [1178] \text{ }^1 71231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 48, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 48, T \in [40, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [1179] \text{ }^{1513221} 61316131513221(12)12^1 231513221(12) 1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [1180] \text{ }^1 6131513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [1181] \text{ }^1 6131513221(12)12^1 231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -110, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [1182] \text{ }^1 1(12)1223151321812^1 3218123218123217123 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [1183] \text{ }^1 232171232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 52, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 52, T \in [40, 52] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
& [1184] \text{ }^1 32171232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
& [1185] \text{ }^1 812321812^1 3218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -102, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [1186] \text{ }^1 812321812^1 321812321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -102, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [1187] \text{ }^1 41413221(12)1223151322^1 1(12)12231513161316 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [1188] \text{ }^1 17123218123218123218123151322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -225, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.52 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1193] 41413221(12)12231513221(12)12¹231513161316+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1191] 2321712321812321812¹32181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[1189] 6131613151322¹⁵¹³²²¹1(12)12231513221 (12) 1 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -109$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1190] 1(12)122315132¹18123218123218123217123+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1192] 321712321812321812321812¹31513221(12)12+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[1194] 1712321812321812¹32181231513221(12)1223+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -289$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.46 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1195] 232171232181232181232¹181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.46 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[1196] 171232181232181232¹181231513221(12)1223+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -361$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.41 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1197] 2321712321812321812321812¹31513221(12)1+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -361$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.41 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[1198] 1712321812321812321812¹31513221(12)1223+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -529$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.34 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

$T = 36$ [1207] 1(12)12231513221(12)12231513221(12)1222²²²¹2 + $\hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -104$, $\det = 4$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43, T_{\max} = 48, T \in [43, 48]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1208] 21(12)12231513221(12)12231513221(12)122³²²¹2 + $\hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 51$, $\Delta = -113$, $\det = 4$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43, T_{\max} = 51, T \in [43, 51]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1209] & \begin{matrix} 2^1 1 \\ 1 \end{matrix} 3221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1210] & \begin{matrix} 221 \\ 1 \end{matrix} 21(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1211] & \begin{matrix} 22^1 1 \\ 1 \end{matrix} 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1212] & \begin{matrix} 1 \\ 22^1 221 \end{matrix} (12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1213] & \begin{matrix} 22^1 21 \\ 1 \end{matrix} 1(12)12231513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1214] & \begin{matrix} 22^1 1 \\ 21 \end{matrix} 21(12)12231513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1215] & \begin{matrix} 3221 \\ 71232181231513221 \end{matrix} (12)12231513221(12)12^1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1216] & 171232181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1217] & \begin{matrix} 21 \\ 181232181232181232181231513221 \end{matrix} (12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1199] & \begin{matrix} 22^1 21 \\ 1 \end{matrix} 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1200] & \begin{matrix} 221 \\ 1 \end{matrix} 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 51, T \in [42, 51]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1201] & 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1202] & 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1218] & 1(12)1223151321812321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1203] & 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1219] & 22^1 181232181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1204] & 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1205] & 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1206] & 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1220] & 22181232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1229] & 21(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1235] & 1(12)12231513218123218123218123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[1230] $222^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1236] $1 \ 8 \ 1232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1228] $21(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1227] $21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1234] $181232181232181232181231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1239] $32^1 2181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1226] $1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1225] $1(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1224] $21(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1223] $1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1222] $1(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1221] $1^{222^1 1} (12) 12231513221 (12) 12231513221 (12) 1222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1237] $1^{32^1 1} 8 1232181232181232181231513221 (12) 1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1232] $71232181231513221 (12) 12231513221 (12) 12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1233] $171232181231513221 (12) 12231513221 (12) 122^1 22 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1238] $2181232181232181232181231513221 (12) 122^1 3 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1231] $1^{222^1 1} (12) 12231513221 (12) 122315132181232171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1246] $1^{21} 8 12^1 32181232181232181231513221 (12) 1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1245] $1^{21} (12) 122315132181232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1247] $221812^1 32181232181232181231513221 (12) 122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1244] $1^{21} 8 12^1 32181232181232181231513221 (12) 122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1243] $171232181231513221 (12) 12231513221 (12) 12^1 222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1242] $2222^1 1 (12) 12231513221 (12) 12231513221 (12) 12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [1240] \overset{221}{1}(12)12231513221(\overset{1}{12})12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& [1241] 21(\overset{221}{12})12231513221(\overset{1}{12})12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& [1272] \overset{1}{1}(12)1223151321812321812321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [1253] \overset{221}{21}(\overset{1}{12})122^1 31513221(\overset{1}{12})12231513221(\overset{221}{12})12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
& [1262] \overset{221}{21}(\overset{1}{12})12231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1263] \overset{221}{21}(\overset{1}{12})122^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
& [1264] \overset{322^1}{71}232181231513221(12)12231513221(\overset{1}{12})12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [1265] 1712^1 32181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
& [1266] 171232181231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1267] 6131513221(\overset{1}{12})12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [1268] 6131513221(\overset{1}{12})12231513221(12)122315132^1 217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1269] $71231513221(12)12231513221(12)122315132^1 16 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1270] $181232181232181232181231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1271] $1812^1 32181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1273] $1^1 8^1 1232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1248] $1(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1274] $181232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1275] $1(12)1223151321812321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1276] $181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1277] $21812^1 32181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1278] $2181232181232181232181231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1279] $322^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1280] $22181232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 25, (n_+, n_-, n_0) =$

$$(1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1252] \quad {}^{21}21(12)12231513221(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1249] \quad {}^{221}1(12)12231513221(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1250] \quad {}^{2221}1(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1251] \quad {}^{2221}1(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1260] \quad {}^{2221}1(12)12231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1259] \quad {}^{22221}1(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1258] \quad {}^{221}21(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1257] \quad {}^{221}21(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1256] \quad {}^{322^1 1}21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1255] \quad {}^{2221}1(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1254] \quad {}^{2221}1(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
& [1261] \quad 1(12)122^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
& [1286] \quad 21^{221}(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [1285] \quad 21(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1287] \quad 1(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [1288] \quad 1(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1289] \quad 21^{3221}(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1290] \quad 21^{221}(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [1291] \quad 21(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1292] \quad 1812^1 31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1293] \quad 22221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1283] \quad 1(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1284] & 1(12)12^{\overset{2221}{1}}231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1294] & 8122^{\overset{1}{1}}31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1295] & 1(12)12^{\overset{2221}{1}}231513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1296] & 21(12)12^{\overset{221}{1}}231513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1297] & 712^{\overset{3221}{1}}32181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1298] & 171232181231513221(12)1223151322^{\overset{1}{1}}1(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1299] & 712^{\overset{1}{1}}31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1300] & 6131513221(12)12231513221(12)1223151322^{\overset{1}{1}}17 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1301] & 712^{\overset{1}{1}}31513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1282] & 1(12)122^{\overset{221}{1}}31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1281] & 1(12)12^{\overset{2221}{1}}231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1327] $1^{12321}1812321^{1812321}181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1328] $1^{12321}1812321^{1812321}1812321^{1812321}181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1306] $21^{21}12^{1231513221}1^{1231513221}1^{1231513221}1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1307] $21^{221}12^{1223151321}1^{12231513221}1^{12231513221}12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[1308] $1^{2221}12^{1223151321}1^{12231513221}1^{12231513221}12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1309] $1^{2221}12^{1223151321}1^{12231513221}1^{12231513221}12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1304] $1^{2221}12^{1223151321}1^{12231513221}1^{12231513221}12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[1310] $21^{3221}12^{12231513221}1^{12231513221}1^{12231513221}1(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1311] $21^{221}12^{1223151321}1^{12231513221}1^{12231513221}12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1312] $21^{221}12^{12231513221}1^{12231513221}1^{12231513221}12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1313] $181231513221^{181231513221}1^{181231513221}1^{181231513221}122^{181231513221}1^{181231513221}1^{181231513221}1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

- [1314] $22221(12)122315132^1 21(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1315] $812^1 231513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1316] $812231513221(12)12231513221(12)122^1 3151315+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1317] $1(12)12231513221(12)1223151321812^1 32171 + \hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1330] $1812321812^1 32181232181231513221(12)122 + \hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1325] $6131513221(12)12231513221(12)122^1 315132217+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1324] $6131513221(12)12231513221(12)122^1 315132171+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1323] $71231513221(12)12231513221(12)122^1 31513161+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1322] $171232181231513221(12)122^1 31513221(12)12222+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1321] $171232^1 181231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [1320] $21(12)12231513221(12)1223151321812^1 32171 + \hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1319] $21(12)122315132^1 21(12)122315132181232171 + \hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) =$

$(1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1318] $1(12)122315132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1331] $1812321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1332] $1(12)122^1 31513218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1333] $1(12)122315132181232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1334] $1812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1335] $1812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1336] $1812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1337] $2181232^1 181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1338] $3221812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1339] $221812321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1340] $221812321812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1329] $1 \overset{321}{8} 12^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1303] $1(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1302] $1(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1326] $71231513221(12)12231513221(12)122^1 31513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1305] $1(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1344] $1 \overset{321}{8} 12321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1345] $1 \overset{21}{8} 123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1346] $1 \overset{21}{8} 1232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1347] $1(12)12^1 231513218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1348] $1(12)1223151321812321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1349] $1 \overset{321}{8} 12321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1350] $1 \overset{321}{8} 1232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1343] $1 \overset{321}{8} 1232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1342] $1(12)122^1 315132181232181232181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1341] $1(12)1223151321812321812321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1351] $1 \overset{21}{8} 123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1352] $1 \overset{21}{8} 1232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1353] $3221812321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1354] $322181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1355] $2218123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1356] $22181232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1375] $21(12)1223151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1384] $71231513221(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1377] \quad 71232181231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1374] \quad 1(12)12231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1373] \quad 1(12)1223151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1372] \quad 812231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1371] \quad 812231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1370] \quad 22221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1369] \quad 181231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1368] \quad 181231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1367] \quad 21(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1366] \quad 21(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1365] \quad 21(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1364] & \begin{matrix} 2221 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1363] & \begin{matrix} 2221 & 1 \\ 1(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1362] & \begin{matrix} 221 & 1 & 221 \\ 21(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1361] & \begin{matrix} 21 & 1 & 21 \\ 21(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1360] & \begin{matrix} 2221 & 1 & 221 \\ 1(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1359] & \begin{matrix} 2221 & 1 & 221 \\ 1(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1358] & \begin{matrix} 221 & 1 & 21 \\ 1(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1357] & \begin{matrix} 2221 & 1 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1378] & \begin{matrix} 3221 \\ 71232^1 181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1379] & \begin{matrix} 171232181231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1380] & \begin{matrix} 1712321812315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1381] $61232181231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 41, \Delta = -95, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1382] $612321812315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 41, \Delta = -95, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1383] $71231513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1387] $6131513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1388] $6131513221(12)12231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1389] $71231513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1390] $71231513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1391] $1(12)12^1 2315132181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1392] $1^3 8^1 12321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1393] $1^3 8^1 123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1394] $1^3 8^1 123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1395] $32218123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1385] & 6131513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1386] & 6131513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1376] & 21(12)12231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1399] & 21(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1400] & 1812315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1401] & 8122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1402] & 712315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1403] & 61315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1404] & 61315132^1 21(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1405] & 712315132^1 21(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1406] & 1(12)122315132181232181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 50, T \in [41, 50]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1407] & \begin{matrix} 2321 \\ 1 \ 8 \ 12^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1408] & \begin{matrix} 2321 \\ 1 \ 8 \ 123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1409] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1410] & \begin{matrix} 21 \\ 1 \ 8 \ 12321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1411] & \begin{matrix} 21 \\ 1(12)122315132181232^1 18123218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1412] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1413] & \begin{matrix} 21 \\ 1 \ 8 \ 12321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1414] & \begin{matrix} 321 \\ 3221812321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1415] & \begin{matrix} 21 \\ 221812321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1398] & \begin{matrix} 21 \\ 21(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1397] & \begin{matrix} 221 \\ 1(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1396] $(12)122^1 3151316131613161316131613151 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.76 \rightarrow 41$, $T_{\max} = 48$, $T \in [41, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1453] $21812321812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 50$, $T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1445] $1812321812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 50$, $T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1455] $22181232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 48$, $T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1454] $322181232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 50$, $T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1448] $1 \overset{321}{8} 1232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 50$, $T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1452] $1 \overset{21}{8} 1232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 48$, $T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1451] $1 \overset{321}{8} 1232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 50$, $T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1450] $1(12)1223151321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 48$, $T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1449] $1 \overset{21}{8} 1232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 48$, $T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1416] $(12)12^1 23151316131613161316131613151 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 48$, $T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1417] $1(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 51$, $\Delta = -113$, $\det = 121$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41$, $T_{\max} = 51$, $T \in [41, 51]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1418] & \begin{matrix} 221 & 1 & 21 \\ 1(12)12231513221(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1419] & \begin{matrix} 21 & 1 & 21 \\ 21(12)12231513221(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1420] & \begin{matrix} 21 & 1 & 21 \\ 21(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1421] & \begin{matrix} 3221 & 1 \\ 21(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1447] & \begin{matrix} 2321 \\ 1 \ 8 \ 1232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1422] & \begin{matrix} 3221 & 1 \\ 21(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1423] & \begin{matrix} 1 \\ 181231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1424] & \begin{matrix} 1 \\ 181231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1425] & \begin{matrix} 1 \\ 812231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1426] & \begin{matrix} 1 \\ 812231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1427] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1428] $1^{2221}(12)12231513221(12)122315132^1181232171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[1429] $21^{221}(12)12231513221(12)12^12315132181232171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1430] $21^{221}(12)12231513221(12)122315132^1181232171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[1431] $71232181231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1432] $71232181231513221(12)1223151322^11(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1433] $17123218123151322^11(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1434] $1712321812^131513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1435] $612321812^131513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 41, \Delta = -95, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1436] $6123218123151322^11(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 41, \Delta = -95, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1437] $71231513221(12)1223151322^11(12)12231513161 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1438] $71231513221(12)122^131513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1439] $6131513221(12)122^131513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1440] & 6131513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1441] & 6131513221(12)122^1 31513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1442] & 6131513221(12)1223151322^1 1(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1443] & 71231513221(12)1223151322^1 1(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1444] & 71231513221(12)122^1 31513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1446] & 1(12)1223151321812321812^1 32181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1468] & 1(12)122315132181232^1 181232181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1478] & 221812321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1457] & 21(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1458] & 21(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1459] & 18123151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1460] & 81223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1461] & 712321812315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1462] & 7123151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1463] & 613151322^1 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1464] & 613151322^1 1(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1465] & 7123151322^1 1(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1466] & 181232181232^1 181232181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1469] & 1^1 8 12321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1470] & 1^1 8 12321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1471] & 1^1 8 12321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1472] & 1(12)122315132^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1473] $1 \overset{321}{8} 12321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1474] $1 \overset{21}{8} 12321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1475] $2181232181232^1 181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[1476] $21812321812321812321812315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1456] $1 \overset{221}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1477] $3221812321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1467] $1 \overset{1}{8} 12321812321812321812315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1479] $1 \overset{221}{(12)} 12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.55 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1480] $21 \overset{21}{(12)} 12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.55 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1481] $21 \overset{3221}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.55 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1482] $181231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -110, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1483] $812231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -110, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1484] & 71232181231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1485] & 71231513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1486] & 6131513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1487] & 6131513221(12)12^1 231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1488] & 71231513221(12)12^1 231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1489] & 1(12)1223151321812^1 32181232181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1490] & 1 \overset{2321}{8} 1232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1491] & 1 \overset{321}{8} 1232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1492] & 1 \overset{321}{8} 1232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1493] & 322181232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1496] & 218123218123218123218123151322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 225, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.52 \rightarrow 41, T_{\max} = 50, T \in [41, 50]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1495] & \begin{matrix} 1 \\ 18123218123218123218123151322 \end{matrix} \overset{1}{1}(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.52 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1494] & \begin{matrix} 3221 \\ 712321812 \end{matrix} \overset{1}{3}1513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.52 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1502] & \begin{matrix} 3221 \\ 3221812321812321812321812 \end{matrix} \overset{1}{3}1513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1501] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812321812321812 \end{matrix} \overset{1}{3}1513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1500] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812321812321812 \end{matrix} \overset{1}{3}1513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1499] & \begin{matrix} 2321 \\ 1 \ 8 \ 12321812321812 \end{matrix} \overset{1}{3}2181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1498] & \begin{matrix} 1 \\ 1(12)122315132 \end{matrix} \overset{1}{1}81232181232181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1497] & \begin{matrix} 3221 \\ 7123218123151322 \end{matrix} \overset{1}{1}(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1505] & \begin{matrix} 3221 \\ 21812321812321812 \end{matrix} \overset{1}{3}2181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.46 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1504] & \begin{matrix} 2321 \\ 1 \ 8 \ 1232181232181232 \end{matrix} \overset{1}{1}81231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 289, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.46 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1503] & \begin{matrix} 1 \\ 1812321812321812 \end{matrix} \overset{1}{3}2181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.46 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1506] $\overset{1}{181232181232181232^1 181231513221(12)1223} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.41 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1507] $\overset{2321}{1 \ 8 \ 12321812321812321812^1 31513221(12)1} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.41 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[1508] $2181232181232181232^1 181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.41 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1509] $\overset{1}{1812321812321812321812^1 31513221(12)1223} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.34 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[1510] $21812321812321812321812^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.34 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

$T = 37$ [1521] $2181232181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1512] $\overset{1}{181232181231513221(12)12231513221} \overset{22^1 1}{(12)12} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1513] $\overset{1}{181232181231513221(12)12231513221(12)1222^1 2} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1514] $\overset{2^1 1}{3221(12)12231513221(12)122315132181232181} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[1515] $\overset{22^1 1}{1(12)12231513221(12)12231513218123218123} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1516] $\overset{22^1 21}{1(12)12231513221(12)1223151321812321812} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [1517] $21^{22^1 1}(12)12231513221(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1518] $413221^1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1519] $32181232181231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1520] $232181232181231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 54, T \in [44, 54]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1511] $181232181231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1522] $413221^1(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1523] $181232181231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1524] $181232181231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1525] $21^{32^1 21}(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1527] $1^{222^1 1}(12)12231513221(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [1528] $21^{21}(12)12231513221(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1529] $32^1 181232181231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -9, (n_+, n_-, n_0) =$

$(1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[1530] $32181232181231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1531] $2181232181231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 49$, $T \in [44, 49]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1532] $413221(12)^1 12231513221(12)^1 12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1526] $1(12)^{221} 12231513221(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1533] $181232181231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.75 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1534] $2181232181231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.75 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1553] $4132^1 21(12)^1 12231513221(12)^1 12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1562] $4132^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1536] $181232181231513221(12)122315132^1 21(12)^1 1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1537] $181232181231513221(12)122315132^1 21(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1538] $1812^1 32181231513221(12)12231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1539] $181232181231513221(12)122315132^1 21(12)12222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1540] $1812^1 32181231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1541] $21(12)12231513221(12)122315132181232181+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1542] $1(12)12231513221(12)1223151321812321812^1 32+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1543] $1(12)122^1 31513221(12)1223151321812321812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1544] $1(12)12231513221(12)122315132181232^1 1812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1545] $21(12)12231513221(12)122315132181232^1 1812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1546] $21(12)122^1 31513221(12)1223151321812321812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1547] $4132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1548] $4132^1 21(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1535] $1812^1 32181231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1552] 2181232181231513221(12)122315132¹21(12)12222+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1549] 6131513221(12)12231513221(12)122315132¹181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1550] 232¹181232181231513221(12)12231513221(12)122+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1551] 21812¹32181231513221(12)12231513221(12)12222+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1554] 612¹31513221(12)12231513221(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1555] 6131513221(12)12231513221(12)122315132¹1812+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1556] 6131513221(12)12231513221(12)122315132¹2181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1557] 71231513221(12)12231513221(12)122315132¹171+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1558] 812231513221(12)12231513221(12)122315132¹16+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1559] 71231513221(12)12231513221(12)122315132¹217+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1560] 4132¹21(12)12231513221(12)12231513218123217+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1561] $4132^1 181232181232181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.44 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1584] $413221(12)12231513221(12)122315132181232^1 17+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1563] $181232181231513221(12)1223151322^1 1(12)1222+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1564] $181232181231513221(12)1223151322^1 1(12)12+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1565] $181232181231513221(12)1223151322^1 1(12)12222+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1566] $21(12)12231513221(12)122315132181232^1 181+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1567] $1(12)122^1 31513221(12)12231513218123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1568] $1(12)12^1 231513221(12)1223151321812321812+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1569] $21(12)122^1 31513221(12)12231513218123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1570] $21(12)12^1 231513221(12)1223151321812321812+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1571] $413221(12)12231513221(12)122315132^1 21(12)1+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1572] $413221(12)12231513221(12)122315132^1 21(12)12+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1573] $32181232181231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1574] $2181232181231513221(12)1223151322^1 1(12)12222+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1575] $413221(12)12231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1576] $1812^1 31513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1577] $61231513221(12)12231513221(12)122315132^1 181+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1578] $8122^1 31513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1579] $6131513221(12)12231513221(12)1223151322^1 181+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1580] $712^1 31513221(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1581] $8122^1 31513221(12)12231513221(12)12231513216+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1582] $712^1 31513221(12)12231513221(12)122315132217+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1583] $71231513221(12)12231513221(12)1223151322^1 17+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) =$

$(1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.22 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1618] $812^1 231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1619] $812231513221(12)12231513221(12)122^1 31513216 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1586] $181232181231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1587] $181232181231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1588] $181232^1 181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1589] $181232181231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1590] $181232^1 181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1591] $1(12)12^1 231513221(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 52$, $T \in [43, 52]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1592] $1(12)122^1 31513221(12)122315132181232181232 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1593] $1(12)12231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[1594] $1(12)122315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1595] & 21(12)12^{21}231513221(12)12231513218123218123+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1596] & 21(12)12231513221(12)1223151321812^{21}321812+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[1597] & 21(12)122315132^{21}21(12)1223151321812321812+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1598] & 41322^1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1599] & 413221(12)12231513221(12)1223151322^1(12)1+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1600] & 41322^1(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1601] & 413221(12)12231513221(12)1223151322^1(12)12+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1602] & 6131513221(12)12231513221(12)122^1315132181+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1603] & 32181232181231513221(12)1223151322^1(12)1222+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1604] & 232181232181231513221(12)122315132^121(12)122+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 54, T \in [43, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1605] & 2181232181231513221(12)122^131513221(12)12222+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1606] 2181232¹181231513221(12)12231513221(12)12222+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [1607] 61316131513221(12)12231513221(12)122¹3151315+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1608] 41322¹1(12)12231513221(12)12231513221(12)1222+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1609] 413221(12)12231513221(12)1223151322¹1(12)1222+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1610] 181231513221(12)12231513221(12)122¹31513161+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1611] 612315132¹21(12)12231513221(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1612] 6131513221(12)12231513221(12)122¹3151321812+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1613] 812¹31513221(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1614] 812231513221(12)12231513221(12)122¹31513161+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1615] 812¹231513221(12)12231513221(12)12231513161+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1616] 6131513221(12)12231513221(12)122¹3151322181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1620] 71231513221(12)12231513221(12)122¹315132217+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$(1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1621] \quad 41322^1 1(12)12231513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1622] \quad 41321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1623] \quad 41321812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1617] \quad 71231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1585] \quad 181232^1 181231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[1626] \quad 4132181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1627] \quad 41321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1628] \quad 4132181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1629] \quad 41321812321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1625] \quad 232181232181231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1624] \quad 1(12)12^1 231513221(12)122315132181232181232 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1643] & 21(12)^{221}12231513221(12)122^13151321812321812+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1630] & 21(12)^{3221}122^131513221(12)^112231513221(12)12231+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 42, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1634] & 181232181231513221(12)^112^1231513221(12)^{221}12+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1635] & 1812321812315132^121(12)^112231513221(12)12222+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1636] & 181232181231513221(12)^112^1231513221(12)12222+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1637] & 21(12)^{3221}12231513221(12)1223151321812^132181+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1638] & 21(12)^{3221}122^131513221(12)122315132181232181+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1639] & 1(12)^{221}12231513221(12)122315132181232^118123+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1640] & 1(12)^{2221}12231513221(12)122^13151321812321812+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1641] & 1(12)^{221}1223151322^11(12)1223151321812321812+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1642] & 21(12)^{21}12231513221(12)122315132181232^118123+\hat{1}\rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1632] $181232181231513221(12)12^1 231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1644] $21(12)1223151322^1 1(12)1223151321812321812+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1645] $413221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1646] $413221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1647] $6131513221(12)12231513221(12)12^1 2315132181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1648] $6131513221(12)122315132^1 21(12)122315132181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1649] $1712321812315132^1 21(12)12231513221(12)122313+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 42, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1650] $171232181231513221(12)12^1 231513221(12)122313+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 42, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1651] $321812^1 32181231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1652] $21812321812315132^1 21(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1653] $2181232181231513221(12)12^1 231513221(12)12222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1654] $61316131513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1655] $61316131513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1656] $413221(12)122^1 31513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1657] $181231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1658] $181231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1659] $6123151322^1 1(12)12231513221(12)122315132181+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1660] $61231513221(12)122^1 31513221(12)122315132181+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1661] $6131513221(12)122315132^1 21(12)1223151321812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1662] $6131513221(12)12231513221(12)12^1 23151321812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1663] $812231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1664] $812231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1665] $6131513221(12)122315132^1 21(12)1223151322181+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [1666] $6131513221(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1667] $71231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1668] $71231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1669] $812231513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1670] $812231513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1671] $71231513221(12)12231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1672] $71231513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1673] $413221(12)122^1 31513221(12)12231513218123217 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1674] $413221(12)12231513221(12)1223151321812^1 3217 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1675] $413218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1676] $413218123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1631] $1812321812315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) =$

$$(1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1633] \overset{1}{1812321812315132^1 21(12)12231513221(12)12+1 \rightarrow 2} \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1677] (12)122^1 31513161316131613161316131613161+1 \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1678] (12)122^1 31513161316131613161316131612+1 \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1679] \overset{3221}{21(12)12231513221(12)122^1 31513221(12)12231+1 \rightarrow 2} \quad T = 37, T_{\max} = 42, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1680] \overset{221}{1(12)122315132^1 21(12)12231513218123218123+1 \rightarrow 2} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1681] \overset{21}{21(12)122315132^1 21(12)12231513218123218123+1 \rightarrow 2} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1682] \overset{1}{413221(12)12231513221(12)122^1 31513221(12)1+1 \rightarrow 2} \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1683] \overset{1}{413221(12)12231513221(12)122^1 31513221(12)12+1 \rightarrow 2} \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1684] \overset{1}{61315132^1 21(12)12231513221(12)122315132181+1 \rightarrow 2} \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1685] \overset{1}{32181232181231513221(12)122^1 31513221(12)1222+1 \rightarrow 2} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1686] \overset{1}{413221(12)12231513221(12)122^1 31513221(12)1222+1 \rightarrow 2} \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1687] $1812315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1688] $61231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1689] $61315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1690] $8122315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1691] $61315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1692] $712315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1693] $8122315132^1 21(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1694] $712315132^1 21(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1695] $41321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1696] $41321812321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1732] $6131513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

- [1747] $4132181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1734] $812231513221(12)1223151322^1 1(12)12231513161+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1733] $81231513221(12)12231513221(12)122^1 315131613+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 54$, $T \in [42, 54]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1736] $6131513221(12)122^1 31513221(12)1223151322181+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1731] $6131513221(12)1223151322^1 1(12)1223151321812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1730] $61231513221(12)122315132^1 21(12)122315132181+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1729] $61231513221(12)12^1 231513221(12)122315132181+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1728] $181231513221(12)122^1 31513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1727] $181231513221(12)1223151322^1 1(12)12231513161+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1726] $413221(12)122315132^1 21(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1725] $413221(12)12^1 231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1724] $61316131513221(12)1223151322^1 1(12)1223151315+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) =$

$(1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1723] $61316131513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1722] $21812321812^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1721] $218123218123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1720] $1712321812^1 31513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 42$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 42$, $T \in [42, 42]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1719] $17123218123151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 42$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 42$, $T \in [42, 42]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1718] $6131513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1717] $6131513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1716] $413221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1715] $413221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1714] $413221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1713] $413221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

[1701] $1812321812^1 31513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1700] $21(12)12^{3221} 231513221(12)12231513221(12)12231+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 42, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1699] $21(12)122315132^{3221} 21(12)12231513221(12)12231+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 42, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1698] $(12)12^1 231513161316131613161316131612+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$

[1697] $(12)12^1 23151316131613161316131613161+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$

[1737] $6131513221(12)1223151322^1 1(12)1223151322181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1738] $71231513221(12)122^1 31513221(12)122315132171+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1739] $71231513221(12)1223151322^1 1(12)122315132171+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1740] $812231513221(12)122^1 31513221(12)12231513216+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1741] $812231513221(12)1223151322^1 1(12)12231513216+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1742] $71231513221(12)122^1 31513221(12)122315132217+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1743] $71231513221(12)1223151322^1 1(12)122315132217+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) =$

$(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

$[1744] \overset{1}{413221}(12)12^1 231513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -121, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

$[1745] \overset{1}{413221}(12)122315132^1 21(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -121, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

$[1746] \overset{1}{4132181232181232^1 181232181231513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -121, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

$[1735] \overset{1}{812231513221}(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

$[1748] \overset{221}{1}(12)12231513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -144, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.64 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

$[1749] \overset{21}{21}(12)12231513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -144, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.64 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

$[1750] \overset{1}{32181232^1 181231513221}(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -144, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.64 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[1759] \overset{1}{613151322^1 1}(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

$[1752] \overset{3221}{21}(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

$[1753] \overset{221}{1}(12)1223151322^1 1(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$[1754] \overset{21}{1}(12)12231513221(12)122315132181232^1 181232 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1755] & 1(12)122315132^1 21(12)122315132181232181232 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1758] & 413221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1751] & 21(12)12231513221(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 42, \Delta = -93, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1760] & 32181232181231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1761] & 232181232181231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1762] & 2321812^1 32181231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1763] & 613161315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1764] & 413221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1765] & 18123151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1766] & 61231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -108, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1767] & 613151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1768] $81231513221^1(12)122315132^121(12)122315131613+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1769] $812315132^121(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1770] $81223151322^11(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1771] $613151322^11(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1772] $7123151322^11(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1773] $81223151322^11(12)12231513221(12)12231513216+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1774] $7123151322^11(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1775] $413221^1(12)12231513221(12)122^131513218123217+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1776] $41321812321812321812^132181231513221(12)1+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1777] $41321812321812321812^132181231513221(12)12+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1756] $21(12)1223151322^11(12)12231513218123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1757] $413221^1(12)12231513221(12)12^1231513221(12)1+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -169, (n_+, n_-, n_0) =$

(1, 37, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 41.59 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1780] $1(12)12231513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 52$, $T \in [42, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1781] $21(12)12231513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 52$, $T \in [42, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1782] $413221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1783] $413221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1784] $6131513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1785] $321812321812315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 52$, $T \in [42, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1779] $21(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 52$, $T \in [42, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1787] $413221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1788] $181231513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1789] $61231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1790] $6131513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1791] & 81231513221(12)^1 12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1792] & 812231513221(12)^1 12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1793] & 6131513221(12)^1 12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1794] & 71231513221(12)^1 12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1795] & 812231513221(12)^1 12^1 231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -108, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1796] & 71231513221(12)^1 12^1 231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1797] & 413221(12)^1 1223151322^1 1(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1798] & 4132181232181232181232^1 181231513221(12)^1 1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1799] & 4132181232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1778] & 21(12)^{3221} 1223151322^1 1(12)^1 12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 42, \Delta = -93, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1786] & 61316131513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 50, T \in [42, 50]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1800] & \begin{matrix} 3221 \\ 21(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -225, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.52 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[1801] & \begin{matrix} 1 \\ 81231513221(12)122^1 31513221(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -225, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.52 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1802] & \begin{matrix} 1 \\ 413221(12)12231513221(12)122315132^1 18123217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -225, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.52 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1803] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1804] & \begin{matrix} 1 \\ 6131613151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1805] & \begin{matrix} 1 \\ 81231513221(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1806] & \begin{matrix} 1 \\ 413221(12)12231513221(12)12^1 231513218123217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1807] & \begin{matrix} 1 \\ 41321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1808] & \begin{matrix} 1 \\ 41321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1809] & \begin{matrix} 221 \\ 1(12)12231513221(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -289, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.46 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1810] & \begin{matrix} 21 \\ 1(12)12231513221(12)1223151321812^1 32181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1811] $1(12)1223151322^1 1(12)122315132181232181232+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1812] $21(12)12231513221(12)12^1 231513218123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1813] $3218123218123151322^1 1(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

[1814] $232181232^1 181231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 373 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$

[1815] $232181232181231513221(12)12^1 231513221(12)122+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$

[1816] $8123151322^1 1(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1817] $1(12)12231513221(12)122315132^1 18123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.43 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1819] $321812321812^1 31513221(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.43 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1818] $21(12)12231513221(12)122315132^1 18123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.43 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1820] $1(12)12231513221(12)122^1 315132181232181232+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.41 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[1821] $2321812321812315132^1 21(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.41 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1822] $81231513221(12)12^1 231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.41 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1823] & 1^{21}(12)12231513221(12)12^{12315132181232181232+\hat{1} \rightarrow 2} \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.34 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[1824] & 23218123218123151322^1 1(12)12231513221(12)122+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.34 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 666 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0} \\
[1826] & 2321812321812^1 31513221(12)12231513221(12)122+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.31 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 799 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0} \\
[1825] & 1^{21}(12)12231513221(12)122315132^1 181232181232+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.31 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
\\
T = 38 & [1833] 21^{22^1 1}(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1835] & 41413221(12)12231513221^1(12)12231513221(12)122^1 2+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1834] & 41413221(12)12231513221^1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1832] & 1^{22^1 21}(12)12231513221(12)12231513221^1(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1831] & 1^{1513221}(12)12231513221^1(12)12231513221(12)122^1 2+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1830] & 1^{31513221}(12)12231513221^1(12)12^1 2+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 55, T \in [45, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1829] & 1^{22^1 31513221}(12)12231513221^1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 58, T \in [45, 58]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1828] & 1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1827] & 21(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1837] & 21(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1836] & 513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.33 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1838] & 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 58, T \in [45, 58] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1839] & 1(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1840] & 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1841] & 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.33 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1842] & 21(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.33 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1843] & 41413221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1845] & 1(12)122315132181232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.75 \rightarrow 44, T_{\max} = 50, T \in [44, 50]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1846] & 512^1 32181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 16, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.75 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 513221 \\
[1844] & 21(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 16, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.75 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& 15132^1 21 \\
[1854] & 71232181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& 1 \\
[1866] & 4132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 52, T \in [44, 52] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& 1 \\
[1855] & 71231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 1 \\
[1856] & 181231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1867] & 61316131513221(12)12231513221(12)122315132^1 16 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -109, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[1853] & 21(12)122^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[1852] & 1(12)122^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 15132^1 21 \\
[1851] & 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& 15132^1 21 \\
[1850] & 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1849] $1(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1848] $1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 58, T \in [44, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1847] $21(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1857] $7122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[1858] $71231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1859] $812231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1860] $71231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1861] $812231513221(12)12231513221(12)122315132^1 217 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[1862] $1(12)1223151321812321812321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1863] $1(12)1223151321812321812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1864] $51232^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1865] $15132^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) =$

(1, 38, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 43.44 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1886] $41413221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1889] $41413221(12)12231513221(12)122315132181232^1 17 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1868] $21(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1869] $1(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1870] $1(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1871] $1(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1872] $1(12)12^1 231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1873] $21(12)12^1 231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1888] $413221(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1874] $712^1 32181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1875] $712^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1876] & 1812^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1877] & 712^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1887] & 41413221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1878] & 712231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1879] & 712^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1880] & 8122^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1881] & 712^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1882] & 71231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1883] & 8122^1 31513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1884] & 812231513221(12)12231513221(12)1223151322^1 17 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1885] & 41413221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 51, T \in [44, 51]
\end{aligned}$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1919] $414132^1 21(12)12231513221(12)^1 12231513221(12)^{21} 12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1904] $21(12)122315132^1 21(12)12231513221(12)^{21} 122315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1890] $21(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1891] $21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1892] $1(12)12231513221(12)^1 12231513221(12)^1 122^1 315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 58, T \in [44, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1893] $1(12)122^1 31513221(12)^1 12231513221(12)^1 12231513 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1928] $414132^1 21(12)12231513221(12)12231513218123217 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1894] $1(12)12231513221(12)^1 12231513221(12)^1 12^1 23151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1895] $1(12)12^1 231513221(12)^1 12231513221(12)^1 1223151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1896] $1(12)12231513221(12)^1 12231513221(12)^{21} 12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1897] $1(12)12231513221(12)^1 1223151322^1 1(12)^{21} 12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1898] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 122315132^1 21(12)122+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1899] $1 \begin{smallmatrix} 151322^1 1 \\ (12) \end{smallmatrix} 12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1900] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1223151322^1 1(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1901] $1 \begin{smallmatrix} 2221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 122^1 315+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1902] $1 \begin{smallmatrix} 2221 \\ (12) \end{smallmatrix} 122315132^1 21(12)12231513221 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 122315+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1927] $61316131513221(12)12231513221(12)122^1 31513216+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1926] $41322^1 1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1925] $414132^1 21(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1924] $41413221(12)12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1223151322^1 1(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1923] $61316131513221(12)12231513221(12)122^1 31513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1922] $151321812^1 32181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1921] $512321812^1 32181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1920] & 41413221(12)12231513221(12)1223151322^1 1(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1918] & 414132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1917] & 41413221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1916] & 1(12)122315132181232181232181232^1 18123151+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1915] & 1(12)122315132181232181232181232^1 18123215+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1914] & 812^1 231513221(12)12231513221(12)122315132217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1913] & 812231513221(12)12231513221(12)122^1 315132217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1912] & 71231513221(12)12231513221(12)122^1 3151322181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1911] & 812^1 231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1910] & 812231513221(12)12231513221(12)122^1 315132171+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1909] & 71231513221(12)12231513221(12)122^1 3151321812+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1908] $7122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1907] $181231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1906] $71231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1905] $71232181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1903] $21(12)12231513221(12)12231513221(12)122^1 315 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1929] $1(12)122^1 31513221(12)12231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1943] $15132181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1942] $151321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1941] $51232181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 64, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1940] $512321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 64, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1939] $315132^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 54, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1938] $1(12)1223151321812321812321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 64, (n_+, n_-, n_0) =$

(1, 38, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

¹[1937] $1(12)122^1 31513218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

¹[1936] $1(12)1223151321812321812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

¹[1935] $1(12)122^1 31513218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

³¹⁵¹³²²¹[1933] $1(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

²³¹⁵¹³²²¹[1932] $1(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

²¹[1931] $1(12)12231513221(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

²¹[1930] $1(12)12^1 231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

^{315132^1 21}[1934] $1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

¹[1958] $71231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

¹[1971] $812231513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

¹[1986] $413221(12)12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1987] & 413221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1988] & 61316131513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1989] & 61316131513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1990] & 41413221(12)12231513221(12)1223151321812^1 3217 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1991] & 4141322^1 1(12)12231513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1992] & 41413221(12)122^1 31513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1948] & 1^{1513221} (12) 122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1949] & 1^{2231513221} (12) 12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1947] & 1^{221} (12) 12231513221(12)122315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1944] & 21^{513221} (12) 12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1945] & 21^{513221} (12) 122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1946] $1(12)12^1 231513221(12)12231513221(12)122315132+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1950] $1(12)12231513221(12)12231513221^1(12)12+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1951] $1(12)122^1 31513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1952] $1(12)12231513221^1(12)12231513221(12)122315+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1953] $1(12)12231513221(12)122^1 31513221(12)122315+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1954] $21(12)12231513221^1(12)12231513221(12)122315+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1955] $21(12)12231513221(12)122^1 31513221(12)122315+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1956] $71232^1 181231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1957] $71232181231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1959] $71231513221(12)122315132^1 21(12)122315132181+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1960] $321812^1 32181231513221(12)12231513221(12)12231+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 43$, $\Delta = -93$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 43$, $T \in [43, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1961] $181231513221(12)12231513221(12)12^1 2315132171+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) =$

(1, 38, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1962] $181231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1963] $71223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1964] $712231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1965] $71231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1966] $71231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1967] $812231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1968] $812231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1969] $71231513221(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1970] $71231513221(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1984] $41413221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1983] $61316131513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1982] & 61316131513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1981] & 1513218123218123218123218123151322^1 1(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1980] & 5123218123218123218123218123151322^1 1(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1979] & 3151321812321812321812321812315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 54, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 54, T \in [43, 54] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1978] & 41413221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1977] & 4141322^1 1(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1976] & 4141322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1975] & 41413221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1974] & 1(12)12^1 231513218123218123218123218123151+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1973] & 1(12)12^1 231513218123218123218123218123215+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1972] & 812231513221(12)12231513221(12)12^1 2315132217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1985] $4141322^1 1(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2018] $41413221(12)12231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2017] $(12)122^1 3151321812321812321812321812315132+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2016] $151321812321812^1 32181232181231513221(12)12+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2015] $512321812321812^1 32181232181231513221(12)12+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2014] $31513218123218123218123218123151322^1 1(12)1+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2013] $3151321812^1 32181232181232181231513221(12)1+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2012] $41413221(12)12231513221(12)122^1 31513221(12)12+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2011] $41413221(12)12231513221(12)122^1 31513221(12)1+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2010] $1(12)122315132181232181232^1 18123218123151+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2009] $1(12)122315132181232181232^1 18123218123215+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2008] $8122315132^1 21(12)12231513221(12)122315132217+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2007] $712315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2006] $8122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2005] $712315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2004] $712231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2003] $1812315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2002] $32181232181231513221(12)122^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 43$, $\Delta = -93$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2001] $712315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2000] $21(12)12231513221(12)12231513221(12)12^1 2315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1999] $1(12)12231513221(12)12231513221(12)12^1 2315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1998] $1(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1997] $1(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

- [1996] $1^{1513221}(12)12231513221(12)122^131513221(12)12+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1995] $1^{221}(12)122315132^121(12)12231513221(12)1223151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1994] $1^{3221}(12)12231513221(12)12231513221(12)12^12315+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1993] $21^{51322^11}(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [2064] $(12)12^123151321812321812321812321812315132+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2051] $812231513221(12)1223151322^11(12)122315132217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2056] $41413221(12)122315132^121(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2054] $41413221(12)12^1231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2067] $413221(12)12^1231513221(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2068] $413221(12)122315132^121(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2069] $61316131513221(12)122^131513221(12)12231513216+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2070] $61316131513221(12)1223151322^11(12)12231513216+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2071] & 41413221(12)122315132^1 21(12)12231513218123217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2072] & 41413221(12)12^1 231513221(12)12231513218123217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2053] & 1(12)1223151321812321812^1 3218123218123151+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2048] & 71231513221(12)122^1 31513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2066] & 41413221(12)122315132^1 21(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2055] & 41413221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2057] & 41413221(12)12^1 231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2058] & 315132181232^1 181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 54, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 54, T \in [43, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2059] & 51232181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2060] & 15132181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2061] & 61316131513221(12)122^1 31513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2062] & 61316131513221(12)1223151322^1 1(12)12231513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[2063] (12)1223151321812321812321812^1 315132 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2047] 812231513221^1 (12)1223151322^1 1 (12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2046] 812231513221^1 (12)122^1 31513221 (12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2045] 71231513221^1 (12)1223151322^1 1 (12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2044] 71231513221^1 (12)122^1 31513221 (12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2043] 712231513221^1 (12)122315132^1 21 (12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2042] 712231513221^1 (12)12^1 231513221 (12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2041] 181231513221^1 (12)122^1 31513221 (12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2040] 181231513221^1 (12)1223151322^1 1 (12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2039] 71231513221^1 (12)122^1 31513221 (12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2038] 71231513221^1 (12)1223151322^1 1 (12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2037] \quad 71232181231513221(12)1223151322^{\overset{1513221}{1}}1(12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2036] \quad 71232181231513221(12)122^{\overset{1513221}{1}}31513221(12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2035] \quad 21(12)12231513221(12)122315132^{\overset{21}{1}}21(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2034] \quad 21(12)12231513221(12)12^{\overset{221}{1}}231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2033] \quad 1(12)12231513221(12)12^{\overset{2221}{1}}231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2032] \quad 1(12)12231513221(12)122315132^{\overset{21}{1}}21(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2031] \quad 1(12) \quad 12^{\overset{1513221}{1}}231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2030] \quad 1(12) \quad 122315132^{\overset{1513221}{1}}21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2029] \quad 1(12) \quad 12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2028] \quad 1(12) \quad 12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2027] \quad 1(12) \quad 122315132^{\overset{1513221}{1}}21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2026] & 1(12) \overset{1513221}{12} \overset{1}{12} \overset{21}{231513221}(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2025] & 1(12)12231513221(12)122 \overset{221}{131513221}(12)1223151 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2024] & 1(12)12231513221(12)1223151322 \overset{1}{1}(12)1223151 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2023] & 1(12)12231513221(12)12231513221(12)12 \overset{21}{1231513} + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2022] & 1(12)122 \overset{3221}{131513221}(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2021] & 1(12)12231513221(12)12231513221(12)122 \overset{1}{1315132} + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2020] & 21(12) \overset{513221}{1223151322} \overset{1}{1}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2019] & 21(12) \overset{513221}{122} \overset{1}{131513221}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2065] & 41413221(12)12 \overset{1}{1231513221}(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2052] & 1(12)1223151321812321812 \overset{1}{13218123218123215} + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2049] & 71231513221(12)1223151322 \overset{1}{1}(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]
\end{aligned}$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2050] $812231513221(12)122^1 31513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2073] $32181232^1 181231513221(12)12231513221(12)12231+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 43, \Delta = -93, \det = 144, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.64 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2102] $41413221(12)12231513221(12)122^1 31513218123217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2074] $1(12)12^{3221} 231513221(12)12231513221(12)122315+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2075] $1(12)122315132^1 21(12)12231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2076] $1(12)1223151322^1 1(12)12231513221(12)1223151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2077] $1(12)12231513221(12)12^1 231513221(12)12+ \hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2078] $1(12)122^{31513221} 31513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2079] $1(12)12231513221(12)122^1 31513221(12)122+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2080] $1(12)12231513221(12)12^1 231513221(12)1222+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2081] $712321812315132^1 21(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [2082] $7123151322^1 1(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2083] $32181232181231513221(12)12^1 231513221(12)12231+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 43, \Delta = -93, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [2084] $18123151322^1 1(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2085] $712231513221(12)12231513221(12)12^1 2315132181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 20 & 3 \end{pmatrix}_{b_0}$
- [2086] $7123151322^1 1(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2087] $81223151322^1 1(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2088] $7123151322^1 1(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2089] $81223151322^1 1(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2090] $1(12)122315132181232^1 18123218123218123215+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2091] $1(12)122315132181232^1 18123218123218123151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2092] $41413221(12)12231513221(12)12^1 231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2093] $41413221(12)12231513221(12)12^1 231513221(12)12+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 169, (n_+, n_-, n_0) =$

(1, 38, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2094] 3151321812321812¹32181232181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2095] 512321812321812321812¹32181231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2096] 151321812321812321812¹32181231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2097] 613161315132¹21(12)12231513221(12)12231513161+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2098] (12)122315132181232181232181232¹1812315132+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2099] 41413221(12)12231513221¹(12)12¹231513221(12)1222+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 169$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2100] 413221¹(12)12231513221(12)122¹315132181232181+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 52$, $T \in [43, 52]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2101] 613161315132¹21(12)12231513221(12)12231513216+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -109$, $\det = 169$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.59 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2117] 812231513221¹(12)12¹231513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2114] 181231513221¹(12)12¹231513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2130] 413221¹(12)1223151322¹1(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 52$, $T \in [43, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2132] $41413221(12)1223151322^1 1(12)12231513218123217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2118] $71231513221(12)12^1 231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2119] $812231513221(12)12^1 231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2120] $1(12)1223151321812^1 3218123218123218123215+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2121] $1(12)1223151321812^1 3218123218123218123151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2113] $321812321812315132^1 21(12)12231513221(12)12231+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 43, \Delta = -93, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[2112] $71231513221(12)12^1 231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2111] $71232181231513221(12)12^1 231513221(12)1+1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2110] $21(12)12231513221(12)1223151322^1 1(12)122315+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2109] $1(12)12231513221(12)1223151322^1 1(12)122315+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2108] $1(12)1223151322^1 1(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2107] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 196, (n_+, n_-, n_0) =$

$(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

$[2106] \overset{1513221}{1} (12) \overset{1}{1223151322} \overset{21}{1} (12) 12231513221 (12) 12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

$[2105] \overset{221}{1} (12) 12231513221 (12) 12 \overset{1}{1} 231513221 (12) 1223151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[2104] \overset{3221}{1} (12) 12231513221 (12) 122315132 \overset{1}{1} 21 (12) 122315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

$[2103] \overset{513221}{21} (12) 12 \overset{1}{1} 231513221 (12) 12231513221 (12) 12222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

$[2122] 41413221 (12) 1223151322 \overset{1}{1} (12) 12231513221 (12) 1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[2123] 41413221 (12) 1223151322 \overset{1}{1} (12) 12231513221 (12) 12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[2124] 315132181232181232 \overset{1}{1} 181232181231513221 (12) 1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

$[2125] 51232181232181232181232 \overset{1}{1} 181231513221 (12) 12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

$[2126] 15132181232181232181232 \overset{1}{1} 181231513221 (12) 12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

$[2127] 61316131513221 (12) 12 \overset{1}{1} 231513221 (12) 12231513161 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

$[2128] (12) 1223151321812321812321812 \overset{1}{1} 321812315132 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 196$, $(n_+, n_-, n_0) =$
 $(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2131] & 61316131513221(12)12^1 231513221(12)12231513216+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -109, \det = 196, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2116] & 71231513221(12)12^1 231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2115] & 712231513221(12)1223151322^1 1(12)122315132181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2129] & 41413221(12)1223151322^1 1(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 196, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.55 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2136] & 1 \begin{pmatrix} 31513221 \\ 12 \end{pmatrix} 122315132^1 21(12)12231513221(12)122+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2135] & 1 \begin{pmatrix} 23151322^1 1 \\ 12 \end{pmatrix} 12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2134] & 1 \begin{pmatrix} 21 \\ 12 \end{pmatrix} 12231513221(12)122^1 31513221(12)12231513+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2133] & 1(12)12231513221(12)12231513221(12)12^1 2315132+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2137] & 712321812^1 31513221(12)12231513221 \begin{pmatrix} 1513221 \\ 12 \end{pmatrix} 1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2138] & 413221(12)12231513221(12)122315132^1 181232181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2139] & 41413221(12)12231513221(12)122315132^1 18123217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2140] $1(12)122315132^1 21(12)12231513221(12)122315132+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2141] $1(12)12231513221(12)1223151322^1 1(12)12231513+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2142] $1(12)12231513221(12)122^1 31513221(12)12+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2143] $1(12)12^1 231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2144] $7123218123151322^1 1(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2145] $1(12)122315132^1 18123218123218123218123215+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2146] $1(12)122315132^1 18123218123218123218123151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2147] $3151321812321812321812^1 32181231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 54, \Delta = -114, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[2154] $41413221(12)12231513221(12)12^1 231513218123217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2153] $6131613151322^1 1(12)12231513221(12)12231513216+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -109, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2152] $413221(12)12231513221(12)12^1 2315132181232181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2151] $(12)122315132181232181232^1 1812321812315132+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.48 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2150] & 6131613151322^1 1(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 256, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[2149] & 151321812321812321812321812^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 256, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2148] & 512321812321812321812321812^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 256, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2156] & 1(12)12231513221(12)1223151322^1 1(12)122315+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2155] & 1(12)12231513221(12)122315132^1 21(12)122315132+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2157] & 1(12)122315132^1 21(12)12231513221(12)122315+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[2158] & 1(12)1223151322^1 1(12)12231513221(12)12231513+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2159] & 1(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2160] & 1(12)12231513221(12)12^1 231513221(12)122+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2161] & 3218123218123151322^1 1(12)12231513221(12)12231+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 43, \Delta = -93, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[2162] & 315132181232181232181232^1 181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 54, \Delta = -114, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 54, T \in [43, 54] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2163] $(12)1223151321812321812^1 321812321812315132+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 289$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.46 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [2164] $1(12)12231513221(12)122^1 31513221(12)122315+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 324$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.43 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2165] $321812321812^1 31513221(12)12231513221(12)12231+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 43$, $\Delta = -93$, $\det = 324$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.43 \rightarrow 43$, $T_{\max} = 43$, $T \in [43, 43]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [2170] $1(12)122315132^1 21(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 361$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [2173] $(12)122315132181232^1 1812321812321812315132+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 361$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2172] $3151321812321812321812321812^1 31513221(12)1+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 361$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [2171] $1(12)1223151322^1 1(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 361$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [2169] $1(12)1223151322^1 1(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 361$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [2168] $1(12)12231513221(12)122^1 31513221(12)1+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 361$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2167] $1(12)12231513221(12)12^1 231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 361$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [2166] $1(12)12231513221(12)122^1 31513221(12)122315132+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 361$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [2174] $(12)1223151321812^1 321812321812321812315132+\hat{1}\rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 400$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.39 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2175] & 1 \overset{1}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 122315132 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 441, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.37 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[2176] & 1 \overset{2231513221}{(12)} 122^1 31513221 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 441, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.37 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[2177] & 1 \overset{231513221}{(12)} 12^1 231513221 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 441, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.37 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2178] & 1 \overset{231513221}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 \overset{1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 441, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.37 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[2179] & 1 \overset{3221}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 484, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.36 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2180] & (12) 122315132^1 1812321812321812321812315132 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 484, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.36 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[2181] & 1 \overset{3221}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 \overset{1}{(12)} 122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 529, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.34 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[2182] & 1 \overset{2231513221}{(12)} 122315132^1 21 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 529, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.34 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[2183] & 1 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 \overset{1}{(12)} 122315132 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 576, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.33 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[2184] & 1 \overset{231513221}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 576, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.33 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[2185] & 1 \overset{2231513221}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 625, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.31 \rightarrow 43, T_{\max} = 58, T \in [43, 58]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[2186] & 1 \quad (12) \quad 12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 676, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.30 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[2187] & 1 \quad (12) \quad 12231513221^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 841, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.27 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0} \\
T = 39 & [2188] \quad 81231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 55, \Delta = -114, \det = \\
& -9, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.22 \rightarrow 46, T_{\max} = 55, T \in [46, 55] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2190] & 1612^1 32181232181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -16, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.75 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2189] & 1(12)122315132181232181232181232181232^1 161+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -16, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.75 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2200] & 61316131513221(12)12231513221(12)122315132^1 217+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2199] & 61316131513221(12)12231513221(12)122315132^1 171+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2198] & 812231513221(12)12231513221(12)122315132^1 2181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2197] & 812231513221(12)12231513221(12)122315132^1 1812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2196] & 181231513221(12)12231513221(12)122315132^1 2181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2195] & 181231513221(12)12231513221(12)122315132^1 1812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2194] $812231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2193] $1812^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2192] $4132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2191] $1^{15132^1 21}(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2201] $1(12)1223151321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2202] $161232^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2206] $1812^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2207] $181231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2209] $8122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2205] $8122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2204] $181231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45$, $T_{\max} = 51$, $T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2203] $1^{1513221}(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) =$

$$(1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2208] \text{ } 1812^1 31513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2210] \text{ } 812231513221(12)12231513221(12)1223151322^1 181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2211] \text{ } 8122^1 31513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2212] \text{ } 61316131513221(12)12231513221(12)1223151322^1 17+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2213] \text{ } 41413221(12)12231513221(12)122315132181232^1 181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2224] \text{ } 812231513221(12)12231513221(12)122^1 3151321812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2232] \text{ } 31612^1 32181232181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 54, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2214] \text{ } 1^{151322^1 1}(12)12231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[2215] \text{ } 812^1 32181231513221(12)12231513221^{31513221}(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2216] \text{ } 41322^1 1(12)12231513221(12)12231513221^{513221}(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2217] \text{ } 413221(12)12231513221(12)12231513221^{5132^1 21}(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[2218] & \begin{matrix} 1 \\ 1812315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2219] & \begin{matrix} 1 \\ 812231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2220] & \begin{matrix} 1 \\ 812^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2221] & \begin{matrix} 1 \\ 181231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2222] & \begin{matrix} 1 \\ 181231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2223] & \begin{matrix} 1 \\ 812^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 55, \Delta = -114, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 55, T \in [45, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2227] & \begin{matrix} 1 \\ 812^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2225] & \begin{matrix} 1 \\ 812^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2228] & \begin{matrix} 1 \\ 61316131513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2229] & \begin{matrix} 1 \\ 61316131513221(12)12231513221(12)122^1 315132217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2230] & \begin{matrix} 1 \\ 1(12)122315132181232181232^1 181232161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2231] $1612321812^1 32181232181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[2233] $414132^1 21(12)12231513221(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 52, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.06 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[2226] $812231513221^1(12)12231513221(12)122^1 3151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2234] $81232181231513221(12)12231513221^1(12)1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2235] $1(12)122^1 315132181232181232181232181232161+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2236] $1(12)1223151321812321812321812^1 32181232161+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2237] $161232181232^1 181232181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2239] $3161232^1 181232181232181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2238] $1612321812321812321812321812315132^1 21(12)12+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2250] $181231513221^1(12)12231513221(12)12^1 23151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2256] $812231513221^1(12)12231513221(12)12^1 23151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2240] $21(12)122315132^1 21(12)12231513221(12)122313+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 44, \Delta = -95, \det = -81, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 513221 \\ [2241] \end{matrix} 21 (12) 12231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 44, \Delta = -95, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ [2242] \end{matrix} 1 (12) 122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ [2243] \end{matrix} 1 (12) 12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2244] \end{matrix} 18123151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2245] \end{matrix} 181231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2246] \end{matrix} 812231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2247] \end{matrix} 812231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2248] \end{matrix} 181231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& [2262] 16123218123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2263] 31612321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 54, \Delta = -113, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 54, T \in [44, 54] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2264] 41413221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 52, T \in [44, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2265] 41413221(12)12231513221(12)1223151321812¹32181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 52$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2266] 4141322¹1(12)12231513221(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 52$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2261] 1(12)12¹2315132181232181232181232161+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2260] 61316131513221(12)12231513221(12)12¹2315132217+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2249] 181231513221¹(12)12231513221(12)12¹23151321812+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2252] 81231513221¹(12)12231513221(12)122315132¹18123+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 55$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2253] 812231513221¹(12)122315132¹21(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2254] 812231513221¹(12)12231513221(12)12¹23151321812+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2255] 812231513221¹(12)122315132¹21(12)1223151322181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2259] 61316131513221(12)122315132¹21(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2258] 61316131513221(12)122315132¹21(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2257] 61316131513221(12)12231513221(12)12¹2315132171+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2251] \text{ } 181231513221^1(12)122315132^121(12)1223151322181+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2273] \text{ } 8122315132^121(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2274] \text{ } 8122315132^121(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2275] \text{ } 1(12)122315132181232181232^1181232181232161+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2276] \text{ } 1612321812321812^132181232181231513221(12)12+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2277] \text{ } 31612321812^132181232181232181231513221(12)1+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 54, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2278] \text{ } 316123218123218123218123218123151322^11(12)1+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 54, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2272] \text{ } 1812315132^121(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2271] \text{ } 1812315132^121(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2270] \text{ } 8122315132^121(12)12231513221(12)122315132181+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2269] \text{ } 181231513221(12)12231513221(12)122^1315132181+\hat{1}\rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2268] $413221(12)12231513221(12)12231513221 \overset{1}{(12)} 1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.76 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2267] $21 \overset{51322^1 1}{(12)} 12231513221(12)12231513221(12)122313+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 44, \Delta = -95, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.76 \rightarrow 44, T_{\max} = 44, T \in [44, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2302] $61316131513221(12)122^1 31513221(12)122315132217+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2279] $21 \overset{513221}{(12)} 122^1 31513221(12)12231513221(12)122313+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 44, \Delta = -95, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 44, T \in [44, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2305] $3161232181232^1 181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 54, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2306] $41413221(12)12^1 231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2307] $41413221(12)122315132^1 21(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2281] $1 \overset{1513221}{(12)} 12^1 231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2282] $1 \overset{1513221}{(12)} 122315132^1 21(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2283] $81232^1 181231513221(12)12231513221 \overset{31513221}{(12)} 1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2284] $81232181231513221(12)12231513221 \overset{3151322^1 1}{(12)} 1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2301] $61316131513221(12)1223151322^1 1(12)122315132217+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2300] & 61316131513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2303] & 1(12)1223151321812321812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2285] & 413221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 56, T \in [44, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2286] & 181231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2287] & 181231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2288] & 812231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2289] & 812231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2290] & 181231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2291] & 181231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2292] & 181231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2293] & 181231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2304] & 161232181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2294] & 81231513221(12)12231513221(12)122^1 31513218123+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 55, \Delta = -114, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 55, T \in [44, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2295] & 812231513221(12)1223151322^1 1(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2296] & 812231513221(12)122^1 31513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2297] & 812231513221(12)122^1 31513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2298] & 812231513221(12)1223151322^1 1(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2299] & 61316131513221(12)122^1 31513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2280] & 21(12)1223151322^1 1(12)12231513221(12)122313+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 44, \Delta = -95, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2324] & 41413221(12)12231513221(12)122^1 315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -169, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.59 \rightarrow 44, T_{\max} = 52, T \in [44, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2308] & 1(12)12231513221(12)122^1 315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2309] & 81232181231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.59 \rightarrow 44, T_{\max} = 56, T \in [44, 56] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2310] $413221(12)12^1 231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2311] $181231513221(12)12231513221(12)12^1 2315132181+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2312] $812231513221^1 1(12)12231513221(12)122315132181+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2313] $181231513221^1 1(12)12231513221(12)1223151321812+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2314] $181231513221^1 1(12)12231513221(12)1223151322181+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2315] $812315132^1 21(12)12231513221(12)12231513218123+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 55, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2316] $81231513221(12)122315132^1 21(12)12231513218123+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 55, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2317] $812231513221^1 1(12)12231513221(12)1223151321812+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2318] $812231513221^1 1(12)12231513221(12)1223151322181+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2319] $613161315132^1 21(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2320] $613161315132^1 21(12)12231513221(12)122315132217+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2321] $1(12)122315132181232^1 181232181232181232161+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) =$

(1, 39, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 43.59 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[2322] 1612321812321812321812¹32181231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.59 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2323] 31612321812321812¹32181232181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 54$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.59 \rightarrow 44$, $T_{\max} = 54$, $T \in [44, 54]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2332] 81231513221(12)¹12231513221(12)12¹231513218123+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 55$, $\Delta = -114$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 55$, $T \in [44, 55]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2325] 21 (12) 12¹231513221(12)12231513221(12)122313+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 44$, $\Delta = -95$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 44$, $T \in [44, 44]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2326] 1¹⁵¹³²²¹(12) 1223151322¹1(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[2327] 413221(12)12231513221(12)¹122315132¹21 (12) 1+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 56$, $T \in [44, 56]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2328] 181231513221(12)1223151322¹1(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2329] 812231513221(12)12¹231513221(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2330] 181231513221(12)12¹231513221(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2340] 41413221(12)1223151322¹1(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 52$, $\Delta = -111$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[2339] 3161232181232181232¹181232181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 54$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 54$, $T \in [44, 54]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2338] & 161232181232181232181232^1 181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2337] & 1(12)1223151321812^1 32181232181232181232161+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0} \\
[2331] & 181231513221(12)12^1 231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2335] & 61316131513221(12)12^1 231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2334] & 812231513221(12)12^1 231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2333] & 812231513221(12)12^1 231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2336] & 61316131513221(12)12^1 231513221(12)122315132217+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2344] & 41413221(12)12231513221(12)122315132^1 181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -225, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.52 \rightarrow 44, T_{\max} = 52, T \in [44, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[2343] & 81231513221(12)122^1 31513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 55, \Delta = -114, \det = -225, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.52 \rightarrow 44, T_{\max} = 55, T \in [44, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2342] & 81232181231513221(12)122^1 31513221 \quad (12) \quad 1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -225, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.52 \rightarrow 44, T_{\max} = 56, T \in [44, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2341] & 1 \quad (12) \quad 12231513221(12)122315132^1 181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -225, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.52 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[2351] & 1612321812321812321812321812^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -256, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 39, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2345] $1^{1513221}(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2346] $81232181231513221(12)1223151322^1 1^{31513221}(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 56$, $T \in [44, 56]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2347] $81231513221(12)^1 1223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 55$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 55$, $T \in [44, 55]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2348] $6131613151322^1 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[2349] $6131613151322^1 1(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2350] $1^{11}(12)122315132^1 181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[2352] $31612321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 54$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 54$, $T \in [44, 54]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2353] $41413221(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 52$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2357] $8123151322^1 1(12)^1 12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 55$, $\Delta = -114$, $\det = -289$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.46 \rightarrow 44$, $T_{\max} = 55$, $T \in [44, 55]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2354] $812321812315132^1 21(12)12231513221^{31513221}(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.46 \rightarrow 44$, $T_{\max} = 56$, $T \in [44, 56]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2355] $413221(12)12231513221(12)1223151322^1 1^{513221}(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.46 \rightarrow 44$, $T_{\max} = 56$, $T \in [44, 56]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2356] $413221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.46 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2358] $3161232181232181232181232^1 181231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.46 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2359] $413221(12)12231513221(12)122^1 31513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -324, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.43 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2363] $31612321812321812321812321812^1 31513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.41 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[2360] $81232181231513221(12)12^1 231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.41 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[2361] $812321812^1 31513221(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.41 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[2362] $81231513221(12)12^1 231513221(12)12231513218123+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 55, \Delta = -114, \det = -361, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.41 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[2364] $8123218123151322^1 1(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.37 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2365] $413221(12)1223151322^1 1(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -484, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.36 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2366] $413221(12)12231513221(12)12^1 231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.34 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$T = 40$ [2367] $1(12)12231513221(12)12231513221(12)1223151315+ \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47, T_{\max} = 52, T \in [47, 52]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2368] $22^1 21(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47, T_{\max} = 52, T \in [47, 52]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2369] $222^1 1(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 52, T \in [47, 52]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2370] $32^1 171232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[2374] $21712^1 32181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.75 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2373] $1(12)122315132181232181232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.75 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2372] $1712^1 32181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.75 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2371] $1712^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.75 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2377] $61316131513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2376] $61316131513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2379] $171232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2375] $1(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2380] $\overset{1}{171232^1 181232181232181232181231513221(12)12+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2381] $\overset{1}{1(12)1223151321812321812321812321812^1 321712+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2382] $\overset{1}{2171232^1 181232181232181232181231513221(12)12+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2383] $\overset{1}{(12)1223151321812321812321812321812321712^1 32+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 56, T \in [46, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[2378] $\overset{1}{61316131513221(12)12231513221(12)122315132^1 2181+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2388] $\overset{1}{61316131513221(12)12231513221(12)1223151322^1 181+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2387] $\overset{1}{712^1 32181231513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -110, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2386] $\overset{1}{2221(12)122^1 31513221(12)12231513221(12)1223151315 + \hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2385] $\overset{21}{21(12)122^1 31513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2384] $\overset{221}{1(12)122^1 31513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2399] $\overset{1}{61316131513221(12)12231513221(12)122^1 3151321812+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[2398] $\overset{1}{61316131513221(12)12231513221(12)122^1 315132181+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2397] $71232181231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2396] $2221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2395] $2221(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2394] $21(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2393] $21(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2400] $61316131513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2391] $1(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2390] $1(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2389] $1(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2401] $1712321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2402] $1712321812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& [2403] \overset{1}{1}(12)122315132181232181232181232^1 1812321712+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 49, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52] \\
& \left(\begin{smallmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{smallmatrix} \right)_{b_0} \\
& [2404] 321712^1 32181232181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.06 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \left(\begin{smallmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{smallmatrix} \right)_{b_0} \\
& [2405] 21712321812^1 32181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 49, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52] \\
& \left(\begin{smallmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{smallmatrix} \right)_{b_0} \\
& [2392] \overset{5132^1 21}{1} (12) 12231513221(12)12231513221(12)1223151+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 49, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52] \\
& \left(\begin{smallmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{smallmatrix} \right)_{b_0} \\
& [2406] \overset{1}{17}12321812321812321812321812315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \left(\begin{smallmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{smallmatrix} \right)_{b_0} \\
& [2411] \overset{1}{1}(12)1223151321812321812321812^1 321812321712+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \left(\begin{smallmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{smallmatrix} \right)_{b_0} \\
& [2410] \overset{1}{1}(12)122^1 3151321812321812321812321812321712+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \left(\begin{smallmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{smallmatrix} \right)_{b_0} \\
& [2412] 32171232^1 181232181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 54, T \in [45, 54] \\
& \left(\begin{smallmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{smallmatrix} \right)_{b_0} \\
& [2413] 2171232181232^1 181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \left(\begin{smallmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{smallmatrix} \right)_{b_0} \\
& [2409] \overset{1}{17}1232181232^1 181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \left(\begin{smallmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{smallmatrix} \right)_{b_0} \\
& [2414] 21712321812321812321812321812315132^1 21(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \left(\begin{smallmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{smallmatrix} \right)_{b_0} \\
& [2408] \overset{1}{17}12321812321812321812321812315132^1 21(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2407] & \begin{matrix} 1 \\ 171232181232^1 181232181232181231513221(12)1+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2426] & \begin{matrix} 1 \\ 61316131513221(12)122315132^1 21(12)1223151321812+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2434] & \begin{matrix} 1 \\ 217123218123218123218123218123151322^1 1(12)12+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2415] & \begin{matrix} 221 \\ 1(12)12231513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2416] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2417] & \begin{matrix} 21 \\ 21(12)12231513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2418] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2419] & \begin{matrix} 1 \\ 2221(12)12231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = \\
& 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2420] & \begin{matrix} 1 \\ 2221(12)12231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = \\
& 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2421] & \begin{matrix} 1 \\ 71232181231513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2422] & \begin{matrix} 1 \\ 71232^1 181231513221(12)12231513221(12)1223151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2423] & \begin{matrix} 1 \\ 71232181231513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 81, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2424] \text{ 61316131513221(12)12231513221(12)12}^1\text{2315132181}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2425] \text{ 61316131513221(12)122315132}^1\text{21(12)122315132181}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2427] \text{ 61316131513221(12)12231513221(12)12}^1\text{23151321812}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2428] \text{ 61316131513221(12)122315132}^1\text{21(12)1223151322181}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2429] \text{ 61316131513221(12)12231513221(12)12}^1\text{23151322181}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2430] \text{ 17123218123218123218123218123151322}^1\text{1(12)1}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2432] \text{ 1(12)12}^1\text{23151321812321812321812321812321712}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2433] \text{ 321712321812321812321812321812315132}^1\text{21(12)1}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2431] \text{ 17123218123218123218123218123151322}^1\text{1(12)12}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2443] \text{ 321712321812}^1\text{32181232181232181231513221(12)1}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.76 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2442] \text{ 3217123218123218123218123218123151322}^1\text{1(12)1}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.76 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2441] $1(12)122315132181232181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2440] $1712321812321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2439] $1712321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2438] $2221(12)122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2437] $21(12)122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2436] $1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2435] $1(12)122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2444] $21712321812321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2445] $(12)122315132181232181232181232^1 171232 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2446] $(12)122^1 315132181232181232181232181232171232 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2462] $171232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2468] $(12)1223151321812321812321812321812^1 32171232 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 56, \quad T \in [45, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2448] \quad 1(12)12231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2449] \quad 1(12)12231513221(12)122315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -110, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2450] \quad 21(12)12231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& [2463] \quad 171232181232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -103, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [2451] \quad 21(12)12231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2452] \quad 2221(12)12231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = \\
& 121, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2464] \quad 1(12)1223151321812321812^1 321812321812321712 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -103, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
& [2467] \quad (12)12^1 2315132181232181232181232171232 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 56, \quad \Delta = -114, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 56, \quad T \in [45, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
& [2465] \quad 32171232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 54, \quad \Delta = -112, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 54, \quad T \in [45, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [2453] \quad 2221(12)12231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = \\
& 121, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& [2454] \quad 71232181231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -110, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2455] $71232181231513221(12)1223151322^11(12)1223151315+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2456] $61316131513221(12)122^131513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2457] $61316131513221(12)1223151322^11(12)122315132181+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2458] $61316131513221(12)122^131513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2459] $61316131513221(12)1223151322^11(12)1223151321812+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2460] $61316131513221(12)1223151322^11(12)1223151322181+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2461] $61316131513221(12)122^131513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2466] $2171232181232181232^1181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2447] $1^{221}(12)12231513221(12)122^131513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2470] $1^{513221}(12)12231513221(12)1223151322^11(12)1223151+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -110, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2480] $321712321812321812^132181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 54, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2482] $(12)122315132181232181232181232^1181232171232+\hat{1}\rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2469] $1(12)1223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2481] $21712321812321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2471] $21(12)1223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2472] $2221(12)1223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2473] $712321812315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -110, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2474] $613161315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[2475] $613161315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[2476] $613161315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[2477] $1712321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2478] $1712321812321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2479] $1(12)122315132181232^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[2496] $(12)1223151321812321812321812^1 32181232171232 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.55 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2494] $32171232181232181232^1 181232181231513221(12)1+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 54$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2483] $1(12)12231513221(12)12^{221}231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2484] $1(12)122^131513221(12)12231513221(12)1223151+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2485] $21(12)12231513221(12)12^{21}231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2486] $2221(12)12231513221(12)12^1231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2487] $71232181231513221(12)12^1231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2488] $61316131513221(12)12^1231513221(12)122315132181+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2489] $61316131513221(12)12^1231513221(12)1223151321812+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2490] $61316131513221(12)12^1231513221(12)1223151322181+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2491] $171232181232181232181232^1 181231513221(12)1+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2492] $171232181232181232181232^1 181231513221(12)12+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2495] $2171232181232181232181232^1 181231513221(12)12+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) =$

$$(1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2493] \overset{1}{1}(12)1223151321812^1 321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

$$[2497] 712321812^1 31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 225, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.52 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2505] 321712321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[2502] \overset{1}{17}12321812321812321812321812^1 31513221(\overset{1}{12})1 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[2501] 6131613151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[2503] \overset{1}{17}12321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[2498] 7123218123151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2499] 6131613151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[2500] 6131613151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[2504] \overset{1}{1}(12)122315132^1 1812321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

$$[2506] 21712321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.48 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

- [2507] $(12)122315132181232181232^1 181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2508] $1 \overset{513221}{(12)} 12231513221 \overset{1}{(12)} 122^1 31513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.46 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [2509] $1 \overset{513221}{(12)} 12^1 231513221 \overset{1}{(12)} 12231513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.46 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [2510] $32171232181232181232181232^1 181231513221 (12)1 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 54$, $\Delta = -112$, $\det = 289$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.46 \rightarrow 45$, $T_{\max} = 54$, $T \in [45, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [2511] $(12)1223151321812321812^1 32181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.46 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [2512] $1 \overset{513221}{(12)} 122315132^1 21 \overset{1}{(12)} 12231513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 324$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.43 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2513] $321712321812321812321812321812^1 31513221 (12)1 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 54$, $\Delta = -112$, $\det = 361$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.41 \rightarrow 45$, $T_{\max} = 54$, $T \in [45, 54]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [2514] $(12)122315132181232^1 181232181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 361$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.41 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2515] $(12)1223151321812^1 32181232181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 400$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.39 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [2517] $(12)122315132^1 181232181232181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 484$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.36 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [2516] $1 \overset{513221}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 484$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.36 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2518] $1 \overset{513221}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 529$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.34 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$T = 41 \quad [2519] \quad 1^{22^1 1} (12)12231513221(12)^1 12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2520] \quad 61231513221(12)12231513221(12)^1 12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2521] \quad 6131513221(12)^1 12231513221(12)12231513221(12)^{22^1 21} 1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2522] \quad 6131513221(12)^1 12231513221(12)12231513221(12)^{22^1 1} 12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2523] \quad 22^1 21(12)12231513221(12)^1 12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2524] \quad 22^1 1(12)12231513221(12)^1 12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 57, T \in [48, 57]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2525] \quad 61231513221(12)^1 12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2526] \quad 6131513221(12)^1 12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2527] \quad 1(12)^1 1223151321812321812321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[2528] \quad 22^1 181232181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2529] \quad 6131513221(12)^1 12231513221(12)12231513221(12)^{32^1 21} 1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2530] $6131513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2531] $222^1 1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2532] $61231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2535] $32^1 2181232181232181232181232181231513221(12)1+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 54, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[2534] $1^1 8^1 1232181232181232181232181231513221(12)1+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 54, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[2533] $6131513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2538] $1(12)122315132181232181232181232181232^1 18122+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2540] $221812^1 32181232181232181232181231513221(12)12+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2536] $6131513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2537] $1^1 8^1 12^1 32181232181232181232181231513221(12)1+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2539] $1^1 8^1 12^1 32181232181232181232181231513221(12)12+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [2541] $6131513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 58$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2542] $612^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2543] $612^1 31513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2544] $6131513221(12)12231513221(12)122315132^1 21(12)1+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2545] $6131513221(12)12231513221(12)122315132^1 21(12)12+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2546] $612^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2547] $6131513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2548] $612^1 31513221(12)12231513221(12)12231513218123217+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2549] $1^1 81232^1 181232181232181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2550] $1(12)1223151321812321812321812321812^1 3218122+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2551] $1^1 81232^1 181232181232181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2552] $(12)1223151321812321812321812321812321^1 81+\hat{1}\rightarrow 2$ $T = 41$, $T_{\max} = 56$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) =$

$(1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 56$, $T \in [47, 56]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2553] $322^1 181232181232181232181232181231513221(12)1+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 54$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2554] $22181232^1 181232181232181232181231513221(12)12+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2555] $1(12)122^1 31513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2556] $21(12)122^1 31513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2557] $61231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2558] $61231513221(12)12231513221(12)122315132^1 21(12)12+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2559] $6131513221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2560] $6131513221(12)12231513221(12)1223151322^1 1(12)12+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2561] $1812^1 32181231513221(12)12231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2562] $2221(12)122^1 31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2563] $61231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2564] \quad 6131513221(12)^1 12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2565] \quad 712^1 32181231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2566] \quad 61231513221(12)12231513221(12)122315132181232^1 17 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2573] \quad 61231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2592] \quad 221812321812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2569] \quad 21(12)^{21} 12^1 231513221(12)^1 12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2570] \quad 21(12)^{21} 12231513221(12)^1 12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2571] \quad 1(12)^{21} 122^1 31513221(12)^1 12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 57, T \in [47, 57]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2572] \quad 612315132^1 21(12)12231513221(12)^1 12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2567] \quad 1(12)^{221} 12231513221(12)^1 12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2574] \quad 612315132^1 21(12)12231513221(12)^1 12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

- [2575] $61231513221(12)12231513221(12)1223151322^11(12)12+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2576] $6131513221(12)12231513221(12)122^131513221(12)1+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2577] $6131513221(12)12231513221(12)122^131513221(12)12+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2578] $181232181231513221(12)12231513221(12)122^13151315+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2579] $2221(12)12231513221(12)12231513221(12)122^131513161 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2580] $2221(12)12^1231513221(12)12231513221(12)12231513161 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2581] $221(12)122^131513221(12)12231513221(12)122315131613 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 57, T \in [47, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2582] $61231513221(12)12231513221(12)1223151322^11(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2583] $612315132^121(12)12231513221(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2584] $6131513221(12)12231513221(12)122^131513221(12)12222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2585] $71232181231513221(12)12231513221(12)122^131513161+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2586] $612315132^121(12)12231513221(12)12231513218123217+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [2587] \quad 1 \quad 8 \quad 12321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2588] \quad 1 \quad 8 \quad 12^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 54, \quad \Delta = -111, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 54, \quad T \in [47, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2589] \quad 1(12)122315132181232181232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [2590] \quad 1 \quad 8 \quad 12321812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2591] \quad 3221812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 54, \quad \Delta = -111, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 54, \quad T \in [47, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2568] \quad 1(12)12^1 231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -106, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2593] \quad 1(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 57, \quad \Delta = -114, \quad \det = -64, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 57, \quad T \in [46, 57] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
& [2594] \quad 221(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 57, \quad \Delta = -114, \quad \det = \\
& -64, \quad (n_+, n_-, n_0) = (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 57, \quad T \in [46, 57] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
& [2595] \quad 1 \quad 8 \quad 12321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -64, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 53, \quad T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2596] \quad 1 \quad 8 \quad 1232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -64, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 53, \quad T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2597] \quad 1 \quad 8 \quad 1232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 54, \quad \Delta = -111, \quad \det = -64, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 54, \quad T \in [46, 54]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2598] & \overset{1}{1}(12)1223151321812321812321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2599] & \overset{1}{1}(12)122^1 31513218123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2600] & \overset{21}{1}81232181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2601] & \overset{21}{1}812321812321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2602] & 322181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2603] & 221812321812321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2604] & 22181232181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2616] & \overset{1}{6}131513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2637] & 2218123218123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2624] & \overset{1}{6}131513221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2623] & \overset{1}{6}1231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2622] & \overset{1}{6}123151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det =
\end{aligned}$$

$$-81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2621] \quad 2221(12)12231513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2620] \quad 2221(12)12231513221(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2619] \quad 181232181231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2618] \quad 181232^1 181231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2615] \quad 6131513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2626] \quad 71232^1 181231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2617] \quad 181232181231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2614] \quad 6131513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2613] \quad 6131513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2612] \quad 6123151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2611] \quad 61231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2610] \ 61231513221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2609] \ 6123151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2608] \ 21(12)12231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2607] \ 21(12)12231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2606] \ 1(12)12231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2605] \ 1(12)12231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2627] \ 71232181231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2628] \ 71232181231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2629] \ 6123151322^1 1(12)12231513221(12)12231513218123217+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2630] \ 61231513221(12)122^1 31513221(12)12231513218123217+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2631] \ 61231513221(12)12231513221(12)1223151321812^1 3217+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2632] \ 1^1 8123218123218123218123151322^1 1(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -81, (n_+, n_-, n_0) =$$

$$(1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2633] \overset{321}{1 \ 8 \ 12321812321812321812321812315132^1 21(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2634] \overset{1}{1(12)12^1 231513218123218123218123218123218122+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2635] \overset{21}{1 \ 8 \ 123218123218123218123218123151322^1 1(12)12+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2636] \overset{321}{3221812321812321812321812321812315132^1 21(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2625] \overset{1}{6131513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2649] \overset{321}{1 \ 8 \ 123218123218123218123218123151322^1 1(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2639] \overset{21}{21(12)122315132^1 21(12)12231513221(12)12231513161+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2640] \overset{3221}{61315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2641] \overset{1}{61231513221(12)12231513221(12)122^1 31513221(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2642] \overset{1}{61231513221(12)12231513221(12)122^1 31513221(12)12+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2643] \overset{2221}{61315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2644] & 61315132^1 21(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2645] & 2221(12)122315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = \\
& -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2646] & 61231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = \\
& -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2647] & 61315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = \\
& -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2648] & 1^2 8 12321812321812^1 32181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2638] & 1(12)122315132^1 21(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2650] & 1^3 8 12321812^1 32181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2651] & 1(12)122315132181232181232^1 18123218123218122+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2652] & 1^2 8 12321812321812^1 32181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2653] & (12)122315132181232181232181232181232^1 1^2 8 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 56, T \in [46, 56] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2654] & (12)122^1 3151321812321812321812321812321^2 8 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 56, T \in [46, 56]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2655] & 3221812321812^1 32181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2656] & 32218123218123218123218123218123151322^1 1(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2657] & 221812321812321812^1 32181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2666] & 61231513221(12)122315132^1 21(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2690] & (12)12^1 23151321812321812321812321812321^{\frac{2321}{8}} 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 56, T \in [46, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2660] & 21(12)12231513221(12)1223151322^1 1(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2661] & 21(12)12231513221(12)122^1 31513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2662] & 1(12)12231513221(12)12231513221(12)122^1 315131613+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2663] & 6131513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 58, T \in [46, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2664] & 61231513221(12)12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2665] & 61231513221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2658] $1^{221}(12)12231513221(12)1223151322^11(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2667] $61231513221(12)12^1231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2668] $6131513221(12)1223151322^11(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2669] $6131513221(12)122^131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [2670] $6131513221(12)122^131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [2671] $6131513221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2672] $181232181231513221(12)1223151322^11(12)1223151315 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2673] $181232181231513221(12)122^131513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2674] $2221(12)12231513221(12)122^131513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2675] $2221(12)12231513221(12)1223151322^11(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2676] $221(12)12231513221(12)12231513221(12)122^1315131613 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 57$, $\Delta = -114$, $\det = -121$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46$, $T_{\max} = 57$, $T \in [46, 57]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2677] $61231513221(12)12^1231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det =$

$$-121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2678] \quad 61231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2679] \quad 6131513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2680] \quad 6131513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2681] \quad 71232181231513221(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[2682] \quad 71232181231513221(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[2683] \quad 61231513221(12)12^1 231513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2684] \quad 61231513221(12)122315132^1 21(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2685] \quad 1^1 8 1232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2686] \quad 1^1 8 1232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2687] \quad 1^1 (12)1223151321812321812^1 3218123218123218122 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2688] \quad 1^1 8 1232181232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2691] & 322181232181232^1 181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2692] & 22181232181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2689] & (12)1223151321812321812321812321812^1 321 \overset{2321}{8} 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 56, T \in [46, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2659] & 1(12)12231513221(12)122^1 31513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2701] & 613151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2703] & 1812321812315132^1 21(12)12231513221(12)1223151315+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2694] & 21(12)1223151322^1 1(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2695] & 1(12)122315132^1 21(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2704] & 2221(12)1223151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2705] & 221(12)12231513221(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2706] & 221(12)122315132^1 21(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2707] $61231513221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2708] $613151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2709] $712321812315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2710] $61231513221(12)12231513221(12)122^1 31513218123217 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -108$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2711] $1^1 812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2712] $1^1 812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 54$, $\Delta = -111$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 54$, $T \in [46, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2713] $1^1 (12)122315132181232^1 18123218123218123218122 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [2714] $1^1 812321812321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2715] $(12)122315132181232181232181232^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 56$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 56$, $T \in [46, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2716] $3221812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 54$, $\Delta = -111$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 54$, $T \in [46, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2717] $221812321812321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2702] $613151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2700] & 61231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2696] & 1(12)12231513221(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2697] & 613151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 58, T \in [46, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2698] & 6131513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 58, T \in [46, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2699] & 61231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2693] & 1(12)1223151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2724] & 6131513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2728] & 221(12)12231513221(12)12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = \\
& -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2723] & 61231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2722] & 61231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2726] & 181232181231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -196, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]
\end{aligned}$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2727] \quad 2221(12)12231513221(12)12^{1231513221(12)12231513161} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2739] \quad 22181232181232181232181232^{1181231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2719] \quad 21(12)12231513221(12)12^{1231513221(12)12231513161} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2718] \quad 1(12)12231513221(12)12^{1231513221(12)12231513161} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2738] \quad 322181232181232181232^{1181232181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2737] \quad (12)1223151321812321812321812^{1321812321 \quad 8 \quad 1} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 56, T \in [46, 56]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2736] \quad 1 \quad 8 \quad 1232181232181232181232^{1181231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2735] \quad 1(12)1223151321812^{13218123218123218123218122} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

$$[2734] \quad 1 \quad 8 \quad 1232181232181232^{1181232181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2733] \quad 1 \quad 8 \quad 1232181232181232181232^{1181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2732] \quad 61231513221(12)1223151322^{11(12)12231513218123217} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2720] $1(12)^{21}12231513221(12)^112231513221(12)12^12315131613+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2731] $71232181231513221(12)12^1231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[2721] $6131513221(12)122^131513221(12)^112231513221(12)^{3221}1+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 58, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2730] $6131513221(12)12^1231513221(12)12231513221(12)12222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2729] $61231513221(12)1223151322^11(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2725] $6131513221(12)^112^1231513221(12)12231513221(12)12+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2740] $1(12)^{21}12231513221(12)^1122^131513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2741] $1812321812^131513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -109, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2742] $221(12)12231513221(12)^1122^131513221(12)122315131613 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2743] $712321812^131513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[2744] $61231513221(12)12231513221(12)122315132^118123217+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -108, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2746] $18123218123151322^11(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -109, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2745] & 1(12)12231513221(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2747] & 221(12)12231513221(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2756] & 221812321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2755] & 3221812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2754] & (12)122315132181232181232^1 1812321812321^{2321} 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 56, T \in [46, 56] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2753] & 1812321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2752] & 1(12)122315132^1 18123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0} \\
[2751] & 1812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2750] & 1812321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2749] & 61231513221(12)12231513221(12)12^1 231513218123217 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2748] & 7123218123151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[2763] & 322181232181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$(1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[2762] \quad (12)1223151321812321812^{2321}8 \ 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 56, T \in [46, 56]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2761] \quad 1 \ 8 \ 12321812321812321812321^{321}181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[2760] \quad 221(12)12231513221^{11}(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2759] \quad 6131513221(12)12^{1231513221}(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[2758] \quad 6131513221(12)12231513221(12)122^{131513221}(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2757] \quad 1^{21}(12)12231513221^{11}(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2764] \quad 6131513221(12)122315132^{121}(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -324, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.43 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[2769] \quad 3221812321812321812321812^{131513221}(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.41 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[2768] \quad (12)122315132181232^{11812321812321812321}8 \ 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.41 \rightarrow 46, T_{\max} = 56, T \in [46, 56]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[2767] \quad 1 \ 8 \ 12321812321812321812^{131513221}(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.41 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[2766] \quad 221(12)12231513221(12)12^{1231513221}(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -361, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 45.41 \rightarrow 46, \quad T_{\max} = 57, \quad T \in [46, 57] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[2765] & 1(12)12231513221(12)12^{21}231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 57, \quad \Delta = -114, \quad \det = -361, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.41 \rightarrow 46, \quad T_{\max} = 57, \quad T \in [46, 57] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[2770] & (12)1223151321812^{2321}321812321812321812321 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 56, \quad \Delta = -113, \quad \det = -400, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.39 \rightarrow 46, \quad T_{\max} = 56, \quad T \in [46, 56] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2771] & 6131513221(12)12231513221(12)12^{3221}231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 58, \quad \Delta = -115, \quad \det = -484, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.36 \rightarrow 46, \quad T_{\max} = 58, \quad T \in [46, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2772] & (12)122315132^{2321}1812321812321812321812321 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 56, \quad \Delta = -113, \quad \det = -484, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.36 \rightarrow 46, \quad T_{\max} = 56, \quad T \in [46, 56] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[2773] & 6131513221(12)1223151322^{3221}1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 58, \quad \Delta = -115, \quad \det = -529, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.34 \rightarrow 46, \quad T_{\max} = 58, \quad T \in [46, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
\\
& T = 42 \quad [3] \quad 71231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = \\
& 4, \quad (n_+, n_-, n_0) = (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[0] & 1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[6] & 71231513221(12)12231513221(12)12231513221(12)1222^12 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[5] & 712231513221(12)12231513221(12)12231513221(12)122^12 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[4] & 22^121(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2] & 71231513221(12)12231513221(12)12231513221(12)122^121 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 712231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 712231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 222^1 1(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 71231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 71231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 71231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 71231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.75 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 612^1 31513221(12)12231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 71231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 71232181231513221(12)12231513221(12)122315132^1 171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 4132^1 21(12)12231513221(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.44 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[27] & 71232181231513221(12)12231513221(12)122315132^1 217 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[20] & 71231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[22] & 7122^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[21] & 2221(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[16] & 71231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[18] & 7122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[17] & 7122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[15] & 21(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[23] & 71231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[14] & 1(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[13] & 4132^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[44] 16131513221(12)12231513221(12)122315132181232¹181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[41] 712¹231513221(12)12231513221(12)12231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[42] 712¹31513221(12)12231513221(12)12231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[43] 71231513221(12)12231513221(12)1223151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[40] 712231513221(12)12231513221(12)122315132¹21(12)1222 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[45] 61231513221(12)12231513221(12)122315132181232¹181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[46] 712¹32181231513221(12)12231513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[47] 71232181231513221(12)12231513221(12)1223151322¹17+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[48] 712¹32181231513221(12)12231513221(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[32] 712231513221(12)12231513221(12)122315132¹21(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[31] 712¹231513221(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[39] 2221(12)122¹31513221(12)12231513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[38] & 71231513221(12)^1 12231513221(12) 1223151322^1 1(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[33] & 712231513221(12)^1 12231513221(12) 122315132^1 21(12)^{21} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[34] & 712^1 231513221(12)^1 12231513221(12) 12231513221(12)^{21} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[30] & 712^1 31513221(12)^1 12231513221(12) 12231513221(12)^{3221} 1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[28] & 1(12)^{221} 122^1 31513221(12)^1 12231513221(12) 122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[35] & 712^1 31513221(12)^1 12231513221(12) 12231513221(12)^{2221} 1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[36] & 71231513221(12)^1 12231513221(12) 1223151322^1 1(12)^{2221} 1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[37] & 712^1 31513221(12)^1 12231513221(12) 12231513221(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[29] & 21(12)^{21} 122^1 31513221(12)^1 12231513221(12) 122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[59] & 71231513221(12)^1 12231513221(12) 122^1 31513221(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[58] & 71231513221(12)^1 12231513221(12) 122^1 31513221(12)^{2221} 1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 2221(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[62] \quad 7122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 712231513221(12)12231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 71231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 161315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 812^1 32181231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 612315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 71232181231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 41322^1 1(12)12231513221(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[70] \quad 71232181231513221(12)12231513221(12)122^1 315132217 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 7122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 712231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) =$$

$$(1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 7122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 712231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 21(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[52] \quad 21(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 1(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[50] \quad 1(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 41321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 2221(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[71] \quad 4132181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 64, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.94 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 71232181231513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[83] $71231513221(12)^1 122315132^1 21(12) 12231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[84] $71231513221(12)^1 12231513221(12) 12^1 231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[85] $2221(12) 12231513221(12)^1 12231513221(12) 12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[86] $2221(12) 12231513221(12)^1 122315132^1 21(12) 122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[87] $712231513221(12) 122^1 31513221(12)^1 12231513221(12) 1222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[88] $71223151322^1 1(12) 12231513221(12)^1 12231513221(12) 1222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[89] $6131513221(12)^1 122315132^1 21(12) 12231513221(12) 122313 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 47$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[90] $6131513221(12)^1 12231513221(12) 12^1 231513221(12) 122313 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 47$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[91] $71231513221(12)^1 12231513221(12) 12^1 231513221(12) 12222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[81] $71231513221(12)^1 12231513221(12) 12^1 231513221(12)^{2221} 1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[92] $71231513221(12)^1 122315132^1 21(12) 12231513221(12) 12222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[93] 16131513221(12)12231513221(12)1223151321812¹32181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[94] 16131513221(12)122¹31513221(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[95] 1613151322¹1(12)12231513221(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[72] 41321812321812321812321812321812315132¹21(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[73] 1²²¹(12)12231513221(12)12231513221(12)12¹2315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[75] 21²¹(12)12231513221(12)12231513221(12)12¹2315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[76] 21²¹(12)12231513221(12)122315132¹21(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[77] 71223151322¹1(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[78] 712231513221(12)122¹31513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[79] 712231513221(12)122¹31513221(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[80] 71223151322¹1(12)12231513221(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[96] 61231513221(12)12231513221(12)1223151321812¹32181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) =$

$(1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[97] $61231513221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[98] $6123151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[99] $71232^1 181231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[100] $71232181231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[101] $71232181231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[102] $71232^1 181231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[103] $71232181231513221(12)12231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[82] $71231513221(12)^1 122315132^1 21(12)12231513221(12)^{2221} 1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[111] $712231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[118] $413221(12)12231513221(12)12231513221(12)122^1 3151316 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[117] $712315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [116] $61315132^1 21(12)12231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 47$, $\Delta = -94$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 47$, $T \in [47, 47]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [115] $712231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [114] $2221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [113] $712315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [105] $41321812321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [106] $413218123218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [107] $1(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [108] $21(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [112] $712315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [109] $712315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [110] $712231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [130] $71231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) =$

$$(1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 413221(12)122^1 31513221(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[124] \quad 71231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[125] \quad 712231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[126] \quad 712231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[127] \quad 712231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[128] \quad 712231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 71231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[119] \quad 4132181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[120] \quad 1(12)12231513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[121] \quad 1(12)12231513221(12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[122] \quad 21(12)12231513221(12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[131] & 71231513221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[132] & 71231513221(12)1223151322^1 1(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[133] & 2221(12)12231513221(12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[134] & 2221(12)12231513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[135] & 712231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[136] & 712231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[137] & 6131513221(12)122^1 31513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 47, \Delta = -94, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[139] & 71231513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[140] & 71231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[138] & 6131513221(12)1223151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 47, \Delta = -94, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[141] & 16131513221(12)122315132^1 21(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]
\end{aligned}$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[151] 71232181231513221(12)122¹31513221(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[150] 71232181231513221(12)1223151322¹1(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[142] 16131513221(12)12¹231513221(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[143] 81232¹181231513221(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[144] 81232181231513221(12)12231513221(12)122¹315131613+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[145] 61231513221(12)122315132¹21(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[146] 61231513221(12)12¹231513221(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[147] 71232181231513221(12)122¹31513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[148] 71232181231513221(12)1223151322¹1(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[123] ²¹21(12)12231513221¹(12)1223151322¹1(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[170] 413221(12)12¹231513221(12)12231513221(12)1223151316 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.59 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[171] 712321812315132¹21(12)12231513221(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[152] & 41321812321812321812^1 32181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[153] & 1^{221}(12)1223151322^1 1^{12231513221}(12)122315132171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[154] & 21^{21}(12)1223151322^1 1^{12231513221}(12)122315132171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[155] & 71231513221(12)12231513221(12)1223151322^1 1^{3221}(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[156] & 7123151322^1 1^{12231513221}(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[157] & 712231513221(12)12231513221(12)12^1 231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[158] & 712231513221(12)12231513221(12)12^1 231513221(12)12+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[159] & 7123151322^1 1^{12231513221}(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[160] & 7123151322^1 1^{12231513221}(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[161] & 2221(12)1223151322^1 1^{12231513221}(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 169, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[162] & 712231513221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 169, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[169] $413221(12)12231513221(12)12231513221(12)12^1 23151316 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[168] $712321812315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[167] $61231513221(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[166] $81232181231513221(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[165] $16131513221(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[164] $7123151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[163] $613151322^1 1(12)12231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 47$, $\Delta = -94$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 47$, $T \in [47, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[179] $71231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[188] $413221(12)12231513221(12)122315132^1 21(12)1223151316 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[189] $71232181231513221(12)12^1 231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[187] $71232181231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[186] $61231513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[185] & 81232181231513221(12)12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[184] & 16131513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[183] & 71231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[182] & 6131513221(12)12^1 231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 47, \Delta = -94, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[181] & 712231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[180] & 2221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[178] & 71231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[172] & 4132181232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[173] & 1(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[174] & 21(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[175] & 71231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[176] $712231513221(12)1223151322^11(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[177] $712231513221(12)1223151322^11(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[192] $61231513221(12)12231513221(12)122315132^1181232181+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -107, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[190] $16131513221(12)12231513221(12)122315132^1181232181+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[191] $81232181231513221(12)122^131513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[194] $712321812^131513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[193] $712321812^131513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[196] $16131513221(12)12231513221(12)12^12315132181232181+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[195] $41321812321812321812321812^132181231513221(12)1+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -110, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[199] $7123218123151322^11(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[200] $7123218123151322^11(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[197] $81232181231513221(12)1223151322^11(12)122315131613+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[198] 61231513221(12)12231513221(12)12¹2315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 256$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[201] 4132181232181232181232181232¹181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[202] 71231513221(12)12¹231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[203] 71231513221(12)12231513221(12)122¹31513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[204] 812321812315132¹21(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[205] 413221(12)12231513221(12)1223151322¹1(12)1223151316 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[206] 413221(12)122315132¹21(12)12231513221(12)1223151316 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[207] 71231513221(12)122315132¹21(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 324$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.43 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[208] 413221(12)12231513221(12)122¹31513221(12)1223151316 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 324$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.43 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[210] 812321812¹31513221(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 361$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.41 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[211] 81232181231513221(12)12¹231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 361$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.41 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[209] 41321812321812321812321812¹31513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 361$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.41 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[212] $8123218123151322^1 1(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 441, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.37 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[213] $71231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 484, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.36 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[214] $413221(12)1223151322^1 1(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 484, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.36 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[215] $71231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.34 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[216] $413221(12)12231513221(12)12^1 231513221(12)1223151316 + \hat{1} \rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.34 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$T = 43$ [229] $22^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[230] $812231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[217] $1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[218] $1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[219] $1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[220] $181231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[221] & 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[222] & 21(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[223] & 81231513221(12)12231513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[224] & 812231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[225] & 812231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[226] & 181231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = \\
& -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[227] & 22^1 1(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = \\
& -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[228] & 22^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = \\
& -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[241] & 222^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = \\
& -9, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[242] & 812231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = \\
& -9, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[231] & 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -9, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 58, T \in [50, 58]
\end{aligned}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[243] \quad 181231513221(12)^1 12231513221(12) 12231513221(12) 12^1 222 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.75 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[273] \quad 51232^1 181232181232181232181232181231513221(12) 1 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -107, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[272] \quad 4132^1 21(12) 12231513221(12)^1 12231513221(12) 12231513217 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 58, \quad \Delta = -113, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 58, \quad T \in [49, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[246] \quad 1(12)^{221} 12231513221(12)^1 12231513221(12) 122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[247] \quad 21(12)^{21} 12231513221(12)^1 12231513221(12) 122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[248] \quad 1(12)^{322^1 1} 12231513221(12)^1 12231513221(12) 122315132181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 58, \quad \Delta = -113, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 58, \quad T \in [49, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[249] \quad 1(12)^1 12231513221(12)^1 12231513221(12) 1223151321812^1 32 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 60, \quad \Delta = -115, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 60, \quad T \in [49, 60]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[250] \quad 1(12)^{221} 12231513221(12)^1 12231513221(12) 122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[251] \quad 21(12)^{21} 12231513221(12)^1 12231513221(12) 122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[252] \quad 1(12)^{221} 12231513221(12)^1 12231513221(12) 122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[253] \quad 21(12)^{21} 12231513221(12)^1 12231513221(12) 122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

- [254] $812231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [255] $1812^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [256] $1(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [257] $21(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [258] $81231513221(12)12231513221(12)12231513221(12)12 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [259] $812231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [260] $812231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [261] $181231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [262] $21(12)12231513221(12)12231513221(12)1223151321812^1 32 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 60$, $T \in [49, 60]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [263] $2221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [264] $2221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

- [265] $6131513221(12)12231513221(12)12231513221(12)122^13151 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [266] $812231513221(12)12231513221(12)122315132^121(12)12222 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [267] $71232181231513221(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [268] $1712^131513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [269] $71232181231513221(12)12231513221(12)122315132^11812 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [270] $7122^131513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [271] $71232181231513221(12)12231513221(12)122315132^12181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [274] $15132^1181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [307] $71232181231513221(12)12231513221(12)1223151322^1181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [306] $712^132181231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [275] $1(12)122^131513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [276] $21(12)122^131513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[277] \quad 1(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[278] \quad 1(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[279] \quad 21(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[280] \quad 1(12)122^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[282] \quad 21(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[281] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[308] \quad 413221(12)12231513221(12)12231513221(12)122315132^1 17 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[284] \quad 8122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[285] \quad 181231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[286] \quad 1(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[287] \quad 1(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[299] $812231513221(12)12231513221(12)12231513221^1(12)12222 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[300] $8122^131513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[301] $712^132181231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[302] $171231513221(12)12231513221(12)122315132181232^1181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[303] $712^132181231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[304] $712^1231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[305] $712231513221(12)12231513221(12)122315132181232^1181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[283] $21(12)122^131513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[348] $151321812^132181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[347] $512321812^132181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[309] $1(12)12^1231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[310] $1(12)12231513221(12)12231513221(12)122^1315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[322] \quad 812^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[323] \quad 181231513221(12)12231513221(12)12231513221^1(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[324] \quad 1812315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[325] \quad 1(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[326] \quad 21(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[327] \quad 812^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[328] \quad 812^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[329] \quad 812231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[330] \quad 812^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[331] \quad 812231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[332] \quad 181231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[333] & 221(12)122^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 58, \quad \Delta = -113, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 58, \quad T \in [49, 58] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[334] & 2221(12)12^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[335] & 2221(12)12231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[336] & 2221(12)12^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[337] & 2221(12)12231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[338] & 6131513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -109, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[339] & 812231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[340] & 812^1 231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[341] & 71232181231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -105, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[342] & 1712315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -105, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[343] & 71232181231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -105, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[344] $7122315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[345] $71232181231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[346] $41322^1 1(12)12231513221(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[350] $1(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.94 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[355] $(12)1223151321812321812321812321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.94 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[349] $1(12)122^1 31513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.94 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[354] $15132181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.94 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[351] $221(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.94 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[352] $21(12)122^1 31513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.94 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[353] $51232181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.94 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[380] $812231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[409] $151321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[382] $181231513221(12)12231513221(12)12^1231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[383] $221(12)12231513221(12)12231513221(12)122315132^118123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[384] $21(12)12^1231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[379] $812231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[378] $812231513221(12)12231513221(12)12^1231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[377] $812231513221(12)122315132^121(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[376] $81231513221(12)12231513221(12)122315132^121(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[375] $21(12)1223151322^11(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[374] $21(12)12231513221(12)122^131513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[373] $1(12)1223151322^11(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[372] $1(12)12231513221(12)122^131513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[360] & 1(12)12^1 231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[359] & 21(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[358] & 21(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[357] & 1(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[356] & 1(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[402] & 71223151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[385] & 2221(12)12231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[386] & 2221(12)12231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[387] & 2221(12)12231513221(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[388] & 2221(12)12231513221(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[389] & 71231513221(12)12231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 48, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [390] $71231513221(12)122315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 48, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 48, T \in [48, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [391] $812231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [392] $812231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [393] $71232181231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [394] $71232181231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [395] $71232^1 181231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [396] $17123151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [397] $171231513221(12)12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [398] $171231513221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [399] $71232181231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [400] $71232181231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [401] $71232^1 181231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[403] $712231513221(12)122^131513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[404] $712231513221(12)12231513221(12)1223151321812^132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[405] $71232181231513221(12)12231513221(12)12^123151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[406] $71232^1181231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[407] $71232181231513221(12)122315132^121(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[408] $512321812321812321812321812321812315132^121(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[381] $181231513221(12)122315132^121(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[422] $8122315132^121(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[410] $1(12)122315132^121(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[433] $1513218123218123218123218123151322^11(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[432] $151321812321812^132181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[431] $512321812321812^132181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[430] 5123218123218123218123218123218123151322¹1(12)1+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[429] 413221(12)12231513221¹(12)12231513221(12)122¹31513217 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[428] 8122315132¹21(12)12231513221(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[427] 712315132¹21(12)12231513221(12)12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 48$, $\Delta = -93$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[426] 61315132¹21(12)12231513221(12)12231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -109$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[425] 2221(12)122315132¹21(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[424] 2221(12)122315132¹21(12)12231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[423] 1812315132¹21(12)12231513221(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[435] (12)122315132181232181232181232181232¹181231513+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[421] 8122315132¹21(12)12231513221(12)12231513221²²²¹(12)1+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[420] 21²²¹(12)12231513221(12)12231513221¹(12)122¹315132181+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[419] $1(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[418] $181231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[417] $8122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[416] $21(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[415] $1(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[414] $21(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[413] $1(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[412] $1(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[411] $21(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[434] $(12)122^1 315132181232181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[449] $21(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[488] $(12)12^1 2315132181232181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$(1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[438] \quad 21(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[439] \quad 21(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[440] \quad 1(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[441] \quad 1(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[442] \quad 1(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[443] \quad 1(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[489] \quad (12)1223151321812321812321812321812^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[444] \quad 21(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[445] \quad 21(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[446] \quad 1(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[447] \quad 1(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[448] & \begin{matrix} 21 & & 1 \\ 21(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[436] & \begin{matrix} 221 & & 1 \\ 1(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[450] & \begin{matrix} & & 1 & & 3221 \\ 812231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[451] & \begin{matrix} 1 & & 1 \\ 181231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[452] & \begin{matrix} 1 & & 1 \\ 181231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[453] & \begin{matrix} & & 1 \\ 1(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[454] & \begin{matrix} & & 1 \\ 1(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[455] & \begin{matrix} & & 1 \\ 21(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[456] & \begin{matrix} & & 1 \\ 21(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[457] & \begin{matrix} & & 1 & & 3221 \\ 81231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[458] & \begin{matrix} & & 1 & & 3221 \\ 81231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix}
\end{aligned}$$

- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [459] $812231513221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [460] $812231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [461] $812231513221(12)1223151322^1 1(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [462] $812231513221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [463] $181231513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [464] $181231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [465] $221(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [466] $2221(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [467] $2221(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [468] $2221(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [469] $2221(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$

$$-121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[482] \quad 712231513221(12)122315132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[483] \quad 71232181231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[484] \quad 71232181231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[485] \quad 413221(12)122^1 31513221(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[486] \quad 51232181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[487] \quad 15132181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[437] \quad 1(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[508] \quad 81223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[490] \quad 1(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[529] \quad (12)122315132181232181232^1 181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[528] \quad 151321812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[527] 512321812321812321812¹32181232181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[526] 413221(12)12¹231513221(12)12231513221(12)12231513217 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[525] 413221(12)12231513221(12)12231513221(12)12¹231513217 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[524] 712321812315132¹21(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[523] 712231513221(12)12231513221(12)122¹315132181232181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[522] 712321812315132¹21(12)12231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[521] 171231513221(12)12231513221(12)122¹315132181232181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[520] 712321812315132¹21(12)12231513221(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[519] 81223151322¹1(12)12231513221(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[518] 7123151322¹1(12)12231513221(12)12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 48$, $\Delta = -93$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[517] 6131513221(12)12231513221(12)1223151322¹1(12)1223151 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -109$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[516] 613151322¹1(12)12231513221(12)12231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -109$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[502] \quad 81223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[501] \quad 812231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[500] \quad 21(12)1223151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[499] \quad 1(12)1223151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[498] \quad 21(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[497] \quad 1(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[496] \quad 1(12)12231513221(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[495] \quad 1(12)122315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[494] \quad 1(12)12231513221(12)12231513221(12)122315132^1 181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[493] \quad 1(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[492] \quad 1(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[491] & 21^{21}(12)1223151322^1 1^{12}(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[506] & 812315132^1 21^{3221}(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[534] & 1^{221}(12)12231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[557] & 413221(12)12231513221(12)122315132^1 21(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[536] & 1^{221}(12)12231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[531] & 21^{21}(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[535] & 21^{21}(12)12231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[537] & 21^{21}(12)12231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[538] & 812231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[539] & 1^{18}1231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[540] & 1^{2221}(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]
\end{aligned}$$

- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [541] $21(12)^{221}12231513221(12)1223151322^11(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [542] $81231513221(12)^112231513221(12)12^1231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [543] $812231513221(12)12^1231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [544] $812231513221(12)^112^1231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [545] $181231513221(12)^112^1231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [546] $221(12)12231513221(12)^112^1231513218123 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [547] $2221(12)12231513221(12)^112^1231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [548] $2221(12)12231513221(12)^112^1231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [549] $6131513221(12)122^131513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [550] $71231513221(12)^112^1231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 48, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 48, T \in [48, 48]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [551] $812231513221(12)^112^1231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[530] & 1(12)12231513221(12)12^{1231513221(12)122315132181+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[552] & 71232181231513221(12)12^{1231513221(12)122315132181+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[532] & 1(12)12231513221(12)122315132^{121(12)122315132181+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[553] & 171231513221(12)1223151322^{11(12)122315132181232181+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[554] & 71232181231513221(12)12^{1231513221(12)1223151321812+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[555] & 712231513221(12)1223151322^{11(12)122315132181232181+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[556] & 71232181231513221(12)12^{1231513221(12)1223151322181+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[533] & 1(12)12231513221(12)12231513221(12)12^{1231513218123+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[560] & (12)1223151321812321812321812^{132181232181231513+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[559] & 15132181232181232181232^{1181232181231513221(12)1+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[558] & 51232181232181232181232^{1181232181231513221(12)1+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 55, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[564] & 221(12)12231513221(12)122^{131513221(12)12231513218123+\hat{1}\rightarrow 2} \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[569] & 712231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[570] & 712321812^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[561] & 1(12)12231513221(12)12231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[562] & 1(12)12231513221(12)122^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[563] & 81231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -225, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[568] & 712321812^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[567] & 171231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[566] & 712321812^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -225, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[565] & 21(12)12231513221(12)12231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[572] & 1(12)12231513221(12)1223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = \\
& -256, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.48 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[571] & 1(12)122315132^1 21(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = \\
& -256, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.48 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [574] $221(12)12231513221(12)1223151322^11(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [575] $21(12)122315132^121(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [576] $7123218123151322^11(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [577] $171231513221(12)12231513221(12)12^12315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [578] $7123218123151322^11(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [579] $712231513221(12)12231513221(12)12^12315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [580] $7123218123151322^11(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [581] $512321812321812321812321812^132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [582] $151321812321812321812321812^132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [583] $(12)122315132181232181232^1181232181232181231513 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [573] $81231513221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [584] $1(12)122315132^121(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -289$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.46 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[585] & \begin{matrix} 3221 & 1 \\ 1(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[586] & \begin{matrix} 1 & 1 \\ 1(12)12231513221(12)122315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 60, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[587] & \begin{matrix} 21 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[588] & \begin{matrix} 3221 \\ 812231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[589] & \begin{matrix} 1 & 3221 \\ 812231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[590] & \begin{matrix} 1 & 3221 \\ 8123151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[591] & \begin{matrix} 1 \\ 221(12)1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[592] & \begin{matrix} 1 \\ 21(12)12231513221(12)122315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 60, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[593] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -109, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[594] & \begin{matrix} 1 \\ 6131513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -109, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[595] & \begin{matrix} 1 \\ 413221(12)12231513221(12)1223151322^1 1(12)12231513217 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix}
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[596] & 413221(12)122315132^1 21(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[597] & 51232181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.46 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[598] & 15132181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.46 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[599] & (12)1223151321812321812^1 32181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[603] & 413221(12)12231513221(12)122^1 31513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.43 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[600] & 1(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.43 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[601] & 812231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.43 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[602] & 6131513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -109, \det = -324, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.43 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[606] & 1(12)12231513221(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.41 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[605] & 1(12)12231513221(12)12231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.41 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[604] & 1(12)12231513221(12)122^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.41 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [607] $81231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [608] $221(12)12231513221(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [609] $21(12)12231513221(12)12231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 60, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [610] $21(12)12231513221(12)122^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 60, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [611] $512321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -107, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [612] $151321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [613] $(12)122315132181232^1 181232181232181232181231513 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [614] $(12)1223151321812^1 32181232181232181232181231513 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -400, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.39 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [618] $21(12)12231513221(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 60, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.37 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [615] $1(12)1223151322^1 1(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 60, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.37 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [616] $1(12)12231513221(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 60, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.37 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [617] $21(12)1223151322^1 1(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 60, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 47.37 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[619] & 1(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -484, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[620] & 812231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -484, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[621] & 6131513221(12)12231513221(12)12^1 231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -109, \det = \\
& -484, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[622] & 413221(12)1223151322^1 1(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = \\
& -484, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[623] & (12)122315132^1 181232181232181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -484, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[627] & 413221(12)12231513221(12)12^1 231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = \\
& -529, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.34 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[626] & 6131513221(12)1223151322^1 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -109, \det = \\
& -529, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.34 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[625] & 812231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -529, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.34 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[624] & 1(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -529, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.34 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[629] & 21(12)12231513221(12)12^1 231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = \\
& -576, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.33 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[628] & 1(12)12231513221(12)12^1 231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = \\
& -576, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.33 \rightarrow 48, T_{\max} = 60, T \in [48, 60]
\end{aligned}$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$T = 44 \quad [632] \quad 61316131513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[631] \quad 61316131513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[630] \quad 61316131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[635] \quad 81232181231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 59, \quad \Delta = -113, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.22 \rightarrow 51, \quad T_{\max} = 59, \quad T \in [51, 59]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[634] \quad 61316131513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.22 \rightarrow 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[633] \quad 61316131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.22 \rightarrow 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[636] \quad 1612^1 32181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.75 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[637] \quad 61316131513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.75 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[638] \quad 161232^1 181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.44 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[639] \quad 71231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -108, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.44 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[640] \quad 1812^1 31513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.44 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

- [641] $61316131513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [642] $61316131513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [643] $181232181231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [644] $181232181231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [645] $61316131513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [646] $4132^1 21(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [647] $4132^1 21(12)12231513221(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [648] $4132^1 21(12)12231513221(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [654] $181232181231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 36$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.22 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [651] $61316131513221(12)12231513221(12)122315132^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.22 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [658] $413221(12)12231513221(12)12231513221(12)122315132^1 218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.22 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [657] $413221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[656] & 61316131513221(12)12231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = \\
& 36, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 56, T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[655] & 1812^1 32181231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 36, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 56, T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[649] & 712^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = \\
& 36, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 56, T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[650] & 181231513221(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 36, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 56, T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[653] & 1812^1 32181231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 36, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 56, T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[652] & 61316131513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = \\
& 36, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 56, T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[670] & 413221(12)12231513221(12)12231513221(12)122315132^1 18 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = \\
& 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[669] & 41322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = \\
& 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[668] & 812^1 32181231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = \\
& 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, T_{\max} = 59, T \in [50, 59] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[667] & 61316131513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = \\
& 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[666] & 181232181231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[665] & 181232181231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det =
\end{aligned}$$

$$49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[664] \quad 61316131513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[673] \quad 41322^1 1(12)12231513221(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[672] \quad 413221(12)12231513221(12)12231513221(12)1223151322^1 18 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[671] \quad 41322^1 1(12)12231513221(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[659] \quad 1612321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[660] \quad (12)122315132181232181232181232181232181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[663] \quad 61316131513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[662] \quad 1812315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[661] \quad 71231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[675] \quad (12)1223151321812321812321812321812321812^1 321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.94 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[674] \quad 161232181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 64, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.94 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[690] 61316131513221(12)12231513221(12)12¹231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[692] 81232181231513221(12)12231513221(12)122315132¹18123 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 59$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[691] 61316131513221(12)122315132¹21(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[693] 413221(12)122¹31513221(12)12231513221(12)12231513218 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[676] 1612321812321812321812321812321812315132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[677] 18123151322¹1(12)12231513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[678] 181231513221(12)122¹31513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[679] 181231513221(12)12231513221(12)1223151321812¹32181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[680] 61316131513221(12)122315132¹21(12)12231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[681] 61316131513221(12)12231513221(12)12¹231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[682] 61316131513221(12)12231513221(12)12¹231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[683] 61316131513221(12)122315132¹21(12)12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det =$

81, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[684] 181232181231513221(12)12231513221(12)12¹23151321812 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[685] 181232181231513221(12)122315132¹21(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[686] 181232¹181231513221(12)12231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[687] 181232¹181231513221(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[688] 181232181231513221(12)12231513221(12)12¹23151322181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[689] 181232181231513221(12)122315132¹21(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[694] 16123218123218123218123218123218123151322¹1(12)1 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[696] (12)122315132181232181232181232181232¹1812321613 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[697] (12)122¹3151321812321812321812321812321812321613 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[698] 712315132¹21(12)12231513221(12)12231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[699] 413221(12)12231513221(12)12231513221(12)122¹315132181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[700] $413221(12)12231513221(12)12231513221(12)122^1315132218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[695] $1612321812321812^132181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[718] $81232181231513221(12)12231513221(12)122^131513218123 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 59$, $\Delta = -113$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 59$, $T \in [49, 59]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[701] $161232181232181232^1181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[704] $71231513221(12)12231513221(12)122315132^121(12)1223151 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -108$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[705] $181231513221(12)122315132^121(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[706] $181231513221(12)12^1231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[707] $61316131513221(12)1223151322^11(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[708] $61316131513221(12)122^131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[709] $61316131513221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[710] $61316131513221(12)122^131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[711] $181232181231513221(12)1223151322^11(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[712] & 181232181231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[713] & 181232181231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[714] & 181232181231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[715] & 61316131513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[716] & 61316131513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[717] & 81232^1 181231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 59, T \in [49, 59] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[702] & (12)12^1 23151321812321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[719] & 413221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[720] & 413221(12)12^1 231513221(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[721] & 413221(12)122315132^1 21(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[722] & 413221(12)122^1 31513221(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[703] & (12)1223151321812321812321812321812^1 321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$(1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[731] $1812321812315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[725] $71231513221(12)12231513221(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[726] $7123151322^1 1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[724] $(12)122315132181232181232181232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[728] $613161315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[729] $613161315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[730] $1812321812315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[723] $1612321812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[732] $613161315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[733] $81232181231513221(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 59$, $\Delta = -113$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 59$, $T \in [49, 59]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[734] $413221(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [735] $413221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [736] $413221(12)12231513221(12)12231513221(12)122^1 31513218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [737] $413221(12)12231513221(12)122^1 31513221(12)12231513218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [738] $413221(12)12^1 231513221(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [739] $413221(12)12231513221(12)12231513221(12)12^1 2315132218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [727] $181231513221(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [741] $(12)1223151321812321812321812^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [740] $161232181232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [743] $181231513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [744] $61316131513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [745] $61316131513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [746] $181232181231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[747] & 181232181231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[748] & 61316131513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[749] & 81232181231513221(12)12231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 59, T \in [49, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[750] & 413221(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[751] & 413221(12)1223151322^1 1(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[752] & 413221(12)12231513221(12)122315132^1 21(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[742] & 71231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[753] & 181231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.52 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[754] & 1812321812^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.52 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[755] & 1812321812^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.52 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[756] & 81232181231513221(12)122^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = \\
& 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.52 \rightarrow 49, T_{\max} = 59, T \in [49, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[757] $413221(12)12231513221(12)122315132^1 21(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.52 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[762] $6131613151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[760] $181231513221(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[761] $6131613151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[759] $(12)122315132181232181232^1 1812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[758] $1612321812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[764] $18123218123151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[765] $6131613151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[766] $81232181231513221(12)1223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[767] $413221(12)12231513221(12)12^1 231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[763] $18123218123151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[769] $(12)1223151321812321812^1 321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[770] & \begin{matrix} 1 \\ 71231513221(12)12231513221(12)122^131513221(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[771] & \begin{matrix} 1 \\ 71231513221(12)12^1231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[772] & \begin{matrix} 1 \\ 812321812315132^121(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 59, T \in [49, 59] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[773] & \begin{matrix} 1 \\ 413221(12)12231513221(12)1223151322^11(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[774] & \begin{matrix} 1 \\ 413221(12)122315132^121(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[775] & \begin{matrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12^1231513218 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[776] & \begin{matrix} 1 \\ 413221(12)12231513221(12)1223151322^11(12)122315132218 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[777] & \begin{matrix} 1 \\ 413221(12)122315132^121(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[768] & \begin{matrix} 1 \\ 161232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[778] & \begin{matrix} 1 \\ 71231513221(12)122315132^121(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 324, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.43 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[779] & \begin{matrix} 1 \\ 413221(12)12231513221(12)122^131513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 324, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.43 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[780] $413221(12)12231513221(12)122^131513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 324, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.43 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[781] $1612321812321812321812321812^131513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[782] $(12)122315132181232^11812321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[783] $81232181231513221(12)12^1231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[784] $812321812^131513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[785] $413221(12)12231513221(12)1223151322^11(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$

[786] $(12)1223151321812^1321812321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 400, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.39 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[787] $8123218123151322^11(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 441, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.37 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[788] $(12)122315132^11812321812321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.36 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[789] $71231513221(12)12231513221(12)12^1231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 484, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.36 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[790] $413221(12)1223151322^11(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.36 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[791] $413221(12)1223151322^11(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.36 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[792] $71231513221(12)12231513221^1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -108, \det = 529, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 48.34 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[793] $413221(12)12231513221(12)12^1231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 58, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 48.34 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[794] $413221(12)12231513221(12)12^1231513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 58, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 48.34 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$T = 45$ [795] $(12)1223151321812321812321812321812321812321712^13 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 51.22 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[796] $1712^132181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50.75 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[797] $21712^132181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50.75 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[798] $171232^1181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[799] $2171232^1181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[800] $81231513221(12)12231513221(12)12231513221^{15132^121}(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[801] $15132^121(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[802] $812231513221(12)12231513221(12)12231513221(12)122^13151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [804] $8122^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.22 \rightarrow 51$, $T_{\max} = 57$, $T \in [51, 57]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [803] $1513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.22 \rightarrow 51$, $T_{\max} = 57$, $T \in [51, 57]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [809] $812^1 31513221(12)12231513221(12)12231513221^{1513221}(12)1 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 57$, $T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [807] $(12)122315132181232181232181232181232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [808] $81231513221(12)12231513221(12)12231513221^{1513221}(12)1 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 57$, $T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [806] $21712321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 57$, $T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [810] $151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 57$, $T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [811] $812^1 31513221(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 62$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 62$, $T \in [51, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [812] $812231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 57$, $T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [813] $812^1 231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 57$, $T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [814] $812^1 31513221(12)12231513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [815] $81231513221(12)12231513221(12)12231513221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det =$

$$-49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.06 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[805] \overset{1}{1712321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.06 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[816] \overset{1}{171232181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.94 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[817] 2171232181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.94 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[818] (12)1223151321812321812321812321812321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.94 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[819] 81231513221(12)12231513221(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.94 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[820] \overset{1}{1712321812321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[821] 21712321812321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[822] 81231513221(12)12231513221(12)122315132^1 21 \overset{1513221}{(12) 1 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[823] 61316131513221(12)12231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 50, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[824] 61316131513221(12)122315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 50, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[825] \overset{1}{81231513221(12)12231513221(12)122315132^1 21(12)12231414 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[826] $81231513221(12)12231513221(12)12231513221(12)12^1 231414 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.84 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[828] $17123218123218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[829] $21712321812321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[830] $217123218123218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[831] $(12)122^1 31513218123218123218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[832] $(12)122315132181232181232181232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[833] $1513221(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[834] $8122315132^1 21(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[827] $1712321812321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[835] $171232181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[836] $2171232181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[837] $(12)12^1 231513218123218123218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.69 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[838] $(12)1223151321812321812321812321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[839] $81231513221(12)12231513221(12)122^1 31513221$ $(12) 1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[840] $81231513221(12)12231513221(12)1223151322^1 1$ $(12) 1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[841] $1513221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[842] $81231513221(12)12231513221(12)12231513221(12)12^1 231513 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 62, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[843] $812231513221(12)12231513221(12)122315132^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[844] $61316131513221(12)1223151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 50, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[845] $61316131513221(12)122^1 31513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 50, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[846] $81231513221(12)12231513221(12)1223151322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[847] $81231513221(12)12231513221(12)122^1 31513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[851] $81231513221(12)122315132^1 21(12)12231513221$ $(12) 1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [852] $812315132^1 21(12)12231513221(12)12231513221^{1513221} (12) 1+\hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [849] $21712321812321812321812^1 32181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [848] $1712321812321812321812^1 32181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [853] $1513221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [854] $1513221(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [855] $812315132^1 21(12)12231513221(12)12231513221(12)12231513 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 62, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [856] $81223151322^1 1(12)12231513221(12)12231513221(12)1223151 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [857] $812231513221(12)12231513221(12)1223151322^1 1(12)1223151 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [858] $613161315132^1 21(12)12231513221(12)12231513221(12)122313 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 50, \Delta = -93, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [859] $81231513221(12)122315132^1 21(12)12231513221(12)12231414 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [860] $812315132^1 21(12)12231513221(12)12231513221(12)12231414 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [850] $(12)122315132181232181232181232^1 18123218123217123+\hat{1}\rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 49.59 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[862] & 2171232181232181232181232^1 181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[863] & (12)1223151321812321812321812^1 3218123218123217123+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[864] & 81231513221(12)12231513221(12)12^1 231513221 \overset{1513221}{(12)} 1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[865] & 1513221(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[866] & 812231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[867] & 61316131513221(12)12^1 231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 50, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 50, T \in [50, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[868] & 81231513221(12)12231513221(12)12^1 231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[861] & 171232181232181232181232^1 181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[869] & 81231513221(12)122^1 31513221(12)12231513221 \overset{1513221}{(12)} 1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -225, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.52 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[870] & 81231513221(12)122^1 31513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -225, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.52 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[871] & 1712321812321812321812321812^1 32181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.48 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [872] 21712321812321812321812321812¹32181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [873] (12)122315132181232181232¹18123218123218123217123+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [874] 81231513221¹(12)1223151322¹1(12)12231513221¹⁵¹³²²¹(12) 1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [875] 81231513221¹(12)12231513221(12)122315132¹21(12)12231513 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 62$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 62$, $T \in [50, 62]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [877] 81231513221¹(12)1223151322¹1(12)12231513221(12)12231414 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [876] 6131613151322¹1(12)12231513221(12)12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 50$, $\Delta = -93$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 50$, $T \in [50, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [878] 171232181232181232181232181232¹181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [879] 2171232181232181232181232181232¹181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [880] (12)1223151321812321812¹3218123218123218123217123+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [881] 8123151322¹1(12)12231513221(12)12231513221¹⁵¹³²²¹(12) 1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [882] 1513221(12)122315132¹21(12)12231513221(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [883] 1513221¹(12)12231513221(12)1223151322¹1(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[884] & 8123151322^1 1(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.46 \rightarrow 50, T_{\max} = 62, T \in [50, 62] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[885] & 812231513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.46 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[886] & 812231513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.46 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[887] & 8123151322^1 1(12)12231513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.46 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[888] & 1513221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -324, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.43 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[889] & 812231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -324, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.43 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[892] & (12)122315132181232^1 18123218123218123218123217123 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[891] & 21712321812321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[893] & 81231513221(12)12^1 231513221(12)12231513221(12)1513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[894] & 81231513221(12)12231513221(12)1223151322^1 1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 62, T \in [50, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[895] & 81231513221(12)122^1 31513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 62, T \in [50, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [896] $81231513221(12)12^1 231513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -361$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.41 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [890] $1712321812321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.41 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [897] $(12)1223151321812^1 3218123218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -400$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.39 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [898] $(12)122315132^1 18123218123218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -484$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.36 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [900] $812231513221(12)12231513221(12)12^1 231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -484$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.36 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [899] $1513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -484$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.36 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [901] $1513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -529$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.34 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [902] $81231513221(12)12^1 231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 62$, $\Delta = -115$, $\det = -529$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.34 \rightarrow 50$, $T_{\max} = 62$, $T \in [50, 62]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$
- [903] $812231513221(12)1223151322^1 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -529$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.34 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [904] $81231513221(12)12231513221(12)122^1 31513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 62$, $\Delta = -115$, $\det = -576$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.33 \rightarrow 50$, $T_{\max} = 62$, $T \in [50, 62]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [905] $81231513221(12)122315132^1 21(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 62$, $\Delta = -115$, $\det = -625$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.31 \rightarrow 50$, $T_{\max} = 62$, $T \in [50, 62]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [906] $81231513221(12)12231513221(12)12^1 231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 62$, $\Delta = -115$, $\det =$

$$-961, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.25 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 681 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$[907] \quad 81231513221(12)1223151322^1 1(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -1024, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.25 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$$

$$\begin{pmatrix} 0 & 32 & 2 \\ 32 & 735 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$$

$$T = 46 \quad [908] \quad 22^1 181232181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53, T_{\max} = 58, T \in [53, 58]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[909] \quad (12)1223151321812321812321812321812321812321812321 \quad 8^{32^1 1} 1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.22 \rightarrow 53, T_{\max} = 58, T \in [53, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[910] \quad (12)12231513218123218123218123218123218123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.22 \rightarrow 53, T_{\max} = 58, T \in [53, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[911] \quad 18 \quad 12^{21} 32181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 16, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.75 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[912] \quad 221812^1 32181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 16, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.75 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[913] \quad 18 \quad 1232^1 181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.44 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[915] \quad (12)1223151321812321812321812321812321812321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.44 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[914] \quad 22181232^1 181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.44 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[916] \quad 18 \quad 12321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.06 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[918] \quad 221812321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.06 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[919] (12)122315132181232181232181232181232181232¹181223 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.06 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[917] (12)122315132181232181232181232181232181232¹1³²¹81 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.06 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[920] ²¹181232181232¹181232181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 64$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.94 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[921] (12)1223151321812321812321812321812321812¹321³²¹81 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.94 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[922] 22181232181232¹181232181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 64$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.94 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[923] (12)1223151321812321812321812321812321812¹32181223 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.94 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[924] ²¹1812321812321812321812321812321812315132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 81$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.84 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[925] 221812321812321812321812321812321812315132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 81$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.84 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[926] ²¹18123218123218123218123218123218123151322¹1(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.76 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[927] ²¹1812321812321812¹32181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.76 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[928] (12)122¹3151321812321812321812321812321812321³²¹81 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.76 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[929] (12)122315132181232181232181232181232¹1812321³²¹81 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) =$

$(1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[930] $2218123218123218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[931] $221812321812321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[932] $(12)122^1 315132181232181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[933] $(12)122315132181232181232181232181232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[937] $22181232181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[936] $(12)1223151321812321812321812321812^1 321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[935] $(12)12^1 23151321812321812321812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[934] $18 \overset{21}{1232181232181232^1 181232181232181231513221(12)1} + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[939] $(12)12^1 2315132181232181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[938] $(12)1223151321812321812321812321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[941] $(12)122315132181232181232181232^1 1812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 169$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.59 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

- [942] $221812321812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 169$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.59 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [943] $(12)122315132181232181232181232^1 181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 169$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.59 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [940] $1812321812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 169$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.59 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [944] $181232181232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 196$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.55 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [945] $(12)1223151321812321812321812^1 321812321812321^{321} 81 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.55 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [946] $22181232181232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 196$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.55 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [947] $(12)1223151321812321812321812^1 32181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.55 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [948] $1812321812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 256$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.48 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [949] $(12)122315132181232181232^1 1812321812321812321^{321} 81 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 256$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.48 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [950] $221812321812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 256$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.48 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [951] $(12)122315132181232181232^1 181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 256$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.48 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [952] $181232181232181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 289$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.46 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[954] $221812321812321812321812321812321^{181231513221}(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 289$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.46 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[955] $(12)1223151321812321812^{132181232181232181232181223} + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.46 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[953] $(12)1223151321812321812^{1321812321812321812321} \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.46 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[956] $\overset{21}{18} 12321812321812321812321812^{131513221}(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 361$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.41 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[957] $(12)122315132181232^{1812321812321812321812321} \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 361$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.41 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[958] $221812321812321812321812321812321812^{131513221}(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 361$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.41 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[959] $(12)122315132181232^{181232181232181232181232181223} + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 361$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.41 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[960] $(12)1223151321812^{1321812321812321812321812321} \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 400$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.39 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[961] $(12)1223151321812^{132181232181232181232181232181223} + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 400$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.39 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[962] $(12)122315132^{1812321812321812321812321812321} \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 484$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.36 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[963] $(12)122315132^{181232181232181232181232181232181223} + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 484$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.36 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[976] $1^{22^1 21} (12) 12231513221(12) 12231513221(12) 12231513221(12) 1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[977] $1^{2221} (12) 12231513221(12) 12231513221(12) 12231513221(12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[978] $21^{3221} (12) 12231513221(12) 12231513221(12) 12231513221(12) 12^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[979] $3221^{2^1 1} (12) 12231513221(12) 12231513221(12) 12231513221(12) 122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$

[980] $21^{22^1 1} (12) 12231513221(12) 12231513221(12) 12231513221(12) 1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[965] $1^{221} (12) 12231513221(12) 12231513221(12) 12231513221(12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[982] $22^1 21 (12) 12231513221(12) 12231513221(12) 12231513221(12) 12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[983] $2221^{1} (12) 12231513221(12) 12231513221(12) 12231513221(12) 1222^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[964] $1^{3221} (12) 12231513221(12) 12231513221(12) 12231513221(12) 1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[985] $71232181231513221(12) 12231513221(12) 12231513221(12) 12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[986] $71232181231513221(12) 12231513221(12) 12231513221(12) 1222^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[981] $21^{221}(12)12231513221(12)12231513221(12)12231513221(12)122^12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[996] $1^{2221}(12)12231513221(12)12231513221(12)12231513221(12)12^122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[989] $1^{21}(12)12231513221(12)12231513221(12)12231513221(12)122^13 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[990] $1^{32^121}(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[991] $21^{21}(12)12231513221(12)12231513221(12)12231513221(12)12^122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[992] $1^{32^121}(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[993] $1^{3221}(12)12231513221(12)12231513221(12)12231513221(12)12^122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[994] $1^{221}(12)12231513221(12)12231513221(12)12231513221(12)122^122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[995] $21^{21}(12)12231513221(12)12231513221(12)12231513221(12)122^122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[987] $1^{32^121}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[997] $1^{222^11}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [998] $^{32^1 21} 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [999] $^{221} 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1000] $^{1} 2221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1001] $^{1} 222^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1002] $^{222^1 1} 71232181231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1003] $71232181231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [988] $^{221} 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1004] $^{221} 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.75 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [1005] $^{21} 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.75 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [1006] $^{1} 2221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.75 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [1007] $71232181231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.75 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1008] $^{1} 4132^1 181232181232181232181232181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1020] & \begin{matrix} 322^1 1 \\ 1 \end{matrix} 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1021] & \begin{matrix} 221 \\ 1 \end{matrix} 21(12)122^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1022] & \begin{matrix} 1 \\ 1 \end{matrix} 2221(12)12231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1023] & \begin{matrix} 221 \\ 1 \end{matrix} 71232181231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1024] & \begin{matrix} 221 \\ 1 \end{matrix} 71232181231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1025] & \begin{matrix} 221 \\ 1 \end{matrix} 71232181231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1046] & \begin{matrix} 1 \\ 1 \end{matrix} 2221(12)122^1 31513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1045] & \begin{matrix} 1 \\ 1 \end{matrix} 2221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1035] & \begin{matrix} 21 \\ 1 \end{matrix} 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1047] & \begin{matrix} 221 \\ 1 \end{matrix} 712^1 32181231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1036] & \begin{matrix} 3221 \\ 1 \end{matrix} 1(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 61, T \in [53, 61]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1043] & 21^{221}(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1044] & 21^{221}(12)12^1 231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1032] & 1^{221}(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1048] & 71232181231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1049] & 712^1 32181231513221(12)12231513221(12)12231513221^{221}(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1050] & 71232181231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1051] & 71232181231513221(12)12231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1052] & 712^1 32181231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1033] & 1^{221}(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1031] & 21^{21}(12)122^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1034] & 21^{21}(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1069] $21(12)12^{21}231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1066] $1(12)12^{221}231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1065] $1(12)12231513221(12)12231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1070] $1(12)122315132^{221}21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1063] $21(12)12231513221(12)12231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1062] $1(12)12^{221}231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1061] $1(12)12231513221(12)12231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1060] $21(12)12^{21}231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1059] $21(12)12231513221(12)12231513221(12)122^131513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1058] $1(12)12231513221(12)12231513221(12)1223151322^11(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1057] $1(12)122^131513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [1056] $1(12)12231513221(12)12231513221(12)122^131513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1055] $1(12)12^1231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1054] $1(12)12231513221(12)12231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1053] $41321812^132181232181232181232181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1071] $1(12)12231513221(12)12231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1072] $21(12)12231513221(12)12231513221(12)122315132^121(12)122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1073] $21(12)122315132^121(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1074] $21(12)12231513221(12)12231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1075] $2221(12)12^1231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1076] $2221(12)12231513221(12)12231513221(12)122^131513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1077] $812^131513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1078] $81231513221(12)12231513221(12)12231513221(12)122^13151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det =$

$$-49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1079] \quad 71232181231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1080] \quad 71232181231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1081] \quad 71232181231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1067] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1068] \quad 21(12)12231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1064] \quad 21(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1082] \quad 4132181232^1 181232181232181232181232181231513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.94 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1083] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.94 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1084] \quad 21(12)12231513221(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.94 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1093] \quad 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1095] \quad 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1101] & \begin{matrix} 221 \\ 21(12)12231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1102] & \begin{matrix} 221 \\ 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1103] & \begin{matrix} 1 \\ 2221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1104] & \begin{matrix} 1 \\ 2221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1105] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1106] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1107] & \begin{matrix} 2221 \\ 71232181231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1108] & \begin{matrix} 2221 \\ 71232^1 181231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1109] & \begin{matrix} 2221 \\ 71232181231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1110] & \begin{matrix} 221 \\ 71232181231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1111] & \begin{matrix} 221 \\ 71232^1 181231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]
\end{aligned}$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1112] $71232181231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1113] $71232181231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1114] $71232^1 181231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1115] $71232181231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1091] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1092] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1090] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1118] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 61$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.76 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1120] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 61$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.76 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1121] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -100$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1122] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -100$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

- [1123] $21^{21}(12)122315132^1 21^1(12)12231513221(12)12231513221^{221}(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1124] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1125] $1(12)122315132^1 21^1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1126] $21^{21}(12)122315132^1 21^1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1127] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1128] $21^{221}(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1129] $2221(12)122315132^1 21^1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1119] $1(12)122315132^1 21^1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1116] $41321812321812321812321812321812321812315132^1 21(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1117] $41321812321812^1 32181232181232181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1157] $81231513221(12)12231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1134] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det =$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1132] & \begin{matrix} 3221 & & 1 & & 221 \\ 1(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 61, \Delta = -112, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1131] & \begin{matrix} 413218123218123218123218123218123218123151322^1 1(12) + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -109, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 10 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[1159] & \begin{matrix} 71232181231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1160] & \begin{matrix} 71232181231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1161] & \begin{matrix} 71232181231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1162] & \begin{matrix} 71232181231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1163] & \begin{matrix} 71232181231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1164] & \begin{matrix} 71232181231513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1156] & \begin{matrix} 2221(12)12231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -103, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1158] & \begin{matrix} 81231513221(12)12231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -109, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1130] & \begin{matrix} 4132181232181232^1 181232181232181232181231513221(12) + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -109, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1179] $21^{21}(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1169] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1170] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1171] $1(12)12^{3221} 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1168] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1173] $21^{21}(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1174] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1175] $21^{21}(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1176] $1(12)12231513221(12)12231513221(12)12^{3221} 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1177] $1(12)12^{3221} 231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1178] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [1167] $1(12)12^{3221}1^{231513221}(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1180] $1(12)12231513221(12)12231513221(12)12^{2221}1^{231513221}(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1181] $21(12)12231513221(12)12231513221(12)122^{3221}1^{31513221}(12)122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1182] $21(12)12231513221(12)122^{3221}1^{31513221}(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1183] $21(12)12231513221(12)12231513221(12)12^{2221}1^{231513221}(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1184] $2221(12)12231513221^{11}(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1185] $81231513221(12)122315132^{11}21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1186] $812315132^{11}21(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1187] $712321812315132^{11}21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [1188] $712321812315132^{11}21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [1189] $712321812315132^{11}21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [1165] $41321812321812321812^{11}32181232181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det =$

$-169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

$[1166] \overset{3221}{1}(12)12231513221(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12^{\overset{221}{221}} + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$[1172] \overset{3221}{1}(12)12231513221(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12^{\overset{21}{21}} + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$[1197] \overset{21}{21}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)12^{\overset{221}{221}} + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1194] \overset{3221}{1}(12)12231513221(12)122315132^{\overset{1}{1}}21(12)12231513221(12)12^{\overset{21}{21}} + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1195] \overset{21}{21}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1196] \overset{221}{1}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)12^{\overset{221}{221}} + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1192] \overset{221}{1}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1198] \overset{3221}{1}(12)12231513221(12)122315132^{\overset{1}{1}}21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1199] \overset{221}{1}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1200] \overset{21}{21}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1201] \overset{2221}{1}(12)12231513221(12)1223151322^{\overset{1}{1}}1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1202] & \begin{matrix} 3221 \\ 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1203] & \begin{matrix} 221 \\ 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1204] & \begin{matrix} 1 \\ 2221(12)12231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1205] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1206] & \begin{matrix} 2221 \\ 71232181231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1207] & \begin{matrix} 221 \\ 71232181231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1208] & \begin{matrix} 1 \\ 71232181231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1193] & \begin{matrix} 21 \\ 1(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1191] & \begin{matrix} 3221 \\ 1(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1190] & \begin{matrix} 221 \\ 4132181232181232181232^1 181232181232181231513221(12) + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1210] & \begin{matrix} 3221 \\ 21(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 61, T \in [52, 61]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1211] & \text{81231513221}(12)122^1 131513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1212] & \text{712321812}^1 131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1213] & \text{712321812}^1 131513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1214] & \text{712321812}^1 131513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1209] & \text{1}(12)12231513221(12)122^1 131513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1215] & \text{41321812321812321812321812}^1 32181232181231513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 47 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1216] & \text{1}(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1217] & \text{21}(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1218] & \text{81231513221}(12)1223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1219] & \text{7123218123151322}^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[1220] & \text{7123218123151322}^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1221] $7123218123151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1231] $8123151322^1 1(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1230] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1227] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1228] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [1229] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1225] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1224] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1226] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [1222] $4132181232181232181232181232^1 181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 50 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1223] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [1232] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det =$

$-324, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.43 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[1233] \overset{3221}{1}(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-324, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.43 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[1234] \overset{3221}{1}(12)12231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-324, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.43 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[1235] 41321812321812321812321812321812^1 32181231513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det =$
 $-361, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.41 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 56 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$[1236] \overset{21}{1}(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-361, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.41 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

$[1237] \overset{3221}{21}(12)12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-361, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.41 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

$[1238] \overset{1}{81}231513221(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det =$
 $-361, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.41 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

$[1239] 4132181232181232181232181232181232^1 181231513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det =$
 $-400, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.39 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 59 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1240] 41321812321812321812321812321812321812^1 31513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det =$
 $-484, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.36 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 65 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

$[1241] \overset{3221}{1}(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-484, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.36 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

$[1242] \overset{3221}{1}(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-484, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.36 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

$[1243] \overset{3221}{1}(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-484, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.36 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1244] & \begin{matrix} 3221 & 1 & 221 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 61, \Delta = -112, \det = \\ -529, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.34 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \end{matrix} \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1246] & \begin{matrix} 3221 & 1 & 221 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 61, \Delta = -112, \det = \\ -529, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.34 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \end{matrix} \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1245] & \begin{matrix} 3221 & 1 & 21 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 61, \Delta = -112, \det = \\ -529, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.34 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \end{matrix} \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
T = 48 & [1248] \begin{matrix} 22^1 1 \\ 21(12)12231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 & T = 48, T_{\max} = 60, \Delta = \\ -103, \det = 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 55, T_{\max} = 60, T \in [55, 60] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1249] & \begin{matrix} 3221 \\ 81232181231513221(12)12231513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2 & T = 48, T_{\max} = 62, \Delta = -112, \det = \\ 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 54, T_{\max} = 62, T \in [54, 62] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1250] & \begin{matrix} 1 & 22^1 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 48, T_{\max} = 61, \Delta = -111, \det = \\ 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 55, T_{\max} = 61, T \in [55, 61] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1251] & \begin{matrix} 1 & 22^1 1 \\ 181232181231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 & T = 48, T_{\max} = 60, \Delta = -103, \det = \\ 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 55, T_{\max} = 60, T \in [55, 60] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1252] & \begin{matrix} 1 & 1 & 22^1 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2 & T = 48, T_{\max} = 61, \Delta = -111, \det = \\ 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 55, T_{\max} = 61, T \in [55, 61] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1253] & \begin{matrix} 1 & 22^1 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 & T = 48, T_{\max} = 61, \Delta = -111, \det = \\ 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 55, T_{\max} = 61, T \in [55, 61] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1247] & \begin{matrix} 22^1 21 \\ 1(12)12231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 & T = 48, T_{\max} = 60, \Delta = -103, \det = \\ 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 55, T_{\max} = 60, T \in [55, 60] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1254] & \begin{matrix} 222^1 1 \\ 1(12)12231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 & T = 48, T_{\max} = 60, \Delta = -103, \det = \\ 9, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 54.22 \rightarrow 55, T_{\max} = 60, T \in [55, 60] \end{matrix}
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[1255] $81232181231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1256] $181232181231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 60, T \in [55, 60]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1257] $413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1258] $(12)122315132181232181232181232181232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.75 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1259] $181232181231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.75 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1260] $(12)122315132181232181232181232181232181232181232^1 3215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1264] $81232181231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1265] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1266] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1267] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1268] $181232181231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1269] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det =$

25, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1270] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1271] $81231513221(12)12231513221(12)12231513221(12)122315132^1 16 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1262] $1(12)122^1 31513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1263] $21(12)122^1 31513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1261] $(12)1223151321812321812321812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1274] $21(12)12^1 231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1276] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1277] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1278] $181232181231513221(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1279] $1812^1 32181231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1280] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1272] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.22 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1273] & \begin{matrix} 2221 \\ 1(12)12^1 231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.22 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1275] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.22 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1297] & \begin{matrix} 1 \\ 812^1 31513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1285] & \begin{matrix} 3221 \\ 812^1 32181231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 62, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1286] & \begin{matrix} 1 & 1 & 21 \\ 413221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1287] & \begin{matrix} 1 & 1 & 21 \\ 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1288] & \begin{matrix} 1 & 221 \\ 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1289] & \begin{matrix} 1 & 221 \\ 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1290] & \begin{matrix} 1 & 21 \\ 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1291] & \begin{matrix} 1 & 21 \\ 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1292] & 181232181231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1293] & 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1294] & 413221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1295] & 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1296] & 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1284] & 21^{221}(12)122315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1298] & 81231513221(12)12231513221(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1299] & 812^1 31513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1300] & 81231513221(12)12231513221(12)12231513221(12)122^1 31513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1281] & (12)122315132181232181232181232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1282] & (12)122315132181232181232181232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -104, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1283] $1(12)122315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1303] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 64, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.94 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1304] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 64, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.94 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1301] $(12)1223151321812321812321812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.94 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1302] $(12)1223151321812321812321812321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 64, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.94 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1305] $1(12)12231513221(12)12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1309] $21(12)12231513221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1310] $21(12)12231513221(12)12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1311] $81232181231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1312] $413221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1313] $181232181231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1314] $181232181231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1315] & 181232^1 181231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1316] & 413221^1(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1317] & 81231513221^1(12)12231513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1318] & 81231513221^1(12)12231513221(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1319] & 81231513221^1(12)12231513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1320] & 81231513221^1(12)12231513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1307] & 1^{2221}(12)1223151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1306] & 1^{2221}(12)12231513221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1308] & 21^{221}(12)1223151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1321] & (12)122315132181232181232181232181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.76 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1322] & (12)122^1 31513218123218123218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.76 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1323] (12)122¹31513218123218123218123218123218123218123151 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -104$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1324] (12)12231513218123218123218123218123218123218123218123151 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -104$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1325] ¹413221(12)12231513221(12)12231513221(12)122¹31513221(12)1 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1326] ¹413221(12)12231513221(12)12231513221(12)122¹31513221(12)12 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1327] ¹413221(12)12231513221(12)12231513221(12)122¹31513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1348] ¹81231513221(12)12231513221(12)122¹31513221(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1332] ²²²¹1(12)12231513221(12)122315132¹21(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1333] ²²²¹1(12)12231513221(12)12¹231513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1334] ²²¹21(12)12231513221(12)122315132¹21(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1335] ²²¹21(12)12231513221(12)12¹231513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1336] ³²²¹81232¹181231513221(12)12231513221(12)12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 62$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 62$, $T \in [53, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1337] ³²²¹81232181231513221(12)12231513221(12)122¹31513221(12)12 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 62$, $\Delta = -112$, $\det =$

$$121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1338] \quad 81232181231513221(12)12231513221(12)12231513221^{3221}1(12)12 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 62, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1339] \quad 413221(12)122315132^{121}(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1340] \quad 413221(12)122^{121}231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1341] \quad 413221(12)122^{131513221}(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1342] \quad 413221(12)122^{131513221}(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1343] \quad 181232181231513221(12)1223151322^{11}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1344] \quad 181232181231513221(12)122^{131513221}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1345] \quad 413221(12)12^{121}231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1346] \quad 413221(12)122315132^{121}(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1347] \quad 413221(12)122^{131513221}(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1331] \quad (12)1223151321812321812321812321812^{13218123218123151} + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1349] $81231513221(12)12231513221(12)1223151322^11(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1350] $81231513221(12)12231513221(12)122^131513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1351] $81231513221(12)12231513221(12)1223151322^11(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1328] $(12)1223151321812321812321812321812^13218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1330] $(12)12^1231513218123218123218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1329] $(12)12^1231513218123218123218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1354] $1(12)12231513221(12)12231513221(12)122^1315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1356] $81232181231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1357] $413221(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1358] $413221(12)12231513221(12)122^131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1359] $413221(12)12231513221(12)12231513221(12)12^1231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [1360] $413221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1361] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1362] $413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1363] $1812321812315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [1364] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1365] $413221(12)12231513221(12)12231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1366] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1367] $413221(12)12^1 231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1368] $812315132^1 21(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1369] $81231513221(12)122315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1370] $81231513221(12)122315132^1 21(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1371] $812315132^1 21(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det =$

169, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.59 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1352] $(12)122315132181232181232181232^1 18123218123218123215 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -106$, $\det =$

169, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.59 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1353] $(12)122315132181232181232181232^1 18123218123218123151 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -104$, $\det =$

169, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.59 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1355] $21^{221}(12)12231513221(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det =$

169, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.59 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1382] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1376] $81232181231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 62$, $\Delta = -112$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 62$, $T \in [53, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1377] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1378] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1379] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1380] $181232181231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[1381] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1375] $21^{221}(12)12231513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1383] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1384] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)12^1 231513221(12)12231513216 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1374] & \begin{matrix} 2221 \\ 1(12)12231513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1372] & (12)1223151321812321812321812^1 3218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1373] & (12)1223151321812321812321812^1 3218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -104, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1385] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[1386] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[1387] & \begin{matrix} 3221 \\ 81232181231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 62, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 62, T \in [53, 62] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1388] & \begin{matrix} 1 & & & 21 \\ 413221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1389] & 1812321812^1 31513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1390] & \begin{matrix} 1 & & & 1 \\ 413221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1391] $81231513221(12)122^1 31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1392] $81231513221(12)122^1 31513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1393] $(12)122315132181232181232^1 18123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1397] $81232181231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1398] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1399] $18123218123151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1400] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1401] $81231513221(12)1223151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1402] $81231513221(12)1223151322^1 1(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1394] $(12)122315132181232181232^1 18123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1396] $21(12)12231513221(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [1395] $1(12)12231513221(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1414] & 8123151322^1 1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1407] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1408] & 413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1409] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1410] & 413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1411] & 413221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1412] & 413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1413] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1406] & 413221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1415] & 8123151322^1 1(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1403] & (12)1223151321812321812^1 3218123218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 60, T \in [53, 60]
\end{aligned}$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[1405] $812321812315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.46 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1404] $(12)1223151321812321812^1 3218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.46 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[1418] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.43 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1417] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.43 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1416] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.43 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1420] $(12)122315132181232^1 18123218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[1421] $81232181231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1422] $812321812^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[1423] $413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1424] $413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1425] $81231513221(12)12^1 231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1426] $81231513221(12)12^1 231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.41 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[1419] $(12)122315132181232^1 18123218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.41 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[1427] $(12)1223151321812^1 3218123218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 400, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.39 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1428] $(12)1223151321812^1 3218123218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 400, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.39 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1429] $8123218123151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.37 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1430] $(12)122315132^1 18123218123218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[1431] $(12)122315132^1 18123218123218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[1432] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[1433] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[1434] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[1435] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.34 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1436] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.34 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[1437] \quad 413221(12)12231513221(12)12^{11}231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.34 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$T = 49 \quad [1438] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1439] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^{11}2 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1440] \quad (12)12231513221(12)12231513221(12)12231513221(12)12^{21}31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 66, T \in [56, 66]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1441] \quad 22^{11}31513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 66, T \in [56, 66]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1442] \quad 31513221(12)12231513221(12)12231513221(12)12231513221(12)12^{11}2 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 65, T \in [56, 65]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1443] \quad 1513221(12)12231513221(12)12231513221(12)12231513221(12)122^{11}2 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1444] \quad 41413221(12)12231513221(12)12231513221(12)12231513221(12)12^{11}2 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1447] \quad 1513221(12)12231513221(12)12231513221(12)12231513221(12)12^{11}22 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.22 \rightarrow 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1445] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.22 \rightarrow 56, T_{\max} = 66, T \in [56, 66]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1446] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^{11}3 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.22 \rightarrow 56, T_{\max} = 66, T \in [56, 66]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1448] $(12)122315132181232181232181232181232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.75 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1458] $81231513221(12)12231513221(12)12231513221(12)122315132^1 217 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1459] $4132^1 21(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 62$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1449] $(12)1223151321812321812321812321812321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1450] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1451] $1(12)12231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1452] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1453] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1454] $1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1455] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1456] $15132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.44 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [1457] $81231513221(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1462] $1513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1463] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1460] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1461] $21(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1487] $41413221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1486] $41413221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1464] $(12)122315132181232181232181232181232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1465] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1466] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1467] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1468] $1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det =$

$-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

$[1469] \overset{21}{1}(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 65, T \in [55, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1470] \overset{221}{1}(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[1471] \overset{221}{1}(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1472] \overset{1513221}{1}(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1473] \overset{151322^1 1}{1}(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[1474] \overset{21}{21}(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[1475] \overset{21}{21}(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1476] \overset{1}{31513221}(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 65, T \in [55, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1477] \overset{1}{1513221}(12)12231513221(12)12231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1478] \overset{1}{151322^1 1}(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[1479] \overset{1}{81231513221}(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1480] & 812^1 31513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1481] & 812^1 31513221(12)12231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1482] & 81231513221(12)12231513221(12)12231513221(12)122^1 315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1483] & 41322^1 1(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1484] & 413221(12)12231513221(12)12231513221(12)122315132181232^1 18 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1485] & 414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1488] & 414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1502] & 41413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1503] & 41413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1489] & (12)1223151321812321812321812321812^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = \\
& -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1490] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1491] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1492] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1493] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1494] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1495] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1496] & 1 \begin{pmatrix} 31513221 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)122315132^1 21(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1497] & 1 \begin{pmatrix} 315132^1 21 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1498] & 1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)1223151322^1 1(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1499] & 231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1500] & 315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1501] & 31513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65]
\end{aligned}$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1519] \quad 41413221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1507] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1504] \quad 1^1 (12)12^1 231513221(12)12231513221(12)12231513221(12)12231513+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1506] \quad 1^1 (12)12^1 231513221(12)12231513221(12)12231513221(12)122315132+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1505] \quad 1^1 (12)12231513221(12)12231513221(12)122315132^1 21(12)1223151+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1520] \quad 4141322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1508] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1509] \quad 1^1 (12) \quad 122^1 31513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1510] \quad 2231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1511] \quad 231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1512] \quad 81231513221(12)12231513221(12)122315132^1 21(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1513] $81231513221(12)12231513221(12)12231513221(12)12^12315132171 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1514] $81231513221(12)12231513221(12)12231513221(12)12^12315132217 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1515] $81231513221(12)12231513221(12)122315132^121(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1516] $413221(12)122^131513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 62, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1517] $41413221(12)122^131513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1518] $4141322^11(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1529] $2231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1530] $1513221(12)12231513221(12)12231513221(12)122^131513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1521] $(12)122315132181232181232181232181232^1181232181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1522] $(12)122^1315132181232181232181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1523] $1^{231513221}(12)12231513221(12)12231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1524] $1(12)12231513221(12)12231513221(12)12231513221(12)12^12315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1525] $1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1526] $1(12)122315132^121(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1527] $21(12)122315132^121(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1528] $(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1551] $81231513221(12)12231513221(12)1223151322^11(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1557] $41413221(12)122315132^121(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1531] $(12)12^12315132181232181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$

[1532] $(12)1223151321812321812321812321812^132181232181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1533] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1534] $1(12)12231513221(12)12231513221(12)122^131513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1535] $1(12)12231513221(12)12231513221(12)1223151322^11(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det =$

$$-121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1536] \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)122^1315132+\hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1537] \overset{2315132^1 21}{1} (12) 12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1538] \overset{3221}{1}(12)122^131513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1539] \overset{21}{1}(12)12231513221(12)12231513221(12)12231513221(12)12^1231513+\hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1540] \overset{3151322^1 1}{1} (12) 12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1541] \overset{221}{1}(12)12231513221(12)12231513221(12)122315132^121(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1542] \overset{1513221}{1} (12) 12^1231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1543] \overset{1513221}{1} (12) 122315132^121(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1544] \overset{21}{21}(12)12231513221(12)12231513221(12)122315132^121(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1545] (12)12^1231513221(12)12231513221(12)12231513221 \overset{2231513221}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1546] \overset{1}{2315132^1 21}(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1547] \quad 3151322^1(12)12231513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1548] \quad 1513221(12)122^131513221(12)12231513221(12)12231513221(12)1222+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1549] \quad 81231513221(12)12231513221(12)122^131513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1550] \quad 81231513221(12)12231513221(12)1223151322^1(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1559] \quad 41413221(12)122315132^121(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1552] \quad 81231513221(12)12231513221(12)122^131513221(12)122315132217+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1558] \quad 41413221(12)12^1231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1554] \quad 413221(12)122315132^121(12)12231513221(12)12231513218123218+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1555] \quad 413221(12)12^1231513221(12)12231513221(12)12231513218123218+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1556] \quad 41413221(12)12^1231513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1553] \quad 413221(12)12231513221(12)12231513221(12)1223151321812^13218+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1585] $41413221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1584] $41413221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1560] $(12)122315132181232181232181232^1 181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[1561] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1562] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1563] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1564] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1565] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1566] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1567] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1568] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1569] $1^{221}(12)12231513221(12)12231513221(12)1223151322^11(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1570] $1^{1513221}(12)12231513221(12)122^131513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1571] $1^{1513221}(12)12231513221(12)12231513221(12)122^131513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1572] $21^{21}(12)1223151322^11(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1573] $21^{21}(12)12231513221(12)12231513221(12)1223151322^11(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1574] $31513221(12)12231513221(12)12231513221(12)122^131513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1575] $1513221(12)12231513221(12)12231513221(12)12^1231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1576] $1513221(12)12^1231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1577] $81231513221(12)122315132^121(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1578] $812315132^121(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1579] $81231513221(12)122315132^121(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [1580] $812315132^1 21(12)12231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1581] $413221(12)12231513221(12)122^1 31513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 62$, $\Delta = -111$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54$, $T_{\max} = 62$, $T \in [54, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1582] $41413221(12)12231513221(12)12231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1583] $41413221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1586] $(12)1223151321812321812321812^1 32181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [1598] $41413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1599] $41413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1587] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1588] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1589] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1590] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1591] $21(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det =$

-196, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1592] $(12)12231513221(12)12231513221(12)12231513221$ $\overset{1}{(12)} \overset{22315132^1 21}{1} + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1593] $22315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1594] $1513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1595] $81231513221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1596] $81231513221(12)12231513221(12)12^1 231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1597] $413221(12)1223151322^1 1(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 62$, $\Delta = -111$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 62$, $T \in [54, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1611] $41413221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -225$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.52 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1612] $41413221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -225$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.52 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1600] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -225$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.52 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1601] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -225$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.52 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1602] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -225$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 231513221^1 \\ [1603] \end{matrix} 1 \begin{matrix} (12) \\ (12) \end{matrix} 12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 513221 \\ [1604] \end{matrix} 1 \begin{matrix} (12) \\ (12) \end{matrix} 12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 31513221 \\ [1605] \end{matrix} 1 \begin{matrix} (12) \\ (12) \end{matrix} 122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ [1606] \end{matrix} 1 \begin{matrix} (12) \\ (12) \end{matrix} 12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1607] \end{matrix} 231513221^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1608] \end{matrix} 81231513221(12)122^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1609] \end{matrix} 81231513221(12)122^1 31513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1610] \end{matrix} 413221(12)12231513221(12)122315132^1 21(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1627] \end{matrix} 41413221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -256, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.48 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1628] \end{matrix} 41413221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -256, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.48 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ [1613] \end{matrix} (12)122315132181232181232^1 181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = \\
& -256, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.48 \rightarrow 54, T_{\max} = 61, T \in [54, 61]
\end{aligned}$$

[1614] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1615] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1616] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1617] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1618] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1619] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1620] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1621] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1622] $231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1623] $31513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1624] $81231513221(12)1223151322^1 1(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1625] $81231513221(12)12231513221^1(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.48 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1626] $413221(12)12231513221(12)12^1231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 62$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.48 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1649] $41413221(12)12231513221(12)12231513221(12)12^1231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1650] $41413221(12)12231513221(12)12231513221(12)12^1231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1629] $(12)1223151321812321812^132181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 526 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$$

[1630] $1(12)12231513221(12)122315132^121(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1631] $1(12)1223151322^11(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1632] $1^{231513221}(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1633] $1(12)12231513221(12)12231513221(12)122315132^121(12)122315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1634] $1(12)122315132^121(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1635] $1(12)12^1231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1636] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1637] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1638] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1639] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1640] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1641] $21(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1642] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1643] $31513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1644] $1513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1645] $1513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1646] $8123151322^1 1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

- [1647] $8123151322^1 1(12)12231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1648] $413221(12)12231513221(12)12231513221(12)122^1 31513218123218 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 62, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1653] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.43 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1654] $1513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.43 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1651] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -324, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.43 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1652] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.43 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1676] $41413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -110, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [1655] $(12)122315132181232^1 181232181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 664 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$
- [1656] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [1657] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [1658] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [1659] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det =$

$$\begin{aligned}
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 231513221 \\ 1 \\ 1660 \end{matrix} 1 (12) 122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ 1 \\ 1661 \end{matrix} 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ 1 \\ 1662 \end{matrix} 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 31513221 \\ 1 \\ 1663 \end{matrix} 1 (12) 12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 31513221 \\ 1 \\ 1664 \end{matrix} 1 (12) 1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ 1 \\ 1665 \end{matrix} 1 (12) 12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1666 \end{matrix} (12)12231513221(12)12231513221(12)12231513221^1 (12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1667 \end{matrix} 2231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1668 \end{matrix} 223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1669 \end{matrix} 31513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1670 \end{matrix} 31513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1671] & 81231513221(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1672] & 81231513221(12)12^1 231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1673] & 413221(12)12231513221(12)12231513221(12)122315132^1 18123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1674] & 413221(12)12231513221(12)1223151322^1 1(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1675] & 41413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1678] & 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -400, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.39 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 139 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1677] & (12)1223151321812^1 32181232181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = -400, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.39 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 739 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0} \\
[1680] & 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1681] & 1(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1679] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[1682] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[1683] & 1 \quad (12) \quad 12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
& 231513221 \\
[1684] & 1 \quad (12) \quad 12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& 31513221 \\
[1685] & 1 \quad (12) \quad 12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
& 31513221 \\
[1686] & 1 \quad (12) \quad 12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& 1 \\
[1687] & (12)12231513221(12)12231513221(12)122315132^1 21 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
& 1 \\
[1688] & 231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& 1 \\
[1689] & 231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
& 1 \\
[1690] & 413221(12)12231513221(12)12231513221(12)12^1 231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& 221 \\
[1694] & 1(12)12231513221(12)12231513221(12)12^1 231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
& 3221 \\
[1692] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[1693] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 66, T \in [54, 66]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1691] & (12)122315132^1 181232181232181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = \\
& -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 901 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0} \\
[1695] & \begin{matrix} 21 \\ 12 \end{matrix} 12231513221(12)12231513221(12)12^1 231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = \\
& -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1696] & (12)122315132^1 21(12)12231513221(12)12231513221 \begin{matrix} 2231513221 \\ (12) \end{matrix} 1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1697] & 1513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = \\
& -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1698] & \begin{matrix} 231513221 \\ 1 \end{matrix} (12) \begin{matrix} 12231513221 \\ (12) \end{matrix} 122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[1699] & \begin{matrix} 3221 \\ 1 \end{matrix} (12)12231513221(12)122^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[1700] & \begin{matrix} 513221 \\ 1 \end{matrix} (12) \begin{matrix} 12231513221 \\ (12) \end{matrix} 12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[1701] & \begin{matrix} 513221 \\ 1 \end{matrix} (12) \begin{matrix} 1223151322^1 \\ (12) \end{matrix} 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1702] & \begin{matrix} 21 \\ 1 \end{matrix} (12)12231513221(12)12^1 231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[1703] & \begin{matrix} 221 \\ 1 \end{matrix} (12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = \\
& -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1704] & \begin{matrix} 21 \\ 21 \end{matrix} (12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = \\
& -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1705] $(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -529$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.34 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$
- [1706] $31513221(12)12231513221(12)1223151322^11(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -529$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.34 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$
- [1707] $1513221(12)12231513221(12)12^1231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -529$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.34 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [1712] $1(12)12231513221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1709] $1(12)12231513221(12)122^131513221(12)12231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1710] $1(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1711] $1(12)12231513221(12)12231513221(12)122^131513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [1708] $1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1713] $231513221(12)12231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1714] $31513221(12)122315132^121(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [1715] $1(12)12231513221(12)12231513221(12)1223151322^11(12)122315132 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -625$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 474 & 42 \\ 2 & 42 & 3 \end{pmatrix}_{b_0}$

[1716] $1^{231513221}(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$

[1717] $1^{21}(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1718] $(12)12231513221(12)122315132^1 21(12)12231513221^{2231513221}(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$

[1719] $2231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$

[1720] $231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 474 & 42 \\ 2 & 42 & 3 \end{pmatrix}_{b_0}$

[1721] $31513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1722] $1^{231513221}(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -676, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.30 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[1723] $1^{513221}(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -676, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.30 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[1724] $(12)12231513221(12)12231513221(12)12231513221^1 (12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -676, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.30 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[1725] $2231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -676, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.30 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[1726] $1^{3221}(12)12231513221(12)12231513221(12)122^1 31513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -729, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.29 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 269 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1727] $1^{513221}(12)12231513221(12)12^1231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -729, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.29 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1728] $(12)12231513221(12)122^131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -729, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.29 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$

[1729] $1^{3221}(12)12231513221(12)12^1231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -784, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.28 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 28 & 2 \\ 28 & 307 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$

[1730] $1^{12231513221}(12)12^1231513221(12)12231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 666 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

[1731] $1^{231513221}(12)12231513221(12)1223151322^11(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1732] $1^{3221}(12)12231513221(12)122315132^121(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 318 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

[1733] $(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$

[1734] $(12)12231513221(12)12231513221(12)12^1231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1735] $2231513221(12)12231513221(12)122315132^121(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1736] $231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 666 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

[1737] $1^{513221}(12)12231513221(12)12231513221(12)12^1231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -900, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.26 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[1738] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 743 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$

[1739] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 681 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1740] $(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$

[1741] $2231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 774 & 54 \\ 2 & 54 & 3 \end{pmatrix}_{b_0}$

[1742] $231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 743 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$

[1743] $31513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 681 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1745] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -1024, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 32 & 2 \\ 32 & 735 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$

[1744] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1024, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 32 & 2 \\ 32 & 799 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$

[1746] $231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1024, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 32 & 2 \\ 32 & 799 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$

[1747] $31513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -1024, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 32 & 2 \\ 32 & 735 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$

[1748] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1089, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.24 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

[1749] $(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1156, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.23 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$

[1750] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.23 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 35 & 2 \\ 35 & 454 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$

[1751] $2231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.23 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 35 & 2 \\ 35 & 1014 & 62 \\ 2 & 62 & 3 \end{pmatrix}_{b_0}$

[1752] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1369, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.21 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 37 & 2 \\ 37 & 517 & 49 \\ 2 & 49 & 3 \end{pmatrix}_{b_0}$

[1753] $2231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1444, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.21 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 38 & 2 \\ 38 & 1177 & 67 \\ 2 & 67 & 3 \end{pmatrix}_{b_0}$

[1754] $2231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1521, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.20 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 39 & 2 \\ 39 & 1247 & 69 \\ 2 & 69 & 3 \end{pmatrix}_{b_0}$

[1755] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1600, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.20 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 40 & 2 \\ 40 & 1239 & 69 \\ 2 & 69 & 3 \end{pmatrix}_{b_0}$

[1756] $231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1600, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.20 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 40 & 2 \\ 40 & 1239 & 69 \\ 2 & 69 & 3 \end{pmatrix}_{b_0}$

[1757] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1681, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.19 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 41 & 2 \\ 41 & 1311 & 71 \\ 2 & 71 & 3 \end{pmatrix}_{b_0}$

[1758] $231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1681, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.19 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 41 & 2 \\ 41 & 1311 & 71 \\ 2 & 71 & 3 \end{pmatrix}_{b_0}$

[1759] $2231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -2401, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.16 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 49 & 2 \\ 49 & 1959 & 87 \\ 2 & 87 & 3 \end{pmatrix}_{b_0}$

[1760] $2231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -2500, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.16 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 50 & 2 \\ 50 & 2049 & 89 \\ 2 & 89 & 3 \end{pmatrix}_{b_0}$

$T = 50$ [1761] $(12)122315132181232181232181232181232181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 16, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.75 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1762] $(12)122315132181232181232181232181232181232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 16, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.75 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1763] $(12)1223151321812321812321812321812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1764] $(12)1223151321812321812321812321812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1765] $81232181231513221(12)12231513221(12)12231513221^{15132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1766] $81231513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1767] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 66, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56, T_{\max} = 66, T \in [56, 66]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1768] $81231513221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1769] $81231513221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1770] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -104, \det = 36, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.22 \rightarrow 56, T_{\max} = 62, T \in [56, 62]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [1771] $(12)122315132181232181232181232181232181232^1181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1772] $(12)122315132181232181232181232181232181232^11812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1773] $414132^121(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1774] $41413221(12)12231513221(12)12231513221(12)122315132181232^118 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1775] $81232181231513221(12)12231513221(12)12231513221^{151322^11}(12)1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1776] $812^132181231513221(12)12231513221(12)12231513221^{1513221}(12)1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1777] $81231513221^{1513221}(12)12231513221(12)12231513221(12)122^1315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1778] $812^131513221^{1513221}(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1779] $413221(12)12231513221(12)12231513221(12)12231513221(12)122^1315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 66$, $T \in [56, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1780] $41322^11(12)12231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 66$, $T \in [56, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1781] $812^131513221^{1513221}(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.06 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1782] $81231513221^{1513221}(12)12231513221(12)12231513221(12)122^13151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 55.06 \rightarrow 56, T_{\max} = 62, T \in [56, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1783] & 81231513221(12)12231513221(12)12231513221(12)122^13151322181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.06 \rightarrow 56, T_{\max} = 62, T \in [56, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1784] & 812^131513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.06 \rightarrow 56, T_{\max} = 62, T \in [56, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1785] & (12)1223151321812321812321812321812321812^132181232171 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 64, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.94 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1786] & (12)1223151321812321812321812321812321812^1321812321712 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 64, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.94 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1787] & 41413221(12)122^131513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1788] & 4141322^11(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1789] & 81232181231513221(12)12231513221(12)122315132^121(12)1 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1790] & 81231513221(12)12231513221(12)12231513221(12)12^12315132181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1791] & 81231513221(12)12231513221(12)122315132^121(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1792] & 81231513221(12)12231513221(12)12231513221(12)12^123151321812 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1793] & 81231513221(12)12231513221(12)122315132^121(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1794] $81231513221(12)12231513221(12)12231513221(12)12^123151322181+\hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1795] $81231513221(12)12231513221(12)122315132^121(12)1223151322181+\hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1796] $(12)122315132181232181232181232^1181232181232171 + \hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1797] $(12)122^1315132181232181232181232181232181232171 + \hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1798] $(12)122315132181232181232181232^11812321812321712 + \hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1799] $(12)122^13151321812321812321812321812321812321712 + \hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1800] $413221(12)12231513221(12)12231513221(12)12231513221(12)12^12315+\hat{1}\rightarrow 2$ $T = 50, T_{\max} = 66, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1801] $(12)12^12315132181232181232181232181232181232171 + \hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1802] $(12)1223151321812321812321812321812^132181232181232171 + \hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1803] $(12)1223151321812321812321812321812^1321812321812321712 + \hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1804] $(12)12^123151321812321812321812321812321812321712 + \hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1805] $41413221(12)12^1231513221(12)12231513221(12)12231513218123218+\hat{1}\rightarrow 2$ $T = 50, T_{\max} = 62, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$

- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [1806] 41413221(12)122315132¹21(12)12231513221(12)12231513218123218+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [1807] 41413221(12)12231513221(12)12231513221(12)1223151321812¹3218+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [1808] 81232181231513221(12)12231513221(12)122¹31513221¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [1809] 81232181231513221(12)12231513221(12)1223151322¹1¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [1810] 81232¹181231513221(12)12231513221(12)12231513221¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
 [1811] 81231513221¹(12)12231513221(12)122¹31513221(12)122315132181¹+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [1812] 81231513221¹(12)12231513221(12)1223151322¹1(12)122315132181¹+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [1813] 413221(12)122¹31513221(12)12231513221¹(12)12231513221(12)122315+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [1814] 81231513221¹(12)12231513221(12)1223151322¹1(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [1815] 81231513221¹(12)12231513221(12)122¹31513221(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [1816] 81231513221¹(12)12231513221(12)1223151322¹1(12)1223151322181+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

- [1817] $81231513221(12)12231513221(12)122^131513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1818] $(12)122315132181232181232181232^1181232181232181232171 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [1819] $(12)122315132181232181232181232^11812321812321812321712 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [1820] $41413221(12)12231513221(12)122^131513221(12)12231513218123218+\hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1821] $81232181231513221(12)122315132^121(12)12231513221(12)1 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1822] $812315132^121(12)12231513221(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1823] $81231513221(12)122315132^121(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1824] $413221(12)12^1231513221(12)12231513221(12)12231513221(12)122315 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1825] $812315132^121(12)12231513221(12)12231513221(12)1223151321812 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1826] $81231513221(12)122315132^121(12)12231513221(12)1223151321812 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1827] $812315132^121(12)12231513221(12)12231513221(12)1223151322181 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1828] $81231513221(12)122315132^121(12)12231513221(12)1223151322181 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det =$

169, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1829] $(12)1223151321812321812321812^1 32181232181232181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[1830] $(12)1223151321812321812321812^1 321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[1831] $41413221(12)1223151322^1 1(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1832] $81232181231513221(12)12231513221(12)12^1 231513221 (12) 1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1833] $81231513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1834] $81231513221(12)12231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1835] $81231513221(12)12231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1836] $41413221(12)12231513221(12)122315132^1 21(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.52 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1837] $81232181231513221(12)122^1 31513221(12)12231513221 (12) 1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.52 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1838] $81231513221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.52 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1839] $81231513221(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.52 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1840] $81231513221(12)122^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.52 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1841] $(12)122315132181232181232^1 181232181232181232181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[1842] $(12)122315132181232181232^1 1812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[1843] $41413221(12)12231513221(12)12^1 231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1844] $81232181231513221(12)1223151322^1 1(12)12231513221^{1513221} (12)1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1845] $81231513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1846] $81231513221(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1847] $81231513221(12)1223151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1848] $(12)1223151321812321812^1 32181232181232181232181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.46 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 526 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$$

[1849] $(12)1223151321812321812^1 321812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.46 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 526 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$$

[1850] $41413221(12)12231513221(12)12231513221(12)122^1 31513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.46 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1851] $812321812315132^1 21(12)12231513221(12)12231513221^{1513221} (12)1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det =$

289, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1852] $8123151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det =$
289, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1853] $413221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det =$
289, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1854] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det =$
289, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1855] $8123151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det =$
289, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1856] $8123151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det =$
289, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1857] $(12)122315132181232^1 181232181232181232181232181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det =$
361, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 664 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[1858] $(12)122315132181232^1 1812321812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det =$
361, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 664 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[1859] $41413221(12)12231513221(12)12231513221(12)122315132^1 18123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det =$
361, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1860] $41413221(12)12231513221(12)1223151322^1 1(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det =$
361, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1861] $81232181231513221(12)12^1 231513221(12)12231513221(12)1513221(12)1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det =$
361, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1862] $812321812^1 31513221(12)12231513221(12)12231513221(12)1513221(12)1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det =$
361, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[1863] & 81231513221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.41 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1864] & 81231513221(12)12^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.41 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1865] & 81231513221(12)12^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.41 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1866] & (12)1223151321812^1 32181232181232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 400, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.39 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 739 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0} \\
[1867] & (12)1223151321812^1 321812321812321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 400, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.39 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 739 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0} \\
[1868] & 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 66, \Delta = -114, \det = 400, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.39 \rightarrow 55, T_{\max} = 66, T \in [55, 66] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 139 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1869] & 41413221(12)12231513221(12)12231513221(12)12^1 231513218123218 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -110, \det = 441, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.37 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1870] & 8123218123151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -106, \det = 441, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.37 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1871] & (12)122315132^1 181232181232181232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 484, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.36 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 901 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0} \\
[1872] & (12)122315132^1 1812321812321812321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 484, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.36 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 901 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0} \\
[1873] & 413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 66, \Delta = -114, \det = 484, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.36 \rightarrow 55, T_{\max} = 66, T \in [55, 66] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1874] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 66, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.34 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$

[1875] $413221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 66, \Delta = -114, \det = 729, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.29 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 269 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1876] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 66, \Delta = -114, \det = 784, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.28 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 28 & 2 \\ 28 & 307 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$

[1877] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 66, \Delta = -114, \det = 841, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.27 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 318 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

[1878] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 66, \Delta = -114, \det = 1225, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.23 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 35 & 2 \\ 35 & 454 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$

[1879] $413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50, T_{\max} = 66, \Delta = -114, \det = 1369, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.21 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 37 & 2 \\ 37 & 517 & 49 \\ 2 & 49 & 3 \end{pmatrix}_{b_0}$

$T = 51$ [1880] $(12)1223151321812321812321812321812321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58, T_{\max} = 63, T \in [58, 63]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1881] $22^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58, T_{\max} = 63, T \in [58, 63]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1882] $(12)122315132181232181232181232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.75 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1883] $(12)122315132181232181232181232181232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.75 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1884] $(12)1223151321812321812321812321812321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.44 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1885] $(12)1223151321812321812321812321812321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det =$

$$-25, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.44 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1886] \quad 513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.44 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1887] \quad (12)122315132181232181232181232181232181232^1 1812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1888] \quad (12)122315132181232181232181232181232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1889] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1890] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1891] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1892] \quad 221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1893] \quad 5132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1894] \quad 513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1895] \quad (12)1223151321812321812321812321812321812^1 321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -64, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.94 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1896] \quad (12)1223151321812321812321812321812321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -64, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.94 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1897] & 1(12)12^{21}1^{1231513221}(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -64, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.94 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1898] & 221(12)12^{21}1^{1231513221}(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -64, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.94 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1899] & 1(12)12231513221(12)12231513221(12)122315132^{21}21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.84 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1900] & 1(12)12231513221(12)12231513221(12)12231513221(12)12^{21}23151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.84 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1901] & 221(12)12231513221(12)12231513221(12)12231513221(12)12^{21}23151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.84 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1902] & 221(12)12231513221(12)12231513221(12)122315132^{21}21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.84 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1903] & (12)122315132181232181232181232181232^{21}181232181232181 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1904] & (12)122^{21}315132181232181232181232181232181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1905] & (12)122^{21}31513218123218123218123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1906] & (12)122315132181232181232181232181232^{21}18123218123218122 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1907] & 51322^{21}1(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1908] $(12)1223151321812321812321812321812^{1\rightarrow 2}32181232181232181^{21} + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1909] $(12)12^{1\rightarrow 2}2315132181232181232181232181232181232181232181^{21} + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1910] $(12)1223151321812321812321812321812^{1\rightarrow 2}3218123218123218122 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1911] $(12)12^{1\rightarrow 2}231513218123218123218123218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1912] $1^{21}(12)12231513221(12)12231513221(12)122^{1\rightarrow 2}31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1913] $1^{21}(12)12231513221(12)12231513221(12)1223151322^{1\rightarrow 2}1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1914] $221(12)12231513221(12)12231513221(12)122^{1\rightarrow 2}31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1915] $221(12)12231513221(12)12231513221(12)1223151322^{1\rightarrow 2}1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1916] $513221(12)12231513221(12)12231513221(12)122315132^{1\rightarrow 2}21(12)1223151 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1917] $(12)122315132181232181232181232^{1\rightarrow 2}181232181232181232181^{21} + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [1918] $(12)122315132181232181232181232^{1\rightarrow 2}18123218123218123218122 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [1919] $1^{21}(12)12231513221(12)122315132^{1\rightarrow 2}21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1920] & \overset{21}{1}(12)122315132\overset{1}{21}(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1921] & \overset{1}{221}(12)12231513221(12)122315132\overset{1}{21}(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1922] & \overset{1}{221}(12)122315132\overset{1}{21}(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1923] & \overset{1}{513221}(12)12231513221(12)12231513221(12)1223151322\overset{1}{1}(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1924] & (12)1223151321812321812321812\overset{21}{32181232181232181232181} + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -196, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.55 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0} \\
[1925] & (12)1223151321812321812321812\overset{1}{3218123218123218123218122} + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -196, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.55 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0} \\
[1926] & \overset{21}{1}(12)12231513221(12)12231513221(12)12\overset{1}{231513221}(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.55 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1927] & \overset{1}{221}(12)12231513221(12)12231513221(12)12\overset{1}{231513221}(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.55 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1928] & \overset{21}{1}(12)12231513221(12)122\overset{1}{31513221}(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -225, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.52 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1929] & \overset{1}{221}(12)12231513221(12)122\overset{1}{31513221}(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -225, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.52 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1930] & (12)122315132181232181232\overset{21}{181232181232181232181232181} + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -256, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.48 \rightarrow 56, T_{\max} = 63, T \in [56, 63]
\end{aligned}$$

$$T_{\min} = 55.27 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 521 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[1954] \quad 513221(12)12231513221^1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -1225, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.23 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{pmatrix} 0 & 35 & 2 \\ 35 & 769 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

$$[1955] \quad 513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -1369, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.21 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{pmatrix} 0 & 37 & 2 \\ 37 & 850 & 58 \\ 2 & 58 & 3 \end{pmatrix}_{b_0}$$

$$T = 52 \quad [0] \quad 22^1(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59, T_{\max} = 64, T \in [59, 64]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12^12 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59, T_{\max} = 64, T \in [59, 64]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 6131513221(12)12231513221(12)12231513221(12)12231513221(12)122^13 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 68, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.22 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 612^131513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.44 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 6131513221(12)12231513221(12)12231513221(12)12231513221(12)12^123 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 68, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.44 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 612^131513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.44 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^131513161 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1(12)122^131513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 61231513221(12)12231513221(12)12231513221(12)122315132^121(12)1 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[9] \text{ 612315132}^1\text{21(12)12231513221(12)12231513221(12)12231513221(12)1}^{21}\text{1}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[10] \text{ 812}^1\text{32181231513221(12)12231513221(12)12231513221(12)1223151315}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[11] \text{ 81232181231513221(12)12231513221(12)12231513221(12)122}^1\text{3151315}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[12] \text{ 221(12)122}^1\text{31513221(12)12231513221(12)12231513221(12)12231513161}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[13] \text{ 221(12)12231513221(12)12231513221(12)12231513221(12)122}^1\text{31513161}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[14] \text{ 612315132}^1\text{21(12)12231513221(12)12231513221(12)12231513221(12)122}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[15] \text{ 61231513221(12)12231513221(12)12231513221(12)122315132}^1\text{21(12)122}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[16] \text{ 1(12)122}^{21}\text{231513221(12)12231513221(12)12231513221(12)12231513161}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[17] \text{ 1(12)122}^1\text{31513221(12)12231513221(12)12231513221(12)122315131613}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 67, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[18] \text{ 61231513221(12)12231513221(12)12231513221(12)1223151322}^1\text{1(12)1}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[19] \text{ 221(12)12}^1\text{231513221(12)12231513221(12)12231513221(12)12231513161}+\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[20] $21(12)122^1 31513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[21] $61231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[22] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[23] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[24] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[25] $61231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[26] $6123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[27] $81232181231513221(12)12231513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[28] $81232181231513221(12)12231513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[29] $221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[30] $221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [31] $21(12)12^1 231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.84 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [32] $6123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.84 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [33] $61231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.84 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [34] $61315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.76 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [35] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [36] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513161+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [37] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 315131613+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [38] $61231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [39] $61231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [40] $81232^1 181231513221(12)12231513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [41] $81232181231513221(12)12231513221(12)1223151322^1 1(12)1223151315+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [42] $81232181231513221(12)12231513221(12)122^1 31513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [43] $221(12)12231513221(12)12231513221(12)12231513221^1(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [44] $221(12)12231513221(12)12231513221(12)122^131513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [45] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [46] $61231513221(12)12^1231513221(12)12231513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [47] $61231513221(12)122315132^121(12)12231513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [48] $1(12)12231513221(12)122315132^121(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [49] $1(12)122315132^121(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [50] $1(12)12231513221(12)12231513221(12)122315132^121(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [51] $61231513221(12)12231513221(12)12231513221(12)122^131513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [52] $61231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [53] $81232181231513221(12)122315132^121(12)12231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[54] $221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[55] $221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[56] $21(12)12231513221(12)12231513221(12)122315132^1 21(12)122315131613+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[57] $6131513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[58] $613151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[59] $61231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[60] $61231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[61] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[62] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315131613+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.55 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[63] $61231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[64] $81232181231513221(12)12231513221(12)12^1 231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

- [65] $221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [66] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315131613+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.55 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [67] $6131513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.55 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [68] $61231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [69] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [70] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [71] $61231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [72] $81232181231513221(12)122^1 31513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -108, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [73] $221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [74] $21(12)12231513221(12)12231513221(12)122^1 31513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [75] $6131513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [76] $61231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det =$

225, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.52 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[77] $1(12)12231513221(12)12231513221^1(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[78] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[79] $1(12)12231513221(12)12231513221(12)12231513221^1(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[80] $61231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[81] $81232181231513221(12)12231513221^1(12)12231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[82] $221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[83] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[84] $21(12)12231513221(12)12231513221(12)12231513221^1(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[85] $61231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[86] $1(12)12231513221^1(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[87] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.46 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[88] \text{ 61231513221(12)12231513221(12)12231513221(12)12}^1\text{231513221(12)1+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 289, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.46 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[89] \text{ 812321812315132}^1\text{21(12)12231513221(12)12231513221(12)1223151315+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -108, \det = 289, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.46 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[90] \text{ 221(12)1223151322}^1\text{1(12)12231513221(12)12231513221(12)12231513161+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 289, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.46 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[91] \text{ 21(12)12231513221(12)122315132}^1\text{21(12)12231513221(12)122315131613+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 67, \Delta = -113, \det = 289, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.46 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[92] \text{ 6131513221(12)12}^1\text{231513221(12)12231513221(12)12231513221(12)1223+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 68, \Delta = -114, \det = 289, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.46 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[93] \text{ 61231513221(12)12231513221(12)12231513221(12)12}^1\text{231513221(12)122+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 289, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.46 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[94] \text{ 6131513221(12)122315132}^1\text{21(12)12231513221(12)12231513221(12)1223+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 68, \Delta = -114, \det = 324, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.43 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[95] \text{ 1(12)12231513221(12)12}^1\text{231513221(12)12231513221(12)12231513161+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.41 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[96] \text{ 1(12)12231513221(12)12231513221(12)12}^1\text{231513221(12)122315131613+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 67, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.41 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[97] \text{ 1(12)12231513221(12)122}^1\text{31513221(12)12231513221(12)122315131613+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 67, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.41 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[98] \text{ 61231513221(12)12231513221(12)1223151322}^1\text{1(12)12231513221(12)1+}\hat{1}\rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 361, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.41 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

- $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
 [99] 812321812¹31513221(12)12231513221(12)12231513221(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [100] 81232181231513221(12)12¹231513221(12)12231513221(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [101] 221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [102] 21(12)12231513221(12)12231513221(12)12¹231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [103] 21(12)12231513221(12)122¹31513221(12)12231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [104] 61231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [107] 8123218123151322¹1(12)12231513221(12)12231513221(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [109] 21(12)12231513221(12)1223151322¹1(12)12231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [108] 21(12)1223151322¹1(12)12231513221(12)12231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [106] 1(12)12231513221(12)1223151322¹1(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [105] 1(12)1223151322¹1(12)12231513221(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$

[110] $6131513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 484$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.36 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[111] $6131513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 529$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.34 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[112] $6131513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 529$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.34 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[113] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 576$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.33 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[114] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 576$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.33 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[115] $6131513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 676$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.30 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[116] $6131513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 729$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.29 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[117] $6131513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 900$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.26 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[118] $6131513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 1089$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.24 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

$T = 53$ [119] $22^1 1(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -4$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60$, $T_{\max} = 65$, $T \in [60, 65]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[120] $712231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -4$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60$, $T_{\max} = 65$, $T \in [60, 65]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [121] $71231513221(12)12231513221(12)12231513221(12)12231513221(12)122^13 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 68$, $T \in [60, 68]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [122] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [123] $7122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [124] $221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [125] $71231513221(12)12231513221(12)12231513221(12)12231513221(12)12^123 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [126] $7122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [127] $612^131513221(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [128] $712^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.22 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [129] $712^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.22 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [130] $712^1231513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.22 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [131] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.06 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [132] $1(12)122^131513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det =$

$$-49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[133] \quad 7122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[134] \quad 712231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[135] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[136] \quad 221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[137] \quad 712231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[138] \quad 7122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[139] \quad 812^1 32181231513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[140] \quad 81232181231513221(12)12231513221(12)12231513221(12)122^1 31513161+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[141] \quad 612315132^1 21(12)12231513221(12)12231513221(12)12231513218123218+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[142] \quad 61231513221(12)12231513221(12)12231513221(12)122315132181232^1 18+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[143] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.94 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.94 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.94 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.94 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[147] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 71223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 712231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 712231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 71223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[155] 81232181231513221(12)12231513221(12)122315132¹21(12)12231513161+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[156] 81232181231513221(12)12231513221(12)12231513221(12)12¹231513161+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[157] 61231513221(12)122¹31513221(12)12231513221(12)12231513218123218+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -81$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[158] 6123151322¹1(12)12231513221(12)12231513221(12)12231513218123218+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -81$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[159] 712315132¹21(12)12231513221(12)12231513221(12)12231513221(12)1223 + $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.76 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[160] 1²¹(12)12231513221(12)12231513221(12)122¹31513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[161] 1²¹(12)12231513221(12)12231513221(12)1223151322¹1(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[162] 712231513221(12)122315132¹21(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[163] 712231513221(12)12¹231513221(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[164] 221(12)12231513221(12)12231513221(12)1223151322¹1(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[165] 221(12)12231513221(12)12231513221(12)122¹31513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

- [166] $712231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [167] $712231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [168] $81232181231513221(12)12231513221(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [169] $81232^1 181231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [170] $81232181231513221(12)12231513221(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [171] $61231513221(12)12^1 231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [172] $61231513221(12)12231513221(12)12231513221(12)1223151321812^1 3218 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [173] $61231513221(12)122315132^1 21(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [174] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [175] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [176] $712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [177] $712231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[178] $221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[179] $221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[180] $71231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[181] $7123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[182] $712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[183] $712231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[184] $81232181231513221(12)122315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[185] $61231513221(12)12231513221(12)122^1 31513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[186] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.55 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[187] $712231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.55 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[188] $221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.55 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

- [189] $71231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.55 \rightarrow 58$, $T_{\max} = 68$, $T \in [58, 68]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [190] $712231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -196$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.55 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [191] $81232181231513221(12)12231513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.55 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [192] $61231513221(12)1223151322^1 1(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.55 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [193] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [194] $712231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [195] $221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [196] $71231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 68$, $T \in [58, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [197] $712231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [198] $81232181231513221(12)122^1 31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -105$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [199] $61231513221(12)12231513221(12)122315132^1 21(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [200] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -256$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[201] \quad 712231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[202] \quad 221(12)12231513221(12)12231513221^1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[203] \quad 712231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[204] \quad 81232181231513221(12)12231513221^1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[205] \quad 61231513221(12)12231513221(12)12^1 231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -106, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[206] \quad 1(12)12231513221^1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[207] \quad 712231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[208] \quad 221(12)12231513221^1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[209] \quad 71231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[210] \quad 712231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[211] \quad 812321812315132^1 21(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

- [212] $61231513221(12)12231513221(12)12231513221(12)122^131513218123218+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [213] $71231513221(12)122315132^121(12)12231513221(12)12231513221(12)1223 + \hat{1}\rightarrow 2$ $T = 53, T_{\max} = 68, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.43 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [214] $1(12)12231513221(12)12^1231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [215] $712231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [216] $221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132171 + \hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [217] $712231513221(12)12231513221(12)1223151322^11(12)12231513221(12)122 + \hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [218] $812321812^131513221(12)12231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [219] $81232181231513221(12)12^1231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [220] $61231513221(12)12231513221(12)12231513221(12)122315132^118123218+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [221] $61231513221(12)12231513221(12)1223151322^11(12)12231513218123218+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [222] $8123218123151322^11(12)12231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.37 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$
- [223] $61231513221(12)12231513221(12)12231513221(12)12^1231513218123218+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -441, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.37 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[224] \quad 71231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -484, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.36 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[225] \quad 71231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.34 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[226] \quad 71231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.34 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[227] \quad 71231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -676, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.30 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[228] \quad 71231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -729, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.29 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[229] \quad 71231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -900, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.26 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[230] \quad 71231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -1089, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.24 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

$$T = 54 \quad [231] \quad 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[232] \quad 81231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[233] \quad 81231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[234] \quad 22^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[235] $22^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[236] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[237] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 68, T \in [61, 68]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[238] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[239] $32^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 68, T \in [61, 68]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[240] $21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 68, T \in [61, 68]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[241] $812231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 68, T \in [61, 68]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[242] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[243] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.75 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[244] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[245] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
 [246] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [247] $1812^1 131513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [248] $81231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [249] $81231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [250] $322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 68, T \in [60, 68]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [251] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 32 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 70, T \in [60, 70]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [252] $221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
 [253] $221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
 [254] $812231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 68, T \in [60, 68]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [255] $81231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [256] $81232181231513221(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [257] $7122^1 31513221(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [258] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [259] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [260] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [261] $3221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 68$, $T \in [60, 68]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [262] $221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [263] $8122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 68$, $T \in [60, 68]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [264] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [265] $712^1 231513221(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [266] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [267] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [268] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta =$

$$-103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[269] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[270] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[271] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[272] \quad 181231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[273] \quad 1812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[274] \quad 812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[275] \quad 81231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[276] \quad 81231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[277] \quad 812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[278] \quad 221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59.06 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[279] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [280] $221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [281] $221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [282] $812^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 68$, $T \in [60, 68]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [283] $812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [284] $81231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [285] $81232181231513221(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [286] $812^1 32181231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [287] $7122315132^1 21(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [288] $712231513221(12)12231513221(12)12231513221(12)122315132181232^1 18 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [289] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [290] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

- [291] $1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [292] $1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [293] $181231513221(12)12231513221(12)12231513221(12)12231513221^{11}(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0}$
- [294] $21(12)122^{21}1^{31}513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [295] $221(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [296] $221(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [297] $1(12)12231513221(12)12231513221(12)12231513221(12)12^{21}1^{23}15132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [298] $1(12)12231513221(12)12231513221(12)122315132^{21}21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [299] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{21}18123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [300] $1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [301] $1(12)12231513221(12)12231513221(12)12231513221(12)12^{21}1^{23}151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

- [302] $1(12)^{21}12231513221(12)^112231513221(12)122315132^121(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [303] $1(12)^{21}12231513221(12)^112231513221(12)122315132^121(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [304] $1(12)^{21}12231513221(12)^112231513221(12)12231513221(12)12^123151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [305] $181231513221(12)122^131513221(12)12231513221(12)^112231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [306] $18123151322^11(12)12231513221(12)12231513221(12)^112231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [307] $81231513221(12)12231513221(12)122315132^121(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [308] $81231513221(12)12231513221(12)12231513221(12)12^1231513221(12)^{2221}1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [309] $81231513221(12)12231513221(12)12231513221(12)12^1231513221(12)^{221}12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [310] $81231513221(12)12231513221(12)122315132^121(12)12231513221(12)^{221}12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [311] $1(12)122^131513221(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [312] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^118123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

- [313] $21(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [314] $221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [315] $221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [316] $221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [317] $221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [318] $81231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [319] $81231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [320] $81232181231513221(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [321] $81232181231513221(12)12231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [322] $71223151322^1 1(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [323] $712231513221(12)122^1 31513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [324] $3221(12)12231513221(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.76 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[325] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 70, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.76 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[326] \quad 8122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.76 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[327] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[328] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[329] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[330] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[331] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[332] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[333] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[334] \quad 181231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[335] \quad 181231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

[336] $\overset{1}{81231513221}(12)12231513221(12)122\overset{2221}{1}31513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

[337] $\overset{1}{81231513221}(12)12231513221(12)1223151322\overset{2221}{1}(12)12231513221(12)1\overset{2221}{\hat{1}}\rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

[338] $\overset{1}{81231513221}(12)12231513221(12)122\overset{221}{1}31513221(12)12231513221(12)12+\hat{1}\rightarrow\hat{2}$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

[339] $81231513221(12)12231513221(12)12231513221^1(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

[340] $\overset{1}{21(12)12231513221(12)12231513221(12)12231513221(12)122^131513218123} + \hat{1} \rightarrow 2 \quad T = 54, \quad T_{\max} = 68, \quad \Delta = -112, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 54, 0), \quad c_{\text{ext}} = 15.0$

[341] $\overset{1}{221}(12)12231513221(12)12231513221(12)1223151322^{\color{red}1}1(12)1223151321812 + \hat{\color{red}1} \rightarrow \color{red}2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

[342] $221(12)12231513221(12)12231513221(12)122^{\overset{1}{\color{red}\uparrow}}131513221(12)1223151321812 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 54, \quad T_{\max} = 66, \quad \Delta = -103, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 54, 0), \quad c_{\text{ext}} = 15.0$

[343] $221(12)12231513221(12)12231513221(12)1223151322^11(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, \quad T_{\max} = 66, \quad \Delta = -103, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 54, 0), \quad c_{\text{ext}} = 15.0$

[344] $221(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

[345] $81231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

[346] $81231513221(12)12231513221(12)122^131513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [347] $81232^1 181231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
 [348] $81232181231513221(12)12231513221(12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
 [349] $81232181231513221(12)12231513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
 [350] $712231513221(12)12231513221(12)12231513221(12)1223151321812^1 3218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [351] $712231513221(12)122315132^1 21(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [352] $712231513221(12)12^1 231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [353] $1^{21}(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132181^1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
 [354] $1^{21}(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181^1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
 [355] $1^{21}(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
 [356] $1^{21}(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [357] $1^{21}(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
 [358] $1^{21}(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta =$

-103 , $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[359] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[360] $181231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[361] $181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[362] $812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[363] $81231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[364] $81231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[365] $812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[366] $3221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[367] $3221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[368] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 181232 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 70$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 70$, $T \in [59, 70]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[369] $21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[370] \quad 221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[371] \quad 221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[372] \quad 221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[373] \quad 221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[374] \quad 81223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[375] \quad 812231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[376] \quad 81231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[377] \quad 812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[378] \quad 81232181231513221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -104, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[379] \quad 712231513221(12)12231513221(12)122^1 31513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -104, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[380] \quad 1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

1159

- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [392] $81231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [393] $81232181231513221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [394] $712231513221(12)1223151322^1 1(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [395] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513218^1 1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [396] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [397] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [398] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [399] $181231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [400] $81231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [401] $81231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [402] $3221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

- [403] $1(12)12231513221(12)12231513221(12)122^1315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [404] $21(12)12231513221(12)12231513221(12)122^131513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [405] $221(12)12231513221(12)122^131513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [406] $221(12)12231513221(12)122^131513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [407] $812231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [408] $81231513221(12)122^131513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [409] $81232181231513221(12)122^131513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [410] $712231513221(12)12231513221(12)122315132^121(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [424] $81232181231513221(12)1223151322^11(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [412] $1(12)122315132^121(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [413] $1(12)12231513221(12)12231513221(12)1223151322^11(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [414] $1(12)12231513221(12)1223151322^11(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [415] $1(12)12231513221(12)1223151322^11(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [416] $181231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [417] $81231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [418] $81231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [419] $21(12)122315132^121(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [420] $21(12)12231513221(12)12231513221(12)1223151322^11(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [421] $221(12)12231513221(12)1223151322^11(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [422] $221(12)12231513221(12)1223151322^11(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [423] $81231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [411] $1(12)12231513221(12)1223151322^11(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [425] $712231513221(12)12231513221(12)12^1231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[426] & \begin{matrix} 21 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[427] & \begin{matrix} 1 \\ 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[428] & \begin{matrix} 21 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[429] & \begin{matrix} 21 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[430] & \begin{matrix} 1 \\ 181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[431] & \begin{matrix} 1 \\ 8123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[432] & \begin{matrix} 1 \\ 8123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[433] & \begin{matrix} 1 \\ 3221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[434] & \begin{matrix} 1 \\ 1(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 70, \Delta = -114, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 70, T \in [59, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[435] & \begin{matrix} 1 \\ 21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[436] & \begin{matrix} 1 \\ 221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]
\end{aligned}$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[437] $221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[438] $812231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[439] $8123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[440] $812321812315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[441] $712231513221(12)12231513221(12)12231513221(12)122^1 31513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[442] $3221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 324, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.43 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[443] $812231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 324, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.43 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[444] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

[445] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.41 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[446] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.41 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[447] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

- [448] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [449] $181231513221(12)12231513221(12)12231513221^1(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [450] $81231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [451] $81231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [452] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [453] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [454] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [455] $21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [456] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [457] $221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [458] $221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

- [459] $81231513221(12)^1 12^1 231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [460] $81232181231513221(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [461] $812321812^1 31513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [462] $712231513221(12)12231513221(12)1223151322^1 1(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [463] $712231513221(12)12231513221(12)12231513221(12)122315132^1 18123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [464] $1(12)12231513221(12)^1 1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [465] $1(12)1223151322^1 1(12)^1 12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [466] $1(12)12231513221(12)^1 12231513221(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [467] $1(12)12231513221(12)^1 122315132^1 21(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [468] $21(12)1223151322^1 1(12)^1 12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [469] $21(12)12231513221(12)^1 1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [470] $8123218123151322^1 1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0} \\
[471] & 712231513221(12)12231513221(12)12231513221(12)12^1 231513218123218 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -104, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.37 \rightarrow 59, T_{\max} = 66, T \in [59, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[472] & 3221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.36 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[473] & 812231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.36 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[474] & 3221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[475] & 3221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[476] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 70, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 70, T \in [59, 70] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[477] & 812231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[478] & 812231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[479] & 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 576, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.33 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[480] & 1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 70, \Delta = -114, \det = 576, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.33 \rightarrow 59, T_{\max} = 70, T \in [59, 70] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[481] & 21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 576, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.33 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [482] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 625, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.31 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$
- [483] $3221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 676, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.30 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [484] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 676, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.30 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [485] $812231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 676, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.30 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [486] $3221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 729, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.29 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [487] $812231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 729, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.29 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [488] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 841, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.27 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [489] $3221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 900, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.26 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [490] $812231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 900, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.26 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [491] $3221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 1089, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.24 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [492] $812231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 1089, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.24 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

- $T = 55$ [493] $81232181231513221(12)12231513221(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -25$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.44 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [494] $81232181231513221(12)12231513221(12)12231513221(12)122315132^11812 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -25$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.44 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [495] $81232181231513221(12)12231513221(12)12231513221(12)122315132^12181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -25$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.44 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [496] $81232181231513221(12)12231513221(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -36$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [497] $812^132181231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [498] $81232181231513221(12)12231513221(12)12231513221(12)122^1315132181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [499] $812^132181231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [500] $81232181231513221(12)12231513221(12)12231513221(12)122^13151321812 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [501] $812^132181231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [502] $81232181231513221(12)12231513221(12)12231513221(12)122^13151322181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61$, $T_{\max} = 67$, $T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [503] $81232181231513221(12)12231513221(12)122315132^121(12)122315132181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.84 \rightarrow 60$, $T_{\max} = 67$, $T \in [60, 67]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [504] $81232181231513221(12)12231513221(12)12231513221(12)12^12315132181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.84 \rightarrow 60$, $T_{\max} = 67$, $T \in [60, 67]$

- [517] $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
 $81232^1 181231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.69 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [518] $81232181231513221(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.59 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [519] $81232181231513221(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.59 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [520] $81232181231513221(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.59 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [521] $81232181231513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.55 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [522] $81232181231513221(12)12231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.55 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [523] $81232181231513221(12)12231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.55 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [524] $81232181231513221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -225, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.52 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [525] $81232181231513221(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -225, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.52 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [526] $81232181231513221(12)122^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -225, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.52 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [527] $81232181231513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -256, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.48 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [528] $81232181231513221(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -256, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.48 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$

- [529] $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
 $81232181231513221(12)1223151322^11(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -256, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.48 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [530] $812321812315132^121(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -289, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.46 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [531] $812321812315132^121(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -289, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.46 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [532] $812321812315132^121(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -289, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.46 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [533] $81232181231513221(12)12^1231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [534] $812321812^131513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [535] $812321812^131513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [536] $81232181231513221(12)12^1231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [537] $81232181231513221(12)12^1231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [538] $812321812^131513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [539] $8123218123151322^11(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -441, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.37 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$
- [540] $8123218123151322^11(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -441, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.37 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

[541] $8123218123151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -441$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 59.37 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

$T = 58$ [542] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[543] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[544] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[545] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[546] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[547] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[548] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[549] $3221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[550] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[551] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[552] \quad 3221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[553] \quad 22^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[554] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[555] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[556] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[557] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[558] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[559] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[560] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[561] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[562] \quad 3221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[563] \quad 32^{\overset{1}{\rightarrow}} 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[564] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\overset{1}{\rightarrow}} 22 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[565] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\overset{21}{\rightarrow}} 222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.75 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[566] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\overset{1}{\rightarrow}} 222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.75 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[577] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{\rightarrow}} 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[568] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{\rightarrow}} 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[569] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{221}{\rightarrow}} 21(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[570] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{\rightarrow}} 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[571] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\overset{221}{\rightarrow}} 23 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[572] \quad 1(12)122^{\overset{221}{\rightarrow}} 31513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[573] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\overset{1}{\rightarrow}} 23 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

1176

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[585] \quad 3221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 36, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.22 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[586] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 36, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.22 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[598] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[588] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[589] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[590] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[591] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[592] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[593] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[594] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[595] \quad 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[596] & 21(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[597] & 21(12)122315132^{21}21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[587] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^{21}31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[599] & 3221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{21}1(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[600] & 221(12)122^{21}31513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[601] & 221(12)12231513221(12)12231513221(12)12231513221(12)122^{21}31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[608] & 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{21}1(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[603] & 1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[604] & 1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[605] & 1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[606] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{21}1(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[618] & \begin{matrix} 2221 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[619] & \begin{matrix} 2221 \\ 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[620] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[621] & \begin{matrix} 3221 \\ 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[622] & \begin{matrix} 221 \\ 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[623] & \begin{matrix} 221 \\ 21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[624] & \begin{matrix} 1 \\ 221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[625] & \begin{matrix} 1 \\ 221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[626] & \begin{matrix} 221 \\ 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.76 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[627] & \begin{matrix} 21 \\ 21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.76 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[628] & \begin{matrix} 1 \\ 3221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.76 \rightarrow 63, T_{\max} = 71, T \in [63, 71]
\end{aligned}$$

1181

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[643] $221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[644] $221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[630] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[634] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[629] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[658] $21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[649] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[650] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[651] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[652] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[653] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

1183

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[662] $3221(12)12231513221(12)12231513221(12)12231513221(12)12^{1231513221}(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[663] $221(12)12231513221(12)122315132^{121}(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 169, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[664] $221(12)122315132^{121}(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 169, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[666] $1(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[665] $1(12)12231513221(12)12231513221(12)12231513221(12)12^{1231513221}(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[667] $1(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[668] $1(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[669] $1(12)12231513221(12)122^{131513221}(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[670] $1(12)12231513221(12)1223151322^{11}(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[671] $21(12)12231513221(12)122^{131513221}(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[672] $21(12)1223151322^{13221}(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [673] $\overset{221}{21(12)12231513221(12)1223151322^11(12)12231513221(12)122+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\overset{1}{3221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1222+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- $\overset{1}{221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- $\overset{1}{1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1223+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- $\overset{21}{1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1222+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- $\overset{21}{1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- $\overset{21}{1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- $\overset{221}{1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1223+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- $\overset{2221}{1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\overset{21}{21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1223+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- $\overset{3221}{21(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2}$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$

- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [684] $21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [685] $3221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [686] $221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [687] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [688] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
 [689] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
 [690] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
 [691] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [692] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [693] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [694] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[695] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[696] & \begin{matrix} 1 \\ 221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[697] & \begin{matrix} 1 \\ 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[698] & \begin{matrix} 21 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[699] & \begin{matrix} 21 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[700] & \begin{matrix} 21 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[701] & \begin{matrix} 221 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[702] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[703] & \begin{matrix} 21 \\ 21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[704] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[705] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70]
\end{aligned}$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[706] $3221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[707] $221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[708] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.43 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[709] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.43 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[710] $3221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.43 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[715] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[712] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[713] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

[714] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

[711] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[716] $1(12)12231513221(12)12231513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[720] & \begin{matrix} 1 \\ 221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[719] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[717] & \begin{matrix} 3221 \\ 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.41 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[718] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.41 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[724] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.37 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[723] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.37 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[722] & \begin{matrix} 1 \\ 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.37 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[721] & \begin{matrix} 1 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.37 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[725] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.36 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[726] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.36 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[727] & \begin{matrix} 1 \\ 3221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.36 \rightarrow 63, T_{\max} = 71, T \in [63, 71]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[728] & \begin{matrix} 221 \\ 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[729] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[731] & \begin{matrix} 21 \\ 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[732] & \begin{matrix} 1 \\ 3221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[733] & \begin{matrix} 1 \\ 3221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 35 & 3 \end{pmatrix}_{b_0} \\
[730] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[734] & \begin{matrix} 1 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 576, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.33 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[735] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 576, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.33 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[736] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 676, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.30 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[737] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 676, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.30 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[738] & \begin{matrix} 1 \\ 3221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 676, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.30 \rightarrow 63, T_{\max} = 71, T \in [63, 71]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[739] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 729, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.29 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[740] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 729, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.29 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[741] & \begin{matrix} 1 \\ 3221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 729, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.29 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[743] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 900, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.26 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[742] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 900, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.26 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[744] & \begin{matrix} 1 \\ 3221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 900, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.26 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[746] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 1089, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.24 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0} \\
[747] & \begin{matrix} 1 \\ 3221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 1089, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.24 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0} \\
[745] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 1089, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.24 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0} \\
T = 59 & [749] \begin{matrix} 22^1 1 \\ 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 66, T_{\max} = 71, T \in [66, 71] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[748] & \begin{matrix} 22^1 21 \\ 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 66, T_{\max} = 71, T \in [66, 71]
\end{aligned}$$

- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [750] 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)1222¹2+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -4$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 66$, $T_{\max} = 71$, $T \in [66, 71]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [752] 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹22+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 65.22 \rightarrow 66$, $T_{\max} = 71$, $T \in [66, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [751] 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)1²²²1+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 65.22 \rightarrow 66$, $T_{\max} = 71$, $T \in [66, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [753] 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹222+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.75 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [754] 4132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [755] 4132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1¹1+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [756] 81232181231513221(12)12231513221(12)12231513221(12)122315132²²²¹21(12)1+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [757] 81232181231513221(12)12231513221(12)12231513221(12)122315132²²¹21(12)12+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [758] 81232181231513221(12)12231513221(12)12231513221(12)122315132¹21(12)12222+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [759] 81232181231513221(12)12231513221(12)12231513221(12)1223151322²²²¹1(12)1+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -36$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.22 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [760] 81232181231513221(12)12231513221(12)12231513221(12)1223151322²²¹1(12)12+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -36$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.22 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [761] 81232181231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)12222+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta =$

-102, $\det = -36$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.22 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[763] $41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[764] $812^1 32181231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[765] $81232181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[766] $812^1 32181231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[767] $81232181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[768] $812^1 32181231513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[769] $81232181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[762] $41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[770] $413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.94 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[771] $413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.94 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[776] $81232181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.84 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[775] & 413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[777] & 81232181231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[778] & 81232181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[779] & 81232181231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[780] & 81232181231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[781] & 81232181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[772] & 413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[773] & 413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[774] & 413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[785] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.69 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[786] & 81232^1 181231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.69 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[796] & 413221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -169, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[797] & 81232181231513221(12)122315132^121(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[798] & 81232181231513221(12)122315132^121(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[799] & 81232181231513221(12)122315132^121(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[804] & 81232181231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[801] & 413221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[802] & 81232181231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[803] & 81232181231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[800] & 413221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[805] & 413221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -225, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[807] & 81232181231513221(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71]
\end{aligned}$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[808] $81232181231513221(12)122^131513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -225$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[809] $81232181231513221(12)122^131513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -225$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[806] $413221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -225$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[815] $81232181231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[816] $81232181231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[810] $413221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[811] $413221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[812] $413221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[813] $413221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[814] $81232181231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[820] $812321812315132^121(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.46 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

- [818] $413221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.46 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [817] $413221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.46 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [819] $812321812315132^121(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.46 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [821] $812321812315132^121(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.46 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [822] $413221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [823] $413221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [824] $413221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [825] $413221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [826] $812321812^131513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [827] $81232181231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [828] $81232181231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [829] $812321812^131513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta =$

-102, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[830] $812321812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[831] $81232181231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

[832] $413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[833] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[834] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[835] $413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[836] $8123218123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

[837] $8123218123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

[838] $8123218123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

[840] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -576$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.33 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[839] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -576$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.33 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$T = 60 \quad [842] \quad 1 \quad (12) \quad 1^{15132^1 21} 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.44 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[841] \quad 1 \quad (12) \quad 1^{12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2} \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.44 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[843] \quad 414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.06 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[846] \quad 1 \quad (12) \quad 1^{151322^1 1} 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.06 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[845] \quad 1 \quad (12) \quad 1^{12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2} \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.06 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[844] \quad 414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.06 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[848] \quad 41413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[847] \quad 41413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[849] \quad 1 \quad (12) \quad 1^{122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2} \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[850] \quad 1 \quad (12) \quad 1^{315132^1 21} 12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[851] \quad 1 \quad (12) \quad 1^{1513221} 12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[852] $41413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[853] $4141322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[854] $41413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[855] $41413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[856] $41413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[857] $4141322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[858] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[859] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[860] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[861] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[862] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[864] $(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221^{231513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 76, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.76 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[863] $1^{31513221} (12) 12231513221(12)12231513221(12)12231513221(12)12231513221^1 (12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 100, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.76 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[874] $(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221^{231513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 76, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[865] $41413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[866] $41413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[867] $41413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[868] $41413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[869] $1^{31513221} (12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[870] $1^{31513221} (12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[871] $1^{31513221} (12) 12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[875] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221^{23151321} (12) 1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 76, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[872] & 1 \text{ (12) } 12^{\overline{1}} 231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[873] & 1 \text{ (12) } 122315132^{\overline{1}} 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[879] & 1 \text{ (12) } 122^{\overline{1}} 31513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 75, T \in [65, 75] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[880] & 1 \text{ (12) } 12231513221(12)122^{\overline{1}} 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[876] & 41413221(12)12231513221(12)122^{\overline{1}} 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[877] & 41413221(12)12231513221(12)122^{\overline{1}} 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[878] & 1(12)12231513221(12)12231513221(12)122315132^{\overline{1}} 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[882] & 41413221(12)1223151322^{\overline{1}} 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 196, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.55 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[884] & 1 \text{ (12) } 1223151322^{\overline{1}} 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.55 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[881] & 41413221(12)1223151322^{\overline{1}} 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 196, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.55 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[883] & 1(12)12231513221(12)12231513221(12)12231513221(12)12^{\overline{1}} 231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.55 \rightarrow 65, T_{\max} = 72, T \in [65, 72]
\end{aligned}$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[894] $41413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 256$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[892] $41413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 256$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[893] $41413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 256$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[891] $41413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 256$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[902] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -105$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[904] $1^{1513221}(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -105$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[905] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21^{231513221}(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 76$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[900] $41413221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[901] $41413221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[903] $1^{31513221}(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 75$, $\Delta = -113$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[906] $41413221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 361$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[907] $41413221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[908] $41413221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[909] $41413221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[910] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[911] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[912] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[913] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[914] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[915] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[916] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[917] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221 \begin{smallmatrix} 231513221 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 76, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[921] $41413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 441$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.37 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[930] $(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 76$, $\Delta = -114$, $\det = 484$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.36 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[932] $(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 76$, $\Delta = -114$, $\det = 529$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.34 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[931] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 75$, $\Delta = -113$, $\det = 529$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.34 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[933] $41413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[935] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -105$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[936] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 75$, $\Delta = -113$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[937] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -105$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[938] $(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 76$, $\Delta = -114$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[934] $41413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[939] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 75$, $\Delta = -113$, $\det = 625$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.31 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[940] $(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 625, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.31 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[941] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 75, \Delta = -113, \det = 676, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.30 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[942] $(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 676, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.30 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[943] $(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 729, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.29 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

[945] $(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 841, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.27 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[946] $(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 841, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.27 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[944] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 75, \Delta = -113, \det = 841, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.27 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[947] $(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 961, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.25 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[948] $(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 1156, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.23 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 62$ [0] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.06 \rightarrow 68, T_{\max} = 74, T \in [68, 74]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.06 \rightarrow 68, T_{\max} = 74, T \in [68, 74]$$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [2] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 64, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.94 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3] $21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 64, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.94 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [4] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [5] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [6] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [7] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [8] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [9] $21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [10] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.69 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [11] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.69 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [12] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.69 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{array}{l} \text{[13]} \quad 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{\hat{1}}1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = \\ 74, \quad \Delta = -107, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0 \end{array}$$

[14] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)1223151315 + \hat{\color{red}1} \rightarrow \color{red}2 \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 169, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

$$\begin{array}{l} \text{[15]} \quad 21(12)12231513221(12)12231513221(12)12231513221^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = \\ 74, \quad \Delta = -107, \quad \det = 169, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0 \end{array}$$

[16] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12\overset{1}{2}31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 196, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

[17] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 196, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

[18] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)122^131513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 225, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

[19] $21(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 225, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

[21] $\overset{1}{1}(12)122315132\overset{1}{2}1(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 256, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

[22] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 256, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

[23] $21(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 256, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

[20] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)1223151322\overset{1}{1}(12)12231513221(12)1223151315 + \overset{1}{\hat{1}} \rightarrow \overset{2}{2} \quad T = 62, \quad T_{\max} = 74, \quad \Delta = -107, \quad \det = 256, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0$

1212

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[35] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 576, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 66.33 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$T = 63$ [36] $612^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.44 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[37] $612^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.44 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[40] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.06 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[39] $612315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.06 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[41] $612315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.06 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[38] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.06 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[43] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.94 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[44] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.94 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[45] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.94 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[42] $21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.94 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

[46] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}231513161 + \hat{1} \rightarrow 2$ $T = 63$, $T_{\max} = 75$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 63, 0)$, $c_{\text{ext}} = 15.0$

[47] $21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[48] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[49] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 63$, $T_{\max} = 77$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 63, 0)$, $c_{\text{ext}} = 15.0$

[50] 61231513221(12)122¹31513221(12)12231513221(12)12231513221¹(12)12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 63$, $T_{\max} = 75$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 63, 0)$, $c_{\text{ext}} = 15.0$

[51] $6123151322^{\hat{1}}1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[52] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[53] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -104, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

[54] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513161 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -104, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

[55] $\overset{1}{1}(12)12231513221\overset{1}{(12)}12231513221(12)12231513221(12)122315132^{\textcolor{red}{1}}21(12)12231513161 + \hat{\textcolor{red}{1}} \rightarrow \textcolor{red}{2} \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -104, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

[56] $61231513221(12)122^{\color{red}1}31513221(12)12231513221(12)12231513221^{\color{blue}1}(12)12231513221^{\color{blue}1}(12)1 + \hat{\color{red}1} \rightarrow \color{red}2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -105, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

[57] $6123151322^{\hat{1}}1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.84 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[58] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -105, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

$$T_{\min} = 67.84 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[59] $1(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.76 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[61] $21(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[62] $\overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\color{red}1}315131613 + \hat{\color{red}1} \rightarrow \color{red}{2} \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.69 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[63] $61231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[64] $61231513221(12)122315132^{\hat{1}}21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[65] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)122^131513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[66] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)1223151322\overset{1}{1}(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -104, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

$$T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[67] $61231513221(12)122315132^{\textcolor{red}{1}}21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{\textcolor{red}{1}} \rightarrow \textcolor{red}{2} \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -105, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

$$T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[68] $61231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -105, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

$$T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[60] $21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[69] $21(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[70] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[71] $61231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[72] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[73] $61231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[74] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[75] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^12315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[76] $61231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[77] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[78] $61231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[91] $61231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[92] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[84] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[93] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[94] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[95] $61231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[96] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[97] $61231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[98] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[99] $21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[100] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[101] $1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.41 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[102] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.41 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[103] $61231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[104] $61231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[105] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[106] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[107] $61231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[108] $61231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[114] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[111] $1(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[110] $21(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[113] $61231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[109] $21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[115] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[116] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[117] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[118] $61231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[112] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[119] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -529, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.34 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[120] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.33 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[121] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.33 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[122] $61231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.33 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[123] & \overset{1}{1}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.33 \rightarrow 68, T_{\max} = 75, T \in [68, 75] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[124] & 61231513221(12)12231513221(12)12231513221(12)1223151322^{\overset{1}{1}}1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.33 \rightarrow 68, T_{\max} = 75, T \in [68, 75] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[125] & \overset{1}{1}(12)1223151322^{\overset{1}{1}}1(12)12231513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -625, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.31 \rightarrow 68, T_{\max} = 77, T \in [68, 77] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[126] & \overset{1}{1}(12)12231513221(12)1223151322^{\overset{1}{1}}1(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -676, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.30 \rightarrow 68, T_{\max} = 77, T \in [68, 77] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[127] & \overset{1}{1}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -841, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.27 \rightarrow 68, T_{\max} = 77, T \in [68, 77] \\
& \begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0} \\
T = 64 & [128] \overset{1}{21}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{1}}171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.44 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[129] & 7122^{\overset{1}{1}}31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.44 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[130] & \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{1}}171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.44 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[131] & 7122^{\overset{1}{1}}31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.44 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[132] & 712^{\overset{1}{1}}231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.22 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[133] & 712^{\overset{1}{1}}231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.22 \rightarrow 70, T_{\max} = 76, T \in [70, 76]
\end{aligned}$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[134] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 69.06 \rightarrow 70, T_{\max} = 76, T \in [70, 76]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[135] \quad 7122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 69.06 \rightarrow 70, T_{\max} = 76, T \in [70, 76]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[136] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 69.06 \rightarrow 70, T_{\max} = 76, T \in [70, 76]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[137] \quad 7122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 69.06 \rightarrow 70, T_{\max} = 76, T \in [70, 76]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[138] \quad 21(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.94 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.94 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.94 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.94 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^12315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 71223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[146] $712231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[147] $712231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[148] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[149] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[150] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[143] $21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[142] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[153] $71223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[152] $712231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[155] $21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.69 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[156] $712231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.69 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

[157] $712231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[158] $\overset{1}{1}(12)1223151322\overset{1}{1}(12)1223151322\overset{1}{1}(12)1223151322\overset{1}{1}(12)1223151322^{\color{red}1}\overset{1}{1}(12)122315132171 + \hat{\textcolor{red}{1}} \rightarrow \textcolor{red}{2}$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[159] $1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

[160] 712231513221(12)122315132¹21(12)12231513221(12)12231513221¹(12)12231513221¹(12)1 + $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[161] 712231513221(12)12¹231513221(12)12231513221(12)12231513221¹(12)12231513221¹(12)1 + $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[154] $21(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

[162] $21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

[163] 712231513221(12)12231513221(12)122¹31513221(12)12231513221¹(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[164] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

[165] 712231513221(12)12231513221(12)122¹31513221(12)12231513221¹(12)12231513221¹(12)1 + $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[166] $21(12)12231513221(12)12231513221(12)12231513221(12)12^{\color{red}1}231513221(12)122315132171 + \hat{\color{red}1} \rightarrow \color{red}2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[167] $712231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.55 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[168] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.55 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[169] $712231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.55 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[170] $21(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 225$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.52 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[171] $712231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 225$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.52 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[172] $1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 225$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.52 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[173] $712231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 225$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.52 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[174] $21(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[175] $21(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[176] $712231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[177] $712231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[178] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[179] & 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[180] & 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[181] & 712231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[183] & 712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.46 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[182] & 21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.46 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[185] & 712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.46 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[184] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.46 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[189] & 712231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.41 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[186] & 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.41 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[187] & 21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.41 \rightarrow 69, T_{\max} = 76, T \in [69, 76]
\end{aligned}$$

1227

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[200] $712231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 441$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.37 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[201] $712231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 441$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.37 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[205] $712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 576$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.33 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[204] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 576$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.33 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[202] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 576$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.33 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[203] $712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 576$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.33 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$T = 65$ [206] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -9$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 71.22 \rightarrow 72, T_{\max} = 78, T \in [72, 78]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[207] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[208] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[209] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 32 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 70.44 \rightarrow 71, T_{\max} = 80, T \in [71, 80]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[210] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[211] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^11812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[212] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^12181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[213] & 1812^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[214] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.22 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[215] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.22 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[216] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^13151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.06 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[217] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^13151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.06 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[218] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.06 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[219] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^13151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.06 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[220] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^13151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.06 \rightarrow 71, T_{\max} = 77, T \in [71, 77] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[221] & 1812315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 70.06 \rightarrow 71, T_{\max} = 77, T \in [71, 77]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[224] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[225] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[222] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[226] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[227] & 181231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[223] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[228] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[230] & 21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[231] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[232] & 21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[233] & 21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77]
\end{aligned}$$

[234] $21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[235] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}23151322181+\hat{1}\rightarrow\hat{2}$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[236] $\overset{1}{1}(12)12231513221\overset{1}{(12)}12231513221(12)12231513221(12)122315132^{\color{red}1}21(12)122315132181 + \hat{\color{red}1} \rightarrow \color{red}2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[237] $\overset{1}{1}(12)12231513221\overset{1}{(12)}12231513221(12)12231513221(12)12231513221(12)12\overset{1}{1}23151321\overset{1}{81} + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[238] $\overset{1}{1}(12)12\overset{1}{1}231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[239] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)122315132\overset{1}{2}1(12)1223151321812+\hat{1}\rightarrow\hat{2} \quad T=65, T_{\max}=77, \Delta=-102, \det=-81, (n_+, n_-, n_0)=(1, 65, 0), c_{\text{ext}}=15.0$

[240] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[241] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12^123151321812+\hat{1}\rightarrow 2 \quad T=65, T_{\max}=77, \Delta=-102, \det=-81, (n_+, n_-, n_0)=(1, 65, 0), c_{\text{ext}}=15.0$

[242] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12^123151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[243] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)122315132^{\color{red}1}21(12)1223151322181+\hat{\color{red}1}\rightarrow\color{red}2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[244] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[245] & \begin{matrix} 1 \\ 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)12+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[246] & \begin{matrix} 1 \\ 181231513221^1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[247] & \begin{matrix} 1 \\ 181231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[229] & \begin{matrix} 1 \\ 1(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[248] & \begin{matrix} 1 \\ 1(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.76 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[249] & \begin{matrix} 1 \\ (12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181232+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.76 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[258] & \begin{matrix} 1 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1223151321812+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[252] & \begin{matrix} 1 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1223151321812+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[253] & \begin{matrix} 1 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1223151322181+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[254] & \begin{matrix} 1 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1223151322181+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[255] & \begin{matrix} 1 \\ (12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181232+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 80, T \in [70, 80]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[251] & 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[257] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[250] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[259] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[260] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[261] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[262] & 181231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[263] & 181231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[256] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[265] & 21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.59 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[266] & 21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.59 \rightarrow 70, T_{\max} = 77, T \in [70, 77]
\end{aligned}$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[267] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70$, $T_{\max} = 80$, $T \in [70, 80]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[268] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[269] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[264] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -169$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[271] $181231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[270] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[272] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -196$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[273] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -196$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[274] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -196$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[275] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -196$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[276] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -196$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[277] & \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.55 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[278] & 181231513221(12)1223151322^{\hat{1}}1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.55 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[279] & \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[280] & \overset{1}{21}(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[281] & 21(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[282] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\hat{1}}315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[283] & \overset{1}{1}(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[284] & \overset{1}{1}(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[285] & \overset{1}{1}(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[286] & 181231513221(12)12231513221(12)122315132^{\hat{1}}21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[296] & \overset{1}{1}(12)12231513221(12)12231513221(12)1223151322^{\hat{1}}1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.48 \rightarrow 70, T_{\max} = 77, T \in [70, 77]
\end{aligned}$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[289] $21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[290] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[291] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[292] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[293] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[294] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[295] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[288] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[287] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[298] $181231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[299] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 47 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[297] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.48 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[301] & 21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[302] & 21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[303] & (12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[304] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[305] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[306] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[307] & 181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[300] & 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[308] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[309] & 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 78, T \in [70, 78]
\end{aligned}$$

[310] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[311] 21(12)12231513221¹(12)12231513221(12)12¹231513221(12)12231513221(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[312] 21(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)1223151321812+¹~~1~~→2 $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[314] $21(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[315] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)122315132181232+\hat{1}\rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[316] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}2315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[317] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[318] $\overset{1}{1}(12)12231513221(\overset{1}{12})122^{\textcolor{red}{1}}31513221(12)12231513221(12)12231513221(12)122315132181 + \hat{\textcolor{red}{1}} \rightarrow \textcolor{red}{2} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[319] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[320] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[321] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12^1231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[322] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[323] & 181231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[324] & 181231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[313] & 21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[335] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[327] & 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[328] & 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[326] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[330] & 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[331] & (12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[332] & (12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 80, T \in [70, 80]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[333] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 797 & 57 \\ 2 & 57 & 3 \end{pmatrix}_{b_0} \\
[334] & 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0} \\
[325] & 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[336] & 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[337] & 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[338] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[339] & 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[340] & 181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[329] & 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[341] & (12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -484, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.36 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[343] & (12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.34 \rightarrow 70, T_{\max} = 80, T \in [70, 80]
\end{aligned}$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[342] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -529$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.34 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[348] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 1031 & 65 \\ 2 & 65 & 3 \end{pmatrix}_{b_0}$$

[346] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[347] $(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 80$, $T \in [70, 80]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[345] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[349] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[350] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[351] $181231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[344] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[352] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -625$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.31 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[353] $(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -625$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.31 \rightarrow 70$, $T_{\max} = 80$, $T \in [70, 80]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[354] $1(12)12231513221(12)1223151322^11(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -676$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.30 \rightarrow 70, T_{\max} = 78, T \in [70, 78]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[355] $(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -676$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.30 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[356] $(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -729$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.29 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

[357] $1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -841$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.27 \rightarrow 70, T_{\max} = 78, T \in [70, 78]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[359] $(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -841$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.27 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[358] $(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -841$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.27 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[360] $(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -961$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.25 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[361] $(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -1156$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.23 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 69$ [362] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^12+\hat{1}\rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -4$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 76, T_{\max} = 81, T \in [76, 81]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[363] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^12+\hat{1}\rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -4$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 76, T_{\max} = 81, T \in [76, 81]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[364] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+^{22}1\hat{1}\rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -4$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 76, T_{\max} = 81, T \in [76, 81]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[365] & \overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[366] & \overset{2^1}{3221}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[367] & \overset{22^1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[368] & \overset{22^1}{21}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[371] & \overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[373] & \overset{222^1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[370] & \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[369] & \overset{1}{21}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[372] & \overset{32^1}{21}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[375] & \overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.75 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[374] & \overset{1}{21}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.75 \rightarrow 75, T_{\max} = 81, T \in [75, 81]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[382] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[381] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[380] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[379] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[378] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[377] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[376] & 21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[388] & 21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[387] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[386] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[385] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[384] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[383] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[393] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[389] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[392] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[390] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[391] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[394] & 21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[400] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[399] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[398] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[397] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[396] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[395] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[408] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[412] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[413] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[414] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[415] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[416] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 8 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[417] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[418] & 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81]
\end{aligned}$$

1248

1249

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[453] & \overset{1}{1}(12)12231513221(12)\overset{1}{1}2231513221(12)122^131513221(12)12231513221(12)12222+\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -225, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.52 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[452] & \overset{1}{1}(12)12231513221(12)\overset{1}{1}2231513221(12)122^131513221(12)12231513221(12)12231513221(12)12+^{\overset{221}{1}}\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -225, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.52 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[451] & \overset{1}{1}(12)12231513221(12)\overset{1}{1}2231513221(12)122^131513221(12)12231513221(12)12231513221(12)1222+\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -225, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.52 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[450] & \overset{1}{21}(12)12231513221(12)\overset{1}{1}2231513221(12)122^131513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -225, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.52 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[463] & \overset{1}{1}(12)12231513221(12)\overset{1}{1}2231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -256, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[464] & \overset{1}{1}(12)122315132^121(12)\overset{1}{1}2231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -256, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[465] & \overset{3221}{21}(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1+^{\overset{1}{1}}\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -256, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[466] & \overset{2221}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12+^{\overset{1}{1}}\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -256, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 47 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[467] & \overset{2221}{1}(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12+^{\overset{1}{1}}\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -256, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[468] & \overset{221}{21}(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12+^{\overset{1}{1}}\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -256, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[469] & \overset{221}{21}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12+^{\overset{1}{1}}\hat{1}\rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -256, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]
\end{aligned}$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 47 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[457] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[458] $21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[459] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[460] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[461] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[462] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[470] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[471] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[472] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[473] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[474] $21(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$
 [498] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
 [499] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
 [500] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
 [501] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
 [502] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [503] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
 [504] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
 [505] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [492] $21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -441$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
 [506] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -110$, $\det = -529$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.34 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$
- $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$
 [508] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -576$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 73.33 \rightarrow 74$, $T_{\max} = 81$, $T \in [74, 81]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[519] & 413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.84 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[520] & 413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.84 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[521] & 413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.76 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[522] & 413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.69 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[523] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.69 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[524] & 413221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.59 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[525] & 413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 196, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.55 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[526] & 413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 225, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.52 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[527] & 413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 256, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.48 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[528] & 413221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 289, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.46 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[531] & 413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.41 \rightarrow 75, T_{\max} = 82, T \in [75, 82]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[530] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 361, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.41 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[529] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 361, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.41 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[533] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 441, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.37 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[532] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 441, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.37 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[534] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 529, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.34 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[535] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 576, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.33 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[536] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 625, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.31 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[537] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 676, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.30 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[538] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 841, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.27 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0} \\
T = 71 & \begin{pmatrix} 15132^1 21 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 71, \quad T_{\max} = 83, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 71, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 76.44 \rightarrow 77, \quad T_{\max} = 83, \quad T \in [77, 83] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[541] & \begin{pmatrix} 151322^1 1 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 71, \quad T_{\max} = 83, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 71, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 76.06 \rightarrow 77, \quad T_{\max} = 83, \quad T \in [77, 83]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[540] $414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 76.06 \rightarrow 77, T_{\max} = 83, T \in [77, 83]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[542] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.94 \rightarrow 76, T_{\max} = 85, T \in [76, 85]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[547] $1^{1513221} (12) 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[546] $1^{1513221} (12) 122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[545] $41413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[544] $41413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[543] $4141322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[550] $(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.76 \rightarrow 76, T_{\max} = 85, T \in [76, 85]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[549] $1^{1513221} (12) 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.76 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[548] $41413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.76 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[554] $1^{1513221} (12) 122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.69 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[553] & 1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[555] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 85, T \in [76, 85] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[556] & (12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 85, T \in [76, 85] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[551] & 41413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[552] & 41413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[557] & 41413221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.59 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[558] & 1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.59 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[559] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.59 \rightarrow 76, T_{\max} = 85, T \in [76, 85] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[560] & 41413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.55 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[561] & 1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.55 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[562] & 41413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -225, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.52 \rightarrow 76, T_{\max} = 83, T \in [76, 83]
\end{aligned}$$

1261

[577] $(12)12231513221(12)12231513221(12)12231513221(12)122^{\overline{1}}31513221(12)12231513221$ $(12)1+\hat{1}\rightarrow 2$ $T = 71$, $T_{\max} = 85$, $\Delta = -112$, $\det = -361$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[576] $1^{1513221}(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1+\hat{1}\rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -104$, $\det = -361$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[575] $1^{1513221}(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -104$, $\det = -361$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[574] $1^{1513221} (12) 12231513221 (12) 1223151322^1 1 (12) 12231513221 (12) 12231513221 (12) 12231513221 (12) 1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$

[573] 41413221(12)12231513221(12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)1¹ + $\hat{1} \rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -108$, $\det = -361$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[584] $(12)12231513221(12)12231513221(12)122315132^{\textcolor{red}{1}}21(12)12231513221(12)12231513221^{31513221}(12)^{-1}+\hat{\textcolor{violet}{1}}\rightarrow \textcolor{red}{2}$ $T = 71$, $T_{\max} = 85$, $\Delta = -112$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[583] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221^{31513221}(12)1+\hat{1}\rightarrow 2$ $T = 71$, $T_{\max} = 85$, $\Delta = -112$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[582] $1^{1513221}(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -104$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[581] $1^{1513221}(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -104$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[580] 41413221(12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -108$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[579] 41413221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)¹12231513221(12)1+ **$\hat{1} \rightarrow 2$** $T = 71$, $T_{\max} = 83$, $\Delta = -108$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^123151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.84 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.84 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.76 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.69 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.69 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.59 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.55 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -225, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.52 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -256, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.48 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.46 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.41 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.41 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.41 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -441, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.37 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -441, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.37 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -529, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.34 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -576, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.33 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -625, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.31 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -676, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.30 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -841, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.27 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$T = 74 \quad [22] \quad 612^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.44 \rightarrow 80, T_{\max} = 86, T \in [80, 86]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513161 + \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.06 \rightarrow 80, T_{\max} = 86, T \in [80, 86]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[24] $612315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 79.06 \rightarrow 80, T_{\max} = 86, T \in [80, 86]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[25] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[26] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[27] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[28] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[29] $6123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[30] $61231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[32] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.76 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[33] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 100, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.76 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[31] $(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.76 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[38] $61231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[39] $61231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[36] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[35] $(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[34] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[37] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[40] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.59 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[42] $61231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 169, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.59 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[41] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.59 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[43] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.55 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[44] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.55 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[45] $61231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 196, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.55 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.52 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.52 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 61231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 225, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.52 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 61231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.48 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.48 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.48 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 61231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.46 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.46 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad (12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.46 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad (12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.41 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 78.41 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1(12)122315132^{121}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1(12)12231513221(12)12231513221(12)122^{131513221}(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 61231513221(12)12231513221(12)1223151322^{11}(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 61231513221(12)12231513221(12)12231513221(12)122315132^{121}(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{131513221}(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 61231513221(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad (12)12231513221(12)12231513221(12)122315132^{121}(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad (12)12231513221(12)12231513221(12)12231513221(12)1223151322^{11}(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{131513221}(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1(12)12231513221(12)12231513221(12)1223151322^{11}(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[66] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.37 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[69] & (12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.36 \rightarrow 79, T_{\max} = 87, T \in [79, 87] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[71] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 529, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.34 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[70] & (12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.34 \rightarrow 79, T_{\max} = 87, T \in [79, 87] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[72] & 61231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 529, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.34 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[73] & (12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 576, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.33 \rightarrow 79, T_{\max} = 87, T \in [79, 87] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[74] & 1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 576, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.33 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[75] & 61231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 576, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.33 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[76] & (12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 625, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.31 \rightarrow 79, T_{\max} = 87, T \in [79, 87] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[77] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 625, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.31 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[78] & 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 625, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.31 \rightarrow 79, T_{\max} = 86, T \in [79, 86]
\end{aligned}$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad (12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 676, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.30 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 676, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.30 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 676, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.30 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad (12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 729, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.29 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad (12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 841, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.27 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad (12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 841, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.27 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 841, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.27 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 841, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.27 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad (12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 961, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.25 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad (12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 1156, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.23 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$$T = 75 \quad [89] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.44 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[90] \text{ 7122}^1\text{31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)}^1\text{12231513221(12)1+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.44 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[91] \text{ 712}^1\text{231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)}^1\text{12231513221(12)1+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.22 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[93] \text{ 7122315132}^1\text{21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)}^1\text{12231513221(12)1+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.06 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[92] \text{ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122}^1\text{315132171+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.06 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[99] \text{ 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132}^1\text{21(12)1+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[98] \text{ 712231513221(12)122}^1\text{31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)}^1\text{12231513221(12)1+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[97] \text{ 71223151322}^1\text{1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)}^1\text{12231513221(12)1+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[96] \text{ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12}^1\text{2315132171+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[95] \text{ 1(12)122}^1\text{31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[94] \text{ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132}^1\text{21(12)122315132171+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[101] \text{ 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322}^1\text{1(12)1+}\hat{\mathbf{1}}\rightarrow\mathbf{2} \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.76 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[100] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.76 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[104] & 712231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.69 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[102] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.69 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[103] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.69 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[105] & 712231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.69 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[106] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.59 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[107] & 712231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.59 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[108] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.55 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[109] & 712231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.55 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[110] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.52 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[111] & 712231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.52 \rightarrow 80, T_{\max} = 87, T \in [80, 87]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[112] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.48 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[113] & 712231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.48 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[115] & 712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.46 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[114] & 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.46 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[121] & 712231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[120] & 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[119] & 712231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[118] & 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[117] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[116] & 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[125] & 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.37 \rightarrow 80, T_{\max} = 87, T \in [80, 87]
\end{aligned}$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[124] \quad 712231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.37 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[123] \quad 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.37 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[122] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.37 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[126] \quad 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -529, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.34 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[127] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -529, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.34 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -576, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.33 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[128] \quad 1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -576, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.33 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[130] \quad 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -625, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.31 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[131] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -625, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.31 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -676, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.30 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -676, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.30 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[134] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -841, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 79.27 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[135] $712231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -841, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 79.27 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$T = 76$ [136] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 3+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 82.22 \rightarrow 83, T_{\max} = 88, T \in [83, 88]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[137] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 81.44 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[138] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 2181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 81.44 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[139] $1812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 81.44 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[140] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 36, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 81.22 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[141] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 81.06 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[142] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 81.06 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[143] $1812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 81.06 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[147] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 18123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 18123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 181231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 8 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad (12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.76 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.76 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.76 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.76 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[165] \quad 181231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[164] \quad 181231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[158] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[159] \quad (12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[160] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[161] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[162] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[163] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[166] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 169, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.59 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[192] & 1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[191] & 1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[190] & 1(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[195] & 181231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[188] & 1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[187] & (12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[186] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[196] & 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 56 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[200] & 1(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.37 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[198] & (12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 441, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.37 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[199] & 1(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.37 \rightarrow 81, T_{\max} = 88, T \in [81, 88]
\end{aligned}$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[197] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[201] $1(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[202] $1(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[203] $181231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[204] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[205] $(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 484$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.36 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[206] $(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 529$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.34 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[207] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 529$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.34 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[208] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 529$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.34 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[209] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 529$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.34 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[213] $181231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 576$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.33 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[212] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 576$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.33 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[211] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 576$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.33 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[210] (12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513218123+ $\hat{1} \rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 576$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.33 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[215] $\overset{1}{1}(12)1223151322\overset{1}{1}1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221812+\hat{\mathbf{1}} \rightarrow \mathbf{2}$ $T =$
76, $T_{\max} = 88$, $\Delta = -101$, $\det = 625$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.31 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[216] $1(12)1223151322\overset{1}{1}1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow\hat{2}$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 625$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.31 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[217] $\overset{1}{18}1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12^{\hat{1}}231513221(12)1+\hat{1}\rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 625, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.31 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

[214] $(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{\color{red}1}\rightarrow\color{red}2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 625$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.31 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[218] $(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 676$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.30 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[220] $1(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 676$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.30 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[219] $1(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 676$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.30 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[221] $\overset{1}{181231513221}(12)12231513221(12)12231513221(12)12231513221(12)12\overset{1}{2}31513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 676$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.30 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 129 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[222] $(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 729, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.29 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

[223] $(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[224] $(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[225] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[226] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[227] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 144 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[228] $(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 961, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.25 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[229] $(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 1156, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.23 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 80$ [233] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 3221+\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86, T_{\max} = 92, T \in [86, 92]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[230] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2+\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 87, T_{\max} = 92, T \in [87, 92]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$22^1 21$ [231] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 87, T_{\max} = 92, T \in [87, 92]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[232] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 87, T_{\max} = 92, T \in [87, 92] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[234] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22+ \\
& \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 86.22 \rightarrow 87, T_{\max} = 92, T \in [87, 92] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[235] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 86.22 \rightarrow 87, T_{\max} = 92, T \in [87, 92] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[236] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+^1 12+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 86.22 \rightarrow 87, T_{\max} = 92, T \in [87, 92] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[237] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222+ \\
& \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 16, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.75 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[238] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12222+ \\
& \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.44 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[239] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.44 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[240] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.44 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[241] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+^1 12+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.44 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[242] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12222+ \\
& \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 36, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.22 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[243] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 36, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.22 \rightarrow 86, T_{\max} = 92, T \in [86, 92]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[254] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 81, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.84 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 9 & 2 \\ 9 & 8 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[250] & \begin{matrix} 1 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222+ \\ \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 81, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.84 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[251] & \begin{matrix} 2221 \\ 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 81, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.84 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[258] & \begin{matrix} 1 \\ 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \\ \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 100, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.76 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[259] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 100, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.76 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[260] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 100, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.76 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[261] & \begin{matrix} 1 \\ (12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.76 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[264] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.69 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[270] & \begin{matrix} 1 \\ (12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.69 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[269] & \begin{matrix} 1 \\ (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.69 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[268] & \begin{matrix} 1 \\ (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.69 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \end{matrix}
\end{aligned}$$

[267] $21^{221}(12)12231513221(12)12^{1231513221}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 121$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[266] ${}^{221}_{21}(12)12231513221(12)122315132^1{}_{21}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 121$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[265] $\overset{2221}{1}(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{\color{red}1}\rightarrow\color{red}2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[263] $\overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322\overset{1}{1}(12)12231513221(12)12222+$
 $\hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[262] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12222+$
 $\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[273] $21^{221}(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 169$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[272] $1^{2221}(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 169, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[271] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221(12)12222+$
 $\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 169, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[274] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12+ \hat{1} \rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[278] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 196, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[277] $21^{221}(12)12231513221(12)12231513221^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 196, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[287] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 289, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.46 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[288] $1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 289, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.46 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[290] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 289, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.46 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[301] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[300] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[297] $21(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$

[299] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[298] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 56 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[295] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 56 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[296] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[294] $1(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$

[293] $1(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[292] $1(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[291] $1(12)122315132^{\hat{1}}21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[303] $1(12)12231513221(12)122315132^{\hat{1}}21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 441$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$

[304] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 441$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[305] $1(12)12231513221(12)12231513221(12)12231513221(12)122^{\hat{1}}231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 441$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[306] $21(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 441$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[307] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 441$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[308] $(12)12231513221(12)12231513221(12)122315132^{\hat{1}}21(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -109$, $\det = 441$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$

[309] $(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{\hat{1}}1(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -109$, $\det = 441$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[302] $1(12)12231513221(12)12231513221(12)1223151322^{\hat{1}}1(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 441$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[310] & (12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 484, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.36 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[314] & (12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 529, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.34 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[313] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 529, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.34 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[312] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 529, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.34 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[311] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 529, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.34 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[318] & (12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 576, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.33 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[317] & 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 576, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.33 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[315] & 1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 576, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.33 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[316] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 576, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.33 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[319] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 625, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.31 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[320] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 625, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.31 \rightarrow 85, T_{\max} = 92, T \in [85, 92]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 99 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[321] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 625, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.31 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 25 & 2 \\ 25 & 99 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[322] & \begin{matrix} 1 \\ (12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 625, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.31 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[323] & \begin{matrix} 1 \\ 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222+ \\ \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 676, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.30 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[324] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 676, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.30 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 26 & 2 \\ 26 & 129 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[325] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 676, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.30 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 26 & 2 \\ 26 & 129 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[326] & \begin{matrix} 1 \\ (12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 676, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.30 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[327] & \begin{matrix} 1 \\ (12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 729, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.29 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[332] & \begin{matrix} 1 \\ (12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 841, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.27 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[331] & \begin{matrix} 1 \\ (12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 841, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.27 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[328] & \begin{matrix} 1 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \\ \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 841, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.27 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[329] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 841, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.27 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \end{matrix}
\end{aligned}$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 144 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[330] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 841, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.27 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 144 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[333] $(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 961, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.25 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[334] $(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 1156, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.23 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 81$ [335] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)-\hat{1} \rightarrow 2$
 $T = 81, T_{\max} = 93, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.44 \rightarrow 87, T_{\max} = 93, T \in [87, 93]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[336] $41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+\hat{1} \rightarrow 2$
 $T = 81, T_{\max} = 93, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.06 \rightarrow 87, T_{\max} = 93, T \in [87, 93]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[337] $413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+\hat{1} \rightarrow 2$
 $T = 81, T_{\max} = 93, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.84 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[338] $413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)+\hat{1} \rightarrow 2$
 $T = 81, T_{\max} = 93, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.76 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[339] $413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+\hat{1} \rightarrow 2$
 $T = 81, T_{\max} = 93, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.69 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[341] $413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+\hat{1} \rightarrow 2$
 $T = 81, T_{\max} = 93, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.69 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[340] $413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)+\hat{1} \rightarrow 2$
 $T = 81, T_{\max} = 93, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.69 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 10 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[342] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+\hat{1} \rightarrow 2$
 $T = 81, T_{\max} = 93, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.59 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

- $$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [343] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.55 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
- [344] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -225, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.52 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
- [345] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -256, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.48 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$
- [346] $413221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -289, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.46 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$
- [348] $413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -361, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.41 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$
- [347] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -361, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.41 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [350] $413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -441, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.37 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$
- [349] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -441, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.37 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [351] $413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -484, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.36 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 65 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$
- [352] $413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -529, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.34 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$
- $$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$
- [353] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -576, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.33 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132¹21(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.84 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[4] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.84 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[6] (12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 100$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.76 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[7] (12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221¹⁵¹³²²¹(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 100$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.76 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[9] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[8] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[13] (12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221¹⁵¹³²²¹(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[12] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322¹⁵¹³²²¹1(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[11] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹31513221¹⁵¹³²²¹(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[10] (12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[15] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)122315132¹⁵¹³²²¹21(12)12231513221¹⁵¹³²²¹(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.59 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

- $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
 [14] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.59 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [17] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221 \overset{1513221}{(12) 1 + \hat{1} \rightarrow 2}$
 $T = 82, T_{\max} = 94, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.55 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [16] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.55 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [19] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221 \overset{1513221}{(12) 1 + \hat{1} \rightarrow 2}$
 $T = 82, T_{\max} = 94, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.52 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [18] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 225, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.52 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [20] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 256, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.48 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [21] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221 \overset{1513221}{(12) 1 + \hat{1} \rightarrow 2}$
 $T = 82, T_{\max} = 94, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.48 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [23] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221 \overset{1513221}{(12) 1 + \hat{1} \rightarrow 2}$
 $T = 82, T_{\max} = 94, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.46 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [22] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 289, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.46 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [24] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.41 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$
- $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
 [25] $(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.41 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[59] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.59 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[60] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.55 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[61] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 225, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.52 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[62] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.48 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[63] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 289, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.46 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[64] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.41 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[65] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.41 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[66] $(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 441, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.37 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[67] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 441, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.37 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[68] $(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 484, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.36 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[69] $(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 529, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.34 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[106] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.41 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[105] (12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.41 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[104] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -361$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.41 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[103] (12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -361$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.41 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[107] (12)12231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -441$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.37 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[108] (12)12231513221(12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -441$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.37 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[109] (12)12231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.37 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[110] (12)12231513221(12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.37 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[111] (12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -484$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.36 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[112] (12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -484$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.36 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 65 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[113] (12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -529$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.34 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

- $$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$
- $[114] \quad (12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -102, \det = -529, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.34 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$
- $[115] \quad (12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -103, \det = -576, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.33 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$
- $[116] \quad (12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -102, \det = -576, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.33 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$
- $[117] \quad (12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -103, \det = -625, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.31 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$
- $[118] \quad (12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -102, \det = -625, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.31 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 99 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$
- $[120] \quad (12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -102, \det = -676, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.30 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 129 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$
- $[119] \quad (12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -103, \det = -676, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.30 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$
- $[122] \quad (12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -102, \det = -729, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.29 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 107 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$
- $[121] \quad (12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -103, \det = -729, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.29 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$
- $[123] \quad (12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -103, \det = -841, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.27 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$
- $[125] \quad (12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 85, T_{\max} = 97, \Delta = -102, \det = -841, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.27 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[139] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132¹21(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 81$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.84 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[138] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹2315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 81$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.84 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[137] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹2315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 81$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.84 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[136] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132¹21(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 81$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.84 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[140] (12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.76 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[141] (12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.76 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[142] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.69 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[146] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.69 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[145] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.69 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[147] (12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.69 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 10 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[144] (12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 90.69 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[158] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.41 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[159] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.41 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[160] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.41 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[161] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.41 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[162] $(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.37 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[164] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.37 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[163] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.37 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[165] $(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.37 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[167] $(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 484, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.36 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 65 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[166] $(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 484, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.36 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[169] $(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 529, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.34 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[168] (12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 529$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.34 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[171] (12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 576$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.33 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[170] (12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 576$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.33 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[173] (12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 625$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.31 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 99 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[172] (12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 625$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.31 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[175] (12)12231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 676$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.30 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 129 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[174] (12)12231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 676$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.30 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[177] (12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 729$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.29 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 107 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[176] (12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 729$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.29 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

[179] (12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 841$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.27 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[178] (12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 841$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.27 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[181] (12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 841$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.27 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 144 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[180] (12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 841$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.27 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 115 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[183] (12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 961$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.25 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 154 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[182] (12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 961$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.25 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[185] (12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 1156$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.23 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 169 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[184] (12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 1156$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.23 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 87$ [187] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.44 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[188] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132¹181+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.44 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[186] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132¹1812+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.44 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[189] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322¹181+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -36$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.22 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[190] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹3151321812+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -49$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.06 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

[illegible]

[224] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221(12)1223151322181 + \color{red}{\hat{1}} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -289$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.46 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[225] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221(12)122315132181 +$
 $\color{red}{\hat{1} \rightarrow 2} \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -289, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.46 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[226] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151321812 + \hat{\mathbf{1}} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -361$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[227] $(12)12231513221(12)12231513221(12)12231513221(12)122^{\overset{1}{\rightarrow}1}31513221(12)12231513221(12)12231513221(12)1223151321812 + \overset{1}{\rightarrow}1 \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -361$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[228] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\color{red}1}231513221(12)12231513221(12)1223151322181 + \color{red}{\hat{1}} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -361$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[229] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -361$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[230] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132181 +$
 $\hat{1} \rightarrow 2 \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -361, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[231] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)122315132181 +$
 $\hat{1} \rightarrow 2 \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -361, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[234] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -441$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.37 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[235] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$
 $T = 87, T_{\max} = 99, \Delta = -100, \det = -441, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.37 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[236] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)122315132181 +$
 $\hat{1} \rightarrow 2 \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -441, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.37 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[illegible]

[illegible]

[illegible]

1.9 $\mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 106, \Delta = -240, \det = 16, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12, T_{\max} = 106, T \in [12, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [1] \quad 1^{11}21^1 + \mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 100, \Delta = -211, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12, T_{\max} = 100, T \in [12, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 2 & 0 & 0 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [2] \quad 1^{11}321^1 + \mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -162, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

1.10 $\mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 170, \Delta = -261, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 170, T \in [13, 170]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 3 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 170, \Delta = -261, \det = 17, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 170, T \in [13, 170]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -6 & 0 & -4 \\ 2 & 0 & 1 & 1 \\ 3 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 101, \Delta = -232, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, T_{\max} = 101, T \in [13, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}21^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 101, \Delta = -232, \det = -14, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 101, T \in [13, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1^{11}221^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -203, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 1^{11}2221^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -174, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{11}22221^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [7] \quad 1^{11}22221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 29, T \in [18, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -4 & 5 & -3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}223131^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \quad 1^{11}2231321^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 19, T_{\max} = 22, T \in [19, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -4 & 5 & -3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 1^131413221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 23, \Delta = -103, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 20, T_{\max} = 23, T \in [20, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

1.11 $su_2 + e_7 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 188, \Delta = -316, \det = 16, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 188, T \in [15, 188]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 3 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 188, \Delta = -316, \det = 23, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 188, T \in [15, 188]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -8 & 0 & -6 \\ 2 & 0 & 1 & 1 \\ 3 & -6 & 1 & -4 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 186, \Delta = -287, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 186, T \in [15, 186]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 186, \Delta = -287, \det = -20, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 186, T \in [15, 186]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -6 & 1 & -4 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1^{11}221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 170, \Delta = -258, \det = 17, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 170, T \in [15, 170]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -6 & 2 & -3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 1^{11}2221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 102, \Delta = -229, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{11}22221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 102, \Delta = -200, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -6 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [7] \quad 1^{11}222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \quad 1^{11}22313^1 21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 103, T \in [19, 103]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 10 & 0 & 8 & 8 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 8 & 0 & 5 & 5 \\ 2 & 8 & -6 & 5 & -3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11}2222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 100, T \in [17, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11}2231321^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 102, \Delta = -160, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 102, T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \quad 1^{11}22222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 6 & 0 & 8 & 8 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 8 & 0 & 7 & 7 \\ 2 & 8 & -6 & 7 & -3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}22313221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 94, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 17.40 \rightarrow 18, T_{\max} = 94, T \in [18, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [13] \quad 1^1 231413221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 12, T_{\max} = 95, \Delta = -130, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.40 \rightarrow 19, T_{\max} = 95, T \in [19, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{11}2231512321^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 13, T_{\max} = 93, \Delta = -127, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.40 \rightarrow 20, T_{\max} = 93, T \in [20, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 5^{11} 1321^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1 \quad T = 13, T_{\max} = 88, \Delta = -125, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.40 \rightarrow 20, T_{\max} = 88, T \in [20, 88]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -4 & 1 & -2 \\ 1 & 5 & 1 & 4 & 4 \\ 2 & 5 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

1.12 $\mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$

- $T = 4$ [0] $1^{11}1^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 4, T_{\max} = 185, \Delta = -288, \det = 14, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 185, T \in [14, 185]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 3 & -3 & -1 & -3 \end{pmatrix}_{b_0}$
- [1] $1^{11}1^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 4, T_{\max} = 185, \Delta = -288, \det = 20, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 185, T \in [14, 185]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -7 & 0 & -5 \\ 2 & 0 & 1 & 1 \\ 3 & -5 & 1 & -3 \end{pmatrix}_{b_0}$
- $T = 5$ [2] $1^{11}21^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 169, \Delta = -259, \det = -5, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 169, T \in [14, 169]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$
- [3] $1^{11}21^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 169, \Delta = -259, \det = -17, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.71 \rightarrow 14, T_{\max} = 169, T \in [14, 169]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -5 & 1 & -3 \end{pmatrix}_{b_0}$
- $T = 6$ [4] $1^{11}221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 101, \Delta = -230, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -5 & 2 & -2 \end{pmatrix}_{b_0}$
- $T = 7$ [5] $1^{11}2221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 101, \Delta = -201, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -5 & 3 & -2 \end{pmatrix}_{b_0}$
- $T = 8$ [6] $1^{11}22221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 8, T_{\max} = 101, \Delta = -172, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -5 & 4 & -1 \end{pmatrix}_{b_0}$
- $T = 9$ [7] $1^{11}222221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 99, T \in [16, 99]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -5 & 5 & -1 \end{pmatrix}_{b_0}$
- $T = 10$ [8] $1^{11}2222221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -5 & 6 & -3 \end{pmatrix}_{b_0}$
- [9] $1^{11}2231321^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 94, \Delta = -132, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -5 & 5 & -1 \end{pmatrix}_{b_0}$

$$T = 11 \quad [10] \quad 1^{11}22313221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}'_7 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 23, \Delta = -103, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20, T_{\max} = 23, T \in [20, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -5 & 6 & -3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [11] \quad 1^{11}231413221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{c}'_7 + \hat{1} \rightarrow 1 \quad T = 12, T_{\max} = 24, \Delta = -102, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21, T_{\max} = 24, T \in [21, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

1.13 $su_2 + \mathfrak{c}_8 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 193, \Delta = -431, \det = 35, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.34 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -12 & 0 & -10 \\ 2 & 0 & 1 & 1 \\ 3 & -10 & 1 & -8 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [1] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 193, \Delta = -402, \det = -32, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.38 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -10 & 1 & -8 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [2] \quad 1^{11}221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 193, \Delta = -373, \det = 29, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.41 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -10 & 2 & -7 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [3] \quad 1^{11}2221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 193, \Delta = -344, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.46 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -10 & 3 & -6 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [4] \quad 1^{11}22221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 192, \Delta = -315, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 192, T \in [19, 192]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -10 & 4 & -5 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11}222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 190, \Delta = -286, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.60 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -10 & 5 & -4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 1^{11}2222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 173, \Delta = -257, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.71 \rightarrow 19, T_{\max} = 173, T \in [19, 173]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -10 & 6 & -3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{11}2222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_8 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 106, \Delta = -228, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.86 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 6 & 0 & 8 & 8 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 8 & 0 & 7 & 7 \\ 2 & 8 & -10 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}22313221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_8 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 151, \Delta = -246, \det = -17, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.71 \rightarrow 20, T_{\max} = 151, T \in [20, 151]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -10 & 6 & -3 \end{pmatrix}_{b_0}$$

1.14 $\textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 102, \Delta = -235, \det = 10, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 102, T \in [12, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 102, \Delta = -235, \det = 14, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 102, T \in [12, 102]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -5 & 0 & -3 \\ 2 & 0 & 1 & 1 \\ 3 & -3 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -206, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -206, \det = -11, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -3 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1^{11}221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -177, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 1^{11}2221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 98, \Delta = -148, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{11}22221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 17, T_{\max} = 45, T \in [17, 45]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -3 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 89, \Delta = -128, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 89, T \in [17, 89]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 6 & 0 & 5 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 5 & 0 & 3 & 3 \\ 2 & 6 & -3 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^13221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 89, \Delta = -128, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 89, T \in [15, 89]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \quad 1^{11}223131^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -3 & 4 & -3 \end{pmatrix}_{b_0}$$

1.15 $su_2 + \hat{1} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -212, \det = -2, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 11^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -212, \det = -3, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -212, \det = -8, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 1 & 1 \\ 3 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [3] \quad 121^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -183, \det = 3, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 \\ 2 & 2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -183, \det = 7, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 2 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [5] \quad 1221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -154, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 96, T \in [13, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 2 & 2 \\ 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -154, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 3 & 3 \\ 2 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [7] \quad 2^1231^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 91, \Delta = -134, \det = 2, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 91, T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 2 & 2 \\ 2 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 12221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 81, \Delta = -125, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 81, T \in [14, 81]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 \\ 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11}2221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 81, \Delta = -125, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 81, T \in [10, 81]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 2 & 4 & 4 \\ 2 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [10] \quad 1\overset{1}{3}221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 16, T_{\max} = 19, T \in [16, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 6 & 5 & 6 \\ 1 & 5 & 3 & 3 \\ 2 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1\overset{1}{3}221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 19, T \in [15, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 \\ 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 213221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 19, T \in [15, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 \\ 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1\overset{11}{2}2221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 13, T \in [10, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 3 & 5 & 5 \\ 2 & 5 & 4 & 4 \\ 2 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [14] \quad 1\overset{11}{2}22221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -67, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 4 & 6 & 6 \\ 2 & 6 & 5 & 5 \\ 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [15] \quad 1\overset{11}{2}222221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 5 & 7 & 7 \\ 2 & 7 & 6 & 6 \\ 2 & 7 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[16] \quad 1\overset{11}{2}2313^1 21^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 10 & 8 & 8 \\ 2 & 8 & 5 & 5 \\ 2 & 8 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[17] \quad 2\overset{2}{3}13221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 5 & 5 \\ 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [18] \quad 31\overset{11}{5}231^{11} 13^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 5 & 5 \\ 2 & 7 & 5 & 5 \end{pmatrix}_{b_0}$$

1.16 $\textcolor{red}{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1\overset{11}{1}1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -210, \det = 2, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -1 & 0 & -1 \\ 2 & 0 & 1 & 1 \\ 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

1.17 $\textcolor{red}{su}_2 + \hat{1} \rightarrow (-2)$

$$T = 4 \quad [0] \quad 12\overset{1}{1}1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 4, T_{\max} = 94, \Delta = -147, \det = 2, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 11, T_{\max} = 94, T \in [11, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

- $T = 5$ [1] $131^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 5$, $T_{\max} = 84$, $\Delta = -126$, $\det = -4$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 12$, $T_{\max} = 84$, $T \in [12, 84]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$
- [2] $1^2 31^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 5$, $T_{\max} = 84$, $\Delta = -126$, $\det = -4$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13$, $T_{\max} = 84$, $T \in [13, 84]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$
- [3] $1^1 31^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 5$, $T_{\max} = 84$, $\Delta = -126$, $\det = -5$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 84$, $T \in [13, 84]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$
- $T = 6$ [4] $1^2 2^1 4^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17$, $T_{\max} = 29$, $T \in [17, 29]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -4 \end{pmatrix}_{b_0}$
- [5] $1^2 4^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$
- [6] $1^2 4^1 1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$
- [7] $2141^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$
- [8] $21^2 41^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17$, $T_{\max} = 29$, $T \in [17, 29]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -4 \end{pmatrix}_{b_0}$
- [9] $1231^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 13$, $\Delta = -98$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$
- [10] $1^1 4^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 5$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.80 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$
- [11] $1^2 4^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 5$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.20 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$
- [12] $1^1 4^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 6$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.67 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$
- [13] $21^1 41^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 6$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.67 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{\overline{1}^2} 4^{\overline{1}^2} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 48 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[15] $1^3 4^{\overline{1}^2} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 29, T \in [16, 29]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$$

[16] $21^{\overline{3}41^2\overline{2}^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 29, T \in [16, 29]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -3 \end{pmatrix}_{b_0}$$

[17] $21^{\overline{1}41^2\overline{2}^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[18] $2^{\overline{1}^1} 3^{\overline{1}^1} 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 18, \Delta = -100, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 18, T \in [12, 18]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{\overline{1}^2} 4^{\overline{1}^2} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{\overline{1}^2} 4^{\overline{1}^2} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 17, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

[21] $1^2 4^{\overline{1}^2} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 17, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15, T_{\max} = 29, T \in [15, 29]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$$

[22] $1^2 4^{\overline{1}^2} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 18, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[23] $21^{\overline{2}41^2\overline{2}^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 18, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$$

[28] $1^4 4^{\overline{1}^3} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.78 \rightarrow 15, T_{\max} = 29, T \in [15, 29]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

[25] $1^2 4^{\overline{1}^3} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$$

- [26] $21^{241^3 2^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [27] $21^{441^3 2^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -2 \end{pmatrix}_{b_0}$
- [24] $1^2 2^{1^3} 4^1 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 108 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [29] $1^2 2^{1^3} 4^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 37$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.68 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [30] $1^3 2^{1^3} 4^1 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 37$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$
- [31] $1^3 2^{1^3} 4^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 38$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [32] $21^3 41^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 38$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [35] $1^3 2^{1^4} 4^1 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 192 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [34] $21^5 41^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.56 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 224 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$
- [33] $1^3 2^{1^4} 4^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [37] $21^3 41^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [36] $1^5 2^{1^4} 4^1 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.56 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [39] $1^3 2^{1^4} 4^1 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 65$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.75 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

[38] $1^4 4^{2^1 1^4} 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 65, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 29, T \in [15, 29]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$$

[41] $21^4 41^{4^2 1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 66, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

[40] $1^4 4^{2^1 1^4} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 66, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

[43] $21^4 41^{5^2 1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

[42] $1^4 4^{2^1 1^5} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

[44] $1^4 4^{2^1 1^5} 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 101, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

[46] $21^5 41^{5^2 1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 102, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 298 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

[45] $1^5 4^{2^1 1^5} 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 102, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 298 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 7$ [53] $1^1 5^{2^1 1^1} 1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[52] $1^1 5^{2^1 1^1} 12 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $22151^{1^2 1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

[50] $1^2 21^5 1^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1 \overset{1^2}{5} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1 \overset{21^2}{5} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1 \overset{1^2}{2^1 1^1} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[56] \quad 21 \overset{1^1}{5} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1 \overset{1^1}{2^1 1^1} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1 \overset{1^2}{5} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[54] \quad 21 \overset{1^1}{2^1 5} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[57] \quad 1 \overset{1^2}{5} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[59] \quad 21 \overset{2^1 1^1}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 30 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1 \overset{1^1}{5} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1 \overset{2^1 1^1}{5} 1^1 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0}$$

- [62] $1^1 \overset{1^2}{5} 1^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.67 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 30 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -3 \end{pmatrix}_{b_0}$
- [63] $21^1 \overset{2^1}{5} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$
- [64] $221^1 \overset{1^1}{5} 1^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$
- [65] $1^1 \overset{1^1}{5} 1^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$
- [66] $1^1 \overset{1^5}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.33 \rightarrow 20$, $T_{\max} = 21$, $T \in [20, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -6 \end{pmatrix}_{b_0}$
- [67] $21^4 \overset{1^1}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.23 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -7 \end{pmatrix}_{b_0}$
- [68] $21^2 \overset{2^1}{5} 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.23 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -7 \end{pmatrix}_{b_0}$
- [69] $1^2 \overset{1^5}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.23 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -7 \end{pmatrix}_{b_0}$
- [70] $1^1 \overset{1^4}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.23 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -7 \end{pmatrix}_{b_0}$
- [71] $21^1 \overset{2^1}{5} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 104 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$
- [73] $21^3 \overset{1^4}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.25 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -7 \end{pmatrix}_{b_0}$
- [74] $1^1 \overset{2^1}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.25 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 32 & 7 & 10 \\ 1 & 7 & 1 & 1 \\ 2 & 10 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[75] \begin{matrix} 1^4 \\ 1^3 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 128 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[72] \begin{matrix} 1^4 \\ 1^1 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.25 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[76] \begin{matrix} 1^3 \\ 1^1 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 48 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[77] \begin{matrix} 21^3 \\ 1^5 \end{matrix} 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.25 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[78] \begin{matrix} 1^3 \\ 1^3 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.25 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[79] \begin{matrix} 1^1 \\ 1^1 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[80] \begin{matrix} 1^4 \\ 1^3 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 20.25 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -7 \end{pmatrix}_{b_0}$$

$$[84] \begin{matrix} 1^2 \\ 1^1 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.53 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[81] \begin{matrix} 1^3 \\ 1^2 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.47 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[82] \begin{matrix} 1^4 \\ 1^2 \end{matrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.88 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 104 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[83] \begin{matrix} 1^4 \\ 1^3 \end{matrix} 2^1 5 1^2 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.76 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -5 \end{pmatrix}_{b_0}$$

- [88] $15 \begin{smallmatrix} 1^2 \\ 1^2 \end{smallmatrix} 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$
- [85] $12 \begin{smallmatrix} 1^4 \\ 1^2 \end{smallmatrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.56 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$
- [86] $1 \begin{smallmatrix} 2^1 1^2 \\ 5 \end{smallmatrix} 1^2 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$
- [87] $21 \begin{smallmatrix} 2^1 1^2 \\ 5 \end{smallmatrix} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.56 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 126 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -4 \end{pmatrix}_{b_0}$
- [95] $13 \begin{smallmatrix} 1^3 \\ 1^3 \end{smallmatrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.26 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -5 \end{pmatrix}_{b_0}$
- [90] $21 \begin{smallmatrix} 1^2 1^2 \\ 1^1 \end{smallmatrix} 5 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$
- [91] $221 \begin{smallmatrix} 3^1 5^1 2^1 \\ 5 \end{smallmatrix} 1^2 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.26 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -5 \end{pmatrix}_{b_0}$
- [92] $11 \begin{smallmatrix} 1^3 \\ 1^1 \end{smallmatrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.32 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 62 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$
- [93] $21 \begin{smallmatrix} 3^1 \\ 5 \end{smallmatrix} 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.58 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$
- [94] $21 \begin{smallmatrix} 1^2 1^2 \\ 1^1 \end{smallmatrix} 5 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.32 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 80 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$
- [89] $221 \begin{smallmatrix} 1^1 5^1 2^1 \\ 5 \end{smallmatrix} 1^2 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$
- [96] $11 \begin{smallmatrix} 1^3 \\ 1^3 \end{smallmatrix} 5 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.58 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$

- [97] $21^{\overset{1}{3}}5^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.26 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -5 \end{pmatrix}_{b_0}$
- [98] $1^{\overset{1}{1}}5^{\overset{1}{1}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 30 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$
- [99] $1^{\overset{1}{2}}5^{\overset{1}{1}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.80 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 44 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$
- [100] $21^{\overset{1}{3}}5^{\overset{1}{2}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.20 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$
- [101] $1^{\overset{1}{3}}2^{\overset{1}{3}}5^{\overset{1}{2}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.20 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$
- [102] $1^{\overset{1}{3}}5^{\overset{1}{2}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.80 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 80 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$
- [103] $21^{\overset{1}{1}}5^{\overset{1}{2}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.80 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 60 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -1 \end{pmatrix}_{b_0}$
- [104] $1^{\overset{1}{2}}5^{\overset{1}{2}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.19 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 60 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$
- [105] $21^{\overset{2}{1}}5^{\overset{2}{1}}1^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.19 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 60 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -2 \end{pmatrix}_{b_0}$
- [106] $21^{\overset{2}{2}}5^{\overset{2}{1}}1^{\overset{1}{3}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.33 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 100 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$
- [107] $1^{\overset{1}{2}}5^{\overset{1}{2}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.33 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 100 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$
- [110] $221^{\overset{2}{2}}5^{\overset{1}{2}}1^{\overset{1}{2}}2^{\overset{1}{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.64 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

- $$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$
- [108] $1^2 \overset{1^2}{5} 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$$
- [109] $21^2 \overset{2^1}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$
- [113] $21^5 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 20.43 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 406 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -7 \end{pmatrix}_{b_0}$$
- [112] $1^1 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.46 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 254 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -4 \end{pmatrix}_{b_0}$$
- [111] $21^1 \overset{2^1}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 82 & 11 & 16 \\ 1 & 11 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$
- [114] $21^5 \overset{1^1}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.46 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 254 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -4 \end{pmatrix}_{b_0}$$
- [115] $221^1 \overset{1^1}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 82 & 11 & 16 \\ 1 & 11 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$
- [116] $221^5 \overset{1^1}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 20.43 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 322 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -7 \end{pmatrix}_{b_0}$$
- [117] $1^1 \overset{1^1}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 46 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$
- [118] $1^5 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 20.43 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 406 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -7 \end{pmatrix}_{b_0}$$
- [123] $1^{\overset{21^4}{5}} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.78 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$
- [119] $1^{\overset{1^4}{5}} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 108 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0} \\
[120] & \begin{matrix} 1^2 \\ 1^2 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 14.44 \rightarrow 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0} \\
[121] & \begin{matrix} 1^4 \\ 1^4 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.78 \rightarrow 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0} \\
[122] & \begin{matrix} 2^1 1^3 \\ 1^5 \end{matrix} 1^2 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 14.44 \rightarrow 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 11 & 16 \\ 1 & 11 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0} \\
[124] & \begin{matrix} 1^4 \\ 1^3 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.32 \rightarrow 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0} \\
[125] & \begin{matrix} 1^3 \\ 1^2 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 14.68 \rightarrow 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0} \\
[137] & \begin{matrix} 1^4 \\ 1^1 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 14.95 \rightarrow 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0} \\
[126] & \begin{matrix} 1^3 \\ 1^3 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 14.95 \rightarrow 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0} \\
[127] & \begin{matrix} 1^5 \\ 1^5 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 19.16 \rightarrow 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -6 \end{pmatrix}_{b_0} \\
[128] & 21^5 \begin{matrix} 1^2 \\ 1^3 \end{matrix} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 17.79 \rightarrow 18, T_{\max} = 21, T \in [18, 21] \\
& \begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 286 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0} \\
[129] & \begin{matrix} 1^5 \\ 1^4 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.63 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 218 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0} \\
[130] & 21^1 \begin{matrix} 2^1 1^3 \\ 1^4 \end{matrix} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 14.95 \rightarrow 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 154 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [131] $1^{\overline{2}1^3} 5^{\overline{1}3} 2^{\overline{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -38$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [132] $21^{\overline{1}1^3} 5^{\overline{1}3} 1^{\overline{2}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -38$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.42 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 102 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [133] $1^{\overline{1}5} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -38$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.68 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 158 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [134] $21^{\overline{1}5} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -38$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.16 \rightarrow 20$, $T_{\max} = 21$, $T \in [20, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -6 \end{pmatrix}_{b_0}$
- [135] $1^{\overline{2}5} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -38$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.79 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 286 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$
- [136] $1^{\overline{1}2} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -38$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.42 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 62 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [142] $1^{\overline{1}3} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 82 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [138] $1^{\overline{3}5} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 322 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -5 \end{pmatrix}_{b_0}$
- [139] $21^{\overline{1}5} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 322 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -5 \end{pmatrix}_{b_0}$
- [140] $1^{\overline{1}5} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.08 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 186 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -3 \end{pmatrix}_{b_0}$
- [141] $21^{\overline{1}1^3} 5^{\overline{1}3} 1^{\overline{2}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 126 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$
- [146] $1^{\overline{2}5} 5^{\overline{1}3} 1^{\overline{2}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -41$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.95 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 248 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[150] \ 1^1 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[143] \ 1^1 \overset{1^2}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.61 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 80 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[152] \ 21^4 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.12 \rightarrow 19, \ T_{\max} = 21, \ T \in [19, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 320 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[148] \ 1^2 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[149] \ 1^1 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 128 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[145] \ 21^4 \overset{1^1}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[151] \ 21^2 \overset{2^1}{5} 1^3 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.61 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 102 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[144] \ 1^4 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.12 \rightarrow 19, \ T_{\max} = 21, \ T \in [19, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 320 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[147] \ 21^2 \overset{2^1}{5} 1^3 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 214 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[156] \ 1^1 \overset{1^3}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -42, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.86 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[155] \ 1^3 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -42, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.43 \rightarrow 18, \ T_{\max} = 21, \ T \in [18, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 282 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$$

[154] $1^1 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -42$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.52 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 154 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$

[153] $1^2 \overset{15}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -42$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.38 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 214 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$

[160] $1^3 \overset{15}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.81 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 246 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$

[159] $1^1 \overset{13}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$

[158] $21^3 \overset{21}{5} 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.81 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 246 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$

[157] $21^3 \overset{21}{5} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 126 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -2 \end{pmatrix}_{b_0}$

[169] $221^4 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.27 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -4 \end{pmatrix}_{b_0}$

[168] $1^4 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.27 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 280 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$

[167] $21^2 \overset{21}{5} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.45 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 180 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$

[166] $1^2 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.27 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 212 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$

[165] $221^2 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$

[164] $21^4 \overset{12}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.27 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 212 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$

- [163] $1^2 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.45 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [170] $1^2 \overset{12}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 100 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [161] $21^4 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.27 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 280 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$
- [162] $21^2 \overset{213}{5} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$
- [171] $1^3 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -45$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.69 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$
- [172] $1^2 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -45$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.80 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$
- [173] $21^4 \overset{13}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -45$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.69 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$
- [174] $21^2 \overset{213}{5} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -45$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 150 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$
- [175] $1^2 \overset{13}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -45$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$
- [177] $1^2 \overset{13}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.39 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [176] $21^3 \overset{213}{5} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.39 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 150 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$
- [179] $1^3 \overset{14}{5} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.17 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$$

[178] $21^{\overset{1}{3}\overset{2}{5}\overset{1}{4}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.17 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 210 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -3 \end{pmatrix}_{b_0}$$

[180] $1^{\overset{1}{3}} \overset{1}{5} 1^{\overset{1}{3}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.72 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

[181] $21^{\overset{1}{3}\overset{2}{5}\overset{1}{3}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.72 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

[182] $221^{\overset{1}{3}} \overset{1}{5} 1^{\overset{1}{3}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.72 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

[183] $21^{\overset{1}{2}\overset{1}{4}} \overset{1}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.27 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 156 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

[184] $1^{\overset{1}{2}} \overset{1}{5} 1^{\overset{1}{4}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.27 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[185] $21^{\overset{1}{2}\overset{1}{4}} \overset{1}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 184 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

[186] $1^{\overset{1}{5}} \overset{1}{5} 1^{\overset{1}{4}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

[187] $1^{\overset{2}{1}\overset{5}{5}} \overset{1}{5} 1^{\overset{1}{4}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

[188] $1^{\overset{1}{5}} \overset{1}{5} 1^{\overset{1}{4}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

[189] $1^{\overset{2}{1}\overset{1}{4}} \overset{1}{5} 1^{\overset{1}{3}} 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.56 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 160 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

[190] $1^{\overset{1}{5}} \overset{1}{5} 1^{\overset{1}{4}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 192 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[191] \ 21 \overset{2^1 1^4}{1} 5 \overset{1^4}{1^4} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 216 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[192] \ 1 \overset{1^3}{1^3} 5 \overset{1^4}{1^4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[193] \ 1 \overset{1^4}{1^4} 5 \overset{1^4}{1^4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[194] \ 1 \overset{1^5}{1^5} 5 \overset{1^4}{1^4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[195] \ 1 \overset{1^2}{1^2} 5 \overset{1^4}{1^4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 104 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[196] \ 1 \overset{1^4}{1^4} 5 \overset{1^4}{1^4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[197] \ 21 \overset{2^1 1^4}{1} 5 \overset{1^1}{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[199] \ 1 \overset{1^4}{1^4} 5 \overset{1^4}{1^4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[198] \ 1 \overset{2^1 1^4}{1} 5 \overset{1^4}{1^4} 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[201] \ 1 \overset{1^3}{1^3} 5 \overset{1^4}{1^4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.53 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 128 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[200] \ 1 \overset{1^5}{1^5} 5 \overset{1^4}{1^4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 188 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[202] \ 1^1 \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -69, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[203] \ 1^2 \overset{1^2}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[204] \ 221^2 \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 182 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[205] \ 21^2 \overset{2^1}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 182 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[206] \ 21^5 \overset{1^1}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.70 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 290 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[207] \ 21^3 \overset{2^1}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 184 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[208] \ 1^1 \overset{1^3}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 154 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[209] \ 1^1 \overset{1^5}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.14 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 218 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[210] \ 1^1 \overset{1^5}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.70 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 290 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[212] \ 1^1 \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.89 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[211] \ 1^1 \overset{1^5}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.39 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 252 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

- [213] $1^2 \overset{1^5}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 216 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [214] $21^4 \overset{2^1 1^4}{5} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 216 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$
- [215] $1^1 \overset{1^4}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 216 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [216] $1^2 \overset{1^3}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$
- [217] $21^2 \overset{2^1 1^4}{5} 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 284 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$
- [218] $21^2 \overset{2^1 1^4}{5} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 212 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$
- [219] $1^2 \overset{1^4}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -74$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.86 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 182 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [220] $21^2 \overset{2^1 1^4}{5} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -74$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.86 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 246 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$
- [221] $1^2 \overset{1^3}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.84 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [222] $1^2 \overset{1^5}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.32 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 248 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [223] $1^2 \overset{1^5}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.89 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 324 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$
- [224] $21^3 \overset{2^1 1^4}{5} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.84 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 212 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$$

[225] $21 \overset{1^2}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 324 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$

[227] $1 \overset{1^5}{2} \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 284 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$

[226] $1 \overset{1^4}{2} \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.05 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 212 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$

[229] $1 \overset{1^4}{2} \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.28 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 246 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$

[228] $21 \overset{2^1 1^4}{4} 5 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.28 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 246 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$

[231] $1 \overset{1^3}{3} \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.03 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$

[230] $21 \overset{2^1 1^4}{3} 5 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.53 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 320 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$

[232] $21 \overset{1^5}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.85 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 450 & 0 & 38 \\ 1 & 0 & -1 & -1 \\ 2 & 38 & -1 & -3 \end{pmatrix}_{b_0}$

[233] $221 \overset{1^5}{3} 51^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.03 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$

[234] $1 \overset{1^5}{3} \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -3 \end{pmatrix}_{b_0}$

[235] $21 \overset{2^1 1^4}{3} 5 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.03 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$

[236] $21 \overset{1^3}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[237] \ 1^3 \overset{15}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.53 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 282 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[238] \ 221 \overset{5}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.85 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 404 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[239] \ 1^5 \overset{15}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.85 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 450 & 0 & 38 \\ 1 & 0 & -1 & -1 \\ 2 & 38 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[244] \ 1^3 \overset{14}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[240] \ 1^4 \overset{15}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.45 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 404 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[241] \ 21 \overset{14}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.45 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 404 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[242] \ 1^3 \overset{15}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 320 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[243] \ 21 \overset{21}{3} \overset{14}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 280 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^4 \overset{15}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 360 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[247] \ 21 \overset{21}{4} \overset{14}{5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 360 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[248] \ 21 \overset{21}{4} \overset{14}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 280 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$$

[245] $1^3 \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.49 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 280 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$

[251] $1^4 \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -82$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.76 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$

[250] $21^4 \overset{2^1 1^4}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -82$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.76 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$

[249] $221^4 \overset{2^1 1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -82$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.76 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$

[252] $1^1 \overset{1^3}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -98$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 158 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$

[257] $1^2 \overset{2^1 1^5}{5} 1^4 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 260 & 19 & 28 \\ 1 & 19 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$

[256] $221^2 \overset{2^1 1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 252 & 19 & 28 \\ 1 & 19 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$

[255] $21^2 \overset{2^1 1^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 252 & 19 & 28 \\ 1 & 19 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$

[254] $1^2 \overset{1^2}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$

[253] $1^4 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$

[259] $1^1 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -101$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 188 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$

[258] $1^5 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -101$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.80 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[262] \ 1^1 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -102, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.63 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 222 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[260] \ 1^2 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -102, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.98 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 298 & 0 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[261] \ 1^1 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -102, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.98 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 298 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[264] \ 1^1 \overset{1^3}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -103, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.48 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 186 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[263] \ 21^3 \overset{2^1}{5} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -103, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.48 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 254 & 19 & 28 \\ 1 & 19 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[265] \ 1^2 \overset{1^3}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -105, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.47 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[266] \ 21^2 \overset{2^1}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -105, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.47 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 286 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[267] \ 1^1 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -106, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.60 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 218 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[268] \ 1^1 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -107, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.76 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 254 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[270] \ 1^2 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -108, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.59 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 216 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[269] \ 21^2 \overset{2^1}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -108, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.59 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 324 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

- [274] $21^{2^{1^5}5^1} + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 366 & 22 & 34 \\ 1 & 22 & 1 & 1 \\ 2 & 34 & 1 & -1 \end{pmatrix}_{b_0}$
- [273] $21^{4^{1^5}5^1} + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 290 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$
- [272] $1^2 \overset{1^5}{5} 1^5 2^1 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 252 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$
- [271] $1^1 \overset{1^4}{5} 1^5 2^1 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 252 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$
- [277] $1^1 \overset{1^5}{5} 1^5 2^1 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 290 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$
- [276] $21^{3^{1^5}5^1} + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.58 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 286 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$
- [275] $1^2 \overset{1^3}{5} 1^5 2^1 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.58 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 214 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$
- [279] $1^1 \overset{1^5}{5} 1^5 2^1 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 330 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$
- [278] $21^{5^{1^5}5^1} + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 330 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$
- [280] $1^2 \overset{1^4}{5} 1^5 2^1 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -113$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 248 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$
- [281] $1^2 \overset{1^5}{5} 1^5 2^1 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -114$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.88 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 286 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$
- [284] $1^3 \overset{1^3}{5} 1^5 2^1 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -115$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 246 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

[282] $21 \overset{2^1 1^5}{3} 5 \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 322 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

[283] $221 \overset{3^1 1^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 322 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

[286] $1 \overset{1^4}{2} \overset{1^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 284 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

[285] $21 \overset{2^1 1^5}{4} 5 \overset{1^2}{1^2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 324 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

[287] $1 \overset{1^5}{2} \overset{1^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.03 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 324 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

[291] $21 \overset{2^1 1^5}{5} 5 \overset{1^2}{1^2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 366 & 22 & 34 \\ 1 & 22 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$$

[288] $1 \overset{1^4}{3} \overset{1^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.85 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 282 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

[289] $21 \overset{3^1 1^5}{3} 5 \overset{1^4}{1^4} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.85 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 362 & 22 & 34 \\ 1 & 22 & 1 & 1 \\ 2 & 34 & 1 & -1 \end{pmatrix}_{b_0}$$

[290] $1 \overset{1^5}{2} \overset{1^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 366 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -2 \end{pmatrix}_{b_0}$$

[292] $1 \overset{1^5}{3} \overset{1^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.02 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 322 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

[293] $21 \overset{2^1 1^5}{3} 5 \overset{1^5}{1^5} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.02 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 406 & 23 & 36 \\ 1 & 23 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$$

[295] $1 \overset{1^4}{3} \overset{1^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 320 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[294] \ 21^4 \overset{21^5}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 362 & 22 & 34 \\ 1 & 22 & 1 & 1 \\ 2 & 34 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[296] \ 1^3 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[297] \ 1^3 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 406 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[298] \ 21^5 \overset{21^5}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 406 & 23 & 36 \\ 1 & 23 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[299] \ 221^4 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 404 & 23 & 36 \\ 1 & 23 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[300] \ 1^4 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 360 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[301] \ 21^4 \overset{21^5}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 404 & 23 & 36 \\ 1 & 23 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[302] \ 1^4 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.35 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 404 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[303] \ 21^4 \overset{21^5}{5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.35 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 450 & 24 & 38 \\ 1 & 24 & 1 & 1 \\ 2 & 38 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[304] \ 21^5 \overset{21^5}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.56 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 450 & 24 & 38 \\ 1 & 24 & 1 & 1 \\ 2 & 38 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[305] \ 1^4 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.56 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 450 & 0 & 38 \\ 1 & 0 & -1 & -1 \\ 2 & 38 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[306] \ 1 \overset{1^5}{5} 1 \overset{5}{2} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.77 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 25 & 1 & 2 \\ 25 & 498 & 0 & 40 \\ 1 & 0 & -1 & -1 \\ 2 & 40 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[307] \ 21 \overset{2^1}{5} 5 \overset{1^5}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.77 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 25 & 1 & 2 \\ 25 & 498 & 25 & 40 \\ 1 & 25 & 1 & 1 \\ 2 & 40 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[308] \ 221 \overset{5}{5} 1 \overset{5}{2} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.77 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 25 & 1 & 2 \\ 25 & 498 & 25 & 40 \\ 1 & 25 & 1 & 1 \\ 2 & 40 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [312] \ 1 \overset{2^1}{6} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[309] \ 1 \overset{2^1}{6} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[310] \ 221 \overset{2^1}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 20 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[311] \ 1 \overset{2^1}{6} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 20 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[318] \ 21 \overset{2^1}{6} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 44 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[317] \ 21 \overset{2^1}{6} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[316] \ 313151 \overset{1^2}{2} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 16, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[315] \ 1 \overset{2^1}{6} 122 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[314] \ 222161 \overset{1^2}{2} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

- [313] $21 \begin{smallmatrix} 2^1 1^1 \\ 1^2 \end{smallmatrix} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 44 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -4 \end{pmatrix}_{b_0}$
- [319] $21 \begin{smallmatrix} 2^1 1^1 \\ 1^2 \end{smallmatrix} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.80 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 30 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$
- [320] $1 \begin{smallmatrix} 2^1 2 \\ 2^1 1^1 \end{smallmatrix} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 30 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -5 \end{pmatrix}_{b_0}$
- [321] $1 \begin{smallmatrix} 2^1 1^1 \\ 1^1 \end{smallmatrix} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.20 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0}$
- [322] $221 \begin{smallmatrix} 2^1 1^1 \\ 1^1 \end{smallmatrix} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.20 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0}$
- [323] $21 \begin{smallmatrix} 2^1 1^1 \\ 1^1 \end{smallmatrix} 6 \ 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0}$
- [324] $21 \begin{smallmatrix} 2^1 1^1 \\ 1^1 \end{smallmatrix} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0}$
- [325] $1 \begin{smallmatrix} 2^1 1^1 \\ 2^1 1^1 \end{smallmatrix} 6 \ 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0}$
- [326] $1 \begin{smallmatrix} 2^1 1^1 \\ 2^1 1^1 \end{smallmatrix} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.67 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 30 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -3 \end{pmatrix}_{b_0}$
- [329] $1 \begin{smallmatrix} 2^1 1^1 \\ 2^1 1^1 \end{smallmatrix} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.29 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$
- [327] $221 \begin{smallmatrix} 2^1 1^1 \\ 1^1 \end{smallmatrix} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.57 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 42 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -4 \end{pmatrix}_{b_0}$
- [328] $1 \begin{smallmatrix} 2^1 1^1 \\ 2^1 1^1 \end{smallmatrix} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.29 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$$

[330] $21^1 \overset{2^1 1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.57 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 42 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -4 \end{pmatrix}_{b_0}$$

[331] $21^1 \overset{2^1 1^1}{6} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

[332] $221^1 \overset{2^1 1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

[333] $21^1 \overset{2^1 1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

[334] $2221^1 \overset{1^1 1^1}{6} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

[342] $21^3 \overset{1^1 1^2}{6} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.25 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -3 \end{pmatrix}_{b_0}$$

[335] $21^1 \overset{2^1 2^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 80 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$

[336] $1^1 \overset{2^1 1^2}{6} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[337] $21^1 \overset{2^1 1^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$$

[338] $21^1 \overset{2^1 1^2}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.25 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$$

[339] $21^3 \overset{2^1 1^2}{6} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.25 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$$

[340] $1^1 \overset{2^1 1^2}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 17.25 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[343] \ 31 \overset{211}{5} \ 13 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 16, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[341] \ 21 \overset{212}{1} \overset{1}{6} \ 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[344] \ 21 \overset{212}{1} \overset{1}{2} \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.53 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 60 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[345] \ 21 \overset{212}{1} \overset{1}{3} \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.47 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 100 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[346] \ 21 \overset{212}{1} \overset{1}{2} \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.89 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[347] \ 1 \overset{212}{211} \overset{1}{6} \ 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.89 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[348] \ 1 \overset{212}{211} \overset{1}{6} \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 126 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[349] \ 21 \overset{212}{2} \overset{1}{6} \ 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.89 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[354] \ 1 \overset{213}{2112} \overset{1}{6} \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.26 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[358] \ 1 \overset{221}{6} \ 1^2 \overset{1}{2} \overset{1}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.47 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[357] \ 21 \overset{212}{1} \overset{1}{4} \overset{1}{6} \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.26 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -5 \end{pmatrix}_{b_0}$$

- [356] $221^{21^2} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.58 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$
- [355] $1^{21^2} 6^{1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.47 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 30 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$
- [350] $1^{21^2} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.32 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 62 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$
- [351] $1^{221^3} 6^{1^2} 1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.26 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -5 \end{pmatrix}_{b_0}$
- [352] $1^{21^3} 6^{1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.58 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 80 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -3 \end{pmatrix}_{b_0}$
- [353] $21^{1^2} 1^6 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.58 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$
- [361] $21^{1^2} 1^6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.20 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$
- [360] $1^{21^3} 6^{1^2} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.20 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 100 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -4 \end{pmatrix}_{b_0}$
- [359] $1^{21^2} 1^6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.80 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 44 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$
- [362] $1^{21^2} 1^6 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.33 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 100 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$
- [364] $21^{2^2} 1^6 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.86 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 148 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -4 \end{pmatrix}_{b_0}$
- [363] $1^{21^2} 1^6 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.19 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 60 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[372] \quad 1 \overset{21^2}{\underset{21^2}{6}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[369] \quad 21 \overset{21^2}{\underset{21^2}{1}} 6 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 42 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[370] \quad 21 \overset{21^2}{\underset{1^1}{1}} 6 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 42 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[371] \quad 221 \overset{21^2}{\underset{3^1}{2}} 6 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.55 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 146 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[367] \quad 2221 \overset{1}{\underset{1}{2}} 6 \overset{2}{\underset{2}{1}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 58 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[365] \quad 21 \overset{21^2}{\underset{1^4}{2}} 6 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.55 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 174 & 14 & 24 \\ 1 & 14 & 1 & 1 \\ 2 & 24 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[373] \quad 21 \overset{21^2}{\underset{1^3}{1}} 6 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[374] \quad 221 \overset{21^2}{\underset{1^3}{1}} 6 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 98 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[375] \quad 21 \overset{21^2}{\underset{1^3}{3}} 6 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.55 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 146 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[376] \quad 1 \overset{221^2}{\underset{1}{6}} 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[377] \quad 21 \overset{21^3}{\underset{1^3}{1}} 6 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.91 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$$

- [378] $221^{2^{1^2} 6^1 1^4} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.55 \rightarrow 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 174 & 14 & 24 \\ 1 & 14 & 1 & 1 \\ 2 & 24 & 1 & -5 \end{pmatrix}_{b_0}$
- [379] $221^{1^{2^1} 6^1 1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.73 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 58 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -1 \end{pmatrix}_{b_0}$
- [368] $21^{1^{2^1} 6^1 1^3} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.91 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 122 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -3 \end{pmatrix}_{b_0}$
- [366] $21^{3^{1^2} 6^1 1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.91 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$
- [382] $21^{3^{1^2} 6^1 1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 23$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.52 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 120 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -4 \end{pmatrix}_{b_0}$
- [383] $21^{1^{2^1} 6^1 1^2} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 23$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.13 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 98 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$
- [381] $21^{1^{2^1} 6^1 1^2} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 23$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.09 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 58 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$
- [380] $221^{1^{2^1} 6^1 1^2} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 23$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.09 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 76 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$
- [384] $21^{2^{1^2} 6^1 1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.67 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 120 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$
- [386] $21^{2^1 6^1 1^{2^2}} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.17 \rightarrow 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 144 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -5 \end{pmatrix}_{b_0}$
- [387] $21^{3^{1^2} 6^1 1^2} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.17 \rightarrow 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 144 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -5 \end{pmatrix}_{b_0}$
- [388] $21^{2^{1^2} 6^1 1^3} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.17 \rightarrow 20$, $T_{\max} = 20$, $T \in [20, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 172 & 14 & 24 \\ 1 & 14 & 1 & 1 \\ 2 & 24 & 1 & -5 \end{pmatrix}_{b_0}$$

[389] $21^2 \overset{21^2}{6} \overset{1^1}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 76 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$

[390] $21^2 \overset{21^2}{6} \overset{1^2}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 76 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$$

[385] $21^1 \overset{21^2}{6} \overset{1^{22^1}}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 76 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$

[394] $221^2 \overset{21^2}{6} \overset{1^1}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.96 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 96 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -2 \end{pmatrix}_{b_0}$$

[393] $21^2 \overset{21^2}{6} \overset{1^1}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.96 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 96 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -2 \end{pmatrix}_{b_0}$$

[392] $221^2 \overset{21^2}{6} \overset{1^3}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 18.24 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 144 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -4 \end{pmatrix}_{b_0}$$

[391] $21^2 \overset{21^2}{6} \overset{1^2}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 18.24 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 144 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -4 \end{pmatrix}_{b_0}$$

[398] $21^2 \overset{21^2}{6} \overset{1^2}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 17.46 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 118 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$$

[395] $2221^2 \overset{21^2}{6} \overset{1^2}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 17.46 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 118 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$$

[396] $21^2 \overset{21^2}{6} \overset{1^{22^1}}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 17.46 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 118 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$$

[397] $221^2 \overset{21^2}{6} \overset{1^2}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 17.46 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 118 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$$

[399] $1 \overset{21^3}{6} \overset{1^1}{1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 46 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[400] \quad 1 \overset{221^1}{6} 1^{32^1} + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 82 & 11 & 16 \\ 1 & 11 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[405] \quad 1 \overset{21^4}{6} 1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[401] \quad 21 \overset{21^4}{6} 12 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[402] \quad 1 \overset{21^1}{6} 1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[403] \quad 21 \overset{21^1}{6} 1^2 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[404] \quad 21 \overset{21^1}{6} 12 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[406] \quad 21 \overset{21^1}{6} 1^4 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[407] \quad 21 \overset{21^1}{6} 1^2 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 180 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[408] \quad 21 \overset{21^1}{6} 1^3 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.32 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 210 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[409] \quad 21 \overset{21^1}{6} 1^2 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 150 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[414] \quad 21 \overset{21^1}{6} 1^3 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

- [410] $1 \overset{21^3}{\underset{21^1}{6}} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [411] $21 \overset{21^3}{\underset{21^1}{3}} 6 \ 12 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$
- [412] $1 \overset{21^3}{\underset{21^1}{6}} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.42 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 62 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [413] $1 \overset{21^3}{\underset{21^1}{6}} 1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [415] $1 \overset{21^3}{\underset{21^1}{6}} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 82 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [419] $21 \overset{21^3}{\underset{1^1}{1}} 6 \ 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 60 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [418] $221 \overset{21^3}{\underset{1^1}{1}} 6 \ 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 100 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [417] $21 \overset{21^3}{\underset{1^1}{1}} 6 \ 1^1 2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 60 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [416] $2221 \overset{21^3}{\underset{1^1}{1}} 6 \ 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 100 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [421] $1 \overset{21^4}{\underset{21^3}{6}} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.98 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 154 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$
- [420] $1 \overset{21^3}{\underset{21^2}{6}} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.61 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 80 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [423] $221 \overset{21^3}{\underset{1^1}{1}} 6 \ 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.58 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 122 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[422] \quad 1 \overset{21^3}{\underset{21^3}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 126 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[424] \quad 21 \overset{21^3}{\underset{1^2}{1}} \overset{1^3}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.58 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 78 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[425] \quad 21 \overset{21^3}{\underset{1^4}{1}} \overset{1^3}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[426] \quad 221 \overset{21^3}{\underset{1^4}{1}} \overset{1^3}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[430] \quad 1 \overset{21^3}{\underset{21^2}{6}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 100 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[431] \quad 1 \overset{21^4}{\underset{21^3}{6}} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.27 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[432] \quad 221 \overset{21^3}{\underset{1^3}{1}} \overset{1^3}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 148 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[433] \quad 1 \overset{221^2}{\underset{1^3}{6}} 1^{321} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[434] \quad 1 \overset{221^4}{\underset{1^3}{6}} 1^{321} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.27 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[429] \quad 1 \overset{21^4}{\underset{21^3}{6}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.27 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 180 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[427] \quad 21 \overset{21^3}{\underset{1^3}{1}} \overset{1^3}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 100 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

- [428] $1 \begin{smallmatrix} 2^1 1^3 \\ 2^1 2^2 \end{smallmatrix} 6 \begin{smallmatrix} 1^4 \\ 1^3 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [435] $1 \begin{smallmatrix} 2^1 1^4 \\ 2^1 1^3 \end{smallmatrix} 6 \begin{smallmatrix} 1^3 \\ 1^2 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.69 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 210 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -3 \end{pmatrix}_{b_0}$
- [436] $1 \begin{smallmatrix} 2^1 1^3 \\ 2^1 2^2 \end{smallmatrix} 6 \begin{smallmatrix} 1^3 \\ 1^2 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.09 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$
- [440] $21 \begin{smallmatrix} 2^1 1^4 \\ 1^3 2^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^3 2^1 \\ 1^3 2^1 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.17 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -3 \end{pmatrix}_{b_0}$
- [438] $21 \begin{smallmatrix} 2^1 1^3 \\ 1^4 \end{smallmatrix} 6 \begin{smallmatrix} 1^4 \\ 1^4 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.17 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$
- [437] $1 \begin{smallmatrix} 2^1 1^3 \\ 2^1 1^3 \end{smallmatrix} 6 \begin{smallmatrix} 1^4 \\ 1^4 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.17 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 210 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -3 \end{pmatrix}_{b_0}$
- [445] $1 \begin{smallmatrix} 2^1 1^3 \\ 2^1 1^3 \end{smallmatrix} 6 \begin{smallmatrix} 1^2 \\ 1^2 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.39 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 150 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$
- [444] $21 \begin{smallmatrix} 2^1 1^3 \\ 1^2 2^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^2 2^1 \\ 1^2 2^1 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [443] $21 \begin{smallmatrix} 2^1 1^3 \\ 1^2 2^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^2 \\ 1^2 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [442] $21 \begin{smallmatrix} 2^1 1^3 \\ 1^1 2^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [439] $21 \begin{smallmatrix} 2^1 1^3 \\ 1^1 2^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.17 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -3 \end{pmatrix}_{b_0}$
- [441] $21 \begin{smallmatrix} 2^1 1^3 \\ 1^4 \end{smallmatrix} 6 \begin{smallmatrix} 1^2 \\ 1^4 \end{smallmatrix} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.39 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[446] 21 \overset{2^1 1^3}{1} \overset{1^2}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.04 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 122 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[447] 21 \overset{2^1 1^3}{1} \overset{1^3}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.72 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[448] 1 \overset{2^1 1^3}{2} \overset{1^3}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.72 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[449] 1 \overset{22^1 3}{6} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.72 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[451] 21 \overset{2^1 1^3}{3} \overset{1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 148 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[452] 21 \overset{2^1 3}{1} \overset{1^3}{6} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 148 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[450] 21 \overset{2^1 1^3}{1} \overset{1^3}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 148 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[453] 22^1 \overset{2^1 1^3}{2} \overset{1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 146 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[458] 21 \overset{2^1 1^3}{2} \overset{1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 120 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[455] 21 \overset{2^1 1^3}{4} \overset{1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.47 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 206 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[454] 21 \overset{2^1 1^3}{2} \overset{1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.65 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 176 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

[456] $221^4 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.45 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 274 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -4 \end{pmatrix}_{b_0}$$

[457] $21^4 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.45 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 274 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -4 \end{pmatrix}_{b_0}$$

[460] $21^4 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.88 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 238 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$$

[459] $21^2 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.28 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 146 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

[462] $21^3 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.37 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 238 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$$

[461] $21^3 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.59 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 174 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

[472] $21^2 6^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.23 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 144 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

[471] $21^4 6^1 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.92 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 21 & 36 \\ 1 & 21 & 1 & 1 \\ 2 & 36 & 1 & -5 \end{pmatrix}_{b_0}$$

[469] $2221^4 6^1 3^2 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.92 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 21 & 36 \\ 1 & 21 & 1 & 1 \\ 2 & 36 & 1 & -5 \end{pmatrix}_{b_0}$$

[468] $21^2 6^1 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.77 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$$

[467] $221^2 6^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.23 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 172 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

[463] $221^2 6^1 1^4 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.92 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 236 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[466] \ 221^2 6^1 3^2 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.23 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 172 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[473] \ 221^4 6^1 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.92 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 21 & 36 \\ 1 & 21 & 1 & 1 \\ 2 & 36 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[464] \ 21^2 6^1 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.77 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 272 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[465] \ 221^4 6^1 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.77 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 308 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[474] \ 221^3 6^1 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.92 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 204 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[470] \ 21^2 6^1 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.23 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 144 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[475] \ 21^4 6^1 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.92 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 21 & 36 \\ 1 & 21 & 1 & 1 \\ 2 & 36 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[476] \ 21^3 6^1 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.92 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 204 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[477] \ 21^4 6^1 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.77 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[478] \ 21^4 6^1 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.77 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 308 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[479] \ 21^2 6^1 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.92 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 204 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$$

- [484] $221^{\overline{2}1^3} \overline{6}^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.53 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 202 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$
- [485] $21^{\overline{4}1^3} \overline{6}^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.19 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 270 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -4 \end{pmatrix}_{b_0}$
- [483] $221^{\overline{4}1^3} \overline{6}^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.25 \rightarrow 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 346 & 20 & 34 \\ 1 & 20 & 1 & 1 \\ 2 & 34 & 1 & -5 \end{pmatrix}_{b_0}$
- [482] $21^{\overline{4}1^3} \overline{6}^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.25 \rightarrow 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 346 & 20 & 34 \\ 1 & 20 & 1 & 1 \\ 2 & 34 & 1 & -5 \end{pmatrix}_{b_0}$
- [481] $21^{\overline{2}1^3} \overline{6}^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.53 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 172 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$
- [480] $21^{\overline{2}1^3} \overline{6}^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.28 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 236 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$
- [486] $21^{\overline{4}1^3} \overline{6}^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.63 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 306 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$
- [487] $21^{\overline{3}1^3} \overline{6}^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.67 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 270 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$
- [488] $21^{\overline{3}1^3} \overline{6}^1 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.63 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 346 & 20 & 34 \\ 1 & 20 & 1 & 1 \\ 2 & 34 & 1 & -4 \end{pmatrix}_{b_0}$
- [489] $21^{\overline{2}1^3} \overline{6}^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.85 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 202 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [490] $21^{\overline{3}1^4} \overline{6}^1 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.63 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 306 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$
- [491] $21^{\overline{3}1^3} \overline{6}^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.85 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 202 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

[492] $21^{21^3} 6 1^{32^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.85 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 202 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$

[493] $221^{32^1} 6 1^4 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.07 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 306 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$

[494] $21^{32^1} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.20 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 234 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$

[495] $221^{32^1} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.20 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 234 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$

[496] $21^{32^1} 6 1^3 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.07 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 306 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$

[497] $21^{32^1} 6 1^3 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.57 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 268 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$

[498] $221^{32^1} 6 1^3 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.57 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 268 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$

[500] $2221^{361} 32^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.57 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 268 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$

[499] $21^{32^1} 6 1^{32^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.57 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 268 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$

[501] $1^{21^4} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.27 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$

[504] $21^{12^1} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 82 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$

[503] $221^{12^1} 6 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 154 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

[502] $21 \overset{2^1 1^4}{1} \overset{1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 82 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[505] $2221 \overset{1^1 1^4}{1} \overset{1^1}{6} \overset{1^1}{2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 154 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

[506] $1 \overset{2^1 1^4}{21} \overset{1^1}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

[510] $1 \overset{2^1 1^4}{21} \overset{1^1}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.56 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[509] $21 \overset{2^1 1^4}{1^3} \overset{1^1}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.56 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

[508] $1 \overset{2^1 1^4}{21^3} \overset{1^1}{6} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.56 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[507] $21 \overset{2^1 1^4}{3} \overset{1^1}{6} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.56 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

[512] $1 \overset{2^1 1^4}{21^2} \overset{1^1}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.38 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 104 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

[511] $21 \overset{2^1 1^4}{1^4} \overset{1^1}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.75 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 280 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

[515] $21 \overset{2^1 1^4}{1^4} \overset{1^1}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.97 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

[514] $1 \overset{2^1 1^4}{21^4} \overset{1^1}{6} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.97 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

[513] $21^4 \overset{2^1 1^4}{6} 12 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.97 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$

[517] $221^1 \overset{2^1 1^4}{6} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.37 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 180 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$

[516] $21^1 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.37 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$

[521] $1 \overset{2^1 1^4}{21^2} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.51 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$

[519] $221^1 \overset{2^1 1^4}{6} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.51 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 210 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$

[518] $21^1 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.51 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$

[520] $1 \overset{221^2}{6} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.51 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 182 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$

[523] $1 \overset{2^1 1^4}{21^3} 6 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.69 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 154 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$

[522] $221^1 \overset{2^1 1^4}{6} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.69 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$

[524] $21^1 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.69 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 154 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$

[525] $21^1 \overset{2^1 1^4}{6} 1^2 2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$

[526] $21^2 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[527] \ 21 \overset{2^1 1^4}{1^2} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 72, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.50 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[528] \ 21 \overset{2^1 1^4}{1^2} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.11 \rightarrow 17, \ T_{\max} = 20, \ T \in [17, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 216 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[529] \ 1 \overset{2^1 1^4}{21^2} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.67 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[530] \ 1 \overset{2^1 1^4}{21^2} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 74, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.86 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 182 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[531] \ 21 \overset{2^1 1^4}{1^3} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.65 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[532] \ 1 \overset{2^1 1^4}{21^3} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.84 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[533] \ 21 \overset{2^1 1^4}{1^4} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.84 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[535] \ 21 \overset{2^1 1^4}{1^2} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 77, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 148 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[534] \ 221 \overset{2^1 1^4}{2^2} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 77, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 208 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[539] \ 21 \overset{2^1 1^4}{1^1} 3^2 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 78, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.82 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[538] \ 21 \overset{2^1 1^4}{1^3} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 78, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.82 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

- [537] $21^{21^4}_1 6^1 1^3 2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [536] $21^{21^4}_1 6^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.28 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 246 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$
- [542] $21^{21^4}_{21^3} 6^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.03 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [541] $221^{21^3}_1 6^1 1^{42^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.03 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$
- [540] $21^{21^4}_1 6^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.03 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [546] $21^{21^4}_1 6^1 1^{42^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$
- [547] $21^{21^4}_{1^3} 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.80 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 176 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [545] $21^{21^4}_{21^3} 6^1 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [544] $21^{21^4}_{1^1} 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$
- [543] $21^{21^4}_{1^4} 6^1 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [548] $21^{21^4}_{21^4} 6^1 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.49 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 280 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$
- [549] $21^{21^4}_{1^4} 6^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 208 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[552] \ 21^2 \overset{2^1 1^4}{6} \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 174 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[551] \ 221^2 \overset{2^1 1^4}{6} \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 238 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[550] \ 1 \overset{2^1 1^4}{6} \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.76 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[554] \ 2221^2 \overset{2^1 1^4}{6} \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 238 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[555] \ 21^2 \overset{2^1 1^4}{6} \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 174 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[553] \ 1 \overset{2^1 1^4}{6} \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.76 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[556] \ 21^3 \overset{2^1 1^4}{6} \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 206 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[559] \ 221^2 \overset{2^1 1^4}{6} \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 272 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[557] \ 21^2 \overset{2^1 1^4}{6} \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 204 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[558] \ 21^4 \overset{2^1 1^4}{6} \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.42 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 276 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[563] \ 221^3 \overset{2^1 1^4}{6} \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.16 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 274 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[562] \ 221^2 \overset{2^1 1^4}{6} \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.16 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 310 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[561] \ 21 \overset{2^1 1^4}{\underset{1^3}{3}} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.16 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 238 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[560] \ 21 \overset{2^1 1^4}{\underset{1^4}{2}} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.16 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 238 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[564] \ 21 \overset{2^1 1^4}{\underset{1^4}{3}} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.39 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 274 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[566] \ 21 \overset{2^1 1^4}{\underset{1^3}{4}} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 312 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[567] \ 21 \overset{2^1 1^4}{\underset{1^3}{2}} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[568] \ 21 \overset{2^1 1^4}{\underset{1^2}{3}} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[565] \ 21 \overset{2^1 1^4}{\underset{1^4}{2}} 6 \ 1^3 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[569] \ 221 \overset{2^1 1^4}{\underset{1^4}{4}} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.90 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 352 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[570] \ 21 \overset{2^1 1^4}{\underset{1^4}{2}} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.36 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 272 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[571] \ 21 \overset{2^1 1^4}{\underset{1^4}{4}} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.90 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 352 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[572] \ 21 \overset{2^1 1^4}{\underset{1^4}{2}} 6 \ 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.60 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 310 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

[573] $21^4 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.60 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 310 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

[574] $21^2 \overset{2^1 1^4}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.60 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 310 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

[575] $221^3 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 308 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

[576] $21^3 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 270 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

[577] $21^3 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 308 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

[578] $21^4 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.82 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 348 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$$

[579] $21^3 \overset{2^1 1^4}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.53 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 306 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

[580] $21^4 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.09 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 390 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -3 \end{pmatrix}_{b_0}$$

[581] $2221^3 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.53 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 346 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$$

[582] $21^3 \overset{2^1 1^4}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.53 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 306 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

[583] $221^4 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.09 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 390 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -3 \end{pmatrix}_{b_0}$$

[584] $221^3 \overset{2^1 1^4}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16.53 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 346 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[585] \ 221 \overset{2^1 1^4}{3} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 388 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[586] \ 21 \overset{2^1 1^4}{3} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 346 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[589] \ 21 \overset{2^1 4}{3} 6 \ 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.04 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[588] \ 21 \overset{2^1 1^4}{3} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.04 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[587] \ 21 \overset{2^1 1^4}{4} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.04 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[591] \ 221 \overset{2^1 1^4}{4} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.32 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 432 & 23 & 38 \\ 1 & 23 & 1 & 1 \\ 2 & 38 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[590] \ 21 \overset{2^1 1^4}{4} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.32 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 432 & 23 & 38 \\ 1 & 23 & 1 & 1 \\ 2 & 38 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[594] \ 221 \overset{2^1 1^4}{4} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.61 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 478 & 24 & 40 \\ 1 & 24 & 1 & 1 \\ 2 & 40 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[593] \ 21 \overset{2^1 4}{4} 6 \ 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.61 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 478 & 24 & 40 \\ 1 & 24 & 1 & 1 \\ 2 & 40 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[592] \ 21 \overset{2^1 1^4}{4} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.61 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 478 & 24 & 40 \\ 1 & 24 & 1 & 1 \\ 2 & 40 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[595] \ 2221 \overset{4^1 4^1}{4} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.61 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 478 & 24 & 40 \\ 1 & 24 & 1 & 1 \\ 2 & 40 & 1 & -3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[610] \ 21 \overset{2^1 1^1}{1} 7 \ 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.29 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[609] \ 21 \overset{221^1}{7} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.29 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[608] \ 1 \overset{221^1}{2^1 1^1} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.29 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[613] \ 1 \overset{221^1}{2^1 1^1} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[612] \ 1 \overset{2221^1}{7} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[611] \ 1 \overset{21^1}{7} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[617] \ 21 \overset{21^1}{1} 7 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[614] \ 21 \overset{221^1}{1} 7 \ 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[618] \ 2221 \overset{2^1 1^1}{1} 7 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[615] \ 221 \overset{2^1 1^1}{1} 7 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[616] \ 22221 \overset{2^1 1^1}{7} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[619] \ 1 \overset{2^1 1^1}{6} 13 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[620] \quad 321 \quad 6 \quad 13 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [622] \quad 21 \quad 8 \quad 12 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[623] \quad 221 \quad 8 \quad 122 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[624] \quad 21 \quad 8 \quad 1222 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[625] \quad 1 \quad 8 \quad 12222 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[626] \quad 22222181 \quad 2^1 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[627] \quad 1 \quad 8 \quad 12 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[621] \quad 1 \quad 8 \quad 1 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[628] \quad 321 \quad 7 \quad 123 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\hat{1}} \rightarrow 2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

1.18 $\mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{\textcolor{red}{11}} 1^{\textcolor{green}{1}} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 + \textcolor{green}{\hat{1}} \rightarrow 1 \quad T = 4, T_{\max} = 99, \Delta = -211, \det = 8, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{11}} 1^{\textcolor{green}{1}} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 + \textcolor{green}{\hat{1}} \rightarrow 1 \quad T = 4, T_{\max} = 99, \Delta = -211, \det = 11, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -4 & 0 & -2 \\ 2 & 0 & 1 & 1 \\ 3 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -182, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -182, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1^{11}221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -153, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 1^{11}2221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 80, \Delta = -124, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 80, T \in [16, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -2 & 3 & -3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^1 1^1 3221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

1.19 $su_2 + so_8 + so_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{green}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -210, \det = -40, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{green}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -181, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

1.20 $su_2 + su_2 + e_6 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^{21} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 100, \Delta = -231, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 100, \Delta = -231, \det = -24, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -231$, $\det = -28$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -231$, $\det = -52$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$T = 6$ [4] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -202$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

[5] $1^{11}1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -202$, $\det = 16$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

1.21 $su_2 + su_2 + e_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 186$, $\Delta = -286$, $\det = -38$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 186$, $T \in [15, 186]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 169$, $\Delta = -257$, $\det = 8$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15$, $T_{\max} = 169$, $T \in [15, 169]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -6 & 0 & -4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

1.22 $su_2 + su_2 + e'_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}'_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 169$, $\Delta = -258$, $\det = -33$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.61 \rightarrow 14$, $T_{\max} = 169$, $T \in [14, 169]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}'_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 101$, $\Delta = -229$, $\det = 6$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -3 & -1 & -4 \end{pmatrix}_{b_0}$$

1.23 $su_2 + su_2 + f_4 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{21} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & -3 & 1 & -1 \end{pmatrix}_{b_0}$$

- [3] $1^{131}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -20$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 16 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [2] $1^{121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -20$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 6 & 4 & 0 & 4 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [1] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -20$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [4] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -23$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [5] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -43$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -63$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 5 & 2 & 1 & 3 \\ 5 & 6 & 4 & 0 & 4 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [7] $1^{131}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -83$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 7 & 2 & 1 & 3 \\ 7 & 16 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- $T = 6$ [9] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 99$, $\Delta = -176$, $\det = 2$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 & -4 \end{pmatrix}_{b_0}$$
- [8] $1^{121}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 99$, $\Delta = -176$, $\det = 2$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 1 & 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 4 & -1 & -1 & -4 \end{pmatrix}_{b_0}$$
- [10] $1^{111}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 99$, $\Delta = -176$, $\det = 12$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 99$, $T \in [13, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

1.24 $\textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow (-1)$

- $T = 4$ [0] $1^{11}1^2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1$ $T = 4$, $T_{\max} = 97$, $\Delta = -182$, $\det = 2$, $(n_+, n_-, n_0) = (1, 4, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & -2 & 0 & 0 \\ 2 & 0 & 1 & 1 \\ 3 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{15}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.1$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 10 & 1 & 3 \\ 10 & 48 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 3 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{14}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 30 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 3 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{17}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 14 & 1 & 3 \\ 14 & 96 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 3 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 16 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 3 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 6 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 3 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{16}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 12 & 1 & 3 \\ 12 & 70 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 3 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 5, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 1 & 1 \\ 3 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.22 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 13, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 6 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 3 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 15, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -2 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 3 & 0 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{13}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 17, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 16 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 3 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{14}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 21, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 30 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 3 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [14] \ 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -153, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 1 & 2 & 0 & 1 \\ 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -153, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 4 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -153, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -153, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 1 \\ 2 & 2 & 0 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [18] \ 1^{11}221^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 79, \Delta = -124, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 79, T \in [10, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

1.25 $\mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -180, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 1 & -1 & 0 \end{pmatrix}_{b_0}$$

1.26 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^{11}1^{21} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{15}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 10 & 2 & 1 & 3 \\ 10 & 48 & 10 & 0 & 10 \\ 2 & 10 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 6 & 4 & 0 & 4 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 16 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{14}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -16$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 11$, $T_{\max} = 98$, $T \in [11, 98]$

$$\begin{pmatrix} 1 & 8 & 2 & 1 & 3 \\ 8 & 30 & 8 & 0 & 8 \\ 2 & 8 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{16}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -16$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 11$, $T_{\max} = 98$, $T \in [11, 98]$

$$\begin{pmatrix} 1 & 12 & 2 & 1 & 3 \\ 12 & 70 & 12 & 0 & 12 \\ 2 & 12 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{17}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -16$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 11$, $T_{\max} = 98$, $T \in [11, 98]$

$$\begin{pmatrix} 1 & 14 & 2 & 1 & 3 \\ 14 & 96 & 14 & 0 & 14 \\ 2 & 14 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{11}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -18$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{11}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -34$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{12}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -50$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 11.04 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 5 & 2 & 1 & 3 \\ 5 & 6 & 4 & 0 & 4 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{13}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -66$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 11.03 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 7 & 2 & 1 & 3 \\ 7 & 16 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{14}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -82$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 11.02 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 9 & 2 & 1 & 3 \\ 9 & 30 & 8 & 0 & 8 \\ 2 & 8 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 6$ [13] $1^{11}1^{21}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 97$, $\Delta = -152$, $\det = 8$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.27 $\text{su}_2 + \text{su}_2 + \text{su}_2 + \text{e}_6 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{11}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{e}_6 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -201$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

1.28 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{121}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 99, \Delta = -175, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.7$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 4 & 0 & 6 & 4 & 0 & 4 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 4 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 99, \Delta = -175, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

1.29 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$

$T = 5$ [1] $1^{121}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$

$$\begin{pmatrix} 1 & 1 & 4 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 4 & 0 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[0] $1^{134}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$

$$\begin{pmatrix} 1 & 7 & 8 & 1 & 3 \\ 7 & 16 & 24 & 0 & 6 \\ 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{134}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$

$$\begin{pmatrix} 1 & 6 & 8 & 1 & 3 \\ 6 & 16 & 24 & 0 & 6 \\ 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{167}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.6$

$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$

$$\begin{pmatrix} 1 & 12 & 14 & 1 & 3 \\ 12 & 70 & 84 & 0 & 12 \\ 14 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{115}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.1$

$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$

$$\begin{pmatrix} 1 & 2 & 10 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{117}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.3$

$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$

$$\begin{pmatrix} 1 & 2 & 14 & 1 & 3 \\ 2 & 0 & 14 & 0 & 2 \\ 14 & 14 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{135}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.9$

$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 16 & 30 & 0 & 6 \\ 10 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{112}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$

- [6] $1^{126}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & -1 & 1 \end{pmatrix}_{b_0}$
- [24] $1^{156}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 6 & 24 & 0 & 4 \\ 12 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [25] $1^{144}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 8 & 8 & 1 & 3 \\ 8 & 30 & 32 & 0 & 8 \\ 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{123}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 6 & 12 & 0 & 4 \\ 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$
- [27] $1^{177}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 14 & 14 & 1 & 3 \\ 14 & 96 & 98 & 0 & 14 \\ 14 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
- [28] $1^{145}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 8 & 10 & 1 & 3 \\ 8 & 30 & 40 & 0 & 8 \\ 10 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$
- [29] $1^{147}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 8 & 14 & 1 & 3 \\ 8 & 30 & 56 & 0 & 8 \\ 14 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
- [30] $1^{133}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 6 & 6 & 1 & 3 \\ 6 & 16 & 18 & 0 & 6 \\ 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$
- [31] $1^{166}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 12 & 12 & 1 & 3 \\ 12 & 70 & 72 & 0 & 12 \\ 12 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [4] $1^{155}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 10 & 10 & 1 & 3 \\ 10 & 48 & 50 & 0 & 10 \\ 10 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{113}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 2 & 6 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$
- [22] $1^{157}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$

- [21] $1^{123}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.3$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 1 & 10 & 14 & 1 & 3 \\ 10 & 48 & 70 & 0 & 10 \\ 14 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 5 & 6 & 1 & 3 \\ 5 & 6 & 12 & 0 & 4 \\ 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$
- [19] $1^{136}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 6 & 12 & 1 & 3 \\ 6 & 16 & 36 & 0 & 6 \\ 12 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [18] $1^{124}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 6 & 16 & 0 & 4 \\ 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$
- [17] $1^{137}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 6 & 14 & 1 & 3 \\ 6 & 16 & 42 & 0 & 6 \\ 14 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{114}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & -1 & 1 \end{pmatrix}_{b_0}$
- [8] $1^{116}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 2 & 12 & 1 & 3 \\ 2 & 0 & 12 & 0 & 2 \\ 12 & 12 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [15] $1^{125}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.6$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 6 & 20 & 0 & 4 \\ 10 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$
- [9] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{146}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 8 & 12 & 1 & 3 \\ 8 & 30 & 48 & 0 & 8 \\ 12 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{127}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$
 $\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 6 & 28 & 0 & 4 \\ 14 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
- [33] $1^{144}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -10$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 1 & 8 & 9 & 1 & 3 \\ 8 & 30 & 32 & 0 & 8 \\ 9 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{112}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 0 & 4 & 0 & 2 \\ 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{133}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 6 & 7 & 1 & 3 \\ 6 & 16 & 18 & 0 & 6 \\ 7 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{122}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 6 & 8 & 0 & 4 \\ 5 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{134}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -24, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 6 & 9 & 1 & 3 \\ 6 & 16 & 24 & 0 & 6 \\ 9 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{112}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -24, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 5 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{123}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -26, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 6 & 12 & 0 & 4 \\ 7 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{113}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -32, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 7 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 7 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{124}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -34, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 6 & 16 & 0 & 4 \\ 9 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{114}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -40, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 9 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 9 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

1.30 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$T = 6 \quad [0] 1^{121}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -151, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 2 & 7 & 2 & 1 & 3 \\ 2 & 0 & 6 & 2 & 0 & 2 \\ 7 & 6 & 16 & 6 & 0 & 6 \\ 2 & 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{1141}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 97$, $\Delta = -151$, $\det = 156$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 11.03 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 1 & 2 & 9 & 2 & 1 & 3 \\ 2 & 0 & 8 & 2 & 0 & 2 \\ 9 & 8 & 30 & 8 & 0 & 8 \\ 2 & 2 & 8 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.31 $su_2 + su_2 + su_2 + su_2 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{1244}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12$, $T_{\max} = 70$, $T \in [12, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 9 & 1 & 3 \\ 4 & 6 & 16 & 16 & 0 & 4 \\ 8 & 16 & 30 & 32 & 0 & 8 \\ 9 & 16 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12$, $T_{\max} = 70$, $T \in [12, 70]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 1 & 3 \\ 3 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 12$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1123}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 12$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 5 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 5 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1334}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 12$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 9 & 1 & 3 \\ 6 & 16 & 18 & 24 & 0 & 6 \\ 6 & 18 & 16 & 24 & 0 & 6 \\ 9 & 24 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{1233}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 16$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 10$, $T_{\max} = 70$, $T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 1 & 3 \\ 4 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 12 & 0 & 4 \\ 6 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{1157}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 16$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 10$, $T_{\max} = 70$, $T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 10 & 14 & 1 & 3 \\ 2 & 0 & 10 & 14 & 0 & 2 \\ 10 & 10 & 48 & 70 & 0 & 10 \\ 14 & 14 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{1667}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 16$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10$, $T_{\max} = 70$, $T \in [10, 70]$

$$\begin{pmatrix} 1 & 12 & 12 & 14 & 1 & 3 \\ 12 & 70 & 72 & 84 & 0 & 12 \\ 12 & 72 & 70 & 84 & 0 & 12 \\ 14 & 84 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 12 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{1124}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 16$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 10$, $T_{\max} = 70$, $T \in [10, 70]$

- [43] $1^{11\overline{16}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 8 & 1 & 3 \\ 2 & 0 & 4 & 8 & 0 & 2 \\ 4 & 4 & 6 & 16 & 0 & 4 \\ 8 & 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [44] $1^{11\overline{47}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 14 & 1 & 3 \\ 2 & 0 & 8 & 14 & 0 & 2 \\ 8 & 8 & 30 & 56 & 0 & 8 \\ 14 & 14 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^{11\overline{23}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{11\overline{22}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [47] $1^{11\overline{56}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 10 & 12 & 1 & 3 \\ 2 & 0 & 10 & 12 & 0 & 2 \\ 10 & 10 & 48 & 60 & 0 & 10 \\ 12 & 12 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [48] $1^{12\overline{25}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 10 & 1 & 3 \\ 4 & 6 & 8 & 20 & 0 & 4 \\ 4 & 8 & 6 & 20 & 0 & 4 \\ 10 & 20 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [39] $1^{11\overline{55}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 10 & 10 & 1 & 3 \\ 2 & 0 & 10 & 10 & 0 & 2 \\ 10 & 10 & 48 & 50 & 0 & 10 \\ 10 & 10 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [38] $1^{12\overline{22}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 8 & 6 & 8 & 0 & 4 \\ 4 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [49] $1^{11\overline{12}}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 0 & 2 \\ 3 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [50] $1^{16\overline{77}}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 12 & 14 & 14 & 1 & 3 \\ 12 & 70 & 84 & 84 & 0 & 12 \\ 14 & 84 & 96 & 98 & 0 & 14 \\ 14 & 84 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 12 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{12\overline{33}}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

- [26] $1^{1125}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 7 & 1 & 3 \\ 4 & 6 & 12 & 12 & 0 & 4 \\ 6 & 12 & 16 & 18 & 0 & 6 \\ 7 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^{1166}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 10 & 1 & 3 \\ 2 & 0 & 4 & 10 & 0 & 2 \\ 4 & 4 & 6 & 20 & 0 & 4 \\ 10 & 10 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [27] $1^{1366}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 12 & 12 & 1 & 3 \\ 6 & 16 & 36 & 36 & 0 & 6 \\ 12 & 36 & 70 & 72 & 0 & 12 \\ 12 & 36 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [28] $1^{1477}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.7$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 14 & 14 & 1 & 3 \\ 8 & 30 & 56 & 56 & 0 & 8 \\ 14 & 56 & 96 & 98 & 0 & 14 \\ 14 & 56 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [37] $1^{1237}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 1 & 3 \\ 4 & 6 & 12 & 28 & 0 & 4 \\ 6 & 12 & 16 & 42 & 0 & 6 \\ 14 & 28 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [41] $1^{1577}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 14 & 14 & 1 & 3 \\ 10 & 48 & 70 & 70 & 0 & 10 \\ 14 & 70 & 96 & 98 & 0 & 14 \\ 14 & 70 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{1144}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 9 & 1 & 3 \\ 2 & 0 & 8 & 8 & 0 & 2 \\ 8 & 8 & 30 & 32 & 0 & 8 \\ 9 & 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{1666}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 12 & 12 & 12 & 1 & 3 \\ 12 & 70 & 72 & 72 & 0 & 12 \\ 12 & 72 & 70 & 72 & 0 & 12 \\ 12 & 72 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 12 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{1246}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 12 & 1 & 3 \\ 4 & 6 & 16 & 24 & 0 & 4 \\ 8 & 16 & 30 & 48 & 0 & 8 \\ 12 & 24 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{1236}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 1 & 3 \\ 4 & 6 & 12 & 24 & 0 & 4 \\ 6 & 12 & 16 & 36 & 0 & 6 \\ 12 & 24 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [52] $1^{1234}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [35] $1^{1337}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 1 & 3 \\ 4 & 6 & 12 & 16 & 0 & 4 \\ 6 & 12 & 16 & 24 & 0 & 6 \\ 8 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [29] $1^{1447}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 14 & 1 & 3 \\ 6 & 16 & 18 & 42 & 0 & 6 \\ 6 & 18 & 16 & 42 & 0 & 6 \\ 14 & 42 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{1445}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 8 & 10 & 1 & 3 \\ 8 & 30 & 32 & 40 & 0 & 8 \\ 8 & 32 & 30 & 40 & 0 & 8 \\ 10 & 40 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [53] $1^{1267}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 12 & 14 & 1 & 3 \\ 4 & 6 & 24 & 28 & 0 & 4 \\ 12 & 24 & 70 & 84 & 0 & 12 \\ 14 & 28 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{1467}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 12 & 14 & 1 & 3 \\ 8 & 30 & 48 & 56 & 0 & 8 \\ 12 & 48 & 70 & 84 & 0 & 12 \\ 14 & 56 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [55] $1^{1355}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 10 & 1 & 3 \\ 6 & 16 & 18 & 30 & 0 & 6 \\ 6 & 18 & 16 & 30 & 0 & 6 \\ 10 & 30 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{1567}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.7$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 12 & 14 & 1 & 3 \\ 10 & 48 & 60 & 70 & 0 & 10 \\ 12 & 60 & 70 & 84 & 0 & 12 \\ 14 & 70 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{1136}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 12 & 1 & 3 \\ 2 & 0 & 6 & 12 & 0 & 2 \\ 6 & 6 & 16 & 36 & 0 & 6 \\ 12 & 12 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [42] $1^{1556}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 10 & 12 & 1 & 3 \\ 10 & 48 & 50 & 60 & 0 & 10 \\ 10 & 50 & 48 & 60 & 0 & 10 \\ 12 & 60 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [56] $1^{1127}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 14 & 1 & 3 \\ 2 & 0 & 4 & 14 & 0 & 2 \\ 4 & 4 & 6 & 28 & 0 & 4 \\ 14 & 14 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [57] $1^{1257}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [58] $1^{1112}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 10 & 14 & 1 & 3 \\ 4 & 6 & 20 & 28 & 0 & 4 \\ 10 & 20 & 48 & 70 & 0 & 10 \\ 14 & 28 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{1177}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 14 & 14 & 1 & 3 \\ 2 & 0 & 14 & 14 & 0 & 2 \\ 14 & 14 & 96 & 98 & 0 & 14 \\ 14 & 14 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{1557}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 10 & 14 & 1 & 3 \\ 10 & 48 & 50 & 70 & 0 & 10 \\ 10 & 50 & 48 & 70 & 0 & 10 \\ 14 & 70 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{1137}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 14 & 1 & 3 \\ 2 & 0 & 6 & 14 & 0 & 2 \\ 6 & 6 & 16 & 42 & 0 & 6 \\ 14 & 14 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{1255}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 10 & 10 & 1 & 3 \\ 4 & 6 & 20 & 20 & 0 & 4 \\ 10 & 20 & 48 & 50 & 0 & 10 \\ 10 & 20 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [63] $1^{1457}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 10 & 14 & 1 & 3 \\ 8 & 30 & 40 & 56 & 0 & 8 \\ 10 & 40 & 48 & 70 & 0 & 10 \\ 14 & 56 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{1356}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 10 & 12 & 1 & 3 \\ 6 & 16 & 30 & 36 & 0 & 6 \\ 10 & 30 & 48 & 60 & 0 & 10 \\ 12 & 36 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [30] $1^{1133}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 1 & 3 \\ 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 16 & 18 & 0 & 6 \\ 6 & 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{1247}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 14 & 1 & 3 \\ 4 & 6 & 16 & 28 & 0 & 4 \\ 8 & 16 & 30 & 56 & 0 & 8 \\ 14 & 28 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{1134}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 8 & 1 & 3 \\ 2 & 0 & 6 & 8 & 0 & 2 \\ 6 & 6 & 16 & 24 & 0 & 6 \\ 8 & 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{1377}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [6] $1^{1357}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 14 & 14 & 1 & 3 \\ 6 & 16 & 42 & 42 & 0 & 6 \\ 14 & 42 & 96 & 96 & 0 & 14 \\ 14 & 42 & 96 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [88] $1^{1227}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 10 & 14 & 1 & 3 \\ 6 & 16 & 30 & 42 & 0 & 6 \\ 10 & 30 & 48 & 70 & 0 & 10 \\ 14 & 42 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{1111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^{1233}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 1 & 3 \\ 4 & 6 & 12 & 12 & 0 & 4 \\ 6 & 12 & 16 & 18 & 0 & 6 \\ 6 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{1144}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 8 & 1 & 3 \\ 2 & 0 & 8 & 8 & 0 & 2 \\ 8 & 8 & 30 & 32 & 0 & 8 \\ 8 & 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [90] $1^{1146}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 12 & 1 & 3 \\ 2 & 0 & 8 & 12 & 0 & 2 \\ 8 & 8 & 30 & 48 & 0 & 8 \\ 12 & 12 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [89] $1^{1235}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 10 & 1 & 3 \\ 4 & 6 & 12 & 20 & 0 & 4 \\ 6 & 12 & 16 & 30 & 0 & 6 \\ 10 & 20 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{1115}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 1 & 3 \\ 2 & 0 & 2 & 10 & 0 & 2 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 10 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [87] $1^{1347}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 8 & 14 & 1 & 3 \\ 6 & 16 & 24 & 42 & 0 & 6 \\ 8 & 24 & 30 & 56 & 0 & 8 \\ 14 & 42 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [86] $1^{1145}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 10 & 1 & 3 \\ 2 & 0 & 8 & 10 & 0 & 2 \\ 8 & 8 & 30 & 40 & 0 & 8 \\ 10 & 10 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [85] $1^{1555}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [84] $1^{12241}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 8 & 1 & 3 \\ 10 & 48 & 50 & 50 & 0 & 10 \\ 10 & 50 & 48 & 50 & 0 & 10 \\ 10 & 50 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [83] $1^{12261}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 1 & 3 \\ 4 & 6 & 8 & 16 & 0 & 4 \\ 4 & 8 & 6 & 16 & 0 & 4 \\ 8 & 16 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [82] $1^{15661}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 12 & 12 & 1 & 3 \\ 10 & 48 & 60 & 60 & 0 & 10 \\ 12 & 60 & 70 & 72 & 0 & 12 \\ 12 & 60 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [81] $1^{12561}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 10 & 12 & 1 & 3 \\ 4 & 6 & 20 & 24 & 0 & 4 \\ 10 & 20 & 48 & 60 & 0 & 10 \\ 12 & 24 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [80] $1^{13551}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 10 & 10 & 1 & 3 \\ 6 & 16 & 30 & 30 & 0 & 6 \\ 10 & 30 & 48 & 50 & 0 & 10 \\ 10 & 30 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [79] $1^{14551}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 10 & 10 & 1 & 3 \\ 8 & 30 & 40 & 40 & 0 & 8 \\ 10 & 40 & 48 & 50 & 0 & 10 \\ 10 & 40 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [78] $1^{11351}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 10 & 1 & 3 \\ 2 & 0 & 6 & 10 & 0 & 2 \\ 6 & 6 & 16 & 30 & 0 & 6 \\ 10 & 10 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [77] $1^{13451}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 8 & 10 & 1 & 3 \\ 6 & 16 & 24 & 30 & 0 & 6 \\ 8 & 24 & 30 & 40 & 0 & 8 \\ 10 & 30 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [76] $1^{11141}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 1 & 3 \\ 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [75] $1^{13331}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 6 & 1 & 3 \\ 6 & 16 & 18 & 18 & 0 & 6 \\ 6 & 18 & 16 & 18 & 0 & 6 \\ 6 & 18 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [74] $1^{12451}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [73] $1^{1266}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 10 & 1 & 3 \\ 4 & 6 & 16 & 20 & 0 & 4 \\ 8 & 16 & 30 & 40 & 0 & 8 \\ 10 & 20 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{117}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 1 & 3 \\ 4 & 6 & 24 & 24 & 0 & 4 \\ 12 & 24 & 70 & 72 & 0 & 12 \\ 12 & 24 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{1346}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 8 & 12 & 1 & 3 \\ 6 & 16 & 24 & 36 & 0 & 6 \\ 8 & 24 & 30 & 48 & 0 & 8 \\ 12 & 36 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{1367}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 12 & 14 & 1 & 3 \\ 6 & 16 & 36 & 42 & 0 & 6 \\ 12 & 36 & 70 & 84 & 0 & 12 \\ 14 & 42 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [72] $1^{1126}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 12 & 1 & 3 \\ 2 & 0 & 4 & 12 & 0 & 2 \\ 4 & 4 & 6 & 24 & 0 & 4 \\ 12 & 12 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [71] $1^{1456}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 10 & 12 & 1 & 3 \\ 8 & 30 & 40 & 48 & 0 & 8 \\ 10 & 40 & 48 & 60 & 0 & 10 \\ 12 & 48 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [70] $1^{1336}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 12 & 1 & 3 \\ 6 & 16 & 18 & 36 & 0 & 6 \\ 6 & 18 & 16 & 36 & 0 & 6 \\ 12 & 36 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [69] $1^{1334}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 8 & 1 & 3 \\ 6 & 16 & 18 & 24 & 0 & 6 \\ 6 & 18 & 16 & 24 & 0 & 6 \\ 8 & 24 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [68] $1^{1444}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 8 & 8 & 1 & 3 \\ 8 & 30 & 32 & 32 & 0 & 8 \\ 8 & 32 & 30 & 32 & 0 & 8 \\ 8 & 32 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{1113}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [22] $1^{1167}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [64] $1^{1344}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 8 & 8 & 1 & 3 \\ 6 & 16 & 24 & 24 & 0 & 6 \\ 8 & 24 & 30 & 32 & 0 & 8 \\ 8 & 24 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [31] $1^{1466}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 12 & 12 & 1 & 3 \\ 8 & 30 & 48 & 48 & 0 & 8 \\ 12 & 48 & 70 & 72 & 0 & 12 \\ 12 & 48 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [32] $1^{1277}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 14 & 14 & 1 & 3 \\ 4 & 6 & 28 & 28 & 0 & 4 \\ 14 & 28 & 96 & 98 & 0 & 14 \\ 14 & 28 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{1244}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 8 & 1 & 3 \\ 4 & 6 & 16 & 16 & 0 & 4 \\ 8 & 16 & 30 & 32 & 0 & 8 \\ 8 & 16 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [25] $1^{1446}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 8 & 12 & 1 & 3 \\ 8 & 30 & 32 & 48 & 0 & 8 \\ 8 & 32 & 30 & 48 & 0 & 8 \\ 12 & 48 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [91] $1^{1222}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 5 & 1 & 3 \\ 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 8 & 6 & 8 & 0 & 4 \\ 5 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [93] $1^{1133}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 28, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 7 & 1 & 3 \\ 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 16 & 18 & 0 & 6 \\ 7 & 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$
- [92] $1^{1111}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 28, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$
- [95] $1^{1122}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 32, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 5 & 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [94] $1^{1234}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 32, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 1 & 3 \\ 4 & 6 & 12 & 16 & 0 & 4 \\ 6 & 12 & 16 & 24 & 0 & 6 \\ 9 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{1223}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 7 & 1 & 3 \\ 4 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 12 & 0 & 4 \\ 7 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{1134}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 10.18 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 9 & 1 & 3 \\ 2 & 0 & 6 & 8 & 0 & 2 \\ 6 & 6 & 16 & 24 & 0 & 6 \\ 9 & 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{1123}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 10.18 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 1 & 3 \\ 2 & 0 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 5 & 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{1123}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 48$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 10.17 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 7 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 0 & 4 \\ 7 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{1224}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 52$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.15 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 1 & 3 \\ 4 & 6 & 8 & 16 & 0 & 4 \\ 4 & 8 & 6 & 16 & 0 & 4 \\ 9 & 16 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{1113}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 60$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 7 & 6 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{1124}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 10.12 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 9 & 1 & 3 \\ 2 & 0 & 4 & 8 & 0 & 2 \\ 4 & 4 & 6 & 16 & 0 & 4 \\ 9 & 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{1114}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 76$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 1 & 3 \\ 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 9 & 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

1.32 $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{black}{so}_8 + \hat{1} \rightarrow (-1)$

$T = 7$ [0] $1^{1111}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{black}{so}_8 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 66$, $\Delta = -121$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 66$, $T \in [12, 66]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{black}{so}_8 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 66$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11$, $T_{\max} = 66$, $T \in [11, 66]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{black}{so}_8 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 66$, $\Delta = -121$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 66, T \in [12, 66]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.33 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$

$$T = 7 \quad [2] \quad 1^{11\bar{3}34}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 9 & 1 & 3 \\ 2 & 0 & 6 & 6 & 8 & 0 & 2 \\ 6 & 6 & 16 & 18 & 24 & 0 & 6 \\ 6 & 6 & 18 & 16 & 24 & 0 & 6 \\ 9 & 8 & 24 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11\bar{1}23}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 6 & 1 & 3 \\ 2 & 0 & 2 & 4 & 6 & 0 & 2 \\ 2 & 2 & 0 & 4 & 6 & 0 & 2 \\ 5 & 4 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11\bar{1}1}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11\bar{1}2}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 1 & 3 \\ 2 & 0 & 2 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 & 0 & 2 \\ 3 & 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11\bar{1}44}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 9 & 1 & 3 \\ 2 & 0 & 2 & 8 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 8 & 0 & 2 \\ 8 & 8 & 8 & 30 & 32 & 0 & 8 \\ 9 & 8 & 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11\bar{2}33}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 7 & 1 & 3 \\ 2 & 0 & 4 & 6 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 12 & 0 & 4 \\ 6 & 6 & 12 & 16 & 18 & 0 & 6 \\ 7 & 6 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{12\bar{3}33}1^1 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 6 & 1 & 3 \\ 4 & 6 & 12 & 12 & 12 & 0 & 4 \\ 6 & 12 & 16 & 18 & 18 & 0 & 6 \\ 6 & 12 & 18 & 16 & 18 & 0 & 6 \\ 6 & 12 & 18 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11\bar{1}44}1^1 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 8 & 1 & 3 \\ 2 & 0 & 2 & 8 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 8 & 0 & 2 \\ 8 & 8 & 8 & 30 & 32 & 0 & 8 \\ 8 & 8 & 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{11224}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 8 & 1 & 3 \\ 2 & 0 & 4 & 4 & 8 & 0 & 2 \\ 4 & 4 & 6 & 8 & 16 & 0 & 4 \\ 4 & 4 & 8 & 6 & 16 & 0 & 4 \\ 8 & 8 & 16 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{11111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{12223}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.3$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 6 & 1 & 3 \\ 4 & 6 & 8 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 8 & 6 & 12 & 0 & 4 \\ 6 & 12 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{1157}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 14 & 1 & 3 \\ 2 & 0 & 2 & 10 & 14 & 0 & 2 \\ 2 & 2 & 0 & 10 & 14 & 0 & 2 \\ 10 & 10 & 10 & 48 & 70 & 0 & 10 \\ 14 & 14 & 14 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{11225}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 10 & 1 & 3 \\ 2 & 0 & 4 & 4 & 10 & 0 & 2 \\ 4 & 4 & 6 & 8 & 20 & 0 & 4 \\ 4 & 4 & 8 & 6 & 20 & 0 & 4 \\ 10 & 10 & 20 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{11344}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 8 & 8 & 1 & 3 \\ 2 & 0 & 6 & 8 & 8 & 0 & 2 \\ 6 & 6 & 16 & 24 & 24 & 0 & 6 \\ 8 & 8 & 24 & 30 & 32 & 0 & 8 \\ 8 & 8 & 24 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{11247}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 8 & 14 & 1 & 3 \\ 2 & 0 & 4 & 8 & 14 & 0 & 2 \\ 4 & 4 & 6 & 16 & 28 & 0 & 4 \\ 8 & 8 & 16 & 30 & 56 & 0 & 8 \\ 14 & 14 & 28 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11135}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.9$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 10 & 1 & 3 \\ 2 & 0 & 2 & 6 & 10 & 0 & 2 \\ 2 & 2 & 0 & 6 & 10 & 0 & 2 \\ 6 & 6 & 6 & 16 & 30 & 0 & 6 \\ 10 & 10 & 10 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{11145}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 10 & 1 & 3 \\ 2 & 0 & 2 & 8 & 10 & 0 & 2 \\ 2 & 2 & 0 & 8 & 10 & 0 & 2 \\ 8 & 8 & 8 & 30 & 40 & 0 & 8 \\ 10 & 10 & 10 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 1 & 3 \\ 2 & 0 & 2 & 2 & 14 & 0 & 2 \\ 2 & 2 & 0 & 2 & 14 & 0 & 2 \\ 2 & 2 & 2 & 0 & 14 & 0 & 2 \\ 14 & 14 & 14 & 14 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 14 & -1 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 1 & 3 \\ 2 & 0 & 2 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 10 & 1 & 3 \\ 2 & 0 & 2 & 10 & 10 & 0 & 2 \\ 2 & 2 & 0 & 10 & 10 & 0 & 2 \\ 10 & 10 & 10 & 48 & 50 & 0 & 10 \\ 10 & 10 & 10 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 10 & 10 & -1 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 14 & 1 & 3 \\ 2 & 0 & 4 & 4 & 14 & 0 & 2 \\ 4 & 4 & 6 & 8 & 28 & 0 & 4 \\ 4 & 4 & 8 & 6 & 28 & 0 & 4 \\ 14 & 14 & 28 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 14 & -1 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 8 & 8 & 0 & 4 \\ 4 & 8 & 6 & 8 & 8 & 0 & 4 \\ 4 & 8 & 8 & 6 & 8 & 0 & 4 \\ 4 & 8 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 14 & 1 & 3 \\ 2 & 0 & 2 & 6 & 14 & 0 & 2 \\ 2 & 2 & 0 & 6 & 14 & 0 & 2 \\ 6 & 6 & 6 & 16 & 42 & 0 & 6 \\ 14 & 14 & 14 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 14 & -1 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 12 & 1 & 3 \\ 2 & 0 & 2 & 12 & 12 & 0 & 2 \\ 2 & 2 & 0 & 12 & 12 & 0 & 2 \\ 12 & 12 & 12 & 70 & 72 & 0 & 12 \\ 12 & 12 & 12 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 12 & 12 & -1 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 12 & 1 & 3 \\ 2 & 0 & 6 & 6 & 12 & 0 & 2 \\ 6 & 6 & 16 & 18 & 36 & 0 & 6 \\ 6 & 6 & 18 & 16 & 36 & 0 & 6 \\ 12 & 12 & 36 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 12 & -1 & 1 \end{pmatrix} b_{\alpha}$$

$$[6] \quad 1^{11167} 1^1 + 511_2 + 511_2 + 511_2 + 511_2 + 511_2 + \hat{1} \rightarrow 1$$

[23] $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 14 & 1 & 3 \\ 2 & 0 & 2 & 12 & 14 & 0 & 2 \\ 2 & 2 & 0 & 12 & 14 & 0 & 2 \\ 12 & 12 & 12 & 70 & 84 & 0 & 12 \\ 14 & 14 & 14 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
 $(1, 7, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 4 & 4 & 6 & 8 & 1 & 3 \\ 4 & 6 & 8 & 12 & 16 & 0 & 4 \\ 4 & 8 & 6 & 12 & 16 & 0 & 4 \\ 6 & 12 & 12 & 16 & 24 & 0 & 6 \\ 8 & 16 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$

[24] $1^{11234}1^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 6 & 8 & 1 & 3 \\ 2 & 0 & 4 & 6 & 8 & 0 & 2 \\ 4 & 4 & 6 & 12 & 16 & 0 & 4 \\ 6 & 6 & 12 & 16 & 24 & 0 & 6 \\ 8 & 8 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$

[25] $1^{11133}1^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 6 & 16 & 18 & 0 & 6 \\ 6 & 6 & 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$

[51] $1^{11146}1^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 1 & 3 \\ 2 & 0 & 2 & 8 & 12 & 0 & 2 \\ 2 & 2 & 0 & 8 & 12 & 0 & 2 \\ 8 & 8 & 8 & 30 & 48 & 0 & 8 \\ 12 & 12 & 12 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$

[26] $1^{12225}1^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.6$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 4 & 4 & 4 & 10 & 1 & 3 \\ 4 & 6 & 8 & 8 & 20 & 0 & 4 \\ 4 & 8 & 6 & 8 & 20 & 0 & 4 \\ 4 & 8 & 8 & 6 & 20 & 0 & 4 \\ 10 & 20 & 20 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$

[27] $1^{11113}1^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 1 & 3 \\ 2 & 0 & 2 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$

[31] $1^{11116}1^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 1 & 3 \\ 2 & 0 & 2 & 2 & 12 & 0 & 2 \\ 2 & 2 & 0 & 2 & 12 & 0 & 2 \\ 2 & 2 & 2 & 0 & 12 & 0 & 2 \\ 12 & 12 & 12 & 12 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 12 & -1 & 1 \end{pmatrix}_{b_0}$

[32] $1^{11246}1^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 8 & 12 & 1 & 3 \\ 2 & 0 & 4 & 8 & 12 & 0 & 2 \\ 4 & 4 & 6 & 16 & 24 & 0 & 4 \\ 8 & 8 & 16 & 30 & 48 & 0 & 8 \\ 12 & 12 & 24 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b$

- [57] $1^{11226}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 12 & 1 & 3 \\ 2 & 0 & 4 & 4 & 12 & 0 & 2 \\ 4 & 4 & 6 & 8 & 24 & 0 & 4 \\ 4 & 4 & 8 & 6 & 24 & 0 & 4 \\ 12 & 12 & 24 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [58] $1^{11115}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 1 & 3 \\ 2 & 0 & 2 & 2 & 10 & 0 & 2 \\ 2 & 2 & 0 & 2 & 10 & 0 & 2 \\ 2 & 2 & 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 10 & 10 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{11156}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 12 & 1 & 3 \\ 2 & 0 & 2 & 2 & 10 & 12 & 0 & 2 \\ 2 & 2 & 0 & 2 & 10 & 12 & 0 & 2 \\ 2 & 2 & 2 & 0 & 10 & 12 & 0 & 2 \\ 10 & 10 & 10 & 10 & 48 & 60 & 0 & 10 \\ 12 & 12 & 12 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{11126}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 12 & 1 & 3 \\ 2 & 0 & 2 & 4 & 12 & 0 & 2 \\ 2 & 2 & 0 & 4 & 12 & 0 & 2 \\ 4 & 4 & 4 & 6 & 24 & 0 & 4 \\ 12 & 12 & 12 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{11134}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 8 & 1 & 3 \\ 2 & 0 & 2 & 6 & 8 & 0 & 2 \\ 2 & 2 & 0 & 6 & 8 & 0 & 2 \\ 6 & 6 & 6 & 16 & 24 & 0 & 6 \\ 8 & 8 & 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [33] $1^{12227}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 14 & 1 & 3 \\ 4 & 6 & 8 & 8 & 28 & 0 & 4 \\ 4 & 8 & 6 & 8 & 28 & 0 & 4 \\ 4 & 8 & 8 & 6 & 28 & 0 & 4 \\ 14 & 28 & 28 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{11244}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 8 & 8 & 1 & 3 \\ 2 & 0 & 4 & 8 & 8 & 0 & 2 \\ 4 & 4 & 6 & 16 & 16 & 0 & 4 \\ 8 & 8 & 16 & 30 & 32 & 0 & 8 \\ 8 & 8 & 16 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [63] $1^{11333}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 6 & 1 & 3 \\ 2 & 0 & 6 & 6 & 6 & 0 & 2 \\ 6 & 6 & 16 & 18 & 18 & 0 & 6 \\ 6 & 6 & 18 & 16 & 18 & 0 & 6 \\ 6 & 6 & 18 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [64] $1^{12224}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 8 & 1 & 3 \\ 4 & 6 & 8 & 8 & 16 & 0 & 4 \\ 4 & 8 & 6 & 8 & 16 & 0 & 4 \\ 4 & 8 & 8 & 6 & 16 & 0 & 4 \\ 8 & 16 & 16 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{11125}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.6$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 10 & 1 & 3 \\ 2 & 0 & 2 & 4 & 10 & 0 & 2 \\ 2 & 2 & 0 & 4 & 10 & 0 & 2 \\ 4 & 4 & 4 & 6 & 20 & 0 & 4 \\ 10 & 10 & 10 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{11123}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.3$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 6 & 1 & 3 \\ 2 & 0 & 2 & 4 & 6 & 0 & 2 \\ 2 & 2 & 0 & 4 & 6 & 0 & 2 \\ 4 & 4 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{11222}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 5 & 1 & 3 \\ 2 & 0 & 4 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 4 & 8 & 6 & 8 & 0 & 4 \\ 5 & 4 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{11127}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.8$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 14 & 1 & 3 \\ 2 & 0 & 2 & 4 & 14 & 0 & 2 \\ 2 & 2 & 0 & 4 & 14 & 0 & 2 \\ 4 & 4 & 4 & 6 & 28 & 0 & 4 \\ 14 & 14 & 14 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{11444}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 8 & 8 & 1 & 3 \\ 2 & 0 & 8 & 8 & 8 & 0 & 2 \\ 8 & 8 & 30 & 32 & 32 & 0 & 8 \\ 8 & 8 & 32 & 30 & 32 & 0 & 8 \\ 8 & 8 & 32 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11255}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 10 & 10 & 1 & 3 \\ 2 & 0 & 4 & 10 & 10 & 0 & 2 \\ 4 & 4 & 6 & 20 & 20 & 0 & 4 \\ 10 & 10 & 20 & 48 & 50 & 0 & 10 \\ 10 & 10 & 20 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{11112}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{11122}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{11245}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.6$

$$\begin{pmatrix} 1 & 2 & 4 & 8 & 10 & 1 & 3 \\ 2 & 0 & 4 & 8 & 10 & 0 & 2 \\ 4 & 4 & 6 & 16 & 20 & 0 & 4 \\ 8 & 8 & 16 & 30 & 40 & 0 & 8 \\ 10 & 10 & 20 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & 10 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 14 & 1 & 3 \\ 2 & 0 & 4 & 6 & 14 & 0 & 2 \\ 4 & 4 & 6 & 12 & 28 & 0 & 4 \\ 6 & 6 & 12 & 16 & 42 & 0 & 6 \\ 14 & 14 & 28 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 14 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 6 & 1 & 3 \\ 2 & 0 & 4 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 8 & 12 & 0 & 4 \\ 4 & 4 & 8 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 6 & -1 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 12 & 1 & 3 \\ 2 & 0 & 2 & 6 & 12 & 0 & 2 \\ 2 & 2 & 0 & 6 & 12 & 0 & 2 \\ 6 & 6 & 6 & 16 & 36 & 0 & 6 \\ 12 & 12 & 12 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 12 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 10 & 1 & 3 \\ 4 & 6 & 8 & 12 & 20 & 0 & 4 \\ 4 & 8 & 6 & 12 & 20 & 0 & 4 \\ 6 & 12 & 12 & 16 & 30 & 0 & 6 \\ 10 & 20 & 20 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & 10 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 14 & 1 & 3 \\ 2 & 0 & 2 & 8 & 14 & 0 & 2 \\ 2 & 2 & 0 & 8 & 14 & 0 & 2 \\ 8 & 8 & 8 & 30 & 56 & 0 & 8 \\ 14 & 14 & 14 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 14 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 12 & 1 & 3 \\ 2 & 0 & 4 & 6 & 12 & 0 & 2 \\ 4 & 4 & 6 & 12 & 24 & 0 & 4 \\ 6 & 6 & 12 & 16 & 36 & 0 & 6 \\ 12 & 12 & 24 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 12 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 10 & 14 & 1 & 3 \\ 2 & 0 & 4 & 10 & 14 & 0 & 2 \\ 4 & 4 & 6 & 20 & 28 & 0 & 4 \\ 10 & 10 & 20 & 48 & 70 & 0 & 10 \\ 14 & 14 & 28 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 10 & 14 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 14 & 1 & 3 \\ 2 & 0 & 2 & 14 & 14 & 0 & 2 \\ 2 & 2 & 0 & 14 & 14 & 0 & 2 \\ 14 & 14 & 14 & 96 & 98 & 0 & 14 \\ 14 & 14 & 14 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 14 & 14 & -1 & 1 \end{pmatrix}$$

1420

$$(1, 7, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 6 & 1 & 3 \\ 4 & 6 & 8 & 12 & 12 & 0 & 4 \\ 4 & 8 & 6 & 12 & 12 & 0 & 4 \\ 6 & 12 & 12 & 16 & 18 & 0 & 6 \\ 6 & 12 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11222}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 4 & 1 & 3 \\ 2 & 0 & 4 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 4 & 8 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11337}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 14 & 1 & 3 \\ 2 & 0 & 6 & 6 & 14 & 0 & 2 \\ 6 & 6 & 16 & 18 & 42 & 0 & 6 \\ 6 & 6 & 18 & 16 & 42 & 0 & 6 \\ 14 & 14 & 42 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11334}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 8 & 1 & 3 \\ 2 & 0 & 6 & 6 & 8 & 0 & 2 \\ 6 & 6 & 16 & 18 & 24 & 0 & 6 \\ 6 & 6 & 18 & 16 & 24 & 0 & 6 \\ 8 & 8 & 24 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11256}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 10 & 12 & 1 & 3 \\ 2 & 0 & 4 & 10 & 12 & 0 & 2 \\ 4 & 4 & 6 & 20 & 24 & 0 & 4 \\ 10 & 10 & 20 & 48 & 60 & 0 & 10 \\ 12 & 12 & 24 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12234}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 9 & 1 & 3 \\ 4 & 6 & 8 & 12 & 16 & 0 & 4 \\ 4 & 8 & 6 & 12 & 16 & 0 & 4 \\ 6 & 12 & 12 & 16 & 24 & 0 & 6 \\ 9 & 16 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{11124}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 8 & 1 & 3 \\ 2 & 0 & 2 & 4 & 8 & 0 & 2 \\ 2 & 2 & 0 & 4 & 8 & 0 & 2 \\ 4 & 4 & 4 & 6 & 16 & 0 & 4 \\ 8 & 8 & 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{11233}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 12 & 0 & 4 \\ 6 & 6 & 12 & 16 & 18 & 0 & 6 \\ 6 & 6 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{12244}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 8 & 8 & 1 & 3 \\ 4 & 6 & 8 & 16 & 16 & 0 & 4 \\ 4 & 8 & 6 & 16 & 16 & 0 & 4 \\ 8 & 16 & 16 & 30 & 32 & 0 & 8 \\ 8 & 16 & 16 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

- [47] $1^{12226}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 12 & 1 & 3 \\ 4 & 6 & 8 & 8 & 24 & 0 & 4 \\ 4 & 8 & 6 & 8 & 24 & 0 & 4 \\ 4 & 8 & 8 & 6 & 24 & 0 & 4 \\ 12 & 24 & 24 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{11235}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 10 & 1 & 3 \\ 2 & 0 & 4 & 6 & 10 & 0 & 2 \\ 4 & 4 & 6 & 12 & 20 & 0 & 4 \\ 6 & 6 & 12 & 16 & 30 & 0 & 6 \\ 10 & 10 & 20 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^{11335}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.7$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 10 & 1 & 3 \\ 2 & 0 & 6 & 6 & 10 & 0 & 2 \\ 6 & 6 & 16 & 18 & 30 & 0 & 6 \\ 6 & 6 & 18 & 16 & 30 & 0 & 6 \\ 10 & 10 & 30 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [44] $1^{11266}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 12 & 12 & 1 & 3 \\ 2 & 0 & 4 & 12 & 12 & 0 & 2 \\ 4 & 4 & 6 & 24 & 24 & 0 & 4 \\ 12 & 12 & 24 & 70 & 72 & 0 & 12 \\ 12 & 12 & 24 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [30] $1^{11345}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.9$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 6 & 8 & 10 & 1 & 3 \\ 2 & 0 & 6 & 8 & 10 & 0 & 2 \\ 6 & 6 & 16 & 24 & 30 & 0 & 6 \\ 8 & 8 & 24 & 30 & 40 & 0 & 8 \\ 10 & 10 & 30 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [74] $1^{12223}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 7 & 1 & 3 \\ 4 & 6 & 8 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 8 & 6 & 12 & 0 & 4 \\ 7 & 12 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$
- [75] $1^{11111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & 0 & 0 & 2 \\ 3 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{11133}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 7 & 1 & 3 \\ 2 & 0 & 2 & 6 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 6 & 16 & 18 & 0 & 6 \\ 7 & 6 & 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$
- [78] $1^{11122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 9 & 1 & 3 \\ 2 & 0 & 2 & 4 & 8 & 0 & 2 \\ 2 & 2 & 0 & 4 & 8 & 0 & 2 \\ 4 & 4 & 4 & 6 & 16 & 0 & 4 \\ 9 & 8 & 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{11114}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 1 & 3 \\ 2 & 0 & 2 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 2 & 0 & 8 & 0 & 2 \\ 9 & 8 & 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$\mathbf{1.34} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \quad 1^{11111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 12, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.35} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \quad 1^{11111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -62, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.36} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow (-1)$$

$$T = 9 \quad [0] \quad 1^{1111111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 10, \Delta = -32, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.37} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow (-1)$$

$$T = 10 \quad [0] \quad 1^{11111111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 10, \Delta = -2, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.38} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow (-1)$$

$$T = 7 \quad [1] \quad 1^{11221}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, T_{\max} = 19, T \in [13, 19]$$

$$\begin{pmatrix} 5 & 7 & 12 & 12 & 1 & 5 \\ 7 & 4 & 12 & 12 & 0 & 4 \\ 12 & 12 & 22 & 24 & 0 & 8 \\ 12 & 12 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[0] $1^{1112}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -4$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 4 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 4 & 0 & 2 \\ 2 & 0 & 2 & 0 & 4 & 0 & 2 \\ 4 & 0 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1113}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 1 & 3 \\ 2 & 0 & 2 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 2 & 6 & 0 & 2 \\ 3 & 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 4 & 0 & 2 \\ 3 & 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11231}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 13 & 18 & 1 & 5 \\ 6 & 4 & 12 & 18 & 0 & 4 \\ 13 & 12 & 22 & 36 & 0 & 8 \\ 18 & 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 7 & 12 & 1 & 5 \\ 6 & 4 & 6 & 12 & 0 & 4 \\ 7 & 6 & 4 & 12 & 0 & 4 \\ 12 & 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11441}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 24 & 25 & 1 & 5 \\ 6 & 4 & 24 & 24 & 0 & 4 \\ 24 & 24 & 94 & 96 & 0 & 16 \\ 25 & 24 & 96 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11231}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 12 & 18 & 1 & 5 \\ 6 & 4 & 12 & 18 & 0 & 4 \\ 12 & 12 & 22 & 36 & 0 & 8 \\ 18 & 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11112}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11271}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 12 & 42 & 1 & 5 \\ 6 & 4 & 12 & 42 & 0 & 4 \\ 12 & 12 & 22 & 84 & 0 & 8 \\ 42 & 42 & 84 & 292 & 0 & 28 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11441}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 24 & 1 & 5 \\ 6 & 4 & 6 & 24 & 0 & 4 \\ 6 & 6 & 4 & 24 & 0 & 4 \\ 24 & 24 & 24 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11441}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 24 & 24 & 1 & 5 \\ 6 & 4 & 24 & 24 & 0 & 4 \\ 24 & 24 & 94 & 96 & 0 & 16 \\ 24 & 24 & 96 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11251}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.6$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 12 & 30 & 1 & 5 \\ 6 & 4 & 12 & 30 & 0 & 4 \\ 12 & 12 & 22 & 60 & 0 & 8 \\ 30 & 30 & 60 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11151}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.1$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 30 & 1 & 5 \\ 6 & 4 & 6 & 30 & 0 & 4 \\ 6 & 6 & 4 & 30 & 0 & 4 \\ 30 & 30 & 30 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{11341}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 18 & 24 & 1 & 5 \\ 6 & 4 & 18 & 24 & 0 & 4 \\ 18 & 18 & 52 & 72 & 0 & 12 \\ 24 & 24 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{11131}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 1 & 3 \\ 2 & 0 & 2 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{11331}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.6$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 18 & 18 & 1 & 5 \\ 6 & 4 & 18 & 18 & 0 & 4 \\ 18 & 18 & 52 & 54 & 0 & 12 \\ 18 & 18 & 54 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11121}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 12 & 1 & 5 \\ 6 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 4 & 12 & 0 & 4 \\ 12 & 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{12221}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 12 & 12 & 12 & 1 & 5 \\ 12 & 22 & 24 & 24 & 0 & 8 \\ 12 & 24 & 22 & 24 & 0 & 8 \\ 12 & 24 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11351}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.9$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 18 & 30 & 1 & 5 \\ 6 & 4 & 18 & 30 & 0 & 4 \\ 18 & 18 & 52 & 90 & 0 & 12 \\ 30 & 30 & 90 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11131}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 18 & 1 & 5 \\ 6 & 4 & 6 & 18 & 0 & 4 \\ 6 & 6 & 4 & 18 & 0 & 4 \\ 18 & 18 & 18 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{12231}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 12 & 12 & 18 & 1 & 5 \\ 12 & 22 & 24 & 36 & 0 & 8 \\ 12 & 24 & 22 & 36 & 0 & 8 \\ 18 & 36 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{11111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{12241}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 12 & 12 & 24 & 1 & 5 \\ 12 & 22 & 24 & 48 & 0 & 8 \\ 12 & 24 & 22 & 48 & 0 & 8 \\ 24 & 48 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11161}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 36 & 1 & 5 \\ 6 & 4 & 6 & 36 & 0 & 4 \\ 6 & 6 & 4 & 36 & 0 & 4 \\ 36 & 36 & 36 & 214 & 0 & 24 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11221}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 12 & 1 & 5 \\ 6 & 4 & 12 & 12 & 0 & 4 \\ 12 & 12 & 22 & 24 & 0 & 8 \\ 12 & 12 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11261}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 36 & 1 & 5 \\ 6 & 4 & 12 & 36 & 0 & 4 \\ 12 & 12 & 22 & 72 & 0 & 8 \\ 36 & 36 & 72 & 214 & 0 & 24 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

- [31] $1^{11241}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 24 & 1 & 5 \\ 6 & 4 & 12 & 24 & 0 & 4 \\ 12 & 12 & 22 & 48 & 0 & 8 \\ 24 & 24 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{11122}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [27] $1^{11171}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 42 & 1 & 5 \\ 6 & 4 & 6 & 42 & 0 & 4 \\ 6 & 6 & 4 & 42 & 0 & 4 \\ 42 & 42 & 42 & 292 & 0 & 28 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$
- [28] $1^{11132}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 4 & 1 & 3 \\ 2 & 0 & 2 & 6 & 4 & 0 & 2 \\ 2 & 2 & 0 & 6 & 4 & 0 & 2 \\ 6 & 6 & 6 & 16 & 12 & 0 & 6 \\ 4 & 4 & 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{11112}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -52$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 1 & 3 \\ 2 & 0 & 2 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 & 0 & 2 \\ 3 & 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{12221}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -52$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 12 & 12 & 12 & 13 & 1 & 5 \\ 12 & 22 & 24 & 24 & 0 & 8 \\ 12 & 24 & 22 & 24 & 0 & 8 \\ 13 & 24 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{11111}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.47 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 7 & 1 & 5 \\ 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 4 & 6 & 0 & 4 \\ 7 & 6 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{11331}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.6$
 $T_{\min} = 10.47 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 18 & 19 & 1 & 5 \\ 6 & 4 & 18 & 18 & 0 & 4 \\ 18 & 18 & 52 & 54 & 0 & 12 \\ 19 & 18 & 54 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{11221}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -88$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 13 & 1 & 5 \\ 6 & 4 & 12 & 12 & 0 & 4 \\ 12 & 12 & 22 & 24 & 0 & 8 \\ 13 & 12 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [37] $1^{11122}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 4 & 0 & 2 \\ 5 & 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{12231}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 12 & 12 & 19 & 1 & 5 \\ 12 & 22 & 24 & 36 & 0 & 8 \\ 12 & 24 & 22 & 36 & 0 & 8 \\ 19 & 36 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11341}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 18 & 25 & 1 & 5 \\ 6 & 4 & 18 & 24 & 0 & 4 \\ 18 & 18 & 52 & 72 & 0 & 12 \\ 25 & 24 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 13 & 1 & 5 \\ 6 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 4 & 12 & 0 & 4 \\ 13 & 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11231}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 12 & 19 & 1 & 5 \\ 6 & 4 & 12 & 18 & 0 & 4 \\ 12 & 12 & 22 & 36 & 0 & 8 \\ 19 & 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{11132}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.35 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 4 & 1 & 3 \\ 2 & 0 & 2 & 6 & 4 & 0 & 2 \\ 2 & 2 & 0 & 6 & 4 & 0 & 2 \\ 7 & 6 & 6 & 16 & 12 & 0 & 6 \\ 4 & 4 & 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12241}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 12 & 12 & 25 & 1 & 5 \\ 12 & 22 & 24 & 48 & 0 & 8 \\ 12 & 24 & 22 & 48 & 0 & 8 \\ 25 & 48 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11131}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 19 & 1 & 5 \\ 6 & 4 & 6 & 18 & 0 & 4 \\ 6 & 6 & 4 & 18 & 0 & 4 \\ 19 & 18 & 18 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{11241}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 12 & 25 & 1 & 5 \\ 6 & 4 & 12 & 24 & 0 & 4 \\ 12 & 12 & 22 & 48 & 0 & 8 \\ 25 & 24 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{11141}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 25 & 1 & 5 \\ 6 & 4 & 6 & 24 & 0 & 4 \\ 6 & 6 & 4 & 24 & 0 & 4 \\ 25 & 24 & 24 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

1.39 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 8 \quad [0] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 4 & 6 & 0 \\ 6 & 0 & 6 & 6 & 3 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 6 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 6 & 4 & 6 & 0 \\ 6 & 6 & 6 & 6 & 3 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 6 & 6 & 7 & 6 & 1 & 5 \\ 6 & 4 & 6 & 6 & 6 & 0 & 4 \\ 6 & 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 6 & 4 & 6 & 0 & 4 \\ 7 & 6 & 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 6 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.40 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [1] \quad 1^{\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{4}\textcolor{red}{1}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 2, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13, T_{\max} = 89, T \in [13, 89]$$

$$\begin{pmatrix} 5 & 19 & 24 & 1 & 5 \\ 19 & 52 & 72 & 0 & 12 \\ 24 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{\textcolor{red}{2}\textcolor{blue}{1}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 2, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13, T_{\max} = 89, T \in [13, 89]$$

$$\begin{pmatrix} 5 & 1 & 12 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 12 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{2}\textcolor{blue}{3}\textcolor{red}{2}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 5 & 6 & 4 & 1 & 3 \\ 5 & 6 & 12 & 8 & 0 & 4 \\ 6 & 12 & 16 & 12 & 0 & 6 \\ 4 & 8 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{1}\textcolor{red}{2}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 1 & 2 & 4 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 4 & 0 & 2 \\ 4 & 0 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{1}\textcolor{red}{1}\textcolor{red}{3}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 3 \\ 3 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\textcolor{red}{1}\textcolor{red}{2}\textcolor{red}{2}} 1^{\textcolor{red}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

- [6] $1^{1223}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 18$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 1 & 3 \\ 3 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.60 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 1 & 3 \\ 4 & 6 & 8 & 12 & 0 & 4 \\ 5 & 8 & 6 & 12 & 0 & 4 \\ 6 & 12 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{1231}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 10.60 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 13 & 18 & 1 & 5 \\ 13 & 22 & 36 & 0 & 8 \\ 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{1142}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 8 & 4 & 1 & 3 \\ 2 & 0 & 8 & 4 & 0 & 2 \\ 8 & 8 & 30 & 16 & 0 & 8 \\ 4 & 4 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{1341}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 24 & 1 & 5 \\ 18 & 52 & 72 & 0 & 12 \\ 24 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [42] $1^{1332}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 6 & 6 & 4 & 1 & 3 \\ 6 & 16 & 18 & 12 & 0 & 6 \\ 6 & 18 & 16 & 12 & 0 & 6 \\ 4 & 12 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{1151}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 30 & 1 & 5 \\ 6 & 4 & 30 & 0 & 4 \\ 30 & 30 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [40] $1^{1441}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 24 & 24 & 1 & 5 \\ 24 & 94 & 96 & 0 & 16 \\ 24 & 96 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [39] $1^{1143}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 8 & 6 & 1 & 3 \\ 2 & 0 & 8 & 6 & 0 & 2 \\ 8 & 8 & 30 & 24 & 0 & 8 \\ 6 & 6 & 24 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [35] $1^{1551}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 30 & 30 & 1 & 5 \\ 30 & 148 & 150 & 0 & 20 \\ 30 & 150 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 20 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [33] $1^{1111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [32] $1^{1242}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 4 & 8 & 4 & 1 & 3 \\ 4 & 6 & 16 & 8 & 0 & 4 \\ 8 & 16 & 30 & 16 & 0 & 8 \\ 4 & 8 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1231}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 22 & 36 & 0 & 8 \\ 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1171}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 6 & 42 & 1 & 5 \\ 6 & 4 & 42 & 0 & 4 \\ 42 & 42 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1113}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1121}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 6 & 12 & 1 & 5 \\ 6 & 4 & 12 & 0 & 4 \\ 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1131}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 6 & 18 & 1 & 5 \\ 6 & 4 & 18 & 0 & 4 \\ 18 & 18 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1222}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 8 & 6 & 8 & 0 & 4 \\ 4 & 8 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1471}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 24 & 42 & 1 & 5 \\ 24 & 94 & 168 & 0 & 16 \\ 42 & 168 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1251}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 12 & 30 & 1 & 5 \\ 12 & 22 & 60 & 0 & 8 \\ 30 & 60 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1162}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 2 & 12 & 4 & 1 & 3 \\ 2 & 0 & 12 & 4 & 0 & 2 \\ 12 & 12 & 70 & 24 & 0 & 12 \\ 4 & 4 & 24 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 12 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1361}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 18 & 36 & 1 & 5 \\ 18 & 52 & 108 & 0 & 12 \\ 36 & 108 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1232}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 4 & 6 & 4 & 1 & 3 \\ 4 & 6 & 12 & 8 & 0 & 4 \\ 6 & 12 & 16 & 12 & 0 & 6 \\ 4 & 8 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

- [20] $1^{1771}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 42 & 42 & 1 & 5 \\ 42 & 292 & 294 & 0 & 28 \\ 42 & 294 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 28 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{1133}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 1 & 3 \\ 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 16 & 18 & 0 & 6 \\ 6 & 6 & 18 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{1261}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 22 & 72 & 0 & 8 \\ 36 & 72 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{1671}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.6$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 36 & 42 & 1 & 5 \\ 36 & 214 & 252 & 0 & 24 \\ 42 & 252 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 24 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{1451}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 24 & 30 & 1 & 5 \\ 24 & 94 & 120 & 0 & 16 \\ 30 & 120 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{1661}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 36 & 36 & 1 & 5 \\ 36 & 214 & 216 & 0 & 24 \\ 36 & 216 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 24 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{1221}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 \\ 12 & 22 & 24 & 0 & 8 \\ 12 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{1331}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 18 & 1 & 5 \\ 18 & 52 & 54 & 0 & 12 \\ 18 & 54 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{1114}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.6$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 1 & 3 \\ 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 8 & 29 & 0 & 7 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{1115}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 1 & 3 \\ 2 & 0 & 2 & 10 & 0 & 2 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 10 & 47 & 0 & 9 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{1351}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.9$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 30 & 1 & 5 \\ 18 & 52 & 90 & 0 & 12 \\ 30 & 90 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{1262}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.9$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 12 & 4 & 1 & 3 \\ 4 & 6 & 24 & 8 & 0 & 4 \\ 12 & 24 & 70 & 24 & 0 & 12 \\ 4 & 8 & 24 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 12 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{1561}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 30 & 36 & 1 & 5 \\ 30 & 148 & 180 & 0 & 20 \\ 36 & 180 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 20 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{1342}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 6 & 8 & 4 & 1 & 3 \\ 6 & 16 & 24 & 12 & 0 & 6 \\ 8 & 24 & 30 & 16 & 0 & 8 \\ 4 & 12 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{1123}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{1241}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 24 & 1 & 5 \\ 12 & 22 & 48 & 0 & 8 \\ 24 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{1152}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 10 & 4 & 1 & 3 \\ 2 & 0 & 10 & 4 & 0 & 2 \\ 10 & 10 & 48 & 20 & 0 & 10 \\ 4 & 4 & 20 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{1132}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 6 & 4 & 1 & 3 \\ 2 & 0 & 6 & 4 & 0 & 2 \\ 6 & 6 & 16 & 12 & 0 & 6 \\ 4 & 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{1122}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{1571}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 30 & 42 & 1 & 5 \\ 30 & 148 & 210 & 0 & 20 \\ 42 & 210 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 20 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{1172}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 14 & 4 & 1 & 3 \\ 2 & 0 & 14 & 4 & 0 & 2 \\ 14 & 14 & 96 & 28 & 0 & 14 \\ 4 & 4 & 28 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 14 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{1161}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 36 & 1 & 5 \\ 6 & 4 & 36 & 0 & 4 \\ 36 & 36 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{1271}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 42 & 1 & 5 \\ 12 & 22 & 84 & 0 & 8 \\ 42 & 84 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

- [55] $1^{1371}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 42 & 1 & 5 \\ 18 & 52 & 126 & 0 & 12 \\ 42 & 126 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$
- [56] $1^{1121}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [57] $1^{1252}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 10 & 4 & 1 & 3 \\ 4 & 6 & 20 & 8 & 0 & 4 \\ 10 & 20 & 48 & 20 & 0 & 10 \\ 4 & 8 & 20 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [37] $1^{1461}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 24 & 36 & 1 & 5 \\ 24 & 94 & 144 & 0 & 16 \\ 36 & 144 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{1141}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 24 & 1 & 5 \\ 6 & 4 & 24 & 0 & 4 \\ 24 & 24 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{1223}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 1 & 3 \\ 4 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 12 & 0 & 4 \\ 6 & 12 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [58] $1^{1121}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 7 & 12 & 1 & 5 \\ 7 & 4 & 12 & 0 & 4 \\ 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{1441}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 24 & 25 & 1 & 5 \\ 24 & 94 & 96 & 0 & 16 \\ 25 & 96 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{1332}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 32$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 6 & 7 & 4 & 1 & 3 \\ 6 & 16 & 18 & 12 & 0 & 6 \\ 7 & 18 & 16 & 12 & 0 & 6 \\ 4 & 12 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [60] $1^{1121}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 32$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 0 & 2 \\ 3 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [62] $1^{1123}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 5 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 5 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [63] $1^{1222}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 38$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

- [65] $1^{1331}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 6 & 8 & 8 & 0 & 4 \\ 5 & 8 & 6 & 8 & 0 & 4 \\ 4 & 8 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [64] $1^{1111}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 19 & 1 & 5 \\ 18 & 52 & 54 & 0 & 12 \\ 19 & 54 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{1221}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 50$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.44 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 13 & 1 & 5 \\ 12 & 22 & 24 & 0 & 8 \\ 13 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{1122}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 5 & 4 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 2 \\ 5 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [68] $1^{1342}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 6 & 9 & 4 & 1 & 3 \\ 6 & 16 & 24 & 12 & 0 & 6 \\ 9 & 24 & 30 & 16 & 0 & 8 \\ 4 & 12 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [69] $1^{1133}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 60$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 7 & 6 & 1 & 3 \\ 2 & 0 & 6 & 6 & 0 & 2 \\ 7 & 6 & 16 & 18 & 0 & 6 \\ 6 & 6 & 18 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [70] $1^{1232}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 62$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 7 & 4 & 1 & 3 \\ 4 & 6 & 12 & 8 & 0 & 4 \\ 7 & 12 & 16 & 12 & 0 & 6 \\ 4 & 8 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [71] $1^{1121}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 13 & 1 & 5 \\ 6 & 4 & 12 & 0 & 4 \\ 13 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [72] $1^{1341}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 25 & 1 & 5 \\ 18 & 52 & 72 & 0 & 12 \\ 25 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [73] $1^{1231}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 74$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 22 & 36 & 0 & 8 \\ 19 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [74] $1^{1132}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 80$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 7 & 4 & 1 & 3 \\ 2 & 0 & 6 & 4 & 0 & 2 \\ 7 & 6 & 16 & 12 & 0 & 6 \\ 4 & 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

- [75] $1^{1143}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 84$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 9 & 6 & 1 & 3 \\ 2 & 0 & 8 & 6 & 0 & 2 \\ 9 & 8 & 30 & 24 & 0 & 8 \\ 6 & 6 & 24 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{1242}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 86$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 9 & 4 & 1 & 3 \\ 4 & 6 & 16 & 8 & 0 & 4 \\ 9 & 16 & 30 & 16 & 0 & 8 \\ 4 & 8 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [77] $1^{1131}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 92$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 19 & 1 & 5 \\ 6 & 4 & 18 & 0 & 4 \\ 19 & 18 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [78] $1^{1241}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 98$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 25 & 1 & 5 \\ 12 & 22 & 48 & 0 & 8 \\ 25 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [79] $1^{1142}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 104$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 9 & 4 & 1 & 3 \\ 2 & 0 & 8 & 4 & 0 & 2 \\ 9 & 8 & 30 & 16 & 0 & 8 \\ 4 & 4 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [80] $1^{1141}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 116$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 25 & 1 & 5 \\ 6 & 4 & 24 & 0 & 4 \\ 25 & 24 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

1.41 $\textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \hat{1} \rightarrow (-1)$

- $T = 7$ [0] $1^{1111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 90$, $\Delta = -130$, $\det = -68$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 11.35 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

1.42 $\textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow (-1)$

- $T = 7$ [0] $1^{1112}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -110$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 1 & 6 & 12 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 12 & 0 & 4 \\ 12 & 0 & 12 & 21 & 0 & 7 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$
- [1] $1^{1111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -110$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$
- [2] $1^{1121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -110$, $\det = -66$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

[12] $1^{11411}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -110$, $\det = -336$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 10.68 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 25 & 6 & 1 & 5 \\ 6 & 4 & 24 & 6 & 0 & 4 \\ 25 & 24 & 94 & 24 & 0 & 16 \\ 6 & 6 & 24 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.43 $\textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow (-1)$

$T = 8$ [0] $1^{11111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 8$, $T_{\max} = 12$, $\Delta = -89$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 3 & 0 & 3 \\ 6 & 0 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.44 $\textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{21} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -4$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1241}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -7$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 5 & 8 & 1 & 3 \\ 5 & 6 & 16 & 0 & 4 \\ 8 & 16 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -7$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 1 & 4 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 4 & 0 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{113}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -9$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 0 & 6 & 0 & 2 \\ 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{144}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.6$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 8 & 8 & 1 & 3 \\ 8 & 30 & 32 & 0 & 8 \\ 8 & 32 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{134}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.4$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 6 & 8 & 1 & 3 \\ 6 & 16 & 24 & 0 & 6 \\ 8 & 24 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{162}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.4$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 12 & 4 & 1 & 3 \\ 12 & 70 & 24 & 0 & 12 \\ 4 & 24 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{125}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 6 & 20 & 0 & 4 \\ 10 & 20 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{116}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.3$

- $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 1 & 2 & 12 & 1 & 3 \\ 2 & 0 & 12 & 0 & 2 \\ 12 & 12 & 69 & 0 & 11 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 11 & -1 & 0 \end{pmatrix}_{b_0}$$
- [28] $1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 5 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{16}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 5 & 36 & 1 & 5 \\ 36 & 214 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 5 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$
- [25] $1^{15}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 5 & 30 & 1 & 5 \\ 30 & 148 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 5 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^{117}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.6$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 1 & 2 & 14 & 1 & 3 \\ 2 & 0 & 14 & 0 & 2 \\ 14 & 14 & 95 & 0 & 13 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 13 & -1 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{112}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{172}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 1 & 14 & 4 & 1 & 3 \\ 14 & 96 & 28 & 0 & 14 \\ 4 & 28 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{163}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 1 & 12 & 6 & 1 & 3 \\ 12 & 70 & 36 & 0 & 12 \\ 6 & 36 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{154}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 1 & 10 & 8 & 1 & 3 \\ 10 & 48 & 40 & 0 & 10 \\ 8 & 40 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{142}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 1 & 8 & 4 & 1 & 3 \\ 8 & 30 & 16 & 0 & 8 \\ 4 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{121}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 22 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 5 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{135}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
- $$\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 16 & 30 & 0 & 6 \\ 10 & 30 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$
- [23] $1^{123}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 6 & 12 & 0 & 4 \\ 6 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{133}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 3 \\ 6 & 16 & 18 & 0 & 6 \\ 6 & 18 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{173}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 14 & 6 & 1 & 3 \\ 14 & 96 & 42 & 0 & 14 \\ 6 & 42 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{124}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 6 & 16 & 0 & 4 \\ 8 & 16 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{143}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 8 & 6 & 1 & 3 \\ 8 & 30 & 24 & 0 & 8 \\ 6 & 24 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{153}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 10 & 6 & 1 & 3 \\ 10 & 48 & 30 & 0 & 10 \\ 6 & 30 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{164}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 12 & 8 & 1 & 3 \\ 12 & 70 & 48 & 0 & 12 \\ 8 & 48 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{174}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.9$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 14 & 8 & 1 & 3 \\ 14 & 96 & 56 & 0 & 14 \\ 8 & 56 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{115}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 10 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{171}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 42 & 1 & 5 \\ 42 & 292 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 5 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{126}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 6 & 24 & 0 & 4 \\ 12 & 24 & 69 & 0 & 11 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 11 & -1 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{145}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 8 & 10 & 1 & 3 \\ 8 & 30 & 40 & 0 & 8 \\ 10 & 40 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{122}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 0 & 4 \\ 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{13}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 18 & 1 & 5 \\ 18 & 52 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 5 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{14}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 24 & 1 & 5 \\ 24 & 94 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 5 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{13}2^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 6 & 4 & 1 & 3 \\ 6 & 16 & 12 & 0 & 6 \\ 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{11}4^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{15}2^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 10 & 4 & 1 & 3 \\ 10 & 48 & 20 & 0 & 10 \\ 4 & 20 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{11}3^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 6 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{14}5^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -13, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 9 & 10 & 1 & 3 \\ 9 & 30 & 40 & 0 & 8 \\ 10 & 40 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{11}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -13, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 1 & 1 & 5 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{13}4^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -19, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 7 & 8 & 1 & 3 \\ 7 & 16 & 24 & 0 & 6 \\ 8 & 24 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{11}2^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -19, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 0 & 4 & 0 & 2 \\ 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{12}3^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -21, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 5 & 6 & 1 & 3 \\ 5 & 6 & 12 & 0 & 4 \\ 6 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{11}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -25, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 7 & 1 & 5 \\ 7 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 5 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -31$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.7$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 5 & 4 & 1 & 3 \\ 5 & 6 & 8 & 0 & 4 \\ 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{144}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -31$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.6$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 9 & 8 & 1 & 3 \\ 9 & 30 & 32 & 0 & 8 \\ 8 & 32 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{133}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -33$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 7 & 6 & 1 & 3 \\ 7 & 16 & 18 & 0 & 6 \\ 6 & 18 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -37$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 22 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 5 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{132}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -43$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 7 & 4 & 1 & 3 \\ 7 & 16 & 12 & 0 & 6 \\ 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{143}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -45$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 9 & 6 & 1 & 3 \\ 9 & 30 & 24 & 0 & 8 \\ 6 & 24 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{131}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -49$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 5 & 19 & 1 & 5 \\ 19 & 52 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 5 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{142}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -55$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 9 & 4 & 1 & 3 \\ 9 & 30 & 16 & 0 & 8 \\ 4 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{141}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -61$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 5 & 25 & 1 & 5 \\ 25 & 94 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 5 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 6$ [54] $1^{111}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 90$, $\Delta = -132$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 13, T_{\max} = 90, T \in [13, 90]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

1.45 $\textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{black}{so}_8 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{11}1^{211} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{black}{so}_8 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 98$, $\Delta = -160$, $\det = 10$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.5$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & 0 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{black}{so}_8 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 98$, $\Delta = -160$, $\det = 46$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 5 & 1 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 6 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{1111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 98, \Delta = -160, \det = 48, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 5 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 2 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{1111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 98, \Delta = -160, \det = 94, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.34 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 5 & 7 & 6 & 1 & 5 \\ 7 & 4 & 6 & 0 & 4 \\ 6 & 6 & 2 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

1.46 $\textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] 1^{11}1^{211} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & 0 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{112}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 15, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 5 & 1 & 12 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 12 & 0 & 21 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{1113}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 21, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 5 & 7 & 18 & 1 & 5 \\ 7 & 4 & 18 & 0 & 4 \\ 18 & 18 & 51 & 0 & 11 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 33, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 5 & 1 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 6 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{1113}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 5 & 6 & 18 & 1 & 5 \\ 6 & 4 & 18 & 0 & 4 \\ 18 & 18 & 51 & 0 & 11 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{1112}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 5 & 6 & 12 & 1 & 5 \\ 6 & 4 & 12 & 0 & 4 \\ 12 & 12 & 21 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] 1^{1411}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 5 & 24 & 6 & 1 & 5 \\ 24 & 94 & 24 & 0 & 16 \\ 6 & 24 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[7] 1^{1211}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 22 & 12 & 0 & 8 \\ 6 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[8] 1^{1312}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -140, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 5 & 19 & 6 & 1 & 5 \\ 19 & 52 & 18 & 0 & 12 \\ 6 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1411}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 177$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.68 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 25 & 6 & 1 & 5 \\ 25 & 94 & 24 & 0 & 16 \\ 6 & 24 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.47 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$T = 7$ [0] $1^{1111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 44$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11$, $T_{\max} = 44$, $T \in [11, 44]$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 3 & 6 & 0 & 3 \\ 6 & 0 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 44$, $\Delta = -119$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11$, $T_{\max} = 44$, $T \in [11, 44]$

$$\begin{pmatrix} 5 & 6 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 3 & 6 & 0 & 3 \\ 6 & 6 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 44$, $\Delta = -119$, $\det = -189$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11$, $T_{\max} = 44$, $T \in [11, 44]$

$$\begin{pmatrix} 5 & 7 & 6 & 6 & 1 & 5 \\ 7 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 3 & 6 & 0 & 3 \\ 6 & 6 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.48 $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$

$T = 12$ [0] $31^{1^{11}2231}13^1 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 7 & 7 \\ 3 & 7 & 6 & 6 \\ 3 & 7 & 6 & 6 \end{pmatrix}_{b_0}$$

1.49 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 107$, $\Delta = -240$, $\det = -45$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13$, $T_{\max} = 107$, $T \in [13, 107]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^{11}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 101$, $\Delta = -211$, $\det = 18$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13$, $T_{\max} = 101$, $T \in [13, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

1.50 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{f}}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -214$, $\det = -37$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -3 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \ 1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 100, \ \Delta = -185, \ \det = 13, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.15 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -3 & 1 & -2 \end{pmatrix}_{b_0}$$

1.51 $su_2 + su_3 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \ 1^{11}1^2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 5, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 10.40 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & -3 & 0 & -1 \\ 2 & 0 & 1 & 1 \\ 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{17}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 14 & 1 & 3 \\ 14 & 95 & 0 & 13 \\ 1 & 0 & -1 & -1 \\ 3 & 13 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{16}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 12 & 1 & 3 \\ 12 & 69 & 0 & 11 \\ 1 & 0 & -1 & -1 \\ 3 & 11 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{15}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 10 & 1 & 3 \\ 10 & 47 & 0 & 9 \\ 1 & 0 & -1 & -1 \\ 3 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 3 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 5 & 1 & 5 \\ 1 & -1 & -1 \\ 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 15 & 0 & 5 \\ 1 & 0 & -1 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{14}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 12.6$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 29 & 0 & 7 \\ 1 & 0 & -1 & -1 \\ 3 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 8, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 1 & 1 \\ 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [9] \ 1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 98, \ \Delta = -162, \ \det = -5, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 98, \ T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & 1 & 1 \\ 2 & 2 & -1 & 1 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [10] \ 1^{11}221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 90, \ \Delta = -133, \ \det = 2, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 15, T_{\max} = 90, T \in [15, 90]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

1.52 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^{21} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{so}_8} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -190, \det = -17, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -3 & 0 & 0 & -1 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{so}_8} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -190, \det = -24, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 2 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 5 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{so}_8} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -190, \det = -29, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11.34 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 1^{11}21^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{so}_8} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 98, \Delta = -161, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

1.53 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^{22} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & -3 & 0 & 0 & -1 \\ 2 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^{12} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}31^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 5 & 18 & 1 & 5 \\ 18 & 51 & 0 & 11 \\ 1 & 0 & -1 & -1 \\ 5 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}51^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 5 & 30 & 1 & 5 \\ 30 & 147 & 0 & 19 \\ 1 & 0 & -1 & -1 \\ 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}21^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 21 & 0 & 7 \\ 1 & 0 & -1 & -1 \\ 5 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 3 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 5 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}41^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 5 & 24 & 1 & 5 \\ 24 & 93 & 0 & 15 \\ 1 & 0 & -1 & -1 \\ 5 & 15 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 5 & 8 & 0 & 3 \\ 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 3 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -21, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [9] 1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -141, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -1 & 1 & -1 \end{pmatrix}_{b_0}$$

1.54 $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{black}{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] 1^{11}1^{111} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -169, \det = 75, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

1.55 $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] 1^{11}1^{112} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -149, \det = 27, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -1 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11}1^{111} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -149, \det = 54, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -149, \det = 54, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 5 & 6 & 6 & 1 & 5 \\ 6 & 3 & 6 & 0 & 3 \\ 6 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.56 $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 7 \quad [0] 1^{11}1^{1111} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 88, \Delta = -128, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 88, T \in [12, 88]$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -1 & -1 & 1 & -1 \end{pmatrix}_{b_0}$$

1.57 $\mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^{11}1^1 + \mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -142, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -1 & -1 \\ 5 & 7 & -1 & -1 \end{pmatrix}_{b_0}$$

1.58 $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 101, \Delta = -212, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 101, \Delta = -183, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -4 & 2 & -2 \end{pmatrix}_{b_0}$$

1.59 $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -186, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 99, \Delta = -157, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

1.60 $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{12}2^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -163, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 12^{12}21^{11} + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 91, \Delta = -134, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 91, T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 2 & 2 \\ 2 & 3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}22^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 91, \Delta = -134, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 91, T \in [11, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 1 & 2 & 2 \\ 2 & 3 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

1.61 $\mathfrak{su}_2 \rightarrow (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^{11}1^1 + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{(-3) \rightarrow (-2)} + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\hat{1} \rightarrow 1} \quad T = 6, \quad T_{\max} = 99, \quad \Delta = -162, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 99, \quad T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^{11}1^1 + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{(-3) \rightarrow (-2)} + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\hat{1} \rightarrow 1} \quad T = 7, \quad T_{\max} = 91, \quad \Delta = -133, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, \quad T_{\max} = 91, \quad T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

1.62 $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 8 \quad [0] \quad 1^{123}1^{131}1^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}_6} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -165, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 3 & 3 \\ 2 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

1.63 $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$

$$T = 9 \quad [0] \quad 1^{12313}1^{21}1^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}_7} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 9, \quad T_{\max} = 103, \quad \Delta = -192, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 16.33 \rightarrow 17, \quad T_{\max} = 103, \quad T \in [17, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 1^{122313}1^{21}1^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}_7} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 10, \quad T_{\max} = 103, \quad \Delta = -163, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 17, \quad T_{\max} = 103, \quad T \in [17, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{123}1^{141321}1^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}_7} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 11, \quad T_{\max} = 104, \quad \Delta = -162, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 18, \quad T_{\max} = 104, \quad T \in [18, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 3 & 3 \\ 2 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

1.64 $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$

$$T = 9 \quad [0] \quad 1^{12313}1^{21}1^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}'_7} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -164, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.65 $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow (-1)$

$$T = 10 \quad [0] \quad 1^1 22313^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow 1 \quad T = 10, \quad T_{\max} = 190, \quad \Delta = -278, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.67 \rightarrow 20, \quad T_{\max} = 190, \quad T \in [20, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \quad 1^1 23^1 1413221^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow 1 \quad T = 12, \quad T_{\max} = 158, \quad \Delta = -248, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 20.80 \rightarrow 21, \quad T_{\max} = 158, \quad T \in [21, 158]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -9 & 0 & -7 \\ 1 & 2 & 0 & 3 & 3 \\ 2 & 2 & -7 & 3 & -3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [0] \quad 1^1 23^1 141413221^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow 1 \quad T = 14, \quad T_{\max} = 109, \quad \Delta = -218, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 22, \quad T_{\max} = 109, \quad T \in [22, 109]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -9 & 0 & -7 \\ 1 & 3 & 0 & 4 & 4 \\ 2 & 3 & -7 & 4 & -2 \end{pmatrix}_{b_0}$$

1.66 $\mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$

$$T = 7 \quad [0] \quad 2^3 3^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 7, \quad T_{\max} = 30, \quad \Delta = -117, \quad \det = -5, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 30, \quad T \in [11, 30]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 3 & 3 \\ 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 2^2 3^1 1^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -90, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 4 & 4 \\ 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 2313^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^1 22313^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -59, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^1 23^1 141321^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[5] \quad 31^1 5^1 13^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[6] \quad 2^1 23^1 141^1 3^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 5 & 5 \\ 2 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{123^1} 1413221^1 + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 6 & 6 \\ 2 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[8] \quad 31^{1^{12231}} 5^{13^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \quad 2^{1^{123^1} 14141} 3^2 + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -30, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 6 & 6 \\ 2 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \quad 321^{3^{121}} 8^{12321^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 13, \quad T_{\max} = 16, \quad \Delta = -87, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [1] \quad 2^{1^{123^1} 1414141} 3^2 + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = 0, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 7 & 7 \\ 2 & 6 & 7 & 7 \end{pmatrix}_{b_0}$$

1.67 $\textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-2)$

$$T = 5 \quad [0] \quad 13^{112^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 5, \quad T_{\max} = 86, \quad \Delta = -128, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 12, \quad T_{\max} = 86, \quad T \in [12, 86]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 123^{112^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 6, \quad T_{\max} = 18, \quad \Delta = -101, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13, \quad T_{\max} = 18, \quad T \in [13, 18]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \quad 31^{2^{11}} 5^{13^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -101, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13, \quad T_{\max} = 20, \quad T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 31^{3^{121}} 6^{12^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -100, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 321^{3^{121}} 7^{12^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 10, \quad T_{\max} = 18, \quad \Delta = -99, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, \quad T_{\max} = 18, \quad T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [6] \quad 3221 \quad 8 \quad 12^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 18, T \in [16, 18]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 2 & 2 \\ 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1 \quad 8 \quad 123^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 18, T \in [16, 18]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [7] \quad 41321 \quad 8 \quad 12^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 18, \Delta = -97, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 2 & 2 \\ 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \quad 512321 \quad 8 \quad 12^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 18, \Delta = -94, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$\mathbf{1.68} \quad \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \quad 2 \quad 3 \quad 21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 12, \Delta = -90, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 3 & 3 \\ 2 & 0 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^1 23^1 13^1 21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -61, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^1 23^1 1413^1 21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[3] \quad 3^1 1^1 5 \quad 13^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -2 & 0 & 0 \\ 3 & 0 & 5 & 5 \\ 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [0] \quad 1^1 23^1 14141413^1 21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 9.0$$

$$\begin{aligned}
[1] \quad & 1^1 2321^1 8^1 123^1 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 16, \Delta = -60, \det = 3, (n_+, n_-, n_0) = \\
& (1, 14, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -2 & 0 & 0 \\ 3 & 0 & 5 & 5 \\ 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

1.69 $\mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$\begin{aligned}
T = 4 \quad & [0] \quad 1^1 1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 179, \Delta = -270, \det = 18, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 13, T_{\max} = 179, T \in [13, 179] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 5 \quad & [1] \quad 1^1 1^1 21^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 112, \Delta = -241, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 13, T_{\max} = 112, T \in [13, 112] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 8 \quad & [2] \quad 1^1 2321^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 15, T_{\max} = 101, T \in [15, 101] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -3 & 1 & -1 \\ 1 & 0 & 1 & 2 & 2 \\ 2 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

1.70 $\mathfrak{su}_3 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$

$$\begin{aligned}
T = 8 \quad & [0] \quad 1^1 2321^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_7 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 103, \Delta = -220, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16, T_{\max} = 103, T \in [16, 103] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -5 & 1 & -3 \\ 1 & 0 & 1 & 2 & 2 \\ 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}
\end{aligned}$$

1.71 $\mathfrak{su}_3 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$

$$\begin{aligned}
T = 8 \quad & [0] \quad 1^1 2321^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}'_7 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 102, \Delta = -192, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.33 \rightarrow 16, T_{\max} = 102, T \in [16, 102] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -4 & 1 & -2 \\ 1 & 0 & 1 & 2 & 2 \\ 2 & -1 & -2 & 2 & -2 \end{pmatrix}_{b_0}
\end{aligned}$$

1.72 $\mathfrak{su}_3 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$$\begin{aligned}
T = 4 \quad & [0] \quad 1^1 1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 133, \Delta = -244, \det = 15, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 133, T \in [13, 133] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 5 \quad & [1] \quad 1^1 1^1 21^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 100, \Delta = -215, \det = -7, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.2 \\
& T_{\min} = 12.14 \rightarrow 13, T_{\max} = 100, T \in [13, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 3 & 1 & -1 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 7 \quad & [2] \quad 1^1 321^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2 \\
& T_{\min} = 14, T_{\max} = 100, T \in [14, 100]
\end{aligned}$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -5 & 0 & -3 \\ 2 & 0 & 3 & 3 \\ 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

1.73 $\mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 3 \quad [1] \quad 1^2 1^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -221, \det = -3, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 2 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^1 1^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -221, \det = -3, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [2] \quad 1^1 21^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -192, \det = 2, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 2^1 231^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \quad 1^1 2^1 321^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 27, T \in [11, 27]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 7 & 5 \\ 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [5] \quad 2^1 3^1 21^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 25, T \in [12, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [6] \quad 1^1 231^1 51321^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 9 & 7 \\ 2 & -1 & 7 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [7] \quad 1^1 231^1 513221^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 10 & 8 \\ 2 & -1 & 8 & 6 \end{pmatrix}_{b_0}$$

1.74 $\mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [1] \quad 1^1 1^1 1^1 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 99, \Delta = -220, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^1 1^1 1^1 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 99, \Delta = -220, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^1 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -191, \det = -5, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

1.75 $su_3 + su_3 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [5] \quad 1^{13}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & -2 & 3 & 0 & 0 \\ 3 & 3 & 6 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{15}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 5 & 1 & 2 \\ 1 & -2 & 5 & 0 & 0 \\ 5 & 5 & 22 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{17}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 7 & 1 & 2 \\ 1 & -2 & 7 & 0 & 0 \\ 7 & 7 & 46 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{22}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 1 \\ 2 & 4 & 1 & 0 & 1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{14}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 4 & 0 & 0 \\ 4 & 4 & 13 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{16}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 6 & 1 & 2 \\ 1 & -2 & 6 & 0 & 0 \\ 6 & 6 & 33 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{23}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 2 & 3 & 1 & 2 \\ 2 & 1 & 6 & 0 & 1 \\ 3 & 6 & 6 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{24}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 2 & 4 & 1 & 2 \\ 2 & 1 & 8 & 0 & 1 \\ 4 & 8 & 13 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [10] \quad 1^{11}21^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -171, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 3 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.76 $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{so}_8} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -199, \det = -36, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{11}21^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{so}_8} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -170, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 3 & 6 & 0 & 3 \\ 3 & 6 & 2 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

1.77 $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}31^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -179, \det = -27, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 3 & 0 & 0 \\ 1 & 1 & -2 & 3 & 0 & 0 \\ 3 & 3 & 3 & 6 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}21^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -179, \det = -27, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 2 & 0 & 0 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 2 & 2 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -179, \det = -27, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.78 $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 98, \Delta = -158, \det = 81, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 0 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

1.79 $\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}21^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -151, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 5 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

1.80 $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^1 2^1 1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 98, \ \Delta = -172, \ \det = -6, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 98, \ T \in [12, 98]$$

$$\begin{pmatrix} 5 & 1 & 5 \\ 1 & -1 & -1 \\ 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \ 1^1 2^2 1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 96, \ \Delta = -143, \ \det = 3, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 96, \ T \in [14, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

1.81 $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \ 1^1 1^2 1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{green}{\mathfrak{so}_8} + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 99, \ \Delta = -171, \ \det = 15, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 99, \ T \in [12, 99]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 11 & 0 & 5 \\ 1 & 0 & -1 & -1 \\ 5 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

1.82 $(-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^1 2^1 1^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_6} + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 123, \ \Delta = -242, \ \det = -15, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 123, \ T \in [14, 123]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \ 1^1 2^1 21^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_6} + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 101, \ \Delta = -213, \ \det = 7, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 3 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^1 2^1 21^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_6} + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 101, \ \Delta = -213, \ \det = 12, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -6 & 0 & -4 \\ 3 & 0 & 6 & 4 \\ 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

1.83 $(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^1 2^1 1^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_7} + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 187, \ \Delta = -297, \ \det = -21, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.90 \rightarrow 15, \ T_{\max} = 187, \ T \in [15, 187]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \ 1^1 2^1 21^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_7} + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 180, \ \Delta = -268, \ \det = 11, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.91 \rightarrow 15, \ T_{\max} = 180, \ T \in [15, 180]$$

$$\begin{pmatrix} 10 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 180$, $\Delta = -268$, $\det = 18$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15$, $T_{\max} = 180$, $T \in [15, 180]$
 $\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -8 & 0 & -6 \\ 3 & 0 & 6 & 4 \\ 3 & -6 & 4 & -2 \end{pmatrix}_{b_0}$

1.84 $(-3) \rightarrow (-2) + \textcolor{blue}{e}'_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}1^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{e}'_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 179$, $\Delta = -269$, $\det = -18$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14$, $T_{\max} = 179$, $T \in [14, 179]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$

$T = 6$ [1] $1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{e}'_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 108$, $\Delta = -240$, $\det = 9$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14$, $T_{\max} = 108$, $T \in [14, 108]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}$

[2] $1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{e}'_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 108$, $\Delta = -240$, $\det = 15$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.13 \rightarrow 15$, $T_{\max} = 108$, $T \in [15, 108]$
 $\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -7 & 0 & -5 \\ 3 & 0 & 6 & 4 \\ 3 & -5 & 4 & -2 \end{pmatrix}_{b_0}$

1.85 $(-3) \rightarrow (-2) + \textcolor{blue}{f}_4 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}1^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -216$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$

$T = 6$ [1] $1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -187$, $\det = 5$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$

[2] $1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -187$, $\det = 9$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -5 & 0 & -3 \\ 3 & 0 & 6 & 4 \\ 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$

1.86 $(-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$T = 4$ [0] $12^{\textcolor{blue}{1}}1^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1$ $T = 4$, $T_{\max} = 98$, $\Delta = -193$, $\det = 3$, $(n_+, n_-, n_0) = (1, 4, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & 1 & -1 \end{pmatrix}_{b_0}$

[2] $1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}1^{\textcolor{blue}{1}} + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1$ $T = 4$, $T_{\max} = 98$, $\Delta = -193$, $\det = 11$, $(n_+, n_-, n_0) = (1, 4, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 98$, $T \in [11, 98]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}_{b_0}$

[1] $21^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1$ $T = 4$, $T_{\max} = 98$, $\Delta = -193$, $\det = 11$, $(n_+, n_-, n_0) = (1, 4, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 98$, $T \in [11, 98]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T = 5 \quad [3] \quad 12^1 21^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 98, T \in [12, 98] \\
& \begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0} \\
& [4] \quad 122^1 1^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 4 \\ 3 & 4 & 0 \end{pmatrix}_{b_0} \\
& [5] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 98, T \in [11, 98] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 6 & 4 \\ 3 & 4 & 2 \end{pmatrix}_{b_0} \\
\\
& T = 14 \quad [0] \quad 2^1 1^1 \quad 8^1 \quad 12321^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
\\
& \mathbf{1.87} \quad (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1) \\
& T = 5 \quad [0] \quad 1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -192, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 99, T \in [12, 99] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0} \\
\\
& T = 6 \quad [1] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0 \\
& T_{\min} = 12, T_{\max} = 99, T \in [12, 99] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 3 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 3 & 3 & -1 & 1 \end{pmatrix}_{b_0} \\
& [2] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99] \\
& \begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -4 & 0 & -2 \\ 3 & 0 & 6 & 4 \\ 3 & -2 & 4 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

2 $\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$ External (5 combos, 6 bases)

2.1 $\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 3 \quad [1] \quad 12^2 1 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 110, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 110, T \in [11, 110]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[0] \quad 212^2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 110, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 110, T \in [11, 110]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

2.2 $\mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 4 \quad [0] \quad 12^2 1^1 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 4, T_{\max} = 99, \Delta = -211, \det = 8, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

2.3 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 5 \quad [0] \quad 1^1 2^2 1^1 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

2.4 $(-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 4 \quad [0] \quad 2^1 12^2 + (-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 4, T_{\max} = 99, \Delta = -221, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & -4 & -2 \\ 2 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

2.5 $\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2)$

$$T = 6 \quad [0] \quad 14^2 122^1 + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2) \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 10, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.5$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 \\ 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

3 $\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$ External (67 combos, 581 bases)

3.1 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 1219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 1 & 4 \\ 6 & 4 & 9 & 3 & 0 & 4 \\ 8 & 9 & 9 & 4 & 0 & 5 \\ 3 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

3.2 $\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 1 & 4 \\ 6 & 4 & 9 & 3 & 0 & 4 \\ 8 & 9 & 9 & 4 & 0 & 5 \\ 3 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

3.3 $\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 12 & 20 & 1 & 7 \\ 12 & 10 & 17 & 0 & 6 \\ 20 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 12 & 16 & 1 & 6 \\ 12 & 13 & 18 & 0 & 7 \\ 16 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 15 & 17 & 1 & 7 \\ 15 & 13 & 18 & 0 & 7 \\ 17 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 15 & 20 & 1 & 7 \\ 15 & 13 & 21 & 0 & 7 \\ 20 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{11} 4^{211} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 10 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{211} 4^{11} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -313, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 10 & 1 & 5 \\ 7 & 5 & 11 & 0 & 5 \\ 10 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

3.4 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2)$

$$T = 10 \quad [0] \quad 1^{11} 5^{11} 1^1 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 & 0 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5^{2^1 2^1 1^1} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 2 & 1 & 3 \\ 5 & 8 & 13 & 4 & 0 & 6 \\ 6 & 13 & 13 & 5 & 0 & 7 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 7 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

3.5 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 5^{11} 1^1 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 1 & 4 & 2 & 0 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

3.6 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 1 & 4 \\ 8 & 9 & 4 & 8 & 0 & 5 \\ 3 & 4 & -3 & 3 & 0 & -1 \\ 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

3.7 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$

$$T = 9 \quad [0] \quad 1^{11} 2^1 3^1 1^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 36, \Delta = -118, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 36, T \in [13, 36]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 1 & 4 \\ 5 & 5 & 5 & 3 & 0 & 5 \\ 3 & 5 & -2 & 2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

3.8 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$

$$T = 7 \quad [0] \quad 1^{11} 3^{11} 1^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 100, \Delta = -173, \det = -350, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{11} 2^1 3^1 1^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}}231^{\overline{11}}2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 98, \ \Delta = -145, \ \det = 275, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 2^{\overline{1}}1^{\overline{1}}41^{\overline{1}}31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{\overline{1}}5^{\overline{211}}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 100, \ \Delta = -138, \ \det = 124, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.61 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [5] \ 1^{\overline{111}}5^{\overline{1231}}12^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -3, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, \ T_{\max} = 23, \ T \in [21, 23]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

3.9 NHC(-2)(-3) + $\mathfrak{e}_6$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{1}}31^{\overline{111}}2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -143, \ \det = 925, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 6 & 4 \\ 8 & 9 & 4 & 0 & 8 & 5 \\ 3 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 3 & 0 & 4 & 4 \\ 4 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

3.10 NHC(-2)(-3) + $\mathfrak{e}_6$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \rightarrow$ (-3)

$$T = 9 \quad [0] \ 1^{\overline{1}}231^{\overline{11}}1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 36, \ \Delta = -118, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, \ T_{\max} = 36, \ T \in [13, 36]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 3 & 4 \\ 5 & 5 & 5 & 0 & 3 & 5 \\ 3 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 4 & 5 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

3.11 NHC(-2)(-3) + $\mathfrak{e}_7$ + (-3) $\rightarrow$ (-2)

$$T = 8 \quad [0] \ 1^{\overline{1}}231^{\overline{11}}2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 102, \ \Delta = -200, \ \det = 425, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

3.12 NHC(-2)(-3) + $\mathfrak{e}'_7$ + (-3) $\rightarrow$ (-2)

$$T = 8 \quad [0] \ 1^{\overline{1}}231^{\overline{11}}2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 101, \ \Delta = -172, \ \det = 350, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

3.13 NHC(-2)(-3) + f₄ + su₂ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^1 3^1 1^{11} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 1 & 4 \\ 8 & 9 & 4 & 8 & 0 & 5 \\ 3 & 4 & -2 & 3 & 0 & 0 \\ 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

3.14 NHC(-2)(-3) + f₄ + su₂→(-3) + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^1 3^1 1^{11} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 1 & 4 \\ 8 & 9 & 4 & 4 & 0 & 5 \\ 3 & 4 & -2 & 2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 2^1 1^1 4^1 1^3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 1 & 6 & 0 & 0 & 3 \\ 5 & 6 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 6 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^1 5^1 1^3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 8 & 1 & 5 \\ 10 & 13 & 12 & 0 & 7 \\ 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

3.15 NHC(-2)(-3) + f₄ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^1 3^1 1^{11} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 97, \Delta = -147, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^1 2^1 3^1 1^{11} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$$

$$\begin{pmatrix} 13 & 20 & 1 & 7 \\ 20 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 2^1 1^1 4^1 1^3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{\overline{1}} 5^{\overline{1}} 13 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

3.16 NHC(-2)(-3) + $\mathfrak{f}_4$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 6 & 4 \\ 8 & 9 & 4 & 0 & 8 & 5 \\ 3 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 3 & 0 & 4 & 4 \\ 4 & 5 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

3.17 NHC(-2)(-3) + $\mathfrak{f}_4$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \rightarrow$ (-3)

$$T = 8 \quad [0] \quad 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 3 & 4 \\ 8 & 9 & 4 & 0 & 4 & 5 \\ 3 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 2 & 0 & 0 & 2 \\ 4 & 5 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 2^{\overline{1}} 1^{\overline{1}} 4 1^{\overline{1}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 1 & 6 & 0 & 0 & 3 \\ 5 & 6 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 6 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{\overline{1}} 5^{\overline{1}} 13 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 8 & 5 \\ 10 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 10 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

3.18 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2)

$$T = 8 \quad [0] \quad 1^{\overline{1}} 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$

[4] $1^{11}31^{14}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$

[5] $1^{11}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$

[6] $1^{11}31^{13}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$

[7] $1^{11}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$

$T = 9$ [8] $1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[9] $1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 16 & 1 & 5 \\ 10 & 13 & 24 & 0 & 7 \\ 16 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[10] $1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 14 & 1 & 5 \\ 10 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[11] $1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 21 & 1 & 5 \\ 10 & 13 & 30 & 0 & 7 \\ 21 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[12] $1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 23 & 1 & 5 \\ 10 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1^{21}1^{14}} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 28 & 1 & 5 \\ 10 & 13 & 40 & 0 & 7 \\ 28 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1^{21}1^{13}} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 30 & 1 & 5 \\ 10 & 13 & 44 & 0 & 7 \\ 30 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1^{21}1^{14}} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 37 & 1 & 5 \\ 10 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

3.19 NHC(-2)(-3) + so₈ + su₂→(-3) + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -95, \quad \det = 130, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 9 & 1 & 7 \\ 17 & 20 & 11 & 0 & 8 \\ 9 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -95, \quad \det = 325, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 9 & 1 & 7 \\ 20 & 25 & 12 & 0 & 9 \\ 9 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

3.20 NHC(-2)(-3) + so₈ + su₂n3mix + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -95, \quad \det = 145, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 24 & 1 & 7 \\ 17 & 20 & 28 & 0 & 8 \\ 24 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

3.21 NHC(-2)(-3) + so₈ + su₃ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -102, \quad \det = 146, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -102, \quad \det = 200, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -102, \quad \det = 209, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{21} 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

3.22 NHC(-2)(-3) + so₈ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^{11} 3 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 13 & 17 & 1 & 7 \\ 17 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 3 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 10 & 16 & 1 & 6 \\ 16 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 3 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 13 & 20 & 1 & 7 \\ 20 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [3] \quad 1^{11} 4 1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{21} 1^1 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{21} 1^1 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{231} 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 4 1^{11} 2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 4 & 7 & 2 & 0 & 4 \\ 3 & 7 & 3 & 2 & 0 & 3 \\ 1 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 2^1 1^4 1^1 31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \quad 1^1 5^1 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 4^1 14^1 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^1 2^1 5^1 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

3.23 NHC(-2)(-3) + so₈ + (-3) → (-2) + su₂

$$T = 8 \quad [0] \quad 1^{111} 31^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 12 & 7 \\ 17 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 10 & 6 \\ 7 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 10 & 16 & 1 & 12 & 6 \\ 16 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 15 & 7 \\ 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 17 & 0 & 13 & 7 \\ 7 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 31^{11} 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 27 & 7 \\ 17 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 33 & 0 & 49 & 13 \\ 7 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + (-3) \rightarrow (-2) + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 13 & 20 & 1 & 27 & 7 \\ 20 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 36 & 0 & 49 & 13 \\ 7 & 9 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[5] \quad & 1^{111}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 434, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.17 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 1 & 72 & 7 \\ 17 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 72 & 84 & 0 & 346 & 34 \\ 7 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{111}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 454, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.15 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 1 & 42 & 7 \\ 17 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 42 & 50 & 0 & 118 & 20 \\ 7 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{111}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 499, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.15 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 1 & 57 & 7 \\ 17 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 57 & 67 & 0 & 217 & 27 \\ 7 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{131}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 525, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.20 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 20 & 1 & 15 & 7 \\ 20 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 20 & 0 & 13 & 7 \\ 7 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0} \\
\\
T = 9 \quad [9] \quad & 1^{11}4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -69, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{12}4 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -229, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{11}4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{12}4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 14 & 5 \\ 10 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{12}4 \quad 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -589, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 21 & 5 \\ 10 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{11}4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0
\end{aligned}$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 23 & 5 \\ 10 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 4 \ 1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -904, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 28 & 5 \\ 10 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 4 \ 1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 30 & 5 \\ 10 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 30 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 4 \ 1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 37 & 5 \\ 10 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

3.24 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \rightarrow$ (-3)

$$T = 8 \quad [0] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 9 & 7 \\ 17 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 11 & 0 & 4 & 4 \\ 7 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 1 & 9 & 7 \\ 20 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 4 & 4 \\ 7 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

3.25 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 8 \quad [0] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 24 & 7 \\ 17 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 28 & 0 & 37 & 11 \\ 7 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

3.26 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 1 & 15 & 7 \\ 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 17 & 0 & 12 & 6 \\ 7 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 1 & 12 & 7 \\ 20 & 25 & 0 & 16 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 9 & 5 \\ 7 & 9 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 16 & 1 & 12 & 6 \\ 16 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [3] $1^{11}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 13 & 14 & 1 & 9 & 7 \\ 14 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 10 & 0 & 4 & 4 \\ 7 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{21}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

3.27 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + (-3) \rightarrow (-2)$

$T = 8$ [0] $1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 1 & 4 \\ 8 & 9 & 8 & 4 & 0 & 5 \\ 6 & 8 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

3.28 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2)$

$T = 8$ [0] $1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 1 & 4 \\ 8 & 9 & 8 & 4 & 0 & 5 \\ 6 & 8 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

3.29 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$T = 8$ [0] $1^{111}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$

[4] $1^{11}31^{14}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$

[5] $1^{11}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$

[6] $1^{11}31^{13}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$

[7] $1^{13}1^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$

$T = 9$ [8] $1^{12}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[9] $1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 16 & 1 & 5 \\ 10 & 13 & 24 & 0 & 7 \\ 16 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[10] $1^{12}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 14 & 1 & 5 \\ 10 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[11] $1^{13}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 21 & 1 & 5 \\ 10 & 13 & 30 & 0 & 7 \\ 21 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[12] $1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 23 & 1 & 5 \\ 10 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overset{2}{1}\overset{1}{4}} 4^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 28 & 1 & 5 \\ 10 & 13 & 40 & 0 & 7 \\ 28 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overset{2}{1}\overset{1}{4}} 4^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 30 & 1 & 5 \\ 10 & 13 & 44 & 0 & 7 \\ 30 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\overset{2}{1}\overset{1}{4}} 4^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 37 & 1 & 5 \\ 10 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

3.30 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_3 + (-3) \rightarrow (-2)$

$$T = 10 \quad [0] \quad 1^{\overset{1}{1}} \overset{1}{5} 1^{\overset{1}{2}} 2^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 3 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overset{1}{1}} \overset{1}{5} 1^{\overset{1}{2}} 2^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overset{1}{1}} \overset{1}{5} 1^{\overset{1}{2}} 2^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 10 & 3 & 0 & 4 \\ 8 & 10 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overset{1}{1}} \overset{1}{5} 1^{\overset{1}{2}} 2^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 85, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 5 & 7 & 8 & 4 & 0 & 6 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overset{1}{1}} \overset{1}{5} 1^{\overset{1}{2}} 2^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 11 & 3 & 0 & 4 \\ 7 & 11 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 181, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 13 & 17 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} \overset{1}{5} 1^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 181, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 15 & 3 & 0 & 4 \\ 9 & 15 & 40 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 196, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 10 & 14 & 38 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{2^1 21} \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 285, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 2 & 1 & 3 \\ 6 & 13 & 12 & 5 & 0 & 7 \\ 5 & 12 & 8 & 4 & 0 & 6 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 316, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 18 & 3 & 0 & 4 \\ 12 & 18 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 381, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 21 & 3 & 0 & 4 \\ 15 & 21 & 88 & 12 & 0 & 18 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1^2 21} \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 2 & 1 & 3 \\ 6 & 13 & 24 & 5 & 0 & 7 \\ 10 & 24 & 38 & 8 & 0 & 12 \\ 2 & 5 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1^2 21} \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 901, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 2 & 1 & 3 \\ 6 & 13 & 36 & 5 & 0 & 7 \\ 15 & 36 & 88 & 12 & 0 & 18 \\ 2 & 5 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

3.31 NHC(-2)(-3) + su₂ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^{11}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 6 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}312^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 3 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{131}1^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 5 & 0 & 4 \\ 6 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{131}1^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 10 & 0 & 4 \\ 12 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1331}1^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 17 & 0 & 4 \\ 15 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1231}1^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 9 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{131}1^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 9 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{131}1^{13}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 18 & 1 & 4 \\ 5 & 4 & 15 & 0 & 4 \\ 18 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 9 & 1 & 4 \\ 5 & 4 & 9 & 0 & 4 \\ 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{112}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 13 & 0 & 4 \\ 12 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 9 & 0 & 4 \\ 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{14}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 18 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{13}1^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 8 & 0 & 4 \\ 6 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 15 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 1 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{13}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 8 & 9 & 1 & 4 \\ 8 & 9 & 12 & 0 & 5 \\ 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 7 & -1 & 1 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 1 \end{pmatrix}_{b_0} \\
[18] \quad & 131^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 8 & 6 & 1 & 4 \\ 8 & 9 & 8 & 0 & 5 \\ 6 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 14 & 0 & 4 \\ 15 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0} \\
[20] \quad & 1^{11}431^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 21 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -1 & 1 \end{pmatrix}_{b_0} \\
[21] \quad & 1^{13}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 9 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[22] \ 1^{114}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{113}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 13 & 0 & 4 \\ 9 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{13}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 23 & 0 & 4 \\ 15 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{114}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 18 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{12}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 18 & 1 & 4 \\ 5 & 4 & 18 & 0 & 4 \\ 18 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{12}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 18 & 0 & 4 \\ 12 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{114}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 24 & 1 & 4 \\ 5 & 4 & 26 & 0 & 4 \\ 24 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}31^{13^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 21 & 1 & 4 \\ 5 & 4 & 19 & 0 & 4 \\ 21 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{13}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 21 & 1 & 4 \\ 5 & 4 & 22 & 0 & 4 \\ 21 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{13}31^{14^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 16 & 0 & 4 \\ 12 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 31^{111}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -409, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$\begin{aligned}
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 1 \end{pmatrix}_{b_0} \\
[33] & 1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 27 & 1 & 4 \\ 5 & 4 & 24 & 0 & 4 \\ 27 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0} \\
[34] & 22^1 1^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 7 & 7 & 7 & 4 \\ 7 & 5 & 8 & 7 & 5 \\ 7 & 8 & 4 & 7 & 4 \\ 7 & 7 & 7 & 5 & 5 \\ 4 & 5 & 4 & 5 & 2 \end{pmatrix}_{b_0} \\
[35] & 1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 17 & 0 & 4 \\ 12 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0} \\
[36] & 1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 24 & 1 & 4 \\ 5 & 4 & 23 & 0 & 4 \\ 24 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0} \\
[39] & 1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 22 & 0 & 4 \\ 15 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0} \\
[37] & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 32 & 0 & 4 \\ 21 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0} \\
[38] & 1^{13}31^{15}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 20 & 0 & 4 \\ 15 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0} \\
[40] & 1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -586, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 27 & 1 & 4 \\ 5 & 4 & 27 & 0 & 4 \\ 27 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[41] & 1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 27 & 0 & 4 \\ 18 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[42] & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -651, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 5 & 30 & 1 & 4 \\ 5 & 4 & 31 & 0 & 4 \\ 30 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0} \\
[43] & 1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 21 & 0 & 4 \\ 15 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0} \\
[44] & 1^{12}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -666, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 30 & 1 & 4 \\ 5 & 4 & 28 & 0 & 4 \\ 30 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0} \\
[45] & 31^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -724, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 3 & 6 & 1 & 2 \\ 3 & 1 & 4 & 6 & 1 & 3 \\ 3 & 4 & 0 & 6 & 1 & 2 \\ 6 & 6 & 6 & 10 & 2 & 6 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 6 & 0 & 1 \end{pmatrix}_{b_0} \\
[46] & 1^{13}31^{16}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 24 & 0 & 4 \\ 18 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0} \\
[47] & 1^{13}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -861, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 33 & 1 & 4 \\ 5 & 4 & 32 & 0 & 4 \\ 33 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0} \\
[48] & 1^{12}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -864, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 26 & 0 & 4 \\ 18 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0} \\
[49] & 1^{13}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -984, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 31 & 0 & 4 \\ 21 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0} \\
[50] & 1^{14}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1006, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 36 & 1 & 4 \\ 5 & 4 & 36 & 0 & 4 \\ 36 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0} \\
[51] & 1^{14}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1024, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 24 & 1 & 3 \\ 4 & 4 & 36 & 0 & 4 \\ 24 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0} \\
[52] & 1^{13}31^{17}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1044, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 21 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[53] $31^{113}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 9 & 1 & 2 \\ 3 & 1 & 4 & 9 & 1 & 3 \\ 3 & 4 & 0 & 9 & 1 & 2 \\ 9 & 9 & 9 & 25 & 3 & 9 \\ 1 & 1 & 1 & 3 & -2 & 0 \\ 2 & 3 & 2 & 9 & 0 & 1 \end{pmatrix}_{b_0}$$

[54] $31^{114}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1984, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 12 & 1 & 2 \\ 3 & 1 & 4 & 12 & 1 & 3 \\ 3 & 4 & 0 & 12 & 1 & 2 \\ 12 & 12 & 12 & 46 & 4 & 12 \\ 1 & 1 & 1 & 4 & -2 & 0 \\ 2 & 3 & 2 & 12 & 0 & 1 \end{pmatrix}_{b_0}$$

[55] $31^{115}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -2929, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$

$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 15 & 1 & 2 \\ 3 & 1 & 4 & 15 & 1 & 3 \\ 3 & 4 & 0 & 15 & 1 & 2 \\ 15 & 15 & 15 & 73 & 5 & 15 \\ 1 & 1 & 1 & 5 & -2 & 0 \\ 2 & 3 & 2 & 15 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [56] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{11}4^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{11}4^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[59] $21^{14}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 13 & 12 & 6 & 7 \\ 6 & 12 & 8 & 5 & 6 \\ 3 & 6 & 5 & 0 & 2 \\ 4 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[60] $22^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 13 & 8 & 6 & 7 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 6 & 3 & 0 & 2 \\ 3 & 7 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{11}4^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 3 & 1 & 3 \\ 6 & 13 & 8 & 0 & 7 \\ 3 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

- [63] $1^{11} 4^{1^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$
- [64] $1^{11} 4^{1^1} 12^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [65] $1^{11} 4^{1^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [66] $1^{12} 4^{1^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$
- [67] $1^1 4^{1^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [68] $1^{11} 4^{1^1} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [69] $21^{11} 41^{11} 12^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 13 & 20 & 6 & 7 \\ 12 & 20 & 26 & 8 & 10 \\ 3 & 6 & 8 & 0 & 2 \\ 4 & 7 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [70] $1^{11} 4^{1^1} 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [72] $1^{13} 4^{1^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [71] $1^{12} 4^{1^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [73] $1^{12} 4^{1^1} 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{11} 4^{2^{11} 1^2} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{11} 4^{2^{11} 1^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 8 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{11} 4^{2^{11} 1^2} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{12} 4^{2^{11} 1^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{14} 1^{11} 22^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{11} 41^{11} 22^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 1 & 2 \\ 1 & 0 & 3 & 4 & 0 & 2 \\ 3 & 3 & 3 & 8 & 0 & 3 \\ 4 & 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[80] $22^{11} 1^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 8 & 5 & 6 \\ 3 & 6 & 5 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{13} 4^{2^{11} 1^2} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[82] $22^{11} 1^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 8 & 3 & 3 \\ 6 & 13 & 20 & 6 & 7 \\ 8 & 20 & 26 & 8 & 10 \\ 3 & 6 & 8 & 0 & 2 \\ 3 & 7 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[83] $21^{11} 1^{11} 41^{14} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 13 & 56 & 6 & 7 \\ 30 & 56 & 212 & 23 & 28 \\ 3 & 6 & 23 & 0 & 2 \\ 4 & 7 & 28 & 2 & 3 \end{pmatrix}_{b_0} \\
[84] \quad & 21^{11}41^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 13 & 32 & 6 & 7 \\ 18 & 32 & 68 & 13 & 16 \\ 3 & 6 & 13 & 0 & 2 \\ 4 & 7 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[85] \quad & 1^{11}41^{11}22^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 8 & 1 & 3 \\ 6 & 13 & 20 & 0 & 7 \\ 8 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[86] \quad & 1^{13}2^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[87] \quad & 1^{12}2^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[88] \quad & 21^{11}41^{13}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 13 & 44 & 6 & 7 \\ 24 & 44 & 130 & 18 & 22 \\ 3 & 6 & 18 & 0 & 2 \\ 4 & 7 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[89] \quad & 1^{13}2^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[90] \quad & 1^{13}2^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 18 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0} \\
[91] \quad & 1^{12}2^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0} \\
[93] \quad & 1^{12}2^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 19 & 25 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[92] \quad & 1^{14}2^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[94] & 1^{1^{2^1 1^3}} 4 \ 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[95] & 1^{1^{2^1 1^2}} 4 \ 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 4 & 19 & 0 & 4 \\ 13 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[96] & 1^{1^{4^2 1^1}} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0} \\
[97] & 1^{1^4 1^{12} 2^2 1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 13 & 24 & 0 & 7 \\ 10 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[98] & 22^{1^2 1^{12} 4^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 10 & 3 & 3 \\ 6 & 13 & 24 & 6 & 7 \\ 10 & 24 & 38 & 10 & 12 \\ 3 & 6 & 10 & 0 & 2 \\ 3 & 7 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[99] & 1^{1^3 2^{1^2}} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0} \\
[100] & 1^{1^4 2^{1^3}} 4 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 17 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0} \\
[101] & 1^{1^3 2^{1^3}} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[102] & 1^{1^4 2^{1^4}} 4 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 20 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[103] & 1^{1^4 2^{1^2}} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[104] & 1^{13} 4^{21} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 447, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 21 & 29 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0} \\
[105] & 1^{13} 4^{21} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 20 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[106] & 22^1 1^{12} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 475, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 13 & 3 & 3 \\ 6 & 13 & 32 & 6 & 7 \\ 13 & 32 & 68 & 13 & 16 \\ 3 & 6 & 13 & 0 & 2 \\ 3 & 7 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[107] & 1^{11} 4^{21} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 26 & 0 & 4 \\ 18 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[108] & 1^{11} 41^{12} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 13 & 1 & 3 \\ 6 & 13 & 32 & 0 & 7 \\ 13 & 32 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[109] & 1^{14} 4^{21} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 20 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0} \\
[110] & 1^1 41^{13} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 553, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 15 & 1 & 3 \\ 6 & 13 & 36 & 0 & 7 \\ 15 & 36 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[111] & 22^1 1^{13} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 15 & 3 & 3 \\ 6 & 13 & 36 & 6 & 7 \\ 15 & 36 & 88 & 15 & 18 \\ 3 & 6 & 15 & 0 & 2 \\ 3 & 7 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[112] & 1^{14} 4^{21} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 23 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0} \\
[113] & 1^{11} 4^{21} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =
\end{aligned}$$

6.0

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 23 & 1 & 3 \\ 4 & 4 & 33 & 0 & 4 \\ 23 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 22^1 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 846, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 18 & 3 & 3 \\ 6 & 13 & 44 & 6 & 7 \\ 18 & 44 & 130 & 18 & 22 \\ 3 & 6 & 18 & 0 & 2 \\ 3 & 7 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{11} 41^{13} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 18 & 1 & 3 \\ 6 & 13 & 44 & 0 & 7 \\ 18 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{11} 41^{14} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 20 & 1 & 3 \\ 6 & 13 & 48 & 0 & 7 \\ 20 & 48 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 22^1 1^{14} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 910, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 20 & 3 & 3 \\ 6 & 13 & 48 & 6 & 7 \\ 20 & 48 & 158 & 20 & 24 \\ 3 & 6 & 20 & 0 & 2 \\ 3 & 7 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 22^1 1^{14} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 23 & 3 & 3 \\ 6 & 13 & 56 & 6 & 7 \\ 23 & 56 & 212 & 23 & 28 \\ 3 & 6 & 23 & 0 & 2 \\ 3 & 7 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{11} 41^{14} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 23 & 1 & 3 \\ 6 & 13 & 56 & 0 & 7 \\ 23 & 56 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{11} 41^{15} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 25 & 1 & 3 \\ 6 & 13 & 60 & 0 & 7 \\ 25 & 60 & 248 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{11} 41^{16} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1876, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 30 & 1 & 3 \\ 6 & 13 & 72 & 0 & 7 \\ 30 & 72 & 358 & 0 & 36 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 36 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{11} 41^{17} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 2513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 35 & 1 & 3 \\ 6 & 13 & 84 & 0 & 7 \\ 35 & 84 & 488 & 0 & 42 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 42 & -1 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 9 \quad [123] \quad 22^1 1^1 5^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
& [125] \quad 1^{11} 5^1 1^4 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = \\
& 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
& [124] \quad 21^1 5^1 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = \\
& 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 10 & 3 & 4 \\ 9 & 10 & 19 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
& [127] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
& [126] \quad 22^1 1^1 5^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = \\
& 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
& [129] \quad 1^{12} 5^1 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
& [128] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
& [130] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
& [131] \quad 1^{11} 5^1 1^2 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = \\
& 6.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0} \\
& [132] \quad 1^{11} 5^1 1^2 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = \\
& 6.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[133] $1 \overset{1}{\underset{11}{5}} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[134] $21 \overset{1}{\underset{11}{5}} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[135] $1 \overset{1}{\underset{11}{5}} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[136] $1 \overset{1}{\underset{11}{5}} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[140] $1 \overset{1}{\underset{11}{5}} 1^2 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[139] $21 \overset{1}{\underset{11}{5}} 1^2 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 1 & 2 \\ 3 & 4 & 5 & 6 & 0 & 4 \\ 2 & 5 & 3 & 4 & 0 & 3 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[138] $1 \overset{1}{\underset{11}{5}} 1^4 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[137] $1 \overset{1}{\underset{11}{5}} 1^2 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 6 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[142] $1 \overset{1}{\underset{11}{5}} 1^3 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[141] & 1^{11} 5^{1^1} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 13 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[145] & 1^{11} 5^{1^4} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 19 & 25 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[144] & 1^{11} 5^{1^1} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[143] & 1^{12} 5^{1^2} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[146] & 1^{1^2} 5^{2^1} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 13 & 5 & 0 & 7 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[147] & 22^{11} 1^{1^1} 5^{1^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 7 & 10 & 19 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[148] & 22^{11} 1^{1^3} 5^{1^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 17 & 22 & 115 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[149] & 21^{11} 5^{1^1} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[150] & 1^{13} 5^{1^4} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{11}5^{12}1^21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{11}5^{12}1^32^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{11}5^{13}1^32^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{11}5^{21}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{11}5^{11}1^21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{11}5^{21}21^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[157] $22^11^{12}5^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 57 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[158] $21^{11}5^{21}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 7 & 2 & 3 \\ 4 & 4 & 10 & 3 & 4 \\ 7 & 10 & 19 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{11}5^{13}1^22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{11}5^{1^1}1^{12}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[161] $22^{11}1^{1^1}5^{1^1}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 36 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{13}5^{1^2}12^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[163] $22^{11}1^{1^1}5^{1^1}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 59 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{13}5^{1^3}12^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{11}5^{2^1}1^{2^1}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{11}5^{1^2}1^{2^1}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{1^1}5^{1^1}1^{12}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[168] $21^{1^2}5^{1^1}12^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -86$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{12} 5^{14} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{12} 5^{21} 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{12} 5^{11} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{11} 5^{13} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{11} 5^{12} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{11} 5^{12} 1^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[175] $21^{12} 5^{21} 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 14 & 20 & 82 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{12} 5^{13} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 13 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{12} 5^{12} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[178] $21^{13}5^{11}12^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[179] $22^{12}5^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 14 & 20 & 82 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{21}5^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 7 & 1 & 3 \\ 6 & 13 & 17 & 0 & 7 \\ 7 & 17 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{12}5^{12}1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{21}5^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{13}5^{11}12^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{13}5^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{12}5^{13}12^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 15 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[185] $21^{12}5^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 10 & 14 & 36 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{1^4} 5^{1^2} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 150 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[188] $22^1 1^3 5^1 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 19 & 26 & 148 & 19 & 24 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{1^2} 5^{1^2} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{1^2} 5^{2^1} 2^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{1^3} 5^{1^2} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{1^1} 5^{1^2} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{1^2} 5^{1^3} 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[194] $22^1 1^3 5^1 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 16 & 24 & 113 & 16 & 21 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{1^3} 5^{1^4} 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{13}_5 1^{13}_5 1^{13}_5 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{12}_5 1^{12}_5 1^{12}_5 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[198] $21^{13}_5 1^{13}_5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 30 & 3 & 4 \\ 21 & 30 & 187 & 21 & 27 \\ 2 & 3 & 21 & 0 & 2 \\ 3 & 4 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{12}_5 1^{12}_5 1^{12}_5 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 21 & 29 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{12}_5 1^{12}_5 1^{12}_5 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{12}_5 1^{12}_5 1^{12}_5 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{13}_5 1^{13}_5 1^{13}_5 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{12}_5 1^{12}_5 1^{12}_5 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 24 & 32 & 234 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{13}_5 1^{13}_5 1^{13}_5 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{\overline{1}^2 \overline{1}^2 \overline{1}^2} 5^{\overline{1}^2} 1^{\overline{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 14 & 1 & 3 \\ 6 & 13 & 34 & 0 & 7 \\ 14 & 34 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{\overline{1}^3} 5^{\overline{1}^2} 1^{\overline{2}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{\overline{1}^3} 5^{\overline{2}^1 \overline{1}^2} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[208] $22^{\overline{1}^3} 5^{\overline{1}^3} 1^{\overline{1}^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -297$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 21 & 30 & 187 & 21 & 27 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[209] $21^{\overline{1}^3} 5^{\overline{2}^1 \overline{1}^1} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 13 & 18 & 59 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{\overline{1}^3} 5^{\overline{1}^3} 1^{\overline{2}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -307$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{\overline{2}^1 \overline{1}^2} 5^{\overline{1}^1} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 13 & 24 & 0 & 7 \\ 10 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{\overline{1}^3} 5^{\overline{1}^3} 1^{\overline{2}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{\overline{1}^3} 5^{\overline{1}^4} 1^{\overline{2}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -321$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 20 & 30 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{13} 5^{2^1 1^2} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 113 & 16 & 21 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{13} 5^{2^1 21^{12}} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 12 & 1 & 3 \\ 6 & 13 & 29 & 0 & 7 \\ 12 & 29 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{13} 5^{1^2} 1^{3^2} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{13} 5^{1^3} 1^{3^2} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{13} 5^{2^1 21^3} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{13} 5^{1^4} 1^{3^2} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 23 & 33 & 232 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{13} 5^{2^1 21^4} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -461, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 26 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 26 & 36 & 281 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 33 & -1 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{13} 5^{2^1 21^{13}} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 21 & 1 & 3 \\ 6 & 13 & 51 & 0 & 7 \\ 21 & 51 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{13} 5^{2^1 21^{13}} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 3 \\ 6 & 13 & 36 & 0 & 7 \\ 15 & 36 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{\overline{1}^{21}1^{13}} 5^{\overline{1}^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -633$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 19 & 1 & 3 \\ 6 & 13 & 46 & 0 & 7 \\ 19 & 46 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{\overline{1}^{21}1^{13}} 5^{\overline{1}^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -657$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 17 & 1 & 3 \\ 6 & 13 & 41 & 0 & 7 \\ 17 & 41 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [225] $1^{\overline{1}^{11}1^{31}} 5^{\overline{1}^1} 12^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{\overline{1}^{11}1^{31}} 5^{\overline{1}^1} 12^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

[227] $1^{\overline{1}^{11}1^{31}} 5^{\overline{1}^1} 12^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 13$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{\overline{1}^{21}1^{11}} 5^{\overline{1}^1} 131^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 24$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 2 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{\overline{1}^{21}1^{11}} 5^{\overline{1}^1} 131^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 26$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 8 & 0 & 5 \\ 5 & 8 & 9 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

[230] $1^{\overline{1}^{11}1^{31}} 5^{\overline{1}^1} 12^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 29$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{\overline{1}^{21}1^{11}} 5^{\overline{1}^1} 131^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 52$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 5 & 13 & 0 & 5 \\ 7 & 13 & 25 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[232] \quad 1^{11} 5^{211} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 62, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 5 & 18 & 0 & 5 \\ 9 & 18 & 49 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[233] \quad 1^{11} 5^{211} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 69, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 4 & 4 & 5 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[234] \quad 1^{11} 5^{211} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 101, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 5 & 5 & 8 & 24 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[235] \quad 1^{12} 5^{211} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 108, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 5 & 6 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[236] \quad 1^{11} 5^{212} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 122, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 5 & 21 & 0 & 5 \\ 12 & 21 & 70 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[237] \quad 1^{11} 5^{212} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 7 & 6 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{11} 5^{212} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 5 & 26 & 0 & 5 \\ 14 & 26 & 106 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{12} 5^{211} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 168, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 6 & 7 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[240] \quad 1^{11} 5^{212} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 189, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 7 & 12 & 57 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{12}5^{211}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 7 & 8 & 12 & 57 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{12}5^{211^2}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 8 & 8 & 10 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{13}5^{211^1}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 7 & 9 & 10 & 52 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{11}5^{211^3}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 10 & 8 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{13}5^{211^1}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 8 & 10 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{11}5^{211^3}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 11 & 9 & 16 & 104 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{12}5^{211^2}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 9 & 9 & 13 & 73 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{13}5^{211^2}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 9 & 10 & 11 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{12}5^{211^2}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 10 & 10 & 16 & 102 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{12} 5^{2^1 1^3} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 11 & 10 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^{13} 5^{2^1 1^2} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 10 & 11 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{12} 5^{2^1 1^3} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 456, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 12 & 11 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{13} 5^{2^1 1^3} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 12 & 12 & 15 & 115 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{13} 5^{2^1 1^2} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 11 & 12 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[255] \quad 1^{13} 5^{2^1 1^3} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 613, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 13 & 13 & 18 & 148 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

3.32 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{13} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 3 & 4 \\ 8 & 9 & 8 & 0 & 4 & 5 \\ 6 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

3.33 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{13} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 3 & 4 \\ 8 & 9 & 8 & 0 & 4 & 5 \\ 6 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & -2 & 0 \\ 4 & 5 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

3.34 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3)→(-2) + $\mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{111}31^{12^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}31^{112^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}31^{112^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}31^{112^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}31^{142^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}31^{122^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111}31^{132^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{131}1^{112^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

3.35 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3)→(-2) + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{2^{11}}5^{131}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 2 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 5^{2^1 1^1} 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -12, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 9 & 9 & 0 & 36 & 12 \\ 2 & 2 & 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 5^{2^1 1^2} 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -24, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 9 & 13 & 13 & 0 & 73 & 17 \\ 2 & 2 & 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 5^{2^1 1^3} 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -28, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 17 & 2 \\ 1 & 1 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 12 & 11 & 17 & 17 & 0 & 124 & 22 \\ 2 & 2 & 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^1 5^{2^1 1^1} 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -33, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 13 & 5 \\ 2 & 5 & 2 & 0 & 10 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 10 & 0 & 25 & 11 \\ 3 & 5 & 4 & -1 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^1 5^{2^1 1^1} 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -53, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 18 & 5 \\ 2 & 5 & 2 & 0 & 14 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 18 & 14 & 0 & 49 & 15 \\ 3 & 5 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^1 5^{2^1 1^2} 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -96, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 26 & 5 \\ 2 & 5 & 2 & 0 & 20 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 26 & 20 & 0 & 106 & 22 \\ 3 & 5 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

3.36 NHC(-2)(-3) + $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \ 1^1 5^{2^1 1^1} 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 47, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 5 & 5 & 0 & 13 & 4 & 5 \\ 2 & 5 & 2 & 0 & 10 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 7 & 13 & 10 & 0 & 25 & 8 & 11 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 5 & 4 & -1 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^1 5^{2^1 1^2} 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 122, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 5 & 5 & 0 & 26 & 4 & 5 \\ 2 & 5 & 2 & 0 & 20 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 26 & 20 & 0 & 106 & 16 & 22 \\ 2 & 4 & 3 & 0 & 16 & 1 & 3 \\ 3 & 5 & 4 & -1 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

3.37 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3)→(-2) + $\mathfrak{su}_2$ →(-3)

$$T = 11 \quad [0] \quad 1^{11} \quad 5^{11} \quad 12^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 4 & 5 \\ 2 & 5 & 2 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 5 & 4 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \quad 5^{11} \quad 12^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \quad 5^{11} \quad 13^{11} \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 8 & 0 & 4 & 5 \\ 5 & 8 & 9 & 0 & 5 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 5 & 7 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \quad 5^{11} \quad 13^{11} \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 2 \\ 3 & 2 & 4 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} \quad 5^{11} \quad 13^{11} \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 4 & 4 & 5 & 11 & 0 & 4 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 7 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} \quad 5^{11} \quad 13^{11} \quad 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 5 & 13 & 0 & 4 & 5 \\ 7 & 13 & 25 & 0 & 8 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 3 \\ 3 & 5 & 11 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} \quad 5^{11} \quad 13^{11} \quad 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 5 & 18 & 0 & 4 & 5 \\ 9 & 18 & 49 & 0 & 11 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 11 & 0 & 1 & 3 \\ 3 & 5 & 15 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} \quad 5^{11} \quad 13^{11} \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 & 2 \\ 5 & 6 & 6 & 21 & 0 & 5 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 5^{21^2} 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 & 2 \\ 7 & 6 & 9 & 36 & 0 & 7 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 7 & 0 & 0 & 2 \\ 2 & 2 & 2 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 5^{21^2} 13^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 5 & 21 & 0 & 4 & 5 \\ 12 & 21 & 70 & 0 & 13 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 13 & 0 & 1 & 3 \\ 3 & 5 & 18 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 5^{21^2} 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 & 2 \\ 8 & 8 & 10 & 50 & 0 & 8 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 8 & 0 & 0 & 2 \\ 2 & 2 & 2 & 14 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 5^{21^2} 13^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 5 & 26 & 0 & 4 & 5 \\ 14 & 26 & 106 & 0 & 16 & 22 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 16 & 0 & 1 & 3 \\ 3 & 5 & 22 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{13} 5^{21^2} 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -271, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 9 & 10 & 11 & 68 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{12} 5^{21^3} 13^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -371, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 & 2 \\ 11 & 10 & 14 & 93 & 0 & 11 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 11 & 0 & 0 & 2 \\ 2 & 2 & 2 & 19 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{13} 5^{21^3} 13^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -476, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 1 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 & 2 \\ 12 & 12 & 15 & 115 & 0 & 12 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 12 & 0 & 0 & 2 \\ 2 & 2 & 2 & 21 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

3.38 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \rightarrow$ (-3) + $\mathfrak{su}_2$

$$T = 12 \quad [0] \ 1^{21^1} 5^{13} 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 5 & 5 & 0 & 4 & 13 & 5 \\ 2 & 5 & 2 & 0 & 3 & 10 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 13 & 10 & 0 & 8 & 25 & 11 \\ 3 & 5 & 4 & -1 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}^{\overline{2}^{\overline{1}^{\overline{2}}}} \ 5 \ 13^{\overline{1}^{\overline{1}^{\overline{4}}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 122, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 5 & 5 & 0 & 4 & 26 & 5 \\ 2 & 5 & 2 & 0 & 3 & 20 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 16 & 3 \\ 14 & 26 & 20 & 0 & 16 & 106 & 22 \\ 3 & 5 & 4 & -1 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

3.39 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 11 \quad [0] \ 1^{\overline{2}^{\overline{1}^{\overline{1}}}} \ 5 \ 13^{\overline{1}^{\overline{1}^{\overline{2}}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -50, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 17 & 5 \\ 2 & 5 & 2 & 0 & 13 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 13 & 0 & 44 & 14 \\ 3 & 5 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{2}^{\overline{1}^{\overline{2}}}} \ 5 \ 13^{\overline{1}^{\overline{1}^{\overline{4}}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -131, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 0 & 23 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 30 & 23 & 0 & 141 & 25 \\ 3 & 5 & 4 & -1 & 25 & 4 \end{pmatrix}_{b_0}$$

3.40 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1}^{\overline{5}^{\overline{1}^{\overline{2}^{\overline{1}}}}}} \ 5 \ 1^{\overline{1}^{\overline{1}^{\overline{2}^{\overline{1}}}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 13, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}^{\overline{1}^{\overline{5}^{\overline{1}^{\overline{2}^{\overline{1}}}}}}} \ 5 \ 12^{\overline{1}^{\overline{1}^{\overline{2}^{\overline{1}}}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 40, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}^{\overline{1}^{\overline{5}^{\overline{1}^{\overline{2}^{\overline{1}}}}}}} \ 5 \ 1^{\overline{1}^{\overline{2}^{\overline{2}^{\overline{1}}}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 60, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 3 & 4 \\ 8 & 10 & 21 & 0 & 6 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}^{\overline{1}^{\overline{5}^{\overline{1}^{\overline{2}^{\overline{1}}}}}}} \ 5 \ 1^{\overline{1}^{\overline{2}^{\overline{2}^{\overline{1}}}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 85, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 5 & 7 & 8 & 0 & 4 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

- [4] $1^{12}5^{11}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 11 & 0 & 3 & 4 \\ 7 & 11 & 21 & 0 & 6 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [5] $1^{13}5^{11}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 15 & 0 & 3 & 4 \\ 9 & 15 & 40 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [6] $1^{12}5^{13}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 13 & 17 & 61 & 0 & 10 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 15 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [7] $1^{12}5^{12}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 10 & 14 & 38 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{2^1 21^{11}}5^{11}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 5 & 1 & 2 & 3 \\ 6 & 13 & 12 & 0 & 5 & 7 \\ 5 & 12 & 8 & 0 & 4 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 7 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{13}5^{12}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 12 & 18 & 61 & 0 & 10 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 15 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [10] $1^{13}5^{13}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 15 & 21 & 88 & 0 & 12 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [11] $1^{2^1 21^{12}}5^{11}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 10 & 1 & 2 & 3 \\ 6 & 13 & 24 & 0 & 5 & 7 \\ 10 & 24 & 38 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 8 & 0 & 0 & 2 \\ 3 & 7 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [12] $1^{2^1 21^{13}}5^{11}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 901, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 2 & 3 \\ 6 & 13 & 36 & 0 & 5 & 7 \\ 15 & 36 & 88 & 0 & 12 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 12 & 0 & 0 & 2 \\ 3 & 7 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

3.41 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{11} 4^{211} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 4^{211} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

3.42 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + (-3)→(-2)

$$T = 10 \quad [0] \quad 1^{11} 51^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 51^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 51^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 10 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 8 & 10 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{5} 1^{\overline{2}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 85, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 5 & 7 & 4 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 1^{\overline{2}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 3 & 11 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 7 & 11 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} \overset{1}{5} 1^{\overline{3}2^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 3 & 17 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 13 & 17 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \overset{1}{5} 1^{\overline{2}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 3 & 15 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 9 & 15 & 8 & 40 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 1^{\overline{2}2^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 196, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 10 & 14 & 8 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{2}1} \overset{1}{5} 1^{\overline{2}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 5 & 1 & 3 \\ 6 & 13 & 5 & 12 & 0 & 7 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 5 & 12 & 4 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13} \overset{1}{5} 1^{\overline{2}2^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 316, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 3 & 18 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 12 & 18 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{13} \overset{1}{5} 1^{\overline{3}2^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 381, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 0 & 4 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 15 & 21 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121^{12}} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 10 & 1 & 3 \\ 6 & 13 & 5 & 24 & 0 & 7 \\ 2 & 5 & 0 & 8 & 0 & 2 \\ 10 & 24 & 8 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{121^{13}} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 901, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 15 & 1 & 3 \\ 6 & 13 & 5 & 36 & 0 & 7 \\ 2 & 5 & 0 & 12 & 0 & 2 \\ 15 & 36 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

3.43 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{1131^{13}2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 1 & 3 \\ 4 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 9 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1131^{11}2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 1 & 4 \\ 8 & 9 & 4 & 4 & 0 & 5 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1231^{11}2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 1 & 3 \\ 4 & 4 & 4 & 14 & 0 & 4 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 9 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1131^{12}2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 1 & 3 \\ 4 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 6 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1131^{11}2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 153, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 1 & 3 \\ 4 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11131^{11}2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 237, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 1 & 4 \\ 5 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 9 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1131^{11}2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 453, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 1 & 3 \\ 4 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 6 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

3.44 NHC(-2)(-3) + su₃ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 3 & 8 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 5 & 0 & 4 \\ 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 17 & 0 & 4 \\ 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{13}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 9 & 1 & 4 \\ 5 & 4 & 9 & 0 & 4 \\ 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 13 & 0 & 4 \\ 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 9 & 0 & 4 \\ 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 8 & 0 & 4 \\ 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 9 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 13 & 0 & 4 \\ 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{12}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 18 & 0 & 4 \\ 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 17 & 0 & 4 \\ 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[17] $31^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -561, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 \\ 3 & 4 & 0 & 3 & 1 \\ 3 & 3 & 3 & 0 & 1 \\ 1 & 1 & 1 & 1 & -2 \\ 2 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -596, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 22 & 0 & 4 \\ 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [19] $1^{21}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[20] $21^{11}41^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 4 \\ 8 & 13 & 12 & 0 & 7 \\ 6 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11}4^{12}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{11}4^{12}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 7 & 0 & 4 \\ 6 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11}4^{12}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{21}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{21}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[26] $22^{11}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[27] $21^{11}4^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 13 & 12 & 6 & 7 \\ 6 & 12 & 7 & 5 & 5 \\ 3 & 6 & 5 & 0 & 2 \\ 4 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{21}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 4 & 10 & 0 & 4 \\ 7 & 10 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{11}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{11}4^{11}22^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{21}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{\overline{2}1^{\overline{1}1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$

[33] $1^{\overline{2}1^{\overline{1}1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$

[34] $1^{\overline{1}} 41^{\overline{1}1} 22^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0}$

[35] $22^{\overline{1}} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 7 & 5 & 5 \\ 3 & 6 & 5 & 0 & 2 \\ 3 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$

[36] $1^{\overline{1}} 41^{\overline{1}1} 22^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 5 & -1 & 2 \end{pmatrix}_{b_0}$

$T = 9$ [37] $21^{\overline{1}} 51^{\overline{1}1} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[38] $1^{\overline{1}} 51^{\overline{1}1} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[39] $1^{\overline{1}} 5^{\overline{2}1^{\overline{1}1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$

[40] $1^{\overline{1}} 5^{\overline{1}} 12^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[41] $21^{\overline{1}} 51^{\overline{1}1} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 1 & 5 & 0 & 3 \\ 4 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

$$[42] \quad 1\overset{1}{5}1\overset{1}{2}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1\overset{2}{1}\overset{2}{5}1\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 5 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[44] \quad 1\overset{2}{1}\overset{2}{5}1\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 13 & 5 & 0 & 7 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 22\overset{1}{1}\overset{1}{5}1\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 2 & 3 & 3 \\ 6 & 13 & 5 & 6 & 7 \\ 2 & 5 & 0 & 2 & 2 \\ 3 & 6 & 2 & 0 & 2 \\ 3 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [46] \quad 1\overset{2}{1}\overset{1}{6}131\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [47] \quad 1\overset{1}{1}\overset{1}{5}131\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 6 \\ 2 & 0 & 2 \\ 6 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[48] \quad 1\overset{1}{1}\overset{1}{5}131\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

3.45 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1\overset{1}{1}131\overset{1}{1}2\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1\overset{1}{1}131\overset{1}{1}2\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -102, \ \det = 209, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -129, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{2^1 1^1}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

3.46 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1^1}} 5 1^{\overline{11}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 13, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 5 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 40, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 5 1^{\overline{2^1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 60, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 6 & 0 & 21 & 9 \\ 3 & 4 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 5 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 85, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 4 & 0 & 8 & 6 \\ 3 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{12}} 5 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 141, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 11 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 11 & 6 & 0 & 21 & 9 \\ 3 & 4 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{12}} 5 1^{\overline{3}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 181, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 17 & 10 & 0 & 61 & 15 \\ 3 & 4 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \overset{1}{5} 1^2 \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 15 & 8 & 0 & 40 & 12 \\ 3 & 4 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 1^2 \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 14 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 14 & 8 & 0 & 38 & 12 \\ 3 & 4 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overset{1}{2} \overset{1}{5}} 1^{\overset{1}{2} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 285, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 5 & 3 \\ 6 & 13 & 5 & 0 & 12 & 7 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 4 & 0 & 8 & 6 \\ 3 & 7 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13} \overset{1}{5} 1^2 \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 316, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 18 & 10 & 0 & 61 & 15 \\ 3 & 4 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{13} \overset{1}{5} 1^3 \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 381, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 21 & 12 & 0 & 88 & 18 \\ 3 & 4 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overset{1}{2} \overset{1}{5}} 1^{\overset{1}{2} \overset{1}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 516, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 10 & 3 \\ 6 & 13 & 5 & 0 & 24 & 7 \\ 2 & 5 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 8 & 0 & 38 & 12 \\ 3 & 7 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overset{1}{2} \overset{1}{5}} 1^{\overset{1}{2} \overset{1}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 901, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 15 & 3 \\ 6 & 13 & 5 & 0 & 36 & 7 \\ 2 & 5 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 36 & 12 & 0 & 88 & 18 \\ 3 & 7 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [13] \quad 1^{\overset{1}{2} \overset{1}{1}} \overset{1}{6} 131 \overset{1}{5} 1^{\overset{1}{2} \overset{1}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 10, (n_+, n_-, n_0) =$$

$$(1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 12 & 6 \\ 2 & 0 & 5 & 2 \\ 12 & 5 & 20 & 12 \\ 6 & 2 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 1^{11} 5^{131} 6^{12^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 11, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

3.47 NHC(-2)(-3) + $\textcolor{blue}{su}_3$ + (-3) $\rightarrow$ (-2) + $\textcolor{blue}{su}_3$

$$T = 8 \quad [0] \ 1^{11} 31^{132^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 9 & 3 \\ 4 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^{112^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 3 & 4 \\ 8 & 9 & 4 & 0 & 4 & 5 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 4 & 5 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 31^{111} 2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 9 & 3 \\ 4 & 4 & 4 & 0 & 14 & 4 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 31^{122^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 31^{112^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 153, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 3 \\ 4 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 31^{112^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 237, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 9 & 4 \\ 5 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 9 & 6 & 0 & 12 & 6 \\ 4 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 31^{111} 2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 453, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

3.48 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{\textcolor{teal}{1}} 31^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 17 & 1 & 7 \\ 20 & 25 & 23 & 0 & 9 \\ 17 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 31^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 17 & 1 & 7 \\ 17 & 20 & 22 & 0 & 8 \\ 17 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 4 \quad 1^{\textcolor{teal}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 10 & 1 & 5 \\ 10 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

3.49 $\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 3 \quad [1] \quad 12^{\textcolor{teal}{2}} 1 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 110, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 110, T \in [11, 110]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[0] \quad 212^{\textcolor{teal}{2}} + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 110, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 110, T \in [11, 110]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

3.50 $\mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 4 \quad [0] \quad 12^{\textcolor{teal}{2}} 1^{\textcolor{teal}{1}} + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 4, T_{\max} = 99, \Delta = -211, \det = 8, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

3.51 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 5 \quad [0] \quad 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{2}} 1^{\textcolor{teal}{1}} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

3.52 $\mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} + \mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -142, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -1 & -1 \\ 5 & 7 & -1 & -1 \end{pmatrix}_{b_0}$$

3.53 $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 101, \Delta = -212, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{c}_6 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{1} \quad T = 7, \quad T_{\max} = 101, \quad \Delta = -183, \quad \det = -13, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$\mathbf{3.54} \quad \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{f}_4 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{(-1)}$$

$$T = 6 \quad [0] \quad 1^{11}2^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{f}_4 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{1} \quad T = 6, \quad T_{\max} = 100, \quad \Delta = -186, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.67 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{f}_4 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{1} \quad T = 7, \quad T_{\max} = 99, \quad \Delta = -157, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 99, \quad T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{3.55} \quad \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{(-1)}$$

$$T = 5 \quad [0] \quad 1^{12}2^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{green}{\hat{1}} \rightarrow \textcolor{green}{1} \quad T = 5, \quad T_{\max} = 98, \quad \Delta = -163, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 & -1 \\ 2 & 2 & 0 & -1 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 12^121^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{green}{\hat{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \quad T_{\max} = 91, \quad \Delta = -134, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 91, \quad T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 3 & 1 & 2 & 2 & 2 \\ 2 & 3 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}22^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{green}{\hat{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \quad T_{\max} = 91, \quad \Delta = -134, \quad \det = 14, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 91, \quad T \in [11, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 1 & 2 & 2 \\ 2 & 3 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{3.56} \quad \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{so}_8 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{(-1)}$$

$$T = 6 \quad [0] \quad 1^{11}2^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{so}_8 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{1} \quad T = 6, \quad T_{\max} = 99, \quad \Delta = -162, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 99, \quad T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^11^1 + \textcolor{red}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 7, \quad T_{\max} = 91, \quad \Delta = -133, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, \quad T_{\max} = 91, \quad T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{3.57} \quad \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$$

$$T = 6 \quad [0] \quad 1^{11}2^11^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 97, \quad \Delta = -151, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.33 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 5 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{3.58} \quad \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{12}2^11^1 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, \quad T_{\max} = 98, \quad \Delta = -172, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.33 \rightarrow 12, \quad T_{\max} = 98, \quad T \in [12, 98]$$

$$\begin{pmatrix} 5 & 1 & 5 \\ 1 & -1 & -1 \\ 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{12}2^11^1 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 96, \quad \Delta = -143, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 96, \quad T \in [14, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$\mathbf{3.59} \quad \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$$

$$T = 6 \quad [0] \quad 1^{11}2^11^1 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 99, \quad \Delta = -171, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 99, \quad T \in [12, 99]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 11 & 0 & 5 \\ 1 & 0 & -1 & -1 \\ 5 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{3.60} \quad (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{12}2^11^1 + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 5, \quad T_{\max} = 123, \quad \Delta = -242, \quad \det = -15, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 123, \quad T \in [14, 123]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{12}2^121^1 + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 101, \quad \Delta = -213, \quad \det = 7, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 3 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}2^121^1 + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 101, \quad \Delta = -213, \quad \det = 12, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -6 & 0 & -4 \\ 3 & 0 & 6 & 4 \\ 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

3.61 $(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 187$, $\Delta = -297$, $\det = -21$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 187, T \in [15, 187]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 180$, $\Delta = -268$, $\det = 11$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 180, T \in [15, 180]$$

$$\begin{pmatrix} 10 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 180$, $\Delta = -268$, $\det = 18$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 15, T_{\max} = 180, T \in [15, 180]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -8 & 0 & -6 \\ 3 & 0 & 6 & 4 \\ 3 & -6 & 4 & -2 \end{pmatrix}_{b_0}$$

3.62 $(-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 179$, $\Delta = -269$, $\det = -18$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14, T_{\max} = 179, T \in [14, 179]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 108$, $\Delta = -240$, $\det = 9$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14, T_{\max} = 108, T \in [14, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 108$, $\Delta = -240$, $\det = 15$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 108, T \in [15, 108]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -7 & 0 & -5 \\ 3 & 0 & 6 & 4 \\ 3 & -5 & 4 & -2 \end{pmatrix}_{b_0}$$

3.63 $(-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -216$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -187$, $\det = 5$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -187$, $\det = 9$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -5 & 0 & -3 \\ 3 & 0 & 6 & 4 \\ 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

3.64 $(-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 12^1 1^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -193, \det = 3, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 2^1 1^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -193, \det = 11, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^1 2^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -193, \det = 11, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [3] \quad 12^1 21^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 122^1 1^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 4 \\ 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 6 & 4 \\ 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [0] \quad 2^1 1^1 8^1 12321^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

3.65 $(-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 4 \quad [0] \quad 2^1 12^2 + (-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 4, T_{\max} = 99, \Delta = -221, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & -4 & -2 \\ 2 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

3.66 $(-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -192, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 3 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -4 & 0 & -2 \\ 3 & 0 & 6 & 4 \\ 3 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$\mathbf{3.67} \quad \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2)$$

$$T=6 \quad [0] \, 14^{\textcolor{red}{2}}122^{\textcolor{blue}{1}} + \textcolor{red}{su}_8 \rightarrow \textcolor{red}{so}_{16} + (\textcolor{blue}{-3}) \rightarrow (\textcolor{blue}{-2}) \quad T=6, \, T_{\max}=99, \, \Delta=-163, \, \det=10, \, (n_+,n_-,n_0)=(1,6,0), \, c_{\text{ext}}=3.5$$

$$T_{\min}=11.40 \rightarrow 12, \, T_{\max}=99, \, T \in [12,99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 \\ 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

4 NHC $(-2)(-3)$ External (1227 combos, 49200 bases)

4.1 NHC $(-2)(-3)$

$$T = 9 \quad [247] \quad 1^1 2231321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -1, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[248] \quad 1^1 2231321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -1, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 10 \quad [367] \quad 1^1 23141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[366] \quad 1^1 31413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -37, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[365] \quad 1^1 31413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -37, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 11 \quad [524] \quad 1^1 314141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[525] \quad 1^1 231512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[526] \quad 1^1 223151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[527] \quad 1^1 231512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[528] \quad 1^1 41413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 11, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[529] \quad 1^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[530] \quad 1^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[531] \quad 1^1 141413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 11, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[532] \quad 1^1 314141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[533] \quad 1^1 5^{12231} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[534] \quad 1^1 223151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 12 \quad [712] \quad 1^1 4131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[711] \quad 1^1 1414141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[710] \quad 1^1 4131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[709] \quad 1^1 1414141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[708] \quad 1^1 3141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[707] \quad 1^1 2321612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 13 \quad [874] \quad 1^1 14141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[881] \quad 1^1 15131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 8 & 8 \\ 3 & 8 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[880] \quad 1^1 22315132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[879] \quad 1^1 14141414131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[878] \quad 1^1 2315132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[877] \quad 1^1 23151315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[876] \quad 1^1 23151315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[875] \quad 1^1 2321 \quad 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[873] \quad 1^1 15131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 14 \quad [1060] \quad 1^1 232161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 15, \Delta = -33, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1058] \quad 1^1 316123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1059] \quad 1^1 2315131 \quad 6 \quad 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1057] \quad 21 \quad 8 \quad 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$[1061] \quad 1^1 141414141 \quad 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 14, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 8 & 8 \\ 3 & 8 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$[1056] \quad 1^1 232161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 15, \Delta = -33, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$[1055] \quad 1^1 16131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 14, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[1054] & 1^1 612231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[1053] & 1^1 216131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[1052] & 1^1 812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[1051] & 1^1 321613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[1050] & 1^1 23151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[1049] & 1^1 161231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[1047] & 1^1 316123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[1048] & 1^1 321613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
T = 15 & [1181] 1^1 2321712315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[1183] & 1^1 2171231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1185] & 1^1 3217123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1186] & 1^1 1316131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}
\end{aligned}$$

[1173] $1^1 3216131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[1174] $1^1 71231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[1175] $1^1 3216131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[1176] $1^1 4131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[1177] $1^1 6^{131} 131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[1178] $1^1 6^{131} 131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[1179] $1^1 4131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[1180] $1^1 2321712315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[1182] $1^1 3217123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[1184] $1^1 712231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

$T = 16$ [1308] $1^1 8^{12231513221^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 16, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[1303] $1^1 23218122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[1307] $1^1 8^{13151321} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[1296] & 1^1 31612321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1297] & 1^1 31612321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[1298] & 1^1 21812231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 16, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[1299] & 1^1 15131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 16, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1302] & 1^1 23218122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
\\
T = 17 & [1401] 1^1 151316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1394] & 1^1 321613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1402] & 1^1 321613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1403] & 1^1 6^{131} 13161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1399] & 1^1 8^{12321} 123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1404] & 1^1 413218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1405] & 1^1 321712321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1398] & 1^1 8^{12321} 123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1406] & 1^1 321712321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1407] & 1^{131} 1^1 6^1 13161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1400] & 1^1 1^1 5^1 1316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1397] & 1^1 413218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1395] & 1^1 1^1 3218^1 1231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[1396] & 1^1 216^1 131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
T = 18 & [1514] 1^1 3216131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1508] & 1^1 1^1 5^1 13218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1526] & 1^{1321} 1^1 8^1 12321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[1525] & 1^{131} 1^1 6^1 13161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1524] & 1^1 1^1 316^1 131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[1523] & 1^1 216^1 1316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1627] & 1 \overset{1}{6} 1316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[1626] & 1^1 41316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[1625] & 1 \overset{1}{6} 1316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[1624] & 1^1 5131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
T = 20 & [1732] 1 \overset{1}{5} 1316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
[1733] & 21 \overset{1}{7} 123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1734] & 1^1 413161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1735] & 1 \overset{1}{8} 123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1736] & 1 \overset{1}{8} 123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1737] & 1 \overset{1}{6} 1316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1738] & 1^1 316123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1739] & 21 \overset{1}{6} 131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

[1731] $1^1 5^{13} 16 13 16 13 16 12 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

$T = 21$ [1844] $1^1 32 21 8 12 32 18 12 31 5 13 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[1836] $21^{1231513221} (12)^{1223141} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[1837] $1^1 32 18 12 31 5 13 21 8 12 32 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 22, T \in [21, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[1838] $1^1 32 18 12 31 5 13 21 8 12 32 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 22, T \in [21, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1839] $1^1 5^{13} 16 13 16 13 16 13 14 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1840] $21^1 6^{13} 16 13 16 13 16 13 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[1841] $1^1 32 21 8 12 32 18 12 31 5 13 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1842] $1^1 6^{12} 32 18 12 32 17 12 32 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1843] $1^1 31 6 12 32 18 12 32 17 12 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[1835] $1^1 5^{13} 16 13 16 13 16 13 14 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[1845] $1^{12321} 8^{12321} 12 32 18 12 32 15 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
[1846] & 21 \overset{1}{6} 1316131613161231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
[1847] & 221 \overset{1}{8} 123218123151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
[1848] & 1 \overset{1}{8} \overset{1}{12321} 1232181231 \overset{1}{5} 1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1849] & 1 \overset{1}{8} \overset{1}{12321} 1232181231 \overset{1}{5} 1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
[1850] & 21 \overset{1}{(12)} \overset{1}{231513221} 1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
[1831] & 1 \overset{1}{8} \overset{1}{1321} 12321812315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1832] & 1 \overset{1}{8} \overset{1}{12321} 1232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1833] & 1^1 13161232181232171231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1834] & 1 \overset{1}{8} \overset{1}{12321} 12321812315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
T = 22 & [1917] 1 \overset{1}{(12)} \overset{1}{1513221} 1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1918] & 1 \overset{1}{8} \overset{1}{12321} 123218123216131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1919] & 21 \overset{1}{(12)} \overset{1}{1513221} 1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1920] & \overset{1}{21} 6 13161316131613141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1921] & \overset{1}{21} \overset{1}{5} 13221 1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1922] & \overset{1}{21} \overset{1}{3} 1513221 122315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -30, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1923] & \overset{1}{1} 41321812321812315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1924] & \overset{1}{21} 7 12321812321712321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1925] & \overset{1}{1} 32171232181232171231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1926] & \overset{1}{1321} 8 123218123151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1927] & \overset{1}{1} \overset{1}{5} 131613161316131 \overset{1}{5} 1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1928] & \overset{1}{1} \overset{1}{3} 1513221 122315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -30, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1929] & \overset{1}{1} 6 12321812321812321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1930] & \overset{1}{1} 31612321812321812231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1931] & \overset{1}{1} 31612321812321812231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1932] & 1^{1231513221} (12) \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1913] & 1^{141321812321812315131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1914] & 1^{1231513221} (12) \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1916] & 1^{123151321} (12) \quad 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
[1915] & 1^{1321} 8^{123218123216131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20} \\
T = 23 & [1993] 1^{1321712321812321812231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1996] & 2218^{12321812321712321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1997] & 1^{1321} 8^{1232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1998] & 217^{12321812321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1999] & 1^{1321} 8^{1232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[2000] & 1^{1315132181232181232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[2001] & 1^1 315132181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[2002] & 1^1 4141413221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[2003] & 1^1 4141413221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[2004] & 1^1 216131613161316131^1 51 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[2005] & 1^1 51321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[2006] & 1^1 51321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[2007] & 1^1 321712321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1995] & 1^1 321812321712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1991] & 1^1 413218123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1994] & 1^1 321812321712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1992] & 1^1 413218123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$T = 24 \quad [2075] \quad 1^1 4132181232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[2073] \quad 221 \overset{1}{8} 12321812321 \overset{1}{1812321} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[2067] \quad 1^1 \overset{1}{5131513221} (12) \quad 1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[2069] \quad 1^1 3161232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[2063] \quad 1^1 612321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[2062] \quad 1^1 \overset{1}{8} 12321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[2066] \quad 1^1 \overset{1}{2321812321} 8 \quad 1232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[2068] \quad 1^1 3161232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[2070] \quad 1^1 513218123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[2077] \quad 1^1 3221812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[2076] \quad 1^1 \overset{1}{8} 1232181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

[2071] $1 \overset{1}{8} 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[2065] $1 \overset{1}{8} 12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 24, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[2074] $1 \overset{1}{4} 132181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[2072] $1 \overset{1}{5} 13218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 24, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[2064] $1 \overset{1}{(12)} 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

$T = 25$ [2138] $1 \overset{1}{4} 1321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[2137] $1 \overset{1}{5} 132181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[2139] $1 \overset{1}{5} 132181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[2136] $1 \overset{1}{3} 2171232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[2135] $1 \overset{1}{3} 2171232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[2134] $1 \overset{1}{4} 1321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

$$\begin{aligned}
& [2133] \quad 1^1 3221 \quad 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
& [2132] \quad 1^1 \quad 21712321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2131] \quad 1^1 231513221 \quad 122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
& [2130] \quad 1^1 16131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
& [2129] \quad 1^1 3221 \quad 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2140] \quad 1^1 321 \quad 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
& [2141] \quad 1^1 31513221 \quad 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 26, T \in [25, 26] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2142] \quad 1^1 321 \quad 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
& [2143] \quad 1^1 \quad 16123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
& [2144] \quad 1^1 31513221 \quad 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 25, T_{\max} = 26, T \in [25, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}
\end{aligned}$$

$$\begin{aligned}
& T = 26 \quad [2188] \quad 16 \overset{1}{1} 1232181232181232171231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2201] \quad 1 \quad (12) \quad 1223151322171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2200] \quad 1 \overset{1}{3} 22181232181232181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2199] \quad 1 \overset{1}{5} 1321812321812321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2198] \quad 1 \overset{1}{4} 13218123218123218123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2197] \quad 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2196] \quad 1 \overset{1}{7} 1231513221(12)1223151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2195] \quad 221 \overset{1}{8} 12321812321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
& [2194] \quad 1 \overset{1}{8} 1232181232181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
& [2193] \quad 1 \overset{1}{5} 1321812321812321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
& [2192] \quad 1 \overset{1}{8} 1232181232181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2191] & 1^{\overline{1}321} 1^{\overline{1}} 8^{\overline{1}} 123218123218123151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2186] & 1^{\overline{1}321} 1^{\overline{1}} 322181232181232181232151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2187] & 1^{\overline{1}} 217123218123218123216131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2189] & 1^{\overline{1}321} 1^{\overline{1}} 8^{\overline{1}} 123218123218123151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2190] & 1^{\overline{1}232181231513221} 1^{\overline{1}} (12) 1223141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
T = 27 & [2250] 2218^{\overline{1}} 123218123218123216131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2241] & 1^{\overline{1}} 1513218123218123218123141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2258] & 1^{\overline{1}} 41413221(12)1223151321812321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
[2257] & 1^{\overline{1}} 4132181232181232181232151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
[2256] & 1^{\overline{1}} 4132181232181232181232151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2255] & 1^{\overline{1}} 161232181232181232181231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27]
\end{aligned}$$

[2244] $1^1 41413221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$

[2243] $1^1 51316131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$

$T = 28$ [2294] $1^1 3161316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$

[2292] $21^1 712321812321812321^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$

[2301] $1^1 5132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -15, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$

[2302] $1^1 5132181232181232181231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -13, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$

[2303] $1^1 61316131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$

[2300] $1^1 5132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -15, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$

[2299] $1^1 51232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$

[2298] $1^1 6123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$

[2297] $221^1 81232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
[2296] & 21\bar{7}12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
[2295] & 1^1\bar{3}16131513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
[2293] & 1^1\bar{3}161316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
T = 29 & [2336] 221\bar{8}12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
[2337] & 1^1\bar{2}32181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
[2338] & 1^1\bar{2}32181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28} \\
[2339] & 221\bar{8}1232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
[2340] & 1\bar{6}1232181232181232181231\bar{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -12, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
[2341] & 1\bar{6}1232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -14, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
[2342] & 1^1\bar{5}132181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
[2343] & 1^1\bar{5}132181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$(1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2344] \quad 1^1 217123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$T = 30 \quad [2381] \quad 1^1 16131513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2374] \quad 1^1 2218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[2377] \quad 1^1 316131513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[2378] \quad 1^1 316131513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2379] \quad 1^1 61231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2380] \quad 1^1 61232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[2375] \quad 1^1 2171232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[2382] \quad 1^1 3221 1(12)1223151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[2383] \quad 1^1 3221 1(12)1223151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$\begin{aligned}
& [2376] \quad 21 \overset{1}{7} 1232181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -10, \quad \det = 1, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28} \\
\\
& T = 31 \quad [2420] \quad 1 \overset{1}{7} 1232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -20, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
\\
& [2421] \quad 1^{\overset{1}{3}1513221} (12) \overset{21}{12231513221(12)12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -27, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
\\
& [2422] \quad 1^{\overset{1}{7}12231513221(12)1223151321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -20, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
\\
& [2423] \quad 221 \overset{1}{8} 1232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -11, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
\\
& [2418] \quad 1^{\overset{1}{4}13221(12)1223151321812321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -26, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
\\
& [2417] \quad 1 \overset{1}{7} 1231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -20, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
\\
& [2416] \quad 1^{\overset{1}{3}1513221} (12) \overset{21}{12231513221(12)12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -27, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
\\
& [2415] \quad 221 \overset{1}{8} 1232181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -9, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
\\
& [2419] \quad 1^{\overset{1}{4}13221(12)1223151321812321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -26, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
\\
& T = 32 \quad [2458] \quad 1^{\overset{1}{5}13221(12)1223151321812321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 32, \quad \Delta = -21, \quad \det = 1, \quad (n_+, n_-, n_0) = \\
& (1, 32, 0), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& [2456] \quad 1^1 1513221(12)1223151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -21, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
& [2457] \quad 1^1 2181232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
& [2455] \quad 1^1 1812231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& [2459] \quad 1^1 32181232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& [2460] \quad 1^1 32181231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
& [2461] \quad 1^1 31513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
& [2462] \quad 1^1 31513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
& [2463] \quad 1^1 2181231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& [2454] \quad 1^1 32181232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 1, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
& T = 33 \quad [2495] \quad 1^1 513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = -1, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}
\end{aligned}$$

$$\begin{aligned}
& [2494] \quad 1^1 (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& [2491] \quad 1^1 (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& [2496] \quad 1^1 (12) \quad 12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& [2493] \quad 1^1 (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& [2492] \quad 1^1 (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
& T = 34 \quad [2512] \quad 21 (12) \quad 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{32} \\
& [2513] \quad 21 (12) \quad 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33} \\
& T = 35 \quad [2535] \quad 1^1 (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& [2536] \quad 1^1 (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33} \\
& [2538] \quad 1^1 (12) \quad 12231513221(12)12231513221 \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33} \\
[2539] & 1^{\textcolor{red}{1}} 232181232181232181231513221(12)12231^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 35, T_{\max} = 35, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33} \\
[2540] & 1^{\textcolor{red}{1}} 232181232181232181231513221(12)12231^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 35, T_{\max} = 35, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& 1^{\textcolor{red}{1}} 316131513221(12)12231513221 \\
[2537] & 1 \quad (12) \quad 1223141^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 35, T_{\max} = 35, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33} \\
& 1^{\textcolor{red}{1}} 3221 \\
T = 36 & [2570] \quad 21 \quad (12) \quad 12231513221(12)122315132181231^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 36, T_{\max} = 36, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& 1^{\textcolor{red}{1}} 13221 \\
[2571] & 1 \quad (12) \quad 12231513221(12)1223151316131^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 36, T_{\max} = 36, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& 1^{\textcolor{red}{1}} 231513221(12)12231513221 \\
[2572] & 1 \quad (12) \quad 12231513161^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 36, T_{\max} = 36, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& 1^{\textcolor{red}{1}} 16131513221(12)12231513221(12)122315131^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 36, T_{\max} = 36, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
[2565] & 1^{\textcolor{red}{1}} 41413221(12)12231513221(12)1223151316131^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 36, T_{\max} = 36, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& 1^{\textcolor{red}{1}} 1513221 \\
[2566] & 1 \quad (12) \quad 12231513221(12)1223151316131^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 36, T_{\max} = 36, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& 1^{\textcolor{red}{1}} 3221 \\
[2567] & 21 \quad (12) \quad 12231513221(12)122315132181231^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 36, T_{\max} = 36, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}
\end{aligned}$$

$$\begin{aligned}
& [2568] \quad 1^{131513221} (12) \quad 12231513221(12)122315132161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& [2569] \quad 1^{141413221} (12)12231513221(12)1223151316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& T = 37 \quad [2591] \quad 1^{1231513221(12)12231513221} (12) \quad 122315132171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& [2592] \quad 1^{131513221} (12) \quad 12231513221(12)1223151322171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& [2593] \quad 1^{132181231513221(12)12231513221} (12) \quad 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& [2594] \quad 1^{171231513221(12)12231513221(12)122315131}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& [2590] \quad 1^{132181231513221(12)12231513221} (12) \quad 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& T = 38 \quad [2620] \quad 1^{181231513221(12)12231513221(12)1223151321}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& [2623] \quad 1^{141413221(12)12231513221(12)122315132181231}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& [2622] \quad 1^{1316131513221(12)12231513221(12)12231513151}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& [2621] \quad 1^{1316131513221(12)12231513221(12)12231513151}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& [2614] \overset{1}{1} 8 1231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -18, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& [2615] \overset{1}{1} 41413221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -23, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& [2616] \overset{1}{1} 8 12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -18, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& [2617] \overset{1}{1} 231513221(12)12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& [2618] \overset{1}{1} 5 13221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& [2619] \overset{1}{1} 5 13221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& T = 39 \quad [2632] \overset{1}{1} 61316131513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -20, \det = \\
& -1, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& [2630] \overset{1}{1} 6 131513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -19, \det = -1, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& [2631] \overset{1}{1} 316131513221(12)12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -20, \det = \\
& -1, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& T = 40 \quad [2641] \overset{1}{1} 32181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -21, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
& \begin{matrix} 1^1 3221 & & 21 \\ [2640] & 21 & (12) & 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] & T = 40, T_{\max} = 40, \Delta = -24, \det = 1, (n_+, n_-, n_0) = \\ & (1, 40, 0), c_{\text{ext}} = 3.0 \end{matrix} \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{38} \\
& [2642] \quad 1^1 32181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -21, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{38} \\
& [2643] \quad 1^1 171231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -18, \det = 1, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{38} \\
& [2644] \quad 1^1 316131513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -18, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{38} \\
& T = 41 \quad [2660] \quad 1^1 1812231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -17, \det = \\
& -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
& [2656] \quad 1^1 16131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -18, \det = \\
& -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
& [2657] \quad 1^1 61231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -19, \det = \\
& -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
& [2658] \quad 1^1 2181231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -17, \det = \\
& -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
& [2659] \quad 1^1 3221 \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -23, \det = -1, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
& [2661] \quad 1^1 3221 \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -23, \det = -1, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41]
\end{aligned}$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$T = 43 \quad [2695] \quad 1^1 8 12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -16, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2696] \quad 1^1 21 8 1231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -16, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2697] \quad 1^1 18 1232181231513221(12)12231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2698] \quad 1^1 41413221(12)12231513221(12)12231513221 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2699] \quad 1^1 41413221(12)12231513221(12)12231513221 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2700] \quad 1^1 413221(12)12231513221 \quad (12) \quad 12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

$$[2701] \quad 1^1 413221(12)12231513221 \quad (12) \quad 12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2702] \quad 1^1 1^1 5 13221 \quad (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2703] \quad 1^1 1^1 5 13221 \quad (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$\begin{aligned}
& T = 44 \quad [2719] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -20, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{43} \\
& [2713] \quad 21 \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -20, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{42} \\
& [2714] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -16, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{42} \\
& [2715] \quad 21 \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -20, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{43} \\
& [2716] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -16, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{42} \\
& [2717] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -12, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{42} \\
& [2718] \quad 1 \overset{1}{4} 1413221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -20, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{42} \\
& [2720] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -20, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{42} \\
& T = 45 \quad [2724] \quad 1 \overset{1}{6} 131513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, \quad T_{\max} = 45, \quad \Delta = -17, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 45, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 45, \quad T_{\max} = 45, \quad T \in [45, 45] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{43} \\
& [2725] \quad 21 \quad 1 \overset{1}{5} 13221(12)12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, \quad T_{\max} = 45, \quad \Delta = -18, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 45, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 45, \quad T_{\max} = 45, \quad T \in [45, 45] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{43}
\end{aligned}$$

$$\begin{aligned}
& \text{[2726]} \quad 1 \quad \text{(12)} \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -19, \det = -1, (n_+, n_-, n_0) = \\
& (1, 45, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 45, T_{\max} = 45, T \in [45, 45] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43} \\
& \text{[2727]} \quad 1 \quad \text{(12)} \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -19, \det = -1, (n_+, n_-, n_0) = \\
& (1, 45, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 45, T_{\max} = 45, T \in [45, 45] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43} \\
& T = 46 \quad \text{[2742]} \quad 1^1 6 131513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = \\
& -16, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44} \\
& \text{[2740]} \quad 1^1 41413221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -18, \det = \\
& 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& \text{[2735]} \quad 1 \quad \text{(12)} \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -17, \det = 1, (n_+, n_-, n_0) = \\
& (1, 46, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44} \\
& \text{[2741]} \quad 1^1 41413221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -18, \det = \\
& 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44} \\
& \text{[2736]} \quad 1 \quad \text{(12)} \quad 12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -14, \det = \\
& 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& \text{[2737]} \quad 1^1 7 1231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -16, \det = \\
& 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44} \\
& \text{[2738]} \quad 21 \quad \text{(12)} \quad 12231513221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -16, \det = \\
& 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44} \\
& \text{[2739]} \quad 1 \quad \text{(12)} \quad 12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -14, \det = \\
& 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}
\end{aligned}$$

$$\begin{aligned}
& T = 47 \quad [2759] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -12, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& [2760] \quad 1^1 6 131513221(12)12231513221(12)12231513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& [2757] \quad 1^1 7 1231513221(12)12231513221(12)12231513221(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& [2756] \quad 1^1 6 131513221(12)12231513221(12)12231513221(12)12231^1 5 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& [2755] \quad 1^1 21 8 1231513221(12)12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& [2754] \quad 1^1 41413221(12)12231513221(12)12231513221(12)122315132161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -16, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& [2753] \quad 1^1 712231513221(12)12231513221(12)12231513221(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& [2758] \quad 1^1 8 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& T = 48 \quad [2771] \quad 1^1 5131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2772] \quad 1^1 7 1231513221(12)12231513221(12)12231513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46}
\end{aligned}$$

$$\begin{aligned}
& [2770] \quad 1 \overset{1^1}{15} 13221(12)12231513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -10, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2769] \quad 1 \overset{1^1}{41} 1413221(12)12231513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2768] \quad 1 \overset{1^1}{7} 1231513221(12)12231513221(12)12231513221(12)12231 \overset{1^1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -10, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2767] \quad 21 \overset{1^1}{8} 1231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2766] \quad 1 \overset{1^1}{8} 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& T = 49 \quad [2783] \quad 1 \overset{1^1}{6} 131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = \\
& -13, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2784] \quad 1 \overset{1^1}{6} 1231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = \\
& -14, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2785] \quad 21 \overset{1^1}{8} 1231513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -13, \det = \\
& -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2786] \quad 1 \overset{1^1}{23} 1513221(12)12231513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = \\
& -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2787] \quad 1 \overset{1^1}{8} 12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -13, \det = \\
& -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}
\end{aligned}$$

$$\begin{aligned}
& [2788] \quad 1 \overset{1}{8} 12231513221(12)12231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2789] \quad 21 \overset{1}{8} 1231513221(12)12231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& T = 50 \quad [2792] \quad 1 \overset{1}{7} 12231513221(12)12231513221(12)12231513221(12)12231513151 \overset{1}{1} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 50, T_{\max} = 50, T \in [50, 50] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{48} \\
& [2793] \quad 1 \overset{1}{7} 1231513221(12)12231513221(12)12231513221(12)12231513151 \overset{1}{1} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 50, T_{\max} = 50, T \in [50, 50] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{48} \\
& T = 51 \quad [2803] \quad 1 \overset{1}{8} 12231513221(12)12231513221(12)12231513221(12)12231513151 \overset{1}{1} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 51, T_{\max} = 51, T \in [51, 51] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{49} \\
& [2802] \quad 21 \overset{1}{8} 1231513221(12)12231513221(12)12231513221(12)12231513151 \overset{1}{1} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 51, T_{\max} = 51, T \in [51, 51] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{49} \\
& T = 53 \quad [2828] \quad 1 \overset{1}{3} 1513221(12)12231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 53, T_{\max} = 53, T \in [53, 53] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{51} \\
& [2827] \quad 1 \overset{1}{3} 1513221(12)12231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 53, T_{\max} = 53, T \in [53, 53] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{51} \\
& T = 54 \quad [2837] \quad 1 \overset{1}{8} 1232181231513221(12)12231513221(12)12231513221(12)1223151321 \overset{1}{1} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 54, T_{\max} = 54, T \in [54, 54] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{52}
\end{aligned}$$

4.2 $\text{NHC}(-2)(-3) + \mathfrak{e}_7$

$T = 10$ [13] $1^{11}2231321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 58, T_{\max} = 100, T \in [58, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 6 & 0 & -2 & 0 \\ 2 & 6 & -1 & 0 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$T = 11$ [15] $1^131413221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7$ $T = 11, T_{\max} = 101, \Delta = -140, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 59, T_{\max} = 101, T \in [59, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 3 & 0 & -5 & -3 \\ 2 & 3 & 2 & -3 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$T = 12$ [20] $1^{11}223151231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7$ $T = 12, T_{\max} = 102, \Delta = -136, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 60, T_{\max} = 102, T \in [60, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 6 & 0 & -2 & 0 \\ 2 & 6 & -1 & 0 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

[19] $1^{11}2231$ $1^15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7$ $T = 12, T_{\max} = 98, \Delta = -135, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 60, T_{\max} = 98, T \in [60, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 5 & 1 & -3 & -1 \\ 2 & 5 & 2 & -1 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

[18] $1^141413221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7$ $T = 12, T_{\max} = 95, \Delta = -131, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 60, T_{\max} = 95, T \in [60, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 3 & 0 & -5 & -3 \\ 2 & 3 & 2 & -3 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$T = 13$ [22] $1^14131513221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7$ $T = 13, T_{\max} = 98, \Delta = -134, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 61, T_{\max} = 98, T \in [61, 98]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 4 & 5 \\ 1 & 4 & -4 & -2 \\ 3 & 5 & -2 & -41 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$T = 14$ [23] $1^{11}22315132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7$ $T = 14, T_{\max} = 97, \Delta = -130, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 62, T_{\max} = 97, T \in [62, 97]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 6 & 0 & -2 & 0 \\ 2 & 6 & -1 & 0 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

[24] $1^15131513221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7$ $T = 14, T_{\max} = 95, \Delta = -129, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 62, T_{\max} = 95, T \in [62, 95]$

$$\begin{pmatrix} 5 & 19 & 5 \\ 19 & 72 & 16 \\ 5 & 16 & -40 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

4.3 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2$

$T = 9$ [31] $1^{11}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9, T_{\max} = 93, \Delta = -133, \det = -1, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 17, T_{\max} = 93, T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 2 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & -3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.4 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 10 \quad [3] \quad 1^{\overline{1}} 23^{\overline{1}} 131^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 66, T_{\max} = 80, T \in [66, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 0 & -1 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -1 & 2 & -1 & -52 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.5 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [3] \quad 1^{\overline{1}} 23^{\overline{1}} 131^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 66, T_{\max} = 80, T \in [66, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -1 & -1 & 2 & -52 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.6 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [3] \quad 1^{\overline{1}} \overline{5}^{\overline{1}} 13^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 3 & 0 & 5 & 3 & 5 \\ 1 & 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 5 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[2] \quad 1^{\overline{11}} \overline{5}^{\overline{1}} 13^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 7 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 5 & 6 & 9 & 7 & 0 & 36 & 7 & 12 \\ 1 & 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 12 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[4] \quad 1^{\overline{12}} \overline{5}^{\overline{1}} 13^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 6 & 5 & 0 & 21 & 5 & 9 \\ 1 & 1 & 2 & 1 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[5] \quad 1^{\overline{11}} \overline{5}^{\overline{1}} 13^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[1] \quad 1^{\overline{13}} \overline{5}^{\overline{1}} 13^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$[0] \begin{smallmatrix} 1^1 \\ 1^{12} 5 \\ 1^{13} 13^1 1^{13} \end{smallmatrix} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{blue}{5}\textcolor{green}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{5}\textcolor{red}{u}_2 + \textcolor{blue}{5}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T = 12 \quad [0] \overset{1}{1} \overset{1}{1} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{blue}{s} \textcolor{green}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{s} \textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$[1] \begin{matrix} 1^1 \\ 1^{13} 5 \\ 1^{12} 13^1 1^{12} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$[2] \quad \begin{array}{c} 1^1 \\ 1^{12} 5 \end{array} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$[3] \quad \begin{array}{c} 1^1 \\ 1^{12} 5 \quad 13^1 1^{11} \\ 1^{11} \end{array} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$[4] \quad 1 \overset{1^1}{\underset{1^{12}}{11}} 5 \overset{1^1}{\underset{1^{12}}{13}} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

[5] $1^{11}_1 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

1570

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 2 & 4 & 3 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.8 NHC(-2)(-3) + $\mathfrak{so}_8$

$$T = 7 \quad [9] \quad 1^{11} 3^{21} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -1, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 14 & 6 \\ 2 & 2 & -1 & 6 & -2 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

4.9 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 13 \quad [1132] \quad 1^{11} 3^1 5^{13} 2^{21} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[1131] \quad 1^{12} 3^1 5^{13} 2^{21} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 26 & 5 \\ 15 & 0 & 26 & 148 & 30 \\ 3 & 0 & 5 & 30 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

4.10 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 13 \quad [12] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 9 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[13] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 11 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 11 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[11] \quad 1^{13} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 3 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 3 & 0 & 1 & 3 \\ 3 & 0 & 3 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

4.11 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 13 \quad [13] \quad 1^{13} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 3 & 0 & 3 & 4 & 3 & 0 & 4 \\ 2 & 0 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[12] $1^{12}3^1 1^1 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 9 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[11] $1^{12}3^1 1^1 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 11 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

4.12 NHC(-2)(-3) + $\mathfrak{su}_2$

$T = 11$ [3662] $1^{13} 1^1 5^1 13^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$T = 12$ [3783] $1^{43} 1^1 5^1 13^1 22^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 19 & 4 & 4 \\ 8 & 0 & 19 & 115 & 27 & 27 \\ 2 & 0 & 4 & 27 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

[3784] $1^{13} 1^1 5^1 13^1 22^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 2 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 5 & 1 & 3 & 5 \\ 2 & 5 & 2 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

[3781] $1^{12321} 6^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

[3782] $1^{231} 5^1 13^1 22^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 7 & 4 \\ 4 & 0 & 7 & 19 & 11 \\ 2 & 0 & 4 & 11 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$T = 13$ [3986] $1^{14} 1^1 4^1 14^1 1^1 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 7 & 6 & 7 \\ 3 & 6 & 4 & 6 \\ 3 & 7 & 6 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

[3987] $1^{12321} 7^1 12^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 14 \quad [4038] \quad 1^{\overline{1^1 3221}} 8^{\overline{1}} 12321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[4039] \quad 1^{\overline{1^1 315131}} 6^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[4037] \quad 1^{\overline{1^1 2321}} 8^{\overline{1^1}} 1231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [4118] \quad 1^{\overline{1^1 1^1 2315131}} 6^{\overline{1^1 1^1}} 131^{\overline{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 4 & 3 & 19 & 11 \\ 2 & 2 & 2 & 11 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[4119] \quad 1^{\overline{1^1 1^1 32315131}} 6^{\overline{1^1 1^1}} 131^{\overline{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 6 & 9 & 83 & 23 \\ 2 & 2 & 2 & 23 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[4120] \quad 1^{\overline{1^1 31}} 6^{\overline{1^1}} 1316131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 4 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[4115] \quad 1^{\overline{1^1 1^1 2315131}} 6^{\overline{1^1 1^1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 3 & 1 & 3 & 5 \\ 2 & 3 & 2 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[4116] \quad 1^{\overline{1^1 2321}} 8^{\overline{1}} 123141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 4 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[4117] \quad 1^{\overline{1^1 31}} 6^{\overline{1^1}} 13161231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 16 \quad [4197] \quad 1^{\overline{1^1 512321}} 8^{\overline{1^1}} 12321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[4198] \ 1^1 5^{1^1} 1321^1 8^{1^1} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -21, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[4199] \ 1^1 1^{1^1} 3151321^1 8^{1^1} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -27, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[4196] \ 1^1 1^{1^1} 6^{1^1} 131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -24, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 5 & -1 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 17 \quad [4236] \ 1^1 1^{1^1} 321^1 8^{1^1} 123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = -26, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[4234] \ 1^1 6^{1^1} 12321^1 8^{1^1} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = -20, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[4235] \ 1^1 5^{1^1} 1316131^1 6^{1^1} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = -19, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [4273] \ 21^1 7^{1^1} 12321^1 8^{1^1} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 18, \ T_{\max} = 18, \ \Delta = -18, \ \det = 1, \ (n_+, n_-, n_0) = (1, 18, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, \ T_{\max} = 18, \ T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$[4274] \ 1^1 1^{1^1} 321^1 8^{1^1} 1232171231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 18, \ T_{\max} = 18, \ \Delta = -25, \ \det = 1, \ (n_+, n_-, n_0) = (1, 18, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, \ T_{\max} = 18, \ T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$T = 19 \quad [4301] \ 1^1 1^{1^1} 321^1 8^{1^1} 12321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 19, \ T_{\max} = 19, \ \Delta = -24, \ \det = -1, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, \ T_{\max} = 19, \ T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[4300] \ 1^1 1^{1^1} 5^{1^1} 13161316131^1 5^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 19, \ T_{\max} = 19, \ \Delta = -13, \ \det = -1, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, \ T_{\max} = 19, \ T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

[4299] $1^{1^1 321} 8^{1^1} 12321^1 8^{1^1} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 4 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

[4298] $221^{1^1} 8^{1^1} 12321^1 8^{1^1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$T = 20$ [4317] $21^{1^1} 6^{1^1} 131613161315^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20, T_{\max} = 20, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 13 & 9 \\ 2 & 4 & 0 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

[4318] $1^{1^1 321} 8^{1^1} 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20, T_{\max} = 20, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 5 & -1 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$T = 21$ [4328] $15^{1^1} 1321812321^1 8^{1^1} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 21, T_{\max} = 21, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

[4329] $1^{1^1 321} 8^{1^1} 1232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 21, T_{\max} = 21, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$T = 23$ [4350] $1^{1^1} 5^{1^1} 132181232181231^1 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 23, T_{\max} = 23, \Delta = -12, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$T = 24$ [4361] $1^{1^1} 6^{1^1} 1232181232181231^1 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 24, T_{\max} = 24, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$T = 25$ [4372] $21^{1^1} 7^{1^1} 1232181232181231^1 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 25, T_{\max} = 25, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 13 & 9 \\ 2 & 4 & 0 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$T = 33 \quad [4415] \quad 1 \quad \begin{matrix} 1^1 & 1^1 \\ 5 & 13221 \end{matrix} \quad (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 33, T_{\max} = 33, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$T = 36 \quad [4420] \quad 1 \quad \begin{matrix} 1^1 & 1^1 \\ 5 & 13221 \end{matrix} \quad (12) \quad 12231513221(12)12231513161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 36, T_{\max} = 36, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$T = 37 \quad [4422] \quad 1 \quad \begin{matrix} 1^1 & 1^1 \\ 5 & 13221 \end{matrix} \quad (12) \quad 12231513221(12)122315132171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 37, T_{\max} = 37, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$T = 38 \quad [4426] \quad 21 \quad \begin{matrix} 1^1 \\ 8 \end{matrix} \quad 1231513221(12)12231513221(12)1223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 38, T_{\max} = 38, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 7 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 7 & 4 & 19 & 11 \\ 2 & 4 & 2 & 11 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[4425] \quad 1 \quad \begin{matrix} 1^1 & 1^1 \\ 5 & 13221 \end{matrix} \quad (12) \quad 12231513221(12)1223151322181^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 38, T_{\max} = 38, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$T = 44 \quad [4432] \quad 1 \quad \begin{matrix} 1^1 & 1^1 \\ 5 & 13221 \end{matrix} \quad (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 44, T_{\max} = 44, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 3 & 4 \\ 3 & 3 & 4 & 6 \\ 3 & 4 & 6 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

4.13 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \quad 1^1 \quad \begin{matrix} 1^{11} \\ 5 \end{matrix} \quad 131^1 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \\ 9 & 14 & 42 & 0 & 12 & 10 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 & 3 \\ 2 & 3 & 10 & 0 & 3 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$\begin{aligned}
[5] \quad & 1^{12} 5^{131} 13^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \\ 11 & 14 & 42 & 0 & 12 & 10 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 & 3 \\ 2 & 3 & 10 & 0 & 3 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7
\end{aligned}$$

$$\begin{aligned}
[3] \quad & 1^{11} 5^{131} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7
\end{aligned}$$

$$\begin{aligned}
[2] \quad & 1^{13} 5^{131} 13^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \\ 14 & 21 & 97 & 0 & 18 & 15 & 21 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 18 & 1 & 1 & 3 & 3 \\ 2 & 3 & 15 & 0 & 3 & 1 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7
\end{aligned}$$

$$\begin{aligned}
[4] \quad & 1^{11} 5^{131} 13^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 3 & 4 \\ 6 & 7 & 9 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 6 & 1 & 1 & 3 & 3 \\ 2 & 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 4 & 7 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7
\end{aligned}$$

$$\begin{aligned}
[1] \quad & 1^{11} 5^{131} 13^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 & 2 \\ 2 & 2 & 4 & 5 & 3 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 3 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6
\end{aligned}$$

4.14 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$\begin{aligned}
T = 11 \quad & [1187] \quad 1^{11} 5^{131} 1321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -21, \det = -1, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 \\ 2 & 2 & 4 & 5 & 3 & 0 & 3 \\ 1 & 1 & 1 & 3 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 \\ 1 & 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7
\end{aligned}$$

$$\begin{aligned}
T = 12 \quad & [1234] \quad 1^{112} 31^1 5^{1321} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 1 & 2 & 4 & 7 & 3 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 & 3 \\ 2 & 4 & 5 & 7 & 13 & 7 & 7 \\ 4 & 7 & 6 & 13 & 18 & 10 & 10 \\ 2 & 3 & 3 & 7 & 10 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8
\end{aligned}$$

$$\begin{aligned}
[1235] \quad & 1^{11} 6^{131} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1236] \ 1^{\overline{1}^1} 6^{1^1} 12321^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -20, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1237] \ 1^{\overline{1}^1} 1^{\overline{1}^2} 31^{\overline{1}^2} 5^{1^2} 1321^{\overline{1}^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 5 & 4 & 7 & 14 & 7 \\ 5 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1238] \ 1^{\overline{1}^2} 31^{\overline{1}^1} 5^{1^1} 1321^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 6 & 3 \\ 2 & 0 & 5 & 7 & 13 & 7 \\ 4 & 0 & 6 & 13 & 18 & 10 \\ 2 & 0 & 3 & 7 & 10 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1232] \ 1^{\overline{1}^1} 4^{1^1} 14141^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -15, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 6 & 5 & 5 & 6 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 3 & 5 \\ 3 & 6 & 5 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[1233] \ 1^{\overline{1}^1} 1^{\overline{1}^1} 31^{\overline{1}^1} 5^{1^1} 1321^{\overline{1}^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 5 & 4 & 7 & 14 & 7 \\ 4 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 14 \quad [1354] \ 1^{\overline{1}^1} 1^{\overline{1}^1} 5^{1^1} 131^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 14, \ \Delta = -19, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 2 & 2 & 1 & 4 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 4 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1353] \ 1^{\overline{1}^1} 6^{1^1} 1315131^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 14, \ \Delta = -19, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[1352] \ 1^{\overline{1}^1} 8^{1^1} 1231^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 14, \ \Delta = -24, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 16 \quad [1384] \ 1^{\overline{1}^1} 1^{\overline{1}^1} 5^{1^1} 1321^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -18, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 2 & 2 & 1 & 4 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 4 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[1385] \ 1^{\overline{1}11} 5 \ 1316131 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -13, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 3 & 4 \\ 1 & 2 & 0 & 3 & 1 & 3 \\ 2 & 3 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1386] \ 1^{\overline{1}} 8 \ 122315131 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -17, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 17 \quad [1391] \ 1^{\overline{1}} 6 \ 123218 \ 1231 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = -17, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 18 \quad [1393] \ 1^{\overline{1}11} 5 \ 132181231 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 18, \ T_{\max} = 18, \ \Delta = -12, \ \det = 1, \ (n_+, n_-, n_0) = (1, 18, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 18, \ T_{\max} = 18, \ T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 3 & 4 \\ 1 & 2 & 0 & 3 & 1 & 3 \\ 2 & 3 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 19 \quad [1398] \ 1^{\overline{1}11} 6 \ 1232181231 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, \ T_{\max} = 19, \ \Delta = -11, \ \det = -1, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, \ T_{\max} = 19, \ T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$T = 22 \quad [1401] \ 21 \ 1^{\overline{1}11} 5 \ 13221 \ 1^{\overline{1}11} 12231 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 22, \ T_{\max} = 22, \ \Delta = -11, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[1400] \ 1 \ 1^{\overline{1}} (12) \ 12231 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 22, \ T_{\max} = 22, \ \Delta = -11, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 2 & 3 & 3 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$T = 25 \quad [1404] \ 1 \ 1^{\overline{1}11} 5 \ 13221 \ 1^{\overline{1}11} 122315131 \ 6 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 25, \ T_{\max} = 25, \ \Delta = -10, \ \det = -1, \ (n_+, n_-, n_0) = (1, 25, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 25, \ T_{\max} = 25, \ T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 2 & 3 & 3 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$T = 28 \quad [1406] \quad 1^{\overset{1}{1}\overset{11}{6}} 1316131513221(12)1223151^{\overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -9, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & 1 & 2 & 0 & 0 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 3 & 4 \\ 1 & 2 & 0 & 3 & 1 & 3 \\ 2 & 3 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

4.15 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 10 \quad [18] \quad 1^{\overset{11}{1}\overset{11}{2}31^{\overset{1}{1}}31^{\overset{11}{1}\overset{11}{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 7 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 2 & 7 & 4 \\ 4 & 0 & 7 & 0 & 7 & 14 & 8 \\ 2 & 0 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

$$[17] \quad 1^{\overset{11}{1}\overset{11}{1}\overset{11}{2}31^{\overset{1}{1}}31^{\overset{11}{1}\overset{11}{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 4 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 7 & 4 & 4 \\ 3 & 7 & 0 & 7 & 14 & 7 & 8 \\ 2 & 4 & 0 & 4 & 7 & 2 & 4 \\ 2 & 4 & -1 & 4 & 8 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

$$T = 11 \quad [26] \quad 1^{\overset{11}{1}\overset{11}{1}\overset{11}{3}141^{\overset{1}{1}}31^{\overset{11}{1}\overset{11}{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 & 4 & 3 \\ 2 & 3 & 1 & 3 & 2 & 4 & 4 \\ 3 & 4 & 0 & 4 & 4 & 5 & 5 \\ 2 & 3 & 0 & 3 & 4 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[27] \quad 1^{\overset{11}{1}\overset{11}{1}\overset{11}{3}1^{\overset{1}{1}}4131^{\overset{11}{1}\overset{11}{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 1 & 3 & 2 & 7 & 4 \\ 4 & 6 & 1 & 6 & 7 & 14 & 8 \\ 2 & 3 & 0 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[28] \quad 1^{\overset{11}{1}\overset{11}{1}\overset{11}{3}1^{\overset{1}{1}}41^{\overset{1}{1}}31^{\overset{11}{1}\overset{11}{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 3 & 4 & 6 & 3 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 5 \\ 4 & 6 & 1 & 6 & 9 & 14 & 8 \\ 2 & 3 & 0 & 3 & 5 & 8 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$T = 14 \quad [30] \quad 1^{\overset{1}{1}\overset{11}{6}} 13151^{\overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -10, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 3 & 2 & 3 & 5 & 3 & 0 & 5 \\ 1 & 1 & 1 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[34] \quad 1^{\overset{1}{1}\overset{11}{6}} 13151^{\overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -10, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 & 2 \\ 1 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 3 & 5 & 5 \\ 2 & 2 & 2 & 3 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[33] \quad 1^{\overset{1}{1}} \overset{1}{6} 131 \overset{1}{5} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[32] \quad 1^{\overset{1}{1}} \overset{1}{6} 131 \overset{1}{5} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 3 & 0 & 1 & 2 \\ 2 & 2 & 3 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & 0 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[31] \quad 1^{\overset{1}{1}} \overset{1}{6} 131 \overset{1}{5} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 1 & 2 \\ 2 & 2 & 3 & 5 & 4 & 3 & 5 \\ 2 & 1 & 2 & 4 & 1 & 1 & 3 \\ 1 & 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 5 & 3 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 15 \quad [35] \quad 1^{\overset{1}{1}} \overset{1}{7} 1231 \overset{1}{5} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 16 \quad [36] \quad 1^{\overset{1}{1}} \overset{1}{8} 12231 \overset{1}{5} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

4.16 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [260] \quad 1^{\overset{1}{1}} \overset{1}{1} 2231 \overset{1}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 11 \quad [276] \quad 1^{\overset{1}{1}} \overset{1}{3} \overset{1}{1} 141321^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -7, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [347] \quad 1^{\overset{1}{1}} \overset{1}{3} \overset{1}{1} 151321^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 10 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[348] \quad 1^{14}3^1 1^3 15 1321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 1 & 20 & 1 & 3 \\ 11 & 16 & 20 & 142 & 12 & 28 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 3 & 28 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[353] \quad 1^{12}3^1 1^1 15 1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 5 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[341] \quad 1^1 3^1 1^1 15 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 1 & 3 \\ 2 & 0 & 5 & 7 & 3 & 7 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 2 & 0 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[346] \quad 1^{12}3^1 1^3 15 1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 7 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[342] \quad 1^4 3^1 1^{11} 5 1321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 16 & 1 & 3 \\ 8 & 0 & 16 & 106 & 11 & 24 \\ 1 & 0 & 1 & 11 & 0 & 2 \\ 2 & 0 & 3 & 24 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[343] \quad 1^{11}3^1 1^2 15 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 4 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[349] \quad 1^{11}2315123^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[344] \quad 1^2 3^1 1^1 5 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 6 & 1 & 3 \\ 4 & 0 & 6 & 18 & 5 & 10 \\ 1 & 0 & 1 & 5 & 0 & 2 \\ 2 & 0 & 3 & 10 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[350] \quad 1^{13}3^1 1^2 15 1321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 8 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[345] \ 1^{11231} \ 1^1 \ 5^1 \ 13^1 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 2 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[351] \ 1^{11} \ 3^1 \ 1^1 \ 5^1 \ 13 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[352] \ 1^{11} \ 23 \ 21 \ 51 \ 3^1 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[340] \ 1^{11} \ 3^1 \ 1^{11} \ 14 \ 14 \ 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 4 & 1 & 3 & 3 & 5 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 0 & 5 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 13 \quad [361] \ 1^{11} \ 3^1 \ 21 \ 61 \ 23 \ 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 14 \quad [363] \ 1^{11} \ 3^1 \ 15 \ 13 \ 15 \ 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 3 & 1 & 5 & 4 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 4 & 5 \\ 1 & 4 & 0 & 4 & 2 & 4 \\ 2 & 5 & -1 & 5 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[362] \ 1^{11} \ 23 \ 21 \ 7^1 \ 12 \ 3^1 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 2 & 4 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 3 & 4 & 2 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [364] \ 1^{11} \ 8^1 \ 12 \ 3^1 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[366] \ 1^{11} \ 31 \ 61^1 \ 23 \ 15 \ 13^1 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) =$$

$$(1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 9 & 6 & 6 & 31 & 15 \\ 3 & 3 & 2 & 15 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[367] \quad 1^{11}3216131513^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[368] \quad 1^{11}3^1 2161315131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[369] \quad 1^{11}3161231513^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[365] \quad 1^{13}3161^{12}231513^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 15 & 12 & 12 & 103 & 27 \\ 3 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 17 \quad [371] \quad 1^{13}151321^1 8^{12} 123^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 2 & 4 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 3 & 4 & 2 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

4.17 NHC(-2)(-3) + su₂ + su₂→(-3) + su₂

$$T = 12 \quad [494] \quad 1^{12}3^1 1^1 5^{12} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[495] \quad 1^{12}31^1 5^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 1 & 7 & 3 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 4 & 7 & 5 & 0 & 4 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.18 NHC(-2)(-3) + su₂ + su₂→(-3) + su₂→(-3)

$$T = 12 \quad [71] \quad 1^{12}3^1 1^1 5^{13} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 6 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[72] \ 1^{12}3^{11}5^{13}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 4 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[73] \ 1^{13}3^{11}5^{13}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & 2 & 3 \\ 1 & 2 & 0 & 0 & 2 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[74] \ 1^{14}3^{11}5^{13}1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 2 & 3 \\ 1 & 2 & 0 & 10 & 2 & 1 & 2 \\ 9 & 14 & 10 & 74 & 8 & 10 & 18 \\ 1 & 1 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 1 & 10 & 1 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.19 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix

$$T = 12 \quad [310] \ 1^{13}3^{11}5^{13}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 1 & 2 & 0 & 3 & 2 \\ 2 & 0 & 3 & 4 & 3 & 5 & 5 \\ 2 & 0 & 2 & 3 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[309] \ 1^{12}3^{11}5^{13}1^{31} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 1 & 3 & 2 \\ 6 & 0 & 9 & 49 & 8 & 18 & 15 \\ 1 & 0 & 1 & 8 & 0 & 3 & 2 \\ 2 & 0 & 3 & 18 & 3 & 5 & 5 \\ 2 & 0 & 2 & 15 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[308] \ 1^{11}3^{11}5^{13}1^{31} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 1 & 3 & 2 \\ 6 & 0 & 6 & 34 & 7 & 15 & 12 \\ 1 & 0 & 1 & 7 & 0 & 3 & 2 \\ 2 & 0 & 3 & 15 & 3 & 5 & 5 \\ 2 & 0 & 2 & 12 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[307] \ 1^{11}3^{11}5^{13}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 3 & 2 \\ 2 & 0 & 4 & 6 & 3 & 7 & 6 \\ 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 7 & 3 & 5 & 5 \\ 2 & 0 & 2 & 6 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.20 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$T = 12 \quad [802] \quad 1^{\overline{1111}1} 3^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 2 & 3 & 6 & 4 & 3 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 3 & 1 & 1 & 6 & 4 & 3 \\ 4 & 6 & 5 & 6 & 14 & 11 & 8 \\ 2 & 4 & 4 & 4 & 11 & 6 & 6 \\ 2 & 3 & 2 & 3 & 8 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[803] \quad 1^{\overline{1111}1} 3^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 4 & 2 \\ 1 & 3 & 1 & 1 & 4 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 7 & 5 \\ 2 & 4 & 4 & 4 & 7 & 6 & 6 \\ 2 & 3 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[804] \quad 1^{\overline{1131}1} 3^{\overline{11}} 5^{\overline{11}} 13^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 3 & 13 & 4 & 3 \\ 1 & 2 & 0 & 1 & 10 & 4 & 2 \\ 1 & 3 & 1 & 1 & 13 & 4 & 3 \\ 9 & 13 & 10 & 13 & 67 & 23 & 17 \\ 2 & 4 & 4 & 4 & 23 & 6 & 6 \\ 2 & 3 & 2 & 3 & 17 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.21 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [308] \quad 1^{\overline{13}1} 3^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 1 & 2 \\ 2 & 0 & 1 & 1 & 4 & 2 & 3 \\ 2 & 0 & 3 & 4 & 5 & 3 & 5 \\ 1 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 0 & 2 & 3 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[307] \quad 1^{\overline{112}1} 3^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 3 & 1 & 2 \\ 6 & 0 & 6 & 34 & 15 & 7 & 12 \\ 2 & 0 & 3 & 15 & 5 & 3 & 5 \\ 1 & 0 & 1 & 7 & 3 & 0 & 2 \\ 2 & 0 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[309] \quad 1^{\overline{123}1} 3^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 3 & 1 & 2 \\ 6 & 0 & 9 & 49 & 18 & 8 & 15 \\ 2 & 0 & 3 & 18 & 5 & 3 & 5 \\ 1 & 0 & 1 & 8 & 3 & 0 & 2 \\ 2 & 0 & 2 & 15 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[310] \quad 1^{\overline{113}1} 3^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 1 & 2 \\ 2 & 0 & 4 & 6 & 7 & 3 & 6 \\ 2 & 0 & 3 & 7 & 5 & 3 & 5 \\ 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.22 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_{2n3\text{mix}}$

$$T = 12 \quad [335] \quad 1^{\overline{11}13} 3^{\overline{11}13} 1^{\overline{11}13} 5^{\overline{11}13} 13^{\overline{11}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 3 & 4 & 13 & 3 \\ 1 & 2 & 0 & 1 & 3 & 10 & 2 \\ 1 & 3 & 1 & 1 & 4 & 13 & 3 \\ 3 & 4 & 3 & 4 & 5 & 21 & 5 \\ 9 & 13 & 10 & 13 & 21 & 67 & 17 \\ 2 & 3 & 2 & 3 & 5 & 17 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[336] \quad 1^{\overline{11}13} 3^{\overline{11}13} 1^{\overline{11}13} 5^{\overline{11}13} 13^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 3 & 6 & 11 & 3 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 3 & 1 & 1 & 6 & 11 & 3 \\ 4 & 6 & 5 & 6 & 14 & 27 & 8 \\ 8 & 11 & 8 & 11 & 27 & 46 & 14 \\ 2 & 3 & 2 & 3 & 8 & 14 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.23 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 16 \quad [1646] \quad 1^{\overline{11}131} 6^{\overline{11}131} 131^{\overline{11}131} 6^{\overline{11}131} 131^{\overline{11}131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & 3 & -1 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 17 \quad [1647] \quad 1^{\overline{11}131} 8^{\overline{11}131} 12321^{\overline{11}131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 0 & 4 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

4.24 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [42] \quad 1^{\overline{11}111} 4^{\overline{11}111} 1^{\overline{11}111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 5 & 3 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.25 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [6] \quad 1^{\overline{11}11} 4^{\overline{11}11} 13^{\overline{11}11} 1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.26 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [6] \quad 1^{\overline{11}11} 4^{\overline{11}11} 13^{\overline{11}11} 1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.27 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & -1 & 3 & 0 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.28 NHC(-2)(-3) + $\mathfrak{su}_3$

$$T = 12 \quad [811] \quad 1^{11} 3^1 5^1 13221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 12, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 4 & -1 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[812] \quad 1^{11} 231^1 5^1 1321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 12, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 4 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 15 \quad [932] \quad 1^{11} 2315131^1 6^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 16 \quad [951] \quad 1^{11} 6^1 131612321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 16, T_{\max} = 16, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 3 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [977] \quad 1^{11} 6^1 13161315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 18, T_{\max} = 18, \Delta = -7, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 3 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 19 \quad [984] \quad 1^{11} 6^1 131613161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 19, T_{\max} = 19, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 5 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[985] \quad 1^{11} 6^1 13161316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 19, T_{\max} = 19, \Delta = -4, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 3 & 5 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 5 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 20 \quad [992] \quad 1^{11} 6^{131} 1316131613141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 20, \quad T_{\max} = 20, \quad \Delta = -3, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 6 & -1 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

4.29 $\text{NHC}(-2)(-3) + \text{su}_3 + \text{su}_2$

$$T = 11 \quad [1628] \quad 1^{11} 4^{13} 13221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -6, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [1690] \quad 1^{11} 5^{14} 13221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 3 & 20 & 5 \\ 12 & 9 & 20 & 67 & 21 \\ 3 & 3 & 5 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[1691] \quad 1^{11} 4^{14} 141321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -5, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 2 & 2 \\ 4 & 5 & 2 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[1692] \quad 1^{11} 5^{13} 13221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 3 & 8 & 5 \\ 4 & 5 & 8 & 11 & 9 \\ 3 & 3 & 5 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [1751] \quad 1^{11} 4^{13} 1414131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 13, \quad T_{\max} = 13, \quad \Delta = -3, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 1 & 3 & 3 & 5 \\ 2 & 4 & 0 & 3 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 14 \quad [1761] \quad 1^{11} 5^{13} 13151321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -3, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 2 & 0 & 6 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 7 & 7 \\ 2 & 2 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 15 \quad [1793] \quad 1^{11} 232161315^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = -2, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1787] \ 1^{11} 6^{11} 123151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = -1, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1788] \ 1^{112} 23151321^{11} 6^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = -1, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 9 & 6 & 6 & 31 & 15 \\ 3 & 3 & 2 & 15 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1789] \ 1^{11} 6^{11} 13151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 3 & 8 & 5 \\ 4 & 5 & 8 & 11 & 9 \\ 3 & 3 & 5 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1790] \ 1^{14} 6^{11} 13151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 3 & 20 & 5 \\ 12 & 9 & 20 & 67 & 21 \\ 3 & 3 & 5 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1791] \ 1^{11} 8^{11} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = -6, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[1792] \ 1^{11} 8^{11} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = -6, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 5 & 4 & 7 & 7 \\ 2 & 2 & 3 & 2 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 16 \quad [1802] \ 1^{112} 6^{11} 1316131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -4, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 0 & 11 & 4 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 11 & 6 & 0 & 34 & 14 \\ 2 & 4 & 2 & -1 & 14 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1803] \ 1^{1131} 6^{11} 1316131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -4, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 2 & 4 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 2 & 4 & 2 & -1 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1801] \ 1^{11} 232171231^{11} 5^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$\begin{aligned}
& T = 17 \quad [1820] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
& [1813] \quad 1^{11} 5^{12} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 2 & 0 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 11 & 0 & 28 & 14 \\ 3 & 4 & 0 & 14 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
& [1814] \quad 1^{11} 2321812231^{11} 5^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
& [1815] \quad 1^{11} 3151316131^{11} 5^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
& [1816] \quad 1^{12} 232181^{11} 2231^{11} 5^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 2 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 13 & 0 & 52 & 18 \\ 3 & 4 & 0 & 18 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
& [1817] \quad 1^{11} 6^{11} 12321712321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 2 & 2 \\ 4 & 5 & 2 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
& [1818] \quad 1^{11} 6^{11} 2321712321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 2 & 11 & 4 \\ 1 & 2 & 0 & 7 & 2 \\ 10 & 11 & 7 & 50 & 18 \\ 3 & 4 & 2 & 18 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
& [1819] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 1 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
& T = 19 \quad [1833] \quad 1^{11} 315132181231^{11} 5^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}
\end{aligned}$$

4.30 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [42] \quad 1^{\overset{1111}{1}} 4^{\overset{31}{1}} 1^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 5 & 3 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.31 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [6] \quad 1^{\overset{11}{1}} 4^{\overset{11}{1}} 13^{\overset{11}{1}} 1^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.32 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 15 \quad [76] \quad 1^{\overset{11}{1}} 3^{\overset{11}{1}} 151321^{\overset{11}{1}} 6^{\overset{11}{1}} 1^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 10 & 2 & 4 \\ 3 & 2 & 8 & 1 & 4 \\ 10 & 8 & 25 & 3 & 13 \\ 2 & 1 & 3 & 0 & 2 \\ 4 & 4 & 13 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

4.33 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [6] \quad 1^{\overset{11}{1}} 4^{\overset{11}{1}} 13^{\overset{11}{1}} 1^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.34 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{\overset{11}{1}} 4^{\overset{11}{1}} 1^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 3 & -1 & 3 & 0 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.35 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 15 \quad [127] \quad 1^{\overset{11}{1}} 3^{\overset{11}{1}} 151321^{\overset{11}{1}} 6^{\overset{11}{1}} 1^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 2 & 10 & 4 \\ 3 & 2 & 1 & 8 & 4 \\ 2 & 1 & 0 & 3 & 2 \\ 10 & 8 & 3 & 25 & 13 \\ 4 & 4 & 2 & 13 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

4.36 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [6] \quad 1^{11} 4^{11} 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.37 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$ + $\mathfrak{su}_2$

$$T = 15 \quad [98] \quad 1^{11} 6^{11} 13^{11} 15^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 & 4 \\ 3 & 2 & 4 & 2 & 6 & 4 & 4 \\ 4 & 5 & 7 & 6 & 10 & 8 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

4.38 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 12 \quad [11] \quad 1^{11} 4^{11} 14^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 2 & 3 & 4 \\ 2 & 5 & 0 & 3 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[10] \quad 1^{11} 4^{11} 14^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 & 3 \\ 2 & 2 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 7 & 6 & 0 & 14 & 7 & 8 \\ 3 & 4 & 3 & 0 & 7 & 2 & 4 \\ 3 & 4 & 3 & -1 & 8 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.39 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 4^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 2 & 0 & 3 \\ 2 & 4 & 2 & 0 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -1 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.40 NHC(-2)(-3)

$$T = 5 \quad [0] \quad 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -5, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, T_{\max} = 95, T \in [9, 95]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 2131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -7, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 95, T \in [11, 95]$$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -7, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$$

- $T_{\min} = 11, T_{\max} = 95, T \in [11, 95]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [3] $21^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.44 \rightarrow 10, T_{\max} = 95, T \in [10, 95]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [4] $21^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.33 \rightarrow 10, T_{\max} = 95, T \in [10, 95]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [5] $131^{11} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -28, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 95, T \in [11, 95]$
 $\begin{pmatrix} 4 & 8 & 4 \\ 8 & 9 & 5 \\ 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [6] $221^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -63, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 95, T \in [10, 95]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- $T = 6$ [7] $1^1 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 29, T \in [9, 29]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- [8] $1^1 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 29, T \in [9, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [9] $1^1 3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 79, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 79, T \in [9, 79]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [13] $21^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 96, \Delta = -136, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [10] $21^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 29, T \in [10, 29]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0}$
- [11] $1^1 1^1 41 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 96, \Delta = -136, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [12] $1^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 29, T \in [10, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [16] $1^1 1^1 41 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 96, \Delta = -136, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [14] $1^1 1^1 1231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 29, T \in [12, 29]$

- $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
 [15] $1^1 4^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
 [17] $13131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 79, \Delta = -124, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 79, T \in [12, 79]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
 [18] $31231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 81, \Delta = -125, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 81, T \in [12, 81]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
 [19] $22141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
 [20] $1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 29, T \in [12, 29]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
 [21] $21321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 29, T \in [12, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
 [22] $1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 29, T \in [12, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
 [23] $2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 29, T \in [10, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
 [24] $21^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 29, T \in [10, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
 [25] $1^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
 [26] $131^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 79, \Delta = -124, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 79, T \in [11, 79]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 \\ 2 & 4 & 3 & 3 \\ 2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 \\ 2 & 4 & 3 & 3 \\ 2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
 [27] $2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.45 \rightarrow 10, T_{\max} = 29, T \in [10, 29]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
 [31] $21^1 41^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$
 $\begin{pmatrix} 4 & 8 & 4 \\ 8 & 13 & 7 \\ 4 & 7 & 3 \end{pmatrix}_{b_0}$
 [30] $31^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 81, \Delta = -125, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$

- $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 81$, $T \in [11, 81]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [29] $131^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 79$, $\Delta = -124$, $\det = 12$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 79$, $T \in [11, 79]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 4 & 4 \\ 1 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [28] $21^1 1^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 12$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [32] $221^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 15$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [34] $31^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 81$, $\Delta = -125$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10$, $T_{\max} = 81$, $T \in [10, 81]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [33] $2^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 43$, $\Delta = -119$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 43$, $T \in [10, 43]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [35] $221^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 23$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [36] $2^1 3^1 1^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 81$, $\Delta = -125$, $\det = 27$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 81$, $T \in [11, 81]$
 $\begin{pmatrix} 0 & 2 & 3 & 2 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [37] $1231^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 29$, $T \in [12, 29]$
 $\begin{pmatrix} 4 & 8 & 4 \\ 8 & 9 & 5 \\ 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [39] $2^1 3^1 1^1 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 81$, $\Delta = -125$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 81$, $T \in [11, 81]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [40] $2^2 3^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 43$, $\Delta = -119$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 43$, $T \in [11, 43]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [41] $21^1 1^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 96$, $T \in [12, 96]$
 $\begin{pmatrix} 4 & 8 & 4 \\ 8 & 9 & 5 \\ 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [42] $21^1 41^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 96$, $T \in [12, 96]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 7 & 4 \\ 2 & 7 & 3 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
- [43] $131^1 31 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6$, $T_{\max} = 79$, $\Delta = -124$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 79$, $T \in [12, 79]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 7 & 4 \\ 2 & 7 & 3 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[38] $13 \overset{1}{2} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 28, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 29, T \in [11, 29]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[44] $231 \overset{1}{2} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 81, \Delta = -125, \det = 32, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 81, T \in [11, 81]$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[45] $31 \overset{1}{3} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 90, \Delta = -132, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 90, T \in [11, 90]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[46] $1321 \overset{1}{3} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 81, \Delta = -125, \det = 63, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 81, T \in [11, 81]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[48] $31 \overset{1}{3} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 90, \Delta = -132, \det = 63, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 90, T \in [11, 90]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 4 & 7 & 4 \\ 3 & 7 & 3 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[47] $141 \overset{1}{2} 22 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 63, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 4 & 7 & 4 \\ 3 & 7 & 3 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[50] $41 \overset{1}{2} 222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 112, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 10 & 13 & 6 \\ 4 & 13 & 9 & 5 \\ 2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[49] $2 \overset{21}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 43, \Delta = -119, \det = 112, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 43, T \in [10, 43]$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 8 & 11 & 6 \\ 4 & 11 & 7 & 5 \\ 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[51] $231 \overset{1}{2} 23 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 85, \Delta = -126, \det = 175, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 85, T \in [11, 85]$

$$\begin{pmatrix} 0 & 5 & 5 & 2 \\ 5 & 8 & 11 & 6 \\ 5 & 11 & 7 & 5 \\ 2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$T = 7$ [54] $13 \overset{1}{2} 221 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -88, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 11, T \in [9, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[55] $13 \overset{1}{2} 21 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 11, T \in [9, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[56] $1 \overset{1}{2} 23131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 13, \Delta = -96, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 13, T \in [9, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[53] $13 \overset{1}{2} 221 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -88, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$

- $T_{\min} = 9, T_{\max} = 11, T \in [9, 11]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [52] $1^1 23131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 13, T \in [9, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [57] $21^1 3221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [59] $1^1 41231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 27, T \in [10, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [60] $123^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [61] $123^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [62] $1^1 4131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 24, T \in [10, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [63] $1^1 4131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 24, T \in [10, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [58] $1^1 41231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 27, T \in [10, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [68] $1^1 1512 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 90, T \in [11, 90]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [65] $131^1 321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 2 & 4 & 4 \\ 2 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [66] $1^1 51^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 90, T \in [11, 90]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [70] $221^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 90, T \in [11, 90]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [64] $131^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$

- $T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [67] $21^{14}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [69] $12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -88, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [72] $131321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 13, \Delta = -96, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 5 & 4 \\ 1 & 5 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
- [71] $312321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 14, \Delta = -98, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [73] $1^{11}4^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [74] $123131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 13, \Delta = -96, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [75] $1^14^1131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [76] $2^{11}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 27, T \in [13, 27]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [77] $141231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 27, T \in [13, 27]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [78] $412231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 27, T \in [13, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [79] $313141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 77, \Delta = -123, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 77, T \in [13, 77]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [80] $132141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 27, T \in [13, 27]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [81] $214131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [82] $131^1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 13, \Delta = -96, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$

- $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 4 & 4 \\ 1 & 4 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [83] $23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [84] $21^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [85] $1231^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 3 \\ 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [86] $21^1 51^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [87] $1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [89] $23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [88] $31^1 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [90] $1231^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [91] $1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [92] $1^1 31^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [93] $21^1 41^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [94] $141^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [95] $31^1 3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$

- $$\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [96] $1^{\overline{131}} 4^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$$
- [97] $21^{\overline{151}} 1^{\overline{1}} 22 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$

$$\begin{pmatrix} 0 & 2 & 3 & 2 \\ 2 & 4 & 5 & 4 \\ 3 & 5 & 3 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$
- [98] $21^{\overline{151}} 1^{\overline{1}} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$$
- [99] $1^{\overline{21}} 5^{\overline{1}} 1^{\overline{1}} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [100] $2141^{\overline{131}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [101] $1321^{\overline{141}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$

$$\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [102] $41^{\overline{12231}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$

$$\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 25 & 9 \\ 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [103] $1^{\overline{221}} 5^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$

$$\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [104] $21^{\overline{151}} 1^{\overline{1}} 22 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 4 & 5 & 4 \\ 2 & 5 & 3 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$
- [105] $141^{\overline{1231}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 27, T \in [11, 27]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$
- [106] $31^{\overline{13141}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 77, T \in [11, 77]$

$$\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$$
- [108] $41^{\overline{1231}} 1^{\overline{1}} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 80, \Delta = -124, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 80, T \in [12, 80]$

$$\begin{pmatrix} 0 & 3 & 4 & 2 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [107] $3131^{\overline{141}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$

$$\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[109] $1 \overset{221}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$

[110] $2 \overset{141}{3} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$

[111] $2221 \overset{151}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 17 & 7 \\ 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$

[112] $1321 \overset{141}{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$

[113] $2 \overset{31}{3} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

[114] $2321 \overset{31}{1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[115] $41 \overset{1}{2} 32 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 34, \Delta = -118, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 34, T \in [12, 34]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$

[116] $412 \overset{11}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 34, \Delta = -118, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 34, T \in [12, 34]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[117] $21 \overset{11}{4} 131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 7 & 4 \\ 2 & 7 & 3 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$

[118] $31 \overset{11}{4} 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[119] $141 \overset{11}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[120] $23131 \overset{2}{1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[121] $41 \overset{1}{2} 31 \overset{1}{3} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 80, \Delta = -124, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 80, T \in [12, 80]$
 $\begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$

- [122] $41^1 1^1 2231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [123] $141^1 1^1 32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [124] $31^1 1^1 413 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 77, \Delta = -123, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [125] $141^1 31^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 77, \Delta = -123, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [126] $2221^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 90, \Delta = -131, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [127] $41^1 1^1 232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 34, \Delta = -118, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 34, T \in [11, 34]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [128] $321^1 41^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 80, \Delta = -124, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 80, T \in [12, 80]$
 $\begin{pmatrix} 0 & 3 & 5 & 2 \\ 3 & 1 & 6 & 3 \\ 5 & 6 & 12 & 6 \\ 2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$
- [129] $2^1 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 14, \Delta = -98, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 6 & 6 \\ 2 & 6 & 1 & 3 \\ 2 & 6 & 3 & 3 \end{pmatrix}_{b_0}$
- [130] $41^1 1^1 322 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$
 $\begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 9 & 6 \\ 3 & 9 & 4 & 4 \\ 2 & 6 & 4 & 3 \end{pmatrix}_{b_0}$
- [131] $14131^1 1^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 77, \Delta = -123, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [132] $12321^1 1^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 14, \Delta = -98, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [133] $41^1 1^1 322 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$
- [134] $41^1 31^1 32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 7, T_{\max} = 80, \Delta = -124, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 80, T \in [12, 80]$
 $\begin{pmatrix} 0 & 4 & 5 & 2 \\ 4 & 10 & 13 & 6 \\ 5 & 13 & 12 & 6 \\ 2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$

[135] $41^{1\bar{1}}2231 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$
 $\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 26 & 29 & 10 \\ 4 & 29 & 25 & 9 \\ 2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$

$T = 8$ [136] $1^{\bar{1}}223131^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 10, \Delta = -67, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[137] $1^{\bar{1}}231321^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 10, \Delta = -68, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[138] $31^{\bar{1}}3221^{\bar{1}}5 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 45, T \in [15, 45]$
 $\begin{pmatrix} 0 & 3 & 5 & 2 \\ 3 & 4 & 4 & 4 \\ 5 & 4 & 2 & 4 \\ 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$

[139] $1^{\bar{1}}223131^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 10, \Delta = -67, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[140] $1^{\bar{1}}412321^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[141] $1^{\bar{1}}412321^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[142] $1^{\bar{1}}141321^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[143] $1^{\bar{1}}141321^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[144] $1^{\bar{1}}314131^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 13, T \in [10, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[147] $21^{\bar{1}}5131^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$

[148] $1^{\bar{1}}15131^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[150] $21^{\bar{1}}5131^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[146] $1^{\bar{1}}322151^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$

- $T_{\min} = 11, T_{\max} = 21, T \in [11, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- $[151] \ 1^1 151231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 21, T \in [11, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- $[152] \ 1^1 413141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- $[153] \ 1^1 4141 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- $[145] \ 21^1 41321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- $[154] \ 1^1 151231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 21, T \in [11, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- $[155] \ 1^1 322151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 21, T \in [11, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- $[149] \ 1^1 15131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- $[157] \ 21^1 5131^1 3 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 3 \\ 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- $[158] \ 1^1 4141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- $[159] \ 1^1 5131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0}$
- $[160] \ 313151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- $[161] \ 21^1 51231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0}$
- $[162] \ 132151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[163] \quad 131\overset{1}{5}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[164] \quad 131\overset{2}{5}1\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[165] \quad 2141\overset{1}{4}1\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 14141\overset{1}{1}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[166] \quad 221\overset{1}{5}131\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 1 & 3 & 3 \\ 3 & 3 & 3 & 3 \\ 2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[176] \quad 151231\overset{1}{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[179] \quad 141412 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[168] \quad 1\overset{1}{5}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[169] \quad 1\overset{1}{5}12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[170] \quad 2\overset{1}{3}21\overset{1}{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[171] \quad 2215131\overset{1}{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[172] \quad 2151231\overset{1}{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[173] \quad 4123141\overset{1}{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$\begin{aligned}
& T_{\min} = 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[174] & 3214141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[175] & 1512231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 21, T \in [14, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
& 1^{11} \quad 1 \\
[167] & 1 \ 4 \ 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 20, T \in [14, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[177] & 1413141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
& 31 \\
[178] & 1 \ 4 \ 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
& 1^1 \\
[182] & 232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \\
[180] & 21^1 5132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 3 & 3 \\ 2 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \\
[181] & 21^1 5132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[183] & 131^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \\
[184] & 314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 24, T \in [12, 24] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 4 & 4 \\ 2 & 0 & 4 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \quad 1^1 \\
[185] & 15132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[186] & 21^1 51231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [187] $1^1 14141^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [188] $1^{21} 5 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [189] $131^1 5 1^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [190] $2^1 3 21^{41} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 6 & 6 \\ 3 & 6 & 5 \end{pmatrix}_{b_0}$
- [191] $1^1 5 1^1 3 2 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [192] $31^1 4 141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$
- [193] $141^1 2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [194] $2^1 3 2151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 23, \Delta = -113, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [198] $13141^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 4 & 4 \\ 1 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [195] $31^1 32151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [196] $131^1 5 1^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [197] $1^1 5 131^1 3 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 35, \Delta = -118, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [200] $221^1 5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

- [199] $22\bar{3}151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 21$, $\Delta = -111$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 21$, $T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [201] $41^1\bar{2}3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 25$, $\Delta = -115$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 25 & 9 \\ 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [202] $151^1\bar{2}231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 21$, $\Delta = -111$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 1 & 6 & 3 \\ 4 & 6 & 17 & 7 \\ 2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$
- [203] $1321^1\bar{5}1^1\bar{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 21$, $\Delta = -111$, $\det = 17$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 4 & 11 & 4 \\ 11 & 26 & 10 \\ 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [204] $31^1\bar{3}221^1\bar{5} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 45$, $\Delta = -119$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 45$, $T \in [13, 45]$
 $\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 3 & 4 & 5 \\ 3 & 4 & 3 & 3 \\ 2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$
- [205] $1^1\bar{4}141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [206] $31^1\bar{3}2151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 45$, $\Delta = -119$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12$, $T_{\max} = 45$, $T \in [12, 45]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [207] $2^1\bar{3}^1\bar{4}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 12$, $\Delta = -89$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [208] $1^1\bar{5}^1\bar{3}21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 21$, $\Delta = -111$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12$, $T_{\max} = 21$, $T \in [12, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [209] $31^1\bar{5}^1\bar{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [210] $1^1\bar{5}^1\bar{3}1^1\bar{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [211] $141^1\bar{3}141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [212] $2151^1\bar{2}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8$, $T_{\max} = 21$, $\Delta = -111$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12$, $T_{\max} = 21$, $T \in [12, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[213] $22\bar{3}151^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 14 & 8 \\ 3 & 8 & 4 \end{pmatrix}_{b_0}$

[219] $41^{\bar{1}}231^{\bar{1}}41 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 25, \Delta = -115, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$
 $\begin{pmatrix} 0 & 3 & 4 & 2 \\ 3 & 10 & 17 & 6 \\ 4 & 17 & 25 & 9 \\ 2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$

[214] $41231^{\bar{1}}41^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 25, \Delta = -115, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$

[215] $1321^{\bar{1}}\bar{5}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 7 & 3 \\ 7 & 26 & 10 \\ 3 & 10 & 3 \end{pmatrix}_{b_0}$

[216] $14131^{\bar{1}}41^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$

[217] $321^{\bar{1}}4141^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 25, \Delta = -115, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$
 $\begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 3 & 2 \\ 5 & 3 & 7 & 5 \\ 2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$

[218] $13221^{\bar{1}}\bar{5}1^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 17 & 7 \\ 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$

[222] $1512\bar{3}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 23, \Delta = -113, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[221] $231^{\bar{1}}3151^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$
 $\begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 \\ 5 & 1 & 32 & 10 \\ 2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$

[220] $1^{\bar{2}}\bar{3}^{\bar{1}}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[223] $141\bar{3}21^{\bar{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[224] $1^{\bar{1}}4^{\bar{2}}1\bar{3}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[225] $2151^{\bar{1}}\bar{3}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

- [226] $21^{11}41412 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -106, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 8 & 11 & 6 \\ 2 & 11 & 7 & 5 \\ 2 & 6 & 5 & 1 \end{pmatrix}_{b_0}$
- [227] $31^1 4 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 8 & 5 \\ 2 & 8 & 4 & 4 \\ 2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [228] $14141^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -106, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 4 & 8 & 4 \\ 8 & 9 & 5 \\ 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [231] $151^1 231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$
 $\begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [229] $151^1 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$
- [230] $41^1 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 25, \Delta = -115, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 25, T \in [12, 25]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [232] $31^1 4 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 13 & 10 & 7 \\ 2 & 10 & 4 & 4 \\ 2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$
- [233] $1321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 23, \Delta = -113, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$
 $\begin{pmatrix} 1 & 8 & 3 \\ 8 & 26 & 10 \\ 3 & 10 & 3 \end{pmatrix}_{b_0}$
- [234] $13221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [235] $31321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$
 $\begin{pmatrix} 1 & 8 & 3 \\ 8 & 26 & 10 \\ 3 & 10 & 3 \end{pmatrix}_{b_0}$
- [236] $151^1 232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 23, \Delta = -113, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [237] $321^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 25, \Delta = -115, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 25, T \in [12, 25]$
 $\begin{pmatrix} 1 & 8 & 3 \\ 8 & 17 & 9 \\ 3 & 9 & 4 \end{pmatrix}_{b_0}$
- [238] $231^1 3151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$
- [239] $1^{31} 5^{11} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 35, \Delta = -118, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[240] $41 \overset{11}{4} 132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 25, T \in [12, 25]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[241] $151 \overset{11}{3} 22 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[242] $215131 \overset{11}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[243] $31 \overset{321}{5} 1 \overset{1}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$

$$\begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 8 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[244] $41 \overset{231}{4} 1 \overset{1}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 25, T \in [12, 25]$

$$\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 13 & 10 & 7 \\ 3 & 10 & 4 & 4 \\ 2 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[245] $21 \overset{2}{11} 4132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 12 & 15 & 8 \\ 4 & 15 & 11 & 7 \\ 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$T = 9$ [246] $1 \overset{1}{223} 13221 \overset{1}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 10, \Delta = -39, \det = 0, (n_+, n_-, n_0) = (1, 9, 1), c_{\text{ext}} = 3.0$

[249] $1413221 \overset{1}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[250] $1413221 \overset{1}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[251] $1 \overset{1}{314} 1321 \overset{1}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -66, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[252] $1 \overset{1}{314} 1321 \overset{1}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -66, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[253] $141 \overset{1}{3221} 15 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 3 & 5 & 2 \\ 3 & 4 & 4 & 4 \\ 5 & 4 & 2 & 4 \\ 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[255] $1 \overset{131}{5} 131 \overset{1}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

- [262] $1^1 1414131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [261] $1^1 2151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [264] $1^1 3151231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [260] $1^1 2151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [259] $1^1 151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [256] $1^1 1512321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [257] $1^1 1512321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [258] $1^1 151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [254] $1^1 1414131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [263] $1^1 3151231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [265] $21^1 512321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [275] $1^1 1612231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [277] $1^1 2216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[278] \quad 21 \overset{1}{6} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[279] \quad 131 \overset{1}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[274] \quad 1 \overset{1}{5} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[273] \quad 1 \overset{1}{5} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[272] \quad 1 \overset{1}{4} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[271] \quad 1 \overset{21}{5} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[270] \quad 1 \overset{1}{4} 132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[269] \quad 1 \overset{1}{4} 132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[268] \quad 1 \overset{21}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[267] \quad 1 \overset{1}{6} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[266] \quad 131 \overset{1}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[280] \quad 21 \overset{1}{4} 14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

- [276] $2315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [284] $1413151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [281] $21^1 513141 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [282] $141^1 3151^2 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0}$
- [283] $4123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [292] $1^{11} 31^{12} 22 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [293] $15123141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [301] $22161231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [300] $2315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [299] $31315131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [298] $41314141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 24, \Delta = -113, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [297] $22216131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [296] $51223141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [295] $1^{21} 6^{11} 1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$

- [294] $21513141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [287] $1 \overset{1}{1^{11}321} 6 \overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [288] $1 \overset{1}{1^{11}3221} 6 \overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [289] $21612231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [285] $1 \overset{1}{1^{11}321} 6 \overset{1}{1} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [290] $1414141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [291] $21 \overset{1}{1^{11}31} 6 \overset{1}{1} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [286] $1 \overset{1}{1^{11}4131} 5 \overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [302] $1 \overset{1}{1^1} 6 \overset{1}{1^1} 1232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [309] $21 \overset{1}{1^1} 6 \overset{1}{1^1} 1232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [303] $1414141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [304] $1 \overset{21}{1^{11}31} 6 \overset{1}{1} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [306] $21 \overset{1}{1^1} 6 \overset{1}{1^1} 12231 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[307] $1 \overset{21}{6} 1231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[308] $21^1 513141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 4 \end{pmatrix}_{b_0}$

[305] $21^1 \overset{1}{6} \overset{31}{12} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[311] $1414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$

[310] $31^1 315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[312] $1 \overset{1}{6} 131^1 3 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 26, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$

[315] $21 \overset{1}{6} 131^1 3 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 26, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

[313] $1 \overset{1}{6} 1231^1 3 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

[314] $31^1 322161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

[316] $221^1 61231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[317] $1 \overset{221}{6} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -107, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[318] $1321^1 5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 1 & 3 \\ 2 & 7 & 3 & 3 \end{pmatrix}_{b_0}$

[319] $41^1 232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 25 & 9 \\ 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$

- [320] $41^1 123151 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [322] $51^1 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 \\ 5 & 2 & 2 & 4 \\ 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [321] $161^1 12231^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$
- [328] $22161^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [327] $141^1 32151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [326] $13151^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [325] $141^1 3151 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [324] $1^1 6^1 12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [323] $1^1 6^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [329] $31^1 413151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 32, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [330] $2221^1 6131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -107, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$
- [331] $1^1 41^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [332] $215131^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [333] $512231^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[334] \ 151231^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[335] \ 41^1 4141^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[338] \ 1^{11} 2^3 3^{21} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[339] \ 141^1 4141 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[340] \ 21612^1 3^2 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[336] \ 31^{11} 5^{31} 13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[337] \ 16131^1 1^1 32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$$

$$\begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 4 & 2 \\ 5 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[341] \ 1^{31} 1^{32} 1^1 6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$$

$$\begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[347] \ 41^1 231^1 5^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 9 & 3 \\ 9 & 51 & 15 \\ 3 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[348] \ 41^1 232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[342] \ 1^{11} 3221^1 6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[343] \ 2161^1 2231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

- [344] $151^1 23141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 6 & 6 \\ 3 & 6 & 4 \end{pmatrix}_{b_0}$
- [345] $41^1 314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 24, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [346] $2161^1 231^1 3 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [349] $141321^1 51^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$
- [350] $412321^1 51^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [351] $2161^1 232 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [352] $41^1 31513 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [353] $1^{2321^1} 6^1 1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [354] $51^1 223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [355] $4131^1 4141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 24, \Delta = -113, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [356] $2231^1 3161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$
 $\begin{pmatrix} 0 & 1 & 7 & 2 \\ 1 & 0 & 6 & 2 \\ 7 & 6 & 32 & 10 \\ 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$
- [357] $3141^1 3151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 32, \Delta = -117, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [360] $41^1 41413 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 24, \Delta = -113, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

$$[362] \quad 1 \begin{smallmatrix} 1 \\ 31^{11}31 \end{smallmatrix} 6 \quad 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 13 & 16 & 7 \\ 3 & 16 & 12 & 6 \\ 2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[358] \quad 1 \begin{smallmatrix} 31^{11}31 \\ 6 \end{smallmatrix} 12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[359] \quad 2216131^{11}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[361] \quad 1315131^{11}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [364] \quad 1^1 231413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 3.0$$

$$[363] \quad 1^1 231413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 3.0$$

$$[377] \quad 21 \begin{smallmatrix} 1 \\ 5 \end{smallmatrix} 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[368] \quad 1^1 4141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[380] \quad 1^1 23215131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[379] \quad 1^1 513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[370] \quad 1^1 23215131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[378] \quad 21 \begin{smallmatrix} 1 \\ 5 \end{smallmatrix} 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[371] \quad 1 \begin{smallmatrix} 1^{1231} \\ 5 \end{smallmatrix} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[372] \quad 1^1 4141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[373] $1^1 31414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -64, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[374] $1^1 23151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[375] $1^1 23151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[376] $1^1 5^{1231} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -61, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[369] $1^1 5^{12321} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$

[394] $1^1 6^{1321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[395] $23^1 151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -62, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[393] $1^1 6^{1321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[392] $1^1 32161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[391] $1^1 414141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$

[390] $21^1 6^{131} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[389] $1^1 6^{12321} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[388] $1^1 41315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[387] $1^1 31612231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[386] $1^1 41315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[385] $21^1 612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[384] $131^1 51321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[383] $1^1 6131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[382] $131^1 51321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[381] $1^1 31612231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[401] $21^1 171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[397] $1^1 51232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[414] $23216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -89, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[409] $16123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[398] $1^1 41322161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[399] $21^1 613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[413] $131\overset{1}{6}1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[412] $15132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$

[411] $15132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[410] $15131\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$

[396] $13216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[408] $1^141413151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[407] $1^141413151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$

[406] $21712231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[405] $22171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[404] $131\overset{1}{6}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[400] $31316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 15, \Delta = -94, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[403] $1\overset{1}{7}12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[402] $1\overset{21}{6}12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

[418] $215131\overset{1}{5}1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[419] $1\overset{1}{5}13151^{\overset{1}{1}}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0}$$

[417] $15131\overset{1}{5}1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[416] $151231\overset{1}{5}1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[415] $512231\overset{1}{5}1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[420] $21^{\overset{1}{1}}15132151^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 7 & 5 \\ 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[434] $161223141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[421] $1^{\overset{1}{1}}\overset{1}{7}^{1321}122 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[423] $1^{\overset{1}{1}}\overset{1}{7}^{13221}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[436] $16123141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[422] $21^{\overset{1}{1}}\overset{1}{7}^{1321}12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[435] $41414141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[424] $512314141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

- [425] $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}^{21}$ $1613141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [426] $151314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [427] $221613141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [428] $221712231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [429] $222171231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [430] $1^{11} 7^{3221} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [431] $216123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [432] $1^{11} 7^{321} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [433] $23216131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -89, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [438] $1^{21} 7^{12231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [444] $1^{21} 6^{13141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [448] $1^{21} 7^{1232} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [447] $31513151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[446] & 21 \overset{1}{7} 12 \overset{1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[437] & 21 \overset{1}{6} 123141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[445] & 1 \overset{21}{7} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[439] & 221 \overset{1}{7} 12 \overset{1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 10 & 4 & 6 \\ 2 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[440] & 21 \overset{1}{7} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[441] & 1 \overset{1}{7} 12 \overset{1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[442] & 1 \overset{1}{5} 131512 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[443] & 21 \overset{1}{5} 132151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[451] & 31 \overset{1}{3} 161231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[453] & 41 \overset{1}{4} 14141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 1 & 3 \\ 2 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[452] & 31 \overset{1}{3} 216131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[450] & 31 \overset{1}{5} 13151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [449] $1 \overset{31}{\underset{31}{6}} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [454] $141^1 1315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [457] $21 \overset{1}{\underset{1}{7}} 1231^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [455] $1 \overset{1}{\underset{1}{7}} 12231^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [456] $1 \overset{1}{\underset{1}{7}} 1231^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [458] $131 \overset{1}{\underset{1}{6}} 131^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [459] $41^1 12315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 42 & 1 & 14 \\ 1 & 1 & -3 & -1 \\ 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$
- [460] $1321^1 61231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [461] $221^1 712231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [462] $1 \overset{221}{\underset{221}{7}} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [463] $221^1 613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [464] $41^1 12321^1 61 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [465] $31 \overset{1}{\underset{31}{6}} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 15, T \in [13, 15]$

- $$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 2 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$
- [466] $1 \overset{31}{\overset{1}{7}} 3221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.56 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$
- [467] $2171^1 2231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.56 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$
- [468] $1^1 1513221^1 5 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.56 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 \\ 5 & 2 & 2 & 4 \\ 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [469] $41^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 5 & 3 \\ 5 & 6 & 6 \\ 3 & 6 & 4 \end{pmatrix}_{b_0}$$
- [471] $132161^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$
- [472] $2^1 \overset{1}{3} \overset{1}{4} 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 2 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$
- [470] $313161^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$
- [473] $2221^1 71231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$
- [474] $31321^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$
- [475] $2^1 \overset{1}{3} \overset{1}{3} 161^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$
- [476] $2161231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$
- [477] $1^1 16131^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [478] $1^1 161231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[479] & 2216131^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[480] & 1612231^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[481] & 231^{1316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.48 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[486] & 221712 \ 3 \ 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.29 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[484] & 316^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.29 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[485] & 1^{11} \ 7 \ 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 8 & 5 \\ 2 & 8 & 4 & 4 \\ 2 & 5 & 4 & 2 \end{pmatrix}_{b_0} \\
[483] & 1^{11} \ 7 \ 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.29 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[487] & 316^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.29 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[482] & 3216^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.29 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[488] & 151^{1232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[489] & 151^{123151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 9 & 3 \\ 9 & 51 & 15 \\ 3 & 15 & 4 \end{pmatrix}_{b_0} \\
[490] & 13161^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.33 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [491] $2161^1 23141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$
- [492] $22171^1 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [493] $1^1 6^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [494] $1^1 7^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$
- [495] $1^1 7^1 12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [496] $1512321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [497] $151321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 8 & 10 \\ 1 & 8 & 0 & 2 \\ 2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$
- [498] $2151321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [499] $13161^1 232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [502] $161^1 223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 42, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [501] $51^1 2314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -109, \det = 42, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [500] $1513221^1 5 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 42, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [503] $321^1 6131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 13 & 6 & 7 \\ 2 & 6 & 1 & 3 \\ 2 & 7 & 3 & 3 \end{pmatrix}_{b_0}$

- [504] $15131^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 1 & 8 & 3 \\ 8 & 17 & 9 \\ 3 & 9 & 4 \end{pmatrix}_{b_0}$
- [505] $321^1 513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 25, \Delta = -113, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [506] $51231^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 1 & 8 & 3 \\ 8 & 17 & 9 \\ 3 & 9 & 4 \end{pmatrix}_{b_0}$
- [507] $31^1 51321^1 5 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 25, \Delta = -113, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$
 $\begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 3 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [508] $14141^1 3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.92 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [509] $13216131^{11} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [510] $41^1 414141 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [511] $1^{31^{11} 31} 6^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 15, \Delta = -94, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [512] $31^{11} 3161313 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -102, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 13 & 16 & 7 \\ 3 & 16 & 12 & 6 \\ 2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [513] $31^{11} 321^1 6^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 19 & 8 & 9 \\ 2 & 8 & 1 & 3 \\ 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$
- [514] $3151^1 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 25, \Delta = -113, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 68 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [515] $21^{11} 414132^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 16 & 19 & 10 \\ 4 & 19 & 15 & 9 \\ 2 & 10 & 9 & 5 \end{pmatrix}_{b_0}$
- [516] $31^{31^{11} 31} 6^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -102, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 6 & 6 & 2 \\ 6 & 22 & 25 & 10 \\ 6 & 25 & 21 & 9 \\ 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$

$T = 11$ [520] $1^1 2231512321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$
 [519] $1^1 2231512321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$
 [522] $1^1 5^{12231} 1321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$
 [523] $1^1 5^{12231} 1321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$
 [517] $1^1 3141413221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -36, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$
 [518] $1^1 3141413221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -36, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$
 [521] $1^1 2314141321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -37, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$
 [540] $1^1 321612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -61, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
 [541] $1^1 413151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
 [539] $1^1 321612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -61, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
 [535] $1^1 413151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
 [536] $1^1 141414131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
 [537] $1^1 6^{12321} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
 [538] $1^1 6^{12321} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
 [542] $1^1 141414131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
 [543] $1^1 1315^{13221} 13221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$
 [544] $1^1 1315^{13221} 13221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$
 [558] $1^1 1316^{12321} 1321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
 [561] $1^1 413161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[546] \quad 221^1 712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[547] \quad 1^1 413161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[548] \quad 1^1 51315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[549] \quad 1^1 321712231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[550] \quad 1^1 321712231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[551] \quad 1^1 6^{131} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[552] \quad 1^1 6^{131} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[553] \quad 1^1 7^{21} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[554] \quad 1^1 413216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[555] \quad 1^1 51315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[556] \quad 1^1 413216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[557] \quad 1^1 31^1 6^{131} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[545] \quad 1^1 315132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[559] $1^{\overline{1}321} 1^{\overline{1}} 7^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[560] $1^{\overline{1}315132151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

[582] $21^{\overline{1}51315131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[562] $232171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 15, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[563] $217123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[564] $21613151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[565] $221812231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[566] $161232151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[567] $222181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[568] $132171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[569] $171223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[570] $1^{\overline{1}414132161} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

- [571] $161314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [572] $16123151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [573] $21^{1^1} 7^{1^1} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- [574] $21^{21} 181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [575] $151322161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [576] $1^{221} 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [577] $131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [578] $1^{21^1} 7^{1^1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [579] $1613151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [580] $1812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [581] $141316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 1 & 3 \\ 2 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [592] $161223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$

- [583] $1 \overset{1}{6} \overset{1}{5} 131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [584] $2161231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [585] $1 \overset{1}{6} \overset{1}{5} 131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [586] $2216131 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [587] $21 \overset{1}{5} 1322161 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [588] $1 \overset{1}{6} \overset{1}{5} 1321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [589] $21 \overset{1}{6} 13151 \overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [590] $1 \overset{1}{6} 13151 \overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 7 & 5 \\ 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [591] $1 \overset{1}{6} 123151 \overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [595] $2161314141 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 23, T \in [17, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [593] $232171231 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 15, \Delta = -88, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [596] $2171223141 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 23, T \in [17, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [594] $5131414141 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 23, T \in [17, 23]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [597] $1612314141 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 23, T \in [17, 23]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[611] & 6122314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 23, T \in [17, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[619] & 21^1 71223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[625] & 1 \begin{matrix} 21^1 \\ 8 \end{matrix} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^1 \\ 2321 \end{matrix} \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[624] & 21^1 \begin{matrix} 1^1 3221 \\ 8 \end{matrix} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[623] & 1 \begin{matrix} 21^1 \\ 1^1 3221 \end{matrix} 8 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[622] & 1 \begin{matrix} 21 \\ 8 \end{matrix} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 21^1 \end{matrix} \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[621] & 21^1 \begin{matrix} 1^1 321 \\ 8 \end{matrix} 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[620] & 21^1 6131^1 513 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[626] & 1 \begin{matrix} 21 \\ 8 \end{matrix} 1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^1 \end{matrix} \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[618] & 221^1 81232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[617] & 1 \begin{matrix} 221^1 \\ 8 \end{matrix} 1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 10 & 4 & 6 \\ 2 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[616] & 31^1 5132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[615] & 21^1 61232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[614] & 21^1 1^1 7 123141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[613] & 21^1 61231^1 51 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[612] & 2221^1 81232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 18 & 5 & 8 \\ 2 & 5 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[627] & 1^1 6131^1 513 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[628] & 21^1 61314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[629] & 21^1 2^1 3^1 8 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[630] & 21^1 1^1 6 131^1 51 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[631] & 31^1 41316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[632] & 31^1 32171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -94, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[633] & 31^1 613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[634] & 31^1 31613141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[635] $31\overset{1}{6}123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[636] $321\overset{1}{6}13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[637] $1\overset{21}{8}1231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$

[638] $221\overset{1}{8}1231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

[639] $21\overset{1}{8}12231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

[640] $1\overset{1}{8}12231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$

[641] $1\overset{221}{8}12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[642] $221\overset{1}{6}131\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[643] $41\overset{1}{2}23216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 15, \Delta = -88, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$

[644] $231\overset{1}{5}13161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[645] $1\overset{221}{8}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[646] $21\overset{221}{8}1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

- [647] $221^1 7123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [648] $41^1 2321^1 71 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [649] $41^1 2321^1 712 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [652] $22181^1 2231^1 3 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$
 $\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$
- [651] $1^{31} 8^{3221} 12 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$
 $\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$
- [650] $1^{31} 8^{3221} 1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$
 $\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 28 & 19 & 10 \\ 3 & 19 & 12 & 6 \\ 2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$
- [653] $2221^1 812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.55 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$
- [654] $412321^1 6131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 15, \Delta = -88, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.55 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$
- [655] $1^{2221} 8^{1231} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.55 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$
- [656] $17^{21} 1231^1 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -23, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [657] $1712231^1 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -23, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [658] $21712231^1 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -23, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$
- [659] $31316131^1 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 16, \Delta = -93, \det = -23, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

- [660] $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$ $22171231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -23, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$
- [661] $41^1 31^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 16, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 1 & 3 \\ 2 & 6 & 3 & 3 \end{pmatrix}_{b_0}$
- [662] $3141^1 316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 16, \Delta = -93, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$
- [663] $21^1 8^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [664] $1^1 8^1 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [665] $4131^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 16, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.29 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [666] $31^1 613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 16, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.29 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [667] $2221812^1 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [668] $1^1 8^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 8 & 5 \\ 2 & 8 & 4 & 4 \\ 2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [677] $222181^1 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [676] $2161^1 231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -102, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [675] $1^1 512321^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -104, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[674] $21^{13221^1} 8^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[673] $2161^{1232151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[672] $1^{151321^1} 6^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[671] $1^{21} 8^{13221^1} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[670] $41^{131513151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[669] $1^{13221^1} 8^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[679] $22221^{181231^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[678] $21^{15131^1} 6^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[681] $21612321^{151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -38, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.16 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[680] $1^{151^{12321}} 6^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -38, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.16 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[684] $161^{1223151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[683] $161^{12314141^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

- [682] $1^{1413221^1} 7^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 43 & 1 & 13 \\ 1 & 1 & -3 & -1 \\ 2 & 13 & -1 & 3 \end{pmatrix}_{b_0}$
- [685] $2171^{1223141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [686] $1^{321^1} 6^1 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$
- [687] $612231^{14141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -106, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [688] $321^{16123151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$
- [689] $1^{16131^1} 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [690] $216131^{14141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -105, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [691] $1^{161231^1} 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -106, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [692] $1^{161^1} 231^1 513 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 18 & 6 & 8 \\ 2 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$
- [693] $61^{12231^1} 513 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.64 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 28 & 7 & 10 \\ 2 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [694] $61^{122314141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 23, \Delta = -106, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 64 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [695] $141316131^{11^1} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 16, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.73 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [696] $61^{1231^1} 5132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 6 & 3 & 2 \\ 6 & 28 & 13 & 10 \\ 3 & 13 & 4 & 4 \\ 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$

$$[697] \ 5131^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 10 & 3 \\ 10 & 21 & 11 \\ 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[698] \ 3221^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -86, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[699] \ 31^1 6131^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 22 & 9 & 10 \\ 2 & 9 & 1 & 3 \\ 2 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[700] \ 32161^1 23151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -93, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[701] \ 1^1 6^1 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -93, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[702] \ 3161^1 223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -146, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [704] \ 1^1 31414141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$$

$$[705] \ 1^1 4141413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$$

$$[706] \ 1^1 4141413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$$

$$[703] \ 1^1 31414141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$$

$$[724] \ 1^1 4131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[722] \ 1^1 414141^1 41 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[721] \ 1^1 513151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[720] \ 1^1 2321712231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[719] \ 1^1 7^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[718] \ 1^1 7^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[717] $1^1 2315132151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[716] $1^1 2321712231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[715] $1^1 513151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$

[714] $1^1 3151315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[713] $1^1 2315132151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[723] $1^1 4131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[727] $1^1 61315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[751] $1^1 8^{221} 12321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[728] $1^1 4141316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[729] $1^1 8^{1321} 1231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[730] $1^1 4131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[731] $1^1 4141316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[732] $1^1 3161232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[733] $1^1 3161232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[734] \quad 1 \overset{1}{5} 13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[735] \quad 1 \overset{1}{5} 13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[736] \quad 2221 \overset{1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[737] \quad 1 \overset{1}{5} 13216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[738] \quad 21 \overset{21}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[739] \quad 1 \overset{1}{8} \overset{1}{3} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[740] \quad 1 \overset{1}{8} \overset{1}{3} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[741] \quad 1 \overset{1}{5} 13216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[742] \quad 1 \overset{1}{6} 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[743] \quad 1 \overset{1}{3} 221812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[744] \quad 1 \overset{1}{5} \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[745] \quad 1 \overset{1}{5} \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

[746] $1^{11}321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[726] $1^{11}4132171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[747] $1^{11}4132171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[748] $21^18^{1321}1231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[749] $23^121712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[750] $21^161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[725] $1^1612315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[758] $1321^1812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[772] $1^1712231^151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[761] $21^1513216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[760] $1^171232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[759] $1^15131513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[763] $21^171231^151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[757] $21^{1\bar{1}}513161231^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[756] $1 \overset{21}{6} 1315131^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$

[755] $1^{\bar{1}}4141322171^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[754] $1^{\bar{1}}2321812231^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[753] $21^{\bar{1}}71232151^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[752] $1321^{\bar{1}}81231^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[764] $131^{\bar{1}}6131^{\bar{1}}51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[765] $1^{\bar{1}}31^{\bar{1}}6131^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[766] $1^{\bar{1}}21^{\bar{1}}5131^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[767] $215131^{\bar{1}}6131^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[768] $13221^{\bar{1}}81231^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[770] $232181232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 16, \Delta = -89, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[771] $21^{\bar{1}}612315131^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[762] & 1^{11} 31321 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 17, T \in [15, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[779] & 1^{11} 4141413161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[778] & 1^{11} 2321 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[777] & 1^{11} 3151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[776] & 1^{11} 2181223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[775] & 1^{11} 321 8 12 3 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[774] & 1^{11} 17123151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[773] & 1^{11} 1712314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[769] & 1^{11} 2218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[788] & 1^{11} 171223151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[787] & 1^{11} 17123151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 7 & 5 \\ 2 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[786] & 1^{11} 51321 7 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[785] & 21712231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[784] & 21 \overset{1}{7} 123151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \left(\begin{smallmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{smallmatrix} \right)_{b_0} \\
[783] & 1 \overset{1}{\overset{1}{7}} \overset{1}{5} 13221 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[782] & 22171231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[781] & 31316131 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[780] & 1 \overset{1}{\overset{1}{7}} \overset{1}{5} 1321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{smallmatrix} \right)_{b_0} \\
[802] & 22181223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{smallmatrix} \right)_{b_0} \\
[789] & 21 \overset{21}{8} 123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{smallmatrix} \right)_{b_0} \\
[790] & 1 \overset{1}{\overset{2}{8}} \overset{21}{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \left(\begin{smallmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{smallmatrix} \right)_{b_0} \\
[791] & 22218123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{smallmatrix} \right)_{b_0} \\
[792] & 21712314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{smallmatrix} \right)_{b_0} \\
[793] & 61231414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{smallmatrix} \right)_{b_0} \\
[794] & 1 \overset{221}{8} 123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[795] $31 \overset{1}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 4 \\ 3 & 4 & 5 \end{pmatrix}_{b_0}$$

[796] $16131414141^{\textcolor{red}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[797] $2321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[798] $17122314141^{\textcolor{red}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[799] $31 \overset{1}{5} 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[800] $1712314141^{\textcolor{red}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[801] $181223141^{\textcolor{red}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[804] $31 \overset{1}{5} 1322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[805] $21^{\textcolor{red}{21}} 8123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[806] $1^{\textcolor{red}{21}} 7123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[807] $21^{\textcolor{red}{1}} 71223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[808] $1^{\textcolor{red}{1}} 7123151^1 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

- [809] $1 \overset{21}{8} 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [810] $21^1 712314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [811] $1 \overset{21}{7} 1232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [803] $21^1 71231 \overset{1}{5} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [812] $31 \overset{1}{6} \overset{1}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 16, \Delta = -88, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix}_{b_0}$
- [818] $321 \overset{1}{7} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -109, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [817] $31^1 316131 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 16, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$
- [816] $4131 \overset{1}{6} 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -108, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [813] $31^1 322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [815] $31^1 321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [814] $1 \overset{31}{8} \overset{321}{1231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [819] $1321 \overset{1}{8} 1231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [824] $231 \overset{1}{5} 132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -109, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[823] \quad & 221^1 71232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -102, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[822] \quad & 221^1 7123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[821] \quad & 221^1 81223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[820] \quad & 1^1 8^{221} 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 19 & 1 & 9 \\ 1 & 1 & -3 & -1 \\ 2 & 9 & -1 & 3 \end{pmatrix}_{b_0} \\
[825] \quad & 41^1 23218122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -106, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.40 \rightarrow 17, T_{\max} = 24, T \in [17, 24] \\
& \begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[826] \quad & 132181^1 2231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.56 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0} \\
[827] \quad & 2^{1^1 4^{1^1} 14141} 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 3 & 5 \\ 2 & 6 & 5 & 6 \end{pmatrix}_{b_0} \\
[828] \quad & 31^{1^1 5^{1^1} 131} 6^{1^1} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[830] \quad & 321^1 51316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.55 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 3 & 6 \\ 1 & 3 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[829] \quad & 2221^1 8123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[831] \quad & 222181231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[833] \quad & 18^{21} 12231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[832] \quad & 1^1 8^{221} 1231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[834] & \begin{matrix} 21 \\ 2181231^1 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[835] & \begin{matrix} 221812231^1 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[836] & \begin{matrix} 231^1 32181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 17, \Delta = -94, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.48 \rightarrow 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[839] & \begin{matrix} 321^1 81231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 17, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.29 \rightarrow 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[838] & \begin{matrix} 15131^1 31^1 6^1 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.29 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[837] & \begin{matrix} 1^{11} 2^1 3^1 21812313 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 17, \Delta = -95, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.29 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 8 & 5 \\ 2 & 8 & 4 & 4 \\ 2 & 5 & 4 & 2 \end{pmatrix}_{b_0} \\
[843] & \begin{matrix} 41^1 316123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 30, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.80 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[842] & \begin{matrix} 41^1 321613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 30, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.80 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[841] & \begin{matrix} 1^1 5131^1 41^1 31^1 6^1 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 30, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.80 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[840] & \begin{matrix} 3132181^1 2231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 30, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.80 \rightarrow 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[844] & \begin{matrix} 313221^1 81231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.50 \rightarrow 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[847] & \begin{matrix} 221712321^1 51^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 24, \Delta = -102, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.16 \rightarrow 17, T_{\max} = 24, T \in [17, 24] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0} \\
[846] & \begin{matrix} 1^1 7^1 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 12, T_{\max} = 24, \Delta = -102, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^{151} 1^{2321} \end{matrix}
\end{aligned}$$

$$[859] \ 71^1 1^2 231 5 132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 74, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.59 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 3 & 2 \\ 7 & 40 & 15 & 12 \\ 3 & 15 & 4 & 4 \\ 2 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[860] \ 61231^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 79, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[861] \ 16131^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 79, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[863] \ 41321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[862] \ 3221^1 7123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[864] \ 413161^1 23151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 93, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.74 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[865] \ 21^{11} 41414132^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 112, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 20 & 23 & 12 \\ 4 & 23 & 19 & 11 \\ 2 & 12 & 11 & 6 \end{pmatrix}_{b_0}$$

$$[866] \ 32171^1 223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 146, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.63 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [872] \ 1^1 314141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[867] \ 1^1 231513151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[868] \ 1^1 223151315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[869] \ 1^1 223151315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[870] \ 1^1 41414141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[871] \ 1^1 41414141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[899] \ 1^1 32161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[901] \ 21613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[900] \ 1^1 32161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[898] \ 1^1 8^1 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

[897] $1^1 8^{13221} 12321^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix} b_0$$

[896] $1^{1315131} 6^{131} + [\text{su}_2 \cdot \text{g}_2]$ $T = 13$, $T_{\max} = 15$, $\Delta = -59$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix} b_0$$

[895] $21^{1\bar{1}2321} 8^{1231} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[894] $1^1 16 123151321^1 + [\text{su}_2 \cdot \text{g}_2]$ $T = 13, T_{\max} = 14, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix} b_0$$

[893] $21^{1^1 2321} 8^{1231} + [\textcolor{red}{\mathfrak{s}}\textcolor{blue}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix} b_0$$

[892] $1^1 31612315131^1 + [\textcolor{red}{\mathfrak{s}u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[891] $1^{1^1 315131} 6^{131^1} + [\text{su}_2 \cdot \text{g}_2]$ $T = 13$, $T_{\max} = 15$, $\Delta = -59$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[890] $1^1 6 1 2 2 3 1 5 1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix} b_0$$

[899] $1 \frac{1}{1^1 2321} 8 1231^1 + [\text{su}_2 \cdot \text{g}_2]$ $T = 13$, $T_{\max} = 15$, $\Delta = -58$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix} b_0$$

[887] $16^1_{1^1}13151321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 13, T_{\max} = 14, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix} b_0$$

[886] $1^1 41321712321^1 + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[885] $1^1 15 13 16 12 32 1^1 + [\text{su}_2 \cdot \text{g}_2]$ $T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[884] $1^1 3 1 6 1 2 3 1 5 1 3 1^1 + [\text{su}_2 \cdot \text{g}_2]$ $T = 13$, $T_{\max} = 15$, $\Delta = -60$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[883] & 1 \overset{1}{5} 131612321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[882] & 1 \overset{1}{8} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[888] & 1 \overset{1}{4} 1321712321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[920] & 13221 \overset{1}{8} 12321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[924] & 1321 \overset{1}{8} 12321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[925] & 1 \overset{1}{5} 132171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[902] & 1 \overset{1}{3} 2171232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[903] & 1 \overset{1}{5} 132171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[904] & 1 \overset{1}{3} 2171232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[919] & 1 \overset{1}{4} 1322181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[923] & 131 \overset{1}{6} 1315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[921] & 1 \overset{1}{4} 1321812231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[905] & 31321 \overset{1}{8} 12321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -66, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[906] $21\overset{1}{6}13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[907] $1\overset{1}{8}123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[908] $1\overset{1}{7}12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[909] $1\overset{1}{6}123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[910] $2\overset{1}{2321}\overset{1}{8}12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[911] $1\overset{1}{8}123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

[912] $1\overset{1}{5}131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[913] $21\overset{1}{6}1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[914] $1\overset{1}{4}1321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[915] $1\overset{1}{5}131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

[916] $1\overset{1}{7}122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

[917] $1\overset{1}{8}12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

- [918] $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$ $1^1 41322181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [922] $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$ $1^1 21712315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
- [930] $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$ $1^1 181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 25, T \in [16, 25]$
- [935] $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$ $1^1 151321613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 25, T \in [16, 25]$
- [949] $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$ $1^1 18122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 25, T \in [16, 25]$
- [945] $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$ $1^1 21^1 5132171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
- [944] $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$ $1^1 2218123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$
- [943] $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 1 & 3 \\ 2 & 3 & 3 & 4 \end{pmatrix}_{b_0}$ $1^1 14132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
- [942] $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$ $1^1 22181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 25, T \in [16, 25]$
- [948] $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$ $1^1 23218123141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
- [941] $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$ $1^1 21^1 5131613141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
- [940] $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$ $1^1 13218123141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
- [939] $1^1 16131513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[938] & 1^1 8 12231 \bar{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[937] & 1^{131} 31 \bar{6} 13161^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[936] & 21^1 6123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[929] & 1^1 21 8 12231 \bar{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[931] & 21^1 7122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[926] & 21^1 1^1 7 12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[927] & 1^1 21 8 12314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[947] & 1^1 61231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[928] & 412321^1 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[932] & 1^{15131} 1^1 \bar{6} 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[933] & 21^1 1^1 8 1231 \bar{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[934] & 1^1 51321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [946] $1^1 41414132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 25, T \in [16, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [960] $1^1 81223151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 7 & 5 \\ 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [959] $2218123151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [958] $22218123151^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [957] $21^1 8^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [956] $1^1 8^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [955] $2181223151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [954] $22181223151^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [953] $1^1 8^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [952] $1^1 8^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [951] $1^1 8^1 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [950] $1^1 8123151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 7 & 5 \\ 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [965] $32161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[970] & 171231414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[961] & 31^1 6^{131} 13161 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -85, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[962] & 18122314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[963] & 1812314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[964] & 23218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[966] & 613141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[967] & 221812314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[968] & 712231414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[969] & 218122314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[983] & 1^{21} 8^{11} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& 1^{11} 5^{1321} \\
& T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[974] & 1^1 8^{21} 1231513 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[973] & 21^1 812231513 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[972] $1^1 8 12231 \overset{1}{5} 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[971] $1^1 8 12314141 \overset{1}{5} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$

[975] $1^1 8 1231 \overset{1}{5} 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[976] $221^1 81231 \overset{1}{5} 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 10 & 4 & 6 \\ 2 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[978] $21^1 1^1 512321 8^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[979] $21^1 8122314141 \overset{1}{5} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[980] $1^1 8^{12} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[981] $21^1 1^1 5^{13} 21 8^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[982] $21^1 6131513151 \overset{1}{5} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[985] $21^1 81231 \overset{1}{5} 132 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 3 & 2 \\ 2 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$

[984] $1^1 8 12231 \overset{1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[977] $21^1 8 1231 \overset{1}{5} 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

- [989] $15131\overset{1}{6}13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [988] $31^1 4132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 17, \Delta = -92, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$
- [990] $321\overset{1}{8}1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [986] $3221\overset{1}{8}123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [987] $321\overset{1}{8}123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [991] $31^1 3218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 17, \Delta = -92, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [996] $1\overset{221}{8}1232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [994] $221^1 812314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [997] $1\overset{1}{8}1231\overset{1}{5}132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [992] $1\overset{221}{8}1231\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [993] $231\overset{1}{5}1322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [995] $221^1 812231\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [998] $21^1 \overset{31}{6}1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -85, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 24 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 4 \end{pmatrix}_{b_0}$$

[1001] $2221^1 81231^1 51 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -22, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[999] $2221^1 81232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -22, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1000] $2231^1 5132181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -22, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 4 & 2 \\ 4 & 4 & 10 & 6 \\ 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1002] $3141^1 32181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 17, \Delta = -92, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.48 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[1003] $41^1 3217123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -30, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.80 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1004] $21^1 8^{151321} 123 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1005] $2221812321^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.16 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[1006] $1^{151^1 2321} 8^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.16 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[1007] $1^{21} 8^{151^1 2321} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.16 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 8 & 10 \\ 1 & 8 & 0 & 2 \\ 2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[1008] $21^{151^1 2321} 8^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.16 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[1012] $1^{15} 8^{13221^1} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1013] $22181^1 2231^1 51 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1011] $161^1 231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -104, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$

[1010] $51^1 2321613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -106, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$

[1009] $1^1 8^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1017] $22181231^1 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.79 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$

[1016] $18^1 1231^1 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.79 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[1014] $21812231^1 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.79 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$

[1015] $1812231^1 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.79 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[1020] $1^1 8^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 64 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

[1022] $1^1 8^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -107, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 28 & 7 & 10 \\ 2 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1021] $2181^1 22314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 40 & 8 & 12 \\ 1 & 8 & 0 & 2 \\ 2 & 12 & 2 & 3 \end{pmatrix}_{b_0}$

[1019] $61^1 2231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -104, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 64 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

[1018] $2181^1 2231513 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -107, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 28 & 7 & 10 \\ 2 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1023] $71^1 2231414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -72, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 89 & 1 & 19 \\ 1 & 1 & -3 & -1 \\ 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1024] $181^1 2231^1 5132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -74, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.59 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 3 & 2 \\ 7 & 40 & 15 & 12 \\ 3 & 15 & 4 & 4 \\ 2 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[1026] $712231^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -79, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1025] $171231^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -79, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1029] $512321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -106, \det = -86, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1030] $3221^1 81223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -86, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1028] $1^1 8^1 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -86, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1027] $151321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -86, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1031] $1513161^1 23151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -93, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.74 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 1 & 12 & 3 \\ 12 & 51 & 15 \\ 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1032] $81^1 2231^1 51322 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -96, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.54 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 8 & 4 & 2 \\ 8 & 54 & 25 & 14 \\ 4 & 25 & 10 & 6 \\ 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[1033] $4141^1 31613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -106, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1034] $6131^1 41414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -119, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1035] $1^1 8^1 123 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -146, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.63 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1036] $322181^1 223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -146, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.63 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1082] & 1^1 \overset{1}{6} \overset{1}{1316131}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1071] & 1^1 \overset{1}{6} \overset{1}{13161231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1062] & 1^1 \overset{1}{321613161231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1063] & 1^1 \overset{1}{413161315131}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1064] & 1^1 \overset{1}{413161315131}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1065] & 1^1 \overset{1}{232171232151}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1066] & 1^1 \overset{1}{7123151321}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[1067] & 1^1 \overset{1}{8123141}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[1068] & 1^1 \overset{1}{8123141}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1069] & 1^1 \overset{1}{613161231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1070] & 1^1 \overset{1}{321712315131}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1081] & 21^1 \overset{1}{7123151321}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1094] & 1^1 \overset{1}{51321812231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[1095] $1^1 5 1321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1093] $1^1 322181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1097] $412321^1 8 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1098] $1^1 5 1316131^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1099] $216^1 131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[1100] $218^1 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1101] $1^1 414132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1102] $1^1 8 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[1103] $1^1 512321 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1104] $1^1 512321 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1105] $1^1 5 1322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1106] $1^1 5 1322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1107] $221^1 8 12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1108] & 1 \overset{1}{15} 8^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1092] & 1^1 322181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1091] & 1^1 31 31 6^{1316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -64, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[1090] & 1^1 217123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1089] & 1^1 316131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1088] & 1^1 316131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1087] & 1^1 414132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1086] & 1^1 413218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1085] & 1 \overset{1}{15} 8^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1084] & 1^1 161232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1096] & 1^1 8^{21} 12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[1120] & 21^1 51322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1119] & 1^1 315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1117] \quad 1^1 7 1231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1122] \quad 2^1 32181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1123] \quad 1^{21} 8^{51321} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1124] \quad 2151321^1 81231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1125] \quad 1^1 712231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1126] \quad 1^1 414141322181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1127] \quad 1^1 513217123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1128] \quad 1321^1 81231^1 51 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1114] \quad 1^1 61316123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1113] \quad 1^1 612321613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1112] \quad 21^1 61232171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1111] \quad 1^{1321} 5^{18} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1110] \quad 2^1 32181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$\begin{aligned}
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[1109] & 21^1 51321812231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1121] & 1^1 81231414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 26, T \in [17, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[1118] & 1^1 151232181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1116] & 1^1 132181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1115] & 1^1 16131613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 26, T \in [17, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1129] & 1^{131} 31 \ 6 \ 1316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -64, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 6 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 5 \end{pmatrix}_{b_0} \\
[1137] & 1^1 31 \ 5132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1135] & 1^1 181231414141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 26, T \in [20, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1134] & 1^1 413161316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1133] & 7123141414141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 26, T \in [20, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1132] & 1^1 232181231 \ 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1131] & 1^1 232181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1130] & 1812231414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 26, T \in [20, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1136] & 2181231414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 26, T \in [20, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1139] & 1^{15131} 6^{13161} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[1140] & 21^{171231513151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1141] & 21^{161316123151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -101, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1138] & 21^{181231414141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[1143] & 31^{132181232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 18, \Delta = -89, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.45 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[1145] & 2161316^{13151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[1144] & 161316^{13151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[1142] & 413218^{123151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[1146] & 31^{3161316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -64, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 2 & 4 \\ 2 & 4 & 4 & 5 \end{pmatrix}_{b_0} \\
[1147] & 321^{15132181231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 18, \Delta = -88, \det = 22, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.55 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 3 & 6 \\ 1 & 3 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[1150] & 41^{132181223151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.80 \rightarrow 18, T_{\max} = 26, T \in [18, 26]
\end{aligned}$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1149] $41^1 1^3 2218123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.80 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1148] $41^1 1^3 21^1 8^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.80 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1151] $161^1 2321613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -103, \det = 42, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.67 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1152] $51^1 23217123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -105, \det = 42, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.67 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1153] $3151^1 232181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 18, \Delta = -89, \det = 44, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.45 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 5 & 10 \\ 1 & 5 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1154] $171^1 2231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 56, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.57 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1155] $181^1 2231414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 72, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 54 & 9 & 14 \\ 1 & 9 & 0 & 2 \\ 2 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[1158] $181231^1 414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 79, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.57 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1157] $1812231^1 414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 79, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.57 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1156] $2181231^1 414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 79, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.57 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1159] $1612321^1 613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -103, \det = 86, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.65 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1161] $21613161^1 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -101, \det = 93, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.74 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[1160] $1613161^1 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -101, \det = 93, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.74 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[1162] $71231^1 41414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 26, \Delta = -102, \det = 119, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$

[1163] $413221^1 8123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 26, \Delta = -107, \det = 137, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.56 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 43 & 1 & 13 \\ 1 & 1 & -2 & 0 \\ 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$

[1164] $4132181^1 223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 26, \Delta = -107, \det = 146, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.63 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$

$T = 15$ [1165] $1^1 23217123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1166] $1^1 23217123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1167] $1^1 32171231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1169] $1^1 414141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1170] $1^1 41316131513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1171] $1^1 41316131513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1172] $1^1 23216131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1168] $1^1 32171231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1213] $1^1 1316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1194] $1^1 3161232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1199] $1^1 3221812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1200] $1^1 4131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1201] $1^1 4131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1202] $1^1 513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1203] $1^1 3221812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1204] $1^1 513221812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

$$\begin{aligned}
[1205] \quad & \overset{1}{221} 8^{123151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1206] \quad & \overset{1}{1} \overset{31}{6} 131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1207] \quad & \overset{1}{1} \overset{31}{6} 131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[1208] \quad & \overset{1}{21} 8^{1223151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1209] \quad & \overset{1}{1} \overset{5}{8}^{1321} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1210] \quad & \overset{1}{1} \overset{3151321}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1211] \quad & \overset{1}{1} \overset{3151321}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1212] \quad & \overset{1}{1} \overset{512321}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1196] \quad & \overset{1}{1} \overset{3218122315131^1}{8} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1214] \quad & \overset{1}{1} \overset{513161315131^1}{8} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1187] \quad & \overset{1}{1} \overset{513221812321^1}{8} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1188] \quad & \overset{1}{1} \overset{512321}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1189] \quad & \overset{1}{1} \overset{513161315131^1}{8} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1190] \quad 1^1 4141321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1191] \quad 1^1 4141321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1197] \quad 1^1 3218122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1195] \quad 1^1 612321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1192] \quad 1^1 81223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[1193] \quad 1^1 3161232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1198] \quad 1^1 8123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[1224] \quad 1^1 4132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1225] \quad 1^1 321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1226] \quad 1^1 6123151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1227] \quad 2218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1228] \quad 1^1 61232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1229] \quad 1^1 613151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[1230] $21^1 61316131^1 51 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1231] $23^1 21812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$

[1232] $16^1 12321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1223] $1^1 4132181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1222] $15^1 13218123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1221] $15^1 13218123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1219] $1^1 3161316123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1218] $31^1 51321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[1217] $1^1 3161316123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1216] $1^1 6^1 1316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1215] $21^1 71232171231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1220] $1^1 6^1 1316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1236] $1^1 6123217123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 27, T \in [18, 27]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1241] & 21^1 613151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[1243] & 1^1 5132218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1242] & 1^1 5132218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1234] & 1^1 61232181232^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1233] & 1^1 51321^1 8^1 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1235] & 1^1 8^1 12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1237] & 1^1 4141414132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1238] & 1^1 7^1 12321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1239] & 21^1 8^1 1231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1240] & 1^1 8^1 1231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1254] & 18123141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1249] & 2221^{11} (12)^{11} 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1256] & 21^{221} (12)^{221} 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& 1^{11} 3221
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1244] & \begin{matrix} 1^{11}3221 \\ 21 \end{matrix} (12) 122222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1257] & \begin{matrix} 1^1 \\ 31 \end{matrix} 61232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.57 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1255] & 22222221(12)12231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1253] & \begin{matrix} 21 \\ 21 \end{matrix} (12) 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^{11}3221 \\ T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1245] & \begin{matrix} 21 \\ 81 \end{matrix} 223141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1252] & \begin{matrix} 221 \\ 1 \end{matrix} (12) 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^{11}3221 \\ T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1251] & \begin{matrix} 1^1 \\ 15 \end{matrix} 13161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.57 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1250] & \begin{matrix} 1^{11}3221 \\ 221 \end{matrix} (12) 12222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1246] & \begin{matrix} 1 \\ 1 \end{matrix} (12) 122222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^{11}3221 \\ T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1247] & \begin{matrix} 21 \\ 1 \end{matrix} (12) 12222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^{11}3221 \\ T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1248] & \begin{matrix} 1^{11}3221 \\ 1 \end{matrix} (12) 1222222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1260] & \begin{matrix} 21^1 \\ 1 \end{matrix} 81231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.50 \rightarrow 19, T_{\max} = 27, T \in [19, 27]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1258] & 21^1 712321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.50 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[1259] & 21^1 812231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.50 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1261] & 512321^1 8123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.45 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[1262] & 151321^1 8123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.45 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[1263] & 221^1 81231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -14, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[1264] & 161^1 23217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -42, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.67 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0} \\
[1265] & 1^{51} 8^{2321} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -42, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.67 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[1266] & 51^1 232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -42, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.67 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[1267] & 2181^1 2231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -56, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1268] & 1^{15131513221} 8^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -56, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1270] & 1712321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -86, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.65 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1269] & 21712321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -86, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.65 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 68 & 11 & 16 \\ 1 & 11 & 0 & 2 \\ 2 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[1271] & 81^1 223141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -90, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.44 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 118 & 1 & 22 \\ 1 & 1 & -3 & -1 \\ 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1272] $4141^1 3218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -106, \det = -106, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.57 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$

[1274] $181231^1 41414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -101, \det = -119, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.45 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 12 & 19 \\ 1 & 12 & 0 & 2 \\ 2 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1273] $812231^1 41414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -101, \det = -119, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.45 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$

[1275] $1513221^1 8123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -102, \det = -137, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.56 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 43 & 1 & 13 \\ 1 & 1 & -2 & 0 \\ 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$

[1277] $51232181^1 223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -104, \det = -146, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.63 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 1 & 15 & 3 \\ 15 & 79 & 19 \\ 3 & 19 & 4 \end{pmatrix}_{b_0}$

[1276] $15132181^1 223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -102, \det = -146, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.63 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 1 & 15 & 3 \\ 15 & 79 & 19 \\ 3 & 19 & 4 \end{pmatrix}_{b_0}$

$T = 16$ [1286] $1^1 8^{123151321} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1278] $1^1 322181231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1279] $1^1 321812231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1290] $1^1 8^{1223151321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1281] $1^1 322181231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1282] $1^1 232181223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1283] $1^1 8^{123151321} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1284] $1^1 51316131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 16, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1289] $1^1 315131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1285] $1^1 51316131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 16, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1288] $1^1 8^{1223151321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1291] $1^1 321812231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1287] $1^1 315131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1280] $1^1 232181223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1313] $1^1 31612321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1314] $1^1 41321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1315] $1^1 13218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1346] & 1^1 7 123151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1343] & 1^1 5 132181231^1 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -73, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1341] & 1^{221} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& 1^{13221} \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1340] & 1^1 31613151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1339] & 21^1 7 12321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1338] & 31^1 6 12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[1337] & 1^1 23218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[1336] & 1^1 32181231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1335] & 1^1 32181231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1334] & 1^1 16 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1333] & 1^1 31613161232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1332] & 1^1 16 13161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[1331] \quad & 221 \overset{1}{8} 1232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1330] \quad & 15 \overset{1}{13} 2181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1344] \quad & 21 \overset{21}{1^1 3221} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1345] \quad & 2221 \overset{1^1 3221}{(12)} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1347] \quad & 1^1 51232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1348] \quad & 15 \overset{1}{13} 2181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1342] \quad & 21 \overset{1}{7} 1232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1349] \quad & 1^1 61315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 28, T \in [19, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1350] \quad & 21 \overset{1}{8} 12321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 28, T \in [19, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1351] \quad & 17 \overset{1}{12} 3217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 28, T \in [19, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1352] \quad & 1^1 51321 8 1232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 28, T \in [19, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1353] \quad & 1^1 61232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 28, T \in [19, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1357] \quad & 2222221(12)1223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 28, T \in [22, 28]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1359] & \begin{matrix} 22221 \\ 21(12)1223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1358] & \begin{matrix} 11 \\ 812314141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1354] & \begin{matrix} 2221 \\ 221(12)1223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1355] & \begin{matrix} 1 \\ 321^1 71232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 19, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1356] & \begin{matrix} 222221 \\ 1(12)1223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1361] & \begin{matrix} 1 \\ 21^1 7123217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.50 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1360] & \begin{matrix} 21^1 \\ 1^1 812321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.50 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1364] & \begin{matrix} 31^1 3221 \\ 221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 20, \Delta = -92, \det = 11, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.45 \rightarrow 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1363] & \begin{matrix} 21 \\ 31^1 3221 \\ 1(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 20, \Delta = -92, \det = 11, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.45 \rightarrow 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[1362] & \begin{matrix} 1^1 \\ 16123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -101, \det = 11, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.45 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[1365] & \begin{matrix} 1^1 \\ 221^1 812321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.57 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[1366] & \begin{matrix} 1^1 \\ 51316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 19, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.26 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1367] $31^{\overline{31}} 6 13161315^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 18, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.80 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 2 & 5 & 4 & 5 \end{pmatrix}_{b_0}$
- [1371] $221(12)12231^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 23, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [1370] $21(12)12231^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 23, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [1369] $2222221(12)12231^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 23, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$
- [1368] $1(12)12231^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 23, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [1373] $231^{\overline{1}} 3221(12)12231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 21, \Delta = -93, \det = 27, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19.48 \rightarrow 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
- [1372] $1(12)12231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 21, \Delta = -93, \det = 27, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19.48 \rightarrow 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [1374] $3221(12)12231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 21, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19.29 \rightarrow 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [1376] $161^{\overline{1}} 232181223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -101, \det = 42, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19.67 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$
- [1375] $1^{\overline{1}} 8^{\overline{1}} 12321^{\overline{1}} 61 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -101, \det = 42, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19.67 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [1377] $2231^{\overline{1}} 3221(12)12231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 21, \Delta = -93, \det = 51, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19.41 \rightarrow 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [1378] $61^{\overline{1}} 2315131613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 56, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19.57 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$
- [1379] $221812321^{\overline{1}} 613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 19.65 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 68 & 11 & 16 \\ 1 & 11 & 0 & 2 \\ 2 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[1380] & 18^{21}12321^1 1613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.65 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1381] & 161232181^1 223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 16, T_{\max} = 28, \Delta = -101, \det = 146, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.63 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1382] & 51321^1 6131613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 152, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.45 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 68 & 1 & 16 \\ 1 & 1 & -2 & 0 \\ 2 & 16 & 0 & 3 \end{pmatrix}_{b_0} \\
[1383] & 81231^1 4141414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 167, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.37 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 218 & 1 & 30 \\ 1 & 1 & -3 & -1 \\ 2 & 30 & -1 & 3 \end{pmatrix}_{b_0} \\
\\
T = 17 & [1393] 1^1 4132181231513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1392] & 1^1 4132181231513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1391] & 1^1 2321613161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1390] & 1^1 3216131613151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1389] & 1^1 2321613161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1388] & 21^1 61316131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 17, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1387] & 1^1 2321712321712321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1386] & 1^1 3216131613151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1385] & 1^1 31 1 6 131613151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1384] & 1^1 31 1 6 131613151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1411] & 1^1 321712321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1412] & 1^1 15 1321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1413] & 1^1 15 1321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1410] & 21^1 613161316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1409] & 1^1 321 1232171231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1414] & 1^1 413218123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	
[1415] $1^1 413218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1408] $1^1 315132181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1416] $1^1 232181231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1417] $1^1 232181231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1418] $21^1 712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1419] $1^1 51316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1427] $1^1 321712321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1426] $1^1 51316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1425] $21^1 6131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1424] $1^1 315132181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1423] $1^1 316131^1 1316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1422] $221^1 812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
[1421] $1^1 31613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	$T = 17, T_{\max} = 18, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1420] & 1^1 \bar{8}^{1321} 1232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1442] & 1^1 \bar{6} 1232181231^1 \bar{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -72, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1428] & 221 \bar{8}^{1321} 12321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1429] & 321 \bar{7}^{1321} 12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[1430] & 221 \bar{8}^{1321} 1232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1431] & 1^1 \bar{8} 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1432] & 21 \bar{8}^{1321} 123151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1433] & 13221 \bar{(12)}^{13221} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -64, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1434] & 1^1 \bar{6} 1315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1435] & 1^1 \bar{6} 12315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 20, T \in [19, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1436] & 221 \bar{(12)}^{13221} 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 20, T \in [19, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1437] & 1^1 \bar{5} 13161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 20, T \in [19, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[1438] $21 \overset{1}{7} 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[1439] $1 \overset{1}{7} 1232161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1440] $1^1 513161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 20, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1441] $1 \overset{1}{6} 1232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -74, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1443] $1 \overset{1}{2} 32181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[1444] $221 \overset{1}{(12)} 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 20, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[1445] $1^1 \overset{1}{6} 5131613113161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 25, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1446] $1 \overset{1}{7} 1232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 25, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1449] $1^1 513161315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 29, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[1448] $1 \overset{1}{7} 123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 25, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1447] $21 \overset{1}{8} 123217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 25, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1452] $21 \overset{1}{(12)} 12222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 24, T \in [21, 24]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$

[1451] $1 \overset{21}{(12)} 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $\overset{1}{1} \overset{1}{5} 13221$

$$\begin{aligned}
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^1 5^{11} 13221 \\ [1453] \end{matrix} 221 \quad (12) \quad 1222 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1450] \end{matrix} 1 \quad (12) \quad 122 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^1 5^{11} 13221 \\ T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^1 5^{11} 13221 \\ [1454] \end{matrix} 1 \quad (12) \quad 122222 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1455] \end{matrix} 1 \quad (12) \quad 12222 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^1 5^{11} 13221 \\ T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^1 5^{11} 13221 \\ [1456] \end{matrix} 2222221(12)1223151^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1457] \end{matrix} 21 \quad (12) \quad 12 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^1 5^{11} 13221 \\ T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1458] \end{matrix} 222221(12)122314141^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1459] \end{matrix} 221(12)122314141^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 2221 \\ [1460] \end{matrix} 21(12)122314141^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^1 \\ [1461] \end{matrix} 322181232181231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.57 \rightarrow 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 22221 \\ [1462] \end{matrix} 1(12)122314141^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[1463] $1^{\overline{1}51321} 1^{\overline{1}} 8 1232171^{\overline{1}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.50 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[1464] $1^{\overline{21}} 8 123217123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.50 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[1465] $21^{\overline{1}} 71232181223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.50 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[1466] $217123218^{\overline{1}} 123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -11, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.45 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0}$

[1467] $21^{\overline{31}} (12)^{\overline{1}3221} 1223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -91, \det = -11, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.45 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 22 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[1468] $17123218^{\overline{1}} 123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -11, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.45 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[1469] $21^{\overline{1}3221} (12)^{\overline{1}3221} 1223141^{\overline{1}} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -91, \det = -11, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.45 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[1470] $221^{\overline{1}} 8123217123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -14, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.57 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[1471] $21^{\overline{321}} 1^{\overline{1}} 8 1232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 24 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 4 \end{pmatrix}_{b_0}$

[1472] $161316131613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -19, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$

[1473] $41^{\overline{3221}} (12)^{\overline{1}3221} 12231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.50 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 1 & 3 \\ 2 & 6 & 3 & 3 \end{pmatrix}_{b_0}$

[1474] $21^{\overline{1}3221} (12)^{\overline{1}3221} 12231^{\overline{1}} 413 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -91, \det = -27, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.48 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$

[1475] $1^{\overline{231}} (12)^{\overline{1}3221} 1223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -27, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 20.48 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 58 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[1476] & \begin{matrix} 1^{13221} \\ 1(12)1223141^1 32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -27, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.48 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[1478] & \begin{matrix} 3221 \\ 413221(12)12231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -28, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.29 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1477] & \begin{matrix} 3221 \\ 3221(12)1223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -28, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.29 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1480] & \begin{matrix} 221 \\ 221(12)12231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.79 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[1481] & \begin{matrix} 2221 \\ 21(12)12231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.79 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[1479] & \begin{matrix} 22221 \\ 1(12)12231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.79 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[1482] & 222221(12)12231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.79 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1483] & 2231^1 413221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.41 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 1 & 15 \\ 1 & 1 & -2 & 0 \\ 2 & 15 & 0 & 3 \end{pmatrix}_{b_0} \\
[1484] & 2231^1 3221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.41 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1485] & \begin{matrix} 1^{13221} \\ 1(12)12231^1 4132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.41 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1486] & 61^1 23216131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.57 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1487] & 71^1 22315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -101, \det = -72, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.50 \rightarrow 21, T_{\max} = 29, T \in [21, 29]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1488] & 223141^1 3221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -83, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.35 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 1 & 19 \\ 1 & 1 & -2 & 0 \\ 2 & 19 & 0 & 3 \end{pmatrix}_{b_0} \\
[1489] & \begin{matrix} 3221 \\ 3221(12)12231^1 41^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -92, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.09 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 22 & 9 & 10 \\ 2 & 9 & 1 & 3 \\ 2 & 10 & 3 & 3 \end{pmatrix}_{b_0} \\
[1491] & 2171232181^1 223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -146, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.63 \rightarrow 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1490] & \begin{matrix} 1 \\ 171232181^1 223151^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -146, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.63 \rightarrow 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1492] & 612321^1 6131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -152, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.45 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \\
& \begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 68 & 1 & 16 \\ 1 & 1 & -2 & 0 \\ 2 & 16 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 18 & [1507] 1^1 22315132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1493] & 1^1 41316131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1494] & 1^1 41316131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1495] & \begin{matrix} 1^1 \\ 13161316131513221^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1496] & 1^1 32161316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1497] & 1^1 \begin{matrix} 2321 \\ 8 \end{matrix} 12321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1498] & \begin{matrix} 1^1 31 \\ 1 \end{matrix} 6 1316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1499] & 1^1 32161316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1500] & 1^1 \begin{matrix} 2321 \\ 8 \end{matrix} 12321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1501] & \begin{matrix} 1^1 31 \\ 1 \end{matrix} 6 1316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1506] & 1^1 22315132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1505] & 1^1 31513218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1504] & \begin{matrix} 1^1 \\ 1 \end{matrix} 5 132181231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1503] & 1^1 31513218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1502] & \begin{matrix} 1^1 \\ 1 \end{matrix} 5 132181231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0 \\
[1530] & \begin{matrix} 1^1 \\ 1 \end{matrix} 5 13218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1529] & \begin{matrix} 1^1 \\ 1 \end{matrix} 5 13218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1547] & 1^1 6123151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19]
\end{aligned}$$

$$\begin{aligned}
[1535] \quad & \overset{1}{2}21\overset{1}{8}12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1534] \quad & \overset{1}{1}613151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1527] \quad & \overset{1}{1}^3161232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1528] \quad & \overset{1}{1}^4132181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1531] \quad & \overset{1}{1}612321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1550] \quad & \overset{21}{1} \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& \overset{1}{1}^513221 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1564] \quad & \overset{21}{1} \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& \overset{1}{1}^513221 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1549] \quad & \overset{1}{1}^3221 \\
& 21 \quad (12) \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1548] \quad & \overset{1}{1}^71232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1553] \quad & \overset{1}{2}21\overset{1}{8}123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1554] \quad & \overset{1}{2}181232161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1555] \quad & \overset{1}{1}^712315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[1556] $1^1 1613161232181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 20, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1551] $1^1 2171232181231^1 51 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 20, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1557] $1^1 413221 21 (12) 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 21, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1558] $1^1 513221 221 (12) 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1559] $1^1 3221 21 (12) 122314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 21, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[1560] $1^1 3161316131513161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1561] $1^1 5132171232181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 21, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1552] $1^1 7122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 20, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1562] $1^1 5132171232181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 21, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1563] $1^1 513221 221 (12) 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1572] $1^1 5131613151322181^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[1566] $1^1 2181232181223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 25, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1565] $1^1 5131613161232171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 30, T \in [21, 30]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1567] & 1 \quad (12) \quad 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1 \\ 31 \ 5 \ 13221 \end{matrix} \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 3 & 3 \\ 2 & 3 & 4 & 4 \\ 2 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[1568] & 21 \quad (12) \quad 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1 \\ 31 \ 5 \ 13221 \end{matrix} \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1569] & 18 \ 12315131613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 30, T \in [21, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1570] & 1^1 6131513218123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 30, T \in [21, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1571] & 21 \ 8 \ 123218123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 25, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1578] & 21 \quad (12) \quad 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1 \\ 31 \ 5 \ 13221 \end{matrix} \\
& T_{\min} = 20.57 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1577] & 21(12)12231414141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1576] & 22221(12)12231414141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1575] & 3221 \quad (12) \quad 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1 \\ 51 \ 13221 \end{matrix} \\
& T_{\min} = 20.57 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 4 \\ 3 & 4 & 5 \end{pmatrix}_{b_0} \\
[1574] & 1(12)12231414141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1573] & 4132181232181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.57 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1579] & 1 \quad (12) \quad 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1 \\ 31 \ 5 \ 13221 \end{matrix}
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 20.57 \rightarrow 21, \quad T_{\max} = 21, \quad T \in [21, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[1582] & \begin{matrix} 21^1 \\ 1 & 8 & 1232181223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 25, \quad \Delta = -98, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.50 \rightarrow 22, \quad T_{\max} = 25, \quad T \in [22, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1581] & \begin{matrix} 21^1 \\ 21^1 812315131613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 30, \quad \Delta = -100, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.50 \rightarrow 22, \quad T_{\max} = 30, \quad T \in [22, 30] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[1580] & \begin{matrix} 21^1 \\ 1 & 8 & 123218123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 25, \quad \Delta = -98, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.50 \rightarrow 22, \quad T_{\max} = 25, \quad T \in [22, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1583] & \begin{matrix} 1^1 513221 \\ 3221 & (12) & 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 21, \quad \Delta = -87, \quad \det = 10, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.60 \rightarrow 21, \quad T_{\max} = 21, \quad T \in [21, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix}_{b_0} \\
[1585] & \begin{matrix} 1^1 \\ 221812321 & 8123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 25, \quad \Delta = -98, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.45 \rightarrow 22, \quad T_{\max} = 25, \quad T \in [22, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[1584] & \begin{matrix} 1 \\ 31^1 413221(12)1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 22, \quad \Delta = -90, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.45 \rightarrow 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[1586] & \begin{matrix} 1^1 3221 \\ 1 & (12) & 122314141^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 22, \quad \Delta = -90, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.45 \rightarrow 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1587] & \begin{matrix} 31^1 3221 \\ 1 & (12) & 122314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 22, \quad \Delta = -90, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.45 \rightarrow 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 22 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[1588] & \begin{matrix} 21^1 \\ 1 & 8 & 123218123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 25, \quad \Delta = -98, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.45 \rightarrow 22, \quad T_{\max} = 25, \quad T \in [22, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[1589] & \begin{matrix} 231^1 513221 \\ 1 & (12) & 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 21, \quad \Delta = -87, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.77 \rightarrow 21, \quad T_{\max} = 21, \quad T \in [21, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[1590] & \begin{matrix} 1^1 51321 \\ 1 & 8 & 1232181^1 22 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 25, \quad \Delta = -98, \quad \det = 14, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.57 \rightarrow 22, \quad T_{\max} = 25, \quad T \in [22, 25] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[1591] & \begin{matrix} 1^1 \\ 221^1 81232181223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 18, \quad T_{\max} = 25, \quad \Delta = -98, \quad \det = 14, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.57 \rightarrow 22, \quad T_{\max} = 25, \quad T \in [22, 25]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1592] $31613161316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -34, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 3 & 5 \\ 2 & 5 & 5 & 6 \end{pmatrix}_{b_0}$

[1593] $3221^{1^1 5 13221} (12) 1223 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 21, \Delta = -87, \det = 20, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.80 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[1594] $2231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 21, \Delta = -87, \det = 21, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.86 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 4 \end{pmatrix}_{b_0}$

[1595] $231^1 413221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -91, \det = 27, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.48 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$

[1596] $3141^1 3221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -90, \det = 27, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.48 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$

[1597] $231^1 41413221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -91, \det = 27, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.48 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$

[1598] $231^1 3221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -91, \det = 27, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.48 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$

[1599] $41^1 3221^{1^1 5 13221} (12) 122 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -106, \det = 30, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.80 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[1600] $2313221(12)12231^1 4141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -91, \det = 47, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.79 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$

[1601] $1^{1^1 3221} (12) 12231^1 4141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -90, \det = 47, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.79 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[1602] $23141^1 3221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -91, \det = 51, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.41 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$

[1603] $1^{1^1 3221} (12) 12231^1 41413 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -90, \det = 51, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.41 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$

[1604] $61^1 231513218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -102, \det = 56, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 21.57 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1605] & 1^{2321^1 513221} (12) 12231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -89, \det = 58, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.41 \rightarrow 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 6 & 12 \\ 1 & 6 & 0 & 2 \\ 2 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1606] & 181^{122315131613151^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 72, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.50 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 54 & 9 & 14 \\ 1 & 9 & 0 & 2 \\ 2 & 14 & 2 & 3 \end{pmatrix}_{b_0} \\
[1607] & 22221(12)12231^1 414141^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 79, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.57 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[1609] & 21^{221}(12)12231^1 414141^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 79, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.57 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[1608] & 1^{2221}(12)12231^1 414141^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 79, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.57 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[1610] & 2314141^{13221}(12)12231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -91, \det = 83, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.35 \rightarrow 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 1 & 19 \\ 1 & 1 & -2 & 0 \\ 2 & 19 & 0 & 3 \end{pmatrix}_{b_0} \\
[1612] & 22181232181^{1223151^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 25, \Delta = -98, \det = 146, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.63 \rightarrow 22, T_{\max} = 25, T \in [22, 25] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1611] & 18^{21} 1232181^{1223151^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 25, \Delta = -98, \det = 146, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.63 \rightarrow 22, T_{\max} = 25, T \in [22, 25] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1613] & 1712321^{16131613151^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 152, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.45 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 128 & 14 & 22 \\ 1 & 14 & 0 & 2 \\ 2 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[1614] & 513161^{123218123151^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -103, \det = 189, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.45 \rightarrow 22, T_{\max} = 30, T \in [22, 30] \\
& \begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 75 & 1 & 17 \\ 1 & 1 & -2 & 0 \\ 2 & 17 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 19 & [1621] 1^{11} 51316131613151321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0 \\
[1619] & 1^{14} 13161316131612321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0 \\
[1623] & 1^{11} 51316131613151321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0 \\
[1622] & 1^{12321} 8^{11} 12321812321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0 \\
[1615] & 1^{1316123218123151321^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

[1616]	$1^1 16 1232181231513221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$	$T = 19, T_{\max} = 19, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$
[1620]	$1^1 316123218123151321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$	$T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$
[1617]	$1^1 413161316131612321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$	$T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$
[1618]	$1^1 315131613161315131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$	$T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$
[1653]	$221^1 (12) 12231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1649]	$1^1 41321812321812231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1648]	$21^1 712321812315131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1647]	$1^1 8 123218123141^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1650]	$1^1 41321812321812231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1646]	$1^1 31613161232181231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1645]	$1^1 8 123218123141^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1651]	$1^1 31613161232181231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1652]	$1^1 16 123218123216131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1644]	$1^1 17 123151321812321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1654]	$1^1 32171232181232151^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1655]	$1^1 32171232181232151^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	$T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
[1656]	$1^1 15 132181232171231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$	$T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1657] & \begin{matrix} 1^1 \\ 15132181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1658] & \begin{matrix} 1^1 \\ 16131612321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1659] & \begin{matrix} 21 \\ 1 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^1 31513221 \\ T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1660] & \begin{matrix} 1^1 \\ 2161316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1661] & \begin{matrix} 1^1 71223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1662] & \begin{matrix} 1^1 51321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1663] & \begin{matrix} 1^1 31513221 \\ 221 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1642] & \begin{matrix} 21 \\ 1 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1^1 31513221 \\ T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1643] & \begin{matrix} 1^1 51321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1667] & \begin{matrix} 1^1 41413221 \\ 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 22, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 22, T \in [21, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1666] & \begin{matrix} 1^1 \\ 18122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 21, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1668] & \begin{matrix} 1^1 \\ 2218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 21, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[1669] \quad & 1^{\overline{151321}} 8^{1232181231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1670] \quad & 1^{\overline{151321}} 8^{1232181231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1671] \quad & 1^{\overline{1513221}} 21^{(12)} 1223141^{1223141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1672] \quad & 1^{\overline{161232171232181231^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1673] \quad & 1^{\overline{21812315132181231^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1674] \quad & 1^{\overline{131613161315132171^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1665] \quad & 1^{\overline{1513221}} 21^{(12)} 1223141^{1223141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1664] \quad & 1^{\overline{141413221}} 1^{(12)} 1223141^{1223141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 22, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 22, T \in [21, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1677] \quad & 1^{\overline{22181232181232151^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1675] \quad & 1^{\overline{132181232181223151^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1676] \quad & 1^{\overline{132181232181223151^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1680] \quad & 1^{\overline{13221}} 1^{(12)} 12231414141^{122314141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 22, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 22, T \in [21, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1679] \quad & 1^{\overline{13221}} 1^{(12)} 12231414141^{122314141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 22, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 22, T \in [21, 22]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1678] & 1^{1^1 3161316131} 6^{13161^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1686] & 1^{1^1 31^5 13221} (12)^{1223141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 22, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 8 & 6 \\ 2 & 0 & 6 & 4 \end{pmatrix}_{b_0} \\
[1685] & 221^{1^1 513221} (12)^{1223151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1684] & 1^{1^1 61316123218123151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1683] & 1^{1^1 71231513218123151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1682] & 1^{21} (12)^{1223151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1681] & 1^{1^1} 8^{123216131613151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1691] & 1^{1^1 31^5 13221} (12)^{1223141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 22, \Delta = -85, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.57 \rightarrow 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 9 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 4 \end{pmatrix}_{b_0} \\
[1689] & 2221(12)1223141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1688] & 1^{221} (12)1223141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1690] & 1^{21} 21(12)1223141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1687] & 1^{1^1} 5123218^{1232181231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 22, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 21.57 \rightarrow 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1692] \quad & 21^1 8123216131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 27, \quad \Delta = -99, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.50 \rightarrow 23, \quad T_{\max} = 27, \quad T \in [23, 27] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[1694] \quad & 31^1 4141413221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 23, \quad \Delta = -89, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, \quad T_{\max} = 23, \quad T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[1695] \quad & 31^1 3221(12)12231414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 23, \quad \Delta = -89, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, \quad T_{\max} = 23, \quad T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1696] \quad & 31^1 413221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 23, \quad \Delta = -89, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, \quad T_{\max} = 23, \quad T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1693] \quad & 31^1 41413221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 23, \quad \Delta = -89, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, \quad T_{\max} = 23, \quad T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 22 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[1697] \quad & 231^1 513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 22, \quad \Delta = -86, \quad \det = -13, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.77 \rightarrow 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[1698] \quad & 3141^1 3221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 23, \quad \Delta = -89, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.48 \rightarrow 23, \quad T_{\max} = 23, \quad T \in [23, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[1699] \quad & 31413221(12)12231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 23, \quad \Delta = -89, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.79 \rightarrow 23, \quad T_{\max} = 23, \quad T \in [23, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1700] \quad & 314141^1 3221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 23, \quad \Delta = -89, \quad \det = -51, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.41 \rightarrow 23, \quad T_{\max} = 23, \quad T \in [23, 23] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 110 & 1 & 22 \\ 1 & 1 & -3 & -1 \\ 2 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[1701] \quad & 2321^1 513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 23, \quad \Delta = -88, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.41 \rightarrow 23, \quad T_{\max} = 23, \quad T \in [23, 23] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 6 & 12 \\ 1 & 6 & 0 & 2 \\ 2 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1702] \quad & 71^1 2231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, \quad T_{\max} = 31, \quad \Delta = -100, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.50 \rightarrow 23, \quad T_{\max} = 31, \quad T \in [23, 31]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1703] $313221(12)12231^1 414141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -79, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.57 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1704] $31414141^1 3221(12)12231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -83, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.35 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 1 & 19 \\ 1 & 1 & -2 & 0 \\ 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

[1705] $4141^1 3221^{1^1 513221} (12) 12 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 31, \Delta = -105, \det = -106, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.57 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1706] $21^{21}(12)12231^1 41414141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -119, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.45 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1707] $2221(12)12231^1 41414141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -119, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.45 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 12 & 19 \\ 1 & 12 & 0 & 2 \\ 2 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1708] $1^{221}(12)12231^1 41414141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -119, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.45 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1709] $1^{1812321^1 6131613151^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -152, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.45 \rightarrow 23, T_{\max} = 27, T \in [23, 27]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 68 & 1 & 16 \\ 1 & 1 & -2 & 0 \\ 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[1710] $21812321^1 6131613151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -152, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.45 \rightarrow 23, T_{\max} = 27, T \in [23, 27]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 128 & 14 & 22 \\ 1 & 14 & 0 & 2 \\ 2 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1711] $1613161^1 23218123151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -189, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22.45 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 75 & 1 & 17 \\ 1 & 1 & -2 & 0 \\ 2 & 17 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 20$ [1716] $1^{13216131613161315131^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1712] $21^1 71232181231513221^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 20, T_{\max} = 20, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1718] $1^{13217123218123151321^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 20, T_{\max} = 21, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1719] $1^{21} (12) 12231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1720] $221^{1^1 2231513221} (12) 12231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1764] & 21 \quad \begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 & 2 & 2 & 1 \end{pmatrix} \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[1765] & 1 \quad \begin{pmatrix} 1 & 5 & 1 & 3 & 2 & 2 & 1 \end{pmatrix} \quad 12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1766] & 1 \quad \begin{pmatrix} 1 & 5 & 1 & 3 & 2 & 2 & 1 \end{pmatrix} \quad 12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1767] & 221812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1754] & 1 \quad \begin{pmatrix} 1 & 3 & 2 & 1 \end{pmatrix} \quad 1232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1769] & 1^1 321812315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1770] & 218123151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1771] & 217123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1750] & 1^1 322181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1751] & 181223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1752] & 1^1 322181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1753] & 1^1 413218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[1780] & 1^1 4141413221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1775] & 1 \quad \begin{matrix} 1 \\ 15 \end{matrix} 13221 \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1778] & 1 \quad \begin{matrix} 1 \\ 15131513221 \end{matrix} (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1779] & 1^1 41413221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1776] & 1 \quad \begin{matrix} 1 \\ 15 \end{matrix} 13221 \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1781] & 1^1 4141413221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1782] & 1^1 414141413221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1783] & 21 \quad \begin{matrix} 1 \\ 15131513221 \end{matrix} (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1784] & 1 \quad \begin{matrix} 1 \\ 15 \end{matrix} 13221 \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1785] & 17 \quad \begin{matrix} 1 \\ 171232171232181231 \end{matrix} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1786] & 18 \quad \begin{matrix} 1 \\ 181231513161316131 \end{matrix} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1787] & 1^1 414141413221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

[1788] $1^{1321812321} 8^{1232161^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1789] $1^{1316131613151322181^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1772] $1^{15131513221} (12)^{12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 23, T \in [22, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1777] $21^{15131513221} (12)^{12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 23, T \in [22, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1773] $21^{1513221} (12)^{12231^1} 5^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1774] $1^{1316131613161232171^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1791] $1^{1812231513218123151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 28, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1794] $31^1 5^{13221} (12)^{12231^1} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 24, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[1793] $31^1 5^{13221} (12)^{122314141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1792] $1^{31^1 5^{13221}} (12)^{12231^1} 5^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1790] $1^{181231513218123151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 28, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1795] $1^{31^1 5^{13221}} (12)^{12231^1} 5^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22.57 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1796] $1^{21} (12)^{122314141414141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 28, \Delta = -99, \det = 7, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 26, T_{\max} = 28, T \in [26, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1797] & \overset{1}{1} 5 13221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.57 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1798] & 221(12)122314141414141^{\textcolor{red}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 7, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 28, T \in [26, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1799] & 21^{\textcolor{red}{1}} 81231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 8, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23.50 \rightarrow 24, T_{\max} = 28, T \in [24, 28] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1800] & \overset{1}{1} 5131513221(12)12231^{\textcolor{red}{1}} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 24, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23.45 \rightarrow 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[1801] & \overset{1}{1} 5 13221(12)12231^{\textcolor{red}{1}} 5 132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -81, \det = 13, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.77 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[1802] & 51321^{\textcolor{red}{1}} 8123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 32, \Delta = -101, \det = 19, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23.26 \rightarrow 24, T_{\max} = 32, T \in [24, 32] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0} \\
[1803] & 231^{\textcolor{red}{1}} 3221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 24, \Delta = -89, \det = 27, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23.48 \rightarrow 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 2 & 11 & 0 & 3 \end{pmatrix}_{b_0} \\
[1804] & \overset{1}{1} 613161316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 29, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23.17 \rightarrow 24, T_{\max} = 27, T \in [24, 27] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[1805] & 61^{\textcolor{red}{1}} 23217123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 32, \Delta = -101, \det = 56, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23.57 \rightarrow 24, T_{\max} = 32, T \in [24, 32] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1806] & 181^{\textcolor{red}{1}} 2231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 72, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23.50 \rightarrow 24, T_{\max} = 28, T \in [24, 28] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1807] & \overset{21}{1}(12)12231^{\textcolor{red}{1}} 4141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 167, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23.37 \rightarrow 24, T_{\max} = 28, T \in [24, 28] \\
& \begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 218 & 1 & 30 \\ 1 & 1 & -3 & -1 \\ 2 & 30 & -1 & 3 \end{pmatrix}_{b_0} \\
[1808] & 221(12)12231^{\textcolor{red}{1}} 4141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 167, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} =
\end{aligned}$$

3.0

$$T_{\min} = 23.37 \rightarrow 24, T_{\max} = 28, T \in [24, 28]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 141 & 14 & 23 \\ 1 & 14 & 0 & 2 \\ 2 & 23 & 2 & 3 \end{pmatrix}_{b_0}$$

¹513221

$$[1809] 414141^1 3221 (12) 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 32, \Delta = -104, \det = 230, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23.45 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 82 & 1 & 18 \\ 1 & 1 & -2 & 0 \\ 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1810] 513221^1 8123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 32, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

3.0

$$T_{\min} = 23.37 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 1 & 18 & 3 \\ 18 & 75 & 21 \\ 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [1822] 1^1 6^{131} 1316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1816] 1^1 8^{1321} 123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1817] 1^1 8^{1321} 123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1818] 1^1 8^{12321} 12321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1819] 1^1 32218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1815] 1^1 6^{131} 1316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1821] 1^1 6^{131} 1316131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1814] 1^1 8^{12321} 12321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1823] 1^1 32218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1824] 21 (12) 1^{12231513221} 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1825] 1^1 41316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1826] 1^1 31612321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1827] 221 8^{131} 1232181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1828] 1^1 31612321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1829] 1^1 32161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1830] 21 (12) 1^{12231513221} 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1813] 1^1 41316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1812] 21 6^{131} 13161316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1811] 1^1 6^{131} 131613161316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1820] 1^1 23218123151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$$

$$[1859] 1^1 (12) 1^{131513221} 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[1854] 1 (12) 1^{122315131} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

¹513221

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1857] & 1^{\overline{1}2321812321} 8^{1232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 21, T_{\max} = 22, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1858] & 1^{\overline{1}612321812321^1} 1^{1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 21, T_{\max} = 22, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1855] & 1^{\overline{1}712321712321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1860] & 1^{\overline{1}513218123218123141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1861] & 21^{\overline{1}513221} (12)^{122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 21, T_{\max} = 22, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1862] & 1^{\overline{1}4132181232181232151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1863] & 21^{\overline{1}71232181232171231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1864] & 21^{\overline{1}513221} (12)^{122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 21, T_{\max} = 22, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1865] & 1^{\overline{1}612321812321812231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1866] & 1^{\overline{1}513218123218123141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1851] & 1^{\overline{1}(12)^{122315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 21, T_{\max} = 22, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1852] & 1^{\overline{1}4132181232181232151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1856] & 1^{131513221} (12) 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1853] & 2218^{1123218123216131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1869] & 15^{113221(12)12231414141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1872] & 1^{15131513221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 24, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 24, T \in [23, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1873] & 31^{1513221} (12) 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 23, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[1874] & 17^{11232181232181231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1875] & 21^{13221} (12) 122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 23, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1876] & 16^{1131513221(12)12231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 23, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1871] & 18^{112321613161316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1870] & 1^{15131513221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 24, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 24, T \in [23, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1868] & 15^{113221(12)12231414141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[1867] $1^1 8 1232171232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1877] $1^1 7123217123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 29, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1878] $1^1 5131613161315132181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 29, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[1879] $1(12)1223141414141414141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 28, \Delta = -98, \det = -7, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 28, T \in [27, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$

[1880] $21(12)1223141414141414141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 28, \Delta = -98, \det = -7, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 28, T \in [27, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$

[1881] $31^1 6 131513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 24, \Delta = -85, \det = -7, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23.57 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[1882] $612321^1 8 123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 33, \Delta = -100, \det = -19, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24.26 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$

[1883] $61^1 23218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 33, \Delta = -100, \det = -56, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24.57 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$

[1884] $2321^1 6131513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 25, \Delta = -88, \det = -58, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24.41 \rightarrow 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 86 & 1 & 18 \\ 1 & 1 & -2 & 0 \\ 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$

[1885] $81^1 223151316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 29, \Delta = -99, \det = -90, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 29, T \in [25, 29]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 54 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$

[1886] $1(12)12231^1 414141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 28, \Delta = -98, \det = -223, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24.31 \rightarrow 25, T_{\max} = 28, T \in [25, 28]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 310 & 1 & 36 \\ 1 & 1 & -3 & -1 \\ 2 & 36 & -1 & 3 \end{pmatrix}_{b_0}$$

[1887] $21(12)12231^1 4141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 28, \Delta = -98, \det = -223, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24.31 \rightarrow 25, T_{\max} = 28, T \in [25, 28]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 193 & 16 & 27 \\ 1 & 16 & 0 & 2 \\ 2 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[1888] $712321^1 61316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 29, \Delta = -99, \det = -236, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24.34 \rightarrow 25, T_{\max} = 29, T \in [25, 29]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 128 & 1 & 22 \\ 1 & 1 & -2 & 0 \\ 2 & 22 & 0 & 3 \end{pmatrix}_{b_0}$$

[1889] $41414141^1 3221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 33, \Delta = -103, \det = -402, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24.39 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 142 & 1 & 24 \\ 1 & 1 & -2 & 0 \\ 2 & 24 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 22$ [1891] $21 \overset{1}{\underset{1}{5}} 13221 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1892] $1^1 413218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1896] $1^1 151316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1897] $1^1 6^{131} 13161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1898] $1 \overset{1}{(12)} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1899] $1^1 \overset{1}{\underset{1}{5}} 13221 123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1900] $1^1 2231513221 (12) 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1901] $1 \overset{1}{(12)} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1902] $1^1 \overset{1}{\underset{1}{5}} 13221 123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1903] $132181232181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1904] $1^1 413218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1905] $1^1 413161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1906] $1 \overset{1}{(12)} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1907] $1^1 \overset{1}{\underset{1}{5}} 13221 1321712321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1908] $1 \overset{1}{(12)} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1909] $1^1 321712321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1910] $21 \overset{1}{\underset{1}{5}} 13221 (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1911] $21 \overset{1}{\underset{1}{5}} 13221 (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1912] $1^1 413161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1890] $21 \overset{1}{\underset{1}{5}} 13221 (12) 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$
[1895] $1^1 6^{131} 13161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1893] $1^1 151316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1894] $1^1 12231513221 (12) 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1946] $1^1 51232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1938] $1^1 16123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[1939] $1^1 1712321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1940] $1^1 1513218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1941] $1^1 15132181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1942] $1^1 132181232171232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1943] $1^1 3221 21 (12) 1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1944] $1^1 21712321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1945] $1^1 22181232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1937] $1^1 2171232181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[1947] $1^1 3221 21 (12) 1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1933] $1^1 21812321712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1934] & 1^1 4141413221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1936] & 1^1 4141413221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1935] & 1^1 5132181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1957] & 1^1 2181232181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1953] & 1^1 5131513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1954] & 1^1 3221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1955] & 1^1 61315132181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1956] & 1^1 5131513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1951] & 1^1 41413221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1952] & 1^1 61231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1950] & 1^1 41413221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1949] & 1^1 6131513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1948] & \overset{1}{1} 7 12 31 5 13 22 1 (12) 12 23 1 \overset{21}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1958] & \overset{1}{21} (12) 12 23 1 5 13 16 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 34, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 34, T \in [25, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1959] & \overset{1}{1} 5 13 16 13 16 13 16 12 32 18 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 29, T \in [25, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1960] & \overset{1}{31} 5 13 22 1 (12) 12 23 1 5 13 15 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 25, \Delta = -82, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[1961] & \overset{1}{1} \overset{1}{5} 13 21 8 12 32 1 12 32 1 7 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 29, T \in [25, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1962] & \overset{1}{1} 8 12 32 1 7 12 32 1 8 12 31 5 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 29, T \in [25, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1963] & 1 (12) 12 23 1 4 14 14 14 14 14 14 1 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 7, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1965] & \overset{1}{31} 6 13 15 13 22 1 (12) 12 23 1 4 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 25, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24.57 \rightarrow 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 9 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 4 \end{pmatrix}_{b_0} \\
[1964] & \overset{1}{31} 5 13 22 1 (12) 12 23 1 5 13 15 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 25, \Delta = -82, \det = 7, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24.57 \rightarrow 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1966] & \overset{1}{21} \overset{1}{8} 12 32 1 7 12 32 1 8 12 31 5 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 8, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25.50 \rightarrow 26, T_{\max} = 29, T \in [26, 29] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1967] & \overset{31}{31} 6 13 16 13 16 13 16 13 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -4, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 4 & 6 \\ 2 & 6 & 6 & 7 \end{pmatrix}_{b_0}$$

[1968] $17123218^{11}123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 19$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 25.26 \rightarrow 26$, $T_{\max} = 29$, $T \in [26, 29]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[1969] $61^{11}231513221^{11}12 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 34$, $\Delta = -101$, $\det = 56$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 25.57 \rightarrow 26$, $T_{\max} = 34$, $T \in [26, 34]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1970] $1812321^{11}61316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 236$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 25.34 \rightarrow 26$, $T_{\max} = 29$, $T \in [26, 29]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 206 & 17 & 28 \\ 1 & 17 & 0 & 2 \\ 2 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1971] $1(12)12231^{11}41414141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 287$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 25.27 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 253 & 18 & 31 \\ 1 & 18 & 0 & 2 \\ 2 & 31 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 23$ [1980] $1^{11}4141413221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -26$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1977] $1^{11}2315132181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1978] $1^{11}6^{11}13161316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -19$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1979] $1^{11}6^{11}13161316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -19$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1973] $1^{11}5^{11}13161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -20$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1981] $1^{11}4141413221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -26$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1982] $1^{11}3151321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 24$, $\Delta = -30$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1983] $1^{11}2321812321712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 24$, $\Delta = -30$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1984] $1^{11}2321712321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1985] $1^{11}8^{11}12321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1986] $1^{11}8^{11}12321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1987] $1^{11}8^{11}1232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1988] $1^{11}8^{11}1232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1989] $1^{11}2321712321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1990] $1^{11}5^{11}13161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -20$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1974] $1^{11}5^{11}13218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -23$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

[1976] $1^{11}2315132181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$

- [1972] $1^1 4131613161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$
- [1975] $1^1 513218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$
- [2008] $1^1 5131513221 (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [2010] $1^1 5131513221 (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [2011] $1^1 321812321 8 1232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [2012] $1^1 61232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [2013] $1^1 217123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$
- [2014] $1^1 221812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [2015] $1^1 6123218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [2016] $1^1 316131513221 (12) 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$
- [2017] $1^1 316131513221 (12) 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [2018] $1^1 21812321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [2019] $1^1 613151321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23, T_{\max} = 24, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2009] & \overset{1}{221} \overset{1}{81232181232181231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2030] & \overset{1}{1}^1 712315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2025] & \overset{1}{1}^1 613161232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2026] & \overset{1}{1}^1 712231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2027] & \overset{1}{1}^1 41413221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2028] & \overset{1}{1}^1 71231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2029] & \overset{1}{1}^1 3221(12)122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 26, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 26, T \in [25, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2024] & \overset{1}{1}^1 6131513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[2031] & \overset{1}{1}^1 3221(12)122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 26, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 26, T \in [25, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2032] & \overset{1}{1}^1 812231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[2020] $1^1 21 8 1231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[2022] $1^1 316131613161315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[2023] $1^1 15 13221 (12) 122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 25, \Delta = -72, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[2021] $1^1 16 131513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 25, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[2036] $1^1 18 123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 29, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 29, T \in [26, 29]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[2033] $1^1 31 5 13221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 26, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[2034] $1^1 21 (12) 122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 31, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[2035] $1^1 16 131513221(12)12231^1 513 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 26, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[2037] $(12)122314141414141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 29, \Delta = -96, \det = -7, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[2038] $1^1 31 6 131513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 26, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25.57 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[2039] $1^1 321 8 1231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 26, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 25.57 \rightarrow 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[2040] & 21^1 8123218123218123151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 23, T_{\max} = 29, \Delta = -97, \det = -8, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26.50 \rightarrow 27, T_{\max} = 29, T \in [27, 29] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[2041] & 2321^1 81231513221(12)12231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 23, T_{\max} = 26, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25.77 \rightarrow 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 11 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 4 \end{pmatrix}_{b_0} \\
[2042] & 21812321^1 8123218123151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 23, T_{\max} = 29, \Delta = -97, \det = -19, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26.26 \rightarrow 27, T_{\max} = 29, T \in [27, 29] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0} \\
[2043] & 1812321^1 8123218123151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 23, T_{\max} = 29, \Delta = -97, \det = -19, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26.26 \rightarrow 27, T_{\max} = 29, T \in [27, 29] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0} \\
[2044] & 71^1 2231513221^{1513221}(12)12 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 23, T_{\max} = 31, \Delta = -99, \det = -72, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26.50 \rightarrow 27, T_{\max} = 31, T \in [27, 31] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[2045] & (12)12231^1 4141414141414141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 23, T_{\max} = 29, \Delta = -96, \det = -359, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26.24 \rightarrow 27, T_{\max} = 29, T \in [27, 29] \\
& \begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 542 & 1 & 48 \\ 1 & 1 & -3 & -1 \\ 2 & 48 & -1 & 3 \end{pmatrix}_{b_0} \\
T = 24 & [2050] 216^{11} 13161316131613161231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 24, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2049] & 1^{12321} 8^{12321} 12321812321812231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2052] & 1^1 41321812321812321712321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2053] & 1^1 41321812321812321712321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2054] & 1^{12321812321} 8^{12321812321} 12321812321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 25, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2055] & 1^1 31612321812321812315131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 25, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2056] & 1^1 31612321812321812315131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 25, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2057] & 15^{11} 131613161316131613141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 24, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2058] & 15^{11} 131613161316131613141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 24, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0 \\
[2059] & 1^{1321} 8^{1321} 12321812321812321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

3.0

$$[2060] \ 1 \overset{1}{8} \overset{1}{12321812321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2061] \ 1 \overset{1}{6} \overset{1}{123218123218123151321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2048] \ 21 \overset{1}{6} \overset{1}{1316131613161316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2047] \ 1 \overset{1}{5131513221} (12) \overset{1}{12231513221} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2046] \ 1 \overset{1}{5131513221} (12) \overset{1}{12231513221} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2051] \ 1 \overset{1}{2321} \overset{1}{8} \overset{1}{12321812321812231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2090] \ 1 \overset{1}{513221} (12) \overset{1}{1223151316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2078] \ 1 \overset{1}{41413221} (12) \overset{1}{1223151316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2079] \ 1 \overset{1}{7123151321812321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2080] \ 21 \overset{1}{3221} (12) \overset{1}{122315132181231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2081] \ 21 \overset{1}{3221} (12) \overset{1}{122315132181231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2082] \ 221 \overset{1}{8123218123218123141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[2083] \ 1 \overset{1}{41413221} (12) \overset{1}{1223151316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2084] \ 1 \overset{1}{6131612321812321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2085] \ 1 \overset{1}{31513221} (12) \overset{1}{122315132161} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[2086] & 1^1 16 131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2087] & 1^1 21 7 1232181232181231^1 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2088] & 1^1 21 7 1232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2089] & 1^1 15 13221 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2101] & 1^1 16 1316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2104] & 1^1 15 13221 1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2103] & 1^1 17 1231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[2102] & 1^1 15131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2100] & 1^1 141413221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2099] & 1^1 17 1231513221(12)12231^1 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2098] & 1^1 1413221(12)122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2097] & 1^1 413221(12)122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2096] & 1^1 2181231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2095] & 1^1 3218123218123151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2094] & 1^1 1812315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2093] & 1^1 3161316131613161232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2092] & 1^1 316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[2091] & 1^1 31513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[2105] & 1^1 1812231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2106] & 1^1 181231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 31, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 31, T \in [27, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[2107] & 1^1 513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 31, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 31, T \in [27, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[2109] & 1^1 32181231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 27, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

3.0

$$T_{\min} = 26.57 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

[2108] $1^1 3161316131513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 27, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 26.57 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[2110] $1^1 513221 21 (12) 122315132181^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 31, \Delta = -98, \det = 8, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 27.50 \rightarrow 28, T_{\max} = 31, T \in [28, 31]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2111] $1^1 513221 21 (12) 12231513221^1 81 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 31, \Delta = -98, \det = 72, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 27.50 \rightarrow 28, T_{\max} = 31, T \in [28, 31]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 54 & 9 & 14 \\ 1 & 9 & 0 & 2 \\ 2 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[2112] $613161^1 23218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 32, \Delta = -99, \det = 317, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 27.32 \rightarrow 28, T_{\max} = 32, T \in [28, 32]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 163 & 1 & 25 \\ 1 & 1 & -2 & 0 \\ 2 & 25 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 25$ [2118] $1^1 231513221 1 (12) 1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 26, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2117] $1^1 316123218123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2119] $21 7^1 123218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2120] $1^1 2321 1 8^1 123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2113] $1^1 51321812321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2121] $1^1 321712321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2114] $1^1 232171232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2122] $1^1 321712321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2123] $21 6^1 131613161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2124] $1^1 2321 1 8^1 123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2115] $1^1 232171232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2125] $1^1 2231513221 1 (12) 122315132161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2126] $1^1 51316131613161316131 5 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[2127] $1^1 16131513221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& [2128] \quad 1^{1231513221} (12) \quad 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0 \\
& [2116] \quad 1^{11} 151321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0 \\
& [2145] \quad 1^{11} 18123151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2146] \quad 1^{132181231513221} (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2154] \quad 1^{132181231513221} (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2153] \quad 1^{11} 171231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2152] \quad 1^{131513221} (12) \quad 1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2151] \quad 1^{13221} (12) \quad 12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2150] \quad 1^{13221} (12) \quad 12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2149] \quad 22181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2148] \quad 1^{123218123218123151322181^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2147] \quad 2218123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2159] & \text{}^1 1^1 71232171232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -75, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2166] & \text{}^1 1^1 61231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -74, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2165] & \text{}^1 1^1 21^1 81231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -69, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2164] & \text{}^1 1^1 1^1 812231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -69, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2163] & \text{}^1 1^1 1^1 513221(12)122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2162] & \text{}^1 1^1 1^1 513221(12)122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2161] & \text{}^1 1^1 1^1 6131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2158] & \text{}^1 1^1 1^1 61316131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -75, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[2160] & \text{}^1 1^1 3221 1(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -78, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2155] & \text{}^1 1^1 1^1 812231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[2156] $1^1 21 8 1231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[2157] $1^1 3221 1 (12) 12231513218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2168] $1^1 513221 1 (12) 122315131613161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 33, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[2167] $1^1 51321812321812315132181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 32, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 32, T \in [28, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[2169] $81^1 2231513218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 32, \Delta = -98, \det = -90, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28.44 \rightarrow 29, T_{\max} = 32, T \in [29, 32]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 54 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$

[2170] $812321^1 613161316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 31, \Delta = -97, \det = -338, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28.27 \rightarrow 29, T_{\max} = 31, T \in [29, 31]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 206 & 1 & 28 \\ 1 & 1 & -2 & 0 \\ 2 & 28 & 0 & 3 \end{pmatrix}_{b_0}$

$T = 26$ [2182] $1^1 3161232181232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2180] $1^1 2321 1^1 8 1232181232181231^1 5 1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2171] $1^1 321 1^1 8 12321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2181] $1^1 17 1231513221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2172] $1^1 2321 1^1 8 1232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2185] $1^1 3221812321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2173] $1^1 3221812321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2174] $1^1 321 1^1 8 12321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2175] $1^1 2231513221 1^1 (12) 1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2176] $1^1 221 8 123218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2177] $1^1 2321 1^1 8 1232181232181231^1 5 1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2179] $1^1 21 6 13 16 13 16 13 16 13 16 13 1^1 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2178] $1^1 3 16 12 32 18 12 32 18 12 32 17 12 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2183] $1^1 23 21 1^1 8 12 32 18 12 32 18 12 32 15 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2184] $1^1 6 12 32 18 12 32 18 12 32 17 12 32 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2206] $1^1 18 12 23 15 13 22 1^1 (12) 12 23 15 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2205] $1^1 15 13 22 1^1 1 (12) 12 23 15 13 21 8 12 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2204] $1^1 15 13 22 1^1 1 (12) 12 23 15 13 21 8 12 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2203] $1^1 7 12 32 17 12 32 18 12 32 18 12 32 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2202] $1^1 4 13 22 1 (12) 12 23 15 13 16 13 16 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2207] $1^1 5 13 16 13 15 13 22 1 (12) 12 23 15 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[2208] $1^1 5 13 16 13 15 13 22 1 (12) 12 23 15 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2209] $1^1 4 14 13 22 1 (12) 12 23 15 13 21 8 12 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2210] $1^1 2 18 12 31 5 13 22 1^1 (12) 12 23 15 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2211] $1^1 3 16 13 15 13 22 1 (12) 12 23 15 13 15 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -52, \det = 2, (n_+, n_-, n_0) =$

$$\begin{aligned}
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2212] & 1^1 316131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -52, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2213] & 1^1 413221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -54, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2214] & 1^1 41413221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -53, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2215] & 1^1 413221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -77, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2216] & 1^1 161316131513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2217] & 1^1 413221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -77, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2218] & 1^1 171231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -72, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2219] & 1^1 513161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -78, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2220] & 1^1 513161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -78, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2221] & 1^1 3221 1 (12) 122315132181232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2222] & 1^1 321812321812321 8 1232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2223] & 1^1 712231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -72, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2224] & 1^1 81232171232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 28, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
T = 27 & [2238] 1^1 8^{1321} 123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2235] & 21^1 712321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -19, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2237] & 1^1 31513221 (12) 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = \\
& 3.0 \\
[2234] & 1^1 51316131513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -24, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2239] & 1^1 8^{1321} 123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = \\
& 3.0 \\
[2240] & 1^1 231513221 (12) 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = \\
& 3.0 \\
[2233] & 21^1 81231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2232] & 1^1 31612321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2231] & 1^1 41321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2230] & 1^1 31513221 (12) 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = \\
& 3.0 \\
[2229] & 1^1 32171232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2228] & 1^1 31612321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2227] & 1^1 51316131513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -24, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2226] & 1^1 231513221 (12) 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = \\
& 3.0 \\
[2225] & 1^1 612321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = \\
& 3.0 \\
[2236] & 1^1 812231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[2260] & 1^1 812321712321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -47, \det = -2, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2262] & 1^1 513221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2263] & 1^1 1513221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2264] & 1^1 16131513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2265] & 1^1 161316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2261] & 1^1 2321812321812321^1 8 1232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2259] & 1^1 1316131513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[2266] & 1^1 1513221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -72, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2267] & 1^1 1513221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -72, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2268] & 1^1 2181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2269] & 1^1 181232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2270] & 1^1 1613161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[2272] $1^1 413221(12)122315132181232161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 27, T_{\max} = 29, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[2273] $1^1 171232181231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 27, T_{\max} = 29, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[2271] $1^1 1812231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 27, T_{\max} = 29, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2274] $1^1 8123217123218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 27, T_{\max} = 33, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 33, T \in [30, 33]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[2275] $1^1 7123218123218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 27, T_{\max} = 33, \Delta = -97, \det = -29, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30.17 \rightarrow 31, T_{\max} = 33, T \in [31, 33]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$

[2276] $61^1 232181231513221(12)1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 27, T_{\max} = 39, \Delta = -100, \det = -56, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30.57 \rightarrow 31, T_{\max} = 39, T \in [31, 39]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$

$T = 28$ [2282] $1^1 151321812321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2288] $1^1 315132181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2283] $1^1 3161316131513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 29, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2281] $1^1 151321812321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2280] $1^1 413218123218123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2279] $1^1 413218123218123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2278] $1^1 321712321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2277] $1^1 321712321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2284] $1^1 3161316131513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 29, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2285] $1^1 21712321812321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2286] $1^1 221812321812321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 28, T_{\max} = 28, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2287] $1^{\overline{1321}} 8^{1232181232181232171231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2291] $1^{\overline{1}} 16^{1316131513221(12)12231513221^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2290] $1^{\overline{1321}} 8^{1232181232181232171231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2289] $1^{\overline{1}} 13^{15132181232181232181232151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2304] $17^{1231513221(12)1223151316131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2305] $1^{\overline{1}} 13^{16131513221(12)1223151322171^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[2306] $1^{\overline{1}} 32^{181231513221(12)12231513151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2310] $1^{\overline{1}} 5^{132181231513221(12)122315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[2311] $1^{\overline{1}} 31^{1613161316131513221(12)12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2312] $1^{\overline{1}} 31^{1613161316131513221(12)12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2313] $21^{1^{\overline{13221}}(12)12231513221(12)12231^{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[2309] $1^{\overline{1}} 5^{132181231513221(12)122315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2308] $1^{\overline{1}} 32^{181231513221(12)12231513151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2307] $1^{\overline{1}} 18^{12321812321812321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2315] & 1^1 171232181231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[2314] & 1^1 2181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2316] & 1^1 1513221(12)122315132181232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2319] & 1^1 513221 (12) 122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 35, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 35, T \in [31, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[2318] & 1^1 51321812321812321 8 1232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 34, T \in [31, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[2317] & 1^1 613161316131513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 35, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 35, T \in [31, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[2320] & 1^1 32181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 31, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30.57 \rightarrow 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[2321] & 1^1 18123218123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 29, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31.17 \rightarrow 32, T_{\max} = 34, T \in [32, 34] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 29 & [2334] 1^1 8^{1321} 1232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2327] & 1^1 1612321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2328] & 1^1 4132181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2329] & 1^1 221812321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -17, \det = 0, (n_+, n_-, n_0) =
\end{aligned}$$

$(1, 29, 1), c_{\text{ext}} = 3.0$
 $[2330] \ 1 \overset{1}{5}132181231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2331] \ 1 \overset{1}{8} \ 12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2332] \ 1 \overset{1}{8} \ 12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2333] \ 1 \overset{1}{3}221812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2326] \ 1 \overset{1}{5}132181231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2335] \ 1 \overset{1}{4}132181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2325] \ 1 \overset{1}{5}13218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2324] \ 1 \overset{1}{5}13218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2322] \ 1 \overset{1}{3}161232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2323] \ 1 \overset{1}{3}161232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 29, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$
 $[2349] \ 21 \overset{1}{(12)} \overset{1}{12231513221(12)1223141}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 30, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\text{min}} = 30, T_{\text{max}} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$
 $[2351] \ 1 \overset{1}{(12)} \overset{1}{122315132181232181231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 30, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\text{min}} = 30, T_{\text{max}} = 30, T \in [30, 30]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
 $[2350] \ 1 \overset{1}{6}1232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 30, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\text{min}} = 30, T_{\text{max}} = 30, T \in [30, 30]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
 $[2353] \ 21 \overset{1}{(12)} \overset{1}{12231513221(12)1223141}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 30, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\text{min}} = 30, T_{\text{max}} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$
 $[2348] \ 1 \overset{1}{8}12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 30, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\text{min}} = 30, T_{\text{max}} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
 $[2347] \ 21 \overset{1}{8}1231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 30, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\text{min}} = 30, T_{\text{max}} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
 $[2346] \ 1 \overset{1}{6}131513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\text{max}} = 30, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2345] & 1^1 61231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -49, \det = -2, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& 1^1 3221 \\
[2352] & 1 (12) 122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -54, \det = -2, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& 1^1 \\
[2354] & 21 81232181231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
& 1^1 \\
[2356] & 17 1232181231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -69, \det = -3, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& 1^1 3221 \quad 21 \\
[2357] & 1 (12) 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -77, \det = -3, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& 1^1 3221 \quad 21 \\
[2358] & 1 (12) 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -77, \det = -3, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2355] & 1^1 321812321812321812315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -74, \det = -3, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& 1^1 \\
[2359] & 18 1232181231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 35, \Delta = -97, \det = -4, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 35, T \in [32, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
& 1^1 513221 \\
[2360] & 1 (12) 122315132181232181^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 35, \Delta = -97, \det = -8, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32.50 \rightarrow 33, T_{\max} = 35, T \in [33, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \\
T = 30 \quad [2361] & 16 123218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -19, \det = 0, (n_+, n_-, n_0) = \\
& (1, 30, 1), c_{\text{ext}} = 3.0 \\
& 1^1 \\
[2362] & 21 712321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -18, \det = 0, (n_+, n_-, n_0) =
\end{aligned}$$

$(1, 30, 1), c_{\text{ext}} = 3.0$
 [2363] $1^{1^1} 41321812321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2364] $1^{1^1} 41321812321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2365] $1^{1^1} 8^{1^1} 123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2366] $1^{1^1} 8^{1^1} 123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2367] $1^{1^1} 15^{1^1} 132181232181232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2368] $1^{1^1} 61232181231513221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2369] $1^{1^1} 316131513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2370] $1^{1^1} 32171232181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2371] $1^{1^1} 32171232181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2372] $1^{1^1} 316131513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2373] $1^{1^1} 15^{1^1} 132181232181232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 [2385] $1^{1^1} 15^{1^1} 13221(12)12231513221^{21}(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$
 [2384] $1^{1^1} 413221(12)122315132181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
 [2387] $1^{1^1} 2321812321812321812315132181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$
 [2388] $1^{1^1} 413221(12)122315132181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
 [2389] $1^{1^1} 413221(12)12231513221^{21}(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$
 [2390] $1^{1^1} 15^{1^1} 13221(12)12231513221^{21}(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

$$\begin{aligned}
[2391] \quad & 1 \overset{1}{7} 1232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2392] \quad & 1 \overset{1}{2} 13221 21 (12) 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -52, \det = 2, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[2393] \quad & 1 \overset{1}{7} 12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -47, \det = 2, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2394] \quad & 1 \overset{1}{2} 13221 21 (12) 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -52, \det = 2, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[2386] \quad & 1 \overset{1}{7} 1231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -47, \det = 2, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2396] \quad & 1 \overset{1}{5} 132181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -77, \det = 3, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2397] \quad & 1 \overset{1}{5} 13221 21 (12) 12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -76, \det = 3, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2398] \quad & 1 \overset{1}{2} 181232181231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -68, \det = 3, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2399] \quad & 1 \overset{1}{5} 132181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -77, \det = 3, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2395] \quad & 1 \overset{1}{5} 13221 21 (12) 12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -76, \det = 3, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
T = 31 \quad & [2400] \quad 1 \overset{1}{5} 1321812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = \\
& (1, 31, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
[2402] & 1 \overset{1}{6} 1232181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2403] & 1 \overset{1}{2} 7123218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2404] & 1 \overset{1}{4} 13218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2405] & 1 \overset{1}{5} 1321812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2406] & 1 \overset{1}{8} 1232181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2407] & 1 \overset{1}{3} 22181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2408] & 1 \overset{1}{3} 22181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2409] & 1 \overset{1}{2} 31513221 \overset{21}{(12)} 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2410] & 1 \overset{1}{2} 31513221 \overset{21}{(12)} 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2411] & 1 \overset{1}{8} 1232181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2412] & 1 \overset{1}{8} 1232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2413] & 1 \overset{1}{8} 1232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2414] & 1 \overset{1}{7} 1232181231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2401] & 221 \overset{1}{8} 12321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0 \\
[2424] & 1 \overset{1}{4} 1413221(12)12231513221 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2426] & 1 \overset{1}{5} 13221 \overset{1}{(12)} 12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2427] & 1 \overset{1}{8} 12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2428] & 1 \overset{1}{5} 13221 \overset{1}{(12)} 12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& [2429] \quad 1 \overset{1}{5} 13221(12)122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2430] \quad 1 \overset{1}{5} 13221(12)122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2431] \quad 21 \overset{1}{8} 1232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2432] \quad 21 \overset{1}{8} 1231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2425] \quad 1 \overset{1}{4} 1413221(12)12231513221 \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2433] \quad 1 \overset{1}{6} 1232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 33, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2434] \quad 21 \overset{1}{(12)} \overset{1}{12231513221(12)1223151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 33, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2435] \quad 21 \overset{1}{(12)} \overset{1}{12231513221(12)1223151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 33, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2436] \quad 1 \overset{1}{8} 1232171232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 33, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& T = 32 \quad [2437] \quad 1 \overset{1}{4} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [2438] \quad 221 \overset{1}{8} 123218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [2439] \quad 1 \overset{1}{4} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [2441] \quad 1 \overset{1}{3} 2181232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
[2442] & 1^{1231513221} (12) \overset{1}{12231513221} (12) 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2443] & 1^{132181231513221} (12) 1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2444] & 1^{132181231513221} (12) 1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2445] & 1^{1513218123218123218123218123141}^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2446] & 1^{1232181232181231513221} (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2447] & 1^{1513218123218123218123218123141}^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2448] & 1^{1612321812321812321812321812231}^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2449] & 1^{1231513221} (12) \overset{1}{12231513221} (12) 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2450] & 1^{161232181232181232181232181231}^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2451] & 1^{1232181232181231513221} (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2452] & 1^{2171232181232181232181232171231}^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2453] & 1^{2181232181231513221} (12) 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2440] & 1^{132181232181231513221} (12) 1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0 \\
[2464] & 1^{1812321712321812321812321812321}^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2466] & 1^{13221} 21 (12) \overset{1}{12231513221} (12) 12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2467] & 1^{1513221} (12) \overset{1}{12231513221} (12) 12231 \overset{1}{51} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2468] & 1^{141413221} (12) 12231513221 (12) 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[2469] & 1^{1513221} 21 (12) \overset{1}{12231513221} (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2470] & 1^{1513221} 21 (12) \overset{1}{12231513221} (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2471] \quad 1 \quad \begin{matrix} 1^1 \\ 1^5 \end{matrix} 13221 \quad (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[2472] \quad 1 \quad \begin{matrix} 1^1 \\ 1^5 \end{matrix} 13221 \quad (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[2465] \quad 21 \quad \begin{matrix} 1^1 \\ 1^5 \end{matrix} 3221 \quad (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2474] \quad 1 \quad \begin{matrix} 1^1 \\ 1^7 \end{matrix} 1232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 34, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2473] \quad 1 \quad \begin{matrix} 1^1 \\ 1^{321812321812321812321} \end{matrix} 8 \quad 1232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 34, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[2475] \quad 1 \quad \begin{matrix} 1^1 \\ 1^{3221} \end{matrix} (12) \quad 12231513221(12)12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 34, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[2476] \quad 61 \quad \begin{matrix} 1^1 \\ 1^{232181232181231513221} \end{matrix} (12) \quad 1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 40, \Delta = -99, \det = 56, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35.57 \rightarrow 36, T_{\max} = 40, T \in [36, 40]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 33 \quad [2478] \quad 1 \quad \begin{matrix} 1^1 \\ 1^{231513221} \end{matrix} (12) \quad 12231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0$$

$$[2479] \quad 1 \quad \begin{matrix} 1^1 \\ 1^5 \end{matrix} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0$$

$$[2481] \quad 1 \quad \begin{matrix} 1^1 \\ 1^5 \end{matrix} 1232181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0$$

$$[2482] \quad 1 \quad \begin{matrix} 1^1 \\ 1^{31513221} \end{matrix} (12) \quad 12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0$$

$$[2483] \quad 1 \quad \begin{matrix} 1^1 \\ 1^{231513221} \end{matrix} (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0$$

$$\begin{aligned}
[2484] & 1 \overset{1}{5} 132181232181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -12, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2485] & 221 \overset{1}{8} 1232181232181232181232171231 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2486] & 1 \overset{1}{6} 123218123218123218123218123141 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2487] & 1 \overset{1}{231513221} (12) 12231513221(12)122314141 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2488] & 21 \overset{1}{7} 12321812321812321812321812231 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2489] & 1 \overset{1}{231513221} (12) 12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2490] & 21 \overset{1}{513221} (12) 12231513221(12)12231513221 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2477] & 1 \overset{1}{5} 132181232181232181232181232151 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2480] & 21 \overset{1}{513221} (12) 12231513221(12)12231513221 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2500] & 1 \overset{1}{5131513221(12)12231513221} (12) 1223141 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 34, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2501] & 1 \overset{1}{5131513221(12)12231513221} (12) 1223141 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 34, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2497] & 1 \overset{1}{2321812321812321812321} 8 1232181 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 34, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2499] & 21 \overset{1}{3221} (12) 12231513221(12)122315132161 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 34, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[2498] & 1 \overset{1}{6} 131513221(12)12231513221(12)12231 \overset{21}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 34, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2502] & 1 \overset{1}{413221(12)12231513221} (12) 12231513161 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 33, T_{\max} = 35, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& [2503] \quad 1 \overset{1}{(12)} \overset{1}{12231513221(12)122315132171^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 35, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2504] \quad 2181232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 35, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2505] \quad 1^1 71232181232181231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 39, T \in [36, 39] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
& [2506] \quad 812321^1 8123218123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 38, \Delta = -95, \det = -41, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36.12 \rightarrow 37, T_{\max} = 38, T \in [37, 38] \\
& \quad \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0} \\
& T = 34 \quad [2507] \quad 161232181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0 \\
& [2508] \quad 16123218123218123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0 \\
& [2509] \quad 22181232181232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0 \\
& [2510] \quad 221812321812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0 \\
& [2511] \quad 217123218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0 \\
& [2523] \quad 1^1 32181232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2514] \quad 1 \overset{1}{(12)} \overset{1}{12231513221(12)12231513221} 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2516] \quad 1 \overset{1}{(12)} \overset{1}{12231513221(12)12231513221} 12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2517] \quad 1 \overset{1}{(12)} \overset{1}{12231513221(12)12231513221} 12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& [2518] \quad 1^{13221} 21 (12) 12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2519] \quad 1^1 41413221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2520] \quad 1^1 41413221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2515] \quad 1^1 16 131513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2522] \quad 1^1 32181232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2521] \quad 1^1 17 1231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2524] \quad 1^{13221} 1 (12) 12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2525] \quad 1^{1413221(12)12231513221} (12) 122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2526] \quad 1^1 18 1231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
& [2527] \quad 1^1 15 13221(12)12231513221(12)12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 36, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2528] \quad 1^1 18 1232181232181231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 40, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 40, T \in [37, 40]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[2529] & 21^1 81232181232181231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 34, T_{\max} = 40, \Delta = -96, \det = 8, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37.50 \rightarrow 38, T_{\max} = 40, T \in [38, 40] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 35 & [2532] 21^1 7123218123218123218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -9, \det = 0, (n_+, n_-, n_0) = \\
& (1, 35, 1), c_{\text{ext}} = 3.0 \\
[2531] & 1^1 231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -24, \det = 0, (n_+, n_-, n_0) = \\
& (1, 35, 1), c_{\text{ext}} = 3.0 \\
[2533] & 21^1 71232181232181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -11, \det = 0, (n_+, n_-, n_0) = \\
& (1, 35, 1), c_{\text{ext}} = 3.0 \\
[2534] & 221^1 8123218123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -13, \det = 0, (n_+, n_-, n_0) = \\
& (1, 35, 1), c_{\text{ext}} = 3.0 \\
[2530] & 1^1 231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -24, \det = 0, (n_+, n_-, n_0) = \\
& (1, 35, 1), c_{\text{ext}} = 3.0 \\
[2549] & 1^1 71231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -45, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2541] & 1^1 6131513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -41, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2542] & 1^1 31513221(12)12231513221(12)12231513161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -49, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2543] & 1^1 6131513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -45, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[2544] & 1^1 1513221(12)12231513221(12)122315132161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -42, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[2545] & 1^1 712231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -45, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2546] & 1^1 41413221(12)12231513221(12)122315132161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -46, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2547] \overset{1}{1} 8 1231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -45, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2548] \overset{1}{1} 8 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -45, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2552] \overset{1}{1} 413221(12)12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 37, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2553] \overset{1}{1} 3221 1(12)12231513221(12)122315131613161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 37, \Delta = -75, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& [2550] \overset{1}{1} 5 13221(12)12231513221(12)122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 37, \Delta = -69, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2551] \overset{1}{1} 8 1231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 37, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2554] \overset{1}{1} 513221 1(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 41, \Delta = -97, \det = -4, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 41, T \in [38, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
& [2555] 81^1 2231513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 41, \Delta = -97, \det = -90, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 41, T \in [39, 41] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 54 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 2 & 14 & 0 & 3 \end{pmatrix}_{b_0} \\
& T = 36 \quad [2561] \overset{1}{1} 31513221 1(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -28, \det = \\
& 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [2564] \overset{1}{1} 6 131513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -21, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [2562] \overset{1}{1} 3221 21(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -27, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [2560] \overset{1}{1} 31513221 1(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -28, \det = 0, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 36, 1), c_{\text{ext}} = 3.0$$

$$[2556] \quad 221 \overset{1}{8} 1232181232181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 36, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0$$

$$[2557] \quad 21 \overset{1}{(12)} 12231513221(12)1223151321812321 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 36, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0$$

$$[2558] \quad 221 \overset{1}{8} 1232181232181232181232181232151 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 36, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0$$

$$[2559] \quad 1 \overset{1}{2231513221(12)12231513221} (12) 12231513161 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 36, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0$$

$$[2563] \quad 1 \overset{1}{231513221} (12) 12231513221(12)122315132161 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 36, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0$$

$$[2581] \quad 1 \overset{1}{8} 12231513221(12)12231513221(12)1223141 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2574] \quad 1 \overset{1}{41413221(12)12231513221(12)1223151322171} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[2575] \quad 21 \overset{1}{8} 1231513221(12)12231513221(12)1223141 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2576] \quad 1 \overset{1}{7} 1231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2577] \quad 1 \overset{1}{5131513221(12)12231513221(12)12231513151} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2578] \quad 1 \overset{1}{7} 1231513221(12)12231513221(12)122314141 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[2579] \quad 1 \overset{1}{1513221} (12) 12231513221(12)1223151322171 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[2580] \quad 1 \overset{1}{31513221(12)12231513221} (12) 122315132171 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2582] & 1^1 8 1231513221(12)12231513221(12)122315^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 38, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2583] & 1^1 413221(12)12231513221(12)122315131613161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 38, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2584] & 1^1 5 13221(12)12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 38, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& T = 37 \quad [2586] 1^1 232181231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0 \\
[2587] & 1^1 7 1231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0 \\
[2588] & 1^1 232181231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0 \\
[2589] & 1^1 231513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0 \\
[2585] & 1^1 2231513221(12)12231513221(12)122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0 \\
[2600] & 1^1 8 12231513221(12)12231513221(12)122315^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 38, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2601] & 1^1 61231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 38, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2602] & 1^1 8 1231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 38, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2595] & 1^1 31513221(12)12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 38, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2596] & 1^1 6 131513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 38, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2597] \ 21 \overset{1}{8} 1231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 38, \ \Delta = -43, \ \det = -2, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 38, \ T_{\max} = 38, \ T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[2598] \ 21 \overset{1}{8} 1231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 38, \ \Delta = -39, \ \det = -2, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 38, \ T_{\max} = 38, \ T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2599] \ 1 \overset{1}{8} 12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 38, \ \Delta = -43, \ \det = -2, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 38, \ T_{\max} = 38, \ T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[2603] \ 1 \overset{1}{5} 13221(12)12231513221(12)122315131613161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 39, \ \Delta = -69, \ \det = -3, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 39, \ T_{\max} = 39, \ T \in [39, 39]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 38 \quad [2610] \ 1^1 41413221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -25, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2608] \ 1^1 \overset{1}{3} 1513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -27, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2604] \ 1 \overset{1}{8} 1231513221(12)12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -19, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2605] \ 1^1 \overset{1}{2} 231513221(12)12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -19, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2606] \ 1^1 \overset{1}{5} 13221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -21, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2607] \ 1^1 \overset{1}{3} 1513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -27, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2609] \ 1 \overset{1}{8} 12231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -19, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2611] \ 1^1 41413221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -25, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2612] \ 1^1 \overset{1}{5} 13221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -21, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2613] \ 21 \overset{1}{8} 1231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -19, \ \det = 0, \ (n_+, n_-, n_0) = (1, 38, 1), \ c_{\text{ext}} = 3.0$$

$$[2624] \ 1^1 61316131513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 39, \ \Delta = -48, \ \det = 2, \ (n_+, n_-, n_0) = (1, 38, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 39, \ T_{\max} = 39, \ T \in [39, 39]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2626] \ 1^1 712231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 39, \ \Delta = -42, \ \det = 2, \ (n_+, n_-, n_0) = (1, 38, 0), \ c_{\text{ext}} = 3.0$$

$$\begin{aligned}
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2625] \quad 1^1 7 1231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 39, \Delta = -42, \det = 2, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2627] \quad 1^1 3221 1(12)12231513221(12)122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 40, \Delta = -74, \det = 3, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& T = 39 \quad [2628] \quad 1^1 316131513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -27, \det = \\
& 0, (n_+, n_-, n_0) = (1, 39, 1), c_{\text{ext}} = 3.0 \\
& [2629] \quad 1^1 61316131513221(12)12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -20, \det = \\
& 0, (n_+, n_-, n_0) = (1, 39, 1), c_{\text{ext}} = 3.0 \\
& [2633] \quad 21^1 8 1231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -41, \det = -2, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2634] \quad 1^1 3221 1(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -53, \det = -2, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2635] \quad 1^1 8 12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -41, \det = -2, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2637] \quad 1^1 413221(12)12231513221(12)122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 41, \Delta = -73, \det = \\
& -3, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& [2636] \quad 1^1 18 1232181231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 41, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
& T = 40 \quad [2638] \quad 1^1 232181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -23, \det = \\
& 0, (n_+, n_-, n_0) = (1, 40, 1), c_{\text{ext}} = 3.0 \\
& [2639] \quad 1^1 232181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -23, \det = \\
& 0, (n_+, n_-, n_0) = (1, 40, 1), c_{\text{ext}} = 3.0 \\
& [2645] \quad 1^1 3221 1(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 41, \Delta = -52, \det = 2, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& [2646] \overset{1}{1} \overset{1}{(12)} \overset{1}{12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 40, T_{\max} = 41, \Delta = -52, \det = 2, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{smallmatrix} \right)_{b_0} \\
& [2647] \overset{1}{1} \overset{1}{8} \overset{1}{1232181231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 40, T_{\max} = 42, \Delta = -72, \det = 3, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{smallmatrix} \right)_{b_0} \\
& [2648] \overset{1}{1} \overset{1}{5} \overset{1}{13221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 40, T_{\max} = 42, \Delta = -76, \det = 3, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0} \\
& [2649] \overset{1}{1} \overset{1}{5} \overset{1}{13221(12)12231513221(12)122315132181232171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 40, T_{\max} = 42, \Delta = -68, \det = 3, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{smallmatrix} \right)_{b_0} \\
& [2650] \overset{1}{1} \overset{1}{5} \overset{1}{13221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 40, T_{\max} = 42, \Delta = -76, \det = 3, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{smallmatrix} \right)_{b_0} \\
& [2651] \overset{1}{1} \overset{1}{5} \overset{1}{13221(12)12231513221(12)122315132181232181^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 40, T_{\max} = 46, \Delta = -96, \det = 4, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 46, T \in [43, 46] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{smallmatrix} \right)_{b_0} \\
& T = 41 \quad [2654] \overset{1}{1} \overset{1}{6} \overset{1}{1231513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 41, T_{\max} = 41, \Delta = \\
& -21, \det = 0, (n_+, n_-, n_0) = (1, 41, 1), c_{\text{ext}} = 3.0 \\
& [2655] \overset{1}{1} \overset{1}{6} \overset{1}{131513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 41, T_{\max} = 41, \Delta = -20, \det = \\
& 0, (n_+, n_-, n_0) = (1, 41, 1), c_{\text{ext}} = 3.0 \\
& [2652] \overset{1}{1} \overset{1}{3} \overset{1}{16131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 41, T_{\max} = 41, \Delta = -26, \det = \\
& 0, (n_+, n_-, n_0) = (1, 41, 1), c_{\text{ext}} = 3.0 \\
& [2653] \overset{1}{1} \overset{1}{3} \overset{1}{16131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 41, T_{\max} = 41, \Delta = -26, \det = \\
& 0, (n_+, n_-, n_0) = (1, 41, 1), c_{\text{ext}} = 3.0 \\
& [2665] \overset{1}{1} \overset{1}{3} \overset{1}{221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 41, T_{\max} = 42, \Delta = -47, \det = -2, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \\
& [2662] \overset{1}{1} \overset{1}{7} \overset{1}{1232181231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 41, T_{\max} = 42, \Delta = -47, \det = \\
& -2, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \\
& [2663] \overset{1}{1} \overset{1}{4} \overset{1}{13221(12)12231513221(12)1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]} \quad T = 41, T_{\max} = 42, \Delta = -51, \det = -2, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2664] & 1 \overset{1}{(12)} \overset{1}{12231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 42, \Delta = -47, \det = -2, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2666] & 1 \overset{1}{8} \overset{1}{1232181231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 43, \Delta = -67, \det = -3, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
T = 42 & [2671] 1 \overset{1}{(12)} \overset{1}{12231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223151321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -27, \det = \\
& 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
[2673] & 1 \overset{1}{7} \overset{1}{1231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223151321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = \\
& 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
[2667] & 1 \overset{1}{(12)} \overset{1}{12231513221(12)} \overset{21}{12231513221(12)} \overset{21}{12231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 42, 1), c_{\text{ext}} = 3.0 \\
[2668] & 1 \overset{1}{(12)} \overset{1}{12231513221(12)} \overset{21}{12231513221(12)} \overset{21}{12231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 42, 1), c_{\text{ext}} = 3.0 \\
[2669] & 1 \overset{1}{7} \overset{1}{12231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223151321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = \\
& 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
[2670] & 1 \overset{1}{7} \overset{1}{1232181231513221(12)} \overset{1}{12231513221(12)} \overset{1}{12231513221} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = \\
& 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
[2672] & 1 \overset{1}{(12)} \overset{1}{12231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223151321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -27, \det = \\
& 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
[2686] & 1 \overset{1}{5} \overset{1}{13221(12)} \overset{1}{12231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = \\
& 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2685] & 1 \overset{1}{5} \overset{1}{13221(12)} \overset{1}{12231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = \\
& 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2684] & 1 \overset{1}{8} \overset{1}{1232181231513221(12)} \overset{1}{12231513221(12)} \overset{1}{122315131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = \\
& 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
T = 43 & [2687] 1 \overset{1}{8} \overset{1}{12231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223151321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = \\
& -18, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
[2689] & 1 \overset{1}{8} \overset{1}{1232181231513221(12)} \overset{1}{12231513221(12)} \overset{1}{12231513221} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = \\
& 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
[2690] & 1 \overset{1}{(12)} \overset{1}{12231513221(12)} \overset{1}{12231513221(12)} \overset{1}{1223141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = \\
& 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
[2691] & 1 \overset{1}{(12)} \overset{1}{12231513221(12)} \overset{1}{1223151321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 43, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& [2692] \quad 1 \quad \overset{1}{1^1 413221(12)12231513221} \quad (12) \quad 12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2693] \quad 1 \quad \overset{1}{1^1 31513221} \quad (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2694] \quad 21 \quad \overset{1}{1^1 81231513221(12)12231513221(12)1223151321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2688] \quad 1 \quad \overset{1}{1^1 32181231513221(12)12231513221(12)122315132181231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2704] \quad 1 \quad \overset{1}{1^1 513221(12)12231513221(12)12231513221(12)1223151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 44, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 44, T_{\max} = 44, T \in [44, 44] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \\
& [2705] \quad 1 \quad \overset{1}{1^1 6131513221(12)12231513221} \quad (12) \quad 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 45, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 45, T_{\max} = 45, T \in [45, 45] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0} \\
& T = 44 \quad [2708] \quad 1 \quad \overset{1}{1^1 31513221(12)12231513221} \quad (12) \quad 12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0 \\
& [2706] \quad 1 \quad \overset{1}{1^1 31513221} \quad (12) \quad 12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0 \\
& [2707] \quad 1 \quad \overset{1}{1^1 31513221} \quad (12) \quad 12231513221(12)12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0 \\
& [2709] \quad 1 \quad \overset{1}{1^1 513221(12)12231513221(12)12231513221(12)1223151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0 \\
& [2710] \quad 1 \quad \overset{1}{1^1 513221(12)12231513221(12)12231513221(12)1223151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0 \\
& [2711] \quad 1 \quad \overset{1}{1^1 31513221} \quad (12) \quad 12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0 \\
& [2712] \quad 1 \quad \overset{1}{1^1 31513221} \quad (12) \quad 12231513221(12)12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0 \\
& [2721] \quad 1 \quad \overset{1}{1^1 71231513221(12)12231513221} \quad (12) \quad 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 46, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 46, T_{\max} = 46, T \in [46, 46] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0} \\
& T = 45 \quad [2723] \quad 21 \quad \overset{1}{1^1 3221} \quad (12) \quad 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 45, 1), c_{\text{ext}} = 3.0 \\
& [2722] \quad 21 \quad \overset{1}{1^1 3221} \quad (12) \quad 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 45, 1), c_{\text{ext}} = 3.0 \\
& [2730] \quad 1 \quad \overset{1}{1^1 81231513221(12)12231513221(12)12231513221(12)12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 47, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0} \\
& [2729] \quad 1 \quad \overset{1}{1^1 3221} \quad (12) \quad 12231513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 47, \Delta = -73, \det =
\end{aligned}$$

$$-3, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$1^1 812231513221(12)12231513221$$

$$[2728] \ 1 \quad (12) \quad 12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 45, T_{\max} = 47, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$1^1 31513221$$

$$T = 46 \quad [2732] \ 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 46, 1), c_{\text{ext}} = 3.0$$

$$1^1 316131513221(12)12231513221(12)12231513221$$

$$[2731] \ 1 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 46, 1), c_{\text{ext}} = 3.0$$

$$1^1 316131513221(12)12231513221(12)12231513221$$

$$[2733] \ 1 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 46, 1), c_{\text{ext}} = 3.0$$

$$1^1 31513221$$

$$[2734] \ 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 46, 1), c_{\text{ext}} = 3.0$$

$$1^1 413221(12)12231513221(12)12231513221$$

$$[2743] \ 1 \quad (12) \quad 122315132181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 48, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$1^1 81231513221(12)12231513221$$

$$[2744] \ 1 \quad (12) \quad 12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 52, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49, T_{\max} = 52, T \in [49, 52]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$1$$

$$[2745] \ 81^1 2231513221(12)12231513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 46, T_{\max} = 52, \Delta = -96, \det = 90, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49.44 \rightarrow 50, T_{\max} = 52, T \in [50, 52]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 54 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

$$1^1 3221$$

$$T = 47 \quad [2752] \ 21 \ (12) \ 12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 3.0$$

$$[2751] \ 1^1 41413221(12)12231513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 3.0$$

$$1^1 513221$$

$$[2750] \ 1 \ (12) \ 12231513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 3.0$$

$$1^1 513221$$

$$[2749] \ 1 \ (12) \ 12231513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 3.0$$

$$1^1$$

$$[2748] \ 16 \ 131513221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 3.0$$

$$1^1 31513221$$

$$[2747] \ 1 \ (12) \ 12231513221(12)12231513221(12)122315132161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 3.0$$

$$1^1 3221$$

$$[2746] \ 21 \ (12) \ 12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 3.0$$

$$1^1$$

$$[2761] \ 15 \ 13221(12)12231513221(12)12231513221(12)122315132181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 47, T_{\max} = 49, \Delta = -67, \det =$$

$$-3, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$T = 48 \quad [2762] \quad 1 \overset{1}{7} 1231513221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 3.0$$

$$[2763] \quad 1 \overset{1}{3} 1513221(12)12231513221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 3.0$$

$$[2764] \quad 1 \overset{1}{3} 2181231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 3.0$$

$$[2765] \quad 1 \overset{1}{3} 2181231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 3.0$$

$$[2773] \quad 1 \overset{1}{3} 1513221(12)12231513221(12)12231513221(12)122315132181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 49, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$T = 49 \quad [2774] \quad 1 \overset{1}{3} 16131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$[2781] \quad 1 \overset{1}{2} 231513221(12)12231513221(12)12231513221(12)122315132181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$[2782] \quad 1 \overset{1}{3} 16131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$[2780] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$[2779] \quad 1 \overset{1}{8} 12231513221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$[2778] \quad 21 \overset{1}{8} 1231513221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$[2777] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$[2776] \quad 1 \overset{1}{4} 1413221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$[2775] \quad 1 \overset{1}{4} 1413221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$$

$$T = 50 \quad [2791] \quad 1 \overset{1}{3} 16131513221(12)12231513221(12)12231513221(12)122315132161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 50, 1), c_{\text{ext}} = 3.0$$

$$[2790] \quad 1 \overset{1}{6} 131513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 50, 1), c_{\text{ext}} = 3.0$$

$$[2794] \quad 1 \overset{1}{3} 221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, T_{\max} = 51, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 51, T_{\max} = 51, T \in [51, 51]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2795] \quad 1 \overset{1}{3} 221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, T_{\max} = 51, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 51, T_{\max} = 51, T \in [51, 51]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2796] \quad 1^{13221} (12) 12231513221(12)12231513221(12)122315132181232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 50, T_{\max} = 52, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 52, T_{\max} = 52, T \in [52, 52] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& T = 51 \quad [2799] \quad 1^{132181231513221}(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2798] \quad 1^{132181231513221}(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2797] \quad 1^{171231513221}(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2801] \quad 1^{1316131513221}(12)12231513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2800] \quad 21^{13221}(12) 12231513221(12)12231513221(12)12231513221(12)12231^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2805] \quad 1^{1413221}(12)12231513221(12)12231513221(12) 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 52, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 52, T_{\max} = 52, T \in [52, 52] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2804] \quad 1^{1413221}(12)12231513221(12)12231513221(12) 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 52, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 52, T_{\max} = 52, T \in [52, 52] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
& [2808] \quad 1^{1413221}(12)12231513221(12)12231513221(12)122315132181232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 53, T_{\max} = 53, T \in [53, 53] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& [2807] \quad 1^{13221}(12)12231513221(12)12231513221(12) 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 53, T_{\max} = 53, T \in [53, 53] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2806] \quad 1^{13221}(12)12231513221(12)12231513221(12) 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 53, T_{\max} = 53, T \in [53, 53] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& T = 52 \quad [2814] \quad 21^{13221}(12) 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2813] \quad 21^{13221}(12) 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2812] \quad 21^{181231513221}(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2811] \quad 1^{1812231513221}(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

[2810] $1^1 16 131513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 52, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0$

[2809] $1^1 61231513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 52, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0$

[2816] $1^1 15 13221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 53, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[2815] $1^1 15 13221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 53, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2817] $1^1 15 13221(12)12231513221(12)12231513221(12)122315132181232181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 54, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 54, T_{\max} = 54, T \in [54, 54]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

$T = 53$ [2823] $1^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2822] $1^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2821] $1^1 21 (12) 12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2826] $1^1 15 13221 (12) 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2820] $1^1 21 (12) 12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2825] $1^1 15 13221 (12) 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2824] $1^1 712231513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2819] $1^1 413221(12)12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2818] $1^1 171231513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[2829] $1^1 81232181231513221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 54, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 54, T_{\max} = 54, T \in [54, 54]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

$T = 54$ [2833] $1^1 41413221(12)12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 54, T_{\max} = 54, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$

[2834] $1^1 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 54, T_{\max} = 54, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$

[2835] $1^1 15 13221 (12) 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 54, T_{\max} = 54, \Delta =$

$$[2850] \overset{1^1}{17}1231513221(12)12231513221(12)12231513221(12)12231513221(\overset{21}{12})12231^{\textcolor{blue}{1}}+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 57, T_{\max} = 57, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 57, 1), c_{\text{ext}} = 3.0$$

$$T = 58 \quad [2862] \overset{1}{1} 8 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231^{\overset{21}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 58, T_{\max} = 58, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 58, 1), c_{\text{ext}} = 3.0$$

[2861] $1^1 4 1 4 1 3 2 2 1 (12) 1 2 2 3 1 5 1 3 2 2 1 (12) 1 2 2 3 1 5 1 3 2 2 1 (12) 1 2 2 3 1 5 1 3 2 2 1 (12) 1 2 2 3 1 5 1 3 2 1 6 1^1 + [\textcolor{red}{\mathfrak{s}u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 58, T_{\max} = 58, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 58, 1), c_{\text{ext}} = 3.0$

$$\begin{array}{l} \text{[2860]} \quad 16^{1^1} 13 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 14 14 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 58, \quad T_{\max} = \\ 58, \quad \Delta = -14, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 58, 1), \quad c_{\text{ext}} = 3.0 \end{array}$$

[2859] $21\ 8\ 1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{g}_2]$ $T = 58, T_{\max} = 58, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 58, 1), c_{\text{ext}} = 3.0$

$$\begin{array}{c} 1 \\ 1 \ 5 \ 13221 \end{array} \\ [2858] \ 1 \ (12) \ 12231513221(12)12231513221(12)12231513221(12)122315132161^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] \quad T = 58, \ T_{\max} = 58, \ \Delta = -11, \ \det = 0, \ (n_+, n_-, n_0) = (1, 58, 1), \ c_{\text{ext}} = 3.0$$

$$[2857] \quad \overset{1}{1}71231513221(12)12231513221(12)12231513221(12)12231513221(12)1223141 \overset{1}{1} + [\textcolor{red}{\mathfrak{s}u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 58, \quad T_{\max} = 58, \quad \Delta = -14, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 58, 1), \quad c_{\text{ext}} = 3.0$$

$$1^1 712231513221(12)12231513221(12)12231513221(12)12231513221$$

$$[2856] \ 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 58, \ T_{\max} = 58, \ \Delta = -14, \ \det =$$

$$0, \ (n_+, n_-, n_0) = (1, 58, 1), \ c_{\text{ext}} = 3.0$$

[2863] $\overset{1}{1}6131513221(12)12231513221(12)12231513221(12)12231513221(12)12231\overset{1}{5}1+[\textcolor{red}{5u}_2\cdot\textcolor{blue}{g}_2] \quad T = 58, T_{\max} = 58, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 58, 1), c_{\text{ext}} = 3.0$

$$T = 59 \quad [2869] \overset{1}{1} 7 1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}_2}, \textcolor{blue}{\mathfrak{g}_2}] \quad T = 59, T_{\max} = 59, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 59, 1), c_{\text{ext}} = 3.0$$

[2868] $1^1 5131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 59, T_{\max} = 59, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 59, 1), c_{\text{ext}} = 3.0$

$$[2867] \quad \overset{1}{21} \overset{1}{8} 1231513221(12)12231513221(12)12231513221(12)12231513221(12)1223141 \overset{1}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 59, \quad T_{\max} = 59, \quad \Delta = -13, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 59, 1), \quad c_{\text{ext}} = 3.0$$

$$[2866] \quad \overset{1}{1} 8 12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223141 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 59, \quad T_{\max} = 59, \quad \Delta = -13, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 59, 1), \quad c_{\text{ext}} = 3.0$$

[2865] $1^4 41413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 59, T_{\max} = 59, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 59, 1), c_{\text{ext}} = 3.0$

$$[2864] \overset{1}{1} \overset{1}{5} \overset{1}{13} \overset{2}{22} 1 (12) 12231513221(12)12231513221(12)12231513221(12)1223151322171^{\textcolor{red}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 59, T_{\max} = 59, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 59, 1), c_{\text{ext}} = 3.0$$

$$[2870] \quad 17^{1^1} 1231513221(12)12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 59, \quad T_{\max} = 59, \quad \Delta = -13, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 59, 1), \quad c_{\text{ext}} = 3.0$$

$$T = 60 \quad [2871] \quad 1812231513221(12)12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] \quad T = 60, T_{\max} = 60, \Delta = -12, \det = 0, (n_+, n_-, n_0) = (1, 60, 1), c_{\text{ext}} = 3.0$$

[2872] $\overset{1}{\textcolor{red}}{\text{1}} \text{ } 8 \text{ } 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231 \overset{1}{\textcolor{blue}}{\text{5}} \text{ } 1 + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\text{g}}_2] \quad T = 60, \quad T_{\max} =$
 $60, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 60, 1), c_{\text{ext}} = 3.0$

[2873] $1^1 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 60, T_{\max} = 60, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 60, 1), c_{\text{ext}} = 3.0$

$$[2874] \quad 16^{1^1} 13 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 15 13 15 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 60, \quad T_{\max} = 60, \quad \Delta = -12, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 60, 1), \quad c_{\text{ext}} = 3.0$$

$$[2875] \quad 21\overset{1}{8}1231513221(12)12231513221(12)12231513221(12)12231513221(12)122314141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 60, \quad T_{\max} = 60, \quad \Delta = -12, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 60, 1), \quad c_{\text{ext}} = 3.0$$

$$T = 61 \quad [2877] \quad 1^1 7 1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 61, T_{\max} = 61, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 61, 1), c_{\text{ext}} = 3.0$$

$$[2876] \quad 1^1 7 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 61, T_{\max} = 61, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 61, 1), c_{\text{ext}} = 3.0$$

$$T = 62 \quad [2879] \quad 21^1 8 1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 62, T_{\max} = 62, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 62, 1), c_{\text{ext}} = 3.0$$

$$[2878] \quad 1^1 8 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 62, T_{\max} = 62, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 62, 1), c_{\text{ext}} = 3.0$$

4.41 NHC(-2)(-3) + $\mathfrak{e}_6$

$$T = 6 \quad [0] \quad 21^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, T_{\max} = 100, \Delta = -193, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, T_{\max} = 100, \Delta = -193, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, T_{\max} = 100, \Delta = -193, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -6 & -4 \\ 3 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 1^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, T_{\max} = 100, \Delta = -193, \det = 33, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 21^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, T_{\max} = 100, \Delta = -193, \det = 131, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 2^1 2^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 21^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & -2 & 0 \\ 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 1^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^1 1^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & -2 & 0 \\ 3 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 1^1 3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

[10] $1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

[11] $1^1 31^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

[12] $1^1 31^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[13] $2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

[14] $21^1 1^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

[15] $1^1 31^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 2 & 3 & 3 & -3 & -1 \\ 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[16] $131^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 1 & 3 & 3 & -3 & -1 \\ 2 & 4 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 8$ [17] $131^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 3 \\ 1 & 4 & 3 & -2 & 0 \\ 2 & 4 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

[18] $1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[19] $1^1 1^1 23131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[20] $1^1 3^1 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 5 & 15 & 5 \\ 15 & 42 & 12 \\ 5 & 12 & 2 \end{pmatrix}_{b_0}$$

[21] $1^1 1^1 2^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

[22] $1^1 1^1 2^1 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[23] $1^1 1^1 3^1 2^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[24] $1^1 2^1 3^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[25] $1^1 1^1 2^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 \\ 1 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[26] $1^1 2^1 3^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15, T_{\max} = 98, T \in [15, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

[27] $1^1 1^1 4^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -163, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -5 & -3 \\ 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[28] $1^1 1^1 2^1 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[29] $1^1 1^1 2^1 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^1 1^1 2^1 3^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & -2 & 0 \\ 2 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[31] $1^1 1^1 4^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -163, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -5 & -3 \\ 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^1 2^1 3^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & -2 & 0 \\ 2 & 0 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

[33] $1^1 2^1 3^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

- $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$
- [34] $1^{1231}1^{131}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$
- [35] $21^{141}1^{131}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -163$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$
- [36] $1^{131}1^{141}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -163$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.71 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 0 & 4 \\ 1 & 0 & -5 & -3 \\ 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$
- [37] $21^{141}1^{131}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -163$, $\det = 103$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 0 & 4 \\ 2 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$
- [38] $31^{112}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -146$, $\det = 215$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$
- $T = 9$ [40] $1^{112}231321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 16.14 \rightarrow 17$, $T_{\max} = 29$, $T \in [17, 29]$
 $\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$
- [39] $1^{112}223131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 16.14 \rightarrow 17$, $T_{\max} = 26$, $T \in [17, 26]$
 $\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$
- [41] $1^{141}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -137$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 15$, $T_{\max} = 99$, $T \in [15, 99]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & -4 & -2 \\ 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [42] $1^{141}1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 15$, $T_{\max} = 95$, $T \in [15, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & -4 & -2 \\ 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [43] $1^{151}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 101$, $\Delta = -158$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & -2 & 0 \\ 3 & 4 & 0 & 1 \end{pmatrix}_{b_0}$
- [45] $1^{131}1^{141}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14.22 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & -3 & -1 \\ 2 & 4 & 0 & -1 & 0 \end{pmatrix}_{b_0}$
- [44] $21^{151}1^{131}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 101$, $\Delta = -158$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14.22 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

- [46] $\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & -2 & 0 \\ 4 & 4 & 0 & 1 \end{pmatrix}_{b_0}$
 $1^1 31^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$
- [47] $21^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & -2 & 0 \\ 3 & 4 & 0 & 1 \end{pmatrix}_{b_0}$
- [48] $2^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$
 $\begin{pmatrix} 10 & 22 & 6 \\ 22 & 42 & 12 \\ 6 & 12 & 3 \end{pmatrix}_{b_0}$
- [49] $1413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 17.14 \rightarrow 18, T_{\max} = 22, T \in [18, 22]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \\ 1 & 3 & -3 & -1 \\ 3 & 4 & -1 & -1 \end{pmatrix}_{b_0}$
- [50] $1^1 3141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 17.14 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$
- [51] $21^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 16, T_{\max} = 93, T \in [16, 93]$
 $\begin{pmatrix} 5 & 15 & 5 \\ 15 & 42 & 12 \\ 5 & 12 & 2 \end{pmatrix}_{b_0}$
- [52] $1^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 16, T_{\max} = 93, T \in [16, 93]$
 $\begin{pmatrix} 5 & 15 & 5 \\ 15 & 42 & 12 \\ 5 & 12 & 2 \end{pmatrix}_{b_0}$
- [53] $1^1 512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 16, T_{\max} = 94, T \in [16, 94]$
 $\begin{pmatrix} 5 & 15 & 5 \\ 15 & 42 & 12 \\ 5 & 12 & 2 \end{pmatrix}_{b_0}$
- [54] $1^1 315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -137, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & 1 \end{pmatrix}_{b_0}$
- [55] $1^1 315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -137, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$
- [56] $1^1 3151^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -138, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 \\ 2 & 7 & -1 & 0 & 1 \end{pmatrix}_{b_0}$
- [57] $31^1 315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -145, \det = 215, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [59] \quad 1^{11} 23215131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 1513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 0 & 2 \\ 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 21^{11} 513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 0 & 2 \\ 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 5^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 4 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 23151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{11} 14141321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{11} 315^{11} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 315^{11} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 131^{11} 51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 6^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 97, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 97, T \in [16, 97]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 6^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 96, \Delta = -134, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 96, T \in [16, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{11} 31^1 51321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 3 \\ 2 & 4 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[70] $31^{11} 316131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 101, \Delta = -142, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[71] $31^{11} 321^1 6131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 101, \Delta = -143, \det = -110, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 4 & 3 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 3 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & -3 & -1 \\ 2 & 6 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[72] $2321^{11} 6131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 101, \Delta = -137, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.62 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 4 & 2 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 2 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & -3 & -1 \\ 2 & 6 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{31} 6^{11} 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 101, \Delta = -142, \det = -215, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.92 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 12$ [74] $1^{11} 31^1 513221^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{11} 6^{11} 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 19.14 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

[76] $1^{11} 413151321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 19.14 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \\ 1 & 3 & -3 & -1 \\ 3 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{11} 321612321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 19.14 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

[78] $1^{11} 6^{11} 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{11} 31^1 6^{11} 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 12, T_{\max} = 27, \Delta = -113, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11} 1^1 31513221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 6 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 4 \\ 2 & 5 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{11} 1^1 613141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 96, \Delta = -133, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 96, T \in [17, 96]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 13$ [82] $1^1 4131612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 1513151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 0 & 2 \\ 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{11} 2321712231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[85] $1^{11} 1^1 71231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

[86] $1^{11} 2315132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{11} 1^1 151316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 94, \Delta = -128, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 17.25 \rightarrow 18, T_{\max} = 94, T \in [18, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 14$ [92] $1^{11} 1^1 812321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[88] $1^{11} 1^1 81231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 4 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$

[89] $1^{11} 15131612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$

$$\begin{aligned}
& T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0} \\
[90] & 1^1 41321712321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0} \\
[91] & 21^1 8^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0} \\
[93] & 1^1 1^{11} 315131^1 6^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
T = 15 & [94] 1^1 1^{11} 2315131^1 6^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[97] & 1^1 1^{11} 232171232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 22.14 \rightarrow 23, T_{\max} = 27, T \in [23, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0} \\
[95] & 1^1 151321712321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 22.14 \rightarrow 23, T_{\max} = 27, T \in [23, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0} \\
[96] & 1^1 413221812321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 15, T_{\max} = 27, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 22.14 \rightarrow 23, T_{\max} = 27, T \in [23, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0} \\
[98] & 1^1 1^{11} 2321^1 8^{11} 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 15, T_{\max} = 27, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 22.14 \rightarrow 23, T_{\max} = 27, T \in [23, 27] \\
& \begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0} \\
T = 16 & [102] 1^1 1^{11} 512321^1 8^{11} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 23.14 \rightarrow 24, T_{\max} = 28, T \in [24, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0} \\
[101] & 1^1 1^1 51321^1 8^{11} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 23.14 \rightarrow 24, T_{\max} = 28, T \in [24, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[100] \ 1^{\textcolor{blue}{1}}4141321812321^{\textcolor{red}{1}\textcolor{green}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 \quad T = 16, \ T_{\max} = 28, \ \Delta = -104, \ \det = 7, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 23.14 \rightarrow 24, \ T_{\max} = 28, \ T \in [24, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{\textcolor{blue}{1}}513221812321^{\textcolor{red}{1}\textcolor{green}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 \quad T = 16, \ T_{\max} = 28, \ \Delta = -100, \ \det = 7, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 23.14 \rightarrow 24, \ T_{\max} = 28, \ T \in [24, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [103] \ 1^{\textcolor{red}{1}\textcolor{green}{1}}232181231513151^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 \quad T = 18, \ T_{\max} = 30, \ \Delta = -102, \ \det = 7, \ (n_+, n_-, n_0) = (1, 18, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 25.14 \rightarrow 26, \ T_{\max} = 30, \ T \in [26, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [104] \ 1^{\textcolor{blue}{1}}5131612321812321^{\textcolor{red}{1}\textcolor{green}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 \quad T = 19, \ T_{\max} = 31, \ \Delta = -101, \ \det = -7, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 26.14 \rightarrow 27, \ T_{\max} = 31, \ T \in [27, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [105] \ 1^{\textcolor{blue}{1}}51321712321812321^{\textcolor{red}{1}\textcolor{green}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 \quad T = 20, \ T_{\max} = 32, \ \Delta = -100, \ \det = 7, \ (n_+, n_-, n_0) = (1, 20, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 27.14 \rightarrow 28, \ T_{\max} = 32, \ T \in [28, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [107] \ 1^{\textcolor{red}{1}\textcolor{green}{1}}232181232181223151^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 \quad T = 21, \ T_{\max} = 29, \ \Delta = -99, \ \det = -7, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 28.14 \rightarrow 29, \ T_{\max} = 29, \ T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{\textcolor{blue}{1}}51321^{\textcolor{red}{1}\textcolor{green}{1}}12321812321^{\textcolor{red}{1}\textcolor{green}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 \quad T = 21, \ T_{\max} = 29, \ \Delta = -99, \ \det = -7, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 28.14 \rightarrow 29, \ T_{\max} = 29, \ T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

4.42 $\text{NHC}(-2)(-3) + \textcolor{red}{e}_6 + \textcolor{red}{e}_6$

$$T = 7 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}}31^{\textcolor{blue}{1}}1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{e}_6 \quad T = 7, \ T_{\max} = 125, \ \Delta = -242, \ \det = -155, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.06 \rightarrow 16, \ T_{\max} = 125, \ T \in [16, 125]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \ 1^{\textcolor{red}{1}}31^{\textcolor{red}{1}\textcolor{blue}{1}}31^{\textcolor{blue}{1}}1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{e}_6 \quad T = 8, \ T_{\max} = 103, \ \Delta = -221, \ \det = 612, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.35 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{11}231^131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 103, \Delta = -193, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 & 0 & 4 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 103, \Delta = -187, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [4] \quad 1^{11}31^151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 12, T_{\max} = 104, \Delta = -158, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 17.45 \rightarrow 18, T_{\max} = 104, T \in [18, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 4 & 0 & 1 & -2 & 0 \\ 2 & 4 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

4.43 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{111}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 103, \Delta = -212, \det = 385, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.06 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -4 & 4 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{11}2^13^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 102, \Delta = -157, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 102, T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 3 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

4.44 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_7$

$$T = 9 \quad [0] \quad 1^{11}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_7 \quad T = 9, T_{\max} = 127, \Delta = -242, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 127, T \in [18, 127]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 1 & 0 & 1 & -5 & -3 \\ 2 & 3 & -1 & -1 & -3 & -4 \end{pmatrix}_{b_0}$$

4.45 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}'_7$

$$T = 9 \quad [0] \quad 1^{11}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}'_7 \quad T = 9, T_{\max} = 104, \Delta = -214, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.24 \rightarrow 17, T_{\max} = 104, T \in [17, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 1 & 0 & 1 & -4 & -2 \\ 2 & 3 & -1 & -1 & -2 & -3 \end{pmatrix}_{b_0}$$

4.46 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{f}_4$

$$T = 7 \quad [0] \ 1^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 7, T_{\max} = 102, \Delta = -216, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \ 1^{11} 3^{11} 2^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 2^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 102, \Delta = -195, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11} 2^{11} 3^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 102, \Delta = -167, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 18 & 16 & 0 & 8 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 2^{11} 3^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 102, \Delta = -167, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 & 0 & 4 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 2^{11} 2^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 101, \Delta = -159, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -5 & -3 \\ 5 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 2^{11} 3^{11} 2^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 2^{11} 3^{11} 2^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \ 1^{11} 3^{11} 5^{11} 13^{11} 2^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 12, T_{\max} = 96, \Delta = -132, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 17, T_{\max} = 96, T \in [17, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 1 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}31^151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 \quad T = 12, T_{\max} = 96, \Delta = -132, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 96, T \in [17, 96]$
 $\begin{pmatrix} 1 & 5 & 1 & 2 & 3 \\ 5 & 18 & 1 & 5 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 8 & -1 & 0 & 1 \end{pmatrix}_{b_0}$

4.47 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 8 \quad [0] \quad 1^{111}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -3 & 4 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{112}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{113}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -3 & 12 & -2 \end{pmatrix}_{b_0}$$

$T = 9 \quad [5] \quad 1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -3 & 4 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{111}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & -3 & 8 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$T = 10 \quad [8] \quad 1^{11}2^321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 \\ 2 & 3 & -1 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[9] $1^{11}2^321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & 0 & -1 \\ 1 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^{221} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 5 & 15 & 1 & 21 & 5 \\ 15 & 42 & 0 & 59 & 12 \\ 1 & 0 & -5 & 0 & -3 \\ 21 & 59 & 0 & 79 & 17 \\ 5 & 12 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

4.48 NHC(-2)(-3) + e₆ + NHC(-2)(-3)

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 21 & 27 & 1 & 9 \\ 27 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 9 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 3 & 3 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \ 1^{11} 2^1 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 4^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 231^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 2 & 6 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \ 1^{11} 5^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -108, \det = -93, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 31^{11} 4131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -205, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.49 $\text{NHC}(-2)(-3) + \textcolor{green}{e}_6 + \text{NHC}(-2)(-3) + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{1111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.50 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.51 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 6 & 3 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 4 & 0 & 12 & 2 \\ 11 & 19 & 0 & 21 & 12 & 73 & 15 \\ 3 & 4 & -4 & 4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -507, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 4 & 0 & 9 & 2 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -4 & 4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -992, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 4 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 4 & 0 & 18 & 2 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -4 & 4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1227, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 4 & 0 & 15 & 2 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -4 & 4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1892, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 4 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -2307, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 33 & 4 \\ 2 & 4 & 0 & 4 & 0 & 21 & 2 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -4 & 4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -3152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 4 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -85, \ \det = 116, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 27 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 31^{112} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 170, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 & 12 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & -4 & 4 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 21 & 27 & 1 & 52 & 9 \\ 27 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 65 & 0 & 123 & 21 \\ 9 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 21 & 27 & 1 & 28 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{11} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 14, \Delta = -92, \det = 440, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 & 6 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.52 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10$ [0] $1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 & 7 \\ 3 & 4 & -4 & 4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 9 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 9 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.53 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.54 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 21 & 27 & 1 & 16 & 9 \\ 27 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 20 & 0 & 10 & 6 \\ 9 & 11 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.55 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 6 & 3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 6 & 9 & 5 \\ 3 & 4 & -4 & 3 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{11} 3^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 & 0 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 705, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 & 6 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.56 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.57 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 & 3 \\ 3 & 4 & -4 & 4 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.58 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 7 \quad [0] \quad 1^{\overline{11}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 101, \Delta = -192, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 101, \Delta = -192, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -6 & -4 \\ 5 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \quad 1^{\overline{11}} 3 1^{\overline{1}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.80 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & -4 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} 3 2 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{11}231^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 & 0 & 4 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -4 & -2 \\ 5 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}231^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 2 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 9 & 4 & 5 \\ 2 & 0 & 4 & -2 & 0 \\ 3 & 0 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{11}3221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{11}4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -5 & -3 \\ 5 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^1231^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.87 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -5 & -3 \\ 5 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [19] $1^{11}4^{11}1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -4 & -2 \\ 5 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [20] $1^{11}31^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 11, T_{\max} = 101, \Delta = -136, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.60 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 12$ [21] $1^{11}31^15^{11}1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 22, T_{\max} = 24, T \in [22, 24]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & -3 \end{pmatrix}_{b_0}$$

[22] $1^{11}31^15^{11}1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 17.40 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 4 & 1 & 0 & 2 \\ 2 & 0 & 3 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{11}31^15^{11}1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.60 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 2 & 4 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 13$ [24] $1^{111}6^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -10, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 18.40 \rightarrow 19, T_{\max} = 25, T \in [19, 25]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{11}31^16^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 13$, $T_{\max} = 27$, $\Delta = -112$, $\det = -30$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 17.20 \rightarrow 18$, $T_{\max} = 27$, $T \in [18, 27]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 10 & 5 & 0 & 6 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & -3 & 1 \end{pmatrix}_{b_0}$

4.59 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$

$T = 9$ $[0] 1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$

4.60 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$

$T = 8$ $[0] 1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 6$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$

[1] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 139$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$

[2] $1^{111}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.71 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -4 & 34 & 0 \end{pmatrix}_{b_0}$

[3] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$

[4] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 213$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$

[5] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 220$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$

[6] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{112} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{113} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{112} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{114} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{113} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 921, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1005$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{115} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1053$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.1$
 $T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -3 & 0 & 15 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & -1 & -2 & 20 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1366$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{114} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1590$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{114} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1650$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [24] $1^1 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 33$, $T_{\max} = 99$, $T \in [33, 99]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 3 & -1 & -2 & -20 \end{pmatrix}_{b_0}$$

[23] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -9$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.11 \rightarrow 17$, $T_{\max} = 94$, $T \in [17, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & -2 & 8 & -3 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.15 \rightarrow 16$, $T_{\max} = 94$, $T \in [16, 94]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -81$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -151$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -165$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.61 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -571$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [34] $1^{11} 2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 17.67 \rightarrow 18$, $T_{\max} = 22$, $T \in [18, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[35] $1^{11} 2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 18$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.11 \rightarrow 17$, $T_{\max} = 22$, $T \in [17, 22]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[36] $1^{111} 3^{221} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 15 & 1 & 13 & 5 \\ 15 & 42 & 0 & 36 & 12 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 36 & 0 & 28 & 10 \\ 5 & 12 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{111} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 16 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 6 & 16 & 0 & 8 & 22 & 10 \\ 3 & 6 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{111} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{1231} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11} 3^{221} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 15 & 1 & 21 & 5 \\ 15 & 42 & 0 & 59 & 12 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 59 & 0 & 79 & 17 \\ 5 & 12 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{111} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 7 & 3 \\ 4 & 10 & 1 & 4 & 16 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 7 & 16 & 0 & 8 & 22 & 10 \\ 3 & 6 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3^{221} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{111} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11} 4^{131} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{112}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 10 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{114}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 18 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 190, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{113}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 14 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{112}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.66 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{114}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{113}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{131}4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{112}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 11 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 11 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{112}2^13^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{112}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 12 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 12 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{11}1^{31}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{113}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 710, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 15 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 15 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{113}2^13^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 740, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{114}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 910, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 19 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 19 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{114}2^13^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 940, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{113}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 980, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 16 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 16 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{114}2^13^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 1420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 20 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 20 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [63] \quad 1^{111} 31^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -106, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.61 NHC(-2)(-3) + $\textcolor{green}{e}_6$ + $\textcolor{orange}{so}_8$ + $\textcolor{red}{su}_2$ + $\textcolor{red}{su}_2$

$$T = 9 \quad [0] \quad 1^{112} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 0 & 6 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & -1 & -2 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -4 & 7 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{11} 3 \ 1^{1311} 4 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 7 & 5 \\ 10 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 7 & 10 & 0 & 9 & 5 & 5 \\ 5 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

4.62 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3 \ 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.63 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

4.64 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{\overline{11}} 3^{\overline{1}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 99$, $\Delta = -137$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.69 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 3 & 5 \\ 2 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 99$, $\Delta = -137$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [3] $1^{\overline{11}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 30$, $\Delta = -116$, $\det = 32$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 1 & 0 & 4 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [4] $1^{\overline{11}} 413^{\overline{1}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -24$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [5] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 5131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 12$, $T_{\max} = 24$, $\Delta = -108$, $\det = 95$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 10 & 1 & 4 & 8 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 6 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

4.65 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 8 \quad [1] \quad 1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 302, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^1 1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.07 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \quad 1^{11} 2^1 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.10 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 30, \Delta = -116, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.49 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{111}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.12 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

4.66 $\text{NHC}(-2)(-3) + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$

$T = 9$ [0] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

4.67 $\text{NHC}(-2)(-3) + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2$

$T = 7$ [0] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2$ $T = 7, T_{\max} = 100, \Delta = -163, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.71 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2$ $T = 7, T_{\max} = 100, \Delta = -163, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.87 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2$ $T = 7, T_{\max} = 100, \Delta = -163, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

- [6] $21^1 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 6 & 10 & 0 & 22 & 8 \\ 3 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$
- [12] $21^1 31^{115} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 3 & 4 & -3 & 9 & -1 \\ 9 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{115} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 3 & 4 & -3 & 15 & -1 \\ 15 & 40 & 15 & 148 & 20 \\ 3 & 5 & -1 & 20 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [5] $21^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{112} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 3 & 4 & -3 & 6 & -1 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [3] $1^{114} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 3 & 4 & -3 & 12 & -1 \\ 12 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{113} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 3 & 4 & -3 & 9 & -1 \\ 9 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [7] $21^1 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 3 & 4 & -3 & 6 & -1 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [8] $21^1 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 3 & 4 & -3 & 12 & -1 \\ 12 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

- [14] $1^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -54$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$
- [15] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -59$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -3 & 6 & -1 \\ 9 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & -1 \end{pmatrix}_{b_0}$
- [16] $1^{12}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$
- [17] $1^{13}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -70$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$
- [18] $1^{14}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -84$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$
- [19] $1^113^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -86$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 \end{pmatrix}_{b_0}$
- [20] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -93$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$
- [21] $1^{15}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -4 & 15 & -1 \end{pmatrix}_{b_0}$
- [22] $1^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -115$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$
- [23] $1^{16}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.19 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -4 & 18 & -1 \end{pmatrix}_{b_0}$
- [24] $1^113^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -146$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -3 & 6 & -1 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & -1 \end{pmatrix}_{b_0}$

- [25] $1^{17}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -150$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.3$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$
- [26] $1^{113}3^11^4 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -158$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 13 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$
- [27] $1^{11}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -162$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 14 & 0 & 37 & 11 \\ 3 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$
- [28] $1^{112}3^11^2 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -174$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -3 & 6 & -1 \\ 8 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [29] $1^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -189$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.31 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$
- [30] $1^{11}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -195$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 11 & 19 & 0 & 73 & 15 \\ 3 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$
- [31] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -203$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -3 & 6 & -1 \\ 7 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{11}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -214$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 14 & 24 & 0 & 121 & 19 \\ 3 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{13}1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -246$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 9 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$
- [34] $1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -267$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.36 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$
- [35] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -304$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 10 & 18 & 0 & 58 & 14 \\ 3 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

- [36] $1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -309$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 12 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$
- [37] $1^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -349$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$
- [38] $21^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -352$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 6 & 8 & 3 & 4 & 4 \\ 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$
- [39] $1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -353$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 10 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$
- [40] $1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -374$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 11 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$
- [41] $131^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -386$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.32 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 12 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & -1 \end{pmatrix}_{b_0}$
- [42] $1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -405$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 13 & 23 & 0 & 100 & 18 \\ 3 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$
- [43] $1^{13}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -450$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.36 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 12 & 22 & 0 & 85 & 17 \\ 3 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$
- [44] $1^{12}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -492$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 16 & 28 & 0 & 154 & 22 \\ 3 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$
- [45] $1^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -498$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 16 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$
- [46] $1^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -543$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 13 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & -1 \end{pmatrix}_{b_0}$

- [47] $1^1 13 1^{115} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -566, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & -3 & 15 & -1 \\ 15 & 40 & 15 & 148 & 20 \\ 3 & 5 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$
- [48] $1^{114} 3^1 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -599, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 15 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [49] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 14 & 26 & 0 & 118 & 20 \\ 3 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$
- [50] $1^{114} 3^1 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -614, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 14 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^{13} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 15 & 27 & 0 & 133 & 21 \\ 3 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$
- [52] $21^{112} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -622, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 12 & 4 \\ 8 & 9 & 4 & 16 & 5 \\ 3 & 4 & -3 & 6 & -1 \\ 12 & 16 & 6 & 22 & 8 \\ 4 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [53] $1^{13} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -774, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 18 & 32 & 0 & 193 & 25 \\ 3 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$
- [54] $1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -837, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 31 & 0 & 172 & 24 \\ 3 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$
- [55] $1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1060, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 20 & 36 & 0 & 238 & 28 \\ 3 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$
- [56] $21^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1072, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 18 & 4 \\ 8 & 9 & 4 & 24 & 5 \\ 3 & 4 & -3 & 9 & -1 \\ 18 & 24 & 9 & 52 & 12 \\ 4 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [57] $21^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1702, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 24 & 4 \\ 8 & 9 & 4 & 32 & 5 \\ 3 & 4 & -3 & 12 & -1 \\ 24 & 32 & 12 & 94 & 16 \\ 4 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [58] $21^{115} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -2512, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 30 & 4 \\ 8 & 9 & 4 & 40 & 5 \\ 3 & 4 & -3 & 15 & -1 \\ 30 & 40 & 15 & 148 & 20 \\ 4 & 5 & -1 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [59] \quad 21^{111} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.67 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & -2 & 7 & 0 \\ 9 & 12 & 7 & 14 & 8 \\ 4 & 5 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & -2 & 3 & 0 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{111} 1^2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[63] \quad 2^1 2 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & -2 & 3 & 0 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{111} 1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.08 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 7 & 0 \\ 6 & 12 & 7 & 14 & 8 \\ 3 & 5 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 2 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{112} 1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[68] \quad 2^1 2 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{111} 1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 7 & 0 \\ 5 & 12 & 7 & 14 & 8 \\ 3 & 5 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{112} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{114} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & -1 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{113} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{14} 3 1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 98, T \in [16, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -4 & 16 & -3 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{11} 3 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{11} 3 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{12} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{14}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{111}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 6 & 0 & 13 & 7 \\ 2 & 3 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{111}1^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{111}3131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{15}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -1 & 10 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{112}31^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 12 & 0 & 58 & 14 \\ 2 & 3 & 2 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{16}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.92 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[89] $1^131^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 3 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

- [90] $1^{12}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 3 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$
- [91] $1^{13}3^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 91$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 6 & 0 & 4 & 4 \\ 2 & 3 & 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$
- [92] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 91$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$
- [93] $1^{17}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 91$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.3$
 $T_{\min} = 14.22 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -1 & 14 & -2 \end{pmatrix}_{b_0}$$
- [94] $1^{113}3^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 106$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 18 & 3 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 18 & 18 & 0 & 133 & 21 \\ 2 & 3 & 2 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$
- [95] $1^{112}3^{13}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$
- [96] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 121$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 3 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$
- [97] $1^{11}3^{12}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{13}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 4 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$
- [99] $1^{114}3^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 0 & 24 & 3 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 24 & 0 & 238 & 28 \\ 2 & 3 & 2 & -4 & 28 & -1 \end{pmatrix}_{b_0}$$
- [100] $1^{12}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0} \\
[101] & 1^{13} 3 2 1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0} \\
[102] & 1^{13} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.78 \rightarrow 13, T_{\max} = 94, T \in [13, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0} \\
[103] & 1^{14} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.09 \rightarrow 14, T_{\max} = 94, T \in [14, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0} \\
[104] & 1^{13} 3 1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.23 \rightarrow 14, T_{\max} = 98, T \in [14, 98] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -4 & 13 & -1 \end{pmatrix}_{b_0} \\
[105] & 1^{12} 3 1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0} \\
[106] & 1^{13} 3 1 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 9 & 0 & 25 & 9 \\ 2 & 3 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0} \\
[107] & 1^{12} 3 1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.08 \rightarrow 14, T_{\max} = 98, T \in [14, 98] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 9 & 0 & 28 & 10 \\ 2 & 3 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0} \\
[108] & 1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.52 \rightarrow 13, T_{\max} = 94, T \in [13, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0} \\
[109] & 1^{14} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 94, T \in [13, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 5 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0} \\
[110] & 1^{11} 3 1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [111] $1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 207$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [112] $1^{12}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 223$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$
- [113] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 226$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [114] $1^{12}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 226$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 4 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$
- [115] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [116] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$
- [117] $1^131^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$
- [118] $1^{14}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [119] $21^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 6 & 8 & 3 & 4 & 4 \\ 4 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [120] $1^{14}3131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 292$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 46 & 12 \\ 2 & 3 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$
- [121] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 301$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{12} 3^{13} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[123] $1^{13} 31^{23} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 15 & 15 & 0 & 85 & 17 \\ 2 & 3 & 2 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

[124] $1^{13} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 5 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[125] $1^{13} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[126] $1^{14} 31^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[127] $1^{12} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

[128] $1^{13} 31^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 358, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 12 & 0 & 49 & 13 \\ 2 & 3 & 2 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

[129] $1^{12} 3^{14} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{15} 3131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 13.65 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 15 & 0 & 73 & 15 \\ 2 & 3 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[131] $1^1 31^{13} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

- [132] $1^{14} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 442$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 6 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$
- [133] $1^{12} 3^{13} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 466$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 6 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$
- [134] $1^{14} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 466$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [135] $1^{11} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 478$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$
- [136] $1^{11} 431^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 481$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 0 & 21 & 3 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 21 & 21 & 0 & 172 & 24 \\ 2 & 3 & 2 & -4 & 24 & -1 \end{pmatrix}_{b_0}$$
- [137] $1^{13} 3^{13} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 483$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [138] $1^{13} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 513$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$
- [139] $1^{13} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 526$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$
- [140] $1^{11} 431^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 553$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 15 & 15 & 0 & 76 & 16 \\ 2 & 3 & 2 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$
- [141] $1^{12} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 568$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$
- [142] $1^{12} 3^{14} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 568$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 7 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

[143] $1^{13} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[144] $1^{14} 31^{23} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 18 & 3 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 18 & 18 & 0 & 118 & 20 \\ 2 & 3 & 2 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

[145] $1^{14} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[146] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 676, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[147] $1^{13} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 681, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 7 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[148] $1^{13} 31^{14} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[149] $1^{11} 31^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 802, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[150] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 826, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 837, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 8 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 898, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 8 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[153] $1^1 3^1 1^5 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1012$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 15 & 0 & 148 & 20 \\ 2 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{12} 3^1 1^3 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1036$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[155] $1^{14} 3^1 1^3 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[156] $1^{14} 3^1 2^1 1^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 1108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 9 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[157] $1^{13} 3^1 1^3 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1138$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[158] $1^{11} 3^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1198$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{13} 3^1 1^6 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1408$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 18 & 0 & 214 & 24 \\ 2 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[160] $1^{12} 3^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1576$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[161] $1^{13} 3^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1822$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[162] $1^{13} 3^1 1^7 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1876$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.3$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 42 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 42 & 21 & 0 & 292 & 28 \\ 2 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[163] $1^{14} 3^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1936$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [164] \quad 1^1 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 2 & 0 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{11} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[166] \quad 1^{11} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[167] \quad 1^{11} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[168] \quad 1^{13} 3^1 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[169] \quad 1^{13} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 20.45 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & -7 \end{pmatrix}_{b_0}$$

$$[170] \quad 1^{13} 3^1 221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 20.45 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 19 & 0 \\ 10 & 10 & 0 & 19 & 104 & 22 \\ 2 & 2 & -1 & 0 & 22 & -7 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{12} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{11} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 2 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

- [176] $1^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [171] $1^{12}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$
- [172] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 19$, $T_{\max} = 21$, $T \in [19, 21]$

$$\begin{pmatrix} 5 & 15 & 12 & 5 \\ 15 & 42 & 36 & 12 \\ 12 & 36 & 28 & 10 \\ 5 & 12 & 10 & -3 \end{pmatrix}_{b_0}$$
- [173] $1^{12}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$
- [174] $1^{14}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 19$, $T_{\max} = 21$, $T \in [19, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & 0 & 4 & -5 \end{pmatrix}_{b_0}$$
- [179] $1^{131}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [178] $1^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [180] $1^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 17$, $T_{\max} = 93$, $T \in [17, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 13 & 0 & 40 & 12 \\ 3 & 4 & -3 & 12 & -2 \end{pmatrix}_{b_0}$$
- [182] $1^{114}231^{231} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 19.56 \rightarrow 20$, $T_{\max} = 25$, $T \in [20, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 2 & 5 & 0 & 22 & 4 \\ 1 & 5 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 22 & 22 & 0 & 134 & 24 \\ 2 & 4 & 3 & -4 & 24 & -6 \end{pmatrix}_{b_0}$$
- [181] $1^{11}231^{231} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 19.56 \rightarrow 20$, $T_{\max} = 25$, $T \in [20, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 10 & 0 \\ 8 & 0 & 0 & 10 & 38 & 12 \\ 2 & 0 & -1 & 0 & 12 & -6 \end{pmatrix}_{b_0}$$
- [183] $1^{11}4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[184] $1^{11}2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{11}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 16.73 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 5 & 15 & 22 & 5 \\ 15 & 42 & 59 & 12 \\ 22 & 59 & 79 & 17 \\ 5 & 12 & 17 & -1 \end{pmatrix}_{b_0}$$

[186] $1^{11}2321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{11}2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{13}2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[190] $1^{11}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{13}231^{12}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 16.93 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 18 & 0 & 94 & 20 \\ 2 & 4 & 3 & -4 & 20 & -3 \end{pmatrix}_{b_0}$$

[198] $1^{12}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{13}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[196] $1^{14}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[195] $1^{15}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 15.38 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & 0 & 10 & -2 \end{pmatrix}_{b_0}$$

[194] $1^{16}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 16.75 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

[193] $1^{1231}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[192] $1^{17}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 18.37 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & 0 & 14 & -5 \end{pmatrix}_{b_0}$$

[191] $1^{11}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{112}231^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 11 & 4 \\ 1 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 11 & 11 & 0 & 32 & 12 \\ 2 & 4 & 3 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[201] $1^{12}3221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 5 & 6 & 0 & 10 & 30 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{1231}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{131^2}4^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 11 & 0 & 28 & 10 \\ 3 & 4 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[203] $1^{11}4^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

- [204] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.59 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [205] $1^1 3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 15 & 13 & 5 \\ 15 & 42 & 36 & 12 \\ 13 & 36 & 28 & 10 \\ 5 & 12 & 10 & 2 \end{pmatrix}_{b_0}$$
- [206] $1^{11} 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [207] $1^{12} 3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 3 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [208] $1^{131^{11}} 4^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 7 & 3 \\ 4 & 4 & 0 & 10 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 10 & 0 & 21 & 9 \\ 3 & 4 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [209] $1^{12} 3221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 20 & 0 \\ 10 & 12 & 0 & 20 & 126 & 24 \\ 2 & 2 & -1 & 0 & 24 & -1 \end{pmatrix}_{b_0}$$
- [210] $1^{131^1} 4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [211] $1^{12} 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$
- [212] $1^1 23131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$
- [213] $1^{11} 23131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 4 & 4 & 0 & 2 & 4 \\ 2 & 4 & 3 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [214] $21^1 41^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

- [215] $1^{\overline{1}1} 4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$
 $\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$
- [216] $1^{\overline{1}1} 231^{\overline{2}31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 6 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & 0 & 8 & -1 \end{pmatrix}_{b_0}$
- [218] $1^{\overline{1}1} 231^{\overline{3}31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 16.96 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 13 & 0 \\ 11 & 0 & 0 & 13 & 74 & 16 \\ 2 & 0 & -1 & 0 & 16 & -3 \end{pmatrix}_{b_0}$
- [217] $1^{\overline{1}3} 4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$
- [219] $1^{\overline{1}2} 2321^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 3 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$
- [220] $1^{\overline{1}4} 2321^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 5 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & -1 & 15 & 1 \end{pmatrix}_{b_0}$
- [221] $1^{\overline{1}4} 4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$
- [222] $1^{\overline{1}1} 3^{\overline{1}1} 221^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 5 & 15 & 21 & 5 \\ 15 & 42 & 59 & 12 \\ 21 & 59 & 79 & 17 \\ 5 & 12 & 17 & 2 \end{pmatrix}_{b_0}$
- [223] $1^{\overline{1}1} 23131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 4 & 0 \\ 3 & 0 & 0 & 4 & 2 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$
- [224] $1^{\overline{1}1} 4^{\overline{1}1} 1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$
- [225] $1^{\overline{1}1} 2^{\overline{1}1} 3^{\overline{1}1} 21^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0} \\
[226] \quad & 1^{1231}331^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 14.60 \rightarrow 15, T_{\max} = 25, T \in [15, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 9 & 0 \\ 2 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0} \\
[227] \quad & 1^{1323}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 4 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0} \\
[228] \quad & 1^{1123}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0} \\
[229] \quad & 1^{154}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.1 \\
& T_{\min} = 14.58 \rightarrow 15, T_{\max} = 93, T \in [15, 93] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0} \\
[230] \quad & 1^{1113}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[231] \quad & 1^{1223}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0} \\
[232] \quad & 1^{1113}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[233] \quad & 1^{1123}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.08 \rightarrow 14, T_{\max} = 21, T \in [14, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 3 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0} \\
[235] \quad & 1^{1231}1131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.72 \rightarrow 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 3 & -4 & 4 & 0 \end{pmatrix}_{b_0} \\
[236] \quad & 1^{1231}1331^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 17.72 \rightarrow 18, T_{\max} = 25, T \in [18, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 3 & -4 & 12 & -4 \end{pmatrix}_{b_0}
\end{aligned}$$

[237] $1^{1231^{14}31^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 3 & -4 & 16 & -8 \end{pmatrix}_{b_0}$$

[234] $1^{1231^{12}31^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 3 & -4 & 8 & -2 \end{pmatrix}_{b_0}$$

[238] $1^{16^14131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[240] $1^{11223131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 8 & 4 \\ 1 & 5 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 8 & 8 & 0 & 14 & 8 \\ 2 & 4 & 3 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

[239] $1^{123131^{112}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[241] $1^{11^131^2}4^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 10 & 0 & 21 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{112^2321^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[243] $1^{14^13221^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 6 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{11^13^2221^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 6 & 4 & 0 & 9 & 20 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{12^23221^{13}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 8 & 7 & 0 & 14 & 54 & 16 \\ 2 & 2 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[246] $1^{112^2321^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[247] $1^{1231}431^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 15.02 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 12 & 0 \\ 12 & 0 & 0 & 12 & 94 & 16 \\ 2 & 0 & -1 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$

[248] $1^{11}4^{131}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{17}4^{131}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[250] $1^{11}4^{131}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{11}2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 3 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{13}3^{221}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 4 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[253] $1^{11}3^{221}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[254] $1^{12}2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{14}2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{13}2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{11} 3^{13} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[258] $1^{11} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{11} 3^{14} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[260] $1^{12} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{14} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -3 & 7 & -1 \\ 6 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[262] $1^{14} 3^{11} 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 16 & 0 \\ 8 & 11 & 0 & 16 & 90 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[263] $1^4 231^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 2 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{11} 231^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 7 & 0 & 14 & 8 \\ 2 & 4 & 3 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

[269] $1^{13} 23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 5 & 0 & 12 & 4 \\ 1 & 5 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 12 & 12 & 0 & 34 & 12 \\ 2 & 4 & 3 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[265] $1^4 23131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

- [266] $1^{11} 3^{12} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [267] $1^{11} 231^{12} 31^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 5 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [268] $1^1 231^{11} 31^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [270] $1^{11} 4^{11} 1^2 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.03 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 4 & 0 & 15 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 15 & 0 & 45 & 13 \\ 3 & 4 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$
- [271] $1^{12} 3^{11} 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 4 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [272] $1^{14} 3^{11} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 5 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [273] $1^1 231^{11} 31^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 4 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [274] $1^{11} 231^{11} 31^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [275] $1^{13} 3^{12} 221^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 15 & 0 \\ 8 & 9 & 0 & 15 & 70 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$
- [276] $1^{13} 2^3 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 6 & -1 \\ 5 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [277] $1^{13} 3^{11} 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{11} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{11} 4^1 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{131} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.78 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{11} 231^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 0 \\ 4 & 9 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{12} 3^1 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{131} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{11} 3^1 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 4 & 0 & 9 & 20 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[286] $1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 4 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 21 & 0 & 101 & 19 \\ 3 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

- [287] $1^{1^1 3 1^1} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 11 & 0 & 21 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [290] $1^{1^2 2 3 2 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 4 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [289] $1^{1^1 1 2 3 1 3} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 4 & 0 \\ 4 & 0 & 0 & 4 & 2 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [292] $1^{1^3 3 2 2 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 6 & 8 & 0 & 11 & 44 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$
- [291] $1^{1^1 1 2 3 1^1 3 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 7 & 0 \\ 6 & 0 & 0 & 7 & 14 & 8 \\ 2 & 0 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$
- [297] $1^{1^2 2 3 2 2 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 6 & 6 & 0 & 10 & 30 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [296] $1^{1^1 1 2 2 3 2 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 5 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [295] $1^{1^2 2 3 2 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [294] $1^{1^4 2 3 2 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 6 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [293] $1^{1^2 2 3 2 2 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [298] $1^{1^1 1 3 2 3 1^1 3 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 5 & 0 & 15 & 4 \\ 1 & 5 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 15 & 0 & 58 & 16 \\ 2 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0} \\
[299] & 1^{12} 231^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0} \\
[300] & 1^{11} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[301] & 1^{14} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[302] & 1^{12} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13.87 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 20 & 0 & 90 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0} \\
[303] & 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0} \\
[304] & 1^{13} 2^3 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0} \\
[305] & 1^{14} 23131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 14.20 \rightarrow 15, T_{\max} = 25, T \in [15, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 16 & 4 \\ 1 & 5 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 16 & 16 & 0 & 62 & 16 \\ 2 & 4 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0} \\
[306] & 1^{12} 23131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 14.20 \rightarrow 15, T_{\max} = 25, T \in [15, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0} \\
[308] & 1^{11} 231^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[307] & 1^{12} 231^{12} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 4 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [309] $1^{13} 4 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -207$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$
- [310] $1^{11} 2 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 5 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [311] $1^{11} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -220$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 6 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [315] $21^{11} 41^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 3 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 4 & 8 & 0 & 10 & 6 \\ 2 & 2 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [314] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [313] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 18 & 0 & 70 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$
- [312] $1^{12} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 14 & 0 & 38 & 12 \\ 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [316] $1^{12} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -227$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [317] $1^{13} 2 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 5 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [318] $1^{14} 231^{13} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -232$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 25 & 0 & 188 & 28 \\ 2 & 4 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$
- [320] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -235$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 15 & 0 & 45 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{12} 3^{13} 4^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 17 & 0 & 61 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[322] $1^{12} 3^{14} 221^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 9 & 7 & 0 & 14 & 54 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{12} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[326] $1^{11} 3^{13} 4^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[327] $1^{13} 3^{11} 4^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 15 & 0 & 40 & 12 \\ 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[323] $1^{12} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[324] $1^1 231^{23} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 7 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[325] $1^{13} 3^{14} 221^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 19 & 0 \\ 11 & 10 & 0 & 19 & 104 & 22 \\ 2 & 2 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[328] $1^{13} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[329] $1^1 231^{12} 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

- [330] $1^{13}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [331] $1^{12}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [332] $1^1231^131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -261$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 5 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [333] $1^{12}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -267$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [334] $1^{12}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -268$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 5 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [335] $1^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -268$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 6 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [336] $1^{14}4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -268$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$
- [337] $1^{12}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [338] $1^{14}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -281$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 6 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$
- [339] $1^{15}23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 14.26 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 5 & 0 & 20 & 4 \\ 1 & 5 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 20 & 20 & 0 & 98 & 20 \\ 2 & 4 & 3 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$
- [340] $1^{11}231^231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 10 & 0 \\ 9 & 0 & 0 & 10 & 38 & 12 \\ 2 & 0 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[341] $1^{12}3131^{115} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 14.26 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[342] $1^{14}2321^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[343] $1^{12}231^{1231^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 14 & 0 & 62 & 16 \\ 2 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[344] $1^{12}3221^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -299$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 6 & 0 & 10 & 30 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[345] $1^{11}23131^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -300$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 4 & 0 \\ 5 & 0 & 0 & 4 & 2 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[346] $1^{11}231^{1231^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -307$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.29 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 10 & 0 & 38 & 12 \\ 2 & 4 & 3 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[347] $1^{14}3221^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -307$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 7 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[348] $1^{11}231^{1331^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.01 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 7 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[349] $1^{14}41^{131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -319$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 19 & 0 & 65 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{12}41^{131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -319$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 16 & 0 & 45 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

- [351] $1^{14} 3^{13} 221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -323$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.66 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 20 & 0 \\ 11 & 12 & 0 & 20 & 126 & 24 \\ 2 & 2 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$
- [352] $1^{14} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -331$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.52 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 6 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [353] $1^{14} 3^{12} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -332$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [354] $1^{14} 231^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -333$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 2 & 5 & 0 & 19 & 4 \\ 1 & 5 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 19 & 19 & 0 & 92 & 20 \\ 2 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$
- [355] $1^{13} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 6 & -1 \\ 6 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [356] $1^1 231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -340$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 6 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [357] $1^1 231^{33} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -345$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 9 & 0 \\ 10 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [358] $1^{13} 3^{13} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [359] $1^{13} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [360] $1^{12} 3^{13} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [361] $1^{13} 3^{13} 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 15 & 0 \\ 9 & 9 & 0 & 15 & 70 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[362] \ 1^{13} 3 \ 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 7 & 8 & 0 & 11 & 44 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{12} 2 \ 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 6 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{14} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -3 & 7 & -1 \\ 7 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[365] \ 1^{13} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -366, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{14} 3 \ 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 16 & 0 \\ 9 & 11 & 0 & 16 & 90 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[367] \ 1^1 231^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 5 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[368] \ 1^{11} 231^1 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 7 & 0 \\ 7 & 0 & 0 & 7 & 14 & 8 \\ 2 & 0 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{13} 4 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 18 & 0 & 61 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[370] \ 1^{13} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[371] \ 1^{12} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 6 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[372] $1^{14}2^3321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[373] $1^1231^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[374] $1^{11}231^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 6 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[375] $1^{116}23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -398, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 2 & 5 & 0 & 24 & 4 \\ 1 & 5 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 24 & 24 & 0 & 142 & 24 \\ 2 & 4 & 3 & -4 & 24 & -1 \end{pmatrix}_{b_0}$$

[376] $1^1231^{13}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 6 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[377] $1^{12}3^1221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 6 & 0 & 10 & 30 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[378] $1^{12}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 7 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[379] $1^{13}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[381] $1^{131^2}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 19 & 0 & 68 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[380] $1^1231^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

- [383] $1^{13} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -450$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 20 & 0 & 70 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$
- [382] $1^{1231} 231^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -450$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 8 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$
- [384] $1^{13} 4 1^{131^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 21 & 0 & 88 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$
- [385] $1^{14} 3 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -467$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [386] $1^{142} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & -1 & 15 & 1 \end{pmatrix}_{b_0}$
- [387] $1^{13} 4 1^{131^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -483$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 24 & 0 & 121 & 21 \\ 3 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$
- [388] $1^{11} 231^{1331^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -487$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 13 & 0 & 74 & 16 \\ 2 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$
- [389] $1^{132} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -492$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 7 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$
- [390] $1^{122} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 7 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$
- [391] $1^{12} 4 1^{131^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -499$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 22 & 0 & 97 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$
- [392] $1^{12} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -508$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 19 & 25 & 0 & 132 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[393] $1^{14} 3^{12} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 7 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[396] $1^{13} 3^2 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 6 & -1 \\ 7 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[394] $1^{13} 3^2 221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 8 & 8 & 0 & 11 & 44 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[395] $1^{13} 3^2 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[397] $1^{14} 3^2 221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 8 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[398] $1^{14} 4^1 1^{31^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 22 & 0 & 90 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[399] $1^{11} 7^2 23131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 2 & 5 & 0 & 28 & 4 \\ 1 & 5 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 28 & 28 & 0 & 194 & 28 \\ 2 & 4 & 3 & -4 & 28 & -1 \end{pmatrix}_{b_0}$$

[400] $1^{12} 231^{14} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 8 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[401] $1^{13} 3^2 221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 15 & 0 \\ 10 & 9 & 0 & 15 & 70 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[402] $1^{11} 231^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -556, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 7 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

- [404] $1^{11}231^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -560$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 9 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$
- [403] $1^{11}231^{12}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -560$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 9 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [405] $1^{13}231^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -567$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 21 & 0 & 142 & 24 \\ 2 & 4 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$
- [406] $31^{111}2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -580$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [407] $1^{14}4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -583$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 4 \\ 3 & 4 & 0 & 24 & 4 & 1 \\ 1 & 0 & -5 & 0 & -3 & 1 \\ 14 & 24 & 0 & 101 & 19 & 3 \\ 3 & 4 & -3 & 19 & 1 & 1 \end{pmatrix}_{b_0}$$
- [410] $1^{14}3^1 221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 20 & 0 \\ 12 & 12 & 0 & 20 & 126 & 24 \\ 2 & 2 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$
- [409] $1^{14}3^1 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [408] $1^{14}2^1 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 8 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [411] $1^{14}2^1 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -603$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -3 & 7 & -1 \\ 8 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$
- [413] $1^{14}3^1 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -611$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 16 & 0 \\ 10 & 11 & 0 & 16 & 90 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$
- [412] $1^{11}231^{12}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -611$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0} \\
[414] & 1^{14} 231^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -623, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 7 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0} \\
[415] & 1^{13} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 8 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0} \\
[416] & 1^{13} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -639, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 13.38 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 23 & 0 & 97 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0} \\
[417] & 1^{12} 231^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 13.24 \rightarrow 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 17 & 0 & 104 & 20 \\ 2 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0} \\
[418] & 1^{11} 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -667, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 13.64 \rightarrow 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 16 & 0 & 122 & 20 \\ 2 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0} \\
[419] & 1^{13} 3 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 9 & 8 & 0 & 11 & 44 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0} \\
[420] & 1^{14} 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -679, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 13.42 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 25 & 0 & 121 & 21 \\ 3 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0} \\
[421] & 1^{13} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 8 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0} \\
[422] & 1^{13} 2 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 6 & -1 \\ 8 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0} \\
[423] & 1^{11} 231^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -695, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 8 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[424] $1^{14} 3^{13} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -731$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 9 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[425] $1^{13} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -774$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 26 & 0 & 130 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[426] $1^{14} 4^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -778$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 28 & 0 & 158 & 24 \\ 3 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[427] $1^1 231^{15} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -788$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 15 & 0 \\ 10 & 30 & 0 & 15 & 148 & 20 \\ 2 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$

[428] $1^1 231^{13} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -800$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 8 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[429] $1^{14} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -832$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -3 & 7 & -1 \\ 9 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[430] $1^1 231^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -836$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[431] $1^{14} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -844$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 27 & 0 & 132 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[432] $1^{14} 3^{13} 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -844$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.52 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 16 & 0 \\ 11 & 11 & 0 & 16 & 90 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[433] $1^{14} 3^{13} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -844$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 9 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[434] $1^{13} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -855$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 21 & 29 & 0 & 169 & 25 \\ 3 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[435] $1^{14} 231^{13} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -871$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[436] $1^{14} 3^{14} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -928$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 10 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[437] $1^{14} 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 9 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[438] $1^{14} 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -952$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 28 & 0 & 254 & 32 \\ 2 & 4 & 3 & -4 & 32 & 0 \end{pmatrix}_{b_0}$$

[439] $1^{11} 231^{13} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -968$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 9 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[440] $1^{12} 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1012$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.31 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 20 & 0 & 158 & 24 \\ 2 & 4 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[441] $1^{14} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -1051$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 30 & 0 & 169 & 25 \\ 3 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[442] $1^{11} 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1051$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 10 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[443] $1^{13} 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 24 & 0 & 202 & 28 \\ 2 & 4 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[444] $1^{11} 231^{13} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1135$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 10 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[445] $1^{11} 231^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1196$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 11 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[446] $1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -1204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 23 & 33 & 0 & 212 & 28 \\ 3 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[447] $1^{1231} 1^{14} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 10 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[448] $1^{1231} 1^{14} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 11 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[449] $1^{11231} 1^{14} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 11 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[450] $1^{1231} 1^{14} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[451] $1^{11231} 1^{14} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1715, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 12 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[452] $1^{11231} 1^{14} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1888, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 13 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [453] $1^1 31^{11} 41^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 4 & 0 & 10 & 6 \\ 2 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[454] $1^{15} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[455] $1^{12} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.20 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 8 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

- [456] $1^{11}31^14131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 35$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$
- [457] $1^{11}5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 39$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & -2 & 2 & 0 \\ 4 & 4 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [458] $1^{14}31^{11}41^331^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 40$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 18 & 0 & 130 & 22 \\ 2 & 4 & 2 & -3 & 22 & 0 \end{pmatrix}_{b_0}$
- [459] $1^131^{11}41^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 40$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 10 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 5 & -1 \\ 6 & 10 & 0 & 5 & 20 & 8 \\ 2 & 4 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$
- [460] $1^{11}5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 43$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & -2 & 2 & 0 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [461] $21^15131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 43$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & -2 & 2 & 0 \\ 3 & 4 & 2 & 1 & 3 \\ 4 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [462] $1^{11}5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 46$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 4 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$
- [463] $1^{11}5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 51$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 6 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0}$
- [464] $1^{11}5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 58$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.17 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & -2 & 2 & 0 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [465] $1^131^{11}4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$
- [466] $1^{12}5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 85$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.19 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 7 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[467] $1^{12}5^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 5 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[468] $1^{13}1^{11}41^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 99, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.59 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -3 & 4 & -1 \\ 5 & 8 & 1 & 4 & 11 & 7 \\ 2 & 4 & 0 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

[469] $1^{12}5^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[470] $1^{13}1^{11}4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[471] $21^{11}5^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.17 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 2 & 3 & -2 & 2 & 0 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[472] $1^{13}1^{11}4131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.24 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 1 & 3 & 5 & 5 \\ 2 & 4 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[473] $1^{12}31^{11}4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[474] $1^{12}31^{11}41^{23}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 16 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 11 & 0 & 52 & 14 \\ 2 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[475] $1^{13}5^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 7 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[476] $1^{13}1^{11}41^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.24 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 10 & 6 \\ 2 & 4 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[477] $1^{13}5^{14}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8$

$$\begin{aligned}
& T_{\min} = 14.21 \rightarrow 15, T_{\max} = 91, T \in [15, 91] \\
& \begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 10 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
[478] & 1^{11}31^{11}4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0} \\
[479] & 1^{13}5^{12}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 14.26 \rightarrow 15, T_{\max} = 91, T \in [15, 91] \\
& \begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 8 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
[480] & 1^{14}31^{11}41^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 16 & 0 & 90 & 18 \\ 2 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0} \\
[481] & 1^{13}5^{13}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 186, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 14.23 \rightarrow 15, T_{\max} = 91, T \in [15, 91] \\
& \begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
[482] & 21^{12}5^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 14.19 \rightarrow 15, T_{\max} = 91, T \in [15, 91] \\
& \begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[483] & 1^{11}31^{11}41^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 211, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 7 & 0 & 21 & 9 \\ 2 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0} \\
[484] & 1^{13}31^{11}41^{23}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 263, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 19 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 19 & 14 & 0 & 77 & 17 \\ 2 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0} \\
[485] & 1^{14}31^{11}41^{23}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 292, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 22 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 22 & 17 & 0 & 108 & 20 \\ 2 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0} \\
[486] & 1^{13}31^{11}41^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 299, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 17 & 13 & 0 & 61 & 15 \\ 2 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0} \\
[487] & 1^{12}31^{11}41^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 308, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 14 & 10 & 0 & 38 & 12 \\ 2 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[488] $21^{13}5^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 328$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 14.21 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [489] $1^{13}31^15^{12}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -12$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 16.33 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[490] $1^{11}31^15^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -12$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[491] $1^{11}31^15^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -20$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[492] $1^{13}31^15^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -27$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 15.37 \rightarrow 16$, $T_{\max} = 23$, $T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[493] $1^{11}31^15^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -27$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[494] $1^{12}31^15^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -67$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.55 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[495] $1^{11}31^15^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -67$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[496] $1^{11}31^15^{12}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -112$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.57 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[497] $1^{11}31\ 5\ 131^{1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[498] $1^{12}31\ 5\ 131^{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -172, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 14.53 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[499] $1^{14}31\ 5\ 131^{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -172, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 6 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[500] $1^{13}31\ 5\ 131^{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.8$

$T_{\min} = 14.58 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[501] $1^{12}31\ 5\ 131^{1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -267, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[502] $1^{13}31\ 5\ 131^{1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -347, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.8$

$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[503] $1^{14}31\ 5\ 131^{1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -387, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

$T = 12$ [504] $1^{11}6\ 131^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 19.33 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 6 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & -3 \end{pmatrix}_{b_0}$$

[506] $1^{11}31\ 5\ 1321^{1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 8 & 2 & 14 & 8 \\ 2 & 0 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[505] $1^{11} 2^3 3^{21} 16131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -107, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 19.14 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$

$$\begin{pmatrix} 0 & 4 & 2 & 1 & 2 & 2 \\ 4 & 10 & 5 & 3 & 4 & 6 \\ 2 & 5 & 0 & 1 & 3 & 2 \\ 1 & 3 & 1 & -3 & 1 & -1 \\ 2 & 4 & 3 & 1 & 1 & 3 \\ 2 & 6 & 2 & -1 & 3 & -3 \end{pmatrix}_{b_0}$$

[509] $1^{1^2} 6^{131} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 3 & 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[507] $1^{1^{11}321} 6^{131} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 5 & 3 & 1 & 5 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 5 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[508] $1^{1^{131}} 6^{131} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[511] $1^{1^2} 6^{131} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 2 & 6 & 2 & 10 & 6 \\ 2 & 2 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[510] $1^{1^1} 6^{131} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[516] $1^{1^{11}321} 6^{131} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.48 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[512] $1^{1^{131}} 6^{131} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 16.37 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 4 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[513] $1^{1^{131}} 6^{131} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.48 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[514] $1^{1^{11}} 6^{131} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 15.48 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[515] \ 1^{11} \overset{1^{13}321}{6} \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16.37 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 4 & 2 & 3 & 1 & 9 & 5 \\ 2 & 2 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[517] \ 1^{11} \overset{1^{11}}{6} \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.29 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[518] \ 1^{11} \overset{1^{11}}{31} \ 5 \ 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 4 & 1 & 2 & 4 \\ 2 & 0 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[521] \ 1^{11} \overset{1^{11}321}{6} \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[522] \ 1^{11} \overset{1^{12}321}{6} \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[523] \ 1^{11} \overset{1^3}{6} \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 5 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[524] \ 31^{11} \overset{1^1}{31} \ 6 \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 17.22 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 1 & 3 \\ 3 & 4 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[520] \ 1^{11} \overset{1^{131}}{6} \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 2 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[519] \ 1^{11} \overset{1^{131}}{6} \ 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 3 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[525] $1^{12} 1^1 31^1 5 1321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 12 & 3 & 34 & 12 \\ 2 & 0 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[526] $1^{11} 1^1 6 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.81 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 6 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 6 & 6 & 13 & 5 & 56 & 14 \\ 2 & 2 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[527] $31^{11} 1^1 31^1 6 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 26, \Delta = -112, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 3 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 3 & 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[528] $1^{12} 1^1 31^1 5 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.84 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 4 & 1 & 2 & 4 \\ 2 & 0 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[529] $1^{11} 1^1 321^1 6 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 24, \Delta = -105, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.52 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 4 & 0 & 3 & 10 & 6 \\ 2 & 2 & 0 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[530] $1^{11} 1^1 6 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.37 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 3 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[531] $1^{12} 1^1 6 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.37 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[532] $1^{12} 1^1 31^1 5 1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 106, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.62 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 8 & 2 & 14 & 8 \\ 2 & 0 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[533] $1^{11} 1^1 6 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 123, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 5 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[534] $1^{11} 1^1 6 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 131, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 4 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[535] $1^{11} \overset{1^4}{6} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 131, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 7 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[537] $1^{11} \overset{1^3}{6} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 163, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 6 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[536] $1^{11} \overset{1^2}{6} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 163, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 5 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[538] $1^{13} 31 \overset{1^{13}}{5} 1321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 171, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 7 & 0 & 16 & 4 & 62 & 16 \\ 2 & 0 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[539] $1^{11} \overset{1^2}{6} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 171, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 6 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 7 & 6 & 13 & 5 & 56 & 14 \\ 2 & 2 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[540] $1^{13} 31 \overset{1^{13}}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 183, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 5 & 0 & 8 & 2 & 14 & 8 \\ 2 & 0 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[541] $1^{11} \overset{1^2}{6} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 187, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 6 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[542] $1^{11} \overset{1^{11} 321^1}{6} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.44 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 4 & 1 & 3 & 11 & 7 \\ 2 & 2 & 0 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[543] $31^{11} 31 \overset{1^{11}}{6} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 211, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 3 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[544] $1^{13} 1^{11} 31 \ 5 \ 1321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 211, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 14.64 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 6 & 0 & 12 & 3 & 34 & 12 \\ 2 & 0 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[545] $1^{13} 1^{11} 6 \ 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 219, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.39 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 7 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[546] $1^{14} 1^{11} 6 \ 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 219, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.39 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 8 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[547] $1^{13} 1^{11} 6 \ 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 235, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 6 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 8 & 6 & 13 & 5 & 56 & 14 \\ 2 & 2 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[549] $1^{11} 1^{12} 321^{11} 6 \ 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 252, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.44 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 6 & 4 & 2 & 3 & 14 & 8 \\ 2 & 2 & 0 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[548] $1^{11} 1^{14} 321^{11} 6 \ 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 252, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 8 & 4 & 4 & 3 & 26 & 10 \\ 2 & 2 & 0 & -1 & 10 & 1 \end{pmatrix}_{b_0}$$

[550] $1^{14} 1^{11} 6 \ 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 267, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.41 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 6 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 9 & 6 & 13 & 5 & 56 & 14 \\ 2 & 2 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[551] $1^{13} 1^{11} 321^{11} 6 \ 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 272, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.49 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 4 & 3 & 3 & 19 & 9 \\ 2 & 2 & 0 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

$T = 13$ [552] $1^{11} 1^{13} 1^{11} 31 \ 6 \ 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

[553] $1^{\overline{1}131} \overline{1}^1 6 \overline{131}^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 3 & 4 & 0 & 4 & 10 & 6 \\ 2 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[554] $1^{\overline{1}131} \overline{1}^1 6 \overline{131}^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[555] $1^{\overline{1}131} \overline{1}^1 6 \overline{131}^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 4 & 4 & 0 & 4 & 10 & 6 \\ 2 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[556] $1^{\overline{1}131} \overline{1}^{11} 6 \overline{131}^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.48 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 1 & 3 & 0 & 2 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[557] $1^{\overline{1}131} \overline{1}^{11} 31 \overline{1}^1 6 \overline{131}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -3 & 6 & 2 \end{pmatrix}_{b_0}$$

[558] $1^{\overline{1}131} \overline{1}^{13} 31 \overline{1}^1 6 \overline{131}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -62, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$T_{\min} = 15.74 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[559] $1^{\overline{1}131} \overline{1}^{11} 6 \overline{131}^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -67, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.43 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[560] $1^{\overline{1}131} \overline{1}^{12} 31 \overline{1}^1 6 \overline{131}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -68, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & -3 & 9 & 2 \end{pmatrix}_{b_0}$$

[561] $1^{\overline{1}131} \overline{1}^{11} 6 \overline{131}^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -75, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 16.81 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 2 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 18 & 0 \\ 13 & 19 & 0 & 18 & 225 & 27 \\ 2 & 3 & -1 & 0 & 27 & 1 \end{pmatrix}_{b_0}$$

[562] $1^{11} 6^{1^{131}} 131^{141^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -99, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 16.37 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 4 & 7 & 0 & 6 & 21 & 9 \\ 2 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[563] $1^{12} 31^{1^{131}} 6^{131^{12}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 15.76 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 2 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[564] $1^{11} 6^{1^{131}} 131^{341^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -123, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 14 & 0 \\ 10 & 15 & 0 & 14 & 133 & 21 \\ 2 & 3 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[565] $1^{11} 6^{1^{131}} 131^{241^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -131, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 10 & 0 \\ 7 & 11 & 0 & 10 & 65 & 15 \\ 2 & 3 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[566] $1^{11} 6^{1^{131}} 131^{141^{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -163, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 16.37 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 5 & 7 & 0 & 6 & 21 & 9 \\ 2 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[567] $1^{13} 31^{1^{131}} 6^{131^{12}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -176, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 15.70 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & -3 & 15 & 2 \end{pmatrix}_{b_0}$$

[568] $1^{14} 31^{1^{131}} 6^{131^{12}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 15.71 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & -3 & 18 & 2 \end{pmatrix}_{b_0}$$

[569] $1^{11} 6^{1^{131}} 131^{441^{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -211, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 16.52 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 2 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 18 & 0 \\ 14 & 19 & 0 & 18 & 225 & 27 \\ 2 & 3 & -1 & 0 & 27 & 1 \end{pmatrix}_{b_0}$$

[570] $1^{11} 6^{1^{131}} 131^{241^{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -219, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} =$

10.5

$$T_{\min} = 16.39 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 10 & 0 \\ 8 & 11 & 0 & 10 & 65 & 15 \\ 2 & 3 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[571] $1^{11} 1^{131} 6^{131} 131^3 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -235, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 16.43 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 14 & 0 \\ 11 & 15 & 0 & 14 & 133 & 21 \\ 2 & 3 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[572] $1^{1331} 1^{131} 6^{131} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 2 & -3 & 18 & 2 \end{pmatrix}_{b_0}$$

[573] $1^{1331} 1^{131} 6^{131} 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -248, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 15.98 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 13 & 3 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 15 & 0 & 97 & 21 \\ 2 & 3 & 2 & -3 & 21 & 2 \end{pmatrix}_{b_0}$$

[574] $1^{1431} 1^{131} 6^{131} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -332, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & -3 & 21 & 2 \end{pmatrix}_{b_0}$$

[575] $1^{1431} 1^{131} 6^{131} 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 15.81 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 2 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

$T = 14$ [576] $1^{11} 1^{15131} 6^{131} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 17.45 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -3 & -1 \\ 3 & 2 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

[577] $1^{11} 1^{15131} 6^{131} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 19, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 17.26 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 0 & 1 & 4 & 4 \\ 2 & 2 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

[578] $1^1 1^{11} 15^{131} 6^{131} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 27, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 17.48 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[579] $1^{11} 1^{11} 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 14, T_{\max} = 21, \Delta = -98, \det = 44, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 17.27 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 0 & 1 & 1 & 3 \\ 2 & 2 & 0 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

4.68 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{e}_6$

$T = 8$ [0] $1^{11} 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8, T_{\max} = 103, \Delta = -212, \det = 385, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.06 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & -1 & 4 & -4 & -3 \end{pmatrix}_{b_0}$$

$T = 10$ [5] $1^{11} 1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 10, T_{\max} = 102, \Delta = -157, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 102, T \in [17, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

4.69 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$

$T = 8$ [0] $1^{11} 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & 4 & -3 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 1^{11} 2^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & 8 & -3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & 12 & -3 & -2 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11} 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & -3 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9, T_{\max} = 101, \Delta = -158, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 \\ 5 & 12 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 8 & -3 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 101$, $\Delta = -158$, $\det = -232$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 14.34 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [8] $1^{11}2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 5$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 16.40 \rightarrow 17$, $T_{\max} = 93$, $T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[9] $1^{11}2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 13$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 16.15 \rightarrow 17$, $T_{\max} = 93$, $T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[10] $1^{11}3 221^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 91$, $\Delta = -129$, $\det = 127$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 14.46 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$

$$\begin{pmatrix} 5 & 15 & 21 & 1 & 5 \\ 15 & 42 & 59 & 0 & 12 \\ 21 & 59 & 79 & 0 & 17 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 12 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

4.70 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$T = 9$ [0] $1^{11}3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 24$, $\Delta = -113$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 24$, $\Delta = -113$, $\det = -210$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.05 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 24$, $\Delta = -113$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 24$, $\Delta = -113$, $\det = -298$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 24$, $\Delta = -113$, $\det = -312$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

13.0

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 19 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 19 & 0 & 73 & 21 & 12 & 15 \\ 3 & 5 & 0 & 21 & 4 & 4 & 4 \\ 2 & 4 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -4 & 15 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & -4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -507, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 2 & 3 \\ 3 & 4 & 0 & 14 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 15 & 9 & 11 \\ 3 & 5 & 0 & 15 & 4 & 4 & 4 \\ 2 & 4 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & -4 & 11 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -992, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 2 & 3 \\ 3 & 4 & 0 & 18 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 18 & 12 & 14 \\ 3 & 5 & 0 & 18 & 4 & 4 & 4 \\ 2 & 4 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -4 & 14 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{121} 31^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 & 2 & 3 \\ 3 & 4 & 0 & 28 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 16 & 28 & 0 & 154 & 30 & 18 & 22 \\ 3 & 5 & 0 & 30 & 4 & 4 & 4 \\ 2 & 4 & 0 & 18 & 4 & 0 & 2 \\ 3 & 4 & -4 & 22 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{121} 31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 2 & 3 \\ 3 & 4 & 0 & 23 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 13 & 23 & 0 & 100 & 24 & 15 & 18 \\ 3 & 5 & 0 & 24 & 4 & 4 & 4 \\ 2 & 4 & 0 & 15 & 4 & 0 & 2 \\ 3 & 4 & -4 & 18 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{131} 31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1892, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 & 2 & 3 \\ 3 & 4 & 0 & 27 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 15 & 27 & 0 & 133 & 27 & 18 & 21 \\ 3 & 5 & 0 & 27 & 4 & 4 & 4 \\ 2 & 4 & 0 & 18 & 4 & 0 & 2 \\ 3 & 4 & -4 & 21 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{131} 31^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -2307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 & 2 & 3 \\ 3 & 4 & 0 & 32 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 18 & 32 & 0 & 193 & 33 & 21 & 25 \\ 3 & 5 & 0 & 33 & 4 & 4 & 4 \\ 2 & 4 & 0 & 21 & 4 & 0 & 2 \\ 3 & 4 & -4 & 25 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{141} 31^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -3152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 & 2 & 3 \\ 3 & 4 & 0 & 36 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 20 & 36 & 0 & 238 & 36 & 24 & 28 \\ 3 & 5 & 0 & 36 & 4 & 4 & 4 \\ 2 & 4 & 0 & 24 & 4 & 0 & 2 \\ 3 & 4 & -4 & 28 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 1 & 9 \\ 27 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 9 & 11 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 31^{121} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 12 & 6 & 8 \\ 2 & 6 & 3 & 0 & 12 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 6 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 8 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 52 & 9 \\ 27 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 65 & 0 & 123 & 21 \\ 9 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 28 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 31^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 440, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 3 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.71 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \text{NHC}(-2)(-3) + \textcolor{brown}{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 6 & 3 & 4 & 4 \\ 3 & 4 & -4 & 4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 9 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 15 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 7 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.72 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.73 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 13 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 12 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

- [7] $1^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 318$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 8 & -2 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 379$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 419$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 430$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 8 & -2 & -1 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 472$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 493$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 12 & -2 & -1 \end{pmatrix}_{b_0}$$
- [13] $1^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 546$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 8 & -2 & -1 \end{pmatrix}_{b_0}$$
- [14] $1^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 738$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$
- [15] $1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 745$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 12 & -2 & -1 \end{pmatrix}_{b_0}$$
- [16] $1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 921$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 12 & -2 & -1 \end{pmatrix}_{b_0}$$
- [17] $1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1005$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 12 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{115} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1053, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.1$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -3 & 15 & 0 & -1 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 20 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1590, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1650, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [23] \ 1^1 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 33, T_{\max} = 99, T \in [33, 99]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & -1 & 0 & -20 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 16.11 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 \\ 6 & 12 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & -3 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$

- [27] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -81$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 \\ 5 & 12 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$
- [28] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$
- [29] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$
- [30] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -151$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [31] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -165$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.61 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -571$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$
- $T = 10$ [34] $1^{111}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 17.67 \rightarrow 18$, $T_{\max} = 22$, $T \in [18, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$
- [35] $1^{111}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 18$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.11 \rightarrow 17$, $T_{\max} = 22$, $T \in [17, 22]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$
- [36] $1^{111}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 15 & 13 & 1 & 5 \\ 15 & 42 & 36 & 0 & 12 \\ 13 & 36 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 12 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 16 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 6 & 16 & 0 & 8 & 22 & 10 \\ 3 & 6 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[39] $1^1231^131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & 0 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11}3^{11}221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 15 & 21 & 1 & 5 \\ 15 & 42 & 59 & 0 & 12 \\ 21 & 59 & 79 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 12 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{111}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 7 & 3 \\ 4 & 10 & 1 & 4 & 16 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 7 & 16 & 0 & 8 & 22 & 10 \\ 3 & 6 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{111}2^13^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{141}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 18 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{121}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 10 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 190, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{131}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 14 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{121}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.66 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{141}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{131}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{131}4^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{121}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 11 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 11 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{121}2^321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 12 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 12 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{121} 2^1 3^{12} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11} 1^{31} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{131} 2^1 3^{11} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 710, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 15 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 15 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{131} 2^1 3^{12} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 740, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{141} 2^1 3^{11} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 910, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 19 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 19 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{141} 2^1 3^{12} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 940, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{131} 2^1 3^{11} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 980, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 16 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 16 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{141} 2^1 3^{12} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 1420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 20 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 20 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [63] \quad 1^{111} 31^1 5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 12, \quad T_{\max} = 24, \quad \Delta = -106, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.73 \rightarrow 16, \quad T_{\max} = 24, \quad T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.74 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{112} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 93, \quad T \in [15, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 0 & 6 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & -1 & 0 & -2 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -114, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -372, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -432, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -4 & 7 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -608, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 4 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{2}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -618, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 2 \ 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{blue}{1}\textcolor{red}{1}} 4 \ 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -102, \ \det = -68, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 7 & 5 \\ 10 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 7 & 10 & 0 & 9 & 5 & 5 \\ 5 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

4.75 $\text{NHC}(-2)(-3) + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 372, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.76 $\text{NHC}(-2)(-3) + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$

$$T = 9 \quad [0] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

16.0

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

4.77 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 28 & 10 \\ 3 & 4 & -4 & 9 & 10 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 21.15 \rightarrow 22, T_{\max} = 92, T \in [22, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & 0 & 8 & -9 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 10 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 10 & 0 & 22 & 12 & 8 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -4 & 8 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -3 & 3 & 3 & -1 \\ 5 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{144} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 60 & 12 \\ 11 & 21 & 0 & 60 & 76 & 16 \\ 3 & 4 & -4 & 12 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{133} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 15 & 3 \\ 3 & 4 & 0 & 17 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 81 & 13 \\ 15 & 27 & 0 & 81 & 133 & 21 \\ 3 & 4 & -4 & 13 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{121} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{124} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 8 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 17 & 31 & 0 & 42 & 172 & 24 \\ 3 & 4 & -4 & 6 & 24 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{131} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{1111} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -3 & 3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[13] $21^1 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -3 & 3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{112} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 10 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 10 & 0 & 22 & 30 & 8 \\ 8 & 14 & 0 & 30 & 37 & 11 \\ 3 & 4 & -4 & 8 & 11 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{13} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -4 & 4 & 13 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 15.23 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

[20] $1^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -4 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 24 & 0 & 42 & 121 & 19 \\ 3 & 4 & -4 & 7 & 19 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{123}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{133}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -4 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{133}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -3 & 3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{124}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 8 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -4 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -4 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.48 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & 0 & 16 & -3 \end{pmatrix}_{b_0}$$

[35] $1^{112}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & -4 & 11 & 10 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{123}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -4 & 6 & 17 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{134}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 12 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 36 & 9 \\ 8 & 16 & 0 & 36 & 46 & 12 \\ 3 & 4 & -4 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{115} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 10 & 20 & 0 & 15 & 73 & 15 \\ 3 & 4 & -4 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -3 & 3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{122} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{113} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.28 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

[42] $1^{125} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 10 & 20 & 0 & 30 & 73 & 15 \\ 3 & 4 & -4 & 6 & 15 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{144} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 16 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 48 & 12 \\ 8 & 16 & 0 & 48 & 46 & 12 \\ 3 & 4 & -4 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{135} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 12 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 45 & 9 \\ 10 & 20 & 0 & 45 & 73 & 15 \\ 3 & 4 & -4 & 9 & 15 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{1111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -3 & 3 & 3 & -1 \\ 4 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{116} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 12 & 24 & 0 & 18 & 106 & 18 \\ 3 & 4 & -4 & 3 & 18 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{113} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{126} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 12 & 24 & 0 & 36 & 106 & 18 \\ 3 & 4 & -4 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{112} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -4 & 3 & 22 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{145} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 16 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 60 & 12 \\ 10 & 20 & 0 & 60 & 73 & 15 \\ 3 & 4 & -4 & 12 & 15 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.81 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$

[53] $1^{136} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 12 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 54 & 9 \\ 12 & 24 & 0 & 54 & 106 & 18 \\ 3 & 4 & -4 & 9 & 18 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{155} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 20 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 0 & 73 & 75 & 15 \\ 10 & 20 & 0 & 75 & 73 & 15 \\ 3 & 4 & -4 & 15 & 15 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{117} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 14 & 28 & 0 & 21 & 145 & 21 \\ 3 & 4 & -4 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{146} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 16 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 72 & 12 \\ 12 & 24 & 0 & 72 & 106 & 18 \\ 3 & 4 & -4 & 12 & 18 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{127} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 14 & 28 & 0 & 42 & 145 & 21 \\ 3 & 4 & -4 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{132} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 54 & 13 \\ 10 & 18 & 0 & 54 & 58 & 14 \\ 3 & 4 & -4 & 13 & 14 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{134} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 329, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & -4 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{111} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 11 & 19 & 0 & 21 & 73 & 15 \\ 3 & 4 & -4 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{111} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 19 & 0 & 33 & 73 & 15 \\ 3 & 4 & -4 & 7 & 15 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{137} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 12 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 63 & 9 \\ 14 & 28 & 0 & 63 & 145 & 21 \\ 3 & 4 & -4 & 9 & 21 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{156} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 20 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 0 & 73 & 90 & 15 \\ 12 & 24 & 0 & 90 & 106 & 18 \\ 3 & 4 & -4 & 15 & 18 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{122} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 54 & 10 \\ 13 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -4 & 10 & 18 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{147} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 16 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 84 & 12 \\ 14 & 28 & 0 & 84 & 145 & 21 \\ 3 & 4 & -4 & 12 & 21 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{113} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 14 & 24 & 0 & 27 & 121 & 19 \\ 3 & 4 & -4 & 4 & 19 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{143} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & -4 & 16 & 17 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{166} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 24 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 24 & 0 & 106 & 108 & 18 \\ 12 & 24 & 0 & 108 & 106 & 18 \\ 3 & 4 & -4 & 18 & 18 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{113} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 57 & 11 \\ 12 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -4 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{157} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 20 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 0 & 73 & 105 & 15 \\ 14 & 28 & 0 & 105 & 145 & 21 \\ 3 & 4 & -4 & 15 & 21 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 405, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 24 & 7 \\ 8 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -4 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 422, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{167} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 24 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 24 & 0 & 106 & 126 & 18 \\ 14 & 28 & 0 & 126 & 145 & 21 \\ 3 & 4 & -4 & 18 & 21 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{14} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 466, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -4 & 4 & 20 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{13} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -4 & 4 & 17 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{177} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 28 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 14 & 28 & 0 & 145 & 147 & 21 \\ 14 & 28 & 0 & 147 & 145 & 21 \\ 3 & 4 & -4 & 21 & 21 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{124} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -4 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{124} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{111} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 39 & 11 \\ 8 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 13 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -4 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{112} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 48 & 11 \\ 10 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -4 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 657, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{123} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 697, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 18 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 75 & 10 \\ 18 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -4 & 10 & 25 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{122} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 702, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{112}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 734, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{12}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 790, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 16 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -4 & 4 & 22 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{124}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 792, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{13}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 819, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 15 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -4 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 837, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

[105] $21^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 884, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 6 & 4 \\ 8 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -3 & 3 & 3 & -1 \\ 6 & 8 & 3 & 4 & 6 & 4 \\ 6 & 8 & 3 & 6 & 4 & 4 \\ 4 & 5 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{112}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 891, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 13 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -4 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{133}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -4 & 13 & 17 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{112}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 930, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 16 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -4 & 7 & 22 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{14}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 17 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -4 & 4 & 24 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{133}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 964, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{134}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 979, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 17 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 93 & 13 \\ 17 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -4 & 13 & 24 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{113}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{124}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 84 & 14 \\ 14 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -4 & 14 & 20 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{144}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1074, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -4 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

[116] $1^{134}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1083, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -4 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

[117] $1^{123}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$

12.3

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 15 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 15 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -4 & 10 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{122} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 60 & 14 \\ 10 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{123} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -4 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{13} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 18 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -4 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{114} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{123} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 72 & 14 \\ 12 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -4 & 14 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{144} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{113} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 15 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{14} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 20 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -4 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{134} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -4 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{124} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{113} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 18 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -4 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{124} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1674, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 20 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 20 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -4 & 10 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{133} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 85 & 87 & 17 \\ 12 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -4 & 17 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{114} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1755, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 17 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{124} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 17 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 17 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -4 & 10 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{134} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1816, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$$

12.8

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 85 & 102 & 17 \\ 14 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -4 & 17 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{114} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 2246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 20 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{144} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 2296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 0 & 118 & 120 & 20 \\ 14 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -4 & 20 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [138] \ 1^{111} 3^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19.50 \rightarrow 20, T_{\max} = 23, T \in [20, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 \\ 2 & 3 & 3 & -4 & 3 & -7 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{111} 3^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 20.45 \rightarrow 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 & 3 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 3 & 2 & -4 & 7 & 3 & -8 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{111} 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{111} 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 0 & -1 \\ 2 & 1 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{11} 3^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{133} 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 6 & 0 & 3 & 16 & 21 & 6 \\ 4 & 7 & 0 & 6 & 21 & 25 & 9 \\ 2 & 2 & -1 & -1 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

- [144] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.52 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$
- [145] $1^{122} 31^{113} 31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 8 & 2 \\ 2 & 4 & 4 & 0 & 12 & 24 & 4 \\ 1 & 4 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 54 & 10 \\ 8 & 24 & 15 & 0 & 54 & 100 & 18 \\ 2 & 4 & 2 & -4 & 10 & 18 & -2 \end{pmatrix}_{b_0}$$
- [146] $1^{114} 31^{123} 31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 18 & 3 \\ 8 & 24 & 18 & 0 & 18 & 118 & 20 \\ 2 & 4 & 2 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$
- [147] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -39$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$
- [148] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$
- [149] $1^{111} 3 21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 2 & 3 & 0 & 4 & 5 & 3 & 5 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$
- [150] $1^{121} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$
- [151] $1^{124} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$
- [152] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$
- [153] $1^{122} 3 21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 3 & 5 & 0 & 5 & 10 & 13 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 7 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 41 & 11 \\ 2 & 2 & -1 & -1 & 0 & 11 & 0 \end{pmatrix}_{b_0}$$

[155] $1^{112} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 4 & 2 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[156] $1^{13} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[157] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[158] $1^{112} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 4 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[160] $1^{122} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[161] $1^{111} 31^{14} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 9 & 27 & 15 & 0 & 15 & 121 & 19 \\ 2 & 4 & 2 & -4 & 3 & 19 & -2 \end{pmatrix}_{b_0}$$

[162] $1^{113} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$T_{\min} = 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 6 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 30 & 4 \\ 10 & 30 & 21 & 0 & 30 & 172 & 24 \\ 2 & 4 & 2 & -4 & 4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 2 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{113} 31^{11} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 12 & 3 \\ 5 & 12 & 15 & 0 & 12 & 49 & 13 \\ 2 & 3 & 3 & -4 & 3 & 13 & -2 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{134} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 5 & 5 \\ 2 & 2 & -1 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{144} 3 \ 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 7 & -1 \\ 5 & 8 & 0 & 4 & 30 & 36 & 8 \\ 6 & 9 & 0 & 7 & 36 & 41 & 11 \\ 2 & 2 & -1 & -1 & 8 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{115} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{134} 3 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 6 & 0 & 3 & 16 & 27 & 6 \\ 5 & 9 & 0 & 7 & 27 & 41 & 11 \\ 2 & 2 & -1 & -1 & 6 & 11 & 0 \end{pmatrix}_{b_0}$$

[180] $1^{125} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -142$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.6$
 $T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[181] $1^{12} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[182] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[183] $1^{144} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 4 & -1 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 8 & -1 \end{pmatrix}_{b_0}$$

[184] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -149$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[185] $1^{123} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[186] $1^{135} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 13.87 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 5 & -1 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -1 & 6 & 10 & -1 \end{pmatrix}_{b_0}$$

[187] $1^{116} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -158$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[188] $1^{13} 3^{21} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 6 & 0 & 25 & 9 \\ 2 & 2 & -1 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

[189] $1^{11} 1^{11} 3^{13} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[190] $1^{126} 3^{21} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

[191] $1^{122} 3^{21} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[192] $1^{145} 3^{21} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.1$

$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 5 & -1 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -1 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

[193] $1^{123} 3^{21} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$

$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[194] $1^{123} 3^{21} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[195] $1^{136} 3^{21} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.1$

$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -1 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

[196] $1^{111} 3^{11} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{violet}{\mathfrak{e}_6} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 9 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

[197] $1^{12} 1^{11} 3 \ 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 5 & 0 & 13 & 7 \\ 2 & 2 & -1 & -1 & 0 & 7 & 0 \end{pmatrix}_{b_0}$$

[198] $1^{122} 1^{11} 3 \ 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 3 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 5 & 10 & 13 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[199] $1^{117} 1^1 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -1 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

[200] $1^{111} 31^{11} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -185, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[201] $1^{133} 1^{11} 3 \ 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.6$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 4 & 6 & 0 & 3 & 16 & 21 & 6 \\ 5 & 7 & 0 & 6 & 21 & 25 & 9 \\ 2 & 2 & -1 & -1 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[202] $1^{127} 1^1 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -1 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

[205] $1^{11} 31^{112} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 18 & 7 \\ 4 & 12 & 6 & 0 & 18 & 22 & 8 \\ 2 & 4 & 2 & -4 & 7 & 8 & -1 \end{pmatrix}_{b_0}$$

[203] $1^{124} 1^{12} 3 \ 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & -1 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

- [204] $1^{137}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -1 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$
- [206] $1^{112}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [207] $1^{112}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -233$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 9 & 12 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$
- [208] $1^{134}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -239$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & 0 \end{pmatrix}_{b_0}$$
- [209] $1^{113}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.52 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$
- [210] $1^{111}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -257$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.20 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 12 & 3 \\ 7 & 21 & 12 & 0 & 12 & 73 & 15 \\ 2 & 4 & 2 & -4 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$
- [211] $1^{112}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -263$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [212] $1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$
- [213] $1^{124}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & -1 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[214] $1^{123} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$

$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[215] $1^{111} 31^{123} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 5 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & -4 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[216] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -305, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[217] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 7 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[218] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[219] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[220] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[221] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[222] $1^{112} 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[223] $1^{114} 1^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[224] $1^1 31^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[225] $1^{113} 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[226] $1^{124} 1^{12} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[227] $1^{113} 31^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 7 & 21 & 15 & 0 & 15 & 85 & 17 \\ 2 & 4 & 2 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

[228] $1^{113} 1^{11} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[229] $1^{11} 31^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -389, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[230] $1^{124} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & -1 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[231] $1^{114} 3^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 8 & 2 \\ 2 & 4 & 4 & 0 & 9 & 24 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 42 & 7 \\ 8 & 24 & 18 & 0 & 42 & 118 & 20 \\ 2 & 4 & 2 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

[232] $1^{124} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[233] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -409$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 6 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[234] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -414$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[235] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[236] $1^{123} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -423$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.3$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[237] $1^{121} 3^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -437$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 12 & 9 & 4 \\ 1 & 4 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 21 & 10 \\ 3 & 9 & 6 & 0 & 21 & 13 & 7 \\ 2 & 4 & 2 & -4 & 10 & 7 & 0 \end{pmatrix}_{b_0}$$

[238] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -457$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 2 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{122} 31^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 30 & 10 \\ 4 & 12 & 9 & 0 & 30 & 28 & 10 \\ 2 & 4 & 2 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{112} 31^{123} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 12 & 3 \\ 6 & 18 & 12 & 0 & 12 & 58 & 14 \\ 2 & 4 & 2 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{134} 3^{111} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -489, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 4 & 6 & 0 & 3 & 16 & 27 & 6 \\ 6 & 9 & 0 & 7 & 27 & 41 & 11 \\ 2 & 2 & -1 & -1 & 6 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{114} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -503, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{122} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{113} 3^{111} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -543, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{113} 3^{112} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[246] \ 1^1 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 12 & 6 & 0 & 12 & 22 & 8 \\ 2 & 4 & 2 & -4 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[247] $1^{123} 1^{111} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -591, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$

$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[248] $1^{124} 1^{111} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & -1 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[249] $1^{124} 1^{111} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[250] $1^{113} 1^{14} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[251] $1^{113} 1^{13} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -653, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[252] $1^{12} 31^{112} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -668, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 6 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 18 & 4 \\ 6 & 18 & 12 & 0 & 18 & 58 & 14 \\ 2 & 4 & 2 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

[253] $1^{111} 31^{111} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -680, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[254] $1^{11} 31^{122} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 15 & 4 \\ 1 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 30 & 8 \\ 5 & 15 & 9 & 0 & 30 & 37 & 11 \\ 2 & 4 & 2 & -4 & 8 & 11 & 0 \end{pmatrix}_{b_0}$$

[255] $1^{122}31^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 42 & 10 \\ 6 & 18 & 12 & 0 & 42 & 58 & 14 \\ 2 & 4 & 2 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

[256] $1^{112}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -725, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 8 & 24 & 15 & 0 & 15 & 100 & 18 \\ 2 & 4 & 2 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

[257] $1^{113}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[258] $1^{133}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.6$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[259] $1^{114}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -761, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[260] $1^{114}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[261] $1^131^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 24 & 8 \\ 4 & 12 & 6 & 0 & 24 & 22 & 8 \\ 2 & 4 & 2 & -4 & 8 & 8 & 0 \end{pmatrix}_{b_0}$$

[262] $1^{134}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -831, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & 0 \end{pmatrix}_{b_0}$$

[263] $1^{113}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -839, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{113} 3^{14} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 8 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[265] \ 1^{112} 31^{14} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -860, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 18 & 3 \\ 10 & 30 & 18 & 0 & 18 & 154 & 22 \\ 2 & 4 & 2 & -4 & 3 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{114} 31^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -869, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 21 & 3 \\ 10 & 30 & 21 & 0 & 21 & 172 & 24 \\ 2 & 4 & 2 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{11} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -893, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 6 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 15 & 4 \\ 5 & 15 & 9 & 0 & 15 & 37 & 11 \\ 2 & 4 & 2 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{113} 31^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -929, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 18 & 3 \\ 9 & 27 & 18 & 0 & 18 & 133 & 21 \\ 2 & 4 & 2 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[269] \ 1^1 31^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 6 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 18 & 4 \\ 6 & 18 & 9 & 0 & 18 & 52 & 12 \\ 2 & 4 & 2 & -4 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{144} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -988, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 4 & -1 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{114} 3^{113} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -991, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{111}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1049, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 15 & 4 \\ 1 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 24 & 7 \\ 5 & 15 & 9 & 0 & 24 & 37 & 11 \\ 2 & 4 & 2 & -4 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{114}3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -1064, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.45 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[274] \ 1^{114}3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -1094, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.56 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{13}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1133, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 18 & 0 & 27 & 133 & 21 \\ 2 & 4 & 2 & -4 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{113}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1145, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 7 & 2 \\ 2 & 4 & 4 & 0 & 9 & 21 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 36 & 7 \\ 7 & 21 & 15 & 0 & 36 & 85 & 17 \\ 2 & 4 & 2 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{114}3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -1273, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{111}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 7 & 2 \\ 2 & 4 & 4 & 0 & 9 & 21 & 4 \\ 1 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 33 & 7 \\ 7 & 21 & 12 & 0 & 33 & 73 & 15 \\ 2 & 4 & 2 & -4 & 7 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{112}31^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1328, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 48 & 11 \\ 6 & 18 & 12 & 0 & 48 & 58 & 14 \\ 2 & 4 & 2 & -4 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{113}31^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1352, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 33 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 21 & 3 \\ 11 & 33 & 21 & 0 & 21 & 193 & 25 \\ 2 & 4 & 2 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

[281] $1^{112}31^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 6 & 2 \\ 2 & 4 & 4 & 0 & 9 & 18 & 4 \\ 1 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 30 & 7 \\ 6 & 18 & 12 & 0 & 30 & 58 & 14 \\ 2 & 4 & 2 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

[282] $1^{114}3^{114}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -1388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[283] $1^{111}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 9 & 2 \\ 2 & 4 & 4 & 0 & 9 & 27 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 42 & 7 \\ 9 & 27 & 15 & 0 & 42 & 121 & 19 \\ 2 & 4 & 2 & -4 & 7 & 19 & 0 \end{pmatrix}_{b_0}$$

[284] $1^131^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 6 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 24 & 4 \\ 8 & 24 & 12 & 0 & 24 & 94 & 16 \\ 2 & 4 & 2 & -4 & 4 & 16 & 0 \end{pmatrix}_{b_0}$$

[285] $1^{113}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1541, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 6 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 21 & 4 \\ 7 & 21 & 12 & 0 & 21 & 73 & 15 \\ 2 & 4 & 2 & -4 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

[286] $1^{114}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1580, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 36 & 4 \\ 1 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 24 & 3 \\ 12 & 36 & 24 & 0 & 24 & 238 & 28 \\ 2 & 4 & 2 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[287] $1^{12}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 6 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 24 & 4 \\ 8 & 24 & 15 & 0 & 24 & 100 & 18 \\ 2 & 4 & 2 & -4 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

[288] $1^{111}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1688, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 39 & 11 \\ 5 & 15 & 9 & 0 & 39 & 37 & 11 \\ 2 & 4 & 2 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

[289] $1^{14}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 12 & 2 \\ 2 & 4 & 4 & 0 & 6 & 36 & 4 \\ 1 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 36 & 4 \\ 12 & 36 & 24 & 0 & 36 & 238 & 28 \\ 2 & 4 & 2 & -4 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{122}31^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2048, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 12 & 0 & 58 & 60 & 14 \\ 6 & 18 & 12 & 0 & 60 & 58 & 14 \\ 2 & 4 & 2 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{112}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2165, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 8 & 2 \\ 2 & 4 & 4 & 0 & 9 & 24 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 39 & 7 \\ 8 & 24 & 15 & 0 & 39 & 100 & 18 \\ 2 & 4 & 2 & -4 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{114}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 10 & 2 \\ 2 & 4 & 4 & 0 & 9 & 30 & 4 \\ 1 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 51 & 7 \\ 10 & 30 & 21 & 0 & 51 & 172 & 24 \\ 2 & 4 & 2 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[293] \quad 1^{113}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 15 & 0 & 27 & 121 & 19 \\ 2 & 4 & 2 & -4 & 4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[294] \quad 1^{133}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 11 & 2 \\ 2 & 4 & 4 & 0 & 6 & 33 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 33 & 4 \\ 11 & 33 & 21 & 0 & 33 & 193 & 25 \\ 2 & 4 & 2 & -4 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[295] \quad 1^{113}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 9 & 2 \\ 2 & 4 & 4 & 0 & 9 & 27 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 45 & 7 \\ 9 & 27 & 18 & 0 & 45 & 133 & 21 \\ 2 & 4 & 2 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[296] \quad 1^{123}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 6 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 30 & 4 \\ 10 & 30 & 18 & 0 & 30 & 154 & 22 \\ 2 & 4 & 2 & -4 & 4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[297] \quad 1^{112}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2840, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 10 & 2 \\ 2 & 4 & 4 & 0 & 9 & 30 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 48 & 7 \\ 10 & 30 & 18 & 0 & 48 & 154 & 22 \\ 2 & 4 & 2 & -4 & 7 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{113}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -3737, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 11 & 2 \\ 2 & 4 & 4 & 0 & 9 & 33 & 4 \\ 1 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 54 & 7 \\ 11 & 33 & 21 & 0 & 54 & 193 & 25 \\ 2 & 4 & 2 & -4 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[299] \ 1^{11}431^{11}431^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -4100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 12 & 2 \\ 2 & 4 & 4 & 0 & 9 & 36 & 4 \\ 1 & 4 & 0 & 0 & 6 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 60 & 7 \\ 12 & 36 & 24 & 0 & 60 & 238 & 28 \\ 2 & 4 & 2 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [300] \ 1^{111}4 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19.60 \rightarrow 20, T_{\max} = 22, T \in [20, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 10 & 3 \\ 11 & 15 & 0 & 10 & 45 & 13 \\ 3 & 4 & -3 & 3 & 13 & -5 \end{pmatrix}_{b_0}$$

$$[301] \ 1^{113}4 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 23 & 7 \\ 9 & 15 & 0 & 23 & 40 & 12 \\ 3 & 4 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[302] \ 1^{11}4 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 5 & 7 & 0 & 6 & 8 & 6 \\ 3 & 4 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[303] \ 1^{12}4 \ 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 7 & 3 \\ 7 & 11 & 0 & 7 & 21 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[304] \ 1^{11}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 3 & 0 \\ 4 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[305] \ 1^{11}4 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 & 3 \\ 4 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[307] \ 1^{11}4 \ 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 5 & 3 \\ 4 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 5 & 7 & 0 & 6 & 8 & 6 \\ 3 & 4 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{112} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 7 \\ 7 & 11 & 0 & 17 & 21 & 9 \\ 3 & 4 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{11} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[309] $1^{11} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.10 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[312] $1^{11} 231^{11} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[311] $1^1 231^{11} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{11} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 6 & 3 \\ 6 & 8 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{121} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[314] $1^{11} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[315] $1^{11} 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 9 & 0 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[316] $1^{11} 1^{31} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[317] $1^{11} 1^{31} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 0 & 13 & 11 & 7 \\ 5 & 7 & 0 & 11 & 8 & 6 \\ 3 & 4 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[318] $1^{122} 1^{31} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 7 & 11 & 0 & 16 & 21 & 9 \\ 3 & 4 & -3 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{111} 4^{131} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{112} 4^{131} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{122} 4^{131} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[322] $1^{113} 4^{131} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[323] $1^{11} 1^{31} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[324] $1^{111} 4^{131} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

11.0

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{114} 4^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -3 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{112} 4^{11} 31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 16 & 22 & 0 & 15 & 97 & 19 \\ 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{11} 4^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{11} 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 7 & 0 \\ 2 & 6 & 0 & 3 & 4 & 9 & 4 \\ 4 & 9 & 0 & 7 & 9 & 14 & 8 \\ 2 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[329] \ 1^1 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[330] \ 1^{111} 4^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[331] \ 1^{11} 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[332] \ 1^{114} 4^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 7 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 25 & 6 \\ 11 & 19 & 0 & 25 & 65 & 15 \\ 3 & 4 & -3 & 6 & 15 & 1 \end{pmatrix}_{b_0}$$

- [333] $1^{122}1^{11}4^{131^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$
- [334] $1^{11}1^{131^{11}}4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [335] $1^{111}1^{131^1}4^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 194$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [336] $1^{121}1^{131^{11}}4^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 194$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 0 & 21 & 24 & 9 \\ 8 & 12 & 0 & 24 & 26 & 10 \\ 3 & 4 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$
- [337] $1^{11}231^{111}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 199$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 7 & 0 \\ 3 & 6 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 7 & 9 & 14 & 8 \\ 2 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$
- [338] $1^{112}1^{131^2}4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 202$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 10 & 3 \\ 10 & 14 & 0 & 10 & 38 & 12 \\ 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [339] $1^{11}1^{131^{11}}4^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.16 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 8 & 3 \\ 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [340] $1^{1231^{12}}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 216$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [341] $1^{112}1^{11}4^{131^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 263$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

[342] $1^{\overline{1}}231^{\overline{12}}31^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[343] $1^{\overline{112}}1^{\overline{1}}31^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 265, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[344] $1^{\overline{111}}1^{\overline{1}}31^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 268, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 10 & 6 \\ 5 & 7 & 0 & 10 & 8 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[345] $1^{\overline{1}}231^{\overline{111}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[346] $1^{\overline{1}}231^{\overline{111}}31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 301, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[347] $1^{\overline{113}}1^{\overline{1}}31^{\overline{14}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 21 & 29 & 0 & 20 & 169 & 25 \\ 3 & 4 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

[349] $1^{\overline{113}}1^{\overline{1}}31^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 20 & 6 \\ 9 & 15 & 0 & 20 & 40 & 12 \\ 3 & 4 & -3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

[348] $1^{\overline{111}}1^{\overline{1}}31^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 16 & 6 \\ 8 & 12 & 0 & 16 & 26 & 10 \\ 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{\overline{114}}1^{\overline{1}}31^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 323, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 31 & 6 \\ 14 & 24 & 0 & 31 & 101 & 19 \\ 3 & 4 & -3 & 6 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[351] \ 1^1 231^{111} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[352] \ 1^1 231^{111} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[353] \ 1^1 231^{123} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 346, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[354] \ 1^{113} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 347, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 15 & 21 & 0 & 15 & 88 & 18 \\ 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[355] \ 1^{112} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 15 & 6 \\ 7 & 11 & 0 & 15 & 21 & 9 \\ 3 & 4 & -3 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[356] \ 1^{11} 231^{123} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 15 & 0 & 10 & 0 & 38 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{113} 4^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{11} 231^{123} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[359] $1^{112} 1^{131^2} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 407, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 13 & 3 \\ 13 & 19 & 0 & 13 & 68 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[360] $1^{113} 1^{131^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 419, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 9 & 15 & 0 & 11 & 40 & 12 \\ 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[361] $1^{112} 1^{131^{11}} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 443, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 21 & 6 \\ 10 & 16 & 0 & 21 & 45 & 13 \\ 3 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[362] $1^{113} 1^{131^{11}} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 26 & 6 \\ 12 & 20 & 0 & 26 & 70 & 16 \\ 3 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[363] $1^{112} 1^{131^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 452, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 10 & 16 & 0 & 11 & 45 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[364] $1^{11} 231^{12} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 15 & 0 & 10 & 0 & 38 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[365] $1^{11} 231^{13} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 13 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 21 & 0 & 13 & 0 & 74 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[366] $1^{113} 1^{131^2} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 13 & 3 \\ 12 & 18 & 0 & 13 & 61 & 15 \\ 3 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[367] $1^{114} 1^{11} 4^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 503, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[368] $1^{1231^{111}31^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 524$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[369] $1^{122^{1131^{11}}4^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 533$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 0 & 21 & 32 & 9 \\ 10 & 16 & 0 & 32 & 45 & 13 \\ 3 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[370] $1^{1231^{13}31^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 549$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[371] $1^{114^{1131^4}4^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 550$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 20 & 28 & 0 & 20 & 158 & 24 \\ 3 & 4 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

[372] $1^{122^{1131^{11}}4^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 568$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 11 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 0 & 21 & 23 & 9 \\ 7 & 11 & 0 & 23 & 21 & 9 \\ 3 & 4 & -3 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[373] $1^{114^{1131^1}4^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 577$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 11 & 19 & 0 & 14 & 65 & 15 \\ 3 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[374] $1^{1231^{13}31^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 621$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[375] $1^{113^{1131^3}4^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 678$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 18 & 26 & 0 & 18 & 130 & 22 \\ 3 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

[376] $1^{1231^{13}31^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 691$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[377] 1^{113} 4^{131^1} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 714, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 20 & 0 & 14 & 70 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[378] 1^{114} 4^{131^3} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 793, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 17 & 25 & 0 & 18 & 121 & 21 \\ 3 & 4 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[379] 1^{114} 4^{131^2} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 802, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 16 & 3 \\ 14 & 22 & 0 & 16 & 90 & 18 \\ 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[380] 1^{113} 4^{131^2} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 813, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 16 & 3 \\ 15 & 23 & 0 & 16 & 97 & 19 \\ 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[381] 1^{11} 231^{13} 31^{113} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 829, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 13 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 21 & 0 & 13 & 0 & 74 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[382] 1^{11} 231^{13} 31^{114} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 951, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 13 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 21 & 0 & 13 & 0 & 74 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[383] 1^{114} 4^{131^1} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 980, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 14 & 24 & 0 & 17 & 101 & 19 \\ 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[384] 1^{114} 4^{131^4} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 1007, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 33 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 23 & 33 & 0 & 23 & 212 & 28 \\ 3 & 4 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[385] \ 1^{11}231^{14}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 1020, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{11}231^{14}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 1074, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 0 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 16 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 27 & 0 & 16 & 0 & 122 & 20 \\ 2 & 4 & -1 & 0 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{11}231^{14}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 1174, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.89 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{114} \ 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 1223, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 19 & 3 \\ 17 & 27 & 0 & 19 & 132 & 22 \\ 3 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[389] \ 1^{114} \ 4 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 1232, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 20 & 30 & 0 & 21 & 169 & 25 \\ 3 & 4 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[390] \ 1^{11}231^{14}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 1420, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 13 & 2 \\ 2 & 4 & 1 & 3 & 0 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 16 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 13 & 27 & 0 & 16 & 0 & 122 & 20 \\ 2 & 4 & -1 & 0 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [391] \ 1^{11} \ 5 \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 18, \ \Delta = -98, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.25 \rightarrow 15, \ T_{\max} = 18, \ T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 3 & 0 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[392] \ 1^{11} \ 5 \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 18, \ \Delta = -98, \ \det = -31, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 18, \ T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[393] \ 1^{11} \ 5 \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 18, \ \Delta = -98, \ \det = -39, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 18, \ T \in [15, 18]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0} \\
[394] \quad & 1^{11} 1^3 31^{11} 41^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 7 & 3 \\ 5 & 11 & 7 & 0 & 7 & 21 & 9 \\ 2 & 4 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0} \\
[395] \quad & 1^{11} 1^3 31^{11} 4131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 4 & 2 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0} \\
[396] \quad & 1^{11} 1^3 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 14.19 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0} \\
[397] \quad & 1^{12} 1^3 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 14.19 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[398] \quad & 1^{12} 1^3 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 14.32 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[399] \quad & 1^{13} 1^3 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -109, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 14.56 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 6 & 0 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
[400] \quad & 1^{12} 1^3 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 14.22 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[401] \quad & 1^{11} 1^3 31^{11} 4131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -139, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0} \\
[402] \quad & 1^{13} 1^3 31^{11} 41^2 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -171, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 11, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 19 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 14 & 3 \\ 9 & 19 & 14 & 0 & 14 & 77 & 17 \\ 2 & 4 & 2 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[403] \ 1^{13} \overset{1}{5} \overset{12}{131}^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -199, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 6 & 0 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[404] \ 1^{13} \overset{1}{31} \overset{11}{41} \overset{1}{31}^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -216, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 13 & 3 \\ 7 & 17 & 13 & 0 & 13 & 61 & 15 \\ 2 & 4 & 2 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[405] \ 1^{13} \overset{1}{5} \overset{13}{131}^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -221, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 6 & 0 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[406] \ 1^{14} \overset{1}{31} \overset{11}{41} \overset{2}{31}^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -229, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 22 & 4 \\ 1 & 4 & 0 & 0 & 3 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 17 & 3 \\ 10 & 22 & 17 & 0 & 17 & 108 & 20 \\ 2 & 4 & 2 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[407] \ 1^{12} \overset{1}{31} \overset{11}{41} \overset{1}{31}^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -234, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 10 & 3 \\ 6 & 14 & 10 & 0 & 10 & 38 & 12 \\ 2 & 4 & 2 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

4.78 $\text{NHC}(-2)(-3) + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11} \overset{1}{31} \overset{1111}{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} \overset{1}{31} \overset{1111}{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 6 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 6 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} \overset{1}{31} \overset{1111}{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 9 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 6 & 4 & 0 & 2 \\ 3 & 4 & -4 & 7 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.79 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{2}} 3 \quad 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 1 & 3 \\ 4 & 9 & 4 & 0 & 16 & 0 & 5 \\ 2 & 4 & -3 & 0 & 6 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 16 & 6 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 0 & 8 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \quad 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}} 3 \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \quad 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{3}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -4 & 7 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \quad 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \quad 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] 1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] 1^{11} 3 1^{11} 4 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 7 & 5 \\ 10 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 7 & 10 & 0 & 9 & 5 & 5 \\ 5 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

4.80 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 10 \quad [0] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.81 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 18.25 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -4 & 3 & 3 & 18 & -6 \end{pmatrix}_{b_0}$$

$$[2] 1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 15 & 7 \\ 5 & 9 & 0 & 12 & 15 & 13 & 7 \\ 3 & 4 & -4 & 6 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{131} 3 1^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 0 & 5 & 17 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 9 & 4 \\ 9 & 17 & 0 & 15 & 49 & 27 & 13 \\ 5 & 9 & 0 & 9 & 27 & 13 & 7 \\ 3 & 4 & -4 & 4 & 13 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 4 & 8 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 10 & 30 & 6 \\ 12 & 22 & 0 & 15 & 30 & 85 & 17 \\ 3 & 4 & -4 & 3 & 6 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1211} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 5 & 9 & 0 & 12 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -4 & 6 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 17.66 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -4 & 4 & 3 & 22 & -5 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -4 & 4 & 3 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1113} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 12 & 3 \\ 3 & 4 & 0 & 4 & 14 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 8 & 14 & 0 & 9 & 37 & 57 & 11 \\ 12 & 22 & 0 & 15 & 57 & 85 & 17 \\ 3 & 4 & -4 & 3 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1222} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 4 & 8 & 0 & 12 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 18 & 28 & 10 \\ 3 & 4 & -4 & 6 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 6 & 4 & 4 \\ 3 & 4 & -4 & 4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.47 \rightarrow 15, \quad T_{\max} = 21, \quad T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1214} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -150, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 11 & 3 \\ 3 & 4 & 0 & 8 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 30 & 6 \\ 5 & 9 & 0 & 12 & 13 & 33 & 7 \\ 11 & 21 & 0 & 30 & 33 & 76 & 16 \\ 3 & 4 & -4 & 6 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -170, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1213} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -194, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 12 & 3 \\ 3 & 4 & 0 & 8 & 9 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 30 & 6 \\ 5 & 9 & 0 & 12 & 13 & 36 & 7 \\ 12 & 22 & 0 & 30 & 36 & 85 & 17 \\ 3 & 4 & -4 & 6 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^1 1^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 9 & 4 \\ 3 & 5 & 0 & 6 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 9 & 13 & 7 \\ 3 & 4 & -4 & 4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1212} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -230, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 5 & 9 & 0 & 12 & 13 & 21 & 7 \\ 7 & 13 & 0 & 18 & 21 & 28 & 10 \\ 3 & 4 & -4 & 6 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.10 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -4 & 3 & 3 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1133} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -248, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 9 & 3 \\ 3 & 4 & 0 & 4 & 12 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 6 & 12 & 0 & 9 & 25 & 36 & 9 \\ 9 & 17 & 0 & 12 & 36 & 49 & 13 \\ 3 & 4 & -4 & 3 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 6 & 10 & 6 \\ 3 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1213} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 24 & 6 \\ 5 & 9 & 0 & 12 & 13 & 27 & 7 \\ 9 & 17 & 0 & 24 & 27 & 49 & 13 \\ 3 & 4 & -4 & 6 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1122} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 4 & 8 & 0 & 6 & 12 & 10 & 6 \\ 3 & 4 & -4 & 3 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1114} 3^1 1^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 33 & 7 \\ 5 & 9 & 0 & 15 & 13 & 33 & 7 \\ 11 & 21 & 0 & 33 & 33 & 76 & 16 \\ 3 & 4 & -4 & 7 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{133} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -280, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 15 & 4 \\ 9 & 17 & 0 & 15 & 49 & 51 & 13 \\ 9 & 17 & 0 & 15 & 51 & 49 & 13 \\ 3 & 4 & -4 & 4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.65 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 9 & 25 & 9 \\ 3 & 4 & -4 & 3 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1431} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 81 & 15 & 16 \\ 12 & 22 & 0 & 81 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 15 & 1 & 3 \\ 3 & 4 & -4 & 16 & 17 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1222} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.68 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 12 & 10 & 6 \\ 3 & 4 & -4 & 6 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1123} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -320, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 4 & 8 & 0 & 6 & 10 & 18 & 6 \\ 6 & 12 & 0 & 9 & 18 & 25 & 9 \\ 3 & 4 & -4 & 3 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{132} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -328, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 0 & 5 & 17 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 12 & 4 \\ 9 & 17 & 0 & 15 & 49 & 39 & 13 \\ 7 & 13 & 0 & 12 & 39 & 28 & 10 \\ 3 & 4 & -4 & 4 & 13 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 16 & 3 \\ 3 & 4 & 0 & 4 & 9 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 5 & 9 & 0 & 6 & 13 & 48 & 7 \\ 16 & 28 & 0 & 18 & 48 & 154 & 22 \\ 3 & 4 & -4 & 3 & 7 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{112}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -4 & 4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{1223}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 8 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 4 & 8 & 0 & 12 & 10 & 18 & 6 \\ 6 & 12 & 0 & 18 & 18 & 25 & 9 \\ 3 & 4 & -4 & 6 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{1122}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -346, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 4 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 4 & 8 & 0 & 6 & 10 & 18 & 6 \\ 7 & 13 & 0 & 9 & 18 & 28 & 10 \\ 3 & 4 & -4 & 3 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{1114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -352, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 12 & 46 & 12 \\ 3 & 4 & -4 & 3 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{1133}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -360, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.89 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 9 & 3 \\ 6 & 12 & 0 & 9 & 25 & 27 & 9 \\ 6 & 12 & 0 & 9 & 27 & 25 & 9 \\ 3 & 4 & -4 & 3 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{1111}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 5 & 3 \\ 3 & 4 & 0 & 4 & 14 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 6 & 3 \\ 8 & 14 & 0 & 9 & 37 & 24 & 11 \\ 5 & 9 & 0 & 6 & 24 & 13 & 7 \\ 3 & 4 & -4 & 3 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{1111}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 9 & 3 \\ 8 & 14 & 0 & 9 & 37 & 39 & 11 \\ 8 & 14 & 0 & 9 & 39 & 37 & 11 \\ 3 & 4 & -4 & 3 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{1124}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 8 & 3 \\ 3 & 4 & 0 & 4 & 8 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 4 & 8 & 0 & 6 & 10 & 24 & 6 \\ 8 & 16 & 0 & 12 & 24 & 46 & 12 \\ 3 & 4 & -4 & 3 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{1123}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{1224}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 8 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 24 & 6 \\ 4 & 8 & 0 & 12 & 10 & 24 & 6 \\ 8 & 16 & 0 & 24 & 24 & 46 & 12 \\ 3 & 4 & -4 & 6 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{1223}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 24 & 6 \\ 4 & 8 & 0 & 12 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 24 & 49 & 13 \\ 3 & 4 & -4 & 6 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{1134}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 6 & 12 & 0 & 9 & 25 & 36 & 9 \\ 8 & 16 & 0 & 12 & 36 & 46 & 12 \\ 3 & 4 & -4 & 3 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{1115}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.1$

$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 10 & 20 & 0 & 15 & 15 & 73 & 15 \\ 3 & 4 & -4 & 3 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{112}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 0 & 5 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 12 & 4 \\ 5 & 9 & 0 & 9 & 13 & 21 & 7 \\ 7 & 13 & 0 & 12 & 21 & 28 & 10 \\ 3 & 4 & -4 & 4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{1125}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 10 & 3 \\ 3 & 4 & 0 & 4 & 8 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 10 & 30 & 6 \\ 10 & 20 & 0 & 15 & 30 & 73 & 15 \\ 3 & 4 & -4 & 3 & 6 & 15 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{1144}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 12 & 3 \\ 8 & 16 & 0 & 12 & 46 & 48 & 12 \\ 8 & 16 & 0 & 12 & 48 & 46 & 12 \\ 3 & 4 & -4 & 3 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1135}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.9$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 10 & 3 \\ 3 & 4 & 0 & 4 & 12 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 6 & 12 & 0 & 9 & 25 & 45 & 9 \\ 10 & 20 & 0 & 15 & 45 & 75 & 15 \\ 3 & 4 & -4 & 3 & 9 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1232}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -490, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 7 & 3 \\ 3 & 4 & 0 & 8 & 22 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 30 & 18 & 6 \\ 12 & 22 & 0 & 30 & 85 & 51 & 17 \\ 7 & 13 & 0 & 18 & 51 & 28 & 10 \\ 3 & 4 & -4 & 6 & 17 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -490, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 0 & 5 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 14 & 26 & 0 & 24 & 18 & 118 & 20 \\ 3 & 4 & -4 & 4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 15 & 7 \\ 5 & 9 & 0 & 9 & 15 & 13 & 7 \\ 3 & 4 & -4 & 4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1116}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 12 & 24 & 0 & 18 & 18 & 106 & 18 \\ 3 & 4 & -4 & 3 & 3 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1112}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 3 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{113}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 21 & 15 & 85 & 17 \\ 3 & 4 & -4 & 4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1214}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 14 & 3 \\ 3 & 4 & 0 & 8 & 9 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 36 & 6 \\ 5 & 9 & 0 & 12 & 13 & 42 & 7 \\ 14 & 26 & 0 & 36 & 42 & 118 & 20 \\ 3 & 4 & -4 & 6 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{1126}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 4 & 8 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 4 & 8 & 0 & 6 & 10 & 36 & 6 \\ 12 & 24 & 0 & 18 & 36 & 106 & 18 \\ 3 & 4 & -4 & 3 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

[60] $1^{1111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -540$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 3 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{122}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -556$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 30 & 10 \\ 7 & 13 & 0 & 12 & 30 & 28 & 10 \\ 3 & 4 & -4 & 4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{1134}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 11 & 3 \\ 3 & 4 & 0 & 4 & 12 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 6 & 12 & 0 & 9 & 25 & 45 & 9 \\ 11 & 21 & 0 & 15 & 45 & 76 & 16 \\ 3 & 4 & -4 & 3 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{1112}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -584$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 8 & 14 & 0 & 9 & 37 & 48 & 11 \\ 10 & 18 & 0 & 12 & 48 & 58 & 14 \\ 3 & 4 & -4 & 3 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{1112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{1222}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -616$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 30 & 10 \\ 7 & 13 & 0 & 18 & 30 & 28 & 10 \\ 3 & 4 & -4 & 6 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1117}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -616$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 14 & 28 & 0 & 21 & 21 & 145 & 21 \\ 3 & 4 & -4 & 3 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{1123} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -640, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 4 & 8 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 4 & 8 & 0 & 6 & 10 & 36 & 6 \\ 14 & 26 & 0 & 18 & 36 & 118 & 20 \\ 3 & 4 & -4 & 3 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1127} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -640, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 4 & 8 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 21 & 3 \\ 4 & 8 & 0 & 6 & 10 & 42 & 6 \\ 14 & 28 & 0 & 21 & 42 & 145 & 21 \\ 3 & 4 & -4 & 3 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1123} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -658, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 4 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 4 & 8 & 0 & 6 & 10 & 24 & 6 \\ 9 & 17 & 0 & 12 & 24 & 49 & 13 \\ 3 & 4 & -4 & 3 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -666, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 13 & 33 & 7 \\ 11 & 21 & 0 & 15 & 33 & 76 & 16 \\ 3 & 4 & -4 & 3 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{113} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.66 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 33 & 4 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -4 & 4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1224} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -740, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 8 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 30 & 6 \\ 4 & 8 & 0 & 12 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 30 & 76 & 16 \\ 3 & 4 & -4 & 6 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.34 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -4 & 3 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1113} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -788, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 5 & 9 & 0 & 6 & 13 & 27 & 7 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -4 & 3 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1223} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -838, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 9 & 3 \\ 3 & 4 & 0 & 8 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 24 & 6 \\ 7 & 13 & 0 & 18 & 28 & 39 & 10 \\ 9 & 17 & 0 & 24 & 39 & 49 & 13 \\ 3 & 4 & -4 & 6 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1113} 3 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -852, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 27 & 7 \\ 5 & 9 & 0 & 15 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 27 & 49 & 13 \\ 3 & 4 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1113} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -872, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 3 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -890, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 4 & 9 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 13 & 39 & 7 \\ 13 & 23 & 0 & 15 & 39 & 100 & 18 \\ 3 & 4 & -4 & 3 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{1224} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 11 & 3 \\ 3 & 4 & 0 & 8 & 13 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 30 & 6 \\ 7 & 13 & 0 & 18 & 28 & 48 & 10 \\ 11 & 21 & 0 & 30 & 48 & 76 & 16 \\ 3 & 4 & -4 & 6 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{1142} 3 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -936, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 7 & 3 \\ 3 & 4 & 0 & 9 & 21 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 21 & 7 \\ 11 & 21 & 0 & 33 & 76 & 48 & 16 \\ 7 & 13 & 0 & 21 & 48 & 28 & 10 \\ 3 & 4 & -4 & 7 & 16 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1122} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 10 & 3 \\ 3 & 4 & 0 & 4 & 13 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 7 & 13 & 0 & 9 & 28 & 42 & 10 \\ 10 & 18 & 0 & 12 & 42 & 58 & 14 \\ 3 & 4 & -4 & 3 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -964, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 15 & 13 & 7 \\ 3 & 4 & -4 & 7 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1124} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 4 & 8 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 10 & 30 & 6 \\ 11 & 21 & 0 & 15 & 30 & 76 & 16 \\ 3 & 4 & -4 & 3 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -988, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 30 & 21 & 172 & 24 \\ 3 & 4 & -4 & 4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1112} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1026, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 10 & 3 \\ 3 & 4 & 0 & 4 & 9 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 5 & 9 & 0 & 6 & 13 & 30 & 7 \\ 10 & 18 & 0 & 12 & 30 & 58 & 14 \\ 3 & 4 & -4 & 3 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{1134} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1066, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 17 & 3 \\ 3 & 4 & 0 & 4 & 17 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 21 & 3 \\ 9 & 17 & 0 & 12 & 49 & 93 & 13 \\ 17 & 31 & 0 & 21 & 93 & 172 & 24 \\ 3 & 4 & -4 & 3 & 13 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1112} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1080, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 21 & 7 \\ 5 & 9 & 0 & 15 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 21 & 28 & 10 \\ 3 & 4 & -4 & 7 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{1113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 3 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{1122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 30 & 10 \\ 7 & 13 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & -4 & 3 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{1224}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1180, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 14 & 3 \\ 3 & 4 & 0 & 8 & 13 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 36 & 6 \\ 7 & 13 & 0 & 18 & 28 & 60 & 10 \\ 14 & 26 & 0 & 36 & 60 & 118 & 20 \\ 3 & 4 & -4 & 6 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{1144}31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 33 & 7 \\ 11 & 21 & 0 & 33 & 76 & 78 & 16 \\ 11 & 21 & 0 & 33 & 76 & 78 & 16 \\ 3 & 4 & -4 & 7 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{1114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{1143}31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1248, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 9 & 3 \\ 3 & 4 & 0 & 9 & 21 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 27 & 7 \\ 11 & 21 & 0 & 33 & 76 & 63 & 16 \\ 9 & 17 & 0 & 27 & 63 & 49 & 13 \\ 3 & 4 & -4 & 7 & 16 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{1143}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1250, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{1124}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 14 & 3 \\ 3 & 4 & 0 & 4 & 18 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 18 & 3 \\ 10 & 18 & 0 & 12 & 58 & 84 & 14 \\ 14 & 26 & 0 & 18 & 84 & 118 & 20 \\ 3 & 4 & -4 & 3 & 14 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{1123}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 15 & 3 \\ 3 & 4 & 0 & 4 & 13 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 18 & 3 \\ 7 & 13 & 0 & 9 & 28 & 63 & 10 \\ 15 & 27 & 0 & 18 & 63 & 133 & 21 \\ 3 & 4 & -4 & 3 & 10 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{1123} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1328, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 27 & 7 \\ 7 & 13 & 0 & 21 & 28 & 39 & 10 \\ 9 & 17 & 0 & 27 & 39 & 49 & 13 \\ 3 & 4 & -4 & 7 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{1123} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1354, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 7 & 13 & 0 & 9 & 28 & 39 & 10 \\ 9 & 17 & 0 & 12 & 39 & 49 & 13 \\ 3 & 4 & -4 & 3 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{1122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 21 & 7 \\ 7 & 13 & 0 & 21 & 28 & 30 & 10 \\ 7 & 13 & 0 & 21 & 30 & 28 & 10 \\ 3 & 4 & -4 & 7 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{1114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 24 & 3 \\ 2 & 4 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 3 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{1133} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 12 & 3 \\ 3 & 4 & 0 & 4 & 17 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 15 & 3 \\ 9 & 17 & 0 & 12 & 49 & 66 & 13 \\ 12 & 22 & 0 & 15 & 66 & 85 & 17 \\ 3 & 4 & -4 & 3 & 13 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{1124} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 11 & 3 \\ 3 & 4 & 0 & 4 & 13 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 7 & 13 & 0 & 9 & 28 & 48 & 10 \\ 11 & 21 & 0 & 15 & 48 & 76 & 16 \\ 3 & 4 & -4 & 3 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{1133} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 27 & 27 & 7 \\ 9 & 17 & 0 & 27 & 49 & 51 & 13 \\ 9 & 17 & 0 & 27 & 51 & 49 & 13 \\ 3 & 4 & -4 & 7 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{1122} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 12 & 3 \\ 10 & 18 & 0 & 12 & 58 & 60 & 14 \\ 10 & 18 & 0 & 12 & 60 & 58 & 14 \\ 3 & 4 & -4 & 3 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{1113}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1520$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 4 & 9 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 13 & 36 & 7 \\ 12 & 22 & 0 & 15 & 36 & 85 & 17 \\ 3 & 4 & -4 & 3 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{1114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1696$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -4 & 3 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{1114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1706$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 14 & 3 \\ 3 & 4 & 0 & 4 & 21 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 18 & 3 \\ 11 & 21 & 0 & 15 & 76 & 96 & 16 \\ 14 & 26 & 0 & 18 & 96 & 118 & 20 \\ 3 & 4 & -4 & 3 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{1123}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 12 & 3 \\ 3 & 4 & 0 & 4 & 18 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 15 & 3 \\ 10 & 18 & 0 & 12 & 58 & 72 & 14 \\ 12 & 22 & 0 & 15 & 72 & 85 & 17 \\ 3 & 4 & -4 & 3 & 14 & 17 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{1113}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 18 & 3 \\ 3 & 4 & 0 & 4 & 9 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 21 & 3 \\ 5 & 9 & 0 & 6 & 13 & 54 & 7 \\ 18 & 32 & 0 & 21 & 54 & 193 & 25 \\ 3 & 4 & -4 & 3 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{1114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 3 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{1133}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1756$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 51 & 13 \\ 9 & 17 & 0 & 12 & 51 & 49 & 13 \\ 3 & 4 & -4 & 3 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{1224}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1772$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 13 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 48 & 10 \\ 7 & 13 & 0 & 30 & 28 & 48 & 10 \\ 11 & 21 & 0 & 48 & 48 & 76 & 16 \\ 3 & 4 & -4 & 10 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{1123} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 12 & 3 \\ 3 & 4 & 0 & 4 & 13 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 7 & 13 & 0 & 9 & 28 & 51 & 10 \\ 12 & 22 & 0 & 15 & 51 & 85 & 17 \\ 3 & 4 & -4 & 3 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{1124} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1826, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 20 & 3 \\ 3 & 4 & 0 & 4 & 13 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 24 & 3 \\ 7 & 13 & 0 & 9 & 28 & 84 & 10 \\ 20 & 36 & 0 & 24 & 84 & 238 & 28 \\ 3 & 4 & -4 & 3 & 10 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{1113} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1836, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 4 & 9 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 5 & 9 & 0 & 6 & 13 & 45 & 7 \\ 15 & 27 & 0 & 18 & 45 & 133 & 21 \\ 3 & 4 & -4 & 3 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{1114} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1850, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 4 & 9 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 5 & 9 & 0 & 6 & 13 & 42 & 7 \\ 14 & 26 & 0 & 18 & 42 & 118 & 20 \\ 3 & 4 & -4 & 3 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{1222} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1852, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 30 & 10 \\ 7 & 13 & 0 & 30 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 30 & 28 & 10 \\ 3 & 4 & -4 & 10 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{1223} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1984, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 39 & 10 \\ 7 & 13 & 0 & 30 & 28 & 39 & 10 \\ 9 & 17 & 0 & 39 & 39 & 49 & 13 \\ 3 & 4 & -4 & 10 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{1134} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1994, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 11 & 3 \\ 3 & 4 & 0 & 4 & 17 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 15 & 3 \\ 9 & 17 & 0 & 12 & 49 & 63 & 13 \\ 11 & 21 & 0 & 15 & 63 & 76 & 16 \\ 3 & 4 & -4 & 3 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{1144} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 78 & 16 \\ 11 & 21 & 0 & 15 & 78 & 76 & 16 \\ 3 & 4 & -4 & 3 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{1134} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2458, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 14 & 3 \\ 3 & 4 & 0 & 4 & 17 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 18 & 3 \\ 9 & 17 & 0 & 12 & 49 & 78 & 13 \\ 14 & 26 & 0 & 18 & 78 & 118 & 20 \\ 3 & 4 & -4 & 3 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{1124} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2506, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 14 & 3 \\ 3 & 4 & 0 & 4 & 13 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 18 & 3 \\ 7 & 13 & 0 & 9 & 28 & 60 & 10 \\ 14 & 26 & 0 & 18 & 60 & 118 & 20 \\ 3 & 4 & -4 & 3 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2618, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 17 & 3 \\ 3 & 4 & 0 & 4 & 9 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 21 & 3 \\ 5 & 9 & 0 & 6 & 13 & 51 & 7 \\ 17 & 31 & 0 & 21 & 51 & 172 & 24 \\ 3 & 4 & -4 & 3 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^{1124} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 17 & 3 \\ 3 & 4 & 0 & 4 & 13 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 21 & 3 \\ 7 & 13 & 0 & 9 & 28 & 72 & 10 \\ 17 & 31 & 0 & 21 & 72 & 172 & 24 \\ 3 & 4 & -4 & 3 & 10 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{1133} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2680, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 15 & 3 \\ 12 & 22 & 0 & 15 & 85 & 87 & 17 \\ 12 & 22 & 0 & 15 & 87 & 85 & 17 \\ 3 & 4 & -4 & 3 & 17 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{1134} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 14 & 3 \\ 3 & 4 & 0 & 4 & 22 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 18 & 3 \\ 12 & 22 & 0 & 15 & 85 & 102 & 17 \\ 14 & 26 & 0 & 18 & 102 & 118 & 20 \\ 3 & 4 & -4 & 3 & 17 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{1114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2970, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 20 & 3 \\ 3 & 4 & 0 & 4 & 9 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 24 & 3 \\ 5 & 9 & 0 & 6 & 13 & 60 & 7 \\ 20 & 36 & 0 & 24 & 60 & 238 & 28 \\ 3 & 4 & -4 & 3 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{1144} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -3880, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.83 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 9 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 6 & 3 & 3 \\ 8 & 14 & 0 & 9 & 37 & 24 & 9 & 11 \\ 5 & 9 & 0 & 6 & 24 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 11 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1121} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 18 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 & 3 \\ 10 & 18 & 0 & 18 & 12 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1131} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 27 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 & 3 \\ 15 & 27 & 0 & 27 & 18 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 & 4 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 & 6 \\ 3 & 4 & -4 & 3 & 3 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1131} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 21 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 & 3 \\ 12 & 22 & 0 & 21 & 15 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{1141}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 706$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 36 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 24 & 3 & 3 \\ 20 & 36 & 0 & 36 & 24 & 238 & 24 & 28 \\ 2 & 4 & 0 & 3 & 3 & 24 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 712$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 724$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{1121}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 796$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 12 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 21 & 6 & 7 \\ 7 & 13 & 0 & 12 & 21 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 6 & 9 & 0 & 2 \\ 3 & 4 & -4 & 4 & 7 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 988$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 9 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1032$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 6 & 15 & 12 & 6 \\ 3 & 4 & -4 & 3 & 3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{1221}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1144$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 28 & 30 & 9 & 10 \\ 7 & 13 & 0 & 12 & 30 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 9 & 0 & 2 \\ 3 & 4 & -4 & 4 & 10 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{1141}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1180$, $(n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 & 3 \\ 17 & 31 & 0 & 30 & 21 & 172 & 21 & 24 \\ 2 & 4 & 0 & 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 15 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 15 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 7 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.84 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111}3^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 2 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 2 & 3 \\ 5 & 15 & 9 & 0 & 9 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13}1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 3 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 669, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 15 & 5 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.85 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -4 & 4 & 3 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -4 & 3 & 3 & 10 & -4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1211} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 15 & 7 \\ 5 & 9 & 0 & 12 & 15 & 12 & 6 \\ 3 & 4 & -4 & 6 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 8 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 21 & 3 & 4 \\ 11 & 19 & 0 & 21 & 73 & 12 & 15 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{113} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 15 & 12 & 48 & 12 \\ 3 & 4 & -4 & 4 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1211} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 5 & 9 & 0 & 12 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 6 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 6 & 7 \\ 11 & 19 & 0 & 33 & 73 & 12 & 15 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -4 & 7 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1121} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -137, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 2 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 33 & 9 & 11 \\ 7 & 13 & 0 & 33 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -4 & 11 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{131} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 17 & 0 & 15 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 5 & 9 & 0 & 6 & 12 & 12 & 6 \\ 3 & 4 & -4 & 3 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1221} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 2 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 54 & 9 & 10 \\ 13 & 23 & 0 & 54 & 100 & 15 & 18 \\ 2 & 4 & 0 & 9 & 15 & 0 & 2 \\ 3 & 4 & -4 & 10 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -219, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 27 & 9 \\ 3 & 4 & -4 & 4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 8 & 14 & 0 & 15 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 9 & 7 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 3 & 4 & -4 & 3 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{114}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 0 & 4 & 21 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 11 & 21 & 0 & 15 & 76 & 18 & 16 \\ 3 & 5 & 0 & 3 & 18 & 3 & 3 \\ 3 & 4 & -4 & 3 & 16 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1221}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 6 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1121}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 13 & 23 & 0 & 15 & 100 & 15 & 18 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1131}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 6 & 12 & 0 & 9 & 25 & 9 & 9 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1221}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 5 & 3 \\ 3 & 4 & 0 & 8 & 13 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 12 & 6 \\ 7 & 13 & 0 & 18 & 28 & 21 & 10 \\ 5 & 9 & 0 & 12 & 21 & 12 & 6 \\ 3 & 4 & -4 & 6 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1131} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 2 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 57 & 9 & 11 \\ 12 & 22 & 0 & 57 & 85 & 15 & 17 \\ 2 & 4 & 0 & 9 & 15 & 0 & 2 \\ 3 & 4 & -4 & 11 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1141} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 8 & 16 & 0 & 12 & 46 & 12 & 12 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 3 \\ 3 & 4 & 0 & 4 & 13 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 7 & 13 & 0 & 9 & 28 & 12 & 10 \\ 3 & 5 & 0 & 3 & 12 & 3 & 3 \\ 3 & 4 & -4 & 3 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 0 & 4 & 17 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 9 & 17 & 0 & 12 & 49 & 15 & 13 \\ 3 & 5 & 0 & 3 & 15 & 3 & 3 \\ 3 & 4 & -4 & 3 & 13 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1221} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 2 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 6 & 6 \\ 7 & 13 & 0 & 18 & 28 & 9 & 10 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -4 & 6 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & -4 & 3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 2 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 24 & 6 & 7 \\ 8 & 14 & 0 & 24 & 37 & 9 & 11 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -4 & 7 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1211} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 18 & 3 & 4 \\ 10 & 18 & 0 & 18 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{121} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 30 & 3 & 4 \\ 16 & 28 & 0 & 30 & 154 & 18 & 22 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 39 & 9 & 11 \\ 8 & 14 & 0 & 39 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -4 & 11 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1121} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 7 & 13 & 0 & 9 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1121} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 10 & 18 & 0 & 12 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{141} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 24 & 3 & 4 \\ 14 & 26 & 0 & 24 & 118 & 18 & 20 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{121} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 24 & 3 & 4 \\ 13 & 23 & 0 & 24 & 100 & 15 & 18 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{131} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 21 & 3 & 4 \\ 12 & 22 & 0 & 21 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1131^1} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 & 3 \\ 18 & 32 & 0 & 21 & 193 & 21 & 25 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 3 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1121^1} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 6 & 3 \\ 7 & 13 & 0 & 9 & 28 & 21 & 10 \\ 5 & 9 & 0 & 6 & 21 & 12 & 6 \\ 3 & 4 & -4 & 3 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1141^1} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 6 & 7 \\ 11 & 21 & 0 & 33 & 76 & 15 & 16 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -4 & 7 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1121^1} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 6 & 7 \\ 7 & 13 & 0 & 21 & 28 & 9 & 10 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -4 & 7 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1121^1} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 2 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 48 & 9 & 11 \\ 10 & 18 & 0 & 48 & 58 & 12 & 14 \\ 2 & 4 & 0 & 9 & 12 & 0 & 2 \\ 3 & 4 & -4 & 11 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1131^1} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 2 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 27 & 6 & 7 \\ 9 & 17 & 0 & 27 & 49 & 12 & 13 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -4 & 7 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1121^1} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 2 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 48 & 6 & 7 \\ 16 & 28 & 0 & 48 & 154 & 18 & 22 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -4 & 7 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1131^1} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 5 & 3 \\ 3 & 4 & 0 & 4 & 17 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 6 & 3 \\ 9 & 17 & 0 & 12 & 49 & 27 & 13 \\ 5 & 9 & 0 & 6 & 27 & 12 & 6 \\ 3 & 4 & -4 & 3 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1141} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 21 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 6 & 3 \\ 11 & 21 & 0 & 15 & 76 & 33 & 16 \\ 5 & 9 & 0 & 6 & 33 & 12 & 6 \\ 3 & 4 & -4 & 3 & 16 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1131} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 9 & 17 & 0 & 12 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{131} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 27 & 3 & 4 \\ 15 & 27 & 0 & 27 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1221} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 2 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 42 & 9 & 10 \\ 10 & 18 & 0 & 42 & 58 & 12 & 14 \\ 2 & 4 & 0 & 9 & 12 & 0 & 2 \\ 3 & 4 & -4 & 10 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1131} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 & 3 \\ 15 & 27 & 0 & 18 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1121} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 2 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 30 & 6 & 7 \\ 10 & 18 & 0 & 30 & 58 & 12 & 14 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -4 & 7 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1121} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 2 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 39 & 6 & 7 \\ 13 & 23 & 0 & 39 & 100 & 15 & 18 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -4 & 7 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1221} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 9 & 10 \\ 7 & 13 & 0 & 30 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -4 & 10 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1131^1}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 12 & 22 & 0 & 15 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{131^1}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 33 & 3 & 4 \\ 18 & 32 & 0 & 33 & 193 & 21 & 25 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{1141^1}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 11 & 21 & 0 & 15 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{141^1}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 30 & 3 & 4 \\ 17 & 31 & 0 & 30 & 172 & 21 & 24 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1131^1}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 2 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 36 & 6 & 7 \\ 12 & 22 & 0 & 36 & 85 & 15 & 17 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -4 & 7 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1221^1}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 60 & 12 & 14 \\ 10 & 18 & 0 & 60 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 12 & 0 & 2 \\ 3 & 4 & -4 & 14 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{141^1}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 36 & 3 & 4 \\ 20 & 36 & 0 & 36 & 238 & 24 & 28 \\ 2 & 4 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & -4 & 4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1141^1}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 & 3 \\ 14 & 26 & 0 & 18 & 118 & 18 & 20 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{1141}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2780$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 24 & 3 & 3 \\ 20 & 36 & 0 & 24 & 238 & 24 & 28 \\ 2 & 4 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & -4 & 3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{1141}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2996$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 2 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 42 & 6 & 7 \\ 14 & 26 & 0 & 42 & 118 & 18 & 20 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -4 & 7 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{1141}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3017$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 & 3 \\ 17 & 31 & 0 & 21 & 172 & 21 & 24 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 3 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1131}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 2 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 45 & 6 & 7 \\ 15 & 27 & 0 & 45 & 133 & 18 & 21 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -4 & 7 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{1131}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3365$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 2 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 54 & 6 & 7 \\ 18 & 32 & 0 & 54 & 193 & 21 & 25 \\ 2 & 4 & 0 & 6 & 21 & 0 & 2 \\ 3 & 4 & -4 & 7 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{1141}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -4373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 2 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 51 & 6 & 7 \\ 17 & 31 & 0 & 51 & 172 & 21 & 24 \\ 2 & 4 & 0 & 6 & 21 & 0 & 2 \\ 3 & 4 & -4 & 7 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{1141}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -5216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 2 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 60 & 6 & 7 \\ 20 & 36 & 0 & 60 & 238 & 24 & 28 \\ 2 & 4 & 0 & 6 & 24 & 0 & 2 \\ 3 & 4 & -4 & 7 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [80] $1^{111}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 44$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{1111} 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{1111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 82, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{1111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 111, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.62 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 & 3 \\ 3 & 6 & 9 & 0 & 6 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 3 & 6 & 0 & 2 \\ 2 & 3 & 3 & -4 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11} 31^{112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 138, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.35 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 12 & 3 & 4 \\ 4 & 12 & 6 & 0 & 12 & 22 & 6 & 8 \\ 1 & 3 & 3 & 0 & 3 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{1111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 224, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.89 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 7 & 0 \\ 4 & 6 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 7 & 9 & 14 & 8 \\ 3 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 239, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{1111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 268, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 269, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{112}1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 9 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 & 3 \\ 4 & 9 & 12 & 0 & 9 & 28 & 9 & 10 \\ 1 & 3 & 3 & 0 & 3 & 9 & 0 & 2 \\ 2 & 3 & 3 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[90] $1^{121}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{11}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 3 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{111}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 363, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 3 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[94] $1^{121}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 374, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 3 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{111}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 704, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111}31^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1065$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 15 & 3 & 4 \\ 5 & 15 & 9 & 0 & 15 & 37 & 9 & 11 \\ 1 & 3 & 3 & 0 & 3 & 9 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{111}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1659$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 21 & 3 & 4 \\ 7 & 21 & 12 & 0 & 21 & 73 & 12 & 15 \\ 1 & 3 & 3 & 0 & 3 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 2145$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 27 & 3 & 4 \\ 9 & 27 & 15 & 0 & 27 & 121 & 15 & 19 \\ 1 & 3 & 3 & 0 & 3 & 15 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [100] $1^{111}4^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -24$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 5 & 7 & 0 & 6 & 8 & 5 & 6 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{111}4^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -48$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 5 & 2 & 3 \\ 4 & 4 & 0 & 5 & 7 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 5 & 7 & 0 & 6 & 8 & 5 & 6 \\ 2 & 3 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

4.86 NHC(-2)(-3) + e₆ + su₂ + su₂ + su₃ + su₂

$T = 10$ [0] $1^{111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 9 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

- [2] $1^{11111}31^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 6 & 3 \\ 8 & 14 & 0 & 9 & 37 & 9 & 24 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 24 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 11 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [3] $1^{1112}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{1113}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{11111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{1113}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 21 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 4 & 3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{1114}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 24 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{11111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{11111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 6 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 \\ 5 & 9 & 0 & 6 & 6 & 6 & 13 \\ 3 & 4 & -4 & 3 & 3 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 796, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 12 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 21 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 21 & 9 & 28 & 10 \\ 3 & 4 & -4 & 4 & 7 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 988, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1032, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1212} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 3 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 9 & 30 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 30 & 9 & 28 & 10 \\ 3 & 4 & -4 & 4 & 10 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1180, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 30 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 4 & 3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2368, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 6 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 7 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.87 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 48, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 4 & 6 & 0 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 228, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1141}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 & 3 \\ 11 & 21 & 0 & 15 & 76 & 18 & 15 & 16 \\ 3 & 5 & 0 & 3 & 18 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 15 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 16 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1121}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 & 3 \\ 7 & 13 & 0 & 9 & 28 & 12 & 9 & 10 \\ 3 & 5 & 0 & 3 & 12 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 9 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 10 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1131}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 & 3 \\ 9 & 17 & 0 & 12 & 49 & 15 & 12 & 13 \\ 3 & 5 & 0 & 3 & 15 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 12 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 13 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.88 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{13} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 2 & 5 & -2 & 3 & 2 & 0 \\ 3 & 5 & 3 & 4 & 3 & 4 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 3 & 5 \\ 2 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & 0 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 3 & 2 & 0 & 4 & 2 & 4 \\ 3 & 3 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 2 & 0 & 4 & 2 & 4 \\ 3 & 3 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 4 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 \\ 1 & 2 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 \\ 2 & 4 & 2 & -4 & 3 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 2 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 6 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -162, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 8 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 12 & 0 \\ 10 & 14 & 0 & 12 & 66 & 16 \\ 3 & 3 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^{12} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 5 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{13} 3^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -257, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{12} 3^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 7 & 10 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -312, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.59 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{112}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 6 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -372, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.59 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 7 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{13}3^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -392, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11}3^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -402, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -522, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.58 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{13}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -522, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 9 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{14}3^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -552, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 24 & 18 & 0 & 118 & 14 & 20 \\ 1 & 3 & 2 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 8 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{13} 3^1 1^{13} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 9 & 0 & 52 & 9 & 12 \\ 1 & 3 & 2 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -4 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{14} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 12 & 0 \\ 11 & 14 & 0 & 12 & 66 & 16 \\ 3 & 3 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11} 3^1 1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -807, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -834, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 10 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{13} 3^1 1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -887, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 15 & 0 & 85 & 12 & 17 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{13} 3^1 1^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -932, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 24 & 12 & 0 & 94 & 12 & 16 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0} \\
[42] \quad & 1^{12}3^11^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -972, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 12 & 0 & 58 & 10 & 14 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0} \\
[43] \quad & 1^{14}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1060, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 12 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 12 & 0 \\ 12 & 14 & 0 & 12 & 66 & 16 \\ 3 & 3 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0} \\
[44] \quad & 1^{11}3^11^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1302, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 12 & 0 & 73 & 11 & 15 \\ 1 & 3 & 2 & 0 & 11 & 0 & 2 \\ 2 & 4 & 2 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0} \\
[45] \quad & 1^{13}3^11^{15}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1337, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.1 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 30 & 15 & 0 & 148 & 15 & 20 \\ 1 & 3 & 2 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0} \\
[46] \quad & 1^{12}3^11^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1737, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 24 & 15 & 0 & 100 & 13 & 18 \\ 1 & 3 & 2 & 0 & 13 & 0 & 2 \\ 2 & 4 & 2 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0} \\
[47] \quad & 1^{14}3^11^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1857, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 30 & 21 & 0 & 172 & 17 & 24 \\ 1 & 3 & 2 & 0 & 17 & 0 & 2 \\ 2 & 4 & 2 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0} \\
[48] \quad & 1^{11}3^11^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1887, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 15 & 0 & 121 & 14 & 19 \\ 1 & 3 & 2 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -4 & 19 & 2 & 0 \end{pmatrix}_{b_0} \\
[49] \quad & 1^{13}3^11^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1922, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 18 & 0 & 133 & 15 & 21 \\ 1 & 3 & 2 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0} \\
[50] \quad & 1^{12}3^11^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2592, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.0
\end{aligned}$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 30 & 18 & 0 & 154 & 16 & 22 \\ 1 & 3 & 2 & 0 & 16 & 0 & 2 \\ 2 & 4 & 2 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{13}3^{11}4^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3047, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 0 & 33 & 3 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 33 & 21 & 0 & 193 & 18 & 25 \\ 1 & 3 & 2 & 0 & 18 & 0 & 2 \\ 2 & 4 & 2 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{14}3^{11}4^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 0 & 36 & 3 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 36 & 24 & 0 & 238 & 20 & 28 \\ 1 & 3 & 2 & 0 & 20 & 0 & 2 \\ 2 & 4 & 2 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [53] \ 1^{12}231^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 5 \\ 14 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 11 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{11}231^{11}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 29 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{12}231^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 14 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11}231^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 27 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{12}231^{11}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 16 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{12}23^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 2 & 0 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11}23^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 2 & 0 \\ 4 & 9 & 0 & 7 & 14 & 5 & 8 \\ 1 & 3 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11}231^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 28 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12}231^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 15 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{12}231^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 504, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 2 & 0 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{11}231^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 7 & 15 & 0 & 10 & 38 & 8 & 12 \\ 1 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [64] \quad 1^{11}3^{11}1^{12}4131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -66, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 1 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [65] \quad 1^{11}3^{11}1^{12}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{13}3^{11}1^{12}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 2 & 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11}3^{11}1^{12}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{13}3^{11}5^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 423, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.61 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 & 2 \\ 1 & 0 & 1 & -3 & 3 & 1 & -1 \\ 6 & 0 & 9 & 3 & 25 & 6 & 9 \\ 1 & 0 & 2 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{14}3^{11}5^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 503, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.64 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 & 2 \\ 1 & 0 & 1 & -3 & 4 & 1 & -1 \\ 7 & 0 & 12 & 4 & 46 & 8 & 12 \\ 1 & 0 & 2 & 1 & 8 & 0 & 2 \\ 2 & 0 & 2 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

4.89 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 329, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 6 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 6 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 464, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 9 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 8 & 2 \\ 5 & 15 & 9 & 0 & 9 & 8 & 37 & 11 \\ 2 & 4 & 2 & -4 & 3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 604, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 6 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 669, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 9 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 9 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 1184, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 5 & 15 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 15 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.90 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 164, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 1 & 3 & 2 & 0 & 2 & 2 & 0 \\ 2 & 4 & 2 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{blue}{1}} 1^{\textcolor{red}{3}} 1^{\textcolor{green}{1}} 3^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 436, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

[1] $1^3 1^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 436$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{123}1^{11}1^{11}31^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{brown}{s}u_2 + \textcolor{red}{s}u_2 \rightarrow (-3) + \textcolor{blue}{s}u_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 7 & 7 & 10 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 7 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 10 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^3 1^3 1^1 1^{12} 3^1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 544, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 6 & 6 & 8 \\ 1 & 3 & 2 & 0 & 6 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 6 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 8 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}1^{11}31^1 + [\textcolor{red}{s}\textcolor{red}{u}_2 - \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) + \textcolor{blue}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 704, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 5 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 5 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 7 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^{12}31^{1+} + [\text{su}_2, \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 1316$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 8 & 8 & 11 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 8 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 11 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{12}3^{11}1^{12}31^1 + [\textcolor{red}{s}\textcolor{red}{u}_2 - \textcolor{red}{g}_2] + \textcolor{blue}{c}_6 + \textcolor{blue}{s}\textcolor{blue}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{blue}{s}\textcolor{blue}{u}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 1616, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 6 & 18 & 12 & 0 & 58 & 10 & 10 & 14 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 10 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 14 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.91 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} + \text{su2n3mix}$

$$T = 11 \quad [0] \quad 1^{23} 1^{13} 1^{111} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{s} \textcolor{brown}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_2 \rightarrow (-3) + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 9 & 5 \\ 14 & 33 & 1 & 0 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 20 & 0 & 0 & 10 & 6 \\ 5 & 11 & -1 & 0 & 6 & 3 \end{pmatrix} b_0$$

4.92 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 0 \\ 2 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 2 & 0 \\ 4 & 5 & 0 & 6 & 9 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 3 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 3 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 328, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 2 & 0 \\ 6 & 8 & 0 & 8 & 22 & 8 & 10 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 3 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 2 & 0 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.93 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{122} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 6 & 12 & 20 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 2 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 3 & 3 \\ 2 & 2 & -1 & -1 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{132} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 3 & -1 \\ 6 & 7 & 0 & 6 & 25 & 18 & 9 \\ 3 & 5 & 0 & 3 & 18 & 11 & 5 \\ 2 & 2 & -1 & -1 & 9 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 3 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 5 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 5 & 3 \\ 2 & 5 & 6 & 0 & 5 & 7 & 5 \\ 2 & 3 & 3 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{113} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 \\ 2 & 2 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

- [8] $1^{11}1^{13}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{11}1^{12}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 3 & -1 \\ 4 & 5 & 0 & 5 & 13 & 13 & 7 \\ 3 & 5 & 0 & 3 & 13 & 11 & 5 \\ 2 & 2 & -1 & -1 & 7 & 5 & -1 \end{pmatrix}_{b_0}$$
- [10] $1^{11}1^{14}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 3 & -1 \\ 4 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -1 & 8 & 5 & -1 \end{pmatrix}_{b_0}$$
- [11] $1^{11}1^{12}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 8 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{11}1^13^11^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 5 & 3 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$
- [13] $1^{11}1^{12}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 6 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{11}1^{12}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 3 & -1 \\ 3 & 5 & 0 & 5 & 13 & 13 & 7 \\ 3 & 5 & 0 & 3 & 13 & 11 & 5 \\ 2 & 2 & -1 & -1 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{11}1^{11}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 5 & 4 & 10 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{11}1^{12}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 7 & 16 & 34 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

- [17] $1^{11} 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -103$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 3 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 3 & 3 \\ 2 & 2 & -1 & -1 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{121} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -118$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -1 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{124} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -120$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & -1 \end{pmatrix}_{b_0}$$
- [20] $1^{121} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 3 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & -1 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{132} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -138$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -1 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$
- [22] $1^{122} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & -1 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$
- [23] $1^{111} 3^1 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -143$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 11 & 7 \\ 2 & 6 & 5 & 0 & 11 & 7 & 5 \\ 2 & 4 & 2 & -4 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{112} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -158$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

- [34] $1^{112}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$
- [35] $1^{131}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 2 & -1 \\ 4 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & -1 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$
- [36] $1^{111}3^11^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 8 & 3 \\ 4 & 12 & 8 & 0 & 8 & 25 & 9 \\ 2 & 4 & 2 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [37] $1^{123}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -247$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{132}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -1 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$
- [39] $1^{112}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [40] $1^{123}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -262$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{141}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 5 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & -1 & 11 & 3 & 0 \end{pmatrix}_{b_0}$$
- [42] $1^{113}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -1 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -1 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 3 & 4 & 0 & 2 & 6 & 10 & 4 \\ 3 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & -1 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 3 & -1 \\ 5 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & -1 & 11 & 5 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{11} 3^1 1^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 17 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 17 & 3 \\ 10 & 30 & 17 & 0 & 17 & 151 & 21 \\ 2 & 4 & 2 & -4 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{14} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 4 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & -1 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.5

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 5 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & -1 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{11} 1^{132} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -366, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 3 & -1 \\ 5 & 7 & 0 & 6 & 25 & 18 & 9 \\ 3 & 5 & 0 & 3 & 18 & 11 & 5 \\ 2 & 2 & -1 & -1 & 9 & 5 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{11} 1^{114} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -370, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{12} 1^{124} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{11} 1^{124} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 6 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{11} 1^{124} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 7 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{11} 1^{131} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 4 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{11} 1^{113} 3^1 1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 11 & 3 \\ 6 & 18 & 11 & 0 & 11 & 55 & 13 \\ 2 & 4 & 2 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{11} 1^{112} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 4 & 2 & -4 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -422, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.19 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 14 & 3 \\ 8 & 24 & 14 & 0 & 14 & 97 & 17 \\ 2 & 4 & 2 & -4 & 3 & 17 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -430, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -1 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -449, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -463, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 5 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -1 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -473, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 5 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -4 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -478, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.0

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 14 & 7 \\ 3 & 9 & 5 & 0 & 14 & 12 & 6 \\ 2 & 4 & 2 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1^{12}} 3^{1^{11}} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 3 & -1 \\ 7 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & -1 & 11 & 5 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1^{11}} 3^{1^{12}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{1^{12}} 3^{1^{11}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 5 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1^{12}} 3^{1^{11}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{1^{12}} 3^{1^{11}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 3 & -1 \\ 6 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -1 & 8 & 5 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{1^{11}} 3^{1^{12}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 3 & -1 \\ 5 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -1 & 8 & 5 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{1^{11}} 3^{1^{11}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -567, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 2 & -1 \\ 5 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & -1 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1^{11}} 3^{1^{11}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{114}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 6 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{142}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 3 & -1 \\ 6 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & -1 & 11 & 5 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{131}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 2 & -1 \\ 6 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & -1 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{113}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -727, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{141}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 6 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -1 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{114}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -763, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{113}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{141}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -769, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 6 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & -1 & 11 & 3 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{111}3^11^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -782, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 20 & 7 \\ 4 & 12 & 8 & 0 & 20 & 25 & 9 \\ 2 & 4 & 2 & -4 & 7 & 9 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{112}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -930, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{111}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1009, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 7 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{112}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1062, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{112}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1068, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 7 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & -1 & 11 & 3 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{112}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 8 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [91] $1^{11}231^13^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 5 & 0 & 5 & 5 \\ 2 & 4 & -1 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{11}23^11^{11}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 9 & 0 & 26 & 10 \\ 2 & 4 & -1 & 0 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{12}3^{11}1^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{12}3^{11}1^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 2 & 0 \\ 2 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{12}3^{11}1^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 21 & 1 & 22 & 7 \\ 21 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 35 & 0 & 33 & 11 \\ 7 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{11}23^{11}1^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 11 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 11 & 0 & 8 & 0 & 21 & 9 \\ 2 & 4 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{11}23^{11}1^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 2 & 0 \\ 4 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{12}3^{11}1^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{12}3^{11}1^{13}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 209, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{11}23^{11}1^{13}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 215, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 9 & 0 & 26 & 10 \\ 2 & 4 & -1 & 0 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{12}3^{11}1^{13}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

[111] $1^1 2 3^1 1^1 2^1 3^1 1^1 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 454$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & 0 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[112] $1^1 2^3 1^1 1^2 3^1 1^1 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{t}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 500$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & 0 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[113] $1^1 23^1 1^1 23^1 1^{12} 3^1 1^{121} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{t}_6 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 586$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 2 & 0 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix} b_0$$

[114] $1^1 2^3 1^1 1^2 3^1 1^3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{t}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 596$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 2 & 0 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{11}23^{11}1231^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 614, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.0$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 12 & 0 & 56 & 14 \\ 2 & 4 & -1 & 0 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{11}23^{11}1231^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 640, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.0$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 12 & 0 & 56 & 14 \\ 2 & 4 & -1 & 0 & 0 & 14 & 1 \end{pmatrix}_b$$

[117] $1^{11}2^{13}1^{12}3^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 688, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.5$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 7 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix} b_0$$

[118] $1^{11}23^11^{12}31^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 896, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.5$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 9 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix} b_0$$

[119] $1^{11}23^11^{12}31^{131} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{s}u_2 + \textcolor{brown}{s}u_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 902, \quad (n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 8 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [120] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 4 1^{\textcolor{red}{1}} 3 1^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [121] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 5 13 1^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 & 1 & 5 & 3 \\ 2 & 5 & 0 & 2 & 5 & 7 & 5 \\ 2 & 3 & -1 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 5 13 1^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 3 & 1 & 1 & 5 & 3 \\ 4 & 0 & 5 & 2 & 5 & 7 & 5 \\ 2 & 0 & 2 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 5 13 1^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 3 & 1 & 1 & 5 & 3 \\ 3 & 0 & 5 & 2 & 5 & 7 & 5 \\ 2 & 0 & 2 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

4.94 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\textcolor{red}{1}} 2 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 9 & 5 \\ 14 & 33 & 1 & 0 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 20 & 0 & 0 & 10 & 6 \\ 5 & 11 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

4.95 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 3 1^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 5 & 0 & 9 & 12 & 5 & 6 \\ 1 & 3 & 3 & 0 & 3 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

4.96 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{14} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 24, T_{\max} = 98, T \in [24, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 11 & 21 & 0 & 18 & 75 & 15 \\ 3 & 4 & -4 & 4 & 15 & -11 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 98, T \in [16, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 3 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 48 & 12 \\ 3 & 4 & -4 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -4 & 7 & 5 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{121}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -4 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -4 & 6 & 5 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 24 & 8 \\ 3 & 4 & -4 & 3 & 8 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 18 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 24 & 14 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -4 & 14 & 5 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{131}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -4 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -4 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{132}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -4 & 9 & 5 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 \\ 3 & 4 & -4 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{132}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 27 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 0 & 133 & 36 & 21 \\ 4 & 8 & 0 & 36 & 9 & 5 \\ 3 & 4 & -4 & 21 & 5 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{141}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -4 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -4 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{151} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 0 & 73 & 15 & 15 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -4 & 16 & 3 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -4 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{161} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 24 & 0 & 106 & 18 & 18 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -4 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -4 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -4 & 10 & 5 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{171}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 14 & 28 & 0 & 145 & 21 & 21 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -4 & 21 & 2 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 57 & 13 \\ 3 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{115}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 487, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -3 & 15 & 0 & -1 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -4 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -4 & 7 & 9 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 48 & 12 \\ 3 & 4 & -4 & 7 & 12 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -4 & 13 & 5 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{132} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 85 & 30 & 17 \\ 4 & 8 & 0 & 30 & 9 & 5 \\ 3 & 4 & -4 & 17 & 5 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -4 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{131} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{131} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{121} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 16 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{141} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -4 & 16 & 6 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{113} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{121} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{122} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -4 & 10 & 9 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114} 31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{141} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{132} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -4 & 13 & 9 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{114} 3 1^{121} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{131} 3 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{131} 3 1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{131} 3 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{141} 3 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{141} 3 1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{141} 3 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [77] $1^{11} 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 0 & -1 \\ 2 & 3 & 0 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 5 & -1 & -4 \end{pmatrix}_{b_0}$$

[79] $1^{11} 3 1^{13} 1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 16, T_{\max} = 68, T \in [16, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 2 & -3 \end{pmatrix}_{b_0}$$

[78] $1^{11} 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{121} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 13 & 4 \\ 5 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 10 & 0 \\ 13 & 15 & 0 & 10 & 38 & 12 \\ 4 & 4 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{12} 31^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 12 & 9 & 0 & 12 & 27 & 9 \\ 2 & 4 & 2 & -4 & 4 & 9 & -3 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{12} 31^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 68, T \in [19, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 12 & 10 \\ 2 & 6 & 3 & 0 & 12 & 3 & 3 \\ 2 & 4 & 2 & -4 & 10 & 3 & -6 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 \\ 5 & 12 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 68, T \in [15, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

- [89] $1^{111} 3^{14} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 9 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [90] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$
- [91] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [92] $1^{121} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [93] $1^{111} 31^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$
- [94] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [95] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [97] $1^{112} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.29 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [99] $1^{121} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

- [98] $1^{111} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{112} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [101] $1^{131} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [102] $1^{141} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [103] $1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -177$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{151} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$
- [106] $1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 12 & 6 \\ 2 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$
- [107] $1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [108] $1^{111} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.0

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 8 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{111} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 7 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{131} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 15 & 0 & 49 & 12 & 13 \\ 1 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 3 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

[114] $1^{112} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 3 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[116] $1^{131} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 11 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{121} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[119] $1^{121} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 9 & 12 & 0 & 28 & 9 & 10 \\ 1 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 3 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{131} 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

[121] $1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -537, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -4 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[122] $1^{121} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 9 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{141} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 14 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$

[124] $1^{121} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{131} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{121} 3^{14} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{11} 31^{111} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -615, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 6 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 2 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{121} 3^{13} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{111} 31^{12} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -762, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 9 & 11 \\ 1 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{121} 3^{12} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 10 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{141} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -771, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{121} 3^{13} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -875, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 11 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{131} 3^{11} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -876, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 12 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{121}1^{14}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -876, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 12 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{111}31^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 12 & 6 \\ 2 & 4 & 2 & -4 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{131}1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -915, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1059, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 12 & 0 & 73 & 12 & 15 \\ 1 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[138] $1^{131}1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{141}1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 15 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{131}1^{12}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 13 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{131}1^{14}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{141}1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{111}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 15 & 0 & 121 & 15 & 19 \\ 1 & 3 & 3 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[144] 1^{131}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 14 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[145] 1^{141}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1643, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[146] 1^{131}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 15 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[147] 1^{141}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 16 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[148] 1^{141}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1920, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[149] 1^{141}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 17 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[150] 1^{141}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 18 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 10 \quad [151] \ 1^{\overline{11}} \ 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 3, \ (n_+, n_-, n_0) = \\
& (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& T_{\min} = 17.33 \rightarrow 18, \ T_{\max} = 24, \ T \in [18, 24] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 0 & 13 & 5 & 7 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & -3 \end{pmatrix}_{b_0} \\
& [152] \ 1^{\overline{11}} \ 4 \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = \\
& 12.0 \\
& T_{\min} = 17.33 \rightarrow 18, \ T_{\max} = 24, \ T \in [18, 24] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & -3 \end{pmatrix}_{b_0} \\
& [153] \ 1^{\overline{1}} \ 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 26, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = \\
& 12.0 \\
& T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24] \\
& \begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0} \\
& [154] \ 1^{\overline{1}} \ 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 27, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = \\
& 12.0 \\
& T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0} \\
& [155] \ 1^{\overline{1}} \ 231^{\overline{1}} \ 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = \\
& 12.0 \\
& T_{\min} = 13.37 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & 0 & 0 & -1 & 0 \end{pmatrix}_{b_0} \\
& [156] \ 1^{\overline{11}} \ 2321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 34, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = \\
& 12.0 \\
& T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 2 & 4 & 0 & 4 & 6 & 4 & 6 \\ 1 & 3 & 0 & 1 & 4 & 0 & 2 \\ 2 & 3 & -1 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0} \\
& [157] \ 1^{\overline{11}} \ 2321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 40, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
& [158] \ 1^{\overline{11}} \ 4 \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 48, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = \\
& 12.0 \\
& T_{\min} = 14.08 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 5 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -3 & 4 & 5 & 0 \end{pmatrix}_{b_0} \\
& [161] \ 1^{\overline{11}} \ 2 \ 3 \ 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 56, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = \\
& 12.0 \\
& T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[159] $1^{11} 3^{221} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[160] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[162] $1^1 231^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 & 0 \\ 4 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[163] $1^{11} 4^{131} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{111} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 7 \\ 5 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

[165] $1^{11} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{141} 3^{221} 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{121} 3^{221} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{112} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{11}1^{11}31^{11}4^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 2 & 3 \\ 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 15 & 0 & 45 & 11 & 13 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[170] $1^{112}2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{121}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 3 & 7 & 0 & 5 & 17 & 7 & 9 \\ 1 & 3 & 0 & 1 & 7 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{111}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 125, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{111}4^{131}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 2 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[175] $1^{131}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 4 & 10 & 0 & 6 & 34 & 10 & 12 \\ 1 & 3 & 0 & 1 & 10 & 0 & 2 \\ 2 & 3 & -1 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{111}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & -3 & 4 & 1 & -1 \\ 3 & 0 & 4 & 4 & 6 & 4 & 6 \\ 2 & 0 & 3 & 1 & 4 & 0 & 2 \\ 2 & 0 & 2 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111} 1^{131} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 158$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111} 23131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 4 & 4 & 0 & 2 & 4 & 4 \\ 1 & 4 & 4 & 0 & 4 & 1 & 3 \\ 2 & 4 & 3 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

[179] $1^1 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.21 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & 0 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[180] $1^{111} 1^{131} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{11} 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 212$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 0 & 0 \\ 4 & 9 & 0 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[182] $1^{121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 215$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{112} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -85$, $\det = 221$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 221$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{121} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 5 & 7 & 0 & 5 & 17 & 7 & 9 \\ 1 & 3 & 0 & 1 & 7 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{111} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 15 & 0 & 45 & 10 & 13 \\ 2 & 4 & 0 & 10 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{111} 4^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{131} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{131} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 3 & 2 \\ 1 & 0 & 1 & -3 & 6 & 1 & -1 \\ 5 & 0 & 10 & 6 & 34 & 10 & 12 \\ 2 & 0 & 3 & 1 & 10 & 0 & 2 \\ 2 & 0 & 2 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 254, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & -3 & 4 & 1 & -1 \\ 4 & 0 & 4 & 4 & 6 & 4 & 6 \\ 2 & 0 & 3 & 1 & 4 & 0 & 2 \\ 2 & 0 & 2 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{131} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 257, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{121} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 269, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 4 & 7 & 0 & 5 & 17 & 7 & 9 \\ 1 & 3 & 0 & 1 & 7 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 271, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[194] 1^{1^{11}31^{11}} 4 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 2 & 3 \\ 4 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[195] 1^{1^{12}2^3 21^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 290, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & 1 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[196] 1^{1^{12}2^3 21^{11}} 3 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 3 & 2 \\ 1 & 0 & 1 & -3 & 5 & 1 & -1 \\ 4 & 0 & 7 & 5 & 17 & 7 & 9 \\ 2 & 0 & 3 & 1 & 7 & 0 & 2 \\ 2 & 0 & 2 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[197] 1^{1^{23}1^{11}31^{11}} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[198] 1^{1^{12}1^{11}2^3 221^{11}} 3 \quad 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[199] 1^{1^{11}1^{12}2^3 21^1} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[200] 1^{1^{11}1^{13}1^{11}} 4 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[201] 1^{1^{14}1^{12}2^3 21^1} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -3 & 4 & 1 & -1 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 2 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{11} 1^{11} 31^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 402, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 3 \\ 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{11} 1^{11} 31^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 2 & 3 \\ 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{11} 231^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 449, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 & 0 \\ 6 & 15 & 0 & 10 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[205] $1^{131} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 479, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 5 & 10 & 0 & 6 & 34 & 10 & 12 \\ 1 & 3 & 0 & 1 & 10 & 0 & 2 \\ 2 & 3 & -1 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{131} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{121} 3^2 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{121} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 586, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{1231} 1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 587, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 & 0 \\ 5 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[210] $1^{131}2^13^{121}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 596, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -3 & 3 & 1 & -1 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 2 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{121}1^{13}3^{11}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 33 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{121}1^{14}3^{11}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 608, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{131}1^{12}2^13^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 614, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 6 & 10 & 0 & 6 & 34 & 10 & 12 \\ 1 & 3 & 0 & 1 & 10 & 0 & 2 \\ 2 & 3 & -1 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{141}1^{11}2^13^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 629, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 6 & 13 & 0 & 7 & 57 & 13 & 15 \\ 1 & 3 & 0 & 1 & 13 & 0 & 2 \\ 2 & 3 & -1 & -1 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{141}1^{11}3^{12}221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 632, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{11}1^{13}231^{13}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 644, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 1 & 3 & 21 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 & 0 \\ 8 & 21 & 0 & 13 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[217] $1^{141}1^{12}2^13^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 650, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{121}1^{121}2^13^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 688, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 3 & 2 \\ 1 & 0 & 1 & -3 & 5 & 1 & -1 \\ 5 & 0 & 7 & 5 & 17 & 7 & 9 \\ 2 & 0 & 3 & 1 & 7 & 0 & 2 \\ 2 & 0 & 2 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{11}231^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 797, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 1 & 3 & 27 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 & 0 \\ 10 & 27 & 0 & 16 & 122 & 0 & 20 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 20 & -1 & 0 \end{pmatrix}_{b_0}$$

[220] $1^1231^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 818, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 & 0 \\ 7 & 18 & 0 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[221] $1^{141}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 3 & 2 \\ 1 & 0 & 1 & -3 & 7 & 1 & -1 \\ 7 & 0 & 13 & 7 & 57 & 13 & 15 \\ 2 & 0 & 3 & 1 & 13 & 0 & 2 \\ 2 & 0 & 2 & -1 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{131}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 902, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 3 & 2 \\ 1 & 0 & 1 & -3 & 6 & 1 & -1 \\ 6 & 0 & 10 & 6 & 34 & 10 & 12 \\ 2 & 0 & 3 & 1 & 10 & 0 & 2 \\ 2 & 0 & 2 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{141}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 7 & 13 & 0 & 7 & 57 & 13 & 15 \\ 1 & 3 & 0 & 1 & 13 & 0 & 2 \\ 2 & 3 & -1 & -1 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[224] $1^1231^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 1007, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 24 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 & 0 \\ 9 & 24 & 0 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[225] $1^{141}3221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1088, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 62 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 62 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{141}3221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 63 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 63 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{141} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 64 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 64 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [228] $1^{13} 5^{13} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 18, T_{\max} = 23, T \in [18, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 3 & 12 & 0 & 4 \\ 2 & 3 & -2 & 6 & 0 & 0 \\ 6 & 12 & 6 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 9 & -1 & -2 \end{pmatrix}_{b_0}$$

[229] $1^{111} 31^{141} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

[230] $1^{12} 5^{13} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 8 & 0 & 4 \\ 2 & 3 & -2 & 4 & 0 & 0 \\ 4 & 8 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{13} 31^{11} 41^{13} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 1 & 5 & 9 & 7 \\ 4 & 6 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{11} 5^{13} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{11} 31^{11} 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -137, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[234] $1^{111} 31^{11} 41^{13} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -251, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 11 & 7 & 0 & 21 & 7 & 9 \\ 1 & 3 & 3 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[235] $1^{111} 31^{11} 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [236] \quad 1^{111} 31^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 12, \quad T_{\max} = 15, \quad \Delta = -86, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.90 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [237] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, \quad T_{\max} = 16, \quad \Delta = -83, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, \quad T_{\max} = 16, \quad \Delta = -83, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 1 & 1 & 3 & 1 & 1 & 3 & 3 \\ 2 & 1 & 3 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{11} 6^{11} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, \quad T_{\max} = 16, \quad \Delta = -83, \quad \det = -34, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.76 \rightarrow 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 4 & 0 & 2 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 1 & 1 & 1 & -3 & 3 & 0 & -1 \\ 3 & 4 & 7 & 3 & 16 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[240] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, \quad T_{\max} = 16, \quad \Delta = -83, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.84 \rightarrow 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[241] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, \quad T_{\max} = 16, \quad \Delta = -83, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.76 \rightarrow 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 3 & 0 & 2 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 2 & 3 & 4 & 2 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[242] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, \quad T_{\max} = 16, \quad \Delta = -83, \quad \det = -112, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.82 \rightarrow 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 4 & 3 & 2 \\ 1 & 1 & 1 & -3 & 2 & 1 & -1 \\ 2 & 3 & 4 & 2 & 5 & 4 & 5 \\ 2 & 1 & 3 & 1 & 4 & 0 & 2 \\ 2 & 2 & 2 & -1 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

4.97 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 4 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.98 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

4.99 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -4 & 3 & 3 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 8 & 14 & 0 & 9 & 36 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -4 & 3 & 10 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1211} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 15 & 6 \\ 5 & 9 & 0 & 12 & 15 & 13 & 7 \\ 3 & 4 & -4 & 6 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -4 & 3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 11 & 19 & 0 & 21 & 12 & 73 & 15 \\ 3 & 4 & -4 & 4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{131} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 17 & 0 & 15 & 48 & 12 & 12 \\ 2 & 4 & 0 & 3 & 12 & 1 & 3 \\ 3 & 4 & -4 & 4 & 12 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 5 & 3 \\ 3 & 4 & 0 & 19 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 19 & 0 & 73 & 12 & 33 & 15 \\ 2 & 4 & 0 & 12 & 0 & 6 & 2 \\ 5 & 9 & 0 & 33 & 6 & 13 & 7 \\ 3 & 4 & -4 & 15 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1211} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 & 12 & 6 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 12 & 6 & 13 & 7 \\ 3 & 4 & -4 & 6 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1112} 3^1 1^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -137, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 7 & 3 \\ 3 & 4 & 0 & 14 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 33 & 11 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 0 & 33 & 9 & 28 & 10 \\ 3 & 4 & -4 & 11 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1211} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -4 & 6 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1212} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 13 & 3 \\ 3 & 4 & 0 & 13 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 9 & 54 & 10 \\ 2 & 4 & 0 & 9 & 0 & 15 & 2 \\ 13 & 23 & 0 & 54 & 15 & 100 & 18 \\ 3 & 4 & -4 & 10 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1211} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 & 9 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -4 & 4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -353, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -4 & 4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -359, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 5 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 4 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 6 & 10 & 6 \\ 3 & 4 & -4 & 3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 11 & 21 & 0 & 15 & 18 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1212} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 & 12 & 6 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 0 & 12 & 6 & 10 & 6 \\ 3 & 4 & -4 & 6 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -425, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -4 & 3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 6 & 12 & 0 & 9 & 9 & 25 & 9 \\ 3 & 4 & -4 & 3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1212} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 5 & 9 & 0 & 12 & 12 & 21 & 6 \\ 7 & 13 & 0 & 18 & 21 & 28 & 10 \\ 3 & 4 & -4 & 6 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 57 & 11 \\ 2 & 4 & 0 & 9 & 0 & 15 & 2 \\ 12 & 22 & 0 & 57 & 15 & 85 & 17 \\ 3 & 4 & -4 & 11 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 8 & 16 & 0 & 12 & 12 & 46 & 12 \\ 3 & 4 & -4 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 7 & 13 & 0 & 9 & 12 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 9 & 17 & 0 & 12 & 15 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1212} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 & 18 & 6 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 7 & 13 & 0 & 18 & 9 & 28 & 10 \\ 3 & 4 & -4 & 6 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 24 & 7 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 8 & 14 & 0 & 24 & 9 & 37 & 11 \\ 3 & 4 & -4 & 7 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -4 & 4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 39 & 11 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 0 & 39 & 9 & 37 & 11 \\ 3 & 4 & -4 & 11 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1112} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -4 & 4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 0 & 5 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 14 & 26 & 0 & 24 & 18 & 118 & 20 \\ 3 & 4 & -4 & 4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 21 & 15 & 85 & 17 \\ 3 & 4 & -4 & 4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -4 & 3 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 12 & 21 & 6 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -4 & 3 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 9 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 33 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 11 & 21 & 0 & 33 & 15 & 76 & 16 \\ 3 & 4 & -4 & 7 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 21 & 7 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 7 & 13 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & -4 & 7 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 48 & 11 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 10 & 18 & 0 & 48 & 12 & 58 & 14 \\ 3 & 4 & -4 & 11 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 27 & 7 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 9 & 17 & 0 & 27 & 12 & 49 & 13 \\ 3 & 4 & -4 & 7 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1112} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 16 & 3 \\ 3 & 4 & 0 & 9 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 48 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 16 & 28 & 0 & 48 & 18 & 154 & 22 \\ 3 & 4 & -4 & 7 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 5 & 9 & 0 & 6 & 12 & 27 & 6 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -4 & 3 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 12 & 33 & 6 \\ 11 & 21 & 0 & 15 & 33 & 76 & 16 \\ 3 & 4 & -4 & 3 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1565, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{1212}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1664, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 9 & 42 & 10 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 10 & 18 & 0 & 42 & 12 & 58 & 14 \\ 3 & 4 & -4 & 10 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{1113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1709, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{1112}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1760, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 10 & 3 \\ 3 & 4 & 0 & 9 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 30 & 7 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 10 & 18 & 0 & 30 & 12 & 58 & 14 \\ 3 & 4 & -4 & 7 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{1112}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1781, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 13 & 3 \\ 3 & 4 & 0 & 9 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 39 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 13 & 23 & 0 & 39 & 15 & 100 & 18 \\ 3 & 4 & -4 & 7 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{1212}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1784, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 9 & 30 & 10 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 0 & 30 & 9 & 28 & 10 \\ 3 & 4 & -4 & 10 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{1113}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1820, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{113}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1865, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 33 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -4 & 4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{114} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 30 & 21 & 172 & 24 \\ 3 & 4 & -4 & 4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1113} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 9 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 36 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 12 & 22 & 0 & 36 & 15 & 85 & 17 \\ 3 & 4 & -4 & 7 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1212} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 12 & 60 & 14 \\ 2 & 4 & 0 & 12 & 0 & 12 & 2 \\ 10 & 18 & 0 & 60 & 12 & 58 & 14 \\ 3 & 4 & -4 & 14 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1114} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -4 & 3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 24 & 3 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1114} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 9 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 42 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 14 & 26 & 0 & 42 & 18 & 118 & 20 \\ 3 & 4 & -4 & 7 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 15 & 3 \\ 3 & 4 & 0 & 9 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 45 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 15 & 27 & 0 & 45 & 18 & 133 & 21 \\ 3 & 4 & -4 & 7 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3365, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 18 & 3 \\ 3 & 4 & 0 & 9 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 54 & 7 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 18 & 32 & 0 & 54 & 21 & 193 & 25 \\ 3 & 4 & -4 & 7 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -4373, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 17 & 3 \\ 3 & 4 & 0 & 9 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 51 & 7 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 17 & 31 & 0 & 51 & 21 & 172 & 24 \\ 3 & 4 & -4 & 7 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -5216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 20 & 3 \\ 3 & 4 & 0 & 9 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 60 & 7 \\ 2 & 4 & 0 & 6 & 0 & 24 & 2 \\ 20 & 36 & 0 & 60 & 24 & 238 & 28 \\ 3 & 4 & -4 & 7 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [80] \quad 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{1111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{1111}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 6 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 6 & 2 \\ 3 & 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{11}31^{121}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 138, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 6 & 12 & 8 \\ 1 & 3 & 3 & 0 & 6 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 12 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 8 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{1111}3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 4 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 6 & 9 & 0 & 7 & 14 & 9 & 8 \\ 4 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111}3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 239, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{1111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 268, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{111}3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 269, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{1112}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 9 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 9 & 2 \\ 4 & 9 & 12 & 0 & 9 & 9 & 28 & 10 \\ 2 & 3 & 3 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[90] $1^{112}3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 300$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 6 \\ 1 & 3 & 3 & 0 & 3 & 0 & 3 \\ 2 & 6 & 3 & 0 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

[92] $1^{111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 320$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 363$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 \\ 2 & 4 & 4 & 0 & 6 & 3 & 9 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 9 \\ 1 & 3 & 3 & 0 & 3 & 0 & 6 \\ 3 & 9 & 6 & 0 & 9 & 6 & 13 \end{pmatrix}_{b_0}$$

[94] $1^{112}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 374$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 3 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 381$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 704$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111}31^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1065$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 5 \\ 2 & 4 & 4 & 0 & 6 & 3 & 15 \\ 1 & 4 & 0 & 0 & 3 & 3 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 15 \\ 1 & 3 & 3 & 0 & 3 & 0 & 9 \\ 5 & 15 & 9 & 0 & 15 & 9 & 37 \end{pmatrix}_{b_0}$$

[98] $1^{111}31^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1659$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 21 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 12 & 2 \\ 7 & 21 & 12 & 0 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & -4 & 4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 2145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 27 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 15 & 2 \\ 9 & 27 & 15 & 0 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & -4 & 4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

$T = 11 \quad [100] \quad 1^{111}4 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 5 & 7 & 0 & 6 & 5 & 8 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{111}31^{11}4 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 5 & 3 \\ 4 & 4 & 0 & 5 & 3 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 \\ 5 & 7 & 0 & 6 & 5 & 8 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.100 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2$

$T = 10 \quad [0] \quad 1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 6 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 24 & 11 \\ 5 & 9 & 0 & 6 & 6 & 24 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1112}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 18 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 2 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

- [4] $1^{\overline{1113}} 3 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 27 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 2 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{\overline{1113}} 3 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 21 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 4 & 2 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{\overline{1114}} 3 1^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 24 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 36 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 2 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{\overline{1111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1112}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 21 & 7 \\ 7 & 13 & 0 & 12 & 9 & 21 & 28 & 10 \\ 3 & 4 & -4 & 4 & 2 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 9 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 15 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 6 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{1122}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 12 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 30 & 10 \\ 7 & 13 & 0 & 12 & 9 & 30 & 28 & 10 \\ 3 & 4 & -4 & 4 & 2 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 30 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 4 & 2 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 15 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 7 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.101 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 6 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 231, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 507, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 651, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 & 3 \\ 5 & 9 & 0 & 6 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1141} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 813, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 & 3 \\ 11 & 21 & 0 & 15 & 18 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1173, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 & 3 \\ 7 & 13 & 0 & 9 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1227, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 & 3 \\ 9 & 17 & 0 & 12 & 15 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.102 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 86, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 2 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 1 & 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 2 & 3 & 3 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 126, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 2 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{121}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 5 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{121}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 146, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{121}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 226, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121}3^{121}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 328, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 8 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{121}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 386, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{121}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 416, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 416, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 6 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 482, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 7 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 626, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{121}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 896, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 914, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 9 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{121}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 1280, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 10 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \ 1^{111}3^{11}15^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, \ T_{\max} = 15, \ \Delta = -58, \ \det = -69, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 & 2 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.103 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2\mathfrak{n}3\mathfrak{mix}$

$$T = 10 \quad [0] \ 1^{111}3^{11}1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 494, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 9 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 9 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

4.104 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{112}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 & 3 \\ 3 & 4 & 0 & 9 & 8 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 12 & 9 & 7 \\ 4 & 8 & 0 & 12 & 9 & 6 & 5 \\ 3 & 5 & 0 & 9 & 6 & 3 & 3 \\ 3 & 4 & -4 & 7 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{114}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & 0 & -1 \\ 13 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -4 & 4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & 0 & -1 \\ 10 & 24 & 9 & 52 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{112}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & 0 & -1 \\ 7 & 16 & 6 & 22 & 0 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 8 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.0

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] 1^{1111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] 1^{122} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 4 & 3 \\ 3 & 4 & 0 & 13 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 12 & 18 & 10 \\ 3 & 5 & 0 & 12 & 3 & 6 & 3 \\ 4 & 8 & 0 & 18 & 6 & 9 & 5 \\ 3 & 4 & -4 & 10 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] 1^{132} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 4 & 3 \\ 3 & 4 & 0 & 17 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 15 & 24 & 13 \\ 3 & 5 & 0 & 15 & 3 & 6 & 3 \\ 4 & 8 & 0 & 24 & 6 & 9 & 5 \\ 3 & 4 & -4 & 13 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[11] 1^{1111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 2 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & -4 & 11 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] 1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & 0 & -1 \\ 11 & 24 & 9 & 52 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[13] 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] 1^{1111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] 1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & 0 & -1 \\ 15 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 7 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & 0 & -1 \\ 14 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{141} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 18 & 15 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 & 3 \\ 2 & 4 & 0 & 15 & 3 & 0 & 2 \\ 3 & 4 & -4 & 16 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 12 & 9 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 & 3 \\ 2 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & -4 & 10 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -933, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 15 & 12 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 & 3 \\ 2 & 4 & 0 & 12 & 3 & 0 & 2 \\ 3 & 4 & -4 & 13 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 12 & 6 \\ 3 & 4 & -4 & 7 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 & 0 \\ 5 & 12 & 7 & 14 & 0 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 477, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 6 \\ 1 & 3 & 3 & 0 & 3 & 0 & 3 \\ 2 & 6 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & -4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 9 & 0 & 13 & 6 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 6 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 774, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 6 & 9 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 7 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \quad 1^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 2 & 3 \\ 4 & 4 & 0 & 4 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 4 & 0 & 1 & 2 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -378, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 2 & 3 \\ 4 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 9 & 0 & 13 & 5 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 3 & 2 \\ 2 & 3 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.105 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{1111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & 3 & -1 \\ 4 & 8 & 3 & 4 & 0 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 6 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 4 & 2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 2 & 2 \\ 3 & 4 & -4 & 4 & 2 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 6 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 18 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 9 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 12 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.106 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 6 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 & 2 \\ 3 & 4 & -4 & 7 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.107 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{\overline{1}} 3 1^{\overline{111}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 98, \ \Delta = -143, \ \det = 925, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 1 & 4 \\ 8 & 9 & 4 & 8 & 0 & 5 \\ 3 & 4 & -3 & 3 & 0 & -1 \\ 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

4.108 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 21^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 22, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & -2 & 2 & 0 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 13^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 34, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & -2 & 2 & 0 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -144, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.09 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 13^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.09 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & -2 & 2 & 0 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 46, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -144, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^1 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -144, \quad \det = 106, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 21^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 170, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.60 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & -2 & 2 & 0 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 1^{11} 23^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, \quad T_{\max} = 32, \quad T \in [16, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 2 & 4 \\ 1 & 5 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 3 & -4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 23^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, \quad T_{\max} = 32, \quad T \in [16, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 2313^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 29, \quad \Delta = -116, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$\begin{aligned}
& T_{\min} = 19, T_{\max} = 29, T \in [19, 29] \\
& \begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & -3 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{11}23^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14, T_{\max} = 32, T \in [14, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & -2 & 2 & 0 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{11}2313^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.38 \rightarrow 14, T_{\max} = 29, T \in [14, 29] \\
& \begin{pmatrix} 5 & 1 & 1 & 5 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 5 & 0 & -1 & 2 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{11}23^{11}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.39 \rightarrow 14, T_{\max} = 32, T \in [14, 32] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{11}23^{11}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13, T_{\max} = 32, T \in [13, 32] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 0 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{11}231^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 29, T \in [13, 29] \\
& \begin{pmatrix} 5 & 14 & 1 & 5 \\ 14 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 5 & 11 & -1 & 3 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{11}2321^{11}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 36, \Delta = -118, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 36, T \in [13, 36] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0} \\
& T = 10 \quad [17] \quad 14^{11}13^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 2 & 0 & 3 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & -2 & 0 & 1 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{11}31^{11}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{11}3^{11}1^{11}4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.93 \rightarrow 14, T_{\max} = 26, T \in [14, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$T = 11 \quad [20] \quad 1^{\overline{13}1} 1^{\overline{15}131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{\overline{13}1} 1^{\overline{15}131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -29, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{\overline{13}151^1} 2^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -115, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 5 & 22 & 1 & 5 \\ 22 & 86 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 5 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [23] \quad 1^{\overline{13}1^{21}} 6^{\overline{1}131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.48 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{13}1^{31}} 6^{\overline{1}131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 28, \Delta = -114, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.94 \rightarrow 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [25] \quad 1^{\overline{13}1^{15}131^1} 6^{\overline{1}131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 17.50 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.109 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{11}1^{11}} 3^{\overline{1}1} 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 1 & 0 & 2 \\ 3 & 3 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

4.110 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{13}1^{21}} 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.69 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 5 & 3 & 0 & 5 \\ 2 & 5 & -2 & 2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [3] $1^{11}23^{11}1^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 1 & 0 & 2 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 4 & 0 & 2 & 0 \\ 2 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [4] $1^{11}4^{11}13^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [5] $1^{11}13^{11}1^{11}5^{11}131^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 10 & 1 & 4 & 8 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 6 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

4.111 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{brown}{\mathfrak{su}}_2$

$T = 9$ [0] $1^{11}3^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & -2 \end{pmatrix}_{b_0}$$

[1] $13^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & -2 & 2 & 3 & 0 \\ 2 & 3 & 2 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 5 & 3 & 0 & 5 \\ 2 & 5 & -2 & 2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 2 & 0 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{12} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 3 & 2 & 0 & 4 & 2 & 4 \\ 3 & 3 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{12} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 2 & 0 & 4 & 2 & 4 \\ 3 & 3 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{12} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -58$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 4 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^11^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{11}3^11^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -87$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 \\ 2 & 4 & 2 & -4 & 2 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^1131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 2 & 6 & 3 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -124$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 6 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{13}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -170$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 8 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -196$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 12 & 0 \\ 10 & 14 & 0 & 12 & 66 & 16 \\ 3 & 3 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -202$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -210, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 5 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{13} 3^1 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -257, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.84 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -282, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{12} 3^1 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -297, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 4 & 12 & 9 & 0 & 7 & 28 & 10 \\ 2 & 4 & 2 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -312, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.59 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 6 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -372, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.59 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 7 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{13}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & 2 & -4 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^11^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 9 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{12}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{14}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 8 & 24 & 18 & 0 & 14 & 118 & 20 \\ 2 & 4 & 2 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{12}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 8 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{13}3^11^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 6 & 18 & 9 & 0 & 9 & 52 & 12 \\ 2 & 4 & 2 & -4 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 10 & 30 & 15 & 0 & 15 & 148 & 20 \\ 2 & 4 & 2 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{12}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1737, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 13 & 2 \\ 8 & 24 & 15 & 0 & 13 & 100 & 18 \\ 2 & 4 & 2 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{14}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1857, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 17 & 2 \\ 10 & 30 & 21 & 0 & 17 & 172 & 24 \\ 2 & 4 & 2 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{11}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1887, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 9 & 27 & 15 & 0 & 14 & 121 & 19 \\ 2 & 4 & 2 & -4 & 2 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{13}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1922, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 9 & 27 & 18 & 0 & 15 & 133 & 21 \\ 2 & 4 & 2 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{12}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 16 & 2 \\ 10 & 30 & 18 & 0 & 16 & 154 & 22 \\ 2 & 4 & 2 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{13}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3047, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 33 & 4 \\ 1 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 18 & 2 \\ 11 & 33 & 21 & 0 & 18 & 193 & 25 \\ 2 & 4 & 2 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{14}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 36 & 4 \\ 1 & 4 & 0 & 0 & 2 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 20 & 2 \\ 12 & 36 & 24 & 0 & 20 & 238 & 28 \\ 2 & 4 & 2 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [53] \ 1^1 231^1 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 5 \\ 14 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 11 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{11} 231^1 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 29 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{12}31^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 14 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{11}231^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 27 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{12}231^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 16 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11}23^{11}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 7 & 0 \\ 1 & 3 & 0 & 2 & 0 & 5 & 2 \\ 4 & 9 & 0 & 7 & 5 & 14 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{12}23^{11}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 3 & 0 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{12}231^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 15 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11}231^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 28 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{12}23^{11}1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 504, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 6 & 0 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{11}23^{11}1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 10 & 0 \\ 1 & 3 & 0 & 2 & 0 & 8 & 2 \\ 7 & 15 & 0 & 10 & 8 & 38 & 12 \\ 2 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [64] \quad 1^{11} 3^1 1^1 4 13 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -66, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 3 \\ 2 & 4 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [65] \quad 1^{11} 3^1 1^1 5 13 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 3^1 1^1 5 13 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 0 & 1 & -3 & 1 & 0 & -1 \\ 1 & 0 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 3^1 1^1 5 13 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 3 & 0 & 1 & 2 & 1 & 3 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 3^1 1^1 5 13 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 59, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 4 & 0 & 3 & 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{11} 3^1 1^1 5 13 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.55 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 2 & 0 & 3 & 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{12} 3^1 1^1 5 13 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 3 & 0 & 6 & 2 & 4 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 3^1 1^1 5 13 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 3 & 0 & 3 & 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 218, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 4 & 0 & 6 & 2 & 4 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{14}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 234, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 6 & 0 & 12 & 4 & 8 & 46 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{13}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 260, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 1 & -3 & 1 & 3 & -1 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 5 & 0 & 9 & 3 & 6 & 25 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{12}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 275, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 5 & 0 & 6 & 2 & 4 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{13}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 423, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 1 & -3 & 1 & 3 & -1 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 6 & 0 & 9 & 3 & 6 & 25 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{14}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 503, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 7 & 0 & 12 & 4 & 8 & 46 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

4.112 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111}3^1 1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 5 & 6 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^1 1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 15 \\ 1 & 4 & 0 & 0 & 2 & 3 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 2 & 8 \\ 1 & 3 & 3 & 0 & 2 & 1 & 9 \\ 5 & 15 & 9 & 0 & 8 & 9 & 37 \\ 2 & 4 & 2 & -4 & 2 & 3 & 11 \end{pmatrix}$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 3 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 4 & 4 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 9 \\ 1 & 4 & 0 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 5 & 2 \\ 2 & 6 & 3 & 0 & 0 & 3 & 9 & 4 \\ 3 & 9 & 6 & 0 & 5 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_b$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 5 & 13 & 7 & 7 \\ 2 & 4 & 2 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 2 \\ 1 & 3 & 2 & 0 & 2 & 2 & 0 \\ 2 & 4 & 2 & -4 & 2 & 3 & 2 \end{pmatrix} b_0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 0 & 2 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 & 2 & 2 \\ 2 & 6 & 3 & 0 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & -4 & 2 & 4 & 2 & 0 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 & 2 \\ 4 & 12 & 9 & 0 & 7 & 28 & 7 & 10 \\ 1 & 3 & 2 & 0 & 2 & 7 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 10 & 2 & 0 \end{pmatrix}$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 6 & 2 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 8 & 2 & 0 \end{pmatrix}$$

2051

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{12}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 & 2 \\ 5 & 15 & 9 & 0 & 8 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}3^{11}1^{12}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1616, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 & 2 \\ 6 & 18 & 12 & 0 & 10 & 58 & 10 & 14 \\ 1 & 3 & 2 & 0 & 2 & 10 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

4.114 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + su2n3mix

$$T = 11 \quad [0] \ 1^{12}3^{11}1^{13}1^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 9 & 5 \\ 14 & 33 & 1 & 0 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 20 & 0 & 0 & 10 & 6 \\ 5 & 11 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

4.115 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{13}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 3 & 3 \\ 1 & 3 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{13}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 2 & 0 \\ 4 & 5 & 0 & 6 & 9 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 3 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 2 & 0 \\ 5 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & 0 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 2 & 0 \\ 6 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & 0 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 2 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 2 & 0 \\ 7 & 8 & 0 & 8 & 22 & 8 & 10 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 3 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 2 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 2 & 0 \\ 8 & 8 & 0 & 8 & 22 & 8 & 10 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 3 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{11} 3^1 15 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 3 & 0 & 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 1 & 2 & 3 & 0 \\ 2 & 3 & -1 & -1 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.116 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\text{su}_2 \rightarrow (-3)$ + $\text{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^3 1^1 1^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \quad 1^2 313^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 1 & 6 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 0 & 0 & 4 & 4 \\ 5 & 0 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^2 3^1 1^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 8 & 5 \\ 14 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 10 & 6 \\ 5 & 11 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1231}1^{1311}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 6 & 5 \\ 14 & 33 & 1 & 14 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 14 & 0 & 4 & 4 \\ 5 & 11 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

4.117 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{113}1^{11}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 2 & 1 \\ 2 & 4 & 2 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13}1^{11}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 436, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 6 \\ 1 & 4 & 0 & 0 & 2 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 4 \\ 2 & 4 & 2 & -4 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{123}1^{11}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 7 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 7 & 2 \\ 4 & 12 & 9 & 0 & 7 & 7 & 28 & 10 \\ 2 & 4 & 2 & -4 & 2 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}1^{11}1^{1231}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 544, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 6 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & 2 & -4 & 2 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113}1^{11}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 704, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113}1^{11}1^{1231}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 8 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 8 & 2 \\ 5 & 15 & 9 & 0 & 8 & 8 & 37 & 11 \\ 2 & 4 & 2 & -4 & 2 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123}1^{11}1^{1231}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1616, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 3 & 2 & 0 & 0 & 2 & 10 & 2 \\ 6 & 18 & 12 & 0 & 10 & 10 & 58 & 14 \\ 2 & 4 & 2 & -4 & 2 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

4.118 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 4 & 2 \\ 2 & 6 & 5 & 0 & 4 & 7 & 5 \\ 2 & 4 & 2 & -4 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -389, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 5 & 15 & 11 & 0 & 9 & 44 & 12 \\ 2 & 4 & 2 & -4 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 4 & 12 & 8 & 0 & 7 & 25 & 9 \\ 2 & 4 & 2 & -4 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -668, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 15 & 8 & 0 & 8 & 36 & 10 \\ 2 & 4 & 2 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1091, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 6 & 18 & 11 & 0 & 10 & 55 & 13 \\ 2 & 4 & 2 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 2 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 7 & 21 & 14 & 0 & 12 & 80 & 16 \\ 2 & 4 & 2 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \quad 1^{11}231^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 19 & 5 \\ 14 & 33 & 1 & 44 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 44 & 0 & 58 & 14 \\ 5 & 11 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -89, \ \det = 60, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 7 & 5 \\ 14 & 33 & 1 & 14 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 14 & 0 & 4 & 4 \\ 5 & 11 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -89, \ \det = 85, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 1 & 9 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 0 & 0 & 10 & 6 \\ 5 & 0 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -89, \ \det = 150, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 5 \\ 14 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 10 & 6 \\ 5 & 11 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.119 NHC(-2)(-3) + $\textcolor{violet}{\mathfrak{e}}_6$ + $\textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3)$ + su2n3mix + $\textcolor{blue}{\mathfrak{su}}_2$

$$T = 11 \quad [0] \ 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -100, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 1 & 5 \\ 14 & 33 & 1 & 20 & 0 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 20 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 11 & -1 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

4.120 NHC(-2)(-3) + $\textcolor{violet}{\mathfrak{e}}_6$ + $\textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3)$ + $\textcolor{blue}{\mathfrak{su}}_3$

$$T = 9 \quad [0] \ 1^{131}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.36 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 5 & 3 & 0 & 5 \\ 2 & 5 & -2 & 2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1131}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.11 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1131}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1131}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 \\ 2 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{13}1^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -363, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [8] $1^{12}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{12}31^{11}1^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 5 \\ 14 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 11 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [10] $1^{11}3^{11}1^{11}41^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}1^{11}4131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -149, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 12$ [12] $1^{11}3^{11}1^{11}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11}3^{11}15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 3 & 0 & 1 & 2 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.121 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 & 3 \\ 1 & 3 & 3 & 0 & 2 & 3 & 1 \\ 2 & 3 & 3 & -4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 2 & 3 & 1 & 3 \\ 2 & 4 & 2 & -4 & 2 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 9 & 7 \\ 3 & 3 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 3 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 \\ 1 & 3 & 3 & 0 & 2 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 4 \\ 2 & 4 & 2 & -4 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 328, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 8 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 8 & 0 & 8 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{11}3^{11}15^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 13, \quad T_{\max} = 15, \quad \Delta = -58, \quad \det = -69, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 3 & 0 & 1 & 2 & 0 & 3 \\ 1 & 3 & 0 & 1 & 2 & 3 & 1 \\ 2 & 3 & -1 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.122 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3$

$$T = 10 \quad [0] \quad 1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 16, \quad \Delta = -96, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 16, \quad T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 249, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 & 3 \\ 1 & 3 & 3 & 0 & 2 & 3 & 0 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 489, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 6 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 \\ 1 & 3 & 3 & 0 & 2 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.123 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$

$$T = 9 \quad [0] \quad 1^{11}23^{11}1^{12}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 36, \quad \Delta = -118, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, \quad T_{\max} = 36, \quad T \in [13, 36]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 1 & 4 \\ 5 & 5 & 5 & 3 & 0 & 5 \\ 3 & 5 & -2 & 2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

4.124 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \text{n3mix}$

$$T = 8 \quad [0] \quad 1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, \quad T_{\max} = 98, \quad T \in [19, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 2 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & -1 & 4 & -6 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -144, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -4 & 5 & 0 \end{pmatrix}_{b_0}$$

- [2] $1^{11}3^11^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 0 & 25 & 9 \\ 2 & 3 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$
- [3] $1^{11}3^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 3 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$
- [4] $1^{13}3^11^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{11}3^11^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 6 & 0 & 7 & 5 \\ 2 & 3 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & -2 & 0 \\ 4 & 3 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{12}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 9 & 4 & 1 & 4 & 5 \\ 4 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 2 & 0 & -2 & 0 \\ 5 & 3 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{11}3^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 3 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{11}3^11^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 5 & 0 & 7 & 5 \\ 2 & 3 & 2 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$
- [11] $1^{12}3^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$
- [12] $1^{12}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 5 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 289, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3^{11} 1^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 5 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{14} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 6 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 354, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{13} 3^{11} 1^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{14} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 6 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 6 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{13} 3^{12} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{14} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 610, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 7 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{13} 3^{11} 1^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 673, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{13} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 676, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{12} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 714, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 8 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

- [43] $1^1 3^1 1^1 1^4 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 970$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^1 1^3 1^1 1^3 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 1016$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^1 2^3 1^1 1^3 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 1254$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 17 & 0 & 128 & 20 \\ 2 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$
- [46] $1^1 4^3 1^1 1^3 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 1334$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 23 & 0 & 208 & 26 \\ 2 & 4 & 2 & -4 & 26 & 0 \end{pmatrix}_{b_0}$$
- [47] $1^1 3^3 1^1 1^3 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 1360$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 20 & 0 & 165 & 23 \\ 2 & 4 & 2 & -4 & 23 & 0 \end{pmatrix}_{b_0}$$
- [48] $1^1 1^1 3^1 1^1 1^4 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 1484$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 17 & 0 & 151 & 21 \\ 2 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$
- [49] $1^1 2^3 1^1 1^4 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 1866$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 20 & 0 & 188 & 24 \\ 2 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$
- [50] $1^1 3^3 1^1 1^4 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 2116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 23 & 0 & 231 & 27 \\ 2 & 4 & 2 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^1 4^3 1^1 1^4 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 2234$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 39 & 4 \\ 1 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 39 & 26 & 0 & 280 & 30 \\ 2 & 4 & 2 & -4 & 30 & 0 \end{pmatrix}_{b_0}$$
- $T = 9$ [52] $1^1 2^3 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 17.29 \rightarrow 18$, $T_{\max} = 32$, $T \in [18, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 13 & 0 & 48 & 14 \\ 2 & 4 & 3 & -4 & 14 & -4 \end{pmatrix}_{b_0}$$
- [53] $1^1 1^1 2^3 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13$, $T_{\max} = 29$, $T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 2 & 4 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

- [54] $1^{1231}1^31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 13 & 21 & 1 & 7 \\ 21 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 7 & 11 & -1 & 3 \end{pmatrix}_{b_0}$
- [55] $1^{1231}1^31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 6 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$
- [56] $1^{1231}1^31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -37$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 4 & 0 \\ 4 & 0 & 0 & 4 & 9 & 5 \\ 2 & 0 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$
- [57] $1^{111}23^1131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 6 & 4 \\ 1 & 5 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 6 & 0 & 8 & 6 \\ 2 & 4 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$
- [58] $1^{123}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 2 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$
- [59] $1^{11}23^1131^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.12 \rightarrow 15$, $T_{\max} = 32$, $T \in [15, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 9 & 0 \\ 7 & 0 & 0 & 9 & 26 & 10 \\ 2 & 0 & -1 & 0 & 10 & -1 \end{pmatrix}_{b_0}$
- [60] $1^{12313}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 13 & 1 & 1 & 7 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 7 & 0 & -1 & 2 \end{pmatrix}_{b_0}$
- [61] $1^{123}1^131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$
- [62] $1^{11}231^{13}1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -87$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 11 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 \\ 8 & 11 & 0 & 8 & 21 & 9 \\ 2 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$
- [63] $1^{113}23^1131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -95$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 2 & 5 & 0 & 17 & 4 \\ 1 & 5 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 17 & 17 & 0 & 78 & 18 \\ 2 & 4 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$
- [64] $1^{11}231^{13}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -95$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 11 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 \\ 4 & 11 & 0 & 8 & 21 & 9 \\ 2 & 4 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$
- [65] $1^{123}1^131^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -97$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

- [66] $1^{123}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.07 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 5 & 0 \\ 5 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [67] $1^{123}1^{11}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 2 & 0 \\ 3 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$
- [68] $1^{112}23^1131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 5 & 0 & 10 & 4 \\ 1 & 5 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 10 & 10 & 0 & 24 & 10 \\ 2 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$
- [69] $1^{112}23^11^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 6 & 18 & 0 & 12 & 56 & 14 \\ 2 & 4 & -1 & 0 & 14 & 0 \end{pmatrix}_{b_0}$$
- [70] $1^{112}23^11^{23}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 12 & 0 \\ 10 & 0 & 0 & 12 & 56 & 14 \\ 2 & 0 & -1 & 0 & 14 & -1 \end{pmatrix}_{b_0}$$
- [71] $1^{123}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 \\ 3 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [72] $1^{123}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$
- [73] $1^{112}23^1131^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 5 & 0 & 0 & 6 & 8 & 6 \\ 2 & 0 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [74] $1^{123}1^{23}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 8 & 0 \\ 8 & 0 & 0 & 8 & 36 & 10 \\ 2 & 0 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$
- [75] $1^{13}23^11^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 20 & 0 & 120 & 22 \\ 2 & 4 & 3 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{1231}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

- [77] $1^{11}23^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -180$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.24 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 3 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$
- [78] $1^{11}231^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -183$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 4 & 12 & 9 & 0 & 26 & 10 \\ 2 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$
- [80] $1^{11}23^{11}1^{11}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -183$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [79] $1^{11}23^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -183$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 5 & 12 & 0 & 9 & 26 & 10 \\ 2 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [81] $1^{11}23^{11}1^{11}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 8 & 12 & 0 & 9 & 26 & 10 \\ 2 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [82] $1^{123}1^{13}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -205$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.68 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 11 & 0 \\ 11 & 0 & 0 & 11 & 72 & 14 \\ 2 & 0 & -1 & 0 & 14 & 0 \end{pmatrix}_{b_0}$$
- [83] $1^{113}23^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 14 & 4 \\ 1 & 5 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 14 & 14 & 0 & 48 & 14 \\ 2 & 4 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$
- [84] $1^{123}1^{13}131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 2 & 0 \\ 4 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$
- [85] $1^{1231}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$
- [86] $1^{123}1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.19 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 \\ 5 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$
- [87] $1^{11}231^{11}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 8 & 18 & 0 & 12 & 56 & 14 \\ 2 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{11}23^11^{12}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -803$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 9 & 18 & 0 & 12 & 56 & 14 \\ 2 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{11}23^11^{12}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -816$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 2 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [112] $1^{11}3^11^14131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 32$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{12}3^11^141^131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 55$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{11}3^11^141^131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 68$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[115] $1^131^{11}413^11^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 76$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 1 & 3 & 5 & 5 \\ 2 & 4 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[116] $1^131^{11}41^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 115$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 4 & -1 \\ 5 & 8 & 0 & 4 & 10 & 6 \\ 2 & 4 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[117] $1^131^{11}413^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 135$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.34 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [118] $1^{11}3^11^15131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -3$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 23$, $T \in [19, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 2 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & -1 & 5 & -3 \end{pmatrix}_{b_0}$$

[119] $1^{12}3^11^15131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 8 & 0 & 3 & 20 & 8 \\ 2 & 3 & -1 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{11}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -29$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.59 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 5 & 0 & 2 & 7 & 5 \\ 2 & 3 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{13}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -48$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 15.58 \rightarrow 16$, $T_{\max} = 23$, $T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 5 & 0 & 11 & 4 & 39 & 11 \\ 2 & 0 & 2 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{13}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -60$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{13}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -83$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{12}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -92$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 8 & 3 & 20 & 8 \\ 2 & 0 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{13}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -96$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 23$, $T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 4 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[126] $1^{11}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -96$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{11}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -179$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.65 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{14}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -227$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.83 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 7 & 0 & 14 & 5 & 64 & 14 \\ 2 & 0 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{12}3^11^5131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -235, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 8 & 3 & 20 & 8 \\ 2 & 0 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13}3^11^5131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -251, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 6 & 0 & 11 & 4 & 39 & 11 \\ 2 & 0 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{11}3^11^5131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -252, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 5 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{12}3^11^5131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 6 & 0 & 8 & 3 & 20 & 8 \\ 2 & 0 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{13}3^11^5131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -444, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 7 & 0 & 11 & 4 & 39 & 11 \\ 2 & 0 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{14}3^11^5131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -480, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 8 & 0 & 14 & 5 & 64 & 14 \\ 2 & 0 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

$T = 12$ [135] $1^{11}3^{121}6^1131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 19.33 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$$

$$\begin{pmatrix} 13 & 4 & 1 & 7 \\ 4 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 7 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{13}3^{121}6^1131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 4 & 2 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{11}3^{1131}6^1131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 28, \Delta = -114, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 3 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{11} 6^{1^{12} 3^{12} 21^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{11} 6^{1^{12} 3^{12} 21^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 6 & 2 & 2 & 2 & 8 & 6 \\ 2 & 2 & 0 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{11} 6^{1^{12} 3^{12} 21^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.70 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 5 & 2 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{11} 6^{1^{12} 3^{12} 21^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.78 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 4 & 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{11} 6^{1^{12} 3^{12} 21^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 96, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.62 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{11} 6^{1^{12} 3^{12} 21^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 147, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.63 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 4 & 0 & 3 & 10 & 6 \\ 2 & 2 & 0 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{11} 6^{1^{12} 3^{12} 21^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 243, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 6 & 4 & 1 & 3 & 11 & 7 \\ 2 & 2 & 0 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{11} 6^{1^{12} 3^{12} 21^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 299, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 4 & 2 & 3 & 14 & 8 \\ 2 & 2 & 0 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$T = 15$ [146] $1^{11} 6^{1^{12} 3^{12} 15131} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 17.63 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 6 & 5 & 1 & 14 & 8 \\ 3 & 3 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{11} 1^{13} 1^{15} 1^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 15, T_{\max} = 18, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 17.83 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

4.125 NHC(-2)(-3) + e₆ + su₂n₃mix + su₂

$T = 9$ [0] $1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 2 & -1 \\ 4 & 6 & 0 & 6 & 20 & 12 & 8 \\ 2 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -1 & 8 & 4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 3 & 5 & 3 \\ 2 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 2 & -1 & -1 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 5 & 0 & 3 & 11 & 18 & 5 \\ 6 & 7 & 0 & 6 & 18 & 25 & 9 \\ 2 & 2 & -1 & -1 & 5 & 9 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 3 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 5 & 6 & 0 & 7 & 5 & 5 \\ 1 & 3 & 3 & 0 & 5 & 1 & 3 \\ 2 & 3 & 3 & -4 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 2 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -1 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{124}3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 4 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -1 & 5 & 8 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{122}3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 5 & -1 \\ 3 & 5 & 0 & 3 & 11 & 13 & 5 \\ 4 & 5 & 0 & 5 & 13 & 13 & 7 \\ 2 & 2 & -1 & -1 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 3 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{142}3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 2 & -1 \\ 8 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & -1 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{132}3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 6 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -1 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{113}3^1 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 0 & 5 & 1 & 3 \\ 2 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{122}3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 5 & -1 \\ 3 & 5 & 0 & 3 & 11 & 13 & 5 \\ 3 & 5 & 0 & 5 & 13 & 13 & 7 \\ 2 & 2 & -1 & -1 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{132}3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 6 & 8 & 0 & 7 & 34 & 16 & 10 \\ 2 & 4 & 0 & 2 & 16 & 6 & 4 \\ 2 & 2 & -1 & -1 & 10 & 4 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 3 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 3 & 5 & 3 \\ 3 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 2 & -1 & -1 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -1 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{142}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 3 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -1 & 9 & 4 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{112}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 3 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & -1 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{123}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 3 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -1 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 4 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -1 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{111}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 5 & 0 & 7 & 11 & 5 \\ 3 & 9 & 6 & 0 & 11 & 13 & 7 \\ 2 & 4 & 2 & -4 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{121}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 2 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 5 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 4 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 6 & 2 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -1 & 5 & 4 & 0 \end{pmatrix}_b$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 4 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 5 & 8 & 0 & 7 & 34 & 16 & 10 \\ 2 & 4 & 0 & 2 & 16 & 6 & 4 \\ 2 & 2 & -1 & -1 & 10 & 4 & 0 \end{pmatrix}_b$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 5 & 0 & 3 & 11 & 18 & 5 \\ 4 & 7 & 0 & 6 & 18 & 25 & 9 \\ 2 & 2 & -1 & -1 & 5 & 9 & 0 \end{pmatrix}_b$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{11} 1^{13} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 4 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{121} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{111} 3^{11} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 8 & 9 \\ 1 & 3 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & 2 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{132} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 4 & 7 & 0 & 4 & 23 & 14 & 7 \\ 3 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -1 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{123} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 5 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -1 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{121} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 4 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{132} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -262, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 4 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -1 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{114} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 5 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & -1 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

- [42] $1^{131}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -1 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [43] $1^{112}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -298$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -1 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^{112}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -303$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -1 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{124}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 5 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & -1 & 5 & 11 & 0 \end{pmatrix}_{b_0}$$
- [46] $1^{122}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 3 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -1 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$
- [47] $1^{132}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -322$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 5 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -1 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$
- [48] $1^{111}3^{114}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -323$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 17 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 30 & 17 & 0 & 151 & 17 & 21 \\ 1 & 3 & 3 & 0 & 17 & 1 & 3 \\ 2 & 4 & 2 & -4 & 21 & 3 & -1 \end{pmatrix}_{b_0}$$
- [49] $1^{142}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -340$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -1 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$
- [50] $1^{112}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -345$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 4 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & -1 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1^{12}} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 5 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & -1 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1^{11}} 1^{23} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -366, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 5 & 0 & 3 & 11 & 18 & 5 \\ 5 & 7 & 0 & 6 & 18 & 25 & 9 \\ 2 & 2 & -1 & -1 & 5 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1^{141}} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -370, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 5 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -1 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1^{12}} 1^{42} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 7 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -1 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1^{142}} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 2 & -1 \\ 6 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & -1 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1^{11}} 1^{42} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 2 & -1 \\ 7 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & -1 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1^{11}} 1^{13} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 4 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1^{11}} 1^{13} 3^1 1^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 11 & 0 & 55 & 11 & 13 \\ 1 & 3 & 3 & 0 & 11 & 1 & 3 \\ 2 & 4 & 2 & -4 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

- [58] $1^{1^{11}} 1^{121} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -395$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$
- [60] $1^{1^{11}} 3^1 1^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -412$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 12 & 9 \\ 2 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 4 & 2 & -4 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$
- [61] $1^{1^{11}} 3^1 1^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -422$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.19 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 24 & 14 & 0 & 97 & 14 & 17 \\ 1 & 3 & 3 & 0 & 14 & 1 & 3 \\ 2 & 4 & 2 & -4 & 17 & 3 & -1 \end{pmatrix}_{b_0}$$
- [62] $1^{1^{11}} 1^{131} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -430$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 5 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & -1 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$
- [63] $1^{1^{11}} 1^{123} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -438$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 4 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -1 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$
- [64] $1^{1^{11}} 1^{121} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -449$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 5 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$
- [65] $1^{1^{11}} 1^{114} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -463$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 5 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -1 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$
- [66] $1^1 3^1 1^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -473$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -4 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$
- [67] $1^{1^{12}} 1^{121} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -478$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -483$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 14 & 6 \\ 3 & 9 & 6 & 0 & 14 & 13 & 7 \\ 2 & 4 & 2 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 7 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & -1 & 5 & 11 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -508$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 6 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -1 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -528$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 5 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -544$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 6 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -548$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 6 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -1 & 5 & 8 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -548$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 5 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -1 & 5 & 8 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -567$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 5 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

- [76] $1^{1^{11}} 1^{131} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -577$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 5 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -1 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [77] $1^{1^{11}} 1^{141} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -604$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 1 & -1 \\ 6 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & -1 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$
- [78] $1^{1^{11}} 1^{124} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -644$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 6 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & -1 & 5 & 11 & 0 \end{pmatrix}_{b_0}$$
- [79] $1^{1^{12}} 1^{113} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -726$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 6 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [80] $1^{1^{11}} 1^{131} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -727$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 6 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & -1 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$
- [81] $1^{1^{12}} 1^{114} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -738$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 6 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -1 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$
- [82] $1^{1^{11}} 1^{141} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -763$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 6 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -1 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$
- [83] $1^{1^{12}} 1^{131} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -768$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 6 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -1 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [84] $1^{1^{11}} 1^{114} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -769$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 6 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & -1 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{111}3^{11}1^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -782$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 12 & 9 & 4 \\ 1 & 4 & 0 & 0 & 8 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 20 & 9 \\ 3 & 9 & 6 & 0 & 20 & 13 & 7 \\ 2 & 4 & 2 & -4 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{112}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -930$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 7 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & -1 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{111}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -1009$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 1 & -1 \\ 7 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & -1 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{112}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -1062$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 7 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -1 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{112}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -1068$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 7 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & -1 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{112}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -1320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 1 & -1 \\ 8 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & -1 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [91] $1^{11}231^{13}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 5 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 & 0 \\ 3 & 5 & 0 & 5 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{11}23^{11}1^{11}31^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 31$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

[illegible]

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[111] $1^1 2^3 3^1 1^{12} 3^1 1^{21} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{c}_6 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 & 0 \\ 7 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 10 & 0 & 1 \end{pmatrix}$$

[112] $1^1 2^1 3^1 1^1 2^1 3^1 1^1 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 500$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 & 0 \\ 8 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 10 & 0 & 1 \end{pmatrix}$$

[113] $1^1 2^3 1^1 1^{12} 3^1 1^{112} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{red}{c}_6 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 586$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{\textcolor{teal}{1}}2^{\textcolor{teal}{1}}3^{\textcolor{teal}{1}}1^{\textcolor{teal}{12}}3^{\textcolor{teal}{11}}3^{\textcolor{teal}{11}} + [\textcolor{red}{\mathfrak{s}}\textcolor{red}{\mathfrak{u}}\textcolor{red}{\mathfrak{2}}^{\textcolor{red}{\mathfrak{g}}}\textcolor{red}{\mathfrak{2}}] + [\textcolor{teal}{\mathfrak{c}}\textcolor{teal}{\mathfrak{6}} + \textcolor{red}{\mathfrak{s}}\textcolor{red}{\mathfrak{u}}\textcolor{red}{\mathfrak{2}} + \textcolor{red}{\mathfrak{s}}\textcolor{red}{\mathfrak{u}}\textcolor{red}{\mathfrak{2}}]$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 596$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix} b_0$$

[115] $1^{11}23^11^{12}31^{131} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{c}_6 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 614, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 & 0 \\ 9 & 18 & 0 & 12 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 14 & 0 & 1 \end{pmatrix}$$

[116] $1^{11}2^31^{12}31^{141} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{s}\textcolor{red}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 640, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.0$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 & 0 \\ 10 & 18 & 0 & 12 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 14 & 0 & 1 \end{pmatrix}$$

[117] $1^{11}23^{11}12^{31}1^{112} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 688, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.5$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 7 & 15 & 0 & 10 & 8 & 38 & 12 \\ 2 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix} b_0$$

[118] $1^{11}23^11^{12}31^{114} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{e}_6 + \textcolor{blue}{s}\textcolor{red}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 9 & 15 & 0 & 10 & 8 & 38 & 12 \\ 2 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{11}23^11^{12}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 902, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 15 & 0 & 10 & 8 & 38 & 12 \\ 2 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [120] \quad 1^{111}3^1 1^1 41^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 4 & 2 & -3 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [121] \quad 1^{111}3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 5 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 5 & 0 & 2 & 7 & 5 & 5 \\ 1 & 3 & 0 & 1 & 5 & 1 & 3 \\ 2 & 3 & -1 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{111}3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 4 & 0 & 5 & 2 & 7 & 5 & 5 \\ 1 & 0 & 3 & 1 & 5 & 1 & 3 \\ 2 & 0 & 2 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{111}3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 3 & 0 & 5 & 2 & 7 & 5 & 5 \\ 1 & 0 & 3 & 1 & 5 & 1 & 3 \\ 2 & 0 & 2 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

4.126 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{su2n3mix} + \text{su}_2 + \text{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{123}1^1 1^1 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 1 & 5 \\ 14 & 33 & 1 & 20 & 0 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 20 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 11 & -1 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

4.127 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{su2n3mix} + \text{su}_2 + \text{su}_3$

$$T = 10 \quad [0] \quad 1^{113}1^1 1^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 9 & 5 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 5 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 6 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.128 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{su2n3mix} + \text{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{113}1^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 5 & 0 & 7 & 4 & 5 \\ 1 & 3 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -389, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 11 & 0 & 44 & 9 & 12 \\ 1 & 3 & 2 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -4 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 7 & 9 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -4 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -668, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 8 & 0 & 36 & 8 & 10 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1091, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 11 & 0 & 55 & 10 & 13 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 14 & 0 & 80 & 12 & 16 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \ 1^{11}231^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 19 & 5 \\ 14 & 33 & 1 & 44 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 44 & 0 & 58 & 14 \\ 5 & 11 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12}31^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 13 & 21 & 1 & 9 & 7 \\ 21 & 33 & 1 & 14 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 14 & 0 & 4 & 4 \\ 7 & 11 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12}31^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 1 & 9 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 0 & 0 & 10 & 6 \\ 5 & 0 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 5 \\ 14 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 10 & 6 \\ 5 & 11 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.129 NHC(-2)(-3) + e₆ + su₂n3mix + su₂→(-3) + su₂

$$T = 11 \quad [0] \ 1^{12}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 5 \\ 14 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 10 & 6 \\ 5 & 11 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.130 NHC(-2)(-3) + e₆ + su₂n3mix + su₂n3mix

$$T = 10 \quad [0] \ 1^{12}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 5 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 2 & 0 \\ 3 & 5 & 0 & 5 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 13 & 21 & 1 & 13 & 7 \\ 21 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 20 & 0 & 10 & 6 \\ 7 & 11 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

4.131 NHC(-2)(-3) + e₆ + su₂n3mix + su₃

$$T = 9 \quad [0] \ 1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 4 & 3 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 5 & 6 & 0 & 7 & 5 & 5 \\ 1 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 3 & 3 & -4 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 7 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

- [4] $1^{11}3^{11}1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 14 & 0 & 120 & 14 & 18 \\ 1 & 3 & 3 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -4 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}3^{11}1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 11 & 0 & 72 & 11 & 14 \\ 1 & 3 & 3 & 0 & 11 & 0 & 2 \\ 2 & 4 & 2 & -4 & 14 & 2 & -1 \end{pmatrix}_{b_0}$$
- [7] $1^{121}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -337, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 8 & 0 & 36 & 8 & 10 \\ 1 & 3 & 3 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 8 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{121}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1^{12}} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1^{11}} 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1^{11}} 1^{121} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 11 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1^{12}} 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1^{12}} 1^{121} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [18] $1^{12} 23^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 & 0 \\ 3 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{1111} 23^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 5 & 0 & 6 & 4 & 4 \\ 1 & 5 & 1 & 0 & 6 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 0 & 8 & 6 & 6 \\ 1 & 4 & 4 & 0 & 6 & 1 & 3 \\ 2 & 4 & 3 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{12} 23^1 1^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{11} 23^1 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 & 0 \\ 5 & 12 & 0 & 9 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 10 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11}231^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 & 0 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 385, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 & 0 \\ 7 & 18 & 0 & 12 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 14 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 & 0 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 679, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 & 0 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 10 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [26] \ 1^{11}3^{11}1^{11}5^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111}3^{11}1^{11}5^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.12 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 5 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 5 & 0 & 2 & 7 & 5 & 5 \\ 1 & 3 & 0 & 1 & 5 & 0 & 2 \\ 2 & 3 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

4.132 NHC(-2)(-3) + $\textcolor{violet}{e}_6$ + su2n3mix + $\textcolor{red}{su}_3$ + $\textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 6 \\ 1 & 4 & 0 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 5 & 9 \\ 1 & 3 & 3 & 0 & 5 & 0 & 3 \\ 2 & 6 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 2 & -4 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

4.133 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$

- $T = 7$ [0] $1^{111}31^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 5 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$
- [1] $1^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 6 & 10 & 0 & 21 & 7 \\ 3 & 4 & -4 & 7 & -2 \end{pmatrix}_{b_0}$
- [2] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$
- [3] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$
- [4] $21^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$
- [5] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -71$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$
- [6] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -80$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$
- [7] $1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -86$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 4 & 8 & 0 & 9 & 5 \\ 3 & 4 & -4 & 5 & -1 \end{pmatrix}_{b_0}$
- [8] $1^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 6 & 12 & 0 & 24 & 8 \\ 3 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$
- [9] $1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 45 & 11 \\ 3 & 4 & -4 & 11 & -1 \end{pmatrix}_{b_0}$
- [10] $1^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -119$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 4 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{15}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 10 & 20 & 0 & 72 & 14 \\ 3 & 4 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 14 & 0 & 36 & 10 \\ 3 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -215, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 7 & 13 & 0 & 27 & 9 \\ 3 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 48 & 12 \\ 3 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{12}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -330, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 10 & 18 & 0 & 57 & 13 \\ 3 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 11 & 21 & 0 & 75 & 15 \\ 3 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{13}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 12 & 22 & 0 & 84 & 16 \\ 3 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[20] $21^{11}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 6 & 8 & 3 & 3 & 3 \\ 4 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{14}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -626, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 14 & 26 & 0 & 117 & 19 \\ 3 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [22] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & -2 & 3 & 0 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{\overline{12}} 3^{\overline{12}} 21^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{13}} 3^{\overline{13}} 21^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{11}} 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 6 & 0 & 12 & 6 \\ 2 & 3 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 2^{\overline{11}} 2^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & -2 & 3 & 0 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11}3131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[33] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -2 & 0 \\ 4 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[36] $1^131^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[37] $1^131^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{12}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [41] $1^13221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 19, T_{\max} = 26, T \in [19, 26]$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

[42] $1^{12}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 32, T \in [16, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 6 & 0 & 2 & 9 & 5 \\ 2 & 3 & -1 & -1 & 5 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{11}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 16.82 \rightarrow 17, T_{\max} = 32, T \in [17, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 2 & 3 & 0 & 5 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{11}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{11}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{12}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 15.14 \rightarrow 16, T_{\max} = 32, T \in [16, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

[48] $1^{11}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{11}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{12}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{11}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{13}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[53] $1^{11}4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{11}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{12}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{13}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{112}231^{13}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 11 & 4 \\ 1 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 11 & 11 & 0 & 31 & 11 \\ 2 & 4 & 3 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{12}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 3 & 5 & 0 & 6 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{123}1^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 3 & 3 \\ 2 & 0 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{123}1^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{11}4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{111}23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 79, T \in [15, 79]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 4 & 4 & 0 & 1 & 3 \\ 2 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{123}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 79, T \in [15, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{12} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{12} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} 3^2 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 7 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11} 3^2 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 3 & 0 & 5 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{11} 2^3 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{12} 23131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 79, T \in [15, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 8 & 4 \\ 1 & 5 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 8 & 8 & 0 & 13 & 7 \\ 2 & 4 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{13} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{12} 231^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 79, T \in [15, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{13} 3^2 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 4 & 7 & 0 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{11} 4^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

- [74] $1^{12} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -123$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [75] $1^{11} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{12} 231^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 2 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [77] $1^{11} 231^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -163$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.36 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 7 & 0 & 13 & 7 \\ 2 & 4 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [78] $1^{13} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -163$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.21 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$
- [79] $1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -171$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 5 & 0 & 6 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$
- [80] $1^{12} 231^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [81] $1^{12} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -188$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$
- [82] $1^{13} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$
- [83] $1^{12} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.19 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [84] $1^{13} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -259$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 7 & 0 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [85] \quad 1^{\overline{1}} 1^{\overline{1}} 4 1321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 2 & 1 & 3 \\ 1 & 2 & -4 & 0 & -2 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 3 & -2 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{\overline{1}} 2 31^{\overline{1}} 41^{\overline{1}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 7 & 0 & 15 & 7 \\ 2 & 4 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{\overline{1}} 31^{\overline{1}} 41^{\overline{1}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 \\ 6 & 8 & 1 & 5 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 5 & 0 & -2 & 0 \\ 4 & 6 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{\overline{1}} 1 31^{\overline{1}} 41^{\overline{1}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{\overline{1}} 5 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -137, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{\overline{1}} 1 31^{\overline{1}} 4131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [92] \quad 1^{\overline{1}} 5 1321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 5 & 15 & 6 & 5 \\ 15 & 42 & 15 & 12 \\ 6 & 15 & 4 & 4 \\ 5 & 12 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{\overline{1}} 5 12321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 24, T \in [21, 24]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{\overline{1}} 1^{\overline{1}} 2 31^{\overline{1}} 5 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 31, \Delta = -116, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 17.40 \rightarrow 18, T_{\max} = 31, T \in [18, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 6 & 0 & 2 & 9 & 5 \\ 2 & 3 & -1 & -1 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} 5 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 31, \Delta = -116, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 31, T \in [16, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{11}315131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 11, T_{\max} = 31, \Delta = -116, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 31, T \in [15, 31]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[96] $1^13151^{11}231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 15.14 \rightarrow 16, T_{\max} = 34, T \in [16, 34]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 7 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$T = 12$ [97] $1^13151321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 4 & 0 & 1 & 1 & 3 \\ 2 & 4 & -1 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[98] $1^16131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 12, T_{\max} = 27, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 27, T \in [18, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 3 & 2 & 6 & 2 & 9 & 5 \\ 2 & 2 & 2 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

[100] $1^16131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 12, T_{\max} = 27, \Delta = -113, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 27, T \in [17, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{11}321^{11}6131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 12, T_{\max} = 28, \Delta = -114, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 28, T \in [17, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[101] $1^16131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 12, T_{\max} = 27, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 15.75 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

[102] $1^16131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 12, T_{\max} = 27, \Delta = -113, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[103] $31^{11}316131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 282, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 16.02 \rightarrow 17, T_{\max} = 71, T \in [17, 71]$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 3 \\ 4 & 2 & 5 & 2 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 2 & 2 & 2 & -2 & 0 \\ 3 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [104] \quad 1^{11} 31^1 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -19, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.26 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{11} 6^1 131^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -28, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 27, T \in [18, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 3 & 4 & 0 & 4 & 9 & 5 \\ 2 & 3 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{11} 31^1 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.16 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 5 & 5 \\ 2 & 3 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{13} 31^1 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -73, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.23 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 6 & 0 & 33 & 11 \\ 2 & 3 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[108] \quad 1^{12} 31^1 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -79, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 16.06 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 5 & 0 & 16 & 8 \\ 2 & 3 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{12} 31^1 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -129, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 16.13 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 29 & 11 \\ 2 & 3 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{13} 31^1 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -187, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.98 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 9 & 0 & 48 & 14 \\ 2 & 3 & 2 & -3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{14} 31^1 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -217, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 15.97 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 14 & 10 & 0 & 73 & 17 \\ 2 & 3 & 2 & -3 & 17 & 2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [112] \quad 1^{11} 5^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 27, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.48 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

4.134 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{121}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 6 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 4 & 3 \\ 3 & 5 & 0 & 6 & 4 & 3 & 3 \\ 3 & 4 & -4 & 5 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & -4 & 6 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{\overline{111}} 3^{\overline{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 0 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 705, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 3 & 3 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.135 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 1 \\ 3 & 4 & -4 & 2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.136 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 0 \\ 3 & 4 & -4 & 2 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.137 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 8 \quad [1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 302, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{\overline{11}} 3^{\overline{1}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 100, \Delta = -150, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.07 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -1 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [8] $1^{11} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 30, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.10 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 30, \Delta = -116, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

4.138 NHC(-2)(-3) + e₆ + su₃ + so₈ + su₂

$T = 9$ [0] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

4.139 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$T = 8$ [0] $1^{14} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.6$

$T_{\min} = 24$, $T_{\max} = 98$, $T \in [24, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 75 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -4 & 15 & 4 & -11 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 16$, $T_{\max} = 98$, $T \in [16, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 78$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.87 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 94$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 103$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 121$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 48 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -4 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -4 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -4 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -4 & 5 & 6 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{131} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 24 & 9 & 8 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -4 & 8 & 3 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{111} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 36 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -4 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 10 & 18 & 0 & 24 & 58 & 14 \\ 3 & 4 & -4 & 5 & 14 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -4 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -4 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{123} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -4 & 5 & 9 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{12} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -4 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{123} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 8 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 36 & 5 \\ 15 & 27 & 0 & 36 & 133 & 21 \\ 3 & 4 & -4 & 5 & 21 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -4 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 18 & 9 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -4 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3^1 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{115} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 10 & 20 & 0 & 15 & 73 & 15 \\ 3 & 4 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 11 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{116} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 12 & 24 & 0 & 18 & 106 & 18 \\ 3 & 4 & -4 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -4 & 5 & 10 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{117} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 14 & 28 & 0 & 21 & 145 & 21 \\ 3 & 4 & -4 & 2 & 21 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 57 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -4 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{115} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 487, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -3 & 0 & 15 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 48 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -4 & 12 & 3 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 21 & 9 \\ 5 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -4 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{131} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 48 & 27 & 12 \\ 5 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & -4 & 12 & 7 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -4 & 5 & 13 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{123} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 30 & 5 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -4 & 5 & 17 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{114} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -4 & 16 & 3 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 30 & 9 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -4 & 9 & 10 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1006$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{123} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1030$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1036$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1456$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1606$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1624$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2110$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2464$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [77] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 4 & 0 & 5 & 5 \\ 2 & 2 & -1 & -1 & -1 & 5 & -4 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 2 & -4 & 2 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 13 & 4 \\ 5 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 10 & 0 \\ 13 & 15 & 0 & 10 & 38 & 12 \\ 4 & 4 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{12} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 27 & 12 & 9 \\ 2 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 4 & 2 & -4 & 9 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{12} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 68, T \in [19, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 12 & 3 \\ 4 & 12 & 9 & 0 & 12 & 28 & 10 \\ 2 & 4 & 2 & -4 & 3 & 10 & -6 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 3 & 3 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

- [89] $1^{111} 3^{11} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 9 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [87] $1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 68$, $T \in [15, 68]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 2 & -4 & 3 & 3 & -2 \end{pmatrix}_{b_0}$
- [88] $1^{11} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [90] $1^{11} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$
- [91] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [92] $1^{112} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [93] $1^{11} 31^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & 2 & -4 & 2 & 8 & -1 \end{pmatrix}_{b_0}$
- [94] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$
- [95] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$
- [96] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$

- [97] $1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.29 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [98] $1^{111} 3^{113} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [99] $1^{112} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$
- [100] $1^{121} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [101] $1^{113} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [102] $1^{114} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [103] $1^{113} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -177$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{115} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$
- [106] $1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 12 & 6 & 6 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[107] $1^{112} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3^{13} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 8 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{111} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{111} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -243$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{111} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -291$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 7 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{113} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -303$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.26 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 5 & 12 & 15 & 0 & 12 & 49 & 13 \\ 2 & 3 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

[114] $1^{121} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{111} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -375$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.0

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 9 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[116] $1^{113}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 11 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{112}3^{111}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{111}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[119] $1^{112}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 4 & 9 & 12 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{11}31^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 3 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[121] $1^{11}31^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -537, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -4 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

[122] $1^{112}3^{111}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 9 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{114}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 14 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$

[124] $1^{112} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{113} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{112} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{113} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -615, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 9 & 3 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

[128] $1^{112} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{111} 31^{12} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -762, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 5 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

[130] $1^{112} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 10 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{114} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -771, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[132] $1^{112} 3^{13} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -875, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 11 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{112} 3^{114} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -876$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 12 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{113} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -876$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 12 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{111} 31^{111} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -906$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 12 & 15 & 6 \\ 3 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{113} 3^{112} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -915$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111} 31^{13} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -1059$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 7 & 21 & 12 & 0 & 12 & 73 & 15 \\ 2 & 4 & 2 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

[138] $1^{113} 3^{113} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{114} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 15 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{113} 3^{112} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1251$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 13 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{113} 3^{114} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1251$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.8

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{111} 31^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 15 & 2 \\ 9 & 27 & 15 & 0 & 15 & 121 & 19 \\ 2 & 4 & 2 & -4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

[144] $1^{113} 3^{13} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 14 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1643, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{113} 3^{14} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 15 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{114} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 16 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1920, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{114} 3^{13} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 17 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{114} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -2496, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 18 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [151] \ 1^{11} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 17.33 \rightarrow 18, \ T_{\max} = 24, \ T \in [18, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 6 & 8 & 0 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & -3 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{11} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 17.33 \rightarrow 18, \ T_{\max} = 24, \ T \in [18, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{11} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 26, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{11} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 27, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{1231} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{111} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 34, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 3 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 3 & -1 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{111} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 40, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{11} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 48, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.08 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -3 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

[161] $1^{111}2^13^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[159] $1^{11}3^1221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[160] $1^1231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 2 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[162] $1^1231^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[163] $1^{11}4^1131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{111}4^11^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[165] $1^{11}3^{11}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{114}3^1221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{112}3^1221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{11}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -3 & 1 & 0 & -1 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{11}1^{31}4^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 10 & 15 & 0 & 11 & 45 & 13 \\ 3 & 4 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[170] $1^{12}1^{21}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 4 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{11}1^{21}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 3 & 0 & 1 & 0 & 7 & 2 \\ 3 & 7 & 0 & 5 & 7 & 17 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{11}1^{21}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 125, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{11}1^{41}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[174] $1^1231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[175] $1^{11}1^{31}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 3 & 0 & 1 & 0 & 10 & 2 \\ 4 & 10 & 0 & 6 & 10 & 34 & 12 \\ 2 & 3 & -1 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{111}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 2 & 0 & 3 & 1 & 0 & 4 & 2 \\ 3 & 0 & 4 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 158, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111}23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 4 & 4 & 0 & 1 & 4 & 3 \\ 1 & 4 & 4 & 0 & 4 & 2 & 4 \\ 2 & 4 & 3 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[179] $1^1231^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 174, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

[180] $1^{111}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{11}231^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 0 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[182] $1^{112}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 215, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{121}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 1 & 0 \\ 4 & 5 & 0 & 6 & 13 & 5 & 7 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{112}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{112}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 224$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 3 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 5 & 7 & 17 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{111}4^11^{31}2^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.85 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 10 & 2 \\ 11 & 15 & 0 & 10 & 45 & 13 \\ 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{111}4^11^{31}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.85 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{113}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 245$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 245$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 1 & -3 & 1 & 6 & -1 \\ 2 & 0 & 3 & 1 & 0 & 10 & 2 \\ 5 & 0 & 10 & 6 & 10 & 34 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 254$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 2 & 0 & 3 & 1 & 0 & 4 & 2 \\ 4 & 0 & 4 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{1131}4^11^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 257$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{112}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 269$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 3 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 5 & 7 & 17 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111}2^13^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 271, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{11}31^{11}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[195] $1^{112}2^13^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 290, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{112}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & -3 & 1 & 5 & -1 \\ 2 & 0 & 3 & 1 & 0 & 7 & 2 \\ 4 & 0 & 7 & 5 & 7 & 17 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{1231}1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[198] $1^{112}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{111}2^13^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{111}4^11^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{114}2^13^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 2 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{11}1^{131}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 402, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{11}1^{131}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 8 & 3 \\ 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{11}231^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 449, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 15 & 0 & 10 & 0 & 38 & 12 \\ 2 & 4 & -1 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[205] $1^{113}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 479, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 3 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 6 & 10 & 34 & 12 \\ 2 & 3 & -1 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{113}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{112}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{112}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 586, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[209] $1^1231^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 587, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[210] $1^{113}2^13^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 596, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -3 & 1 & 3 & -1 \\ 2 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{113}2^13^{13}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 33 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{112}2^13^{14}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 608, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{113}2^13^{12}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 614, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 3 & 0 & 1 & 0 & 10 & 2 \\ 6 & 10 & 0 & 6 & 10 & 34 & 12 \\ 2 & 3 & -1 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{114}2^13^{11}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 629, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 3 & 0 & 1 & 0 & 13 & 2 \\ 6 & 13 & 0 & 7 & 13 & 57 & 15 \\ 2 & 3 & -1 & -1 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{114}2^13^{11}221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 632, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{11}231^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 644, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 13 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 21 & 0 & 13 & 0 & 74 & 16 \\ 2 & 4 & -1 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[217] $1^{114}2^13^{12}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 650, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{112}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 688, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & -3 & 1 & 5 & -1 \\ 2 & 0 & 3 & 1 & 0 & 7 & 2 \\ 5 & 0 & 7 & 5 & 7 & 17 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{11}231^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 797, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 16 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 27 & 0 & 16 & 0 & 122 & 20 \\ 2 & 4 & -1 & 0 & -1 & 20 & 0 \end{pmatrix}_{b_0}$$

[220] $1^{11}231^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 818, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[221] $1^{114}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 1 & -3 & 1 & 7 & -1 \\ 2 & 0 & 3 & 1 & 0 & 13 & 2 \\ 7 & 0 & 13 & 7 & 13 & 57 & 15 \\ 2 & 0 & 2 & -1 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{113}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 902, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 1 & -3 & 1 & 6 & -1 \\ 2 & 0 & 3 & 1 & 0 & 10 & 2 \\ 6 & 0 & 10 & 6 & 10 & 34 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{114}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 3 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 3 & 0 & 1 & 0 & 13 & 2 \\ 7 & 13 & 0 & 7 & 13 & 57 & 15 \\ 2 & 3 & -1 & -1 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{11}231^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 1007, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[225] $1^{114}3^1221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1088, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 62 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 62 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{114}3^1221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 63 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 63 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{114} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -85$, $\det = 1760$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 64 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 64 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [228] $1^{13} 5^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 18$, $T_{\max} = 23$, $T \in [18, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 6 & 0 & 25 & 9 \\ 3 & 4 & 0 & -1 & 9 & -2 \end{pmatrix}_{b_0}$$

[229] $1^{111} 31^1 41^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -26$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.46 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 4 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[230] $1^{12} 5^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -52$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.54 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 0 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{13} 1^{11} 41^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -80$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 1 & 5 & 9 & 7 \\ 4 & 6 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{11} 5^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -82$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{11} 31^{11} 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -137$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.30 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[234] $1^{111} 31^{11} 41^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -251$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 5 & 11 & 7 & 0 & 7 & 21 & 9 \\ 2 & 4 & 2 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[235] $1^{111} 31^{11} 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -314$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [236] \quad 1^{111} 31^1 5 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -86, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [237] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -32, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{11} 6^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.76 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & 1 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 7 & 3 & 0 & 16 & 8 \\ 2 & 2 & 2 & -1 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[240] \quad 1^{11} 6^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.84 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[241] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.76 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 2 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[242] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & -3 & 1 & 2 & -1 \\ 2 & 1 & 3 & 1 & 0 & 4 & 2 \\ 2 & 3 & 4 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & -1 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

4.140 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 4 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.141 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & -1 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

4.142 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -4 & 3 & 3 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -4 & 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{12}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 36 & 9 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 & 3 \\ 2 & 4 & 0 & 9 & 3 & 1 & 3 \\ 3 & 4 & -4 & 10 & 3 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1121} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 12 & 15 & 6 \\ 4 & 8 & 0 & 12 & 10 & 12 & 6 \\ 5 & 9 & 0 & 15 & 12 & 13 & 7 \\ 3 & 4 & -4 & 6 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -4 & 2 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 21 & 4 \\ 11 & 19 & 0 & 12 & 21 & 73 & 15 \\ 3 & 4 & -4 & 2 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{131} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 48 & 15 & 12 & 12 \\ 3 & 5 & 0 & 15 & 4 & 3 & 4 \\ 2 & 4 & 0 & 12 & 3 & 1 & 3 \\ 3 & 4 & -4 & 12 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1121} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 5 & 9 & 0 & 6 & 12 & 13 & 7 \\ 3 & 4 & -4 & 2 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 19 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 6 & 2 \\ 11 & 19 & 0 & 12 & 73 & 33 & 15 \\ 5 & 9 & 0 & 6 & 33 & 13 & 7 \\ 3 & 4 & -4 & 2 & 15 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1112} 3^1 1^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -137, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 7 & 3 \\ 3 & 4 & 0 & 4 & 14 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 33 & 11 \\ 7 & 13 & 0 & 9 & 33 & 28 & 10 \\ 3 & 4 & -4 & 2 & 11 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 9 & 17 & 0 & 12 & 15 & 49 & 13 \\ 3 & 4 & -4 & 2 & 4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1121} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 12 & 6 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -4 & 6 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1122} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 13 & 3 \\ 3 & 4 & 0 & 4 & 13 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 15 & 2 \\ 7 & 13 & 0 & 9 & 28 & 54 & 10 \\ 13 & 23 & 0 & 15 & 54 & 100 & 18 \\ 3 & 4 & -4 & 2 & 10 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -4 & 6 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{121} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 12 & 9 & 9 \\ 3 & 5 & 0 & 12 & 4 & 3 & 4 \\ 2 & 4 & 0 & 9 & 3 & 1 & 3 \\ 3 & 4 & -4 & 9 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -353, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 5 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 8 & 14 & 0 & 9 & 15 & 37 & 11 \\ 3 & 4 & -4 & 2 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -359, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 4 \\ 7 & 13 & 0 & 9 & 12 & 28 & 10 \\ 3 & 4 & -4 & 2 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 6 & 10 & 6 \\ 3 & 4 & -4 & 2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -4 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 11 & 21 & 0 & 18 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1122} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 4 & 8 & 0 & 6 & 12 & 10 & 6 \\ 3 & 4 & -4 & 2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -425, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -4 & 2 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 9 & 25 & 9 \\ 3 & 4 & -4 & 2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1122} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 7 & 3 \\ 3 & 4 & 0 & 9 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 12 & 21 & 6 \\ 4 & 8 & 0 & 12 & 10 & 18 & 6 \\ 7 & 13 & 0 & 21 & 18 & 28 & 10 \\ 3 & 4 & -4 & 6 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 12 & 3 \\ 3 & 4 & 0 & 4 & 14 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 15 & 2 \\ 8 & 14 & 0 & 9 & 37 & 57 & 11 \\ 12 & 22 & 0 & 15 & 57 & 85 & 17 \\ 3 & 4 & -4 & 2 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 12 & 46 & 12 \\ 3 & 4 & -4 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1122} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 4 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 4 & 8 & 0 & 6 & 10 & 18 & 6 \\ 7 & 13 & 0 & 9 & 18 & 28 & 10 \\ 3 & 4 & -4 & 2 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 15 & 6 \\ 2 & 4 & 0 & 6 & 1 & 6 & 3 \\ 5 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 6 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 8 & 3 \\ 3 & 4 & 0 & 4 & 9 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 6 & 13 & 24 & 7 \\ 8 & 14 & 0 & 9 & 24 & 37 & 11 \\ 3 & 4 & -4 & 2 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 5 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 18 & 4 \\ 10 & 18 & 0 & 12 & 18 & 58 & 14 \\ 3 & 4 & -4 & 2 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 16 & 3 \\ 3 & 4 & 0 & 4 & 5 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 30 & 4 \\ 16 & 28 & 0 & 18 & 30 & 154 & 22 \\ 3 & 4 & -4 & 2 & 4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 39 & 11 \\ 8 & 14 & 0 & 9 & 39 & 37 & 11 \\ 3 & 4 & -4 & 2 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 9 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 2 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 12 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 2 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 13 & 3 \\ 3 & 4 & 0 & 4 & 5 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 24 & 4 \\ 13 & 23 & 0 & 15 & 24 & 100 & 18 \\ 3 & 4 & -4 & 2 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 14 & 3 \\ 3 & 4 & 0 & 4 & 5 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 24 & 4 \\ 14 & 26 & 0 & 18 & 24 & 118 & 20 \\ 3 & 4 & -4 & 2 & 4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 5 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 21 & 4 \\ 12 & 22 & 0 & 15 & 21 & 85 & 17 \\ 3 & 4 & -4 & 2 & 4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 21 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -4 & 2 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 21 & 6 \\ 2 & 4 & 0 & 6 & 1 & 9 & 3 \\ 7 & 13 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & -4 & 6 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 33 & 7 \\ 11 & 21 & 0 & 15 & 33 & 76 & 16 \\ 3 & 4 & -4 & 2 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -4 & 2 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 8 & 14 & 0 & 9 & 37 & 48 & 11 \\ 10 & 18 & 0 & 12 & 48 & 58 & 14 \\ 3 & 4 & -4 & 2 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 5 & 9 & 0 & 6 & 13 & 27 & 7 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -4 & 2 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 16 & 3 \\ 3 & 4 & 0 & 4 & 9 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 48 & 7 \\ 16 & 28 & 0 & 18 & 48 & 154 & 22 \\ 3 & 4 & -4 & 2 & 7 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 27 & 6 \\ 2 & 4 & 0 & 6 & 1 & 12 & 3 \\ 9 & 17 & 0 & 27 & 12 & 49 & 13 \\ 3 & 4 & -4 & 6 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 9 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 33 & 6 \\ 2 & 4 & 0 & 6 & 1 & 15 & 3 \\ 11 & 21 & 0 & 33 & 15 & 76 & 16 \\ 3 & 4 & -4 & 6 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 2 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{113} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 15 & 3 \\ 3 & 4 & 0 & 4 & 5 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 27 & 4 \\ 15 & 27 & 0 & 18 & 27 & 133 & 21 \\ 3 & 4 & -4 & 2 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1122} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 10 & 3 \\ 3 & 4 & 0 & 4 & 13 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 7 & 13 & 0 & 9 & 28 & 42 & 10 \\ 10 & 18 & 0 & 12 & 42 & 58 & 14 \\ 3 & 4 & -4 & 2 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1113} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 2 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1112} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 10 & 3 \\ 3 & 4 & 0 & 4 & 9 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 5 & 9 & 0 & 6 & 13 & 30 & 7 \\ 10 & 18 & 0 & 12 & 30 & 58 & 14 \\ 3 & 4 & -4 & 2 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 4 & 9 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 39 & 7 \\ 13 & 23 & 0 & 15 & 39 & 100 & 18 \\ 3 & 4 & -4 & 2 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1122} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 30 & 10 \\ 7 & 13 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & -4 & 2 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1113} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 2 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{113} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 18 & 3 \\ 3 & 4 & 0 & 4 & 5 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 33 & 4 \\ 18 & 32 & 0 & 21 & 33 & 193 & 25 \\ 3 & 4 & -4 & 2 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{1114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1889$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 15 \\ 11 & 21 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 2 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{114}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1928$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 17 & 3 \\ 3 & 4 & 0 & 4 & 5 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 30 & 4 \\ 17 & 31 & 0 & 21 & 30 & 172 & 24 \\ 3 & 4 & -4 & 2 & 4 & 24 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1113}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 4 & 9 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 36 & 7 \\ 12 & 22 & 0 & 15 & 36 & 85 & 17 \\ 3 & 4 & -4 & 2 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1122}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2624$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 60 & 14 \\ 10 & 18 & 0 & 12 & 60 & 58 & 14 \\ 3 & 4 & -4 & 2 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{114}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2636$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 20 & 3 \\ 3 & 4 & 0 & 4 & 5 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 3 & 5 & 0 & 3 & 4 & 36 & 4 \\ 20 & 36 & 0 & 24 & 36 & 238 & 28 \\ 3 & 4 & -4 & 2 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1114}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2720$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 18 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -4 & 2 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{1114}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2780$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 24 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 2 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{1114}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2996$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 4 & 9 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 42 & 7 \\ 14 & 26 & 0 & 18 & 42 & 118 & 20 \\ 3 & 4 & -4 & 2 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 21 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 2 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 4 & 9 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 45 & 7 \\ 15 & 27 & 0 & 18 & 45 & 133 & 21 \\ 3 & 4 & -4 & 2 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3365, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 18 & 3 \\ 3 & 4 & 0 & 4 & 9 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 5 & 9 & 0 & 6 & 13 & 54 & 7 \\ 18 & 32 & 0 & 21 & 54 & 193 & 25 \\ 3 & 4 & -4 & 2 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -4373, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 17 & 3 \\ 3 & 4 & 0 & 4 & 9 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 5 & 9 & 0 & 6 & 13 & 51 & 7 \\ 17 & 31 & 0 & 21 & 51 & 172 & 24 \\ 3 & 4 & -4 & 2 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -5216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 20 & 3 \\ 3 & 4 & 0 & 4 & 9 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 24 & 2 \\ 5 & 9 & 0 & 6 & 13 & 60 & 7 \\ 20 & 36 & 0 & 24 & 60 & 238 & 28 \\ 3 & 4 & -4 & 2 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [80] \quad 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{1111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0} \\
[83] \quad & 1^{1111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 111, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.0 \\
& T_{\min} = 13.62 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 6 & 3 \\ 3 & 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 2 & 3 & 7 & -1 \end{pmatrix}_{b_0} \\
[84] \quad & 1^{11} 31^{121} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 138, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.5 \\
& T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 12 & 8 \\ 2 & 6 & 3 & 0 & 3 & 12 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 8 & 4 & -1 \end{pmatrix}_{b_0} \\
[85] \quad & 1^{1111} 3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 224, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.0 \\
& T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 7 & 0 \\ 4 & 6 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 7 & 9 & 14 & 8 \\ 3 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0} \\
[86] \quad & 1^{111} 3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 239, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.0 \\
& T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[87] \quad & 1^{1111} 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 268, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.0 \\
& T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0} \\
[88] \quad & 1^{111} 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 269, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.0 \\
& T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0} \\
[89] \quad & 1^{1112} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 273, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.0 \\
& T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 9 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 9 & 3 \\ 4 & 9 & 12 & 0 & 9 & 9 & 28 & 10 \\ 2 & 3 & 3 & -4 & 2 & 3 & 10 & -1 \end{pmatrix}_{b_0} \\
[90] \quad & 1^{112} 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 296, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.5 \\
& T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 4 \\ 2 & 4 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{111}3^{111}3^{111}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 320, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 363, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 \\ 2 & 6 & 3 & 0 & 3 & 4 & 9 \\ 3 & 9 & 6 & 0 & 6 & 9 & 13 \\ 2 & 4 & 2 & -4 & 2 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{111}3^{111}3^{111}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 374, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 3 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{111}3^{111}3^{111}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 381, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{111}3^{111}3^{111}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 704, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{111}31^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1065, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 5 \\ 2 & 4 & 4 & 0 & 3 & 6 & 15 \\ 1 & 4 & 0 & 0 & 3 & 3 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 9 \\ 2 & 6 & 3 & 0 & 3 & 4 & 15 \\ 5 & 15 & 9 & 0 & 9 & 15 & 37 \\ 2 & 4 & 2 & -4 & 2 & 4 & 11 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{111}31^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1659, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 12 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 21 & 4 \\ 7 & 21 & 12 & 0 & 12 & 21 & 73 & 15 \\ 2 & 4 & 2 & -4 & 2 & 4 & 15 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111}31^{114}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 2145$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 15 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 27 & 4 \\ 9 & 27 & 15 & 0 & 15 & 27 & 121 & 19 \\ 2 & 4 & 2 & -4 & 2 & 4 & 19 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ $[100] 1^{111}4 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -24$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 5 & 7 & 0 & 5 & 6 & 8 & 6 \\ 3 & 4 & -3 & 2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{111}4 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -48$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 5 & 3 \\ 4 & 4 & 0 & 3 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 5 & 7 & 0 & 5 & 6 & 8 & 6 \\ 3 & 4 & -3 & 2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.143 NHC(-2)(-3) + $\mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$

$T = 10$ $[0] 1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 9 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 220$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 6 & 3 \\ 8 & 14 & 0 & 9 & 9 & 37 & 24 & 11 \\ 5 & 9 & 0 & 6 & 6 & 24 & 13 & 7 \\ 3 & 4 & -4 & 2 & 3 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1112}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 274$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & -4 & 2 & 4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

- [4] $1^{\overline{1113}} 31^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & -4 & 2 & 4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 6 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 6 & 3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{\overline{1113}} 31^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 21 & 15 & 85 & 17 \\ 3 & 4 & -4 & 2 & 4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{\overline{1114}} 31^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 24 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & -4 & 2 & 4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{\overline{1111}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1112}} 31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 12 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 21 & 7 \\ 7 & 13 & 0 & 9 & 12 & 21 & 28 & 10 \\ 3 & 4 & -4 & 2 & 4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{1111}} 31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 6 & 15 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 15 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 6 & 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 12 & 4 \\ 7 & 13 & 0 & 9 & 12 & 28 & 30 & 10 \\ 7 & 13 & 0 & 9 & 12 & 30 & 28 & 10 \\ 3 & 4 & -4 & 2 & 4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 30 & 21 & 172 & 24 \\ 3 & 4 & -4 & 2 & 4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 3 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 7 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.144 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$[1] \ 1^{11}3^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 126, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 & 2 & 3 \\ 1 & 3 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 5 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 146, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 226, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 328, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 8 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 386, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 416, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 416, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 6 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 7 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 626, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 9 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 10 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{111}3^{11}15^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 2 & 0 & 1 & 2 & 2 & 2 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

4.146 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ n3mix

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 5 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 9 & 4 \\ 3 & 9 & 5 & 0 & 5 & 9 & 12 & 6 \\ 2 & 4 & 2 & -4 & 2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.147 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 & 3 \\ 3 & 4 & 0 & 8 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 12 & 6 & 5 \\ 5 & 9 & 0 & 12 & 13 & 9 & 7 \\ 3 & 5 & 0 & 6 & 9 & 3 & 3 \\ 3 & 4 & -4 & 5 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -3 & 0 & 12 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 13 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 4 & -4 & 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 1 & 3 \\ 4 & 9 & 4 & 0 & 24 & 0 & 5 \\ 2 & 4 & -3 & 0 & 9 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 10 & 24 & 9 & 0 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.5

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 1 & 3 \\ 4 & 9 & 4 & 0 & 16 & 0 & 5 \\ 2 & 4 & -3 & 0 & 6 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 7 & 16 & 6 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{1111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 9 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -4 & 2 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 0 & 5 & 13 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 12 & 6 & 3 \\ 7 & 13 & 0 & 12 & 28 & 18 & 10 \\ 4 & 8 & 0 & 6 & 18 & 9 & 5 \\ 3 & 4 & -4 & 3 & 10 & 5 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{132}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 0 & 5 & 17 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 15 & 6 & 3 \\ 9 & 17 & 0 & 15 & 49 & 24 & 13 \\ 4 & 8 & 0 & 6 & 24 & 9 & 5 \\ 3 & 4 & -4 & 3 & 13 & 5 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{1111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 2 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 1 & 3 \\ 4 & 9 & 4 & 0 & 24 & 0 & 5 \\ 2 & 4 & -3 & 0 & 9 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 11 & 24 & 9 & 0 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 12 & 6 \\ 3 & 4 & -4 & 2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -3 & 0 & 12 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 15 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -3 & 0 & 12 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{141} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 18 & 3 & 3 \\ 11 & 21 & 0 & 18 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 12 & 3 & 3 \\ 7 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{131} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -933, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 15 & 3 & 3 \\ 9 & 17 & 0 & 15 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & -4 & 2 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 1 & 3 \\ 4 & 9 & 4 & 0 & 12 & 0 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 12 & 7 & 0 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 477, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 3 & 6 & 9 & 0 & 6 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 3 & 6 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 774, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 9 & 7 \\ 2 & 6 & 3 & 0 & 3 & 9 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \quad 1^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 1^{13} 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 4 & 0 & 3 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 1^{13} 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 3 & 4 & 0 & 2 & 1 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11} 1^{13} 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -378, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 2 & 3 \\ 4 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 3 & 2 \\ 6 & 9 & 0 & 5 & 13 & 5 & 7 \\ 2 & 3 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.148 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{1111} 1^{1111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 6 & 9 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 2 & 4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 18 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 15 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.149 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 & 3 \\ 5 & 9 & 0 & 9 & 13 & 6 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.150 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{131}} 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.36 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 3 & 5 \\ 2 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.11 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{13}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 2 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{13}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -193, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 \\ 2 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -363, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 \\ 2 & 4 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 1 & 3 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{13}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -508, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 \\ 2 & 4 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 3 & 9 & 6 & 0 & 12 & 5 \\ 1 & 3 & 2 & 0 & 5 & 0 \\ 2 & 4 & 2 & -4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \ 1^{123}1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 16, \ \Delta = -96, \ \det = 106, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.30 \rightarrow 14, \ T_{\max} = 16, \ T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 \\ 2 & 4 & -1 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123}1^{13}1^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -95, \ \det = 149, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 5 \\ 14 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 11 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 1^{11}3^{11}1^141^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11}3^{11}1^14131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -149, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [12] \quad 1^{11}3^{11}1^15131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11}3^{11}1^15131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.151 NHC(-2)(-3) + $\textcolor{violet}{e}_6$ + $\textcolor{brown}{su}_3$ + $\textcolor{red}{su}_2 \rightarrow (-3)$ + $\textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 \\ 1 & 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 1 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 1 \\ 2 & 4 & 2 & -4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 7 & 5 & 14 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{112}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 9 & 15 & 0 & 10 & 8 & 38 & 12 \\ 3 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 15 & 0 & 10 & 8 & 38 & 12 \\ 3 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{111}3^{11}15^{11}131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 2 & 1 \\ 2 & 3 & -1 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.152 NHC(-2)(-3) + e₆ + su₃ + su₂ → (-3) + su₃

$$T = 10 \quad [0] \quad 1^{111}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 \\ 1 & 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 0 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.153 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -107, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 32, \quad T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -112, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 79, \quad T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 5 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 2 & 5 & 6 & 0 & 5 & 7 & 5 \\ 2 & 3 & 3 & -4 & 2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -179, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 32, \quad T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 7 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 32, \quad T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^1 4^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -265, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 79, \quad T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 14 & 2 \\ 9 & 27 & 14 & 0 & 14 & 120 & 18 \\ 2 & 4 & 2 & -4 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -292, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 79, \quad T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^1 3^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -328, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 79, \quad T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 11 & 2 \\ 7 & 21 & 11 & 0 & 11 & 72 & 14 \\ 2 & 4 & 2 & -4 & 2 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -332, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 32, \quad T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^{11} 1^1 2^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -337, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

12.0

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 8 & 2 \\ 5 & 15 & 8 & 0 & 8 & 36 & 10 \\ 2 & 4 & 2 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 8 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 11 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [18] \quad 1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111}23^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 4 & 6 & 4 \\ 1 & 5 & 1 & 0 & 4 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 4 & 4 & 0 & 1 & 6 & 3 \\ 2 & 6 & 6 & 0 & 6 & 8 & 6 \\ 2 & 4 & 3 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{112}23^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 9 & 0 & 26 & 10 \\ 2 & 4 & -1 & 0 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{123}1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112}23^{11}1^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 385, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 18 & 0 & 12 & 0 & 56 & 14 \\ 2 & 4 & -1 & 0 & -1 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{123}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 679, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & 0 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [26] \quad 1^{11} 3^1 1^1 5^{11} 13^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 2 & 2 \\ 2 & 0 & 2 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 3^1 1^1 5^{11} 13^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.12 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 & 0 & 5 & 2 \\ 2 & 5 & 0 & 2 & 5 & 7 & 5 \\ 2 & 3 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

4.154 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^1 1^{11} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 5 & 3 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

4.155 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{13} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 92$, $\Delta = -130$, $\det = -279$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 92$, $T \in [14, 92]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.21 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [17] $1^{11}2^321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 16$, $\Delta = -96$, $\det = 19$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.26 \rightarrow 14$, $T_{\max} = 16$, $T \in [14, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -3 & 1 & 0 & -1 \\ 1 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{12}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -94$, $\det = 29$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 13.59 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{11}4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 69$, $\Delta = -121$, $\det = 30$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.20 \rightarrow 15$, $T_{\max} = 69$, $T \in [15, 69]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{11}41^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 69$, $\Delta = -121$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 69$, $T \in [15, 69]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{111}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 16$, $\Delta = -96$, $\det = 75$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 16$, $T \in [14, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[22] $1^1231^131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 96$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{11}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -94$, $\det = 104$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111}2^13^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^1221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 33 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 33 & 0 & 78 & 16 \\ 5 & 7 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{11}4^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{111}2^13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^1221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[29] $1^1231^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^1221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [31] $1^{111}31^15^1131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 17, \Delta = -95, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [32] \quad 1^1 \overset{1}{6} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -40, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.20 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 3 & 0 \\ 1 & 1 & 1 & -3 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

4.156 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{111} \overset{1}{3} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \overset{1}{3} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 2 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.157 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$

$$T = 9 \quad [0] \quad 1^{121} \overset{1}{3} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 9 & 3 \\ 4 & 8 & 0 & 6 & 9 & 12 & 5 \\ 5 & 9 & 0 & 9 & 12 & 13 & 7 \\ 3 & 4 & -4 & 3 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{114} \overset{1}{3} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 13 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} \overset{1}{3} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{3} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -4 & 3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} \overset{1}{3} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.0

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -4 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{113} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 9 & 0 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 0 & 6 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & -1 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 14 & 0 & 9 & 36 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -4 & 2 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 0 & 5 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 12 & 3 \\ 4 & 8 & 0 & 6 & 9 & 18 & 5 \\ 7 & 13 & 0 & 12 & 18 & 28 & 10 \\ 3 & 4 & -4 & 3 & 5 & 10 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{123} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 0 & 5 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 9 & 24 & 5 \\ 9 & 17 & 0 & 15 & 24 & 49 & 13 \\ 3 & 4 & -4 & 3 & 5 & 13 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -4 & 2 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 11 & 24 & 9 & 0 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 9 & 6 \\ 3 & 5 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -4 & 2 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 15 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 14 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 18 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 0 & 7 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 7 & 0 & 0 & 14 & 8 \\ 3 & 5 & 0 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 39, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 126, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 182, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 31^{111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 477, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 6 & 3 \\ 2 & 6 & 3 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{1111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -100$, $\det = 486$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 3 & 3 & 0 & 3 & 0 & 6 & 2 \\ 3 & 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{1111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -100$, $\det = 774$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 9 & 3 \\ 3 & 9 & 6 & 0 & 6 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [31] $1^{11}4^{11}31^{11}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -36$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}4^{11}31^{11}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -108$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.22 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11}4^{11}31^{11}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -210$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 3 & 0 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{11}4^{11}31^{11}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -378$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.25 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.158 NHC(-2)(-3) + $\textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{1111}3^{1111}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 228, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 3 & 9 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 6 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 2 & 2 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 15 & 3 \\ 11 & 21 & 0 & 18 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 2 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 2 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 15 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 2 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

4.159 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

4.160 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 1 & 2 & 3 & 0 & 2 & 2 & 0 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.161 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 0 & 3 & 3 \\ 3 & 5 & -1 & -1 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 3 & 3 \\ 3 & 5 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{111} 3^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 321, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.162 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{violet}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{violet}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

4.163 NHC(-2)(-3) + $\mathfrak{e}_6$ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^1 3^1 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 100, \Delta = -173, \det = -350, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{11} 231^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 231^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 2^1 1^1 41^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{1^{2^1 1^1}} 5^{131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 100, \Delta = -138, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [5] \quad 1^{1^{1^1 1^1} 2^{31}} 5^{12^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

4.164 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{131^{111} 2^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 6 & 4 \\ 8 & 9 & 4 & 0 & 8 & 5 \\ 3 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 3 & 0 & 4 & 4 \\ 4 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

4.165 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{1^{13} 2^{31} 1^{12} 1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 36, \Delta = -118, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 36, T \in [13, 36]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 3 & 4 \\ 5 & 5 & 5 & 0 & 3 & 5 \\ 3 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 4 & 5 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

4.166 $\text{NHC}(-2)(-3) + \mathfrak{e}_7$

$$T = 7 \quad [0] \quad 2^{13} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 7, T_{\max} = 102, \Delta = -220, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -4 & -2 \\ 3 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1^{13} 3^{21} 1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 7, T_{\max} = 102, \Delta = -220, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -5 & -3 \\ 2 & 2 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1^{13} 3^{21} 1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 7, T_{\max} = 102, \Delta = -220, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -4 & -2 \\ 3 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 21^{1^{13} 3^{21} 1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 7, T_{\max} = 102, \Delta = -220, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -4 & -2 \\ 4 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [4] \quad 1^1 1^3 2^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & -3 & -1 \\ 3 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^1 2^1 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -193, \ \det = 23, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.35 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -5 & -3 \\ 2 & 3 & -1 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^1 2^1 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -193, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.17 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -5 & -3 \\ 2 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 3 2 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -191, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.08 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -4 & -2 \\ 2 & 2 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 2 3 1 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -199, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.91 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 2 3 1^1 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -199, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & -4 & -2 \\ 2 & 0 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 2 3 1^1 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -199, \ \det = 68, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -4 & -2 \\ 2 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 3 1^1 2 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -201, \ \det = 341, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -5 & -3 \\ 2 & 3 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [12] \ 1^1 1^1 2 2 3 1 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 9, \ T_{\max} = 102, \ \Delta = -170, \ \det = -11, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 17.91 \rightarrow 18, \ T_{\max} = 102, \ T \in [18, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -3 & -1 \\ 2 & 5 & -1 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [14] \ 1^1 4 1 3 2 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 10, \ T_{\max} = 102, \ \Delta = -161, \ \det = 11, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 18.91 \rightarrow 19, \ T_{\max} = 102, \ T \in [19, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \\ 1 & 3 & -5 & -3 \\ 3 & 4 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [17] \ 2 1^1 5 1 3 2 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 11, \ T_{\max} = 102, \ \Delta = -156, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19.91 \rightarrow 20, \ T_{\max} = 102, \ T \in [20, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & -2 & 0 \\ 3 & 7 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^1 5 1 3 2 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 \quad T = 11, \ T_{\max} = 102, \ \Delta = -156, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19.91 \rightarrow 20, \ T_{\max} = 102, \ T \in [20, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & -2 & 0 \\ 3 & 7 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [21] \quad 131 \overset{1}{5} 13221^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 \quad T = 12, T_{\max} = 98, \Delta = -135, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 20.91 \rightarrow 21, T_{\max} = 98, T \in [21, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -3 & -1 \\ 2 & 5 & -1 & -1 & -3 \end{pmatrix}_{b_0}$$

4.167 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_7 + \textcolor{violet}{e}_6$

$$T = 9 \quad [0] \quad 1^{11} 2 \overset{1}{3} 21^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{e}_6 \quad T = 9, T_{\max} = 127, \Delta = -242, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 127, T \in [18, 127]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & -3 & -1 & -4 \end{pmatrix}_{b_0}$$

4.168 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_7 + \textcolor{violet}{f}_4$

$$T = 8 \quad [0] \quad 1^{11} 3 \overset{1}{2} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 131, \Delta = -243, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.11 \rightarrow 17, T_{\max} = 131, T \in [17, 131]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & -2 & -3 & -3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{11} 3 \overset{1}{2} 21^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 104, \Delta = -214, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.41 \rightarrow 17, T_{\max} = 104, T \in [17, 104]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 2 \overset{1}{3} 21^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 104, \Delta = -216, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.24 \rightarrow 17, T_{\max} = 104, T \in [17, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

4.169 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_7 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{11} 3 \overset{1}{2} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 5 \\ 7 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3 \overset{1}{2} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \quad 1^{111} 2 \overset{1}{3} 21^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{11}2^3 21^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + [\text{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.35 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 0 & 2 \\ 2 & 3 & -1 & -3 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{11}2^3 21^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + [\text{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^1 221^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + [\text{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 5 \\ 7 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^1 221^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + [\text{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.06 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

[7] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + [\text{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 101, \Delta = -149, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

4.170 $\text{NHC}(-2)(-3) + \mathbf{e}_7 + \text{NHC}(-2)(-3) + \text{su}_2$

$T = 10 \quad [0] \quad 1^{111}3^1 21^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + [\text{su}_2 \cdot \mathbf{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 7 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 5 & -1 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^1 21^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + [\text{su}_2 \cdot \mathbf{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 548, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 5 & -1 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$T = 11 \quad [2] \quad 1^{111}2^3 21^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + [\text{su}_2 \cdot \mathbf{g}_2] + \text{su}_2 \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 3 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & -12 \end{pmatrix}_{b_0}$$

- [3] $1^{1112}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 6 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$
- [4] $1^{111}3^{111}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 12 & 5 \\ 7 & 8 & 1 & 5 & 15 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 11 & 0 \\ 12 & 15 & 0 & 11 & 24 & 12 \\ 5 & 6 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{1114}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.24 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 12 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 12 & -3 \end{pmatrix}_{b_0}$$
- [6] $1^{111}3^{111}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 7 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -155, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.94 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -190, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [9] $1^{1113}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 15.10 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 9 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$
- [10] $1^{1114}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -267, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 15.17 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 12 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 24, \quad \Delta = -111, \quad \det = -452, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.80 \rightarrow 15, \quad T_{\max} = 24, \quad T \in [15, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1231} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 48, \quad \Delta = -119, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, \quad T_{\max} = 48, \quad T \in [16, 48]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

4.171 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_{2 \rightarrow (-3)}$

$$T = 10 \quad [0] \quad 1^{11} 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -143, \quad \det = 304, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.79 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 3 & -1 \\ 4 & 5 & 0 & 3 & 1 & 3 \\ 5 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1231} 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 11, \quad T_{\max} = 75, \quad \Delta = -122, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.09 \rightarrow 16, \quad T_{\max} = 75, \quad T \in [16, 75]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 2 & -2 \\ 2 & 6 & 0 & 3 & 4 & 1 & 3 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

4.172 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -149, \quad \det = 103, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.03 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 4 & 0 & -2 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -2 & 5 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1111} 2^3 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 75, \quad \Delta = -122, \quad \det = -33, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.27 \rightarrow 19, \quad T_{\max} = 75, \quad T \in [19, 75]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 3 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 59, \quad \Delta = -120, \quad \det = -82, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.12 \rightarrow 16, \quad T_{\max} = 59, \quad T \in [16, 59]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 4 & 0 & -1 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

4.173 NHC(-2)(-3) + e₇ + s₀₈

$$T = 8 \quad [0] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} \quad T = 8, \ T_{\max} = 103, \ \Delta = -219, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.33 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} \quad T = 8, \ T_{\max} = 103, \ \Delta = -219, \ \det = 127, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.39 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{11} 3 2 2 1^{11} + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -190, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 2 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -192, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.82 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 9 & 4 & 5 \\ 2 & 0 & 4 & -4 & -2 \\ 3 & 0 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 2 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -192, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -4 & -2 \\ 3 & 6 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 2 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -190, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.47 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 2 3 1^{11} 3 1^{11} + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -198, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & -2 & -2 \end{pmatrix}_{b_0}$$

4.174 NHC(-2)(-3) + e₇ + s₀₈ + s_{u2}

$$T = 9 \quad [0] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} + \textcolor{red}{s}_{u2} \quad T = 9, \ T_{\max} = 103, \ \Delta = -189, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}_{08} + \textcolor{red}{s}_{u2} \quad T = 9, \ T_{\max} = 103, \ \Delta = -189, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -261$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -2 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -2 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^{11} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 28$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.71 \rightarrow 18$, $T_{\max} = 103$, $T \in [18, 103]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & -2 & 0 & -3 \end{pmatrix}_{b_0}$$

[5] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 112$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 2 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -168$, $\det = 124$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.68 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 102$, $\Delta = -160$, $\det = 139$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.51 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 308$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 532$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} =$

15.5

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 8 & -2 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 12 & -2 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -2 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 692, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.46 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 16 & -2 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -2 & 24 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 8 & -2 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{113} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 1248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 12 & -2 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -2 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 1648, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.42 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 16 & -2 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -2 & 24 & -1 \end{pmatrix}_{b_0}$$

4.175 NHC(-2)(-3) + $\textcolor{green}{e}_7$ + $\textcolor{orange}{so}_8$ + $\textcolor{red}{su}_2$ + $\textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & 0 & -3 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \ 1^{111} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 95, \Delta = -132, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 95, T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 94, \ \Delta = -130, \ \det = -132, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 94, \ T \in [16, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 95, \ \Delta = -132, \ \det = -200, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, \ T_{\max} = 95, \ T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

4.176 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 103, \ \Delta = -192, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -2 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

4.177 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 103, \ \Delta = -198, \ \det = -294, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \ 1^{11} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 103, \ \Delta = -171, \ \det = 96, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.83 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 103, \ \Delta = -171, \ \det = 192, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 103, \ \Delta = -169, \ \det = 245, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.73 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

4.178 NHC(-2)(-3) + e₇ + su₂

$$T = 8 \quad [0] 1^{111} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.65 \rightarrow 18, T_{\max} = 102, T \in [18, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -4 & 3 & -2 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & -2 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.71 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[2] 1^{112} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[4] 2^{11} 23^{111} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -4 & 3 & -2 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[6] 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[7] 1^{13} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -4 & 3 & -2 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] 1^{14} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}1^1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 102, \ \Delta = -190, \ \det = 113, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.83 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -4 & 3 & -2 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{15} 3^{21} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{c}_7 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 102, \ \Delta = -190, \ \det = 137, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.1$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & -3 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -3 & 10 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3^{21} 1^1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 102, \, \Delta = -190, \, \det = 139, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 11.0$
 $T_{\min} = 14.20 \rightarrow 15, \, T_{\max} = 102, \, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$

[13] $1^{16}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_7 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 6 & -3 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -3 & 12 & -2 \end{pmatrix}_{b_0}$

[14] $1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$

[15] $1^{17}3^{21}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_7 + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 7 & -3 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -3 & 14 & -2 \end{pmatrix}_{b_0}$

[16] $1^{12} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & -2 \end{pmatrix}_{b_0}$

[17] $1^{11} 1^{13} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$

[18] $1^{11} 1^{14} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, \quad T_{\max} = 102, \quad \Delta = -190, \quad \det = 265, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$
 $T_{\min} = 14.72 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$

[19] $1^{13}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 8$, $T_{\max} = 102$, $\Delta = -190$, $\det = 327$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.27 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & -2 \end{pmatrix}_{b_0}$

[illegible]

$$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [31] \quad 1^{11} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 2 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{112} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19, T_{\max} = 101, T \in [19, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 3 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -2 & 4 & -5 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{114} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 101, T \in [21, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 5 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -2 & 4 & -7 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.95 \rightarrow 18, T_{\max} = 101, T \in [18, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 4 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -2 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{12} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.35 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & -1 & -3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{112} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

- [40] $1^1 2^1 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$
- [41] $1^1 1^1 3^1 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.24 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$
- [43] $1^{11} 1^1 3^1 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$
- [42] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$
- [44] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -57$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.68 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$
- [45] $1^{12} 1^1 3^1 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [46] $1^{13} 2^1 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.60 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 4 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$
- [47] $1^{13} 1^1 3^1 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.57 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [48] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [49] $1^{12} 1^1 3^1 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 3 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [50] $1^{14} 1^1 3^1 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{\overline{1}} 23131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{\overline{1}} 231^{\overline{2}} 31^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 6 & -2 \\ 6 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -2 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{\overline{15}} 3221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -2 & 10 & -2 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{\overline{11}} 2^{\overline{11}} 321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.53 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{\overline{11}} 2^{\overline{11}} 321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{\overline{12}} 2^{\overline{11}} 321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -113, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{\overline{11}} 3^{\overline{11}} 221^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 3 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{\overline{11}} 3^{\overline{11}} 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{\overline{16}} 3^{\overline{11}} 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 15.19 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -2 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{\overline{11}} 2^{\overline{12}} 321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

- [61] $1^{11}23131^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -146$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 15.18 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 3 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$
- [62] $1^{11}231^{331}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$
- [63] $1^{17}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -2 & 14 & -2 \end{pmatrix}_{b_0}$$
- [64] $1^{13}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 4 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$
- [65] $1^{12}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 3 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$
- [66] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.42 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$
- [67] $1^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [68] $1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -180$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [69] $1^{12}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [70] $1^{13}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -191$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 4 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{11} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.61 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[72] $1^{14} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 5 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{1231} 431^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.43 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 12 & -2 \\ 12 & 0 & 0 & 12 & 94 & 16 \\ 2 & 0 & -1 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

[74] $1^{11} 3^{12} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 4 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{11} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[76] $1^{1231} 1131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{14} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 5 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{11} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -233, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.55 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{1231} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 4 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{123131} 113 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 15.27 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[81] $1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -253, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 4 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.36 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{12} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 4 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{12} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.36 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{11} 2^3 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 5 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{14} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{11} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.49 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 5 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{12} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{14} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & -3 \\ 6 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

[90] $1^{14} 231^{12} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 4 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{13} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{14} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^2 221^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 6 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[94] $1^{14} 23131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.32 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[95] $1^{13} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 5 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[96] $1^{13} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -398, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.52 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 5 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{12} 3^2 221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.44 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 5 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{12} 231^{231} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -422, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 6 & -2 \\ 7 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[100] $1^{12} 2^3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

[101] $1^{12} 1^{13} 3 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.36 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[102] $1^1 231^1 31^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -455, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.94 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 5 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[103] $1^1 231^{12} 31^{14} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{12} 1^{14} 3 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.53 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[105] $1^{14} 2^3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -498, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 6 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

[106] $1^1 231^{13} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.84 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 6 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[107] $1^{14} 1^{11} 3 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -541, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.32 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 6 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[108] $1^{12} 2^3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 6 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

[109] $1^{13} 1^{13} 2^3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -561, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[110] $1^{12} 1^{13} 3 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -565, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 6 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[111] \ 1^1 23131^{115} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 102, \ \Delta = -169, \ \det = -566, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.1$$

$$T_{\min} = 15.34 \rightarrow 16, \ T_{\max} = 102, \ T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[112] \ 1^1 231^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 102, \ \Delta = -169, \ \det = -585, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.70 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 5 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{14} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -600, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[114] \ 1^1 231^3 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 102, \ \Delta = -169, \ \det = -617, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 9 & -2 \\ 10 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{13} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -619, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.40 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{13} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -626, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.31 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 6 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{14} 2 \ 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 102, \ \Delta = -163, \ \det = -641, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & -3 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^1 231^{12} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 102, \ \Delta = -169, \ \det = -657, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.66 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[120] \ 1^1 231^{12} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 102, \ \Delta = -169, \ \det = -700, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{12} 3 \ 221^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -700, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 7 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

- [121] $1^{12}31^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -705$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 6 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [122] $1^{13}3221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -774$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [123] $1^{12}31^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -782$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 6 & -2 \\ 8 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [124] $1^{13}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -788$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 7 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$
- [125] $1^{14}3221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -837$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [127] $1^{12}31^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -840$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 8 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$
- [126] $1^{14}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -840$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 7 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$
- [128] $1^{13}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -843$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.34 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 7 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$
- [129] $31^{11}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -171$, $\det = -922$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -5 & 1 & -3 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [130] $1^{14}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -960$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & -3 \\ 8 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$
- [131] $1^{12}31^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -981$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 7 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[132] $1^{13} 3^{14} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -1046$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 8 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[133] $1^{14} 3^{14} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -1060$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[134] $1^{14} 3^{13} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -1125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 8 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[135] $1^1 231^{15} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 14.88 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 15 & -2 \\ 10 & 30 & 0 & 15 & 148 & 20 \\ 2 & 4 & -1 & -2 & 20 & -1 \end{pmatrix}_{b_0}$$

[136] $1^1 231^{13} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.69 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 8 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[137] $1^1 231^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.66 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[138] $1^{14} 3^{14} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -1396$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 9 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[139] $1^1 231^{13} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1405$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[140] $1^1 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1465$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 9 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[141] $1^1 231^{14} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1932$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 10 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[142] $1^1 231^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -2241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 11 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[143] \ 1^1 231^{14} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -2392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

4.179 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 5 \\ 7 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 548, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 5 & -1 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \ 1^{111} 23 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 4 \\ 1 & 3 & 0 & 1 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{121} 23 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 1 & 4 \\ 1 & 3 & 0 & 1 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.29 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 12 & 5 \\ 7 & 8 & 1 & 5 & 15 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 11 & 0 \\ 12 & 15 & 0 & 11 & 24 & 12 \\ 5 & 6 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{141} 23 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16.24 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & 1 & -3 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 12 & 4 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 3 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 5 \\ 7 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -155, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -190, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{131} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 9 & 4 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{141} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -267, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & 1 & -3 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 12 & 4 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^3 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1231} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 48, \Delta = -119, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 48, T \in [16, 48]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

4.180 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} 3^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 103, \Delta = -189, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

- [1] $1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$
- [2] $1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -261$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & -2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$
- [3] $1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & -2 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$
- $T = 10$ [4] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 28$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.71 \rightarrow 18$, $T_{\max} = 103$, $T \in [18, 103]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & -2 & 0 & -3 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 112$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -168$, $\det = 124$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.68 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 102$, $\Delta = -160$, $\det = 139$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.51 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$
- [8] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 308$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{121}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 532, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 8 & -2 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{131}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 12 & -2 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -2 & 18 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{141}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 692, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.46 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 16 & -2 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -2 & 24 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{121}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 8 & -2 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{131}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 1248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 12 & -2 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -2 & 18 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{141}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 1648, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.42 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 16 & -2 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -2 & 24 & -1 \end{pmatrix}_{b_0}$$

4.181 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$

$T = 10 \quad [0] \quad 1^{111}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.58 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$T = 11 \quad [1] \quad 1^{1111}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 95, \Delta = -132, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 95, T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 94, \ \Delta = -130, \ \det = -132, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 94, \ T \in [16, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 1^{11} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 95, \ \Delta = -132, \ \det = -200, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, \ T_{\max} = 95, \ T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

4.182 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{132} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -19, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 15.42 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 3 & 6 & 0 & 3 & 16 & 12 & 6 \\ 3 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -3 & 6 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -51, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.73 \rightarrow 17, \ T_{\max} = 101, \ T \in [17, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -59, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.19 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -3 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.07 \rightarrow 17, \ T_{\max} = 101, \ T \in [17, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{144} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.53 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 4 & -3 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & 8 & -3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.24 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -3 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -3 & 4 & 8 & -2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 3 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -3 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$t_{\text{in}} = 14.40 \rightarrow 15, T_{\text{m}} \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 3 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -3 & 4 & 6 & -2 \end{pmatrix} b_0$$

[15] $1^{11} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -3 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{133} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & 6 & -2 \end{pmatrix}_{b_0}$$

[17] $1^{12} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -3 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[19] $1^{111} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[20] $1^{133} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & 6 & -2 \end{pmatrix}_{b_0}$$

[21] $1^{123} 3^{112} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -3 & 4 & 6 & -2 \end{pmatrix}_{b_0}$$

[22] $1^{124} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -3 & 4 & 8 & -2 \end{pmatrix}_{b_0}$$

[23] $1^{122} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

[24] $1^{134}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -274$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -3 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[25] $1^{115}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.76 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 5 & -3 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -3 & 2 & 10 & -2 \end{pmatrix}_{b_0}$$

[26] $1^{125}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -298$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 14.79 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 5 & -3 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -3 & 4 & 10 & -2 \end{pmatrix}_{b_0}$$

[27] $1^{144}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 4 & -3 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & 8 & -2 \end{pmatrix}_{b_0}$$

[28] $1^{135}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -328$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.9$
 $T_{\min} = 14.84 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 5 & -3 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -3 & 6 & 10 & -2 \end{pmatrix}_{b_0}$$

[29] $1^{112}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -330$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.76 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[30] $1^{116}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -346$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 6 & -3 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -3 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

[31] $1^{126}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -364$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 6 & -3 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -3 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

[32] $1^{145}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -370$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 5 & -3 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -3 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

[33] $1^{136} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -394$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 14.92 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 6 & -3 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -3 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -411$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 14.24 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{134} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -419$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -3 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[36] $1^{117} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 7 & -3 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -3 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.25 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{127} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -442$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 7 & -3 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -3 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

[39] $1^{123} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -458$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -3 & 4 & 6 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{137} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -472$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 7 & -3 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -3 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$

[49] $1^{122} 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

[50] $1^{114} 1^{11} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -811, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[51] $1^{113} 1^{12} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

[52] $1^{123} 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -935, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -3 & 4 & 6 & -2 \end{pmatrix}_{b_0}$$

[53] $1^{124} 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -3 & 4 & 8 & -2 \end{pmatrix}_{b_0}$$

[54] $1^{113} 1^{14} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

[55] $1^{113} 1^{13} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1021, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

[56] $1^{133} 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & 6 & -2 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[58] $1^{134} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -3 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[59] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[60] $1^{144} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 4 & -3 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & 8 & -2 \end{pmatrix}_{b_0}$$

[61] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [62] $1^{132} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 17.11 \rightarrow 18, T_{\max} = 93, T \in [18, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 16 & 12 & 6 \\ 3 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -2 & 6 & 4 & -4 \end{pmatrix}_{b_0}$$

[63] $1^1 231^1 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 19.50 \rightarrow 20, T_{\max} = 100, T \in [20, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

[64] $1^1 231^1 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 18, T_{\max} = 100, T \in [18, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & 0 & -2 \\ 3 & 0 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 4 & 0 & -4 \end{pmatrix}_{b_0}$$

- [65] $1^1 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 26$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 17.31 \rightarrow 18$, $T_{\max} = 94$, $T \in [18, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 0 & -4 \end{pmatrix}_{b_0}$$
- [66] $1^{11} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 4 & 3 & 0 & 5 & 3 & 6 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$
- [67] $1^{14} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 8 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 42 & 12 \\ 2 & 2 & -1 & -2 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$
- [68] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 37$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.41 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$
- [69] $1^{144} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 42$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.81 \rightarrow 17$, $T_{\max} = 93$, $T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -3 \end{pmatrix}_{b_0}$$
- [70] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 44$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [71] $1^{11} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 49$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.24 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$
- [72] $1^{134} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 60$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 8 & -2 \\ 3 & 6 & 0 & 3 & 16 & 27 & 6 \\ 5 & 9 & 0 & 8 & 27 & 42 & 12 \\ 2 & 2 & -1 & -2 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$
- [73] $1^{111} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 61$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.46 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 5 & 3 & 6 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{121} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{12} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[76] $1^{124} 3^{113} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{1112} 3^{111} 23^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{112} 2^{113} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{1111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[81] $1^{112} 3^{1123} 23^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 16.02 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

[80] $1^{112} 2^{111} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{1231^{12}31^{114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 99, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 5 & 0 & 6 & 6 \\ 2 & 2 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{111} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{112} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{1231^{11}31^{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{123} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{13} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[90] $1^{11} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

12.0

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{112} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{123} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{1231} 3^{14} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 17, T_{\max} = 100, T \in [17, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & 0 & 16 & -3 \end{pmatrix}_{b_0}$$

[94] $1^{122} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.53 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[95] $1^{13} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 7 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 7 & 0 & 26 & 10 \\ 2 & 2 & -1 & -2 & 0 & 10 & -1 \end{pmatrix}_{b_0}$$

[96] $1^{113} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{124} 3^{112} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 8 & -2 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{134} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 8 & -2 \\ 4 & 6 & 0 & 3 & 16 & 27 & 6 \\ 6 & 9 & 0 & 8 & 27 & 42 & 12 \\ 2 & 2 & -1 & -2 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{111} 1^{12} 3 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{123} 1^1 3 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 3 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{123} 1^{11} 3 \ 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 6 & 0 & 14 & 8 \\ 2 & 2 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{133} 1^{111} 2 \ 3 \ 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & 3 & -3 \\ 4 & 0 & 9 & 3 & 25 & 27 & 9 \\ 4 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{133} 1^{11} 3 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{1231} 1^{11} 31^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & 3 & -2 \\ 3 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 0 & 0 & 3 & 6 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{114} 1^{11} 3 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{114} 1^1 3 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[109] $1^{112}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[108] $1^{112}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

[107] $1^{133}3^1221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[110] $1^{132}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 6 & 2 \\ 1 & 0 & 1 & -5 & 3 & 2 & -3 \\ 4 & 0 & 9 & 3 & 25 & 18 & 9 \\ 3 & 0 & 6 & 2 & 18 & 10 & 6 \\ 2 & 0 & 2 & -3 & 9 & 6 & -1 \end{pmatrix}_{b_0}$$

[111] $1^{12}3^1221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[112] $1^{124}3^1221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[113] $1^{111}3^1221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[114] $1^{112}3^1221^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 195, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[115] $1^{123} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 196$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[116] $1^{134} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[117] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 94$, $T \in [16, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

[118] $1^{115} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 208$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.1$

$T_{\min} = 15.04 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -2 \end{pmatrix}_{b_0}$$

[119] $1^{123} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 211$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{123} 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 212$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[121] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 214$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[122] $1^{122} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 214$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[123] $1^{12}2^13^{112}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 218$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[124] $1^{125}3^{11}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 220$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 5 & -2 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -2 \end{pmatrix}_{b_0}$$

[125] $1^{231}1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[126] $1^{144}3^{11}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 232$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -2 \end{pmatrix}_{b_0}$$

[129] $1^{135}3^{11}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.9$

$T_{\min} = 15.17 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 5 & -2 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -2 & 6 & 10 & -2 \end{pmatrix}_{b_0}$$

[128] $1^{114}2^13^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[127] $1^{113}2^13^{14}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[130] $1^{116}3^{11}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 252$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 15.21 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

[131] $1^{113} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 256$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[132] $1^{112} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 258$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[133] $1^{113} 2^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 260$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.66 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 4 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[134] $1^{112} 2^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 260$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[135] $1^{113} 2^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 264$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{126} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 264$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.24 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 6 & -2 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

[137] $1^{112} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 268$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[138] $1^{145} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 268$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.25 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 5 & -2 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -2 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

[139] $1^{113}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 278$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.42 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[140] $1^{124}3^{11}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 280$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 8 & -2 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

[141] $1^{136}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 284$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.30 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 6 & -2 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -2 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

[142] $1^{114}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 294$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[143] $1^{122}2^{124}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 296$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.68 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 6 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[144] $1^{117}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 304$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -2 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

[145] $1^{127}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 316$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.37 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -2 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

[146] $1^{112}3^{114}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 316$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 7 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[147] $1^{1231}1^{131}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 319, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

[148] $1^{111}2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 324, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[149] $1^{134}1^{11} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

[150] $1^{134}1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 330, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -5 & 3 & 4 & -3 \\ 4 & 0 & 9 & 3 & 25 & 36 & 9 \\ 6 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -3 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[151] $1^{137}1^{11} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$

$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 7 & -2 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -2 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$

[152] $1^{112}1^{11} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 337, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[153] $1^{112}1^{11} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 337, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[154] $1^{113}1^{111} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[155] $1^{113}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 346, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[156] $1^{1231}12^131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[157] $1^{111}1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[158] $1^{122}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 3 & 0 & 6 & 2 & 10 & 12 & 6 \\ 3 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[159] $1^{122}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 362, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 3 & 0 & 6 & 2 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[160] $1^{123}1^{11}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 370, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[161] $1^{113}1^{11}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[162] $1^{111}1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[163] $1^{13}2^1 3^{113} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 388$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -5 & 0 & 3 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -3 & 0 & 9 & -1 \end{pmatrix}_{b_0}$$

[164] $1^{124} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 406$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 8 & -2 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

[165] $1^{112} 3^{113} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 409$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[166] $1^{114} 2^1 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 410$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[167] $1^{114} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 411$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 2 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[168] $1^{114} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 412$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[169] $1^{114} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 419$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[170] $1^{112} 3^{112} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 432$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{112} 3^{13} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 441, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.44 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 6 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{112} 3^{12} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 448, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.34 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{1231} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 448, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.75 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{124} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 448, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.32 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 8 & -2 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{113} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 452, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.62 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{112} 3^3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 458, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.82 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{112} 3^3 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.34 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{231} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 477, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[179] $1^{113} 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 480, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[181] $1^{111} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 488, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$

$$T_{\min} = 14.80 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 5 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[180] $1^{122} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 488, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$

$$T_{\min} = 14.41 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[182] $1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 493, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[184] $1^{12} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[183] $1^{231} 1^{12} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[185] $1^{13} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 506, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[186] $1^{113} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$

$$T_{\min} = 14.46 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[187] $1^{144}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 522$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & 4 & -3 \\ 6 & 0 & 12 & 4 & 46 & 48 & 12 \\ 7 & 0 & 12 & 4 & 48 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[188] $1^{123}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 532$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 3 & 0 & 6 & 2 & 10 & 18 & 6 \\ 6 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[189] $1^{124}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 534$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[190] $1^{1231}1^{11}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 549$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.66 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[191] $1^{1231}1^{12}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 582$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[192] $1^{114}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 584$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[193] $1^{1231}1^{11}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 592$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.65 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[194] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 608$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[195] $1^{113} 3^{11} 221 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 621$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.30 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[196] $1^{14} 2^3 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 626$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 12 & 2 \\ 1 & 0 & 1 & -5 & 0 & 4 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 0 & 12 & 4 & 0 & 46 & 12 \\ 2 & 0 & 2 & -3 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[197] $1^{111} 2^3 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 642$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.55 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 4 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[199] $1^{122} 3^3 221^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 652$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[198] $1^{114} 2^3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 652$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.52 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[200] $1^{114} 3^3 221^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 657$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[201] $1^{113} 2^3 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 658$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 3 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[202] $1^{112} 2^3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 666$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[203] $1^{113}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 674$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[204] $1^{123}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 690$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 3 & 0 & 6 & 2 & 10 & 18 & 6 \\ 5 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[205] $1^{113}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 724$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[206] $1^{124}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 746$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 6 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[207] $1^{123}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 763$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[208] $1^{124}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 770$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 8 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[209] $1^{123}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 776$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 5 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[210] $1^{122}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 786$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[211] $1^{124} 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 792, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$T_{\min} = 14.40 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[212] $1^{113} 1^{14} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 832, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$T_{\min} = 14.49 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[213] $1^{231} 1^{113} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 836, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$

$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[214] $1^{113} 1^{13} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 837, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$T_{\min} = 14.41 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[215] $1^{143} 1^{122} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 840, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$

$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -5 & 4 & 3 & -3 \\ 6 & 0 & 12 & 4 & 46 & 36 & 12 \\ 5 & 0 & 9 & 3 & 36 & 25 & 9 \\ 2 & 0 & 2 & -3 & 12 & 9 & -1 \end{pmatrix}_{b_0}$$

[216] $1^{132} 1^{114} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 842, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$T_{\min} = 14.44 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -5 & 0 & 3 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -3 & 0 & 9 & -1 \end{pmatrix}_{b_0}$$

[217] $1^{113} 1^{12} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 868, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$T_{\min} = 14.31 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[218] $1^{112} 1^{113} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 884, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$

$T_{\min} = 14.41 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[219] $1^{114}2^13^{124}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 896$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 3 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[220] $1^{1231}13^{112}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 901$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[221] $1^{114}3^{111}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 913$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[222] $1^{112}2^13^{114}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 914$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[223] $1^{133}2^13^{123}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 946$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & 3 & -3 \\ 5 & 0 & 9 & 3 & 25 & 27 & 9 \\ 6 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & 9 & -1 \end{pmatrix}_{b_0}$$

[224] $1^{133}3^{111}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 964$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[225] $1^{122}2^13^{123}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 986$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 5 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[226] $1^{113}3^{113}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 997$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[228] $1^{113} 3^{14} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1008, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 8 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[227] $1^{122} 2^{122} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1008, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[229] $1^{114} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1024, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[230] $1^{112} 2^{123} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1026, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[231] $1^{231} 1^{43} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1066, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[232] $1^{134} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1083, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

[233] $1^{124} 2^{113} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1092, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 7 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[234] $1^{231} 1^{33} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1117, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[235] $1^{144}2^13^{122}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & 4 & -3 \\ 6 & 0 & 12 & 4 & 46 & 48 & 12 \\ 6 & 0 & 12 & 4 & 48 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[236] $1^{231}1^{331}1^{13}3 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[238] $1^{134}2^13^{124}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -5 & 3 & 4 & -3 \\ 5 & 0 & 9 & 3 & 25 & 36 & 9 \\ 8 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -3 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[237] $1^{133}2^13^{122}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & 3 & -3 \\ 5 & 0 & 9 & 3 & 25 & 27 & 9 \\ 5 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & 9 & -1 \end{pmatrix}_{b_0}$$

[239] $1^{113}2^13^{113}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1236, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[240] $1^{114}1^{13}3^{113}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[241] $1^{144}1^{111}3^{111}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1292, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -1 \end{pmatrix}_{b_0}$$

[242] $1^{114}1^{12}3^{112}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[243] $1^{114}2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[244] $1^{112}2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[245] $1^{114} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1404, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[246] $1^{124}2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1418, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 4 & 0 & 6 & 2 & 10 & 24 & 6 \\ 7 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[247] $1^{113}2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 3 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[248] $1^{114} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1561, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[249] $1^{123}2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1578, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 6 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[250] $1^{113}2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1610, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[251] $1^{1231}1^{431}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1684, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[252] $1^{114}1^{14}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1708, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[253] $1^{123}2^{124}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 7 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[254] $1^{134}2^{123}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1746, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -5 & 3 & 4 & -3 \\ 5 & 0 & 9 & 3 & 25 & 36 & 9 \\ 7 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -3 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[255] $1^{1231}1^{431}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[256] $1^{114}1^{14}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 2034, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[257] $1^{124}2^{124}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 2376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 4 & 0 & 6 & 2 & 10 & 24 & 6 \\ 8 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

4.183 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$T = 10 \quad [0] 1^{11}1^{111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.58 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1111} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 11, T_{\max} = 95, \Delta = -132, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 95, T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 11, T_{\max} = 94, \Delta = -130, \det = -132, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 11, T_{\max} = 95, \Delta = -132, \det = -200, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, T_{\max} = 95, T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

4.184 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$

$$T = 10 \quad [0] \quad 1^{1112} 3 \quad 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 19.80 \rightarrow 20, T_{\max} = 93, T \in [20, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 2 & 4 & -7 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3 \quad 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 0 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1211} 3 \quad 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 1 & -3 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 4 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1211} 3 \quad 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$[111] \text{ } 1^{1122} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 356, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & 4 & -2 \end{pmatrix} b_0$$

$$[12] \quad 1^{1111} 3 \, 21^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 10, \, T_{\max} = 93, \, \Delta = -130, \, \det = 360, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & 2 & -2 \end{pmatrix} b_0$$

$$[13] \quad 1^{1113} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 \\ 3 & 6 & 0 & 3 & 6 & 6 & 16 \\ 2 & 2 & -1 & -3 & 2 & 2 & 6 \end{pmatrix} b_0$$

[14] $1^{1122}3^{21}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -130$, $\det = 430$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & 4 & -2 \end{pmatrix} b_0$$

[15] $1^{1114}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 & -2 \end{pmatrix} b_0$$

[16] $1^{1115}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 572, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.1$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 5 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 10 & 48 & 10 \\ 2 & 2 & -1 & -3 & 2 & 2 & 10 & -2 \end{pmatrix} b_0$$

$$[17] \quad 1^{1112} 3^{21} 1^{12} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 624, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 \\ 4 & 4 & 0 & 2 & 4 & 4 & 6 \\ 2 & 2 & -1 & -3 & 2 & 2 & 4 \end{pmatrix} b_0$$

[18] $1^{1111} 3^{21} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 656, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

[26] $1^{1122} 3^{211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_7 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{teal}{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 1384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & 4 & -2 \end{pmatrix} b_0$$

[27] $1^{1114} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 1524, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 \end{pmatrix} b_0$$

[28] $1^{1114} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 1680, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 8 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 \end{pmatrix} b_0$$

[29] $1^{1114} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -130$, $\det = 2080$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 6 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 \end{pmatrix} b_0$$

[30] $1^{1114} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 2132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 & -2 \end{pmatrix} b_0$$

$$T = 11 \quad [31] \quad 1^{111} 3^{11} 221^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & 2 & -1 \end{pmatrix} b,$$

[32] $1^{1112}3^{221}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 20.25 \rightarrow 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 \\ 5 & 4 & 0 & 2 & 4 & 4 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & -7 \end{pmatrix} b_0$$

[34] $1^1 2^3 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -103, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.5$

$$T_{\min} = 22.33 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 0 & 0 & -9 \end{pmatrix}_{b_0}$$

[33] $1^{1211} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{1111} 2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3^{1111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{1211} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 3 & 2 \\ 3 & 3 & 0 & 5 & 0 & 3 & 6 & 6 \\ 2 & 2 & -1 & -2 & 0 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{1121} 3^{131} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 3 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{113} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 0 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 0 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{\text{1112}} 2^{\text{3}} 3^{\text{21}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -103, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 6 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 3 & 6 & -1 \end{pmatrix} b_0$$

[42] $1^2 \overset{1111}{3} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -103, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 0 & 0 & -2 \end{pmatrix} b_0$$

[44] $1^{1121}3^{221}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 6 & 5 & 14 & 5 & 8 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 8 & 2 & -1 \end{pmatrix} b_0$$

[43] $1^{1112} 3^{221} 1^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{e}_7 + \textcolor{blue}{s}\textcolor{red}{u}_2 + \textcolor{blue}{s}\textcolor{red}{u}_2 + \textcolor{blue}{s}\textcolor{red}{u}_2 \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 \\ 5 & 5 & 0 & 6 & 5 & 5 & 14 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 \end{pmatrix} b_0$$

[45] $1^{113} 3^{111} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -130, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 7 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 0 & 2 & 10 & -1 \end{pmatrix} b_0$$

[46] $1^{111} 3^{221} + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{5u}_2 + \textcolor{red}{5u}_2 + \textcolor{blue}{5u}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -101, \quad \det = -170, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.0$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & -1 \end{pmatrix} b_0$$

[47] $1^{112} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -101, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.5$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 0 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 0 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{112} 3^{221} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 0 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{1113}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -216, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 3 & 9 & -2 \end{pmatrix}_{b_0}$$

[50] $1^{1111}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -216, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{1112}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -220, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{1212}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -230, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 8 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 8 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{1113}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -232, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

[53] $1^{1111}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -232, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{1112}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -242, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{1112}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -256, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.44 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{1122}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{1122}2^1321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.49 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 12 & 6 \\ 3 & 0 & 6 & 2 & 6 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[59] $1^{1111}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.28 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[60] $1^{1113}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 & 2 \\ 3 & 6 & 0 & 3 & 6 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[61] $1^{1112}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -300, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{11}2^1321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -328, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.66 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 0 & 3 & 1 & 0 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{1112}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -336, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[64] $1^{1122} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -346, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{1113} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -348, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{1114} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -352, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[67] $1^{1114} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -380, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{1113} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -388, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 9 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[69] $1^{1114} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -408, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[70] $1^{1112} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 6 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{1115}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 15.06 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 10 & 48 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -2 \end{pmatrix}_{b_0}$$

[72] $1^{1113}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 7 & 2 \\ 4 & 7 & 0 & 7 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{1112}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.51 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{1113}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -436, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 9 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{1112}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -504, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 0 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 0 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[76] $1^{1112}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -512, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{1111}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{1113}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{1121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -588, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 3 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{1112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -592, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{1113} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -616, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 7 & 2 \\ 7 & 7 & 0 & 7 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[81] $1^{1114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -616, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{1122} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -620, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 12 & 6 \\ 3 & 0 & 6 & 2 & 6 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{1122} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -624, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 6 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{1114} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -666, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

- [86] $1^{1114} 3^{22111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -674, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.32 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$
- [87] $1^{1231111} 3^{11111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -109, \det = -696, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$
- [88] $1^{1114} 3^{22111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -736, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 5 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$
- [89] $1^{1112} 3^{22111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -746, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [90] $1^{1113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -760, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 9 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$
- [91] $1^{1113} 3^{22111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -788, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [92] $1^{1122} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -792, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 6 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$
- [93] $1^{1111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -840, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$[101] \ 1^{1122} 3^{111} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1132, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{1114} 3^{221} 1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1232, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{11142} 3 \ 21^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_7 + \textcolor{brown}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -1236, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{1114} 3^{221} 1^{14} + [\textcolor{red}{s}2 \cdot \textcolor{blue}{g}2] + \textcolor{green}{e}7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1248, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 9 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{1114} 1^{14} 3 \ 221^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}\textcolor{green}{7} + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1376, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 3^{\textcolor{red}{1}} 2\textcolor{red}{1}^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}\textcolor{red}{7} + \textcolor{red}{s}\textcolor{red}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -1416, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 9 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{1114} 3^{221} 1^{11} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1488, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 9 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 2^{\textcolor{red}{1}} 3^{\textcolor{red}{1}} 2\textcolor{red}{1}^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}\textcolor{red}{7} + \textcolor{red}{s}\textcolor{red}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -1560, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 6 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{1114} 3^{12} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1696, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{1114} 3^{13} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1744, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 8 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{1114} 3^{13} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1744, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{1114} 3^{12} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1824, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.32 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 7 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

4.185 NHC(-2)(-3) + $\textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 11 \quad [0] \ 1^{1112} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -100, \ \det = -64, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.25 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 1 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 0 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 0 & 2 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -100, \ \det = -524, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^{11} 321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -100, \ \det = -592, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.24 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.187 NHC(-2)(-3) + e₇ + su₂ + su₂ + su₂ + su₃

$$T = 11 \quad [0] \quad 1^{1111} 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -202, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \quad 1^{1111} 2 \quad 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 3 & 0 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3 \quad 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1112} 2 \quad 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 244, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 3 & 3 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1114} 2 \quad 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 472, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1113} 2 \quad 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 628, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11141} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

4.188 NHC(-2)(-3) + $\textcolor{violet}{e}_7$ + $\textcolor{brown}{su}_2$ + $\textcolor{red}{su}_2$ + $\textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{113} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 6 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 4 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 10 & 3 \\ 2 & 1 & 1 & 2 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 8 & -2 \\ 4 & 6 & 0 & 4 & 10 & 24 & 6 \\ 10 & 12 & 0 & 8 & 24 & 46 & 12 \\ 3 & 3 & -1 & -2 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{113}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 808, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 7 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{112}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 824, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{114}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1312, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [19] \quad 1^{1231}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{123}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{123}1^{123}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{12}3^{11}1^{12}3^{11}2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.29 \rightarrow 16, \ T_{\max} = 25, \ T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -411, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.78 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

4.189 $\text{NHC}(-2)(-3) + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$

$$T = 11 \quad [0] \ 1^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}2^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -256, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 & 3 \\ 4 & 6 & 0 & 4 & 6 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 3 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}2^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -400, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}2^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -576, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}2^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -960, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1112} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -1040, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.190 NHC(-2)(-3) + $\textcolor{red}{e}_7$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2 \rightarrow (-3)$ + $\textcolor{blue}{su}_3$

$$T = 11 \quad [0] \ 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.191 NHC(-2)(-3) + $\textcolor{red}{e}_7$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2 \text{n3mix}$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 86, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.44 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 0 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1211} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 174, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.45 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 4 & 6 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1121} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 356, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 3 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 & 3 \\ 2 & 2 & -1 & -3 & 2 & 4 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1112} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 384, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.54 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 408, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [12] \quad 1^{12311} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.192 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 & 1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & 2 & 2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{131} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 9 & -2 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 & 12 \\ 3 & 4 & -1 & -2 & 0 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 420, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 464, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 589, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 1088, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -56, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -65, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 23 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -84, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

- [14] $1^{1111} 3^{12} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{1111} 2^1 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [16] $1^{111} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [17] $1^{111} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -149, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{131} 3^{111} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 15.39 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$
- [19] $1^{1111} 2^1 3^{112} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -246, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.15 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$
- [20] $1^{131} 3^{111} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 14.90 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 14 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & -1 & 0 & 17 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{1111} 2^1 3^{111} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{121} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -332, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 & 4 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 11 & -1 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 & 13 \\ 3 & 4 & -1 & -1 & 0 & 13 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -341, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 & 1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1231} 3^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{121} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -380, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -408, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 & 1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 2^{122} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -432, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{112} 2^{111} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -434, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 & 3 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{1231}1^{11}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \quad T_{\max} = 38, \quad \Delta = -118, \quad \det = -455, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$

[30] $1^{1231}1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11$, $T_{\max} = 38$, $\Delta = -118$, $\det = -455$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix} b_0$$

$$[31] \quad 1^{1111} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}}2^{\textcolor{red}{1}}3^{\textcolor{red}{1}}21^{\textcolor{red}{1}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}7 + \textcolor{red}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_3 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 3 & 2 & -1 \end{pmatrix} b_C$$

[33] $1^{1131}2^33^{21}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -626, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 & 3 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 9 & 2 & -1 \end{pmatrix} b_0$$

[34] $1^{1131}2^33^{21}1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -668, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 & 3 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{1141}2^33^{21}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -674, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 & 3 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 12 & 2 & -1 \end{pmatrix} b_0$$

[36] $1^{1111}3^{221} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{c}7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -896, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix} b_0$$

[37] $1^{114}2^33^{21}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -1268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 & 3 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

4.193 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{1111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 0 \\ 2 & 2 & -1 & -3 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -202, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \quad 1^{1111} 23 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1112} 23 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 244, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 6 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 6 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 6 \\ 1 & 3 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1114} 23 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 472, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1113} 23 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 628, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 & 9 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11114} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$4.194 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \ 1^{11113} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.195 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$$

$$T = 11 \quad [0] \ 1^{11113} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{11111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -91, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11111} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 228, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$4.196 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

[9] $1^{11} 1^{13} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 1^{14} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 8 & -2 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{12} 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{12} 1^{13} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{12} 1^{14} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1048, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 8 & -2 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [14] $1^{12} 23^1 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 18, T_{\max} = 100, T \in [18, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -2 & 0 & 2 & -4 \end{pmatrix}_{b_0}$$

[15] $1^{12} 23^1 1^3 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 16.20 \rightarrow 17, T_{\max} = 100, T \in [17, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 9 & 2 & -2 \\ 10 & 0 & 0 & 9 & 52 & 9 & 12 \\ 1 & 0 & 0 & 2 & 9 & 0 & 2 \\ 2 & 0 & -1 & -2 & 12 & 2 & -3 \end{pmatrix}_{b_0}$$

[16] $1^{12} 23^1 1^3 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.20 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

[17] $1^{12} 23^1 1^3 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & -2 \\ 2 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[27] $1^1 231^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -141, \quad \det = 313, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$

[27] $1^1 231^{11} 3^1 1^1 + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{c}}_7 + \textcolor{brown}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -141, \quad \det = 313, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$

$$T_{\min} = 14.77 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 4 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{12}231^{11}31^{12} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}\textcolor{teal}{g}_7 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 6 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^{11}1^{11}31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^1 2 3^1 1^2 3 1^1 + [\textcolor{blue}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{red}{\mathbf{c}}_7 + \textcolor{brown}{\mathbf{su}}_2 + \textcolor{blue}{\mathbf{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 480, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & -2 \\ 4 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

[31] $1^1 2 3 1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 5 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{123}1^{12}31^{13} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{e}}_7 + \textcolor{brown}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 540, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & -2 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

[33] $1^{12}23^11^{12}31^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}\textcolor{teal}{g} + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 828, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & -2 \\ 5 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{12}23^11^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 848, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & -2 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

4.197 NHC(-2)(-3) + e₇ + su₂ + su₂→(-3) + su₂

$$T = 10 \quad [0] \ 1^{131}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & 2 & -2 \\ 6 & 9 & 0 & 6 & 25 & 9 & 9 \\ 2 & 3 & 0 & 2 & 9 & 1 & 3 \\ 3 & 3 & -1 & -2 & 9 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{122}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{121}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{121}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 8 & 3 \\ 2 & 1 & 1 & 2 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 6 & -2 \\ 4 & 6 & 0 & 4 & 10 & 18 & 6 \\ 8 & 9 & 0 & 6 & 18 & 25 & 9 \\ 3 & 3 & -1 & -2 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^{11} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{12} 1^{12} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 1312, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{12} 1^{12} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 1752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [19] $1^{12} 231^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{12} 231^{11} 31^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[21] $1^{12} 231^{11} 31^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[22] $1^{12} 231^{11} 31^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -408, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 15.29 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[23] $1^{12} 231^{11} 31^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -411, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.78 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 \\ 3 & 3 & -1 & -2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

4.200 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{12311}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12311}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 6 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

4.201 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 11 \quad [0] \quad 1^{12311}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.202 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{121}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{112}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 4 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 3 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12} 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 5 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 6 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{12} 3^1 1^{12} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & 0 & -2 \\ 4 & 12 & 0 & 6 & 22 & 6 & 0 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} 3^1 1^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.08 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12} 3^1 1^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & 0 & -1 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12} 3^1 1^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -572, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & 0 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^1 1^{12} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & 0 & -2 \\ 5 & 12 & 0 & 6 & 22 & 6 & 0 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

4.203 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, \quad T_{\max} = 25, \quad T \in [15, 25]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.204 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 9 \quad [0] \quad 1^{141} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -46, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & 2 & -3 \\ 4 & 8 & 0 & 4 & 30 & 12 & 8 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 8 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -65, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.09 \rightarrow 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 2 & 0 & -4 & 2 & 2 & -2 \\ 3 & 3 & 0 & 2 & 1 & 3 & 3 \\ 4 & 3 & -1 & -2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -78, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 3 & 2 & 0 & 1 & 0 & 3 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -95, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 2 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -95, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.17 \rightarrow 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & 2 & -3 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 & 2 \\ 4 & 7 & 0 & 4 & 7 & 23 & 7 & 7 \\ 2 & 2 & -1 & -3 & 2 & 7 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{131} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -130, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.46 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & 2 & -3 \\ 3 & 6 & 0 & 3 & 16 & 9 & 6 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 6 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{123} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.54 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 6 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -3 & 4 & 7 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{142}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 3 & -3 \\ 4 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -3 & 8 & 5 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{121}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -3 & 4 & 3 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{124}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 15.16 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 5 & -3 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -3 & 4 & 9 & -3 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{122}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & -3 & 4 & 5 & -2 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{132}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -3 & 6 & 5 & -2 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

[31] $1^{1^{11}} 1^{132} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -623, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 4 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -3 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

[32] $1^{1^{11}} 1^{132} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -692, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 4 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -3 & 6 & 5 & -2 \end{pmatrix}_{b_0}$$

[33] $1^{1^{12}} 1^{112} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{1^{11}} 1^{141} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -751, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 5 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -3 & 8 & 3 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{1^{11}} 1^{124} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -820, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 5 & -3 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -3 & 4 & 9 & -2 \end{pmatrix}_{b_0}$$

[36] $1^{1^{12}} 1^{131} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -830, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 5 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -3 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{1^{12}} 1^{142} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 3 & -3 \\ 6 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -3 & 8 & 5 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{1^{11}} 1^{142} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -880, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 3 & -3 \\ 5 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -3 & 8 & 5 & -2 \end{pmatrix}_{b_0}$$

[39] $1^{113}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -899, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -3 & 2 & 7 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{141}3^{12}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 6 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -3 & 8 & 3 & -2 \end{pmatrix}_{b_0}$$

[41] $1^{113}3^{12}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -3 & 2 & 7 & -2 \end{pmatrix}_{b_0}$$

[42] $1^{114}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 5 & -3 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{114}3^{12}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 5 & -3 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [44] $1^{123}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{123}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -2 & 0 & 5 & -2 \end{pmatrix}_{b_0}$$

[46] $1^{123}1^{11}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[47] $1^1 23^1 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & -2 \\ 2 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[48] $1^1 23^1 1^{11} 31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & 2 & -2 \\ 3 & 0 & 0 & 3 & 4 & 3 & 4 \\ 2 & 0 & 0 & 2 & 3 & 0 & 2 \\ 2 & 0 & -1 & -2 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

[49] $1^1 23^1 131^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -2 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

[50] $1^1 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 5 & -1 & 6 \\ 7 & 10 & 0 & 5 & 8 & 6 & 6 \\ 4 & 6 & -1 & -1 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

[51] $1^1 23^1 1^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[52] $1^1 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 303, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -2 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

[53] $1^1 23^1 1^{12} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 322, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.19 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 8 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -2 & 0 & 10 & -2 \end{pmatrix}_{b_0}$$

[54] $1^1 23^1 1^{11} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[55] $1^1 23^1 1^{11} 31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

[illegible]

4.205 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 21^{\textcolor{red}{1}} + [\textcolor{red}{s}u_{\textcolor{red}{2}} \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}g_7 + \textcolor{red}{s}u_{\textcolor{red}{2}} + \textcolor{red}{s}u_{\textcolor{red}{2}} + \textcolor{blue}{s}u_{\textcolor{blue}{2}} \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 2 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 0 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1211} 3^{1211} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 174, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & 1 & -3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 4 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 4 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[2] \quad 1^{1211} 3^{1211} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 356, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & 1 & -3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 4 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 4 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[3] \quad 1^{1121} 3^{1211} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c} \textcolor{red}{g}_7 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 384, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 5 & 2 & -2 \end{pmatrix} b_0$$

[4] $1^{11111}3^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[5] \quad 1^{1111} 3^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 428, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[6] \quad 1^{1111} 3^{1211} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 486, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[7] \quad 1^{1112} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 588, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 4 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1121} 3^{121} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 656, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1121} 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 768, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1112} 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 838, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 4 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1121} 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 964, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [12] \quad 1^1 23^1 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.206 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_{2 \rightarrow (-3)}$

$$T = 11 \quad [0] \quad 1^1 23^1 1^{13} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.207 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

4.208 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -113, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & -2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 6 & -2 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & -2 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{112} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{113} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.79 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{131} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{141} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{151} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 15.19 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 15 & -2 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -2 & 20 & -2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -641, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

14.0

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{141} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -2577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{141} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -3008, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [29] \ 1^{111} 2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 24, T_{\max} = 100, T \in [24, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -10 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{122} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 32.40 \rightarrow 33, T_{\max} = 100, T \in [33, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 0 & 1 & -5 & 2 & 0 & -3 \\ 3 & 0 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 6 & -1 & -19 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112} 2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 17.67 \rightarrow 18, T_{\max} = 100, T \in [18, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 2 & 6 & 0 & 2 & 6 & 9 & 5 \\ 2 & 3 & -1 & -3 & 3 & 5 & -4 \end{pmatrix}_{b_0}$$

$$[32] \ 1^1 231^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.35 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 3 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[34] \ 1^1 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 18.30 \rightarrow 19, T_{\max} = 100, T \in [19, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 0 & -1 & -5 \end{pmatrix}_{b_0}$$

[35] $1^{131}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 100, T \in [19, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 4 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -5 \end{pmatrix}_{b_0}$$

[36] $1^{122}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.7$

$T_{\min} = 15.38 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 3 & 6 & 0 & 2 & 10 & 12 & 6 \\ 2 & 6 & 0 & 2 & 12 & 9 & 5 \\ 2 & 3 & -1 & -3 & 6 & 5 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{1231}1^{31}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.34 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{1231}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -2 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

[42] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

[44] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -3 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{1231}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 0 & -2 \\ 4 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$$

[46] $1^{142}1^{12}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 6 & 12 & 0 & 4 & 46 & 24 & 12 \\ 2 & 6 & 0 & 2 & 24 & 9 & 5 \\ 2 & 3 & -1 & -3 & 12 & 5 & -2 \end{pmatrix}_{b_0}$$

[47] $1^{121}1^{11}2^13^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & 0 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{111}1^{12}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{132}1^{11}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 1 & 9 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 4 & 9 & 0 & 3 & 25 & 18 & 9 \\ 2 & 6 & 0 & 2 & 18 & 9 & 5 \\ 2 & 3 & -1 & -3 & 9 & 5 & -2 \end{pmatrix}_{b_0}$$

[50] $1^{111}1^{12}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & -3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{111} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 6 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 179, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[55] $1^{142} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 5 & 12 & 0 & 4 & 46 & 24 & 12 \\ 2 & 6 & 0 & 2 & 24 & 9 & 5 \\ 2 & 3 & -1 & -3 & 12 & 5 & -2 \end{pmatrix}_{b_0}$$

[57] $1^{111} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{121} 3^{221} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{121} 3^{221} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 8 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -3 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{121} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -2 \end{pmatrix}_{b_0}$$

[63] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[64] $1^{131} 3^{221} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 214, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[65] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 217, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

[66] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & -3 \\ 2 & 0 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 3 & -1 & -2 \end{pmatrix}_{b_0}$$

[67] $1^{13}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 15.62 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 0 & 2 \\ 1 & 0 & 1 & -5 & 3 & 0 & -3 \\ 5 & 0 & 9 & 3 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 9 & -1 & -2 \end{pmatrix}_{b_0}$$

[68] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 243, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{111}3^1221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 7 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{141}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.31 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[71] $1^1231^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[72] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{151}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$

$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 15 & -1 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 6 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[76] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 365, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -3 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{112}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 391, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{131}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 406, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.78 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 1 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{112}3^12^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 434, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & -3 \\ 3 & 0 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 3 & -1 & -2 \end{pmatrix}_{b_0}$$

[83] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 452, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{131}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 469, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{121}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 478, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{141}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{121}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 0 & 1 & -5 & 2 & 0 & -3 \\ 4 & 0 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 6 & -1 & -2 \end{pmatrix}_{b_0}$$

[88] $1^{121}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 499, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 5 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{1231}1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 508, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

[90] $1^{121}1^{11}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 526, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{141}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 548, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 15.36 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -5 & 4 & 1 & -3 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -3 & 12 & 2 & -2 \end{pmatrix}_{b_0}$$

[92] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 562, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{131}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 584, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

[94] $1^{121}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 590, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 9 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{141}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 14 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{141}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 718, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -5 & 4 & 1 & -3 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 2 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{121}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 736, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{121}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{131}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{121} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 814$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{131} 2^1 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 850$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[102] $1^{121} 3^{12} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 864$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 10 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{131} 2^1 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 907$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 6 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{1231} 2^{1231} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -148$, $\det = 937$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 0 & -2 \\ 5 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$$

[105] $1^{121} 2^1 3^{121} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 942$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[106] $1^{131} 2^1 3^{121} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 994$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 2 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[107] $1^{121} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 1006$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 11 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{141} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 1006$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{121} 3^{14} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 1016, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 12 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{131} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 1016, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 12 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{131} 2^3 3^{14} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1028, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 7 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{141} 2^3 3^{13} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -5 & 4 & 1 & -3 \\ 7 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -3 & 12 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{141} 2^3 3^{12} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{131} 3^{12} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 1174, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{1231} 3^{13} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 1300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & 0 & -2 \\ 7 & 18 & 0 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 12 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{141} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 1454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 15 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{131} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1456, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{131} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1494, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.72 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 13 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{141} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1544, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -5 & 4 & 1 & -3 \\ 8 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{231} 1^{431} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 1597, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.02 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 24 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & 0 & -2 \\ 9 & 24 & 0 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{131} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1606, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{141} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1624, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.78 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{131} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1840, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 14 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{131} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2054, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 15 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{141} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2110, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{141} 3^{112} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 2136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 16 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{141} 3^{114} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 2464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{141} 3^{113} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 2686, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 17 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{141} 3^{114} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 3104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 18 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

4.209 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 156, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 \\ 2 & 2 & -1 & -3 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -2 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 589, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1088, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \quad 1^{11} \quad 3 \quad 221^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix} b_0$$

$$[12] \quad 1^{111} 3^{221} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -65, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix} b_0$$

$$[13] \quad 1^{1111} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 \end{pmatrix} b_0$$

[14] $1^{1111}3^{221}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix} b_0$$

$$[15] \quad 1^{1111} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & -1 \end{pmatrix} b_0$$

[16] $1^{111} 3^{221} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 & -2 \end{pmatrix} b_0$$

$$[17] \quad 1^{111} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -149, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & -1 & 0 & 9 & 0 \end{pmatrix} b_0$$

[18] $1^{113} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.8$

$$T_{\min} = 15.39 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -246, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, \quad T_{\max} = 25, \quad T \in [16, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -269, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.90 \rightarrow 15, \quad T_{\max} = 23, \quad T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 14 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & -1 & 0 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -290, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 25, \quad T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -332, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, \quad T_{\max} = 23, \quad T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 11 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & -1 & 0 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -341, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, \quad T_{\max} = 23, \quad T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1111} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.81 \rightarrow 15, \quad T_{\max} = 23, \quad T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1231} 3^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 38, \quad \Delta = -118, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.05 \rightarrow 16, \quad T_{\max} = 38, \quad T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{112} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.83 \rightarrow 15, \quad T_{\max} = 23, \quad T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1111} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -408, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.77 \rightarrow 15, \quad T_{\max} = 23, \quad T \in [15, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -432, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.96 \rightarrow 15, \quad T_{\max} = 25, \quad T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -434, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.72 \rightarrow 15, \quad T_{\max} = 25, \quad T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1231} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 38, \quad \Delta = -118, \quad \det = -455, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.09 \rightarrow 16, \quad T_{\max} = 38, \quad T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1111} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, \quad T_{\max} = 23, \quad T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.93 \rightarrow 15, \quad T_{\max} = 25, \quad T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -626, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.73 \rightarrow 15, \quad T_{\max} = 25, \quad T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1113} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -668, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{111} 4^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -674, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{111} 1^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -896, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{111} 4^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -1268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

4.210 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{111} 1^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 0 \\ 2 & 2 & -1 & -3 & 2 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 1^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -202, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \ 1^{111} 1^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 1^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11112} 2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -82$, $\det = 244$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11114} 2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -82$, $\det = 472$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11113} 2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -82$, $\det = 628$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.8$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 & 9 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11114} 2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -82$, $\det = 724$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.77 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

4.211 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 25$, $\Delta = -112$, $\det = -128$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.88 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.212 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 11$ [0] $1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 38$, $\Delta = -118$, $\det = -284$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 15.20 \rightarrow 16$, $T_{\max} = 38$, $T \in [16, 38]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{\overline{11111}} 2 \overline{3} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -91, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3 \overline{2} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 228, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.213 NHC(-2)(-3) + $\textcolor{red}{e}_7$ + $\textcolor{blue}{su}_2$ + $\textcolor{red}{su}_3$ + $\textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{121}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{121}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 100, \ \Delta = -142, \ \det = 1408, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{12} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -152, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.16 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 0 & 2 & -2 \\ 4 & 12 & 0 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -296, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.08 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 2 & -2 \\ 2 & 6 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 3^{11} 3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 59, \ \Delta = -120, \ \det = -364, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, \ T_{\max} = 59, \ T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & 0 & -1 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -572, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -1196, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 0 & 2 & -2 \\ 5 & 12 & 0 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

4.214 $\text{NHC}(-2)(-3) + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$

$$T = 11 \quad [0] \ 1^{11} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.215 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + su2n3mix

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

4.216 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 10 \quad [1] \quad 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & 0 & -2 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 4 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \quad 1^{111} 2^1 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 \\ 1 & 0 & 1 & -5 & 1 & 1 & 0 \\ 3 & 0 & 3 & 1 & 1 & 3 & 0 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 2^1 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -95, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 \\ 1 & 0 & 1 & -5 & 1 & 1 & 0 \\ 2 & 0 & 3 & 1 & 1 & 3 & 0 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 48, \Delta = -119, \det = -159, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.87 \rightarrow 16, T_{\max} = 48, T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 2^1 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -162, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{121} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -194, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & 0 & -3 \\ 4 & 0 & 6 & 2 & 10 & 6 & 0 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 48, \Delta = -119, \det = -231, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 48, T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -303, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 1 & -3 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 6 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1211} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -807, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 1 & -3 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 6 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1311} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1095, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1411} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1167, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.26 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & 1 & -3 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{1311}2^321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -1518, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$

$T_{\min} = 15.23 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{1411}2^321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -2418, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & 1 & -3 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

4.217 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 11$ [0] $1^{1111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$T = 12$ [1] $1^{1111}2^321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 12, T_{\max} = 16, \Delta = -91, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 12, T_{\max} = 16, \Delta = -89, \det = 228, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.218 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 8$ [0] $1^{131}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 102, \Delta = -193, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 2 & 0 & -4 & -2 \\ 3 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{131}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 102, \Delta = -193, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & -4 & 2 & -2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $21^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\max} = 102$, $\Delta = -193$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & -4 & 2 & -2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [3] $1^{12}3^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & -2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{12}313^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 102$, $\Delta = -171$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & -3 & -1 \\ 3 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{12}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 14.86 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & -2 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{12}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 102$, $\Delta = -171$, $\det = -129$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{12}321^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 102$, $\Delta = -173$, $\det = -595$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 14.76 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

4.219 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10$, $T_{\max} = 100$, $\Delta = -143$, $\det = 304$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.79 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 1 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [1] $1^{12}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 75$, $\Delta = -122$, $\det = -440$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 75$, $T \in [16, 75]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

4.220 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 103, \Delta = -192, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -4 & 2 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

4.221 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -4 & 2 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -316, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.44 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & -2 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.38 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -456, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.39 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.39 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 8 & -2 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -584, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -828, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.40 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -1048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.40 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 8 & -2 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [14] \quad 1^1 23^1 131^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 18, \quad T_{\max} = 100, \quad T \in [18, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 0 & -2 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 2 & 0 & -4 \end{pmatrix}_{b_0}$$

[24] $1^{123}1^{131}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 3 & -2 \\ 1 & 0 & 0 & 2 & 0 & 3 & 2 \\ 6 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[25] $1^{123}1^{231}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 6 & -2 \\ 1 & 0 & 0 & 2 & 0 & 6 & 2 \\ 8 & 0 & 0 & 6 & 6 & 22 & 8 \\ 2 & 0 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{123}1^{1131}1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.89 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{1231}1^{131}1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 313, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 4 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{1231}1^{131}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 6 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{123}1^{1131}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{123}1^{1231}1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 4 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{1231}1^{1231}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 5 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{123}1^{1231}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 540, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{123}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 828, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{123}1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

4.222 NHC(-2)(-3) + $\textcolor{violet}{e}_7$ + $\textcolor{brown}{su}_2 \rightarrow (-3)$ + $\textcolor{red}{su}_2$ + $\textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{123}1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & 2 & -2 \\ 6 & 9 & 0 & 6 & 25 & 9 & 9 \\ 2 & 3 & 0 & 2 & 9 & 1 & 3 \\ 3 & 3 & -1 & -2 & 9 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{122}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{121}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 10 & 3 \\ 2 & 1 & 1 & 2 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 8 & -2 \\ 4 & 6 & 0 & 4 & 10 & 24 & 6 \\ 10 & 12 & 0 & 8 & 24 & 46 & 12 \\ 3 & 3 & -1 & -2 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 808, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 7 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 824, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1312, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [19] \quad 1^1 231^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^1 23^{11} 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 3 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^1 23^{11} 1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 0 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{123}1^{12}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.29 \rightarrow 16, \ T_{\max} = 25, \ T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1231}1^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -411, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.78 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

4.223 NHC(-2)(-3) + $\textcolor{green}{e}_7$ + $\textcolor{blue}{su}_2 \rightarrow (-3)$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2$

$$T = 11 \quad [0] \ 1^{111}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1211}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -256, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 6 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -400, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -576, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1112}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -960, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -1040, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.224 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$T = 11$ [0] $1^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.225 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{11} 23^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 27, \Delta = -114, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 231^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 27, \Delta = -114, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 6 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

4.226 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$T = 11$ [0] $1^{11} 23^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.227 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$T = 10$ [0] $1^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1}231}1^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & 0 & -1 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1}23}1^{\overline{1}1}31^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -572, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1}23}1^{\overline{1}2}31^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$$

4.228 NHC(-2)(-3) + $\textcolor{violet}{e}_7$ + $\textcolor{brown}{su}_2 \rightarrow (-3)$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_3$ + $\textcolor{blue}{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{1}1}1^{\overline{1}1}3^{\overline{1}1}21^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.229 NHC(-2)(-3) + $\textcolor{violet}{e}_7$ + $\textcolor{brown}{su}_2 \rightarrow (-3)$ + $\textcolor{brown}{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{1}23}1^{\overline{1}3}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 3 & -1 \\ 2 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}231}1^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 2 & -1 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 3 & 0 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}23}1^{\overline{1}3}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & -1 \\ 2 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}231}1^{\overline{1}3}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

4.230 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 231^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 4 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 2 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

4.231 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^1 23^1 13^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 3 & 2 & -1 \\ 2 & 0 & 0 & 3 & 1 & 2 & 3 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 3 & 0 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 2 & -1 \\ 2 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.232 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$

$$T = 11 \quad [0] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 2 & -1 \\ 2 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 4 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 & 2 \\ 3 & 5 & 0 & 3 & 2 & 1 & 3 \\ 3 & 6 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

4.233 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 81, \Delta = -123, \det = -312, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 81, T \in [16, 81]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

4.234 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$

$T = 10$ [0] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 4 & 0 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.79 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.235 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix} + \mathfrak{su}_2$

$T = 11$ [0] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

4.236 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 34, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 4 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 34, \Delta = -117, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.237 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -4 & 2 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \ 1^{12} 23^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18, T_{\max} = 101, T \in [18, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 0 & -2 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -2 & 2 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} 3^1 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} 3^1 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -150, \det = 237, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

4.238 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -2 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{123}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -572$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.07 \rightarrow 16$, $T_{\max} = 70$, $T \in [16, 70]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 \\ 2 & 4 & -1 & -2 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

[11] $1^{123}1^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -1196$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 15.07 \rightarrow 16$, $T_{\max} = 70$, $T \in [16, 70]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 12 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 6 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 \\ 5 & 12 & 0 & 6 & 6 & 0 & 22 \\ 2 & 4 & -1 & -2 & 2 & -1 & 8 \end{pmatrix}_{b_0}$$

4.239 NHC(-2)(-3) + e₇ + su₂→(-3) + su₃ + su₂ + su₂

$T = 11$ [0] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 25$, $\Delta = -112$, $\det = -128$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.88 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 0 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & -2 & 2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

4.240 NHC(-2)(-3) + e₇ + su₂→(-3) + su₃ + su₂→(-3)

$T = 11$ [0] $1^{123}1^{13}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 81$, $\Delta = -123$, $\det = -312$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.13 \rightarrow 16$, $T_{\max} = 81$, $T \in [16, 81]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

4.241 NHC(-2)(-3) + e₇ + su₂→(-3) + su₃ + su₃

$T = 10$ [0] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 60$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 15.20 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 0 & 5 \\ 2 & 4 & -4 & 2 & 0 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

4.242 NHC(-2)(-3) + e₇ + su₂n₃mix

$T = 8$ [0] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 8$, $T_{\max} = 102$, $\Delta = -193$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 14.63 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & -4 & -2 \\ 4 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 8$, $T_{\max} = 102$, $\Delta = -193$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 15.04 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & -3 \\ 6 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -3 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[12] \text{ } 1^{12} 3^{12} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 102, \Delta = -193, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 6 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -3 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[13] \text{ } 1^{12} 3^{12} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 102, \Delta = -193, \det = 1004, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & -3 \\ 7 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -3 & 9 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [14] \text{ } 1^{12} 23^{11} 1^{11} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[15] \text{ } 1^{12} 231^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 \\ 6 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & -3 & -1 \\ 4 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[16] \text{ } 1^{12} 231^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 4 & -2 \\ 4 & 0 & 0 & 4 & 9 & 5 \\ 2 & 0 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[17] \text{ } 1^{12} 23^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & -2 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[18] \text{ } 1^{12} 231^{11} 3^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 4 & -2 \\ 6 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[19] \text{ } 1^{12} 2313^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 0 & 0 & -3 & -1 \\ 4 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[20] \text{ } 1^{12} 23^{11} 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \text{ } 1^{12} 23^{11} 1^{11} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 11 & -2 \\ 11 & 0 & 0 & 11 & 72 & 14 \\ 2 & 0 & -1 & -2 & 14 & -1 \end{pmatrix}_{b_0}$$

[33] $1^1 23^1 1^1 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & -2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[34] $1^1 23^1 1^1 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -449$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.92 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 5 & -2 \\ 6 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[35] $1^1 23^1 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -572$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & -2 \\ 5 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

[36] $1^1 23^1 1^2 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -704$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 8 & -2 \\ 9 & 0 & 0 & 8 & 36 & 10 \\ 2 & 0 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[37] $1^1 23^1 1^2 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -753$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & -2 \\ 6 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[38] $1^1 23^1 1^1 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -759$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 5 & -2 \\ 7 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[39] $1^1 23^1 1^2 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -912$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.68 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & -2 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[40] $1^1 23^1 1^2 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -964$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & -2 \\ 7 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[41] $1^1 23^1 1^2 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -1017$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & -2 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

4.243 NHC(-2)(-3) + e₇ + su2n3mix + su₂

$T = 9$ [0] $1^{1143^1} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 4 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -3 & 3 & 8 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.09 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 3 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 0 & -3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 3 & 0 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{131}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.17 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 2 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -3 & 7 & 2 & -3 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 14.46 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 3 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -3 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.54 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 6 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -3 & 7 & 4 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{124}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 4 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -3 & 5 & 8 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -3 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{142}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 15.16 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 2 & -3 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 3 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -3 & 9 & 4 & -3 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{122}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 4 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -3 & 5 & 4 & -2 \end{pmatrix}_{b_0}$$

[13] $1^{121}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 2 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{123}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 3 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -3 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^{12}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 4 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{122}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -3 & 5 & 4 & -2 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 3 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -3 & 5 & 4 & -2 \end{pmatrix}_{b_0}$$

[26] $1^{1^{11}} 1^{132} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -518, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 5 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -3 & 7 & 4 & -2 \end{pmatrix}_{b_0}$$

[27] $1^{1^{12}} 1^{142} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 2 & -3 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -3 & 9 & 4 & -2 \end{pmatrix}_{b_0}$$

[28] $1^{1^{12}} 1^{142} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 2 & -3 \\ 7 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -3 & 9 & 4 & -2 \end{pmatrix}_{b_0}$$

[29] $1^{1^{11}} 1^{141} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.55 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 1 & -3 \\ 5 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

[30] $1^{1^{11}} 1^{121} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -609, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

[31] $1^{1^{11}} 1^{113} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -623, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 4 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -3 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

[32] $1^{1^{11}} 1^{123} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -692, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 4 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -3 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$

[33] $1^{1^{12}} 1^{121} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{114}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -751, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 5 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -3 & 3 & 8 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{142}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -820, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 2 & -3 \\ 6 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -3 & 9 & 4 & -2 \end{pmatrix}_{b_0}$$

[36] $1^{112}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -830, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 5 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -3 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{124}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 6 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -3 & 5 & 8 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{124}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -880, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 5 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -3 & 5 & 8 & -2 \end{pmatrix}_{b_0}$$

[39] $1^{131}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -899, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 5 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -3 & 7 & 2 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{114}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 6 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -3 & 3 & 8 & -2 \end{pmatrix}_{b_0}$$

[41] $1^{131}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 6 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -3 & 7 & 2 & -2 \end{pmatrix}_{b_0}$$

[42] $1^{141}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 1 & -3 \\ 6 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{141}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 1 & -3 \\ 7 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [44] $1^{123}1^{131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{1231}1^{131}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 4 & 0 & -2 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 5 & 0 & -2 \end{pmatrix}_{b_0}$$

[46] $1^{123}1^{11}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 6 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{123}1^{131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 3 & -2 \\ 2 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[49] $1^{123}1^{131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 0 & -2 \\ 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$

[50] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 5 & -1 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 4 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.88 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 303, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 4 & 0 & -2 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 5 & 0 & -1 \end{pmatrix}_{b_0}$$

[53] $1^{1231}1^{1231}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 322, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.19 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 10 & 0 & -2 \end{pmatrix}_{b_0}$$

[54] $1^{1231}1^{11}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{1231}1^{11}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{1231}1^{1231}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 562, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{1231}1^{1231}1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 9 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 10 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{123}1^{12}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 718, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{123}1^{12}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 744, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 7 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 10 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{123}1^{12}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 850, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 8 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 10 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{123}1^{12}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 942, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{123}1^{12}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 994, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

4.244 NHC(-2)(-3) + e₇ + su₂n₃mix + su₂ + su₂

$$T = 10 \quad [0] \ 1^{11}1^{11}3^{121}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 0 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}1^{12}3^{121}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 174, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 3 & 4 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}1^{12}3^{121}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{1112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 838$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 2 & -3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 6 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 3 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{1211}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 964$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$T = 11$ [12] $1^{1231}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 25$, $\Delta = -112$, $\det = -290$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

4.245 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2\mathfrak{n}3\text{mix} + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{1231}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 27$, $\Delta = -114$, $\det = -168$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15$, $T_{\max} = 27$, $T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

4.246 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2\mathfrak{n}3\text{mix} + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 181$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.13 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 6 & 0 & -2 \end{pmatrix}_{b_0}$$

4.247 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2\mathfrak{n}3\text{mix} + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{1231}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 100$, $\Delta = -144$, $\det = 16$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1231}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 100$, $\Delta = -144$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 4 & 0 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{\textcolor{teal}{1}}231^{\textcolor{teal}{1}}3^{\textcolor{teal}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 0 & 2 \\ 4 & 6 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\textcolor{teal}{1}}231^{\textcolor{teal}{1}}3^{\textcolor{teal}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\textcolor{teal}{1}}231^{\textcolor{teal}{1}}3^{\textcolor{teal}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.79 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.248 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$T = 11 \quad [0] \quad 1^{\textcolor{teal}{1}}231^{\textcolor{teal}{1}}3^{\textcolor{teal}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

4.249 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11 \quad [0] \quad 1^{\textcolor{teal}{1}}231^{\textcolor{teal}{1}}3^{\textcolor{teal}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 4 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\textcolor{teal}{1}}231^{\textcolor{teal}{1}}3^{\textcolor{teal}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.250 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{su2n3mix} + \text{su2n3mix}$

$T = 10 \quad [0] \quad 1^{\textcolor{teal}{1}}231^{\textcolor{teal}{1}}3^{\textcolor{teal}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & -1 \\ 4 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.251 NHC(-2)(-3) + e₇ + su₂n₃mix + su₃

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 4 & 3 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -977, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -1280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \quad 1^{123} 3^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 3 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 6 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{123} 3^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 223, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.23 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{12}31^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -150$, $\det = 358$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 4 & 0 & -2 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 5 & -1 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{12}3^{11}31^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 682$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 6 & -1 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{11}31^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 1075$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 10 & -1 & -2 \end{pmatrix}_{b_0}$$

4.252 NHC(-2)(-3) + $\mathfrak{e}_7$ + su2n3mix + su_3 + su_2

$T = 10$ [0] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 181$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.13 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 6 & 0 & -2 \end{pmatrix}_{b_0}$$

4.253 NHC(-2)(-3) + $\mathfrak{e}_7$ + su_3

$T = 8$ [0] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 68$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & -1 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -4 & -2 \\ 3 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{12}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

- [4] $1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.13 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -3 & 5 & -3 \end{pmatrix}_{b_0}$$
- [5] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 170$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -4 & -2 \\ 4 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 261$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$
- [7] $1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 358$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.61 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -3 & 5 & -2 \end{pmatrix}_{b_0}$$
- $T = 9$ [8] $1^{13}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -9$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 102$, $T \in [19, 102]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & -4 \end{pmatrix}_{b_0}$$
- [9] $1^{12}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 15.10 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 2 & 6 & 0 & 2 & 9 & 5 \\ 2 & 3 & -1 & -3 & 5 & -2 \end{pmatrix}_{b_0}$$
- [10] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 16.73 \rightarrow 17$, $T_{\max} = 102$, $T \in [17, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & -3 \\ 2 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$
- [11] $1^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [12] $1^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -57$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.68 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [13] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & -1 & -2 \end{pmatrix}_{b_0}$$

- [14] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -71$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$
- [15] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{12} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -86$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$
- [17] $1^{13} 2^3 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 4 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$
- [18] $1^{13} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -2 & 5 & -2 \end{pmatrix}_{b_0}$$
- [19] $1^{12} 31^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.08 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [20] $1^{12} 31^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 3 & 0 & 0 & 3 & 3 & 3 \\ 2 & 0 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [21] $1^{12} 31^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -119$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.57 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [22] $1^{12} 31^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [23] $1^{11} 2^3 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -129$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [24] $1^{12} 2^3 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 14.81 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 3 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -3 & 5 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.76 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -3 & -1 \\ 4 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{11} 2^3 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 3 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{12} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

[28] $1^1 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.19 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 2 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[29] $1^1 231^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{13} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{12} 2^3 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 4 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -3 & 5 & -1 \end{pmatrix}_{b_0}$$

[32] $1^{13} 2^3 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 5 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -3 & 8 & -1 \end{pmatrix}_{b_0}$$

[33] $1^1 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.254 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 101, \Delta = -149, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -4 & 0 & 3 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -2 & -1 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1111} 2321^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 75, \Delta = -122, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.27 \rightarrow 19, T_{\max} = 75, T \in [19, 75]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 4 \\ 1 & 3 & 0 & 1 & 3 & 4 & 0 \\ 2 & 3 & -1 & -3 & 2 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.12 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 3 & 4 & -1 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

4.255 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{black}{\mathfrak{so}}_8$

$$T = 9 \quad [0] \quad 1^{11} 321^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 103, \Delta = -198, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 1^{11} 2321^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -171, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.83 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 & 5 \\ 2 & 0 & 4 & -4 & 0 & -2 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 0 & 5 & -2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 2321^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -171, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 & -2 \\ 4 & 12 & 0 & 4 & 9 & 5 & 5 \\ 3 & 6 & -1 & -2 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -169, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

4.256 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$

$$T = 9 \quad [0] \quad 1^{11} 321^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -113, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{115} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 15.19 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 15 & -2 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -2 & 20 & -2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{121} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -641, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -932, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

13.5

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -932, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -993, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{114} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1433, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1961, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1988, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -2577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{114} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -3008, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [29] $1^{111} 23 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 24, T_{\max} = 100, T \in [24, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -10 \end{pmatrix}_{b_0}$$

[30] $1^{122} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 32.40 \rightarrow 33, T_{\max} = 100, T \in [33, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & -1 & 6 & -19 \end{pmatrix}_{b_0}$$

[31] $1^{121} 23 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 17.67 \rightarrow 18, T_{\max} = 100, T \in [18, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 6 & 0 & 2 & 9 & 6 & 5 \\ 1 & 3 & 0 & 1 & 6 & 1 & 3 \\ 2 & 3 & -1 & -3 & 5 & 3 & -4 \end{pmatrix}_{b_0}$$

[32] $1^{123} 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[33] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.35 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{12} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 18.30 \rightarrow 19, T_{\max} = 100, T \in [19, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & -1 & 0 & -5 \end{pmatrix}_{b_0}$$

[35] $1^{113} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 100, T \in [19, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 4 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -5 \end{pmatrix}_{b_0}$$

- [36] $1^{122}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.7$
 $T_{\min} = 15.38 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 6 & 0 & 2 & 9 & 12 & 5 \\ 3 & 6 & 0 & 2 & 12 & 10 & 6 \\ 2 & 3 & -1 & -3 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$
- [37] $1^{1231}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 101, \Delta = -148, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.34 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$
- [38] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [39] $1^{11}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -140, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [40] $1^{1231}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 101, \Delta = -148, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$
- [41] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [42] $1^{11}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -140, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.03 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$
- [43] $1^{12}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.43 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$
- [44] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 3 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{1231} 1^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & -1 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{124} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 1 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 6 & 0 & 2 & 9 & 24 & 5 \\ 6 & 12 & 0 & 4 & 24 & 46 & 12 \\ 2 & 3 & -1 & -3 & 5 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{112} 1^{11} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{111} 1^{12} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{111} 1^{11} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & -3 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{123} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 6 & 0 & 2 & 9 & 18 & 5 \\ 4 & 9 & 0 & 3 & 18 & 25 & 9 \\ 2 & 3 & -1 & -3 & 5 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{111} 1^1 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{111} 1^{11} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 6 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{121} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 \\ 2 & 4 & 0 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 0 \\ 2 & 2 & -1 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 179, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{124} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 1 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 6 & 0 & 2 & 9 & 24 & 5 \\ 5 & 12 & 0 & 4 & 24 & 46 & 12 \\ 2 & 3 & -1 & -3 & 5 & 12 & -2 \end{pmatrix}_{b_0}$$

[55] $1^{121} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[59] $1^{112} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{112} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 8 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

- [61] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 196$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 14.69 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 4 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$
- [62] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 203$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 15.21 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$
- [63] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 205$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [64] $1^{113}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 214$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$
- [65] $1^1231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 101$, $\Delta = -148$, $\det = 217$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$
- [66] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 221$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 15.10 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$
- [67] $1^{132}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 226$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 15.62 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -5 & 0 & 3 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -3 & -1 & 9 & -2 \end{pmatrix}_{b_0}$$
- [68] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 243$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.77 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [69] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 246$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 7 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{114} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.31 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{1231} 1^{11} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3^{111} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{115} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 15 & -1 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{111} 3^{112} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{112} 2^1 3^{14} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{121} 2^1 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 365, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 3 & 0 & 6 & 2 & 9 & 6 & 5 \\ 2 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{112} 2^1 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 373$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{121}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 391$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{113}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 406$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 11 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 429$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 434$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[83] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 452$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{113}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 469$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 478$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{114}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{12}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

[88] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 499, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{1231}1^{131}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 508, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[90] $1^{112}2^13^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 526, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{114}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 548, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.36 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

[92] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 562, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 584, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$

[94] $1^{112} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 590, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 9 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{114} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 14 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{114} 2^1 3^{112} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 718, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{112} 3^{112} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 736, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{112} 3^{114} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{112} 3^{113} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 814, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{113} 2^1 3^{112} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 850, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{112} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 864, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.70 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 10 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{113} 2 \ 3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -142, \ \det = 907, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{123} 1^{12} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -148, \ \det = 937, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.02 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & -1 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{112} 2 \ 3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -142, \ \det = 942, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{113} 2 \ 3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -142, \ \det = 994, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{114} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 1006, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{112} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 1006, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.70 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 11 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{113} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 1016, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 12 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{112} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 1016, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.70 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 12 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1028, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.91 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1147, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.02 \rightarrow 16, \quad T_{\max} = 100, \quad T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1150, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{113} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1174, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.76 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 4 \\ 3 & 4 & 1 & 3 & 18 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 & 1 \\ 11 & 18 & 0 & 9 & 52 & 12 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{1231} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 1300, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 15.02 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & -1 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{114} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1454, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 & 4 \\ 3 & 4 & 1 & 3 & 27 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 & 1 \\ 15 & 27 & 0 & 17 & 123 & 21 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{113} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1456, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 & 4 \\ 3 & 4 & 1 & 3 & 18 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 & 1 \\ 12 & 18 & 0 & 9 & 52 & 12 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{113} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1494, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.72 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 & 4 \\ 3 & 4 & 1 & 3 & 21 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 & 1 \\ 13 & 21 & 0 & 14 & 75 & 17 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[119] \quad & 1^{114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1544, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.94 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 2 & 12 & -1 \end{pmatrix}_{b_0} \\
[120] \quad & 1^{1231} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 1597, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 15.02 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & -1 & 16 & -2 \end{pmatrix}_{b_0} \\
[121] \quad & 1^{113} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1606, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8 \\
& T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0} \\
[122] \quad & 1^{114} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1624, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.78 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0} \\
[123] \quad & 1^{113} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1840, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8 \\
& T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 14 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0} \\
[124] \quad & 1^{113} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2054, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8 \\
& T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 15 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{114} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2110, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.76 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{114} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2136, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.73 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 16 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0} \\
[127] \quad & 1^{114} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2464, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.75 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{114} 3^{113} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 2686, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.72 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 17 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{114} 3^{114} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 3104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.72 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 18 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

4.257 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 68, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 79, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 112, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 156, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.18 \rightarrow 16, \ T_{\max} = 100, \ T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & -1 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 221, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.18 \rightarrow 16, \ T_{\max} = 100, \ T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -2 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 413, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 589, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1088, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \ 1^{11} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -65, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 23 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 2 \ 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 \\ 2 & 2 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -149, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.39 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111} 2 \ 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -246, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{113} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 14 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & -1 & 0 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1111} 2 \ 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{112} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -332, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 11 & -1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 & 13 \\ 3 & 4 & -1 & -1 & 0 & 13 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -341, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{123} 3^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{112} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -380, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -408, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -432, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -434, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1231} 1^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -455, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -485, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1111} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{1113} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -626, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{1113} 2^1 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -668, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{1114} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -674, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{1111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -896, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{\overline{1114}} 2^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -1268, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$4.258 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 11 \quad [0] \ 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -109, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & -1 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{111}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -109, \ \det = -202, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \ 1^{\overline{11111}} 2^{\overline{1}} 23^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 24, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{111}} 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -80, \ \det = 166, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.95 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11112}} 2^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 244, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.72 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11114}} 2^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 472, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11113} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 628, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11114} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 724, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.77 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$4.259 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \ 1^{11113} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.260 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 11 \quad [0] \ 1^{11111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 38, \ \Delta = -118, \ \det = -284, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 38, \ T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{11111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, \ T_{\max} = 16, \ \Delta = -91, \ \det = 108, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11111} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, \ T_{\max} = 16, \ \Delta = -89, \ \det = 228, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.261 NHC(-2)(-3) + e₇ + su₃ + su₂ + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{12} 3^1 1^{12} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.08 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1231}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -572, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{123}1^{12}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

4.262 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$

$$T = 11 \quad [0] \ 1^{111}3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.263 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$

$$T = 10 \quad [0] \ 1^{111}3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

4.264 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_3$

$$T = 10 \quad [1] \ 1^{111}3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -4 & 0 & 3 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 101, \ \Delta = -148, \ \det = 192, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 101, \ \Delta = -148, \ \det = 280, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -14, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 0 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 0 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -95, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.11 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 0 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 0 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 48, \ \Delta = -119, \ \det = -159, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.87 \rightarrow 16, \ T_{\max} = 48, \ T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -162, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -194, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.11 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 0 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 0 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 48, \ \Delta = -119, \ \det = -231, \ (n_+, n_-, n_0) =$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 12 & 3 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & 12 & 2 & -2 \end{pmatrix}_{b_0}$$

4.265 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{1111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 38, \ \Delta = -118, \ \det = -284, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 38, \ T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{\overline{1111}} 2 \ 321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 16, \ \Delta = -91, \ \det = 108, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 16, \ \Delta = -89, \ \det = 228, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.266 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.90 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -192, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.75 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \ 1^{\overline{1}} 23^{\overline{1}} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 16, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 18, \ T_{\max} = 101, \ T \in [18, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -2 & -1 & 2 & -4 \end{pmatrix}_{b_0}$$

[4] $1^{12}3^11^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 2 & -4 \end{pmatrix}_{b_0}$$

[5] $1^{12}3^11^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 101, \Delta = -150, \det = 237, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

4.267 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$T = 10 \quad [0] 1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{123}1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 \\ 4 & 12 & 0 & 6 & 0 & 6 & 22 \\ 2 & 4 & -1 & -2 & -1 & 2 & 8 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.08 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -572, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{123}1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 6 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & -1 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

$$4.268 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 11 \quad [0] \ 1^{111}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 0 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

4.269 NHC(-2)(-3) + e₇ + su₃ + su₂→(-3) + su₂→(-3)

$$T = 11 \quad [0] \quad 1^{11} 23^1 1^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 81, \Delta = -123, \det = -312, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 81, T \in [16, 81]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.270 NHC(-2)(-3) + e₇ + su₃ + su₂→(-3) + su₃

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & 2 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

4.271 NHC(-2)(-3) + e₇ + su₃ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -977, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -1280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 103, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.03 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 223, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.23 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -150, \ \det = 358, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.03 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -2 & -1 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 682, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.05 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 1075, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.03 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 8 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -2 & -1 & 10 & -2 \end{pmatrix}_{b_0}$$

4.272 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -142, \ \det = 181, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, \ T_{\max} = 100, \ T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 6 & 0 & -2 \end{pmatrix}_{b_0}$$

4.273 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 102, \ \Delta = -178, \ \det = -48, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -71$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -3 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -209$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -297$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.18 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [5] $1^{121} 23^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 27$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 17.78 \rightarrow 18$, $T_{\max} = 101$, $T \in [18, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 6 & 0 & 2 & 9 & 6 & 5 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 5 & 2 & -4 \end{pmatrix}_{b_0}$$

[6] $1^{11} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 31$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.10 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & -3 \\ 1 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{111} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.25 \rightarrow 19$, $T_{\max} = 101$, $T \in [19, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[8] $1^{12} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 57$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{111} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 135$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.20 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 3 & 0 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 & 1 \\ 3 & 4 & -1 & -1 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

[10] $1^1 231^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 102$, $\Delta = -157$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.28 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 1^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 196$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 223$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.23 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^1 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 246$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 4 \\ 3 & 4 & 1 & 3 & 6 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 & 1 \\ 3 & 6 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^1 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 102$, $\Delta = -157$, $\det = 342$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.27 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^1 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 384$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 4 \\ 3 & 4 & 1 & 3 & 6 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 & 1 \\ 4 & 6 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

4.274 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 10$ [1] $1^{11} 1^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 101$, $\Delta = -148$, $\det = 63$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 0 & 3 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -2 & -1 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 280, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \quad 1^{111} 2 \, 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 70, \quad \Delta = -121, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, \quad T_{\max} = 70, \quad T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 2 \, 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 70, \quad \Delta = -121, \quad \det = -95, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.11 \rightarrow 16, \quad T_{\max} = 70, \quad T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3 \, 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 48, \quad \Delta = -119, \quad \det = -159, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.87 \rightarrow 16, \quad T_{\max} = 48, \quad T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 2 \, 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 70, \quad \Delta = -121, \quad \det = -162, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, \quad T_{\max} = 70, \quad T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112} 2 \, 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 70, \quad \Delta = -121, \quad \det = -194, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.11 \rightarrow 16, \quad T_{\max} = 70, \quad T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 6 & 2 & 6 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111} 3 \, 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 48, \quad \Delta = -119, \quad \det = -231, \quad (n_+, n_-, n_0) =$$

$(1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 48, T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -303, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -402, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 2 & 6 & -2 \end{pmatrix} b_0$$

[12] $1^{1112}2^33^{21}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -807$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.5$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 1 & 3 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

[13] $1^{1113}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -1095$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.8$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 2 & 9 & -2 \end{pmatrix} b_0$$

[14] $1^{1114}2^33^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -1167, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.26 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{1113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -1518$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.8$

$$T_{\min} = 15.23 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 2 & 9 & -2 \end{pmatrix} b_0$$

[16] $1^{1114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -2418, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

4.275 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{1111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{\overline{1111}} 2 \ 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 16, \Delta = -91, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 228, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.276 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 0 & 4 & 5 \\ 2 & 4 & -4 & 0 & 0 & 2 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 3 & 5 & -2 & -1 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

4.277 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 102, \Delta = -157, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.28 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \ 1^{\overline{111}} 2 \ 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 94, \Delta = -130, \det = -297, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 \\ 2 & 3 & -1 & -3 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 11, T_{\max} = 92, \Delta = -128, \det = -405, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.39 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.278 NHC(-2)(-3) + e₇ + (-3)→(-2)

$T = 8$ [0] $1^1 231^{11} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 102, \Delta = -200, \det = 425, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

4.279 NHC(-2)(-3) + e'₇

$T = 7$ [0] $2^1 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -4 & -2 \\ 2 & 2 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $21^{11} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$T = 8$ [4] $1^1 13221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 8, T_{\max} = 101, \Delta = -163, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & -2 & 0 \\ 3 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11} 2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 8, T_{\max} = 101, \Delta = -165, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -4 & -2 \\ 2 & 3 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11} 2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 8, T_{\max} = 101, \Delta = -165, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -4 & -2 \\ 2 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 \bar{3} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -163, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 23131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.08 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 231^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 41, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.05 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & -3 & -1 \\ 2 & 0 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 231^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.71 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 31^1 2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -173, \ \det = 278, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -4 & -2 \\ 2 & 3 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [12] \ 1^1 1^1 223131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 9, \ T_{\max} = 99, \ \Delta = -142, \ \det = -9, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 17, \ T_{\max} = 99, \ T \in [17, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 \\ 2 & 5 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \ 1^1 4^1 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 18, \ T_{\max} = 94, \ T \in [18, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \\ 1 & 3 & -4 & -2 \\ 3 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [15] \ 21^1 5^1 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 11, \ T_{\max} = 92, \ \Delta = -128, \ \det = -9, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, \ T_{\max} = 92, \ T \in [19, 92]$$

$$\begin{pmatrix} 5 & 17 & 5 \\ 17 & 56 & 14 \\ 5 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^1 5^1 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 11, \ T_{\max} = 92, \ \Delta = -128, \ \det = -9, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, \ T_{\max} = 92, \ T \in [19, 92]$$

$$\begin{pmatrix} 5 & 17 & 5 \\ 17 & 56 & 14 \\ 5 & 14 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [16] \ 131^1 5^1 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 12, \ T_{\max} = 24, \ \Delta = -107, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 20, \ T_{\max} = 24, \ T \in [20, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 \\ 2 & 5 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

4.280 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{e}_6$

$$T = 9 \quad [0] \ 1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 104, \ \Delta = -214, \ \det = -103, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.24 \rightarrow 17, \ T_{\max} = 104, \ T \in [17, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

4.281 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 103, \ \Delta = -215, \ \det = 122, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \ 1^{11} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{f}_4 \quad T = 9, \ T_{\max} = 103, \ \Delta = -186, \ \det = -42, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.43 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{f}_4 \quad T = 9, \ T_{\max} = 103, \ \Delta = -188, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.33 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

4.282 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \ T_{\max} = 99, \ \Delta = -142, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 5 \\ 7 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \ T_{\max} = 99, \ \Delta = -142, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.06 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \ 1^{111} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 27, \ \Delta = -115, \ \det = 18, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.56 \rightarrow 15, \ T_{\max} = 27, \ T \in [15, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 27, \ \Delta = -115, \ \det = 27, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 27, \ T \in [15, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 0 & 2 \\ 2 & 3 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11}2^321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 27, \Delta = -115, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14, T_{\max} = 27, T \in [14, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & -2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^1221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 21 & 27 & 1 & 9 \\ 27 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 9 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^1221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[7] $1^1231^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 265, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.283 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{e}}'_7 + \text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_2$

$T = 10 \quad [0] \quad 1^{111}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 7 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 452, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$T = 11 \quad [2] \quad 1^{111}3^1221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -116, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 21 & 27 & 1 & 27 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^1221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -356, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 21 & 27 & 1 & 28 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{1231}1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -440, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.18 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.284 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 27, \Delta = -115, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 27, T \in [14, 27]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 3 & 0 \\ 4 & 5 & 0 & 3 & 1 & 3 \\ 5 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{11} 23^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -94, \det = -355, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 2 & -1 \\ 2 & 6 & 0 & 3 & 4 & 1 & 3 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.285 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 4 & 0 & -1 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 & 0 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

4.286 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 102, \Delta = -164, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 9 & 4 & 5 \\ 2 & 0 & 4 & -3 & -1 \\ 3 & 0 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 102, \Delta = -164, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}3221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11}231^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

4.287 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$

$T = 9 \quad [0] \quad 1^{11}321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11}321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.51 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111}321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111}321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$T = 10 \quad [4] \quad 1^{11}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 16.87 \rightarrow 17, T_{\max} = 95, T \in [17, 95]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

[5] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.53 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}2^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 421, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 8 & -1 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{114}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 501, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 16 & -1 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -1 & 24 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{113}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 505, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 12 & -1 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -1 & 18 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 606, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 8 & -1 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{113}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 994$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.8$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 12 & -1 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -1 & 18 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{114}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 1294$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 16 & -1 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -1 & 24 & 0 \end{pmatrix}_{b_0}$$

4.288 $\text{NHC}(-2)(-3) + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{111}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 132$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$T = 11$ [1] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -104$, $\det = -84$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -102$, $\det = -104$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -104$, $\det = -150$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.289 $\text{NHC}(-2)(-3) + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 102$, $\Delta = -164$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.290 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 1^{11} 2 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 2 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

4.291 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 2 2 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{12} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{13} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{14} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.02 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{111} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.94 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{15} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 14.02 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -2 & 10 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{16} 3^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 14.19 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -2 & 12 & -2 \end{pmatrix}_{b_0}$$

[14] $1^{17} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -2 & 14 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{12} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.61 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{13} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{12} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{14} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[22] $21^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 6 & 8 & 3 & 4 & 4 \\ 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{12} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 405, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{13} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 600, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{13} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 774, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 837, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1060, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [32] $1^{12} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 20, T_{\max} = 93, T \in [20, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 3 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -1 & 4 & -6 \end{pmatrix}_{b_0}$$

[36] $1^{11} 23 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{\overline{1}14} 3 221^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 20$, $T_{\max} = 93$, $T \in [20, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -1 & 4 & -6 \end{pmatrix}_{b_0}$$

[34] $1^{\overline{1}13} 3 221^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 17$, $T_{\max} = 93$, $T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -1 & 4 & -3 \end{pmatrix}_{b_0}$$

[33] $1^{\overline{1}12} 2 3 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

[37] $1^{\overline{1}12} 3 221^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{\overline{1}11} 2 3 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{\overline{1}11} 2 3 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{\overline{1}11} 2 3 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{\overline{1}11} 3 221^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{\overline{1}1} 231^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{\overline{1}1} 3 221^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

- [44] $1^{12}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{12}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [46] $1^{13}2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 4 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$
- [47] $1^{13}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [48] $1^{12}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$
- [49] $1^{14}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$
- [50] $1^{12}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 3 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^{15}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -1 & 10 & -1 \end{pmatrix}_{b_0}$$
- [52] $1^{12}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$
- [53] $1^{12}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.47 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & -1 \\ 6 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$
- [54] $1^{16}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{11}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{11}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{12}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{11}3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 3 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{11}3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{17}3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -1 & 14 & -2 \end{pmatrix}_{b_0}$$

[61] $1^{12}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.07 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 3 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{12}31^{33} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.72 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 9 & -1 \\ 9 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{12}31^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[64] $1^{13}3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -121, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 4 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{12}2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{11}3^{12} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{11}2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{14}2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 5 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{12}3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{12}31^{431^{114}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.89 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 12 & -1 \\ 12 & 0 & 0 & 12 & 94 & 16 \\ 2 & 0 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{13}2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{12}31^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.78 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{11}3^{13} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.78 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{11}3^{14} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{11} 3^{12} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -169, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 4 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{12} 231^{11} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{14} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 5 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{12} 23131^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.55 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{11} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{12} 231^{131} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 4 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[81] $1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 4 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{12} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 4 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{12} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{11}2^3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 5 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{12}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{11}3^{13}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{12}3^{12}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{14}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & -2 \\ 6 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{12}31^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 4 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{13}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{14}3^{11}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{14}3^{11}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -289, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 6 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{14}23131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.62 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[95] $1^{12} 3^{13} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{13} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 5 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{13} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 5 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{12} 231^{13} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & -1 \\ 7 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{12} 3^{12} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 5 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{12} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{13} 3^{12} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{12} 231^{12} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{12} 231^{13} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.11 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 5 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{12} 3^{14} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.69 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{14}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.52 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 6 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[106] $1^1 23131^{15} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -427, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[107] $1^1 231^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 6 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{14}3^1 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 6 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{13}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{12}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 6 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{12}3^1 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{12}3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[113] $1^1 231^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -478, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 5 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[114] $1^1 231^3 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.13 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 9 & -1 \\ 10 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[115] $1^{13} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[116] $1^{12} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -505, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 6 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[117] $1^{13} 2 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -514, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & -2 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{12} 231^{12} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{12} 231^{13} 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 6 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{12} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 7 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[121] $1^{12} 231^{12} 31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{13} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -603, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[123] $1^{12} 231^{13} 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & -1 \\ 8 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[124] $1^{13} 2 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 7 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[125] $1^{14} 3^{13} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[126] $1^{14} 3^{12} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -676, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 7 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[127] $1^{13} 3^{13} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -679, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[128] $1^1 231^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -688, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.95 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 8 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[129] $31^{11} 2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -751, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -4 & 1 & -2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{14} 2^3 3^{14} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & -2 \\ 8 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[131] $1^1 231^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -802, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 7 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[132] $1^{14} 3^{14} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[133] $1^{13} 3^{14} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -841, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 8 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[134] $1^{13} 3^{13} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -904, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 8 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[135] $1^1 231^{15} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1012$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 13.97 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 15 & -1 \\ 10 & 30 & 0 & 15 & 148 & 20 \\ 2 & 4 & -1 & -1 & 20 & 0 \end{pmatrix}_{b_0}$$

[136] $1^1 231^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1036$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 8 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[137] $1^1 231^{13} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[138] $1^{14} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -1120$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 9 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[139] $1^1 231^{13} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1138$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.75 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[140] $1^1 231^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1198$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 9 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[141] $1^1 231^{14} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.79 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 10 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[142] $1^1 231^{14} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1822$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 11 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[143] $1^1 231^{14} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1936$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.75 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

4.292 $\text{NHC}(-2)(-3) + \text{e}'_7 + \text{su}_2 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2]$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 147$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.80 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 5 \\ 7 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 452$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.80 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [2] $1^{11} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -116$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.93 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 21 & 27 & 1 & 1 & 9 \\ 27 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 9 & 11 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -356$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.93 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 21 & 27 & 1 & 28 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{1231} 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -440$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.18 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.293 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -139$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.51 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -213$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

- $T = 10$ [4] $1^{11}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.87 \rightarrow 17, T_{\max} = 95, T \in [17, 95]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$
- [5] $1^{111}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.53 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{1231}131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 100, \Delta = -140, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -1 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$
- [7] $1^{11}3^2 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 94, \Delta = -132, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{111}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{121}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 421, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 8 & -1 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{141}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 501, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 16 & -1 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -1 & 24 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{131}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 505, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$
 $T_{\min} = 14.54 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 12 & -1 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -1 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 95, \ \Delta = -134, \ \det = 606, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.49 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 8 & -1 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{131} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 95, \ \Delta = -134, \ \det = 994, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.50 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 12 & -1 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -1 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{141} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 95, \ \Delta = -134, \ \det = 1294, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 16 & -1 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -1 & 24 & 0 \end{pmatrix}_{b_0}$$

4.294 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^2 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 132, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.64 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \ 1^{111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -104, \ \det = -84, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^2 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -102, \ \det = -104, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -104, \ \det = -150, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.56 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.295 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{132} 3^2 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -9, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 16.11 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 16 & 12 & 6 \\ 3 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -2 & 6 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.02 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{144} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.81 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{124} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{12} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{124} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.91 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.28 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{123} 3^{112} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -196$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{134} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.02 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[24] $1^{115} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.04 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -2 \end{pmatrix}_{b_0}$$

[25] $1^{122} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -214$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.36 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{125} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -220$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 14.09 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 5 & -2 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -2 \end{pmatrix}_{b_0}$$

[27] $1^{144} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -232$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

12.8

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{113} 3^{21} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{c}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -493, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{124} 3^{21} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{c}}_7' + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_b$$

[48] $1^{124} 3^{211} + [\textcolor{blue}{\mathfrak{s}u_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{c}}'_7 + \textcolor{brown}{\mathfrak{s}u_2} + \textcolor{red}{\mathfrak{s}u_2}$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -534$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_b$$

[49] $1^{122} 3^{111} 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix} b_0$$

[50] $1^{114} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -657$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{113} 3^{211} + [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{blue}{\mathbf{g}_2}] + \textcolor{red}{\mathbf{c}}' + \textcolor{orange}{\mathbf{su}_2} + \textcolor{red}{\mathbf{su}_2}$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -724$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{123} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -763, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_b$$

[53] $1^{124} 3^{111} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -792, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_b$$

[54] $1^{113} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{113} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -837, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{133} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{134} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -1083, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

[59] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -1273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[60] $1^{144} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -1292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -1 \end{pmatrix}_{b_0}$$

[61] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -1404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [62] $1^{1231} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 20.50 \rightarrow 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -1 & 0 & 0 & -7 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{134} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 8 & -1 \\ 3 & 6 & 0 & 3 & 16 & 27 & 6 \\ 5 & 9 & 0 & 8 & 27 & 42 & 12 \\ 2 & 2 & -1 & -1 & 6 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{1231} 3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 21.45 \rightarrow 22, T_{\max} = 23, T \in [22, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 0 & -1 \\ 3 & 0 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 4 & 0 & -8 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{134} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 8 & -1 \\ 4 & 6 & 0 & 3 & 16 & 27 & 6 \\ 6 & 9 & 0 & 8 & 27 & 42 & 12 \\ 2 & 2 & -1 & -1 & 6 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{111} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 4 & 3 & 0 & 5 & 3 & 6 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{121} 3^{211} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16.80 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{14} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 42 & 12 \\ 2 & 2 & -1 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{1231} 3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{111} 2^3 3^{211} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{1231^{12}31^{114}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 37, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 5 & 3 & 6 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{124} 3^{113} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{12} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{123} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 58, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{112} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{123} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 22, T \in [16, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{111} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 5 & 0 & 6 & 6 \\ 2 & 2 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{124} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[85] \ 1^1 231^{11} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 77, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{112} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{123} 3^{11} 221 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{13} 3^{111} 221 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{111} 3^{11} 221 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{112} 3^{11} 221 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{112} 3^{11} 21 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 95, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{122} 3^{11} 221 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{1231} 1431^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$

[94] $1^{133} 2^3 21 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & 3 & -2 \\ 4 & 0 & 9 & 3 & 25 & 27 & 9 \\ 4 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 9 & -1 \end{pmatrix}_{b_0}$$

[95] $1^{113} 3^{11} 221 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

12.8

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{13} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 7 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 7 & 0 & 26 & 10 \\ 2 & 2 & -1 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{123} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{123} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 3 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[100] $1^{111} 3^{12} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{133} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 117, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{114} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[103] $1^{133} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{12} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 123, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 6 & 0 & 14 & 8 \\ 2 & 2 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{124} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{132} 2^3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 6 & 2 \\ 1 & 0 & 1 & -4 & 3 & 2 & -2 \\ 4 & 0 & 9 & 3 & 25 & 18 & 9 \\ 3 & 0 & 6 & 2 & 18 & 10 & 6 \\ 2 & 0 & 2 & -2 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{1231} 11 31^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 0 & 0 & 3 & 6 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{112} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{134} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 134, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.58 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{111} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 135, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{115} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.1$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{112} 2^3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{112} 3^{124} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

[114] $1^{125} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 142$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[115] $1^{121} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 144$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[116] $1^{144} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.81 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 4 & -1 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 8 & -1 \end{pmatrix}_{b_0}$$

[117] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 149$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{123} 3^{112} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 150$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{135} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 152$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.9$

$T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 5 & -1 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -1 & 6 & 10 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{116} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 158$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[121] $1^{112}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 162$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.04 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[122] $1^{126}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

[124] $1^{123}131^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[123] $1^{122}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[125] $1^{145}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 5 & -1 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -1 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

[126] $1^{112}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 167$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[127] $1^{114}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 172$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[128] $1^{136}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -1 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

[129] $1^{12} 2^1 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{117} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 184$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 15.24 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -1 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

[131] $1^{1231} 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 185$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[132] $1^{127} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 190$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -1 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

[133] $1^{113} 2^1 3^{14} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 192$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[134] $1^{137} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 200$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.38 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -1 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$

[135] $1^{113} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 202$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[136] $1^{113} 2^1 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[137] $1^{112} 3^{14} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 206$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[139] $1^{114} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 207$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[138] $1^{112} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 207$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[140] $1^{124} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 208$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[141] $1^{113} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 209$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[142] $1^{112} 3^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 211$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[143] $1^{111} 2^3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 212$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 4 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[144] $1^{134} 2^3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 232$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -4 & 3 & 4 & -2 \\ 4 & 0 & 9 & 3 & 25 & 36 & 9 \\ 6 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -2 & 9 & 12 & 0 \end{pmatrix}_{b_0}$$

[145] $1^{134} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 239$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & 0 \end{pmatrix}_{b_0}$$

[146] $1^{122} 2^{124} 3^{11} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 6 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[147] $1^{112} 3^{114} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 248$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 7 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[148] $1^{113} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 253$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[149] $1^{1231} 13^{111} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 257$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.20 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[150] $1^{112} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 263$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{112} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 263$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{111} 2^{111} 3^{11} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 264$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{113}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 275$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{122}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 280$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 3 & 0 & 6 & 2 & 10 & 12 & 6 \\ 3 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[155] $1^{111}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 280$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[156] $1^{123}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 282$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[157] $1^1231^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 284$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[158] $1^{122}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 288$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 3 & 0 & 6 & 2 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{124}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 292$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[160] $1^{112}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 303$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[161] $1^{124} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 307$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[162] $1^{132} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 311$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -2 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

[164] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 312$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 2 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[163] $1^{114} 2^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 312$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[165] $1^{111} 2^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 315$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[167] $1^{112} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 319$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[166] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 319$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[168] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 328$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[169] $1^{112} 3^{12} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 338$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[170] $1^{112} 3^{13} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 347$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 6 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[171] $1^{112} 3^{12} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 352$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[172] $1^{113} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 356$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[173] $1^{112} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 359$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[174] $1^1 231^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 368$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[175] $1^{114} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 371$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[176] $1^{113} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 372$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[177] $1^{1231} 1^{13} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 380$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.75 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[178] $1^{124} 3^{12} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 380$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 381$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[180] $1^{144} 2^{123} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 388$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & 4 & -2 \\ 6 & 0 & 12 & 4 & 46 & 48 & 12 \\ 7 & 0 & 12 & 4 & 48 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 12 & 0 \end{pmatrix}_{b_0}$$

[181] $1^{1231} 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 389$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[182] $1^{114} 2^{13} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 395$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[184] $1^{112} 2^{122} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 396$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[183] $1^{111} 2^{124} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 396$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 5 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[185] $1^{124} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 404$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[186] $1^{12} 2^3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 420$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[187] $1^{123} 2^3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 423$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 3 & 0 & 6 & 2 & 10 & 18 & 6 \\ 6 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[188] $1^{1231} 111 31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 437$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[190] $1^{114} 2^3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 464$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[189] $1^{1231} 111 31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 464$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[191] $1^1 231^{12} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 464$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[192] $1^{113} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 493$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[193] $1^{111}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 496$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[194] $1^{14}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 500$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 12 & 2 \\ 1 & 0 & 1 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 0 & 12 & 4 & 0 & 46 & 12 \\ 2 & 0 & 2 & -2 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[195] $1^{114}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 503$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[196] $1^{114}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 506$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[197] $1^{122}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 508$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[198] $1^{111}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 523$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 4 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[199] $1^{113}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 531$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 3 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[200] $1^{113}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 539$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[201] $1^{112}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 540$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[202] $1^{123}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 547$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 3 & 0 & 6 & 2 & 10 & 18 & 6 \\ 5 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[203] $1^{113}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 564$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[204] $1^{124}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 580$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 6 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[205] $1^{123}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 591$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[206] $1^{124}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 608$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[207] $1^{124}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 612$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 8 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[208] $1^{123}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 620$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 5 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[209] $1^{12} 2^1 3^{114} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 640$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[210] $1^{113} 3^{14} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 648$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[211] $1^{143} 2^1 3^{122} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 652$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -4 & 4 & 3 & -2 \\ 6 & 0 & 12 & 4 & 46 & 36 & 12 \\ 5 & 0 & 9 & 3 & 36 & 25 & 9 \\ 2 & 0 & 2 & -2 & 12 & 9 & 0 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3^{13} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 653$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[213] $1^{1231} 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 680$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[214] $1^{132} 3^{114} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 683$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -2 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

[215] $1^{113} 3^{12} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 690$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[216] $1^{112} 3^{113} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 719$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[217] $1^{114}2^13^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 720$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 3 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[218] $1^1231^{13}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 725$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[219] $1^{114}3^1221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 727$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[221] $1^{112}2^13^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 744$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[220] $1^{133}3^1221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 744$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[222] $1^{133}2^13^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 755$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & 3 & -2 \\ 5 & 0 & 9 & 3 & 25 & 27 & 9 \\ 6 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[223] $1^{113}3^1221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 793$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[224] $1^{114}3^1221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 794$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[225] $1^{122}2^13^{123}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 800$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 5 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[226] $1^{113}3^{114}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 802$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 8 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[227] $1^{122}2^13^{122}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 816$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[228] $1^{134}3^{111}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 831$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & 0 \end{pmatrix}_{b_0}$$

[229] $1^{112}2^13^{123}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 835$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[230] $1^1231^{14}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 860$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.93 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[231] $1^{124}2^13^{113}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 868$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 7 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[232] $1^1231^{13}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 869$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[233] $1^{144}2^13^{122}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 880, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & 4 & -2 \\ 6 & 0 & 12 & 4 & 46 & 48 & 12 \\ 6 & 0 & 12 & 4 & 48 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 12 & 0 \end{pmatrix}_{b_0}$$

[234] $1^1231^{13}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 929, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[235] $1^{133}2^13^{122}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & 3 & -2 \\ 5 & 0 & 9 & 3 & 25 & 27 & 9 \\ 5 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[236] $1^{134}2^13^{124}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -4 & 3 & 4 & -2 \\ 5 & 0 & 9 & 3 & 25 & 36 & 9 \\ 8 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -2 & 9 & 12 & 0 \end{pmatrix}_{b_0}$$

[237] $1^{144}1^{111}3^{111}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 4 & -1 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 8 & 0 \end{pmatrix}_{b_0}$$

[238] $1^{114}1^{13}3^{113}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 991, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[239] $1^{113}1^{113}2^13^{113}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1000, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[240] $1^{114}1^{12}3^{112}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[241] $1^{114}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}'_7+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 1055$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[242] $1^{114}3^1221^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}'_7+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 1094$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[243] $1^{1124}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}'_7+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 1120$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[244] $1^{124}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}'_7+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 1136$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 4 & 0 & 6 & 2 & 10 & 24 & 6 \\ 7 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[245] $1^{113}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}'_7+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 1228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 3 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[246] $1^{114}3^1221^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}'_7+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 1243$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[247] $1^{123}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}'_7+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 6 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[248] $1^{113}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}'_7+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 1307$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[249] $1^{1231}1^{431}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 1352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[250] $1^{114}1^{14}3^{221}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 1360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[251] $1^{123}2^{124}3^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1388, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 7 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[252] $1^{134}2^{123}3^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1395, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -4 & 3 & 4 & -2 \\ 5 & 0 & 9 & 3 & 25 & 36 & 9 \\ 7 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -2 & 9 & 12 & 0 \end{pmatrix}_{b_0}$$

[253] $1^{1231}1^{431}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 1580, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[254] $1^{114}2^{114}3^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1644, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[255] $1^{124}2^{124}3^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1920, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 4 & 0 & 6 & 2 & 10 & 24 & 6 \\ 8 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

4.296 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{so}_8$

$T = 10 \quad [0] \quad 1^{111}1^{111}3^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -104, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.297 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1112} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 19.25 \rightarrow 20, T_{\max} = 22, T \in [20, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -7 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1211} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1211} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$[111] \text{ } 1^{1122} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 280, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix} b_0$$

$$[13] \quad 1^{1111} 3^{211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 296, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & 2 & -1 \end{pmatrix} b_0$$

$$[12] \quad 1^{1113} 3^{121} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 296, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 3 & 6 & 0 & 3 & 6 & 6 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix} b_0$$

[14] $1^{1122}3^{21}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 346, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix} b_0$$

[15] $1^{1114}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -102, \det = 352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 4 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 \end{pmatrix} b_0$$

[16] $1^{1115}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 22, \Delta = -102, \det = 424, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$

$$T_{\min} = 14.06 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 10 & 48 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -2 \end{pmatrix} b_0$$

[17] $1^{1112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -102, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 \\ 4 & 4 & 0 & 2 & 4 & 4 & 6 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix} b_0$$

[18] $1^{1111} 3^{211} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7' + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -102, \det = 540, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$[26] \ 1^{1122} 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 1132, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.35 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1114} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 1232, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1114} 1^{14} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 1376, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1114} 1^{12} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 1696, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1114} 1^{13} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 1744, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \ 1^1 231^{111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -81, \ \det = -556, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.77 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.298 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 11 \quad [0] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 14, \ \Delta = -81, \ \det = -166, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.95 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.299 NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{113} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 6 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -1 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 4 & -1 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 2 & -1 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 4 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 222, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 8 & 3 \\ 2 & 1 & 1 & 2 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 6 & -1 \\ 4 & 6 & 0 & 4 & 10 & 18 & 6 \\ 8 & 9 & 0 & 6 & 18 & 25 & 9 \\ 3 & 3 & -1 & -1 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 236, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 749, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1068, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{114}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1418, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [19] $1^{1231}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -329, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.300 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$

$T = 11$ [0] $1^{111}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 3 & -1 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

4.301 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$

$T = 10$ [0] $1^{111}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 71, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 \\ 2 & 2 & -1 & -2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{1211}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$[9] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 632, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 0 & 3 & 5 & 5 & 11 & 5 \\ 2 & 2 & -1 & -2 & 2 & 2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 687, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 3 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 & 3 \\ 2 & 2 & -1 & -2 & 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 792, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 5 & 11 & 5 \\ 2 & 2 & -1 & -2 & 2 & 2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [12] \ 1^{123} 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -231, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.302 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 56, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 65, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 92, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 128, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{131} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 485, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 11 \quad [11] \quad 1^{\overline{111}} 3 \quad 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -44, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 14.0 \\
& T_{\min} = 13.82 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0} \\
& [12] \quad 1^{\overline{111}} 2 \quad 3 \quad 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -52, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 14.0 \\
& T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0} \\
& [13] \quad 1^{\overline{111}} 2 \quad 3 \quad 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -69, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 14.0 \\
& T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0} \\
& [14] \quad 1^{\overline{111}} 3 \quad 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -72, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 14.0 \\
& T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0} \\
& [15] \quad 1^{\overline{111}} 3 \quad 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -101, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 14.0 \\
& T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
& [16] \quad 1^{\overline{131}} 3 \quad 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -209, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 14.8 \\
& T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 14 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & 0 & 0 & 17 & 1 \end{pmatrix}_{b_0} \\
& [17] \quad 1^{\overline{111}} 2 \quad 3 \quad 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -231, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 14.0 \\
& T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0} \\
& [18] \quad 1^{\overline{121}} 3 \quad 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -254, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 14.5 \\
& T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 11 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & 0 & 0 & 13 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[19] $1^{1111}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{111}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{121}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{1231}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -90, \det = -300, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{1111}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -320, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{1121}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -327, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 & 3 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{1231}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -90, \det = -363, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{1111}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -381, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1141}2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -447, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 & 3 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{1131}2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -453, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 & 3 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{1131}2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 & 3 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}3^1 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -704, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{1141}2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -954, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 & 3 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

4.303 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$

$T = 11$ [0] $1^{1111}3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.304 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{1111}3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.305 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 11 \quad [0] \ 1^{1111} 3^{21} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.306 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{13} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 4 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -259, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.52 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.45 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -369, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.46 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -418, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 8 & -1 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -478, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -675, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.47 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -850, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.47 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 8 & -1 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [14] \quad 1^1 23^1 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 42, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.10 \rightarrow 16, \quad T_{\max} = 26, \quad T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & 2 & -1 \\ 7 & 0 & 0 & 6 & 22 & 6 & 8 \\ 1 & 0 & 0 & 2 & 6 & 0 & 2 \\ 2 & 0 & -1 & -1 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^1 23^1 1^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 47, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.21 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^1 23^1 1^1 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 51, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 2 & -1 \\ 4 & 0 & 0 & 3 & 4 & 3 & 4 \\ 1 & 0 & 0 & 2 & 3 & 0 & 2 \\ 2 & 0 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^1 23^1 1^1 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 71, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.49 \rightarrow 16, \ T_{\max} = 26, \ T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 2 & -1 \\ 6 & 0 & 0 & 3 & 4 & 3 & 4 \\ 1 & 0 & 0 & 2 & 3 & 0 & 2 \\ 2 & 0 & -1 & -1 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^1 23^1 1^1 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 81, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.64 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 2 & -1 \\ 5 & 0 & 0 & 3 & 4 & 3 & 4 \\ 1 & 0 & 0 & 2 & 3 & 0 & 2 \\ 2 & 0 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^1 231^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 25, \ \Delta = -113, \ \det = 87, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.84 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^1 231^1 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 25, \ \Delta = -113, \ \det = 98, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.22 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 5 & 0 \\ 5 & 0 & 0 & 5 & 8 & 6 \\ 3 & 0 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^1 2313^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 25, \ \Delta = -113, \ \det = 98, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.22 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^1 23^1 1^1 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.12 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 2 & -1 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^1 23^1 1^2 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 162, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & 2 & -1 \\ 8 & 0 & 0 & 6 & 22 & 6 & 8 \\ 1 & 0 & 0 & 2 & 6 & 0 & 2 \\ 2 & 0 & -1 & -1 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

4.307 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 8, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, \ T_{\max} = 22, \ T \in [15, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & 2 & -1 \\ 6 & 9 & 0 & 6 & 25 & 9 & 9 \\ 2 & 3 & 0 & 2 & 9 & 1 & 3 \\ 3 & 3 & -1 & -1 & 9 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 52, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.54 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 4 & -1 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 62, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 2 & -1 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 93, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{121} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 118, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 2 & -1 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 184, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 222, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.3$$

$$T_{\min} = 13.49 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 8 & 3 \\ 2 & 1 & 1 & 2 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 6 & -1 \\ 4 & 6 & 0 & 4 & 10 & 18 & 6 \\ 8 & 9 & 0 & 6 & 18 & 25 & 9 \\ 3 & 3 & -1 & -1 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 236, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 749, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 1068, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{114}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 1418, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [19] $1^{1231}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -329, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.308 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3$

$T = 11$ [0] $1^{111}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 3 & -1 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

4.309 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{1231}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 6 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

4.310 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \mathbf{n}3 \mathbf{mix}$

$T = 11$ [0] $1^{1231}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$T = 11 \quad [7] \quad 1^{1231}1^{1231}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -122, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.26 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & 0 & -1 \\ 4 & 12 & 0 & 6 & 22 & 6 & 0 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 8 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1231}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -239, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.17 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 2 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1231}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 11, \quad T_{\max} = 15, \quad \Delta = -92, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 & 0 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1231}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -452, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 2 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1231}1^{1231}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -959, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & 0 & -1 \\ 5 & 12 & 0 & 6 & 22 & 6 & 0 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 8 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

4.312 NHC(-2)(-3) + e'_7 + su_2 + su_2 → (-3) + su_3 + su_2

$$T = 11 \quad [0] \quad 1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -105, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

4.313 NHC(-2)(-3) + e'_7 + su_2 + su2n3mix

$$T = 9 \quad [0] \quad 1^{141}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -22, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, \quad T_{\max} = 95, \quad T \in [15, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 & 2 \\ 4 & 8 & 0 & 4 & 30 & 12 & 8 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 & 3 \\ 2 & 2 & -1 & -2 & 8 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.15 \rightarrow 15, \quad T_{\max} = 95, \quad T \in [15, 95]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -63, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 3 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -2 & 2 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -79, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 \\ 2 & 2 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{131} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -99, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.59 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.58 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 6 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -2 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{142} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 3 & -2 \\ 4 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -2 & 8 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{121} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{124} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -162, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 5 & -2 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -2 & 4 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{122} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

12.0

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^1 23^1 131^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -1 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^1 23^1 1^1 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 2 & -1 \\ 3 & 0 & 0 & 3 & 4 & 3 & 4 \\ 2 & 0 & 0 & 2 & 3 & 0 & 2 \\ 2 & 0 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^1 23^1 1^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 2 & -1 \\ 2 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^1 231^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 143, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 5 & 0 & 0 \\ 7 & 10 & 0 & 5 & 8 & 6 & 6 \\ 4 & 6 & -1 & 0 & 6 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^1 23^1 1^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 230, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^1 231^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 242, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[53] \ 1^1 23^1 1^{12} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 259, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -1 & 0 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^1 23^1 1^{11} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 282, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

4.314 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + **su2n3mix** + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 71, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1211} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 4 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1211} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 286, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 4 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 315, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 401, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 4 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 536, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 632, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 687, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 4 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [12] $1^123^11^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -231, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.315 NHC(-2)(-3) + $\mathfrak{c}'_7$ + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^123^11^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.316 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3mix}$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 150, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, \quad T_{\max} = 26, \quad T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

4.317 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -94, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.70 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -103, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, \quad T_{\max} = 99, \quad T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 & 1 \\ 2 & 2 & -1 & -2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -126, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.81 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 & 1 \\ 10 & 12 & 0 & 6 & 22 & 8 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -127, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.02 \rightarrow 15, \quad T_{\max} = 99, \quad T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 & 1 \\ 3 & 8 & 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 & 1 \\ 3 & 5 & -1 & 4 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 & 1 \\ 4 & 8 & 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 & 1 \\ 3 & 5 & -1 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.76 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 & 1 \\ 3 & 6 & 0 & 3 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.07 \rightarrow 15, \quad T_{\max} = 99, \quad T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[7] \text{ } 1^{112} 3 \text{ } 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{c}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \, T_{\max} = 99, \, \Delta = -141, \, \det = -181, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.17 \rightarrow 15, \, T_{\max} = 99, \, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{121}3^{21}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$
 $T_{\min} = 13.91 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$
 $T_{\min} = 13.72 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \text{ } 1^{11} 1^{13} 1^{32} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{c}}'_7 + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -214, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.11 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[11] \text{ } 1^{11} 1^{11} 3 \text{ } 21^1 + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{red}{\mathbf{e}}'_7 + \textcolor{red}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -243, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.77 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{141} 3^{21} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.31 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$

[13] $1^{111} 3\ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$

[15] $1^{111} 1^{12} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$

[14] $1^{151} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 14.49 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 15 & -1 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$

[16] $1^{112} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{141} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -2110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{141} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -2464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [29] \ 1^{1231} 31^{111} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 17, T_{\max} = 58, T \in [17, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & -1 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{131} 2 \ 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 21.84 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 4 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & -8 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 3 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{122} 2 \ 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 3 & 6 & 0 & 2 & 10 & 12 & 6 \\ 2 & 6 & 0 & 2 & 12 & 9 & 5 \\ 2 & 3 & -1 & -2 & 6 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{121} 2 \ 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 17.78 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 0 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{1231} 31^{111} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{142}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 16.19 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 6 & 12 & 0 & 4 & 46 & 24 & 12 \\ 2 & 6 & 0 & 2 & 24 & 9 & 5 \\ 2 & 3 & -1 & -2 & 12 & 5 & -3 \end{pmatrix}_{b_0}$$

[36] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{111}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[37] $1^1231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 58, T \in [16, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -1 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{111}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{112}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -2 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[42] $1^12^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{142}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

15.2

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 5 & 12 & 0 & 4 & 46 & 24 & 12 \\ 2 & 6 & 0 & 2 & 24 & 9 & 5 \\ 2 & 3 & -1 & -2 & 12 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{1231} 1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 0 & -1 \\ 4 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{132} 2^1 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 1 & 9 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 4 & 9 & 0 & 3 & 25 & 18 & 9 \\ 2 & 6 & 0 & 2 & 18 & 9 & 5 \\ 2 & 3 & -1 & -2 & 9 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{121} 1^{11} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{111} 1^{12} 2^1 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{111} 1^{11} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 & 0 \\ 6 & 9 & 0 & 8 & 15 & 9 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{11} 1^{11} 2^1 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & -2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{111} 1^{11} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{112} 1^{11} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{121} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 140, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.11 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{111} 3^{13} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 140, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{121} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 140, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 8 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{111} 3^{13} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 140, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{112} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 141, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 145, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{131} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 155, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{112} 3^{12} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 156, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.77 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -2 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

- [60] $1^{121}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 157$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$
- [61] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 165$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$
- [62] $1^{13}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 171$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 0 & 2 \\ 1 & 0 & 1 & -4 & 3 & 0 & -2 \\ 5 & 0 & 9 & 3 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 9 & -1 & -1 \end{pmatrix}_{b_0}$$
- [63] $1^{141}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.77 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [64] $1^{1231}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 177$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$
- [65] $1^{11}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 181$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & -2 \\ 2 & 0 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$
- [66] $1^{111}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 187$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 7 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{111}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 192$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [68] $1^{151}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{1231} 1^{11} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.08 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 243, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{121} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.06 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 6 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{112} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -2 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{121} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{121} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{112} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 306, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{131}2^13^{221}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 323, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 11 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{111}2^13^{111}2^13^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 3 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{141}2^13^{11}2^13^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 354, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{111}2^13^{11}2^13^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & -2 \\ 3 & 0 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

[81] $1^{131}2^13^{11}2^13^{21}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & -2 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{121}2^13^{11}2^13^{21}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.04 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{121}2^13^{11}2^13^{21}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{121}2^13^{11}2^13^{21}1^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 0 & 1 & -4 & 2 & 0 & -2 \\ 4 & 0 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{121}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 405$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 5 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{121}2^13^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 411$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.91 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{1231}1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 411$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.13 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{141}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 412$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -4 & 4 & 1 & -2 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -2 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{111}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 459$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{131}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 460$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.22 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{121}2^13^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 467$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 9 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{141}2^13^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 512$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.94 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 14 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{141}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 552, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -4 & 4 & 1 & -2 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 2 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -2 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{121}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{131}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 588, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{121}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 588, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{121}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 635, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{131}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & -2 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{121}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 688, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 10 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{131}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 729, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 6 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[101] $1^{1231}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 762$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.12 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 0 & -1 \\ 5 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[102] $1^{121}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 764$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{141}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 771$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.02 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{131}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 795$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 2 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{121}1^{13}3^1221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 803$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 11 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{131}1^{11}3^1221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 812$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 12 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{121}1^{14}3^1221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 812$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 12 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{131}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 832$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.00 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 7 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[109] $1^{141}1^{12}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 900$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.80 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{141}2^321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 909, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -4 & 4 & 1 & -2 \\ 7 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -2 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[111] $1^{131}3^1221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 915, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{231}13^131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 1059, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & 0 & -1 \\ 7 & 18 & 0 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[113] $1^{131}3^1221^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{141}3^1221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 15 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{131}3^1221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1195, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 13 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{141}2^321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 1240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -4 & 4 & 1 & -2 \\ 8 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -2 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[117] $1^{131}3^1221^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1251, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{141} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1260, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{1231} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 58, \Delta = -120, \det = 1302, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 24 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & 0 & -1 \\ 9 & 24 & 0 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{131} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 14 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{141} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1643, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{131} 3^{14} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1643, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 15 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{141} 3^{12} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1708, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 16 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{141} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1920, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{141} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 2147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 17 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{141} 3^{14} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 2480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 18 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

4.318 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 56, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 65, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 92, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 128, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 341, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 344, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 380, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1111} 3 \, 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1111} 3 \, 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 485, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1111} 3 \, 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \quad 1^{111} 3 \, 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1111} 2 \, 3 \, 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1111} 2 \, 3 \, 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3 \, 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{111}} 3 \ 221^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -101, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{\overline{113}} 3 \ 221^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -209, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.97 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 14 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & 0 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{1111}} 2 \ 3 \ 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -231, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{112}} 3 \ 221^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -254, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 11 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & 0 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{1111}} 3 \ 221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -268, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.97 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{111}} 3 \ 221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -269, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{112}} 3 \ 221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -296, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.99 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{1231}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 15, \ \Delta = -90, \ \det = -300, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{1111}3221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -320, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{1112}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -327, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{1231}1^{11}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 15, \Delta = -90, \det = -363, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{1111}3221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -381, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1114}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -447, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{1113}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -453, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{1113}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}3221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -704, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{1114}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -954, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

4.319 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 11$ [0] $1^{1111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.320 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.321 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 11$ [0] $1^{1111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.322 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{121}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 & 0 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{123}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 16, \Delta = -93, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{123}1^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 16, \Delta = -93, \det = -959, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 0 & 2 & -1 \\ 5 & 12 & 0 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.323 NHC(-2)(-3) + e'_7 + su_2 + su_3 + $\text{su}_2 \rightarrow (-3)$ + su_2

$T = 11$ [0] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.324 NHC(-2)(-3) + e'_7 + su_2 + su_3 + $\text{su}_2\text{n3mix}$

$T = 10$ [0] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

4.325 NHC(-2)(-3) + e'_7 + su_2 + su_3 + su_3

$T = 10$ [0] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \quad 1^{111} 2^1 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 0 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 2^1 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 0 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 2^1 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 2^1 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -146, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 0 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 4 & 0 & 6 & 2 & 10 & 6 & 0 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -182, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.26 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111} 2^1 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -234, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{121} 2^1 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 1 & 3 & 0 & 1 & 6 & 3 & 0 \\ 2 & 3 & -1 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1211} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -621, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.34 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 1 & -2 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 6 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1411} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -801, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & 1 & -2 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 12 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1311} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -810, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.39 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & 1 & -2 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 9 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1311} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -1179, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.36 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & 1 & -2 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 9 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1411} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -1854, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.34 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & 1 & -2 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 12 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.326 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{1111} 3^2 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 15, \quad \Delta = -90, \quad \det = -228, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.327 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 13^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -165, \quad \det = 56, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.93 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & -3 & 2 & -1 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 2 & 0 & -3 & -1 \\ 3 & 3 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $21^{11}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & -3 & 2 & -1 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [3] $1^{12}23^1131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & -1 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{12}2313^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & -2 & 0 \\ 3 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{12}3^11^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & -1 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{12}231^13^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{12}2321^{11}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -145, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

4.328 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{11}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 27, \Delta = -115, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.87 \rightarrow 14, T_{\max} = 27, T \in [14, 27]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 1 & 0 & 2 \\ 3 & 3 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [1] $1^{12}23^11^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -94, \det = -355, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.18 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.329 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_{2 \rightarrow (-3)}$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{so}_8 \quad T = 9, T_{\max} = 102, \Delta = -164, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -3 & 2 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 2 & -2 & -1 \end{pmatrix}_{b_0}$$

4.330 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_{2 \rightarrow (-3)}$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{13} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -3 & 2 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{123} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 4 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -205$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$

[7] $1^{12}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -259$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$

[8] $1^{11}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -298$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$

[9] $1^{11}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -369$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$

[10] $1^{14}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -418$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 8 & -1 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$

[11] $1^{12}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -478$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$

[12] $1^{12}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -675$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$

[13] $1^{12}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -850$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 8 & -1 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$

$T = 10$ [14] $1^{123}1^{23}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 42$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 15.10 \rightarrow 16$, $T_{\max} = 26$, $T \in [16, 26]$

[15] $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 6 \\ 1 & 0 & 0 & 2 & 0 & 6 \\ 7 & 0 & 0 & 6 & 6 & 22 \\ 2 & 0 & -1 & -1 & 2 & 8 \end{pmatrix}_{b_0}$ $1^{12}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & -1 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 \\ 2 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$

[16] $1^{12}3^{11}1^{11}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 3 & -1 \\ 1 & 0 & 0 & 2 & 0 & 3 & 2 \\ 4 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$

[17] $1^{12}3^{11}1^{11}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 71, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.49 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 3 & -1 \\ 1 & 0 & 0 & 2 & 0 & 3 & 2 \\ 6 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 2 & 4 & -2 \end{pmatrix}_{b_0}$

[18] $1^{12}3^{11}1^{11}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.64 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 3 & -1 \\ 1 & 0 & 0 & 2 & 0 & 3 & 2 \\ 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$

[19] $1^{12}31^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 25, \Delta = -113, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$
 $\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$

[20] $1^{12}31^{13}1^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 25, \Delta = -113, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$
 $\begin{pmatrix} 1 & 1 & 1 & 2 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 5 & 0 \\ 5 & 0 & 0 & 5 & 8 & 6 \\ 3 & 0 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$

[21] $1^{12}31^{13}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 25, \Delta = -113, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$
 $\begin{pmatrix} 1 & 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$

[22] $1^{12}3^{11}1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.12 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & -1 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$

[23] $1^{12}3^{11}2^{31}1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 162, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & -1 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

4.331 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & 2 & -1 \\ 6 & 9 & 0 & 6 & 25 & 9 & 9 \\ 2 & 3 & 0 & 2 & 9 & 1 & 3 \\ 3 & 3 & -1 & -1 & 9 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{122} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 4 & -1 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 2 & -1 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 2 & -1 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{123} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 222, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 8 & 3 \\ 2 & 1 & 1 & 2 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 6 & -1 \\ 4 & 6 & 0 & 4 & 10 & 18 & 6 \\ 8 & 9 & 0 & 6 & 18 & 25 & 9 \\ 3 & 3 & -1 & -1 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 674, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 749, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{112}3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 1068, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 1418, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [19] $1^{1231}1^{11}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -329, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.332 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$

$T = 11$ [0] $1^{111}3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & -1 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.333 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{1231}1^{11}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 6 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 2 & -1 \\ 6 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & -1 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{123}1^{123}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -122, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.26 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 4 & 12 & 0 & 6 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{123}1^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -239, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.17 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{123}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, \quad T_{\max} = 15, \quad \Delta = -92, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 & 0 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{123}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -452, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{123}1^{123}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -93, \quad \det = -959, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

$$4.336 \quad \text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$$

$$T = 11 \quad [0] \quad 1^{111}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -105, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$4.337 \quad \text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \quad 1^{123}1^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 30, \quad \Delta = -116, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.50 \rightarrow 15, \quad T_{\max} = 30, \quad T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 3 & 0 \\ 2 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12313^{11}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.11 \rightarrow 15, \ T_{\max} = 30, \ T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 2 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 3 & 0 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12311^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 167, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 30, \ T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 2 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12311^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 200, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 30, \ T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

4.338 NHC(-2)(-3) + $\mathbf{e}'_7$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$

$$T = 11 \quad [0] \ 1^{12311^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -86, \ \det = -164, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 4 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

4.339 NHC(-2)(-3) + $\mathbf{e}'_7$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \ 1^{12311^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 15, \ \Delta = -89, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 3 & 2 & 0 \\ 2 & 0 & 0 & 3 & 1 & 2 & 3 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 3 & 0 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12311^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 15, \ \Delta = -89, \ \det = -300, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 2 & 0 \\ 2 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.340 NHC(-2)(-3) + $\mathbf{e}'_7$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \mathbf{n} \mathbf{3} \mathbf{mix}$

$$T = 11 \quad [0] \ 1^{12311^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 15, \ \Delta = -89, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 2 & 0 \\ 2 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12311^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 15, \ \Delta = -89, \ \det = -344, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 4 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 3 & 0 \\ 2 & 4 & 0 & 2 & 0 & 2 & 2 \\ 3 & 5 & 0 & 3 & 2 & 1 & 3 \\ 3 & 6 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

4.341 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^{\overline{1}23^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -95, \quad \det = -251, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

4.342 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 10 \quad [0] \quad 1^{\overline{1}23^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 30, \quad \Delta = -116, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14, \quad T_{\max} = 30, \quad T \in [14, 30]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}23^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 30, \quad \Delta = -116, \quad \det = 18, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.22 \rightarrow 16, \quad T_{\max} = 30, \quad T \in [16, 30]$$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 4 & 0 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}23^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 30, \quad \Delta = -116, \quad \det = 83, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, \quad T_{\max} = 30, \quad T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}23^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 30, \quad \Delta = -116, \quad \det = 118, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.12 \rightarrow 15, \quad T_{\max} = 30, \quad T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}23^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 30, \quad \Delta = -116, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.88 \rightarrow 14, \quad T_{\max} = 30, \quad T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.343 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \mathfrak{n}3\text{mix} + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{\overline{1}23^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -86, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.88 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

4.344 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_{2\text{n}3\text{mix}} + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{123}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 4 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{123}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -89, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.345 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11}3^{121}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -3 & 2 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{121}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{121}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{123}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{123}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 74, \Delta = -122, \det = 193, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 74, T \in [15, 74]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

4.346 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11}3^{121}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & 3 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1}}231^{\overline{1}}3^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1}}23^{\overline{1}}1^{\overline{1}}31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & 3 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1}}23^{\overline{1}}1^{\overline{1}}31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -959, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & 6 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

$$4.347 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 11 \quad [0] \ 1^{\overline{1}}1^{\overline{1}}3^{\overline{1}}21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 0 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & -1 & 2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$4.348 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \ 1^{\overline{1}}23^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -95, \det = -251, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$4.349 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3$$

$$T = 10 \quad [0] \ 1^{\overline{1}}1^{\overline{1}}3^{\overline{1}}21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 2 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

4.350 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix

$$T = 8 \quad [0] \ 1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.70 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & -3 & -1 \\ 4 & 3 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 89, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 9 & 4 & 1 & 4 & 5 \\ 4 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 2 & 0 & -3 & -1 \\ 5 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 99, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 145, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.54 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 190, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 195, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.61 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 4 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{14} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 249, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.67 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 5 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 267, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.60 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 304, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -165, \ \det = 422, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 5 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -2 & 7 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 449, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 5 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 6 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 6 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -2 & 7 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 825, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [14] $1^{12}3^{11}1^{11}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{12}31^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 \\ 6 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & -2 & 0 \\ 4 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{12}31^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 4 & -1 \\ 4 & 0 & 0 & 4 & 9 & 5 \\ 2 & 0 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{12}31^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 4 & -1 \\ 6 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{12}31^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.11 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & -1 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{12}31^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & -1 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

4.351 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 4 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -2 & 3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 3 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 2 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -2 & 7 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 3 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 6 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -2 & 7 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 4 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -2 & 5 & 8 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{1123} 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -154, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$
 $T_{\min} = 13.51 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{142}3^{11}21^1 + [\textcolor{blue}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{red}{\mathbf{e}}'_7 + \textcolor{orange}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & 2 & -2 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 3 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -2 & 9 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[10] \quad \overset{11}{1} \overset{122}{3} \overset{1}{2} \overset{1}{1} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}'_7 + \textcolor{orange}{s} \textcolor{red}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 4 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -2 & 5 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3^{21} 1^{1+} [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}'_7 + \textcolor{orange}{s} u_2 + \textcolor{red}{s} u_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -187, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{123}3^{21}1^{1+}[\textcolor{red}{\mathbf{su}_2}\cdot\textcolor{blue}{\mathbf{g}_2}]+\textcolor{green}{\mathbf{c}}'_7+\textcolor{orange}{\mathbf{su}_2}+\textcolor{red}{\mathbf{su}_2}$ $T=9$, $T_{\max}=95$, $\Delta=-135$, $\det=-191$, $(n_+, n_-, n_0)=(1, 9, 0)$, $c_{\text{ext}}=12.8$
 $T_{\min}=13.64 \rightarrow 14$, $T_{\max}=95$, $T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 3 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -2 & 5 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^{11} 2^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}'_7 + \textcolor{blue}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.60 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & -2 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 2 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 3^{12} 2^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -243, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.77 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 4 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{11}3^{11}2^{11}+[\textcolor{red}{s}u_2\cdot\textcolor{blue}{g}_2]+[\textcolor{red}{c}'_7+\textcolor{red}{s}u_2+\textcolor{red}{s}u_2] \quad T=9, \, T_{\max}=95, \, \Delta=-135, \, \det=-244, \, (n_+, n_-, n_0)=(1, 9, 0), \, c_{\text{ext}}=12.0$
 $T_{\min}=13.48 \rightarrow 14, \, T_{\max}=95, \, T \in [14, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$

$$[16] \quad 1^{12} 3^{21} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -246, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.58 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -2 & 5 & 4 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -274, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$
 $T_{\min} = 13.52 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{121}3^{121}1^{1+}[\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\text{g}}_2] + \textcolor{green}{c}'_7 + \textcolor{orange}{\text{su}}_2 + \textcolor{red}{\text{su}}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -279, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$
 $T_{\min} = 13.55 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & -2 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{11} 1^{132} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 4 & 7 & 0 & 4 & 23 & 14 & 7 \\ 3 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -2 & 7 & 4 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{12} 3^{12} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 5 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -2 & 5 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{13} 2^3 1^{21} + [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{blue}{\mathbf{g}_2}] + \textcolor{green}{c'_7} + \textcolor{orange}{\mathbf{su}_2} + \textcolor{red}{\mathbf{su}_2} \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -346, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.64 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 4 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -2 & 7 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{131} 3^{21} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -379, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.62 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{12} 3^{11} 2^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}'_7 + \textcolor{brown}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -382, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 3^{11} 2^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}'_7 + \textcolor{red}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -387, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^{122} 2^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 3 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -2 & 5 & 4 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{11} 1^{132} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 5 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -2 & 7 & 4 & -1 \end{pmatrix}_{b_0}$

[27] $1^{14}2^31^{21}1^{11}+[\textcolor{blue}{5}\textcolor{red}{u}2\cdot\textcolor{blue}{9}2]+[\textcolor{red}{c}'_7+\textcolor{blue}{5}\textcolor{red}{u}2+\textcolor{red}{5}\textcolor{blue}{u}2] \quad T=9, \, T_{\max}=95, \, \Delta=-135, \, \det=-454, \, (n_+, n_-, n_0)=(1, 9, 0), \, c_{\text{ext}}=12.5$
 $T_{\min}=13.70 \rightarrow 14, \, T_{\max}=95, \, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & 2 & -2 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -2 & 9 & 4 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{12} 3^{14} 21^{21} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & 2 & -2 \\ 7 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -2 & 9 & 4 & -1 \end{pmatrix}_{b_0}$

[29] $1^{141}3^{121}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -487, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & 1 & -2 \\ 5 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11} 1^{12} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -502, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & -2 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11} 3^{21} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -508, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.49 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 4 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[32] $1^{11} 1^{123} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$
 $T_{\min} = 13.55 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 4 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -2 & 5 & 6 & -1 \end{pmatrix}_{b_0}$

$$[33] \ 1^{\overline{11}} 1^{11} 3^{\overline{21}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -607, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.50 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 5 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -2 & 3 & 8 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{12} 3^{12} 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, \, T_{\max} = 95, \, \Delta = -135, \, \det = -607, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 12.0$
 $T_{\min} = 13.52 \rightarrow 14, \, T_{\max} = 95, \, T \in [14, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & -2 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$

$$T = 10 \quad [44] \quad 1^1 23^1 1^{11} 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 6 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^1 23^1 1^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^1 231^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 4 & 0 & -1 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 5 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^1 23^1 131^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 0 & -1 \\ 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^1 23^1 1^1 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 3 & -1 \\ 2 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^1 23^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^1 231^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 143, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 4 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^1 23^1 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 230, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^1 231^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 242, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & 0 & -1 \\ 8 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 10 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{11} 23^1 1^{12} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 764, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{11} 23^1 1^{12} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 795, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

4.352 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 71, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 0 & 1 & -2 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & 1 & -2 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 3 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 286, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & 1 & -2 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 1 & 2 & 0 & 1 & 3 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 315, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & 1 & -2 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 & 5 \\ 2 & 2 & 0 & 1 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 5 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$T = 11 \quad [12] \quad 1^{123}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -231, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

4.353 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{123}1^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

4.354 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111}1^{11}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

4.355 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{123}1^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14, T_{\max} = 30, T \in [14, 30]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{123}1^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 30, T \in [16, 30]$$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 4 & 0 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{123}1^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 30, T \in [14, 30]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 0 & 2 \\ 4 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{123}1^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{123}1^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 30, T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.356 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{su2n3mix} + \text{su}_2 \rightarrow (-3) + \text{su}_2$

$$T = 11 \quad [0] \quad 1^{123}1^{13}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -86, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.88 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

4.357 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{su2n3mix} + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{123}1^{13}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 15, \quad \Delta = -89, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 4 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{123}1^{13}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 15, \quad \Delta = -89, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.358 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{su2n3mix} + \text{su2n3mix}$

$$T = 10 \quad [0] \quad 1^{123}1^{13}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 30, \quad \Delta = -116, \quad \det = 83, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, \quad T_{\max} = 30, \quad T \in [14, 30]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 0 \\ 4 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.359 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{su2n3mix} + \text{su}_3$

$$T = 9 \quad [0] \quad 1^{111}3^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 4 & 3 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.90 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121}3^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -424, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.97 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{121}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -806, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -1056, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [7] $1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 3 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 74, \Delta = -122, \det = 292, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 74, T \in [15, 74]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 4 & 0 & -1 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{123}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 877, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & 0 & -1 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

4.360 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

4.361 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -2 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -3 & -1 \\ 4 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

- $T = 9$ $[8] 1^1 3 2 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -142, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 18, T_{\max} = 99, T \in [18, 99]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$
- [9] $1^{12} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 14.54 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 6 & 0 & 2 & 9 & 5 \\ 2 & 3 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$
- [10] $1^{12} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 99, T \in [16, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [11] $1^{11} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{11} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{12} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$
- [14] $1^{11} 3 2 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -142, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.05 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [15] $1^{11} 3 2 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -142, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{13} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 4 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [17] $1^{12} 3 2 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -142, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

- [18] $1^{13}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$
- [20] $1^{12}31^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 3 & 3 \\ 2 & 0 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$
- [19] $1^{12}31^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [21] $1^{12}31^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -93$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.81 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [22] $1^{12}31^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -102$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.12 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [23] $1^{11}2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -106$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{12}2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.93 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 3 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -2 & 5 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{11}3^2 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -2 & 0 \\ 4 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{11}2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 3 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$
- [27] $1^{12}3^2 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -169$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [28] $1^{12}31^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.27 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{11}231^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.04 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{13}3^{11}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{12}2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.75 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 4 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -2 & 5 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{13}2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 5 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11}231^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -282$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.04 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.362 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$T = 10$ $[0] 1^{111}3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10$, $T_{\max} = 69$, $\Delta = -121$, $\det = 82$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15$, $T_{\max} = 69$, $T \in [15, 69]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ $[1] 1^{111}3 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 15$, $\Delta = -92$, $\det = -61$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.28 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 3 & 4 & -1 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

4.363 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 9$ $[0] 1^{111}3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 1^{11} 2^1 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 2^1 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

4.364 NHC(-2)(-3) + $\textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -178$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 14.07 \rightarrow 15$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 \\ 2 & 4 & 0 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 0 \\ 2 & 2 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -181$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 14.17 \rightarrow 15$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 \\ 2 & 4 & 0 & 2 & 5 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 \\ 2 & 2 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.91 \rightarrow 14$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.72 \rightarrow 14$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -214$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.11 \rightarrow 15$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -243$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T \in [14, 99]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{114} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.31 \rightarrow 15$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.72 \rightarrow 14$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{115} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 14.49 \rightarrow 15$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 15 & -1 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{121} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -760, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{113} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -760, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -814, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1006, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{113} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{113} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{113} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1606, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -2110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -2464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [29] $1^{1231} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 17, T_{\max} = 58, T \in [17, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[30] $1^{113} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 21.84 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 4 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & -8 \end{pmatrix}_{b_0}$$

[31] $1^{111} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[32] $1^{122} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.7$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 6 & 0 & 2 & 9 & 12 & 5 \\ 3 & 6 & 0 & 2 & 12 & 10 & 6 \\ 2 & 3 & -1 & -2 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$

[33] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 17.78 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & -1 & 0 & -4 \end{pmatrix}_{b_0}$$

[34] $1^{1231} 1^{31} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{124} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 16.19 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 1 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 6 & 0 & 2 & 9 & 24 & 5 \\ 6 & 12 & 0 & 4 & 24 & 46 & 12 \\ 2 & 3 & -1 & -2 & 5 & 12 & -3 \end{pmatrix}_{b_0}$$

[36] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{111} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{1231} 1^{31} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 58, T \in [16, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 3 & 0 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{111} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.12 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 3 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -2 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{124} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 1 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 6 & 0 & 2 & 9 & 24 & 5 \\ 5 & 12 & 0 & 4 & 24 & 46 & 12 \\ 2 & 3 & -1 & -2 & 5 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{112} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{1231} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{123} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 6 & 0 & 2 & 9 & 18 & 5 \\ 4 & 9 & 0 & 3 & 18 & 25 & 9 \\ 2 & 3 & -1 & -2 & 5 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{112} 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{111} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 6 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{111} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & -2 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{111}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{121}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{112}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 8 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{112}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{111}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{111}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{121}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{111}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{113}3^{11}221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 155, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{121}2^13^{12}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 4 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{112}2^13^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[61] $1^{111}2^13^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{13}2^13^{112}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -2 & -1 & 9 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{114}3^{11}221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^1231^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 177, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{111}2^13^{111}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{111}3^{112}221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 187, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 7 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{115} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^1 231^{11} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.08 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{111} 3^{12} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 243, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{112} 2^1 3^{14} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.06 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{121} 2^1 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 3 & 0 & 6 & 2 & 9 & 6 & 5 \\ 2 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112} 2^1 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{112} 2^1 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{121} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 306, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{113} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 323, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 11 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{111} 2^1 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{114} 2^1 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 354, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{112} 3^{112} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[81] $1^{113} 2^1 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{112} 2^1 3^{112} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.04 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{112}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{112}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{112}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{112}3^1221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 411, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[87] $1^1231^{11}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 411, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{114}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 412, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{111}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{113}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{112} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 467, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 9 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{114} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 14 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{114} 3^{112} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 552, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 2 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{112} 3^{112} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{112} 3^{114} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 588, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{113} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 588, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{112} 3^{113} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 635, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3^{112} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{112} 3^{11} 221^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 688, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 10 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{113} 2^3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 729, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 1 & 0 & 3 & 1 & 9 & 2 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

[101] $1^{1231} 1231^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 762, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[102] $1^{112} 2^3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 764, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{114} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 771, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{113} 2^3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 795, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{112} 3^{11} 221^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 803, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 11 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{112} 3^{11} 221^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 812, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 12 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{113} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 812, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 12 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{113}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 832$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.00 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[109] $1^{114}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 900$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.80 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{114}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 909$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.16 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[111] $1^{113}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 915$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.91 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{1231}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 1059$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.12 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[113] $1^{113}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 1136$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{114}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 1163$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.86 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 15 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{113}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 1195$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 13 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{114} 2^1 3^{14} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 1240, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{113} 3^{14} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1251, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.87 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{114} 3^{12} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1260, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{123} 1^{14} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 58, \ \Delta = -120, \ \det = 1302, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.12 \rightarrow 15, \ T_{\max} = 58, \ T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{113} 3^{13} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1472, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.82 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 14 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{114} 3^{13} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1643, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{113} 3^{14} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1643, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.82 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 15 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{114} 3^{12} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1708, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 16 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{114} 3^{14} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1920, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.89 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{114} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 2147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 17 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{114} 3^{14} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 2480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 18 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

4.365 NHC(-2)(-3) + e'_7 + su_3 + su_2 + su_2

$T = 10 \quad [0] 1^{11} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & -1 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{113} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 485, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \ 1^{111} 2 3 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 2 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 2 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -69, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -101, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 8 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 & 9 \\ 3 & 4 & -1 & 0 & 0 & 9 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -209, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 14 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 & 17 \\ 3 & 4 & -1 & 0 & 0 & 17 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 2 \ 3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -231, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -254, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 11 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 & 13 \\ 3 & 4 & -1 & 0 & 0 & 13 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112} 3 \ 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -296, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -300, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -320, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1112} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -327, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -363, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1111} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -381, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1114} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -447, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1113} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -453, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{1113}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -704, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{1114}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -954, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

4.366 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 11$ [0] $1^{1111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.367 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.368 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 11$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.369 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{11}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.17 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1231} 1^{11} 3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1231} 1^{11} 3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1231} 1^{12} 3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -959, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.370 \quad \text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$$

$$T = 11 \quad [0] \ 1^{111} 3^{11} 2^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$4.371 \quad \text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_{2n3\text{mix}}$$

$$T = 10 \quad [0] \ 1^{111} 3^{11} 2^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

$$4.372 \quad \text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$$

$$T = 10 \quad [0] \ 1^{111} 3^{11} 2^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 2^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

15.0

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & 0 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 0 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 2 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & 0 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 0 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 2 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -146, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & 0 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 0 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -182, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.26 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -234, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.374 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{12}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12}31^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 74, \Delta = -122, \det = 193, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 74, T \in [15, 74]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

4.375 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 287, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 495, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 528, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 807, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{123}1^{12}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -122, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.26 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 6 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 4 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & -1 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -239, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.17 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 15, \ \Delta = -92, \ \det = -296, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}23^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}23^{11}1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -959, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 6 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & -1 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

4.376 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{11}1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 0 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

4.377 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \ 1^{11}23^{11}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -95, \det = -251, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.378 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{11}1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & 2 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

4.379 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \mathfrak{n}3 \text{mix}$

$$T = 9 \quad [0] \ 1^{11}1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -806, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -1056, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [7] $1^1 23^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

[8] $1^1 23^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[9] $1^1 231^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 74, \Delta = -122, \det = 292, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 74, T \in [15, 74]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

[10] $1^1 23^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{11}23^{11}1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 80$, $\Delta = -123$, $\det = 877$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.13 \rightarrow 15$, $T_{\max} = 80$, $T \in [15, 80]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -1 & -1 & 10 & -1 \end{pmatrix}_{b_0}$$

4.380 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$

$T = 10$ [0] $1^{111}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 150$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

4.381 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$

$T = 9$ [0] $1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -39$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.08 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{12}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -57$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 14.05 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -246$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.29 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -384$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.05 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [5] $1^{121}23^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 80$, $\Delta = -123$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 27.67 \rightarrow 28$, $T_{\max} = 80$, $T \in [28, 80]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 6 & 0 & 2 & 9 & 6 & 5 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -2 & 5 & 2 & -14 \end{pmatrix}_{b_0}$$

[6] $1^{11}2^3 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & -2 \\ 1 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}2^3 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 18, T_{\max} = 80, T \in [18, 80]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 2 & -4 \end{pmatrix}_{b_0}$$

[8] $1^{12}3^2 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{11}2^3 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{12}31^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 135, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^2 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11}2^3 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.10 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11}2^3 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^2 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 195, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

14.0

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 2^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 279, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.382 NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \ 1^{111} 2^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 2^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$(1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1113} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -1179, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1114} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -1854, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$4.383 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 11 \quad [0] \quad 1^{1111} 3^2 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.384 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \quad 1^{1111} 3^2 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 0 & 4 & 5 \\ 2 & 4 & -3 & 0 & 0 & 2 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 3 & 5 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.385 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$$

$$T = 10 \quad [0] \quad 1^{1111} 3^2 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -216, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 \\ 2 & 3 & -1 & -2 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{11}221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 23, \Delta = -100, \det = -321, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.55 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.386 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + (-3) \rightarrow (-2)$

$T = 8 \quad [0] \quad 1^1231^{11}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 101, \Delta = -172, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

4.387 $\text{NHC}(-2)(-3) + \mathfrak{e}_8$

$T = 8 \quad [0] \quad 1^{11}3^{11}221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 \quad T = 8, T_{\max} = 191, \Delta = -306, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 17.96 \rightarrow 18, T_{\max} = 191, T \in [18, 191]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -8 & -6 \\ 2 & 2 & -1 & -6 & -4 \end{pmatrix}_{b_0}$$

4.388 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] \quad 1^{11}3^{11}221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + [\textcolor{brown}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 172, \Delta = -256, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 172, T \in [19, 172]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 5 \\ 7 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & -6 & -4 \\ 5 & 6 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + [\textcolor{brown}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 172, \Delta = -256, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.74 \rightarrow 19, T_{\max} = 172, T \in [19, 172]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 4 & -5 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & -5 & 5 & -4 \end{pmatrix}_{b_0}$$

4.389 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$T = 11 \quad [0] \quad 1^{111}3^{11}221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + [\textcolor{brown}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -271, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 7 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 5 & -4 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -4 & 6 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + [\textcolor{brown}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -836, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 5 & -4 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -4 & 6 & -3 \end{pmatrix}_{b_0}$$

4.390 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 11 \quad [0] \ 1^{\overline{11}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 109, \ \Delta = -235, \ \det = -166, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.89 \rightarrow 19, \ T_{\max} = 109, \ T \in [19, 109]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 4 & 0 & -5 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -5 & 5 & -1 & -4 \end{pmatrix}_{b_0}$$

4.391 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{11}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 192, \ \Delta = -305, \ \det = -190, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.24 \rightarrow 20, \ T_{\max} = 192, \ T \in [20, 192]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 & -6 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & -1 & -6 & 0 & -6 & 0 \end{pmatrix}_{b_0}$$

4.392 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 190, \ \Delta = -275, \ \det = 251, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.30 \rightarrow 20, \ T_{\max} = 190, \ T \in [20, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 & -6 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

4.393 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{12}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 189, \ \Delta = -276, \ \det = -48, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 18.83 \rightarrow 19, \ T_{\max} = 189, \ T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 0 & -6 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & -1 & -6 & 0 & -5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 189, \ \Delta = -276, \ \det = -81, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.09 \rightarrow 19, \ T_{\max} = 189, \ T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 0 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & -1 & -6 & 0 & -5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 189, \ \Delta = -276, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 18, \ T_{\max} = 189, \ T \in [18, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 \\ 2 & 2 & -1 & -6 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{12}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 189, \ \Delta = -276, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 18.10 \rightarrow 19, \ T_{\max} = 189, \ T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 0 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 & 0 \\ 2 & 2 & -1 & -6 & 4 & -5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{13}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 189, \ \Delta = -276, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 18.22 \rightarrow 19, \ T_{\max} = 189, \ T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & -6 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & -5 \end{pmatrix}_{b_0}$$

4.394 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 11 \quad [0] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 106, \ \Delta = -226, \ \det = -271, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 5 \\ 7 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 0 & -4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & -4 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 106, \ \Delta = -226, \ \det = -836, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 5 & -4 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -4 & 6 & -3 \end{pmatrix}_{b_0}$$

4.395 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 10 \quad [0] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 190, \ \Delta = -275, \ \det = 251, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.30 \rightarrow 20, \ T_{\max} = 190, \ T \in [20, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

4.396 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{132} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 49, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.3$$

$$T_{\min} = 18.49 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 2 & -6 \\ 3 & 6 & 0 & 3 & 16 & 12 & 6 \\ 3 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -6 & 6 & 4 & -5 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 89, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.09 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -6 & 0 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 18.11 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -6 & 4 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 140, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.26 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 0 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & 0 & -5 \end{pmatrix}_{b_0}$$

[4] $1^{124} 3^{113} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 142, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.27 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[5] $1^{144} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.62 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & 4 & -6 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & 8 & -5 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 18.04 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[7] $1^{13} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 3 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -6 & 0 & 6 & -5 \end{pmatrix}_{b_0}$$

[8] $1^{112} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3^{112} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 18.26 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[10] $1^{122} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.20 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 2 & -6 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -6 & 4 & 4 & -5 \end{pmatrix}_{b_0}$$

[11] $1^{113} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 18.24 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 3 & -6 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -6 & 6 & 6 & -5 \end{pmatrix}_{b_0}$$

[20] $1^{122} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 414, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 2 & -6 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -6 & 4 & 4 & -5 \end{pmatrix}_{b_0}$$

[21] $1^{124} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 424, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.36 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[22] $1^{134} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.40 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 4 & -6 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -6 & 6 & 8 & -5 \end{pmatrix}_{b_0}$$

[23] $1^{115} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$

$$T_{\min} = 18.40 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 5 & -6 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -6 & 2 & 10 & -5 \end{pmatrix}_{b_0}$$

[24] $1^{112} 3^{114} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[25] $1^{125} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 532, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.6$

$$T_{\min} = 18.42 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 5 & -6 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -6 & 4 & 10 & -5 \end{pmatrix}_{b_0}$$

[26] $1^{144} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 568, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & 4 & -6 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & 8 & -5 \end{pmatrix}_{b_0}$$

[27] $1^{135} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.9$

$$T_{\min} = 18.45 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 5 & -6 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -6 & 6 & 10 & -5 \end{pmatrix}_{b_0}$$

[28] $1^{116} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 18.46 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 6 & -6 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -6 & 2 & 12 & -5 \end{pmatrix}_{b_0}$$

[29] $1^{112} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 633, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 18.11 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & -6 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[31] $1^{126} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 664, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.47 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 6 & -6 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -6 & 4 & 12 & -5 \end{pmatrix}_{b_0}$$

[32] $1^{145} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.1$

$$T_{\min} = 18.47 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & 5 & -6 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -6 & 8 & 10 & -5 \end{pmatrix}_{b_0}$$

[33] $1^{134} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 689, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.22 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 4 & -6 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -6 & 6 & 8 & -5 \end{pmatrix}_{b_0}$$

[34] $1^{123} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 722, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$

$$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 3 & -6 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -6 & 4 & 6 & -5 \end{pmatrix}_{b_0}$$

[35] $1^{136} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.1$

$$T_{\min} = 18.49 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 6 & -6 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -6 & 6 & 12 & -5 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 769, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 18.26 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[37] $1^{117} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 784, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$

$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 7 & -6 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -6 & 2 & 14 & -5 \end{pmatrix}_{b_0}$$

[38] $1^{112} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 808, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[39] $1^{114} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 819, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[40] $1^{127} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 820, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 7 & -6 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -6 & 4 & 14 & -5 \end{pmatrix}_{b_0}$$

[41] $1^{112} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 859, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[42] $1^{137} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 880, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.1$

$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 7 & -6 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -6 & 6 & 14 & -5 \end{pmatrix}_{b_0}$$

[43] $1^{113} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 912, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[44] $1^{113} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 941, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[45] $1^{124} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 960, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[46] $1^{124} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1054, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[47] $1^{122} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 2 & -6 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -6 & 4 & 4 & -5 \end{pmatrix}_{b_0}$$

[48] $1^{114} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[49] $1^{113} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[50] $1^{123} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1451, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 3 & -6 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -6 & 4 & 6 & -5 \end{pmatrix}_{b_0}$$

[51] $1^{124} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1528, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[52] $1^{113} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1568, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.25 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[53] $1^{113} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1573, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.18 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[54] $1^{133} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1844, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.6$

$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 3 & -6 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -6 & 6 & 6 & -5 \end{pmatrix}_{b_0}$$

[55] $1^{114} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[56] $1^{134} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 2091, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 4 & -6 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -6 & 6 & 8 & -5 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 2401, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[58] $1^{144} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 2508, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & 4 & -6 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & 8 & -5 \end{pmatrix}_{b_0}$$

[59] $1^{114} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 2644, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.19 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

4.397 NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [1] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -32, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.13 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & -1 & -6 & 0 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{1112} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -32, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 23.12 \rightarrow 24, \ T_{\max} = 106, \ T \in [24, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 2 & 4 & -10 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1211} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -56, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.14 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & 1 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 4 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1211} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -114, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.14 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & 1 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 4 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -138, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 18.14 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 3 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 0 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 0 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -314, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.17 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 1 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 0 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -372, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.13 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

[14] $1^{1122}3^{221}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -682$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & 4 & -5 \end{pmatrix} b_0$$

[15] $1^{1114}3^{221}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -800$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 18.34 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 2 & 8 & -5 \end{pmatrix} b_0$$

[16] $1^{1112}3^{221}1 + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_8 + \textcolor{orange}{5}\textcolor{blue}{u}_2 + \textcolor{red}{5}\textcolor{blue}{u}_2 + \textcolor{blue}{5}\textcolor{blue}{u}_2$ $T = 11, T_{\max} = 106, \Delta = -216, \det = -960, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 \\ 4 & 4 & 0 & 2 & 4 & 4 & 6 \\ 2 & 2 & -1 & -6 & 2 & 2 & -5 \end{pmatrix} b_0$$

[17] $1^{1111}3^{221} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_8 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -1004$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & 2 & -5 \end{pmatrix} b_0$$

[18] $1^{1115}3^{221}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 106, \Delta = -216, \det = -1016, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.1$

$$T_{\min} = 18.41 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 5 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 10 & 48 & 10 \\ 2 & 2 & -1 & -6 & 2 & 2 & 10 & -5 \end{pmatrix} b_0$$

[19] $1^{1112}3^{221}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -1120$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 18.12 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 \\ 3 & 4 & 0 & 2 & 4 & 4 & 6 \\ 2 & 2 & -1 & -6 & 2 & 2 & 4 \end{pmatrix} b_0$$

[20] $1^{1114} 3^{221} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 106, \Delta = -216, \det = -1322, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 2 & 8 & -5 \end{pmatrix} b_0$$

$$[21] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}\textcolor{violet}{2}} 3^{\textcolor{blue}{1}} 221^{\textcolor{blue}{1}} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{violet}{g}_2] + \textcolor{brown}{c}_8 + \textcolor{blue}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -1402, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

[29] $1^{1114} 3^{221} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -3232$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 6 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -6 & 2 & 2 & 8 \end{pmatrix} b_0$$

[30] $1^{1114} 3^{221} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 106, \Delta = -216, \det = -3296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 2 & 8 & -5 \end{pmatrix} b_0$$

4.398 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 3 \quad 221^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{c}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 12, \quad T_{\max} = 106, \quad \Delta = -186, \quad \det = 112, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 1 & 2 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 0 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 0 & 2 & 2 & 4 & -5 \end{pmatrix} b_0$$

[1] $1^{11111} 3^{221} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12$, $T_{\max} = 106$, $\Delta = -186$, $\det = 812$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 18.10 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 2 & 0 \\ 2 & 2 & -1 & -6 & 2 & 2 & 2 & -5 \end{pmatrix} b,$$

[2] $1^{11111}3^{221}1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 106, \Delta = -186, \det = 928, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.10 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & 2 & -5 \end{pmatrix} b,$$

[3] $1^{11111} 3^{221} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 106, \Delta = -186, \det = 1376, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & 2 & 2 & -5 \end{pmatrix} b,$$

4.399 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 12 \quad [0] \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{3} \overset{1}{2} \overset{1}{2} \overset{1}{1} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_8 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_3 \quad T = 12, T_{\max} = 106, \Delta = -195, \det = 212, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -6 & 2 & 2 & -1 & -6 \end{pmatrix}_{b_0}$$

[1] $1^{1111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, \ T_{\max} = 106, \ \Delta = -195, \ \det = 310, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$

$T_{\min} = 18.69 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.400 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 11 \quad [0] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -104, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$

$T_{\min} = 18.54 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -5 & 0 & 0 & -4 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -121, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$

$T_{\min} = 18.67 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -172, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$

$T_{\min} = 18.63 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -240, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$

$T_{\min} = 19.12 \rightarrow 20, \ T_{\max} = 106, \ T \in [20, 106]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -6 & 2 & 2 & -1 & -6 \end{pmatrix}_{b_0}$$

[4] $1^{131} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -365, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.8$

$T_{\min} = 18.87 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 9 & 9 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 & 12 \\ 3 & 4 & -1 & -5 & 0 & 12 & -4 & 4 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -629, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$

$T_{\min} = 18.52 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -648, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -716, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 18.58 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 6 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -5 & 0 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -760, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -901, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.53 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -1664, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.51 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.401 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 12 \quad [0] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 106, \ \Delta = -195, \ \det = 212, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 20, \ T_{\max} = 106, \ T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 0 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & -1 & 2 & -6 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 106, \ \Delta = -195, \ \det = 310, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.69 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.402 NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 0 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 187, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.87 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 \\ 2 & 2 & -1 & -6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.64 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 6 & -5 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 306, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{112} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 18.84 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -6 & 2 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 342, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 18.71 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -6 & 2 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.60 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111} 3^{221} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 447, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

14.0

$$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[9] $1^{131} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 450, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.68 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[10] $1^{111} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 560, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[11] $1^{141} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.75 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 578, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[13] $1^{112} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 731, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 18.46 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -6 & 2 & 3 & -5 \end{pmatrix}_{b_0}$$

[14] $1^{151} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 738, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.1$

$$T_{\min} = 18.80 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 15 & -5 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -5 & 20 & -4 \end{pmatrix}_{b_0}$$

[15] $1^{121} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[16] $1^{121} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

15.0

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[26] $1^{141} 3^{114} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 171, \Delta = -255, \det = 4640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

4.403 NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$T = 11$ [0] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -104, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 0 & 0 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -121, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.67 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & -4 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^{112} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -172, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.63 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -240, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 19.12 \rightarrow 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 & -6 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & -1 & -6 \end{pmatrix}_{b_0}$$

[4] $1^{113} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -365, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$

$$T_{\min} = 18.87 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 9 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -5 & 0 & 12 & -4 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -629, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -648, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -716, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 18.58 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 6 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -5 & 0 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -760, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -901, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.53 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -1664, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.51 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.404 NHC(-2)(-3) + $\mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 12 \quad [0] \ 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 106, \ \Delta = -195, \ \det = 212, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 20, \ T_{\max} = 106, \ T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & -1 & 2 & -6 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 106, \ \Delta = -195, \ \det = 310, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.69 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.405 NHC(-2)(-3) + e₈ + su₂ + su₃ + su₃

$$T = 11 \quad [1] \ 1^{111} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 108, \Delta = -234, \det = -291, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.56 \rightarrow 20, T_{\max} = 108, T \in [20, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -5 & 0 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 108, \Delta = -234, \det = -427, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.95 \rightarrow 19, T_{\max} = 108, T \in [19, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 0 & -5 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -5 & 4 & -1 & -4 \end{pmatrix}_{b_0}$$

4.406 NHC(-2)(-3) + e₈ + su₃

$$T = 9 \quad [0] \ 1^1 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.75 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & -6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -6 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -7 & -5 \\ 3 & 4 & -1 & -5 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 18.69 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & -6 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -6 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.83 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & -6 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -6 & 5 & -5 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -257, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -7 & -5 \\ 4 & 4 & -1 & -5 & -4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 18.46 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & -6 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -6 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.43 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & -6 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -6 & 5 & -5 \end{pmatrix}_{b_0}$$

4.407 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 11 \quad [0] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 109, \Delta = -235, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.89 \rightarrow 19, T_{\max} = 109, T \in [19, 109]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 4 & -5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 3 & 4 & -1 & -5 & -1 & 5 & -4 \end{pmatrix}_{b_0}$$

4.408 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 0 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 187, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.87 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -6 & -1 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.64 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 6 & -5 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 306, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{121} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 18.84 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & -6 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -6 & 3 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 342, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 18.71 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & -6 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -6 & 3 & 2 & -5 \end{pmatrix}_{b_0}$$

- [7] $1^{112}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 360$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 18.60 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$
- [6] $1^{111}3^{113}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 360$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.50 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$
- [8] $1^{111}3^{111}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 447$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.54 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$
- [9] $1^{113}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 450$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 18.68 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$
- [10] $1^{111}3^{111}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 560$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.51 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$
- [11] $1^{114}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 576$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$
- [12] $1^{111}3^{112}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 578$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.50 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$
- [13] $1^{121}3^{111}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 731$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 18.46 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & -6 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -6 & 3 & 2 & -5 \end{pmatrix}_{b_0}$$
- [14] $1^{115}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 738$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.1$
 $T_{\min} = 18.80 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 15 & -5 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -5 & 20 & -4 \end{pmatrix}_{b_0}$$

[15] $1^{112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[16] $1^{112} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1448, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[18] $1^{112} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1448, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[19] $1^{112} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1530, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1946, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[21] $1^{113} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 2210, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[22] $1^{113} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 2736, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[23] $1^{114} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 3026, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

14.8

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[24] $1^{114} 3^{112} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 171, \Delta = -255, \det = 3080, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[25] $1^{114} 3^{113} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 171, \Delta = -255, \det = 3978, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3^{114} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 171, \Delta = -255, \det = 4640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

4.409 NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$T = 11$ [0] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -104, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 0 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -121, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.67 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^{112} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -172, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.63 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -240, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 19.12 \rightarrow 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 \\ 2 & 2 & -1 & -6 & -1 & 2 & -6 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -365, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 18.87 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 9 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -5 & 0 & 12 & -4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -629, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.52 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -648, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -716, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 18.58 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 6 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -5 & 0 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -760, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -901, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.53 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -1664, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.51 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.410 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 12 \quad [0] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 106, \ \Delta = -195, \ \det = 212, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 20, \ T_{\max} = 106, \ T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 1 & -6 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 0 \\ 2 & 2 & -1 & -6 & -1 & 2 & 2 & -6 \end{pmatrix}_{b_0}$$

[1] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 106$, $\Delta = -195$, $\det = 310$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 18.69 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.411 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 11$ [1] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 108$, $\Delta = -234$, $\det = -291$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 19.56 \rightarrow 20$, $T_{\max} = 108$, $T \in [20, 108]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -5 & 0 & 3 & -5 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 108$, $\Delta = -234$, $\det = -427$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 18.95 \rightarrow 19$, $T_{\max} = 108$, $T \in [19, 108]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & -1 & 4 & -4 \end{pmatrix}_{b_0}$$

4.412 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$

$T = 10$ [1] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 181$, $\Delta = -264$, $\det = 314$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 18.89 \rightarrow 19$, $T_{\max} = 181$, $T \in [19, 181]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & -5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -5 & -1 & -4 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 181$, $\Delta = -264$, $\det = 450$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 19$, $T_{\max} = 181$, $T \in [19, 181]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -5 & 3 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 181$, $\Delta = -264$, $\det = 704$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 18.78 \rightarrow 19$, $T_{\max} = 181$, $T \in [19, 181]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -5 & 3 & -4 \end{pmatrix}_{b_0}$$

4.413 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 11$ [1] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 108$, $\Delta = -234$, $\det = -291$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 19.56 \rightarrow 20$, $T_{\max} = 108$, $T \in [20, 108]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 0 & -5 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 3 & 0 & -5 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 108$, $\Delta = -234$, $\det = -427$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 18.95 \rightarrow 19$, $T_{\max} = 108$, $T \in [19, 108]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & -1 & 4 & -4 \end{pmatrix}_{b_0}$$

4.414 NHC(-2)(-3) + f₄

$T = 6$ [0] $21^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 6$, $T_{\max} = 99$, $\Delta = -167$, $\det = 13$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 6$, $T_{\max} = 99$, $\Delta = -167$, $\det = 13$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 6$, $T_{\max} = 99$, $\Delta = -167$, $\det = 21$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -5 & -3 \\ 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 6$, $T_{\max} = 99$, $\Delta = -167$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 99$, $T \in [13, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

[4] $21^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 6$, $T_{\max} = 99$, $\Delta = -167$, $\det = 103$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 7$ [5] $2231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{11} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[7] $21^1 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -10$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 13 & 20 & 7 \\ 20 & 30 & 10 \\ 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[8] $1^1 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -5 & -3 \\ 3 & 5 & -3 & 2 \end{pmatrix}_{b_0}$$

[9] $1^1 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$

$T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{\overline{1}1} 3 2 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{1}1} 1 3 2 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 5 & 13 & 5 \\ 13 & 30 & 10 \\ 5 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overline{1}1} 1 3 2 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{1}1} 1 3 1 3 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overline{1}1} 1 3 2 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\overline{1}1} 1^{\overline{1}1} 3 2 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 10 & 18 & 6 \\ 18 & 30 & 10 \\ 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{\overline{1}1} 1 2 3 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -5 & -3 \\ 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{\overline{1}1} 1^{\overline{1}1} 3 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{\overline{1}1} 1^{\overline{1}1} 3 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 2^{\overline{1}1} 2 3 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 10 & 19 & 6 \\ 19 & 30 & 10 \\ 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 2 1^{\overline{1}1} 3 2 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 13 & 20 & 7 \\ 20 & 25 & 9 \\ 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{\overline{1}1} 1^{\overline{1}1} 3 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 2 & 3 & 3 & -2 & 0 \\ 2 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 2 \overset{1^{11}}{2} 2 3 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 131 \overset{1^{11}}{3} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 1 & 3 & 3 & -2 & 0 \\ 2 & 4 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 31 \overset{1^{11}}{3} 231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -147, \det = -215, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [25] \quad 2 \overset{1^1}{3} 141 \overset{1^{11}}{3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 1 \overset{1^1}{3} 221 \overset{1^{11}}{3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 6 & 3 \\ 5 & 6 & 17 & 7 \\ 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1 \overset{1^1}{3} 3221 \overset{1^{11}}{3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -5 & -3 \\ 3 & 6 & -3 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 131 \overset{1^1}{3} 21 \overset{1^{11}}{3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 18 & 16 & 8 \\ 5 & 16 & 12 & 6 \\ 3 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[31] \quad 1 \overset{1^1}{3} 3221 \overset{1^{11}}{3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1 \overset{1^1}{3} 2321 \overset{1^{11}}{3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1 \overset{1^1}{3} 3221 \overset{1^{11}}{3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[35] \quad 1 \overset{1^{11}}{3} 23131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[36] \quad 1 \overset{1^{12}}{3} 321 \overset{1^{11}}{3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

- [32] $1^{11}23131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 \\ 1 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$
- [33] $1^123131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 3 & 0 & -1 & 2 \end{pmatrix}_{b_0}$
- [34] $1^12321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [37] $1^13221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -110$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & 0 & 2 \\ 3 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [38] $1^13221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -110$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 4 & 3 & 9 & 4 \\ 3 & 1 & 6 & 3 \\ 9 & 6 & 17 & 7 \\ 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$
- [39] $1^1231^131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 14$, $T_{\max} = 35$, $T \in [14, 35]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$
- [40] $1^14131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -137$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -4 & -2 \\ 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [41] $1^13221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -110$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 0 & 2 \\ 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [44] $12321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 19$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 0 \end{pmatrix}_{b_0}$
- [43] $1^{11}41231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 19$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [42] $1^14131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -137$, $\det = 19$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [46] $1^{11}231^131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 4 & 9 & 8 & 4 \\ 9 & 18 & 16 & 8 \\ 8 & 16 & 12 & 6 \\ 4 & 8 & 6 & 3 \end{pmatrix}_{b_0}$
- [45] $1^{11}2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$

- $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 3 & 3 \\ 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$
- [47] $1^1 231^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 23$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 1 & 1 & 5 & 3 \\ 1 & -2 & 1 & 0 \\ 5 & 1 & 12 & 6 \\ 3 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [48] $1^1 231^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$
- [49] $1^1 231^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 3 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [50] $21^1 41^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -137$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [51] $1^1 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -137$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 0 & 4 \\ 1 & 0 & -4 & -2 \\ 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [52] $1^1 41^1 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [53] $21^1 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -137$, $\det = 75$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 0 & 4 \\ 2 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [54] $1^1 41^1 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 13 & 0 & 7 \\ 1 & 0 & -4 & -2 \\ 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$
- [55] $2^1 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 86$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 10 & 24 & 6 \\ 24 & 49 & 13 \\ 6 & 13 & 3 \end{pmatrix}_{b_0}$
- [56] $31^1 12321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 56$, $\Delta = -120$, $\det = 152$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 56$, $T \in [13, 56]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [57] $41^1 12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 369$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 10 & 9 & -3 & 1 \end{pmatrix}_{b_0}$
- [58] $1^1 412321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & -3 & -1 \\ 3 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[59] $1^1 141321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & -3 & -1 \\ 3 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[60] $1^1 231^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 10 & 12 & 6 \\ 12 & 13 & 7 \\ 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[62] $1^1 1^1 51231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[61] $1^1 1^1 15131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[63] $21^{11} 5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 13 & 14 & 7 \\ 14 & 13 & 7 \\ 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[64] $1^1 31^{11} 4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[65] $1^1 31^{11} 4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[66] $21^{11} 5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 10 & 5 \\ 10 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[67] $2^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 6 & 6 & 6 \\ 3 & 6 & 0 & 2 \\ 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[68] $21^1 51^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 13 & 35 & 7 \\ 35 & 90 & 18 \\ 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[69] $1^1 3221^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 17 & 5 & 7 \\ 1 & 0 & 5 & -2 & 0 \\ 2 & 0 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

[70] $1^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 23 & 5 \\ 23 & 90 & 18 \\ 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[71] $1^1 3221^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 1 & 5 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 \\ 2 & 8 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [72] \quad 1^1 31^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^1 31^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^1 51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & 0 & 2 \\ 3 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[75] \quad 21^1 51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & 0 & 2 \\ 3 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^1 1512321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & 0 & 2 \\ 3 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^1 3151^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 7 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^1 3151^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 3 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 31^{11} 315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 10, T_{\max} = 47, \Delta = -119, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 47, T \in [15, 47]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [80] \quad 1^1 31^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^1 31^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^1 6^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[84] \quad 1 \overset{1^1 31}{\underset{1^1}{6}} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 23, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[83] \quad 1 \overset{1^1 321}{\underset{1^1}{6}} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[85] \quad 1 \overset{1^1}{\underset{1^1}{3}} 151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 4 & 4 \\ 2 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 31 \overset{1^1}{\underset{1^1}{3}} 16131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 31, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15, T_{\max} = 31, T \in [15, 31]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[87] \quad 1 \overset{1^1}{\underset{1^1}{3}} 321^1 61231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 1 & 4 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1 \overset{1^1}{\underset{1^1}{3}} 321^1 61231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -55, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 5 & 0 & 6 \\ 1 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 6 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[89] \quad 31 \overset{1^1}{\underset{1^1}{3}} 321^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 0 & 4 & 3 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 3 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & -2 & 0 \\ 2 & 6 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[90] \quad 2321 \overset{1^1}{\underset{1^1}{6}} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -111, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 4 & 2 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 2 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & -2 & 0 \\ 2 & 6 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[92] \quad 31 \overset{1^1}{\underset{1^1}{3}} 161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 34, T \in [16, 34]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[91] \quad 1 \overset{31^{11} 31}{\underset{1^1}{6}} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 31, \Delta = -116, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 31, T \in [16, 31]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [93] \quad 1 \overset{1^1 2321}{\underset{1^1}{6}} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[94] $1^{131} 6^{131} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{131} 6^{131} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 12, T_{\max} = 24, \Delta = -107, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 3 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{1321} 7^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 13$ [97] $1^{12321} 7^{12321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 16, \Delta = -81, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[98] $1^1 15^{131} 6^{131} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 25, \Delta = -102, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{13221} 8^{13221} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{13221} 8^{13221} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{1321} 8^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[102] $31^{1321} 8^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 30, \Delta = -115, \det = -32, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 17, T_{\max} = 30, T \in [17, 30]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[103] $31^{13221} 8^{13221} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 30, \Delta = -115, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 17.13 \rightarrow 18, T_{\max} = 30, T \in [18, 30]$

$$\begin{pmatrix} 0 & 5 & 3 & 1 & 2 \\ 5 & 18 & 6 & 4 & 8 \\ 3 & 6 & 0 & 1 & 2 \\ 1 & 4 & 1 & -2 & 0 \\ 2 & 8 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 14$ [106] $1^{12321} 8^{12321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 14, T_{\max} = 17, \Delta = -80, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{\overline{1}1315131} 6^{11} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[104] $1^{\overline{1}13221} 8^{11} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 14, T_{\max} = 17, \Delta = -80, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{\overline{1}1321} 8^{11} 123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 17.33 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 3 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 15$ [109] $1^{\overline{1}1512321} 8^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -12, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 18.33 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{\overline{1}1} 15^{11} 13218^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -12, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 18.33 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$T = 16$ [110] $1^{\overline{1}13151321} 8^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 16, T_{\max} = 19, \Delta = -80, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 18$ [111] $1^{\overline{1}13221} (12)^{11} 12231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 8, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 20.50 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0}$$

4.415 NHC(-2)(-3) + $\textcolor{green}{f}_4$ + $\textcolor{violet}{e}_6$

$T = 7$ [0] $1^{\overline{1}11} 3^{11} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 102, \Delta = -216, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$T = 8$ [1] $1^{\overline{1}1} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 5 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 321^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 5 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 1^{13} 1^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -195, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11} 231^1 31^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 102, \Delta = -167, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 18 & 16 & 0 & 8 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 231^1 31^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 102, \Delta = -167, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 4 & 3 & 0 & -2 & 0 \\ 2 & 4 & 3 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3221^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -159, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -5 & -3 \\ 5 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 2321^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 2321^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \ 1^{13} 31^1 51321^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 12, T_{\max} = 96, \Delta = -132, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 17, T_{\max} = 96, T \in [17, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 4 & 0 & 1 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

4.416 $\text{NHC}(-2)(-3) + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 + \textcolor{blue}{s}\textcolor{red}{u}_2$

$$T = 8 \quad [0] \ 1^{111} 31^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 + \textcolor{blue}{s}\textcolor{red}{u}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112} 31^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 + \textcolor{blue}{s}\textcolor{red}{u}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -4 & 8 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -4 & 12 & -2 \end{pmatrix}_{b_0}$$

$T = 9 \quad [5] \quad 1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & -3 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & -2 & 7 & 0 \\ 5 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -3 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$T = 10 \quad [8] \quad 1^{112}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[9] $1^{112}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 16.15 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[10] $1^{11}3221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.46 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 5 & 15 & 1 & 21 & 5 \\ 15 & 42 & 0 & 59 & 12 \\ 1 & 0 & -5 & 0 & -3 \\ 21 & 59 & 0 & 79 & 17 \\ 5 & 12 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

4.417 $\text{NHC}(-2)(-3) + \textcolor{green}{f}_4 + \textcolor{orange}{e}_7$

$T = 8 \quad [0] \quad 1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_7 \quad T = 8, T_{\max} = 131, \Delta = -243, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 16.11 \rightarrow 17, T_{\max} = 131, T \in [17, 131]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & -4 & -2 \\ 3 & 5 & -3 & -2 & -3 \end{pmatrix}_{b_0}$$

$T = 9 \quad [1] \quad 1^{11}3221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}_7 \quad T = 9, T_{\max} = 104, \Delta = -214, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 16.41 \rightarrow 17, T_{\max} = 104, T \in [17, 104]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & -3 & -1 \\ 3 & 6 & -3 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 2 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}'_7 \quad T = 9, T_{\max} = 104, \Delta = -216, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.24 \rightarrow 17, T_{\max} = 104, T \in [17, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -5 & -3 \\ 2 & 3 & -1 & 0 & -3 & -3 \end{pmatrix}_{b_0}$$

4.418 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{e}'_7$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3 2 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}'_7 \quad T = 8, T_{\max} = 103, \Delta = -215, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & -3 & -1 \\ 3 & 5 & -3 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \ 1^{\overline{11}} 3 2 2 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}'_7 \quad T = 9, T_{\max} = 103, \Delta = -186, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & -2 & 0 \\ 3 & 6 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 2 3 2 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{e}'_7 \quad T = 9, T_{\max} = 103, \Delta = -188, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -4 & -2 \\ 2 & 3 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

4.419 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4$

$$T = 7 \quad [0] \ 1^{\overline{11}} 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 101, \Delta = -190, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -5 & -3 \\ 5 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3 2 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -5 & -3 \\ 5 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -169, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 18 & 16 & 0 & 8 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3 2 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 6 & 0 & 3 \\ 5 & 6 & 17 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 7 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3 2 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & 0 & 2 \\ 3 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 2 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \quad 1^{11} 3 1^{11} 5 1 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 12, T_{\max} = 24, \Delta = -106, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 43 & 1 & 13 \\ 1 & 1 & -3 & -1 \\ 5 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

4.420 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -3 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -3 & 16 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{115}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 100, \Delta = -160, \det = 1192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -2 & 0 & 15 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 0 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 13.92 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11}321^{115} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 0 & 15 & 3 \\ 1 & 4 & 0 & 1 & 0 & 15 & 2 \\ 1 & 1 & 1 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 15 & 5 & 0 & 73 & 15 \\ 2 & 3 & 2 & -1 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{11}321^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 4 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 4 & 0 & 46 & 12 \\ 2 & 3 & 2 & -1 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{11}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 0 & 9 & 3 \\ 1 & 4 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 3 & 0 & 25 & 9 \\ 2 & 3 & 2 & -1 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 2 & 0 & 10 & 6 \\ 2 & 3 & 2 & -1 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^{117} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 1 & 0 & 21 & 3 \\ 1 & 4 & 0 & 1 & 0 & 21 & 2 \\ 1 & 1 & 1 & -3 & 0 & 7 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 21 & 7 & 0 & 145 & 21 \\ 2 & 3 & 2 & -1 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}321^{116} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.6$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 1 & 0 & 18 & 3 \\ 1 & 4 & 0 & 1 & 0 & 18 & 2 \\ 1 & 1 & 1 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 18 & 6 & 0 & 106 & 18 \\ 2 & 3 & 2 & -1 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{111}321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -165$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 13.61 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$

$T = 10$ [15] $1^{11}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 19$, $T_{\max} = 22$, $T \in [19, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 0 & 0 & -5 \end{pmatrix}_{b_0}$

[16] $1^{11}3221^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 88$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 6 & 0 & 6 & 11 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 8 & 2 \\ 5 & 11 & 0 & 8 & 15 & 9 \\ 3 & 6 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$

[17] $1^{11}3221^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 103$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 6 & 0 & 6 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 13 & 2 \\ 7 & 17 & 0 & 13 & 36 & 14 \\ 3 & 6 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$

[18] $1^{14}3221^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 103$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.4$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 6 & 0 & 6 & 38 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 27 & 2 \\ 18 & 38 & 0 & 27 & 211 & 31 \\ 3 & 6 & -3 & 2 & 31 & 1 \end{pmatrix}_{b_0}$

[19] $1^{12}3221^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 208$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.9$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 6 & 0 & 6 & 22 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 16 & 2 \\ 10 & 22 & 0 & 16 & 66 & 18 \\ 3 & 6 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$

[20] $1^{12}3221^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 343$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.9$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 6 & 0 & 6 & 28 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 21 & 2 \\ 12 & 28 & 0 & 21 & 103 & 23 \\ 3 & 6 & -3 & 2 & 23 & 1 \end{pmatrix}_{b_0}$

[21] $1^{13}3221^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 408$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 6 & 0 & 6 & 33 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 24 & 2 \\ 15 & 33 & 0 & 24 & 151 & 27 \\ 3 & 6 & -3 & 2 & 27 & 1 \end{pmatrix}_{b_0}$

4.421 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$

$T = 9$ [0] $1^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 92$, $\Delta = -130$, $\det = -544$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.4$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 92$, $T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 0 & -3 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \quad 1^{\overline{1}} \overline{3} 2 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 1 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 1 \\ 2 & 3 & 2 & -1 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1}} \overline{3} 2 1^{\overline{1112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.9$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & 2 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 1 & 6 & 3 \\ 2 & 6 & 6 & 2 & 0 & 6 & 10 & 6 \\ 2 & 3 & 2 & -1 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.422} \quad \mathbf{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \quad 1^{\overline{111}} \overline{3} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.423} \quad \mathbf{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 8 \quad [0] \quad 1^{\overline{11}} \overline{3} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.424} \quad \mathbf{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$$

$$T = 9 \quad [0] \quad 1^{\overline{111}} \overline{3} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.425} \quad \mathbf{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \mathbf{n} \mathbf{3} \mathbf{mix}$$

$$T = 8 \quad [0] \quad 1^{\overline{111}} \overline{3} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.426 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -169, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \ 1^{\overline{11}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 10 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -1 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.427 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 31, T \in [13, 31]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 31, T \in [13, 31]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 \\ 3 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{11}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 20 & 0 & 25 & 9 \\ 5 & 7 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 4 & 5 \\ 9 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 0 & 2 \\ 5 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 31^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 3 & 3 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 31^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

4.428 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{111} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

4.429 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{111} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 4 & 0 & 12 & 2 \\ 11 & 19 & 0 & 21 & 12 & 73 & 15 \\ 3 & 4 & -3 & 4 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 4 & 0 & 9 & 2 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -3 & 4 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 4 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 4 & 0 & 18 & 2 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -3 & 4 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -956, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 4 & 0 & 15 & 2 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -3 & 4 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 4 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -3 & 4 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{113} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 33 & 4 \\ 2 & 4 & 0 & 4 & 0 & 21 & 2 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -3 & 4 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{114} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -2456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 4 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -3 & 4 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

4.430 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 9 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 3 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.431 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.432 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 1 \\ 3 & 4 & -3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 9 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.433 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \text{su2n3mix}$

$$T = 9 \quad [0] \quad 1^{\overline{111}1^1} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 5 & 2 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 4 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}1^1} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 4 & 0 & 8 & 2 \\ 7 & 12 & 0 & 12 & 8 & 25 & 9 \\ 3 & 4 & -3 & 4 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{\overline{11}1^1} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.434 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \text{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}1^1} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}1^1} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 6 & 3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 6 & 9 & 5 \\ 3 & 4 & -3 & 3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}1^1} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}1^1} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}1^1} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.435 NHC(-2)(-3) + f₄ + s₀₈

$$T = 7 \quad [0] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -5 & -3 \\ 5 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \quad 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -4 & -2 \\ 5 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

- [10] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$
- [12] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 0 & -2 & 0 \end{pmatrix}_{b_0}$
- [13] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 4 & 0 & 0 & 2 \\ 2 & 2 & 3 & -3 & 2 & -1 \end{pmatrix}_{b_0}$
- [14] $1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 14, T_{\max} = 32, T \in [14, 32]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 2 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$
- [15] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & 2 & -2 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 15.15 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 6 & 0 & 3 \\ 5 & 6 & 17 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 7 & -2 & -1 \end{pmatrix}_{b_0}$
- [17] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 13 & 12 & 24 & 7 \\ 12 & 10 & 21 & 6 \\ 24 & 21 & 42 & 12 \\ 7 & 6 & 12 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & 1 & 3 \\ 3 & 6 & -3 & 3 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 5 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 1 & 12 & 5 \\ 1 & -2 & 1 & 0 \\ 12 & 1 & 25 & 9 \\ 5 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 3 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^1 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 8 & 1 & 5 \\ 8 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 5 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 4^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 99, \Delta = -136, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^1 221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 4 & 4 \\ 2 & 2 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^1 231^1 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 1 & 5 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 99, \Delta = -136, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -4 & -2 \\ 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [29] \ 1^1 4^1 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -3 & -1 \\ 5 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [30] \ 1^{11} 31^1 5^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [31] \ 1^{\overline{1}1} 31^{\overline{1}1} 51321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 2 & 4 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

4.436 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

4.437 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

- [7] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 291$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 331$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 365$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 370$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 540$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{113}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 581$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 709$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{115}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 765$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.3$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -2 & 0 & 15 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 0 & -2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 3 \ 1^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 781, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3 \ 1^{\overline{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114} 3 \ 1^{\overline{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114} 3 \ 1^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [23] \ 1^{11} 3 \ 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3 \ 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -2 & 7 & -3 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3 \ 21^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 \\ 1 & 4 & 0 & 1 & 0 & 6 \\ 1 & 1 & 1 & -3 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 \\ 2 & 6 & 6 & 2 & 0 & 10 \\ 2 & 3 & 2 & -1 & -2 & 6 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 3 \ 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 7 & 2 \\ 5 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

- [27] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$
- [28] $1^{11}321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 \\ 1 & 1 & 1 & -3 & 0 & 1 \\ 1 & 0 & 0 & 0 & -4 & 0 \\ 1 & 3 & 3 & 1 & 0 & 1 \\ 2 & 3 & 2 & -1 & -2 & 3 \end{pmatrix}_{b_0}$$
- [29] $1^{111}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [30] $1^{11}321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 2 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [31] $1^{111}31^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$
- [32] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 \\ 2 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{111}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [34] $1^{14}321^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 20 & 2 \\ 16 & 30 & 0 & 20 & 158 & 24 \\ 3 & 5 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{12}321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 10 & 2 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 5 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{11}321^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -102$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 4 & 1 & 0 & 6 \\ 1 & 4 & 0 & 1 & 0 & 6 \\ 1 & 1 & 1 & -3 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 \\ 3 & 6 & 6 & 2 & 0 & 10 \\ 2 & 3 & 2 & -1 & -2 & 6 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -137, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -190, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 0 & 9 & 3 \\ 1 & 4 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 9 & 3 & 0 & 25 & 9 \\ 2 & 3 & 2 & -1 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11}321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 4 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 12 & 4 & 0 & 46 & 12 \\ 2 & 3 & 2 & -1 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{11}321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -310, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 19 & 2 \\ 11 & 25 & 0 & 19 & 92 & 20 \\ 3 & 5 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -310, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 11 & 2 \\ 7 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{11}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -350, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 15 & 2 \\ 9 & 20 & 0 & 15 & 58 & 16 \\ 3 & 5 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{13}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -390, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 17 & 2 \\ 13 & 25 & 0 & 17 & 104 & 20 \\ 3 & 5 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -435, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{12}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -460, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 14 & 2 \\ 10 & 20 & 0 & 14 & 62 & 16 \\ 3 & 5 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{12}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -740, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 18 & 2 \\ 12 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{14}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -860, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 24 & 2 \\ 18 & 35 & 0 & 24 & 202 & 28 \\ 3 & 5 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{13}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -910, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 21 & 2 \\ 15 & 30 & 0 & 21 & 142 & 24 \\ 3 & 5 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{12}321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -940, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 22 & 2 \\ 14 & 30 & 0 & 22 & 134 & 24 \\ 3 & 5 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{13}321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -1350, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 25 & 2 \\ 17 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{14}321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -1540, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 28 & 2 \\ 20 & 40 & 0 & 28 & 254 & 32 \\ 3 & 5 & -3 & 2 & 32 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \ 1^{11}3221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 30, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 6 & 6 & 0 & 6 & 6 \\ 2 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 5 & 0 & 3 & 5 \\ 3 & 6 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{111}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -81, \ \det = 34, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 22 & 5 \\ 12 & 26 & 1 & 48 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 48 & 0 & 86 & 18 \\ 5 & 10 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{11} 3^{11} 221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 44, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 6 & 6 & 0 & 11 & 6 \\ 2 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 8 & 0 & 15 & 9 \\ 3 & 6 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{11} 3^{11} 221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 49, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 6 & 0 & 6 & 11 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 8 & 3 \\ 5 & 11 & 0 & 8 & 15 & 9 \\ 3 & 6 & -3 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11} 3^{11} 221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 61, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 6 & 0 & 6 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 13 & 3 \\ 7 & 17 & 0 & 13 & 36 & 14 \\ 3 & 6 & -3 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{111} 2^1 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -81, \ \det = 70, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 14 & 5 \\ 12 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 28 & 0 & 26 & 10 \\ 5 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{114} 2^1 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -81, \ \det = 94, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 78 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 78 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -106, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.56 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{12} 3^{11} 221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 6 & 6 & 0 & 22 & 6 \\ 2 & 6 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 22 & 16 & 0 & 66 & 18 \\ 3 & 6 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{12} 3^{11} 221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 106, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 6 & 0 & 6 & 22 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 16 & 3 \\ 10 & 22 & 0 & 16 & 66 & 18 \\ 3 & 6 & -3 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11} 3^{11} 221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 110, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 6 & 6 & 0 & 17 & 6 \\ 2 & 6 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 17 & 13 & 0 & 36 & 14 \\ 3 & 6 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{112} 2^1 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -81, \ \det = 126, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{111}2^321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 131$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 13 & 5 \\ 12 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 28 & 0 & 26 & 10 \\ 5 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{113}2^321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 146$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 56 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{12}3^221^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 154$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 6 & 0 & 6 & 34 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 26 & 3 \\ 14 & 34 & 0 & 26 & 150 & 28 \\ 3 & 6 & -3 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{112}2^321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 199$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 25 & 5 \\ 12 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 56 & 0 & 110 & 20 \\ 5 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{13}3^221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 201$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 6 & 0 & 6 & 33 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 24 & 3 \\ 15 & 33 & 0 & 24 & 151 & 27 \\ 3 & 6 & -3 & 3 & 27 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{13}3^221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 6 & 6 & 0 & 33 & 6 \\ 2 & 6 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 33 & 24 & 0 & 151 & 27 \\ 3 & 6 & 2 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{12}3^221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 214$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 6 & 0 & 6 & 28 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 21 & 3 \\ 12 & 28 & 0 & 21 & 103 & 23 \\ 3 & 6 & -3 & 3 & 23 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{12}3^221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 230$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 6 & 6 & 0 & 28 & 6 \\ 2 & 6 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 28 & 21 & 0 & 103 & 23 \\ 3 & 6 & 2 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{114}2^321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 294$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{112}2^321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 299$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{112}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{112}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 56 & 0 & 110 & 20 \\ 5 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{131}4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 317, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{14}3^{11}221^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 334, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 6 & 0 & 6 & 44 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 32 & 3 \\ 20 & 44 & 0 & 32 & 270 & 36 \\ 3 & 6 & -3 & 3 & 36 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{113}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 395, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 57 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{13}3^{11}221^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 6 & 0 & 6 & 39 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 29 & 3 \\ 17 & 39 & 0 & 29 & 204 & 32 \\ 3 & 6 & -3 & 3 & 32 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{114}2^3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 419, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 79 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 79 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{113}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 58 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{11}4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{113}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 38 & 5 \\ 12 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 84 & 0 & 250 & 30 \\ 5 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{113} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 47 & 5 \\ 12 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 104 & 0 & 390 & 38 \\ 5 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{114} 2^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 586, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 80 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 80 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{114} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 619, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 132 & 0 & 626 & 48 \\ 5 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{113} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 686, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 48 & 5 \\ 12 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 104 & 0 & 390 & 38 \\ 5 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{114} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 719, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 69 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{114} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 132 & 0 & 626 & 48 \\ 5 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{114} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 70 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

4.438 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 9$ [0] $1^{112} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$

$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 0 & 6 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & 0 & -2 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -3 & 7 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$4.439 \quad \text{NHC}(-2)(-3) + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -222, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.440 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

4.441 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -83, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

4.442 NHC(-2)(-3) + f₄ + s₀8 + s_u2→(-3)

$$T = 8 \quad [0] \quad 1^{11}3^{11}1^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}3^{11}1^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 10 & 13 & 0 & 4 & 6 \\ 5 & 13 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 6 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 1^{11}4^{11}13^{11}1^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}23^{11}1^{11}3^{11}1^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -90, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 1 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 \\ 2 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}3^{11}4^{11}1^1 + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 202, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [9] \quad 1^{\overline{11}} 4 13^{\overline{1}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [10] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 5 13 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 26 & 1 & 20 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

4.443 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -2 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 4 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 4 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -2 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.444 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.445 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 171, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 221, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{11}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 233, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.59 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [6] \ 1^{111}3^{11}1^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 94, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.87 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 20 & 5 \\ 12 & 26 & 1 & 44 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 44 & 0 & 70 & 16 \\ 5 & 10 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.446 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

4.447 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -145, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, \ T_{\max} = 98, \ T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 34 & 0 & 57 & 13 \\ 5 & 8 & -3 & 13 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$

$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 6 & 2 & 0 & 9 & 5 \\ 2 & 3 & 2 & -1 & -2 & 5 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & -3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 10 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 \\ 2 & 2 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 0 & 7 & 13 & 7 \\ 3 & 5 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 82, \Delta = -124, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 82, T \in [14, 82]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [15] \ 1^{11} 2 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -90, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 12 & 1 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 12 & 1 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 0 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3 2 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 6 & 0 & 6 & 11 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 8 & 3 \\ 5 & 11 & 0 & 8 & 14 & 8 \\ 3 & 6 & -3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3 2 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

4.448 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 8 & 3 & 0 & 0 & 4 \\ 3 & 5 & 0 & -2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

15.2

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$4.449 \quad \text{NHC}(-2)(-3) + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

$$4.450 \quad \text{NHC}(-2)(-3) + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2$$

$$T = 7 \quad [1] 1^{112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 3 & 4 & -2 & 6 & 0 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[0] 21^1 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 3 & 4 & -2 & 6 & 0 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 97, T \in [15, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 10 & 0 & 22 & 8 \\ 3 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[6] 21^1 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -2 & 6 & 0 \\ 9 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{15}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[13] $131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{16}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.4$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{17}3^11 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[19] $1^1 13^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -2 & 6 & 0 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{113}3^11^4 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 13 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{11}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 14 & 0 & 37 & 11 \\ 3 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{112}3^11^2 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -2 & 6 & 0 \\ 8 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{11}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 19 & 0 & 73 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -2 & 6 & 0 \\ 7 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{11}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 24 & 0 & 121 & 19 \\ 3 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[27] $1^1 13^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 9 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 18 & 0 & 58 & 14 \\ 3 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{113}3^11^3 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 12 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{14}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{113}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 10 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[33] $21^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 6 & 8 & 3 & 4 & 4 \\ 4 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{113}3^11^2 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 11 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[35] $1^113^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -2 & 12 & 0 \\ 12 & 32 & 12 & 94 & 16 \\ 3 & 5 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 23 & 0 & 100 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{13}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 22 & 0 & 85 & 17 \\ 3 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{12}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 28 & 0 & 154 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{114}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -2 & 12 & 0 \\ 16 & 32 & 12 & 94 & 16 \\ 3 & 5 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{114}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -418, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -2 & 12 & 0 \\ 13 & 32 & 12 & 94 & 16 \\ 3 & 5 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[41] $1^1 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & -2 & 15 & 0 \\ 15 & 40 & 15 & 148 & 20 \\ 3 & 5 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{14}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -466, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 26 & 0 & 118 & 20 \\ 3 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{114}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -466, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -2 & 12 & 0 \\ 15 & 32 & 12 & 94 & 16 \\ 3 & 5 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{114}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -2 & 12 & 0 \\ 14 & 32 & 12 & 94 & 16 \\ 3 & 5 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{13}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 27 & 0 & 133 & 21 \\ 3 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[46] $21^{112}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -486, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 3 & 12 & 4 \\ 8 & 9 & 4 & 16 & 5 \\ 3 & 4 & -2 & 6 & 0 \\ 12 & 16 & 6 & 22 & 8 \\ 4 & 5 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{13}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -603, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 32 & 0 & 193 & 25 \\ 3 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{14}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -652, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 31 & 0 & 172 & 24 \\ 3 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{14}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -826, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 36 & 0 & 238 & 28 \\ 3 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -836, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 18 & 4 \\ 8 & 9 & 4 & 24 & 5 \\ 3 & 4 & -2 & 9 & 0 \\ 18 & 24 & 9 & 52 & 12 \\ 4 & 5 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 21^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -1326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 24 & 4 \\ 8 & 9 & 4 & 32 & 5 \\ 3 & 4 & -2 & 12 & 0 \\ 24 & 32 & 12 & 94 & 16 \\ 4 & 5 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 21^{115}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -1956, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 30 & 4 \\ 8 & 9 & 4 & 40 & 5 \\ 3 & 4 & -2 & 15 & 0 \\ 30 & 40 & 15 & 148 & 20 \\ 4 & 5 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [53] \quad 1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 10 & 16 & 26 & 6 \\ 16 & 25 & 40 & 9 \\ 26 & 40 & 63 & 15 \\ 6 & 9 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 2231^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 5 & 13 & 9 & 5 \\ 13 & 30 & 20 & 10 \\ 9 & 20 & 13 & 7 \\ 5 & 10 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[57] \quad 2231^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 10 & 19 & 12 & 6 \\ 19 & 30 & 20 & 10 \\ 12 & 20 & 13 & 7 \\ 6 & 10 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{14}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 18.57 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -3 & 16 & -6 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 18 & 12 & 6 \\ 18 & 30 & 20 & 10 \\ 12 & 20 & 13 & 7 \\ 6 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -2 & 0 \\ 5 & 10 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 5 & 12 & 11 & 5 \\ 12 & 25 & 20 & 9 \\ 11 & 20 & 13 & 7 \\ 5 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{12} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 6 & 4 \\ 2 & 2 & 2 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{11} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{11} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 2 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{12} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 4 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{12} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{11} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -2 & 0 \\ 5 & 9 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 5 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{12} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{11} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 5 & 12 & 19 & 5 \\ 12 & 25 & 40 & 9 \\ 19 & 40 & 63 & 15 \\ 5 & 9 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{11} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

- [72] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$
- [73] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$
- [76] $1^{11}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 4 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$
- [77] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$
- [78] $1^{14}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 0 & 3 & -3 \end{pmatrix}_{b_0}$
- [74] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [75] $1^{11}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 10 & 0 & 33 & 11 \\ 2 & 2 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$
- [79] $1^{16}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.4$
 $T_{\min} = 15.75 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$
- [80] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$
- [81] $1^{15}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.3$
 $T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 20$, $T \in [15, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & 0 & 10 & -2 \end{pmatrix}_{b_0}$

- [82] $1^{14}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$
- [83] $1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [84] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [85] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$
- [86] $1^{13}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$
- [87] $1^{12}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 6 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 0 & 13 & -1 \end{pmatrix}_{b_0}$$
- [88] $1^{17}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 17.37 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & 0 & 14 & -5 \end{pmatrix}_{b_0}$$
- [89] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 33$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [90] $1^{11}31^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [91] $1^{111}31^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 41$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 6 & 0 & 13 & 7 \\ 2 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [92] $1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 47$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 9 & 0 & 12 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 12 & 0 & 14 & 8 \\ 3 & 5 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{111}3131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{111}1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{111}1321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{112}31^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 0 & 58 & 14 \\ 2 & 3 & 2 & -3 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{111}1321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 5 & 12 & 10 & 5 \\ 12 & 25 & 20 & 9 \\ 10 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{113}31^331^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 18 & 3 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 18 & 18 & 0 & 133 & 21 \\ 2 & 3 & 2 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{11}1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{13}1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 20 & 0 & 74 & 16 \\ 3 & 5 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[101] \ 1231^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 9 & 0 & 8 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 8 & 0 & 4 & 4 \\ 3 & 5 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{11}1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{12}1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^1 31^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 3 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^1 31^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 4 & 4 \\ 2 & 3 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[108] \ 1^1 13 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 2 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{112} 3131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{12} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 15 & 0 & 38 & 12 \\ 3 & 5 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{114} 31^4 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 28, T \in [15, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 0 & 24 & 3 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 24 & 0 & 238 & 28 \\ 2 & 3 & 2 & -3 & 28 & -2 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 3 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{112} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{11}3^{13}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 5 & 0 & 10 & 33 & 11 \\ 2 & 2 & -1 & 0 & 11 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{11}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 5 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{11}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 15 & 0 & 32 & 12 \\ 3 & 5 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{11}3^{14}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 25 & 0 & 92 & 20 \\ 3 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{11}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 3 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{11}3^{13}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 58 & 16 \\ 3 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{11}3^{13}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{14}3^{13}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 7 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 0 \end{pmatrix}_{b_0}$$

[123] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{11} 3^{14} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[126] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{11} 3^{12} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{13} 31^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{13} 31^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[129] $1^{11} 31^{13} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 9 & 0 & 25 & 9 \\ 2 & 3 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[130] $1^{13} 21^{11} 3^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 9 & 3 \\ 1 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 3 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & -1 & 9 & -1 \end{pmatrix}_{b_0}$$

[131] $1^{12} 31^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{11} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 9 & 9 & 0 & 28 & 10 \\ 2 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[133] $1^{11} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

- [134] $1^{14}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$
- [135] $1^{13}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [136] $1^{13}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [137] $1^{12}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 5 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [138] $1^{12}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 9 & 3 \\ 1 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 5 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$
- [139] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 151$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [140] $1^{12}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 167$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [141] $1^{12}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 170$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 20 & 0 & 62 & 16 \\ 3 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$
- [145] $1^{12}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [144] $1^{12}321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$
- [143] $1^{12}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{14}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 6 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{14}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 41 & 11 \\ 2 & 2 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{13}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 7 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[150] $1^{12}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 7 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[151] $1^{14}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 5 & 0 & 10 & 33 & 11 \\ 2 & 2 & -1 & 0 & 11 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{11}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{14}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{12}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{13} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 7 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{13} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{13} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 9 & 3 \\ 1 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 4 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[158] $1^1 13 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.20 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 4 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[159] $1^{14} 3 13 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.20 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 46 & 12 \\ 2 & 3 & 2 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

[160] $21^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 13 & 7 \\ 7 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

[161] $1^1 3 1^{12} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{13} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 25 & 0 & 104 & 20 \\ 3 & 5 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{12} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

- [165] $1^{14}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 230$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 30 & 0 & 158 & 24 \\ 3 & 5 & -3 & 24 & 1 \end{pmatrix}_{b_0}$
- [166] $1^{11}3^{12}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 15 & 0 & 85 & 17 \\ 2 & 3 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$
- [167] $1^{12}3^{11}21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 25 & 0 & 94 & 20 \\ 3 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$
- [168] $1^{14}3^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$
- [169] $1^{11}3^{14}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 252$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [170] $1^{13}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 252$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$
- [171] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 252$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$
- [172] $2^12^131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 257$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$
- [173] $1^{11}3^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 261$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 12 & 0 & 49 & 13 \\ 2 & 3 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$
- [174] $1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 267$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 6 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$
- [175] $1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 267$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{12} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{12} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 6 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{12} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{15} 3131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 15 & 0 & 73 & 15 \\ 2 & 3 & 2 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[180] $1^1 1321^{15} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 15 & 3 \\ 1 & 4 & 0 & 1 & 15 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 5 & 15 & 15 & 5 & 73 & 15 \\ 2 & 3 & 2 & -1 & 15 & -1 \end{pmatrix}_{b_0}$$

[181] $1^{12} 321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 30 & 0 & 134 & 24 \\ 3 & 5 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{12} 321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 6 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[183] $1^1 321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 5 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[186] $1^{12} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 8 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{14} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{14} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{13} 31^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[189] $1^{14} 31^3 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 345, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 0 & 21 & 3 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 21 & 21 & 0 & 172 & 24 \\ 2 & 3 & 2 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

[188] $1^{13} 321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 345, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 30 & 0 & 142 & 24 \\ 3 & 5 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{13} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{13} 321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{14} 321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 8 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{12} 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 6 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{14} 321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[195] $1^{11} 31^{12} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

- [200] $1^{13}321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$
- [199] $1^{13}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$
- [198] $1^{13}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 8 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$
- [197] $1^{14}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$
- [196] $1^{12}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 7 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$
- [201] $1^{13}31^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 395$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$
- [202] $1^1321^{116} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 398$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.4$
 $T_{\min} = 13.30 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 1 & 18 & 3 \\ 1 & 4 & 0 & 1 & 18 & 2 \\ 1 & 1 & 1 & -3 & 6 & -1 \\ 6 & 18 & 18 & 6 & 106 & 18 \\ 2 & 3 & 2 & -1 & 18 & -1 \end{pmatrix}_{b_0}$
- [203] $1^{114}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 401$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 15 & 0 & 76 & 16 \\ 2 & 3 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$
- [204] $1^{14}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 420$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 35 & 0 & 202 & 28 \\ 3 & 5 & -3 & 28 & 2 \end{pmatrix}_{b_0}$
- [205] $1^{12}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 427$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 7 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$
- [206] $1^{13}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{13}321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{12}31^{12}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{14}31^{2}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 18 & 3 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 18 & 18 & 0 & 118 & 20 \\ 2 & 3 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

[210] $2^12^131^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 452, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{13}321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 35 & 0 & 188 & 28 \\ 3 & 5 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{12}321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 9 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{14}321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{14}321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{12}321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 8 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{14}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{13} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 512, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 7 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[218] $1^1 13 21^{117} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 1 & 21 & 3 \\ 1 & 4 & 0 & 1 & 21 & 2 \\ 1 & 1 & 1 & -3 & 7 & -1 \\ 7 & 21 & 21 & 7 & 145 & 21 \\ 2 & 3 & 2 & -1 & 21 & -1 \end{pmatrix}_{b_0}$$

[219] $1^1 31^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[220] $1^{13} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 9 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[221] $31^{111} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 8 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{12} 14 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 9 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 590, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 40 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 40 & 0 & 254 & 32 \\ 3 & 5 & -3 & 32 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{14} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[226] $1^{12} 14 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{14}3^{11}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{11}31^{13}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 623, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 627, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 8 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 8 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 9 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{13}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 10 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[233] $2^{11}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 777, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 6 & 2 \\ 2 & 1 & 4 & 1 & 9 & 3 \\ 2 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 6 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{14}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 787, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 9 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[235] $1^{13}31^{15}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 15 & 0 & 148 & 20 \\ 2 & 4 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

[236] $1^{12}31^{13}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{14} 3^{13} 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 812, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 10 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{14} 3^{11} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 827, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 9 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{14} 31^{13} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 836, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{13} 31^{13} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 871, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{11} 31^{14} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 931, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{14} 3^{12} 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 972, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 10 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{14} 3^{13} 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1027, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 11 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[244] $1^1 31^{16} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1096, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.4$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 18 & 0 & 214 & 24 \\ 2 & 4 & 2 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

[245] $1^{12} 31^{14} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[246] $2^1 2^1 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 8 & 2 \\ 2 & 1 & 4 & 1 & 12 & 3 \\ 2 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 8 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

- [247] $1^{13}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1403$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$
- [248] $1^131^{17}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1460$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 42 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 42 & 21 & 0 & 292 & 28 \\ 2 & 4 & 2 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$
- [249] $1^{14}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1480$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$
- [250] $1^{115}2^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 1817$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.3$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 10 & 2 \\ 2 & 1 & 4 & 1 & 15 & 3 \\ 2 & 4 & 0 & 1 & 15 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 10 & 15 & 15 & 5 & 73 & 15 \\ 2 & 3 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$
- [251] $1^{116}2^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 2532$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.4$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 12 & 2 \\ 2 & 1 & 4 & 1 & 18 & 3 \\ 2 & 4 & 0 & 1 & 18 & 2 \\ 1 & 1 & 1 & -3 & 6 & -1 \\ 12 & 18 & 18 & 6 & 106 & 18 \\ 2 & 3 & 2 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$
- [252] $1^{117}2^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 3377$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 14 & 2 \\ 2 & 1 & 4 & 1 & 21 & 3 \\ 2 & 4 & 0 & 1 & 21 & 2 \\ 1 & 1 & 1 & -3 & 7 & -1 \\ 14 & 21 & 21 & 7 & 145 & 21 \\ 2 & 3 & 2 & -1 & 21 & 1 \end{pmatrix}_{b_0}$$
- $T = 9$ [253] $1^{112}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$
- [254] $1^{112}3^1221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 16 & 0 & 130 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$
- [255] $1^{113}3^1221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 9 & 0 & 20 & 10 \\ 2 & 2 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$
- [257] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$
- [256] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 2 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[258] \ 1^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 13 & 0 & 40 & 12 \\ 3 & 4 & -2 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[261] \ 1^{11} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{11} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[262] \ 1^{11} 3221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 & 6 \\ 3 & 6 & 0 & 5 & 2 \\ 2 & 6 & 5 & 3 & 5 \\ 3 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{11} 3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[263] \ 1^{11} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[265] \ 1^{11} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{11} 3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{11} 231^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 5 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 0 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

- [269] $1^{1231}1^{31}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 3 \\ 5 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$
- [268] $1^{12}231^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 60 & 0 & 258 & 30 \\ 3 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$
- [271] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 11 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [270] $1^{11}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [272] $1^{12}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$
- [275] $1^{15}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 15.38 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & -1 \end{pmatrix}_{b_0}$$
- [274] $1^{14}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$
- [273] $1^{12}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [281] $1^{11}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [280] $1^{11}4^{11}1231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$
- [279] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 7 & 3 \\ 4 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 21 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [278] $1^{13}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{17} 4^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 18.37 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -2 & 21 & -4 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{16} 4^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.4$$

$$T_{\min} = 16.75 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -2 & 18 & -2 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{14} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{13} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[284] \ 21^1 4^{11} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{13} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{12} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{11} 3^{11} 221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 6 & 0 & 17 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 17 & 0 & 36 & 14 \\ 3 & 6 & -3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[289] \ 2^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 10 & 24 & 12 & 6 \\ 24 & 49 & 26 & 13 \\ 12 & 26 & 13 & 7 \\ 6 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{13} 3^{11} 221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 6 & 6 & 27 & 6 \\ 3 & 6 & 0 & 19 & 2 \\ 13 & 27 & 19 & 108 & 22 \\ 3 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[294] \ 1^{11} 3^{11} 221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{11} 3 \ 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[291] \ 1^{11} 3 \ 221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 13 & 0 & 101 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[292] \ 1^{14} 2 \ 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[293] \ 1^{11} 3 \ 221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 8 & 0 & 31 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[295] \ 1^{12} 2 \ 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[298] \ 1^{14} 3 \ 221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[299] \ 1^{11} 2 \ 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[296] \ 1^{13} 2 \ 3 \ 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[297] \ 1^{14} 3 \ 221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[302] \ 1^{11} 41^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 & 3 \\ 6 & 13 & 0 & 8 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 8 & 0 & 4 & 4 \\ 3 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

- [300] $1^{11}2^321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [301] $1^{11}4^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [303] $1^{12}31^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$
- [304] $1^{11}3^1221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -57$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 6 & 6 & 11 & 6 \\ 3 & 6 & 0 & 8 & 2 \\ 5 & 11 & 8 & 15 & 9 \\ 3 & 6 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [305] $1^{11}4^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 10 & 0 & 21 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [306] $1^{13}2^321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 5 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$
- [307] $1^{13}2^3321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [309] $1^{13}3^1221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$
- [310] $1^{13}3^1221^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 295 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$
- [308] $1^{12}2^3321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [312] $1^{11}4^14131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{11} 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{11} 1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 3 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{12} 3 221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 14 & 0 & 54 & 16 \\ 2 & 2 & 3 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{12} 3 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 14 & 8 \\ 2 & 2 & 3 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{11} 1^{12} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{11} 3 221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 6 & 6 & 23 & 6 \\ 3 & 6 & 0 & 18 & 2 \\ 9 & 23 & 18 & 67 & 19 \\ 3 & 6 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[321] $1^{11} 231^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 7 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 13 & 7 \\ 3 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{13} 231^{12} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 29 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 75 & 0 & 418 & 38 \\ 3 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{11} 231^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 5 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 15 & 0 & 13 & 7 \\ 3 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{12} 2 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 8 & 0 \\ 5 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{11} 231^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{11} 3^{112} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\ T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 8 & 0 & 31 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{11} 231^{131} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\ T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 1 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 0 & 0 & 13 & 7 \\ 3 & 0 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{11} 3^{11} 221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\ T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 6 & 6 & 17 & 6 \\ 3 & 6 & 0 & 13 & 2 \\ 7 & 17 & 13 & 36 & 14 \\ 3 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{11} 4^{131} 4^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\ T_{\min} = 13.06 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 15 & 0 & 45 & 13 \\ 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[328] \quad 1^{11} 231^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\ T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 7 & 0 & 14 & 8 \\ 2 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[327] \quad 1^{14} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\ T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[329] \quad 1^{14} 2^{11} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\ T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 6 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & -1 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[331] \quad 1^{11} 3^{114} 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\ T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 13 & 0 & 101 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[330] \quad 1^{13} 3^{114} 221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 30 & 3 \\ 2 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 29 & 0 & 465 & 41 \\ 3 & 3 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[335] \quad 1^{11} 231^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\ T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 11 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 30 & 0 & 63 & 15 \\ 3 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[336] \quad 1^{11} 23131^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\ T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 1 & 8 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 0 & 0 & 18 & 8 \\ 3 & 0 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[332] \ 1^{12} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 4 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[333] \ 1^{11} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[334] \ 1^{11} 231^{11} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 13 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 30 & 0 & 63 & 15 \\ 3 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[338] \ 1^{12} 3^2 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[339] \ 1^{12} 3^2 221^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 16 & 0 & 130 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[337] \ 1^{12} 3^2 221^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 10 & 0 & 30 & 12 \\ 2 & 2 & 3 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[341] \ 1^{12} 4^1 131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -107, \ \det = -107, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[340] \ 1^{13} 3^2 221^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -107, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 19 & 0 & 165 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[342] \ 1^{13} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 5 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[343] \ 1^{12} 3^2 221^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 6 & 0 & 34 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 34 & 0 & 150 & 28 \\ 3 & 6 & -3 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[344] \ 1^{13} 4^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -107, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[345] \ 1^{12} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[346] \ 1^{12} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[347] \ 1^{12} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[348] \ 1^{12} 3^{221} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 6 & 6 & 22 & 6 \\ 3 & 6 & 0 & 16 & 2 \\ 10 & 22 & 16 & 66 & 18 \\ 3 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[349] \ 1^{11} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[350] \ 1^1 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[351] \ 1^{11} 3^{14} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 21 & 0 & 101 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[352] \ 1^{12} 3^{131} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 11 & 0 & 21 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[353] \ 1^{11} 3^{221} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 8 & 0 & 31 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[354] \ 1^{13} 2^{31} 1^{14} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 36 & 3 \\ 5 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 105 & 0 & 778 & 52 \\ 3 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

- [355] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 6 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 13 & 7 \\ 3 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [356] $1^{13} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 9 & 0 \\ 6 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [357] $1^{13} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [358] $1^{14} 3^{13} 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 21 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$
- [359] $1^{14} 3^{13} 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 6 & 6 & 38 & 6 \\ 3 & 6 & 0 & 27 & 2 \\ 18 & 38 & 27 & 211 & 31 \\ 3 & 6 & 2 & 31 & 3 \end{pmatrix}_{b_0}$$
- [360] $1^{14} 231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 25 & 0 & 188 & 28 \\ 2 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$
- [361] $1^{14} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 10 & 0 \\ 7 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$
- [368] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [362] $1^{11} 231^{11} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 12 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 30 & 0 & 63 & 15 \\ 3 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$
- [363] $1^1 231^{12} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 10 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [364] $1^{12} 231^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 30 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$
- [365] $1^1 231^{12} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 14 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{13} 231^{13} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 32 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 90 & 0 & 583 & 45 \\ 3 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[367] \ 1^{12} \overset{1}{4} \overset{31}{4} \overset{1}{1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 20 & 0 & 90 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{13} \overset{1}{4} \overset{131}{1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[370] \ 1^{12} \overset{1}{2} \overset{3}{3} \overset{21}{1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[371] \ 1^{14} \overset{1}{3} \overset{221}{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 31 & 3 \\ 2 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 32 & 0 & 526 & 44 \\ 3 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[372] \ 1^{12} \overset{1}{2} \overset{3}{3} \overset{21}{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 5 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[374] \ 1^{13} \overset{1}{3} \overset{221}{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[373] \ 1^{13} \overset{1}{2} \overset{3}{3} \overset{21}{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[375] \ 21^{11} \overset{1}{41} \overset{11}{31} \overset{1}{1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 3 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 4 & 8 & 0 & 10 & 6 \\ 2 & 2 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[376] \ 1^{13} \overset{1}{3} \overset{221}{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 19 & 0 & 104 & 22 \\ 2 & 2 & 3 & -3 & 22 & 2 \end{pmatrix}_{b_0}$$

- [378] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 18 & 0 & 70 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [377] $1^{12} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 38 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [379] $1^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [380] $1^{13} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [381] $1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$
- [384] $1^{13} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 15 & 0 & 40 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [382] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 15 & 0 & 45 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [383] $1^{12} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 17 & 0 & 61 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [385] $1^{13} 3^{11} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 6 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [386] $1^{13} 3^{11} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 11 & 0 & 44 & 14 \\ 2 & 2 & 3 & -3 & 14 & 2 \end{pmatrix}_{b_0}$$
- [387] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[388] $1^{14} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[389] $1^{14} 4^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{12} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{12} 3^{13} 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 16 & 0 & 130 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[391] $1^{14} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{12} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 8 & 0 \\ 6 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{12} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 5 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{11} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 6 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{14} 3^{11} 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[397] $1^{12} 3^{11} 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 6 & 6 & 28 & 6 \\ 3 & 6 & 0 & 21 & 2 \\ 12 & 28 & 21 & 103 & 23 \\ 3 & 6 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

- [398] $1^{114}231^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -209$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 2 & 5 & 0 & 19 & 4 \\ 1 & 5 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 19 & 19 & 0 & 92 & 20 \\ 2 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$
- [399] $1^{13}3221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -211$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 295 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$
- [400] $1^{13}3221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -211$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 15 & 0 & 70 & 18 \\ 2 & 2 & 3 & -3 & 18 & 2 \end{pmatrix}_{b_0}$$
- [401] $1^{12}231^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 14 & 0 & 62 & 16 \\ 2 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$
- [402] $1^{12}231^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 60 & 0 & 258 & 30 \\ 3 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$
- [403] $1^{11}231^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 20 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$
- [404] $1^{13}3221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -225$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 6 & 6 & 33 & 6 \\ 3 & 6 & 0 & 24 & 2 \\ 15 & 33 & 24 & 151 & 27 \\ 3 & 6 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$
- [405] $1^{14}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$
- [406] $1^{14}1^{11}231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 5 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 12 & 0 & 8 & 6 \\ 3 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [407] $1^{14}3221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$
- [408] $1^{13}3221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -233$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 19 & 0 & 165 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$
- [409] $1^{14}41^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -235$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 19 & 0 & 65 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{11}231^{12}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 10 & 0 & 38 & 12 \\ 2 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{11}3^{13}221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{14}3^{11}221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 12 & 0 & 62 & 16 \\ 2 & 2 & 3 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{14}2^3 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 8 & 15 & 0 & 13 & 97 & 21 \\ 2 & 3 & -1 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{13}2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{12}4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 16 & 0 & 45 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[416] \ 1^1 231^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[420] \ 1^{13}2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 12 & 0 \\ 8 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[417] \ 1^{12}2^3 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 6 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[418] \ 1^{12}2^3 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 11 & 0 \\ 8 & 0 & 9 & 11 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

- [419] $1^{13}2^13^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 6 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [421] $1^{14}2^13^121^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [422] $1^{13}3^1221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -263$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$
- [423] $1^{14}2^13^121^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -263$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & -1 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$
- [424] $1^{14}2^13^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 7 & 0 \\ 7 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [425] $1^{14}231^{12}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 13 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [426] $1^{14}231^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 11 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [427] $1^{13}2^13^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -266$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [428] $1^{13}2^13^121^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -281$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [429] $1^{12}3^1221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -282$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 16 & 0 & 130 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$
- [430] $1^{11}231^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

- [431] $1^1 231^{231} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$
- [432] $1^{12} 231^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 26 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 75 & 0 & 393 & 37 \\ 3 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$
- [433] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 18 & 0 & 61 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [434] $1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 9 & 0 \\ 7 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [435] $1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 6 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [436] $1^{12} 3^1 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$
- [437] $1^{11} 41^{11} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 8 & 3 \\ 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 20 & 0 & 26 & 10 \\ 3 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [438] $1^1 231^{12} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 12 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [439] $1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 8 & 0 \\ 7 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [440] $1^{12} 4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 19 & 0 & 68 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [441] $1^{14} 3^1 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 16 & 0 & 90 & 20 \\ 2 & 2 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

[442] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 20 & 0 & 70 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[443] $1^{12} 2 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[447] $1^{12} 231^{12} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 23 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 60 & 0 & 258 & 30 \\ 3 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[444] $1^{13} 2 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 7 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[445] $1^{12} 2 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 10 & 0 \\ 8 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[446] $1^{12} 2 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 13 & 0 \\ 9 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

[448] $1^{13} 4 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -347, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 21 & 0 & 88 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[449] $1^{12} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[450] $1^{13} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[451] $1^{14} 3 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 24 & 0 & 295 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

- [452] $1^{13} 3^{12} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 19 & 0 & 165 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$
- [455] $1^{11} 3^{14} 221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 20 & 0 & 126 & 24 \\ 2 & 2 & 3 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$
- [453] $1^{12} 3^{14} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$
- [454] $1^{14} 3^{14} 221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 32 & 3 \\ 2 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 32 & 0 & 526 & 44 \\ 3 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$
- [458] $1^{12} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 11 & 0 \\ 9 & 0 & 9 & 11 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [457] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 24 & 0 & 121 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [456] $1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 7 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [459] $1^{11} 231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 13 & 0 & 74 & 16 \\ 2 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$
- [461] $1^{13} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 7 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [460] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 22 & 0 & 97 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$
- [462] $1^{14} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 9 & 15 & 0 & 13 & 97 & 21 \\ 2 & 3 & -1 & 0 & 21 & 2 \end{pmatrix}_{b_0} \\
[464] & 1^{1231}1431^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 20 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0} \\
[463] & 1^{12231}1231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 24 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 60 & 0 & 258 & 30 \\ 3 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0} \\
[465] & 1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7 \\
& T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 25 & 0 & 132 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0} \\
[466] & 1^{14}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 22 & 0 & 90 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0} \\
[467] & 1^{14}3221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 26 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[470] & 1^{11231}1231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7 \\
& T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 19 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0} \\
[469] & 1^{11231}1431^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 25 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0} \\
[468] & 1^{11231}1231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7 \\
& T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 17 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0} \\
[471] & 1^{142}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & -1 & 0 & 18 & 2 \end{pmatrix}_{b_0} \\
[472] & 1^{142}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 8 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0} \\
[473] & 1^{122}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 8 & 0 \\ 8 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[474] $31^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[475] $1^{13}231^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 21 & 0 & 142 & 24 \\ 2 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[476] $1^{13}2321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 12 & 0 \\ 9 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[477] $1^{13}231^{13}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 33 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 90 & 0 & 583 & 45 \\ 3 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[478] $1^{13}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[479] $1^{14}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 24 & 0 & 101 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[480] $1^{14}41^{12}231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 10 & 3 \\ 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 24 & 0 & 38 & 12 \\ 3 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[481] $1^{14}2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 7 & 0 \\ 8 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[482] $1^{13}2321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 9 & 0 \\ 8 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[483] $1^{11}231^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 18 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[484] $1^{13}231^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[485] $1^{13} 3^{14} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -461, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 19 & 0 & 165 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[486] $1^{14} 3^{13} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -462, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[487] $1^{13} 4^{11} 1^{31} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 23 & 0 & 97 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[488] $1^{12} 231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 17 & 0 & 104 & 20 \\ 2 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[489] $1^{14} 3^{13} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -502, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[490] $1^{13} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 8 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[491] $1^{11} 231^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[492] $1^{14} 4^{11} 1^{31} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 25 & 0 & 121 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[493] $1^{11} 231^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 16 & 0 & 122 & 20 \\ 2 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[494] $1^{14} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 10 & 0 \\ 9 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[495] $1^{14} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 13 & 0 \\ 10 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

[499] $1^1 231^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 17 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[500] $1^{12} 231^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 27 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 75 & 0 & 393 & 37 \\ 3 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[497] $1^{13} 231^{14} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 37 & 3 \\ 5 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 105 & 0 & 778 & 52 \\ 3 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[496] $1^1 231^{15} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 25 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 75 & 0 & 373 & 35 \\ 3 & 7 & -1 & 35 & 2 \end{pmatrix}_{b_0}$$

[498] $1^1 231^{13} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 19 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[502] $1^{14} 3 221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 27 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[501] $1^{14} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[504] $1^{13} 2 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 9 & 0 \\ 9 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[505] $1^{13} 2 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 12 & 0 \\ 10 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[503] $1^{12} 231^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 31 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[506] $1^{13} 31^3 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 26 & 0 & 130 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[507] $1^{14} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 28 & 0 & 158 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$

[508] $1^{14} 231^{13} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -604$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 18 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$

[509] $1^{14} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -608$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 7 & 0 \\ 9 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$

[510] $1^{13} 231^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -624$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 34 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 90 & 0 & 583 & 45 \\ 3 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$

[511] $1^{14} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -635$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 27 & 0 & 132 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$

[512] $1^{14} 3^{22} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -638$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$

[513] $1^{13} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -651$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 29 & 0 & 169 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$

[514] $1^{14} 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -664$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$

[515] $1^{11} 41^{12} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -688$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 6 & 1 & 13 & 3 \\ 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 32 & 0 & 68 & 16 \\ 3 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$

[516] $1^{14} 231^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -696$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 28 & 0 & 254 & 32 \\ 2 & 4 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$

[517] $1^{11} 231^{13} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -704$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$

[518] $1^{14} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -728$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 10 & 0 \\ 10 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{11}231^{14}31^{11} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -764, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 26 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[520] $1^{12}231^{13}31^{13} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -764, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 28 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 75 & 0 & 393 & 37 \\ 3 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[521] $1^{14}2^3 3 21^{13} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 13 & 0 \\ 11 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

[522] $1^{12}231^{14}31^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 20 & 0 & 158 & 24 \\ 2 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[523] $1^{14}1^{13}231^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -788, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 15 & 3 \\ 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 36 & 0 & 88 & 18 \\ 3 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[524] $1^{14}4^1 1^{11} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 30 & 0 & 169 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[525] $1^{11}231^{13}31^{13} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 23 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[526] $1^{13}231^{14}31^1 + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -836, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 24 & 0 & 202 & 28 \\ 2 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[528] $1^1 231^{14}31^{12} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[527] $1^{11}231^{13}31^{14} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 24 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[529] $1^{12}231^{13}31^{14} + [\text{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 29 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 75 & 0 & 393 & 37 \\ 3 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

- [530] $1^{14} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -915$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 33 & 0 & 212 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$
- [531] $1^{13} 231^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -924$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 38 & 3 \\ 5 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 105 & 0 & 778 & 52 \\ 3 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$
- [532] $1^{12} 231^{14} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -944$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 32 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$
- [533] $1^{12} 31^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -984$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 23 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$
- [534] $1^{12} 31^{14} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1024$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 24 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$
- [535] $1^{11} 231^{14} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1044$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 27 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$
- [536] $1^{13} 231^{14} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 39 & 3 \\ 5 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 105 & 0 & 778 & 52 \\ 3 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$
- [537] $1^{11} 41^{13} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -1218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 6 & 1 & 18 & 3 \\ 6 & 13 & 0 & 44 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 44 & 0 & 130 & 22 \\ 3 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$
- [538] $1^{12} 231^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 33 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$
- [539] $1^{11} 231^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 28 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$
- [540] $1^{14} 41^{14} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -1278$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 6 & 1 & 20 & 3 \\ 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 48 & 0 & 158 & 24 \\ 3 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$
- [541] $1^{11} 231^{14} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1364$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 29 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$
- [542] $1^{12} 231^{14} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$$\begin{aligned}
& T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 34 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0} \\
[543] & 1^{11} 41^{14} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1888, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 6 & 1 & 23 & 3 \\ 6 & 13 & 0 & 56 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 56 & 0 & 212 & 28 \\ 3 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0} \\
T = 10 & [544] 1^{11} 51^{11} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 3 & 3 & 0 & 4 & 10 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[545] & 1^{12} 5^{14} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 16 & 5 \\ 9 & 13 & 20 & 7 \\ 16 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[546] & 231^{11} 51^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[547] & 1^{11} 5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.26 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[548] & 1^{11} 31^1 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0} \\
[549] & 1^{12} 5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.29 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 8 & 5 \\ 9 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[555] & 21^{11} 5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 5 & 5 \\ 7 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[550] & 1^{11} 51231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[554] & 1^{13} 51231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14.37 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

[557] $1 \overset{1}{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[553] $1 \overset{1}{12} 5 1231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$

[552] $1 \overset{1}{12} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 12 & 5 \\ 9 & 13 & 20 & 7 \\ 12 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[551] $1 \overset{1}{13} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 5 & 9 & 18 & 5 \\ 9 & 13 & 30 & 7 \\ 18 & 30 & 61 & 15 \\ 5 & 7 & 15 & 2 \end{pmatrix}_{b_0}$

[556] $1^1 3221^1 51^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 1 & 4 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 17 & 5 & 0 & 7 \\ 1 & 0 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 7 & 0 & 0 & 0 \end{pmatrix}_{b_0}$

[558] $1 \overset{1}{11} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 7 & 5 \\ 9 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[559] $1 \overset{1}{12} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 20 & 7 \\ 15 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[560] $1 \overset{1}{12} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 13 & 5 \\ 9 & 13 & 20 & 7 \\ 13 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[561] $1 \overset{1}{1} 5^1 1^1 231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 23 & 1 & 5 \\ 23 & 90 & 0 & 18 \\ 1 & 0 & -2 & 0 \\ 5 & 18 & 0 & 3 \end{pmatrix}_{b_0}$

[562] $1 \overset{1}{12} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 14 & 5 \\ 9 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[563] $21 \overset{1}{11} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[564] $1 \overset{1}{13} 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 91, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 19 & 5 \\ 9 & 13 & 30 & 7 \\ 19 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[565] \ 1^{11} 5^1 1^{231} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 5 & 0 & 5 & 18 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[566] \ 1^{11} 5^1 3^{231} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 10 & 11 & 0 & 13 & 126 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[568] \ 1^{13} 5^{14} 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 22 & 5 \\ 9 & 13 & 30 & 7 \\ 22 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[567] \ 1^{11} 5^1 2^{231} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 8 & 0 & 9 & 60 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[569] \ 1^{13} 5^{12} 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 20 & 5 \\ 9 & 13 & 30 & 7 \\ 20 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[570] \ 1^{14} 3^{11} 4^{11} 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 16 & 0 & 90 & 18 \\ 2 & 4 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[571] \ 1^{13} 5^1 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[572] \ 21^{12} 5^1 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 142, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 14 & 5 \\ 10 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[573] \ 1^{11} 3^{11} 4^{11} 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 7 & 0 & 21 & 9 \\ 2 & 4 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[574] \ 1^{12} 5^1 1^{231} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 7 & 0 & 6 & 30 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[575] \ 1^{13} 3^{11} 4^{11} 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 17 & 13 & 0 & 61 & 15 \\ 2 & 4 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[576] \ 1^{12}31^{11}41^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 10 & 0 & 38 & 12 \\ 2 & 4 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[577] \ 21^{13}5^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 232, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 21 & 5 \\ 10 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[578] \ 1^{13}5^{11}1^{23}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 46 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[579] \ 1^{12}5^{12}231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 10 & 0 & 10 & 78 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[580] \ 1^{12}5^{13}1^{32}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 332, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 11 & 13 & 0 & 14 & 150 & 22 \\ 2 & 2 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[581] \ 1^{13}221^{11}51^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 382, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 4 & 2 \\ 4 & 18 & 1 & 5 & 20 & 8 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 5 & 0 & -2 & 5 & 0 \\ 4 & 20 & 0 & 5 & 18 & 8 \\ 2 & 8 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[582] \ 1^{13}5^{12}231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 9 & 12 & 0 & 11 & 100 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[583] \ 1^{13}5^{13}1^{32}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 572, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 15 & 0 \\ 12 & 15 & 0 & 15 & 178 & 24 \\ 2 & 2 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[584] \ 1^{13}221^{12}51^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 772, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 8 & 2 \\ 4 & 18 & 1 & 5 & 40 & 8 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 5 & 0 & -2 & 10 & 0 \\ 8 & 40 & 0 & 10 & 78 & 16 \\ 2 & 8 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[585] \ 1^{13}221^{13}51^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 1422, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 12 & 2 \\ 4 & 18 & 1 & 5 & 60 & 8 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 5 & 0 & -2 & 15 & 0 \\ 12 & 60 & 0 & 15 & 178 & 24 \\ 2 & 8 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [586] \quad 1^{11} 31^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -81, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[587] $1^{11}315^1131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 2 \end{pmatrix} b_0$$

[588] $1^{11}31^15\ 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & 0 & 0 & 2 \end{pmatrix} b_0$$

[589] $1^{12}31^5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -44$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.7$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[590] $1^{11}31^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[591] $1^{14}31^5131^{1+}[\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{s}u_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[592] $1^{14}31^5131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -104, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[593] $1^{12}31^5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -120, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.7$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[594] $1^{13}31^15^{12}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -128$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[595] $1^{12}31^15^131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2}$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -188$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.7$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[596] $1^{13}31^15131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -240$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[597] $1^{13}431^15131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -260$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.85 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [599] $1^{11}6^{11}321^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 5 & 3 & 1 & 5 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[598] $1^{11}6^{11}321^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[601] $1^{11}6^{11}321^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 20$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.60 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[600] $1^{11}6^{11}321^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 20$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.60 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[603] $1^{11}6^{11}321^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 44$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[602] $1^{11}6^{11}321^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 44$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[606] $1^{11}6^{11}321^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.58 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 3 & 5 & 0 & 11 & 7 \\ 2 & 2 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[605] $1^{11}6^{11}321^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.58 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[604] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[607] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 4 & 0 & 3 & 10 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[608] $1^{11} 6^{11} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 5 & 11 & 0 & 43 & 13 \\ 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[609] $1^{11} 6^{11} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 4 & 8 & 0 & 24 & 10 \\ 2 & 2 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[610] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 5 & 6 & 0 & 19 & 9 \\ 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[611] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 140, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 4 & 1 & 3 & 11 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[612] $31^{11} 31^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 3 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[613] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 164, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 4 & 4 & 3 & 26 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[614] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 180, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 4 & 2 & 3 & 14 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[615] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 188, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.68 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 4 & 3 & 3 & 19 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[616] $1^{11} \overset{1}{6} \overset{1}{131} \overset{1}{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 204, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 6 & 9 & 0 & 34 & 12 \\ 2 & 2 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[617] $1^{11} \overset{1}{6} \overset{1}{131} \overset{1}{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 248, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 7 & 12 & 0 & 55 & 15 \\ 2 & 2 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[618] $1^{11} \overset{1}{6} \overset{1}{131} \overset{1}{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 260, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.66 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 8 & 15 & 0 & 82 & 18 \\ 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

4.451 $\text{NHC}(-2)(-3) + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{e}_6$

$T = 8 \quad [0] \ 1^{11} \overset{1}{3} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 0 & 4 & -4 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11} \overset{2}{3} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 0 & 8 & -4 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11} \overset{3}{3} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 0 & 12 & -4 & -2 \end{pmatrix}_{b_0}$$

$T = 9 \quad [5] \ 1^{11} \overset{1}{3} \overset{1}{21} \overset{1}{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 5 & 0 & 3 & -2 & 0 \\ 3 & 5 & -3 & 4 & 0 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{11} \overset{1}{3} \overset{1}{21} \overset{1}{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 2 & 3 \\ 4 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & -2 & 0 \\ 3 & 5 & -3 & 8 & 0 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11} \overset{1}{3} \overset{1}{21} \overset{1}{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 9 & 0 & 8 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & -2 & 0 \\ 3 & 5 & -3 & 4 & 0 & -1 \end{pmatrix}_{b_0}$$

4.452 NHC(-2)(-3) + f₄ + su₂ + f₄

$$T = 8 \quad [0] \ 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 8 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 12 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 16 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{115} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 1192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -2 & 15 & 0 & 0 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 20 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{11} 3^1 2^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 10 & 7 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 2^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 63 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 15 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 2^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 4 & 1 & 21 & 0 & 3 \\ 1 & 4 & 0 & 1 & 21 & 0 & 2 \\ 1 & 1 & 1 & -3 & 7 & 0 & -1 \\ 7 & 21 & 21 & 7 & 145 & 0 & 21 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 21 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 2^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.6$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 4 & 1 & 18 & 0 & 3 \\ 1 & 4 & 0 & 1 & 18 & 0 & 2 \\ 1 & 1 & 1 & -3 & 6 & 0 & -1 \\ 6 & 18 & 18 & 6 & 106 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 18 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 2^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 \\ 1 & 4 & 0 & 1 & 3 & 0 \\ 1 & 1 & 1 & -3 & 1 & 0 \\ 1 & 3 & 3 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & -5 \\ 2 & 3 & 2 & -1 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 3 2 1^{\overline{1} \overline{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 6 & 0 & 3 \\ 1 & 4 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 2 & 6 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 6 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 3 2 1^{\overline{1} \overline{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 1 & 9 & 0 & 3 \\ 1 & 4 & 0 & 1 & 9 & 0 & 2 \\ 1 & 1 & 1 & -3 & 3 & 0 & -1 \\ 3 & 9 & 9 & 3 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 9 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 3 2 1^{\overline{1} \overline{4}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 1 & 12 & 0 & 3 \\ 1 & 4 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & 1 & -3 & 4 & 0 & -1 \\ 4 & 12 & 12 & 4 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 12 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 3 2 1^{\overline{1} \overline{5}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 1 & 15 & 0 & 3 \\ 1 & 4 & 0 & 1 & 15 & 0 & 2 \\ 1 & 1 & 1 & -3 & 5 & 0 & -1 \\ 5 & 15 & 15 & 5 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 15 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{1} \overline{1} \overline{1}} \overline{1}^{\overline{1}} 3 2 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -165, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [15] \ 1^{\overline{1} \overline{1} \overline{1}} \overline{1}^{\overline{1}} 3 2 2 1^{\overline{1} \overline{2} \overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 6 & 0 & 17 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 17 & 0 & 36 & 13 & 14 \\ 2 & 6 & 0 & 13 & 0 & 2 \\ 3 & 6 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

4.453 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{1} \overline{1} \overline{1} \overline{1}} \overline{1}^{\overline{1}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 0 & 4 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{\overline{1}} \overline{1}^{\overline{1} \overline{1} \overline{1}} 3 2 1^{\overline{1} \overline{1} \overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 3 \\ 1 & 1 & 1 & -3 & 1 & 0 & 1 \\ 1 & 3 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 1 & 3 & 0 & 1 \\ 2 & 3 & 2 & -1 & 3 & -3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 3 2 1^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.9$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 6 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & 2 & -1 \\ 1 & 3 & 3 & 1 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 2 & 6 & 0 & 10 & 6 \\ 2 & 3 & 2 & -1 & 3 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.454 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.4$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 3 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.455 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$T = 9$ [0] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 19 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 21 & 12 & 15 \\ 3 & 5 & 0 & 21 & 4 & 4 & 4 \\ 2 & 4 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -3 & 15 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -356$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & -3 & 7 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -396$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 2 & 3 \\ 3 & 4 & 0 & 14 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 15 & 9 & 11 \\ 3 & 5 & 0 & 15 & 4 & 4 & 4 \\ 2 & 4 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & -3 & 11 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{121} 3 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -776$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 2 & 3 \\ 3 & 4 & 0 & 18 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 18 & 12 & 14 \\ 3 & 5 & 0 & 18 & 4 & 4 & 4 \\ 2 & 4 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -3 & 14 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121} 3^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 & 2 & 3 \\ 3 & 4 & 0 & 28 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 30 & 18 & 22 \\ 3 & 5 & 0 & 30 & 4 & 4 & 4 \\ 2 & 4 & 0 & 18 & 4 & 0 & 2 \\ 3 & 4 & -3 & 22 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121} 3^1 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -956, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 2 & 3 \\ 3 & 4 & 0 & 23 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 24 & 15 & 18 \\ 3 & 5 & 0 & 24 & 4 & 4 & 4 \\ 2 & 4 & 0 & 15 & 4 & 0 & 2 \\ 3 & 4 & -3 & 18 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{131} 3^1 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 & 2 & 3 \\ 3 & 4 & 0 & 27 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 27 & 18 & 21 \\ 3 & 5 & 0 & 27 & 4 & 4 & 4 \\ 2 & 4 & 0 & 18 & 4 & 0 & 2 \\ 3 & 4 & -3 & 21 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{131} 3^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 & 2 & 3 \\ 3 & 4 & 0 & 32 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 33 & 21 & 25 \\ 3 & 5 & 0 & 33 & 4 & 4 & 4 \\ 2 & 4 & 0 & 21 & 4 & 0 & 2 \\ 3 & 4 & -3 & 25 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{141} 3^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -2456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 & 2 & 3 \\ 3 & 4 & 0 & 36 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 36 & 24 & 28 \\ 3 & 5 & 0 & 36 & 4 & 4 & 4 \\ 2 & 4 & 0 & 24 & 4 & 0 & 2 \\ 3 & 4 & -3 & 28 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$4.456 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \ 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 9 & 6 & 5 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 5 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.457 NHC(-2)(-3) + f₄ + su₂ + so₈

$$T = 8 \quad [0] \ 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 13 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

- [10] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 365$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 370$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 540$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 581$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 709$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{115} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 765$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -2 & 15 & 0 & 0 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 20 & -2 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 781$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 924$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1044$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [25] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -2$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 18$, $T_{\max} = 21$, $T \in [18, 21]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 7 & -2 & -3 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -2$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 6 & 0 & 3 \\ 1 & 4 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 2 & 6 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 6 & -2 & -3 \end{pmatrix}_{b_0}$$

[23] $1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -2$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 2 & 3 \\ 4 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & 0 & 2 \\ 3 & 5 & -3 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 63 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 9 & 0 & 8 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{14}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 30 & 0 & 158 & 20 & 24 \\ 2 & 5 & 0 & 20 & 0 & 2 \\ 3 & 5 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{12}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 38 & 10 & 12 \\ 2 & 5 & 0 & 10 & 0 & 2 \\ 3 & 5 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{11}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -102$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 1 & 6 & 0 & 3 \\ 1 & 4 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 3 & 6 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 6 & -2 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -117$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 10 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 10 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 7 & 8 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & -3 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 1 & 9 & 0 & 3 \\ 1 & 4 & 0 & 1 & 9 & 0 & 2 \\ 1 & 1 & 1 & -3 & 3 & 0 & -1 \\ 4 & 9 & 9 & 3 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 9 & -2 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{11}321^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 1 & 12 & 0 & 3 \\ 1 & 4 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & 1 & -3 & 4 & 0 & -1 \\ 5 & 12 & 12 & 4 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{11}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 0 & 2 \\ 3 & 5 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 92 & 19 & 20 \\ 2 & 5 & 0 & 19 & 0 & 2 \\ 3 & 5 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11}321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 15 & 16 \\ 2 & 5 & 0 & 15 & 0 & 2 \\ 3 & 5 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{13}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -390, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 25 & 0 & 104 & 17 & 20 \\ 2 & 5 & 0 & 17 & 0 & 2 \\ 3 & 5 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{12}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 62 & 14 & 16 \\ 2 & 5 & 0 & 14 & 0 & 2 \\ 3 & 5 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{12}321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 25 & 0 & 94 & 18 & 20 \\ 2 & 5 & 0 & 18 & 0 & 2 \\ 3 & 5 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{14}3^{11}321^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -860, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 35 & 0 & 202 & 24 & 28 \\ 2 & 5 & 0 & 24 & 0 & 2 \\ 3 & 5 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{13}3^{11}321^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -910, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 30 & 0 & 142 & 21 & 24 \\ 2 & 5 & 0 & 21 & 0 & 2 \\ 3 & 5 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{12}3^{11}321^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -940, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 30 & 0 & 134 & 22 & 24 \\ 2 & 5 & 0 & 22 & 0 & 2 \\ 3 & 5 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{13}3^{11}321^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 35 & 0 & 188 & 25 & 28 \\ 2 & 5 & 0 & 25 & 0 & 2 \\ 3 & 5 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{14}3^{11}321^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 5 & 0 & 40 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 40 & 0 & 254 & 28 & 32 \\ 2 & 5 & 0 & 28 & 0 & 2 \\ 3 & 5 & -3 & 32 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [55] $1^{111}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 22 & 5 \\ 12 & 26 & 1 & 48 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 48 & 0 & 86 & 18 \\ 5 & 10 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{11}3^{11}3221^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 6 & 0 & 17 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 17 & 0 & 36 & 13 & 14 \\ 2 & 6 & 0 & 13 & 1 & 3 \\ 3 & 6 & -3 & 14 & 3 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{11}4^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{121}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{111}2^3 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 13 & 5 \\ 12 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 28 & 0 & 26 & 10 \\ 5 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[60] 1^{131} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 56 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

$$[61] 1^{121} 3 221^{141} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 6 & 0 & 34 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 34 & 0 & 150 & 26 & 28 \\ 2 & 6 & 0 & 26 & 1 & 3 \\ 3 & 6 & -3 & 28 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[62] 1^{121} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 199, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 25 & 5 \\ 12 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 56 & 0 & 110 & 20 \\ 5 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[63] 1^{141} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

$$[64] 1^{121} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 299, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[65] 1^{121} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 56 & 0 & 110 & 20 \\ 5 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[66] 1^{121} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[67] 1^{131} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 317, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[68] 1^{131} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 395, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 57 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

$$[69] 1^{141} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 419, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 79 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 79 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{131} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 58 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{131} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 38 & 5 \\ 12 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 84 & 0 & 250 & 30 \\ 5 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{11} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{131} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 47 & 5 \\ 12 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 104 & 0 & 390 & 38 \\ 5 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{141} 2^3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 586, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 80 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 80 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{141} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 619, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 132 & 0 & 626 & 48 \\ 5 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{131} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 686, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 48 & 5 \\ 12 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 104 & 0 & 390 & 38 \\ 5 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{141} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 719, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 69 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{141} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 132 & 0 & 626 & 48 \\ 5 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{141} 2^3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 70 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

4.458 NHC(-2)(-3) + $\mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$

$T = 9$ $[0] 1^{112} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & 0 & 0 & -2 & 8 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -3 & 7 & 27 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 4 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{112} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

4.459 NHC(-2)(-3) + f₄ + su₂ + so₈ + su₂ → (-3)

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 3 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 4 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.460 NHC(-2)(-3) + f₄ + su₂ + so₈ + su_{2n3mix}

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

4.461 NHC(-2)(-3) + f₄ + su₂ + so₈ + su₃

$$T = 9 \quad [0] \ 1^{111} 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{s}\textcolor{blue}{u}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{s}\textcolor{blue}{u}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

4.462 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{111} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 10 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 0 & 22 & 12 & 8 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -3 & 8 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 \\ 5 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -2 & 3 & 3 & 0 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 21^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -2 & 3 & 3 & 0 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{133} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 15 & 3 \\ 3 & 4 & 0 & 17 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 81 & 13 \\ 15 & 27 & 0 & 81 & 133 & 21 \\ 3 & 4 & -3 & 13 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{121} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{124} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 8 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 17 & 31 & 0 & 42 & 172 & 24 \\ 3 & 4 & -3 & 6 & 24 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{131} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 10 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 0 & 22 & 30 & 8 \\ 8 & 14 & 0 & 30 & 37 & 11 \\ 3 & 4 & -3 & 8 & 11 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{133} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -3 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{112} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

[17] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 13.19 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 24 & 0 & 42 & 121 & 19 \\ 3 & 4 & -3 & 7 & 19 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -3 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{133}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -3 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{133}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -3 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{124}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 8 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -3 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{134}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 12 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 36 & 9 \\ 8 & 16 & 0 & 36 & 46 & 12 \\ 3 & 4 & -3 & 9 & 12 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{115}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 10 & 20 & 0 & 15 & 73 & 15 \\ 3 & 4 & -3 & 3 & 15 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{125}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 10 & 20 & 0 & 30 & 73 & 15 \\ 3 & 4 & -3 & 6 & 15 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{144}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 16 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 48 & 12 \\ 8 & 16 & 0 & 48 & 46 & 12 \\ 3 & 4 & -3 & 12 & 12 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{123}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -3 & 6 & 17 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{112} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & -3 & 11 & 10 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{135} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 12 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 45 & 9 \\ 10 & 20 & 0 & 45 & 73 & 15 \\ 3 & 4 & -3 & 9 & 15 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{116} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 12 & 24 & 0 & 18 & 106 & 18 \\ 3 & 4 & -3 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.66 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{131} 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{126} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.9$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 12 & 24 & 0 & 36 & 106 & 18 \\ 3 & 4 & -3 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{145} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 16 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 60 & 12 \\ 10 & 20 & 0 & 60 & 73 & 15 \\ 3 & 4 & -3 & 12 & 15 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{136} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 12 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 54 & 9 \\ 12 & 24 & 0 & 54 & 106 & 18 \\ 3 & 4 & -3 & 9 & 18 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{117} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 14 & 28 & 0 & 21 & 145 & 21 \\ 3 & 4 & -3 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{155} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 20 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 75 & 15 \\ 10 & 20 & 0 & 75 & 73 & 15 \\ 3 & 4 & -3 & 15 & 15 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{1111} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 \\ 4 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{146} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.28 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 16 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 72 & 12 \\ 12 & 24 & 0 & 72 & 106 & 18 \\ 3 & 4 & -3 & 12 & 18 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{127} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 14 & 28 & 0 & 42 & 145 & 21 \\ 3 & 4 & -3 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{113} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{137} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 12 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 63 & 9 \\ 14 & 28 & 0 & 63 & 145 & 21 \\ 3 & 4 & -3 & 9 & 21 & -1 \end{pmatrix}_{b_0}$$

[53] $1^{112} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -3 & 3 & 22 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{156} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 20 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 90 & 15 \\ 12 & 24 & 0 & 90 & 106 & 18 \\ 3 & 4 & -3 & 15 & 18 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$

[57] $1^{147}3^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 16 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 84 & 12 \\ 14 & 28 & 0 & 84 & 145 & 21 \\ 3 & 4 & -3 & 12 & 21 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{166}3^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 24 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 24 & 0 & 106 & 108 & 18 \\ 12 & 24 & 0 & 108 & 106 & 18 \\ 3 & 4 & -3 & 18 & 18 & -1 \end{pmatrix}_{b_0}$$

[59] $1^{157}3^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 20 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 105 & 15 \\ 14 & 28 & 0 & 105 & 145 & 21 \\ 3 & 4 & -3 & 15 & 21 & -1 \end{pmatrix}_{b_0}$$

[60] $1^{134}3^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & -3 & 9 & 16 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{132}3^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 54 & 13 \\ 10 & 18 & 0 & 54 & 58 & 14 \\ 3 & 4 & -3 & 13 & 14 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{167}3^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 24 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 24 & 0 & 106 & 126 & 18 \\ 14 & 28 & 0 & 126 & 145 & 21 \\ 3 & 4 & -3 & 18 & 21 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{11}3^11^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 11 & 19 & 0 & 21 & 73 & 15 \\ 3 & 4 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{111}3^11^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 19 & 0 & 33 & 73 & 15 \\ 3 & 4 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{112}3^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{11}3^11^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 14 & 24 & 0 & 27 & 121 & 19 \\ 3 & 4 & -3 & 4 & 19 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{143} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & -3 & 16 & 17 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{177} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 28 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 28 & 0 & 145 & 147 & 21 \\ 14 & 28 & 0 & 147 & 145 & 21 \\ 3 & 4 & -3 & 21 & 21 & -1 \end{pmatrix}_{b_0}$$

[69] $1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{122} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 54 & 10 \\ 13 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -3 & 10 & 18 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{114} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{113} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 57 & 11 \\ 12 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -3 & 11 & 17 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{111} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 317, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 24 & 7 \\ 8 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -3 & 7 & 16 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{12}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{114}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -3 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{14}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -3 & 4 & 20 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{13}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{124}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 380, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -3 & 6 & 20 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 39 & 11 \\ 8 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -3 & 11 & 11 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{12} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 13 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112} 3^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 48 & 11 \\ 10 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -3 & 11 & 14 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{122} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{123} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 543, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 18 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 75 & 10 \\ 18 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -3 & 10 & 25 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{122} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{113} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{112} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{123} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -3 & 10 & 13 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{124}3^11^{111} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -3 & 10 & 16 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{123}3^11^{114} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 16 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -3 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{13}3^11^{113} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 15 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -3 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{113}3^11^{14} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{113}3^11^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 653, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

[102] $21^{1111}3^11 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 6 & 4 \\ 8 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -2 & 3 & 3 & 0 \\ 6 & 8 & 3 & 4 & 6 & 4 \\ 6 & 8 & 3 & 6 & 4 & 4 \\ 4 & 5 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{112}3^11^{113} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 695, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 13 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{133}3^11^{112} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -3 & 13 & 17 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{112}3^11^{114} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 16 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -3 & 7 & 22 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{14}3^11^{113} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 17 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -3 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{134}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -3 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{134}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 759, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 17 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 93 & 13 \\ 17 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -3 & 13 & 24 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{124}31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 764, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 84 & 14 \\ 14 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -3 & 14 & 20 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{113}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{114}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 794, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{144}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 824, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -3 & 16 & 20 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{134}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 831, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -3 & 13 & 16 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{123}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 863, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 15 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 15 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -3 & 10 & 21 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{122}31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 60 & 14 \\ 10 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -3 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{13}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 879, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 18 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -3 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{123}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 879, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -3 & 10 & 17 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{114}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 890, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{123}3^11^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 954, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 72 & 14 \\ 12 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -3 & 14 & 17 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{144}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 988, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -3 & 16 & 16 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{114}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{113}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1064, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 15 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -3 & 7 & 21 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{143}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1080, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 20 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -3 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{114}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1094, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{134}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -3 & 13 & 20 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{124}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -3 & 10 & 20 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{113} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1269$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 18 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -3 & 7 & 25 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{124} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 20 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 20 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -3 & 10 & 28 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{133} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1320$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 87 & 17 \\ 12 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -3 & 17 & 17 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{114} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1367$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 17 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -3 & 7 & 24 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{124} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1376$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 17 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 17 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -3 & 10 & 24 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{134} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1410$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 102 & 17 \\ 14 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -3 & 17 & 20 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{114} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1750$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 20 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -3 & 7 & 28 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{144} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1780$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 120 & 20 \\ 14 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -3 & 20 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [135] $1^{111} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 & 0 \\ 2 & 1 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[136] $1^1 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 0 & 0 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 1 & 0 \\ 2 & 3 & 0 & 4 & 5 & 3 & 5 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & 0 & 4 & 8 & 1 \end{pmatrix} b_C$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 3 & 5 & 0 & 5 & 10 & 13 & 7 \\ 2 & 2 & -1 & 0 & 4 & 7 & 1 \end{pmatrix} b_C$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 7 & 3 \\ 3 & 4 & 7 & 0 & 7 & 16 & 8 \\ 2 & 2 & 2 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 8 & 9 & 0 & 9 & 40 & 12 \\ 2 & 2 & 2 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{1^{111}} 3^{21^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{1^{12}31^{11}31^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 4 & 2 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{1^4 31^{11} 31^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 6 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 30 & 4 \\ 10 & 30 & 21 & 0 & 30 & 172 & 24 \\ 2 & 4 & 2 & -3 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{1^4 3^{11} 3^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 41 & 11 \\ 2 & 2 & -1 & 0 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{1^4 3^{11} 3^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[151] $1^{1^4 3^{11} 3^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{1^{12} 3^{21} 3^{21}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[152] $1^{1^{11} 3^{21} 3^{21}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{1^{11} 3^{21} 3^{21}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 4 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

- [154] $1^{112} 3^{11} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 8 & 6 & 26 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [156] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & -3 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [157] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 2 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [158] $1^{111} 3^{11} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 5 & 0 & 10 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 30 & 8 \\ 9 & 20 & 0 & 30 & 58 & 16 \\ 3 & 5 & -3 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$
- [160] $1^{113} 3^{11} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [161] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [159] $1^{111} 3^{112} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [162] $1^{122} 3^{111} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 3 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 5 & 10 & 13 & 7 \\ 2 & 2 & -1 & 0 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$
- [164] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -99$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$
- [163] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -99$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 5 & 5 \\ 2 & 2 & -1 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{123} 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & 0 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{123} 3^{111} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{12} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{12} 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 8 & 3 \\ 4 & 6 & 8 & 0 & 8 & 26 & 10 \\ 2 & 2 & 2 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -109$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{112} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -109$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{124} 3^{112} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -114$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & 0 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{123} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -114$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{122} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -114$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{13} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 6 & 0 & 25 & 9 \\ 2 & 2 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{111} 3^{11} 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 5 & 0 & 15 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 56 & 12 \\ 11 & 25 & 0 & 56 & 92 & 20 \\ 3 & 5 & -3 & 12 & 20 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{11} 3^{11} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 11 & 4 \\ 7 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{11} 3^{11} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 19 & 4 \\ 11 & 25 & 0 & 19 & 92 & 20 \\ 3 & 5 & -3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{112} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 4 & 6 & 8 & 0 & 6 & 26 & 10 \\ 2 & 2 & 2 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{12} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 5 & 0 & 13 & 7 \\ 2 & 2 & -1 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{11} 3^{11} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 15 & 4 \\ 9 & 20 & 0 & 15 & 58 & 16 \\ 3 & 5 & -3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{14} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 13 & 3 \\ 7 & 11 & 13 & 0 & 13 & 81 & 17 \\ 2 & 2 & 2 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{12} 3^{11} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 11 & 3 \\ 5 & 7 & 11 & 0 & 11 & 45 & 13 \\ 2 & 2 & 2 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{112} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 8 & 6 & 26 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{112} 3^{13} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -149$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 8 & 6 & 26 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{113} 3^{12} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -149$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 12 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 12 & 9 & 61 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{111} 3^{21} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -154$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 11 & 8 \\ 2 & 3 & 4 & 0 & 11 & 5 & 5 \\ 2 & 2 & 2 & -3 & 8 & 5 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113} 3^{21} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 13 & 5 \\ 4 & 7 & 6 & 0 & 13 & 25 & 9 \\ 2 & 2 & 2 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{112} 3^{21} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 3 & 5 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{124} 3^{21} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 40 & 10 \\ 6 & 10 & 10 & 0 & 40 & 58 & 14 \\ 2 & 2 & 2 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

- [192] $1^{113} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [191] $1^{112} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [194] $1^{123} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & 0 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$
- [193] $1^{124} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & 0 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$
- [195] $1^{111} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 5 & 0 & 10 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 16 & 8 \\ 5 & 10 & 0 & 16 & 14 & 8 \\ 3 & 5 & -3 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$
- [199] $1^{111} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [196] $1^{111} 3 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 18 & 8 \\ 3 & 4 & 7 & 0 & 18 & 16 & 8 \\ 2 & 2 & 2 & -3 & 8 & 8 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{111} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{13} 3 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 12 & 3 \\ 6 & 9 & 12 & 0 & 12 & 61 & 15 \\ 2 & 2 & 2 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{111} 3 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -210$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 5 & 0 & 10 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 23 & 8 \\ 7 & 15 & 0 & 23 & 32 & 12 \\ 3 & 5 & -3 & 8 & 12 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{112} 3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -214$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 7 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 22 & 8 \\ 4 & 6 & 8 & 0 & 22 & 26 & 10 \\ 2 & 2 & 2 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{112} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -214$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{112} 3 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -214$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 22 & 5 \\ 6 & 8 & 14 & 0 & 22 & 70 & 16 \\ 2 & 2 & 2 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111} 31^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 5 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{114} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{112} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{112} 3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 7 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{113}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -234$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 0 & 1 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 4 & 7 & 6 & 0 & 7 & 25 & 9 \\ 2 & 2 & 2 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{113}321^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -234$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 6 & 9 & 12 & 0 & 9 & 61 & 15 \\ 2 & 2 & 2 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{112}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 10 & 5 \\ 3 & 5 & 5 & 0 & 10 & 13 & 7 \\ 2 & 2 & 2 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{113}321^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 12 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 12 & 9 & 61 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{112}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{114}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 6 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{112}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{114}31^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -250$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 8 & 2 \\ 2 & 4 & 4 & 0 & 9 & 24 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 42 & 7 \\ 8 & 24 & 18 & 0 & 42 & 118 & 20 \\ 2 & 4 & 2 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{123}321^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 10 & 2 \\ 1 & 2 & 0 & 0 & 14 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 80 & 16 \\ 7 & 10 & 15 & 0 & 80 & 88 & 18 \\ 2 & 2 & 2 & -3 & 16 & 18 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{112} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{12} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 18 & 4 \\ 12 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{113} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{124} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & 0 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 31^{12} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 7 & 21 & 15 & 0 & 15 & 85 & 17 \\ 2 & 4 & 2 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{124} 3^{112} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{113} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{113} 3^{114} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 12 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 10 & 9 & 0 & 12 & 9 & 61 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{114} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{124} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{123} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{13} 3 \ 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 15 & 3 \\ 7 & 10 & 15 & 0 & 15 & 88 & 18 \\ 2 & 2 & 2 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{131} 31^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{111} 3 \ 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 5 & 0 & 15 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 45 & 12 \\ 9 & 20 & 0 & 45 & 58 & 16 \\ 3 & 5 & -3 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{111} 3 \ 21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 5 & 0 & 25 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 92 & 75 & 20 \\ 9 & 20 & 0 & 75 & 58 & 16 \\ 3 & 5 & -3 & 20 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{134} 3 \ 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 7 & 0 \\ 4 & 6 & 0 & 3 & 16 & 27 & 6 \\ 6 & 9 & 0 & 7 & 27 & 41 & 11 \\ 2 & 2 & -1 & 0 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{113} 3 \ 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{11} 31^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -301, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{11} 3^{11} 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 8 & 11 & 0 & 13 & 11 & 81 & 17 \\ 2 & 2 & -1 & 0 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{12} 3^{11} 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 20 & 7 \\ 4 & 6 & 8 & 0 & 20 & 26 & 10 \\ 2 & 2 & 2 & -3 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{13} 3^{11} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 6 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{12} 3^{11} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -309, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{11} 3^{12} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -309, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{13} 3^{11} 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 33 & 9 \\ 5 & 8 & 9 & 0 & 33 & 40 & 12 \\ 2 & 2 & 2 & -3 & 9 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{11} 3^{12} 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 7 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 33 & 8 \\ 6 & 9 & 12 & 0 & 33 & 61 & 15 \\ 2 & 2 & 2 & -3 & 8 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{11} 3^{11} 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

10.2

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 5 & 0 & 25 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 92 & 94 & 20 \\ 11 & 25 & 0 & 94 & 92 & 20 \\ 3 & 5 & -3 & 20 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{111} 3^{21} 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 5 & 0 & 15 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 34 & 12 \\ 7 & 15 & 0 & 34 & 32 & 12 \\ 3 & 5 & -3 & 12 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{113} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 5 & 8 & 9 & 0 & 8 & 40 & 12 \\ 2 & 2 & 2 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{121} 3^{11} 1^{11} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 12 & 9 & 4 \\ 1 & 4 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 21 & 10 \\ 3 & 9 & 6 & 0 & 21 & 13 & 7 \\ 2 & 4 & 2 & -3 & 10 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{114} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 2 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{122} 3^{11} 1^{11} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 30 & 10 \\ 4 & 12 & 9 & 0 & 30 & 28 & 10 \\ 2 & 4 & 2 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{14} 3^{21} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 16 & 3 \\ 8 & 12 & 16 & 0 & 16 & 110 & 20 \\ 2 & 2 & 2 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{112} 3^{11} 1^{23} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 12 & 3 \\ 6 & 18 & 12 & 0 & 12 & 58 & 14 \\ 2 & 4 & 2 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{114} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{111}321^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 5 & 0 & 20 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 60 & 16 \\ 9 & 20 & 0 & 60 & 58 & 16 \\ 3 & 5 & -3 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{12}321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 22 & 4 \\ 14 & 30 & 0 & 22 & 134 & 24 \\ 3 & 5 & -3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{122}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[254] $1^{114}321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 1 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 8 & 12 & 16 & 0 & 12 & 110 & 20 \\ 2 & 2 & 2 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{122}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 15 & 7 \\ 3 & 5 & 5 & 0 & 15 & 13 & 7 \\ 2 & 2 & 2 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{123}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 30 & 7 \\ 6 & 9 & 12 & 0 & 30 & 61 & 15 \\ 2 & 2 & 2 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{123}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 20 & 7 \\ 4 & 7 & 6 & 0 & 20 & 25 & 9 \\ 2 & 2 & 2 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{112}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 18 & 5 \\ 5 & 7 & 11 & 0 & 18 & 45 & 13 \\ 2 & 2 & 2 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{112} 3^{21} 1^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 7 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 29 & 8 \\ 5 & 7 & 11 & 0 & 29 & 45 & 13 \\ 2 & 2 & 2 & -3 & 8 & 13 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{113} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{112} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 14 & 5 \\ 4 & 6 & 8 & 0 & 14 & 26 & 10 \\ 2 & 2 & 2 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{124} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & 0 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[262] $1^{112} 3^{21} 1^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 7 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 36 & 8 \\ 6 & 8 & 14 & 0 & 36 & 70 & 16 \\ 2 & 2 & 2 & -3 & 8 & 16 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{123} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & 0 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{124} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{13} 3^{21} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 25 & 4 \\ 17 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{114}3^{21}1^{24} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 12 & 2 \\ 1 & 2 & 0 & 0 & 7 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 44 & 8 \\ 8 & 12 & 16 & 0 & 44 & 110 & 20 \\ 2 & 2 & 2 & -3 & 8 & 20 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{113}3^{21}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{113}3^{21}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{122}3^{21}1^{23} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 36 & 10 \\ 5 & 7 & 11 & 0 & 36 & 45 & 13 \\ 2 & 2 & 2 & -3 & 10 & 13 & 1 \end{pmatrix}_{b_0}$$

[272] $1^{122}3^{21}1^{43} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 0 & 14 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 58 & 16 \\ 5 & 7 & 11 & 0 & 58 & 45 & 13 \\ 2 & 2 & 2 & -3 & 16 & 13 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{113}3^{21}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[274] $1^{114}3^{21}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 6 & 10 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & 2 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[275] $1^{123}1^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 6 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 18 & 4 \\ 6 & 18 & 12 & 0 & 18 & 58 & 14 \\ 2 & 4 & 2 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{114}3^{21}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 0 & 1 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 11 & 2 \\ 7 & 11 & 13 & 0 & 11 & 81 & 17 \\ 2 & 2 & 2 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{133} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{133} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 27 & 9 \\ 4 & 7 & 6 & 0 & 27 & 25 & 9 \\ 2 & 2 & 2 & -3 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[276] $1^{134} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 6 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 34 & 9 \\ 5 & 9 & 7 & 0 & 34 & 41 & 11 \\ 2 & 2 & 2 & -3 & 9 & 11 & 1 \end{pmatrix}_{b_0}$$

[277] $1^{111} 31^{111} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{122} 31^{112} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 42 & 10 \\ 6 & 18 & 12 & 0 & 42 & 58 & 14 \\ 2 & 4 & 2 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{124} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 40 & 7 \\ 8 & 12 & 16 & 0 & 40 & 110 & 20 \\ 2 & 2 & 2 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{113} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{112} 31^{13} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 8 & 24 & 15 & 0 & 15 & 100 & 18 \\ 2 & 4 & 2 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{113} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[286] $1^{114} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 7 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{114} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 9 & 11 & 0 & 13 & 11 & 81 & 17 \\ 2 & 2 & -1 & 0 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{113} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 17 & 5 \\ 5 & 8 & 9 & 0 & 17 & 40 & 12 \\ 2 & 2 & 2 & -3 & 5 & 12 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{114} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{113} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 9 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[291] $1^{123} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 11 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 54 & 13 \\ 6 & 9 & 12 & 0 & 54 & 61 & 15 \\ 2 & 2 & 2 & -3 & 13 & 15 & 1 \end{pmatrix}_{b_0}$$

[292] $1^{114} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[293] $1^{114} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -574, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 20 & 5 \\ 6 & 10 & 10 & 0 & 20 & 58 & 14 \\ 2 & 2 & 2 & -3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[294] \ 1^{134} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -579, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 4 & 0 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & 0 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[295] \ 1^{123} 3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -614, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 8 & 2 \\ 1 & 2 & 0 & 0 & 8 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 34 & 10 \\ 5 & 8 & 9 & 0 & 34 & 40 & 12 \\ 2 & 2 & 2 & -3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[296] \ 1^{114} 31^{13} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 21 & 3 \\ 10 & 30 & 21 & 0 & 21 & 172 & 24 \\ 2 & 4 & 2 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[298] \ 1^{113} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[299] \ 1^{122} 3 21^{144} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 8 & 2 \\ 1 & 2 & 0 & 0 & 14 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 72 & 16 \\ 6 & 8 & 14 & 0 & 72 & 70 & 16 \\ 2 & 2 & 2 & -3 & 16 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[297] \ 1^{122} 3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 28 & 10 \\ 4 & 6 & 8 & 0 & 28 & 26 & 10 \\ 2 & 2 & 2 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[300] \ 1^{113} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -629, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 8 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[301] \ 1^{113} 3 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 7 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 40 & 8 \\ 7 & 10 & 15 & 0 & 40 & 88 & 18 \\ 2 & 2 & 2 & -3 & 8 & 18 & 1 \end{pmatrix}_{b_0}$$

[302] $2 \overset{1^{1111}}{2} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[303] $1 \overset{1^1}{1^{23}} 3 21^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 25 & 7 \\ 5 & 8 & 9 & 0 & 25 & 40 & 12 \\ 2 & 2 & 2 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[304] $1 \overset{1^{111}}{1^{44}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 4 & 0 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 8 & 1 \end{pmatrix}_{b_0}$$

[305] $1 \overset{1^{13}}{1^{13}} 31^{\overline{13}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -691, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 18 & 3 \\ 9 & 27 & 18 & 0 & 18 & 133 & 21 \\ 2 & 4 & 2 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

[306] $1 \overset{1^1}{1^{11}} 31^{\overline{112}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -693, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 6 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 15 & 4 \\ 5 & 15 & 9 & 0 & 15 & 37 & 11 \\ 2 & 4 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[307] $1 \overset{1^1}{1^{24}} 3 21^{\overline{134}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 11 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 72 & 13 \\ 8 & 12 & 16 & 0 & 72 & 110 & 20 \\ 2 & 2 & 2 & -3 & 13 & 20 & 1 \end{pmatrix}_{b_0}$$

[308] $1 \overset{1^1}{1^{22}} 3 21^{\overline{133}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 47 & 13 \\ 5 & 7 & 11 & 0 & 47 & 45 & 13 \\ 2 & 2 & 2 & -3 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

[309] $1 \overset{1^{13}}{1^{14}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{133}321^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 9 & 2 \\ 1 & 2 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 51 & 12 \\ 6 & 9 & 12 & 0 & 51 & 61 & 15 \\ 2 & 2 & 2 & -3 & 12 & 15 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{114}321^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -724, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 10 & 11 & 0 & 13 & 11 & 81 & 17 \\ 2 & 2 & -1 & 0 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[312] $1^{134}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -734, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 40 & 9 \\ 6 & 10 & 10 & 0 & 40 & 58 & 14 \\ 2 & 2 & 2 & -3 & 9 & 14 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{113}321^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -764, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 21 & 5 \\ 6 & 9 & 12 & 0 & 21 & 61 & 15 \\ 2 & 2 & 2 & -3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

[314] $1^{114}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[315] $1^{114}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 8 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[316] $1^{114}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[317] $1^{113}321^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 25 & 5 \\ 7 & 10 & 15 & 0 & 25 & 88 & 18 \\ 2 & 2 & 2 & -3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$$

[318] $1^{111}31^{112}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -811, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 15 & 4 \\ 1 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 24 & 7 \\ 5 & 15 & 9 & 0 & 24 & 37 & 11 \\ 2 & 4 & 2 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[319] \ 1^{144} 3^{21} 1^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 11 & 2 \\ 1 & 2 & 0 & 0 & 10 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 70 & 14 \\ 7 & 11 & 13 & 0 & 70 & 81 & 17 \\ 2 & 2 & 2 & -3 & 14 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[320] \ 1^{114} 3^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -829, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 11 & 11 & 0 & 13 & 11 & 81 & 17 \\ 2 & 2 & -1 & 0 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{131} 1^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -853, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 18 & 0 & 27 & 133 & 21 \\ 2 & 4 & 2 & -3 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{123} 3^{21} 1^{24} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -854, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 50 & 10 \\ 7 & 10 & 15 & 0 & 50 & 88 & 18 \\ 2 & 2 & 2 & -3 & 10 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{134} 3^{21} 1^{24} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -854, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 68 & 12 \\ 8 & 12 & 16 & 0 & 68 & 110 & 20 \\ 2 & 2 & 2 & -3 & 12 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[324] \ 1^{113} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -859, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 7 & 2 \\ 2 & 4 & 4 & 0 & 9 & 21 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 36 & 7 \\ 7 & 21 & 15 & 0 & 36 & 85 & 17 \\ 2 & 4 & 2 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{124} 3^{21} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 30 & 7 \\ 6 & 10 & 10 & 0 & 30 & 58 & 14 \\ 2 & 2 & 2 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{114} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[326] $1^{124} 3^{21} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 35 & 7 \\ 7 & 11 & 13 & 0 & 35 & 81 & 17 \\ 2 & 2 & 2 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

[328] $1^{114} 3^{21} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -949, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 9 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[329] $1^{133} 3^{21} 1^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 10 & 2 \\ 1 & 2 & 0 & 0 & 12 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 75 & 15 \\ 7 & 10 & 15 & 0 & 75 & 88 & 18 \\ 2 & 2 & 2 & -3 & 15 & 18 & 1 \end{pmatrix}_{b_0}$$

[331] $1^{123} 3^{21} 1^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 42 & 10 \\ 6 & 9 & 12 & 0 & 42 & 61 & 15 \\ 2 & 2 & 2 & -3 & 10 & 15 & 1 \end{pmatrix}_{b_0}$$

[330] $1^{134} 3^{21} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 50 & 12 \\ 6 & 10 & 10 & 0 & 50 & 58 & 14 \\ 2 & 2 & 2 & -3 & 12 & 14 & 1 \end{pmatrix}_{b_0}$$

[332] $1^{111} 3^{11} 1^{13} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -1000, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 7 & 2 \\ 2 & 4 & 4 & 0 & 9 & 21 & 4 \\ 1 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 33 & 7 \\ 7 & 21 & 12 & 0 & 33 & 73 & 15 \\ 2 & 4 & 2 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[333] $1^{133} 3^{21} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1004, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 42 & 12 \\ 5 & 8 & 9 & 0 & 42 & 40 & 12 \\ 2 & 2 & 2 & -3 & 12 & 12 & 1 \end{pmatrix}_{b_0}$$

[334] $1^{114} 3^{21} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1004, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 0 & 4 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 24 & 5 \\ 7 & 11 & 13 & 0 & 24 & 81 & 17 \\ 2 & 2 & 2 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[335] $1^{112}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1018, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 48 & 11 \\ 6 & 18 & 12 & 0 & 48 & 58 & 14 \\ 2 & 4 & 2 & -3 & 11 & 14 & 1 \end{pmatrix}_{b_0}$$

[336] $1^{113}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1020, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 33 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 21 & 3 \\ 11 & 33 & 21 & 0 & 21 & 193 & 25 \\ 2 & 4 & 2 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

[337] $1^{114}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1029, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[338] $2^1 2^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1034, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 4 & 2 \\ 2 & 1 & 4 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 2 \\ 1 & 1 & 1 & -3 & 1 & 2 & -1 \\ 2 & 3 & 3 & 1 & 1 & 6 & 3 \\ 4 & 6 & 6 & 2 & 6 & 10 & 6 \\ 2 & 3 & 2 & -1 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[340] $1^{123}321^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1044, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 11 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 65 & 13 \\ 7 & 10 & 15 & 0 & 65 & 88 & 18 \\ 2 & 2 & 2 & -3 & 13 & 18 & 1 \end{pmatrix}_{b_0}$$

[339] $1^{114}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1044, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 10 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[341] $1^{112}31^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1046, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 6 & 2 \\ 2 & 4 & 4 & 0 & 9 & 18 & 4 \\ 1 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 30 & 7 \\ 6 & 18 & 12 & 0 & 30 & 58 & 14 \\ 2 & 4 & 2 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

[342] $1^{124}321^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 11 & 2 \\ 1 & 2 & 0 & 0 & 8 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 48 & 10 \\ 7 & 11 & 13 & 0 & 48 & 81 & 17 \\ 2 & 2 & 2 & -3 & 10 & 17 & 1 \end{pmatrix}_{b_0}$$

[343] $1^{134}321^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1094$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 12 & 2 \\ 1 & 2 & 0 & 0 & 15 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 100 & 18 \\ 8 & 12 & 16 & 0 & 100 & 110 & 20 \\ 2 & 2 & 2 & -3 & 18 & 20 & 1 \end{pmatrix}_{b_0}$$

[344] $1^{134}321^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 11 & 2 \\ 1 & 2 & 0 & 0 & 12 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 72 & 15 \\ 7 & 11 & 13 & 0 & 72 & 81 & 17 \\ 2 & 2 & 2 & -3 & 15 & 17 & 1 \end{pmatrix}_{b_0}$$

[345] $1^{114}31^{143}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 36 & 4 \\ 1 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 24 & 3 \\ 12 & 36 & 24 & 0 & 24 & 238 & 28 \\ 2 & 4 & 2 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

[346] $1^{113}1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1197$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 6 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 21 & 4 \\ 7 & 21 & 12 & 0 & 21 & 73 & 15 \\ 2 & 4 & 2 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[347] $1^{123}1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 6 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 24 & 4 \\ 8 & 24 & 15 & 0 & 24 & 100 & 18 \\ 2 & 4 & 2 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[348] $1^{114}321^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 28 & 5 \\ 8 & 12 & 16 & 0 & 28 & 110 & 20 \\ 2 & 2 & 2 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

[349] $1^{144}321^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 12 & 2 \\ 1 & 2 & 0 & 0 & 13 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 96 & 17 \\ 8 & 12 & 16 & 0 & 96 & 110 & 20 \\ 2 & 2 & 2 & -3 & 17 & 20 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{111}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 39 & 11 \\ 5 & 15 & 9 & 0 & 39 & 37 & 11 \\ 2 & 4 & 2 & -3 & 11 & 11 & 1 \end{pmatrix}_{b_0}$$

[351] $1^{133}321^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 63 & 15 \\ 6 & 9 & 12 & 0 & 63 & 61 & 15 \\ 2 & 2 & 2 & -3 & 15 & 15 & 1 \end{pmatrix}_{b_0}$$

[352] $1^{14}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 12 & 2 \\ 2 & 4 & 4 & 0 & 6 & 36 & 4 \\ 1 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 36 & 4 \\ 12 & 36 & 24 & 0 & 36 & 238 & 28 \\ 2 & 4 & 2 & -3 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

[353] $1^{144}321^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 10 & 2 \\ 1 & 2 & 0 & 0 & 10 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 60 & 14 \\ 6 & 10 & 10 & 0 & 60 & 58 & 14 \\ 2 & 2 & 2 & -3 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

[354] $1^{134}321^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 59 & 12 \\ 7 & 11 & 13 & 0 & 59 & 81 & 17 \\ 2 & 2 & 2 & -3 & 12 & 17 & 1 \end{pmatrix}_{b_0}$$

[355] $2^1 2^1 31^{1122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & 2 & -1 \\ 4 & 6 & 6 & 2 & 10 & 12 & 6 \\ 4 & 6 & 6 & 2 & 12 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[356] $1^{124}321^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 56 & 10 \\ 8 & 12 & 16 & 0 & 56 & 110 & 20 \\ 2 & 2 & 2 & -3 & 10 & 20 & 1 \end{pmatrix}_{b_0}$$

[357] $1^{133}321^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 90 & 18 \\ 7 & 10 & 15 & 0 & 90 & 88 & 18 \\ 2 & 2 & 2 & -3 & 18 & 18 & 1 \end{pmatrix}_{b_0}$$

[358] $1^{122}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 12 & 0 & 58 & 60 & 14 \\ 6 & 18 & 12 & 0 & 60 & 58 & 14 \\ 2 & 4 & 2 & -3 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

[359] $1^{112}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1669, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 8 & 2 \\ 2 & 4 & 4 & 0 & 9 & 24 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 39 & 7 \\ 8 & 24 & 15 & 0 & 39 & 100 & 18 \\ 2 & 4 & 2 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[360] $2^{\overline{1113}} 2^1 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1684$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 6 & 2 \\ 2 & 1 & 4 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 1 & 3 & 9 & 2 \\ 1 & 1 & 1 & -3 & 1 & 3 & -1 \\ 2 & 3 & 3 & 1 & 1 & 9 & 3 \\ 6 & 9 & 9 & 3 & 9 & 25 & 9 \\ 2 & 3 & 2 & -1 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[361] $1^{\overline{1143}} 31^{\overline{1133}} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1741$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 10 & 2 \\ 2 & 4 & 4 & 0 & 9 & 30 & 4 \\ 1 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 51 & 7 \\ 10 & 30 & 21 & 0 & 51 & 172 & 24 \\ 2 & 4 & 2 & -3 & 7 & 24 & 1 \end{pmatrix}_{b_0}$$

[362] $1^{\overline{1133}} 31^{\overline{1143}} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1813$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 15 & 0 & 27 & 121 & 19 \\ 2 & 4 & 2 & -3 & 4 & 19 & 1 \end{pmatrix}_{b_0}$$

[363] $1^{\overline{1343}} 321^{\overline{134}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1814$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 12 & 2 \\ 1 & 2 & 0 & 0 & 12 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 84 & 15 \\ 8 & 12 & 16 & 0 & 84 & 110 & 20 \\ 2 & 2 & 2 & -3 & 15 & 20 & 1 \end{pmatrix}_{b_0}$$

[364] $1^{\overline{1443}} 321^{\overline{133}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1904$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 83 & 17 \\ 7 & 11 & 13 & 0 & 83 & 81 & 17 \\ 2 & 2 & 2 & -3 & 17 & 17 & 1 \end{pmatrix}_{b_0}$$

[365] $1^{\overline{1133}} 31^{\overline{1133}} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1916$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 9 & 2 \\ 2 & 4 & 4 & 0 & 9 & 27 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 45 & 7 \\ 9 & 27 & 18 & 0 & 45 & 133 & 21 \\ 2 & 4 & 2 & -3 & 7 & 21 & 1 \end{pmatrix}_{b_0}$$

[366] $1^{\overline{1333}} 31^{\overline{1143}} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1917$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 11 & 2 \\ 2 & 4 & 4 & 0 & 6 & 33 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 33 & 4 \\ 11 & 33 & 21 & 0 & 33 & 193 & 25 \\ 2 & 4 & 2 & -3 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

[367] $1^{\overline{1233}} 31^{\overline{1143}} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2076$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 6 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 30 & 4 \\ 10 & 30 & 18 & 0 & 30 & 154 & 22 \\ 2 & 4 & 2 & -3 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

[368] $1^{\overline{1123}} 31^{\overline{1143}} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 10 & 2 \\ 2 & 4 & 4 & 0 & 9 & 30 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 48 & 7 \\ 10 & 30 & 18 & 0 & 48 & 154 & 22 \\ 2 & 4 & 2 & -3 & 7 & 22 & 1 \end{pmatrix}_{b_0}$$

[369] $1^{144} 3^{21} 1^{44} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -2304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 16 & 0 & 110 & 112 & 20 \\ 8 & 12 & 16 & 0 & 112 & 110 & 20 \\ 2 & 2 & 2 & -3 & 20 & 20 & 1 \end{pmatrix}_{b_0}$$

[370] $2 \ 2 \ 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -2594, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 8 & 2 \\ 2 & 1 & 4 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 1 & 3 & 12 & 2 \\ 1 & 1 & 1 & -3 & 1 & 4 & -1 \\ 2 & 3 & 3 & 1 & 1 & 12 & 3 \\ 8 & 12 & 12 & 4 & 12 & 46 & 12 \\ 2 & 3 & 2 & -1 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[371] $1^{13} 31^{14} 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2875, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 11 & 2 \\ 2 & 4 & 4 & 0 & 9 & 33 & 4 \\ 1 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 54 & 7 \\ 11 & 33 & 21 & 0 & 54 & 193 & 25 \\ 2 & 4 & 2 & -3 & 7 & 25 & 1 \end{pmatrix}_{b_0}$$

[372] $1^{14} 31^{14} 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -3134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 12 & 2 \\ 2 & 4 & 4 & 0 & 9 & 36 & 4 \\ 1 & 4 & 0 & 0 & 6 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 60 & 7 \\ 12 & 36 & 24 & 0 & 60 & 238 & 28 \\ 2 & 4 & 2 & -3 & 7 & 28 & 1 \end{pmatrix}_{b_0}$$

[373] $2 \ 2 \ 31^{15} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -3764, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 10 & 2 \\ 2 & 1 & 4 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 1 & 3 & 15 & 2 \\ 1 & 1 & 1 & -3 & 1 & 5 & -1 \\ 2 & 3 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 15 & 5 & 15 & 73 & 15 \\ 2 & 3 & 2 & -1 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [374] $1^{11} 231^{11} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 13 & 3 \\ 5 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 30 & 0 & 0 & 63 & 15 \\ 3 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[375] $1^{11} 231^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 11 & 3 \\ 5 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 30 & 0 & 0 & 63 & 15 \\ 3 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[376] $1^{11} 231^{11} 31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 12 & 3 \\ 5 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 30 & 0 & 0 & 63 & 15 \\ 3 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[377] $1^{12} 231^{12} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 23 & 3 \\ 5 & 13 & 1 & 0 & 60 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 23 & 60 & 0 & 0 & 258 & 30 \\ 3 & 7 & -1 & 0 & 30 & 3 \end{pmatrix}_{b_0}$$

[378] $1^{13}231^{13}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 268, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 34 & 3 \\ 5 & 13 & 1 & 0 & 90 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 34 & 90 & 0 & 0 & 583 & 45 \\ 3 & 7 & -1 & 0 & 45 & 3 \end{pmatrix}_{b_0}$$

[379] $1^{12}231^{12}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 24 & 3 \\ 5 & 13 & 1 & 0 & 60 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 24 & 60 & 0 & 0 & 258 & 30 \\ 3 & 7 & -1 & 0 & 30 & 3 \end{pmatrix}_{b_0}$$

4.463 $\text{NHC}(-2)(-3) + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{f}_4$

$T = 9$ $[0] 1^{1111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{f}_4$ $T = 9, T_{\max} = 92, \Delta = -130, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

4.464 $\text{NHC}(-2)(-3) + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{so}_8$

$T = 9$ $[0] 1^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$

$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 1 & 3 \\ 4 & 9 & 4 & 0 & 16 & 0 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 16 & 6 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 0 & 8 & -2 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -3 & 7 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{o}}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{o}}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{o}}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{112}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{o}}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$4.465 \quad \text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{\overline{124}} 3^{\overline{1}} 1^{\overline{13}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 2 & 3 \\ 3 & 4 & 0 & 8 & 31 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 42 & 4 & 6 \\ 17 & 31 & 0 & 42 & 172 & 17 & 24 \\ 2 & 3 & 0 & 4 & 17 & 0 & 2 \\ 3 & 4 & -3 & 6 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{122}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 4 & 6 \\ 4 & 8 & 0 & 12 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{121}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 4 & 6 \\ 5 & 9 & 0 & 12 & 13 & 5 & 7 \\ 2 & 3 & 0 & 4 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$\begin{aligned}
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[4] & 1^{13}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 17 & 0 & 15 & 49 & 9 & 13 \\ 2 & 3 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 4 & 13 & 2 & 1 \end{pmatrix}_{b_0} \\
[5] & 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 \\ 4 & 8 & 0 & 6 & 10 & 4 & 6 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0} \\
[6] & 1^{13}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0} \\
[7] & 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[8] & 1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 2 & 3 \\ 3 & 4 & 0 & 8 & 22 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 4 & 6 \\ 12 & 22 & 0 & 30 & 85 & 12 & 17 \\ 2 & 3 & 0 & 4 & 12 & 0 & 2 \\ 3 & 4 & -3 & 6 & 17 & 2 & 1 \end{pmatrix}_{b_0} \\
[10] & 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 2 & 3 \\ 3 & 4 & 0 & 8 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 4 & 6 \\ 7 & 13 & 0 & 18 & 28 & 7 & 10 \\ 2 & 3 & 0 & 4 & 7 & 0 & 2 \\ 3 & 4 & -3 & 6 & 10 & 2 & 1 \end{pmatrix}_{b_0} \\
[9] & 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 2 & 3 \\ 8 & 14 & 0 & 9 & 37 & 8 & 11 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -3 & 3 & 11 & 2 & 1 \end{pmatrix}_{b_0} \\
[11] & 1^{12}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7
\end{aligned}$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 28 & 7 & 10 \\ 2 & 3 & 0 & 3 & 7 & 0 & 2 \\ 3 & 4 & -3 & 4 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 5 & 7 \\ 2 & 3 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 \\ 5 & 9 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 8 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 24 & 4 & 6 \\ 9 & 17 & 0 & 24 & 49 & 9 & 13 \\ 2 & 3 & 0 & 4 & 9 & 0 & 2 \\ 3 & 4 & -3 & 6 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 9 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 5 & 7 \\ 11 & 21 & 0 & 33 & 76 & 11 & 16 \\ 2 & 3 & 0 & 5 & 11 & 0 & 2 \\ 3 & 4 & -3 & 7 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 2 & 3 \\ 3 & 4 & 0 & 8 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 4 & 6 \\ 11 & 21 & 0 & 30 & 76 & 11 & 16 \\ 2 & 3 & 0 & 4 & 11 & 0 & 2 \\ 3 & 4 & -3 & 6 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 2 & 3 \\ 7 & 13 & 0 & 9 & 28 & 7 & 10 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -3 & 3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 2 & 3 \\ 3 & 4 & 0 & 8 & 26 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 36 & 4 & 6 \\ 14 & 26 & 0 & 36 & 118 & 14 & 20 \\ 2 & 3 & 0 & 4 & 14 & 0 & 2 \\ 3 & 4 & -3 & 6 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 5 & 7 \\ 5 & 9 & 0 & 15 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 1^{12} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 23 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 2 & 3 \\ 13 & 23 & 0 & 15 & 100 & 13 & 18 \\ 2 & 3 & 0 & 2 & 13 & 0 & 2 \\ 3 & 4 & -3 & 3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 2 & 3 \\ 9 & 17 & 0 & 12 & 49 & 9 & 13 \\ 2 & 3 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -3 & 3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 2 & 3 \\ 3 & 4 & 0 & 9 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 5 & 7 \\ 9 & 17 & 0 & 27 & 49 & 9 & 13 \\ 2 & 3 & 0 & 5 & 9 & 0 & 2 \\ 3 & 4 & -3 & 7 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 18 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 2 & 3 \\ 10 & 18 & 0 & 12 & 58 & 10 & 14 \\ 2 & 3 & 0 & 2 & 10 & 0 & 2 \\ 3 & 4 & -3 & 3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 1^{14} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 2 & 3 \\ 11 & 21 & 0 & 15 & 76 & 11 & 16 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -3 & 3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 5 & 7 \\ 7 & 13 & 0 & 21 & 28 & 7 & 10 \\ 2 & 3 & 0 & 5 & 7 & 0 & 2 \\ 3 & 4 & -3 & 7 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{124} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 2 & 3 \\ 3 & 4 & 0 & 13 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 48 & 7 & 10 \\ 11 & 21 & 0 & 48 & 76 & 11 & 16 \\ 2 & 3 & 0 & 7 & 11 & 0 & 2 \\ 3 & 4 & -3 & 10 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{122} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 7 & 10 \\ 7 & 13 & 0 & 30 & 28 & 7 & 10 \\ 2 & 3 & 0 & 7 & 7 & 0 & 2 \\ 3 & 4 & -3 & 10 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 2 & 3 \\ 3 & 4 & 0 & 13 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 39 & 7 & 10 \\ 9 & 17 & 0 & 39 & 49 & 9 & 13 \\ 2 & 3 & 0 & 7 & 9 & 0 & 2 \\ 3 & 4 & -3 & 10 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{113}3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 22 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 2 & 3 \\ 12 & 22 & 0 & 15 & 85 & 12 & 17 \\ 2 & 3 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{133}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 51 & 9 & 13 \\ 9 & 17 & 0 & 51 & 49 & 9 & 13 \\ 2 & 3 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -3 & 13 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{113}3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 4 & 32 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 2 & 3 \\ 18 & 32 & 0 & 21 & 193 & 18 & 25 \\ 2 & 3 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{134}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 2 & 3 \\ 3 & 4 & 0 & 17 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 63 & 9 & 13 \\ 11 & 21 & 0 & 63 & 76 & 11 & 16 \\ 2 & 3 & 0 & 9 & 11 & 0 & 2 \\ 3 & 4 & -3 & 13 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{113}3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 4 & 27 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 2 & 3 \\ 15 & 27 & 0 & 18 & 133 & 15 & 21 \\ 2 & 3 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{144}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 78 & 11 & 16 \\ 11 & 21 & 0 & 78 & 76 & 11 & 16 \\ 2 & 3 & 0 & 11 & 11 & 0 & 2 \\ 3 & 4 & -3 & 16 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{114}3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 26 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 2 & 3 \\ 14 & 26 & 0 & 18 & 118 & 14 & 20 \\ 2 & 3 & 0 & 2 & 14 & 0 & 2 \\ 3 & 4 & -3 & 3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 4^1 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 4 & 31 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 2 & 3 \\ 17 & 31 & 0 & 21 & 172 & 17 & 24 \\ 2 & 3 & 0 & 2 & 17 & 0 & 2 \\ 3 & 4 & -3 & 3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} 4^1 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 4 & 36 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 24 & 2 & 3 \\ 20 & 36 & 0 & 24 & 238 & 20 & 28 \\ 2 & 3 & 0 & 2 & 20 & 0 & 2 \\ 3 & 4 & -3 & 3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{11} 3^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 247, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 2 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^1 3^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 6 & 3 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} 3^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 517, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11} 3^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 648, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 24 & 10 \\ 6 & 15 & 9 & 0 & 24 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{11} 3^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 912, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 15 & 5 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.466 NHC(-2)(-3) + f₄ + su₂ + su₂ + su₂→(-3) + su₃

$$T = 10 \quad [0] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 5 & 6 & 7 \\ 2 & 3 & 0 & 3 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 6 & 5 & 12 & 6 \\ 3 & 4 & -3 & 3 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 5 & 6 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 5 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 5 & 6 & 7 \\ 2 & 3 & 0 & 5 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.467 NHC(-2)(-3) + f₄ + su₂ + su₂ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 & 4 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 & 4 \\ 2 & 4 & 0 & 3 & 1 & 5 & 3 & 3 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 & 5 \\ 3 & 4 & -3 & 4 & 3 & 5 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{1211} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 10 & 6 \\ 5 & 9 & 0 & 12 & 13 & 11 & 7 \\ 4 & 7 & 0 & 10 & 11 & 7 & 5 \\ 3 & 4 & -3 & 6 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{1122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 4 & 8 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 11 & 3 \\ 4 & 8 & 0 & 6 & 10 & 22 & 6 \\ 9 & 16 & 0 & 11 & 22 & 44 & 12 \\ 3 & 4 & -3 & 3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{1131} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 4 & 3 \\ 3 & 4 & 0 & 4 & 12 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 5 & 3 \\ 6 & 12 & 0 & 9 & 25 & 15 & 9 \\ 4 & 7 & 0 & 5 & 15 & 7 & 5 \\ 3 & 4 & -3 & 3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{1113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 0 & 5 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 6 & 4 \\ 5 & 9 & 0 & 9 & 13 & 11 & 7 \\ 4 & 7 & 0 & 6 & 11 & 7 & 5 \\ 3 & 4 & -3 & 4 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{1313} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 0 & 5 & 17 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 6 & 4 \\ 9 & 17 & 0 & 15 & 49 & 21 & 13 \\ 4 & 7 & 0 & 6 & 21 & 7 & 5 \\ 3 & 4 & -3 & 4 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[6] 1^{1221} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 9 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 22 & 6 \\ 5 & 9 & 0 & 12 & 13 & 26 & 7 \\ 9 & 16 & 0 & 22 & 26 & 44 & 12 \\ 3 & 4 & -3 & 6 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[7] 1^{1221} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 10 & 6 \\ 4 & 8 & 0 & 12 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 10 & 7 & 5 \\ 3 & 4 & -3 & 6 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[8] 1^{1131} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 7 & 3 \\ 3 & 4 & 0 & 4 & 17 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 8 & 3 \\ 9 & 17 & 0 & 12 & 49 & 36 & 13 \\ 7 & 12 & 0 & 8 & 36 & 25 & 9 \\ 3 & 4 & -3 & 3 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 17 & 7 \\ 5 & 9 & 0 & 15 & 13 & 17 & 7 \\ 6 & 11 & 0 & 17 & 17 & 20 & 8 \\ 3 & 4 & -3 & 7 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 7 & 12 & 0 & 12 & 8 & 25 & 9 \\ 3 & 4 & -3 & 4 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 1 & 14 & 3 \\ 11 & 20 & 0 & 18 & 14 & 69 & 15 \\ 3 & 4 & -3 & 4 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 0 & 5 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 6 & 4 \\ 7 & 13 & 0 & 12 & 28 & 16 & 10 \\ 4 & 7 & 0 & 6 & 16 & 7 & 5 \\ 3 & 4 & -3 & 4 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 5 & 3 \\ 4 & 8 & 0 & 6 & 10 & 10 & 6 \\ 4 & 7 & 0 & 5 & 10 & 7 & 5 \\ 3 & 4 & -3 & 3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 8 & 3 \\ 5 & 9 & 0 & 6 & 13 & 17 & 7 \\ 6 & 11 & 0 & 8 & 17 & 20 & 8 \\ 3 & 4 & -3 & 3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 1 & 11 & 3 \\ 9 & 16 & 0 & 15 & 11 & 44 & 12 \\ 3 & 4 & -3 & 4 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 8 & 3 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & -3 & 3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -276, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 5 & 3 \\ 2 & 4 & 0 & 3 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -316, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 4 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 5 & 3 \\ 5 & 9 & 0 & 6 & 13 & 11 & 7 \\ 4 & 7 & 0 & 5 & 11 & 7 & 5 \\ 3 & 4 & -3 & 3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{1221}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -326, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 4 & 3 \\ 3 & 4 & 0 & 8 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 10 & 6 \\ 7 & 13 & 0 & 18 & 28 & 16 & 10 \\ 4 & 7 & 0 & 10 & 16 & 7 & 5 \\ 3 & 4 & -3 & 6 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{1122}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -356, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 8 & 3 \\ 4 & 8 & 0 & 6 & 10 & 16 & 6 \\ 6 & 11 & 0 & 8 & 16 & 20 & 8 \\ 3 & 4 & -3 & 3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{1121}} 3^{\overline{1}} 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 4 & 3 \\ 3 & 4 & 0 & 4 & 18 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 5 & 3 \\ 10 & 18 & 0 & 12 & 58 & 22 & 14 \\ 4 & 7 & 0 & 5 & 22 & 7 & 5 \\ 3 & 4 & -3 & 3 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{1112}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -416, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 8 & 3 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 6 & 11 & 0 & 8 & 8 & 20 & 8 \\ 3 & 4 & -3 & 3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{1122}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -426, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 6 & 3 \\ 3 & 4 & 0 & 9 & 13 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 17 & 7 \\ 7 & 13 & 0 & 21 & 28 & 25 & 10 \\ 6 & 11 & 0 & 17 & 25 & 20 & 8 \\ 3 & 4 & -3 & 7 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -456, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 11 & 7 \\ 5 & 9 & 0 & 15 & 13 & 11 & 7 \\ 4 & 7 & 0 & 11 & 11 & 7 & 5 \\ 3 & 4 & -3 & 7 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1131}3^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 4 & 3 \\ 3 & 4 & 0 & 4 & 27 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 5 & 3 \\ 15 & 27 & 0 & 18 & 133 & 33 & 21 \\ 4 & 7 & 0 & 5 & 33 & 7 & 5 \\ 3 & 4 & -3 & 3 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{1112}3^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 14 & 3 \\ 2 & 4 & 0 & 3 & 1 & 14 & 3 \\ 12 & 21 & 0 & 14 & 14 & 80 & 16 \\ 3 & 4 & -3 & 3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1121}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -514, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 8 & 3 \\ 7 & 13 & 0 & 9 & 28 & 28 & 10 \\ 7 & 12 & 0 & 8 & 28 & 25 & 9 \\ 3 & 4 & -3 & 3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 8 & 3 \\ 5 & 9 & 0 & 6 & 13 & 20 & 7 \\ 7 & 12 & 0 & 8 & 20 & 25 & 9 \\ 3 & 4 & -3 & 3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{1121}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 4 & 3 \\ 3 & 4 & 0 & 4 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 5 & 3 \\ 7 & 13 & 0 & 9 & 28 & 16 & 10 \\ 4 & 7 & 0 & 5 & 16 & 7 & 5 \\ 3 & 4 & -3 & 3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{1114}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -546, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 4 & 3 \\ 3 & 4 & 0 & 9 & 21 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 11 & 7 \\ 11 & 21 & 0 & 33 & 76 & 26 & 16 \\ 4 & 7 & 0 & 11 & 26 & 7 & 5 \\ 3 & 4 & -3 & 7 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{1122}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -546, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 6 & 3 \\ 3 & 4 & 0 & 4 & 13 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 8 & 3 \\ 7 & 13 & 0 & 9 & 28 & 25 & 10 \\ 6 & 11 & 0 & 8 & 25 & 20 & 8 \\ 3 & 4 & -3 & 3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{1113}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 11 & 3 \\ 2 & 4 & 0 & 3 & 1 & 11 & 3 \\ 8 & 15 & 0 & 11 & 11 & 39 & 11 \\ 3 & 4 & -3 & 3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{\overline{1141}3^{\overline{114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2} \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -616, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 4 & 3 \\ 3 & 4 & 0 & 4 & 36 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 24 & 5 & 3 \\ 20 & 36 & 0 & 24 & 238 & 44 & 28 \\ 4 & 7 & 0 & 5 & 44 & 7 & 5 \\ 3 & 4 & -3 & 3 & 28 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{\overline{1112}3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2} \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -620, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 11 & 3 \\ 2 & 4 & 0 & 3 & 1 & 11 & 3 \\ 9 & 16 & 0 & 11 & 11 & 44 & 12 \\ 3 & 4 & -3 & 3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{\overline{1121}3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2} \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -626, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 4 & 3 \\ 3 & 4 & 0 & 9 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 11 & 7 \\ 7 & 13 & 0 & 21 & 28 & 16 & 10 \\ 4 & 7 & 0 & 11 & 16 & 7 & 5 \\ 3 & 4 & -3 & 7 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{\overline{1131}3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2} \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -656, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 4 & 3 \\ 3 & 4 & 0 & 9 & 17 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 11 & 7 \\ 9 & 17 & 0 & 27 & 49 & 21 & 13 \\ 4 & 7 & 0 & 11 & 21 & 7 & 5 \\ 3 & 4 & -3 & 7 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{\overline{1131}3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2} \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -716, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 4 & 3 \\ 3 & 4 & 0 & 4 & 17 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 5 & 3 \\ 9 & 17 & 0 & 12 & 49 & 21 & 13 \\ 4 & 7 & 0 & 5 & 21 & 7 & 5 \\ 3 & 4 & -3 & 3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{\overline{1114}3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2} \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -720, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 14 & 3 \\ 2 & 4 & 0 & 3 & 1 & 14 & 3 \\ 10 & 19 & 0 & 14 & 14 & 64 & 14 \\ 3 & 4 & -3 & 3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{\overline{1112}3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2} \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -810, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 11 & 3 \\ 5 & 9 & 0 & 6 & 13 & 26 & 7 \\ 9 & 16 & 0 & 11 & 26 & 44 & 12 \\ 3 & 4 & -3 & 3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{\overline{1131}3^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2} \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -836, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 4 & 3 \\ 3 & 4 & 0 & 4 & 22 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 5 & 3 \\ 12 & 22 & 0 & 15 & 85 & 27 & 17 \\ 4 & 7 & 0 & 5 & 27 & 7 & 5 \\ 3 & 4 & -3 & 3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{\overline{1113}3^1}1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -840, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 30 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 20 & 3 \\ 2 & 4 & 0 & 3 & 1 & 20 & 3 \\ 17 & 30 & 0 & 20 & 20 & 165 & 23 \\ 3 & 4 & -3 & 3 & 3 & 23 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{\overline{1141}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -856, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 4 & 3 \\ 3 & 4 & 0 & 4 & 21 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 5 & 3 \\ 11 & 21 & 0 & 15 & 76 & 26 & 16 \\ 4 & 7 & 0 & 5 & 26 & 7 & 5 \\ 3 & 4 & -3 & 3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{\overline{1221}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 16 & 10 \\ 7 & 13 & 0 & 30 & 28 & 16 & 10 \\ 4 & 7 & 0 & 16 & 16 & 7 & 5 \\ 3 & 4 & -3 & 10 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{\overline{1113}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -964, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 14 & 3 \\ 5 & 9 & 0 & 6 & 13 & 32 & 7 \\ 11 & 20 & 0 & 14 & 32 & 69 & 15 \\ 3 & 4 & -3 & 3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{\overline{1113}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -984, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 14 & 3 \\ 2 & 4 & 0 & 3 & 1 & 14 & 3 \\ 11 & 20 & 0 & 14 & 14 & 69 & 15 \\ 3 & 4 & -3 & 3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{\overline{1114}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -986, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 4 & 9 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 17 & 3 \\ 5 & 9 & 0 & 6 & 13 & 38 & 7 \\ 13 & 24 & 0 & 17 & 38 & 100 & 18 \\ 3 & 4 & -3 & 3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{\overline{1122}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -1020, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 11 & 3 \\ 7 & 13 & 0 & 9 & 28 & 37 & 10 \\ 9 & 16 & 0 & 11 & 37 & 44 & 12 \\ 3 & 4 & -3 & 3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{\overline{1113}3^1}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -1076, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 17 & 3 \\ 2 & 4 & 0 & 3 & 1 & 17 & 3 \\ 14 & 25 & 0 & 17 & 17 & 111 & 19 \\ 3 & 4 & -3 & 3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{114131^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 4 & 3 \\ 3 & 4 & 0 & 4 & 31 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 5 & 3 \\ 17 & 31 & 0 & 21 & 172 & 38 & 24 \\ 4 & 7 & 0 & 5 & 38 & 7 & 5 \\ 3 & 4 & -3 & 3 & 24 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{114131^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 4 & 3 \\ 3 & 4 & 0 & 4 & 26 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 5 & 3 \\ 14 & 26 & 0 & 18 & 118 & 32 & 20 \\ 4 & 7 & 0 & 5 & 32 & 7 & 5 \\ 3 & 4 & -3 & 3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{114131^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 4 & 39 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 26 & 3 \\ 2 & 4 & 0 & 3 & 1 & 26 & 3 \\ 22 & 39 & 0 & 26 & 26 & 280 & 30 \\ 3 & 4 & -3 & 3 & 3 & 30 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114131^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 17 & 3 \\ 2 & 4 & 0 & 3 & 1 & 17 & 3 \\ 13 & 24 & 0 & 17 & 17 & 100 & 18 \\ 3 & 4 & -3 & 3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{114131^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 4 & 34 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 23 & 3 \\ 2 & 4 & 0 & 3 & 1 & 23 & 3 \\ 19 & 34 & 0 & 23 & 23 & 208 & 26 \\ 3 & 4 & -3 & 3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{114131^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 4 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 20 & 3 \\ 2 & 4 & 0 & 3 & 1 & 20 & 3 \\ 16 & 29 & 0 & 20 & 20 & 148 & 22 \\ 3 & 4 & -3 & 3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \quad 1^{1131^{111}3^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 5 \\ 1 & 4 & 0 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 2 & 6 & 3 & 0 & 4 & 6 & 5 \\ 2 & 6 & 3 & 0 & 6 & 4 & 5 \\ 2 & 5 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111131^{11}3^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 5 \\ 1 & 4 & 0 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 1 & 3 & 3 & 0 & 1 & 6 & 4 \\ 3 & 9 & 6 & 0 & 6 & 13 & 9 \\ 2 & 5 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 2 & -3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{111}31^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 262, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 5 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 9 & 7 \\ 2 & 5 & 4 & 0 & 5 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{111}31^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 752, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 5 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 15 & 9 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 9 & 7 \\ 2 & 5 & 4 & 0 & 9 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 7 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$4.468 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 9 \quad [0] \ 1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 3 & 4 \\ 11 & 19 & 0 & 21 & 73 & 12 & 15 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 4 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 8 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 6 & 7 \\ 11 & 19 & 0 & 33 & 73 & 12 & 15 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -3 & 7 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 5 & 9 & 0 & 6 & 12 & 12 & 6 \\ 3 & 4 & -3 & 3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -3 & 4 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 27 & 9 \\ 3 & 4 & -3 & 4 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 8 & 14 & 0 & 15 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 4 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 4 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -3 & 3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{121} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -331, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 13 & 23 & 0 & 15 & 100 & 15 & 18 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -338, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 5 & 3 \\ 3 & 4 & 0 & 8 & 13 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 12 & 6 \\ 7 & 13 & 0 & 18 & 28 & 21 & 10 \\ 5 & 9 & 0 & 12 & 21 & 12 & 6 \\ 3 & 4 & -3 & 6 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & -3 & 3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 2 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 24 & 6 & 7 \\ 8 & 14 & 0 & 24 & 37 & 9 & 11 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -3 & 7 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{121} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 3 & 4 \\ 10 & 18 & 0 & 18 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 4 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 30 & 3 & 4 \\ 16 & 28 & 0 & 30 & 154 & 18 & 22 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -3 & 4 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & -3 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 39 & 9 & 11 \\ 8 & 14 & 0 & 39 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -3 & 11 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1121} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 7 & 13 & 0 & 9 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1121} 31^{12} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 10 & 18 & 0 & 12 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{121} 31^{113} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 24 & 3 & 4 \\ 13 & 23 & 0 & 24 & 100 & 15 & 18 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 4 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{131} 31^{112} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 3 & 4 \\ 12 & 22 & 0 & 21 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 4 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1131} 31^{14} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 & 3 \\ 18 & 32 & 0 & 21 & 193 & 21 & 25 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -3 & 3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1121} 31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 6 & 3 \\ 7 & 13 & 0 & 9 & 28 & 21 & 10 \\ 5 & 9 & 0 & 6 & 21 & 12 & 6 \\ 3 & 4 & -3 & 3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1121} 31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 6 & 7 \\ 7 & 13 & 0 & 21 & 28 & 9 & 10 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -3 & 7 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1121} 31^{122} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -962, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 2 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 48 & 9 & 11 \\ 10 & 18 & 0 & 48 & 58 & 12 & 14 \\ 2 & 4 & 0 & 9 & 12 & 0 & 2 \\ 3 & 4 & -3 & 11 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1131} 31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 2 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 6 & 7 \\ 9 & 17 & 0 & 27 & 49 & 12 & 13 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -3 & 7 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1121} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 2 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 48 & 6 & 7 \\ 16 & 28 & 0 & 48 & 154 & 18 & 22 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -3 & 7 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1131} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 5 & 3 \\ 3 & 4 & 0 & 4 & 17 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 6 & 3 \\ 9 & 17 & 0 & 12 & 49 & 27 & 13 \\ 5 & 9 & 0 & 6 & 27 & 12 & 6 \\ 3 & 4 & -3 & 3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1141} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 21 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 6 & 3 \\ 11 & 21 & 0 & 15 & 76 & 33 & 16 \\ 5 & 9 & 0 & 6 & 33 & 12 & 6 \\ 3 & 4 & -3 & 3 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1131} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 9 & 17 & 0 & 12 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{131} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 27 & 3 & 4 \\ 15 & 27 & 0 & 27 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -3 & 4 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1221} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 2 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 42 & 9 & 10 \\ 10 & 18 & 0 & 42 & 58 & 12 & 14 \\ 2 & 4 & 0 & 9 & 12 & 0 & 2 \\ 3 & 4 & -3 & 10 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1131} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 & 3 \\ 15 & 27 & 0 & 18 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -3 & 3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1121} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 2 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 30 & 6 & 7 \\ 10 & 18 & 0 & 30 & 58 & 12 & 14 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -3 & 7 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{122\bar{1}} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 9 & 10 \\ 7 & 13 & 0 & 30 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -3 & 10 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{112\bar{1}} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 2 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 39 & 6 & 7 \\ 13 & 23 & 0 & 39 & 100 & 15 & 18 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -3 & 7 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{113\bar{1}} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 12 & 22 & 0 & 15 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{13\bar{1}} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 33 & 3 & 4 \\ 18 & 32 & 0 & 33 & 193 & 21 & 25 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -3 & 4 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{14\bar{1}} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 30 & 3 & 4 \\ 17 & 31 & 0 & 30 & 172 & 21 & 24 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -3 & 4 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{113\bar{1}} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 2 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 36 & 6 & 7 \\ 12 & 22 & 0 & 36 & 85 & 15 & 17 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -3 & 7 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{122\bar{1}} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 60 & 12 & 14 \\ 10 & 18 & 0 & 60 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 12 & 0 & 2 \\ 3 & 4 & -3 & 14 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{14\bar{1}} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 2 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 60 & 6 & 7 \\ 20 & 36 & 0 & 60 & 238 & 24 & 28 \\ 2 & 4 & 0 & 6 & 24 & 0 & 2 \\ 3 & 4 & -3 & 7 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{111} 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111} 3 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 7 & 4 & 4 \\ 5 & 10 & 0 & 7 & 14 & 7 & 8 \\ 2 & 5 & 0 & 4 & 7 & 1 & 3 \\ 3 & 5 & -3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 1 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 2 & 3 \\ 1 & 2 & 1 & 0 & 2 & 0 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 4 \\ 2 & 2 & 2 & -3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 3 21^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 15 & 4 \\ 2 & 5 & 0 & 4 & 2 & 15 & 4 \\ 9 & 20 & 0 & 15 & 15 & 57 & 15 \\ 3 & 5 & -3 & 4 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1111} 3 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 3 \\ 1 & 2 & 0 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 5 & 7 \\ 2 & 3 & 4 & 0 & 4 & 7 & 4 \\ 2 & 2 & 2 & -3 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1111} 3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 20 & 5 \\ 9 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 0 & 63 & 15 \\ 5 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{111} 3 21^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 11 & 4 & 4 \\ 7 & 15 & 0 & 11 & 32 & 11 & 12 \\ 2 & 5 & 0 & 4 & 11 & 1 & 3 \\ 3 & 5 & -3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1111} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 2 & 2 & 2 & -3 & 2 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{12}321^{1131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 18 & 4 & 4 \\ 12 & 25 & 0 & 18 & 94 & 18 & 20 \\ 2 & 5 & 0 & 4 & 18 & 1 & 3 \\ 3 & 5 & -3 & 4 & 20 & 3 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111}321^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 16 & 7 & 8 \\ 5 & 10 & 0 & 16 & 14 & 7 & 8 \\ 2 & 5 & 0 & 7 & 7 & 1 & 3 \\ 3 & 5 & -3 & 8 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{13}321^{1141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 25 & 4 & 4 \\ 17 & 35 & 0 & 25 & 188 & 25 & 28 \\ 2 & 5 & 0 & 4 & 25 & 1 & 3 \\ 3 & 5 & -3 & 4 & 28 & 3 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{111}321^{1121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 390, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 5 & 0 & 10 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 23 & 7 & 8 \\ 7 & 15 & 0 & 23 & 32 & 11 & 12 \\ 2 & 5 & 0 & 7 & 11 & 1 & 3 \\ 3 & 5 & -3 & 8 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111}321^{1231} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 630, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 2 & 3 \\ 3 & 5 & 0 & 15 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 45 & 11 & 12 \\ 9 & 20 & 0 & 45 & 58 & 15 & 16 \\ 2 & 5 & 0 & 11 & 15 & 1 & 3 \\ 3 & 5 & -3 & 12 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{12}321^{1141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 22 & 4 & 4 \\ 14 & 30 & 0 & 22 & 134 & 22 & 24 \\ 2 & 5 & 0 & 4 & 22 & 1 & 3 \\ 3 & 5 & -3 & 4 & 24 & 3 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{111}321^{1221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 720, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 34 & 11 & 12 \\ 7 & 15 & 0 & 34 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 11 & 1 & 3 \\ 3 & 5 & -3 & 12 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{1111}321^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 7 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 2 & 2 & 2 & -3 & 5 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{111}3^{11}21^{1331} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 840, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 2 & 3 & 3 \\ 3 & 5 & 0 & 20 & 20 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 & 3 \\ 9 & 20 & 0 & 58 & 60 & 15 & 16 & 16 \\ 9 & 20 & 0 & 60 & 58 & 15 & 16 & 16 \\ 2 & 5 & 0 & 15 & 15 & 1 & 3 & 3 \\ 3 & 5 & -3 & 16 & 16 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

4.469 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{111}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 5 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 2 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 2 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 5 & 6 \\ 2 & 3 & 0 & 2 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 2 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 5 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 6 & 5 & 7 \\ 5 & 9 & 0 & 15 & 13 & 6 & 5 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 5 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.470 NHC(-2)(-3) + f₄ + su₂ + su₂→(-3)

$$T = 8 \quad [0] \quad 1^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 6 & 9 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 10 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 0 & 22 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 21^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 3 & 4 & -2 & 3 & 2 & 0 \\ 3 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 3 & 4 & -2 & 3 & 2 & 0 \\ 3 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 \\ 5 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$\begin{aligned}
& T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[9] & \begin{matrix} 1^1 \\ 13^1 1^{11} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20] \\
& \begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 \\ 3 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0} \\
[10] & \begin{matrix} 1^1 \\ 1^{11} 3^1 1^1 \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.12 \rightarrow 13, T_{\max} = 20, T \in [13, 20] \\
& \begin{pmatrix} 1 & 4 & 2 & 4 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 \\ 4 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0} \\
[12] & \begin{matrix} 1^1 \\ 1^{11} 3^1 1^{11} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0} \\
[11] & \begin{matrix} 1^1 \\ 1^{11} 3^1 1^{14} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.14 \rightarrow 13, T_{\max} = 20, T \in [13, 20] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 24 & 0 & 121 & 14 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 19 & 2 & 0 \end{pmatrix}_{b_0} \\
[14] & \begin{matrix} 1^1 \\ 1^{11} 3^1 1^{12} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0} \\
[13] & \begin{matrix} 1^1 \\ 1^{11} 3^1 1^{13} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0} \\
[15] & \begin{matrix} 1^1 \\ 1^{12} 3^1 1^{11} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0} \\
[16] & \begin{matrix} 1^1 \\ 1^{13} 3^1 1^{11} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 9 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [17] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 11 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$
- [19] $21^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 3 & 4 \\ 8 & 9 & 4 & 8 & 4 & 5 \\ 3 & 4 & -2 & 3 & 2 & 0 \\ 6 & 8 & 3 & 4 & 3 & 4 \\ 3 & 4 & 2 & 3 & 0 & 2 \\ 4 & 5 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 12 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$
- [22] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 14 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 15 & 21 \\ 2 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$
- [25] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 882, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 18 & 25 \\ 2 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{14}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 31 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 17 & 24 \\ 2 & 3 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{14}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 20 & 28 \\ 2 & 3 & 0 & 20 & 0 & 2 \\ 3 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [28] $1^{11}3^{12}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 11 & 5 \\ 6 & 5 & 1 & 10 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 10 & 0 & 18 & 8 \\ 5 & 5 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{12}3^{12}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 12 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{11}3^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 \\ 2 & 6 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 6 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^{12}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 16 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{12}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 1 & 5 \\ 6 & 5 & 1 & 0 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 5 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{11}3^{12}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 6 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11}3^{11}1^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0} \\
[35] \quad & 1^{12}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 13 & 0 & 44 & 14 \\ 3 & 6 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0} \\
[36] \quad & 1^{11}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[37] \quad & 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 2 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0} \\
[38] \quad & 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0} \\
[39] \quad & 1^{13}3^{12}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 38 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0} \\
[40] \quad & 1^{12}3^{12}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 22 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0} \\
[41] \quad & 1^{111}3^{12}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 7 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0} \\
[42] \quad & 1^{13}3^{12}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 28 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[44] \quad & 1^{112}3^{12}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 6 & 1 & 8 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0} \\
[43] \quad & 1^{12} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -145, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 28 & 5 \\ 6 & 5 & 1 & 27 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 27 & 0 & 125 & 21 \\ 5 & 5 & -1 & 21 & 3 \end{pmatrix}_{b_0} \\
[45] \quad & 1^{14} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.75 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 34 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[46] \quad & 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -159, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 17 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0} \\
[48] \quad & 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 13 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[47] \quad & 1^{14} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 44 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[49] \quad & 1^{13} 3^1 1^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.04 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 2 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 5 & 0 & 8 & 6 \\ 3 & 6 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0} \\
[50] \quad & 1^{13} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 49 & 5 \\ 6 & 5 & 1 & 51 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 51 & 0 & 421 & 39 \\ 5 & 5 & -1 & 39 & 3 \end{pmatrix}_{b_0} \\
[51] \quad & 1^{13} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 19 & 5 \\ 6 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 21 & 0 & 61 & 15 \\ 5 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[52] \quad & 1^{13} 3^1 1^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{14}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 25 & 5 \\ 6 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 28 & 0 & 110 & 20 \\ 5 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{12}3^121^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 44 & 5 \\ 6 & 5 & 1 & 44 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 44 & 0 & 326 & 34 \\ 5 & 5 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{12}3^121^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 33 & 5 \\ 6 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 34 & 0 & 186 & 26 \\ 5 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{12}3^11^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 7 & 10 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.74 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 18 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{12}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -266, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 14 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{13}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -274, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 8 & 6 & 0 & 35 & 6 \\ 2 & 6 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 35 & 22 & 0 & 134 & 24 \\ 3 & 6 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 6 & 1 & 23 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0} \\
[62] \quad & 1^{13}31^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 10 & 0 & 38 & 12 \\ 3 & 6 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0} \\
[63] \quad & 1^{13}31^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -306, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0} \\
[64] \quad & 1^{11}3^{11}1^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -311, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0} \\
[65] \quad & 1^{12}3^{12}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -369, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 20 & 5 \\ 6 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 61 & 15 \\ 5 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[66] \quad & 1^{14}3^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 18 & 0 & 118 & 14 & 20 \\ 1 & 3 & 2 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0} \\
[68] \quad & 1^{13}3^{11}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -395, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 39 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0} \\
[67] \quad & 1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -395, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 29 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[69] \quad & 1^{12}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -426, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 34 & 5 \\ 6 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 34 & 0 & 186 & 26 \\ 5 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0} \\
[71] \quad & 1^{14}3^{11}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 5 & 6 & 1 & 55 & 5 \\ 6 & 5 & 1 & 58 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 55 & 58 & 0 & 530 & 44 \\ 5 & 5 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{12}3^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 24 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12}3^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -454, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 26 & 5 \\ 6 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 28 & 0 & 110 & 20 \\ 5 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13}1^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -481, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.07 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 9 & 0 & 52 & 9 & 12 \\ 1 & 3 & 2 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{14}3^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -486, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 35 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{13}1^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.14 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 8 & 6 & 0 & 30 & 6 \\ 2 & 6 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 30 & 15 & 0 & 88 & 18 \\ 3 & 6 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{12}31^{12}3^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 8 & 6 & 0 & 30 & 6 \\ 2 & 6 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 30 & 18 & 0 & 94 & 20 \\ 3 & 6 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{11}31^{12}3^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.02 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 8 & 6 & 0 & 25 & 6 \\ 2 & 6 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 25 & 14 & 0 & 62 & 16 \\ 3 & 6 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{13}3^121^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -549, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 50 & 5 \\ 6 & 5 & 1 & 51 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 51 & 0 & 421 & 39 \\ 5 & 5 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{14}3^121^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -566, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 45 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{11}3^{11}1^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{13}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -629, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 30 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{14}31^{13}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -664, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 31 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 31 & 0 & 272 & 34 \\ 3 & 6 & 3 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{13}3^{11}1^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 15 & 0 & 85 & 12 & 17 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{13}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -689, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 40 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{13}3^{11}1^{14}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 12 & 0 & 94 & 12 & 16 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{12}3^{11}1^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -746, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 12 & 0 & 58 & 10 & 14 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{13}3^{14}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 8 & 6 & 0 & 40 & 6 \\ 2 & 6 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 20 & 0 & 158 & 24 \\ 3 & 6 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{14}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 36 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{12}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -874, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 56 & 5 \\ 6 & 5 & 1 & 58 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 58 & 0 & 530 & 44 \\ 5 & 5 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{11}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -874, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 8 & 6 & 0 & 35 & 6 \\ 2 & 6 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 35 & 19 & 0 & 122 & 22 \\ 3 & 6 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{12}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -934, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 46 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{13}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 8 & 6 & 0 & 45 & 6 \\ 2 & 6 & 1 & 0 & 27 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 45 & 27 & 0 & 214 & 30 \\ 3 & 6 & 3 & -3 & 30 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{11}3^11^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1011, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 12 & 0 & 73 & 11 & 15 \\ 1 & 3 & 2 & 0 & 11 & 0 & 2 \\ 2 & 4 & 2 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{12}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 8 & 6 & 0 & 40 & 6 \\ 2 & 6 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 23 & 0 & 164 & 26 \\ 3 & 6 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{13}3^11^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1041, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 15 & 0 & 148 & 15 & 20 \\ 1 & 3 & 2 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{13}3^11^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 25 & 0 & 248 & 30 \\ 3 & 6 & 3 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{11}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 8 & 6 & 0 & 45 & 6 \\ 2 & 6 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 45 & 24 & 0 & 202 & 28 \\ 3 & 6 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{12}3^11^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$\begin{aligned}
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 15 & 0 & 100 & 13 & 18 \\ 1 & 3 & 2 & 0 & 13 & 0 & 2 \\ 2 & 4 & 2 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0} \\
[99] \quad & 1^{14}3^{11}1^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 21 & 0 & 172 & 17 & 24 \\ 1 & 3 & 2 & 0 & 17 & 0 & 2 \\ 2 & 4 & 2 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0} \\
[100] \quad & 1^{11}3^{11}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 15 & 0 & 121 & 14 & 19 \\ 1 & 3 & 2 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -3 & 19 & 2 & 1 \end{pmatrix}_{b_0} \\
[101] \quad & 1^{13}3^{11}1^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1471, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 18 & 0 & 133 & 15 & 21 \\ 1 & 3 & 2 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0} \\
[102] \quad & 1^{14}31^{14}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 24 & 3 \\ 4 & 8 & 6 & 0 & 60 & 6 \\ 2 & 6 & 1 & 0 & 36 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 24 & 60 & 36 & 0 & 382 & 40 \\ 3 & 6 & 3 & -3 & 40 & 2 \end{pmatrix}_{b_0} \\
[103] \quad & 1^{12}31^{14}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 28 & 0 & 254 & 32 \\ 3 & 6 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0} \\
[104] \quad & 1^{13}31^{14}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.00 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 22 & 3 \\ 4 & 8 & 6 & 0 & 55 & 6 \\ 2 & 6 & 1 & 0 & 32 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 55 & 32 & 0 & 314 & 36 \\ 3 & 6 & 3 & -3 & 36 & 1 \end{pmatrix}_{b_0} \\
[105] \quad & 1^{12}3^{11}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2006, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 18 & 0 & 154 & 16 & 22 \\ 1 & 3 & 2 & 0 & 16 & 0 & 2 \\ 2 & 4 & 2 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0} \\
[106] \quad & 1^{13}3^{11}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 0 & 33 & 3 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 33 & 21 & 0 & 193 & 18 & 25 \\ 1 & 3 & 2 & 0 & 18 & 0 & 2 \\ 2 & 4 & 2 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0} \\
[107] \quad & 1^{14}3^{11}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 0 & 36 & 3 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 36 & 24 & 0 & 238 & 20 & 28 \\ 1 & 3 & 2 & 0 & 20 & 0 & 2 \\ 2 & 4 & 2 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [108] \quad 1^{11} 4 13^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{11} 4 13^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{11} 4 13^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{11} 4 1^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{14} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 1 & 3 \\ 3 & 4 & 0 & 19 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 65 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 15 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{11} 4 1^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 7 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{12} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 11 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{13} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 15 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 40 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\textcolor{red}{11}} 1^{\textcolor{blue}{13}} 3^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 128, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 12 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\textcolor{red}{12}} 1^{\textcolor{blue}{13}} 3^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 176, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 1 & 3 \\ 3 & 4 & 0 & 16 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 45 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\textcolor{red}{14}} 1^{\textcolor{blue}{13}} 3^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 176, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 1 & 3 \\ 3 & 4 & 0 & 24 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 101 & 0 & 19 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 19 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\textcolor{red}{13}} 1^{\textcolor{blue}{13}} 3^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 192, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 1 & 3 \\ 3 & 4 & 0 & 20 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 70 & 0 & 16 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 16 & 0 & 2 \end{pmatrix}_{b_0}$$

$$4.471 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$$

$$T = 9 \quad [0] \ 1^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} 3^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 3 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$$4.472 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} 3^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 200, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} 3^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 600, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 9 & 6 & 7 \\ 2 & 3 & 0 & 5 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 9 & 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.473 NHC(-2)(-3) + f₄ + su₂ + su₂→(-3) + so₈

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 3 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 4 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

4.474 NHC(-2)(-3) + f₄ + su₂ + su₂→(-3) + su₂

$$T = 9 \quad [1] \ 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 8 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 12 & 6 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 4 & 8 & 0 & 12 & 4 & 10 & 6 \\ 3 & 4 & -3 & 6 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 17 & 3 \\ 3 & 4 & 0 & 8 & 3 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 42 & 6 \\ 2 & 3 & 0 & 4 & 0 & 17 & 2 \\ 17 & 31 & 0 & 42 & 17 & 172 & 24 \\ 3 & 4 & -3 & 6 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 12 & 6 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 5 & 9 & 0 & 12 & 5 & 13 & 7 \\ 3 & 4 & -3 & 6 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 3 & 0 & 9 & 2 \\ 9 & 17 & 0 & 15 & 9 & 49 & 13 \\ 3 & 4 & -3 & 4 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 4 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 6 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 1 & 3 \\ 3 & 4 & -3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13}3^{111}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12}3^{11}1^2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 12 & 3 \\ 3 & 4 & 0 & 8 & 3 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 30 & 6 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 12 & 22 & 0 & 30 & 12 & 85 & 17 \\ 3 & 4 & -3 & 6 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 8 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 18 & 6 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 7 & 13 & 0 & 18 & 7 & 28 & 10 \\ 3 & 4 & -3 & 6 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 3 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 9 & 3 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 14 & 0 & 9 & 8 & 37 & 11 \\ 3 & 4 & -3 & 3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 3 & 0 & 7 & 2 \\ 7 & 13 & 0 & 12 & 7 & 28 & 10 \\ 3 & 4 & -3 & 4 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 \\ 5 & 9 & 0 & 9 & 5 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{123}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 8 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 24 & 6 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 9 & 17 & 0 & 24 & 9 & 49 & 13 \\ 3 & 4 & -3 & 6 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 9 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 33 & 7 \\ 2 & 3 & 0 & 5 & 0 & 11 & 2 \\ 11 & 21 & 0 & 33 & 11 & 76 & 16 \\ 3 & 4 & -3 & 7 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 11 & 3 \\ 3 & 4 & 0 & 8 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 30 & 6 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 11 & 21 & 0 & 30 & 11 & 76 & 16 \\ 3 & 4 & -3 & 6 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 9 & 3 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 13 & 0 & 9 & 7 & 28 & 10 \\ 3 & 4 & -3 & 3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{124}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 14 & 3 \\ 3 & 4 & 0 & 8 & 3 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 36 & 6 \\ 2 & 3 & 0 & 4 & 0 & 14 & 2 \\ 14 & 26 & 0 & 36 & 14 & 118 & 20 \\ 3 & 4 & -3 & 6 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 15 & 7 \\ 2 & 3 & 0 & 5 & 0 & 5 & 2 \\ 5 & 9 & 0 & 15 & 5 & 13 & 7 \\ 3 & 4 & -3 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 3 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 15 & 3 \\ 2 & 3 & 0 & 2 & 0 & 13 & 2 \\ 13 & 23 & 0 & 15 & 13 & 100 & 18 \\ 3 & 4 & -3 & 3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 12 & 3 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 9 & 17 & 0 & 12 & 9 & 49 & 13 \\ 3 & 4 & -3 & 3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 27 & 7 \\ 2 & 3 & 0 & 5 & 0 & 9 & 2 \\ 9 & 17 & 0 & 27 & 9 & 49 & 13 \\ 3 & 4 & -3 & 7 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 1^{13} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 3 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 12 & 3 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 10 & 18 & 0 & 12 & 10 & 58 & 14 \\ 3 & 4 & -3 & 3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 15 & 3 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 21 & 0 & 15 & 11 & 76 & 16 \\ 3 & 4 & -3 & 3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 21 & 7 \\ 2 & 3 & 0 & 5 & 0 & 7 & 2 \\ 7 & 13 & 0 & 21 & 7 & 28 & 10 \\ 3 & 4 & -3 & 7 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 11 & 3 \\ 3 & 4 & 0 & 13 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 48 & 10 \\ 2 & 3 & 0 & 7 & 0 & 11 & 2 \\ 11 & 21 & 0 & 48 & 11 & 76 & 16 \\ 3 & 4 & -3 & 10 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 13 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 30 & 10 \\ 2 & 3 & 0 & 7 & 0 & 7 & 2 \\ 7 & 13 & 0 & 30 & 7 & 28 & 10 \\ 3 & 4 & -3 & 10 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 9 & 3 \\ 3 & 4 & 0 & 13 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 39 & 10 \\ 2 & 3 & 0 & 7 & 0 & 9 & 2 \\ 9 & 17 & 0 & 39 & 9 & 49 & 13 \\ 3 & 4 & -3 & 10 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1133}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 3 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 15 & 3 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 12 & 22 & 0 & 15 & 12 & 85 & 17 \\ 3 & 4 & -3 & 3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{133}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 9 & 3 \\ 3 & 4 & 0 & 17 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 9 & 51 & 13 \\ 2 & 3 & 0 & 9 & 0 & 9 & 2 \\ 9 & 17 & 0 & 51 & 9 & 49 & 13 \\ 3 & 4 & -3 & 13 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{113}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 3 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 21 & 3 \\ 2 & 3 & 0 & 2 & 0 & 18 & 2 \\ 18 & 32 & 0 & 21 & 18 & 193 & 25 \\ 3 & 4 & -3 & 3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{134}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 11 & 3 \\ 3 & 4 & 0 & 17 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 9 & 63 & 13 \\ 2 & 3 & 0 & 9 & 0 & 11 & 2 \\ 11 & 21 & 0 & 63 & 11 & 76 & 16 \\ 3 & 4 & -3 & 13 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{113}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 3 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 18 & 3 \\ 2 & 3 & 0 & 2 & 0 & 15 & 2 \\ 15 & 27 & 0 & 18 & 15 & 133 & 21 \\ 3 & 4 & -3 & 3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{144}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 11 & 3 \\ 3 & 4 & 0 & 21 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 11 & 78 & 16 \\ 2 & 3 & 0 & 11 & 0 & 11 & 2 \\ 11 & 21 & 0 & 78 & 11 & 76 & 16 \\ 3 & 4 & -3 & 16 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{114}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 3 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 18 & 3 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 14 & 26 & 0 & 18 & 14 & 118 & 20 \\ 3 & 4 & -3 & 3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{114}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 3 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 21 & 3 \\ 2 & 3 & 0 & 2 & 0 & 17 & 2 \\ 17 & 31 & 0 & 21 & 17 & 172 & 24 \\ 3 & 4 & -3 & 3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1604, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 3 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 24 & 3 \\ 2 & 3 & 0 & 2 & 0 & 20 & 2 \\ 20 & 36 & 0 & 24 & 20 & 238 & 28 \\ 3 & 4 & -3 & 3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{11}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 247, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 9 \\ 1 & 4 & 0 & 0 & 3 & 2 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 1 & 3 & 3 & 0 & 1 & 2 & 6 \\ 1 & 3 & 2 & 0 & 2 & 0 & 5 \\ 3 & 9 & 6 & 0 & 6 & 5 & 13 \\ 2 & 4 & 2 & -3 & 3 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 472, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 2 & 6 & 3 & 0 & 4 & 3 & 6 \\ 1 & 3 & 2 & 0 & 3 & 0 & 3 \\ 2 & 6 & 3 & 0 & 6 & 3 & 4 \\ 2 & 4 & 2 & -3 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 517, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 \\ 2 & 4 & 4 & 0 & 6 & 3 & 9 \\ 1 & 4 & 0 & 0 & 3 & 2 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 2 & 6 & 3 & 0 & 4 & 3 & 9 \\ 1 & 3 & 2 & 0 & 3 & 0 & 5 \\ 3 & 9 & 6 & 0 & 9 & 5 & 13 \\ 2 & 4 & 2 & -3 & 4 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 648, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 24 & 10 \\ 6 & 15 & 9 & 0 & 24 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{11}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 912, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 3 \\ 2 & 4 & 4 & 0 & 9 & 3 & 9 \\ 1 & 4 & 0 & 0 & 6 & 2 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 3 & 9 & 6 & 0 & 13 & 5 & 15 \\ 1 & 3 & 2 & 0 & 5 & 0 & 5 \\ 3 & 9 & 6 & 0 & 15 & 5 & 13 \\ 2 & 4 & 2 & -3 & 7 & 2 & 7 \end{pmatrix}_{b_0}$$

4.475 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 \\ 3 & 4 & 0 & 5 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 0 \\ 3 & 4 & -3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 0 \\ 3 & 4 & -3 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 9 & 5 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 6 \\ 2 & 3 & 0 & 2 & 0 & 2 & 5 \\ 2 & 4 & 0 & 3 & 2 & 1 & 6 \\ 5 & 9 & 0 & 6 & 5 & 6 & 12 \\ 3 & 4 & -3 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 6 & 5 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 15 & 6 & 7 \\ 2 & 3 & 0 & 5 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 15 & 5 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.476 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 & 4 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 4 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 5 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 5 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 475, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 5 & 9 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 9 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 7 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.477 NHC(-2)(-3) + f₄ + su₂ + su₂→(-3) + su₃

$$T = 9 \quad [0] \quad 1^{121}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 12 & 6 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 5 & 9 & 0 & 12 & 5 & 12 & 6 \\ 3 & 4 & -3 & 6 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 3 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 6 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 11 & 12 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 4 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 4 & 8 & 0 & 6 & 4 & 9 & 5 \\ 3 & 4 & -3 & 3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[7] \quad & 1^{11} 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 4 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 12 & 7 \\ 2 & 3 & 0 & 5 & 0 & 4 & 2 \\ 4 & 8 & 0 & 12 & 4 & 9 & 5 \\ 3 & 4 & -3 & 7 & 2 & 5 & 0 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{11} 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 12 & 6 \\ 3 & 4 & -3 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{11} 1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 8 & 9 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 2 & 4 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 9 & 3 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 13 & 0 & 9 & 7 & 27 & 9 \\ 3 & 4 & -3 & 3 & 2 & 9 & 1 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{12} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.7 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 9 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 4 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[16] $1^{13}1^13^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 9 & 12 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 4 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{12}1^13^11^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 28 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 16 & 18 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 & 2 \\ 2 & 4 & 0 & 18 & 2 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{14}1^13^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 11 & 15 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 4 & 0 & 15 & 2 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{12}1^13^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 10 & 12 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 & 2 \\ 2 & 4 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{12}1^13^11^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 23 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 13 & 15 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 & 2 \\ 2 & 4 & 0 & 15 & 2 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{13}1^13^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 12 & 15 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 2 & 4 & 0 & 15 & 2 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{13}1^13^11^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 2 & 3 \\ 3 & 4 & 0 & 27 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 15 & 18 & 21 \\ 2 & 3 & 0 & 15 & 0 & 2 & 2 \\ 2 & 4 & 0 & 18 & 2 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{14}1^13^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 14 & 18 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 2 & 4 & 0 & 18 & 2 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13}1^13^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 2 & 3 \\ 3 & 4 & 0 & 32 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 18 & 21 & 25 \\ 2 & 3 & 0 & 18 & 0 & 2 & 2 \\ 2 & 4 & 0 & 21 & 2 & 0 & 2 \\ 3 & 4 & -3 & 25 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{14}1^13^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 31 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 17 & 21 & 24 \\ 2 & 3 & 0 & 17 & 0 & 2 & 2 \\ 2 & 4 & 0 & 21 & 2 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14}1^13^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 2 & 3 \\ 3 & 4 & 0 & 36 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 20 & 24 & 28 \\ 2 & 3 & 0 & 20 & 0 & 2 & 2 \\ 2 & 4 & 0 & 24 & 2 & 0 & 2 \\ 3 & 4 & -3 & 28 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11}1^{11}4^{13}1^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11}1^{11}1^13^11^14^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 2 & 3 \\ 3 & 4 & 0 & 7 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 5 & 0 & 0 & 2 \\ 3 & 4 & -2 & 6 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11}1^{11}1^13^11^14^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 2 & 3 \\ 3 & 4 & 0 & 12 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 0 & 8 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 8 & 0 & 0 & 2 \\ 3 & 4 & -2 & 10 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$4.478 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$$

$$T = 10 \quad [0] \quad 1^{11}1^{11}1^13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 3 \\ 2 & 4 & 0 & 3 & 2 & 0 & 3 \\ 3 & 5 & 0 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^{11}1^13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 1 \\ 3 & 4 & -3 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 6 & 5 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 6 & 5 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 6 & 15 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 6 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 7 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.479 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + su2n3mix

$$T = 8 \quad [1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 14 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 17 & 11 \\ 4 & 7 & 0 & 17 & 7 & 5 \\ 3 & 4 & -3 & 11 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 19 & 10 \\ 5 & 8 & 0 & 19 & 12 & 6 \\ 3 & 4 & -3 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{153}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 8 & 3 \\ 3 & 4 & 0 & 20 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 55 & 15 \\ 8 & 15 & 0 & 55 & 39 & 11 \\ 3 & 4 & -3 & 15 & 11 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 23 & 7 \\ 8 & 15 & 0 & 23 & 39 & 11 \\ 3 & 4 & -3 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 4 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{131}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 15 & 9 \\ 4 & 7 & 0 & 15 & 7 & 5 \\ 3 & 4 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{142}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 6 & 3 \\ 3 & 4 & 0 & 31 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 59 & 24 \\ 6 & 11 & 0 & 59 & 20 & 8 \\ 3 & 4 & -3 & 24 & 8 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{142}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 16 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 32 & 12 \\ 6 & 11 & 0 & 32 & 20 & 8 \\ 3 & 4 & -3 & 12 & 8 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 13 & 22 & 0 & 14 & 97 & 17 \\ 3 & 4 & -3 & 3 & 17 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{123}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 14 & 25 & 0 & 34 & 111 & 19 \\ 3 & 4 & -3 & 6 & 19 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{143}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 8 & 3 \\ 3 & 4 & 0 & 26 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 68 & 20 \\ 8 & 15 & 0 & 68 & 39 & 11 \\ 3 & 4 & -3 & 20 & 11 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 12 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 51 & 9 \\ 13 & 24 & 0 & 51 & 100 & 18 \\ 3 & 4 & -3 & 9 & 18 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{121}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 23 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 28 & 18 \\ 4 & 7 & 0 & 28 & 7 & 5 \\ 3 & 4 & -3 & 18 & 5 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{143}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 13 & 24 & 0 & 21 & 100 & 18 \\ 3 & 4 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 8 & 0 & 9 & 12 & 6 \\ 3 & 4 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 43 & 10 \\ 10 & 19 & 0 & 43 & 64 & 14 \\ 3 & 4 & -3 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{154}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 20 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 70 & 15 \\ 10 & 19 & 0 & 70 & 64 & 14 \\ 3 & 4 & -3 & 15 & 14 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{132}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 12 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 24 & 9 \\ 6 & 11 & 0 & 24 & 20 & 8 \\ 3 & 4 & -3 & 9 & 8 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 22 & 6 \\ 9 & 16 & 0 & 22 & 44 & 12 \\ 3 & 4 & -3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{143}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 16 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 44 & 12 \\ 8 & 15 & 0 & 44 & 39 & 11 \\ 3 & 4 & -3 & 12 & 11 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 7 \\ 6 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 14 & 7 \\ 5 & 8 & 0 & 14 & 12 & 6 \\ 3 & 4 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 6 & 11 & 0 & 16 & 20 & 8 \\ 3 & 4 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 10 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{131}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 36 & 13 \\ 7 & 12 & 0 & 36 & 25 & 9 \\ 3 & 4 & -3 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{133}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 11 & 20 & 0 & 18 & 69 & 15 \\ 3 & 4 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 12 & 0 & 12 & 25 & 9 \\ 3 & 4 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{133}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 12 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 33 & 9 \\ 8 & 15 & 0 & 33 & 39 & 11 \\ 3 & 4 & -3 & 9 & 11 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{112}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{144}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 16 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 56 & 12 \\ 10 & 19 & 0 & 56 & 64 & 14 \\ 3 & 4 & -3 & 12 & 14 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{131}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 4 & 3 \\ 3 & 4 & 0 & 32 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 39 & 25 \\ 4 & 7 & 0 & 39 & 7 & 5 \\ 3 & 4 & -3 & 25 & 5 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{123}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 16 & 0 & 15 & 44 & 12 \\ 3 & 4 & -3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 13 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 34 & 10 \\ 8 & 15 & 0 & 34 & 39 & 11 \\ 3 & 4 & -3 & 10 & 11 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{132}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 6 & 3 \\ 3 & 4 & 0 & 22 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 42 & 17 \\ 6 & 11 & 0 & 42 & 20 & 8 \\ 3 & 4 & -3 & 17 & 8 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{123}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 22 & 6 \\ 8 & 15 & 0 & 22 & 39 & 11 \\ 3 & 4 & -3 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{134}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 12 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 42 & 9 \\ 10 & 19 & 0 & 42 & 64 & 14 \\ 3 & 4 & -3 & 9 & 14 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 8 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{121}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 18 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 22 & 14 \\ 4 & 7 & 0 & 22 & 7 & 5 \\ 3 & 4 & -3 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 16 & 10 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 25 & 10 \\ 6 & 11 & 0 & 25 & 20 & 8 \\ 3 & 4 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{142}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 4 & 0 & 21 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 59 & 16 \\ 9 & 16 & 0 & 59 & 44 & 12 \\ 3 & 4 & -3 & 16 & 12 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 28 & 6 \\ 11 & 20 & 0 & 28 & 69 & 15 \\ 3 & 4 & -3 & 6 & 15 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 57 & 13 \\ 10 & 19 & 0 & 57 & 64 & 14 \\ 3 & 4 & -3 & 13 & 14 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{124}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 28 & 6 \\ 10 & 19 & 0 & 28 & 64 & 14 \\ 3 & 4 & -3 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{112}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 362, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 18 & 31 & 0 & 20 & 188 & 24 \\ 3 & 4 & -3 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{121}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 28 & 10 \\ 7 & 12 & 0 & 28 & 25 & 9 \\ 3 & 4 & -3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{114}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 10 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 20 & 7 \\ 7 & 12 & 0 & 20 & 25 & 9 \\ 3 & 4 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{124}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 8 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 40 & 6 \\ 16 & 29 & 0 & 40 & 148 & 22 \\ 3 & 4 & -3 & 6 & 22 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 21 & 13 \\ 4 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{133}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 17 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 45 & 13 \\ 8 & 15 & 0 & 45 & 39 & 11 \\ 3 & 4 & -3 & 13 & 11 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 438, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 3 \\ 3 & 4 & 0 & 17 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 33 & 13 \\ 6 & 11 & 0 & 33 & 20 & 8 \\ 3 & 4 & -3 & 13 & 8 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 26 & 16 \\ 4 & 7 & 0 & 26 & 7 & 5 \\ 3 & 4 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{112}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 15 & 26 & 0 & 17 & 128 & 20 \\ 3 & 4 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{131}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 473, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 27 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 33 & 21 \\ 4 & 7 & 0 & 33 & 7 & 5 \\ 3 & 4 & -3 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 13 & 24 & 0 & 34 & 100 & 18 \\ 3 & 4 & -3 & 6 & 18 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{112}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 515, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 26 & 7 \\ 9 & 16 & 0 & 26 & 44 & 12 \\ 3 & 4 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{144}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 10 & 3 \\ 3 & 4 & 0 & 21 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 71 & 16 \\ 10 & 19 & 0 & 71 & 64 & 14 \\ 3 & 4 & -3 & 16 & 14 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{131}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 27 & 17 \\ 4 & 7 & 0 & 27 & 7 & 5 \\ 3 & 4 & -3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{142}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 4 & 0 & 21 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 41 & 16 \\ 6 & 11 & 0 & 41 & 20 & 8 \\ 3 & 4 & -3 & 16 & 8 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{142}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 4 & 0 & 26 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 50 & 20 \\ 6 & 11 & 0 & 50 & 20 & 8 \\ 3 & 4 & -3 & 20 & 8 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{132}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 555, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 48 & 13 \\ 9 & 16 & 0 & 48 & 44 & 12 \\ 3 & 4 & -3 & 13 & 12 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{143}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 8 & 3 \\ 3 & 4 & 0 & 21 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 56 & 16 \\ 8 & 15 & 0 & 56 & 39 & 11 \\ 3 & 4 & -3 & 16 & 11 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 577, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 11 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 32 & 7 \\ 11 & 20 & 0 & 32 & 69 & 15 \\ 3 & 4 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 38 & 7 \\ 13 & 24 & 0 & 38 & 100 & 18 \\ 3 & 4 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{122}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 37 & 10 \\ 9 & 16 & 0 & 37 & 44 & 12 \\ 3 & 4 & -3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{141}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 4 & 3 \\ 3 & 4 & 0 & 36 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 44 & 28 \\ 4 & 7 & 0 & 44 & 7 & 5 \\ 3 & 4 & -3 & 28 & 5 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{143}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 727, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 74 & 16 \\ 11 & 20 & 0 & 74 & 69 & 15 \\ 3 & 4 & -3 & 16 & 15 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{141}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 32 & 20 \\ 4 & 7 & 0 & 32 & 7 & 5 \\ 3 & 4 & -3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 13 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 14 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{123}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 807, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 46 & 10 \\ 11 & 20 & 0 & 46 & 69 & 15 \\ 3 & 4 & -3 & 10 & 15 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{141}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 4 & 3 \\ 3 & 4 & 0 & 31 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 38 & 24 \\ 4 & 7 & 0 & 38 & 7 & 5 \\ 3 & 4 & -3 & 24 & 5 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{133}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 60 & 13 \\ 11 & 20 & 0 & 60 & 69 & 15 \\ 3 & 4 & -3 & 13 & 15 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{113}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 865, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 17 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -3 & 3 & 23 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{113}3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 20 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -3 & 3 & 27 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{124}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 928, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 55 & 10 \\ 13 & 24 & 0 & 55 & 100 & 18 \\ 3 & 4 & -3 & 10 & 18 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{114}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1062, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 16 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{144}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1068, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 13 & 3 \\ 3 & 4 & 0 & 21 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 89 & 16 \\ 13 & 24 & 0 & 89 & 100 & 18 \\ 3 & 4 & -3 & 16 & 18 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{134}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1083, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 17 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 72 & 13 \\ 13 & 24 & 0 & 72 & 100 & 18 \\ 3 & 4 & -3 & 13 & 18 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{114}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 19 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{114}3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 26 & 3 \\ 22 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -3 & 3 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [92] $1^{124}3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 8 & 0 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 8 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & 0 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

- [93] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 2 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 3 & 3 \\ 2 & 2 & -1 & 0 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$
- [94] $1^{11}3^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 5 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 5 & 3 \\ 2 & 5 & 6 & 0 & 5 & 7 & 5 \\ 2 & 3 & 3 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$
- [95] $1^{11}3^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 13 & 3 \\ 6 & 17 & 13 & 0 & 13 & 60 & 14 \\ 2 & 4 & 2 & -3 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 3 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [98] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 2 & 0 \\ 6 & 10 & 0 & 10 & 58 & 16 & 14 \\ 2 & 3 & 0 & 2 & 16 & 3 & 3 \\ 2 & 2 & -1 & 0 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$
- [97] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 7 & 16 & 34 & 10 \\ 2 & 2 & -1 & 0 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$
- [99] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 \\ 2 & 2 & -1 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 6 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & 0 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{11}3^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [102] $1^{11}3^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 5 & 3 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 5 & 4 & 10 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 3 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 3 & 3 \\ 2 & 2 & -1 & 0 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{\overline{111}} 31^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 4 & 3 \\ 2 & 4 & 3 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 3 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{\overline{121}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & 0 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{\overline{111}} 313^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 4 & 3 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 3 & 2 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{\overline{121}} 31^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 11 & 4 \\ 1 & 4 & 0 & 0 & 9 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 25 & 10 \\ 4 & 11 & 7 & 0 & 25 & 20 & 8 \\ 2 & 4 & 2 & -3 & 10 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{\overline{111}} 31^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 0 & 6 & 4 & 3 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 6 & 0 & 13 & 9 & 7 \\ 2 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 3 & 2 & -3 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{\overline{122}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & 0 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{\overline{121}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 3 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & 0 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{11}3^11^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 5 & 4 \\ 2 & 5 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{11}3^11^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 11 & 7 \\ 2 & 6 & 5 & 0 & 11 & 7 & 5 \\ 2 & 4 & 2 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{11}3^12^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{11}3^12^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{11}3^12^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 1 & 1 & 11 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 2 & 0 \\ 9 & 11 & 0 & 13 & 81 & 18 & 17 \\ 2 & 3 & 0 & 2 & 18 & 3 & 3 \\ 2 & 2 & -1 & 0 & 17 & 3 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{11}3^12^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 5 & 4 & 10 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{12}3^12^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 7 & 16 & 34 & 10 \\ 2 & 2 & -1 & 0 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{11}3^11^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 8 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 12 & 7 \\ 3 & 8 & 4 & 0 & 12 & 9 & 5 \\ 2 & 4 & 2 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{11}3^11^{14}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.2

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 & 2 \\ 2 & 4 & 4 & 0 & 27 & 5 & 4 \\ 1 & 4 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 15 & 0 & 121 & 24 & 19 \\ 2 & 5 & 4 & 0 & 24 & 4 & 4 \\ 2 & 4 & 2 & -3 & 19 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{13} 3^{11} 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 3 & 0 \\ 4 & 7 & 0 & 6 & 25 & 18 & 9 \\ 3 & 5 & 0 & 3 & 18 & 11 & 5 \\ 2 & 2 & -1 & 0 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{11} 3^{11} 21^{12} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 9 & 7 & 35 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{13} 3^{11} 21^{12} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 2 & 0 \\ 6 & 8 & 0 & 9 & 40 & 13 & 12 \\ 2 & 3 & 0 & 2 & 13 & 3 & 3 \\ 2 & 2 & -1 & 0 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{13} 3^{11} 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 2 & 0 \\ 4 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & 0 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{12} 3^{11} 3^1 1^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 6 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 14 & 4 \\ 5 & 14 & 10 & 0 & 14 & 37 & 11 \\ 2 & 4 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{\overline{11}1}3^{\overline{11}1}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=13, \ \Delta=-88, \ \det=-175, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=10.2$$

$$T_{\min}=12.03 \rightarrow 13, \ T_{\max}=13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 7 & 3 \\ 4 & 11 & 7 & 0 & 7 & 20 & 8 \\ 2 & 4 & 2 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{\overline{11}1}3^{\overline{11}1}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=13, \ \Delta=-88, \ \det=-181, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=10.2$$

$$T_{\min}=11.96 \rightarrow 12, \ T_{\max}=13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 8 & 3 \\ 4 & 12 & 8 & 0 & 8 & 25 & 9 \\ 2 & 4 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{\overline{12}}3^{\overline{12}}2^{\overline{1}}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=12, \ \Delta=-82, \ \det=-185, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=11.0$$

$$T_{\min}=11.96 \rightarrow 12, \ T_{\max}=12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & 0 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{\overline{12}}3^{\overline{12}}2^{\overline{1}}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=12, \ \Delta=-82, \ \det=-192, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=10.2$$

$$T_{\min}=11.75 \rightarrow 12, \ T_{\max}=12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{\overline{11}1}3^{\overline{12}3}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=13, \ \Delta=-88, \ \det=-196, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=10.2$$

$$T_{\min}=12.29 \rightarrow 13, \ T_{\max}=13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 10 & 3 \\ 6 & 17 & 10 & 0 & 10 & 48 & 12 \\ 2 & 4 & 2 & -3 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{\overline{13}3}3^{\overline{12}}2^{\overline{1}}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=12, \ \Delta=-82, \ \det=-205, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=10.2$$

$$T_{\min}=11.89 \rightarrow 12, \ T_{\max}=12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 10 & 9 & 51 & 13 \\ 2 & 2 & -1 & 0 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{\overline{12}3}3^{\overline{11}3}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=13, \ \Delta=-88, \ \det=-209, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=10.2$$

$$T_{\min}=11.93 \rightarrow 12, \ T_{\max}=13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 10 & 3 \\ 5 & 14 & 10 & 0 & 10 & 37 & 11 \\ 2 & 4 & 2 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{\overline{14}1}3^{\overline{11}3}2^{\overline{1}}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=12, \ \Delta=-82, \ \det=-209, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=11.2$$

$$T_{\min}=11.81 \rightarrow 12, \ T_{\max}=12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 5 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & 0 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{\overline{12}1}3^{\overline{11}3}2^{\overline{1}}1^{\overline{1}}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2 \quad T=9, \ T_{\max}=12, \ \Delta=-82, \ \det=-214, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=10.7$$

$$T_{\min}=11.79 \rightarrow 12, \ T_{\max}=12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & 0 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

- [138] $1^{121}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & 0 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$
- [139] $1^{121}3^11^13^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 5 & 4 \\ 1 & 4 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 13 & 10 \\ 2 & 5 & 4 & 0 & 13 & 4 & 4 \\ 2 & 4 & 2 & -3 & 10 & 4 & 1 \end{pmatrix}_{b_0}$$
- [140] $1^{122}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 3 & 4 & 0 & 2 & 6 & 10 & 4 \\ 3 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & 0 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$
- [141] $1^{142}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & 0 & 11 & 5 & 1 \end{pmatrix}_{b_0}$$
- [142] $1^{123}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & 0 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$
- [143] $1^{112}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 9 & 7 & 35 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [144] $1^{124}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 8 & 0 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 7 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & 0 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$
- [145] $1^{111}3^11^14^13^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 17 & 3 \\ 10 & 30 & 17 & 0 & 17 & 151 & 21 \\ 2 & 4 & 2 & -3 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$
- [146] $1^{121}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 4 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & 0 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{124}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 8 & 0 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 6 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & 0 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{124}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{121}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 5 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & 0 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{111}3^11^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 5 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 5 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[151] $1^{113}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 9 & 10 & 0 & 13 & 10 & 74 & 16 \\ 2 & 2 & -1 & 0 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[152] $1^{113}3^11^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 6 & 2 \\ 2 & 4 & 4 & 0 & 9 & 17 & 4 \\ 1 & 4 & 0 & 0 & 6 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 30 & 7 \\ 6 & 17 & 13 & 0 & 30 & 60 & 14 \\ 2 & 4 & 2 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{131}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -278$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 2 & 0 \\ 4 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & 0 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{112}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{131}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -297$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

11.0

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 2 & 0 \\ 7 & 8 & 0 & 9 & 40 & 13 & 12 \\ 2 & 3 & 0 & 2 & 13 & 3 & 3 \\ 2 & 2 & -1 & 0 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{11}3^1 1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 11 & 3 \\ 6 & 18 & 11 & 0 & 11 & 55 & 13 \\ 2 & 4 & 2 & -3 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[157] \ 1^1 31^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 8 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 8 & 4 \\ 3 & 8 & 4 & 0 & 8 & 9 & 5 \\ 2 & 4 & 2 & -3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{11} 31^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 4 & 0 & 21 & 5 & 4 \\ 1 & 4 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 12 & 0 & 73 & 19 & 15 \\ 2 & 5 & 4 & 0 & 19 & 4 & 4 \\ 2 & 4 & 2 & -3 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{13} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & 0 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{11} 3^1 1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 4 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{14} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 5 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & 0 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{11} 3^1 1^{13} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 14 & 3 \\ 8 & 24 & 14 & 0 & 14 & 97 & 17 \\ 2 & 4 & 2 & -3 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{\overline{112}3^121^{\overline{11}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{\overline{114}3^11^{\overline{12}3^11^{\overline{1}}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 26 & 4 \\ 1 & 4 & 0 & 0 & 3 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 19 & 3 \\ 9 & 26 & 19 & 0 & 19 & 135 & 21 \\ 2 & 4 & 2 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{\overline{112}3^121^{\overline{12}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{\overline{124}3^121^{\overline{11}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{\overline{111}3^11^{\overline{12}3^11^{\overline{1}}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 & 2 \\ 2 & 4 & 4 & 0 & 15 & 5 & 4 \\ 1 & 4 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 14 & 11 \\ 2 & 5 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 2 & -3 & 11 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{\overline{1}3^11^{\overline{111}3^11^{\overline{1}}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -369, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 5 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{\overline{11}3^11^{\overline{111}3^11^{\overline{1}}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 14 & 7 \\ 3 & 9 & 5 & 0 & 14 & 12 & 6 \\ 2 & 4 & 2 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{\overline{131}3^121^{\overline{112}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 2 & 0 \\ 5 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & 0 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{\overline{142}3^121^{\overline{11}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 3 & 0 \\ 5 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & 0 & 8 & 5 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{141}3^{12}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 & 2 \\ 2 & 4 & 4 & 0 & 24 & 5 & 4 \\ 1 & 4 & 0 & 0 & 18 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 18 & 0 & 118 & 26 & 20 \\ 2 & 5 & 4 & 0 & 26 & 4 & 4 \\ 2 & 4 & 2 & -3 & 20 & 4 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{142}3^12^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -390, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 3 & 0 \\ 6 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & 0 & 8 & 5 & 1 \end{pmatrix}_{b_0}$$

[175] $1^{112}3^12^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{114}3^12^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 11 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 7 & 11 & 0 & 11 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{113}3^12^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{141}3^12^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 2 & 0 \\ 7 & 10 & 0 & 10 & 58 & 16 & 14 \\ 2 & 3 & 0 & 2 & 16 & 3 & 3 \\ 2 & 2 & -1 & 0 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{131}3^12^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 2 & 0 \\ 5 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & 0 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{113}3^12^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 10 & 9 & 51 & 13 \\ 2 & 2 & -1 & 0 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{113}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 11 & 4 \\ 4 & 11 & 7 & 0 & 11 & 20 & 8 \\ 2 & 4 & 2 & -3 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{114}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 6 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{123}31^{123}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -454$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 20 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 13 & 3 \\ 7 & 20 & 13 & 0 & 13 & 71 & 15 \\ 2 & 4 & 2 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{142}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -462$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 3 & 0 \\ 6 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & 0 & 11 & 5 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{122}31^{111}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 14 & 4 \\ 1 & 4 & 0 & 0 & 9 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 34 & 10 \\ 5 & 14 & 10 & 0 & 34 & 37 & 11 \\ 2 & 4 & 2 & -3 & 10 & 11 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{114}3^{11}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -469$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 14 & 0 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 9 & 12 & 0 & 14 & 12 & 96 & 18 \\ 2 & 2 & -1 & 0 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{113}31^{123}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 23 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 16 & 3 \\ 8 & 23 & 16 & 0 & 16 & 100 & 18 \\ 2 & 4 & 2 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{141}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -522$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 6 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & 0 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -541$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

- [190] $1^{1^{12}} 1^{131} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -542$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 2 & 0 \\ 6 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & 0 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$
- [191] $1^{1^{11}} 1^{114} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -544$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [192] $1^{1^{12}} 1^{113} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -557$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [193] $1^{1^{11}} 1^{141} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -574$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 6 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & 0 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$
- [194] $1^{1^{21}} 1^{123} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -586$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 4 & 0 & 18 & 5 & 4 \\ 1 & 4 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 12 & 0 & 58 & 18 & 14 \\ 2 & 5 & 4 & 0 & 18 & 4 & 4 \\ 2 & 4 & 2 & -3 & 14 & 4 & 1 \end{pmatrix}_{b_0}$$
- [195] $1^{1^{31}} 1^{123} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -596$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 4 & 0 & 21 & 5 & 4 \\ 1 & 4 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 15 & 0 & 85 & 22 & 17 \\ 2 & 5 & 4 & 0 & 22 & 4 & 4 \\ 2 & 4 & 2 & -3 & 17 & 4 & 1 \end{pmatrix}_{b_0}$$
- [196] $1^{1^{12}} 1^{113} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -597$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 10 & 9 & 51 & 13 \\ 2 & 2 & -1 & 0 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{1^{11}} 1^{111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -600$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 11 & 4 \\ 1 & 4 & 0 & 0 & 6 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 18 & 7 \\ 4 & 11 & 7 & 0 & 18 & 20 & 8 \\ 2 & 4 & 2 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{1^{11}} 1^{111} 3^1 1^{111} 3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -601$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 20 & 7 \\ 4 & 12 & 8 & 0 & 20 & 25 & 9 \\ 2 & 4 & 2 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{112}3^11^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 26 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 16 & 3 \\ 9 & 26 & 16 & 0 & 16 & 117 & 19 \\ 2 & 4 & 2 & -3 & 3 & 19 & 0 \end{pmatrix}_{b_0}$$

[200] $1^{112}3^11^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -649, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 14 & 4 \\ 1 & 4 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 24 & 7 \\ 5 & 14 & 10 & 0 & 24 & 37 & 11 \\ 2 & 4 & 2 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{112}3^11^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 32 & 4 \\ 1 & 4 & 0 & 0 & 3 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 19 & 3 \\ 11 & 32 & 19 & 0 & 19 & 175 & 23 \\ 2 & 4 & 2 & -3 & 3 & 23 & 0 \end{pmatrix}_{b_0}$$

[202] $1^{112}1^{141}3^12^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 2 & 0 \\ 8 & 10 & 0 & 10 & 58 & 16 & 14 \\ 2 & 3 & 0 & 2 & 16 & 3 & 3 \\ 2 & 2 & -1 & 0 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{112}1^{113}3^12^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -692, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{112}1^{114}3^12^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 14 & 0 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 10 & 12 & 0 & 14 & 12 & 96 & 18 \\ 2 & 2 & -1 & 0 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{111}1^{114}3^12^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -724, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 11 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 8 & 11 & 0 & 11 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{111}1^{114}3^12^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 7 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{114}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -765$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{141}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -797$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 7 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & 0 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{121}3^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -821$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 & 2 \\ 2 & 4 & 4 & 0 & 24 & 5 & 4 \\ 1 & 4 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 15 & 0 & 100 & 23 & 18 \\ 2 & 5 & 4 & 0 & 23 & 4 & 4 \\ 2 & 4 & 2 & -3 & 18 & 4 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{113}3^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -871$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 29 & 4 \\ 1 & 4 & 0 & 0 & 3 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 19 & 3 \\ 10 & 29 & 19 & 0 & 19 & 152 & 22 \\ 2 & 4 & 2 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{121}3^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -926$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 & 2 \\ 2 & 4 & 4 & 0 & 30 & 5 & 4 \\ 1 & 4 & 0 & 0 & 18 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 18 & 0 & 154 & 28 & 22 \\ 2 & 5 & 4 & 0 & 28 & 4 & 4 \\ 2 & 4 & 2 & -3 & 22 & 4 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{114}3^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -929$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 32 & 4 \\ 1 & 4 & 0 & 0 & 3 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 22 & 3 \\ 11 & 32 & 22 & 0 & 22 & 193 & 25 \\ 2 & 4 & 2 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{114}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -965$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 11 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 9 & 11 & 0 & 11 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{114}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -980$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 8 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{131}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 & 2 \\ 2 & 4 & 4 & 0 & 27 & 5 & 4 \\ 1 & 4 & 0 & 0 & 18 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 18 & 0 & 133 & 27 & 21 \\ 2 & 5 & 4 & 0 & 27 & 4 & 4 \\ 2 & 4 & 2 & -3 & 21 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{131}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 35 & 4 \\ 1 & 4 & 0 & 0 & 3 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 22 & 3 \\ 12 & 35 & 22 & 0 & 22 & 216 & 26 \\ 2 & 4 & 2 & -3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{141}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 & 2 \\ 2 & 4 & 4 & 0 & 30 & 5 & 4 \\ 1 & 4 & 0 & 0 & 21 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 21 & 0 & 172 & 31 & 24 \\ 2 & 5 & 4 & 0 & 31 & 4 & 4 \\ 2 & 4 & 2 & -3 & 24 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{114}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 3 & 38 & 4 \\ 1 & 4 & 0 & 0 & 3 & 25 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 25 & 3 \\ 13 & 38 & 25 & 0 & 25 & 263 & 29 \\ 2 & 4 & 2 & -3 & 3 & 29 & 1 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{131}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 & 2 \\ 2 & 4 & 4 & 0 & 33 & 5 & 4 \\ 1 & 4 & 0 & 0 & 21 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 33 & 21 & 0 & 193 & 32 & 25 \\ 2 & 5 & 4 & 0 & 32 & 4 & 4 \\ 2 & 4 & 2 & -3 & 25 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[220] \ 1^{141}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 & 2 \\ 2 & 4 & 4 & 0 & 36 & 5 & 4 \\ 1 & 4 & 0 & 0 & 24 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 36 & 24 & 0 & 238 & 36 & 28 \\ 2 & 5 & 4 & 0 & 36 & 4 & 4 \\ 2 & 4 & 2 & -3 & 28 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [221] \ 1^{12}231^{11}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 21 & 3 \\ 5 & 13 & 1 & 0 & 54 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 21 & 54 & 0 & 0 & 217 & 27 \\ 3 & 7 & -1 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{13}231^{12}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 32 & 3 \\ 5 & 13 & 1 & 0 & 84 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 32 & 84 & 0 & 0 & 520 & 42 \\ 3 & 7 & -1 & 0 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{11}4 \ 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{11} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{11} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{12} 23^1 1^{11} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -61, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 22 & 3 \\ 5 & 13 & 1 & 0 & 54 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 22 & 54 & 0 & 0 & 217 & 27 \\ 3 & 7 & -1 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{121} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{122} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 8 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{131} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 3 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{112} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{111} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{111} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 9 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{121} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 3 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{121} 4^{1^3 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 17 & 9 \\ 6 & 9 & 0 & 17 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{111} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{122} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 5 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{122} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{112} 4^{1^3 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 10 & 6 \\ 5 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{112} 4^{1^3 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 16 & 6 \\ 8 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{111} 4^{1^3 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 11 & 6 \\ 6 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{111} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{111}1^{13^11^1}4^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 3 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{112}1^{13^11^1}4^{13^11^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{112}1^{13^11^1}4^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 8 & 11 & 0 & 8 & 21 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{111}1^{13^11^1}4^11^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{122}1^{13^11^1}4^11^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 45 & 22 & 13 \\ 5 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -2 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{122}1^{13^11^1}4^11^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 25 & 9 \\ 8 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{113}1^{13^11^1}4^{13^11^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{121}1^{13^11^1}4^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 225, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 8 & 9 \\ 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{122}1^{13^11^1}4^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 254, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 16 & 9 \\ 5 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & -2 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[251] \ 1^{114} 4^{11} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[252] \ 1^{112} 4^{11} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 277, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[253] \ 1^{131} 4^{11} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 281, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 40 & 11 & 12 \\ 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[254] \ 1^{141} 4^{11} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 305, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 65 & 14 & 15 \\ 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[255] \ 1^{112} 4^{11} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 11 & 16 & 0 & 11 & 45 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[256] \ 1^{121} 4^{11} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 45 & 11 & 13 \\ 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[257] \ 1^{113} 4^{11} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 349, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 10 & 15 & 0 & 11 & 40 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[258] \ 1^{113} 4^{11} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 421, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{131} 1^{131} 1^1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 70 & 14 & 16 \\ 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{141} 1^{131} 1^1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 101 & 17 & 19 \\ 3 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{114} 1^{131} 1^1 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 525, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 19 & 0 & 14 & 65 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{114} 1^{11} 4^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 565, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{113} 1^{131} 1^1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 13 & 20 & 0 & 14 & 70 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{114} 1^{131} 1^1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 15 & 24 & 0 & 17 & 101 & 19 \\ 3 & 4 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [265] $1^{111} 31^1 413^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{111} 31^1 141^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{112} 31^{11} 41^1 31^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 10 & 3 \\ 7 & 14 & 10 & 0 & 10 & 38 & 12 \\ 2 & 4 & 2 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{141}31^{11}41^13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 & 2 \\ 2 & 4 & 4 & 0 & 20 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 20 & 16 & 0 & 90 & 16 & 18 \\ 2 & 3 & 3 & 0 & 16 & 1 & 3 \\ 2 & 4 & 2 & -2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{111}31^{11}41^13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 11 & 7 & 0 & 21 & 7 & 9 \\ 2 & 3 & 3 & 0 & 7 & 1 & 3 \\ 2 & 4 & 2 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{114}31^{11}41^13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 20 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 16 & 3 \\ 9 & 20 & 16 & 0 & 16 & 90 & 18 \\ 2 & 4 & 2 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[271] $1^{113}31^{11}41^13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 13 & 3 \\ 8 & 17 & 13 & 0 & 13 & 61 & 15 \\ 2 & 4 & 2 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{121}31^{11}41^13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 4 & 0 & 14 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 14 & 10 & 0 & 38 & 10 & 12 \\ 2 & 3 & 3 & 0 & 10 & 1 & 3 \\ 2 & 4 & 2 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{131}31^{11}41^13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 17 & 13 & 0 & 61 & 13 & 15 \\ 2 & 3 & 3 & 0 & 13 & 1 & 3 \\ 2 & 4 & 2 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

4.480 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{so}_8$

$T = 9 \quad [0] \quad 1^{113}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

4.481 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 4 & 7 & 0 & 6 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & -3 & 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1211}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -106$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 12 & 6 \\ 4 & 7 & 0 & 10 & 7 & 11 & 5 \\ 5 & 9 & 0 & 12 & 11 & 13 & 7 \\ 3 & 4 & -3 & 6 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1221}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 22 & 6 & 6 \\ 9 & 16 & 0 & 22 & 44 & 11 & 12 \\ 2 & 4 & 0 & 6 & 11 & 1 & 3 \\ 3 & 4 & -3 & 6 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1311}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 2 & 3 \\ 3 & 4 & 0 & 12 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 15 & 9 & 9 \\ 4 & 7 & 0 & 15 & 7 & 5 & 5 \\ 2 & 4 & 0 & 9 & 5 & 1 & 3 \\ 3 & 4 & -3 & 9 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1131}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -146$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 0 & 5 & 7 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 15 & 4 \\ 4 & 7 & 0 & 6 & 7 & 21 & 5 \\ 9 & 17 & 0 & 15 & 21 & 49 & 13 \\ 3 & 4 & -3 & 4 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -146$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 9 & 4 \\ 4 & 7 & 0 & 6 & 7 & 11 & 5 \\ 5 & 9 & 0 & 9 & 11 & 13 & 7 \\ 3 & 4 & -3 & 4 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{1221}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 5 & 3 \\ 3 & 4 & 0 & 8 & 16 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 22 & 12 & 6 \\ 9 & 16 & 0 & 22 & 44 & 26 & 12 \\ 5 & 9 & 0 & 12 & 26 & 13 & 7 \\ 3 & 4 & -3 & 6 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1212}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -156$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 12 & 6 \\ 4 & 7 & 0 & 10 & 7 & 10 & 5 \\ 4 & 8 & 0 & 12 & 10 & 10 & 6 \\ 3 & 4 & -3 & 6 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1311}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 2 & 3 \\ 3 & 4 & 0 & 17 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 36 & 12 & 13 \\ 7 & 12 & 0 & 36 & 25 & 8 & 9 \\ 2 & 4 & 0 & 12 & 8 & 1 & 3 \\ 3 & 4 & -3 & 13 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{1121}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 11 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 15 & 7 \\ 6 & 11 & 0 & 17 & 20 & 17 & 8 \\ 5 & 9 & 0 & 15 & 17 & 13 & 7 \\ 3 & 4 & -3 & 7 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1121}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 12 & 0 & 12 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 4 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{1121}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 3 & 4 \\ 11 & 20 & 0 & 18 & 69 & 14 & 15 \\ 2 & 4 & 0 & 3 & 14 & 1 & 3 \\ 3 & 4 & -3 & 4 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{1121}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 0 & 5 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 12 & 4 \\ 4 & 7 & 0 & 6 & 7 & 16 & 5 \\ 7 & 13 & 0 & 12 & 16 & 28 & 10 \\ 3 & 4 & -3 & 4 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1211}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 6 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 & 5 \\ 2 & 4 & 0 & 6 & 5 & 1 & 3 \\ 3 & 4 & -3 & 6 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1121}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 6 & 7 \\ 6 & 11 & 0 & 17 & 20 & 8 & 8 \\ 2 & 4 & 0 & 6 & 8 & 1 & 3 \\ 3 & 4 & -3 & 7 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1211}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 16 & 0 & 15 & 44 & 11 & 12 \\ 2 & 4 & 0 & 3 & 11 & 1 & 3 \\ 3 & 4 & -3 & 4 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1111}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 & 3 \\ 7 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & -3 & 3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 6 & 3 \\ 4 & 7 & 0 & 5 & 7 & 11 & 5 \\ 5 & 9 & 0 & 6 & 11 & 13 & 7 \\ 3 & 4 & -3 & 3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1212} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 18 & 6 \\ 4 & 7 & 0 & 10 & 7 & 16 & 5 \\ 7 & 13 & 0 & 18 & 16 & 28 & 10 \\ 3 & 4 & -3 & 6 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1221} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 2 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 16 & 6 & 6 \\ 6 & 11 & 0 & 16 & 20 & 8 & 8 \\ 2 & 4 & 0 & 6 & 8 & 1 & 3 \\ 3 & 4 & -3 & 6 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 10 & 3 \\ 3 & 4 & 0 & 4 & 7 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 12 & 3 \\ 4 & 7 & 0 & 5 & 7 & 22 & 5 \\ 10 & 18 & 0 & 12 & 22 & 58 & 14 \\ 3 & 4 & -3 & 3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 & 3 \\ 6 & 11 & 0 & 8 & 20 & 8 & 8 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 11 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 21 & 7 \\ 6 & 11 & 0 & 17 & 20 & 25 & 8 \\ 7 & 13 & 0 & 21 & 25 & 28 & 10 \\ 3 & 4 & -3 & 7 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 5 & 3 \\ 3 & 4 & 0 & 9 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 15 & 7 \\ 4 & 7 & 0 & 11 & 7 & 11 & 5 \\ 5 & 9 & 0 & 15 & 11 & 13 & 7 \\ 3 & 4 & -3 & 7 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 15 & 3 \\ 3 & 4 & 0 & 4 & 7 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 18 & 3 \\ 4 & 7 & 0 & 5 & 7 & 33 & 5 \\ 15 & 27 & 0 & 18 & 33 & 133 & 21 \\ 3 & 4 & -3 & 3 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{1121}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 & 3 \\ 12 & 21 & 0 & 14 & 80 & 14 & 16 \\ 2 & 4 & 0 & 3 & 14 & 1 & 3 \\ 3 & 4 & -3 & 3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1112}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -514, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 9 & 3 \\ 7 & 12 & 0 & 8 & 25 & 28 & 9 \\ 7 & 13 & 0 & 9 & 28 & 28 & 10 \\ 3 & 4 & -3 & 3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{1111}3^11^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 0 & 4 & 12 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 6 & 3 \\ 7 & 12 & 0 & 8 & 25 & 20 & 9 \\ 5 & 9 & 0 & 6 & 20 & 13 & 7 \\ 3 & 4 & -3 & 3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{1112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 4 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 9 & 3 \\ 4 & 7 & 0 & 5 & 7 & 16 & 5 \\ 7 & 13 & 0 & 9 & 16 & 28 & 10 \\ 3 & 4 & -3 & 3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{1112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -546, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 9 & 3 \\ 6 & 11 & 0 & 8 & 20 & 25 & 8 \\ 7 & 13 & 0 & 9 & 25 & 28 & 10 \\ 3 & 4 & -3 & 3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{1114}3^11^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -546, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 11 & 3 \\ 3 & 4 & 0 & 9 & 7 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 33 & 7 \\ 4 & 7 & 0 & 11 & 7 & 26 & 5 \\ 11 & 21 & 0 & 33 & 26 & 76 & 16 \\ 3 & 4 & -3 & 7 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{1131}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 & 3 \\ 8 & 15 & 0 & 11 & 39 & 11 & 11 \\ 2 & 4 & 0 & 3 & 11 & 1 & 3 \\ 3 & 4 & -3 & 3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 20 & 3 \\ 3 & 4 & 0 & 4 & 7 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 24 & 3 \\ 4 & 7 & 0 & 5 & 7 & 44 & 5 \\ 20 & 36 & 0 & 24 & 44 & 238 & 28 \\ 3 & 4 & -3 & 3 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -620, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 & 3 \\ 9 & 16 & 0 & 11 & 44 & 11 & 12 \\ 2 & 4 & 0 & 3 & 11 & 1 & 3 \\ 3 & 4 & -3 & 3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 7 & 3 \\ 3 & 4 & 0 & 9 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 21 & 7 \\ 4 & 7 & 0 & 11 & 7 & 16 & 5 \\ 7 & 13 & 0 & 21 & 16 & 28 & 10 \\ 3 & 4 & -3 & 7 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 9 & 3 \\ 3 & 4 & 0 & 9 & 7 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 27 & 7 \\ 4 & 7 & 0 & 11 & 7 & 21 & 5 \\ 9 & 17 & 0 & 27 & 21 & 49 & 13 \\ 3 & 4 & -3 & 7 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 4 & 7 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 12 & 3 \\ 4 & 7 & 0 & 5 & 7 & 21 & 5 \\ 9 & 17 & 0 & 12 & 21 & 49 & 13 \\ 3 & 4 & -3 & 3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1141} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 & 3 \\ 10 & 19 & 0 & 14 & 64 & 14 & 14 \\ 2 & 4 & 0 & 3 & 14 & 1 & 3 \\ 3 & 4 & -3 & 3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -810, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 5 & 3 \\ 3 & 4 & 0 & 4 & 16 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 6 & 3 \\ 9 & 16 & 0 & 11 & 44 & 26 & 12 \\ 5 & 9 & 0 & 6 & 26 & 13 & 7 \\ 3 & 4 & -3 & 3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -836, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 4 & 7 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 15 & 3 \\ 4 & 7 & 0 & 5 & 7 & 27 & 5 \\ 12 & 22 & 0 & 15 & 27 & 85 & 17 \\ 3 & 4 & -3 & 3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1131}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -840, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 & 3 \\ 17 & 30 & 0 & 20 & 165 & 20 & 23 \\ 2 & 4 & 0 & 3 & 20 & 1 & 3 \\ 3 & 4 & -3 & 3 & 23 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1114}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 4 & 7 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 15 & 3 \\ 4 & 7 & 0 & 5 & 7 & 26 & 5 \\ 11 & 21 & 0 & 15 & 26 & 76 & 16 \\ 3 & 4 & -3 & 3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1212}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 7 & 3 \\ 3 & 4 & 0 & 13 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 16 & 30 & 10 \\ 4 & 7 & 0 & 16 & 7 & 16 & 5 \\ 7 & 13 & 0 & 30 & 16 & 28 & 10 \\ 3 & 4 & -3 & 10 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1131}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 20 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 6 & 3 \\ 11 & 20 & 0 & 14 & 69 & 32 & 15 \\ 5 & 9 & 0 & 6 & 32 & 13 & 7 \\ 3 & 4 & -3 & 3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1131}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 & 3 \\ 11 & 20 & 0 & 14 & 69 & 14 & 15 \\ 2 & 4 & 0 & 3 & 14 & 1 & 3 \\ 3 & 4 & -3 & 3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1131}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -986, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 2 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 38 & 6 & 7 \\ 13 & 24 & 0 & 38 & 100 & 17 & 18 \\ 2 & 4 & 0 & 6 & 17 & 1 & 3 \\ 3 & 4 & -3 & 7 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1122}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1020, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 7 & 3 \\ 3 & 4 & 0 & 4 & 16 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 9 & 3 \\ 9 & 16 & 0 & 11 & 44 & 37 & 12 \\ 7 & 13 & 0 & 9 & 37 & 28 & 10 \\ 3 & 4 & -3 & 3 & 12 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1131}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1076, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 & 3 \\ 14 & 25 & 0 & 17 & 111 & 17 & 19 \\ 2 & 4 & 0 & 3 & 17 & 1 & 3 \\ 3 & 4 & -3 & 3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{1114}3^11^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 17 & 3 \\ 3 & 4 & 0 & 4 & 7 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 21 & 3 \\ 4 & 7 & 0 & 5 & 7 & 38 & 5 \\ 17 & 31 & 0 & 21 & 38 & 172 & 24 \\ 3 & 4 & -3 & 3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{1114}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 4 & 7 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 18 & 3 \\ 4 & 7 & 0 & 5 & 7 & 32 & 5 \\ 14 & 26 & 0 & 18 & 32 & 118 & 20 \\ 3 & 4 & -3 & 3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{1141}3^11^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 2 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 26 & 3 & 3 \\ 22 & 39 & 0 & 26 & 280 & 26 & 30 \\ 2 & 4 & 0 & 3 & 26 & 1 & 3 \\ 3 & 4 & -3 & 3 & 30 & 3 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{1141}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 & 3 \\ 13 & 24 & 0 & 17 & 100 & 17 & 18 \\ 2 & 4 & 0 & 3 & 17 & 1 & 3 \\ 3 & 4 & -3 & 3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{1141}3^11^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 2 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 23 & 3 & 3 \\ 19 & 34 & 0 & 23 & 208 & 23 & 26 \\ 2 & 4 & 0 & 3 & 23 & 1 & 3 \\ 3 & 4 & -3 & 3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{1141}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1664, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 & 3 \\ 16 & 29 & 0 & 20 & 148 & 20 & 22 \\ 2 & 4 & 0 & 3 & 20 & 1 & 3 \\ 3 & 4 & -3 & 3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [55] $1^{1131}1^{11}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 6 \\ 1 & 4 & 0 & 0 & 3 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 5 & 6 \\ 2 & 5 & 4 & 0 & 5 & 4 & 5 \\ 2 & 6 & 3 & 0 & 6 & 5 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{1111}3^11^{13}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 6 & 3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 6 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111}31^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 5 & 9 & 4 \\ 2 & 5 & 4 & 0 & 5 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111}31^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 752, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 9 & 15 & 7 \\ 2 & 5 & 4 & 0 & 9 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 15 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

4.482 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_{2 \rightarrow (-3)}$

$$T = 10 \quad [0] \quad 1^{111}3^11^13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 2 & 3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^11^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 175, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 5 & 3 & 4 \\ 2 & 5 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^11^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 475, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 9 & 5 & 7 \\ 2 & 5 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 5 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.483 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 8 & 0 & 9 & 12 & 5 & 6 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -116, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 & 3 \\ 10 & 17 & 0 & 11 & 55 & 11 & 13 \\ 2 & 4 & 0 & 3 & 11 & 0 & 2 \\ 3 & 4 & -3 & 3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^1}1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 4 & 7 & 0 & 6 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}3^1}1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 & 3 \\ 3 & 4 & 0 & 4 & 11 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 & 3 \\ 6 & 11 & 0 & 8 & 20 & 9 & 8 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 3 & 4 & -3 & 3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}3^1}1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 & 3 \\ 4 & 7 & 0 & 5 & 7 & 6 & 5 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 4 & -3 & 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}3^1}1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 14 & 6 & 7 \\ 5 & 8 & 0 & 14 & 12 & 5 & 6 \\ 2 & 4 & 0 & 6 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}3^1}1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 12 & 0 & 12 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 0 & 2 \\ 3 & 4 & -3 & 4 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{131}3^1}1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 3 & 4 \\ 11 & 20 & 0 & 18 & 69 & 14 & 15 \\ 2 & 4 & 0 & 3 & 14 & 0 & 2 \\ 3 & 4 & -3 & 4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}3^1}1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}3^1}1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 6 & 3 \\ 4 & 7 & 0 & 5 & 7 & 11 & 5 \\ 5 & 9 & 0 & 6 & 11 & 12 & 6 \\ 3 & 4 & -3 & 3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 & 3 \\ 7 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 0 & 2 \\ 3 & 4 & -3 & 3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1213}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 16 & 0 & 15 & 44 & 11 & 12 \\ 2 & 4 & 0 & 3 & 11 & 0 & 2 \\ 3 & 4 & -3 & 4 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 6 & 7 \\ 4 & 7 & 0 & 11 & 7 & 5 & 5 \\ 2 & 4 & 0 & 6 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 20 & 6 & 7 \\ 7 & 12 & 0 & 20 & 25 & 8 & 9 \\ 2 & 4 & 0 & 6 & 8 & 0 & 2 \\ 3 & 4 & -3 & 7 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

4.484 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 15.51 \rightarrow 16, \ T_{\max} = 28, \ T \in [16, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{132}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 3 & 4 & -3 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.11 \rightarrow 14, \ T_{\max} = 28, \ T \in [14, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 48 & 12 \\ 3 & 4 & -3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -3 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 24 & 8 \\ 3 & 4 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{131}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{132}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -3 & 9 & 5 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{122}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 18 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 24 & 14 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -3 & 14 & 5 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{141}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 \\ 3 & 4 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{132}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 27 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 36 & 21 \\ 4 & 8 & 0 & 36 & 9 & 5 \\ 3 & 4 & -3 & 21 & 5 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112}3^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{151}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 15 & 15 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -3 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -3 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{14}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -3 & 16 & 3 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{161}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 24 & 0 & 106 & 18 & 18 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{171}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.69 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 28 & 0 & 145 & 21 & 21 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -3 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{13}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -3 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -3 & 10 & 5 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{115}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -2 & 15 & 0 & 0 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 57 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 48 & 12 \\ 3 & 4 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -3 & 13 & 5 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{132}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 30 & 17 \\ 4 & 8 & 0 & 30 & 9 & 5 \\ 3 & 4 & -3 & 17 & 5 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{131} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{141} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -3 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{121} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{121} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -3 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{131} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{131} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{141} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{141} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1643, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{141} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [76] $1^{12} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 38 & 10 & 12 \\ 2 & 5 & 0 & 10 & 1 & 3 \\ 3 & 5 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 63 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{111} 3 1^{13} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 1 & 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{13} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11} 3 1^{13} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 4 & 3 \\ 1 & 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 23 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 \\ 1 & 2 & 0 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 4 \\ 2 & 3 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & -3 & 3 & 4 \end{pmatrix}_{b_0}$$

[81] $1^{11} 3 1^{13} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 \\ 1 & 4 & 0 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 8 & 9 & 0 & 9 & 39 & 11 \\ 2 & 2 & 2 & -3 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{11} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 7 & 3 \\ 3 & 4 & 7 & 0 & 7 & 15 & 7 \\ 2 & 2 & 2 & -3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 & 7 \\ 9 & 13 & 1 & 0 & 7 & 1 \\ 1 & 1 & -3 & 0 & -1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 & 1 \\ 2 & 2 & -1 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 14 & 1 & 28 & 7 \\ 14 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 58 & 14 \\ 7 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{11} 3 1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 13 & 7 \\ 3 & 5 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 1 & 3 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 10 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 10 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{11} 3^{111} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{13} 3^{13} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{112} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{111} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 18 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3^{111} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 & 2 \\ 1 & 2 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 5 & 7 \\ 1 & 1 & 3 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{141} 3^{13} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 66 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 66 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{112}3^{111}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{11}3^{11}21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 7 & 8 \\ 1 & 1 & 3 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{11}3^{11}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 2 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 4 & 5 \\ 1 & 1 & 3 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{121}3^{11}21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 18 & 0 & 94 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{12}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 8 & 3 \\ 4 & 6 & 8 & 0 & 8 & 25 & 9 \\ 2 & 2 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{11}3^{11}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 7 & 8 \\ 2 & 5 & 0 & 7 & 1 & 3 \\ 3 & 5 & -3 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{141}3^{11}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 7 & 0 & 41 & 16 & 11 \\ 2 & 3 & 4 & 0 & 16 & 4 & 4 \\ 2 & 2 & 2 & -3 & 11 & 4 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{11}3^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{11}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 11 & 4 \\ 7 & 15 & 0 & 11 & 31 & 11 \\ 3 & 5 & -3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{11} 3^{11} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 19 & 4 \\ 11 & 25 & 0 & 19 & 91 & 19 \\ 3 & 5 & -3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 13 & 14 & 1 & 14 & 7 \\ 14 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 15 & 0 & 13 & 7 \\ 7 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 14 & 8 \\ 3 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{11} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 15 & 4 \\ 9 & 20 & 0 & 15 & 57 & 15 \\ 3 & 5 & -3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{112} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -161$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 4 & 6 & 8 & 0 & 6 & 25 & 9 \\ 2 & 2 & 2 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 12 & 6 \\ 2 & 4 & 2 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{13} 3^{11} 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -175$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 25 & 0 & 104 & 17 & 20 \\ 2 & 5 & 0 & 17 & 1 & 3 \\ 3 & 5 & -3 & 20 & 3 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{121} 3^{114} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 52 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{131} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{131} 3 \ 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{111} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{111} 3 \ 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 11 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{121} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{111} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{131} 31^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 15 & 0 & 49 & 12 & 13 \\ 1 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 3 & 3 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{112} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 3 & 5 & 5 & 0 & 5 & 12 & 6 \\ 2 & 2 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{11} 3 \ 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 92 & 19 & 20 \\ 2 & 5 & 0 & 19 & 1 & 3 \\ 3 & 5 & -3 & 20 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{11} 3 \ 21^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 1 & 3 \\ 3 & 5 & -3 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{111} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{14} 3^{21} 1^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 10 & 14 \\ 1 & 1 & 3 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{121} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{112} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{111} 3^{21} 1^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 11 & 8 \\ 2 & 3 & 4 & 0 & 11 & 4 & 4 \\ 2 & 2 & 2 & -3 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{111} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 0 & 4 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 11 & 5 \\ 3 & 4 & 7 & 0 & 11 & 15 & 7 \\ 2 & 2 & 2 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{11} 3^{21} 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 15 & 16 \\ 2 & 5 & 0 & 15 & 1 & 3 \\ 3 & 5 & -3 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{111} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{12} 3^{21} 1^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 8 & 10 \\ 1 & 1 & 3 & 0 & 8 & 0 & 2 \\ 2 & 2 & 2 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{121} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 9 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 3 & 3 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[144] $1^{12}321^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 62 & 14 & 16 \\ 2 & 5 & 0 & 14 & 1 & 3 \\ 3 & 5 & -3 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{131}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{121}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{121}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 14 & 0 & 62 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{112}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 10 & 5 \\ 3 & 5 & 5 & 0 & 10 & 12 & 6 \\ 2 & 2 & 2 & -3 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$

[149] $1^{131}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 13 & 9 \\ 2 & 3 & 4 & 0 & 13 & 4 & 4 \\ 2 & 2 & 2 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{13}321^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 9 & 12 \\ 1 & 1 & 3 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -314$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.31 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{121}31^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -332$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 9 & 12 & 0 & 28 & 9 & 10 \\ 1 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 3 & 3 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{123}321^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 7 & 1 & 2 \\ 1 & 2 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 11 & 13 \\ 1 & 1 & 3 & 0 & 11 & 0 & 2 \\ 2 & 2 & 2 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{131}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 25 & 0 & 188 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{121}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -356$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 10 & 7 \\ 2 & 3 & 4 & 0 & 10 & 4 & 4 \\ 2 & 2 & 2 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{131}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -381$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 13 & 0 & 74 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{121}321^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -386$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 14 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 22 & 16 \\ 2 & 3 & 4 & 0 & 22 & 4 & 4 \\ 2 & 2 & 2 & -3 & 16 & 4 & 1 \end{pmatrix}_{b_0}$$

[158] $1^131^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{121}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{131}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{121}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[162] 1^{121} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[163] 1^{141} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[164] 1^{141} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[165] 1^{121} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[166] 1^{11} 31^{111} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[167] 1^{14} 3 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 18 & 35 & 0 & 202 & 24 & 28 & 28 \\ 2 & 5 & 0 & 24 & 1 & 3 & 3 \\ 3 & 5 & -3 & 28 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[168] 1^{141} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 & 4 \\ 2 & 4 & 1 & 0 & 16 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 27 & 16 & 0 & 122 & 20 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[169] 1^{121} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[170] 1^{11} 31^{111} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 6 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 2 & -3 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[171] 1^{112} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 14 & 5 \\ 4 & 6 & 8 & 0 & 14 & 25 & 9 \\ 2 & 2 & 2 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{12} 3 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 25 & 0 & 94 & 18 & 20 \\ 2 & 5 & 0 & 18 & 1 & 3 \\ 3 & 5 & -3 & 20 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{131} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{121} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{141} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{13} 3 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 9 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 12 & 15 \\ 1 & 1 & 3 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{13} 3 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -565, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 30 & 0 & 142 & 21 & 24 \\ 2 & 5 & 0 & 21 & 1 & 3 \\ 3 & 5 & -3 & 24 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{121} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{131} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{121} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 42 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 42 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

- [181] $1^{131} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -581$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 21 & 0 & 142 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$
- [182] $1^{121} 3 21^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -586$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 14 & 10 \\ 2 & 3 & 4 & 0 & 14 & 4 & 4 \\ 2 & 2 & 2 & -3 & 10 & 4 & 1 \end{pmatrix}_{b_0}$$
- [183] $1^{111} 3 1^{12} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -587$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 9 & 11 \\ 1 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$
- [184] $1^{131} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -591$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 104 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$
- [185] $1^{121} 3 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -596$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 11 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 18 & 13 \\ 2 & 3 & 4 & 0 & 18 & 4 & 4 \\ 2 & 2 & 2 & -3 & 13 & 4 & 1 \end{pmatrix}_{b_0}$$
- [186] $1^{14} 3 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -636$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 13 & 17 \\ 1 & 1 & 3 & 0 & 13 & 0 & 2 \\ 2 & 2 & 2 & -3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$
- [187] $1^{12} 3 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -650$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 30 & 0 & 134 & 22 & 24 \\ 2 & 5 & 0 & 22 & 1 & 3 \\ 3 & 5 & -3 & 24 & 3 & 2 \end{pmatrix}_{b_0}$$
- [192] $1^{131} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$
- [191] $1^{131} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$
- [190] $1^{121} 3 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{141} 3 \ 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 67 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 67 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{121} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{13} 3 \ 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -671, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 15 & 18 \\ 1 & 1 & 3 & 0 & 15 & 0 & 2 \\ 2 & 2 & 2 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{111} 31^{111} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -699, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 12 & 6 \\ 2 & 4 & 2 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{131} 3 \ 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{131} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{111} 31^{13} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 12 & 0 & 73 & 12 & 15 \\ 1 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{131} 3 \ 21^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 17 & 12 \\ 2 & 3 & 4 & 0 & 17 & 4 & 4 \\ 2 & 2 & 2 & -3 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{141} 3 \ 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 20 & 0 & 158 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{13} 3^{11} 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -895$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 35 & 0 & 188 & 25 & 28 \\ 2 & 5 & 0 & 25 & 1 & 3 \\ 3 & 5 & -3 & 28 & 3 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{141} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{131} 3^{11} 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{131} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{141} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{1111} 2^2 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -901$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{141} 3^{11} 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -926$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 20 & 14 \\ 2 & 3 & 4 & 0 & 20 & 4 & 4 \\ 2 & 2 & 2 & -3 & 14 & 4 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{131} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -936$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{141} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -936$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 57 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{14} 3 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -936$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 0 & 12 & 1 & 2 \\ 1 & 2 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 16 & 0 & 110 & 16 & 20 \\ 1 & 1 & 3 & 0 & 16 & 0 & 2 \\ 2 & 2 & 2 & -3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

[210] $1^{14} 3 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -950$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 5 & 0 & 40 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 40 & 0 & 254 & 28 & 32 \\ 2 & 5 & 0 & 28 & 1 & 3 \\ 3 & 5 & -3 & 32 & 3 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{141} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -966$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 28 & 0 & 254 & 32 \\ 3 & 4 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{111} 31^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -1007$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 15 & 0 & 121 & 15 & 19 \\ 1 & 3 & 3 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -3 & 19 & 2 & 0 \end{pmatrix}_{b_0}$$

[213] $1^{141} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1016$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 24 & 0 & 202 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{131} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1016$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{131} 3 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 21 & 15 \\ 2 & 3 & 4 & 0 & 21 & 4 & 4 \\ 2 & 2 & 2 & -3 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{141} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 68 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{131} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{131} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.0

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{141} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{131} 3^{21} 1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 25 & 18 \\ 2 & 3 & 4 & 0 & 25 & 4 & 4 \\ 2 & 2 & 2 & -3 & 18 & 4 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{131} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{131} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{141} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{141} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{141} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{141} 3^{21} 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 11 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 24 & 17 \\ 2 & 3 & 4 & 0 & 24 & 4 & 4 \\ 2 & 2 & 2 & -3 & 17 & 4 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{141} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 69 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{141} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{141} 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{141} 3 \ 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 16 & 0 & 110 & 28 & 20 \\ 2 & 3 & 4 & 0 & 28 & 4 & 4 \\ 2 & 2 & 2 & -3 & 20 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{141} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{141} 3 \ 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 70 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{141} 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [234] \ 1^{111} 2 \ 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 1 & 0 \\ 2 & 4 & 0 & 4 & 6 & 4 & 6 \\ 1 & 3 & 0 & 1 & 4 & 0 & 2 \\ 2 & 3 & -1 & 0 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{11} 3 \ 221^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 6 & 0 & 17 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 17 & 0 & 36 & 13 & 14 \\ 2 & 6 & 0 & 13 & 2 & 4 \\ 3 & 6 & -3 & 14 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{12} 3 \ 221^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

11.7

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 6 & 0 & 34 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 34 & 0 & 150 & 26 & 28 \\ 2 & 6 & 0 & 26 & 2 & 4 \\ 3 & 6 & -3 & 28 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{\overline{111}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{\overline{111}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 117, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{\overline{112}} 3 \ 221^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 3 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 11 & 3 \\ 10 & 11 & 0 & 11 & 49 & 13 \\ 3 & 3 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{\overline{111}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{\overline{111}} 3 \ 221^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 6 & 8 & 6 \\ 2 & 3 & 5 & 0 & 8 & 5 & 5 \\ 2 & 2 & 3 & -3 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{\overline{1}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{\overline{1}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 222, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 2 & 3 \\ 4 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{\overline{111}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{112} 3^{11} 221^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 1 & 0 & 5 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 6 & 16 & 6 \\ 4 & 6 & 10 & 0 & 16 & 29 & 11 \\ 2 & 2 & 3 & -3 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{11} 4^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 3 \\ 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{11} 4^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 338, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 2 & 3 \\ 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{121} 3^{11} 221^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 & 2 \\ 1 & 2 & 1 & 0 & 10 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 10 & 0 & 30 & 16 & 12 \\ 2 & 3 & 5 & 0 & 16 & 5 & 5 \\ 2 & 2 & 3 & -3 & 12 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{131} 3^{11} 221^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 1 & 0 & 15 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 15 & 0 & 70 & 24 & 18 \\ 2 & 3 & 5 & 0 & 24 & 5 & 5 \\ 2 & 2 & 3 & -3 & 18 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[251] \ 1^{141} 3^{11} 221^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 1116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 1 & 0 & 20 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 20 & 0 & 126 & 32 & 24 \\ 2 & 3 & 5 & 0 & 32 & 5 & 5 \\ 2 & 2 & 3 & -3 & 24 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [252] \ 1^{12} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[253] \ 1^{11} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

4.485 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

4.486 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 11 & 19 & 0 & 21 & 12 & 73 & 15 \\ 3 & 4 & -3 & 4 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -3 & 3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 5 & 3 \\ 3 & 4 & 0 & 19 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 12 & 33 & 15 \\ 2 & 4 & 0 & 12 & 0 & 6 & 2 \\ 5 & 9 & 0 & 33 & 6 & 13 & 7 \\ 3 & 4 & -3 & 15 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1211} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -3 & 6 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 4 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 & 9 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -3 & 4 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -3 & 4 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -3 & 4 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -3 & 3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1212} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 5 & 9 & 0 & 12 & 12 & 21 & 6 \\ 7 & 13 & 0 & 18 & 21 & 28 & 10 \\ 3 & 4 & -3 & 6 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -3 & 3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 24 & 7 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 8 & 14 & 0 & 24 & 9 & 37 & 11 \\ 3 & 4 & -3 & 7 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -3 & 4 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & -3 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 39 & 11 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 0 & 39 & 9 & 37 & 11 \\ 3 & 4 & -3 & 11 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -723, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -3 & 3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -738, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -3 & 3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1112} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -744, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -3 & 4 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -779, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 21 & 15 & 85 & 17 \\ 3 & 4 & -3 & 4 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -828, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -3 & 3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -845, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 12 & 21 & 6 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -3 & 3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -941, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 21 & 7 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 7 & 13 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & -3 & 7 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1112} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 48 & 11 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 10 & 18 & 0 & 48 & 12 & 58 & 14 \\ 3 & 4 & -3 & 11 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 16 & 3 \\ 3 & 4 & 0 & 9 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 48 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 16 & 28 & 0 & 48 & 18 & 154 & 22 \\ 3 & 4 & -3 & 7 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 27 & 7 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 9 & 17 & 0 & 27 & 12 & 49 & 13 \\ 3 & 4 & -3 & 7 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 5 & 9 & 0 & 6 & 12 & 27 & 6 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -3 & 3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 12 & 33 & 6 \\ 11 & 21 & 0 & 15 & 33 & 76 & 16 \\ 3 & 4 & -3 & 3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -3 & 3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -3 & 4 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1212} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 9 & 42 & 10 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 10 & 18 & 0 & 42 & 12 & 58 & 14 \\ 3 & 4 & -3 & 10 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -3 & 3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 10 & 3 \\ 3 & 4 & 0 & 9 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 30 & 7 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 10 & 18 & 0 & 30 & 12 & 58 & 14 \\ 3 & 4 & -3 & 7 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1212} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 9 & 30 & 10 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 0 & 30 & 9 & 28 & 10 \\ 3 & 4 & -3 & 10 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 13 & 3 \\ 3 & 4 & 0 & 9 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 39 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 13 & 23 & 0 & 39 & 15 & 100 & 18 \\ 3 & 4 & -3 & 7 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -3 & 3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 33 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -3 & 4 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 30 & 21 & 172 & 24 \\ 3 & 4 & -3 & 4 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 9 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 36 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 12 & 22 & 0 & 36 & 15 & 85 & 17 \\ 3 & 4 & -3 & 7 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1212} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 12 & 60 & 14 \\ 2 & 4 & 0 & 12 & 0 & 12 & 2 \\ 10 & 18 & 0 & 60 & 12 & 58 & 14 \\ 3 & 4 & -3 & 14 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -3 & 4 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -3 & 3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 24 & 3 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -3 & 3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 9 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 42 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 14 & 26 & 0 & 42 & 18 & 118 & 20 \\ 3 & 4 & -3 & 7 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -3 & 3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 15 & 3 \\ 3 & 4 & 0 & 9 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 45 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 15 & 27 & 0 & 45 & 18 & 133 & 21 \\ 3 & 4 & -3 & 7 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1113} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 18 & 3 \\ 3 & 4 & 0 & 9 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 54 & 7 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 18 & 32 & 0 & 54 & 21 & 193 & 25 \\ 3 & 4 & -3 & 7 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 17 & 3 \\ 3 & 4 & 0 & 9 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 51 & 7 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 17 & 31 & 0 & 51 & 21 & 172 & 24 \\ 3 & 4 & -3 & 7 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 20 & 3 \\ 3 & 4 & 0 & 9 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 60 & 7 \\ 2 & 4 & 0 & 6 & 0 & 24 & 2 \\ 20 & 36 & 0 & 60 & 24 & 238 & 28 \\ 3 & 4 & -3 & 7 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 3 \quad 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 3 \quad 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 7 & 4 \\ 2 & 5 & 0 & 4 & 1 & 7 & 3 \\ 5 & 10 & 0 & 7 & 7 & 14 & 8 \\ 3 & 5 & -3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111} 3 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 4 & 3 & 4 \\ 1 & 2 & 1 & 0 & 2 & 3 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 3 \quad 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 15 & 4 & 4 \\ 9 & 20 & 0 & 15 & 57 & 15 & 15 \\ 2 & 5 & 0 & 4 & 15 & 2 & 4 \\ 3 & 5 & -3 & 4 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111} 3 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 20 & 5 \\ 9 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 0 & 63 & 15 \\ 5 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111} 3 \quad 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 4 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 4 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{11} \overset{1}{3} 21^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 & 5 \\ 3 & 5 & 0 & 5 & 5 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 11 & 4 & 4 \\ 2 & 5 & 0 & 4 & 1 & 11 & 3 & 3 \\ 7 & 15 & 0 & 11 & 11 & 32 & 12 & 12 \\ 3 & 5 & -3 & 4 & 3 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{1111} \overset{1}{3} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 3 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 2 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{12} \overset{1}{3} 21^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 & 5 \\ 3 & 5 & 0 & 5 & 5 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 18 & 4 & 4 \\ 2 & 5 & 0 & 4 & 1 & 18 & 3 & 3 \\ 12 & 25 & 0 & 18 & 18 & 94 & 20 & 20 \\ 3 & 5 & -3 & 4 & 3 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111} \overset{1}{3} 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 & 5 \\ 3 & 5 & 0 & 10 & 5 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 7 & 16 & 8 & 8 \\ 2 & 5 & 0 & 7 & 1 & 7 & 3 & 3 \\ 5 & 10 & 0 & 16 & 7 & 14 & 8 & 8 \\ 3 & 5 & -3 & 8 & 3 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{13} \overset{1}{3} 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 & 5 \\ 3 & 5 & 0 & 5 & 5 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 25 & 4 & 4 \\ 2 & 5 & 0 & 4 & 1 & 25 & 3 & 3 \\ 17 & 35 & 0 & 25 & 25 & 188 & 28 & 28 \\ 3 & 5 & -3 & 4 & 3 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{111} \overset{1}{3} 21^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 390, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 & 5 \\ 3 & 5 & 0 & 10 & 5 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 7 & 23 & 8 & 8 \\ 2 & 5 & 0 & 7 & 1 & 11 & 3 & 3 \\ 7 & 15 & 0 & 23 & 11 & 32 & 12 & 12 \\ 3 & 5 & -3 & 8 & 3 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111} \overset{1}{3} 21^{1213} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 630, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 9 & 3 & 5 \\ 3 & 5 & 0 & 15 & 5 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 11 & 45 & 12 & 12 \\ 2 & 5 & 0 & 11 & 1 & 15 & 3 & 3 \\ 9 & 20 & 0 & 45 & 15 & 58 & 16 & 16 \\ 3 & 5 & -3 & 12 & 3 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{12} \overset{1}{3} 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 22 & 4 \\ 2 & 5 & 0 & 4 & 1 & 22 & 3 \\ 14 & 30 & 0 & 22 & 22 & 134 & 24 \\ 3 & 5 & -3 & 4 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{1111}321^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 720, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 11 & 34 & 12 \\ 2 & 5 & 0 & 11 & 1 & 11 & 3 \\ 7 & 15 & 0 & 34 & 11 & 32 & 12 \\ 3 & 5 & -3 & 12 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1111}321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 752, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 3 & 4 & 0 & 7 & 7 & 5 \\ 2 & 2 & 2 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1111}321^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 840, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 15 & 60 & 16 \\ 2 & 5 & 0 & 15 & 1 & 15 & 3 \\ 9 & 20 & 0 & 60 & 15 & 58 & 16 \\ 3 & 5 & -3 & 16 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [71] \quad 1^{1111}3221^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -52, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 6 & 0 & 17 & 6 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 17 & 0 & 36 & 13 & 38 & 14 \\ 2 & 6 & 0 & 13 & 2 & 13 & 4 \\ 7 & 17 & 0 & 38 & 13 & 36 & 14 \\ 3 & 6 & -3 & 14 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1111}3221^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -468, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 6 & 8 & 6 \\ 2 & 3 & 5 & 0 & 8 & 5 & 5 \\ 2 & 3 & 5 & 0 & 8 & 8 & 6 \\ 2 & 2 & 3 & -3 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$4.487 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \quad 1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 3 \\ 2 & 3 & 0 & 3 & 2 & 3 & 0 \\ 3 & 4 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 3 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 2 \\ 2 & 3 & 0 & 2 & 2 & 2 & 0 \\ 3 & 4 & -3 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^1 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 3 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 2 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 5 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 6 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 2 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^1 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 15 & 5 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 & 7 \\ 5 & 9 & 0 & 15 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.488 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{1213}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 4 & 6 \\ 5 & 9 & 0 & 12 & 12 & 5 & 6 \\ 2 & 3 & 0 & 4 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^1 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 3 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^1 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 12 & 11 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 2 & 3 & 0 & 11 & 2 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1213}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 8 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 & 2 \\ 2 & 3 & 0 & 8 & 2 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 5 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 2 & 3 \\ 7 & 13 & 0 & 9 & 27 & 7 & 9 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -3 & 3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 9 & 7 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 & 2 \\ 2 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 12 & 9 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 2 & 3 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{121}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 18 & 16 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 2 & 3 & 0 & 16 & 2 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 15 & 11 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 2 & 3 & 0 & 11 & 2 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 12 & 10 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 2 & 3 & 0 & 10 & 2 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{121}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 15 & 13 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 2 & 3 & 0 & 13 & 2 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{131} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1346, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 15 & 12 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 2 & 3 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{131} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 18 & 15 & 21 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 2 & 3 & 0 & 15 & 2 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{141} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 18 & 14 & 20 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 2 & 3 & 0 & 14 & 2 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{131} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1826, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 2 & 3 \\ 3 & 4 & 0 & 32 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 21 & 18 & 25 \\ 2 & 4 & 0 & 21 & 0 & 2 & 2 \\ 2 & 3 & 0 & 18 & 2 & 0 & 2 \\ 3 & 4 & -3 & 25 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 21 & 17 & 24 \\ 2 & 4 & 0 & 21 & 0 & 2 & 2 \\ 2 & 3 & 0 & 17 & 2 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{141} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 24 & 20 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 & 2 \\ 2 & 3 & 0 & 20 & 2 & 0 & 2 \\ 3 & 4 & -3 & 28 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11} 4^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 7 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 5 & 0 & 6 \\ 2 & 4 & 0 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 6 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -192, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 8 & 0 & 10 \\ 2 & 4 & 0 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 10 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$4.489 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$$

$$T = 10 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 6 \\ 2 & 4 & 0 & 3 & 0 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 3 \\ 3 & 5 & 0 & 6 & 3 & 3 & 4 \\ 3 & 4 & -3 & 4 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 1 \\ 3 & 4 & -3 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 \\ 3 & 4 & 0 & 5 & 4 & 3 & 9 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 \\ 2 & 4 & 0 & 3 & 0 & 2 & 6 \\ 2 & 3 & 0 & 3 & 2 & 0 & 5 \\ 5 & 9 & 0 & 9 & 6 & 5 & 13 \\ 3 & 4 & -3 & 4 & 2 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 & 3 \\ 5 & 9 & 0 & 6 & 12 & 5 & 6 & 6 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & -3 & 3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 5 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 5 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 15 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 7 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.490 NHC(-2)(-3) + f₄ + su₂ + su₃ + su_{2n3mix}

$$T = 9 \quad [0] \quad 1^{\overline{111}3^{\overline{111}}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 5 & 8 & 0 & 9 & 5 & 12 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}3^{\overline{111}}} 1^{\overline{112}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 11 & 3 \\ 2 & 4 & 0 & 3 & 0 & 11 & 2 \\ 10 & 17 & 0 & 11 & 11 & 55 & 13 \\ 3 & 4 & -3 & 3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}3^{\overline{111}}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 4 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{112}3^{\overline{111}}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 5 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 8 & 3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 6 & 11 & 0 & 8 & 9 & 20 & 8 \\ 3 & 4 & -3 & 3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}3^{\overline{111}}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 5 & 3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 4 & 7 & 0 & 5 & 6 & 7 & 5 \\ 3 & 4 & -3 & 3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}3^{\overline{111}}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 14 & 7 \\ 2 & 4 & 0 & 6 & 0 & 5 & 2 \\ 5 & 8 & 0 & 14 & 5 & 12 & 6 \\ 3 & 4 & -3 & 7 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{111}3^{\overline{111}}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 7 & 12 & 0 & 12 & 8 & 25 & 9 \\ 3 & 4 & -3 & 4 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 14 & 2 \\ 11 & 20 & 0 & 18 & 14 & 69 & 15 \\ 3 & 4 & -3 & 4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 5 & 3 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 4 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 5 & 3 \\ 5 & 9 & 0 & 6 & 12 & 11 & 6 \\ 4 & 7 & 0 & 5 & 11 & 7 & 5 \\ 3 & 4 & -3 & 3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 8 & 3 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 7 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & -3 & 3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 0 & 11 & 2 \\ 9 & 16 & 0 & 15 & 11 & 44 & 12 \\ 3 & 4 & -3 & 4 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 11 & 7 \\ 2 & 4 & 0 & 6 & 0 & 5 & 2 \\ 4 & 7 & 0 & 11 & 5 & 7 & 5 \\ 3 & 4 & -3 & 7 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 20 & 7 \\ 2 & 4 & 0 & 6 & 0 & 8 & 2 \\ 7 & 12 & 0 & 20 & 8 & 25 & 9 \\ 3 & 4 & -3 & 7 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$4.491 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -119$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -3 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -126$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 & 0 \\ 10 & 24 & 9 & 52 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 & 0 \\ 7 & 16 & 6 & 22 & 0 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -182$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.9$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 4 & 3 \\ 3 & 4 & 0 & 13 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 12 & 18 & 10 \\ 3 & 5 & 0 & 12 & 3 & 6 & 3 \\ 4 & 8 & 0 & 18 & 6 & 9 & 5 \\ 3 & 4 & -3 & 10 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{1111}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -3 & 3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{132}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 4 & 3 \\ 3 & 4 & 0 & 17 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 15 & 24 & 13 \\ 3 & 5 & 0 & 15 & 3 & 6 & 3 \\ 4 & 8 & 0 & 24 & 6 & 9 & 5 \\ 3 & 4 & -3 & 13 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 2 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 & 0 \\ 11 & 24 & 9 & 52 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 & 0 \\ 15 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & -3 & 7 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -3 & 3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 & 0 \\ 14 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{141} 1^{111} 31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 18 & 15 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 & 3 \\ 2 & 4 & 0 & 15 & 3 & 0 & 2 \\ 3 & 4 & -3 & 16 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{111} 1^{111} 31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{121} 1^{111} 31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 12 & 9 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 & 3 \\ 2 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & -3 & 10 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{131} 1^{111} 31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 15 & 12 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 & 3 \\ 2 & 4 & 0 & 12 & 3 & 0 & 2 \\ 3 & 4 & -3 & 13 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{111} 1^{111} 31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 12 & 6 \\ 3 & 4 & -3 & 7 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.492 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{s}u_2 + (-3) \rightarrow (-2)$

$T = 8 \quad [0] \quad 1^3 1^{11} 2^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 1 & 4 \\ 8 & 9 & 4 & 8 & 0 & 5 \\ 3 & 4 & -2 & 3 & 0 & 0 \\ 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

4.493 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3)$

$T = 7 \quad [1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 3 & 4 & -2 & 2 & 0 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[0] $21^1 3^1 1^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$

- $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 3 & 4 & -2 & 2 & 0 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$
- $[2] \ 13^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & -2 & 2 & 0 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$
- $[3] \ 1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- $[4] \ 21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -170$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & -2 & 2 & 0 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$
- $T = 8 \quad [5] \ 1^{11}313^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 2$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 15$, $T_{\max} = 35$, $T \in [15, 35]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 5 & 0 & 4 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & -1 \end{pmatrix}_{b_0}$$
- $[6] \ 21^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 30 & 13 & 10 \\ 9 & 13 & 4 & 4 \\ 7 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$
- $[7] \ 13^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 13 & 6 & 5 \\ 13 & 30 & 13 & 10 \\ 6 & 13 & 4 & 4 \\ 5 & 10 & 4 & 2 \end{pmatrix}_{b_0}$$
- $[8] \ 1^{11}3^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$
- $[9] \ 13^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 4 & 4 \\ 5 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$
- $[10] \ 1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 5 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$
- $[11] \ 1^{13}31^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$
- $[12] \ 1^{13}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 63$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0} \\
[13] & 1^1 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 8 & 6 & 0 & 6 \\ 2 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & -3 & 2 \end{pmatrix}_{b_0} \\
[14] & 1^1 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[15] & 21^1 1^1 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 4 & 4 \\ 7 & 9 & 4 & 3 \end{pmatrix}_{b_0} \\
[16] & 1^1 321^1 1^1 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 45, T \in [12, 45] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0} \\
T = 9 & [18] 21^1 1^1 413^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 0 & 0 & 4 \\ 2 & 7 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 3 & -2 & 0 & -2 \end{pmatrix}_{b_0} \\
[17] & 1^1 1^1 4 13^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 9 & 0 & 0 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & -1 \end{pmatrix}_{b_0} \\
[19] & 1^1 1^1 4 13^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0} \\
[20] & 1^1 1^1 23^1 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 3 & 3 \\ 5 & 18 & 1 & 10 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 10 & 0 & 4 & 4 \\ 3 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0} \\
[21] & 1^1 1^1 2313^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 2 \\ 3 & 0 & -1 & 2 & 2 \end{pmatrix}_{b_0} \\
[22] & 1^1 231^1 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 8 & 6 & 0 & 6 \\ 2 & 6 & 2 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 4 & -3 & 2 \end{pmatrix}_{b_0} \\
[23] & 21^1 1^1 41^1 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 0 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & -2 & 0 & 1 \end{pmatrix}_{b_0} \\
[24] \quad & 1^1 23^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 3 & 3 \\ 5 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 9 & 0 & 4 & 4 \\ 3 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0} \\
[25] \quad & 1^1 23^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[26] \quad & 1^1 231^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 2 & 3 \\ 5 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0} \\
[27] \quad & 1^1 3^1 1^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 4 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0} \\
[28] \quad & 1^1 2321^1 1^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0} \\
T = 10 \quad [29] \quad & 1^1 15^1 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 8 & 5 \\ 9 & 13 & 12 & 7 \\ 8 & 12 & 10 & 6 \\ 5 & 7 & 6 & 3 \end{pmatrix}_{b_0} \\
[31] \quad & 21^1 1^1 5^1 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 13 & 14 & 12 & 7 \\ 14 & 13 & 12 & 7 \\ 12 & 12 & 10 & 6 \\ 7 & 7 & 6 & 3 \end{pmatrix}_{b_0} \\
[30] \quad & 21^1 1^1 5^1 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 10 & 8 & 5 \\ 10 & 13 & 12 & 7 \\ 8 & 12 & 10 & 6 \\ 5 & 7 & 6 & 3 \end{pmatrix}_{b_0} \\
[32] \quad & 1^1 31^1 413^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0} \\
[33] \quad & 1^1 3^1 1^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 2 & 0 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^1 3^1 1^1 4 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 60, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [35] \ 1^1 3 1^1 5 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -8, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^1 3^1 1 5 1^1 2 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -10, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 8 & 1 & 4 & 3 \\ 8 & 38 & 1 & 20 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 20 & 0 & 10 & 6 \\ 3 & 12 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[37] \ 1^1 3^1 1 5 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -20, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.60 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^1 3^1 1 5 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -21, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.67 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^1 3 1 5 1^1 2 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 14, \ \Delta = -85, \ \det = -84, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.90 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 8 & 1 & 2 & 3 \\ 8 & 38 & 1 & 8 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 8 & 0 & 0 & 2 \\ 3 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [40] \ 1^1 3^1 6^1 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 20, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.60 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[41] \ 1^1 3^1 6^1 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 16, \ \Delta = -88, \ \det = 52, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.15 \rightarrow 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^1 3^1 2 1^1 6 1 2 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 15, \ \Delta = -83, \ \det = 84, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.90 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 10 & 5 & 0 & 4 & 6 \\ 1 & 5 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 4 & 2 & 0 & 0 & 2 \\ 2 & 6 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [43] \quad 1^{\overline{11}321} 8^{11} 123^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 16, \Delta = -81, \det = -20, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [44] \quad 1^{\overline{11}321} 8^{11} 123^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 17, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{\overline{11}3221} 8^{11} 123^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 17, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [46] \quad 1^{\overline{11}321} 8^{11} 123^1 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$4.494 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$$

$$T = 8 \quad [0] \quad 1^{\overline{111}3^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$$4.495 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2$$

$$T = 9 \quad [0] \quad 1^{\overline{1111}3^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.496 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$$

$$T = 9 \quad [0] \quad 1^{\overline{1111}3^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 0 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$4.497 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2$$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 200, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 600, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 2 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$4.498 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 174, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{\overline{13}} 3^{\overline{11}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 22, \ \Delta = -111, \ \det = -59, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 5 & 13 & 6 & 1 & 5 \\ 13 & 30 & 13 & 0 & 10 \\ 6 & 13 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 22, \ \Delta = -111, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}31^13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 5 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^11^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -187, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \ 1^{11}4^{11}31^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 2, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 17, \ T_{\max} = 22, \ T \in [17, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}23^11^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 14, \ \Delta = -90, \ \det = 20, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 2 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 4 & 0 & 2 & 0 \\ 2 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}3^11^14^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 202, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.63 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \ 1^{11}3^11^15^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 66, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 26 & 1 & 20 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

4.499 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^11^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -222, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$4.500 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_3$$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$4.501 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$$

$$T = 8 \quad [0] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & -3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 10 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 10 & 0 & 6 & 22 & 8 \\ 3 & 4 & -3 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 21^{13} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 3 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 3 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 3 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 3 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 5 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 5 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 72, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.49 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 13^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 98, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 3 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 4 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 4 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 24 & 0 & 14 & 121 & 19 \\ 3 & 4 & -3 & 2 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 202, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 14 & 0 & 8 & 37 & 11 \\ 3 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 202, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.79 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 19 & 0 & 11 & 73 & 15 \\ 3 & 4 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 232, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.55 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 0 & 9 & 49 & 13 \\ 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 21 & 0 & 11 & 76 & 16 \\ 3 & 4 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 18 & 0 & 10 & 58 & 14 \\ 3 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[19] $21^{111}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 6 & 4 \\ 8 & 9 & 4 & 4 & 8 & 5 \\ 3 & 4 & -2 & 2 & 3 & 0 \\ 3 & 4 & 2 & 0 & 3 & 2 \\ 6 & 8 & 3 & 3 & 4 & 4 \\ 4 & 5 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 13 & 23 & 0 & 13 & 100 & 18 \\ 3 & 4 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 22 & 0 & 12 & 85 & 17 \\ 3 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 16 & 28 & 0 & 16 & 154 & 22 \\ 3 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 26 & 0 & 14 & 118 & 20 \\ 3 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$

11.0

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 15 & 27 & 0 & 15 & 133 & 21 \\ 3 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 882, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 18 & 32 & 0 & 18 & 193 & 25 \\ 3 & 4 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 17 & 31 & 0 & 17 & 172 & 24 \\ 3 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 1212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 20 & 36 & 0 & 20 & 238 & 28 \\ 3 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [28] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 11 & 5 \\ 6 & 5 & 1 & 10 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 10 & 0 & 18 & 8 \\ 5 & 5 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 \\ 2 & 6 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 6 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 12 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 16 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 1 & 5 \\ 6 & 5 & 1 & 0 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 5 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11}3^{11}1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -61, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.54 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 3 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -61, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 6 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{12}31^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 13 & 0 & 44 & 14 \\ 3 & 6 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11}3^{11}1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 3 \\ 2 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11}3^{11}1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 3 \\ 2 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11}3^{11}1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 2 & 6 & 3 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{13}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 38 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -94, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 22 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 7 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{13}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 28 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{11}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 8 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 28 & 5 \\ 6 & 5 & 1 & 27 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 27 & 0 & 125 & 21 \\ 5 & 5 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 34 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 17 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{12}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 13 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{14}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 44 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{13}3^11^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 2 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 5 & 0 & 8 & 6 \\ 3 & 6 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{13}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 19 & 5 \\ 6 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 21 & 0 & 61 & 15 \\ 5 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{13}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -195, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 49 & 5 \\ 6 & 5 & 1 & 51 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 51 & 0 & 421 & 39 \\ 5 & 5 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{13} 3^1 1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{14} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 25 & 5 \\ 6 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 28 & 0 & 110 & 20 \\ 5 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 33 & 5 \\ 6 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 34 & 0 & 186 & 26 \\ 5 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{12} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 44 & 5 \\ 6 & 5 & 1 & 44 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 44 & 0 & 326 & 34 \\ 5 & 5 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{12} 3^1 1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 4 & 12 & 9 & 0 & 7 & 28 & 10 \\ 2 & 4 & 2 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 31^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 18 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 14 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{13} 31^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 8 & 6 & 0 & 35 & 6 \\ 2 & 6 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 35 & 22 & 0 & 134 & 24 \\ 3 & 6 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 23 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{13} 3^1 1^{12} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 10 & 0 & 38 & 12 \\ 3 & 6 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{13} 3^1 1^{12} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -306, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 \\ 1 & 4 & 0 & 0 & 2 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 6 \\ 4 & 12 & 6 & 0 & 6 & 22 \\ 2 & 4 & 2 & -3 & 2 & 8 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{11} 3^1 1^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -311, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 \\ 1 & 4 & 0 & 0 & 2 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 \\ 3 & 9 & 6 & 0 & 5 & 13 \\ 2 & 4 & 2 & -3 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{12} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -369, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 20 & 5 \\ 6 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 61 & 15 \\ 5 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{14} 3^1 1^{12} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 8 & 24 & 18 & 0 & 14 & 118 & 20 \\ 2 & 4 & 2 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -395, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 29 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -395, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 39 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{12} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -426, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 34 & 5 \\ 6 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 34 & 0 & 186 & 26 \\ 5 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 24 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 55 & 5 \\ 6 & 5 & 1 & 55 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 55 & 55 & 0 & 530 & 44 \\ 5 & 5 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -454, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 26 & 5 \\ 6 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 28 & 0 & 110 & 20 \\ 5 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13} 3^1 1^{13} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -481, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.07 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 \\ 1 & 4 & 0 & 0 & 2 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 9 \\ 6 & 18 & 9 & 0 & 9 & 52 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{14} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -486, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 35 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13} 31^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.14 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 8 & 6 & 0 & 30 & 6 \\ 2 & 6 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 30 & 15 & 0 & 88 & 18 \\ 3 & 6 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{12} 31^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 8 & 6 & 0 & 30 & 6 \\ 2 & 6 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 30 & 18 & 0 & 94 & 20 \\ 3 & 6 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{11} 31^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.02 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 8 & 6 & 0 & 25 & 6 \\ 2 & 6 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 25 & 14 & 0 & 62 & 16 \\ 3 & 6 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -549, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 50 & 5 \\ 6 & 5 & 1 & 51 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 51 & 0 & 421 & 39 \\ 5 & 5 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{14}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -566, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 45 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{11}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 15 & 9 & 0 & 8 & 37 & 11 \\ 2 & 4 & 2 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{13}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -629, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 30 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{14}3^11^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -664, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 31 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 31 & 0 & 272 & 34 \\ 3 & 6 & 3 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{13}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 7 & 21 & 15 & 0 & 12 & 85 & 17 \\ 2 & 4 & 2 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{13}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -689, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 40 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{13}3^11^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 8 & 24 & 12 & 0 & 12 & 94 & 16 \\ 2 & 4 & 2 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{12}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -746, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 6 & 18 & 12 & 0 & 10 & 58 & 14 \\ 2 & 4 & 2 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{13}3^11^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 8 & 6 & 0 & 40 & 6 \\ 2 & 6 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 20 & 0 & 158 & 24 \\ 3 & 6 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 36 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{14}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -874, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 56 & 5 \\ 6 & 5 & 1 & 58 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 58 & 0 & 530 & 44 \\ 5 & 5 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{11}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -874, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 8 & 6 & 0 & 35 & 6 \\ 2 & 6 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 35 & 19 & 0 & 122 & 22 \\ 3 & 6 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{14}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -934, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 46 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{13}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 8 & 6 & 0 & 45 & 6 \\ 2 & 6 & 1 & 0 & 27 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 45 & 27 & 0 & 214 & 30 \\ 3 & 6 & 3 & -3 & 30 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{11}3^11^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1011, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 11 & 2 \\ 7 & 21 & 12 & 0 & 11 & 73 & 15 \\ 2 & 4 & 2 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{12}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 8 & 6 & 0 & 40 & 6 \\ 2 & 6 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 23 & 0 & 164 & 26 \\ 3 & 6 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{13}3^11^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1041, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 10 & 30 & 15 & 0 & 15 & 148 & 20 \\ 2 & 4 & 2 & -3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{13}3^11^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 25 & 0 & 248 & 30 \\ 3 & 6 & 3 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{11}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 8 & 6 & 0 & 45 & 6 \\ 2 & 6 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 45 & 24 & 0 & 202 & 28 \\ 3 & 6 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{12}3^{11}3^{13}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 13 & 2 \\ 8 & 24 & 15 & 0 & 13 & 100 & 18 \\ 2 & 4 & 2 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{14}3^{11}3^{13}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 17 & 2 \\ 10 & 30 & 21 & 0 & 17 & 172 & 24 \\ 2 & 4 & 2 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{11}3^{11}3^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 9 & 27 & 15 & 0 & 14 & 121 & 19 \\ 2 & 4 & 2 & -3 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{13}3^{11}3^{13}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1471, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 9 & 27 & 18 & 0 & 15 & 133 & 21 \\ 2 & 4 & 2 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{14}3^{11}3^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 24 & 3 \\ 4 & 8 & 6 & 0 & 60 & 6 \\ 2 & 6 & 1 & 0 & 36 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 24 & 60 & 36 & 0 & 382 & 40 \\ 3 & 6 & 3 & -3 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{12}3^{11}3^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 28 & 0 & 254 & 32 \\ 3 & 6 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{13}3^{11}3^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.00 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 22 & 3 \\ 4 & 8 & 6 & 0 & 55 & 6 \\ 2 & 6 & 1 & 0 & 32 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 55 & 32 & 0 & 314 & 36 \\ 3 & 6 & 3 & -3 & 36 & 1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{12}3^{11}3^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2006, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 16 & 2 \\ 10 & 30 & 18 & 0 & 16 & 154 & 22 \\ 2 & 4 & 2 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{13}3^{11}3^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 33 & 4 \\ 1 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 18 & 2 \\ 11 & 33 & 21 & 0 & 18 & 193 & 25 \\ 2 & 4 & 2 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{14}3^{11}1431^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -2486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 36 & 4 \\ 1 & 4 & 0 & 0 & 2 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 20 & 2 \\ 12 & 36 & 24 & 0 & 20 & 238 & 28 \\ 2 & 4 & 2 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [108] $1^{11}413^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{11}413^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{11}413^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{11}413^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{11}413^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 0 & 8 & 6 \\ 3 & 4 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{14}413^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 19 & 0 & 0 & 65 & 15 \\ 3 & 4 & -2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{12}413^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 0 & 0 & 21 & 9 \\ 3 & 4 & -2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{13} 4^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 0 & 40 & 12 \\ 3 & 4 & -2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{11} 4^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 0 & 26 & 10 \\ 3 & 4 & -2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{12} 4^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 16 & 0 & 0 & 45 & 13 \\ 3 & 4 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{14} 4^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 24 & 0 & 0 & 101 & 19 \\ 3 & 4 & -2 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{13} 4^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 20 & 0 & 0 & 70 & 16 \\ 3 & 4 & -2 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

4.502 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{f}_4$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 2 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

4.503 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 3 & 2 & 2 \\ 5 & 9 & 0 & 5 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 3 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 6 & 4 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.504 NHC(-2)(-3) + f₄ + su₂→(-3) + su₂ + so₈

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 4 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

4.505 NHC(-2)(-3) + f₄ + su₂→(-3) + su₂ + su₂

$$T = 9 \quad [0] \quad 1^{124}3^{11}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 17 & 3 \\ 3 & 4 & 0 & 3 & 8 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 17 & 2 \\ 4 & 8 & 0 & 4 & 10 & 42 & 6 \\ 17 & 31 & 0 & 17 & 42 & 172 & 24 \\ 3 & 4 & -3 & 2 & 6 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 4 & 8 & 0 & 4 & 10 & 12 & 6 \\ 4 & 8 & 0 & 4 & 12 & 10 & 6 \\ 3 & 4 & -3 & 2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 4 & 8 & 0 & 4 & 10 & 12 & 6 \\ 5 & 9 & 0 & 5 & 12 & 13 & 7 \\ 3 & 4 & -3 & 2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 3 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 9 & 17 & 0 & 9 & 15 & 49 & 13 \\ 3 & 4 & -3 & 2 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 10 & 6 & 6 \\ 2 & 4 & 0 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 3 & 8 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 4 & 8 & 0 & 4 & 10 & 30 & 6 \\ 12 & 22 & 0 & 12 & 30 & 85 & 17 \\ 3 & 4 & -3 & 2 & 6 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 3 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 4 & 8 & 0 & 4 & 10 & 18 & 6 \\ 7 & 13 & 0 & 7 & 18 & 28 & 10 \\ 3 & 4 & -3 & 2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 4 & 0 & 2 & 1 & 9 & 3 \\ 8 & 14 & 0 & 8 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 3 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 7 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 4 \\ 7 & 13 & 0 & 7 & 12 & 28 & 10 \\ 3 & 4 & -3 & 2 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 3 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 5 & 9 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 \\ 5 & 9 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 3 & 8 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 4 & 8 & 0 & 4 & 10 & 24 & 6 \\ 9 & 17 & 0 & 9 & 24 & 49 & 13 \\ 3 & 4 & -3 & 2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 3 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 11 & 2 \\ 5 & 9 & 0 & 5 & 13 & 33 & 7 \\ 11 & 21 & 0 & 11 & 33 & 76 & 16 \\ 3 & 4 & -3 & 2 & 7 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 3 & 8 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 4 & 8 & 0 & 4 & 10 & 30 & 6 \\ 11 & 21 & 0 & 11 & 30 & 76 & 16 \\ 3 & 4 & -3 & 2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 4 & 0 & 2 & 1 & 9 & 3 \\ 7 & 13 & 0 & 7 & 9 & 28 & 10 \\ 3 & 4 & -3 & 2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 3 & 8 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 14 & 2 \\ 4 & 8 & 0 & 4 & 10 & 36 & 6 \\ 14 & 26 & 0 & 14 & 36 & 118 & 20 \\ 3 & 4 & -3 & 2 & 6 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 5 & 2 \\ 5 & 9 & 0 & 5 & 13 & 15 & 7 \\ 5 & 9 & 0 & 5 & 15 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 13 & 2 \\ 2 & 4 & 0 & 2 & 1 & 15 & 3 \\ 13 & 23 & 0 & 13 & 15 & 100 & 18 \\ 3 & 4 & -3 & 2 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 4 & 0 & 2 & 1 & 12 & 3 \\ 9 & 17 & 0 & 9 & 12 & 49 & 13 \\ 3 & 4 & -3 & 2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{113}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 3 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 9 & 2 \\ 5 & 9 & 0 & 5 & 13 & 27 & 7 \\ 9 & 17 & 0 & 9 & 27 & 49 & 13 \\ 3 & 4 & -3 & 2 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 4 & 0 & 2 & 1 & 12 & 3 \\ 10 & 18 & 0 & 10 & 12 & 58 & 14 \\ 3 & 4 & -3 & 2 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 4 & 0 & 2 & 1 & 15 & 3 \\ 11 & 21 & 0 & 11 & 15 & 76 & 16 \\ 3 & 4 & -3 & 2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{112}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 3 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 7 & 2 \\ 5 & 9 & 0 & 5 & 13 & 21 & 7 \\ 7 & 13 & 0 & 7 & 21 & 28 & 10 \\ 3 & 4 & -3 & 2 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{124}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 11 & 3 \\ 3 & 4 & 0 & 3 & 13 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 11 & 2 \\ 7 & 13 & 0 & 7 & 28 & 48 & 10 \\ 11 & 21 & 0 & 11 & 48 & 76 & 16 \\ 3 & 4 & -3 & 2 & 10 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{122}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 7 & 2 \\ 7 & 13 & 0 & 7 & 28 & 30 & 10 \\ 7 & 13 & 0 & 7 & 30 & 28 & 10 \\ 3 & 4 & -3 & 2 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 1^{12} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 9 & 3 \\ 3 & 4 & 0 & 3 & 13 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 9 & 2 \\ 7 & 13 & 0 & 7 & 28 & 39 & 10 \\ 9 & 17 & 0 & 9 & 39 & 49 & 13 \\ 3 & 4 & -3 & 2 & 10 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 1^{13} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 4 & 0 & 2 & 1 & 15 & 3 \\ 12 & 22 & 0 & 12 & 15 & 85 & 17 \\ 3 & 4 & -3 & 2 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 1^{13} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 18 & 2 \\ 2 & 4 & 0 & 2 & 1 & 21 & 3 \\ 18 & 32 & 0 & 18 & 21 & 193 & 25 \\ 3 & 4 & -3 & 2 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11} 1^{13} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 9 & 2 \\ 9 & 17 & 0 & 9 & 49 & 51 & 13 \\ 9 & 17 & 0 & 9 & 51 & 49 & 13 \\ 3 & 4 & -3 & 2 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11} 1^{14} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 11 & 3 \\ 3 & 4 & 0 & 3 & 17 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 11 & 2 \\ 9 & 17 & 0 & 9 & 49 & 63 & 13 \\ 11 & 21 & 0 & 11 & 63 & 76 & 16 \\ 3 & 4 & -3 & 2 & 13 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11} 1^{14} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 15 & 2 \\ 2 & 4 & 0 & 2 & 1 & 18 & 3 \\ 15 & 27 & 0 & 15 & 18 & 133 & 21 \\ 3 & 4 & -3 & 2 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} 1^{14} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 11 & 3 \\ 3 & 4 & 0 & 3 & 21 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 11 & 2 \\ 11 & 21 & 0 & 11 & 76 & 78 & 16 \\ 11 & 21 & 0 & 11 & 78 & 76 & 16 \\ 3 & 4 & -3 & 2 & 16 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11} 1^{14} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 4 & 0 & 2 & 1 & 18 & 3 \\ 14 & 26 & 0 & 14 & 18 & 118 & 20 \\ 3 & 4 & -3 & 2 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{114}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1434, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 17 & 2 \\ 2 & 4 & 0 & 2 & 1 & 21 & 3 \\ 17 & 31 & 0 & 17 & 21 & 172 & 24 \\ 3 & 4 & -3 & 2 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{114}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1604, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 20 & 2 \\ 2 & 4 & 0 & 2 & 1 & 24 & 3 \\ 20 & 36 & 0 & 20 & 24 & 238 & 28 \\ 3 & 4 & -3 & 2 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{111}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 247, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 & 5 \\ 1 & 3 & 3 & 0 & 2 & 1 & 6 \\ 3 & 9 & 6 & 0 & 5 & 6 & 13 \\ 2 & 4 & 2 & -3 & 2 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13}3^11^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 472, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{113}3^11^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 517, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 5 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 5 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{111}31^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 648, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 24 & 10 \\ 6 & 15 & 9 & 0 & 24 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111}3^11^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 912, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 5 & 15 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.506 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 5 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 7 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 5 & 6 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 3 & 3 & 6 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 5 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 7 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 5 & 2 & 2 \\ 5 & 9 & 0 & 5 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 5 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 2 & 6 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 7 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 5 \\ 1 & 4 & 0 & 0 & 2 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 \\ 1 & 3 & 3 & 0 & 2 & 1 & 4 \\ 2 & 5 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 5 & 4 \\ 2 & 5 & 4 & 0 & 3 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 475, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 3 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 9 & 7 \\ 2 & 5 & 4 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

4.508 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 4 & 8 & 0 & 4 & 10 & 12 & 6 \\ 5 & 9 & 0 & 5 & 12 & 12 & 6 \\ 3 & 4 & -3 & 2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 11 & 19 & 0 & 11 & 73 & 12 & 15 \\ 2 & 4 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 10 & 6 & 6 \\ 2 & 4 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -3 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -166, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 \\ 4 & 8 & 0 & 4 & 6 & 9 & 5 \\ 3 & 4 & -3 & 2 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 4 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 3 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 4 & 2 \\ 5 & 9 & 0 & 5 & 13 & 12 & 7 \\ 4 & 8 & 0 & 4 & 12 & 9 & 5 \\ 3 & 4 & -3 & 2 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 \\ 5 & 9 & 0 & 5 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 8 & 14 & 0 & 8 & 37 & 9 & 11 \\ 2 & 4 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -3 & 2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 5 & 13 & 6 & 7 \\ 2 & 4 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 4 & 0 & 2 & 1 & 9 & 3 \\ 7 & 13 & 0 & 7 & 9 & 27 & 9 \\ 3 & 4 & -3 & 2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 13 & 0 & 7 & 28 & 9 & 10 \\ 2 & 4 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -3 & 2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13}1^31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 9 & 17 & 0 & 9 & 49 & 12 & 13 \\ 2 & 4 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12}1^31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 3 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 16 & 2 & 2 \\ 16 & 28 & 0 & 16 & 154 & 18 & 22 \\ 2 & 4 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{14}1^31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 11 & 21 & 0 & 11 & 76 & 15 & 16 \\ 2 & 4 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 2 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12}1^31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 10 & 2 & 2 \\ 10 & 18 & 0 & 10 & 58 & 12 & 14 \\ 2 & 4 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12}1^31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 3 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 13 & 2 & 2 \\ 13 & 23 & 0 & 13 & 100 & 15 & 18 \\ 2 & 4 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{13}1^31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 3 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 12 & 22 & 0 & 12 & 85 & 15 & 17 \\ 2 & 4 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{13}1^31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 3 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 15 & 2 & 2 \\ 15 & 27 & 0 & 15 & 133 & 18 & 21 \\ 2 & 4 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 2 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{14}1^31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 14 & 26 & 0 & 14 & 118 & 18 & 20 \\ 2 & 4 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 2 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13} 1^1 3^1 1^{14} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 3 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 18 & 2 & 2 \\ 18 & 32 & 0 & 18 & 193 & 21 & 25 \\ 2 & 4 & 0 & 2 & 21 & 0 & 2 \\ 3 & 4 & -3 & 2 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{14} 1^1 3^1 1^{13} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 3 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 17 & 2 & 2 \\ 17 & 31 & 0 & 17 & 172 & 21 & 24 \\ 2 & 4 & 0 & 2 & 21 & 0 & 2 \\ 3 & 4 & -3 & 2 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14} 1^1 3^1 1^{14} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 3 & 36 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 20 & 2 & 2 \\ 20 & 36 & 0 & 20 & 238 & 24 & 28 \\ 2 & 4 & 0 & 2 & 24 & 0 & 2 \\ 3 & 4 & -3 & 2 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11} 1^1 4 \quad 13^1 1^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 0 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 1^1 3^1 1^1 4 \quad 1^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 0 & 8 & 5 & 6 \\ 2 & 4 & 0 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 0 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 1^1 3^1 1^1 4 \quad 1^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 12 & 0 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 0 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$4.509 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_2$$

$$T = 10 \quad [0] \quad 1^{11} 1^1 3^1 1^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 2 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 2 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 5 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 6 & 13 & 7 & 7 \\ 3 & 4 & -3 & 2 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 & 5 & 2 \\ 5 & 9 & 0 & 5 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 2 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 15 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.510 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}3^{\overline{11}}}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 4 & 0 & 2 & 2 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 & 3 \\ 1 & 2 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & -3 & 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}3^{\overline{1}}}1^{\overline{1}3^{\overline{1}}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 & 3 & 3 \\ 2 & 6 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 & 3 & 3 \\ 2 & 5 & 3 & 0 & 1 & 3 & 3 & 3 \\ 3 & 6 & 3 & -3 & 3 & 1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}3^{\overline{1}}}1^{\overline{1}3^{\overline{1}}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -156, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.511 NHC(-2)(-3) + f₄ + su₂→(-3) + su₂n3mix

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 4 \\ 2 & 6 & 5 & 0 & 4 & 7 \\ 2 & 4 & 2 & -3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 2 & 2 & 0 & 0 & 3 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 2 & 3 & 2 & -3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 8 & 6 & 0 & 25 & 6 \\ 2 & 6 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 25 & 16 & 0 & 69 & 17 \\ 3 & 6 & 3 & -3 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 \\ 1 & 4 & 0 & 0 & 2 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 3 \\ 2 & 5 & 4 & 0 & 3 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 \\ 2 & 4 & 4 & 0 & 3 & 8 \\ 1 & 4 & 0 & 0 & 2 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 4 \\ 3 & 8 & 4 & 0 & 4 & 9 \\ 2 & 4 & 2 & -3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 8 & 0 & 21 & 9 \\ 3 & 6 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 12 & 0 & 41 & 13 \\ 3 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{13}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 2 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 6 & 17 & 13 & 0 & 10 & 60 & 14 \\ 2 & 4 & 2 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{13}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{12}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 5 & 15 & 11 & 0 & 9 & 44 & 12 \\ 2 & 4 & 2 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 4 & 12 & 8 & 0 & 7 & 25 & 9 \\ 2 & 4 & 2 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 2 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 6 & 2 \\ 4 & 11 & 7 & 0 & 6 & 20 & 8 \\ 2 & 4 & 2 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{12}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -465, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 14 & 10 & 0 & 8 & 37 & 11 \\ 2 & 4 & 2 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{13}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 15 & 8 & 0 & 8 & 36 & 10 \\ 2 & 4 & 2 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -844, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 6 & 18 & 11 & 0 & 10 & 55 & 13 \\ 2 & 4 & 2 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -968, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 2 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 7 & 21 & 14 & 0 & 12 & 80 & 16 \\ 2 & 4 & 2 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

4.512 NHC(-2)(-3) + f₄ + su₂→(-3) + su_{2n3mix} + su₂

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 & 2 \\ 2 & 5 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 2 & 4 & 1 & 3 \\ 2 & 4 & 2 & -3 & 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 175, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 2 & 5 & 4 & 0 & 3 & 4 & 5 & 4 \\ 2 & 6 & 3 & 0 & 3 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 475, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 5 & 2 \\ 2 & 5 & 4 & 0 & 3 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 5 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

4.513 NHC(-2)(-3) + f₄ + su₂→(-3) + su₃

$$T = 8 \quad [0] \quad 1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 35, \quad T \in [14, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 & 4 \\ 3 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 6 & 12 & 0 & 6 & 24 & 8 & 8 \\ 3 & 4 & -3 & 2 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 & 1 \\ 3 & 5 & 0 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 98, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.22 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 9 & 5 & 5 \\ 3 & 4 & -3 & 2 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 118, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -3 & 2 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 128, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 35, \quad T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & -3 & 2 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 208, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 278, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 0 & 7 & 27 & 9 \\ 3 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 328, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 0 & 9 & 48 & 12 \\ 3 & 4 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{13}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 13 & 6 & 1 & 5 \\ 13 & 30 & 13 & 0 & 10 \\ 6 & 13 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 10 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -52, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}31^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 6 & 1 & 12 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 25 & 9 \\ 5 & 5 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 6 & 1 & 6 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 4 & 4 \\ 5 & 5 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}31^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 \\ 2 & 6 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 6 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}3^{11}1^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}31^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 13 & 0 & 43 & 13 \\ 3 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11}3^{11}1^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -149, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{13}1^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 2 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 5 & 0 & 7 & 5 \\ 3 & 6 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{13}1^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -284, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.42 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11}31^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -306, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 0 & 21 & 9 \\ 3 & 6 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11}3^{11}1^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -394, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [21] \ 1^{11}4^{11}13^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11}3^{11}1^14^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.06 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 2 & 3 \\ 4 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & -2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [23] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 8 & 1 & 5 & 7 \\ 9 & 13 & 12 & 0 & 7 & 8 \\ 8 & 12 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -3 & -1 & 3 \\ 5 & 7 & 6 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

4.514 NHC(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_3$ + $\mathbf{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.515 NHC(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_3$ + $\mathbf{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 3 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 4 & 2 \\ 5 & 9 & 0 & 5 & 12 & 12 & 6 \\ 4 & 8 & 0 & 4 & 12 & 10 & 6 \\ 3 & 4 & -3 & 2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 4 & 8 & 0 & 4 & 6 & 10 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 11 & 19 & 0 & 11 & 12 & 73 & 15 \\ 3 & 4 & -3 & 2 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 9 & 6 & 5 \\ 2 & 4 & 0 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 4 & 8 & 0 & 4 & 9 & 12 & 5 \\ 5 & 9 & 0 & 5 & 12 & 13 & 7 \\ 3 & 4 & -3 & 2 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 5 & 12 & 6 & 6 \\ 2 & 4 & 0 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 4 & 0 & 2 & 0 & 9 & 2 \\ 8 & 14 & 0 & 8 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -476, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 13 & 0 & 7 & 27 & 9 & 9 \\ 2 & 4 & 0 & 2 & 9 & 1 & 3 \\ 3 & 4 & -3 & 2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1123}3^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 4 & 0 & 2 & 0 & 9 & 2 \\ 7 & 13 & 0 & 7 & 9 & 28 & 10 \\ 3 & 4 & -3 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1123}3^{114} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 16 & 2 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 16 & 28 & 0 & 16 & 18 & 154 & 22 \\ 3 & 4 & -3 & 2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1123}3^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 9 & 17 & 0 & 9 & 12 & 49 & 13 \\ 3 & 4 & -3 & 2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1123}3^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 11 & 21 & 0 & 11 & 15 & 76 & 16 \\ 3 & 4 & -3 & 2 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1123}3^{112} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 10 & 18 & 0 & 10 & 12 & 58 & 14 \\ 3 & 4 & -3 & 2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1123}3^{113} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 13 & 2 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 13 & 23 & 0 & 13 & 15 & 100 & 18 \\ 3 & 4 & -3 & 2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1123}3^{112} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 12 & 22 & 0 & 12 & 15 & 85 & 17 \\ 3 & 4 & -3 & 2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1123}3^{113} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{s}u_3 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 15 & 2 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 15 & 27 & 0 & 15 & 18 & 133 & 21 \\ 3 & 4 & -3 & 2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^{14} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 14 & 26 & 0 & 14 & 18 & 118 & 20 \\ 3 & 4 & -3 & 2 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 1^{13} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 18 & 2 \\ 2 & 4 & 0 & 2 & 0 & 21 & 2 \\ 18 & 32 & 0 & 18 & 21 & 193 & 25 \\ 3 & 4 & -3 & 2 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 1^{14} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 17 & 2 \\ 2 & 4 & 0 & 2 & 0 & 21 & 2 \\ 17 & 31 & 0 & 17 & 21 & 172 & 24 \\ 3 & 4 & -3 & 2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 1^{14} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 20 & 2 \\ 2 & 4 & 0 & 2 & 0 & 24 & 2 \\ 20 & 36 & 0 & 20 & 24 & 238 & 28 \\ 3 & 4 & -3 & 2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11} 1^{11} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 0 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 1^{11} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 0 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 1^{11} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 0 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

4.516 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{113}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{violet}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 3 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 5 & 6 & 9 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 & 2 \\ 5 & 9 & 0 & 5 & 12 & 6 & 6 & 6 \\ 2 & 4 & 0 & 2 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 6 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 6 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 5 & 6 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 5 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 5 & 6 & 13 & 15 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 7 & 1 \end{pmatrix} b_0$$

4.517 $\mathbf{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 8 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 3 & 2 \\ 4 & 8 & 0 & 4 & 9 & 6 & 5 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 4 & -3 & 2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -274, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -354, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -544, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.518 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^13^{11}1^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 45, \ \Delta = -119, \ \det = 450, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 45, \ T \in [12, 45]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 1 & 4 \\ 8 & 9 & 4 & 4 & 0 & 5 \\ 3 & 4 & -2 & 2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \ 2^11^14^13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 14, \ \Delta = -89, \ \det = 40, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 1 & 6 & 0 & 0 & 3 \\ 5 & 6 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 6 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \ 1^15^13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -40, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.80 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 8 & 1 & 5 \\ 10 & 13 & 12 & 0 & 7 \\ 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

4.519 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \text{n3mix}$

$$T = 7 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -139, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.09 \rightarrow 15, \ T_{\max} = 97, \ T \in [15, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 5 & 0 \\ 7 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 3 & 4 & -2 & 5 & 0 \\ 5 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[1] $21^13^{111}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 3 & 4 & -2 & 5 & 0 \\ 5 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^13^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 15.65 \rightarrow 16, T_{\max} = 97, T \in [16, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 23 & 0 & 120 & 18 \\ 3 & 4 & -3 & 18 & -3 \end{pmatrix}_{b_0}$$

[5] $1^13^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 18 & 0 & 72 & 14 \\ 3 & 4 & -3 & 14 & -1 \end{pmatrix}_{b_0}$$

[4] $1^13^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 8 & 0 & 12 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[6] $1^13^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 13 & 0 & 36 & 10 \\ 3 & 4 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[7] $1^13^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 \\ 3 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 5 & 0 \\ 6 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[9] $13^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 5 & 0 \\ 5 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 \\ 3 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 15 & 0 & 39 & 11 \\ 3 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -151, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -190, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 19 & 0 & 64 & 14 \\ 3 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -211, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 17 & 0 & 55 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -240, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 16 & 0 & 44 & 12 \\ 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -259, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 22 & 0 & 97 & 17 \\ 3 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -295, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.05 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 27 & 0 & 151 & 21 \\ 3 & 4 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -320, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & -2 & 5 & 0 \\ 9 & 12 & 5 & 12 & 6 \\ 4 & 5 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -331, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 20 & 0 & 69 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -354, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 21 & 0 & 80 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -424, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 24 & 0 & 100 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -456, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 26 & 0 & 128 & 20 \\ 3 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 25 & 0 & 111 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -546, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 31 & 0 & 188 & 24 \\ 3 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 29 & 0 & 148 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -655, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 30 & 0 & 165 & 23 \\ 3 & 4 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -799, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 35 & 0 & 231 & 27 \\ 3 & 4 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 19 & 34 & 0 & 208 & 26 \\ 3 & 4 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -1054, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 22 & 39 & 0 & 280 & 30 \\ 3 & 4 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [30] \quad 1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 19.45 \rightarrow 20, T_{\max} = 22, T \in [20, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 3 & 0 & 7 & 15 & 7 \\ 2 & 2 & -1 & 0 & 7 & -7 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{13}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 25 & 9 \\ 2 & 3 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 3 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11} 3^1 1^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 6 & 0 & 7 & 5 \\ 2 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 13 & 9 & 1 & 7 \\ 9 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{12} 31^1 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 2 & 4 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 8 & 0 & 15 & 7 \\ 2 & 3 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 25 & 12 & 1 & 9 \\ 12 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11} 31^1 31^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{11} 3^1 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 7 & 5 \\ 2 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

- [44] $1^{11}3^1 1^3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 4 & 4 \\ 2 & 3 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{12} 1^{11} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 7 & 6 & 0 & 11 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$
- [48] $1^{12} 1^{13} 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{12} 1^{13} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 5 & 7 & 0 & 9 & 35 & 11 \\ 2 & 2 & -1 & 0 & 11 & 0 \end{pmatrix}_{b_0}$$
- [47] $1^{11} 31^1 1^3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [50] $1^{11} 1^{13} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 5 & 5 & 0 & 8 & 23 & 9 \\ 2 & 2 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$
- [49] $1^{13} 1^{11} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 7 & 10 & 0 & 13 & 74 & 16 \\ 2 & 2 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^{11} 1^{12} 31^1 1^3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 10 & 0 & 37 & 11 \\ 2 & 3 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$
- [52] $1^{13} 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & 0 & 7 & 0 \end{pmatrix}_{b_0}$$
- [53] $1^{11} 1^{13} 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{11} 1^{12} 313^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 7 & 0 & 15 & 7 \\ 2 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{12} 3^{11} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 7 & 8 & 0 & 12 & 56 & 14 \\ 2 & 2 & -1 & 0 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{14} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{13} 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 5 \\ 2 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 3^{12} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 16 & 0 & 100 & 18 \\ 2 & 3 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{13} 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{12} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{13} 3^{11} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

- [64] $1^{12}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 13 & 0 & 60 & 14 \\ 2 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{13}3^{12}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{13}3^{12}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 6 & 9 & 0 & 10 & 51 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$
- [68] $1^{11}3^{12}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 6 & 5 & 0 & 8 & 23 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$
- [69] $1^{114}3^{13}3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 0 & 22 & 3 \\ 1 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 22 & 22 & 0 & 193 & 25 \\ 2 & 3 & 2 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$
- [70] $1^{113}3^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 10 & 0 & 32 & 10 \\ 2 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [71] $1^{112}3^{12}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [72] $1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [73] $1^{12}3^{12}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 6 & 7 & 0 & 9 & 35 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$
- [74] $1^{14}3^{12}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 8 & 12 & 0 & 14 & 96 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{13}3^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 7 & 0 & 31 & 9 \\ 2 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{11}31^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{14}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 6 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{11}31^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 13 & 13 & 0 & 60 & 14 \\ 2 & 3 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{12}31^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 11 \\ 2 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{14}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 7 & 11 & 0 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{12}3^{11}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 8 & 8 & 0 & 12 & 56 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{13}3^{11}21^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{13}3^{11}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 8 & 10 & 0 & 13 & 74 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{14}3^{13}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 13 & 13 & 0 & 55 & 13 \\ 2 & 3 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{13}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 355, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{13}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{14}3^{11}21^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{12}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 7 & 0 & 9 & 35 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{13}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 380, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 9 & 0 & 10 & 51 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0} \\
[95] & 1^{14}3^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 398, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.38 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 19 & 3 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 19 & 19 & 0 & 135 & 21 \\ 2 & 3 & 2 & -3 & 21 & 0 \end{pmatrix}_{b_0} \\
[96] & 1^{13}3^{13}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.20 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 20 & 10 & 0 & 65 & 13 \\ 2 & 4 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0} \\
[97] & 1^{14}3^{13}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.41 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 16 & 16 & 0 & 89 & 17 \\ 2 & 3 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0} \\
[98] & 1^{14}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 7 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0} \\
[99] & 1^{14}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 7 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
[100] & 1^{14}3^{11}23^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0} \\
[101] & 1^{14}3^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 483, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0} \\
[102] & 1^{14}3^{12}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 9 & 12 & 0 & 14 & 96 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0} \\
[103] & 1^{13}3^{13}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.11 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0} \\
[105] & 1^{13}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [104] $1^{13}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{14}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 8 & 11 & 0 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$
- [107] $1^{13}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 9 & 10 & 0 & 13 & 74 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$
- [108] $1^{12}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$
- [109] $1^{14}31^{12}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 19 & 0 & 135 & 21 \\ 2 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$
- [110] $1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 563, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 9 & 0 & 10 & 51 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$
- [111] $1^{12}31^{12}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 20 & 13 & 0 & 71 & 15 \\ 2 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [112] $1^{13}31^{12}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 23 & 16 & 0 & 100 & 18 \\ 2 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$
- [113] $1^{13}31^{14}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 13 & 0 & 111 & 17 \\ 2 & 4 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$
- [114] $1^{14}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 8 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [115] $1^{14}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 9 & 11 & 0 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{14}3^{12}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 752, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 10 & 12 & 0 & 14 & 96 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[117] $1^3 1^{14} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 755, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{11} 31^{13} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 767, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 23 & 13 & 0 & 88 & 16 \\ 2 & 4 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{11} 31^{13} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{12} 31^{13} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 967, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 17 & 0 & 128 & 20 \\ 2 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{14} 31^{13} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1007, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 23 & 0 & 208 & 26 \\ 2 & 4 & 2 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{12} 31^{13} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1008, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 16 & 0 & 117 & 19 \\ 2 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{13} 31^{13} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1040, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 20 & 0 & 165 & 23 \\ 2 & 4 & 2 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{11} 31^{14} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 29 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 29 & 16 & 0 & 140 & 20 \\ 2 & 4 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

[125] $1^{13} 31^{13} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 29 & 4 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 29 & 19 & 0 & 152 & 22 \\ 2 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{11} 31^{14} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 17 & 0 & 151 & 21 \\ 2 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{14}31^{13}3^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 32 & 4 \\ 1 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 32 & 22 & 0 & 193 & 25 \\ 2 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{12}3^{11}1^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 20 & 0 & 188 & 24 \\ 2 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{12}31^{14}3^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1460, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 32 & 4 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 32 & 19 & 0 & 175 & 23 \\ 2 & 4 & 2 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13}3^{11}1^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 23 & 0 & 231 & 27 \\ 2 & 4 & 2 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{14}3^{11}1^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1707, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 39 & 4 \\ 1 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 39 & 26 & 0 & 280 & 30 \\ 2 & 4 & 2 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{13}31^{14}3^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1707, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 35 & 4 \\ 1 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 35 & 22 & 0 & 216 & 26 \\ 2 & 4 & 2 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{14}31^{14}3^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1848, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 38 & 4 \\ 1 & 4 & 0 & 0 & 25 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 38 & 25 & 0 & 263 & 29 \\ 2 & 4 & 2 & -3 & 29 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [134] $1^{12}31^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$

$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 3 & 3 \\ 5 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 2 \\ 3 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{11}231^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 5 & 5 \\ 2 & 4 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{13}231^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 31 & 3 \\ 5 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 84 & 0 & 520 & 42 \\ 3 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

[137] $1^{12}231^{12}31^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 24 & 3 \\ 5 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 69 & 0 & 340 & 34 \\ 3 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[138] $1^1 23^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 0 & 0 & 4 & 4 \\ 3 & 0 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{11} 23^1 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 6 & 4 \\ 1 & 5 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 0 & 8 & 6 \\ 2 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[140] $1^1 3^1 1^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 8 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[141] $1^1 4^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{12} 23^1 1^{11} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 54 & 0 & 217 & 27 \\ 3 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[143] $1^{12} 231^{11} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 50 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 50 & 0 & 185 & 25 \\ 3 & 7 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{12} 231^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 19 & 3 \\ 5 & 13 & 1 & 50 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 50 & 0 & 185 & 25 \\ 3 & 7 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{12} 231^{11} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 54 & 0 & 217 & 27 \\ 3 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{11} 4^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[147] $1^1 3^1 1^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{11} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 7 & 0 & 8 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [150] $1^{11}231^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 12 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0}$
- [149] $1^123^{11}1^131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 4 & 3 \\ 5 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 4 & 4 \\ 3 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$
- [152] $1^{112}2313^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 5 & 0 & 9 & 4 \\ 1 & 5 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 9 & 0 & 19 & 9 \\ 2 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$
- [151] $1^{11}231^{11}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0}$
- [154] $21^{11}41^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 2 & 6 & 0 & 4 & 4 \\ 2 & 2 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [153] $1^{11}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [155] $1^{12}231^{11}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 20 & 3 \\ 5 & 13 & 1 & 50 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 50 & 0 & 185 & 25 \\ 3 & 7 & -1 & 25 & 3 \end{pmatrix}_{b_0}$
- [158] $1^{11}231^{11}231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 18 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$
- [157] $1^{12}413^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [156] $1^{12}231^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 12 & 0 & 41 & 13 \\ 2 & 4 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$
- [159] $1^{13}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 4 & 1 & 4 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$
- [161] $1^123^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

- [160] $1^{13}231^{12}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 8 & 3 \\ 5 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 24 & 0 & 37 & 11 \\ 3 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$
- [162] $1^{11}4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [163] $1^{13}4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [164] $1^{123}1^{11}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 11 & 3 \\ 5 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 24 & 0 & 37 & 11 \\ 3 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$
- [165] $1^{1231}1^{13}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 7 & 3 \\ 5 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 20 & 0 & 25 & 9 \\ 3 & 7 & -1 & 9 & 2 \end{pmatrix}_{b_0}$
- [166] $1^{1231}1^{13}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 9 & 3 \\ 5 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 25 & 9 \\ 3 & 7 & -1 & 9 & 2 \end{pmatrix}_{b_0}$
- [167] $1^{1123}1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 14 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 39 & 0 & 107 & 19 \\ 3 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$
- [168] $1^{1123}1^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 0 & 26 & 10 \\ 2 & 4 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$
- [169] $1^{14}4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$
- [170] $1^{12}1^{13}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 11 & 0 & 21 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [171] $1^{1123}1^{11}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 17 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 39 & 0 & 107 & 19 \\ 3 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$
- [173] $1^{1231}1^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$

$$\begin{aligned}
& T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 8 & 3 \\ 5 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 25 & 9 \\ 3 & 7 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[174] & 1^{11}231^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 8 & 0 & 21 & 9 \\ 2 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0} \\
[172] & 1^{11}3^{11}4^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 12 & 0 & 26 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0} \\
[175] & 1^{12}23^{11}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 1 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 0 & 0 & 37 & 11 \\ 3 & 0 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[176] & 1^{12}4^{11}13^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0} \\
[177] & 1^{11}231^{11}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[178] & 1^{12}23^{11}12^{13}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 13 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0} \\
[179] & 1^{11}231^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 13 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[180] & 1^{12}23^{11}1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 9 & 3 \\ 5 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 24 & 0 & 37 & 11 \\ 3 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[181] & 1^{12}23^{11}1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 10 & 3 \\ 5 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 24 & 0 & 37 & 11 \\ 3 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[182] & 1^{13}23^{11}1^{12}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 32 & 3 \\ 5 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 84 & 0 & 520 & 42 \\ 3 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0} \\
[183] & 1^{11}23^{11}13^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 1 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 0 & 0 & 44 & 12 \\ 3 & 0 & -1 & 12 & 2 \end{pmatrix}_{b_0} \\
[184] & 1^{11}23^{11}1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 39 & 0 & 107 & 19 \\ 3 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0} \\
[185] & 1^{11}231^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 14 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[186] & 1^{11}231^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 39 & 0 & 107 & 19 \\ 3 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0} \\
[187] & 1^{13}4^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 15 & 0 & 40 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0} \\
[188] & 1^{13}4^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0} \\
[189] & 1^{12}231^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 25 & 3 \\ 5 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 69 & 0 & 340 & 34 \\ 3 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0} \\
[190] & 1^{12}4^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 16 & 0 & 45 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0} \\
[191] & 1^{12}231^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 16 & 0 & 84 & 18 \\ 2 & 4 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0} \\
[192] & 1^{11}231^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.47 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 0 & 56 & 14 \\ 2 & 4 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0} \\
[193] & 1^{14}4^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 19 & 0 & 65 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0} \\
[194] & 1^{14}4^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0} \\
[195] & 1^{11}231^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

[196] $1^{12}3^{11}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 14 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$

[197] $1^{12}3^{11}1^{12}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 17 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$

[198] $1^{13}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 20 & 0 & 70 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$

[199] $1^{12}3^{11}1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$

[201] $1^{12}3^{11}1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$

[200] $1^{12}3^{11}1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 26 & 3 \\ 5 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 69 & 0 & 340 & 34 \\ 3 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$

[202] $1^{11}23^{11}1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 20 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$

[203] $1^{12}23^{11}1^{12}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 27 & 3 \\ 5 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 69 & 0 & 340 & 34 \\ 3 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$

[204] $1^{14}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 24 & 0 & 101 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$

[205] $1^{11}23^{11}1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$

[206] $1^{11}23^{11}1^{12}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$

$T = 10$ [207] $1^{\frac{1}{1}} 5 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 12 & 7 \\ 9 & 12 & 10 & 6 \\ 5 & 7 & 6 & 3 \end{pmatrix}_{b_0}$

[209] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 13 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$

[208] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 13 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -2 & 2 & 0 \\ 2 & 3 & 1 & 2 & 1 & 3 \\ 2 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$

[211] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 13 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$

[210] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 7 & 0 & 21 & 9 \\ 2 & 4 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$

[212] $1^{\frac{1}{1}} 5 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 17 & 5 \\ 9 & 13 & 22 & 7 \\ 17 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0}$

[214] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 1^{\frac{1}{1}} 3 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 5 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$

[213] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 1^{\frac{1}{1}} 3 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$

[216] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 1^{\frac{1}{1}} 3 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$

[215] $1^{\frac{1}{1}} 5 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 14 & 5 \\ 9 & 13 & 22 & 7 \\ 14 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0}$

[218] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 13 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$

[217] $1^{\frac{1}{1}} 3^{\frac{1}{1}} 1^{\frac{1}{1}} 4 13 1^{\frac{1}{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 1 & 3 & 5 & 5 \\ 2 & 4 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[219] \ 1 \overset{1^2}{\overset{1^1}{5}} 13 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 16 & 5 \\ 9 & 13 & 22 & 7 \\ 16 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[220] \ 1 \overset{1^1}{\overset{1^1}{5}} 13 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 22 & 7 \\ 15 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[221] \ 1 \overset{1^1}{\overset{1^2}{5}} 13 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 20 & 5 \\ 9 & 13 & 32 & 7 \\ 20 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^1 31 \overset{1^1}{1^1} 41 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 5 & 8 & 0 & 4 & 10 & 6 \\ 2 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[223] \ 1 \overset{1^4}{\overset{1^2}{5}} 13 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 24 & 5 \\ 9 & 13 & 32 & 7 \\ 24 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[224] \ 21 \overset{1^1}{\overset{1^1}{5}} 13 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 15 & 5 \\ 10 & 13 & 22 & 7 \\ 15 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 1^1 31 \overset{1^1}{1^1} 413 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[226] \ 1 \overset{1^1}{\overset{1^3}{5}} 13 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 26 & 5 \\ 9 & 13 & 42 & 7 \\ 26 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1 \overset{1^1}{\overset{1^2}{5}} 13 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 123, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 32 & 7 \\ 21 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[228] \ 1^1 31 \overset{1^1}{1^1} 413 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 12 & 0 & 49 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[229] \ 1 \overset{1^3}{\overset{1^1}{5}} 13 \overset{1^1}{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 23 & 5 \\ 9 & 13 & 32 & 7 \\ 23 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{12}5^{12}13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 22 & 5 \\ 9 & 13 & 32 & 7 \\ 22 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{11}31^{11}413^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{12}31^{11}413^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 14 & 10 & 0 & 38 & 12 \\ 2 & 4 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{13}5^{13}13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 179, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 27 & 5 \\ 9 & 13 & 42 & 7 \\ 27 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{12}31^{11}413^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[235] $1^{14}31^{11}413^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 16 & 0 & 90 & 18 \\ 2 & 4 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{13}31^{11}413^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 17 & 13 & 0 & 61 & 15 \\ 2 & 4 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[237] $21^{12}5^{12}13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 218, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 22 & 5 \\ 10 & 13 & 32 & 7 \\ 22 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{13}5^{13}13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 28 & 5 \\ 9 & 13 & 42 & 7 \\ 28 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{13}5^{14}13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 30 & 5 \\ 9 & 13 & 42 & 7 \\ 30 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{13}5^{13}13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 235, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 29 & 5 \\ 9 & 13 & 42 & 7 \\ 29 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[241] \ 21^{13}513^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 372, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 29 & 5 \\ 10 & 13 & 42 & 7 \\ 29 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [242] \ 1^{11}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -18, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.78 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 5 & 0 & 2 & 7 & 5 \\ 2 & 3 & -1 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{11}31^1513^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -24, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 2 & 0 & 4 & 3 & 5 & 5 \\ 2 & 0 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{11}31^1513^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -33, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 2 & 4 & 0 & 3 & 5 & 5 \\ 2 & 3 & -1 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{13}31^1513^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -41, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 10 & 0 & 5 & 33 & 11 \\ 2 & 3 & -1 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{13}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -44, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.82 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{12}31^1513^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -46, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 3 & 7 & 0 & 4 & 16 & 8 \\ 2 & 3 & -1 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{11}3^11^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -68, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.82 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 3 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{11}31^1513^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -80, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.75 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 0 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{13}31^1513^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -104, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 5 & 0 & 10 & 5 & 33 & 11 \\ 2 & 0 & 2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{12}3^{11}5^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -260$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.85 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 6 & 0 & 8 & 3 & 20 & 8 \\ 2 & 0 & 2 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{12}3^{11}5^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -284$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.80 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 6 & 0 & 7 & 4 & 16 & 8 \\ 2 & 0 & 2 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{13}3^{11}5^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -308$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.86 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 7 & 0 & 11 & 4 & 39 & 11 \\ 2 & 0 & 2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{14}3^{11}5^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -324$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.93 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 8 & 0 & 14 & 5 & 64 & 14 \\ 2 & 0 & 2 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{13}3^{11}5^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -368$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 7 & 0 & 10 & 5 & 33 & 11 \\ 2 & 0 & 2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{14}3^{11}5^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -420$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.79 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 8 & 0 & 13 & 6 & 56 & 14 \\ 2 & 0 & 2 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [267] $1^{11}3^{11}21^{11}6^{11}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -82$, $\det = 32$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 2 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{11}3^{11}21^{11}6^{11}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 16$, $\Delta = -88$, $\det = 40$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 15$, $T_{\max} = 16$, $T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 3 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{11}3^{11}21^{11}6^{11}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -82$, $\det = 44$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{11}3^{11}21^{11}6^{11}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -82$, $\det = 56$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 2 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[271] $1^{\overline{11}} 6^{1^{\overline{11}} 3^{\overline{1}} 2^{\overline{1}}}$ $131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 15, \Delta = -82, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{\overline{11}} 6^{1^{\overline{11}} 3^{\overline{1}} 2^{\overline{1}}}$ $131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 15, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.78 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 4 & 0 & 3 & 10 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{\overline{11}} 6^{1^{\overline{11}} 3^{\overline{1}} 2^{\overline{1}}}$ $131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 15, \Delta = -82, \det = 176, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 4 & 1 & 3 & 11 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{\overline{11}} 6^{1^{\overline{12}} 3^{\overline{1}} 2^{\overline{1}}}$ $131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 15, \Delta = -82, \det = 212, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 4 & 2 & 3 & 14 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$T = 14$ [275] $1^{\overline{1}} 8^{1^{\overline{1}}}$ $123^{\overline{1} \overline{1} \overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 14, T_{\max} = 17, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{\overline{1}} 8^{1^{\overline{1}}}$ $123^{\overline{1} \overline{1} \overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 14, T_{\max} = 17, \Delta = -80, \det = 24, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 16.83 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 3 & 2 & 5 & 2 & 7 & 5 \\ 2 & 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{\overline{1}} 8^{1^{\overline{1} 3 2 2 1^{\overline{1}}}}$ $123^{\overline{1} \overline{1} \overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 14, T_{\max} = 17, \Delta = -80, \det = 104, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 5 & 8 & 0 & 31 & 11 \\ 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[278] $1^{\overline{1}} 8^{1^{\overline{1} 3 2 2 1^{\overline{1}}}}$ $123^{\overline{1} \overline{1} \overline{1} 2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 14, T_{\max} = 17, \Delta = -80, \det = 156, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 6 & 11 & 0 & 50 & 14 \\ 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$T = 15$ [279] $1^{\overline{1}} 8^{1^{\overline{1} 3 2 1}}$ $123^{\overline{1} \overline{1} \overline{1}} 141^{\overline{1} \overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 15, T_{\max} = 18, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 17.83 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

4.520 NHC(-2)(-3) + f₄ + su2n3mix + f₄

$$T = 8 \quad [0] \ 1^{\overline{111}3^{\overline{11}}1^{\overline{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 360, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 6 & -3 & -1 \end{pmatrix}_{b_0}$$

4.521 NHC(-2)(-3) + f₄ + su2n3mix + NHC(-2)(-3)

$$T = 9 \quad [0] \ 1^{\overline{11}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.04 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 8 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 9 & 5 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 5 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 12 & 8 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 & 4 \\ 2 & 4 & 0 & 8 & 4 & 0 & 2 \\ 3 & 4 & -3 & 9 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \ 1^{\overline{11}3^{\overline{11}}4^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -12, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

4.522 NHC(-2)(-3) + f₄ + su2n3mix + so₈

$$T = 8 \quad [0] \ 1^{\overline{11}3^{\overline{11}}1^{\overline{12}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 171, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^{\overline{11}}1^{\overline{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 221, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 6 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 6 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}3^{\overline{11}}1^{\overline{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 6 & -2 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{111} 31^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -240, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 32, \quad T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{11} 1^{13} 1^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -108, \quad \det = 233, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.59 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [6] \quad 1^{111} 31^1 5^1 13^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 12, \quad T_{\max} = 15, \quad \Delta = -82, \quad \det = 94, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.87 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 20 & 5 \\ 12 & 26 & 1 & 44 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 44 & 0 & 70 & 16 \\ 5 & 10 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.523 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -212, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

4.524 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$

$$T = 8 \quad [1] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 43, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, \quad T_{\max} = 20, \quad T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 6 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 43, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.63 \rightarrow 14, \quad T_{\max} = 20, \quad T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 6 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.58 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 7 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 17 & 5 \\ 8 & 14 & 0 & 17 & 37 & 11 \\ 3 & 4 & -3 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 20, \quad T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 19 & 6 \\ 7 & 13 & 0 & 19 & 28 & 10 \\ 3 & 4 & -3 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{135}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 15 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 55 & 11 \\ 10 & 20 & 0 & 55 & 73 & 15 \\ 3 & 4 & -3 & 11 & 15 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{131}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 23 & 11 \\ 5 & 9 & 0 & 23 & 13 & 7 \\ 3 & 4 & -3 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{113}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{113}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 15 & 5 \\ 6 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{124}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 32 & 8 \\ 8 & 16 & 0 & 32 & 46 & 12 \\ 3 & 4 & -3 & 8 & 12 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{124}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 17 & 3 \\ 3 & 4 & 0 & 11 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 59 & 8 \\ 17 & 31 & 0 & 59 & 172 & 24 \\ 3 & 4 & -3 & 8 & 24 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 22 & 0 & 97 & 14 & 17 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -3 & 17 & 3 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{132}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 25 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 0 & 111 & 34 & 19 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -3 & 19 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{134}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 15 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 68 & 11 \\ 14 & 26 & 0 & 68 & 118 & 20 \\ 3 & 4 & -3 & 11 & 20 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{143}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 6 & 3 \\ 3 & 4 & 0 & 24 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 51 & 18 \\ 6 & 12 & 0 & 51 & 25 & 9 \\ 3 & 4 & -3 & 18 & 9 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^11^{13} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 7 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 28 & 5 \\ 13 & 23 & 0 & 28 & 100 & 18 \\ 3 & 4 & -3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^11^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 8 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{143}3^11^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 4 & 0 & 24 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 21 & 18 \\ 3 & 5 & 0 & 21 & 4 & 4 \\ 3 & 4 & -3 & 18 & 4 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{142}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 7 & 3 \\ 3 & 4 & 0 & 19 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 43 & 14 \\ 7 & 13 & 0 & 43 & 28 & 10 \\ 3 & 4 & -3 & 14 & 10 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{145}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 19 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 70 & 14 \\ 10 & 20 & 0 & 70 & 73 & 15 \\ 3 & 4 & -3 & 14 & 15 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{123}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 24 & 8 \\ 6 & 12 & 0 & 24 & 25 & 9 \\ 3 & 4 & -3 & 8 & 9 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 22 & 12 \\ 4 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -3 & 12 & 6 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{134}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 15 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 44 & 11 \\ 8 & 16 & 0 & 44 & 46 & 12 \\ 3 & 4 & -3 & 11 & 12 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 13 & 7 \\ 3 & 4 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 14 & 6 \\ 5 & 9 & 0 & 14 & 13 & 7 \\ 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 16 & 8 \\ 4 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & -3 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 0 & 55 & 11 & 13 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -3 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{133}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -3 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{133}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 15 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 33 & 11 \\ 6 & 12 & 0 & 33 & 25 & 9 \\ 3 & 4 & -3 & 11 & 9 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{121}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{144}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 8 & 3 \\ 3 & 4 & 0 & 19 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 56 & 14 \\ 8 & 16 & 0 & 56 & 46 & 12 \\ 3 & 4 & -3 & 14 & 12 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 18 & 3 \\ 3 & 4 & 0 & 7 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 39 & 5 \\ 18 & 32 & 0 & 39 & 193 & 25 \\ 3 & 4 & -3 & 5 & 25 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{123}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -3 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{132}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 15 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 34 & 11 \\ 7 & 13 & 0 & 34 & 28 & 10 \\ 3 & 4 & -3 & 11 & 10 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{123}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 11 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 42 & 8 \\ 12 & 22 & 0 & 42 & 85 & 17 \\ 3 & 4 & -3 & 8 & 17 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{132}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 15 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 22 & 11 \\ 4 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -3 & 11 & 6 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{143}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 6 & 3 \\ 3 & 4 & 0 & 19 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 42 & 14 \\ 6 & 12 & 0 & 42 & 25 & 9 \\ 3 & 4 & -3 & 14 & 9 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{131}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 22 & 5 \\ 10 & 18 & 0 & 22 & 58 & 14 \\ 3 & 4 & -3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 25 & 8 \\ 7 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{124}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 59 & 12 \\ 11 & 21 & 0 & 59 & 76 & 16 \\ 3 & 4 & -3 & 12 & 16 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{132}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 28 & 15 \\ 4 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -3 & 15 & 6 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{143}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 9 & 3 \\ 3 & 4 & 0 & 19 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 57 & 14 \\ 9 & 17 & 0 & 57 & 49 & 13 \\ 3 & 4 & -3 & 14 & 13 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{142}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 19 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 28 & 14 \\ 4 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -3 & 14 & 6 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{121}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 362, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 31 & 0 & 188 & 20 & 24 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -3 & 24 & 3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{112}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 28 & 9 \\ 7 & 13 & 0 & 28 & 28 & 10 \\ 3 & 4 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{141}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{142}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 4 & 3 \\ 3 & 4 & 0 & 29 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 0 & 148 & 40 & 22 \\ 4 & 8 & 0 & 40 & 10 & 6 \\ 3 & 4 & -3 & 22 & 6 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{111}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 20 & 9 \\ 5 & 9 & 0 & 20 & 13 & 7 \\ 3 & 4 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{113}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 21 & 5 \\ 9 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{133}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 15 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 45 & 11 \\ 9 & 17 & 0 & 45 & 49 & 13 \\ 3 & 4 & -3 & 11 & 13 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{123}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 438, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 11 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 33 & 8 \\ 9 & 17 & 0 & 33 & 49 & 13 \\ 3 & 4 & -3 & 8 & 13 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{114}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 26 & 5 \\ 11 & 21 & 0 & 26 & 76 & 16 \\ 3 & 4 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{121}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 26 & 0 & 128 & 17 & 20 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -3 & 20 & 3 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{113}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 473, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 7 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 33 & 5 \\ 15 & 27 & 0 & 33 & 133 & 21 \\ 3 & 4 & -3 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{121}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{142}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 24 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 34 & 18 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -3 & 18 & 6 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{121}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 515, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 26 & 12 \\ 5 & 9 & 0 & 26 & 13 & 7 \\ 3 & 4 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{144}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 11 & 3 \\ 3 & 4 & 0 & 19 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 71 & 14 \\ 11 & 21 & 0 & 71 & 76 & 16 \\ 3 & 4 & -3 & 14 & 16 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 27 & 5 \\ 12 & 22 & 0 & 27 & 85 & 17 \\ 3 & 4 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{124}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 11 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 50 & 8 \\ 14 & 26 & 0 & 50 & 118 & 20 \\ 3 & 4 & -3 & 8 & 20 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 11 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 41 & 8 \\ 11 & 21 & 0 & 41 & 76 & 16 \\ 3 & 4 & -3 & 8 & 16 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{123}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 555, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 16 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 48 & 12 \\ 9 & 17 & 0 & 48 & 49 & 13 \\ 3 & 4 & -3 & 12 & 13 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 15 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 56 & 11 \\ 11 & 21 & 0 & 56 & 76 & 16 \\ 3 & 4 & -3 & 11 & 16 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 577, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{131}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 20 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 32 & 15 \\ 5 & 9 & 0 & 32 & 13 & 7 \\ 3 & 4 & -3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{141}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 5 & 3 \\ 3 & 4 & 0 & 24 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 38 & 18 \\ 5 & 9 & 0 & 38 & 13 & 7 \\ 3 & 4 & -3 & 18 & 7 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{122}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 37 & 12 \\ 7 & 13 & 0 & 37 & 28 & 10 \\ 3 & 4 & -3 & 12 & 10 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{114}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 20 & 3 \\ 3 & 4 & 0 & 7 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 44 & 5 \\ 20 & 36 & 0 & 44 & 238 & 28 \\ 3 & 4 & -3 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{134}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 727, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 20 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 74 & 15 \\ 11 & 21 & 0 & 74 & 76 & 16 \\ 3 & 4 & -3 & 15 & 16 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{114}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 32 & 5 \\ 14 & 26 & 0 & 32 & 118 & 20 \\ 3 & 4 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{141}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{131}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{132}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 807, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 7 & 3 \\ 3 & 4 & 0 & 20 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 46 & 15 \\ 7 & 13 & 0 & 46 & 28 & 10 \\ 3 & 4 & -3 & 15 & 10 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{114}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 7 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 38 & 5 \\ 17 & 31 & 0 & 38 & 172 & 24 \\ 3 & 4 & -3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{133}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 4 & 0 & 20 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 60 & 15 \\ 9 & 17 & 0 & 60 & 49 & 13 \\ 3 & 4 & -3 & 15 & 13 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{131}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 865, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 30 & 0 & 165 & 20 & 23 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -3 & 23 & 3 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{131}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -3 & 27 & 3 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{142}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 928, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 7 & 3 \\ 3 & 4 & 0 & 24 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 55 & 18 \\ 7 & 13 & 0 & 55 & 28 & 10 \\ 3 & 4 & -3 & 18 & 10 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{141}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1062, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{144}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1068, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 11 & 3 \\ 3 & 4 & 0 & 24 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 89 & 18 \\ 11 & 21 & 0 & 89 & 76 & 16 \\ 3 & 4 & -3 & 18 & 16 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{143}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1083, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 9 & 3 \\ 3 & 4 & 0 & 24 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 72 & 18 \\ 9 & 17 & 0 & 72 & 49 & 13 \\ 3 & 4 & -3 & 18 & 13 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{141}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 19 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{141}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 22 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 1 & 3 \\ 3 & 4 & -3 & 30 & 3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [92] $1^{142}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 2 & 0 \\ 8 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & 0 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

- [93] $1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 3 & 0 & 2 & 3 & 5 & 3 \\ 2 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 2 & -1 & 0 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [94] $1^{11}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 6 & 0 & 7 & 5 & 5 \\ 1 & 3 & 3 & 0 & 5 & 1 & 3 \\ 2 & 3 & 3 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$
- [95] $1^{11}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 13 & 0 & 60 & 13 & 14 \\ 1 & 3 & 3 & 0 & 13 & 1 & 3 \\ 2 & 4 & 2 & -3 & 14 & 3 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 3 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [98] $1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 6 & 8 & 0 & 7 & 34 & 16 & 10 \\ 2 & 4 & 0 & 2 & 16 & 6 & 4 \\ 2 & 2 & -1 & 0 & 10 & 4 & 1 \end{pmatrix}_{b_0}$$
- [97] $1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 3 & 16 & 3 \\ 6 & 10 & 0 & 10 & 16 & 58 & 14 \\ 2 & 2 & -1 & 0 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$
- [99] $1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 6 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & 0 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{11}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 4 & 2 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$
- [102] $1^{11}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 0 & 5 & 1 & 3 \\ 2 & 4 & 2 & -3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 21^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 1 & 0 \\ 3 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 21^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 3 & 0 & 2 & 3 & 5 & 3 \\ 3 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 2 & -1 & 0 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{\overset{1}{111}} 31^{\overset{1}{3}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 3 & 3 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{\overset{1}{112}} 3^{\overset{1}{1}} 21^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & 0 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{\overset{1}{111}} 313^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 & 3 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 3 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{\overset{1}{112}} 31^{\overset{1}{11}} 3^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 11 & 12 & 4 \\ 1 & 4 & 0 & 0 & 7 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 25 & 8 \\ 4 & 12 & 9 & 0 & 25 & 28 & 10 \\ 2 & 4 & 2 & -3 & 8 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{\overset{1}{111}} 31^{\overset{1}{3}} 1^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 0 & 4 & 6 & 3 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 4 & 9 & 4 \\ 3 & 6 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & 2 & -3 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{\overset{1}{112}} 3^{\overset{1}{1}} 21^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 2 & 0 \\ 4 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & 0 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{\overset{1}{112}} 3^{\overset{1}{1}} 21^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 3 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & 0 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{11}3^1 1^{11}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 4 \\ 2 & 6 & 3 & 0 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{11}3^1 1^{11}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 11 & 5 \\ 3 & 9 & 6 & 0 & 11 & 13 & 7 \\ 2 & 4 & 2 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{11}3^1 2^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{12}3^1 2^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 1 & 0 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 2 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{12}3^1 2^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 13 & 0 \\ 2 & 3 & 0 & 2 & 3 & 18 & 3 \\ 9 & 11 & 0 & 13 & 18 & 81 & 17 \\ 2 & 2 & -1 & 0 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{11}3^1 2^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 1 & 0 \\ 4 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{13}3^1 2^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 5 & 8 & 0 & 7 & 34 & 16 & 10 \\ 2 & 4 & 0 & 2 & 16 & 6 & 4 \\ 2 & 2 & -1 & 0 & 10 & 4 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{11}3^1 1^{11}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 8 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 12 & 5 \\ 3 & 9 & 6 & 0 & 12 & 13 & 7 \\ 2 & 4 & 2 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{11}3^1 1^{14}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.2

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 5 & 27 & 4 \\ 1 & 4 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 24 & 4 \\ 9 & 27 & 15 & 0 & 24 & 121 & 19 \\ 2 & 4 & 2 & -3 & 4 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{11} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{12} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 6 & 0 \\ 3 & 5 & 0 & 3 & 11 & 18 & 5 \\ 4 & 7 & 0 & 6 & 18 & 25 & 9 \\ 2 & 2 & -1 & 0 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{12} 3^{11} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 1 & 0 \\ 6 & 7 & 0 & 9 & 35 & 7 & 11 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & 0 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{12} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 1 & 0 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{13} 3^{11} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 9 & 0 \\ 2 & 3 & 0 & 2 & 3 & 13 & 3 \\ 6 & 8 & 0 & 9 & 13 & 40 & 12 \\ 2 & 2 & -1 & 0 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{11} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{13} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 4 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & 0 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{12} 3^{11} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 & 2 \\ 2 & 4 & 4 & 0 & 14 & 6 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 14 & 11 \\ 2 & 6 & 3 & 0 & 14 & 4 & 4 \\ 2 & 4 & 2 & -3 & 11 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{\overline{11}1}3^{\overline{11}3}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 7 & 8 \\ 1 & 3 & 3 & 0 & 7 & 1 & 3 \\ 2 & 4 & 2 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{\overline{11}1}3^{\overline{1}1}1^{\overline{11}3}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 8 & 9 \\ 1 & 3 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & 2 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{\overline{12}3}3^{\overline{1}1}2^{\overline{1}1}1^{\overline{12}3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -185, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 5 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & 0 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{\overline{12}3}3^{\overline{1}1}2^{\overline{1}1}1^{\overline{12}3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 1 & 0 \\ 4 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{\overline{11}1}3^{\overline{12}3}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 10 & 0 & 48 & 10 & 12 \\ 1 & 3 & 3 & 0 & 10 & 1 & 3 \\ 2 & 4 & 2 & -3 & 12 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{\overline{13}1}3^{\overline{1}1}2^{\overline{1}1}1^{\overline{13}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 1 & 0 \\ 6 & 9 & 0 & 10 & 51 & 9 & 13 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{\overline{14}3}3^{\overline{1}1}2^{\overline{1}1}1^{\overline{14}3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 5 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & 0 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{\overline{12}1}3^{\overline{11}3}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 14 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 10 & 11 \\ 1 & 3 & 3 & 0 & 10 & 1 & 3 \\ 2 & 4 & 2 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{\overline{12}3}3^{\overline{1}1}2^{\overline{1}1}1^{\overline{12}3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & 0 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

- [138] $1^{112}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & 0 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [139] $1^{112}3^11^13^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 5 & 12 & 4 \\ 1 & 4 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 13 & 4 \\ 4 & 12 & 9 & 0 & 13 & 28 & 10 \\ 2 & 4 & 2 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$
- [141] $1^{124}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 7 & 0 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 5 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & 0 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$
- [140] $1^{122}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 3 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & 0 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$
- [142] $1^{132}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 5 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & 0 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$
- [143] $1^{121}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 1 & 0 \\ 7 & 7 & 0 & 9 & 35 & 7 & 11 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & 0 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$
- [144] $1^{142}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 2 & 0 \\ 7 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & 0 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$
- [145] $1^{111}3^11^{14}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 17 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 17 & 0 & 151 & 17 & 21 \\ 1 & 3 & 3 & 0 & 17 & 1 & 3 \\ 2 & 4 & 2 & -3 & 21 & 3 & 0 \end{pmatrix}_{b_0}$$
- [146] $1^{112}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 4 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & 0 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{142}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 2 & 0 \\ 6 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & 0 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{142}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 7 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & 0 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{112}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 5 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & 0 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{111}3^11^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[151] $1^{131}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 1 & 0 \\ 9 & 10 & 0 & 13 & 74 & 10 & 16 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & 0 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[152] $1^{131}3^11^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 3 & 2 \\ 2 & 4 & 4 & 0 & 17 & 9 & 4 \\ 1 & 4 & 0 & 0 & 13 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 13 & 0 & 60 & 30 & 14 \\ 3 & 9 & 6 & 0 & 30 & 13 & 7 \\ 2 & 4 & 2 & -3 & 14 & 7 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{113}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 4 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & 0 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{121}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 1 & 0 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{113}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.0

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 9 & 0 \\ 2 & 3 & 0 & 2 & 3 & 13 & 3 \\ 7 & 8 & 0 & 9 & 13 & 40 & 12 \\ 2 & 2 & -1 & 0 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{11}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 11 & 0 & 55 & 11 & 13 \\ 1 & 3 & 3 & 0 & 11 & 1 & 3 \\ 2 & 4 & 2 & -3 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

[157] $1^131^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 8 & 6 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 8 & 5 \\ 2 & 6 & 3 & 0 & 8 & 4 & 4 \\ 2 & 4 & 2 & -3 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{11}1^{12}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 4 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & 0 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[158] $1^{11}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 5 & 21 & 4 \\ 1 & 4 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 19 & 4 \\ 7 & 21 & 12 & 0 & 19 & 73 & 15 \\ 2 & 4 & 2 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[160] $1^{11}3^11^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 12 & 9 \\ 2 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 4 & 2 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{11}1^{14}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 5 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & 0 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{13}1^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 1 & 0 \\ 5 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11}3^11^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 14 & 0 & 97 & 14 & 17 \\ 1 & 3 & 3 & 0 & 14 & 1 & 3 \\ 2 & 4 & 2 & -3 & 17 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{1^{11}} 1^{121} 3^1 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 1 & 0 \\ 5 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{1^{11}} 1^{141} 31^{12} 3^1 1^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 26 & 3 & 4 \\ 1 & 4 & 0 & 0 & 19 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 26 & 19 & 0 & 135 & 19 & 21 \\ 1 & 3 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & 2 & -3 & 21 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{1^{12}} 1^{121} 3^1 21^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 1 & 0 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{1^{11}} 1^{142} 3^1 21^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 6 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & 0 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{1^{11}} 1^{111} 31^{12} 3^1 1^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 5 & 15 & 4 \\ 1 & 4 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 14 & 4 \\ 5 & 15 & 9 & 0 & 14 & 37 & 11 \\ 2 & 4 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[169] \ 1^1 3^1 1^{111} 31^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -369, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{11} 3^1 1^{111} 31^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 14 & 6 \\ 3 & 9 & 6 & 0 & 14 & 13 & 7 \\ 2 & 4 & 2 & -3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{1^{12}} 1^{113} 3^1 21^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 5 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & 0 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{1^{11}} 1^{124} 3^1 21^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 4 & 0 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 5 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & 0 & 5 & 8 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114}3^{12}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 5 & 24 & 4 \\ 1 & 4 & 0 & 0 & 4 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 26 & 4 \\ 8 & 24 & 18 & 0 & 26 & 118 & 20 \\ 2 & 4 & 2 & -3 & 4 & 20 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{112}3^{12}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -390, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 4 & 0 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 6 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & 0 & 5 & 8 & 1 \end{pmatrix}_{b_0}$$

[175] $1^{112}3^{12}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 1 & 0 \\ 6 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{11}3^{12}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 1 & 0 \\ 7 & 11 & 0 & 11 & 71 & 11 & 15 \\ 1 & 2 & 0 & 1 & 11 & 0 & 2 \\ 2 & 2 & -1 & 0 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111}3^{12}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 1 & 0 \\ 5 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111}3^{12}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 3 & 16 & 3 \\ 7 & 10 & 0 & 10 & 16 & 58 & 14 \\ 2 & 2 & -1 & 0 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{111}3^{12}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 5 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & 0 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{111}3^{12}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 1 & 0 \\ 7 & 9 & 0 & 10 & 51 & 9 & 13 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{11}3^{12}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 11 & 6 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 11 & 8 \\ 2 & 6 & 3 & 0 & 11 & 4 & 4 \\ 2 & 4 & 2 & -3 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{141}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 1 & 0 \\ 6 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{121}3^{11}23^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -454$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 20 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 20 & 13 & 0 & 71 & 13 & 15 \\ 1 & 3 & 3 & 0 & 13 & 1 & 3 \\ 2 & 4 & 2 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{124}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -462$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 7 & 0 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 6 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & 0 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{122}3^{11}1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 4 & 2 \\ 2 & 4 & 4 & 0 & 14 & 12 & 4 \\ 1 & 4 & 0 & 0 & 10 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 34 & 11 \\ 4 & 12 & 9 & 0 & 34 & 28 & 10 \\ 2 & 4 & 2 & -3 & 11 & 10 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{141}3^{11}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -469$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 1 & 0 \\ 9 & 12 & 0 & 14 & 96 & 12 & 18 \\ 1 & 2 & 0 & 1 & 12 & 0 & 2 \\ 2 & 2 & -1 & 0 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{131}3^{11}23^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 23 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 23 & 16 & 0 & 100 & 16 & 18 \\ 1 & 3 & 3 & 0 & 16 & 1 & 3 \\ 2 & 4 & 2 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{114}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -522$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 6 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & 0 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{131}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -541$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 1 & 0 \\ 6 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

- [190] $1^{1^{12}} 1^{113} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -542$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 6 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & 0 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [191] $1^{1^{11}} 1^{141} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -544$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 1 & 0 \\ 6 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$
- [192] $1^{1^{12}} 1^{131} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -557$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 1 & 0 \\ 6 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$
- [193] $1^{1^{11}} 1^{114} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -574$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 6 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & 0 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [194] $1^{1^{12}} 31^{123} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -586$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 5 & 18 & 4 \\ 1 & 4 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 18 & 4 \\ 6 & 18 & 12 & 0 & 18 & 58 & 14 \\ 2 & 4 & 2 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$
- [195] $1^{1^{13}} 31^{123} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -596$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 5 & 21 & 4 \\ 1 & 4 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 22 & 4 \\ 7 & 21 & 15 & 0 & 22 & 85 & 17 \\ 2 & 4 & 2 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$
- [196] $1^{1^{12}} 1^{131} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -597$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 1 & 0 \\ 8 & 9 & 0 & 10 & 51 & 9 & 13 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{1^{11}} 31^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -600$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 11 & 9 & 4 \\ 1 & 4 & 0 & 0 & 7 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 18 & 8 \\ 3 & 9 & 6 & 0 & 18 & 13 & 7 \\ 2 & 4 & 2 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{1^{11}} 3^1 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -601$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 12 & 9 & 4 \\ 1 & 4 & 0 & 0 & 8 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 20 & 9 \\ 3 & 9 & 6 & 0 & 20 & 13 & 7 \\ 2 & 4 & 2 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{121}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 26 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 26 & 16 & 0 & 117 & 16 & 19 \\ 1 & 3 & 3 & 0 & 16 & 1 & 3 \\ 2 & 4 & 2 & -3 & 19 & 3 & 0 \end{pmatrix}_{b_0}$$

[200] $1^{121}31^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -649, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 3 & 2 \\ 2 & 4 & 4 & 0 & 14 & 9 & 4 \\ 1 & 4 & 0 & 0 & 10 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 24 & 11 \\ 3 & 9 & 6 & 0 & 24 & 13 & 7 \\ 2 & 4 & 2 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{121}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 0 & 32 & 3 & 4 \\ 1 & 4 & 0 & 0 & 19 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 32 & 19 & 0 & 175 & 19 & 23 \\ 1 & 3 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & 2 & -3 & 23 & 3 & 0 \end{pmatrix}_{b_0}$$

[202] $1^{112}1^{14}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 3 & 16 & 3 \\ 8 & 10 & 0 & 10 & 16 & 58 & 14 \\ 2 & 2 & -1 & 0 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{112}1^{131}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -692, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 1 & 0 \\ 7 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{112}1^{141}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 1 & 0 \\ 10 & 12 & 0 & 14 & 96 & 12 & 18 \\ 1 & 2 & 0 & 1 & 12 & 0 & 2 \\ 2 & 2 & -1 & 0 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{111}1^{141}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -724, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 1 & 0 \\ 8 & 11 & 0 & 11 & 71 & 11 & 15 \\ 1 & 2 & 0 & 1 & 11 & 0 & 2 \\ 2 & 2 & -1 & 0 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{111}1^{141}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 1 & 0 \\ 7 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{1^{12}} 1^{141} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -765$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 1 & 0 \\ 7 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{1^{12}} 1^{114} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -797$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 7 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & 0 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{1^{12}} 3^1 1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -821$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 5 & 24 & 4 \\ 1 & 4 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 23 & 4 \\ 8 & 24 & 15 & 0 & 23 & 100 & 18 \\ 2 & 4 & 2 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{1^{12}} 3^1 1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -871$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 29 & 3 & 4 \\ 1 & 4 & 0 & 0 & 19 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 29 & 19 & 0 & 152 & 19 & 22 \\ 1 & 3 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & 2 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{1^{12}} 3^1 1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -926$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 5 & 30 & 4 \\ 1 & 4 & 0 & 0 & 4 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 28 & 4 \\ 10 & 30 & 18 & 0 & 28 & 154 & 22 \\ 2 & 4 & 2 & -3 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{1^{12}} 3^1 1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -929$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 0 & 32 & 3 & 4 \\ 1 & 4 & 0 & 0 & 22 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 32 & 22 & 0 & 193 & 22 & 25 \\ 1 & 3 & 3 & 0 & 22 & 1 & 3 \\ 2 & 4 & 2 & -3 & 25 & 3 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{1^{12}} 1^{141} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -965$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 1 & 0 \\ 9 & 11 & 0 & 11 & 71 & 11 & 15 \\ 1 & 2 & 0 & 1 & 11 & 0 & 2 \\ 2 & 2 & -1 & 0 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{1^{12}} 1^{141} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -980$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 1 & 0 \\ 8 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{113}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 5 & 27 & 4 \\ 1 & 4 & 0 & 0 & 4 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 27 & 4 \\ 9 & 27 & 18 & 0 & 27 & 133 & 21 \\ 2 & 4 & 2 & -3 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{131}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 0 & 35 & 3 & 4 \\ 1 & 4 & 0 & 0 & 22 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 35 & 22 & 0 & 216 & 22 & 26 \\ 1 & 3 & 3 & 0 & 22 & 1 & 3 \\ 2 & 4 & 2 & -3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{114}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 5 & 30 & 4 \\ 1 & 4 & 0 & 0 & 4 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 31 & 4 \\ 10 & 30 & 21 & 0 & 31 & 172 & 24 \\ 2 & 4 & 2 & -3 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{141}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 1 & 2 \\ 2 & 4 & 4 & 0 & 38 & 3 & 4 \\ 1 & 4 & 0 & 0 & 25 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 38 & 25 & 0 & 263 & 25 & 29 \\ 1 & 3 & 3 & 0 & 25 & 1 & 3 \\ 2 & 4 & 2 & -3 & 29 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{113}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 11 & 2 \\ 2 & 4 & 4 & 0 & 5 & 33 & 4 \\ 1 & 4 & 0 & 0 & 4 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 32 & 4 \\ 11 & 33 & 21 & 0 & 32 & 193 & 25 \\ 2 & 4 & 2 & -3 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[220] \ 1^{114}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 12 & 2 \\ 2 & 4 & 4 & 0 & 5 & 36 & 4 \\ 1 & 4 & 0 & 0 & 4 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 36 & 4 \\ 12 & 36 & 24 & 0 & 36 & 238 & 28 \\ 2 & 4 & 2 & -3 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [221] \ 1^{12}23^{11}1^{11}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 1 & 3 \\ 5 & 13 & 1 & 54 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 54 & 0 & 217 & 0 & 27 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 27 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{13}23^{11}1^{12}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 32 & 1 & 3 \\ 5 & 13 & 1 & 84 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 32 & 84 & 0 & 520 & 0 & 42 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 42 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{11}4^{11}13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{11} 1^{13} 1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{11} 1^{13} 1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{12} 23^1 1^{11} 31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -61, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 22 & 1 & 3 \\ 5 & 13 & 1 & 54 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 54 & 0 & 217 & 0 & 27 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 27 & 0 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{112} 1^{11} 4^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{122} 1^{11} 4^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{113} 1^{11} 4^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{121} 1^{11} 4^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{111} 1^{11} 4^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{111} 1^{13} 1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 12 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{112} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{111} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{112} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 17 & 7 \\ 7 & 11 & 0 & 17 & 21 & 9 \\ 3 & 4 & -2 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{122} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{122} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{121} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 10 & 6 \\ 5 & 7 & 0 & 10 & 8 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{121} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 16 & 10 \\ 5 & 7 & 0 & 16 & 8 & 6 \\ 3 & 4 & -2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{111} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 11 & 7 \\ 5 & 7 & 0 & 11 & 8 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{111} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{111} 1^{13^1 1^1} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{121} 4^{13^1 1^1} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{121} 1^{13^1 1^1} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 0 & 21 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{111} 1^{13^1 1^1} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{122} 1^{13^1 1^1} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 22 & 6 \\ 10 & 16 & 0 & 22 & 45 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{122} 1^{13^1 1^1} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 25 & 10 \\ 7 & 11 & 0 & 25 & 21 & 9 \\ 3 & 4 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{131} 4^{13^1 1^1} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{112} 1^{13^1 1^1} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 225, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 8 & 3 \\ 7 & 11 & 0 & 8 & 21 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{122}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 254, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 16 & 6 \\ 7 & 11 & 0 & 16 & 21 & 9 \\ 3 & 4 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{141}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{121}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 277, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{113}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 281, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 11 & 3 \\ 9 & 15 & 0 & 11 & 40 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{114}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 305, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 14 & 3 \\ 11 & 19 & 0 & 14 & 65 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{121}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 16 & 0 & 45 & 11 & 13 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{112}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 11 & 3 \\ 10 & 16 & 0 & 11 & 45 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{131}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 349, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 15 & 0 & 40 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{131}1^{13^1}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 421, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{113}1^{13^11^1}4^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 14 & 3 \\ 12 & 20 & 0 & 14 & 70 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{114}1^{13^11^1}4^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 17 & 3 \\ 14 & 24 & 0 & 17 & 101 & 19 \\ 3 & 4 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{141}1^{13^11^1}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 525, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 19 & 0 & 65 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{141}1^{11}4^113^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 565, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{131}1^{13^11^1}4^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 20 & 0 & 70 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{141}1^{13^11^1}4^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 24 & 0 & 101 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [265] $1^{111}1^{11}31^1413^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{111}1^{11}31^141^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 4 & 2 & -2 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{121}31^{11}41^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -48, (n_+, n_-, n_0) =$

$$(1, 11, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 14 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 14 & 10 & 0 & 38 & 10 & 12 \\ 1 & 3 & 3 & 0 & 10 & 1 & 3 \\ 2 & 4 & 2 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[268] \quad 1^{114}31^{114}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 20 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 16 & 3 \\ 8 & 20 & 16 & 0 & 16 & 90 & 18 \\ 2 & 4 & 2 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[269] \quad 1^{111}31^{114}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 7 & 3 \\ 5 & 11 & 7 & 0 & 7 & 21 & 9 \\ 2 & 4 & 2 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{141}31^{114}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 20 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 16 & 0 & 90 & 16 & 18 \\ 1 & 3 & 3 & 0 & 16 & 1 & 3 \\ 2 & 4 & 2 & -2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[273] \quad 1^{131}31^{114}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 17 & 13 & 0 & 61 & 13 & 15 \\ 1 & 3 & 3 & 0 & 13 & 1 & 3 \\ 2 & 4 & 2 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[272] \quad 1^{113}31^{114}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 13 & 3 \\ 7 & 17 & 13 & 0 & 13 & 61 & 15 \\ 2 & 4 & 2 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[271] \quad 1^{112}31^{114}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 10 & 3 \\ 6 & 14 & 10 & 0 & 10 & 38 & 12 \\ 2 & 4 & 2 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

4.525 NHC(-2)(-3) + f₄ + su₂n₃mix + su₂ + so₈

$$T = 9 \quad [0] \quad 1^{113}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

4.526 NHC(-2)(-3) + f₄ + su2n3mix + su₂ + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & -3 & 5 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 11 & 5 \\ 4 & 8 & 0 & 10 & 10 & 12 & 6 \\ 5 & 9 & 0 & 11 & 12 & 13 & 7 \\ 3 & 4 & -3 & 5 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 2 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 22 & 11 & 12 \\ 4 & 8 & 0 & 22 & 10 & 6 & 6 \\ 2 & 4 & 0 & 11 & 6 & 1 & 3 \\ 3 & 4 & -3 & 12 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 2 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 15 & 5 & 5 \\ 6 & 12 & 0 & 15 & 25 & 9 & 9 \\ 2 & 4 & 0 & 5 & 9 & 1 & 3 \\ 3 & 4 & -3 & 5 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 9 & 3 \\ 3 & 4 & 0 & 7 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 21 & 5 \\ 3 & 5 & 0 & 6 & 4 & 15 & 4 \\ 9 & 17 & 0 & 21 & 15 & 49 & 13 \\ 3 & 4 & -3 & 5 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 5 & 3 \\ 3 & 4 & 0 & 7 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 11 & 5 \\ 3 & 5 & 0 & 6 & 4 & 9 & 4 \\ 5 & 9 & 0 & 11 & 9 & 13 & 7 \\ 3 & 4 & -3 & 5 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 5 & 3 \\ 3 & 4 & 0 & 16 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 22 & 26 & 12 \\ 4 & 8 & 0 & 22 & 10 & 12 & 6 \\ 5 & 9 & 0 & 26 & 12 & 13 & 7 \\ 3 & 4 & -3 & 12 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 10 & 5 \\ 4 & 8 & 0 & 10 & 10 & 12 & 6 \\ 4 & 8 & 0 & 10 & 12 & 10 & 6 \\ 3 & 4 & -3 & 5 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1131}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 36 & 8 & 9 \\ 9 & 17 & 0 & 36 & 49 & 12 & 13 \\ 2 & 4 & 0 & 8 & 12 & 1 & 3 \\ 3 & 4 & -3 & 9 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{1211}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 17 & 8 \\ 5 & 9 & 0 & 17 & 13 & 15 & 7 \\ 5 & 9 & 0 & 17 & 15 & 13 & 7 \\ 3 & 4 & -3 & 8 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1131}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 12 & 8 & 9 \\ 3 & 5 & 0 & 12 & 4 & 3 & 4 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 9 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{1311}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 18 & 14 & 15 \\ 3 & 5 & 0 & 18 & 4 & 3 & 4 \\ 2 & 4 & 0 & 14 & 3 & 1 & 3 \\ 3 & 4 & -3 & 15 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{1121}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 7 & 3 \\ 3 & 4 & 0 & 7 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 16 & 5 \\ 3 & 5 & 0 & 6 & 4 & 12 & 4 \\ 7 & 13 & 0 & 16 & 12 & 28 & 10 \\ 3 & 4 & -3 & 5 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1121}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 5 & 5 \\ 4 & 8 & 0 & 10 & 10 & 6 & 6 \\ 2 & 4 & 0 & 5 & 6 & 1 & 3 \\ 3 & 4 & -3 & 5 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1211}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 2 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 8 & 8 \\ 5 & 9 & 0 & 17 & 13 & 6 & 7 \\ 2 & 4 & 0 & 8 & 6 & 1 & 3 \\ 3 & 4 & -3 & 8 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1211}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 15 & 11 & 12 \\ 3 & 5 & 0 & 15 & 4 & 3 & 4 \\ 2 & 4 & 0 & 11 & 3 & 1 & 3 \\ 3 & 4 & -3 & 12 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 9 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 & 3 \\ 2 & 4 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 11 & 5 \\ 2 & 4 & 0 & 5 & 1 & 6 & 3 \\ 5 & 9 & 0 & 11 & 6 & 13 & 7 \\ 3 & 4 & -3 & 5 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{1122}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 16 & 5 \\ 4 & 8 & 0 & 10 & 10 & 18 & 6 \\ 7 & 13 & 0 & 16 & 18 & 28 & 10 \\ 3 & 4 & -3 & 5 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{1221}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 2 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 16 & 8 & 8 \\ 4 & 8 & 0 & 16 & 10 & 6 & 6 \\ 2 & 4 & 0 & 8 & 6 & 1 & 3 \\ 3 & 4 & -3 & 8 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1112}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 10 & 3 \\ 3 & 4 & 0 & 7 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 22 & 5 \\ 2 & 4 & 0 & 5 & 1 & 12 & 3 \\ 10 & 18 & 0 & 22 & 12 & 58 & 14 \\ 3 & 4 & -3 & 5 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{1211}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 8 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 8 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{1212}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 7 & 3 \\ 3 & 4 & 0 & 11 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 25 & 8 \\ 5 & 9 & 0 & 17 & 13 & 21 & 7 \\ 7 & 13 & 0 & 25 & 21 & 28 & 10 \\ 3 & 4 & -3 & 8 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{11113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -456, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 11 & 5 \\ 5 & 9 & 0 & 11 & 13 & 15 & 7 \\ 5 & 9 & 0 & 11 & 15 & 13 & 7 \\ 3 & 4 & -3 & 5 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -476, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 15 & 3 \\ 3 & 4 & 0 & 7 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 33 & 5 \\ 2 & 4 & 0 & 5 & 1 & 18 & 3 \\ 15 & 27 & 0 & 33 & 18 & 133 & 21 \\ 3 & 4 & -3 & 5 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{1211}} 3^{\overline{1}} 1^{\overline{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 12 & 21 & 0 & 80 & 14 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 & 3 \\ 2 & 4 & 0 & 14 & 3 & 1 & 3 \\ 3 & 4 & -3 & 16 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{\overline{1112}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 28 & 9 \\ 2 & 4 & 0 & 8 & 1 & 9 & 3 \\ 7 & 13 & 0 & 28 & 9 & 28 & 10 \\ 3 & 4 & -3 & 9 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{11113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 5 & 3 \\ 3 & 4 & 0 & 12 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 20 & 9 \\ 2 & 4 & 0 & 8 & 1 & 6 & 3 \\ 5 & 9 & 0 & 20 & 6 & 13 & 7 \\ 3 & 4 & -3 & 9 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{1112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -536, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 16 & 5 \\ 2 & 4 & 0 & 5 & 1 & 9 & 3 \\ 7 & 13 & 0 & 16 & 9 & 28 & 10 \\ 3 & 4 & -3 & 5 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{1114}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 11 & 3 \\ 3 & 4 & 0 & 7 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 26 & 5 \\ 5 & 9 & 0 & 11 & 13 & 33 & 7 \\ 11 & 21 & 0 & 26 & 33 & 76 & 16 \\ 3 & 4 & -3 & 5 & 7 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\overline{1212}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 25 & 8 \\ 2 & 4 & 0 & 8 & 1 & 9 & 3 \\ 7 & 13 & 0 & 25 & 9 & 28 & 10 \\ 3 & 4 & -3 & 8 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 11 & 11 & 11 \\ 2 & 4 & 0 & 11 & 1 & 3 & 3 \\ 2 & 4 & 0 & 11 & 3 & 1 & 3 \\ 3 & 4 & -3 & 11 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1114} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 20 & 3 \\ 3 & 4 & 0 & 7 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 44 & 5 \\ 2 & 4 & 0 & 5 & 1 & 24 & 3 \\ 20 & 36 & 0 & 44 & 24 & 238 & 28 \\ 3 & 4 & -3 & 5 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1211} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -620, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 & 3 \\ 2 & 4 & 0 & 11 & 3 & 1 & 3 \\ 3 & 4 & -3 & 12 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 7 & 3 \\ 3 & 4 & 0 & 7 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 16 & 5 \\ 5 & 9 & 0 & 11 & 13 & 21 & 7 \\ 7 & 13 & 0 & 16 & 21 & 28 & 10 \\ 3 & 4 & -3 & 5 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 21 & 5 \\ 5 & 9 & 0 & 11 & 13 & 27 & 7 \\ 9 & 17 & 0 & 21 & 27 & 49 & 13 \\ 3 & 4 & -3 & 5 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 21 & 5 \\ 2 & 4 & 0 & 5 & 1 & 12 & 3 \\ 9 & 17 & 0 & 21 & 12 & 49 & 13 \\ 3 & 4 & -3 & 5 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1411} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 14 & 14 & 14 \\ 2 & 4 & 0 & 14 & 1 & 3 & 3 \\ 2 & 4 & 0 & 14 & 3 & 1 & 3 \\ 3 & 4 & -3 & 14 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1211} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -810, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 5 & 3 \\ 3 & 4 & 0 & 16 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 26 & 12 \\ 2 & 4 & 0 & 11 & 1 & 6 & 3 \\ 5 & 9 & 0 & 26 & 6 & 13 & 7 \\ 3 & 4 & -3 & 12 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{1113}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -836$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 27 & 5 \\ 2 & 4 & 0 & 5 & 1 & 15 & 3 \\ 12 & 22 & 0 & 27 & 15 & 85 & 17 \\ 3 & 4 & -3 & 5 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{1311}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -840$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 17 & 30 & 0 & 165 & 20 & 20 & 23 \\ 2 & 4 & 0 & 20 & 1 & 3 & 3 \\ 2 & 4 & 0 & 20 & 3 & 1 & 3 \\ 3 & 4 & -3 & 23 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{1114}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -856$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 26 & 5 \\ 2 & 4 & 0 & 5 & 1 & 15 & 3 \\ 11 & 21 & 0 & 26 & 15 & 76 & 16 \\ 3 & 4 & -3 & 5 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{1122}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 16 & 16 & 5 \\ 7 & 13 & 0 & 16 & 28 & 30 & 10 \\ 7 & 13 & 0 & 16 & 30 & 28 & 10 \\ 3 & 4 & -3 & 5 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{1311}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -964$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 5 & 3 \\ 3 & 4 & 0 & 20 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 32 & 15 \\ 2 & 4 & 0 & 14 & 1 & 6 & 3 \\ 5 & 9 & 0 & 32 & 6 & 13 & 7 \\ 3 & 4 & -3 & 15 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{1311}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -984$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 & 3 \\ 2 & 4 & 0 & 14 & 3 & 1 & 3 \\ 3 & 4 & -3 & 15 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{1411}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -986$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 5 & 2 & 3 \\ 3 & 4 & 0 & 24 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 38 & 17 & 18 \\ 5 & 9 & 0 & 38 & 13 & 6 & 7 \\ 2 & 4 & 0 & 17 & 6 & 1 & 3 \\ 3 & 4 & -3 & 18 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{1212}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1020$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 7 & 3 \\ 3 & 4 & 0 & 16 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 37 & 12 \\ 2 & 4 & 0 & 11 & 1 & 9 & 3 \\ 7 & 13 & 0 & 37 & 9 & 28 & 10 \\ 3 & 4 & -3 & 12 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1311311^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1076, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 14 & 25 & 0 & 111 & 17 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 & 3 \\ 2 & 4 & 0 & 17 & 3 & 1 & 3 \\ 3 & 4 & -3 & 19 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114311^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 17 & 3 \\ 3 & 4 & 0 & 7 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 38 & 5 \\ 2 & 4 & 0 & 5 & 1 & 21 & 3 \\ 17 & 31 & 0 & 38 & 21 & 172 & 24 \\ 3 & 4 & -3 & 5 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1114311^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 32 & 5 \\ 2 & 4 & 0 & 5 & 1 & 18 & 3 \\ 14 & 26 & 0 & 32 & 18 & 118 & 20 \\ 3 & 4 & -3 & 5 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1411311^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 22 & 39 & 0 & 280 & 26 & 26 & 30 \\ 2 & 4 & 0 & 26 & 1 & 3 & 3 \\ 2 & 4 & 0 & 26 & 3 & 1 & 3 \\ 3 & 4 & -3 & 30 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1411311^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 17 & 17 & 18 \\ 2 & 4 & 0 & 17 & 1 & 3 & 3 \\ 2 & 4 & 0 & 17 & 3 & 1 & 3 \\ 3 & 4 & -3 & 18 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1411311^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 19 & 34 & 0 & 208 & 23 & 23 & 26 \\ 2 & 4 & 0 & 23 & 1 & 3 & 3 \\ 2 & 4 & 0 & 23 & 3 & 1 & 3 \\ 3 & 4 & -3 & 26 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1411311^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 16 & 29 & 0 & 148 & 20 & 20 & 22 \\ 2 & 4 & 0 & 20 & 1 & 3 & 3 \\ 2 & 4 & 0 & 20 & 3 & 1 & 3 \\ 3 & 4 & -3 & 22 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \quad 1^{1131111311^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 5 & 4 \\ 2 & 6 & 3 & 0 & 5 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 5 & 6 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{111}3^11^{11}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{111}3^11^{11}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 9 & 4 \\ 2 & 6 & 3 & 0 & 5 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{111}3^11^{11}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 9 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 9 & 15 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.527 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su2n3mix} + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] \quad 1^{111}3^11^{11}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 2 & 3 \\ 1 & 3 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{11}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 3 & 4 \\ 2 & 6 & 3 & 0 & 5 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 475, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.528 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su2n3mix} + \mathbf{su}_2 + \mathbf{su}_3$

$T = 9 \quad [0] \quad 1^{111}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 8 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 9 & 5 & 6 \\ 3 & 5 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 6 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}1} 3^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 17 & 0 & 55 & 11 & 11 & 13 \\ 2 & 4 & 0 & 11 & 1 & 3 & 3 \\ 2 & 4 & 0 & 11 & 3 & 0 & 2 \\ 3 & 4 & -3 & 13 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 5 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{121}1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 & 3 \\ 3 & 4 & 0 & 11 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 9 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 & 3 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 3 & 4 & -3 & 8 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 6 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 & 3 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -3 & 5 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 14 & 5 & 6 \\ 5 & 9 & 0 & 14 & 13 & 6 & 7 \\ 2 & 4 & 0 & 5 & 6 & 0 & 2 \\ 3 & 4 & -3 & 6 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{131}1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 18 & 14 & 15 \\ 3 & 5 & 0 & 18 & 4 & 3 & 4 \\ 2 & 4 & 0 & 14 & 3 & 0 & 2 \\ 3 & 4 & -3 & 15 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 12 & 8 & 9 \\ 3 & 5 & 0 & 12 & 4 & 3 & 4 \\ 2 & 4 & 0 & 8 & 3 & 0 & 2 \\ 3 & 4 & -3 & 9 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 & 3 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 5 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 11 & 5 \\ 2 & 4 & 0 & 5 & 1 & 6 & 3 \\ 5 & 9 & 0 & 11 & 6 & 12 & 6 \\ 3 & 4 & -3 & 5 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 0 & 2 \\ 3 & 4 & -3 & 9 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1213}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 15 & 11 & 12 \\ 3 & 5 & 0 & 15 & 4 & 3 & 4 \\ 2 & 4 & 0 & 11 & 3 & 0 & 2 \\ 3 & 4 & -3 & 12 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 5 & 5 \\ 5 & 9 & 0 & 11 & 13 & 6 & 7 \\ 2 & 4 & 0 & 5 & 6 & 0 & 2 \\ 3 & 4 & -3 & 5 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 2 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 20 & 8 & 9 \\ 5 & 9 & 0 & 20 & 13 & 6 & 7 \\ 2 & 4 & 0 & 8 & 6 & 0 & 2 \\ 3 & 4 & -3 & 9 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.529 NHC(-2)(-3) + f_4 + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 4 & 5 \\ 1 & 3 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -85, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.39 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 4 & 2 & 3 \\ 1 & 4 & 0 & 0 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 2 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{12}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 8 & 6 & 0 & 25 & 6 \\ 2 & 6 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 25 & 16 & 0 & 69 & 17 \\ 3 & 6 & 3 & -3 & 17 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{13}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 4 & 5 \\ 1 & 3 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{13}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 8 & 0 & 21 & 9 \\ 3 & 6 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 12 & 0 & 41 & 13 \\ 3 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{13}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 13 & 0 & 60 & 10 & 14 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{13}1^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{12}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 11 & 0 & 44 & 9 & 12 \\ 1 & 3 & 2 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 7 & 9 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 6 & 8 \\ 1 & 3 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{12}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -465, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 14 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{13}1^{11}1^{12}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -520, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 8 & 0 & 36 & 8 & 10 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}3^{11}1^{12}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -844, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.02 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 11 & 0 & 55 & 10 & 13 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}3^{11}1^{12}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -968, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.00 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 14 & 0 & 80 & 12 & 16 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

4.530 NHC(-2)(-3) + f₄ + su₂n₃mix + su₂ → (-3) + su₂

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 4 & 2 & 1 \\ 2 & 4 & 2 & -3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 175, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 0 & 5 & 3 & 4 \\ 2 & 4 & 2 & -3 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 475, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 \\ 2 & 4 & 4 & 0 & 5 & 3 & 9 \\ 1 & 4 & 0 & 0 & 4 & 2 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 3 & 9 \\ 1 & 3 & 2 & 0 & 3 & 0 & 5 \\ 3 & 9 & 6 & 0 & 9 & 5 & 13 \\ 2 & 4 & 2 & -3 & 4 & 2 & 7 \end{pmatrix}_{b_0}$$

4.531 NHC(-2)(-3) + f₄ + su₂n₃mix + su₂n₃mix

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -40, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 11 & 4 \\ 1 & 4 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 13 & 5 \\ 4 & 11 & 7 & 0 & 13 & 20 & 8 \\ 2 & 4 & 2 & -3 & 5 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^11^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 4 \\ 1 & 4 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 7 & 5 \\ 2 & 5 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 2 & -3 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^11^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 8 & 4 \\ 3 & 9 & 5 & 0 & 8 & 12 & 6 \\ 2 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 5 & 12 & 4 \\ 1 & 4 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 12 & 4 \\ 4 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 4 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \ 1^{111}3^11^{14}1^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

4.532 NHC(-2)(-3) + f₄ + su₂n3mix + su₃

$$T = 8 \quad [0] \ 1^{141}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 19 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 29 & 14 \\ 5 & 9 & 0 & 29 & 12 & 6 \\ 3 & 4 & -3 & 14 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 9 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 16 & 9 \\ 4 & 8 & 0 & 16 & 9 & 5 \\ 3 & 4 & -3 & 9 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 6 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

- [5] $1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 142$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{11}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 27 & 0 & 151 & 17 & 21 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 197$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 9 & 5 \\ 3 & 4 & -3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 23 & 11 \\ 5 & 9 & 0 & 23 & 12 & 6 \\ 3 & 4 & -3 & 11 & 6 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 227$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 12 & 6 \\ 3 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 12 & 6 \\ 3 & 4 & -3 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{12}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 332$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{11}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 332$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 22 & 0 & 97 & 14 & 17 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

- [15] $1^{111}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 368$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{112}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 382$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 27 & 9 \\ 3 & 4 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{111}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 403$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 0 & 55 & 11 & 13 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{131}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 443$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{141}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 560$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{121}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 635$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{121}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 832$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$
- [22] $1^{121}3^11^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 908$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 31 & 0 & 188 & 20 & 24 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{131}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 908$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{121}3^11^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 923$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 26 & 0 & 128 & 17 & 20 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

- [25] $1^{141}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1187$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$
- [26] $1^{131}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1267$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$
- [27] $1^{131}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1520$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 30 & 0 & 165 & 20 & 23 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 3 & 4 & -3 & 23 & 2 & 1 \end{pmatrix}_{b_0}$$
- [28] $1^{131}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1667$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 3 & 4 & -3 & 27 & 2 & 1 \end{pmatrix}_{b_0}$$
- [29] $1^{141}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1708$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$
- [30] $1^{141}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 2123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 19 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 3 & 4 & -3 & 26 & 2 & 1 \end{pmatrix}_{b_0}$$
- [31] $1^{141}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 2432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 22 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 3 & 4 & -3 & 30 & 2 & 1 \end{pmatrix}_{b_0}$$
- $T = 9$ [32] $1^{111}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 54 & 0 & 217 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$
- [33] $1^{11}31^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 4 & 5 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$
- [34] $1^{121}31^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 & 3 \\ 2 & 4 & 3 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 7 & 8 & 0 & 15 & 7 & 7 \\ 1 & 3 & 3 & 0 & 7 & 0 & 2 \\ 2 & 3 & 3 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 6 & 0 & 7 & 5 & 5 \\ 1 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 3 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{121}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 44 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{111}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 9 & 1 & 9 & 7 \\ 9 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 7 & 0 & 4 & 4 \\ 7 & 5 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{121}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 84 & 0 & 520 & 42 \\ 5 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{111}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111}31^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{111}31^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 3 & 3 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{111}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{111}31^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 4 \\ 2 & 6 & 3 & 0 & 5 & 3 & 3 \\ 2 & 4 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{111}3^11^14^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 14 & 0 & 120 & 14 & 18 \\ 1 & 3 & 3 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{121}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 24 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 11 & 0 & 72 & 11 & 14 \\ 1 & 3 & 3 & 0 & 11 & 0 & 2 \\ 2 & 4 & 2 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{11}3^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 8 & 0 & 36 & 8 & 10 \\ 1 & 3 & 3 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 16 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{112}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 17 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{121}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{112}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111}31^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{121}3^{11}21^{12}+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=13, \Delta=-91, \det=-384, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$
 $T_{\min}=12.03 \rightarrow 13, T_{\max}=13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 45 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{111}31^{11}31^{11}1^1+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=15, \Delta=-97, \det=-430, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$

$T_{\min}=12.33 \rightarrow 13, T_{\max}=15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 7 & 8 \\ 1 & 3 & 3 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^{11}21^1+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=13, \Delta=-91, \det=-464, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$
 $T_{\min}=12.23 \rightarrow 13, T_{\max}=13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{121}3^{11}21^{11}+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=13, \Delta=-91, \det=-584, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$
 $T_{\min}=12.08 \rightarrow 13, T_{\max}=13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 35 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{121}3^{11}21^{12}+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=13, \Delta=-91, \det=-608, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$
 $T_{\min}=12.02 \rightarrow 13, T_{\max}=13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{121}3^{11}21^1+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=13, \Delta=-91, \det=-608, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$
 $T_{\min}=12.18 \rightarrow 13, T_{\max}=13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{111}31^{12}31^1+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=15, \Delta=-97, \det=-679, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$

$T_{\min}=12.29 \rightarrow 13, T_{\max}=15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 10 & 0 & 48 & 10 & 12 \\ 1 & 3 & 3 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{121}3^{11}21^{11}+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=13, \Delta=-91, \det=-768, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$
 $T_{\min}=12.06 \rightarrow 13, T_{\max}=13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 36 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111}31^{13}31^1+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=15, \Delta=-97, \det=-886, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$

$T_{\min}=12.28 \rightarrow 13, T_{\max}=15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 23 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 23 & 13 & 0 & 88 & 13 & 16 \\ 1 & 3 & 3 & 0 & 13 & 0 & 2 \\ 2 & 4 & 2 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{111}31^{14}31^1+[\mathbf{su}_2 \cdot \mathbf{g}_2]+\mathbf{f}_4+\mathbf{su}_2+\mathbf{su}_3 \quad T=9, T_{\max}=15, \Delta=-97, \det=-1051, (n_+, n_-, n_0)=(1, 9, 0), c_{\text{ext}}=11.2$

$T_{\min}=12.28 \rightarrow 13, T_{\max}=15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 29 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 29 & 16 & 0 & 140 & 16 & 20 \\ 1 & 3 & 3 & 0 & 16 & 0 & 2 \\ 2 & 4 & 2 & -3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [65] \quad 1^{11} 4^{11} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 3 \\ 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 0 & 13 & 6 & 7 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{111} 4^{11} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{111} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 12 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{111} 4^{11} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [73] \quad 1^{11} 5^{11} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 14 & 1 & 5 \\ 9 & 13 & 22 & 0 & 7 \\ 14 & 22 & 33 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{12} \overset{1^1}{5} 13 \overset{1^1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 20 & 1 & 5 \\ 9 & 13 & 32 & 0 & 7 \\ 20 & 32 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13} \overset{1^1}{5} 13 \overset{1^1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -145, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 26 & 1 & 5 \\ 9 & 13 & 42 & 0 & 7 \\ 26 & 42 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

4.533 NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} \overset{1^1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 9 & 6 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 3 & 5 & 0 & 9 & 3 & 4 & 4 \\ 3 & 4 & -3 & 6 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \overset{1^1}{3} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 17 & 0 & 55 & 11 & 11 & 13 \\ 2 & 4 & 0 & 11 & 0 & 3 & 2 \\ 2 & 4 & 0 & 11 & 3 & 1 & 3 \\ 3 & 4 & -3 & 13 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} \overset{1^1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 6 & 5 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 5 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{121} \overset{1^1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 2 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 9 & 8 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 8 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} \overset{1^1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 2 & 4 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} \overset{1^1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 14 & 6 \\ 2 & 4 & 0 & 5 & 0 & 6 & 2 \\ 5 & 9 & 0 & 14 & 6 & 13 & 7 \\ 3 & 4 & -3 & 6 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 & 3 \\ 3 & 4 & 0 & 12 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 12 & 9 \\ 2 & 4 & 0 & 8 & 0 & 3 & 2 \\ 3 & 5 & 0 & 12 & 3 & 4 & 4 \\ 3 & 4 & -3 & 9 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{131}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 & 3 \\ 3 & 4 & 0 & 20 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 18 & 15 \\ 2 & 4 & 0 & 14 & 0 & 3 & 2 \\ 3 & 5 & 0 & 18 & 3 & 4 & 4 \\ 3 & 4 & -3 & 15 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 & 5 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 2 & 4 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & -3 & 5 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 5 & 5 \\ 5 & 9 & 0 & 11 & 12 & 6 & 6 \\ 2 & 4 & 0 & 5 & 6 & 1 & 3 \\ 3 & 4 & -3 & 5 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 4 & 0 & 8 & 0 & 3 & 2 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 9 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{121}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 & 3 \\ 3 & 4 & 0 & 16 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 15 & 12 \\ 2 & 4 & 0 & 11 & 0 & 3 & 2 \\ 3 & 5 & 0 & 15 & 3 & 4 & 4 \\ 3 & 4 & -3 & 12 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 11 & 5 \\ 2 & 4 & 0 & 5 & 0 & 6 & 2 \\ 5 & 9 & 0 & 11 & 6 & 13 & 7 \\ 3 & 4 & -3 & 5 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 5 & 3 \\ 3 & 4 & 0 & 12 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 20 & 9 \\ 2 & 4 & 0 & 8 & 0 & 6 & 2 \\ 5 & 9 & 0 & 20 & 6 & 13 & 7 \\ 3 & 4 & -3 & 9 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.534 NHC(-2)(-3) + f₄ + su₂n₃mix + su₃ + su₃

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 4 & 3 \\ 3 & 4 & 0 & 7 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 10 & 5 \\ 3 & 5 & 0 & 6 & 3 & 6 & 3 \\ 4 & 8 & 0 & 10 & 6 & 9 & 5 \\ 3 & 4 & -3 & 5 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 2 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 9 & 8 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 0 & 2 \\ 3 & 4 & -3 & 8 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 5 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.535 NHC(-2)(-3) + f₄ + su₃

$$T = 7 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^{131} 1^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{12} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 5 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 10 & 0 & 21 & 7 \\ 3 & 4 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{12}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 8 & 0 & 9 & 5 \\ 3 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{13}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 12 & 0 & 24 & 8 \\ 3 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 45 & 11 \\ 3 & 4 & -3 & 11 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{15}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 20 & 0 & 72 & 14 \\ 3 & 4 & -3 & 14 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{11}131^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 4 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{11}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 14 & 0 & 36 & 10 \\ 3 & 4 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 13 & 0 & 27 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 48 & 12 \\ 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 18 & 0 & 57 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 21 & 0 & 75 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{13}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -372, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 22 & 0 & 84 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[20] $21^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 6 & 8 & 3 & 3 & 3 \\ 4 & 5 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{14}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -487, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 26 & 0 & 117 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [22] $1^1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 15, T_{\max} = 35, T \in [15, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 \\ 1 & 4 & 0 & 1 & 0 \\ 1 & 1 & 1 & -3 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

[23] $1^1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 35, T \in [15, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 10 & 0 & 13 & 7 \\ 3 & 5 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 35, T \in [15, 35]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 5 & 13 & 9 & 5 \\ 13 & 30 & 20 & 10 \\ 9 & 20 & 12 & 6 \\ 5 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -3 & -1 \\ 5 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 35, T \in [15, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 5 & 3 \\ 2 & 2 & 2 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 5 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 10 & 18 & 12 & 6 \\ 18 & 30 & 20 & 10 \\ 12 & 20 & 12 & 6 \\ 6 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 5 & 0 & 1 & 3 \\ 3 & 5 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} 3 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{11}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 10 & 0 & 32 & 10 \\ 2 & 2 & 2 & -3 & 10 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

[37] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{111}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 6 & 0 & 12 & 6 \\ 2 & 3 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[44] $1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{111}3131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 83, T \in [14, 83]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[45] $2231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 10 & 19 & 12 & 6 \\ 19 & 30 & 20 & 10 \\ 12 & 20 & 12 & 6 \\ 6 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 10 & 0 & 13 & 7 \\ 3 & 5 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 7 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{11} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 0 & 15 & 7 \\ 2 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[51] \ 13 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 2 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{11} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 15 & 0 & 31 & 11 \\ 3 & 5 & -3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{11} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 25 & 0 & 91 & 19 \\ 3 & 5 & -3 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{11} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 57 & 15 \\ 3 & 5 & -3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{11} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 3 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{12} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

- [57] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [58] $1^{13}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [59] $1^{13}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 163$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 0 & 24 & 8 \\ 2 & 2 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$
- [60] $1^{13}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 163$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$
- [61] $1^{12}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [62] $1^{13}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$
- [63] $1^{12}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 188$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 0 & 25 & 9 \\ 2 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [64] $1^{12}3^{11}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 0 & 44 & 12 \\ 2 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 219$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [66] $1^{13}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 283$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 0 & 39 & 11 \\ 2 & 2 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{111}2231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 353$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{13}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 60 & 14 \\ 2 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[69] $2^2 2^1 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [70] $1^{13}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & 2 & 4 \\ 3 & 6 & -3 & 4 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{11}3221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 6 & 0 & 23 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 23 & 0 & 66 & 18 \\ 3 & 6 & -3 & 18 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{11}2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{11}3221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 9 & 0 & 19 & 9 \\ 2 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{11}2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{11}231^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 11 & 4 \\ 1 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 11 & 0 & 31 & 11 \\ 2 & 4 & 3 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{11}2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{13}4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{11} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{11} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{12} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{11} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{12} 3^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 1 & 1 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 0 & 0 & 12 & 6 \\ 3 & 0 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{12} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 11 & 0 & 49 & 13 \\ 3 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{12} 3^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 14.17 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{11} 2^{3131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 14.17 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 4 & 4 & 0 & 1 & 3 \\ 2 & 4 & 3 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{12} 3^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 1 & 5 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 0 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{11} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{11} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 6 & 0 & 11 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 11 & 0 & 14 & 8 \\ 3 & 6 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{11} 3^{11} 221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 6 & 0 & 17 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 17 & 0 & 35 & 13 \\ 3 & 6 & -3 & 13 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{11} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{11} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 5 \\ 2 & 2 & 3 & -3 & 5 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{11} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 8 & 0 & 30 & 10 \\ 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{11} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 3 \\ 5 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 3 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{11} 23131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 8 & 4 \\ 1 & 5 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 8 & 8 & 0 & 13 & 7 \\ 2 & 4 & 3 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$

[96] $1^{11} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{13} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 14 & 0 & 74 & 16 \\ 3 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{12} 3^{11} 221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 14 & 0 & 53 & 15 \\ 2 & 2 & 3 & -3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{\overline{1}1} \overline{4} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{\overline{1}1} \overline{4} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{\overline{1}1} \overline{4} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{\overline{1}1} \overline{2} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{\overline{1}1} \overline{2} \ 3 \ 2 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 13 & 7 \\ 2 & 2 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{\overline{1}1} \overline{2} \ 3 \ 2 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{\overline{1}1} \overline{2} \ 3 \ 2 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 7 & 0 & 25 & 9 \\ 2 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{\overline{1}1} \overline{2} \ 3 \ 2 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 24 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{\overline{1}1} \overline{2} \ 3 \ 2 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 5 & 1 & 7 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 12 & 6 \\ 3 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{\overline{1}1} \overline{2} \ 3 \ 2 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 5 & 1 & 5 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 15 & 0 & 12 & 6 \\ 3 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{\overline{1}1} \overline{2} \ 3 \ 2 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 10 & 0 & 29 & 11 \\ 2 & 2 & 3 & -3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 11 & 0 & 49 & 13 \\ 3 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11} 231^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 7 & 0 & 13 & 7 \\ 2 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{12} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 4 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{12} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 5 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{12} 231^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 5 & 1 & 6 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 12 & 6 \\ 3 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{13} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 74 & 16 \\ 3 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{13} 3^{11} 221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 19 & 0 & 103 & 21 \\ 2 & 2 & 3 & -3 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{13} 3^{11} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 11 & 0 & 43 & 13 \\ 2 & 2 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{13} 3^{11} 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 15 & 0 & 69 & 17 \\ 2 & 2 & 3 & -3 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{14} 41^{11} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 6 & 1 & 5 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 12 & 0 & 7 & 5 \\ 3 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [121] \quad 1^1 31^1 41^1 31^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 13 & 21 & 1 & 7 \\ 21 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 7 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^1 231^1 41^1 31^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 7 & 0 & 15 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^1 11 31^1 41^1 31^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^1 51^1 31^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^1 11 31^1 4131^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^1 51^1 1231^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 23 & 1 & 5 \\ 23 & 90 & 0 & 18 \\ 1 & 0 & -3 & -1 \\ 5 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [127] \quad 1^1 11 31^1 5131^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^1 11 31^1 5131^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^1 3151^1 231^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 8 & 1 & 1 & 3 \\ 8 & 38 & 1 & 0 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 12 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [131] \quad 1^1 6^1 131^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^1 11 6^1 131^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{\overline{1}} \ 6 \ 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{\overline{1}} \ 6 \ 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{\overline{1}} \ 321^{\overline{1}} 61231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 10 & 5 & 0 & 4 & 6 \\ 1 & 5 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 4 & 3 & 0 & 0 & 2 \\ 2 & 6 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[135] \ 31^{\overline{1}} \ 316 \ 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 17, \Delta = -95, \det = 193, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 5 & 8 & 8 & 5 \\ 8 & 6 & 9 & 6 \\ 8 & 9 & 5 & 5 \\ 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [136] \ 1^{\overline{1}} \ 6 \ 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [137] \ 1^{\overline{1}} \ 8 \ 1231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 16, T_{\max} = 19, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 18.60 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

4.536 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{1}} \ 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -169, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 3 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \ 1^{\overline{1}} \ 321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 10 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

4.537 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$T = 9 \quad [0] \quad 1^{111} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 0 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 6 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 4 & 3 \\ 3 & 5 & 0 & 6 & 4 & 3 & 3 \\ 3 & 4 & -3 & 5 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & -3 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.538 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$

$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 34 & 0 & 57 & 13 \\ 5 & 8 & -3 & 13 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 6 & 0 & 3 \\ 1 & 4 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 2 & 6 & 6 & 2 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 5 & -2 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 5 & 0 & 3 & 0 & 2 \\ 3 & 5 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 0 & 3 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & -3 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 2 & -2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{11}} 1^{\overline{11}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 3 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{11}} 1^{\overline{11}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{11}} 1^{\overline{11}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 13 & 7 & 7 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{\overline{11}} 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 82$, $\Delta = -124$, $\det = -277$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 82$, $T \in [14, 82]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 0 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [15] $1^{\overline{11}} 2^{\overline{11}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 14$, $\Delta = -90$, $\det = 45$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 5 & 1 & 12 & 1 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 12 & 1 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 0 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{\overline{11}} 1^{\overline{11}} 3 2 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 64$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 6 & 0 & 11 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 11 & 0 & 14 & 8 & 8 \\ 2 & 6 & 0 & 8 & 1 & 3 \\ 3 & 6 & -3 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{\overline{11}} 1^{\overline{11}} 3 2 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 98$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.61 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

4.539 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$

$T = 9$ [0] $1^{\overline{11}} 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

4.540 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.541 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 8$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 15.51 \rightarrow 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -3 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 75$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 82$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

- [6] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 48 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -3 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{121}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 137$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

- [15] $1^{121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -3 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{131}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 24 & 9 & 8 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 159$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 36 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -3 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -3 & 5 & 9 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{122}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 10 & 18 & 0 & 24 & 58 & 14 \\ 3 & 4 & -3 & 5 & 14 & 0 \end{pmatrix}_{b_0}$$
- [22] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 171$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 178$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

- [25] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 192$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$
- [27] $1^{123}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 197$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 8 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 36 & 5 \\ 15 & 27 & 0 & 36 & 133 & 21 \\ 3 & 4 & -3 & 5 & 21 & -1 \end{pmatrix}_{b_0}$$
- [28] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [29] $1^{115}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 203$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 10 & 20 & 0 & 15 & 73 & 15 \\ 3 & 4 & -3 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$
- [30] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 210$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 18 & 9 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -3 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$
- [31] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 226$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{14}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 232$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 11 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -3 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{116}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 12 & 24 & 0 & 18 & 106 & 18 \\ 3 & 4 & -3 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$
- [34] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

- [35] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 243$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [37] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 266$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{117}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 275$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.69 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 14 & 28 & 0 & 21 & 145 & 21 \\ 3 & 4 & -3 & 2 & 21 & -1 \end{pmatrix}_{b_0}$$
- [39] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 292$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [40] $1^{13}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 294$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -3 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 306$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$
- [42] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 312$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -3 & 5 & 10 & 0 \end{pmatrix}_{b_0}$$
- [43] $1^{115}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 338$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -2 & 0 & 15 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 0 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$
- [44] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 340$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

- [45] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 371$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{121}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 381$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 57 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -3 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$
- [47] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 411$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [48] $1^{131}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 48 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$
- [49] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [50] $1^{121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 450$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 21 & 9 \\ 5 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$
- [51] $1^{131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 463$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 48 & 27 & 12 \\ 5 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$
- [52] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 489$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -3 & 5 & 13 & 0 \end{pmatrix}_{b_0}$$
- [53] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 490$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [54] $1^{123}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 30 & 5 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -3 & 5 & 17 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 507$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 576$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 579$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 587$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 588$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{112}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 588$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{131}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 607$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 626$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 635$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

- [65] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 656$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 30 & 9 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 770$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [67] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 771$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$
- [68] $1^{123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 798$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$
- [69] $1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [70] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 915$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [71] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1136$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$
- [72] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1251$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$
- [73] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [74] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1643$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

- [75] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1920$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$
- $T = 9$ [76] $1^{12}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 10 & 3 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 5 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [77] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$
- [78] $1^{111}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 \\ 2 & 3 & 3 & -3 & 2 & 3 \end{pmatrix}_{b_0}$$
- [79] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 3 \\ 2 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$
- [80] $1^{11}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 \\ 2 & 3 & 3 & -3 & 2 & 4 \end{pmatrix}_{b_0}$$
- [81] $1^{11}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [82] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$
- [83] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 23 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$
- [84] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$
- [85] $1^{13}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 39 & 9 & 11 \\ 1 & 1 & 3 & 0 & 9 & 1 & 3 \\ 2 & 2 & 2 & -3 & 11 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11} 3 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 15 & 7 & 7 \\ 1 & 1 & 3 & 0 & 7 & 1 & 3 \\ 2 & 2 & 2 & -3 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{11} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 2 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 1 & 3 & 0 & 4 & 1 & 3 \\ 2 & 2 & 2 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{112} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 13 & 14 & 1 & 28 & 7 \\ 14 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 58 & 14 \\ 7 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{11} 31^{12311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & 2 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{11} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 13 & 7 & 7 \\ 2 & 5 & 0 & 7 & 2 & 4 \\ 3 & 5 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{11} 3 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 & 2 \\ 1 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

- [95] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [96] $1^{111} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [97] $1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{12} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 5 & 2 \\ 3 & 5 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [99] $1^{112} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$
- [100] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [101] $1^{114} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 66 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 66 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [102] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 18 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [103] $1^{111} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{11} 3^{11} 21^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 4 & 2 \\ 2 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{11} 3^{11} 21^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 7 & 2 \\ 3 & 4 & 7 & 0 & 7 & 16 & 8 \\ 2 & 2 & 2 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{12} 3^{11} 21^{121} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 25 & 8 & 9 \\ 1 & 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 2 & 2 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{112} 3^{11} 21^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 18 & 0 & 94 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{11} 3^{11} 21^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 7 & 3 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{114} 3^{11} 21^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 16 & 4 \\ 5 & 9 & 7 & 0 & 16 & 41 & 11 \\ 2 & 2 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11} 3^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{11} 3^{11} 21^{141} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 91 & 19 & 19 \\ 2 & 5 & 0 & 19 & 2 & 4 \\ 3 & 5 & -3 & 19 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{11} 3^{11} 21^{121} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 31 & 11 & 11 \\ 2 & 5 & 0 & 11 & 2 & 4 \\ 3 & 5 & -3 & 11 & 4 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 14 & 1 & 14 & 7 \\ 14 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 15 & 0 & 13 & 7 \\ 7 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 14 & 8 \\ 3 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{11} 3 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 57 & 15 & 15 \\ 2 & 5 & 0 & 15 & 2 & 4 \\ 3 & 5 & -3 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{121} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 & 2 \\ 1 & 2 & 0 & 0 & 8 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 25 & 6 & 9 \\ 1 & 2 & 1 & 0 & 6 & 0 & 2 \\ 2 & 2 & 2 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 6 & 6 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 4 & 2 & -3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{13} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 17 & 3 \\ 13 & 25 & 0 & 17 & 104 & 20 \\ 3 & 5 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[122] $1^{113} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[123] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[124] $1^{111} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 11 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{112} 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 52 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{113} 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{112} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{113} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 5 & 12 & 15 & 0 & 12 & 49 & 13 \\ 2 & 3 & 3 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{121} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 12 & 5 & 6 \\ 1 & 2 & 1 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{111} 3 \ 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 19 & 3 \\ 11 & 25 & 0 & 19 & 92 & 20 \\ 3 & 5 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{111} 3 \ 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 11 & 3 \\ 7 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -3 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{112} 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{14} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 10 & 2 \\ 6 & 10 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & 2 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{111} 3^{21} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 15 & 11 & 7 \\ 2 & 3 & 4 & 0 & 11 & 5 & 5 \\ 2 & 2 & 2 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{121} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{111} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 0 & 4 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 11 & 4 \\ 3 & 4 & 7 & 0 & 11 & 16 & 8 \\ 2 & 2 & 2 & -3 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{11} 3^{21} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 15 & 3 \\ 9 & 20 & 0 & 15 & 58 & 16 \\ 3 & 5 & -3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{111} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{12} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 8 & 2 \\ 4 & 6 & 8 & 0 & 8 & 26 & 10 \\ 2 & 2 & 2 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{112} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{111} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 9 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[144] $1^{112}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 14 & 3 \\ 10 & 20 & 0 & 14 & 62 & 16 \\ 3 & 5 & -3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{113}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{112}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{112}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 14 & 0 & 62 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{121}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 12 & 10 & 6 \\ 2 & 3 & 4 & 0 & 10 & 5 & 5 \\ 2 & 2 & 2 & -3 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$

[149] $1^{113}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 13 & 4 \\ 4 & 7 & 6 & 0 & 13 & 25 & 9 \\ 2 & 2 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{13}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 9 & 2 \\ 5 & 8 & 9 & 0 & 9 & 40 & 12 \\ 2 & 2 & 2 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{111}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -314$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.31 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{112}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -332$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 4 & 9 & 12 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{12}321^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 11 & 2 \\ 5 & 7 & 11 & 0 & 11 & 45 & 13 \\ 2 & 2 & 2 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{113}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 25 & 0 & 188 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{112}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -356$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 10 & 4 \\ 3 & 5 & 5 & 0 & 10 & 13 & 7 \\ 2 & 2 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{113}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -381$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 13 & 0 & 74 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{112}321^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -386$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 22 & 4 \\ 6 & 8 & 14 & 0 & 22 & 70 & 16 \\ 2 & 2 & 2 & -3 & 4 & 16 & 1 \end{pmatrix}_{b_0}$$

[158] $1^{131}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 6 & 3 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

[161] $1^{114}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{114}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{112}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

- [162] $1^{112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [163] $1^{112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$
- [164] $1^{113} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$
- [165] $1^{112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$
- [166] $1^{113} 3^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -417$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$
- [167] $1^{14} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 24 & 3 \\ 18 & 35 & 0 & 24 & 202 & 28 \\ 3 & 5 & -3 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$
- [169] $1^{112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [168] $1^{114} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 27 & 16 & 0 & 122 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$
- [170] $1^{113} 3^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 9 & 3 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [171] $1^{121} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -481$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 25 & 14 & 9 \\ 2 & 3 & 4 & 0 & 14 & 5 & 5 \\ 2 & 2 & 2 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{12} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 18 & 3 \\ 12 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -3 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{114} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.36 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{112} 3^{113} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{113} 3^{112} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{13} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -551$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 12 & 2 \\ 6 & 9 & 12 & 0 & 12 & 61 & 15 \\ 2 & 2 & 2 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

[177] $1^{13} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -565$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 21 & 3 \\ 15 & 30 & 0 & 21 & 142 & 24 \\ 3 & 5 & -3 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{112} 3^{114} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 42 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 42 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3^{111} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{112} 3^{112} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

- [181] $1^{113}321^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -581$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 21 & 0 & 142 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$
- [182] $1^{112}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -586$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 14 & 4 \\ 4 & 6 & 8 & 0 & 14 & 26 & 10 \\ 2 & 2 & 2 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$
- [183] $1^{111}31^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -587$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 5 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$
- [184] $1^{113}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -591$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 104 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$
- [185] $1^{112}321^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -596$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 18 & 4 \\ 5 & 7 & 11 & 0 & 18 & 45 & 13 \\ 2 & 2 & 2 & -3 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$
- [186] $1^{14}321^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -636$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 13 & 2 \\ 7 & 11 & 13 & 0 & 13 & 81 & 17 \\ 2 & 2 & 2 & -3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$
- [187] $1^{12}321^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -650$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 22 & 3 \\ 14 & 30 & 0 & 22 & 134 & 24 \\ 3 & 5 & -3 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$
- [190] $1^{113}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$
- [192] $1^{114}321^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 67 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 67 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [188] $1^{112}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{113} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{112} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{13} 3 \ 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -671, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 \\ 1 & 2 & 0 & 0 & 3 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 1 & 3 & 0 & 0 & 15 \\ 7 & 10 & 15 & 0 & 15 & 88 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{111} 31^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -699, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 15 & 6 \\ 3 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & -3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{113} 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{113} 3 \ 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{111} 31^{13} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 7 & 21 & 12 & 0 & 12 & 73 & 15 \\ 2 & 4 & 2 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{113} 3 \ 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 17 & 4 \\ 5 & 8 & 9 & 0 & 17 & 40 & 12 \\ 2 & 2 & 2 & -3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{114} 3 \ 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 20 & 0 & 158 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{13} 3^{114} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -895$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 25 & 3 \\ 17 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -3 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{114} 3^{111} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{114} 3^{112} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{113} 3^{112} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{113} 3^{114} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[205] $2^{1111} 2^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -901$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{114} 3^{112} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -926$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 20 & 4 \\ 6 & 10 & 10 & 0 & 20 & 58 & 14 \\ 2 & 2 & 2 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{114} 3^{112} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -936$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 57 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{14} 3^{114} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -936$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 16 & 2 \\ 8 & 12 & 16 & 0 & 16 & 110 & 20 \\ 2 & 2 & 2 & -3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{\overline{113}} 3 \ 21^{\overline{11}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -936, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.05 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

$$[210] \ 1^{\overline{114}} 3 \ 21^{\overline{114}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -950, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 28 & 3 \\ 20 & 40 & 0 & 28 & 254 & 32 \\ 3 & 5 & -3 & 3 & 32 & 2 \end{pmatrix}_{b_0}$$

$$[211] \ 1^{\overline{114}} 3 \ 21^{\overline{14}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -966, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 28 & 0 & 254 & 32 \\ 3 & 4 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{\overline{111}} 31^{\overline{14}} 31^{\overline{1}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -1007, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 15 & 2 \\ 9 & 27 & 15 & 0 & 15 & 121 & 19 \\ 2 & 4 & 2 & -3 & 2 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{\overline{113}} 3 \ 21^{\overline{13}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1016, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[213] \ 1^{\overline{114}} 3 \ 21^{\overline{13}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1016, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 24 & 0 & 202 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{\overline{113}} 3 \ 21^{\overline{113}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1111, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 21 & 4 \\ 6 & 9 & 12 & 0 & 21 & 61 & 15 \\ 2 & 2 & 2 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{\overline{113}} 3 \ 21^{\overline{12}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.01 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{\overline{113}} 3 \ 21^{\overline{11}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.04 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{\overline{114}} 3 \ 21^{\overline{13}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

12.2

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 68 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{114} 3^{113} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3^{11} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 25 & 4 \\ 7 & 10 & 15 & 0 & 25 & 88 & 18 \\ 2 & 2 & 2 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^{114} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{113} 3^{114} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{114} 3^{112} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3^{112} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{114} 3^{114} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{114} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 0 & 4 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 24 & 4 \\ 7 & 11 & 13 & 0 & 24 & 81 & 17 \\ 2 & 2 & 2 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{114} 3^{113} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 69 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{114} 3 \ 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{114} 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{114} 3 \ 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 28 & 4 \\ 8 & 12 & 16 & 0 & 28 & 110 & 20 \\ 2 & 2 & 2 & -3 & 4 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{114} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{114} 3 \ 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 70 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{114} 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [234] \ 1^{111} 2 \ 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 3 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 3 & -1 & 0 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{11} 3 \ 221^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 6 & 0 & 6 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 2 & 13 & 4 \\ 7 & 17 & 0 & 13 & 36 & 14 \\ 3 & 6 & -3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{12} 3 \ 221^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

11.7

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 6 & 0 & 6 & 34 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 2 & 26 & 4 \\ 14 & 34 & 0 & 26 & 150 & 28 \\ 3 & 6 & -3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{\overline{111}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 117, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{\overline{121}} 3 \ 221^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 1 & 1 & 11 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 11 & 0 & 49 & 11 & 13 \\ 2 & 3 & 0 & 11 & 1 & 3 \\ 3 & 3 & -1 & 13 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{\overline{111}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{\overline{111}} 3 \ 221^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 8 & 5 \\ 2 & 3 & 5 & 0 & 8 & 6 & 6 \\ 2 & 2 & 3 & -3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{\overline{1}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{\overline{1}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 222, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{\overline{111}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{121} 3^{221} 1^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 & 2 \\ 1 & 2 & 1 & 0 & 10 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 10 & 0 & 29 & 16 & 11 \\ 2 & 3 & 5 & 0 & 16 & 6 & 6 \\ 2 & 2 & 3 & -3 & 11 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{11} 4^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{11} 4^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 338, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 8 & 3 \\ 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{112} 3^{221} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 1 & 0 & 5 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 16 & 5 \\ 4 & 6 & 10 & 0 & 16 & 30 & 12 \\ 2 & 2 & 3 & -3 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{113} 3^{221} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 1 & 0 & 5 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 24 & 5 \\ 6 & 9 & 15 & 0 & 24 & 70 & 18 \\ 2 & 2 & 3 & -3 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[251] \ 1^{114} 3^{221} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 1116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 1 & 0 & 5 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 32 & 5 \\ 8 & 12 & 20 & 0 & 32 & 126 & 24 \\ 2 & 2 & 3 & -3 & 5 & 24 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [252] \ 1^{12} 5^{131} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[253] \ 1^{11} 5^{131} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

4.542 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

4.543 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 21 & 4 \\ 11 & 19 & 0 & 12 & 21 & 73 & 15 \\ 3 & 4 & -3 & 2 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 19 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 6 & 2 \\ 11 & 19 & 0 & 12 & 73 & 33 & 15 \\ 5 & 9 & 0 & 6 & 33 & 13 & 7 \\ 3 & 4 & -3 & 2 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{1121} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 12 & 6 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -3 & 6 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -3 & 6 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] 1^{121} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 12 & 9 & 9 \\ 3 & 5 & 0 & 12 & 4 & 3 & 4 \\ 2 & 4 & 0 & 9 & 3 & 1 & 3 \\ 3 & 4 & -3 & 9 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] 1^{111} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] 1^{111} 3 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 5 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 8 & 14 & 0 & 9 & 15 & 37 & 11 \\ 3 & 4 & -3 & 2 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] 1^{112} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 4 \\ 7 & 13 & 0 & 9 & 12 & 28 & 10 \\ 3 & 4 & -3 & 2 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] 1^{111} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -3 & 6 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -3 & 2 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 7 & 3 \\ 3 & 4 & 0 & 9 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 12 & 21 & 6 \\ 4 & 8 & 0 & 12 & 10 & 18 & 6 \\ 7 & 13 & 0 & 21 & 18 & 28 & 10 \\ 3 & 4 & -3 & 6 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 15 & 6 \\ 2 & 4 & 0 & 6 & 1 & 6 & 3 \\ 5 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & -3 & 6 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 8 & 3 \\ 3 & 4 & 0 & 4 & 9 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 6 & 13 & 24 & 7 \\ 8 & 14 & 0 & 9 & 24 & 37 & 11 \\ 3 & 4 & -3 & 2 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 5 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 18 & 4 \\ 10 & 18 & 0 & 12 & 18 & 58 & 14 \\ 3 & 4 & -3 & 2 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 16 & 3 \\ 3 & 4 & 0 & 4 & 5 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 30 & 4 \\ 16 & 28 & 0 & 18 & 30 & 154 & 22 \\ 3 & 4 & -3 & 2 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 39 & 11 \\ 8 & 14 & 0 & 9 & 39 & 37 & 11 \\ 3 & 4 & -3 & 2 & 11 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -723, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 9 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -3 & 2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1112} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -738, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 12 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -3 & 2 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -744, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 13 & 3 \\ 3 & 4 & 0 & 4 & 5 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 24 & 4 \\ 13 & 23 & 0 & 15 & 24 & 100 & 18 \\ 3 & 4 & -3 & 2 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{113} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -779, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 5 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 21 & 4 \\ 12 & 22 & 0 & 15 & 21 & 85 & 17 \\ 3 & 4 & -3 & 2 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -828, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 21 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -3 & 2 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1112} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -845, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 21 & 6 \\ 2 & 4 & 0 & 6 & 1 & 9 & 3 \\ 7 & 13 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & -3 & 6 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1112} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -941, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -3 & 2 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1112} 31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 8 & 14 & 0 & 9 & 37 & 48 & 11 \\ 10 & 18 & 0 & 12 & 48 & 58 & 14 \\ 3 & 4 & -3 & 2 & 11 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 5 & 9 & 0 & 6 & 13 & 27 & 7 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -3 & 2 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 16 & 3 \\ 3 & 4 & 0 & 4 & 9 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 48 & 7 \\ 16 & 28 & 0 & 18 & 48 & 154 & 22 \\ 3 & 4 & -3 & 2 & 7 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 27 & 6 \\ 2 & 4 & 0 & 6 & 1 & 12 & 3 \\ 9 & 17 & 0 & 27 & 12 & 49 & 13 \\ 3 & 4 & -3 & 6 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 9 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 33 & 6 \\ 2 & 4 & 0 & 6 & 1 & 15 & 3 \\ 11 & 21 & 0 & 33 & 15 & 76 & 16 \\ 3 & 4 & -3 & 6 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -3 & 2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 15 & 3 \\ 3 & 4 & 0 & 4 & 5 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 27 & 4 \\ 15 & 27 & 0 & 18 & 27 & 133 & 21 \\ 3 & 4 & -3 & 2 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1122} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 10 & 3 \\ 3 & 4 & 0 & 4 & 13 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 7 & 13 & 0 & 9 & 28 & 42 & 10 \\ 10 & 18 & 0 & 12 & 42 & 58 & 14 \\ 3 & 4 & -3 & 2 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 18 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -3 & 2 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 10 & 3 \\ 3 & 4 & 0 & 4 & 9 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 5 & 9 & 0 & 6 & 13 & 30 & 7 \\ 10 & 18 & 0 & 12 & 30 & 58 & 14 \\ 3 & 4 & -3 & 2 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1122} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 30 & 10 \\ 7 & 13 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & -3 & 2 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 4 & 9 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 39 & 7 \\ 13 & 23 & 0 & 15 & 39 & 100 & 18 \\ 3 & 4 & -3 & 2 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -3 & 2 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 18 & 3 \\ 3 & 4 & 0 & 4 & 5 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 33 & 4 \\ 18 & 32 & 0 & 21 & 33 & 193 & 25 \\ 3 & 4 & -3 & 2 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 17 & 3 \\ 3 & 4 & 0 & 4 & 5 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 30 & 4 \\ 17 & 31 & 0 & 21 & 30 & 172 & 24 \\ 3 & 4 & -3 & 2 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 4 & 9 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 36 & 7 \\ 12 & 22 & 0 & 15 & 36 & 85 & 17 \\ 3 & 4 & -3 & 2 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1122} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 60 & 14 \\ 10 & 18 & 0 & 12 & 60 & 58 & 14 \\ 3 & 4 & -3 & 2 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 20 & 3 \\ 3 & 4 & 0 & 4 & 5 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 3 & 5 & 0 & 3 & 4 & 36 & 4 \\ 20 & 36 & 0 & 24 & 36 & 238 & 28 \\ 3 & 4 & -3 & 2 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 18 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -3 & 2 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 24 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -3 & 2 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 4 & 9 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 42 & 7 \\ 14 & 26 & 0 & 18 & 42 & 118 & 20 \\ 3 & 4 & -3 & 2 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 21 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -3 & 2 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 4 & 9 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 45 & 7 \\ 15 & 27 & 0 & 18 & 45 & 133 & 21 \\ 3 & 4 & -3 & 2 & 7 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1113} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 18 & 3 \\ 3 & 4 & 0 & 4 & 9 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 5 & 9 & 0 & 6 & 13 & 54 & 7 \\ 18 & 32 & 0 & 21 & 54 & 193 & 25 \\ 3 & 4 & -3 & 2 & 7 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 17 & 3 \\ 3 & 4 & 0 & 4 & 9 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 5 & 9 & 0 & 6 & 13 & 51 & 7 \\ 17 & 31 & 0 & 21 & 51 & 172 & 24 \\ 3 & 4 & -3 & 2 & 7 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 20 & 3 \\ 3 & 4 & 0 & 4 & 9 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 24 & 2 \\ 5 & 9 & 0 & 6 & 13 & 60 & 7 \\ 20 & 36 & 0 & 24 & 60 & 238 & 28 \\ 3 & 4 & -3 & 2 & 7 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 7 & 3 \\ 2 & 5 & 0 & 4 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 7 & 14 & 8 \\ 3 & 5 & -3 & 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111} 3^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 1 & 1 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 3 & 3 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 & 0 \\ 2 & 2 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 3^{111} 21^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 20 & 0 & 57 & 15 & 15 & 15 \\ 2 & 5 & 0 & 15 & 2 & 4 & 4 \\ 2 & 5 & 0 & 15 & 4 & 2 & 4 \\ 3 & 5 & -3 & 15 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111} 3^{111} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 20 & 5 \\ 9 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 0 & 63 & 15 \\ 5 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111} 3^{111} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 1 & 1 & 3 & 0 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 7 & 4 & 5 & 5 \\ 2 & 2 & 2 & -3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{11} \overset{1}{3} 21^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 11 & 3 \\ 2 & 5 & 0 & 4 & 2 & 11 & 4 \\ 7 & 15 & 0 & 11 & 11 & 32 & 12 \\ 3 & 5 & -3 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{111} \overset{1}{3} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 1 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 3 & 7 & 4 \\ 1 & 2 & 1 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 7 & 3 & 5 & 5 \\ 2 & 2 & 2 & -3 & 4 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{12} \overset{1}{3} 21^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 18 & 3 \\ 2 & 5 & 0 & 4 & 2 & 18 & 4 \\ 12 & 25 & 0 & 18 & 18 & 94 & 20 \\ 3 & 5 & -3 & 3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111} \overset{1}{3} 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 7 & 7 & 3 \\ 5 & 10 & 0 & 7 & 14 & 16 & 8 \\ 5 & 10 & 0 & 7 & 16 & 14 & 8 \\ 3 & 5 & -3 & 3 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{13} \overset{1}{3} 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 25 & 3 \\ 2 & 5 & 0 & 4 & 2 & 25 & 4 \\ 17 & 35 & 0 & 25 & 25 & 188 & 28 \\ 3 & 5 & -3 & 3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{111} \overset{1}{3} 21^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 390, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 5 & 0 & 5 & 10 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 7 & 11 & 3 \\ 5 & 10 & 0 & 7 & 14 & 23 & 8 \\ 7 & 15 & 0 & 11 & 23 & 32 & 12 \\ 3 & 5 & -3 & 3 & 8 & 12 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111} \overset{1}{3} 21^{1123} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 630, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 9 & 3 \\ 3 & 5 & 0 & 5 & 15 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 11 & 15 & 3 \\ 7 & 15 & 0 & 11 & 32 & 45 & 12 \\ 9 & 20 & 0 & 15 & 45 & 58 & 16 \\ 3 & 5 & -3 & 3 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{12} \overset{1}{3} 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 22 & 3 \\ 2 & 5 & 0 & 4 & 2 & 22 & 4 \\ 14 & 30 & 0 & 22 & 22 & 134 & 24 \\ 3 & 5 & -3 & 3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{111}321^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 720$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 11 & 11 & 3 \\ 7 & 15 & 0 & 11 & 32 & 34 & 12 \\ 7 & 15 & 0 & 11 & 34 & 32 & 12 \\ 3 & 5 & -3 & 3 & 12 & 12 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{111}321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 752$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 7 & 7 & 4 \\ 2 & 3 & 4 & 0 & 7 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 4 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{111}321^{1133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 840$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.00 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 15 & 15 & 3 \\ 9 & 20 & 0 & 15 & 58 & 60 & 16 \\ 9 & 20 & 0 & 15 & 60 & 58 & 16 \\ 3 & 5 & -3 & 3 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [71] $1^{111}3221^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -52$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 6 & 0 & 6 & 17 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 0 & 2 & 13 & 13 & 4 \\ 7 & 17 & 0 & 13 & 36 & 38 & 14 \\ 7 & 17 & 0 & 13 & 38 & 36 & 14 \\ 3 & 6 & -3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{111}3221^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -468$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 5 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 8 & 8 & 5 \\ 2 & 3 & 5 & 0 & 8 & 6 & 8 & 6 \\ 2 & 3 & 5 & 0 & 8 & 8 & 6 & 6 \\ 2 & 2 & 3 & -3 & 5 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

4.544 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 392$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 6 & 5 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 5 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 6 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 2 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 5 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.545 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{1123}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 12 & 5 & 6 \\ 4 & 8 & 0 & 12 & 10 & 4 & 6 \\ 2 & 3 & 0 & 5 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 19 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 11 & 19 & 0 & 12 & 73 & 11 & 15 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -3 & 2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1123}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 & 2 \\ 8 & 14 & 0 & 9 & 37 & 8 & 11 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -3 & 2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 9 & 7 & 9 \\ 2 & 4 & 0 & 9 & 1 & 2 & 3 \\ 2 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & -3 & 9 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 & 2 \\ 7 & 13 & 0 & 9 & 28 & 7 & 10 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -3 & 2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{112} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 4 & 28 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 16 & 28 & 0 & 18 & 154 & 16 & 22 \\ 2 & 3 & 0 & 2 & 16 & 0 & 2 \\ 3 & 4 & -3 & 2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 9 & 17 & 0 & 12 & 49 & 9 & 13 \\ 2 & 3 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -3 & 2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 11 & 21 & 0 & 15 & 76 & 11 & 16 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -3 & 2 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 18 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 10 & 18 & 0 & 12 & 58 & 10 & 14 \\ 2 & 3 & 0 & 2 & 10 & 0 & 2 \\ 3 & 4 & -3 & 2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 4 & 7 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 0 & 2 \\ 5 & 7 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 3^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -192, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 4 & 12 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 0 & 2 \\ 8 & 12 & 0 & 8 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

$$4.546 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$$

$$T = 10 \quad [0] \ 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 3 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 6 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 9 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 5 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 6 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 5 & 15 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 15 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.547 NHC(-2)(-3) + f₄ + su₃ + su₂ + su2n3mix

$$T = 9 \quad [0] \quad 1^{\overline{111}3^{\overline{1}}1^{\overline{111}}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 8 & 0 & 5 & 9 & 12 & 6 \\ 3 & 4 & -3 & 2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}3^{\overline{1}}1^{\overline{12}}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 11 & 2 \\ 2 & 4 & 0 & 3 & 1 & 11 & 3 \\ 10 & 17 & 0 & 11 & 11 & 55 & 13 \\ 3 & 4 & -3 & 2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}3^{\overline{1}}1^{\overline{11}}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 4 & 7 & 0 & 5 & 6 & 7 & 5 \\ 3 & 4 & -3 & 2 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{112}3^{\overline{1}}1^{\overline{11}}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 6 & 11 & 0 & 9 & 8 & 20 & 8 \\ 3 & 4 & -3 & 3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}3^{\overline{1}}1^{\overline{11}}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 5 & 3 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}3^{\overline{1}}1^{\overline{11}}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 5 & 2 \\ 5 & 9 & 0 & 6 & 13 & 14 & 7 \\ 5 & 8 & 0 & 5 & 14 & 12 & 6 \\ 3 & 4 & -3 & 2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{113}3^{\overline{1}}1^{\overline{11}}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 14 & 2 \\ 3 & 5 & 0 & 3 & 4 & 18 & 4 \\ 11 & 20 & 0 & 14 & 18 & 69 & 15 \\ 3 & 4 & -3 & 2 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}3^1} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 8 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 4 \\ 7 & 12 & 0 & 8 & 12 & 25 & 9 \\ 3 & 4 & -3 & 2 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 2 & 4 & 0 & 3 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 5 & 7 & 5 \\ 3 & 4 & -3 & 2 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 11 & 6 \\ 2 & 4 & 0 & 6 & 1 & 5 & 3 \\ 4 & 7 & 0 & 11 & 5 & 7 & 5 \\ 3 & 4 & -3 & 6 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 8 & 2 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & -3 & 2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}3^1} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 5 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 11 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 9 & 16 & 0 & 11 & 15 & 44 & 12 \\ 3 & 4 & -3 & 2 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 4 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 5 & 2 \\ 5 & 9 & 0 & 6 & 13 & 11 & 7 \\ 4 & 7 & 0 & 5 & 11 & 7 & 5 \\ 3 & 4 & -3 & 2 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}3^1} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 8 & 2 \\ 5 & 9 & 0 & 6 & 13 & 20 & 7 \\ 7 & 12 & 0 & 8 & 20 & 25 & 9 \\ 3 & 4 & -3 & 2 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

4.548 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}3^1} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -119$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 4 & -3 & 2 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -126$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 13.05 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{113}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 1 & 3 \\ 4 & 9 & 4 & 0 & 24 & 0 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 10 & 24 & 9 & 0 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$

$T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 1 & 3 \\ 4 & 9 & 4 & 0 & 16 & 0 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 7 & 16 & 6 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -182$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.9$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 0 & 5 & 13 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 12 & 6 & 3 \\ 7 & 13 & 0 & 12 & 28 & 18 & 10 \\ 4 & 8 & 0 & 6 & 18 & 9 & 5 \\ 3 & 4 & -3 & 3 & 10 & 5 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{1111}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -3 & 2 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{132}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 0 & 5 & 17 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 15 & 6 & 3 \\ 9 & 17 & 0 & 15 & 49 & 24 & 13 \\ 4 & 8 & 0 & 6 & 24 & 9 & 5 \\ 3 & 4 & -3 & 3 & 13 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 2 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 1 & 3 \\ 4 & 9 & 4 & 0 & 24 & 0 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 11 & 24 & 9 & 0 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 12 & 6 \\ 3 & 4 & -3 & 2 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 15 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{141}31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 18 & 3 & 3 \\ 11 & 21 & 0 & 18 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{1111}31^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{121}31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 12 & 3 & 3 \\ 7 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{131}31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 15 & 3 & 3 \\ 9 & 17 & 0 & 15 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{1111}31^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & -3 & 2 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

4.549 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 \rightarrow (-3)$

$T = 8 \quad [0] \quad 1^{13}31^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 24 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{12}31^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_3 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$

$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 48 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 1^{13} 3^{12} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^{12} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^{12} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

12.4

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 6 & 1 & 12 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 25 & 9 \\ 5 & 5 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 \\ 2 & 6 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 6 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 2 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{12}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 13 & 0 & 43 & 13 \\ 3 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{13}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 2 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 5 & 0 & 7 & 5 \\ 3 & 6 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{13}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 \\ 2 & 4 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 1 & 3 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 0 & 21 & 9 \\ 3 & 6 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 5 & 6 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [21] \quad 1^{11} 4^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^{11} 1^{11} 4^{11} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [23] \quad 1^{11} 5^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 8 & 5 \\ 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 10 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.550 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_3 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{o}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}}\mathfrak{o}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.551 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_3 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 5 & 12 & 6 \\ 2 & 3 & 0 & 5 & 0 & 4 & 2 \\ 4 & 8 & 0 & 12 & 4 & 10 & 6 \\ 3 & 4 & -3 & 6 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{13} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\mathfrak{u}_3 + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}}\mathfrak{u}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 5 & 6 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 1 & 3 \\ 3 & 4 & -3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 3 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 14 & 0 & 9 & 8 & 37 & 11 \\ 3 & 4 & -3 & 2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 7 & 9 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 4 & 0 & 9 & 2 & 1 & 3 \\ 3 & 4 & -3 & 9 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 13 & 0 & 9 & 7 & 28 & 10 \\ 3 & 4 & -3 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 9 & 17 & 0 & 12 & 9 & 49 & 13 \\ 3 & 4 & -3 & 2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 1^{11} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 3 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 2 & 3 & 0 & 2 & 0 & 16 & 2 \\ 16 & 28 & 0 & 18 & 16 & 154 & 22 \\ 3 & 4 & -3 & 2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 21 & 0 & 15 & 11 & 76 & 16 \\ 3 & 4 & -3 & 2 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 3 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 10 & 18 & 0 & 12 & 10 & 58 & 14 \\ 3 & 4 & -3 & 2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 3 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 2 & 3 & 0 & 2 & 0 & 13 & 2 \\ 13 & 23 & 0 & 15 & 13 & 100 & 18 \\ 3 & 4 & -3 & 2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 3 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 12 & 22 & 0 & 15 & 12 & 85 & 17 \\ 3 & 4 & -3 & 2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 3 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 2 & 3 & 0 & 2 & 0 & 15 & 2 \\ 15 & 27 & 0 & 18 & 15 & 133 & 21 \\ 3 & 4 & -3 & 2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 3 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 14 & 26 & 0 & 18 & 14 & 118 & 20 \\ 3 & 4 & -3 & 2 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 3 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 21 & 2 \\ 2 & 3 & 0 & 2 & 0 & 18 & 2 \\ 18 & 32 & 0 & 21 & 18 & 193 & 25 \\ 3 & 4 & -3 & 2 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 3 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 21 & 2 \\ 2 & 3 & 0 & 2 & 0 & 17 & 2 \\ 17 & 31 & 0 & 21 & 17 & 172 & 24 \\ 3 & 4 & -3 & 2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 3 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 24 & 2 \\ 2 & 3 & 0 & 2 & 0 & 20 & 2 \\ 20 & 36 & 0 & 24 & 20 & 238 & 28 \\ 3 & 4 & -3 & 2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{\overline{111}} 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -96, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 0 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 5 & 0 & 8 & 6 \\ 3 & 4 & -2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{\overline{111}} 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -192, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 8 & 3 \\ 3 & 4 & 0 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 8 & 0 & 26 & 10 \\ 3 & 4 & -2 & 2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

4.552 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 1 \\ 3 & 4 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 5 \\ 3 & 4 & 0 & 4 & 3 & 5 & 9 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 6 \\ 2 & 3 & 0 & 2 & 0 & 3 & 5 \\ 3 & 5 & 0 & 3 & 3 & 4 & 9 \\ 5 & 9 & 0 & 6 & 5 & 9 & 13 \\ 3 & 4 & -3 & 2 & 2 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 5 & 6 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 1 & 3 \\ 2 & 4 & 0 & 6 & 2 & 3 & 1 \\ 3 & 4 & -3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 \\ 3 & 4 & 0 & 4 & 3 & 4 & 9 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 6 \\ 2 & 3 & 0 & 2 & 0 & 2 & 5 \\ 2 & 4 & 0 & 3 & 2 & 1 & 6 \\ 5 & 9 & 0 & 6 & 5 & 6 & 13 \\ 3 & 4 & -3 & 2 & 2 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 6 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 5 & 13 & 15 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.553 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{123}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 & 4 \\ 3 & 4 & 0 & 8 & 3 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 4 & 6 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 & 3 \\ 3 & 4 & -3 & 5 & 2 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 0 & 5 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 5 & 0 & -1 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 & 4 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & -3 & 2 & 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 6 & 5 & 12 & 6 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

4.554 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \mathfrak{n}_3 \text{mix}$

$$T = 8 \quad [0] \ 1^{\overline{114}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 & 3 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 29 & 6 & 6 & 6 \\ 10 & 19 & 0 & 29 & 64 & 14 & 14 & 14 \\ 3 & 4 & -3 & 6 & 14 & 1 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{123}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 6 & 11 & 0 & 9 & 20 & 8 \\ 3 & 4 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 16 & 5 \\ 7 & 12 & 0 & 16 & 25 & 9 \\ 3 & 4 & -3 & 5 & 9 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 16 & 27 & 0 & 17 & 151 & 21 \\ 3 & 4 & -3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 10 & 5 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 23 & 6 \\ 8 & 15 & 0 & 23 & 39 & 11 \\ 3 & 4 & -3 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 17 & 6 \\ 6 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{112}3^11^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 11 & 6 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{112}3^11^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{\overline{111}3^11^{\overline{13}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 13 & 22 & 0 & 14 & 97 & 17 \\ 3 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{\overline{111}3^11^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{\overline{121}3^11^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 16 & 9 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{\overline{111}3^11^{\overline{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 10 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{\overline{113}3^11^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 8 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{\overline{114}3^11^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 10 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{\overline{112}3^11^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{\overline{112}3^11^{\overline{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 12 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{112}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 18 & 31 & 0 & 20 & 188 & 24 \\ 3 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 11 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{112}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 923, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 15 & 26 & 0 & 17 & 128 & 20 \\ 3 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 13 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 14 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -3 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 17 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -3 & 2 & 23 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1667, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 20 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -3 & 2 & 27 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{114}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1708, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 16 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{114}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 19 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -3 & 2 & 26 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{114}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 22 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -3 & 2 & 30 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [32] \quad 1^{112} 3^{12} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 54 & 0 & 217 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11} 31^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 3 & 8 & 4 & 0 & 4 & 9 & 5 \\ 2 & 4 & 2 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112} 31^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 7 & 3 \\ 2 & 4 & 3 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 3 & 7 & 8 & 0 & 7 & 15 & 7 \\ 2 & 3 & 3 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{111} 3^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 5 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 2 & 5 & 6 & 0 & 5 & 7 & 5 \\ 2 & 3 & 3 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{112} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 44 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{112} 3^{11} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 84 & 0 & 520 & 42 \\ 5 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{111} 31^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 2 & 5 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{111} 31^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 4 & 3 \\ 2 & 4 & 3 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 2 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 3 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11}31^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 5 & 3 \\ 2 & 5 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{11}3^{11}1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 14 & 2 \\ 9 & 27 & 14 & 0 & 14 & 120 & 18 \\ 2 & 4 & 2 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{112}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 24 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{11}3^{11}1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 11 & 2 \\ 7 & 21 & 11 & 0 & 11 & 72 & 14 \\ 2 & 4 & 2 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{11}3^{11}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 8 & 2 \\ 5 & 15 & 8 & 0 & 8 & 36 & 10 \\ 2 & 4 & 2 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 16 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

- [51] $1^{1^{12}}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 17 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$
- [52] $1^{1^{12}}3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$
- [53] $1^{1^{12}}3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$
- [54] $1^{1^{11}}3^1 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -354$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 5 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 9 & 6 \\ 2 & 5 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$
- [55] $1^{1^{12}}3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -384$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 45 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [56] $1^{1^{11}}3^1 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -430$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 4 & 11 & 7 & 0 & 7 & 20 & 8 \\ 2 & 4 & 2 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [57] $1^{1^{12}}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{1^{12}}3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -584$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 35 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$
- [60] $1^{1^{12}}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -608$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.18 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$
- [59] $1^{1^{12}}3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -608$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [61] $1^{1^{11}}3^1 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -679$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 10 & 2 \\ 6 & 17 & 10 & 0 & 10 & 48 & 12 \\ 2 & 4 & 2 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

¹¹²

$$[62] \ 1^{112} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 36 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{111} 31^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 23 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 13 & 2 \\ 8 & 23 & 13 & 0 & 13 & 88 & 16 \\ 2 & 4 & 2 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{111} 31^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -1051, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 29 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 16 & 2 \\ 10 & 29 & 16 & 0 & 16 & 140 & 20 \\ 2 & 4 & 2 & -3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [65] \ 1^{11} 4 \ 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{111} 4 \ 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 6 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{13} 3^1 1^1 4 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} 3^1 1^1 4 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 8 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{111} 4 \ 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{11} 3^1 1^1 4 \ 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111} 1^{13} 1^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 9 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 1^{11} 4 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [73] $1^{11} 5 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 14 & 5 & 7 \\ 9 & 13 & 0 & 22 & 7 & 1 \\ 1 & 0 & -3 & 0 & -1 & 1 \\ 14 & 22 & 0 & 33 & 11 & 1 \\ 5 & 7 & -1 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{12} 5 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 & 7 \\ 9 & 13 & 0 & 32 & 7 & 1 \\ 1 & 0 & -3 & 0 & -1 & 1 \\ 20 & 32 & 0 & 70 & 16 & 3 \\ 5 & 7 & -1 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

4.555 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_2$

$T = 9$ [0] $1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 8 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 5 & 8 & 0 & 5 & 12 & 9 & 6 \\ 3 & 5 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -3 & 2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 3 & 2 \\ 10 & 17 & 0 & 11 & 55 & 11 & 13 \\ 2 & 4 & 0 & 3 & 11 & 1 & 3 \\ 3 & 4 & -3 & 2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 4 & 7 & 0 & 5 & 7 & 6 & 5 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -3 & 2 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 6 & 11 & 0 & 9 & 20 & 8 & 8 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}3^1}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 4 & 7 & 0 & 6 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & -3 & 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}3^1}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 6 & 2 \\ 5 & 8 & 0 & 5 & 12 & 14 & 6 \\ 5 & 9 & 0 & 6 & 14 & 13 & 7 \\ 3 & 4 & -3 & 2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}3^1}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 3 \\ 3 & 4 & 0 & 4 & 12 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 3 & 2 \\ 7 & 12 & 0 & 8 & 25 & 12 & 9 \\ 3 & 5 & 0 & 3 & 12 & 4 & 4 \\ 3 & 4 & -3 & 2 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}3^1}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 0 & 4 & 20 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 3 & 2 \\ 11 & 20 & 0 & 14 & 69 & 18 & 15 \\ 3 & 5 & 0 & 3 & 18 & 4 & 4 \\ 3 & 4 & -3 & 2 & 15 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}3^1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 4 & 7 & 0 & 5 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & -3 & 2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{111}3^1}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 11 & 6 & 6 \\ 4 & 7 & 0 & 11 & 7 & 5 & 5 \\ 2 & 4 & 0 & 6 & 5 & 1 & 3 \\ 3 & 4 & -3 & 6 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}3^1}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 3 & 2 \\ 7 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{112}3^1}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 0 & 4 & 16 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 3 & 2 \\ 9 & 16 & 0 & 11 & 44 & 15 & 12 \\ 3 & 5 & 0 & 3 & 15 & 4 & 4 \\ 3 & 4 & -3 & 2 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 6 & 2 \\ 4 & 7 & 0 & 5 & 7 & 11 & 5 \\ 5 & 9 & 0 & 6 & 11 & 13 & 7 \\ 3 & 4 & -3 & 2 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 0 & 4 & 12 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 6 & 2 \\ 7 & 12 & 0 & 8 & 25 & 20 & 9 \\ 5 & 9 & 0 & 6 & 20 & 13 & 7 \\ 3 & 4 & -3 & 2 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

4.556 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_3$

$T = 9$ [0] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 5 & 8 & 0 & 5 & 12 & 5 & 6 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 1 & 3 \\ 4 & 9 & 4 & 0 & 12 & 0 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 12 & 5 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 6 & 3 \\ 4 & 7 & 0 & 6 & 7 & 10 & 5 \\ 4 & 8 & 0 & 6 & 10 & 9 & 5 \\ 3 & 4 & -3 & 3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1211}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 6 & 11 & 0 & 9 & 20 & 8 & 8 \\ 2 & 4 & 0 & 3 & 8 & 0 & 2 \\ 3 & 4 & -3 & 3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 4 & 7 & 0 & 6 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

4.557 NHC(-2)(-3) + f₄ + su₃ + su₃

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{11} 3^1 2^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 2^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 10 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 11 & 0 & 31 & 11 \\ 3 & 4 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 3 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 1^{11} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3 1^{11} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{12} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 5 & 2 \\ 3 & 5 & 5 & 0 & 5 & 12 & 6 \\ 2 & 2 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 4 & 2 \\ 2 & 3 & 4 & 0 & 4 & 4 & 4 \\ 2 & 2 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 & 5 \\ 9 & 13 & 1 & 15 & 7 & 7 \\ 1 & 1 & -3 & 0 & -1 & -1 \\ 9 & 15 & 0 & 12 & 6 & 6 \\ 5 & 7 & -1 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 7 & 3 \\ 5 & 10 & 0 & 7 & 13 & 7 \\ 3 & 5 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 & 5 \\ 9 & 13 & 1 & 15 & 7 & 7 \\ 1 & 1 & -3 & 0 & -1 & -1 \\ 10 & 15 & 0 & 12 & 6 & 6 \\ 5 & 7 & -1 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 11 & 3 \\ 7 & 15 & 0 & 11 & 31 & 11 \\ 3 & 5 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 15 & 3 \\ 9 & 20 & 0 & 15 & 57 & 15 \\ 3 & 5 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & -3 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [28] \ 1^{11} 4 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.558 NHC(-2)(-3) + $\mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$

$T = 9$ [0] $1^{11} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 9 & 0 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & 0 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{12} 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

14.9

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 0 & 5 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 12 & 3 \\ 4 & 8 & 0 & 6 & 9 & 18 & 5 \\ 7 & 13 & 0 & 12 & 18 & 28 & 10 \\ 3 & 4 & -3 & 3 & 5 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3 1^{\overline{12}} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{s} \textcolor{brown}{u}_3 + \textcolor{red}{s} \textcolor{red}{u}_3 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 14 & 0 & 9 & 36 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -3 & 2 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 152$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 0 & 5 & 8 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 9 & 24 & 5 \\ 9 & 17 & 0 & 15 & 24 & 49 & 13 \\ 3 & 4 & -3 & 3 & 5 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{12}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{s} \textcolor{blue}{u}_3 + \textcolor{red}{s} \textcolor{red}{u}_3 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13 \ 2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 11 & 24 & 9 & 0 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{brown}{s}u_3 + \textcolor{red}{s}u_3 + \textcolor{blue}{s}u_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

$$15.2 \quad T_{\min} = 12.23 \rightarrow 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 9 & 6 \\ 3 & 5 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -3 & 2 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, \quad T_{\max} = 14, \quad \Delta = -95, \quad \det = -399, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

$$13.2 \quad T_{\min} = 12.53 \rightarrow 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 \\ 3 & 4 & -3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{4}} 3^{\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \textcolor{blue}{g}_2] + \textcolor{blue}{f}_4 + \textcolor{red}{s}\textcolor{red}{u}_3 + \textcolor{red}{s}\textcolor{red}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 9, \quad T_{\max} = 14, \quad \Delta = -95, \quad \det = -426, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

$$14.2 \quad T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 15 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{111}1} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{\overline{114}1^{\overline{12}}} 3^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 14 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{114}1^{\overline{1}}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 18 & 15 & 76 & 16 \\ 3 & 4 & -3 & 3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{111}1^{\overline{1}}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{112}1^{\overline{1}}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -3 & 3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{113}1^{\overline{1}}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -3 & 3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{111}1^{\overline{1}}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -3 & 2 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

4.559 NHC(-2)(-3) + f₄ + su₃ + su₃ + su₂→(-3)

$$T = 9 \quad [0] \ 1^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 4 & 8 & 0 & 6 & 9 & 4 & 5 \\ 2 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 6 & 12 & 5 & 6 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

4.560 NHC(-2)(-3) + f₄ + su₃ + su₃ + su₂n3mix

$$T = 9 \quad [0] \ 1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 5 & 8 & 0 & 5 & 5 & 12 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 0 & 5 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 5 & 0 & 0 & 12 & 6 \\ 3 & 5 & 0 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 0 & 5 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 10 & 5 \\ 4 & 7 & 0 & 6 & 10 & 7 & 5 \\ 3 & 4 & -3 & 3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 6 & 11 & 0 & 9 & 8 & 20 & 8 \\ 3 & 4 & -3 & 3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

4.561 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$T = 9$ [0] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 21, \Delta = -104, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 0 & 3 & 3 \\ 3 & 5 & 0 & -1 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 21, \Delta = -104, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 3 & 3 \\ 3 & 5 & 0 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 21, \Delta = -104, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.562 NHC(-2)(-3) + $\mathfrak{f}_4$ + (-3)→(-2)

$T = 7$ [0] $1^{\overline{1}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 97, \Delta = -147, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [1] $1^{\overline{1}} 23^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$

$$\begin{pmatrix} 13 & 20 & 1 & 7 \\ 20 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 2^1 1^1 4 1^1 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^1 2^1 1^1 5 13 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

4.563 NHC(-2)(-3) + $\mathbf{f}_4$ + (-3) $\rightarrow$ (-2) + $\mathbf{su}_2$

$$T = 8 \quad [0] \quad 1^1 3 1^1 1^1 2^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + (-3) \rightarrow (-2) + \mathbf{su}_2 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 6 & 4 \\ 8 & 9 & 4 & 0 & 8 & 5 \\ 3 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 3 & 0 & 4 & 4 \\ 4 & 5 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

4.564 NHC(-2)(-3) + $\mathbf{f}_4$ + (-3) $\rightarrow$ (-2) + $\mathbf{su}_2 \rightarrow$ (-3)

$$T = 8 \quad [0] \quad 1^1 3 1^1 1^1 2^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 3 & 4 \\ 8 & 9 & 4 & 0 & 4 & 5 \\ 3 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 2 & 0 & 0 & 2 \\ 4 & 5 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 2^1 1^1 4 1^1 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 1 & 6 & 0 & 0 & 3 \\ 5 & 6 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 6 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^1 2^1 1^1 5 13 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 8 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 10 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.565 NHC(-2)(-3) + NHC(-2)(-3)

$$T = 7 \quad [0] \quad 1^1 3 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 3 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3 1^{\overline{1}} + [\textcolor{red}{5} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{red}{5} \textcolor{blue}{u}_2 \cdot \textcolor{red}{g}_2] \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -94, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 12, \quad T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[3] $1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$

$$[5] \quad 1^1 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 3 & 5 & 4 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[4] $21^{11}31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{red}{u}_2 \cdot \textcolor{red}{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 6 & 5 & 4 & 3 & 3 \\ 4 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$

$$[6] \quad \begin{matrix} 1^1 \\ 131^{111} \end{matrix} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{blue}{5}\textcolor{red}{u}_2 \cdot \textcolor{red}{g}_2] \quad T=7, \, T_{\max}=12, \, \Delta=-94, \, \det=-32, \, (n_+, n_-, n_0)=(1, 7, 0), \, c_{\text{ext}}=6.0$$

$$T_{\min}=11.38 \rightarrow 12, \, T_{\max}=12, \, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[7] $21^{11}31^{11} + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{5}u_2 \cdot \textcolor{orange}{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 5 & 4 & 4 \\ 6 & 5 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$

[8] $21^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{brown}{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 5 & 4 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$

[9] $21^{\textcolor{brown}{1}1}31^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

$$T = 8 \quad [10] \quad 1 \overset{1}{\underset{1}{4}} 1 \overset{1}{\underset{1}{4}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s} \textcolor{blue}{u}_2 \cdot \textcolor{red}{g}_2] \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 10, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s} \textcolor{green}{u}_2 \cdot \textcolor{orange}{g}_2] \quad T = 8, \ T_{\max} = 10, \ \Delta = -66, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 21^{11} 4^1 1^1 + [\textcolor{red}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{u}}_2 \cdot \textcolor{red}{\mathfrak{g}}_2] \quad T = 8, \, T_{\max} = 11, \, \Delta = -85, \, \det = 16, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \, T_{\max} = 11, \, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[13] $21^1 41^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 221^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 12 & 5 \\ 6 & 8 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 21^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 31^{11}31^{11}3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 6 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 3 & 3 & 1 & 4 & 3 \\ 3 & 6 & 3 & 4 & 3 & 3 \\ 2 & 4 & 2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [18] \quad 21^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 21^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 21^{11}51^{11}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 6 & 6 & 5 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 221^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 5 & 6 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11}4131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -64, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 3 & 4 \\ 3 & 1 & 2 & 3 \\ 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

- [24] $1 \overset{1}{5} 1^{\overset{1}{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [25] $21 \overset{1}{5} 1^{\overset{1}{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 4 & 6 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$
- [26] $21 \overset{1}{5} 1^{\overset{1}{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [27] $1 \overset{1}{5} 1^{\overset{1}{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [28] $1 \overset{1}{5} 1^{\overset{1}{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [29] $221 \overset{1}{5} 1^{\overset{1}{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 5 & 2 & 3 \\ 6 & 13 & 12 & 5 & 7 \\ 5 & 12 & 8 & 5 & 6 \\ 2 & 5 & 5 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- $T = 10$ [30] $1 \overset{1}{5} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [31] $1 \overset{1}{5} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & 2 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$
- [32] $1 \overset{1}{5} 1231^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$
- [33] $2 \overset{1}{3} 1 \overset{1}{5} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 5 & 5 & 3 & 5 \\ 3 & 5 & 2 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$
- [34] $22 \overset{1}{3} 151^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 243, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 8 \\ 5 & 13 & 9 & 12 & 7 \\ 5 & 13 & 12 & 8 & 6 \\ 3 & 8 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [35] \quad 1^{11} 413151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & -2 \\ 2 & 2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 41^{11} 23151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 8 & 1 & 3 \\ 9 & 51 & 48 & 0 & 15 \\ 8 & 48 & 42 & 1 & 14 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 15 & 14 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [37] \quad 1^{11} 5^{131} 6^{131} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

4.566 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

4.567 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{11} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.568 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.569 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 19 & 4 \\ 3 & 5 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 21 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -507, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 14 & 4 \\ 3 & 5 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 15 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -992, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 \\ 3 & 5 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 5 & 4 & 0 & 28 & 4 \\ 3 & 5 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 30 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1227, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 5 & 4 & 0 & 23 & 4 \\ 3 & 5 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 24 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1892, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -2307, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -3152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 36 & 4 \\ 3 & 5 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

4.570 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 44, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 429, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.571 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.572 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 8 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 6 & 0 & 9 & 5 & 5 \\ 3 & 4 & 3 & 3 & -4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 4 & 2 & -4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.573 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.574 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 6 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.575 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 31, T \in [13, 31]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 31, T \in [13, 31]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

4.576 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{113} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{114} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -2456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 36 & 4 \\ 3 & 5 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$4.578 \quad \text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \ 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 3 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$4.579 \quad \text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.580 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.581 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \mathbf{n}3 \mathbf{mix}$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 12 & 4 \\ 3 & 5 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

4.582 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 5 & 0 & 8 & 4 \\ 2 & 4 & 1 & 4 & 0 & 6 & 3 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

13.2

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{s}\textcolor{blue}{u}_3$ $T = 9, T_{\max} = 15, \Delta = -96, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 & 0 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

4.583 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{so}_8$

$T = 8$ [0] $21^{11}31^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 15 & 7 \\ 17 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 13 & 6 \\ 15 & 17 & 13 & 12 & 6 \\ 7 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 1 & 0 \\ 9 & 17 & 1 & 12 & 6 \\ 5 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 1 & 0 \\ 12 & 17 & 1 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 13 & 6 \\ 12 & 17 & 13 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11} 1 3 1^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 13 & 6 \\ 9 & 17 & 13 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[10] $21^{111} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 13 & 6 \\ 12 & 17 & 13 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [11] $1^{11} 4 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

4.584 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{e}_6$

$T = 9$ [0] $1^{111} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{e}_6$ $T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.585 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{f}_4$

$T = 9$ [0] $1^{111} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{f}_4$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.586 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{red}{s}\textcolor{blue}{o}_8$

$T = 9$ [0] $1^{11} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 9, T_{\max} = 13, \Delta = -92, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -2 & 1 & 0 & 0 \\ 9 & 17 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{s}\textcolor{blue}{o}_8 + \textcolor{red}{s}\textcolor{blue}{o}_8$ $T = 9, T_{\max} = 13, \Delta = -92, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

4.587 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -64, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{11} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 13 & 10 & 0 & 75 & 17 \\ 2 & 2 & 2 & 2 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 2 & 2 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 4 & 6 & 6 & 5 & 0 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 5 & 4 & 0 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 5^{13} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 5 & 6 & 9 & 7 & 0 & 36 & 12 \\ 2 & 2 & 2 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{13} 5^{12} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 1 & 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 11 & 9 & 0 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{12} \overset{1}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 1 & 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 10 & 8 & 0 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0}$

[9] $1^{12} \overset{1}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 1 & 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 14 & 11 & 0 & 93 & 19 \\ 2 & 2 & 2 & 2 & -1 & 19 & 3 \end{pmatrix}_{b_0}$

[12] $1^{12} \overset{1}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 2 & 5 & 2 & 3 & 8 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$

[10] $1^{12} \overset{1}{5} 1^3 231^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 24 & 3 \\ 3 & 5 & 5 & 0 & 45 & 5 \\ 2 & 5 & 2 & 1 & 33 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 24 & 45 & 33 & 0 & 349 & 39 \\ 3 & 5 & 4 & -1 & 39 & 4 \end{pmatrix}_{b_0}$

[13] $1^{12} \overset{1}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 3 & 12 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[14] $1^{12} \overset{1}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 3 & 16 & 4 \\ 2 & 3 & 3 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[15] $1^{12} \overset{1}{5} 1^1 231^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 2 & 1 & 11 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 15 & 11 & 0 & 37 & 13 \\ 3 & 5 & 4 & -1 & 13 & 4 \end{pmatrix}_{b_0}$

[16] $1^{12} \overset{1}{5} 1^2 231^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 1 & 22 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 16 & 30 & 22 & 0 & 154 & 26 \\ 3 & 5 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$

[11] $1^{12} \overset{1}{5} 131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{green}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 2 & 3 & 4 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

$$[17] \quad 1^{13} \overset{1}{5} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 15 & 2 \\ 1 & 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 9 & 12 & 15 & 12 & 0 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 2 \overset{1}{3} 151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 9 & 3 \\ 4 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 3 & 12 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 2 \overset{1}{3} 151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 15 & 3 \\ 4 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 3 & 20 & 4 \\ 2 & 3 & 3 & 0 & 10 & 2 \\ 15 & 25 & 20 & 10 & 98 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 2 \overset{1}{3} 151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 12 & 3 \\ 4 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 3 & 16 & 4 \\ 2 & 3 & 3 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 2 \overset{1}{3} 151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 2 & 3 & 4 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 2 \overset{1}{3} 151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 6 & 3 \\ 4 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 3 & 8 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 2 \overset{1}{3} 151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 18 & 3 \\ 4 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 3 & 24 & 4 \\ 2 & 3 & 3 & 0 & 12 & 2 \\ 18 & 30 & 24 & 12 & 142 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 2 \overset{1}{3} 151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 21 & 3 \\ 4 & 5 & 5 & 3 & 35 & 5 \\ 3 & 5 & 2 & 3 & 28 & 4 \\ 2 & 3 & 3 & 0 & 14 & 2 \\ 21 & 35 & 28 & 14 & 194 & 28 \\ 3 & 5 & 4 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [25] \quad 41^{11} 231 \overset{1}{5} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 216, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 8 & 1 & 8 & 3 \\ 9 & 51 & 48 & 0 & 48 & 15 \\ 8 & 48 & 42 & 1 & 44 & 14 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 48 & 44 & 0 & 42 & 14 \\ 3 & 15 & 14 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

4.588 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -34, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 5 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \quad 2^{1122} 3^{11} 15^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 6 & 6 & 3 \\ 4 & 5 & 5 & 3 & 10 & 10 & 5 \\ 3 & 5 & 2 & 3 & 8 & 8 & 4 \\ 2 & 3 & 3 & 0 & 4 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 16 & 8 \\ 6 & 10 & 8 & 4 & 16 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 2^{1111} 3^{11} 15^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 & 3 \\ 4 & 5 & 5 & 3 & 5 & 5 & 5 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 & 4 \\ 3 & 5 & 4 & 2 & 2 & 4 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 2^{1115} 3^{11} 15^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 15 & 3 \\ 4 & 5 & 5 & 3 & 5 & 25 & 5 \\ 3 & 5 & 2 & 3 & 4 & 20 & 4 \\ 2 & 3 & 3 & 0 & 2 & 10 & 2 \\ 3 & 5 & 4 & 2 & 2 & 20 & 4 \\ 15 & 25 & 20 & 10 & 20 & 98 & 20 \\ 3 & 5 & 4 & 2 & 4 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 2^{1131} 3^{11} 15^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 9 & 3 & 3 \\ 4 & 5 & 5 & 3 & 15 & 5 & 5 \\ 3 & 5 & 2 & 3 & 12 & 4 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 & 12 \\ 3 & 5 & 4 & 2 & 12 & 2 & 4 \\ 3 & 5 & 4 & 2 & 12 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 2^{1141} 3^{11} 15^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 12 & 3 & 3 \\ 4 & 5 & 5 & 3 & 20 & 5 & 5 \\ 3 & 5 & 2 & 3 & 16 & 4 & 4 \\ 2 & 3 & 3 & 0 & 8 & 2 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 & 16 \\ 3 & 5 & 4 & 2 & 16 & 2 & 4 \\ 3 & 5 & 4 & 2 & 16 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 2^{1121} 3^{11} 15^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 6 & 3 & 3 \\ 4 & 5 & 5 & 3 & 10 & 5 & 5 \\ 3 & 5 & 2 & 3 & 8 & 4 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 & 8 \\ 3 & 5 & 4 & 2 & 8 & 2 & 4 \\ 3 & 5 & 4 & 2 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

4.589 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [6] \quad 1^{13} 5^{13} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 15 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 12 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 9 & 12 & 15 & 12 & 0 & 115 & 12 & 21 \\ 1 & 1 & 2 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[7] \quad & 1^{12} \overset{1}{5} 13^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 8 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 6 & 8 & 10 & 8 & 0 & 50 & 8 & 14 \\ 1 & 1 & 2 & 1 & 0 & 8 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 14 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[8] \quad & 1^{11} \overset{1}{5} 13^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 5 & 4 & 0 & 11 & 4 & 7 \\ 1 & 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 7 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

4.590 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$

$$\begin{aligned}
T = 11 \quad & [0] \quad 1^{11} \overset{1}{5} 13^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

4.591 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$\begin{aligned}
T = 12 \quad & [6] \quad 1^{13} \overset{1}{5} 13^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 15 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 9 & 12 & 15 & 12 & 0 & 12 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[7] \quad & 1^{12} \overset{1}{5} 13^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 6 & 8 & 10 & 8 & 0 & 8 & 50 & 14 \\ 1 & 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[8] \quad & 1^{11} \overset{1}{5} 13^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 4 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

4.592 NHC(-2)(-3) + NHC(-2)(-3) + su2n3mix

$$\begin{aligned}
T = 11 \quad & [1] \quad 1^{12} \overset{1}{5} 13^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 12 & 9 & 0 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{13} \overset{1}{5} \overset{1}{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{brown}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 0 & 16 & 2 \\ 1 & 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 9 & 11 & 16 & 12 & 0 & 117 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} \overset{1}{5} \overset{1}{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{brown}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -8, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 0 & 13 & 2 \\ 1 & 2 & 0 & 1 & 0 & 17 & 2 \\ 1 & 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 10 & 13 & 17 & 13 & 0 & 141 & 23 \\ 2 & 2 & 2 & 2 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

4.593 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{11} \overset{1}{5} \overset{1}{1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{brown}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 21 \overset{1}{1} \overset{1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{brown}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^1 \overset{1}{5} \overset{1}{1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{brown}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 21, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 21 \overset{1}{1} \overset{1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{brown}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 34, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 21 \overset{1}{1} \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{brown}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 105, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 3 \\ 4 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.594 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{11} \overset{1}{5} \overset{1}{1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{brown}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} \ 5^{\overline{1}} \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -15, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 10 & 8 & 6 & 0 & 21 & 9 \\ 3 & 4 & 3 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{12}} \ 5^{\overline{1}} \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -23, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 3 & 0 & 17 & 4 \\ 2 & 3 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 17 & 13 & 10 & 0 & 61 & 15 \\ 3 & 4 & 3 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} \ 5^{\overline{1}} \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -44, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 4 & 0 & 8 & 6 \\ 3 & 4 & 3 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{12}} \ 5^{\overline{1}} \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -50, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \\ 2 & 3 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 10 & 8 & 0 & 38 & 12 \\ 3 & 4 & 3 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{13}} \ 5^{\overline{1}} \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 3 & 0 & 21 & 4 \\ 2 & 3 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 12 & 21 & 15 & 12 & 0 & 88 & 18 \\ 3 & 4 & 3 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

4.595 NHC(-2)(-3) + $\mathfrak{so}_8$

$$T = 6 \quad [1] \ 1^{\overline{11}} \ 3 \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 6, \ T_{\max} = 96, \ \Delta = -143, \ \det = 6, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 96, \ T \in [11, 96]$$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 21 \ 1^{\overline{11}} \ 3 \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 6, \ T_{\max} = 96, \ \Delta = -143, \ \det = 6, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 96, \ T \in [11, 96]$$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} \ 3 \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 6, \ T_{\max} = 96, \ \Delta = -143, \ \det = 16, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.25 \rightarrow 11, \ T_{\max} = 96, \ T \in [11, 96]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -4 & -2 \\ 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 13 \ 1^{\overline{11}} \ 3 \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 6, \ T_{\max} = 96, \ \Delta = -143, \ \det = 19, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 7.0$$

- $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 96$, $T \in [12, 96]$
 $\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [4] $1^1 1^1 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 6$, $T_{\max} = 96$, $\Delta = -143$, $\det = 21$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 96$, $T \in [12, 96]$
 $\begin{pmatrix} 5 & 11 & 5 \\ 11 & 20 & 8 \\ 5 & 8 & 2 \end{pmatrix}_{b_0}$
- [5] $1^1 1^1 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 6$, $T_{\max} = 96$, $\Delta = -143$, $\det = 25$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 10 & 15 & 6 \\ 15 & 20 & 8 \\ 6 & 8 & 3 \end{pmatrix}_{b_0}$
- [6] $21^1 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 6$, $T_{\max} = 96$, $\Delta = -143$, $\det = 29$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 13 & 17 & 7 \\ 17 & 20 & 8 \\ 7 & 8 & 3 \end{pmatrix}_{b_0}$
- [7] $21^1 1^1 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 6$, $T_{\max} = 96$, $\Delta = -143$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 10 & 16 & 6 \\ 16 & 20 & 8 \\ 6 & 8 & 3 \end{pmatrix}_{b_0}$
- [8] $21^1 1^1 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 6$, $T_{\max} = 96$, $\Delta = -143$, $\det = 75$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 13 & 20 & 7 \\ 20 & 25 & 9 \\ 7 & 9 & 3 \end{pmatrix}_{b_0}$
- $T = 7$ [10] $21^1 3^1 2^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -5$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [13] $1^1 4^1 1^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [12] $1^1 1^1 4^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 9 & 8 & 5 \\ 4 & 8 & 6 & 4 \\ 3 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [11] $13^1 1^1 3^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [14] $21^1 3^1 2^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [16] $1^1 1^1 3^1 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [15] $1^1 1^1 3^1 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 3 \end{pmatrix}_{b_0}$
- [20] $1^1 1^1 3^1 2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$

- $T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11}3131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [18] $1^1 1321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [17] $1^1 1^1 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$
- [21] $1^1 1321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [22] $21^1 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 92, T \in [12, 92]$
 $\begin{pmatrix} 10 & 12 & 6 \\ 12 & 13 & 7 \\ 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [23] $1^1 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 92, T \in [12, 92]$
 $\begin{pmatrix} 10 & 12 & 6 \\ 12 & 13 & 7 \\ 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [27] $1^1 1321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{11} 321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [24] $1^{11} 3131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [26] $1^{11} 1321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [28] $1^1 31^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & 1 \end{pmatrix}_{b_0}$
- [30] $1^1 1^1 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

- $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 4 & 3 & 7 & 4 \\ 3 & 1 & 5 & 3 \\ 7 & 5 & 9 & 5 \\ 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [29] $1^1 1^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 5 & 9 & 5 \\ 9 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$
- [32] $21^1 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$
 $\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [31] $1^1 4^1 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$
 $\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [33] $12^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [36] $1^1 1^1 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [35] $21^{11} 41^{11} 2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 2 & 4 \\ 2 & 7 & 3 & 2 & 3 \\ 2 & 2 & 2 & -2 & 0 \\ 2 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [37] $21^{11} 41^{11} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 9 & 8 & 5 \\ 6 & 8 & 6 & 4 \\ 4 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [34] $1^1 1^1 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 3 & 3 \\ 2 & 2 & -1 & 3 & 2 \end{pmatrix}_{b_0}$
- [38] $12^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$
- [39] $31^1 231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 77$, $\Delta = -123$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 77$, $T \in [12, 77]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [40] $1^1 31^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [41] $22^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^1 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 3 & 3 \\ 2 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 21^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 92, T \in [12, 92]$$

$$\begin{pmatrix} 13 & 14 & 7 \\ 14 & 13 & 7 \\ 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^1 1^1 31^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 11 & 5 \\ 2 & 3 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 131^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 10 & 13 & 6 \\ 5 & 13 & 12 & 6 \\ 3 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[46] \quad 2^1 2^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 4 & 3 & 8 & 4 \\ 3 & 1 & 5 & 3 \\ 8 & 5 & 9 & 5 \\ 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^1 31^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 71, T \in [11, 71]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[48] \quad 21^{11} 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^1 1^1 31^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 4 & 7 & 8 & 4 \\ 7 & 10 & 13 & 6 \\ 8 & 13 & 12 & 6 \\ 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^1 1^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 92, T \in [12, 92]$$

$$\begin{pmatrix} 5 & 10 & 5 \\ 10 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 21^{11} 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[52] \quad 21^{11} 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

- [54] $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$ $1^1 31^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 4 & 8 & 8 & 4 \\ 8 & 13 & 16 & 7 \\ 8 & 16 & 12 & 6 \\ 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [55] $2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [56] $2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [57] $2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -68$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [58] $131^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -75$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 13 & 16 & 7 \\ 5 & 16 & 12 & 6 \\ 3 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [59] $21^1 1^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -75$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 13 & 20 & 7 \\ 20 & 25 & 9 \\ 7 & 9 & 3 \end{pmatrix}_{b_0}$
- [60] $2^1 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 30$, $\Delta = -117$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 30$, $T \in [11, 30]$
 $\begin{pmatrix} 4 & 6 & 6 & 4 \\ 6 & 5 & 8 & 5 \\ 6 & 8 & 4 & 4 \\ 4 & 5 & 4 & 3 \end{pmatrix}_{b_0}$
- [61] $31^1 1^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 77$, $\Delta = -123$, $\det = -95$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 77$, $T \in [12, 77]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [62] $2^1 3^1 1^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 77$, $\Delta = -123$, $\det = -98$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 77$, $T \in [12, 77]$
 $\begin{pmatrix} 4 & 8 & 7 & 4 \\ 8 & 10 & 9 & 6 \\ 7 & 9 & 4 & 4 \\ 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$
- [63] $31^1 1^1 31^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 90$, $\Delta = -130$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$
 $\begin{pmatrix} 9 & 12 & 12 & 5 \\ 12 & 13 & 16 & 7 \\ 12 & 16 & 12 & 6 \\ 5 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [64] $31^1 1^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 77$, $\Delta = -123$, $\det = -152$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 77$, $T \in [12, 77]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [65] $221^1 1^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -173$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 2 & 4 \\ 3 & 7 & 3 & 2 & 3 \\ 1 & 2 & 2 & -2 & 0 \\ 2 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$

$T = 8$ [67] $21^1 4131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 23, \Delta = -113, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 4 \\ 2 & 1 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[66] $131^1 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 3 \\ 1 & 4 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[69] $1^1 3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -4 & -2 \\ 3 & 6 & -2 & 4 \end{pmatrix}_{b_0}$

[68] $1^1 3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$

[70] $1^1 2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 3 & 4 & -1 & 4 \end{pmatrix}_{b_0}$

[74] $1^1 2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0}$

[73] $1^1 2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 3 \\ 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[72] $21^1 41^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$

[71] $1^1 3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & -2 & 3 \end{pmatrix}_{b_0}$

[75] $1^1 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$

[79] $1^1 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 1 & 3 \\ 3 & 6 & 3 & 3 \end{pmatrix}_{b_0}$

[77] $1^1 23131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 0 & 2 \\ 2 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$

- [76] $1^1 23131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$
- [78] $41^1 2231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 9 & -2 & 2 \end{pmatrix}_{b_0}$
- [80] $1^1 231^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 10$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [81] $1^1 1^1 4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [82] $1^1 1^1 4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$
- [83] $1^{11} 1^1 2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 3 & 4 \\ 7 & 10 & 5 & 6 \\ 3 & 5 & 1 & 3 \\ 4 & 6 & 3 & 3 \end{pmatrix}_{b_0}$
- [86] $1^1 4131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [85] $1^1 1^1 4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$
- [84] $1^{11} 1^1 41231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [89] $1^1 1^1 41231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [88] $1^1 1^1 4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [87] $1^{11} 1^1 2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 4 & 2 \\ 2 & 0 & 4 & 4 & 4 \\ 2 & 0 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [90] $1^1 231^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 0 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^1 14131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & -2 & 0 \\ 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{11} 231^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^1 3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 1 & 3 \\ 3 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^1 41231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & -2 & 0 \\ 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{11} 231^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 4 \\ 1 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{11} 2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 21^{11} 3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 3 & 5 & 4 & 4 \\ 2 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 13141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{11} 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^1 4131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{11} 23131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 2 & 0 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^1 231^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

- [99] $1^{1231}1^{31}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$
- [105] $1^{1231}1^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$
- [104] $141^{1231}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 1 & 4 \\ 2 & 1 & -2 & 0 \\ 3 & 4 & 0 & 2 \end{pmatrix}_{b_0}$
- [106] $1^{112321}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [107] $1^{112321}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 3 & 5 & 4 & 4 \\ 2 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [109] $1^{141}1^{231}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [110] $21^{141}1^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 2 \end{pmatrix}_{b_0}$
- [108] $1^{1131}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [111] $1231^{131}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [114] $1^{131}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$
- [112] $1^{131}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$
- [113] $1^{141}1^{231}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 2 & 4 \\ 1 & 2 & -2 & 0 \\ 3 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[115] $131^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 5 & 10 & 5 \\ 10 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[116] $1^{11}41^{1231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[117] $2^3 21^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[118] $31^{11}2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 9 & 12 & 4 & 5 \\ 12 & 10 & 5 & 6 \\ 4 & 5 & 1 & 3 \\ 5 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[119] $31^{11}31^{141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 1 & 6 & 4 & 3 \\ 6 & 13 & 6 & 7 \\ 4 & 6 & 0 & 2 \\ 3 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[120] $21^{11}4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 0 & 4 \\ 2 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[121] $2^3 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[122] $31^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{141^1}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{141^1}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 3 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[125] $2^3 1^{141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 13 & 6 & 7 \\ 3 & 6 & 0 & 2 \\ 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[126] $2^3 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[127] \quad 2^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 4 & 4 & 2 & 4 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[128] \quad 31^1 1^1 2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 31^1 31^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 2 & 4 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[130] \quad 31^1 1^1 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[131] \quad 41^1 1^1 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 10 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [132] \quad 1^1 5^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 21^1 5^1 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^1 4^1 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 5 \\ 4 & 2 & 4 \\ 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^1 4^1 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[136] \quad 2^1 3^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^1 1^1 314131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 151^1 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 10 & 21 & 6 \\ 9 & 21 & 42 & 12 \\ 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^1 5^1 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[143] $21^{1^1}4141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[144] $14141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 3 \end{pmatrix}_{b_0}$$

[139] $15^{1^1}131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[140] $14141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{1^1}4141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{1^1}413141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[148] $14^{1^1}141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[146] $21^{1^1}5132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 10 & 9 & 6 \\ 6 & 9 & 7 & 5 \\ 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[147] $14141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[153] $15^{1^1}131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[152] $21^{1^1}4141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

[154] $15^{1^1}1231^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{\overline{1}} 4^{1\overline{1}} 1321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{\overline{1}} 1^{\overline{1}} 322151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[149] \ 21^{\overline{1}} 5^{1\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[155] \ 21^{\overline{1}} 5^{1\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{\overline{1}} 1^{\overline{1}} 41^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[157] \ 2141^{\overline{1}} 41^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{\overline{1}} 31^{\overline{1}} 4131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{\overline{1}} 31^{\overline{1}} 4131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 3 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[160] \ 21^{\overline{1}} 5^{1\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[161] \ 21^{\overline{1}} 51^{\overline{1}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 4 & 11 & 3 & 4 \\ 11 & 26 & 8 & 10 \\ 3 & 8 & 0 & 2 \\ 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[162] \ 2^{\overline{1}} 3^{1\overline{1}} 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 6 & 6 & 6 \\ 3 & 6 & 1 & 3 \\ 3 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{\overline{1}} 1^{\overline{1}} 4141^{\overline{1}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

- $$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$
- [164] $1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [165] $41^{\overline{1}} 231^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 0 & 3 & 4 & 1 & 2 \\ 3 & 10 & 17 & 0 & 6 \\ 4 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 6 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$
- [166] $1^{\overline{1}} 3221^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 1 & 9 & 3 \\ 1 & -2 & 1 & 0 \\ 9 & 1 & 42 & 12 \\ 3 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$
- [167] $1^{\overline{1}} 41^{\overline{1}} 3141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 3 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$
- [168] $1^{\overline{1}} 41^{\overline{1}} 3141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 7 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [169] $1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 7 & 2 & 3 \\ 7 & 26 & 8 & 10 \\ 2 & 8 & 0 & 2 \\ 3 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$
- [170] $31^{\overline{1}} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 9 & 0 & 3 & 3 \\ 2 & 7 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$
- [171] $2^{\overline{1}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 5 & 1 & 4 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$
- [172] $31^{\overline{1}} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 5 & 8 & 0 & 5 \\ 2 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 5 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$
- [173] $2^{\overline{1}} 3^{\overline{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$
- [174] $31^{\overline{1}} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 13 & 10 & 0 & 7 \\ 2 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$
- [175] $2^{\overline{1}} 3^{\overline{1}} 21^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$\begin{aligned}
& T_{\min} = 12.70 \rightarrow 13, \quad T_{\max} = 22, \quad T \in [13, 22] \\
& \begin{pmatrix} 1 & 8 & 3 & 3 \\ 8 & 26 & 8 & 10 \\ 3 & 8 & 0 & 2 \\ 3 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[176] & 1^1 3221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -109, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.64 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21] \\
& \begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 43 & 1 & 13 \\ 1 & 1 & -3 & -1 \\ 3 & 13 & -1 & 3 \end{pmatrix}_{b_0} \\
[177] & 1^{11} 22 3 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -109, \quad \det = -107, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.61 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21] \\
& \begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 14 & 13 & 8 \\ 5 & 13 & 7 & 5 \\ 3 & 8 & 5 & 3 \end{pmatrix}_{b_0} \\
T = 10 & [178] 1^1 414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -83, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 2 & 4 & 4 \\ 4 & 5 & 5 \\ 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[179] & 1^1 315 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 7, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 3 & 4 & -1 & 4 \end{pmatrix}_{b_0} \\
[180] & 1^1 315 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 3 \\ 3 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[181] & 1^1 414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -83, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[182] & 21^1 5131^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -107, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.56 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22] \\
& \begin{pmatrix} 4 & 5 & 1 & 4 \\ 5 & 5 & 2 & 5 \\ 1 & 2 & -2 & 0 \\ 4 & 5 & 0 & 3 \end{pmatrix}_{b_0} \\
[183] & 1^1 3151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0} \\
[184] & 1^{11} 3151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 0 & 2 \\ 2 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0} \\
[185] & 1^1 414 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.45 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22] \\
& \begin{pmatrix} 4 & 3 & 1 & 4 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 4 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[186] & 1^1 41315 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -107, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.45 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22] \\
& \begin{pmatrix} 5 & 1 & 5 \\ 1 & -2 & 0 \\ 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[187] & 1^{141} 1 4 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 12, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 15, \quad T_{\max} = 22, \quad T \in [15, 22]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{\overline{1}}414131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[189] $1^{\overline{11}}315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{\overline{11}}414141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 4 & 4 & 4 & 4 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{\overline{1}}414141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[192] $2^{\overline{11}}315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 10 & 5 & 6 \\ 3 & 5 & 1 & 3 \\ 4 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{\overline{1}}414141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{\overline{1}}3151^{\overline{1}}231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 7 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{\overline{1}}3151^{\overline{1}}231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & 0 & 2 \\ 2 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[196] $15131^{\overline{1}}41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{\overline{1}}41^{\overline{1}}32151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[198] $1414141^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{\overline{1}}41321^{\overline{1}}51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 2 & 1 & 1 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0} \\
[200] & 31^{11}315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.73 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
T = 11 & \quad [201] \quad 1^{11}3151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 3 & 5 & -1 & 5 \end{pmatrix}_{b_0} \\
[202] & 1^{11}3151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 2 & 4 \\ 3 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
[205] & 1^{11}31^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[204] & 1^{11}6^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[203] & 1^{11}6^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[207] & 1^{11}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 5 & 5 \\ 2 & 4 & -1 & 5 & 4 \end{pmatrix}_{b_0} \\
[206] & 2^13^{11}151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[208] & 31^{11}316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.40 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
[209] & 31^{11}321^16131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.40 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 7 & 4 & 3 \\ 7 & 19 & 8 & 9 \\ 4 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0} \\
[210] & 31^{11}3161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.73 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[211] $1^{31\overline{11}31} 1^{\overline{1}} 6^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 11$, $T_{\max} = 15$, $\Delta = -92$, $\det = -89$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[212] $1^{\overline{1}} 4141^{\overline{1}} 3151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 11$, $T_{\max} = 23$, $\Delta = -106$, $\det = -127$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 14.54 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 8 & 1 & 2 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 2 \end{pmatrix}_{b_0}$

[213] $41^{\overline{11}} 4^{\overline{1}} 14141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 11$, $T_{\max} = 23$, $\Delta = -102$, $\det = -145$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 2 & 5 & 1 & 4 \\ 3 & 5 & 1 & 1 & 3 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 4 & 3 & 0 & 2 \end{pmatrix}_{b_0}$

$T = 12$ [214] $1^{\overline{1}} 31^{\overline{1}} 5^{\overline{1}} 13221^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 5 \\ 3 & 0 & 5 & 6 \end{pmatrix}_{b_0}$

[215] $1^{\overline{1}} 6^{\overline{1}2321} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 14$, $\Delta = -58$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$

[216] $1^{\overline{1}31} 1^{\overline{1}} 6^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 14$, $\Delta = -63$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$

[220] $1^{\overline{1}} 31^{\overline{1}} 5^{\overline{1}} 13221^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 6 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 6 & 6 \\ 2 & 5 & -1 & 6 & 5 \end{pmatrix}_{b_0}$

[219] $1^{\overline{1}1321} 1^{\overline{1}} 7^{\overline{1}} 1231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 15$, $\Delta = -84$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[218] $1^{\overline{1}131} 1^{\overline{1}} 6^{\overline{1}} 13141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 15$, $\Delta = -83$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$

[217] $1^{\overline{1}131} 1^{\overline{1}} 6^{\overline{1}} 13141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 15$, $\Delta = -83$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$

[221] $31^{\overline{11}} 31613141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 16$, $\Delta = -91$, $\det = 89$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 15.73 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

$$\begin{aligned}
T = 13 \quad [222] \quad 2 \overset{1^1}{3} 2 \overset{1^1}{1} 8 1231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[223] \quad 1 \overset{1^1}{8} \quad 1231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & -3 \end{pmatrix}_{b_0} \\
[224] \quad 1 \overset{1^1}{8} \quad 1231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[225] \quad 1 \overset{1^1}{8} \quad 1231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[226] \quad 31 \overset{1^1}{3} 21 \overset{1^1}{8} 1231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 17, \Delta = -91, \det = -20, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 16.40 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
[227] \quad 31 \overset{1^1}{3} 22 \overset{1^1}{1} 8 1231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 17, \Delta = -91, \det = -29, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 16.59 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
\begin{pmatrix} 1 & 9 & 4 & 3 \\ 9 & 34 & 10 & 12 \\ 4 & 10 & 1 & 3 \\ 3 & 12 & 3 & 3 \end{pmatrix}_{b_0} \\
[228] \quad 31 \overset{1^1}{6} \quad 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 15.91 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
T = 14 \quad [229] \quad 1 \overset{1^1}{8} \quad 123141^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[230] \quad 1 \overset{1^1}{8} \quad 123141^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0} \\
[231] \quad 2 \overset{1^1}{3} 2 \overset{1^1}{1} 8 12321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0} \\
T = 15 \quad [232] \quad 1 \overset{1^1}{6} \quad 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 15, T_{\max} = 16, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 16, T_{\max} = 16, T \in [16, 16]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[233] $1^{\overline{1}1512321} 8^{1231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

$T = 18$ [234] $1^{\overline{1}3221} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 18, T_{\max} = 20, \Delta = -62, \det = 5, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 19.60 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$

$T = 19$ [235] $3221^{\overline{1}513221} (12) 12231^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 19, T_{\max} = 22, \Delta = -85, \det = -44, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 21.91 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$

4.596 NHC(-2)(-3) + so₈ + e₆

$T = 7$ [0] $1^{\overline{1}1} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 101, \Delta = -192, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -6 & -4 \\ 5 & 9 & -4 & 0 \end{pmatrix}_{b_0}$

[1] $1^{\overline{1}1} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 101, \Delta = -192, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -6 & -4 \\ 5 & 8 & -4 & 0 \end{pmatrix}_{b_0}$

$T = 8$ [2] $1^{\overline{1}1} 31^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 16.80 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & 2 & -4 & -4 \end{pmatrix}_{b_0}$

[3] $1^{\overline{1}1} 321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & 0 \end{pmatrix}_{b_0}$

[4] $1^{\overline{1}1} 321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & 0 \end{pmatrix}_{b_0}$

[5] $1^{\overline{1}1} 321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$

[6] $1^{\overline{1}1} 31^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^1 31^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 350$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.79 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [8] $1^{11} 231^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 4 & 3 & 0 & -2 & 0 \\ 2 & 4 & 3 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[9] $1^1 3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -4 & -2 \\ 5 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11} 2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -137$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11} 2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -137$, $\det = -37$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 102$, $\Delta = -162$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.57 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -5 & -3 \\ 5 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.87 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 3 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[14] $1^1 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 102$, $\Delta = -162$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.42 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -5 & -3 \\ 5 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [15] $1^1 4^1 1321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 18$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -4 & -2 \\ 5 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [16] $1^1 31^1 51321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 12$, $T_{\max} = 24$, $\Delta = -108$, $\det = 3$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 22$, $T_{\max} = 24$, $T \in [22, 24]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & -3 \end{pmatrix}_{b_0}$$

4.597 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \ T_{\max} = 99, \ \Delta = -142, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.76 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.598 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 9 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 140, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 141, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 59 & 124 & 0 & 613 & 45 \\ 5 & 9 & -4 & 45 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 213, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 220, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.53 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 241, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 36 & 0 & 49 & 13 \\ 5 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 318, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 18 & 40 & 0 & 58 & 14 \\ 5 & 9 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 379, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 34 & 72 & 0 & 202 & 26 \\ 5 & 9 & -4 & 26 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{113}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 27 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 27 & 60 & 0 & 133 & 21 \\ 5 & 9 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 26 & 56 & 0 & 118 & 20 \\ 5 & 9 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 36 & 80 & 0 & 238 & 28 \\ 5 & 9 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 51 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 51 & 108 & 0 & 457 & 39 \\ 5 & 9 & -4 & 39 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 921, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 35 & 76 & 0 & 217 & 27 \\ 5 & 9 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1005, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 43 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 43 & 92 & 0 & 325 & 33 \\ 5 & 9 & -4 & 33 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{115}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1053, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.1$
 $T_{\min} = 13.52 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 100 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 45 & 100 & 0 & 373 & 35 \\ 5 & 9 & -4 & 35 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 68 & 144 & 0 & 814 & 52 \\ 5 & 9 & -4 & 52 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1366$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 44 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 44 & 96 & 0 & 346 & 34 \\ 5 & 9 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1590$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 60 & 128 & 0 & 634 & 46 \\ 5 & 9 & -4 & 46 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1650$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 52 & 112 & 0 & 478 & 40 \\ 5 & 9 & -4 & 40 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [24] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 33$, $T_{\max} = 99$, $T \in [33, 99]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 3 \\ 5 & 13 & 16 & 0 & 12 & 7 \\ 5 & 16 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 7 & 6 & -4 & 6 & -19 \end{pmatrix}_{b_0}$$

[25] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -9$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.11 \rightarrow 17$, $T_{\max} = 94$, $T \in [17, 94]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 7 & 0 \\ 6 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -2 & 0 & 8 & -3 \end{pmatrix}_{b_0}$$

[26] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.15 \rightarrow 16$, $T_{\max} = 94$, $T \in [16, 94]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

[27] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -81$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 7 & 0 \\ 5 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -2 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -165, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -571, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [34] $1^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 17.67 \rightarrow 18, T_{\max} = 22, T \in [18, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[35] $1^{11}3221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 15 & 1 & 13 & 5 \\ 15 & 42 & 0 & 36 & 12 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 36 & 0 & 28 & 10 \\ 5 & 12 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11}231^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{11}3221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 15 & 1 & 21 & 5 \\ 15 & 42 & 0 & 59 & 12 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 59 & 0 & 79 & 17 \\ 5 & 12 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11}41^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{\overline{1}1\overline{3}1\overline{1}1} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{\overline{1}1\overline{3}1\overline{1}1} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

4.599 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 9 \quad [0] 1^{\overline{1}1\overline{2}3\overline{1}1\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 26 & 5 \\ 12 & 25 & 0 & 16 & 56 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 36 & 6 \\ 26 & 56 & 0 & 36 & 118 & 20 \\ 5 & 9 & -4 & 6 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{1}1\overline{1}1\overline{3}1\overline{1}1} 3^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{1}1\overline{1}1\overline{3}1\overline{1}1} 3^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{1}1\overline{1}1\overline{3}1\overline{1}1} 3^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 17 & 5 \\ 12 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 20 & 0 & 13 & 27 & 7 \\ 17 & 36 & 0 & 27 & 49 & 13 \\ 5 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{1}1\overline{1}1\overline{3}1\overline{1}1} 3^{\overline{1}1\overline{3}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -4 & 7 & 27 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{1}1\overline{1}1\overline{3}1\overline{1}1} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 9 & 5 \\ 12 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 20 & 0 & 13 & 15 & 7 \\ 9 & 20 & 0 & 15 & 13 & 7 \\ 5 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{1}1\overline{1}1\overline{3}1\overline{1}1} 3^{\overline{1}1\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 17 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 36 & 0 & 49 & 51 & 13 \\ 17 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{112}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{\overline{111}} 3^{\overline{111}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{\overline{11}} 3^{\overline{111}} 4^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 7 & 5 \\ 10 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 7 & 10 & 0 & 9 & 5 & 5 \\ 5 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

4.600 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.601 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 8 & 5 \\ 12 & 25 & 0 & 20 & 16 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 20 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 9 & 5 \\ 5 & 9 & -4 & 7 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

4.602 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 99$, $\Delta = -137$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.69 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 3 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 99$, $\Delta = -137$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [3] $1^{11}4^{11}13^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -24$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

4.603 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$

$T = 8$ [1] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 95$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 9 & 5 \\ 5 & 9 & -4 & 5 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 302$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.86 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 9 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 330$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.68 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 36 & 0 & 48 & 12 \\ 5 & 9 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -196$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -354$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.07 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 3 \\ 5 & 13 & 16 & 0 & 12 & 7 \\ 5 & 16 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 9 & 5 \\ 3 & 7 & 6 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

4.604 NHC(-2)(-3) + so₈ + e₆ + su₃ + su₂

$T = 9$ [0] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 9 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 9 & 20 & 0 & 12 & 13 & 7 \\ 5 & 9 & -4 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

4.605 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_7$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 \quad T = 8, T_{\max} = 103, \Delta = -219, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & -4 & -2 \\ 3 & 5 & -2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 \quad T = 8, T_{\max} = 103, \Delta = -219, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.39 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -5 & -3 \\ 2 & 2 & -1 & 0 & -3 & -3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{\overline{11}} 3 221^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 \quad T = 9, T_{\max} = 103, \Delta = -190, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 6 & 0 & -3 & -1 \\ 3 & 6 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 231^{\overline{11}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 \quad T = 9, T_{\max} = 103, \Delta = -198, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.64 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 3 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & -2 & -2 \end{pmatrix}_{b_0}$$

4.606 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 103, \Delta = -189, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 103, \Delta = -189, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 103, \Delta = -189, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.34 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -4 & 3 & -2 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 103, \Delta = -189, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -4 & 3 & -2 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{11} 231^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -168, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$4.607 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$$

$$T = 10 \quad [0] \quad 1^{11} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$4.608 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 103, \Delta = -192, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$4.609 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_3$$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 103, \Delta = -198, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$4.610 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7$$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -4 & -2 \\ 2 & 2 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{11} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 6 & 0 & -2 & 0 \\ 3 & 6 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}231^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 3 & 0 & -3 & -1 \\ 2 & 0 & 3 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

4.611 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$

$T = 9$ [0] $1^{11}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -139$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14.51 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & -1 & 0 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -213$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -3 & 3 & -1 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -3 & 3 & -1 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^{11}231^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 94$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

4.612 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10$ [0] $1^{11}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 132$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

4.613 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 102$, $\Delta = -164$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.614 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 102, \ \Delta = -170, \ \det = -245, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

4.615 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4$

$$T = 7 \quad [0] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 7, \ T_{\max} = 100, \ \Delta = -166, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -5 & -3 \\ 5 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 7, \ T_{\max} = 100, \ \Delta = -166, \ \det = -85, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -5 & -3 \\ 5 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \ 1^{11} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 0 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.09 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.09 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -4 & -2 \\ 5 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 51, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -145$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 99$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -145$, $\det = 275$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [13] $1^{11}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 6 & 0 & 0 & 2 \\ 3 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 15.15 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 6 & 0 & 3 \\ 5 & 6 & 17 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 7 & -2 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 5 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & -2 & 0 \\ 2 & 3 & 2 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{11}4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 9$, $T_{\max} = 99$, $\Delta = -136$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11}231^131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 1 & 5 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^1 1^{131^1} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 99$, $\Delta = -136$, $\det = -127$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -4 & -2 \\ 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$

$T = 10$ [20] $1^1 4^1 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 22$, $\Delta = -108$, $\det = 12$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -3 & -1 \\ 5 & 4 & -1 & 3 \end{pmatrix}_{b_0}$

4.616 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$T = 9$ [0] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -395$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$

4.617 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 8$ [0] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 14$, $T_{\max} = 98$, $T \in [14, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$

[1] $1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 94$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -3 & 34 & 1 \end{pmatrix}_{b_0}$

[2] $1^{113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 109$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 59 & 124 & 0 & 613 & 45 \\ 5 & 9 & -3 & 45 & 1 \end{pmatrix}_{b_0}$

[3] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 111$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$

[4] $1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 165$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$

[5] $1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$

- [6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 36 & 0 & 49 & 13 \\ 5 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$
- [7] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 40 & 0 & 58 & 14 \\ 5 & 9 & -3 & 14 & 1 \end{pmatrix}_{b_0}$
- [8] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 291$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$
- [9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 331$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 34 & 72 & 0 & 202 & 26 \\ 5 & 9 & -3 & 26 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 365$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 27 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 27 & 60 & 0 & 133 & 21 \\ 5 & 9 & -3 & 21 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 370$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 26 & 56 & 0 & 118 & 20 \\ 5 & 9 & -3 & 20 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 540$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 36 & 80 & 0 & 238 & 28 \\ 5 & 9 & -3 & 28 & 1 \end{pmatrix}_{b_0}$
- [15] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 581$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 51 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 51 & 108 & 0 & 457 & 39 \\ 5 & 9 & -3 & 39 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 709$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 35 & 76 & 0 & 217 & 27 \\ 5 & 9 & -3 & 27 & 1 \end{pmatrix}_{b_0}$

- [17] $1^{115}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 98, \Delta = -136, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 100 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 45 & 100 & 0 & 373 & 35 \\ 5 & 9 & -3 & 35 & 1 \end{pmatrix}_{b_0}$
- [18] $1^{113}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 98, \Delta = -136, \det = 781, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 43 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 43 & 92 & 0 & 325 & 33 \\ 5 & 9 & -3 & 33 & 1 \end{pmatrix}_{b_0}$
- [19] $1^{114}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 98, \Delta = -136, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 68 & 144 & 0 & 814 & 52 \\ 5 & 9 & -3 & 52 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 98, \Delta = -136, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 44 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 44 & 96 & 0 & 346 & 34 \\ 5 & 9 & -3 & 34 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{114}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 98, \Delta = -136, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 60 & 128 & 0 & 634 & 46 \\ 5 & 9 & -3 & 46 & 1 \end{pmatrix}_{b_0}$
- [22] $1^{114}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 98, \Delta = -136, \det = 1276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 1 & 52 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 52 & 112 & 0 & 478 & 40 \\ 5 & 9 & -3 & 40 & 1 \end{pmatrix}_{b_0}$
- $T = 9$ [23] $1^{11}321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 \\ 1 & 4 & 0 & 1 & 0 & 6 \\ 1 & 1 & 1 & -2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 2 & 6 & 6 & 2 & 0 & 10 \\ 2 & 3 & 2 & 0 & -3 & 6 \end{pmatrix}_{b_0}$
- [24] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$
 $\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -2 & 7 & -3 \end{pmatrix}_{b_0}$
- [25] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$
- [26] $1^{111}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 2 & -3 & 8 & -2 \end{pmatrix}_{b_0}$
- [27] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 5 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{\overline{11}} 3^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{\overline{12}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 5 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{\overline{14}} 3^{\overline{11}} 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 20 & 0 & 158 & 24 \\ 3 & 5 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 4 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 7 & 4 & 0 & 16 & 8 \\ 2 & 3 & 2 & 0 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{\overline{11}} 3^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[38] 1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[39] 1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[40] 1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[41] 1^{11} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 0 & 10 & 3 \\ 1 & 4 & 0 & 1 & 0 & 10 & 2 \\ 1 & 1 & 1 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 10 & 5 & 0 & 33 & 11 \\ 2 & 3 & 2 & 0 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[42] 1^{11} 3 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 4 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 13 & 13 & 6 & 0 & 56 & 14 \\ 2 & 3 & 2 & 0 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[44] 1^{11} 3 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 19 & 0 & 92 & 20 \\ 3 & 5 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[43] 1^{11} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 12 \\ 3 & 5 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[45] 1^{11} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 58 & 16 \\ 3 & 5 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[46] 1^{13} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -390, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 25 & 17 & 0 & 104 & 20 \\ 3 & 5 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] 1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{12} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 14 & 0 & 62 & 16 \\ 3 & 5 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{12} 3^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 25 & 18 & 0 & 94 & 20 \\ 3 & 5 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{14} 3^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -860, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 35 & 24 & 0 & 202 & 28 \\ 3 & 5 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{13} 3^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -910, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 30 & 21 & 0 & 142 & 24 \\ 3 & 5 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{12} 3^{21} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -940, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 30 & 22 & 0 & 134 & 24 \\ 3 & 5 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{13} 3^{21} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 0 & 0 & 25 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 35 & 25 & 0 & 188 & 28 \\ 3 & 5 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{14} 3^{21} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 5 & 5 & 0 & 40 & 5 \\ 2 & 5 & 0 & 0 & 28 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 40 & 28 & 0 & 254 & 32 \\ 3 & 5 & 2 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \ 1^{11} 3^{22} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 6 & 0 & 6 & 6 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 5 & 2 \\ 3 & 6 & 0 & 5 & 3 & 5 \\ 3 & 6 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11} 3^{22} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 6 & 0 & 6 & 11 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 8 & 2 \\ 5 & 11 & 0 & 8 & 15 & 9 \\ 3 & 6 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{11} 1^{11} 31^{11} 4 \quad 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{12} 3^{11} 221^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 6 & 0 & 6 & 22 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 16 & 2 \\ 10 & 22 & 0 & 16 & 66 & 18 \\ 3 & 6 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{11} 1^{11} 3^{11} 221^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 6 & 0 & 6 & 17 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 13 & 2 \\ 8 & 17 & 0 & 13 & 36 & 14 \\ 3 & 6 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{13} 1^{11} 3^{11} 221^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 6 & 0 & 6 & 33 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 24 & 2 \\ 15 & 33 & 0 & 24 & 151 & 27 \\ 3 & 6 & -2 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{12} 1^{11} 3^{11} 221^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 230, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 6 & 0 & 6 & 28 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 21 & 2 \\ 13 & 28 & 0 & 21 & 103 & 23 \\ 3 & 6 & -2 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{11} 1^{11} 31^{11} 4 \quad 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 317, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 1^{11} 31^{11} 4 \quad 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

4.618 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{112} 31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 26 & 5 \\ 12 & 25 & 0 & 16 & 56 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 36 & 6 \\ 26 & 56 & 0 & 36 & 118 & 20 \\ 5 & 9 & -3 & 6 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 17 & 5 \\ 12 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 13 & 27 & 7 \\ 17 & 36 & 0 & 27 & 49 & 13 \\ 5 & 9 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -3 & 7 & 27 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 9 & 5 \\ 12 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 13 & 15 & 7 \\ 9 & 20 & 0 & 15 & 13 & 7 \\ 5 & 9 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 17 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 36 & 0 & 49 & 51 & 13 \\ 17 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -3 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -482$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

4.619 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -222, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 6 & 5 \\ 12 & 25 & 0 & 20 & 12 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 13 & 9 & 7 \\ 6 & 12 & 0 & 9 & 4 & 4 \\ 5 & 9 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 6 & 5 \\ 12 & 25 & 0 & 36 & 12 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 36 & 0 & 49 & 17 & 13 \\ 6 & 12 & 0 & 17 & 4 & 4 \\ 5 & 9 & -3 & 13 & 4 & 1 \end{pmatrix}_{b_0}$$

4.620 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \ 1^{111}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

4.621 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$

$$T = 9 \quad [0] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -83, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 8 & 5 \\ 12 & 25 & 0 & 20 & 16 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 9 & 5 \\ 5 & 9 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

4.622 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 174, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 22, \ \Delta = -111, \ \det = -59, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 22, \ \Delta = -111, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 10 & 13 & 0 & 4 & 6 \\ 5 & 13 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 6 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^1 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -187, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \ 1^{11} 4^{11} 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 2, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 17, \ T_{\max} = 22, \ T \in [17, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 4^{11} 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 202, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.63 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [8] \ 1^{11} 4^{11} 13^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 14, \ \Delta = -81, \ \det = -14, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

4.623 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 9 & 5 \\ 12 & 25 & 0 & 12 & 20 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 4 & 9 & 4 \\ 9 & 20 & 0 & 9 & 13 & 7 \\ 5 & 9 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 17 & 5 \\ 12 & 25 & 0 & 12 & 36 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 4 & 17 & 4 \\ 17 & 36 & 0 & 17 & 49 & 13 \\ 5 & 9 & -3 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

4.624 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.625 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \text{n3mix}$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 23 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 23 & 48 & 0 & 89 & 17 \\ 5 & 9 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 15 & 5 \\ 12 & 25 & 0 & 32 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 32 & 0 & 37 & 11 \\ 5 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{111}31^{131}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{11}4^{131}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 233, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

4.626 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11}3^{1111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

4.627 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{11}3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 34 & 0 & 57 & 13 \\ 5 & 8 & -3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 9 & 5 \\ 5 & 9 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 36 & 0 & 48 & 12 \\ 5 & 9 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11}3^{121}1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 2 & 0 & 9 & 5 \\ 2 & 3 & 2 & 0 & -3 & 5 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 2 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 10 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -53$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 5 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 13 & 7 \\ 3 & 5 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 82$, $\Delta = -124$, $\det = -277$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 82$, $T \in [14, 82]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 3 \\ 5 & 13 & 16 & 0 & 12 & 7 \\ 5 & 16 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 9 & 5 \\ 3 & 7 & 6 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

4.628 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 9 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 9 & 20 & 0 & 12 & 13 & 7 \\ 5 & 9 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

4.629 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.630 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$

$$T = 8 \quad [0] \quad 21^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 15 & 7 \\ 17 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 13 & 6 \\ 15 & 17 & 13 & 12 & 6 \\ 7 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 3 & 5 & 0 & 1 & 3 & 3 \\ 3 & 4 & -2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 4 & 0 & 1 & 0 & 2 \\ 3 & 4 & -2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 1 & 0 \\ 12 & 17 & 1 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 13 & 6 \\ 12 & 17 & 13 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 13 & 6 \\ 9 & 17 & 13 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[10] $2 1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 13 & 6 \\ 12 & 17 & 13 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [12] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 10 & 7 \\ 8 & 10 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[11] $2 1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 10 & 7 \\ 8 & 10 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 21^{11} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 15 & 9 & 7 \\ 14 & 13 & 14 & 10 & 7 \\ 15 & 14 & 13 & 10 & 7 \\ 9 & 10 & 10 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

4.631 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{e}_6$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.632 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.633 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 9 & 17 & 0 & 1 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

4.634 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 7 \quad [0] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 97, \Delta = -142, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 97, \Delta = -142, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -4 & -2 \\ 5 & 8 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \ 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 10 & 11 & 16 & 6 \\ 11 & 10 & 17 & 6 \\ 16 & 17 & 25 & 9 \\ 6 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{\overline{11}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 3 & 3 \\ 6 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.41 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 5 & 8 & 12 & 5 \\ 8 & 10 & 17 & 6 \\ 12 & 17 & 25 & 9 \\ 5 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 35, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 13 & 12 & 19 & 7 \\ 12 & 10 & 17 & 6 \\ 19 & 17 & 25 & 9 \\ 7 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 35, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 43, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 5 & 8 & 1 & 5 \\ 8 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 5 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 67, \ \Delta = -121, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 67, \ T \in [13, 67]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 11 & 5 \\ 3 & 6 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 3 & 3 \\ 5 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -4 & -2 \\ 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[16] $2^2 2^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 13 & 12 & 20 & 7 \\ 12 & 10 & 17 & 6 \\ 20 & 17 & 25 & 9 \\ 7 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[19] $21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 3 & 3 \\ 7 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[20] $1^1 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [21] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 6 & 0 & 1 & 3 \\ 3 & 6 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{11}4131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11}4131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[24] $1^1 4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[25] $1^1 231^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 \\ 1 & 0 & 3 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 4 & 4 \\ 2 & 2 & 3 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11} 2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[28] $1^1 41^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 1 & 2 & 4 \\ 2 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[29] $1^1 4^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[30] $1^1 4^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [31] $1^1 4^{1321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -2 & 0 \\ 5 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{11} 4^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^1 4^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

[34] $21^{11}4141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 30$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11}4141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 32$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[35] $14141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 32$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 14 & 17 & 8 \\ 9 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [37] $1^{11}414131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -6$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 2 & 4 & 2 & 4 \\ 4 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 4 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{11}414141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 11$, $T_{\max} = 23$, $\Delta = -102$, $\det = -165$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 1 & 4 & 1 & 0 & 3 \\ 2 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

4.635 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$

$T = 9$ [0] $1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 9 & 17 & 0 & 1 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [2] $1^{11}41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 6 & 5 \\ 9 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 13 & 10 & 7 \\ 6 & 10 & 0 & 10 & 4 & 4 \\ 5 & 7 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

4.636 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 24$, $\Delta = -113$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_8 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -135, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \ 1^{11} 4 1^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_8 \quad T = 11, \ T_{\max} = 23, \ \Delta = -102, \ \det = -21, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 14 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

4.637 NHC(-2)(-3) + so₈ + so₈ + su₂

$$T = 8 \quad [0] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 2, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, \ T_{\max} = 22, \ T \in [15, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 9 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.71 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 54 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 54 & 108 & 0 & 457 & 39 \\ 5 & 9 & -2 & 39 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 13 & 7 \\ 5 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 36 & 72 & 0 & 202 & 26 \\ 5 & 9 & -2 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 20 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 33 & 0 & 49 & 13 \\ 5 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 59 & 124 & 0 & 613 & 45 \\ 5 & 9 & -2 & 45 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 39 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 39 & 76 & 0 & 217 & 27 \\ 5 & 9 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 36 & 0 & 49 & 13 \\ 5 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 39 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 39 & 67 & 0 & 217 & 27 \\ 5 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 36 & 0 & 49 & 13 \\ 5 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 40 & 0 & 58 & 14 \\ 5 & 9 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 48 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 48 & 84 & 0 & 346 & 34 \\ 5 & 8 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 4 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 12 & 5 \\ 4 & 12 & 0 & 12 & 10 & 6 \\ 3 & 6 & -1 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{11} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 12 & 19 & 1 & 7 \\ 12 & 10 & 17 & 0 & 6 \\ 19 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 7 & 6 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{11} 321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 4 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 7 & 4 & 0 & 16 & 8 \\ 2 & 3 & 2 & 0 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 20 & 0 & 26 & 10 \\ 5 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{12} 321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 10 & 2 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 5 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{14} 321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 20 & 2 \\ 16 & 30 & 0 & 20 & 158 & 24 \\ 3 & 5 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -93, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 10 & 0 & 5 & 5 \\ 5 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0} \\
[74] \quad & 1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[73] \quad & 1^{11} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 3 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[75] \quad & 1^{11} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 6 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[76] \quad & 1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 5 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 12 & 5 \\ 5 & 12 & 0 & 12 & 10 & 6 \\ 3 & 6 & -1 & 5 & 6 & 1 \end{pmatrix}_{b_0} \\
[77] \quad & 1^{11} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[78] \quad & 1^{11} 3 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 4 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 13 & 6 & 0 & 56 & 14 \\ 2 & 3 & 2 & 0 & -2 & 14 & 1 \end{pmatrix}_{b_0} \\
[79] \quad & 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 24 & 0 & 38 & 12 \\ 5 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0} \\
[80] \quad & 1^{11} 3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 11 & 2 \\ 7 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[81] \quad & 1^{13} 3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 17 & 2 \\ 14 & 25 & 0 & 17 & 104 & 20 \\ 3 & 5 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [82] $1^{13}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 17 & 2 \\ 13 & 25 & 0 & 17 & 104 & 20 \\ 3 & 5 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [83] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 10 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [84] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 34 & 0 & 77 & 17 \\ 5 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [85] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [86] $1^{12}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 14 & 2 \\ 10 & 20 & 0 & 14 & 62 & 16 \\ 3 & 5 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [87] $1^{11}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 11 & 2 \\ 8 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [88] $1^{13}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 27 & 44 & 0 & 130 & 22 \\ 5 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [89] $1^{12}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 14 & 2 \\ 11 & 20 & 0 & 14 & 62 & 16 \\ 3 & 5 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [90] $1^{14}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 1 & 33 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 33 & 54 & 0 & 197 & 27 \\ 5 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$
- [91] $1^{12}3^{11}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 18 & 2 \\ 12 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{11} 1^{13} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -495, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 15 & 2 \\ 10 & 20 & 0 & 15 & 58 & 16 \\ 3 & 5 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{14} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -594, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 24 & 2 \\ 18 & 35 & 0 & 24 & 202 & 28 \\ 3 & 5 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{13} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -619, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 21 & 2 \\ 15 & 30 & 0 & 21 & 142 & 24 \\ 3 & 5 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{14} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -654, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 24 & 2 \\ 19 & 35 & 0 & 24 & 202 & 28 \\ 3 & 5 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{12} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -706, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 18 & 2 \\ 13 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{13} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 21 & 2 \\ 16 & 30 & 0 & 21 & 142 & 24 \\ 3 & 5 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{13} 3 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -895, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 25 & 2 \\ 17 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{12} 3 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -966, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 22 & 2 \\ 15 & 30 & 0 & 22 & 134 & 24 \\ 3 & 5 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{14} 3 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1046, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 28 & 2 \\ 20 & 40 & 0 & 28 & 254 & 32 \\ 3 & 5 & -2 & 2 & 32 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{13} 1^{11} 3 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1195$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 25 & 2 \\ 18 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{14} 1^{11} 3 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1266$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 28 & 2 \\ 21 & 40 & 0 & 28 & 254 & 32 \\ 3 & 5 & -2 & 2 & 32 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [103] $1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 31$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{11} 1^{11} 31^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 61$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 14 & 1 & 14 & 8 \\ 5 & 7 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{11} 1^{12} 31^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 66$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 6 & 6 & 1 & 1 & 8 & 6 \\ 3 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{11} 1^{11} 31^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 69$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{11} 1^{12} 31^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 88$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 18 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 2 & -1 \\ 18 & 24 & 2 & 42 & 14 \\ 5 & 7 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{11} 1^{11} 31^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 109$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 8 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 8 & 10 & 1 & 6 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{11} 1^{11} 31^3 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 115$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 8 & 4 & 3 & 2 & 18 & 8 \\ 3 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[110] 1^{\overline{11}} 4 131^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 1 & 2 & 2 & 4 \\ 3 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[111] 1^{\overline{111}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 4 & 2 & 1 & 2 & 2 & 4 \\ 3 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[112] 1^{\overline{111}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 6 & 3 & 2 & 2 & 8 & 6 \\ 3 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[113] 1^{\overline{111}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 9 & 6 & 4 & 2 & 26 & 10 \\ 3 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[114] 1^{\overline{114}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 210, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 12 & 12 & 2 & 2 & 38 & 12 \\ 3 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[115] 1^{\overline{131}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[116] 1^{\overline{131}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[117] 1^{\overline{111}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 2 & 2 & 6 & 6 \\ 3 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{11} 1^{111} 31^2 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 259, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 7 & 5 & 3 & 2 & 14 & 8 \\ 3 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{11} 1^{11} 31^{11} 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{11} 1^{11} 31^{11} 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[121] $1^{11} 1^{11} 31^{11} 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 17 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 17 & 24 & 1 & 39 & 13 \\ 5 & 7 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

4.638 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 9$ [0] $1^{11} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 39 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 39 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 45 & 5 \\ 11 & 20 & 0 & 0 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 45 & 84 & 0 & 0 & 346 & 34 \\ 5 & 8 & -2 & 0 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 10 & 5 \\ 12 & 25 & 0 & 36 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 36 & 0 & 49 & 27 & 13 \\ 10 & 20 & 0 & 27 & 13 & 7 \\ 5 & 9 & -2 & 13 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 18 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 36 & 0 & 49 & 51 & 13 \\ 18 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 17 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 36 & 0 & 49 & 51 & 13 \\ 17 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 17 & 5 \\ 12 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 13 & 27 & 7 \\ 17 & 36 & 0 & 27 & 49 & 13 \\ 5 & 9 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3 \ 1^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3 \ 1^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 48 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 48 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 27 & 7 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 36 & 5 \\ 11 & 20 & 0 & 0 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 36 & 67 & 0 & 0 & 217 & 27 \\ 5 & 8 & -2 & 0 & 27 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 18 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -262, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 29 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 29 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 9 & 5 \\ 12 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 13 & 15 & 7 \\ 9 & 20 & 0 & 15 & 13 & 7 \\ 5 & 9 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 45 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 72 & 7 \\ 45 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 27 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 27 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 46 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 46 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{111} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 36 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 36 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 27 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 27 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 37 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 37 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

4.639 NHC(-2)(-3) + so₈ + so₈ + su₂ + su₂ → (-3)

$$T = 9 \quad [0] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 6 & 5 \\ 12 & 25 & 0 & 20 & 12 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 13 & 9 & 7 \\ 6 & 12 & 0 & 9 & 4 & 4 \\ 5 & 9 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 6 & 5 \\ 12 & 25 & 0 & 36 & 12 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 36 & 0 & 49 & 17 & 13 \\ 6 & 12 & 0 & 17 & 4 & 4 \\ 5 & 9 & -2 & 13 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 6 & 5 \\ 12 & 25 & 0 & 36 & 12 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 36 & 0 & 49 & 17 & 13 \\ 6 & 12 & 0 & 17 & 4 & 4 \\ 5 & 9 & -2 & 13 & 4 & 2 \end{pmatrix}_{b_0}$$

4.640 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix}$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & -2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 16 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 16 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

4.641 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 8 & 5 \\ 12 & 25 & 0 & 20 & 16 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 9 & 5 \\ 5 & 9 & -2 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 1 & 5 \\ 12 & 25 & 0 & 76 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 35 & 76 & 0 & 217 & 0 & 27 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 27 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 1 & 5 \\ 12 & 25 & 0 & 36 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 36 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 10 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 1 & 5 \\ 12 & 25 & 0 & 56 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 26 & 56 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 1 & 5 \\ 12 & 25 & 0 & 72 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 34 & 72 & 0 & 202 & 0 & 26 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 26 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 10 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 10 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 26 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 51 & 1 & 5 \\ 12 & 25 & 0 & 108 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 51 & 108 & 0 & 457 & 0 & 39 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 39 & -1 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 43 & 1 & 5 \\ 12 & 25 & 0 & 92 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 43 & 92 & 0 & 325 & 0 & 33 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 33 & -1 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 1 & 5 \\ 12 & 25 & 0 & 112 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 52 & 112 & 0 & 478 & 0 & 40 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 40 & -1 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 1 & 5 \\ 12 & 25 & 0 & 144 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 68 & 144 & 0 & 814 & 0 & 52 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 52 & -1 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 1 & 5 \\ 12 & 25 & 0 & 128 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 60 & 128 & 0 & 634 & 0 & 46 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 46 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 6 & 5 \\ 9 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 6 & 10 & 0 & 9 & 4 & 4 \\ 5 & 7 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 6 & 5 \\ 9 & 13 & 0 & 10 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 5 & 7 & 5 \\ 6 & 10 & 0 & 7 & 4 & 4 \\ 5 & 7 & -2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 6 & 5 \\ 9 & 13 & 0 & 24 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 24 & 0 & 38 & 16 & 12 \\ 6 & 10 & 0 & 16 & 4 & 4 \\ 5 & 7 & -2 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

4.642 NHC(-2)(-3) + so₈ + so₈ + su₂ → (-3)

$$T = 8 \quad [0] \quad 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 3 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 31^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 10 & 13 & 0 & 4 & 6 \\ 5 & 13 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 6 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{\overline{1}} 413^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 1 & 3 \\ 3 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1}} 413^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 4 \quad 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 4 \quad 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

4.643 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 1 & 5 \\ 11 & 20 & 0 & 11 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -153$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 18 & 5 \\ 12 & 25 & 0 & 12 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 4 & 17 & 4 \\ 18 & 36 & 0 & 17 & 49 & 13 \\ 5 & 9 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -153$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 9 & 5 \\ 12 & 25 & 0 & 12 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 4 & 9 & 4 \\ 9 & 20 & 0 & 9 & 13 & 7 \\ 5 & 9 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -163$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 17 & 5 \\ 12 & 25 & 0 & 12 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 4 & 17 & 4 \\ 17 & 36 & 0 & 17 & 49 & 13 \\ 5 & 9 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

4.644 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 6 & 5 \\ 12 & 25 & 0 & 12 & 12 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 4 & 6 & 4 \\ 6 & 12 & 0 & 6 & 4 & 4 \\ 5 & 9 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

4.645 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$T = 9$ [0] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -294$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.646 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix}$

$T = 8$ [0] $1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 11 & 0 & 4 & 4 \\ 5 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 24 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 24 & 48 & 0 & 89 & 17 \\ 5 & 9 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 25 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 25 & 45 & 0 & 100 & 18 \\ 5 & 8 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.82 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 28 & 0 & 37 & 11 \\ 5 & 8 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 161, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 23 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 48 & 0 & 89 & 17 \\ 5 & 9 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 15 & 5 \\ 12 & 25 & 0 & 32 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 32 & 0 & 37 & 11 \\ 5 & 9 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -12, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 4 & 3 \\ 1 & 4 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 2 & 0 & 4 & 4 \\ 2 & 3 & 2 & 0 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -40, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 18 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 3 & 5 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 7 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 26 & 10 \\ 3 & 5 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$

[12] $1^{111}3^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$

[13] $1^{11}3^{11}21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 18 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 13 & 2 \\ 9 & 18 & 0 & 13 & 48 & 14 \\ 3 & 5 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$

[14] $1^{12}3^{11}21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 23 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 16 & 2 \\ 12 & 23 & 0 & 16 & 84 & 18 \\ 3 & 5 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$

[15] $1^{12}3^{11}21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 28 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 20 & 2 \\ 14 & 28 & 0 & 20 & 120 & 22 \\ 3 & 5 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$

$T = 10$ [16] $1^{111}3^{11}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 13 & 5 & 1 & 1 & 7 \\ 5 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & -2 & 0 \\ 7 & 3 & 0 & 0 & 3 \end{pmatrix}_{b_0}$

[17] $1^{11}3^{11}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 4 & 3 & 2 & 0 & 4 & 4 \\ 3 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

[18] $1^{11}4^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$

[19] $1^{11}3^{11}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 3 & 1 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 11 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 14 & 1 & 14 & 8 \\ 5 & 7 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{11} 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

4.647 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_2$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 1 & 5 \\ 11 & 20 & 0 & 28 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 28 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 10 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 10 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

4.648 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_3$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.649 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 67, T \in [15, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 34 & 0 & 57 & 13 \\ 5 & 8 & -2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 9 & 5 \\ 5 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 67, T \in [12, 67]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 36 & 0 & 48 & 12 \\ 5 & 9 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 36 & 0 & 48 & 12 \\ 5 & 9 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \ 1^{11} 3^1 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 10 & 13 & 0 & 0 & 6 \\ 5 & 13 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 6 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

- [10] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 0 & 1 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 1 \\ 1 & 1 & 3 & 0 & 1 & 0 \end{pmatrix}_{b_0}$
- [12] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 1 & 3 \\ 4 & 10 & 1 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 & -1 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$
- [16] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 3 & -1 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 10 & 0 & 4 & 4 \\ 5 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -166$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 0 & 7 & 13 & 7 \\ 3 & 5 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

13.0

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 3 \\ 5 & 13 & 16 & 0 & 12 & 7 \\ 5 & 16 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 12 & 0 & 9 & 5 \\ 3 & 7 & 6 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [21] \quad 1^{\overline{11}1} 4^{\overline{11}31} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 14 & 10 & 2 & 8 \\ 10 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 8 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{\overline{11}1} 4^{\overline{11}31} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 1 & 0 & 1 & -2 & 0 \\ 5 & 7 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{11}1} 4^{\overline{11}31} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [24] \quad 1^{\overline{11}1} 4^{\overline{11}1} 4^{\overline{11}1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 2 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{11}1} 4^{\overline{11}1} 4^{\overline{11}1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 & 3 \\ 1 & 3 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{11}1} 4^{\overline{11}1} 4^{\overline{11}1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 5 & 0 & 3 & 1 & 3 \\ 3 & 5 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.650 NHC(-2)(-3) + so₈ + so₈ + su₃ + su₂

$$T = 9 \quad [0] \quad 1^{\overline{11}1} 3^{\overline{11}1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 9 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 9 & 20 & 0 & 12 & 13 & 7 \\ 5 & 9 & -2 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}1} 3^{\overline{11}1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 28 & 5 \\ 12 & 25 & 0 & 16 & 60 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 36 & 5 \\ 28 & 60 & 0 & 36 & 133 & 21 \\ 5 & 9 & -2 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -127$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 10 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 10 & 20 & 0 & 12 & 13 & 7 \\ 5 & 9 & -2 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 10 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -133$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 19 & 5 \\ 12 & 25 & 0 & 16 & 40 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 24 & 5 \\ 19 & 40 & 0 & 24 & 58 & 14 \\ 5 & 9 & -2 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -190$, $(n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 20 & 5 \\ 12 & 25 & 0 & 16 & 40 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 24 & 5 \\ 20 & 40 & 0 & 24 & 58 & 14 \\ 5 & 9 & -2 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 9 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & -1 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 30 & 5 \\ 12 & 25 & 0 & 16 & 60 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 36 & 5 \\ 30 & 60 & 0 & 36 & 133 & 21 \\ 5 & 9 & -2 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 29 & 5 \\ 12 & 25 & 0 & 16 & 60 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 36 & 5 \\ 29 & 60 & 0 & 36 & 133 & 21 \\ 5 & 9 & -2 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 38 & 5 \\ 12 & 25 & 0 & 16 & 80 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 48 & 5 \\ 38 & 80 & 0 & 48 & 238 & 28 \\ 5 & 9 & -2 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{114}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -442, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 40 & 5 \\ 12 & 25 & 0 & 16 & 80 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 48 & 5 \\ 40 & 80 & 0 & 48 & 238 & 28 \\ 5 & 9 & -2 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{114}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -510, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 39 & 5 \\ 12 & 25 & 0 & 16 & 80 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 48 & 5 \\ 39 & 80 & 0 & 48 & 238 & 28 \\ 5 & 9 & -2 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{112}} 3 \ 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{111}} 3 \ 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{111}} 3 \ 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \ 1^{\overline{111}} 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 165, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 9 & 5 \\ 9 & 13 & 0 & 10 & 14 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 4 & 9 & 4 \\ 9 & 14 & 0 & 9 & 13 & 7 \\ 5 & 7 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{\overline{111}} 4 \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 261, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 6 & 5 \\ 9 & 13 & 0 & 10 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 4 & 7 & 4 \\ 6 & 10 & 0 & 7 & 5 & 5 \\ 5 & 7 & -2 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{\overline{111}} 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 480, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 15 & 5 \\ 9 & 13 & 0 & 10 & 24 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 4 & 16 & 4 \\ 15 & 24 & 0 & 16 & 38 & 12 \\ 5 & 7 & -2 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

4.651 NHC(-2)(-3) + so₈ + so₈ + su₃ + su₂→(-3)

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.652 NHC(-2)(-3) + so₈ + so₈ + su₃ + su_{2n3mix}

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 7 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 7 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.653 NHC(-2)(-3) + so₈ + so₈ + su₃ + su₃

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 1 & 5 \\ 12 & 25 & 0 & 16 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 9 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 9 & 20 & 0 & 12 & 12 & 6 \\ 5 & 9 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 10 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 10 & 20 & 0 & 12 & 12 & 6 \\ 5 & 9 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 1 & 5 \\ 12 & 25 & 12 & 0 & 0 & 9 \\ 6 & 12 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 3 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 1 & 3 \\ 5 & 13 & 16 & 0 & 12 & 0 & 7 \\ 5 & 16 & 12 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 12 & 12 & 0 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -2 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

4.654 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 7 \quad [1] \ 1 \overset{1^{11}}{3} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 8 & 5 \\ 12 & 25 & 16 & 9 \\ 8 & 16 & 10 & 6 \\ 5 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[0] \ 21 \overset{1^{11}}{3} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1 \overset{1^{11}}{11} 3 \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1 \overset{1^{13}}{13} 3 \overset{1^{14}}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 80 & 6 \\ 16 & 25 & 124 & 9 \\ 80 & 124 & 613 & 45 \\ 6 & 9 & 45 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1 \overset{1^{12}}{11} 3 \overset{1^{11}}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 14 & 6 \\ 15 & 20 & 17 & 8 \\ 14 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1 \overset{1^{12}}{11} 3 \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 11 & 5 \\ 12 & 25 & 20 & 9 \\ 11 & 20 & 13 & 7 \\ 5 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1 \overset{1^{14}}{14} 3 \overset{1^{12}}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 74 & 6 \\ 16 & 25 & 112 & 9 \\ 74 & 112 & 478 & 40 \\ 6 & 9 & 40 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1 \overset{1^{12}}{11} 3 \overset{1^{11}}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1 \overset{1^{12}}{11} 3 \overset{1^{11}}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 37 & 6 \\ 16 & 25 & 56 & 9 \\ 37 & 56 & 118 & 20 \\ 6 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1 \overset{1^{13}}{13} 3 \overset{1^{12}}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 20.33 \rightarrow 21, T_{\max} = 22, T \in [21, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 10 & 0 & 22 & 8 \\ 3 & 4 & -2 & 8 & -8 \end{pmatrix}_{b_0}$$

$$[10] \ 1 \overset{1^1}{1} 3 \overset{1^{12}}{1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 10 & 0 & 22 & 8 \\ 3 & 4 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -13, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 12.54 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -2 & 0 \\ 5 & 9 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -17, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 23 & 6 \\ 16 & 25 & 36 & 9 \\ 23 & 36 & 49 & 13 \\ 6 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -17, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.8 \\ T_{\min} = 12.41 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 10 & 16 & 47 & 6 \\ 16 & 25 & 76 & 9 \\ 47 & 76 & 217 & 27 \\ 6 & 9 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 21^{13} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -20, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.8 \\ T_{\min} = 12.20 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 10 & 16 & 84 & 6 \\ 16 & 25 & 128 & 9 \\ 84 & 128 & 634 & 46 \\ 6 & 9 & 46 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -20, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0 \\ T_{\min} = 12.20 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 10 & 16 & 84 & 6 \\ 16 & 25 & 128 & 9 \\ 84 & 128 & 634 & 46 \\ 6 & 9 & 46 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 13.09 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 30 & 5 \\ 11 & 20 & 50 & 8 \\ 30 & 50 & 118 & 20 \\ 5 & 8 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 17 & 8 \\ 10 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -2 & 0 \\ 5 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -25, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 17 & 8 \\ 12 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -27, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.5 \\ T_{\min} = 11.22 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 35 & 6 \\ 16 & 25 & 56 & 9 \\ 35 & 56 & 118 & 20 \\ 6 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -27, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 10.48 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[23] $1 \overset{1^{11}}{3} 1 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 17 & 8 \\ 9 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

[24] $1 \overset{1^3}{1^{12}} 3 1 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 12 & 29 & 5 \\ 12 & 25 & 56 & 9 \\ 29 & 56 & 118 & 20 \\ 5 & 9 & 20 & 0 \end{pmatrix}_{b_0}$$

[25] $21 \overset{1}{1^3} 3 1 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 13 & 17 & 12 & 7 \\ 17 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[26] $1 \overset{1^2}{1^{11}} 3 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 16 & 8 \\ 10 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[27] $1 \overset{1^{11}}{1^3} 3 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 1 & 6 \\ 15 & 20 & 0 & 8 \\ 1 & 0 & -2 & 0 \\ 6 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[36] $1 \overset{1^1}{1^3} 3 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[39] $1 \overset{1^1}{1^5} 3 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & -1 \end{pmatrix}_{b_0}$$

[38] $1 \overset{1^1}{1^6} 3 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -2 & 18 & -2 \end{pmatrix}_{b_0}$$

[37] $1 \overset{1^1}{1^7} 3 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 16.37 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -2 & 21 & -4 \end{pmatrix}_{b_0}$$

[35] $1 \overset{1^{12}}{1^{11}} 3 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[34] $1 \overset{1^{14}}{1^3} 3 1 \overset{1^{13}}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 59 & 5 \\ 12 & 25 & 124 & 9 \\ 59 & 124 & 613 & 45 \\ 5 & 9 & 45 & 2 \end{pmatrix}_{b_0}$$

- [33] $1^{12}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{12}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & -2 \end{pmatrix}_{b_0}$
- [31] $1^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 20 & 0 & 94 & 16 \\ 3 & 4 & -2 & 16 & -2 \end{pmatrix}_{b_0}$
- [30] $1^{112}3^11^2 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 46 & 6 \\ 16 & 25 & 72 & 9 \\ 46 & 72 & 202 & 26 \\ 6 & 9 & 26 & 3 \end{pmatrix}_{b_0}$
- [29] $1^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 0 \end{pmatrix}_{b_0}$
- [40] $1^{113}3^11^4 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -33$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 60 & 5 \\ 12 & 25 & 124 & 9 \\ 60 & 124 & 613 & 45 \\ 5 & 9 & 45 & 2 \end{pmatrix}_{b_0}$
- [41] $1^{112}3^11^3 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -33$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 21 & 5 \\ 12 & 25 & 40 & 9 \\ 21 & 40 & 58 & 14 \\ 5 & 9 & 14 & 1 \end{pmatrix}_{b_0}$
- [42] $1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -35$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 27 & 6 \\ 15 & 20 & 33 & 8 \\ 27 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [43] $1^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -35$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & -2 \end{pmatrix}_{b_0}$
- [44] $1^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -35$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 17 & 8 \\ 13 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$
- [45] $1^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -35$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[46] $21^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 15 & 7 \\ 17 & 20 & 17 & 8 \\ 15 & 17 & 13 & 7 \\ 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 50 & 6 \\ 16 & 25 & 76 & 9 \\ 50 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 36 & 9 \\ 18 & 36 & 49 & 13 \\ 5 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 15 & 0 & 52 & 12 \\ 3 & 4 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[50] $21^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 25 & 56 & 9 \\ 36 & 56 & 118 & 20 \\ 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 25 & 56 & 9 \\ 36 & 56 & 118 & 20 \\ 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 10 & 0 & 22 & 8 \\ 3 & 4 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 22 & 8 \\ 3 & 4 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[54] $131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 8 & 5 \\ 11 & 20 & 16 & 8 \\ 8 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 20 & 5 \\ 11 & 20 & 33 & 8 \\ 20 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

[56] $131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 44 & 5 \\ 11 & 20 & 84 & 8 \\ 44 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$$

[57] $131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[59] $21^{12} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 92 & 9 \\ 60 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 11 & 6 \\ 15 & 20 & 16 & 8 \\ 11 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 72 & 9 \\ 36 & 72 & 202 & 26 \\ 5 & 9 & 26 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 35 & 6 \\ 15 & 20 & 50 & 8 \\ 35 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 92 & 9 \\ 60 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 16 & 8 \\ 13 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[65] $1^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 36 & 9 \\ 17 & 36 & 49 & 13 \\ 5 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 10 & 5 \\ 12 & 25 & 20 & 9 \\ 10 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 12 & 47 & 5 \\ 12 & 25 & 92 & 9 \\ 47 & 92 & 325 & 33 \\ 5 & 9 & 33 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{113} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 69 & 6 \\ 16 & 25 & 108 & 9 \\ 69 & 108 & 457 & 39 \\ 6 & 9 & 39 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 16 & 8 \\ 9 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[70] $1^1 1^3 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 15 & 0 & 52 & 12 \\ 3 & 4 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 58 & 6 \\ 16 & 25 & 92 & 9 \\ 58 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{11} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 39 & 5 \\ 11 & 20 & 67 & 8 \\ 39 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{11} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 54 & 5 \\ 12 & 25 & 108 & 9 \\ 54 & 108 & 457 & 39 \\ 5 & 9 & 39 & 1 \end{pmatrix}_{b_0}$$

[74] $1^1 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 15 & 0 & 52 & 12 \\ 3 & 4 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[77] $21^{11} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 20 & 50 & 8 \\ 36 & 50 & 118 & 20 \\ 6 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 70 & 6 \\ 16 & 25 & 112 & 9 \\ 70 & 112 & 478 & 40 \\ 6 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 48 & 6 \\ 16 & 25 & 76 & 9 \\ 48 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[80] $21^{11} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 48 & 6 \\ 16 & 25 & 76 & 9 \\ 48 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{11} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 63 & 6 \\ 16 & 25 & 96 & 9 \\ 63 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 23 & 6 \\ 15 & 20 & 33 & 8 \\ 23 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$

[83] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 48 & 5 \\ 11 & 20 & 84 & 8 \\ 48 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$

[84] $21^{112} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 54 & 7 \\ 20 & 25 & 72 & 9 \\ 54 & 72 & 202 & 26 \\ 7 & 9 & 26 & 3 \end{pmatrix}_{b_0}$

[85] $131^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 40 & 9 \\ 18 & 40 & 58 & 14 \\ 5 & 9 & 14 & 2 \end{pmatrix}_{b_0}$

[86] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$

[87] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$

[88] $1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 20 & 0 & 94 & 16 \\ 3 & 4 & -2 & 16 & -2 \end{pmatrix}_{b_0}$

[89] $1^{112} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 20 & 0 & 94 & 16 \\ 3 & 4 & -2 & 16 & 0 \end{pmatrix}_{b_0}$

[90] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 49 & 6 \\ 16 & 25 & 76 & 9 \\ 49 & 76 & 217 & 27 \\ 6 & 9 & 27 & 3 \end{pmatrix}_{b_0}$

[91] $21^{113} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 96 & 9 \\ 60 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[92] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 96 & 9 \\ 60 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[93] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 48 & 6 \\ 15 & 20 & 67 & 8 \\ 48 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 12 & 72 & 5 \\ 12 & 25 & 144 & 9 \\ 72 & 144 & 814 & 52 \\ 5 & 9 & 52 & 0 \end{pmatrix}_{b_0}$

[95] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 31 & 5 \\ 12 & 25 & 60 & 9 \\ 31 & 60 & 133 & 21 \\ 5 & 9 & 21 & 2 \end{pmatrix}_{b_0}$

[96] $1^131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 35 & 5 \\ 11 & 20 & 67 & 8 \\ 35 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$

[97] $1^131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 17 & 5 \\ 11 & 20 & 33 & 8 \\ 17 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$

[98] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 61 & 6 \\ 15 & 20 & 84 & 8 \\ 61 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$

[99] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 1 \end{pmatrix}_{b_0}$

[100] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$

[101] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 92 & 6 \\ 16 & 25 & 144 & 9 \\ 92 & 144 & 814 & 52 \\ 6 & 9 & 52 & 3 \end{pmatrix}_{b_0}$

[102] $1^131^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 34 & 5 \\ 12 & 25 & 72 & 9 \\ 34 & 72 & 202 & 26 \\ 5 & 9 & 26 & 2 \end{pmatrix}_{b_0}$

[103] $1^131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 20 & 0 & 94 & 16 \\ 3 & 4 & -2 & 16 & 0 \end{pmatrix}_{b_0}$

[104] $1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$

[105] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 35 & 5 \\ 12 & 25 & 72 & 9 \\ 35 & 72 & 202 & 26 \\ 5 & 9 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 20 & 5 \\ 12 & 25 & 40 & 9 \\ 20 & 40 & 58 & 14 \\ 5 & 9 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{114} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 73 & 6 \\ 16 & 25 & 112 & 9 \\ 73 & 112 & 478 & 40 \\ 6 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 19 & 5 \\ 11 & 20 & 33 & 8 \\ 19 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 29 & 5 \\ 11 & 20 & 50 & 8 \\ 29 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 26 & 5 \\ 11 & 20 & 50 & 8 \\ 26 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 26 & 6 \\ 15 & 20 & 33 & 8 \\ 26 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.64 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{114} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 81 & 6 \\ 16 & 25 & 128 & 9 \\ 81 & 128 & 634 & 46 \\ 6 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{11} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.64 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 39 & 5 \\ 12 & 25 & 76 & 9 \\ 39 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

- [117] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 28 & 5 \\ 12 & 25 & 56 & 9 \\ 28 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$
- [118] $1^{11}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -112$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$
- [119] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -112$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [120] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -113$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 19 & 5 \\ 12 & 25 & 40 & 9 \\ 19 & 40 & 58 & 14 \\ 5 & 9 & 14 & 2 \end{pmatrix}_{b_0}$
- [121] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -114$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 18 & 5 \\ 11 & 20 & 33 & 8 \\ 18 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [122] $1^{111}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -114$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 45 & 5 \\ 11 & 20 & 84 & 8 \\ 45 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$
- [123] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -115$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [124] $1^{11}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -122$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 26 & 5 \\ 12 & 25 & 56 & 9 \\ 26 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$
- [125] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -125$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 24 & 6 \\ 15 & 20 & 33 & 8 \\ 24 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [126] $1^131^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -128$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 27 & 5 \\ 12 & 25 & 60 & 9 \\ 27 & 60 & 133 & 21 \\ 5 & 9 & 21 & 1 \end{pmatrix}_{b_0}$
- [127] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -128$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 62 & 6 \\ 16 & 25 & 96 & 9 \\ 62 & 96 & 346 & 34 \\ 6 & 9 & 34 & 3 \end{pmatrix}_{b_0}$
- [128] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -130$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 36 & 6 \\ 15 & 20 & 50 & 8 \\ 36 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[129] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 38 & 5 \\ 11 & 20 & 67 & 8 \\ 38 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$

[130] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 39 & 6 \\ 15 & 20 & 50 & 8 \\ 39 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$

[131] $1^{111} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 47 & 5 \\ 11 & 20 & 84 & 8 \\ 47 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$

[132] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 61 & 6 \\ 16 & 25 & 96 & 9 \\ 61 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[133] $21^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 27 & 7 \\ 20 & 25 & 36 & 9 \\ 27 & 36 & 49 & 13 \\ 7 & 9 & 13 & 3 \end{pmatrix}_{b_0}$

[134] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 25 & 6 \\ 15 & 20 & 33 & 8 \\ 25 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$

[135] $21^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 27 & 7 \\ 17 & 20 & 33 & 8 \\ 27 & 33 & 49 & 13 \\ 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$

[136] $21^{12} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 72 & 6 \\ 16 & 25 & 112 & 9 \\ 72 & 112 & 478 & 40 \\ 6 & 9 & 40 & 3 \end{pmatrix}_{b_0}$

[137] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 72 & 6 \\ 16 & 25 & 112 & 9 \\ 72 & 112 & 478 & 40 \\ 6 & 9 & 40 & 3 \end{pmatrix}_{b_0}$

[138] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 27 & 5 \\ 12 & 25 & 56 & 9 \\ 27 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$

[139] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 27 & 5 \\ 11 & 20 & 50 & 8 \\ 27 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$

[140] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 36 & 5 \\ 11 & 20 & 67 & 8 \\ 36 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$

- [141] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -142$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 28 & 5 \\ 11 & 20 & 50 & 8 \\ 28 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$
- [142] $1^{111}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -142$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 46 & 5 \\ 11 & 20 & 84 & 8 \\ 46 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$
- [143] $21^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$
- [144] $1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$
- [145] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -147$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [146] $1^{113}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -148$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 53 & 5 \\ 12 & 25 & 108 & 9 \\ 53 & 108 & 457 & 39 \\ 5 & 9 & 39 & 2 \end{pmatrix}_{b_0}$
- [147] $1^{111}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -150$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 62 & 6 \\ 15 & 20 & 84 & 8 \\ 62 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$
- [148] $1^{111}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -155$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 49 & 6 \\ 15 & 20 & 67 & 8 \\ 49 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$
- [151] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -156$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$
- [149] $1^{111}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -156$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 37 & 5 \\ 11 & 20 & 67 & 8 \\ 37 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$
- [150] $1^{114}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -156$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 1 \end{pmatrix}_{b_0}$
- [152] $1^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -160$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -160$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[154] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -164$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 51 & 5 \\ 12 & 25 & 108 & 9 \\ 51 & 108 & 457 & 39 \\ 5 & 9 & 39 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -170$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 37 & 6 \\ 15 & 20 & 50 & 8 \\ 37 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -170$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 38 & 6 \\ 15 & 20 & 50 & 8 \\ 38 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{11} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -170$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 63 & 6 \\ 15 & 20 & 84 & 8 \\ 63 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -172$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -173$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 46 & 5 \\ 12 & 25 & 92 & 9 \\ 46 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -173$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 30 & 5 \\ 12 & 25 & 60 & 9 \\ 30 & 60 & 133 & 21 \\ 5 & 9 & 21 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{11} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -175$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 51 & 6 \\ 15 & 20 & 67 & 8 \\ 51 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -176$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -179$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.65 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

- [164] $1^{113}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -181$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 52 & 5 \\ 12 & 25 & 108 & 9 \\ 52 & 108 & 457 & 39 \\ 5 & 9 & 39 & 2 \end{pmatrix}_{b_0}$
- [165] $1^{111}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -185$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 50 & 6 \\ 15 & 20 & 67 & 8 \\ 50 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$
- [166] $1^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -187$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$
- [168] $1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$
- [167] $1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 1 \end{pmatrix}_{b_0}$
- [170] $21^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -190$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 24 & 6 \\ 16 & 20 & 33 & 8 \\ 24 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [169] $21^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -190$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 72 & 7 \\ 17 & 20 & 84 & 8 \\ 72 & 84 & 346 & 34 \\ 7 & 8 & 34 & 3 \end{pmatrix}_{b_0}$
- [171] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -193$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 28 & 5 \\ 12 & 25 & 60 & 9 \\ 28 & 60 & 133 & 21 \\ 5 & 9 & 21 & 2 \end{pmatrix}_{b_0}$
- [172] $1^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -195$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$
- [173] $21^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -198$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 42 & 7 \\ 17 & 20 & 50 & 8 \\ 42 & 50 & 118 & 20 \\ 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$
- [174] $131^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -198$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 80 & 9 \\ 36 & 80 & 238 & 28 \\ 5 & 9 & 28 & 1 \end{pmatrix}_{b_0}$
- [175] $21^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -200$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 13 & 7 \\ 7 & 9 & 7 & 3 \end{pmatrix}_{b_0}$

- [176] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [177] $1^{114} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 71 & 5 \\ 12 & 25 & 144 & 9 \\ 71 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$$
- [178] $1^{113} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 29 & 5 \\ 12 & 25 & 60 & 9 \\ 29 & 60 & 133 & 21 \\ 5 & 9 & 21 & 2 \end{pmatrix}_{b_0}$$
- [179] $1^{113} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 38 & 5 \\ 12 & 25 & 76 & 9 \\ 38 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$
- [180] $1^{11} 3^1 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 35 & 5 \\ 12 & 25 & 76 & 9 \\ 35 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$
- [181] $21^{13} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 57 & 7 \\ 17 & 20 & 67 & 8 \\ 57 & 67 & 217 & 27 \\ 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$
- [182] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [183] $1^{14} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [184] $1^{12} 3^1 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 43 & 5 \\ 12 & 25 & 92 & 9 \\ 43 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$
- [185] $1^{11} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$
- [186] $1^{12} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [187] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{113} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 92 & 9 \\ 45 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{114} 3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -242, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 64 & 5 \\ 12 & 25 & 128 & 9 \\ 64 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[190] $1^{112} 3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{113} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{113} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -257, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 44 & 5 \\ 12 & 25 & 92 & 9 \\ 44 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 37 & 5 \\ 12 & 25 & 76 & 9 \\ 37 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 76 & 9 \\ 36 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{114} 3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 68 & 5 \\ 12 & 25 & 144 & 9 \\ 68 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{114} 3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[198] $1^{112} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

- [200] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -272$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$
- [199] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -272$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$
- [201] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -274$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 70 & 5 \\ 12 & 25 & 144 & 9 \\ 70 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$
- [202] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -275$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$
- [203] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -278$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 40 & 5 \\ 12 & 25 & 80 & 9 \\ 40 & 80 & 238 & 28 \\ 5 & 9 & 28 & 2 \end{pmatrix}_{b_0}$
- [204] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -288$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$
- [205] $1^1 3^1 1^{15} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -288$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.1$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 100 & 9 \\ 45 & 100 & 373 & 35 \\ 5 & 9 & 35 & 1 \end{pmatrix}_{b_0}$
- [206] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -293$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 37 & 5 \\ 12 & 25 & 80 & 9 \\ 37 & 80 & 238 & 28 \\ 5 & 9 & 28 & 2 \end{pmatrix}_{b_0}$
- [207] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -293$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 69 & 5 \\ 12 & 25 & 144 & 9 \\ 69 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$
- [208] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -304$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$
- [209] $1^1 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -322$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 44 & 5 \\ 12 & 25 & 96 & 9 \\ 44 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$
- [211] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -323$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{13}3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -323$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{14}3^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -326$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 56 & 5 \\ 12 & 25 & 112 & 9 \\ 56 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{12}3^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -332$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{14}3^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -332$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{14}3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -333$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 39 & 5 \\ 12 & 25 & 80 & 9 \\ 39 & 80 & 238 & 28 \\ 5 & 9 & 28 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{14}3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -338$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 38 & 5 \\ 12 & 25 & 80 & 9 \\ 38 & 80 & 238 & 28 \\ 5 & 9 & 28 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{14}3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -338$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 48 & 5 \\ 12 & 25 & 96 & 9 \\ 48 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{14}3^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -345$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 63 & 5 \\ 12 & 25 & 128 & 9 \\ 63 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{12}3^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -347$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{13}3^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -347$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[221] $21^{12}3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -350$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 30 & 7 \\ 20 & 25 & 40 & 9 \\ 30 & 40 & 58 & 14 \\ 7 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

- [222] $1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -352$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$
- [223] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -354$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 60 & 5 \\ 12 & 25 & 128 & 9 \\ 60 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$
- [224] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -362$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 42 & 7 \\ 20 & 25 & 56 & 9 \\ 42 & 56 & 118 & 20 \\ 7 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$
- [225] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -363$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [227] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -364$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [226] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -364$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.04 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [228] $1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -368$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$
- [229] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -371$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [230] $1^{12} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -374$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 52 & 5 \\ 12 & 25 & 112 & 9 \\ 52 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$
- [231] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -380$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [232] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -398$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 62 & 5 \\ 12 & 25 & 128 & 9 \\ 62 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$
- [233] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -401$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 61 & 5 \\ 12 & 25 & 128 & 9 \\ 61 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -401, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 96 & 9 \\ 45 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[235] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -409, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 47 & 5 \\ 12 & 25 & 96 & 9 \\ 47 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[236] $1^{114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -413, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 55 & 5 \\ 12 & 25 & 112 & 9 \\ 55 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$

[237] $1^{12}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$

[238] $1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$

[239] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 46 & 5 \\ 12 & 25 & 96 & 9 \\ 46 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[240] $1^{13}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -432, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$

[241] $1^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -435, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$

[242] $1^{114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -437, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 53 & 5 \\ 12 & 25 & 112 & 9 \\ 53 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$

[243] $1^{114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 54 & 5 \\ 12 & 25 & 112 & 9 \\ 54 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$

[244] $1^{14}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$

- [245] $1^{13}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -476$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [246] $21^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -488$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 69 & 7 \\ 20 & 25 & 92 & 9 \\ 69 & 92 & 325 & 33 \\ 7 & 9 & 33 & 3 \end{pmatrix}_{b_0}$$
- [247] $1^{12}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -491$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [248] $1^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -492$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [250] $1^{13}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -499$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [249] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -499$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [251] $1^{12}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -515$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [252] $1^{12}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -528$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [253] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -539$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [254] $1^{13}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -540$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [255] $1^{13}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -572$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [256] $21^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -578$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 96 & 7 \\ 20 & 25 & 128 & 9 \\ 96 & 128 & 634 & 46 \\ 7 & 9 & 46 & 3 \end{pmatrix}_{b_0}$$

$$[257] \ 1^{14} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -592, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[258] \ 21^{11} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 45 & 7 \\ 20 & 25 & 60 & 9 \\ 45 & 60 & 133 & 21 \\ 7 & 9 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{12} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{13} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[261] \ 1^{13} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[262] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -620, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[263] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -651, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{12} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[265] \ 21^{11} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -692, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 57 & 7 \\ 20 & 25 & 76 & 9 \\ 57 & 76 & 217 & 27 \\ 7 & 9 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{13} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -707, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -716, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -752, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -763, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -780, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[271] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -800, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -803, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -864, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -923, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -944, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 22 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[276] $21^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -950, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 60 & 7 \\ 20 & 25 & 80 & 9 \\ 60 & 80 & 238 & 28 \\ 7 & 9 & 28 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -976, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 22 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[278] $21^{14} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -998, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 84 & 7 \\ 20 & 25 & 112 & 9 \\ 84 & 112 & 478 & 40 \\ 7 & 9 & 40 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1011, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[281] $21^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 72 & 7 \\ 20 & 25 & 96 & 9 \\ 72 & 96 & 346 & 34 \\ 7 & 9 & 34 & 3 \end{pmatrix}_{b_0}$$

[282] $1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 24 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[283] $21^{115} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 75 & 7 \\ 20 & 25 & 100 & 9 \\ 75 & 100 & 373 & 35 \\ 7 & 9 & 35 & 3 \end{pmatrix}_{b_0}$$

$T = 8$ [284] $1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 4 & 0 & -1 \end{pmatrix}_{b_0}$$

[285] $21^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 5 & 5 \\ 6 & 9 & 5 & 0 \end{pmatrix}_{b_0}$$

[286] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 4 & 0 & 10 & 6 \\ 3 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

- [290] $1^1 31^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [291] $1^1 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 11 & 5 \\ 9 & 13 & 14 & 7 \\ 11 & 14 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [292] $1^1 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -2 & 4 & 3 \end{pmatrix}_{b_0}$
- [293] $1^1 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 5 & 5 \\ 5 & 9 & 5 & 1 \end{pmatrix}_{b_0}$
- [294] $1^{11} 31^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [295] $1^{11} 31^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 13 & 7 \\ 3 & 6 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [296] $1^{11} 31^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 1 & 4 \\ 7 & 10 & 13 & 0 & 6 \\ 8 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 6 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [297] $1^{12} 321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 10 & 2 \\ 8 & 15 & 10 & 38 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [298] $1^{14} 321^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 20 & 2 \\ 17 & 30 & 20 & 158 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [299] $1^{11} 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [300] $1^{11} 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & 2 \end{pmatrix}_{b_0}$

[301] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 16 & 5 \\ 9 & 13 & 20 & 7 \\ 16 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[302] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 9 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 15 & 6 \\ 9 & 12 & 15 & 13 & 7 \\ 4 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$

[303] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$

[306] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$

[310] $21^{14}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[309] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 28 & 6 \\ 12 & 13 & 34 & 7 \\ 28 & 34 & 77 & 17 \\ 6 & 7 & 17 & 3 \end{pmatrix}_{b_0}$

[308] $1^{14}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 14 & 7 \\ 12 & 14 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$

[307] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[305] $21^{14}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 28 & 6 \\ 12 & 13 & 34 & 7 \\ 28 & 34 & 77 & 17 \\ 6 & 7 & 17 & 3 \end{pmatrix}_{b_0}$

[304] $21^{14}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 14 & 7 \\ 12 & 14 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$

[311] $1^{14}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$

[313] $21^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 9 & 10 & 7 & 14 & 8 \\ 4 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[312] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[314] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[315] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 6 & 3 \\ 1 & 4 & 0 & 3 & 6 & 2 \\ 2 & 3 & 3 & 2 & 6 & 2 \\ 3 & 6 & 6 & 6 & 10 & 6 \\ 2 & 3 & 2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[317] $1^{12}31^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[318] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{11}31^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 19 & 5 \\ 9 & 13 & 24 & 7 \\ 19 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[322] $1^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 14 & 7 \\ 9 & 14 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[325] $1^{12}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 8 & 5 \\ 9 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[323] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 20 & 6 \\ 12 & 13 & 24 & 7 \\ 20 & 24 & 38 & 12 \\ 6 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[324] $21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 20 & 6 \\ 12 & 13 & 24 & 7 \\ 20 & 24 & 38 & 12 \\ 6 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[328] $1^{11}3131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 10 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[326] $1^{11}31^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[327] $1^{12}31^{12}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 6 & 6 & 0 & 6 & 10 & 6 \\ 2 & 3 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[329] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 2 & 5 & 4 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[338] $141^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 12 & 5 \\ 9 & 13 & 20 & 7 \\ 12 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 10 & 5 \\ 9 & 13 & 14 & 7 \\ 10 & 14 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[336] $141^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 5 & 9 & 18 & 5 \\ 9 & 13 & 30 & 7 \\ 18 & 30 & 61 & 15 \\ 5 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[335] $141^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 5 & 9 & 24 & 5 \\ 9 & 13 & 40 & 7 \\ 24 & 40 & 110 & 20 \\ 5 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

[334] $141^{15} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 5 & 9 & 30 & 5 \\ 9 & 13 & 50 & 7 \\ 30 & 50 & 173 & 25 \\ 5 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

- [333] $141^{16}_{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 15.75 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 5 & 9 & 36 & 5 \\ 9 & 13 & 60 & 7 \\ 36 & 60 & 250 & 30 \\ 5 & 7 & 30 & -1 \end{pmatrix}_{b_0}$
- [332] $141^{17}_{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.3$
 $T_{\min} = 17.38 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 5 & 9 & 42 & 5 \\ 9 & 13 & 70 & 7 \\ 42 & 70 & 341 & 35 \\ 5 & 7 & 35 & -3 \end{pmatrix}_{b_0}$
- [331] $141^{11}_{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [330] $21^{11}_{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 5 & 5 \\ 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [340] $1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 33$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 2 & 4 \\ 3 & 5 & -2 & 4 & 3 \end{pmatrix}_{b_0}$
- [339] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 33$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [341] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$
- [342] $1^{13} 31^{11} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$
- [343] $1^{11} 31^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 25 & 3 \\ 4 & 10 & 1 & 60 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 60 & 0 & 346 & 34 \\ 3 & 6 & -1 & 34 & 2 \end{pmatrix}_{b_0}$
- [344] $131^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 15 & 6 \\ 5 & 12 & 15 & 13 & 7 \\ 3 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [345] $1^{12} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 6 & 0 & 6 & 10 & 6 \\ 2 & 3 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [346] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 15 & 6 \\ 8 & 12 & 15 & 13 & 7 \\ 4 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{11}31^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 7 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 12 & 6 \\ 7 & 12 & 12 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[349] $1^{11}31^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 11 & 6 & 5 \\ 1 & 3 & 0 & 6 & 1 & 3 \\ 2 & 3 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[348] $1^{12}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 10 & 2 \\ 9 & 15 & 10 & 38 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{11}321^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 15 & 0 & 34 & 12 \\ 3 & 5 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{11}31^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[352] $1^{11}31^{11}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[353] $1^{11}41^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 7 & 5 \\ 9 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{11}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 25 & 0 & 92 & 20 \\ 3 & 5 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{11}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{11}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[357] $1^{11}31^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 12 & 6 \\ 5 & 12 & 12 & 10 & 6 \\ 3 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[358] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -2 & 8 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{11}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -2 & 8 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{111}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 12 & 6 \\ 8 & 12 & 12 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[365] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 5 & 5 \\ 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{12}31^{12}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[363] $1^{114}31^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 3 & 12 & 3 \\ 1 & 4 & 0 & 3 & 12 & 2 \\ 2 & 3 & 3 & 2 & 12 & 2 \\ 6 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & 2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[361] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[367] $1^{111}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 19 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[366] $21^{12}321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 18 & 20 & 14 & 62 & 16 \\ 4 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[369] $1^{12}31^{11}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

- [368] $1^{13}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 6 & 0 & 4 & 4 \\ 2 & 3 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [370] $1^{11}1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 11 & 6 & 5 \\ 2 & 3 & 0 & 6 & 1 & 3 \\ 2 & 3 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$
- [371] $1^{14}31^{11}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 15 & 0 & 76 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$
- [372] $1^{13}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [373] $1^{13}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 3 & 5 & 4 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [374] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 13 & 14 & 15 & 7 \\ 14 & 13 & 14 & 7 \\ 15 & 14 & 13 & 7 \\ 7 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [377] $1^{11}1^{11}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 10 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [375] $1^{13}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 5 & 10 & 7 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [376] $1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 20 & 7 \\ 15 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$
- [378] $1^{112}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [379] $1^{11}1^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [380] $1^{13}31^{11}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 12 & 0 & 49 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[381] $1^{13}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[382] $1^{11}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 58 & 16 \\ 3 & 5 & -2 & 16 & 3 \end{pmatrix}_{b_0}$$

[383] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[384] $1^{11}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 15 & 0 & 32 & 12 \\ 3 & 5 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

[385] $1^{13}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 17 & 2 \\ 13 & 25 & 17 & 104 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[389] $1^{13}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[388] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 9 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[387] $1^{11}31^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 21 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[386] $1^{14}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 6 & 12 & 0 & 12 & 46 & 12 \\ 2 & 3 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[391] $1^{14}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 24 & 7 \\ 15 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[390] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 5 & 9 & 13 & 5 \\ 9 & 13 & 20 & 7 \\ 13 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[392] \quad & 1^{12}31^{13}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 5 & 6 & 0 & 6 & 10 & 6 \\ 2 & 3 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[393] \quad & 21^{11}41^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 13 & 14 & 51 & 7 \\ 14 & 13 & 54 & 7 \\ 51 & 54 & 197 & 27 \\ 7 & 7 & 27 & 3 \end{pmatrix}_{b_0} \\
[394] \quad & 1^{11}31^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0} \\
[395] \quad & 1^{13}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[396] \quad & 1^{11}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 11 & 0 & 65 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[397] \quad & 1^{11}31^{23}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 14 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[399] \quad & 1^{12}31^{13}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 4 & 6 & 0 & 6 & 10 & 6 \\ 2 & 3 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[398] \quad & 1^{11}31^{13}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 11 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[400] \quad & 1^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 14 & 5 \\ 9 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[401] \quad & 1^{11}41^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 18 & 5 \\ 9 & 13 & 24 & 7 \\ 18 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[402] \quad & 1^{13}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 8 & 2 \\ 5 & 10 & 8 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [403] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [404] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [406] $1^{12}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [405] $1^{13}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [408] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 6 & 10 & 7 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [407] $1^{14}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 15 & 0 & 76 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$
- [409] $2231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 1 & 4 & 1 & 4 & 3 \\ 2 & 4 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 3 & 0 \\ 3 & 4 & 4 & 3 & 5 & 5 \\ 2 & 3 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$
- [411] $1^{13}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 86$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 5 & 9 & 0 & 9 & 25 & 9 \\ 2 & 3 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [410] $1^{11}31^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 86$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 16 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [412] $21^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$
- [414] $1^{12}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 89$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 9 & 0 & 28 & 10 \\ 2 & 3 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

- [413] $1^{11}31^331^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 20 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$
- [415] $1^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 19 & 5 \\ 9 & 13 & 30 & 7 \\ 19 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$
- [418] $1^{11}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 11 & 2 \\ 7 & 15 & 11 & 32 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [417] $1^{11}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 17 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 17 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [416] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 16 & 5 \\ 9 & 13 & 24 & 7 \\ 16 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$
- [419] $1^{12}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$
- [420] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 17 & 5 \\ 9 & 13 & 24 & 7 \\ 17 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$
- [421] $21^{13}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 27 & 30 & 21 & 142 & 24 \\ 4 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [423] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [422] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$
- [425] $1^{11}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 5 & 10 & 0 & 33 & 11 \\ 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [424] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[426] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[427] $1^1 4^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 34 & 7 \\ 21 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$

[428] $1^{11}3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0}$

[429] $1^{12}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$

[430] $1^{11}31^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 15 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0}$

[431] $1^{13}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 22 & 5 \\ 9 & 13 & 30 & 7 \\ 22 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$

[432] $1^{11}3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 15 & 2 \\ 9 & 20 & 15 & 58 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[433] $1^{14}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 25 & 5 \\ 9 & 13 & 40 & 7 \\ 25 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$

[434] $1^{12}3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 10 & 20 & 14 & 62 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[435] $1^{12}3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 62 & 16 \\ 3 & 5 & -2 & 16 & 3 \end{pmatrix}_{b_0}$

[436] $1^{12}3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[437] $1^{12}3^{112}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 18 & 0 & 166 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[438] $1^{14}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[439] $1^{12}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[440] $2^12^131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 17 & 2 \\ 15 & 25 & 17 & 104 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[441] $1^{13}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 21 & 0 & 205 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[442] $1^{11}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 15 & 0 & 32 & 12 \\ 3 & 5 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

[443] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 15 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 15 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[444] $1^{13}4^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 20 & 5 \\ 9 & 13 & 30 & 7 \\ 20 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[445] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[446] $2^12^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 3 & 5 & 4 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[447] $1^{13}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 12 & 0 & 49 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[448] $21^{11}41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 13 & 14 & 24 & 7 \\ 14 & 13 & 24 & 7 \\ 24 & 24 & 38 & 12 \\ 7 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[449] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.35 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[450] $1^{11}321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 11 & 0 & 65 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[452] $1^{12}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[451] $1^{12}321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[453] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 10 & 3 \\ 4 & 10 & 13 & 24 & 6 \\ 5 & 13 & 12 & 27 & 6 \\ 10 & 24 & 27 & 49 & 13 \\ 3 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[454] $31^{11}231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 4 & 2 \\ 3 & 1 & 4 & 1 & 4 & 3 \\ 3 & 4 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 3 & 0 \\ 4 & 4 & 4 & 3 & 5 & 5 \\ 2 & 3 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[455] $1^{13}31^{13}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 6 & 9 & 0 & 9 & 25 & 9 \\ 2 & 3 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[456] $1^{13}41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[457] $1^{13}31^{13}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 134$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 7 & 9 & 0 & 9 & 25 & 9 \\ 2 & 3 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[458] $21^{11}41^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 134$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 13 & 14 & 42 & 7 \\ 14 & 13 & 44 & 7 \\ 42 & 44 & 130 & 22 \\ 7 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[459] \quad & 131^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 9 & 3 \\ 4 & 10 & 13 & 24 & 6 \\ 5 & 13 & 12 & 27 & 6 \\ 9 & 24 & 27 & 49 & 13 \\ 3 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0} \\
[460] \quad & 1^{14}3^{11}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 28 & 0 & 366 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[461] \quad & 1^{13}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 17 & 2 \\ 14 & 25 & 17 & 104 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[462] \quad & 21^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 14 & 5 \\ 10 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[463] \quad & 21^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 16 & 5 \\ 10 & 13 & 24 & 7 \\ 16 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[466] \quad & 1^{14}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0} \\
[465] \quad & 1^{13}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0} \\
[464] \quad & 1^{13}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0} \\
[467] \quad & 1^4 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 27 & 5 \\ 9 & 13 & 44 & 7 \\ 27 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[468] \quad & 1^{12}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0} \\
[469] \quad & 1^{14}3^{11}21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 24 & 2 \\ 18 & 35 & 24 & 202 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [470] $21^{11}41^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 152$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 13 & 14 & 33 & 7 \\ 14 & 13 & 34 & 7 \\ 33 & 34 & 77 & 17 \\ 7 & 7 & 17 & 3 \end{pmatrix}_{b_0}$
- [471] $1^{13}321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 153$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 25 & 0 & 313 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$
- [472] $1^{14}31^{13}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 7 & 12 & 0 & 12 & 46 & 12 \\ 2 & 3 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$
- [473] $1^{13}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 12 & 0 & 49 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$
- [474] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [475] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 159$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 16 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 16 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [476] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 22 & 5 \\ 9 & 13 & 34 & 7 \\ 22 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$
- [478] $2^12^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 6 & 10 & 7 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [477] $1^{12}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 25 & 0 & 94 & 20 \\ 3 & 5 & -2 & 20 & 3 \end{pmatrix}_{b_0}$
- [479] $1^{11}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 58 & 16 \\ 3 & 5 & -2 & 16 & 3 \end{pmatrix}_{b_0}$
- [480] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 163$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 25 & 5 \\ 9 & 13 & 34 & 7 \\ 25 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$
- [482] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 168$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$

$$\begin{aligned}
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0} \\
[481] & 1^{114}3131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 12 & 0 & 46 & 12 \\ 2 & 3 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[483] & 21^{14}321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 5 & 3 & 36 & 4 \\ 5 & 5 & 5 & 40 & 5 \\ 3 & 5 & 0 & 28 & 2 \\ 36 & 40 & 28 & 254 & 32 \\ 4 & 5 & 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[485] & 1^{11}321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 11 & 2 \\ 8 & 15 & 11 & 32 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[484] & 1^{13}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[486] & 1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 26 & 5 \\ 9 & 13 & 40 & 7 \\ 26 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0} \\
[487] & 1^{12}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0} \\
[488] & 1^{12}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0} \\
[489] & 1^{13}1^{13}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[490] & 2^{111}2231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0} \\
[491] & 1^{12}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[494] & 1^{14}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[493] $1^{11} 1^{11} 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 25 & 0 & 92 & 20 \\ 3 & 5 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

[492] $1^{13} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[496] $1^{12} 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 30 & 0 & 134 & 24 \\ 3 & 5 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

[495] $1^{12} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 14 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[497] $1^1 4 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 33 & 5 \\ 9 & 13 & 54 & 7 \\ 33 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[498] $1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 23 & 5 \\ 9 & 13 & 34 & 7 \\ 23 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[500] $1^{13} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[499] $1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 24 & 5 \\ 9 & 13 & 34 & 7 \\ 24 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[501] $1^{12} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 11 & 20 & 14 & 62 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[502] $1^1 31^{14} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[503] $1^{12} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[504] $1^{14}31^{11}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 10 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[505] $2^1 2^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 8 & 2 \\ 6 & 10 & 8 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[506] $1^{13}31^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[507] $1^{12}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 18 & 2 \\ 12 & 25 & 18 & 94 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[508] $1^{12}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{14}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 27 & 5 \\ 9 & 13 & 40 & 7 \\ 27 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[510] $1^{14}31^{11}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 8 & 12 & 0 & 12 & 46 & 12 \\ 2 & 3 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[511] $1^{14}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 28 & 5 \\ 9 & 13 & 40 & 7 \\ 28 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[512] $1^{13}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 15 & 30 & 21 & 142 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[513] $1^{13}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[514] $1^{12}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 18 & 0 & 166 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[515] $1^{14}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 222$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 16 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[516] $1^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 28 & 5 \\ 9 & 13 & 44 & 7 \\ 28 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[517] $21^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 232$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 21 & 5 \\ 10 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[518] $1^{12}3^{13}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[519] $1^{13}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 30 & 0 & 142 & 24 \\ 3 & 5 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

[520] $1^{14}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 24 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[521] $1^{13}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[522] $1^{12}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[523] $1^{11}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 15 & 2 \\ 10 & 20 & 15 & 58 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[524] $1^{13}31^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[525] $1^{14}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[526] $1^{12}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[527] $1^{114}31^{131}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 15 & 0 & 76 & 16 \\ 2 & 3 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[528] $1^{133}1^{113}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[529] $1^{13}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[530] $1^{13}3^{12}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 205 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[531] $1^{12}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 25 & 0 & 94 & 20 \\ 3 & 5 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

[532] $1^{13}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[533] $1^{13}31^{123}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[534] $1^{11}31^{123}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[535] $1^{12}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 22 & 2 \\ 14 & 30 & 22 & 134 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[536] $1^{11}31^{123}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0} \\
[537] & 1^{13} 31^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0} \\
[538] & 1^{14} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0} \\
[539] & 1^{12} 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 14 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0} \\
[540] & 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 29 & 5 \\ 9 & 13 & 44 & 7 \\ 29 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[541] & 1^{14} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 24 & 2 \\ 19 & 35 & 24 & 202 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[542] & 1^{13} 31^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[543] & 1^{13} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0} \\
[544] & 1^{14} 4 31^{23} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 18 & 3 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 18 & 0 & 118 & 20 \\ 2 & 3 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0} \\
[545] & 21^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 23 & 5 \\ 10 & 13 & 34 & 7 \\ 23 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0} \\
[546] & 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 34 & 5 \\ 9 & 13 & 54 & 7 \\ 34 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0} \\
[547] & 1^{14} 31^{11} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[548] $1^{11} 3^{11} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 19 & 2 \\ 12 & 25 & 19 & 92 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[549] $1^{14} 31^{13} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[550] $1^{12} 3^{12} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 297, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[551] $1^{14} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 31 & 5 \\ 9 & 13 & 44 & 7 \\ 31 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[552] $1^{13} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 30 & 5 \\ 9 & 13 & 44 & 7 \\ 30 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[553] $2^{11} 2^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 5 & 2 \\ 2 & 1 & 4 & 1 & 7 & 3 \\ 2 & 4 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -2 & 4 & 0 \\ 5 & 7 & 7 & 4 & 16 & 8 \\ 2 & 3 & 2 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[554] $1^{12} 31^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[555] $1^{12} 31^{12} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[556] $1^{13} 3^{11} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 35 & 0 & 188 & 28 \\ 3 & 5 & -2 & 28 & 3 \end{pmatrix}_{b_0}$$

[557] $1^{13} 3^{11} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 25 & 0 & 313 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[558] $2^{11} 2^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[559] $2 \ 2 \ 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 12 & 2 \\ 9 & 15 & 12 & 34 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[560] $1^{12} 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 18 & 0 & 166 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[561] $1^{12} 3 \ 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 18 & 2 \\ 13 & 25 & 18 & 94 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[562] $1^{14} 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 3 \\ 2 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 28 & 0 & 366 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[563] $1^{12} 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 30 & 0 & 134 & 24 \\ 3 & 5 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

[564] $1^{11} 31^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[565] $1^{13} 3 \ 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 16 & 30 & 21 & 142 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[566] $1^{14} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 16 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[567] $1^{12} 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 14 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[568] $1^{13} 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 25 & 2 \\ 17 & 35 & 25 & 188 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[569] $1^{13} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[570] $1^{14}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 5 & 5 & 40 & 5 \\ 3 & 5 & 0 & 28 & 2 \\ 20 & 40 & 28 & 254 & 32 \\ 3 & 5 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[571] $1^{14}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[572] $2^12^131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 12 & 20 & 14 & 62 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[573] $1^{11}31^{13}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 348$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[574] $21^{14}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 358$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 28 & 5 \\ 10 & 13 & 40 & 7 \\ 28 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[575] $1^{14}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 360$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[576] $1^{14}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 363$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 35 & 5 \\ 9 & 13 & 54 & 7 \\ 35 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[577] $2^12^131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 364$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 24 & 2 \\ 21 & 35 & 24 & 202 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[578] $1^{13}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 368$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[579] $2^12^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 369$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 11 & 2 \\ 9 & 15 & 11 & 32 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[580] $1^131^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0} \\
[581] \quad & 1^{13} 31^{14} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0} \\
[582] \quad & 1^{13} 3^{13} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 21 & 0 & 205 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[583] \quad & 1^{14} 31^{12} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0} \\
[584] \quad & 1^{14} 31^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0} \\
[585] \quad & 1^{14} 3^{12} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 24 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0} \\
[586] \quad & 1^{13} 3^{13} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0} \\
[587] \quad & 1^{14} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0} \\
[588] \quad & 1^{14} 3^{11} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 40 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 40 & 0 & 254 & 32 \\ 3 & 5 & -2 & 32 & 3 \end{pmatrix}_{b_0} \\
[589] \quad & 1^{11} 31^{14} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 408, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0} \\
[590] \quad & 1^{14} 3^{11} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 408, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & -2 & 20 & 2 \end{pmatrix}_{b_0} \\
[591] \quad & 31^{11} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[592] $1^{14} 3^{13} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 412$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 36 & 5 \\ 9 & 13 & 54 & 7 \\ 36 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$

[593] $1^{13} 3^{11} 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 420$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$

[594] $1^{12} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$

[595] $1^{12} 31^{12} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$

[596] $1^{14} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 435$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 37 & 5 \\ 9 & 13 & 54 & 7 \\ 37 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$

[597] $1^{12} 31^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$

[598] $1^{12} 3^{11} 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 439$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 22 & 2 \\ 15 & 30 & 22 & 134 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$

[599] $1^{11} 31^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 444$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$

[600] $1^{14} 3^{13} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 446$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 16 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$

[601] $1^{13} 3^{11} 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 447$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 35 & 0 & 188 & 28 \\ 3 & 5 & -2 & 28 & 3 \end{pmatrix}_{b_0}$

[602] $1^{13} 31^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 448$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[603] $1^{13}3^{14}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 3 \\ 2 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 25 & 0 & 313 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[604] $1^{13}31^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[605] $1^{13}31^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[606] $21^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 470, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 30 & 5 \\ 10 & 13 & 44 & 7 \\ 30 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[607] $2^{114}2^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 16 & 2 \\ 12 & 20 & 16 & 62 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[608] $1^{14}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[609] $1^{13}3^{14}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[610] $1^{14}3^{13}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 27 & 3 \\ 2 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 28 & 0 & 366 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[611] $1^{11}31^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[612] $1^{13}3^{14}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 21 & 0 & 205 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[613] $1^{13}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 515, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 25 & 2 \\ 18 & 35 & 25 & 188 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[614] $1^{14}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 519, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 5 & 5 & 40 & 5 \\ 3 & 5 & 0 & 28 & 2 \\ 21 & 40 & 28 & 254 & 32 \\ 3 & 5 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[616] $1^{13}31^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[615] $21^{15}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 35 & 5 \\ 10 & 13 & 50 & 7 \\ 35 & 50 & 173 & 25 \\ 5 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

[617] $1^{14}31^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[618] $1^{14}31^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[619] $1^{14}3^{13}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 538, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[620] $1^{11}31^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[621] $2^{113}2^231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 6 & 2 \\ 2 & 1 & 4 & 1 & 9 & 3 \\ 2 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -2 & 3 & 0 \\ 6 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[622] $1^{14}3^{13}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 558, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[623] $1^{13}31^{15}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 30 & 15 & 0 & 148 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[624] $1^{12}31^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[625] $1^{14}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 564$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[626] $1^{14}321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 584$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[627] $1^{13}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 604$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[628] $2^2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 605$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 7 & 2 \\ 2 & 1 & 4 & 1 & 10 & 3 \\ 2 & 4 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -2 & 5 & 0 \\ 7 & 10 & 10 & 5 & 33 & 11 \\ 2 & 3 & 2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[629] $2^2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 625$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 15 & 2 \\ 12 & 20 & 15 & 58 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[630] $2^2 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 649$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 18 & 30 & 21 & 142 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[631] $1^{12}31^{13}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 660$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[632] $1^{14}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 662$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[633] $1^{11}31^{14}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 664$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[634] $1^{11}31^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 664$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[635] $1^{14}321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 678$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 28 & 3 \\ 2 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 28 & 0 & 366 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[636] \ 21^{14} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 37 & 5 \\ 10 & 13 & 54 & 7 \\ 37 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[637] \ 1^{14} 3 \ 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 706, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[638] \ 21^{16} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 42 & 5 \\ 10 & 13 & 60 & 7 \\ 42 & 60 & 250 & 30 \\ 5 & 7 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[639] \ 1^{12} 31^{13} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[640] \ 1^{13} 31^{13} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[641] \ 2 \ 2 \ 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 18 & 2 \\ 15 & 25 & 18 & 94 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[642] \ 1^{12} 31^{13} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[643] \ 1^{13} 1^{16} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 36 & 18 & 0 & 214 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[644] \ 1^{11} 31^{14} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 792, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[645] \ 1^{11} 31^{14} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 792, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[646] \ 1^{13} 31^{13} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 828, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0} \\
[647] & 1^{14}31^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[648] & 2^2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 856, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 8 & 2 \\ 2 & 1 & 4 & 1 & 12 & 3 \\ 2 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -2 & 4 & 0 \\ 8 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0} \\
[650] & 1^{12}31^{14}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 864, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0} \\
[649] & 1^{12}31^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 864, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0} \\
[651] & 1^{13}31^{13}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 892, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0} \\
[652] & 1^{13}31^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[653] & 2^2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 929, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 19 & 2 \\ 15 & 25 & 19 & 92 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[654] & 1^{14}31^{13}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0} \\
[655] & 21^{17}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.3 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 49 & 5 \\ 10 & 13 & 70 & 7 \\ 49 & 70 & 341 & 35 \\ 5 & 7 & 35 & 3 \end{pmatrix}_{b_0} \\
[656] & 1^{13}31^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[657] & 2^2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1000, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 9 & 2 \\ 2 & 1 & 4 & 1 & 13 & 3 \\ 2 & 4 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -2 & 6 & 0 \\ 9 & 13 & 13 & 6 & 56 & 14 \\ 2 & 3 & 2 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[658] $1^{14}31^{13}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1012$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[659] $1^{14}31^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1024$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[661] $1^{14}31^{13}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1044$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[660] $1^131^{17}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1044$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.3$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 42 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 42 & 21 & 0 & 292 & 28 \\ 2 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[663] $1^{12}31^{14}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1056$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[662] $1^{12}31^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1056$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[664] $2^114 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 1076$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 24 & 3 \\ 4 & 5 & 5 & 40 & 5 \\ 3 & 5 & 0 & 28 & 2 \\ 24 & 40 & 28 & 254 & 32 \\ 3 & 5 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[665] $1^{12}31^{14}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[666] $2^114 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 1176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 22 & 2 \\ 18 & 30 & 22 & 134 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[667] $2^114 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 1225$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 25 & 2 \\ 21 & 35 & 25 & 188 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[669] $1^131^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

[illegible]

$$\begin{aligned}
& T = 9 \quad [679] \ 1^{\overline{111}31^3} \ 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 8 & 4 & 5 & 18 & 8 \\ 3 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
& [680] \ 1^{\overline{11}} 4 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
& [681] \ 1^{\overline{11}} 2321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 6 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
& [682] \ 21^{\overline{14}} 1^{\overline{13}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 1 & 2 & 4 \\ 2 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
& [683] \ 21^{\overline{14}} 1^{\overline{14}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 1 & 15 & 4 \\ 2 & 1 & -2 & 2 & 0 \\ 12 & 15 & 2 & 54 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0} \\
& [684] \ 1^{\overline{11}} 2321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
& [685] \ 1^{\overline{11}} 2321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -2 & 3 & 3 \end{pmatrix}_{b_0} \\
& [687] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 2 & 4 & 4 & 4 \\ 4 & 4 & 3 & 4 \\ 4 & 3 & 1 & 3 \\ 4 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
& [686] \ 1^{\overline{11}} 4 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 4 & 5 & 1 & 5 & 5 \\ 3 & 4 & -1 & 5 & 3 \end{pmatrix}_{b_0} \\
& [688] \ 1^{\overline{11}} 2321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0} \\
& [689] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[690] $1^1 4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 3 & 1 & 5 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[692] $1^1 41^{11} 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 5 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 5 & 6 & 2 & 2 & 8 & 6 \\ 2 & 4 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$

[693] $1^{11} 2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 2 & 1 & 8 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 22 & 10 \\ 3 & 4 & -1 & 10 & 4 \end{pmatrix}_{b_0}$

[691] $2^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 13 & 6 & 6 & 7 \\ 3 & 6 & 0 & 4 & 2 \\ 3 & 6 & 4 & 2 & 4 \\ 3 & 7 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[694] $1^1 4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$

[695] $1^{11} 2^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[696] $21^1 41^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

[697] $1^1 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

[698] $1^{11} 4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -17$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 3 & 4 & 1 & 2 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$

[699] $1^1 4^1 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$

[700] $1^{11} 4^1 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 7 & 1 & 9 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[701] $1^{\overline{11}1}4^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[702] $1^{\overline{11}1}4^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 3 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[703] $1^{\overline{11}1}4^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 2 & 1 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[706] $1^{\overline{11}1}4^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[705] $21^{\overline{11}1}4131^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 1 & 7 & 4 \\ 2 & 1 & -2 & 2 & 0 \\ 4 & 7 & 2 & 6 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[704] $1^{\overline{11}1}4131^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[707] $1^{\overline{11}1}4131^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[708] $1^{\overline{11}1}4^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[709] $21^{\overline{11}1}41^{\overline{11}1}31^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 3 & 2 & 6 & 1 & 5 & 5 \\ 2 & 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[710] $1^{\overline{11}1}4^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[711] $21^{\overline{11}1}41^{\overline{11}1}31^{\overline{11}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 0 & 3 & 0 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[712] $1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 4 & 2 \\ 4 & 5 & 4 & 5 & 5 \\ 3 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[713] $1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$

[714] $1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 4 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 4 & 5 & 1 & 5 & 5 \\ 3 & 4 & -1 & 5 & 3 \end{pmatrix}_{b_0}$

[715] $21^{11} 41^3 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 1 & 13 & 4 \\ 2 & 1 & -2 & 2 & 0 \\ 10 & 13 & 2 & 36 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$

[716] $1^{13} 2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$

[717] $1^{12} 2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 2 & 1 & 12 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 42 & 14 \\ 3 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$

[719] $1^{11} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$

[718] $2^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 13 & 6 & 8 & 7 \\ 3 & 6 & 0 & 3 & 2 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 7 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[720] $1^{11} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 4 & 3 \\ 3 & 0 & 4 & 2 & 4 \\ 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[721] $1^1 4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[724] $1^{11} 2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 4 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[725] $1^{114}2321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 8 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[726] $1^{1112}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 4 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[723] $1^{111}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

[722] $1^{111}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[727] $1^{113}2321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 6 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[728] $1^{1311}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 8 & 0 & 13 & 7 & 7 \\ 3 & 4 & -1 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

[729] $1^{11}41^{1231} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 & 3 & 3 \\ 6 & 13 & 0 & 8 & 7 & 7 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 0 & 4 & 4 & 4 \\ 3 & 7 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[730] $21^{11}41^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 4 & 1 & 9 & 4 & 4 \\ 2 & 1 & -2 & 2 & 0 & 0 \\ 6 & 9 & 2 & 12 & 8 & 8 \\ 3 & 4 & 0 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[731] $21^{11}41^{231} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 & 3 \\ 3 & 4 & 1 & 11 & 4 & 4 \\ 2 & 1 & -2 & 2 & 0 & 0 \\ 8 & 11 & 2 & 22 & 10 & 10 \\ 3 & 4 & 0 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

[732] $1^{13}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

[733] $1^{13}2321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 20 & 0 & 130 & 24 \\ 3 & 4 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[734] $1^{11} 4^{131^2} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 10 & 0 & 21 & 9 \\ 3 & 4 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[735] $1^{11} 4^{131} 14 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 6 & 1 & 32 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[736] $1^{11} 4^{131} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[737] $1^{11} 4^{131} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[739] $1^{12} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & -2 & 9 & 3 \end{pmatrix}_{b_0}$$

[738] $1^{11} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 7 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[742] $1^{13} 4^{131^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 4 & 1 & 18 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[744] $1^{11} 4^{131} 11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[740] $1^{12} 4^{131} 11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 5 & 8 & 1 & 11 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[743] $1^{13} 4^{131^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[741] $1^{11} 4^{131^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 2 & 1 & 2 & 4 \\ 3 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[745] \ 1^{11} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[746] \ 2^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 5 & 5 & 7 & 5 \\ 3 & 5 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[747] \ 1^{12} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 13 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[748] \ 1^{11} 31^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 4 & 4 & 3 & 2 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[749] \ 1^{12} 31^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 3 & 1 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[751] \ 1^{11} 31^{12} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 6 & 3 & 4 & 8 & 6 \\ 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[750] \ 2^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 3 & 6 & 3 \\ 6 & 13 & 6 & 14 & 7 \\ 3 & 6 & 0 & 7 & 2 \\ 6 & 14 & 7 & 14 & 8 \\ 3 & 7 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[752] \ 1^{11} 31^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[753] \ 1^{11} 31^{13} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 0 & 2 & 6 & 2 \\ 3 & 2 & 0 & 6 & 2 \\ 9 & 6 & 6 & 26 & 10 \\ 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[754] \ 1^{11} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 3 & 3 & 2 & 4 \\ 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[755] $1^{\overline{1}131^{\overline{1}1}} 4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[756] $1^{\overline{1}1} 4^{\overline{1}1} 131^{\overline{1}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 8 & 12 & 2 & 29 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$

[757] $1^{\overline{1}1} 4^{\overline{1}1} 131^{\overline{1}3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 5 & 1 & 18 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[758] $1^{\overline{1}14} 2321^{\overline{1}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$

[759] $1^{\overline{1}1} 4^{\overline{1}1} 131^{\overline{1}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[760] $1^{\overline{1}1} 4^{\overline{1}1} 131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 9 & 1 & 14 & 8 \\ 3 & 4 & -1 & 8 & 3 \end{pmatrix}_{b_0}$

[761] $1^{\overline{1}11} 31^{\overline{1}1} 4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 4 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 4 & 2 & 3 & 2 & 4 \\ 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[762] $1^{\overline{1}11} 2^{\overline{1}1} 3^{\overline{1}1} 21^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 3 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$

[763] $1^{\overline{1}1} 31^{\overline{1}1} 4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

[764] $1^{\overline{1}12} 2^{\overline{1}1} 3^{\overline{1}1} 21^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 9 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[765] $31^{\overline{1}1} 31^{\overline{1}1} 41^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 6 & 4 & 8 & 3 \\ 6 & 13 & 6 & 20 & 7 \\ 4 & 6 & 0 & 8 & 2 \\ 8 & 20 & 8 & 26 & 10 \\ 3 & 7 & 2 & 10 & 2 \end{pmatrix}_{b_0}$

[766] $1^{13}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 7 & 12 & 1 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[768] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 3 & 2 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[767] $1^{11}4^{11}231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 6 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 12 & 1 & 9 & 7 \\ 3 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[769] $1^{12}2^3 3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 2 & 1 & 12 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 42 & 14 \\ 3 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$

[771] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 8 & 9 & 7 & 17 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[770] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 6 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[773] $1^{12}4^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 14 & 17 & 1 & 62 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$

[772] $31^{11}31^{11}41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 4 & 4 & 2 & 3 & 4 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 4 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$

[775] $1^{12}2^3 3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 8 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 8 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[774] $1^{12}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 1 & 2 & 7 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$

[776] $1^{12}2^3 3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 2 & -2 & 12 & 3 \end{pmatrix}_{b_0}$

[777] $1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 21 & 9 \\ 3 & 4 & -1 & 9 & 3 \end{pmatrix}_{b_0}$

[779] $1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 4 & 0 & 3 & 12 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$

[778] $1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 5 & 0 & 3 & 12 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$

[780] $1^{13} 2^3 3^{21} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -85$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 15 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$

[781] $1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -87$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 8 & 11 & 1 & 22 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[784] $1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$

[783] $1^{13} 3^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 3 & 3 & 2 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$

[782] $1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 7 & 1 & 9 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[785] $1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 12 & 1 & 27 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$

[786] $1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -95$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 12 & 15 & 1 & 46 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$

[789] $1^{12} 4^{11} 13^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 9 & 13 & 2 & 32 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[788] $1^{11} 4^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[787] $1^{12} 4^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 14 & 1 & 39 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[791] $2^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 & 2 \\ 3 & 4 & 4 & 2 & 3 & 4 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 4 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[790] $1^{11} 4^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 7 & 5 & 5 & 14 & 8 \\ 3 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[792] $1^{11} 4^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 5 & 4 & 4 & 6 & 6 \\ 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[793] $1^{11} 4^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 4 & 2 & 6 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[794] $1^{13} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

[795] $1^{11} 4^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[796] $1^{11} 4^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 3 & 3 & 2 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[797] $1^{13} 4^{13} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 1 & 3 & 17 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[798] $31^{13} 131^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$\begin{aligned}
& T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 6 & 4 & 5 & 3 \\ 6 & 13 & 6 & 12 & 7 \\ 4 & 6 & 0 & 5 & 2 \\ 5 & 12 & 5 & 8 & 6 \\ 3 & 7 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[799] & 21^{11} 41^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 3 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 4 & 8 & 0 & 10 & 6 \\ 2 & 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0} \\
[800] & 1^{14} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 20 & 0 & 106 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0} \\
[801] & 1^{13} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 20 & 0 & 130 & 24 \\ 3 & 4 & -1 & 24 & 4 \end{pmatrix}_{b_0} \\
[802] & 1^{13} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & -2 & 15 & 3 \end{pmatrix}_{b_0} \\
[803] & 1^{12} 4 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 8 & 13 & 1 & 29 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0} \\
[804] & 1^{14} 4 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 10 & 16 & 2 & 50 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0} \\
[805] & 1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 15 & 0 & 40 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[806] & 1^{12} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 4 & 2 & 14 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0} \\
[807] & 1^{12} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 38 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[808] & 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 18 & 0 & 70 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0} \\
[809] & 1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8
\end{aligned}$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 11 & 15 & 2 & 44 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[810] \ 1^{12} 4^{11} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 4 & 2 & 2 & 10 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[811] \ 1^{11} 4^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 3 & 2 & 2 & 8 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[813] \ 1^{11} 4^{13} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 15 & 0 & 45 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[814] \ 1^{13} 4^{11} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 12 & 17 & 3 & 58 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[812] \ 1^{12} 4^{13} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 17 & 0 & 61 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[815] \ 21^{11} 41^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 5 & 2 \\ 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 3 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 5 & 4 & 8 & 1 & 11 & 7 \\ 2 & 2 & 3 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[816] \ 1^{11} 4^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 7 & 3 \\ 4 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 7 & 9 & 6 & 14 & 8 \\ 3 & 4 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[820] \ 1^{11} 4^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 10 & 12 & 2 & 30 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[819] \ 1^{11} 4^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[818] \ 1^{12} 4^{13} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 3 & 2 & 15 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[817] $1^{112}2^13^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 5 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[823] $1^{12}2^13^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 14 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$

[822] $1^{12}2^13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 10 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[821] $1^{14}2^13^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & -2 & 18 & 3 \end{pmatrix}_{b_0}$

[824] $1^{14}4^1131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -138$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 15 & 21 & 4 & 92 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$

[825] $1^{14}4^11^13^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -139$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 14 & 19 & 3 & 74 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$

[826] $1^{13}4^11^12^13^131^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -142$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 17 & 21 & 2 & 92 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$

[828] $1^{12}4^11^12^13^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 12 & 16 & 2 & 49 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$

[829] $1^{13}4^11^12^13^12^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 14 & 18 & 2 & 65 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[827] $1^{11}4^11^12^13^12^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 5 & 2 & 24 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[830] $1^{14}2^13^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 2 & 1 & 28 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 28 & 0 & 266 & 34 \\ 3 & 4 & -1 & 34 & 4 \end{pmatrix}_{b_0}$

[831] $1^{12} 4^{1^{12}31^3} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -149$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 18 & 22 & 2 & 101 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[832] $1^{11} 4^{1^{11}31^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 7 & 5 & 3 & 14 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[833] $1^{141} 1^{11} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 3 & 6 & 0 & 2 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[834] $1^{11} 4^{1^{11}31^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 4 & 1 & 3 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[836] $1^{12} 4^{1^{11}31^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 6 & 0 & 4 & 22 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[835] $1^{11} 4^{1^{11}31^{11}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[839] $1^{13} 4^{1^{11}31^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -154$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 15 & 1 & 41 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[837] $1^{11} 4^{1^{12}31^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -154$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 5 & 2 & 12 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[838] $1^{14} 4^{1^{14}31^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -154$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 6 & 2 & 38 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[841] $1^{12} 4^{1^{12}31^{11}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 3 & 2 & 2 & 8 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[844] $1^{12} 2^3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 6 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[843] $1^{13}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 12 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[842] $1^{13}2^3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 2 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

[840] $1^{13}2^3 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 20 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

[846] $1^{14}1^{11} 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 5 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 8 & 6 \\ 3 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[845] $1^{12}4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 45 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[847] $1^{12}31^2 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 15 & 19 & 2 & 72 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[848] $31^1 31^{11} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 4 & 4 & 2 & 6 & 4 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 2 & 0 \\ 3 & 6 & 3 & 2 & 4 & 4 \\ 2 & 4 & 2 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[849] $1^{13}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

[850] $1^{14}2^3 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$

[851] $1^{13}4^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 17 & 1 & 50 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[852] $1^{12}31^3 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 5 & 3 & 24 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[853] $1^{12} 4^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 5 & 1 & 3 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[855] $2^3 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[854] $2^3 1^{14} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 5 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 3 & 0 \\ 5 & 6 & 6 & 3 & 13 & 7 \\ 2 & 3 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[856] $1^{14} 4^{13} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 17 & 22 & 3 & 99 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[857] $1^{13} 4^{11} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 11 & 17 & 2 & 53 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[858] $1^{13} 4^{13} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 15 & 20 & 3 & 79 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[859] $1^{11} 4^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 4 & 2 & 3 & 16 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[860] $1^{13} 4^{11} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 6 & 2 & 3 & 20 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[861] $1^{12} 4^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 16 & 1 & 46 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[862] $1^{11} 4^{13} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 6 & 2 & 22 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[863] $1^{12}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 11 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[864] $1^{11}4^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 9 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 12 & 1 & 27 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[865] $1^{13}2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 16 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[866] $1^{11}41^{11}231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 9 & 3 \\ 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 20 & 1 & 27 & 11 \\ 3 & 7 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[867] $1^{11}4^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 4 & 3 & 3 & 21 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[868] $1^{11}4^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 5 & 1 & 4 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[869] $1^{11}32^3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 7 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[870] $1^{13}4^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -193$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 18 & 0 & 61 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[873] $1^{14}1^{12}231^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -195$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 9 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 9 & 12 & 3 & 4 & 31 & 11 \\ 2 & 4 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[871] $1^{14}1^{11}231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -195$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 4 & 6 & 1 & 2 & 5 & 5 \\ 2 & 4 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[872] $1^{14}4^{13}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -195$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 20 & 25 & 3 & 130 & 24 \\ 3 & 4 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[874] $1^{12} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -205$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 15 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$

[875] $1^{13} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -205$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 13 & 18 & 1 & 62 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$

[876] $1^{11} 4^{14} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 7 & 2 & 36 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$

[877] $1^{14} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 20 & 0 & 106 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$

[878] $1^{11} 41^{11} 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 8 & 3 \\ 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 20 & 0 & 26 & 10 \\ 3 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[880] $1^{13} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 7 & 8 & 0 & 5 & 36 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$

[879] $1^{12} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 8 & 7 & 0 & 5 & 36 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$

[881] $1^{14} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 13 & 19 & 2 & 69 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[883] $1^{13} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 19 & 24 & 1 & 122 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

[882] $1^{14} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 18 & 24 & 4 & 117 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$

[884] $1^{14} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 12 & 19 & 1 & 66 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$

[885] $1^{13} 4^{1^{13} 31^2} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 18 & 23 & 3 & 106 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[886] $1^{14} 2^1 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & -2 & 21 & 3 \end{pmatrix}_{b_0}$$

[887] $1^{13} 4^{1^{13} 31^3} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 21 & 26 & 3 & 139 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[888] $1^{12} 4^{1^{11} 31^4} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 20 & 25 & 1 & 133 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[889] $1^{13} 4^{1^{13} 31^1} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 70 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[890] $1^{14} 4^{1^{11}} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 12 & 21 & 1 & 77 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[891] $1^{13} 4^{1^{11} 31^3} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 16 & 21 & 1 & 89 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[892] $1^{12} 4^{1^{11} 31^2} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 19 & 0 & 68 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[893] $1^{14} 2^1 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 22 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

[894] $1^{14} 2^1 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 18 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[895] $1^{12} 4^{1^{11} 31^2} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -230$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 14 & 19 & 1 & 69 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

- [896] $1^{12} 4^{1^{12} 31^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 5 & 2 & 3 & 16 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$
- [897] $1^{1^{13} 31^3} 4^{1^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -238$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 6 & 3 & 34 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$
- [899] $1^{13} 4^{1^{13} 31^3} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -238$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 21 & 0 & 88 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$
- [898] $1^{14} 4^{1^{11}} 131^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -238$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 14 & 21 & 3 & 85 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$
- [900] $1^{11} 4^{1^{12} 31^2} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -239$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 6 & 3 & 20 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$
- [902] $1^{13} 4^{1^1} 131^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 6 & 3 & 3 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [901] $1^{13} 4^{1^{11} 31^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 7 & 1 & 4 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [903] $1^{12} 4^{1^{11} 31^3} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 17 & 22 & 1 & 98 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$
- [904] $1^{13} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 13 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$
- [906] $1^{14} 4^{1^1} 131^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 8 & 2 & 4 & 34 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$
- [905] $1^{11} 4^{1^{12} 31^3} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 5 & 2 & 4 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[907] $1^{13} 4^{11} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -253, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 24 & 0 & 121 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[908] $1^{14} 2^3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 9 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[909] $1^{14} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$

[910] $1^{12} 4^{14} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 5 & 4 & 3 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[911] $1^{13} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 6 & 4 & 3 & 32 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[912] $1^{12} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 22 & 0 & 97 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[913] $1^{12} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 5 & 3 & 3 & 21 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[914] $1^{12} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 6 & 1 & 4 & 23 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[915] $1^{14} 4^{11} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 90 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[916] $1^{14} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 13 & 21 & 2 & 80 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[917] $1^{12} 1^{14} 31^4 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -266$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 25 & 0 & 132 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

[918] $1^{14} 1^{14} 31^2 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -266$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 21 & 27 & 4 & 148 & 26 \\ 3 & 4 & -1 & 26 & 3 \end{pmatrix}_{b_0}$

[919] $1^{14} 1^{12} 231^1 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -270$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 6 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 6 & 12 & 0 & 4 & 22 & 8 \\ 2 & 4 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$

[920] $1^{13} 1^{12} 31^1 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -270$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 14 & 20 & 2 & 74 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$

[921] $1^{11} 1^{13} 31^3 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 10 & 5 & 3 & 4 & 33 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$

[922] $1^{12} 1^{14} 3 21^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 16 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$

[923] $1^{14} 1^{12} 321^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 2 & -2 & 24 & 3 \end{pmatrix}_{b_0}$

[924] $1^{14} 1^{14} 31^4 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -274$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 27 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 27 & 33 & 4 & 228 & 32 \\ 3 & 4 & -1 & 32 & 3 \end{pmatrix}_{b_0}$

[925] $1^{14} 1^{12} 231^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 11 & 3 \\ 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 24 & 1 & 39 & 13 \\ 3 & 7 & -1 & 13 & 3 \end{pmatrix}_{b_0}$

[926] $1^{13} 1^{13} 3 21^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -277$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 21 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$

[927] $1^{14} 1^{13} 31^1 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -281$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 24 & 0 & 101 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$

[928] $1^{13} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -282$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 13 & 20 & 1 & 71 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[929] $1^{14} 4^{12} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -282$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 16 & 22 & 2 & 94 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[930] $1^{14} 4^{13} 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -283$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 7 & 3 & 48 & 14 \\ 3 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[931] $1^{14} 4^{14} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -285$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 24 & 30 & 4 & 185 & 29 \\ 3 & 4 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[932] $1^{13} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 8 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[933] $1^{13} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 17 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[934] $1^{14} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 7 & 9 & 1 & 5 & 41 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[935] $1^{11} 4^{12} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -300$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 11 & 6 & 2 & 5 & 44 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[936] $1^{14} 1^{12} 231^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 10 & 3 \\ 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 24 & 0 & 38 & 12 \\ 3 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[937] $1^{14} 4^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 15 & 22 & 1 & 91 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[938] $1^{11} 4^{13} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 7 & 3 & 30 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

- [939] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -312$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 9 & 9 & 0 & 6 & 52 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [940] $1^{11} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -318$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 7 & 4 & 32 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$
- [941] $1^{14} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -319$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 19 & 25 & 2 & 125 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$
- [942] $1^{13} 2^3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 26 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 26 & 0 & 28 & 322 & 36 \\ 3 & 0 & 3 & 36 & 4 \end{pmatrix}_{b_0}$$
- [943] $1^{13} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 14 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$
- [944] $1^{14} 2^3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -325$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 27 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 32 & 3 \\ 27 & 0 & 32 & 382 & 40 \\ 3 & 0 & 3 & 40 & 4 \end{pmatrix}_{b_0}$$
- [945] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -325$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 7 & 4 & 48 & 14 \\ 3 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$
- [946] $1^{14} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -326$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 22 & 28 & 2 & 162 & 26 \\ 3 & 4 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$
- [948] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -327$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 17 & 23 & 2 & 101 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$
- [947] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -327$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 23 & 0 & 97 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$
- [949] $1^{14} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -341$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 17 & 24 & 3 & 110 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[950] $1^{12} 4^{1^{12} 31^2} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 6 & 2 & 4 & 26 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[951] $1^{12} 4^{1^{11} 31^3} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 9 & 7 & 1 & 5 & 37 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[952] $1^{11} 4^{1^{13} 31^4} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 12 & 6 & 3 & 5 & 49 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[953] $1^{13} 4^{1^{12} 31^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 7 & 2 & 4 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[954] $1^{14} 4^{13} 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 8 & 3 & 4 & 39 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[955] $1^{13} 4^{1^{12} 31^4} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 23 & 29 & 2 & 173 & 27 \\ 3 & 4 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[956] $1^{14} 4^{1^{13} 31^3} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 25 & 0 & 121 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[957] $1^{13} 4^{1^{12} 31^3} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -354$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 20 & 26 & 2 & 134 & 24 \\ 3 & 4 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[958] $1^{14} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -357$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 19 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[959] $1^{13} 4^{1^{11} 31^2} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -363$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 16 & 23 & 1 & 98 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[960] $1^{11} 4^{1^{14} 31^2} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -363$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 8 & 3 & 44 & 14 \\ 3 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[961] $1^{13} 4^{11} 31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -368$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 8 & 8 & 1 & 5 & 37 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[962] $1^{14} 4^{11} 31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -371$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 18 & 25 & 1 & 122 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[963] $1^{14} 2^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 23 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

[964] $1^{14} 4^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 15 & 24 & 1 & 102 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[965] $1^{13} 2^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 22 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

[966] $1^{13} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 18 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[967] $1^{12} 4^{11} 31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 6 & 3 & 4 & 31 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[968] $1^{14} 4^{11} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -393$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 16 & 24 & 2 & 105 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[970] $1^{14} 4^{11} 31^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -402$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 21 & 28 & 1 & 159 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[969] $1^{13} 4^{11} 31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -402$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 26 & 0 & 130 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[971] $1^{13} 4^{11} 31^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 11 & 10 & 0 & 7 & 72 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[972] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 10 & 11 & 0 & 7 & 72 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[973] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 28 & 0 & 158 & 24 \\ 3 & 4 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[974] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 8 & 4 & 62 & 16 \\ 3 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[975] \ 1^{14} 4 \ 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 8 & 4 & 4 & 46 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[976] \ 1^{12} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 10 & 6 & 4 & 4 & 38 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[977] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 20 & 27 & 3 & 141 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[978] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 19 & 26 & 1 & 131 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[981] \ 1^{11} 4 \ 2 \ 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 10 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[979] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 7 & 3 & 4 & 33 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[980] \ 1^{12} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 11 & 8 & 1 & 6 & 55 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[982] $1^{11}1^{13}31^3 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 8 & 4 & 42 & 14 \\ 3 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[983] $1^{14}1^{13}31^2 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 27 & 0 & 132 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[984] $1^{13}1^{11}31^4 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 22 & 29 & 1 & 170 & 26 \\ 3 & 4 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[985] $31^{11}31^{11}41^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 6 & 2 \\ 3 & 4 & 4 & 2 & 9 & 4 \\ 3 & 4 & 0 & 1 & 6 & 2 \\ 1 & 2 & 1 & -2 & 3 & 0 \\ 6 & 9 & 6 & 3 & 13 & 7 \\ 2 & 4 & 2 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[987] $1^{14}1^{12}31^1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 8 & 9 & 2 & 5 & 44 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[986] $1^{12}1^{12}31^3 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 10 & 7 & 2 & 5 & 40 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[988] $1^{13}1^{13}31^4 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -447, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 29 & 0 & 169 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[989] $1^4 41^{12} 231^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 8 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 8 & 12 & 2 & 4 & 26 & 10 \\ 2 & 4 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[990] $1^{13}1^{14}31^1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 7 & 4 & 4 & 40 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[991] $1^{14}1^{13}31^3 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -457, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 23 & 30 & 3 & 178 & 28 \\ 3 & 4 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[992] $1^{14}1^{11}31^2 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$\begin{aligned}
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 9 & 10 & 1 & 6 & 55 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0} \\
[993] & 1^{14} 41^{13} 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -470, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 9 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 9 & 18 & 0 & 6 & 52 & 12 \\ 2 & 4 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0} \\
[994] & 1^{14} 4^{13} 31^4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -470, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 26 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 26 & 33 & 3 & 221 & 31 \\ 3 & 4 & -1 & 31 & 3 \end{pmatrix}_{b_0} \\
[995] & 1^{14} 41^{12} 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -475, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 7 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 7 & 12 & 1 & 4 & 23 & 9 \\ 2 & 4 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0} \\
[997] & 1^{14} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 1 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 20 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0} \\
[996] & 1^{14} 2^3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 1 & 2 & 28 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 32 & 3 \\ 28 & 0 & 32 & 382 & 40 \\ 3 & 0 & 3 & 40 & 4 \end{pmatrix}_{b_0} \\
[998] & 1^{14} 41^{12} 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 1 & 13 & 3 \\ 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 32 & 0 & 68 & 16 \\ 3 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0} \\
[999] & 1^{13} 4^{11} 31^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 10 & 9 & 1 & 6 & 53 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0} \\
[1000] & 1^{14} 4^{12} 31^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -490, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 19 & 27 & 2 & 136 & 24 \\ 3 & 4 & -1 & 24 & 3 \end{pmatrix}_{b_0} \\
[1001] & 1^{13} 4^{12} 31^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 9 & 8 & 2 & 5 & 40 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0} \\
[1002] & 1^{14} 4^{11} 31^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 18 & 27 & 1 & 133 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[1004] & 1^{14} 2^3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0
\end{aligned}$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 24 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 24 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[1003] \ 1^{12} 4 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 11 & 7 & 3 & 5 & 45 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1005] \ 1^{11} 4 \ 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -514, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 9 & 4 & 56 & 16 \\ 3 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1006] \ 1^{11} 41^{12} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -521, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 14 & 3 \\ 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 14 & 32 & 1 & 69 & 17 \\ 3 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1007] \ 1^{11} 4 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -534, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 5 & 0 \\ 15 & 9 & 5 & 58 & 16 \\ 3 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1008] \ 1^{12} 4 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 12 & 8 & 2 & 6 & 58 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1009] \ 1^{14} 4 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 12 & 12 & 0 & 8 & 94 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1010] \ 1^{14} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -541, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 30 & 0 & 169 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1011] \ 1^{12} 4 \ 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 12 & 7 & 4 & 5 & 52 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1012] \ 1^{14} 1^{13} 231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 13 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 13 & 18 & 4 & 6 & 68 & 16 \\ 2 & 4 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1013] \ 1^{14} 1^{13} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 15 & 3 \\ 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 36 & 0 & 88 & 18 \\ 3 & 7 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1014] \ 1^{14} 4 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -557, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 22 & 30 & 2 & 173 & 27 \\ 3 & 4 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1015] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 9 & 9 & 3 & 5 & 49 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1016] \ 1^{14} 1^3 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 16 & 3 \\ 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 16 & 36 & 1 & 89 & 19 \\ 3 & 7 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1017] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 10 & 8 & 3 & 5 & 45 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1018] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -585, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 21 & 30 & 1 & 170 & 26 \\ 3 & 4 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1019] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -594, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 25 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 25 & 33 & 2 & 216 & 30 \\ 3 & 4 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[1020] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 12 & 10 & 1 & 7 & 73 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1022] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 11 & 11 & 1 & 7 & 73 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1021] \ 1^{12} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 13 & 8 & 3 & 6 & 63 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1023] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 33 & 0 & 212 & 28 \\ 3 & 4 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[1024] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -632, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 10 & 10 & 2 & 6 & 58 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[1025] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -632$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 11 & 9 & 2 & 6 & 56 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[1026] $1^{14} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -644$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 10 & 9 & 4 & 5 & 56 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[1027] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -644$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 11 & 8 & 4 & 5 & 52 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[1028] $1^{14} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -646$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 24 & 33 & 1 & 213 & 29 \\ 3 & 4 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[1029] $1^{11} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -659$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 5 & 0 \\ 17 & 10 & 5 & 72 & 18 \\ 3 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1030] $1^{12} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -680$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 14 & 8 & 4 & 6 & 70 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1031] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 12 & 9 & 3 & 6 & 61 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[1032] $1^{14} 4^1 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -750$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 12 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 12 & 24 & 0 & 8 & 94 & 16 \\ 2 & 4 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1033] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -764$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 13 & 10 & 2 & 7 & 76 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1034] $1^{14} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -768$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 11 & 10 & 3 & 6 & 63 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[1035] $1^{14} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -776$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 13 & 12 & 1 & 8 & 95 & 17 \\ 2 & 2 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

[1036] $1^{14} 4^1 1^{1231^3} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -812$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 12 & 11 & 2 & 7 & 76 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1037] $1^{13} 4^1 1^{1431^3} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -824$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 13 & 9 & 4 & 6 & 68 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1038] $1^{141} 13^{231} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -835$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 10 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 10 & 18 & 1 & 6 & 53 & 13 \\ 2 & 4 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[1039] $1^{11} 41^{13} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -852$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 18 & 3 \\ 6 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 44 & 0 & 130 & 22 \\ 3 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[1040] $1^{14} 4^1 1^{1431^2} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -872$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 12 & 10 & 4 & 6 & 70 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1041] $1^{141} 13^{231} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -875$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 12 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 12 & 18 & 3 & 6 & 61 & 15 \\ 2 & 4 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[1043] $1^{13} 4^1 1^{1331^4} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 14 & 10 & 3 & 7 & 81 & 17 \\ 2 & 2 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

[1042] $1^{141} 14^{231} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 20 & 3 \\ 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 48 & 0 & 158 & 24 \\ 3 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[1044] $1^{11} 41^{13} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -948$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 19 & 3 \\ 6 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 19 & 44 & 1 & 131 & 23 \\ 3 & 7 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[1045] $1^{14} 4^1 1^{1331^3} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -968$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 13 & 11 & 3 & 7 & 81 & 17 \\ 2 & 2 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

[1046] $1^{141} 13^{231} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -970$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 11 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 11 & 18 & 2 & 6 & 56 & 14 \\ 2 & 4 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[1047] $1^{14} 41^{14} 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -976$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 21 & 3 \\ 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 21 & 48 & 1 & 159 & 25 \\ 3 & 7 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[1048] $1^{14} 4^{13} 31^4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -984$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 14 & 12 & 2 & 8 & 98 & 18 \\ 2 & 2 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[1049] $1^{13} 4^{14} 31^4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -996$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 15 & 10 & 4 & 7 & 88 & 18 \\ 2 & 2 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[1050] $1^{14} 4^{14} 31^3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -1092$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 14 & 11 & 4 & 7 & 88 & 18 \\ 2 & 2 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[1051] $1^{14} 4^{13} 31^4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -1160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 15 & 12 & 3 & 8 & 103 & 19 \\ 2 & 2 & 0 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

[1052] $1^{14} 41^{14} 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 13 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 13 & 24 & 1 & 8 & 95 & 17 \\ 2 & 4 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

[1053] $1^{14} 4^{14} 31^4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -1304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 16 & 12 & 4 & 8 & 110 & 20 \\ 2 & 2 & 0 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

[1054] $1^{11} 41^{14} 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1321$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 23 & 3 \\ 6 & 13 & 0 & 56 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 56 & 0 & 212 & 28 \\ 3 & 7 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[1055] $1^{14} 41^{14} 231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1470$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 16 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 16 & 24 & 4 & 8 & 110 & 20 \\ 2 & 4 & 0 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

[1056] $1^{11} 41^{14} 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1473$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 24 & 3 \\ 6 & 13 & 0 & 56 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 24 & 56 & 1 & 213 & 29 \\ 3 & 7 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[1057] $1^{14} 41^{14} 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1570$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 14 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 14 & 24 & 2 & 8 & 98 & 18 \\ 2 & 4 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[1058] $1^{14}41^{14}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1635$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 15 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 15 & 24 & 3 & 8 & 103 & 19 \\ 2 & 4 & 0 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [1059] $231^{11}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1060] $1^{11}51^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 8$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 26 & 10 \\ 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[1061] $1^{11}51^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 17$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[1062] $1^{11}51^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 21$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[1063] $231^{11}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 27$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 5 & 7 & 6 & 9 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1064] $1^{11}41^{11}3141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 33$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 5 & 7 \\ 1 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 5 & 3 & 0 & 1 & 3 \\ 2 & 7 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1065] $1^{14}41^{14}31^{14}41^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 39$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 26 & 10 \\ 3 & 12 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[1066] $1^{11}51^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 41$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 225 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[1067] $1^{11}51^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 43$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 95 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[1068] $1^{12}51^{12}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 46$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.5$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1069] \ 1^1 5^1 1^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 2 & 1 & 3 \\ 7 & 26 & 8 & 0 & 10 \\ 2 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 10 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1070] \ 1^1 1^1 41^1 31^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 49, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 13 & 6 & 0 & 17 & 7 \\ 1 & 6 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 17 & 7 & 0 & 21 & 9 \\ 2 & 7 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1071] \ 2^1 3^1 1^1 5^1 1^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 58, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 7 & 11 & 8 & 22 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1072] \ 1^1 1^1 5^1 1^1 3^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 295 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[1073] \ 1^1 1^1 5^1 1^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 10 & 0 & 43 & 13 \\ 3 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1074] \ 2^1 3^1 1^1 5^1 1^1 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 9 & 15 & 10 & 41 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1075] \ 1^1 1^1 5^1 1^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1076] \ 1^1 1^1 5^1 1^1 2^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 77, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 141 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[1077] \ 1^1 1^1 3221^1 1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 1 & 13 & 3 \\ 9 & 43 & 1 & 65 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 65 & 0 & 95 & 19 \\ 3 & 13 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1078] \ 31^1 4^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 & 2 \\ 3 & 13 & 10 & 0 & 9 & 7 \\ 2 & 10 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 9 & 7 & 0 & 5 & 5 \\ 2 & 7 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1079] \ 1^1 1^1 5^1 1^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 13 & 0 & 66 & 16 \\ 3 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1080] \ 1^{12} 5^1 1^{23} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 17 & 0 & 126 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1081] \ 1^{14} 1^{11} 31^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 6 & 2 \\ 3 & 13 & 6 & 0 & 27 & 7 \\ 1 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 27 & 9 & 0 & 49 & 13 \\ 2 & 7 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1082] \ 1^{14} 1^{11} 3141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 1 & 8 & 3 \\ 8 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 40 & 0 & 38 & 12 \\ 3 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1083] \ 31^{11} 4141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 3 & 2 \\ 3 & 13 & 1 & 9 & 15 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 9 & 0 & 3 & 9 & 3 \\ 3 & 15 & 0 & 9 & 13 & 7 \\ 2 & 7 & -1 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1084] \ 1^{14} 1^{11} 3141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 13 & 6 & 0 & 15 & 7 \\ 1 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 15 & 5 & 0 & 13 & 7 \\ 2 & 7 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1085] \ 1^{13} 5^1 1^{23} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 16 & 0 & 95 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1086] \ 1^{12} 5^1 1^{23} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 24 & 0 & 270 & 32 \\ 3 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[1088] \ 1^{14} 1^{11} 31^{24} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 186, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 1 & 22 & 3 \\ 8 & 38 & 1 & 104 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 104 & 0 & 278 & 32 \\ 3 & 12 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[1087] \ 1^{12} 5^1 1^{23} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 186, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 20 & 0 & 178 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1089] \ 31^{11} 4141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 190, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 4 & 2 \\ 3 & 13 & 1 & 9 & 15 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 9 & 0 & 3 & 9 & 3 \\ 4 & 15 & 0 & 9 & 13 & 7 \\ 2 & 7 & -1 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1090] \ 31^{11} 4141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 193, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$\begin{aligned}
& T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 2 & 1 & 3 & 2 \\ 3 & 13 & 10 & 0 & 15 & 7 \\ 2 & 10 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 15 & 9 & 0 & 13 & 7 \\ 2 & 7 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1091] & 1^1 41^{11} 3141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 195, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 1 & 9 & 3 \\ 8 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 40 & 0 & 38 & 12 \\ 3 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[1094] & 1^{12} 5^1 1^3 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 29 & 3 \\ 2 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 31 & 0 & 470 & 42 \\ 3 & 3 & -1 & 42 & 3 \end{pmatrix}_{b_0} \\
[1093] & 1^{11} 41^{11} 3141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 13 & 6 & 0 & 20 & 7 \\ 1 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 20 & 8 & 0 & 26 & 10 \\ 2 & 7 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1092] & 1^{11} 41^{11} 31^2 41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 10 & 2 \\ 3 & 13 & 6 & 0 & 44 & 7 \\ 1 & 6 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 44 & 16 & 0 & 138 & 22 \\ 2 & 7 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0} \\
[1096] & 1^{14} 41^{11} 31^2 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 211, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 1 & 23 & 3 \\ 8 & 38 & 1 & 104 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 104 & 0 & 278 & 32 \\ 3 & 12 & -1 & 32 & 3 \end{pmatrix}_{b_0} \\
[1095] & 1^{13} 5^1 1^3 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 211, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 20 & 0 & 163 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0} \\
[1097] & 1^{12} 5^1 1^3 231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[1098] & 1^1 41^{11} 31^1 41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 239, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 1 & 15 & 3 \\ 8 & 38 & 1 & 72 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 72 & 0 & 130 & 22 \\ 3 & 12 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[1099] & 1^1 3221^{11} 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 1 & 9 & 3 \\ 9 & 43 & 1 & 45 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 45 & 0 & 43 & 13 \\ 3 & 13 & -1 & 13 & 3 \end{pmatrix}_{b_0} \\
[1100] & 1^1 41^{11} 31^1 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 1 & 16 & 3 \\ 8 & 38 & 1 & 72 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 72 & 0 & 130 & 22 \\ 3 & 12 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[1101] & 1^{13} 5^1 1^2 231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8
\end{aligned}$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 23 & 0 & 221 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[1102] \ 1^{13} 5^1 1^2 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 321, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 27 & 0 & 321 & 35 \\ 3 & 3 & -1 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[1103] \ 1^{11} 41^{11} 31^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 7 & 2 \\ 3 & 13 & 6 & 0 & 32 & 7 \\ 1 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 32 & 12 & 0 & 70 & 16 \\ 2 & 7 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1104] \ 31^{11} 4 \ 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 5 & 2 \\ 3 & 13 & 10 & 0 & 24 & 7 \\ 2 & 10 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 24 & 16 & 0 & 38 & 12 \\ 2 & 7 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1105] \ 1^{13} 5^1 1^2 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 391, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 27 & 3 \\ 2 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 30 & 0 & 403 & 39 \\ 3 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[1106] \ 1^{13} 5^1 1^2 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 403, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 31 & 3 \\ 2 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 34 & 0 & 535 & 45 \\ 3 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[1107] \ 1^{11} 3221^{12} 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 440, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 1 & 22 & 3 \\ 9 & 43 & 1 & 110 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 110 & 0 & 270 & 32 \\ 3 & 13 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[1108] \ 1^1 3221^{12} 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 536, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 1 & 18 & 3 \\ 9 & 43 & 1 & 90 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 90 & 0 & 178 & 26 \\ 3 & 13 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1109] \ 1^{11} 3221^{13} 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 970, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 1 & 31 & 3 \\ 9 & 43 & 1 & 155 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 155 & 0 & 535 & 45 \\ 3 & 13 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[1110] \ 1^1 3221^{13} 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 1 & 27 & 3 \\ 9 & 43 & 1 & 135 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 135 & 0 & 403 & 39 \\ 3 & 13 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1111] \ 1^{11} 31^1 5^1 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 6 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[1112] $1^{11}3151321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 3 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 14 & 0 & 46 & 16 \\ 3 & 5 & -1 & 16 & 5 \end{pmatrix}_{b_0}$

[1113] $1^{12}3151321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 3 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 28 & 0 & 190 & 32 \\ 3 & 5 & -1 & 32 & 5 \end{pmatrix}_{b_0}$

[1114] $1^{11}3151321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 \\ 3 & 0 & 5 & 3 & 5 \\ 3 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$

[1115] $1^{13}3151321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 5 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 27 & 5 \\ 6 & 15 & 0 & 27 & 79 & 21 \\ 2 & 4 & -1 & 5 & 21 & 4 \end{pmatrix}_{b_0}$

[1116] $1^{12}3151321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 18 & 5 \\ 4 & 10 & 0 & 18 & 34 & 14 \\ 2 & 4 & -1 & 5 & 14 & 4 \end{pmatrix}_{b_0}$

[1117] $1^{11}3151321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 9 & 5 \\ 2 & 5 & 0 & 9 & 7 & 7 \\ 2 & 4 & -1 & 5 & 7 & 4 \end{pmatrix}_{b_0}$

[1118] $1^{14}3151321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 5 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 36 & 5 \\ 8 & 20 & 0 & 36 & 142 & 28 \\ 2 & 4 & -1 & 5 & 28 & 4 \end{pmatrix}_{b_0}$

[1119] $1^{11}3151321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 7 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0}$

[1120] $1^{11}3151321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 14 & 4 \\ 9 & 0 & 14 & 46 & 16 \\ 3 & 0 & 4 & 16 & 5 \end{pmatrix}_{b_0}$

[1121] $1^{12}3151321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 15 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$

[1123] $1^{11}3151321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 19 & 4 \\ 12 & 0 & 19 & 79 & 21 \\ 3 & 0 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1122] $1^{11}31 \ 5 \ 1321^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 14 & 4 \\ 10 & 0 & 14 & 46 & 16 \\ 3 & 0 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[1125] $1^{12}31 \ 5 \ 1321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 118, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 16 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[1124] $1^{12}31 \ 5 \ 1321^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 118, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 28 & 4 \\ 18 & 0 & 28 & 190 & 32 \\ 3 & 0 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

[1126] $1^{13}31 \ 5 \ 1321^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 138, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 32 & 4 \\ 22 & 0 & 32 & 282 & 38 \\ 3 & 0 & 4 & 38 & 5 \end{pmatrix}_{b_0}$$

[1127] $1^{11}31 \ 5 \ 1321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 147, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 19 & 4 \\ 13 & 0 & 19 & 79 & 21 \\ 3 & 0 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1128] $1^{12}31 \ 5 \ 1321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 183, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 17 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[1129] $1^{12}31 \ 5 \ 1321^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 213, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 28 & 4 \\ 19 & 0 & 28 & 190 & 32 \\ 3 & 0 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

[1130] $1^{13}31 \ 5 \ 1321^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 233, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 32 & 4 \\ 23 & 0 & 32 & 282 & 38 \\ 3 & 0 & 4 & 38 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [1133] $1^{11}31 \ 6 \ 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[1134] $1^{11}31\ 5\ 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 22 & 5 \\ 12 & 0 & 22 & 88 & 24 \\ 3 & 0 & 5 & 24 & 6 \end{pmatrix}_{b_0}$

[1135] $1^{11}31\ 5\ 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 16 & 5 \\ 9 & 0 & 16 & 50 & 18 \\ 3 & 0 & 5 & 18 & 6 \end{pmatrix}_{b_0}$

[1136] $1^{12}31\ 5\ 13221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 32 & 5 \\ 18 & 0 & 32 & 206 & 36 \\ 3 & 0 & 5 & 36 & 6 \end{pmatrix}_{b_0}$

[1137] $1^{12}31\ 5\ 13221^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 3 & 1 & 6 & 12 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 20 & 6 \\ 4 & 12 & 0 & 20 & 38 & 16 \\ 2 & 5 & -1 & 6 & 16 & 5 \end{pmatrix}_{b_0}$

[1139] $1^{13}31\ 5\ 13221^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 3 & 1 & 6 & 18 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 30 & 6 \\ 6 & 18 & 0 & 30 & 88 & 24 \\ 2 & 5 & -1 & 6 & 24 & 5 \end{pmatrix}_{b_0}$

[1140] $1^{14}31\ 5\ 13221^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 3 & 1 & 6 & 24 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 40 & 6 \\ 8 & 24 & 0 & 40 & 158 & 32 \\ 2 & 5 & -1 & 6 & 32 & 5 \end{pmatrix}_{b_0}$

[1138] $1^{11}31\ 5\ 13221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 6 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 10 & 6 \\ 2 & 6 & 0 & 10 & 8 & 8 \\ 2 & 5 & -1 & 6 & 8 & 5 \end{pmatrix}_{b_0}$

[1141] $1^{11}31\ 6\ 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 14, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$

[1142] $1^{12}31\ 6\ 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 14, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 4 & 7 & 5 & 1 & 18 & 10 \\ 2 & 3 & 2 & -1 & 10 & 4 \end{pmatrix}_{b_0}$

$T = 19$ [1143] $1^{11}3221\ (12)\ 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 19, T_{\max} = 20, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1144] \quad 1^{11}3221 \quad (12) \quad 12231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -32, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1145] \quad 1^{12}3221 \quad (12) \quad 12231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -32, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 4 & 7 & 5 & 1 & 18 & 10 \\ 2 & 3 & 2 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

4.655 NHC(-2)(-3) + so₈ + su₂ + e₆

$$T = 8 \quad [0] \quad 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 139, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 140, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 141, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 45 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 213, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.53 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 318, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 26 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 39 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 921, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1005, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 33 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{115} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1053, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.1$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 35 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 52 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1590, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 46 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1650, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 40 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [23] \ 1^{11} 3^1 1^{11} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3^1 1^{11} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -571, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.656 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \ 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -372$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}} 3^{\overline{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}} 3^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -608$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}} 3^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -618$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

4.657 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{\overline{111}} 3^{\overline{1111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 372$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.658 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3$

$T = 9$ [0] $1^{\overline{111}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.659 NHC(-2)(-3) + so₈ + su₂ + f₄

$$T = 8 \quad [0] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 45 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 26 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 39 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{115}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 35 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 781, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 33 & -3 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 52 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 46 & -3 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 40 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{111}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.660 NHC(-2)(-3) + so₈ + su₂ + f₄ + su₂

$T = 9$ [0] $1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

14.2

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$4.661 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -222, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 6 & 5 \\ 12 & 25 & 20 & 0 & 12 & 9 \\ 9 & 20 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 4 & 4 \\ 5 & 9 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.662 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix}$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

4.663 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -83, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.664 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 6 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 54 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 36 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 45 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 60 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 45 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 93$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 107$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 39 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 39 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 137$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.04 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 48 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 48 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 237$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 35 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 29 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 29 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 302, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 45 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 45 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 342, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 33 & 0 & 49 & 13 \\ 5 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 53 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 53 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 39 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 39 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{114} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 902, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 38 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1038, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 45 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1058, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 62 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 62 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -2 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1070, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 61 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 61 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -2 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 54 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [61] $1^{13} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 18 & 1 & 5 \\ 9 & 13 & 30 & 0 & 7 \\ 18 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 4 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 12 & 5 \\ 4 & 12 & 0 & 12 & 10 & 6 \\ 3 & 6 & -1 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

- [65] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.36 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$
- [66] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$
- [67] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$
- [68] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -93$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 5 & -2 & 2 \end{pmatrix}_{b_0}$
- [70] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [69] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [71] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -117$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 12 & 5 \\ 5 & 12 & 0 & 12 & 10 & 6 \\ 3 & 6 & -1 & 5 & 6 & 1 \end{pmatrix}_{b_0}$
- [72] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 10 & 0 & 14 & 7 & 8 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$
- [73] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -186$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 15 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 15 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 12 & -2 & 2 \end{pmatrix}_{b_0}$
- [74] $1^{11}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -199$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 15 & 0 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 0 & 2 \\ 3 & 5 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$
- [75] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.0

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 10 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{12} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 21 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 21 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{111} 3 1^{11} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{12} 3 2 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 20 & 0 & 62 & 14 & 16 \\ 2 & 5 & 0 & 14 & 0 & 2 \\ 3 & 5 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{11} 3 2 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 0 & 2 \\ 3 & 5 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{13} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 27 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{14} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 33 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 33 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{12} 3 2 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 25 & 0 & 94 & 18 & 20 \\ 2 & 5 & 0 & 18 & 0 & 2 \\ 3 & 5 & -2 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{11} 3 2 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -495, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 20 & 0 & 58 & 15 & 16 \\ 2 & 5 & 0 & 15 & 0 & 2 \\ 3 & 5 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{13} 3 2 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -619, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 30 & 0 & 142 & 21 & 24 \\ 2 & 5 & 0 & 21 & 0 & 2 \\ 3 & 5 & -2 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{12} 3^{11} 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -706, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 25 & 0 & 94 & 18 & 20 \\ 2 & 5 & 0 & 18 & 0 & 2 \\ 3 & 5 & -2 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{13} 3^{11} 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -895, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 35 & 0 & 188 & 25 & 28 \\ 2 & 5 & 0 & 25 & 0 & 2 \\ 3 & 5 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{12} 3^{11} 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -966, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 30 & 0 & 134 & 22 & 24 \\ 2 & 5 & 0 & 22 & 0 & 2 \\ 3 & 5 & -2 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{14} 3^{11} 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1046, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 5 & 0 & 40 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 40 & 0 & 254 & 28 & 32 \\ 2 & 5 & 0 & 28 & 0 & 2 \\ 3 & 5 & -2 & 32 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{13} 3^{11} 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 35 & 0 & 188 & 25 & 28 \\ 2 & 5 & 0 & 25 & 0 & 2 \\ 3 & 5 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [90] \ 1^{11} 4^{11} 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{11} 4^{11} 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 14 & 1 & 14 & 8 \\ 5 & 7 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{11} 4^{11} 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{11} 4^{11} 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 2 & 2 & 4 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{11} 4^{11} 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

12.0

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 2 & 3 \\ 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 2 & 1 & 2 & 2 & 4 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{11} 4^{11131^2} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 2 & 3 \\ 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 6 & 3 & 2 & 8 & 2 & 6 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{11} 4^{11131^1} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 5 & 4 & 2 & 6 & 2 & 6 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{11} 4^{11131^2} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 259, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 2 & 3 \\ 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 1 & -2 & 3 & 0 & 0 \\ 7 & 5 & 3 & 14 & 2 & 8 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

4.665 NHC(-2)(-3) + so₈ + su₂ + so₈ + su₂

$T = 9$ [0] $1^{11} 3^{111} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 39 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 39 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^{111} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 45 & 5 \\ 11 & 20 & 0 & 0 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 45 & 84 & 0 & 0 & 346 & 34 \\ 5 & 8 & -2 & 0 & 34 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 10 & 5 \\ 12 & 25 & 36 & 0 & 20 & 9 \\ 17 & 36 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 27 & 0 & 13 & 7 \\ 5 & 9 & 13 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 18 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 18 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 18 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{111} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 48 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 48 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 36 & 5 \\ 11 & 20 & 0 & 0 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 36 & 67 & 0 & 0 & 217 & 27 \\ 5 & 8 & -2 & 0 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 18 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -262, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 29 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 29 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 45 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 10 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 27 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 27 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 45 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 45 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{\overline{111}} 3 1^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 10 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{\overline{111}} 3 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{\overline{111}} 3 1^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{\overline{111}} 3 1^{\overline{113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 36 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 10 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{\overline{111}} 3 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{\overline{111}} 3 1^{\overline{113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 38 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 38 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{\overline{111}} 3 1^{\overline{114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 47 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 47 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{\overline{111}} 3 1^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 28 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 28 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{\overline{111}} 3 1^{\overline{114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -530, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 46 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 46 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{111} 3^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -532, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 36 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 36 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111} 3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -550, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 27 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 27 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 3^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -616, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 37 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 37 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

4.666 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 18 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 6 & 5 \\ 12 & 25 & 20 & 0 & 12 & 9 \\ 9 & 20 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 9 & 0 & 4 & 4 \\ 5 & 9 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -217, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.667 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & -2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 10 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 16 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 16 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

4.668 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 8 & 5 \\ 12 & 25 & 60 & 0 & 16 & 9 \\ 28 & 60 & 133 & 0 & 36 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 36 & 0 & 9 & 5 \\ 5 & 9 & 21 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 1 & 5 \\ 12 & 25 & 36 & 0 & 0 & 9 \\ 17 & 36 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 10 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 8 & 5 \\ 12 & 25 & 40 & 0 & 16 & 9 \\ 19 & 40 & 58 & 0 & 24 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 24 & 0 & 9 & 5 \\ 5 & 9 & 14 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 1 & 5 \\ 12 & 25 & 72 & 0 & 0 & 9 \\ 34 & 72 & 202 & 0 & 0 & 26 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 10 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 10 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 26 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 1 & 5 \\ 12 & 25 & 108 & 0 & 0 & 9 \\ 51 & 108 & 457 & 0 & 0 & 39 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 29 & 1 & 8 & 5 \\ 12 & 25 & 60 & 0 & 16 & 9 \\ 29 & 60 & 133 & 0 & 36 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 36 & 0 & 9 & 5 \\ 5 & 9 & 21 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 8 & 5 \\ 12 & 25 & 80 & 0 & 16 & 9 \\ 38 & 80 & 238 & 0 & 48 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 48 & 0 & 9 & 5 \\ 5 & 9 & 28 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 1 & 5 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 68 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 39 & 1 & 8 & 5 \\ 12 & 25 & 80 & 0 & 16 & 9 \\ 39 & 80 & 238 & 0 & 48 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 48 & 0 & 9 & 5 \\ 5 & 9 & 28 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 19 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 6 & 5 \\ 9 & 13 & 14 & 0 & 10 & 7 \\ 9 & 14 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 9 & 0 & 4 & 4 \\ 5 & 7 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 6 & 5 \\ 9 & 13 & 10 & 0 & 10 & 7 \\ 6 & 10 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 7 & 0 & 4 & 4 \\ 5 & 7 & 5 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 15 & 1 & 6 & 5 \\ 9 & 13 & 24 & 0 & 10 & 7 \\ 15 & 24 & 38 & 0 & 16 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 16 & 0 & 4 & 4 \\ 5 & 7 & 12 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.669 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 1 & 6 & 6 \\ 15 & 20 & 0 & 0 & 8 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 & 6 \\ 15 & 20 & 17 & 0 & 8 & 8 \\ 12 & 17 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 7 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}} 3^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 48 & 6 & 6 \\ 15 & 20 & 0 & 67 & 8 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 48 & 67 & 0 & 217 & 27 & 27 \\ 6 & 8 & 0 & 27 & 3 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{133}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 15 & 3 \\ 3 & 4 & 0 & 17 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 81 & 13 \\ 15 & 27 & 0 & 81 & 133 & 21 \\ 3 & 4 & -2 & 13 & 21 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{111}} 3^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 35 & 6 & 6 \\ 15 & 20 & 17 & 50 & 8 & 8 \\ 12 & 17 & 13 & 42 & 7 & 7 \\ 35 & 50 & 42 & 118 & 20 & 20 \\ 6 & 8 & 7 & 20 & 3 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 16 & 23 & 13 & 6 & 6 \\ 16 & 25 & 36 & 20 & 9 & 9 \\ 23 & 36 & 49 & 27 & 13 & 13 \\ 13 & 20 & 27 & 13 & 7 & 7 \\ 6 & 9 & 13 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

- [13] $1^{11} 1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{11} 1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [14] $1^{121} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{124} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 8 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 17 & 31 & 0 & 42 & 172 & 24 \\ 3 & 4 & -2 & 6 & 24 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 8 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 12 & 48 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 48 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$
- [19] $1^{11} 1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 13 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [20] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [21] $1^{133} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 16 & 3 \\ 3 & 4 & 0 & 17 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 81 & 13 \\ 16 & 27 & 0 & 81 & 133 & 21 \\ 3 & 4 & -2 & 13 & 21 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 13 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 13 & 7 \\ 6 & 8 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[26] $1^1 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 16 & 23 & 23 & 6 \\ 16 & 25 & 36 & 36 & 9 \\ 23 & 36 & 49 & 51 & 13 \\ 23 & 36 & 51 & 49 & 13 \\ 6 & 9 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 35 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 35 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

- [33] $1^{11} 1^{12} 3 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 18 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 54 & 14 \\ 10 & 17 & 0 & 54 & 49 & 13 \\ 3 & 4 & -2 & 14 & 13 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{11} 1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [35] $1^{13} 1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 0 & 25 & 9 \\ 3 & 4 & -2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [37] $1^{11} 1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$
- [36] $1^{11} 1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 4 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{11} 1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{13} 1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [43] $1^{12} 1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [42] $1^{12} 1^{12} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 8 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$
- [41] $1^{11} 1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{11} 1^{12} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{123} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 11 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{114} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 33 & 7 \\ 12 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 9 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{14} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 12 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{123} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 27 & 10 \\ 7 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 39 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 39 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{122} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 54 & 10 \\ 14 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 45 & 7 \\ 15 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 27 & 9 \\ 8 & 13 & 0 & 27 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 0 & 76 & 16 \\ 3 & 4 & -2 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 7 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 6 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -2 & 10 & 7 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -2 & 6 & 17 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 13 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{112} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & -2 & 11 & 10 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 14 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 24 & 11 \\ 6 & 9 & 0 & 24 & 13 & 7 \\ 3 & 4 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 19 & 0 & 0 & 73 & 15 \\ 3 & 4 & -2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 62 & 6 \\ 15 & 20 & 17 & 84 & 8 \\ 12 & 17 & 13 & 72 & 7 \\ 62 & 84 & 72 & 346 & 34 \\ 6 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{13} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 27 & 4 \\ 16 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 48 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 13 & 17 & 13 & 57 & 7 \\ 48 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{134} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 22 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 81 & 17 \\ 12 & 21 & 0 & 81 & 76 & 16 \\ 3 & 4 & -2 & 17 & 16 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 36 & 6 \\ 15 & 20 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 36 & 50 & 0 & 118 & 20 \\ 6 & 8 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

- [83] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 14 & 0 & 0 & 37 & 11 \\ 3 & 4 & -2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$
- [85] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$
- [87] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$
- [86] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$
- [88] $1^{133} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$
- [89] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 0 & 49 & 13 \\ 3 & 4 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$
- [90] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$
- [91] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$
- [92] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$
- [93] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 30 & 6 \\ 13 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -2 & 6 & 17 & 2 \end{pmatrix}_{b_0}$$
- [94] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 49 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 49 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{123} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 17 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 17 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 24 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{132} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 8 & 13 & 0 & 39 & 28 & 10 \\ 3 & 4 & -2 & 13 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{13} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 10 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{112} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{112} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{124} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 17 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 72 & 10 \\ 17 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{11} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{143} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 10 & 3 \\ 3 & 4 & 0 & 21 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 63 & 16 \\ 10 & 17 & 0 & 63 & 49 & 13 \\ 3 & 4 & -2 & 16 & 13 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{12} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{134} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & -2 & 9 & 16 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 38 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 38 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{13} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 11 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 48 & 11 \\ 10 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{113} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 57 & 11 \\ 13 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -2 & 11 & 17 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{12} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 17 & 3 \\ 3 & 4 & 0 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 17 & 28 & 0 & 0 & 154 & 22 \\ 3 & 4 & -2 & 0 & 22 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{123} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overline{111}} 3 \ 1^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 161, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 63 & 6 \\ 15 & 20 & 17 & 84 & 8 \\ 12 & 17 & 13 & 72 & 7 \\ 63 & 84 & 72 & 346 & 34 \\ 6 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overline{144}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 60 & 12 \\ 11 & 21 & 0 & 60 & 76 & 16 \\ 3 & 4 & -2 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overline{12}} 3 \ 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 164, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.66 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{111}} 3 \ 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{12}} 3 \ 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 18 & 4 \\ 11 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -2 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{\overline{134}} 3 \ 1^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 167, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 18 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 93 & 13 \\ 18 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -2 & 13 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{\overline{12}} 3 \ 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 168, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.71 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 0 & 58 & 14 \\ 3 & 4 & -2 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{\overline{14}} 3 \ 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 168, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.71 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 26 & 0 & 0 & 118 & 20 \\ 3 & 4 & -2 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{\overline{111}} 3 \ 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 169, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 37 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 37 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{\overline{111}} 3 \ 1^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 171, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 51 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 51 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

- [126] $1^{124} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$
- [125] $1^{132} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 54 & 13 \\ 10 & 18 & 0 & 54 & 58 & 14 \\ 3 & 4 & -2 & 13 & 14 & 2 \end{pmatrix}_{b_0}$$
- [127] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$
- [128] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$
- [129] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 177, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$
- [132] $1^{124} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 13 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$
- [131] $1^{123} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 10 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [130] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 23 & 0 & 0 & 100 & 18 \\ 3 & 4 & -2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$
- [134] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 22 & 0 & 0 & 85 & 17 \\ 3 & 4 & -2 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$
- [133] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 50 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 50 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$
- [136] $1^{112} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 33 & 11 \\ 8 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & -2 & 11 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{133} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 66 & 13 \\ 13 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -2 & 13 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{143} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & -2 & 16 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{124} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 10 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{123} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{124} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 15 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -2 & 6 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{113} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 18 & 3 \\ 16 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{122} 3 \ 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 8 & 3 \\ 3 & 4 & 0 & 18 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 42 & 14 \\ 8 & 13 & 0 & 42 & 28 & 10 \\ 3 & 4 & -2 & 14 & 10 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{122} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 54 & 10 \\ 13 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 19 & 3 \\ 3 & 4 & 0 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 19 & 32 & 0 & 0 & 193 & 25 \\ 3 & 4 & -2 & 0 & 25 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{113} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 15 & 4 \\ 9 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{14} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 18 & 3 \\ 3 & 4 & 0 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 31 & 0 & 0 & 172 & 24 \\ 3 & 4 & -2 & 0 & 24 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 16 & 3 \\ 3 & 4 & 0 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 27 & 0 & 0 & 133 & 21 \\ 3 & 4 & -2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{123} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 14 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[154] $21^{114} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 72 & 7 \\ 17 & 20 & 17 & 84 & 8 \\ 15 & 17 & 13 & 72 & 7 \\ 72 & 84 & 72 & 346 & 34 \\ 7 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{114} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{113} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 57 & 11 \\ 12 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -2 & 11 & 17 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 21 & 3 \\ 3 & 4 & 0 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 21 & 36 & 0 & 0 & 238 & 28 \\ 3 & 4 & -2 & 0 & 28 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{113} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 57 & 11 \\ 13 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -2 & 11 & 17 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 11 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[159] $21^{1111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 27 & 27 & 7 \\ 20 & 25 & 36 & 36 & 9 \\ 27 & 36 & 49 & 51 & 13 \\ 27 & 36 & 51 & 49 & 13 \\ 7 & 9 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 24 & 7 \\ 8 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[163] $21^{112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 42 & 7 \\ 17 & 20 & 17 & 50 & 8 \\ 15 & 17 & 13 & 42 & 7 \\ 42 & 50 & 42 & 118 & 20 \\ 7 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{11} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{12} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -2 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{13} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 19 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 33 & 4 \\ 19 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{123} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 19 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 75 & 10 \\ 19 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -2 & 10 & 25 & 2 \end{pmatrix}_{b_0}$$

[168] $21^{1111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 27 & 15 & 7 \\ 20 & 25 & 36 & 20 & 9 \\ 27 & 36 & 49 & 27 & 13 \\ 15 & 20 & 27 & 13 & 7 \\ 7 & 9 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[170] $1^{11} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 10 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[171] $1^{132} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 11 & 3 \\ 3 & 4 & 0 & 22 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 72 & 17 \\ 11 & 18 & 0 & 72 & 58 & 14 \\ 3 & 4 & -2 & 17 & 14 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{122} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 54 & 10 \\ 14 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{14} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 1 & 5 & 5 & 3 \\ 34 & 0 & 8 & 8 & 4 \\ 10 & -4 & 0 & 0 & -2 \\ 58 & 0 & 10 & 12 & 6 \\ 58 & 0 & 12 & 10 & 6 \\ 34 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{113}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{12}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 24 & 4 \\ 14 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 33 & 7 \\ 12 & 19 & 0 & 33 & 73 & 15 \\ 3 & 4 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{124}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -2 & 6 & 20 & 2 \end{pmatrix}_{b_0}$$

[181] $21^{113}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 57 & 7 \\ 17 & 20 & 17 & 67 & 8 \\ 15 & 17 & 13 & 57 & 7 \\ 57 & 67 & 57 & 217 & 27 \\ 7 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{13}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{123}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{122}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 8 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[188] 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[187] 1^{111} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 39 & 11 \\ 9 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -2 & 11 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[189] 1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 24 & 7 \\ 9 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[190] 1^{124} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[191] 1^{111} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 12 & 19 & 0 & 33 & 73 & 15 \\ 3 & 4 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[192] 1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[194] 1^{124} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 13 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[193] 1^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 11 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[195] 1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[196] 1^{133} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -2 & 13 & 17 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{132} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 11 & 3 \\ 3 & 4 & 0 & 17 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 54 & 13 \\ 11 & 18 & 0 & 54 & 58 & 14 \\ 3 & 4 & -2 & 13 & 14 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{111} 3^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 39 & 11 \\ 8 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -2 & 11 & 11 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{123} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 7 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{122} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 42 & 10 \\ 11 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -2 & 10 & 14 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 8 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{133} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 10 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{143} 3^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 13 & 3 \\ 3 & 4 & 0 & 26 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 102 & 20 \\ 13 & 22 & 0 & 102 & 85 & 17 \\ 3 & 4 & -2 & 20 & 17 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{112} 3^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 48 & 11 \\ 11 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{124} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 13 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{124} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 36 & 6 \\ 15 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -2 & 6 & 20 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{123} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 345, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 19 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 75 & 10 \\ 19 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -2 & 10 & 25 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 10 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 21 & 3 \\ 18 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{113} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 15 & 3 \\ 13 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{143} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 13 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 81 & 16 \\ 13 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & -2 & 16 & 17 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 13 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{112} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 48 & 11 \\ 10 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{122}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{13}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 16 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{113}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 15 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{12}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 12 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -2 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{134}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -2 & 13 & 20 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{144}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -2 & 16 & 20 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{133}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 27 & 9 \\ 7 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -2 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{114}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{113}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 10 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{123}3^11^{14}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 389, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 18 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 75 & 10 \\ 18 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -2 & 10 & 25 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{133} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 51 & 13 \\ 10 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 15 & 3 \\ 12 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 24 & 7 \\ 9 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{122} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -2 & 10 & 14 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{112} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 398, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 12 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{114} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 11 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{13} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 401, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 13 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{113} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 407, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 12 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{14} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 410, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 15 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{112} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 24 & 6 \\ 10 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{124} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{134} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 12 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 9 & 16 & 0 & 36 & 46 & 12 \\ 3 & 4 & -2 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 12 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{144} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 78 & 16 \\ 12 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -2 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{14} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 18 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 449, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 15 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 452, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 18 & 3 \\ 15 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 15 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 14 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 3 & 25 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 10 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 477, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 11 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 17 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 48 & 7 \\ 17 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -2 & 7 & 22 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{122} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 11 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 60 & 14 \\ 11 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -2 & 14 & 14 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{1^{12}} 1^{14} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 494, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 16 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{1^{11}} 1^{134} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 497, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 45 & 9 \\ 12 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & -2 & 9 & 16 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{1^{11}} 1^{12} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 13 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{1^{11}} 1^{12} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[271] $21^{1^{111}} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 15 & 15 & 7 \\ 20 & 25 & 20 & 20 & 9 \\ 15 & 20 & 13 & 15 & 7 \\ 15 & 20 & 15 & 13 & 7 \\ 7 & 9 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{1^{13}} 1^{14} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 17 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{1^{13}} 1^{133} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 509, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -2 & 13 & 17 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{1^{11}} 1^{112} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 514, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 16 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -2 & 7 & 22 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{1^{11}} 1^{112} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 11 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{1^{11}} 1^{14} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 17 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{133} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

[278] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 30 & 6 \\ 12 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{124} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 15 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 84 & 14 \\ 15 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -2 & 14 & 20 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{114} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 42 & 7 \\ 15 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[281] $1^{12} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 17 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{134} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 539, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 17 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 93 & 13 \\ 17 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -2 & 13 & 24 & 2 \end{pmatrix}_{b_0}$$

[283] $1^{124} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 84 & 14 \\ 14 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -2 & 14 & 20 & 2 \end{pmatrix}_{b_0}$$

[284] $1^{113} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{111} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 39 & 11 \\ 9 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -2 & 11 & 11 & 2 \end{pmatrix}_{b_0}$$

[287] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 565, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{144} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -2 & 16 & 20 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -2 & 13 & 16 & 2 \end{pmatrix}_{b_0}$$

[290] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 15 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 12 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 15 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{112} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 48 & 11 \\ 11 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 63 & 13 \\ 12 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -2 & 13 & 16 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 16 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 15 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{124} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 598, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 22 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 22 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -2 & 10 & 28 & 2 \end{pmatrix}_{b_0}$$

[299] $1^{112} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 602, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 30 & 7 \\ 11 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{123} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 13 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 72 & 14 \\ 13 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -2 & 14 & 17 & 2 \end{pmatrix}_{b_0}$$

[301] $1^{124} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 16 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 16 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

[302] $1^{123} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 16 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 16 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{134} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 614, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 15 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 78 & 13 \\ 15 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -2 & 13 & 20 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{124} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 48 & 10 \\ 12 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{112} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 39 & 7 \\ 14 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{123} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 15 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 15 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{123}3^{1122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 60 & 14 \\ 10 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -2 & 14 & 14 & 2 \end{pmatrix}_{b_0}$$

[308] $1^{113}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 14 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{123}3^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{13}3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 18 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{114}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{114}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 15 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{113}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 16 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{12}3^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 638, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 17 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{13}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 16 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{113}3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 19 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 3 & 25 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{122} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 42 & 10 \\ 11 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -2 & 10 & 14 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{114} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 17 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{123} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 13 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{113} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 36 & 7 \\ 13 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{122} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 30 & 10 \\ 8 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{112} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 655, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 14 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{133} 3^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 13 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 87 & 17 \\ 13 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -2 & 17 & 17 & 2 \end{pmatrix}_{b_0}$$

[324] $1^{113} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 14 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[325] $1^{124} 3^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 15 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 84 & 14 \\ 15 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -2 & 14 & 20 & 2 \end{pmatrix}_{b_0}$$

[326] $1^{123} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 72 & 14 \\ 12 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -2 & 14 & 17 & 2 \end{pmatrix}_{b_0}$$

[327] $1^{144} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -2 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

[329] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[328] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 18 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[330] $1^{112} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 690, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 18 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -2 & 7 & 22 & 2 \end{pmatrix}_{b_0}$$

[331] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 704, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 21 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 3 & 25 & 2 \end{pmatrix}_{b_0}$$

[334] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 17 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[333] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[332] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 14 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[335] $1^{124} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 712, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 19 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 19 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

- [336] $1^{123} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 715, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 39 & 10 \\ 10 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$
- [337] $1^{112} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 17 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -2 & 7 & 22 & 2 \end{pmatrix}_{b_0}$$
- [338] $1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 720, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 20 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 3 & 25 & 2 \end{pmatrix}_{b_0}$$
- [339] $1^{113} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 13 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$
- [340] $1^{114} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 723, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 21 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 21 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$
- [341] $1^{14} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 18 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$
- [342] $1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 18 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$
- [343] $1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 19 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 19 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$
- [344] $1^{112} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 17 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$
- [345] $1^{134} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 755, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 18 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 93 & 13 \\ 18 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -2 & 13 & 24 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{113} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 764, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 15 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{12} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 18 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

[348] $1^{14} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 774, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 20 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[349] $1^{13} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 783, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 18 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$

[350] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[351] $1^{123} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 787, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 16 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 63 & 10 \\ 16 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

[352] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 15 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[353] $1^{134} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 794, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -2 & 13 & 20 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{144} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 15 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 120 & 20 \\ 15 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -2 & 20 & 20 & 2 \end{pmatrix}_{b_0}$$

[355] $1^{113} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 812, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 16 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 45 & 7 \\ 16 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[356] $1^{124} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

[357] $1^{113} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 817, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 19 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 54 & 7 \\ 19 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

[358] $1^{114} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 818, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 20 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

[359] $1^{113} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 830, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 19 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 19 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

[360] $1^{114} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 843, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 18 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 51 & 7 \\ 18 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

[361] $1^{133} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 849, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 13 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 66 & 13 \\ 13 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -2 & 13 & 17 & 2 \end{pmatrix}_{b_0}$$

[362] $1^{124} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 850, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 21 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 84 & 10 \\ 21 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -2 & 10 & 28 & 2 \end{pmatrix}_{b_0}$$

[363] $1^{111} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 857, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 19 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 19 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[365] $1^{112} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 19 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 19 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{134} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 15 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 102 & 17 \\ 15 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -2 & 17 & 20 & 2 \end{pmatrix}_{b_0}$$

[366] $1^{114} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 870, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 15 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[367] $1^{114} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 874, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 16 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[368] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 884, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 18 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[369] $1^{14} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 21 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 21 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[370] $1^{113} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 909, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 18 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 16 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[372] $1^{113} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 20 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[373] $1^{124} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 934, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 20 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 20 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -2 & 10 & 28 & 2 \end{pmatrix}_{b_0}$$

[374] $1^{124} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 938, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 15 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 15 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

[375] $1^{133} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 51 & 13 \\ 10 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

[376] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 941, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 18 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[377] $1^{133} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 944, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 87 & 17 \\ 12 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -2 & 17 & 17 & 2 \end{pmatrix}_{b_0}$$

[378] $1^{123} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 945, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 51 & 10 \\ 13 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[379] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 17 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[380] $1^{113} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 972, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 17 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[381] $1^{114} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 979, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 17 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

[383] $1^{124} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 17 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 17 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

[382] $1^{113} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 16 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[385] $1^{114} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 994, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 21 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 60 & 7 \\ 21 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

[384] $1^{124} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 994, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 21 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 21 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -2 & 10 & 28 & 2 \end{pmatrix}_{b_0}$$

[386] $1^{144} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1002, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 15 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 96 & 16 \\ 15 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -2 & 16 & 20 & 2 \end{pmatrix}_{b_0}$$

[387] $1^{134} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1003, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 63 & 13 \\ 12 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -2 & 13 & 16 & 2 \end{pmatrix}_{b_0}$$

[388] $1^{134} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1004, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 102 & 17 \\ 14 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -2 & 17 & 20 & 2 \end{pmatrix}_{b_0}$$

[389] $1^{122} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1008, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 60 & 14 \\ 11 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -2 & 14 & 14 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{112} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1013, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 20 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{113} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1017, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 21 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 21 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 21 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{14} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1026, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 21 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 21 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{123} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 13 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 72 & 14 \\ 13 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -2 & 14 & 17 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1061, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 21 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{124} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1076, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 18 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 18 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

[397] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1077, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 19 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[398] $1^{124} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1082, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 60 & 10 \\ 15 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

[399] $1^{13} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1097, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 21 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 21 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{13} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 22 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 22 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 20 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[402] $1^{114} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 20 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

[403] $1^{124} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 18 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 72 & 10 \\ 18 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[404] 1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 22 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[405] 1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 24 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 24 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[406] 1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1177, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 19 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 19 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[407] 1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 24 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 24 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[408] 1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 22 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 22 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[409] 1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 20 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[410] 1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 23 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[411] 1^{144} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 78 & 16 \\ 12 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -2 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[412] 1^{134} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 15 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 78 & 13 \\ 15 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -2 & 13 & 20 & 2 \end{pmatrix}_{b_0}$$

[413] $1^{114} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 20 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{114} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 18 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

[415] $1^{144} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 120 & 20 \\ 14 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -2 & 20 & 20 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{14} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1314, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 23 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 23 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{114} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 19 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 19 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

[418] $1^{14} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 24 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 24 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[419] $1^{133} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 13 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 87 & 17 \\ 13 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -2 & 17 & 17 & 2 \end{pmatrix}_{b_0}$$

[420] $1^{134} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 15 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 102 & 17 \\ 15 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -2 & 17 & 20 & 2 \end{pmatrix}_{b_0}$$

[421] $1^{114} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1554, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 23 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 23 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[422] \ 1^{114} 3 \ 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 1586, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 21 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 21 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{114} 3 \ 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 1686, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 22 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 22 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[424] \ 1^{144} 3 \ 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 1904, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.72 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 15 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 26 & 0 & 118 & 120 & 20 \\ 15 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -2 & 20 & 20 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [425] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.670 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 10 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -232, \ (n_+, n_-, n_0) =$$

$$[13] \ 1^{\overline{111}} 3 \ 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -522, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 28 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 28 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{111}} 3 \ 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -550, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 27 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 27 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

4.671 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 4 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 0 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

4.672 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 22, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 23 & 8 & 6 \\ 16 & 25 & 36 & 12 & 9 \\ 23 & 36 & 49 & 17 & 13 \\ 8 & 12 & 17 & 4 & 4 \\ 6 & 9 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & -2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 3 \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_2 + \textcolor{red}{s}u_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[4] $1^1 1^3 1^1 1^1 + [\textcolor{red}{\mathbf{s}}\textcolor{blue}{\mathbf{u}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{s}}\textcolor{blue}{\mathbf{o}}_8 + \textcolor{orange}{\mathbf{s}}\textcolor{blue}{\mathbf{u}}_2 + \textcolor{red}{\mathbf{s}}\textcolor{blue}{\mathbf{u}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$

$$[5] \ 1^{\textcolor{brown}{1}} 1^{\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{brown}{o}_8 + \textcolor{brown}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 8 & 6 \\ 15 & 20 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 11 & 0 & 4 & 4 \\ 6 & 8 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{brown}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}o_8 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 \rightarrow (-3)] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 13 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\textcolor{brown}{1}} 1^{\textcolor{brown}{1}} 3^{\textcolor{brown}{1}} 1^{\textcolor{brown}{1}} + [\textcolor{blue}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{green}{o}\textcolor{green}{g} + \textcolor{orange}{s}\textcolor{orange}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3)] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{\textcolor{brown}{1}}1^{\textcolor{blue}{1}}3^{\textcolor{red}{1}}1^{\textcolor{red}{1}}+[\textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{s}\mathfrak{o}}_8+\textcolor{brown}{\mathfrak{s}\mathfrak{u}}_2+\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2\rightarrow(-3) \quad T=8, T_{\max}=11, \Delta=-85, \det=89, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=9.0$
 $T_{\min}=10.74\rightarrow 11, T_{\max}=11, T\in[11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

$$[10] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 119, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.71 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[111] \overset{11}{1} \overset{12}{2} \overset{3}{3} 1^1 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 6 & 9 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 8 & 12 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 9 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11} 1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 11 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[19] $21^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 27 & 9 & 7 \\ 20 & 25 & 36 & 12 & 9 \\ 27 & 36 & 49 & 17 & 13 \\ 9 & 12 & 17 & 4 & 4 \\ 7 & 9 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11} 1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$\begin{aligned}
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 2 \end{pmatrix}_{b_0} \\
[22] & 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0} \\
[23] & 1^{11} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 19 & 0 & 73 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0} \\
[24] & 21^{11} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 13 & 20 & 15 & 9 & 7 \\ 20 & 25 & 20 & 12 & 9 \\ 15 & 20 & 13 & 9 & 7 \\ 9 & 12 & 9 & 4 & 4 \\ 7 & 9 & 7 & 4 & 3 \end{pmatrix}_{b_0} \\
[25] & 1^{12} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 23 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0} \\
[26] & 1^{13} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 12 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 2 \end{pmatrix}_{b_0} \\
[27] & 1^{12} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 28 & 0 & 154 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0} \\
[28] & 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 399, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 9 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 2 \end{pmatrix}_{b_0} \\
[30] & 1^{14} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 14 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[29] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 11 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 511, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 27 & 0 & 133 & 15 & 21 \\ 2 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 23 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 23 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 12 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 32 & 0 & 193 & 18 & 25 \\ 2 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 654, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 28 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 28 & 0 & 154 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 670, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 31 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 31 & 0 & 172 & 17 & 24 \\ 2 & 3 & 0 & 17 & 0 & 2 \\ 3 & 4 & -2 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 26 & 0 & 118 & 14 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 811, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 27 & 0 & 133 & 15 & 21 \\ 2 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 854, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 36 & 0 & 238 & 20 & 28 \\ 2 & 3 & 0 & 20 & 0 & 2 \\ 3 & 4 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 969, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 32 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 32 & 0 & 193 & 18 & 25 \\ 2 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1034, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 31 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 31 & 0 & 172 & 17 & 24 \\ 2 & 3 & 0 & 17 & 0 & 2 \\ 3 & 4 & -2 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 2 & 3 \\ 3 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 21 & 36 & 0 & 238 & 20 & 28 \\ 2 & 3 & 0 & 20 & 0 & 2 \\ 3 & 4 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [44] $1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 3 & 2 & 2 \\ 3 & 5 & 3 & 4 & 3 & 4 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & -2 & 4 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 1 & 1 & 4 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 0 & 10 & 4 & 6 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & -1 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & -2 & 11 & 3 \end{pmatrix}_{b_0}$$

[48] $21^{11}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 3 & 4 \\ 5 & 5 & 5 & 10 & 3 & 5 \\ 3 & 5 & 0 & 7 & 2 & 2 \\ 9 & 10 & 7 & 14 & 5 & 8 \\ 3 & 3 & 2 & 5 & 0 & 2 \\ 4 & 5 & 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{12}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 2 & 3 \\ 3 & 5 & 5 & 15 & 3 & 5 \\ 3 & 5 & 0 & 10 & 2 & 2 \\ 8 & 15 & 10 & 38 & 8 & 12 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 5 & 2 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{11}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 2 & 3 \\ 2 & 1 & 1 & 7 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 6 & 9 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 3 & 3 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{13}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 4 & 2 & 2 \\ 2 & 5 & 4 & 2 & 2 & 4 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 5 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 \\ 3 & 5 & 5 & 20 & 3 & 5 \\ 3 & 5 & 0 & 15 & 2 & 2 \\ 9 & 20 & 15 & 58 & 9 & 16 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 5 & 2 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{11}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \\ 3 & 3 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{14}3^121^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 25 & 2 & 3 \\ 2 & 1 & 1 & 28 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 25 & 28 & 0 & 366 & 24 & 36 \\ 2 & 2 & 0 & 24 & 0 & 2 \\ 3 & 3 & -1 & 36 & 2 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{11}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11} 3^{12} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 21^{12} 3^{12} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 3 & 4 \\ 5 & 5 & 5 & 20 & 3 & 5 \\ 3 & 5 & 0 & 14 & 2 & 2 \\ 18 & 20 & 14 & 62 & 10 & 16 \\ 3 & 3 & 2 & 10 & 0 & 2 \\ 4 & 5 & 2 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{13} 3^{12} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 2 & 3 \\ 2 & 1 & 1 & 17 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 17 & 0 & 121 & 14 & 21 \\ 2 & 2 & 0 & 14 & 0 & 2 \\ 3 & 3 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{12} 3^{12} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 2 & 3 \\ 2 & 1 & 1 & 10 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 8 & 12 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 3 & 3 & -1 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{12} 3^{12} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 3^{12} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & -2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 3^{12} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 5 & 5 & 10 & 3 & 5 \\ 3 & 5 & 0 & 7 & 2 & 2 \\ 5 & 10 & 7 & 14 & 5 & 8 \\ 2 & 3 & 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{11} 3^{12} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 2 & 3 \\ 2 & 1 & 1 & 11 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 11 & 0 & 65 & 10 & 15 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 3 & 3 & -1 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{13} 3^{12} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -88, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 2 & 3 \\ 2 & 1 & 1 & 25 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 25 & 0 & 313 & 22 & 33 \\ 2 & 2 & 0 & 22 & 0 & 2 \\ 3 & 3 & -1 & 33 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{13} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 1 & 1 & 13 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 13 & 0 & 61 & 10 & 15 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 3 & 3 & -1 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 2 & 3 \\ 3 & 5 & 5 & 15 & 3 & 5 \\ 3 & 5 & 0 & 11 & 2 & 2 \\ 7 & 15 & 11 & 32 & 7 & 12 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 5 & 2 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 2 & 3 \\ 2 & 1 & 1 & 7 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 7 & 0 & 21 & 6 & 9 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 3 & 3 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 2 & 3 \\ 2 & 1 & 1 & 21 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 21 & 0 & 205 & 18 & 27 \\ 2 & 2 & 0 & 18 & 0 & 2 \\ 3 & 3 & -1 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{14} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 2 & 3 \\ 2 & 1 & 1 & 20 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 20 & 0 & 158 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 3 & 3 & -1 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{12} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 1 & 1 & 18 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 18 & 0 & 166 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 3 & 3 & -1 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & -2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 1 & 1 & 14 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 14 & 0 & 90 & 12 & 18 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 3 & 3 & -1 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & -2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{11}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 2 & 3 \\ 3 & 5 & 5 & 20 & 3 & 5 \\ 3 & 5 & 0 & 14 & 2 & 2 \\ 10 & 20 & 14 & 62 & 10 & 16 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 5 & 2 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13}3^{11}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -157, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & -2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -168, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 2 & 3 \\ 2 & 1 & 1 & 7 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 7 & 0 & 21 & 6 & 9 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 3 & 3 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{12}3^{11}21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 2 & 3 \\ 3 & 5 & 5 & 30 & 3 & 5 \\ 3 & 5 & 0 & 22 & 2 & 2 \\ 14 & 30 & 22 & 134 & 14 & 24 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 5 & 2 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{12}3^{11}21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 2 & 3 \\ 2 & 1 & 1 & 10 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 10 & 0 & 38 & 8 & 12 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 3 & 3 & -1 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{12}3^{11}21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -198, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 \\ 3 & 5 & 5 & 25 & 3 & 5 \\ 3 & 5 & 0 & 18 & 2 & 2 \\ 12 & 25 & 18 & 94 & 12 & 20 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 5 & 2 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{13}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -229, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & -2 & 17 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{13}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & -2 & 21 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{13}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 2 & 3 \\ 2 & 1 & 1 & 13 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 13 & 0 & 61 & 10 & 15 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 3 & 3 & -1 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{13}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 2 & 3 \\ 2 & 1 & 1 & 17 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 17 & 0 & 121 & 14 & 21 \\ 2 & 2 & 0 & 14 & 0 & 2 \\ 3 & 3 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{12}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 2 & 3 \\ 2 & 1 & 1 & 14 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 14 & 0 & 90 & 12 & 18 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 3 & 3 & -1 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{14}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 2 & 3 \\ 2 & 1 & 1 & 24 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 24 & 0 & 250 & 20 & 30 \\ 2 & 2 & 0 & 20 & 0 & 2 \\ 3 & 3 & -1 & 30 & 2 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{14}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{12}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 1 & 1 & 10 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 10 & 0 & 38 & 8 & 12 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 3 & 3 & -1 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{14}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 1 & 1 & 16 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 16 & 0 & 90 & 12 & 18 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 3 & 3 & -1 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{12}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 2 & 3 \\ 2 & 1 & 1 & 21 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 21 & 0 & 205 & 18 & 27 \\ 2 & 2 & 0 & 18 & 0 & 2 \\ 3 & 3 & -1 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{14} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -384, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 2 & 3 \\ 2 & 1 & 1 & 28 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 26 & 28 & 0 & 366 & 24 & 36 \\ 2 & 2 & 0 & 24 & 0 & 2 \\ 3 & 3 & -1 & 36 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{14} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -384, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & -2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{14} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{14} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 2 & 3 \\ 2 & 1 & 1 & 20 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 20 & 0 & 158 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 3 & 3 & -1 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{13} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -424, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.99 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 2 & 3 \\ 2 & 1 & 1 & 13 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 13 & 0 & 61 & 10 & 15 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 3 & 3 & -1 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{13} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -472, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 2 & 3 \\ 2 & 1 & 1 & 17 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 17 & 0 & 121 & 14 & 21 \\ 2 & 2 & 0 & 14 & 0 & 2 \\ 3 & 3 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{14} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 2 & 3 \\ 2 & 1 & 1 & 24 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 24 & 0 & 250 & 20 & 30 \\ 2 & 2 & 0 & 20 & 0 & 2 \\ 3 & 3 & -1 & 30 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{14} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -640, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 1 & 1 & 20 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 20 & 0 & 158 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 3 & 3 & -1 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

4.673 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

4.674 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 4 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 4 & 2 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 11 & 11 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 3 & 0 & 11 & 2 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 9 & 9 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 3 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 5 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 8 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 2 & 3 & 0 & 8 & 2 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 7 & 7 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 10 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 & 2 \\ 2 & 3 & 0 & 10 & 2 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -468, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 12 & 12 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 2 & 3 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -488, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 14 & 14 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 2 & 3 & 0 & 14 & 2 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.675 NHC(-2)(-3) + so₈ + su₂ + su₂ → (-3) + su_{2n3mix}

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 4 & 3 \\ 3 & 4 & 0 & 14 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 8 & 17 & 11 \\ 2 & 3 & 0 & 8 & 0 & 4 & 2 \\ 4 & 7 & 0 & 17 & 4 & 7 & 5 \\ 3 & 4 & -2 & 11 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 8 & 3 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 22 & 6 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 9 & 16 & 0 & 22 & 9 & 44 & 12 \\ 3 & 4 & -2 & 6 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -86, \ (n_+, n_-, n_0) =$$

$(1, 9, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 8 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 10 & 6 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 4 & 7 & 0 & 10 & 4 & 7 & 5 \\ 3 & 4 & -2 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 1^3 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 0 & 4 & 2 \\ 4 & 7 & 0 & 6 & 4 & 7 & 5 \\ 3 & 4 & -2 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{111}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{sg}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 5 & 3 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 4 & 7 & 0 & 5 & 4 & 7 & 5 \\ 3 & 4 & -2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112}3^{11}1^1 + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{blue}{o}_8 + \textcolor{brown}{5}\textcolor{blue}{u}_2 + \textcolor{red}{5}\textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{blue}{5}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 8 & 3 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 11 & 0 & 8 & 6 & 20 & 8 \\ 3 & 4 & -2 & 3 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -238, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 11 & 3 \\ 3 & 4 & 0 & 8 & 3 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 28 & 6 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 11 & 20 & 0 & 28 & 11 & 69 & 15 \\ 3 & 4 & -2 & 6 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad \begin{array}{c} 1^1 \\ 1^{11} 3^{11} 1^{11} \end{array} + [\mathfrak{s} \mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{s} \mathfrak{o}_8 + \mathfrak{s} \mathfrak{u}_2 + \mathfrak{s} \mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s} \mathfrak{u}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -251, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 5 & 11 & 7 \\ 2 & 3 & 0 & 5 & 0 & 4 & 2 \\ 4 & 7 & 0 & 11 & 4 & 7 & 5 \\ 3 & 4 & -2 & 7 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{124}3^{111}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 16 & 3 \\ 3 & 4 & 0 & 8 & 3 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 40 & 6 \\ 2 & 3 & 0 & 4 & 0 & 16 & 2 \\ 16 & 29 & 0 & 40 & 16 & 148 & 22 \\ 3 & 4 & -2 & 6 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^{111}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{black}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{blue}{s}\textcolor{blue}{u}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 3 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 8 & 3 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 12 & 0 & 8 & 7 & 25 & 9 \\ 3 & 4 & -2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 4 & 3 \\ 3 & 4 & 0 & 21 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 11 & 26 & 16 \\ 2 & 3 & 0 & 11 & 0 & 4 & 2 \\ 4 & 7 & 0 & 26 & 4 & 7 & 5 \\ 3 & 4 & -2 & 16 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 4 & 3 \\ 3 & 4 & 0 & 18 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 10 & 22 & 14 \\ 2 & 3 & 0 & 10 & 0 & 4 & 2 \\ 4 & 7 & 0 & 22 & 4 & 7 & 5 \\ 3 & 4 & -2 & 14 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 6 & 3 \\ 3 & 4 & 0 & 13 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 7 & 25 & 10 \\ 2 & 3 & 0 & 7 & 0 & 6 & 2 \\ 6 & 11 & 0 & 25 & 6 & 20 & 8 \\ 3 & 4 & -2 & 10 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 4 & 3 \\ 3 & 4 & 0 & 13 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 7 & 16 & 10 \\ 2 & 3 & 0 & 7 & 0 & 4 & 2 \\ 4 & 7 & 0 & 16 & 4 & 7 & 5 \\ 3 & 4 & -2 & 10 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 4 & 3 \\ 3 & 4 & 0 & 17 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 9 & 21 & 13 \\ 2 & 3 & 0 & 9 & 0 & 4 & 2 \\ 4 & 7 & 0 & 21 & 4 & 7 & 5 \\ 3 & 4 & -2 & 13 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 13 & 3 \\ 3 & 4 & 0 & 8 & 3 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 34 & 6 \\ 2 & 3 & 0 & 4 & 0 & 13 & 2 \\ 13 & 24 & 0 & 34 & 13 & 100 & 18 \\ 3 & 4 & -2 & 6 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{132}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 6 & 3 \\ 3 & 4 & 0 & 17 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 9 & 33 & 13 \\ 2 & 3 & 0 & 9 & 0 & 6 & 2 \\ 6 & 11 & 0 & 33 & 6 & 20 & 8 \\ 3 & 4 & -2 & 13 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 6 & 3 \\ 3 & 4 & 0 & 21 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 11 & 41 & 16 \\ 2 & 3 & 0 & 11 & 0 & 6 & 2 \\ 6 & 11 & 0 & 41 & 6 & 20 & 8 \\ 3 & 4 & -2 & 16 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 3 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 11 & 3 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 9 & 16 & 0 & 11 & 9 & 44 & 12 \\ 3 & 4 & -2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 4 & 3 \\ 3 & 4 & 0 & 22 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 12 & 27 & 17 \\ 2 & 3 & 0 & 12 & 0 & 4 & 2 \\ 4 & 7 & 0 & 27 & 4 & 7 & 5 \\ 3 & 4 & -2 & 17 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{14}23^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 6 & 3 \\ 3 & 4 & 0 & 26 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 14 & 50 & 20 \\ 2 & 3 & 0 & 14 & 0 & 6 & 2 \\ 6 & 11 & 0 & 50 & 6 & 20 & 8 \\ 3 & 4 & -2 & 20 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11}23^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 14 & 3 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 12 & 21 & 0 & 14 & 12 & 80 & 16 \\ 3 & 4 & -2 & 3 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 3 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 14 & 3 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 20 & 0 & 14 & 11 & 69 & 15 \\ 3 & 4 & -2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 4 & 3 \\ 3 & 4 & 0 & 26 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 14 & 32 & 20 \\ 2 & 3 & 0 & 14 & 0 & 4 & 2 \\ 4 & 7 & 0 & 32 & 4 & 7 & 5 \\ 3 & 4 & -2 & 20 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11}43^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 3 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 17 & 3 \\ 2 & 3 & 0 & 2 & 0 & 13 & 2 \\ 13 & 24 & 0 & 17 & 13 & 100 & 18 \\ 3 & 4 & -2 & 3 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 3 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 17 & 3 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 14 & 25 & 0 & 17 & 14 & 111 & 19 \\ 3 & 4 & -2 & 3 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11}43^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 3 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 20 & 3 \\ 2 & 3 & 0 & 2 & 0 & 16 & 2 \\ 16 & 29 & 0 & 20 & 16 & 148 & 22 \\ 3 & 4 & -2 & 3 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

4.676 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + su2n3mix

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 20 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 12 & 17 & 13 & 24 & 7 \\ 20 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 20 & 6 \\ 15 & 20 & 0 & 28 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 28 & 0 & 37 & 11 \\ 6 & 8 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{144} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 16 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 68 & 12 \\ 13 & 24 & 0 & 68 & 100 & 18 \\ 3 & 4 & -2 & 12 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 16 & 0 & 0 & 44 & 12 \\ 3 & 4 & -2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 12 & 6 \\ 3 & 4 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 14 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 17 & 11 \\ 4 & 7 & 0 & 17 & 7 & 5 \\ 3 & 4 & -2 & 11 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 0 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 7 & 0 & 0 & 7 & 5 \\ 3 & 4 & -2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 4 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -2 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 0 & 25 & 9 \\ 3 & 4 & -2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 20 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 13 & 17 & 13 & 24 & 7 \\ 20 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 12 & 4 \\ 7 & 12 & 0 & 12 & 25 & 9 \\ 3 & 4 & -2 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 14 & 25 & 0 & 34 & 111 & 19 \\ 3 & 4 & -2 & 6 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 0 & 21 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 89 & 16 \\ 14 & 24 & 0 & 89 & 100 & 18 \\ 3 & 4 & -2 & 16 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 21 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 12 & 17 & 13 & 24 & 7 \\ 21 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 23 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 23 & 0 & 100 & 28 & 18 \\ 4 & 7 & 0 & 28 & 7 & 5 \\ 3 & 4 & -2 & 18 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 14 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 17 & 11 \\ 4 & 7 & 0 & 17 & 7 & 5 \\ 3 & 4 & -2 & 11 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 17 & 15 & 24 & 7 \\ 17 & 20 & 17 & 28 & 8 \\ 15 & 17 & 13 & 24 & 7 \\ 24 & 28 & 24 & 37 & 11 \\ 7 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 12 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 42 & 9 \\ 11 & 20 & 0 & 42 & 69 & 15 \\ 3 & 4 & -2 & 9 & 15 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{121}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 7 & 11 & 0 & 16 & 20 & 8 \\ 3 & 4 & -2 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 22 & 6 \\ 9 & 16 & 0 & 22 & 44 & 12 \\ 3 & 4 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 22 & 6 \\ 9 & 16 & 0 & 22 & 44 & 12 \\ 3 & 4 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 8 & 3 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{124}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 8 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 40 & 6 \\ 17 & 29 & 0 & 40 & 148 & 22 \\ 3 & 4 & -2 & 6 & 22 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 17 & 7 \\ 6 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -2 & 7 & 8 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 14 & 7 \\ 5 & 8 & 0 & 14 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 11 & 3 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 28 & 6 \\ 12 & 20 & 0 & 28 & 69 & 15 \\ 3 & 4 & -2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 36 & 13 \\ 7 & 12 & 0 & 36 & 25 & 9 \\ 3 & 4 & -2 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -2 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 14 & 3 \\ 11 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 7 & 3 \\ 3 & 4 & 0 & 21 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 44 & 16 \\ 7 & 12 & 0 & 44 & 25 & 9 \\ 3 & 4 & -2 & 16 & 9 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 4 & 3 \\ 3 & 4 & 0 & 32 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 32 & 0 & 193 & 39 & 25 \\ 4 & 7 & 0 & 39 & 7 & 5 \\ 3 & 4 & -2 & 25 & 5 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 11 & 20 & 0 & 18 & 69 & 15 \\ 3 & 4 & -2 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 14 & 7 \\ 5 & 8 & 0 & 14 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 12 & 0 & 12 & 25 & 9 \\ 3 & 4 & -2 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11}2^13^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{11}1^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11}1^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -2 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{121}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 16 & 0 & 15 & 44 & 12 \\ 3 & 4 & -2 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 28 & 6 \\ 11 & 20 & 0 & 28 & 69 & 15 \\ 3 & 4 & -2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 14 & 24 & 0 & 21 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{132}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 6 & 3 \\ 3 & 4 & 0 & 22 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 42 & 17 \\ 6 & 11 & 0 & 42 & 20 & 8 \\ 3 & 4 & -2 & 17 & 8 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 25 & 10 \\ 6 & 11 & 0 & 25 & 20 & 8 \\ 3 & 4 & -2 & 10 & 8 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{11}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 12 & 0 & 12 & 25 & 9 \\ 3 & 4 & -2 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 16 & 6 \\ 6 & 11 & 0 & 16 & 20 & 8 \\ 3 & 4 & -2 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{134}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 12 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 51 & 9 \\ 13 & 24 & 0 & 51 & 100 & 18 \\ 3 & 4 & -2 & 9 & 18 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 22 & 6 \\ 9 & 15 & 0 & 22 & 39 & 11 \\ 3 & 4 & -2 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{121}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 23 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 23 & 0 & 100 & 28 & 18 \\ 4 & 7 & 0 & 28 & 7 & 5 \\ 3 & 4 & -2 & 18 & 5 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 4 & 0 & 21 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 59 & 16 \\ 9 & 16 & 0 & 59 & 44 & 12 \\ 3 & 4 & -2 & 16 & 12 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{11}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 60 & 13 \\ 12 & 20 & 0 & 60 & 69 & 15 \\ 3 & 4 & -2 & 13 & 15 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 18 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 22 & 14 \\ 4 & 7 & 0 & 22 & 7 & 5 \\ 3 & 4 & -2 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 26 & 7 \\ 9 & 16 & 0 & 26 & 44 & 12 \\ 3 & 4 & -2 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{11} 1^{13} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 15 & 9 \\ 4 & 7 & 0 & 15 & 7 & 5 \\ 3 & 4 & -2 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 16 & 10 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -2 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 25 & 10 \\ 6 & 11 & 0 & 25 & 20 & 8 \\ 3 & 4 & -2 & 10 & 8 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 34 & 6 \\ 13 & 24 & 0 & 34 & 100 & 18 \\ 3 & 4 & -2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 28 & 6 \\ 11 & 20 & 0 & 28 & 69 & 15 \\ 3 & 4 & -2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{112} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 7 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 20 & 7 \\ 8 & 12 & 0 & 20 & 25 & 9 \\ 3 & 4 & -2 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 12 & 20 & 0 & 18 & 69 & 15 \\ 3 & 4 & -2 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 10 & 16 & 0 & 15 & 44 & 12 \\ 3 & 4 & -2 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{141} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 16 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 20 & 12 \\ 4 & 7 & 0 & 20 & 7 & 5 \\ 3 & 4 & -2 & 12 & 5 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{11} 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 12 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 24 & 9 \\ 6 & 11 & 0 & 24 & 20 & 8 \\ 3 & 4 & -2 & 9 & 8 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 8 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 40 & 6 \\ 16 & 29 & 0 & 40 & 148 & 22 \\ 3 & 4 & -2 & 6 & 22 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 20 & 7 \\ 7 & 12 & 0 & 20 & 25 & 9 \\ 3 & 4 & -2 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 28 & 10 \\ 7 & 12 & 0 & 28 & 25 & 9 \\ 3 & 4 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 37 & 10 \\ 10 & 16 & 0 & 37 & 44 & 12 \\ 3 & 4 & -2 & 10 & 12 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 20 & 7 \\ 7 & 12 & 0 & 20 & 25 & 9 \\ 3 & 4 & -2 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 21 & 13 \\ 4 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & -2 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 3 \\ 3 & 4 & 0 & 17 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 33 & 13 \\ 6 & 11 & 0 & 33 & 20 & 8 \\ 3 & 4 & -2 & 13 & 8 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 7 & 3 \\ 3 & 4 & 0 & 17 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 36 & 13 \\ 7 & 12 & 0 & 36 & 25 & 9 \\ 3 & 4 & -2 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 26 & 16 \\ 4 & 7 & 0 & 26 & 7 & 5 \\ 3 & 4 & -2 & 16 & 5 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{124}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 14 & 24 & 0 & 34 & 100 & 18 \\ 3 & 4 & -2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 16 & 10 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -2 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{131}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 27 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 27 & 0 & 133 & 33 & 21 \\ 4 & 7 & 0 & 33 & 7 & 5 \\ 3 & 4 & -2 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 18 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 22 & 14 \\ 4 & 7 & 0 & 22 & 7 & 5 \\ 3 & 4 & -2 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 9 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{124}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 13 & 24 & 0 & 34 & 100 & 18 \\ 3 & 4 & -2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 28 & 10 \\ 7 & 12 & 0 & 28 & 25 & 9 \\ 3 & 4 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 10 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 26 & 7 \\ 9 & 16 & 0 & 26 & 44 & 12 \\ 3 & 4 & -2 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 27 & 17 \\ 4 & 7 & 0 & 27 & 7 & 5 \\ 3 & 4 & -2 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 13 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 46 & 10 \\ 11 & 20 & 0 & 46 & 69 & 15 \\ 3 & 4 & -2 & 10 & 15 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 4 & 0 & 21 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 41 & 16 \\ 6 & 11 & 0 & 41 & 20 & 8 \\ 3 & 4 & -2 & 16 & 8 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 6 & 3 \\ 3 & 4 & 0 & 17 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 33 & 13 \\ 6 & 11 & 0 & 33 & 20 & 8 \\ 3 & 4 & -2 & 13 & 8 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 4 & 0 & 26 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 50 & 20 \\ 6 & 11 & 0 & 50 & 20 & 8 \\ 3 & 4 & -2 & 20 & 8 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 48 & 13 \\ 9 & 16 & 0 & 48 & 44 & 12 \\ 3 & 4 & -2 & 13 & 12 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 4 & 3 \\ 3 & 4 & 0 & 32 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 32 & 0 & 193 & 39 & 25 \\ 4 & 7 & 0 & 39 & 7 & 5 \\ 3 & 4 & -2 & 25 & 5 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 26 & 7 \\ 10 & 16 & 0 & 26 & 44 & 12 \\ 3 & 4 & -2 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 11 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 32 & 7 \\ 11 & 20 & 0 & 32 & 69 & 15 \\ 3 & 4 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 38 & 7 \\ 13 & 24 & 0 & 38 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{112}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 16 & 26 & 0 & 17 & 128 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 13 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{122}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 37 & 10 \\ 9 & 16 & 0 & 37 & 44 & 12 \\ 3 & 4 & -2 & 10 & 12 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{122}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 37 & 10 \\ 9 & 16 & 0 & 37 & 44 & 12 \\ 3 & 4 & -2 & 10 & 12 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 11 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 21 & 13 \\ 4 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & -2 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{142}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 482, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 6 & 3 \\ 3 & 4 & 0 & 26 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 26 & 0 & 118 & 50 & 20 \\ 6 & 11 & 0 & 50 & 20 & 8 \\ 3 & 4 & -2 & 20 & 8 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{143}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 74 & 16 \\ 11 & 20 & 0 & 74 & 69 & 15 \\ 3 & 4 & -2 & 16 & 15 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{141}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 502, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 4 & 3 \\ 3 & 4 & 0 & 36 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 36 & 0 & 238 & 44 & 28 \\ 4 & 7 & 0 & 44 & 7 & 5 \\ 3 & 4 & -2 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{123}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 505, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 46 & 10 \\ 12 & 20 & 0 & 46 & 69 & 15 \\ 3 & 4 & -2 & 10 & 15 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 32 & 7 \\ 12 & 20 & 0 & 32 & 69 & 15 \\ 3 & 4 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{134}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 0 & 17 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 72 & 13 \\ 14 & 24 & 0 & 72 & 100 & 18 \\ 3 & 4 & -2 & 13 & 18 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{141}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 32 & 20 \\ 4 & 7 & 0 & 32 & 7 & 5 \\ 3 & 4 & -2 & 20 & 5 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 9 & 3 \\ 3 & 4 & 0 & 21 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 59 & 16 \\ 9 & 16 & 0 & 59 & 44 & 12 \\ 3 & 4 & -2 & 16 & 12 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 13 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{134}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 13 & 3 \\ 3 & 4 & 0 & 17 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 72 & 13 \\ 13 & 24 & 0 & 72 & 100 & 18 \\ 3 & 4 & -2 & 13 & 18 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{132}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 9 & 3 \\ 3 & 4 & 0 & 17 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 48 & 13 \\ 9 & 16 & 0 & 48 & 44 & 12 \\ 3 & 4 & -2 & 13 & 12 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 557, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 14 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{131}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 4 & 3 \\ 3 & 4 & 0 & 27 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 27 & 0 & 133 & 33 & 21 \\ 4 & 7 & 0 & 33 & 7 & 5 \\ 3 & 4 & -2 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{142}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 6 & 3 \\ 3 & 4 & 0 & 21 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 41 & 16 \\ 6 & 11 & 0 & 41 & 20 & 8 \\ 3 & 4 & -2 & 16 & 8 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{114}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 38 & 7 \\ 14 & 24 & 0 & 38 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{123}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 46 & 10 \\ 11 & 20 & 0 & 46 & 69 & 15 \\ 3 & 4 & -2 & 10 & 15 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{141}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 4 & 3 \\ 3 & 4 & 0 & 31 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 31 & 0 & 172 & 38 & 24 \\ 4 & 7 & 0 & 38 & 7 & 5 \\ 3 & 4 & -2 & 24 & 5 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{131}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 27 & 17 \\ 4 & 7 & 0 & 27 & 7 & 5 \\ 3 & 4 & -2 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 26 & 16 \\ 4 & 7 & 0 & 26 & 7 & 5 \\ 3 & 4 & -2 & 16 & 5 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 605, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 60 & 13 \\ 11 & 20 & 0 & 60 & 69 & 15 \\ 3 & 4 & -2 & 13 & 15 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 17 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -2 & 3 & 23 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{113}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 20 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -2 & 3 & 27 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 11 & 3 \\ 3 & 4 & 0 & 17 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 60 & 13 \\ 11 & 20 & 0 & 60 & 69 & 15 \\ 3 & 4 & -2 & 13 & 15 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{124}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 55 & 10 \\ 13 & 24 & 0 & 55 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{124}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 55 & 10 \\ 14 & 24 & 0 & 55 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 15 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{144}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 13 & 3 \\ 3 & 4 & 0 & 21 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 89 & 16 \\ 13 & 24 & 0 & 89 & 100 & 18 \\ 3 & 4 & -2 & 16 & 18 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{143}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 11 & 3 \\ 3 & 4 & 0 & 21 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 74 & 16 \\ 11 & 20 & 0 & 74 & 69 & 15 \\ 3 & 4 & -2 & 16 & 15 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 752, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 14 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 16 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -2 & 3 & 22 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{134}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 17 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 72 & 13 \\ 13 & 24 & 0 & 72 & 100 & 18 \\ 3 & 4 & -2 & 13 & 18 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 18 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -2 & 3 & 23 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{113}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 797, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 21 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -2 & 3 & 27 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{144}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 808, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 13 & 3 \\ 3 & 4 & 0 & 21 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 89 & 16 \\ 13 & 24 & 0 & 89 & 100 & 18 \\ 3 & 4 & -2 & 16 & 18 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{141}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 822, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 26 & 0 & 118 & 32 & 20 \\ 4 & 7 & 0 & 32 & 7 & 5 \\ 3 & 4 & -2 & 20 & 5 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{141}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 834, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 4 & 3 \\ 3 & 4 & 0 & 36 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 21 & 36 & 0 & 238 & 44 & 28 \\ 4 & 7 & 0 & 44 & 7 & 5 \\ 3 & 4 & -2 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{141}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 899, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 4 & 3 \\ 3 & 4 & 0 & 31 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 31 & 0 & 172 & 38 & 24 \\ 4 & 7 & 0 & 38 & 7 & 5 \\ 3 & 4 & -2 & 24 & 5 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 19 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -2 & 3 & 26 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 17 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -2 & 3 & 22 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 997, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 26 & 3 \\ 22 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -2 & 3 & 30 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 20 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -2 & 3 & 26 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 26 & 3 \\ 23 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -2 & 3 & 30 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [150] $1^{11}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 0 & 10 & 20 & 6 \\ 8 & 9 & 0 & 20 & 35 & 11 \\ 3 & 3 & -1 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{111}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 14 & 8 \\ 3 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & -2 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{122}3^121^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 41 & 13 \\ 5 & 7 & 9 & 0 & 41 & 35 & 11 \\ 2 & 2 & 2 & -2 & 13 & 11 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{11}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 5 & 13 & 5 \\ 3 & 5 & 0 & 4 & 9 & 2 \\ 2 & 5 & 4 & 2 & 9 & 4 \\ 7 & 13 & 9 & 9 & 26 & 10 \\ 3 & 5 & 2 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{11}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 5 & 0 & 10 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 10 & 0 & 14 & 19 & 8 \\ 6 & 13 & 0 & 19 & 24 & 10 \\ 3 & 5 & -2 & 8 & 10 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{131}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 8 & 3 \\ 2 & 1 & 1 & 13 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 13 & 0 & 61 & 47 & 15 \\ 8 & 9 & 0 & 47 & 35 & 11 \\ 3 & 3 & -1 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{112}3^121^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 7 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 25 & 8 \\ 5 & 7 & 9 & 0 & 25 & 35 & 11 \\ 2 & 2 & 2 & -2 & 8 & 11 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{121}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 23 & 13 \\ 3 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & -2 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -29, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 5 & 10 & 6 \\ 2 & 2 & 2 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{112} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 8 & 3 \\ 2 & 1 & 1 & 11 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 11 & 0 & 65 & 49 & 15 \\ 8 & 9 & 0 & 49 & 35 & 11 \\ 3 & 3 & -1 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{131} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 8 & 3 \\ 2 & 1 & 1 & 13 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 13 & 0 & 61 & 47 & 15 \\ 8 & 9 & 0 & 47 & 35 & 11 \\ 3 & 3 & -1 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -34, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 8 & 9 & 0 & 9 & 35 & 11 \\ 3 & 3 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{123} 3^1 21^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -37, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 11 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 59 & 13 \\ 7 & 10 & 13 & 0 & 59 & 74 & 16 \\ 2 & 2 & 2 & -2 & 13 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[163] \ 1^1 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 5 & 0 & 5 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 6 & 4 \\ 4 & 8 & 0 & 6 & 8 & 6 \\ 3 & 5 & -2 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{13} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 5 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 10 & 4 \\ 6 & 13 & 0 & 10 & 24 & 10 \\ 3 & 5 & -2 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{132} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 0 & 15 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 57 & 18 \\ 5 & 7 & 9 & 0 & 57 & 35 & 11 \\ 2 & 2 & 2 & -2 & 18 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{131} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 6 & 0 & 25 & 17 & 9 \\ 3 & 4 & 5 & 0 & 17 & 10 & 6 \\ 2 & 2 & 2 & -2 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{1^1} 1^{12} 3^1 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 6 & 3 \\ 3 & 5 & 0 & 25 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 25 & 0 & 94 & 48 & 20 \\ 6 & 13 & 0 & 48 & 24 & 10 \\ 3 & 5 & -2 & 20 & 10 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{1^1} 1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 8 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 29 & 9 \\ 8 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{1^1} 1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 9 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 29 & 9 \\ 9 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{1^1} 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 \\ 15 & 16 & 0 & 16 & 116 & 20 \\ 3 & 3 & -1 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{1^1} 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 9 & 9 & 0 & 9 & 35 & 11 \\ 3 & 3 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{1^1} 1^{13} 3^1 21^{142} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 5 & 0 & 25 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 25 & 0 & 92 & 49 & 20 \\ 6 & 13 & 0 & 49 & 24 & 10 \\ 3 & 5 & -2 & 20 & 10 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{1^1} 1^{13} 3^1 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 7 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 36 & 8 \\ 7 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & -2 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{1^1} 1^{142} 3^1 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 0 & 10 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 10 & 0 & 58 & 46 & 14 \\ 5 & 7 & 9 & 0 & 46 & 35 & 11 \\ 2 & 2 & 2 & -2 & 14 & 11 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{1^1} 1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 6 & 5 \\ 2 & 2 & 4 & 0 & 6 & 4 & 4 \\ 2 & 2 & 2 & -2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{12}3^{11}21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 7 & 9 & 0 & 9 & 35 & 11 \\ 2 & 2 & 2 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{124}3^{11}21^{134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 2 & 0 & 0 & 11 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 77 & 13 \\ 9 & 13 & 17 & 0 & 77 & 127 & 21 \\ 2 & 2 & 2 & -2 & 13 & 21 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{11}3^{11}21^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 5 & 0 & 20 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 58 & 24 & 16 \\ 4 & 8 & 0 & 24 & 8 & 6 \\ 3 & 5 & -2 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 0 & 4 & 10 & 6 \\ 2 & 2 & 2 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{112}3^{11}21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 10 & 3 \\ 2 & 1 & 1 & 7 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 36 & 9 \\ 10 & 12 & 0 & 36 & 56 & 14 \\ 3 & 3 & -1 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{132}3^{11}21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 10 & 3 \\ 2 & 1 & 1 & 17 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 17 & 0 & 121 & 84 & 21 \\ 10 & 12 & 0 & 84 & 56 & 14 \\ 3 & 3 & -1 & 21 & 14 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{131}3^{11}21^{141} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 0 & 15 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 32 & 18 \\ 3 & 4 & 5 & 0 & 32 & 10 & 6 \\ 2 & 2 & 2 & -2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{11}3^{11}21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 5 & 0 & 10 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 10 & 0 & 14 & 12 & 8 \\ 4 & 8 & 0 & 12 & 8 & 6 \\ 3 & 5 & -2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{121}3^{11}21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 8 & 3 \\ 2 & 1 & 1 & 10 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 38 & 12 \\ 8 & 9 & 0 & 38 & 35 & 11 \\ 3 & 3 & -1 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{11} 1^{122} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 11 & 3 \\ 2 & 1 & 1 & 10 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 48 & 12 \\ 11 & 12 & 0 & 48 & 56 & 14 \\ 3 & 3 & -1 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{11} 1^{131} 3^1 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 8 & 3 \\ 2 & 1 & 1 & 17 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 17 & 0 & 121 & 67 & 21 \\ 8 & 9 & 0 & 67 & 35 & 11 \\ 3 & 3 & -1 & 21 & 11 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{11} 1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 8 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 7 & 0 & 21 & 29 & 9 \\ 8 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{11} 1^{113} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 5 & 0 & 15 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 15 & 0 & 32 & 29 & 12 \\ 6 & 13 & 0 & 29 & 24 & 10 \\ 3 & 5 & -2 & 12 & 10 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{11} 1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 1 & 10 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 68 & 12 \\ 15 & 16 & 0 & 68 & 116 & 20 \\ 3 & 3 & -1 & 12 & 20 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{11} 1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 1 & 3 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 19 & 3 \\ 16 & 19 & 0 & 19 & 151 & 23 \\ 3 & 3 & -1 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{12} 1^{113} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 22 & 3 \\ 2 & 1 & 1 & 3 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 23 & 3 \\ 22 & 23 & 0 & 23 & 243 & 29 \\ 3 & 3 & -1 & 3 & 29 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{11} 1^{121} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 8 & 3 \\ 2 & 1 & 1 & 10 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 10 & 0 & 38 & 38 & 12 \\ 8 & 9 & 0 & 38 & 35 & 11 \\ 3 & 3 & -1 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{11} 1^{112} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 10 & 12 & 0 & 12 & 56 & 14 \\ 3 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{1^1} 1^{12} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 5 & 0 & 15 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 15 & 0 & 32 & 18 & 12 \\ 4 & 8 & 0 & 18 & 8 & 6 \\ 3 & 5 & -2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{1^{12}} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 16 & 3 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 \\ 16 & 16 & 0 & 16 & 116 & 20 \\ 3 & 3 & -1 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{1^1} 1^{12} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 12 & 10 \\ 2 & 2 & 4 & 0 & 12 & 4 & 4 \\ 2 & 2 & 2 & -2 & 10 & 4 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{1^1} 1^{12} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 3 \\ 3 & 5 & 0 & 20 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 58 & 39 & 16 \\ 6 & 13 & 0 & 39 & 24 & 10 \\ 3 & 5 & -2 & 16 & 10 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{1^{41}} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 8 & 3 \\ 2 & 1 & 1 & 20 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 20 & 0 & 158 & 76 & 24 \\ 8 & 9 & 0 & 76 & 35 & 11 \\ 3 & 3 & -1 & 24 & 11 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{1^{12}} 1^{21} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 8 & 3 \\ 2 & 1 & 1 & 18 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 18 & 0 & 166 & 78 & 24 \\ 8 & 9 & 0 & 78 & 35 & 11 \\ 3 & 3 & -1 & 24 & 11 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{1^1} 1^{23} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 12 & 3 \\ 6 & 8 & 12 & 0 & 12 & 56 & 14 \\ 2 & 2 & 2 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{1^1} 1^{12} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 5 & 0 & 20 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 20 & 0 & 62 & 24 & 16 \\ 4 & 8 & 0 & 24 & 8 & 6 \\ 3 & 5 & -2 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{114}3^{121}21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 2 \\ 1 & 2 & 0 & 0 & 7 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 47 & 8 \\ 9 & 13 & 17 & 0 & 47 & 127 & 21 \\ 2 & 2 & 2 & -2 & 8 & 21 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{112}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 16 & 3 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 52 & 9 \\ 16 & 16 & 0 & 52 & 116 & 20 \\ 3 & 3 & -1 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{113}3^{121}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 13 & 3 \\ 7 & 10 & 13 & 0 & 13 & 74 & 16 \\ 2 & 2 & 2 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{111}3^{121}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 23 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 17 & 4 \\ 11 & 23 & 0 & 17 & 78 & 18 \\ 3 & 5 & -2 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{123}3^{121}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 21 & 3 \\ 2 & 1 & 1 & 10 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 98 & 12 \\ 21 & 23 & 0 & 98 & 243 & 29 \\ 3 & 3 & -1 & 12 & 29 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{113}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 16 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 16 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{112}3^{121}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 5 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & -2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{111}3^{121}21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 7 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 21 & 8 \\ 4 & 5 & 8 & 0 & 21 & 23 & 9 \\ 2 & 2 & 2 & -2 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{112}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 15 & 3 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 52 & 9 \\ 15 & 16 & 0 & 52 & 116 & 20 \\ 3 & 3 & -1 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[211] 1^{1^1} 1^{13} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 10 & 3 \\ 2 & 1 & 1 & 13 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 13 & 0 & 61 & 60 & 15 \\ 10 & 12 & 0 & 60 & 56 & 14 \\ 3 & 3 & -1 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[213] 1^{1^1} 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 9 & 5 \\ 3 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & -2 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[212] 1^{1^1} 1^{13} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 28 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 21 & 4 \\ 13 & 28 & 0 & 21 & 116 & 22 \\ 3 & 5 & -2 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[214] 1^{1^{12}} 1^{13} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 18 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[215] 1^{1^{12}} 1^{13} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 8 & 3 \\ 2 & 1 & 1 & 25 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 25 & 0 & 313 & 107 & 33 \\ 8 & 9 & 0 & 107 & 35 & 11 \\ 3 & 3 & -1 & 33 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[216] 1^{1^{11}} 1^{142} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 10 & 3 \\ 2 & 1 & 1 & 16 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 16 & 0 & 90 & 72 & 18 \\ 10 & 12 & 0 & 72 & 56 & 14 \\ 3 & 3 & -1 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[217] 1^{1^{12}} 1^{124} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 28 & 3 \\ 2 & 1 & 1 & 10 & 30 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 128 & 12 \\ 28 & 30 & 0 & 128 & 416 & 38 \\ 3 & 3 & -1 & 12 & 38 & 3 \end{pmatrix}_{b_0}$$

$$[218] 1^{1^1} 1^{122} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 1 & 10 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 48 & 12 \\ 10 & 12 & 0 & 48 & 56 & 14 \\ 3 & 3 & -1 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{1^1} 1^{121} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -197$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 13 & 7 \\ 3 & 4 & 5 & 0 & 13 & 10 & 6 \\ 2 & 2 & 2 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{1^1} 1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -199$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 8 & 3 \\ 2 & 1 & 1 & 14 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 14 & 0 & 90 & 58 & 18 \\ 8 & 9 & 0 & 58 & 35 & 11 \\ 3 & 3 & -1 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{1^2} 1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -202$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 8 & 3 \\ 2 & 1 & 1 & 14 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 14 & 0 & 90 & 58 & 18 \\ 8 & 9 & 0 & 58 & 35 & 11 \\ 3 & 3 & -1 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{1^1} 1^{12} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -202$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 5 & 0 & 30 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 30 & 0 & 134 & 58 & 24 \\ 6 & 13 & 0 & 58 & 24 & 10 \\ 3 & 5 & -2 & 24 & 10 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{1^1} 1^{13} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 18 & 15 \\ 2 & 2 & 4 & 0 & 18 & 4 & 4 \\ 2 & 2 & 2 & -2 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{1^1} 1^{114} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 1 & 7 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 66 & 9 \\ 18 & 22 & 0 & 66 & 192 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{1^2} 1^{112} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 20 & 3 \\ 2 & 1 & 1 & 7 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 68 & 9 \\ 20 & 20 & 0 & 68 & 200 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{1^1} 1^{112} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 11 & 12 & 0 & 12 & 56 & 14 \\ 3 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{1^1} 1^{111} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 13 & 5 \\ 4 & 5 & 8 & 0 & 13 & 23 & 9 \\ 2 & 2 & 2 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{141}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 8 & 3 \\ 2 & 1 & 1 & 20 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 20 & 0 & 158 & 76 & 24 \\ 8 & 9 & 0 & 76 & 35 & 11 \\ 3 & 3 & -1 & 24 & 11 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{131}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 8 & 3 \\ 2 & 1 & 1 & 21 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 21 & 0 & 205 & 87 & 27 \\ 8 & 9 & 0 & 87 & 35 & 11 \\ 3 & 3 & -1 & 27 & 11 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{131}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 5 & 0 & 25 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 25 & 0 & 94 & 30 & 20 \\ 4 & 8 & 0 & 30 & 8 & 6 \\ 3 & 5 & -2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[232] $1^{142}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 10 & 3 \\ 2 & 1 & 1 & 20 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 20 & 0 & 158 & 96 & 24 \\ 10 & 12 & 0 & 96 & 56 & 14 \\ 3 & 3 & -1 & 24 & 14 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{122}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 10 & 3 \\ 2 & 1 & 1 & 10 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 10 & 0 & 38 & 48 & 12 \\ 10 & 12 & 0 & 48 & 56 & 14 \\ 3 & 3 & -1 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{113}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 17 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 17 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{131}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 16 & 3 \\ 8 & 11 & 16 & 0 & 16 & 103 & 19 \\ 2 & 2 & 2 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{121}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 18 & 10 \\ 3 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & -2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{1^1} 1^{12} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 5 & 0 & 30 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 30 & 0 & 134 & 36 & 24 \\ 4 & 8 & 0 & 36 & 8 & 6 \\ 3 & 5 & -2 & 24 & 6 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{1^{12}} 1^{124} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 1 & 10 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 88 & 12 \\ 20 & 22 & 0 & 88 & 192 & 26 \\ 3 & 3 & -1 & 12 & 26 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{1^1} 1^{14} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 0 & 0 & 3 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 17 & 3 \\ 9 & 13 & 17 & 0 & 17 & 127 & 21 \\ 2 & 2 & 2 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{1^{11}} 1^{114} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 19 & 3 \\ 2 & 1 & 1 & 7 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 66 & 9 \\ 19 & 22 & 0 & 66 & 192 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{1^{12}} 1^{112} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 12 & 3 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 12 & 12 & 0 & 12 & 56 & 14 \\ 3 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{1^{11}} 1^{114} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 18 & 3 \\ 2 & 1 & 1 & 3 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 22 & 3 \\ 18 & 22 & 0 & 22 & 192 & 26 \\ 3 & 3 & -1 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{1^{11}} 1^{114} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 23 & 3 \\ 2 & 1 & 1 & 3 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 26 & 3 \\ 23 & 26 & 0 & 26 & 292 & 32 \\ 3 & 3 & -1 & 3 & 32 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{1^{12}} 1^{123} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 1 & 10 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 78 & 12 \\ 16 & 19 & 0 & 78 & 151 & 23 \\ 3 & 3 & -1 & 12 & 23 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{1^{11}} 1^{132} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 10 & 3 \\ 2 & 1 & 1 & 13 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 13 & 0 & 61 & 60 & 15 \\ 10 & 12 & 0 & 60 & 56 & 14 \\ 3 & 3 & -1 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{12} 1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 8 & 3 \\ 2 & 1 & 1 & 17 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 17 & 0 & 121 & 67 & 21 \\ 8 & 9 & 0 & 67 & 35 & 11 \\ 3 & 3 & -1 & 21 & 11 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{11} 1^{113} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 17 & 3 \\ 2 & 1 & 1 & 3 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 19 & 3 \\ 17 & 19 & 0 & 19 & 151 & 23 \\ 3 & 3 & -1 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[247] $1^1 1^{121} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 26 & 10 \\ 4 & 5 & 8 & 0 & 26 & 23 & 9 \\ 2 & 2 & 2 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{11} 1^{113} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 21 & 3 \\ 2 & 1 & 1 & 7 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 75 & 9 \\ 21 & 23 & 0 & 75 & 243 & 29 \\ 3 & 3 & -1 & 9 & 29 & 3 \end{pmatrix}_{b_0}$$

[249] $1^1 1^{113} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 0 & 1 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 7 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{11} 1^{141} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 10 & 0 & 58 & 26 & 14 \\ 3 & 4 & 5 & 0 & 26 & 10 & 6 \\ 2 & 2 & 2 & -2 & 14 & 6 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{11} 1^{141} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 8 & 3 \\ 2 & 1 & 1 & 20 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 20 & 0 & 158 & 76 & 24 \\ 8 & 9 & 0 & 76 & 35 & 11 \\ 3 & 3 & -1 & 24 & 11 & 3 \end{pmatrix}_{b_0}$$

[252] $1^1 1^{112} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 7 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 32 & 8 \\ 6 & 8 & 12 & 0 & 32 & 56 & 14 \\ 2 & 2 & 2 & -2 & 8 & 14 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{131}3^{11}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 8 & 3 \\ 2 & 1 & 1 & 17 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 17 & 0 & 121 & 67 & 21 \\ 8 & 9 & 0 & 67 & 35 & 11 \\ 3 & 3 & -1 & 21 & 11 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{122}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 23 & 7 \\ 5 & 7 & 9 & 0 & 23 & 35 & 11 \\ 2 & 2 & 2 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{121}3^{11}21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 34 & 13 \\ 4 & 5 & 8 & 0 & 34 & 23 & 9 \\ 2 & 2 & 2 & -2 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{123}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 5 & 0 & 5 & 33 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 24 & 4 \\ 16 & 33 & 0 & 24 & 164 & 26 \\ 3 & 5 & -2 & 4 & 26 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{122}3^{11}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 18 & 7 \\ 4 & 6 & 6 & 0 & 18 & 20 & 8 \\ 2 & 2 & 2 & -2 & 7 & 8 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{142}3^{11}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 10 & 3 \\ 2 & 1 & 1 & 24 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 24 & 0 & 250 & 120 & 30 \\ 10 & 12 & 0 & 120 & 56 & 14 \\ 3 & 3 & -1 & 30 & 14 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{143}3^{11}21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 24 & 20 \\ 2 & 2 & 4 & 0 & 24 & 4 & 4 \\ 2 & 2 & 2 & -2 & 20 & 4 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{123}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 17 & 3 \\ 2 & 1 & 1 & 10 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 78 & 12 \\ 17 & 19 & 0 & 78 & 151 & 23 \\ 3 & 3 & -1 & 12 & 23 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{11} 1^{113} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 \\ 1 & 2 & 0 & 0 & 1 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 \\ 6 & 9 & 10 & 0 & 9 & 51 \\ 2 & 2 & 2 & -2 & 2 & 13 \end{pmatrix}_{b_0}$$

[262] $1^{12} 1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 26 & 3 \\ 2 & 1 & 1 & 7 & 27 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 91 & 9 \\ 26 & 27 & 0 & 91 & 359 & 35 \\ 3 & 3 & -1 & 9 & 35 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{11} 1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 9 & 0 & 40 & 22 & 12 \\ 3 & 4 & 5 & 0 & 22 & 10 & 6 \\ 2 & 2 & 2 & -2 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{11} 1^{131} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 39 & 15 \\ 4 & 5 & 8 & 0 & 39 & 23 & 9 \\ 2 & 2 & 2 & -2 & 15 & 9 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{12} 1^{113} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 22 & 3 \\ 2 & 1 & 1 & 7 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 75 & 9 \\ 22 & 23 & 0 & 75 & 243 & 29 \\ 3 & 3 & -1 & 9 & 29 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{11} 1^{113} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 15 & 3 \\ 13 & 15 & 0 & 15 & 83 & 17 \\ 3 & 3 & -1 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{11} 1^{131} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 27 & 15 \\ 3 & 4 & 5 & 0 & 27 & 10 & 6 \\ 2 & 2 & 2 & -2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{11} 1^{132} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 10 & 3 \\ 2 & 1 & 1 & 17 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 17 & 0 & 121 & 84 & 21 \\ 10 & 12 & 0 & 84 & 56 & 14 \\ 3 & 3 & -1 & 21 & 14 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{123}3^121^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 38 & 10 \\ 6 & 9 & 10 & 0 & 38 & 51 & 13 \\ 2 & 2 & 2 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{124}3^121^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 24 & 3 \\ 2 & 1 & 1 & 10 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 108 & 12 \\ 24 & 26 & 0 & 108 & 292 & 32 \\ 3 & 3 & -1 & 12 & 32 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{113}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 18 & 3 \\ 2 & 1 & 1 & 3 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 19 & 3 \\ 18 & 19 & 0 & 19 & 151 & 23 \\ 3 & 3 & -1 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{131}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 8 & 3 \\ 2 & 1 & 1 & 21 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 21 & 0 & 205 & 87 & 27 \\ 8 & 9 & 0 & 87 & 35 & 11 \\ 3 & 3 & -1 & 27 & 11 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{114}3^121^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 23 & 3 \\ 2 & 1 & 1 & 7 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 82 & 9 \\ 23 & 26 & 0 & 82 & 292 & 32 \\ 3 & 3 & -1 & 9 & 32 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{112}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 16 & 5 \\ 5 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & -2 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{114}3^121^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 0 & 1 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 9 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & -2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{124}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 18 & 3 \\ 2 & 1 & 1 & 10 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 88 & 12 \\ 18 & 22 & 0 & 88 & 192 & 26 \\ 3 & 3 & -1 & 12 & 26 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{11} 1^{141} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 8 & 3 \\ 2 & 1 & 1 & 24 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 24 & 0 & 250 & 96 & 30 \\ 8 & 9 & 0 & 96 & 35 & 11 \\ 3 & 3 & -1 & 30 & 11 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{11} 1^{113} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 7 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 43 & 8 \\ 8 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & -2 & 8 & 19 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{112} 1^{142} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 10 & 3 \\ 2 & 1 & 1 & 20 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 20 & 0 & 158 & 96 & 24 \\ 10 & 12 & 0 & 96 & 56 & 14 \\ 3 & 3 & -1 & 24 & 14 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{112} 1^{114} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 24 & 3 \\ 2 & 1 & 1 & 3 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 26 & 3 \\ 24 & 26 & 0 & 26 & 292 & 32 \\ 3 & 3 & -1 & 3 & 32 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{11} 1^{123} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 33 & 7 \\ 7 & 10 & 13 & 0 & 33 & 74 & 16 \\ 2 & 2 & 2 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{112} 1^{114} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 24 & 3 \\ 2 & 1 & 1 & 7 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 82 & 9 \\ 24 & 26 & 0 & 82 & 292 & 32 \\ 3 & 3 & -1 & 9 & 32 & 3 \end{pmatrix}_{b_0}$$

[283] $1^{112} 1^{113} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.00 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 15 & 3 \\ 14 & 15 & 0 & 15 & 83 & 17 \\ 3 & 3 & -1 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{11} 1^{112} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 20 & 5 \\ 6 & 8 & 12 & 0 & 20 & 56 & 14 \\ 2 & 2 & 2 & -2 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{11} 1^{141} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 16 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 52 & 20 \\ 4 & 5 & 8 & 0 & 52 & 23 & 9 \\ 2 & 2 & 2 & -2 & 20 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{122} 3^{121} 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 11 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 52 & 13 \\ 6 & 8 & 12 & 0 & 52 & 56 & 14 \\ 2 & 2 & 2 & -2 & 13 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{122} 3^{121} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 32 & 10 \\ 5 & 7 & 9 & 0 & 32 & 35 & 11 \\ 2 & 2 & 2 & -2 & 10 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{114} 3^{121} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 19 & 3 \\ 2 & 1 & 1 & 3 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 22 & 3 \\ 19 & 22 & 0 & 22 & 192 & 26 \\ 3 & 3 & -1 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[289] \ 1^{124} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 23 & 3 \\ 2 & 1 & 1 & 10 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 108 & 12 \\ 23 & 26 & 0 & 108 & 292 & 32 \\ 3 & 3 & -1 & 12 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{131} 3^{121} 21^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 15 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 47 & 18 \\ 4 & 5 & 8 & 0 & 47 & 23 & 9 \\ 2 & 2 & 2 & -2 & 18 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[291] \ 1^{124} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 19 & 3 \\ 2 & 1 & 1 & 10 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 88 & 12 \\ 19 & 22 & 0 & 88 & 192 & 26 \\ 3 & 3 & -1 & 12 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[292] \ 1^{132} 3^{121} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 9 & 0 & 40 & 39 & 12 \\ 5 & 7 & 9 & 0 & 39 & 35 & 11 \\ 2 & 2 & 2 & -2 & 12 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[293] \ 1^{113} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 19 & 5 \\ 6 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & -2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{141}3^121^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 12 & 4 & 2 \\ 1 & 2 & 0 & 0 & 16 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 36 & 20 \\ 3 & 4 & 5 & 0 & 36 & 10 & 6 \\ 2 & 2 & 2 & -2 & 20 & 6 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{142}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 10 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 10 & 0 & 58 & 36 & 14 \\ 4 & 6 & 6 & 0 & 36 & 20 & 8 \\ 2 & 2 & 2 & -2 & 14 & 8 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{122}3^121^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 8 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 40 & 10 \\ 6 & 8 & 12 & 0 & 40 & 56 & 14 \\ 2 & 2 & 2 & -2 & 10 & 14 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{114}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 1 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 8 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{114}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 28 & 3 \\ 2 & 1 & 1 & 7 & 30 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 98 & 9 \\ 28 & 30 & 0 & 98 & 416 & 38 \\ 3 & 3 & -1 & 9 & 38 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{141}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 13 & 0 & 81 & 31 & 17 \\ 3 & 4 & 5 & 0 & 31 & 10 & 6 \\ 2 & 2 & 2 & -2 & 17 & 6 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{141}3^121^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 8 & 3 \\ 2 & 1 & 1 & 28 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 26 & 28 & 0 & 366 & 116 & 36 \\ 8 & 9 & 0 & 116 & 35 & 11 \\ 3 & 3 & -1 & 36 & 11 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{142}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 10 & 3 \\ 2 & 1 & 1 & 20 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 20 & 0 & 158 & 96 & 24 \\ 10 & 12 & 0 & 96 & 56 & 14 \\ 3 & 3 & -1 & 24 & 14 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{141}3^{12}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 8 & 3 \\ 2 & 1 & 1 & 24 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 24 & 0 & 250 & 96 & 30 \\ 8 & 9 & 0 & 96 & 35 & 11 \\ 3 & 3 & -1 & 30 & 11 & 3 \end{pmatrix}_{b_0}$$

[304] $1^{123}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 28 & 7 \\ 6 & 9 & 10 & 0 & 28 & 51 & 13 \\ 2 & 2 & 2 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{132}3^{11}21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 7 & 2 \\ 1 & 2 & 0 & 0 & 12 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 48 & 15 \\ 5 & 7 & 9 & 0 & 48 & 35 & 11 \\ 2 & 2 & 2 & -2 & 15 & 11 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{132}3^{11}21^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 8 & 2 \\ 1 & 2 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 60 & 15 \\ 6 & 8 & 12 & 0 & 60 & 56 & 14 \\ 2 & 2 & 2 & -2 & 15 & 14 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{124}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 13 & 2 \\ 1 & 2 & 0 & 0 & 5 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 43 & 7 \\ 9 & 13 & 17 & 0 & 43 & 127 & 21 \\ 2 & 2 & 2 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{114}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -628, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 1 & 3 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 22 & 3 \\ 20 & 22 & 0 & 22 & 192 & 26 \\ 3 & 3 & -1 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{123}3^{11}21^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 11 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 70 & 13 \\ 8 & 11 & 16 & 0 & 70 & 103 & 19 \\ 2 & 2 & 2 & -2 & 13 & 19 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{132}3^{11}21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 8 & 2 \\ 1 & 2 & 0 & 0 & 15 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 72 & 18 \\ 6 & 8 & 12 & 0 & 72 & 56 & 14 \\ 2 & 2 & 2 & -2 & 18 & 14 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{142}3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 7 & 2 \\ 1 & 2 & 0 & 0 & 13 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 13 & 0 & 81 & 55 & 17 \\ 5 & 7 & 9 & 0 & 55 & 35 & 11 \\ 2 & 2 & 2 & -2 & 17 & 11 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{142}3^1 21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 0 & 12 & 8 & 2 \\ 1 & 2 & 0 & 0 & 16 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 80 & 20 \\ 6 & 8 & 12 & 0 & 80 & 56 & 14 \\ 2 & 2 & 2 & -2 & 20 & 14 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{113}3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -717, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 23 & 5 \\ 7 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{142}3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 0 & 12 & 7 & 2 \\ 1 & 2 & 0 & 0 & 16 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 64 & 20 \\ 5 & 7 & 9 & 0 & 64 & 35 & 11 \\ 2 & 2 & 2 & -2 & 20 & 11 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{124}3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 33 & 7 \\ 7 & 11 & 11 & 0 & 33 & 71 & 15 \\ 2 & 2 & 2 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{124}3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 52 & 10 \\ 8 & 12 & 14 & 0 & 52 & 96 & 18 \\ 2 & 2 & 2 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{123}3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -780, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 46 & 10 \\ 7 & 10 & 13 & 0 & 46 & 74 & 16 \\ 2 & 2 & 2 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{113}3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 27 & 5 \\ 8 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & -2 & 5 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[318] \quad 1^{124}3^1 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 11 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 54 & 10 \\ 8 & 11 & 16 & 0 & 54 & 103 & 19 \\ 2 & 2 & 2 & -2 & 10 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{124}3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -852, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 38 & 7 \\ 8 & 12 & 14 & 0 & 38 & 96 & 18 \\ 2 & 2 & 2 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[320] \quad 1^{114}3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 26 & 5 \\ 8 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & -2 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[321] \quad 1^{114}3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 0 & 4 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 30 & 5 \\ 9 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & -2 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[322] \quad 1^{124}3^1 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 6 & 13 & 2 \\ 1 & 2 & 0 & 0 & 8 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 60 & 10 \\ 9 & 13 & 17 & 0 & 60 & 127 & 21 \\ 2 & 2 & 2 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [323] \quad 1^{13}3^1 1^5 13221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 9.0$$

$$[324] \quad 1^{13}3^1 1^5 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 8 & 4 \\ 3 & 0 & 5 & 3 & 8 & 5 \\ 5 & 0 & 8 & 8 & 15 & 9 \\ 3 & 0 & 4 & 5 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{11}3^1 1^5 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 19 & 8 & 4 \\ 13 & 0 & 19 & 79 & 36 & 21 \\ 5 & 0 & 8 & 36 & 15 & 9 \\ 3 & 0 & 4 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{11}3^1 1^5 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) =$$

$$(1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 7 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 9 & 8 & 4 \\ 7 & 0 & 9 & 23 & 20 & 11 \\ 5 & 0 & 8 & 20 & 15 & 9 \\ 3 & 0 & 4 & 11 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{11}3^{11}1^5 1321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 14 & 8 & 4 \\ 10 & 0 & 14 & 46 & 28 & 16 \\ 5 & 0 & 8 & 28 & 15 & 9 \\ 3 & 0 & 4 & 16 & 9 & 5 \end{pmatrix}_{b_0}$$

$$4.677 \quad \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{so}_8$$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & -2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$4.678 \quad \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 2 & 3 \\ 3 & 4 & 0 & 14 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 17 & 8 & 11 \\ 4 & 7 & 0 & 17 & 7 & 4 & 5 \\ 2 & 3 & 0 & 8 & 4 & 0 & 2 \\ 3 & 4 & -2 & 11 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 8 & 16 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 22 & 4 & 6 \\ 9 & 16 & 0 & 22 & 44 & 9 & 12 \\ 2 & 3 & 0 & 4 & 9 & 0 & 2 \\ 3 & 4 & -2 & 6 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 10 & 4 & 6 \\ 4 & 7 & 0 & 10 & 7 & 4 & 5 \\ 2 & 3 & 0 & 4 & 4 & 0 & 2 \\ 3 & 4 & -2 & 6 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 4 & 7 & 0 & 6 & 7 & 4 & 5 \\ 2 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -2 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 2 & 3 \\ 4 & 7 & 0 & 5 & 7 & 4 & 5 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -2 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 0 & 4 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 2 & 3 \\ 6 & 11 & 0 & 8 & 20 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -2 & 3 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 2 & 3 \\ 3 & 4 & 0 & 8 & 20 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 28 & 4 & 6 \\ 11 & 20 & 0 & 28 & 69 & 11 & 15 \\ 2 & 3 & 0 & 4 & 11 & 0 & 2 \\ 3 & 4 & -2 & 6 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 11 & 5 & 7 \\ 4 & 7 & 0 & 11 & 7 & 4 & 5 \\ 2 & 3 & 0 & 5 & 4 & 0 & 2 \\ 3 & 4 & -2 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 2 & 3 \\ 3 & 4 & 0 & 8 & 29 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 40 & 4 & 6 \\ 16 & 29 & 0 & 40 & 148 & 16 & 22 \\ 2 & 3 & 0 & 4 & 16 & 0 & 2 \\ 3 & 4 & -2 & 6 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 2 & 3 \\ 7 & 12 & 0 & 8 & 25 & 7 & 9 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -2 & 3 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 2 & 3 \\ 3 & 4 & 0 & 21 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 26 & 11 & 16 \\ 4 & 7 & 0 & 26 & 7 & 4 & 5 \\ 2 & 3 & 0 & 11 & 4 & 0 & 2 \\ 3 & 4 & -2 & 16 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 2 & 3 \\ 3 & 4 & 0 & 18 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 22 & 10 & 14 \\ 4 & 7 & 0 & 22 & 7 & 4 & 5 \\ 2 & 3 & 0 & 10 & 4 & 0 & 2 \\ 3 & 4 & -2 & 14 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 2 & 3 \\ 3 & 4 & 0 & 13 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 25 & 7 & 10 \\ 6 & 11 & 0 & 25 & 20 & 6 & 8 \\ 2 & 3 & 0 & 7 & 6 & 0 & 2 \\ 3 & 4 & -2 & 10 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 2 & 3 \\ 3 & 4 & 0 & 13 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 16 & 7 & 10 \\ 4 & 7 & 0 & 16 & 7 & 4 & 5 \\ 2 & 3 & 0 & 7 & 4 & 0 & 2 \\ 3 & 4 & -2 & 10 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 2 & 3 \\ 3 & 4 & 0 & 17 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 21 & 9 & 13 \\ 4 & 7 & 0 & 21 & 7 & 4 & 5 \\ 2 & 3 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & -2 & 13 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{124}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 2 & 3 \\ 3 & 4 & 0 & 8 & 24 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 34 & 4 & 6 \\ 13 & 24 & 0 & 34 & 100 & 13 & 18 \\ 2 & 3 & 0 & 4 & 13 & 0 & 2 \\ 3 & 4 & -2 & 6 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 2 & 3 \\ 3 & 4 & 0 & 17 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 33 & 9 & 13 \\ 6 & 11 & 0 & 33 & 20 & 6 & 8 \\ 2 & 3 & 0 & 9 & 6 & 0 & 2 \\ 3 & 4 & -2 & 13 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{142}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 2 & 3 \\ 3 & 4 & 0 & 21 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 41 & 11 & 16 \\ 6 & 11 & 0 & 41 & 20 & 6 & 8 \\ 2 & 3 & 0 & 11 & 6 & 0 & 2 \\ 3 & 4 & -2 & 16 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 16 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 2 & 3 \\ 9 & 16 & 0 & 11 & 44 & 9 & 12 \\ 2 & 3 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -2 & 3 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{131}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 2 & 3 \\ 3 & 4 & 0 & 22 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 27 & 12 & 17 \\ 4 & 7 & 0 & 27 & 7 & 4 & 5 \\ 2 & 3 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -2 & 17 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{142}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 2 & 3 \\ 3 & 4 & 0 & 26 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 50 & 14 & 20 \\ 6 & 11 & 0 & 50 & 20 & 6 & 8 \\ 2 & 3 & 0 & 14 & 6 & 0 & 2 \\ 3 & 4 & -2 & 20 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 2 & 3 \\ 12 & 21 & 0 & 14 & 80 & 12 & 16 \\ 2 & 3 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -2 & 3 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 20 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 2 & 3 \\ 11 & 20 & 0 & 14 & 69 & 11 & 15 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -2 & 3 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{141}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 2 & 3 \\ 3 & 4 & 0 & 26 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 32 & 14 & 20 \\ 4 & 7 & 0 & 32 & 7 & 4 & 5 \\ 2 & 3 & 0 & 14 & 4 & 0 & 2 \\ 3 & 4 & -2 & 20 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 24 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 2 & 3 \\ 13 & 24 & 0 & 17 & 100 & 13 & 18 \\ 2 & 3 & 0 & 2 & 13 & 0 & 2 \\ 3 & 4 & -2 & 3 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 25 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 2 & 3 \\ 14 & 25 & 0 & 17 & 111 & 14 & 19 \\ 2 & 3 & 0 & 2 & 14 & 0 & 2 \\ 3 & 4 & -2 & 3 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 4 & 29 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 2 & 3 \\ 16 & 29 & 0 & 20 & 148 & 16 & 22 \\ 2 & 3 & 0 & 2 & 16 & 0 & 2 \\ 3 & 4 & -2 & 3 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

4.679 NHC(-2)(-3) + so₈ + su₂ + su2n3mix + su2n3mix

$$T = 9 \quad [0] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 8 & 3 \\ 6 & 11 & 0 & 8 & 20 & 23 & 8 \\ 7 & 12 & 0 & 8 & 23 & 25 & 9 \\ 3 & 4 & -2 & 3 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 9 & 5 \\ 4 & 7 & 0 & 10 & 9 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 11 & 11 & 7 \\ 4 & 7 & 0 & 11 & 7 & 9 & 5 \\ 4 & 7 & 0 & 11 & 9 & 7 & 5 \\ 3 & 4 & -2 & 7 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 4 & 7 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 8 & 3 \\ 4 & 7 & 0 & 5 & 7 & 15 & 5 \\ 7 & 12 & 0 & 8 & 15 & 25 & 9 \\ 3 & 4 & -2 & 3 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 5 & 3 \\ 4 & 7 & 0 & 5 & 7 & 9 & 5 \\ 4 & 7 & 0 & 5 & 9 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 4 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 14 & 3 \\ 4 & 7 & 0 & 5 & 7 & 26 & 5 \\ 12 & 21 & 0 & 14 & 26 & 80 & 16 \\ 3 & 4 & -2 & 3 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1211} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 16 & 16 & 10 \\ 4 & 7 & 0 & 16 & 7 & 9 & 5 \\ 4 & 7 & 0 & 16 & 9 & 7 & 5 \\ 3 & 4 & -2 & 10 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 4 & 7 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 11 & 3 \\ 4 & 7 & 0 & 5 & 7 & 20 & 5 \\ 9 & 16 & 0 & 11 & 20 & 44 & 12 \\ 3 & 4 & -2 & 3 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

4.680 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

- [1] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & 0 & 13 & 2 \end{pmatrix}_{b_0}$
- [2] $1^{141} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 24 & 12 \\ 5 & 9 & 0 & 24 & 12 & 6 \\ 3 & 4 & -2 & 12 & 6 & 1 \end{pmatrix}_{b_0}$
- [3] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 3 & 4 & -2 & 9 & 9 & 1 \end{pmatrix}_{b_0}$
- [4] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [5] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 25$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 0 & -1 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 24 & 0 & 121 & 15 & 19 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 5 & 5 \\ 12 & 25 & 20 & 16 & 9 & 9 \\ 9 & 20 & 13 & 12 & 7 & 7 \\ 8 & 16 & 12 & 9 & 5 & 5 \\ 5 & 9 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & -2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^{21} 1^3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 & 5 \\ 11 & 20 & 16 & 0 & 8 & 8 \\ 10 & 16 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 8 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 & 5 \\ 11 & 20 & 17 & 0 & 8 & 8 \\ 10 & 17 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 8 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 & 5 \\ 11 & 20 & 0 & 17 & 8 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 10 & 17 & 0 & 12 & 6 & 6 \\ 5 & 8 & 0 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11} 1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 11 & 13 & 6 & 6 \\ 15 & 20 & 16 & 17 & 8 & 8 \\ 11 & 16 & 10 & 12 & 6 & 6 \\ 13 & 17 & 12 & 12 & 6 & 6 \\ 6 & 8 & 6 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1^3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 1 & 3 \\ 3 & 4 & 0 & 19 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 15 & -1 & 1 \end{pmatrix}_{b_0}$$

[22] $1^1 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

- [21] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 45 & -1 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 10 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 8 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 & 6 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 60 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 45 & -1 & 2 \end{pmatrix}_{b_0}$
- [27] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 25 & 18 & 9 \\ 5 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$
- [28] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 3 & -1 & 1 \end{pmatrix}_{b_0}$
- [29] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 37 & 8 & 5 \\ 12 & 25 & 80 & 16 & 9 \\ 37 & 80 & 238 & 48 & 28 \\ 8 & 16 & 48 & 9 & 5 \\ 5 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [31] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 10 & 8 & 5 \\ 12 & 25 & 20 & 16 & 9 \\ 10 & 20 & 13 & 12 & 7 \\ 8 & 16 & 12 & 9 & 5 \\ 5 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[41] $21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 13 & 15 & 7 \\ 15 & 17 & 15 & 12 & 6 \\ 7 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 8 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -2 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 17 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 28 & 8 & 5 \\ 12 & 25 & 60 & 16 & 9 \\ 28 & 60 & 133 & 36 & 21 \\ 8 & 16 & 36 & 9 & 5 \\ 5 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 19 & 8 & 5 \\ 12 & 25 & 40 & 16 & 9 \\ 19 & 40 & 58 & 24 & 14 \\ 8 & 16 & 24 & 9 & 5 \\ 5 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[51] \ 1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

- [54] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 75$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [55] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 76$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [56] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 77$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [57] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 78$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 20 & 8 & 5 \\ 12 & 25 & 40 & 16 & 9 \\ 20 & 40 & 58 & 24 & 14 \\ 8 & 16 & 24 & 9 & 5 \\ 5 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$
- [58] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$
- [59] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$
- [60] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [61] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 82$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [62] $1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 82$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$
- [63] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 83$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 4 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
- [64] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 10 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 10 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 14 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 85, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 85, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{13} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 86, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 48 & 12 \\ 3 & 4 & -2 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 89, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -2 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{12} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 93, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.44 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 11 & 1 & 6 \\ 15 & 20 & 16 & 0 & 8 \\ 11 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.55 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 20 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 20 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{11} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 24 & 0 & 121 & 15 & 19 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

- [77] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$
- [78] $1^{121} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$
- [80] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [75] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$
- [74] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$
- [79] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [81] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 18 & 9 \\ 5 & 9 & 0 & 18 & 12 & 6 \\ 3 & 4 & -2 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$
- [82] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 48 & 1 & 5 & 8 \\ 11 & 20 & 84 & 0 & 8 & 8 \\ 48 & 84 & 346 & 0 & 34 & 34 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 8 & 34 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [83] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 9 & 7 \\ 4 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$
- [84] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 12 & 21 & 0 & 18 & 75 & 15 \\ 3 & 4 & -2 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$
- [85] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -2 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{121} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 12 & 14 \\ 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -2 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{111} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 17 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{12} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 23 & 1 & 6 \\ 15 & 20 & 33 & 0 & 8 \\ 23 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 9 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -2 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 30 & 8 & 5 \\ 12 & 25 & 60 & 16 & 9 \\ 30 & 60 & 133 & 36 & 21 \\ 8 & 16 & 36 & 9 & 5 \\ 5 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 29 & 8 & 5 \\ 12 & 25 & 60 & 16 & 9 \\ 29 & 60 & 133 & 36 & 21 \\ 8 & 16 & 36 & 9 & 5 \\ 5 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -2 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 24 & 0 & 121 & 15 & 19 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 26 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 26 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 \\ 3 & 4 & -2 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 13 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 13 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{12} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 1 & 3 \\ 3 & 4 & 0 & 28 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 28 & 0 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 22 & -1 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 28 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{112} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 20 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 20 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -2 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 1 & 3 \\ 3 & 4 & 0 & 18 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -1 & 2 \end{pmatrix}_{b_0}$$

[114] $21^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 35 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -1 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 38 & 8 & 5 \\ 12 & 25 & 80 & 16 & 9 \\ 38 & 80 & 238 & 48 & 28 \\ 8 & 16 & 48 & 9 & 5 \\ 5 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 1 & 3 \\ 3 & 4 & 0 & 23 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 23 & 0 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 18 & -1 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 29 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 29 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 166$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 166$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 168$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 170$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 10 & 17 & 0 & 15 & 48 & 12 \\ 3 & 4 & -2 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 15 & 16 \\ 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -2 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{112} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{122} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -2 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{113} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 46 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{14} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 1 & 3 \\ 3 & 4 & 0 & 26 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{13} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 1 & 3 \\ 3 & 4 & 0 & 22 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 17 & -1 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{112} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[143] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 15 & 12 & 7 \\ 20 & 25 & 20 & 16 & 9 \\ 15 & 20 & 13 & 12 & 7 \\ 12 & 16 & 12 & 9 & 5 \\ 7 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{132} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 3 & 4 & -2 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 40 & 8 & 5 \\ 12 & 25 & 80 & 16 & 9 \\ 40 & 80 & 238 & 48 & 28 \\ 8 & 16 & 48 & 9 & 5 \\ 5 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{131} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

- [147] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 14 & -1 & 2 \end{pmatrix}_{b_0}$
- [148] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$
- [149] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 27 & 9 \\ 3 & 4 & -2 & 4 & 9 & 1 \end{pmatrix}_{b_0}$
- [150] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 19 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$
- [151] $1^1 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 17 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$
- [152] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 19 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$
- [153] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 5 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 10 & 5 & 1 \end{pmatrix}_{b_0}$
- [154] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 39 & 8 & 5 \\ 12 & 25 & 80 & 16 & 9 \\ 39 & 80 & 238 & 48 & 28 \\ 8 & 16 & 48 & 9 & 5 \\ 5 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$
- [155] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 53 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 53 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$
- [156] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$
- [157] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 45 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 45 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 210$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 6 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -2 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[160] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 210$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -2 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 211$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 213$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 214$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 221$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 221$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{131} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 15 & 17 \\ 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{132} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{131} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 71 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 71 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -1 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 18 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 25 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 25 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 27 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

- [178] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 0 & 9 & 24 & 8 \\ 3 & 4 & -2 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$
- [179] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$
- [180] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 38 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 38 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$
- [181] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 24 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [182] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$
- [184] $21^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 27 & 15 & 7 \\ 17 & 20 & 33 & 17 & 8 \\ 27 & 33 & 49 & 27 & 13 \\ 15 & 17 & 27 & 12 & 6 \\ 7 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [183] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 30 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 30 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$
- [185] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [187] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 1 & 3 \\ 3 & 4 & 0 & 27 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 27 & 0 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 21 & -1 & 1 \end{pmatrix}_{b_0}$$
- [186] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$$
- [188] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 47 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 47 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 24 & 13 \\ 5 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -2 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{141} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 31 & 0 & 172 & 21 & 24 \\ 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -2 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{132} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[195] $21^{112} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 30 & 12 & 7 \\ 20 & 25 & 40 & 16 & 9 \\ 30 & 40 & 58 & 24 & 14 \\ 12 & 16 & 24 & 9 & 5 \\ 7 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 57 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 38 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 36 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 36 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

- [199] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$
- [200] $21^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 24 & 12 & 6 \\ 16 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [201] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 27 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 27 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [202] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 28 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 28 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [203] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 1 & 3 \\ 3 & 4 & 0 & 32 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 32 & 0 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 25 & -1 & 1 \end{pmatrix}_{b_0}$$
- [204] $1^{132} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 8 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -2 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$
- [205] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 46 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 46 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [206] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -2 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$
- [207] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$
- [208] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$
- [209] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 301, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -2 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 52 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{141} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 18 & 20 \\ 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 28 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 1 & 3 \\ 3 & 4 & 0 & 31 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 31 & 0 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 37 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 37 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{131} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -2 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 48 & 12 \\ 3 & 4 & -2 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{13}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -2 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{113}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 29 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 29 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 30 & 17 \\ 4 & 8 & 0 & 30 & 9 & 5 \\ 3 & 4 & -2 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 22 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 30 & 17 \\ 4 & 8 & 0 & 30 & 9 & 5 \\ 3 & 4 & -2 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -2 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -2 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 11 & 18 & 0 & 12 & 57 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 45 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 56 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 56 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{13} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 40 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 40 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -2 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -2 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 395$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{14} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 1 & 3 \\ 3 & 4 & 0 & 36 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 36 & 0 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{14} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 & 9 \\ 12 & 25 & 144 & 0 & 9 & 52 \\ 68 & 144 & 814 & 0 & 52 & -1 \\ 1 & 0 & 0 & -3 & -1 & 2 \\ 5 & 9 & 52 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 & 9 \\ 12 & 25 & 96 & 0 & 9 & 34 \\ 44 & 96 & 346 & 0 & 34 & -1 \\ 1 & 0 & 0 & -3 & -1 & 2 \\ 5 & 9 & 34 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 406$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -2 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 411$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 416$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 416$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{12} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 416$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{12} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 416$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{121} 3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{131} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 8 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -2 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 45 & 12 & 7 \\ 20 & 25 & 60 & 16 & 9 \\ 45 & 60 & 133 & 36 & 21 \\ 12 & 16 & 36 & 9 & 5 \\ 7 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 37 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{132} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -2 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{113} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 37 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 44 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 36 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{113} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

- [260] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 70 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 70 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -1 & 2 \end{pmatrix}_{b_0}$$
- [261] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 48 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 48 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [262] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -2 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [263] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 10 & 17 & 0 & 27 & 48 & 12 \\ 3 & 4 & -2 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$
- [264] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$
- [265] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$
- [266] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$
- [267] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -2 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$
- [268] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$
- [269] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$
- [270] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -2 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[271] 1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 69 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 69 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[272] 1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 498, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -2 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[273] 1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 63 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 63 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[274] 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[275] 1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[276] 1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[277] 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[278] 1^{141} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[279] 1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 39 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 39 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[280] 1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 38 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

- [281] $1^{132}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -2 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$
- [282] $1^{14}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$
- [283] $1^{121}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$
- [284] $1^{141}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -2 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$
- [285] $1^{113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$
- [286] $1^{131}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$
- [290] $1^{141}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$
- [287] $1^{121}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$
- [288] $1^{141}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$
- [289] $1^{121}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$
- [291] $21^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 60 & 12 & 7 \\ 20 & 25 & 80 & 16 & 9 \\ 60 & 80 & 238 & 48 & 28 \\ 12 & 16 & 48 & 9 & 5 \\ 7 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[292] \ 1^{114} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 45 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[293] \ 1^{131} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[294] \ 1^{114} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 645, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 55 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 55 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[295] \ 1^{131} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[296] \ 1^{114} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 62 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 62 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[297] \ 1^{114} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 47 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 47 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[298] \ 1^{114} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 669, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 61 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 61 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[304] \ 1^{121} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[300] \ 1^{131} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[301] \ 1^{121} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[302] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{141} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 53 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 53 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{132} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -2 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{141} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 730, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -2 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 54 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 19 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 19 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[353] \ 1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 0 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[354] \ 21^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 9 & 7 \\ 8 & 10 & 9 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[355] \ 1^{12} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[356] \ 1^{111} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 6 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 9 & 7 \\ 8 & 10 & 9 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[358] \ 21^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 6 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[359] \ 21^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 15 & 9 & 7 \\ 14 & 13 & 14 & 10 & 7 \\ 15 & 14 & 13 & 9 & 7 \\ 9 & 10 & 9 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 7 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 9 & 14 & 13 & 9 & 7 \\ 7 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 8 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 8 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[363] \ 21^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 9 & 7 \\ 14 & 13 & 10 & 10 & 7 \\ 9 & 10 & 5 & 7 & 5 \\ 9 & 10 & 7 & 4 & 4 \\ 7 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{111} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 6 & 8 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

[364] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$

[365] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 9 & 6 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 9 & 14 & 13 & 9 & 7 \\ 6 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$

[366] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[367] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 12 & 20 & 8 & 6 \\ 12 & 13 & 24 & 10 & 7 \\ 20 & 24 & 38 & 16 & 12 \\ 8 & 10 & 16 & 4 & 4 \\ 6 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$

[368] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 12 & 20 & 8 & 6 \\ 12 & 13 & 24 & 10 & 7 \\ 20 & 24 & 38 & 16 & 12 \\ 8 & 10 & 16 & 4 & 4 \\ 6 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$

[369] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 10 & 7 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 10 & 14 & 13 & 9 & 7 \\ 7 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$

[370] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 10 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 10 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$

[371] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 10 & 0 & 4 & 4 \\ 5 & 7 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

[372] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 7 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[373] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -97$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

[383] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -166$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 14 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[385] $1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -175$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 25 & 1 & 5 \\ 9 & 13 & 40 & 0 & 7 \\ 25 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0}$

[386] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -190$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 18 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 18 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$

[387] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -195$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 27 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$

[388] $1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 33 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 33 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$

[389] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 16 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 16 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[390] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 17 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 17 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[391] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 16 & 6 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 16 & 24 & 38 & 16 & 12 \\ 6 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$

[392] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 22 & 1 & 5 \\ 9 & 13 & 30 & 0 & 7 \\ 22 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[393] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 16 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 16 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$

[394] $21^{111} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 24 & 9 & 7 \\ 14 & 13 & 24 & 10 & 7 \\ 24 & 24 & 38 & 16 & 12 \\ 9 & 10 & 16 & 4 & 4 \\ 7 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[395] $1^{13} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 20 & 1 & 5 \\ 9 & 13 & 30 & 0 & 7 \\ 20 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[396] $1^{111} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 17 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 17 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[397] $1^{13} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 21 & 1 & 5 \\ 9 & 13 & 30 & 0 & 7 \\ 21 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[398] $21^{111} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 7 & 7 & 5 \\ 10 & 13 & 10 & 10 & 7 \\ 7 & 10 & 5 & 7 & 5 \\ 7 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[399] $1^{12} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 22 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 22 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[400] $1^{12} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 25 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 25 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[401] $1^{14} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 26 & 1 & 5 \\ 9 & 13 & 40 & 0 & 7 \\ 26 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

[402] $21^{111} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 16 & 7 & 5 \\ 10 & 13 & 24 & 10 & 7 \\ 16 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[403] $1^{12} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 23 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 37 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [415] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 1 & 2 & 5 \\ 4 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 5 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.681 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6$

$$T = 9 \quad [0] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.682 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.683 NHC(-2)(-3) + so₈ + su₂ + su₃ + so₈

$$T = 9 \quad [0] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 28 & 8 & 1 & 5 \\ 12 & 25 & 60 & 16 & 0 & 9 \\ 28 & 60 & 133 & 36 & 0 & 21 \\ 8 & 16 & 36 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 10 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 10 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] 1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 19 & 8 & 1 & 5 \\ 12 & 25 & 40 & 16 & 0 & 9 \\ 19 & 40 & 58 & 24 & 0 & 14 \\ 8 & 16 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 20 & 8 & 1 & 5 \\ 12 & 25 & 40 & 16 & 0 & 9 \\ 20 & 40 & 58 & 24 & 0 & 14 \\ 8 & 16 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 10 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 10 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 26 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 30 & 8 & 1 & 5 \\ 12 & 25 & 60 & 16 & 0 & 9 \\ 30 & 60 & 133 & 36 & 0 & 21 \\ 8 & 16 & 36 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 29 & 8 & 1 & 5 \\ 12 & 25 & 60 & 16 & 0 & 9 \\ 29 & 60 & 133 & 36 & 0 & 21 \\ 8 & 16 & 36 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 38 & 8 & 1 & 5 \\ 12 & 25 & 80 & 16 & 0 & 9 \\ 38 & 80 & 238 & 48 & 0 & 28 \\ 8 & 16 & 48 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 40 & 8 & 1 & 5 \\ 12 & 25 & 80 & 16 & 0 & 9 \\ 40 & 80 & 238 & 48 & 0 & 28 \\ 8 & 16 & 48 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 39 & 8 & 1 & 5 \\ 12 & 25 & 80 & 16 & 0 & 9 \\ 39 & 80 & 238 & 48 & 0 & 28 \\ 8 & 16 & 48 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 19 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 6 & 1 & 5 \\ 9 & 13 & 14 & 10 & 0 & 7 \\ 9 & 14 & 13 & 9 & 0 & 7 \\ 6 & 10 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

4.684 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 6 \\ 2 & 4 & 0 & 3 & 0 & 3 \\ 3 & 5 & 0 & 6 & 3 & 4 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 \\ 3 & 4 & -2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 6 \\ 2 & 4 & 0 & 3 & 0 & 3 \\ 3 & 5 & 0 & 6 & 3 & 4 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

4.685 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$\begin{aligned}
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0} \\
[5] & 1^{11} 31^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0} \\
[6] & 1^{11} 31^{13} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0} \\
[7] & 1^{11} 31^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0} \\
& T = 9 \quad [8] \quad 1^{2^1 1^{11}} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[9] & 1^{11} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 16 & 1 & 5 \\ 10 & 13 & 24 & 0 & 7 \\ 16 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[10] & 1^{2^1 1^{12}} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 14 & 1 & 5 \\ 10 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[11] & 1^{2^1 1^{13}} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 21 & 1 & 5 \\ 10 & 13 & 30 & 0 & 7 \\ 21 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[12] & 1^{2^1 1^{12}} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 23 & 1 & 5 \\ 10 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0} \\
[13] & 1^{2^1 1^{14}} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -904, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 28 & 1 & 5 \\ 10 & 13 & 40 & 0 & 7 \\ 28 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[14] \ 1^{11} 1^{21} 1^{13} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -1164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 30 & 1 & 5 \\ 10 & 13 & 44 & 0 & 7 \\ 30 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 1^{21} 1^{14} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -1709, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 37 & 1 & 5 \\ 10 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

4.686 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 7 \quad [1] \ 1^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 4 & 4 \\ 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 21^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 4 & 4 \\ 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 13^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 4 & 4 \\ 5 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -35, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.60 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 13^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 6 & 5 \\ 11 & 20 & 11 & 8 \\ 6 & 11 & 4 & 4 \\ 5 & 8 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -50, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 8 & 6 \\ 15 & 20 & 11 & 8 \\ 8 & 11 & 4 & 4 \\ 6 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 21^{13} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -57, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 13 & 17 & 9 & 7 \\ 17 & 20 & 11 & 8 \\ 9 & 11 & 4 & 4 \\ 7 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 21^{13} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -102, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 20 & 11 & 8 \\ 8 & 11 & 4 & 4 \\ 6 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{13} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -123, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 4 & 4 \\ 7 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [9] \ 21^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 3 & 3 & 5 \\ 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 5 & 2 & 3 \\ 4 & 5 & 9 & 4 & 5 \\ 2 & 2 & 4 & 0 & 2 \\ 3 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^1 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 1^1 3^1 1^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 7 & 3 & 4 \\ 3 & 1 & 5 & 2 & 3 \\ 7 & 5 & 9 & 4 & 5 \\ 3 & 2 & 4 & 0 & 2 \\ 4 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^1 1^1 3^1 1^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 10 & 1 & 8 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 4 \\ 3 & 6 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

- [20] $1^{11}3^{11}2^{11}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -95, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{11}3^{11}2^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [22] $1^{11}3^{11}1^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$
- [23] $13^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 3 \\ 4 & 10 & 13 & 4 & 6 \\ 5 & 13 & 12 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 3 & 6 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$
- [24] $1^{11}3^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [25] $1^{11}3^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 3 & 4 \\ 7 & 10 & 13 & 4 & 6 \\ 8 & 13 & 12 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 4 & 6 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$
- [26] $1^{11}3^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 8 & 6 & 0 & 6 \\ 2 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$
- [27] $13^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 10 & 13 & 8 & 6 \\ 5 & 13 & 12 & 9 & 6 \\ 3 & 8 & 9 & 4 & 4 \\ 3 & 6 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$
- [29] $1^{11}3^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 5 & 4 \\ 7 & 10 & 13 & 8 & 6 \\ 8 & 13 & 12 & 9 & 6 \\ 5 & 8 & 9 & 4 & 4 \\ 4 & 6 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$
- [28] $1^{11}3^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [30] $21^{11}3^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [31] $31^{11}23^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 1 & 5 & 4 & 2 & 3 \\ 5 & 5 & 5 & 3 & 5 \\ 4 & 5 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

- [32] $31^{11}23^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 5 & 8 & 8 & 5 \\ 8 & 6 & 9 & 6 \\ 8 & 9 & 5 & 5 \\ 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$
- [33] $1^1 321^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -95, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$
- $T = 9$ [36] $21^1 41^1 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^1 413^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$
- [35] $1^1 1^3 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^1 413^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^1 413^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$
- [39] $21^1 41^1 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 0 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$
- [40] $1^1 413^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$
- [41] $1^1 1^3 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$
- [42] $1^1 41^1 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 6 & 1 & 1 & 3 \\ 6 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0} \\
[43] & 1^1 41^1 23^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0} \\
[44] & 1^1 23^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 2 & 3 \\ 2 & 4 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0} \\
[45] & 1^1 41^1 31^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.14 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 4 & 4 & 2 & 0 & 4 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0} \\
[46] & 31^1 3^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.11 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 2 & 2 \\ 3 & 4 & 4 & 2 & 3 & 4 \\ 3 & 4 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 3 & 2 & 1 & 0 & 2 \\ 2 & 4 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
T = 10 & [47] 21^1 513^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[48] & 21^1 513^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 4 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[50] & 1^1 5123^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 4 & 4 \\ 3 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0} \\
[49] & 1^1 513^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[51] & 1^1 413^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0} \\
[52] & 1^1 413^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0
\end{aligned}$$

8.0

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[53] \ 21^1 51^1 23^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 37, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 11 & 3 & 3 & 4 \\ 11 & 26 & 8 & 8 & 10 \\ 3 & 8 & 0 & 3 & 2 \\ 3 & 8 & 3 & 1 & 3 \\ 4 & 10 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{13^1 21^1} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 2 & 2 & 3 \\ 7 & 26 & 8 & 8 & 10 \\ 2 & 8 & 0 & 3 & 2 \\ 2 & 8 & 3 & 1 & 3 \\ 3 & 10 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[56] \ 1^1 41^1 3^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 & 2 \\ 3 & 13 & 6 & 0 & 9 & 7 \\ 1 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 9 & 3 & 0 & 4 & 4 \\ 2 & 7 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{11^1 3^1} 4^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 5 & 8 & 0 & 3 & 5 \\ 2 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 5 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{13^1 1^1} 4^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 1 & 2 \\ 3 & 13 & 1 & 9 & 5 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 9 & 0 & 3 & 3 & 3 \\ 1 & 5 & 0 & 3 & 0 & 2 \\ 2 & 7 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[58] \ 1^1 41^1 3^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 1 & 5 & 3 \\ 8 & 38 & 1 & 24 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 24 & 0 & 13 & 7 \\ 3 & 12 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{13^1 1^1} 4^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 1 & 2 \\ 3 & 13 & 10 & 0 & 5 & 7 \\ 2 & 10 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 5 & 3 & 0 & 0 & 2 \\ 2 & 7 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [61] \ 21^1 513^1 1^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 4 & 5 & 1 & 2 & 4 \\ 5 & 5 & 2 & 3 & 5 \\ 1 & 2 & -2 & 1 & 0 \\ 2 & 3 & 1 & 0 & 2 \\ 4 & 5 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[60] \ 1^1 315^1 13^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 4 & 4 \\ 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^1 1^1 413^1 1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 2 & 5 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 0 & 2 \\ 5 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^1 1^1 315^1 1^1 13^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 3 & 4 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^1 3^1 1^1 51^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^1 3^1 1^1 51^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 \\ 3 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^1 413^1 21^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -106, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 & 2 \\ 4 & 2 & 1 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 4 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 15^1 13^1 1^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 & 2 \\ 3 & 4 & 1 & 2 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 1 & 0 & 2 \\ 2 & 4 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [68] \quad 1^1 1^1 3^1 21^1 6^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^1 1^1 3^1 21^1 6^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [70] \quad 23^1 21^1 8^1 123^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

4.687 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{e}_6$

$$T = 9 \quad [0] \quad 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.69 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 5 & 0 & 2 & -2 & 0 \\ 3 & 5 & -2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 1 & 3 \\ 4 & 10 & 13 & 8 & 0 & 6 \\ 5 & 13 & 12 & 9 & 0 & 6 \\ 3 & 8 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

4.688 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 6 & 1 & 5 \\ 11 & 20 & 11 & 0 & 8 \\ 6 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 3 \\ 4 & 10 & 13 & 4 & 0 & 6 \\ 5 & 13 & 12 & 5 & 0 & 6 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 1 & 3 \\ 4 & 10 & 13 & 8 & 0 & 6 \\ 5 & 13 & 12 & 9 & 0 & 6 \\ 3 & 8 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.689 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 6 & 1 & 9 & 5 \\ 11 & 20 & 11 & 0 & 17 & 8 \\ 6 & 11 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 9 & 0 & 13 & 7 \\ 5 & 8 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 9 & 5 \\ 12 & 25 & 12 & 0 & 20 & 9 \\ 6 & 12 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 9 & 0 & 13 & 7 \\ 5 & 9 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 17 & 5 \\ 12 & 25 & 12 & 0 & 36 & 9 \\ 6 & 12 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 17 & 0 & 49 & 13 \\ 5 & 9 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.690} \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$$

$$T = 9 \quad [0] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 6 & 1 & 9 & 5 \\ 11 & 20 & 11 & 0 & 17 & 8 \\ 6 & 11 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 9 & 0 & 12 & 6 \\ 5 & 8 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.691} \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{green}{so}_8$$

$$T = 8 \quad [0] \ 1^{111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 5 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^1 4 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

4.692 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^1 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 1 & 5 \\ 11 & 20 & 0 & 11 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 9 & 5 \\ 12 & 25 & 12 & 0 & 20 & 9 \\ 6 & 12 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 9 & 0 & 13 & 7 \\ 5 & 9 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 18 & 5 \\ 12 & 25 & 12 & 0 & 36 & 9 \\ 6 & 12 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 17 & 0 & 49 & 13 \\ 5 & 9 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 17 & 5 \\ 12 & 25 & 12 & 0 & 36 & 9 \\ 6 & 12 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 17 & 0 & 49 & 13 \\ 5 & 9 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

4.693 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^1 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 6 & 5 \\ 12 & 25 & 12 & 0 & 12 & 9 \\ 6 & 12 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 6 & 0 & 4 & 4 \\ 5 & 9 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.694 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.695 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 8 & 23 & 6 \\ 16 & 25 & 12 & 36 & 9 \\ 8 & 12 & 4 & 17 & 4 \\ 23 & 36 & 17 & 49 & 13 \\ 6 & 9 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 8 & 12 & 6 \\ 15 & 20 & 11 & 17 & 8 \\ 8 & 11 & 4 & 9 & 4 \\ 12 & 17 & 9 & 13 & 7 \\ 6 & 8 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 8 & 1 & 6 \\ 15 & 20 & 11 & 0 & 8 \\ 8 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 8 & 13 & 6 \\ 15 & 20 & 11 & 17 & 8 \\ 8 & 11 & 4 & 9 & 4 \\ 13 & 17 & 9 & 13 & 7 \\ 6 & 8 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

- [8] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 5 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 14 & 0 & 8 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 9 & 16 & 0 & 8 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [17] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 0 & 9 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 21 & 0 & 11 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[19] $21^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 9 & 27 & 7 \\ 20 & 25 & 12 & 36 & 9 \\ 9 & 12 & 4 & 17 & 4 \\ 27 & 36 & 17 & 49 & 13 \\ 7 & 9 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 18 & 0 & 10 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 9 & 14 & 0 & 8 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 8 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 12 & 19 & 0 & 11 & 73 & 15 \\ 3 & 4 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[24] $21^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 9 & 15 & 7 \\ 20 & 25 & 12 & 20 & 9 \\ 9 & 12 & 4 & 9 & 4 \\ 15 & 20 & 9 & 13 & 7 \\ 7 & 9 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{12}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 13 & 23 & 0 & 13 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 22 & 0 & 12 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 16 & 28 & 0 & 16 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 399, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 10 & 17 & 0 & 9 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 11 & 18 & 0 & 10 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 26 & 0 & 14 & 118 & 20 \\ 3 & 4 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 12 & 21 & 0 & 11 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 511, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 15 & 27 & 0 & 15 & 133 & 21 \\ 3 & 4 & -2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 14 & 23 & 0 & 13 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

- [34] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 18 & 32 & 0 & 18 & 193 & 25 \\ 3 & 4 & -2 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 13 & 22 & 0 & 12 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 654, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 17 & 28 & 0 & 16 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 670, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 17 & 31 & 0 & 17 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 15 & 26 & 0 & 14 & 118 & 20 \\ 3 & 4 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 811, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 16 & 27 & 0 & 15 & 133 & 21 \\ 3 & 4 & -2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [40] $1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 854, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 20 & 36 & 0 & 20 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [41] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 969, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 3 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 19 & 32 & 0 & 18 & 193 & 25 \\ 3 & 4 & -2 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1034, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 18 & 31 & 0 & 17 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{14}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 21 & 36 & 0 & 20 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [44] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & -2 & 4 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 4 & 2 \\ 5 & 4 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & -2 & 11 & 3 \end{pmatrix}_{b_0}$$

[48] $21^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 9 & 4 \\ 5 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 0 & 5 & 2 \\ 9 & 10 & 7 & 5 & 14 & 8 \\ 4 & 5 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 8 & 15 & 10 & 8 & 38 & 12 \\ 3 & 5 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 6 & 2 \\ 6 & 7 & 0 & 6 & 21 & 9 \\ 3 & 3 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{11} 1^3 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11} 1^3 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 0 & 2 & 15 & 2 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 9 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{11} 1^3 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{11} 1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 25 & 3 \\ 2 & 1 & 1 & 2 & 28 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 24 & 2 \\ 25 & 28 & 0 & 24 & 366 & 36 \\ 3 & 3 & -1 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{11} 1^3 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

[57] $21^{12} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 18 & 4 \\ 5 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 18 & 20 & 14 & 10 & 62 & 16 \\ 4 & 5 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{11} 1^3 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{11} 1^{13} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 1 & 1 & 2 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 14 & 2 \\ 14 & 17 & 0 & 14 & 121 & 21 \\ 3 & 3 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 8 & 2 \\ 8 & 10 & 0 & 8 & 38 & 12 \\ 3 & 3 & -1 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{12} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -69, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & -2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 5 & 10 & 7 & 5 & 14 & 8 \\ 3 & 5 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{12} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 24 & 3 \\ 2 & 1 & 1 & 2 & 25 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 22 & 2 \\ 24 & 25 & 0 & 22 & 313 & 33 \\ 3 & 3 & -1 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{11} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 12 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 10 & 2 \\ 12 & 11 & 0 & 10 & 65 & 15 \\ 3 & 3 & -1 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{13} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 10 & 2 \\ 10 & 13 & 0 & 10 & 61 & 15 \\ 3 & 3 & -1 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 7 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -108, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 6 & 2 \\ 7 & 7 & 0 & 6 & 21 & 9 \\ 3 & 3 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{11} 1^{13} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 19 & 3 \\ 2 & 1 & 1 & 2 & 21 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 18 & 2 \\ 19 & 21 & 0 & 18 & 205 & 27 \\ 3 & 3 & -1 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{12} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 18 & 3 \\ 2 & 1 & 1 & 2 & 18 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 16 & 2 \\ 18 & 18 & 0 & 16 & 166 & 24 \\ 3 & 3 & -1 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{14} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 1 & 2 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 16 & 2 \\ 16 & 20 & 0 & 16 & 158 & 24 \\ 3 & 3 & -1 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & -2 & 10 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{12} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & -2 & 14 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{12} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 12 & 2 \\ 13 & 14 & 0 & 12 & 90 & 18 \\ 3 & 3 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{12} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 10 & 20 & 14 & 10 & 62 & 16 \\ 3 & 5 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{13} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & -2 & 25 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 6 & 2 \\ 8 & 7 & 0 & 6 & 21 & 9 \\ 3 & 3 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{12} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 0 & 2 & 22 & 2 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 14 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 8 & 2 \\ 9 & 10 & 0 & 8 & 38 & 12 \\ 3 & 3 & -1 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{12} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -198, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 0 & 2 & 18 & 2 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 12 & 25 & 18 & 12 & 94 & 20 \\ 3 & 5 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{13} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -229, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & -2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{13} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -237, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & -2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{11} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 1 & 2 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 14 & 2 \\ 15 & 17 & 0 & 14 & 121 & 21 \\ 3 & 3 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 10 & 2 \\ 11 & 13 & 0 & 10 & 61 & 15 \\ 3 & 3 & -1 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{12} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 12 & 2 \\ 14 & 14 & 0 & 12 & 90 & 18 \\ 3 & 3 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{11} 1^4 3^1 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 21 & 3 \\ 2 & 1 & 1 & 2 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 20 & 2 \\ 21 & 24 & 0 & 20 & 250 & 30 \\ 3 & 3 & -1 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{11} 1^4 3^1 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{12} 1^2 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 8 & 2 \\ 10 & 10 & 0 & 8 & 38 & 12 \\ 3 & 3 & -1 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{11} 1^4 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 1 & 2 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 12 & 2 \\ 13 & 16 & 0 & 12 & 90 & 18 \\ 3 & 3 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{12} 1^3 3^1 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 1 & 2 & 21 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 18 & 2 \\ 20 & 21 & 0 & 18 & 205 & 27 \\ 3 & 3 & -1 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{11} 1^4 3^1 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & -2 & 28 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{12} 1^4 3^1 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 26 & 3 \\ 2 & 1 & 1 & 2 & 28 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 24 & 2 \\ 26 & 28 & 0 & 24 & 366 & 36 \\ 3 & 3 & -1 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{11} 1^4 3^1 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{11} 1^4 3^1 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 17 & 3 \\ 2 & 1 & 1 & 2 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 16 & 2 \\ 17 & 20 & 0 & 16 & 158 & 24 \\ 3 & 3 & -1 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{13}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.8$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 12 & 3 \\ 2 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 10 & 2 \\ 12 & 13 & 0 & 10 & 61 & 15 \\ 3 & 3 & -1 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{13}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -472$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.8$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 1 & 2 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 14 & 2 \\ 16 & 17 & 0 & 14 & 121 & 21 \\ 3 & 3 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{14}3^{12}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 22 & 3 \\ 2 & 1 & 1 & 2 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 20 & 2 \\ 22 & 24 & 0 & 20 & 250 & 30 \\ 3 & 3 & -1 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{14}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -640$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 18 & 3 \\ 2 & 1 & 1 & 2 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 16 & 2 \\ 18 & 20 & 0 & 16 & 158 & 24 \\ 3 & 3 & -1 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [98] $1^{11}3^{11}15131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 6 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[99] $1^{11}3^{11}5131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 3 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

[100] $1^{11}3^{11}15131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 7$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 7 & 3 \\ 6 & 8 & 0 & 7 & 22 & 10 \\ 3 & 4 & -1 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[102] $1^{12}3^{11}5131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 10$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 9 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[101] \quad & 1^{12} 3^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 11 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0} \\
[103] \quad & 1^{13} 3^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 2 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[104] \quad & 1^{11} 3^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 2 & 1 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 & 4 \\ 3 & 4 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[105] \quad & 1^{12} 3^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 10 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0} \\
[106] \quad & 1^{13} 3^1 5^1 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 2 & 1 & 3 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 17 & 3 \\ 14 & 20 & 0 & 17 & 130 & 24 \\ 3 & 4 & -1 & 3 & 24 & 4 \end{pmatrix}_{b_0} \\
[107] \quad & 1^{11} 3^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 3 & 8 \\ 1 & 1 & 0 & 3 & 0 & 3 \\ 2 & 4 & 0 & 8 & 3 & 6 \\ 2 & 3 & -1 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[108] \quad & 1^{12} 3^1 5^1 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 13 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0} \\
[109] \quad & 1^{13} 3^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 2 & 1 & 3 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 13 & 3 \\ 10 & 16 & 0 & 13 & 70 & 18 \\ 3 & 4 & -1 & 3 & 18 & 4 \end{pmatrix}_{b_0} \\
[110] \quad & 1^{12} 3^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 34, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 2 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 10 & 3 \\ 8 & 12 & 0 & 10 & 42 & 14 \\ 3 & 4 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[111] & 1^{11} 3^{11} 1^5 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 7 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[112] & 1^{12} 3^{11} 1^5 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 8 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 8 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[113] & 1^{14} 3^{11} 1^5 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 2 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 20 & 3 \\ 16 & 24 & 0 & 20 & 174 & 28 \\ 3 & 4 & -1 & 3 & 28 & 4 \end{pmatrix}_{b_0} \\
[114] & 1^{12} 3^{11} 1^5 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 91, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 9 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[115] & 1^{13} 3^{11} 1^5 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 21 & 3 \\ 20 & 0 & 24 & 21 & 214 & 30 \\ 3 & 0 & 3 & 3 & 30 & 4 \end{pmatrix}_{b_0} \\
[116] & 1^{12} 3^{11} 1^5 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 14 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0} \\
[117] & 1^{12} 3^{11} 1^5 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 143, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 15 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0} \\
[118] & 1^{12} 3^{11} 1^5 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 154, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 10 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[119] & 1^{13} 3^{11} 1^5 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 187, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.5
\end{aligned}$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 11 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{13}3^{11}5 \ 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 197, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 21 & 3 \\ 21 & 0 & 24 & 21 & 214 & 30 \\ 3 & 0 & 3 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{14}3^{11}5 \ 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 203, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 27 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 32 & 3 \\ 2 & 0 & 3 & 1 & 28 & 3 \\ 27 & 0 & 32 & 28 & 382 & 40 \\ 3 & 0 & 3 & 3 & 40 & 4 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{13}3^{11}5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 214, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 20 & 3 \\ 2 & 0 & 3 & 1 & 17 & 3 \\ 16 & 0 & 20 & 17 & 130 & 24 \\ 3 & 0 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{14}3^{11}5 \ 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 226, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 28 & 3 \\ 2 & 0 & 3 & 1 & 24 & 3 \\ 22 & 0 & 28 & 24 & 266 & 34 \\ 3 & 0 & 3 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{13}3^{11}5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 305, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 20 & 3 \\ 2 & 0 & 3 & 1 & 17 & 3 \\ 17 & 0 & 20 & 17 & 130 & 24 \\ 3 & 0 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{14}3^{11}5 \ 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 375, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 28 & 3 \\ 2 & 0 & 3 & 1 & 24 & 3 \\ 23 & 0 & 28 & 24 & 266 & 34 \\ 3 & 0 & 3 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{14}3^{11}5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 427, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.00 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 20 & 3 \\ 19 & 0 & 24 & 20 & 174 & 28 \\ 3 & 0 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

4.696 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 1 & 5 \\ 11 & 20 & 0 & 11 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

$$4.697 \quad \text{NHC}(-2)(-3) + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 10 & 4 & 6 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -2 & 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 11 & 21 & 0 & 11 & 76 & 11 & 16 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -2 & 2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -2 & 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -192, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 3 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 9 & 17 & 0 & 9 & 49 & 9 & 13 \\ 2 & 3 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -2 & 2 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -208, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 5 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -2 & 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 3 & 14 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 8 & 14 & 0 & 8 & 37 & 8 & 11 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -2 & 2 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 3 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 13 & 0 & 7 & 28 & 7 & 10 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -2 & 2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 3 & 18 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 & 2 \\ 10 & 18 & 0 & 10 & 58 & 10 & 14 \\ 2 & 3 & 0 & 2 & 10 & 0 & 2 \\ 3 & 4 & -2 & 2 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 3 & 22 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 12 & 22 & 0 & 12 & 85 & 12 & 17 \\ 2 & 3 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -2 & 2 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 14 & 26 & 0 & 14 & 118 & 14 & 20 \\ 2 & 3 & 0 & 2 & 14 & 0 & 2 \\ 3 & 4 & -2 & 2 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [14] \ 1^{12} 3^{11} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 10 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 10 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

4.698 NHC(-2)(-3) + so₈ + su₂ → (-3) + su₂ + su₂n3mix

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 4 & 3 \\ 3 & 4 & 0 & 3 & 14 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 4 & 2 \\ 8 & 14 & 0 & 8 & 37 & 17 & 11 \\ 4 & 7 & 0 & 4 & 17 & 7 & 5 \\ 3 & 4 & -2 & 2 & 11 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 3 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 4 & 8 & 0 & 4 & 10 & 22 & 6 \\ 9 & 16 & 0 & 9 & 22 & 44 & 12 \\ 3 & 4 & -2 & 2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 4 & 8 & 0 & 4 & 10 & 10 & 6 \\ 4 & 7 & 0 & 4 & 10 & 7 & 5 \\ 3 & 4 & -2 & 2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 3 & 5 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 4 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 4 & 7 & 0 & 4 & 6 & 7 & 5 \\ 3 & 4 & -2 & 2 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 4 & 0 & 2 & 1 & 5 & 3 \\ 4 & 7 & 0 & 4 & 5 & 7 & 5 \\ 3 & 4 & -2 & 2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 4 & 0 & 2 & 1 & 8 & 3 \\ 6 & 11 & 0 & 6 & 8 & 20 & 8 \\ 3 & 4 & -2 & 2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 3 & 8 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 4 & 8 & 0 & 4 & 10 & 28 & 6 \\ 11 & 20 & 0 & 11 & 28 & 69 & 15 \\ 3 & 4 & -2 & 2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 3 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 4 & 2 \\ 5 & 9 & 0 & 5 & 13 & 11 & 7 \\ 4 & 7 & 0 & 4 & 11 & 7 & 5 \\ 3 & 4 & -2 & 2 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 16 & 3 \\ 3 & 4 & 0 & 3 & 8 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 16 & 2 \\ 4 & 8 & 0 & 4 & 10 & 40 & 6 \\ 16 & 29 & 0 & 16 & 40 & 148 & 22 \\ 3 & 4 & -2 & 2 & 6 & 22 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 4 & 0 & 2 & 1 & 8 & 3 \\ 7 & 12 & 0 & 7 & 8 & 25 & 9 \\ 3 & 4 & -2 & 2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14}1^31^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -299, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 4 & 3 \\ 3 & 4 & 0 & 3 & 21 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 4 & 2 \\ 11 & 21 & 0 & 11 & 76 & 26 & 16 \\ 4 & 7 & 0 & 4 & 26 & 7 & 5 \\ 3 & 4 & -2 & 2 & 16 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12}1^31^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 4 & 3 \\ 3 & 4 & 0 & 3 & 18 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 4 & 2 \\ 10 & 18 & 0 & 10 & 58 & 22 & 14 \\ 4 & 7 & 0 & 4 & 22 & 7 & 5 \\ 3 & 4 & -2 & 2 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{12}1^31^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 6 & 3 \\ 3 & 4 & 0 & 3 & 13 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 6 & 2 \\ 7 & 13 & 0 & 7 & 28 & 25 & 10 \\ 6 & 11 & 0 & 6 & 25 & 20 & 8 \\ 3 & 4 & -2 & 2 & 10 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{12}1^31^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -339, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 4 & 3 \\ 3 & 4 & 0 & 3 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 4 & 2 \\ 7 & 13 & 0 & 7 & 28 & 16 & 10 \\ 4 & 7 & 0 & 4 & 16 & 7 & 5 \\ 3 & 4 & -2 & 2 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{13}1^31^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -355, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 4 & 3 \\ 3 & 4 & 0 & 3 & 17 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 4 & 2 \\ 9 & 17 & 0 & 9 & 49 & 21 & 13 \\ 4 & 7 & 0 & 4 & 21 & 7 & 5 \\ 3 & 4 & -2 & 2 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}1^31^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -356, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 13 & 3 \\ 3 & 4 & 0 & 3 & 8 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 13 & 2 \\ 4 & 8 & 0 & 4 & 10 & 34 & 6 \\ 13 & 24 & 0 & 13 & 34 & 100 & 18 \\ 3 & 4 & -2 & 2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13}1^31^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -412, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 6 & 3 \\ 3 & 4 & 0 & 3 & 17 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 6 & 2 \\ 9 & 17 & 0 & 9 & 49 & 33 & 13 \\ 6 & 11 & 0 & 6 & 33 & 20 & 8 \\ 3 & 4 & -2 & 2 & 13 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{14}1^31^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -416, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 6 & 3 \\ 3 & 4 & 0 & 3 & 21 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 6 & 2 \\ 11 & 21 & 0 & 11 & 76 & 41 & 16 \\ 6 & 11 & 0 & 6 & 41 & 20 & 8 \\ 3 & 4 & -2 & 2 & 16 & 8 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{1123}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 4 & 0 & 2 & 1 & 11 & 3 \\ 9 & 16 & 0 & 9 & 11 & 44 & 12 \\ 3 & 4 & -2 & 2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{1313}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 4 & 3 \\ 3 & 4 & 0 & 3 & 22 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 4 & 2 \\ 12 & 22 & 0 & 12 & 85 & 27 & 17 \\ 4 & 7 & 0 & 4 & 27 & 7 & 5 \\ 3 & 4 & -2 & 2 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{1423}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 6 & 3 \\ 3 & 4 & 0 & 3 & 26 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 6 & 2 \\ 14 & 26 & 0 & 14 & 118 & 50 & 20 \\ 6 & 11 & 0 & 6 & 50 & 20 & 8 \\ 3 & 4 & -2 & 2 & 20 & 8 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{1123}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 4 & 0 & 2 & 1 & 14 & 3 \\ 12 & 21 & 0 & 12 & 14 & 80 & 16 \\ 3 & 4 & -2 & 2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{1133}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 4 & 0 & 2 & 1 & 14 & 3 \\ 11 & 20 & 0 & 11 & 14 & 69 & 15 \\ 3 & 4 & -2 & 2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{1413}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 4 & 3 \\ 3 & 4 & 0 & 3 & 26 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 4 & 2 \\ 14 & 26 & 0 & 14 & 118 & 32 & 20 \\ 4 & 7 & 0 & 4 & 32 & 7 & 5 \\ 3 & 4 & -2 & 2 & 20 & 5 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{1143}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 13 & 2 \\ 2 & 4 & 0 & 2 & 1 & 17 & 3 \\ 13 & 24 & 0 & 13 & 17 & 100 & 18 \\ 3 & 4 & -2 & 2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{1133}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 4 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 4 & 0 & 2 & 1 & 17 & 3 \\ 14 & 25 & 0 & 14 & 17 & 111 & 19 \\ 3 & 4 & -2 & 2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 1^4 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 4 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 16 & 2 \\ 2 & 4 & 0 & 2 & 1 & 20 & 3 \\ 16 & 29 & 0 & 16 & 20 & 148 & 22 \\ 3 & 4 & -2 & 2 & 3 & 22 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [29] \ 1^{11} 3^{11} 1^5 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 5 & 3 \\ 2 & 2 & 1 & 3 & 8 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 7 & 6 & 3 \\ 6 & 8 & 0 & 7 & 22 & 19 & 10 \\ 5 & 7 & 0 & 6 & 19 & 14 & 8 \\ 3 & 4 & -1 & 3 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3^{11} 1^5 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 5 & 3 \\ 2 & 2 & 1 & 3 & 4 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 2 & 7 & 4 \\ 5 & 7 & 0 & 6 & 7 & 14 & 8 \\ 3 & 4 & -1 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{13} 3^{11} 1^5 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 5 & 3 \\ 2 & 2 & 1 & 3 & 16 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 13 & 6 & 3 \\ 10 & 16 & 0 & 13 & 70 & 33 & 18 \\ 5 & 7 & 0 & 6 & 33 & 14 & 8 \\ 3 & 4 & -1 & 3 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11} 3^{11} 1^5 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 13 & 3 \\ 2 & 2 & 1 & 3 & 4 & 19 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 16 & 3 \\ 2 & 4 & 0 & 3 & 2 & 19 & 4 \\ 13 & 19 & 0 & 16 & 19 & 110 & 22 \\ 3 & 4 & -1 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{12} 3^{11} 1^5 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 5 & 3 \\ 2 & 2 & 1 & 3 & 12 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 10 & 6 & 3 \\ 8 & 12 & 0 & 10 & 42 & 26 & 14 \\ 5 & 7 & 0 & 6 & 26 & 14 & 8 \\ 3 & 4 & -1 & 3 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

4.699 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 21^1 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 10 & 16 & 8 & 8 & 6 \\ 16 & 25 & 12 & 12 & 9 \\ 8 & 12 & 4 & 6 & 4 \\ 8 & 12 & 6 & 4 & 4 \\ 6 & 9 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 10 & 16 & 8 & 8 & 6 \\ 16 & 25 & 12 & 12 & 9 \\ 8 & 12 & 4 & 6 & 4 \\ 8 & 12 & 6 & 4 & 4 \\ 6 & 9 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[2] \quad & 13^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 52, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.38 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 12 & 6 & 6 & 5 \\ 12 & 25 & 12 & 12 & 9 \\ 6 & 12 & 4 & 6 & 4 \\ 6 & 12 & 6 & 4 & 4 \\ 5 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[3] \quad & 1^13^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 76, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[4] \quad & 21^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 192, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 13 & 20 & 9 & 9 & 7 \\ 20 & 25 & 12 & 12 & 9 \\ 9 & 12 & 4 & 6 & 4 \\ 9 & 12 & 6 & 4 & 4 \\ 7 & 9 & 4 & 4 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 10 \quad [5] \quad & 1^14^{11}13^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = \\
& 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[6] \quad & 1^14^{11}13^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 1 & 3 \\ 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 0 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 12 \quad [7] \quad & 1^13^{11}15^{11}13^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -31, \quad \det = \\
& 2, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 3 & 0 & 3 & 4 & 3 & 4 \\ 2 & 0 & 3 & 3 & 1 & 3 \\ 3 & 0 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

4.700 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$\begin{aligned}
T = 9 \quad [0] \quad & 1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -87, \quad \det = \\
& -132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0 \\
& T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 12 & 6 & 6 & 1 & 5 \\ 12 & 25 & 12 & 12 & 0 & 9 \\ 6 & 12 & 4 & 6 & 0 & 4 \\ 6 & 12 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

4.701 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$\begin{aligned}
T = 9 \quad [0] \quad & 1^{12}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = \\
& -8, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.5 \\
& T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 10 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & -2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -468$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$

$T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 12 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & -2 & 2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{14}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -488$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 14 & 26 & 0 & 14 & 14 & 118 & 20 \\ 3 & 4 & -2 & 2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$T = 13$ [14] $1^{12}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 13$, $\Delta = -1$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 10.5$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 10 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

4.702 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix

$T = 8$ [0] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 13 & 0 & 8 & 36 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{13}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 115$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{14}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 19 & 0 & 10 & 64 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{12}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 136$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 11 & 0 & 6 & 20 & 8 \\ 3 & 4 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 15 & 0 & 8 & 39 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 1^{13} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 17 & 0 & 10 & 55 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13} 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{14} 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 13 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12} 1^{13} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{13} 1^{13} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & -2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{14}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 16 & 29 & 0 & 16 & 148 & 22 \\ 3 & 4 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11}4^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{14}4^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 & 3 \\ 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{12}4^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{11}4^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -1 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{11}4^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{12}4^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -1 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11}4^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 1 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{11}4^{13}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 0 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 8 & 2 & 4 & 0 & 34 & 10 \\ 2 & 2 & 0 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [30] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 3 & 0 & 3 & 4 & 17 & 4 \\ 12 & 0 & 15 & 17 & 65 & 17 \\ 3 & 0 & 3 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 3^1 1^5 13^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 2 & 1 & 3 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 5 & 7 & 0 & 6 & 14 & 8 \\ 3 & 4 & -1 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 2 & 1 & 3 & 19 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 \\ 13 & 19 & 0 & 16 & 110 & 22 \\ 3 & 4 & -1 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 162, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 2 & 0 & 3 & 1 & 13 & 3 \\ 13 & 0 & 15 & 13 & 76 & 18 \\ 3 & 0 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

4.703 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 8 & 3 \\ 3 & 4 & 0 & 3 & 7 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 8 & 2 \\ 4 & 7 & 0 & 4 & 7 & 17 & 5 \\ 8 & 14 & 0 & 8 & 17 & 37 & 11 \\ 3 & 4 & -2 & 2 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 4 & 3 \\ 3 & 4 & 0 & 3 & 16 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 4 & 2 \\ 9 & 16 & 0 & 9 & 44 & 22 & 12 \\ 4 & 8 & 0 & 4 & 22 & 10 & 6 \\ 3 & 4 & -2 & 2 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 4 & 7 & 0 & 4 & 7 & 10 & 5 \\ 4 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & -2 & 2 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 7 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 3 & 2 \\ 4 & 7 & 0 & 4 & 7 & 6 & 5 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -2 & 2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 7 & 0 & 4 & 7 & 5 & 5 \\ 2 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & -2 & 2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{121}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 6 & 11 & 0 & 6 & 20 & 8 & 8 \\ 2 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & -2 & 2 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 4 & 3 \\ 3 & 4 & 0 & 3 & 20 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 4 & 2 \\ 11 & 20 & 0 & 11 & 69 & 28 & 15 \\ 4 & 8 & 0 & 4 & 28 & 10 & 6 \\ 3 & 4 & -2 & 2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 4 & 7 & 0 & 4 & 7 & 11 & 5 \\ 5 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & -2 & 2 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{142}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 4 & 3 \\ 3 & 4 & 0 & 3 & 29 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 4 & 2 \\ 16 & 29 & 0 & 16 & 148 & 40 & 22 \\ 4 & 8 & 0 & 4 & 40 & 10 & 6 \\ 3 & 4 & -2 & 2 & 22 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 12 & 0 & 7 & 25 & 8 & 9 \\ 2 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & -2 & 2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{114}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 4 & 7 & 0 & 4 & 7 & 26 & 5 \\ 11 & 21 & 0 & 11 & 26 & 76 & 16 \\ 3 & 4 & -2 & 2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 10 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 4 & 7 & 0 & 4 & 7 & 22 & 5 \\ 10 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & -2 & 2 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 7 & 3 \\ 3 & 4 & 0 & 3 & 11 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 7 & 2 \\ 6 & 11 & 0 & 6 & 20 & 25 & 8 \\ 7 & 13 & 0 & 7 & 25 & 28 & 10 \\ 3 & 4 & -2 & 2 & 8 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 4 & 7 & 0 & 4 & 7 & 16 & 5 \\ 7 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & -2 & 2 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 4 & 7 & 0 & 4 & 7 & 21 & 5 \\ 9 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & -2 & 2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 4 & 3 \\ 3 & 4 & 0 & 3 & 24 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 4 & 2 \\ 13 & 24 & 0 & 13 & 100 & 34 & 18 \\ 4 & 8 & 0 & 4 & 34 & 10 & 6 \\ 3 & 4 & -2 & 2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 9 & 3 \\ 3 & 4 & 0 & 3 & 11 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 9 & 2 \\ 6 & 11 & 0 & 6 & 20 & 33 & 8 \\ 9 & 17 & 0 & 9 & 33 & 49 & 13 \\ 3 & 4 & -2 & 2 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 11 & 3 \\ 3 & 4 & 0 & 3 & 11 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 11 & 2 \\ 6 & 11 & 0 & 6 & 20 & 41 & 8 \\ 11 & 21 & 0 & 11 & 41 & 76 & 16 \\ 3 & 4 & -2 & 2 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 9 & 16 & 0 & 9 & 44 & 11 & 12 \\ 2 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & -2 & 2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 4 & 7 & 0 & 4 & 7 & 27 & 5 \\ 12 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & -2 & 2 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 14 & 3 \\ 3 & 4 & 0 & 3 & 11 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 14 & 2 \\ 6 & 11 & 0 & 6 & 20 & 50 & 8 \\ 14 & 26 & 0 & 14 & 50 & 118 & 20 \\ 3 & 4 & -2 & 2 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 12 & 21 & 0 & 12 & 80 & 14 & 16 \\ 2 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & -2 & 2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 11 & 20 & 0 & 11 & 69 & 14 & 15 \\ 2 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & -2 & 2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 14 & 2 \\ 4 & 7 & 0 & 4 & 7 & 32 & 5 \\ 14 & 26 & 0 & 14 & 32 & 118 & 20 \\ 3 & 4 & -2 & 2 & 5 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 2 & 2 \\ 13 & 24 & 0 & 13 & 100 & 17 & 18 \\ 2 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & -2 & 2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 14 & 25 & 0 & 14 & 111 & 17 & 19 \\ 2 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & -2 & 2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 2 & 2 \\ 16 & 29 & 0 & 16 & 148 & 20 & 22 \\ 2 & 4 & 0 & 2 & 20 & 1 & 3 \\ 3 & 4 & -2 & 2 & 22 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [29] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 2 & 3 \\ 2 & 2 & 1 & 3 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 7 & 0 & 6 & 14 & 7 & 8 \\ 2 & 4 & 0 & 3 & 7 & 2 & 4 \\ 3 & 4 & -1 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}3^{11}15^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 6 & 3 \\ 2 & 2 & 1 & 3 & 7 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 7 & 3 \\ 5 & 7 & 0 & 6 & 14 & 19 & 8 \\ 6 & 8 & 0 & 7 & 19 & 22 & 10 \\ 3 & 4 & -1 & 3 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11}3^{11}15^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 10 & 3 \\ 2 & 2 & 1 & 3 & 7 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 13 & 3 \\ 5 & 7 & 0 & 6 & 14 & 33 & 8 \\ 10 & 16 & 0 & 13 & 33 & 70 & 18 \\ 3 & 4 & -1 & 3 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11}3^{11}15^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 2 & 3 \\ 2 & 2 & 1 & 3 & 19 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 & 3 \\ 13 & 19 & 0 & 16 & 110 & 19 & 22 \\ 2 & 4 & 0 & 3 & 19 & 2 & 4 \\ 3 & 4 & -1 & 3 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11}3^{11}15^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 8 & 3 \\ 2 & 2 & 1 & 3 & 7 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 10 & 3 \\ 5 & 7 & 0 & 6 & 14 & 26 & 8 \\ 8 & 12 & 0 & 10 & 26 & 42 & 14 \\ 3 & 4 & -1 & 3 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

4.704 NHC(-2)(-3) + so₈ + su₂ → (-3) + su₃

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -2 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 6 & 10 & 5 \\ 11 & 20 & 11 & 17 & 8 \\ 6 & 11 & 4 & 9 & 4 \\ 10 & 17 & 9 & 12 & 6 \\ 5 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 6 & 1 & 5 \\ 11 & 20 & 11 & 0 & 8 \\ 6 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[6] $13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 6 & 9 & 5 \\ 11 & 20 & 11 & 17 & 8 \\ 6 & 11 & 4 & 9 & 4 \\ 9 & 17 & 9 & 12 & 6 \\ 5 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 8 & 12 & 6 \\ 15 & 20 & 11 & 17 & 8 \\ 8 & 11 & 4 & 9 & 4 \\ 12 & 17 & 9 & 12 & 6 \\ 6 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[8] $21^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 9 & 15 & 7 \\ 17 & 20 & 11 & 17 & 8 \\ 9 & 11 & 4 & 9 & 4 \\ 15 & 17 & 9 & 12 & 6 \\ 7 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[10] $21^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 8 & 12 & 6 \\ 16 & 20 & 11 & 17 & 8 \\ 8 & 11 & 4 & 9 & 4 \\ 12 & 17 & 9 & 12 & 6 \\ 6 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 0 & 7 & 27 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 0 & 9 & 48 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [14] \quad 1^{14} 41^{13} 1^1 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 161, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 & 1 & 2 \\ 3 & 4 & 1 & 2 & 3 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

4.705 NHC(-2)(-3) + so₈ + su₂→(-3) + su₃ + f₄

$$T = 9 \quad [0] \quad 1^{11} \overset{1}{3} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 6 & 9 & 1 & 5 \\ 11 & 20 & 11 & 17 & 0 & 8 \\ 6 & 11 & 4 & 9 & 0 & 4 \\ 9 & 17 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.706 NHC(-2)(-3) + so₈ + su₂→(-3) + su₃ + so₈

$$T = 9 \quad [0] \quad 1^{11} \overset{1}{3} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.707 NHC(-2)(-3) + so₈ + su₂→(-3) + su₃ + su₂

$$T = 12 \quad [0] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 3 & 0 & 6 & 2 \\ 2 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 6 & 5 & 0 & 9 & 7 \\ 3 & 4 & 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 3 & 0 & 10 & 2 \\ 2 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 15 & 10 & 9 & 0 & 41 & 13 \\ 3 & 4 & 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 3 & 0 & 11 & 4 \\ 2 & 3 & 0 & 3 & 0 & 8 & 2 \\ 2 & 3 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 11 & 8 & 7 & 0 & 22 & 10 \\ 3 & 4 & 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

4.708 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $(-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 9 & 1 & 7 \\ 17 & 20 & 11 & 0 & 8 \\ 9 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 9 & 1 & 7 \\ 20 & 25 & 12 & 0 & 9 \\ 9 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

4.709 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \mathfrak{n}3 \text{mix}$

$$T = 7 \quad [0] \ 1^{12} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 32 & 9 \\ 17 & 32 & 37 & 11 \\ 5 & 9 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 21^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 20 & 6 \\ 16 & 20 & 28 & 8 \\ 20 & 28 & 37 & 11 \\ 6 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 13^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 24 & 5 \\ 11 & 20 & 45 & 8 \\ 24 & 45 & 100 & 18 \\ 5 & 8 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 31 & 6 \\ 16 & 25 & 48 & 9 \\ 31 & 48 & 89 & 17 \\ 6 & 9 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{13} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 10 & 16 & 20 & 6 \\ 16 & 25 & 32 & 9 \\ 20 & 32 & 37 & 11 \\ 6 & 9 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 10 & 16 & 20 & 6 \\ 16 & 25 & 32 & 9 \\ 20 & 32 & 37 & 11 \\ 6 & 9 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 33 & 6 \\ 15 & 20 & 45 & 8 \\ 33 & 45 & 100 & 18 \\ 6 & 8 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 25 & 5 \\ 11 & 20 & 45 & 8 \\ 25 & 45 & 100 & 18 \\ 5 & 8 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{12}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 13 & 17 & 39 & 7 \\ 17 & 20 & 45 & 8 \\ 39 & 45 & 100 & 18 \\ 7 & 8 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 7 & 5 \\ 11 & 20 & 11 & 8 \\ 7 & 11 & 4 & 4 \\ 5 & 8 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}3^{11}4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 17, T_{\max} = 24, T \in [17, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 23 & 0 & 120 & 18 \\ 3 & 4 & -2 & 18 & -4 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 20 & 6 \\ 15 & 20 & 28 & 8 \\ 20 & 28 & 37 & 11 \\ 6 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 24 & 5 \\ 12 & 25 & 48 & 9 \\ 24 & 48 & 89 & 17 \\ 5 & 9 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 16 & 5 \\ 11 & 20 & 28 & 8 \\ 16 & 28 & 37 & 11 \\ 5 & 8 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 9 & 6 \\ 15 & 20 & 11 & 8 \\ 9 & 11 & 4 & 4 \\ 6 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 15 & 5 \\ 11 & 20 & 28 & 8 \\ 15 & 28 & 37 & 11 \\ 5 & 8 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11}3^{11}3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 18 & 0 & 72 & 14 \\ 3 & 4 & -2 & 14 & 0 \end{pmatrix}_{b_0}$$

- [18] $1^1 1^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 12 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [19] $1^1 1^3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -51$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 36 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -55$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 10 & 15 & 21 & 6 \\ 15 & 20 & 28 & 8 \\ 21 & 28 & 37 & 11 \\ 6 & 8 & 11 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -59$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 \\ 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$
- [22] $13^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 5 & 12 & 23 & 5 \\ 12 & 25 & 48 & 9 \\ 23 & 48 & 89 & 17 \\ 5 & 9 & 17 & 2 \end{pmatrix}_{b_0}$
- [23] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -63$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 13 & 17 & 24 & 7 \\ 17 & 20 & 28 & 8 \\ 24 & 28 & 37 & 11 \\ 7 & 8 & 11 & 3 \end{pmatrix}_{b_0}$
- [24] $1^1 1^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 8 & 0 & 12 & 6 \\ 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [25] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -65$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 5 & 12 & 16 & 5 \\ 12 & 25 & 32 & 9 \\ 16 & 32 & 37 & 11 \\ 5 & 9 & 11 & 2 \end{pmatrix}_{b_0}$
- [26] $13^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 5 & 12 & 15 & 5 \\ 12 & 25 & 32 & 9 \\ 15 & 32 & 37 & 11 \\ 5 & 9 & 11 & 2 \end{pmatrix}_{b_0}$
- [27] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 \\ 3 & 4 & -2 & 8 & 2 \end{pmatrix}_{b_0}$
- [29] $1^1 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -91$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$

- $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 13 & 0 & 36 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$
- [30] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -93$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 13 & 20 & 36 & 7 \\ 20 & 25 & 48 & 9 \\ 36 & 48 & 89 & 17 \\ 7 & 9 & 17 & 3 \end{pmatrix}_{b_0}$
- [31] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 15 & 0 & 39 & 11 \\ 3 & 4 & -2 & 11 & 1 \end{pmatrix}_{b_0}$
- [32] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 18 & 0 & 72 & 14 \\ 3 & 4 & -2 & 14 & 1 \end{pmatrix}_{b_0}$
- [33] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -112$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -115$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 23 & 0 & 120 & 18 \\ 3 & 4 & -2 & 18 & 0 \end{pmatrix}_{b_0}$
- [35] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -131$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 19 & 0 & 64 & 14 \\ 3 & 4 & -2 & 14 & 1 \end{pmatrix}_{b_0}$
- [36] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -139$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 11 & 0 & 20 & 8 \\ 3 & 4 & -2 & 8 & 2 \end{pmatrix}_{b_0}$
- [37] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -155$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 55 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -160$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -176$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 44 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 22 & 0 & 97 & 17 \\ 3 & 4 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 15 & 0 & 39 & 11 \\ 3 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{11} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 27 & 0 & 151 & 21 \\ 3 & 4 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 17 & 0 & 55 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[44] \quad 21^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -233, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 13 & 20 & 24 & 7 \\ 20 & 25 & 32 & 9 \\ 24 & 32 & 37 & 11 \\ 7 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 20 & 0 & 69 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{14} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 19 & 0 & 64 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{12} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 16 & 0 & 44 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 80 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{11} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 22 & 0 & 97 & 17 \\ 3 & 4 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{14} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 24 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 27 & 0 & 151 & 21 \\ 3 & 4 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{13} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 26 & 0 & 128 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 20 & 0 & 69 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{12} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -363, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 21 & 0 & 80 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -363, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 25 & 0 & 111 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{12} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -395, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.03 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 31 & 0 & 188 & 24 \\ 3 & 4 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{14} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 24 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{12} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -460, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 26 & 0 & 128 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{14} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 29 & 0 & 148 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{13} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 30 & 0 & 165 & 23 \\ 3 & 4 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 25 & 0 & 111 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 1^{12} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -547, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 31 & 0 & 188 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{11} 1^{13} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -579, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 35 & 0 & 231 & 27 \\ 3 & 4 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{14} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 34 & 0 & 208 & 26 \\ 3 & 4 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{14} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -635, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.79 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 29 & 0 & 148 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 1^{13} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -636, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.82 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 30 & 0 & 165 & 23 \\ 3 & 4 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{14} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -763, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 22 & 39 & 0 & 280 & 30 \\ 3 & 4 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{13} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -763, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 35 & 0 & 231 & 27 \\ 3 & 4 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{14} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -812, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.79 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 34 & 0 & 208 & 26 \\ 3 & 4 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{14} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -979, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 39 & 0 & 280 & 30 \\ 3 & 4 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [72] \quad 21^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 14, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 5 & 5 & 13 & 5 \\ 3 & 5 & 0 & 9 & 2 \\ 12 & 13 & 9 & 26 & 10 \\ 4 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 3^1 1^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -94, \quad \det = 14, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 1 & 9 & 3 \\ 4 & 10 & 1 & 20 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 37 & 11 \\ 3 & 6 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[73] & 1^{13} 3^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 11 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 12 & 0 & 39 & 11 \\ 2 & 3 & 3 & -2 & 11 & 1 \end{pmatrix}_{b_0} \\
& 1^1 \\
[74] & 1^{12} 3^{12} 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 0 & 12 & 2 \\ 10 & 18 & 12 & 56 & 14 \\ 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[75] & 1^{13} 3^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[76] & 1^3 1^{12} 21^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 12 & 2 \\ 10 & 18 & 12 & 56 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[77] & 1^{11} 3^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 4 & 3 \\ 4 & 10 & 1 & 8 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 4 & 4 \\ 3 & 6 & -1 & 4 & 2 \end{pmatrix}_{b_0} \\
[78] & 1^{12} 3^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 7 & 0 & 15 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0} \\
[79] & 1^{11} 3^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0} \\
[80] & 1^{11} 1^3 1^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 8 & 0 & 25 & 9 \\ 2 & 3 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0} \\
[81] & 1^{11} 1^3 1^3 1^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 10 & 1 & 16 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 20 & 8 \\ 3 & 6 & -1 & 8 & 2 \end{pmatrix}_{b_0} \\
[82] & 1^{11} 1^3 1^3 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 7 & 0 & 15 & 7 \\ 2 & 3 & 2 & -2 & 7 & 0 \end{pmatrix}_{b_0} \\
[83] & 1^{11} 1^3 1^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 10 & 1 & 20 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 37 & 11 \\ 3 & 6 & -1 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [84] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 4 & 3 \\ 4 & 10 & 13 & 8 & 6 \\ 5 & 13 & 12 & 9 & 6 \\ 4 & 8 & 9 & 4 & 4 \\ 3 & 6 & 6 & 4 & 1 \end{pmatrix}_{b_0}$
- [85] $1^{11}3^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 23 & 3 \\ 4 & 10 & 1 & 52 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 52 & 0 & 257 & 29 \\ 3 & 6 & -1 & 29 & 1 \end{pmatrix}_{b_0}$
- [87] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 6 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$
- [89] $1^{11}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 18 & 0 & 48 & 14 \\ 3 & 5 & -2 & 14 & 2 \end{pmatrix}_{b_0}$
- [88] $1^{13}3^{11}1^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$
- [86] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [92] $1^{13}3^{11}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 11 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 12 & 0 & 39 & 11 \\ 2 & 3 & 3 & -2 & 11 & 1 \end{pmatrix}_{b_0}$
- [91] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 6 & 0 & 7 & 5 \\ 2 & 3 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$
- [90] $1^{12}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 2 & 4 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 8 & 0 & 15 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [98] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 3 & 3 \\ 2 & 2 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [97] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$

[96] $1^1 1^1 3^1 2^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 4 & 3 \\ 1 & 4 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 2 & 4 & 4 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[95] $1^1 3^1 1^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 4 & 0 & 9 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{11} 1^1 3^1 1^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[93] $1^1 3^1 1^1 2^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[99] $2^1 1^1 3^1 2^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 5 & 5 & 18 & 5 \\ 3 & 5 & 0 & 13 & 2 \\ 15 & 18 & 13 & 48 & 14 \\ 4 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{14} 3^1 1^1 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 14 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 14 & 15 & 0 & 64 & 14 \\ 2 & 3 & 3 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{11} 3^1 1^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 2 & 3 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{11} 3^1 1^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 6 & 4 \\ 7 & 10 & 13 & 8 & 6 \\ 8 & 13 & 12 & 9 & 6 \\ 6 & 8 & 9 & 4 & 4 \\ 4 & 6 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[103] $1^1 3^1 2^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 10 & 2 \\ 9 & 13 & 10 & 24 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{11} 3^1 2^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 7 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[105] $1^1 3^1 2^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 8 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 8 & 6 \\ 3 & 5 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

- [107] $1^{11}3^{12}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 35 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$
- [106] $1^{11}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 5 & 0 & 13 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 13 & 0 & 24 & 10 \\ 3 & 5 & -2 & 10 & 3 \end{pmatrix}_{b_0}$
- [109] $21^{12}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 5 & 5 & 23 & 5 \\ 3 & 5 & 0 & 16 & 2 \\ 21 & 23 & 16 & 84 & 18 \\ 4 & 5 & 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [108] $1^{11}3^{12}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 9 & 2 \\ 8 & 13 & 9 & 26 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [110] $1^{11}3^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 5 & 0 & 7 & 5 \\ 2 & 3 & 2 & -2 & 5 & 0 \end{pmatrix}_{b_0}$
- [111] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 7 & 3 \\ 4 & 10 & 13 & 16 & 6 \\ 5 & 13 & 12 & 17 & 6 \\ 7 & 16 & 17 & 20 & 8 \\ 3 & 6 & 6 & 8 & 1 \end{pmatrix}_{b_0}$
- [112] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 21 & 3 \\ 4 & 10 & 1 & 52 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 52 & 0 & 257 & 29 \\ 3 & 6 & -1 & 29 & 2 \end{pmatrix}_{b_0}$
- [114] $1^{11}3^{12}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 23 & 0 & 243 & 29 \\ 3 & 3 & -1 & 29 & 2 \end{pmatrix}_{b_0}$
- [113] $1^{11}3^{12}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 2 & 5 & 0 & 6 & 2 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [117] $1^{12}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 16 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$
- [116] $1^{11}3^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$
- [118] $1^{12}3^{12}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{12} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 6 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{11} 31^{13} 3^{11} 12 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 13 & 3 \\ 4 & 10 & 1 & 28 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 28 & 0 & 69 & 15 \\ 3 & 6 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{11} 313^{11} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 10 & 1 & 16 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 16 & 0 & 20 & 8 \\ 3 & 6 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{13} 3^{12} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 13 & 2 \\ 12 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{12} 31^{13} 3^{11} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 10 & 10 & 0 & 37 & 11 \\ 2 & 3 & 2 & -2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{14} 3^{11} 11 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 17 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{14} 3^{11} 11 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 17 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{11} 31^2 3^{11} 12 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 18 & 3 \\ 4 & 10 & 1 & 40 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 40 & 0 & 148 & 22 \\ 3 & 6 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[126] \ 13^1 21^{11} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 2 & 5 & 0 & 6 & 2 \\ 4 & 8 & 6 & 8 & 6 \\ 3 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{11} 31^3 3^{11} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 4 & 4 \\ 2 & 3 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{11} 31^3 3^{11} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 13 & 0 & 60 & 14 \\ 2 & 3 & 2 & -2 & 14 & 0 \end{pmatrix}_{b_0}$$

[128] $1^{11}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 13 & 0 & 87 & 17 \\ 3 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{14}3^{11}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 14 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 15 & 0 & 64 & 14 \\ 2 & 3 & 3 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{11}31^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{12}3^{11}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 9 & 0 & 20 & 8 \\ 2 & 3 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{13}31^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 2 & 4 & 3 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 10 & 11 & 0 & 32 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{13}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 11 & 0 & 39 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{11}31^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 11 & 4 \\ 7 & 10 & 13 & 16 & 6 \\ 8 & 13 & 12 & 17 & 6 \\ 11 & 16 & 17 & 20 & 8 \\ 4 & 6 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[135] $1^13^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 3 & 5 & 0 & 6 & 2 \\ 4 & 8 & 6 & 8 & 6 \\ 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[137] $1^{12}3^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 9 & 0 & 20 & 8 \\ 2 & 3 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{11}3^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 7 & 5 \\ 2 & 3 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{12}3^{11}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 308 & 32 \\ 3 & 3 & -1 & 32 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 7 & 3 \\ 4 & 10 & 1 & 16 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 16 & 0 & 20 & 8 \\ 3 & 6 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 22 & 3 \\ 4 & 10 & 1 & 52 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 52 & 0 & 257 & 29 \\ 3 & 6 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{11}13^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 9 & 2 \\ 7 & 13 & 9 & 26 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{11}1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 13 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 26 & 10 \\ 3 & 5 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{11}1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 5 & 0 & 7 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{12}3^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 2 & 4 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 8 & 0 & 15 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{13}3^{11}1^{11}3^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 11 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 12 & 0 & 39 & 11 \\ 2 & 3 & 3 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[145] $13^{11}1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 3 \\ 4 & 10 & 13 & 16 & 6 \\ 5 & 13 & 12 & 17 & 6 \\ 6 & 16 & 17 & 20 & 8 \\ 3 & 6 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{12}3^{11}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 20 & 0 & 200 & 26 \\ 3 & 3 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{11}1^{11}3^{11}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 8 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{11}1^{11}3^{11}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 3 & 5 & 0 & 9 & 2 \\ 7 & 13 & 9 & 26 & 10 \\ 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[151] & 1^{112}313^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 7 & 0 & 15 & 7 \\ 2 & 3 & 2 & -2 & 7 & 0 \end{pmatrix}_{b_0} \\
[153] & 1^{111}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 7 & 8 & 10 & 4 \\ 7 & 10 & 13 & 16 & 6 \\ 8 & 13 & 12 & 17 & 6 \\ 10 & 16 & 17 & 20 & 8 \\ 4 & 6 & 6 & 8 & 2 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[152] & 1^{113}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 3 & 5 & 0 & 10 & 2 \\ 6 & 13 & 10 & 24 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[154] & 1^{113}31^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 16 & 0 & 100 & 18 \\ 2 & 3 & 2 & -2 & 18 & 0 \end{pmatrix}_{b_0} \\
[155] & 1^{111}31^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 16 & 3 \\ 4 & 10 & 1 & 40 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 40 & 0 & 148 & 22 \\ 3 & 6 & -1 & 22 & 2 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[156] & 1^{113}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[157] & 1^{123}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 56 & 14 \\ 3 & 5 & -2 & 14 & 2 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[158] & 1^{13}3^{12}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 27 & 0 & 359 & 35 \\ 3 & 3 & -1 & 35 & 2 \end{pmatrix}_{b_0} \\
[159] & 1^{111}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.51 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 11 & 3 \\ 4 & 10 & 1 & 28 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 28 & 0 & 69 & 15 \\ 3 & 6 & -1 & 15 & 2 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[160] & 1^{123}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 12 & 0 & 56 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[161] & 1^{113}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 11 & 0 & 42 & 12 \\ 2 & 2 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[162] $1^1 3^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 4 & 0 & 9 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{12} 1^{11} 3^1 2^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 99$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 5 & 0 & 7 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{11} 3^1 2^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 17 & 3 \\ 4 & 10 & 1 & 40 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 40 & 0 & 148 & 22 \\ 3 & 6 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{11} 3^1 2^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 9 & 0 & 35 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{11} 3^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 12 & 3 \\ 4 & 10 & 1 & 28 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 28 & 0 & 69 & 15 \\ 3 & 6 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[169] $2^1 1^{12} 3^1 2^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 5 & 5 & 28 & 5 \\ 3 & 5 & 0 & 20 & 2 \\ 24 & 28 & 20 & 120 & 22 \\ 4 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[170] $1^4 1^3 2^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 14 & 2 \\ 12 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[167] $1^1 1^{11} 3^1 2^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 9 \\ 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[168] $1^2 1^3 2^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 10 & 2 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[171] $1^{11} 1^{12} 3^1 2^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 8 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{12} 1^3 2^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 16 & 2 \\ 14 & 23 & 16 & 84 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[172] & 1^{11}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 10 & 10 & 0 & 32 & 10 \\ 2 & 3 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0} \\
[174] & 1^{11}3^{11}2^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 10 & 2 \\ 7 & 13 & 10 & 24 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[180] & 1^{12}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[179] & 1^{13}1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[178] & 1^{13}1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[177] & 1^{12}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[176] & 1^{11}3^{11}2^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 18 & 0 & 48 & 14 \\ 3 & 5 & -2 & 14 & 3 \end{pmatrix}_{b_0} \\
[175] & 1^{13}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 17 & 13 & 0 & 60 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0} \\
[181] & 1^{13}3^{11}2^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 19 & 0 & 151 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[182] & 1^{11}3^{11}2^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 6 & 0 & 20 & 8 \\ 2 & 2 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0} \\
[183] & 1^{11}3^{11}2^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 10 & 3 \\ 1 & 4 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -2 & 4 & 0 \\ 5 & 10 & 10 & 4 & 32 & 10 \\ 2 & 3 & 2 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{13}3^{121^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 83 & 17 \\ 3 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{12}3^{121^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 13 & 0 & 87 & 17 \\ 3 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{113}3^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 10 & 10 & 0 & 32 & 10 \\ 2 & 3 & 2 & -2 & 10 & 0 \end{pmatrix}_{b_0}$$

[187] $1^{12}3^{121^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 13 & 2 \\ 11 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{131}3^{123^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 7 & 0 & 31 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{113}3^{121^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 23 & 0 & 78 & 18 \\ 3 & 5 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{111}3^{121^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 11 & 0 & 39 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{12}3^{121^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.03 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 9 & 0 & 35 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{113}3^{121^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 0 & 13 & 2 \\ 9 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{14}3^{121^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 26 & 0 & 292 & 32 \\ 3 & 3 & -1 & 32 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{13}3^{112}3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

- $$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [195] $1^1 3^1 1^{14} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 0 \end{pmatrix}_{b_0}$$
- [196] $1^1 1^{12} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 159$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 0 & 16 & 2 \\ 12 & 23 & 16 & 84 & 18 \\ 3 & 5 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$
- [198] $1^1 1^{11} 3^{11} 1^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [199] $1^1 1^{14} 31^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 16 & 0 & 89 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [200] $1^1 1^{12} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 16 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$
- [201] $1^1 1^{11} 3^{12} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 28 & 0 & 116 & 22 \\ 3 & 5 & -2 & 22 & 3 \end{pmatrix}_{b_0}$$
- [197] $1^1 1^{13} 3^{11} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [202] $1^1 1^{12} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 13 & 2 \\ 9 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [203] $1^1 1^{13} 31^{13} 1^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 13 & 0 & 60 & 14 \\ 2 & 3 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [206] $1^1 1^{12} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 164$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 8 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [205] $1^1 1^{12} 3^{12} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 164$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 18 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{13}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 7 & 0 & 34 & 10 \\ 2 & 2 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{14}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 15 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{13}3^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{12}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 16 & 2 \\ 13 & 23 & 16 & 84 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{14}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 12 & 0 & 56 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{14}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 18 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{12}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 23 & 0 & 84 & 18 \\ 3 & 5 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{12}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 16 & 2 \\ 12 & 23 & 16 & 84 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{11}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 13 & 2 \\ 10 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{13}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 14 & 2 \\ 9 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{13}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 14 & 2 \\ 11 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[217] \ 1^1 3^1 1^2 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 14 & 7 & 0 & 31 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[222] \ 1^1 4^3 1^1 11^3 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 17 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[221] \ 1^1 4^3 1^1 11^3 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 17 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[220] \ 1^1 4^3 1^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 14 & 0 & 64 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[219] \ 1^1 11^3 1^1 11^3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 1^1 11^3 1^1 11^3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 11 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[226] \ 1^1 1^2 3^1 21^1 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[227] \ 1^1 3^1 1^1 11^3 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[223] \ 1^1 3^1 1^1 11^3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[224] \ 1^1 2^3 1^1 11^3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 14 & 10 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[225] \ 1^1 1^4 3^1 21^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$\begin{aligned}
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 8 & 0 & 52 & 12 \\ 2 & 2 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[228] & 1^{114} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 13 & 13 & 0 & 55 & 13 \\ 2 & 3 & 2 & -2 & 13 & 0 \end{pmatrix}_{b_0} \\
[230] & 1^{11} 1^{14} 3^{12} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 22 & 0 & 192 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0} \\
[229] & 1^{112} 1^{12} 3^{12} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 20 & 0 & 200 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0} \\
[231] & 1^{12} 1^{13} 3^{12} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 14 & 2 \\ 10 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[232] & 1^{111} 1^{13} 3^{12} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 23 & 0 & 243 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0} \\
[233] & 1^{114} 3^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 16 & 0 & 89 & 17 \\ 2 & 3 & 2 & -2 & 17 & 0 \end{pmatrix}_{b_0} \\
[234] & 1^{11} 1^{13} 3^{12} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 0 & 17 & 2 \\ 11 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[236] & 1^{13} 1^{12} 3^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0} \\
[235] & 1^{14} 3^{11} 1^{23} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0} \\
[237] & 1^{114} 3^{11} 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 13 & 0 & 55 & 13 \\ 2 & 3 & 2 & -2 & 13 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

- [239] $1^{12}3^{12}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 14 \\ 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$
- [240] $1^{12}3^{12}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 15 & 0 & 83 & 17 \\ 2 & 2 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$
- [238] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$
- [241] $1^{12}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 227$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 11 & 0 & 39 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$
- [242] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 15 & 0 & 83 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$
- [243] $1^{11}3^{11}1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$
- [244] $1^{11}431^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 19 & 2 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 19 & 19 & 0 & 135 & 21 \\ 2 & 3 & 2 & -2 & 21 & 0 \end{pmatrix}_{b_0}$
- [245] $13^121^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 11 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [246] $1^{13}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 17 & 13 & 0 & 60 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$
- [247] $1^{11}431^{21}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 27 & 3 \\ 2 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 30 & 0 & 416 & 38 \\ 3 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0}$
- [248] $1^{12}3^{11}1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[249] & 1^{12}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[250] & 1^{12}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 12 & 0 & 56 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0} \\
[251] & 1^{12}3^{12}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 28 & 0 & 120 & 22 \\ 3 & 5 & -2 & 22 & 3 \end{pmatrix}_{b_0} \\
[252] & 1^{12}3^{11}1^{12}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0} \\
[253] & 1^{13}3^{11}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0} \\
[254] & 1^{13}3^{11}1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0} \\
[255] & 1^{12}3^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 10 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0} \\
[257] & 1^{11}3^{12}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 13 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[256] & 1^{13}3^{12}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 26 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 27 & 0 & 359 & 35 \\ 3 & 3 & -1 & 35 & 3 \end{pmatrix}_{b_0} \\
[258] & 1^{12}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 16 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0} \\
[259] & 1^{13}3^{12}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 14 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 19 & 0 & 151 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 13 \\ 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{114}31^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 16 & 0 & 89 & 17 \\ 2 & 3 & 2 & -2 & 17 & 0 \end{pmatrix}_{b_0}$$

[262] $1^{14}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 18 & 2 \\ 14 & 23 & 18 & 80 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{13}3^{11}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 17 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{12}3^{11}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 0 & 20 & 2 \\ 14 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{14}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 64 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{13}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 18 & 2 \\ 12 & 23 & 18 & 80 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{11}3^{11}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 12 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{12}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 33 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 33 & 0 & 164 & 26 \\ 3 & 5 & -2 & 26 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{13}31^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 10 & 0 & 65 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{11}3^{11}1^{13}3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[270] \ 1^13^{13}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 20 & 10 & 0 & 65 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{12}1^{11}3^{12}2^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 305, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 13 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[274] \ 1^{14}3^{11}2^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.03 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 18 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[275] \ 1^31^{13}2^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 18 & 2 \\ 13 & 23 & 18 & 80 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{12}1^32^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 14 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{13}3^{11}1^{12}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{11}3^{11}1^{12}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{14}3^{11}2^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 26 & 0 & 292 & 32 \\ 3 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{12}1^{13}3^{11}2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 15 & 0 & 83 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[282] $1^{12}3^{12}21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 16 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[281] $1^{14}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 11 & 0 & 71 & 15 \\ 2 & 2 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[284] $1^{13}3^{12}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 348$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[283] $1^{11}3^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 348$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 17 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{13}3^{12}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 359$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 23 & 0 & 243 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{12}3^{12}21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 359$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 15 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{14}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 360$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 22 & 0 & 192 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{13}3^{11}3^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 360$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{13}3^{11}3^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 360$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{14}3^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 372$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 26 & 19 & 0 & 135 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[292] $1^{12}3^{11}1^{23}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 380$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0} \\
[291] \quad & 1^{11}3^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 380, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 17 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[293] \quad & 1^{12}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 19 & 0 & 151 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[294] \quad & 31^{11}23^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.18 \rightarrow 12, T_{\max} = 14, T \in [12, 14] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 4 & 2 \\ 3 & 1 & 4 & 1 & 4 & 3 \\ 3 & 4 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 4 & 4 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 4 & 1 \end{pmatrix}_{b_0} \\
[295] \quad & 1^{11}3^{12}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 3 & 5 & 0 & 24 & 2 \\ 16 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[296] \quad & 1^{13}3^{12}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 19 \\ 2 & 2 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0} \\
[297] \quad & 1^{13}3^{12}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 401, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 21 & 2 \\ 17 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[298] \quad & 1^{11}3^{11}123^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0} \\
[299] \quad & 1^{11}3^{12}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 21 & 2 \\ 14 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[300] \quad & 1^{13}1^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0} \\
[301] \quad & 1^{12}3^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 20 & 13 & 0 & 71 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0} \\
[302] \quad & 1^{14}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 18 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[303] & 1^{11}3^{11}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0} \\
[305] & 1^{13}31^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 23 & 16 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0} \\
[304] & 1^{13}1^{14}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0} \\
[306] & 1^{12}1^{14}3^{21}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 28 & 3 \\ 2 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 416 & 38 \\ 3 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0} \\
[307] & 1^{13}1^{13}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 21 & 2 \\ 16 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[308] & 1^{12}1^{13}3^{21}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 21 & 2 \\ 15 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[309] & 1^{14}3^{12}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 18 \\ 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0} \\
[310] & 1^{14}2^{13}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 20 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0} \\
[311] & 1^{12}13^{12}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 16 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0} \\
[312] & 1^{13}3^{11}1^{12}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0} \\
[313] & 1^{13}31^{14}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0
\end{aligned}$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 26 & 13 & 0 & 111 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[314] \ 1^{12}3^{11}1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 476, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[315] \ 1^{13}3^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 26 & 13 & 0 & 111 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[316] \ 1^{14}3^{11}1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[317] \ 1^{12}3^{12}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 512, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 22 & 0 & 192 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[318] \ 1^{14}3^{12}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 26 & 0 & 292 & 32 \\ 3 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[319] \ 1^{12}3^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 13 & 0 & 71 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[320] \ 1^{12}3^{12}2^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 539, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 17 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{13}3^{12}2^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 539, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 19 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{12}3^{11}1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{13}3^{14}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{11}3^{13}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 23 & 13 & 0 & 88 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[324] \ 1^{14}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 21 \\ 2 & 2 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{11}3^{11}1331^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{11}3^{11}1331^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[329] \ 1^{13}3^{11}1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{12}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 18 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[330] \ 1^{12}3^{11}1231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[331] \ 1^{14}31^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 26 & 19 & 0 & 135 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[334] \ 1^{14}3^{11}1231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[332] \ 1^{13}31^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 23 & 16 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[333] \ 1^{13}1^{14}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[335] \ 1^{11}31^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 23 & 13 & 0 & 88 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 17 & 0 & 128 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\min_{\mathbf{z}} \left(\begin{array}{cccccc} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 33 & 23 & 0 & 208 & 26 \\ 2 & 4 & 2 & -2 & 26 & 1 \end{array} \right)$$

$$T_{\min} = 11 \ 19 \rightarrow 12 \quad T_{\max} = 13 \quad T \in [12 \ 13]$$

$$\min = 11.15 \quad 7.12, \quad 1 \text{ m}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 17 & 0 & 128 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}.$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}.$$

$$T_{\min} = 11.19 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_b$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\min \begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 30 & 17 & 0 & 151 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}$$

$$T_{\min} = 11.10 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 17 & 2 \\ 2 & 4 & 4 & 0 & 39 & 4 \\ 1 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 39 & 26 & 0 & 280 & 30 \\ 2 & 4 & 2 & -2 & 30 & 1 \end{pmatrix}_{b_0}$$

[391] $1^{14}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 1852$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 16 & 2 \\ 2 & 4 & 4 & 0 & 39 & 4 \\ 1 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 39 & 26 & 0 & 280 & 30 \\ 2 & 4 & 2 & -2 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [392] $1^{11}3^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 3 \\ 2 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 4 & 3 & 2 & 4 & 4 \\ 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[393] $1^{11}4^{11}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[394] $1^{11}3^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 3 & 1 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[396] $1^{11}4^{11}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[395] $1^{11}4^{11}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[399] $1^{11}4^{11}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[397] $1^{11}3^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 4 & 2 \\ 5 & 5 & 4 & 5 & 5 \\ 3 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[398] $21^{11}4^{11}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 4 & 2 & 6 & 1 & 5 & 5 \\ 2 & 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[402] $1^{11}3^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 7 & 0 & 8 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[400] $1^{\overline{1}13^{\overline{1}1}1^{\overline{1}}}4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$

[401] $1^{\overline{1}13^{\overline{1}1}1^{\overline{1}}}4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$

[403] $1^{\overline{1}11}4^{\overline{1}13^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$

[405] $1^{\overline{1}11}4^{\overline{1}13^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[404] $21^{\overline{1}11}4^{\overline{1}13^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 2 & 6 & 0 & 4 & 4 \\ 2 & 2 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$

[409] $1^{\overline{1}11}4^{\overline{1}13^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 7 & 3 \\ 4 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 7 & 8 & 1 & 14 & 8 \\ 3 & 4 & -1 & 8 & 3 \end{pmatrix}_{b_0}$

[408] $1^{\overline{1}11}4^{\overline{1}11}23^{\overline{1}1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$

[407] $1^{\overline{1}11}4^{\overline{1}13^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[406] $1^{\overline{1}13^{\overline{1}1}1^{\overline{1}}}4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[410] $1^{\overline{1}12}4^{\overline{1}13^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 8 & 1 & 11 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[411] $1^{\overline{1}12}4^{\overline{1}13^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 6 & 1 & 32 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{11} 4 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 4 & 1 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[424] \ 1^{13} 4 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 8 & 12 & 1 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[425] \ 1^{12} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 11 & 0 & 21 & 9 \\ 3 & 4 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[426] \ 1^{11} 4 13^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 5 & 1 & 18 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[427] \ 1^{11} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[428] \ 1^{12} 4 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[429] \ 1^{12} 4 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 4 & 1 & 2 & 7 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[430] \ 1^{11} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 3 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[431] \ 1^{11} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 4 & 2 & 6 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[434] \ 1^{12} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 5 & 0 & 3 & 12 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

- [432] $1^{12} 4 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 13 & 1 & 29 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$
- [433] $1^{11} 4^1 1^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 3 & 2 & 2 & 8 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [435] $1^{14} 4 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 16 & 1 & 47 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$
- [436] $1^{13} 4^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 15 & 0 & 40 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$
- [437] $1^{13} 4^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 15 & 1 & 41 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$
- [438] $1^4 41^{11} 23^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 6 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 8 & 6 \\ 3 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$
- [439] $1^{12} 4 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 2 & 2 & 10 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [440] $1^{12} 4 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 4 & 3 & 2 & 15 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$
- [441] $1^{13} 4 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -180$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 6 & 1 & 3 & 17 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$
- [442] $1^4 41^{11} 23^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 5 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 5 & 6 & 1 & 2 & 5 & 5 \\ 2 & 4 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$
- [443] $1^{12} 4^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -193$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 12 & 16 & 1 & 46 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[445] $1^{11} 4^{12311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -197$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 5 & 2 & 12 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[444] $1^{12} 4^{1311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -197$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 45 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[446] $1^{14} 4^{112311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 4 & 6 & 0 & 2 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[448] $1^{13} 4^{1311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 17 & 1 & 50 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[447] $1^{14} 4^{1311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 19 & 0 & 65 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[449] $1^{12} 4^{11311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 5 & 1 & 3 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[450] $1^{11} 4^{13311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -238$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 6 & 2 & 22 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[451] $1^{14} 4^{11311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -250$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 4 & 2 \\ 3 & 4 & 4 & 2 & 6 & 4 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 2 & 0 \\ 4 & 6 & 3 & 2 & 4 & 4 \\ 2 & 4 & 2 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[452] $1^{14} 4^{11311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -253$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 13 & 19 & 1 & 66 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[453] $1^{13} 4^{13112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 6 & 2 & 3 & 20 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[454] $1^{11} 4^{14311} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -261$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 7 & 2 & 36 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[455] $1^{11}4^{11}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -265$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 9 & 3 \\ 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 26 & 10 \\ 3 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[456] $1^{13}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -282$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 20 & 0 & 70 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[457] $1^{12}1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 5 & 2 & 3 & 16 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[458] $1^{14}4^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 13 & 21 & 1 & 77 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[459] $1^{12}4^{14}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -300$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 9 & 5 & 4 & 3 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[461] $1^{12}4^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 5 & 3 & 3 & 21 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[460] $1^{13}4^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 6 & 3 & 3 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[462] $1^{13}4^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 14 & 20 & 1 & 71 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[463] $1^{13}4^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 7 & 1 & 4 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[464] $1^{13}4^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 6 & 4 & 3 & 32 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

- [465] $1^{14}4^{11}13^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -348$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 8 & 2 & 4 & 34 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$
- [466] $1^{14}1^{12}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -353$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 11 & 3 \\ 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 24 & 0 & 38 & 12 \\ 3 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$
- [467] $1^{14}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -365$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 24 & 0 & 101 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$
- [469] $1^{14}1^{12}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 7 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 7 & 12 & 0 & 4 & 22 & 8 \\ 2 & 4 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$
- [468] $1^{14}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 8 & 9 & 1 & 5 & 41 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$
- [470] $1^{13}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -420$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 7 & 2 & 4 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$
- [471] $1^{14}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -425$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 16 & 24 & 1 & 102 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$
- [472] $1^{14}4^{11}13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 8 & 3 & 4 & 39 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$
- [473] $1^{14}1^{12}23^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 9 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 9 & 12 & 2 & 4 & 26 & 10 \\ 2 & 4 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$
- [474] $1^{13}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -488$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 7 & 3 & 4 & 33 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$
- [475] $1^{14}4^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 8 & 4 & 4 & 46 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[476] $1^{13} 1^{14} 3^1 1^1 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 10 & 7 & 4 & 4 & 40 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[477] $1^{14} 1^{12} 3^1 1^1 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -548$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 9 & 9 & 2 & 5 & 44 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{14} 1^{12} 23^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -549$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 8 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 8 & 12 & 1 & 4 & 23 & 9 \\ 2 & 4 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[479] $1^{11} 1^{14} 1^{12} 23^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -580$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 14 & 3 \\ 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 32 & 0 & 68 & 16 \\ 3 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[480] $1^{14} 1^{12} 23^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -640$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 16 & 3 \\ 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 36 & 0 & 88 & 18 \\ 3 & 7 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[481] $1^{14} 1^{13} 3^1 1^1 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -660$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 10 & 9 & 3 & 5 & 49 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[482] $1^{14} 1^{13} 23^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -684$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 10 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 10 & 18 & 0 & 6 & 52 & 12 \\ 2 & 4 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[483] $1^{14} 1^{13} 3^1 1^1 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -740$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 11 & 9 & 4 & 5 & 56 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[484] $1^{14} 1^{13} 23^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -989$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 11 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 11 & 18 & 1 & 6 & 53 & 13 \\ 2 & 4 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[485] $1^{11} 1^{14} 1^{13} 23^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -993$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 19 & 3 \\ 6 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 44 & 0 & 130 & 22 \\ 3 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[486] $1^{14} 1^{14} 23^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -1025$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 21 & 3 \\ 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 48 & 0 & 158 & 24 \\ 3 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[487] \ 1^1 41^{14} 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -1044, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 13 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 13 & 24 & 0 & 8 & 94 & 16 \\ 2 & 4 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[488] \ 1^1 41^{13} 23^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -1064, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 12 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 12 & 18 & 2 & 6 & 56 & 14 \\ 2 & 4 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[489] \ 1^{11} 41^{14} 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -1504, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 24 & 3 \\ 6 & 13 & 0 & 56 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 56 & 0 & 212 & 28 \\ 3 & 7 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[490] \ 1^1 41^{14} 23^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -1509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 14 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 14 & 24 & 1 & 8 & 95 & 17 \\ 2 & 4 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[491] \ 1^1 41^{14} 23^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -1744, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 15 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 15 & 24 & 2 & 8 & 98 & 18 \\ 2 & 4 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [493] \ 1^1 41^3 13^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 4, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 2 & 5 \\ 4 & 2 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 5 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[492] \ 21^1 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 4, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 5 & 5 & 7 & 5 \\ 3 & 5 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 4 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[494] \ 1^{11} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 6, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 9 & 10 & 9 & 22 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[495] \ 1^{11} 5^{13} 13^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 7, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[496] \ 21^1 5^{13} 13^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 4 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[497] \ 1^{11} 41^3 13^{11} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 15, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 13 & 5 \\ 4 & 2 & 11 & 4 \\ 13 & 11 & 31 & 13 \\ 5 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[499] \ 1^{11} 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 7 & 10 & 9 & 22 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[498] \ 1^{11} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 7 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[500] \ 1^{11} 5^{12} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 10 & 9 & 22 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[503] \ 1^{31} 4^{11} 141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 & 2 \\ 3 & 13 & 1 & 9 & 5 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 9 & 0 & 3 & 3 & 3 \\ 2 & 5 & 0 & 3 & 0 & 2 \\ 2 & 7 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[501] \ 1^{11} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[502] \ 1^{11} 5^{12} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 13 & 7 \\ 3 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[505] \ 1^{41} 1^{13} 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 1 & 12 & 3 \\ 8 & 38 & 1 & 56 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 56 & 0 & 81 & 17 \\ 3 & 12 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[504] \ 1^{41} 1^{13} 1^{11} 141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 1 & 6 & 3 \\ 8 & 38 & 1 & 24 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 24 & 0 & 13 & 7 \\ 3 & 12 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[506] \ 1^{11} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[507] \ 1^{11} 4^{13} 13^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 6 & 6 \\ 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[508] $1^{11} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[509] $1^{11} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 8 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[511] $1^{11} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[510] $1^{12} \overset{1}{5} 123^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 28 & 10 \\ 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[512] $1^{11} \overset{1}{4} 13^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 4 & 3 \\ 3 & 3 & 1 & 5 & 3 \\ 5 & 4 & 5 & 6 & 6 \\ 3 & 3 & 3 & 6 & 4 \end{pmatrix}_{b_0}$

[513] $1^{12} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 10 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[514] $1^{11} \overset{1}{5} 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 13 & 0 & 79 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[515] $1^{12} \overset{1}{4} 13^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 10 & 2 & 60 & 16 \\ 3 & 3 & 0 & 16 & 4 \end{pmatrix}_{b_0}$

[517] $1^{11} \overset{1}{5} 1^{223} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 20 & 0 & 201 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$

[516] $21^{11} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[518] $21^{12} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[519] \ 1^{11}41^{13}1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 76, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 6 & 2 \\ 3 & 13 & 6 & 0 & 26 & 7 \\ 1 & 6 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 26 & 10 & 0 & 49 & 13 \\ 2 & 7 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[520] \ 1^{12}5^{11}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 83, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[522] \ 1^{12}5^{13}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 87, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 9 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[521] \ 1^{12}5^{11}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 87, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 9 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[523] \ 1^{12}5^{14}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 90, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 12 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[524] \ 1^{12}5^{12}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 94, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.70 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[526] \ 1^{15}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 10 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[525] \ 1^{12}413^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 8 & 2 & 30 & 12 \\ 3 & 3 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[527] \ 1^{11}413^{12}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 105, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 1 & 7 & 3 \\ 6 & 6 & 7 & 12 & 8 \\ 3 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[528] \ 1^{12}5^{12}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 106, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[529] $1^{12}5^{13}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 107, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 11 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[530] $1^{11}41^{13}13^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 13 & 6 & 0 & 14 & 7 \\ 1 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 14 & 6 & 0 & 13 & 7 \\ 2 & 7 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$

[531] $1^{13}5^{11}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 9 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[532] $1^{12}5^{11}123^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 16 & 0 & 108 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$

[533] $1^{13}4^{13}13^{121}14 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 14 & 3 & 112 & 22 \\ 3 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$

[534] $21^{12}5^{13}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 143, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[536] $1^{13}5^{14}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 12 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[535] $1^{13}5^{12}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 10 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[537] $1^{11}4^{31}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 2 & 1 & 3 & 2 \\ 3 & 13 & 10 & 0 & 14 & 7 \\ 2 & 10 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 14 & 10 & 0 & 13 & 7 \\ 2 & 7 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$

[538] $1^{13}5^{13}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[539] $1^{13}5^{11}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 11 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[540] $1^{13}4^{11}13^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 10 & 3 & 40 & 14 \\ 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

[541] $1^{13}4^{11}13^{12}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 12 & 3 & 70 & 18 \\ 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[542] $1^{13}5^{12}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 12 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[543] $1^{13}5^{12}1^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 23 & 0 & 244 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[544] $1^{13}5^{14}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 14 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[545] $1^{13}5^{13}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 13 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[546] $1^{13}5^{11}1^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 215, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 19 & 0 & 143 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[547] $1^{12}5^{13}1^{32}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 28 & 3 \\ 2 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 436 & 40 \\ 3 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$

[548] $21^{13}5^{13}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 13 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[549] $21^{13}5^{13}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 241, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[550] $1^{14}4^{13}13^{12}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 14 & 4 & 84 & 20 \\ 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[551] \ 1^{14} 4 13^{12} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 16 & 4 & 126 & 24 \\ 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[552] \ 1^{13} 5 1^2 23^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 26 & 0 & 293 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[553] \ 1^{13} 5 1^3 23^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 415, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 30 & 3 \\ 2 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 33 & 0 & 499 & 43 \\ 3 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [554] \ 1^{14} 1^1 3^{11} 15 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 9 & 5 \\ 1 & -2 & 3 & 0 \\ 9 & 3 & 15 & 9 \\ 5 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[555] \ 1^{11} 3^{11} 15 13 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 0 & 7 & 5 & 5 \\ 2 & 3 & -1 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[556] \ 1^{11} 3 15 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 11 & 4 \\ 3 & 5 & 0 & 11 & 14 & 8 \\ 2 & 3 & -1 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[557] \ 1^{11} 3 15 13^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 7 & 3 \\ 5 & 0 & 7 & 14 & 8 \\ 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[558] \ 1^{11} 3 15 13^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 2 & 1 & 7 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[560] \ 1^{11} 3^{11} 15 13 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[559] \ 1^{12} 3 15 13^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 11 & 0 & 32 & 12 \\ 3 & 4 & -1 & 12 & 4 \end{pmatrix}_{b_0}$$

[561] $1^1 41^1 3^1 1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -79$, $\det = -39$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 4 & 1 & 2 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 3 & 3 & 0 & 1 & 1 & 3 \\ 2 & 4 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[562] $1^{12} 3^1 1^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -45$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 1 & 15 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 65 & 17 \\ 3 & 4 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[563] $1^1 413^1 1^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -79$, $\det = -46$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13.65 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 5 & 1 & 6 & 5 \\ 1 & -2 & 1 & 0 \\ 6 & 1 & 3 & 5 \\ 5 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[564] $1^{11} 31^1 51^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -53$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 7 & 3 \\ 6 & 0 & 7 & 14 & 8 \\ 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[565] $1^{13} 31^1 51^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -61$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 2 & 1 & 19 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 19 & 0 & 110 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

[566] $1^{11} 31^1 1^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -72$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 11 & 3 \\ 9 & 0 & 11 & 39 & 13 \\ 3 & 0 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[568] $1^{13} 31^1 51^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -75$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 2 & 1 & 23 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 23 & 0 & 167 & 27 \\ 3 & 4 & -1 & 27 & 4 \end{pmatrix}_{b_0}$$

[567] $1^{14} 31^1 51^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -75$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 1 & 23 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 23 & 0 & 152 & 26 \\ 3 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

[569] $1^1 413^1 1^1 51^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -79$, $\det = -83$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 5 & 1 & 10 & 5 \\ 1 & -2 & 1 & 0 \\ 10 & 1 & 12 & 8 \\ 5 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

[570] $1^{12} 31^1 51^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -85$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 12 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[571] $1^{14} 31^1 51^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -101$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} =$

8.0

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 2 & 1 & 27 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 27 & 0 & 217 & 31 \\ 3 & 4 & -1 & 31 & 4 \end{pmatrix}_{b_0}$$

[572] $1^1 41^{13} 1^1 51^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 13 & 5 \\ 1 & -2 & 3 & 0 \\ 13 & 3 & 24 & 12 \\ 5 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[573] $1^1 413^{11} 1^1 51^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -118, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 14 & 5 \\ 1 & -2 & 1 & 0 \\ 14 & 1 & 27 & 11 \\ 5 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[574] $1^{11} 3^{11} 1^1 513^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -125, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 11 & 3 \\ 10 & 0 & 1 & 39 & 13 \\ 3 & 0 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[575] $1^{12} 3^{11} 1^1 513^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 13 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[576] $1^{12} 3^{11} 1^1 513^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 12 & 0 & 15 & 65 & 17 \\ 3 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[577] $1^{12} 3^{11} 1^1 513^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -189, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 16 & 0 & 19 & 125 & 23 \\ 3 & 0 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[578] $1^1 41^{13} 1^1 51^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -206, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 17 & 5 \\ 1 & -2 & 3 & 0 \\ 17 & 3 & 39 & 15 \\ 5 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

[579] $1^{13} 3^{11} 1^1 513^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -212, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 15 & 0 & 19 & 110 & 22 \\ 3 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[580] $1^{13} 3^{11} 1^1 513^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -229, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 22 & 0 & 27 & 259 & 33 \\ 3 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$$

[581] $1^{12}31^1513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -229, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 14 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[582] $1^{13}31^1513^{12}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -245, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 18 & 0 & 23 & 167 & 27 \\ 3 & 0 & 3 & 27 & 4 \end{pmatrix}_{b_0}$$

[583] $1^{12}31^1513^{12}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 17 & 0 & 19 & 125 & 23 \\ 3 & 0 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[584] $1^{14}31^1513^{12}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -293, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 24 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 31 & 3 \\ 24 & 0 & 31 & 317 & 37 \\ 3 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$$

[585] $1^{13}31^1513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -309, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 16 & 0 & 19 & 110 & 22 \\ 3 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[587] $1^{13}31^1513^{13}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 23 & 0 & 27 & 259 & 33 \\ 3 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$$

[586] $1^{13}31^1513^{12}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 19 & 0 & 23 & 167 & 27 \\ 3 & 0 & 3 & 27 & 4 \end{pmatrix}_{b_0}$$

[588] $1^{14}31^1513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -365, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 18 & 0 & 23 & 152 & 26 \\ 3 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0}$$

[590] $1^{14}31^1513^{14}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 29 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 35 & 3 \\ 29 & 0 & 35 & 441 & 43 \\ 3 & 0 & 3 & 43 & 4 \end{pmatrix}_{b_0}$$

[589] $1^{14}31^1513^{12}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 21 & 0 & 27 & 31 & 4 \\ 3 & 0 & 3 & 31 & 4 \end{pmatrix}_{b_0}$$

$$[591] 1^{14} 3^{11} 5 131^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -456, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 25 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 31 & 3 \\ 25 & 0 & 31 & 37 & 4 \\ 3 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [592] 1^{13} 3^{11} 5 1321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 8 & 4 \\ 5 & 0 & 8 & 15 & 9 \\ 3 & 0 & 4 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[593] 1^{11} 3^{121} 6 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[594] 1^{11} 3^{121} 6 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[595] 1^{11} 3^{121} 6 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 8 & 1 & 37 & 13 \\ 3 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[596] 1^{11} 3^{121} 6 131^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 4 & 8 & 0 & 24 & 10 \\ 2 & 2 & 2 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[597] 1^{11} 3^{121} 6 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 7 & 1 & 22 & 10 \\ 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[599] 1^{11} 3^{121} 6 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.98 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 5 & 6 & 0 & 19 & 9 \\ 2 & 2 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[598] 1^{11} 3^{121} 6 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.98 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 11 & 0 & 55 & 15 \\ 3 & 3 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[600] 1^{11} 3^{11} 5 1321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 22 & 4 \\ 15 & 0 & 22 & 119 & 25 \\ 3 & 0 & 4 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[601] \ 1^{11} \ 6^{1^1 3^1 21^1} \ 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 141, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 15 & 1 & 116 & 22 \\ 3 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[603] \ 1^{11} \ 6^{1^1 3^1 21^1} \ 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 157, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 6 & 9 & 0 & 34 & 12 \\ 2 & 2 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[602] \ 1^{11} \ 6^{1^1 3^1 21^1} \ 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 157, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 11 & 1 & 56 & 16 \\ 3 & 3 & 0 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[604] \ 1^{11} \ 6^{1^{12} 3^1 21^1} \ 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 173, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 22 & 19 & 2 & 203 & 29 \\ 3 & 3 & 0 & 29 & 4 \end{pmatrix}_{b_0}$$

$$[605] \ 1^{11} \ 6^{1^{12} 3^1 21^1} \ 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 11 & 2 & 59 & 17 \\ 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[606] \ 1^{11} \ 6^{1^1 3^1 21^1} \ 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 7 & 12 & 0 & 55 & 15 \\ 2 & 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[607] \ 1^{11} \ 6^{1^{12} 3^1 21^1} \ 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 221, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 15 & 2 & 119 & 23 \\ 3 & 3 & 0 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[608] \ 1^{11} \ 6^{1^1 3^1 21^1} \ 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 229, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 8 & 15 & 0 & 82 & 18 \\ 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

4.710 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 24 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 24 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 18 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 15 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 23 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 23 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 15 & 1 & 5 \\ 12 & 25 & 32 & 0 & 9 \\ 15 & 32 & 37 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 1 & 3 \\ 4 & 10 & 13 & 16 & 0 & 6 \\ 5 & 13 & 12 & 17 & 0 & 6 \\ 6 & 16 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

4.711 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{f}_4 + \text{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 9 & 5 \\ 11 & 20 & 28 & 0 & 17 & 8 \\ 15 & 28 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 24 & 0 & 13 & 7 \\ 5 & 8 & 11 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

4.712 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 24 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 24 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 7 & 1 & 5 \\ 11 & 20 & 11 & 0 & 8 \\ 7 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 24 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 24 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 5 & 11 & 25 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 25 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 5 & 11 & 16 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 16 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 15 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 5 & 12 & 23 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 23 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 5 & 12 & 15 & 1 & 5 \\ 12 & 25 & 32 & 0 & 9 \\ 15 & 32 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [8] $1^13^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 & 0 \\ 2 & 4 & 4 & 2 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{12}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 5 & 5 & 18 & 0 & 5 \\ 2 & 5 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^13^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 7 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 5 & 5 & 13 & 0 & 5 \\ 2 & 5 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^13^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}3^{121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 5 & 0 & 18 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 18 & 0 & 48 & 13 & 14 \\ 2 & 5 & 0 & 13 & 0 & 2 \\ 3 & 5 & -2 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{12}3^{121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 23 & 0 & 84 & 16 & 18 \\ 2 & 5 & 0 & 16 & 0 & 2 \\ 3 & 5 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}3^{121}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 28 & 0 & 120 & 20 & 22 \\ 2 & 5 & 0 & 20 & 0 & 2 \\ 3 & 5 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

4.713 NHC(-2)(-3) + so₈ + su_{2n3mix} + so₈ + su₂

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 1 & 5 \\ 11 & 20 & 28 & 0 & 0 & 8 \\ 15 & 28 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & 11 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 10 & 5 \\ 11 & 20 & 28 & 0 & 17 & 8 \\ 15 & 28 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 24 & 0 & 13 & 7 \\ 5 & 8 & 11 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 16 & 1 & 9 & 5 \\ 11 & 20 & 28 & 0 & 17 & 8 \\ 16 & 28 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 13 & 7 \\ 5 & 8 & 11 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 9 & 5 \\ 11 & 20 & 28 & 0 & 17 & 8 \\ 15 & 28 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 13 & 7 \\ 5 & 8 & 11 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

4.714 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 7 & 1 & 9 & 5 \\ 11 & 20 & 11 & 0 & 17 & 8 \\ 7 & 11 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 9 & 0 & 12 & 6 \\ 5 & 8 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.715 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 20 & 12 & 6 \\ 15 & 20 & 28 & 17 & 8 \\ 20 & 28 & 37 & 24 & 11 \\ 12 & 17 & 24 & 13 & 7 \\ 6 & 8 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 20 & 1 & 6 \\ 15 & 20 & 28 & 0 & 8 \\ 20 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{144}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 9 & 3 \\ 3 & 4 & 0 & 24 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 68 & 18 \\ 9 & 16 & 0 & 68 & 46 & 12 \\ 3 & 4 & -2 & 18 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 16 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 7 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 17 & 5 \\ 8 & 14 & 0 & 17 & 37 & 11 \\ 3 & 4 & -2 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 7 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

- [8] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 12 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 20 & 13 & 6 \\ 15 & 20 & 28 & 17 & 8 \\ 20 & 28 & 37 & 24 & 11 \\ 13 & 17 & 24 & 13 & 7 \\ 6 & 8 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 12 & 9 \\ 4 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{132} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 25 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 34 & 19 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -2 & 19 & 6 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{144} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 11 & 3 \\ 3 & 4 & 0 & 24 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 89 & 18 \\ 11 & 21 & 0 & 89 & 76 & 16 \\ 3 & 4 & -2 & 18 & 16 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 21 & 12 & 6 \\ 15 & 20 & 28 & 17 & 8 \\ 21 & 28 & 37 & 24 & 11 \\ 12 & 17 & 24 & 13 & 7 \\ 6 & 8 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$
- [14] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 7 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 28 & 5 \\ 13 & 23 & 0 & 28 & 100 & 18 \\ 3 & 4 & -2 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 17 & 5 \\ 9 & 14 & 0 & 17 & 37 & 11 \\ 3 & 4 & -2 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$
- [16] $21^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 24 & 15 & 7 \\ 17 & 20 & 28 & 17 & 8 \\ 24 & 28 & 37 & 24 & 11 \\ 15 & 17 & 24 & 13 & 7 \\ 7 & 8 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$
- [17] $1^{133} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 7 & 3 \\ 3 & 4 & 0 & 20 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 42 & 15 \\ 7 & 12 & 0 & 42 & 25 & 9 \\ 3 & 4 & -2 & 15 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & -2 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 16 & 8 \\ 4 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & -2 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 22 & 12 \\ 5 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -2 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 22 & 12 \\ 4 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -2 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 3 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{142} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 4 & 3 \\ 3 & 4 & 0 & 29 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 29 & 0 & 148 & 40 & 22 \\ 4 & 8 & 0 & 40 & 10 & 6 \\ 3 & 4 & -2 & 22 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 13 & 7 \\ 3 & 4 & -2 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 14 & 6 \\ 5 & 9 & 0 & 14 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 11 & 12 \\ 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{132} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 28 & 15 \\ 4 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -2 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{131} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 14 & 15 \\ 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{114} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 44 & 9 \\ 12 & 21 & 0 & 44 & 76 & 16 \\ 3 & 4 & -2 & 9 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 18 & 3 \\ 3 & 4 & 0 & 7 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 39 & 5 \\ 18 & 32 & 0 & 39 & 193 & 25 \\ 3 & 4 & -2 & 5 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -2 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 14 & 6 \\ 6 & 9 & 0 & 14 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 6 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -2 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 5 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & -2 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -2 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{13} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 28 & 15 \\ 5 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 4 & 0 & 24 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 21 & 18 \\ 3 & 5 & 0 & 21 & 4 & 4 \\ 3 & 4 & -2 & 18 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{12} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 11 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 42 & 8 \\ 12 & 22 & 0 & 42 & 85 & 17 \\ 3 & 4 & -2 & 8 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 25 & 8 \\ 8 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & -2 & 8 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{122} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 16 & 8 \\ 5 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & -2 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{143} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 7 & 3 \\ 3 & 4 & 0 & 24 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 51 & 18 \\ 7 & 12 & 0 & 51 & 25 & 9 \\ 3 & 4 & -2 & 18 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{132} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 15 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 39 & 22 & 11 \\ 4 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -2 & 11 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 7 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 28 & 5 \\ 14 & 23 & 0 & 28 & 100 & 18 \\ 3 & 4 & -2 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{124} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 59 & 12 \\ 11 & 21 & 0 & 59 & 76 & 16 \\ 3 & 4 & -2 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{133} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 9 & 3 \\ 3 & 4 & 0 & 20 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 60 & 15 \\ 9 & 17 & 0 & 60 & 49 & 13 \\ 3 & 4 & -2 & 15 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 22 & 5 \\ 10 & 18 & 0 & 22 & 58 & 14 \\ 3 & 4 & -2 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{121} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 26 & 12 \\ 6 & 9 & 0 & 26 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{11} 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 15 & 5 \\ 7 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & -2 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 1^{11} 2^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 25 & 8 \\ 7 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & -2 & 8 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{11} 1^{11} 2^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 5 & 3 \\ 3 & 4 & 0 & 24 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 34 & 18 \\ 5 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 1^{11} 2^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 28 & 15 \\ 4 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 20 & 9 \\ 5 & 9 & 0 & 20 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -2 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 1^{11} 2^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 44 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -2 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 1^{11} 2^1 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 20 & 5 \\ 9 & 16 & 0 & 20 & 46 & 12 \\ 3 & 4 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{123} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 24 & 8 \\ 7 & 12 & 0 & 24 & 25 & 9 \\ 3 & 4 & -2 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{142} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 4 & 3 \\ 3 & 4 & 0 & 29 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 40 & 22 \\ 4 & 8 & 0 & 40 & 10 & 6 \\ 3 & 4 & -2 & 22 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{112} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 28 & 9 \\ 7 & 13 & 0 & 28 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{111} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 20 & 9 \\ 6 & 9 & 0 & 20 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{122} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 7 & 3 \\ 3 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 44 & 37 & 12 \\ 7 & 13 & 0 & 37 & 28 & 10 \\ 3 & 4 & -2 & 12 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{111} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 20 & 9 \\ 5 & 9 & 0 & 20 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{113} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 21 & 5 \\ 9 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 11 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 33 & 8 \\ 9 & 17 & 0 & 33 & 49 & 13 \\ 3 & 4 & -2 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{113} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 10 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{11} 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 5 \\ 11 & 21 & 0 & 26 & 76 & 16 \\ 3 & 4 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{11} 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 24 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 34 & 18 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 8 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & -2 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{11} 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 7 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 33 & 5 \\ 15 & 27 & 0 & 33 & 133 & 21 \\ 3 & 4 & -2 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{11} 1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 7 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 22 & 5 \\ 11 & 18 & 0 & 22 & 58 & 14 \\ 3 & 4 & -2 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11} 1^{13} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 11 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{11} 1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11} 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 24 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 34 & 18 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 28 & 9 \\ 8 & 13 & 0 & 28 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{121} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 26 & 12 \\ 5 & 9 & 0 & 26 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{113} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 27 & 5 \\ 12 & 22 & 0 & 27 & 85 & 17 \\ 3 & 4 & -2 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{132} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 8 & 3 \\ 3 & 4 & 0 & 20 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 46 & 15 \\ 8 & 13 & 0 & 46 & 28 & 10 \\ 3 & 4 & -2 & 15 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{124} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 11 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 41 & 8 \\ 11 & 21 & 0 & 41 & 76 & 16 \\ 3 & 4 & -2 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 11 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 33 & 8 \\ 10 & 17 & 0 & 33 & 49 & 13 \\ 3 & 4 & -2 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{124} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 11 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 50 & 8 \\ 14 & 26 & 0 & 50 & 118 & 20 \\ 3 & 4 & -2 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{123} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 16 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 48 & 12 \\ 9 & 17 & 0 & 48 & 49 & 13 \\ 3 & 4 & -2 & 12 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{113} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 19 & 3 \\ 3 & 4 & 0 & 7 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 39 & 5 \\ 19 & 32 & 0 & 39 & 193 & 25 \\ 3 & 4 & -2 & 5 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{121} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 44 & 26 & 12 \\ 5 & 9 & 0 & 26 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{131} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[99] 1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 20 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 32 & 15 \\ 5 & 9 & 0 & 32 & 13 & 7 \\ 3 & 4 & -2 & 15 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[100] 1^{141} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 5 & 3 \\ 3 & 4 & 0 & 24 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 38 & 18 \\ 5 & 9 & 0 & 38 & 13 & 7 \\ 3 & 4 & -2 & 18 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[101] 1^{121} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 26 & 0 & 128 & 17 & 20 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 20 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[102] 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[103] 1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 37 & 12 \\ 8 & 13 & 0 & 37 & 28 & 10 \\ 3 & 4 & -2 & 12 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[104] 1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 37 & 12 \\ 7 & 13 & 0 & 37 & 28 & 10 \\ 3 & 4 & -2 & 12 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[105] 1^{141} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 14 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[106] 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 21 & 5 \\ 10 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[107] 1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 482, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 11 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 50 & 8 \\ 15 & 26 & 0 & 50 & 118 & 20 \\ 3 & 4 & -2 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[108] 1^{134} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 20 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 74 & 15 \\ 11 & 21 & 0 & 74 & 76 & 16 \\ 3 & 4 & -2 & 15 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[109] 1^{11} 1^{14} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 502, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 20 & 3 \\ 3 & 4 & 0 & 7 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 44 & 5 \\ 20 & 36 & 0 & 44 & 238 & 28 \\ 3 & 4 & -2 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[110] 1^{11} 1^{13} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 505, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 7 & 3 \\ 3 & 4 & 0 & 20 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 46 & 15 \\ 7 & 13 & 0 & 46 & 28 & 10 \\ 3 & 4 & -2 & 15 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[111] 1^{11} 1^{13} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 5 & 3 \\ 3 & 4 & 0 & 20 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 32 & 15 \\ 5 & 9 & 0 & 32 & 13 & 7 \\ 3 & 4 & -2 & 15 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[112] 1^{11} 1^{14} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 9 & 3 \\ 3 & 4 & 0 & 24 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 72 & 18 \\ 9 & 17 & 0 & 72 & 49 & 13 \\ 3 & 4 & -2 & 18 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[113] 1^{11} 1^{14} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 32 & 5 \\ 14 & 26 & 0 & 32 & 118 & 20 \\ 3 & 4 & -2 & 5 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[114] 1^{11} 1^{12} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 59 & 12 \\ 12 & 21 & 0 & 59 & 76 & 16 \\ 3 & 4 & -2 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[115] 1^{11} 1^{14} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[116] 1^{11} 1^{14} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 10 & 3 \\ 3 & 4 & 0 & 24 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 72 & 18 \\ 10 & 17 & 0 & 72 & 49 & 13 \\ 3 & 4 & -2 & 18 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[117] 1^{11} 1^{12} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 16 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 48 & 12 \\ 10 & 17 & 0 & 48 & 49 & 13 \\ 3 & 4 & -2 & 12 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[118] 1^{11} 1^{13} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 557, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{113} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 7 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 33 & 5 \\ 16 & 27 & 0 & 33 & 133 & 21 \\ 3 & 4 & -2 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{124} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 11 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 41 & 8 \\ 12 & 21 & 0 & 41 & 76 & 16 \\ 3 & 4 & -2 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{141} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 5 & 3 \\ 3 & 4 & 0 & 24 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 38 & 18 \\ 5 & 9 & 0 & 38 & 13 & 7 \\ 3 & 4 & -2 & 18 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{132} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 7 & 3 \\ 3 & 4 & 0 & 20 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 46 & 15 \\ 7 & 13 & 0 & 46 & 28 & 10 \\ 3 & 4 & -2 & 15 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{114} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 7 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 38 & 5 \\ 17 & 31 & 0 & 38 & 172 & 24 \\ 3 & 4 & -2 & 5 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{113} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 27 & 5 \\ 13 & 22 & 0 & 27 & 85 & 17 \\ 3 & 4 & -2 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 5 \\ 12 & 21 & 0 & 26 & 76 & 16 \\ 3 & 4 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{133} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 605, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 4 & 0 & 20 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 60 & 15 \\ 9 & 17 & 0 & 60 & 49 & 13 \\ 3 & 4 & -2 & 15 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{131} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 30 & 0 & 165 & 20 & 23 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -2 & 23 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{133} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 10 & 3 \\ 3 & 4 & 0 & 20 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 60 & 15 \\ 10 & 17 & 0 & 60 & 49 & 13 \\ 3 & 4 & -2 & 15 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{131} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -2 & 27 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{142} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 7 & 3 \\ 3 & 4 & 0 & 24 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 55 & 18 \\ 7 & 13 & 0 & 55 & 28 & 10 \\ 3 & 4 & -2 & 18 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{142} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 7 & 3 \\ 3 & 4 & 0 & 24 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 55 & 18 \\ 7 & 13 & 0 & 55 & 28 & 10 \\ 3 & 4 & -2 & 18 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{131} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{144} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 11 & 3 \\ 3 & 4 & 0 & 24 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 89 & 18 \\ 11 & 21 & 0 & 89 & 76 & 16 \\ 3 & 4 & -2 & 18 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{134} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 20 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 74 & 15 \\ 12 & 21 & 0 & 74 & 76 & 16 \\ 3 & 4 & -2 & 15 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{141} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 752, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{141} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -2 & 22 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{143} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 9 & 3 \\ 3 & 4 & 0 & 24 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 72 & 18 \\ 9 & 17 & 0 & 72 & 49 & 13 \\ 3 & 4 & -2 & 18 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{131} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 30 & 0 & 165 & 20 & 23 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -2 & 23 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{131} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 797, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 21 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -2 & 27 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{144} 3^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 808, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 12 & 3 \\ 3 & 4 & 0 & 24 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 89 & 18 \\ 12 & 21 & 0 & 89 & 76 & 16 \\ 3 & 4 & -2 & 18 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{114} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 822, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 32 & 5 \\ 15 & 26 & 0 & 32 & 118 & 20 \\ 3 & 4 & -2 & 5 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{114} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 834, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 21 & 3 \\ 3 & 4 & 0 & 7 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 44 & 5 \\ 21 & 36 & 0 & 44 & 238 & 28 \\ 3 & 4 & -2 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{114} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 899, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 18 & 3 \\ 3 & 4 & 0 & 7 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 38 & 5 \\ 18 & 31 & 0 & 38 & 172 & 24 \\ 3 & 4 & -2 & 5 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{141} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -2 & 26 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{141} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -2 & 22 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{141} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 997, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 22 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 1 & 3 \\ 3 & 4 & -2 & 30 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[148] \ 1^{141} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -2 & 26 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{141} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 23 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 1 & 3 \\ 3 & 4 & -2 & 30 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [150] \ 1^{113} 3^{121} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 5 & 3 \\ 2 & 1 & 1 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 20 & 11 \\ 5 & 4 & 0 & 20 & 10 & 6 \\ 3 & 3 & -1 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{111} 3^{121} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 14 & 6 \\ 3 & 4 & 7 & 0 & 14 & 16 & 8 \\ 2 & 2 & 2 & -2 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{122} 3^{121} 1^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 41 & 11 \\ 5 & 7 & 11 & 0 & 41 & 45 & 13 \\ 2 & 2 & 2 & -2 & 11 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{113} 3^{121} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 2 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 3 & 5 & 0 & 9 & 4 & 2 \\ 7 & 13 & 9 & 26 & 9 & 10 \\ 2 & 5 & 4 & 9 & 2 & 4 \\ 3 & 5 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{113} 3^{121} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 11 & 3 \\ 2 & 1 & 1 & 9 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 47 & 11 \\ 11 & 13 & 0 & 47 & 61 & 15 \\ 3 & 3 & -1 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{113} 3^{121} 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 5 & 0 & 13 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 19 & 10 \\ 5 & 10 & 0 & 19 & 14 & 8 \\ 3 & 5 & -2 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{113} 3^{121} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 5 & 6 \\ 1 & 1 & 3 & 0 & 5 & 1 & 3 \\ 2 & 2 & 2 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{112} 3^{121} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 23 & 6 \\ 5 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{\overline{121}^1} 3^{\overline{122}} 21^{\overline{122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -29, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 9 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 25 & 11 \\ 3 & 4 & 7 & 0 & 25 & 16 & 8 \\ 2 & 2 & 2 & -2 & 11 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{\overline{111}^1} 3^{\overline{112}} 21^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 12 & 3 \\ 2 & 1 & 1 & 9 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 49 & 11 \\ 12 & 11 & 0 & 49 & 65 & 15 \\ 3 & 3 & -1 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{\overline{113}^1} 3^{\overline{121}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 1 & 9 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 47 & 11 \\ 10 & 13 & 0 & 47 & 61 & 15 \\ 3 & 3 & -1 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{\overline{111}^1} 3^{\overline{121}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -34, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 2 & 3 \\ 2 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 9 & 11 \\ 2 & 3 & 0 & 9 & 1 & 3 \\ 3 & 3 & -1 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{\overline{132}^1} 3^{\overline{121}} 21^{\overline{133}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -37, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 0 & 13 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 59 & 16 \\ 5 & 7 & 11 & 0 & 59 & 45 & 13 \\ 2 & 2 & 2 & -2 & 16 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{\overline{13}^1} 3^{\overline{121}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 5 & 0 & 8 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 6 & 6 \\ 2 & 5 & 0 & 6 & 2 & 4 \\ 3 & 5 & -2 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{\overline{13}^1} 3^{\overline{121}} 21^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 5 & 0 & 13 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 10 & 10 \\ 2 & 5 & 0 & 10 & 2 & 4 \\ 3 & 5 & -2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{\overline{113}^1} 3^{\overline{121}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 17 & 6 \\ 4 & 7 & 6 & 0 & 17 & 25 & 9 \\ 2 & 2 & 2 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{\overline{123}^1} 3^{\overline{121}} 21^{\overline{124}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 57 & 11 \\ 7 & 10 & 15 & 0 & 57 & 88 & 18 \\ 2 & 2 & 2 & -2 & 11 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{1^1} 1^{12} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 48 & 10 \\ 12 & 25 & 0 & 48 & 94 & 20 \\ 3 & 5 & -2 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{1^1} 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 6 & 3 \\ 2 & 1 & 1 & 9 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 29 & 11 \\ 6 & 7 & 0 & 29 & 21 & 9 \\ 3 & 3 & -1 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{1^1} 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -63, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 6 & 3 \\ 2 & 1 & 1 & 9 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 9 & 0 & 35 & 29 & 11 \\ 6 & 7 & 0 & 29 & 21 & 9 \\ 3 & 3 & -1 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{1^1} 1^{12} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 2 & 3 \\ 2 & 1 & 1 & 16 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 16 & 0 & 116 & 16 & 20 \\ 2 & 3 & 0 & 16 & 1 & 3 \\ 3 & 3 & -1 & 20 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{1^1} 1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 2 & 3 \\ 2 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 9 & 0 & 35 & 9 & 11 \\ 2 & 3 & 0 & 9 & 1 & 3 \\ 3 & 3 & -1 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{1^1} 1^{13} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 49 & 10 \\ 11 & 25 & 0 & 49 & 92 & 20 \\ 3 & 5 & -2 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{1^1} 1^{13} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 0 & 13 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 36 & 16 \\ 3 & 4 & 7 & 0 & 36 & 16 & 8 \\ 2 & 2 & 2 & -2 & 16 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{1^1} 1^{13} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 24 & 6 \\ 9 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & -2 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{1^1} 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 6 & 4 \\ 2 & 3 & 4 & 0 & 6 & 5 & 5 \\ 2 & 2 & 2 & -2 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{123}3^1 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 7 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 9 & 11 \\ 1 & 1 & 3 & 0 & 9 & 1 & 3 \\ 2 & 2 & 2 & -2 & 11 & 3 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{142}3^1 21^{143} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 0 & 13 & 7 & 2 \\ 1 & 2 & 0 & 0 & 17 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 77 & 21 \\ 5 & 7 & 11 & 0 & 77 & 45 & 13 \\ 2 & 2 & 2 & -2 & 21 & 13 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{124}3^1 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 46 & 11 \\ 6 & 10 & 10 & 0 & 46 & 58 & 14 \\ 2 & 2 & 2 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{111}3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{121}3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 6 & 3 \\ 2 & 1 & 1 & 12 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 36 & 14 \\ 6 & 7 & 0 & 36 & 21 & 9 \\ 3 & 3 & -1 & 14 & 9 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{123}3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 14 & 3 \\ 2 & 1 & 1 & 12 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 84 & 14 \\ 14 & 17 & 0 & 84 & 121 & 21 \\ 3 & 3 & -1 & 14 & 21 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{113}3^1 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 32 & 6 \\ 7 & 10 & 15 & 0 & 32 & 88 & 18 \\ 2 & 2 & 2 & -2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{113}3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 12 & 6 \\ 5 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & -2 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{112}3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 8 & 3 \\ 2 & 1 & 1 & 9 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 38 & 11 \\ 8 & 10 & 0 & 38 & 38 & 12 \\ 3 & 3 & -1 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{122}3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 8 & 3 \\ 2 & 1 & 1 & 12 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 12 & 0 & 56 & 48 & 14 \\ 8 & 10 & 0 & 48 & 38 & 12 \\ 3 & 3 & -1 & 14 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 14 & 3 \\ 2 & 1 & 1 & 9 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 67 & 11 \\ 14 & 17 & 0 & 67 & 121 & 21 \\ 3 & 3 & -1 & 11 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 7 & 3 \\ 2 & 1 & 1 & 9 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 29 & 11 \\ 7 & 7 & 0 & 29 & 21 & 9 \\ 3 & 3 & -1 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{11} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 29 & 10 \\ 7 & 15 & 0 & 29 & 32 & 12 \\ 3 & 5 & -2 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{11} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 8 & 3 \\ 2 & 1 & 1 & 16 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 16 & 0 & 116 & 68 & 20 \\ 8 & 10 & 0 & 68 & 38 & 12 \\ 3 & 3 & -1 & 20 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{11} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 2 & 3 \\ 2 & 1 & 1 & 19 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 19 & 0 & 151 & 19 & 23 \\ 2 & 3 & 0 & 19 & 1 & 3 \\ 3 & 3 & -1 & 23 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{12} 3^{11} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 2 & 3 \\ 2 & 1 & 1 & 23 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 23 & 0 & 243 & 23 & 29 \\ 2 & 3 & 0 & 23 & 1 & 3 \\ 3 & 3 & -1 & 29 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 9 & 3 \\ 2 & 1 & 1 & 9 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 38 & 11 \\ 9 & 10 & 0 & 38 & 38 & 12 \\ 3 & 3 & -1 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 12 & 14 \\ 2 & 3 & 0 & 12 & 1 & 3 \\ 3 & 3 & -1 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{11}3^{12}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 18 & 6 \\ 7 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & -2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$

[195] $1^{112}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 2 & 3 \\ 2 & 1 & 1 & 16 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 16 & 0 & 116 & 16 & 20 \\ 2 & 3 & 0 & 16 & 1 & 3 \\ 3 & 3 & -1 & 20 & 3 & 3 \end{pmatrix}_{b_0}$

[196] $1^{11}3^{12}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 12 & 4 \\ 4 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$

[197] $1^{11}3^{12}21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 39 & 10 \\ 9 & 20 & 0 & 39 & 58 & 16 \\ 3 & 5 & -2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$

[198] $1^{114}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 1 & 9 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 76 & 11 \\ 16 & 20 & 0 & 76 & 158 & 24 \\ 3 & 3 & -1 & 11 & 24 & 3 \end{pmatrix}_{b_0}$

[199] $1^{112}3^{11}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 18 & 3 \\ 2 & 1 & 1 & 9 & 18 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 78 & 11 \\ 18 & 18 & 0 & 78 & 166 & 24 \\ 3 & 3 & -1 & 11 & 24 & 3 \end{pmatrix}_{b_0}$

[200] $1^{123}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 24 & 6 \\ 10 & 20 & 0 & 24 & 62 & 16 \\ 3 & 5 & -2 & 6 & 16 & 3 \end{pmatrix}_{b_0}$

[201] $1^{123}3^{11}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 12 & 14 \\ 1 & 1 & 3 & 0 & 12 & 1 & 3 \\ 2 & 2 & 2 & -2 & 14 & 3 & 2 \end{pmatrix}_{b_0}$

[202] $1^{141}3^{11}21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 13 & 4 & 2 \\ 1 & 2 & 0 & 0 & 17 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 47 & 21 \\ 3 & 4 & 7 & 0 & 47 & 16 & 8 \\ 2 & 2 & 2 & -2 & 21 & 8 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{121}3^{121}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 6 & 3 \\ 2 & 1 & 1 & 16 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 16 & 0 & 116 & 52 & 20 \\ 6 & 7 & 0 & 52 & 21 & 9 \\ 3 & 3 & -1 & 20 & 9 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{131}3^{131}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 13 & 16 \\ 1 & 1 & 3 & 0 & 13 & 1 & 3 \\ 2 & 2 & 2 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{111}3^{111}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 23 & 0 & 78 & 17 & 18 \\ 2 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & -2 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{131}3^{131}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 6 & 3 \\ 2 & 1 & 1 & 19 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 19 & 0 & 151 & 59 & 23 \\ 6 & 7 & 0 & 59 & 21 & 9 \\ 3 & 3 & -1 & 23 & 9 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{132}3^{111}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 8 & 3 \\ 2 & 1 & 1 & 23 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 23 & 0 & 243 & 98 & 29 \\ 8 & 10 & 0 & 98 & 38 & 12 \\ 3 & 3 & -1 & 29 & 12 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{111}3^{111}21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 21 & 9 \\ 3 & 4 & 7 & 0 & 21 & 16 & 8 \\ 2 & 2 & 2 & -2 & 9 & 8 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{121}3^{111}21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 & 2 \\ 1 & 2 & 0 & 0 & 9 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 7 & 11 \\ 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & -2 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{121}3^{111}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 6 & 3 \\ 2 & 1 & 1 & 16 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 16 & 0 & 116 & 52 & 20 \\ 6 & 7 & 0 & 52 & 21 & 9 \\ 3 & 3 & -1 & 20 & 9 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{123}3^{11}21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 10 & 3 \\ 2 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 60 & 14 \\ 10 & 13 & 0 & 60 & 61 & 15 \\ 3 & 3 & -1 & 14 & 15 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{11}3^{11}21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 28 & 0 & 116 & 21 & 22 \\ 2 & 5 & 0 & 21 & 2 & 4 \\ 3 & 5 & -2 & 22 & 4 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{111}3^{11}21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 9 & 6 \\ 2 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{112}3^{11}21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 24 & 3 \\ 2 & 1 & 1 & 9 & 25 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 107 & 11 \\ 24 & 25 & 0 & 107 & 313 & 33 \\ 3 & 3 & -1 & 11 & 33 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{131}3^{11}21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 6 & 3 \\ 2 & 1 & 1 & 19 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 19 & 0 & 151 & 59 & 23 \\ 6 & 7 & 0 & 59 & 21 & 9 \\ 3 & 3 & -1 & 23 & 9 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{124}3^{11}21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 13 & 3 \\ 2 & 1 & 1 & 12 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 72 & 14 \\ 13 & 16 & 0 & 72 & 90 & 18 \\ 3 & 3 & -1 & 14 & 18 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{122}3^{11}21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 8 & 3 \\ 2 & 1 & 1 & 12 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 48 & 14 \\ 8 & 10 & 0 & 48 & 38 & 12 \\ 3 & 3 & -1 & 14 & 12 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{142}3^{11}21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 28 & 8 & 3 \\ 2 & 1 & 1 & 30 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 28 & 30 & 0 & 416 & 128 & 38 \\ 8 & 10 & 0 & 128 & 38 & 12 \\ 3 & 3 & -1 & 38 & 12 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{112}3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 13 & 6 \\ 3 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{112}3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 13 & 3 \\ 2 & 1 & 1 & 9 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 58 & 11 \\ 13 & 14 & 0 & 58 & 90 & 18 \\ 3 & 3 & -1 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{112}3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 14 & 3 \\ 2 & 1 & 1 & 9 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 58 & 11 \\ 14 & 14 & 0 & 58 & 90 & 18 \\ 3 & 3 & -1 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{112}3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 5 & 0 & 13 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 58 & 10 \\ 14 & 30 & 0 & 58 & 134 & 24 \\ 3 & 5 & -2 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{113}3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 18 & 4 \\ 6 & 9 & 12 & 0 & 18 & 61 & 15 \\ 2 & 2 & 2 & -2 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{121}3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 6 & 3 \\ 2 & 1 & 1 & 20 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 20 & 0 & 200 & 68 & 26 \\ 6 & 7 & 0 & 68 & 21 & 9 \\ 3 & 3 & -1 & 26 & 9 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{141}3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 6 & 3 \\ 2 & 1 & 1 & 22 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 22 & 0 & 192 & 66 & 26 \\ 6 & 7 & 0 & 66 & 21 & 9 \\ 3 & 3 & -1 & 26 & 9 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{121}3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 2 & 3 \\ 2 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 12 & 0 & 56 & 12 & 14 \\ 2 & 3 & 0 & 12 & 1 & 3 \\ 3 & 3 & -1 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{111}3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 13 & 9 \\ 2 & 3 & 4 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & -2 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{114}3^{112}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 18 & 3 \\ 2 & 1 & 1 & 9 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 76 & 11 \\ 18 & 20 & 0 & 76 & 158 & 24 \\ 3 & 3 & -1 & 11 & 24 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{113}3^{113}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 19 & 3 \\ 2 & 1 & 1 & 9 & 21 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 87 & 11 \\ 19 & 21 & 0 & 87 & 205 & 27 \\ 3 & 3 & -1 & 11 & 27 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{123}3^{113}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 30 & 6 \\ 12 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & -2 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[232] $1^{122}3^{111}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 9 & 3 \\ 2 & 1 & 1 & 12 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 48 & 14 \\ 9 & 10 & 0 & 48 & 38 & 12 \\ 3 & 3 & -1 & 14 & 12 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{124}3^{112}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 16 & 3 \\ 2 & 1 & 1 & 12 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 96 & 14 \\ 16 & 20 & 0 & 96 & 158 & 24 \\ 3 & 3 & -1 & 14 & 24 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{131}3^{121}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 6 & 3 \\ 2 & 1 & 1 & 19 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 19 & 0 & 151 & 59 & 23 \\ 6 & 7 & 0 & 59 & 21 & 9 \\ 3 & 3 & -1 & 23 & 9 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{133}3^{141}21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 16 & 19 \\ 1 & 1 & 3 & 0 & 16 & 1 & 3 \\ 2 & 2 & 2 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{112}3^{112}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 18 & 6 \\ 4 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{1\bar{2}}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 5 & 0 & 8 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 36 & 6 \\ 14 & 30 & 0 & 36 & 134 & 24 \\ 3 & 5 & -2 & 6 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{1\bar{2}}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 8 & 3 \\ 2 & 1 & 1 & 22 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 22 & 0 & 192 & 88 & 26 \\ 8 & 10 & 0 & 88 & 38 & 12 \\ 3 & 3 & -1 & 26 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{1\bar{2}}3^121^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -260, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 13 & 1 & 2 \\ 1 & 2 & 0 & 0 & 17 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 17 & 21 \\ 1 & 1 & 3 & 0 & 17 & 1 & 3 \\ 2 & 2 & 2 & -2 & 21 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{1\bar{4}}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -261, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 6 & 3 \\ 2 & 1 & 1 & 22 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 22 & 0 & 192 & 66 & 26 \\ 6 & 7 & 0 & 66 & 21 & 9 \\ 3 & 3 & -1 & 26 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{1\bar{4}}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 1 & 1 & 22 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 22 & 0 & 192 & 22 & 26 \\ 2 & 3 & 0 & 22 & 1 & 3 \\ 3 & 3 & -1 & 26 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{1\bar{2}}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 2 & 3 \\ 2 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 12 & 0 & 56 & 12 & 14 \\ 2 & 3 & 0 & 12 & 1 & 3 \\ 3 & 3 & -1 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{1\bar{4}}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -265, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 2 & 3 \\ 2 & 1 & 1 & 26 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 23 & 26 & 0 & 292 & 26 & 32 \\ 2 & 3 & 0 & 26 & 1 & 3 \\ 3 & 3 & -1 & 32 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{1\bar{3}}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -266, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 8 & 3 \\ 2 & 1 & 1 & 19 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 19 & 0 & 151 & 78 & 23 \\ 8 & 10 & 0 & 78 & 38 & 12 \\ 3 & 3 & -1 & 23 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{1\bar{2}}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 11 & 3 \\ 2 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 60 & 14 \\ 11 & 13 & 0 & 60 & 61 & 15 \\ 3 & 3 & -1 & 14 & 15 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{112}3^{112}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 1 & 9 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 67 & 11 \\ 16 & 17 & 0 & 67 & 121 & 21 \\ 3 & 3 & -1 & 11 & 21 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{111}3^{111}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 2 & 3 \\ 2 & 1 & 1 & 19 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 19 & 0 & 151 & 19 & 23 \\ 2 & 3 & 0 & 19 & 1 & 3 \\ 3 & 3 & -1 & 23 & 3 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{112}3^{112}21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 26 & 9 \\ 4 & 6 & 8 & 0 & 26 & 26 & 10 \\ 2 & 2 & 2 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{111}3^{111}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 6 & 3 \\ 2 & 1 & 1 & 23 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 23 & 0 & 243 & 75 & 29 \\ 6 & 7 & 0 & 75 & 21 & 9 \\ 3 & 3 & -1 & 29 & 9 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{111}3^{111}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 & 2 \\ 1 & 2 & 0 & 0 & 13 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 10 & 16 \\ 1 & 2 & 1 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{114}3^{114}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 26 & 6 \\ 6 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & -2 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{111}3^{111}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 17 & 3 \\ 2 & 1 & 1 & 9 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 76 & 11 \\ 17 & 20 & 0 & 76 & 158 & 24 \\ 3 & 3 & -1 & 11 & 24 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{121}3^{121}21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 32 & 14 \\ 3 & 4 & 7 & 0 & 32 & 16 & 8 \\ 2 & 2 & 2 & -2 & 14 & 8 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{113}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 1 & 9 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 67 & 11 \\ 15 & 17 & 0 & 67 & 121 & 21 \\ 3 & 3 & -1 & 11 & 21 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{122}3^{11}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 23 & 11 \\ 3 & 5 & 5 & 0 & 23 & 13 & 7 \\ 2 & 2 & 2 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{112}3^{11}21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 34 & 9 \\ 5 & 7 & 11 & 0 & 34 & 45 & 13 \\ 2 & 2 & 2 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{123}3^{11}21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 5 & 0 & 33 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 33 & 0 & 164 & 24 & 26 \\ 2 & 5 & 0 & 24 & 2 & 4 \\ 3 & 5 & -2 & 26 & 4 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{124}3^{11}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 21 & 3 \\ 2 & 1 & 1 & 12 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 120 & 14 \\ 21 & 24 & 0 & 120 & 250 & 30 \\ 3 & 3 & -1 & 14 & 30 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{143}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 24 & 4 \\ 8 & 12 & 16 & 0 & 24 & 110 & 20 \\ 2 & 2 & 2 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{122}3^{11}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 6 & 0 & 20 & 18 & 8 \\ 3 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & -2 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{132}3^{11}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 8 & 3 \\ 2 & 1 & 1 & 19 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 19 & 0 & 151 & 78 & 23 \\ 8 & 10 & 0 & 78 & 38 & 12 \\ 3 & 3 & -1 & 23 & 12 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{1^1} 1^{131} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 10 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 9 & 13 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & -2 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{1^{12}} 1^{131} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 6 & 3 \\ 2 & 1 & 1 & 27 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 26 & 27 & 0 & 359 & 91 & 35 \\ 6 & 7 & 0 & 91 & 21 & 9 \\ 3 & 3 & -1 & 35 & 9 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{1^1} 1^{113} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 22 & 6 \\ 5 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{1^1} 1^{113} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 39 & 9 \\ 6 & 9 & 12 & 0 & 39 & 61 & 15 \\ 2 & 2 & 2 & -2 & 9 & 15 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{1^{12}} 1^{131} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 6 & 3 \\ 2 & 1 & 1 & 23 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 23 & 0 & 243 & 75 & 29 \\ 6 & 7 & 0 & 75 & 21 & 9 \\ 3 & 3 & -1 & 29 & 9 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{1^{11}} 1^{131} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 15 & 0 & 83 & 15 & 17 \\ 2 & 3 & 0 & 15 & 1 & 3 \\ 3 & 3 & -1 & 17 & 3 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{1^1} 1^{113} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 27 & 6 \\ 6 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & -2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{1^{11}} 1^{123} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 15 & 3 \\ 2 & 1 & 1 & 12 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 84 & 14 \\ 15 & 17 & 0 & 84 & 121 & 21 \\ 3 & 3 & -1 & 14 & 21 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{132}3^121^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 6 & 2 \\ 1 & 2 & 0 & 0 & 10 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 38 & 13 \\ 4 & 6 & 8 & 0 & 38 & 26 & 10 \\ 2 & 2 & 2 & -2 & 13 & 10 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{142}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 8 & 3 \\ 2 & 1 & 1 & 26 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 26 & 0 & 292 & 108 & 32 \\ 8 & 10 & 0 & 108 & 38 & 12 \\ 3 & 3 & -1 & 32 & 12 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{131}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 1 & 1 & 19 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 19 & 0 & 151 & 19 & 23 \\ 2 & 3 & 0 & 19 & 1 & 3 \\ 3 & 3 & -1 & 23 & 3 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{113}3^121^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 1 & 9 & 21 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 87 & 11 \\ 20 & 21 & 0 & 87 & 205 & 27 \\ 3 & 3 & -1 & 11 & 27 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{141}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 6 & 3 \\ 2 & 1 & 1 & 26 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 23 & 26 & 0 & 292 & 82 & 32 \\ 6 & 7 & 0 & 82 & 21 & 9 \\ 3 & 3 & -1 & 32 & 9 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{121}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 16 & 11 \\ 2 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & -2 & 11 & 5 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{141}3^121^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 & 2 \\ 1 & 2 & 0 & 0 & 17 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 13 & 21 \\ 1 & 2 & 1 & 0 & 13 & 0 & 2 \\ 2 & 2 & 2 & -2 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{142}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 8 & 3 \\ 2 & 1 & 1 & 22 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 22 & 0 & 192 & 88 & 26 \\ 8 & 10 & 0 & 88 & 38 & 12 \\ 3 & 3 & -1 & 26 & 12 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{11} 1^{14} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 21 & 3 \\ 2 & 1 & 1 & 9 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 96 & 11 \\ 21 & 24 & 0 & 96 & 250 & 30 \\ 3 & 3 & -1 & 11 & 30 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{11} 1^{131} 3^1 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 0 & 16 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 43 & 19 \\ 3 & 4 & 7 & 0 & 43 & 16 & 8 \\ 2 & 2 & 2 & -2 & 19 & 8 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{112} 1^{141} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 2 & 3 \\ 2 & 1 & 1 & 26 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 26 & 0 & 292 & 26 & 32 \\ 2 & 3 & 0 & 26 & 1 & 3 \\ 3 & 3 & -1 & 32 & 3 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{112} 1^{124} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 18 & 3 \\ 2 & 1 & 1 & 12 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 96 & 14 \\ 18 & 20 & 0 & 96 & 158 & 24 \\ 3 & 3 & -1 & 14 & 24 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{11} 1^{132} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 33 & 16 \\ 3 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & -2 & 16 & 7 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{112} 1^{141} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 6 & 3 \\ 2 & 1 & 1 & 26 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 26 & 0 & 292 & 82 & 32 \\ 6 & 7 & 0 & 82 & 21 & 9 \\ 3 & 3 & -1 & 32 & 9 & 3 \end{pmatrix}_{b_0}$$

[283] $1^{112} 1^{131} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.00 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 2 & 3 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 15 & 0 & 83 & 15 & 17 \\ 2 & 3 & 0 & 15 & 1 & 3 \\ 3 & 3 & -1 & 17 & 3 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{11} 1^{114} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 52 & 9 \\ 8 & 12 & 16 & 0 & 52 & 110 & 20 \\ 2 & 2 & 2 & -2 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{11} 1^{122} 3^1 21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 0 & 12 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 52 & 14 \\ 5 & 7 & 11 & 0 & 52 & 45 & 13 \\ 2 & 2 & 2 & -2 & 14 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{1^{11}} 1^{121} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 20 & 14 \\ 2 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & -2 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{1^{11}} 1^{122} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 32 & 11 \\ 4 & 6 & 8 & 0 & 32 & 26 & 10 \\ 2 & 2 & 2 & -2 & 11 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{1^{11}} 1^{141} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 2 & 3 \\ 2 & 1 & 1 & 22 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 22 & 0 & 192 & 22 & 26 \\ 2 & 3 & 0 & 22 & 1 & 3 \\ 3 & 3 & -1 & 26 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[289] \ 1^{1^{11}} 1^{142} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 8 & 3 \\ 2 & 1 & 1 & 26 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 23 & 26 & 0 & 292 & 108 & 32 \\ 8 & 10 & 0 & 108 & 38 & 12 \\ 3 & 3 & -1 & 32 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{1^{11}} 1^{113} 3^1 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 47 & 9 \\ 7 & 10 & 15 & 0 & 47 & 88 & 18 \\ 2 & 2 & 2 & -2 & 9 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[291] \ 1^{1^{11}} 1^{142} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 8 & 3 \\ 2 & 1 & 1 & 22 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 22 & 0 & 192 & 88 & 26 \\ 8 & 10 & 0 & 88 & 38 & 12 \\ 3 & 3 & -1 & 26 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[292] \ 1^{1^{11}} 1^{123} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 39 & 11 \\ 5 & 8 & 9 & 0 & 39 & 40 & 12 \\ 2 & 2 & 2 & -2 & 11 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[293] \ 1^{1^{11}} 1^{131} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 19 & 13 \\ 2 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & -2 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{114}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 36 & 6 \\ 8 & 12 & 16 & 0 & 36 & 110 & 20 \\ 2 & 2 & 2 & -2 & 6 & 20 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{124}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 6 & 0 & 20 & 36 & 8 \\ 6 & 10 & 10 & 0 & 36 & 58 & 14 \\ 2 & 2 & 2 & -2 & 8 & 14 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{122}3^121^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 6 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 40 & 14 \\ 4 & 6 & 8 & 0 & 40 & 26 & 10 \\ 2 & 2 & 2 & -2 & 14 & 10 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{141}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 & 2 \\ 1 & 2 & 0 & 0 & 14 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 12 & 18 \\ 1 & 2 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{141}3^121^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 28 & 6 & 3 \\ 2 & 1 & 1 & 30 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 28 & 30 & 0 & 416 & 98 & 38 \\ 6 & 7 & 0 & 98 & 21 & 9 \\ 3 & 3 & -1 & 38 & 9 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{114}3^121^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 31 & 6 \\ 7 & 11 & 13 & 0 & 31 & 81 & 17 \\ 2 & 2 & 2 & -2 & 6 & 17 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{114}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 26 & 3 \\ 2 & 1 & 1 & 9 & 28 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 116 & 11 \\ 26 & 28 & 0 & 116 & 366 & 36 \\ 3 & 3 & -1 & 11 & 36 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{124}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 17 & 3 \\ 2 & 1 & 1 & 12 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 96 & 14 \\ 17 & 20 & 0 & 96 & 158 & 24 \\ 3 & 3 & -1 & 14 & 24 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{114}3^{112}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 22 & 3 \\ 2 & 1 & 1 & 9 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 96 & 11 \\ 22 & 24 & 0 & 96 & 250 & 30 \\ 3 & 3 & -1 & 11 & 30 & 3 \end{pmatrix}_{b_0}$$

[303] $1^{132}3^{11}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 28 & 13 \\ 3 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & -2 & 13 & 7 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{123}3^{11}21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 48 & 11 \\ 6 & 9 & 12 & 0 & 48 & 61 & 15 \\ 2 & 2 & 2 & -2 & 11 & 15 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{123}3^{11}21^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 60 & 14 \\ 6 & 9 & 12 & 0 & 60 & 61 & 15 \\ 2 & 2 & 2 & -2 & 14 & 15 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{142}3^{11}21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 13 & 5 & 2 \\ 1 & 2 & 0 & 0 & 17 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 43 & 21 \\ 3 & 5 & 5 & 0 & 43 & 13 & 7 \\ 2 & 2 & 2 & -2 & 21 & 7 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{141}3^{112}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -628, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 2 & 3 \\ 2 & 1 & 1 & 22 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 22 & 0 & 192 & 22 & 26 \\ 2 & 3 & 0 & 22 & 1 & 3 \\ 3 & 3 & -1 & 26 & 3 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{132}3^{11}21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 7 & 2 \\ 1 & 2 & 0 & 0 & 16 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 70 & 19 \\ 5 & 7 & 11 & 0 & 70 & 45 & 13 \\ 2 & 2 & 2 & -2 & 19 & 13 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{123}3^{11}21^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 10 & 2 \\ 1 & 2 & 0 & 0 & 12 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 72 & 14 \\ 7 & 10 & 15 & 0 & 72 & 88 & 18 \\ 2 & 2 & 2 & -2 & 14 & 18 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{124}3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 55 & 11 \\ 7 & 11 & 13 & 0 & 55 & 81 & 17 \\ 2 & 2 & 2 & -2 & 11 & 17 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{124}3^1 21^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 12 & 2 \\ 1 & 2 & 0 & 0 & 12 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 80 & 14 \\ 8 & 12 & 16 & 0 & 80 & 110 & 20 \\ 2 & 2 & 2 & -2 & 14 & 20 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{131}3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -717, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 23 & 16 \\ 2 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & -2 & 16 & 5 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{124}3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 64 & 11 \\ 8 & 12 & 16 & 0 & 64 & 110 & 20 \\ 2 & 2 & 2 & -2 & 11 & 20 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{142}3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 11 & 0 & 71 & 33 & 15 \\ 3 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & -2 & 15 & 7 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{142}3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 6 & 2 \\ 1 & 2 & 0 & 0 & 14 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 52 & 18 \\ 4 & 6 & 8 & 0 & 52 & 26 & 10 \\ 2 & 2 & 2 & -2 & 18 & 10 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{132}3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -780, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 13 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 46 & 16 \\ 4 & 6 & 8 & 0 & 46 & 26 & 10 \\ 2 & 2 & 2 & -2 & 16 & 10 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{131}3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 11 & 3 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 27 & 19 \\ 2 & 3 & 4 & 0 & 27 & 5 & 5 \\ 2 & 2 & 2 & -2 & 19 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[318] \quad 1^{132}3^{11}21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 11 & 6 & 2 \\ 1 & 2 & 0 & 0 & 16 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 54 & 19 \\ 4 & 6 & 8 & 0 & 54 & 26 & 10 \\ 2 & 2 & 2 & -2 & 19 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{142}3^{11}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -852, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 14 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 38 & 18 \\ 3 & 5 & 5 & 0 & 38 & 13 & 7 \\ 2 & 2 & 2 & -2 & 18 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[320] \quad 1^{141}3^{11}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 0 & 0 & 14 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 26 & 18 \\ 2 & 3 & 4 & 0 & 26 & 5 & 5 \\ 2 & 2 & 2 & -2 & 18 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[321] \quad 1^{141}3^{11}21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 2 & 0 & 13 & 3 & 2 \\ 1 & 2 & 0 & 0 & 17 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 30 & 21 \\ 2 & 3 & 4 & 0 & 30 & 5 & 5 \\ 2 & 2 & 2 & -2 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[322] \quad 1^{142}3^{11}21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 13 & 6 & 2 \\ 1 & 2 & 0 & 0 & 17 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 60 & 21 \\ 4 & 6 & 8 & 0 & 60 & 26 & 10 \\ 2 & 2 & 2 & -2 & 21 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [323] \quad 1^{111}3^{11}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 4 & 3 \\ 6 & 0 & 7 & 14 & 7 & 8 \\ 2 & 0 & 4 & 7 & 2 & 4 \\ 3 & 0 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{121}3^{11}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 4 & 3 \\ 12 & 0 & 15 & 76 & 15 & 18 \\ 2 & 0 & 4 & 15 & 2 & 4 \\ 3 & 0 & 3 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{111}3^{11}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 11 & 8 & 3 \\ 9 & 0 & 11 & 39 & 31 & 13 \\ 6 & 0 & 8 & 31 & 22 & 10 \\ 3 & 0 & 3 & 13 & 10 & 4 \end{pmatrix}_{b_0}$$

[326] $1^{111}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 11 & 4 & 3 \\ 9 & 0 & 11 & 39 & 11 & 13 \\ 2 & 0 & 4 & 11 & 2 & 4 \\ 3 & 0 & 3 & 13 & 4 & 4 \end{pmatrix}_{b_0}$$

[327] $1^{111}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 4 & 3 \\ 5 & 0 & 7 & 14 & 7 & 8 \\ 3 & 0 & 4 & 7 & 2 & 4 \\ 3 & 0 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

[328] $1^{131}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 19 & 4 & 3 \\ 16 & 0 & 19 & 110 & 19 & 22 \\ 2 & 0 & 4 & 19 & 2 & 4 \\ 3 & 0 & 3 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[329] $1^{111}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 2 & 1 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & 2 & 4 \\ 3 & 4 & -1 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

[330] $1^{113}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 2 & 1 & 7 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 33 & 8 \\ 10 & 16 & 0 & 33 & 70 & 18 \\ 3 & 4 & -1 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[331] $1^{111}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 2 & 1 & 7 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 19 & 8 \\ 6 & 8 & 0 & 19 & 22 & 10 \\ 3 & 4 & -1 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

[332] $1^{121}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 6 & 3 \\ 2 & 2 & 1 & 15 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 39 & 17 \\ 6 & 8 & 0 & 39 & 22 & 10 \\ 3 & 4 & -1 & 17 & 10 & 4 \end{pmatrix}_{b_0}$$

[333] $1^{121}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 35, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 2 & 1 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 15 & 17 \\ 2 & 4 & 0 & 15 & 2 & 4 \\ 3 & 4 & -1 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

[334] $1^{131}3^11^15^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 35, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 2 & 1 & 19 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 19 & 22 \\ 2 & 4 & 0 & 19 & 2 & 4 \\ 3 & 4 & -1 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[335] \quad 1^{111}31^1 5 \, 13^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 8 & 3 \\ 5 & 0 & 7 & 14 & 19 & 8 \\ 7 & 0 & 8 & 19 & 22 & 10 \\ 3 & 0 & 3 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[336] \quad 1^{113}31^1 5 \, 13^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 5 & 0 & 7 & 14 & 33 & 8 \\ 13 & 0 & 16 & 33 & 70 & 18 \\ 3 & 0 & 3 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[337] \quad 1^{112}31^1 5 \, 13^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 3 \\ 2 & 2 & 1 & 7 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 26 & 8 \\ 8 & 12 & 0 & 26 & 42 & 14 \\ 3 & 4 & -1 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[339] \quad 1^{122}31^1 5 \, 13^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 8 & 3 \\ 2 & 2 & 1 & 11 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 11 & 0 & 32 & 38 & 12 \\ 8 & 12 & 0 & 38 & 42 & 14 \\ 3 & 4 & -1 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[338] \quad 1^{112}31^1 15 \, 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 8 & 3 \\ 2 & 2 & 1 & 11 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 11 & 0 & 39 & 42 & 13 \\ 8 & 12 & 0 & 42 & 42 & 14 \\ 3 & 4 & -1 & 13 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[340] \quad 1^{123}31^1 15 \, 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 10 & 3 \\ 2 & 2 & 1 & 15 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 69 & 17 \\ 10 & 16 & 0 & 69 & 70 & 18 \\ 3 & 4 & -1 & 17 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[341] \quad 1^{112}31^1 5 \, 13^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 12 & 3 \\ 5 & 0 & 7 & 14 & 26 & 8 \\ 10 & 0 & 12 & 26 & 42 & 14 \\ 3 & 0 & 3 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[342] \quad 1^{141}31^1 15 \, 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 2 & 1 & 27 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 27 & 0 & 217 & 27 & 31 \\ 2 & 4 & 0 & 27 & 2 & 4 \\ 3 & 4 & -1 & 31 & 4 & 4 \end{pmatrix}_{b_0}$$

[343] $1^{122}3^{11}5^{11}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 8 & 3 \\ 2 & 2 & 1 & 15 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 54 & 17 \\ 8 & 12 & 0 & 54 & 42 & 14 \\ 3 & 4 & -1 & 17 & 14 & 4 \end{pmatrix}_{b_0}$$

$T = 13$ [346] $1^{113}3^{11}5^{13}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 19 & 4 \\ 5 & 0 & 8 & 15 & 36 & 9 \\ 13 & 0 & 19 & 36 & 79 & 21 \\ 3 & 0 & 4 & 9 & 21 & 5 \end{pmatrix}_{b_0}$$

[344] $1^{13}3^{11}5^{11}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 5 & 4 \\ 5 & 0 & 8 & 15 & 8 & 9 \\ 3 & 0 & 5 & 8 & 3 & 5 \\ 3 & 0 & 4 & 9 & 5 & 5 \end{pmatrix}_{b_0}$$

[345] $1^{11}3^{11}5^{11}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 9 & 4 \\ 5 & 0 & 8 & 15 & 20 & 9 \\ 7 & 0 & 9 & 20 & 23 & 11 \\ 3 & 0 & 4 & 9 & 11 & 5 \end{pmatrix}_{b_0}$$

[347] $1^{11}3^{11}5^{12}13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 14 & 4 \\ 5 & 0 & 8 & 15 & 28 & 9 \\ 10 & 0 & 14 & 28 & 46 & 16 \\ 3 & 0 & 4 & 9 & 16 & 5 \end{pmatrix}_{b_0}$$

4.716 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 1 & 5 \\ 11 & 20 & 0 & 28 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 28 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

4.717 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 2 & 3 \\ 3 & 4 & 0 & 7 & 14 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 17 & 4 & 5 \\ 8 & 14 & 0 & 17 & 37 & 8 & 11 \\ 2 & 3 & 0 & 4 & 8 & 0 & 2 \\ 3 & 4 & -2 & 5 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 2 & 3 \\ 3 & 4 & 0 & 16 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 22 & 9 & 12 \\ 4 & 8 & 0 & 22 & 10 & 4 & 6 \\ 2 & 3 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & -2 & 12 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -86, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 4 & 5 \\ 4 & 8 & 0 & 10 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 4 & 0 & 2 \\ 3 & 4 & -2 & 5 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 6 & 4 & 5 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 3 & 0 & 4 & 3 & 0 & 2 \\ 3 & 4 & -2 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 4 & 5 \\ 2 & 4 & 0 & 5 & 1 & 2 & 3 \\ 2 & 3 & 0 & 4 & 2 & 0 & 2 \\ 3 & 4 & -2 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 6 & 8 \\ 2 & 4 & 0 & 8 & 1 & 2 & 3 \\ 2 & 3 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & -2 & 8 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -238, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 2 & 3 \\ 3 & 4 & 0 & 20 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 28 & 11 & 15 \\ 4 & 8 & 0 & 28 & 10 & 4 & 6 \\ 2 & 3 & 0 & 11 & 4 & 0 & 2 \\ 3 & 4 & -2 & 15 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -251, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 4 & 5 \\ 5 & 9 & 0 & 11 & 13 & 5 & 7 \\ 2 & 3 & 0 & 4 & 5 & 0 & 2 \\ 3 & 4 & -2 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{142}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 4 & 2 & 3 \\ 3 & 4 & 0 & 29 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 40 & 16 & 22 \\ 4 & 8 & 0 & 40 & 10 & 4 & 6 \\ 2 & 3 & 0 & 16 & 4 & 0 & 2 \\ 3 & 4 & -2 & 22 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -287, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 8 & 7 & 9 \\ 2 & 4 & 0 & 8 & 1 & 2 & 3 \\ 2 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & -2 & 9 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 2 & 3 \\ 3 & 4 & 0 & 7 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 4 & 5 \\ 11 & 21 & 0 & 26 & 76 & 11 & 16 \\ 2 & 3 & 0 & 4 & 11 & 0 & 2 \\ 3 & 4 & -2 & 5 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 2 & 3 \\ 3 & 4 & 0 & 7 & 18 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 22 & 4 & 5 \\ 10 & 18 & 0 & 22 & 58 & 10 & 14 \\ 2 & 3 & 0 & 4 & 10 & 0 & 2 \\ 3 & 4 & -2 & 5 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 1^{22} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 2 & 3 \\ 3 & 4 & 0 & 11 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 25 & 6 & 8 \\ 7 & 13 & 0 & 25 & 28 & 7 & 10 \\ 2 & 3 & 0 & 6 & 7 & 0 & 2 \\ 3 & 4 & -2 & 8 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 2 & 3 \\ 3 & 4 & 0 & 7 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 4 & 5 \\ 7 & 13 & 0 & 16 & 28 & 7 & 10 \\ 2 & 3 & 0 & 4 & 7 & 0 & 2 \\ 3 & 4 & -2 & 5 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 7 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 21 & 4 & 5 \\ 9 & 17 & 0 & 21 & 49 & 9 & 13 \\ 2 & 3 & 0 & 4 & 9 & 0 & 2 \\ 3 & 4 & -2 & 5 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 1^{23} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 2 & 3 \\ 3 & 4 & 0 & 24 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 34 & 13 & 18 \\ 4 & 8 & 0 & 34 & 10 & 4 & 6 \\ 2 & 3 & 0 & 13 & 4 & 0 & 2 \\ 3 & 4 & -2 & 18 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 1^{23} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 2 & 3 \\ 3 & 4 & 0 & 11 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 33 & 6 & 8 \\ 9 & 17 & 0 & 33 & 49 & 9 & 13 \\ 2 & 3 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -2 & 8 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{12} 1^{24} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 2 & 3 \\ 3 & 4 & 0 & 11 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 41 & 6 & 8 \\ 11 & 21 & 0 & 41 & 76 & 11 & 16 \\ 2 & 3 & 0 & 6 & 11 & 0 & 2 \\ 3 & 4 & -2 & 8 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{124}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 11 & 9 & 12 \\ 2 & 4 & 0 & 11 & 1 & 2 & 3 \\ 2 & 3 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -2 & 12 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 2 & 3 \\ 3 & 4 & 0 & 7 & 22 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 27 & 4 & 5 \\ 12 & 22 & 0 & 27 & 85 & 12 & 17 \\ 2 & 3 & 0 & 4 & 12 & 0 & 2 \\ 3 & 4 & -2 & 5 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 2 & 3 \\ 3 & 4 & 0 & 11 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 50 & 6 & 8 \\ 14 & 26 & 0 & 50 & 118 & 14 & 20 \\ 2 & 3 & 0 & 6 & 14 & 0 & 2 \\ 3 & 4 & -2 & 8 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 21 & 0 & 80 & 14 & 12 & 16 \\ 2 & 4 & 0 & 14 & 1 & 2 & 3 \\ 2 & 3 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -2 & 16 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 14 & 11 & 15 \\ 2 & 4 & 0 & 14 & 1 & 2 & 3 \\ 2 & 3 & 0 & 11 & 2 & 0 & 2 \\ 3 & 4 & -2 & 15 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{141}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 2 & 3 \\ 3 & 4 & 0 & 7 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 32 & 4 & 5 \\ 14 & 26 & 0 & 32 & 118 & 14 & 20 \\ 2 & 3 & 0 & 4 & 14 & 0 & 2 \\ 3 & 4 & -2 & 5 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{141}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 17 & 13 & 18 \\ 2 & 4 & 0 & 17 & 1 & 2 & 3 \\ 2 & 3 & 0 & 13 & 2 & 0 & 2 \\ 3 & 4 & -2 & 18 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{131}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 17 & 14 & 19 \\ 2 & 4 & 0 & 17 & 1 & 2 & 3 \\ 2 & 3 & 0 & 14 & 2 & 0 & 2 \\ 3 & 4 & -2 & 19 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{141}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 20 & 16 & 22 \\ 2 & 4 & 0 & 20 & 1 & 2 & 3 \\ 2 & 3 & 0 & 16 & 2 & 0 & 2 \\ 3 & 4 & -2 & 22 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 13$ [28] $1^{111}3^{11}5^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 2 & 3 \\ 2 & 2 & 1 & 7 & 8 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 19 & 6 & 8 \\ 6 & 8 & 0 & 19 & 22 & 7 & 10 \\ 2 & 3 & 0 & 6 & 7 & 1 & 3 \\ 3 & 4 & -1 & 8 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{111}3^{11}5^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 2 & 3 \\ 2 & 2 & 1 & 7 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 6 & 8 \\ 2 & 4 & 0 & 7 & 2 & 3 & 4 \\ 2 & 3 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -1 & 8 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{111}3^{11}5^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 2 & 3 \\ 2 & 2 & 1 & 7 & 16 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 33 & 6 & 8 \\ 10 & 16 & 0 & 33 & 70 & 13 & 18 \\ 2 & 3 & 0 & 6 & 13 & 1 & 3 \\ 3 & 4 & -1 & 8 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{131}3^{11}5^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 2 & 3 \\ 2 & 2 & 1 & 19 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 19 & 16 & 22 \\ 2 & 4 & 0 & 19 & 2 & 3 & 4 \\ 2 & 3 & 0 & 16 & 3 & 1 & 3 \\ 3 & 4 & -1 & 22 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{112}3^{11}5^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 2 & 3 \\ 2 & 2 & 1 & 7 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 26 & 6 & 8 \\ 8 & 12 & 0 & 26 & 42 & 10 & 14 \\ 2 & 3 & 0 & 6 & 10 & 1 & 3 \\ 3 & 4 & -1 & 8 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

4.718 NHC(-2)(-3) + so₈ + su₂n₃mix + su₂ + su₂n₃mix

$T = 9$ [0] $1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 23 & 8 \\ 2 & 4 & 0 & 8 & 1 & 8 & 3 \\ 7 & 12 & 0 & 23 & 8 & 25 & 9 \\ 3 & 4 & -2 & 8 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 9 & 5 \\ 4 & 8 & 0 & 10 & 10 & 10 & 6 \\ 4 & 7 & 0 & 9 & 10 & 7 & 5 \\ 3 & 4 & -2 & 5 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}3^{\overline{11}1^{\overline{11}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 4 & 3 \\ 3 & 4 & 0 & 7 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 9 & 5 \\ 5 & 9 & 0 & 11 & 13 & 11 & 7 \\ 4 & 7 & 0 & 9 & 11 & 7 & 5 \\ 3 & 4 & -2 & 5 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}3^{\overline{11}1^{\overline{11}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 15 & 5 \\ 2 & 4 & 0 & 5 & 1 & 8 & 3 \\ 7 & 12 & 0 & 15 & 8 & 25 & 9 \\ 3 & 4 & -2 & 5 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}3^{\overline{11}1^{\overline{11}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 7 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 9 & 5 \\ 2 & 4 & 0 & 5 & 1 & 5 & 3 \\ 4 & 7 & 0 & 9 & 5 & 7 & 5 \\ 3 & 4 & -2 & 5 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1112}3^{\overline{11}1^{\overline{12}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 26 & 5 \\ 2 & 4 & 0 & 5 & 1 & 14 & 3 \\ 12 & 21 & 0 & 26 & 14 & 80 & 16 \\ 3 & 4 & -2 & 5 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{112}3^{\overline{11}1^{\overline{11}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 4 & 3 \\ 3 & 4 & 0 & 7 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 9 & 5 \\ 7 & 13 & 0 & 16 & 28 & 16 & 10 \\ 4 & 7 & 0 & 9 & 16 & 7 & 5 \\ 3 & 4 & -2 & 5 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1112}3^{\overline{11}1^{\overline{11}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 20 & 5 \\ 2 & 4 & 0 & 5 & 1 & 11 & 3 \\ 9 & 16 & 0 & 20 & 11 & 44 & 12 \\ 3 & 4 & -2 & 5 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [8] \quad 1^{\overline{1112}3^{\overline{11}5^{\overline{13}1^{\overline{11}}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 10 & 3 \\ 2 & 2 & 1 & 7 & 4 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 32 & 8 \\ 2 & 4 & 0 & 7 & 2 & 15 & 4 \\ 10 & 15 & 0 & 32 & 15 & 65 & 17 \\ 3 & 4 & -1 & 8 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

4.719 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 36 & 8 & 10 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 19 & 0 & 64 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 39 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 55 & 10 & 13 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 381, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 440, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 80 & 12 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 440, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{13}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 591, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 14 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{14}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 724, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [14] \quad 1^{11}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 30, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{14}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 34, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 3 \\ 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{11}4^{13}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{11}4^{13}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{12}4^{13}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 13 & 0 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11}4^{13}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11}4^{13}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 1 & 3 \\ 2 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{11}4^{13}1^{11}1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 1 & 3 \\ 2 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 10 & 6 & 1 & 32 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{11}4^{13}1^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 1 & 3 \\ 2 & 0 & 1 & 5 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 8 & 5 & 1 & 18 & 0 & 8 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{11}4^{13}1^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 1 & 3 \\ 2 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 6 & 4 & 1 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{12}4^{13}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 & 0 \\ 4 & 4 & 1 & 2 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{13}4^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 & 0 \\ 5 & 6 & 1 & 3 & 17 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12}4^{13}1^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 & 0 \\ 5 & 4 & 2 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13}4^{13}1^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 337, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 & 0 \\ 6 & 6 & 2 & 3 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{14}4^{13}1^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 388, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 & 0 \\ 7 & 8 & 2 & 4 & 34 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [30] \quad 1^{12}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 12 & 0 & 15 & 65 & 17 & 17 \\ 3 & 0 & 3 & 17 & 4 & 4 \\ 3 & 0 & 3 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 2 & 1 & 7 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 8 \\ 2 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & -1 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 2 & 1 & 19 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 16 & 22 \\ 2 & 3 & 0 & 16 & 1 & 3 \\ 3 & 4 & -1 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 162, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 13 & 0 & 15 & 76 & 13 & 18 \\ 2 & 0 & 3 & 13 & 1 & 3 \\ 3 & 0 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

4.720 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 8 & 3 \\ 3 & 4 & 0 & 7 & 3 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 17 & 5 \\ 2 & 3 & 0 & 4 & 0 & 8 & 2 \\ 8 & 14 & 0 & 17 & 8 & 37 & 11 \\ 3 & 4 & -2 & 5 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{122} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 4 & 3 \\ 3 & 4 & 0 & 16 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 9 & 22 & 12 \\ 2 & 3 & 0 & 9 & 0 & 4 & 2 \\ 4 & 8 & 0 & 22 & 4 & 10 & 6 \\ 3 & 4 & -2 & 12 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 7 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 4 & 8 & 0 & 10 & 4 & 10 & 6 \\ 3 & 4 & -2 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 6 & 5 \\ 2 & 3 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -2 & 5 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 5 & 2 & 1 & 3 \\ 3 & 4 & -2 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 8 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 & 2 \\ 2 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & -2 & 8 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{132} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 4 & 3 \\ 3 & 4 & 0 & 20 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 11 & 28 & 15 \\ 2 & 3 & 0 & 11 & 0 & 4 & 2 \\ 4 & 8 & 0 & 28 & 4 & 10 & 6 \\ 3 & 4 & -2 & 15 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 5 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & -2 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{142}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 4 & 3 \\ 3 & 4 & 0 & 29 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 16 & 40 & 22 \\ 2 & 3 & 0 & 16 & 0 & 4 & 2 \\ 4 & 8 & 0 & 40 & 4 & 10 & 6 \\ 3 & 4 & -2 & 22 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 7 & 8 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & -2 & 9 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 26 & 5 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 11 & 21 & 0 & 26 & 11 & 76 & 16 \\ 3 & 4 & -2 & 5 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 10 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 10 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & -2 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 7 & 3 \\ 3 & 4 & 0 & 11 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 25 & 8 \\ 2 & 3 & 0 & 6 & 0 & 7 & 2 \\ 7 & 13 & 0 & 25 & 7 & 28 & 10 \\ 3 & 4 & -2 & 8 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 7 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & -2 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 9 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & -2 & 5 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 4 & 3 \\ 3 & 4 & 0 & 24 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 13 & 34 & 18 \\ 2 & 3 & 0 & 13 & 0 & 4 & 2 \\ 4 & 8 & 0 & 34 & 4 & 10 & 6 \\ 3 & 4 & -2 & 18 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 9 & 3 \\ 3 & 4 & 0 & 11 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 0 & 9 & 2 \\ 9 & 17 & 0 & 33 & 9 & 49 & 13 \\ 3 & 4 & -2 & 8 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 11 & 3 \\ 3 & 4 & 0 & 11 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 41 & 8 \\ 2 & 3 & 0 & 6 & 0 & 11 & 2 \\ 11 & 21 & 0 & 41 & 11 & 76 & 16 \\ 3 & 4 & -2 & 8 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & -2 & 12 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 12 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & -2 & 5 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 14 & 3 \\ 3 & 4 & 0 & 11 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 50 & 8 \\ 2 & 3 & 0 & 6 & 0 & 14 & 2 \\ 14 & 26 & 0 & 50 & 14 & 118 & 20 \\ 3 & 4 & -2 & 8 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 21 & 0 & 80 & 12 & 14 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 2 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & -2 & 16 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 11 & 14 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & -2 & 15 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 32 & 5 \\ 2 & 3 & 0 & 4 & 0 & 14 & 2 \\ 14 & 26 & 0 & 32 & 14 & 118 & 20 \\ 3 & 4 & -2 & 5 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{14} 1^3 1^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 13 & 17 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 & 2 \\ 2 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & -2 & 18 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{13} 1^3 1^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 14 & 17 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 2 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & -2 & 19 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{14} 1^3 1^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 16 & 20 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 & 2 \\ 2 & 4 & 0 & 20 & 2 & 1 & 3 \\ 3 & 4 & -2 & 22 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [27] \ 1^{11} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 10 & 3 \\ 2 & 2 & 1 & 7 & 3 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 1 & 13 & 3 \\ 10 & 16 & 0 & 33 & 13 & 70 & 18 \\ 3 & 4 & -1 & 8 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 1^3 1^1 1^5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 2 & 3 \\ 2 & 2 & 1 & 7 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 7 & 8 \\ 2 & 3 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 7 & 3 & 2 & 4 \\ 3 & 4 & -1 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 1^3 1^1 1^5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 6 & 3 \\ 2 & 2 & 1 & 7 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 19 & 8 \\ 2 & 3 & 0 & 6 & 1 & 7 & 3 \\ 6 & 8 & 0 & 19 & 7 & 22 & 10 \\ 3 & 4 & -1 & 8 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{13} 1^3 1^1 1^5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 2 & 3 \\ 2 & 2 & 1 & 19 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 16 & 19 & 22 \\ 2 & 3 & 0 & 16 & 1 & 3 & 3 \\ 2 & 4 & 0 & 19 & 3 & 2 & 4 \\ 3 & 4 & -1 & 22 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11} 2^3 1^1 1^5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) =$$

$$(1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 8 & 3 \\ 2 & 2 & 1 & 7 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 26 & 8 \\ 2 & 3 & 0 & 6 & 1 & 10 & 3 \\ 8 & 12 & 0 & 26 & 10 & 42 & 14 \\ 3 & 4 & -1 & 8 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

4.721 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + su2n3mix

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{132} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 15 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 39 & 42 & 11 \\ 9 & 16 & 0 & 42 & 44 & 12 \\ 3 & 4 & -2 & 11 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 23 & 8 \\ 7 & 12 & 0 & 23 & 25 & 9 \\ 3 & 4 & -2 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 21 & 5 \\ 10 & 17 & 0 & 21 & 55 & 13 \\ 3 & 4 & -2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 5 \\ 4 & 7 & 0 & 9 & 7 & 5 \\ 3 & 4 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 14 & 5 \\ 6 & 11 & 0 & 14 & 20 & 8 \\ 3 & 4 & -2 & 5 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 15 & 5 \\ 7 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 19 & 5 \\ 8 & 15 & 0 & 19 & 39 & 11 \\ 3 & 4 & -2 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{114} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 24 & 5 \\ 10 & 19 & 0 & 24 & 64 & 14 \\ 3 & 4 & -2 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 11 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 48 & 8 \\ 14 & 25 & 0 & 48 & 111 & 19 \\ 3 & 4 & -2 & 8 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 11 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 22 & 8 \\ 6 & 11 & 0 & 22 & 20 & 8 \\ 3 & 4 & -2 & 8 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 31 & 8 \\ 9 & 16 & 0 & 31 & 44 & 12 \\ 3 & 4 & -2 & 8 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 11 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 30 & 8 \\ 8 & 15 & 0 & 30 & 39 & 11 \\ 3 & 4 & -2 & 8 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 11 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 38 & 8 \\ 10 & 19 & 0 & 38 & 64 & 14 \\ 3 & 4 & -2 & 8 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 20 & 5 \\ 9 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 5 \\ 12 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 11 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 39 & 8 \\ 11 & 20 & 0 & 39 & 69 & 15 \\ 3 & 4 & -2 & 8 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 485, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 25 & 5 \\ 11 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & -2 & 5 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{124}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 16 & 3 \\ 3 & 4 & 0 & 11 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 56 & 8 \\ 16 & 29 & 0 & 56 & 148 & 22 \\ 3 & 4 & -2 & 8 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 4^1 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 30 & 5 \\ 13 & 24 & 0 & 30 & 100 & 18 \\ 3 & 4 & -2 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 7 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 31 & 5 \\ 14 & 25 & 0 & 31 & 111 & 19 \\ 3 & 4 & -2 & 5 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{12} 4^1 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 11 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 47 & 8 \\ 13 & 24 & 0 & 47 & 100 & 18 \\ 3 & 4 & -2 & 8 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 4^1 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 807, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 7 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 36 & 5 \\ 16 & 29 & 0 & 36 & 148 & 22 \\ 3 & 4 & -2 & 5 & 22 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [23] \ 1^{11} 4^1 13^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 4^1 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 4 & 3 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 2 & 2 \\ 4 & 3 & 1 & 2 & 2 & 4 \\ 3 & 2 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{12} 1^{11} 4^1 13^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 1^{11} 4^1 13^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 1^{11} 4^1 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 4 & 2 \\ 6 & 4 & 1 & 4 & 8 & 6 \\ 3 & 2 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 4 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 0 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 4 13^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{111} 4 13^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11} 4 13^{11} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 8 & 3 \\ 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 6 & 2 \\ 8 & 5 & 1 & 6 & 18 & 8 \\ 3 & 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{112} 4 13^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{112} 4 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 193, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 2 & 2 & 4 & 10 & 6 \\ 2 & 2 & 0 & 0 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112} 4 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 1 & 2 & 4 & 7 & 5 \\ 2 & 2 & 0 & 0 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{113} 4 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 1 & 3 & 6 & 17 & 7 \\ 2 & 2 & 0 & 0 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{113} 4^{13} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 2 & 3 & 6 & 20 & 8 \\ 2 & 2 & 0 & 0 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{114} 4^{13} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 601, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 2 & 4 & 8 & 34 & 10 \\ 2 & 2 & 0 & 0 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [38] $1^{111} 3^{11} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 11 & 3 \\ 5 & 0 & 7 & 14 & 25 & 8 \\ 10 & 0 & 11 & 25 & 39 & 13 \\ 3 & 0 & 3 & 8 & 13 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{112} 3^{11} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 2 & 1 & 7 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 32 & 8 \\ 10 & 15 & 0 & 32 & 65 & 17 \\ 3 & 4 & -1 & 8 & 17 & 4 \end{pmatrix}_{b_0}$$

4.722 NHC(-2)(-3) + $\textcolor{green}{so}_8$ + $\textcolor{red}{su}_{2n3\text{mix}}$ + $\textcolor{red}{su}_{2n3\text{mix}}$ + $\textcolor{red}{su}_2$

$T = 9$ [0] $1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 2 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 23 & 8 & 8 \\ 7 & 12 & 0 & 23 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 8 & 1 & 3 \\ 3 & 4 & -2 & 8 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 10 & 5 \\ 4 & 7 & 0 & 9 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 10 & 10 & 6 \\ 3 & 4 & -2 & 5 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 11 & 5 \\ 4 & 7 & 0 & 9 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 11 & 13 & 7 \\ 3 & 4 & -2 & 5 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -303, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 2 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 15 & 5 & 5 \\ 7 & 12 & 0 & 15 & 25 & 8 & 9 \\ 2 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & -2 & 5 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}3^{\overline{11}}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 5 & 5 \\ 4 & 7 & 0 & 9 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1121}3^{\overline{11}}1^{\overline{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 2 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 5 & 5 \\ 12 & 21 & 0 & 26 & 80 & 14 & 16 \\ 2 & 4 & 0 & 5 & 14 & 1 & 3 \\ 3 & 4 & -2 & 5 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1112}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 16 & 5 \\ 4 & 7 & 0 & 9 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 16 & 28 & 10 \\ 3 & 4 & -2 & 5 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1121}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 20 & 5 & 5 \\ 9 & 16 & 0 & 20 & 44 & 11 & 12 \\ 2 & 4 & 0 & 5 & 11 & 1 & 3 \\ 3 & 4 & -2 & 5 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [8] \quad 1^{\overline{1121}3^{\overline{11}}1^{\overline{15}}13^{\overline{1}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 2 & 3 \\ 2 & 2 & 1 & 7 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 32 & 7 & 8 \\ 10 & 15 & 0 & 32 & 65 & 15 & 17 \\ 2 & 4 & 0 & 7 & 15 & 2 & 4 \\ 3 & 4 & -1 & 8 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

4.723 NHC(-2)(-3) + so₈ + su_{2n3mix} + su₃

$$T = 8 \quad [0] \quad 1^{\overline{11}3^{\overline{1}}1^{\overline{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 24 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 24 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}3^{\overline{1}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 24 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 24 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

- [2] $1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [3] $1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 7 & 1 & 5 \\ 11 & 20 & 11 & 0 & 8 \\ 7 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$
- [4] $1^{14}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 19 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 64 & 15 & 14 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{141}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 42$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 19 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 19 & 0 & 64 & 29 & 14 \\ 5 & 9 & 0 & 29 & 12 & 6 \\ 3 & 4 & -2 & 14 & 6 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 0 & 12 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 49$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 7 & 9 & 5 \\ 11 & 20 & 11 & 17 & 8 \\ 7 & 11 & 4 & 9 & 4 \\ 9 & 17 & 9 & 12 & 6 \\ 5 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{112}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 27 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{11}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 25 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 25 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 7 & 10 & 5 \\ 11 & 20 & 11 & 17 & 8 \\ 7 & 11 & 4 & 9 & 4 \\ 10 & 17 & 9 & 12 & 6 \\ 5 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{112}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 9 & 5 \\ 3 & 4 & -2 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 9 & 12 & 6 \\ 15 & 20 & 11 & 17 & 8 \\ 9 & 11 & 4 & 9 & 4 \\ 12 & 17 & 9 & 12 & 6 \\ 6 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 16 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 16 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 9 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 16 & 9 \\ 4 & 8 & 0 & 16 & 9 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 15 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 16 & 1 & 5 \\ 12 & 25 & 32 & 0 & 9 \\ 16 & 32 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 23 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 23 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 27 & 0 & 151 & 17 & 21 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 15 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 39 & 12 & 11 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 12 & 15 & 1 & 5 & 5 \\ 12 & 25 & 32 & 0 & 9 & 9 \\ 15 & 32 & 37 & 0 & 11 & 11 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 11 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 12 & 6 \\ 3 & 4 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 9 & 5 \\ 3 & 4 & -2 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 9 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 39 & 23 & 11 \\ 5 & 9 & 0 & 23 & 12 & 6 \\ 3 & 4 & -2 & 11 & 6 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 12 & 6 \\ 3 & 4 & -2 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 12 & 6 \\ 3 & 4 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 12 & 6 \\ 3 & 4 & -2 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 97 & 14 & 17 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 19 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 64 & 29 & 14 \\ 5 & 9 & 0 & 29 & 12 & 6 \\ 3 & 4 & -2 & 14 & 6 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 27 & 0 & 151 & 17 & 21 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 39 & 23 & 11 \\ 5 & 9 & 0 & 23 & 12 & 6 \\ 3 & 4 & -2 & 11 & 6 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 27 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 22 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 3 & 4 & -2 & 30 & 2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{141} 3^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 1952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 3 & 4 & -2 & 26 & 2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{141} 3^{11} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 2208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 23 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 3 & 4 & -2 & 30 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [75] $1^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{12} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 14 & 0 & 4 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 8 & 14 & 11 & 39 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{13} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -133, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 3 & 18 & 0 & 4 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 10 & 18 & 13 & 62 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

4.724 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.725 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + (-3) \rightarrow (-2)$

$T = 8$ [0] $1^{11} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 24 & 1 & 7 \\ 17 & 20 & 28 & 0 & 8 \\ 24 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

4.726 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3$

$T = 7$ [0] $1^{12} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$

- $T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 71$, $T \in [15, 71]$
 $\begin{pmatrix} 5 & 11 & 18 & 5 \\ 11 & 20 & 34 & 8 \\ 18 & 34 & 57 & 13 \\ 5 & 8 & 13 & -1 \end{pmatrix}_{b_0}$
- [2] $21^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14$, $T_{\max} = 71$, $T \in [14, 71]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 12 & 6 \\ 6 & 9 & 6 & 0 \end{pmatrix}_{b_0}$
- [1] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14$, $T_{\max} = 71$, $T \in [14, 71]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 12 & 6 \\ 6 & 9 & 6 & 0 \end{pmatrix}_{b_0}$
- [4] $1^{\overline{11}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 10 & 15 & 25 & 6 \\ 15 & 20 & 34 & 8 \\ 25 & 34 & 57 & 13 \\ 6 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [3] $1^{\overline{11}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 11 & 19 & 5 \\ 11 & 20 & 34 & 8 \\ 19 & 34 & 57 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [5] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 8 & 5 \\ 12 & 25 & 16 & 9 \\ 8 & 16 & 9 & 5 \\ 5 & 9 & 5 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 10 & 16 & 23 & 6 \\ 16 & 25 & 36 & 9 \\ 23 & 36 & 48 & 12 \\ 6 & 9 & 12 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{\overline{11}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -27$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 11 & 5 \\ 12 & 25 & 20 & 9 \\ 11 & 20 & 12 & 6 \\ 5 & 9 & 6 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{\overline{11}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 71$, $T \in [14, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 10 & 0 & 21 & 7 \\ 3 & 4 & -2 & 7 & -1 \end{pmatrix}_{b_0}$
- [9] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 5 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 17 & 8 \\ 10 & 17 & 12 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{15} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 71, T \in [15, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 72 & 14 \\ 3 & 4 & -2 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 71, T \in [13, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 24 & 8 \\ 3 & 4 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{12} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 9 & 5 \\ 3 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{14} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 71, T \in [14, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 45 & 11 \\ 3 & 4 & -2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 71, T \in [11, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^3 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 17 & 8 \\ 9 & 17 & 12 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 17 & 8 \\ 12 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 5 & 0 & 3 & 3 \\ 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 1 & 6 \\ 15 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 17 & 8 \\ 13 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 36 & 9 \\ 18 & 36 & 48 & 12 \\ 5 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 71, T \in [13, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 21 & 7 \\ 3 & 4 & -2 & 7 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 13 & 17 & 15 & 7 \\ 17 & 20 & 17 & 8 \\ 15 & 17 & 12 & 6 \\ 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[28] $1^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 71, T \in [13, 71]$

$$\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 12 & 6 \\ 5 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$

[29] $1^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 36 & 9 \\ 17 & 36 & 48 & 12 \\ 5 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 5 & 12 & 10 & 5 \\ 12 & 25 & 20 & 9 \\ 10 & 20 & 12 & 6 \\ 5 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[31] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 13 & 20 & 12 & 7 \\ 20 & 25 & 16 & 9 \\ 12 & 16 & 9 & 5 \\ 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[33] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 20 & 17 & 8 \\ 12 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 9 & 5 \\ 3 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 36 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 27 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 24 & 8 \\ 3 & 4 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 45 & 11 \\ 3 & 4 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 48 & 12 \\ 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 36 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{12}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 57 & 13 \\ 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 27 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 21 & 0 & 75 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[45] $21^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.03 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 13 & 20 & 27 & 7 \\ 20 & 25 & 36 & 9 \\ 27 & 36 & 48 & 12 \\ 7 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 48 & 12 \\ 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{13}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 22 & 0 & 84 & 16 \\ 3 & 4 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[48] $21^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 12 & 6 \\ 7 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{12}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 18 & 0 & 57 & 13 \\ 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 75 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 26 & 0 & 117 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{13}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 71, T \in [11, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 22 & 0 & 84 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -508, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 71, T \in [11, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 26 & 0 & 117 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [54] $1^{11}4^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 18, T_{\max} = 23, T \in [18, 23]$

$$\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 10 & 7 \\ 9 & 10 & 4 & 4 \\ 5 & 7 & 4 & -3 \end{pmatrix}_{b_0}$$

[55] $21^{11}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 4 & 4 \\ 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{11}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[57] $13^11^13^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 & 3 \\ 1 & 4 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 2 & 3 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[58] $1^{11} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 0 & 7 & 5 \\ 7 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{11} 3131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{11} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{11} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{11} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 4 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{11} 31^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{11} 1131^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 & 3 \\ 1 & 4 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 2 & 3 & 2 \\ 2 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{11} 11321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 3 & 7 & 3 & 4 \\ 3 & 1 & 5 & 3 & 3 \\ 7 & 5 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{11} 11321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 4 & 3 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{11} 31^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 6 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 11 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & -1 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

[69] $1^1 1^1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[70] $1^1 1^1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[71] $1^1 1^1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[72] $1^1 1^1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{13} 3 1^1 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 4 & 9 & 0 & 9 & 24 & 8 \\ 2 & 3 & -1 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[74] $1^1 1^2 3 2 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{11} 1^1 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[76] $1^1 1^2 3 2 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 5 & 4 & 1 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{11} 1^1 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 5 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[80] $2^1 2^1 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 4 & 3 & 8 & 3 & 4 \\ 3 & 1 & 5 & 3 & 3 \\ 8 & 5 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{11} 1^1 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & 1 & 3 \\ 3 & 5 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[83] \ 21^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 9 & 10 & 7 & 13 & 7 \\ 4 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 12 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 1 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{11} 2^2 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{11} 31^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{11} 1^{11} 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{11} 1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 6 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{11} 1^{11} 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{11} 1^{11} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 4 & 4 \\ 5 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[97] \quad 21^{11} 1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 6 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{11} 1^{11} 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{11} 1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{11} 1^{11} 31^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 6 & 0 & 12 & 6 \\ 2 & 3 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{11} 1^{11} 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$\begin{aligned}
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 3 & -1 & 1 \end{pmatrix}_{b_0} \\
[102] & 1^{11}31^{13}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[103] & 1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 4 & 3 \\ 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[104] & 1^{11}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 13.17 \rightarrow 14, T_{\max} = 20, T \in [14, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 2 & -2 & 2 & -1 \end{pmatrix}_{b_0} \\
[107] & 1^{11}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0} \\
[105] & 1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 5 & 9 & 8 & 5 \\ 9 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0} \\
[106] & 1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 8 & 8 & 7 & 4 \\ 8 & 13 & 16 & 12 & 7 \\ 8 & 16 & 12 & 12 & 6 \\ 7 & 12 & 12 & 9 & 5 \\ 4 & 7 & 6 & 5 & 2 \end{pmatrix}_{b_0} \\
[109] & 1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 1 \end{pmatrix}_{b_0} \\
[110] & 1^{12}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 6 & 0 & 9 & 5 \\ 2 & 3 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0} \\
[108] & 1^{11}31^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[111] & 1^{11}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 7 & 8 & 1 & 4 \\ 7 & 10 & 13 & 0 & 6 \\ 8 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 6 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[112] & 1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 23, T \in [12, 23]
\end{aligned}$$

$$\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 12 \overset{11}{3} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[114] \ 31^{\overset{11}{1}} 231^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -102, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 & 3 \\ 5 & 5 & 5 & 5 & 5 \\ 4 & 5 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 2 \ 2 \ 31^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 51, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overset{11}{1}} 321^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 51, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 2 & 5 & 4 & 1 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overset{11}{1}} 321^{\overset{14}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 25 & 0 & 91 & 19 \\ 3 & 5 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overset{11}{1}} 3 \ 21^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overset{12}{1}} 31^{\overset{11}{1}} 31^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.44 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 6 & 0 & 6 & 9 & 5 \\ 2 & 3 & -1 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{\overset{11}{1}} 321^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.79 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 0 & 4 & 2 \\ 3 & 5 & 4 & 1 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{\overset{11}{1}} 31^{\overset{11}{1}} 31^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{\overset{11}{1}} 31^{\overset{11}{1}} 31^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{\overset{11}{1}} 321^{\overset{12}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 7 & 0 & 15 & 7 \\ 2 & 2 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 10 & 0 & 13 & 7 \\ 3 & 5 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{12} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 37 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 5 & 9 & 7 & 5 \\ 9 & 13 & 10 & 7 \\ 7 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[126] $2 1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 4 & 4 \\ 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[128] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 0 & 7 & 2 \\ 6 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{11} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 57 & 15 \\ 3 & 5 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 0 & 7 & 2 \\ 5 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{11} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 15 & 0 & 31 & 11 \\ 3 & 5 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{12} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0} \\
[135] & 131^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 20, T \in [13, 20] \\
& \begin{pmatrix} 1 & 5 & 5 & 4 & 3 \\ 5 & 13 & 16 & 12 & 7 \\ 5 & 16 & 12 & 12 & 6 \\ 4 & 12 & 12 & 9 & 5 \\ 3 & 7 & 6 & 5 & 1 \end{pmatrix}_{b_0} \\
[134] & 1^{11}321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 3 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & 0 & 5 & 0 \end{pmatrix}_{b_0} \\
[137] & 1^{11}321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 5 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[138] & 1^{14}31^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.6 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 6 & 12 & 0 & 12 & 45 & 11 \\ 2 & 3 & -1 & 2 & 11 & 1 \end{pmatrix}_{b_0} \\
[140] & 1^{13}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0} \\
[139] & 1^{13}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 13 & 0 & 60 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0} \\
[141] & 1^{12}31^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 4 & 6 & 0 & 6 & 9 & 5 \\ 2 & 3 & -1 & 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[142] & 1^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[143] & 1^{13}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 24 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0} \\
[144] & 1^{13}31^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 5 & 9 & 0 & 9 & 24 & 8 \\ 2 & 3 & -1 & 2 & 8 & 1 \end{pmatrix}_{b_0} \\
[145] & 1^{13}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{12}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 0 & 24 & 8 \\ 2 & 2 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[148] $21^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[149] $21^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 4 & 4 \\ 7 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

[150] $1^131^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 10 & 0 & 37 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{12}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 25 & 9 \\ 2 & 2 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{12}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 11 & 0 & 44 & 12 \\ 2 & 2 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[155] $2^12^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 3 & 5 & 4 & 1 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 13 & 0 & 60 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 188, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{13}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 200, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.14 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 9 & 0 & 39 & 11 \\ 2 & 2 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[159] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 211, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.08 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 6 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 212, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.34 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[161] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 249, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{13}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 252, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.10 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 12 & 0 & 60 & 14 \\ 2 & 2 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[163] \ 2 \ 2 \ 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 384, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.09 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [164] \ 21^{14}1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -9, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.56 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 7 & 2 & 5 \\ 7 & 8 & 1 & 6 \\ 2 & 1 & -2 & 0 \\ 5 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{13}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{14}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -20, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{\overline{1}} 4^{11} 131^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & -2 & 0 \\ 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[168] $1^{\overline{11}} 4^{11} 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0}$

[169] $1^{\overline{1}} 4^{11} 1231^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 5 & 2 & 5 \\ 1 & 2 & -2 & 0 \\ 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$

[170] $1^{\overline{1}} 4^{11} 131^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[171] $1^{\overline{11}} 4^{11} 131^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 3 & 4 & 1 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 \end{pmatrix}_{b_0}$

[172] $1^{\overline{11}} 4^{11} 31^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 7 & 1 & 8 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$

[173] $1^{\overline{11}} 4^{11} 31^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 4 & 2 & 5 \\ 4 & 1 & 1 & 3 \\ 2 & 1 & -2 & 0 \\ 5 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[174] $1^{\overline{1}} 4^{11} 31^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 14 & 10 & 8 \\ 10 & 5 & 5 \\ 8 & 5 & 4 \end{pmatrix}_{b_0}$

[175] $1^{\overline{1}} 4^{11} 1231^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 2 & 6 & 2 & 4 \\ 6 & 8 & 1 & 6 \\ 2 & 1 & -2 & 0 \\ 4 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[176] $1^{\overline{11}} 4^{11} 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$

[177] $1^{\overline{11}} 4^{11} 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$

[178] $1^{\overline{11}} 4^{11} 31^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 4 & -1 & 0 & 2 \end{pmatrix}_{b_0}$

[179] $1^{\overline{1}} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 2 & 0 & 3 & 2 \\ 1 & 0 & -2 & 1 & 0 \\ 3 & 3 & 1 & 1 & 3 \\ 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[180] $1^{\overline{1}} 4^{11} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 13 & 9 & 1 & 7 \\ 9 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 7 & 5 & 0 & 3 \end{pmatrix}_{b_0}$

[181] $1^{\overline{1}} 4^{11} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$

[182] $1^{\overline{1}} 4^{11} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 2 & 0 & 3 & 2 \\ 1 & 0 & -2 & 1 & 0 \\ 4 & 3 & 1 & 1 & 3 \\ 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[183] $1^{\overline{1}} 4^{11} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$

[184] $1^{\overline{1}} 4^{11} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -39$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 4 & 5 & 5 \\ 4 & 1 & 3 & 3 \\ 5 & 3 & 1 & 3 \\ 5 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

[185] $1^{\overline{1}} 4^{11} 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 0 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$

[186] $1^{\overline{1}} 4^{11} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 7 & 1 & 5 \\ 7 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$

[187] $1^{\overline{1}} 4^{11} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$

[188] $1^{\overline{1}} 4^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$

[191] $1^{\overline{1}} 4^{11} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -57$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 2 & 1 & 1 & 3 \\ 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[190] $1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 4 & 1 & 17 & 7 \\ 3 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{11} 4^{11} 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{11} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 3 & 1 & 7 & 5 \\ 3 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[198] $31^{11} 31^{11} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 3 & 3 & 3 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 1 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 3 & 2 & 2 & 7 & 5 \\ 2 & 2 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

- [201] $2^{\overline{1}41^{\overline{1}}}1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -186$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 & 2 \\ 3 & 4 & 4 & 2 & 3 & 4 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 4 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$
- [202] $1^{\overline{1}41^{\overline{1}}}231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -205$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 3 & 6 & 0 & 2 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$
- [203] $1^{\overline{1}41^{\overline{1}}}231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 6 & 1 & 5 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 7 & 5 \\ 3 & 7 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$
- [204] $1^{\overline{1}41^{\overline{1}}}231^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -250$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 4 & 6 & 1 & 2 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- $T = 10$ [205] $21^{\overline{1}1}5131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [206] $1^{\overline{1}1}3221^{\overline{1}5}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 10 & 21 & 0 & 6 \\ 9 & 21 & 42 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$
- [207] $1^{\overline{1}1}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 10$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$
- [209] $1^{\overline{1}1}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 18$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 2 & 5 & 3 & 1 & 3 \\ 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$
- [208] $1^{\overline{1}1}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 18$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 9 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 9 & 0 & 11 & 7 \\ 3 & 5 & -2 & 7 & 3 \end{pmatrix}_{b_0}$$
- [211] $1^{\overline{1}1}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 20$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$
- [210] $1^{\overline{1}1}51231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 20$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{11}4141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{11}4141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 4 & 5 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{11}4141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & 1 & 3 \\ 3 & 5 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{11}5131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[216] $21^{11}4141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{11}4141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{11}41^{11}31^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 13 & 6 & 0 & 12 & 7 \\ 1 & 6 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 12 & 4 & 0 & 9 & 5 \\ 2 & 7 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{11}4141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 4 & 5 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 5 & 3 & 1 & 3 \\ 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[220] $31^{11}4141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 5 & 8 & 0 & 9 & 5 \\ 2 & 8 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 9 & 0 & 12 & 6 \\ 2 & 5 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[221] $2^13^1151^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{11}41^{11}31^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 25 & 9 \\ 3 & 12 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} 231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 104$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 7 & 2 & 1 & 3 \\ 7 & 26 & 8 & 0 & 10 \\ 2 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 10 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [224] $1^{\overline{11}} 41^{\overline{1}} 4141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -18$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 4 \\ 3 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 \\ 4 & 3 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{\overline{11}} 41^{\overline{1}} 4141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -33$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.79 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{\overline{1}} 41^{\overline{1}} 3151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -86$, $\det = -84$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{\overline{11}} 41^{\overline{1}} 4141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -97$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{\overline{11}} 41^{\overline{1}} 4141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -97$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.85 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 1 & 4 & 1 & 0 & 3 \\ 2 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [229] $1^{\overline{1}} 6^{\overline{1}} 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 12$, $T_{\max} = 14$, $\Delta = -63$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

4.727 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{e}_6$

$T = 8$ [1] $1^{\overline{11}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 95$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.96 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 302, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 330, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 5 & 0 & 3 & -2 & 0 \\ 3 & 5 & -2 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3 2 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.88 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 100, \ \Delta = -150, \ \det = -354, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.07 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

4.728 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -931, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

4.729 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_7$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_7 \quad T = 9, \quad T_{\max} = 103, \quad \Delta = -198, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, \quad T_{\max} = 103, \quad T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 1 & 0 & -5 & -3 \\ 2 & 2 & -1 & 0 & -1 & -3 & -3 \end{pmatrix}_{b_0}$$

4.730 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}'_7$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}'_7 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -170, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 1 & 0 & -4 & -2 \\ 2 & 2 & -1 & 0 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

4.731 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 19, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, \quad T_{\max} = 98, \quad T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 74, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 196, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 235, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 259, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.84 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 32, \quad T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11}3^{11}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 3 & 0 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -5 \\ 2 & 2 & 2 & 0 & 2 & -3 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}3^{11}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 10 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}3^{11}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}3^{11}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}3^{11}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 5 & 5 & 10 & 0 & 5 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 5 & 10 & 7 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}3^{11}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 82, \Delta = -124, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 82, T \in [14, 82]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

4.732 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.733 NHC(-2)(-3) + so₈ + su₃ + f₄ + su₂ → (-3)

$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.734 NHC(-2)(-3) + so₈ + su₃ + so₈

$T = 8 \quad [0] \quad 1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 67, T \in [15, 67]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12, T_{\max} = 67, T \in [12, 67]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [9] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 10 & 13 & 0 & 0 & 6 \\ 5 & 13 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 6 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 0 & 5 \\ 2 & 5 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 3 & 0 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 10 & 1 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 0 & 11 & 5 \\ 3 & 6 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 1 & 4 & 3 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -166$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 10 & 0 & 13 & 7 & 7 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 7 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [21] $1^{11}31^{11}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 66$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 14 & 10 & 2 & 8 \\ 10 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 8 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{11}31^{11}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 84$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 1 & 0 & 1 & -2 & 0 \\ 5 & 7 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11}31^{11}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 129$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [24] $1^{11}4^11^14^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -69$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

4.735 NHC(-2)(-3) + so₈ + su₃ + so₈ + su₂

$T = 9$ [0] $1^{11}31^{11}4^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 28 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 28 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 10 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

14.5

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 19 & 5 \\ 12 & 25 & 16 & 0 & 40 & 9 \\ 8 & 16 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 40 & 24 & 0 & 58 & 14 \\ 5 & 9 & 5 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 20 & 5 \\ 12 & 25 & 16 & 0 & 40 & 9 \\ 8 & 16 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 40 & 24 & 0 & 58 & 14 \\ 5 & 9 & 5 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 9 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & -1 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 30 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 30 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 29 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 29 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 8 & 1 & 38 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 38 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$

[19] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 8 & 1 & 40 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 40 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$

[20] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 8 & 1 & 39 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 39 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$

[21] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$

[22] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$

[23] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$

$T = 10$ [24] $1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 9 & 6 & 1 & 9 & 5 \\ 9 & 13 & 10 & 0 & 14 & 7 \\ 6 & 10 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 13 & 7 \\ 5 & 7 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$

[25] $1^{111} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 9 & 6 & 1 & 6 & 5 \\ 9 & 13 & 10 & 0 & 10 & 7 \\ 6 & 10 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 7 & 0 & 5 & 5 \\ 5 & 7 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$

[26] $1^{111} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 15 & 5 \\ 9 & 13 & 10 & 0 & 24 & 7 \\ 6 & 10 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 16 & 0 & 38 & 12 \\ 5 & 7 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

4.736 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.737 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 7 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 7 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.738 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 6 & 5 \\ 12 & 25 & 12 & 0 & 12 & 9 \\ 6 & 12 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 7 & 0 & 4 & 4 \\ 5 & 9 & 4 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 168, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & 5 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

4.739 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{11} 3 \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 14 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 14 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 24 & 6 \\ 9 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -2 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{123} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -2 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 25, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 20 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}} 3 \ 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{11}} 3 \ 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 17 & 24 & 0 & 15 & 121 & 19 \\ 3 & 4 & -2 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{11}} 3 \ 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 13 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 13 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -2 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 59 & 124 & 0 & 613 & 45 \\ 5 & 9 & -1 & 45 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 19 & 0 & 0 & 73 & 15 \\ 3 & 4 & -2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 10 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.52 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.42 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 18 & 5 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.42 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 60 & 124 & 0 & 613 & 45 \\ 5 & 9 & -1 & 45 & 2 \end{pmatrix}_{b_0}$$

- [29] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 49$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 37 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 37 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$
- [31] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [33] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 36 & 0 & 49 & 13 \\ 5 & 9 & -1 & 13 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [35] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [37] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$

- [40] $21^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 12 & 15 & 6 \\ 15 & 17 & 15 & 13 & 7 \\ 7 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [41] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [42] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 65$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$
- [43] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 18 & 9 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 9 & 6 & 1 \end{pmatrix}_{b_0}$
- [44] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [45] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$
- [46] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 9 & 6 \\ 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$
- [47] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 69$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [48] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 69$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 49 & 13 \\ 3 & 4 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$
- [49] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 70$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 8 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 8 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [52] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 71$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 19 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{113} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 71, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 28 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{13} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 71, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 77, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{112} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 78, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 20 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 7 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{121} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 6 & 5 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{111} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 11 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{113} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 8 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{\overline{11}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{\overline{11}} 3^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 0 & 37 & 11 \\ 3 & 4 & -2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{\overline{111}} 3^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 36 & 0 & 49 & 13 \\ 5 & 9 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{\overline{13}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -2 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{\overline{131}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 12 & 12 \\ 3 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{\overline{1}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{\overline{12}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -2 & -1 & 10 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 20 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 20 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{111} 3 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{112} 3 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 15 & 24 & 0 & 15 & 121 & 19 \\ 3 & 4 & -2 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -2 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111} 3 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{112} 3 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 7 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 14 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -2 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{121} 3 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 18 & 6 \\ 7 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 48 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 48 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 75 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -2 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 6 & 6 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 12 & 5 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 9 & 9 \\ 3 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 11 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{11} 3^{12} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 36 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 15 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 30 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

- [103] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [104] $1^{122} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 12 & 5 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 16 & 24 & 0 & 15 & 121 & 19 \\ 3 & 4 & -2 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$
- [102] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 29 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$
- [106] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 26 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 26 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [108] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [107] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$
- [109] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$
- [111] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 28 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 56 & 0 & 118 & 20 \\ 5 & 9 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [112] $1^{12} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 16 & 3 \\ 3 & 4 & 0 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 0 & 154 & 22 \\ 3 & 4 & -2 & -1 & 22 & 1 \end{pmatrix}_{b_0}$$
- [110] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 20 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 40 & 0 & 58 & 14 \\ 5 & 9 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 18 & 9 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 36 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[116] $21^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 34 & 72 & 0 & 202 & 26 \\ 5 & 9 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 0 & 58 & 14 \\ 3 & 4 & -2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 11 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 72 & 0 & 202 & 26 \\ 5 & 9 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 38 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 23 & 0 & 0 & 100 & 18 \\ 3 & 4 & -2 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 29 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 11 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 48 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -2 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 18 & 9 \\ 5 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 7 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{121} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{14} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 26 & 0 & 0 & 118 & 20 \\ 3 & 4 & -2 & -1 & 20 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 46 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 92 & 0 & 325 & 33 \\ 5 & 9 & -1 & 33 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 56 & 0 & 118 & 20 \\ 5 & 9 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 0 & 85 & 17 \\ 3 & 4 & -2 & -1 & 17 & 1 \end{pmatrix}_{b_0}$$

[143] $21^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 13 & 7 \\ 7 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

- [144] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 7 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -2 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$
- [145] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 40 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [146] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [147] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 40 & 0 & 58 & 14 \\ 5 & 9 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [148] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [149] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$
- [151] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [150] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [152] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [153] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 18 & 5 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$
- [154] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 39 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[155] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 53 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 53 & 108 & 0 & 457 & 39 \\ 5 & 9 & -1 & 39 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 45 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[159] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 12 & 21 & 6 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[160] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 21 & 9 \\ 6 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 8 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 7 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

- [165] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$
- [172] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 8 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$
- [166] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [167] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [168] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 6 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [169] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 13 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -2 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [170] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 12 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -2 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [171] $1^{113} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 10 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [173] $1^{114} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 71 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 71 & 144 & 0 & 814 & 52 \\ 5 & 9 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [174] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [175] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[176] 1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 27 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 56 & 0 & 118 & 20 \\ 5 & 9 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[179] 1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 10 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[178] 1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[177] 1^{131} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 24 & 9 & 8 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[181] 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[180] 1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 38 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[182] 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[183] 1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 30 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 30 & 60 & 0 & 133 & 21 \\ 5 & 9 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[184] 21^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 17 & 15 & 27 & 7 \\ 17 & 20 & 17 & 33 & 8 \\ 15 & 17 & 12 & 27 & 6 \\ 27 & 33 & 27 & 49 & 13 \\ 7 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[185] 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{13} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 27 & 0 & 0 & 133 & 21 \\ 3 & 4 & -2 & -1 & 21 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 51 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 51 & 108 & 0 & 457 & 39 \\ 5 & 9 & -1 & 39 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 47 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 47 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 24 & 5 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 12 & 15 & 6 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 7 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[195] $21^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 30 & 7 \\ 20 & 25 & 16 & 40 & 9 \\ 12 & 16 & 9 & 24 & 5 \\ 30 & 40 & 24 & 58 & 14 \\ 7 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 38 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 76 & 0 & 217 & 27 \\ 5 & 9 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 57 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[198] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 36 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 76 & 0 & 217 & 27 \\ 5 & 9 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 27 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 28 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[202] $21^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 24 & 6 \\ 16 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 18 & 3 \\ 3 & 4 & 0 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 18 & 32 & 0 & 0 & 193 & 25 \\ 3 & 4 & -2 & -1 & 25 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 46 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{123} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 39 & 9 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 11 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{12} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 10 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[208] 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[209] 1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 301, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[210] 1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 12 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[213] 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 28 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 60 & 0 & 133 & 21 \\ 5 & 9 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[211] 1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[212] 1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 52 & 108 & 0 & 457 & 39 \\ 5 & 9 & -1 & 39 & 2 \end{pmatrix}_{b_0}$$

$$[214] 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[216] 1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[215] 1^{141} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 17 & 3 \\ 3 & 4 & 0 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 31 & 0 & 0 & 172 & 24 \\ 3 & 4 & -2 & -1 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[217] 1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 37 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 320$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 7 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 326$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 21 & 9 \\ 5 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 331$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 27 & 12 \\ 5 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 333$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 43 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 43 & 92 & 0 & 325 & 33 \\ 5 & 9 & -1 & 33 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 334$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 13 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 341$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 343$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 349$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 29 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 60 & 0 & 133 & 21 \\ 5 & 9 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 356$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 30 & 5 \\ 13 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -2 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 356$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 30 & 5 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -2 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 366$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 366$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{12} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 371$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 372$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 57 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 381$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 92 & 0 & 325 & 33 \\ 5 & 9 & -1 & 33 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 56 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 56 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 40 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 40 & 80 & 0 & 238 & 28 \\ 5 & 9 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 386$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 21 & 9 \\ 5 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{123} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 390$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 11 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & -2 \\ 10 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 10 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 12 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 44 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 44 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{14} 3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 20 & 3 \\ 3 & 4 & 0 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 20 & 36 & 0 & 0 & 238 & 28 \\ 3 & 4 & -2 & -1 & 28 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{114} 3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 68 & 144 & 0 & 814 & 52 \\ 5 & 9 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{112} 3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 14 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 9 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 30 & 9 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 45 & 7 \\ 20 & 25 & 16 & 60 & 9 \\ 12 & 16 & 9 & 36 & 5 \\ 45 & 60 & 36 & 133 & 21 \\ 7 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 10 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 37 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 80 & 0 & 238 & 28 \\ 5 & 9 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 37 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 76 & 0 & 217 & 27 \\ 5 & 9 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 76 & 0 & 217 & 27 \\ 5 & 9 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

- [258] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 44 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 44 & 92 & 0 & 325 & 33 \\ 5 & 9 & -1 & 33 & 2 \end{pmatrix}_{b_0}$$
- [259] $1^{131} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$
- [260] $1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 70 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 70 & 144 & 0 & 814 & 52 \\ 5 & 9 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [261] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 48 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 48 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [262] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 48 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$
- [263] $1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 48 & 27 & 12 \\ 5 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$
- [264] $1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$
- [265] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [266] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 12 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$
- [267] $1^{122} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 30 & 9 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$
- [268] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 11 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 488$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 491$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 30 & 9 \\ 8 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 493$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 69 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 69 & 144 & 0 & 814 & 52 \\ 5 & 9 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 498$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 63 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 63 & 128 & 0 & 634 & 46 \\ 5 & 9 & -1 & 46 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 503$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 15 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[276] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 528$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 52 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 528$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 60 & 128 & 0 & 634 & 46 \\ 5 & 9 & -1 & 46 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 535$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 13 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

- [279] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 39 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 39 & 80 & 0 & 238 & 28 \\ 5 & 9 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [280] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 38 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 80 & 0 & 238 & 28 \\ 5 & 9 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [281] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$
- [282] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 14 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$
- [283] $1^{112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 17 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$
- [284] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 14 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$
- [285] $1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$
- [286] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 11 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$
- [287] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [288] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 13 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [289] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 12 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[290] 1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[291] 21^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 12 & 60 & 7 \\ 20 & 25 & 16 & 80 & 9 \\ 12 & 16 & 9 & 48 & 5 \\ 60 & 80 & 48 & 238 & 28 \\ 7 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[292] 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[293] 1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 10 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[294] 1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 645, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 55 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 55 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[295] 1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[296] 1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 62 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 62 & 128 & 0 & 634 & 46 \\ 5 & 9 & -1 & 46 & 2 \end{pmatrix}_{b_0}$$

$$[297] 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 47 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 47 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[298] 1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 669, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 61 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 61 & 128 & 0 & 634 & 46 \\ 5 & 9 & -1 & 46 & 2 \end{pmatrix}_{b_0}$$

$$[304] 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 13 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[300] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[301] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 12 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[302] $1^{112} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 17 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{112} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 20 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{112} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 14 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 53 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 53 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{123} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 10 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 46 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{114} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 730, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 12 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{112} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 23 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

[351] $1^{\overline{11}4}1^{\overline{14}}31^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 24 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 24 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [352] $1^{\overline{11}1}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 7 & 19 & 5 \\ 9 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 19 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{\overline{11}}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[358] $21^{\overline{11}}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 12 & 6 \\ 12 & 13 & 10 & 14 & 7 \\ 8 & 10 & 4 & 9 & 4 \\ 12 & 14 & 9 & 13 & 7 \\ 6 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{\overline{11}}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 12 & 6 \\ 12 & 13 & 10 & 14 & 7 \\ 8 & 10 & 4 & 9 & 4 \\ 12 & 14 & 9 & 13 & 7 \\ 6 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[356] $21^{\overline{11}}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 6 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{\overline{11}1}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 6 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{\overline{12}1}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[359] $21^{\overline{11}1}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 9 & 15 & 7 \\ 14 & 13 & 10 & 14 & 7 \\ 9 & 10 & 4 & 9 & 4 \\ 15 & 14 & 9 & 13 & 7 \\ 7 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{\overline{11}1}41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

[illegible]

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 7 & 7 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 7 & 10 & 4 & 7 & 4 \\ 7 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[384] \ 1^{13} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[383] \ 1^{12} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 14 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[382] \ 1^{12} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[385] \ 1^{14} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{111} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 7 & 18 & 5 \\ 9 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 18 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{13} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 16 & 5 \\ 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[389] \ 1^{14} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 33 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 33 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[390] \ 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 17 & 5 \\ 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[391] \ 1^{111} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 16 & 5 \\ 9 & 13 & 10 & 24 & 7 \\ 6 & 10 & 4 & 16 & 4 \\ 16 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[392] \ 1^{13} 4^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -229, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[393] \ 21^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -254, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 24 & 7 \\ 14 & 13 & 10 & 24 & 7 \\ 9 & 10 & 4 & 16 & 4 \\ 24 & 24 & 16 & 38 & 12 \\ 7 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[394] \ 1^{111} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -254, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 7 & 16 & 5 \\ 9 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 16 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[395] \ 1^{13} 4^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -265, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[396] \ 1^{111} 4^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -270, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 7 & 17 & 5 \\ 9 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 17 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[397] \ 1^{13} 4^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -286, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[398] \ 21^{111} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -300, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 7 & 5 \\ 10 & 13 & 10 & 10 & 7 \\ 7 & 10 & 4 & 7 & 4 \\ 7 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[399] \ 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -325, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[400] \ 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[401] \ 1^{14} 4^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -346, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[402] \ 21 \overset{1^1}{1^{11}} 41 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 16 & 5 \\ 10 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 16 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[403] \ 1 \overset{1^{12}}{1^{12}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 23 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[404] \ 1 \overset{1^{13}}{1^{13}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 24 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[405] \ 1 \overset{1^{11}}{1^{13}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[406] \ 1 \overset{1^{13}}{1^{14}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[407] \ 1 \overset{1^{14}}{1^{14}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[408] \ 1 \overset{1^{11}}{1^{14}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 34 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[409] \ 1 \overset{1^{12}}{1^{13}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -579, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[410] \ 1 \overset{1^{14}}{1^{13}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 31 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[411] \ 1 \overset{1^{13}}{1^{13}} 4 \overset{1^1}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -654, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 30 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[412] $1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -734$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 35 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[413] $1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -881$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 36 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[414] $1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -950$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [415] $1^{11} 31^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -62$, $\det = 12$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 4 & 1 & 2 & 5 \\ 4 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 5 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.740 NHC(-2)(-3) + $\textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6$

$T = 9$ [0] $1^{11} 31^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 31^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 31^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11} 31^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.741 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.742 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 28 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

- [4] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 10 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 19 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 20 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 9 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & -1 & 20 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 30 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 29 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 38 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 40 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 39 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 19 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \ 1^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 9 & 1 & 5 \\ 9 & 13 & 10 & 14 & 0 & 7 \\ 6 & 10 & 4 & 9 & 0 & 4 \\ 9 & 14 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

4.743 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

4.744 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 9 & 7 \\ 17 & 20 & 17 & 11 & 8 \\ 15 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 7 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[10] $21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 16 & 12 & 8 & 6 \\ 16 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [13] $1^{11}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [14] $1^{14}1^{13}1^{11}5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 161, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 & 2 \\ 3 & 4 & 1 & 2 & 0 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 0 & 2 \\ 2 & 4 & 0 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

4.745 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.746 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{green}{so}_8$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.747 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 12 \quad [0] \ 1^{11} \overset{1}{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 4 & 2 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1}{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \\ 2 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 6 & 0 & 5 & 9 & 7 \\ 3 & 4 & 2 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} \overset{1}{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 3 & 0 & 3 & 15 & 4 \\ 2 & 3 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 9 & 3 \\ 8 & 15 & 10 & 0 & 9 & 41 & 13 \\ 3 & 4 & 2 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} \overset{1}{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 3 & 0 & 3 & 11 & 4 \\ 2 & 3 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 7 & 3 \\ 6 & 11 & 8 & 0 & 7 & 22 & 10 \\ 3 & 4 & 2 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

4.748 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 8 \quad [0] \ 1^{11} \overset{1}{3} 1^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1}{3} 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} \overset{1}{3} 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 24 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 48 & 0 & 89 & 17 \\ 5 & 9 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} \overset{1}{3} 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{14} \overset{1}{3} 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 11 & 19 & 0 & 15 & 64 & 14 \\ 3 & 4 & -2 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 29 & 6 \\ 10 & 19 & 0 & 29 & 64 & 14 \\ 3 & 4 & -2 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 6 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 25 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 16 & 9 \\ 5 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 10 & 5 \\ 5 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 9 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 6 & 11 & 0 & 9 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 16 & 5 \\ 7 & 12 & 0 & 16 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 16 & 5 \\ 12 & 25 & 0 & 32 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 32 & 0 & 37 & 11 \\ 5 & 9 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 23 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 48 & 0 & 89 & 17 \\ 5 & 9 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{111}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 16 & 27 & 0 & 17 & 151 & 21 \\ 3 & 4 & -2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 9 & 15 & 0 & 12 & 39 & 11 \\ 3 & 4 & -2 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 17 & 6 \\ 7 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -2 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 13 & 22 & 0 & 14 & 97 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 29 & 6 \\ 11 & 19 & 0 & 29 & 64 & 14 \\ 3 & 4 & -2 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 17 & 27 & 0 & 17 & 151 & 21 \\ 3 & 4 & -2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 23 & 6 \\ 9 & 15 & 0 & 23 & 39 & 11 \\ 3 & 4 & -2 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 16 & 9 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 10 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 8 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 23 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -2 & 2 & 30 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [75] \quad 1^{11} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{12} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 14 & 4 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 11 & 0 & 39 & 13 \\ 3 & 4 & 2 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{13} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -133, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 18 & 13 & 0 & 62 & 16 \\ 3 & 4 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.749 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_{2n3\text{mix}} + \textcolor{green}{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 7 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 7 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.750 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 7 & 12 & 0 & 9 & 24 & 8 \\ 3 & 4 & -2 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 3 \\ 3 & 4 & 0 & 10 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 21 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 7 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 11 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 11 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{12} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 0 & 9 & 5 \\ 3 & 4 & -2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 21 & 0 & 0 & 75 & 15 \\ 3 & 4 & -2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[11] $1 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

- [14] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & -2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 36 & 0 & 48 & 12 \\ 5 & 9 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [19] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 5 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [23] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [22] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 48 & 12 \\ 3 & 4 & -2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 9 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 12 & 6 \\ 5 & 9 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 0 & 36 & 10 \\ 3 & 4 & -2 & -1 & 10 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 36 & 0 & 48 & 12 \\ 5 & 9 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 0 & 27 & 9 \\ 3 & 4 & -2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 0 & 57 & 13 \\ 3 & 4 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{14} 3^{11} 3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 26 & 0 & 0 & 117 & 19 \\ 3 & 4 & -2 & -1 & 19 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{13} 3^{11} 3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 0 & 84 & 16 \\ 3 & 4 & -2 & -1 & 16 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3^{11} 3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{111} 3^{11} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[39] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 12 & 6 \\ 7 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111} 3^{11} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{111} 3^{11} 3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [42] $1^4 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 5 \\ 12 & 25 & 12 & 12 & 9 \\ 6 & 12 & 4 & 7 & 4 \\ 6 & 12 & 7 & 4 & 4 \\ 5 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 3 & 0 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 8 & 8 & 7 & 1 & 4 \\ 8 & 13 & 16 & 12 & 0 & 7 \\ 8 & 16 & 12 & 12 & 0 & 6 \\ 7 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 7 & 6 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[46] \quad & 1^{11} 1^1 31^1 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 2 \\ 1 & 3 & 0 & 3 & 0 & 3 & 2 \\ 3 & 3 & 0 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[45] \quad & 1^{11} 1^1 31^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 10 & 1 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 0 & 12 & 6 \\ 3 & 6 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0} \\
[47] \quad & 1^{11} 1^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0} \\
[48] \quad & 21^{11} 1^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 13 & 20 & 9 & 9 & 7 \\ 20 & 25 & 12 & 12 & 9 \\ 9 & 12 & 4 & 7 & 4 \\ 9 & 12 & 7 & 4 & 4 \\ 7 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0} \\
[49] \quad & 1^{11} 1^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0} \\
[50] \quad & 1^1 31^1 1^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 5 & 1 & 1 & 3 \\ 5 & 13 & 16 & 0 & 0 & 7 \\ 5 & 16 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0} \\
[51] \quad & 1^{11} 1^1 31^1 1^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 9 & 6 & 0 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0} \\
[52] \quad & 1^{11} 1^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 10 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0} \\
[53] \quad & 1^{11} 1^1 31^1 1^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 2 \\ 1 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[54] \quad & 1^{11} 1^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

4.751 NHC(-2)(-3) + so₈ + su₃ + su₃ + so₈

$T = 9$ [0] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^{11}41^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 1 & 5 \\ 12 & 25 & 12 & 12 & 0 & 9 \\ 6 & 12 & 4 & 7 & 0 & 4 \\ 6 & 12 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.752 NHC(-2)(-3) + so₈ + su₃ + su₃ + su₃

$T = 9$ [0] $1^{111}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 10 & 5 \\ 12 & 25 & 0 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 0 & 12 & 6 \\ 5 & 9 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 17 & 5 \\ 12 & 25 & 0 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 36 & 0 & 0 & 48 & 12 \\ 5 & 9 & -1 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & -1 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.753 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{11} 4^{211} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{211} 4^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

4.754 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^{11} 31^{12} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 13 & 17 & 1 & 7 \\ 17 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 10 & 16 & 1 & 6 \\ 16 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 13 & 20 & 1 & 7 \\ 20 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [3] \quad 1^{11} 4^{12} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] 1^{\overline{1}} 2^{11} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 4 & 7 & 2 & 0 & 4 \\ 3 & 7 & 3 & 2 & 0 & 3 \\ 1 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] 2^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] 1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[10] 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] 1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

4.755 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 8 \quad [0] 1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 12 & 7 \\ 17 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 10 & 6 \\ 7 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.46 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 10 & 16 & 1 & 12 & 6 \\ 16 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 15 & 7 \\ 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 17 & 0 & 13 & 7 \\ 7 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 299, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 27 & 7 \\ 17 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 33 & 0 & 49 & 13 \\ 7 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 325, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 1 & 27 & 7 \\ 20 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 36 & 0 & 49 & 13 \\ 7 & 9 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 434, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.17 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 72 & 7 \\ 17 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 72 & 84 & 0 & 346 & 34 \\ 7 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 454, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 42 & 7 \\ 17 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 42 & 50 & 0 & 118 & 20 \\ 7 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 499, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 57 & 7 \\ 17 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 57 & 67 & 0 & 217 & 27 \\ 7 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{13}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 525, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 1 & 15 & 7 \\ 20 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 20 & 0 & 13 & 7 \\ 7 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \ 1^{11}2^{11}4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}2^{11}4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -229, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -364, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 14 & 5 \\ 10 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -589, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 21 & 5 \\ 10 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -709, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 23 & 5 \\ 10 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -904, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 28 & 5 \\ 10 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -1164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 30 & 5 \\ 10 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 30 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -1709, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 37 & 5 \\ 10 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

4.756 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^1 2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \rightarrow \textcolor{red}{(-3)} \quad T = 8, \ T_{\max} = 13, \ \Delta = -95, \ \det = 130, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 9 & 7 \\ 17 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 11 & 0 & 4 & 4 \\ 7 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 1^{11} 2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \rightarrow \textcolor{red}{(-3)} \quad T = 8, \ T_{\max} = 13, \ \Delta = -95, \ \det = 325, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 1 & 9 & 7 \\ 20 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 4 & 4 \\ 7 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

4.757 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3)→(-2) + su2n3mix

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 24 & 7 \\ 17 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 28 & 0 & 37 & 11 \\ 7 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

4.758 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3)→(-2) + $\mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 1 & 15 & 7 \\ 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 17 & 0 & 12 & 6 \\ 7 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 1 & 12 & 7 \\ 20 & 25 & 0 & 16 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 9 & 5 \\ 7 & 9 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 1 & 12 & 6 \\ 16 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 1 & 9 & 7 \\ 14 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 10 & 0 & 4 & 4 \\ 7 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

4.759 NHC(-2)(-3) + $\mathfrak{so}_8$ + su3n2mix

$$T = 7 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 10 & 16 & 13 & 6 \\ 16 & 20 & 17 & 8 \\ 13 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 13 & 14 & 10 & 7 \\ 14 & 13 & 10 & 7 \\ 10 & 10 & 4 & 4 \\ 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 4 \\ 5 & 5 & 4 & 5 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 1 & 3 \\ 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.760 NHC(-2)(-3) + $\mathfrak{su}_2$

$$T = 6 \quad [0] \ 1^2 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 9 & 16 & 5 \\ 7 & 16 & 28 & 10 \\ 3 & 5 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 5 & 4 \\ 5 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^1 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 22 & 8 \\ 3 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 22 & 8 \\ 3 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9, T_{\max} = 22, T \in [9, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.56 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^1 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 15 & 4 \\ 13 & 15 & 52 & 12 \\ 3 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 21^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.45 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 21^1 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 19 & 4 \\ 12 & 19 & 73 & 15 \\ 3 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.45 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.45 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.45 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 9 & 20 & 5 \\ 11 & 20 & 37 & 11 \\ 3 & 5 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^1 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 13, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 28 & 5 \\ 12 & 28 & 85 & 17 \\ 3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^1 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^1 1^1 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 20 & 4 \\ 13 & 20 & 94 & 16 \\ 3 & 4 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{12} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 8 & 4 \\ 8 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^1 1^1 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 22 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^1 1^2 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 22 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^1 1^1 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 15, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 15 & 4 \\ 10 & 15 & 52 & 12 \\ 3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 15, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.33 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 9 & 4 \\ 9 & 9 & 13 & 7 \\ 3 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 17, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

- $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 12 & 5 \\ 6 & 12 & 13 & 7 \\ 3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$
- [24] $21^1 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 17$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 12 & 5 \\ 6 & 12 & 13 & 7 \\ 3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$
- [25] $2131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 19$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$
- [26] $1^1 3^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 19$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$
- [27] $1^1 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 19$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 15 & 4 \\ 12 & 15 & 52 & 12 \\ 3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$
- [28] $1^1 13^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 19$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$
- [29] $1^1 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 9 & 12 & 5 \\ 5 & 12 & 13 & 7 \\ 3 & 5 & 7 & 2 \end{pmatrix}_{b_0}$
- [30] $21^1 13^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 5 & 4 \\ 6 & 5 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [31] $1^1 1^2 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 21$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 15 & 4 \\ 11 & 15 & 52 & 12 \\ 3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$
- [32] $1^1 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 9 & 4 \\ 5 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [36] $1^1 1^{12} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 24 & 5 \\ 12 & 24 & 58 & 14 \\ 3 & 5 & 14 & 0 \end{pmatrix}_{b_0}$
- [34] $1^1 13^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 16 & 4 \\ 8 & 16 & 46 & 12 \\ 3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$
- [33] $21^2 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 24 & 5 \\ 12 & 24 & 58 & 14 \\ 3 & 5 & 14 & 0 \end{pmatrix}_{b_0}$

- [35] $21^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 10 & 4 \\ 12 & 10 & 22 & 8 \\ 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 8 & 4 \\ 7 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$
- [39] $1^131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 94 & 16 \\ 3 & 4 & 16 & 0 \end{pmatrix}_{b_0}$
- [38] $1^131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 94 & 16 \\ 3 & 4 & 16 & 0 \end{pmatrix}_{b_0}$
- [37] $1^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 8 & 4 \\ 5 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$
- [41] $1^{113}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 9 & 32 & 5 \\ 17 & 32 & 100 & 18 \\ 3 & 5 & 18 & -1 \end{pmatrix}_{b_0}$
- [44] $1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 30, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [43] $1^131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 30, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 94 & 16 \\ 3 & 4 & 16 & 1 \end{pmatrix}_{b_0}$
- [42] $1^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 30, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$
- [45] $21^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [48] $1^131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 34, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 16 & 5 \\ 6 & 16 & 22 & 8 \\ 3 & 5 & 8 & 1 \end{pmatrix}_{b_0}$
- [47] $2131^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 34, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 16 & 5 \\ 6 & 16 & 22 & 8 \\ 3 & 5 & 8 & 1 \end{pmatrix}_{b_0}$
- [46] $1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 34, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 9.59 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^4 1^3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 35, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 9 & 40 & 5 \\ 17 & 40 & 172 & 24 \\ 3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{12} 1^3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 35, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 37 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^1 1^3 1^{15} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 35, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 20 & 4 \\ 10 & 20 & 73 & 15 \\ 3 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{12} 1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.56 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 9 & 4 \\ 7 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 21^1 1^3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 21 & 4 \\ 18 & 21 & 76 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 21^{13} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 15 & 4 \\ 18 & 15 & 52 & 12 \\ 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^2 1^3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 38, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 9 & 24 & 5 \\ 10 & 24 & 58 & 14 \\ 3 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{13} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 39, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 12 & 4 \\ 7 & 12 & 25 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 1^{13} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 40, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 28 & 5 \\ 15 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 1^3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 13 & 4 \\ 7 & 13 & 28 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{12} 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 9 & 20 & 5 \\ 10 & 20 & 37 & 11 \\ 3 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{13} 1^{11} 3^1 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

- [58] $21^3 31^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 36 & 5 \\ 18 & 36 & 133 & 21 \\ 3 & 5 & 21 & 0 \end{pmatrix}_{b_0}$
- [63] $21^{11} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 14 & 4 \\ 9 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0}$
- [62] $1^3 1^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 19 & 4 \\ 11 & 19 & 73 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [64] $21^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 45, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.64 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 9 & 4 \\ 9 & 9 & 13 & 7 \\ 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [65] $1^1 1^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 45, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 9 & 20 & 5 \\ 8 & 20 & 37 & 11 \\ 3 & 5 & 11 & 1 \end{pmatrix}_{b_0}$
- [66] $1^1 1^3 1^{16} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 46, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 24 & 4 \\ 12 & 24 & 106 & 18 \\ 3 & 4 & 18 & 0 \end{pmatrix}_{b_0}$
- [67] $21^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 50, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 20 & 4 \\ 24 & 20 & 94 & 16 \\ 4 & 4 & 16 & 1 \end{pmatrix}_{b_0}$
- [73] $1^{11} 1^3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0}$
- [72] $1^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [71] $1^1 1^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 121 & 19 \\ 3 & 4 & 19 & 1 \end{pmatrix}_{b_0}$
- [70] $1^{11} 1^3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0}$
- [69] $21^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [68] $1^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 52, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 9 & 20 & 5 \\ 9 & 20 & 37 & 11 \\ 3 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[75] \ 21^{131}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 52, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 9 & 20 & 5 \\ 9 & 20 & 37 & 11 \\ 3 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[76] \ 21^{113}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 53, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 17 & 4 \\ 15 & 17 & 49 & 13 \\ 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 54, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 16 & 4 \\ 9 & 16 & 46 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 55, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[79] \ 21^{123}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 56, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[80] \ 21^{113}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 56, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.64 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 13 & 4 \\ 12 & 13 & 28 & 10 \\ 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[81] \ 2131^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 9 & 24 & 5 \\ 9 & 24 & 52 & 12 \\ 3 & 5 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{112}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 13 & 4 \\ 8 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{113}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 9 & 24 & 5 \\ 9 & 24 & 52 & 12 \\ 3 & 5 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{113}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

- [86] $21^{111}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 9 & 4 \\ 8 & 9 & 12 & 5 \\ 9 & 12 & 13 & 7 \\ 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$
- [87] $131^{117} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 28 & 4 \\ 14 & 28 & 145 & 21 \\ 3 & 4 & 21 & -1 \end{pmatrix}_{b_0}$
- [88] $131^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 60, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 17 & 4 \\ 9 & 17 & 49 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [90] $1^{114}31^4 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 62, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 24 & 3 \\ 4 & 9 & 48 & 5 \\ 24 & 48 & 238 & 28 \\ 3 & 5 & 28 & 0 \end{pmatrix}_{b_0}$
- [89] $21^431^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 62, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 24 & 3 \\ 4 & 9 & 48 & 5 \\ 24 & 48 & 238 & 28 \\ 3 & 5 & 28 & 0 \end{pmatrix}_{b_0}$
- [91] $1^{12}31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 65, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [92] $1^{11}31^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 66, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 19 & 4 \\ 12 & 19 & 73 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [93] $131^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 66, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 58 & 14 \\ 3 & 4 & 14 & 2 \end{pmatrix}_{b_0}$
- [97] $21^{15}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 4 & 5 & 30 & 4 \\ 5 & 4 & 25 & 4 \\ 30 & 25 & 148 & 20 \\ 4 & 4 & 20 & 0 \end{pmatrix}_{b_0}$
- [96] $131^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 36 & 5 \\ 15 & 36 & 133 & 21 \\ 3 & 5 & 21 & 2 \end{pmatrix}_{b_0}$
- [95] $21^{112}31^2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 18 & 4 \\ 8 & 9 & 24 & 5 \\ 18 & 24 & 58 & 14 \\ 4 & 5 & 14 & 2 \end{pmatrix}_{b_0}$
- [94] $1^{113}31^2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 9 & 32 & 5 \\ 16 & 32 & 100 & 18 \\ 3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$
- [98] $1^{12}31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 69, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 13 & 4 \\ 9 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{12}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 71, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.63 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[100] $1^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 74, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 46 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[101] $131^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 76, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 9 & 8 & 5 \\ 6 & 8 & 4 & 4 \\ 4 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 77, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 9 & 28 & 5 \\ 14 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{11}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 79, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 24 & 4 \\ 15 & 24 & 121 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{11}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 80, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[104] $131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 80, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 9 & 28 & 5 \\ 11 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

[106] $131^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 21 & 4 \\ 11 & 21 & 76 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

[108] $21^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 14 & 4 \\ 15 & 14 & 37 & 11 \\ 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[107] $131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 9 & 32 & 5 \\ 13 & 32 & 100 & 18 \\ 3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$$

[109] $21^{113}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 4 & 8 & 27 & 4 \\ 8 & 9 & 36 & 5 \\ 27 & 36 & 133 & 21 \\ 4 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{11}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 84, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[111] $1^{13}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 86$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 19 & 4 \\ 14 & 19 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[112] $1^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 86$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$

[113] $1^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 86$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 16 & 4 \\ 11 & 16 & 46 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[114] $1^{13}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 89$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 23 & 4 \\ 13 & 23 & 100 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$

[115] $1^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 90$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 46 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[116] $21^{16}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 90$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.2$
 $T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 4 & 5 & 36 & 4 \\ 5 & 4 & 30 & 4 \\ 36 & 30 & 214 & 24 \\ 4 & 4 & 24 & 0 \end{pmatrix}_{b_0}$

[117] $1^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 90$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$

[118] $1^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 91$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 32 & 5 \\ 15 & 32 & 100 & 18 \\ 3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$

[119] $21^231^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 91$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 32 & 5 \\ 15 & 32 & 100 & 18 \\ 3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$

[120] $2131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 94$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 32 & 5 \\ 12 & 32 & 94 & 16 \\ 3 & 5 & 16 & 0 \end{pmatrix}_{b_0}$

[121] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 94$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 32 & 5 \\ 12 & 32 & 94 & 16 \\ 3 & 5 & 16 & 0 \end{pmatrix}_{b_0}$

[122] $21^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 96$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[123] $1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 96$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 9 & 28 & 5 \\ 13 & 28 & 73 & 15 \\ 3 & 5 & 15 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{113}3^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 97, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 28 & 5 \\ 12 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

[125] $21^{131}1^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 97, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 28 & 5 \\ 12 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{112}3^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 99, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 121 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{112}3^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 99, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 22 & 4 \\ 12 & 22 & 85 & 17 \\ 3 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{114}3^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 101, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 22 & 3 \\ 4 & 9 & 44 & 5 \\ 22 & 44 & 193 & 25 \\ 3 & 5 & 25 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{132}3^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 17 & 4 \\ 11 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[129] $21^{114}3^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 4 & 8 & 36 & 4 \\ 8 & 9 & 48 & 5 \\ 36 & 48 & 238 & 28 \\ 4 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

[130] $21^{123}1^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 28 & 4 \\ 18 & 28 & 154 & 22 \\ 3 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{122}3^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 106, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[133] $21^{123}1^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 107, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.64 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 13 & 4 \\ 9 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[135] $1^{14}3^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 110, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 9 & 48 & 5 \\ 20 & 48 & 238 & 28 \\ 3 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{14}3^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 110, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 28 & 4 \\ 16 & 28 & 154 & 22 \\ 3 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

- [136] $1^{13}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 111$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 121 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$
- [138] $1^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 114$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 17 & 4 \\ 12 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$
- [139] $1^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 114$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 18 & 4 \\ 13 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$
- [137] $1^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 114$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$
- [140] $1^{11}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 115$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.09 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 121 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$
- [141] $1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 115$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.09 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 76 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$
- [142] $1^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 116$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.69 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$
- [143] $21^{17}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 116$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.3$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 4 & 5 & 42 & 4 \\ 5 & 4 & 35 & 4 \\ 42 & 35 & 292 & 28 \\ 4 & 4 & 28 & 0 \end{pmatrix}_{b_0}$
- [144] $1^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 119$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.05 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 23 & 4 \\ 14 & 23 & 100 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$
- [145] $1^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 122$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 9 & 40 & 5 \\ 20 & 40 & 154 & 22 \\ 3 & 5 & 22 & 1 \end{pmatrix}_{b_0}$
- [147] $131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 125$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 9 & 36 & 5 \\ 14 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$
- [148] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 125$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 36 & 5 \\ 18 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$
- [146] $21^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 125$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$

$$\begin{aligned}
& T_{\min} = 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 19 & 4 \\ 21 & 19 & 73 & 15 \\ 4 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[149] & 21^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 126, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 18 & 4 \\ 18 & 18 & 58 & 14 \\ 4 & 4 & 14 & 3 \end{pmatrix}_{b_0} \\
[150] & 21^{12}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 131, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 23 & 4 \\ 15 & 23 & 100 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[151] & 1^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 131, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 22 & 4 \\ 13 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0} \\
[152] & 21^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 132, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 18 & 4 \\ 12 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0} \\
[153] & 1^3 1^{11}4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 133, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.05 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 44 & 5 \\ 18 & 44 & 193 & 25 \\ 3 & 5 & 25 & 1 \end{pmatrix}_{b_0} \\
[154] & 1^{12} 1^3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 134, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 26 & 4 \\ 14 & 26 & 118 & 20 \\ 3 & 4 & 20 & 2 \end{pmatrix}_{b_0} \\
[155] & 1^{13} 1^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 136, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 27 & 4 \\ 15 & 27 & 133 & 21 \\ 3 & 4 & 21 & 2 \end{pmatrix}_{b_0} \\
[156] & 21^3 31^{11}4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 136, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.21 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 9 & 44 & 5 \\ 21 & 44 & 193 & 25 \\ 3 & 5 & 25 & 1 \end{pmatrix}_{b_0} \\
[157] & 1^{11}4^3 31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 136, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.21 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 9 & 44 & 5 \\ 21 & 44 & 193 & 25 \\ 3 & 5 & 25 & 1 \end{pmatrix}_{b_0} \\
[158] & 131^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 136, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 8 & 12 & 4 \\ 8 & 9 & 16 & 5 \\ 12 & 16 & 22 & 8 \\ 4 & 5 & 8 & 2 \end{pmatrix}_{b_0} \\
[159] & 1^2 1^{11}4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 138, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 9 & 40 & 5 \\ 16 & 40 & 154 & 22 \\ 3 & 5 & 22 & 1 \end{pmatrix}_{b_0} \\
[161] & 21^{11}2 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 139, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 8 & 15 & 4 \\ 8 & 9 & 20 & 5 \\ 15 & 20 & 37 & 11 \\ 4 & 5 & 11 & 2 \end{pmatrix}_{b_0} \\
[162] & 2131^{11}5 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 139, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1
\end{aligned}$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 40 & 5 \\ 15 & 40 & 148 & 20 \\ 3 & 5 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{115} 3^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 139, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 40 & 5 \\ 15 & 40 & 148 & 20 \\ 3 & 5 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{12} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 141, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 23 & 4 \\ 15 & 23 & 100 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 141, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{12} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 146, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 28 & 4 \\ 17 & 28 & 154 & 22 \\ 3 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{114} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 152, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 9 & 36 & 5 \\ 17 & 36 & 121 & 19 \\ 3 & 5 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{114} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 152, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 36 & 5 \\ 15 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[168] \ 21^1 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 152, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 36 & 5 \\ 15 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[166] \ 21^{14} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 152, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 16 & 4 \\ 12 & 16 & 46 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{114} 3^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 153, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 9 & 40 & 5 \\ 19 & 40 & 154 & 22 \\ 3 & 5 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[173] \ 21^{12} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 155, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 22 & 4 \\ 21 & 22 & 85 & 17 \\ 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{13} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 155, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{12} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 155, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 23 & 4 \\ 16 & 23 & 100 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 159, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[176] $1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 161, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 9 & 36 & 5 \\ 16 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$

[175] $1^{12}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 161, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 23 & 4 \\ 17 & 23 & 100 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[177] $1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 166, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 40 & 5 \\ 18 & 40 & 154 & 22 \\ 3 & 5 & 22 & 2 \end{pmatrix}_{b_0}$

[178] $21^{23}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 166, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 40 & 5 \\ 18 & 40 & 154 & 22 \\ 3 & 5 & 22 & 2 \end{pmatrix}_{b_0}$

[179] $1^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[180] $21^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 170, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 26 & 4 \\ 24 & 26 & 118 & 20 \\ 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$

[181] $1^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 32 & 4 \\ 18 & 32 & 193 & 25 \\ 3 & 4 & 25 & 2 \end{pmatrix}_{b_0}$

[182] $221^{11}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[183] $1^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 22 & 4 \\ 15 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$

[184] $21^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 4 & 5 & 27 & 4 \\ 5 & 4 & 24 & 4 \\ 27 & 24 & 121 & 19 \\ 4 & 4 & 19 & 2 \end{pmatrix}_{b_0}$

[185] $1^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 174, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 27 & 4 \\ 16 & 27 & 133 & 21 \\ 3 & 4 & 21 & 2 \end{pmatrix}_{b_0}$

- [186] $1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 174$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.03 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 26 & 4 \\ 15 & 26 & 118 & 20 \\ 3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$
- [187] $1^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 174$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.03 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 28 & 4 \\ 18 & 28 & 154 & 22 \\ 3 & 4 & 22 & 2 \end{pmatrix}_{b_0}$
- [189] $21^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 179$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$
- [188] $1^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 179$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$
- [190] $1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 185$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 31 & 4 \\ 17 & 31 & 172 & 24 \\ 3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$
- [191] $1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 194$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 28 & 4 \\ 19 & 28 & 154 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [192] $21^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 200$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 23 & 4 \\ 24 & 23 & 100 & 18 \\ 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [193] $1^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 204$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 27 & 4 \\ 17 & 27 & 133 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$
- [195] $1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 206$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 26 & 4 \\ 16 & 26 & 118 & 20 \\ 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$
- [194] $1^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 206$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 28 & 4 \\ 20 & 28 & 154 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [196] $21^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 208$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 24 & 4 \\ 8 & 9 & 32 & 5 \\ 24 & 32 & 100 & 18 \\ 4 & 5 & 18 & 2 \end{pmatrix}_{b_0}$
- [197] $1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 215$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 32 & 4 \\ 19 & 32 & 193 & 25 \\ 3 & 4 & 25 & 2 \end{pmatrix}_{b_0}$
- [198] $21^{15}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 224$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.1$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 20 & 4 \\ 15 & 20 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 226, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 27 & 4 \\ 18 & 27 & 133 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 230, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 26 & 4 \\ 17 & 26 & 118 & 20 \\ 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 231, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 31 & 4 \\ 18 & 31 & 172 & 24 \\ 3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{14}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 234, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 36 & 4 \\ 20 & 36 & 238 & 28 \\ 3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[203] $131^{13}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 236, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 4 & 8 & 18 & 4 \\ 8 & 9 & 24 & 5 \\ 18 & 24 & 52 & 12 \\ 4 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

[204] $21^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 236, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 22 & 4 \\ 15 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 240, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 27 & 4 \\ 19 & 27 & 133 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 246, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 118 & 20 \\ 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 251, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 32 & 4 \\ 20 & 32 & 193 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[208] $21^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 259, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 4 & 8 & 21 & 4 \\ 8 & 9 & 28 & 5 \\ 21 & 28 & 73 & 15 \\ 4 & 5 & 15 & 2 \end{pmatrix}_{b_0}$$

[209] $21^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 261, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 4 & 5 & 27 & 4 \\ 5 & 4 & 27 & 4 \\ 27 & 27 & 133 & 21 \\ 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[210] $21^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 267, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[211] $21^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 267, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 27 & 4 \\ 18 & 27 & 133 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{14}31^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 269, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 31 & 4 \\ 19 & 31 & 172 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[213] \ 21^{13}31^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 272, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 32 & 4 \\ 21 & 32 & 193 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[214] \ 21^{14}31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 278, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 30 & 4 \\ 5 & 4 & 28 & 4 \\ 30 & 28 & 154 & 22 \\ 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 279, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 32 & 4 \\ 21 & 32 & 193 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 21^{114}31^3 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 283, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.49 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 4 & 8 & 33 & 4 \\ 8 & 9 & 44 & 5 \\ 33 & 44 & 193 & 25 \\ 4 & 5 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{14}31^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 286, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 36 & 4 \\ 21 & 36 & 238 & 28 \\ 3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{14}31^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 299, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 31 & 4 \\ 20 & 31 & 172 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 299, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 32 & 4 \\ 22 & 32 & 193 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[220] \ 221^{12}3 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 306, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 6 & 6 & 10 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[221] \ 21^{13}31^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 308, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 30 & 4 \\ 5 & 4 & 31 & 4 \\ 30 & 31 & 172 & 24 \\ 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 21^{16}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 312, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 24 & 4 \\ 18 & 24 & 106 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{14}31^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 321, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 31 & 4 \\ 21 & 31 & 172 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{14}3^{12}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 330, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 36 & 4 \\ 22 & 36 & 238 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$

[225] $21^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 356, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 26 & 4 \\ 18 & 26 & 118 & 20 \\ 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$

[226] $1^{14}3^{13}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 366, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 4 & 36 & 4 \\ 23 & 36 & 238 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$

[227] $21^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 371, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 33 & 4 \\ 5 & 4 & 32 & 4 \\ 33 & 32 & 193 & 25 \\ 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$

[228] $131^{14}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 376, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 24 & 4 \\ 8 & 9 & 32 & 5 \\ 24 & 32 & 94 & 16 \\ 4 & 5 & 16 & 2 \end{pmatrix}_{b_0}$

[229] $21^{114}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 388, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 30 & 4 \\ 8 & 9 & 40 & 5 \\ 30 & 40 & 154 & 22 \\ 4 & 5 & 22 & 2 \end{pmatrix}_{b_0}$

[230] $1^{14}3^{14}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 394, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 36 & 4 \\ 24 & 36 & 238 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$

[231] $21^{17}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 416, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 28 & 4 \\ 21 & 28 & 145 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[232] $21^{114}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 419, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 27 & 4 \\ 8 & 9 & 36 & 5 \\ 27 & 36 & 121 & 19 \\ 4 & 5 & 19 & 2 \end{pmatrix}_{b_0}$

[233] $21^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 419, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 31 & 4 \\ 21 & 31 & 172 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[234] $21^{14}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 450, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 36 & 4 \\ 5 & 4 & 36 & 4 \\ 36 & 36 & 238 & 28 \\ 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$

[235] $21^{14}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 456, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 24 & 3 \\ 4 & 4 & 36 & 4 \\ 24 & 36 & 238 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$

[236] $221^{113}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 531, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 0 & 3 & 3 & 9 & 2 \\ 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 9 & 9 & 25 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$

[237] $131^{115}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 556, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1$

$$\begin{aligned}
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 8 & 30 & 4 \\ 8 & 9 & 40 & 5 \\ 30 & 40 & 148 & 20 \\ 4 & 5 & 20 & 2 \end{pmatrix}_{b_0} \\
[238] & 221^{\overline{1}4}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 846, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 0 & 3 & 3 & 12 & 2 \\ 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[239] & 221^{\overline{1}5}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 1251, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 0 & 3 & 3 & 15 & 2 \\ 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 15 & 15 & 73 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
\\
& T = 7 \quad [240] \quad 1^{\overline{1}}321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9, T_{\max} = 10, T \in [9, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0} \\
[241] & 1^{\overline{1}}3131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0} \\
[242] & 1^{\overline{1}1}321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0} \\
[243] & 1^{\overline{1}}41^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 19, T \in [10, 19] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[244] & 1^{\overline{1}1}321^{\overline{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 6 & 0 & 56 & 14 \\ 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0} \\
[245] & 21^{\overline{1}}41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 19, T \in [10, 19] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[246] & 1^{\overline{1}2}321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[247] & 1^{\overline{1}}321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[248] & 1^{\overline{1}}321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9, T_{\max} = 10, T \in [9, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[251] $21^1 3^1 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{11} 3^1 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[249] $1^1 3^1 1^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[254] $21^1 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[253] $1^1 4^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[252] $1^1 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 11 & 4 \\ 7 & 11 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[257] $1^1 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 19, T \in [10, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[256] $1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[255] $2^1 2^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[264] $1^{11} 3^1 1^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[263] $1^1 3^1 1^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[262] $1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[261] \quad 1^1 1^1 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[260] \quad 1^1 1^1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[259] \quad 1^1 1^1 1^1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[258] \quad 2^1 2^1 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[265] \quad 1^1 1^1 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[266] \quad 2 1^1 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 7 & 4 \\ 6 & 7 & 8 & 6 \\ 4 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[267] \quad 2 1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 11 & 4 \\ 12 & 11 & 28 & 10 \\ 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[269] \quad 2 1^{14} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 25 & 4 \\ 24 & 25 & 132 & 22 \\ 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[268] \quad 1^1 3 1^1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[271] \quad 1^{13} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 7 & 4 \\ 7 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[274] \quad 2 1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 11 & 4 \\ 9 & 11 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

- [276] $1^{131}1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 3 & 3 & 6 & 4 & 4 \\ 2 & 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$
- [273] $1^{11}1^{11}41 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [272] $1^{12}1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 11 & 4 \\ 9 & 11 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [270] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [275] $1^{12}31^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 3 & 6 & 6 & 10 & 6 \\ 2 & 3 & 3 & 6 & 2 \end{pmatrix}_{b_0}$
- [277] $1^{13}321^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 61 & 15 \\ 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$
- [278] $1^{12}321^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 26 & 10 \\ 2 & 2 & -1 & 10 & 3 \end{pmatrix}_{b_0}$
- [280] $1^{14}321^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 110 & 20 \\ 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$
- [279] $1^{11}321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 5 & 5 \\ 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$
- [284] $221^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [285] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 8 & 4 \\ 5 & 8 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [286] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

- [287] $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
 $1 \overset{21}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [288] $1 \overset{1}{3} 1 \overset{11}{3} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$
- [289] $1 \overset{1^3}{1^2} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [283] $1 \overset{1^2}{1^1} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [291] $1 \overset{1}{1} 1 \overset{11}{3} 1 \overset{11}{3} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [292] $1 \overset{1}{1^1} 4 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [282] $131 \overset{1}{1} 1 \overset{11}{3} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 3 \\ 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [281] $131 \overset{1}{1} 1 \overset{11}{3} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [290] $1 \overset{1^2}{1^1} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 8 & 4 \\ 7 & 8 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [293] $21 \overset{1}{1} 1 \overset{11}{4} 1^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 13 & 12 & 7 \\ 6 & 12 & 8 & 6 \\ 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [294] $21 \overset{1^2}{1^1} 1 \overset{11}{4} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [295] $1 \overset{1}{1} 3 2 1 \overset{11}{1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.33 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[297] $1^{11} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{11} 3 1^{12} 2 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 5 & 0 & 33 & 11 \\ 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{14} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 90 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{11} 3 1^{13} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 3 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[299] $2 3 1^{12} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 3 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 10 & 6 \\ 3 & 2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{11} 4 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[303] $3 1^{11} 2 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 10 & 3 \\ 5 & 5 & 15 & 5 \\ 10 & 15 & 38 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[305] $2 2 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[306] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[307] $2 1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[302] $2 2 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 13 & 8 & 7 \\ 3 & 8 & 4 & 4 \\ 3 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[304] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[308] \quad 1 \overset{1^4}{4} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 101 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[309] \quad 21 \overset{1^1}{4} 321^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[310] \quad 1^{\overset{1}{1}} 31^{\overset{2}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 22 & 8 \\ 2 & 3 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[311] \quad 1^{\overset{1}{1}} 31^{\overset{1}{1}} 21^{\overset{2}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 4 & 0 & 16 & 8 \\ 2 & 2 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[312] \quad 21 \overset{1^3}{4} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 18 & 4 \\ 18 & 18 & 70 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[321] \quad 1 \overset{1^{11}}{3} 21^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[316] \quad 1^{\overset{1^2}{1}} 41^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[317] \quad 1^{\overset{1^2}{1}} 41^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[315] \quad 1 \overset{1^1}{4} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[319] \quad 21 \overset{1^1}{4} 1^{\overset{13}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[320] \quad 1^{\overset{1^1}{1}} 321^{\overset{12}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

[314] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[322] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 \\ 1 & 2 & 0 & 4 \\ 5 & 3 & 4 & 5 \\ 2 & 2 & 2 & 5 \end{pmatrix}_{b_0}$

[323] $21^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[324] $1^{13}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[325] $1^{11}3131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$

[313] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[318] $1^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$
 $\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$

[326] $2^332 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 2 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[327] $131^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 2 & 3 & 6 & 4 & 4 \\ 2 & 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$

[328] $1^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 5 & 4 \\ 6 & 5 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$

[329] $1^{12}31^{11}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$

[330] $21^{11}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 32 & 12 \\ 4 & 5 & 12 & 4 \end{pmatrix}_{b_0}$

- [331] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$
- [332] $1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.24 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$
- [333] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.24 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$
- [334] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 8 & 4 \\ 9 & 8 & 13 & 7 \\ 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [335] $1^{11}3131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$
- [336] $21^141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 9 & 4 \\ 5 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [338] $1^{11}31^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 6 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [340] $1^{14}31^{11}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 9 & 18 & 15 & 76 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [341] $1^{12}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.67 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$
- [342] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.67 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$
- [343] $2^13^12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 4 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 4 & 4 & 6 & 10 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [344] $1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 9 & 4 \\ 5 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

- [345] $1^1 31^{11} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [346] $31^{11} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [337] $1^1 31^{13} 31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 9 & 18 & 9 & 52 & 12 \\ 2 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [339] $1^1 31^{14} 31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 12 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [349] $1^{11} 3^{11} 21^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 45 & 13 \\ 2 & 2 & -1 & 13 & 3 \end{pmatrix}_{b_0}$
- [350] $1^3 1^3 41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$
- [351] $1^1 1^1 41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 11 & 4 \\ 6 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [347] $1^1 1^1 4^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 16 & 4 \\ 8 & 16 & 46 & 12 \\ 3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$
- [348] $21^1 41^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 16 & 4 \\ 8 & 16 & 46 & 12 \\ 3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$
- [352] $1^{12} 1^1 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [353] $1^{11} 1^1 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [354] $2^1 2^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[355] $21^{11}41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 27 & 4 \\ 5 & 4 & 29 & 4 \\ 27 & 29 & 169 & 25 \\ 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{11}31^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 9 & 3 \\ 5 & 6 & 9 & 13 & 7 \\ 2 & 3 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[357] $141^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[358] $221^{11}41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 3 & 12 & 3 \\ 6 & 8 & 12 & 34 & 10 \\ 2 & 2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[359] $141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[360] $21^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[361] $21^{12}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{11}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[363] $1^{12}31^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{14}31^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 8 & 18 & 15 & 76 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[365] $141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{12}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[367] \quad 131^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[369] \quad 1^{12}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 11 & 4 \\ 10 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[370] \quad 1^141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 12 & 4 \\ 7 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[368] \quad 1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 12 & 4 \\ 11 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[371] \quad 1^{12}321^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 8 & 0 & 70 & 16 \\ 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[375] \quad 2141^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 20 & 4 \\ 10 & 20 & 73 & 15 \\ 3 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[373] \quad 21^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 15 & 4 \\ 10 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[376] \quad 1^141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 20 & 4 \\ 10 & 20 & 73 & 15 \\ 3 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[377] \quad 21^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 25 & 4 \\ 17 & 25 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[378] \quad 1^{11}31^{11}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[379] \quad 21^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[372] \quad 1^{13}31^{11}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 8 & 15 & 12 & 49 & 13 \\ 2 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[374] \ 21^{1^3} 41^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[387] \ 131^{1^1} 31^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 6 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[383] \ 31^{1^2} 31^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[384] \ 31^{1^3} 131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[382] \ 1^2 4^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[386] \ 1^2 4^{1^1} 1^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 21 & 4 \\ 18 & 21 & 76 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[381] \ 21^{1^1} 41^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 30 & 4 \\ 5 & 4 & 33 & 4 \\ 30 & 33 & 212 & 28 \\ 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{1^3} 3^{1^1} 21^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 88 & 18 \\ 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[380] \ 2^3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[385] \ 2^2 31^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[389] \ 1^{1^2} 3131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 1 & 4 & 0 & 6 & 2 \\ 3 & 6 & 6 & 10 & 6 \\ 2 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

- [390] $2 \overset{1^{11}}{3} 1^{\overset{1}{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [391] $2 \overset{1^{14}}{3} 1^{\overset{1}{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 11 & 2 \\ 3 & 1 & 3 & 11 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 11 & 11 & 14 & 65 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [392] $1 \overset{1}{4} 1^{\overset{1}{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 15 & 4 \\ 8 & 15 & 40 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [393] $1 \overset{1^2}{3} 2 1^{\overset{1}{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$
- [394] $1 \overset{1^4}{3} 2 1^{\overset{1}{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [395] $2 2 1^{\overset{1}{1}4} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 6 & 3 \\ 3 & 2 & 6 & 4 & 4 \\ 2 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$
- [396] $1 \overset{1^4}{3} 2 1^{\overset{1}{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$
- [397] $1 \overset{1}{3} 1^{\overset{1}{1}4} 2 1^{\overset{1}{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [404] $1 \overset{1^4}{3} 1^{\overset{1}{1}4} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 21 & 4 \\ 17 & 21 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [398] $2 \overset{1^{11}}{2} 3 1^{\overset{1}{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [399] $1 \overset{1}{3} 1^{\overset{1}{1}3} 3 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$
- [400] $1 \overset{1}{3} 1^{\overset{1}{1}3} 1^{\overset{1}{1}3} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[401] \ 1^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 11 & 4 \\ 7 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[402] \ 21^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 21 & 4 \\ 17 & 21 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[403] \ 1^{13}31^131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.60 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[405] \ 21^141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 11 & 4 \\ 7 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[406] \ 1^{12}31^{11}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.51 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 12 & 3 \\ 7 & 9 & 12 & 28 & 10 \\ 2 & 3 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[407] \ 21^141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.10 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 13 & 4 \\ 7 & 13 & 28 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[408] \ 1^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.10 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 13 & 4 \\ 7 & 13 & 28 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[409] \ 1^{13}3131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[418] \ 2^332 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 11, \ \Delta = -88, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 11, \ T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 15 & 4 \\ 13 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{12}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[412] \ 1^1 \overset{13}{3} 21^{\overset{13}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[413] \ 2^1 \overset{11}{3} 1^{\overset{11}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[414] \ 1^1 \overset{13}{2} \overset{11}{3} 21^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[415] \ 31^{\overset{11}{13}} 231^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.36 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[416] \ 1^{\overset{13}{12}} \overset{11}{4} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 11 & 4 \\ 9 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[417] \ 131^{\overset{11}{13}} 1^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 4 & 12 & 9 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[421] \ 1^{\overset{13}{13}} 31^{\overset{11}{11}} 1^{\overset{14}{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.64 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 9 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[419] \ 131^{\overset{11}{13}} 1^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.64 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 9 & 3 \\ 3 & 6 & 9 & 13 & 7 \\ 2 & 3 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[420] \ 21^{\overset{12}{11}} \overset{11}{4} 1^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.47 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 15 & 4 \\ 15 & 15 & 45 & 13 \\ 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[422] \ 1^{\overset{21}{11}} \overset{11}{4} 1^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.64 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 9 & 4 \\ 9 & 9 & 13 & 7 \\ 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[428] \ 1^{\overset{11}{11}} \overset{13}{4} 1^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -46, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[427] \ 21^{\overset{11}{11}} \overset{11}{4} 1^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -46, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[429] \quad 1 \overset{1}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 24 & 4 \\ 12 & 24 & 106 & 18 \\ 3 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[430] \quad 2 \overset{1}{2} 3 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[431] \quad 1 \overset{12}{1} \overset{4}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[432] \quad 1 \overset{11}{1} \overset{4}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[423] \quad 21 \overset{1}{4} 1 \overset{16}{6} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 24 & 4 \\ 12 & 24 & 106 & 18 \\ 3 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[424] \quad 1 \overset{11}{1} \overset{4}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 18 & 4 \\ 15 & 18 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[425] \quad 21 \overset{4}{4} 1 \overset{12}{6} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[426] \quad 2 \overset{12}{2} 3 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[433] \quad 1 \overset{12}{1} \overset{4}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 11 & 4 \\ 8 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[434] \quad 1 \overset{12}{1} \overset{4}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[440] \quad 1 \overset{11}{1} \overset{3}{3} 2 \overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[441] \quad 21 \overset{11}{1} \overset{4}{4} 1 \overset{12}{6} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 22 & 4 \\ 21 & 22 & 97 & 19 \\ 4 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[439] $1^{11}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$

[438] $1^{11}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$

[437] $1^{14}31^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 1 \end{pmatrix}_{b_0}$

[436] $1^{11}31^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 1 \end{pmatrix}_{b_0}$

[435] $2^2 2^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[442] $1^{11}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[443] $221^{11}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 2 & 4 \\ 3 & 7 & 3 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$

[444] $1^{12}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[448] $1^{12}41^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 17 & 4 \\ 14 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[447] $1^{11}41^{12}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 12 & 4 \\ 9 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$

[445] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 9 & 3 \\ 4 & 6 & 9 & 13 & 7 \\ 2 & 3 & 3 & 7 & 2 \end{pmatrix}_{b_0}$

[446] $21^{12}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 5 & 30 & 5 \\ 24 & 30 & 134 & 24 \\ 4 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[452] $1^2 1^4 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[451] $1^4 1^4 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[450] $131^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[453] $1^2 1^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[449] $1^{13} 131^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 6 & 15 & 12 & 49 & 13 \\ 2 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[456] $1^{11} 131^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[454] $1^{14} 3131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 46 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[455] $131^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 1 & 4 & 0 & 9 & 2 \\ 4 & 9 & 9 & 28 & 10 \\ 2 & 3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[459] $1^2 1^3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[458] $1^{11} 131^{11} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[457] $1^{12} 3131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[461] $131^{11} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 5 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[460] $1 \overset{21}{4} \overset{11}{1} \overset{13}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 17 & 4 \\ 15 & 17 & 49 & 13 \\ 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[466] $1 \overset{11}{4} \overset{13}{1} \overset{11}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 15 & 4 \\ 12 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[467] $21 \overset{1}{2} \overset{1}{4} \overset{1}{1} \overset{12}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[469] $1 \overset{11}{4} \overset{13}{1} \overset{11}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[468] $1 \overset{1}{4} \overset{1}{1} \overset{14}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 19 & 4 \\ 10 & 19 & 65 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[462] $21 \overset{11}{4} \overset{11}{1} \overset{11}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 12 & 4 \\ 12 & 12 & 26 & 10 \\ 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[463] $1 \overset{12}{4} \overset{1}{1} \overset{1}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[464] $21 \overset{11}{3} \overset{1}{4} \overset{11}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[465] $1 \overset{12}{4} \overset{13}{1} \overset{1}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[471] $31 \overset{11}{1} \overset{11}{2} \overset{1}{31} \overset{1}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[470] $1 \overset{1}{4} \overset{1}{1} \overset{13}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[479] $1 \overset{1}{3} \overset{12}{3} \overset{1}{31} \overset{1}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$$\begin{aligned}
& T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 5 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[477] \quad & 1^{11} 31^{13} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.79 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 2 \end{pmatrix}_{b_0} \\
[480] \quad & 1^{12} 31^{23} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.14 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0} \\
[481] \quad & 131^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.71 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 4 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[482] \quad & 2^1 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 11, \quad \Delta = -88, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 11, \quad T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[483] \quad & 21^2 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.36 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 15 & 4 \\ 11 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[484] \quad & 131^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.64 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 12 & 3 \\ 4 & 9 & 12 & 28 & 10 \\ 2 & 3 & 3 & 10 & 2 \end{pmatrix}_{b_0} \\
[485] \quad & 1^{12} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.36 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[474] \quad & 21^{11} 1^4 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.64 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 13 & 4 \\ 12 & 13 & 28 & 10 \\ 4 & 4 & 10 & 3 \end{pmatrix}_{b_0} \\
[478] \quad & 21^3 41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.36 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[472] \quad & 1^{11} 41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.36 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 15 & 4 \\ 11 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[475] \quad & 21^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 4 & 12 & 4 \\ 8 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[476] $1^{\overline{21}1^2} 1^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$

[473] $21^{\overline{11}41^1} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 12 & 4 \\ 8 & 13 & 20 & 7 \\ 12 & 20 & 26 & 10 \\ 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[486] $31^{\overline{11}231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$

[490] $1^{\overline{1}4^1} 1^{\overline{11}} 4^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 16 & 4 \\ 9 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$

[489] $1^{\overline{11}31^2} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$

[491] $1^{\overline{13}31^1} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 7 & 15 & 12 & 49 & 13 \\ 2 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$

[492] $1^{\overline{11}31^2} 31^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 8 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$

[493] $1^{\overline{11}31^1} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 4 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$

[488] $1^{\overline{1}4^1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 19, T \in [15, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 28 & 4 \\ 14 & 28 & 145 & 21 \\ 3 & 4 & 21 & -1 \end{pmatrix}_{b_0}$

[487] $21^{\overline{1}41^{\overline{17}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 19, T \in [15, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 28 & 4 \\ 14 & 28 & 145 & 21 \\ 3 & 4 & 21 & -1 \end{pmatrix}_{b_0}$

[494] $1^{\overline{11}} 1^{\overline{13}} 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 17 & 4 \\ 9 & 17 & 49 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$

[495] $1^{\overline{12}31^1} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 6 & 12 & 9 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$

- [496] $21 \overset{11}{4} 1 \overset{13}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 17 & 4 \\ 9 & 17 & 49 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [498] $1 \overset{13}{1} \overset{11}{4} 1 \overset{1}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$
- [497] $21 \overset{11}{1} \overset{13}{4} 1 \overset{13}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$
- [499] $21 \overset{13}{1} \overset{11}{4} 1 \overset{13}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 26 & 4 \\ 24 & 26 & 130 & 22 \\ 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [500] $31 \overset{11}{1} \overset{13}{3} 1 \overset{1}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [501] $21 \overset{11}{1} \overset{13}{4} 1 \overset{14}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 27 & 4 \\ 5 & 4 & 30 & 4 \\ 27 & 30 & 169 & 25 \\ 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [502] $21 \overset{11}{1} \overset{11}{4} 1 \overset{14}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 24 & 4 \\ 21 & 24 & 101 & 19 \\ 4 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [503] $1 \overset{12}{1} \overset{13}{3} 1 \overset{13}{1} \overset{13}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [504] $1 \overset{14}{4} 1 \overset{12}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 25 & 4 \\ 15 & 25 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [507] $1 \overset{13}{1} \overset{12}{3} 21 \overset{11}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [513] $1 \overset{13}{1} \overset{13}{3} 21 \overset{11}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [512] $1 \overset{14}{1} \overset{13}{3} 21 \overset{13}{+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[511] $1^{11}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[510] $221^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 8 & 6 \\ 3 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[509] $1^{12}31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[508] $1^{13}3^{14}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[505] $1^{11}3^{12}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[506] $1^{13}3^{12}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 20 & 5 \\ 13 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[518] $1^{12}31^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[517] $1^{12}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 5 & 12 & 9 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[515] $1^{11}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 20 & 5 \\ 10 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[519] $21^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[516] $1^{12}3131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[514] $21^{11}41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 19 & 4 \\ 18 & 19 & 68 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[520] $1 \overset{1^2}{4} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -69, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 19 & 4 \\ 11 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[525] $21 \overset{1^2}{3} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[521] $2 \overset{1^{12}}{2} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[522] $221 \overset{1^{12}}{1} 41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 3 & 6 & 2 \\ 3 & 3 & 3 & 10 & 3 \\ 5 & 6 & 10 & 20 & 8 \\ 2 & 2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[523] $1 \overset{1^3}{4} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 22 & 4 \\ 13 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[524] $21 \overset{1^1}{1} 41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 16 & 4 \\ 15 & 16 & 45 & 13 \\ 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[529] $21 \overset{1^2}{1} 41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 27 & 4 \\ 24 & 27 & 132 & 22 \\ 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[527] $1 \overset{1^2}{1} 2 \overset{1^2}{3} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[526] $1 \overset{1^2}{1} 31^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 12 & 3 \\ 6 & 9 & 12 & 28 & 10 \\ 2 & 3 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[530] $1 \overset{1^4}{1} 321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[528] $1 \overset{1^2}{1} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[532] $221 \overset{1^1}{1} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 3 & 4 & 2 \\ 3 & 3 & 3 & 8 & 3 \\ 4 & 4 & 8 & 10 & 6 \\ 2 & 2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[531] $21^{11}41^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 20 & 4 \\ 18 & 20 & 70 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[533] $141^{12}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 22 & 4 \\ 12 & 22 & 90 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[536] $31^{11}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 7 & 3 \\ 5 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[535] $1^{13}41^{12}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 15 & 4 \\ 10 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[534] $21^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 4 & 11 & 4 \\ 8 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[537] $1^{11}31^331^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[540] $2^3 3^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 5 & 2 \\ 2 & 1 & 4 & 5 & 3 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[538] $2^2 2^3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[541] $2^3 3^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[542] $2^3 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 6 & 25 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[539] $21^{12}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[543] $21^{11}41^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$

$$\begin{aligned}
& T_{\min} = 10.48 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 23 & 4 \\ 21 & 23 & 97 & 19 \\ 4 & 4 & 19 & 3 \end{pmatrix}_{b_0} \\
[544] \quad & 1^{12}31^{11}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -77, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.48 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 12 & 3 \\ 5 & 9 & 12 & 28 & 10 \\ 2 & 3 & 3 & 10 & 2 \end{pmatrix}_{b_0} \\
[548] \quad & 1^{13}3131^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.27 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[551] \quad & 1^{12}41^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.57 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[552] \quad & 231^{11}3 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -94, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.16 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[550] \quad & 1^{13}31^231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.27 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[545] \quad & 1^{11}31^231^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.27 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[546] \quad & 21^141^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.57 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[547] \quad & 1^{13}41^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.57 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 15 & 4 \\ 12 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[549] \quad & 1^{12}31^331^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.27 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0} \\
[553] \quad & 21^{13}41^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -80, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.40 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[554] \quad & 1^131^{13}31^2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -80, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.40 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 8 & 18 & 9 & 52 & 12 \\ 2 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[557] \quad & 1^{13}41^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -105, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 15 & 4 \\ 11 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[556] \quad 1^{11} 21^4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 21 & 4 \\ 11 & 21 & 76 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[555] \quad 1^{11} 1^4 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 21 & 4 \\ 11 & 21 & 76 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[558] \quad 1^{14} 1^{12} 3 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 20 & 5 \\ 13 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[559] \quad 1^{11} 1^{12} 3 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[560] \quad 1^{11} 3 1^{13} 3 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 11 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[561] \quad 1^{12} 1^{12} 3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[562] \quad 1^{14} 1^{12} 3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[563] \quad 1^{14} 1^4 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 19 & 4 \\ 11 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[564] \quad 21^4 1^4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 19 & 4 \\ 11 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[565] \quad 1^4 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 20 & 4 \\ 11 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[567] \quad 1^3 1^4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 25 & 4 \\ 14 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[566] $1^{13}31^{11}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 8 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[569] $1^{11}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[568] $1^{12}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[570] $1^{14}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 7 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[571] $221^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 6 & 8 & 3 \\ 6 & 13 & 20 & 7 \\ 8 & 20 & 26 & 10 \\ 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[572] $1^{13}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 6 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[573] $1^4 141^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 28 & 4 \\ 16 & 28 & 158 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[574] $21^2 41^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[575] $1^{12} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[576] $1^{12} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 16 & 4 \\ 11 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[577] $1^{11}31^4 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[578] $1^{13} 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

- [579] $1^{11}31^231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -91$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.58 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$
- [584] $1^{13}31^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -92$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.09 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$
- [580] $21^{14}41^{11}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -92$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 30 & 4 \\ 8 & 13 & 56 & 7 \\ 30 & 56 & 212 & 28 \\ 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$
- [581] $1^{14}31^{11}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -92$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.52 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 10 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$
- [583] $21^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -92$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 22 & 4 \\ 15 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [582] $21^{12}41^{11}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -92$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 18 & 4 \\ 8 & 13 & 32 & 7 \\ 18 & 32 & 68 & 16 \\ 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$
- [587] $1^{12}31^131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -95$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.05 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$
- [585] $1^{11}31^231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -95$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 1 \end{pmatrix}_{b_0}$
- [586] $1^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -95$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.47 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$
- [588] $21^{13}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [591] $131^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 6 & 18 & 9 & 52 & 12 \\ 2 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [590] $2^13^12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0}$
- [589] $1^14^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[592] \ 2 \ 2 \ 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[594] \ 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[593] \ 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 18 & 4 \\ 13 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[596] \ 1^{12} 31^2 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[595] \ 1^{14} 31^2 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[597] \ 2 \ 2 \ 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[598] \ 1^{12} 31^4 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[599] \ 31^{12} 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 17 & 3 \\ 5 & 5 & 25 & 5 \\ 17 & 25 & 104 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[600] \ 31^{12} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 8 & 3 \\ 5 & 5 & 10 & 5 \\ 8 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[602] \ 31^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 4 & 2 \\ 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 2 \\ 4 & 4 & 4 & 5 & 5 \\ 2 & 3 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[601] \ 1^{13} 31^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 7 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[603] \ 1^{13} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[611] \ 1 \overset{1^2}{4} \overset{1^1}{1} \overset{1^3}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 23 & 4 \\ 13 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[604] \ 2 \overset{1^1}{3} \overset{1^4}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 46 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[606] \ 31 \overset{1^2}{12} 231 \overset{1^4}{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 3 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[607] \ 21 \overset{1^3}{13} 41 \overset{1^2}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 8 & 24 & 4 \\ 8 & 13 & 44 & 7 \\ 24 & 44 & 130 & 22 \\ 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[608] \ 1 \overset{1^1}{31} \overset{1^4}{13} 31 \overset{1^3}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 11 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[609] \ 1 \overset{1^1}{31} \overset{1^3}{13} 31 \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 7 & 18 & 9 & 52 & 12 \\ 2 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[610] \ 1 \overset{1^1}{13} 1 \overset{1^3}{31} \overset{1^2}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[613] \ 1 \overset{1^1}{12} \overset{1^1}{3} 21 \overset{1^3}{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 25 & 5 \\ 13 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[612] \ 1 \overset{1^4}{13} 131 \overset{1^2}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 46 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[605] \ 131 \overset{1^2}{12} 31 \overset{1^4}{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 8 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[615] \ 1 \overset{1^2}{12} \overset{1^2}{4} 1 \overset{1^1}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[614] \ 21 \overset{1^2}{2} 4 \overset{1^1}{1} \overset{1^2}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[616] $1^{14}41^{12}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 19 & 4 \\ 12 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[618] $1^{11}31^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 7 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[620] $1^{13}3131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[619] $1^{21}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 13 & 4 \\ 9 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[617] $131^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 5 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[621] $1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[623] $1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[622] $21^{34}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[624] $1^{11}43131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 1 & 4 & 0 & 12 & 2 \\ 6 & 12 & 12 & 46 & 12 \\ 2 & 3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[625] $1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 19 & 4 \\ 14 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[626] $1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 21 & 4 \\ 16 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[627] $1^{11}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 25 & 5 \\ 13 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[628] \quad 2 \quad 2 \quad 31 \overset{1^{12}}{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[629] \quad 1 \overset{1^2}{1^{12}} 3 21 \overset{1^3}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[630] \quad 1 \overset{1^1}{1^3} 4 1 \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[631] \quad 1 \overset{1^3}{1^{11}} 4 1 \overset{1^3}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 26 & 4 \\ 15 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[632] \quad 21 \overset{1^{11}}{1^4} 4 1 \overset{1^{13}}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[633] \quad 1 \overset{1^4}{1^3} 4 1 \overset{1^1}{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[635] \quad 1 \overset{1^1}{1^4} 4 1 \overset{1^{14}}{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 24 & 4 \\ 13 & 24 & 101 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[636] \quad 21 \overset{1^1}{1^4} 4 1 \overset{1^{13}}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[637] \quad 1 \overset{1^2}{1^3} 3 1 \overset{1^3}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[634] \quad 1 \overset{1^4}{1^4} 3 1 \overset{1^1}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[638] \quad 1 \overset{1^4}{1^4} 3 1 \overset{1^1}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 8 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[639] \ 1^4 4^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -116, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.69 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 29 & 4 \\ 17 & 29 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[640] \ 1^{13} 3^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -116, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.24 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[641] \ 1^{14} 3^1 1^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -117, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 9 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[642] \ 2^2 2^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -118, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[643] \ 2^2 2^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -118, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[650] \ 2^1 3^1 4^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -119, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[646] \ 1^{12} 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -119, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 25 & 5 \\ 16 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[645] \ 1^{13} 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -119, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[647] \ 1^{12} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -119, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 22 & 4 \\ 17 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[644] \ 1^{12} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -119, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[648] \ 1^{12} 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -119, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[649] \ 3^1 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -119, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.51 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 7 & 2 \\ 7 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[651] \ 1^{14} 4^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -120, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[653] \ 21^4 4^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -121, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.37 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 25 & 4 \\ 19 & 25 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[652] \ 1^{12} 4^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -121, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.37 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 25 & 4 \\ 19 & 25 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[656] \ 21^{12} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -123, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 19 & 4 \\ 13 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[655] \ 1^{11} 31^{12} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -123, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 6 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[654] \ 21^{13} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -123, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[661] \ 1^{12} 3^{13} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -124, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[660] \ 1^{11} 31^{13} 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -124, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.58 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[659] \ 2^1 3^1 1^{13} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -124, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.13 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 9 & 2 \\ 3 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 11 & 2 \\ 9 & 9 & 11 & 40 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[658] \ 2^1 31^{12} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -124, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.23 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 6 & 2 \\ 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[657] \ 1^{12} 3^{13} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -124, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[662] \ 1^{14} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -126, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.63 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 19 & 4 \\ 14 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[665] $1^{13}3^{14}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[663] $1^{12}31^{13}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[664] $1^{13}3131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[666] $1^{11}3^{13}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[668] $1^{13}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[667] $1^{11}31^{14}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[669] $21^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 4 & 16 & 4 \\ 11 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[670] $1^{12}3^{14}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[673] $21^241^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[671] $221^{12}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 6 & 10 & 3 \\ 6 & 13 & 24 & 7 \\ 10 & 24 & 38 & 12 \\ 3 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[672] $1^{14}41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[674] $131^{12}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 7 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[675] $21^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 4 & 15 & 4 \\ 11 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[676] $1^{13}31^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 11 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[677] $1^{12}31^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[680] $1^{11}321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[679] $131^{12}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 6 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[678] $1^{12}31^{12}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 9 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[681] $1^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 20 & 4 \\ 13 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[682] $2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[685] $1^{12}31^{12}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[684] $1^{13}31^{13}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[683] $1^{11}31^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[686] $1^{13}31^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[687] \quad 2 \overset{1}{\underset{1^{15}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 15 & 10 & 73 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[689] \quad 2 \overset{1^{12}}{\underset{1^{11}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 6 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[690] \quad 2 \overset{1^{12}}{2} 2 \overset{1^{14}}{3} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[688] \quad 21 \overset{1^2}{\underset{1^{11}}{1^3}} 41 \overset{1^2}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 18 & 4 \\ 13 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[691] \quad 1 \overset{1^2}{\underset{1^{11}}{4}} 1 \overset{1^4}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 27 & 4 \\ 15 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[692] \quad 2 \overset{1^{11}}{3} 1 \overset{1^3}{1} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 5 & 2 \\ 3 & 1 & 3 & 5 & 3 \\ 3 & 3 & 0 & 5 & 2 \\ 2 & 3 & 5 & 8 & 6 \\ 5 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[694] \quad 1 \overset{1^3}{\underset{1^1}{1^3}} 41 \overset{1^3}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -146, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.55 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[693] \quad 1 \overset{1^3}{\underset{1^2}{1^4}} 41 \overset{1^3}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -146, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.55 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 22 & 4 \\ 15 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[695] \quad 1 \overset{1^2}{1^3} 321 \overset{1^3}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 30 & 5 \\ 17 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[696] \quad 1 \overset{1^4}{1^3} 321 \overset{1^3}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 5 & 30 & 5 \\ 19 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[697] \quad 1 \overset{1^4}{1^3} 3131 \overset{1^3}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 46 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[698] $1^{13}4^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.53 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[699] $1^{13}31^{43}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0}$

[700] $1^{14}31^{33}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$

[701] $1^{14}4^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$

[702] $21^14^11^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$

[703] $21^{13}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[707] $131^{14}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 8 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$

[708] $1^{12}3^{13}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 9 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[709] $1^{13}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$

[706] $1^131^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 10 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$

[704] $1^{14}3^{12}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[705] $21^{14}1^41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 16 & 4 \\ 12 & 16 & 46 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[710] $1^{1\bar{1}4}31^131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 1 & 4 & 0 & 15 & 2 \\ 7 & 15 & 15 & 76 & 16 \\ 2 & 3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[711] $1^{1\bar{1}4}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[712] $1^{1\bar{1}1}31^{1\bar{3}}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 10 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[713] $1^{1\bar{1}1}31^{1\bar{3}}31^{1\bar{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[714] $1^{1\bar{3}}41^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 23 & 4 \\ 15 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[715] $21^{1\bar{3}}321^{1\bar{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[716] $21^241^{1\bar{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 23 & 4 \\ 15 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[718] $31^{1\bar{2}}231^{1\bar{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 3 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[717] $1^341^{1\bar{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 30 & 4 \\ 17 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[719] $2^31^{1\bar{1}2}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 7 & 2 \\ 3 & 1 & 3 & 7 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 7 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[720] $1^{1\bar{2}1}31^{1\bar{4}}31^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[721] $2^2231^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[722] $1^{12}3^{12}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[725] $1^{12}3^{12}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 30 & 5 \\ 16 & 30 & 134 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[727] $31^{12}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 3 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[728] $1^{12}31^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 8 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[729] $1^{12}31^{231}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[730] $1^{12}3^{14}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[731] $1^{14}3^{13}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[732] $1^{14}4^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 25 & 4 \\ 17 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[723] $1^{12}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 7 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[724] $2^{11}3^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 8 & 7 & 21 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[726] $21^{13}4^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 25 & 4 \\ 17 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[733] $131^{231}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 18 & 3 \\ 1 & 4 & 0 & 18 & 2 \\ 8 & 18 & 18 & 118 & 20 \\ 2 & 3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[737] $1^1 31^{14} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 9 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[736] $1^1 31^{14} 31^{12} 31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 12 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[735] $1^1 31^{14} 31^{12} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 9 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[734] $1^1 31^{14} 31^{12} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[738] $1^1 41^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 33 & 4 \\ 19 & 33 & 212 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[739] $1^1 31^{13} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 23 & 4 \\ 16 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[741] $1321^{11} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[742] $31^{11} 31^{11} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[740] $21^{13} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 4 & 29 & 4 \\ 20 & 29 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[743] $2^1 2^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 98 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[744] $1^1 31^{13} 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[745] $1^1 31^{14} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$\begin{aligned}
& T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 25 & 4 \\ 18 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[746] & 1^{12}31^331^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0} \\
[747] & 31^{13}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 34 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0} \\
[748] & 1^{11}31^431^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0} \\
[749] & 2^131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[750] & 1^{14}31^131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0} \\
[752] & 1^{14}31^231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[751] & 1^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 23 & 4 \\ 17 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0} \\
[754] & 1^{12}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 30 & 5 \\ 17 & 30 & 134 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0} \\
[760] & 1^{13}31^{12}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 10 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[759] & 1^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[758] & 31^{13}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 10 & 2 \\ 10 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[757] & 1^{13}31^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 8 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[753] $131^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 7 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 24 & 4 \\ 15 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[755] $1^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 24 & 4 \\ 15 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[756] $1^{13}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[762] $31^{12}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 32 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[761] $31^{12}231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 14 & 3 \\ 5 & 5 & 20 & 5 \\ 14 & 20 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[763] $1^{12}31^231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[764] $1^{12}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 30 & 5 \\ 18 & 30 & 134 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[765] $1^{13}31^331^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[766] $2^13^12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 0 & 12 & 2 \\ 12 & 18 & 12 & 106 & 18 \\ 2 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[767] $1^{14}3131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 46 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[771] $1^{14}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 28 & 4 \\ 20 & 28 & 158 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[768] $21^441^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 28 & 4 \\ 20 & 28 & 158 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[769] \ 1^{13} 4^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[770] \ 21^{11} 3^{13} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[772] \ 1^{13} 3^{14} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[773] \ 1^{13} 31^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -191, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[774] \ 1^{13} 4^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -194, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 26 & 4 \\ 19 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[775] \ 1^{11} 31^{13} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 9 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[776] \ 1^{13} 31^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 9 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[779] \ 231^{13} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 9 & 2 \\ 3 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 9 & 6 & 25 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[778] \ 1^{14} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[777] \ 1^{13} 3^{12} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[782] \ 21^{14} 4^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 4 & 28 & 4 \\ 20 & 28 & 158 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[783] \ 1^{14} 31^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$\begin{aligned}
& T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 10 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0} \\
[781] & 1^{13}31^{13}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[780] & 1^{14}31^{12}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 11 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0} \\
[784] & 1^{14}31^431^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0} \\
[787] & 21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.66 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 19 & 4 \\ 14 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[785] & 1^{11}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 8 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[786] & 1^{11}31^{14}31^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 13 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0} \\
[790] & 21^4 4 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 29 & 4 \\ 21 & 29 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[789] & 1^{12}31^{13}31^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 12 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[788] & 1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 29 & 4 \\ 21 & 29 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[791] & 221^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 6 & 13 & 3 \\ 6 & 13 & 32 & 7 \\ 13 & 32 & 68 & 16 \\ 3 & 7 & 16 & 3 \end{pmatrix}_{b_0} \\
[792] & 1^{13}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 5 & 35 & 5 \\ 19 & 35 & 188 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0} \\
[793] & 21^2 4 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 27 & 4 \\ 17 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[794] $1^{14}4^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 27 & 4 \\ 17 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[795] $1^{12}3^{13}3^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[796] $1^{14}4^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[797] $1^{14}3^{12}2^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[798] $2^{13}4^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 26 & 4 \\ 18 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[799] $1^{12}3^{14}3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[800] $2^2 2^{13}3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[801] $2^{13}4^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 20 & 4 \\ 14 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[802] $1^{13}3^{14}2^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[803] $1^{13}3^{14}2^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 35 & 5 \\ 20 & 35 & 188 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[805] $13^{15}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 10 & 30 & 15 & 148 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[804] $1^2 4^{15}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 20 & 4 \\ 15 & 20 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[806] \ 21^{1^3}41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 25 & 4 \\ 18 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[809] \ 1^{1^3}31^{4^3}1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[808] \ 1^{1^4}31^{1^3}1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[807] \ 21^{1^3}41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 4 & 23 & 4 \\ 16 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[811] \ 1^{1^3}321^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 5 & 35 & 5 \\ 21 & 35 & 188 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[810] \ 21^{1^4}41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 4 & 22 & 4 \\ 16 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[812] \ 1^{1^4}41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 27 & 4 \\ 18 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[814] \ 2^2 2^3 31^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[813] \ 1^{1^4}321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[817] \ 1^{1^3}31^{2^3}1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[816] \ 1^{1^4}31^{3^3}1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[815] \ 221^{1^3}41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$\begin{aligned}
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 6 & 15 & 3 \\ 6 & 13 & 36 & 7 \\ 15 & 36 & 88 & 18 \\ 3 & 7 & 18 & 3 \end{pmatrix}_{b_0} \\
[818] & 131^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 8 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[819] & 2^1 2^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[820] & 2^1 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 21 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 14 & 21 & 14 & 145 & 21 \\ 2 & 3 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[821] & 21^{14}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[822] & 2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0} \\
[823] & 2^1 2^1 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0} \\
[824] & 2^1 2^1 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0} \\
[825] & 31^{14}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 3 & 2 & 0 & 13 & 2 \\ 13 & 6 & 13 & 56 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[826] & 1^{12}31^3 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0} \\
[827] & 1^{13}3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[828] & 2^1 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 11 & 3 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 11 & 9 & 40 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[830] $1^{13}31^331^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[829] $1^{14}31^231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[831] $1^{14}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[832] $1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -245, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 27 & 4 \\ 19 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[833] $21^341^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 30 & 4 \\ 20 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[835] $1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 30 & 4 \\ 20 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[834] $31^{13}231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 35 & 5 \\ 24 & 35 & 202 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$$

[836] $21^{14}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[837] $1^{12}31^431^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[838] $1^{14}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 5 & 40 & 5 \\ 23 & 40 & 254 & 32 \\ 3 & 5 & 32 & 4 \end{pmatrix}_{b_0}$$

[839] $1^{12}31^{13}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 11 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[841] $1^{14}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[840] $1^{14}3^{14}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 5 & 40 & 5 \\ 24 & 40 & 254 & 32 \\ 3 & 5 & 32 & 4 \end{pmatrix}_{b_0}$$

[842] $2^{112}3^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 8 & 10 & 10 & 38 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[843] $131^{133}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 9 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[844] $131^{143}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 9 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[845] $1^{11}31^{14}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 12 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[846] $1^4 4^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[851] $1^{14}31^{13}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[847] $1^{14}4^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 30 & 4 \\ 21 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[848] $2^2 2^{13}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[849] $21^{14}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[850] $1^{13}31^{23}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[852] $131^{13}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 10 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

- [853] $1^{12}31^{13}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 9 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [854] $1^{11}231^{12}3 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 8 & 2 \\ 3 & 1 & 3 & 8 & 3 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 8 & 7 & 21 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [855] $31^{14}231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 16 & 3 \\ 5 & 5 & 20 & 5 \\ 16 & 20 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$
- [856] $31^{13}231^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 3 & 2 & 0 & 11 & 2 \\ 11 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [857] $1^{12}31^{13}31^2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 10 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [858] $31^{12}31^{11}3 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 3 & 9 & 2 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [859] $21^441^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 4 & 33 & 4 \\ 23 & 33 & 212 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$
- [860] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 4 & 33 & 4 \\ 23 & 33 & 212 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$
- [861] $1^{14}321^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [862] $1^{12}31^431^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$
- [863] $1^{14}31^431^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [864] $1^{13}31^431^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

- [865] $1^{13}31^331^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -296$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.09 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$
- [866] $1^{13}31^{13}31^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -299$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.34 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 13 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$
- [867] $1^{11}31^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -299$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 10 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$
- [868] $1^{11}31^{14}31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -299$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.44 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 11 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$
- [869] $1^{14}31^231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -304$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.07 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$
- [870] $2^131^{14}3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -304$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 12 & 2 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 12 & 8 & 46 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [871] $1^{14}31^331^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -311$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.04 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$
- [874] $131^{16}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -312$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.2$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 12 & 36 & 18 & 214 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$
- [872] $1^14^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -312$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.2$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 24 & 4 \\ 18 & 24 & 106 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [873] $1^{14}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -312$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [875] $31^{13}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -315$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 15 & 3 \\ 5 & 5 & 20 & 5 \\ 15 & 20 & 58 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$
- [876] $2^12^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -316$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.3$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 194 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$

[877] $21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[878] $1^{13}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 10 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[879] $21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 23 & 3 \\ 4 & 4 & 33 & 4 \\ 23 & 33 & 212 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[880] $2^3 3^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 11 & 65 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[881] $2^2 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 134 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[882] $2^2 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -327, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[883] $31^{13}231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 12 & 2 \\ 12 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[884] $1^{13}31^{13}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 12 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[885] $1^{13}31^{13}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[886] $31^{13}231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 3 & 2 & 0 & 13 & 2 \\ 13 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[887] $1^{14}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 11 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[888] $1^{13}31^{13}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 11 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

- [889] $21^{14}41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 4 & 27 & 4 \\ 19 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [890] $21^{14}41^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 30 & 4 \\ 21 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [892] $1^{13}31^{431}^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0}$
- [891] $31^{13}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 21 & 3 \\ 5 & 5 & 30 & 5 \\ 21 & 30 & 142 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$
- [895] $1^{12}31^{14}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 14 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$
- [893] $131^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 10 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$
- [894] $1^{14}31^{231}^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$
- [896] $221^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -366, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 6 & 18 & 3 \\ 6 & 13 & 44 & 7 \\ 18 & 44 & 130 & 22 \\ 3 & 7 & 22 & 3 \end{pmatrix}_{b_0}$
- [897] $1^{14}31^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -368, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 14 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$
- [898] $31^{13}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 18 & 3 \\ 5 & 5 & 25 & 5 \\ 18 & 25 & 94 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$
- [899] $1^{14}31^{431}^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0}$
- [900] $1^{14}31^{331}^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$
- [901] $221^{14}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 6 & 20 & 3 \\ 6 & 13 & 48 & 7 \\ 20 & 48 & 158 & 24 \\ 3 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[902] $2^1 2^1 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 188 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[903] $2^1 3^1 1^{12} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 10 & 2 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 10 & 10 & 38 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[904] $1^{14} 31^{13} 31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 12 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[905] $31^{13} 31^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -387, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 3 & 12 & 2 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[906] $1^{14} 31^{13} 31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -392, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 13 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[907] $2^1 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 0 & 12 & 2 \\ 10 & 13 & 12 & 61 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[908] $2^1 2^1 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[910] $1^{12} 31^{14} 31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 13 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[909] $1^{12} 31^{14} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 11 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[911] $31^{14} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 14 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 3 & 2 & 0 & 14 & 2 \\ 14 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[912] $2^1 2^1 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 24 & 3 \\ 4 & 5 & 40 & 5 \\ 24 & 40 & 254 & 32 \\ 3 & 5 & 32 & 4 \end{pmatrix}_{b_0}$$

[913] $1^{13} 31^{14} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -407, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

[914] $31^{15}231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 20 & 3 \\ 5 & 5 & 25 & 5 \\ 20 & 25 & 98 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

[916] $131^{17}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 42 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 14 & 42 & 21 & 292 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[915] $1^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 28 & 4 \\ 21 & 28 & 145 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[917] $131^{14}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -419, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 11 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[918] $1^{12}31^{14}31^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -420, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 12 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[920] $2^1 3^1 1^{12}3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 12 & 2 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 13 & 2 \\ 12 & 12 & 13 & 61 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[921] $1^{14}31^3 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[919] $2^1 3^1 1^{15}3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 15 & 2 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 15 & 15 & 10 & 73 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[923] $2^1 3^1 1^{12}3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 14 & 2 \\ 3 & 1 & 3 & 14 & 3 \\ 2 & 3 & 0 & 16 & 2 \\ 14 & 14 & 16 & 90 & 18 \\ 2 & 3 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[922] $2^1 3^1 1^{13}3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 11 & 2 \\ 3 & 1 & 3 & 11 & 3 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 11 & 9 & 40 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[924] $1^{14}31^4 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -452, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[925] $131^{14}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 12 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0} \\
[926] \quad & 31^{14}31^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 3 & 3 & 15 & 2 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0} \\
[927] \quad & 31^{14}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 19 & 3 \\ 5 & 5 & 25 & 5 \\ 19 & 25 & 92 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0} \\
[928] \quad & 1^{13}31^{14}31^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 15 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[929] \quad & 1^{13}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 12 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[930] \quad & 1^{13}31^{14}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -515, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 13 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[931] \quad & 1^{13}31^{14}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -515, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 14 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[932] \quad & 31^{14}231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -519, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 3 & 2 & 0 & 15 & 2 \\ 15 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[933] \quad & 1^{14}31^431^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -520, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0} \\
[934] \quad & 2^13^12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 16 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 12 & 16 & 14 & 90 & 18 \\ 2 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[935] \quad & 1^{14}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 13 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0} \\
[936] \quad & 221^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -567, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 6 & 23 & 3 \\ 6 & 13 & 56 & 7 \\ 23 & 56 & 212 & 28 \\ 3 & 7 & 28 & 3 \end{pmatrix}_{b_0} \\
[937] \quad & 221^{15}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 6 & 25 & 3 \\ 6 & 13 & 60 & 7 \\ 25 & 60 & 248 & 30 \\ 3 & 7 & 30 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [938] $31^{16}231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -580, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 30 & 5 \\ 24 & 30 & 142 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$
- [941] $1^{14}31^{14}31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 16 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$
- [940] $1^{14}31^{14}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 14 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$
- [939] $31^{14}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 16 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 3 & 2 & 0 & 16 & 2 \\ 16 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$
- [942] $31^{14}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 28 & 3 \\ 5 & 5 & 40 & 5 \\ 28 & 40 & 254 & 32 \\ 3 & 5 & 32 & 3 \end{pmatrix}_{b_0}$
- [943] $31^{14}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 30 & 5 \\ 22 & 30 & 134 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$
- [944] $1^{14}31^{14}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 15 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$
- [945] $1^{11}231^{14}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -639, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 14 & 2 \\ 3 & 1 & 3 & 14 & 3 \\ 2 & 3 & 0 & 11 & 2 \\ 14 & 14 & 11 & 65 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [946] $1^{12}231^{13}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -639, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 13 & 2 \\ 3 & 1 & 3 & 13 & 3 \\ 2 & 3 & 0 & 12 & 2 \\ 13 & 13 & 12 & 61 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [947] $31^{14}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 25 & 3 \\ 5 & 5 & 35 & 5 \\ 25 & 35 & 188 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$
- [948] $31^{17}231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -775, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 28 & 3 \\ 5 & 5 & 35 & 5 \\ 28 & 35 & 194 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$
- [949] $1^{13}231^{13}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 15 & 2 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 0 & 15 & 2 \\ 15 & 15 & 15 & 88 & 18 \\ 2 & 3 & 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [950] $221^{16}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -802, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 6 & 30 & 3 \\ 6 & 13 & 72 & 7 \\ 30 & 72 & 358 & 36 \\ 3 & 7 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[951] \quad 2 \overset{1^{14}}{3} 1^{13} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -879, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 17 & 2 \\ 3 & 1 & 3 & 17 & 3 \\ 2 & 3 & 0 & 18 & 2 \\ 17 & 17 & 18 & 121 & 21 \\ 2 & 3 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[952] \quad 2 \overset{1^{12}}{3} 1^{14} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -924, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 16 & 2 \\ 3 & 1 & 3 & 16 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 16 & 16 & 14 & 90 & 18 \\ 2 & 3 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[953] \quad 221 \overset{1^7}{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -1075, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 35 & 3 \\ 6 & 13 & 84 & 7 \\ 35 & 84 & 488 & 42 \\ 3 & 7 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[954] \quad 2 \overset{1^{13}}{3} 1^{14} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 18 & 2 \\ 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 0 & 17 & 2 \\ 18 & 18 & 17 & 121 & 21 \\ 2 & 3 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[955] \quad 2 \overset{1^{14}}{3} 1^{14} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 20 & 2 \\ 3 & 1 & 3 & 20 & 3 \\ 2 & 3 & 0 & 20 & 2 \\ 20 & 20 & 20 & 158 & 24 \\ 2 & 3 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [956] \quad 1^1 \overset{1^1}{2} 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[957] \quad 1^1 \overset{1^1}{3} 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 9, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9, T_{\max} = 9, T \in [9, 9]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[961] \quad 1^1 2 3 1 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[962] \quad 1^{11} \overset{1^1}{2} 3 2 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 17 & 9 \\ 2 & 3 & -1 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[963] \quad 1^{13} \overset{1^1}{2} 3 2 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 8 & 0 & 13 & 97 & 21 \\ 2 & 0 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[964] \quad 1^1 \overset{1^1}{2} 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 0 & 1 & 1 & 3 \\ 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[965] $1^{\overline{1}} 4 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 3 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[958] $1^{\overline{11}} 23131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[960] $1^{\overline{1}} 231^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 4 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[959] $1^{\overline{12}} 3 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[974] $1^{\overline{1}} \begin{smallmatrix} 5 \\ 1,1,1 \end{smallmatrix} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[979] $21^{\overline{14}} 1^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

[980] $221^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[981] $131^{\overline{1}} 4 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[982] $1^{\overline{1}} \begin{smallmatrix} 5 \\ 1,1,1 \end{smallmatrix} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[978] $1^{\overline{11}} 4 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[977] $1^{\overline{1}} 4 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 5 & 3 & 5 \\ 3 & 1 & 3 \\ 5 & 3 & 5 \end{pmatrix}_{b_0}$$

[969] $21^{\overline{1}} \begin{smallmatrix} 5 \\ 1,1 \end{smallmatrix} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[968] \quad 1 \overset{1}{5} 1^{\overset{1}{2}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[967] \quad 4 1^{\overset{1}{1}} 2^{\overset{1}{1}} 3^{\overset{1}{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[966] \quad 1^{\overset{1}{1}} 4 1 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[976] \quad 1^{\overset{1}{1}} 4 1 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 7 & 5 \\ 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[970] \quad 1^{\overset{1}{1}} 4 1 2 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 3 & 5 \\ 3 & 1 & 3 \\ 5 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[975] \quad 1^{\overset{1}{1}} 1^{\overset{1}{1}} 2^{\overset{1}{1}} 3^{\overset{1}{1}} 2 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 4 & 2 \\ 2 & 0 & 4 & 6 & 6 \\ 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[971] \quad 1^{\overset{1}{1}} 4 1^{\overset{1}{1}} 2 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 4 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[972] \quad 2 1^{\overset{1}{1}} 4 1^{\overset{1}{1}} 4 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 16 & 4 \\ 14 & 16 & 60 & 16 \\ 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[973] \quad 1^{\overset{1}{1}} 4 1 2 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[983] \quad 1^{\overset{1}{1}} 3 1^{\overset{1}{1}} 4^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 4 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[984] \quad 1^{\overset{1}{1}} 5^{\overset{1}{1}} 1^{\overset{1}{1}} 3^{\overset{1}{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[985] \quad 1^{\overset{1}{1}} 5^{\overset{1}{1}} 1^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 10 & 4 \\ 10 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[987] $21 \overset{1^4}{1} \overset{1^1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[986] $1 \overset{1^3}{1^3, 1^1} \overset{1^1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[988] $21 \overset{1^{11}}{1} \overset{1^1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 10 & 4 \\ 9 & 10 & 19 & 9 \\ 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[989] $1 \overset{1^1}{1^{11}} 2^3 2^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[990] $21 \overset{1^1}{1} \overset{1^1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 10 & 4 \\ 9 & 10 & 19 & 9 \\ 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[991] $2^3 3 14 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[992] $21 \overset{1^1}{1} \overset{1^1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[993] $1 \overset{1^{11}}{1^1} 4 13 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

[994] $1 \overset{1^{11}}{1^1} \overset{1^{31}}{4} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 22 & 10 \\ 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[995] $1 \overset{1^{11}}{1^1} \overset{1^{31}}{4} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 8 & 4 \\ 7 & 8 & 14 & 8 \\ 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[996] $1 \overset{1^1}{1^1, 1^1} \overset{1^1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1002] \begin{smallmatrix} 1^1 3 1^1 \\ 1 \quad 4 \quad 1^1 \end{smallmatrix} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 7, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.57 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\left(\begin{smallmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{smallmatrix} \right)_{b_0}$$

$$\begin{aligned}
& [1003] \ 1 \overset{1}{4} \ 131 \overset{1}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.57 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[1001] \begin{smallmatrix} 1^1 & 3 & 1^1 \\ 1 & 4 & 1^1 \end{smallmatrix} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$[1006] \overset{1}{2} 3 141^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{array}{l} \text{[1005]} \quad 1^1 14131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\ \left(\begin{array}{cccc} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{array} \right)_{b_0} \end{array}$$

[1004] $1^1 41231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[997] $1^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} 3^{\textcolor{green}{1}} 4^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{u}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

[998] $1^{\textcolor{red}{1}}141231^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$

[999] $1^1 4 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 3 & 2 \\ 4 & 3 & 8 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

$$[1000] \ 1 \overset{1}{1} 4 \overset{1}{1} 3 \overset{1}{1} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.57 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1009] \text{ } ^1 1^3 2^3 21^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 70 & 18 \\ 2 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1007] \overset{1}{1} \overset{1}{4} 2 \overset{4}{3} 2 \overset{1}{1} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{u}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 126 & 24 \\ 2 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[1008] $1^{1^1} 4^{11} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[1014] $1^{1^1} 1^1 2^3 21^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 6 & 6 \\ 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1013] $1^{1^1} 5^{1^1} 1^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 7 & 4 \\ 6 & 7 & 8 & 6 \\ 4 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[1012] $1^{1^1} 1^1 2^3 21^{1^1} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 30 & 12 \\ 2 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0}$$

[1011] $31^{1^1} 31^{1^1} 41^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1010] $1^{1^1} 1^1 2^3 21^{1^1} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 34 & 12 \\ 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1017] $1^{1^3} 5^{1^1} 1^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1016] $1^{1^1} 1^{1^4} 5^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 16 & 4 \\ 14 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1015] $1^{1^1} 1^1 5^{1^1} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[1023] $41^{1^1} 1^1 3^{1^1} 22 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 4 & 2 \\ 3 & 5 & 8 & 8 & 5 \\ 3 & 8 & 4 & 8 & 4 \\ 4 & 8 & 8 & 10 & 6 \\ 2 & 5 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[1022] $1^{1^1} 1^1 4^{1^1} 13^{1^1} 4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 62 & 16 \\ 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1021] \ 1^{13} 4^{11} 131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 10, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 12 & 4 \\ 9 & 12 & 34 & 12 \\ 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1020] \ 1^{11} 4^{12} 31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 10, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 18 & 4 \\ 16 & 18 & 74 & 18 \\ 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1019] \ 1^{12} 4^{11} 131^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 10, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 14 & 8 \\ 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1018] \ 1^{11} 4^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 10, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 2 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1025] \ 1^{12} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 11, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 49 & 15 \\ 2 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1037] \ 21^{14} 5^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 11, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 28 & 4 \\ 22 & 28 & 193 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1043] \ 21^{11} 5^{11} 1^{11} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 11, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 0 & 3 & 3 & 2 \\ 2 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 4 & 3 \\ 2 & 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1044] \ 1^{12} 5^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 11, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1045] \ 1^{11} 5^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 11, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1040] \ 1^{11} 2^3 3 21^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 11, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 17 & 9 \\ 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1039] \ 21^{12} 5^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 11, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[1038] $221^{1^2}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 9 & 3 \\ 6 & 13 & 22 & 7 \\ 9 & 22 & 36 & 12 \\ 3 & 7 & 12 & 3 \end{pmatrix}_{b_0}$

[1026] $1^{1^3}51^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 19 & 4 \\ 12 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1027] $21^{1^2}51^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 22 & 4 \\ 18 & 22 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1028] $2221^{1^1}51^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 17 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 7 & 0 & 2 \end{pmatrix}_{b_0}$

[1029] $221^{1^4}51^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 28 & 4 \\ 22 & 28 & 193 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$

[1030] $21^{1^2}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 13 & 4 \\ 9 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1031] $1^{1^1}5^{1^4}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 28 & 4 \\ 22 & 28 & 193 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$

[1024] $1^{1^1}5^{1^4}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 13 & 4 \\ 12 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1032] $21^151^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[1033] $1^{1^1}5^{1^4}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 22 & 4 \\ 18 & 22 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1034] $1^{221^1}51^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[1041] $1^{1^2}2^321^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 49 & 15 \\ 2 & 3 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1035] \quad 1 \overset{1}{\underset{21}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1036] \quad 221 \overset{1}{\underset{21}{5}} 1 \overset{1}{\underset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1042] \quad 1 \overset{1}{\underset{1}{\underset{1}{1}}} 12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1048] \quad 1 \overset{1}{\underset{4}{1}} 131 \overset{1}{\underset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1047] \quad 21 \overset{1}{\underset{5}{1}} 1 \overset{1}{\underset{1}{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 & 2 \\ 2 & 3 & 3 & 2 & 3 \\ 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1046] \quad 1 \overset{1}{\underset{1}{\underset{5}{1}}} 1 \overset{1}{\underset{1}{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1059] \quad 21 \overset{1}{\underset{1}{\underset{5}{1}}} 1 \overset{1}{\underset{1}{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 90 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[1058] \quad 221 \overset{1}{\underset{5}{1}} 1 \overset{1}{\underset{1}{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1049] \quad 1 \overset{1}{\underset{1}{\underset{1}{\underset{2}{3}}}} 21 \overset{1}{\underset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1050] \quad 1 \overset{1}{\underset{1}{\underset{1}{\underset{1}{\underset{1}{5}}}}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1051] \quad 1 \overset{1}{\underset{1}{\underset{1}{\underset{1}{\underset{1}{1}}}}} 4 \overset{1}{\underset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[1052] $1 \overset{1}{\underset{12}{5}} 12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[1053] $221 \overset{1}{\underset{11}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 2 & 3 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[1054] $1 \overset{11}{\underset{11}{23}} 1^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 7 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 8 & 4 \end{pmatrix}_{b_0}$

[1055] $1 \overset{112}{3} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[1056] $1 \overset{14}{\underset{11,12}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 90 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$

[1057] $1 \overset{11}{\underset{11}{23}} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[1067] $2 \overset{11}{3} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1064] $1 \overset{1}{\underset{11,11}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$

[1063] $31 \overset{1}{\underset{11}{4}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 5 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 6 & 5 & 8 & 6 \\ 2 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1062] $21 \overset{13}{\underset{11}{4}} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 25 & 4 \\ 20 & 25 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1061] $2 \overset{141}{3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[1060] $21 \overset{1}{\underset{11}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0} \\
[1065] & 31^{11} 31^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1066] & 1^{11} 5^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 25 & 4 \\ 20 & 25 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[1068] & 3131^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 13 & 8 & 7 \\ 3 & 8 & 4 & 4 \\ 3 & 7 & 4 & 3 \end{pmatrix}_{b_0} \\
[1074] & 1^{11} 4^{11} 131^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 9 & 4 \\ 7 & 9 & 17 & 9 \\ 3 & 4 & 9 & 4 \end{pmatrix}_{b_0} \\
[1073] & 21^{11} 41^{23} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 1 & 10 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 10 & 0 & 20 & 8 \\ 2 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0} \\
[1072] & 31^{14} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1071] & 1^{12} 4^{12} 31^3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 17 & 4 \\ 15 & 17 & 65 & 17 \\ 3 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[1070] & 31^{11} 31^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[1069] & 21^{11} 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 4 & 4 \\ 2 & 4 & 0 & 4 & 3 \end{pmatrix}_{b_0} \\
[1076] & 1^{11} 2321^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[1075] & 2^1 3^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 5 & 7 & 5 \\ 5 & 7 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[1098] & 21^{24} 5^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5
\end{aligned}$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1095] \ 1^{13} 2^3 31^{11} 4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 13 & 0 & 97 & 21 \\ 2 & 3 & -1 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1096] \ 21^{15} 51^{11} 22 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 5 & 6 & 4 \\ 2 & 5 & 3 & 4 & 3 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1097] \ 321^{14} 41^{11} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 5 & 3 & 2 \\ 3 & 1 & 6 & 3 & 3 \\ 5 & 6 & 12 & 5 & 6 \\ 3 & 3 & 5 & 1 & 3 \\ 2 & 3 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1094] \ 1^{13} 4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 5 & 1 & 25 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1099] \ 31^{13} 31^3 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 19 & 3 \\ 5 & 5 & 26 & 5 \\ 19 & 26 & 122 & 22 \\ 3 & 5 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1085] \ 1^{12} 4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 5 & 2 \\ 5 & 5 & 14 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1087] \ 1^{15} 12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[1077] \ 1^{11} 5^{14} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 19 & 4 \\ 16 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1078] \ 1^{11} 41231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1079] \ 221^{15} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[1080] \ 1^{11} 31^{11} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 3 & 1 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1081] \ 1^4 4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1082] \quad 1 \overset{1}{\underset{1^1}{5}} 1^{\overset{1}{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1083] \quad 1 \overset{1^2}{\underset{1^2, 1^1}{5}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1084] \quad 1 \overset{1^2}{\underset{1^1, 1^1}{5}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 13 & 4 \\ 8 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1088] \quad 1 \overset{1^{11} 31^1}{\underset{1^1}{4}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 2 & 1 & 5 & 5 \\ 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1089] \quad 1 \overset{1^1}{\underset{1^1}{4}} 131^{\overset{1}{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 8 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1090] \quad 1 \overset{1}{\underset{1^1, 1^{13}}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[1091] \quad 1 \overset{1^3}{\underset{1^1}{4}} 131^{\overset{1}{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 5 & 2 \\ 7 & 5 & 24 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1092] \quad 1 \overset{1^{11}}{\underset{1^1}{4}} 131^{\overset{1}{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1093] \quad 1 \overset{1^{11}}{\underset{1^2, 1^2}{5}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 16 & 4 \\ 14 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1086] \quad 1 \overset{1^{11}}{\underset{1^1}{4}} 1231^{\overset{1}{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1100] \quad 21 \overset{1^{11}}{\underset{1^1}{4}} 131^{\overset{1}{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 12 & 8 \\ 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[1106] $221^{11}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 2 & 3 \\ 6 & 13 & 5 & 7 \\ 2 & 5 & 1 & 3 \\ 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$

[1105] $1^{12}2321^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 8 & 0 & 10 & 74 & 18 \\ 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1104] $1^{131}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

[1103] $41^{1231}13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 4 & 3 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 3 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 10 & 6 \\ 2 & 6 & 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1102] $1^{1131}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 2 & 2 \\ 4 & 2 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[1101] $1^{11}1^{131}41^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 20 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1115] $1^{11}5^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[1116] $1^{11}5^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[1107] $31^{131}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 7 & 3 \\ 6 & 13 & 14 & 7 \\ 7 & 14 & 14 & 8 \\ 3 & 7 & 8 & 3 \end{pmatrix}_{b_0}$

[1108] $1^{11}5^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$

[1109] $41^{12231}1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$

[1110] $21^{34}5^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 25 & 4 \\ 19 & 25 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1111] \ 21^{1^3} 5^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 16 & 4 \\ 11 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1112] \ 1^{11} 5^{1^4} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 22 & 4 \\ 18 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1113] \ 21^{3^1} 5^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1114] \ 1^{11} 5^{1^3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1125] \ 1^{11} 4^{1^1 31^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1124] \ 1^{1^1 31^1} 4^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 4 & 2 \\ 6 & 4 & 14 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1123] \ 31^{1^1 3141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1122] \ 1^{11} 4^{1^2 31^2} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 15 & 4 \\ 13 & 15 & 49 & 15 \\ 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1121] \ 1^1 4^{1^2} 131^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 4 & 2 \\ 6 & 4 & 14 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1120] \ 1^{1^2} 4^{1^2 31^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 11 & 4 \\ 9 & 11 & 25 & 11 \\ 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1119] \ 1^{11} 41^{1^2} 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1118] \ 31^{1^1 31^4} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 15 & 2 \\ 15 & 9 & 15 & 121 & 19 \\ 2 & 2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[1117] \ 1^{11} 4^{13} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 4 & 2 \\ 4 & 4 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1126] \ 1^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1133] \ 21^{11} 41^{13} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1127] \ 21^3 5^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1128] \ 1^{11} 4^{12} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 30 & 12 \\ 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1129] \ 1^{13} 5^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1130] \ 1^{12} 4^{12} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1131] \ 1^{11} 5^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 11 & 4 \\ 6 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1132] \ 21^3 5^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 11 & 4 \\ 6 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1134] \ 1^{12} 4^{11} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 37 & 13 \\ 3 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[1135] \ 1^4 41^{12} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[1136] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 5 & 5 \\ 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$

[1137] $1^{11} 4^{12} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 4 & 1 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[1138] $1^{11} 4^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[1139] $2^1 3^{11} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$

[1145] $21^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 13 & 4 \\ 9 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1140] $221^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[1141] $1^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 22 & 4 \\ 17 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1142] $1^{11} 5^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1143] $1^{11} 231^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 7 & 3 \\ 4 & 0 & 7 & 14 & 8 \\ 2 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$

[1151] $21^{11} 3^{12} 5^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1150] $21^{13} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 22 & 4 \\ 17 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1149] $1^{12} 2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[1148] & 1^{11} 5^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[1147] & 1^{11} 5^{1^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[1146] & 221^3 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 22 & 4 \\ 17 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[1144] & 21^1 51^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[1153] & 1^{1^2} 5^{1^2} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[1152] & 21^1 51^{1^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[1163] & 2^{1^2} 3^{1^2} 41^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 5 & 6 & 5 & 20 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[1162] & 1^{11} 5^{1^3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1161] & 21^{1^2} 4131^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 1 & 12 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 7 & 12 & 3 & 31 & 11 \\ 2 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0} \\
[1160] & 21^{1^2} 41^1 31^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 1 & 14 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 14 & 3 & 41 & 13 \\ 2 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1159] & 31^{1^2} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 9 & 2 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 7 & 2 \\ 9 & 12 & 7 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1158] $1^{11} \overset{1^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1157] $21^{11} 41^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 8 & 1 & 11 & 7 \\ 2 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[1156] $21^2 \overset{1^1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1155] $21^2 \overset{1^3}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 19 & 4 \\ 14 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1154] $21^{11} 4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 6 & 1 & 5 & 5 \\ 2 & 4 & 0 & 5 & 3 \end{pmatrix}_{b_0}$

[1165] $21^4 \overset{1^1}{5} 1^{13} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 25 & 3 \\ 4 & 4 & 36 & 4 \\ 25 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$

[1166] $1 \overset{1^1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[1167] $21^{13} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 25 & 3 \\ 4 & 4 & 36 & 4 \\ 25 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$

[1168] $21^{11} 41^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 8 & 0 & 10 & 6 \\ 2 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[1169] $2 \overset{1^1}{3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1164] $21 \overset{1^1}{5} 1^{11} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[1170] $1^{12} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[1172] $1^{13} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 24 & 4 \\ 21 & 24 & 130 & 24 \\ 3 & 4 & 24 & 4 \end{pmatrix}_{b_0}$$

[1173] $1^{13} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 15 & 4 \\ 12 & 15 & 49 & 15 \\ 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[1174] $1^{12} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 25 & 4 \\ 22 & 25 & 141 & 25 \\ 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[1175] $21^{11} 41^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 18 & 10 \\ 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[1171] $1^{14} 5^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 23 & 4 \\ 14 & 23 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1176] $21^{12} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 4 & 23 & 4 \\ 16 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1177] $1^{11} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1178] $31^{11} 31^{14} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 4 & 2 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 4 & 2 \\ 4 & 5 & 4 & 5 & 5 \\ 2 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1179] $1^{12} 23 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[1180] $15^1 1^{12} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1181] $21^{12} 5^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1182] $221^1 5^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1183] \quad 1 \overset{1}{5} 1 \overset{1}{1} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1190] \quad 21 \overset{1}{1} 5 1 \overset{1}{1} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1189] \quad 21 \overset{1}{1} 41 \overset{2}{3} 1 \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 28 & 12 \\ 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1184] \quad 141 \overset{1}{3} 1 \overset{1}{1} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1185] \quad 2 \overset{1}{1} 41 \overset{1}{1} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 3 & 6 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1186] \quad 1 \overset{1}{3} 4 \overset{1}{1} 13 \overset{1}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 65 & 17 \\ 3 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1187] \quad 21 \overset{1}{1} 1 \overset{1}{5} 1 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1188] \quad 2 \overset{1}{1} 3 \overset{1}{1} 141 \overset{1}{1} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1200] \quad 1 \overset{1}{1} 41 \overset{2}{2} 31 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 6 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1197] \quad 1 \overset{1}{1} 5 \overset{1}{1} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 15 & 4 \\ 8 & 15 & 40 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1191] \quad 1 \overset{1}{1} 41 \overset{2}{2} 31 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 24 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1192] \quad 1 \overset{1}{1} 4 \overset{1}{1} 131 \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 3 & 2 & 8 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1193] \ 21 \overset{1}{\underset{1}{5}} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 15 & 4 \\ 8 & 15 & 40 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1194] \ 1^{14} 4 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 6 & 2 & 44 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1195] \ 1^{11} 4^{131^2} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 24 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1198] \ 1^{12} 4 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 24 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1202] \ 31^{11} 31^{341^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 31 & 5 \\ 22 & 31 & 170 & 26 \\ 3 & 5 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1203] \ 1^{11} 4^{131^2} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 6 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1204] \ 1^{12} 4^{131^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 6 & 0 & 20 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1205] \ 1^{12} 4^{131^1} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 7 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1206] \ 1^{14} 4 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 48 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1199] \ 1^{11} 4^{131^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 5 & 0 & 20 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1201] \ 31^{11} 4 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1196] $1^{\overline{1}^{12}31^1} 4^{\overline{1}^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 48 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1213] $1^{\overline{1}^4} 4^{\overline{1}^4} 231^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 16 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 32 & 0 & 190 & 24 \\ 2 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1211] $1^{\overline{1}^4} 4^{\overline{1}^3} 231^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 12 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 24 & 0 & 106 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1212] $2^{\overline{1}^{11}} 3^{\overline{1}^1} 141^{\overline{1}^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[1210] $1^{\overline{1}^4} 4^{\overline{1}^2} 231^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 8 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 16 & 0 & 46 & 12 \\ 2 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1214] $1^{\overline{1}^4} 4^{\overline{1}^1} 231^{\overline{1}^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 6 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 8 & 2 & 14 & 8 \\ 2 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[1215] $1^{\overline{1}^4} 4^{\overline{1}^2} 231^{\overline{1}^{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 12 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 16 & 4 & 62 & 16 \\ 2 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1216] $1^{\overline{1}^{11}31^2} 4^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 14 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[1217] $1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{1}^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1218] $221^{\overline{1}^{12}51^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 16 & 3 \\ 6 & 13 & 39 & 7 \\ 16 & 39 & 113 & 21 \\ 3 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[1219] $1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{1}^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 11 & 4 \\ 7 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[1220] $1^{\overline{1}^{11}31^1} 4^{\overline{1}^1} 1^{\overline{1}^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 5 & 5 \\ 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0} \\
[1208] & 1^{141} 1^{11} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 8 & 0 & 10 & 6 \\ 2 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0} \\
[1207] & 1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 4 & 2 \\ 5 & 4 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[1209] & 1^{11} 1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 7 & 1 & 53 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1221] & 1^{11} 1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.82 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 6 & 2 \\ 7 & 6 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0} \\
[1222] & 1^{13} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -100, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.45 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[1223] & 1^{13} 1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.82 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 34 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[1224] & 1^{14} 1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 81 & 19 \\ 3 & 4 & 19 & 4 \end{pmatrix}_{b_0} \\
[1225] & 1^{12} 1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.82 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 7 & 1 & 37 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0} \\
[1226] & 1^{12} 1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.82 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 5 & 1 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0} \\
[1227] & 31^{11} 3141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.36 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1235] & 1^{11} 1^{12} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -100, \quad \det = 46, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.43 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1228] $1^{1^3 3 1^1} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 54 & 16 \\ 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$

[1236] $1^{1^4} 5 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 26 & 4 \\ 16 & 26 & 150 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1229] $1^{1^2} 5 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1230] $2^1 3 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[1231] $1^{1^3 3 1^3} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 21 & 4 \\ 18 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$

[1232] $2^1 3 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[1233] $1^{1^3 3 1^2} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 18 & 4 \\ 15 & 18 & 70 & 18 \\ 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$

[1234] $1^{1^4} 5 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 32 & 4 \\ 20 & 32 & 234 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$

[1237] $1^{1^3} 4 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 5 & 2 & 28 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[1238] $2^1 3 141^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[1239] $1^{1^2} 4 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 4 & 2 & 16 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1240] $1^{1^3} 4 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 7 & 2 \\ 8 & 7 & 32 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[1241] & 1^{11} 4 1^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 5 & 2 \\ 6 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[1242] & 1^{11} 4 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 5 & 2 \\ 6 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[1243] & 1^{11} 4^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 5 & 2 \\ 6 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[1245] & 1^{12} 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 11 & 0 & 32 & 12 \\ 2 & 4 & -1 & 12 & 4 \end{pmatrix}_{b_0} \\
[1244] & 1^{12} 4 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 6 & 1 & 23 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0} \\
[1246] & 21^{13} 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 20 & 4 \\ 14 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[1247] & 1^{14} 1^{12} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[1248] & 1^{11} 4^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[1249] & 1^{12} 4^{13} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 22 & 4 \\ 19 & 22 & 106 & 22 \\ 3 & 4 & 22 & 4 \end{pmatrix}_{b_0} \\
[1250] & 1^{12} 2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[1251] & 1^{12} 2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[1252] & 1^{13} 2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1253] $1^{13}2^321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1254] $1^{14}5^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 29 & 4 \\ 18 & 29 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

[1255] $31^{11}31^{14}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 6 & 2 \\ 6 & 3 & 6 & 13 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1256] $31^{11}31^{13}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 3 & 2 & 0 & 12 & 2 \\ 12 & 7 & 12 & 73 & 15 \\ 2 & 2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1257] $21^15^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1258] $21^15^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1259] $1^{12}5^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1260] $1^{14}5^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1261] $1^{11}4^{12}131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 6 & 2 \\ 7 & 6 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1262] $1^{11}1^{11}31^14^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 11 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[1265] $1^{112}31^14^11^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 32 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1264] $1^{11}41^{1231^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 11 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[1263] $1^{11}4^{1131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 6 & 1 & 35 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$

[1270] $1^{12}4^{1331^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 19 & 4 \\ 16 & 19 & 77 & 19 \\ 3 & 4 & 19 & 4 \end{pmatrix}_{b_0}$

[1269] $1^{15}1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1268] $1^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 23 & 4 \\ 14 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1267] $1^{12}2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1266] $1^{12}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1273] $2151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1274] $1^{13}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1271] $31^{11}3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

[1272] $1^{112}4^{1231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$

[1282] $1^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 26 & 4 \\ 16 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1281] \quad 21 \overset{1^4}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 23 & 4 \\ 15 & 23 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1280] \quad 21 \overset{1^1}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1279] \quad 1^{12} \overset{1^1}{\underset{1^1, 1^1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1278] \quad 1 \overset{1^1}{\underset{21^{12}}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1277] \quad 1 \overset{1^{11} 31^1}{\underset{1^1}{4}} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 21 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1276] \quad 21 \overset{1^1}{\underset{1^1}{5}} 1^{12} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1275] \quad 1^{12} \overset{1^4}{\underset{1^1, 1^1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1283] \quad 1 \overset{1^1}{\underset{1^{12}}{5}} 1^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1284] \quad 221 \overset{1^1}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1285] \quad 31 \overset{1^1}{\underset{1^1}{3}} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1286] \quad 31 \overset{1^1}{\underset{1^1}{3}} 1^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 9 & 2 \\ 9 & 5 & 9 & 37 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1287] \quad 31 \overset{1^1}{\underset{1^1}{3}} 1^1 41^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 9 & 2 \\ 9 & 9 & 9 & 49 & 13 \\ 2 & 2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1288] \quad 1 \overset{1}{\underset{1^1, 1^1}{5}} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1289] \quad 31 \overset{1}{\underset{1^1, 1^1}{31}} 1^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 9 & 3 \\ 6 & 13 & 18 & 7 \\ 9 & 18 & 22 & 10 \\ 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1290] \quad 1 \overset{1}{\underset{1^{13}}{5}} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1291] \quad 2 \overset{1}{\underset{1^1, 1^1}{3}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 5 & 12 & 5 \\ 8 & 12 & 22 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1292] \quad 141 \overset{1}{\underset{1^1, 1^1}{31}} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1298] \quad 21 \overset{1}{\underset{1^1, 1^1}{2}} 41 \overset{1}{\underset{1^1, 1^1}{31}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 1 & 14 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 14 & 0 & 32 & 10 \\ 2 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1297] \quad 21 \overset{1}{\underset{1^1, 1^1}{2}} 41 \overset{1}{\underset{1^1, 1^1}{31}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 18 & 0 & 64 & 14 \\ 2 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1293] \quad 21 \overset{1}{\underset{1^1, 1^1}{5}} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 20 & 4 \\ 13 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1294] \quad 21 \overset{1}{\underset{1^1, 1^1}{5}} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 17 & 4 \\ 11 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1295] \quad 1 \overset{1}{\underset{1^1, 1^1}{5}} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1296] \quad 1 \overset{1}{\underset{1^1, 1^1}{5}} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1299] $221^{11}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 7 & 3 \\ 6 & 13 & 17 & 7 \\ 7 & 17 & 19 & 9 \\ 3 & 7 & 9 & 3 \end{pmatrix}_{b_0}$

[1301] $1^{14}4^{1^4 31^2}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 22 & 4 \\ 18 & 22 & 106 & 22 \\ 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$

[1300] $1^{13}4^{1^4 31^1}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 86 & 20 \\ 3 & 4 & 20 & 4 \end{pmatrix}_{b_0}$

[1311] $1^{11}4^{1^4 31^1}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 5 & 1 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[1312] $1^{13}4^{1^3}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 9 & 1 & 53 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$

[1313] $1^{11}4^{1^4}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 6 & 3 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$

[1314] $1^{11}4^{1^3 31^1}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$

[1310] $221^{11}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 8 & 6 \\ 3 & 7 & 6 & 3 \end{pmatrix}_{b_0}$

[1309] $1^{11}4^{1^4 31^2}1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 7 & 2 \\ 9 & 7 & 34 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1308] $1^{13}4^{1^2}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 8 & 1 & 37 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$

[1302] $3131^{11}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 8 & 6 \\ 3 & 7 & 6 & 3 \end{pmatrix}_{b_0}$

[1303] $1^{11}4^{1^3}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 7 & 5 & 3 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1304] $1^{1^3}4^{13}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 48 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1305] $1^{1^4}4^{12}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 7 & 2 \\ 9 & 7 & 34 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1306] $1^{1^2}4^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 5 & 2 \\ 7 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[1307] $1^{1^4}4^{12}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1319] $1^{1^2}4^{1^3}1^{31^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 7 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[1315] $1^{1^2}4^{1^3}1^{31^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 9 & 0 & 64 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1316] $1^{1^4}4^{1^3}1^{31^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1318] $1^{1^2}5^{1^4}1^{12}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1317] $1^{1^4}4^{1^3}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1320] $1^{1^2}5^{1^4}1^{1^4}1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 17 & 4 \\ 14 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1323] $21^{1^2}4^{1^4}3^{1^3}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 18 & 1 & 65 & 15 \\ 2 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1322] $21^{1^2}4^{1^4}1^{1^3}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 1 & 12 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 12 & 1 & 23 & 9 \\ 2 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[1321] $2^1 3^{12} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[1329] $1^{14} 4^{13} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 62 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1325] $21^2 5^{11} 1^{12} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1324] $21^{12} 5^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1332] $1^{11} 41^{12} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 4 & 2 & 14 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[1331] $1^{12} 4^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 8 & 0 & 46 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1330] $1^{11} 4^{12} 31^{11} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 4 & 2 & 14 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[1328] $1^{12} 4^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 5 & 2 & 16 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[1327] $1^{12} 41^{12} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 8 & 0 & 46 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1326] $1^{11} 4^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 62 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1337] $21^2 5^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 17 & 4 \\ 12 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1336] $1^{12} \overset{1^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 17 & 4 \\ 12 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1335] $1^{14} \overset{1^4 31^4}{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 28 & 4 \\ 24 & 28 & 174 & 28 \\ 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$

[1334] $21 \overset{1^1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 23 & 4 \\ 18 & 23 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1333] $1^{12} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 23 & 4 \\ 18 & 23 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1338] $1^{11} \overset{1^{11} 31^2}{4} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 8 & 1 & 69 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1339] $1^{11} \overset{1^{11} 31^2}{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 5 & 1 & 21 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$

[1340] $31 \overset{1^{12} 31^2}{1} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 7 & 3 \\ 5 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$

[1341] $1^{11} \overset{1^{12} 31^2}{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$

[1342] $1^{11} \overset{1^{12} 31^1}{4} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 9 & 2 \\ 11 & 9 & 60 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[1343] $1 \overset{1^1}{5} 1^{12} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 4 & 11 & 4 \\ 8 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[1344] $1 \overset{1^2}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1345] $21 \overset{1^2}{4} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 1 & 12 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 12 & 2 & 26 & 10 \\ 2 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[1346] $1^{14} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 25 & 4 \\ 21 & 25 & 137 & 25 \\ 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[1347] $2^1 3^{13} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[1348] $1^{12} 5^{14} 1^{13,11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1349] $21^{12} 41^{23} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 16 & 2 & 50 & 14 \\ 2 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1350] $1^{12} 5^{13} 1^{12,11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1351] $1^{13} 2^1 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[1352] $31^{13} 1^1 2^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 14 & 3 \\ 5 & 5 & 19 & 5 \\ 14 & 19 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1353] $31^{11} 4^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1354] $1^{14} 5^{13} 1^{11,11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 27 & 4 \\ 16 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1356] $21^{12} 41^{23} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 16 & 0 & 46 & 12 \\ 2 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1357] $41^{12} 2^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 8 & 2 \\ 4 & 10 & 5 & 20 & 6 \\ 2 & 5 & 0 & 10 & 2 \\ 8 & 20 & 10 & 38 & 12 \\ 2 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1355] \ 21^3 \overset{1}{5} 1^{\overset{1}{1} 12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1359] \ 1 \overset{1}{5} 1^{\overset{1}{1} 13} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1358] \ 1^{\overset{1}{1} 12} \overset{1}{5} 1^{\overset{1}{1} 1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1360] \ 1^{\overset{1}{1} 41} \overset{1}{1} 231^{\overset{1}{1} 1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 8 & 1 & 11 & 7 \\ 2 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1361] \ 1 \overset{1}{5} 1^{\overset{1}{1} 13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 81, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1362] \ 1^{\overset{1}{1} 11} \overset{1}{4} 131^{\overset{1}{1} 13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 83, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 7 & 2 \\ 9 & 7 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1363] \ 21^{\overset{1}{1} 3} \overset{1}{5} 1^{\overset{1}{1} 1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 83, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.35 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 23 & 3 \\ 4 & 4 & 33 & 4 \\ 23 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[1364] \ 1^{\overset{1}{1} 11} \overset{1}{4} 131^{\overset{1}{1} 13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 83, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 9 & 2 \\ 11 & 9 & 58 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1365] \ 1^{\overset{1}{1} 11} \overset{1}{4} 1^{\overset{1}{1} 12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 83, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 6 & 1 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1366] \ 1^{\overset{1}{1} 13} \overset{1}{2} 321^{\overset{1}{1} 12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 83, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1367] \ 1^{\overset{1}{1} 12} \overset{1}{4} 1^{\overset{1}{1} 1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 83, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 6 & 1 & 21 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1368] \ 1^{\overset{1}{1} 12} \overset{1}{4} 1^{\overset{1}{1} 4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 83, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 9 & 1 & 69 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1369] $1^{1^{12}31^2} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 44 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1370] $1^{1^{11}31^2} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 7 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1371] $1^{1^4 41^2 231^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 6 & 1 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1372] $21^{1^2} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1382] $1^{1^{11}31^1} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 18 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1376] $1^{1^{11}31^1} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 5 & 2 & 24 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[1383] $41^{1^4 2231^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[1384] $1^{1^4} 4^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 8 & 2 & 58 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1381] $1^{1^2} 4^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 6 & 2 & 26 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[1377] $1^{1^4 41^1 231^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 18 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1380] $1^{1^{11}31^1} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 7 & 2 & 56 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1379] $1^{\overline{1}^{\overline{1}^3\overline{3}^1\overline{1}^1}} 4^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1378] $1^{\overline{1}^{\overline{1}^3\overline{4}^1\overline{3}^1\overline{4}^1}} 4^{\overline{1}^{\overline{1}^1\overline{1}^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 25 & 3 \\ 3 & 4 & 29 & 4 \\ 25 & 29 & 185 & 29 \\ 3 & 4 & 29 & 4 \end{pmatrix}_{b_0}$

[1373] $1^{\overline{1}^{\overline{1}^1\overline{4}^1\overline{2}^1\overline{3}^1\overline{1}^2}} 4^{\overline{1}^{\overline{1}^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1374] $1^{\overline{1}^{\overline{1}^3\overline{4}^1\overline{3}^1\overline{2}^1}} 4^{\overline{1}^{\overline{1}^1\overline{1}^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 23 & 4 \\ 19 & 23 & 113 & 23 \\ 3 & 4 & 23 & 4 \end{pmatrix}_{b_0}$

[1375] $1^{\overline{1}^{\overline{1}^2\overline{3}^1\overline{2}^1}} 4^{\overline{1}^{\overline{1}^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1385] $1^{\overline{1}^{\overline{1}^3}} \overline{5}^{\overline{1}^1} 1^{\overline{1}^{\overline{1}^1, \overline{1}^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1387] $221^{\overline{1}^{\overline{1}^1\overline{5}^1\overline{1}^3}} 5^{\overline{1}^{\overline{1}^3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1386] $41^{\overline{1}^{\overline{1}^1\overline{1}^1}} \overline{3}^{\overline{1}^1} 22 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 4 & 7 & 4 \\ 3 & 8 & 7 & 5 & 5 \\ 2 & 6 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[1388] $21^{\overline{1}^{\overline{1}^1\overline{5}^1\overline{1}^3}} \overline{1}^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1389] $1^{\overline{1}^{\overline{1}^2}} \overline{5}^{\overline{1}^2} 1^{\overline{1}^{\overline{1}^1, \overline{1}^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1390] $221^{\overline{1}^{\overline{1}^1\overline{2}^1\overline{5}^1\overline{1}^2}} 5^{\overline{1}^{\overline{1}^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1391] $21^{\overline{2}^{\overline{1}^2}} \overline{5}^{\overline{1}^{\overline{1}^2}} 1^{\overline{1}^{\overline{1}^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1392] \ 21^{1^4} 5^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 150 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1393] \ 1^{12} 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1394] \ 1^{13} 2^2 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1395] \ 21^{13} 41^{13} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 26 & 3 & 141 & 23 \\ 2 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[1396] \ 2^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 5 & 7 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1400] \ 1^{13} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 58 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1401] \ 1^{11} 4^1 13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 8 & 2 \\ 10 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1398] \ 1^{11} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 6 & 2 & 38 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1397] \ 1^{12} 4^1 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 7 & 2 & 40 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1399] \ 1^{11} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 8 & 2 \\ 10 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1402] \ 1^{13} 2^2 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[1403] $1^{13} 4^{1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 26 & 4 \\ 22 & 26 & 146 & 26 \\ 3 & 4 & 26 & 4 \end{pmatrix}_{b_0}$

[1407] $21^{2^3} 5^{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 23 & 4 \\ 16 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1406] $3131^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 8 & 3 \\ 6 & 13 & 20 & 7 \\ 8 & 20 & 26 & 10 \\ 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[1405] $31^{2^3} 31^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 16 & 3 \\ 6 & 13 & 32 & 7 \\ 16 & 32 & 74 & 18 \\ 3 & 7 & 18 & 3 \end{pmatrix}_{b_0}$

[1404] $1^{12} 5^{1^3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1414] $1^{13} 31^{1^1} 4^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1413] $1^{12} 4^{1^{11}} 31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 \\ 1 & 1 & -2 & 1 \\ 8 & 8 & 1 & 49 \end{pmatrix}_{b_0}$

[1412] $1^{11} 4^{1^{12}} 31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 7 & 2 \\ 9 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1411] $1^{12} 4^{1^{11}} 31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 \\ 1 & 0 & 1 & 7 \\ 1 & 1 & -2 & 1 \\ 7 & 7 & 1 & 33 \end{pmatrix}_{b_0}$

[1410] $1^{1^1} 5^{1^2} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1409] $31^{11} 31^{1^1} 41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 9 & 3 \\ 5 & 5 & 12 & 5 \\ 9 & 12 & 22 & 10 \\ 3 & 5 & 10 & 3 \end{pmatrix}_{b_0}$

[1408] $1^{1^2} 5^{1^3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1417] $21^{12}41^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 1 & 14 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 14 & 2 & 36 & 12 \\ 2 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1416] $21^{11}51^{12}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 14 & 4 \\ 10 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1415] $21^{12}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 14 & 4 \\ 10 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1420] $1^{12}5^{14}1^{13,12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 23 & 4 \\ 18 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1419] $1^{12}5^{14}1^{14,12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 26 & 4 \\ 20 & 26 & 150 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1418] $21^{12}45^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 26 & 4 \\ 20 & 26 & 150 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1422] $2^{11}3^{41}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 5 & 17 & 5 \\ 11 & 17 & 46 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

[1423] $1^{11}5^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[1424] $1^{12}5^{13}1^{13,12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 23 & 4 \\ 17 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1425] $21^{13}4131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 18 & 4 & 68 & 16 \\ 2 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1426] $41^{11}232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 10 & 6 \\ 2 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1427] $21^{11}51^{13}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[1428] $21^{\overset{1^2}{\underset{1^1}{3}}51^{\overset{1^2}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 23 & 4 \\ 17 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1421] $21^{\overset{1^3}{1^1}41^{\overset{2^3}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 22 & 4 & 100 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1429] $2^{\overset{1^2}{1^1}}3141^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[1431] $31^{\overset{1^1}{1^1}31^{\overset{2^1}{1^1}}41^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 17 & 3 \\ 5 & 5 & 24 & 5 \\ 17 & 24 & 98 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

[1430] $1^{\overset{1^4}{1^2,1^1}}51^{\overset{1^3}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 30 & 4 \\ 18 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

[1432] $1^{\overset{1^3}{1^2,1^1}}51^{\overset{1^3}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 27 & 4 \\ 16 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1434] $2^{\overset{1^3}{1^1}}3141^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1435] $31^{\overset{1^2}{1^1}3141^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 8 & 3 \\ 5 & 5 & 10 & 5 \\ 8 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[1436] $1^{\overset{1^2}{1^1}31^{\overset{2^1}{1^1}}41^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1437] $1^{\overset{1^2}{1^1}31^{\overset{2^1}{1^1}}41^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 6 & 2 \\ 11 & 6 & 32 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1433] $1^{\overset{1^1}{1^1}31^{\overset{2^1}{1^1}}41^{\overset{1^2}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 6 & 1 & 23 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[1438] $1^{\overset{1^2}{1^1}31^{\overset{1^2}{1^1}}51^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1439] \ 21^{1^2} 51^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1440] \ 41^{1^2} 231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1448] \ 1^{1^3} 4 \ 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 10 & 0 & 64 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1445] \ 1^{1^2} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 9 & 0 & 64 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1446] \ 1^{1^2} 4 \ 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 80 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1447] \ 1^{1^2} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 10 & 0 & 84 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1444] \ 1^{1^2} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 5 & 2 & 24 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1449] \ 21^{1^2} 41^{2^3} 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 16 & 1 & 47 & 13 \\ 2 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1450] \ 1^{1^3} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 11 & 0 & 84 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1451] \ 21^{1^2} 41^{1^3} 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 1 & 14 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 14 & 1 & 33 & 11 \\ 2 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1443] \ 1^{1^2} 41^{3^2} 231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 10 & 0 & 84 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1441] \ 31^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 6 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 5 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1442] \ 1^{13} 4 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 7 & 2 & 28 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1452] \ 1^{11} 1^{13} 31^{11} 4^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 5 & 3 & 29 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1453] \ 31^{11} 31^{14} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 7 & 2 \\ 7 & 5 & 7 & 21 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1454] \ 1^{12} 5^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 26 & 4 \\ 19 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1455] \ 1^{12} 5^{14} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 26 & 4 \\ 20 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1456] \ 21^{13} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 26 & 4 \\ 19 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1457] \ 21^{13} 5^{14} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 29 & 4 \\ 21 & 29 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1458] \ 221^{13} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 26 & 4 \\ 19 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1460] \ 1^{12} 5^{14} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 29 & 4 \\ 22 & 29 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1459] \ 21^{13} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 29 & 4 \\ 22 & 29 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1461] \ 31^{11} 31^{13} 341^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 16 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 3 & 2 & 0 & 16 & 2 \\ 16 & 15 & 16 & 161 & 23 \\ 2 & 2 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[1464] $31^{11}31^{14}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 3 & 2 & 0 & 8 & 2 \\ 8 & 7 & 8 & 33 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1462] $221^{13}51^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 23 & 3 \\ 6 & 13 & 56 & 7 \\ 23 & 56 & 232 & 30 \\ 3 & 7 & 30 & 3 \end{pmatrix}_{b_0}$$

[1463] $21^{13}51^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 4 & 27 & 4 \\ 19 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1465] $1^{11}4^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 7 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1466] $21^{13}51^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 30 & 4 \\ 21 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

[1467] $1^{11}4^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 9 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1468] $1^{11}4^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 6 & 2 \\ 11 & 6 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1469] $1^{11}51^{13}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1473] $21^{13}51^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 27 & 4 \\ 17 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1474] $1^{13}5^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1472] $1^{13}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 18 & 4 \\ 13 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1470] $21^{1^3}_{1^1} 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 24 & 4 \\ 15 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1471] $1^{1^3}_{1^3, 1^1} 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 30 & 4 \\ 18 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$

[1478] $21^{1^4}_{1^1} 5^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 32 & 4 \\ 24 & 32 & 234 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$

[1479] $1^{1^2}_{1^4, 1^4} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 32 & 4 \\ 24 & 32 & 234 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$

[1477] $1^{1^2}_{1^3} 2^3 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1475] $1^{1^4}_{1^3} 2^3 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1476] $221^{1^1}_{1^4} 5^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 32 & 4 \\ 24 & 32 & 234 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$

[1480] $1^{1^2}_{1^1} 4^{1^2} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 44 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1481] $1^{1^4}_{1^3, 1^1} 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 33 & 4 \\ 20 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$

[1482] $31^{1^1}_{1^1} 31^{1^1} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 17 & 5 \\ 12 & 17 & 46 & 14 \\ 3 & 5 & 14 & 3 \end{pmatrix}_{b_0}$

[1483] $1^{1^4}_{1^1} 4^{1^2} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 44 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1484] $1^{1^1}_{1^1} 41^{1^3} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 8 & 1 & 67 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1485] $1^{11} 4^{12} 13^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 8 & 2 \\ 11 & 8 & 44 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1486] $1^{11} 4^{12} 31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 60 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1487] $1^{11} 4^{11} 31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 8 & 1 & 67 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1488] $21^{13} 41^{13} 13^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 20 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 20 & 4 & 82 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1489] $1^{14} 41^{32} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 60 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1490] $1^{13} 4^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 8 & 1 & 35 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1491] $2^3 3^{12} 141^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1492] $1^{12} 4^{12} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 7 & 6 & 3 & 31 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1493] $1^{13} 2^3 3^{13} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1494] $1^{12} 4^{12} 2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1495] $1^5 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 36 & 4 \\ 22 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[1496] \ 1 \overset{1}{\underset{13}{5}} 1^4 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1499] \ 2221 \overset{11}{\underset{13}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 1 & 20 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 20 & 0 & 18 & 8 \\ 2 & 8 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1500] \ 1 \overset{11}{\underset{13}{5}} 1^2 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 10 & 2 \\ 13 & 10 & 76 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1497] \ 1 \overset{11}{\underset{13}{5}} 1^3 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 6 & 3 & 43 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1498] \ 41 \overset{11}{\underset{13}{5}} 31^1 32 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 5 & 4 & 2 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 12 & 6 \\ 4 & 12 & 12 & 10 & 6 \\ 2 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1501] \ 2 \overset{11}{\underset{13}{5}} 3^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 5 & 17 & 5 \\ 11 & 17 & 44 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1502] \ 2 \overset{11}{\underset{13}{5}} 3 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1503] \ 1 \overset{11}{\underset{13}{5}} 41^2 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 6 & 2 & 36 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1504] \ 1 \overset{11}{\underset{13}{5}} 1^2 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 6 & 2 & 36 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1505] \ 1 \overset{11}{\underset{13}{5}} 1^3 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1506] \ 1 \overset{11}{\underset{13}{5}} 41^1 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1507] \ 1 \overset{13}{\underset{1^2, 1^1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

- [1508] $1^{11} 1^{12} 31^2 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [1509] $21^2 51^1 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -100, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [1510] $1^{12} 1^{12} 31^1 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 6 & 2 & 24 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$
- [1511] $1^{13} 4^{12} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 8 & 2 & 40 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$
- [1512] $1^{12} 1^{11} 31^2 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 7 & 1 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$
- [1513] $1^{11} 1^{13} 31^1 4^1 1^4 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 7 & 3 & 61 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1514] $2^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 7 & 8 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1516] $1^{14} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [1517] $1^{14} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [1515] $221^{12} 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -100, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 14 & 3 \\ 6 & 13 & 34 & 7 \\ 14 & 34 & 82 & 18 \\ 3 & 7 & 18 & 3 \end{pmatrix}_{b_0}$
- [1518] $1^3 1^3 1^{13} 2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -100, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [1519] $31^{11} 31^2 41^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 3 & 2 & 0 & 12 & 2 \\ 12 & 11 & 12 & 85 & 17 \\ 2 & 2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[1520] \ 1^{13} 4^{1^{11} 31^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 11 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1521] \ 1^{12} 4^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 7 & 3 & 45 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1522] \ 1^{13} 4^{1^{11} 31^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 9 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1523] \ 1^{11} 4^{1^{12} 31^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 7 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1524] \ 1^{13} 4^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 9 & 2 & 56 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1525] \ 1^{41} 2^{23} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 40 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1526] \ 1^{11} 4^{13} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 9 & 2 \\ 12 & 9 & 56 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1527] \ 1^{13} 4^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 40 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1528] \ 1^{11} 4^{1^{12} 31^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 8 & 2 & 72 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1529] \ 1^{11} 4^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 9 & 2 \\ 12 & 9 & 56 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1530] \ 1^{13} 4^{14} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 10 & 2 & 76 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1531] $1^{1^4 31^1} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 56 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1532] $1^{11} 41^2 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 8 & 2 \\ 11 & 8 & 40 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1533] $1^{1^3} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 10 & 1 & 67 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1534] $1^{1^2} 4^{1^4} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 8 & 3 & 63 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1535] $1^{11} 4^{1^2 31^2} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 8 & 2 \\ 11 & 8 & 40 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1536] $1^{1^3} 5^{1^3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1537] $1^{221} 5^{1^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 10 & 3 \\ 6 & 13 & 24 & 7 \\ 10 & 24 & 38 & 12 \\ 3 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[1541] $31^{11} 31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 3 & 2 & 0 & 10 & 2 \\ 10 & 7 & 10 & 49 & 13 \\ 2 & 2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1540] $1^{1^3} 5^{1^4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 21 & 4 \\ 16 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1539] $1^1 5^1 1^{13} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 4 & 15 & 4 \\ 11 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1538] $31^{11} 31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 16 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 3 & 2 & 0 & 16 & 2 \\ 16 & 11 & 16 & 141 & 21 \\ 2 & 2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[1542] $1^{1^4} 4^{11} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 10 & 2 \\ 13 & 10 & 72 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1544] \ 21^{13}51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1545] \ 2^{14}3^{41}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 9 & 11 & 10 & 69 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1546] \ 1^{12}4^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 47 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1547] \ 1^{12}41^{22}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 47 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1548] \ 21^{13}51^{13}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1543] \ 1^{12}4^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 10 & 1 & 87 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1549] \ 1^{12}4^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 7 & 2 & 36 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1550] \ 2^{13}41^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 9 & 9 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1551] \ 1^{14}4^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 72 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1552] \ 1^{13}4^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 72 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1553] \ 2^3 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1554] $1^{12} 4^{1^{11} 31^2} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 9 & 1 & 65 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1555] $1^{13} 4^{1^{13} 31^2} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 54 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1556] $1^{11} 4^{1^{13} 31^1} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 8 & 2 \\ 11 & 8 & 38 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1557] $31^{13} 1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 12 & 3 \\ 6 & 13 & 26 & 7 \\ 12 & 26 & 46 & 14 \\ 3 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[1558] $21^{1^2} 51^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1559] $221^{1^1} 251^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1560] $1^{1^3} 5^{1^2} 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1565] $15^{1^2} 1^{1^3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 18 & 4 \\ 13 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1562] $1^{11} 4^{1^{13} 31^1} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 10 & 2 \\ 13 & 10 & 70 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1563] $31^{11} 31^{1^3} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 13 & 2 \\ 13 & 9 & 13 & 89 & 17 \\ 2 & 2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1564] $2^3 141^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1561] $2^{1^{11} 41^1} 3^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 6 & 10 & 38 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1566] \ 1^{13} \overset{1^3}{5} \overset{1^3, 1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1567] \ 21^3 \overset{1^1}{5} \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1568] \ 1^{12} \overset{1^1}{4} \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 9 & 2 & 72 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1569] \ 1^{12} \overset{1^1}{4} \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 8 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1570] \ 1^{11} \overset{1^1}{4} \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 9 & 2 \\ 12 & 9 & 52 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1571] \ 41^{12} \overset{1^1}{2} \overset{1^1}{3} \overset{1^1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 8 & 2 \\ 4 & 10 & 5 & 24 & 6 \\ 2 & 5 & 0 & 8 & 2 \\ 8 & 24 & 8 & 46 & 12 \\ 2 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1572] \ 21^{13} \overset{1^1}{4} \overset{1^1}{2} \overset{1^1}{3} \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 22 & 0 & 84 & 16 \\ 2 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1573] \ 31^{11} \overset{1^1}{4} \overset{1^1}{1} \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 11 & 2 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 10 & 2 \\ 11 & 15 & 10 & 45 & 13 \\ 2 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1574] \ 31^{11} \overset{1^1}{4} \overset{1^1}{1} \overset{1^1}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 5 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1575] \ 21^{13} \overset{1^1}{4} \overset{1^1}{2} \overset{1^1}{3} \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 26 & 0 & 132 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1576] \ 1^{13} \overset{1^4}{5} \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1577] $1^{11} 1^{13} 31^2 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 6 & 3 & 41 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1578] $1^1 1^{13} 31^3 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 7 & 2 \\ 13 & 7 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1579] $1^{11} 41^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 6 & 3 & 41 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1580] $1^{12} 1^{13} 31^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 6 & 3 & 29 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1581] $1^{13} 1^{13} 31^3 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 13 & 13 & 0 & 132 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1582] $1^{11} 1^{14} 31^1 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 7 & 4 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1583] $1^{13} 1^{13} 31^3 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 11 & 0 & 84 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1584] $21^2 51^{13} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 27 & 4 \\ 18 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1585] $1^{13} 1^3 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1586] $1^{13} 1^4 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 27 & 4 \\ 20 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1587] $21^4 51^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 27 & 4 \\ 20 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

- [1588] $21^{13}4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 18 & 3 & 61 & 15 \\ 2 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1589] $21^{13}41^331^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 24 & 3 & 115 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1592] $1^{11}41^1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$
- [1594] $1^{12}4131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 9 & 7 & 4 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$
- [1595] $31^{11}31^441^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 19 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 3 & 2 & 0 & 19 & 2 \\ 19 & 17 & 19 & 225 & 27 \\ 2 & 2 & 2 & 27 & 2 \end{pmatrix}_{b_0}$
- [1596] $1^{13}4^131^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 12 & 0 & 106 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$
- [1597] $31^{11}31^241^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 11 & 2 \\ 11 & 9 & 11 & 65 & 15 \\ 2 & 2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1598] $1^{14}4131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 10 & 2 & 58 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$
- [1599] $1^{11}1^{14}31^14^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$
- [1600] $1^{11}1^{12}31^34^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$
- [1601] $1^{13}41^3231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 12 & 0 & 106 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$
- [1593] $221^{12}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 12 & 3 \\ 6 & 13 & 29 & 7 \\ 12 & 29 & 57 & 15 \\ 3 & 7 & 15 & 3 \end{pmatrix}_{b_0}$

[1591] $21^{1^3 5 1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 20, \Delta = -100, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1590] $1^{1^2 3 1^3} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 7 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[1602] $1^{1^2 3 1^3} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 11 & 2 \\ 15 & 11 & 96 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1603] $21^{2^4 5 1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 20, \Delta = -100, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 30 & 4 \\ 20 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$

[1604] $1^{1^2 3 1^3} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 8 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$

[1605] $1^{1^2 2^3 2 1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$

[1606] $1^{1^3} 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 20, \Delta = -100, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1607] $21^{1^3 5 1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 20, \Delta = -100, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 4 & 27 & 4 \\ 19 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1608] $1^{1^4} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 11 & 1 & 69 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1609] $41^{1^2 2 2 3 1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$

[1610] $21^{1^3 4 1 3 1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 18 & 2 & 56 & 14 \\ 2 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[1611] $21^{1^3 4 1^4 3 1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 26 & 2 & 136 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0} \\
[1613] & 1^{241}1^{12}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 9 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 16 & 1 & 47 & 13 \\ 2 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1614] & 1^{241}1^{12}231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 11 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 16 & 3 & 55 & 15 \\ 2 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0} \\
[1612] & 21^{13}41^331^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 24 & 0 & 106 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1615] & 1^{13}23141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0} \\
[1619] & 1^{14}1^{11}31^14^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 13 & 1 & 113 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1620] & 1^{13}1^{11}31^24^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 9 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1621] & 1^{11}1^{13}31^24^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 7 & 3 & 57 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0} \\
[1618] & 1^{13}1^{12}4131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 8 & 3 & 45 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1616] & 1^{14}1^{11}31^14^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 12 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0} \\
[1617] & 1^{11}1^{12}31^34^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 9 & 2 \\ 13 & 9 & 56 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[1622] & 1^{11}41^3231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 8 & 2 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1623] $41^{11}2^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 6 & 2 \\ 4 & 10 & 5 & 16 & 6 \\ 2 & 5 & 0 & 7 & 2 \\ 6 & 16 & 7 & 21 & 9 \\ 2 & 6 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1624] $1^{13}4^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 8 & 2 & 38 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1625] $21^{11}51^{13}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 18 & 4 \\ 13 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1626] $1^{12}4^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 7 & 2 & 36 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1627] $1^{11}4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 8 & 2 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1628] $1^{14}4131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 11 & 2 & 76 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1630] $41^{13}2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[1631] $1^{11}41^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 10 & 2 \\ 14 & 10 & 74 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1632] $1^{11}4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 10 & 2 \\ 14 & 10 & 74 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1633] $21^{13}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 18 & 4 \\ 13 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1634] $31^{11}31^{11}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 7 & 2 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 2 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1629] $1^{12}4131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 8 & 4 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1635] $21^3 \overset{1^2}{5} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 27 & 4 \\ 19 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1636] $1^{\overline{14}} 2^3 21^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[1637] $1^{\overline{13}} \overset{1^3}{5} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 27 & 4 \\ 19 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1638] $1^{\overline{14}} 2^3 21^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[1639] $1^{\overline{11}} \overset{1^3}{31} 4^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 38 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1643] $1^{\overline{14}} 4^{\overline{1}} 131^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 12 & 2 & 98 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1644] $31^{\overline{13}} 31^{\overline{4}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 3 & 2 & 0 & 15 & 2 \\ 15 & 13 & 15 & 133 & 21 \\ 2 & 2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[1645] $1^{\overline{14}} 1^{\overline{2}} 231^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 8 & 2 \\ 14 & 8 & 54 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1642] $1^{\overline{11}} \overset{1^{12}}{31} 4^{\overline{1}} 1^{\overline{4}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 9 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1641] $1^{\overline{14}} \overset{1^{12}}{31} 4^{\overline{1}} 1^{\overline{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 8 & 2 \\ 14 & 8 & 54 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1640] $1^{\overline{11}} \overset{1^{13}}{31} 4^{\overline{1}} 1^{\overline{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 8 & 2 \\ 14 & 8 & 54 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1646] $21^{13}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1647] $1^{13}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 27 & 4 \\ 20 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1648] $1^{12}41^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 7 & 3 & 41 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$

[1649] $1^{12}41^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 9 & 1 & 65 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1650] $41^1322 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 4 & 7 & 4 \\ 3 & 7 & 7 & 5 & 5 \\ 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[1651] $1^{11}41^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 8 & 3 & 77 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$

[1652] $141^{13}31^13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1653] $23141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 98 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$

[1654] $23141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1655] $21^{13}41^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 1 & 20 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 20 & 1 & 67 & 15 \\ 2 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1656] $41^1322 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 3 & 7 & 2 \\ 3 & 5 & 8 & 15 & 5 \\ 3 & 8 & 4 & 15 & 4 \\ 7 & 15 & 15 & 33 & 11 \\ 2 & 5 & 4 & 11 & 3 \end{pmatrix}_{b_0}$

[1657] $21^{13}41^431^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 26 & 1 & 133 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1658] $31^{13}3141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 34 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[1659] $21^{25}1^{13}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1660] $21^{13}51^{12}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1662] $1^{13}4^{12}1^{11}31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 10 & 1 & 65 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1661] $21^{45}1^{13}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 30 & 4 \\ 22 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

[1666] $1^141^{3231}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 9 & 2 \\ 15 & 9 & 70 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1667] $1^{13}31^34^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 9 & 2 \\ 15 & 9 & 70 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1668] $1^{12}41^{3231}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 10 & 1 & 85 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1669] $1^{12}4^{13}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 10 & 1 & 85 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1670] $1^{12}4^{11}31^34^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 11 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1671] $31^{11}31^{341}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 14 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 3 & 2 & 0 & 14 & 2 \\ 14 & 11 & 14 & 109 & 19 \\ 2 & 2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[1663] $1^{11}4^{14}31^14^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 9 & 2 \\ 13 & 9 & 52 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1664] \ 1^{13} 4^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 9 & 3 & 61 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1665] \ 1^{13} 5^{14} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 30 & 4 \\ 22 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1672] \ 31^{12} 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 32 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1673] \ 1^{13} 4^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 10 & 2 \\ 16 & 10 & 90 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1674] \ 1^{13} 5^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1676] \ 221^{13} 5^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1675] \ 21^{13} 5^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1677] \ 21^{13} 41^{13} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 22 & 3 & 93 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1678] \ 21^{13} 41^{13} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 20 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 20 & 3 & 75 & 17 \\ 2 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1679] \ 1^{12} 41^{23} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 8 & 2 & 50 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1680] \ 1^{12} 4^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 8 & 2 & 50 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1681] $1^{13} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 9 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1682] $2^1 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 5 & 22 & 5 \\ 14 & 22 & 74 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[1683] $2^1 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 5 & 22 & 5 \\ 14 & 22 & 74 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[1688] $1^{13} 4^{14} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 10 & 3 & 81 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1687] $1^{11} 4^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 9 & 2 \\ 13 & 9 & 50 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1689] $1^{14} 4^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 9 & 2 \\ 15 & 9 & 68 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1690] $1^{13} 4^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 12 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1684] $1^{11} 4^{12} 2^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 9 & 2 \\ 13 & 9 & 50 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1686] $1^{13} 4^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 11 & 1 & 85 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1685] $1^{12} 4^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 8 & 3 & 57 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1691] $31^{11} 31^4 4^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 17 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 3 & 2 & 0 & 17 & 2 \\ 17 & 13 & 17 & 165 & 23 \\ 2 & 2 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[1693] $1^{13} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 30 & 4 \\ 22 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

[1692] $21^3 \overset{1^4}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 4 & 33 & 4 \\ 23 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[1694] $2^{\overset{1^1}{1^1} 41^1} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 7 & 13 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1695] $31^{\overset{1^1}{1^1} 41^1} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 8 & 2 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1696] $1^{\overset{1^3}{1^4} 2321^4} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1697] $31^{\overset{1^1}{1^1} 31^4 41^1} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 18 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 3 & 2 & 0 & 18 & 2 \\ 18 & 15 & 18 & 193 & 25 \\ 2 & 2 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[1698] $1^{\overset{1^3}{1^2} 31^1} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 9 & 3 & 77 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1699] $1^{\overset{1^3}{1^1} 31^2} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 11 & 2 \\ 15 & 11 & 86 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1701] $1^{\overset{1^3}{1^1} 31^2} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 10 & 2 \\ 14 & 10 & 66 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1702] $31^{\overset{1^4}{1^2} 31^1 41^1} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 13 & 3 \\ 5 & 5 & 17 & 5 \\ 13 & 17 & 44 & 14 \\ 3 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[1703] $1^{\overset{1^4}{1^1} 41^1} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 10 & 2 \\ 16 & 10 & 86 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1704] $1^{\overset{1^4}{1^2} 31^2} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 9 & 2 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1705] $1^{\overset{1^4}{1^1} 41^1} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 10 & 2 \\ 14 & 10 & 66 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1706] \ 1^{13} 4^{1^{12}31^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 10 & 2 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1707] \ 1^{11} 4^{1^{12}31^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 10 & 2 \\ 15 & 10 & 76 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1700] \ 31^{1^{12}31^4} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 28 & 3 \\ 5 & 5 & 38 & 5 \\ 28 & 38 & 254 & 32 \\ 3 & 5 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[1708] \ 1^{14} 2321^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[1719] \ 1^{11} 4^{1^{12}31^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 11 & 2 \\ 16 & 11 & 96 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1713] \ 1^{13} 4^{1^{13}31^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 13 & 0 & 132 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1714] \ 1^{11} 4^{1^{12}31^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 9 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1715] \ 21^{1^3} 51^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 33 & 4 \\ 24 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[1716] \ 1^{13} 41^4 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 14 & 0 & 160 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1717] \ 1^{13} 4^{1^2} 131^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 9 & 8 & 4 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1718] \ 1^{12} 4^{1^{14}31^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 7 & 4 & 48 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1712] $1^{1^3 1^4} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 14 & 0 & 160 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1711] $1^{1^4 1^3 1^3} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 13 & 14 & 0 & 132 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1709] $1^{1^3} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 33 & 4 \\ 24 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[1710] $1^{1^4 1^3 1^3} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 15 & 0 & 160 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1720] $1^{1^2 1^3 1^2} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 7 & 3 & 41 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1723] $1^{1^2 1^3 1^2} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 10 & 2 & 90 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1721] $1^{2 4 1^{12} 2 3 1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 10 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 16 & 2 & 50 & 14 \\ 2 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1722] $1^{1^3 1^2 1^1} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 11 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1724] $1^{1^4 1^3 1^1} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 11 & 2 \\ 15 & 11 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1725] $1^{1^4 1^3 1^2} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 11 & 1 & 69 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1726] $2 1^{1^3 4 1^{13} 1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 20 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 20 & 2 & 70 & 16 \\ 2 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1727] $2 1^{1^3 4 1^3 3 1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 24 & 2 & 110 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1728] $1^{11} 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 10 & 2 & 116 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1729] $21^{13} 41^3 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 24 & 1 & 107 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1730] $1^{11} 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 8 & 4 & 84 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1731] $1^{14} 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 10 & 2 & 56 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1732] $2^{12} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 8 & 11 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1733] $21^{13} 41^3 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 22 & 1 & 85 & 17 \\ 2 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1734] $1^{12} 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 8 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1735] $1^{11} 41^4 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 10 & 2 & 116 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1736] $1^{13} 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 9 & 2 \\ 16 & 9 & 72 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1737] $1^{12} 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 10 & 1 & 87 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1738] $1^{11} 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 8 & 3 & 75 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

- [1739] $1^{11}41^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 8 & 3 & 75 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1740] $1^{13}41^{13}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 8 & 3 & 43 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$
- [1741] $21^451^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 4 & 36 & 4 \\ 26 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$
- [1742] $221^451^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 4 & 36 & 4 \\ 26 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$
- [1743] $1^{13}51^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 4 & 36 & 4 \\ 26 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$
- [1744] $1^{13}41^{11}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 10 & 1 & 67 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1745] $1^{14}4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 10 & 3 & 63 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1747] $1^{11}41^{13}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 52 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$
- [1746] $1^{11}41^{14}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 52 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$
- [1748] $221^{13}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 21 & 3 \\ 6 & 13 & 51 & 7 \\ 21 & 51 & 187 & 27 \\ 3 & 7 & 27 & 3 \end{pmatrix}_{b_0}$
- [1749] $1^{11}41^{13}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 9 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1752] $1^{12}41^{13}31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 11 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1751] $1^{\overline{1}1^{\overline{3}31^{\overline{4}}}}4^{\overline{1}3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 10 & 2 \\ 17 & 10 & 90 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1750] $21^{\overline{1}3}41^{\overline{2}31^{\overline{1}2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 22 & 2 & 88 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1753] $1^{\overline{1}41^{\overline{4}231^{\overline{1}3}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 11 & 2 \\ 18 & 11 & 112 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1754] $1^{\overline{1}1^{\overline{3}31^{\overline{4}}}}4^{\overline{1}4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 11 & 2 \\ 18 & 11 & 112 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1755] $1^{\overline{1}3}4131^{\overline{1}4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 9 & 4 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1756] $1^{\overline{1}2}1^{\overline{1}4}31^{\overline{1}1}4^{\overline{1}3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 8 & 4 & 64 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1757] $1^{\overline{1}3}1^{\overline{1}2}31^{\overline{1}2}4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 9 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1758] $1^{\overline{1}1^{\overline{1}4}31^{\overline{1}3}}4^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 9 & 2 \\ 16 & 9 & 68 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1759] $1^{\overline{1}2}1^{\overline{1}1}31^{\overline{1}4}4^{\overline{1}4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 12 & 1 & 135 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1760] $1^{\overline{1}2}41^{\overline{4}231^{\overline{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 12 & 1 & 135 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1762] $41^{\overline{1}2}231^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[1763] $41^{\overline{1}3}232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 12 & 2 \\ 4 & 10 & 5 & 36 & 6 \\ 2 & 5 & 0 & 12 & 2 \\ 12 & 36 & 12 & 106 & 18 \\ 2 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1761] \ 1^{14} 4 \ 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 12 & 1 & 87 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1764] \ 1^{12} 4 \ 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 8 & 3 & 55 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1765] \ 1^{12} 41^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 8 & 3 & 55 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1766] \ 1^{14} 4 \ 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 231, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 11 & 2 & 72 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1767] \ 1^{12} 4 \ 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 231, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 9 & 2 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1768] \ 1^{14} 4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 11 & 3 & 81 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1769] \ 2 \ 3 \ 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1770] \ 1^{13} 4 \ 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 11 & 1 & 85 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1771] \ 1^{221} 5 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 6 & 15 & 3 \\ 6 & 13 & 36 & 7 \\ 15 & 36 & 88 & 18 \\ 3 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1775] \ 21^{14} 4131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 24 & 2 & 98 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1776] \ 1^{14} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 13 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1777] & 2221^{12}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 18 & 1 & 40 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 40 & 0 & 78 & 16 \\ 2 & 8 & -1 & 16 & 2 \end{pmatrix}_{b_0} \\
[1774] & 1^{11}4^{1331^3}1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 10 & 2 \\ 15 & 10 & 66 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[1773] & 1^{13}4^{1331^1}1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 9 & 3 & 57 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0} \\
[1772] & 2^13^{14}141^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1778] & 2^13^{14}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0} \\
[1779] & 2^13^{16}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.2 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0} \\
[1780] & 1^{13}4^{14}131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 10 & 4 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1781] & 1^{12}4^{1431^1}1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 9 & 4 & 84 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1783] & 1^341^{13}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 13 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 24 & 1 & 107 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1782] & 1^{14}4^{1131^2}1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 14 & 1 & 135 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[1785] & 1^{14}4^{14}131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 12 & 3 & 103 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1786] $31^{12}31^341^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 23 & 3 \\ 5 & 5 & 31 & 5 \\ 23 & 31 & 164 & 26 \\ 3 & 5 & 26 & 3 \end{pmatrix}_{b_0}$
- [1787] $31^{123}31^241^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 18 & 3 \\ 5 & 5 & 24 & 5 \\ 18 & 24 & 94 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$
- [1788] $1^{14}31^34^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 10 & 2 \\ 17 & 10 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [1789] $1^{11}41^2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 10 & 2 \\ 15 & 10 & 64 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [1790] $1^{14}41^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 10 & 2 \\ 17 & 10 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [1791] $1^{11}31^24^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 10 & 2 \\ 15 & 10 & 64 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [1792] $1^{13}41^3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 12 & 1 & 107 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1784] $1^{13}31^34^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 12 & 1 & 107 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1793] $1^{11}31^34^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 11 & 2 \\ 16 & 11 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [1794] $1^{11}41^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 11 & 2 \\ 16 & 11 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [1795] $1^{12}41^3231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 10 & 2 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$
- [1796] $21^{14}41^131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 26 & 1 & 113 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1797] $1^{14}31^34^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 12 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1798] $1^{13} 4^1 1^2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 10 & 2 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1799] $1^{12} 4^1 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 10 & 2 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1800] $1^{11} 4^1 1^4 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 12 & 2 \\ 17 & 12 & 106 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1801] $1^{13} 4^1 1^4 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 13 & 1 & 133 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1802] $1^{12} 4^1 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 9 & 3 & 73 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1805] $1^{14} 4^1 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 11 & 2 \\ 18 & 11 & 104 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1803] $1^{12} 4^1 1^4 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 11 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1804] $1^{14} 4^1 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 13 & 2 & 116 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1807] $21^{14} 41^3 31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 30 & 0 & 160 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1808] $1^{13} 4^1 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 10 & 3 & 75 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1809] $21^{14} 41 31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 24 & 4 & 110 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1810] $2^{1^{13}4^{11}} 3^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 10 & 12 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1806] $21^{1^4 4^{14}} 3^{1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 20 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 20 & 32 & 4 & 206 & 28 \\ 2 & 4 & 0 & 28 & 3 \end{pmatrix}_{b_0}$

[1811] $1^{1^3} 4^{1^{11} 3^{14}} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 259$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 11 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$

[1812] $1^{1^2} 4^{1^{13} 3^{12}} 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 10 & 3 & 95 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1813] $1^{1^4} 4^{1^3} 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 15 & 0 & 160 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1814] $1^{1^{11} 3^{14}} 4^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 10 & 2 \\ 16 & 10 & 70 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1815] $1^{1^{11} 3^{14}} 4^{1^3} 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 263$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 11 & 2 \\ 16 & 11 & 80 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1816] $1^{1^3} 4^{1^{12} 3^{12}} 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 263$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 11 & 2 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1817] $1^{1^{11} 3^{14}} 4^{1^4} 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 9 & 4 & 104 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1818] $1^{1^2} 4^{1^4} 1^{13} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 10 & 4 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[1819] $1^{1^2} 4^{1^{14} 3^{12}} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 8 & 4 & 62 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[1820] $1^{1^4} 4^{1^4} 2^{31} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 16 & 0 & 190 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0} \\
[1821] & 1^{12}41^2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 8 & 4 & 62 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0} \\
[1822] & 1^{14}4^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 16 & 0 & 190 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0} \\
[1823] & 2^13^{14}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 5 & 27 & 5 \\ 17 & 27 & 112 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0} \\
[1826] & 1^{14}4^{11}3^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 12 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0} \\
[1827] & 1^{11}4^{13}3^{14}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 10 & 3 & 121 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[1824] & 1^{11}41^4231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 10 & 3 & 121 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[1825] & 1^{14}4^{13}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 10 & 3 & 61 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0} \\
[1828] & 1^{13}4^{13}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 11 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1829] & 41^{12}2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 1 & 9 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0} \\
[1830] & 21^{14}4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 24 & 3 & 103 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1831] & 1^{13}4^{12}3^{12}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 12 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1832] $2^1 3^{13} 14^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 271$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1833] $1^{11} 4^{14} 31^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 12 & 2 \\ 17 & 12 & 100 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$

[1834] $21^{14} 41^{43} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 32 & 0 & 190 & 24 \\ 2 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$

[1835] $221^{13} 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 275$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 19 & 3 \\ 6 & 13 & 46 & 7 \\ 19 & 46 & 148 & 24 \\ 3 & 7 & 24 & 3 \end{pmatrix}_{b_0}$

[1836] $1^{13} 4^{13} 31^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 275$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 9 & 3 & 57 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1837] $1^{14} 31^{34} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 10 & 2 \\ 18 & 10 & 86 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1838] $41^{13} 2231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$

[1839] $31^{12} 31^4 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 31 & 3 \\ 5 & 5 & 43 & 5 \\ 31 & 43 & 320 & 36 \\ 3 & 5 & 36 & 3 \end{pmatrix}_{b_0}$

[1840] $31^{12} 31^4 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 16 & 3 \\ 5 & 5 & 22 & 5 \\ 16 & 22 & 74 & 18 \\ 3 & 5 & 18 & 3 \end{pmatrix}_{b_0}$

[1841] $1^{12} 4^{12} 31^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 10 & 2 & 90 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1842] $1^{11} 4^{14} 31^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 10 & 2 \\ 16 & 10 & 66 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1843] $1^{13} 4^{14} 31^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 280$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 9 & 4 & 64 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[1844] $1^{14} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 11 & 2 & 72 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1845] $1^{13} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 10 & 2 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1846] $31^{14} 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 16 & 3 \\ 5 & 5 & 20 & 5 \\ 16 & 20 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1847] $221^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 17 & 3 \\ 6 & 13 & 41 & 7 \\ 17 & 41 & 115 & 21 \\ 3 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[1848] $1^{11} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 11 & 2 \\ 17 & 11 & 86 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1849] $1^{13} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 12 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1850] $1^{12} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 9 & 3 & 73 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1851] $2^{14} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 287, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 12 & 13 & 92 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1852] $2^{11} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 5 & 27 & 5 \\ 17 & 27 & 110 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[1853] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 13 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1856] $2^{12} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 11 & 9 & 14 & 77 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1855] $1^{12} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 9 & 4 & 80 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1854] $1^{14}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 11 & 4 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1857] $1^{12}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 11 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1858] $1^{11}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 12 & 2 \\ 18 & 12 & 106 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1859] $1^{11}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 13 & 2 \\ 19 & 13 & 130 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1860] $1^{11}4^{14}2^{31}1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 13 & 2 \\ 19 & 13 & 130 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1861] $1^{14}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 11 & 2 \\ 19 & 11 & 104 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1862] $1^{14}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 11 & 3 & 77 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1863] $1^{13}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 13 & 1 & 133 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1864] $21^{14}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 26 & 4 & 128 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1865] $21^{14}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 30 & 4 & 176 & 26 \\ 2 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

[1866] $1^{13}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 16 & 14 & 1 & 161 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1867] $1^{14}4^{13}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 14 & 1 & 133 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$

[1868] $1^{12}41^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 10 & 3 & 93 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1869] $1^{13}41^4231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 16 & 14 & 1 & 161 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$

[1870] $1^{11}4^{14}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 11 & 2 \\ 17 & 11 & 80 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1871] $1^{13}4^{13}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 10 & 3 & 73 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$

[1872] $1^{12}4^{13}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 10 & 3 & 93 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1875] $1^{12}4^{12}31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 308$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 12 & 2 & 138 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1874] $1^{13}4^{12}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 308$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 11 & 2 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1877] $1^{14}4^{12}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 308$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 12 & 2 & 90 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1876] $1^{13}4^{14}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 308$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 10 & 4 & 82 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1873] $1^{12}41^4231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 308$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 12 & 2 & 138 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1878] $21^{14}41^231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 311$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86$

[1881] $1^{14}4^{14}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 0 & 12 & 2 \\ 20 & 12 & 126 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[1882] $1^{14}4^{14}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 12 & 4 & 110 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1880] $1^{12}4^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 10 & 4 & 102 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1879] $1^{14}4^{14}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 0 & 12 & 2 \\ 20 & 12 & 126 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[1883] $1^{14}4^{11}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 15 & 1 & 161 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$

[1884] $31^{13}3141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 15 & 3 \\ 5 & 5 & 20 & 5 \\ 15 & 20 & 58 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$

[1885] $2^3 141^{17} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 194 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$

[1886] $21^{14}4^{14}1^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 26 & 2 & 116 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1887] $1^{44}1^{14}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 17 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 32 & 1 & 191 & 25 \\ 2 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0}$

[1888] $21^{14}4^{14}2^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 1 & 28 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 28 & 4 & 150 & 24 \\ 2 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$

[1889] $1^{12}4^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 323, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 11 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$

[1890] $1^{14}4^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 323, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 12 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1892] $31^{12}31^341^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 26 & 3 \\ 5 & 5 & 36 & 5 \\ 26 & 36 & 218 & 30 \\ 3 & 5 & 30 & 3 \end{pmatrix}_{b_0}$

[1891] $31^{123}31^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 21 & 3 \\ 5 & 5 & 29 & 5 \\ 21 & 29 & 136 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$

[1893] $1^{12}1^{14}31^34^12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 9 & 4 & 80 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1894] $2^13^141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1895] $1^{14}1^{11}31^44^11 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 13 & 1 & 113 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1896] $1^{13}41^3231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 12 & 2 & 110 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1897] $1^{13}1^{14}31^14^11 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 11 & 4 & 104 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1898] $1^{13}1^{12}31^34^13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 12 & 2 & 110 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1899] $1^{14}1^{12}31^24^13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 13 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1900] $1^{11}1^{14}31^34^13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 12 & 2 \\ 18 & 12 & 98 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$

[1901] $1^{11}41^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 12 & 2 \\ 18 & 12 & 98 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$

[1902] $31^{13}31^141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 17 & 3 \\ 5 & 5 & 22 & 5 \\ 17 & 22 & 74 & 18 \\ 3 & 5 & 18 & 3 \end{pmatrix}_{b_0}$

[1903] $1^{13} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 11 & 3 & 93 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1905] $21^{14} 41^{13} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 335, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 26 & 3 & 121 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$

[1907] $1^{11} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 335, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 11 & 2 \\ 18 & 11 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1904] $21^{14} 41^{43} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 335, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 19 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 19 & 32 & 3 & 199 & 27 \\ 2 & 4 & 0 & 27 & 3 \end{pmatrix}_{b_0}$

[1906] $1^{13} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 335, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 11 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1909] $41^{14} 232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 2 & 16 & 2 \\ 4 & 10 & 5 & 48 & 6 \\ 2 & 5 & 0 & 16 & 2 \\ 16 & 48 & 16 & 190 & 24 \\ 2 & 6 & 2 & 24 & 3 \end{pmatrix}_{b_0}$

[1908] $1^{12} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 10 & 3 & 95 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1910] $1^{11} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 13 & 2 \\ 19 & 13 & 120 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[1911] $1^{14} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 13 & 3 & 121 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$

[1912] $1^{13} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 10 & 3 & 75 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$

[1913] $1^{13} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 13 & 2 & 136 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1914] $1^{14} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 14 & 2 & 138 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0} \\
[1915] & 21^{14}41^331^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 30 & 1 & 161 & 23 \\ 2 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0} \\
[1916] & 21^{14}41^431^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 32 & 1 & 191 & 25 \\ 2 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0} \\
[1917] & 1^{13}4^{13}31^2 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 12 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[1918] & 1^{14}4^{13}31^2 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 11 & 3 & 77 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0} \\
[1919] & 1^{14}4^{12}31^3 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 12 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1921] & 41^{12}2^3 3^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 2 & 10 & 2 \\ 4 & 10 & 5 & 28 & 6 \\ 2 & 5 & 0 & 11 & 2 \\ 10 & 28 & 11 & 65 & 15 \\ 2 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1920] & 1^{14}4^{11}31^4 4^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 14 & 1 & 135 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[1924] & 1^{12}4^{14}31^3 4^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 10 & 4 & 100 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[1925] & 1^{14}4^{14}31^1 4^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 11 & 4 & 84 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1923] & 1^{12}41^3231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 10 & 4 & 100 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[1922] & 1^{13}4^{14}31^2 4^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 10 & 4 & 80 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1926] $1^{341}1^{3231}1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 16 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 24 & 4 & 122 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1927] $1^{341}1^{3231}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 14 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 24 & 2 & 110 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [1928] $1^{12}1^{1331}4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 11 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1929] $21^{1441}2^{31}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 28 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 28 & 2 & 138 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1930] $21^{1441}4^{31}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 32 & 2 & 194 & 26 \\ 2 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$
- [1931] $1^{13}1^{1331}4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 12 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [1932] $21^{1441}3^{31}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 30 & 3 & 169 & 25 \\ 2 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0}$
- [1933] $21^{1441}2^{31}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 1 & 28 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 28 & 3 & 143 & 23 \\ 2 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$
- [1934] $1^{11}1^{1431}4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 12 & 2 \\ 19 & 12 & 100 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$
- [1935] $2^{112}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 13 & 10 & 17 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1936] $2^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 12 & 11 & 15 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1937] $1^{14}1^{1131}4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 16 & 15 & 1 & 161 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1938] $1^{13} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 11 & 3 & 93 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1939] $21^{14} 41^3 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 380, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 30 & 2 & 164 & 24 \\ 2 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1940] $1^{12} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 383, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 11 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1941] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 383, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 13 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1942] $1^{12} 41^4 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 12 & 3 & 143 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1943] $1^{12} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 12 & 3 & 143 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1944] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 12 & 3 & 95 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1945] $1^{14} 41^4 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 16 & 1 & 191 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[1946] $1^{14} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 16 & 1 & 191 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[1947] $1^{14} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 12 & 4 & 104 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1948] $1^{11} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 0 & 13 & 2 \\ 20 & 13 & 120 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

- [1949] $1^{13} 4^{1^{12}31^4} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 391$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 13 & 2 & 136 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1950] $1^{13} 4^{1^{14}31^2} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 391$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 11 & 4 & 100 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [1951] $1^3 41^{13} 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 15 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 24 & 3 & 115 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1952] $2^3 3 141^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1953] $2^3 3 1^{14} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 402$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 5 & 32 & 5 \\ 20 & 32 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$
- [1954] $1^{13} 4^{1^{13}31^4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 403$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 11 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1955] $1^{13} 4^{1^{13}31^3} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 12 & 3 & 115 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1956] $1^{11} 4^{1^{14}31^4} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 0 & 14 & 2 \\ 21 & 14 & 144 & 26 \\ 3 & 2 & 26 & 4 \end{pmatrix}_{b_0}$
- [1957] $1^{13} 41^3 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 12 & 3 & 115 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1958] $1^{11} 41^4 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 0 & 14 & 2 \\ 21 & 14 & 144 & 26 \\ 3 & 2 & 26 & 4 \end{pmatrix}_{b_0}$
- [1961] $1^{13} 4^{1^{12}31^4} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 407$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 17 & 14 & 2 & 164 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [1960] $1^{13} 41^4 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 407$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 17 & 14 & 2 & 164 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1959] $1^{14} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 407, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 14 & 2 & 136 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1962] $1^{14} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 408, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 11 & 4 & 84 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1963] $1^{14} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 412, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 13 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1964] $1^{13} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 415, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 12 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1965] $31^{15} 3141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 415, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 20 & 3 \\ 5 & 5 & 25 & 5 \\ 20 & 25 & 98 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

[1966] $1^{14} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 415, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 13 & 4 & 128 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1967] $41^{12} 2^3 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 12 & 2 \\ 4 & 10 & 5 & 32 & 6 \\ 2 & 5 & 0 & 14 & 2 \\ 12 & 32 & 14 & 90 & 18 \\ 2 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1968] $1^{14} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 13 & 2 & 116 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1969] $1^{14} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 423, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 15 & 2 & 164 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1970] $1^{13} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 13 & 3 & 141 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1971] $1^{12} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 11 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1972] $1^{14} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 12 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1973] $41^{13} 2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[1974] $1^{14} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 14 & 3 & 143 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1975] $1^{13} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 11 & 4 & 100 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1976] $2221^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 1 & 60 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 60 & 0 & 178 & 24 \\ 2 & 8 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[1977] $31^{13} 31^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 439, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 29 & 5 \\ 22 & 29 & 134 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$$

[1978] $2^1 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 13 & 13 & 16 & 121 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1979] $1^{13} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 12 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1980] $41^{14} 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$

[1982] $1^{12} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 452, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 12 & 4 & 150 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1981] $1^{12} 41^4 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 452, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 12 & 4 & 150 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

- [1983] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 452$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 12 & 4 & 102 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [1984] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 456$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 14 & 2 & 138 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1985] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 467$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 13 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1986] $31^{13} 31^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 471$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 20 & 3 \\ 5 & 5 & 27 & 5 \\ 20 & 27 & 110 & 22 \\ 3 & 5 & 22 & 3 \end{pmatrix}_{b_0}$
- [1987] $1^{13} 41^3 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 472$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 12 & 4 & 122 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1988] $1^{13} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 472$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 12 & 4 & 122 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1989] $1^{13} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 475$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 13 & 3 & 141 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$
- [1990] $31^{14} 3141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 480$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 19 & 3 \\ 5 & 5 & 25 & 5 \\ 19 & 25 & 92 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$
- [1991] $1^{14} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 484$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 17 & 15 & 2 & 164 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [1992] $41^{13} 2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 485$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$
- [1993] $1^{14} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 488$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 13 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1994] $2^1 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 492$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 14 & 12 & 18 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1995] $1^{14} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 495$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 12 & 4 & 104 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1996] $31^{14} 31^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 496$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 21 & 3 \\ 5 & 5 & 27 & 5 \\ 21 & 27 & 112 & 22 \\ 3 & 5 & 22 & 3 \end{pmatrix}_{b_0}$$

[1999] $31^{13} 31^3 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 499$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 27 & 3 \\ 5 & 5 & 36 & 5 \\ 27 & 36 & 214 & 30 \\ 3 & 5 & 30 & 3 \end{pmatrix}_{b_0}$$

[2000] $1^{14} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 499$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 14 & 3 & 141 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1998] $1^{13} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 499$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 18 & 14 & 3 & 169 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[1997] $1^{13} 41^4 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 499$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 18 & 14 & 3 & 169 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[2001] $1^{13} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 500$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 13 & 4 & 148 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[2002] $1^{14} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 504$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 16 & 2 & 194 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

[2003] $1^{14} 41^4 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 504$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 16 & 2 & 194 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

[2004] $1^{14} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 504$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 13 & 3 & 121 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[2005] $31^{13} 31^4 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 511$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 32 & 3 \\ 5 & 5 & 43 & 5 \\ 32 & 43 & 314 & 36 \\ 3 & 5 & 36 & 3 \end{pmatrix}_{b_0}$$

[2006] $1^{13} 4^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 511$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 12 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2007] \ 1^{14} 4^{1^{14} 31^2} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 14 & 4 & 150 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2008] \ 1^{441} 1^{14} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 1 & 18 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 32 & 2 & 194 & 26 \\ 2 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[2009] \ 1^{14} 4^{1^{13} 31^3} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 523, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 15 & 3 & 169 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2010] \ 41^{13} 2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[2011] \ 1^{14} 4^{1^{14} 31^3} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 543, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 13 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2012] \ 1^{14} 4^{1^{13} 31^4} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 14 & 3 & 143 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2013] \ 1^{13} 4^{1^{14} 31^4} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 551, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 13 & 4 & 148 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2014] \ 31^{13} 31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 25 & 3 \\ 5 & 5 & 34 & 5 \\ 25 & 34 & 182 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2015] \ 31^{16} 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 30 & 5 \\ 24 & 30 & 142 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2016] \ 41^{13} 2^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 14 & 2 \\ 4 & 10 & 5 & 40 & 6 \\ 2 & 5 & 0 & 15 & 2 \\ 14 & 40 & 15 & 133 & 21 \\ 2 & 6 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2017] \ 1^{14} 4^{1^{14} 31^4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 13 & 4 & 128 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

- [2018] $1^{13} 1^{14} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 583$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 19 & 14 & 4 & 176 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$
- [2019] $1^{14} 1^{14} 31^3 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 583$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 14 & 4 & 148 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [2020] $1^{13} 41^4 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 583$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 19 & 14 & 4 & 176 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$
- [2021] $1^{14} 1^{13} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 584$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 18 & 15 & 3 & 169 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$
- [2022] $2^1 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 591$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 2 & 2 & 0 & 19 & 2 \\ 15 & 14 & 19 & 156 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2023] $1^{14} 1^{13} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 612$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 19 & 16 & 3 & 199 & 27 \\ 2 & 2 & 0 & 27 & 3 \end{pmatrix}_{b_0}$
- [2024] $1^{14} 41^4 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 612$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 19 & 16 & 3 & 199 & 27 \\ 2 & 2 & 0 & 27 & 3 \end{pmatrix}_{b_0}$
- [2025] $1^{14} 1^{14} 31^3 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 615$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 15 & 4 & 176 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$
- [2026] $1^{14} 1^{14} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 632$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 14 & 4 & 150 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [2027] $31^{13} 31^3 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 639$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 30 & 3 \\ 5 & 5 & 41 & 5 \\ 30 & 41 & 274 & 34 \\ 3 & 5 & 34 & 3 \end{pmatrix}_{b_0}$
- [2028] $1^4 41^{14} 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 640$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 19 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 19 & 32 & 3 & 199 & 27 \\ 2 & 4 & 0 & 27 & 3 \end{pmatrix}_{b_0}$
- [2029] $31^{13} 31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 651$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 5 & 35 & 3 \\ 5 & 5 & 48 & 5 \\ 35 & 48 & 386 & 40 \\ 3 & 5 & 40 & 3 \end{pmatrix}_{b_0} \\
[2030] & 41^{15}2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 140 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 140 & 0 & 698 & 50 \\ 2 & 10 & -1 & 50 & 3 \end{pmatrix}_{b_0} \\
[2031] & 31^{14}31^241^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 26 & 3 \\ 5 & 5 & 34 & 5 \\ 26 & 34 & 182 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0} \\
[2032] & 1^{14}4^131^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 676, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 19 & 15 & 4 & 176 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0} \\
[2033] & 1^{44}1^{14}231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 20 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 20 & 32 & 4 & 206 & 28 \\ 2 & 4 & 0 & 28 & 3 \end{pmatrix}_{b_0} \\
[2034] & 31^{14}31^141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 32 & 5 \\ 24 & 32 & 154 & 26 \\ 3 & 5 & 26 & 3 \end{pmatrix}_{b_0} \\
[2036] & 1^{14}4^131^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 712, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 20 & 16 & 4 & 206 & 28 \\ 2 & 2 & 0 & 28 & 3 \end{pmatrix}_{b_0} \\
[2035] & 1^{14}41^4231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 712, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 20 & 16 & 4 & 206 & 28 \\ 2 & 2 & 0 & 28 & 3 \end{pmatrix}_{b_0} \\
[2037] & 41^{14}2231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 716, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0} \\
[2038] & 41^{14}2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0} \\
[2039] & 31^{17}3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 775, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 28 & 3 \\ 5 & 5 & 35 & 5 \\ 28 & 35 & 194 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0} \\
[2040] & 31^{14}31^341^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 31 & 3 \\ 5 & 5 & 41 & 5 \\ 31 & 41 & 272 & 34 \\ 3 & 5 & 34 & 3 \end{pmatrix}_{b_0} \\
[2041] & 41^{13}2^13^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 2 & 16 & 2 \\ 4 & 10 & 5 & 44 & 6 \\ 2 & 5 & 0 & 18 & 2 \\ 16 & 44 & 18 & 166 & 24 \\ 2 & 6 & 2 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2042] $41^{14}2^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 836, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 2 & 18 & 2 \\ 4 & 10 & 5 & 52 & 6 \\ 2 & 5 & 0 & 19 & 2 \\ 18 & 52 & 19 & 225 & 27 \\ 2 & 6 & 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2043] $31^{14}31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 856, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 36 & 3 \\ 5 & 5 & 48 & 5 \\ 36 & 48 & 382 & 40 \\ 3 & 5 & 40 & 3 \end{pmatrix}_{b_0}$

[2044] $31^{14}31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 864, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 29 & 3 \\ 5 & 5 & 39 & 5 \\ 29 & 39 & 236 & 32 \\ 3 & 5 & 32 & 3 \end{pmatrix}_{b_0}$

[2045] $41^{14}2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 885, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$

[2046] $41^{14}2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 18 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$

[2047] $41^{16}2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 24 & 2 \\ 4 & 26 & 1 & 168 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 168 & 0 & 1006 & 60 \\ 2 & 10 & -1 & 60 & 3 \end{pmatrix}_{b_0}$

[2048] $31^{14}31^3 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 34 & 3 \\ 5 & 5 & 46 & 5 \\ 34 & 46 & 338 & 38 \\ 3 & 5 & 38 & 3 \end{pmatrix}_{b_0}$

[2049] $31^{14}31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1056, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 39 & 3 \\ 5 & 5 & 53 & 5 \\ 39 & 53 & 460 & 44 \\ 3 & 5 & 44 & 3 \end{pmatrix}_{b_0}$

[2050] $41^{14}2^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 1184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 2 & 20 & 2 \\ 4 & 10 & 5 & 56 & 6 \\ 2 & 5 & 0 & 22 & 2 \\ 20 & 56 & 22 & 266 & 30 \\ 2 & 6 & 2 & 30 & 3 \end{pmatrix}_{b_0}$

[2051] $41^{17}2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 28 & 2 \\ 4 & 26 & 1 & 196 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 196 & 0 & 1370 & 70 \\ 2 & 10 & -1 & 70 & 3 \end{pmatrix}_{b_0}$

$T = 9$ [2052] $1^1 4 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[2055] $1^1 4 14 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$

[2056] $1^1 5 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 1^1

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 3 & 5 \\ 2 & 3 & 1 & 3 \\ 3 & 5 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[2057] \quad 2 \quad 3 \quad 21^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[2058] \quad 1^{1+} 412321^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[2059] \quad 1^{1+} 51231^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2060] \quad 1^{1+} 314131^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2061] \quad 1^{1+} 31^{1+} 4131^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2054] \quad 14141^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2053] \quad 1^{1+} 412321^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[2071] \quad 21^{1+} 5131^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2066] \quad 1^{1+} 4131^{1+} 41^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2067] \quad 1^{1+} 5131^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2068] \quad 1^{1+} 51^{1+} 321^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 19 & 9 \\ 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[2069] \quad 1^{1+} 413141^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[2070] $1^1 5^{12} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2065] $21^1 5^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 3 & 5 \\ 3 & 3 & 1 & 3 \\ 4 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[2072] $151^1 2^1 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2073] $1^1 1^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -75$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

[2074] $1321^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 10 & 6 \\ 2 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[2075] $21^2 5^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2076] $1^1 5^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[2077] $1^{11} 322151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2078] $1^1 5^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[2079] $1^1 5^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2080] $1^2 5^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2064] $31^1 1^1 4 14 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 3 & 0 & 9 & 13 & 7 \\ 2 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[2062] $15 12 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2063] $1^{11} 5^{13} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 7 & 4 \\ 7 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[2081] $23 21^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 5 & 10 & 5 \\ 7 & 10 & 19 & 9 \\ 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[2082] $31 4 14 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 5 & 1 & 0 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 5 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[2083] $13 15^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

[2097] $22 3 15 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 2 & 4 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[2089] $15 13 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[2104] $1^{11} 4 13 14 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[2105] $2 3 15 1^{11} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2106] $15 13 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2096] $1^1 5^{11} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[2095] $1^1 3221^{11} 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 18 & 8 \\ 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2094] $21^{11} 5^{11} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2093] $2^1 3^{11} 151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2092] $21^1 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2100] $2^1 32151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2101] $1^{11} 322151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[2102] $1^{11} 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[2103] $2^1 3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2086] $1^1 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[2099] $1^{11} 5^{11} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2084] $1^1 5^{11} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2098] $31^1 4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[2087] $1^1 322151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[2088] $1^2 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 11 & 5 \\ 6 & 11 & 22 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

[2108] $1^1 5^{12} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[2107] $21^1 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[2091] $1^1 41^1 3141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[2090] $21^1 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[2085] $1^1 5^{11} 132 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2109] $1^{11} 2231^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 3 & 1 & 8 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 8 & 0 & 15 & 9 \\ 2 & 5 & -1 & 9 & 5 \end{pmatrix}_{b_0}$$

[2110] $231^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[2112] $1^{1^1} 5^{1^1} 1231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[2113] $1^{1^1} 5^{1^1} 131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2114] $1^{1^2} 5^{1^2} 131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 8 & 5 \\ 6 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$

[2115] $2^1 3^1 1^1 5^{1^2} 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 5 & 11 & 5 \\ 7 & 11 & 22 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$

[2111] $1^{1^2} 5^{1^2} 131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2118] $1^{1^1} 223131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$

[2117] $1^{1^1} 231321^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$

[2116] $1^{1^1} 223131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 5 & 4 \\ 2 & 0 & 5 & 3 & 5 \\ 2 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$

[2119] $1^{1^3} 5^{1^3} 1231^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 61 & 15 \\ 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$

[2120] $1^{1^1} 5^{1^3} 131^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$

[2121] $1^{1^1} 5^{1^2} 131^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$

[2122] $1^{1^2} 5^{1^2} 1231^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 26 & 10 \\ 2 & 2 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2123] \ 1^{\overline{11}} 5^{11} 1231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -10, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 5 & 5 \\ 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2124] \ 1^{\overline{1}} 5^{11} 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -10, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2125] \ 2^{\overline{141}} 3^{11} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -10, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 4 & 2 & 4 \\ 2 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2126] \ 31^{\overline{1}} 4^{11} 141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -10, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 9 & 0 & 13 & 7 \\ 2 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2127] \ 1^{\overline{1}} 5^{11} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -12, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2128] \ 2321^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -12, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2129] \ 1^{\overline{1}} 5^{11} 1231^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -12, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2131] \ 21^{\overline{1}} 5^{11} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -12, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2130] \ 1^{\overline{1}} 5^{11} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -12, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 8 & 5 \\ 4 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2132] \ 22^{\overline{11}} 3^{11} 151^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -13, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 4 & 13 & 4 \\ 6 & 8 & 13 & 35 & 11 \\ 2 & 2 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2137] \ 1^{\overline{11}} 314131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -14, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0} \\
[2138] & 1^{11} 412321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 6 & 6 \\ 3 & 3 & 6 & 5 \end{pmatrix}_{b_0} \\
[2134] & 1^{11} 3141^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0} \\
[2140] & 31^{11} 3151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[2139] & 1^{11} 5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[2133] & 1^{11} 41^{11} 2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 4 & 3 \\ 6 & 4 & 10 & 8 \\ 3 & 3 & 8 & 5 \end{pmatrix}_{b_0} \\
[2135] & 1^{11} 412321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 6 & 6 \\ 3 & 3 & 6 & 5 \end{pmatrix}_{b_0} \\
[2136] & 1^{11} 41^{11} 2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 4 & 3 \\ 6 & 4 & 10 & 8 \\ 3 & 3 & 8 & 5 \end{pmatrix}_{b_0} \\
[2144] & 21^{11} 5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[2146] & 2^1 3^1 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 5 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2147] & 1^{11} 151^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[2143] & 1^{11} 5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0} \\
[2142] & 21^{12} 5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 9 & 9 & 69 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2141] \quad 31 \quad 4 \quad 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 7 & 4 \\ 2 & 0 & 7 & 5 & 5 \\ 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2148] \quad 2 \quad 3 \quad 151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2149] \quad 21^{11} \quad 5 \quad 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2145] \quad 2 \quad 3 \quad 2151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2153] \quad 141^{11} \quad 31^{11} \quad 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 13 & 1 & 12 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 12 & 0 & 10 & 6 \\ 2 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2151] \quad 1 \quad 5 \quad 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2152] \quad 21^{11} \quad 3 \quad 5 \quad 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 19 & 5 \\ 13 & 19 & 65 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[2154] \quad 1 \quad 41^{11} \quad 11 \quad 31^{11} \quad 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 17 & 2 \\ 3 & 13 & 1 & 68 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 68 & 0 & 346 & 34 \\ 2 & 7 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[2155] \quad 2 \quad 3 \quad 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2156] \quad 1^{11} \quad 5 \quad 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2157] \quad 21^{11} \quad 3 \quad 5 \quad 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 5 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2158] $21^{11}51231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[2159] $1^1 \overset{1^3}{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 19 & 5 \\ 13 & 19 & 65 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$

[2160] $1^{11} \overset{1^3}{5} 1^3 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 12 & 0 & 155 & 23 \\ 2 & 2 & -1 & 23 & 3 \end{pmatrix}_{b_0}$

[2161] $1^1 \overset{1^3}{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 5 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2162] $21^1 \overset{1^3}{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2163] $1^1 \overset{1^3}{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2150] $1^1 \overset{1^3}{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2164] $1^{11} 2231^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -36$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 8 & 4 \\ 4 & 0 & 8 & 15 & 9 \\ 2 & 0 & 4 & 9 & 5 \end{pmatrix}_{b_0}$

[2173] $1^1 \overset{1^3}{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 19 & 5 \\ 10 & 19 & 65 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$

[2174] $21^1 \overset{1^3}{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2175] $1^{11} 322151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[2172] $1^1 \overset{1^3}{5} 1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2171] \ 1^{11} \overset{1^3}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2170] \ 21^3 \overset{1^{11}}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2169] \ 1^{11} 413141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2168] \ 1^{11} \overset{1^3}{5} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2167] \ 1^{13} \overset{1^3}{5} 1231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2166] \ 1^{11} \overset{1^3}{5} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 5 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2165] \ 1^{14} \overset{1^3}{41} \overset{1^3}{3141}^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2176] \ 1^5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 13 & 5 \\ 6 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[2177] \ 1^5 \overset{1^3}{131}^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2178] \ 1^5 \overset{1^{11}}{131}^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2179] \ 31^1 321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 3 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 3 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & 0 & 2 \\ 2 & 6 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[2183] & 2 \overset{1^{11}}{3} \overset{1^3}{5} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2182] & 1 \overset{1^3}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[2181] & 2 \overset{1^{11}}{3} \overset{1^1}{5} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[2180] & 1 \overset{1^2}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2184] & 151^{12} 2231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 18 & 6 & 55 & 8 \\ 1 & 6 & 0 & 19 & 2 \\ 11 & 55 & 19 & 165 & 25 \\ 2 & 8 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2186] & 21^{12} \overset{1^4}{5} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2185] & 21^{12} \overset{1^2}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 5 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2191] & 1 \overset{1^2}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[2187] & 2 \overset{1^{41}}{3} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 6 & 6 & 6 \\ 3 & 6 & 3 & 5 \\ 3 & 6 & 5 & 5 \end{pmatrix}_{b_0} \\
[2188] & 1 \overset{1^2}{5} \overset{1^2}{5} 1^{12} 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2189] & 31^{12} 4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 13 & 7 \\ 2 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2190] \quad 1^1 \bar{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 16 & 5 \\ 8 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2192] \quad 21^2 \bar{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2193] \quad 1^1 \bar{5} 1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2194] \quad 1^1 \bar{3} 221^1 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[2195] \quad 1^1 \bar{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2196] \quad 1^{11} \bar{5} 1^{12} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 31 & 11 \\ 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2197] \quad 1^{11} \bar{4} 1^1 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 26 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2201] \quad 1^{12} \bar{5} 131^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2200] \quad 1^{11} \bar{5} 1^2 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 81 & 17 \\ 2 & 2 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2199] \quad 1^{12} \bar{5} 131^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2198] \quad 31^2 \bar{3} 1^1 \bar{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[2204] $1^{1^3}_{1^2} 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 16 & 5 \\ 11 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$

[2205] $1^{1^2}_{1^1} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[2203] $1^{1^1} 1^1 3221^{15} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 5 & 0 & 8 & 31 & 11 \\ 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2212] $1^{1^1} 1^1 5 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2211] $1^{1^2} 1^1 5 1^1 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$

[2210] $1^{1^1} 1^1 321^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$

[2209] $1^{1^2} 1^1 5 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2207] $1^{1^1} 1^1 4131^{14} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 7 & 0 & 21 & 9 \\ 2 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$

[2206] $1^{1^1}_{1^1} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[2208] $1^{1^3}_{1^1} 1^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[2202] $1^{1^3}_{1^2} 1^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[2219] $2^1 3^1 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$\begin{aligned}
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 19 & 5 \\ 12 & 19 & 65 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0} \\
[2218] & 1^1 3221^1 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 5 & 0 & 5 & 18 & 8 \\ 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2217] & 1^{11} 51^1 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2216] & 2^1 3^1 2151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0} \\
[2215] & 31^1 4^1 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 7 & 0 & 5 & 5 \\ 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0} \\
[2214] & 1^1 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 13 & 5 \\ 9 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0} \\
[2213] & 1^1 5^1 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0} \\
[2220] & 1^{11} 31^1 4131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 6 & 1 & 14 & 8 \\ 2 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0} \\
[2230] & 2^{13} 3^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 7 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2229] & 21^{12} 5^1 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2234] & 2^1 3^1 21^1 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2233] & 21^{12} 51^1 3^1 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 13 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2232] $21^{12}51^2231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 13 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$
- [2231] $21^{11}51231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [2221] $31^{12}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 18 & 4 \\ 7 & 0 & 18 & 58 & 14 \\ 2 & 0 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$
- [2222] $31^{11}4141^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 6 & 0 & 9 & 13 & 7 \\ 2 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [2223] $21^{12}51231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$
- [2235] $21^{11}51^1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [2224] $21^{12}51^3231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 15 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$
- [2225] $221^{11}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 2 & 2 \\ 3 & 6 & 2 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$
- [2226] $21^{11}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [2227] $23^1151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [2228] $31^{11}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 4 & 0 & 9 & 13 & 7 \\ 2 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [2236] $1^151^1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$$

[2237] $21^{11}5^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$

[2238] $1^{11}5^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$

[2239] $23^{11}15^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 5 & 8 & 5 \\ 5 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$

[2240] $2321^{11}51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 7 & 6 & 30 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2241] $1^{11}3221^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 31 & 11 \\ 2 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$

[2242] $23^{12}15^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2246] $1^{12}5^{13}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$

[2245] $223^{11}151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 7 & 4 \\ 3 & 2 & 7 & 5 & 5 \\ 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[2244] $2315^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$

[2243] $1^{12}5^{13}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$

[2251] $21^{11}5^{12}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[2254] $21^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2253] $1^{12} 5^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 13 & 5 \\ 8 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$$

[2249] $1^{11} 5^{12} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2255] $1^{11} 41^{12} 2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 1 & 6 & 3 \\ 7 & 6 & 16 & 10 \\ 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$$

[2247] $1^{11} 41^{12} 2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 1 & 6 & 3 \\ 7 & 6 & 16 & 10 \\ 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$$

[2248] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2250] $2^{11} 3^{11} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 1 & 5 & 3 \\ 3 & 3 & 5 & 6 & 6 \\ 2 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

[2252] $1^{11} 5^{12} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2265] $21^{13} 51^{11} 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 20 & 9 & 67 & 15 \\ 2 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2273] $1^{12} 5^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 16 & 5 \\ 10 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

[2256] $1^{14} 5^{12} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2272] $2^3 3^2 15^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2271] \ 21^4 5^{11} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 7 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2270] \ 2^1 3^{141} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 6 & 12 & 6 \\ 6 & 12 & 18 & 10 \\ 3 & 6 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[2269] \ 2^1 3^{111} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2257] \ 21^2 5^{11} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2258] \ 21^{11} 51^{12} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2259] \ 31^1 4^{111} 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 7 & 4 \\ 5 & 0 & 7 & 5 & 5 \\ 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2260] \ 21^{11} 51^1 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 8 & 5 & 11 & 7 \\ 2 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2268] \ 21^2 5^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 16 & 5 \\ 10 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2261] \ 1^1 5^{12} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2266] \ 1^{13} 5^{11} 131^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2262] \ 1^{13} 5^1 1^1 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2263] $2 \overset{11}{3} 2151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2267] $21 \overset{11}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2264] $1 \overset{14}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 7 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2274] $2 \overset{1241}{3} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 1 & 6 & 3 \\ 4 & 5 & 6 & 14 & 8 \\ 2 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[2289] $1 \overset{11}{5} 1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2288] $1 \overset{11}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 4 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2282] $1 \overset{14}{5} 1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2283] $1 \overset{12}{5} 1^2 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 25 & 5 \\ 16 & 25 & 114 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[2284] $1 \overset{1321^2}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2285] $1 \overset{11321^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 114 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[2286] $1 \overset{11}{4} 1^2 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 13 & 0 & 81 & 17 \\ 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2287] \ 1^1 5^{12} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 21 & 5 \\ 10 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[2291] \ 1^1 5^{14} 1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2290] \ 1^{12} 41^{13} 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 11 & 0 & 45 & 13 \\ 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2281] \ 1^{11} 4131^{24} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 11 & 0 & 65 & 15 \\ 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2280] \ 1^1 5^{14} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 27 & 5 \\ 14 & 27 & 130 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[2279] \ 1^{11} 5^{13} 1^2 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 10 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2278] \ 1^1 5^{12} 1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2277] \ 1^{12} 3221^{15} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 11 & 2 \\ 6 & 0 & 11 & 50 & 14 \\ 2 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2276] \ 1^{11} 3221^{25} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 13 & 0 & 101 & 19 \\ 2 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2275] \ 1^1 5^{14} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2298] \ 2^1 3^1 2151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2297] \ 1^1 41^{11} 3141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 20 & 0 & 26 & 10 \\ 2 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2296] \quad 31^1 4141^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 5 & 0 & 9 & 13 & 7 \\ 2 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2295] \quad 2^1 3^1 151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2294] \quad 1^1 5^1 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2293] \quad 1^1 5^1 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 6 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2299] \quad 1^1 5^1 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2300] \quad 31^1 4^1 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 7 & 4 \\ 3 & 0 & 7 & 5 & 5 \\ 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2292] \quad 31^2 4^1 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 5 & 1 & 25 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 25 & 0 & 101 & 19 \\ 2 & 5 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2301] \quad 2^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 6 & 11 & 6 \\ 6 & 11 & 15 & 9 \\ 3 & 6 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[2313] \quad 1^1 12^1 5^1 1231^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 66 & 16 \\ 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2308] \quad 1^1 12^1 5^1 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2303] \quad 1^1 5^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2304] \quad 2 \overset{1^1 1^1}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2305] \quad 21 \overset{1^4}{1^2} 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2316] \quad 1^1 5 1231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2315] \quad 1^1 \overset{1^3}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 6 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2314] \quad 1^1 \overset{1^4}{1^2} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 10 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2312] \quad 31 \overset{1^1}{1^1} 5 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2311] \quad 1^1 \overset{1^1}{1^2} 5 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2306] \quad 1^1 \overset{1^3}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 24 & 5 \\ 12 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2310] \quad 1^1 \overset{1^4}{1^2} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 8 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2309] \quad 1^1 \overset{1^1}{1^1} 5^2 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 114 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[2307] \quad 21^3 \overset{1^1}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 6 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2302] \quad 1^3 \quad 1^5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2317] \quad 221^{11} 5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 4 & 3 & 8 & 4 \\ 1 & 3 & 0 & 5 & 2 \\ 4 & 8 & 5 & 11 & 7 \\ 2 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2323] \quad 31^1 4^1 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 7 & 0 & 5 & 5 \\ 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2322] \quad 2^1 3^1 2151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2321] \quad 1^1 1^1 1^2 \quad 1^5 1^3 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2320] \quad 31^1 1^1 1^1 \quad 4^1 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 5 & 1 & 16 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 16 & 0 & 38 & 12 \\ 2 & 5 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2319] \quad 31^1 4^1 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 14 & 0 & 26 & 10 \\ 2 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2318] \quad 31^3 31^1 51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 21 & 4 \\ 18 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2337] \quad 1^1 1^1 1^1 \quad 1^5 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 18 & 5 \\ 9 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[2341] \quad 2^1 3^1 21^1 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2342] \quad 1^1 1^4 \quad 1^5 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 27 & 5 \\ 18 & 27 & 130 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[2340] $2 \overset{11}{3} 1 \overset{11}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2334] $2 \overset{11}{3} 1 \overset{12}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2335] $1 \overset{12}{5} 1 \overset{12}{3} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 5 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2336] $1^1 3 2 2 1^2 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 9 & 0 & 10 & 78 & 16 \\ 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2333] $2 1^1 5 1 \overset{13}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 18 & 5 \\ 9 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[2339] $1^1 3 2 2 1^1 5 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 0 & 5 & 18 & 8 \\ 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2343] $2 1^4 5 1 \overset{13}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 27 & 5 \\ 18 & 27 & 130 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[2344] $2 \overset{11}{3} 1 \overset{11}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2345] $1 \overset{11}{5} 1 \overset{12}{5} 1 \overset{12}{3} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2338] $1^{11} 4 1 3 1^1 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 7 & 0 & 21 & 9 \\ 2 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[2324] $1 \overset{11}{5} 1 \overset{12}{5} 1^2 2 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2325] $31 \overset{1^{11}}{4} 141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 7 & 4 \\ 4 & 0 & 7 & 5 & 5 \\ 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[2326] $2 \overset{1^{11}}{3} 2151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[2327] $2 \overset{1^{12}}{3} 1 \overset{1^4}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2328] $1 \overset{1^1}{1^1} 5 \overset{1^2}{1^1} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2329] $2 \overset{1^1}{3} 21^2 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2330] $1 \overset{1^{12}}{5} 1 \overset{1^{11}}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[2331] $1 \overset{1^3}{1^1} 5 1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2332] $1 \overset{1^3}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 6 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2349] $1 \overset{1^{12}}{5} 321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 20 & 5 \\ 10 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$

[2351] $1 \overset{1^{12}}{1^2} 3141^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 7 & 1 & 17 & 9 \\ 2 & 3 & 0 & 9 & 4 \end{pmatrix}_{b_0}$

[2353] $1 \overset{1^4}{1^1} 5 \overset{1^1}{1^1} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 20 & 5 \\ 14 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$

[2352] $1 \overset{1^1}{1^1} 3221^2 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$\begin{aligned}
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 9 & 0 & 13 & 101 & 19 \\ 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2346] & 1^{12} 4131^{14} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 10 & 0 & 38 & 12 \\ 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[2347] & 1^{12} 5^{12} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[2348] & 1^{12} 5^{12} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2350] & 1^{12} 5^{14} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 9 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2354] & 1^{12} 41^{12} 2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 8 & 3 \\ 8 & 8 & 26 & 12 \\ 3 & 3 & 12 & 5 \end{pmatrix}_{b_0} \\
[2355] & 1^{14} 1^{12} 2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 8 & 3 \\ 8 & 8 & 26 & 12 \\ 3 & 3 & 12 & 5 \end{pmatrix}_{b_0} \\
[2357] & 2^3 3^{13} 1^3 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 6 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2362] & 1^{11} 41^{13} 31^{14} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 70 & 16 \\ 2 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0} \\
[2361] & 1^{11} 321^{11} 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2360] & 1^{11} 5^{14} 1^{12} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2359] & 1^{11} 41^{13} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 26 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2358] $22 \overset{1^{12}}{3} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 3 & 4 & 2 \\ 3 & 3 & 4 & 14 & 4 \\ 6 & 4 & 14 & 26 & 10 \\ 2 & 2 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[2356] $21 \overset{1^3}{\overset{1^2}{5}} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2365] $1 \overset{1^1}{4} 1^1 2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 8 & 3 \\ 10 & 8 & 34 & 14 \\ 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[2364] $1 \overset{1^1}{4} 1^1 2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 8 & 3 \\ 10 & 8 & 34 & 14 \\ 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[2363] $1 \overset{1^3}{\overset{1^2}{1^1}} 5 131^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[2374] $31 \overset{1^1}{\overset{1^1}{3}} 2151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[2373] $2 \overset{1^{13}}{3} 1 \overset{1^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2372] $21 \overset{1^{11}}{\overset{1^1}{5}} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2371] $21 \overset{1^4}{\overset{1^1}{5}} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 11 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2370] $2 \overset{1^1}{3} 21^3 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 13 & 11 & 13 & 126 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2369] $31 \overset{1^2}{\overset{1^2}{3}} 21^3 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 29 & 3 \\ 5 & 5 & 40 & 5 \\ 29 & 40 & 287 & 35 \\ 3 & 5 & 35 & 3 \end{pmatrix}_{b_0}$$

[2368] $1 \overset{1^1}{\overset{1^1}{5}} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2367] $1^{11} \overset{1^4}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2366] $1^{11} \overset{1^4}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 11 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2375] $21^{12} \overset{1^1}{5} 1^{13} 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2376] $21^{12} \overset{1^1}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2381] $21^{12} \overset{1^1}{5} 1^3 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 14 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2382] $1^{12} \overset{1^1}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 5 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2378] $2^{11} \overset{1^2}{3} \overset{1^1}{4} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 4 & 8 & 6 & 14 & 8 \\ 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[2377] $2^3 3^{11} 1^1 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2383] $2^3 3^{11} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 6 & 18 & 6 \\ 9 & 18 & 43 & 15 \\ 3 & 6 & 15 & 5 \end{pmatrix}_{b_0}$$

[2384] $1^{12} \overset{1^3}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2385] $21^{12} \overset{1^{13}}{5} 1^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 10 & 18 & 13 & 73 & 17 \\ 2 & 4 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2389] $21^{12}51^13^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 14 & 7 & 33 & 11 \\ 2 & 4 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2390] $1^15^{14}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 18 & 5 \\ 12 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[2380] $31^{11}31^151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 5 & 8 & 5 \\ 6 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[2391] $31^15^{12}13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[2388] $1^{12}5^{13}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2379] $21^{12}5^{13}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2386] $21^35^{12}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2387] $21^15^{12}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 5 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2392] $2^13^{11}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 1 & 9 & 3 \\ 5 & 4 & 9 & 20 & 10 \\ 2 & 2 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[2396] $41^{1231}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 44 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 44 & 0 & 70 & 16 \\ 2 & 10 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[2397] $1^{12}5^{13}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2395] $1^1 5 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 18 & 6 & 60 & 8 \\ 1 & 6 & 0 & 15 & 2 \\ 15 & 60 & 15 & 178 & 24 \\ 2 & 8 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2398] $1^1 5 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 24 & 5 \\ 16 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2399] $1^1 3221^{11} 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 18 & 6 & 20 & 8 \\ 1 & 6 & 0 & 5 & 2 \\ 5 & 20 & 5 & 18 & 8 \\ 2 & 8 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2400] $1^1 3221^{12} 51^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 18 & 6 & 40 & 8 \\ 1 & 6 & 0 & 10 & 2 \\ 10 & 40 & 10 & 78 & 16 \\ 2 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2394] $1^{12} 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2393] $1^1 5 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 18 & 5 \\ 10 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[2401] $2^1 3 15 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 5 & 16 & 5 \\ 10 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

[2402] $22^1 3 151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 3 & 6 & 2 \\ 3 & 3 & 4 & 11 & 4 \\ 5 & 6 & 11 & 21 & 9 \\ 2 & 2 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[2410] $21^{12} 5 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2409] $1^{12} 3221^{11} 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 11 & 0 & 50 & 14 \\ 2 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[2411] $1^1 5 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 21 & 5 \\ 14 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[2412] $1^{11} \overset{12}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2413] $22 \ 3 \ 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 14 & 13 & 8 \\ 5 & 13 & 9 & 7 \\ 3 & 8 & 7 & 4 \end{pmatrix}_{b_0}$

[2404] $1^{13} \overset{13}{5} 1^{12} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 20 & 5 \\ 13 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$

[2405] $1^{11} \overset{14}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2406] $1^{13} \overset{13}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 18 & 5 \\ 11 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$

[2407] $1^{11} \overset{14}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2408] $1^{11} \overset{13}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2403] $1^{11} \overset{11}{5} 1^{12} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$

[2420] $1^1 3221^1 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 7 & 0 & 5 & 18 & 8 \\ 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2421] $1^{13} \overset{13}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 7 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2422] $21^2 \overset{12}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

$$\begin{aligned}
[2423] \quad & 31^{11}4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 24 & 0 & 38 & 12 \\ 2 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[2419] \quad & 1^{12}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2425] \quad & 1^{14}1^{11}31^{34}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 13 & 2 \\ 3 & 13 & 1 & 56 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 56 & 0 & 230 & 28 \\ 2 & 7 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[2426] \quad & 1^{11}51^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2427] \quad & 1^{11}51^{13}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 126 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2428] \quad & 21^{35}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 7 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2429] \quad & 1^{13}51^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 115 & 21 \\ 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0} \\
[2418] \quad & 1^{13}221^{35}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 15 & 2 \\ 13 & 0 & 15 & 178 & 24 \\ 2 & 0 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2417] \quad & 1^{11}51^{13}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2416] \quad & 1^{11}3221^{15}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 6 & 0 & 8 & 31 & 11 \\ 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2415] \quad & 1^{11}51^{13}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 13 & 11 & 13 & 126 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2414] \quad & 23151^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2424] \ 1^{13} 5^{13} 131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2430] \ 1^{12} 5^{12} 231^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 12 & 0 & 130 & 22 \\ 2 & 2 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2431] \ 1^{14} 5^{14} 131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 21 & 4 \\ 16 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2446] \ 1^{13} 5^{32} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 25 & 5 \\ 12 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2447] \ 1^{13} 5^{12} 1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2432] \ 1^{13} 4131^{14} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 13 & 0 & 61 & 15 \\ 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2433] \ 1^{13} 5^{12} 1231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[2434] \ 1^{13} 5^{13} 131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 7 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2435] \ 1^{11} 3221^3 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 18 & 2 \\ 13 & 0 & 18 & 211 & 27 \\ 2 & 0 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[2439] \ 21^{11} 5^{13} 131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 6 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2438] \ 1^{12} 5^{12} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2437] $1 \overset{1^4}{\underset{1^{11}}{5}} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 8 & 9 & 14 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2442] $1 \overset{1^3}{\underset{1^{13}}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2443] $1 \overset{1^1}{\underset{1^1}{5}} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 6 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2444] $1 \overset{1^3}{\underset{1^1}{5}} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 29 & 5 \\ 14 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

[2445] $1 \overset{1^2}{\underset{1^{13}}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2441] $1 \overset{1^1}{\underset{1^1}{4}} 131^{241^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 11 & 0 & 65 & 15 \\ 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[2436] $1 \overset{1^4}{\underset{1^1}{5}} 1231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2440] $1 \overset{1^3}{\underset{1^1}{5}} 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2448] $1 \overset{1^2}{\underset{1^1}{5}} 1^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 12 & 11 & 13 & 126 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2450] $1 \overset{1^4}{\underset{1^3}{5}} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 7 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2451] $2 \overset{1^{12}}{\underset{1^1}{3}} 1 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 5 & 13 & 5 \\ 8 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[2452] \ 2 \overset{13}{3} \overset{14}{1} 5 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 5 & 27 & 5 \\ 17 & 27 & 130 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[2453] \ 1 \overset{12}{1} \overset{11}{5} \overset{1}{1} 23 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2454] \ 2 \overset{11}{3} 21 \overset{11}{5} 1 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 5 & 15 & 5 \\ 10 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[2449] \ 1 \overset{12}{1} \overset{1}{5} \overset{1}{1} 23 \overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[2455] \ 22 \overset{11}{3} 15 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 3 & 4 & 2 \\ 3 & 3 & 4 & 9 & 4 \\ 4 & 4 & 9 & 11 & 7 \\ 2 & 2 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2456] \ 1 \overset{13}{1} \overset{12}{3} 22 \overset{12}{1} \overset{12}{5} 1 \overset{2}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 18 & 6 & 55 & 8 \\ 1 & 6 & 0 & 19 & 2 \\ 13 & 55 & 19 & 165 & 25 \\ 2 & 8 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2461] \ 21 \overset{12}{1} \overset{12}{5} 123 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2468] \ 1 \overset{13}{1} \overset{13}{5} 13 \overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2467] \ 1 \overset{21}{5} \overset{12}{1} 13 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2466] \ 1 \overset{13}{1} \overset{13}{5} 13 \overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2465] \ 1 \overset{12}{1} \overset{12}{5} 13 \overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2464] $21^{2^{11}}5^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2463] $21^{3^{11}}5^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2462] $1^{11}5^{12}132 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2457] $1^{12}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 21 & 5 \\ 12 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[2458] $1^{13}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 24 & 5 \\ 15 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2459] $21^{3^{11}}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 24 & 5 \\ 15 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2460] $21^{2^{11}}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 21 & 5 \\ 12 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[2469] $2^3 3^{11} 21^{25} 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 10 & 7 & 11 & 65 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2470] $2^3 3^{12} 2151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2471] $21^{12} 51^{12} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 30 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2475] $1^{11} 41^{13} 31^{24} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 16 & 0 & 138 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2473] $1^{12} \overset{1}{5} 1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2474] $1^{14} \overset{1}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 8 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2472] $1^{14} \overset{1}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 32 & 5 \\ 16 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$

[2476] $1^{14} \overset{1}{5} 1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$

[2477] $1^{13} \overset{1}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 21 & 5 \\ 13 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$

[2478] $21^{14} \overset{1}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 8 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2479] $1^{12} \overset{1}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 5 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2480] $1^{12} \overset{1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 6 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2481] $1^{13} \overset{1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2482] $1^{12} \overset{1}{4} 1^2 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 110 & 20 \\ 2 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$

[2483] $1^{12} \overset{1}{5} 1^2 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 12 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2484] $1^{13} \overset{1^4}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 12 & 11 & 10 & 89 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2485] $1^{13} \overset{1^1}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2486] $1^{12} 31^{14} 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 10 & 1 & 39 & 13 \\ 2 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[2487] $1^{13} \overset{1^4}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 9 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2488] $21^4 \overset{1^{13}}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 12 & 11 & 10 & 89 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2489] $21^1 \overset{1^{13}}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2490] $1^{11} 41^2 31^{14} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 17 & 0 & 149 & 23 \\ 2 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[2491] $2^1 \overset{1^{11}}{3} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 1 & 13 & 3 \\ 7 & 5 & 13 & 42 & 14 \\ 2 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[2495] $2^1 \overset{1^{13}}{3} \overset{1^1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 6 & 12 & 9 & 34 & 12 \\ 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[2497] $21^{12} 51^{12} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2492] $2^1 321^2 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2493] \quad 2 \overset{1}{3} 2151^{\overset{13}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2494] \quad 1 \overset{21}{5} 131^{\overset{12}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2496] \quad 31 \overset{1}{4} 141^{\overset{13}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 21 & 0 & 61 & 15 \\ 2 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2498] \quad 31 \overset{1}{5} 13 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 6 & 25 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2499] \quad 2 \overset{1}{3} 21^{\overset{11}{2}} 51^{\overset{13}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 5 & 25 & 5 \\ 17 & 25 & 114 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[2500] \quad 21^{\overset{12}{1}} 51^{\overset{2}{2}} 231^{\overset{13}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2501] \quad 1 \overset{1}{5} 1 \overset{13}{3} 2 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2502] \quad 31 \overset{1}{1} 4 141^{\overset{11}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 16 & 4 \\ 6 & 0 & 16 & 38 & 12 \\ 2 & 0 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2503] \quad 31^{\overset{12}{2}} 4 141^{\overset{13}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 18 & 4 \\ 8 & 0 & 18 & 58 & 14 \\ 2 & 0 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2504] \quad 1^{\overset{12}{1}} 5 1^{\overset{3}{2}} 231^{\overset{13}{2}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 15 & 0 & 218 & 28 \\ 2 & 2 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2505] \quad 21^{\overset{12}{1}} 5 131^{\overset{11}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2506] \quad 21^{\overset{12}{1}} 5 131^{\overset{13}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$\begin{aligned}
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2507] & 1^{14} 41^{11} 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 20 & 0 & 26 & 10 \\ 2 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[2508] & 2^{12} 3^{13} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2509] & 31^{11} 321^{15} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0} \\
[2510] & 2^{12} 3^{13} 21^{15} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 38 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2511] & 2^{14} 3^{14} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 8 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2512] & 1^{13} 3221^{12} 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 6 & 55 & 8 \\ 1 & 6 & 0 & 19 & 2 \\ 12 & 55 & 19 & 165 & 25 \\ 2 & 8 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2513] & 22^1 3^{13} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 3 & 6 & 2 \\ 3 & 3 & 4 & 21 & 4 \\ 9 & 6 & 21 & 61 & 15 \\ 2 & 2 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[2521] & 1^{14} 41^{22} 2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 1 & 10 & 3 \\ 13 & 10 & 56 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0} \\
[2522] & 1^{12} 41^{13} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 11 & 0 & 45 & 13 \\ 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0} \\
[2514] & 1^{13} 5^{14} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 10 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2515] & 1^{11} 41^{22} 2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 1 & 10 & 3 \\ 13 & 10 & 56 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[2516] $1^{1^2}_{1^1} 5^{14} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 23 & 5 \\ 12 & 23 & 81 & 19 \\ 3 & 5 & 19 & 4 \end{pmatrix}_{b_0}$

[2517] $1^{1^2}_{1^1} 5^{14} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2518] $1^{1^1 321^2}_{1^1} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2519] $1^{1^4}_{1^3} 5^{14} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 11 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2520] $1^{1^3}_{1^1} 5^{14} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2523] $1^{1^2}_{1^1} 5^{14} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2524] $1^{1^1 41^2 3141^{11}}_{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 13 & 0 & 81 & 17 \\ 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[2538] $21^{1^1}_{1^1} 5^{14} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[2526] $41^{1^1 23141^{11}}_{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[2527] $1^{1^2 4131^{14} 141^{11}}_{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 10 & 0 & 38 & 12 \\ 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$

[2528] $1^{1^4}_{1^2} 5^{14} 1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2529] $1^{1^3}51^2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 11 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2530] $1^{1^2}3221^251^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 16 & 2 \\ 10 & 0 & 16 & 130 & 22 \\ 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2531] $1^{1^2}321^251^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 30 & 5 \\ 16 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$

[2532] $1^{1^2}5131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 6 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2533] $1^{1^4}5131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2534] $1^{1^4}51^2231^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 30 & 5 \\ 20 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$

[2535] $151^{1^1}2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 6 & 20 & 8 \\ 1 & 6 & 0 & 5 & 2 \\ 4 & 20 & 5 & 18 & 8 \\ 2 & 8 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2536] $1^{1^2}51231^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2525] $1^{1^2}4131^241^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 14 & 0 & 90 & 18 \\ 2 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$

[2539] $1^{1^2}51^2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2537] $2^13^121^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 6 & 24 & 6 \\ 12 & 24 & 78 & 20 \\ 3 & 6 & 20 & 5 \end{pmatrix}_{b_0}$

[2545] $2^13^121^{1^1}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 14 & 10 & 15 & 133 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2544] $21^{13}5^{14}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 11 & 11 & 10 & 89 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2543] $2321^{15}51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 11 & 12 & 11 & 100 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2542] $231^{14}51^{13}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 7 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2541] $1^{11}41^{12}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 1 & 10 & 3 \\ 11 & 10 & 44 & 16 \\ 3 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

[2540] $1^{12}41^{12}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 1 & 10 & 3 \\ 11 & 10 & 44 & 16 \\ 3 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

[2552] $1^{13}51^{12}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2550] $1^{12}51^{13}1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2549] $31^{13}4141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 27 & 4 \\ 11 & 0 & 27 & 133 & 21 \\ 2 & 0 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[2548] $31^{11}24141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 25 & 4 \\ 9 & 0 & 25 & 101 & 19 \\ 2 & 0 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[2547] $1^{12}51^{12}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2546] $1^{11}41^{11}31^{14}11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 70 & 16 \\ 2 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[2551] $1^{13}221^{25}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 10 & 0 & 10 & 78 & 16 \\ 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2553] $1^{11}3221^{125}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 6 & 45 & 8 \\ 1 & 6 & 0 & 13 & 2 \\ 12 & 45 & 13 & 101 & 19 \\ 2 & 8 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2554] $2^{11}3^{41}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 6 & 17 & 6 \\ 9 & 17 & 36 & 14 \\ 3 & 6 & 14 & 5 \end{pmatrix}_{b_0}$$

[2557] $1^{12}41^{131}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 97 & 19 \\ 2 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[2555] $1^{13}5^{12}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2558] $1^{11}5^{14}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2559] $1^{11}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2560] $1^{13}5^{13}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2561] $1^{13}5^{13}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 10 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2562] $41^{231}4^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 5 & 3 & 3 & 2 \\ 5 & 13 & 10 & 9 & 7 \\ 3 & 10 & 4 & 7 & 4 \\ 3 & 9 & 7 & 5 & 5 \\ 2 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[2556] $21^{313}5^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 10 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2563] $21^{2^{13}}5^{11}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2564] $1^{1^1}5^{1^1}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 25 & 5 \\ 13 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$

[2565] $1^{12}5^{1^1}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2566] $1^{13}3221^{1^1}51^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 14 & 0 & 75 & 17 \\ 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[2567] $21^{1^{12}}5^{1^1}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2568] $2^3 3^{1^1}5^{1^1}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2569] $1^{12}3221^{1^1}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 11 & 2 \\ 7 & 0 & 11 & 50 & 14 \\ 2 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2570] $1^{11}5^{1^1}1^{11}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2571] $31^{1^{12}}4^{1^1}141^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 5 & 1 & 23 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 23 & 0 & 77 & 17 \\ 2 & 5 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[2572] $31^{1^{11}}4^{1^1}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 5 & 1 & 16 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 16 & 0 & 38 & 12 \\ 2 & 5 & -1 & 12 & 3 \end{pmatrix}_{b_0}$

[2573] $2^3 3^{1^1}5^{1^1}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2574] $21^{1^1}5^{1^1}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2584] \quad 21^{12} 51^1 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 8 & 16 & 10 & 50 & 14 \\ 2 & 4 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2578] \quad 1^4 41^{11} 31^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 7 & 2 \\ 3 & 13 & 1 & 32 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 70 & 16 \\ 2 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2577] \quad 221^{12} 5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 8 & 2 \\ 3 & 4 & 3 & 16 & 4 \\ 1 & 3 & 0 & 10 & 2 \\ 8 & 16 & 10 & 50 & 14 \\ 2 & 4 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2576] \quad 2^{11} 3^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 7 & 4 \\ 2 & 4 & 1 & 7 & 3 \\ 4 & 7 & 7 & 12 & 8 \\ 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2579] \quad 1^2 5^{13} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2575] \quad 21^{12} 51^2 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2580] \quad 21^{12} 5^{12} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2581] \quad 21^{13} 51231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2582] \quad 2^3 3^{12} 151^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2583] \quad 1^3 5^{13} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 23 & 5 \\ 13 & 23 & 81 & 19 \\ 3 & 5 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[2585] \quad 2^3 21^{12} 41^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 1 & 10 & 3 \\ 6 & 6 & 10 & 30 & 12 \\ 2 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[2586] $1^{12} 31^1 4^{11} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 14 & 0 & 26 & 10 \\ 2 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[2589] $1^{13} 1^5 1^2 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 5 & 30 & 5 \\ 19 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$$

[2594] $1^{11} 3221^{15} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 7 & 0 & 8 & 31 & 11 \\ 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2593] $1^{14} 1^{11} 5^1 1^2 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2592] $21^2 5^{11} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2591] $1^{11} 5^1 1^2 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 30 & 5 \\ 17 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$$

[2590] $1^{11} 5^1 1^2 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2588] $1^{14} 1^5 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 23 & 5 \\ 14 & 23 & 81 & 19 \\ 3 & 5 & 19 & 4 \end{pmatrix}_{b_0}$$

[2587] $1^{11} 5^1 1^2 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2597] $2^3 3^{11} 21^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2595] $31^{12} 32151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 8 & 3 \\ 5 & 5 & 10 & 5 \\ 8 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

- [2596] $1^{11} 5^1 1^3 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [2598] $41^{11} 231^{13} 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 7 & 2 \\ 4 & 26 & 1 & 44 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 44 & 0 & 70 & 16 \\ 2 & 10 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$
- [2600] $1^{11} 3221^{11} 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 6 & 2 \\ 4 & 18 & 6 & 25 & 8 \\ 1 & 6 & 0 & 8 & 2 \\ 6 & 25 & 8 & 31 & 11 \\ 2 & 8 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [2599] $151^{11} 2231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 18 & 6 & 25 & 8 \\ 1 & 6 & 0 & 8 & 2 \\ 5 & 25 & 8 & 31 & 11 \\ 2 & 8 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [2602] $31^{11} 4141^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 18 & 4 \\ 9 & 0 & 18 & 58 & 14 \\ 2 & 0 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$
- [2601] $1^{11} 41^2 31^2 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 21 & 0 & 241 & 29 \\ 2 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$
- [2604] $1^{13} 41^2 3141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 19 & 0 & 145 & 23 \\ 2 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$
- [2603] $1^{13} 21^3 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$
- [2605] $1^{11} 5^1 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 6 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$
- [2606] $1^{13} 5^1 1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [2607] $1^{12} 41^3 3141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 205 & 27 \\ 2 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$
- [2610] $21^{12} 5^1 13^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 26 & 5 \\ 14 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0} \\
[2611] & 1^4 41^{11} 31^2 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 10 & 2 \\ 3 & 13 & 1 & 44 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 44 & 0 & 138 & 22 \\ 2 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[2612] & 1^{12} 3221^2 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 16 & 0 & 130 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[2613] & 1^{11} 5^1 1^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2614] & 1^{11} 5^1 1^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2615] & 31^{12} 31^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 16 & 5 \\ 12 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0} \\
[2616] & 1^4 41^{11} 31^3 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 14 & 2 \\ 3 & 13 & 1 & 56 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 56 & 0 & 230 & 28 \\ 2 & 7 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[2617] & 1^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2618] & 31^{11} 4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 6 & 2 \\ 3 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 24 & 0 & 38 & 12 \\ 2 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[2619] & 1^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2620] & 1^1 5^1 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 30 & 5 \\ 18 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0} \\
[2608] & 1^1 5^1 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 26 & 5 \\ 14 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[2609] $2^{\overset{1^4}{1^4} \overset{1^1}{4} 1} 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 16 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 8 & 16 & 12 & 62 & 16 \\ 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[2621] $1^{\overset{1^3}{1^3} \overset{1^2}{5} 1^2} 231^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 193 & 27 \\ 2 & 2 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[2622] $1^{\overset{1^1}{1^1} \overset{1^3}{3} 221^2} 51^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 10 & 0 & 13 & 101 & 19 \\ 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2623] $1^{\overset{1^1}{1^1} \overset{1^3}{3} 51^2} 231^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 11 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2624] $1^{\overset{1^4}{1^4} \overset{1^1}{5} 1^2} 231^{\overset{1^3}{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 25 & 5 \\ 16 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2625] $1^{\overset{1^3}{1^3} \overset{1^2}{5} 1^3} 131^{\overset{1^3}{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2626] $1^{\overset{1^4}{1^4} \overset{1^2}{5} 1^3} 131^{\overset{1^3}{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2627] $1^{\overset{1^2}{1^2} \overset{1^1}{5} 1^2} 231^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2628] $1^{\overset{1^2}{1^2} \overset{1^1}{5} 1^2} 231^{\overset{1^3}{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2630] $1^{\overset{1^3}{1^3} \overset{1^1}{5} 1^2} 231^{\overset{1^2}{1^2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2629] $1^{\overset{1^3}{1^3} \overset{1^2}{5} 1^2} 131^{\overset{1^2}{1^2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2632] $1^{11}41^22321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 86 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$

[2631] $1^{11}41^22321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 86 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$

[2633] $1^{12}41^12321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 74 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$

[2637] $21^{12}51^1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 66 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2634] $21^{12}51^2231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2635] $21^{13}51^3231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 19 & 2 \\ 2 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 19 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$

[2636] $1^{11}5^{13}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 9 & 7 & 14 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2638] $31^{12}4141^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 18 & 4 \\ 10 & 0 & 18 & 58 & 14 \\ 2 & 0 & 4 & 14 & 3 \end{pmatrix}_{b_0}$

[2639] $1^{11}5^{11}1^2231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 8 & 8 & 14 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2640] $1^{12}5^{12}1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2641] $2^33^12151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2642] $1^{11}41^33141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 166 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$

[2643] $1^{1^{12}321^3} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 5 & 40 & 5 \\ 22 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$

[2644] $1^{1^{13}} 1^1 231^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$

[2645] $1^{1^4} 1^3 231^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 5 & 40 & 5 \\ 26 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$

[2646] $1^{1^{12}4131^3} 41^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 166 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$

[2647] $1^{1^2} 1^5 131^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 7 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2648] $1^{1^3} 3221^{15} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 14 & 2 \\ 8 & 0 & 14 & 75 & 17 \\ 2 & 0 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2649] $1^{1^{11}321^3} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2650] $1^{1^{12}1^{12}} 1^5 1^3 2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2651] $1^{1^1} 2321^3 51^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 14 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2652] $1^{1^3} 41^1 3141^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 14 & 0 & 70 & 16 \\ 2 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$

[2653] $1^{1^3} 3221^2 51^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 19 & 2 \\ 11 & 0 & 19 & 165 & 25 \\ 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2654] $1^{1^3} 1^3 1^5 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 24 & 5 \\ 15 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0} \\
[2655] & 1^{11} 41^{12} 2321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 12 & 3 \\ 12 & 12 & 58 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0} \\
[2656] & 1^{13} 41^{12} 2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 12 & 3 \\ 12 & 12 & 58 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0} \\
[2664] & 1^{11} 5^{14} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2657] & 1^{11} 5^{14} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 9 & 14 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2658] & 22^1 3^{111} 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 3 & 4 & 2 \\ 3 & 5 & 8 & 9 & 5 \\ 3 & 8 & 4 & 9 & 4 \\ 4 & 9 & 9 & 11 & 7 \\ 2 & 5 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[2659] & 1^{13} 5^{12} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 26 & 5 \\ 15 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0} \\
[2660] & 21^3 5^{11} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2663] & 2^1 3^{13} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 1 & 11 & 3 \\ 7 & 8 & 11 & 44 & 14 \\ 2 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[2661] & 21^4 5^{11} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 9 & 14 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2662] & 1^{11} 5^{13} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2665] & 1^{12} 5^{14} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 26 & 5 \\ 16 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[2672] $1^{12}3221^{12}51^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 18 & 6 & 50 & 8 \\ 1 & 6 & 0 & 16 & 2 \\ 13 & 50 & 16 & 130 & 22 \\ 2 & 8 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2673] $21^4 5^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2674] $1^{11}41^{13}31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 16 & 0 & 138 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2666] $1^{12}5^{14}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2667] $1^{12}5^{12}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2668] $2^3 3^{13} 15^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 5 & 18 & 5 \\ 11 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[2669] $21^2 5^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2670] $21^{12}51^{12}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2671] $1^{12}5^{14}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2679] $31^2 4^{11} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 5 & 1 & 25 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 25 & 0 & 101 & 19 \\ 2 & 5 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[2681] $31^3 4^{12} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 41 & 0 & 269 & 31 \\ 2 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[2675] $2^3 3^{14} 2151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[2676] \quad 1^{11} 5^1 1^4 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[2680] \quad 1^{14} 31^1 4^1 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 28 & 0 & 110 & 20 \\ 2 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2677] \quad 31^2 4^1 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 5 & 1 & 32 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 32 & 0 & 158 & 24 \\ 2 & 5 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2678] \quad 31^3 4^1 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 34 & 0 & 194 & 26 \\ 2 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[2682] \quad 1^{12} 5^1 1^3 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2683] \quad 31^1 4^1 141^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 16 & 4 \\ 7 & 0 & 16 & 38 & 12 \\ 2 & 0 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2684] \quad 1^{13} 4131^1 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 13 & 0 & 61 & 15 \\ 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2685] \quad 1^{12} 5^1 1^3 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 15 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2686] \quad 1^{14} 3221^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 17 & 0 & 106 & 20 \\ 2 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2687] \quad 1^{13} 5^1 1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2688] \quad 21^2 5^1 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2689] $1^{11}5^{12}1^{14}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 30 & 5 \\ 15 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

[2690] $1^{12}5^{12}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2691] $1^{11}5^{14}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 7 & 14 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2692] $1^{12}5^{12}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2693] $1^{12}5^{32}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2694] $21^{13}5^{13}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 10 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2695] $1^{12}5^{13}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2697] $1^{12}5^{11}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 7 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2698] $1^{14}5^{14}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 32 & 5 \\ 20 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[2699] $22^3 3^{12} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 10 & 3 \\ 6 & 14 & 26 & 8 \\ 10 & 26 & 42 & 14 \\ 3 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

[2700] $1^{12}5^{14}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 11 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2701] \ 2 \ 3 \ 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 6 & 30 & 6 \\ 15 & 30 & 123 & 25 \\ 3 & 6 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[2702] \ 21^{112} 5 \ 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 7 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2696] \ 21^{14} 5 \ 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 32 & 5 \\ 20 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[2703] \ 1^{12} 41^2 2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 70 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[2704] \ 1^{141} 2321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 70 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[2705] \ 1^{14} 5 \ 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 29 & 5 \\ 18 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[2712] \ 1^{13} 5 \ 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 29 & 5 \\ 17 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[2708] \ 1^{13} 5 \ 1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2706] \ 21^{13} 5 \ 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2707] \ 21^{13} 5 \ 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 29 & 5 \\ 17 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[2709] \ 22 \ 3 \ 151^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 4 & 28 & 4 \\ 12 & 8 & 28 & 110 & 20 \\ 2 & 2 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[2710] \quad & 1 \overset{1^3}{\underset{1^2}{5}} 131^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 8 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2711] \quad & 1 \overset{1^3}{\underset{1^3}{5}} 131^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2713] \quad & 221^{1^2 5} 131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 7 & 2 \\ 3 & 4 & 3 & 14 & 4 \\ 1 & 3 & 0 & 7 & 2 \\ 7 & 14 & 7 & 33 & 11 \\ 2 & 4 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2720] \quad & 1 \overset{1^3}{\underset{1^3}{5}} 1231^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2721] \quad & 1 \overset{1^1}{\underset{1^1}{5}} 131^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2722] \quad & 1 \overset{1^3}{\underset{1^3}{5}} 321^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 35 & 5 \\ 18 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0} \\
[2723] \quad & 1 \overset{1^3}{\underset{1^3}{5}} 1^3 231^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 25 & 3 \\ 3 & 5 & 40 & 5 \\ 25 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0} \\
[2724] \quad & 1^{1^1} 41^2 31^{1^1} 41^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 17 & 0 & 149 & 23 \\ 2 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[2725] \quad & 2 \overset{1^2}{\underset{1^2}{3}} 1 \overset{1^3}{\underset{1^3}{5}} 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2726] \quad & 1 \overset{1^2}{\underset{1^2}{5}} 1^3 231^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 14 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[2727] \quad & 1^{1^2} 3221^3 51^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 21 & 2 \\ 14 & 0 & 21 & 250 & 30 \\ 2 & 0 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[2719] \quad & 1^{1^2} 3221^3 51^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 21 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[2729] $2^1 3^{12} 15^4 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2730] $1^1 3221^2 51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 11 & 0 & 10 & 78 & 16 \\ 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2731] $1^1 3^4 4131^2 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 17 & 0 & 121 & 21 \\ 2 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[2732] $1^1 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2718] $1^{12} 5^1 3^2 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 13 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2717] $1^1 5^1 3^2 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 5 & 40 & 5 \\ 23 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$$

[2716] $21^1 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2715] $1^1 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2714] $1^1 3221^3 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 15 & 2 \\ 14 & 0 & 15 & 178 & 24 \\ 2 & 0 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2728] $1^{13} 5^1 1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2733] $1^{13} 41^1 31^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 130 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2734] $1^{12} 5^{14} 1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2735] $1^{14} 5^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2736] $2^1 3^{13} 15^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 5 & 21 & 5 \\ 13 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[2737] $1^{12} 4131^{24} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 14 & 0 & 90 & 18 \\ 2 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[2741] $1^{11} 5^1 1^3 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2742] $21^{13} 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2740] $1^{12} 5^1 3^{23} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 5 & 40 & 5 \\ 24 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$$

[2743] $2^1 3^{13} 15^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 11 & 9 & 14 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2744] $1^{11} 321^{12} 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2745] $1^{12} 41^2 31^{14} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 20 & 0 & 186 & 26 \\ 2 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[2746] $1^{11} 5^1 1^{12} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 9 & 8 & 14 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2739] $1^{12} 41^2 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 16 & 0 & 110 & 20 \\ 2 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0} \\
[2738] & 1^{11} 3 2 2 1^{15} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 0 & 8 & 31 & 11 \\ 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2754] & 1^{11} 5 1 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2753] & 2^{12} 3 1 5 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2752] & 2^{12} 3 2 1 5 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2751] & 1^{12} 4 1^{13} 3 1^{24} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 19 & 0 & 173 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0} \\
[2750] & 1^{13} 5 1 3 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2749] & 2^{13} 3 1 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2748] & 2 1^{13} 5 1 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2747] & 2 1^{13} 5 1 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2755] & 2^{11} 3 2 1^{15} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2756] & 1^{14} 4 1^{11} 3 1^{14} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 1 & 8 & 2 \\ 3 & 13 & 1 & 32 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 32 & 0 & 70 & 16 \\ 2 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[2757] $1^{13}5^{11}3^{23}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 18 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 18 & 0 & 295 & 33 \\ 2 & 2 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[2758] $321^{11}4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 8 & 7 & 3 \\ 8 & 17 & 16 & 9 \\ 7 & 16 & 12 & 8 \\ 3 & 9 & 8 & 4 \end{pmatrix}_{b_0}$$

[2762] $2^13^{12}21^{25}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 12 & 8 & 14 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2761] $2^{15}3^{11}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 2 & 4 & 20 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 10 & 20 & 15 & 98 & 20 \\ 2 & 4 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[2760] $2^13^{12}15^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2759] $1^{12}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2763] $2^13^{12}21^{15}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 5 & 20 & 5 \\ 13 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

[2764] $1^{12}41^131^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 97 & 19 \\ 2 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[2765] $22^13^{12}151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 3 & 6 & 2 \\ 3 & 3 & 4 & 16 & 4 \\ 7 & 6 & 16 & 36 & 12 \\ 2 & 2 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[2766] $1^{13}5^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2767] $1^{12}3221^{15}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 11 & 2 \\ 8 & 0 & 11 & 50 & 14 \\ 2 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2768] $1^{11}5^{14}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2769] $1^{11}5^{12}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2770] $1^{11}5^{14}3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 15 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2775] $31^{11}4^{13}141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 16 & 4 \\ 8 & 0 & 16 & 38 & 12 \\ 2 & 0 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[2776] $1^13221^{12}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 9 & 2 \\ 4 & 18 & 6 & 40 & 8 \\ 1 & 6 & 0 & 10 & 2 \\ 9 & 40 & 10 & 78 & 16 \\ 2 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2771] $1^141^{11}31^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 11 & 2 \\ 3 & 13 & 1 & 44 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 44 & 0 & 138 & 22 \\ 2 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2772] $1^{11}5^{12}3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 13 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2773] $1^{11}5^{13}1^231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2774] $21^{13}5^{14}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 14 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2777] $1^15^{12}1^231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 30 & 5 \\ 16 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

[2778] $2^13^{11}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 6 & 23 & 6 \\ 12 & 23 & 67 & 19 \\ 3 & 6 & 19 & 5 \end{pmatrix}_{b_0}$$

[2779] $1^{13}5^{321^2}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 10 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2780] $1^{14}1^{32321^{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 14 & 3 \\ 19 & 14 & 120 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$

[2781] $1^{11}41^{32321^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 14 & 3 \\ 19 & 14 & 120 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$

[2782] $1^{11}5^{1231^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 8 & 14 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2783] $1^{11}3221^{351^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 18 & 2 \\ 14 & 0 & 18 & 211 & 27 \\ 2 & 0 & 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2784] $1^{11}5^{13231^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 14 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2785] $1^{11}3221^{1351^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 17 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2786] $2^{11241}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 1 & 14 & 3 \\ 8 & 7 & 14 & 54 & 16 \\ 2 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$

[2794] $31^{111}4^{141^{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 16 & 4 \\ 9 & 0 & 16 & 38 & 12 \\ 2 & 0 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[2791] $1^{13}5^{131^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2792] $2321^{351^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 15 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2795] $21^{13}1^5131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2796] \quad 21^{13}51^3231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 15 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2797] \quad 1^{13}5131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 9 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2798] \quad 2^3 3^{13} 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2787] \quad 1^{13}5131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2788] \quad 21^3 5^{11} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 9 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2789] \quad 21^{13}51^1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 13 & 13 & 19 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2790] \quad 21^{13}51231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2793] \quad 31^3 4141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 27 & 4 \\ 12 & 0 & 27 & 133 & 21 \\ 2 & 0 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2810] \quad 2^3 3^{13} 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2805] \quad 1^{13}41^33141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 24 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[2806] \quad 1^{14}4131^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 90 & 18 \\ 2 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[2807] $1^{12} \overset{13}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2808] $1^{1321} \overset{3}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2809] $1^{11} \overset{12}{5} 1^{1231} \overset{14}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 10 & 9 & 17 & 106 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2804] $1^{13} \overset{12}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 9 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2799] $1^{12} \overset{11}{5} 1^{2231} \overset{11}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2800] $2^3 3 21^{251} \overset{11}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 11 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2801] $2^3 3 21^{151} \overset{13}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 9 & 10 & 10 & 63 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2803] $1^{12} \overset{13}{5} 1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 9 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2802] $1^{11} \overset{12}{5} 1^{2231} \overset{13}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 5 & 35 & 5 \\ 19 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$

[2811] $31^{13} \overset{1}{4} 141^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 5 & 1 & 30 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 30 & 0 & 130 & 22 \\ 2 & 5 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

[2812] $321^{11} \overset{14}{4} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 8 & 10 & 3 \\ 8 & 17 & 24 & 9 \\ 10 & 24 & 30 & 12 \\ 3 & 9 & 12 & 4 \end{pmatrix}_{b_0}$

[2813] $1^{13} \overset{14}{4} 1^{2321} \overset{13}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$\begin{aligned}
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 1 & 14 & 3 \\ 15 & 14 & 88 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0} \\
[2814] & 1^{13} 41^2 2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 1 & 14 & 3 \\ 15 & 14 & 88 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0} \\
[2824] & 21^{13} 51^2 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 10 & 22 & 12 & 88 & 18 \\ 2 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2821] & 21^{13} 51^2 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 12 & 12 & 11 & 100 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2822] & 21^{13} 51^2 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2823] & 2^3 3^{12} 151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2820] & 41^{12} 23141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0} \\
[2825] & 151^{12} 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 18 & 6 & 40 & 8 \\ 1 & 6 & 0 & 10 & 2 \\ 8 & 40 & 10 & 78 & 16 \\ 2 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2826] & 1^{12} 5^{14} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2819] & 21^{13} 51231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2817] & 21^{13} 5^{14} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 12 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2816] & 21^{13} 5^{12} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2815] $21^{13}51^2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 16 & 16 & 23 & 236 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$
- [2818] $21^{13}51^13^12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 14 & 2 \\ 2 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 14 & 26 & 18 & 148 & 24 \\ 2 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [2827] $21^{13}5^1131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [2828] $1^{12}3221^251^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 16 & 2 \\ 11 & 0 & 16 & 130 & 22 \\ 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$
- [2829] $1^15^11^231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 30 & 5 \\ 17 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$
- [2830] $1^{11}5^1131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 9 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [2831] $1^{13}5^1131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$
- [2832] $1^{11}5^11^2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [2833] $21^25^1131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$
- [2834] $231^{11}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 35 & 0 & 33 & 11 \\ 2 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$
- [2835] $2^13^15^11^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$
- [2836] $1^{12}5^1131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2837] \ 2^{12} 5^{13} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2838] \ 31^1 4^{13} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 21 & 0 & 61 & 15 \\ 2 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2842] \ 1^{13} 5^{12} 1231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2839] \ 1^{12} 5^{13} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 8 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2841] \ 1^{14} 41^{13} 31^{14} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 21 & 0 & 169 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2840] \ 1^{11} 41^{23} 31^{24} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 21 & 0 & 241 & 29 \\ 2 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2854] \ 2^1 3^{14} 21^{16} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 6 & 36 & 6 \\ 18 & 36 & 178 & 30 \\ 3 & 6 & 30 & 5 \end{pmatrix}_{b_0}$$

$$[2855] \ 1^{12} 5^{14} 1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2853] \ 1^{13} 221^{45} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 20 & 2 \\ 18 & 0 & 20 & 318 & 32 \\ 2 & 0 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[2852] \ 1^{12} 5^{12} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2851] \ 1^{12} 321^{13} 5^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 12 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2850] $2^{1^{12}1^{11}4^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 11 & 4 \\ 2 & 4 & 1 & 10 & 3 \\ 6 & 11 & 10 & 30 & 12 \\ 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[2849] $1^{12}4^{13}1^{34}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 18 & 0 & 166 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[2848] $1^{12}5^{14}1^{23}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 13 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2847] $1^{11}3^{22}1^{25}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 11 & 0 & 13 & 101 & 19 \\ 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2843] $1^15^{14}1^{23}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 30 & 5 \\ 18 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

[2844] $1^{11}4^{13}3^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 22 & 0 & 258 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[2845] $15^{13}2^{23}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 18 & 6 & 80 & 8 \\ 1 & 6 & 0 & 27 & 2 \\ 16 & 80 & 27 & 346 & 36 \\ 2 & 8 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[2846] $1^15^{13}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 10 & 12 & 18 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2860] $1^{13}5^{13}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2861] $1^{11}5^{14}1^{23}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2859] $1^15^{14}1^{23}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 5 & 35 & 5 \\ 22 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$$

[2858] $2^{\overline{12}} 3^{\overline{12}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2857] $1^{\overline{12}} 5^{\overline{12}} 1^{\overline{23}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 35 & 5 \\ 20 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$$

[2856] $1^{\overline{14}} 5^{\overline{14}} 1^{\overline{13}} 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 10 & 17 & 128 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2867] $1^{\overline{12}} 4^{\overline{12}} 2^{\overline{23}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 1 & 14 & 3 \\ 17 & 14 & 100 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[2862] $1^{\overline{11}} 5^{\overline{13}} 1^{\overline{23}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2863] $1^{\overline{13}} 5^{\overline{14}} 1^{\overline{12}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 11 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2864] $2^{\overline{12}} 3^{\overline{12}} 1^{\overline{15}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2865] $21^{\overline{4}} 5^{\overline{11}} 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 10 & 17 & 128 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2869] $31^{\overline{11}} 4^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 25 & 4 \\ 10 & 0 & 25 & 101 & 19 \\ 2 & 0 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[2868] $1^{\overline{11}} 4^{\overline{12}} 2^{\overline{23}} 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 1 & 14 & 3 \\ 17 & 14 & 100 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[2866] $1^{\overline{13}} 3^{\overline{22}} 1^{\overline{25}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 19 & 0 & 165 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[2870] $1^{\overline{12}} 3^{\overline{22}} 1^{\overline{45}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 26 & 2 \\ 18 & 0 & 26 & 410 & 38 \\ 2 & 0 & 2 & 38 & 3 \end{pmatrix}_{b_0}$$

[2871] $1^{14}4131^{241^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 20 & 0 & 158 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[2873] $1^{12}51^{2231^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 12 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2874] $1^{12}5^{12}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 9 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2872] $1^{14}321^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 40 & 5 \\ 20 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$$

[2875] $151^1 3^1 22 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 4 & 7 & 4 \\ 3 & 7 & 7 & 5 & 5 \\ 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[2876] $1^{12}41^3 31^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 25 & 0 & 305 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[2877] $1^{13}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 8 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2878] $1^{12}41^1 31^3 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 23 & 0 & 273 & 31 \\ 2 & 3 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[2879] $21^1 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 8 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2883] $1^1 5^{12} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 5 & 35 & 5 \\ 21 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$$

[2880] $31^1 4^{15} 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 5 & 1 & 35 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 35 & 0 & 173 & 25 \\ 2 & 5 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[2881] $2^1 3^{15} 2151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 98 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[2884] $1^{11} 1^3 51^{12} 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 11 & 9 & 17 & 106 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2885] $1^1 1^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 98 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[2886] $1^{12} 5^{13} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2882] $1^{13} 5^1 1^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2887] $1^{13} 5^{14} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2888] $31^{13} 32151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 34 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[2889] $21^3 5^{13} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 11 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2890] $21^{13} 5^{14} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 13 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2891] $21^4 5^{13} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 13 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2892] $1^{13} 5^{13} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 11 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2893] $1^{12} 5^1 1^3 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2894] \quad 31^4 4^{11} 141^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 36 & 4 \\ 15 & 0 & 36 & 238 & 28 \\ 2 & 0 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2895] \quad 1^{13} 5^{14} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 13 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2899] \quad 2^3 1^4 5^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 5 & 23 & 5 \\ 14 & 23 & 81 & 19 \\ 3 & 5 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[2896] \quad 2^3 1^4 5^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 10 & 7 & 14 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2897] \quad 1^{12} 5^{14} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2898] \quad 31^1 4^{12} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 5 & 1 & 23 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 23 & 0 & 77 & 17 \\ 2 & 5 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2900] \quad 22^3 151^{15} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 3 & 10 & 2 \\ 3 & 3 & 4 & 35 & 4 \\ 15 & 10 & 35 & 173 & 25 \\ 2 & 2 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2904] \quad 1^{13} 5^{13} 1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2903] \quad 1^3 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2902] \quad 1^{13} 3221^{15} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 14 & 2 \\ 9 & 0 & 14 & 75 & 17 \\ 2 & 0 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2905] \quad 1^{13} 321^3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$\begin{aligned}
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 5 & 45 & 5 \\ 24 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0} \\
[2906] & 1^{13} 3221^3 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 24 & 2 \\ 15 & 0 & 24 & 295 & 33 \\ 2 & 0 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2907] & 1^{13} 41^2 3141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 19 & 0 & 145 & 23 \\ 2 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[2908] & 1^{13} 4131^3 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 205 & 27 \\ 2 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[2909] & 1^{12} 41^3 3141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 205 & 27 \\ 2 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[2910] & 1^{11} 321^3 51^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2901] & 1^{11} 51^2 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 11 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2911] & 221^{13} 5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 12 & 2 \\ 3 & 4 & 3 & 24 & 4 \\ 1 & 3 & 0 & 15 & 2 \\ 12 & 24 & 15 & 115 & 21 \\ 2 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2912] & 2^3 3^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 5 & 32 & 5 \\ 20 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0} \\
[2913] & 21^{13} 5131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2914] & 21^{13} 5131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 11 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2915] & 21^3 5^{12} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2916] $2 \overset{1^6 1^1}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 2 & 4 & 24 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 12 & 24 & 18 & 142 & 24 \\ 2 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$
- [2917] $1 \overset{1^4}{1^1} 5 1^2 231^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 9 & 17 & 106 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2918] $1 \overset{1^2 3 2 1^2}{5} 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [2919] $1 \overset{1^2}{1^3} 5 1^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 14 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2920] $3 1 \overset{1^3}{1^3} 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 18 & 3 \\ 5 & 5 & 24 & 5 \\ 18 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$
- [2921] $1 \overset{1^2}{1^2} 4 1^2 3 1^2 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 24 & 0 & 286 & 32 \\ 2 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$
- [2922] $1 \overset{1^3}{1^2} 5 1^3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2923] $1 \overset{1^4}{1^3} 5 1^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 16 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2924] $2 1 \overset{1^1}{1^3} 5 1^3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2925] $1 \overset{1^4}{1^3} 5 1^3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2927] $1 \overset{1^4}{1^3} 4 1 3 1^2 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 17 & 0 & 121 & 21 \\ 2 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$
- [2929] $1 \overset{1^3}{1^3} 4 1^2 3 1^1 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$\begin{aligned}
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 23 & 0 & 229 & 29 \\ 2 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0} \\
[2928] & 1^{11} 3^{22} 1^4 5^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 23 & 2 \\ 18 & 0 & 23 & 361 & 35 \\ 2 & 0 & 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2926] & 1^{13} 5^{14} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2932] & 1^{12} 5^{13} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2931] & 1^{12} 5^{13} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 9 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2930] & 1^1 3^{22} 1^3 5^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 15 & 2 \\ 15 & 0 & 15 & 178 & 24 \\ 2 & 0 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2933] & 1^{13} 5^{13} 1^3 2^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 15 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2935] & 31^3 4^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 34 & 4 \\ 13 & 0 & 34 & 194 & 26 \\ 2 & 0 & 4 & 26 & 3 \end{pmatrix}_{b_0} \\
[2934] & 31^4 4^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 43 & 4 \\ 16 & 0 & 43 & 317 & 33 \\ 2 & 0 & 4 & 33 & 3 \end{pmatrix}_{b_0} \\
[2936] & 21^4 5^{12} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2942] & 21^{13} 5^{12} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2941] & 1^{11} 5^{11} 1^3 2^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 13 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2940] $1^{12}5^{14}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2939] $1^{12}3221^{15}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 11 & 2 \\ 9 & 0 & 11 & 50 & 14 \\ 2 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2938] $2^3 3 21^{35}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 16 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 16 & 11 & 18 & 166 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2937] $22^3 3 151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 8 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 4 & 18 & 4 \\ 8 & 8 & 18 & 50 & 14 \\ 2 & 2 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[2944] $1^{14}41^{12}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 16 & 3 \\ 16 & 16 & 106 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[2943] $2^{12}3^{41}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 1 & 18 & 3 \\ 10 & 8 & 18 & 86 & 20 \\ 2 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[2945] $1^{12}41^{13}31^{24}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 19 & 0 & 173 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[2946] $1^{13}5^{12}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2956] $21^{13}5^{13}231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 18 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2950] $1^{12}5^{12}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2947] $21^{13}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2948] $21^{21^{12}5131^{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2949] $21^{1351^{1231^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 63 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2955] $31^{31^{14}4141^{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 27 & 4 \\ 13 & 0 & 27 & 133 & 21 \\ 2 & 0 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[2954] $23^{11^{13}51^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 15 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2951] $21^{1351^3 32} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 12 & 24 & 15 & 115 & 21 \\ 2 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2952] $1^{1341^1 31^{241^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 22 & 0 & 214 & 28 \\ 2 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[2953] $1^{1341^1 31^{141^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 130 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2957] $1^{1241^2 2321^{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 1 & 16 & 3 \\ 20 & 16 & 142 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[2958] $1^{1441^3 3141^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 27 & 0 & 301 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[2959] $1^{1141^4 31^{141^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 27 & 0 & 397 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$$

[2960] $21^{1351^2 5131^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2961] $1^{1341^4 3141^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 29 & 0 & 385 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$$

[2962] $22 \overset{1^{12}}{3} 151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 3 & 10 & 2 \\ 3 & 3 & 4 & 20 & 4 \\ 9 & 10 & 20 & 68 & 16 \\ 2 & 2 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[2963] $1^{13} \overset{1^2}{5} 1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2964] $1 \overset{1^2}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 9 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2965] $1^{12} 41^{13} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 27 & 0 & 397 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$

[2968] $1^{12} 41^2 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 20 & 0 & 186 & 26 \\ 2 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$

[2966] $31^{12} 321^{25} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 30 & 5 \\ 22 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$

[2967] $1^{12} \overset{1^4}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 9 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2969] $1^{11} \overset{1^2}{5} 1^{22} 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 12 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2973] $1^{11} \overset{1^2}{5} 1^{22} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 5 & 40 & 5 \\ 21 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$

[2972] $2^{13} \overset{1^4}{3} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 1 & 15 & 3 \\ 9 & 9 & 15 & 70 & 18 \\ 2 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$

[2970] $1^{13} 3221^{25} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 19 & 2 \\ 12 & 0 & 19 & 165 & 25 \\ 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2971] $1^{12} \overset{1^1}{5} 1^{32} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2975] & 1^1 3221^1 51^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 18 & 6 & 60 & 8 \\ 1 & 6 & 0 & 15 & 2 \\ 14 & 60 & 15 & 178 & 24 \\ 2 & 8 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2974] & 1^1 5^1 1^3 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 25 & 3 \\ 3 & 5 & 45 & 5 \\ 25 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0} \\
[2977] & 2^1 3 21^2 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 5 & 30 & 5 \\ 20 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0} \\
[2976] & 2^1 3 15^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 5 & 26 & 5 \\ 16 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0} \\
[2982] & 21^1 5^1 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 9 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2980] & 2^1 3 15^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2979] & 1^{11} 5^1 1^2 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 12 & 12 & 21 & 180 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2978] & 1^{14} 3221^1 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 17 & 2 \\ 10 & 0 & 17 & 106 & 20 \\ 2 & 0 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2981] & 1^{13} 5^1 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 9 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2987] & 1^{11} 41^3 31^2 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 26 & 0 & 374 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[2986] & 1^{12} 41^4 31^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 30 & 0 & 454 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2985] $2 \overset{1^{14}}{3} 1 \overset{1^3}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 29 & 5 \\ 18 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$
- [2984] $31 \overset{1^{13}}{2} 4 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 5 & 1 & 39 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 39 & 0 & 229 & 29 \\ 2 & 5 & -1 & 29 & 3 \end{pmatrix}_{b_0}$
- [2983] $1 \overset{1^{13}321^3}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 13 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [2988] $2 \overset{1^{11}41}{3} 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 6 & 29 & 6 \\ 15 & 29 & 108 & 24 \\ 3 & 6 & 24 & 5 \end{pmatrix}_{b_0}$
- [2989] $1^{12}41^32321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 1 & 16 & 3 \\ 20 & 16 & 138 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$
- [2990] $1 \overset{1^{13}}{5} 1^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2993] $2 \overset{1^{13}}{3} 1 \overset{1^1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2992] $1^{12} \overset{1^1}{5} 1^{23} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [2991] $1^{11} \overset{1^2}{5} 1^{32} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 14 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [2994] $221^{13}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 10 & 2 \\ 3 & 4 & 3 & 20 & 4 \\ 1 & 3 & 0 & 9 & 2 \\ 10 & 20 & 9 & 67 & 15 \\ 2 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2999] $2 \overset{1^{11}}{3} 21^{25} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 12 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2998] $31^{12}321^{15} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 15 & 3 \\ 5 & 5 & 20 & 5 \\ 15 & 20 & 63 & 17 \\ 3 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[2997] $22 \overset{1}{3} 3 151^{\overset{1}{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 15 & 3 \\ 6 & 14 & 39 & 8 \\ 15 & 39 & 97 & 21 \\ 3 & 8 & 21 & 4 \end{pmatrix}_{b_0}$$

[2996] $2 \overset{1}{3} 3 151^{\overset{1}{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 12 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2995] $2 \overset{1}{3} 3 151^{\overset{1}{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 13 & 10 & 17 & 128 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3001] $21^{\overset{1}{3}} 51^{\overset{1}{3}} 231^{\overset{1}{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3003] $2 \overset{1}{3} 21^{\overset{1}{41}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 6 & 42 & 6 \\ 21 & 42 & 243 & 35 \\ 3 & 6 & 35 & 5 \end{pmatrix}_{b_0}$$

[3002] $1^{\overset{1}{2}} 51^{\overset{1}{3}} 131^{\overset{1}{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3000] $21^{\overset{1}{3}} 51^{\overset{1}{3}} 231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 16 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3004] $1^{\overset{1}{3}} 41^{\overset{1}{2}} 2321^{\overset{1}{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 1 & 16 & 3 \\ 18 & 16 & 118 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$$

[3005] $1^{\overset{1}{2}} 51^{\overset{1}{3}} 231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 14 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[3006] $1^{\overset{1}{2}} 51^{\overset{1}{3}} 231^{\overset{1}{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 5 & 45 & 5 \\ 26 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$$

[3007] $1^{\overset{1}{3}} 51^{\overset{1}{4}} 131^{\overset{1}{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

- [3008] $1^{14}51^{13}231^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 28 & 3 \\ 3 & 5 & 45 & 5 \\ 28 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$
- [3009] $21^{13}51^{13}131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3010] $1^{11}3221^{25}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 12 & 0 & 13 & 101 & 19 \\ 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3011] $1^{11}51^{22}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3012] $1^{13}51^{13}131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3013] $231^{14}51^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -220$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 9 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3014] $1^{11}3221^{12}51^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 18 & 6 & 45 & 8 \\ 1 & 6 & 0 & 13 & 2 \\ 11 & 45 & 13 & 101 & 19 \\ 2 & 8 & 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3015] $1^{13}4131^{44}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -223$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 25 & 0 & 313 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$
- [3016] $1^{13}51^{13}1231^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -223$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [3017] $1^{14}3221^{35}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -223$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 27 & 2 \\ 16 & 0 & 27 & 346 & 36 \\ 2 & 0 & 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [3018] $1^{13}51^{13}131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -223$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 10 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3026] $1^{14}41^{23}141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 22 & 0 & 186 & 26 \\ 2 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0} \\
[3019] & 1^{12}3221^3 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 21 & 2 \\ 15 & 0 & 21 & 250 & 30 \\ 2 & 0 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3025] & 1^{13}51^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 15 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3024] & 1^{13}3221^3 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 295 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0} \\
[3023] & 1^{12}3221^{12} 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 6 & 50 & 8 \\ 1 & 6 & 0 & 16 & 2 \\ 12 & 50 & 16 & 130 & 22 \\ 2 & 8 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3022] & 1^{12}41^4 3141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 26 & 0 & 330 & 34 \\ 2 & 3 & -1 & 34 & 3 \end{pmatrix}_{b_0} \\
[3021] & 151^{12}2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 18 & 6 & 50 & 8 \\ 1 & 6 & 0 & 16 & 2 \\ 10 & 50 & 16 & 130 & 22 \\ 2 & 8 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3020] & 1^{13}51^3 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 27 & 3 \\ 3 & 5 & 45 & 5 \\ 27 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0} \\
[3027] & 31^{14}4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 5 & 1 & 37 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 37 & 0 & 197 & 27 \\ 2 & 5 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[3031] & 1^{11}51^4 2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3030] & 1^{12}51^4 3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 16 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3029] & 1^{12}51^3 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 15 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3028] & 1^{11}3221^3 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 18 & 2 \\ 15 & 0 & 18 & 211 & 27 \\ 2 & 0 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3035] & 1^{14} 3221^2 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 22 & 0 & 206 & 28 \\ 2 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[3039] & 31^{11} 4^{11} 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 25 & 4 \\ 11 & 0 & 25 & 101 & 19 \\ 2 & 0 & 4 & 19 & 3 \end{pmatrix}_{b_0} \\
[3038] & 1^{12} 41^2 31^3 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 28 & 0 & 410 & 38 \\ 2 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0} \\
[3037] & 1^{11} 5^{14} 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 16 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3036] & 1^{12} 5^{12} 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 5 & 40 & 5 \\ 22 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0} \\
[3032] & 2^{13} 3^{13} 21^{15} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 5 & 25 & 5 \\ 16 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0} \\
[3033] & 1^{13} 321^2 5^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 11 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3034] & 1^{12} 5^{14} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3040] & 22^3 3^{13} 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 4 & 23 & 4 \\ 10 & 8 & 23 & 75 & 17 \\ 2 & 2 & 4 & 17 & 3 \end{pmatrix}_{b_0} \\
[3041] & 31^{12} 4^{12} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -230, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 5 & 1 & 32 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 32 & 0 & 158 & 24 \\ 2 & 5 & -1 & 24 & 3 \end{pmatrix}_{b_0} \\
[3042] & 2^{13} 3^{13} 151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3043] \quad & 21^{13}51^1231^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 86 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3044] \quad & 21^{13}51^2231^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 13 & 13 & 14 & 125 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3045] \quad & 21^{13}51^2231^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 15 & 15 & 20 & 193 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3046] \quad & 21^{13}51^3231^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 17 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3047] \quad & 1^{12}41^331^241^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 29 & 0 & 429 & 39 \\ 2 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0} \\
[3048] \quad & 1^{13}1^{13}51^132 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3049] \quad & 1^{12}3221^251^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 16 & 2 \\ 12 & 0 & 16 & 130 & 22 \\ 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3050] \quad & 1^{12}51^2231^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3051] \quad & 2^131^3151^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 11 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3052] \quad & 2^144121^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 1 & 16 & 3 \\ 10 & 11 & 16 & 90 & 20 \\ 2 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0} \\
[3053] \quad & 1^{13}51^3131^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3058] $2 \overset{1^{17} 4^1}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 14 & 2 \\ 2 & 2 & 4 & 28 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 14 & 28 & 21 & 194 & 28 \\ 2 & 4 & 3 & 28 & 4 \end{pmatrix}_{b_0}$

[3055] $21 \overset{1^{12}}{3} 5 131 \overset{1^4}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[3056] $2 \overset{1^{12}}{3} 21 \overset{1^5}{5} 1 \overset{1^2}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 66 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3054] $1 \overset{1^{12}}{5} 1 \overset{1^{14}}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3057] $1 \overset{1^3}{12} 5 131 \overset{1^4}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[3059] $1 \overset{1^{14}}{3} 3221 \overset{1^3}{5} 1 \overset{1^3}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 18 & 6 & 80 & 8 \\ 1 & 6 & 0 & 27 & 2 \\ 19 & 80 & 27 & 346 & 36 \\ 2 & 8 & 2 & 36 & 3 \end{pmatrix}_{b_0}$

[3065] $31 \overset{1^{16}}{4} 141 \overset{1^1}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 5 & 1 & 42 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 42 & 0 & 250 & 30 \\ 2 & 5 & -1 & 30 & 3 \end{pmatrix}_{b_0}$

[3064] $1 \overset{1^1}{5} 1 \overset{1^{16}}{3} 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$

[3063] $2 \overset{1^{12}}{3} 3 21 \overset{1^3}{5} 1 \overset{1^1}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 27 & 3 \\ 4 & 5 & 40 & 5 \\ 27 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$

[3062] $2 \overset{1^{16}}{3} 3 2151 \overset{1^1}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$

[3061] $31 \overset{1^{13}}{3} 4 141 \overset{1^1}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 48 & 0 & 358 & 36 \\ 2 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$

[3060] $31 \overset{1^{14}}{4} 141 \overset{1^1}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 28 & 0 & 110 & 20 \\ 2 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3071] \ 1^{13}5^{14}1231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[3070] \ 1^{13}5^{14}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 11 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3069] \ 1^{13}5^{14}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3068] \ 1^{14}3221^{25}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 22 & 2 \\ 13 & 0 & 22 & 206 & 28 \\ 2 & 0 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[3067] \ 1^{12}41^{13}1^{34}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 23 & 0 & 273 & 31 \\ 2 & 3 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[3066] \ 1^{11}5^{12}231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 13 & 12 & 21 & 180 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3073] \ 1^{14}41^{23}1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 26 & 0 & 278 & 32 \\ 2 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[3072] \ 2^{13}3^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 15 & 4 \\ 2 & 4 & 1 & 13 & 3 \\ 8 & 15 & 13 & 56 & 16 \\ 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[3075] \ 1^{13}5^{14}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3074] \ 1^{13}5^{12}231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 5 & 40 & 5 \\ 23 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[3078] \ 1^{11}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3080] $31^4 1^{13} 4 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 5 & 1 & 57 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 57 & 0 & 517 & 43 \\ 2 & 5 & -1 & 43 & 3 \end{pmatrix}_{b_0}$$

[3083] $1^{11} 5^1 1^3 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 14 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[3082] $22^3 151^{16} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 18 & 2 \\ 1 & 0 & 3 & 12 & 2 \\ 3 & 3 & 4 & 42 & 4 \\ 18 & 12 & 42 & 250 & 30 \\ 2 & 2 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[3081] $1^{14} 4131^2 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 20 & 0 & 158 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[3085] $1^{12} 5^1 1^3 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 13 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3086] $1^{13} 3221^4 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 29 & 2 \\ 19 & 0 & 29 & 465 & 41 \\ 2 & 0 & 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[3087] $1^{13} 41^3 31^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 28 & 0 & 358 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[3077] $1^{13} 3221^1 51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 14 & 2 \\ 10 & 0 & 14 & 75 & 17 \\ 2 & 0 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3076] $1^{13} 41^3 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 24 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[3079] $1^{14} 321^3 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 5 & 50 & 5 \\ 26 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0}$$

[3084] $1^{14} 4131^3 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 24 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[3091] $1^{13} 4131^3 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -247$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 21 & 0 & 205 & 27 \\ 2 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3090] \ 1^{1^2 3 2 1^3} 5^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 13 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[3089] \ 1^4 3 2 2 1^4 5 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 20 & 2 \\ 19 & 0 & 20 & 318 & 32 \\ 2 & 0 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[3088] \ 1^{1^4} 4 1^1 3 1^1 4 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 169 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3094] \ 1^{1^2 4} 5^{1^4} 1 3 1^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[3093] \ 1^{1^2 4} 5^{1^4} 1^2 2 3 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 14 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[3092] \ 1^{1^4} 5^{1^4} 1^2 2 3 1^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 5 & 40 & 5 \\ 24 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[3096] \ 2 1^3 5^{1^3} 1 3 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[3095] \ 1^{1^3 3} 5^{1^3} 1 3 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[3097] \ 2 2 1^{1^3} 5 1 3 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -255, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 11 & 2 \\ 3 & 4 & 3 & 22 & 4 \\ 1 & 3 & 0 & 12 & 2 \\ 11 & 22 & 12 & 88 & 18 \\ 2 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[3098] \ 1^{1^4} 4 1^1 3 1^2 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -255, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 25 & 0 & 261 & 31 \\ 2 & 3 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[3099] \ 2^3 3 2 1^{2^5} 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -255, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 13 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3100] & 1^{12} 41^3 31^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -255, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 25 & 0 & 305 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0} \\
[3101] & 21^{13} 51^2 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 14 & 14 & 17 & 156 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3102] & 1^{13} 41^1 31^3 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 26 & 0 & 322 & 34 \\ 2 & 3 & -1 & 34 & 3 \end{pmatrix}_{b_0} \\
[3103] & 31^{14} 4141^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 36 & 4 \\ 16 & 0 & 36 & 238 & 28 \\ 2 & 0 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[3104] & 1^{13} 51^4 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3105] & 31^{12} 4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 41 & 0 & 269 & 31 \\ 2 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0} \\
[3106] & 1^{12} 51^4 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 12 & 18 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3107] & 1^{12} 3221^{12} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 18 & 6 & 50 & 8 \\ 1 & 6 & 0 & 16 & 2 \\ 11 & 50 & 16 & 130 & 22 \\ 2 & 8 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3108] & 2^3 21^{25} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 13 & 13 & 14 & 125 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3109] & 21^{12} 415131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 12 & 18 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3110] & 1^{14} 3221^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 18 & 6 & 80 & 8 \\ 1 & 6 & 0 & 27 & 2 \\ 17 & 80 & 27 & 346 & 36 \\ 2 & 8 & 2 & 36 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3111] $1^{12}41^231^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 286 & 32 \\ 2 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$

[3112] $1^{13}41^231^241^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 27 & 0 & 337 & 35 \\ 2 & 3 & -1 & 35 & 3 \end{pmatrix}_{b_0}$

[3113] $1^{13}3221^451^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 29 & 0 & 465 & 41 \\ 2 & 3 & -1 & 41 & 3 \end{pmatrix}_{b_0}$

[3114] $1^{14}41^22321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 18 & 3 \\ 19 & 18 & 140 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$

[3115] $1^{11}51^2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 14 & 12 & 21 & 180 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$

[3116] $1^{12}41^32321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 23 & 3 \\ 2 & 1 & 18 & 3 \\ 23 & 18 & 180 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$

[3117] $1^{13}51^{13}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3121] $1^{14}41^43141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 32 & 0 & 446 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$

[3120] $1^{12}41^231^441^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 32 & 0 & 558 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$

[3119] $1^{12}51^{12}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 12 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3118] $1^{14}51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 12 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$

[3122] $31^44141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 5 & 1 & 50 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 50 & 0 & 410 & 38 \\ 2 & 5 & -1 & 38 & 3 \end{pmatrix}_{b_0}$

- [3123] $1^{13}41^32321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -267$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 1 & 18 & 3 \\ 21 & 18 & 160 & 30 \\ 3 & 3 & 30 & 5 \end{pmatrix}_{b_0}$
- [3124] $1^{13}41^22321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -267$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 1 & 18 & 3 \\ 21 & 18 & 160 & 30 \\ 3 & 3 & 30 & 5 \end{pmatrix}_{b_0}$
- [3125] $21^{13}45131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -267$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 14 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3126] $1^{13}5^{14}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -267$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 14 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3128] $41^{12}231^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 26 & 1 & 72 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 72 & 0 & 186 & 26 \\ 2 & 10 & -1 & 26 & 3 \end{pmatrix}_{b_0}$
- [3127] $151^{12}2231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 9 & 2 \\ 4 & 18 & 6 & 45 & 8 \\ 1 & 6 & 0 & 13 & 2 \\ 9 & 45 & 13 & 101 & 19 \\ 2 & 8 & 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3129] $31^{13}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -270$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 5 & 1 & 30 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 30 & 0 & 130 & 22 \\ 2 & 5 & -1 & 22 & 3 \end{pmatrix}_{b_0}$
- [3133] $1^{13}5^{12}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 11 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [3132] $23^{13}151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3131] $21^{13}5^{13}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 11 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [3130] $31^{11}24141^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 25 & 4 \\ 12 & 0 & 25 & 101 & 19 \\ 2 & 0 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [3134] $1^{13}41^231^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 23 & 0 & 229 & 29 \\ 2 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0} \\
[3137] & 1^{13}41^131^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 22 & 0 & 214 & 28 \\ 2 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[3135] & 1^{11}51^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 15 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[3136] & 1^{13}51^3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3138] & 321^{12}4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 8 & 14 & 3 \\ 8 & 17 & 32 & 9 \\ 14 & 32 & 54 & 16 \\ 3 & 9 & 16 & 4 \end{pmatrix}_{b_0} \\
[3139] & 1^{12}41^131^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 27 & 0 & 397 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0} \\
[3140] & 151^{13}2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 6 & 60 & 8 \\ 1 & 6 & 0 & 15 & 2 \\ 12 & 60 & 15 & 178 & 24 \\ 2 & 8 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3141] & 1^13221^{13}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 18 & 6 & 60 & 8 \\ 1 & 6 & 0 & 15 & 2 \\ 13 & 60 & 15 & 178 & 24 \\ 2 & 8 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3142] & 41^{13}23141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0} \\
[3143] & 1^{12}41^431^241^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 34 & 0 & 602 & 46 \\ 2 & 3 & -1 & 46 & 3 \end{pmatrix}_{b_0} \\
[3144] & 1^151^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 27 & 3 \\ 3 & 5 & 50 & 5 \\ 27 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0} \\
[3145] & 1^{11}51^2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 15 & 12 & 21 & 180 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3146] & 1^{12}51^3231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 14 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3147] $1^{13} 5^{14} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3149] $2^3 3^{13} 151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 13 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[3148] $31^{14} 32151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 16 & 3 \\ 5 & 5 & 20 & 5 \\ 16 & 20 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[3150] $1^{11} 3221^{12} 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 18 & 6 & 45 & 8 \\ 1 & 6 & 0 & 13 & 2 \\ 10 & 45 & 13 & 101 & 19 \\ 2 & 8 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3152] $2^3 3^{13} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 1 & 19 & 3 \\ 11 & 10 & 19 & 104 & 22 \\ 2 & 2 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[3151] $1^{12} 41^3 31^3 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 33 & 0 & 577 & 45 \\ 2 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[3153] $1^{14} 3221^{13} 51^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 18 & 2 \\ 4 & 18 & 6 & 80 & 3 \\ 1 & 6 & 0 & 27 & 2 \\ 18 & 80 & 27 & 346 & 36 \\ 2 & 8 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[3154] $1^{13} 3221^{13} 51^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -285, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 19 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[3156] $2^3 3^{14} 151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 9 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3158] $1^{12} 3221^4 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 26 & 2 \\ 19 & 0 & 26 & 410 & 38 \\ 2 & 0 & 2 & 38 & 3 \end{pmatrix}_{b_0}$$

[3155] $2^3 3^{11} 351^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 16 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 16 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[3157] $2^3 3^{12} 21^{15} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 86 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[3159] \ 1^{11} 3^4 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 34 & 4 \\ 14 & 0 & 34 & 194 & 26 \\ 2 & 0 & 4 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3160] \ 1^{11} 3221^4 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 23 & 2 \\ 19 & 0 & 23 & 361 & 35 \\ 2 & 0 & 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[3161] \ 1^{14} 3221^1 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 17 & 2 \\ 11 & 0 & 17 & 106 & 20 \\ 2 & 0 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3162] \ 1^{11} 5^1 3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 25 & 2 \\ 15 & 15 & 25 & 278 & 32 \\ 2 & 2 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[3165] \ 1^{11} 5^1 3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 16 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[3166] \ 1^{12} 5^1 2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 13 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3167] \ 1^{13} 3221^3 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 24 & 2 \\ 16 & 0 & 24 & 295 & 33 \\ 2 & 0 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[3168] \ 1^{13} 5^1 3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 15 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3164] \ 1^{13} 3221^2 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 19 & 2 \\ 13 & 0 & 19 & 165 & 25 \\ 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3163] \ 2^3 3^1 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 11 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3169] \ 2^3 3^1 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 12 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3170] $1^{13}5^{13}131^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3171] $1^{12}3221^{13}51^4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 18 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 18 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3172] $31^{14}3151^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 32 & 5 \\ 24 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[3173] $1^{13}41^431^{14}1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 33 & 0 & 517 & 43 \\ 2 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0}$$

[3174] $1^{13}4131^441^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 25 & 0 & 313 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[3175] $1^{13}41^42321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 24 & 3 \\ 2 & 1 & 20 & 3 \\ 24 & 20 & 214 & 34 \\ 3 & 3 & 34 & 5 \end{pmatrix}_{b_0}$$

[3176] $1^{12}3221^251^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 16 & 2 \\ 13 & 0 & 16 & 130 & 22 \\ 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3177] $1^{12}5^{12}1^3231^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 15 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3178] $1^{13}5^{13}1^3231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 16 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3179] $1^{11}5^{14}1^3231^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 17 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[3181] $1^{11}3221^351^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 18 & 2 \\ 16 & 0 & 18 & 211 & 27 \\ 2 & 0 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3180] $1^{12}5^{12}1^3231^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$\begin{aligned}
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 28 & 3 \\ 3 & 5 & 50 & 5 \\ 28 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0} \\
[3184] & 1^{14} 41^3 3141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 27 & 0 & 301 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0} \\
[3183] & 1^{13} 5^{12} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 14 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[3182] & 1^{13} 41^4 3141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 29 & 0 & 385 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0} \\
[3185] & 1^{13} 41^1 31^4 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 454 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0} \\
[3186] & 1^{13} 5^4 1^3 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 17 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3187] & 1^{12} 41^4 31^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 454 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0} \\
[3188] & 1^{12} 5^1 1^2 231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 13 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3189] & 31^2 4^{14} 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 5 & 1 & 46 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 46 & 0 & 314 & 34 \\ 2 & 5 & -1 & 34 & 3 \end{pmatrix}_{b_0} \\
[3190] & 1^{13} 5^{14} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3193] & 1^{14} 3221^4 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 32 & 2 \\ 20 & 0 & 32 & 526 & 44 \\ 2 & 0 & 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[3191] & 1^{12} 41^2 31^3 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 28 & 0 & 410 & 38 \\ 2 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3192] $1^{14}4131^441^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 28 & 0 & 366 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[3194] $1^{12}5^12231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 14 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[3195] $1^{12}41^42321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 26 & 3 \\ 2 & 1 & 20 & 3 \\ 26 & 20 & 230 & 36 \\ 3 & 3 & 36 & 5 \end{pmatrix}_{b_0}$$

[3196] $1^{14}41^331^441^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 31 & 0 & 417 & 39 \\ 2 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

[3199] $1^15^13^12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 194 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[3198] $31^14^141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 5 & 1 & 49 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 49 & 0 & 341 & 35 \\ 2 & 5 & -1 & 35 & 3 \end{pmatrix}_{b_0}$$

[3197] $2^13^12151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 194 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[3200] $1^{14}41^32321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 1 & 20 & 3 \\ 22 & 20 & 186 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$$

[3201] $1^{13}5^13^1231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 29 & 3 \\ 3 & 5 & 50 & 5 \\ 29 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0}$$

[3202] $21^13^15^1131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3203] $1^{12}3221^351^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 21 & 2 \\ 16 & 0 & 21 & 250 & 30 \\ 2 & 0 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3204] $1^{13}5^13^1131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3206] $1^{14}3221^351^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 27 & 0 & 346 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$

[3205] $1^{12}5^13^1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 16 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$

[3207] $2^13^15^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3208] $22^13^151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 21 & 2 \\ 1 & 0 & 3 & 14 & 2 \\ 3 & 3 & 4 & 49 & 4 \\ 21 & 14 & 49 & 341 & 35 \\ 2 & 2 & 4 & 35 & 3 \end{pmatrix}_{b_0}$

[3209] $1^{12}41^331^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 29 & 0 & 429 & 39 \\ 2 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}$

[3211] $1^{12}5^12^1231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 15 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3216] $1^{11}5^13^1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 25 & 2 \\ 16 & 15 & 25 & 278 & 32 \\ 2 & 2 & 2 & 32 & 3 \end{pmatrix}_{b_0}$

[3212] $1^{15}5^13^1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 30 & 3 \\ 3 & 5 & 50 & 5 \\ 30 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0}$

[3215] $231^{12}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 5 & 1 & 10 & 2 \\ 5 & 33 & 1 & 70 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 70 & 0 & 138 & 22 \\ 2 & 11 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

[3214] $2^114^14^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 2 & 4 & 19 & 4 \\ 2 & 4 & 1 & 16 & 3 \\ 10 & 19 & 16 & 90 & 20 \\ 2 & 4 & 3 & 20 & 4 \end{pmatrix}_{b_0}$

[3213] $41^{12}231^141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 72 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 72 & 0 & 186 & 26 \\ 2 & 10 & -1 & 26 & 3 \end{pmatrix}_{b_0}$

[3210] $31^44^1141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 43 & 4 \\ 17 & 0 & 43 & 317 & 33 \\ 2 & 0 & 4 & 33 & 3 \end{pmatrix}_{b_0}$$

[3217] $2^1 3^1 21^{14} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 5 & 30 & 5 \\ 19 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

[3218] $1^{12} 41^3 31^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 37 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 37 & 0 & 749 & 51 \\ 2 & 3 & -1 & 51 & 3 \end{pmatrix}_{b_0}$$

[3219] $1^{13} 5^1 1^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3220] $1^{12} 5^1 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 17 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3221] $22^1 3^1 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 3 & 10 & 2 \\ 3 & 3 & 4 & 30 & 4 \\ 13 & 10 & 30 & 128 & 22 \\ 2 & 2 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[3224] $1^{14} 4131^3 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 24 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[3222] $1^{12} 5^{321^3} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 14 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3223] $1^{13} 41^3 31^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 32 & 0 & 490 & 42 \\ 2 & 3 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

[3225] $1^{14} 41^2 2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 1 & 20 & 3 \\ 22 & 20 & 182 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$$

[3226] $1^{13} 41^2 31^3 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 31 & 0 & 469 & 41 \\ 2 & 3 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

[3227] $2^1 3^1 151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 14 & 12 & 18 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3228] $22^1 3^1 151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 11 & 2 \\ 1 & 0 & 3 & 10 & 2 \\ 3 & 3 & 4 & 25 & 4 \\ 11 & 10 & 25 & 93 & 19 \\ 2 & 2 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3229] \ 1^{14} 2^3 15 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 13 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[3230] \ 1^{12} 2^3 21^3 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 17 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 19 & 2 \\ 17 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3231] \ 1^{13} 5^{14} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[3232] \ 1^{12} 41^4 31^3 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 38 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 38 & 0 & 774 & 52 \\ 2 & 3 & -1 & 52 & 3 \end{pmatrix}_{b_0}$$

$$[3233] \ 1^{13} 41^3 2321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 24 & 3 \\ 2 & 1 & 20 & 3 \\ 24 & 20 & 202 & 34 \\ 3 & 3 & 34 & 5 \end{pmatrix}_{b_0}$$

$$[3234] \ 1^{14} 41^1 31^3 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 29 & 0 & 377 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$$

$$[3237] \ 1^{14} 41^2 31^1 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 26 & 0 & 278 & 32 \\ 2 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[3235] \ 1^{14} 41^2 31^2 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 30 & 0 & 394 & 38 \\ 2 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

$$[3236] \ 1^{13} 41^1 31^3 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 26 & 0 & 322 & 34 \\ 2 & 3 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[3238] \ 1^{11} 3221^{13} 51^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 16 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3240] \ 1^{14} 3221^2 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 22 & 2 \\ 14 & 0 & 22 & 206 & 28 \\ 2 & 0 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[3241] \ 1^{12} 5^1 2231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$\begin{aligned}
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 14 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3239] & 1^{13} 41^3 31^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 28 & 0 & 358 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[3242] & 1^{14} 41^1 31^2 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -347, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 25 & 0 & 261 & 31 \\ 2 & 3 & -1 & 31 & 3 \end{pmatrix}_{b_0} \\
[3243] & 1^{12} 51^1 31^2 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -347, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 15 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[3244] & 22^1 3^{14} 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 20 & 3 \\ 6 & 14 & 52 & 8 \\ 20 & 52 & 174 & 28 \\ 3 & 8 & 28 & 4 \end{pmatrix}_{b_0} \\
[3245] & 2^1 3^{13} 21^2 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 23 & 3 \\ 4 & 5 & 35 & 5 \\ 23 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0} \\
[3246] & 2^{14} 3^{41} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 1 & 20 & 3 \\ 12 & 12 & 20 & 126 & 24 \\ 2 & 2 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[3247] & 1^{13} 5^{14} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 15 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3248] & 21^4 5^{13} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 15 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3249] & 1^{11} 51^3 231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 25 & 2 \\ 17 & 15 & 25 & 278 & 32 \\ 2 & 2 & 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[3250] & 1^{14} 3221^3 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 27 & 2 \\ 17 & 0 & 27 & 346 & 36 \\ 2 & 0 & 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[3251] & 2^1 3^{12} 21^2 51^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 14 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3252] $31^{213}4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 5 & 1 & 39 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 39 & 0 & 229 & 29 \\ 2 & 5 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[3253] $1^{13}41^231^241^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 27 & 0 & 337 & 35 \\ 2 & 3 & -1 & 35 & 3 \end{pmatrix}_{b_0}$$

[3254] $1^{12}41^231^441^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 32 & 0 & 558 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[3255] $1^{14}321^351^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 15 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[3256] $2^13^151^{35}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 17 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 17 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3257] $1^{13}51^3231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 15 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3258] $31^34141^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -367, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 34 & 4 \\ 15 & 0 & 34 & 194 & 26 \\ 2 & 0 & 4 & 26 & 3 \end{pmatrix}_{b_0}$$

[3259] $31^34141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -370, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 5 & 1 & 55 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 55 & 0 & 461 & 41 \\ 2 & 5 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

[3261] $1^{11}51^3231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 25 & 2 \\ 18 & 15 & 25 & 278 & 32 \\ 2 & 2 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[3260] $1^{12}51^2231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 15 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3262] $31^14141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 5 & 1 & 37 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 37 & 0 & 197 & 27 \\ 2 & 5 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[3264] & 1^{13}3221^251^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -379, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 19 & 2 \\ 14 & 0 & 19 & 165 & 25 \\ 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3265] & 1^{12}51^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -379, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 16 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[3263] & 1^{13}3221^451^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -379, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 29 & 2 \\ 20 & 0 & 29 & 465 & 41 \\ 2 & 0 & 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[3266] & 2^3 3^{14} 151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3267] & 31^{13}321^51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 19 & 3 \\ 5 & 5 & 25 & 5 \\ 19 & 25 & 97 & 21 \\ 3 & 5 & 21 & 3 \end{pmatrix}_{b_0} \\
[3268] & 1^{12}41^331^341^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 33 & 0 & 577 & 45 \\ 2 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0} \\
[3269] & 1^{12}41^431^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 34 & 0 & 602 & 46 \\ 2 & 3 & -1 & 46 & 3 \end{pmatrix}_{b_0} \\
[3270] & 1^{11}3221^451^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 23 & 2 \\ 20 & 0 & 23 & 361 & 35 \\ 2 & 0 & 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[3271] & 1^{14}41^43141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 32 & 0 & 446 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0} \\
[3272] & 1^{14}41^431^141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 36 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 36 & 0 & 586 & 46 \\ 2 & 3 & -1 & 46 & 3 \end{pmatrix}_{b_0} \\
[3273] & 1^{13}41^231^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 35 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 35 & 0 & 625 & 47 \\ 2 & 3 & -1 & 47 & 3 \end{pmatrix}_{b_0} \\
[3274] & 1^{13}41^431^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 37 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 37 & 0 & 673 & 49 \\ 2 & 3 & -1 & 49 & 3 \end{pmatrix}_{b_0}$$

[3275] $1^{12}51^2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 16 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3276] $1^{13}51^3231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 16 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3277] $1^{14}3221^451^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 32 & 0 & 526 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[3278] $1^{13}41^42321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 27 & 3 \\ 2 & 1 & 22 & 3 \\ 27 & 22 & 256 & 38 \\ 3 & 3 & 38 & 5 \end{pmatrix}_{b_0}$$

[3279] $223151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -397, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 3 & 12 & 2 \\ 3 & 3 & 4 & 27 & 4 \\ 12 & 12 & 27 & 115 & 21 \\ 2 & 2 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[3281] $31^{13}321^351^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 33 & 3 \\ 5 & 5 & 45 & 5 \\ 33 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$$

[3280] $1^{13}41^131^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 454 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$

[3282] $1^{13}3221^351^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 24 & 2 \\ 17 & 0 & 24 & 295 & 33 \\ 2 & 0 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[3283] $1^{12}51^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 17 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[3285] $1^{13}41^331^341^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 36 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 36 & 0 & 646 & 48 \\ 2 & 3 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[3284] $1^{13}321^351^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 15 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[3286] $1^{12}3221^451^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 26 & 2 \\ 20 & 0 & 26 & 410 & 38 \\ 2 & 0 & 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[3287] & 1^{14} 4 13 1^4 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 28 & 0 & 366 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[3288] & 3 1^{14} 4 14 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 20 & 2 \\ 2 & 5 & 1 & 64 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 64 & 0 & 638 & 48 \\ 2 & 5 & -1 & 48 & 3 \end{pmatrix}_{b_0} \\
[3289] & 1^{14} 4 1^3 23 2 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 25 & 3 \\ 2 & 1 & 22 & 3 \\ 25 & 22 & 228 & 36 \\ 3 & 3 & 36 & 5 \end{pmatrix}_{b_0} \\
[3290] & 1^{12} 5 1^3 23 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 16 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[3292] & 1^{13} 5 1^3 23 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 17 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3291] & 1^{12} 3 2 2 1^3 5 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 21 & 2 \\ 17 & 0 & 21 & 250 & 30 \\ 2 & 0 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3293] & 3 1^3 4 14 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 48 & 0 & 358 & 36 \\ 2 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[3294] & 1^{14} 4 1^1 3 1^4 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -415, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 33 & 0 & 517 & 43 \\ 2 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0} \\
[3295] & 3 1^{15} 3 2 1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -415, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 20 & 3 \\ 5 & 5 & 25 & 5 \\ 20 & 25 & 98 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0} \\
[3296] & 2 3 1^4 5 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -415, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 13 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3297] & 1^{13} 4 1^4 3 1^1 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -415, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 33 & 0 & 517 & 43 \\ 2 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0} \\
[3298] & 3 2 1^{12} 4 14 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 8 & 17 & 3 \\ 8 & 17 & 40 & 9 \\ 17 & 40 & 84 & 20 \\ 3 & 9 & 20 & 4 \end{pmatrix}_{b_0}$$

[3299] $1^{12} 5^{14} 1^3 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 18 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[3300] $1^{14} 41^3 31^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 35 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 35 & 0 & 557 & 45 \\ 2 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[3301] $1^{13} 5^{14} 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 18 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3302] $151^{13} 2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 15 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[3303] $1^{13} 3221^{13} 51^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 18 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 18 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[3304] $2^3 3^{14} 15^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 2 & 2 & 0 & 19 & 2 \\ 15 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3305] $1^{14} 41^3 31^1 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 31 & 0 & 417 & 39 \\ 2 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

[3306] $1^{13} 41^2 31^3 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 31 & 0 & 469 & 41 \\ 2 & 3 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

[3307] $2^3 3^{12} 21^2 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 15 & 14 & 17 & 156 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3308] $1^{14} 41^2 31^3 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 34 & 0 & 534 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[3309] $2^3 3^{14} 15^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 14 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3310] $1^{12} 41^3 31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$\begin{aligned}
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 24 & 2 \\ 1 & 1 & 1 & 37 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 37 & 0 & 749 & 51 \\ 2 & 3 & -1 & 51 & 3 \end{pmatrix}_{b_0} \\
[3312] & 1^{13} 41^3 31^2 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 32 & 0 & 490 & 42 \\ 2 & 3 & -1 & 42 & 3 \end{pmatrix}_{b_0} \\
[3311] & 1^{13} 2^3 21^3 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 30 & 3 \\ 4 & 5 & 45 & 5 \\ 30 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0} \\
[3314] & 1^{14} 41^1 31^3 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 29 & 0 & 377 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0} \\
[3313] & 31^4 4^{11} 141^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 43 & 4 \\ 18 & 0 & 43 & 317 & 33 \\ 2 & 0 & 4 & 33 & 3 \end{pmatrix}_{b_0} \\
[3315] & 41^{14} 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0} \\
[3316] & 1^{12} 5^1 1^3 231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -447, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 17 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[3317] & 1^{14} 3221^2 51^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -447, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 22 & 2 \\ 15 & 0 & 22 & 206 & 28 \\ 2 & 0 & 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3318] & 1^{13} 5^1 1^3 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 16 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3319] & 1^{14} 41^2 31^2 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 30 & 0 & 394 & 38 \\ 2 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0} \\
[3320] & 31^4 4^{13} 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 19 & 2 \\ 2 & 5 & 1 & 57 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 57 & 0 & 517 & 43 \\ 2 & 5 & -1 & 43 & 3 \end{pmatrix}_{b_0} \\
[3321] & 1^{12} 41^4 31^3 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 24 & 2 \\ 1 & 1 & 1 & 38 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 38 & 0 & 774 & 52 \\ 2 & 3 & -1 & 52 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3322] $31^{13}321^{25}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -455, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 26 & 3 \\ 5 & 5 & 35 & 5 \\ 26 & 35 & 202 & 30 \\ 3 & 5 & 30 & 3 \end{pmatrix}_{b_0}$

[3323] $2^1 3^{12} 21^{35} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 18 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 2 & 2 & 0 & 20 & 2 \\ 18 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$

[3324] $1^{11} 3221^{13} 51^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -461, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 15 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3325] $1^{14} 3221^{45} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -463, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 32 & 2 \\ 21 & 0 & 32 & 526 & 44 \\ 2 & 0 & 2 & 44 & 3 \end{pmatrix}_{b_0}$

[3326] $1^{12} 3221^{13} 51^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 17 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0}$

[3327] $1^{14} 41^{43} 2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -470, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 28 & 3 \\ 2 & 1 & 24 & 3 \\ 28 & 24 & 286 & 40 \\ 3 & 3 & 40 & 5 \end{pmatrix}_{b_0}$

[3328] $1^{13} 41^{33} 1^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 24 & 2 \\ 1 & 1 & 1 & 40 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 40 & 0 & 826 & 54 \\ 2 & 3 & -1 & 54 & 3 \end{pmatrix}_{b_0}$

[3329] $1^{14} 321^{31} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 16 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$

[3330] $1^{13} 41^{43} 1^3 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -475, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 24 & 2 \\ 1 & 1 & 1 & 41 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 41 & 0 & 853 & 55 \\ 2 & 3 & -1 & 55 & 3 \end{pmatrix}_{b_0}$

[3331] $1^{12} 51^{32} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -479, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 18 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0}$

[3332] $1^{14} 3221^{35} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -479, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 27 & 2 \\ 18 & 0 & 27 & 346 & 36 \\ 2 & 0 & 2 & 36 & 3 \end{pmatrix}_{b_0}$

[3333] $1^{13} 51^{32} 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 17 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0}$

[3335] $151^{13}2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 13 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3334] $41^{13}231^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 26 & 1 & 100 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 100 & 0 & 358 & 36 \\ 2 & 10 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[3336] $22^13151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 14 & 2 \\ 1 & 0 & 3 & 12 & 2 \\ 3 & 3 & 4 & 32 & 4 \\ 14 & 12 & 32 & 150 & 24 \\ 2 & 2 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[3337] $31^{14}41^{11}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -490, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 5 & 1 & 46 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 46 & 0 & 314 & 34 \\ 2 & 5 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

[3339] $1^{13}41^{23}31^{44}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -499, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 35 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 35 & 0 & 625 & 47 \\ 2 & 3 & -1 & 47 & 3 \end{pmatrix}_{b_0}$$

[3338] $321^{13}4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -499, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 8 & 21 & 3 \\ 8 & 17 & 48 & 9 \\ 21 & 48 & 124 & 24 \\ 3 & 9 & 24 & 4 \end{pmatrix}_{b_0}$$

[3341] $1^{13}3221^{13}51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 17 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[3340] $1^{13}3221^{13}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 16 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[3342] $2^1321^{25}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -502, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 26 & 3 \\ 4 & 5 & 40 & 5 \\ 26 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$$

[3343] $1^{12}51^{13}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -503, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 19 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[3344] $1^{14}41^{43}31^{24}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 40 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 40 & 0 & 750 & 52 \\ 2 & 3 & -1 & 52 & 3 \end{pmatrix}_{b_0}$$

[3345] $1^{11}3221^{13}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 14 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3346] $1^{13}3221^451^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 29 & 2 \\ 21 & 0 & 29 & 465 & 41 \\ 2 & 0 & 2 & 41 & 3 \end{pmatrix}_{b_0}$

[3347] $1^{13}3221^351^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 24 & 2 \\ 18 & 0 & 24 & 295 & 33 \\ 2 & 0 & 2 & 33 & 3 \end{pmatrix}_{b_0}$

[3348] $1^{12}41^431^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 27 & 2 \\ 1 & 1 & 1 & 42 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 42 & 0 & 970 & 58 \\ 2 & 3 & -1 & 58 & 3 \end{pmatrix}_{b_0}$

[3349] $1^{13}51^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 18 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0}$

[3350] $1^{13}41^431^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 37 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 37 & 0 & 673 & 49 \\ 2 & 3 & -1 & 49 & 3 \end{pmatrix}_{b_0}$

[3351] $1^{14}41^231^441^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 38 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 38 & 0 & 698 & 50 \\ 2 & 3 & -1 & 50 & 3 \end{pmatrix}_{b_0}$

[3352] $22^13^151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -519, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 25 & 3 \\ 6 & 14 & 65 & 8 \\ 25 & 65 & 273 & 35 \\ 3 & 8 & 35 & 4 \end{pmatrix}_{b_0}$

[3353] $1^{12}3221^451^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -519, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 26 & 2 \\ 21 & 0 & 26 & 410 & 38 \\ 2 & 0 & 2 & 38 & 3 \end{pmatrix}_{b_0}$

[3354] $1^{13}41^331^341^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 36 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 36 & 0 & 646 & 48 \\ 2 & 3 & -1 & 48 & 3 \end{pmatrix}_{b_0}$

[3355] $1^{14}41^431^141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 36 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 36 & 0 & 586 & 46 \\ 2 & 3 & -1 & 46 & 3 \end{pmatrix}_{b_0}$

[3356] $1^{14}41^131^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -523, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 33 & 0 & 517 & 43 \\ 2 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0}$

[3357] $1^{14}41^331^341^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -523, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 39 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 39 & 0 & 721 & 51 \\ 2 & 3 & -1 & 51 & 3 \end{pmatrix}_{b_0}$

[3358] $1^{13}51^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 17 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[3359] $1^{13}51^4231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 19 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[3360] $151^{13}2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 14 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3361] $1^{14}41^331^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -547, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 35 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 35 & 0 & 557 & 45 \\ 2 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[3362] $1^{14}41^231^341^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 34 & 0 & 534 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[3363] $1^{13}41^431^441^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 27 & 2 \\ 1 & 1 & 1 & 45 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 45 & 0 & 1057 & 61 \\ 2 & 3 & -1 & 61 & 3 \end{pmatrix}_{b_0}$$

[3364] $1^{12}3221^{13}51^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 16 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3365] $2^3321^{12}51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -567, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 19 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 2 & 2 & 0 & 21 & 2 \\ 19 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3366] $1^{13}51^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 18 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[3367] $31^{14}321^{15}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -575, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 23 & 3 \\ 5 & 5 & 30 & 5 \\ 23 & 30 & 139 & 25 \\ 3 & 5 & 25 & 3 \end{pmatrix}_{b_0}$$

[3368] $31^34141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -580, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 5 & 1 & 55 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 55 & 0 & 461 & 41 \\ 2 & 5 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

[3369] $1^{12}3221^{13}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 15 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3370] $1^{13}41^331^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 25 & 2 \\ 1 & 1 & 1 & 40 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 40 & 0 & 826 & 54 \\ 2 & 3 & -1 & 54 & 3 \end{pmatrix}_{b_0}$$

[3371] $231^{13}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -599, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 15 & 2 \\ 5 & 33 & 1 & 105 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 105 & 0 & 313 & 33 \\ 2 & 11 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[3372] $1^{13}5^13^{231}14 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 19 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[3373] $1^{14}3221^351^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 27 & 2 \\ 19 & 0 & 27 & 346 & 36 \\ 2 & 0 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[3374] $22^13^151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -607, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 3 & 14 & 2 \\ 3 & 3 & 4 & 34 & 4 \\ 15 & 14 & 34 & 176 & 26 \\ 2 & 2 & 4 & 26 & 3 \end{pmatrix}_{b_0}$$

[3375] $1^{13}41^331^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -607, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 25 & 2 \\ 1 & 1 & 1 & 41 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 41 & 0 & 853 & 55 \\ 2 & 3 & -1 & 55 & 3 \end{pmatrix}_{b_0}$$

[3377] $1^{14}3221^451^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 32 & 2 \\ 22 & 0 & 32 & 526 & 44 \\ 2 & 0 & 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[3376] $1^{14}41^431^341^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 25 & 2 \\ 1 & 1 & 1 & 44 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 44 & 0 & 938 & 58 \\ 2 & 3 & -1 & 58 & 3 \end{pmatrix}_{b_0}$$

[3378] $1^{14}41^331^441^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -615, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 25 & 2 \\ 1 & 1 & 1 & 43 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 43 & 0 & 909 & 57 \\ 2 & 3 & -1 & 57 & 3 \end{pmatrix}_{b_0}$$

[3379] $1^{13}5^13^{231}14 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -632, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 20 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[3380] $41^{13}231^141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 100 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 100 & 0 & 358 & 36 \\ 2 & 10 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[3381] $1^{14}41^231^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 38 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 38 & 0 & 698 & 50 \\ 2 & 3 & -1 & 50 & 3 \end{pmatrix}_{b_0}$$

- [3382] $1^{13}3221^451^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -643, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 29 & 2 \\ 22 & 0 & 29 & 465 & 41 \\ 2 & 0 & 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [3383] $1^{14}41^431^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 40 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 40 & 0 & 750 & 52 \\ 2 & 3 & -1 & 52 & 3 \end{pmatrix}_{b_0}$
- [3384] $31^{14}44141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -646, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 21 & 2 \\ 2 & 5 & 1 & 64 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 64 & 0 & 638 & 48 \\ 2 & 5 & -1 & 48 & 3 \end{pmatrix}_{b_0}$
- [3385] $2321^{14}351^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -646, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 33 & 3 \\ 4 & 5 & 50 & 5 \\ 33 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0}$
- [3386] $1^{14}41^331^341^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -655, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 39 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 39 & 0 & 721 & 51 \\ 2 & 3 & -1 & 51 & 3 \end{pmatrix}_{b_0}$
- [3387] $1^{13}41^431^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -691, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 28 & 2 \\ 1 & 1 & 1 & 45 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 1057 & 61 \\ 2 & 3 & -1 & 61 & 3 \end{pmatrix}_{b_0}$
- [3388] $1^{14}41^431^441^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -712, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 28 & 2 \\ 1 & 1 & 1 & 48 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 48 & 0 & 1150 & 64 \\ 2 & 3 & -1 & 64 & 3 \end{pmatrix}_{b_0}$
- [3389] $223151^{16} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -728, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 30 & 3 \\ 6 & 14 & 78 & 8 \\ 30 & 78 & 394 & 42 \\ 3 & 8 & 42 & 4 \end{pmatrix}_{b_0}$
- [3390] $31^{14}41^3251^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 30 & 3 \\ 5 & 5 & 40 & 5 \\ 30 & 40 & 258 & 34 \\ 3 & 5 & 34 & 3 \end{pmatrix}_{b_0}$
- [3391] $41^{14}231^141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 26 & 1 & 128 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 128 & 0 & 586 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$
- [3392] $1^{14}41^331^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -755, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 26 & 2 \\ 1 & 1 & 1 & 43 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 43 & 0 & 909 & 57 \\ 2 & 3 & -1 & 57 & 3 \end{pmatrix}_{b_0}$
- [3393] $1^{14}3221^451^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 32 & 2 \\ 23 & 0 & 32 & 526 & 44 \\ 2 & 0 & 2 & 44 & 3 \end{pmatrix}_{b_0}$
- [3394] $1^{14}41^431^341^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -760, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$\begin{aligned}
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 26 & 2 \\ 1 & 1 & 1 & 44 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 44 & 0 & 938 & 58 \\ 2 & 3 & -1 & 58 & 3 \end{pmatrix}_{b_0} \\
[3395] \quad & 321^{13}4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -774, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 8 & 24 & 3 \\ 8 & 17 & 56 & 9 \\ 24 & 56 & 166 & 28 \\ 3 & 9 & 28 & 4 \end{pmatrix}_{b_0} \\
[3396] \quad & 31^{14}321^{35}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -775, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 37 & 3 \\ 5 & 5 & 50 & 5 \\ 37 & 50 & 419 & 43 \\ 3 & 5 & 43 & 3 \end{pmatrix}_{b_0} \\
[3397] \quad & 321^{14}4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -814, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 8 & 28 & 3 \\ 8 & 17 & 64 & 9 \\ 28 & 64 & 222 & 32 \\ 3 & 9 & 32 & 4 \end{pmatrix}_{b_0} \\
[3398] \quad & 1^{14}41^431^441^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -868, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 29 & 2 \\ 1 & 1 & 1 & 48 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 48 & 0 & 1150 & 64 \\ 2 & 3 & -1 & 64 & 3 \end{pmatrix}_{b_0} \\
& 1^{17} \\
[3399] \quad & 22 \ 3 \ 151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -975, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 35 & 3 \\ 6 & 14 & 91 & 8 \\ 35 & 91 & 537 & 49 \\ 3 & 8 & 49 & 4 \end{pmatrix}_{b_0} \\
[3400] \quad & 231^{14}3151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 5 & 1 & 20 & 2 \\ 5 & 33 & 1 & 140 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 140 & 0 & 558 & 44 \\ 2 & 11 & -1 & 44 & 3 \end{pmatrix}_{b_0} \\
[3401] \quad & 41^{14}231^441^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -996, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 128 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 128 & 0 & 586 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0} \\
[3402] \quad & 321^{14}4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 8 & 31 & 3 \\ 8 & 17 & 72 & 9 \\ 31 & 72 & 276 & 36 \\ 3 & 9 & 36 & 4 \end{pmatrix}_{b_0} \\
[3403] \quad & 231^{15}3151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1479, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 5 & 1 & 25 & 2 \\ 5 & 33 & 1 & 175 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 175 & 0 & 873 & 55 \\ 2 & 11 & -1 & 55 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \\
T = 10 \quad [3405] \quad & 1^1 5 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& 1^1 \\
[3406] \quad & 1^1 31 5 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
& 1^1 \\
[3407] \quad & 1^1 4 14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[3404] \ 1^1 5^1 1321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[3408] \ 1^{13} 31^1 5^1 131^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 8 & 0 & 13 & 97 & 21 \\ 2 & 0 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[3409] \ 1^1 3151231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[3410] \ 1^{11} 3151231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3411] \ 1^1 31^1 5^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 0 & 1 & 1 & 3 \\ 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3412] \ 1^1 4131^1 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[3413] \ 1^1 5^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3414] \ 1^1 41^1 4^1 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3422] \ 1^{11} 6^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3421] \ 1^1 6^1 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -77, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3420] \ 1^1 6^1 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -77, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[3423] $1^{11}315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 4 & 2 \\ 2 & 0 & 4 & 6 & 6 \\ 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3424] $1^{11}2231321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 10, \Delta = -8, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

[3419] $216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -76, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[3415] $2161^1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[3416] $1^{11}2231321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 10, \Delta = -8, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0}$$

[3417] $1^1413151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -78, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[3418] $1^{11}4132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[3425] $1^{11}315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[3431] $161232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[3432] $232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[3433] $1^14132151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[3434] $232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3430] \quad 2^1 3 2 1^1 6 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 5 & 4 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[3429] \quad 2^1 3 2 1^1 6 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3428] \quad 1^1 4 1 3 2 1 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3427] \quad 1^1 4 1 3 2 1 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3426] \quad 2^1 3 2 1^1 6 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 7 & 7 & 7 & 42 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3436] \quad 1^1 2 3 1^1 5 1 3 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 30 & 12 \\ 2 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3437] \quad 1^1 3 3 1^1 5 1 3 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 70 & 18 \\ 2 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[3438] \quad 1^1 3 1^1 4 1 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 6 & 6 \\ 2 & 2 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[3439] \quad 1^1 4 3 1^1 5 1 3 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 126 & 24 \\ 2 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[3440] \quad 1^1 1^1 3 1^1 5 1 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 6 & 6 \\ 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[3435] \quad 1^1 1^1 3 1^1 5 1 3 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 34 & 12 \\ 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[34441] & 1^{11} 3141321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[34442] & 1^{11} 3141321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \\
[34443] & 41^{11} 31^{11} 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 5 & 1 & 4 & 2 \\ 5 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 32 & 0 & 26 & 10 \\ 2 & 12 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[34445] & 1^{12} 31^1 5^{11} 131^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 49 & 15 \\ 2 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0} \\
[34444] & 2^1 3^{11} 15131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[34446] & 1^{11} 31^1 5^{11} 131^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 17 & 9 \\ 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[34448] & 1^{11} 2^3 3^{21} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[34447] & 2^1 3^{11} 15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[34450] & 1^{11} 3151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[34449] & 1^{11} 3151^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 9 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 27 & 11 \\ 2 & 4 & -1 & 11 & 4 \end{pmatrix}_{b_0} \\
[34451] & 1^{11} 31^1 5^{12} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[3452] $1^{11}31^1 1^1 5^1 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3453] $1^{11}3151231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[3455] $2^1 3^1 2161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 5 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3454] $1^1 16123^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[3457] $1^{11}31^1 5^1 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3456] $1^{11}31^1 41321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 8 & 4 \\ 6 & 8 & 19 & 11 \\ 3 & 4 & 11 & 6 \end{pmatrix}_{b_0}$$

[3458] $1^{11}4132151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[3460] $2^1 3^1 2161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[3461] $2^1 3^1 21^1 61^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 9 & 7 & 10 & 63 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3462] $1^{12}31^1 5^1 131^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 8 & 0 & 10 & 74 & 18 \\ 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3459] $2^1 3^1 15131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3463] $41^1 31^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 32 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 32 & 0 & 38 & 12 \\ 2 & 10 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[3464] & 41^{1\bar{1}31^3}4141^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 124 & 0 & 578 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0} \\
[3465] & 1^{\bar{1}}41321^{\bar{1}}51^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 2 & 6 & 4 \\ 6 & 6 & 6 \\ 4 & 6 & 5 \end{pmatrix}_{b_0} \\
[3466] & 1^{\bar{1}}41^{\bar{1}}32151^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0} \\
[3467] & 2^{\bar{1}}3^{1\bar{1}}15131^{\bar{1}\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[3468] & 41^{\bar{1}}4141^{\bar{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[3469] & 1^{\bar{1}}41321^{\bar{1}}51^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[3472] & 1^{\bar{1}}41321^{\bar{1}}51^{\bar{1}\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 2 & 2 \\ 6 & 2 & 4 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[3471] & 2^{\bar{1}}321^{\bar{1}}61^{\bar{1}\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 6 & 6 & 28 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[3470] & 21^{\bar{1}}61^{\bar{1}}2^{\bar{1}}32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 12 & 6 & 28 & 10 \\ 2 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[3473] & 1^{\bar{1}2}3151^{\bar{1}}231^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 13 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 13 & 0 & 49 & 15 \\ 2 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0} \\
[3474] & 1^{\bar{1}2}31^{\bar{1}}5131^{\bar{1}\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[3475] & 1^{\bar{1}}6^{2\bar{1}}21^{\bar{1}}1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[3476] $2^1 3^1 2161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 31$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[3477] $2^1 3^1 2161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 32$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[3478] $1^{11} 41321^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 32$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 3 & 0 & 6 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[3479] $1^{11} 3151^1 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 33$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 27 & 11 \\ 2 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[3480] $1^{12} 31^1 5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 33$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[3481] $41^1 231^1 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 34$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 50 & 14 \\ 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[3482] $1^{12} 31^1 5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 35$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[3483] $2^1 3^1 2161^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3484] $21^{11} 61^1 2^1 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 14 & 9 & 43 & 13 \\ 2 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3485] $21^{11} 612^1 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 8 & 5 & 11 & 7 \\ 2 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[3486] $2^1 3^1 21^1 61^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 11 & 10 & 12 & 103 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3488] $1^{12}41^{13}2151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 6 & 0 & 20 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[3487] $1^{11}41321^{25}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 4 & 0 & 20 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[3489] $1^{11}1^{11}61232 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[3490] $41^{11}2321^{15}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 48 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 48 & 0 & 86 & 18 \\ 2 & 10 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[3496] $2^{12}32161^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3492] $1^{14}41321^{25}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 3 & 2 \\ 10 & 3 & 16 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[3493] $2^{11}32161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[3494] $2^{11}321^{12}61^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 6 & 9 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3495] $1^{11}41^{13}2151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 5 & 2 \\ 6 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[3491] $2^{11}321^{11}61^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3497] $1^{11}41321^{25}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 4 & 2 \\ 9 & 4 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[3498] $1^{13}315131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3499] \quad 1^{1^2} 3 1^1 5 13 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3500] \quad 1^{1^2} 3 1^1 5 13 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3501] \quad 1^{1^3} 3 1^1 5 13 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3502] \quad 1^{1^2} 3 1^1 5 13 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3503] \quad 1^{1^1} 3 1^1 5 1^{12} 23 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 27 & 11 \\ 2 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[3504] \quad 1^{1^2} 3 1^1 5 1^{12} 23 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 1 & 13 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 13 & 0 & 49 & 15 \\ 2 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[3505] \quad 2^1 3^1 21^1 6^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 5 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3506] \quad 2^1 3^1 15 13 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3507] \quad 2^1 3^1 15 13 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[3509] \quad 1^{1^1} 4 1^1 3 2 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 5 & 0 & 22 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[3508] \quad 1^{1^2} 4 13 2 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 5 & 0 & 14 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[3512] \quad 2^{\overline{12}} 3^{\overline{14}} 21^{\overline{1}} 61^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 12 & 9 & 14 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3511] \quad 1^{\overline{11}} 41321^{\overline{11}} 51^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 8 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[3510] \quad 1^{\overline{14}} 41321^{\overline{3}} 51^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 4 & 2 \\ 14 & 4 & 36 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[3513] \quad 1^{\overline{11}} 41321^{\overline{3}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 5 & 2 \\ 13 & 5 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[3514] \quad 1^{\overline{11}} 6^{\overline{11}} 12^{\overline{12}} 3^{\overline{12}} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 71, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[3515] \quad 1^{\overline{13}} 31^{\overline{1}} 5131^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 77, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[3516] \quad 2^{\overline{11}} 3^{\overline{11}} 21^{\overline{13}} 61^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 10 & 9 & 11 & 79 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3517] \quad 2^{\overline{12}} 3^{\overline{13}} 21^{\overline{11}} 61^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[3518] \quad 1^{\overline{12}} 2^{\overline{12}} 3^{\overline{11}} 61^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 38 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3519] \quad 41^{\overline{11}} 231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 81, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 9 & 3 \\ 9 & 51 & 48 & 15 \\ 9 & 48 & 42 & 14 \\ 3 & 15 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[3520] \quad 1^{\overline{13}} 31^{\overline{14}} 5131^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[3521] \quad 1^{\overline{13}} 31^{\overline{15}} 1^{\overline{1}} 231^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$\begin{aligned}
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 1 & 17 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 17 & 0 & 79 & 19 \\ 2 & 4 & -1 & 19 & 4 \end{pmatrix}_{b_0} \\
[3522] & 1^{12}3151^1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 13 & 3 \\ 7 & 0 & 13 & 49 & 15 \\ 2 & 0 & 3 & 15 & 4 \end{pmatrix}_{b_0} \\
[3527] & 1^{11}41^132151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 18 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0} \\
[3523] & 41^{11}314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[3524] & 41^{11}231^151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 8 & 3 \\ 9 & 51 & 48 & 15 \\ 8 & 48 & 42 & 14 \\ 3 & 15 & 14 & 4 \end{pmatrix}_{b_0} \\
[3525] & 1^{11}41^1321^251^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 6 & 0 & 42 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0} \\
[3526] & 41^{11}232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[3528] & 1^{13}31^15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0} \\
[3529] & 1^141321^251^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 4 & 2 \\ 12 & 4 & 22 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[3530] & 1^{13}31^15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0} \\
[3532] & 1^{13}41321^151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 7 & 0 & 26 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[3531] & 2^132161^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[3534] & 2^1321^161^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 95, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 8 & 13 & 86 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[3533] \quad 2 \overset{1^{12}}{3} \overset{1^2}{21} 6 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 95, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3535] \quad 41^{\overset{1}{11}} 31 \overset{1}{5} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 10 & 6 \\ 2 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[3536] \quad 1^{\overset{1}{11}} 41321^{\overset{1}{51}} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 99, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 14 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[3537] \quad 1^{\overset{1}{11}} \overset{2}{3} \overset{1}{21} 6 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3538] \quad 41^{\overset{1}{11}} 314141^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[3539] \quad 2 \overset{1^{12}}{3} \overset{1^2}{21} 6 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 6 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[3542] \quad 2 \overset{1^{11}}{3} \overset{1^2}{21} 6 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 9 & 8 & 10 & 59 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3541] \quad 1^{\overset{1}{12}} 41321^{\overset{2}{51}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 6 & 0 & 30 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3540] \quad 31^{\overset{1}{11}} 41^{\overset{1}{13}} 151^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 8 & 2 \\ 3 & 4 & 1 & 12 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 12 & 0 & 26 & 10 \\ 2 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[3543] \quad 51^{\overset{1}{11}} 2231^{\overset{1}{41}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 11 & 3 \\ 7 & 7 & 14 & 7 \\ 11 & 14 & 24 & 12 \\ 3 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3544] \quad 1^{\overset{1}{11}} 41321^{\overset{2}{51}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 107, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 5 & 2 \\ 11 & 5 & 22 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3546] $2^1 3^1 21^1 61^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 7 & 12 & 66 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3545] $1^{11} 41^1 321^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 24 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[3548] $1^{13} 31^1 5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[3547] $1^{13} 31^1 5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[3549] $2^1 3^1 21^1 61^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 7 & 9 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[3550] $1^{12} 41^1 321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 7 & 0 & 34 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$

[3551] $1^{12} 3151^2 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 \\ 10 & 0 & 18 & 114 & 22 \\ 2 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$

[3552] $1^{13} 31^1 5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[3553] $1^{14} 31^1 5131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[3554] $1^{13} 3151^2 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 1 & 22 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 22 & 0 & 154 & 26 \\ 2 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$

[3555] $1^{13} 3151^1 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 1 & 17 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 17 & 0 & 79 & 19 \\ 2 & 4 & -1 & 19 & 4 \end{pmatrix}_{b_0}$

[3556] $1^{12} 3151^1 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 13 & 3 \\ 8 & 0 & 13 & 49 & 15 \\ 2 & 0 & 3 & 15 & 4 \end{pmatrix}_{b_0} \\
[3557] & 51^{1\bar{1}223141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 119, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 6 & 3 \\ 7 & 7 & 7 & 7 \\ 6 & 7 & 4 & 6 \\ 3 & 7 & 6 & 4 \end{pmatrix}_{b_0} \\
[3558] & 1^{14}41321^{151^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 119, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 9 & 0 & 42 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0} \\
[3559] & 1^{11}41^{132151^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 119, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[3560] & 41^{1\bar{1}31^24141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 26 & 1 & 92 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 92 & 0 & 314 & 34 \\ 2 & 10 & -1 & 34 & 3 \end{pmatrix}_{b_0} \\
[3561] & 31^{141^{11}3151^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 5 & 1 & 8 & 2 \\ 5 & 8 & 1 & 16 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 16 & 0 & 26 & 10 \\ 2 & 6 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[3563] & 1^{14}31^5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3564] & 2^3 3^{13}15131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3562] & 1^{14}31^5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3565] & 41^4 4^{11}141^{11}3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 4 & 10 & 4 \\ 6 & 6 & 10 & 14 & 8 \\ 2 & 3 & 4 & 8 & 3 \end{pmatrix}_{b_0} \\
[3566] & 1^{12}41321^{351^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 7 & 0 & 54 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0} \\
[3567] & 1^4 141321^{351^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 5 & 2 \\ 16 & 5 & 42 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[3568] & 41^{131^{11}4141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 1 & 9 & 2 \\ 5 & 38 & 1 & 72 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 72 & 0 & 130 & 22 \\ 2 & 12 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[3569] $51^{11}223141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 9 & 3 \\ 7 & 7 & 12 & 7 \\ 9 & 12 & 16 & 10 \\ 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[3570] $1^{11}41321^{15}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 6 & 2 \\ 11 & 6 & 24 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3571] $2^13^{13}15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[3572] $4131^{11}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 40 & 0 & 38 & 12 \\ 2 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[3573] $41^{11}31^{11}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 9 & 2 \\ 4 & 26 & 1 & 60 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 60 & 0 & 130 & 22 \\ 2 & 10 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[3574] $41^{12}23151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 18 & 3 \\ 9 & 51 & 96 & 15 \\ 18 & 96 & 174 & 28 \\ 3 & 15 & 28 & 4 \end{pmatrix}_{b_0}$$

[3575] $1^{11}41321^{35}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 6 & 2 \\ 15 & 6 & 44 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[3576] $1^{13}3151^{1231}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 17 & 3 \\ 9 & 0 & 17 & 79 & 19 \\ 2 & 0 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[3577] $1^{14}3151^{1231}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 1 & 21 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 21 & 0 & 117 & 23 \\ 2 & 4 & -1 & 23 & 4 \end{pmatrix}_{b_0}$$

[3578] $4131^{11}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 6 & 2 \\ 5 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 40 & 0 & 38 & 12 \\ 2 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[3579] $1^{14}315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 155, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3580] $41^{12}232151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[3581] $41^{12}231\overset{1}{5}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 16 & 3 \\ 9 & 51 & 96 & 15 \\ 16 & 96 & 174 & 28 \\ 3 & 15 & 28 & 4 \end{pmatrix}_{b_0}$$

[3584] $2\overset{12}{3}21\overset{14}{6}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 13 & 11 & 15 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3585] $1^{12}3151^2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 \\ 11 & 0 & 18 & 114 & 22 \\ 2 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[3586] $1^{13}41^1321^151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 9 & 0 & 50 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[3582] $3141^{11}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 8 & 1 & 10 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 10 & 0 & 8 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[3583] $1^{14}3151^2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 2 & 1 & 26 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 26 & 0 & 202 & 30 \\ 2 & 4 & -1 & 30 & 4 \end{pmatrix}_{b_0}$$

[3588] $41^{11}31^34141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 124 & 0 & 578 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[3587] $1^{11}41321^251^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 6 & 2 \\ 13 & 6 & 28 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[3589] $1^{14}31\overset{13}{5}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3594] $1^{12}41^1321^251^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 13 & 8 & 0 & 56 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[3593] $1^{14}31\overset{14}{5}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3592] $1^{12}2\overset{11}{3}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3591] $1^{13}41321^251^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 8 & 0 & 44 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[3590] $1^{11}41^1321^251^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 7 & 2 \\ 14 & 7 & 44 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[3595] $41^231^{11}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 5 & 1 & 14 & 2 \\ 5 & 38 & 1 & 104 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 104 & 0 & 278 & 32 \\ 2 & 12 & -1 & 32 & 3 \end{pmatrix}_{b_0}$

[3597] $1^141321^351^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 6 & 2 \\ 18 & 6 & 52 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[3596] $2^13^{12}21^161^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 12 & 10 & 14 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3598] $2^13^{12}21^161^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 8 & 12 & 64 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3599] $1^{13}431^15131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[3600] $1^{13}3151^1231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 17 & 3 \\ 10 & 0 & 17 & 79 & 19 \\ 2 & 0 & 3 & 19 & 4 \end{pmatrix}_{b_0}$

[3602] $1^{12}3151^2231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 \\ 12 & 0 & 18 & 114 & 22 \\ 2 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$

[3601] $1^{14}31^15131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[3603] $2^13^{12}21^161^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 199, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 9 & 13 & 82 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3604] $41^{13}31^14141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 26 & 1 & 60 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 60 & 0 & 130 & 22 \\ 2 & 10 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

- [3605] $41^{12}23151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 17 & 3 \\ 9 & 51 & 96 & 15 \\ 17 & 96 & 174 & 28 \\ 3 & 15 & 28 & 4 \end{pmatrix}_{b_0}$
- [3606] $1^{14}41^{13}21^{15}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 11 & 0 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$
- [3607] $1^{12}41^{13}21^{35}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 17 & 9 & 0 & 86 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [3608] $2315131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$
- [3609] $1^{14}3151^{23}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 2 & 1 & 26 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 26 & 0 & 202 & 30 \\ 2 & 4 & -1 & 30 & 4 \end{pmatrix}_{b_0}$
- [3610] $1^{13}3151^{23}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 22 & 3 \\ 12 & 0 & 22 & 154 & 26 \\ 2 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0}$
- [3611] $2315131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [3612] $1^{11}41321^{25}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 219, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 7 & 2 \\ 15 & 7 & 38 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$
- [3613] $41^{11}31^{24}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 26 & 1 & 92 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 92 & 0 & 314 & 34 \\ 2 & 10 & -1 & 34 & 3 \end{pmatrix}_{b_0}$
- [3614] $1^{11}41^{13}21^{35}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 8 & 2 \\ 18 & 8 & 72 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [3615] $1^{13}41321^{35}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 9 & 0 & 70 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$
- [3616] $51^{12}223141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 12 & 3 \\ 7 & 7 & 14 & 7 \\ 12 & 14 & 22 & 12 \\ 3 & 7 & 12 & 4 \end{pmatrix}_{b_0}$
- [3617] $1^{11}41321^{35}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 7 & 2 \\ 17 & 7 & 50 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3618] & 1^{14}41321^251^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 10 & 0 & 62 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0} \\
[3619] & 2^1315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0} \\
[3620] & 41^{13}1^{11}4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 5 & 1 & 10 & 2 \\ 5 & 38 & 1 & 72 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 72 & 0 & 130 & 22 \\ 2 & 12 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[3621] & 1^{11}41^{13}21^251^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 8 & 2 \\ 16 & 8 & 50 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3622] & 1^{13}41^{13}21^251^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 10 & 0 & 74 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[3623] & 1^{13}3151^3231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 \\ 15 & 0 & 27 & 259 & 33 \\ 2 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0} \\
[3624] & 41^{13}231^151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 27 & 3 \\ 9 & 51 & 144 & 15 \\ 27 & 144 & 394 & 42 \\ 3 & 15 & 42 & 4 \end{pmatrix}_{b_0} \\
[3625] & 41^{12}2321^151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 260, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 76 & 0 & 210 & 28 \\ 2 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[3626] & 1^{13}3151^2231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 263, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 22 & 3 \\ 13 & 0 & 22 & 154 & 26 \\ 2 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0} \\
[3628] & 41^{13}232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0} \\
[3627] & 41^{13}231^151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 24 & 3 \\ 9 & 51 & 144 & 15 \\ 24 & 144 & 394 & 42 \\ 3 & 15 & 42 & 4 \end{pmatrix}_{b_0} \\
[3629] & 1^{11}41321^351^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 8 & 2 \\ 19 & 8 & 60 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[3630] $1^{14}3151^2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 26 & 3 \\ 14 & 0 & 26 & 202 & 30 \\ 2 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$

[3631] $1^{13}3151^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 318, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 \\ 16 & 0 & 27 & 259 & 33 \\ 2 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$

[3633] $1^{14}41321^351^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 319, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 11 & 0 & 90 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[3632] $1^{13}41^1321^351^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 319, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 18 & 11 & 0 & 106 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[3635] $1^{11}41^1321^351^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 0 & 9 & 2 \\ 20 & 9 & 78 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$

[3634] $1^{14}41^1321^251^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 12 & 0 & 96 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[3636] $51^{12}2231^141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 17 & 3 \\ 7 & 7 & 21 & 7 \\ 17 & 21 & 52 & 18 \\ 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$

[3638] $41^{13}231^51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 26 & 3 \\ 9 & 51 & 144 & 15 \\ 26 & 144 & 394 & 42 \\ 3 & 15 & 42 & 4 \end{pmatrix}_{b_0}$

[3637] $51^{12}2231^141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 20 & 3 \\ 7 & 7 & 26 & 7 \\ 20 & 26 & 82 & 22 \\ 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$

[3639] $41^{13}231^51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 369, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 25 & 3 \\ 9 & 51 & 144 & 15 \\ 25 & 144 & 394 & 42 \\ 3 & 15 & 42 & 4 \end{pmatrix}_{b_0}$

[3640] $51^{12}223141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 374, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 15 & 3 \\ 7 & 7 & 19 & 7 \\ 15 & 19 & 40 & 16 \\ 3 & 7 & 16 & 4 \end{pmatrix}_{b_0}$

[3641] $1^{14}3151^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 383, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 31 & 3 \\ 17 & 0 & 31 & 317 & 37 \\ 2 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$

[3642] $41^{14}231^51^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 396, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 9 & 36 & 3 \\ 9 & 51 & 192 & 15 \\ 36 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0} \\
[3643] & 51^{1^3 223141^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 399, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 18 & 3 \\ 7 & 7 & 21 & 7 \\ 18 & 21 & 52 & 18 \\ 3 & 7 & 18 & 4 \end{pmatrix}_{b_0} \\
[3644] & 41^{1^3 2321^2 51^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 404, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 124 & 0 & 570 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0} \\
[3645] & 1^{11} 41^1 321^3 51^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 428, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 0 & 10 & 2 \\ 22 & 10 & 88 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[3646] & 1^{14} 41^1 321^3 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 19 & 13 & 0 & 130 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0} \\
[3648] & 41^{1^4 232151^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0} \\
[3647] & 41^{1^4 23151^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 32 & 3 \\ 9 & 51 & 192 & 15 \\ 32 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0} \\
[3649] & 41^{1^4 2321^3 51^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 28 & 2 \\ 4 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 172 & 0 & 1106 & 64 \\ 2 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0} \\
[3650] & 41^{1^3 2321^1 51^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 524, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 104 & 0 & 390 & 38 \\ 2 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0} \\
[3651] & 41^{1^4 23151^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 561, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 35 & 3 \\ 9 & 51 & 192 & 15 \\ 35 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0} \\
[3652] & 41^{1^4 23151^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 585, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 33 & 3 \\ 9 & 51 & 192 & 15 \\ 33 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0} \\
[3653] & 41^{1^4 23151^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 624, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 34 & 3 \\ 9 & 51 & 192 & 15 \\ 34 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0} \\
[3654] & 51^{1^3 2231^1 41^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 665, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 23 & 3 \\ 7 & 7 & 28 & 7 \\ 23 & 28 & 92 & 24 \\ 3 & 7 & 24 & 4 \end{pmatrix}_{b_0} \\
[3655] & 51^{1^3 223141^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 689, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 21 & 3 \\ 7 & 7 & 26 & 7 \\ 21 & 26 & 76 & 22 \\ 3 & 7 & 22 & 4 \end{pmatrix}_{b_0} \\
[3656] & 51^{13} 2231^{14} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 815, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 26 & 3 \\ 7 & 7 & 33 & 7 \\ 26 & 33 & 128 & 28 \\ 3 & 7 & 28 & 4 \end{pmatrix}_{b_0} \\
[3657] & 41^{14} 2321^{15} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 836, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 132 & 0 & 626 & 48 \\ 2 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0} \\
[3658] & 41^{14} 2321^{25} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 860, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 24 & 2 \\ 4 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 152 & 0 & 846 & 56 \\ 2 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0} \\
\\
T = 11 & [3661] 1^{13} 1513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 4.0 \\
[3659] & 1^{12} 231513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 11, 2), c_{\text{ext}} = 4.0 \\
[3660] & 1^{12} 23151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 4.0 \\
[3669] & 1^{13} 151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 5 & 3 \\ 2 & 0 & 3 & 7 & 5 \end{pmatrix}_{b_0} \\
[3670] & 1^{43} 151321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 16 & 3 \\ 8 & 0 & 16 & 106 & 24 \\ 2 & 0 & 3 & 24 & 5 \end{pmatrix}_{b_0} \\
[3663] & 1^{16} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[3664] & 1^{11} 6^{1321} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0} \\
[3665] & 1^{11} 6^{1321} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \\
[3666] & 1^{14} 14141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 5 & 6 \\ 3 & 5 & 3 & 5 \\ 3 & 6 & 5 & 6 \end{pmatrix}_{b_0} \\
[3667] & 1^{16} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
[3668] & 1^1 4 14 1^1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 5 & 6 \\ 2 & 5 & 3 & 5 \\ 3 & 6 & 5 & 6 \end{pmatrix}_{b_0} \\
[3673] & 1^1 1^1 3 1^1 5 13 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & 1 & 2 & 4 \\ 2 & 4 & 2 & 4 & 5 \end{pmatrix}_{b_0} \\
[3674] & 1^1 1^1 5 13 2 15 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -74, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0} \\
[3671] & 1^1 1^1 5 13 1^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0} \\
[3672] & 1^1 1^1 5 13 1^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[3682] & 1^1 1^1 7 12 23 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -75, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[3676] & 1^1 1^1 23 14 13 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -7, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[3675] & 1^1 1^1 6 12 3 14 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -76, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
[3683] & 1^1 1^1 5 13 1^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -73, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[3677] & 1^1 1^1 5 13 2 15 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -74, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[3678] & 1^1 1^1 3 14 13 2 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -6, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[3679] & 3 1^1 3 1^1 6 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 2 & 4 & 6 & 6 \\ 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[3680] \quad 21^1 71231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -75, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3681] \quad 1^{11} 31413221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 11, \Delta = -6, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[3686] \quad 2^1 3151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[3685] \quad 1^1 3151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 7 & 7 \\ 2 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[3684] \quad 2^1 3151321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[3687] \quad 1^2 3151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 10 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 10 & 0 & 34 & 14 \\ 2 & 4 & -1 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[3688] \quad 1^3 3151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 1 & 15 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 79 & 21 \\ 2 & 4 & -1 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[3689] \quad 1^4 3151321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 142 & 28 \\ 2 & 4 & -1 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[3690] \quad 1^2 3151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 18 & 10 \\ 2 & 0 & 3 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[3691] \quad 1^4 3151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 78 & 20 \\ 2 & 0 & 3 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[3692] \quad 2^1 3151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[3693] \quad 2 \overset{12}{3} 151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[3694] \quad 31 \overset{1}{5} 13151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3695] \quad 1 \overset{11}{3} 14141^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 5 & 0 & 6 & 6 \\ 2 & 4 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[3696] \quad 2 \overset{1}{3} 21^1 6131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 9 & 3 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 9 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3697] \quad 1 \overset{11}{3} 31^1 5 \overset{12}{13} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 11 & 3 \\ 6 & 0 & 11 & 53 & 17 \\ 2 & 0 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[3705] \quad 1 \overset{12}{11} 31^1 5 \overset{11}{13} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 4 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[3706] \quad 1 \overset{11}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[3698] \quad 1 \overset{11}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 5 & 4 & 7 & 7 \\ 2 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[3699] \quad 1 \overset{11}{2} 3151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[3700] \quad 1 \overset{11}{2} 3215131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[3701] \quad 1 \overset{11}{2} 3151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 5 & 4 \\ 2 & 0 & 5 & 3 & 5 \\ 2 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[3702] & 1^{11}23215131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 5 & 4 \\ 2 & 0 & 5 & 3 & 5 \\ 2 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[3703] & 1^{11}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 5 & 4 & 7 & 7 \\ 2 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0} \\
[3704] & 1^{11}31414131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 4 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 5 \end{pmatrix}_{b_0} \\
[3707] & 2^1 3^1 216131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[3709] & 2^1 3^1 216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[3708] & 1^{11}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0} \\
[3710] & 316131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -63, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 2 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[3712] & 1^{12}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 5 & 7 & 6 & 18 & 10 \\ 2 & 3 & 3 & 10 & 5 \end{pmatrix}_{b_0} \\
[3711] & 1^{11}31^1 414131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 9 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 9 & 1 & 23 & 11 \\ 2 & 4 & 0 & 11 & 5 \end{pmatrix}_{b_0} \\
[3714] & 1^{12}23151^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 3 & 1 & 15 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 15 & 0 & 53 & 17 \\ 2 & 5 & -1 & 17 & 5 \end{pmatrix}_{b_0} \\
[3715] & 1^{12}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 7 & 6 & 18 & 10 \\ 2 & 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[3713] $1^{11}23151^{1231^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 10 & 4 \\ 5 & 0 & 10 & 28 & 12 \\ 2 & 0 & 4 & 12 & 5 \end{pmatrix}_{b_0}$

[3716] $2^13^1216131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[3718] $1^{11}31^151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 9 & 5 & 21 & 11 \\ 2 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$

[3717] $1^{11}31^151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 9 & 5 & 21 & 11 \\ 2 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$

[3719] $2^13^1216131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[3720] $31^15^113151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$

[3721] $1^{11}2321^15131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 12 & 4 \\ 6 & 0 & 12 & 38 & 14 \\ 2 & 0 & 4 & 14 & 5 \end{pmatrix}_{b_0}$

[3726] $1^{12}31^151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 5 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$

[3722] $1^{12}31^151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 7 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$

[3723] $1^{12}2321^15131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 17 & 0 & 67 & 19 \\ 2 & 5 & -1 & 19 & 5 \end{pmatrix}_{b_0}$

[3724] $1^{11}31^151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 13 & 4 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 13 & 6 & 43 & 15 \\ 2 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$

[3725] $1^{11}31^151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 13 & 4 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 13 & 6 & 43 & 15 \\ 2 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[3727] $2^1 3^1 21^1 6131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -35$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 2 & 2 & 0 & 11 & 2 \\ 10 & 16 & 11 & 101 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3728] $1^{12} 31^1 51321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -36$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[3729] $31^1 513151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -46$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

[3730] $2^1 3^1 21^1 6131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -48$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 9 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 9 & 10 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[3732] $1^{12} 31^1 4141^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -33$, $\det = -50$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 1 & 14 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 14 & 1 & 55 & 17 \\ 2 & 4 & 0 & 17 & 5 \end{pmatrix}_{b_0}$$

[3731] $1^{13} 31^1 51321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -50$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 6 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

[3734] $1^{13} 31^1 51321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -51$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3733] $1^{12} 31^1 51321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -51$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3737] $1^{12} 23151^{12} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -31$, $\det = -54$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 15 & 4 \\ 7 & 0 & 15 & 53 & 17 \\ 2 & 0 & 4 & 17 & 5 \end{pmatrix}_{b_0}$$

[3735] $1^{13} 23151^{12} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -31$, $\det = -54$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 1 & 20 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 20 & 0 & 88 & 22 \\ 2 & 5 & -1 & 22 & 5 \end{pmatrix}_{b_0}$$

[3736] $1^{13} 31^1 51321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -54$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 8 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

[3739] $1^{1^2}3151321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -55, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3738] $1^{1^2}3151321^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -55, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

[3742] $2^1321^{1^1}6131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 8 & 7 & 32 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3740] $2^{1^2}3216131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[3741] $2^{1^2}3216131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3743] $1^{1^3}2321^{1^5}5131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 1 & 22 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 22 & 0 & 106 & 24 \\ 2 & 5 & -1 & 24 & 5 \end{pmatrix}_{b_0}$$

[3745] $1^{1^4}3151321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 10 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[3746] $1^{1^2}2321^{1^5}5131^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 17 & 4 \\ 8 & 0 & 17 & 67 & 19 \\ 2 & 0 & 4 & 19 & 5 \end{pmatrix}_{b_0}$$

[3744] $1^{1^2}3151321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 8 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[3750] $1^{1^2}23151321^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[3749] $1^{1^3}3151321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[3747] $1^{1^4}23151321^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[3748] \quad 1^{13} 31 \bar{5} 13151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[3751] \quad 1^{12} 31 \bar{5} 1321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[3752] \quad 1^{13} 31 \bar{5} 1321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 19 & 13 & 113 & 25 \\ 2 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[3753] \quad 1^{13} 31 \bar{5} 1321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 19 & 13 & 113 & 25 \\ 2 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[3754] \quad 1^{13} 23151^{12} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 20 & 4 \\ 9 & 0 & 20 & 88 & 22 \\ 2 & 0 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[3755] \quad 1^{14} 23151^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 3 & 1 & 25 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 25 & 0 & 133 & 27 \\ 2 & 5 & -1 & 27 & 5 \end{pmatrix}_{b_0}$$

$$[3756] \quad 1^{14} 31 \bar{5} 1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 9 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[3757] \quad 1^{14} 31 \bar{5} 1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -106, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 11 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[3758] \quad 1^{14} 31 \bar{5} 1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[3759] \quad 2 \bar{3} 21^{12} 6131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 11 & 8 & 51 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[3760] \quad 2 \bar{3} 21^{12} 6131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 13 & 3 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 13 & 14 & 95 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3761] $2^1 3^{13} 216131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3762] $1^{14} 31^1 51321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -129, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 11 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[3763] $1^{13} 23151^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -129, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 25 & 4 \\ 12 & 0 & 25 & 163 & 29 \\ 2 & 0 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

[3764] $1^{14} 31^1 51321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -132, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[3765] $2^1 3^{13} 216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -136, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[3766] $2^1 3^{12} 21^1 6131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 12 & 11 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3767] $1^{14} 23151^2 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 30 & 4 \\ 14 & 0 & 30 & 218 & 34 \\ 2 & 0 & 4 & 34 & 5 \end{pmatrix}_{b_0}$$

[3769] $2^1 3^{14} 216131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[3768] $2^1 3^{14} 216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3770] $2^1 3^{13} 21^1 6131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -220, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 1 & 2 & 17 & 3 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 17 & 18 & 157 & 27 \\ 2 & 3 & 2 & 27 & 4 \end{pmatrix}_{b_0}$$

[3771] $2^1 3^{14} 216131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3772] $2^1 3^{13} 21^1 6131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -248, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 15 & 12 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3773] $2^1 3^{13} 21^1 6131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -262, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 16 & 15 & 124 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[3774] $2^1 3^{14} 21^1 6131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -342, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 1 & 2 & 18 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 14 & 18 & 13 & 130 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[3775] $2^1 3^{14} 21^1 6131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -398, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 1 & 2 & 20 & 3 \\ 1 & 2 & 0 & 19 & 2 \\ 16 & 20 & 19 & 194 & 30 \\ 2 & 3 & 2 & 30 & 4 \end{pmatrix}_{b_0}$$

[3776] $2^1 3^{14} 21^1 6131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -398, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 1 & 2 & 19 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 15 & 19 & 16 & 159 & 27 \\ 2 & 3 & 2 & 27 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [3777] $1^{14} 231^1 5^1 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[3778] $1^{12} 231^1 5^1 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.5$

[3779] $1^{11} 231^1 5^1 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.0$

[3780] $1^{14} 231^1 5^1 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[3795] $1^{11} 1^{13} 2321^1 6^1 131^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 122 & 28 \\ 3 & 3 & 28 & 6 \end{pmatrix}_{b_0}$$

[3796] $1^{11} 7^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

[3785] $1^{11} 6^{131} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & -1 & 3 & 5 \end{pmatrix}_{b_0}$$

[3786] $1^{11} 31^1 6^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

[3787] $1^{\overline{1}131} 6^{11} 13141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[3788] $1^{\overline{1}131} 6^{11} 131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 8 \\ 2 & 3 & 2 & 8 & 5 \end{pmatrix}_{b_0}$

[3789] $21^{\overline{1}1} 712321^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[3790] $1^{\overline{1}231} 6^{11} 131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 26 & 12 \\ 2 & 3 & 2 & 12 & 5 \end{pmatrix}_{b_0}$

[3791] $1^{\overline{1}1} 6^{11} 131^{\overline{1}2321} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 7 & 7 \\ 3 & 3 & 7 & 6 \end{pmatrix}_{b_0}$

[3792] $1^{\overline{1}1} 51315131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[3793] $1^{\overline{1}1} 6^{11} 131^{\overline{1}32321} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 110 & 26 \\ 3 & 3 & 26 & 6 \end{pmatrix}_{b_0}$

[3794] $1^{\overline{1}1} 6^{11} 131^{\overline{1}2321} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 6 & 3 \\ 8 & 6 & 30 & 14 \\ 3 & 3 & 14 & 6 \end{pmatrix}_{b_0}$

[3804] $1^{\overline{1}1} 612231^{\overline{1}1} 51^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[3805] $1^{\overline{1}1} 613151^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$

[3797] $21^{\overline{1}1} 613151^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$

[3798] $1^{\overline{1}1} 6^{11} 131^{\overline{1}2321} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 2 & 1 & 2 & 4 \\ 2 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[3799] \quad 1^{11} 6 131 5 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[3800] \quad 1^{11} 6 131 5 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[3801] \quad 1^{11} 6 131 5 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[3802] \quad 1^{11} 6 1231 5 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[3803] \quad 1^{11} 6 1231 5 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3827] \quad 1^{11} 314141321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$$[3806] \quad 1^{11} 231 5 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0}$$

$$[3807] \quad 1^{11} 5 131 6 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3808] \quad 221 8 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3809] \quad 1^{11} 5 1321 6 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3810] \quad 1^{11} 6 1314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
[3811] & 21\overset{1}{8}12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[3812] & 1\overset{1}{7}1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
[3813] & 1\overset{1}{6}1232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -73, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[3814] & 1\overset{1}{5}^{11}2231^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 2 & 6 & 5 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 6 & 4 & 8 & 8 \\ 2 & 5 & 2 & 8 & 6 \end{pmatrix}_{b_0} \\
[3815] & 1\overset{1}{1}223151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0} \\
[3816] & 1\overset{1}{4}31513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 24 & 0 & 158 & 32 \\ 2 & 5 & -1 & 32 & 6 \end{pmatrix}_{b_0} \\
[3817] & 1\overset{1}{3}31513221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 1 & 18 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 18 & 0 & 88 & 24 \\ 2 & 5 & -1 & 24 & 6 \end{pmatrix}_{b_0} \\
[3818] & 1\overset{1}{2}31513221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 3 & 1 & 12 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 12 & 0 & 38 & 16 \\ 2 & 5 & -1 & 16 & 6 \end{pmatrix}_{b_0} \\
[3819] & 1\overset{1}{1}31513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 6 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 8 & 8 \\ 2 & 5 & -1 & 8 & 6 \end{pmatrix}_{b_0} \\
[3820] & 1\overset{1}{1}31513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 4 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0} \\
[3821] & 1\overset{1}{5}^{11}2231^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 2 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0} \\
[3822] \quad & 1^{11} 231512321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \\
[3823] \quad & 1^{11} 223151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \\
[3824] \quad & 1^{11} 231512321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0} \\
[3825] \quad & 1^{11} 314141321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[3826] \quad & 1^{11} 231 \quad 5 \quad 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0} \\
[3828] \quad & 1^{31} 5 \quad 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 13 & 4 \\ 6 & 0 & 13 & 57 & 19 \\ 2 & 0 & 4 & 19 & 6 \end{pmatrix}_{b_0} \\
[3830] \quad & 1^{112} 2321^1 \quad 6 \quad 131^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 9 & 3 \\ 12 & 9 & 67 & 21 \\ 3 & 3 & 21 & 6 \end{pmatrix}_{b_0} \\
[3831] \quad & 1^{11} 31513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0} \\
[3829] \quad & 1^{11} 2321^1 \quad 6 \quad 131^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 6 & 3 \\ 6 & 6 & 22 & 12 \\ 3 & 3 & 12 & 6 \end{pmatrix}_{b_0} \\
[3834] \quad & 321^1 \quad 6 \quad 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 13 & 7 \\ 2 & 3 & 0 & 7 & 3 \end{pmatrix}_{b_0} \\
[3833] \quad & 231^1 \quad 5 \quad 13161^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 2 & 4 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3832] $1^{11} 31 \bar{5} 132161^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[3835] $1^{11} 6^{12} 131^{2321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 9 & 3 \\ 10 & 9 & 57 & 19 \\ 3 & 3 & 19 & 6 \end{pmatrix}_{b_0}$

[3836] $1^{11} 41^{13} 151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 2 & 3 & 7 & 7 \\ 2 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$

[3837] $1^{11} 321612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$

[3838] $1^{11} 321612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$

[3839] $1^{12} 231 \bar{5} 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 9 & 6 & 22 & 12 \\ 2 & 4 & 3 & 12 & 6 \end{pmatrix}_{b_0}$

[3840] $1^{11} 413151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$

[3841] $1^{11} 31 \bar{5} 13221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 3 & 2 & 11 & 5 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 11 & 5 & 25 & 13 \\ 2 & 5 & 2 & 13 & 6 \end{pmatrix}_{b_0}$

[3842] $1^{11} 413151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$

[3843] $1^{11} 2321^{11} 6^{13} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$

[3844] $1^2 6^{11} 2321^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 7 & 4 \\ 6 & 7 & 19 & 11 \\ 3 & 4 & 11 & 6 \end{pmatrix}_{b_0}$

[3845] $1^{12} 231 \bar{5} 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 5 & 9 & 6 & 22 & 12 \\ 2 & 4 & 3 & 12 & 6 \end{pmatrix}_{b_0}$

[3846] $1^{\overline{11}2321} \begin{smallmatrix} 1 & 6 & 131 \end{smallmatrix}^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 4 & 7 & 7 \\ 2 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$

[3847] $1^{\overline{12}223151} \begin{smallmatrix} 1 & 231 \end{smallmatrix}^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 4 & 1 & 17 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 17 & 0 & 57 & 19 \\ 2 & 6 & -1 & 19 & 6 \end{pmatrix}_{b_0}$

[3848] $1^{\overline{11}2321} \begin{smallmatrix} 1 & 6 & 131 \end{smallmatrix}^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$

[3849] $1^{\overline{11}31513221} \begin{smallmatrix} 1 & 2 \end{smallmatrix}^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 3 & 2 & 11 & 5 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 11 & 5 & 25 & 13 \\ 2 & 5 & 2 & 13 & 6 \end{pmatrix}_{b_0}$

[3850] $1^{\overline{11}2321} \begin{smallmatrix} 1 & 6 & 131 \end{smallmatrix}^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 2 & 6 & 4 \\ 5 & 6 & 11 & 9 \\ 3 & 4 & 9 & 6 \end{pmatrix}_{b_0}$

[3851] $1^{\overline{11}2321} \begin{smallmatrix} 1 & 6 & 131 \end{smallmatrix}^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 106 & 26 \\ 3 & 3 & 26 & 6 \end{pmatrix}_{b_0}$

[3852] $1^{\overline{12}31513221} \begin{smallmatrix} 1 & 2 \end{smallmatrix}^{\overline{3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 7 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$

[3853] $1^{\overline{12}31513221} \begin{smallmatrix} 1 & 2 \end{smallmatrix}^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 5 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$

[3854] $1^{\overline{11}31513221} \begin{smallmatrix} 1 & 2 \end{smallmatrix}^{\overline{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 2 & 16 & 5 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 16 & 6 & 52 & 18 \\ 2 & 5 & 2 & 18 & 6 \end{pmatrix}_{b_0}$

[3855] $1^{\overline{11}31513221} \begin{smallmatrix} 1 & 2 \end{smallmatrix}^{\overline{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 2 & 16 & 5 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 16 & 6 & 52 & 18 \\ 2 & 5 & 2 & 18 & 6 \end{pmatrix}_{b_0}$

[3857] $1^{\overline{12}31513221} \begin{smallmatrix} 1 & 2 \end{smallmatrix}^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 6 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$

[3858] $1^{\overline{11}41^13151321} \begin{smallmatrix} 1 & 1 \end{smallmatrix}^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 5 & 4 & 3 & 13 & 9 \\ 2 & 2 & 2 & 9 & 5 \end{pmatrix}_{b_0} \\
[3856] & 1^2 \overset{11}{6}^{2321} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 10 & 4 \\ 7 & 10 & 32 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0} \\
[3860] & 1^1 \overset{11}{6}^{2321} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 9 & 4 \\ 6 & 9 & 22 & 12 \\ 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \\
[3861] & 1^1 321^1 612321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 4 & 4 & 18 & 10 \\ 2 & 2 & 2 & 10 & 5 \end{pmatrix}_{b_0} \\
[3859] & 1^1 321^1 612321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 9 & 4 \\ 7 & 9 & 27 & 13 \\ 3 & 4 & 13 & 6 \end{pmatrix}_{b_0} \\
[3863] & 1^1 41^1 3151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 8 & 4 \\ 7 & 8 & 20 & 12 \\ 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \\
[3864] & 1^2 \overset{12}{6}^{2321} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 5 & 4 & 5 & 21 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0} \\
[3862] & 1^1 \overset{12}{6}^{2321} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 4 & 4 & 4 & 13 & 9 \\ 2 & 2 & 2 & 9 & 5 \end{pmatrix}_{b_0} \\
[3869] & 1^1 31^1 513221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 2 & 17 & 5 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 17 & 9 & 67 & 21 \\ 2 & 5 & 2 & 21 & 6 \end{pmatrix}_{b_0} \\
[3865] & 1^1 3223151^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 4 & 1 & 23 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 23 & 0 & 97 & 25 \\ 2 & 6 & -1 & 25 & 6 \end{pmatrix}_{b_0} \\
[3866] & 1^1 3223151^1 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 17 & 5 \\ 7 & 0 & 17 & 57 & 19 \\ 2 & 0 & 5 & 19 & 6 \end{pmatrix}_{b_0} \\
[3867] & 1^1 3231^1 51321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 8 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0} \\
[3868] & 1^1 3231^1 51321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 6 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0}$$

[3870] $1^{13}2315^{12}1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0}$$

[3873] $316^{11}123151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[3871] $315^{11}1321^{11}61^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 6 & 6 & 28 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[3872] $321^{11}6^{11}13151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[3875] $1^{12}32161^{12}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 10 & 4 \\ 8 & 10 & 30 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0}$$

[3874] $1^{11}2321^{11}6^{11}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 10 & 4 \\ 8 & 10 & 30 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0}$$

[3883] $1^{11}2321^{11}6^{11}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$$

[3884] $1^{12}2321^{12}6^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$$

[3876] $1^{12}2321^{12}6^{12}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 4 & 5 & 19 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[3877] $1^{12}2321^{12}6^{12}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 4 & 6 & 27 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0}$$

[3878] $1^{11}32161^{11}2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 4 & 5 & 19 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[3879] $1^3 \overset{13}{6} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 6 & 6 & 43 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$

[3880] $1^2 \overset{13}{6} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 6 & 5 & 31 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0}$

[3881] $1^3 \overset{12}{6} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 13 & 4 \\ 11 & 13 & 64 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$

[3882] $1^4 1^{13} 151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 11 & 4 \\ 8 & 11 & 35 & 15 \\ 3 & 4 & 15 & 6 \end{pmatrix}_{b_0}$

[3887] $1^3 1^2 31513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 8 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$

[3888] $1^4 1^3 31513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 10 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$

[3885] $1^{11} 41^4 3151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 4 & 6 & 22 & 12 \\ 2 & 2 & 2 & 12 & 5 \end{pmatrix}_{b_0}$

[3886] $1^2 \overset{12}{6} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 2 & 12 & 4 \\ 10 & 12 & 50 & 18 \\ 3 & 4 & 18 & 6 \end{pmatrix}_{b_0}$

[3891] $1^3 1^3 31513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 9 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$

[3889] $1^{12} 31513221^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 2 & 22 & 5 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 22 & 10 & 106 & 26 \\ 2 & 5 & 2 & 26 & 6 \end{pmatrix}_{b_0}$

[3890] $1^{12} 31513221^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 3 & 2 & 22 & 5 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 22 & 10 & 106 & 26 \\ 2 & 5 & 2 & 26 & 6 \end{pmatrix}_{b_0}$

[3892] $3151321^1 61^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 18 & 6 & 4 & 8 \\ 2 & 6 & 0 & 1 & 2 \\ 1 & 4 & 1 & 0 & 2 \\ 2 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[3893] $321^{1^1}6^{1^1}13151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 13 & 1 & 5 & 7 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 1 & 3 \\ 2 & 7 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[3898] $1^{1^3}4^{1^3}231^15^{1^3}1321^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 19 & 16 & 122 & 28 \\ 2 & 4 & 3 & 28 & 6 \end{pmatrix}_{b_0}$$

[3897] $1^{1^2}4^{1^2}231^15^{1^2}1321^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 1 & 16 & 3 \\ 9 & 19 & 16 & 122 & 28 \\ 2 & 4 & 3 & 28 & 6 \end{pmatrix}_{b_0}$$

[3894] $1^{1^2}4^{1^2}31^15^{1^2}13221^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 2 & 19 & 4 \\ 9 & 15 & 19 & 115 & 27 \\ 2 & 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

[3895] $1^{1^3}4^{1^3}31^15^{1^3}13221^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 2 & 19 & 4 \\ 10 & 15 & 19 & 115 & 27 \\ 2 & 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

[3896] $1^{1^3}2^{1^3}231^15^{1^3}1321^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 23 & 5 \\ 9 & 0 & 23 & 97 & 25 \\ 2 & 0 & 5 & 25 & 6 \end{pmatrix}_{b_0}$$

[3899] $1^{1^2}2^{1^2}2321^{1^1}6^{1^2}131^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 14 & 4 \\ 13 & 14 & 72 & 22 \\ 3 & 4 & 22 & 6 \end{pmatrix}_{b_0}$$

[3900] $1^{1^1}2^{1^2}2321^{1^1}6^{1^1}131^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 2 & 14 & 4 \\ 10 & 14 & 57 & 19 \\ 3 & 4 & 19 & 6 \end{pmatrix}_{b_0}$$

[3901] $1^{1^1}2^{1^3}2321^{1^1}6^{1^1}131^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 6 & 6 & 7 & 34 & 14 \\ 2 & 2 & 2 & 14 & 5 \end{pmatrix}_{b_0}$$

[3902] $1^{1^3}2^{1^3}2321^{1^1}6^{1^1}131^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 6 & 9 & 58 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3903] $1^{1^1}2^{1^2}2321^{1^1}6^{1^1}131^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 13 & 4 \\ 12 & 13 & 58 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$$

- [3904] $1^3 \overset{1^{12}2321}{6} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 16 & 4 \\ 12 & 16 & 85 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0}$
- [3905] $1^1 \overset{1^{13}21^{16}12321^{13}}{6} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 6 & 7 & 39 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
- [3906] $1^1 \overset{1^{13}2321^1}{6} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 6 & 6 & 37 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
- [3907] $1^2 \overset{1^{13}2321^1}{6} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 9 & 6 & 7 & 49 & 17 \\ 2 & 2 & 2 & 17 & 5 \end{pmatrix}_{b_0}$
- [3908] $1^1 \overset{1^{12}321^{16}12321^{12}}{6} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 14 & 4 \\ 11 & 14 & 62 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$
- [3909] $1^1 \overset{1^{12}41^13151321^{12}}{6} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 6 & 6 & 32 & 14 \\ 2 & 2 & 2 & 14 & 5 \end{pmatrix}_{b_0}$
- [3910] $1^2 \overset{1^{12}2321}{6} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 15 & 4 \\ 11 & 15 & 69 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0}$
- [3911] $1^2 \overset{1^{13}2321}{6} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 7 & 6 & 8 & 44 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0}$
- [3912] $1^1 \overset{1^{14}1^13151321^{13}}{6} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 13 & 4 \\ 11 & 13 & 53 & 19 \\ 3 & 4 & 19 & 6 \end{pmatrix}_{b_0}$
- [3913] $321^1 \overset{1^{11}}{6} 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 8 & 0 & 26 & 10 \\ 2 & 3 & 0 & 10 & 3 \end{pmatrix}_{b_0}$
- [3914] $1^1 \overset{1^3}{4} 31513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 2 & 24 & 4 \\ 11 & 16 & 24 & 158 & 32 \\ 2 & 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$
- [3915] $1^1 \overset{1^4}{4} 31513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 2 & 24 & 4 \\ 12 & 16 & 24 & 158 & 32 \\ 2 & 3 & 4 & 32 & 6 \end{pmatrix}_{b_0} \\
[3918] & 1^{12}321^{16}1^{2321^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 16 & 4 \\ 14 & 16 & 88 & 24 \\ 3 & 4 & 24 & 6 \end{pmatrix}_{b_0} \\
[3919] & 1^{12}32161^{2321^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 6 & 8 & 42 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0} \\
[3916] & 1^2^{13}6^{13}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 10 & 6 & 10 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0} \\
[3917] & 1^{13}6^{13}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 6 & 8 & 42 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0} \\
[3927] & 1^{12}41^{13}151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 6 & 9 & 47 & 17 \\ 2 & 2 & 2 & 17 & 5 \end{pmatrix}_{b_0} \\
[3920] & 1^4^{13}6^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 19 & 4 \\ 16 & 19 & 135 & 29 \\ 3 & 4 & 29 & 6 \end{pmatrix}_{b_0} \\
[3921] & 1^2^{13}6^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 17 & 4 \\ 14 & 17 & 99 & 25 \\ 3 & 4 & 25 & 6 \end{pmatrix}_{b_0} \\
[3922] & 1^{11}41^{13}151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 6 & 9 & 59 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0} \\
[3923] & 1^{12}6^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 15 & 4 \\ 12 & 15 & 67 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0} \\
[3924] & 1^3^{14}6^{13}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 11 & 8 & 8 & 79 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0} \\
[3925] & 1^2^{14}6^{13}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 10 & 8 & 7 & 63 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0} \\
[3926] & 1^{13}32161^{2321^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 15 & 4 \\ 12 & 15 & 67 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0}$$

[3928] $1^{\overline{1}51321^{\overline{1}}}$ $6^{\overline{1}}$ $1^{\overline{1}}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 75$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.96 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 & 2 \\ 7 & 19 & 1 & 6 & 9 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 1 & 3 \\ 2 & 9 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[3934] $1^{\overline{1}3^{\overline{1}3}2321^{\overline{1}}}$ $6^{\overline{1}3}$ $131^{\overline{1}3}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 18 & 4 \\ 15 & 18 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

[3935] $1^{\overline{1}1}41^{\overline{1}3}151321^{\overline{1}4}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 16 & 4 \\ 12 & 16 & 74 & 22 \\ 3 & 4 & 22 & 6 \end{pmatrix}_{b_0}$$

[3929] $1^{\overline{1}2^{\overline{1}3}2321^{\overline{1}}}$ $6^{\overline{1}3}$ $131^{\overline{1}3}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 18 & 4 \\ 15 & 18 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

[3930] $1^{\overline{2}^{\overline{1}4}2321^{\overline{1}}}$ $6^{\overline{1}3}$ $131^{\overline{1}2}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 8 & 8 & 58 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3931] $1^{\overline{1}1}41^{\overline{2}3}151321^{\overline{1}4}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 16 & 4 \\ 14 & 16 & 86 & 24 \\ 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$$

[3932] $1^{\overline{1}4^{\overline{1}4}2321^{\overline{1}}}$ $6^{\overline{1}3}$ $131^{\overline{1}2}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 10 & 8 & 10 & 90 & 22 \\ 2 & 2 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[3933] $1^{\overline{1}3^{\overline{1}3}2321^{\overline{1}}}$ $6^{\overline{1}3}$ $131^{\overline{1}2}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 6 & 9 & 52 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3937] $1^{\overline{1}2^{\overline{1}3}2321^{\overline{1}}}$ $6^{\overline{1}3}$ $131^{\overline{1}3}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 80$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 16 & 4 \\ 13 & 16 & 79 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0}$$

[3936] $1^{\overline{1}3^{\overline{1}4}2321^{\overline{1}}}$ $6^{\overline{1}3}$ $131^{\overline{1}2}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 80$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 8 & 9 & 72 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

[3938] $1^{\overline{2}^{\overline{1}2}2321^{\overline{1}}}$ $6^{\overline{1}3}$ $131^{\overline{1}3}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 81$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 17 & 4 \\ 14 & 17 & 95 & 25 \\ 3 & 4 & 25 & 6 \end{pmatrix}_{b_0}$$

[3939] $41^{\overline{1}1}31513151^{\overline{1}}$ + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -80$, $\det = 84$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.86 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[3941] $1^2 \overset{14}{6} \overset{2321}{131^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 11 & 2 \\ 9 & 8 & 11 & 77 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$$

[3942] $1^3 \overset{14}{6} \overset{2321}{131^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 10 & 8 & 12 & 93 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$$

[3940] $1^1 \overset{13}{6} \overset{2321^1}{131^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 18 & 4 \\ 16 & 18 & 109 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

[3946] $1^1 \overset{13}{321^1} \overset{161^1}{2321^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

[3943] $1^3 \overset{12}{6} \overset{2321^1}{131^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 21 & 4 \\ 16 & 21 & 146 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

[3944] $1^1 \overset{12}{6} \overset{2321^1}{131^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 19 & 4 \\ 14 & 19 & 106 & 26 \\ 3 & 4 & 26 & 6 \end{pmatrix}_{b_0}$$

[3945] $1^1 \overset{13}{321^1} \overset{1612321^{13}}{2321^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

[3947] $1^1 \overset{12}{321^1} \overset{1612321^{14}}{2321^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 9 & 8 & 10 & 70 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

[3948] $1^1 \overset{13}{41^1} \overset{13151321^{13}}{2321^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 8 & 9 & 61 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0}$$

[3949] $1^2 \overset{13}{6} \overset{2321}{131^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 20 & 4 \\ 15 & 20 & 124 & 28 \\ 3 & 4 & 28 & 6 \end{pmatrix}_{b_0}$$

[3950] $1^2 \overset{12}{6} \overset{2321^1}{131^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 20 & 4 \\ 15 & 20 & 124 & 28 \\ 3 & 4 & 28 & 6 \end{pmatrix}_{b_0}$$

[3951] $1^3 \overset{13}{6} \overset{2321^1}{131^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 20 & 4 \\ 18 & 20 & 145 & 31 \\ 3 & 4 & 31 & 6 \end{pmatrix}_{b_0}$$

[3952] $1^{11}321^1 61^1 2321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 8 & 9 & 75 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$$

[3953] $1^{13} 2^{13} 2321^1 6^{13} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 103, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 19 & 4 \\ 17 & 19 & 125 & 29 \\ 3 & 4 & 29 & 6 \end{pmatrix}_{b_0}$$

[3954] $1^{12} 41^2 3151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 106, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 8 & 9 & 73 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$$

[3955] $1^{13} 2^{13} 2321^1 4^{13} 6^{13} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 106, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 22 & 4 \\ 17 & 22 & 164 & 32 \\ 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$$

[3956] $1^{13} 2^{13} 2321^1 3^{13} 6^{13} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 21 & 4 \\ 16 & 21 & 142 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

[3957] $1^{14} 2^{14} 2321^1 3^{14} 6^{14} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 11 & 2 \\ 12 & 8 & 11 & 98 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

[3958] $1^{14} 2^{14} 2321^1 1^{14} 6^{14} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 8 & 9 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

[3959] $1^{14} 2^{14} 2321^1 2^{14} 6^{14} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 111, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 11 & 8 & 10 & 80 & 22 \\ 2 & 2 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[3960] $1^{15} 51321^1 31^{15} 6^{15} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 9 & 0 & 34 & 12 \\ 2 & 3 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[3961] $61^{11} 22315^{11} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 124, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 6 & 2 & 6 & 2 \\ 6 & 28 & 7 & 30 & 10 \\ 2 & 7 & 0 & 6 & 2 \\ 6 & 30 & 6 & 28 & 10 \\ 2 & 10 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[3962] $1^{12} 321^1 61^1 2321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 8 & 12 & 96 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

[3963] $1^{14} 2^{14} 2321^1 2^{14} 6^{14} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 13 & 2 \\ 12 & 8 & 13 & 103 & 25 \\ 2 & 2 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[3964] \ 1^3 \overset{14}{6} \ 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 23 & 4 \\ 19 & 23 & 184 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0}$$

$$[3965] \ 1^{13} 41^2 3151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 10 & 9 & 91 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[3966] \ 1^1 \overset{14}{6} \overset{2321}{1} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 8 & 12 & 87 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[3967] \ 1^4 \overset{14}{6} \overset{2321}{1} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 132, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 2 & 24 & 4 \\ 20 & 24 & 206 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[3968] \ 1^{12} 41^2 3151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 132, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 8 & 12 & 94 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[3969] \ 1^{13} 321^1 61^1 2321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 133, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 21 & 4 \\ 18 & 21 & 145 & 31 \\ 3 & 4 & 31 & 6 \end{pmatrix}_{b_0}$$

$$[3970] \ 1^1 \overset{13}{6} \overset{2321}{1} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 137, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 21 & 4 \\ 17 & 21 & 136 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[3971] \ 1^4 \overset{13}{6} \overset{2321}{1} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 140, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 2 & 24 & 4 \\ 20 & 24 & 202 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[3972] \ 1^2 \overset{13}{6} \overset{2321}{1} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 144, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 22 & 4 \\ 18 & 22 & 154 & 32 \\ 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$$

$$[3973] \ 1^3 \overset{13}{6} \overset{2321}{1} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 145, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 23 & 4 \\ 19 & 23 & 176 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0}$$

$$[3974] \ 1^2 \overset{14}{6} \overset{2321}{1} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 2 & 24 & 4 \\ 21 & 24 & 196 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[3975] \ 1^{12} 41^3 3151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 167, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 15 & 10 & 12 & 132 & 28 \\ 2 & 2 & 2 & 28 & 5 \end{pmatrix}_{b_0}$$

[3977] $1^{\overline{14}2321^1} 6^{131^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 170$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 23 & 3 \\ 2 & 2 & 26 & 4 \\ 23 & 26 & 244 & 40 \\ 3 & 4 & 40 & 6 \end{pmatrix}_{b_0}$$

[3976] $1^{\overline{14}321^1 61^1 2321^{\overline{13}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 170$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 24 & 4 \\ 19 & 24 & 178 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0}$$

[3978] $1^{\overline{14}2321^1} 6^{131^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 171$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 2 & 25 & 4 \\ 22 & 25 & 218 & 38 \\ 3 & 4 & 38 & 6 \end{pmatrix}_{b_0}$$

[3979] $1^{\overline{13}41^2 3151321^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 175$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 13 & 10 & 12 & 112 & 26 \\ 2 & 2 & 2 & 26 & 5 \end{pmatrix}_{b_0}$$

[3981] $1^{\overline{13}41^3 3151321^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 204$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 16 & 12 & 12 & 154 & 30 \\ 2 & 2 & 2 & 30 & 5 \end{pmatrix}_{b_0}$$

[3980] $1^{\overline{14}321^1 61^1 2321^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 204$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 2 & 26 & 4 \\ 22 & 26 & 220 & 38 \\ 3 & 4 & 38 & 6 \end{pmatrix}_{b_0}$$

[3982] $1^{\overline{14}41^2 3151321^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 208$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 14 & 12 & 12 & 134 & 28 \\ 2 & 2 & 2 & 28 & 5 \end{pmatrix}_{b_0}$$

[3983] $3221^{\overline{11}} 613151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 263$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.65 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 1 & 7 & 2 \\ 7 & 26 & 1 & 28 & 10 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 28 & 0 & 26 & 10 \\ 2 & 10 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[3984] $3161^{\overline{11}} 223151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 463$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.63 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 9 & 1 & 9 & 2 \\ 9 & 34 & 1 & 36 & 12 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 36 & 0 & 34 & 12 \\ 2 & 12 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$T = 13$ [3985] $1^{\overline{11}2321} 7^{12321^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -30$, $\det = 0$, $(n_+, n_-, n_0) = (1, 13, 1)$, $c_{\text{ext}} = 4.0$

[3996] $1^{\overline{11}321} 8^{1231^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -54$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[3997] $1^{\overline{11}} 612315131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -50$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[3998] \quad \begin{array}{c} 1^1 \\ 21^1 812321^1 + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{green}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \\ 1^1 \end{array} \quad T = 13, \quad T_{\max} = 14, \quad \Delta = -47, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[3988] \quad 1 \overset{1}{\overset{1}{\underset{5}{6}}} \overset{1}{131} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 13, \, T_{\max} = 14, \, \Delta = -49, \, \det = -2, \, (n_+, n_-, n_0) = (1, 13, 0), \, c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \, T_{\max} = 14, \, T \in [14, 14]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$\begin{array}{l}
\text{[3989]} \quad \overset{1^1}{1} 6^1 13^1 15^1 13^1 \overset{1^1}{1} + [\textcolor{red}{5}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{5}\textcolor{blue}{u}_2 \quad T = 13, \quad T_{\max} = 14, \quad \Delta = -49, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 4.0 \\
T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14] \\
\left(\begin{array}{cccccc} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{array} \right)_{b_0}
\end{array}$$

[3990] $1^1 1^{3221} 8 \quad 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

$$[3991] \overset{1^1}{15} \overset{1^1}{1316131}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, \, T_{\max} = 14, \, \Delta = -49, \, \det = -2, \, (n_+, n_-, n_0) = (1, 13, 0), \, c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \, T_{\max} = 14, \, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[3992] \ 1^1 \overset{13221}{8} \ 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, \ T_{\max} = 14, \ \Delta = -54, \ \det = -2, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

[3993] $15^{1^1}1316^{1^1}131^1 + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2$ $T = 13, T_{\max} = 14, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[3994] $1^1 1^1 3 1 6 1 2 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 14, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\left(\begin{smallmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{smallmatrix} \right)_{b_0}$

$$[3995] \quad 1^1 3216131^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 13, \quad T_{\max} = 14, \quad \Delta = -50, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[4003] \begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix} \begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix} \begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[4004] $\overset{1}{1}712231\overset{1}{5}1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

$$[4005] \ 1^1 \overset{1^1}{7} 12231 \overset{1^1}{5} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}\textcolor{teal}{u}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[3999] & 1^{\overline{1}} 7^{11} 1231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[4000] & 21^{\overline{1}} 7^{11} 1231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0} \\
[4001] & 21^{\overline{1}} 7^{11} 1231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[4002] & 1^{\overline{1}} 7^{11} 1231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[4013] & 31^{\overline{1}} 321^{\overline{1}} 8 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -62, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 2 & 4 & 6 & 6 \\ 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[4014] & 1^{\overline{1}} 2321612321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[4015] & 1^{\overline{1}} 4131513221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[4006] & 1^{\overline{1}} 3141414131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 5 & 1 & 4 & 6 \\ 2 & 5 & 0 & 6 & 6 \end{pmatrix}_{b_0} \\
[4007] & 1^{\overline{1}} 5^{11} 13221^{\overline{1}} 7 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[4008] & 1^{\overline{1}} 7^{11} 12314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -73, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
[4009] & 1^{\overline{1}} 5^{11} 1321^{\overline{1}} 7 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[4010] $1^{11}4131513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$

[4011] $1^{14}1^{13}1513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 4 & 2 & 4 & 8 & 8 \\ 2 & 2 & 3 & 8 & 6 \end{pmatrix}_{b_0}$

[4012] $1^{13}171232151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 15, \Delta = -71, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[4017] $2^13^121712321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$

[4018] $2^13^121712321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$

[4019] $2^13^121712321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$

[4016] $2^13^121712321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$

[4021] $1^{14}1^{13}1513221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 3 & 10 & 5 \\ 7 & 10 & 24 & 14 \\ 3 & 5 & 14 & 7 \end{pmatrix}_{b_0}$

[4020] $1^{11}1^{13}151315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$

[4022] $1^{14}1^{13}1513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 3 & 14 & 5 \\ 8 & 14 & 44 & 18 \\ 3 & 5 & 18 & 7 \end{pmatrix}_{b_0}$

[4023] $1^{11}1^{13}1513221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 6 & 4 & 8 & 26 & 14 \\ 2 & 2 & 3 & 14 & 6 \end{pmatrix}_{b_0}$

[4024] $2^13^121812^13^12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 5 & 3 & 16 & 5 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 16 & 7 & 37 & 13 \\ 2 & 5 & 2 & 13 & 4 \end{pmatrix}_{b_0} \\
[4025] & 1^{12}31^1414141^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -3, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 1 & 18 & 5 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 18 & 1 & 71 & 21 \\ 2 & 5 & 0 & 21 & 6 \end{pmatrix}_{b_0} \\
[4027] & 1^{12}41^131513221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 8 & 6 & 12 & 56 & 20 \\ 2 & 2 & 3 & 20 & 6 \end{pmatrix}_{b_0} \\
[4026] & 1^{11}41^231513221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 10 & 6 & 12 & 68 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4028] & 1^{11}41^131513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 3 & 20 & 5 \\ 12 & 20 & 90 & 26 \\ 3 & 5 & 26 & 7 \end{pmatrix}_{b_0} \\
[4029] & 1^{13}41^131513221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 9 & 8 & 12 & 70 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4030] & 1^{14}2^321812^32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 5 & 3 & 32 & 5 \\ 2 & 3 & 0 & 14 & 2 \\ 12 & 32 & 14 & 154 & 26 \\ 2 & 5 & 2 & 26 & 4 \end{pmatrix}_{b_0} \\
[4031] & 1^{12}41^231513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 12 & 8 & 16 & 110 & 28 \\ 2 & 2 & 3 & 28 & 6 \end{pmatrix}_{b_0} \\
[4032] & 1^{13}41^231513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -83, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 13 & 10 & 16 & 128 & 30 \\ 2 & 2 & 3 & 30 & 6 \end{pmatrix}_{b_0} \\
T = 14 & [4036] 1^{1^{12315131}}6^{131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 4.0 \\
[4033] & 1^{1^{2321}}8^{12321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 4.0 \\
[4034] & 1^{1^1}4^1414141^141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 4.0 \\
[4035] & 1^{12}31^141414141^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 4.5 \\
[4047] & 1^{1^1}7^12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[4048] & 1^{1^{1321}}8^{123141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[4049] $31^1 1^1 321 8 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -35$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 4 & 2 & 5 & 7 & 7 \\ 2 & 2 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

[4040] $23218 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 1 & 2 & 1 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[4041] $1^1 1^1 32171231 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -49$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

[4042] $1^1 1^1 7 122315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -48$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[4043] $1^1 1^1 41316131 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -48$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

[4044] $216 1316 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -46$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

[4045] $1^1 1^1 321 8^{11} 123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -53$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & -1 & 3 & 5 \end{pmatrix}_{b_0}$$

[4046] $1^1 1^1 6 123216131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -49$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[4059] $1^1 1^1 5131 6^{11} 13151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 16$, $\Delta = -72$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[4060] $21^1 1^1 8 12231 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 16$, $\Delta = -68$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[4061] $21^1 1^1 8 12231 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 16$, $\Delta = -68$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 3 & 7 & 5 \\ 5 & 7 & 12 & 10 \\ 3 & 5 & 10 & 7 \end{pmatrix}_{b_0}$$

$$[4062] \quad 1^{11} 6 13 15 13 15 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 16, \Delta = -72, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[4063] \quad 1^{11} 7^{2321} 12 32 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$$[4064] \quad 1^{11} 8 12 23 14 14 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 16, \Delta = -72, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 & 1 \\ 1 & 2 & 0 & 1 & 3 & 4 \\ 2 & 3 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[4065] \quad 1^1 15 13 22 1^1 8 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[4067] \quad 21^1 8^{1232151} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 16, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[4068] \quad 1^{11} 23 15 13 15 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[4069] \quad 1^{11} 23 15 13 15 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[4071] \quad 1^{113} 2^3 3^{21} 12 32 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[4072] \quad 2^3 21^4 8^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 8 & 20 & 142 & 28 \\ 2 & 2 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[4073] \quad 2^3 21^4 8^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 6 & 15 & 79 & 21 \\ 2 & 2 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[4074] \quad 2^3 21^4 8^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 4 & 10 & 34 & 14 \\ 2 & 2 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[4075] \quad 1^{112} 2^{321} 8^{12} 12321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 4 & 10 & 34 & 14 \\ 2 & 2 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[4076] \quad 2^{11} 3^{11} 218^{11} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 7 & 7 \\ 2 & 2 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[4077] \quad 1^{114} 2^{321} 8^{14} 12321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[4078] \quad 1^{111} 2^{321} 8^{11} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[4079] \quad 2^{111} 3^{11} 218^{11} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 1 & 2 & 5 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 5 & 6 & 13 & 9 \\ 2 & 3 & 3 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[4083] \quad 1^{111} 31612315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[4084] \quad 1^{111} 32161315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[4080] \quad 1^{111} 7^{2321} 12321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 3 & 11 & 5 \\ 6 & 11 & 26 & 14 \\ 3 & 5 & 14 & 7 \end{pmatrix}_{b_0}$$

$$[4081] \quad 1^{111} 32161315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[4082] \quad 1^{111} 31612315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[4085] \quad 2^{112} 3^{11} 218^{12} 12321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 14, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 1 & 2 & 9 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 9 & 11 & 44 & 16 \\ 2 & 3 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

- [4087] $1^{13}21^{16}1315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 19$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 4 & 4 & 18 & 10 \\ 2 & 2 & 2 & 10 & 5 \end{pmatrix}_{b_0}$
- [4086] $1^{13}21^{16}1315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 19$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 9 & 4 \\ 7 & 9 & 27 & 13 \\ 3 & 4 & 13 & 6 \end{pmatrix}_{b_0}$
- [4088] $2^1 3^1 21^8 12321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 20$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 13 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 13 & 16 & 93 & 23 \\ 2 & 3 & 3 & 23 & 5 \end{pmatrix}_{b_0}$
- [4089] $1^{112} 7^{2321} 12321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 14$, $\Delta = -1$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 3 & 17 & 5 \\ 10 & 17 & 66 & 22 \\ 3 & 5 & 22 & 7 \end{pmatrix}_{b_0}$
- [4090] $2^1 3^1 21^8 12321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 26$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 1 & 2 & 17 & 3 \\ 1 & 2 & 1 & 21 & 3 \\ 13 & 17 & 21 & 160 & 30 \\ 2 & 3 & 3 & 30 & 5 \end{pmatrix}_{b_0}$
- [4092] $1^{11} 3161^{12} 2315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 33$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 4 & 6 & 27 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0}$
- [4091] $1^{12} 3161^{12} 2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 33$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$
- [4093] $1^{12} 3161^{12} 2315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 51$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.98 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 13 & 4 \\ 12 & 13 & 58 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$
- [4096] $1^{11} 3161^{12} 2315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 54$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 6 & 6 & 37 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
- [4094] $1^{11} 321^{16} 1315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 54$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 6 & 7 & 39 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
- [4095] $1^{12} 321^{16} 1315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 54$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 14 & 4 \\ 11 & 14 & 62 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$
- [4097] $1^{12} 3161^{12} 2315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 6 & 9 & 52 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$
- [4098] $1^{13} 3161^{12} 2315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 80$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 16 & 4 \\ 13 & 16 & 79 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0} \\
[4099] & 1^{13}3161^12315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 18 & 4 \\ 16 & 18 & 109 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0} \\
[4102] & 1^{14}3161^12315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 19 & 4 \\ 14 & 19 & 106 & 26 \\ 3 & 4 & 26 & 6 \end{pmatrix}_{b_0} \\
[4100] & 1^{12}321^161315131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 9 & 8 & 10 & 70 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0} \\
[4101] & 1^{13}321^161315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0} \\
[4103] & 1^{12}3161^12315131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 8 & 9 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0} \\
[4104] & 1^{13}3161^12315131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 8 & 12 & 87 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0} \\
[4105] & 1^{14}3161^12315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 137, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 21 & 4 \\ 17 & 21 & 136 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0} \\
\\
T = 15 & [4112] 1^{11}1^{13}22315131^6 131^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0 \\
[4113] & 1^{112}22315131^6 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0 \\
[4114] & 1^{13}3161^1231513221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.8 \\
[4106] & 1^{111}22315131^6 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0 \\
[4107] & 1^{214}22315131^6 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.5 \\
[4108] & 1^{212}22315131^6 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.5 \\
[4109] & 1^{313}22315131^6 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.8 \\
[4110] & 1^{414}22315131^6 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0 \\
[4111] & 1^{131}6 131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0 \\
[4125] & 1^1 8 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[4133] & 1^1 6 1232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0}$$

[4145] $1^{\overline{1}1}8^{11}1231414141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 17, \Delta = -71, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[4146] $1^{\overline{1}1}8^{11}12321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 3 & 6 & 6 & 14 & 10 \\ 2 & 4 & 3 & 10 & 6 \end{pmatrix}_{b_0}$$

[4147] $1^{\overline{1}1}321613151321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

[4148] $1^{\overline{1}1}2315131^{\overline{1}1}6^{11}131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

[4149] $21^{\overline{1}1}8^{11}12321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

[4150] $1^{\overline{1}1}2315131^{\overline{1}1}6^{11}131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

[4152] $1^{\overline{1}1}1^{12}6^{12}2315131^{\overline{1}1}131^{\overline{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -5, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 5 & 6 & 45 & 17 \\ 2 & 2 & 2 & 17 & 6 \end{pmatrix}_{b_0}$$

[4153] $1^{\overline{1}1}1^{12}2315131^{\overline{1}1}6^{11}131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -6, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 4 & 3 & 8 & 8 \\ 2 & 3 & 2 & 8 & 6 \end{pmatrix}_{b_0}$$

[4154] $31^{\overline{1}1}6^{11}1316131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 16, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 6 & 1 & 8 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 8 & 0 & 6 & 6 \\ 2 & 6 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

[4157] $1^{\overline{1}2}1^{12}2315131^{\overline{1}1}6^{11}131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 5 & 6 & 5 & 25 & 13 \\ 2 & 3 & 2 & 13 & 6 \end{pmatrix}_{b_0}$$

[4155] $1^{\overline{1}1}321^{\overline{1}1}613151321^{\overline{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 6 & 4 & 22 & 12 \\ 2 & 3 & 2 & 12 & 6 \end{pmatrix}_{b_0}$$

[4156] $1^{11} 8^{2321} 12321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 10 & 4 \\ 1 & 2 & 1 & 7 & 3 \\ 4 & 10 & 7 & 30 & 14 \\ 2 & 4 & 3 & 14 & 6 \end{pmatrix}_{b_0}$$

[4159] $21^{11} 8^{2321} 12321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 3 & 12 & 5 \\ 7 & 12 & 34 & 16 \\ 3 & 5 & 16 & 7 \end{pmatrix}_{b_0}$$

[4160] $1^{12} 6^{2315131} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 4 & 2 \\ 4 & 6 & 4 & 17 & 11 \\ 2 & 3 & 2 & 11 & 6 \end{pmatrix}_{b_0}$$

[4158] $1^{11} 8^{2321} 12321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 10 & 4 \\ 1 & 2 & 1 & 7 & 3 \\ 5 & 10 & 7 & 30 & 14 \\ 2 & 4 & 3 & 14 & 6 \end{pmatrix}_{b_0}$$

[4165] $31^{31} 6^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 16, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 3 & 4 \\ 2 & 4 & 2 & 5 & 4 \\ 2 & 3 & 5 & 3 & 5 \\ 2 & 4 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[4161] $1^{13} 6^{2315131} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 9 & 5 & 40 & 16 \\ 2 & 3 & 2 & 16 & 6 \end{pmatrix}_{b_0}$$

[4162] $1^{12} 6^{2315131} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 5 & 7 & 6 & 28 & 14 \\ 2 & 3 & 2 & 14 & 6 \end{pmatrix}_{b_0}$$

[4163] $1^3 6^{12} 2315131 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 9 & 6 & 52 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

[4164] $1^{13} 6^{2315131} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 9 & 6 & 52 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

[4166] $1^2 6^{12} 2315131 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -18, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 8 & 6 & 38 & 16 \\ 2 & 3 & 2 & 16 & 6 \end{pmatrix}_{b_0}$$

[4167] $31^{11} 6^{11} 1316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 16, \Delta = -33, \det = -21, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 6 & 1 & 8 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 8 & 1 & 7 & 7 \\ 2 & 6 & 0 & 7 & 5 \end{pmatrix}_{b_0} \\
[4173] & 1^{12}3161^{12}3151321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 9 & 6 & 46 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4168] & 1^{12}321^{16}13151321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 10 & 6 & 50 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4169] & 1^{11}321^{16}13151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 9 & 7 & 48 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4170] & 1^{12}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 15 & 4 \\ 1 & 2 & 1 & 12 & 3 \\ 6 & 15 & 12 & 74 & 22 \\ 2 & 4 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4171] & 1^{13}2315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 6 & 9 & 7 & 43 & 17 \\ 2 & 3 & 2 & 17 & 6 \end{pmatrix}_{b_0} \\
[4172] & 1^{11}3161^{12}3151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 9 & 6 & 46 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4174] & 1^{11}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -29, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 15 & 4 \\ 1 & 2 & 1 & 12 & 3 \\ 7 & 15 & 12 & 74 & 22 \\ 2 & 4 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4175] & 21^{11}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -29, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 3 & 18 & 5 \\ 11 & 18 & 76 & 24 \\ 3 & 5 & 24 & 7 \end{pmatrix}_{b_0} \\
[4177] & 1^{13}2315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 11 & 9 & 72 & 22 \\ 2 & 3 & 2 & 22 & 6 \end{pmatrix}_{b_0} \\
[4176] & 1^{13}2315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 13 & 9 & 108 & 26 \\ 2 & 3 & 2 & 26 & 6 \end{pmatrix}_{b_0} \\
[4179] & 1^{14}2315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 12 & 8 & 74 & 22 \\ 2 & 3 & 2 & 22 & 6 \end{pmatrix}_{b_0} \\
[4180] & 1^3 \overset{14}{6} \overset{2315131}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 12 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0} \\
[4178] & 1^3 \overset{13}{6} \overset{2315131}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 12 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0} \\
[4185] & 1^{\textcolor{green}{12}} 3161^{\textcolor{red}{1}} 23151321^{\textcolor{blue}{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 12 & 9 & 82 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0} \\
[4181] & 1^{\textcolor{red}{13}} 3161^{\textcolor{green}{1}} 23151321^{\textcolor{blue}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 12 & 9 & 82 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0} \\
[4182] & 1^2 \overset{13}{6} \overset{2315131}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 14 & 9 & 97 & 25 \\ 2 & 3 & 2 & 25 & 6 \end{pmatrix}_{b_0} \\
[4183] & 1^3 \overset{13}{6} \overset{2315131}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 15 & 9 & 115 & 27 \\ 2 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0} \\
[4184] & 1^{\textcolor{red}{13}} 321^{\textcolor{green}{1}} 613151321^{\textcolor{blue}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 13 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0} \\
[4187] & 21^{\textcolor{red}{1}} \overset{13}{8} \overset{2321}{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 24 & 5 \\ 15 & 24 & 138 & 32 \\ 3 & 5 & 32 & 7 \end{pmatrix}_{b_0} \\
[4186] & 1^{\textcolor{red}{1}} \overset{13}{8} \overset{2321}{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 20 & 4 \\ 1 & 2 & 1 & 17 & 3 \\ 9 & 20 & 17 & 138 & 30 \\ 2 & 4 & 3 & 30 & 6 \end{pmatrix}_{b_0} \\
[4188] & 31^{\textcolor{red}{12}} \overset{31}{6} \overset{1316131}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -53, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15.98 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 6 & 1 & 16 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 16 & 1 & 31 & 13 \\ 2 & 6 & 0 & 13 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[4190] $1^{\overline{14}} 3^{14} 6^{2315131} 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 15 & 12 & 136 & 30 \\ 2 & 3 & 2 & 30 & 6 \end{pmatrix}_{b_0}$

[4189] $1^{\overline{14}} 3161^{\overline{1}} 23151321^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 15 & 9 & 109 & 27 \\ 2 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0}$

[4191] $1^{\overline{14}} 3^{14} 6^{2315131} 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 16 & 12 & 158 & 32 \\ 2 & 3 & 2 & 32 & 6 \end{pmatrix}_{b_0}$

$T = 16$ [4194] $1^{\overline{1}} 23151321^{\overline{1}} 8^{12321^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 2), c_{\text{ext}} = 4.0$

[4192] $1^{\overline{1}} 23151321^{\overline{1}} 8^{1231^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$

[4193] $1^{\overline{1}} 3151321^{\overline{1}} 8^{12321^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$

[4195] $1^{\overline{1}} 3151321^{\overline{1}} 8^{12321^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$

[4201] $1^{\overline{1}} 6^{12321812231^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4203] $1^{\overline{1}} 6^{123151316131^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4204] $1^{\overline{1}} 6^{1232181231^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[4205] $1^{\overline{1}} 8^{1232161^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4200] $1^{\overline{1}} 4132181231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

[4202] $21^{\overline{1}} 6^{1231613151^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$

[4206] $1^{\overline{1}} 8^{123151^{\overline{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 18, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[4215] & 1^{\overline{11}} 2321712315131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[4207] & 1^{\overline{11}} 2321712315131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[4208] & 1^{\overline{11}} 3216131612321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[4209] & 1^{\overline{11}} 3217123151321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[4210] & 1^{\overline{1}} 81231513151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 18, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[4211] & 1^{\overline{11}} 3217123151321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[4212] & 1^{\overline{1}} 712321613151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 18, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[4213] & 1^{\overline{11}} 3216131612321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[4214] & 1^{\overline{1}} 812231513151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 18, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[4216] & 2^{\overline{13}} 321812315131^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 6 & 27 & 6 \\ 15 & 27 & 115 & 27 \\ 3 & 6 & 27 & 6 \end{pmatrix}_{b_0} \\
[4217] & 31^{\overline{13}} 51321812321^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0} \\
[4218] & 2^{\overline{14}} 321812315131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 6 & 36 & 6 \\ 20 & 36 & 206 & 36 \\ 3 & 6 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[4219] \quad 31 \quad 5 \quad 1321812321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[4220] \quad 31 \quad 5 \quad 1321812321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[4221] \quad 2 \quad 3 \quad 21812315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 11 & 9 \\ 3 & 6 & 9 & 6 \end{pmatrix}_{b_0}$$

$$[4222] \quad 2 \quad 3 \quad 21812315131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 6 & 18 & 6 \\ 10 & 18 & 50 & 18 \\ 3 & 6 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[4223] \quad 1^1 321^1 6131612321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 5 & 4 & 7 & 27 & 13 \\ 2 & 2 & 3 & 13 & 6 \end{pmatrix}_{b_0}$$

$$[4224] \quad 1^{11} 321^1 6131612321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 7 & 6 & 11 & 55 & 19 \\ 2 & 2 & 3 & 19 & 6 \end{pmatrix}_{b_0}$$

$$[4225] \quad 1^{12} 321^1 6131612321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 3 & 19 & 5 \\ 11 & 19 & 87 & 25 \\ 3 & 5 & 25 & 7 \end{pmatrix}_{b_0}$$

$$[4226] \quad 1^{13} 321^1 6131612321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 25 & 5 \\ 15 & 25 & 151 & 33 \\ 3 & 5 & 33 & 7 \end{pmatrix}_{b_0}$$

$$[4227] \quad 1^{11} 321^1 613161^{12} 2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 11 & 8 & 13 & 91 & 25 \\ 2 & 2 & 3 & 25 & 6 \end{pmatrix}_{b_0}$$

$$[4228] \quad 1^{13} 321^1 613161^{12} 2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 25 & 5 \\ 15 & 25 & 151 & 33 \\ 3 & 5 & 33 & 7 \end{pmatrix}_{b_0}$$

$$[4229] \quad 1^{14} 321^1 613161^{12} 2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 85, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 3 & 31 & 5 \\ 19 & 31 & 227 & 41 \\ 3 & 5 & 41 & 7 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [4233] \quad 1^{11} 6 \quad 13161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$$

$$[4230] \quad 1^{12321} \quad 8 \quad 123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$$

[4231] $1^{\overline{1}1} 8^{\overline{1}2321} 123216131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$

[4232] $1^{\overline{1}13} 8^{\overline{1}23151321} 12321^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$

[4243] $1^{\overline{1}1} 7^{\overline{1}1} 123151316131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4238] $1^{\overline{1}1} 5^{\overline{1}1} 132181231^{\overline{1}1} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[4239] $1^{\overline{1}1} 5^{\overline{1}1} 132181231^{\overline{1}1} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4240] $1^{\overline{1}1} 21^{\overline{1}1} 7^{\overline{1}1} 12321^{\overline{1}1} 8^{\overline{1}1} 1231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[4241] $1^{\overline{1}1} 6^{\overline{1}1} 123218123141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$

[4242] $1^{\overline{1}1} 316131^{\overline{1}1} 6^{\overline{1}1} 13161^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[4237] $1^{\overline{1}1} 5^{\overline{1}1} 132181232151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4244] $1^{\overline{1}1} 51321^{\overline{1}1} 8^{\overline{1}1} 1232161^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 19, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 5 & 5 \\ 2 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$

[4245] $1^{\overline{1}1} 7^{\overline{1}1} 123217123151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 19, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[4246] $1^{\overline{1}1} 31612321712321^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$

[4247] $1^{\overline{1}1} 31513161315131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 6 & 6 & 9 & 38 & 16 \\ 2 & 2 & 3 & 16 & 6 \end{pmatrix}_{b_0} \\
[4259] & 1^{12}3161^12321712321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 3 & 19 & 5 \\ 12 & 19 & 94 & 26 \\ 3 & 5 & 26 & 7 \end{pmatrix}_{b_0} \\
[4261] & 1^{12}232181^122315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 10 & 6 & 12 & 68 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4262] & 1^{12}3161^12321712321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 10 & 8 & 15 & 102 & 26 \\ 2 & 2 & 3 & 26 & 6 \end{pmatrix}_{b_0} \\
[4263] & 1^{13}3161^12321712321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 3 & 25 & 5 \\ 16 & 25 & 158 & 34 \\ 3 & 5 & 34 & 7 \end{pmatrix}_{b_0} \\
[4264] & 1^{14}232181^122315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 16.96 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 24 & 5 \\ 15 & 24 & 138 & 32 \\ 3 & 5 & 32 & 7 \end{pmatrix}_{b_0} \\
[4265] & 1^{14}3161^12321712321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 3 & 29 & 5 \\ 17 & 29 & 200 & 38 \\ 3 & 5 & 38 & 7 \end{pmatrix}_{b_0} \\
[4266] & 1^{13}232181^122315131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 16.97 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 12 & 8 & 16 & 110 & 28 \\ 2 & 2 & 3 & 28 & 6 \end{pmatrix}_{b_0} \\
\\
T = 18 & [4267] 1^{11}8^1232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0 \\
[4268] & 1^{11}321^18^12321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0 \\
[4269] & 1^{11}2321^18^12321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 2), c_{\text{ext}} = 4.0 \\
[4270] & 1^{11}31^16^131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0 \\
[4271] & 1^{11}31^16^13161316^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0 \\
[4272] & 1^{11}2321^18^1232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0 \\
[4284] & 1^{11}7^1232161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 18, T_{\max} = 19, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[4275] $1^{1^3 3221} (12) 12231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 26 & 12 \\ 2 & 3 & 2 & 12 & 5 \end{pmatrix}_{b_0}$

[4276] $1^{1^1} 61315132181231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4277] $1^{1^1} 61232181232151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4278] $1^{1^1} 5132181232161^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -41, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4279] $1^{1^1} 6123218123151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -41, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[4280] $1^{1^1 3221} (12) 12231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0}$

[4281] $1^{1^1 3221} (12) 12231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 8 \\ 2 & 3 & 2 & 8 & 5 \end{pmatrix}_{b_0}$

[4282] $1^{1^1} 81223151316131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4283] $22181232181231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 19, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[4287] $1^{1^1} 61316131613151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$

[4285] $1^{1^1} 7123218123151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[4286] $1^{\overline{1}} 7^{1^1} 12321 8^{1^1} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[4289] $1^{\overline{1}} 7^{1^1} 1232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[4288] $1^{\overline{1}} 5^{1^1} 1321 8^{1^1} 12321 71^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$T = 19$ [4291] $1^{\overline{1}} 3^{1^1} 21 8^{1^1} 12321 8^{1^1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4295] $1^{\overline{1}} 3^{1^1} 6^{1^1} 1316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4297] $1^{\overline{1}} 2^{1^1} 321 8^{1^1} 12321 8^{1^1} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4294] $1^{\overline{1}} 1^{1^1} 2^{1^1} 321 8^{1^1} 12321 1712321^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4293] $1^{\overline{1}} 1^{1^1} 2^{1^1} 321 8^{1^1} 12321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4292] $1^{\overline{1}} 1^{1^1} 2^{1^1} 321 8^{1^1} 12321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4290] $1^{\overline{1}} 1^{1^1} 3^{1^1} 21 8^{1^1} 12321 8^{1^1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4296] $1^{\overline{1}} 1^{1^1} 2^{1^1} 321 8^{1^1} 12321 8^{1^1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 2), c_{\text{ext}} = 4.0$

[4306] $221^{\overline{1}} 5^{1^1} 13221 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 20, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

[4304] $1^{\overline{1}} 2^{1^1} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 20, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[4303] $21^{\overline{1}} 7^{1^1} 1232181231 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 20, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$$

[4302] $1^{\overline{1}} 7^{1^1} 12315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 20, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[4305] $1^{1^1} 6^{1^1} 13161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19$, $T_{\max} = 20$, $\Delta = -46$, $\det = -2$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 20$, $T_{\max} = 20$, $T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[4308] $21^{1^1} 8^{1^1} 12321^1 8^{1^1} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19$, $T_{\max} = 21$, $\Delta = -67$, $\det = -3$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 21$, $T_{\max} = 21$, $T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[4307] $21^{1^1} 8^{1^1} 12321^1 8^{1^1} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19$, $T_{\max} = 21$, $\Delta = -67$, $\det = -3$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 21$, $T_{\max} = 21$, $T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[4310] $1^{1^1} 8^{1^1} 12315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19$, $T_{\max} = 21$, $\Delta = -69$, $\det = -4$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 21$, $T_{\max} = 21$, $T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[4309] $1^{1^1} 3^{1^1} 151321812315131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19$, $T_{\max} = 19$, $\Delta = 0$, $\det = -4$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 19$, $T_{\max} = 19$, $T \in [19, 19]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

[4311] $31^{31} 6^{1^1} 1316131^1 6^{1^1} 131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19$, $T_{\max} = 19$, $\Delta = -3$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 19$, $T_{\max} = 19$, $T \in [19, 19]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 4 & 5 \\ 2 & 5 & 3 & 6 & 5 \\ 2 & 4 & 6 & 4 & 6 \\ 2 & 5 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

$T = 20$ [4312] $1^{1^3} 3^{1^1} 6^{1^1} 1^1 23218123151321^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20$, $T_{\max} = 20$, $\Delta = 0$, $\det = 0$, $(n_+, n_-, n_0) = (1, 20, 1)$, $c_{\text{ext}} = 4.0$

[4313] $1^{1^1} 6^{1^1} 1316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20$, $T_{\max} = 20$, $\Delta = -18$, $\det = 0$, $(n_+, n_-, n_0) = (1, 20, 1)$, $c_{\text{ext}} = 4.0$

[4314] $1^{1^1} 8^{1^1} 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20$, $T_{\max} = 20$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 20, 1)$, $c_{\text{ext}} = 4.0$

[4315] $1^{1^1} 8^{1^1} 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20$, $T_{\max} = 20$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 20, 1)$, $c_{\text{ext}} = 4.0$

[4316] $15^{1^1} 131613161316^{1^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20$, $T_{\max} = 20$, $\Delta = -18$, $\det = 0$, $(n_+, n_-, n_0) = (1, 20, 1)$, $c_{\text{ext}} = 4.0$

[4319] $1^{1^1} 8^{1^1} 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20$, $T_{\max} = 21$, $\Delta = -45$, $\det = 2$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 21$, $T_{\max} = 21$, $T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[4320] $1^1 \bar{5}^{11} 13221$
 $21 \quad (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 20, T_{\max} = 21, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$

[4321] $1^1 \bar{8}^{11} 123216131613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 20, T_{\max} = 22, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

$T = 21$ [4324] $1^1 \bar{8}^{11} 1232181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 21, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4323] $1^1 \bar{8}^{11} 12321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 2), c_{\text{ext}} = 4.0$

[4322] $1^1 \bar{8}^{11} 12321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4326] $1^1 \bar{5}^{11} 1321812321 \bar{8}^{11} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4327] $1^1 \bar{8}^{11} 1232181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 21, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4325] $1^1 \bar{5}^{11} 1321812321 \bar{8}^{11} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4336] $1^1 \bar{7}^{11} 1232171232181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 22, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4332] $1^1 \bar{8}^{11} 1231513161316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 22, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4333] $1^1 \bar{5}^{11} 13221$
 $1 \quad (12) \quad 122314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 22, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$

[4334] $1 \quad (12) \quad 12231 \bar{5}^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$
 $1^1 \bar{5}^{11} 13221$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$

[4335] $1^1 \bar{5}^{11} 13221$
 $21 \quad (12) \quad 12231 \bar{5}^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[4331] $1^{11} (12) 1223151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$

$$1^1 5 13221$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[4330] $21^{11} (12) 1223151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$

$$1^1 5 13221$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[4337] $1^{11} 8 1231513218123151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 21, T_{\max} = 23, \Delta = -68, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$

$$1^1 8 1231513218123151^1$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$T = 22$ [4339] $1^{11} 8^{12321} 123218123216131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 22, 2), c_{\text{ext}} = 4.0$

[4340] $1^{11} 8^{1321} 123218123216131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 22, T_{\max} = 22, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 4.0$

[4341] $1^{11} 5 13161316131613151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 22, T_{\max} = 22, \Delta = -12, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 4.0$

[4338] $1^{11} 6 123218123218^{1321} 12321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 22, T_{\max} = 22, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 4.0$

[4342] $21^{13221} (12) 12231513161^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 4.0$

$$1^1 3221$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[4343] $1^{1321812321} 8^{11} 1232171^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 22, T_{\max} = 23, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 4.0$

$$1^1 321812321$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[4344] $1^{11} 8 12321613161316131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 22, T_{\max} = 23, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 4.0$

$$1^1 8 12321613161316131^1$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[4345] $1^{11} 4141413221(12)1223151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 22, T_{\max} = 23, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 4.0$

$$1^1 4141413221(12)1223151^1$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 23 \quad [4346] \quad 1^1 \overset{1^1}{8} \overset{1^1}{1232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 23, 2), c_{\text{ext}} = 4.0 \\
& [4347] \quad 21 \overset{1^1}{7} \overset{1^1}{12321812321} \overset{1^1}{8} \overset{1^1}{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0 \\
& [4348] \quad 1^1 \overset{1^1}{8} \overset{1^1}{1232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0 \\
& [4349] \quad 21 \overset{1^1}{6} \overset{1^1}{131613161316131} \overset{1^1}{5} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0 \\
& [4353] \quad 21 \overset{1^1}{(12)} \overset{1^1}{1223151321} \overset{1^1}{7} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4352] \quad 1^1 \overset{1^1}{(12)} \overset{1^1}{12231513161} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4351] \quad 1^1 \overset{1^1}{5131513221} \overset{1^1}{(12)} \overset{1^1}{12231} \overset{1^1}{5} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 24, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4354] \quad 1^1 \overset{1^1}{712321} \overset{1^1}{8} \overset{1^1}{123218123151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 25, \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0} \\
& [4355] \quad 1^1 \overset{1^1}{8} \overset{1^1}{123217123218123151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 25, \Delta = -67, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
& T = 24 \quad [4356] \quad 221 \overset{1^1}{8} \overset{1^1}{12321812321} \overset{1^1}{8} \overset{1^1}{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0 \\
& [4357] \quad 1^1 \overset{1^1}{8} \overset{1^1}{12321812321} \overset{1^1}{8} \overset{1^1}{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0 \\
& [4358] \quad 1^1 \overset{1^1}{8} \overset{1^1}{12321812321812231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0 \\
& [4359] \quad 1^1 \overset{1^1}{8} \overset{1^1}{12321812321} \overset{1^1}{8} \overset{1^1}{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 2), c_{\text{ext}} = 4.0 \\
& [4360] \quad 1^1 \overset{1^1}{8} \overset{1^1}{12321812321812231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 2), c_{\text{ext}} = 4.0 \\
& [4362] \quad 1^1 \overset{1^1}{6} \overset{1^1}{131513221(12)} \overset{1^1}{122314141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 25, T_{\max} = 25, T \in [25, 25]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[4363] \quad 1^1 5 13221 (12) 122315131 6 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -40, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4364] \quad 1^1 5 13221 (12) 122315131 6 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -40, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[4365] \quad 21 (12) 12231513221 8 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -44, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4366] \quad 1^1 413221 (12) 1223151321 7 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -44, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4367] \quad 1^1 18 12321 8 123218123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 24, \quad T_{\max} = 26, \quad \Delta = -66, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[4368] \quad 1^1 8 12321 8 123218123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 24, \quad T_{\max} = 26, \quad \Delta = -66, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[4369] \quad 1^1 5 1321812321 (12) 12321 8 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 24, \quad T_{\max} = 26, \quad \Delta = -66, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 25 \quad [4370] \quad 1^1 321 8 123218123218123141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -22, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 25, 1), \quad c_{\text{ext}} = 4.0$$

$$[4371] \quad 1^1 2321 8 123218123218123141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -24, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 25, 2), \quad c_{\text{ext}} = 4.0$$

$$[4378] \quad 1^1 413221 (12) 12231513221 8 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -43, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[4376] $1^{\overline{1}} 7^{11} 1231513221(12)122314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$

[4377] $1^{\overline{1}} 8^{11} 12315132181232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4373] $1^{\overline{1}} 1^{13221} 122315131613161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4374] $1^{\overline{1}} 1^{1513221} 122315132171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$

[4375] $1^{\overline{1}} 1^{1513221} 122315132171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4379] $21^{\overline{1}} 1^{1513221} 122315132181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 27, \Delta = -67, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

$T = 26$ [4382] $1^{\overline{1}} 1^{1321} 8^{11} 123218123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 4.0$

[4384] $1^{\overline{1}} 1^{12321} 8^{11} 1232181232181232151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 2), c_{\text{ext}} = 4.0$

[4380] $1^{\overline{1}} 1^{15132181232181232181231^{\overline{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 4.0$

[4381] $1^{\overline{1}} 1^{1321} 8^{11} 1232181232181232151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 4.0$

[4383] $1^{\overline{1}} 1^{151321812321812321812321^{\overline{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 26, 2), c_{\text{ext}} = 4.0$

[4391] $21^{\overline{1}} 1^{181231513221} 1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 5 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 5 & 4 & 15 & 9 \\ 2 & 3 & 2 & 9 & 5 \end{pmatrix}_{b_0}$

$$\begin{aligned}
[4390] \quad & 1^1 5 13221(12)12231513221 8 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -38, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[4389] \quad & 1^1 5 1316131513221(12)12231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -42, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[4388] \quad & 1^1 6 131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -42, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[4387] \quad & 1^1 6 1316131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -44, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0} \\
[4386] \quad & 1^1 8 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -42, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0} \\
[4385] \quad & 1^1 5 13221(12)12231513221 8 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -38, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0} \\
T = 27 \quad [4392] \quad & 1^1 6 12321812321812321 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 27, T_{\max} = 27, \Delta = -16, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 4.0 \\
[4395] \quad & 1^1 5 13221(12)122315131613161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 27, T_{\max} = 28, \Delta = -39, \det = -2, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[4394] \quad & 1^1 6 1316131513221(12)12231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 27, T_{\max} = 28, \Delta = -39, \det = -2, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
[4396] \quad & 1^1 7 1231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 27, T_{\max} = 28, \Delta = -41, \det = -2, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[4393] $1^{\overline{1}} 8 1232171232181232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 27, T_{\max} = 28, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

$T = 28$ [4397] $1^{\overline{1}} 5 132181232181232181231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 28, T_{\max} = 28, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 4.0$

[4400] $1^{\overline{1}} 6 13161316131513221(12)12231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 28, T_{\max} = 29, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4398] $1^{\overline{1}} 5 132181231513221(12)12231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 28, T_{\max} = 29, \Delta = -41, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

[4402] $1^{\overline{1}} 3221 (12) 1223151321812321 7 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 28, T_{\max} = 29, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4399] $1^{\overline{1}} 321812321812321 8 12321 8 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 28, T_{\max} = 29, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[4401] $1^{\overline{1}} 8 12231513221(12)12231513151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 28, T_{\max} = 29, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

$T = 29$ [4403] $1^{\overline{1}} 6 1232181232181232181231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 29, T_{\max} = 29, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 4.0$

[4404] $1^{\overline{1}} 7 1232181231513221(12)1223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 29, T_{\max} = 30, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$

[4405] $1^{\overline{1}} 8 12321 8 123218123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 29, T_{\max} = 31, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

$T = 30$ [4406] $21^{\overline{1}} 7 1232181232181232181231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 30, T_{\max} = 30, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& [4407] \quad 1 \overset{1}{5} 13221(12)1223151321812321 \overset{1}{7} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 30, T_{\max} = 31, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4408] \quad 1 \overset{1}{7} 1232181231513221(12)12231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 30, T_{\max} = 31, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
& [4409] \quad 1 \overset{1}{5} 13221(12)1223151321812321 \overset{1}{8} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 30, T_{\max} = 32, \Delta = -66, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
& T = 31 \quad [4410] \quad 1 \overset{1}{8} 1232181232181232181231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 31, T_{\max} = 31, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 31, 2), c_{\text{ext}} = 4.0 \\
& [4411] \quad 221 \overset{1}{8} 1232181232181232181231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 31, T_{\max} = 31, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 4.0 \\
& [4412] \quad 21 \overset{1}{8} 1232181231513221(12)12231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 31, T_{\max} = 32, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0} \\
& T = 32 \quad [4413] \quad 1 \overset{1}{6} 12321812321812321812321 \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 32, T_{\max} = 32, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 32, 2), c_{\text{ext}} = 4.0 \\
& [4414] \quad 21 \overset{1}{5} 13221(12)12231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 32, T_{\max} = 33, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \quad \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
& T = 33 \quad [4416] \quad 1 \overset{1}{7} 1232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 33, T_{\max} = 34, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& T = 35 \quad [4417] \quad 1 \overset{1}{3} 221(12)12231513221(12)1223151321 \overset{1}{8} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 35, T_{\max} = 36, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4418] \quad 1 \overset{1}{8} 1232181232181231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 35, T_{\max} = 37, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 36 \quad [4419] \quad 221^1 8^{1^2} 1232181232181232181232181231^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 36, T_{\max} = 36, \Delta = -6, \det = 0, (n_+, n_-, n_0) = (1, 36, 2), c_{\text{ext}} = 4.0$$

$$[4421] \quad 1^{1^4} 413221(12)12231513221 \quad 122315132181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 36, T_{\max} = 37, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$T = 37 \quad [4424] \quad 1^1 5^{1^1} 13221(12)12231513221(12)122315132181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 37, T_{\max} = 38, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[4423] \quad 1^{1^1} 5^{1^1} 13221(12)12231513221 \quad 122315132181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 37, T_{\max} = 38, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 40 \quad [4427] \quad 1^{1^1} 3221 \quad 1(12)12231513221(12)1223151321812321^1 81^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 40, T_{\max} = 41, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 41 \quad [4428] \quad 1^{1^1} 8^{1^1} 1232181231513221(12)12231513221(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 41, T_{\max} = 42, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 42 \quad [4430] \quad 1^{1^1} 8^{1^1} 1232181231513221(12)12231513221(12)12231^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 42, T_{\max} = 43, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[4429] \quad 1^{1^1} 5^{1^1} 13221(12)12231513221(12)1223151321812321^1 81^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 42, T_{\max} = 43, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 44 \quad [4431] \quad 1^{1^1} 5^{1^1} 13221 \quad 1(12)12231513221(12)12231513221(12)12231^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 44, T_{\max} = 44, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 4.0$$

$$T = 46 \quad [4433] \quad 1 \quad \begin{matrix} 1^1 3221(12)12231513221 \\ (12) \end{matrix} \quad 12231513221(12)122315132181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 46, \quad T_{\max} = 47, \quad \Delta = -42, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 46, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 47, \quad T_{\max} = 47, \quad T \in [47, 47]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$T = 47 \quad [4434] \quad 1 \quad \begin{matrix} 1^1 5^1 13221 \\ (12) \end{matrix} \quad 12231513221(12)12231513221(12)12231513161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -9, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 48 \quad [4435] \quad 1 \quad \begin{matrix} 1^1 5^1 13221 \\ (12) \end{matrix} \quad 12231513221(12)12231513221(12)122315132171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 48, \quad T_{\max} = 48, \quad \Delta = -8, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 48, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 49 \quad [4436] \quad 21^1 8^1 1231513221(12)12231513221(12)12231513221(12)1223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -7, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 4.0$$

$$[4437] \quad 1 \quad \begin{matrix} 1^1 5^1 13221 \\ (12) \end{matrix} \quad 12231513221(12)12231513221(12)1223151322181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -7, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 4.0$$

4.761 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{e}_6$

$$T = 7 \quad [0] \quad 1^1 1^1 3^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -163, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.71 \rightarrow 16, \quad T_{\max} = 100, \quad T \in [16, 100]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 28 & 0 & 10 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 10 & -4 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^1 3^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -163, \quad \det = -23, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.87 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^1 3^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -163, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 8 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 21^1 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -163, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 12 & 32 & 94 & 12 & 16 \\ 3 & 4 & 12 & -3 & -1 \\ 3 & 5 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 21^1 31^{15} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -163, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 11.1$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 15 & 40 & 148 & 15 & 20 \\ 3 & 4 & 15 & -3 & -1 \\ 3 & 5 & 20 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -163, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -3 & -1 \\ 3 & 5 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{15} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -163, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 11.1$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 15 & 40 & 148 & 15 & 20 \\ 3 & 4 & 15 & -3 & -1 \\ 3 & 5 & 20 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -3 & -1 \\ 3 & 5 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 9 & 24 & 4 & 5 \\ 9 & 24 & 52 & 9 & 12 \\ 3 & 4 & 9 & -3 & -1 \\ 3 & 5 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 21^131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 9 & 24 & 4 & 5 \\ 9 & 24 & 52 & 9 & 12 \\ 3 & 4 & 9 & -3 & -1 \\ 3 & 5 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 21^131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -3 & -1 \\ 3 & 5 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 21^131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -3 & -1 \\ 3 & 5 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{114}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 12 & 32 & 94 & 12 & 16 \\ 3 & 4 & 12 & -3 & -1 \\ 3 & 5 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -54, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.15 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112}3^11^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -59, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.95 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 85 & 0 & 17 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 17 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{12}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -60, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{13}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{11}3^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 2 & 4 & 3 & -3 & -1 \\ 3 & 5 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{15}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 15 & -4 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{16}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.19 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 12 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 18 & -4 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{11}3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -146, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 2 & 4 & 6 & -3 & -1 \\ 3 & 5 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{17}3^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 14 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 21 & -4 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^14 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 17 & 40 & 172 & 0 & 24 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 24 & -4 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{11}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{112}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 24 & 0 & 5 \\ 10 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.31 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 8 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 19 & -4 & 0 \end{pmatrix}_{b_0}$$

[33] $13^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 9 & 24 & 4 & 5 \\ 9 & 24 & 52 & 9 & 12 \\ 2 & 4 & 9 & -3 & -1 \\ 3 & 5 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{13}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{12}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -309, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

[38] $21^{11}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -3 & -1 \\ 4 & 5 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -353, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 11 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{113}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -374, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 13 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 18 & -4 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{13}3^11^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -386, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 12 & 32 & 94 & 12 & 16 \\ 2 & 4 & 12 & -3 & -1 \\ 3 & 5 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{12}3^11^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -405, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 18 & -4 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{13}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 17 & -4 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{12}3^11^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -492, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 22 & -4 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{114}3^11^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -498, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 48 & 0 & 5 \\ 20 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{114}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -543, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 14 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 19 & -4 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{13}3^11^5 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -566, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 15 & 40 & 148 & 15 & 20 \\ 2 & 4 & 15 & -3 & -1 \\ 3 & 5 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{114}3^11^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -599, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 18 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 25 & -4 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{14}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{114}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -614, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 16 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 22 & -4 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{13}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[52] $21^{112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -622, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 9 & 16 & 4 & 5 \\ 12 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -3 & -1 \\ 4 & 5 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{13}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -774, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 18 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 25 & -4 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{14}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -837, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 17 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 24 & -4 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{14}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1060, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 20 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[56] $21^{113}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1072, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 9 & 24 & 4 & 5 \\ 18 & 24 & 52 & 9 & 12 \\ 3 & 4 & 9 & -3 & -1 \\ 4 & 5 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[57] $21^{114}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1702, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 9 & 32 & 4 & 5 \\ 24 & 32 & 94 & 12 & 16 \\ 3 & 4 & 12 & -3 & -1 \\ 4 & 5 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[58] $21^{115}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -2512, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 9 & 40 & 4 & 5 \\ 30 & 40 & 148 & 15 & 20 \\ 3 & 4 & 15 & -3 & -1 \\ 4 & 5 & 20 & -1 & 0 \end{pmatrix}_{b_0}$$

$T = 8$ [59] $1^13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.14 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 5 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & -2 \end{pmatrix}_{b_0}$$

[60] $1^13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 5 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[61] $2^22^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 5 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{11} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & -3 & -1 \\ 2 & 2 & -1 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & -3 & -1 \\ 2 & 2 & -1 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{14} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 98$, $T \in [16, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 6 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 16 & -4 & -3 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{11} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{11} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 91$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 2 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 4 & -4 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{11} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{13} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 5 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 13 & -4 & -1 \end{pmatrix}_{b_0}$$

[70] $1^{12} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{112} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 193$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 9 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$

[72] $1^{11} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 196$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 3 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{12}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 4 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{13}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 12 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{14}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{13}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{11}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{13}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 7 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 17 & -4 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{12}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{114}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 4 & 18 & 0 & 3 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 18 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & 20 & -4 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{13}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{11}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 802, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & 3 & -3 & -1 \\ 2 & 0 & 2 & 3 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 2 \\ 2 & 0 & 4 & 6 & 4 & 6 \\ 1 & 0 & 1 & 4 & -3 & -1 \\ 2 & 0 & 2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{11}4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & -3 & -1 \\ 2 & 3 & -1 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [99] \ 1^{11}5^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

4.762 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{e}_6$

$$T = 8 \quad [0] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 103, \Delta = -212, \det = 385, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.06 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

4.763 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 8 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 102, \ \Delta = -186, \ \det = 739, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.21 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 12 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{11} 3^1 2^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{f}_4 \quad T = 9, \ T_{\max} = 101, \ \Delta = -158, \ \det = -20, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 0 & -3 & -2 \end{pmatrix}_{b_0}$$

4.764 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -147, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -210, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 2 & 3 \\ 4 & 9 & 12 & 0 & 8 & 4 & 5 \\ 5 & 12 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 5 & 7 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -298, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -312, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 2 & 3 \\ 3 & 4 & 19 & 0 & 5 & 4 & 4 \\ 11 & 19 & 73 & 0 & 21 & 12 & 15 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 21 & 0 & 4 & 4 & 4 \\ 2 & 4 & 12 & 0 & 4 & 0 & 2 \\ 3 & 4 & 15 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -452, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -507, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 2 & 3 \\ 3 & 4 & 14 & 0 & 5 & 4 & 4 \\ 8 & 14 & 37 & 0 & 15 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 15 & 0 & 4 & 4 & 4 \\ 2 & 4 & 9 & 0 & 4 & 0 & 2 \\ 3 & 4 & 11 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -992, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 2 & 3 \\ 3 & 4 & 18 & 0 & 5 & 4 & 4 \\ 10 & 18 & 58 & 0 & 18 & 12 & 14 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 18 & 0 & 4 & 4 & 4 \\ 2 & 4 & 12 & 0 & 4 & 0 & 2 \\ 3 & 4 & 14 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{121} 3 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 & 2 & 3 \\ 3 & 4 & 28 & 0 & 5 & 4 & 4 \\ 16 & 28 & 154 & 0 & 30 & 18 & 22 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 30 & 0 & 4 & 4 & 4 \\ 2 & 4 & 18 & 0 & 4 & 0 & 2 \\ 3 & 4 & 22 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{121} 3 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1227, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 & 2 & 3 \\ 3 & 4 & 23 & 0 & 5 & 4 & 4 \\ 13 & 23 & 100 & 0 & 24 & 15 & 18 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 24 & 0 & 4 & 4 & 4 \\ 2 & 4 & 15 & 0 & 4 & 0 & 2 \\ 3 & 4 & 18 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{131} 3 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -1892, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 & 2 & 3 \\ 3 & 4 & 27 & 0 & 5 & 4 & 4 \\ 15 & 27 & 133 & 0 & 27 & 18 & 21 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 27 & 0 & 4 & 4 & 4 \\ 2 & 4 & 18 & 0 & 4 & 0 & 2 \\ 3 & 4 & 21 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{131} 3 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -2307, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 & 2 & 3 \\ 3 & 4 & 32 & 0 & 5 & 4 & 4 \\ 18 & 32 & 193 & 0 & 33 & 21 & 25 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 33 & 0 & 4 & 4 & 4 \\ 2 & 4 & 21 & 0 & 4 & 0 & 2 \\ 3 & 4 & 25 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{141} 3 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -3152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 & 2 & 3 \\ 3 & 4 & 36 & 0 & 5 & 4 & 4 \\ 20 & 36 & 238 & 0 & 36 & 24 & 28 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 36 & 0 & 4 & 4 & 4 \\ 2 & 4 & 24 & 0 & 4 & 0 & 2 \\ 3 & 4 & 28 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \ 1^{11} 3 1^{121} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 170, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 3 & 2 \\ 4 & 12 & 6 & 22 & 0 & 12 & 6 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 12 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 6 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & 8 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3 1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 440, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.765 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$$

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 7 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$4.766 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.767 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$$

$$T = 8 \quad [0] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{113}31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 45 & -4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 8 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 26 & -4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 12 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 16 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113}3^11^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 39 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 921, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113}3^11^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1005, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 33 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{115}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1053, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.1$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 20 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114}3^11^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 52 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114}3^11^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1590, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 46 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114}3^11^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1650, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 40 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [23] \ 1^{11}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

[24] $1^{111}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -151$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{111}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -571$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [27] $1^{111}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 60$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 3 \\ 4 & 10 & 1 & 16 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 16 & 0 & 22 & 8 & 10 \\ 2 & 4 & 0 & 8 & -2 & 0 \\ 3 & 6 & -1 & 10 & 0 & 1 \end{pmatrix}_{b_0}$$

4.768 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$

$T = 9$ [0] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -114$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -372$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 4 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}} 3 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -608$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}} 3 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -618$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

4.769 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10$ [0] $1^{\overline{111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 372$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.770 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$

$T = 9$ [0] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.771 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$

$T = 8$ [0] $1^{132}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.3$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 27 & 0 & 28 & 10 \\ 3 & 4 & 9 & -4 & 10 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{112}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$

$$T_{\min} = 21.15 \rightarrow 22, T_{\max} = 92, T \in [22, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 3 & -4 & 8 & -9 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{121} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 10 & 0 & 5 & 4 \\ 6 & 10 & 22 & 0 & 12 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 8 & -4 & 4 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^{21} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 5 & 3 \\ 4 & 9 & 16 & 0 & 12 & 5 \\ 7 & 16 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 21 & 0 & 13 & 7 \\ 3 & 5 & 10 & -4 & 7 & -3 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 41$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 3 & -4 & 4 & -3 \end{pmatrix}_{b_0}$$

[5] $1^{144}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 42$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 11 & 3 \\ 3 & 4 & 16 & 0 & 21 & 4 \\ 8 & 16 & 46 & 0 & 60 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 60 & 0 & 76 & 16 \\ 3 & 4 & 12 & -4 & 16 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{133}31^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.6$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 27 & 4 \\ 9 & 17 & 49 & 0 & 81 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 81 & 0 & 133 & 21 \\ 3 & 4 & 13 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 3 & -4 & 7 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{124}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 8 & 0 & 31 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 42 & 0 & 172 & 24 \\ 3 & 4 & 6 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[13] $21^131^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -3 & 3 & -1 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{1111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -3 & 3 & -1 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 3 & -4 & 14 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{11}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 10 & 0 & 14 & 4 \\ 6 & 10 & 22 & 0 & 30 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 30 & 0 & 37 & 11 \\ 3 & 4 & 8 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 15.23 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 3 & -4 & 21 & -3 \end{pmatrix}_{b_0}$$

[20] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 3 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 24 & 42 & 0 & 121 & 19 \\ 3 & 4 & 7 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{133}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 9 & 3 \\ 3 & 4 & 12 & 0 & 17 & 4 \\ 6 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 36 & 0 & 49 & 13 \\ 3 & 4 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{133}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 12 & 0 & 12 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 2 & 4 & 3 & -3 & 3 & -1 \\ 4 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{114}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.48 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 3 & -4 & 28 & -3 \end{pmatrix}_{b_0}$$

[35] $1^{112}3^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 \\ 8 & 14 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 33 & 0 & 28 & 10 \\ 3 & 4 & 11 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{123}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 6 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{134}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 12 & 0 & 16 & 4 \\ 6 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 36 & 0 & 46 & 12 \\ 3 & 4 & 9 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{115}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 15 & 0 & 73 & 15 \\ 3 & 4 & 3 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 2 & 4 & 3 & -3 & 3 & -1 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.28 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 3 & -4 & 15 & -2 \end{pmatrix}_{b_0}$$

[42] $1^{125}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 30 & 0 & 73 & 15 \\ 3 & 4 & 6 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{144}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 16 & 0 & 16 & 4 \\ 8 & 16 & 46 & 0 & 48 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 48 & 0 & 46 & 12 \\ 3 & 4 & 12 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{1111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 9 & 12 & 0 & 12 & 5 \\ 5 & 12 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 5 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{135}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 12 & 0 & 20 & 4 \\ 6 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 45 & 0 & 73 & 15 \\ 3 & 4 & 9 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{116}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 18 & 0 & 106 & 18 \\ 3 & 4 & 3 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 3 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{126}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 36 & 0 & 106 & 18 \\ 3 & 4 & 6 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{11}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{112}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 3 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{145}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 16 & 0 & 20 & 4 \\ 8 & 16 & 46 & 0 & 60 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 60 & 0 & 73 & 15 \\ 3 & 4 & 12 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.81 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 3 & -4 & 19 & -2 \end{pmatrix}_{b_0}$$

[53] $1^{136}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 12 & 0 & 24 & 4 \\ 6 & 12 & 25 & 0 & 54 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 54 & 0 & 106 & 18 \\ 3 & 4 & 9 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{155}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 20 & 0 & 20 & 4 \\ 10 & 20 & 73 & 0 & 75 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 75 & 0 & 73 & 15 \\ 3 & 4 & 15 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{117}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 21 & 0 & 145 & 21 \\ 3 & 4 & 3 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{146}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 16 & 0 & 24 & 4 \\ 8 & 16 & 46 & 0 & 72 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 72 & 0 & 106 & 18 \\ 3 & 4 & 12 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{127}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 28 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 42 & 0 & 145 & 21 \\ 3 & 4 & 6 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 18 & 4 \\ 9 & 17 & 49 & 0 & 54 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 54 & 0 & 58 & 14 \\ 3 & 4 & 13 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{134}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 329, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 11 & 3 \\ 3 & 4 & 12 & 0 & 21 & 4 \\ 6 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 45 & 0 & 76 & 16 \\ 3 & 4 & 9 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{11}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 21 & 0 & 73 & 15 \\ 3 & 4 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 33 & 0 & 73 & 15 \\ 3 & 4 & 7 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{137}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 14 & 3 \\ 3 & 4 & 12 & 0 & 28 & 4 \\ 6 & 12 & 25 & 0 & 63 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 63 & 0 & 145 & 21 \\ 3 & 4 & 9 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{156}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 20 & 0 & 24 & 4 \\ 10 & 20 & 73 & 0 & 90 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 90 & 0 & 106 & 18 \\ 3 & 4 & 15 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{122}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 7 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{147}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 14 & 3 \\ 3 & 4 & 16 & 0 & 28 & 4 \\ 8 & 16 & 46 & 0 & 84 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 84 & 0 & 145 & 21 \\ 3 & 4 & 12 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[69] $1^{11}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 24 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 24 & 27 & 0 & 121 & 19 \\ 3 & 4 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{143}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 \\ 11 & 21 & 76 & 0 & 81 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 81 & 0 & 85 & 17 \\ 3 & 4 & 16 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{166}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 24 & 0 & 24 & 4 \\ 12 & 24 & 106 & 0 & 108 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 108 & 0 & 106 & 18 \\ 3 & 4 & 18 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 3 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{157}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 20 & 0 & 28 & 4 \\ 10 & 20 & 73 & 0 & 105 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 105 & 0 & 145 & 21 \\ 3 & 4 & 15 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{113}3^11^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 8 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 405, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 5 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{112}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{114}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 422, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 3 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{167}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 24 & 0 & 28 & 4 \\ 12 & 24 & 106 & 0 & 126 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 126 & 0 & 145 & 21 \\ 3 & 4 & 18 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{14}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 466, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{13}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{113}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{177}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 28 & 0 & 28 & 4 \\ 14 & 28 & 145 & 0 & 147 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 147 & 0 & 145 & 21 \\ 3 & 4 & 21 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{124}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{124}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{111}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 8 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{12}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{112}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 8 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 657, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{123}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 697, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 18 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 7 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{122}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 702, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 7 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{112}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 734, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{12}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 790, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{124}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 792, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{13}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 819, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 837, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[105] $21^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 884, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 6 & 4 \\ 8 & 9 & 8 & 4 & 8 & 5 \\ 6 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -3 & 3 & -1 \\ 6 & 8 & 6 & 3 & 4 & 4 \\ 4 & 5 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{112}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 891, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{133}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 9 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{112}3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 930, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 5 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{14}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{133}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 964, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 9 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{134}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 979, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 17 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 9 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{113}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{124}3^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 10 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{114}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{144}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1074, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 11 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[116] $1^{134}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1083, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 9 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{123} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{122} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 10 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{13} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{123} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{114} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{123} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 10 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{144} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 11 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{113} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 9 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{12} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1674, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 20 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 12 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1755, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 7 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{134} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1816, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 12 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 2246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{144} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 2296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 14 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [138] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19.50 \rightarrow 20, T_{\max} = 23, T \in [20, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 \\ 2 & 3 & 3 & 3 & -4 & 3 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{111} 3^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 & 0 & 3 \\ 1 & 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 3 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{113} 1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{122} 3^1 1^{13} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 8 & 2 \\ 2 & 4 & 4 & 12 & 0 & 24 & 4 \\ 1 & 4 & 0 & 9 & 0 & 15 & 2 \\ 4 & 12 & 9 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 54 & 0 & 100 & 18 \\ 2 & 4 & 2 & 10 & -4 & 18 & -2 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{114} 3^1 1^{12} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 1 & 3 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 18 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[143] $1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 3 & 5 \\ 1 & 1 & 0 & 4 & -3 & 1 & -1 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 2 & -1 & 5 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[144] $1^{112}3^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[145] $1^{111}3^{11}431^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & 3 & -4 & 19 & -2 \end{pmatrix}_{b_0}$$

[146] $1^{14}3^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 6 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 30 & 0 & 172 & 24 \\ 2 & 4 & 2 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[147] $1^{113}3^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.05 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 3 & 0 & 15 & 3 \\ 1 & 3 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 12 & 0 & 49 & 13 \\ 2 & 3 & 3 & 3 & -4 & 13 & -2 \end{pmatrix}_{b_0}$$

[148] $1^{111}3^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -179$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 3 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[149] $1^{111}3^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -185$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[150] $1^{11}3^{11}1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 0 & 12 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 18 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 18 & 0 & 22 & 8 \\ 2 & 4 & 2 & 7 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

[151] $1^{112}3^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -233$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 12 & 3 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 3 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{\overline{11}1}31^{\overline{13}31^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -257, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.20 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & 3 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{\overline{11}1}31^{\overline{12}31^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -284, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.89 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & 3 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{\overline{1}31^{\overline{11}1}31^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{\overline{11}3}31^{\overline{12}31^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 15 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & 3 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{\overline{11}3}31^{\overline{11}31^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -389, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{\overline{11}4}31^{\overline{12}2}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -392, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 8 & 2 \\ 2 & 4 & 4 & 9 & 0 & 24 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 18 & 42 & 0 & 118 & 20 \\ 2 & 4 & 2 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{\overline{12}1}31^{\overline{11}1}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -437, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 3 & 2 \\ 2 & 4 & 4 & 12 & 0 & 9 & 4 \\ 1 & 4 & 0 & 9 & 0 & 6 & 2 \\ 4 & 12 & 9 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 21 & 0 & 13 & 7 \\ 2 & 4 & 2 & 10 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{\overline{12}2}31^{\overline{12}31^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{\overline{12}2}31^{\overline{11}1}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 0 & 12 & 4 \\ 1 & 4 & 0 & 9 & 0 & 9 & 2 \\ 4 & 12 & 9 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 30 & 0 & 28 & 10 \\ 2 & 4 & 2 & 10 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[161] \ 1^1 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -584, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.89 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 12 & 0 & 22 & 8 \\ 2 & 4 & 2 & 4 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{12} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -668, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 18 & 0 & 58 & 14 \\ 2 & 4 & 2 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{111} 31^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -680, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 13 & 7 \\ 2 & 4 & 2 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{11} 31^{122} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -704, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 0 & 15 & 4 \\ 1 & 4 & 0 & 6 & 0 & 9 & 2 \\ 4 & 12 & 6 & 22 & 0 & 30 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 30 & 0 & 37 & 11 \\ 2 & 4 & 2 & 8 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{122} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -716, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 0 & 18 & 4 \\ 1 & 4 & 0 & 9 & 0 & 12 & 2 \\ 4 & 12 & 9 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 42 & 0 & 58 & 14 \\ 2 & 4 & 2 & 10 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{112} 31^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -725, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[167] \ 1^1 31^{122} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -800, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 0 & 12 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 4 & 12 & 6 & 22 & 0 & 24 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 24 & 0 & 22 & 8 \\ 2 & 4 & 2 & 8 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{12} 31^{14} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -860, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 1 & 3 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 18 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & 3 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{114} 31^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -869, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 1 & 3 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{11}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -893, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 15 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{113}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -929, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 1 & 3 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{13}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -944, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 9 & 18 & 0 & 52 & 12 \\ 2 & 4 & 2 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{111}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1049, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 0 & 15 & 4 \\ 1 & 4 & 0 & 6 & 0 & 9 & 2 \\ 3 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 24 & 0 & 37 & 11 \\ 2 & 4 & 2 & 7 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{13}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1133, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 27 & 0 & 133 & 21 \\ 2 & 4 & 2 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{113}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1145, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 7 & 2 \\ 2 & 4 & 4 & 9 & 0 & 21 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 15 & 36 & 0 & 85 & 17 \\ 2 & 4 & 2 & 7 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{111}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 7 & 2 \\ 2 & 4 & 4 & 9 & 0 & 21 & 4 \\ 1 & 4 & 0 & 6 & 0 & 12 & 2 \\ 3 & 9 & 6 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 33 & 0 & 73 & 15 \\ 2 & 4 & 2 & 7 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{112}31^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1328, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 6 & 2 \\ 2 & 4 & 4 & 15 & 0 & 18 & 4 \\ 1 & 4 & 0 & 9 & 0 & 12 & 2 \\ 5 & 15 & 9 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 48 & 0 & 58 & 14 \\ 2 & 4 & 2 & 11 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{113}31^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1352, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 3 & 0 & 33 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 1 & 3 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 21 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & 3 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{112}31^{112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1364, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 6 & 2 \\ 2 & 4 & 4 & 9 & 0 & 18 & 4 \\ 1 & 4 & 0 & 6 & 0 & 12 & 2 \\ 3 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 30 & 0 & 58 & 14 \\ 2 & 4 & 2 & 7 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{111}31^{114}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1409, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 9 & 2 \\ 2 & 4 & 4 & 9 & 0 & 27 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 42 & 0 & 121 & 19 \\ 2 & 4 & 2 & 7 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[181] \ 1^131^{114}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1448, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 12 & 24 & 0 & 94 & 16 \\ 2 & 4 & 2 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{111}31^{113}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1541, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 21 & 0 & 73 & 15 \\ 2 & 4 & 2 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{114}31^{143}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1580, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 3 & 0 & 36 & 4 \\ 1 & 4 & 0 & 3 & 0 & 24 & 2 \\ 1 & 3 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{123}31^{113}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1604, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 24 & 0 & 100 & 18 \\ 2 & 4 & 2 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{111}31^{122}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1688, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 5 & 2 \\ 2 & 4 & 4 & 15 & 0 & 15 & 4 \\ 1 & 4 & 0 & 9 & 0 & 9 & 2 \\ 5 & 15 & 9 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 39 & 0 & 37 & 11 \\ 2 & 4 & 2 & 11 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{143}31^{114}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1784, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 0 & 36 & 4 \\ 1 & 4 & 0 & 3 & 0 & 24 & 2 \\ 2 & 6 & 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 36 & 0 & 238 & 28 \\ 2 & 4 & 2 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{122}31^{122}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -2048, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 6 & 2 \\ 2 & 4 & 4 & 18 & 0 & 18 & 4 \\ 1 & 4 & 0 & 12 & 0 & 12 & 2 \\ 6 & 18 & 12 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 60 & 0 & 58 & 14 \\ 2 & 4 & 2 & 14 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 10 \quad [197] \quad 1^{12} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 20, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 11 & 4 \\ 3 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 7 & 0 & 21 & 9 \\ 3 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0} \\
& [198] \quad 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 11.0 \\
& T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0} \\
& [199] \quad 1^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 11.0 \\
& T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0} \\
& [200] \quad 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 11.0 \\
& T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 3 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0} \\
& T = 11 \quad [201] \quad 1^{11} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -16, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 14.25 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0} \\
& [202] \quad 1^{11} 5^{12} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = \\
& 11.0 \\
& T_{\min} = 14.16 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0} \\
& [203] \quad 1^{11} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = \\
& 11.0 \\
& T_{\min} = 14.33 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0} \\
& [204] \quad 1^{11} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = \\
& 11.0 \\
& T_{\min} = 14.19 \rightarrow 15, T_{\max} = 18, T \in [15, 18] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0} \\
& [205] \quad 1^{12} 5^{13} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = \\
& 11.5
\end{aligned}$$

$$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 7 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{12}_{1^1} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 5 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{13}_{1^1} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -109, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[208] \ 1^{12}_{1^1} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 6 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[209] \ 1^{13}_{1^1} 5^{12} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -199, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[210] \ 1^{13}_{1^1} 5^{13} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -221, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

4.772 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11}_{1^1} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 6 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}_{1^1} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 6 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}_{1^1} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 9 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & 7 & -4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.773 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -3 & 3 & 0 & -1 \\ 4 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -3 & 3 & 0 & -1 \\ 3 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$4.774 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2$$

$$T = 10 \quad [0] 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$4.775 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$$

$$T = 9 \quad [0] 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 18.25 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 3 & -4 & 3 & 18 & -6 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 15 & 7 \\ 5 & 9 & 12 & 0 & 15 & 13 & 7 \\ 3 & 4 & 6 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 5 & 0 & 17 & 9 & 4 \\ 3 & 5 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 27 & 13 \\ 5 & 9 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & 4 & -4 & 13 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] 1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 12 & 3 \\ 3 & 4 & 4 & 0 & 8 & 22 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 30 & 6 \\ 12 & 22 & 15 & 0 & 30 & 85 & 17 \\ 3 & 4 & 3 & -4 & 6 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1211} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -4 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -58, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -58, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 17.66 \rightarrow 18, \ T_{\max} = 21, \ T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 5 & 0 & 4 & 28 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 16 & 28 & 30 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -4 & 3 & 22 & -5 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.50 \rightarrow 17, \ T_{\max} = 21, \ T \in [17, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 3 & -4 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 14 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 15 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 3 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1113} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 12 & 3 \\ 3 & 4 & 4 & 0 & 14 & 22 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 57 & 11 \\ 12 & 22 & 15 & 0 & 57 & 85 & 17 \\ 3 & 4 & 3 & -4 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1222} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 7 & 3 \\ 3 & 4 & 8 & 0 & 8 & 13 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 18 & 6 \\ 7 & 13 & 18 & 0 & 18 & 28 & 10 \\ 3 & 4 & 6 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3^1 1^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 6 & 4 \\ 3 & 5 & 6 & 0 & 6 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.47 \rightarrow 15, \quad T_{\max} = 21, \quad T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 15 & 7 \\ 5 & 12 & 6 & 0 & 15 & 13 & 7 \\ 3 & 5 & 3 & -4 & 7 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1214} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -150, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 11 & 3 \\ 3 & 4 & 8 & 0 & 9 & 21 & 4 \\ 4 & 8 & 10 & 0 & 12 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 33 & 7 \\ 11 & 21 & 30 & 0 & 33 & 76 & 16 \\ 3 & 4 & 6 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -170, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1213} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -194, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 12 & 3 \\ 3 & 4 & 8 & 0 & 9 & 22 & 4 \\ 4 & 8 & 10 & 0 & 12 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 36 & 7 \\ 12 & 22 & 30 & 0 & 36 & 85 & 17 \\ 3 & 4 & 6 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 5 & 0 & 5 & 9 & 4 \\ 3 & 5 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1212} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -230, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 7 & 3 \\ 3 & 4 & 8 & 0 & 9 & 13 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 21 & 7 \\ 7 & 13 & 18 & 0 & 21 & 28 & 10 \\ 3 & 4 & 6 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 3 & -4 & 3 & 25 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{1111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -232$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{1133}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 9 & 3 \\ 3 & 4 & 4 & 0 & 12 & 17 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 36 & 9 \\ 9 & 17 & 12 & 0 & 36 & 49 & 13 \\ 3 & 4 & 3 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{1112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 10 & 6 \\ 3 & 4 & 3 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{1213}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 9 & 3 \\ 3 & 4 & 8 & 0 & 9 & 17 & 4 \\ 4 & 8 & 10 & 0 & 12 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 27 & 7 \\ 9 & 17 & 24 & 0 & 27 & 49 & 13 \\ 3 & 4 & 6 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{1122}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 4 & 8 & 6 & 0 & 12 & 10 & 6 \\ 3 & 4 & 3 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{133}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 51 & 13 \\ 9 & 17 & 15 & 0 & 51 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{1114}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 11 & 3 \\ 3 & 4 & 9 & 0 & 9 & 21 & 4 \\ 5 & 9 & 13 & 0 & 15 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 33 & 7 \\ 11 & 21 & 33 & 0 & 33 & 76 & 16 \\ 3 & 4 & 7 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{1431}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 2 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 & 4 \\ 11 & 21 & 76 & 0 & 81 & 15 & 16 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 81 & 0 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 15 & 1 & 3 \\ 3 & 4 & 16 & -4 & 17 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.65 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 6 & 12 & 9 & 0 & 9 & 25 & 9 \\ 3 & 4 & 3 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1222} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.68 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 12 & 6 \\ 4 & 8 & 12 & 0 & 10 & 12 & 6 \\ 3 & 4 & 6 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1123} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -320, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 12 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 18 & 6 \\ 6 & 12 & 9 & 0 & 18 & 25 & 9 \\ 3 & 4 & 3 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{132} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -328, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 5 & 0 & 17 & 13 & 4 \\ 3 & 5 & 4 & 0 & 15 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 39 & 13 \\ 7 & 13 & 12 & 0 & 39 & 28 & 10 \\ 3 & 4 & 4 & -4 & 13 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1112} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 4 & 0 & 9 & 28 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 48 & 7 \\ 16 & 28 & 18 & 0 & 48 & 154 & 22 \\ 3 & 4 & 3 & -4 & 7 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -340, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 5 & 0 & 4 & 23 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 23 & 24 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1223} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.84 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 6 & 3 \\ 3 & 4 & 8 & 0 & 8 & 12 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 18 & 6 \\ 6 & 12 & 18 & 0 & 18 & 25 & 9 \\ 3 & 4 & 6 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{1122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -346, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 7 & 3 \\ 3 & 4 & 4 & 0 & 8 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 18 & 6 \\ 7 & 13 & 9 & 0 & 18 & 28 & 10 \\ 3 & 4 & 3 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{1114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -352, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 16 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 8 & 16 & 12 & 0 & 12 & 46 & 12 \\ 3 & 4 & 3 & -4 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{1133} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -360, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.89 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 27 & 9 \\ 6 & 12 & 9 & 0 & 27 & 25 & 9 \\ 3 & 4 & 3 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{1111} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 5 & 3 \\ 3 & 4 & 4 & 0 & 14 & 9 & 4 \\ 2 & 4 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 24 & 11 \\ 5 & 9 & 6 & 0 & 24 & 13 & 7 \\ 3 & 4 & 3 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{1124} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 8 & 3 \\ 3 & 4 & 4 & 0 & 8 & 16 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 24 & 6 \\ 8 & 16 & 12 & 0 & 24 & 46 & 12 \\ 3 & 4 & 3 & -4 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{1111} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 39 & 11 \\ 8 & 14 & 9 & 0 & 39 & 37 & 11 \\ 3 & 4 & 3 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{112} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -386, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{1224} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -400, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 8 & 3 \\ 3 & 4 & 8 & 0 & 8 & 16 & 4 \\ 4 & 8 & 10 & 0 & 12 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 24 & 6 \\ 8 & 16 & 24 & 0 & 24 & 46 & 12 \\ 3 & 4 & 6 & -4 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{1223}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -412$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 9 & 3 \\ 3 & 4 & 8 & 0 & 8 & 17 & 4 \\ 4 & 8 & 10 & 0 & 12 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 24 & 6 \\ 9 & 17 & 24 & 0 & 24 & 49 & 13 \\ 3 & 4 & 6 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{1134}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 16 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 36 & 9 \\ 8 & 16 & 12 & 0 & 36 & 46 & 12 \\ 3 & 4 & 3 & -4 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{1115}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$

$T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 20 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 10 & 20 & 15 & 0 & 15 & 73 & 15 \\ 3 & 4 & 3 & -4 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{112}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 5 & 0 & 9 & 13 & 4 \\ 3 & 5 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 12 & 0 & 21 & 28 & 10 \\ 3 & 4 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{1125}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 10 & 3 \\ 3 & 4 & 4 & 0 & 8 & 20 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 30 & 6 \\ 10 & 20 & 15 & 0 & 30 & 73 & 15 \\ 3 & 4 & 3 & -4 & 6 & 15 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{1144}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -472$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 12 & 0 & 46 & 48 & 12 \\ 8 & 16 & 12 & 0 & 48 & 46 & 12 \\ 3 & 4 & 3 & -4 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{1135}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -488$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.9$

$T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 10 & 3 \\ 3 & 4 & 4 & 0 & 12 & 20 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 45 & 9 \\ 10 & 20 & 15 & 0 & 45 & 73 & 15 \\ 3 & 4 & 3 & -4 & 9 & 15 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{1232}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -490$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 7 & 3 \\ 3 & 4 & 8 & 0 & 22 & 13 & 4 \\ 4 & 8 & 10 & 0 & 30 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 30 & 0 & 85 & 51 & 17 \\ 7 & 13 & 18 & 0 & 51 & 28 & 10 \\ 3 & 4 & 6 & -4 & 17 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -490, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 5 & 0 & 4 & 26 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 14 & 26 & 24 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1112} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 3 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1116} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 24 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 12 & 24 & 18 & 0 & 18 & 106 & 18 \\ 3 & 4 & 3 & -4 & 3 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 21 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1214} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 14 & 3 \\ 3 & 4 & 8 & 0 & 9 & 26 & 4 \\ 4 & 8 & 10 & 0 & 12 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 42 & 7 \\ 14 & 26 & 36 & 0 & 42 & 118 & 20 \\ 3 & 4 & 6 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1126} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -536, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 12 & 3 \\ 3 & 4 & 4 & 0 & 8 & 24 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 36 & 6 \\ 12 & 24 & 18 & 0 & 36 & 106 & 18 \\ 3 & 4 & 3 & -4 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -556$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 30 & 10 \\ 7 & 13 & 12 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{1134}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 11 & 3 \\ 3 & 4 & 4 & 0 & 12 & 21 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 45 & 9 \\ 11 & 21 & 15 & 0 & 45 & 76 & 16 \\ 3 & 4 & 3 & -4 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{1112}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -584$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 10 & 3 \\ 3 & 4 & 4 & 0 & 14 & 18 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 48 & 11 \\ 10 & 18 & 12 & 0 & 48 & 58 & 14 \\ 3 & 4 & 3 & -4 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{1112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1117}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -616$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 28 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 14 & 28 & 21 & 0 & 21 & 145 & 21 \\ 3 & 4 & 3 & -4 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{1222}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -616$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 30 & 10 \\ 7 & 13 & 18 & 0 & 30 & 28 & 10 \\ 3 & 4 & 6 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{1124}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -640$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 14 & 3 \\ 3 & 4 & 4 & 0 & 8 & 26 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 36 & 6 \\ 14 & 26 & 18 & 0 & 36 & 118 & 20 \\ 3 & 4 & 3 & -4 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{1127}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -640$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 14 & 3 \\ 3 & 4 & 4 & 0 & 8 & 28 & 4 \\ 2 & 4 & 1 & 0 & 6 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 42 & 6 \\ 14 & 28 & 21 & 0 & 42 & 145 & 21 \\ 3 & 4 & 3 & -4 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

[69] $1^{1123}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -658$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 9 & 3 \\ 3 & 4 & 4 & 0 & 8 & 17 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 24 & 6 \\ 9 & 17 & 12 & 0 & 24 & 49 & 13 \\ 3 & 4 & 3 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1114}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -666$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 21 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 33 & 7 \\ 11 & 21 & 15 & 0 & 33 & 76 & 16 \\ 3 & 4 & 3 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{113}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.66 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 5 & 0 & 4 & 32 & 4 \\ 3 & 5 & 4 & 0 & 3 & 33 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 32 & 33 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1224}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -740$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 11 & 3 \\ 3 & 4 & 8 & 0 & 8 & 21 & 4 \\ 4 & 8 & 10 & 0 & 12 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 30 & 6 \\ 11 & 21 & 30 & 0 & 30 & 76 & 16 \\ 3 & 4 & 6 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{1112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -746$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 21 & 7 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -746$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.47 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{1113}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -788$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 27 & 7 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 3 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1223}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -838$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 9 & 3 \\ 3 & 4 & 8 & 0 & 13 & 17 & 4 \\ 4 & 8 & 10 & 0 & 18 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 39 & 10 \\ 9 & 17 & 24 & 0 & 39 & 49 & 13 \\ 3 & 4 & 6 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1113} 3^1 1^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -852, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 9 & 3 \\ 3 & 4 & 9 & 0 & 9 & 17 & 4 \\ 5 & 9 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 27 & 7 \\ 9 & 17 & 27 & 0 & 27 & 49 & 13 \\ 3 & 4 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -872, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 3 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1112} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -890, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 13 & 3 \\ 3 & 4 & 4 & 0 & 9 & 23 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 39 & 7 \\ 13 & 23 & 15 & 0 & 39 & 100 & 18 \\ 3 & 4 & 3 & -4 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{1224} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 11 & 3 \\ 3 & 4 & 8 & 0 & 13 & 21 & 4 \\ 4 & 8 & 10 & 0 & 18 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 48 & 10 \\ 11 & 21 & 30 & 0 & 48 & 76 & 16 \\ 3 & 4 & 6 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1113} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{1142} 3^1 1^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -936, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 7 & 3 \\ 3 & 4 & 9 & 0 & 21 & 13 & 4 \\ 5 & 9 & 13 & 0 & 33 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 48 & 16 \\ 7 & 13 & 21 & 0 & 48 & 28 & 10 \\ 3 & 4 & 7 & -4 & 16 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1122} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 10 & 3 \\ 3 & 4 & 4 & 0 & 13 & 18 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 42 & 10 \\ 10 & 18 & 12 & 0 & 42 & 58 & 14 \\ 3 & 4 & 3 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{1111} 3^1 1^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -964, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 15 & 7 \\ 5 & 9 & 15 & 0 & 15 & 13 & 7 \\ 3 & 4 & 7 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1124} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 11 & 3 \\ 3 & 4 & 4 & 0 & 8 & 21 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 30 & 6 \\ 11 & 21 & 15 & 0 & 30 & 76 & 16 \\ 3 & 4 & 3 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -988, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 30 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1026, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 4 & 0 & 9 & 18 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 30 & 7 \\ 10 & 18 & 12 & 0 & 30 & 58 & 14 \\ 3 & 4 & 3 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{1134} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1066, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 17 & 3 \\ 3 & 4 & 4 & 0 & 17 & 31 & 4 \\ 2 & 4 & 1 & 0 & 12 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 93 & 13 \\ 17 & 31 & 21 & 0 & 93 & 172 & 24 \\ 3 & 4 & 3 & -4 & 13 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1080, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 7 & 3 \\ 3 & 4 & 9 & 0 & 9 & 13 & 4 \\ 5 & 9 & 13 & 0 & 15 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 21 & 7 \\ 7 & 13 & 21 & 0 & 21 & 28 & 10 \\ 3 & 4 & 7 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{1113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 3 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{1122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 30 & 10 \\ 7 & 13 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 3 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{1224} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1180, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 14 & 3 \\ 3 & 4 & 8 & 0 & 13 & 26 & 4 \\ 4 & 8 & 10 & 0 & 18 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 60 & 10 \\ 14 & 26 & 36 & 0 & 60 & 118 & 20 \\ 3 & 4 & 6 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{1143} 31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 78 & 16 \\ 11 & 21 & 33 & 0 & 78 & 76 & 16 \\ 3 & 4 & 7 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{1114} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 4 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 11 & 21 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{1143} 31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1248, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 9 & 3 \\ 3 & 4 & 9 & 0 & 21 & 17 & 4 \\ 5 & 9 & 13 & 0 & 33 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 63 & 16 \\ 9 & 17 & 27 & 0 & 63 & 49 & 13 \\ 3 & 4 & 7 & -4 & 16 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{114} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1250, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{1124} 31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 14 & 3 \\ 3 & 4 & 4 & 0 & 18 & 26 & 4 \\ 2 & 4 & 1 & 0 & 12 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 84 & 14 \\ 14 & 26 & 18 & 0 & 84 & 118 & 20 \\ 3 & 4 & 3 & -4 & 14 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{1123} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 15 & 3 \\ 3 & 4 & 4 & 0 & 13 & 27 & 4 \\ 2 & 4 & 1 & 0 & 9 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 63 & 10 \\ 15 & 27 & 18 & 0 & 63 & 133 & 21 \\ 3 & 4 & 3 & -4 & 10 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{1123} 31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1328, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 9 & 3 \\ 3 & 4 & 9 & 0 & 13 & 17 & 4 \\ 5 & 9 & 13 & 0 & 21 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 39 & 10 \\ 9 & 17 & 27 & 0 & 39 & 49 & 13 \\ 3 & 4 & 7 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{1123} 31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1354, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 17 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 39 & 10 \\ 9 & 17 & 12 & 0 & 39 & 49 & 13 \\ 3 & 4 & 3 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{1114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 4 & 1 & 0 & 3 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 3 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{1122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 30 & 10 \\ 7 & 13 & 21 & 0 & 30 & 28 & 10 \\ 3 & 4 & 7 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{1133} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 12 & 3 \\ 3 & 4 & 4 & 0 & 17 & 22 & 4 \\ 2 & 4 & 1 & 0 & 12 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 66 & 13 \\ 12 & 22 & 15 & 0 & 66 & 85 & 17 \\ 3 & 4 & 3 & -4 & 13 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{1124} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 11 & 3 \\ 3 & 4 & 4 & 0 & 13 & 21 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 48 & 10 \\ 11 & 21 & 15 & 0 & 48 & 76 & 16 \\ 3 & 4 & 3 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{1133} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 27 & 0 & 49 & 51 & 13 \\ 9 & 17 & 27 & 0 & 51 & 49 & 13 \\ 3 & 4 & 7 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{1122} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 60 & 14 \\ 10 & 18 & 12 & 0 & 60 & 58 & 14 \\ 3 & 4 & 3 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{1113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 12 & 3 \\ 3 & 4 & 4 & 0 & 9 & 22 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 36 & 7 \\ 12 & 22 & 15 & 0 & 36 & 85 & 17 \\ 3 & 4 & 3 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[108] \quad 1^{1114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 14 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 3 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{1144}3^11^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1706$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 14 & 3 \\ 3 & 4 & 4 & 0 & 21 & 26 & 4 \\ 2 & 4 & 1 & 0 & 15 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 96 & 16 \\ 14 & 26 & 18 & 0 & 96 & 118 & 20 \\ 3 & 4 & 3 & -4 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{1113}3^11^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 18 & 3 \\ 3 & 4 & 4 & 0 & 9 & 32 & 4 \\ 2 & 4 & 1 & 0 & 6 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 54 & 7 \\ 18 & 32 & 21 & 0 & 54 & 193 & 25 \\ 3 & 4 & 3 & -4 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{1123}3^11^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 12 & 3 \\ 3 & 4 & 4 & 0 & 18 & 22 & 4 \\ 2 & 4 & 1 & 0 & 12 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 72 & 14 \\ 12 & 22 & 15 & 0 & 72 & 85 & 17 \\ 3 & 4 & 3 & -4 & 14 & 17 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{1114}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 3 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{1133}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1756$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 51 & 13 \\ 9 & 17 & 12 & 0 & 51 & 49 & 13 \\ 3 & 4 & 3 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{1224}3^11^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1772$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 11 & 3 \\ 3 & 4 & 13 & 0 & 13 & 21 & 4 \\ 7 & 13 & 28 & 0 & 30 & 48 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 48 & 10 \\ 11 & 21 & 48 & 0 & 48 & 76 & 16 \\ 3 & 4 & 10 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{1123}3^11^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1816$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 12 & 3 \\ 3 & 4 & 4 & 0 & 13 & 22 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 51 & 10 \\ 12 & 22 & 15 & 0 & 51 & 85 & 17 \\ 3 & 4 & 3 & -4 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

[116] $1^{1124}3^11^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1826$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 20 & 3 \\ 3 & 4 & 4 & 0 & 13 & 36 & 4 \\ 2 & 4 & 1 & 0 & 9 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 84 & 10 \\ 20 & 36 & 24 & 0 & 84 & 238 & 28 \\ 3 & 4 & 3 & -4 & 10 & 28 & 0 \end{pmatrix}_{b_0}$$

[117] $1^{1113}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1836$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 4 & 0 & 9 & 27 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 45 & 7 \\ 15 & 27 & 18 & 0 & 45 & 133 & 21 \\ 3 & 4 & 3 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{1114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1850$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 14 & 3 \\ 3 & 4 & 4 & 0 & 9 & 26 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 42 & 7 \\ 14 & 26 & 18 & 0 & 42 & 118 & 20 \\ 3 & 4 & 3 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{1222}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1852$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 30 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 30 & 10 \\ 7 & 13 & 30 & 0 & 30 & 28 & 10 \\ 3 & 4 & 10 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{1223}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1984$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 9 & 3 \\ 3 & 4 & 13 & 0 & 13 & 17 & 4 \\ 7 & 13 & 28 & 0 & 30 & 39 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 39 & 10 \\ 9 & 17 & 39 & 0 & 39 & 49 & 13 \\ 3 & 4 & 10 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

[121] $1^{1134}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1994$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 11 & 3 \\ 3 & 4 & 4 & 0 & 17 & 21 & 4 \\ 2 & 4 & 1 & 0 & 12 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 63 & 13 \\ 11 & 21 & 15 & 0 & 63 & 76 & 16 \\ 3 & 4 & 3 & -4 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{1144}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -2412$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 78 & 16 \\ 11 & 21 & 15 & 0 & 78 & 76 & 16 \\ 3 & 4 & 3 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

[123] $1^{1134}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -2458$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 14 & 3 \\ 3 & 4 & 4 & 0 & 17 & 26 & 4 \\ 2 & 4 & 1 & 0 & 12 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 78 & 13 \\ 14 & 26 & 18 & 0 & 78 & 118 & 20 \\ 3 & 4 & 3 & -4 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

[124] $1^{1124}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -2506$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 14 & 3 \\ 3 & 4 & 4 & 0 & 13 & 26 & 4 \\ 2 & 4 & 1 & 0 & 9 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 60 & 10 \\ 14 & 26 & 18 & 0 & 60 & 118 & 20 \\ 3 & 4 & 3 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{1114} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2618, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 17 & 3 \\ 3 & 4 & 4 & 0 & 9 & 31 & 4 \\ 2 & 4 & 1 & 0 & 6 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 51 & 7 \\ 17 & 31 & 21 & 0 & 51 & 172 & 24 \\ 3 & 4 & 3 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^{1124} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 17 & 3 \\ 3 & 4 & 4 & 0 & 13 & 31 & 4 \\ 2 & 4 & 1 & 0 & 9 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 72 & 10 \\ 17 & 31 & 21 & 0 & 72 & 172 & 24 \\ 3 & 4 & 3 & -4 & 10 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{1133} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2680, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 87 & 17 \\ 12 & 22 & 15 & 0 & 87 & 85 & 17 \\ 3 & 4 & 3 & -4 & 17 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{1134} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 14 & 3 \\ 3 & 4 & 4 & 0 & 22 & 26 & 4 \\ 2 & 4 & 1 & 0 & 15 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 102 & 17 \\ 14 & 26 & 18 & 0 & 102 & 118 & 20 \\ 3 & 4 & 3 & -4 & 17 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{1114} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2970, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 4 & 0 & 9 & 36 & 4 \\ 2 & 4 & 1 & 0 & 6 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 60 & 7 \\ 20 & 36 & 24 & 0 & 60 & 238 & 28 \\ 3 & 4 & 3 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{1144} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -3880, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 18 & 0 & 118 & 120 & 20 \\ 14 & 26 & 18 & 0 & 120 & 118 & 20 \\ 3 & 4 & 3 & -4 & 20 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [131] \quad 1^{1111} 3^1 1^{12} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 24 & 7 \\ 5 & 15 & 9 & 9 & 0 & 24 & 37 & 11 \\ 2 & 4 & 2 & 3 & -4 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{1111} 3^1 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 556, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & 3 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[133] $1^{11}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 652$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 9 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[134] $1^{11}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 880$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 6 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[135] $1^{1111}31^{122}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 952$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 9 & 0 & 37 & 39 & 11 \\ 5 & 15 & 9 & 9 & 0 & 39 & 37 & 11 \\ 2 & 4 & 2 & 3 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

[136] $1^{1111}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 964$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 9 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[137] $1^{1111}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1816$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 15 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & 7 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.776 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$T = 10$ [0] $1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 372$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.777 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

- [1] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [2] $1^{1111}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 24 & 9 & 11 \\ 5 & 9 & 6 & 0 & 24 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 9 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 11 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [3] $1^{1121}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 18 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{1131}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 27 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 27 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{1131}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 22 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 & 3 \\ 12 & 22 & 21 & 0 & 15 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{1141}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 36 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 36 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 24 & 3 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$(1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 9 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 15 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 15 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & 7 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.778 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ $[0] 1^{11}1^31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 329$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}1^31^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 464$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 9 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{13}1^{11}1^31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 604$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}1^31^{11}1^31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 669$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}1^31^{11}1^31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 1184$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 5 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 5 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.779 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 9$ $[0] 1^{11}1^31^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 16.86 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 3 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{11}1^31^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 16.71 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 3 & -4 & 3 & 10 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -65, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1211} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 15 & 7 \\ 5 & 9 & 12 & 0 & 15 & 12 & 6 \\ 3 & 4 & 6 & -4 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -92, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -101, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 & 4 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 19 & 21 & 0 & 73 & 12 & 15 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -107, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 15 & 0 & 12 & 48 & 12 \\ 3 & 4 & 4 & -4 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1211} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 6 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 2 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 & 4 \\ 5 & 9 & 13 & 0 & 33 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 19 & 33 & 0 & 73 & 12 & 15 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1121} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -137, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 2 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 & 4 \\ 8 & 14 & 37 & 0 & 33 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 33 & 0 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 11 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{131} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1121} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 5 & 9 & 6 & 0 & 12 & 12 & 6 \\ 3 & 4 & 3 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -3 & 3 & 0 & -1 \\ 4 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1221} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 2 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 & 4 \\ 7 & 13 & 28 & 0 & 54 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 54 & 0 & 100 & 15 & 18 \\ 2 & 4 & 9 & 0 & 15 & 0 & 2 \\ 3 & 4 & 10 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -219, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -245, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 27 & 9 \\ 3 & 4 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -284, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -3 & 3 & 0 & -1 \\ 3 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -353, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 15 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -359, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 9 & 7 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 3 & 4 & 3 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 2 & 3 \\ 4 & 9 & 12 & 0 & 12 & 4 & 5 \\ 5 & 12 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 5 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1121}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{114} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 & 3 \\ 3 & 4 & 4 & 0 & 21 & 5 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 18 & 16 \\ 3 & 5 & 3 & 0 & 18 & 3 & 3 \\ 3 & 4 & 3 & -4 & 16 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1221} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 2 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 6 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1121} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -425, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 15 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1131} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 9 & 9 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1221} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 5 & 3 \\ 3 & 4 & 8 & 0 & 13 & 9 & 4 \\ 4 & 8 & 10 & 0 & 18 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 21 & 10 \\ 5 & 9 & 12 & 0 & 21 & 12 & 6 \\ 3 & 4 & 6 & -4 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1131} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 2 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 & 4 \\ 8 & 14 & 37 & 0 & 57 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 57 & 0 & 85 & 15 & 17 \\ 2 & 4 & 9 & 0 & 15 & 0 & 2 \\ 3 & 4 & 11 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1141}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 12 & 0 & 46 & 12 & 12 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 & 3 \\ 3 & 4 & 4 & 0 & 13 & 5 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 12 & 10 \\ 3 & 5 & 3 & 0 & 12 & 3 & 3 \\ 3 & 4 & 3 & -4 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 & 3 \\ 3 & 4 & 4 & 0 & 17 & 5 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 15 & 13 \\ 3 & 5 & 3 & 0 & 15 & 3 & 3 \\ 3 & 4 & 3 & -4 & 13 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1221}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 2 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 9 & 10 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 6 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 3 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 2 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 & 4 \\ 5 & 9 & 13 & 0 & 24 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 24 & 0 & 37 & 9 & 11 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{121}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 & 4 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{121}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 & 4 \\ 3 & 5 & 4 & 0 & 30 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 28 & 30 & 0 & 154 & 18 & 22 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 2 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 & 4 \\ 8 & 14 & 37 & 0 & 39 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 39 & 0 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 11 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{121} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 & 4 \\ 3 & 5 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 24 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{141} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 & 4 \\ 3 & 5 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 24 & 0 & 118 & 18 & 20 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{131} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 & 4 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 21 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1131} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 2 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 21 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 3 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{112\bar{1}} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 4 & 0 & 13 & 9 & 4 \\ 2 & 4 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 21 & 10 \\ 5 & 9 & 6 & 0 & 21 & 12 & 6 \\ 3 & 4 & 3 & -4 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{114\bar{1}} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 2 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 & 4 \\ 5 & 9 & 13 & 0 & 33 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 15 & 16 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{112\bar{1}} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 & 4 \\ 5 & 9 & 13 & 0 & 21 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 9 & 10 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{112\bar{1}} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 2 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 & 4 \\ 8 & 14 & 37 & 0 & 48 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 48 & 0 & 58 & 12 & 14 \\ 2 & 4 & 9 & 0 & 12 & 0 & 2 \\ 3 & 4 & 11 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{112\bar{1}} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 2 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 & 4 \\ 5 & 9 & 13 & 0 & 48 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 28 & 48 & 0 & 154 & 18 & 22 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{113\bar{1}} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 2 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 & 4 \\ 5 & 9 & 13 & 0 & 27 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 27 & 0 & 49 & 12 & 13 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{113\bar{1}} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 5 & 3 \\ 3 & 4 & 4 & 0 & 17 & 9 & 4 \\ 2 & 4 & 1 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 27 & 13 \\ 5 & 9 & 6 & 0 & 27 & 12 & 6 \\ 3 & 4 & 3 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{114\bar{1}} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 21 & 9 & 4 \\ 2 & 4 & 1 & 0 & 15 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 33 & 16 \\ 5 & 9 & 6 & 0 & 33 & 12 & 6 \\ 3 & 4 & 3 & -4 & 16 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1131^1} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{131^1} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 & 4 \\ 3 & 5 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1221^1} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 2 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 & 4 \\ 7 & 13 & 28 & 0 & 42 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 42 & 0 & 58 & 12 & 14 \\ 2 & 4 & 9 & 0 & 12 & 0 & 2 \\ 3 & 4 & 10 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1131^1} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 18 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1121^1} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 2 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 & 4 \\ 5 & 9 & 13 & 0 & 30 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 30 & 0 & 58 & 12 & 14 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1121^1} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 2 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 & 4 \\ 5 & 9 & 13 & 0 & 39 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 39 & 0 & 100 & 15 & 18 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1221^1} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 2 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 & 4 \\ 7 & 13 & 28 & 0 & 30 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 10 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1131^1} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{131}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1865$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 & 4 \\ 3 & 5 & 4 & 0 & 33 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 33 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{1141}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1889$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{141}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1928$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 & 4 \\ 3 & 5 & 4 & 0 & 30 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 30 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1131}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 2 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 & 4 \\ 5 & 9 & 13 & 0 & 36 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 36 & 0 & 85 & 15 & 17 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1221}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2624$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 2 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 & 4 \\ 10 & 18 & 58 & 0 & 60 & 12 & 14 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 60 & 0 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 12 & 0 & 2 \\ 3 & 4 & 14 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{141}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2636$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 & 4 \\ 3 & 5 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1141}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2720$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 18 & 0 & 118 & 18 & 20 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{1141}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2780$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 24 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 3 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1141} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 2 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 & 4 \\ 5 & 9 & 13 & 0 & 42 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 42 & 0 & 118 & 18 & 20 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1141} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 21 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 3 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1131} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 2 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 & 4 \\ 5 & 9 & 13 & 0 & 45 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 45 & 0 & 133 & 18 & 21 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1131} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3365, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 2 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 & 4 \\ 5 & 9 & 13 & 0 & 54 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 54 & 0 & 193 & 21 & 25 \\ 2 & 4 & 6 & 0 & 21 & 0 & 2 \\ 3 & 4 & 7 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1141} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -4373, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 2 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 & 4 \\ 5 & 9 & 13 & 0 & 51 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 51 & 0 & 172 & 21 & 24 \\ 2 & 4 & 6 & 0 & 21 & 0 & 2 \\ 3 & 4 & 7 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1141} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -5216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 2 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 & 4 \\ 5 & 9 & 13 & 0 & 60 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 60 & 0 & 238 & 24 & 28 \\ 2 & 4 & 6 & 0 & 24 & 0 & 2 \\ 3 & 4 & 7 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [80] \quad 1^{1111} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 4 & 1 & 9 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 14 & 7 & 9 & 8 \\ 2 & 3 & 0 & 7 & -2 & 3 & 0 \\ 3 & 6 & 0 & 9 & 3 & 4 & 4 \\ 3 & 4 & -1 & 8 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 111, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.62 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{11}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 138$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 12 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{112}131^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 273$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 12 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 9 & 12 & 9 & 0 & 28 & 9 & 10 \\ 1 & 3 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{11}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 300$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{111}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 363$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{111}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1065$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 15 & 0 & 37 & 9 & 11 \\ 1 & 3 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{111}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1659$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 12 & 21 & 0 & 73 & 12 & 15 \\ 1 & 3 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{111}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 2145$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 15 & 27 & 0 & 121 & 15 & 19 \\ 1 & 3 & 3 & 3 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

4.780 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{111}31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 9 & 24 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 24 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 11 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 21 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 24 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

- [8] $1^{11111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 \\ 3 & 4 & 3 & -4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$
- [9] $1^{11111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{1112}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 21 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 7 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{11111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{1212}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 9 & 30 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 30 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 30 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{11111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{111111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 2368$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 7 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.781 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$T = 10$ [0] $1^{111111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 12 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 15 & 6 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 6 & 3 & 0 & 2 \\ 3 & 5 & 7 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 507$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 651$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{114111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 813$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 18 & 15 & 16 \\ 3 & 5 & 3 & 0 & 18 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 16 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{1121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 12 & 9 & 10 \\ 3 & 5 & 3 & 0 & 12 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 10 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{1131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 15 & 12 & 13 \\ 3 & 5 & 3 & 0 & 15 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 13 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.782 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ $[0] 13^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 4 & 3 & 3 & 4 \\ 2 & 5 & 3 & -2 & 2 & 0 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 0 & 2 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 3 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & 0 & 0 & 2 & -4 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 4 & -2 & 0 \\ 3 & 3 & -1 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 2 & 3 \\ 2 & 1 & 1 & 5 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 9 & 6 & 7 \\ 2 & 2 & 0 & 6 & -2 & 0 \\ 3 & 3 & -1 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

[6] $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -257$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

[7] $\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -297$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

[8] $\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 4 & 12 & 9 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 7 & 0 & 0 & 2 \\ 2 & 4 & 2 & 10 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

[9] $\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 \\ 2 & 4 & 2 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -402$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

[10] $\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 3 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 & 2 \\ 8 & 24 & 18 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 14 & 0 & 0 & 2 \\ 2 & 4 & 2 & 20 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -552$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

[11] $\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 \\ 6 & 18 & 9 & 52 & 0 & 9 & 12 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 9 & 0 & 0 & 2 \\ 2 & 4 & 2 & 12 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -617$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

[12] $\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 \\ 5 & 15 & 9 & 37 & 0 & 8 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 8 & 0 & 0 & 2 \\ 2 & 4 & 2 & 11 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -807$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

[13] $\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 7 & 21 & 15 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 12 & 0 & 0 & 2 \\ 2 & 4 & 2 & 17 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -887$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

[14] $\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 7 & 21 & 15 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 12 & 0 & 0 & 2 \\ 2 & 4 & 2 & 17 & -4 & 2 & 0 \end{pmatrix}_{b_0}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -932$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 1 & 2 \\ 2 & 4 & 4 & 33 & 0 & 3 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 & 2 \\ 11 & 33 & 21 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 18 & 0 & 0 & 2 \\ 2 & 4 & 2 & 25 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{14}3^{11}1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 3 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 & 2 \\ 12 & 36 & 24 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 20 & 0 & 0 & 2 \\ 2 & 4 & 2 & 28 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.783 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$

$T = 10 \quad [0] \quad 1^{111}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 9 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 \\ 1 & 3 & 3 & 1 & 0 & 2 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 5 \\ 3 & 9 & 6 & 6 & 0 & 5 & 13 \\ 2 & 4 & 2 & 3 & -4 & 2 & 7 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 15 \\ 1 & 4 & 0 & 3 & 0 & 2 & 9 \\ 1 & 3 & 3 & 1 & 0 & 2 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 8 \\ 5 & 15 & 9 & 9 & 0 & 8 & 37 \\ 2 & 4 & 2 & 3 & -4 & 2 & 11 \end{pmatrix}_{b_0}$$

[2] $1^{13}1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^{11}1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 669, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 9 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 5 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 15 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.784 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] \quad 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{13}1^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 436, (n_+, n_-, n_0) =$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 & 2 \\ 4 & 12 & 9 & 28 & 0 & 7 & 7 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 7 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 7 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 10 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 6 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 6 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 6 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 8 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 5 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 5 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 7 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 & 2 \\ 5 & 15 & 9 & 37 & 0 & 8 & 8 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 8 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 8 & 0 & 0 & 2 & 2 \\ 2 & 4 & 2 & 11 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 18 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 & 2 & 2 \\ 6 & 18 & 12 & 58 & 0 & 10 & 10 & 14 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 10 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 10 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 14 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.56 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & 3 & -4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & -4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{121}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 2 & 2 & 3 \\ 2 & 1 & 1 & 5 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 9 & 6 & 5 & 7 \\ 2 & 2 & 0 & 6 & -2 & 2 & 0 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 3 & -1 & 7 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.786 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2\textcolor{blue}{n}3\textcolor{blue}{mix}$

$$T = 9 \quad [0] \ 1^{111}3^{121}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -34, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 & 5 \\ 1 & 1 & 0 & 4 & -3 & 2 & -1 \\ 2 & 3 & 0 & 5 & 2 & 3 & 3 \\ 2 & 2 & -1 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -48, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.58 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 3 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -78, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 1 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -143, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 5 & 2 \\ 3 & 9 & 6 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 5 & 11 & 0 & 7 & 5 \\ 2 & 4 & 2 & 7 & -4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -242, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 1 & 3 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^{11}1^{14}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -323, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.77 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 17 & 2 \\ 1 & 3 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 17 & 17 & 0 & 151 & 21 \\ 2 & 4 & 2 & 3 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111}3^{11}1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -395, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 11 & 2 \\ 1 & 3 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 11 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -412$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 12 & 0 & 25 & 9 \\ 2 & 4 & 2 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}1^{13}31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -422$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 13.19 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 14 & 2 \\ 1 & 3 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 14 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & 3 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{13}3^{11}1^{11}31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -473$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -483$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 5 & 2 \\ 3 & 9 & 6 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 14 & 0 & 12 & 6 \\ 2 & 4 & 2 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2}$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -782$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 0 & 12 & 4 \\ 1 & 4 & 0 & 6 & 0 & 8 & 2 \\ 3 & 9 & 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 20 & 0 & 25 & 9 \\ 2 & 4 & 2 & 7 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

4.787 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_3$

$T = 10$ [0] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3}$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 494$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 9 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

4.788 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$

$T = 8$ [0] $1^{14}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3}$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.6$

$T_{\min} = 24$, $T_{\max} = 98$, $T \in [24, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 18 & 0 & 75 & 15 \\ 3 & 4 & 4 & -4 & 15 & -11 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3}$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 16$, $T_{\max} = 98$, $T \in [16, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & -4 & 3 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.87 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 3 & 4 & 9 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.03 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 3 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 5 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 48 & 12 \\ 3 & 4 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 8 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 7 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 11 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 6 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 8 & 4 \\ 10 & 18 & 58 & 0 & 24 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 14 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 6 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{13}2^13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 6 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 0 & 27 & 9 \\ 3 & 4 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 12 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{132}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 8 & 4 \\ 15 & 27 & 133 & 0 & 36 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 36 & 0 & 9 & 5 \\ 3 & 4 & 21 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{112}3^12^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 2 & 3 \\ 4 & 9 & 24 & 0 & 4 & 5 \\ 10 & 24 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 14 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{141}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 8 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{151}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 20 & 0 & 4 & 4 \\ 10 & 20 & 73 & 0 & 15 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 15 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{14}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 11 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 5 & 11 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 16 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{161}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 12 & 24 & 106 & 0 & 18 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 18 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{13}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 15 & 36 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 21 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{171}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 14 & 28 & 145 & 0 & 21 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 21 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 0 & 57 & 13 \\ 3 & 4 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{115}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 487, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 20 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{113}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 48 & 12 \\ 3 & 4 & 7 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 18 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{132}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 9 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{132}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 8 & 4 \\ 12 & 22 & 85 & 0 & 30 & 17 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 30 & 0 & 9 & 5 \\ 3 & 4 & 17 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{121}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114}3^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 2 & 3 \\ 4 & 9 & 48 & 0 & 4 & 5 \\ 20 & 48 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 5 & 28 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{121}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{131}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{121}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 16 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{131}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{141} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 11 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 0 & 84 & 16 \\ 3 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{121} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 13 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 14 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 5 & 25 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 11 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{114} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{131} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 4 & 4 \\ 15 & 27 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 21 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{131} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 4 & 4 \\ 18 & 32 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 25 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{141} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 4 & 4 \\ 14 & 26 & 118 & 0 & 18 & 20 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 20 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{141} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 4 & 4 \\ 17 & 31 & 172 & 0 & 21 & 24 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 24 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{141} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 4 \\ 20 & 36 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 4 & 28 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [77] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 0 & 5 \\ 1 & 1 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 5 & -1 & -1 & -4 \end{pmatrix}_{b_0}$$

[79] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

[78] $1^{111} 31^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 16, T_{\max} = 68, T \in [16, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 2 & -3 \end{pmatrix}_{b_0}$$

[80] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

[81] $1^{12} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

12.0

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 12 & 0 & 27 & 9 \\ 2 & 4 & 2 & 4 & -4 & 9 & -3 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{12}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 68, T \in [19, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 & 2 \\ 2 & 4 & 4 & 12 & 0 & 6 & 4 \\ 1 & 4 & 0 & 9 & 0 & 3 & 2 \\ 4 & 12 & 9 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 12 & 0 & 3 & 3 \\ 2 & 4 & 2 & 10 & -4 & 3 & -6 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 & 3 \\ 2 & 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 4 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 68, T \in [15, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 3 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{111}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 1 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 14 & 7 & 8 \\ 2 & 3 & 0 & 7 & -2 & 0 \\ 3 & 4 & -1 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{11}31^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 4 & 2 & 8 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -177, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{13}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 & 3 \\ 5 & 12 & 15 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 12 & 0 & 0 & 2 \\ 2 & 3 & 3 & 13 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 & 3 \\ 3 & 6 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 3 & 3 & 7 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 4 & 2 & 7 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{121}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 & 3 \\ 4 & 9 & 12 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 9 & 0 & 0 & 2 \\ 2 & 3 & 3 & 10 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[94] $1^{131}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 \\ 2 & 4 & 2 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[95] $1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -537, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[96] $1^{11}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -615, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 9 & 0 & 3 & 3 \\ 2 & 4 & 2 & 7 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{111}31^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -762, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 3 & 2 \\ 5 & 15 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 9 & 0 & 0 & 2 \\ 2 & 4 & 2 & 11 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 12 & 6 \\ 2 & 4 & 2 & 7 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111}31^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1059, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 3 & 2 \\ 7 & 21 & 12 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 12 & 0 & 0 & 2 \\ 2 & 4 & 2 & 15 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{green}{1}} 31^{\textcolor{red}{1}\textcolor{blue}{4}} 31^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -1302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 3 & 2 \\ 9 & 27 & 15 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 19 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [101] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 2^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 21^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 34, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 4 & 6 \\ 1 & 1 & 0 & 4 & -3 & 1 & -1 \\ 1 & 3 & 0 & 4 & 1 & 0 & 2 \\ 2 & 3 & -1 & 6 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

4.789 $\text{NHC}(-2)(-3) + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 31^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + [\textcolor{blue}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 470, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.790 $\text{NHC}(-2)(-3) + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{o}_8$

$$T = 9 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 31^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{o}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 31^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{o}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 31^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{o}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 31^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{c}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{o}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -931, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.791 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{111} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 3 & -4 & 3 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 36 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 3 & -4 & 10 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1211} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 12 & 15 & 6 \\ 5 & 9 & 12 & 0 & 15 & 13 & 7 \\ 3 & 4 & 6 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 3 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 19 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 11 & 19 & 21 & 0 & 12 & 73 & 15 \\ 3 & 4 & 4 & -4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{131} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 48 & 12 & 12 \\ 2 & 4 & 3 & 0 & 12 & 1 & 3 \\ 3 & 4 & 4 & -4 & 12 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1211} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 8 & 0 & 4 & 9 & 4 \\ 4 & 8 & 10 & 0 & 6 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 12 & 0 & 6 & 13 & 7 \\ 3 & 4 & 6 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 5 & 3 \\ 3 & 4 & 19 & 0 & 4 & 9 & 4 \\ 11 & 19 & 73 & 0 & 12 & 33 & 15 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 6 & 2 \\ 5 & 9 & 33 & 0 & 6 & 13 & 7 \\ 3 & 4 & 15 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1112} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -137, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 7 & 3 \\ 3 & 4 & 14 & 0 & 4 & 13 & 4 \\ 8 & 14 & 37 & 0 & 9 & 33 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 7 & 13 & 33 & 0 & 9 & 28 & 10 \\ 3 & 4 & 11 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1131} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1211} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 12 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 4 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1212} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 13 & 3 \\ 3 & 4 & 13 & 0 & 4 & 23 & 4 \\ 7 & 13 & 28 & 0 & 9 & 54 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 15 & 2 \\ 13 & 23 & 54 & 0 & 15 & 100 & 18 \\ 3 & 4 & 10 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -219, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 27 & 9 & 9 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 14 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 15 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 5 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 5 & 3 \\ 4 & 9 & 12 & 0 & 4 & 12 & 5 \\ 5 & 12 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 12 & 15 & 0 & 6 & 13 & 7 \\ 3 & 5 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 4 & 8 & 6 & 0 & 6 & 10 & 6 \\ 3 & 4 & 3 & -4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 11 & 21 & 15 & 0 & 18 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1212} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 4 & 3 \\ 3 & 4 & 8 & 0 & 4 & 8 & 4 \\ 4 & 8 & 10 & 0 & 6 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 4 & 8 & 12 & 0 & 6 & 10 & 6 \\ 3 & 4 & 6 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1112} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 3 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 6 & 12 & 9 & 0 & 9 & 25 & 9 \\ 3 & 4 & 3 & -4 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1212} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 7 & 3 \\ 3 & 4 & 8 & 0 & 9 & 13 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 12 & 21 & 6 \\ 7 & 13 & 18 & 0 & 21 & 28 & 10 \\ 3 & 4 & 6 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 12 & 3 \\ 3 & 4 & 14 & 0 & 4 & 22 & 4 \\ 8 & 14 & 37 & 0 & 9 & 57 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 15 & 2 \\ 12 & 22 & 57 & 0 & 15 & 85 & 17 \\ 3 & 4 & 11 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 16 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 8 & 16 & 12 & 0 & 12 & 46 & 12 \\ 3 & 4 & 3 & -4 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 7 & 13 & 9 & 0 & 12 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 9 & 3 \\ 3 & 4 & 4 & 0 & 5 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 9 & 17 & 12 & 0 & 15 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1212} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 7 & 3 \\ 3 & 4 & 8 & 0 & 4 & 13 & 4 \\ 4 & 8 & 10 & 0 & 6 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 7 & 13 & 18 & 0 & 9 & 28 & 10 \\ 3 & 4 & 6 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 8 & 3 \\ 3 & 4 & 9 & 0 & 4 & 14 & 4 \\ 5 & 9 & 13 & 0 & 6 & 24 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 8 & 14 & 24 & 0 & 9 & 37 & 11 \\ 3 & 4 & 7 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 5 & 0 & 4 & 28 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 16 & 28 & 30 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{111} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 8 & 3 \\ 3 & 4 & 14 & 0 & 4 & 14 & 4 \\ 8 & 14 & 37 & 0 & 9 & 39 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 8 & 14 & 39 & 0 & 9 & 37 & 11 \\ 3 & 4 & 11 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 3 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 5 & 0 & 4 & 26 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 14 & 26 & 24 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 5 & 0 & 4 & 23 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 23 & 24 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 21 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 3 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 21 & 6 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 11 & 3 \\ 3 & 4 & 9 & 0 & 4 & 21 & 4 \\ 5 & 9 & 13 & 0 & 6 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 11 & 21 & 33 & 0 & 15 & 76 & 16 \\ 3 & 4 & 7 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 13 & 4 \\ 5 & 9 & 13 & 0 & 6 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 7 & 13 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 7 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 10 & 3 \\ 3 & 4 & 14 & 0 & 4 & 18 & 4 \\ 8 & 14 & 37 & 0 & 9 & 48 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 10 & 18 & 48 & 0 & 12 & 58 & 14 \\ 3 & 4 & 11 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 4 & 17 & 4 \\ 5 & 9 & 13 & 0 & 6 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 9 & 17 & 27 & 0 & 12 & 49 & 13 \\ 3 & 4 & 7 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 16 & 3 \\ 3 & 4 & 9 & 0 & 4 & 28 & 4 \\ 5 & 9 & 13 & 0 & 6 & 48 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 16 & 28 & 48 & 0 & 18 & 154 & 22 \\ 3 & 4 & 7 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 27 & 6 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 3 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 21 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 33 & 6 \\ 11 & 21 & 15 & 0 & 33 & 76 & 16 \\ 3 & 4 & 3 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1212} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 10 & 3 \\ 3 & 4 & 13 & 0 & 4 & 18 & 4 \\ 7 & 13 & 28 & 0 & 9 & 42 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 10 & 18 & 42 & 0 & 12 & 58 & 14 \\ 3 & 4 & 10 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1113} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 3 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 10 & 3 \\ 3 & 4 & 9 & 0 & 4 & 18 & 4 \\ 5 & 9 & 13 & 0 & 6 & 30 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 10 & 18 & 30 & 0 & 12 & 58 & 14 \\ 3 & 4 & 7 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 13 & 3 \\ 3 & 4 & 9 & 0 & 4 & 23 & 4 \\ 5 & 9 & 13 & 0 & 6 & 39 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 13 & 23 & 39 & 0 & 15 & 100 & 18 \\ 3 & 4 & 7 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1212} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 7 & 3 \\ 3 & 4 & 13 & 0 & 4 & 13 & 4 \\ 7 & 13 & 28 & 0 & 9 & 30 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 7 & 13 & 30 & 0 & 9 & 28 & 10 \\ 3 & 4 & 10 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1820, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 3 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{113}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1865, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 5 & 0 & 4 & 32 & 4 \\ 3 & 5 & 4 & 0 & 3 & 33 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 32 & 33 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{1114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1889, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 4 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1928, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 17 & 31 & 30 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 12 & 3 \\ 3 & 4 & 9 & 0 & 4 & 22 & 4 \\ 5 & 9 & 13 & 0 & 6 & 36 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 12 & 22 & 36 & 0 & 15 & 85 & 17 \\ 3 & 4 & 7 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1212}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 10 & 3 \\ 3 & 4 & 18 & 0 & 4 & 18 & 4 \\ 10 & 18 & 58 & 0 & 12 & 60 & 14 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 12 & 2 \\ 10 & 18 & 60 & 0 & 12 & 58 & 14 \\ 3 & 4 & 14 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1114}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 4 & 1 & 9 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 14 & 7 & 9 & 8 \\ 2 & 3 & 0 & 7 & -2 & 3 & 0 \\ 3 & 6 & 0 & 9 & 3 & 4 & 4 \\ 3 & 4 & -1 & 8 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{11}1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 9 \\ 1 & 3 & 3 & 1 & 0 & 3 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 \\ 3 & 6 & 9 & 6 & 0 & 6 & 13 \\ 2 & 3 & 3 & 3 & -4 & 2 & 7 \end{pmatrix}_{b_0}$$

[82] $1^{11}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 138, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 3 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 12 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 12 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 8 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{11}1^{12}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 12 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 9 & 2 \\ 4 & 9 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{11}1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 363, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 9 & 6 & 9 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{11}1^{11}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1065, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 9 & 2 \\ 5 & 15 & 9 & 15 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & 4 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{11}1^{11}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1659, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 12 & 2 \\ 7 & 21 & 12 & 21 & 0 & 12 & 73 & 15 \\ 2 & 4 & 2 & 4 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{11}1^{11}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 2145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 15 & 2 \\ 9 & 27 & 15 & 27 & 0 & 15 & 121 & 19 \\ 2 & 4 & 2 & 4 & -4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

4.792 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 6 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 24 & 11 \\ 5 & 9 & 6 & 0 & 6 & 24 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1112}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 21 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{1114}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 706$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 24 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 712$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 724$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{1112}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 796$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 21 & 7 \\ 7 & 13 & 12 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 988$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1032$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 & 15 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 6 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{1122}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1144$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 30 & 10 \\ 7 & 13 & 12 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1180$, $(n_+, n_-, n_0) =$

(1, 10, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 21 & 3 \\ 17 & 31 & 30 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{\overline{11111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{\overline{11111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 2368$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 15 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 7 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.793 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{\overline{11111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 5 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 12 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 15 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 3 & 2 \\ 5 & 12 & 15 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 3 & 6 & 0 & 2 \\ 3 & 5 & 7 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{11111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 507$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{11111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 651$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 & 3 \\ 5 & 9 & 6 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 813$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 21 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 & 3 \\ 11 & 21 & 15 & 0 & 18 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{1121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1173$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 & 3 \\ 7 & 13 & 9 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{1131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1227$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.8$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 17 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 & 3 \\ 9 & 17 & 12 & 0 & 15 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.794 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{111}3^11^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 86$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 126$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 2 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 130$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 1 & 9 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 14 & 7 & 5 & 8 \\ 2 & 3 & 0 & 7 & -2 & 2 & 0 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & -1 & 8 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 296$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.795 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_{2n3mix}$

$T = 10$ [0] $1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 5 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

4.796 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$T = 9$ [0] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 & 3 \\ 3 & 4 & 9 & 0 & 8 & 5 & 4 \\ 5 & 9 & 13 & 0 & 12 & 9 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 9 & 6 & 5 \\ 3 & 5 & 9 & 0 & 6 & 3 & 3 \\ 3 & 4 & 7 & -4 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 5 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 4 & 5 \\ 14 & 36 & 121 & 0 & 15 & 15 & 19 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 5 & 19 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 12 & 15 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 3 & 2 \\ 2 & 4 & 12 & 0 & 3 & 0 & 2 \\ 3 & 5 & 15 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 5 & 11 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{122}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -267$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.7$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 4 & 3 \\ 3 & 4 & 13 & 0 & 5 & 8 & 4 \\ 7 & 13 & 28 & 0 & 12 & 18 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 12 & 0 & 3 & 6 & 5 \\ 4 & 8 & 18 & 0 & 6 & 9 & 5 \\ 3 & 4 & 10 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{132}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -294$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 4 & 3 \\ 3 & 4 & 17 & 0 & 5 & 8 & 4 \\ 9 & 17 & 49 & 0 & 15 & 24 & 13 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 15 & 0 & 3 & 6 & 3 \\ 4 & 8 & 24 & 0 & 6 & 9 & 5 \\ 3 & 4 & 13 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -402$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 4 & 11 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -420$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 15 & 18 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 5 & 18 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -510$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -555, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 21 & 25 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 3 & 2 \\ 2 & 4 & 21 & 0 & 3 & 0 & 2 \\ 3 & 5 & 25 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -645, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 3 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -651, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -690, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 18 & 22 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 3 & 2 \\ 2 & 4 & 18 & 0 & 3 & 0 & 2 \\ 3 & 5 & 22 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{141} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 2 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 & 4 \\ 11 & 21 & 76 & 0 & 18 & 15 & 16 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 18 & 0 & 3 & 3 & 3 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 4 & 16 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -789, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 2 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 & 4 \\ 7 & 13 & 28 & 0 & 12 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 12 & 0 & 3 & 3 & 3 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 4 & 10 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{131} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 2 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 & 4 \\ 9 & 17 & 49 & 0 & 15 & 12 & 13 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 15 & 0 & 3 & 3 & 3 \\ 2 & 4 & 12 & 0 & 3 & 0 & 2 \\ 3 & 4 & 13 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 12 & 6 \\ 3 & 4 & 7 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 477, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & 4 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{111}} 31^{\overline{11}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 486, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 & 3 & 3 \\ 3 & 6 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 6 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{111}} 31^{\overline{11}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 774, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 6 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 9 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & 7 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.797 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 5 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 12 & 5 \\ 5 & 12 & 13 & 0 & 6 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 6 & 2 \\ 5 & 12 & 15 & 0 & 6 & 6 & 13 & 7 \\ 3 & 5 & 7 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 228, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 231, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 18 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 15 & 0 & 18 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 9 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 15 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 12 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.798 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 7 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.799 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{e}_6$ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^1 3^1 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 1 & 4 \\ 8 & 9 & 8 & 4 & 0 & 5 \\ 6 & 8 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

4.800 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$

$$T = 7 \quad [0] \quad 21^1 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -2 & 0 \\ 3 & 5 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -2 & 0 \\ 3 & 5 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 97, T \in [15, 97]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 28 & 0 & 10 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 10 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 8 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 21^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112}3^11^3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 85 & 0 & 17 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 17 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{15}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 15 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{13}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 2 & 4 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{16}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.4$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 12 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 18 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{17}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 14 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 21 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{13}1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 2 & 4 & 6 & -2 & 0 \\ 3 & 5 & 8 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{113}3^11^4 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 17 & 40 & 172 & 0 & 24 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 24 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 24 & 0 & 5 \\ 10 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 14 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 15 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 8 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 11 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 19 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 13^11^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 9 & 24 & 4 & 5 \\ 9 & 24 & 52 & 9 & 12 \\ 2 & 4 & 9 & -2 & 0 \\ 3 & 5 & 12 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{13}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{12}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 21 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 11 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 15 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 21^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -2 & 0 \\ 4 & 5 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{113}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 13 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 18 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 131^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 12 & 32 & 94 & 12 & 16 \\ 2 & 4 & 12 & -2 & 0 \\ 3 & 5 & 16 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{12}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 18 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{12}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 22 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{114}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 48 & 0 & 5 \\ 20 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 28 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -418, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 14 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 19 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[41] \ 131^{115} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 15 & 40 & 148 & 15 & 20 \\ 2 & 4 & 15 & -2 & 0 \\ 3 & 5 & 20 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{14}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -466, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11}4^1 3^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -466, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 18 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 25 & -3 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{11}4^1 3^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 16 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 22 & -3 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$

[46] $21^{11} 3^1 1^{12} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -486, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 9 & 16 & 4 & 5 \\ 12 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -2 & 0 \\ 4 & 5 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{13} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -603, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 18 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 25 & -3 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{14} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -652, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 17 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 24 & -3 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -826, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 20 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

[50] $21^{11} 3^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -836, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 9 & 24 & 4 & 5 \\ 18 & 24 & 52 & 9 & 12 \\ 3 & 4 & 9 & -2 & 0 \\ 4 & 5 & 12 & 0 & 1 \end{pmatrix}_{b_0}$$

[51] $21^{11} 4^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -1326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 9 & 32 & 4 & 5 \\ 24 & 32 & 94 & 12 & 16 \\ 3 & 4 & 12 & -2 & 0 \\ 4 & 5 & 16 & 0 & 1 \end{pmatrix}_{b_0}$$

[52] $21^{11} 5^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -1956, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 9 & 40 & 4 & 5 \\ 30 & 40 & 148 & 15 & 20 \\ 3 & 4 & 15 & -2 & 0 \\ 4 & 5 & 20 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [53] $1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 5 & 13 & 9 & 5 \\ 13 & 30 & 20 & 10 \\ 9 & 20 & 13 & 7 \\ 5 & 10 & 7 & -1 \end{pmatrix}_{b_0}$$

- [55] $2 \overset{1}{2} 3 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 7$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 10 & 19 & 12 & 6 \\ 19 & 30 & 20 & 10 \\ 12 & 20 & 13 & 7 \\ 6 & 10 & 7 & 1 \end{pmatrix}_{b_0}$
- [54] $1 \overset{1}{4} 3 \overset{1}{1} 3 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 7$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 18.57 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 6 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 16 & -3 & -6 \end{pmatrix}_{b_0}$
- [56] $1 \overset{1}{1} 3 2 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 10 & 18 & 12 & 6 \\ 18 & 30 & 20 & 10 \\ 12 & 20 & 13 & 7 \\ 6 & 10 & 7 & 3 \end{pmatrix}_{b_0}$
- [57] $1 \overset{1}{1} 3 2 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 15$, $T_{\max} = 20$, $T \in [15, 20]$
 $\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -2 & 0 \\ 5 & 10 & 0 & 0 \end{pmatrix}_{b_0}$
- [58] $1 \overset{1}{1} 3 2 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 3 & -3 & 1 \end{pmatrix}_{b_0}$
- [59] $1 \overset{1}{1} 3 2 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [60] $1 \overset{1}{2} 3 2 \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 8 & 10 \\ 1 & 1 & 0 & 8 & -2 & 0 \\ 2 & 2 & -1 & 10 & 0 & -1 \end{pmatrix}_{b_0}$
- [61] $1 \overset{1}{1} 3 2 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & -3 & 1 \end{pmatrix}_{b_0}$
- [62] $1 \overset{1}{1} 3 2 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & -3 & 2 \end{pmatrix}_{b_0}$
- [65] $1 \overset{1}{1} 3 2 \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 4 & 0 & 16 & 7 & 8 \\ 1 & 1 & 0 & 7 & -2 & 0 \\ 2 & 2 & -1 & 8 & 0 & 0 \end{pmatrix}_{b_0}$
- [63] $1 \overset{1}{1} 3 2 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & -1 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{12}3^{11}21^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 7 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 45 & 11 & 13 \\ 1 & 1 & 0 & 11 & -2 & 0 \\ 2 & 2 & -1 & 13 & 0 & -1 \end{pmatrix}_{b_0}$$

[67] $1^{11}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{11}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 8 & -3 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{13}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 2 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{13}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{13}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 5 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 13 & -3 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{12}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 9 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 3 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 4 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{11}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 12 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{14}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{11}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 12 & -3 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{11}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{13}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 7 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{12}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{114}31^{21}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 4 & 18 & 0 & 3 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 18 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 20 & -3 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{131}1^{41}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 16 & -3 & 0 \end{pmatrix}_{b_0}$$

[87] $31^{11}231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{11}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 623, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 7 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 15 & -3 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{13}31^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 10 & 30 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 20 & -3 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{12}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 8 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 18 & -3 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{14}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 836, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 10 & 30 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 24 & -3 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{13}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 871, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 9 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{11}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 931, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 27 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 19 & -3 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{13}31^{16}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1096, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.4$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 12 & 36 & 18 & 214 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 24 & -3 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{12}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 10 & 30 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 22 & -3 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{13}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 33 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 11 & 33 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 25 & -3 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{13}31^{17}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1460, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 1 & 2 \\ 2 & 4 & 4 & 42 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 14 & 42 & 21 & 292 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 28 & -3 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{14}31^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1480$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 \\ 12 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [99] $1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 2 \\ 2 & 0 & 4 & 6 & 4 & 6 \\ 1 & 0 & 1 & 4 & -2 & 0 \\ 2 & 0 & 2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{11}31^14^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -11$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & -2 & 0 \\ 2 & 3 & -1 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [102] $1^{11}51^1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 12$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & 10 & 4 & 6 \\ 1 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & -1 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{11}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 19$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.26 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

4.801 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{e}_6$

$T = 8$ [0] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 102$, $\Delta = -186$, $\det = 299$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 14.20 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 102$, $\Delta = -186$, $\det = 464$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.7$
 $T_{\min} = 14.21 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 8 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 102$, $\Delta = -186$, $\det = 739$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.21 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 12 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3 2 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{c}_6 \quad T = 9, \quad T_{\max} = 101, \quad \Delta = -158, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.40 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 3 & 0 & -2 & 0 \\ 3 & 5 & 4 & -3 & 0 & -2 \end{pmatrix}_{b_0}$$

4.802 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 100, \quad \Delta = -160, \quad \det = 232, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 100, \quad \Delta = -160, \quad \det = 352, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.9$$

$$T_{\min} = 13.36 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 8 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 100, \quad \Delta = -160, \quad \det = 552, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 12 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 100, \quad \Delta = -160, \quad \det = 832, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 16 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{115} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 100, \quad \Delta = -160, \quad \det = 1192, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 20 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3 2 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -5, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 15.40 \rightarrow 16, \quad T_{\max} = 93, \quad T \in [16, 93]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 10 & 7 & -3 & -1 \end{pmatrix}_{b_0}$$

4.803 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 92, \quad \Delta = -130, \quad \det = -544, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 92, \quad T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

4.804 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 8 & 4 & 0 & 4 & 5 \\ 3 & 8 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.805 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 2 & 3 \\ 3 & 4 & 19 & 0 & 5 & 4 & 4 \\ 11 & 19 & 73 & 0 & 21 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 21 & 0 & 4 & 4 & 4 \\ 2 & 4 & 12 & 0 & 4 & 0 & 2 \\ 3 & 4 & 15 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 2 & 3 \\ 3 & 4 & 14 & 0 & 5 & 4 & 4 \\ 8 & 14 & 37 & 0 & 15 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 15 & 0 & 4 & 4 & 4 \\ 2 & 4 & 9 & 0 & 4 & 0 & 2 \\ 3 & 4 & 11 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{121}} 3^{\overline{1}} 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 2 & 3 \\ 3 & 4 & 18 & 0 & 5 & 4 & 4 \\ 10 & 18 & 58 & 0 & 18 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 18 & 0 & 4 & 4 & 4 \\ 2 & 4 & 12 & 0 & 4 & 0 & 2 \\ 3 & 4 & 14 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{121}} 3^{\overline{1}} 1^{\overline{141}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 & 2 & 3 \\ 3 & 4 & 28 & 0 & 5 & 4 & 4 \\ 16 & 28 & 154 & 0 & 30 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 30 & 0 & 4 & 4 & 4 \\ 2 & 4 & 18 & 0 & 4 & 0 & 2 \\ 3 & 4 & 22 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121} 3^1 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -956, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 & 2 & 3 \\ 3 & 4 & 23 & 0 & 5 & 4 & 4 \\ 13 & 23 & 100 & 0 & 24 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 24 & 0 & 4 & 4 & 4 \\ 2 & 4 & 15 & 0 & 4 & 0 & 2 \\ 3 & 4 & 18 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{131} 3^1 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 & 2 & 3 \\ 3 & 4 & 27 & 0 & 5 & 4 & 4 \\ 15 & 27 & 133 & 0 & 27 & 18 & 21 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 27 & 0 & 4 & 4 & 4 \\ 2 & 4 & 18 & 0 & 4 & 0 & 2 \\ 3 & 4 & 21 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{131} 3^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 & 2 & 3 \\ 3 & 4 & 32 & 0 & 5 & 4 & 4 \\ 18 & 32 & 193 & 0 & 33 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 33 & 0 & 4 & 4 & 4 \\ 2 & 4 & 21 & 0 & 4 & 0 & 2 \\ 3 & 4 & 25 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{141} 3^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -2456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 & 2 & 3 \\ 3 & 4 & 36 & 0 & 5 & 4 & 4 \\ 20 & 36 & 238 & 0 & 36 & 24 & 28 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 36 & 0 & 4 & 4 & 4 \\ 2 & 4 & 24 & 0 & 4 & 0 & 2 \\ 3 & 4 & 28 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.806 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 3 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.807 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{113}31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 109, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0 \\ T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 45 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 111, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2 \\ T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 165, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2 \\ T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 174, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2 \\ T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2 \\ T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 240, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.7 \\ T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 8 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 291, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2 \\ T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 331, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2 \\ T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 336, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.7 \\ T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 26 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 365, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0 \\ T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 12 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 370, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 16 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 39 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{115} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 20 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 781, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 33 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3^1 1^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 52 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 46 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 40 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [23] \quad 1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 18, \quad T_{\max} = 21, \quad T \in [18, 21]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 7 & -2 & -3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 15.11 \rightarrow 16, \quad T_{\max} = 21, \quad T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 3 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 26, \quad \Delta = -115, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 26, \quad T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 26, \quad \Delta = -115, \quad \det = -115, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 26, \quad T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -190, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 14 & 0 & 7 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 3 & 5 & 8 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \quad T_{\max} = 26, \quad \Delta = -115, \quad \det = -435, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 26, \quad T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [30] \quad 1^{11}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -81, \quad \det = 34, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 22 & 5 \\ 12 & 26 & 1 & 48 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 48 & 0 & 86 & 18 \\ 5 & 10 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

4.808 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{green}{\mathfrak{su}}_2$

$$T = 9 \quad [0] \quad 1^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -106, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.7$$

$$T_{\min} = 15, \quad T_{\max} = 21, \quad T \in [15, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -106, \quad \det = -90, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

14.2

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 4 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$4.809 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 8 & 4 & 0 & 4 & 5 \\ 3 & 8 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.810 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + su2n3mix

$$T = 9 \quad [0] \ 1^{111} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

4.811 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & 0 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.812 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{1}1} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 10 & 0 & 5 & 4 \\ 6 & 10 & 22 & 0 & 12 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 8 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}11} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 5 & 3 \\ 4 & 9 & 16 & 0 & 12 & 5 \\ 7 & 16 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 21 & 0 & 13 & 7 \\ 3 & 5 & 10 & -3 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}11} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 3 & -3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}11} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -2 & 3 & 0 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 21^{\overline{1}1} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -2 & 3 & 0 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1}33} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 27 & 4 \\ 9 & 17 & 49 & 0 & 81 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 81 & 0 & 133 & 21 \\ 3 & 4 & 13 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1}1} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1}11} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 3 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1}21} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1}24} 3^{\overline{1}1} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 8 & 0 & 31 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 42 & 0 & 172 & 24 \\ 3 & 4 & 6 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{11}1} 3^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}2} 3^{\overline{12}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 3 & -3 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{11}3} 3^{\overline{12}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 10 & 0 & 14 & 4 \\ 6 & 10 & 22 & 0 & 30 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 30 & 0 & 37 & 11 \\ 3 & 4 & 8 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{13}3} 3^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{11}1} 3^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{11}2} 3^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{\overline{11}3} 3^{\overline{13}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 3 & -3 & 21 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{12}2} 3^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{11}3} 3^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{11}2} 3^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.19 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 3 & -3 & 11 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 24 & 42 & 0 & 121 & 19 \\ 3 & 4 & 7 & -3 & 19 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{123}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{133}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 9 & 3 \\ 3 & 4 & 12 & 0 & 17 & 4 \\ 6 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 36 & 0 & 49 & 13 \\ 3 & 4 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{114}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{133}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 12 & 0 & 12 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{124}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{134}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 12 & 0 & 16 & 4 \\ 6 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 36 & 0 & 46 & 12 \\ 3 & 4 & 9 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{115}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 15 & 0 & 73 & 15 \\ 3 & 4 & 3 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 \\ 4 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{125}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 30 & 0 & 73 & 15 \\ 3 & 4 & 6 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{144}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 16 & 0 & 16 & 4 \\ 8 & 16 & 46 & 0 & 48 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 48 & 0 & 46 & 12 \\ 3 & 4 & 12 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{114}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 3 & -3 & 28 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{123}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 6 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{112}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 \\ 8 & 14 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 33 & 0 & 28 & 10 \\ 3 & 4 & 11 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{135}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 12 & 0 & 20 & 4 \\ 6 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 45 & 0 & 73 & 15 \\ 3 & 4 & 9 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{116}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 18 & 0 & 106 & 18 \\ 3 & 4 & 3 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.66 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 3 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{131}1^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{126}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.9$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 36 & 0 & 106 & 18 \\ 3 & 4 & 6 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{145}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 16 & 0 & 20 & 4 \\ 8 & 16 & 46 & 0 & 60 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 60 & 0 & 73 & 15 \\ 3 & 4 & 12 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{136}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 12 & 0 & 24 & 4 \\ 6 & 12 & 25 & 0 & 54 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 54 & 0 & 106 & 18 \\ 3 & 4 & 9 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{117}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 21 & 0 & 145 & 21 \\ 3 & 4 & 3 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{155}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 20 & 0 & 20 & 4 \\ 10 & 20 & 73 & 0 & 75 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 75 & 0 & 73 & 15 \\ 3 & 4 & 15 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{1111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 9 & 12 & 0 & 12 & 5 \\ 5 & 12 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 5 & 7 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{146}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.28 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 16 & 0 & 24 & 4 \\ 8 & 16 & 46 & 0 & 72 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 72 & 0 & 106 & 18 \\ 3 & 4 & 12 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{127}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 28 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 42 & 0 & 145 & 21 \\ 3 & 4 & 6 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 3 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{137}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 14 & 3 \\ 3 & 4 & 12 & 0 & 28 & 4 \\ 6 & 12 & 25 & 0 & 63 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 63 & 0 & 145 & 21 \\ 3 & 4 & 9 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{112}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 3 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{156}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 20 & 0 & 24 & 4 \\ 10 & 20 & 73 & 0 & 90 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 90 & 0 & 106 & 18 \\ 3 & 4 & 15 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 3 & -3 & 19 & -2 \end{pmatrix}_{b_0}$$

[57] $1^{147}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 14 & 3 \\ 3 & 4 & 16 & 0 & 28 & 4 \\ 8 & 16 & 46 & 0 & 84 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 84 & 0 & 145 & 21 \\ 3 & 4 & 12 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{166}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 24 & 0 & 24 & 4 \\ 12 & 24 & 106 & 0 & 108 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 108 & 0 & 106 & 18 \\ 3 & 4 & 18 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[59] $1^{157}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 20 & 0 & 28 & 4 \\ 10 & 20 & 73 & 0 & 105 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 105 & 0 & 145 & 21 \\ 3 & 4 & 15 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[60] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 11 & 3 \\ 3 & 4 & 12 & 0 & 21 & 4 \\ 6 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 45 & 0 & 76 & 16 \\ 3 & 4 & 9 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{132}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 18 & 4 \\ 9 & 17 & 49 & 0 & 54 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 54 & 0 & 58 & 14 \\ 3 & 4 & 13 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{167}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 24 & 0 & 28 & 4 \\ 12 & 24 & 106 & 0 & 126 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 126 & 0 & 145 & 21 \\ 3 & 4 & 18 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 21 & 0 & 73 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{111}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 33 & 0 & 73 & 15 \\ 3 & 4 & 7 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{143}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 \\ 11 & 21 & 76 & 0 & 81 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 81 & 0 & 85 & 17 \\ 3 & 4 & 16 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 24 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 24 & 27 & 0 & 121 & 19 \\ 3 & 4 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{177}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 28 & 0 & 28 & 4 \\ 14 & 28 & 145 & 0 & 147 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 147 & 0 & 145 & 21 \\ 3 & 4 & 21 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 282$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{122}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 284$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 7 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 3 & -3 & 25 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{113}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 312$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 8 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{111}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 317$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 5 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 319$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 319$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 324$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.21 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 3 & -3 & 22 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{12}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 324$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 338$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 359$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{14}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 360$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 372$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{13}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 375$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{124}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 380$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 381$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{124}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 8 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{12}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 480$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 503$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112}3^11^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 8 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{122}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{123}3^11^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 543, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 18 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 7 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{122}3^11^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 7 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{112}3^11^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{123}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{124}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{12}3^11^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{13}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{\overline{13}3}31^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{\overline{13}3}31^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 653, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[102] $21^{\overline{11}1}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 6 & 4 \\ 8 & 9 & 8 & 4 & 8 & 5 \\ 6 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -2 & 3 & 0 \\ 6 & 8 & 6 & 3 & 4 & 4 \\ 4 & 5 & 4 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{\overline{12}3}31^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 695, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{\overline{13}3}31^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 9 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{\overline{12}3}31^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 5 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{\overline{14}3}31^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{\overline{13}3}31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 9 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{\overline{134}3}31^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 759, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 17 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 9 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{\overline{124}3}31^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 764, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 10 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{\overline{13}3} 3^{\overline{1}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 765, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{\overline{14}3} 3^{\overline{1}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 794, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.51 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\overline{14}4} 3^{\overline{1}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 824, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.55 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 11 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overline{13}4} 3^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 831, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 9 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overline{12}3} 3^{\overline{1}13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 863, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 11.53 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overline{12}2} 3^{\overline{1}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 868, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.53 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 10 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overline{13}3} 3^{\overline{1}14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 879, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overline{12}3} 3^{\overline{1}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 879, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 11.51 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{14}4} 3^{\overline{1}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 890, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.51 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{12}3} 3^{\overline{1}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 954, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 11.52 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 10 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{144}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 988$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 11 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{114}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 991$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{113}3^11^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1064$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{14}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1080$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{114}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1094$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{134}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1122$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 9 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{124}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{113}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1269$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{124}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 20 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{133}3^11^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1320$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 12 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{\overline{1}14}31^{\overline{1}13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1367$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{\overline{1}24}31^{\overline{1}13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1376$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 7 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{\overline{1}34}31^{\overline{1}22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1410$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 12 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{\overline{1}14}31^{\overline{1}14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1750$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{\overline{1}44}31^{\overline{1}22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1780$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 14 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [135] $1^{\overline{1}11}321^{\overline{1}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 & 0 & 3 \\ 1 & 1 & 0 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{\overline{1}1}321^{\overline{1}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 5 & 4 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{\overline{1}11}321^{\overline{1}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 3 & 5 \\ 1 & 1 & 0 & 4 & -2 & 1 & 0 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 2 & -1 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{\overline{1}1}321^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{11}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 3 & 0 & 7 & 2 \\ 1 & 1 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{12}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{14}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 6 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 21 & 30 & 0 & 172 & 24 \\ 2 & 4 & 2 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 4 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{111}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 5 & 10 & 0 & 20 & 5 \\ 5 & 10 & 14 & 0 & 30 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 30 & 0 & 58 & 16 \\ 3 & 5 & 8 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{12}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 3 & 0 & 8 & 2 \\ 1 & 1 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{11}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -120$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 2 & 0 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 12 \\ 3 & 5 & 4 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{11} 3^{11} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -120$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 2 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 19 & 0 & 92 & 20 \\ 3 & 5 & 4 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{112} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 6 \\ 1 & 2 & 0 & 1 & 0 & 8 \\ 1 & 2 & 1 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 4 & 6 & 8 & 6 & 0 & 26 \\ 2 & 2 & 2 & 2 & -3 & 10 \end{pmatrix}_{b_0}$$

[151] $1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -139$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[152] $1^{11} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 2 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 58 & 16 \\ 3 & 5 & 4 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{12} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 11 & 2 \\ 1 & 1 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{14} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & 3 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{111} 3^{11} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -154$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 3 & 4 & 5 & 0 & 11 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 11 & 0 & 16 & 8 \\ 2 & 2 & 2 & 5 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{113} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 6 & 2 \\ 2 & 3 & 4 & 5 & 0 & 13 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 13 & 0 & 25 & 9 \\ 2 & 2 & 2 & 5 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{112} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[158] $1^{\overline{111}1^1}321^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 5 & 10 & 0 & 10 & 5 \\ 5 & 10 & 14 & 0 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 16 & 0 & 14 & 8 \\ 3 & 5 & 8 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{\overline{13}1^1}321^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 3 & 0 & 12 & 2 \\ 1 & 1 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[160] $1^{\overline{111}1^1}321^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 5 & 5 \\ 2 & 2 & 2 & 5 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{\overline{111}1^1}321^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 5 & 10 & 0 & 15 & 5 \\ 5 & 10 & 14 & 0 & 23 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 23 & 0 & 32 & 12 \\ 3 & 5 & 8 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{\overline{112}1^1}321^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 14 & 2 \\ 2 & 3 & 4 & 5 & 0 & 22 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 14 & 22 & 0 & 70 & 16 \\ 2 & 2 & 2 & 5 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{\overline{111}1^1}31^{\overline{123}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{\overline{113}1^1}321^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 9 & 0 & 61 & 15 \\ 2 & 2 & 2 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{\overline{113}1^1}321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 7 & 0 & 25 & 9 \\ 2 & 2 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{\overline{112}1^1}321^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 5 & 2 \\ 2 & 3 & 4 & 5 & 0 & 10 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 10 & 0 & 13 & 7 \\ 2 & 2 & 2 & 5 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{114}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 8 & 2 \\ 2 & 4 & 4 & 9 & 0 & 24 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 18 & 42 & 0 & 118 & 20 \\ 2 & 4 & 2 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{12}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 2 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 25 & 18 & 0 & 94 & 20 \\ 3 & 5 & 4 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{113}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 15 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & 3 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{13}321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 3 & 0 & 15 & 2 \\ 1 & 1 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{13}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{113}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{113}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 8 & 0 & 40 & 12 \\ 2 & 2 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{121}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 3 & 2 \\ 2 & 4 & 4 & 12 & 0 & 9 & 4 \\ 1 & 4 & 0 & 9 & 0 & 6 & 2 \\ 4 & 12 & 9 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 21 & 0 & 13 & 7 \\ 2 & 4 & 2 & 10 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{122}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 0 & 12 & 4 \\ 1 & 4 & 0 & 9 & 0 & 9 & 2 \\ 4 & 12 & 9 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 30 & 0 & 28 & 10 \\ 2 & 4 & 2 & 10 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{14}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 3 & 0 & 16 & 2 \\ 1 & 1 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{112}3^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -346$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{12}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -360$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 30 & 22 & 0 & 134 & 24 \\ 3 & 5 & 4 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{114}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -374$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 16 & 2 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 12 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{112}3^{11}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 11 & 2 \\ 2 & 3 & 4 & 5 & 0 & 18 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 18 & 0 & 45 & 13 \\ 2 & 2 & 2 & 5 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{112}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -414$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 6 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 3 & 4 & 5 & 0 & 14 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 14 & 0 & 26 & 10 \\ 2 & 2 & 2 & 5 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{13}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 2 & 0 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 35 & 25 & 0 & 188 & 28 \\ 3 & 5 & 4 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{114}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -494$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{12}3^{11}21^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -508$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 18 & 0 & 58 & 14 \\ 2 & 4 & 2 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{114}3^{21}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 2 & 1 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 11 & 0 & 81 & 17 \\ 2 & 2 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{111}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 15 & 0 & 13 & 7 \\ 2 & 4 & 2 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{122}3^{11}1^{12}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -532$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 0 & 18 & 4 \\ 1 & 4 & 0 & 9 & 0 & 12 & 2 \\ 4 & 12 & 9 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 42 & 0 & 58 & 14 \\ 2 & 4 & 2 & 10 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{112}3^{11}1^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -549$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113}3^{21}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -554$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 9 & 2 \\ 2 & 3 & 4 & 5 & 0 & 17 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 17 & 0 & 40 & 12 \\ 2 & 2 & 2 & 5 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{114}3^{21}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -574$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 10 & 2 \\ 2 & 3 & 4 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 20 & 0 & 58 & 14 \\ 2 & 2 & 2 & 5 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{114}3^{11}1^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -621$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 1 & 3 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 21 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{113}3^{11}1^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -691$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 1 & 3 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & 3 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111}3^{11}1^{12}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -693$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 15 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{113} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -764, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 9 & 2 \\ 1 & 2 & 0 & 4 & 0 & 12 & 2 \\ 2 & 3 & 4 & 5 & 0 & 21 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 21 & 0 & 61 & 15 \\ 2 & 2 & 2 & 5 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{113} 3^{11} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 15 & 2 \\ 1 & 2 & 0 & 4 & 0 & 15 & 2 \\ 2 & 3 & 4 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 25 & 0 & 88 & 18 \\ 2 & 2 & 2 & 5 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{111} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -811, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 0 & 15 & 4 \\ 1 & 4 & 0 & 6 & 0 & 9 & 2 \\ 3 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 24 & 0 & 37 & 11 \\ 2 & 4 & 2 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{13} 31^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -853, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 27 & 0 & 133 & 21 \\ 2 & 4 & 2 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{113} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -859, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 7 & 2 \\ 2 & 4 & 4 & 9 & 0 & 21 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 15 & 36 & 0 & 85 & 17 \\ 2 & 4 & 2 & 7 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{111} 31^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1000, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 7 & 2 \\ 2 & 4 & 4 & 9 & 0 & 21 & 4 \\ 1 & 4 & 0 & 6 & 0 & 12 & 2 \\ 3 & 9 & 6 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 33 & 0 & 73 & 15 \\ 2 & 4 & 2 & 7 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{114} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1004, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 11 & 2 \\ 1 & 2 & 0 & 4 & 0 & 13 & 2 \\ 2 & 3 & 4 & 5 & 0 & 24 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 24 & 0 & 81 & 17 \\ 2 & 2 & 2 & 5 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{112} 31^{122} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1018, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 6 & 2 \\ 2 & 4 & 4 & 15 & 0 & 18 & 4 \\ 1 & 4 & 0 & 9 & 0 & 12 & 2 \\ 5 & 15 & 9 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 48 & 0 & 58 & 14 \\ 2 & 4 & 2 & 11 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{113}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1020$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 3 & 0 & 33 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 1 & 3 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & 3 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{112}31^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1046$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 6 & 2 \\ 2 & 4 & 4 & 9 & 0 & 18 & 4 \\ 1 & 4 & 0 & 6 & 0 & 12 & 2 \\ 3 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 30 & 0 & 58 & 14 \\ 2 & 4 & 2 & 7 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{114}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 3 & 0 & 36 & 4 \\ 1 & 4 & 0 & 3 & 0 & 24 & 2 \\ 1 & 3 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{113}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1197$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 21 & 0 & 73 & 15 \\ 2 & 4 & 2 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{123}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 24 & 0 & 100 & 18 \\ 2 & 4 & 2 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{114}321^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 0 & 2 & 3 & 0 & 12 & 2 \\ 1 & 2 & 0 & 4 & 0 & 16 & 2 \\ 2 & 3 & 4 & 5 & 0 & 28 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 28 & 0 & 110 & 20 \\ 2 & 2 & 2 & 5 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{111}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 5 & 2 \\ 2 & 4 & 4 & 15 & 0 & 15 & 4 \\ 1 & 4 & 0 & 9 & 0 & 9 & 2 \\ 5 & 15 & 9 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 39 & 0 & 37 & 11 \\ 2 & 4 & 2 & 11 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{14}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 0 & 36 & 4 \\ 1 & 4 & 0 & 3 & 0 & 24 & 2 \\ 2 & 6 & 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 36 & 0 & 238 & 28 \\ 2 & 4 & 2 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{122}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1568$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 6 & 2 \\ 2 & 4 & 4 & 18 & 0 & 18 & 4 \\ 1 & 4 & 0 & 12 & 0 & 12 & 2 \\ 6 & 18 & 12 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 60 & 0 & 58 & 14 \\ 2 & 4 & 2 & 14 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{12}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1669$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 8 & 2 \\ 2 & 4 & 4 & 9 & 0 & 24 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 39 & 0 & 100 & 18 \\ 2 & 4 & 2 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{14}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1741$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 10 & 2 \\ 2 & 4 & 4 & 9 & 0 & 30 & 4 \\ 1 & 4 & 0 & 6 & 0 & 21 & 2 \\ 3 & 9 & 6 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 21 & 51 & 0 & 172 & 24 \\ 2 & 4 & 2 & 7 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{11}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1813$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 15 & 27 & 0 & 121 & 19 \\ 2 & 4 & 2 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{13}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1916$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 9 & 2 \\ 2 & 4 & 4 & 9 & 0 & 27 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 45 & 0 & 133 & 21 \\ 2 & 4 & 2 & 7 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{13}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1917$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 6 & 0 & 33 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 6 & 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 33 & 0 & 193 & 25 \\ 2 & 4 & 2 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{12}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2076$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 6 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 18 & 30 & 0 & 154 & 22 \\ 2 & 4 & 2 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{12}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 10 & 2 \\ 2 & 4 & 4 & 9 & 0 & 30 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 18 & 48 & 0 & 154 & 22 \\ 2 & 4 & 2 & 7 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{13}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2875$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 11 & 2 \\ 2 & 4 & 4 & 9 & 0 & 33 & 4 \\ 1 & 4 & 0 & 6 & 0 & 21 & 2 \\ 3 & 9 & 6 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 54 & 0 & 193 & 25 \\ 2 & 4 & 2 & 7 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{14}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -3134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 12 & 2 \\ 2 & 4 & 4 & 9 & 0 & 36 & 4 \\ 1 & 4 & 0 & 6 & 0 & 24 & 2 \\ 3 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 60 & 0 & 238 & 28 \\ 2 & 4 & 2 & 7 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

4.813 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{\overline{111}1^1} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 & 0 \\ 3 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 0 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

4.814 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{112}3^1} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}1^1} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}1^1} 3^1 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}1^1} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 & 0 \\ 4 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}1^1} 3^1 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}1^1} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 & 0 \\ 3 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{111}1^1} 3^1 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.2

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

4.815 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 2 & 3 \\ 3 & 4 & 8 & 0 & 8 & 3 & 4 \\ 4 & 8 & 10 & 0 & 12 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{124} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 2 & 3 \\ 3 & 4 & 8 & 0 & 31 & 3 & 4 \\ 4 & 8 & 10 & 0 & 42 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 42 & 0 & 172 & 17 & 24 \\ 2 & 3 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & 6 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 3 & 4 \\ 4 & 8 & 10 & 0 & 12 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 5 & 7 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 6 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 3 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 9 & 13 \\ 2 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[5] \quad & 1^{112}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7 \\
& T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 4 & 6 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{13}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 3 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{111}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.50 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{123}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5 \\
& T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 2 & 3 \\ 3 & 4 & 8 & 0 & 22 & 3 & 4 \\ 4 & 8 & 10 & 0 & 30 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 30 & 0 & 85 & 12 & 17 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 6 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{122}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 2 & 3 \\ 3 & 4 & 8 & 0 & 13 & 3 & 4 \\ 4 & 8 & 10 & 0 & 18 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 7 & 10 \\ 2 & 3 & 4 & 0 & 7 & 0 & 2 \\ 3 & 4 & 6 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{111}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 3 & 4 \\ 2 & 4 & 1 & 0 & 9 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 8 & 11 \\ 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 3 & 4 & 3 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{123}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7 \\
& T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 3 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 7 & 10 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{113}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 3 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[13] $1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{123}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 2 & 3 \\ 3 & 4 & 8 & 0 & 17 & 3 & 4 \\ 4 & 8 & 10 & 0 & 24 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 24 & 0 & 49 & 9 & 13 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 6 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{114}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 2 & 3 \\ 3 & 4 & 9 & 0 & 21 & 3 & 4 \\ 5 & 9 & 13 & 0 & 33 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 33 & 0 & 76 & 11 & 16 \\ 2 & 3 & 5 & 0 & 11 & 0 & 2 \\ 3 & 4 & 7 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{124}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 2 & 3 \\ 3 & 4 & 8 & 0 & 21 & 3 & 4 \\ 4 & 8 & 10 & 0 & 30 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 30 & 0 & 76 & 11 & 16 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & 6 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{112}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 3 & 4 \\ 2 & 4 & 1 & 0 & 9 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 7 & 10 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 3 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{124}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 2 & 3 \\ 3 & 4 & 8 & 0 & 26 & 3 & 4 \\ 4 & 8 & 10 & 0 & 36 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 36 & 0 & 118 & 14 & 20 \\ 2 & 3 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & 6 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 3 & 4 \\ 5 & 9 & 13 & 0 & 15 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 5 & 0 & 2 \\ 3 & 4 & 7 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{112}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 23 & 3 & 4 \\ 2 & 4 & 1 & 0 & 15 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 13 & 18 \\ 2 & 3 & 2 & 0 & 13 & 0 & 2 \\ 3 & 4 & 3 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{113}3^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 22 & 3 & 4 \\ 2 & 4 & 1 & 0 & 15 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 12 & 17 \\ 2 & 3 & 2 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{134}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 2 & 3 \\ 3 & 4 & 17 & 0 & 17 & 3 & 4 \\ 9 & 17 & 49 & 0 & 51 & 9 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 51 & 0 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 13 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{113}3^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 2 & 3 \\ 3 & 4 & 4 & 0 & 32 & 3 & 4 \\ 2 & 4 & 1 & 0 & 21 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 18 & 25 \\ 2 & 3 & 2 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{134}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 2 & 3 \\ 3 & 4 & 17 & 0 & 21 & 3 & 4 \\ 9 & 17 & 49 & 0 & 63 & 9 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 63 & 0 & 76 & 11 & 16 \\ 2 & 3 & 9 & 0 & 11 & 0 & 2 \\ 3 & 4 & 13 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{113}3^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 4 & 0 & 27 & 3 & 4 \\ 2 & 4 & 1 & 0 & 18 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 15 & 21 \\ 2 & 3 & 2 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{144}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 2 & 3 \\ 3 & 4 & 21 & 0 & 21 & 3 & 4 \\ 11 & 21 & 76 & 0 & 78 & 11 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 78 & 0 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 11 & 0 & 2 \\ 3 & 4 & 16 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{114}3^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 26 & 3 & 4 \\ 2 & 4 & 1 & 0 & 18 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 14 & 20 \\ 2 & 3 & 2 & 0 & 14 & 0 & 2 \\ 3 & 4 & 3 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{114}3^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 31 & 3 & 4 \\ 2 & 4 & 1 & 0 & 21 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 17 & 24 \\ 2 & 3 & 2 & 0 & 17 & 0 & 2 \\ 3 & 4 & 3 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{114}3^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -1604, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 4 & 0 & 36 & 3 & 4 \\ 2 & 4 & 1 & 0 & 24 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 20 & 28 \\ 2 & 3 & 2 & 0 & 20 & 0 & 2 \\ 3 & 4 & 3 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \ 1^{111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 247, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 3 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^13^11^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 472, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11}3^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 517, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 648, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 6 & 3 \\ 4 & 8 & 6 & 15 & 0 & 15 & 6 \\ 2 & 6 & 1 & 9 & 0 & 9 & 3 \\ 6 & 15 & 9 & 22 & 0 & 24 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 24 & 0 & 22 & 10 \\ 3 & 6 & 3 & 10 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{111}3^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 912, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 15 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 5 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 7 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.816 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 6 & 5 & 12 & 6 \\ 3 & 4 & 3 & -3 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 3 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 5 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 5 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.817 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + su2n3mix

$$T = 9 \quad [0] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 6 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1211} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 4 & 3 \\ 3 & 4 & 8 & 0 & 9 & 7 & 4 \\ 4 & 8 & 10 & 0 & 12 & 10 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 11 & 7 \\ 4 & 7 & 10 & 0 & 11 & 7 & 5 \\ 3 & 4 & 6 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 9 & 3 \\ 3 & 4 & 4 & 0 & 8 & 16 & 4 \\ 2 & 4 & 1 & 0 & 6 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 22 & 6 \\ 9 & 16 & 11 & 0 & 22 & 44 & 12 \\ 3 & 4 & 3 & -3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{131}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 4 & 3 \\ 3 & 4 & 4 & 0 & 12 & 7 & 4 \\ 2 & 4 & 1 & 0 & 9 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 9 & 0 & 25 & 15 & 9 \\ 4 & 7 & 5 & 0 & 15 & 7 & 5 \\ 3 & 4 & 3 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 5 & 0 & 9 & 7 & 4 \\ 3 & 5 & 4 & 0 & 9 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 6 & 0 & 11 & 7 & 5 \\ 3 & 4 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{131}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 5 & 0 & 17 & 7 & 4 \\ 3 & 5 & 4 & 0 & 15 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 21 & 13 \\ 4 & 7 & 6 & 0 & 21 & 7 & 5 \\ 3 & 4 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{121}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 9 & 3 \\ 3 & 4 & 8 & 0 & 9 & 16 & 4 \\ 4 & 8 & 10 & 0 & 12 & 22 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 26 & 7 \\ 9 & 16 & 22 & 0 & 26 & 44 & 12 \\ 3 & 4 & 6 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{122}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 7 & 4 \\ 4 & 8 & 10 & 0 & 12 & 10 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 10 & 6 \\ 4 & 7 & 10 & 0 & 10 & 7 & 5 \\ 3 & 4 & 6 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 7 & 3 \\ 3 & 4 & 4 & 0 & 17 & 12 & 4 \\ 2 & 4 & 1 & 0 & 12 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 36 & 13 \\ 7 & 12 & 8 & 0 & 36 & 25 & 9 \\ 3 & 4 & 3 & -3 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 11 & 4 \\ 5 & 9 & 13 & 0 & 15 & 17 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 17 & 7 \\ 6 & 11 & 17 & 0 & 17 & 20 & 8 \\ 3 & 4 & 7 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 20 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 14 & 3 \\ 11 & 20 & 18 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}3^{\overline{1}}1^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 12 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{112}3^{\overline{1}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 7 & 4 \\ 2 & 4 & 1 & 0 & 6 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 10 & 6 \\ 4 & 7 & 5 & 0 & 10 & 7 & 5 \\ 3 & 4 & 3 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{121}3^{\overline{1}}1^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 5 & 0 & 13 & 7 & 4 \\ 3 & 5 & 4 & 0 & 12 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 16 & 10 \\ 4 & 7 & 6 & 0 & 16 & 7 & 5 \\ 3 & 4 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{\overline{1112}3^{\overline{1}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 11 & 4 \\ 2 & 4 & 1 & 0 & 6 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 17 & 7 \\ 6 & 11 & 8 & 0 & 17 & 20 & 8 \\ 3 & 4 & 3 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{\overline{1121}3^{\overline{1}}1^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 16 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 11 & 3 \\ 9 & 16 & 15 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{\overline{1111}3^{\overline{1}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 12 & 8 & 0 & 8 & 25 & 9 \\ 3 & 4 & 3 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{\overline{1111}3^{\overline{1}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 2 & 4 & 1 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 5 & 0 & 5 & 7 & 5 \\ 3 & 4 & 3 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{\overline{1111}3^{\overline{1}}1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 4 & 3 \\ 3 & 4 & 4 & 0 & 9 & 7 & 4 \\ 2 & 4 & 1 & 0 & 6 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 11 & 7 \\ 4 & 7 & 5 & 0 & 11 & 7 & 5 \\ 3 & 4 & 3 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{11221}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 4 & 3 \\ 3 & 4 & 8 & 0 & 13 & 7 & 4 \\ 4 & 8 & 10 & 0 & 18 & 10 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 16 & 10 \\ 4 & 7 & 10 & 0 & 16 & 7 & 5 \\ 3 & 4 & 6 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11223}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 11 & 4 \\ 2 & 4 & 1 & 0 & 6 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 16 & 6 \\ 6 & 11 & 8 & 0 & 16 & 20 & 8 \\ 3 & 4 & 3 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1121}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 4 & 3 \\ 3 & 4 & 4 & 0 & 18 & 7 & 4 \\ 2 & 4 & 1 & 0 & 12 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 22 & 14 \\ 4 & 7 & 5 & 0 & 22 & 7 & 5 \\ 3 & 4 & 3 & -3 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11123}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 11 & 4 \\ 2 & 4 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 6 & 11 & 8 & 0 & 8 & 20 & 8 \\ 3 & 4 & 3 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11223}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 6 & 3 \\ 3 & 4 & 9 & 0 & 13 & 11 & 4 \\ 5 & 9 & 13 & 0 & 21 & 17 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 25 & 10 \\ 6 & 11 & 17 & 0 & 25 & 20 & 8 \\ 3 & 4 & 7 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{11113}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 4 & 3 \\ 3 & 4 & 9 & 0 & 9 & 7 & 4 \\ 5 & 9 & 13 & 0 & 15 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 11 & 7 \\ 4 & 7 & 11 & 0 & 11 & 7 & 5 \\ 3 & 4 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1131}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 4 & 3 \\ 3 & 4 & 4 & 0 & 27 & 7 & 4 \\ 2 & 4 & 1 & 0 & 18 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 33 & 21 \\ 4 & 7 & 5 & 0 & 33 & 7 & 5 \\ 3 & 4 & 3 & -3 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11123}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 14 & 3 \\ 12 & 21 & 14 & 0 & 14 & 80 & 16 \\ 3 & 4 & 3 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11213}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 28 & 10 \\ 7 & 12 & 8 & 0 & 28 & 25 & 9 \\ 3 & 4 & 3 & -3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111213}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 12 & 4 \\ 2 & 4 & 1 & 0 & 6 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 20 & 7 \\ 7 & 12 & 8 & 0 & 20 & 25 & 9 \\ 3 & 4 & 3 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11213}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -536, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 4 & 3 \\ 3 & 4 & 4 & 0 & 13 & 7 & 4 \\ 2 & 4 & 1 & 0 & 9 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 16 & 10 \\ 4 & 7 & 5 & 0 & 16 & 7 & 5 \\ 3 & 4 & 3 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{111413}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 4 & 3 \\ 3 & 4 & 9 & 0 & 21 & 7 & 4 \\ 5 & 9 & 13 & 0 & 33 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 33 & 0 & 76 & 26 & 16 \\ 4 & 7 & 11 & 0 & 26 & 7 & 5 \\ 3 & 4 & 7 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11213}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 6 & 3 \\ 3 & 4 & 4 & 0 & 13 & 11 & 4 \\ 2 & 4 & 1 & 0 & 9 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 25 & 10 \\ 6 & 11 & 8 & 0 & 25 & 20 & 8 \\ 3 & 4 & 3 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -564, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 15 & 4 \\ 2 & 4 & 1 & 0 & 3 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 11 & 3 \\ 8 & 15 & 11 & 0 & 11 & 39 & 11 \\ 3 & 4 & 3 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{111413}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -616, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 4 & 3 \\ 3 & 4 & 4 & 0 & 36 & 7 & 4 \\ 2 & 4 & 1 & 0 & 24 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 44 & 28 \\ 4 & 7 & 5 & 0 & 44 & 7 & 5 \\ 3 & 4 & 3 & -3 & 28 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 16 & 4 \\ 2 & 4 & 1 & 0 & 3 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 11 & 3 \\ 9 & 16 & 11 & 0 & 11 & 44 & 12 \\ 3 & 4 & 3 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{111} 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -626, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 4 & 3 \\ 3 & 4 & 9 & 0 & 13 & 7 & 4 \\ 5 & 9 & 13 & 0 & 21 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 16 & 10 \\ 4 & 7 & 11 & 0 & 16 & 7 & 5 \\ 3 & 4 & 7 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{111} 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -656, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 4 & 3 \\ 3 & 4 & 9 & 0 & 17 & 7 & 4 \\ 5 & 9 & 13 & 0 & 27 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 21 & 13 \\ 4 & 7 & 11 & 0 & 21 & 7 & 5 \\ 3 & 4 & 7 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{111} 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 4 & 3 \\ 3 & 4 & 4 & 0 & 17 & 7 & 4 \\ 2 & 4 & 1 & 0 & 12 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 21 & 13 \\ 4 & 7 & 5 & 0 & 21 & 7 & 5 \\ 3 & 4 & 3 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111} 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 19 & 4 \\ 2 & 4 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 14 & 3 \\ 10 & 19 & 14 & 0 & 14 & 64 & 14 \\ 3 & 4 & 3 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{111} 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -810, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 16 & 4 \\ 2 & 4 & 1 & 0 & 6 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 26 & 7 \\ 9 & 16 & 11 & 0 & 26 & 44 & 12 \\ 3 & 4 & 3 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{111} 1^{111} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -836, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 4 & 3 \\ 3 & 4 & 4 & 0 & 22 & 7 & 4 \\ 2 & 4 & 1 & 0 & 15 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 27 & 17 \\ 4 & 7 & 5 & 0 & 27 & 7 & 5 \\ 3 & 4 & 3 & -3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{111} 1^{111} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -840, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 30 & 4 \\ 2 & 4 & 1 & 0 & 3 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 20 & 3 \\ 17 & 30 & 20 & 0 & 20 & 165 & 23 \\ 3 & 4 & 3 & -3 & 3 & 23 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111} 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 4 & 3 \\ 3 & 4 & 4 & 0 & 21 & 7 & 4 \\ 2 & 4 & 1 & 0 & 15 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 26 & 16 \\ 4 & 7 & 5 & 0 & 26 & 7 & 5 \\ 3 & 4 & 3 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{112213}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 4 & 3 \\ 3 & 4 & 13 & 0 & 13 & 7 & 4 \\ 7 & 13 & 28 & 0 & 30 & 16 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 16 & 10 \\ 4 & 7 & 16 & 0 & 16 & 7 & 5 \\ 3 & 4 & 10 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{1113}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 20 & 4 \\ 2 & 4 & 1 & 0 & 6 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 32 & 7 \\ 11 & 20 & 14 & 0 & 32 & 69 & 15 \\ 3 & 4 & 3 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{1113}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 4 & 20 & 4 \\ 2 & 4 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 14 & 3 \\ 11 & 20 & 14 & 0 & 14 & 69 & 15 \\ 3 & 4 & 3 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{1113}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -986, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 13 & 3 \\ 3 & 4 & 4 & 0 & 9 & 24 & 4 \\ 2 & 4 & 1 & 0 & 6 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 38 & 7 \\ 13 & 24 & 17 & 0 & 38 & 100 & 18 \\ 3 & 4 & 3 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{11223}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1020, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 16 & 4 \\ 2 & 4 & 1 & 0 & 9 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 37 & 10 \\ 9 & 16 & 11 & 0 & 37 & 44 & 12 \\ 3 & 4 & 3 & -3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{1113}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1076, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 25 & 4 \\ 2 & 4 & 1 & 0 & 3 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 17 & 3 \\ 14 & 25 & 17 & 0 & 17 & 111 & 19 \\ 3 & 4 & 3 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{1113}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 4 & 3 \\ 3 & 4 & 4 & 0 & 31 & 7 & 4 \\ 2 & 4 & 1 & 0 & 21 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 38 & 24 \\ 4 & 7 & 5 & 0 & 38 & 7 & 5 \\ 3 & 4 & 3 & -3 & 24 & 5 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{1113}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 4 & 3 \\ 3 & 4 & 4 & 0 & 26 & 7 & 4 \\ 2 & 4 & 1 & 0 & 18 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 32 & 20 \\ 4 & 7 & 5 & 0 & 32 & 7 & 5 \\ 3 & 4 & 3 & -3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

4.818 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 & 4 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 21 & 0 & 73 & 12 & 15 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 2 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 & 4 \\ 5 & 9 & 13 & 0 & 33 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 33 & 0 & 73 & 12 & 15 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 5 & 9 & 6 & 0 & 12 & 12 & 6 \\ 3 & 4 & 3 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 27 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

- [8] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 15 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{121}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{1121}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{1221}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 5 & 3 \\ 3 & 4 & 8 & 0 & 13 & 9 & 4 \\ 4 & 8 & 10 & 0 & 13 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 21 & 10 \\ 5 & 9 & 12 & 0 & 21 & 12 & 6 \\ 3 & 4 & 6 & -3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{1111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 3 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{1111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 2 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 & 4 \\ 5 & 9 & 13 & 0 & 24 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 24 & 0 & 37 & 9 & 11 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{121} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 & 4 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{121} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 & 4 \\ 3 & 5 & 4 & 0 & 30 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 30 & 0 & 154 & 18 & 22 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 7 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 2 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 & 4 \\ 8 & 14 & 37 & 0 & 39 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 39 & 0 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 11 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{121} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 & 4 \\ 3 & 5 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 24 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{131} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 & 4 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 21 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1131} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 2 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 3 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1121} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 4 & 0 & 13 & 9 & 4 \\ 2 & 4 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 21 & 10 \\ 5 & 9 & 6 & 0 & 21 & 12 & 6 \\ 3 & 4 & 3 & -3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1121} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 & 4 \\ 5 & 9 & 13 & 0 & 21 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 9 & 10 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1121} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -962, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 2 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 & 4 \\ 8 & 14 & 37 & 0 & 48 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 48 & 0 & 58 & 12 & 14 \\ 2 & 4 & 9 & 0 & 12 & 0 & 2 \\ 3 & 4 & 11 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1121} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 2 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 & 4 \\ 5 & 9 & 13 & 0 & 48 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 48 & 0 & 154 & 18 & 22 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1131} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 2 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 & 4 \\ 5 & 9 & 13 & 0 & 27 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 12 & 13 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1131} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -989, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 5 & 3 \\ 3 & 4 & 4 & 0 & 17 & 9 & 4 \\ 2 & 4 & 1 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 27 & 13 \\ 5 & 9 & 6 & 0 & 27 & 12 & 6 \\ 3 & 4 & 3 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1141} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1005, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 21 & 9 & 4 \\ 2 & 4 & 1 & 0 & 15 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 33 & 16 \\ 5 & 9 & 6 & 0 & 33 & 12 & 6 \\ 3 & 4 & 3 & -3 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1131^1} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{131^1} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 & 4 \\ 3 & 5 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1221^1} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 2 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 & 4 \\ 7 & 13 & 28 & 0 & 42 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 42 & 0 & 58 & 12 & 14 \\ 2 & 4 & 9 & 0 & 12 & 0 & 2 \\ 3 & 4 & 10 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1131^1} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1121^1} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 2 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 & 4 \\ 5 & 9 & 13 & 0 & 30 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 30 & 0 & 58 & 12 & 14 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1221^1} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 2 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 & 4 \\ 7 & 13 & 28 & 0 & 30 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 10 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1121^1} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 2 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 & 4 \\ 5 & 9 & 13 & 0 & 39 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 39 & 0 & 100 & 15 & 18 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1131^1} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{131} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 & 4 \\ 3 & 5 & 4 & 0 & 33 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 33 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{141} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 & 4 \\ 3 & 5 & 4 & 0 & 30 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 30 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1131} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 2 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 & 4 \\ 5 & 9 & 13 & 0 & 36 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 36 & 0 & 85 & 15 & 17 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1221} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 2 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 & 4 \\ 10 & 18 & 58 & 0 & 60 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 60 & 0 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 12 & 0 & 2 \\ 3 & 4 & 14 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{141} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 & 4 \\ 3 & 5 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1141} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 18 & 20 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1141} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 3 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1141} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 2 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 & 4 \\ 5 & 9 & 13 & 0 & 42 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 42 & 0 & 118 & 18 & 20 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{1141}31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 3 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{1131}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 2 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 & 4 \\ 5 & 9 & 13 & 0 & 45 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 45 & 0 & 133 & 18 & 21 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{1131}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2621$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 2 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 & 4 \\ 5 & 9 & 13 & 0 & 54 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 54 & 0 & 193 & 21 & 25 \\ 2 & 4 & 6 & 0 & 21 & 0 & 2 \\ 3 & 4 & 7 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{1141}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -3405$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 2 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 & 4 \\ 5 & 9 & 13 & 0 & 51 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 51 & 0 & 172 & 21 & 24 \\ 2 & 4 & 6 & 0 & 21 & 0 & 2 \\ 3 & 4 & 7 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{1141}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -4064$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 2 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 & 4 \\ 5 & 9 & 13 & 0 & 60 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 60 & 0 & 238 & 24 & 28 \\ 2 & 4 & 6 & 0 & 24 & 0 & 2 \\ 3 & 4 & 7 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [54] $1^{11}321^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 40$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 & 5 \\ 2 & 5 & 2 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 7 & 8 \\ 2 & 5 & 4 & 0 & 7 & 1 & 3 \\ 3 & 5 & 4 & -3 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{1111}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{11}321^{1113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 5 & 5 & 0 & 5 & 20 & 5 \\ 2 & 5 & 2 & 0 & 4 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 15 & 4 \\ 9 & 20 & 15 & 0 & 15 & 57 & 15 \\ 3 & 5 & 4 & -3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 321^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 4 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & 3 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 321^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 & 5 \\ 2 & 5 & 2 & 0 & 11 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 11 & 12 \\ 2 & 5 & 4 & 0 & 11 & 1 & 3 \\ 3 & 5 & 4 & -3 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111} 321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 3 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{12} 321^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 340, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 & 5 \\ 2 & 5 & 2 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 25 & 18 & 0 & 94 & 18 & 20 \\ 2 & 5 & 4 & 0 & 18 & 1 & 3 \\ 3 & 5 & 4 & -3 & 20 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{111} 321^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 360, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 5 & 10 & 0 & 10 & 5 & 5 \\ 5 & 10 & 14 & 0 & 16 & 7 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 16 & 0 & 14 & 7 & 8 \\ 2 & 5 & 7 & 0 & 7 & 1 & 3 \\ 3 & 5 & 8 & -3 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{13} 321^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 & 5 \\ 2 & 5 & 2 & 0 & 25 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 35 & 25 & 0 & 188 & 25 & 28 \\ 2 & 5 & 4 & 0 & 25 & 1 & 3 \\ 3 & 5 & 4 & -3 & 28 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{111} 321^{1121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 390, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 5 & 10 & 0 & 15 & 5 & 5 \\ 5 & 10 & 14 & 0 & 23 & 7 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 23 & 0 & 32 & 11 & 12 \\ 2 & 5 & 7 & 0 & 11 & 1 & 3 \\ 3 & 5 & 8 & -3 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{12} 321^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 640, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 & 5 \\ 2 & 5 & 2 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 30 & 22 & 0 & 134 & 22 & 24 \\ 2 & 5 & 4 & 0 & 22 & 1 & 3 \\ 3 & 5 & 4 & -3 & 24 & 3 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{1111}3^{11}2^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 7 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 7 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & 5 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

4.819 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] \quad 1^{1111}3^{11}1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^{11}1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 5 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 2 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 5 & 6 \\ 2 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 6 & 5 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^{11}1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 & 3 & 4 \\ 5 & 9 & 13 & 0 & 15 & 6 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 6 & 5 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.820 NHC(-2)(-3) + su₂ + f₄ + su₂→(-3)

$$T = 8 \quad [0] \quad 1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 3 & 4 \\ 6 & 12 & 25 & 0 & 6 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 3 & 4 \\ 6 & 10 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 8 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 4 & 5 \\ 3 & 8 & 4 & 3 & 3 & 4 \\ 3 & 4 & 3 & -2 & 2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 4 & 5 \\ 3 & 8 & 4 & 3 & 3 & 4 \\ 3 & 4 & 3 & -2 & 2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 2 & 3 \\ 4 & 9 & 16 & 0 & 4 & 5 \\ 7 & 16 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 7 & 0 & 0 & 2 \\ 3 & 5 & 10 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 3 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 3 \\ 4 & 9 & 8 & 4 & 4 & 5 \\ 3 & 8 & 4 & 3 & 3 & 4 \\ 2 & 4 & 3 & -2 & 2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3^{11} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 3 & 4 \\ 14 & 24 & 121 & 0 & 14 & 19 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 19 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 3^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 8 & 14 & 37 & 0 & 8 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^{11} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 \\ 11 & 19 & 73 & 0 & 11 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 7 & 13 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 10 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 9 & 17 & 49 & 0 & 9 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 13 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

- [17] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 11 & 21 & 76 & 0 & 11 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 16 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 10 & 18 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 14 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [19] $21^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 3 & 4 \\ 8 & 9 & 8 & 4 & 4 & 5 \\ 6 & 8 & 4 & 3 & 3 & 4 \\ 3 & 4 & 3 & -2 & 2 & 0 \\ 3 & 4 & 3 & 2 & 0 & 2 \\ 4 & 5 & 4 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 \\ 13 & 23 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 12 & 22 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 17 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [22] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 \\ 16 & 28 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 16 & 0 & 0 & 2 \\ 3 & 4 & 22 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 \\ 14 & 26 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 20 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 \\ 15 & 27 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [25] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 882, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 3 & 4 \\ 18 & 32 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 25 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{14}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 3 & 4 \\ 17 & 31 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 24 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{14}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 3 & 4 \\ 20 & 36 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 28 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 11 & 5 \\ 6 & 5 & 1 & 10 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 10 & 0 & 18 & 8 \\ 5 & 5 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{11}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 8 & 6 & 5 & 0 & 6 \\ 2 & 6 & 1 & 4 & 0 & 3 \\ 2 & 5 & 4 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 16 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^{11}1^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{12}31^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 8 & 6 & 20 & 0 & 6 \\ 2 & 6 & 1 & 13 & 0 & 3 \\ 8 & 20 & 13 & 44 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{11}3^{11}1^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{13}1^{13}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 2 & 6 & 1 & 5 & 0 & 3 \\ 4 & 10 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0} \\
[35] \quad & 1^1 3^1 1^1 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[36] \quad & 1^{12} 3^1 1^1 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 \\ 4 & 12 & 9 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 7 & 0 & 0 & 2 \\ 2 & 4 & 2 & 10 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[37] \quad & 1^{11} 3^1 1^1 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 8 & 6 & 15 & 0 & 6 \\ 2 & 6 & 1 & 9 & 0 & 3 \\ 6 & 15 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 10 & -3 & 2 \end{pmatrix}_{b_0} \\
[38] \quad & 1^{13} 3^1 1^{23} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -274, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 8 & 6 & 35 & 0 & 6 \\ 2 & 6 & 1 & 22 & 0 & 3 \\ 14 & 35 & 22 & 134 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 24 & -3 & 1 \end{pmatrix}_{b_0} \\
[39] \quad & 1^{13} 3^1 1^{23} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 8 & 6 & 20 & 0 & 6 \\ 2 & 6 & 1 & 10 & 0 & 3 \\ 8 & 20 & 10 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 12 & -3 & 1 \end{pmatrix}_{b_0} \\
[40] \quad & 1^{13} 3^1 1^{23} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -306, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & 2 & 8 & -3 & 2 & 0 \end{pmatrix}_{b_0} \\
[41] \quad & 1^{11} 3^1 1^1 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -311, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 \\ 2 & 4 & 2 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[42] \quad & 1^{14} 3^1 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 3 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 & 2 \\ 8 & 24 & 18 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 14 & 0 & 0 & 2 \\ 2 & 4 & 2 & 20 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[43] \quad & 1^{13} 3^1 1^{13} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -481, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.07 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 8 & 6 & 40 & 0 & 6 \\ 2 & 6 & 1 & 20 & 0 & 3 \\ 16 & 40 & 20 & 158 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 24 & -3 & 1 \end{pmatrix}_{b_0} \\
[53] \quad & 1^{11}31^{13}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -874, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 8 & 6 & 35 & 0 & 6 \\ 2 & 6 & 1 & 19 & 0 & 3 \\ 14 & 35 & 19 & 122 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 22 & -3 & 1 \end{pmatrix}_{b_0} \\
[54] \quad & 1^{13}31^{13}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 8 & 6 & 45 & 0 & 6 \\ 2 & 6 & 1 & 27 & 0 & 3 \\ 18 & 45 & 27 & 214 & 0 & 30 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 30 & -3 & 2 \end{pmatrix}_{b_0} \\
[55] \quad & 1^{11}3^{11}1^{13}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1011, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 & 2 \\ 7 & 21 & 12 & 73 & 0 & 11 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 11 & 0 & 0 & 2 \\ 2 & 4 & 2 & 15 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[56] \quad & 1^{12}31^{13}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 8 & 6 & 40 & 0 & 6 \\ 2 & 6 & 1 & 23 & 0 & 3 \\ 16 & 40 & 23 & 164 & 0 & 26 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 26 & -3 & 1 \end{pmatrix}_{b_0} \\
[57] \quad & 1^{13}3^{11}1^{15}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1041, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3 \\
& T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 10 & 30 & 15 & 148 & 0 & 15 & 20 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 20 & -3 & 2 & 0 \end{pmatrix}_{b_0} \\
[58] \quad & 1^{13}31^{15}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3 \\
& T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 8 & 6 & 50 & 0 & 6 \\ 2 & 6 & 1 & 25 & 0 & 3 \\ 20 & 50 & 25 & 248 & 0 & 30 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 30 & -3 & 1 \end{pmatrix}_{b_0} \\
[59] \quad & 1^{11}31^{14}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 8 & 6 & 45 & 0 & 6 \\ 2 & 6 & 1 & 24 & 0 & 3 \\ 18 & 45 & 24 & 202 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 28 & -3 & 1 \end{pmatrix}_{b_0} \\
[60] \quad & 1^{12}3^{11}1^{13}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 8 & 24 & 15 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 13 & 0 & 0 & 2 \\ 2 & 4 & 2 & 18 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[61] \quad & 1^{14}3^{11}1^{13}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 3 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 & 2 \\ 10 & 30 & 21 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 17 & 0 & 0 & 2 \\ 2 & 4 & 2 & 24 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[62] \quad & 1^{11}3^{11}1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1466, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 9 & 27 & 15 & 121 & 0 & 14 & 19 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 14 & 0 & 0 & 2 \\ 2 & 4 & 2 & 19 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[63] \quad & 1^{13}3^{11}1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1471, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 & 2 \\ 9 & 27 & 18 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 21 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[64] \quad & 1^{14}31^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1584, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 24 & 1 & 3 \\ 4 & 8 & 6 & 60 & 0 & 6 \\ 2 & 6 & 1 & 36 & 0 & 3 \\ 24 & 60 & 36 & 382 & 0 & 40 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 40 & -3 & 2 \end{pmatrix}_{b_0} \\
[65] \quad & 1^{12}31^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1624, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 8 & 6 & 50 & 0 & 6 \\ 2 & 6 & 1 & 28 & 0 & 3 \\ 20 & 50 & 28 & 254 & 0 & 32 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 32 & -3 & 1 \end{pmatrix}_{b_0} \\
[66] \quad & 1^{13}31^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1714, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.00 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 22 & 1 & 3 \\ 4 & 8 & 6 & 55 & 0 & 6 \\ 2 & 6 & 1 & 32 & 0 & 3 \\ 22 & 55 & 32 & 314 & 0 & 36 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 36 & -3 & 1 \end{pmatrix}_{b_0} \\
[67] \quad & 1^{12}3^{11}1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2006, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 3 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 & 2 \\ 10 & 30 & 18 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 16 & 0 & 0 & 2 \\ 2 & 4 & 2 & 22 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[68] \quad & 1^{13}3^{11}1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2346, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 1 & 2 \\ 2 & 4 & 4 & 33 & 0 & 3 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 & 2 \\ 11 & 33 & 21 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 18 & 0 & 0 & 2 \\ 2 & 4 & 2 & 25 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[69] \quad & 1^{14}3^{11}1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2486, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 3 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 & 2 \\ 12 & 36 & 24 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 20 & 0 & 0 & 2 \\ 2 & 4 & 2 & 28 & -3 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

4.821 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

4.822 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 5 & 9 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 9 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.823 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.824 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [1] \quad 1^{\overline{122}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 4 & 3 \\ 3 & 4 & 8 & 0 & 3 & 8 & 4 \\ 4 & 8 & 10 & 0 & 4 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 4 & 2 \\ 4 & 8 & 12 & 0 & 4 & 10 & 6 \\ 3 & 4 & 6 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{124} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -24, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 4 & 8 & 0 & 3 & 31 & 4 \\ 4 & 8 & 10 & 0 & 4 & 42 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 17 & 2 \\ 17 & 31 & 42 & 0 & 17 & 172 & 24 \\ 3 & 4 & 6 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 8 & 0 & 3 & 9 & 4 \\ 4 & 8 & 10 & 0 & 4 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 9 & 12 & 0 & 5 & 13 & 7 \\ 3 & 4 & 6 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 3^{11} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 3 & 17 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 15 & 0 & 9 & 49 & 13 \\ 3 & 4 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 & 4 \\ 4 & 8 & 10 & 0 & 4 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 1 & 3 \\ 3 & 4 & 6 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3^{11} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 12 & 3 \\ 3 & 4 & 8 & 0 & 3 & 22 & 4 \\ 4 & 8 & 10 & 0 & 4 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 12 & 22 & 30 & 0 & 12 & 85 & 17 \\ 3 & 4 & 6 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 3 & 14 & 4 \\ 2 & 4 & 1 & 0 & 2 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 8 & 14 & 9 & 0 & 8 & 37 & 11 \\ 3 & 4 & 3 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 7 & 3 \\ 3 & 4 & 8 & 0 & 3 & 13 & 4 \\ 4 & 8 & 10 & 0 & 4 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 7 & 2 \\ 7 & 13 & 18 & 0 & 7 & 28 & 10 \\ 3 & 4 & 6 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 3 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 12 & 0 & 7 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -234, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 3 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -254, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -284, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 9 & 3 \\ 3 & 4 & 8 & 0 & 3 & 17 & 4 \\ 4 & 8 & 10 & 0 & 4 & 24 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 9 & 17 & 24 & 0 & 9 & 49 & 13 \\ 3 & 4 & 6 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 11 & 3 \\ 3 & 4 & 9 & 0 & 3 & 21 & 4 \\ 5 & 9 & 13 & 0 & 5 & 33 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 11 & 2 \\ 11 & 21 & 33 & 0 & 11 & 76 & 16 \\ 3 & 4 & 7 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{124}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 4 & 8 & 0 & 3 & 21 & 4 \\ 4 & 8 & 10 & 0 & 4 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 11 & 2 \\ 11 & 21 & 30 & 0 & 11 & 76 & 16 \\ 3 & 4 & 6 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 3 & 13 & 4 \\ 2 & 4 & 1 & 0 & 2 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 13 & 9 & 0 & 7 & 28 & 10 \\ 3 & 4 & 3 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 4 & 8 & 0 & 3 & 26 & 4 \\ 4 & 8 & 10 & 0 & 4 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 14 & 2 \\ 14 & 26 & 36 & 0 & 14 & 118 & 20 \\ 3 & 4 & 6 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 3 & 9 & 4 \\ 5 & 9 & 13 & 0 & 5 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 5 & 2 \\ 5 & 9 & 15 & 0 & 5 & 13 & 7 \\ 3 & 4 & 7 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 3 & 23 & 4 \\ 2 & 4 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 13 & 23 & 15 & 0 & 13 & 100 & 18 \\ 3 & 4 & 3 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 3 & 17 & 4 \\ 2 & 4 & 1 & 0 & 2 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 9 & 17 & 12 & 0 & 9 & 49 & 13 \\ 3 & 4 & 3 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{113}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 3 & 17 & 4 \\ 5 & 9 & 13 & 0 & 5 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 9 & 2 \\ 9 & 17 & 27 & 0 & 9 & 49 & 13 \\ 3 & 4 & 7 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 3 & 18 & 4 \\ 2 & 4 & 1 & 0 & 2 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 10 & 2 \\ 10 & 18 & 12 & 0 & 10 & 58 & 14 \\ 3 & 4 & 3 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{112}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 3 & 13 & 4 \\ 5 & 9 & 13 & 0 & 5 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 7 & 2 \\ 7 & 13 & 21 & 0 & 7 & 28 & 10 \\ 3 & 4 & 7 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 3 & 21 & 4 \\ 2 & 4 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 11 & 21 & 15 & 0 & 11 & 76 & 16 \\ 3 & 4 & 3 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{124}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 11 & 3 \\ 3 & 4 & 13 & 0 & 3 & 21 & 4 \\ 7 & 13 & 28 & 0 & 7 & 48 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 11 & 2 \\ 11 & 21 & 48 & 0 & 11 & 76 & 16 \\ 3 & 4 & 10 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{122}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 7 & 3 \\ 3 & 4 & 13 & 0 & 3 & 13 & 4 \\ 7 & 13 & 28 & 0 & 7 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 7 & 2 \\ 7 & 13 & 30 & 0 & 7 & 28 & 10 \\ 3 & 4 & 10 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{123}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 9 & 3 \\ 3 & 4 & 13 & 0 & 3 & 17 & 4 \\ 7 & 13 & 28 & 0 & 7 & 39 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 9 & 2 \\ 9 & 17 & 39 & 0 & 9 & 49 & 13 \\ 3 & 4 & 10 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 3 & 22 & 4 \\ 2 & 4 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 12 & 22 & 15 & 0 & 12 & 85 & 17 \\ 3 & 4 & 3 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{113}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 3 & 32 & 4 \\ 2 & 4 & 1 & 0 & 2 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 18 & 2 \\ 18 & 32 & 21 & 0 & 18 & 193 & 25 \\ 3 & 4 & 3 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{133}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 9 & 3 \\ 3 & 4 & 17 & 0 & 3 & 17 & 4 \\ 9 & 17 & 49 & 0 & 9 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 9 & 2 \\ 9 & 17 & 51 & 0 & 9 & 49 & 13 \\ 3 & 4 & 13 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{134}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 11 & 3 \\ 3 & 4 & 17 & 0 & 3 & 21 & 4 \\ 9 & 17 & 49 & 0 & 9 & 63 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 11 & 2 \\ 11 & 21 & 63 & 0 & 11 & 76 & 16 \\ 3 & 4 & 13 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{113}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 3 & 27 & 4 \\ 2 & 4 & 1 & 0 & 2 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 15 & 2 \\ 15 & 27 & 18 & 0 & 15 & 133 & 21 \\ 3 & 4 & 3 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{144}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 11 & 3 \\ 3 & 4 & 21 & 0 & 3 & 21 & 4 \\ 11 & 21 & 76 & 0 & 11 & 78 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 11 & 2 \\ 11 & 21 & 78 & 0 & 11 & 76 & 16 \\ 3 & 4 & 16 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 3 & 26 & 4 \\ 2 & 4 & 1 & 0 & 2 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 14 & 26 & 18 & 0 & 14 & 118 & 20 \\ 3 & 4 & 3 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{114}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 3 & 31 & 4 \\ 2 & 4 & 1 & 0 & 2 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 17 & 2 \\ 17 & 31 & 21 & 0 & 17 & 172 & 24 \\ 3 & 4 & 3 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{114}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 3 & 36 & 4 \\ 2 & 4 & 1 & 0 & 2 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 20 & 2 \\ 20 & 36 & 24 & 0 & 20 & 238 & 28 \\ 3 & 4 & 3 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{111}3^{11}1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 247, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 6 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 3 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^13^11^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11}3^11^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 517, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 9 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{111}3^1 1^{11}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 648, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 6 & 3 \\ 4 & 8 & 6 & 15 & 0 & 15 & 6 \\ 2 & 6 & 1 & 9 & 0 & 9 & 3 \\ 6 & 15 & 9 & 22 & 0 & 24 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 24 & 0 & 22 & 10 \\ 3 & 6 & 3 & 10 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111}3^1 1^{11}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 912, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 5 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 15 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 7 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.825 NHC(-2)(-3) + su₂ + f₄ + su₂ → (-3) + su₂ + su₃

$$T = 10 \quad [0] \quad 1^{11}3^1 1^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 5 & 6 & 12 & 6 \\ 3 & 4 & 3 & -3 & 2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 6 & 0 & 5 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 9 & 4 & 4 \\ 5 & 9 & 13 & 0 & 5 & 15 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 15 & 0 & 5 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.826 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 4 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 2 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 4 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 5 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 475, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 4 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 9 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 5 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 9 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 7 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.827 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 8 & 0 & 3 & 9 & 4 \\ 4 & 8 & 10 & 0 & 4 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 9 & 12 & 0 & 5 & 12 & 6 \\ 3 & 4 & 6 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 & 4 \\ 11 & 19 & 73 & 0 & 11 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 2 & 2 \\ 2 & 4 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 & 4 \\ 4 & 8 & 10 & 0 & 4 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 5 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}2^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 3 & 8 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 4 & 8 & 6 & 0 & 4 & 9 & 5 \\ 3 & 4 & 3 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 5 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}2^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 3 & 8 & 4 \\ 5 & 9 & 13 & 0 & 5 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 4 & 8 & 12 & 0 & 4 & 9 & 5 \\ 3 & 4 & 7 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 12 & 6 \\ 3 & 4 & 3 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 & 4 \\ 8 & 14 & 37 & 0 & 8 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 8 & 0 & 0 & 2 & 2 \\ 2 & 4 & 9 & 0 & 2 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 & 4 \\ 5 & 9 & 13 & 0 & 5 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 3 & 13 & 4 \\ 2 & 4 & 1 & 0 & 2 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 13 & 9 & 0 & 7 & 27 & 9 \\ 3 & 4 & 3 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 & 4 \\ 7 & 13 & 28 & 0 & 7 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 2 & 2 \\ 2 & 4 & 9 & 0 & 2 & 0 & 2 \\ 3 & 4 & 10 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{121}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 & 4 \\ 16 & 28 & 154 & 0 & 16 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 16 & 0 & 0 & 2 & 2 \\ 2 & 4 & 18 & 0 & 2 & 0 & 2 \\ 3 & 4 & 22 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 & 4 \\ 9 & 17 & 49 & 0 & 9 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 2 & 2 \\ 2 & 4 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 13 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 & 4 \\ 11 & 21 & 76 & 0 & 11 & 15 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 2 & 2 \\ 2 & 4 & 15 & 0 & 2 & 0 & 2 \\ 3 & 4 & 16 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 & 4 \\ 10 & 18 & 58 & 0 & 10 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 10 & 0 & 0 & 2 & 2 \\ 2 & 4 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 14 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{121}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 & 4 \\ 13 & 23 & 100 & 0 & 13 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 13 & 0 & 0 & 2 & 2 \\ 2 & 4 & 15 & 0 & 2 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{131}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 & 4 \\ 12 & 22 & 85 & 0 & 12 & 15 & 17 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 12 & 0 & 0 & 2 & 2 \\ 2 & 4 & 15 & 0 & 2 & 0 & 2 \\ 3 & 4 & 17 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{131}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 & 4 \\ 15 & 27 & 133 & 0 & 15 & 18 & 21 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 15 & 0 & 0 & 2 & 2 \\ 2 & 4 & 18 & 0 & 2 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{141}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 & 4 \\ 14 & 26 & 118 & 0 & 14 & 18 & 20 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 14 & 0 & 0 & 2 & 2 \\ 2 & 4 & 18 & 0 & 2 & 0 & 2 \\ 3 & 4 & 20 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{131}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 2 & 3 \\ 3 & 4 & 32 & 0 & 3 & 4 & 4 \\ 18 & 32 & 193 & 0 & 18 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 18 & 0 & 0 & 2 & 2 \\ 2 & 4 & 21 & 0 & 2 & 0 & 2 \\ 3 & 4 & 25 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 2 & 3 \\ 3 & 4 & 31 & 0 & 3 & 4 & 4 \\ 17 & 31 & 172 & 0 & 17 & 21 & 24 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 17 & 0 & 0 & 2 & 2 \\ 2 & 4 & 21 & 0 & 2 & 0 & 2 \\ 3 & 4 & 24 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{141}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 2 & 3 \\ 3 & 4 & 36 & 0 & 3 & 4 & 4 \\ 20 & 36 & 238 & 0 & 20 & 24 & 28 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 20 & 0 & 0 & 2 & 2 \\ 2 & 4 & 24 & 0 & 2 & 0 & 2 \\ 3 & 4 & 28 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$4.828 \quad \text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$$

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 2 & 0 & 3 \\ 3 & 5 & 6 & 0 & 3 & 3 & 4 \\ 3 & 4 & 4 & -3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}1}3^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}1}3^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}1}3^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 6 & 0 & 5 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}1}3^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}1}3^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 5 & 6 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 6 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 7 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.829 NHC(-2)(-3) + su₂ + f₄ + su₂n3mix

$$T = 8 \quad [0] \ 1^{\overline{11}1}3^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 43, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}1}3^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 43, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.63 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 16 & 8 & 0 & 25 & 9 \\ 3 & 5 & 3 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{12}1}3^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 19 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 19 & 0 & 12 & 6 \\ 3 & 4 & 10 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 14 & 0 & 7 & 4 \\ 8 & 14 & 37 & 0 & 17 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 17 & 0 & 7 & 5 \\ 3 & 4 & 11 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{153}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 8 & 3 \\ 3 & 4 & 20 & 0 & 15 & 4 \\ 10 & 20 & 73 & 0 & 55 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 55 & 0 & 39 & 11 \\ 3 & 4 & 15 & -3 & 11 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 15 & 4 \\ 5 & 9 & 13 & 0 & 23 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 23 & 0 & 39 & 11 \\ 3 & 4 & 7 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 3 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 \\ 3 & 4 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{131}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 7 & 4 \\ 6 & 12 & 25 & 0 & 15 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 15 & 0 & 7 & 5 \\ 3 & 4 & 9 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{142}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 6 & 3 \\ 3 & 4 & 31 & 0 & 11 & 4 \\ 17 & 31 & 172 & 0 & 59 & 24 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 59 & 0 & 20 & 8 \\ 3 & 4 & 24 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{142}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 6 & 3 \\ 3 & 4 & 16 & 0 & 11 & 4 \\ 8 & 16 & 46 & 0 & 32 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 32 & 0 & 20 & 8 \\ 3 & 4 & 12 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 22 & 14 & 0 & 97 & 17 \\ 3 & 4 & 3 & -3 & 17 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{123}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 25 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 34 & 0 & 111 & 19 \\ 3 & 4 & 6 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{143}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 82$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 8 & 3 \\ 3 & 4 & 26 & 0 & 15 & 4 \\ 14 & 26 & 118 & 0 & 68 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 68 & 0 & 39 & 11 \\ 3 & 4 & 20 & -3 & 11 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 83$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 13 & 3 \\ 3 & 4 & 12 & 0 & 24 & 4 \\ 6 & 12 & 25 & 0 & 51 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 51 & 0 & 100 & 18 \\ 3 & 4 & 9 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{121}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 103$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 23 & 0 & 7 & 4 \\ 13 & 23 & 100 & 0 & 28 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 28 & 0 & 7 & 5 \\ 3 & 4 & 18 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{143}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 24 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 21 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 118$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 10 & 0 & 10 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 6 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 19 & 4 \\ 7 & 13 & 28 & 0 & 43 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 43 & 0 & 64 & 14 \\ 3 & 4 & 10 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{154}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 130$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.3$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 20 & 0 & 19 & 4 \\ 10 & 20 & 73 & 0 & 70 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 70 & 0 & 64 & 14 \\ 3 & 4 & 15 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{132}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 138$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 12 & 0 & 11 & 4 \\ 6 & 12 & 25 & 0 & 24 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 24 & 0 & 20 & 8 \\ 3 & 4 & 9 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 22 & 0 & 44 & 12 \\ 3 & 4 & 6 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{143}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 16 & 0 & 15 & 4 \\ 8 & 16 & 46 & 0 & 44 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 44 & 0 & 39 & 11 \\ 3 & 4 & 12 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 \\ 5 & 9 & 13 & 0 & 17 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 8 \\ 3 & 4 & 7 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 14 & 0 & 12 & 6 \\ 3 & 4 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{122}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 11 & 4 \\ 4 & 8 & 10 & 0 & 16 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 16 & 0 & 20 & 8 \\ 3 & 4 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 12 & 4 \\ 9 & 17 & 49 & 0 & 36 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 36 & 0 & 25 & 9 \\ 3 & 4 & 13 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 5 & 9 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 18 & 0 & 69 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{113}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 12 & 0 & 15 & 4 \\ 6 & 12 & 25 & 0 & 33 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 33 & 0 & 39 & 11 \\ 3 & 4 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{112}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{144}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 16 & 0 & 19 & 4 \\ 8 & 16 & 46 & 0 & 56 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 56 & 0 & 64 & 14 \\ 3 & 4 & 12 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{131}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 4 & 3 \\ 3 & 4 & 32 & 0 & 7 & 4 \\ 18 & 32 & 193 & 0 & 39 & 25 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 39 & 0 & 7 & 5 \\ 3 & 4 & 25 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 15 & 0 & 44 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 15 & 4 \\ 7 & 13 & 28 & 0 & 34 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 34 & 0 & 39 & 11 \\ 3 & 4 & 10 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{132}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 6 & 3 \\ 3 & 4 & 22 & 0 & 11 & 4 \\ 12 & 22 & 85 & 0 & 42 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 42 & 0 & 20 & 8 \\ 3 & 4 & 17 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{123}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 15 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 22 & 0 & 39 & 11 \\ 3 & 4 & 6 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{134}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 12 & 0 & 19 & 4 \\ 6 & 12 & 25 & 0 & 42 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 42 & 0 & 64 & 14 \\ 3 & 4 & 9 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{11}1^{13}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 11 & 0 & 39 & 11 \\ 3 & 4 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{121}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 7 & 4 \\ 10 & 18 & 58 & 0 & 22 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 22 & 0 & 7 & 5 \\ 3 & 4 & 14 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 28 & 0 & 16 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 10 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 11 & 4 \\ 7 & 13 & 28 & 0 & 25 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 25 & 0 & 20 & 8 \\ 3 & 4 & 10 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{142}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 9 & 3 \\ 3 & 4 & 21 & 0 & 16 & 4 \\ 11 & 21 & 76 & 0 & 59 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 59 & 0 & 44 & 12 \\ 3 & 4 & 16 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 28 & 0 & 69 & 15 \\ 3 & 4 & 6 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 19 & 4 \\ 9 & 17 & 49 & 0 & 57 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 57 & 0 & 64 & 14 \\ 3 & 4 & 13 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{124}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 19 & 4 \\ 4 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 28 & 0 & 64 & 14 \\ 3 & 4 & 6 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{112}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 362, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 31 & 20 & 0 & 188 & 24 \\ 3 & 4 & 3 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{121}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 28 & 0 & 28 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 28 & 0 & 25 & 9 \\ 3 & 4 & 10 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{\overline{114}3^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 370, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 14 & 0 & 64 & 14 \\ 3 & 4 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{\overline{111}3^{\overline{1}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 372, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 20 & 0 & 25 & 9 \\ 3 & 4 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{\overline{124}3^{\overline{1}}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 372, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 16 & 3 \\ 3 & 4 & 8 & 0 & 29 & 4 \\ 4 & 8 & 10 & 0 & 40 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 40 & 0 & 148 & 22 \\ 3 & 4 & 6 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{\overline{131}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 393, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 7 & 4 \\ 9 & 17 & 49 & 0 & 21 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 21 & 0 & 7 & 5 \\ 3 & 4 & 13 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{\overline{112}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 395, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{\overline{133}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 417, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 8 & 3 \\ 3 & 4 & 17 & 0 & 15 & 4 \\ 9 & 17 & 49 & 0 & 45 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 45 & 0 & 39 & 11 \\ 3 & 4 & 13 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{\overline{132}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 438, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 6 & 3 \\ 3 & 4 & 17 & 0 & 11 & 4 \\ 9 & 17 & 49 & 0 & 33 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 33 & 0 & 20 & 8 \\ 3 & 4 & 13 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{\overline{141}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 463, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 21 & 0 & 7 & 4 \\ 11 & 21 & 76 & 0 & 26 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 26 & 0 & 7 & 5 \\ 3 & 4 & 16 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{\overline{112}3^{\overline{1}}1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 467, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 26 & 17 & 0 & 128 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{\overline{131}3^{\overline{1}}1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -109, \quad \det = 473, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 7 & 4 \\ 15 & 27 & 133 & 0 & 33 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 33 & 0 & 7 & 5 \\ 3 & 4 & 21 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{124}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 14 & 0 & 80 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 34 & 0 & 100 & 18 \\ 3 & 4 & 6 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{124}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 515, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 5 & 9 & 13 & 0 & 26 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 26 & 0 & 44 & 12 \\ 3 & 4 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{144}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 10 & 3 \\ 3 & 4 & 21 & 0 & 19 & 4 \\ 11 & 21 & 76 & 0 & 71 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 71 & 0 & 64 & 14 \\ 3 & 4 & 16 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{131}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 7 & 4 \\ 12 & 22 & 85 & 0 & 27 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 27 & 0 & 7 & 5 \\ 3 & 4 & 17 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{142}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 6 & 3 \\ 3 & 4 & 21 & 0 & 11 & 4 \\ 11 & 21 & 76 & 0 & 41 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 41 & 0 & 20 & 8 \\ 3 & 4 & 16 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{142}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 6 & 3 \\ 3 & 4 & 26 & 0 & 11 & 4 \\ 14 & 26 & 118 & 0 & 50 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 50 & 0 & 20 & 8 \\ 3 & 4 & 20 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{132}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 555, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 16 & 4 \\ 9 & 17 & 49 & 0 & 48 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 48 & 0 & 44 & 12 \\ 3 & 4 & 13 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{143}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 8 & 3 \\ 3 & 4 & 21 & 0 & 15 & 4 \\ 11 & 21 & 76 & 0 & 56 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 56 & 0 & 39 & 11 \\ 3 & 4 & 16 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 577, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 20 & 4 \\ 5 & 9 & 13 & 0 & 32 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 32 & 0 & 69 & 15 \\ 3 & 4 & 7 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 38 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 38 & 0 & 100 & 18 \\ 3 & 4 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{122}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 16 & 4 \\ 7 & 13 & 28 & 0 & 37 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 37 & 0 & 44 & 12 \\ 3 & 4 & 10 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{141}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 4 & 3 \\ 3 & 4 & 36 & 0 & 7 & 4 \\ 20 & 36 & 238 & 0 & 44 & 28 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 44 & 0 & 7 & 5 \\ 3 & 4 & 28 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{143}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 727, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 20 & 4 \\ 11 & 21 & 76 & 0 & 74 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 74 & 0 & 69 & 15 \\ 3 & 4 & 16 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{141}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 4 & 3 \\ 3 & 4 & 26 & 0 & 7 & 4 \\ 14 & 26 & 118 & 0 & 32 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 32 & 0 & 7 & 5 \\ 3 & 4 & 20 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 100 & 18 \\ 3 & 4 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 17 & 0 & 111 & 19 \\ 3 & 4 & 3 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{123}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 807, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 20 & 4 \\ 7 & 13 & 28 & 0 & 46 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 46 & 0 & 69 & 15 \\ 3 & 4 & 10 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{141}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 4 & 3 \\ 3 & 4 & 31 & 0 & 7 & 4 \\ 17 & 31 & 172 & 0 & 38 & 24 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 38 & 0 & 7 & 5 \\ 3 & 4 & 24 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 20 & 4 \\ 9 & 17 & 49 & 0 & 60 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 60 & 0 & 69 & 15 \\ 3 & 4 & 13 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 865, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 20 & 0 & 165 & 23 \\ 3 & 4 & 3 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 3 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{124}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 928, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 24 & 4 \\ 7 & 13 & 28 & 0 & 55 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 55 & 0 & 100 & 18 \\ 3 & 4 & 10 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{114}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1062, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{144}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1068, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 13 & 3 \\ 3 & 4 & 21 & 0 & 24 & 4 \\ 11 & 21 & 76 & 0 & 89 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 89 & 0 & 100 & 18 \\ 3 & 4 & 16 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{134}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1083, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 24 & 4 \\ 9 & 17 & 49 & 0 & 72 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 72 & 0 & 100 & 18 \\ 3 & 4 & 13 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{114}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 19 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{114}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 1 & 0 & 26 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 3 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 8 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 2 \\ 3 & 9 & 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 12 & 0 & 9 & 5 \\ 2 & 4 & 2 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0} \\
[102] & 1^{\overline{11}1}31^{\overline{14}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 & 2 \\ 2 & 4 & 4 & 27 & 0 & 5 & 4 \\ 1 & 4 & 0 & 15 & 0 & 4 & 2 \\ 9 & 27 & 15 & 121 & 0 & 24 & 19 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 24 & 0 & 4 & 4 \\ 2 & 4 & 2 & 19 & -3 & 4 & 1 \end{pmatrix}_{b_0} \\
[103] & 1^{\overline{12}3}1^{\overline{11}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 14 & 4 \\ 1 & 4 & 0 & 3 & 0 & 10 & 2 \\ 2 & 6 & 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 14 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0} \\
[104] & 1^{\overline{11}1}31^{\overline{11}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 11 & 4 \\ 1 & 4 & 0 & 3 & 0 & 7 & 2 \\ 1 & 3 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0} \\
[105] & 1^{\overline{11}1}3^{\overline{1}}1^{\overline{11}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 1 & 3 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0} \\
[106] & 1^{\overline{11}1}31^{\overline{12}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 17 & 4 \\ 1 & 4 & 0 & 3 & 0 & 10 & 2 \\ 1 & 3 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 10 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & 3 & -3 & 12 & 0 \end{pmatrix}_{b_0} \\
[107] & 1^{\overline{12}3}1^{\overline{11}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 14 & 4 \\ 1 & 4 & 0 & 3 & 0 & 10 & 2 \\ 1 & 3 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 10 & 0 & 37 & 11 \\ 2 & 4 & 2 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0} \\
[108] & 1^{\overline{12}1}31^{\overline{11}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 & 2 \\ 2 & 4 & 4 & 12 & 0 & 5 & 4 \\ 1 & 4 & 0 & 9 & 0 & 4 & 2 \\ 4 & 12 & 9 & 28 & 0 & 13 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 13 & 0 & 4 & 4 \\ 2 & 4 & 2 & 10 & -3 & 4 & 1 \end{pmatrix}_{b_0} \\
[109] & 1^{\overline{11}1}3^{\overline{1}}1^{\overline{14}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 17 & 2 \\ 1 & 3 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 17 & 17 & 0 & 151 & 21 \\ 2 & 4 & 2 & 3 & -3 & 21 & 0 \end{pmatrix}_{b_0} \\
[110] & 1^{\overline{11}1}31^{\overline{11}3}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 5 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 2 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{113}31^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 6 & 2 \\ 2 & 4 & 4 & 9 & 0 & 17 & 4 \\ 1 & 4 & 0 & 6 & 0 & 13 & 2 \\ 3 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 13 & 30 & 0 & 60 & 14 \\ 2 & 4 & 2 & 7 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{111}3^11^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 11 & 2 \\ 1 & 3 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 11 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & 3 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[113] $1^131^{111}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 8 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 2 & 6 & 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 8 & 0 & 9 & 5 \\ 2 & 4 & 2 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{111}31^{13}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 & 2 \\ 2 & 4 & 4 & 21 & 0 & 5 & 4 \\ 1 & 4 & 0 & 12 & 0 & 4 & 2 \\ 7 & 21 & 12 & 73 & 0 & 19 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 19 & 0 & 4 & 4 \\ 2 & 4 & 2 & 15 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{113}3^11^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 12 & 0 & 25 & 9 \\ 2 & 4 & 2 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{111}3^11^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 14 & 2 \\ 1 & 3 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 14 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & 3 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

[117] $1^{114}31^{12}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 26 & 4 \\ 1 & 4 & 0 & 3 & 0 & 19 & 2 \\ 1 & 3 & 3 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 19 & 19 & 0 & 135 & 21 \\ 2 & 4 & 2 & 3 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{111}31^{12}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 & 2 \\ 2 & 4 & 4 & 15 & 0 & 5 & 4 \\ 1 & 4 & 0 & 9 & 0 & 4 & 2 \\ 5 & 15 & 9 & 37 & 0 & 14 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 2 & 11 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[119] $1^13^11^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -369, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{11}3^{11}1^{11}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 5 & 2 \\ 3 & 9 & 6 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 14 & 0 & 12 & 6 \\ 2 & 4 & 2 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[121] $1^{14}3^{11}1^{12}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 & 2 \\ 2 & 4 & 4 & 24 & 0 & 5 & 4 \\ 1 & 4 & 0 & 18 & 0 & 4 & 2 \\ 8 & 24 & 18 & 118 & 0 & 26 & 20 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 26 & 0 & 4 & 4 \\ 2 & 4 & 2 & 20 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{11}3^{11}1^{11}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 11 & 4 \\ 1 & 4 & 0 & 3 & 0 & 7 & 2 \\ 2 & 6 & 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 11 & 0 & 20 & 8 \\ 2 & 4 & 2 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{12}3^{11}1^{23}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 20 & 4 \\ 1 & 4 & 0 & 3 & 0 & 13 & 2 \\ 1 & 3 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 20 & 13 & 13 & 0 & 71 & 15 \\ 2 & 4 & 2 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{122}3^{11}1^{11}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 0 & 14 & 4 \\ 1 & 4 & 0 & 9 & 0 & 10 & 2 \\ 4 & 12 & 9 & 28 & 0 & 34 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 34 & 0 & 37 & 11 \\ 2 & 4 & 2 & 10 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{113}3^{11}1^{12}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 23 & 4 \\ 1 & 4 & 0 & 3 & 0 & 16 & 2 \\ 1 & 3 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 23 & 16 & 16 & 0 & 100 & 18 \\ 2 & 4 & 2 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{121}3^{11}1^{23}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -586, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 & 2 \\ 2 & 4 & 4 & 18 & 0 & 5 & 4 \\ 1 & 4 & 0 & 12 & 0 & 4 & 2 \\ 6 & 18 & 12 & 58 & 0 & 18 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 18 & 0 & 4 & 4 \\ 2 & 4 & 2 & 14 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{131}3^{11}1^{23}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 & 2 \\ 2 & 4 & 4 & 21 & 0 & 5 & 4 \\ 1 & 4 & 0 & 15 & 0 & 4 & 2 \\ 7 & 21 & 15 & 85 & 0 & 22 & 17 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 22 & 0 & 4 & 4 \\ 2 & 4 & 2 & 17 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{111}3^{11}1^{11}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 0 & 11 & 4 \\ 1 & 4 & 0 & 6 & 0 & 7 & 2 \\ 3 & 9 & 6 & 13 & 0 & 18 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 18 & 0 & 20 & 8 \\ 2 & 4 & 2 & 7 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{112}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 26 & 4 \\ 1 & 4 & 0 & 3 & 0 & 16 & 2 \\ 1 & 3 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 16 & 16 & 0 & 117 & 19 \\ 2 & 4 & 2 & 3 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[129] $1^{111}3^11^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 0 & 12 & 4 \\ 1 & 4 & 0 & 6 & 0 & 8 & 2 \\ 3 & 9 & 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 20 & 0 & 25 & 9 \\ 2 & 4 & 2 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{112}31^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -649, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 0 & 14 & 4 \\ 1 & 4 & 0 & 6 & 0 & 10 & 2 \\ 3 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 24 & 0 & 37 & 11 \\ 2 & 4 & 2 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{112}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 3 & 0 & 32 & 4 \\ 1 & 4 & 0 & 3 & 0 & 19 & 2 \\ 1 & 3 & 3 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 32 & 19 & 19 & 0 & 175 & 23 \\ 2 & 4 & 2 & 3 & -3 & 23 & 0 \end{pmatrix}_{b_0}$$

[133] $1^{121}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 & 2 \\ 2 & 4 & 4 & 24 & 0 & 5 & 4 \\ 1 & 4 & 0 & 15 & 0 & 4 & 2 \\ 8 & 24 & 15 & 100 & 0 & 23 & 18 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 23 & 0 & 4 & 4 \\ 2 & 4 & 2 & 18 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{113}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -871, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 29 & 4 \\ 1 & 4 & 0 & 3 & 0 & 19 & 2 \\ 1 & 3 & 3 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 29 & 19 & 19 & 0 & 152 & 22 \\ 2 & 4 & 2 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{121}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -926, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 & 2 \\ 2 & 4 & 4 & 30 & 0 & 5 & 4 \\ 1 & 4 & 0 & 18 & 0 & 4 & 2 \\ 10 & 30 & 18 & 154 & 0 & 28 & 22 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 28 & 0 & 4 & 4 \\ 2 & 4 & 2 & 22 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{114}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -929, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 3 & 0 & 32 & 4 \\ 1 & 4 & 0 & 3 & 0 & 22 & 2 \\ 1 & 3 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 32 & 22 & 22 & 0 & 193 & 25 \\ 2 & 4 & 2 & 3 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{131}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 & 2 \\ 2 & 4 & 4 & 27 & 0 & 5 & 4 \\ 1 & 4 & 0 & 18 & 0 & 4 & 2 \\ 9 & 27 & 18 & 133 & 0 & 27 & 21 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 27 & 0 & 4 & 4 \\ 2 & 4 & 2 & 21 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{1\bar{1}3}31^{14}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 3 & 0 & 35 & 4 \\ 1 & 4 & 0 & 3 & 0 & 22 & 2 \\ 1 & 3 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 35 & 22 & 22 & 0 & 216 & 26 \\ 2 & 4 & 2 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{14}31^{13}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 & 2 \\ 2 & 4 & 4 & 30 & 0 & 5 & 4 \\ 1 & 4 & 0 & 21 & 0 & 4 & 2 \\ 10 & 30 & 21 & 172 & 0 & 31 & 24 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 31 & 0 & 4 & 4 \\ 2 & 4 & 2 & 24 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{14}31^{14}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 3 & 0 & 38 & 4 \\ 1 & 4 & 0 & 3 & 0 & 25 & 2 \\ 1 & 3 & 3 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 38 & 25 & 25 & 0 & 263 & 29 \\ 2 & 4 & 2 & 3 & -3 & 29 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{13}31^{14}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 & 2 \\ 2 & 4 & 4 & 33 & 0 & 5 & 4 \\ 1 & 4 & 0 & 21 & 0 & 4 & 2 \\ 11 & 33 & 21 & 193 & 0 & 32 & 25 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 32 & 0 & 4 & 4 \\ 2 & 4 & 2 & 25 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{14}31^{14}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 & 2 \\ 2 & 4 & 4 & 36 & 0 & 5 & 4 \\ 1 & 4 & 0 & 24 & 0 & 4 & 2 \\ 12 & 36 & 24 & 238 & 0 & 36 & 28 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 36 & 0 & 4 & 4 \\ 2 & 4 & 2 & 28 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [143] $1^{11}1^{13}1^14^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{13}1^14^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 \\ 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.830 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{so}_8$

$T = 9$ [0] $1^{11}3^11^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

4.831 NHC(-2)(-3) + su₂ + f₄ + su2n3mix + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1221} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 5 & 3 \\ 3 & 4 & 8 & 0 & 7 & 9 & 4 \\ 4 & 8 & 10 & 0 & 10 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 11 & 5 \\ 5 & 9 & 12 & 0 & 11 & 13 & 7 \\ 3 & 4 & 6 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1221} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 2 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 22 & 0 & 44 & 11 & 12 \\ 2 & 4 & 6 & 0 & 11 & 1 & 3 \\ 3 & 4 & 6 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1311} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 2 & 3 \\ 3 & 4 & 12 & 0 & 7 & 4 & 4 \\ 6 & 12 & 25 & 0 & 15 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 15 & 0 & 7 & 5 & 5 \\ 2 & 4 & 9 & 0 & 5 & 1 & 3 \\ 3 & 4 & 9 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 5 & 0 & 7 & 9 & 4 \\ 3 & 5 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 11 & 5 \\ 5 & 9 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1131} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 5 & 0 & 7 & 17 & 4 \\ 3 & 5 & 4 & 0 & 6 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 21 & 5 \\ 9 & 17 & 15 & 0 & 21 & 49 & 13 \\ 3 & 4 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1221} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 5 & 3 \\ 3 & 4 & 8 & 0 & 16 & 9 & 4 \\ 4 & 8 & 10 & 0 & 22 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 22 & 0 & 44 & 26 & 12 \\ 5 & 9 & 12 & 0 & 26 & 13 & 7 \\ 3 & 4 & 6 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1212} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 8 & 4 \\ 4 & 8 & 10 & 0 & 10 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 10 & 5 \\ 4 & 8 & 12 & 0 & 10 & 10 & 6 \\ 3 & 4 & 6 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1311}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 2 & 3 \\ 3 & 4 & 17 & 0 & 12 & 4 & 4 \\ 9 & 17 & 49 & 0 & 36 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 36 & 0 & 25 & 8 & 9 \\ 2 & 4 & 12 & 0 & 8 & 1 & 3 \\ 3 & 4 & 13 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{1121}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 5 & 3 \\ 3 & 4 & 9 & 0 & 11 & 9 & 4 \\ 5 & 9 & 13 & 0 & 17 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 17 & 8 \\ 5 & 9 & 15 & 0 & 17 & 13 & 7 \\ 3 & 4 & 7 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{1311}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 & 4 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 18 & 0 & 69 & 14 & 15 \\ 2 & 4 & 3 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1111}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1121}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 5 & 0 & 7 & 13 & 4 \\ 3 & 5 & 4 & 0 & 6 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 16 & 5 \\ 7 & 13 & 12 & 0 & 16 & 28 & 10 \\ 3 & 4 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{1211}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 2 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 & 4 \\ 4 & 8 & 10 & 0 & 10 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 5 & 5 \\ 2 & 4 & 6 & 0 & 5 & 1 & 3 \\ 3 & 4 & 6 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 2 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 & 4 \\ 5 & 9 & 13 & 0 & 17 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 8 & 8 \\ 2 & 4 & 6 & 0 & 8 & 1 & 3 \\ 3 & 4 & 7 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1211}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 15 & 0 & 44 & 11 & 12 \\ 2 & 4 & 3 & 0 & 11 & 1 & 3 \\ 3 & 4 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 3 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 3 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 9 & 4 \\ 2 & 4 & 1 & 0 & 5 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 11 & 5 \\ 5 & 9 & 6 & 0 & 11 & 13 & 7 \\ 3 & 4 & 3 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 7 & 3 \\ 3 & 4 & 8 & 0 & 7 & 13 & 4 \\ 4 & 8 & 10 & 0 & 10 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 16 & 5 \\ 7 & 13 & 18 & 0 & 16 & 28 & 10 \\ 3 & 4 & 6 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1221} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 2 & 3 \\ 3 & 4 & 8 & 0 & 11 & 4 & 4 \\ 4 & 8 & 10 & 0 & 16 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 16 & 0 & 20 & 8 & 8 \\ 2 & 4 & 6 & 0 & 8 & 1 & 3 \\ 3 & 4 & 6 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 10 & 3 \\ 3 & 4 & 4 & 0 & 7 & 18 & 4 \\ 2 & 4 & 1 & 0 & 5 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 22 & 5 \\ 10 & 18 & 12 & 0 & 22 & 58 & 14 \\ 3 & 4 & 3 & -3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 2 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 8 & 8 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 3 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 7 & 3 \\ 3 & 4 & 9 & 0 & 11 & 13 & 4 \\ 5 & 9 & 13 & 0 & 17 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 25 & 8 \\ 7 & 13 & 21 & 0 & 25 & 28 & 10 \\ 3 & 4 & 7 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{11113}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -456, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 5 & 3 \\ 3 & 4 & 9 & 0 & 7 & 9 & 4 \\ 5 & 9 & 13 & 0 & 11 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 11 & 5 \\ 5 & 9 & 15 & 0 & 11 & 13 & 7 \\ 3 & 4 & 7 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{1113}3^11^{\overline{13}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -476, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 15 & 3 \\ 3 & 4 & 4 & 0 & 7 & 27 & 4 \\ 2 & 4 & 1 & 0 & 5 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 33 & 5 \\ 15 & 27 & 18 & 0 & 33 & 133 & 21 \\ 3 & 4 & 3 & -3 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{1121}3^11^{\overline{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 14 & 0 & 80 & 14 & 16 \\ 2 & 4 & 3 & 0 & 14 & 1 & 3 \\ 3 & 4 & 3 & -3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{\overline{1112}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 13 & 4 \\ 2 & 4 & 1 & 0 & 8 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 28 & 9 \\ 7 & 13 & 9 & 0 & 28 & 28 & 10 \\ 3 & 4 & 3 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{11113}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 4 & 0 & 12 & 9 & 4 \\ 2 & 4 & 1 & 0 & 8 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 20 & 9 \\ 5 & 9 & 6 & 0 & 20 & 13 & 7 \\ 3 & 4 & 3 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{1112}3^11^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -536, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 7 & 3 \\ 3 & 4 & 4 & 0 & 7 & 13 & 4 \\ 2 & 4 & 1 & 0 & 5 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 16 & 5 \\ 7 & 13 & 9 & 0 & 16 & 28 & 10 \\ 3 & 4 & 3 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{1112}3^11^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 7 & 3 \\ 3 & 4 & 4 & 0 & 11 & 13 & 4 \\ 2 & 4 & 1 & 0 & 8 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 25 & 8 \\ 7 & 13 & 9 & 0 & 25 & 28 & 10 \\ 3 & 4 & 3 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\overline{1114}3^11^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 11 & 3 \\ 3 & 4 & 9 & 0 & 7 & 21 & 4 \\ 5 & 9 & 13 & 0 & 11 & 33 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 26 & 5 \\ 11 & 21 & 33 & 0 & 26 & 76 & 16 \\ 3 & 4 & 7 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11131} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 11 & 0 & 39 & 11 & 11 \\ 2 & 4 & 3 & 0 & 11 & 1 & 3 \\ 3 & 4 & 3 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11143} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 20 & 3 \\ 3 & 4 & 4 & 0 & 7 & 36 & 4 \\ 2 & 4 & 1 & 0 & 5 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 44 & 5 \\ 20 & 36 & 24 & 0 & 44 & 238 & 28 \\ 3 & 4 & 3 & -3 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11143} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -620, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 11 & 12 \\ 2 & 4 & 3 & 0 & 11 & 1 & 3 \\ 3 & 4 & 3 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11123} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 7 & 3 \\ 3 & 4 & 9 & 0 & 7 & 13 & 4 \\ 5 & 9 & 13 & 0 & 11 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 16 & 5 \\ 7 & 13 & 21 & 0 & 16 & 28 & 10 \\ 3 & 4 & 7 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11133} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 9 & 3 \\ 3 & 4 & 9 & 0 & 7 & 17 & 4 \\ 5 & 9 & 13 & 0 & 11 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 21 & 5 \\ 9 & 17 & 27 & 0 & 21 & 49 & 13 \\ 3 & 4 & 7 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11133} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 9 & 3 \\ 3 & 4 & 4 & 0 & 7 & 17 & 4 \\ 2 & 4 & 1 & 0 & 5 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 21 & 5 \\ 9 & 17 & 12 & 0 & 21 & 49 & 13 \\ 3 & 4 & 3 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11141} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 14 & 0 & 64 & 14 & 14 \\ 2 & 4 & 3 & 0 & 14 & 1 & 3 \\ 3 & 4 & 3 & -3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11213} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -810, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 5 & 3 \\ 3 & 4 & 4 & 0 & 16 & 9 & 4 \\ 2 & 4 & 1 & 0 & 11 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 26 & 12 \\ 5 & 9 & 6 & 0 & 26 & 13 & 7 \\ 3 & 4 & 3 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11131^1 1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -836, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 12 & 3 \\ 3 & 4 & 4 & 0 & 7 & 22 & 4 \\ 2 & 4 & 1 & 0 & 5 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 27 & 5 \\ 12 & 22 & 15 & 0 & 27 & 85 & 17 \\ 3 & 4 & 3 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1131^1 1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -840, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 30 & 20 & 0 & 165 & 20 & 23 \\ 2 & 4 & 3 & 0 & 20 & 1 & 3 \\ 3 & 4 & 3 & -3 & 23 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1114^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 11 & 3 \\ 3 & 4 & 4 & 0 & 7 & 21 & 4 \\ 2 & 4 & 1 & 0 & 5 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 26 & 5 \\ 11 & 21 & 15 & 0 & 26 & 76 & 16 \\ 3 & 4 & 3 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{12123^1 1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 7 & 3 \\ 3 & 4 & 13 & 0 & 7 & 13 & 4 \\ 7 & 13 & 28 & 0 & 16 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 16 & 0 & 7 & 16 & 5 \\ 7 & 13 & 30 & 0 & 16 & 28 & 10 \\ 3 & 4 & 10 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1131^1 1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -964, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 20 & 9 & 4 \\ 2 & 4 & 1 & 0 & 14 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 32 & 15 \\ 5 & 9 & 6 & 0 & 32 & 13 & 7 \\ 3 & 4 & 3 & -3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1131^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -984, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 14 & 15 \\ 2 & 4 & 3 & 0 & 14 & 1 & 3 \\ 3 & 4 & 3 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1141^1 1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 2 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 & 4 \\ 5 & 9 & 13 & 0 & 38 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 38 & 0 & 100 & 17 & 18 \\ 2 & 4 & 6 & 0 & 17 & 1 & 3 \\ 3 & 4 & 7 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{11223^1 1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1020, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 7 & 3 \\ 3 & 4 & 4 & 0 & 16 & 13 & 4 \\ 2 & 4 & 1 & 0 & 11 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 37 & 12 \\ 7 & 13 & 9 & 0 & 37 & 28 & 10 \\ 3 & 4 & 3 & -3 & 12 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{11\bar{1}4\bar{3}1^1\bar{1}^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1076, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 17 & 0 & 111 & 17 & 19 \\ 2 & 4 & 3 & 0 & 17 & 1 & 3 \\ 3 & 4 & 3 & -3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{11\bar{1}4\bar{3}1^1\bar{1}^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 17 & 3 \\ 3 & 4 & 4 & 0 & 7 & 31 & 4 \\ 2 & 4 & 1 & 0 & 5 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 38 & 5 \\ 17 & 31 & 21 & 0 & 38 & 172 & 24 \\ 3 & 4 & 3 & -3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{11\bar{1}4\bar{3}1^1\bar{1}^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 14 & 3 \\ 3 & 4 & 4 & 0 & 7 & 26 & 4 \\ 2 & 4 & 1 & 0 & 5 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 32 & 5 \\ 14 & 26 & 18 & 0 & 32 & 118 & 20 \\ 3 & 4 & 3 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11\bar{1}4\bar{3}1^1\bar{1}^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 2 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 & 4 \\ 2 & 4 & 1 & 0 & 26 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 22 & 39 & 26 & 0 & 280 & 26 & 30 \\ 2 & 4 & 3 & 0 & 26 & 1 & 3 \\ 3 & 4 & 3 & -3 & 30 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11\bar{1}4\bar{3}1^1\bar{1}^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 17 & 0 & 100 & 17 & 18 \\ 2 & 4 & 3 & 0 & 17 & 1 & 3 \\ 3 & 4 & 3 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11\bar{1}4\bar{3}1^1\bar{1}^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 2 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 19 & 34 & 23 & 0 & 208 & 23 & 26 \\ 2 & 4 & 3 & 0 & 23 & 1 & 3 \\ 3 & 4 & 3 & -3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11\bar{1}4\bar{3}1^1\bar{1}^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 2 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 20 & 0 & 148 & 20 & 22 \\ 2 & 4 & 3 & 0 & 20 & 1 & 3 \\ 3 & 4 & 3 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \quad 1^{11\bar{1}31^{11}\bar{1}4\bar{3}1^1\bar{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 5 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 5 & 4 \\ 2 & 6 & 3 & 6 & 0 & 5 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1111}31^{1113}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 4 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 4 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 3 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1111}31^{1113}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 9 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 4 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1111}31^{1113}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 9 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 15 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 7 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

4.832 $\text{NHC}(-2)(-3) + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_{2\text{n}3\text{mix}} + \mathbf{su}_{2 \rightarrow (-3)}$

$$T = 10 \quad [0] \quad 1^{1113}1^{113}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_{2 \rightarrow (-3)} \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 4 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{113}1^{1113}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_{2 \rightarrow (-3)} \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1113}1^{113}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_{2 \rightarrow (-3)} \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 475, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 9 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 5 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 7 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.833 $\text{NHC}(-2)(-3) + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_{2\text{n}3\text{mix}} + \mathbf{su}_3$

$$T = 9 \quad [0] \quad 1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 12 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^1 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 11 & 0 & 55 & 11 & 13 \\ 2 & 4 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 3 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^1 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}} 3^1 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 & 3 \\ 3 & 4 & 4 & 0 & 11 & 5 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 9 & 8 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 3 & 4 & 3 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^1 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 0 & 7 & 5 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 6 & 5 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 3 & 4 & 3 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^1 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 & 4 \\ 5 & 9 & 13 & 0 & 14 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 14 & 0 & 12 & 5 & 6 \\ 2 & 4 & 6 & 0 & 5 & 0 & 2 \\ 3 & 4 & 7 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{131}} 3^1 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 & 4 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 18 & 0 & 69 & 14 & 15 \\ 2 & 4 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & 4 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^1 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^1 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 9 & 4 \\ 2 & 4 & 1 & 0 & 5 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 11 & 5 \\ 5 & 9 & 6 & 0 & 11 & 12 & 6 \\ 3 & 4 & 3 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 3 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1211}} 3^{\overline{111}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 15 & 0 & 44 & 11 & 12 \\ 2 & 4 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 & 4 \\ 5 & 9 & 13 & 0 & 11 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 5 & 5 \\ 2 & 4 & 6 & 0 & 5 & 0 & 2 \\ 3 & 4 & 7 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 & 4 \\ 5 & 9 & 13 & 0 & 20 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 20 & 0 & 25 & 8 & 9 \\ 2 & 4 & 6 & 0 & 8 & 0 & 2 \\ 3 & 4 & 7 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

4.834 NHC(-2)(-3) + $\mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 15.51 \rightarrow 16, \ T_{\max} = 28, \ T \in [16, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & -3 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1321}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 3 & 4 & 9 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.11 \rightarrow 14, \ T_{\max} = 28, \ T \in [14, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 3 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 3 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 48 & 12 \\ 3 & 4 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 8 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 11 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{121}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 6 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{131}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 6 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{132}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 6 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 8 & 4 \\ 10 & 18 & 58 & 0 & 24 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 14 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{141}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 8 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 12 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 0 & 27 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{132}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 8 & 4 \\ 15 & 27 & 133 & 0 & 36 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 36 & 0 & 9 & 5 \\ 3 & 4 & 21 & -3 & 5 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 2 & 3 \\ 4 & 9 & 24 & 0 & 4 & 5 \\ 10 & 24 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 14 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{151}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 20 & 0 & 4 & 4 \\ 10 & 20 & 73 & 0 & 15 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 11 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{161}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 12 & 24 & 106 & 0 & 18 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 & 0 \\ 3 & 5 & 16 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 5 & 11 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{171}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.69 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 14 & 28 & 145 & 0 & 21 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{123}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{13}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{115}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 20 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 15 & 36 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 21 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 0 & 57 & 13 \\ 3 & 4 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 48 & 12 \\ 3 & 4 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{132}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 9 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 18 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{132}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 8 & 4 \\ 12 & 22 & 85 & 0 & 30 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 30 & 0 & 9 & 5 \\ 3 & 4 & 17 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{121}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114}3^11^{41} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 2 & 3 \\ 4 & 9 & 48 & 0 & 4 & 5 \\ 20 & 48 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 5 & 28 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{131}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{141}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 11 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{131}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{121}3^11^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 16 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{113}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 0 & 84 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{114}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 14 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{121} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 13 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{114} 3^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 5 & 25 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{141} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 11 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{114} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{131} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{131} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 4 & 4 \\ 15 & 27 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{131} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 4 & 4 \\ 18 & 32 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 25 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{141} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 4 & 4 \\ 14 & 26 & 118 & 0 & 18 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 20 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{141} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1643$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 4 & 4 \\ 17 & 31 & 172 & 0 & 21 & 24 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 24 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{141} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1920$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 4 \\ 20 & 36 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 4 & 28 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [76] $1^{111} 31^{13} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{13} 321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 5 & 4 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{11} 31^{13} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 & 3 \\ 2 & 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{11} 321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 3 & 0 & 7 & 2 \\ 1 & 1 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 7 & 0 & 15 & 7 \\ 2 & 2 & 2 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{111} 321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{13} 321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 9 & 0 & 39 & 11 \\ 2 & 2 & 2 & 3 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{11} 31^{13} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 3 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{11} 321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11} 1^1 3 2 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 6 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{11} 3 1^{12} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 4 & 2 & 8 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 13 & 7 \\ 3 & 5 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{11} 1^1 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 3 & 2 \\ 2 & 3 & 4 & 5 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{12} 3 2 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 3 & 0 & 8 & 2 \\ 1 & 1 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 8 & 0 & 25 & 9 \\ 2 & 2 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{11} 3 2 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 14 & 0 & 7 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 7 & 0 & 1 & 3 \\ 3 & 5 & 8 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{11} 3 1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{11} 3 2 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 2 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 19 & 0 & 91 & 19 \\ 3 & 5 & 4 & -3 & 19 & 2 \end{pmatrix}_{b_0}$$

- [93] $1^{11}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 2 & 0 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 31 & 11 \\ 3 & 5 & 4 & -3 & 11 & 2 \end{pmatrix}_{b_0}$$
- [94] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 7 & 0 & 3 \\ 5 & 9 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$
- [95] $1^{11}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 2 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 57 & 15 \\ 3 & 5 & 4 & -3 & 15 & 2 \end{pmatrix}_{b_0}$$
- [96] $1^{112}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -161$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [97] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{131}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -188$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 & 3 \\ 5 & 12 & 15 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 12 & 0 & 0 & 2 \\ 2 & 3 & 3 & 13 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$
- [99] $1^{112}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -191$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [100] $1^{111}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 3 & 4 & 5 & 0 & 11 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 11 & 0 & 15 & 7 \\ 2 & 2 & 2 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [101] $1^{111}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 4 & 4 \\ 2 & 2 & 2 & 5 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [102] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 & 3 \\ 3 & 6 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 3 & 3 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{112}3^{11}21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$

$T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 5 & 2 \\ 2 & 3 & 4 & 5 & 0 & 10 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 10 & 0 & 12 & 6 \\ 2 & 2 & 2 & 5 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -314$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.31 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 4 & 2 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{121}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -332$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 & 3 \\ 4 & 9 & 12 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 9 & 0 & 0 & 2 \\ 2 & 3 & 3 & 10 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{131}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 \\ 2 & 4 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -417$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 9 & 0 & 3 & 3 \\ 2 & 4 & 2 & 7 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{112}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -481$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 6 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 3 & 4 & 5 & 0 & 14 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 14 & 0 & 25 & 9 \\ 2 & 2 & 2 & 5 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{111}31^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -587$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 3 & 2 \\ 5 & 15 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 9 & 0 & 0 & 2 \\ 2 & 4 & 2 & 11 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -699$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

11.2

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 15 & 0 & 12 & 6 \\ 2 & 4 & 2 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\overline{111}} 3^{\overline{13}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 3 & 2 \\ 7 & 21 & 12 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 12 & 0 & 0 & 2 \\ 2 & 4 & 2 & 15 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overline{111}} 3^{\overline{14}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -1007, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 3 & 2 \\ 9 & 27 & 15 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 19 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [114] \ 1^{\overline{111}} 2^{\overline{11}} 3^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 4 & 6 \\ 1 & 1 & 0 & 4 & -2 & 1 & 0 \\ 1 & 3 & 0 & 4 & 1 & 0 & 2 \\ 2 & 3 & -1 & 6 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

4.835 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.836 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 3^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 19 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 11 & 19 & 21 & 0 & 12 & 73 & 15 \\ 3 & 4 & 4 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 3^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 3 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 3^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 5 & 3 \\ 3 & 4 & 19 & 0 & 4 & 9 & 4 \\ 11 & 19 & 73 & 0 & 12 & 33 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 6 & 2 \\ 5 & 9 & 33 & 0 & 6 & 13 & 7 \\ 3 & 4 & 15 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overset{1}{12}} 3^{\overset{1}{11}} 3^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 12 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 3^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overset{1}{12}} 3^{\overset{1}{11}} 3^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 27 & 9 & 9 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 3^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 14 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 15 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{112}} 3 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{112}} 3 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -331, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 3 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{121}} 3 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -338, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 7 & 3 \\ 3 & 4 & 8 & 0 & 9 & 13 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 12 & 21 & 6 \\ 7 & 13 & 18 & 0 & 21 & 28 & 10 \\ 3 & 4 & 6 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -381, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -573, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{\overline{111}} 3 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -605, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 8 & 3 \\ 3 & 4 & 9 & 0 & 4 & 14 & 4 \\ 5 & 9 & 13 & 0 & 6 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 8 & 14 & 24 & 0 & 9 & 37 & 11 \\ 3 & 4 & 7 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 5 & 0 & 4 & 28 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 16 & 28 & 30 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 7 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 8 & 3 \\ 3 & 4 & 14 & 0 & 4 & 14 & 4 \\ 8 & 14 & 37 & 0 & 9 & 39 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 8 & 14 & 39 & 0 & 9 & 37 & 11 \\ 3 & 4 & 11 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 3 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 5 & 0 & 4 & 23 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 23 & 24 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 21 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 3 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 21 & 6 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 13 & 4 \\ 5 & 9 & 13 & 0 & 6 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 7 & 13 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 7 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1112} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -962, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 10 & 3 \\ 3 & 4 & 14 & 0 & 4 & 18 & 4 \\ 8 & 14 & 37 & 0 & 9 & 48 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 10 & 18 & 48 & 0 & 12 & 58 & 14 \\ 3 & 4 & 11 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1113} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 4 & 17 & 4 \\ 5 & 9 & 13 & 0 & 6 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 9 & 17 & 27 & 0 & 12 & 49 & 13 \\ 3 & 4 & 7 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1112} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 16 & 3 \\ 3 & 4 & 9 & 0 & 4 & 28 & 4 \\ 5 & 9 & 13 & 0 & 6 & 48 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 16 & 28 & 48 & 0 & 18 & 154 & 22 \\ 3 & 4 & 7 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1113} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -989, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 27 & 6 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 3 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1114} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1005, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 21 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 33 & 6 \\ 11 & 21 & 15 & 0 & 33 & 76 & 16 \\ 3 & 4 & 3 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{\overline{1113}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{\overline{1113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{\overline{1212}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 10 & 3 \\ 3 & 4 & 13 & 0 & 4 & 18 & 4 \\ 7 & 13 & 28 & 0 & 9 & 42 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 10 & 18 & 42 & 0 & 12 & 58 & 14 \\ 3 & 4 & 10 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{\overline{1113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 3 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{\overline{1112}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 10 & 3 \\ 3 & 4 & 9 & 0 & 4 & 18 & 4 \\ 5 & 9 & 13 & 0 & 6 & 30 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 10 & 18 & 30 & 0 & 12 & 58 & 14 \\ 3 & 4 & 7 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{\overline{1212}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 7 & 3 \\ 3 & 4 & 13 & 0 & 4 & 13 & 4 \\ 7 & 13 & 28 & 0 & 9 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 7 & 13 & 30 & 0 & 9 & 28 & 10 \\ 3 & 4 & 10 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{\overline{1112}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 13 & 3 \\ 3 & 4 & 9 & 0 & 4 & 23 & 4 \\ 5 & 9 & 13 & 0 & 6 & 39 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 13 & 23 & 39 & 0 & 15 & 100 & 18 \\ 3 & 4 & 7 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{\overline{1113}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 3 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 5 & 0 & 4 & 32 & 4 \\ 3 & 5 & 4 & 0 & 3 & 33 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 32 & 33 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 17 & 31 & 30 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 12 & 3 \\ 3 & 4 & 9 & 0 & 4 & 22 & 4 \\ 5 & 9 & 13 & 0 & 6 & 36 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 12 & 22 & 36 & 0 & 15 & 85 & 17 \\ 3 & 4 & 7 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1212} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 10 & 3 \\ 3 & 4 & 18 & 0 & 4 & 18 & 4 \\ 10 & 18 & 58 & 0 & 12 & 60 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 12 & 2 \\ 10 & 18 & 60 & 0 & 12 & 58 & 14 \\ 3 & 4 & 14 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 14 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 3 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1114} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 4 & 1 & 0 & 3 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 3 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 14 & 3 \\ 3 & 4 & 9 & 0 & 4 & 26 & 4 \\ 5 & 9 & 13 & 0 & 6 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 14 & 26 & 42 & 0 & 18 & 118 & 20 \\ 3 & 4 & 7 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 17 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 3 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1113} 3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 15 & 3 \\ 3 & 4 & 9 & 0 & 4 & 27 & 4 \\ 5 & 9 & 13 & 0 & 6 & 45 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 15 & 27 & 45 & 0 & 18 & 133 & 21 \\ 3 & 4 & 7 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1113} 3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 18 & 3 \\ 3 & 4 & 9 & 0 & 4 & 32 & 4 \\ 5 & 9 & 13 & 0 & 6 & 54 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 18 & 32 & 54 & 0 & 21 & 193 & 25 \\ 3 & 4 & 7 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1114} 3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 17 & 3 \\ 3 & 4 & 9 & 0 & 4 & 31 & 4 \\ 5 & 9 & 13 & 0 & 6 & 51 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 17 & 31 & 51 & 0 & 21 & 172 & 24 \\ 3 & 4 & 7 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114} 3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 20 & 3 \\ 3 & 4 & 9 & 0 & 4 & 36 & 4 \\ 5 & 9 & 13 & 0 & 6 & 60 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 24 & 2 \\ 20 & 36 & 60 & 0 & 24 & 238 & 28 \\ 3 & 4 & 7 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{111} 3^{21} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 5 & 5 & 0 & 5 & 10 & 5 \\ 2 & 5 & 2 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 7 & 3 \\ 5 & 10 & 7 & 0 & 7 & 14 & 8 \\ 3 & 5 & 4 & -3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1111} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 1 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11} 3^{21} 1^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 & 5 \\ 2 & 5 & 2 & 0 & 15 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 15 & 0 & 57 & 15 & 15 \\ 2 & 5 & 4 & 0 & 15 & 2 & 4 \\ 3 & 5 & 4 & -3 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 321^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 4 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & 3 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 321^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 5 & 5 & 0 & 5 & 15 & 5 \\ 2 & 5 & 2 & 0 & 4 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 11 & 3 \\ 7 & 15 & 11 & 0 & 11 & 32 & 12 \\ 3 & 5 & 4 & -3 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111} 321^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 3 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & 2 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{12} 321^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 340, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 5 & 5 & 0 & 5 & 25 & 5 \\ 2 & 5 & 2 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 18 & 3 \\ 12 & 25 & 18 & 0 & 18 & 94 & 20 \\ 3 & 5 & 4 & -3 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{111} 321^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 360, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 5 & 10 & 0 & 5 & 10 & 5 \\ 5 & 10 & 14 & 0 & 7 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 7 & 0 & 1 & 7 & 3 \\ 5 & 10 & 16 & 0 & 7 & 14 & 8 \\ 3 & 5 & 8 & -3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{13} 321^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 5 & 5 & 0 & 5 & 35 & 5 \\ 2 & 5 & 2 & 0 & 4 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 25 & 3 \\ 17 & 35 & 25 & 0 & 25 & 188 & 28 \\ 3 & 5 & 4 & -3 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{111} 321^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 390, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 5 & 10 & 0 & 5 & 15 & 5 \\ 5 & 10 & 14 & 0 & 7 & 23 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 7 & 0 & 1 & 11 & 3 \\ 7 & 15 & 23 & 0 & 11 & 32 & 12 \\ 3 & 5 & 8 & -3 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{12} 321^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 640, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 5 & 5 & 0 & 5 & 30 & 5 \\ 2 & 5 & 2 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 22 & 3 \\ 14 & 30 & 22 & 0 & 22 & 134 & 24 \\ 3 & 5 & 4 & -3 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 7 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & 5 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

4.837 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] \quad 1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 & 5 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 6 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 15 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.838 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 3 & 4 \\ 4 & 8 & 10 & 0 & 12 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 12 & 5 & 6 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 6 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 8 & 4 & 0 & 4 & 5 \\ 3 & 8 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 3 & 4 \\ 4 & 8 & 10 & 0 & 6 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 3 & 4 \\ 11 & 19 & 73 & 0 & 12 & 11 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 2 & 3 & 11 & 0 & 2 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 4 & 5 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 2 & 4 & 5 & 0 & 2 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 3 & 4 \\ 5 & 9 & 13 & 0 & 12 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 9 & 4 & 5 \\ 2 & 3 & 5 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 5 & 6 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -386, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 3 & 4 \\ 8 & 14 & 37 & 0 & 9 & 8 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 & 2 \\ 2 & 3 & 8 & 0 & 2 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -416, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 3 & 4 \\ 5 & 9 & 13 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 3 & 4 \\ 2 & 4 & 1 & 0 & 9 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 27 & 7 & 9 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 3 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -626, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 3 & 4 \\ 7 & 13 & 28 & 0 & 9 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 & 2 \\ 2 & 3 & 7 & 0 & 2 & 0 & 2 \\ 3 & 4 & 10 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{121}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 3 & 4 \\ 16 & 28 & 154 & 0 & 18 & 16 & 22 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 2 & 3 & 16 & 0 & 2 & 0 & 2 \\ 3 & 4 & 22 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 3 & 4 \\ 9 & 17 & 49 & 0 & 12 & 9 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 2 & 3 & 9 & 0 & 2 & 0 & 2 \\ 3 & 4 & 13 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{141}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 3 & 4 \\ 11 & 21 & 76 & 0 & 15 & 11 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 2 & 3 & 11 & 0 & 2 & 0 & 2 \\ 3 & 4 & 16 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 3 & 4 \\ 10 & 18 & 58 & 0 & 12 & 10 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 2 & 3 & 10 & 0 & 2 & 0 & 2 \\ 3 & 4 & 14 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{121}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 3 & 4 \\ 13 & 23 & 100 & 0 & 15 & 13 & 18 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 2 & 3 & 13 & 0 & 2 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{131}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 3 & 4 \\ 12 & 22 & 85 & 0 & 15 & 12 & 17 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 2 & 3 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 17 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{131}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 2 & 3 \\ 3 & 4 & 27 & 0 & 4 & 3 & 4 \\ 15 & 27 & 133 & 0 & 18 & 15 & 21 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 2 & 3 & 15 & 0 & 2 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{141}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 2 & 3 \\ 3 & 4 & 26 & 0 & 4 & 3 & 4 \\ 14 & 26 & 118 & 0 & 18 & 14 & 20 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 2 & 3 & 14 & 0 & 2 & 0 & 2 \\ 3 & 4 & 20 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 1^{13} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 2 & 3 \\ 3 & 4 & 32 & 0 & 4 & 3 & 4 \\ 18 & 32 & 193 & 0 & 21 & 18 & 25 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 & 2 \\ 2 & 3 & 18 & 0 & 2 & 0 & 2 \\ 3 & 4 & 25 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 2 & 3 \\ 3 & 4 & 31 & 0 & 4 & 3 & 4 \\ 17 & 31 & 172 & 0 & 21 & 17 & 24 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 & 2 \\ 2 & 3 & 17 & 0 & 2 & 0 & 2 \\ 3 & 4 & 24 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{141} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 3 & 4 \\ 20 & 36 & 238 & 0 & 24 & 20 & 28 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 24 & 0 & 0 & 2 & 2 \\ 2 & 3 & 20 & 0 & 2 & 0 & 2 \\ 3 & 4 & 28 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.839 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 3 \\ 3 & 5 & 6 & 0 & 3 & 3 & 4 \\ 3 & 4 & 4 & -3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 1 \\ 3 & 4 & 3 & -3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 \\ 3 & 4 & 5 & 0 & 4 & 3 & 9 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 6 \\ 2 & 3 & 3 & 0 & 2 & 0 & 5 \\ 5 & 9 & 9 & 0 & 6 & 5 & 13 \\ 3 & 4 & 4 & -3 & 2 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 5 & 6 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 3 & 0 & 6 & 2 & 1 \\ 3 & 4 & 3 & -3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 3 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 5 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 5 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 15 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 7 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.840 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + su2n3mix

$$T = 9 \quad [0] \ 1^{\overline{111}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 8 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 8 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}3^1}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 11 & 2 \\ 10 & 17 & 11 & 0 & 11 & 55 & 13 \\ 3 & 4 & 3 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 4 & 7 & 6 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 6 & 3 \\ 3 & 4 & 4 & 0 & 5 & 11 & 4 \\ 2 & 4 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 6 & 11 & 8 & 0 & 9 & 20 & 8 \\ 3 & 4 & 3 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 7 & 4 \\ 2 & 4 & 1 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 4 & 7 & 5 & 0 & 6 & 7 & 5 \\ 3 & 4 & 3 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 8 & 4 \\ 5 & 9 & 13 & 0 & 6 & 14 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 5 & 2 \\ 5 & 8 & 14 & 0 & 5 & 12 & 6 \\ 3 & 4 & 7 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 20 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 14 & 2 \\ 11 & 20 & 18 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 12 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 2 & 4 & 1 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 4 & 7 & 5 & 0 & 5 & 7 & 5 \\ 3 & 4 & 3 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 4 & 3 \\ 3 & 4 & 4 & 0 & 9 & 7 & 4 \\ 2 & 4 & 1 & 0 & 6 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 11 & 6 \\ 4 & 7 & 5 & 0 & 11 & 7 & 5 \\ 3 & 4 & 3 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 7 & 12 & 8 & 0 & 8 & 25 & 9 \\ 3 & 4 & 3 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 16 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 11 & 2 \\ 9 & 16 & 15 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 4 & 7 & 4 \\ 5 & 9 & 13 & 0 & 6 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 5 & 2 \\ 4 & 7 & 11 & 0 & 5 & 7 & 5 \\ 3 & 4 & 7 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 12 & 4 \\ 5 & 9 & 13 & 0 & 6 & 20 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 8 & 2 \\ 7 & 12 & 20 & 0 & 8 & 25 & 9 \\ 3 & 4 & 7 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

4.841 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 3 & 2 \\ 2 & 4 & 12 & 0 & 3 & 0 & 2 \\ 3 & 5 & 15 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 5 & 11 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 4 & 3 \\ 3 & 4 & 13 & 0 & 5 & 8 & 4 \\ 7 & 13 & 28 & 0 & 12 & 18 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 12 & 0 & 3 & 6 & 3 \\ 4 & 8 & 18 & 0 & 6 & 9 & 5 \\ 3 & 4 & 10 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 3 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{132} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 4 & 3 \\ 3 & 4 & 17 & 0 & 5 & 8 & 4 \\ 9 & 17 & 49 & 0 & 15 & 24 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 15 & 0 & 3 & 6 & 3 \\ 4 & 8 & 24 & 0 & 6 & 9 & 5 \\ 3 & 4 & 13 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 5 & 18 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3^1 1^4 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^1 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 3 & 2 \\ 2 & 4 & 21 & 0 & 3 & 0 & 2 \\ 3 & 5 & 25 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 3 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^1 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.2

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 3 & 2 \\ 2 & 4 & 18 & 0 & 3 & 0 & 2 \\ 3 & 5 & 22 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{141} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 2 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 & 4 \\ 11 & 21 & 76 & 0 & 18 & 15 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 18 & 0 & 3 & 3 & 3 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 4 & 16 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{121} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 2 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 & 4 \\ 7 & 13 & 28 & 0 & 12 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 12 & 0 & 3 & 3 & 3 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 4 & 10 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{131} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 2 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 & 4 \\ 9 & 17 & 49 & 0 & 15 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 15 & 0 & 3 & 3 & 3 \\ 2 & 4 & 12 & 0 & 3 & 0 & 2 \\ 3 & 4 & 13 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 12 & 6 \\ 3 & 4 & 7 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.842 NHC(-2)(-3) + $\textcolor{red}{su}_2$ + $\textcolor{orange}{f}_4$ + (-3)→(-2)

$T = 8$ [0] $1^{131} 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 1 & 4 \\ 8 & 9 & 8 & 4 & 0 & 5 \\ 6 & 8 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

4.843 NHC(-2)(-3) + $\textcolor{red}{su}_2$ + NHC(-2)(-3)

$T = 8$ [0] $1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{orange}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -64, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 5 & 32 & 5 & 0 & 5 \\ 16 & 32 & 174 & 24 & 0 & 28 \\ 2 & 5 & 24 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 28 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 \bar{5}^1 \bar{1}^3 231^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 2 & 1 & 3 \\ 3 & 5 & 45 & 5 & 0 & 5 \\ 24 & 45 & 349 & 33 & 0 & 39 \\ 2 & 5 & 33 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 39 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 \bar{5}^1 \bar{1}^1 231^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 5 & 15 & 5 & 0 & 5 \\ 8 & 15 & 37 & 11 & 0 & 13 \\ 2 & 5 & 11 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 13 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 \bar{5}^1 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 5 & 8 & 5 & 0 & 5 \\ 4 & 8 & 9 & 6 & 0 & 7 \\ 2 & 5 & 6 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 7 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[13] \ 1^1 \bar{5}^1 131^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 5 & 16 & 5 & 0 & 5 \\ 8 & 16 & 42 & 12 & 0 & 14 \\ 2 & 5 & 12 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 14 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[16] \ 1^1 \bar{5}^1 131^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 5 & 24 & 5 & 0 & 5 \\ 12 & 24 & 97 & 18 & 0 & 21 \\ 2 & 5 & 18 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 21 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[15] \ 1^1 \bar{5}^1 \bar{1}^2 231^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 5 & 30 & 5 & 0 & 5 \\ 16 & 30 & 154 & 22 & 0 & 26 \\ 2 & 5 & 22 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 26 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[17] \ 1^1 \bar{5}^1 131^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 15 & 21 & 97 & 0 & 18 & 21 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 18 & 1 & 1 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[20] \ 2 \ 3 \ 1 \bar{5}^1 \bar{1}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 2 & 3 \\ 4 & 5 & 20 & 5 & 3 & 5 \\ 12 & 20 & 62 & 16 & 8 & 16 \\ 3 & 5 & 16 & 2 & 3 & 4 \\ 2 & 3 & 8 & 3 & 0 & 2 \\ 3 & 5 & 16 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[19] \ 2 \ 3 \ 1 \bar{5}^1 \bar{1}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 2 & 3 \\ 4 & 5 & 10 & 5 & 3 & 5 \\ 6 & 10 & 14 & 8 & 4 & 8 \\ 3 & 5 & 8 & 2 & 3 & 4 \\ 2 & 3 & 4 & 3 & 0 & 2 \\ 3 & 5 & 8 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 2 \quad 3 \quad 1 \overset{111}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 2 & 3 & 4 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 5 & 4 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 2 \quad 3 \quad 1 \overset{171}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 & 2 & 3 \\ 4 & 5 & 35 & 5 & 3 & 5 \\ 21 & 35 & 194 & 28 & 14 & 28 \\ 3 & 5 & 28 & 2 & 3 & 4 \\ 2 & 3 & 14 & 3 & 0 & 2 \\ 3 & 5 & 28 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 2 \quad 3 \quad 1 \overset{151}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 2 & 3 \\ 4 & 5 & 25 & 5 & 3 & 5 \\ 15 & 25 & 98 & 20 & 10 & 20 \\ 3 & 5 & 20 & 2 & 3 & 4 \\ 2 & 3 & 10 & 3 & 0 & 2 \\ 3 & 5 & 20 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 2 \quad 3 \quad 1 \overset{131}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 2 & 3 \\ 4 & 5 & 15 & 5 & 3 & 5 \\ 9 & 15 & 34 & 12 & 6 & 12 \\ 3 & 5 & 12 & 2 & 3 & 4 \\ 2 & 3 & 6 & 3 & 0 & 2 \\ 3 & 5 & 12 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 2 \quad 3 \quad 1 \overset{161}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 2 & 3 \\ 4 & 5 & 30 & 5 & 3 & 5 \\ 18 & 30 & 142 & 24 & 12 & 24 \\ 3 & 5 & 24 & 2 & 3 & 4 \\ 2 & 3 & 12 & 3 & 0 & 2 \\ 3 & 5 & 24 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [25] \quad 41 \overset{111}{1} \overset{1}{1} 231 \overset{1}{1} 51 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 216, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 8 & 8 & 1 & 3 \\ 9 & 51 & 48 & 48 & 0 & 15 \\ 8 & 48 & 42 & 44 & 0 & 14 \\ 8 & 48 & 44 & 42 & 1 & 14 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 15 & 14 & 14 & -1 & 4 \end{pmatrix}_{b_0}$$

4.844 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1 \overset{111}{1} \overset{1}{1} 3 \overset{111}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -34, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 1 & 3 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \quad 2 \quad 3 \quad 1 \overset{1112}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 6 & 3 \\ 4 & 5 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 4 & 2 & 8 & 4 \\ 3 & 5 & 4 & 2 & 3 & 8 & 4 \\ 2 & 3 & 2 & 3 & 0 & 4 & 2 \\ 6 & 10 & 8 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 2 \quad 3 \quad 1 \overset{1114}{1} \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 12 & 3 \\ 4 & 5 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 4 & 2 & 16 & 4 \\ 3 & 5 & 4 & 2 & 3 & 16 & 4 \\ 2 & 3 & 2 & 3 & 0 & 8 & 2 \\ 12 & 20 & 16 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 2 \ 3 \ 15 1^{\overset{1}{11}5} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 15 & 3 \\ 4 & 5 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 4 & 2 & 20 & 4 \\ 3 & 5 & 4 & 2 & 3 & 20 & 4 \\ 2 & 3 & 2 & 3 & 0 & 10 & 2 \\ 15 & 25 & 20 & 20 & 10 & 98 & 20 \\ 3 & 5 & 4 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 2 \ 3 \ 15 1^{\overset{1}{11}3} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 9 & 3 \\ 4 & 5 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 4 & 2 & 12 & 4 \\ 3 & 5 & 4 & 2 & 3 & 12 & 4 \\ 2 & 3 & 2 & 3 & 0 & 6 & 2 \\ 9 & 15 & 12 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 2 \ 3 \ 15 1^{\overset{1}{12}2} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 2 & 6 & 3 \\ 4 & 5 & 10 & 5 & 3 & 10 & 5 \\ 6 & 10 & 14 & 8 & 4 & 16 & 8 \\ 3 & 5 & 8 & 2 & 3 & 8 & 4 \\ 2 & 3 & 4 & 3 & 0 & 4 & 2 \\ 6 & 10 & 16 & 8 & 4 & 14 & 8 \\ 3 & 5 & 8 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 2 \ 3 \ 15 1^{\overset{1}{11}1} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 & 3 \\ 4 & 5 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 2 & 4 & 2 & 4 & 4 \\ 3 & 5 & 4 & 2 & 3 & 4 & 4 \\ 2 & 3 & 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 4 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

4.845 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [6] \ 1^{\overset{1}{12}5} 1^{\overset{1}{21}} 13^{\overset{1}{1}} 1^{\overset{1}{21}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \\ 10 & 14 & 42 & 0 & 12 & 10 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 & 3 \\ 2 & 3 & 10 & 0 & 3 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overset{1}{13}5} 1^{\overset{1}{31}} 13^{\overset{1}{1}} 1^{\overset{1}{31}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \\ 15 & 21 & 97 & 0 & 18 & 15 & 21 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 18 & 1 & 1 & 3 & 3 \\ 2 & 3 & 15 & 0 & 3 & 1 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overset{1}{11}5} 1^{\overset{1}{13}} 13^{\overset{1}{11}} 1^{\overset{1}{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 3 & 4 \\ 5 & 7 & 9 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 6 & 1 & 1 & 3 & 3 \\ 2 & 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 4 & 7 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

4.846 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 7 \quad [0] \ 1^{\overset{1}{11}3} 3^{\overset{1}{41}} 1^{\overset{1}{41}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 80 & 6 \\ 16 & 25 & 124 & 9 \\ 80 & 124 & 613 & 45 \\ 6 & 9 & 45 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 8 & 5 \\ 12 & 25 & 16 & 9 \\ 8 & 16 & 10 & 6 \\ 5 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 21^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 14 & 6 \\ 15 & 20 & 17 & 8 \\ 14 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 74 & 6 \\ 16 & 25 & 112 & 9 \\ 74 & 112 & 478 & 40 \\ 6 & 9 & 40 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 10 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 5 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 37 & 6 \\ 16 & 25 & 56 & 9 \\ 37 & 56 & 118 & 20 \\ 6 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 20.33 \rightarrow 21, T_{\max} = 22, T \in [21, 22]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 9 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & -8 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

- [12] $1^{\overline{1}1} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & -2 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{\overline{1}\overline{1}3} 3^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 10 & 16 & 47 & 6 \\ 16 & 25 & 76 & 9 \\ 47 & 76 & 217 & 27 \\ 6 & 9 & 27 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{\overline{1}\overline{1}1} 3^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 23 & 6 \\ 16 & 25 & 36 & 9 \\ 23 & 36 & 49 & 13 \\ 6 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$
- [15] $21^{\overline{3}1} 3^{\overline{1}\overline{1}4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 10 & 16 & 84 & 6 \\ 16 & 25 & 128 & 9 \\ 84 & 128 & 634 & 46 \\ 6 & 9 & 46 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{\overline{1}\overline{1}4} 3^{\overline{1}31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 10 & 16 & 84 & 6 \\ 16 & 25 & 128 & 9 \\ 84 & 128 & 634 & 46 \\ 6 & 9 & 46 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{\overline{1}1} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -2 & 0 \\ 5 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$
- [18] $1^{\overline{1}1} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 17 & 8 \\ 10 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$
- [19] $1^{\overline{1}\overline{1}1} 3^{\overline{1}12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 11 & 30 & 5 \\ 11 & 20 & 50 & 8 \\ 30 & 50 & 118 & 20 \\ 5 & 8 & 20 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{\overline{1}1} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 17 & 8 \\ 12 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$
- [21] $1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$
- [22] $1^{\overline{1}\overline{1}2} 3^{\overline{1}11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 35 & 6 \\ 16 & 25 & 56 & 9 \\ 35 & 56 & 118 & 20 \\ 6 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$
- [23] $1^{\overline{1}1} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 17 & 8 \\ 9 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

- [24] $1^{112}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 12 & 29 & 5 \\ 12 & 25 & 56 & 9 \\ 29 & 56 & 118 & 20 \\ 5 & 9 & 20 & 0 \end{pmatrix}_{b_0}$$
- [25] $21^{131}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 13 & 17 & 12 & 7 \\ 17 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [26] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 16 & 8 \\ 10 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [27] $1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 1 & 6 \\ 15 & 20 & 0 & 8 \\ 1 & 0 & -2 & 0 \\ 6 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$
- [28] $1^{112}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 46 & 6 \\ 16 & 25 & 72 & 9 \\ 46 & 72 & 202 & 26 \\ 6 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$
- [29] $1^{11}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 13 & 20 & 94 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & -2 \end{pmatrix}_{b_0}$$
- [30] $1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$
- [31] $1^{14}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{11}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 4 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [33] $1^{13}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{12}3^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 8 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & -2 \end{pmatrix}_{b_0}$$
- [35] $1^{17}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 16.37 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 14 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & -4 \end{pmatrix}_{b_0}$$

[36] $1^{16}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 12 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{12}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{15}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{41}1^13^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 59 & 5 \\ 12 & 25 & 124 & 9 \\ 59 & 124 & 613 & 45 \\ 5 & 9 & 45 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{113}1^{41}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 15 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{112}1^{41}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 17 & -2 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{11}1^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{11}1^{41}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & -2 \end{pmatrix}_{b_0}$$

[45] $1^{11}1^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 17 & 8 \\ 13 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{111}1^{41}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 27 & 6 \\ 15 & 20 & 33 & 8 \\ 27 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[46] $21^{11}1^13^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 15 & 7 \\ 17 & 20 & 17 & 8 \\ 15 & 17 & 13 & 7 \\ 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

- [47] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -38$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 50 & 6 \\ 16 & 25 & 76 & 9 \\ 50 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [48] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -41$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 36 & 9 \\ 18 & 36 & 49 & 13 \\ 5 & 9 & 13 & 2 \end{pmatrix}_{b_0}$
- [49] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 10 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 0 \end{pmatrix}_{b_0}$
- [50] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 0 \end{pmatrix}_{b_0}$
- [51] $21^{11} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 25 & 56 & 9 \\ 36 & 56 & 118 & 20 \\ 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$
- [52] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 1 \end{pmatrix}_{b_0}$
- [53] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 25 & 56 & 9 \\ 36 & 56 & 118 & 20 \\ 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$
- [54] $1^1 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 44 & 5 \\ 11 & 20 & 84 & 8 \\ 44 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$
- [55] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 20 & 5 \\ 11 & 20 & 33 & 8 \\ 20 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [56] $1^1 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 8 & 5 \\ 11 & 20 & 16 & 8 \\ 8 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [58] $1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.58 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$
- [57] $1^1 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

- [64] $21^{21}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 92 & 9 \\ 60 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$
- [60] $1^{113}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 92 & 9 \\ 60 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$
- [61] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 35 & 6 \\ 15 & 20 & 50 & 8 \\ 35 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$
- [62] $1^{112}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 72 & 9 \\ 36 & 72 & 202 & 26 \\ 5 & 9 & 26 & 1 \end{pmatrix}_{b_0}$$
- [63] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 16 & 8 \\ 13 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$
- [59] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 11 & 6 \\ 15 & 20 & 16 & 8 \\ 11 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$
- [65] $1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 36 & 9 \\ 17 & 36 & 49 & 13 \\ 5 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$
- [66] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{113}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 12 & 47 & 5 \\ 12 & 25 & 92 & 9 \\ 47 & 92 & 325 & 33 \\ 5 & 9 & 33 & 0 \end{pmatrix}_{b_0}$$
- [68] $1^{113}31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 69 & 6 \\ 16 & 25 & 108 & 9 \\ 69 & 108 & 457 & 39 \\ 6 & 9 & 39 & 3 \end{pmatrix}_{b_0}$$
- [69] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 16 & 8 \\ 9 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [70] $1^{131}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 12 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{113} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -62, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.10 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 58 & 6 \\ 16 & 25 & 92 & 9 \\ 58 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -64, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 39 & 5 \\ 11 & 20 & 67 & 8 \\ 39 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{113} 3^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -65, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 5 \\ 12 & 25 & 108 & 9 \\ 54 & 108 & 457 & 39 \\ 5 & 9 & 39 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{121} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.55 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[77] \ 21^{111} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.11 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 20 & 50 & 8 \\ 36 & 50 & 118 & 20 \\ 6 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{114} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 70 & 6 \\ 16 & 25 & 112 & 9 \\ 70 & 112 & 478 & 40 \\ 6 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[79] \ 21^{111} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -74, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.08 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 48 & 6 \\ 16 & 25 & 76 & 9 \\ 48 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{113} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -74, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.08 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 48 & 6 \\ 16 & 25 & 76 & 9 \\ 48 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -75, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 23 & 6 \\ 15 & 20 & 33 & 8 \\ 23 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -75, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.08 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 63 & 6 \\ 16 & 25 & 96 & 9 \\ 63 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 48 & 5 \\ 11 & 20 & 84 & 8 \\ 48 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$

[84] $1^{11} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 40 & 9 \\ 18 & 40 & 58 & 14 \\ 5 & 9 & 14 & 2 \end{pmatrix}_{b_0}$

[85] $21^{112} 31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 54 & 7 \\ 20 & 25 & 72 & 9 \\ 54 & 72 & 202 & 26 \\ 7 & 9 & 26 & 3 \end{pmatrix}_{b_0}$

[86] $1^{141} 1^3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 16 & 20 & 94 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & -2 \end{pmatrix}_{b_0}$

[87] $1^{12} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 2 \end{pmatrix}_{b_0}$

[88] $1^{121} 1^3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 94 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 0 \end{pmatrix}_{b_0}$

[89] $1^{12} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 7 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$

[90] $1^{113} 1^{31} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 49 & 6 \\ 16 & 25 & 76 & 9 \\ 49 & 76 & 217 & 27 \\ 6 & 9 & 27 & 3 \end{pmatrix}_{b_0}$

[91] $21^{113} 13^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 96 & 9 \\ 60 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[92] $1^{114} 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 96 & 9 \\ 60 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[93] $1^{111} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 48 & 6 \\ 15 & 20 & 67 & 8 \\ 48 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$

[94] $1^{114} 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 12 & 72 & 5 \\ 12 & 25 & 144 & 9 \\ 72 & 144 & 814 & 52 \\ 5 & 9 & 52 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 17 & 5 \\ 11 & 20 & 33 & 8 \\ 17 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{11} 1^{13} 3 1^4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 17 & 40 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{13} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 35 & 5 \\ 11 & 20 & 67 & 8 \\ 35 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{11} 1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 61 & 6 \\ 15 & 20 & 84 & 8 \\ 61 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{11} 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 8 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{11} 1^{14} 3 1^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 92 & 6 \\ 16 & 25 & 144 & 9 \\ 92 & 144 & 814 & 52 \\ 6 & 9 & 52 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{11} 1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{21} 1^3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 34 & 5 \\ 12 & 25 & 72 & 9 \\ 34 & 72 & 202 & 26 \\ 5 & 9 & 26 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{12} 1^{21} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{11} 1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 15 & 20 & 94 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{11} 1^{12} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 10 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

- [106] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -98$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 24 & 0 & 5 \\ 10 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$
- [107] $1^{114}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 73 & 6 \\ 16 & 25 & 112 & 9 \\ 73 & 112 & 478 & 40 \\ 6 & 9 & 40 & 2 \end{pmatrix}_{b_0}$
- [108] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 19 & 5 \\ 11 & 20 & 33 & 8 \\ 19 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [109] $1^{112}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 26 & 5 \\ 11 & 20 & 50 & 8 \\ 26 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$
- [110] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 29 & 5 \\ 11 & 20 & 50 & 8 \\ 29 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$
- [111] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -105$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 26 & 6 \\ 15 & 20 & 33 & 8 \\ 26 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [112] $1^{114}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 81 & 6 \\ 16 & 25 & 128 & 9 \\ 81 & 128 & 634 & 46 \\ 6 & 9 & 46 & 2 \end{pmatrix}_{b_0}$
- [113] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [114] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.64 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$
- [115] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.64 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$
- [116] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 39 & 5 \\ 12 & 25 & 76 & 9 \\ 39 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [117] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 28 & 5 \\ 12 & 25 & 56 & 9 \\ 28 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{11} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$

[119] $1^{13} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}$

[120] $1^{112} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 8 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & -2 & 1 \end{pmatrix}_{b_0}$

[121] $1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 45 & 5 \\ 11 & 20 & 84 & 8 \\ 45 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$

[122] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 18 & 5 \\ 11 & 20 & 33 & 8 \\ 18 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$

[123] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 7 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$

[124] $1^1 3^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 26 & 5 \\ 12 & 25 & 56 & 9 \\ 26 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$

[125] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 24 & 6 \\ 15 & 20 & 33 & 8 \\ 24 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$

[126] $1^1 3^1 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 27 & 5 \\ 12 & 25 & 60 & 9 \\ 27 & 60 & 133 & 21 \\ 5 & 9 & 21 & 1 \end{pmatrix}_{b_0}$

[127] $1^{114} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 62 & 6 \\ 16 & 25 & 96 & 9 \\ 62 & 96 & 346 & 34 \\ 6 & 9 & 34 & 3 \end{pmatrix}_{b_0}$

[128] $1^{111} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 38 & 5 \\ 11 & 20 & 67 & 8 \\ 38 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$

[129] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 36 & 6 \\ 15 & 20 & 50 & 8 \\ 36 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 47 & 5 \\ 11 & 20 & 84 & 8 \\ 47 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 39 & 6 \\ 15 & 20 & 50 & 8 \\ 39 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 61 & 6 \\ 16 & 25 & 96 & 9 \\ 61 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[133] \ 21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 27 & 7 \\ 20 & 25 & 36 & 9 \\ 27 & 36 & 49 & 13 \\ 7 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 25 & 6 \\ 15 & 20 & 33 & 8 \\ 25 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[135] \ 21^{11} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 17 & 27 & 7 \\ 17 & 20 & 33 & 8 \\ 27 & 33 & 49 & 13 \\ 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{114} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 72 & 6 \\ 16 & 25 & 112 & 9 \\ 72 & 112 & 478 & 40 \\ 6 & 9 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[136] \ 21^{21} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 72 & 6 \\ 16 & 25 & 112 & 9 \\ 72 & 112 & 478 & 40 \\ 6 & 9 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 27 & 5 \\ 12 & 25 & 56 & 9 \\ 27 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 46 & 5 \\ 11 & 20 & 84 & 8 \\ 46 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 28 & 5 \\ 11 & 20 & 50 & 8 \\ 28 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 36 & 5 \\ 11 & 20 & 67 & 8 \\ 36 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 27 & 5 \\ 11 & 20 & 50 & 8 \\ 27 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

[143] $21^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{113}31^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 16 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 62 & 6 \\ 15 & 20 & 84 & 8 \\ 62 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 49 & 6 \\ 15 & 20 & 67 & 8 \\ 49 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 37 & 5 \\ 11 & 20 & 67 & 8 \\ 37 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 10 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

- [153] $1^{13} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}$
- [154] $1^{31} 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 51 & 5 \\ 12 & 25 & 108 & 9 \\ 51 & 108 & 457 & 39 \\ 5 & 9 & 39 & 2 \end{pmatrix}_{b_0}$
- [155] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 38 & 6 \\ 15 & 20 & 50 & 8 \\ 38 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$
- [156] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 37 & 6 \\ 15 & 20 & 50 & 8 \\ 37 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$
- [157] $1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 63 & 6 \\ 15 & 20 & 84 & 8 \\ 63 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$
- [158] $1^{12} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$
- [159] $1^{113} 3^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$
- [160] $1^{113} 3^1 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 46 & 5 \\ 12 & 25 & 92 & 9 \\ 46 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$
- [161] $1^{111} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 51 & 6 \\ 15 & 20 & 67 & 8 \\ 51 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$
- [162] $1^{13} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 9 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [163] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$
- [164] $1^{113} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 14 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 50 & 6 \\ 15 & 20 & 67 & 8 \\ 50 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[168] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 11 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[169] $21^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 72 & 7 \\ 17 & 20 & 84 & 8 \\ 72 & 84 & 346 & 34 \\ 7 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

[170] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 24 & 6 \\ 16 & 20 & 33 & 8 \\ 24 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -193, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 11 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 12 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$

[174] $21^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 42 & 7 \\ 17 & 20 & 50 & 8 \\ 42 & 50 & 118 & 20 \\ 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[173] $13 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 80 & 9 \\ 36 & 80 & 238 & 28 \\ 5 & 9 & 28 & 1 \end{pmatrix}_{b_0}$$

[175] $21^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 13 & 7 \\ 7 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 11 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{114} 3^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 22 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 22 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 13 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 38 & 5 \\ 12 & 25 & 76 & 9 \\ 38 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{11} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 35 & 5 \\ 12 & 25 & 76 & 9 \\ 35 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[181] $21^{13} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 57 & 7 \\ 17 & 20 & 67 & 8 \\ 57 & 67 & 217 & 27 \\ 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{11} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{14} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{21} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 43 & 5 \\ 12 & 25 & 92 & 9 \\ 43 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{12} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

- [188] $1^{113}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -240$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 92 & 9 \\ 45 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$
- [189] $1^{114}31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -242$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 64 & 5 \\ 12 & 25 & 128 & 9 \\ 64 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$
- [191] $1^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -251$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0}$
- [190] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -251$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$
- [192] $1^{13}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -252$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$
- [193] $1^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -257$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 13 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [194] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -260$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 37 & 5 \\ 12 & 25 & 76 & 9 \\ 37 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [195] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -261$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 76 & 9 \\ 36 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [196] $1^{41}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -262$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 68 & 5 \\ 12 & 25 & 144 & 9 \\ 68 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$
- [197] $1^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -267$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [198] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -268$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$
- [199] $1^{12}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -272$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{14}3^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{14}3^{14}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 20 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{11}3^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{14}3^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -278, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 48 & 0 & 5 \\ 20 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{14}3^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[205] $1^3 1^{15} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 100 & 9 \\ 45 & 100 & 373 & 35 \\ 5 & 9 & 35 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{14}3^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 18 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{14}3^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 14 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{11}3^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[209] $1^3 1^{14}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -322, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 44 & 5 \\ 12 & 25 & 96 & 9 \\ 44 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{14}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{13}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{114}3^{14}1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 56 & 5 \\ 12 & 25 & 112 & 9 \\ 56 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{14}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{12}3^{121}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{114}3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -333, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 18 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{114}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 16 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{114}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 48 & 5 \\ 12 & 25 & 96 & 9 \\ 48 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{114}3^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -345, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 63 & 5 \\ 12 & 25 & 128 & 9 \\ 63 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{13}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{12}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 14 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[221] $21^{112}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -350, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 30 & 7 \\ 20 & 25 & 40 & 9 \\ 30 & 40 & 58 & 14 \\ 7 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

- [222] $1^{11} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$
- [223] $1^{31} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 60 & 5 \\ 12 & 25 & 128 & 9 \\ 60 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$
- [224] $21^{12} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -362, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 42 & 7 \\ 20 & 25 & 56 & 9 \\ 42 & 56 & 118 & 20 \\ 7 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$
- [225] $1^{13} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -363, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$
- [227] $1^{12} 3^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 13 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$
- [226] $1^{12} 3^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [228] $1^{11} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -368, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 18 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$
- [229] $1^{13} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$
- [230] $1^{21} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -374, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 52 & 5 \\ 12 & 25 & 112 & 9 \\ 52 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$
- [231] $1^{13} 3^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$
- [232] $1^{14} 3^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -398, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 19 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 19 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$
- [234] $1^{14} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -401, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 17 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{114}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -401, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 96 & 9 \\ 45 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{114}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -409, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 47 & 5 \\ 12 & 25 & 96 & 9 \\ 47 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{114}3^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -413, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 55 & 5 \\ 12 & 25 & 112 & 9 \\ 55 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{12}3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 17 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{14}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{114}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 46 & 5 \\ 12 & 25 & 96 & 9 \\ 46 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{13}3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -432, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 18 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{12}3^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -435, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{114}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -437, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 16 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{114}3^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 54 & 5 \\ 12 & 25 & 112 & 9 \\ 54 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{14}3^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 17 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{13}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[246] $21^{113}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -488, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 69 & 7 \\ 20 & 25 & 92 & 9 \\ 69 & 92 & 325 & 33 \\ 7 & 9 & 33 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{12}3^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -491, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{14}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -492, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{14}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{13}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 16 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{12}3^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -515, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{12}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -528, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 18 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{14}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -539, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{13}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -540, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 15 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{13}3^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -572, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[256] $21^{114}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -578, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 96 & 7 \\ 20 & 25 & 128 & 9 \\ 96 & 128 & 634 & 46 \\ 7 & 9 & 46 & 3 \end{pmatrix}_{b_0}$$

$$[257] \ 1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -592, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 20 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[258] \ 21^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 45 & 7 \\ 20 & 25 & 60 & 9 \\ 45 & 60 & 133 & 21 \\ 7 & 9 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{12} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 19 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 19 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[261] \ 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[262] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -620, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[263] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -651, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 18 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{12} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 20 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[265] \ 21^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -692, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 57 & 7 \\ 20 & 25 & 76 & 9 \\ 57 & 76 & 217 & 27 \\ 7 & 9 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -707, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -716, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 17 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{13}3^{121}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -752, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 20 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{13}3^{141}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -763, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{14}3^{141}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -780, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[271] $1^{14}3^{111}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -800, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 21 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{14}3^{121}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -803, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 19 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{13}3^{131}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -864, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 21 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{14}3^{131}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -923, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 20 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{13}3^{141}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -944, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 22 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[276] $21^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -950, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 60 & 7 \\ 20 & 25 & 80 & 9 \\ 60 & 80 & 238 & 28 \\ 7 & 9 & 28 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{14}3^{121}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -976, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 22 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[278] $21^{114}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -998, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 84 & 7 \\ 20 & 25 & 112 & 9 \\ 84 & 112 & 478 & 40 \\ 7 & 9 & 40 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{14}3^{13}1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -1011$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 21 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$

[280] $1^{14}3^{13}1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -1120$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 23 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$

[281] $21^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -1122$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 72 & 7 \\ 20 & 25 & 96 & 9 \\ 72 & 96 & 346 & 34 \\ 7 & 9 & 34 & 3 \end{pmatrix}_{b_0}$

[282] $1^{14}3^{13}1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -1232$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 24 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$

[283] $21^{115}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -1400$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.1$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 75 & 7 \\ 20 & 25 & 100 & 9 \\ 75 & 100 & 373 & 35 \\ 7 & 9 & 35 & 3 \end{pmatrix}_{b_0}$

$T = 8$ [284] $21^141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 2$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 15$, $T_{\max} = 20$, $T \in [15, 20]$
 $\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 5 & 5 \\ 6 & 9 & 5 & 0 \end{pmatrix}_{b_0}$

[285] $1^{11}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 2$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 15$, $T_{\max} = 20$, $T \in [15, 20]$
 $\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 5 & 5 \\ 6 & 9 & 5 & 0 \end{pmatrix}_{b_0}$

[286] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 5 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[287] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 6$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 4 & 0 & 10 & 6 \\ 3 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$

[289] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 \\ 1 & 2 & 0 & 3 & 0 \\ 1 & 1 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 & -4 \end{pmatrix}_{b_0}$

[288] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -4 \end{pmatrix}_{b_0}$

- [291] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$
- [290] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$
- [293] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 5 & 5 \\ 5 & 9 & 5 & 1 \end{pmatrix}_{b_0}$
- [292] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -2 & 3 \end{pmatrix}_{b_0}$
- [295] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [294] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 13 & 7 \\ 3 & 6 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [297] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 1 & 4 \\ 7 & 10 & 13 & 0 & 6 \\ 8 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 6 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [296] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 2 \\ 3 & 5 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [299] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 9 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 15 & 6 \\ 9 & 12 & 15 & 13 & 7 \\ 4 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [298] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 90 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$
- [300] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 17$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.59 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$
- [303] $21^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[304] \ 21^{\overline{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 14 & 7 \\ 12 & 14 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[302] \ 21^{\overline{11}} 4^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 28 & 6 \\ 12 & 13 & 34 & 7 \\ 28 & 34 & 77 & 17 \\ 6 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[301] \ 1^{\overline{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[305] \ 1^{\overline{12}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 28 & 6 \\ 12 & 13 & 34 & 7 \\ 28 & 34 & 77 & 17 \\ 6 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[306] \ 1^{\overline{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 14 & 7 \\ 12 & 14 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[307] \ 1^{\overline{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[308] \ 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[309] \ 1^{\overline{12}} 3^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 6 & 3 & 3 \\ 1 & 4 & 0 & 6 & 3 & 2 \\ 3 & 6 & 6 & 10 & 6 & 6 \\ 2 & 3 & 3 & 6 & 2 & 2 \\ 2 & 3 & 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[310] \ 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[311] \ 1^{\overline{12}} 3^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 3 & 6 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[312] \ 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{\overline{11}} 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{\overline{11}} 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{\overline{11}} 1^{\overline{11}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 14 & 7 \\ 9 & 14 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[316] $1^{\overline{11}} 1^{\overline{11}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 101 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{\overline{11}} 1^{\overline{11}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{\overline{11}} 1^{\overline{11}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 20 & 6 \\ 12 & 13 & 24 & 7 \\ 20 & 24 & 38 & 12 \\ 6 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[319] $21^{\overline{11}} 1^{\overline{11}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 20 & 6 \\ 12 & 13 & 24 & 7 \\ 20 & 24 & 38 & 12 \\ 6 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[320] $1^{\overline{11}} 1^{\overline{11}} 31^{\overline{12}} 31^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 10 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{\overline{11}} 1^{\overline{11}} 31^{\overline{12}} 31^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{\overline{12}} 31^{\overline{14}} 1^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 6 & 0 & 10 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[323] $1^{\overline{11}} 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 & 4 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[329] \quad 21^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 5 & 5 \\ 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[331] \quad 1^1 41^{17} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.3$$

$$T_{\min} = 17.38 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 5 & 9 & 42 & 5 \\ 9 & 13 & 70 & 7 \\ 42 & 70 & 341 & 35 \\ 5 & 7 & 35 & -3 \end{pmatrix}_{b_0}$$

$$[332] \quad 1^1 41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 12 & 5 \\ 9 & 13 & 20 & 7 \\ 12 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[330] \quad 1^1 41^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 5 & 9 & 30 & 5 \\ 9 & 13 & 50 & 7 \\ 30 & 50 & 173 & 25 \\ 5 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^1 41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 5 & 9 & 24 & 5 \\ 9 & 13 & 40 & 7 \\ 24 & 40 & 110 & 20 \\ 5 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^1 41^{16} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.75 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 5 & 9 & 36 & 5 \\ 9 & 13 & 60 & 7 \\ 36 & 60 & 250 & 30 \\ 5 & 7 & 30 & -1 \end{pmatrix}_{b_0}$$

$$[327] \quad 1^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[328] \quad 1^1 41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 5 & 9 & 18 & 5 \\ 9 & 13 & 30 & 7 \\ 18 & 30 & 61 & 15 \\ 5 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[333] \quad 1^1 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[334] \quad 1^{11} 31^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

- [336] $1^{13}31^{11}31^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [335] $1^{11}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$
- [337] $1^{11}31^{43}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 25 & 3 \\ 4 & 10 & 1 & 60 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 60 & 0 & 346 & 34 \\ 3 & 6 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [338] $131^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 15 & 6 \\ 5 & 12 & 15 & 13 & 7 \\ 3 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$
- [339] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 15 & 6 \\ 8 & 12 & 15 & 13 & 7 \\ 4 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$
- [340] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 7 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 12 & 6 \\ 7 & 12 & 12 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [341] $1^{12}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 10 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$
- [342] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 40$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 11 & 5 \\ 2 & 3 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [343] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 41$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [344] $1^{11}31^{11}31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 42$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 5 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [345] $1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 2 \end{pmatrix}_{b_0} \\
[346] & 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 12 & 6 \\ 5 & 12 & 12 & 10 & 6 \\ 3 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[348] & 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 8 & -2 & 3 \end{pmatrix}_{b_0} \\
[347] & 1^{11} 321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 8 & -2 & 3 \end{pmatrix}_{b_0} \\
[353] & 1^{12} 31^{12} 31^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0} \\
[352] & 21^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20] \\
& \begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 5 & 5 \\ 7 & 9 & 5 & 2 \end{pmatrix}_{b_0} \\
[351] & 1^{11} 431^{11} 31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 4 & 12 & 3 & 3 \\ 1 & 4 & 0 & 12 & 3 & 2 \\ 6 & 12 & 12 & 46 & 12 & 12 \\ 2 & 3 & 3 & 12 & 2 & 2 \\ 2 & 3 & 2 & 12 & 2 & 1 \end{pmatrix}_{b_0} \\
[350] & 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 12 & 6 \\ 8 & 12 & 12 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[349] & 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.58 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 5 & -2 & 2 \end{pmatrix}_{b_0} \\
[354] & 1^{11} 31^3 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 19 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0} \\
[355] & 1^{12} 31^{11} 31^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 7 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0} \\
[356] & 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 2 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 4 & -2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [357] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 11 & 5 \\ 2 & 3 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [358] $1^{14}31^{11}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 18 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [359] $1^{13}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 3 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [360] $21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 14 & 15 & 7 \\ 14 & 13 & 14 & 7 \\ 15 & 14 & 13 & 7 \\ 7 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [361] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 & 4 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$
- [362] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 5 & 10 & 14 & 7 & 8 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$
- [363] $1^{111}3131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 10 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [364] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$
- [367] $1^{13}31^{11}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 15 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$
- [365] $1^{112}3131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 6 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [366] $1^{11}31^{11}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [369] $1^{14}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$\begin{aligned}
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 46 & 12 & 12 \\ 2 & 3 & 0 & 12 & 2 & 2 \\ 2 & 3 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0} \\
[368] & 1^{11} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 9 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[370] & 1^{13} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 5 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0} \\
[371] & 1^{11} 31^{13} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 21 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0} \\
[372] & 1^{12} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 6 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0} \\
[373] & 1^1 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 24 & 7 \\ 15 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[374] & 1^{12} 31^{13} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 6 & 0 & 10 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0} \\
[376] & 1^{11} 31^{12} 31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 8 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0} \\
[375] & 21^{11} 41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 13 & 14 & 51 & 7 \\ 14 & 13 & 54 & 7 \\ 51 & 54 & 197 & 27 \\ 7 & 7 & 27 & 3 \end{pmatrix}_{b_0} \\
[379] & 1^{11} 31^{23} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 14 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[377] & 1^{12} 31^{13} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 10 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0} \\
[378] & 1^{11} 31^{13} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 1 & 11 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[380] $1^{11}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 70 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[381] $1^{12}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[382] $1^{11}3^{11}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 5 & 10 & 14 & 8 & 8 \\ 3 & 5 & 8 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[383] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

[384] $1^{11}3^{11}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 6 & 10 & 14 & 7 & 8 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[385] $1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[386] $1^{14}31^{11}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 8 & 18 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[387] $1^{13}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[388] $1^{11}1^{11}31^{23}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 16 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[389] $1^{13}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 25 & 9 & 9 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[390] $21^{11}4^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

- [392] $1^{11231^1 1^1 31^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 9 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$
- [391] $1^{11131^3 31^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 20 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$
- [393] $1^{1341^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 8 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$
- [395] $1^{11131^1 1^1 31^{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 17 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 17 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [394] $1^{1141^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$
- [396] $1^{1141^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$
- [397] $1^{11131^1 1^1 31^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0}$
- [398] $1^{11131^1 1^1 31^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 3 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$
- [399] $1^1 4^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 34 & 7 \\ 21 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$
- [400] $1^{11131^1 31^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 4 & 4 & 7 & 16 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 8 & -2 & 2 \end{pmatrix}_{b_0}$
- [401] $1^{11131^1 231^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 1 & 15 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[402] & 1^{12}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 4 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0} \\
[403] & 1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0} \\
[404] & 1^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 10 & 19 & 65 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0} \\
[405] & 1^{11}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 5 & 15 & 0 & 5 \\ 8 & 15 & 32 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 12 & -2 & 3 \end{pmatrix}_{b_0} \\
[406] & 2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 & 4 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[407] & 1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0} \\
[408] & 1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 7 & 8 & 15 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 15 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0} \\
[409] & 1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0} \\
[410] & 1^{13}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 15 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0} \\
[411] & 21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 13 & 14 & 24 & 7 \\ 14 & 13 & 24 & 7 \\ 24 & 24 & 38 & 12 \\ 7 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[412] & 1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 4 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{12}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 4 & 5 & 5 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[414] \ 31^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 129, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.32 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 4 & 1 & 2 \\ 3 & 1 & 4 & 4 & 1 & 3 \\ 3 & 4 & 0 & 4 & 1 & 2 \\ 4 & 4 & 4 & 5 & 3 & 5 \\ 1 & 1 & 1 & 3 & -2 & 0 \\ 2 & 3 & 2 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 129, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 5 & 10 & 3 \\ 4 & 10 & 13 & 24 & 6 \\ 5 & 13 & 12 & 27 & 6 \\ 10 & 24 & 27 & 49 & 13 \\ 3 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[416] \ 1^{13}31^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 129, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.44 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 25 & 9 & 9 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[417] \ 1^{13}41^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 131, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[418] \ 1^{13}31^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 134, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.42 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 9 & 0 & 25 & 9 & 9 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[419] \ 21^{11}41^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 134, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 13 & 14 & 42 & 7 \\ 14 & 13 & 44 & 7 \\ 42 & 44 & 130 & 22 \\ 7 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[420] \ 131^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 136, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 3 \\ 4 & 10 & 13 & 24 & 6 \\ 5 & 13 & 12 & 27 & 6 \\ 9 & 24 & 27 & 49 & 13 \\ 3 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[421] \ 21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 142, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 16 & 5 \\ 10 & 13 & 24 & 7 \\ 16 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[422] \ 21^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 142, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 14 & 5 \\ 10 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[423] \ 1^131^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.22 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 12 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

- [424] $1^{14}31^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$
- [425] $1^{13}31^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$
- [426] $1^1 4^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 27 & 5 \\ 9 & 13 & 44 & 7 \\ 27 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$
- [427] $1^{12}31^{11}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$
- [428] $21^{11} 41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 14 & 33 & 7 \\ 14 & 13 & 34 & 7 \\ 33 & 34 & 77 & 17 \\ 7 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$
- [429] $1^{14}31^{13}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 12 & 0 & 46 & 12 & 12 \\ 2 & 3 & 0 & 12 & 2 & 2 \\ 2 & 3 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$
- [430] $1^{13}31^{11}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 7 & 15 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$
- [431] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [432] $1^{11} 1^{13} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 16 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 16 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [433] $1^{12} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$
- [434] $2^1 2^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 5 & 5 \\ 6 & 10 & 14 & 7 & 8 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$
- [435] $1^{11} 1^{13} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 5 & 20 & 0 & 5 \\ 10 & 20 & 58 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 16 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[436] \ 1^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 15 & 25 & 132 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[437] \ 1^{11}43131^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 12 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[438] \ 1^{11}321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 5 & 15 & 5 & 5 \\ 8 & 15 & 32 & 11 & 12 \\ 3 & 5 & 11 & 0 & 2 \\ 3 & 5 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[439] \ 1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[440] \ 1^{12}31^{11}31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 6 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[441] \ 1^{13}1^{13}31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 8 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[442] \ 1^{12}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[443] \ 1^{14}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 7 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[444] \ 1^441^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 33 & 5 \\ 9 & 13 & 54 & 7 \\ 33 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[445] \ 1^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[446] \ 1^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[447] \ 1^{12} 3^{11} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 5 & 20 & 5 & 5 \\ 11 & 20 & 62 & 14 & 16 \\ 3 & 5 & 14 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[448] \ 1^{13} 31^{1431} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 11 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[449] \ 1^{12} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 5 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[450] \ 1^{14} 31^{11} 31^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 10 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[451] \ 2^{121} 2^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 5 & 5 \\ 6 & 10 & 14 & 8 & 8 \\ 3 & 5 & 8 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[452] \ 1^{13} 31^{11} 31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 8 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[453] \ 1^{12} 3^{11} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 5 & 6 & 8 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[454] \ 1^{14} 41^{11} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 14 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[455] \ 1^{14} 31^{11} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 12 & 0 & 46 & 12 & 12 \\ 2 & 3 & 0 & 12 & 2 & 2 \\ 2 & 3 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[456] \ 1^{14} 41^{11} 14^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[457] \ 1^{13} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 6 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[458] \ 1^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 11 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[459] \ 21^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 21 & 5 \\ 10 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[460] \ 1^{11}321^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 5 & 20 & 5 & 5 \\ 10 & 20 & 58 & 15 & 16 \\ 3 & 5 & 15 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[461] \ 1^{13}31^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[462] \ 1^{11}431^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 7 & 15 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[463] \ 1^{13}31^{11}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 7 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[464] \ 1^{11}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 5 & 25 & 0 & 5 \\ 13 & 25 & 94 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 20 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[465] \ 1^{11}31^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[466] \ 1^{13}31^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 7 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[467] \ 1^{13}31^{12}31^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[468] \ 1^{11}31^{12}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 7 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[469] $1^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$

$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

[470] $1^{13}31^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$

$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 7 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[471] $1^{11}431^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 4 & 18 & 0 & 3 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 18 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

[472] $1^{13}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$

$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 6 & 8 & 9 & 40 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[474] $1^{14}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 13 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

[473] $21^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 23 & 5 \\ 10 & 13 & 34 & 7 \\ 23 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[475] $1^{14}31^{11}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 8 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[476] $1^{14}31^{11}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 9 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[477] $1^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$

$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 17 & 29 & 169 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$

$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

[480] $1^{12}31^{12}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 9 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[479] \ 1^{12}31^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[481] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 5 & 15 & 5 & 5 \\ 9 & 15 & 34 & 12 & 12 \\ 3 & 5 & 12 & 0 & 2 \\ 3 & 5 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[482] \ 1^{12}3 \ 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 5 & 25 & 5 & 5 \\ 13 & 25 & 94 & 18 & 20 \\ 3 & 5 & 18 & 0 & 2 \\ 3 & 5 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[483] \ 1^{11}31^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[484] \ 2 \ 2 \ 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 5 & 5 \\ 12 & 20 & 62 & 14 & 16 \\ 3 & 5 & 14 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[485] \ 1^{11}31^{13}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 10 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[486] \ 21^{14}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 358, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 28 & 5 \\ 10 & 13 & 40 & 7 \\ 28 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[487] \ 1^{14}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[488] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 5 & 15 & 5 & 5 \\ 9 & 15 & 32 & 11 & 12 \\ 3 & 5 & 11 & 0 & 2 \\ 3 & 5 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[489] \ 1^{13}31^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[490] \ 1^{13}31^{14}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 10 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[491] $1^{14}31^{12}31^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 12 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

[492] $1^{14}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 9 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

[493] $1^{11}31^{14}31^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 408$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.31 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 13 & 27 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[494] $31^{11}1^{12}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 409$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[495] $1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 412$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 17 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[496] $1^{12}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 7 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[497] $1^{12}31^{12}31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[498] $1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 435$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 19 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[499] $1^{12}31^{13}31^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 12 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[500] $1^{11}31^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 444$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 7 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[501] $1^131^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 448$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

- [502] $1^{13}31^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 9 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [503] $1^{13}31^{12}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 8 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$
- [504] $21^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 470, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 30 & 5 \\ 10 & 13 & 44 & 7 \\ 30 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$
- [505] $2^{11}2^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 5 & 5 \\ 12 & 20 & 62 & 16 & 16 \\ 3 & 5 & 16 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$
- [506] $1^{11}31^{13}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 9 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$
- [507] $1^{13}31^{12}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$
- [508] $21^{15}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 35 & 5 \\ 10 & 13 & 50 & 7 \\ 35 & 50 & 173 & 25 \\ 5 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$
- [509] $1^{14}31^{12}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 11 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$
- [510] $1^{14}31^{12}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 10 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$
- [511] $1^{11}31^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$
- [514] $1^{14}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

- [513] $1^{12}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 564$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 10 & 30 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$
- [512] $1^{13}31^{15}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 564$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.1$
 $T_{\min} = 11.35 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 8 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$
- [515] $1^{13}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 604$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 9 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$
- [516] $2^1 2^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 625$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 5 & 5 \\ 12 & 20 & 58 & 15 & 16 \\ 3 & 5 & 15 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$
- [517] $2^1 2^1 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 649$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 5 & 30 & 5 & 5 \\ 18 & 30 & 142 & 21 & 24 \\ 3 & 5 & 21 & 0 & 2 \\ 3 & 5 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$
- [518] $1^{12}31^{13}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 660$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 11 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$
- [519] $1^{11}31^{14}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 664$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 12 & 27 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$
- [520] $1^{11}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 664$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 27 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$
- [521] $21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 688$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 37 & 5 \\ 10 & 13 & 54 & 7 \\ 37 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$
- [522] $21^{16}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 718$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 42 & 5 \\ 10 & 13 & 60 & 7 \\ 42 & 60 & 250 & 30 \\ 5 & 7 & 30 & 3 \end{pmatrix}_{b_0}$$
- [523] $1^{12}31^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 724$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 12 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0} \\
[535] & 1^{13}31^{13}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 11 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0} \\
[536] & 1^{141} 2^2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 929, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 5 & 25 & 5 & 5 \\ 15 & 25 & 92 & 19 & 20 \\ 3 & 5 & 19 & 0 & 2 \\ 3 & 5 & 20 & 2 & 3 \end{pmatrix}_{b_0} \\
[537] & 1^{14}31^{13}31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 14 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 14 & 30 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_{b_0} \\
[538] & 21^{17} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.3 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 49 & 5 \\ 10 & 13 & 70 & 7 \\ 49 & 70 & 341 & 35 \\ 5 & 7 & 35 & 3 \end{pmatrix}_{b_0} \\
[539] & 1^{13}31^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 33 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 11 & 33 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0} \\
[540] & 1^{14}31^{13}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1012, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 12 & 30 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_{b_0} \\
[541] & 1^{14}31^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 \\ 12 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0} \\
[542] & 1^{14}31^{13}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 13 & 30 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_{b_0} \\
[543] & 1^{13} 31^{17} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.3 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 14 & 1 & 2 \\ 2 & 4 & 4 & 42 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 14 & 42 & 21 & 292 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0} \\
[544] & 1^{12}31^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1056, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 11 & 30 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0} \\
[545] & 1^{12}31^{14}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1056, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 \\ 16 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[557] $1^{14}31^{14}31^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1600, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 \\ 15 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [558] $1^{11}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 6 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[559] $21^{14}1^{31}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 2 & 1 & 4 \\ 2 & 2 & 0 & 0 & 2 \\ 2 & 1 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[560] $1^{11}4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 3 & 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[561] $1^{11}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[562] $1^{11}4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[563] $1^{11}41^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[564] $1^{11}4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 3 & 1 & 5 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[565] $1^{11}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 2 & 1 & 8 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 22 & 10 \\ 3 & 4 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[566] $1^{11}4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[567] $1^{11}4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[568] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 9 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[569] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 2 & 1 & 0 & 1 & 2 \\ 3 & 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[570] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[571] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[572] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[573] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 2 & 1 & 0 & 2 & 2 \\ 3 & 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[574] \ 1^{\overline{121}} 2^{\overline{321}} 2^{\overline{21}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 8 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 16 & 12 \\ 2 & 4 & 0 & 16 & 4 & 4 \\ 2 & 3 & -1 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[575] \ 1^{\overline{131}} 2^{\overline{321}} 3^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 24 & 18 \\ 2 & 4 & 0 & 24 & 4 & 4 \\ 2 & 3 & -1 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[576] \ 1^{\overline{111}} 2^{\overline{321}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 8 & 6 \\ 2 & 4 & 0 & 8 & 4 & 4 \\ 2 & 3 & -1 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[577] \ 1^{\overline{141}} 2^{\overline{321}} 4^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 16 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 32 & 24 \\ 2 & 4 & 0 & 32 & 4 & 4 \\ 2 & 3 & -1 & 24 & 4 & 3 \end{pmatrix}_{b_0}$$

[578] $1^{11} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 6 & -2 & 3 \end{pmatrix}_{b_0}$$

[579] $1^{11} 4^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 4 & 2 & 2 & 1 & 4 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[580] $1^{11} 4^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 3 & 1 & 2 \\ 3 & 3 & 2 & 1 & 4 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[581] $1^{11} 4^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[582] $2^1 3^1 1^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 5 & 7 & 8 & 5 & 6 \\ 3 & 5 & 5 & 0 & 2 \\ 3 & 5 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[583] $1^{11} 4^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 2 & 4 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[584] $1^{11} 4^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 0 & 3 & 2 & 2 \\ 3 & 3 & 2 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[585] $1^{11} 4^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 4 & 2 & 2 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[586] $31^{11} 31^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 6 & 8 & 4 & 3 \\ 6 & 13 & 20 & 6 & 7 \\ 8 & 20 & 26 & 8 & 10 \\ 4 & 6 & 8 & 0 & 2 \\ 3 & 7 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[587] $31^{11} 31^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 3 & 2 & 4 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 4 & 2 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[588] $1^{11} 4^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 & 3 \\ 2 & 0 & 4 & 2 & 2 \\ 5 & 4 & 6 & 4 & 6 \\ 3 & 2 & 4 & 0 & 2 \\ 3 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[589] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 0 & 4 & 1 & 2 \\ 5 & 4 & 6 & 2 & 6 \\ 1 & 1 & 2 & -2 & 0 \\ 3 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[590] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 5 & 2 & 5 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[591] \ 31^{\overline{131}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 6 & 5 & 4 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 8 & 5 & 6 \\ 4 & 6 & 5 & 0 & 2 \\ 3 & 7 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[592] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 7 & 1 & 3 \\ 2 & 0 & 5 & 1 & 2 \\ 7 & 5 & 12 & 2 & 8 \\ 1 & 1 & 2 & -2 & 0 \\ 3 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[593] \ 31^{\overline{131}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 2 & 4 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 6 & 3 & 4 & 2 & 4 \\ 1 & 2 & 1 & 2 & -2 & 0 \\ 2 & 4 & 2 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[594] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 5 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 5 & 1 & 13 & 3 & 7 \\ 1 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & 0 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[595] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 9 & 1 & 3 \\ 2 & 0 & 6 & 1 & 2 \\ 9 & 6 & 20 & 3 & 10 \\ 1 & 1 & 3 & -2 & 0 \\ 3 & 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[596] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 7 & 6 & 1 & 23 & 4 & 9 \\ 1 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & 0 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[597] \ 31^{\overline{1131}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 1 & 2 \\ 3 & 4 & 4 & 9 & 2 & 4 \\ 3 & 4 & 0 & 6 & 1 & 2 \\ 6 & 9 & 6 & 13 & 3 & 7 \\ 1 & 2 & 1 & 3 & -2 & 0 \\ 2 & 4 & 2 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [598] \ 1^{\overline{1}} 51^{\overline{1}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 26 & 10 \\ 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[599] $1^{11}5^{11}1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[600] $1^{11}2^13^115^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 9 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[601] $1^{11}41^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 26 & 10 \\ 3 & 12 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[602] $1^{11}5^{11}1^{21}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 225 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[603] $1^{11}5^{11}1^{11}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 95 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [604] $1^{11}31^{11}5^{11}1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 6 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[605] $1^{11}31^{11}5^{11}1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 9 & 7 \\ 2 & 5 & 0 & 9 & 5 & 5 \\ 2 & 4 & -1 & 7 & 5 & 4 \end{pmatrix}_{b_0}$$

[606] $1^{41}31^{11}5^{11}1321^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 2 & 1 & 20 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 20 & 0 & 142 & 36 & 28 \\ 2 & 5 & 0 & 36 & 5 & 5 \\ 2 & 4 & -1 & 28 & 5 & 4 \end{pmatrix}_{b_0}$$

[607] $1^{31}31^{11}5^{11}1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 2 & 1 & 15 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 15 & 0 & 79 & 27 & 21 \\ 2 & 5 & 0 & 27 & 5 & 5 \\ 2 & 4 & -1 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

[608] $1^{21}31^{11}5^{11}1321^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 2 & 1 & 10 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 18 & 14 \\ 2 & 5 & 0 & 18 & 5 & 5 \\ 2 & 4 & -1 & 14 & 5 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [609] \quad 1^{41} 31^1 5 13221^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 3 & 1 & 24 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 24 & 0 & 158 & 40 & 32 \\ 2 & 6 & 0 & 40 & 6 & 6 \\ 2 & 5 & -1 & 32 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[610] \quad 1^{31} 31^1 5 13221^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 3 & 1 & 18 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 18 & 0 & 88 & 30 & 24 \\ 2 & 6 & 0 & 30 & 6 & 6 \\ 2 & 5 & -1 & 24 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[611] \quad 1^{21} 31^1 5 13221^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 3 & 1 & 12 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 12 & 0 & 38 & 20 & 16 \\ 2 & 6 & 0 & 20 & 6 & 6 \\ 2 & 5 & -1 & 16 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[612] \quad 1^{11} 31^1 5 13221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 6 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 8 & 10 & 8 \\ 2 & 6 & 0 & 10 & 6 & 6 \\ 2 & 5 & -1 & 8 & 6 & 5 \end{pmatrix}_{b_0}$$

4.847 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113} 31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 45 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 26 & -4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 39 & -4 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 921, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1005, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 33 & -4 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{115}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1053, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.1$
 $T_{\min} = 13.52 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 35 & -4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 52 & -4 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1590, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 46 & -4 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1650, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 40 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -571, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.848 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2$

$T = 9$ [0] $1^{112}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

4.849 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{violet}{e}_6 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.850 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.851 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 31^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 45 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

- [4] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 165$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 291$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 331$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 26 & -3 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 365$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 370$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 540$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 581$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 39 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 709$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{115}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 765$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 35 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 781$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 33 & -3 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 924$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 52 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1044$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 46 & -3 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 40 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{111}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.852 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

4.853 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 6 & 5 \\ 12 & 25 & 20 & 0 & 12 & 9 \\ 9 & 20 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 4 & 4 \\ 5 & 9 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.854 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

4.855 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.856 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8$

$$T = 8 \quad [0] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 6 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{113} 3 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 54 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] 1^{112} 3 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 36 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] 1^{113} 3 1^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 45 & -2 & 2 \end{pmatrix}_{b_0}$$

[illegible]

- [42] $1^{115} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 477$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 35 & -2 & 1 \end{pmatrix}_{b_0}$
- [43] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 485$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 38 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [44] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 497$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [45] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 557$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -2 & 2 \end{pmatrix}_{b_0}$
- [46] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 621$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 45 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -2 & 2 \end{pmatrix}_{b_0}$
- [47] $1^{114} 3 1^{41} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 662$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 52 & -2 & 2 \end{pmatrix}_{b_0}$
- [48] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 689$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 37 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [49] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 693$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 36 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [50] $1^{114} 3 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 710$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 56 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 56 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -2 & 2 \end{pmatrix}_{b_0}$
- [51] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 722$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$
- [52] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 786$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 48 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 48 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$

$$[53] \ 1^{114} 3 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 846, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 63 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 63 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{114} 3 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 882, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{114} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 902, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 1038, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 45 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{114} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 1058, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.82 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 55 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 55 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 1070, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 47 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 47 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{114} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 1154, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{114} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 1206, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 54 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [61] \ 1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -21, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 18 & 1 & 5 \\ 9 & 13 & 30 & 0 & 7 \\ 18 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -24, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 12 & 0 & 10 & 12 & 6 \\ 5 & 12 & 0 & 12 & 11 & 5 \\ 3 & 6 & -1 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -33, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{12}41^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.36 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{11}41^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{11}41^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -93$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 12 & 0 & 6 \\ 5 & 12 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{111}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -117$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 10 & 12 & 6 \\ 5 & 12 & 0 & 12 & 11 & 5 \\ 3 & 6 & -1 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{111}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 14 & 0 & 7 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 3 & 5 & 8 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{11}41^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -186$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 15 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 15 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 10 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 10 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 21 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 21 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{11}321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 5 & 15 & 0 & 5 & 5 \\ 8 & 15 & 32 & 0 & 11 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 11 & 0 & 0 & 2 \\ 3 & 5 & 12 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 27 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 33 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 33 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11}321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -495, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 5 & 20 & 0 & 5 & 5 \\ 10 & 20 & 58 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 15 & 0 & 0 & 2 \\ 3 & 5 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{12}321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -706, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 5 & 25 & 0 & 5 & 5 \\ 13 & 25 & 94 & 0 & 18 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 18 & 0 & 0 & 2 \\ 3 & 5 & 20 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [82] $1^{11}31^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 0 & 3 & 1 & 1 & 2 \\ 3 & 3 & 2 & 1 & 2 & 4 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 2 & 0 & -2 & 0 \\ 3 & 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 4 & 2 & 2 & 1 & 2 & 4 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 2 & 0 & -2 & 0 \\ 3 & 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 2 & 3 \\ 2 & 0 & 4 & 1 & 1 & 2 \\ 5 & 4 & 6 & 2 & 2 & 6 \\ 1 & 1 & 2 & -2 & 0 & 0 \\ 2 & 1 & 2 & 0 & -2 & 0 \\ 3 & 2 & 6 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

4.857 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & 0 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 39 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 39 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 45 & 5 \\ 11 & 20 & 0 & 0 & 84 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 84 & 0 & 0 & 346 & 34 \\ 5 & 8 & 0 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

- [5] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & 0 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{1111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 10 & 5 \\ 12 & 25 & 36 & 0 & 20 & 9 \\ 17 & 36 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 27 & 0 & 13 & 7 \\ 5 & 9 & 13 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [7] $1^{1111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 18 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 18 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{1111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 18 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{1111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 48 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 9 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 48 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$(1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 10 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 36 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 10 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 38 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 38 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 47 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 9 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 47 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 28 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 28 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{111} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -530, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 46 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 9 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 46 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{111} 3 \ 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -532, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 36 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111} 3 \ 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -550, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 3 \ 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -616, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 37 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 37 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

4.858 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & 0 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 6 & 5 \\ 12 & 25 & 20 & 0 & 12 & 9 \\ 9 & 20 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 9 & 0 & 4 & 4 \\ 5 & 9 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 18 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -217, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.859 NHC(-2)(-3) + su₂ + so₈ + so₈ + su_{2n3mix}

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & 0 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 10 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 16 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

4.860 NHC(-2)(-3) + su₂ + so₈ + so₈ + su₃

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 1 & 5 \\ 12 & 25 & 76 & 0 & 0 & 9 \\ 35 & 76 & 217 & 0 & 0 & 27 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & 0 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 1 & 5 \\ 12 & 25 & 36 & 0 & 0 & 9 \\ 17 & 36 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

- [4] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 10 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 1 & 5 \\ 11 & 20 & 16 & 0 & 0 & 8 \\ 8 & 16 & 10 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 1 & 5 \\ 12 & 25 & 56 & 0 & 0 & 9 \\ 26 & 56 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 1 & 5 \\ 12 & 25 & 72 & 0 & 0 & 9 \\ 34 & 72 & 202 & 0 & 0 & 26 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 10 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -222, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 9 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -244, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 1 & 5 \\ 11 & 20 & 50 & 0 & 0 & 8 \\ 26 & 50 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -293, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 1 & 5 \\ 11 & 20 & 33 & 0 & 0 & 8 \\ 17 & 33 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{113} 3 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -295, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 1 & 5 \\ 12 & 25 & 108 & 0 & 0 & 9 \\ 51 & 108 & 457 & 0 & 0 & 39 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -315, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 1 & 5 \\ 12 & 25 & 92 & 0 & 0 & 9 \\ 43 & 92 & 325 & 0 & 0 & 33 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -330, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 1 & 5 \\ 12 & 25 & 112 & 0 & 0 & 9 \\ 52 & 112 & 478 & 0 & 0 & 40 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -442, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 1 & 5 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 68 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 1 & 5 \\ 12 & 25 & 128 & 0 & 0 & 9 \\ 60 & 128 & 634 & 0 & 0 & 46 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -539, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 19 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 18 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \ 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 6 & 5 \\ 9 & 13 & 14 & 0 & 10 & 7 \\ 9 & 14 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 9 & 0 & 4 & 4 \\ 5 & 7 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 6 & 5 \\ 9 & 13 & 10 & 0 & 10 & 7 \\ 6 & 10 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 7 & 0 & 4 & 4 \\ 5 & 7 & 5 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 15 & 1 & 6 & 5 \\ 9 & 13 & 24 & 0 & 10 & 7 \\ 15 & 24 & 38 & 0 & 16 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 16 & 0 & 4 & 4 \\ 5 & 7 & 12 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.861 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$

$$T = 8 \quad [0] \ 1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 20 & 13 & 15 & 7 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & 0 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & 0 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 1 & 6 \\ 15 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 48 & 6 \\ 15 & 20 & 0 & 67 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 48 & 67 & 0 & 217 & 27 \\ 6 & 8 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{133} 3^1 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 27 & 4 \\ 9 & 17 & 49 & 0 & 81 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 81 & 0 & 133 & 21 \\ 3 & 4 & 13 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3^1 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 35 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 35 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 16 & 23 & 13 & 6 \\ 16 & 25 & 36 & 20 & 9 \\ 23 & 36 & 49 & 27 & 13 \\ 13 & 20 & 27 & 13 & 7 \\ 6 & 9 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{124} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 8 & 0 & 31 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 42 & 0 & 172 & 24 \\ 3 & 4 & 6 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 48 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 48 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 4 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[19] $1^{133} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 16 & 3 \\ 3 & 4 & 17 & 0 & 27 & 4 \\ 10 & 17 & 49 & 0 & 81 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 81 & 0 & 133 & 21 \\ 3 & 4 & 13 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 13 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 13 & 7 \\ 6 & 8 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{1111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 16 & 23 & 23 & 6 \\ 16 & 25 & 36 & 36 & 9 \\ 23 & 36 & 49 & 51 & 13 \\ 23 & 36 & 51 & 49 & 13 \\ 6 & 9 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{113} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 6 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 35 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 35 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 4 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{123} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 18 & 0 & 17 & 4 \\ 10 & 18 & 58 & 0 & 54 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 54 & 0 & 49 & 13 \\ 3 & 4 & 14 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [35] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -2 & 3 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{13} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 0 & 25 & 9 \\ 3 & 4 & 0 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$
- [37] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [43] $1^{12} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & 0 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$
- [41] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [42] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [44] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{114} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 6 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{123} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{14} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{123} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 28 & 0 & 27 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 10 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{111} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{111} 3 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 39 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 39 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{111} 3 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{113} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 6 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{122} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 7 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{124} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 5 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 5 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{122} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 7 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 27 & 0 & 28 & 10 \\ 3 & 4 & 9 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & 0 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 0 & 76 & 16 \\ 3 & 4 & 0 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 13 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 10 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 6 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{14} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 6 & 3 \\ 3 & 4 & 14 & 0 & 9 & 4 \\ 8 & 14 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 24 & 0 & 13 & 7 \\ 3 & 4 & 11 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{112} 3 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 \\ 8 & 14 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 33 & 0 & 28 & 10 \\ 3 & 4 & 11 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 5 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 62 & 6 \\ 15 & 20 & 17 & 84 & 8 \\ 12 & 17 & 13 & 72 & 7 \\ 62 & 84 & 72 & 346 & 34 \\ 6 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 19 & 0 & 0 & 73 & 15 \\ 3 & 4 & 0 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{131} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 4 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 48 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 13 & 17 & 13 & 57 & 7 \\ 48 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{134} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 22 & 0 & 21 & 4 \\ 12 & 22 & 85 & 0 & 81 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 81 & 0 & 76 & 16 \\ 3 & 4 & 17 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 0 & 37 & 11 \\ 3 & 4 & 0 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 36 & 6 \\ 15 & 20 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 36 & 50 & 0 & 118 & 20 \\ 6 & 8 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

- [85] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 12 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$
- [87] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [86] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$
- [88] $1^{133} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 12 & 0 & 17 & 4 \\ 7 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 36 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [89] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 0 & 49 & 13 \\ 3 & 4 & 0 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [90] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & 0 & -2 & 10 & 2 \end{pmatrix}_{b_0}$
- [91] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [92] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [93] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 5 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 6 & -2 & 17 & 2 \end{pmatrix}_{b_0}$
- [94] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 13 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$
- [95] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 49 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 49 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{123} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 24 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{132} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 8 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 39 & 0 & 28 & 10 \\ 3 & 4 & 13 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{131} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{112} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{112} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{124} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 8 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{11} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{143} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 10 & 3 \\ 3 & 4 & 21 & 0 & 17 & 4 \\ 11 & 21 & 76 & 0 & 63 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 63 & 0 & 49 & 13 \\ 3 & 4 & 16 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{134} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 12 & 0 & 21 & 4 \\ 7 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 45 & 0 & 76 & 16 \\ 3 & 4 & 9 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 38 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 38 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{122} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 9 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 6 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{113} 3 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 13 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 8 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 17 & 3 \\ 3 & 4 & 0 & 0 & 28 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 0 & 0 & 154 & 22 \\ 3 & 4 & 0 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 63 & 6 \\ 15 & 20 & 17 & 84 & 8 \\ 12 & 17 & 13 & 72 & 7 \\ 63 & 84 & 72 & 346 & 34 \\ 6 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[116] $1^{144} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 16 & 0 & 21 & 4 \\ 9 & 16 & 46 & 0 & 60 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 60 & 0 & 76 & 16 \\ 3 & 4 & 12 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[118] $1^{12} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 4 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{134} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 18 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 9 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 0 & 0 & 118 & 20 \\ 3 & 4 & 0 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 0 & 0 & 58 & 14 \\ 3 & 4 & 0 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 37 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 37 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[124] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 51 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 51 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[126] $1^{132} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 18 & 4 \\ 9 & 17 & 49 & 0 & 54 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 54 & 0 & 58 & 14 \\ 3 & 4 & 13 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

- [125] $1^{124} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 8 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [128] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [127] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [129] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 177, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [132] $1^{12} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 0 & 0 & 100 & 18 \\ 3 & 4 & 0 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [131] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [130] $1^{124} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [133] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 50 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 50 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$
- [134] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 0 & 0 & 85 & 17 \\ 3 & 4 & 0 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [136] $1^{112} 3 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 8 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 \\ 9 & 14 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 33 & 0 & 28 & 10 \\ 3 & 4 & 11 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [135] $1^{133} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 9 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[138] 1^{143} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 \\ 11 & 21 & 76 & 0 & 81 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 81 & 0 & 85 & 17 \\ 3 & 4 & 16 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[137] 1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[140] 1^{124} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[139] 1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[141] 1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[142] 1^{124} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 15 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[143] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[144] 1^{113} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 3 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[145] 1^{122} 3 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 8 & 3 \\ 3 & 4 & 18 & 0 & 13 & 4 \\ 10 & 18 & 58 & 0 & 42 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 42 & 0 & 28 & 10 \\ 3 & 4 & 14 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[146] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{123}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 7 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{13}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 19 & 3 \\ 3 & 4 & 0 & 0 & 32 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 0 & 0 & 193 & 25 \\ 3 & 4 & 0 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 4 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{13}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 16 & 3 \\ 3 & 4 & 0 & 0 & 27 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 0 & 0 & 133 & 21 \\ 3 & 4 & 0 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{14}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 18 & 3 \\ 3 & 4 & 0 & 0 & 31 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 0 & 0 & 172 & 24 \\ 3 & 4 & 0 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{123}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[154] $21^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 72 & 7 \\ 17 & 20 & 17 & 84 & 8 \\ 15 & 17 & 13 & 72 & 7 \\ 72 & 84 & 72 & 346 & 34 \\ 7 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{114}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{113}3^11^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 9 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{113} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 8 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 21 & 3 \\ 3 & 4 & 0 & 0 & 36 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 0 & 0 & 238 & 28 \\ 3 & 4 & 0 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[160] $21^{1111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 27 & 27 & 7 \\ 20 & 25 & 36 & 36 & 9 \\ 27 & 36 & 49 & 51 & 13 \\ 27 & 36 & 51 & 49 & 13 \\ 7 & 9 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 5 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[163] $21^{112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 42 & 7 \\ 17 & 20 & 17 & 50 & 8 \\ 15 & 17 & 13 & 42 & 7 \\ 42 & 50 & 42 & 118 & 20 \\ 7 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{11} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{12} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{13} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 4 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{123} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 19 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 8 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[168] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[169] $21^{111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 27 & 15 & 7 \\ 20 & 25 & 36 & 20 & 9 \\ 27 & 36 & 49 & 27 & 13 \\ 15 & 20 & 27 & 13 & 7 \\ 7 & 9 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[171] $1^{132} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 11 & 3 \\ 3 & 4 & 22 & 0 & 18 & 4 \\ 12 & 22 & 85 & 0 & 72 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 72 & 0 & 58 & 14 \\ 3 & 4 & 17 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{122} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 8 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{14} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{122} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{113} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{124}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{12}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 4 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{111}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 6 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 19 & 33 & 0 & 73 & 15 \\ 3 & 4 & 7 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[181] $21^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 57 & 7 \\ 17 & 20 & 17 & 67 & 8 \\ 15 & 17 & 13 & 57 & 7 \\ 57 & 67 & 57 & 217 & 27 \\ 7 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{123}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 8 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{13}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 5 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{111}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 9 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 8 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[188] 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[189] 1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 5 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[190] 1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[191] 1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 19 & 33 & 0 & 73 & 15 \\ 3 & 4 & 7 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[192] 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[194] 1^{124} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[193] 1^{113} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[195] 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[196] 1^{133} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 10 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[197] 1^{132} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 18 & 4 \\ 10 & 17 & 49 & 0 & 54 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 54 & 0 & 58 & 14 \\ 3 & 4 & 13 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{123} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 5 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{111} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 8 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{122} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 7 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{122} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[202] \ 1^{133} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 12 & 0 & 17 & 4 \\ 7 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 36 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{143} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 13 & 3 \\ 3 & 4 & 26 & 0 & 22 & 4 \\ 14 & 26 & 118 & 0 & 102 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 102 & 0 & 85 & 17 \\ 3 & 4 & 20 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{112} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 11 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 8 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[205] \ 1^{112} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{124} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 5 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{124} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{123} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 345, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 19 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 7 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{114} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 3 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 3 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{143} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 13 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 \\ 12 & 21 & 76 & 0 & 81 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 81 & 0 & 85 & 17 \\ 3 & 4 & 16 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 8 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 6 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 365$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 369$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 378$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 378$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 10 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 382$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 12 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.6$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 12 & 4 \\ 7 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 388$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{13} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 389$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.3$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 18 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 7 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 392$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.6$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 5 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{124} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 6 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{134} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 12 & 0 & 16 & 4 \\ 7 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 36 & 0 & 46 & 12 \\ 3 & 4 & 9 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{144} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 11 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 449, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{114} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 452, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 3 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{13} 3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{13} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{113} 3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{113} 3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{123} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{113} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 477, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 6 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{122} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 11 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 10 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{14} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 494, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{134} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 497, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 12 & 0 & 21 & 4 \\ 7 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 45 & 0 & 76 & 16 \\ 3 & 4 & 9 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{112} 3^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[270] $21^{1111} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 15 & 15 & 7 \\ 20 & 25 & 20 & 20 & 9 \\ 15 & 20 & 13 & 15 & 7 \\ 15 & 20 & 15 & 13 & 7 \\ 7 & 9 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{12} 3^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{14} 3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{133} 3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 509, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 9 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{112} 3^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 514, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 5 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{112} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{133} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 9 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{14} 3^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[278] $1^{124} 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 5 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{124} 1^{111} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 15 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 10 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{114} 1^{111} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 26 & 7 \\ 6 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[281] $1^{12} 1^{14} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{134} 1^{113} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 539, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 17 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 9 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[283] $1^{124} 1^{111} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 10 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[284] $1^{113} 1^{112} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{111} 1^{111} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 9 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{114} 1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[287] $1^{113} 1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 565, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{144} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 11 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[290] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 9 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 9 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 9 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{124} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 598, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 22 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[299] $1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 602, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 6 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{123} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 13 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 10 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[301] $1^{124} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 16 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[302] $1^{123} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 16 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{134} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 614, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 9 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{124} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 8 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{123} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{112} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 6 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{122} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 10 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[308] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{123} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{12} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 638, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{13} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{122} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 8 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 6 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 8 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 655, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{133} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 13 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 12 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[324] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[325] $1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 15 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 11 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[327] $1^{144} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 11 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[326] $1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 10 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{114} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[329] \ 1^{113} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[330] \ 1^{112} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 690, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 5 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[331] \ 1^{113} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 704, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 21 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[332] \ 1^{114} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[333] \ 1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[334] \ 1^{113} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[335] \ 1^{124} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 712, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 19 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 7 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[336] \ 1^{123} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 715, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 8 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[337] \ 1^{112} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 5 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[338] 1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 720, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[339] 1^{113} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[340] 1^{114} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 723, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 21 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[341] 1^{14} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[342] 1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[343] 1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[344] 1^{13} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[345] 1^{134} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 755, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 18 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 10 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[346] 1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 764, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[347] 1^{12} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[348] \ 1^{14} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 774, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[349] \ 1^{13} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 783, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[350] \ 1^{14} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[351] \ 1^{123} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 787, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 16 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 8 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[352] \ 1^{14} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[353] \ 1^{134} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 794, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 9 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[354] \ 1^{144} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 15 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 14 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[355] \ 1^{113} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 812, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 6 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[356] \ 1^{124} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{113} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 817, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 19 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 6 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{12} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 818, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[359] \ 1^{13} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 830, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{14} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 843, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 6 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{133} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 849, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 10 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[362] \ 1^{124} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 850, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 21 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 8 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{13} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 857, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{12} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[365] \ 1^{134} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 15 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 12 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{114} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 870, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[367] $1^{114} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 874, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[368] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 884, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[369] $1^{14} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[370] $1^{113} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 909, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[372] $1^{14} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[373] $1^{124} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 934, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 20 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[374] $1^{124} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 938, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[375] $1^{133} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 10 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[376] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 941, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

- [377] $1^{133} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 944, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 12 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [378] $1^{123} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 945, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 8 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [379] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [380] $1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 972, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [381] $1^{114} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 979, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [382] $1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [383] $1^{124} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 7 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [384] $1^{124} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 994, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 21 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [385] $1^{114} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 994, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 21 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 6 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [386] $1^{144} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1002, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 15 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 12 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[387] $1^{134} 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1003, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 10 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[388] $1^{134} 1^{111} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1004, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 12 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[389] $1^{122} 1^{111} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1008, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 11 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{13} 1^{12} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1013, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{113} 1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1017, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 21 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{114} 1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 21 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{14} 1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1026, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{123} 1^{111} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 13 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 11 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{114} 1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1061, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 21 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[406] $1^{113} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1177$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 19 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[407] $1^{114} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1190$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 24 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{14} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[409] $1^{113} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1213$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[410] $1^{114} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1216$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 23 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[411] $1^{114} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1228$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 12 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[412] $1^{134} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 10 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[413] $1^{114} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1254$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{114} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1263$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[415] $1^{144} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 14 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{14} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1314, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 23 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{114} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 19 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[418] $1^{14} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 24 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[419] $1^{133} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 13 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 13 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[420] $1^{134} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 15 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 13 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[421] $1^{114} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1554, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 23 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[422] $1^{114} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 21 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[423] $1^{114} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1686, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 22 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[424] $1^{144} 1^{111} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1904, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 15 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 15 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [425] $1^{111} 3 1^{131} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.862 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & 0 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 10 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & 0 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 36 & 5 \\ 11 & 20 & 0 & 0 & 67 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 0 & 0 & 217 & 27 \\ 5 & 8 & 0 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}} 3 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 19 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}} 3 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 18 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & 0 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}} 3 1^{\overline{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 27 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & 0 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{111}} 3 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}} 3 1^{\overline{122}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 10 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{111}} 3 1^{\overline{122}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{111}} 3 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{111}} 3 1^{\overline{122}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 28 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 28 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{\overline{111}} 3 1^{\overline{122}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -550, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

4.863 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & 0 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

4.864 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 23 & 8 & 6 \\ 16 & 25 & 36 & 12 & 9 \\ 23 & 36 & 49 & 17 & 13 \\ 8 & 12 & 17 & 4 & 4 \\ 6 & 9 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 8 & 6 \\ 15 & 20 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 11 & 0 & 4 & 4 \\ 6 & 8 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 13 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 3 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 5 & 8 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 3 & 4 \\ 7 & 12 & 25 & 0 & 6 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 8 & 14 & 37 & 0 & 8 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 3 & 4 \\ 9 & 16 & 46 & 0 & 8 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 7 & 13 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 6 & 9 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 9 & 17 & 49 & 0 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 11 & 21 & 76 & 0 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[19] $21^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 27 & 9 & 7 \\ 20 & 25 & 36 & 12 & 9 \\ 27 & 36 & 49 & 17 & 13 \\ 9 & 12 & 17 & 4 & 4 \\ 7 & 9 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 10 & 18 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11} 1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 9 & 14 & 37 & 0 & 8 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 8 & 13 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 \\ 12 & 19 & 73 & 0 & 11 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 21^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 15 & 9 & 7 \\ 20 & 25 & 20 & 12 & 9 \\ 15 & 20 & 13 & 9 & 7 \\ 9 & 12 & 9 & 4 & 4 \\ 7 & 9 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 \\ 13 & 23 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 12 & 22 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{12} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 \\ 16 & 28 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 16 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 399, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 10 & 17 & 49 & 0 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 \\ 14 & 26 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 11 & 18 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

- [31] $1^{11} 1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 12 & 21 & 76 & 0 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [32] $1^{11} 1^{13} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 511, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 \\ 15 & 27 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [33] $1^{11} 1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 \\ 14 & 23 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{11} 1^{13} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 3 & 4 \\ 18 & 32 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^{11} 1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 13 & 22 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^{11} 1^{12} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 654, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 \\ 17 & 28 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 16 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{11} 1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 670, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 3 & 4 \\ 17 & 31 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^{11} 1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 \\ 15 & 26 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{11} 1^{13} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 811, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 \\ 16 & 27 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{14}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 854, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 3 & 4 \\ 20 & 36 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 969, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 3 & 4 \\ 19 & 32 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{14}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1034, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 3 & 4 \\ 18 & 31 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{14}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 3 & 4 \\ 21 & 36 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [44] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 5 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 0 \\ 3 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 1 & 1 & 4 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 0 & 10 & 4 & 6 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & -1 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 \\ 2 & 3 & 1 & 4 & 0 \\ 2 & 2 & 4 & 2 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 2 & 3 \\ 2 & 1 & 1 & 7 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 6 & 9 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 3 & 3 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{11} 1^1 3^1 21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 2 & 5 & 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{11} 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{11} 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 6 & 0 & 3 \\ 4 & 5 & 6 & 9 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 7 & -2 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{11} 3^1 21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 3 \\ 3 & 5 & 10 & 5 & 3 & 5 \\ 5 & 10 & 14 & 7 & 5 & 8 \\ 3 & 5 & 7 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 5 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.865 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{orange}{\mathfrak{so}}_8$

$T = 9 \quad [0] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & 0 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.866 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$

$T = 9 \quad [0] \quad 1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 3 & 4 \\ 4 & 8 & 10 & 0 & 4 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 3 & 4 \\ 12 & 22 & 85 & 0 & 12 & 12 & 17 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 & 2 \\ 2 & 3 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{14}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 3 & 4 \\ 14 & 26 & 118 & 0 & 14 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 & 2 \\ 2 & 3 & 14 & 0 & 2 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.867 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_{2n3\text{mix}}$

$T = 9 \quad [0] 1^{111}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 4 & 3 \\ 3 & 4 & 14 & 0 & 3 & 7 & 4 \\ 8 & 14 & 37 & 0 & 8 & 17 & 11 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 4 & 2 \\ 4 & 7 & 17 & 0 & 4 & 7 & 5 \\ 3 & 4 & 11 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{122}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 9 & 3 \\ 3 & 4 & 8 & 0 & 3 & 16 & 4 \\ 4 & 8 & 10 & 0 & 4 & 22 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 9 & 16 & 22 & 0 & 9 & 44 & 12 \\ 3 & 4 & 6 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{121}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 4 & 3 \\ 3 & 4 & 8 & 0 & 3 & 7 & 4 \\ 4 & 8 & 10 & 0 & 4 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 4 & 2 \\ 4 & 7 & 10 & 0 & 4 & 7 & 5 \\ 3 & 4 & 6 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 3 & 7 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 4 & 7 & 6 & 0 & 4 & 7 & 5 \\ 3 & 4 & 4 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 3 & 7 & 4 \\ 2 & 4 & 1 & 0 & 2 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 4 & 7 & 5 & 0 & 4 & 7 & 5 \\ 3 & 4 & 3 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{112}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 3 & 11 & 4 \\ 2 & 4 & 1 & 0 & 2 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 6 & 2 \\ 6 & 11 & 8 & 0 & 6 & 20 & 8 \\ 3 & 4 & 3 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{123}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 4 & 8 & 0 & 3 & 20 & 4 \\ 4 & 8 & 10 & 0 & 4 & 28 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 11 & 2 \\ 11 & 20 & 28 & 0 & 11 & 69 & 15 \\ 3 & 4 & 6 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 3 & 7 & 4 \\ 5 & 9 & 13 & 0 & 5 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 4 & 7 & 11 & 0 & 4 & 7 & 5 \\ 3 & 4 & 7 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{124} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 16 & 3 \\ 3 & 4 & 8 & 0 & 3 & 29 & 4 \\ 4 & 8 & 10 & 0 & 4 & 40 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 16 & 2 \\ 16 & 29 & 40 & 0 & 16 & 148 & 22 \\ 3 & 4 & 6 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 3 & 12 & 4 \\ 2 & 4 & 1 & 0 & 2 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 12 & 8 & 0 & 7 & 25 & 9 \\ 3 & 4 & 3 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{141} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 4 & 3 \\ 3 & 4 & 21 & 0 & 3 & 7 & 4 \\ 11 & 21 & 76 & 0 & 11 & 26 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 4 & 2 \\ 4 & 7 & 26 & 0 & 4 & 7 & 5 \\ 3 & 4 & 16 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{121} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 4 & 3 \\ 3 & 4 & 18 & 0 & 3 & 7 & 4 \\ 10 & 18 & 58 & 0 & 10 & 22 & 14 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 4 & 2 \\ 4 & 7 & 22 & 0 & 4 & 7 & 5 \\ 3 & 4 & 14 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 6 & 3 \\ 3 & 4 & 13 & 0 & 3 & 11 & 4 \\ 7 & 13 & 28 & 0 & 7 & 25 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 6 & 2 \\ 6 & 11 & 25 & 0 & 6 & 20 & 8 \\ 3 & 4 & 10 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 4 & 3 \\ 3 & 4 & 13 & 0 & 3 & 7 & 4 \\ 7 & 13 & 28 & 0 & 7 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 4 & 2 \\ 4 & 7 & 16 & 0 & 4 & 7 & 5 \\ 3 & 4 & 10 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{131} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 4 & 3 \\ 3 & 4 & 17 & 0 & 3 & 7 & 4 \\ 9 & 17 & 49 & 0 & 9 & 21 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 4 & 2 \\ 4 & 7 & 21 & 0 & 4 & 7 & 5 \\ 3 & 4 & 13 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 13 & 3 \\ 3 & 4 & 8 & 0 & 3 & 24 & 4 \\ 4 & 8 & 10 & 0 & 4 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 13 & 2 \\ 13 & 24 & 34 & 0 & 13 & 100 & 18 \\ 3 & 4 & 6 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{132}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 6 & 3 \\ 3 & 4 & 17 & 0 & 3 & 11 & 4 \\ 9 & 17 & 49 & 0 & 9 & 33 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 6 & 2 \\ 6 & 11 & 33 & 0 & 6 & 20 & 8 \\ 3 & 4 & 13 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 6 & 3 \\ 3 & 4 & 21 & 0 & 3 & 11 & 4 \\ 11 & 21 & 76 & 0 & 11 & 41 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 6 & 2 \\ 6 & 11 & 41 & 0 & 6 & 20 & 8 \\ 3 & 4 & 16 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 3 & 16 & 4 \\ 2 & 4 & 1 & 0 & 2 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 9 & 16 & 11 & 0 & 9 & 44 & 12 \\ 3 & 4 & 3 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{131}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 4 & 3 \\ 3 & 4 & 22 & 0 & 3 & 7 & 4 \\ 12 & 22 & 85 & 0 & 12 & 27 & 17 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 4 & 2 \\ 4 & 7 & 27 & 0 & 4 & 7 & 5 \\ 3 & 4 & 17 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{142}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 6 & 3 \\ 3 & 4 & 26 & 0 & 3 & 11 & 4 \\ 14 & 26 & 118 & 0 & 14 & 50 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 6 & 2 \\ 6 & 11 & 50 & 0 & 6 & 20 & 8 \\ 3 & 4 & 20 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 3 & 21 & 4 \\ 2 & 4 & 1 & 0 & 2 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 12 & 21 & 14 & 0 & 12 & 80 & 16 \\ 3 & 4 & 3 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 3 & 20 & 4 \\ 2 & 4 & 1 & 0 & 2 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 11 & 20 & 14 & 0 & 11 & 69 & 15 \\ 3 & 4 & 3 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 1^{14} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 4 & 3 \\ 3 & 4 & 26 & 0 & 3 & 7 & 4 \\ 14 & 26 & 118 & 0 & 14 & 32 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 4 & 2 \\ 4 & 7 & 32 & 0 & 4 & 7 & 5 \\ 3 & 4 & 20 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 1^{14} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 3 & 24 & 4 \\ 2 & 4 & 1 & 0 & 2 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 13 & 24 & 17 & 0 & 13 & 100 & 18 \\ 3 & 4 & 3 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 1^{13} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 3 & 25 & 4 \\ 2 & 4 & 1 & 0 & 2 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 14 & 25 & 17 & 0 & 14 & 111 & 19 \\ 3 & 4 & 3 & -2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 1^{14} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 16 & 3 \\ 3 & 4 & 4 & 0 & 3 & 29 & 4 \\ 2 & 4 & 1 & 0 & 2 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 16 & 2 \\ 16 & 29 & 20 & 0 & 16 & 148 & 22 \\ 3 & 4 & 3 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

4.868 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 8 \quad [0] \ 1^{11} 1^{13} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 20 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 12 & 17 & 13 & 24 & 7 \\ 20 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 20 & 6 \\ 15 & 20 & 0 & 28 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 28 & 0 & 37 & 11 \\ 6 & 8 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{14} 1^{13} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 16 & 0 & 24 & 4 \\ 9 & 16 & 46 & 0 & 68 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 68 & 0 & 100 & 18 \\ 3 & 4 & 12 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 1^{13} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 0 & 12 & 6 \\ 3 & 4 & 0 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 0 & 44 & 12 \\ 3 & 4 & 0 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^{11} 3^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 14 & 0 & 7 & 4 \\ 8 & 14 & 37 & 0 & 17 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 17 & 0 & 7 & 5 \\ 3 & 4 & 11 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 0 & 0 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 0 & 0 & 7 & 5 \\ 3 & 4 & 0 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 0 & 25 & 9 \\ 3 & 4 & 0 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 20 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 13 & 17 & 13 & 24 & 7 \\ 20 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 4 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{12} 3^{11} 3^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 25 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 25 & 34 & 0 & 111 & 19 \\ 3 & 4 & 6 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{14} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 24 & 4 \\ 11 & 21 & 76 & 0 & 89 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 89 & 0 & 100 & 18 \\ 3 & 4 & 16 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 21 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 12 & 17 & 13 & 24 & 7 \\ 21 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 67, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 23 & 0 & 7 & 4 \\ 13 & 23 & 100 & 0 & 28 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 28 & 0 & 7 & 5 \\ 3 & 4 & 18 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 69, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 14 & 0 & 7 & 4 \\ 9 & 14 & 37 & 0 & 17 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 17 & 0 & 7 & 5 \\ 3 & 4 & 11 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 21^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 76, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 13 & 17 & 15 & 24 & 7 \\ 17 & 20 & 17 & 28 & 8 \\ 15 & 17 & 13 & 24 & 7 \\ 24 & 28 & 24 & 37 & 11 \\ 7 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{133}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 80, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.85 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 12 & 0 & 20 & 4 \\ 7 & 12 & 25 & 0 & 42 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 42 & 0 & 69 & 15 \\ 3 & 4 & 9 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 80, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 3 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{122}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 11 & 4 \\ 4 & 8 & 10 & 0 & 16 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 16 & 0 & 20 & 8 \\ 3 & 4 & 6 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{121}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 10 & 0 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{122}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 96, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 22 & 0 & 44 & 12 \\ 3 & 4 & 6 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{122}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 96, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 5 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 22 & 0 & 44 & 12 \\ 3 & 4 & 6 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 105, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 3 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 109$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 113$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 3 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{124}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 114$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 8 & 0 & 29 & 4 \\ 4 & 8 & 10 & 0 & 40 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 40 & 0 & 148 & 22 \\ 3 & 4 & 6 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 115$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 \\ 5 & 9 & 13 & 0 & 17 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 17 & 0 & 20 & 8 \\ 3 & 4 & 7 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 14 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 3 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 28 & 0 & 69 & 15 \\ 3 & 4 & 6 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 12 & 4 \\ 9 & 17 & 49 & 0 & 36 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 36 & 0 & 25 & 9 \\ 3 & 4 & 13 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 142$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 5 & 9 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 3 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 3 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 7 & 3 \\ 3 & 4 & 21 & 0 & 12 & 4 \\ 12 & 21 & 76 & 0 & 44 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 44 & 0 & 25 & 9 \\ 3 & 4 & 16 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 4 & 3 \\ 3 & 4 & 32 & 0 & 7 & 4 \\ 18 & 32 & 193 & 0 & 39 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 39 & 0 & 7 & 5 \\ 3 & 4 & 25 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 18 & 0 & 69 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 6 & 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 14 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11} 1^{13} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 6 & 9 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{12} 1^{13} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 5 & 8 & 10 & 0 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{123}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 15 & 0 & 44 & 12 \\ 3 & 4 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{111}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{123}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 5 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 28 & 0 & 69 & 15 \\ 3 & 4 & 6 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{14}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 24 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 21 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{132}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 185$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 6 & 3 \\ 3 & 4 & 22 & 0 & 11 & 4 \\ 12 & 22 & 85 & 0 & 42 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 42 & 0 & 20 & 8 \\ 3 & 4 & 17 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{122}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 11 & 4 \\ 8 & 13 & 28 & 0 & 25 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 25 & 0 & 20 & 8 \\ 3 & 4 & 10 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{113}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{122}3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 190$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 11 & 4 \\ 5 & 8 & 10 & 0 & 16 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 16 & 0 & 20 & 8 \\ 3 & 4 & 6 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{134}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 192$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 12 & 0 & 24 & 4 \\ 7 & 12 & 25 & 0 & 51 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 51 & 0 & 100 & 18 \\ 3 & 4 & 9 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{123}3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 194$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 15 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 22 & 0 & 39 & 11 \\ 3 & 4 & 6 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{121}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 4 & 3 \\ 3 & 4 & 23 & 0 & 7 & 4 \\ 14 & 23 & 100 & 0 & 28 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 28 & 0 & 7 & 5 \\ 3 & 4 & 18 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{142}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 9 & 3 \\ 3 & 4 & 21 & 0 & 16 & 4 \\ 11 & 21 & 76 & 0 & 59 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 59 & 0 & 44 & 12 \\ 3 & 4 & 16 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{133}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 20 & 4 \\ 9 & 17 & 49 & 0 & 60 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 60 & 0 & 69 & 15 \\ 3 & 4 & 13 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 7 & 4 \\ 10 & 18 & 58 & 0 & 22 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 22 & 0 & 7 & 5 \\ 3 & 4 & 14 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{112}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 6 & 9 & 13 & 0 & 26 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 26 & 0 & 44 & 12 \\ 3 & 4 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{131}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 7 & 4 \\ 7 & 12 & 25 & 0 & 15 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 15 & 0 & 7 & 5 \\ 3 & 4 & 9 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 28 & 0 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 10 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 11 & 4 \\ 7 & 13 & 28 & 0 & 25 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 25 & 0 & 20 & 8 \\ 3 & 4 & 10 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 28 & 0 & 69 & 15 \\ 3 & 4 & 6 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 5 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 34 & 0 & 100 & 18 \\ 3 & 4 & 6 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 20 & 0 & 25 & 9 \\ 3 & 4 & 7 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{13}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 18 & 0 & 69 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{12}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 15 & 0 & 44 & 12 \\ 3 & 4 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{141}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 16 & 0 & 7 & 4 \\ 9 & 16 & 46 & 0 & 20 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 20 & 0 & 7 & 5 \\ 3 & 4 & 12 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{132}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 12 & 0 & 11 & 4 \\ 7 & 12 & 25 & 0 & 24 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 24 & 0 & 20 & 8 \\ 3 & 4 & 9 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{124}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 16 & 3 \\ 3 & 4 & 8 & 0 & 29 & 4 \\ 4 & 8 & 10 & 0 & 40 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 29 & 40 & 0 & 148 & 22 \\ 3 & 4 & 6 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{121}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 28 & 0 & 28 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 28 & 0 & 25 & 9 \\ 3 & 4 & 10 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 6 & 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 20 & 0 & 25 & 9 \\ 3 & 4 & 7 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{122}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 16 & 4 \\ 7 & 13 & 28 & 0 & 37 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 37 & 0 & 44 & 12 \\ 3 & 4 & 10 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{11} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 20 & 0 & 25 & 9 \\ 3 & 4 & 7 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{11} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 7 & 4 \\ 9 & 17 & 49 & 0 & 21 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 21 & 0 & 7 & 5 \\ 3 & 4 & 13 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{11} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{11} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 6 & 3 \\ 3 & 4 & 17 & 0 & 11 & 4 \\ 9 & 17 & 49 & 0 & 33 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 33 & 0 & 20 & 8 \\ 3 & 4 & 13 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{11} 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 12 & 4 \\ 10 & 17 & 49 & 0 & 36 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 36 & 0 & 25 & 9 \\ 3 & 4 & 13 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{11} 1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 21 & 0 & 7 & 4 \\ 11 & 21 & 76 & 0 & 26 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 26 & 0 & 7 & 5 \\ 3 & 4 & 16 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 34 & 0 & 100 & 18 \\ 3 & 4 & 6 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{11} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 8 & 13 & 28 & 0 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 10 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{11} 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 7 & 4 \\ 15 & 27 & 133 & 0 & 33 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 33 & 0 & 7 & 5 \\ 3 & 4 & 21 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 1^{12} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 7 & 4 \\ 11 & 18 & 58 & 0 & 22 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 22 & 0 & 7 & 5 \\ 3 & 4 & 14 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{11} 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 11 & 0 & 39 & 11 \\ 3 & 4 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{11} 1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 14 & 0 & 80 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 34 & 0 & 100 & 18 \\ 3 & 4 & 6 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 8 & 13 & 28 & 0 & 28 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 28 & 0 & 25 & 9 \\ 3 & 4 & 10 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 5 & 9 & 13 & 0 & 26 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 26 & 0 & 44 & 12 \\ 3 & 4 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 7 & 4 \\ 12 & 22 & 85 & 0 & 27 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 27 & 0 & 7 & 5 \\ 3 & 4 & 17 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 20 & 4 \\ 8 & 13 & 28 & 0 & 46 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 46 & 0 & 69 & 15 \\ 3 & 4 & 10 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 6 & 3 \\ 3 & 4 & 21 & 0 & 11 & 4 \\ 11 & 21 & 76 & 0 & 41 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 41 & 0 & 20 & 8 \\ 3 & 4 & 16 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 6 & 3 \\ 3 & 4 & 17 & 0 & 11 & 4 \\ 10 & 17 & 49 & 0 & 33 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 33 & 0 & 20 & 8 \\ 3 & 4 & 13 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{142}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 6 & 3 \\ 3 & 4 & 26 & 0 & 11 & 4 \\ 14 & 26 & 118 & 0 & 50 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 50 & 0 & 20 & 8 \\ 3 & 4 & 20 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{132}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 16 & 4 \\ 9 & 17 & 49 & 0 & 48 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 48 & 0 & 44 & 12 \\ 3 & 4 & 13 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{131}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 4 & 3 \\ 3 & 4 & 32 & 0 & 7 & 4 \\ 19 & 32 & 193 & 0 & 39 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 39 & 0 & 7 & 5 \\ 3 & 4 & 25 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{112}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 5 & 9 & 13 & 0 & 26 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 26 & 0 & 44 & 12 \\ 3 & 4 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 3 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 20 & 4 \\ 5 & 9 & 13 & 0 & 32 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 32 & 0 & 69 & 15 \\ 3 & 4 & 7 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{114}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 38 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 38 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{112}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 17 & 0 & 128 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 14 & 0 & 80 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{122}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 16 & 4 \\ 7 & 13 & 28 & 0 & 37 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 37 & 0 & 44 & 12 \\ 3 & 4 & 10 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{1^{11}2^2}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 16 & 4 \\ 8 & 13 & 28 & 0 & 37 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 37 & 0 & 44 & 12 \\ 3 & 4 & 10 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{1^{11}4}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 14 & 0 & 64 & 14 \\ 3 & 4 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{1^{11}3}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 7 & 4 \\ 10 & 17 & 49 & 0 & 21 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 21 & 0 & 7 & 5 \\ 3 & 4 & 13 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{1^{11}2}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 482, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 6 & 3 \\ 3 & 4 & 26 & 0 & 11 & 4 \\ 15 & 26 & 118 & 0 & 50 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 50 & 0 & 20 & 8 \\ 3 & 4 & 20 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{1^{11}3}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 20 & 4 \\ 11 & 21 & 76 & 0 & 74 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 74 & 0 & 69 & 15 \\ 3 & 4 & 16 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{1^{11}4}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 502, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 4 & 3 \\ 3 & 4 & 36 & 0 & 7 & 4 \\ 20 & 36 & 238 & 0 & 44 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 44 & 0 & 7 & 5 \\ 3 & 4 & 28 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{1^{11}2}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 505, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 20 & 4 \\ 7 & 13 & 28 & 0 & 46 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 46 & 0 & 69 & 15 \\ 3 & 4 & 10 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{1^{11}3}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 20 & 4 \\ 5 & 9 & 13 & 0 & 32 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 32 & 0 & 69 & 15 \\ 3 & 4 & 7 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{1^{11}4}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 24 & 4 \\ 9 & 17 & 49 & 0 & 72 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 72 & 0 & 100 & 18 \\ 3 & 4 & 13 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{1^{11}3}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 4 & 3 \\ 3 & 4 & 26 & 0 & 7 & 4 \\ 14 & 26 & 118 & 0 & 32 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 32 & 0 & 7 & 5 \\ 3 & 4 & 20 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{1^{11}} 1^{1^{12}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 9 & 3 \\ 3 & 4 & 21 & 0 & 16 & 4 \\ 12 & 21 & 76 & 0 & 59 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 59 & 0 & 44 & 12 \\ 3 & 4 & 16 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{1^{11}} 1^{1^{14}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 24 & 4 \\ 10 & 17 & 49 & 0 & 72 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 72 & 0 & 100 & 18 \\ 3 & 4 & 13 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{1^{11}} 1^{1^{14}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 17 & 0 & 100 & 18 \\ 3 & 4 & 3 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{1^{11}} 1^{1^{12}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 16 & 4 \\ 10 & 17 & 49 & 0 & 48 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 48 & 0 & 44 & 12 \\ 3 & 4 & 13 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{1^{11}} 1^{1^{13}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 557, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 25 & 17 & 0 & 111 & 19 \\ 3 & 4 & 3 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{1^{11}} 1^{1^{13}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 3 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{1^{11}} 1^{1^{13}} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 7 & 4 \\ 16 & 27 & 133 & 0 & 33 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 33 & 0 & 7 & 5 \\ 3 & 4 & 21 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{1^{11}} 1^{1^{12}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 6 & 3 \\ 3 & 4 & 21 & 0 & 11 & 4 \\ 12 & 21 & 76 & 0 & 41 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 41 & 0 & 20 & 8 \\ 3 & 4 & 16 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{1^{11}} 1^{1^{14}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 38 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 38 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{1^{11}} 1^{1^{13}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 20 & 4 \\ 7 & 13 & 28 & 0 & 46 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 46 & 0 & 69 & 15 \\ 3 & 4 & 10 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{141}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 4 & 3 \\ 3 & 4 & 31 & 0 & 7 & 4 \\ 17 & 31 & 172 & 0 & 38 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 38 & 0 & 7 & 5 \\ 3 & 4 & 24 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{131}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 7 & 4 \\ 13 & 22 & 85 & 0 & 27 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 27 & 0 & 7 & 5 \\ 3 & 4 & 17 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 21 & 0 & 7 & 4 \\ 12 & 21 & 76 & 0 & 26 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 26 & 0 & 7 & 5 \\ 3 & 4 & 16 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 605, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 20 & 4 \\ 9 & 17 & 49 & 0 & 60 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 60 & 0 & 69 & 15 \\ 3 & 4 & 13 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{133}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 30 & 20 & 0 & 165 & 23 \\ 3 & 4 & 3 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 20 & 4 \\ 10 & 17 & 49 & 0 & 60 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 60 & 0 & 69 & 15 \\ 3 & 4 & 13 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{133}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 3 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{124}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 24 & 4 \\ 7 & 13 & 28 & 0 & 55 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 55 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{124}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 24 & 4 \\ 7 & 13 & 28 & 0 & 55 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 55 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{133}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 25 & 17 & 0 & 111 & 19 \\ 3 & 4 & 3 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{144}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 13 & 3 \\ 3 & 4 & 21 & 0 & 24 & 4 \\ 11 & 21 & 76 & 0 & 89 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 89 & 0 & 100 & 18 \\ 3 & 4 & 16 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{144}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 20 & 4 \\ 12 & 21 & 76 & 0 & 74 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 74 & 0 & 69 & 15 \\ 3 & 4 & 16 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{144}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 752, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 17 & 0 & 100 & 18 \\ 3 & 4 & 3 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{144}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 3 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{144}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 24 & 4 \\ 9 & 17 & 49 & 0 & 72 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 72 & 0 & 100 & 18 \\ 3 & 4 & 13 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{144}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 30 & 20 & 0 & 165 & 23 \\ 3 & 4 & 3 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{144}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 797, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 21 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 3 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{144}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 808, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 13 & 3 \\ 3 & 4 & 21 & 0 & 24 & 4 \\ 12 & 21 & 76 & 0 & 89 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 89 & 0 & 100 & 18 \\ 3 & 4 & 16 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{141}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 822, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 26 & 0 & 7 & 4 \\ 15 & 26 & 118 & 0 & 32 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 32 & 0 & 7 & 5 \\ 3 & 4 & 20 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{141}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 834, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 4 & 3 \\ 3 & 4 & 36 & 0 & 7 & 4 \\ 21 & 36 & 238 & 0 & 44 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 44 & 0 & 7 & 5 \\ 3 & 4 & 28 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{\overline{114}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 899, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 4 & 3 \\ 3 & 4 & 31 & 0 & 7 & 4 \\ 18 & 31 & 172 & 0 & 38 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 38 & 0 & 7 & 5 \\ 3 & 4 & 24 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{\overline{114}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 3 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{\overline{114}} 3^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 3 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{\overline{114}} 3^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 997, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 1 & 0 & 26 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 3 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{\overline{114}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 3 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{\overline{114}} 3^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 23 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 1 & 0 & 26 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 3 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [150] $1^{\overline{113}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 0 & 10 & 20 & 6 \\ 8 & 9 & 0 & 20 & 35 & 11 \\ 3 & 3 & -1 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{\overline{113}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 7 & 3 \\ 3 & 5 & 5 & 5 & 13 & 5 \\ 2 & 5 & 2 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 2 \\ 7 & 13 & 9 & 9 & 26 & 10 \\ 3 & 5 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{\overline{113}} 3^{\overline{122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 5 & 10 & 0 & 13 & 5 \\ 5 & 10 & 14 & 0 & 19 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 19 & 0 & 24 & 10 \\ 3 & 5 & 8 & -2 & 10 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 3 & 0 & 5 & 2 \\ 1 & 1 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[154] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 5 & 5 & 0 & 8 & 5 \\ 2 & 5 & 2 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 8 & 6 \\ 3 & 5 & 4 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{11}3^121^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 5 & 5 & 0 & 13 & 5 \\ 2 & 5 & 2 & 0 & 10 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 10 & 0 & 24 & 10 \\ 3 & 5 & 4 & -2 & 10 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 8 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 29 & 9 \\ 8 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[157] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 9 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 29 & 9 \\ 9 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 6 & 0 & 4 & 4 \\ 2 & 2 & 2 & 5 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{12}3^121^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{112}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 10 & 3 \\ 2 & 1 & 1 & 7 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 36 & 9 \\ 10 & 12 & 0 & 36 & 56 & 14 \\ 3 & 3 & -1 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{111}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 4 & 0 & 10 & 6 \\ 2 & 2 & 2 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 5 & 10 & 0 & 8 & 5 \\ 5 & 10 & 14 & 0 & 12 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 8 & 6 \\ 3 & 5 & 8 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

[163] $1^{1^1} 1^{12} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 3 & 0 & 12 & 2 \\ 1 & 1 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 12 & 0 & 56 & 14 \\ 2 & 2 & 2 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[164] $1^{1^{12}} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 16 & 3 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 52 & 9 \\ 16 & 16 & 0 & 52 & 116 & 20 \\ 3 & 3 & -1 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{1^1} 1^{13} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 13 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{1^{11}} 1^{13} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 23 & 5 \\ 2 & 5 & 2 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 23 & 17 & 0 & 78 & 18 \\ 3 & 5 & 4 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{1^{11}} 1^{13} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 16 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 16 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{1^{12}} 1^{12} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 7 & 0 & 35 & 11 \\ 2 & 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[169] $1^{1^{11}} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 15 & 3 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 52 & 9 \\ 15 & 16 & 0 & 52 & 116 & 20 \\ 3 & 3 & -1 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{1^{11}} 1^{13} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 5 & 5 & 0 & 28 & 5 \\ 2 & 5 & 2 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 28 & 21 & 0 & 116 & 22 \\ 3 & 5 & 4 & -2 & 22 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{\overline{111}1}3^{\overline{111}}21^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 5 & 2 \\ 2 & 3 & 4 & 5 & 0 & 9 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 9 & 0 & 10 & 6 \\ 2 & 2 & 2 & 5 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{\overline{112}}3^{\overline{112}}21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 18 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{\overline{114}}3^{\overline{112}}21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 1 & 7 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 66 & 9 \\ 18 & 22 & 0 & 66 & 192 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{\overline{112}}3^{\overline{113}}21^{\overline{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 20 & 3 \\ 2 & 1 & 1 & 7 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 68 & 9 \\ 20 & 20 & 0 & 68 & 200 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{\overline{111}}3^{\overline{112}}21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 3 & 4 & 5 & 0 & 13 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 8 & 13 & 0 & 23 & 9 \\ 2 & 2 & 2 & 5 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{\overline{113}}3^{\overline{112}}21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 17 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 17 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{\overline{13}}3^{\overline{114}}21^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 16 & 2 \\ 1 & 1 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 16 & 16 & 0 & 103 & 19 \\ 2 & 2 & 2 & 3 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{\overline{14}}3^{\overline{114}}21^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 13 & 2 \\ 1 & 2 & 0 & 3 & 0 & 17 & 2 \\ 1 & 1 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 17 & 17 & 0 & 127 & 21 \\ 2 & 2 & 2 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{\overline{114}}3^{\overline{112}}21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$\begin{aligned}
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 19 & 3 \\ 2 & 1 & 1 & 7 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 66 & 9 \\ 19 & 22 & 0 & 66 & 192 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0} \\
& [180] \ 1^{\overline{113}3^121^{\overline{113}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 21 & 3 \\ 2 & 1 & 1 & 7 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 75 & 9 \\ 21 & 23 & 0 & 75 & 243 & 29 \\ 3 & 3 & -1 & 9 & 29 & 3 \end{pmatrix}_{b_0} \\
& [181] \ 1^{\overline{113}3^121^{\overline{13}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 13 & 10 & 0 & 74 & 16 \\ 2 & 2 & 2 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [182] \ 1^{\overline{123}3^121^{\overline{114}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 33 & 5 \\ 2 & 5 & 2 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 33 & 24 & 0 & 164 & 26 \\ 3 & 5 & 4 & -2 & 26 & 3 \end{pmatrix}_{b_0} \\
& [183] \ 1^{\overline{113}3^121^{\overline{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 10 & 9 & 0 & 51 & 13 \\ 2 & 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0} \\
& [184] \ 1^{\overline{113}3^121^{\overline{114}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 26 & 3 \\ 2 & 1 & 1 & 7 & 27 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 91 & 9 \\ 26 & 27 & 0 & 91 & 359 & 35 \\ 3 & 3 & -1 & 9 & 35 & 3 \end{pmatrix}_{b_0} \\
& [185] \ 1^{\overline{113}3^121^{\overline{113}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 22 & 3 \\ 2 & 1 & 1 & 7 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 75 & 9 \\ 22 & 23 & 0 & 75 & 243 & 29 \\ 3 & 3 & -1 & 9 & 29 & 3 \end{pmatrix}_{b_0} \\
& [186] \ 1^{\overline{114}3^121^{\overline{113}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 23 & 3 \\ 2 & 1 & 1 & 7 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 82 & 9 \\ 23 & 26 & 0 & 82 & 292 & 32 \\ 3 & 3 & -1 & 9 & 32 & 3 \end{pmatrix}_{b_0} \\
& [187] \ 1^{\overline{112}3^121^{\overline{112}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 9 & 2 \\ 2 & 3 & 4 & 5 & 0 & 16 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 16 & 0 & 35 & 11 \\ 2 & 2 & 2 & 5 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{114}3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 1 & 0 & 17 & 2 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 17 & 13 & 0 & 127 & 21 \\ 2 & 2 & 2 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{112}3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 24 & 3 \\ 2 & 1 & 1 & 7 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 82 & 9 \\ 24 & 26 & 0 & 82 & 292 & 32 \\ 3 & 3 & -1 & 9 & 32 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{112}3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 12 & 2 \\ 2 & 3 & 4 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 20 & 0 & 56 & 14 \\ 2 & 2 & 2 & 5 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{113}3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 9 & 2 \\ 1 & 2 & 0 & 4 & 0 & 10 & 2 \\ 2 & 3 & 4 & 5 & 0 & 19 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 10 & 19 & 0 & 51 & 13 \\ 2 & 2 & 2 & 5 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{114}3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 14 & 12 & 0 & 96 & 18 \\ 2 & 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{112}3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 28 & 3 \\ 2 & 1 & 1 & 7 & 30 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 98 & 9 \\ 28 & 30 & 0 & 98 & 416 & 38 \\ 3 & 3 & -1 & 9 & 38 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{113}3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -717, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 13 & 2 \\ 2 & 3 & 4 & 5 & 0 & 23 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 13 & 23 & 0 & 74 & 16 \\ 2 & 2 & 2 & 5 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{113}3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 0 & 2 & 3 & 0 & 11 & 2 \\ 1 & 2 & 0 & 4 & 0 & 16 & 2 \\ 2 & 3 & 4 & 5 & 0 & 27 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 16 & 27 & 0 & 103 & 19 \\ 2 & 2 & 2 & 5 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{11} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -884$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 0 & 2 & 3 & 0 & 12 & 2 \\ 1 & 2 & 0 & 4 & 0 & 14 & 2 \\ 2 & 3 & 4 & 5 & 0 & 26 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 14 & 26 & 0 & 96 & 18 \\ 2 & 2 & 2 & 5 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{11} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -1125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 9 & 2 \\ 1 & 0 & 2 & 3 & 0 & 13 & 2 \\ 1 & 2 & 0 & 4 & 0 & 17 & 2 \\ 2 & 3 & 4 & 5 & 0 & 30 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 17 & 30 & 0 & 127 & 21 \\ 2 & 2 & 2 & 5 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

4.869 NHC(-2)(-3) + $\textcolor{red}{su}_2$ + $\textcolor{orange}{so}_8$ + $\textcolor{red}{su}_2$ + $\textcolor{blue}{su}_2$ + $\textcolor{orange}{so}_8$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & 0 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

4.870 NHC(-2)(-3) + $\textcolor{red}{su}_2$ + $\textcolor{orange}{so}_8$ + $\textcolor{red}{su}_2$ + $\textcolor{blue}{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 2 & 3 \\ 3 & 4 & 14 & 0 & 7 & 3 & 4 \\ 8 & 14 & 37 & 0 & 17 & 8 & 11 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 17 & 0 & 7 & 4 & 5 \\ 2 & 3 & 8 & 0 & 4 & 0 & 2 \\ 3 & 4 & 11 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$

$T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 2 & 3 \\ 3 & 4 & 8 & 0 & 16 & 3 & 4 \\ 4 & 8 & 10 & 0 & 22 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 22 & 0 & 44 & 9 & 12 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 6 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -86$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$

$T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 2 & 3 \\ 3 & 4 & 8 & 0 & 7 & 3 & 4 \\ 4 & 8 & 10 & 0 & 10 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 4 & 5 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 3 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 6 & 0 & 7 & 4 & 5 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 7 & 3 & 4 \\ 2 & 4 & 1 & 0 & 5 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 4 & 5 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 2 & 3 \\ 3 & 4 & 4 & 0 & 11 & 3 & 4 \\ 2 & 4 & 1 & 0 & 8 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 6 & 8 \\ 2 & 3 & 2 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 2 & 3 \\ 3 & 4 & 8 & 0 & 20 & 3 & 4 \\ 4 & 8 & 10 & 0 & 28 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 28 & 0 & 69 & 11 & 15 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & 6 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 7 & 3 & 4 \\ 5 & 9 & 13 & 0 & 11 & 5 & 7 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 4 & 5 \\ 2 & 3 & 5 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 16 & 2 & 3 \\ 3 & 4 & 8 & 0 & 29 & 3 & 4 \\ 4 & 8 & 10 & 0 & 40 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 40 & 0 & 148 & 16 & 22 \\ 2 & 3 & 4 & 0 & 16 & 0 & 2 \\ 3 & 4 & 6 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 3 & 4 \\ 2 & 4 & 1 & 0 & 8 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 8 & 0 & 25 & 7 & 9 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 3 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 2 & 3 \\ 3 & 4 & 21 & 0 & 7 & 3 & 4 \\ 11 & 21 & 76 & 0 & 26 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 26 & 0 & 7 & 4 & 5 \\ 2 & 3 & 11 & 0 & 4 & 0 & 2 \\ 3 & 4 & 16 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 2 & 3 \\ 3 & 4 & 18 & 0 & 7 & 3 & 4 \\ 10 & 18 & 58 & 0 & 22 & 10 & 14 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 22 & 0 & 7 & 4 & 5 \\ 2 & 3 & 10 & 0 & 4 & 0 & 2 \\ 3 & 4 & 14 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[12] \quad & 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 2 & 3 \\ 3 & 4 & 13 & 0 & 11 & 3 & 4 \\ 7 & 13 & 28 & 0 & 25 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 25 & 0 & 20 & 6 & 8 \\ 2 & 3 & 7 & 0 & 6 & 0 & 2 \\ 3 & 4 & 10 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 2 & 3 \\ 3 & 4 & 13 & 0 & 7 & 3 & 4 \\ 7 & 13 & 28 & 0 & 16 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 4 & 5 \\ 2 & 3 & 7 & 0 & 4 & 0 & 2 \\ 3 & 4 & 10 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 2 & 3 \\ 3 & 4 & 17 & 0 & 7 & 3 & 4 \\ 9 & 17 & 49 & 0 & 21 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 21 & 0 & 7 & 4 & 5 \\ 2 & 3 & 9 & 0 & 4 & 0 & 2 \\ 3 & 4 & 13 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 2 & 3 \\ 3 & 4 & 8 & 0 & 24 & 3 & 4 \\ 4 & 8 & 10 & 0 & 34 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 34 & 0 & 100 & 13 & 18 \\ 2 & 3 & 4 & 0 & 13 & 0 & 2 \\ 3 & 4 & 6 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{132}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 9 & 1 & 6 & 2 & 3 \\ 3 & 4 & 17 & 0 & 11 & 3 & 4 \\ 9 & 17 & 49 & 0 & 33 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 33 & 0 & 20 & 6 & 8 \\ 2 & 3 & 9 & 0 & 6 & 0 & 2 \\ 3 & 4 & 13 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 11 & 1 & 6 & 2 & 3 \\ 3 & 4 & 21 & 0 & 11 & 3 & 4 \\ 11 & 21 & 76 & 0 & 41 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 41 & 0 & 20 & 6 & 8 \\ 2 & 3 & 11 & 0 & 6 & 0 & 2 \\ 3 & 4 & 16 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 16 & 3 & 4 \\ 2 & 4 & 1 & 0 & 11 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 11 & 0 & 44 & 9 & 12 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{131}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 2 & 3 \\ 3 & 4 & 22 & 0 & 7 & 3 & 4 \\ 12 & 22 & 85 & 0 & 27 & 12 & 17 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 27 & 0 & 7 & 4 & 5 \\ 2 & 3 & 12 & 0 & 4 & 0 & 2 \\ 3 & 4 & 17 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[20] \quad 1^{1423^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 6 & 2 & 3 \\ 3 & 4 & 26 & 0 & 11 & 3 & 4 \\ 14 & 26 & 118 & 0 & 50 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 50 & 0 & 20 & 6 & 8 \\ 2 & 3 & 14 & 0 & 6 & 0 & 2 \\ 3 & 4 & 20 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1123^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 3 & 4 \\ 2 & 4 & 1 & 0 & 14 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 14 & 0 & 80 & 12 & 16 \\ 2 & 3 & 2 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1133^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 20 & 3 & 4 \\ 2 & 4 & 1 & 0 & 14 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 14 & 0 & 69 & 11 & 15 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 3 & 4 & 3 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1413^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 4 & 2 & 3 \\ 3 & 4 & 26 & 0 & 7 & 3 & 4 \\ 14 & 26 & 118 & 0 & 32 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 32 & 0 & 7 & 4 & 5 \\ 2 & 3 & 14 & 0 & 4 & 0 & 2 \\ 3 & 4 & 20 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1143^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 24 & 3 & 4 \\ 2 & 4 & 1 & 0 & 17 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 17 & 0 & 100 & 13 & 18 \\ 2 & 3 & 2 & 0 & 13 & 0 & 2 \\ 3 & 4 & 3 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1133^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 25 & 3 & 4 \\ 2 & 4 & 1 & 0 & 17 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 14 & 25 & 17 & 0 & 111 & 14 & 19 \\ 2 & 3 & 2 & 0 & 14 & 0 & 2 \\ 3 & 4 & 3 & -2 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1143^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 2 & 3 \\ 3 & 4 & 4 & 0 & 29 & 3 & 4 \\ 2 & 4 & 1 & 0 & 20 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 20 & 0 & 148 & 16 & 22 \\ 2 & 3 & 2 & 0 & 16 & 0 & 2 \\ 3 & 4 & 3 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

4.871 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_{2n3\text{mix}} + \textcolor{blue}{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{1123^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 7 & 3 \\ 3 & 4 & 4 & 0 & 11 & 12 & 4 \\ 2 & 4 & 1 & 0 & 8 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 23 & 8 \\ 7 & 12 & 8 & 0 & 23 & 25 & 9 \\ 3 & 4 & 3 & -2 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1211}1^1}3^{\overline{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 7 & 4 \\ 4 & 8 & 10 & 0 & 10 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 9 & 5 \\ 4 & 7 & 10 & 0 & 9 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}1^1}3^{\overline{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 7 & 4 \\ 5 & 9 & 13 & 0 & 11 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 9 & 5 \\ 4 & 7 & 11 & 0 & 9 & 7 & 5 \\ 3 & 4 & 7 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}1^1}3^{\overline{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -303, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 7 & 3 \\ 3 & 4 & 4 & 0 & 7 & 12 & 4 \\ 2 & 4 & 1 & 0 & 5 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 15 & 5 \\ 7 & 12 & 8 & 0 & 15 & 25 & 9 \\ 3 & 4 & 3 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}1^1}3^{\overline{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -308, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 7 & 4 \\ 2 & 4 & 1 & 0 & 5 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 9 & 5 \\ 4 & 7 & 5 & 0 & 9 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1112}1^1}3^{\overline{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -438, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 12 & 3 \\ 3 & 4 & 4 & 0 & 7 & 21 & 4 \\ 2 & 4 & 1 & 0 & 5 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 26 & 5 \\ 12 & 21 & 14 & 0 & 26 & 80 & 16 \\ 3 & 4 & 3 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1211}1^1}3^{\overline{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -448, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 7 & 4 \\ 7 & 13 & 28 & 0 & 16 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 9 & 5 \\ 4 & 7 & 16 & 0 & 9 & 7 & 5 \\ 3 & 4 & 10 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1112}1^1}3^{\overline{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -589, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 9 & 3 \\ 3 & 4 & 4 & 0 & 7 & 16 & 4 \\ 2 & 4 & 1 & 0 & 5 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 20 & 5 \\ 9 & 16 & 11 & 0 & 20 & 44 & 12 \\ 3 & 4 & 3 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

4.872 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{141} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 16 & 0 & 9 & 4 \\ 9 & 16 & 46 & 0 & 24 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 24 & 0 & 12 & 6 \\ 3 & 4 & 12 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{132} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 3 & 4 & 9 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 5 \\ 12 & 25 & 20 & 16 & 9 \\ 9 & 20 & 13 & 12 & 7 \\ 8 & 16 & 12 & 9 & 5 \\ 5 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 4 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 17 & 24 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3 \ 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & 0 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & 0 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 11 & 13 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 13 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 1 & 3 \\ 3 & 4 & 19 & 0 & 0 & 4 \\ 11 & 19 & 73 & 0 & 0 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 5 & 0 & 0 & 4 \\ 3 & 5 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{113} 3^1 1^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 & 5 \\ 12 & 25 & 124 & 0 & 9 & 9 \\ 59 & 124 & 613 & 0 & 45 & 45 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 45 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 8 & 0 & 5 & 4 \\ 5 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 6 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 & 6 \\ 15 & 20 & 17 & 0 & 8 & 8 \\ 12 & 17 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 6 & 8 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 & 5 \\ 11 & 20 & 16 & 17 & 8 & 8 \\ 10 & 16 & 10 & 12 & 6 & 6 \\ 10 & 17 & 12 & 12 & 6 & 6 \\ 5 & 8 & 6 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 5 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 6 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 15 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 & 5 \\ 11 & 20 & 17 & 17 & 8 & 8 \\ 10 & 17 & 13 & 15 & 7 & 7 \\ 9 & 17 & 15 & 12 & 6 & 6 \\ 5 & 8 & 7 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 1 & 3 \\ 4 & 9 & 36 & 0 & 0 & 5 \\ 14 & 36 & 121 & 0 & 0 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

- [31] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 9 & 12 & 0 & 0 & 5 \\ 5 & 12 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -2 & -1 & 1 \end{pmatrix}_{b_0}$
- [33] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$
- [35] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [37] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [41] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$
- [40] $21^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 13 & 15 & 7 \\ 15 & 17 & 15 & 12 & 6 \\ 7 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 65, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 3 & 4 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 1 & 3 \\ 3 & 4 & 17 & 0 & 0 & 4 \\ 9 & 17 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^1 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 70, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 71, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 1 & 3 \\ 4 & 9 & 20 & 0 & 0 & 5 \\ 8 & 20 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 71, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 1 & 3 \\ 4 & 9 & 28 & 0 & 0 & 5 \\ 11 & 28 & 73 & 0 & 0 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

- [52] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [53] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [54] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [55] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [56] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [57] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 1 & 3 \\ 4 & 9 & 24 & 0 & 0 & 5 \\ 10 & 24 & 58 & 0 & 0 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 14 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [58] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 7 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 11 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [61] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 8 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 5 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 1 & 3 \\ 3 & 4 & 14 & 0 & 0 & 4 \\ 8 & 14 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 9 & 0 & 0 & 4 \\ 5 & 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{13} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 4 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 1 & 3 \\ 3 & 4 & 13 & 0 & 0 & 4 \\ 7 & 13 & 28 & 0 & 0 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 20 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 20 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 11 & 1 & 6 \\ 15 & 20 & 16 & 0 & 8 \\ 11 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 15 & 24 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 7 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 11 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 14 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 97, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.18 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 12 & 0 & 9 & 4 \\ 7 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 18 & 0 & 12 & 6 \\ 3 & 4 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 98, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 48 & 1 & 5 & 8 \\ 11 & 20 & 84 & 0 & 8 & 7 \\ 48 & 84 & 346 & 0 & 34 & 1 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 8 & 34 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 18 & 0 & 75 & 15 \\ 3 & 4 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 7 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 7 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 11 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 & 8 \\ 11 & 20 & 67 & 0 & 8 & 27 \\ 35 & 67 & 217 & 0 & 27 & -1 \\ 1 & 0 & 0 & -3 & -1 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 1 & 6 & 8 \\ 15 & 20 & 33 & 0 & 8 & 23 \\ 23 & 33 & 49 & 0 & 13 & -1 \\ 1 & 0 & 0 & -3 & -1 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 & 8 \\ 11 & 20 & 16 & 0 & 8 & 9 \\ 9 & 16 & 10 & 0 & 6 & 1 \\ 1 & 0 & 0 & -3 & -1 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 15 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3^1 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 1 & 3 \\ 4 & 9 & 36 & 0 & 0 & 5 \\ 15 & 36 & 133 & 0 & 0 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 21 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 3 & 4 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 16 & 24 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 6 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 1 & 3 \\ 4 & 9 & 32 & 0 & 0 & 5 \\ 13 & 32 & 100 & 0 & 0 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 26 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 26 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 13 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 13 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 12 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{12} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 1 & 3 \\ 3 & 4 & 28 & 0 & 0 & 4 \\ 16 & 28 & 154 & 0 & 0 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 28 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 2 & 3 \\ 4 & 9 & 24 & 0 & 4 & 5 \\ 10 & 24 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 1 & 3 \\ 3 & 4 & 18 & 0 & 0 & 4 \\ 10 & 18 & 58 & 0 & 0 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{112}31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 & 5 \\ 12 & 25 & 72 & 0 & 9 & 9 \\ 34 & 72 & 202 & 0 & 26 & 26 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 26 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[115] $21^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 & 6 \\ 16 & 20 & 16 & 17 & 8 & 8 \\ 12 & 16 & 10 & 12 & 6 & 6 \\ 12 & 17 & 12 & 12 & 6 & 6 \\ 6 & 8 & 6 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 12 & 6 & 6 \\ 15 & 20 & 33 & 17 & 8 & 8 \\ 23 & 33 & 49 & 27 & 13 & 13 \\ 12 & 17 & 27 & 12 & 6 & 6 \\ 6 & 8 & 13 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{14}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 11 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 & 5 \\ 11 & 20 & 33 & 0 & 8 & 8 \\ 17 & 33 & 49 & 0 & 13 & 13 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 8 & 13 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 5 \\ 10 & 20 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 5 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 1 & 3 \\ 4 & 9 & 40 & 0 & 0 & 5 \\ 16 & 40 & 154 & 0 & 0 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{12} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 1 & 3 \\ 3 & 4 & 23 & 0 & 0 & 4 \\ 13 & 23 & 100 & 0 & 0 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 29 & 1 & 5 & 8 \\ 11 & 20 & 50 & 0 & 8 & 20 \\ 29 & 50 & 118 & 0 & 20 & 2 \\ 1 & 0 & 0 & -3 & -1 & 2 \\ 5 & 8 & 20 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 & 8 \\ 11 & 20 & 50 & 0 & 8 & 20 \\ 26 & 50 & 118 & 0 & 20 & 2 \\ 1 & 0 & 0 & -3 & -1 & 2 \\ 5 & 8 & 20 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 11 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 11 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 5 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{122} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 6 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{122} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{111} 3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 7 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{143} 3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 1 & 3 \\ 3 & 4 & 26 & 0 & 0 & 4 \\ 14 & 26 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{122} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{133} 3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 1 & 3 \\ 3 & 4 & 22 & 0 & 0 & 4 \\ 12 & 22 & 85 & 0 & 0 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{133} 3^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 46 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 21^{111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 15 & 12 & 7 \\ 20 & 25 & 20 & 16 & 9 \\ 15 & 20 & 13 & 12 & 7 \\ 12 & 16 & 12 & 9 & 5 \\ 7 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 185$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 7 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 3 & 4 & 9 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 1 & 3 \\ 4 & 9 & 48 & 0 & 0 & 5 \\ 20 & 48 & 238 & 0 & 0 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 28 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 188$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 5 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 191$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 191$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 12 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[151] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 196$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 196$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 199$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{122} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 201$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

- [154] $1^{114} 3 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 1 & 3 \\ 4 & 9 & 44 & 0 & 0 & 5 \\ 18 & 44 & 193 & 0 & 0 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [155] $1^{113} 3 1^{21} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 5 \\ 16 & 32 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [158] $1^{111} 3 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 45 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 45 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [157] $1^{111} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [156] $1^{13} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [159] $1^{121} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [160] $1^{112} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 6 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [161] $1^{12} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [162] $1^{122} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 8 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [163] $1^{11} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 7 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [164] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 12 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{131} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 10 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{132} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 8 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 6 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{131} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 5 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 13 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{114} 3^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 22 & 1 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 5 \\ 22 & 44 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 5 & 25 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 18 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 25 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 25 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 27 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 6 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 10 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 24 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 38 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 38 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 6 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{113} 3 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 15 & 36 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 21 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[183] $21^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 27 & 15 & 7 \\ 17 & 20 & 33 & 17 & 8 \\ 27 & 33 & 49 & 27 & 13 \\ 15 & 17 & 27 & 12 & 6 \\ 7 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 18 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{113} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 1 & 3 \\ 3 & 4 & 27 & 0 & 0 & 4 \\ 15 & 27 & 133 & 0 & 0 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 47 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 47 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 9 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{141} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 & 3 \\ 3 & 4 & 31 & 0 & 4 & 4 \\ 17 & 31 & 172 & 0 & 21 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 7 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 9 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[195] \ 21^{112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 30 & 12 & 7 \\ 20 & 25 & 40 & 16 & 9 \\ 30 & 40 & 58 & 24 & 14 \\ 12 & 16 & 24 & 9 & 5 \\ 7 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 38 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 57 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[198] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 36 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 36 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[200] $21^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 24 & 12 & 6 \\ 16 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 28 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 28 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 27 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 27 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 1 & 3 \\ 3 & 4 & 32 & 0 & 0 & 4 \\ 18 & 32 & 193 & 0 & 0 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 46 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 46 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 8 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 11 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[207] 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 10 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[208] 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[209] 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 301, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[210] 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 12 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[213] 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 14 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[211] 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[212] 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 & 3 \\ 3 & 4 & 26 & 0 & 4 & 4 \\ 14 & 26 & 118 & 0 & 18 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[214] 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 6 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[216] 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 1 & 3 \\ 3 & 4 & 31 & 0 & 0 & 4 \\ 17 & 31 & 172 & 0 & 0 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[215] 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 37 & 1 & 5 & 3 \\ 11 & 20 & 67 & 0 & 8 & 8 \\ 37 & 67 & 217 & 0 & 27 & 9 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 8 & 27 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{131} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 7 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 48 & 12 \\ 3 & 4 & 7 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 & 3 \\ 12 & 25 & 92 & 0 & 9 & 9 \\ 43 & 92 & 325 & 0 & 33 & 3 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 9 & 33 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 13 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 9 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 8 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{113} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[226] $1^{132} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 8 & 4 \\ 13 & 22 & 85 & 0 & 30 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 30 & 0 & 9 & 5 \\ 3 & 4 & 17 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{132} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 8 & 4 \\ 12 & 22 & 85 & 0 & 30 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 30 & 0 & 9 & 5 \\ 3 & 4 & 17 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 9 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 9 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 12 & 0 & 57 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 45 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 2 & 3 \\ 4 & 9 & 48 & 0 & 4 & 5 \\ 20 & 48 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 5 & 28 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{114} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 56 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 56 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{131} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 9 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{132} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 11 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 10 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 10 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 12 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{114} 3 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 & 5 \\ 12 & 25 & 144 & 0 & 9 & 9 \\ 68 & 144 & 814 & 0 & 52 & 52 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 52 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{114} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 & 5 \\ 12 & 25 & 96 & 0 & 9 & 9 \\ 44 & 96 & 346 & 0 & 34 & 34 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 34 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 1 & 3 \\ 3 & 4 & 36 & 0 & 0 & 4 \\ 20 & 36 & 238 & 0 & 0 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{141} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 11 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 14 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 8 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{141} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 9 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{121} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 16 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 45 & 12 & 7 \\ 20 & 25 & 60 & 16 & 9 \\ 45 & 60 & 133 & 36 & 21 \\ 12 & 16 & 36 & 9 & 5 \\ 7 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 14 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 10 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 37 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

- [257] $1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 36 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$
- [258] $1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 13 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [259] $1^{113} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 15 & 0 & 84 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$
- [260] $1^{114} 3 1^{21} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 20 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [261] $1^{114} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 48 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 48 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [262] $1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [263] $1^{113} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 48 & 12 \\ 3 & 4 & 7 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [264] $1^{121} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 13 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [265] $1^{121} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 8 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [266] $1^{13} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 12 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [267] $1^{122} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{121} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 11 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[269] \ 1^{121} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 9 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{122} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 8 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{114} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 18 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{121} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 498, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 8 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{114} 3 \ 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 63 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 63 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[274] \ 1^{14} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 15 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{114} 3 \ 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{114} 3 \ 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{14} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 13 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 11 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{114} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 5 & 25 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{114} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 14 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{121} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 17 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{141} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 14 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 15 & 0 & 84 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 11 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

[298] $1^{114}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 669$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 17 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 17 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

$$[303] \ 1^{\overline{131}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 13 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{131} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 10 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

$$[300] \ 1^{121} 3 \ 1^{14} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 17 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[304] $1^{141} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 11 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

$$[301] \ 1^{121} 3 \ 1^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{red}{u}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 12 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[302] \ 1^{121} 3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 20 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{121} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 680$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 14 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

[306] $1^{114}3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 717$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 16 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

[307] $1^{132} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 722$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 10 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -2 & 9 & 1 \end{pmatrix}_{b_0}$

$$[308] \ 1^{114} 3^{121} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 724, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{141} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 730$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 12 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$

$$[310] \ 1^{121} 3 \ 1^{13} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 744, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.06 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 16 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[311] \ 1^{114} 3 \ 1^{21} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 756, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 54 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[312] \ 1^{121} 3 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 776, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.06 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 15 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{141} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 790$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 13 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$

[314] $1^{1\overline{3}1}3^{1\overline{1}1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 12 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

[317] $1^{121} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 18 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

[316] $1^{131} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 11 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

[315] $1^{121} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 19 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 19 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$

[318] $1^{131}3^{13} + [\textcolor{red}{5}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{green}{u}_2 + \textcolor{orange}{5}\textcolor{orange}{o}_8 + \textcolor{red}{5}\textcolor{red}{u}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 816, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 4 & 4 \\ 15 & 27 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 12 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 4 & 4 \\ 14 & 26 & 118 & 0 & 18 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 4 & 4 \\ 18 & 32 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 75 & 12 & 7 \\ 20 & 25 & 100 & 16 & 9 \\ 75 & 100 & 373 & 60 & 35 \\ 12 & 16 & 60 & 9 & 5 \\ 7 & 9 & 35 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 13 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 16 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 14 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 13 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 15 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 4 & 4 \\ 21 & 31 & 172 & 0 & 21 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[349] $1^{141} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 4 \\ 22 & 36 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{141} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 4 \\ 23 & 36 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[351] $1^{141} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 4 \\ 24 & 36 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [352] $1^{111} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 21 & 0 & 7 & 4 \\ 13 & 21 & 101 & 0 & 31 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 31 & 0 & 7 & 5 \\ 3 & 4 & 19 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[353] $1^{111} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{111} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 6 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[355] $21^{111} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 6 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[356] $21^{111} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 9 & 7 \\ 8 & 10 & 9 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{111} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 9 & 7 \\ 8 & 10 & 9 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[358] $1^{121} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[359] \ 21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 15 & 9 & 7 \\ 14 & 13 & 14 & 10 & 7 \\ 15 & 14 & 13 & 9 & 7 \\ 9 & 10 & 9 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 7 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 9 & 14 & 13 & 9 & 7 \\ 7 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 3 & 4 \\ 6 & 10 & 21 & 0 & 8 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[362] \ 21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 9 & 7 \\ 14 & 13 & 10 & 10 & 7 \\ 9 & 10 & 5 & 7 & 5 \\ 9 & 10 & 7 & 4 & 4 \\ 7 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 8 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[365] \ 1^4 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 6 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 9 & 14 & 13 & 9 & 7 \\ 6 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[368] \ 1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 20 & 8 & 6 \\ 12 & 13 & 24 & 10 & 7 \\ 20 & 24 & 38 & 16 & 12 \\ 8 & 10 & 16 & 4 & 4 \\ 6 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[367] \ 21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 20 & 8 & 6 \\ 12 & 13 & 24 & 10 & 7 \\ 20 & 24 & 38 & 16 & 12 \\ 8 & 10 & 16 & 4 & 4 \\ 6 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 5 & 8 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[370] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 5 & 8 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[371] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[372] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -94, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 4 & 7 & 8 & 0 & 5 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[373] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -97, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 10 & 0 & 7 & 4 \\ 6 & 10 & 21 & 0 & 15 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 15 & 0 & 7 & 5 \\ 3 & 4 & 9 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[374] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -109, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 7 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 5 & 7 & 5 \\ 7 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[375] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 10 & 17 & 61 & 0 & 13 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[376] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 15 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 15 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[377] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 4 & 7 & 8 & 0 & 5 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[378] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -139, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 11 & 0 & 3 & 4 \\ 6 & 11 & 21 & 0 & 7 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[379] \ 1^{\overline{111}}_1 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 15 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 15 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[380] \ 1^{\overline{111}}_{1^{31}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 11 & 18 & 70 & 0 & 14 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[381] \ 1^{\overline{111}}_{1^{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 7 & 0 & 7 & 4 \\ 4 & 7 & 8 & 0 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[382] \ 1^{\overline{12}}_{1^{21}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 8 & 14 & 38 & 0 & 10 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[383] \ 1^{\overline{13}}_{1^{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 15 & 0 & 3 & 4 \\ 8 & 15 & 40 & 0 & 9 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[384] \ 1^{\overline{12}}_{1^1} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 21 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 21 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[385] \ 1^{\overline{14}}_{1^{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 \\ 10 & 19 & 65 & 0 & 11 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{\overline{111}}_{1^{31}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 7 & 4 \\ 11 & 18 & 70 & 0 & 26 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 26 & 0 & 7 & 5 \\ 3 & 4 & 16 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{\overline{13}}_{1^1} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 27 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[389] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 3 & 4 \\ 7 & 12 & 26 & 0 & 8 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[388] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 33 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 33 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[390] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 15 & 0 & 3 & 4 \\ 9 & 15 & 45 & 0 & 11 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 7 & 12 & 26 & 0 & 8 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 8 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 3 & 4 \\ 14 & 24 & 121 & 0 & 18 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 7 & 4 \\ 7 & 12 & 26 & 0 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 10 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[394] $21^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 24 & 9 & 7 \\ 14 & 13 & 24 & 10 & 7 \\ 24 & 24 & 38 & 16 & 12 \\ 9 & 10 & 16 & 4 & 4 \\ 7 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[395] $1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 10 & 18 & 61 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 15 & 0 & 7 & 4 \\ 9 & 15 & 45 & 0 & 21 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 21 & 0 & 7 & 5 \\ 3 & 4 & 13 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[397] $1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 12 & 21 & 88 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[398] \ 21^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 7 & 5 \\ 10 & 13 & 10 & 10 & 7 \\ 7 & 10 & 5 & 7 & 5 \\ 7 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[399] \ 1^{12} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 3 & 4 \\ 9 & 16 & 45 & 0 & 10 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[400] \ 1^{12} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 25 & 0 & 3 & 4 \\ 15 & 25 & 132 & 0 & 19 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 19 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[401] \ 1^{14} 4^{11} 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 12 & 22 & 90 & 0 & 14 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[402] \ 21^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 16 & 7 & 5 \\ 10 & 13 & 24 & 10 & 7 \\ 16 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[403] \ 1^{12} 4^{11} 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 \\ 11 & 19 & 68 & 0 & 13 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[404] \ 1^{12} 4^{11} 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 13 & 22 & 97 & 0 & 16 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 16 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[405] \ 1^{13} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 20 & 0 & 3 & 4 \\ 11 & 20 & 70 & 0 & 12 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[406] \ 1^{14} 4^{11} 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 25 & 0 & 3 & 4 \\ 14 & 25 & 121 & 0 & 17 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[407] \ 1^{14} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -454, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 \\ 16 & 28 & 158 & 0 & 20 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[408] \ 1^{14} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 3 & 4 \\ 13 & 24 & 101 & 0 & 14 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[409] \ 1^{13} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -579, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 \\ 13 & 23 & 97 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{13} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -651, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 29 & 0 & 3 & 4 \\ 17 & 29 & 169 & 0 & 21 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 21 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{13} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -654, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 \\ 15 & 26 & 130 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{14} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -734, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 \\ 15 & 27 & 132 & 0 & 17 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{14} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -881, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 30 & 0 & 3 & 4 \\ 17 & 30 & 169 & 0 & 20 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{14} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -950, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 2 & 3 \\ 3 & 4 & 33 & 0 & 3 & 4 \\ 19 & 33 & 212 & 0 & 23 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 23 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [415] \ 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 2 & 1 & 5 \\ 4 & 1 & 1 & 1 & 3 \\ 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 5 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

4.873 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.874 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.875 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 1 & 5 \\ 12 & 25 & 76 & 0 & 0 & 9 \\ 35 & 76 & 217 & 0 & 0 & 27 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & 0 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 1 & 5 \\ 12 & 25 & 36 & 0 & 0 & 9 \\ 17 & 36 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 1 & 5 \\ 11 & 20 & 16 & 0 & 0 & 8 \\ 8 & 16 & 10 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 1 & 5 \\ 12 & 25 & 56 & 0 & 0 & 9 \\ 26 & 56 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 1 & 5 \\ 12 & 25 & 72 & 0 & 0 & 9 \\ 34 & 72 & 202 & 0 & 0 & 26 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 10 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 9 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 1 & 5 \\ 11 & 20 & 50 & 0 & 0 & 8 \\ 26 & 50 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 1 & 5 \\ 11 & 20 & 33 & 0 & 0 & 8 \\ 17 & 33 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 1 & 5 \\ 12 & 25 & 108 & 0 & 0 & 9 \\ 51 & 108 & 457 & 0 & 0 & 39 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 1 & 5 \\ 12 & 25 & 92 & 0 & 0 & 9 \\ 43 & 92 & 325 & 0 & 0 & 33 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 1 & 5 \\ 12 & 25 & 112 & 0 & 0 & 9 \\ 52 & 112 & 478 & 0 & 0 & 40 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 1 & 5 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 68 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 1 & 5 \\ 12 & 25 & 128 & 0 & 0 & 9 \\ 60 & 128 & 634 & 0 & 0 & 46 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 19 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 & 1 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 18 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 6 & 1 & 5 \\ 9 & 13 & 14 & 10 & 0 & 7 \\ 9 & 14 & 13 & 9 & 0 & 7 \\ 6 & 10 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$4.876 \quad \text{NHC}(-2)(-3) + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{su}_2$$

$$T = 9 \quad [0] \quad 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 4 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.877 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{13} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0} \\
[7] & 1^{131} 1^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 525, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.20 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0} \\
T = 9 & [8] 1^{121} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -229, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[9] & 1^{112} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 16 & 1 & 5 \\ 10 & 13 & 24 & 0 & 7 \\ 16 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[10] & 1^{121} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 14 & 1 & 5 \\ 10 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[11] & 1^{121} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -589, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 21 & 1 & 5 \\ 10 & 13 & 30 & 0 & 7 \\ 21 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[12] & 1^{112} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 23 & 1 & 5 \\ 10 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0} \\
[13] & 1^{112} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 28 & 1 & 5 \\ 10 & 13 & 40 & 0 & 7 \\ 28 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0} \\
[14] & 1^{112} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 30 & 1 & 5 \\ 10 & 13 & 44 & 0 & 7 \\ 30 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[15] & 1^{112} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 37 & 1 & 5 \\ 10 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

4.878 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2$

- $T = 7$ [1] $1^{13} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$
- [0] $1^{12} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 4 & 6 & 4 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [2] $1^{112} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 5 & 9 & 13 & 39 & 7 \\ 17 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$
- [3] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 4 \end{pmatrix}_{b_0}$
- [5] $1^{123} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 16 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$
- [4] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 4 & 3 & 4 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [6] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [7] $21^{131} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 4 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [10] $1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.67 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.67 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -13, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.46 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -13, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.46 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{124} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -13, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.77 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 21 & 3 \\ 3 & 4 & 8 & 31 & 4 \\ 5 & 8 & 10 & 42 & 6 \\ 21 & 31 & 42 & 172 & 24 \\ 3 & 4 & 6 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{122} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -14, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 5 & 5 & 4 & 15 & 4 \\ 10 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{11} 3^{122} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.33 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{112} 3^{121} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{112} 3^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{112} 3^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 5 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{121} 3^{132} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{133} 3^{123} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 18 & 3 \\ 3 & 4 & 17 & 27 & 4 \\ 11 & 17 & 49 & 81 & 13 \\ 18 & 27 & 81 & 133 & 21 \\ 3 & 4 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{112} 3^{111} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 13 & 12 & 7 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{133} 3^{132} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 11 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 5 & 5 & 4 & 12 & 4 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{122} 3^1 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{111} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{121} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[36] $1^1 1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{113} 3^1 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 10 & 14 & 37 & 57 & 11 \\ 14 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[41] $1^{111} 3^1 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{122} 3^1 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 8 & 13 & 28 & 54 & 10 \\ 16 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[43] $1^1 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{111} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{113} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 8 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{12} 3 1^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 5 & 28 & 4 \\ 5 & 5 & 4 & 30 & 4 \\ 20 & 28 & 30 & 154 & 22 \\ 3 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{14} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 14 & 21 & 18 & 76 & 16 \\ 3 & 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{133} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 19 & 3 \\ 3 & 4 & 17 & 27 & 4 \\ 12 & 17 & 49 & 81 & 13 \\ 19 & 27 & 81 & 133 & 21 \\ 3 & 4 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{12} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 8 & 0 & 10 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 7 & 8 & 10 & 24 & 6 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 8 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 9 & 0 & 13 & 7 \\ 3 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[61] $21^{113} 31^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 14 & 22 & 4 \\ 9 & 14 & 37 & 57 & 11 \\ 15 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{123} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{123} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{112} 3^1 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 11 & 14 & 37 & 48 & 11 \\ 12 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{122} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{11} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 12 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{12} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 5 & 5 & 4 & 18 & 4 \\ 14 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{111} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 8 & 9 & 13 & 24 & 7 \\ 10 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{121} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{14} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 13 & 21 & 18 & 76 & 16 \\ 3 & 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 5 & 9 & 13 & 7 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{12} 3 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 5 & 5 & 4 & 24 & 4 \\ 17 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{11} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 7 & 9 & 13 & 7 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{12} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 10 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{123} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 11 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{122} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 11 & 13 & 28 & 42 & 10 \\ 13 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.62 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{113} 3 1^4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[84] $21^{111} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 17 & 4 \\ 9 & 9 & 13 & 27 & 7 \\ 15 & 17 & 27 & 49 & 13 \\ 4 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 12 & 14 & 37 & 39 & 11 \\ 10 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{133} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 5 & 9 & 13 & 36 & 7 \\ 16 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{11} 1^3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{132} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 8 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 12 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 9 & 14 & 37 & 39 & 11 \\ 11 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{124} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 8 & 31 & 4 \\ 5 & 8 & 10 & 42 & 6 \\ 20 & 31 & 42 & 172 & 24 \\ 3 & 4 & 6 & 24 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.56 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 9 & 0 & 13 & 7 \\ 3 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{112} 3^1 1^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 10 & 14 & 37 & 33 & 11 \\ 10 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[100] $21^{112} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 21 & 4 \\ 5 & 4 & 5 & 22 & 4 \\ 6 & 5 & 4 & 21 & 4 \\ 21 & 22 & 21 & 85 & 17 \\ 4 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{11} 31^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[102] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[114] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[111] $1^{112} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[110] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{112} 3^1 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{111} 3^1 1^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[106] $1^{111} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{112}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{13}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[112] $1^{11}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[115] $21^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 9 & 9 & 6 & 13 & 7 \\ 4 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[104] $21^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{11}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[116] $1^{112}3^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 11 & 14 & 37 & 33 & 11 \\ 11 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[117] $1^{144}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -49$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 16 & 21 & 4 \\ 11 & 16 & 46 & 60 & 12 \\ 13 & 21 & 60 & 76 & 16 \\ 3 & 4 & 12 & 16 & 3 \end{pmatrix}_{b_0}$$

[118] $1^{13}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -49$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 10 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[121] $1^{112}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[119] $1^{112}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{112} 3^1 1^{23} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -50, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.69 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.77 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 11 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.69 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 5 & 9 & 13 & 24 & 7 \\ 10 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -54, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 13 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -54, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 14 & 0 & 37 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -54, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 14 & 0 & 37 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -54, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[133] $1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[134] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 8 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[132] $1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[135] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[136] $21^{123} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 8 & 22 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[137] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[138] $1^{13} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[139] $1^{112} 3 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 12 & 14 & 37 & 33 & 11 \\ 11 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[140] $1^{133} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 10 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[141] $1^{13} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[142] $1^{112} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[143] $1^{112} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{134} 3 1^{43} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 10 & 12 & 25 & 36 & 9 \\ 11 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 11 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{124} 3 1^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 7 & 8 & 10 & 24 & 6 \\ 11 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{112} 3 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 9 & 14 & 37 & 33 & 11 \\ 8 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 10 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[150] $21^{1131} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 15 & 17 & 12 & 49 & 13 \\ 4 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 7 & 9 & 13 & 24 & 7 \\ 9 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{13} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{112} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 5 & 9 & 13 & 48 & 7 \\ 20 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{111} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 5 & 9 & 13 & 33 & 7 \\ 13 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[157] $21^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 5 & 9 & 4 \\ 6 & 5 & 4 & 9 & 4 \\ 9 & 9 & 9 & 13 & 7 \\ 4 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{122} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{13} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{113} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 11 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{114} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 13 & 33 & 7 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{123} 3^1 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 18 & 17 & 4 \\ 11 & 18 & 58 & 54 & 14 \\ 11 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

[163] $1^{112} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 5 & 9 & 13 & 30 & 7 \\ 13 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[166] $21^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 12 & 13 & 9 & 28 & 10 \\ 4 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{12} 3^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 11 & 14 & 37 & 33 & 11 \\ 10 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{12} 3^1 1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 10 & 14 & 37 & 33 & 11 \\ 9 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{12} 3^1 1^{24} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 11 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 14 & 0 & 37 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 14 & 0 & 37 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{13} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 11 & 14 & 37 & 57 & 11 \\ 16 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{12} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 11 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{11} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 9 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[178] $1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 13 & 0 & 28 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{14} 3^1 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{113} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[184] $21^{122} 31^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 24 & 4 \\ 5 & 4 & 18 & 26 & 4 \\ 18 & 18 & 58 & 84 & 14 \\ 24 & 26 & 84 & 118 & 20 \\ 4 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{122} 3^1 1^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{113} 3 1^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{114} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 13 & 33 & 7 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{112} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 8 & 9 & 13 & 30 & 7 \\ 13 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[191] $21^{111} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{133} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 11 & 17 & 49 & 51 & 13 \\ 13 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 13 & 0 & 28 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{11} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 10 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{112} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 10 & 14 & 37 & 48 & 11 \\ 11 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{123} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{123} 3 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 18 & 17 & 4 \\ 12 & 18 & 58 & 54 & 14 \\ 12 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{122} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 12 & 18 & 58 & 60 & 14 \\ 14 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 7 & 8 & 10 & 30 & 6 \\ 15 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.64 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 13 & 0 & 28 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 14 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[210] $21^{132} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{11} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 12 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{133} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 10 & 12 & 25 & 36 & 9 \\ 13 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{113} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 9 & 14 & 37 & 57 & 11 \\ 13 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{113} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 9 & 14 & 37 & 57 & 11 \\ 15 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[205] $21^{111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3^1 1^{24} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{12} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 14 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{123} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 14 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{123} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 18 & 17 & 4 \\ 13 & 18 & 58 & 54 & 14 \\ 13 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 8 & 9 & 13 & 27 & 7 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 11 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 16 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{122} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 10 & 13 & 28 & 42 & 10 \\ 12 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{113} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 8 & 9 & 13 & 36 & 7 \\ 16 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[224] $21^{123} 31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 3 \\ 4 & 4 & 18 & 17 & 4 \\ 12 & 18 & 58 & 54 & 14 \\ 12 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{11} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 10 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[232] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{123} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 22 & 3 \\ 3 & 4 & 13 & 32 & 4 \\ 8 & 13 & 28 & 75 & 10 \\ 22 & 32 & 75 & 193 & 25 \\ 3 & 4 & 10 & 25 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 4 & 5 & 4 & 21 & 4 \\ 14 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[235] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 6 & 9 & 13 & 24 & 7 \\ 11 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{122} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 8 & 13 & 28 & 42 & 10 \\ 13 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 19 & 0 & 73 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 28 & 39 & 10 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{111} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 8 & 9 & 13 & 33 & 7 \\ 15 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 5 & 9 & 13 & 24 & 7 \\ 9 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{122} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{11} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 15 & 19 & 21 & 73 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{124} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 6 & 8 & 10 & 36 & 6 \\ 17 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{122} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 9 & 13 & 28 & 54 & 10 \\ 17 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{14} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 46 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{11} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 11 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{14} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 17 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{132} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 17 & 18 & 4 \\ 10 & 17 & 49 & 54 & 13 \\ 11 & 18 & 54 & 58 & 14 \\ 3 & 4 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{134} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 22 & 21 & 4 \\ 13 & 22 & 85 & 81 & 17 \\ 13 & 21 & 81 & 76 & 16 \\ 3 & 4 & 17 & 16 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 19 & 0 & 73 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{133} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 13 & 17 & 49 & 66 & 13 \\ 15 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[256] $1^{133} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 14 & 22 & 85 & 87 & 17 \\ 16 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{143} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 46 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{111} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 7 & 9 & 13 & 24 & 7 \\ 12 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{111} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 8 & 9 & 13 & 24 & 7 \\ 11 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[266] $21^{112} 31^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 21 & 4 \\ 5 & 4 & 17 & 22 & 4 \\ 15 & 17 & 49 & 66 & 13 \\ 21 & 22 & 66 & 85 & 17 \\ 4 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{113} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[268] $21^{111} 31^{143} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 15 & 4 \\ 5 & 4 & 21 & 17 & 4 \\ 18 & 21 & 76 & 63 & 16 \\ 15 & 17 & 63 & 49 & 13 \\ 4 & 4 & 16 & 13 & 3 \end{pmatrix}_{b_0}$$

[267] $21^{111} 31^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 14 & 9 & 4 \\ 9 & 14 & 37 & 24 & 11 \\ 6 & 9 & 24 & 13 & 7 \\ 3 & 4 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{122} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 8 & 13 & 28 & 54 & 10 \\ 15 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{13} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 17 & 0 & 49 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[289] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{112} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 8 & 9 & 13 & 39 & 7 \\ 17 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[291] $1^{123} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 11 & 13 & 28 & 51 & 10 \\ 16 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[292] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 7 & 8 & 10 & 24 & 6 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{122} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 17 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 10 & 13 & 28 & 54 & 10 \\ 17 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[298] $1^{132} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 17 & 18 & 4 \\ 11 & 17 & 49 & 54 & 13 \\ 12 & 18 & 54 & 58 & 14 \\ 3 & 4 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{124} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 17 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 13 & 18 & 58 & 84 & 14 \\ 17 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 14 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[300] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[297] $1^{134} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 22 & 21 & 4 \\ 14 & 22 & 85 & 81 & 17 \\ 14 & 21 & 81 & 76 & 16 \\ 3 & 4 & 17 & 16 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 13 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[307] $1^{112} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 9 & 14 & 37 & 48 & 11 \\ 13 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[306] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 8 & 9 & 13 & 24 & 7 \\ 12 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[305] $1^{12} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 12 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[304] $1^{112} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 12 & 14 & 37 & 48 & 11 \\ 14 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[303] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

- [301] $1^{133} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 12 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$
- [309] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.66 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$
- [310] $21^{112} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$
- [311] $21^{111} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 14 & 14 & 4 \\ 9 & 14 & 37 & 39 & 11 \\ 9 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$
- [314] $1^{133} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 12 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$
- [313] $1^{134} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 9 & 12 & 25 & 45 & 9 \\ 14 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$
- [312] $1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 6 & 9 & 13 & 24 & 7 \\ 9 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$
- [315] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.66 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$
- [316] $1^{123} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [317] $1^{13} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 16 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$
- [318] $1^{123} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [319] $1^{133} 3 1^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 11 & 17 & 49 & 66 & 13 \\ 16 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[320] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{112} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[324] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 49 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{143} 3 \ 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 15 & 3 \\ 3 & 4 & 21 & 22 & 4 \\ 15 & 21 & 76 & 81 & 16 \\ 15 & 22 & 81 & 85 & 17 \\ 3 & 4 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{132} 3 \ 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 18 & 4 \\ 12 & 17 & 49 & 54 & 13 \\ 13 & 18 & 54 & 58 & 14 \\ 3 & 4 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[336] \ 1^{124} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 10 & 13 & 28 & 48 & 10 \\ 14 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[335] \ 1^{133} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 11 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[334] \ 1^{123} 3 \ 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 13 & 3 \\ 3 & 4 & 18 & 17 & 4 \\ 14 & 18 & 58 & 54 & 14 \\ 13 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[333] \ 1^{12} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 18 & 0 & 58 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[332] \ 21^{111} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 12 & 4 \\ 5 & 4 & 9 & 13 & 4 \\ 9 & 9 & 13 & 21 & 7 \\ 12 & 13 & 21 & 28 & 10 \\ 4 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[331] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 7 & 9 & 13 & 24 & 7 \\ 10 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[330] $1^{113} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 7 & 9 & 13 & 36 & 7 \\ 14 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[329] $1^{113} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 9 & 14 & 37 & 57 & 11 \\ 14 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[328] $1^{12} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 18 & 0 & 58 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[327] $21^{12} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 18 & 4 \\ 5 & 4 & 5 & 18 & 4 \\ 6 & 5 & 4 & 18 & 4 \\ 18 & 18 & 18 & 58 & 14 \\ 4 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[326] $1^{123} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 10 & 14 & 37 & 57 & 11 \\ 15 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[325] $1^{114} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 13 & 33 & 7 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[338] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{114} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[345] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 49 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[344] $1^{13} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 13 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[343] $1^{113} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[341] $1^{122} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 9 & 13 & 28 & 54 & 10 \\ 16 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[340] $1^{112} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 5 & 9 & 13 & 39 & 7 \\ 16 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{111} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 7 & 9 & 13 & 33 & 7 \\ 14 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[348] $21^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 15 & 4 \\ 5 & 4 & 5 & 14 & 4 \\ 6 & 5 & 4 & 15 & 4 \\ 15 & 14 & 15 & 37 & 11 \\ 4 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[347] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[346] $21^{113} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 30 & 4 \\ 5 & 4 & 5 & 31 & 4 \\ 6 & 5 & 4 & 30 & 4 \\ 30 & 31 & 30 & 172 & 24 \\ 4 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{124} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 17 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 10 & 13 & 28 & 60 & 10 \\ 17 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 6 & 9 & 13 & 24 & 7 \\ 10 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[352] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 10 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{114} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 5 & 9 & 13 & 42 & 7 \\ 18 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[355] $21^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.69 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 9 & 4 \\ 5 & 4 & 9 & 9 & 4 \\ 9 & 9 & 13 & 15 & 7 \\ 9 & 9 & 15 & 13 & 7 \\ 4 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{111} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 7 & 9 & 13 & 33 & 7 \\ 15 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{112} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 7 & 9 & 13 & 39 & 7 \\ 15 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{13} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -145$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 4 & 5 & 4 & 21 & 4 \\ 15 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[358] $1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -145$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{12} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -145$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.73 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 13 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -145$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{124} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -148$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 7 & 8 & 10 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -148$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{124} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{134} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 20 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 10 & 17 & 49 & 93 & 13 \\ 20 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{113} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 4 & 5 & 4 & 21 & 4 \\ 14 & 19 & 21 & 73 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[367] $1^{123} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[373] $1^{111} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 7 & 9 & 13 & 24 & 7 \\ 11 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[369] $1^{123} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{13} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 15 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[371] $1^{133} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 13 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[372] $1^{124} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 8 & 13 & 28 & 48 & 10 \\ 15 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[374] $1^{134} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 13 & 17 & 49 & 63 & 13 \\ 14 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[375] $1^{113} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 16 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 10 & 14 & 37 & 57 & 11 \\ 16 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[377] $1^{123} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 8 & 13 & 28 & 51 & 10 \\ 16 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[376] $21^{111} 31^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 12 & 4 \\ 5 & 4 & 17 & 13 & 4 \\ 15 & 17 & 49 & 39 & 13 \\ 12 & 13 & 39 & 28 & 10 \\ 4 & 4 & 13 & 10 & 3 \end{pmatrix}_{b_0}$$

[378] $1^{12} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 23 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[379] $1^{12} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 14 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[380] $1^{14} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 18 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[386] $1^{123} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 13 & 18 & 58 & 72 & 14 \\ 14 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[385] $1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[384] $1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[383] $1^{14} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 18 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[382] $1^{133} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 12 & 17 & 49 & 66 & 13 \\ 14 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[381] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 14 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[387] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 5 & 9 & 13 & 30 & 7 \\ 12 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[388] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.64 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[390] $1^{13} 3^1 1^{23} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[389] $1^{13} 3^1 1^{43} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[391] $21^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[393] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 9 & 14 & 37 & 39 & 11 \\ 10 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[394] $21^{12} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[392] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 12 & 14 & 37 & 39 & 11 \\ 11 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[398] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[397] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 16 & 21 & 4 \\ 12 & 16 & 46 & 60 & 12 \\ 15 & 21 & 60 & 76 & 16 \\ 3 & 4 & 12 & 16 & 3 \end{pmatrix}_{b_0}$$

[396] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[395] $1^{122} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 14 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[399] $1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[400] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[402] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[401] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[404] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 22 & 0 & 85 & 17 \\ 3 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[405] $1^{14} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 16 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[403] $1^{143} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 21 & 22 & 4 \\ 13 & 21 & 76 & 81 & 16 \\ 14 & 22 & 81 & 85 & 17 \\ 3 & 4 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[406] $1^{144} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 17 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 15 & 21 & 76 & 96 & 16 \\ 17 & 26 & 96 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[407] $1^{14} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 17 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

- [408] $1^{113} 3 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 5 & 9 & 13 & 36 & 7 \\ 15 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$
- [409] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$
- [410] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$
- [411] $1^{113} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 19 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 5 & 9 & 13 & 45 & 7 \\ 19 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$
- [414] $1^{13} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 4 & 5 & 4 & 27 & 4 \\ 17 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$
- [413] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.66 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$
- [412] $1^{12} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 15 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [418] $21^{123} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 21 & 22 & 15 & 85 & 17 \\ 4 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$
- [419] $1^{111} 3 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 10 & 14 & 37 & 39 & 11 \\ 11 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$
- [420] $21^{122} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$
- [417] $21^{111} 3 1^{133} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 15 & 15 & 4 \\ 5 & 4 & 17 & 17 & 4 \\ 15 & 17 & 49 & 51 & 13 \\ 15 & 17 & 51 & 49 & 13 \\ 4 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$
- [416] $1^{13} 3 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 4 & 5 & 4 & 21 & 4 \\ 16 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{113} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[421] \ 1^{114} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[422] \ 1^{114} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 13 & 33 & 7 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{112} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 7 & 9 & 13 & 30 & 7 \\ 12 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[424] \ 1^{12} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 23 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[425] \ 1^{14} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 21 & 0 & 76 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[426] \ 1^{144} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 16 & 21 & 4 \\ 11 & 16 & 46 & 60 & 12 \\ 14 & 21 & 60 & 76 & 16 \\ 3 & 4 & 12 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[433] \ 1^{12} 3 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 17 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[431] \ 1^{124} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 5 & 8 & 10 & 36 & 6 \\ 18 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[435] \ 1^{111} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 9 & 14 & 37 & 39 & 11 \\ 9 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[434] \ 1^{123} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[432] \ 1^{133} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[430] \ 21^{123} 31^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 14 & 18 & 4 \\ 9 & 14 & 37 & 48 & 11 \\ 12 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[429] \ 1^{111} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 12 & 14 & 37 & 39 & 11 \\ 12 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[428] \ 1^{112} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 8 & 9 & 13 & 30 & 7 \\ 14 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[427] \ 1^{133} 3 1^{144} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 10 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[440] \ 1^{112} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 5 & 9 & 13 & 30 & 7 \\ 11 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[439] \ 1^{113} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[438] \ 1^{122} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[437] \ 1^{112} 3 1^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 5 & 9 & 13 & 48 & 7 \\ 19 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[436] \ 1^{123} 3 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 19 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 8 & 13 & 28 & 63 & 10 \\ 19 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[441] \ 1^{113} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[442] $1^{124} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 11 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[443] $21^{111} 31^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 12 & 4 \\ 5 & 4 & 13 & 13 & 4 \\ 12 & 13 & 28 & 30 & 10 \\ 12 & 13 & 30 & 28 & 10 \\ 4 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[444] $1^{134} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 12 & 17 & 49 & 63 & 13 \\ 13 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[445] $1^{143} 3 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 21 & 22 & 4 \\ 14 & 21 & 76 & 81 & 16 \\ 15 & 22 & 81 & 85 & 17 \\ 3 & 4 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[448] $1^{134} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 8 & 12 & 25 & 45 & 9 \\ 13 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

[450] $1^{112} 3 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 10 & 14 & 37 & 48 & 11 \\ 14 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[451] $21^{112} 31^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 14 & 4 \\ 9 & 9 & 13 & 24 & 7 \\ 15 & 14 & 24 & 37 & 11 \\ 4 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[446] $1^{114} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[449] $1^{123} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[447] $1^{112} 3 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 11 & 14 & 37 & 48 & 11 \\ 13 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[453] $1^{112} 3 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 14 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[452] $1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[454] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -193, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 23 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[455] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[456] $1^{124} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 11 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[457] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 15 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[458] $1^{124} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 18 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 6 & 8 & 10 & 36 & 6 \\ 18 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[459] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[460] $1^{134} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 10 & 12 & 25 & 36 & 9 \\ 12 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[461] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 22 & 0 & 85 & 17 \\ 3 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[462] $1^{112} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 9 & 14 & 37 & 48 & 11 \\ 11 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[463] $1^{134} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 12 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[464] $1^{123} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 11 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[465] $1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 16 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[466] $1^{124} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 5 & 8 & 10 & 36 & 6 \\ 16 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[467] $21^{112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.66 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[468] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 13 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[469] $1^{133} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[470] $1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 21 & 0 & 76 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[471] $1^{143} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 16 & 3 \\ 3 & 4 & 21 & 22 & 4 \\ 15 & 21 & 76 & 81 & 16 \\ 16 & 22 & 81 & 85 & 17 \\ 3 & 4 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[472] $1^{122} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 11 & 13 & 28 & 42 & 10 \\ 14 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[473] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[474] $21^{12} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -206, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 24 & 26 & 18 & 118 & 20 \\ 4 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[476] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 17 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[475] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 11 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[479] $21^{113} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[480] $21^{134} 31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 15 & 3 \\ 4 & 4 & 22 & 21 & 4 \\ 15 & 22 & 85 & 81 & 17 \\ 15 & 21 & 81 & 76 & 16 \\ 3 & 4 & 17 & 16 & 3 \end{pmatrix}_{b_0}$$

[481] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[482] $1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[478] $21^{112} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 18 & 4 \\ 5 & 4 & 13 & 18 & 4 \\ 12 & 13 & 28 & 42 & 10 \\ 18 & 18 & 42 & 58 & 14 \\ 4 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[477] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[483] $1^{134} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 8 & 12 & 25 & 45 & 9 \\ 15 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

[484] $1^{114} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[485] $1^{112} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 9 & 14 & 37 & 48 & 11 \\ 12 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[486] $1^{114} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 7 & 9 & 13 & 42 & 7 \\ 17 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[487] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[489] $1^{112} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 10 & 14 & 37 & 48 & 11 \\ 12 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[491] $1^{144} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 14 & 21 & 76 & 96 & 16 \\ 16 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[488] $1^{13} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 22 & 0 & 85 & 17 \\ 3 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[490] $1^{122} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 8 & 13 & 28 & 42 & 10 \\ 12 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[500] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 5 & 9 & 13 & 39 & 7 \\ 15 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[496] $1^{111} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 10 & 14 & 37 & 39 & 11 \\ 10 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[497] $1^{113} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

[498] $1^{124} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 9 & 13 & 28 & 48 & 10 \\ 13 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[499] $1^{123} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 9 & 13 & 28 & 63 & 10 \\ 17 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[495] $1^{124} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 5 & 8 & 10 & 36 & 6 \\ 17 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[501] $1^{134} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 19 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 11 & 17 & 49 & 93 & 13 \\ 19 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[502] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[503] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[492] $1^{12} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 16 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[493] $1^{14} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 4 & 5 & 4 & 30 & 4 \\ 20 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[494] $1^{111} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 11 & 14 & 37 & 39 & 11 \\ 11 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[506] $1^{123} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 21 & 3 \\ 3 & 4 & 13 & 32 & 4 \\ 8 & 13 & 28 & 75 & 10 \\ 21 & 32 & 75 & 193 & 25 \\ 3 & 4 & 10 & 25 & 3 \end{pmatrix}_{b_0}$$

[505] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[504] $1^{122} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 8 & 13 & 28 & 42 & 10 \\ 11 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[508] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 28 & 0 & 154 & 22 \\ 3 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[507] $1^{124} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[510] $1^{124} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 17 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 11 & 18 & 58 & 84 & 14 \\ 17 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[511] $1^{113} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[509] $1^{124} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 16 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 12 & 18 & 58 & 84 & 14 \\ 16 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[512] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[513] $1^{134} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 10 & 17 & 49 & 63 & 13 \\ 14 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[514] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[515] $1^{134} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 11 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[516] $1^{123} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 11 & 13 & 28 & 39 & 10 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[517] $1^{124} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 15 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[518] $21^{133} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 12 & 17 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 12 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[519] $1^{134} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 17 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 16 & 22 & 85 & 102 & 17 \\ 17 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[520] $1^{124} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 14 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[523] $21^{13} 31^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 27 & 27 & 18 & 133 & 21 \\ 4 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[522] $21^{112} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.66 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[524] $1^{13} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 4 & 5 & 4 & 27 & 4 \\ 18 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[525] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 13 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[521] $1^{113} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[526] $1^{134} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 9 & 12 & 25 & 45 & 9 \\ 15 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

[527] $1^{12} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 17 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[528] $1^{122} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 11 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[529] $1^{113} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 13 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[530] $1^{122} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 10 & 13 & 28 & 42 & 10 \\ 13 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[531] \ 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -243, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 13 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[536] \ 1^{134} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 8 & 12 & 25 & 45 & 9 \\ 14 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[535] \ 1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 17 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[534] \ 1^{113} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 5 & 9 & 13 & 36 & 7 \\ 14 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[533] \ 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[532] \ 1^{113} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 21 & 32 & 21 & 193 & 25 \\ 3 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[537] \ 1^{12} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 28 & 0 & 154 & 22 \\ 3 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[538] \ 1^{122} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 12 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[539] \ 1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 28 & 39 & 10 \\ 10 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[540] \ 1^{133} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 10 & 17 & 49 & 66 & 13 \\ 14 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[541] \ 21^{14} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 32 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 33 & 32 & 21 & 193 & 25 \\ 4 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

[542] $1^{13} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 4 & 5 & 4 & 33 & 4 \\ 20 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[543] $1^{13} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 20 & 32 & 21 & 193 & 25 \\ 3 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

[544] $1^{134} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 11 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[545] $1^{112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 10 & 14 & 37 & 48 & 11 \\ 13 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[546] $1^{112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 11 & 14 & 37 & 48 & 11 \\ 14 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[547] $1^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 3 & 5 & 4 & 27 & 4 \\ 19 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[548] $1^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 3 & 5 & 4 & 27 & 4 \\ 17 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[549] $1^{124} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 13 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[551] $1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 13 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[550] $1^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[555] $1^{14} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 26 & 0 & 118 & 20 \\ 3 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[552] $1^{123} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 28 & 39 & 10 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[556] $21^{123} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[553] $1^{122} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[554] $21^{113} 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 27 & 4 \\ 5 & 4 & 5 & 27 & 4 \\ 6 & 5 & 4 & 27 & 4 \\ 27 & 27 & 27 & 133 & 21 \\ 4 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[558] $1^{114} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 5 & 9 & 13 & 42 & 7 \\ 17 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[557] $1^{112} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 7 & 9 & 13 & 30 & 7 \\ 13 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[559] $1^{112} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 12 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[560] $1^{13} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 3 & 5 & 4 & 27 & 4 \\ 18 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[561] $1^{122} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 13 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[562] $1^{123} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 22 & 3 \\ 3 & 4 & 13 & 32 & 4 \\ 9 & 13 & 28 & 75 & 10 \\ 22 & 32 & 75 & 193 & 25 \\ 3 & 4 & 10 & 25 & 3 \end{pmatrix}_{b_0}$$

[563] $21^{122} 31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 18 & 4 \\ 5 & 4 & 14 & 18 & 4 \\ 15 & 14 & 37 & 48 & 11 \\ 18 & 18 & 48 & 58 & 14 \\ 4 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[564] $1^{122} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[565] $1^{124} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 14 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[566] $1^{123} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 10 & 13 & 28 & 51 & 10 \\ 15 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[567] $1^{113} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[568] $1^{144} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 16 & 16 & 4 \\ 11 & 16 & 46 & 48 & 12 \\ 12 & 16 & 48 & 46 & 12 \\ 3 & 4 & 12 & 12 & 3 \end{pmatrix}_{b_0}$$

[569] $1^{134} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -270, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 12 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[570] $1^{114} 3 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 20 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 7 & 9 & 13 & 51 & 7 \\ 20 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[571] $1^{112} 3 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 16 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 7 & 9 & 13 & 39 & 7 \\ 16 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[572] $1^{113} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[573] $1^{134} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 16 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 15 & 22 & 85 & 102 & 17 \\ 16 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[574] $1^{12} 3 1^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 5 & 28 & 4 \\ 3 & 5 & 4 & 30 & 4 \\ 20 & 28 & 30 & 154 & 22 \\ 3 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

- [575] $21^{112}31^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 24 & 4 \\ 5 & 4 & 9 & 26 & 4 \\ 9 & 9 & 13 & 42 & 7 \\ 24 & 26 & 42 & 118 & 20 \\ 4 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$
- [576] $1^{123}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$
- [577] $1^{122}31^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 10 & 13 & 28 & 42 & 10 \\ 14 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$
- [578] $1^{123}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 28 & 39 & 10 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$
- [579] $1^{133}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 10 & 17 & 49 & 51 & 13 \\ 11 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$
- [580] $1^{113}31^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 21 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 7 & 9 & 13 & 54 & 7 \\ 21 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$
- [581] $1^{113}31^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 7 & 9 & 13 & 36 & 7 \\ 15 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$
- [583] $1^{112}31^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 7 & 9 & 13 & 30 & 7 \\ 14 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$
- [582] $1^{123}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 28 & 39 & 10 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$
- [584] $21^{113}31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$
- [585] $1^{113}31^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 22 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 5 & 9 & 13 & 54 & 7 \\ 22 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$
- [586] $1^{12}31^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -281, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 5 & 28 & 4 \\ 4 & 5 & 4 & 30 & 4 \\ 19 & 28 & 30 & 154 & 22 \\ 3 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[587] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 4 & 5 & 4 & 30 & 4 \\ 21 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[588] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 4 & 5 & 4 & 27 & 4 \\ 19 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[590] \ 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 27 & 0 & 133 & 21 \\ 3 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[589] \ 1^{123} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 19 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 10 & 13 & 28 & 63 & 10 \\ 19 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[591] \ 21^{13} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 6 & 24 & 4 \\ 5 & 4 & 5 & 23 & 4 \\ 6 & 5 & 4 & 24 & 4 \\ 24 & 23 & 24 & 100 & 18 \\ 4 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[592] \ 21^{12} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 9 & 23 & 4 \\ 6 & 9 & 13 & 39 & 7 \\ 15 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[594] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 26 & 0 & 118 & 20 \\ 3 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[593] \ 1^{122} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 11 & 18 & 58 & 60 & 14 \\ 12 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[595] \ 21^{12} 31^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -290, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 24 & 4 \\ 5 & 4 & 17 & 26 & 4 \\ 15 & 17 & 49 & 78 & 13 \\ 24 & 26 & 78 & 118 & 20 \\ 4 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[596] \ 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 5 & 9 & 13 & 45 & 7 \\ 18 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[599] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 3 & 5 & 4 & 30 & 4 \\ 19 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[598] $1^{124} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -292$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 8 & 13 & 28 & 48 & 10 \\ 14 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[600] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -292$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[601] $1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -292$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

[597] $21^{113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -292$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[602] $1^{133} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -293$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.6$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 10 & 17 & 49 & 66 & 13 \\ 13 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[603] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -294$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[604] $1^{124} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -295$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 21 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 10 & 13 & 28 & 72 & 10 \\ 21 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

[605] $21^{124} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -296$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.73 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 4 & 8 & 26 & 4 \\ 6 & 8 & 10 & 36 & 6 \\ 18 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[606] $21^{122} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -296$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 15 & 4 \\ 5 & 4 & 14 & 14 & 4 \\ 15 & 14 & 37 & 39 & 11 \\ 15 & 14 & 39 & 37 & 11 \\ 4 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[607] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -297$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[608] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -298$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[609] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -301, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 6 & 9 & 13 & 39 & 7 \\ 17 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[610] $21^{123} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -302, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 8 & 17 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[611] $1^{14} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 3 & 5 & 4 & 30 & 4 \\ 21 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[612] $1^{134} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -305, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 19 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 10 & 17 & 49 & 93 & 13 \\ 19 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[614] $1^{13} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 21 & 32 & 0 & 193 & 25 \\ 3 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[613] $21^{112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 18 & 4 \\ 5 & 4 & 9 & 18 & 4 \\ 9 & 9 & 13 & 30 & 7 \\ 18 & 18 & 30 & 58 & 14 \\ 4 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[616] $1^{14} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 3 & 5 & 4 & 30 & 4 \\ 20 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[615] $1^{114} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 19 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[617] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -309, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 6 & 9 & 13 & 39 & 7 \\ 15 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[621] $21^{122} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 12 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[620] $1^{114} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[619] $21^{144} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 16 & 21 & 4 \\ 12 & 16 & 46 & 60 & 12 \\ 15 & 21 & 60 & 76 & 16 \\ 3 & 4 & 12 & 16 & 3 \end{pmatrix}_{b_0}$$

[618] $1^{123} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 16 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 12 & 18 & 58 & 72 & 14 \\ 16 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[622] $1^{134} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -311, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 17 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 10 & 17 & 49 & 78 & 13 \\ 17 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

[624] $1^{113} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 18 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 7 & 9 & 13 & 45 & 7 \\ 18 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[623] $1^{113} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[625] $1^{114} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 5 & 9 & 13 & 42 & 7 \\ 16 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[629] $1^{114} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[628] $1^{12} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 5 & 28 & 4 \\ 4 & 5 & 4 & 30 & 4 \\ 20 & 28 & 30 & 154 & 22 \\ 3 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[627] $21^{114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 9 & 21 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[626] $21^{112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 12 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[631] $1^{124} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -319, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 8 & 13 & 28 & 48 & 10 \\ 13 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[630] $1^{124} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -319, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 10 & 13 & 28 & 48 & 10 \\ 15 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0} \\
[632] \quad & 21^{114} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0} \\
[634] \quad & 1^{114} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 21 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 5 & 9 & 13 & 51 & 7 \\ 21 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0} \\
[633] \quad & 1^{133} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 10 & 17 & 49 & 51 & 13 \\ 10 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0} \\
[635] \quad & 1^{144} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 16 & 16 & 4 \\ 11 & 16 & 46 & 48 & 12 \\ 11 & 16 & 48 & 46 & 12 \\ 3 & 4 & 12 & 12 & 3 \end{pmatrix}_{b_0} \\
[641] \quad & 1^{134} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 21 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 12 & 17 & 49 & 93 & 13 \\ 21 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0} \\
[640] \quad & 1^{123} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 8 & 13 & 28 & 51 & 10 \\ 15 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0} \\
[639] \quad & 1^{14} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 31 & 0 & 172 & 24 \\ 3 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0} \\
[636] \quad & 1^{14} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 23 & 3 \\ 3 & 4 & 5 & 36 & 4 \\ 4 & 5 & 4 & 36 & 4 \\ 23 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[637] \quad & 1^{123} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0} \\
[638] \quad & 1^{13} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 4 & 5 & 4 & 33 & 4 \\ 21 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[642] \quad & 1^{114} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[643] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -327, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[644] $1^{134} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 18 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 13 & 17 & 49 & 78 & 13 \\ 18 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

[645] $21^{113} 31^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 27 & 4 \\ 5 & 4 & 13 & 27 & 4 \\ 12 & 13 & 28 & 63 & 10 \\ 27 & 27 & 63 & 133 & 21 \\ 4 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[646] $1^{112} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 20 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 7 & 9 & 13 & 48 & 7 \\ 20 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[647] $1^{124} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 11 & 18 & 58 & 84 & 14 \\ 16 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[648] $1^{123} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -329, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 8 & 13 & 28 & 51 & 10 \\ 13 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[650] $1^{133} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 11 & 17 & 49 & 51 & 13 \\ 12 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[649] $1^{133} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 11 & 17 & 49 & 66 & 13 \\ 15 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[651] $21^{13} 31^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 30 & 31 & 21 & 172 & 24 \\ 4 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[654] $1^{112} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 6 & 9 & 13 & 39 & 7 \\ 16 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[652] $1^{124} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 9 & 13 & 28 & 60 & 10 \\ 16 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[655] $1^{123} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 18 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 8 & 13 & 28 & 63 & 10 \\ 18 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[656] $1^{113} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 14 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[657] $21^{112} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 21 & 4 \\ 5 & 4 & 9 & 22 & 4 \\ 9 & 9 & 13 & 36 & 7 \\ 21 & 22 & 36 & 85 & 17 \\ 4 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[653] $1^{123} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 9 & 13 & 28 & 51 & 10 \\ 14 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[658] $1^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -335, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 28 & 39 & 10 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[659] $1^{122} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 12 & 18 & 58 & 60 & 14 \\ 13 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[660] $21^{133} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 12 & 12 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[663] $1^{134} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 21 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 11 & 17 & 49 & 93 & 13 \\ 21 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[664] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 7 & 9 & 13 & 39 & 7 \\ 17 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[665] $1^{122} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 11 & 18 & 58 & 60 & 14 \\ 11 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[661] $1^{134} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 11 & 17 & 49 & 63 & 13 \\ 15 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[662] $1^{144} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 16 & 16 & 4 \\ 12 & 16 & 46 & 48 & 12 \\ 12 & 16 & 48 & 46 & 12 \\ 3 & 4 & 12 & 12 & 3 \end{pmatrix}_{b_0}$$

[666] $21^{113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[667] $1^{133} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 11 & 17 & 49 & 66 & 13 \\ 14 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[668] $1^{133} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 13 & 17 & 49 & 66 & 13 \\ 16 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[669] $1^{113} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -346, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 16 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[670] $1^{13} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -346, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 22 & 32 & 0 & 193 & 25 \\ 3 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[671] $1^{124} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 18 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 12 & 18 & 58 & 84 & 14 \\ 18 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[672] $1^{124} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 9 & 13 & 28 & 48 & 10 \\ 14 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[674] $1^{113} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 5 & 9 & 13 & 45 & 7 \\ 17 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[673] $1^{122} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 13 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 14 & 18 & 58 & 60 & 14 \\ 13 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[675] $1^{133} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 12 & 17 & 49 & 51 & 13 \\ 13 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0} \\
[676] & 21^{112} 31^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 12 & 21 & 4 \\ 5 & 4 & 13 & 22 & 4 \\ 12 & 13 & 28 & 51 & 10 \\ 21 & 22 & 51 & 85 & 17 \\ 4 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0} \\
[677] & 1^{123} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 11 & 18 & 58 & 72 & 14 \\ 13 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0} \\
[679] & 1^{124} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -360, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 18 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 13 & 18 & 58 & 84 & 14 \\ 18 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0} \\
[678] & 1^{114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -360, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 18 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 7 & 9 & 13 & 42 & 7 \\ 18 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0} \\
[680] & 1^{144} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -361, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 17 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 13 & 21 & 76 & 96 & 16 \\ 17 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0} \\
[681] & 1^{133} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -361, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 12 & 17 & 49 & 66 & 13 \\ 15 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0} \\
[683] & 1^{114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 6 & 9 & 13 & 42 & 7 \\ 16 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0} \\
[682] & 1^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 20 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 6 & 9 & 13 & 48 & 7 \\ 20 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[684] & 1^{113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 16 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 7 & 9 & 13 & 36 & 7 \\ 16 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0} \\
[685] & 1^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 19 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 6 & 9 & 13 & 48 & 7 \\ 19 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[686] & 1^{123} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 8 & 13 & 28 & 63 & 10 \\ 17 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[687] $1^{13}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 3 & 5 & 4 & 33 & 4 \\ 21 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[688] $21^{124}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -368, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[689] $21^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -368, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[690] $1^{14}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 21 & 31 & 0 & 172 & 24 \\ 3 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[691] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 15 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[692] $1^{124}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 23 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 9 & 13 & 28 & 84 & 10 \\ 23 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0}$$

[693] $1^{13}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 3 & 5 & 4 & 33 & 4 \\ 22 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[694] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[695] $1^{123}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 11 & 18 & 58 & 72 & 14 \\ 14 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[696] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[697] $1^{134}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 20 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 11 & 17 & 49 & 93 & 13 \\ 20 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[698] $1^{124} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -372, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 9 & 13 & 28 & 48 & 10 \\ 15 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[699] $1^{114} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -374, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 19 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[700] $1^{124} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 17 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 12 & 18 & 58 & 84 & 14 \\ 17 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[701] $1^{114} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -378, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 1 & 24 & 3 \\ 23 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

[702] $21^{122} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[703] $1^{123} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -381, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 18 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 9 & 13 & 28 & 63 & 10 \\ 18 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[704] $1^{123} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 14 & 18 & 58 & 72 & 14 \\ 16 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[705] $1^{133} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 16 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 12 & 17 & 49 & 66 & 13 \\ 16 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[706] $1^{114} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 22 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 4 & 5 & 4 & 33 & 4 \\ 22 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[707] $1^{114} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 20 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[708] $1^{123} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 8 & 13 & 28 & 51 & 10 \\ 14 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[709] $1^{134} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -385, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 10 & 17 & 49 & 63 & 13 \\ 13 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[710] $1^{113} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -387, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 21 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 5 & 9 & 13 & 54 & 7 \\ 21 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

[711] $1^{113} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 23 & 36 & 0 & 238 & 28 \\ 3 & 4 & 0 & 28 & 3 \end{pmatrix}_{b_0}$$

[712] $1^{124} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -394, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 18 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 8 & 13 & 28 & 60 & 10 \\ 18 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[713] $1^{133} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 11 & 17 & 49 & 51 & 13 \\ 11 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[714] $1^{114} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -397, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 20 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[715] $21^{134} 31^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -398, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 21 & 3 \\ 4 & 4 & 17 & 31 & 4 \\ 12 & 17 & 49 & 93 & 13 \\ 21 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[716] $1^{122} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 12 & 18 & 58 & 60 & 14 \\ 12 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[717] $1^{114} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 1 & 24 & 3 \\ 22 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

[723] $1^{114} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 24 & 3 \\ 3 & 4 & 5 & 36 & 4 \\ 4 & 5 & 4 & 36 & 4 \\ 24 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[722] $21^{122} 31^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 21 & 4 \\ 5 & 4 & 18 & 22 & 4 \\ 18 & 18 & 58 & 72 & 14 \\ 21 & 22 & 72 & 85 & 17 \\ 4 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[721] \ 1^{134} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 18 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 14 & 22 & 85 & 102 & 17 \\ 18 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[720] \ 21^{14} 31^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 36 & 4 \\ 5 & 4 & 4 & 36 & 4 \\ 3 & 4 & 1 & 24 & 3 \\ 36 & 36 & 24 & 238 & 28 \\ 4 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[718] \ 1^{114} 3 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 1 & 24 & 3 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[719] \ 1^{123} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 12 & 18 & 58 & 72 & 14 \\ 14 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[724] \ 21^{134} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -408, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 12 & 21 & 4 \\ 9 & 12 & 25 & 45 & 9 \\ 15 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[725] \ 1^{123} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -414, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.74 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 13 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[727] \ 1^{133} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -416, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 13 & 22 & 85 & 87 & 17 \\ 14 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[726] \ 21^{112} 31^{124} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -416, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.71 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 24 & 4 \\ 5 & 4 & 13 & 26 & 4 \\ 12 & 13 & 28 & 60 & 10 \\ 24 & 26 & 60 & 118 & 20 \\ 4 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[728] \ 1^{124} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -418, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.73 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 18 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 10 & 13 & 28 & 60 & 10 \\ 18 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[729] \ 1^{114} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -418, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.73 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[730] \ 1^{114} 3 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -418, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 1 & 24 & 3 \\ 23 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

[731] $1^{123} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -420, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 16 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 10 & 13 & 28 & 51 & 10 \\ 16 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[732] $1^{134} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 11 & 17 & 49 & 63 & 13 \\ 13 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[733] $21^{114} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 36 & 4 \\ 5 & 4 & 5 & 36 & 4 \\ 6 & 5 & 4 & 36 & 4 \\ 36 & 36 & 36 & 238 & 28 \\ 4 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[734] $21^{113} 31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 24 & 4 \\ 5 & 4 & 9 & 23 & 4 \\ 9 & 9 & 13 & 39 & 7 \\ 24 & 23 & 39 & 100 & 18 \\ 4 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[735] $1^{114} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 5 & 9 & 13 & 51 & 7 \\ 20 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[736] $21^{123} 31^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 18 & 3 \\ 4 & 4 & 13 & 27 & 4 \\ 9 & 13 & 28 & 63 & 10 \\ 18 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[738] $1^{144} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 13 & 21 & 76 & 96 & 16 \\ 16 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[737] $1^{123} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 9 & 13 & 28 & 51 & 10 \\ 16 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[740] $221^{111} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -432, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[739] $1^{123} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -432, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 19 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 9 & 13 & 28 & 63 & 10 \\ 19 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[741] $1^{113} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -435, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 6 & 9 & 13 & 45 & 7 \\ 17 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[742] $1^{122} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 13 & 18 & 58 & 60 & 14 \\ 13 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[743] $1^{133} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 12 & 17 & 49 & 51 & 13 \\ 12 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[744] $1^{122} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 14 & 18 & 58 & 60 & 14 \\ 14 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[745] $1^{134} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 13 & 17 & 49 & 63 & 13 \\ 15 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[748] $1^{114} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 6 & 9 & 13 & 42 & 7 \\ 17 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[747] $1^{134} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 17 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 12 & 17 & 49 & 78 & 13 \\ 17 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

[746] $1^{123} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 9 & 13 & 28 & 51 & 10 \\ 15 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[749] $21^{114} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -440, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 30 & 4 \\ 5 & 4 & 5 & 28 & 4 \\ 6 & 5 & 4 & 30 & 4 \\ 30 & 28 & 30 & 154 & 22 \\ 4 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[750] $1^{14} 31^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -441, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 5 & 36 & 4 \\ 3 & 5 & 4 & 36 & 4 \\ 23 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[751] $1^{14} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 36 & 0 & 238 & 28 \\ 3 & 4 & 0 & 28 & 3 \end{pmatrix}_{b_0}$$

[752] $1^{133} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 13 & 17 & 49 & 51 & 13 \\ 13 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[753] $21^{124} 31^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 18 & 3 \\ 4 & 4 & 18 & 26 & 4 \\ 12 & 18 & 58 & 84 & 14 \\ 18 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[754] $1^{113} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 19 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 7 & 9 & 13 & 45 & 7 \\ 19 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[755] $21^{114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 24 & 3 \\ 4 & 4 & 4 & 36 & 4 \\ 3 & 4 & 1 & 24 & 3 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

[756] $21^{134} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 12 & 16 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 12 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[760] $1^{134} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 12 & 17 & 49 & 63 & 13 \\ 14 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[758] $1^{144} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 18 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 14 & 21 & 76 & 96 & 16 \\ 18 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[759] $1^{14} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 5 & 36 & 4 \\ 3 & 5 & 4 & 36 & 4 \\ 24 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[757] $1^{123} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 12 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[761] $1^{144} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 13 & 21 & 76 & 78 & 16 \\ 14 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$$

[762] $1^{114} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 18 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 6 & 9 & 13 & 42 & 7 \\ 18 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[763] $1^{113} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -460, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 22 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 7 & 9 & 13 & 54 & 7 \\ 22 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

[764] $1^{114} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -462, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 1 & 24 & 3 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

[765] $1^{14} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -463, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[766] $21^{15} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -464, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 20 & 15 & 73 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[769] $1^{14} 3^1 1^{24} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 21 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 7 & 9 & 13 & 51 & 7 \\ 21 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[768] $21^{12} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 18 & 4 \\ 5 & 4 & 18 & 18 & 4 \\ 18 & 18 & 58 & 60 & 14 \\ 18 & 18 & 60 & 58 & 14 \\ 4 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[767] $1^{14} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 16 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 10 & 17 & 49 & 78 & 13 \\ 16 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

[770] $1^{14} 3^1 1^{33} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -469, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 17 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 14 & 21 & 76 & 96 & 16 \\ 17 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[771] $1^{14} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 5 & 9 & 13 & 51 & 7 \\ 19 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[772] $1^{14} 3^1 1^{44} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -474, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 18 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 15 & 21 & 76 & 96 & 16 \\ 18 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[773] $1^{14} 3^1 1^{23} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -475, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 11 & 17 & 49 & 63 & 13 \\ 14 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[774] $1^{14} 3^1 1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 11 & 17 & 49 & 78 & 13 \\ 16 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

[775] $21^{14} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 33 & 4 \\ 5 & 4 & 5 & 32 & 4 \\ 6 & 5 & 4 & 33 & 4 \\ 33 & 32 & 33 & 193 & 25 \\ 4 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[776] $21^{124}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -478, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 8 & 21 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[777] $21^{122}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 3 \\ 4 & 4 & 18 & 18 & 4 \\ 12 & 18 & 58 & 60 & 14 \\ 12 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[778] $1^{114}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 24 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 5 & 9 & 13 & 60 & 7 \\ 24 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

[779] $1^{123}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -490, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 13 & 18 & 58 & 72 & 14 \\ 16 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[780] $21^{114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -490, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 30 & 4 \\ 5 & 4 & 9 & 28 & 4 \\ 9 & 9 & 13 & 48 & 7 \\ 30 & 28 & 48 & 154 & 22 \\ 4 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[781] $1^{113}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -490, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 20 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 6 & 9 & 13 & 54 & 7 \\ 20 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

[783] $21^{122}31^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -494, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 21 & 24 & 4 \\ 5 & 4 & 22 & 26 & 4 \\ 21 & 22 & 85 & 102 & 17 \\ 24 & 26 & 102 & 118 & 20 \\ 4 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[782] $21^{123}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -494, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 13 & 17 & 4 \\ 9 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[784] $1^{134}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 18 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 11 & 17 & 49 & 78 & 13 \\ 18 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

[785] $21^{113}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 15 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[786] $1^{133}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 14 & 22 & 85 & 87 & 17 \\ 15 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

[787] $21^{114}31^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 36 & 4 \\ 5 & 4 & 13 & 36 & 4 \\ 12 & 13 & 28 & 84 & 10 \\ 36 & 36 & 84 & 238 & 28 \\ 4 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0}$$

[788] $1^{113} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -500, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 18 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 6 & 9 & 13 & 45 & 7 \\ 18 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[789] $1^{124} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -502, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 24 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 8 & 13 & 28 & 84 & 10 \\ 24 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0}$$

[790] $1^{133} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 13 & 22 & 85 & 87 & 17 \\ 13 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

[791] $1^{114} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 24 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 7 & 9 & 13 & 60 & 7 \\ 24 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

[792] $1^{113} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -507, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 19 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 6 & 9 & 13 & 45 & 7 \\ 19 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[793] $1^{144} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -507, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 14 & 21 & 76 & 78 & 16 \\ 15 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$$

[794] $1^{124} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -509, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 8 & 13 & 28 & 60 & 10 \\ 17 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[795] $1^{124} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -510, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 8 & 13 & 28 & 60 & 10 \\ 16 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[796] $1^{124} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -511, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 21 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 8 & 13 & 28 & 72 & 10 \\ 21 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

[797] $21^{125} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -512, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 20 & 30 & 73 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[798] $21^{133} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -512, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 17 & 22 & 4 \\ 12 & 17 & 49 & 66 & 13 \\ 15 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[799] $1^{124} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -514, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 22 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 8 & 13 & 28 & 84 & 10 \\ 22 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0}$$

[800] $1^{114} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -517, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 19 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 6 & 9 & 13 & 51 & 7 \\ 19 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[801] $21^{114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -520, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[802] $1^{134} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -525, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 12 & 17 & 49 & 63 & 13 \\ 15 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[803] $1^{124} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -529, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 9 & 13 & 28 & 60 & 10 \\ 17 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[804] $1^{134} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -530, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 13 & 22 & 85 & 102 & 17 \\ 16 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[805] $1^{124} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -535, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 20 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 9 & 13 & 28 & 72 & 10 \\ 20 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

[807] $1^{133} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 16 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 15 & 22 & 85 & 87 & 17 \\ 16 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

[806] $1^{144} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 13 & 21 & 76 & 78 & 16 \\ 13 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$$

[808] $21^{144} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -560, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 3 \\ 4 & 4 & 16 & 16 & 4 \\ 12 & 16 & 46 & 48 & 12 \\ 12 & 16 & 48 & 46 & 12 \\ 3 & 4 & 12 & 12 & 3 \end{pmatrix}_{b_0}$$

[809] $1^{124} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -565, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 8 & 23 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 8 & 13 & 28 & 84 & 10 \\ 23 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0} \\
[810] & 21^{113}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -568, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 21 & 3 \\ 4 & 4 & 9 & 32 & 4 \\ 6 & 9 & 13 & 54 & 7 \\ 21 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0} \\
[811] & 21^{123}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -568, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 18 & 22 & 4 \\ 12 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0} \\
[812] & 1^{124}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -570, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 24 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 9 & 13 & 28 & 84 & 10 \\ 24 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0} \\
[813] & 1^{114}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 23 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 5 & 9 & 13 & 60 & 7 \\ 23 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0} \\
[814] & 21^{113}31^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -572, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 12 & 30 & 4 \\ 5 & 4 & 13 & 31 & 4 \\ 12 & 13 & 28 & 72 & 10 \\ 30 & 31 & 72 & 172 & 24 \\ 4 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0} \\
[815] & 1^{124}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -575, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 19 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 8 & 13 & 28 & 72 & 10 \\ 19 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0} \\
[816] & 21^{113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -576, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 9 & 27 & 4 \\ 5 & 4 & 9 & 27 & 4 \\ 9 & 9 & 13 & 45 & 7 \\ 27 & 27 & 45 & 133 & 21 \\ 4 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0} \\
[817] & 1^{134}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -579, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 17 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 11 & 17 & 49 & 78 & 13 \\ 17 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0} \\
[818] & 21^{124}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -580, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 13 & 21 & 4 \\ 9 & 13 & 28 & 48 & 10 \\ 15 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0} \\
[820] & 21^{122}31^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 21 & 21 & 4 \\ 5 & 4 & 22 & 22 & 4 \\ 21 & 22 & 85 & 87 & 17 \\ 21 & 22 & 87 & 85 & 17 \\ 4 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0} \\
[819] & 21^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0} \\
[821] & 1^{113}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 21 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 6 & 9 & 13 & 54 & 7 \\ 21 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0} \\
[822] & 21^{113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -588, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 4 & 9 & 27 & 4 \\ 6 & 9 & 13 & 45 & 7 \\ 18 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0} \\
[823] & 21^{135} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -592, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.9 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 12 & 20 & 4 \\ 9 & 12 & 25 & 45 & 9 \\ 15 & 20 & 45 & 73 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0} \\
[824] & 21^{124} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -598, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 24 & 3 \\ 4 & 4 & 13 & 36 & 4 \\ 9 & 13 & 28 & 84 & 10 \\ 24 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0} \\
[826] & 1^{133} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 14 & 22 & 85 & 87 & 17 \\ 14 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0} \\
[825] & 1^{124} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 20 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 8 & 13 & 28 & 72 & 10 \\ 20 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0} \\
[827] & 1^{134} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -606, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 14 & 22 & 85 & 102 & 17 \\ 16 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0} \\
[829] & 21^{123} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 13 & 22 & 4 \\ 9 & 13 & 28 & 51 & 10 \\ 15 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0} \\
[828] & 21^{114} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0} \\
[830] & 1^{124} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -610, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 18 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 9 & 13 & 28 & 60 & 10 \\ 18 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0} \\
[831] & 1^{134} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -616, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 18 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 12 & 17 & 49 & 78 & 13 \\ 18 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0} \\
[832] & 1^{144} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -616, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 14 & 21 & 76 & 78 & 16 \\ 14 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [834] $1^{114} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -622, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 20 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 6 & 9 & 13 & 51 & 7 \\ 20 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$
- [833] $1^{113} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -622, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 22 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 6 & 9 & 13 & 54 & 7 \\ 22 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$
- [835] $1^{114} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -624, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 22 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 6 & 9 & 13 & 60 & 7 \\ 22 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$
- [837] $21^{116} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 24 & 18 & 106 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$
- [836] $21^{113} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 9 & 30 & 4 \\ 5 & 4 & 9 & 31 & 4 \\ 9 & 9 & 13 & 51 & 7 \\ 30 & 31 & 51 & 172 & 24 \\ 4 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$
- [838] $21^{122} 31^{144} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -644, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 24 & 24 & 4 \\ 5 & 4 & 26 & 26 & 4 \\ 24 & 26 & 118 & 120 & 20 \\ 24 & 26 & 120 & 118 & 20 \\ 4 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$
- [840] $1^{134} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 17 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 15 & 22 & 85 & 102 & 17 \\ 17 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$
- [839] $1^{134} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 18 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 16 & 22 & 85 & 102 & 17 \\ 18 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$
- [841] $21^{114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 4 & 9 & 26 & 4 \\ 6 & 9 & 13 & 42 & 7 \\ 18 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$
- [842] $1^{134} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -650, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 17 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 14 & 22 & 85 & 102 & 17 \\ 17 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$
- [843] $1^{144} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 17 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 16 & 26 & 118 & 120 & 20 \\ 17 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$
- [844] $1^{144} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 15 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 15 & 21 & 76 & 76 & 16 \\ 15 & 21 & 76 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$$

[845] $1^{133} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -664$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.6$
 $T_{\min} = 9.75 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 15 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 15 & 22 & 85 & 85 & 17 \\ 15 & 22 & 85 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

[846] $21^{133} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -668$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 3 \\ 4 & 4 & 17 & 17 & 4 \\ 12 & 17 & 49 & 49 & 13 \\ 12 & 17 & 49 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[847] $1^{114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -669$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 21 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 6 & 9 & 13 & 51 & 7 \\ 21 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[848] $1^{124} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -676$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 21 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 9 & 13 & 28 & 72 & 10 \\ 21 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

[849] $21^{126} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -688$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 4 & 8 & 24 & 4 \\ 6 & 8 & 10 & 36 & 6 \\ 18 & 24 & 36 & 106 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[850] $1^{133} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -696$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 16 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 16 & 22 & 85 & 85 & 17 \\ 16 & 22 & 85 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

[851] $21^{145} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -704$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.1$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 16 & 20 & 4 \\ 12 & 16 & 46 & 60 & 12 \\ 15 & 20 & 60 & 73 & 15 \\ 3 & 4 & 12 & 15 & 3 \end{pmatrix}_{b_0}$$

[852] $21^{144} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -718$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 18 & 3 \\ 4 & 4 & 21 & 26 & 4 \\ 15 & 21 & 76 & 96 & 16 \\ 18 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[853] $1^{134} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -730$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.75 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 18 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 15 & 22 & 85 & 102 & 17 \\ 18 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[854] $1^{144} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -736$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.78 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 18 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 17 & 26 & 118 & 120 & 20 \\ 18 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$$

[855] $1^{114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -759$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 23 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 6 & 9 & 13 & 60 & 7 \\ 23 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

- [856] $21^{114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -766, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 33 & 4 \\ 5 & 4 & 9 & 32 & 4 \\ 9 & 9 & 13 & 54 & 7 \\ 33 & 32 & 54 & 193 & 25 \\ 4 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$
- [857] $21^{136}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -768, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 18 & 3 \\ 4 & 4 & 12 & 24 & 4 \\ 9 & 12 & 25 & 54 & 9 \\ 18 & 24 & 54 & 106 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$
- [858] $1^{144}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -804, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 16 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 16 & 26 & 118 & 120 & 20 \\ 16 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$$
- [859] $21^{134}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -814, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 17 & 21 & 4 \\ 12 & 17 & 49 & 63 & 13 \\ 15 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$
- [860] $1^{114}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -836, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 24 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 6 & 9 & 13 & 60 & 7 \\ 24 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$
- [861] $21^{155}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -848, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 15 & 3 \\ 4 & 4 & 20 & 20 & 4 \\ 15 & 20 & 73 & 75 & 15 \\ 15 & 20 & 75 & 73 & 15 \\ 3 & 4 & 15 & 15 & 3 \end{pmatrix}_{b_0}$$
- [862] $21^{117}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -848, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 28 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 21 & 28 & 21 & 145 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$
- [863] $21^{114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -862, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 21 & 3 \\ 4 & 4 & 9 & 31 & 4 \\ 6 & 9 & 13 & 51 & 7 \\ 21 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$
- [864] $21^{124}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -878, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 18 & 3 \\ 4 & 4 & 13 & 26 & 4 \\ 9 & 13 & 28 & 60 & 10 \\ 18 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$
- [865] $21^{146}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -880, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 18 & 3 \\ 4 & 4 & 16 & 24 & 4 \\ 12 & 16 & 46 & 72 & 12 \\ 18 & 24 & 72 & 106 & 18 \\ 3 & 4 & 12 & 18 & 3 \end{pmatrix}_{b_0}$$
- [866] $21^{124}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -884, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 21 & 3 \\ 4 & 4 & 13 & 31 & 4 \\ 9 & 13 & 28 & 72 & 10 \\ 21 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$
- [867] $21^{133}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -896, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 15 & 3 \\ 4 & 4 & 22 & 22 & 4 \\ 15 & 22 & 85 & 87 & 17 \\ 15 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$
- [868] $21^{127}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -896, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 21 & 3 \\ 4 & 4 & 8 & 28 & 4 \\ 6 & 8 & 10 & 42 & 6 \\ 21 & 28 & 42 & 145 & 21 \\ 3 & 4 & 6 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[869] \ 1^{144} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -900, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 17 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 17 & 26 & 118 & 120 & 20 \\ 17 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[870] \ 21^{134} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -902, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 18 & 3 \\ 4 & 4 & 17 & 26 & 4 \\ 12 & 17 & 49 & 78 & 13 \\ 18 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[871] \ 21^{114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -954, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 36 & 4 \\ 5 & 4 & 9 & 36 & 4 \\ 9 & 9 & 13 & 60 & 7 \\ 36 & 36 & 60 & 238 & 28 \\ 4 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[872] \ 1^{144} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -964, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 18 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 18 & 26 & 118 & 120 & 20 \\ 18 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[873] \ 21^{114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -966, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 24 & 3 \\ 4 & 4 & 9 & 36 & 4 \\ 6 & 9 & 13 & 60 & 7 \\ 24 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[874] \ 21^{137} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -976, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 21 & 3 \\ 4 & 4 & 12 & 28 & 4 \\ 9 & 12 & 25 & 63 & 9 \\ 21 & 28 & 63 & 145 & 21 \\ 3 & 4 & 9 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[875] \ 21^{134} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1016, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 18 & 3 \\ 4 & 4 & 22 & 26 & 4 \\ 15 & 22 & 85 & 102 & 17 \\ 18 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[876] \ 21^{144} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1020, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 15 & 3 \\ 4 & 4 & 21 & 21 & 4 \\ 15 & 21 & 76 & 78 & 16 \\ 15 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[877] \ 21^{156} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1024, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.4$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 18 & 3 \\ 4 & 4 & 20 & 24 & 4 \\ 15 & 20 & 73 & 90 & 15 \\ 18 & 24 & 90 & 106 & 18 \\ 3 & 4 & 15 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[878] \ 21^{147} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1088, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 21 & 3 \\ 4 & 4 & 16 & 28 & 4 \\ 12 & 16 & 46 & 84 & 12 \\ 21 & 28 & 84 & 145 & 21 \\ 3 & 4 & 12 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[879] \ 21^{166} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 18 & 3 \\ 4 & 4 & 24 & 24 & 4 \\ 18 & 24 & 106 & 108 & 18 \\ 18 & 24 & 108 & 106 & 18 \\ 3 & 4 & 18 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[880] \ 21^{157} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 15 & 21 & 3 \\ 4 & 4 & 20 & 28 & 4 \\ 15 & 20 & 73 & 105 & 15 \\ 21 & 28 & 105 & 145 & 21 \\ 3 & 4 & 15 & 21 & 3 \end{pmatrix}_{b_0} \\
[881] \quad & 21^{144}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 18 & 3 \\ 4 & 4 & 26 & 26 & 4 \\ 18 & 26 & 118 & 120 & 20 \\ 18 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0} \\
[882] \quad & 21^{167}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1408, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 21 & 3 \\ 4 & 4 & 24 & 28 & 4 \\ 18 & 24 & 106 & 126 & 18 \\ 21 & 28 & 126 & 145 & 21 \\ 3 & 4 & 18 & 21 & 3 \end{pmatrix}_{b_0} \\
[883] \quad & 21^{177}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1616, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 21 & 21 & 3 \\ 4 & 4 & 28 & 28 & 4 \\ 21 & 28 & 145 & 147 & 21 \\ 21 & 28 & 147 & 145 & 21 \\ 3 & 4 & 21 & 21 & 3 \end{pmatrix}_{b_0} \\
T = 8 \quad [884] \quad & 1^{11}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[885] \quad & 1^{13}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 \\ 2 & 3 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0} \\
[886] \quad & 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[887] \quad & 1^{12}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 0 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 28 & 10 \\ 2 & 3 & -1 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[888] \quad & 1^{11}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -75, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[889] \quad & 1^{11}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -75, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[890] \quad & 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[892] \quad & 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 4 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[905] & 1^{112} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0} \\
[904] & 1^{112} 31^{21} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 12 & 3 \\ 7 & 12 & 0 & 12 & 58 & 14 \\ 2 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0} \\
[903] & 1^{112} 31^{21} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 7 & 12 & 0 & 12 & 58 & 14 \\ 2 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0} \\
[906] & 2 \overset{1}{\underset{1^{11}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
[907] & 1^{112} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 5 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0} \\
[908] & 1^{112} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 21 & 7 \\ 5 & 9 & 0 & 21 & 28 & 10 \\ 2 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[909] & 2 \overset{1}{\underset{1^{12}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 1 & 6 & 3 \\ 4 & 6 & 4 & 6 & 10 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[910] & 2 \overset{1^{12}}{\underset{1^{11}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 7 & 3 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 1 & 7 & 3 \\ 6 & 7 & 8 & 7 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[911] & 1^{112} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 21 & 7 \\ 5 & 9 & 0 & 21 & 28 & 10 \\ 2 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[913] & 1^{111} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 15 & 7 \\ 4 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[912] $2 \overset{1}{\underset{1^{122}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 4 & 2 \\ 2 & 1 & 3 & 6 & 6 & 3 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 4 & 6 & 4 & 10 & 12 & 6 \\ 4 & 6 & 4 & 12 & 10 & 6 \\ 2 & 3 & 2 & 6 & 6 & 3 \end{pmatrix}_{b_0}$

[914] $2 \overset{1}{\underset{1^{111}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 4 & 5 & 5 & 8 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$

[915] $2 \overset{1}{\underset{1^{113}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 9 & 3 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 1 & 9 & 3 \\ 6 & 9 & 6 & 9 & 25 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$

[916] $2 \overset{1}{\underset{1^{123}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 6 & 2 \\ 2 & 1 & 3 & 6 & 9 & 3 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 4 & 6 & 4 & 10 & 18 & 6 \\ 6 & 9 & 6 & 18 & 25 & 9 \\ 2 & 3 & 2 & 6 & 9 & 3 \end{pmatrix}_{b_0}$

[917] $2 \overset{1}{\underset{1^{123}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 10 & 2 \\ 2 & 1 & 3 & 6 & 13 & 3 \\ 2 & 3 & 0 & 4 & 12 & 2 \\ 4 & 6 & 4 & 10 & 26 & 6 \\ 10 & 13 & 12 & 26 & 61 & 15 \\ 2 & 3 & 2 & 6 & 15 & 3 \end{pmatrix}_{b_0}$

[918] $2 \overset{1}{\underset{1^{122}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 6 & 2 \\ 2 & 1 & 3 & 6 & 8 & 3 \\ 2 & 3 & 0 & 4 & 7 & 2 \\ 4 & 6 & 4 & 10 & 16 & 6 \\ 6 & 8 & 7 & 16 & 21 & 9 \\ 2 & 3 & 2 & 6 & 9 & 3 \end{pmatrix}_{b_0}$

[919] $2 \overset{1}{\underset{1^{114}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 1 & 12 & 3 \\ 8 & 12 & 8 & 12 & 46 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$

[920] $2 \overset{1}{\underset{1^{124}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 8 & 2 \\ 2 & 1 & 3 & 6 & 12 & 3 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 4 & 6 & 4 & 10 & 24 & 6 \\ 8 & 12 & 8 & 24 & 46 & 12 \\ 2 & 3 & 2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$

[921] $2 \overset{1}{\underset{1^{115}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 10 & 15 & 10 & 15 & 73 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$

[922] $2 \overset{1}{\underset{1^{112}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[923] \quad 2 \overset{1^{11}}{\underset{1^{123}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 8 & 2 \\ 2 & 1 & 3 & 6 & 11 & 3 \\ 2 & 3 & 0 & 4 & 9 & 2 \\ 4 & 6 & 4 & 10 & 22 & 6 \\ 8 & 11 & 9 & 22 & 40 & 12 \\ 2 & 3 & 2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[924] \quad 2 \overset{1^1}{\underset{1^{116}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 1 & 18 & 3 \\ 12 & 18 & 12 & 18 & 106 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[925] \quad 2 \overset{1^{12}}{\underset{1^{112}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 8 & 10 & 10 & 10 & 38 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[926] \quad 2 \overset{1^{11}}{\underset{1^{113}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 11 & 3 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 40 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[927] \quad 2 \overset{1^{12}}{\underset{1^{124}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 12 & 2 \\ 2 & 1 & 3 & 6 & 16 & 3 \\ 2 & 3 & 0 & 4 & 14 & 2 \\ 4 & 6 & 4 & 10 & 32 & 6 \\ 12 & 16 & 14 & 32 & 90 & 18 \\ 2 & 3 & 2 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[928] \quad 2 \overset{1^1}{\underset{1^{117}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 3 & 21 & 3 \\ 2 & 3 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 1 & 21 & 3 \\ 14 & 21 & 14 & 21 & 145 & 21 \\ 2 & 3 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[929] \quad 2 \overset{1^{11}}{\underset{1^{124}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 10 & 2 \\ 2 & 1 & 3 & 6 & 14 & 3 \\ 2 & 3 & 0 & 4 & 11 & 2 \\ 4 & 6 & 4 & 10 & 28 & 6 \\ 10 & 14 & 11 & 28 & 65 & 15 \\ 2 & 3 & 2 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[930] \quad 2 \overset{1^{11}}{\underset{1^{114}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 14 & 3 \\ 2 & 3 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 1 & 14 & 3 \\ 10 & 14 & 11 & 14 & 65 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[931] \quad 2 \overset{1^{12}}{\underset{1^{113}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 1 & 13 & 3 \\ 10 & 13 & 12 & 13 & 61 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[932] \begin{smallmatrix} 1^{12} \\ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \\ 1^{114} \end{smallmatrix} \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 920, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 16 & 3 \\ 2 & 3 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 1 & 16 & 3 \\ 12 & 16 & 14 & 16 & 90 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [935] \begin{smallmatrix} 1^{11} 231^{131} 1^{11} \\ + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 7 & 0 & 14 & 8 \\ 2 & 0 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[942] \begin{smallmatrix} 1^{133} 2 \ 3 \ 21^{34} \\ + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.6$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 84 & 18 \\ 8 & 13 & 0 & 84 & 97 & 21 \\ 2 & 3 & -1 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[938] \begin{smallmatrix} 1^{111} 2 \ 3 \ 21^{11} \\ + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 4 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[937] \begin{smallmatrix} 1^{111} 2 \ 3 \ 21^{21} \\ + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 17 & 12 & 9 \\ 2 & 4 & 0 & 12 & 6 & 6 \\ 2 & 3 & -1 & 9 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[936] \begin{smallmatrix} 1^{133} 2 \ 3 \ 21^{123} \\ + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.6$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 6 & 11 & 9 & 49 & 60 & 15 \\ 8 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[940] \begin{smallmatrix} 1^{112} 2 \ 3 \ 21^{13} \\ + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 2 & 0 & 1 & 1 & 10 & 3 \\ 6 & 0 & 6 & 10 & 34 & 12 \\ 2 & 0 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[941] \begin{smallmatrix} 1^{112} \\ 1 \ 3 \ 21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 3 & 1 & 1 & 4 & 3 \\ 2 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[934] \begin{smallmatrix} 1^{111} 231^{131} 1^{11} \\ + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 4 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 7 & 4 \\ 3 & 7 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[933] \begin{smallmatrix} 1^{11} 23131^{111} \\ + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \end{smallmatrix} \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 4 & 0 & 2 & 4 \\ 2 & 0 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[943] $1^{114}2^13^121^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$

[939] $1^{112}2^13^121^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$

[944] $1^{1^1}5^{1^1}1^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$

[945] $1^{11}2^13^121^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[946] $1^{111}2^13^121^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$

[947] $1^{1^1}5^{1^1}1^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 8 & 5 & 6 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$

[948] $21^{1^1}5^{1^1}1^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[949] $1^{1^1}5^{1^1}1^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[950] $1^{1^1}5^{1^1}1^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$

[951] $1^{1^1}5^{1^1}1^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 4 \end{pmatrix}_{b_0}$

[952] $21^{1^1}5^{1^1}1^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[954] \quad 1^{122} 2^3 21^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 1 & 8 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 40 & 12 \\ 6 & 9 & 0 & 40 & 49 & 15 \\ 2 & 3 & -1 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[953] \quad 1^{122} 2^3 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 6 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[955] \quad 1^1 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[956] \quad 1^{11} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 2 & 0 & 1 & 1 & 7 & 3 \\ 4 & 0 & 5 & 7 & 17 & 9 \\ 2 & 0 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[959] \quad 1^{114} 2^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 3 & 1 & 1 & 15 & 3 \\ 9 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[957] \quad 1^{132} 2^3 21^{43} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 13 & 9 & 2 \\ 8 & 0 & 13 & 97 & 71 & 21 \\ 6 & 0 & 9 & 71 & 49 & 15 \\ 2 & 0 & 2 & 21 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[958] \quad 1^{112} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 3 & 1 & 1 & 7 & 3 \\ 5 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[960] \quad 1^{111} 2^3 1^3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 7 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & 2 & 4 \\ 2 & 4 & -1 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[961] \quad 1^{111} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 5 & 2 \\ 2 & 0 & 4 & 6 & 12 & 6 \\ 4 & 0 & 5 & 12 & 17 & 9 \\ 2 & 0 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[962] \quad 1^{112} 2^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 9 & 2 \\ 2 & 0 & 4 & 6 & 20 & 6 \\ 6 & 0 & 9 & 20 & 49 & 15 \\ 2 & 0 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[963] $1^{112}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[964] $1^{113}2321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[965] $1^{122}321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 2 & 0 & 1 & 1 & 14 & 3 \\ 8 & 0 & 10 & 14 & 74 & 18 \\ 2 & 0 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[966] $1^{114}2321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 3 & 1 & 1 & 14 & 3 \\ 10 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[969] $1^{131}2321^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 24 & 18 \\ 2 & 4 & 0 & 24 & 6 & 6 \\ 2 & 3 & -1 & 18 & 6 & 4 \end{pmatrix}_{b_0}$$

[978] $1^{132}2321^{32} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 1 & 12 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 48 & 18 \\ 4 & 8 & 0 & 48 & 30 & 12 \\ 2 & 3 & -1 & 18 & 12 & 4 \end{pmatrix}_{b_0}$$

[977] $1^{111}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 8 & 6 \\ 2 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[976] $1^{141}2321^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 16 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 32 & 24 \\ 2 & 4 & 0 & 32 & 6 & 6 \\ 2 & 3 & -1 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

[975] $1^{142}2321^{42} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 1 & 16 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 64 & 24 \\ 4 & 8 & 0 & 64 & 30 & 12 \\ 2 & 3 & -1 & 24 & 12 & 4 \end{pmatrix}_{b_0}$$

[974] $1^{143}2321^{43} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 1 & 1 & 16 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 96 & 24 \\ 6 & 12 & 0 & 96 & 70 & 18 \\ 2 & 3 & -1 & 24 & 18 & 4 \end{pmatrix}_{b_0}$$

[973] $1^{112}2^3 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 10 & 2 \\ 2 & 0 & 4 & 6 & 24 & 6 \\ 8 & 0 & 10 & 24 & 74 & 18 \\ 2 & 0 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[972] $1^{144}2^3 21^{44} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 1 & 16 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 128 & 24 \\ 8 & 16 & 0 & 128 & 126 & 24 \\ 2 & 3 & -1 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

[971] $1^{122}2^3 21^{22} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 32 & 12 \\ 4 & 8 & 0 & 32 & 30 & 12 \\ 2 & 3 & -1 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[970] $1^{121}2^3 21^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 8 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 16 & 12 \\ 2 & 4 & 0 & 16 & 6 & 6 \\ 2 & 3 & -1 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

[967] $1^{133}2^3 21^{33} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 1 & 12 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 72 & 18 \\ 6 & 12 & 0 & 72 & 70 & 18 \\ 2 & 3 & -1 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[968] $1^{111}2^3 21^{33} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 6 & 2 \\ 6 & 0 & 6 & 34 & 36 & 12 \\ 6 & 0 & 6 & 36 & 34 & 12 \\ 2 & 0 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[980] $1^{111}2^3 21^{32} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 5 & 2 \\ 6 & 0 & 6 & 34 & 26 & 12 \\ 4 & 0 & 5 & 26 & 17 & 9 \\ 2 & 0 & 2 & 12 & 9 & 4 \end{pmatrix}_{b_0}$$

[979] $1^{112}2^3 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 20 & 6 \\ 6 & 9 & 0 & 20 & 49 & 15 \\ 2 & 3 & -1 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[981] $1^{122}2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

[982] $1^{123}2^3 21^{24} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 1 & 8 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 56 & 12 \\ 8 & 13 & 0 & 56 & 97 & 21 \\ 2 & 3 & -1 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[983] $1^{12}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

[984] $1^{12}1^3 2^3 3 21^{32} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 5 & 2 \\ 6 & 0 & 9 & 49 & 31 & 15 \\ 4 & 0 & 5 & 31 & 17 & 9 \\ 2 & 0 & 2 & 15 & 9 & 4 \end{pmatrix}_{b_0}$$

[985] $1^{12}2^3 3 21^{34} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 10 & 2 \\ 6 & 0 & 6 & 34 & 52 & 12 \\ 8 & 0 & 10 & 52 & 74 & 18 \\ 2 & 0 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[986] $1^{12}2^3 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 6 & 7 & 5 & 17 & 24 & 9 \\ 7 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[987] $1^{12}2^3 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 6 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[989] $1^{12}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 4 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[990] $1^{11}1^3 2^3 3 21^{22} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 5 & 2 \\ 4 & 0 & 5 & 17 & 19 & 9 \\ 4 & 0 & 5 & 19 & 17 & 9 \\ 2 & 0 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[988] $1^{12}2^3 3 21^{33} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 9 & 2 \\ 6 & 0 & 9 & 49 & 51 & 15 \\ 6 & 0 & 9 & 51 & 49 & 15 \\ 2 & 0 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[991] $1^{13}2^3 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[992] $1^{12}2^3 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[993] $1^{122}2^13^121^{34} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 10 & 2 \\ 6 & 0 & 9 & 49 & 62 & 15 \\ 8 & 0 & 10 & 62 & 74 & 18 \\ 2 & 0 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[995] $1^{13}2^13^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[996] $1^{13}2^13^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[997] $1^{114}2^13^121^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[994] $1^{112}2^13^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 6 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[998] $1^{113}2^13^121^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 28 & 6 \\ 8 & 13 & 0 & 28 & 97 & 21 \\ 2 & 3 & -1 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[999] $1^{12}2^13^121^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[1000] $1^{114}2^13^121^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 4 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1001] $1^{111}2^13^121^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[1002] $1^{111}2^13^121^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 4 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1003] \ 1^{\overline{114}2\overline{13}21^{\overline{112}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 9 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1004] \ 1^{\overline{112}2\overline{13}21^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -37, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 5 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1005] \ 1^{\overline{113}2\overline{13}21^{\overline{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -37, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1006] \ 1^{\overline{111}231^{\overline{131}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 7 & 3 \\ 2 & 0 & 4 & 2 & 7 & 4 \\ 4 & 0 & 7 & 7 & 14 & 8 \\ 2 & 0 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1007] \ 1^{\overline{123}2\overline{13}21^{\overline{122}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -42, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 9 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1008] \ 1^{\overline{122}2\overline{13}21^{\overline{112}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -42, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 5 & 7 & 5 & 17 & 24 & 9 \\ 6 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1010] \ 1^{\overline{122}2\overline{13}21^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -43, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 5 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1009] \ 1^{\overline{123}2\overline{13}21^{\overline{113}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -43, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1012] \ 1^{\overline{112}2\overline{13}21^{\overline{124}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -46, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 10 & 2 \\ 4 & 0 & 5 & 17 & 38 & 9 \\ 8 & 0 & 10 & 38 & 74 & 18 \\ 2 & 0 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1011] \ 1^{\overline{112}2\overline{13}21^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -46, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 12 & 6 \\ 4 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[1013] $1^{112}2^1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 12 & 6 \\ 6 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[1014] $1^{112}2^1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 4 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[1015] $1^{113}2^1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1017] $1^{114}2^1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 11 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1016] $1^{113}2^1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 9 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1018] $1^{132}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

[1019] $1^{132}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

[1021] $1^{112}2^1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1022] $1^{123}2^1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1020] $1^{111}2^1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1023] \ 1^{122} 2^3 21^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 4 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1024] \ 1^{122} 2^3 21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -63, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1025] \ 1^{122} 2^3 21^{44} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 10 & 2 \\ 8 & 0 & 10 & 74 & 76 & 18 \\ 8 & 0 & 10 & 76 & 74 & 18 \\ 2 & 0 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1026] \ 1^{134} 2^3 21^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1027] \ 1^{144} 2^3 21^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 11 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 10 & 14 & 10 & 74 & 86 & 18 \\ 11 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1028] \ 1^{114} 2^3 21^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1029] \ 1^{113} 2^3 21^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1030] \ 1^{123} 2^3 21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 6 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1031] \ 1^{123} 2^3 21^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1032] \ 1^{132} 2^3 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1033] \ 1^{122} 2^3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -77, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 6 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1034] \ 1^{142} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1035] \ 1^{142} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1036] \ 1^{144} 2^3 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 8 & 14 & 10 & 74 & 86 & 18 \\ 9 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1037] \ 1^{114} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 8 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1038] \ 1^{144} 2^3 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -83, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 9 & 14 & 10 & 74 & 86 & 18 \\ 10 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1039] \ 1^{122} 2^3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -85, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1040] \ 1^{124} 2^3 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 9 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1041] \ 1^{114} 2^3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 8 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1042] \ 1^{133} 2^3 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.6$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 9 & 11 & 9 & 49 & 60 & 15 \\ 10 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1043] $1^{113}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 6 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1044] $1^{113}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1045] $1^{114}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 8 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1046] $1^{113}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 4 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1047] $1^{123}2^3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 5 & 7 & 5 & 17 & 26 & 9 \\ 6 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1048] $1^{113}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 20 & 6 \\ 7 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1049] $1^{142}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

[1050] $1^{114}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 9 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1051] $1^{134}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 7 & 10 & 6 & 34 & 52 & 12 \\ 9 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1052] $1^{112}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 7 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1053] $1^{134}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 10 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1054] $1^{112}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 5 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1055] $1^{124}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1056] $1^{142}2^3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

[1057] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1058] $1^{123}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 7 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1059] $1^{133}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 8 & 11 & 9 & 49 & 60 & 15 \\ 9 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1061] $1^{114}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 10 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1060] $1^{123}2^3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 26 & 9 \\ 6 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1062] $1^{122}2^3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 4 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[1063] $1^{123}2^3 3^{21}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 40 & 12 \\ 6 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1064] $1^{123}2^3 3^{21}1^{24} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 4 & 4 & 6 & 20 & 6 \\ 9 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1066] $1^{123}2^3 3^{21}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 5 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1065] $1^{123}2^3 3^{21}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 40 & 12 \\ 7 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1069] $1^{134}2^3 3^{21}1^{23} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 6 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1068] $1^{122}2^3 3^{21}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[1067] $1^{133}2^3 3^{21}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 10 & 10 & 3 \\ 1 & 2 & 0 & 6 & 6 & 2 \\ 6 & 10 & 6 & 34 & 36 & 12 \\ 7 & 10 & 6 & 36 & 34 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1070] $1^{124}2^3 3^{21}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 8 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1071] $1^{112}2^3 3^{21}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1072] $1^{122}2^3 3^{21}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 7 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1073] $1^{14}2^3 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 2 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

[1074] $1^{12}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 5 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1076] $1^{14}2^3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1075] $1^{14}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 6 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1077] $1^{13}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 9 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1078] $1^{14}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1079] $1^{13}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 8 & 8 & 30 & 40 & 12 \\ 9 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1080] $1^{14}2^3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 7 & 8 & 8 & 30 & 56 & 12 \\ 11 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1081] $1^{13}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1082] $1^{14}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1083] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 7 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1084] $1^{123}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1086] $1^{114}2^3 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 2 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1085] $1^{114}2^3 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 4 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1087] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 7 & 5 & 17 & 31 & 9 \\ 9 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1088] $1^{113}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 7 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1089] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 9 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1090] $1^{113}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1094] $1^{123}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 7 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1093] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 6 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1092] $1^{123}2^3 3^{21}1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 5 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1091] $1^{134}2^3 3^{21}1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 8 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1095] $1^{134}2^3 3^{21}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 9 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1096] $1^{114}2^3 3^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1097] $1^{123}2^3 3^{21}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 7 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1098] $1^{134}2^3 3^{21}1^{33} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 9 & 12 & 12 & 70 & 84 & 18 \\ 10 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1099] $1^{124}2^3 3^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 10 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1100] $1^{122}2^3 3^{21}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 6 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1102] $1^{123}2^3 3^{21}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1101] $1^{133}2^3 3^{21}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 6 & 11 & 9 & 49 & 51 & 15 \\ 7 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1104] $1^{134}2^3 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 7 & 10 & 6 & 34 & 52 & 12 \\ 10 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1103] $1^{134}2^3 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 9 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1105] $1^{134}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 9 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1106] $1^{134}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 9 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1107] $1^{133}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1108] $1^{134}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 8 & 12 & 12 & 70 & 84 & 18 \\ 9 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1109] $1^{134}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 11 & 9 & 49 & 62 & 15 \\ 8 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1110] $1^{124}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 8 & 8 & 30 & 56 & 12 \\ 10 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1114] $1^{124}2^3 3 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 5 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[1112] $1^{113}2^3 3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 8 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1113] $1^{134}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1115] $1^{124}2^3 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 7 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[1111] $1^{124}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 10 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1116] $1^{133}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.6$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1117] $1^{134}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -234$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 8 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1119] $1^{134}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 8 & 12 & 12 & 70 & 84 & 18 \\ 10 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1118] $1^{114}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 9 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1121] $1^{123}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 10 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1122] $1^{123}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 8 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1120] $1^{113}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1123] $1^{124}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -253, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 9 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1124] $1^{114}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -255, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 9 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1125] $1^{124}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 9 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1126] $1^{124}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 9 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1127] $1^{124}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1128] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 10 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1129] $1^{134}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 9 & 12 & 12 & 70 & 84 & 18 \\ 11 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1130] $1^{134}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 9 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1131] $1^{134}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1132] $1^{124}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 8 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1133] $1^{134}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1134] $1^{124}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 8 & 8 & 30 & 56 & 12 \\ 9 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1136] $1^{133}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 10 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1135] $1^{133}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 8 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1137] $1^{124}2^3 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1138] $1^{144}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 8 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1139] $1^{124}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 11 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1140] $1^{114}2^3 3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1141] $1^{144}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1142] $1^{133}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 11 & 9 & 49 & 51 & 15 \\ 9 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1143] $1^{123}2^33^{21}1^{23} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1144] $1^{114}2^33^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1145] $1^{134}2^33^{21}1^{23} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 8 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1146] $1^{133}2^33^{21}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 11 & 9 & 49 & 51 & 15 \\ 7 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1147] $1^{133}2^33^{21}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1149] $1^{134}2^33^{21}1^{34} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 10 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[1148] $1^{134}2^33^{21}1^{34} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 8 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[1150] $1^{134}2^33^{21}1^{23} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -373, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 6 & 11 & 9 & 49 & 71 & 15 \\ 9 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1151] $1^{134}2^33^{21}1^{23} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 9 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1152] $1^{114}2^33^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 11 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1153] $1^{144}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 9 & 15 & 13 & 97 & 99 & 21 \\ 10 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1154] $1^{114}2^3 3 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1155] $1^{124}2^3 3 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 8 & 8 & 30 & 64 & 12 \\ 11 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[1156] $1^{144}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -437, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 11 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 10 & 15 & 13 & 97 & 99 & 21 \\ 11 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1158] $1^{133}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 9 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1157] $1^{144}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 8 & 14 & 10 & 74 & 76 & 18 \\ 8 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1159] $1^{134}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -453, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1160] $1^{134}2^3 3 21^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 9 & 12 & 12 & 70 & 96 & 18 \\ 11 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[1161] $1^{134}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 8 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1163] $1^{144}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 9 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1162] $1^{144}2^3 21^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 10 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1164] $1^{124}2^3 21^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[1165] $1^{144}2^3 21^{144} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 12 & 2 \\ 1 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 0 & 16 & 16 & 2 \\ 11 & 16 & 16 & 126 & 128 & 24 \\ 12 & 16 & 16 & 128 & 126 & 24 \\ 2 & 3 & 2 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

[1166] $1^{144}2^3 21^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 9 & 15 & 13 & 97 & 99 & 21 \\ 9 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1167] $1^{134}2^3 21^{134} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 9 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[1168] $1^{144}2^3 21^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 10 & 15 & 13 & 97 & 99 & 21 \\ 10 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1169] $1^{144}2^3 21^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 11 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 11 & 15 & 13 & 97 & 99 & 21 \\ 11 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1170] $1^{144}2^3 21^{144} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -780, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 12 & 2 \\ 1 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 0 & 16 & 16 & 2 \\ 12 & 16 & 16 & 126 & 128 & 24 \\ 12 & 16 & 16 & 128 & 126 & 24 \\ 2 & 3 & 2 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ [1174] $1^{111}314131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1171] $1^{11}4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 5 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[1172] $1^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 4 & 4 & 5 \\ 3 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 2 & 4 \\ 3 & 5 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

[1173] $1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[1175] $1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 4 & 5 & 4 \\ 3 & 4 & 0 & 5 & 5 & 5 \\ 2 & 3 & 0 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

[1176] $1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 3 & 0 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1177] $1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 4 & 6 & 1 & 6 & 14 & 8 \\ 2 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[1178] $1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -49, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[1179] $1^{\overline{1}} 1^{\overline{1}} 2^{\overline{1}} 2^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -6, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 3 & 1 & 8 & 5 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 8 & 0 & 15 & 8 & 9 \\ 2 & 5 & 0 & 8 & 3 & 5 \\ 2 & 5 & -1 & 9 & 5 & 5 \end{pmatrix}_{b_0}$$

[1180] $1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 4 & 0 & 5 & 4 & 5 \\ 2 & 3 & 1 & 4 & 2 & 4 \\ 2 & 3 & 0 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[1181] $1^{\overline{1}} 1^{\overline{1}} 2^{\overline{1}} 2^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -6, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 8 & 4 \\ 2 & 0 & 5 & 3 & 8 & 5 \\ 4 & 0 & 8 & 8 & 15 & 9 \\ 2 & 0 & 4 & 5 & 9 & 5 \end{pmatrix}_{b_0}$$

[1182] $1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 4 & 6 & 3 \\ 1 & 1 &$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 6 & 7 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 4 & 6 & 1 & 14 & 16 & 8 \\ 5 & 7 & 1 & 16 & 17 & 9 \\ 2 & 3 & 0 & 8 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1184] \ 1^{111} 31^{111} 4131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 3 & 1 & 2 & 7 & 4 \\ 4 & 6 & 1 & 7 & 14 & 8 \\ 2 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1185] \ 1^{112} 31^{111} 41^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 3 & 1 & 2 & 11 & 4 \\ 7 & 10 & 1 & 11 & 39 & 13 \\ 2 & 3 & 0 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[1186] \ 231^{111} 3151^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 428, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 5 & 2 \\ 5 & 33 & 1 & 35 & 35 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 35 & 0 & 33 & 35 & 11 \\ 5 & 35 & 0 & 35 & 33 & 11 \\ 2 & 11 & -1 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1189] \ 1^{11} 3151231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 4 & 0 & 2 & 4 \\ 2 & 0 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1188] \ 1^{111} 3151231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1191] \ 1^{11} 31^{111} 41321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 6 & 5 & 6 \\ 2 & 2 & 3 & 5 & 3 & 5 \\ 2 & 2 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1190] \ 1^{11} 61231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -47, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1192] \ 1^{111} 3151^{111} 231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 2 & 0 & 4 & 2 & 9 & 4 \\ 5 & 0 & 9 & 9 & 27 & 11 \\ 2 & 0 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1194] \ 1^{111} 31^{111} 41321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 3 \\ 2 & 2 & 5 & 8 & 4 \\ 4 & 5 & 7 & 12 & 7 \\ 6 & 8 & 12 & 19 & 11 \\ 3 & 4 & 7 & 11 & 6 \end{pmatrix}_{b_0}$$

$$[1193] \ 1^{111} 3151^{111} 231^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

5.0

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 2 & 0 & 4 & 2 & 9 & 4 \\ 6 & 0 & 9 & 9 & 27 & 11 \\ 2 & 0 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1196] 1^{12}3151^{22}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 0 & 18 & 0 & 114 & 22 \\ 2 & 0 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1195] 1^{114}3151^{22}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 2 & 1 & 4 & 26 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 26 & 4 \\ 11 & 26 & 0 & 26 & 202 & 30 \\ 2 & 4 & -1 & 4 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[1197] 1^{13}3151^{22}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 0 & 22 & 0 & 154 & 26 \\ 2 & 0 & 3 & 0 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1198] 2^1 3^{111} 15131^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 2 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 4 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1199] 2^1 3^{113} 15131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 8 & 2 \\ 2 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1200] 1^{112}3151^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 4 & 13 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 13 & 4 \\ 6 & 13 & 0 & 13 & 49 & 15 \\ 2 & 4 & -1 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1201] 1^{113}3151^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 17 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 17 & 4 \\ 7 & 17 & 0 & 17 & 79 & 19 \\ 2 & 4 & -1 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[1202] 2^1 3^{111} 15131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1203] 2^1 3^{112} 15131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -93, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 3 & 5 & 2 \\ 2 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0} \\
[1204] \quad & 1^{114}3151^{3231^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -107, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 31 & 3 \\ 2 & 0 & 4 & 2 & 31 & 4 \\ 17 & 0 & 31 & 31 & 317 & 37 \\ 2 & 0 & 3 & 4 & 37 & 4 \end{pmatrix}_{b_0} \\
[1205] \quad & 1^{112}3151^{1231^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -107, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 2 & 0 & 4 & 2 & 13 & 4 \\ 7 & 0 & 13 & 13 & 49 & 15 \\ 2 & 0 & 3 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[1206] \quad & 1^{113}3151^{2231^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 22 & 3 \\ 2 & 0 & 4 & 2 & 22 & 4 \\ 12 & 0 & 22 & 22 & 154 & 26 \\ 2 & 0 & 3 & 4 & 26 & 4 \end{pmatrix}_{b_0} \\
[1207] \quad & 1^{112}3151^{1231^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -133, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 2 & 0 & 4 & 2 & 13 & 4 \\ 8 & 0 & 13 & 13 & 49 & 15 \\ 2 & 0 & 3 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[1208] \quad & 2^1 3^{123} 15131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.3 \\
& T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 8 & 2 \\ 2 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0} \\
[1209] \quad & 2^1 3^{112} 15131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0} \\
[1210] \quad & 1^{113}3151^{2231^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -182, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 22 & 3 \\ 2 & 0 & 4 & 2 & 22 & 4 \\ 13 & 0 & 22 & 22 & 154 & 26 \\ 2 & 0 & 3 & 4 & 26 & 4 \end{pmatrix}_{b_0} \\
[1211] \quad & 2^1 3^{114} 15131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 10 & 2 \\ 2 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0} \\
[1212] \quad & 2^1 3^{122} 15131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -220, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 6 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1213] $2^1 3^{122} 15131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -220$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 5 & 2 \\ 2 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[1214] $2^1 3^{113} 15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -241$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.8$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 8 & 2 \\ 2 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1215] $2^1 3^{113} 15131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -268$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.8$

$T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 9 & 2 \\ 2 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1216] $2^1 3^{123} 15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -313$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.3$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 8 & 2 \\ 2 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1217] $1^{114} 3151^{2231} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -344$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.5$

$T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 26 & 3 \\ 2 & 0 & 4 & 2 & 26 & 4 \\ 14 & 0 & 26 & 26 & 202 & 30 \\ 2 & 0 & 3 & 4 & 30 & 4 \end{pmatrix}_{b_0}$$

[1218] $2^1 3^{124} 15131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -344$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 11 & 2 \\ 2 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1219] $2^1 3^{134} 15131^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -436$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 10 & 2 \\ 2 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1220] $2^1 3^{114} 15131^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -436$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 12 & 2 \\ 2 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1221] $2^1 3^{114} 15131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -437$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 11 & 2 \\ 2 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 1 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1222] \ 2 \ 3 \ 15131^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -508, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 10 & 2 \\ 2 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1223] \ 2 \ 3 \ 15131^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -516, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 8 & 2 \\ 2 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1224] \ 2 \ 3 \ 15131^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -796, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 10 & 2 \\ 2 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 10 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1225] \ 1^{13} 31 \ 5 \ 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$$

$$[1231] \ 1^{31} 31 \ 5 \ 13221^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$$

$$[1230] \ 1^{111} 31 \ 5 \ 13221^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[1229] \ 1^{114} 31 \ 5 \ 13221^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[1228] \ 1^{13} 231 \ 5 \ 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$$

$$[1227] \ 1^{11} 31 \ 5 \ 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[1226] \ 1^{12} 31 \ 5 \ 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.5$$

$$[1239] \ 1^{111} \ 5 \ 131 \ 5 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 4 & 4 & 5 \\ 3 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 2 & 4 \\ 3 & 5 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1240] \ 1^{11} \ 5 \ 131 \ 5 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 5 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1243] \ 1^{111} 314141^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 5 & 0 & 6 & 5 & 6 \\ 2 & 4 & 1 & 5 & 3 & 5 \\ 2 & 4 & 0 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1241] \ 1^{112} 2321^{11} 5131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 3 & 1 & 5 & 17 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 17 & 5 \\ 6 & 17 & 0 & 17 & 67 & 19 \\ 2 & 5 & -1 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

[1242] $1^{11}712231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -45, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1244] $1^{13}31^151321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 11 & 3 & 3 \\ 2 & 0 & 5 & 7 & 22 & 7 & 7 \\ 6 & 0 & 11 & 22 & 53 & 17 & 17 \\ 2 & 0 & 3 & 7 & 17 & 5 & 5 \end{pmatrix}_{b_0}$

[1245] $1^{11}31^151321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$

[1246] $1^{12}31^151321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 7 & 6 & 0 & 18 & 10 \\ 2 & 3 & 3 & 0 & 10 & 5 \end{pmatrix}_{b_0}$

[1247] $1^{12}31^{13}51321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 5 & 4 & 7 & 23 & 7 \\ 8 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$

[1248] $1^{111}31^{14}414131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -3, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 4 & 1 & 3 & 10 & 5 \\ 4 & 9 & 1 & 10 & 23 & 11 \\ 2 & 4 & 0 & 5 & 11 & 5 \end{pmatrix}_{b_0}$

[1249] $1^{112}31^151321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 5 & 4 & 7 & 23 & 7 \\ 6 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$

[1265] $2^13^15151321^{144} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 12 & 2 \\ 2 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 1 & 20 & 20 & 3 \\ 12 & 16 & 20 & 142 & 144 & 28 \\ 12 & 16 & 20 & 144 & 142 & 28 \\ 2 & 3 & 3 & 28 & 28 & 5 \end{pmatrix}_{b_0}$

[1266] $1^{33}31^151321^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 2 & 1 & 15 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 15 & 0 & 79 & 81 & 21 \\ 6 & 15 & 0 & 81 & 79 & 21 \\ 2 & 4 & -1 & 21 & 21 & 5 \end{pmatrix}_{b_0}$

[1267] $2^13^15151321^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 3 & 2 \\ 2 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 1 & 15 & 5 & 3 \\ 9 & 12 & 15 & 79 & 27 & 21 \\ 3 & 4 & 5 & 27 & 7 & 7 \\ 2 & 3 & 3 & 21 & 7 & 5 \end{pmatrix}_{b_0}$

$$\begin{aligned}
[1268] \quad & 1^{22} 31 \overset{1}{5} 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 2 & 1 & 10 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 36 & 14 \\ 4 & 10 & 0 & 36 & 34 & 14 \\ 2 & 4 & -1 & 14 & 14 & 5 \end{pmatrix}_{b_0} \\
[1269] \quad & 2 \overset{111}{3} 151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0} \\
[1250] \quad & 2 \overset{112}{3} 151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0} \\
[1251] \quad & 1^{11} 31 \overset{1}{5} 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 9 & 7 \\ 2 & 5 & 0 & 9 & 7 & 7 \\ 2 & 4 & -1 & 7 & 7 & 5 \end{pmatrix}_{b_0} \\
[1252] \quad & 1^{44} 31 \overset{1}{5} 1321^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 2 & 1 & 20 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 20 & 0 & 142 & 144 & 28 \\ 8 & 20 & 0 & 144 & 142 & 28 \\ 2 & 4 & -1 & 28 & 28 & 5 \end{pmatrix}_{b_0} \\
[1253] \quad & 1^{14} 31 \overset{1}{5} 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 5 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 36 & 7 \\ 8 & 20 & 0 & 36 & 142 & 28 \\ 2 & 4 & -1 & 7 & 28 & 5 \end{pmatrix}_{b_0} \\
[1254] \quad & 1^{34} 31 \overset{1}{5} 1321^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 2 & 1 & 15 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 15 & 0 & 79 & 108 & 21 \\ 8 & 20 & 0 & 108 & 142 & 28 \\ 2 & 4 & -1 & 21 & 28 & 5 \end{pmatrix}_{b_0} \\
[1255] \quad & 1^{13} 31 \overset{1}{5} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 5 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 27 & 7 \\ 6 & 15 & 0 & 27 & 79 & 21 \\ 2 & 4 & -1 & 7 & 21 & 5 \end{pmatrix}_{b_0} \\
[1256] \quad & 2 \overset{122}{3} 151321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 6 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0} \\
[1257] \quad & 1^{12} 31 \overset{1}{5} 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 18 & 7 \\ 4 & 10 & 0 & 18 & 34 & 14 \\ 2 & 4 & -1 & 7 & 14 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[1258] \quad & 2^1 3^{141} 151321^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 3 & 2 \\ 2 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 1 & 20 & 5 & 3 \\ 12 & 16 & 20 & 142 & 36 & 28 \\ 3 & 4 & 5 & 36 & 7 & 7 \\ 2 & 3 & 3 & 28 & 7 & 5 \end{pmatrix}_{b_0} \\
[1259] \quad & 2^1 3^{142} 151321^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 6 & 2 \\ 2 & 1 & 2 & 16 & 8 & 3 \\ 1 & 2 & 1 & 20 & 10 & 3 \\ 12 & 16 & 20 & 142 & 72 & 28 \\ 6 & 8 & 10 & 72 & 34 & 14 \\ 2 & 3 & 3 & 28 & 14 & 5 \end{pmatrix}_{b_0} \\
[1260] \quad & 1^{24} 31^1 51321^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 2 & 1 & 10 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 72 & 14 \\ 8 & 20 & 0 & 72 & 142 & 28 \\ 2 & 4 & -1 & 14 & 28 & 5 \end{pmatrix}_{b_0} \\
[1261] \quad & 2^1 3^{143} 151321^{143} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 9 & 2 \\ 2 & 1 & 2 & 16 & 12 & 3 \\ 1 & 2 & 1 & 20 & 15 & 3 \\ 12 & 16 & 20 & 142 & 108 & 28 \\ 9 & 12 & 15 & 108 & 79 & 21 \\ 2 & 3 & 3 & 28 & 21 & 5 \end{pmatrix}_{b_0} \\
[1262] \quad & 2^1 3^{132} 151321^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 6 & 2 \\ 2 & 1 & 2 & 12 & 8 & 3 \\ 1 & 2 & 1 & 15 & 10 & 3 \\ 9 & 12 & 15 & 79 & 54 & 21 \\ 6 & 8 & 10 & 54 & 34 & 14 \\ 2 & 3 & 3 & 21 & 14 & 5 \end{pmatrix}_{b_0} \\
[1263] \quad & 2^1 3^{133} 151321^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 9 & 2 \\ 2 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 1 & 15 & 15 & 3 \\ 9 & 12 & 15 & 79 & 81 & 21 \\ 9 & 12 & 15 & 81 & 79 & 21 \\ 2 & 3 & 3 & 21 & 21 & 5 \end{pmatrix}_{b_0} \\
[1264] \quad & 1^{23} 31^1 51321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 1 & 10 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 54 & 14 \\ 6 & 15 & 0 & 54 & 79 & 21 \\ 2 & 4 & -1 & 14 & 21 & 5 \end{pmatrix}_{b_0} \\
[1270] \quad & 1^{11} 31^1 51321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 2 & 4 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 4 & 1 & 2 & 9 & 4 \\ 4 & 9 & 5 & 9 & 21 & 11 \\ 2 & 4 & 2 & 4 & 11 & 5 \end{pmatrix}_{b_0} \\
[1271] \quad & 1^{12} 31^1 51321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 2 & 2 & 4 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 4 & 1 & 2 & 9 & 4 \\ 5 & 9 & 5 & 9 & 21 & 11 \\ 2 & 4 & 2 & 4 & 11 & 5 \end{pmatrix}_{b_0} \\
[1272] \quad & 1^{12} 31^1 51321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 13 & 7 \\ 4 & 7 & 6 & 13 & 18 & 10 \\ 2 & 3 & 3 & 7 & 10 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[1273] $1^{\overline{111}}3151321^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 2 & 2 & 5 & 5 & 4 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 5 & 4 & 7 & 9 & 7 \\ 3 & 5 & 4 & 9 & 7 & 7 \\ 2 & 4 & 2 & 7 & 7 & 5 \end{pmatrix}_{b_0}$

[1274] $1^{\overline{123}}3151321^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 18$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 10 & 0 & 34 & 14 \\ 2 & 3 & 3 & 0 & 14 & 5 \end{pmatrix}_{b_0}$

[1275] $1^{\overline{111}}3151321^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 4 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$

[1276] $1^{\overline{111}}3151321^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 2 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$

[1277] $1^{\overline{112}}3151321^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 5 & 4 & 7 & 28 & 7 \\ 9 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$

[1278] $1^{\overline{112}}3151321^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 5 & 4 & 7 & 28 & 7 \\ 9 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$

[1283] $1^{\overline{123}}3151321^{\overline{122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 1 & 10 & 11 & 3 \\ 5 & 8 & 10 & 34 & 44 & 14 \\ 6 & 11 & 11 & 44 & 53 & 17 \\ 2 & 3 & 3 & 14 & 17 & 5 \end{pmatrix}_{b_0}$

[1279] $1^{\overline{124}}3151321^{\overline{123}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 62 & 14 \\ 10 & 15 & 16 & 62 & 106 & 24 \\ 2 & 3 & 3 & 14 & 24 & 5 \end{pmatrix}_{b_0}$

[1280] $1^{\overline{112}}31^{\overline{14141}}31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -3$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 14 & 4 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 4 & 1 & 3 & 15 & 5 \\ 7 & 14 & 1 & 15 & 55 & 17 \\ 2 & 4 & 0 & 5 & 17 & 5 \end{pmatrix}_{b_0}$

[1281] $1^{\overline{111}}3151321^{\overline{123}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 13 & 4 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 9 & 5 & 21 & 32 & 11 \\ 7 & 13 & 6 & 32 & 43 & 15 \\ 2 & 4 & 2 & 11 & 15 & 5 \end{pmatrix}_{b_0}$

[1282] $1^{13}31\ 5\ 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 11 & 0 & 53 & 17 \\ 2 & 3 & 3 & 0 & 17 & 5 \end{pmatrix}_{b_0}$

[1285] $1^{13}31\ 5\ 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 11 & 0 & 53 & 17 \\ 2 & 3 & 3 & 0 & 17 & 5 \end{pmatrix}_{b_0}$

[1286] $1^{111}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 5 & 4 & 7 & 14 & 7 \\ 4 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0}$

[1287] $1^{111}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 5 & 4 & 7 & 14 & 7 \\ 5 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0}$

[1284] $1^{113}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 2 & 4 & 5 & 7 & 22 & 7 \\ 7 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$

[1288] $1^{112}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 34$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$

[1289] $1^{133}31\ 5\ 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 38$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 15 & 0 & 79 & 21 \\ 2 & 3 & 3 & 0 & 21 & 5 \end{pmatrix}_{b_0}$

[1290] $1^{111}31\ 5\ 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 40$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$

[1292] $1^{111}31\ 5\ 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 42$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 2 & 2 & 9 & 9 & 4 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 9 & 5 & 21 & 23 & 11 \\ 4 & 9 & 5 & 23 & 21 & 11 \\ 2 & 4 & 2 & 11 & 11 & 5 \end{pmatrix}_{b_0}$

[1291] $1^{123}31\ 5\ 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 42$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 4 & 18 & 4 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 4 & 1 & 2 & 18 & 4 \\ 8 & 18 & 10 & 18 & 90 & 22 \\ 2 & 4 & 2 & 4 & 22 & 5 \end{pmatrix}_{b_0}$

[1293] $1^{113}23151^1231^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 43$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} =$

5.0

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 3 & 1 & 5 & 20 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 20 & 5 \\ 7 & 20 & 0 & 20 & 88 & 22 \\ 2 & 5 & -1 & 5 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[1294] \ 1^{12} 23151^{12} 231^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 15 & 4 \\ 2 & 0 & 5 & 3 & 15 & 5 \\ 7 & 0 & 15 & 15 & 53 & 17 \\ 2 & 0 & 4 & 5 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[1295] \ 1^{13} 315^{13} 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 2 & 2 & 4 & 18 & 4 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 4 & 1 & 2 & 18 & 4 \\ 9 & 18 & 10 & 18 & 90 & 22 \\ 2 & 4 & 2 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[1296] \ 1^{14} 315^{13} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 53, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 16 & 0 & 106 & 24 \\ 2 & 3 & 3 & 0 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[1297] \ 1^{12} 315^{12} 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 5 & 4 & 7 & 28 & 7 \\ 8 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[1298] \ 1^{13} 315^{13} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 2 & 4 & 5 & 7 & 27 & 7 \\ 9 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[1299] \ 1^{14} 315^{13} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 61, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 31 & 7 \\ 10 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[1300] \ 1^{123} 315^{112} 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 33 & 10 \\ 8 & 11 & 11 & 33 & 53 & 17 \\ 2 & 3 & 3 & 10 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[1301] \ 1^{112} 2321^{113} 15131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 17 & 4 \\ 2 & 0 & 5 & 3 & 17 & 5 \\ 8 & 0 & 17 & 17 & 67 & 19 \\ 2 & 0 & 4 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[1302] \ 1^{12} 315^{12} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 5 & 4 & 7 & 23 & 7 \\ 7 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

[1303] $1^{112}31\ 5\ 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 67$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 5 & 4 & 7 & 23 & 7 \\ 7 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$

[1305] $1^{112}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 70$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 7 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$

[1304] $1^{112}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 70$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 5 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$

[1306] $1^{111}31\ 5\ 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 75$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 2 & 9 & 13 & 4 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 9 & 5 & 21 & 32 & 11 \\ 6 & 13 & 6 & 32 & 43 & 15 \\ 2 & 4 & 2 & 11 & 15 & 5 \end{pmatrix}_{b_0}$

[1307] $1^{111}31\ 5\ 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 80$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 2 & 2 & 9 & 9 & 4 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 9 & 5 & 21 & 23 & 11 \\ 4 & 9 & 5 & 23 & 21 & 11 \\ 2 & 4 & 2 & 11 & 11 & 5 \end{pmatrix}_{b_0}$

[1309] $1^{113}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 82$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 22 & 7 \\ 8 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$

[1310] $1^{122}31\ 5\ 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 82$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 5 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0}$

[1308] $1^{114}31\ 5\ 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 82$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 1 & 5 & 20 & 3 \\ 2 & 4 & 5 & 7 & 36 & 7 \\ 12 & 16 & 20 & 36 & 142 & 28 \\ 2 & 3 & 3 & 7 & 28 & 5 \end{pmatrix}_{b_0}$

[1311] $1^{124}31\ 5\ 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 87$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 1 & 6 & 16 & 3 \\ 4 & 7 & 6 & 18 & 46 & 10 \\ 9 & 15 & 16 & 46 & 106 & 24 \\ 2 & 3 & 3 & 10 & 24 & 5 \end{pmatrix}_{b_0}$

[1313] $1^{113}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 88$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 22 & 7 \\ 7 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$

[1312] $1^{112}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 88$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[1314] \ 1^{124}31^15 \ 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 1 & 6 & 16 & 3 \\ 4 & 7 & 6 & 18 & 46 & 10 \\ 10 & 15 & 16 & 46 & 106 & 24 \\ 2 & 3 & 3 & 10 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[1315] \ 1^{112}31^15 \ 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 93, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 9 & 5 & 21 & 37 & 11 \\ 7 & 14 & 9 & 37 & 58 & 18 \\ 2 & 4 & 2 & 11 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[1318] \ 1^{123}31^15 \ 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 1 & 10 & 15 & 3 \\ 5 & 8 & 10 & 34 & 54 & 14 \\ 9 & 12 & 15 & 54 & 79 & 21 \\ 2 & 3 & 3 & 14 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[1316] \ 1^{122}31^15 \ 1321^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 2 & 2 & 14 & 14 & 4 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 6 & 14 & 9 & 58 & 60 & 18 \\ 7 & 14 & 9 & 60 & 58 & 18 \\ 2 & 4 & 2 & 18 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[1317] \ 1^{123}31^15 \ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 33 & 10 \\ 7 & 11 & 11 & 33 & 53 & 17 \\ 2 & 3 & 3 & 10 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[1319] \ 1^{112}31^15 \ 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 111, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 9 & 5 & 21 & 37 & 11 \\ 6 & 14 & 9 & 37 & 58 & 18 \\ 2 & 4 & 2 & 11 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[1320] \ 1^{124}31^15 \ 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 115, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 62 & 14 \\ 9 & 15 & 16 & 62 & 106 & 24 \\ 2 & 3 & 3 & 14 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[1321] \ 1^{114}23151^2231^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 118, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 30 & 4 \\ 2 & 0 & 5 & 3 & 30 & 5 \\ 14 & 0 & 30 & 30 & 218 & 34 \\ 2 & 0 & 4 & 5 & 34 & 5 \end{pmatrix}_{b_0}$$

$$[1322] \ 1^{114}31^15 \ 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 121, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 31 & 7 \\ 9 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[1323] \ 1^{133}31^15 \ 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 125, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 1 & 11 & 11 & 3 \\ 6 & 11 & 11 & 53 & 55 & 17 \\ 7 & 11 & 11 & 55 & 53 & 17 \\ 2 & 3 & 3 & 17 & 17 & 5 \end{pmatrix}_{b_0}$$

[1324] $1^{122}31 \overset{12}{5} 1321^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 138, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 2 & 2 & 14 & 18 & 4 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 14 & 9 & 58 & 74 & 18 \\ 8 & 18 & 10 & 74 & 90 & 22 \\ 2 & 4 & 2 & 18 & 22 & 5 \end{pmatrix}_{b_0}$$

[1325] $1^{113}31 \overset{12}{5} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 150, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 5 & 7 & 27 & 7 \\ 8 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[1326] $1^{112}31 \overset{13}{5} 1321^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 158, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 2 & 2 & 9 & 18 & 4 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 9 & 5 & 21 & 46 & 11 \\ 9 & 18 & 10 & 46 & 90 & 22 \\ 2 & 4 & 2 & 11 & 22 & 5 \end{pmatrix}_{b_0}$$

[1327] $1^{114}31 \overset{12}{5} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 162, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 31 & 7 \\ 9 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[1328] $1^{113}31 \overset{13}{5} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 168, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 5 & 7 & 27 & 7 \\ 9 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[1329] $1^{112}31 \overset{12}{5} 1321^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 188, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 2 & 2 & 9 & 18 & 4 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 9 & 5 & 21 & 46 & 11 \\ 8 & 18 & 10 & 46 & 90 & 22 \\ 2 & 4 & 2 & 11 & 22 & 5 \end{pmatrix}_{b_0}$$

[1330] $1^{114}31 \overset{13}{5} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 31 & 7 \\ 10 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[1331] $1^{114}31 \overset{13}{5} 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 262, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 5 & 7 & 36 & 7 \\ 11 & 16 & 20 & 36 & 142 & 28 \\ 2 & 3 & 3 & 7 & 28 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [1333] $1^1 4 1414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.0$

[1332] $1^{112}31^1 414141^1 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.5$

[1337] $1^1 6 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{aligned}
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1346] & 1^{112} 321^1 612321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 11 & 3 \\ 2 & 2 & 5 & 14 & 4 \\ 4 & 5 & 7 & 22 & 7 \\ 11 & 14 & 22 & 62 & 20 \\ 3 & 4 & 7 & 20 & 6 \end{pmatrix}_{b_0} \\
[1345] & 1^{111} 321^1 612321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 0 & 3 & 7 & 2 \\ 2 & 2 & 3 & 3 & 13 & 5 \\ 7 & 6 & 7 & 13 & 39 & 15 \\ 2 & 2 & 2 & 5 & 15 & 5 \end{pmatrix}_{b_0} \\
[1347] & 1^{114} 321^1 61^1 2321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 19 & 3 \\ 2 & 2 & 5 & 24 & 4 \\ 4 & 5 & 7 & 36 & 7 \\ 19 & 24 & 36 & 178 & 34 \\ 3 & 4 & 7 & 34 & 6 \end{pmatrix}_{b_0} \\
[1348] & 1^{113} 321^1 61^1 2321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -54, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 18 & 3 \\ 2 & 2 & 5 & 21 & 4 \\ 4 & 5 & 7 & 33 & 7 \\ 18 & 21 & 33 & 145 & 31 \\ 3 & 4 & 7 & 31 & 6 \end{pmatrix}_{b_0} \\
[1349] & 1^{113} 32161^1 2321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 12 & 3 \\ 2 & 2 & 5 & 15 & 4 \\ 4 & 5 & 7 & 23 & 7 \\ 12 & 15 & 23 & 67 & 21 \\ 3 & 4 & 7 & 21 & 6 \end{pmatrix}_{b_0} \\
[1350] & 1^{114} 321^1 61^1 2321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -196, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 22 & 3 \\ 2 & 2 & 5 & 26 & 4 \\ 4 & 5 & 7 & 42 & 7 \\ 22 & 26 & 42 & 220 & 38 \\ 3 & 4 & 7 & 38 & 6 \end{pmatrix}_{b_0} \\
T = 14 & [1351] 1^1 \overset{11}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 5.0 \\
[1360] & 1^1 \overset{11}{7} 1231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1359] & 1^1 \overset{11}{7} 1231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1358] & 1^1 \overset{11}{7} 1231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 5 & 4 & 5 \\ 2 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
[1357] & 1^{\overline{11}} 7^{11} 1231^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
[1356] & 21^{\overline{11}} 7^{11} 1231^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 2 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0} \\
[1355] & 1^{\overline{11}} 7^{11} 12231^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 5 \end{pmatrix}_{b_0} \\
[1362] & 1^{\overline{11}} 5^{11} 1321^{\overline{11}} 7^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1361] & 1^{\overline{11}} 7^{11} 1232151^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -41, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1369] & 2^{\overline{141}} 3^{141} 21712321^{\overline{141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 3 & 2 \\ 2 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 1 & 20 & 5 & 3 \\ 12 & 16 & 20 & 142 & 36 & 28 \\ 3 & 4 & 5 & 36 & 7 & 7 \\ 2 & 3 & 3 & 28 & 7 & 5 \end{pmatrix}_{b_0} \\
[1363] & 2^{\overline{111}} 3^{111} 21712321^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0} \\
[1368] & 2^{\overline{142}} 3^{142} 21712321^{\overline{142}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 6 & 2 \\ 2 & 1 & 2 & 16 & 8 & 3 \\ 1 & 2 & 1 & 20 & 10 & 3 \\ 12 & 16 & 20 & 142 & 72 & 28 \\ 6 & 8 & 10 & 72 & 34 & 14 \\ 2 & 3 & 3 & 28 & 14 & 5 \end{pmatrix}_{b_0} \\
[1367] & 2^{\overline{122}} 3^{122} 21712321^{\overline{122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 6 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0} \\
[1366] & 2^{\overline{132}} 3^{132} 21712321^{\overline{132}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 6 & 2 \\ 2 & 1 & 2 & 12 & 8 & 3 \\ 1 & 2 & 1 & 15 & 10 & 3 \\ 9 & 12 & 15 & 79 & 54 & 21 \\ 6 & 8 & 10 & 54 & 34 & 14 \\ 2 & 3 & 3 & 21 & 14 & 5 \end{pmatrix}_{b_0}$$

[1365] $2 \overset{1^{121}}{3} 21712321^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 3 & 2 \\ 2 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 1 & 10 & 5 & 3 \\ 6 & 8 & 10 & 34 & 18 & 14 \\ 3 & 4 & 5 & 18 & 7 & 7 \\ 2 & 3 & 3 & 14 & 7 & 5 \end{pmatrix}_{b_0}$$

[1364] $2 \overset{1^{131}}{3} 21712321^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 3 & 2 \\ 2 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 1 & 15 & 5 & 3 \\ 9 & 12 & 15 & 79 & 27 & 21 \\ 3 & 4 & 5 & 27 & 7 & 7 \\ 2 & 3 & 3 & 21 & 7 & 5 \end{pmatrix}_{b_0}$$

$T = 15$ [1370] $1 \overset{1^{131}}{6} 1316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$

[1380] $1 \overset{1^1}{8} 12231 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[1381] $21 \overset{1^1}{8} 1231 \overset{1^{11}}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 5 & 6 & 5 \\ 2 & 2 & 3 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$

[1379] $1 \overset{21^1}{8} 1231 \overset{1^{11}}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[1378] $21 \overset{1^1}{8} 1231 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 3 & 2 & 3 & 5 & 2 & 5 \\ 1 & 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[1377] $21 \overset{1^1}{8} 1231 \overset{1^{11}}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 4 & 2 & 2 \\ 4 & 3 & 4 & 11 & 5 & 7 \\ 1 & 1 & 2 & 5 & 1 & 3 \\ 2 & 2 & 2 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[1376] $221 \overset{1^1}{8} 1231 \overset{1^{11}}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 3 & 0 & 12 & 8 \\ 3 & 2 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

[1375] $1 \overset{1^1}{8} 12231 \overset{1^{11}}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1374] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 8 12231 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$$

5.0

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 5 & 4 & 5 \\ 2 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1372] \ 21^{\overline{1}} \overline{1}^{\overline{1}} 8 1231 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1371] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 8 12231 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1373] \ 21^{\overline{1}} \overline{1}^{\overline{1}} 8 12231 \ 5 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$$

5.0

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 2 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1382] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 13221 \ 8 \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [1383] \ 1^{\overline{1}} \overline{1}^{\overline{1}} \overline{1}^{\overline{1}} 5 1321 \ 8 \ 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 5.0$$

$$[1387] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 6 1316 13151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} =$$

5.0

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1388] \ 1^{\overline{1}} \overline{1}^{\overline{1}} \overline{1}^{\overline{1}} 6 1316 13151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} =$$

5.0

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 2 & 0 & 5 & 3 & 5 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1389] \ 31^{\overline{1}} \overline{1}^{\overline{1}} \overline{1}^{\overline{1}} \overline{1}^{\overline{1}} 6 \ 1316131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} =$$

5.5

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 5 & 2 \\ 2 & 6 & 1 & 8 & 16 & 6 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 8 & 1 & 7 & 17 & 7 \\ 5 & 16 & 1 & 17 & 31 & 13 \\ 2 & 6 & 0 & 7 & 13 & 5 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [1390] \quad 1^{1^1} 6^{1^1} 12321^1 8^{1^1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

$$[1392] \quad 1^{1^1} 8^{1^1} 1231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 18, \Delta = -40, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [1395] \quad 1^{1^1} 8^{1^1} 12321^1 8^{1^1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 19, 2), c_{\text{ext}} = 5.0$$

$$[1394] \quad 1^{1^1} 8^{1^1} 12321^1 8^{1^1} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 5.0$$

$$[1397] \quad 31^{31} 6^{1^1} 1316131^1 6^{1^1} 131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 5.0$$

$$[1396] \quad 1^{1^1} 5^{1^1} 13161316131^1 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -12, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 5.0$$

$$[1399] \quad 1^{1^1} 7^{1^1} 12321^1 8^{1^1} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, T_{\max} = 20, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 23 \quad [1402] \quad 1^{1^1} 5^{1^1} 132181232181231^1 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$$

$$T = 24 \quad [1403] \quad 1^{1^1} 6^{1^1} 1232181232181231^1 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 5.0$$

$$T = 25 \quad [1405] \quad 1^{1^1} 8^{1^1} 12321^1 8^{1^1} 123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -36, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 2 & 0 & 5 & 2 & 5 \\ 1 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 33 \quad [1407] \quad 1^{1^1} 5^{1^1} 13221^{1^1} 12231513221^{1^1} 12231^1 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 33, T_{\max} = 33, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 5.0$$

$$T = 36 \quad [1408] \quad 1^{1^1} 5^{1^1} 13221^{1^1} 12231513221^{1^1} 122315131^1 6^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 36, T_{\max} = 36, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 5.0$$

$$T = 44 \quad [1409] \quad 1^{1^1} 5^{1^1} 13221^{1^1} (12) 12231513221^{1^1} 12231^1 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 44, T_{\max} = 44, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 5.0$$

4.879 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

4.880 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 6 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.881 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 13 & 27 & 0 & 7 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.882 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 9 & 5 \\ 11 & 20 & 17 & 33 & 0 & 17 & 8 \\ 9 & 17 & 13 & 27 & 0 & 15 & 7 \\ 17 & 33 & 27 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 27 & 0 & 13 & 7 \\ 5 & 8 & 7 & 13 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

4.883 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 18.25 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & 3 & -4 & 18 & -6 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 5 & 9 & 0 & 17 & 4 \\ 3 & 5 & 4 & 9 & 0 & 15 & 4 \\ 5 & 9 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 27 & 0 & 49 & 13 \\ 3 & 4 & 4 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 4 & 3 \\ 3 & 4 & 4 & 9 & 0 & 8 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 12 & 0 & 10 & 6 \\ 3 & 4 & 3 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 17.66 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 5 & 4 & 0 & 28 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 30 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 3 & -4 & 22 & -5 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 4 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3 1^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 14 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 15 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 3 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -170, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{112}} 3 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 5 & 9 & 0 & 5 & 4 \\ 3 & 5 & 4 & 9 & 0 & 6 & 4 \\ 5 & 9 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 9 & 0 & 4 & 4 \\ 3 & 4 & 4 & 7 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1113}} 3 1^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.10 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & 3 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1112}} 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{1113}} 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -296, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1112}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 16 & 3 \\ 3 & 4 & 4 & 9 & 0 & 28 & 4 \\ 2 & 4 & 1 & 6 & 0 & 18 & 3 \\ 5 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 48 & 0 & 154 & 22 \\ 3 & 4 & 3 & 7 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 5 & 4 & 0 & 23 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 24 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & 3 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 14 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 24 & 0 & 37 & 11 \\ 3 & 4 & 3 & 7 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1115}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.1$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 15 & 15 & 0 & 73 & 15 \\ 3 & 4 & 3 & 3 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 5 & 9 & 0 & 13 & 4 \\ 3 & 5 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 21 & 0 & 28 & 10 \\ 3 & 4 & 4 & 7 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -490, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 5 & 4 & 0 & 26 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 24 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 5 & 9 & 0 & 9 & 4 \\ 3 & 5 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 4 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 18 & 18 & 0 & 106 & 18 \\ 3 & 4 & 3 & 3 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 21 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 3 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 21 & 21 & 0 & 145 & 21 \\ 3 & 4 & 3 & 3 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 21 & 4 \\ 2 & 4 & 1 & 6 & 0 & 15 & 3 \\ 5 & 9 & 6 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 33 & 0 & 76 & 16 \\ 3 & 4 & 3 & 7 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{113} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.66 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 3 & 0 & 33 & 4 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 3 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.34 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 7 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -788, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 1 & 6 & 0 & 12 & 3 \\ 5 & 9 & 6 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 3 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -872, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1112} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -890, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 13 & 3 \\ 3 & 4 & 4 & 9 & 0 & 23 & 4 \\ 2 & 4 & 1 & 6 & 0 & 15 & 3 \\ 5 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 39 & 0 & 100 & 18 \\ 3 & 4 & 3 & 7 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{114} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -988, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 18 & 3 \\ 3 & 4 & 4 & 9 & 0 & 32 & 4 \\ 2 & 4 & 1 & 6 & 0 & 21 & 3 \\ 5 & 9 & 6 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 54 & 0 & 193 & 25 \\ 3 & 4 & 3 & 7 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{1114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{1114}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1836$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 15 & 3 \\ 3 & 4 & 4 & 9 & 0 & 27 & 4 \\ 2 & 4 & 1 & 6 & 0 & 18 & 3 \\ 5 & 9 & 6 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 45 & 0 & 133 & 21 \\ 3 & 4 & 3 & 7 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{1114}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1850$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 14 & 3 \\ 3 & 4 & 4 & 9 & 0 & 26 & 4 \\ 2 & 4 & 1 & 6 & 0 & 18 & 3 \\ 5 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 42 & 0 & 118 & 20 \\ 3 & 4 & 3 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{1114}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -2618$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 17 & 3 \\ 3 & 4 & 4 & 9 & 0 & 31 & 4 \\ 2 & 4 & 1 & 6 & 0 & 21 & 3 \\ 5 & 9 & 6 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 51 & 0 & 172 & 24 \\ 3 & 4 & 3 & 7 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{1114}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -2970$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 20 & 3 \\ 3 & 4 & 4 & 9 & 0 & 36 & 4 \\ 2 & 4 & 1 & 6 & 0 & 24 & 3 \\ 5 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 60 & 0 & 238 & 28 \\ 3 & 4 & 3 & 7 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

4.884 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{so}_8$

$T = 10$ $[0] 1^{111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 372$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 9 & 5 \\ 11 & 20 & 17 & 33 & 0 & 17 & 8 \\ 9 & 17 & 13 & 27 & 0 & 15 & 7 \\ 17 & 33 & 27 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 27 & 0 & 13 & 7 \\ 5 & 8 & 7 & 13 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

4.885 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$

$T = 10$ $[0] 1^{111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 6 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

- [1] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 & \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [2] $1^{1111}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 220$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 14 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 24 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 24 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$
- [3] $1^{1121}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 274$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{1131}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 454$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 27 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 660$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & 3 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{1131}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 676$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 22 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 21 & 15 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{1141}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 706$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 36 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 36 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 24 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 712$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{11111}} 3 1^{\overline{11111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 724$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1121}} 3 1^{\overline{11111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 796$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 12 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 21 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 21 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{1111}} 3 1^{\overline{11111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 988$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 9 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1032$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 3 & 3 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{1141}} 3 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1180$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 31 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 30 & 21 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.886 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$

$T = 9$ [0] $1^{\overline{11}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 16.86 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 3 & -4 & 3 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 16.71 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & 3 & -4 & 10 & -4 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1113}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.32 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 48 & 12 \\ 3 & 4 & 4 & 3 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{1121}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 27 & 9 \\ 3 & 4 & 4 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{1111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -341$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -359$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 3 & 4 & 3 & 7 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -485$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{1111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -729$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 12 & 6 \\ 3 & 4 & 3 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.887 NHC(-2)(-3) + su₂ + su₂ + e₆ + su₃ + su₂

$T = 10$ [0] $1^{1111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 6 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 9 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & 7 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 3 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 220$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 9 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 24 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 24 & 0 & 9 & 37 & 11 \\ 3 & 4 & 3 & 7 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 274$, $(n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 18 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & 3 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1113} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 27 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & 3 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1113} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 21 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 21 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & 3 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 36 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 24 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & 3 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 12 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 21 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & 7 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 988, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 9 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1032, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1180, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 30 & 4 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 21 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 & 3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 21 & 2 & 2 \\ 17 & 31 & 30 & 21 & 0 & 21 & 172 & 24 & 0 \\ 3 & 4 & 4 & 3 & -4 & 2 & 24 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.888 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 228, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & 3 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 651$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.889 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4$

$T = 8$ [0] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 & 3 \\ 3 & 4 & 4 & 3 & -3 & 1 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 & 3 \\ 3 & 4 & 3 & 3 & -3 & 1 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 135$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 & 3 \\ 3 & 4 & 4 & 7 & -3 & 1 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 149$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 & 3 \\ 3 & 4 & 3 & 7 & -3 & 1 & 1 \end{pmatrix}_{b_0}$$

4.890 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$

$T = 9$ [0] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 & 8 \\ 9 & 17 & 13 & 12 & 0 & 7 & 7 \\ 8 & 16 & 12 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 & 3 \\ 5 & 8 & 7 & 6 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 & 8 \\ 9 & 17 & 13 & 27 & 0 & 7 & 7 \\ 17 & 33 & 27 & 49 & 0 & 13 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 & 3 \\ 5 & 8 & 7 & 13 & -3 & 1 & 1 \end{pmatrix}_{b_0}$$

4.891 $\mathbf{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -3 & 2 & 1 \end{pmatrix} b_0$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{u}_2 + \textcolor{orange}{s} \textcolor{orange}{u}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 3 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 4 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad \begin{array}{c} 1 \\ 1 \\ 1 \end{array} \begin{array}{c} 1 \\ 1 \\ 1 \end{array} \begin{array}{c} 1 \\ 1 \\ 1 \end{array} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 \\ 5 & 9 & 6 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.892 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overset{1^1}{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}\textcolor{brown}{3}}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{brown}{s}\textcolor{blue}{u}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{red}{s}\textcolor{blue}{u}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \overset{1}{1} \overset{111}{3} \overset{1111}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{violet}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 5 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 & 0 \\ 3 & 4 & 4 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad \begin{array}{c} 1^1 \\ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{violet}{su}_3 \\ (1, 10, 0), c_{\text{ext}} = 13.2 \end{array} \quad T = 10, T_{\text{max}} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) =$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 6 & 0 & 5 & 12 & 6 \\ 3 & 4 & 3 & 3 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 5 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 6 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.893 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{\overline{111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 5 & 9 & 0 & 7 & 4 \\ 3 & 5 & 4 & 9 & 0 & 6 & 4 \\ 5 & 9 & 9 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 11 & 0 & 7 & 5 \\ 3 & 4 & 4 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 20 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 18 & 14 & 0 & 69 & 15 \\ 3 & 4 & 4 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 12 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 6 & 3 \\ 3 & 4 & 4 & 9 & 0 & 11 & 4 \\ 2 & 4 & 1 & 6 & 0 & 8 & 3 \\ 5 & 9 & 6 & 13 & 0 & 17 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 17 & 0 & 20 & 8 \\ 3 & 4 & 3 & 7 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 16 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 15 & 11 & 0 & 44 & 12 \\ 3 & 4 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{111}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 3 & 0 & 17 & 3 \\ 2 & 4 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 17 & 0 & 100 & 18 \\ 3 & 4 & 3 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{1114}3^11^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 1 & 3 & 0 & 23 & 3 \\ 2 & 4 & 3 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 19 & 34 & 23 & 23 & 0 & 208 & 26 \\ 3 & 4 & 3 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{1114}3^11^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 1 & 3 & 0 & 20 & 3 \\ 2 & 4 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 20 & 20 & 0 & 148 & 22 \\ 3 & 4 & 3 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

4.894 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 9 & 0 & 27 & 9 \\ 3 & 4 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}3^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -269, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 4 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -320, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -381, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 12 & 6 \\ 3 & 4 & 3 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.895 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 2 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 12 & 5 & 6 \\ 2 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 2 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 & 2 \\ 2 & 3 & 2 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.896 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 10, \Delta = -34, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 3 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

4.897 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 1 & 6 \\ 15 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

- [9] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 4 & 5 & 4 & 12 & 0 & 4 \\ 8 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 13 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 13 & 7 \\ 6 & 8 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$
- [11] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 4 & 5 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$
- [14] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 5 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$
- [16] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$
- [17] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 4 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$
- [19] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$
- [20] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 39 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 39 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 5 & 8 & 10 & 12 & 0 & 6 \\ 6 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 7 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 13 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

- [31] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$
- [32] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$
- [33] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 9 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 36 & 6 \\ 15 & 20 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 36 & 50 & 0 & 118 & 20 \\ 6 & 8 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$
- [34] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 14 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 14 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 & 3 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 6 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 6 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$
- [40] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 6 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$
- [41] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 5 & 8 & 10 & 30 & 0 & 6 \\ 13 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 4 & 9 & 0 & 4 \\ 6 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 24 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{13} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 10 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{112} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{112} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 9 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{12} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 9 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 38 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 38 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{13} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 11 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{12} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 8 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{12} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 1 & 3 \\ 3 & 4 & 5 & 18 & 0 & 4 \\ 4 & 5 & 4 & 18 & 0 & 4 \\ 11 & 18 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 37 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 37 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 51 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 51 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 5 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 177, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 6 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{112} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 1 & 3 \\ 3 & 4 & 14 & 13 & 0 & 4 \\ 9 & 14 & 37 & 33 & 0 & 11 \\ 8 & 13 & 33 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{11} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 5 & 14 & 0 & 4 \\ 4 & 5 & 4 & 15 & 0 & 4 \\ 9 & 14 & 15 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{113} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 1 & 3 \\ 3 & 4 & 14 & 22 & 0 & 4 \\ 9 & 14 & 37 & 57 & 0 & 11 \\ 13 & 22 & 57 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 3 & 4 & 1 & 12 & 0 & 3 \\ 11 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[63] $21^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 42 & 7 \\ 17 & 20 & 17 & 50 & 8 \\ 15 & 17 & 13 & 42 & 7 \\ 42 & 50 & 42 & 118 & 20 \\ 7 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 14 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 10 & 14 & 15 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{122} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 5 & 8 & 10 & 12 & 0 & 6 \\ 5 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 9 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 5 & 8 & 10 & 18 & 0 & 6 \\ 8 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 14 & 0 & 4 \\ 5 & 9 & 13 & 24 & 0 & 7 \\ 9 & 14 & 24 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 3 & 4 & 1 & 12 & 0 & 3 \\ 10 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{114} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 13 & 33 & 0 & 7 \\ 13 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 6 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{13} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 1 & 3 \\ 3 & 4 & 5 & 22 & 0 & 4 \\ 3 & 5 & 4 & 21 & 0 & 4 \\ 16 & 22 & 21 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{113} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 1 & 3 \\ 3 & 4 & 9 & 22 & 0 & 4 \\ 5 & 9 & 13 & 36 & 0 & 7 \\ 15 & 22 & 36 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{12} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 5 & 18 & 0 & 4 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 12 & 18 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 10 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 1 & 3 \\ 3 & 4 & 9 & 14 & 0 & 4 \\ 6 & 9 & 13 & 24 & 0 & 7 \\ 9 & 14 & 24 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{112} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 398, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 1 & 3 \\ 3 & 4 & 9 & 18 & 0 & 4 \\ 5 & 9 & 13 & 30 & 0 & 7 \\ 12 & 18 & 30 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 11 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{13} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 401, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 1 & 3 \\ 3 & 4 & 5 & 26 & 0 & 4 \\ 3 & 5 & 4 & 24 & 0 & 4 \\ 17 & 26 & 24 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 522$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 18 & 0 & 4 \\ 5 & 9 & 13 & 30 & 0 & 7 \\ 11 & 18 & 30 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{12} 3^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 1 & 3 \\ 3 & 4 & 5 & 23 & 0 & 4 \\ 3 & 5 & 4 & 24 & 0 & 4 \\ 17 & 23 & 24 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{111} 3^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 560$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 1 & 3 \\ 3 & 4 & 14 & 14 & 0 & 4 \\ 9 & 14 & 37 & 39 & 0 & 11 \\ 9 & 14 & 39 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{113} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 564$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 13 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{112} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 579$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 1 & 3 \\ 3 & 4 & 9 & 23 & 0 & 4 \\ 5 & 9 & 13 & 39 & 0 & 7 \\ 15 & 23 & 39 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{12} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 584$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 5 & 23 & 0 & 4 \\ 3 & 5 & 4 & 24 & 0 & 4 \\ 15 & 23 & 24 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{112} 3^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 588$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 1 & 3 \\ 3 & 4 & 14 & 18 & 0 & 4 \\ 9 & 14 & 37 & 48 & 0 & 11 \\ 11 & 18 & 48 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{113} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 596$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 15 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{12} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 596$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 1 & 3 \\ 3 & 4 & 5 & 23 & 0 & 4 \\ 3 & 5 & 4 & 24 & 0 & 4 \\ 16 & 23 & 24 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 602$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 1 & 3 \\ 3 & 4 & 9 & 18 & 0 & 4 \\ 6 & 9 & 13 & 30 & 0 & 7 \\ 11 & 18 & 30 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 14 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 15 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 1 & 3 \\ 3 & 4 & 9 & 26 & 0 & 4 \\ 5 & 9 & 13 & 42 & 0 & 7 \\ 17 & 26 & 42 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{122} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 1 & 3 \\ 3 & 4 & 13 & 18 & 0 & 4 \\ 8 & 13 & 28 & 42 & 0 & 10 \\ 11 & 18 & 42 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{113} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 1 & 3 \\ 3 & 4 & 9 & 22 & 0 & 4 \\ 6 & 9 & 13 & 36 & 0 & 7 \\ 13 & 22 & 36 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 8 & 13 & 28 & 30 & 0 & 10 \\ 8 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{113} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 1 & 3 \\ 3 & 4 & 9 & 22 & 0 & 4 \\ 5 & 9 & 13 & 36 & 0 & 7 \\ 14 & 22 & 36 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 1 & 18 & 0 & 3 \\ 18 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 13 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 704, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 1 & 3 \\ 3 & 4 & 4 & 32 & 0 & 4 \\ 2 & 4 & 1 & 21 & 0 & 3 \\ 21 & 32 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 25 & -2 & 2 \end{pmatrix} b_0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 14 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 1 & 18 & 0 & 3 \\ 17 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 21 & -2 & 2 \end{pmatrix}_b$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 8 & 13 & 28 & 39 & 0 & 10 \\ 10 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 13 & -2 & 2 \end{pmatrix} b_{\alpha}$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 32 & 0 & 4 \\ 2 & 4 & 1 & 21 & 0 & 3 \\ 20 & 32 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 25 & -2 & 2 \end{pmatrix}_b$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 21 & 1 & 3 \\ 3 & 4 & 9 & 31 & 0 & 4 \\ 5 & 9 & 13 & 51 & 0 & 7 \\ 21 & 31 & 51 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 24 & -2 & 2 \end{pmatrix}_b$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 1 & 3 \\ 3 & 4 & 9 & 27 & 0 & 4 \\ 5 & 9 & 13 & 45 & 0 & 7 \\ 18 & 27 & 45 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 21 & -2 & 2 \end{pmatrix}_b$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 1 & 3 \\ 3 & 4 & 5 & 27 & 0 & 4 \\ 3 & 5 & 4 & 27 & 0 & 4 \\ 17 & 27 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 21 & -2 & 2 \end{pmatrix}_b$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 1 & 3 \\ 3 & 4 & 5 & 27 & 0 & 4 \\ 3 & 5 & 4 & 27 & 0 & 4 \\ 19 & 27 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 21 & -2 & 2 \end{pmatrix}_b$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 1 & 3 \\ 3 & 4 & 5 & 27 & 0 & 4 \\ 3 & 5 & 4 & 27 & 0 & 4 \\ 18 & 27 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 21 & -2 & 2 \end{pmatrix}_b$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 1 & 3 \\ 3 & 4 & 5 & 28 & 0 & 4 \\ 3 & 5 & 4 & 30 & 0 & 4 \\ 20 & 28 & 30 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{133} 1^{111} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 849, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 1 & 3 \\ 3 & 4 & 17 & 22 & 0 & 4 \\ 10 & 17 & 49 & 66 & 0 & 13 \\ 13 & 22 & 66 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{14} 1^{21} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 1 & 3 \\ 3 & 4 & 5 & 31 & 0 & 4 \\ 3 & 5 & 4 & 30 & 0 & 4 \\ 19 & 31 & 30 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{14} 1^{21} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 874, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 1 & 3 \\ 3 & 4 & 9 & 26 & 0 & 4 \\ 5 & 9 & 13 & 42 & 0 & 7 \\ 16 & 26 & 42 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{14} 1^{41} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 884, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 1 & 18 & 0 & 3 \\ 18 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{14} 1^{41} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 1 & 3 \\ 3 & 4 & 5 & 31 & 0 & 4 \\ 3 & 5 & 4 & 30 & 0 & 4 \\ 21 & 31 & 30 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{14} 1^{21} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 1 & 18 & 0 & 3 \\ 16 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{14} 1^{31} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 1 & 3 \\ 3 & 4 & 5 & 31 & 0 & 4 \\ 3 & 5 & 4 & 30 & 0 & 4 \\ 20 & 31 & 30 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{133} 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 1 & 3 \\ 3 & 4 & 17 & 17 & 0 & 4 \\ 10 & 17 & 49 & 51 & 0 & 13 \\ 10 & 17 & 51 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{123} 1^{111} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 945, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 1 & 3 \\ 3 & 4 & 13 & 22 & 0 & 4 \\ 8 & 13 & 28 & 51 & 0 & 10 \\ 13 & 22 & 51 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 1 & 18 & 0 & 3 \\ 17 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 972, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 1 & 3 \\ 3 & 4 & 9 & 27 & 0 & 4 \\ 5 & 9 & 13 & 45 & 0 & 7 \\ 17 & 27 & 45 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{122} 3 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1008, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 1 & 3 \\ 3 & 4 & 18 & 18 & 0 & 4 \\ 11 & 18 & 58 & 60 & 0 & 14 \\ 11 & 18 & 60 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1017, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 21 & 1 & 3 \\ 3 & 4 & 9 & 32 & 0 & 4 \\ 5 & 9 & 13 & 54 & 0 & 7 \\ 21 & 32 & 54 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{123} 3 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 1 & 3 \\ 3 & 4 & 18 & 22 & 0 & 4 \\ 11 & 18 & 58 & 72 & 0 & 14 \\ 13 & 22 & 72 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1061, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 2 & 4 & 1 & 21 & 0 & 3 \\ 21 & 31 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1077, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 2 & 4 & 1 & 21 & 0 & 3 \\ 19 & 31 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1097, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 1 & 3 \\ 3 & 4 & 5 & 32 & 0 & 4 \\ 3 & 5 & 4 & 33 & 0 & 4 \\ 21 & 32 & 33 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 1 & 3 \\ 3 & 4 & 5 & 32 & 0 & 4 \\ 3 & 5 & 4 & 33 & 0 & 4 \\ 22 & 32 & 33 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$[160] \ 1^{114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 1554, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 23 & 1 & 3 \\ 3 & 4 & 9 & 36 & 0 & 4 \\ 5 & 9 & 13 & 60 & 0 & 7 \\ 23 & 36 & 60 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

$$4.898 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$$

$$T = 9 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 18 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 10 & 17 & 13 & 27 & 0 & 7 \\ 18 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -248, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 13 & 27 & 0 & 7 \\ 19 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -258, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 18 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 13 & 27 & 0 & 7 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 27 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 27 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 27 & 1 & 5 \\ 11 & 20 & 17 & 50 & 0 & 8 \\ 10 & 17 & 13 & 42 & 0 & 7 \\ 27 & 50 & 42 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 13 & 27 & 0 & 7 \\ 18 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 \ 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 28 & 1 & 5 \\ 11 & 20 & 17 & 50 & 0 & 8 \\ 9 & 17 & 13 & 42 & 0 & 7 \\ 28 & 50 & 42 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

4.899 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 6 & 0 & 3 & 4 \\ 4 & 5 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & 0 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.900 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [1] \ 1^{111} 31^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 3 & 1 & 6 & 3 \\ 4 & 6 & 0 & 9 & 6 & 13 & 7 \\ 2 & 3 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{11} 4 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 3 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 31^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$[12] \ 1^{\overline{111}} 31^{\overline{111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 3 & 2 \\ 1 & 1 & 1 & 6 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 15 & 7 \\ 4 & 6 & 0 & 15 & 13 & 15 & 7 \\ 3 & 6 & 0 & 15 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{112}} 31^{\overline{11}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 5 & 2 \\ 1 & 1 & 1 & 3 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 9 & 3 \\ 3 & 6 & 0 & 6 & 13 & 21 & 7 \\ 5 & 9 & 0 & 9 & 21 & 28 & 10 \\ 2 & 3 & -1 & 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{112}} 31^{\overline{11}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 1 & 1 & 3 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 9 & 12 & 4 \\ 4 & 6 & 0 & 9 & 13 & 21 & 7 \\ 5 & 9 & 0 & 12 & 21 & 28 & 10 \\ 2 & 3 & -1 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{111}} 31^{\overline{111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 15 & 7 \\ 4 & 6 & 0 & 15 & 13 & 15 & 7 \\ 4 & 6 & 0 & 15 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{\overline{112}} 31^{\overline{11}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 5 & 2 \\ 1 & 1 & 1 & 6 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 21 & 7 \\ 4 & 6 & 0 & 15 & 13 & 21 & 7 \\ 5 & 9 & 0 & 21 & 21 & 28 & 10 \\ 2 & 3 & -1 & 7 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [19] \ 1^{\overline{1}} \overset{1}{5} \overset{1}{1^{11}, 1^{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 \\ 2 & 3 & 5 & 1 & 2 & 3 \\ 3 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{1}} \overset{1}{5} \overset{1}{1^{11}, 1^{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 \\ 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{1}} \overset{1}{5} \overset{1}{1^{11}, 1^{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 \\ 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{1}} \overset{1}{5} \overset{1}{1^{11}, 1^{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 4 & 6 & 8 & 5 & 0 & 6 \\ 2 & 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{11},1^{\overline{11}}} \quad 1^{\overline{11},1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 5 & 6 & 8 & 5 & 5 & 6 \\ 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{11},1^{\overline{11}}} \quad 1^{\overline{11},1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{11},1^{\overline{11}}} \quad 1^{\overline{11},1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [29] \quad 1^{\overline{1111}} 31^{\overline{141}} 131^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 3 & 4 & 0 & 5 & 4 & 9 & 5 \\ 2 & 3 & 1 & 4 & 2 & 7 & 4 \\ 4 & 6 & 1 & 9 & 7 & 14 & 8 \\ 2 & 3 & 0 & 5 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

4.901 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 131^{\overline{11}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & 1 & 3 & 6 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 6 & 3 & 3 \\ 4 & 6 & 0 & 6 & 13 & 15 & 6 & 7 \\ 3 & 6 & 0 & 6 & 15 & 13 & 6 & 7 \\ 2 & 3 & 0 & 3 & 6 & 6 & 1 & 3 \\ 2 & 3 & -1 & 3 & 7 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1121}} 31^{\overline{1111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 5 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 9 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 9 & 12 & 9 & 4 \\ 3 & 6 & 0 & 9 & 13 & 21 & 15 & 7 \\ 5 & 9 & 0 & 12 & 21 & 28 & 21 & 10 \\ 4 & 6 & 0 & 9 & 15 & 21 & 13 & 7 \\ 2 & 3 & -1 & 4 & 7 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{\overline{1111}} 131^{\overline{141}} 131^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 7.0$$

$$[3] \quad 1^{\overline{11},1^{\overline{11}}} \quad 1^{\overline{11},1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -10, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11},1^{\overline{11}}} \quad 1^{\overline{11},1^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -10, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 0 & 4 \\ 5 & 6 & 8 & 5 & 5 & 0 & 6 \\ 2 & 3 & 5 & 1 & 2 & 0 & 3 \\ 2 & 3 & 5 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [5] \quad 1^{11} 6^{11} 131^1 5^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 7.0$$

$$[6] \quad 1^{11} 6^{11} 131^1 5^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 7.0$$

4.902 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 5^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 8.0$$

4.903 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 5^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 11, 2), c_{\text{ext}} = 9.0$$

4.904 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{11} 11^{11} 31^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 2 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 2 & 9 & 4 & 4 \\ 6 & 9 & 0 & 9 & 22 & 9 & 10 \\ 2 & 4 & 0 & 4 & 9 & 2 & 4 \\ 3 & 4 & -1 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

4.905 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & -1 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 3 & 4 \\ 1 & 2 & 0 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 0 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 1 & 3 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 2 & 3 \\ 3 & 6 & 0 & 6 & 13 & 5 & 7 \\ 1 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 3 & -1 & 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$[18] \ 1^{22}3^11^151321^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 7.0$$

$$[19] \ 1^{33}3^11^151321^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 7.6$$

$$[20] \ 1^{111}3151^{123}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 13 & 4 \\ 3 & 0 & 5 & 3 & 13 & 5 \\ 9 & 0 & 13 & 13 & 43 & 15 \\ 3 & 0 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[21] \ 2^13^11513^11^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 12, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 7 & 2 & 3 \\ 2 & 2 & 0 & 5 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 19 & 5 & 9 \\ 4 & 7 & 5 & 19 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{112}3^1151^{1231}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 55, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 1 & 2 \\ 1 & 2 & 1 & 4 & 13 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 13 & 3 & 4 \\ 6 & 13 & 0 & 13 & 49 & 11 & 15 \\ 1 & 3 & 0 & 3 & 11 & 1 & 3 \\ 2 & 4 & -1 & 4 & 15 & 3 & 4 \end{pmatrix}_{b_0}$$

4.906 NHC(-2)(-3) + su₂ + su₂ + su₂ → (-3) + su₂

$$T = 10 \quad [0] \ 1^{1111}31^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 2 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 2 & 9 & 4 & 4 \\ 6 & 9 & 0 & 9 & 22 & 9 & 10 \\ 2 & 4 & 0 & 4 & 9 & 2 & 4 \\ 3 & 4 & -1 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

4.907 NHC(-2)(-3) + su₂ + su₂ + su₂n3mix

$$T = 12 \quad [4] \ 1^{132}3^11^151321^{1132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 6.8$$

$$[24] \ 1^{441}3^11^151321^{1441} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 8.0$$

$$[19] \ 1^{133}3^11^151321^{1133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 6.8$$

$$[18] \ 1^{121}3^11^151321^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 6.5$$

$$[17] \ 1^{141}3^11^151321^{1141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 7.0$$

$$[16] \ 1^{134}3^11^151321^{1134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 6.8$$

$$[15] \ 1^{142}3^11^151321^{1142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 7.0$$

$$[14] \ 1^{242}3^11^151321^{1242} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 7.5$$

$$[13] \ 1^{143}3^11^151321^{1143} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 7.0$$

4.908 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

- $T = 8$ $[0] 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$
- [1] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$
- [2] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$
- [4] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 & 4 \\ 3 & 5 & 4 & 6 & 3 & 4 \\ 4 & 5 & 6 & 4 & 3 & 4 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$
- [3] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 0 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 4 \\ 4 & 5 & 6 & 4 & 3 & 4 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$
- [8] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 4 & 5 & 4 & 12 & 0 & 4 \\ 8 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 16 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & 0 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 9 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 3 & 4 & 1 & 9 & 0 & 3 \\ 8 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 6 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 6 & 9 & 13 & 21 & 0 & 7 \\ 8 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

4.909 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9$ [0] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 8 & 0 & 9 & 4 \\ 5 & 9 & 13 & 12 & 0 & 15 & 7 \\ 5 & 8 & 12 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 15 & 12 & 0 & 13 & 7 \\ 3 & 4 & 7 & 6 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 0 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 6 & 0 & 13 & 7 \\ 3 & 4 & 0 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 3 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 8 & 0 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 12 & 0 & 13 & 7 \\ 3 & 4 & 0 & 6 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 6 & 3 \\ 3 & 4 & 8 & 9 & 0 & 9 & 4 \\ 5 & 8 & 10 & 12 & 0 & 12 & 6 \\ 6 & 9 & 12 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 12 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 5 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 9 & 0 & 13 & 7 \\ 3 & 4 & 0 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 3 & 7 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 9 & 0 & 28 & 10 \\ 3 & 4 & 0 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 9 & 0 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 15 & 0 & 13 & 7 \\ 3 & 4 & 0 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 1 & 3 & 0 & 9 & 3 \\ 3 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -264, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 6 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 6 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -271, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 3 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1112} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 6 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -348, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 3 & 4 & 1 & 6 & 0 & 6 & 3 \\ 6 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1112} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 3 & 4 & 1 & 6 & 0 & 9 & 3 \\ 6 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

4.910 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 0 & 3 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1112} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 36, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 12 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 3 & 0 & 1 & 9 & 3 \\ 8 & 13 & 12 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 6 & 7 \\ 3 & 4 & 3 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & 3 & -1 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1121} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 8 & 13 & 0 & 9 & 0 & 28 & 21 & 10 \\ 6 & 9 & 0 & 6 & 0 & 21 & 13 & 7 \\ 3 & 4 & 0 & 3 & -1 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 6 & 0 & 1 & 6 & 3 \\ 6 & 9 & 6 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 3 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 6 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 3 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{11111}} 3^{\overline{11111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 3 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

4.911 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10 \quad [1] 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 0 & 3 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[0] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

4.912 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9 \quad [0] 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & 0 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 3 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 5 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & 0 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 3 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.913 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 3 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 4 \\ 3 & 4 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.914 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 7 \quad [0] \quad 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 5 & 5 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.46 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 4 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

- [26] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 37 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [27] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 9 & 3 & 4 \\ 6 & 9 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
- [28] $21^{11}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 3 & 4 \\ 12 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [29] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 37 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [30] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 11 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [32] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 37 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [31] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 8 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [34] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 10 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [33] $1^{14}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 11 & 16 & 46 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$
- [36] $1^{14}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$
- [35] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 9 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

- [37] $1^{14}1^{13}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -127$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 15 & 19 & 73 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$
- [38] $21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$
- [39] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -138$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 10 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$
- [40] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -138$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 73 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$
- [41] $21^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -141$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 14 & 3 & 4 \\ 15 & 14 & 37 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 \\ 4 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [42] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -153$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$
- [43] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -171$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 11 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$
- [44] $21^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -181$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 13 & 3 & 4 \\ 9 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [45] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -182$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$
- [46] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -183$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$
- [47] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -186$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 12 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{1^4}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{1^3}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -193, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 13 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[50] $21^{1^4}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -206, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[51] $21^{1^2}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 18 & 3 & 4 \\ 18 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 4 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{1^3}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 13 & 22 & 85 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

[53] $21^{1^2}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 23 & 3 & 4 \\ 15 & 23 & 100 & 13 & 18 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{1^2}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -222, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 13 & 21 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[55] $21^{1^2}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 12 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{1^2}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 100 & 13 & 18 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{1^3}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 14 & 21 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[58] $21^{1^2}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$

$$\begin{aligned}
& T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 22 & 3 & 4 \\ 21 & 22 & 85 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 \\ 4 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0} \\
[59] & 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 85 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0} \\
[60] & 1^{13} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 16 & 23 & 100 & 13 & 18 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[62] & 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -266, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[61] & 1^{14} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -266, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 17 & 23 & 100 & 13 & 18 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[63] & 21^{12} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 26 & 3 & 4 \\ 24 & 26 & 118 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 \\ 4 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0} \\
[64] & 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 15 & 22 & 85 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0} \\
[65] & 21^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 17 & 3 & 4 \\ 12 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[66] & 221^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 0 \\ 3 & 4 & 0 & 3 & 0 \\ 3 & 3 & 3 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0} \\
[67] & 21^{15} 3^{11} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 15 & 20 & 73 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[68] & 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -297, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 85 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[69] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 19 & 28 & 154 & 16 & 22 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[70] $21^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 23 & 3 & 4 \\ 24 & 23 & 100 & 13 & 18 \\ 3 & 3 & 13 & 0 & 2 \\ 4 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 154 & 16 & 22 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 16 & 26 & 118 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -343, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 133 & 15 & 21 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 17 & 26 & 118 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 18 & 27 & 133 & 15 & 21 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[76] $21^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 22 & 3 & 4 \\ 15 & 22 & 85 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 19 & 27 & 133 & 15 & 21 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 118 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

[79] $21^{16}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 18 & 24 & 106 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[80] \quad & 21^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 21 & 3 & 4 \\ 15 & 21 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[81] \quad & 21^{13}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -441, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 27 & 3 & 4 \\ 5 & 4 & 27 & 3 & 4 \\ 27 & 27 & 133 & 15 & 21 \\ 3 & 3 & 15 & 0 & 2 \\ 4 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[82] \quad & 1^{12}1^{14}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -446, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 19 & 31 & 172 & 17 & 24 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0} \\
[83] \quad & 21^{13}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 27 & 3 & 4 \\ 18 & 27 & 133 & 15 & 21 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[85] \quad & 21^{14}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 28 & 3 & 4 \\ 30 & 28 & 154 & 16 & 22 \\ 3 & 3 & 16 & 0 & 2 \\ 4 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[84] \quad & 21^{13}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 32 & 3 & 4 \\ 21 & 32 & 193 & 18 & 25 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0} \\
[86] \quad & 1^{13}1^{13}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 32 & 3 & 4 \\ 21 & 32 & 193 & 18 & 25 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0} \\
[88] \quad & 1^{13}1^{14}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -501, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 20 & 31 & 172 & 17 & 24 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0} \\
[87] \quad & 1^{14}1^{13}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -501, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 22 & 2 & 3 \\ 3 & 4 & 32 & 3 & 4 \\ 22 & 32 & 193 & 18 & 25 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0} \\
[89] \quad & 21^{13}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -516, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 31 & 3 & 4 \\ 30 & 31 & 172 & 17 & 24 \\ 3 & 3 & 17 & 0 & 2 \\ 4 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[90] $21^{17}3^11^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 28 & 3 & 4 \\ 21 & 28 & 145 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{14}3^11^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -538, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 21 & 31 & 172 & 17 & 24 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[92] $21^{14}3^11^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -586, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 26 & 3 & 4 \\ 18 & 26 & 118 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{14}3^11^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -613, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 23 & 2 & 3 \\ 3 & 4 & 36 & 3 & 4 \\ 23 & 36 & 238 & 20 & 28 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[94] $21^{14}3^11^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -621, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 33 & 3 & 4 \\ 5 & 4 & 32 & 3 & 4 \\ 33 & 32 & 193 & 18 & 25 \\ 3 & 3 & 18 & 0 & 2 \\ 4 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{14}3^11^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -662, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 24 & 2 & 3 \\ 3 & 4 & 36 & 3 & 4 \\ 24 & 36 & 238 & 20 & 28 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[96] $21^{14}3^11^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -701, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 31 & 3 & 4 \\ 21 & 31 & 172 & 17 & 24 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[97] $21^{14}3^11^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -756, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 36 & 3 & 4 \\ 5 & 4 & 36 & 3 & 4 \\ 36 & 36 & 238 & 20 & 28 \\ 3 & 3 & 20 & 0 & 2 \\ 4 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[98] $21^{14}3^11^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -766, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 24 & 2 & 3 \\ 4 & 4 & 36 & 3 & 4 \\ 24 & 36 & 238 & 20 & 28 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 8$ [99] $1^13^1131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[100] $1^13^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

[illegible]

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^1 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 10 & 3 & 5 \\ 5 & 10 & 14 & 4 & 8 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^1 1^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 3 & 5 & 2 & 2 & 4 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^1 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 5 & 10 & 3 & 5 \\ 7 & 10 & 14 & 4 & 8 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[116] \quad 21^{12} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^1 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 5 & 15 & 3 & 5 \\ 8 & 15 & 34 & 6 & 12 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^1 1^1 3^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 2 & 1 & 9 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 22 & 10 \\ 3 & 4 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^1 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 5 & 10 & 3 & 5 \\ 6 & 10 & 14 & 4 & 8 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 10 & 3 & 5 \\ 5 & 10 & 14 & 5 & 8 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^1 1^1 3^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^1 1^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 7 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[133] & 1^{14} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 10 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[136] & 1^{14} 3^{12} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 5 & 20 & 3 & 5 \\ 12 & 20 & 62 & 8 & 16 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0} \\
[135] & 1^{11} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 5 & 15 & 3 & 5 \\ 8 & 15 & 32 & 7 & 12 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0} \\
[137] & 1^{11} 3^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & -1 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[139] & 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[138] & 1^{13} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 9 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[140] & 1^{11} 3^{12} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 5 & 20 & 3 & 5 \\ 10 & 20 & 58 & 9 & 16 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0} \\
[141] & 1^{13} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 5 & 15 & 3 & 5 \\ 10 & 15 & 32 & 7 & 12 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0} \\
[143] & 1^{13} 3^{12} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 9 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[142] & 1^{11} 3^{12} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[144] $1^{12}1^{13}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 5 & 15 & 3 & 5 \\ 9 & 15 & 32 & 7 & 12 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

[145] $1^{12}1^{13}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[146] $1^{14}1^{13}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[147] $1^{13}1^{13}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 11 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[148] $1^{12}1^{13}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 5 & 20 & 3 & 5 \\ 11 & 20 & 58 & 9 & 16 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0}$$

[149] $1^{12}1^{13}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 9 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[150] $21^{13}1^{13}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[151] $1^{14}1^{13}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 5 & 20 & 3 & 5 \\ 13 & 20 & 58 & 9 & 16 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0}$$

[152] $1^{12}1^{13}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 5 & 25 & 3 & 5 \\ 13 & 25 & 92 & 11 & 20 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 5 & 20 & 2 & 4 \end{pmatrix}_{b_0}$$

[154] $21^{12}1^{13}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[153] $1^{13}1^{13}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

[illegible]

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 14 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 15 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_b$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 14 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 5 & 30 & 3 & 5 \\ 16 & 30 & 134 & 14 & 24 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 5 & 24 & 2 & 4 \end{pmatrix}_b$$

$$m_{\text{min}} = 9.89 \rightarrow 10, 7$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$2] 1^{12} 3^{14} 21^{14} + [5u_2$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_b$$

$$3] 1^{12} 3^{13} 21^{14} + [\text{su}_2, \dots]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 5 & 30 & 3 & 5 \\ 17 & 30 & 134 & 14 & 24 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 5 & 24 & 2 & 4 \end{pmatrix}_b$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 14 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$5] \ 21^{13}3^{12}21^{12} + [51^{13}3^{12}21^{12}]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[176] $1^{14} 1^{12} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 5 & 30 & 3 & 5 \\ 18 & 30 & 134 & 14 & 24 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 5 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

[177] $1^{13} 1^{14} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 16 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 15 & 16 & 16 & 94 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[178] $1^{14} 1^{14} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 16 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 16 & 16 & 16 & 94 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[179] $1^{13} 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[180] $1^{13} 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 16 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{12} 1^{14} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 18 & 20 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[182] $1^{13} 1^{13} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[183] $1^{14} 1^{13} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 357, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[184] $1^{13} 1^{14} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 17 & 18 & 20 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[185] $21^{14} 1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 1 & 3 & 20 & 3 \\ 2 & 3 & 1 & 24 & 3 \\ 20 & 20 & 24 & 174 & 28 \\ 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[186] $1^{14} 1^{14} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 18 & 18 & 20 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0} \\
[187] & 21^{143^1} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 18 & 18 & 20 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0} \\
[188] & 1^{143^1} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 19 & 3 \\ 2 & 1 & 3 & 20 & 3 \\ 2 & 3 & 1 & 24 & 3 \\ 19 & 20 & 24 & 174 & 28 \\ 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0} \\
[189] & 1^{143^1} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 20 & 3 \\ 2 & 3 & 1 & 24 & 3 \\ 20 & 20 & 24 & 174 & 28 \\ 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0} \\
T = 9 & [191] 1^{123^1} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[190] & 1^{1123^1} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[193] & 1^{141^1} 23^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 4 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 4 & 0 & 4 \end{pmatrix}_{b_0} \\
[192] & 1^{114123^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[194] & 23^{1141^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 4 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 5 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[195] & 1^{114123^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[196] & 1^{14123^1} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[197] & 1^{11}2313^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 \\ 3 & 0 & 5 & 3 & 5 \\ 3 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[198] & 1^{11}23^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 2 & 3 \\ 2 & 0 & 4 & 2 & 2 & 4 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 3 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[199] & 1^{11}41^{11}23^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[200] & 1^{11}4123^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[201] & 1^{11}4123^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 4 & 0 & 4 \end{pmatrix}_{b_0} \\
[202] & 1^{11}41^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 0 & 32 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[203] & 1^{11}41^{11}23^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 0 & 3 & 0 & 2 \\ 5 & 3 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[204] & 23^{11}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 3 & 2 & 2 \\ 1 & 0 & 2 & 3 & 1 & 2 \\ 2 & 2 & 0 & 4 & 2 & 2 \\ 3 & 3 & 4 & 5 & 3 & 5 \\ 2 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[205] & 23^{11}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 4 & 5 \\ 3 & 5 & 2 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 5 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[206] & 1^{11}41^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 5 & 0 & 2 \\ 8 & 5 & 24 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 10 & 0 & 4 \end{pmatrix}_{b_0} \\
[207] & 1^{12}41^223^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 0 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[208] & 23^{112}141^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 4 & 5 \\ 6 & 10 & 14 & 6 & 8 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 5 & 8 & 3 & 4 \end{pmatrix}_{b_0} \\
[209] & 23^{111}141^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 4 & 5 \\ 6 & 10 & 14 & 6 & 8 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 5 & 8 & 3 & 4 \end{pmatrix}_{b_0} \\
[210] & 1^{11}41^123^{111}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 6 & 1 & 3 \\ 2 & 0 & 5 & 0 & 2 \\ 6 & 5 & 12 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 8 & 0 & 4 \end{pmatrix}_{b_0} \\
[211] & 1^{11}41^123^{111}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 4 & 1 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[212] & 1^{11}41^123^{112}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 1 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 7 & 4 & 12 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 8 & 0 & 4 \end{pmatrix}_{b_0} \\
[213] & 23^{113}141^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 5 & 15 & 4 & 5 \\ 9 & 15 & 34 & 9 & 12 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 5 & 12 & 3 & 4 \end{pmatrix}_{b_0} \\
[214] & 1^{11}41^223^{111}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 9 & 1 & 3 \\ 2 & 0 & 7 & 0 & 2 \\ 9 & 7 & 34 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 12 & 0 & 4 \end{pmatrix}_{b_0} \\
[215] & 1^{12}41^323^{111}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 0 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 16 & 0 & 3 \end{pmatrix}_{b_0} \\
[216] & 23^{112}141^{12}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 & 2 \\ 1 & 0 & 2 & 6 & 1 & 2 \\ 2 & 2 & 0 & 8 & 2 & 2 \\ 6 & 6 & 8 & 26 & 6 & 10 \\ 2 & 1 & 2 & 6 & 0 & 2 \\ 2 & 2 & 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[217] \quad 23^{1^{14}}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 4 & 5 \\ 12 & 20 & 62 & 12 & 16 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 5 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[218] \quad 1^{11}41^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 8 & 6 & 1 & 33 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[219] \quad 1^{13}41^{323}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 0 & 106 & 0 & 18 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[220] \quad 1^{11}41^{123}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 8 & 6 & 18 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 10 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[221] \quad 23^{1^{12}}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 5 & 15 & 4 & 5 \\ 9 & 15 & 32 & 9 & 12 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 5 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[222] \quad 1^{14}1^{23}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 10 & 6 & 30 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[223] \quad 1^{11}41^{323}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 11 & 8 & 1 & 67 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[224] \quad 23^{1^{15}}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 5 & 25 & 4 & 5 \\ 15 & 25 & 98 & 15 & 20 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 5 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[225] \quad 1^{14}1^{323}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 13 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 13 & 8 & 60 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 16 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[226] \quad 1^{13}41^{423}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -166, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 1 & 14 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 15 & 14 & 0 & 160 & 0 & 22 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 22 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{12}41^{223^1}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 9 & 8 & 1 & 47 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 13 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{11}41^{123^1}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 1 & 3 \\ 2 & 0 & 7 & 0 & 2 \\ 10 & 7 & 28 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{16}23^1141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 5 & 30 & 4 & 5 \\ 18 & 30 & 142 & 18 & 24 \\ 3 & 4 & 18 & 1 & 3 \\ 3 & 5 & 24 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{11}41^{223^1}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 11 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 11 & 8 & 40 & 0 & 14 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{14}41^{223^1}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 1 & 3 \\ 2 & 0 & 7 & 0 & 2 \\ 12 & 7 & 40 & 0 & 14 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{14}41^{423^1}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 1 & 2 \\ 1 & 0 & 1 & 16 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 16 & 16 & 0 & 190 & 0 & 24 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 24 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{13}23^1141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 4 & 5 \\ 12 & 20 & 58 & 12 & 16 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 5 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{12}41^{323^1}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -223, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 12 & 10 & 1 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 17 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{11}41^{323^1}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 14 & 10 & 74 & 0 & 18 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 18 & 0 & 4 \end{pmatrix}_{b_0}$$

[236] $23^{1^{17}}141^{1^{17}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 & 3 \\ 4 & 5 & 35 & 4 & 5 \\ 21 & 35 & 194 & 21 & 28 \\ 3 & 4 & 21 & 1 & 3 \\ 3 & 5 & 28 & 3 & 4 \end{pmatrix}_{b_0}$$

[237] $1^{12}41^{22}23^{1^{12}}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 10 & 8 & 2 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{12}41^{42}23^{1^{11}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 15 & 12 & 1 & 135 & 0 & 21 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 21 & 0 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{41}41^{23}1^{14}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 14 & 8 & 54 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 16 & 0 & 4 \end{pmatrix}_{b_0}$$

[240] $1^{41}41^{32}1^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 1 & 3 \\ 2 & 0 & 9 & 0 & 2 \\ 15 & 9 & 70 & 0 & 18 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 18 & 0 & 4 \end{pmatrix}_{b_0}$$

[241] $1^{13}41^{32}1^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 13 & 12 & 1 & 107 & 0 & 19 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

[242] $23^{1^{14}}141^{1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 5 & 25 & 4 & 5 \\ 15 & 25 & 92 & 15 & 20 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 5 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

[243] $1^{11}41^{22}23^{1^{13}}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 1 & 3 \\ 2 & 0 & 9 & 0 & 2 \\ 13 & 9 & 50 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 16 & 0 & 4 \end{pmatrix}_{b_0}$$

[244] $1^{41}41^{42}23^{1^{13}}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 1 & 3 \\ 2 & 0 & 11 & 0 & 2 \\ 18 & 11 & 112 & 0 & 22 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 22 & 0 & 4 \end{pmatrix}_{b_0}$$

[245] $1^{12}41^{32}23^{1^{12}}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 13 & 10 & 2 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 18 & 0 & 3 \end{pmatrix}_{b_0} \\
[246] & 1^{13}41^{423}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 1 & 2 \\ 1 & 0 & 1 & 14 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 16 & 14 & 1 & 161 & 0 & 23 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 23 & 0 & 3 \end{pmatrix}_{b_0} \\
[247] & 1^{11}41^{323}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -373, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 16 & 1 & 3 \\ 2 & 0 & 11 & 0 & 2 \\ 16 & 11 & 84 & 0 & 20 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 20 & 0 & 4 \end{pmatrix}_{b_0} \\
[248] & 1^{14}1^{323}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 17 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 17 & 10 & 84 & 0 & 20 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 20 & 0 & 4 \end{pmatrix}_{b_0} \\
[249] & 1^{11}41^{223}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 15 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 15 & 10 & 64 & 0 & 18 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 18 & 0 & 4 \end{pmatrix}_{b_0} \\
[250] & 1^{12}41^{423}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -418, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 16 & 12 & 2 & 138 & 0 & 22 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 22 & 0 & 3 \end{pmatrix}_{b_0} \\
[251] & 1^{14}41^{423}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -423, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 1 & 2 \\ 1 & 0 & 1 & 16 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 17 & 16 & 1 & 191 & 0 & 25 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 25 & 0 & 3 \end{pmatrix}_{b_0} \\
[252] & 1^{11}41^{423}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -429, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 19 & 1 & 3 \\ 2 & 0 & 13 & 0 & 2 \\ 19 & 13 & 130 & 0 & 24 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 24 & 0 & 4 \end{pmatrix}_{b_0} \\
[253] & 1^{13}41^{323}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -430, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 14 & 12 & 2 & 110 & 0 & 20 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 20 & 0 & 3 \end{pmatrix}_{b_0} \\
[254] & 1^{14}1^{423}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -480, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 20 & 1 & 3 \\ 2 & 0 & 12 & 0 & 2 \\ 20 & 12 & 126 & 0 & 24 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 24 & 0 & 4 \end{pmatrix}_{b_0} \\
[255] & 1^{11}41^{323}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 18 & 1 & 3 \\ 2 & 0 & 12 & 0 & 2 \\ 18 & 12 & 98 & 0 & 22 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 22 & 0 & 4 \end{pmatrix}_{b_0} \\
[256] & 1^{13}41^{423}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -530, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 1 & 2 \\ 1 & 0 & 1 & 14 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 17 & 14 & 2 & 164 & 0 & 24 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 24 & 0 & 3 \end{pmatrix}_{b_0} \\
[257] & 1^{11}41^{423}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -612, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 21 & 1 & 3 \\ 2 & 0 & 14 & 0 & 2 \\ 21 & 14 & 144 & 0 & 26 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 26 & 0 & 4 \end{pmatrix}_{b_0} \\
[258] & 1^{14}41^{423}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -626, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 1 & 2 \\ 1 & 0 & 1 & 16 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 18 & 16 & 2 & 194 & 0 & 26 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 26 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 10 & [259] 1^{11}22313^{121}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 5.0 \\
[261] & 1^{14}413^{121}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 5.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[262] & 1^{11}23^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 3 & 1 & 5 & 3 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 3 & 5 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 5 & -1 & 5 & 3 & 5 \end{pmatrix}_{b_0} \\
[263] & 31^{13}1^{15}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 9 & 5 & 7 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[264] & 1^{11}5^{11}131^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 9 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 4 \end{pmatrix}_{b_0} \\
[265] & 1^{11}3^{11}14131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 15, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 3 & 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 0 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[266] & 1^{11}3^{11}141^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 18, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 4 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 5 & 3 & 5 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[267] $31^{23}1^151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 20$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 12 & 14 & 42 & 10 & 14 \\ 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[268] $1^{12}5131^23^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 24$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 42 & 0 & 14 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

[269] $31^{33}1^151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 35$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 18 & 21 & 97 & 15 & 21 \\ 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[270] $1^{13}5131^33^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 39$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 18 & 21 & 97 & 0 & 21 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 21 & 0 & 4 \end{pmatrix}_{b_0}$$

[271] $31^1_513^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 65$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 2 & 4 \\ 3 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[272] $31^1_513^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 74$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 2 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 14 & 8 \\ 3 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[273] $31^1_{13}513^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 89$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 2 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 34 & 12 \\ 3 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[274] $231^{113}1^{15}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -60$, $\det = 275$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 3 & 2 \\ 5 & 33 & 1 & 35 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 35 & 0 & 33 & 20 & 11 \\ 3 & 20 & 0 & 20 & 10 & 6 \\ 2 & 11 & -1 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [275] $1^{11}23^1141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 11$, $\Delta = -9$, $\det = 0$, $(n_+, n_-, n_0) = (1, 11, 1)$, $c_{\text{ext}} = 5.0$

[277] $1^{13}31^1513^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 13 & 1 & 2 \\ 8 & 0 & 13 & 97 & 11 & 21 \\ 1 & 0 & 1 & 11 & 0 & 2 \\ 2 & 0 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0} \\
[278] & 1^{12}3151^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 3 & 1 & 18 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 74 & 20 \\ 3 & 5 & -1 & 20 & 5 \end{pmatrix}_{b_0} \\
[279] & 1^{13}1513^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & 3 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[280] & 1^{13}1513^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[281] & 1^{12}31513^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 6 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0} \\
[282] & 1^{13}15131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -3, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[283] & 1^{11}31513^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 2 \\ 2 & 0 & 4 & 6 & 3 & 6 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0} \\
[284] & 1^{11}31513^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 1 & 2 \\ 6 & 0 & 6 & 34 & 7 & 12 \\ 1 & 0 & 1 & 7 & 0 & 2 \\ 2 & 0 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0} \\
[291] & 1^{13}31513^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 12 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 18 \\ 1 & 1 & 0 & 9 & 0 & 2 \\ 2 & 3 & -1 & 18 & 2 & 4 \end{pmatrix}_{b_0} \\
[288] & 1^{12}3^{11}151^{22}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 & 3 \\ 12 & 0 & 18 & 114 & 16 & 22 \\ 1 & 0 & 3 & 16 & 1 & 3 \\ 2 & 0 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{14}31513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 8 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{11}3^{1151}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 6 & 0 & 9 & 27 & 8 & 11 \\ 1 & 0 & 3 & 8 & 1 & 3 \\ 2 & 0 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[287] \quad 1^{11}31513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 6 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{12}31513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 8 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 6 & 12 \\ 1 & 1 & 0 & 6 & 0 & 2 \\ 2 & 3 & -1 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[293] \quad 2^3 1513^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 2 & 2 & 0 & 10 & 1 & 2 \\ 8 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[294] \quad 2^3 1513^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 2 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{12}31513^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 4 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[289] \quad 1^{14}31513^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 1 & 16 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 24 \\ 1 & 1 & 0 & 12 & 0 & 2 \\ 2 & 3 & -1 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[295] \quad 1^{11}3^{1151}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 4 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[296] \quad 1^{12}31513^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 5 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[297] \quad 1^{11}3^{11}15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 2 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{11}31413^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -8, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 3 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[300] \quad 1^{12}31513^{11}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 1 & 2 \\ 6 & 0 & 9 & 49 & 8 & 15 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 2 & 0 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[299] \quad 1^{11}31513^{11}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 1 & 2 \\ 4 & 0 & 5 & 17 & 5 & 9 \\ 1 & 0 & 1 & 5 & 0 & 2 \\ 2 & 0 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[301] \quad 1^{14}31513^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 9 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{14}31513^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 10 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[303] \quad 1^{11}3^{11}151231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 4 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[304] \quad 1^{11}3^{11}151^{12}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 2 & 1 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 9 & 0 & 27 & 8 & 11 \\ 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & -1 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[305] \quad 1^{11} 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 2 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[306] \quad 1^{11} 3^{15} 1^{12} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 \\ 3 & 0 & 5 & 3 & 5 \\ 3 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[307] \quad 1^{12} 3^{15} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 1 & 2 \\ 8 & 0 & 10 & 74 & 10 & 18 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 2 & 0 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[308] \quad 1^{11} 3^{15} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 3 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[309] \quad 1^{12} 3^{15} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 6 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[310] \quad 1^{12} 3^{15} 1^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 7 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[311] \quad 1^{12} 3^{15} 1^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 6 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^{12} 3^{15} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 5 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[315] \quad 1^{12} 3^{15} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$\begin{aligned}
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0} \\
[314] & 1^{11} 3^{15} 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -43, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 13 & 4 \\ 9 & 0 & 13 & 43 & 15 \\ 3 & 0 & 4 & 15 & 5 \end{pmatrix}_{b_0} \\
[313] & 1^{12} 3^{11} 5^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 5 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0} \\
[316] & 1^{12} 3^{11} 5^{11} 23^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -60, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 2 & 1 & 13 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 13 & 0 & 49 & 11 & 15 \\ 1 & 3 & 0 & 11 & 1 & 3 \\ 2 & 4 & -1 & 15 & 3 & 4 \end{pmatrix}_{b_0} \\
[317] & 1^{13} 3^{11} 5^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 10 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0} \\
[318] & 1^{13} 3^{11} 5^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 9 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0} \\
[319] & 31^{11} 41^{13} 1^{15} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -80, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 8 & 3 & 2 \\ 3 & 4 & 1 & 12 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 3 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & 0 & 10 & 3 & 3 \end{pmatrix}_{b_0} \\
[320] & 1^{13} 3^{11} 5^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 9 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0} \\
[321] & 1^{13} 3^{11} 5^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -87, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 8 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0} \\
[322] & 1^{13} 3^{11} 5^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -92, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 8 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{13}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -97, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 7 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{12}3^{15}1^{12}23^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 18 & 4 \\ 12 & 0 & 18 & 74 & 20 \\ 3 & 0 & 4 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{13}3^{15}1^{12}23^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -115, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 2 & 1 & 22 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 22 & 0 & 154 & 19 & 26 \\ 1 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & -1 & 26 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{13}3^{15}1^{12}23^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -115, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 2 & 1 & 17 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 17 & 0 & 79 & 14 & 19 \\ 1 & 3 & 0 & 14 & 1 & 3 \\ 2 & 4 & -1 & 19 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[327] \quad 41^{11}3^{15}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -122, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 1 & 2 \\ 4 & 10 & 5 & 12 & 0 & 6 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 4 & 12 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 6 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[328] \quad 1^{14}3^{11}5^{13}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -138, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 0 & 16 & 1 & 2 \\ 12 & 16 & 16 & 126 & 12 & 24 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 2 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[329] \quad 1^{14}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -138, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 10 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[330] \quad 1^{14}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -147, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 9 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[331] \quad 1^{14}3^{11}5^{13}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -147, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 0 & 16 & 1 & 2 \\ 11 & 16 & 16 & 126 & 12 & 24 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 2 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[332] \ 1^{14}3^{11}5^{13}13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -158, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 11 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[333] \ 3^{11}1^{14}1^{11}3^{15}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -162, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 1 & 12 & 3 \\ 7 & 12 & 1 & 24 & 8 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 24 & 0 & 42 & 14 \\ 3 & 8 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[334] \ 1^{14}3^{11}5^{13}13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -162, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 9 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[335] \ 1^{14}3^{11}5^{13}13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -167, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 10 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[336] \ 1^{14}3^{11}5^{12}23^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -210, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 2 & 1 & 26 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 26 & 0 & 202 & 22 & 30 \\ 1 & 3 & 0 & 22 & 1 & 3 \\ 2 & 4 & -1 & 30 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[337] \ 3^{11}1^{14}1^{11}3^{15}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -227, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 1 & 7 & 3 \\ 7 & 12 & 1 & 14 & 8 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 14 & 0 & 12 & 8 \\ 3 & 8 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [338] \ 1^{14}3^{11}5^{13}13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 6.0$$

$$[339] \ 1^{11}3^{14}1^{14}13^{12}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -7, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 5.0$$

$$[354] \ 1^{11}3^{12}1^{21}1^{13}6^{13}13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 2, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[355] \ 1^{11}3^{12}1^{21}1^{13}6^{13}13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 3, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{14}3^{11}5^{14}1321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 1 & 20 & 1 & 3 \\ 12 & 16 & 20 & 142 & 12 & 28 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 3 & 28 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[359] \ 1^{12}3^{11}5^{12}1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 6 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{13}3^{11}5^{13}1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 9 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{11}3^{11}5^{11}1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[356] \ 1^{33}3^{11}5^{12}1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 11 & 1 & 3 \\ 6 & 0 & 11 & 53 & 8 & 17 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 2 & 0 & 3 & 17 & 2 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [370] \ 1^{12}3^{16}1^{23}15^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, \ T_{\max} = 15, \ \Delta = -1, \ \det = -4, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 9 & 9 & 61 & 21 \\ 3 & 3 & 2 & 21 & 6 \end{pmatrix}_{b_0}$$

4.915 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6$

$$T = 9 \quad [0] \ 1^{13}3^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -10, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.60 \rightarrow 18, \ T_{\max} = 21, \ T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 5 & 0 & 2 & -2 & 0 \\ 3 & 5 & 0 & 2 & 0 & -4 \end{pmatrix}_{b_0}$$

4.916 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 72, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.49 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 5 & 9 & 13 & 5 & 0 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

4.917 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 200, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 600, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 5 & 0 & 9 & 6 & 7 \\ 2 & 3 & 5 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.918 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{so}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 13 & 9 & 0 & 7 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.919 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 3 & 0 & 8 & 4 \\ 5 & 9 & 13 & 5 & 0 & 12 & 7 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 4 & 0 & 10 & 6 \\ 3 & 4 & 7 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[2] \quad & 1^{13}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 3 & 0 & 17 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 9 & 0 & 49 & 13 \\ 3 & 4 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0} \\
[3] \quad & 1^{112}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.7 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 8 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 4 & 0 & 10 & 6 \\ 3 & 4 & 3 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0} \\
[4] \quad & 1^{13}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 3 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0} \\
[5] \quad & 1^{111}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{111}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 3 & 0 & 14 & 4 \\ 2 & 4 & 1 & 2 & 0 & 9 & 3 \\ 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 8 & 0 & 37 & 11 \\ 3 & 4 & 3 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{12}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.7 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 3 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 7 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{11}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 3 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 5 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{111}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 13 & 7 \\ 3 & 4 & 3 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[10] \quad 1^{114}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 11 & 3 \\ 3 & 4 & 9 & 3 & 0 & 21 & 4 \\ 5 & 9 & 13 & 5 & 0 & 33 & 7 \\ 2 & 3 & 5 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 11 & 0 & 76 & 16 \\ 3 & 4 & 7 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{112}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 3 & 0 & 13 & 4 \\ 2 & 4 & 1 & 2 & 0 & 9 & 3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 7 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 3 & 0 & 9 & 4 \\ 5 & 9 & 13 & 5 & 0 & 15 & 7 \\ 2 & 3 & 5 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 5 & 0 & 13 & 7 \\ 3 & 4 & 7 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{112}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 3 & 0 & 23 & 4 \\ 2 & 4 & 1 & 2 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 13 & 0 & 100 & 18 \\ 3 & 4 & 3 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{113}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 3 & 0 & 17 & 4 \\ 2 & 4 & 1 & 2 & 0 & 12 & 3 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 9 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{113}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 3 & 0 & 17 & 4 \\ 5 & 9 & 13 & 5 & 0 & 27 & 7 \\ 2 & 3 & 5 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 9 & 0 & 49 & 13 \\ 3 & 4 & 7 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{112}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 1 & 2 & 0 & 12 & 3 \\ 2 & 3 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 10 & 0 & 58 & 14 \\ 3 & 4 & 3 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 1 & 2 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 11 & 0 & 76 & 16 \\ 3 & 4 & 3 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1123}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 3 & 0 & 13 & 4 \\ 5 & 9 & 13 & 5 & 0 & 21 & 7 \\ 2 & 3 & 5 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 7 & 0 & 28 & 10 \\ 3 & 4 & 7 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{113}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 3 & 0 & 22 & 4 \\ 2 & 4 & 1 & 2 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 12 & 0 & 85 & 17 \\ 3 & 4 & 3 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 3 & 0 & 32 & 4 \\ 2 & 4 & 1 & 2 & 0 & 21 & 3 \\ 2 & 3 & 2 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 18 & 0 & 193 & 25 \\ 3 & 4 & 3 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 1 & 2 & 0 & 18 & 3 \\ 2 & 3 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 15 & 0 & 133 & 21 \\ 3 & 4 & 3 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 3 & 0 & 26 & 4 \\ 2 & 4 & 1 & 2 & 0 & 18 & 3 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 14 & 0 & 118 & 20 \\ 3 & 4 & 3 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{114}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 3 & 0 & 31 & 4 \\ 2 & 4 & 1 & 2 & 0 & 21 & 3 \\ 2 & 3 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 17 & 0 & 172 & 24 \\ 3 & 4 & 3 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 3 & 0 & 36 & 4 \\ 2 & 4 & 1 & 2 & 0 & 24 & 3 \\ 2 & 3 & 2 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 20 & 0 & 238 & 28 \\ 3 & 4 & 3 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$4.920 \quad \text{NHC}(-2)(-3) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$$

$$T = 10 \quad [0] \quad 1^{113}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}3^1 1^{\overline{11}}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}3^1 1^{\overline{11}}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 5 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 2 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}3^1 1^{\overline{11}}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 5 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & 2 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}3^1 1^{\overline{11}}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 2 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1111}3^1 1^{\overline{11}}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 9 & 4 & 4 \\ 5 & 9 & 13 & 5 & 0 & 15 & 6 & 7 \\ 2 & 3 & 5 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 5 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 2 & 0 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.921 $\text{NHC}(-2)(-3) + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{s}u_3$

$$T = 9 \quad [0] \quad 1^{\overline{11}3^1 1^{\overline{11}}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}3^1 1^{\overline{11}}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{s}u_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 3 & 0 & 5 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 8 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 4 & 0 & 9 & 5 \\ 3 & 4 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 3 & 0 & 8 & 4 \\ 5 & 9 & 13 & 5 & 0 & 12 & 7 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 4 & 0 & 9 & 5 \\ 3 & 4 & 7 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 12 & 6 \\ 3 & 4 & 3 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 & 4 \\ 5 & 9 & 13 & 5 & 0 & 6 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 2 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -476, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 3 & 0 & 13 & 4 \\ 2 & 4 & 1 & 2 & 0 & 9 & 3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 7 & 0 & 27 & 9 \\ 3 & 4 & 3 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.922} \quad \text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$$

$$T = 10 \quad [0] \quad 1^{112}3^{11}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{magenta}{\mathfrak{su}}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 5 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 2 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 5 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 5 & 0 & 6 & 15 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 2 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 5 & 0 & 6 & 13 & 7 \\ 3 & 4 & 7 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.923 NHC(-2)(-3) + $\mathbf{su}_2$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.70 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

- $$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [3] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 8 & 6 \\ 15 & 20 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 11 & 0 & 4 & 4 \\ 6 & 8 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$
- [4] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 4 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [5] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 13 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$
- [7] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 3 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 5 & 9 & 13 & 5 & 0 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 8 & 3 & 0 & 4 \\ 5 & 8 & 10 & 4 & 0 & 6 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 6 & 9 & 13 & 5 & 0 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{11}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 9 & 14 & 37 & 8 & 0 & 11 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 13 & 3 & 0 & 4 \\ 8 & 13 & 28 & 7 & 0 & 10 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 399, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 10 & 17 & 49 & 9 & 0 & 13 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 18 & 3 & 0 & 4 \\ 11 & 18 & 58 & 10 & 0 & 14 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 1^{13} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 22 & 3 & 0 & 4 \\ 13 & 22 & 85 & 12 & 0 & 17 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [16] $1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 4 & 0 & 3 \\ 2 & 2 & 4 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 4 & -2 & 3 \end{pmatrix}_{b_0}$$

[17] $1^1 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{11} 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[19] $1^1 1^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 6 & 0 & 3 \\ 4 & 5 & 6 & 9 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 7 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^{11} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 5 & 10 & 3 & 5 & 5 \\ 5 & 10 & 14 & 5 & 7 & 8 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 5 & 7 & 2 & 0 & 2 \\ 3 & 5 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [21] \quad 1^{11} 3^{11} 513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 24, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 8 & 6 \\ 1 & 1 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 8 & 3 & 4 & 4 \\ 2 & 3 & -1 & 6 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

4.924 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 1 & 5 \\ 11 & 20 & 0 & 11 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 11 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -163, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 13 & 9 & 0 & 7 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

4.925 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -108, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -116, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 1 & 2 & 0 & 2 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -208, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 3 & 4 \\ 5 & 9 & 13 & 5 & 0 & 5 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 2 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.926 NHC(-2)(-3) + su₂ + su₂→(-3) + so₈ + su2n3mix

$$T = 9 \quad [0] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 3 & 0 & 7 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 4 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 4 & 1 & 2 & 0 & 5 & 3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 4 & 0 & 7 & 5 \\ 3 & 4 & 3 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 3 & 0 & 11 & 4 \\ 2 & 4 & 1 & 2 & 0 & 8 & 3 \\ 2 & 3 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 8 & 6 & 0 & 20 & 8 \\ 3 & 4 & 3 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 3 & 0 & 7 & 4 \\ 5 & 9 & 13 & 5 & 0 & 11 & 7 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 4 & 0 & 7 & 5 \\ 3 & 4 & 7 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 1 & 2 & 0 & 8 & 3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 7 & 0 & 25 & 9 \\ 3 & 4 & 3 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 3 & 0 & 16 & 4 \\ 2 & 4 & 1 & 2 & 0 & 11 & 3 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 11 & 9 & 0 & 44 & 12 \\ 3 & 4 & 3 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 1 & 2 & 0 & 14 & 3 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 14 & 12 & 0 & 80 & 16 \\ 3 & 4 & 3 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 3 & 0 & 20 & 4 \\ 2 & 4 & 1 & 2 & 0 & 14 & 3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 14 & 11 & 0 & 69 & 15 \\ 3 & 4 & 3 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 1 & 2 & 0 & 17 & 3 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 17 & 13 & 0 & 100 & 18 \\ 3 & 4 & 3 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 3 & 0 & 25 & 4 \\ 2 & 4 & 1 & 2 & 0 & 17 & 3 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 14 & 25 & 17 & 14 & 0 & 111 & 19 \\ 3 & 4 & 3 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 3 & 0 & 29 & 4 \\ 2 & 4 & 1 & 2 & 0 & 20 & 3 \\ 2 & 3 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 16 & 29 & 20 & 16 & 0 & 148 & 22 \\ 3 & 4 & 3 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

4.927 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$T = 9$ [4] $1^{113}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 1 & 3 \\ 2 & 3 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{13}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 3 & -1 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{113}3^{12}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 13 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 10 & 15 & 34 & 6 & 45 & 12 \\ 2 & 3 & 6 & 0 & 9 & 2 \\ 13 & 20 & 45 & 9 & 58 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{112}3^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 8 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 7 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[0] $1^{112}3^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 8 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 4 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^{121} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 4 & 5 & 2 & 2 & 7 & 4 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 6 & 10 & 7 & 5 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{23} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 4 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 21^{121} 3^{121} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 5 & 3 \\ 3 & 1 & 3 & 12 & 5 & 3 \\ 2 & 3 & 1 & 16 & 6 & 3 \\ 12 & 12 & 16 & 70 & 28 & 18 \\ 5 & 5 & 6 & 28 & 9 & 7 \\ 3 & 3 & 3 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3^{121} 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 10 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 8 & 15 & 34 & 6 & 45 & 12 \\ 2 & 3 & 6 & 0 & 9 & 2 \\ 10 & 20 & 45 & 9 & 58 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{23} 3^{121} 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 14 & 3 \\ 3 & 5 & 15 & 3 & 25 & 5 \\ 8 & 15 & 34 & 6 & 57 & 12 \\ 2 & 3 & 6 & 0 & 11 & 2 \\ 14 & 25 & 57 & 11 & 92 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^{121} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^{121} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 4 & 2 & 4 & 2 & 6 & 4 \\ 6 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{13} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 2 & 2 & 4 & 2 & 10 & 4 \\ 9 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{122} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 4 & 2 & 4 & 2 & 10 & 4 \\ 8 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13} 3^1 21^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 12 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 9 & 15 & 34 & 6 & 45 & 12 \\ 2 & 3 & 6 & 0 & 9 & 2 \\ 12 & 20 & 45 & 9 & 58 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 3^1 21^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 8 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 5 & 10 & 14 & 4 & 22 & 8 \\ 2 & 3 & 4 & 0 & 7 & 2 \\ 8 & 15 & 22 & 7 & 32 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 3^1 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 1 & 2 & 16 & 3 \\ 2 & 3 & 2 & 1 & 12 & 3 \\ 13 & 12 & 16 & 12 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13} 3^1 21^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 7 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 12 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{13} 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 5 & 0 & 2 & 2 & 4 \\ 3 & 5 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1^1} 1^3 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.71 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1^2} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 2 & 2 & 4 & 2 & 12 & 4 \\ 10 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1^{11}} 1^3 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 & 4 \\ 3 & 2 & 4 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1^{24}} 1^{12} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 16 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 4 & 20 & 3 \\ 4 & 2 & 4 & 2 & 20 & 4 \\ 16 & 14 & 20 & 20 & 106 & 22 \\ 3 & 3 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1^{13}} 1^{122} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 15 & 3 \\ 2 & 1 & 3 & 10 & 14 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 9 & 10 & 12 & 42 & 68 & 14 \\ 15 & 14 & 20 & 68 & 106 & 22 \\ 3 & 3 & 3 & 14 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1^{24}} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 12 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 4 & 2 & 4 & 2 & 12 & 4 \\ 12 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^2} 1^{112} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 1 & 2 & 16 & 3 \\ 12 & 3 & 2 & 1 & 12 & 3 \\ 12 & 12 & 16 & 12 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1^{11}} 1^{113} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 & 4 \\ 3 & 4 & 0 & 4 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1^{32}} 1^{112} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 12 & 3 \\ 2 & 1 & 3 & 9 & 12 & 3 \\ 2 & 3 & 1 & 14 & 16 & 3 \\ 11 & 9 & 14 & 49 & 60 & 15 \\ 12 & 12 & 16 & 60 & 70 & 18 \\ 3 & 3 & 3 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 3 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 6 & 10 & 0 & 4 & 14 & 8 \\ 3 & 5 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 & 3 \\ 2 & 2 & 1 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 22 & 9 & 10 \\ 3 & 4 & 0 & 9 & 2 & 4 \\ 3 & 4 & -1 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 2 & 6 & 3 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 3 & 6 & 0 & 6 & 5 & 13 & 7 \\ 2 & 3 & -1 & 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 13^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 2 & 5 & 2 & 2 & 7 & 4 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 5 & 10 & 7 & 5 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 4 & 5 & 6 & 9 & 7 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{13} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 7 & 5 & 6 & 9 & 7 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 2 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 9 & 4 \\ 6 & 9 & 0 & 9 & 22 & 10 \\ 3 & 4 & -1 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{121}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 4 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{124}3^121^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 14 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 4 & 2 & 4 & 2 & 16 & 4 \\ 14 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{123}3^121^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 2 & 2 & 4 & 2 & 16 & 4 \\ 14 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{121}3^121^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 22 & 8 \\ 2 & 3 & 4 & 0 & 7 & 2 \\ 8 & 15 & 22 & 7 & 32 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{133}3^121^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 4 & 18 & 3 \\ 2 & 2 & 4 & 2 & 18 & 4 \\ 15 & 15 & 18 & 18 & 97 & 21 \\ 3 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{111}3^121^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 8 & 15 & 0 & 7 & 32 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{112}3^121^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 11 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 7 & 5 & 6 & 9 & 22 & 7 \\ 11 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{111}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 3 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 10 & 0 & 5 & 14 & 8 \\ 3 & 5 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{13} 1^{11} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 11 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 4 & 5 & 6 & 9 & 22 & 7 \\ 11 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{11} 1^3 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 5 & 10 & 3 & 5 & 5 \\ 5 & 10 & 14 & 4 & 8 & 8 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 3 & 5 & 8 & 2 & 2 & 4 \\ 3 & 5 & 8 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{12} 1^{11} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{13} 1^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 9 & 15 & 0 & 6 & 34 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{22} 1^3 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 4 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11} 1^{11} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^3 1^3 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -30, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.87 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 1^3 1^3 1^3 1^3 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 2 & 6 & 3 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 4 & 6 & 0 & 6 & 5 & 13 & 7 \\ 2 & 3 & -1 & 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 1^3 1^3 1^3 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 1 & 3 \\ 2 & 3 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{113} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 4 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{112} 3^{121} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 8 & 7 & 10 & 25 & 34 & 11 \\ 12 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{112} 3^{121} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 3 & 3 & 2 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{112} 3^{121} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 12 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{113} 3^{121} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 5 & 10 & 3 & 10 & 5 \\ 5 & 10 & 14 & 4 & 16 & 8 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 10 & 16 & 4 & 14 & 8 \\ 3 & 5 & 8 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{112} 3^{121} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 10 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 9 & 7 & 10 & 25 & 34 & 11 \\ 10 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{113} 3^{121} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 10 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 7 & 10 & 14 & 4 & 22 & 8 \\ 2 & 3 & 4 & 0 & 7 & 2 \\ 10 & 15 & 22 & 7 & 32 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{123} 1^1 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 4 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{122} 1^1 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 4 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{11} 1^1 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[68] $1^4 1^{14} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 2 & 20 & 3 \\ 2 & 3 & 1 & 4 & 24 & 3 \\ 2 & 2 & 4 & 2 & 24 & 4 \\ 20 & 20 & 24 & 24 & 174 & 28 \\ 3 & 3 & 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{14} 1^1 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 12 & 20 & 0 & 8 & 62 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{11} 1^1 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{12} 1^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{11} 1^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{12} 1^1 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 6 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{13} 3^{12} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 3 & 3 & 2 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{13} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 7 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{13} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 8 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{13} 3^{12} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{14} 3^{12} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{13} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 9 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 5 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{12} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 11 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 5 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{12} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[80] $1^{11}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[84] $1^{122}3^121^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 9 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 22 & 8 \\ 2 & 3 & 4 & 0 & 7 & 2 \\ 9 & 15 & 22 & 7 & 32 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[83] $1^{132}3^121^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 2 & 6 & 3 \\ 3 & 5 & 10 & 3 & 10 & 5 \\ 7 & 10 & 14 & 4 & 16 & 8 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 10 & 16 & 4 & 14 & 8 \\ 3 & 5 & 8 & 2 & 8 & 4 \end{pmatrix}_{b_0}$

[85] $1^{111}3^121^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 3 & 7 & 4 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 6 & 10 & 7 & 5 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$

[86] $1^{124}3^121^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 4 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[87] $1^{112}3^121^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 11 & 20 & 0 & 9 & 58 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[88] $1^{134}3^121^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 2 & 15 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 7 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[89] $1^{112}3^121^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 9 & 15 & 0 & 7 & 32 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[90] $1^{112}3^121^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{14} 1^1 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 5 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{14} 1^{114} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 20 & 3 \\ 2 & 1 & 3 & 3 & 20 & 3 \\ 2 & 3 & 1 & 2 & 24 & 3 \\ 3 & 3 & 2 & 1 & 20 & 3 \\ 20 & 20 & 24 & 20 & 174 & 28 \\ 3 & 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{12} 1^1 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.79 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{33} 1^1 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 9 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 7 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{33} 1^{111} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 9 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 7 & 5 & 6 & 9 & 17 & 7 \\ 9 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^2 1^{111} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 8 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 4 & 5 & 6 & 9 & 17 & 7 \\ 8 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{22} 1^1 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 6 & 3 \\ 3 & 5 & 15 & 3 & 10 & 5 \\ 8 & 15 & 34 & 6 & 24 & 12 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 6 & 10 & 24 & 4 & 14 & 8 \\ 3 & 5 & 12 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{11} 1^{12} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[102] \quad & 1^{133} 1^{124} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 15 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 9 & 8 & 8 & 22 & 48 & 10 \\ 15 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0} \\
[101] \quad & 1^{13} 1^{12} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0} \\
[99] \quad & 1^{24} 1^{122} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 10 & 16 & 3 \\ 2 & 1 & 3 & 10 & 14 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 10 & 10 & 12 & 42 & 68 & 14 \\ 16 & 14 & 20 & 68 & 106 & 22 \\ 3 & 3 & 3 & 14 & 22 & 4 \end{pmatrix}_{b_0} \\
[98] \quad & 1^{11} 1^{12} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 3 & 2 & 4 & 2 & 12 & 4 \\ 9 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0} \\
[103] \quad & 1^4 1^{112} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -66, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 4 & 16 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 4 & 5 & 6 & 9 & 34 & 7 \\ 16 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0} \\
[108] \quad & 1^{12} 1^{11} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0} \\
[107] \quad & 1^{14} 1^{11} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 10 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
[109] \quad & 1^{21} 1^{112} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 7 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 6 & 5 & 6 & 9 & 16 & 7 \\ 7 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0} \\
[106] \quad & 1^{13} 1^{11} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[105] $1^{11}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 7 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[104] $1^{13}3^{112}21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 9 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 5 & 6 & 9 & 16 & 7 \\ 9 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[110] $1^{11}3^{112}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 1 & 8 & 3 \\ 7 & 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[111] $1^{13}3^{112}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 1 & 8 & 3 \\ 9 & 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[112] $1^{11}3^{11}3^{11}21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 10 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 5 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 10 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[113] $1^{23}3^{11}3^{11}21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 9 & 3 \\ 3 & 5 & 15 & 3 & 15 & 5 \\ 8 & 15 & 34 & 6 & 36 & 12 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 15 & 36 & 6 & 34 & 12 \\ 3 & 5 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[114] $1^{22}3^{11}3^{11}21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 13 & 3 \\ 3 & 5 & 15 & 3 & 25 & 5 \\ 8 & 15 & 34 & 6 & 57 & 12 \\ 2 & 3 & 6 & 0 & 11 & 2 \\ 13 & 25 & 57 & 11 & 92 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[115] $21^{132}3^{11}21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 10 & 3 \\ 3 & 1 & 3 & 17 & 10 & 3 \\ 2 & 3 & 1 & 22 & 12 & 3 \\ 17 & 17 & 22 & 137 & 78 & 25 \\ 10 & 10 & 12 & 78 & 42 & 14 \\ 3 & 3 & 3 & 25 & 14 & 4 \end{pmatrix}_{b_0}$$

[116] $1^{23}3^{11}3^{11}21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 4 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{123} 1^{11} 3^{121} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 4 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 10 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{14} 1^{11} 3^{121} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{12} 1^{11} 3^{121} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{134} 1^{11} 3^{121} 1^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 7 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{133} 1^{11} 3^{121} 1^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 14 & 3 \\ 3 & 5 & 15 & 3 & 25 & 5 \\ 9 & 15 & 34 & 6 & 57 & 12 \\ 2 & 3 & 6 & 0 & 11 & 2 \\ 14 & 25 & 57 & 11 & 92 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{13} 1^{114} 3^{121} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 15 & 3 \\ 2 & 1 & 3 & 5 & 16 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 4 & 5 & 6 & 9 & 32 & 7 \\ 15 & 16 & 16 & 32 & 94 & 20 \\ 3 & 3 & 3 & 7 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{122} 1^{11} 3^{121} 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 3 \\ 3 & 5 & 10 & 3 & 10 & 5 \\ 6 & 10 & 14 & 4 & 16 & 8 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 10 & 16 & 4 & 14 & 8 \\ 3 & 5 & 8 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{134} 1^{113} 3^{121} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 14 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 7 & 5 & 6 & 9 & 27 & 7 \\ 14 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{13} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -77, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 13 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 4 & 5 & 6 & 9 & 27 & 7 \\ 13 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^{14} 3^{12} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -78, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 10 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{12} 3^{12} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -80, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{13} 3^{12} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 9 & 14 & 0 & 49 & 15 \\ 3 & 3 & 3 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{13} 3^{12} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 12 & 20 & 0 & 9 & 58 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{13} 3^{12} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 14 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 5 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 3 & 3 & 2 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{13} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[134] $1^{14} 1^{11} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 9 & 14 & 0 & 49 & 15 \\ 3 & 3 & 3 & 0 & 15 & 4 \end{pmatrix}_{b_0}$

[132] $1^{13} 1^{11} 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 14 & 25 & 0 & 11 & 92 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[135] $1^{24} 1^{13} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 10 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 10 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[136] $1^{14} 1^{13} 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 2 & 22 & 3 \\ 2 & 3 & 2 & 1 & 17 & 3 \\ 18 & 17 & 22 & 17 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$

[137] $1^{24} 1^{13} 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 2 & 12 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 8 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 12 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[138] $1^{11} 1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 5 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 4 & 5 & 6 & 9 & 11 & 7 \\ 5 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$

[139] $1^{32} 1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 6 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 7 & 5 & 6 & 9 & 11 & 7 \\ 6 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$

[140] $1^{13} 1^{11} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[141] $1^1 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 9 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[142] & 1^{121} 1^{112} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 9 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 8 & 7 & 10 & 25 & 34 & 11 \\ 9 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0} \\
[143] & 1^{12} 1^{11} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 8 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
[144] & 1^{13} 1^{11} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 9 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
[145] & 1^{13} 1^3 3^{121} 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[146] & 1^{24} 1^{12} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 18 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 4 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[147] & 1^{13} 1^{11} 3^{121} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -93, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 3 & 17 & 3 \\ 2 & 3 & 1 & 2 & 22 & 3 \\ 2 & 3 & 2 & 1 & 17 & 3 \\ 17 & 17 & 22 & 17 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0} \\
[148] & 1^{34} 1^{11} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -93, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 13 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 7 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 13 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[149] & 1^{43} 1^3 3^{121} 1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 9 & 3 \\ 3 & 5 & 15 & 3 & 15 & 5 \\ 10 & 15 & 34 & 6 & 36 & 12 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 15 & 36 & 6 & 34 & 12 \\ 3 & 5 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[153] $1^{12}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 14 & 25 & 0 & 12 & 94 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[154] $1^{12}3^{12}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 1 & 8 & 3 \\ 8 & 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[155] $1^{12}3^{12}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 13 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 13 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[156] $1^{24}3^{12}21^{23} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 13 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 13 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[152] $1^{23}3^{12}21^{24} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 11 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[150] $1^{12}3^{12}21^{24} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 13 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 6 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 13 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[151] $1^{12}3^{12}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 1 & 8 & 3 \\ 8 & 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[157] $1^{33}3^{12}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 9 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 7 & 5 & 6 & 9 & 16 & 7 \\ 9 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$

[158] $1^{11}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 8 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[159] \quad 1^{112} 3^{121} 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -99, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 7 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 4 & 5 & 6 & 9 & 16 & 7 \\ 7 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[160] \quad 1^{12} 3^{121} 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 3 & 2 & 4 & 2 & 12 & 4 \\ 10 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[161] \quad 1^{14} 3^{121} 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 3 & 2 & 4 & 2 & 12 & 4 \\ 12 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[162] \quad 1^{123} 3^{121} 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 14 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 10 & 10 & 12 & 42 & 54 & 14 \\ 14 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[163] \quad 1^{143} 3^{121} 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 13 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 12 & 10 & 12 & 42 & 54 & 14 \\ 13 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[164] \quad 1^{12} 3^{121} 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{14} 3^{121} 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 3 & 3 & 2 & 1 & 10 & 3 \\ 12 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[166] \quad 1^{123} 3^{121} 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 9 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[167] $1^{122} 1^{11} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 11 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[168] $1^{111} 1^{11} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 10 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[169] $1^{13} 1^{11} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 10 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[170] $1^3 1^{11} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[174] $1^{44} 1^3 1^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 12 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 10 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 12 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[175] $1^{33} 1^3 1^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 2 & 11 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 9 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 11 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[176] $1^{22} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 11 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 9 & 8 & 8 & 22 & 32 & 10 \\ 11 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$

[173] $1^{24} 1^{11} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 10 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 6 & 5 & 6 & 9 & 17 & 7 \\ 10 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$

[171] $1^{13} 1^{22} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 11 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 7 & 8 & 8 & 22 & 32 & 10 \\ 11 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

[172] $1^{\overline{111}} 3^{\overline{1}} 21^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 7 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 5 & 5 & 6 & 9 & 17 & 7 \\ 7 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[177] $1^{\overline{34}} 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{134}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 15 & 3 \\ 3 & 5 & 15 & 3 & 25 & 5 \\ 9 & 15 & 34 & 6 & 57 & 12 \\ 2 & 3 & 6 & 0 & 11 & 2 \\ 15 & 25 & 57 & 11 & 92 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[178] $1^{\overline{24}} 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 15 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 4 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[179] $1^{\overline{3}} 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 2 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[180] $1^{\overline{13}} 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 9 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 5 & 5 & 6 & 9 & 17 & 7 \\ 9 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{\overline{22}} 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 8 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 6 & 5 & 6 & 9 & 17 & 7 \\ 8 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[182] $1^{\overline{13}} 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 3 & 2 & 4 & 2 & 12 & 4 \\ 11 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[184] $1^{\overline{12}} 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{122}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 10 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 7 & 7 & 10 & 25 & 34 & 11 \\ 10 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[183] $1^{144} 1^{112} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 10 & 7 & 10 & 25 & 34 & 11 \\ 12 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[185] $1^{14} 1^{113} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 16 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 4 & 5 & 6 & 9 & 33 & 7 \\ 16 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[186] $1^{12} 1^{113} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 1 & 13 & 3 \\ 12 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[187] $1^{12} 1^{11} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 13 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 5 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 13 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[188] $1^{21} 1^{111} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 5 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 6 & 5 & 6 & 9 & 11 & 7 \\ 5 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[189] $1^{14} 1^{11} 3^1 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 15 & 25 & 0 & 11 & 92 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[190] $1^{24} 1^{11} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 13 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 4 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[191] $21^{123} 3^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 17 & 3 \\ 3 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 12 & 12 & 16 & 70 & 100 & 18 \\ 17 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

[192] $1^{34} 1^3 1^1 21^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 12 & 3 \\ 3 & 5 & 20 & 3 & 20 & 5 \\ 11 & 20 & 62 & 8 & 64 & 16 \\ 2 & 3 & 8 & 0 & 8 & 2 \\ 12 & 20 & 64 & 8 & 62 & 16 \\ 3 & 5 & 16 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{113} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 15 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 7 & 7 & 10 & 25 & 51 & 11 \\ 15 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[194] \quad 1^{12} 3^{121} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[193] \quad 1^{12} 3^{121} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[197] \quad 1^{113} 3^{121} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -125, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 3 & 3 & 2 & 1 & 15 & 3 \\ 16 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[196] \quad 1^{11} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -125, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[198] \quad 1^{12} 3^{121} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 15 & 25 & 0 & 12 & 94 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[199] \quad 1^{111} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 10 & 7 & 10 & 25 & 37 & 11 \\ 12 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[200] \quad 1^{111} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 11 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 7 & 7 & 10 & 25 & 37 & 11 \\ 11 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}$$

[201] $1^{14}3^121^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[202] $1^{123}3^121^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 12 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[203] $1^{133}3^121^{133} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 9 & 3 \\ 3 & 5 & 15 & 3 & 15 & 5 \\ 9 & 15 & 34 & 6 & 36 & 12 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 15 & 36 & 6 & 34 & 12 \\ 3 & 5 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[204] $1^{114}3^121^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 20 & 3 \\ 2 & 1 & 3 & 7 & 20 & 3 \\ 2 & 3 & 1 & 10 & 24 & 3 \\ 7 & 7 & 10 & 25 & 68 & 11 \\ 20 & 20 & 24 & 68 & 174 & 28 \\ 3 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[206] $1^{13}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

[205] $1^{12}3^121^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 2 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 16 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[207] $1^{112}3^121^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 4 & 18 & 3 \\ 3 & 2 & 4 & 2 & 18 & 4 \\ 14 & 15 & 18 & 18 & 97 & 21 \\ 3 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[208] $1^{122}3^121^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 6 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 6 & 5 & 6 & 9 & 11 & 7 \\ 6 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[209] $1^{124}3^121^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[210] $1^{111}3^{11}21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 5 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 5 & 5 & 6 & 9 & 11 & 7 \\ 5 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$

[211] $1^{12}3^{13}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 3 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 17 & 30 & 0 & 14 & 134 & 24 \\ 3 & 5 & 0 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[212] $21^{111}3^{11}21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 5 & 5 & 6 & 9 & 11 & 7 \\ 5 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$

[213] $1^{123}3^{11}21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 14 & 15 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 14 & 12 & 16 & 70 & 84 & 18 \\ 15 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$

[214] $1^{123}3^{11}21^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 16 & 17 & 3 \\ 2 & 1 & 3 & 14 & 17 & 3 \\ 2 & 3 & 1 & 20 & 22 & 3 \\ 16 & 14 & 20 & 106 & 122 & 22 \\ 17 & 17 & 22 & 122 & 137 & 25 \\ 3 & 3 & 3 & 22 & 25 & 4 \end{pmatrix}_{b_0}$

[215] $1^{112}3^{11}21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 13 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 4 & 5 & 6 & 9 & 28 & 7 \\ 13 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$

[216] $1^{112}3^{11}21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 13 & 3 \\ 2 & 1 & 3 & 9 & 12 & 3 \\ 2 & 3 & 1 & 14 & 16 & 3 \\ 11 & 9 & 14 & 49 & 60 & 15 \\ 13 & 12 & 16 & 60 & 70 & 18 \\ 3 & 3 & 3 & 15 & 18 & 4 \end{pmatrix}_{b_0}$

[217] $1^{111}3^{11}21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 9 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 6 & 5 & 6 & 9 & 17 & 7 \\ 9 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$

[218] $1^{111}3^{11}21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) =$

$$\begin{aligned}
& (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 5 & 8 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 5 & 5 & 6 & 9 & 17 & 7 \\ 8 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0} \\
& [219] \quad 1^{143} 3^{122} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -157, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 10 & 9 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 10 & 7 & 10 & 25 & 27 & 11 \\ 9 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0} \\
& [220] \quad 1^{12} 1^{111} 3^{122} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -157, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 8 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 7 & 7 & 10 & 25 & 27 & 11 \\ 8 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0} \\
& [221] \quad 1^{22} 1^{113} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -160, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 14 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 8 & 7 & 10 & 25 & 51 & 11 \\ 14 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0} \\
& [222] \quad 1^{22} 1^{113} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -160, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 12 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 6 & 5 & 6 & 9 & 27 & 7 \\ 12 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0} \\
& [223] \quad 1^{22} 1^{112} 3^{121} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -162, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 8 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 6 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0} \\
& [224] \quad 1^{12} 1^{112} 3^{121} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -162, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 5 & 8 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0} \\
& [225] \quad 1^{33} 1^{123} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -163, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 13 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 9 & 8 & 8 & 22 & 40 & 10 \\ 13 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0} \\
& [226] \quad 1^{14} 1^{123} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -163, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 14 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 7 & 8 & 8 & 22 & 40 & 10 \\ 14 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[227] $1^{123}3^{121}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 9 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 6 & 5 & 6 & 9 & 16 & 7 \\ 9 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[228] $1^{111}3^{121}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 7 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 5 & 6 & 9 & 16 & 7 \\ 7 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[229] $1^{134}3^{121}1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 12 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 7 & 5 & 6 & 9 & 22 & 7 \\ 12 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[230] $1^{12}3^{121}1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 10 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 4 & 5 & 6 & 9 & 22 & 7 \\ 10 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[231] $1^{122}3^{121}1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 3 & 2 & 4 & 2 & 16 & 4 \\ 12 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[232] $1^{24}3^{121}1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 8 & 8 & 8 & 22 & 32 & 10 \\ 12 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

[233] $1^{22}3^{121}1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 10 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 8 & 8 & 8 & 22 & 32 & 10 \\ 10 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

[234] $1^{111}3^{121}1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 9 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 7 & 7 & 10 & 25 & 34 & 11 \\ 9 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[235] \quad & 1^{12} 1^{13} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0} \\
[236] \quad & 1^{13} 1^{14} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 1 & 18 & 3 \\ 17 & 18 & 20 & 18 & 130 & 24 \\ 3 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0} \\
[237] \quad & 1^{24} 1^{11} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 8 & 7 & 10 & 25 & 37 & 11 \\ 12 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0} \\
[238] \quad & 1^{13} 1^{22} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 13 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 9 & 10 & 12 & 42 & 56 & 14 \\ 13 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0} \\
[239] \quad & 1^{11} 1^{22} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 9 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 7 & 8 & 8 & 22 & 32 & 10 \\ 9 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0} \\
[240] \quad & 1^{33} 1^{11} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 11 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 9 & 7 & 10 & 25 & 37 & 11 \\ 11 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0} \\
[241] \quad & 1^{13} 1^{12} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0} \\
[242] \quad & 1^4 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 16 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0} \\
[243] \quad & 1^{34} 1^3 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 9 & 2 & 12 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 9 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 12 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[244] & 1^{\overline{144}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{134}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 12 & 16 & 3 \\ 2 & 1 & 3 & 9 & 14 & 3 \\ 2 & 3 & 1 & 14 & 20 & 3 \\ 12 & 9 & 14 & 49 & 74 & 15 \\ 16 & 14 & 20 & 74 & 106 & 22 \\ 3 & 3 & 3 & 15 & 22 & 4 \end{pmatrix}_{b_0} \\
[245] & 1^{\overline{124}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 16 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 6 & 5 & 6 & 9 & 34 & 7 \\ 16 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0} \\
[246] & 1^{\overline{123}} 1^{\overline{123}} 3^{\overline{1}} 21^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 15 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 8 & 8 & 8 & 22 & 48 & 10 \\ 15 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0} \\
[247] & 1^{\overline{113}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 1 & 13 & 3 \\ 13 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[248] & 21^{\overline{113}} 3^{\overline{1}} 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 17 & 3 \\ 3 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 5 & 5 & 6 & 9 & 39 & 7 \\ 17 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0} \\
[249] & 1^{\overline{14}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 1 & 13 & 3 \\ 14 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[250] & 1^{\overline{134}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 16 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 9 & 7 & 10 & 25 & 51 & 11 \\ 16 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0} \\
[251] & 1^{\overline{112}} 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[252] & 1^{\overline{14}} 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 3 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 18 & 30 & 0 & 14 & 134 & 24 \\ 3 & 5 & 0 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{123} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 12 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 11 & 10 & 12 & 42 & 54 & 14 \\ 12 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{123} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 16 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 16 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[255] \quad 1^{113} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 14 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[256] \quad 1^{111} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 8 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 9 & 7 & 10 & 25 & 27 & 11 \\ 8 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[257] \quad 1^{124} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[258] \quad 1^{124} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 20 & 3 \\ 2 & 1 & 3 & 8 & 20 & 3 \\ 2 & 3 & 1 & 8 & 24 & 3 \\ 8 & 8 & 8 & 22 & 64 & 10 \\ 20 & 20 & 24 & 64 & 174 & 28 \\ 3 & 3 & 3 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[259] \quad 1^{112} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 11 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 8 & 7 & 10 & 25 & 34 & 11 \\ 11 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[260] \quad 1^{13} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 4 & 18 & 3 \\ 3 & 2 & 4 & 2 & 18 & 4 \\ 15 & 15 & 18 & 18 & 97 & 21 \\ 3 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[261] \quad 1^{1^3} 1^{12} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 15 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 4 & 5 & 6 & 9 & 34 & 7 \\ 15 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[262] \quad 1^{1^{33}} 1^{12} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.87 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 11 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 9 & 7 & 10 & 25 & 34 & 11 \\ 11 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[263] \quad 1^{1^{14}} 1^{11} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[264] \quad 1^{1^{12}} 1^{11} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 13 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[265] \quad 1^{1^3} 1^{12} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[266] \quad 1^{1^{22}} 1^{12} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -198, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 10 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 6 & 5 & 6 & 9 & 22 & 7 \\ 10 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[267] \quad 1^{1^{14}} 1^{12} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -198, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 12 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 12 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[268] \quad 21^{112} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[269] \quad 1^{1^{23}} 1^{11} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 14 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 6 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{14} 3^{12} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[271] \quad 1^{13} 3^{12} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[272] \quad 1^{13} 3^{12} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 3 & 2 & 4 & 2 & 16 & 4 \\ 13 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[273] \quad 1^{14} 3^{12} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 18 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 4 & 5 & 6 & 9 & 38 & 7 \\ 18 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[274] \quad 1^{12} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 8 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 7 & 8 & 8 & 22 & 24 & 10 \\ 8 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[275] \quad 1^{32} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 8 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 9 & 8 & 8 & 22 & 24 & 10 \\ 8 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[276] \quad 1^{13} 3^{12} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 3 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[277] \quad 1^3 3^{12} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 1 & 13 & 3 \\ 13 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[278] \quad 1^{113} 3^{114} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -213, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 1 & 13 & 3 \\ 14 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[279] \quad 1^{112} 3^{122} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -214, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 10 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 8 & 7 & 10 & 25 & 34 & 11 \\ 10 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[280] \quad 1^{112} 3^{122} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -214, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 9 & 7 & 10 & 25 & 34 & 11 \\ 12 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[281] \quad 1^{123} 3^{113} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 3 & 2 & 4 & 2 & 16 & 4 \\ 14 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[282] \quad 1^{143} 3^{113} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 4 & 18 & 3 \\ 3 & 2 & 4 & 2 & 18 & 4 \\ 16 & 15 & 18 & 18 & 97 & 21 \\ 3 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[283] \quad 1^{122} 3^{112} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -220, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 9 & 8 & 8 & 22 & 32 & 10 \\ 12 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{143} 3^{114} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -220, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 12 & 3 \\ 3 & 5 & 20 & 3 & 20 & 5 \\ 12 & 20 & 62 & 8 & 64 & 16 \\ 2 & 3 & 8 & 0 & 8 & 2 \\ 12 & 20 & 64 & 8 & 62 & 16 \\ 3 & 5 & 16 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{113} 3^{112} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -220, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 12 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 4 & 5 & 6 & 9 & 27 & 7 \\ 12 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[286] $1^{12} 1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 10 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 7 & 8 & 8 & 22 & 32 & 10 \\ 10 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

[287] $1^{14} 1^{12} 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 14 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 14 & 20 & 0 & 106 & 22 \\ 3 & 3 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[288] $1^{32} 1^{123} 3^1 21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 14 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 13 & 12 & 16 & 70 & 84 & 18 \\ 14 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[289] $1^{34} 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 18 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 9 & 7 & 10 & 25 & 61 & 11 \\ 18 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

[290] $1^{24} 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 15 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 6 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[291] $1^{23} 1^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 13 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 6 & 5 & 6 & 9 & 28 & 7 \\ 13 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[292] $1^{33} 1^{112} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 13 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 9 & 7 & 10 & 25 & 44 & 11 \\ 13 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

[293] $1^{34} 1^{111} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 9 & 7 & 10 & 25 & 37 & 11 \\ 12 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}$$

[294] $1^{23} 1^{111} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 11 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 8 & 7 & 10 & 25 & 37 & 11 \\ 11 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[295] \quad 1^{1^2} 1^{12} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -242, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 12 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 4 & 5 & 6 & 9 & 28 & 7 \\ 12 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[296] \quad 1^{11} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -243, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 9 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 9 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[297] \quad 1^{133} 1^{111} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -244, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 9 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 9 & 7 & 10 & 25 & 27 & 11 \\ 9 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{122} 1^{111} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -244, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 8 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 8 & 7 & 10 & 25 & 27 & 11 \\ 8 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[299] \quad 1^{12} 1^{113} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 14 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 7 & 7 & 10 & 25 & 51 & 11 \\ 14 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[300] \quad 1^{23} 1^{122} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -248, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 11 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 8 & 8 & 8 & 22 & 32 & 10 \\ 11 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[301] \quad 1^{13} 1^{112} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -251, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 13 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 7 & 7 & 10 & 25 & 44 & 11 \\ 13 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{33} 1^{122} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 9 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 9 & 8 & 8 & 22 & 24 & 10 \\ 9 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

[303] $1^{13}3^{12}21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 4 & 20 & 3 \\ 3 & 2 & 4 & 2 & 20 & 4 \\ 15 & 14 & 20 & 20 & 106 & 22 \\ 3 & 3 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[304] $1^{11}3^{122}21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 7 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 7 & 8 & 8 & 22 & 24 & 10 \\ 7 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

[305] $1^{13}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[306] $1^{13}3^{12}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

[307] $1^{14}3^{12}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 18 & 3 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 1 & 18 & 3 \\ 18 & 18 & 20 & 18 & 130 & 24 \\ 3 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[308] $1^{23}3^{114}21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 19 & 3 \\ 2 & 1 & 3 & 7 & 20 & 3 \\ 2 & 3 & 1 & 10 & 24 & 3 \\ 8 & 7 & 10 & 25 & 68 & 11 \\ 19 & 20 & 24 & 68 & 174 & 28 \\ 3 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[309] $1^{13}3^{14}21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 19 & 3 \\ 2 & 1 & 3 & 2 & 20 & 3 \\ 2 & 3 & 1 & 4 & 24 & 3 \\ 3 & 2 & 4 & 2 & 24 & 4 \\ 19 & 20 & 24 & 24 & 174 & 28 \\ 3 & 3 & 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[310] $1^{14}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[311] $1^{13}3^{11}21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^{124} 3^{123} 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -280, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 14 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 10 & 10 & 12 & 42 & 56 & 14 \\ 14 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[313] \quad 1^{144} 3^{123} 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -280, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 14 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 12 & 10 & 12 & 42 & 56 & 14 \\ 14 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[314] \quad 21^{112} 3^{123} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -280, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 10 & 3 \\ 3 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 10 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[315] \quad 1^{112} 3^{123} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -283, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 14 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 14 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[316] \quad 1^{122} 3^{123} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 8 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 8 & 8 & 8 & 22 & 24 & 10 \\ 8 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[317] \quad 1^{144} 3^{123} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -291, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[318] \quad 1^{114} 3^{123} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -292, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 4 & 20 & 3 \\ 3 & 2 & 4 & 2 & 20 & 4 \\ 16 & 14 & 20 & 20 & 106 & 22 \\ 3 & 3 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{112} 3^{123} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 16 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[320] \quad 1^{113} 3^{123} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -301, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 15 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 4 & 5 & 6 & 9 & 33 & 7 \\ 15 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{113} 3^{121} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 18 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 4 & 5 & 6 & 9 & 39 & 7 \\ 18 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{113} 3^{121} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 15 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 34 & 7 \\ 15 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{114} 3^{121} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 17 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 6 & 5 & 6 & 9 & 38 & 7 \\ 17 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[324] \ 1^{123} 3^{121} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 9 & 10 & 12 & 42 & 54 & 14 \\ 12 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{113} 3^{121} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 4 & 22 & 3 \\ 3 & 2 & 4 & 2 & 22 & 4 \\ 17 & 17 & 22 & 22 & 137 & 25 \\ 3 & 3 & 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{112} 3^{121} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 11 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 11 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{123} 3^{121} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 18 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 9 & 10 & 12 & 42 & 78 & 14 \\ 18 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{112} 3^{121} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 15 & 3 \\ 2 & 1 & 3 & 9 & 14 & 3 \\ 2 & 3 & 1 & 14 & 20 & 3 \\ 11 & 9 & 14 & 49 & 74 & 15 \\ 15 & 14 & 20 & 74 & 106 & 22 \\ 3 & 3 & 3 & 15 & 22 & 4 \end{pmatrix}_{b_0}$$

[329] $1^{123}3^{121}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 11 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 6 & 5 & 6 & 9 & 22 & 7 \\ 11 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[330] $1^{113}3^{121}1^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 15 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 8 & 7 & 10 & 25 & 51 & 11 \\ 15 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[331] $1^{114}3^{121}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 15 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[332] $1^{112}3^{121}1^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 16 & 3 \\ 2 & 1 & 3 & 7 & 14 & 3 \\ 2 & 3 & 1 & 10 & 20 & 3 \\ 7 & 7 & 10 & 25 & 54 & 11 \\ 16 & 14 & 20 & 54 & 106 & 22 \\ 3 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}$$

[333] $1^{123}3^{121}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 15 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 15 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

[334] $1^{124}3^{121}1^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 16 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 8 & 7 & 10 & 25 & 51 & 11 \\ 16 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[335] $1^{133}3^{121}1^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 19 & 3 \\ 2 & 1 & 3 & 12 & 20 & 3 \\ 2 & 3 & 1 & 16 & 24 & 3 \\ 13 & 12 & 16 & 70 & 112 & 18 \\ 19 & 20 & 24 & 112 & 174 & 28 \\ 3 & 3 & 3 & 18 & 28 & 4 \end{pmatrix}_{b_0}$$

[336] $1^{124}3^{121}1^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 17 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 11 & 10 & 12 & 42 & 76 & 14 \\ 17 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[337] $1^{1^{12}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 12 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 28 & 7 \\ 12 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$

[338] $1^{1^{12}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 10 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 10 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$

[339] $1^{1^{22}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 8 & 8 & 8 & 22 & 40 & 10 \\ 12 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$

[340] $1^{1^{44}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 16 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 12 & 10 & 12 & 42 & 66 & 14 \\ 16 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$

[341] $1^{1^{22}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 8 & 7 & 10 & 25 & 44 & 11 \\ 12 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$

[342] $1^{1^{24}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 12 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 6 & 5 & 6 & 9 & 22 & 7 \\ 12 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$

[343] $1^{1^3}3^{1^{14}}21^{1^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 19 & 3 \\ 2 & 1 & 3 & 3 & 20 & 3 \\ 2 & 3 & 1 & 2 & 24 & 3 \\ 2 & 3 & 2 & 1 & 20 & 3 \\ 19 & 20 & 24 & 20 & 174 & 28 \\ 3 & 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$

[344] $1^{1^{13}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 19 & 3 \\ 2 & 1 & 3 & 7 & 20 & 3 \\ 2 & 3 & 1 & 10 & 24 & 3 \\ 7 & 7 & 10 & 25 & 68 & 11 \\ 19 & 20 & 24 & 68 & 174 & 28 \\ 3 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$

[345] $1^{1^{12}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 10 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 9 & 10 & 12 & 42 & 44 & 14 \\ 10 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[346] $1^{122}3^{121}1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 10 & 10 & 12 & 42 & 56 & 14 \\ 12 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[347] $1^{122}3^{121}1^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 16 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 12 & 12 & 16 & 70 & 88 & 18 \\ 16 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

[348] $1^{14}3^{121}1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 20 & 3 \\ 2 & 1 & 3 & 2 & 20 & 3 \\ 2 & 3 & 1 & 4 & 24 & 3 \\ 3 & 2 & 4 & 2 & 24 & 4 \\ 20 & 20 & 24 & 24 & 174 & 28 \\ 3 & 3 & 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[349] $1^{114}3^{121}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 3 & 20 & 3 \\ 2 & 3 & 1 & 2 & 24 & 3 \\ 2 & 3 & 2 & 1 & 20 & 3 \\ 20 & 20 & 24 & 20 & 174 & 28 \\ 3 & 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[350] $1^{122}3^{121}1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 13 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 11 & 10 & 12 & 42 & 56 & 14 \\ 13 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[351] $1^{14}3^{121}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 17 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 17 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

[352] $1^{113}3^{121}1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 15 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 6 & 5 & 6 & 9 & 33 & 7 \\ 15 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[353] $1^{113}3^{121}1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 13 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 6 & 5 & 6 & 9 & 27 & 7 \\ 13 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[354] $1^{1^{14}} 1^{1^{13}} 3^1 21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 14 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 5 & 5 & 6 & 9 & 27 & 7 \\ 14 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[355] $1^{1^{12}} 1^{1^{13}} 3^1 21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 12 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 7 & 7 & 10 & 25 & 44 & 11 \\ 12 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

[356] $1^{1^{24}} 1^{1^{23}} 3^1 21^{1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 16 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 8 & 8 & 8 & 22 & 48 & 10 \\ 16 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

[357] $1^{1^{23}} 1^{1^{23}} 3^1 21^{1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 15 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 12 & 12 & 16 & 70 & 84 & 18 \\ 15 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[358] $1^{1^{14}} 1^{1^{13}} 3^1 21^{1^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & 0 & 25 & 4 \end{pmatrix}_{b_0}$$

[359] $1^{1^{12}} 1^{1^{13}} 3^1 21^{1^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 16 & 3 \\ 2 & 1 & 3 & 7 & 14 & 3 \\ 2 & 3 & 1 & 10 & 20 & 3 \\ 9 & 7 & 10 & 25 & 54 & 11 \\ 16 & 14 & 20 & 54 & 106 & 22 \\ 3 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}$$

[360] $1^{1^{14}} 1^{1^{13}} 3^1 21^{1^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 18 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 7 & 7 & 10 & 25 & 61 & 11 \\ 18 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

[361] $1^{1^{24}} 1^{1^{24}} 3^1 21^{1^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 20 & 3 \\ 2 & 1 & 3 & 12 & 20 & 3 \\ 2 & 3 & 1 & 16 & 24 & 3 \\ 12 & 12 & 16 & 70 & 112 & 18 \\ 20 & 20 & 24 & 112 & 174 & 28 \\ 3 & 3 & 3 & 18 & 28 & 4 \end{pmatrix}_{b_0}$$

[362] $1^{1^{14}} 1^{1^{13}} 3^1 21^{1^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 18 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 4 & 22 & 3 \\ 3 & 2 & 4 & 2 & 22 & 4 \\ 18 & 17 & 22 & 22 & 137 & 25 \\ 3 & 3 & 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[363] $1^{124} 1^{112} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 14 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 6 & 5 & 6 & 9 & 28 & 7 \\ 14 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[364] $1^{14} 1^{112} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 16 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 34 & 7 \\ 16 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

[365] $1^{14} 1^{112} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 14 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 28 & 7 \\ 14 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[366] $1^{12} 1^{122} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 9 & 10 & 12 & 42 & 56 & 14 \\ 12 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[367] $21^{123} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 13 & 3 \\ 3 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 10 & 10 & 12 & 42 & 54 & 14 \\ 13 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[368] $1^{14} 1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -366, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 1 & 18 & 3 \\ 18 & 18 & 20 & 18 & 130 & 24 \\ 3 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[369] $1^{13} 1^{12} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[370] $1^{13} 1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 1 & 18 & 3 \\ 17 & 18 & 20 & 18 & 130 & 24 \\ 3 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[372] $1^{14} 1^{114} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -378, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 16 & 3 \\ 2 & 1 & 3 & 5 & 16 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 32 & 7 \\ 16 & 16 & 16 & 32 & 94 & 20 \\ 3 & 3 & 3 & 7 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[371] \quad 1^{124} 3^{121} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 16 & 3 \\ 2 & 1 & 3 & 5 & 16 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 6 & 5 & 6 & 9 & 32 & 7 \\ 16 & 16 & 16 & 32 & 94 & 20 \\ 3 & 3 & 3 & 7 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[373] \quad 1^{14} 3^{121} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[376] \quad 1^{113} 3^{121} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 13 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 28 & 7 \\ 13 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[375] \quad 1^{123} 3^{121} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 13 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 7 & 8 & 8 & 22 & 40 & 10 \\ 13 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[374] \quad 1^{134} 3^{121} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 14 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 9 & 8 & 8 & 22 & 40 & 10 \\ 14 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[378] \quad 1^{122} 3^{121} 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 11 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 12 & 10 & 12 & 42 & 44 & 14 \\ 11 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[377] \quad 1^{123} 3^{121} 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 11 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 10 & 10 & 12 & 42 & 44 & 14 \\ 11 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[379] \quad 1^{114} 3^{121} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 17 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 4 & 5 & 6 & 9 & 38 & 7 \\ 17 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[380] \quad 1^{112} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 12 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 5 & 5 & 6 & 9 & 27 & 7 \\ 12 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[381] \quad 1^{114} 3^{121} 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -398, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 18 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[384] \quad 1^{111} 3^{122} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 9 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 9 & 10 & 12 & 42 & 44 & 14 \\ 9 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[383] \quad 1^{124} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 16 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 16 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[382] \quad 1^{134} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 16 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 9 & 8 & 8 & 22 & 48 & 10 \\ 16 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[386] \quad 1^{112} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -406, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 14 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 9 & 7 & 10 & 25 & 44 & 11 \\ 14 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[385] \quad 1^{124} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -406, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 14 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 8 & 7 & 10 & 25 & 44 & 11 \\ 14 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[390] \quad 1^{113} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 14 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 6 & 5 & 6 & 9 & 27 & 7 \\ 14 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[389] $1^{1^{13}}3^{1^{12}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 13 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 5 & 5 & 6 & 9 & 27 & 7 \\ 13 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[388] $1^{1^{24}}3^{1^{14}}21^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 18 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 6 & 5 & 6 & 9 & 39 & 7 \\ 18 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[387] $21^{1^{22}3^{1^{12}}21^{1^{12}}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 10 & 3 \\ 3 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 10 & 10 & 12 & 42 & 44 & 14 \\ 10 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[392] $1^{1^{23}}3^{1^{12}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -418, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 13 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 10 & 10 & 12 & 42 & 54 & 14 \\ 13 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[391] $1^{1^{44}}3^{1^{12}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -418, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 14 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 12 & 10 & 12 & 42 & 54 & 14 \\ 14 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[393] $1^{1^{13}}3^{1^{12}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -427, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 15 & 3 \\ 2 & 1 & 3 & 5 & 16 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 32 & 7 \\ 15 & 16 & 16 & 32 & 94 & 20 \\ 3 & 3 & 3 & 7 & 20 & 4 \end{pmatrix}_{b_0}$$

[394] $1^{1^{24}}3^{1^{14}}21^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 19 & 3 \\ 2 & 1 & 3 & 8 & 20 & 3 \\ 2 & 3 & 1 & 8 & 24 & 3 \\ 7 & 8 & 8 & 22 & 64 & 10 \\ 19 & 20 & 24 & 64 & 174 & 28 \\ 3 & 3 & 3 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

[395] $1^{1^{23}}3^{1^{12}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 13 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 8 & 7 & 10 & 25 & 44 & 11 \\ 13 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

[396] $1^{1^3}3^{1^{14}}21^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -437, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 17 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 4 & 5 & 6 & 9 & 39 & 7 \\ 17 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[397] $1^{134} 3^{112} 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 16 & 3 \\ 2 & 1 & 3 & 9 & 14 & 3 \\ 2 & 3 & 1 & 14 & 20 & 3 \\ 11 & 9 & 14 & 49 & 74 & 15 \\ 16 & 14 & 20 & 74 & 106 & 22 \\ 3 & 3 & 3 & 15 & 22 & 4 \end{pmatrix}_{b_0}$$

[398] $1^{123} 3^{124} 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 17 & 3 \\ 2 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 8 & 8 & 8 & 22 & 56 & 10 \\ 17 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

[399] $1^{124} 3^{114} 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 20 & 3 \\ 2 & 1 & 3 & 7 & 20 & 3 \\ 2 & 3 & 1 & 10 & 24 & 3 \\ 8 & 7 & 10 & 25 & 68 & 11 \\ 20 & 20 & 24 & 68 & 174 & 28 \\ 3 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[400] $1^{123} 3^{122} 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 13 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 10 & 10 & 12 & 42 & 56 & 14 \\ 13 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[402] $1^{114} 3^{124} 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 20 & 3 \\ 2 & 1 & 3 & 8 & 20 & 3 \\ 2 & 3 & 1 & 8 & 24 & 3 \\ 7 & 8 & 8 & 22 & 64 & 10 \\ 20 & 20 & 24 & 64 & 174 & 28 \\ 3 & 3 & 3 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

[401] $1^{114} 3^{114} 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 20 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & 0 & 28 & 4 \end{pmatrix}_{b_0}$$

[403] $1^{112} 3^{123} 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 12 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 7 & 8 & 8 & 22 & 40 & 10 \\ 12 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[404] $1^{122} 3^{123} 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 10 & 10 & 12 & 42 & 54 & 14 \\ 12 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[405] \quad & 1^{1^{23}} 1^{12} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -459, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 15 & 3 \\ 2 & 1 & 3 & 7 & 14 & 3 \\ 2 & 3 & 1 & 10 & 20 & 3 \\ 8 & 7 & 10 & 25 & 54 & 11 \\ 15 & 14 & 20 & 54 & 106 & 22 \\ 3 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0} \\
[406] \quad & 1^{1^{13}} 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -459, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 15 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 9 & 10 & 12 & 42 & 66 & 14 \\ 15 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0} \\
[407] \quad & 1^{1^{33}} 1^{122} 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -460, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 13 & 15 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 13 & 12 & 16 & 70 & 88 & 18 \\ 15 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0} \\
[408] \quad & 1^{1^{33}} 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -467, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 13 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 11 & 10 & 12 & 42 & 54 & 14 \\ 13 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0} \\
[409] \quad & 1^{1^{34}} 1^{123} 3^1 21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -467, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 13 & 16 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 13 & 12 & 16 & 70 & 84 & 18 \\ 16 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0} \\
[410] \quad & 1^{1^{34}} 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -467, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 14 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 11 & 10 & 12 & 42 & 54 & 14 \\ 14 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0} \\
[411] \quad & 21^{1^{13} 3^1 21^{112}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -468, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 13 & 3 \\ 3 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 5 & 5 & 6 & 9 & 27 & 7 \\ 13 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0} \\
[412] \quad & 1^{1^{34}} 1^{122} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -472, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 14 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 11 & 10 & 12 & 42 & 56 & 14 \\ 14 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0} \\
[413] \quad & 1^{1^{44}} 1^{122} 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -472, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 16 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 14 & 12 & 16 & 70 & 88 & 18 \\ 16 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[414] \quad 1^{1^{23}3^1 21^{144}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -475, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 17 & 3 \\ 2 & 1 & 3 & 14 & 17 & 3 \\ 2 & 3 & 1 & 20 & 22 & 3 \\ 15 & 14 & 20 & 106 & 122 & 22 \\ 17 & 17 & 22 & 122 & 137 & 25 \\ 3 & 3 & 3 & 22 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[415] \quad 1^{1^{23}3^1 21^{133}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -480, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 13 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 12 & 12 & 16 & 70 & 72 & 18 \\ 13 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[416] \quad 1^{1^{24}3^1 21^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -482, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 14 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 8 & 8 & 8 & 22 & 40 & 10 \\ 14 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[417] \quad 1^{1^{23}3^1 21^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -482, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 13 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 8 & 8 & 8 & 22 & 40 & 10 \\ 13 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[418] \quad 1^{1^{22}3^1 21^{133}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -482, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 14 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 12 & 12 & 16 & 70 & 84 & 18 \\ 14 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[419] \quad 1^{1^4 3^1 21^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -490, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 20 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 4 & 5 & 6 & 9 & 44 & 7 \\ 20 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[420] \quad 1^{1^{22}3^1 21^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 10 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 10 & 10 & 12 & 42 & 44 & 14 \\ 10 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[421] \quad 1^{1^{44}3^1 21^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 12 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 12 & 10 & 12 & 42 & 44 & 14 \\ 12 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[422] \quad & 1^{13} 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -496, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 17 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 9 & 10 & 12 & 42 & 76 & 14 \\ 17 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0} \\
[423] \quad & 1^{13} 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 17 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 7 & 7 & 10 & 25 & 61 & 11 \\ 17 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0} \\
[424] \quad & 21^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 3 \\ 3 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 5 & 5 & 6 & 9 & 33 & 7 \\ 15 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0} \\
[425] \quad & 1^{144} 1^{123} 3^1 21^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -514, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 16 & 18 & 3 \\ 2 & 1 & 3 & 14 & 17 & 3 \\ 2 & 3 & 1 & 20 & 22 & 3 \\ 16 & 14 & 20 & 106 & 122 & 22 \\ 18 & 17 & 22 & 122 & 137 & 25 \\ 3 & 3 & 3 & 22 & 25 & 4 \end{pmatrix}_{b_0} \\
[426] \quad & 1^{12} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -517, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 5 & 14 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 5 & 5 & 6 & 9 & 33 & 7 \\ 14 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0} \\
[427] \quad & 1^{24} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -520, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 16 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 6 & 5 & 6 & 9 & 33 & 7 \\ 16 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0} \\
[428] \quad & 1^{24} 1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -522, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 20 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 6 & 5 & 6 & 9 & 44 & 7 \\ 20 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0} \\
[429] \quad & 1^{34} 1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -523, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 18 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 11 & 10 & 12 & 42 & 78 & 14 \\ 18 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0} \\
[430] \quad & 1^{33} 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -523, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 15 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 11 & 10 & 12 & 42 & 66 & 14 \\ 15 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[431] $1^{123} 1^{113} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -525, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 17 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 8 & 7 & 10 & 25 & 61 & 11 \\ 17 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

[432] $1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 11 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 11 & 10 & 12 & 42 & 44 & 14 \\ 11 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[433] $1^{144} 1^{123} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -534, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 16 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 14 & 12 & 16 & 70 & 84 & 18 \\ 16 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[434] $1^{133} 1^{123} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 15 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 13 & 12 & 16 & 70 & 84 & 18 \\ 15 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[435] $1^{134} 1^{122} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 14 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 13 & 12 & 16 & 70 & 72 & 18 \\ 14 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[436] $1^{123} 1^{122} 3^1 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 15 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 12 & 12 & 16 & 70 & 88 & 18 \\ 15 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

[437] $1^{124} 1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -558, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 18 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 6 & 5 & 6 & 9 & 38 & 7 \\ 18 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[438] $1^{124} 1^{112} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -562, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 16 & 3 \\ 2 & 1 & 3 & 7 & 14 & 3 \\ 2 & 3 & 1 & 10 & 20 & 3 \\ 8 & 7 & 10 & 25 & 54 & 11 \\ 16 & 14 & 20 & 54 & 106 & 22 \\ 3 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}$$

[439] $1^{13} 1^{124} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 15 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 15 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$

[440] $1^{23} 1^{124} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 15 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 8 & 8 & 8 & 22 & 48 & 10 \\ 15 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$

[441] $1^{22} 1^{123} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -598, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 10 & 14 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 10 & 10 & 12 & 42 & 66 & 14 \\ 14 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$

[442] $21^{123} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 13 & 3 \\ 3 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 8 & 8 & 8 & 22 & 40 & 10 \\ 13 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$

[443] $1^{13} 1^{123} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -603, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 17 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 9 & 10 & 12 & 42 & 78 & 14 \\ 17 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$

[444] $1^{14} 1^{124} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 18 & 3 \\ 2 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 7 & 8 & 8 & 22 & 56 & 10 \\ 18 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$

[446] $1^{23} 1^{124} 3^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 19 & 3 \\ 2 & 1 & 3 & 12 & 20 & 3 \\ 2 & 3 & 1 & 16 & 24 & 3 \\ 12 & 12 & 16 & 70 & 112 & 18 \\ 19 & 20 & 24 & 112 & 174 & 28 \\ 3 & 3 & 3 & 18 & 28 & 4 \end{pmatrix}_{b_0}$

[445] $1^{22} 1^{122} 3^1 21^{133} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 12 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 12 & 12 & 16 & 70 & 72 & 18 \\ 12 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$

[448] $1^{34} 1^{122} 3^1 21^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 16 & 3 \\ 2 & 1 & 3 & 14 & 14 & 3 \\ 2 & 3 & 1 & 20 & 20 & 3 \\ 15 & 14 & 20 & 106 & 108 & 22 \\ 16 & 14 & 20 & 108 & 106 & 22 \\ 3 & 3 & 3 & 22 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[447] \quad 1^{1^{13}} 3^1 21^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -612, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 15 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 5 & 5 & 6 & 9 & 33 & 7 \\ 15 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[449] \quad 1^{1^{14}} 3^1 21^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -613, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 16 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 5 & 5 & 6 & 9 & 33 & 7 \\ 16 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[450] \quad 1^{1^{14}} 3^1 21^{1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -627, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 19 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 4 & 5 & 6 & 9 & 44 & 7 \\ 19 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[451] \quad 1^{1^{44}} 3^1 21^{1^{134}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -630, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 18 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 14 & 12 & 16 & 70 & 100 & 18 \\ 18 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[452] \quad 21^{1^{23}} 3^1 21^{1^{23}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -632, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 15 & 3 \\ 3 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 10 & 10 & 12 & 42 & 66 & 14 \\ 15 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[453] \quad 1^{1^{34}} 3^1 21^{1^{134}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 16 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 13 & 12 & 16 & 70 & 88 & 18 \\ 16 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[454] \quad 1^{1^{24}} 3^1 21^{1^{134}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -642, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 18 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 12 & 12 & 16 & 70 & 100 & 18 \\ 18 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[455] \quad 1^{1^{12}} 3^1 21^{1^{23}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -643, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 14 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 9 & 10 & 12 & 42 & 66 & 14 \\ 14 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[456] \quad 1^{1^{24}} 3^1 21^{1^{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -648, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 20 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 9 & 10 & 12 & 42 & 88 & 14 \\ 20 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[457] \quad 1^{13} 3^{14} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 17 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 5 & 5 & 6 & 9 & 39 & 7 \\ 17 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[458] \quad 1^{23} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -658, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 17 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 10 & 10 & 12 & 42 & 78 & 14 \\ 17 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[459] \quad 1^{24} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -664, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 18 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 8 & 7 & 10 & 25 & 61 & 11 \\ 18 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[460] \quad 1^{24} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 16 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 8 & 8 & 8 & 22 & 48 & 10 \\ 16 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[461] \quad 1^{34} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 20 & 3 \\ 2 & 1 & 3 & 12 & 20 & 3 \\ 2 & 3 & 1 & 16 & 24 & 3 \\ 13 & 12 & 16 & 70 & 112 & 18 \\ 20 & 20 & 24 & 112 & 174 & 28 \\ 3 & 3 & 3 & 18 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[462] \quad 1^{24} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -680, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 18 & 3 \\ 2 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 8 & 8 & 8 & 22 & 56 & 10 \\ 18 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[463] \quad 1^{13} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 17 & 3 \\ 2 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 7 & 8 & 8 & 22 & 56 & 10 \\ 17 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[464] \quad 1^{24} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -694, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 16 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 10 & 10 & 12 & 42 & 66 & 14 \\ 16 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[465] $1^{133} 1^{122} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 13 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 13 & 12 & 16 & 70 & 72 & 18 \\ 13 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[466] $1^{134} 1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -712, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 20 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 11 & 10 & 12 & 42 & 88 & 14 \\ 20 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

[468] $21^{124} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 18 & 3 \\ 3 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 10 & 10 & 12 & 42 & 76 & 14 \\ 18 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[467] $21^{124} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 18 & 3 \\ 3 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 8 & 8 & 8 & 22 & 56 & 10 \\ 18 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

[469] $1^{114} 1^{113} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -733, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 18 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 5 & 5 & 6 & 9 & 39 & 7 \\ 18 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[470] $1^{144} 1^{122} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 14 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 14 & 12 & 16 & 70 & 72 & 18 \\ 14 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[471] $1^{134} 1^{124} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 18 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 11 & 10 & 12 & 42 & 76 & 14 \\ 18 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[473] $1^{133} 1^{123} 3^1 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -747, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 17 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 13 & 12 & 16 & 70 & 100 & 18 \\ 17 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

[472] $1^{113} 1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -747, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 17 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 38 & 7 \\ 17 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[474] \ 1^{114} 3^{121} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -778, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 18 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 38 & 7 \\ 18 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[476] \ 1^{124} 3^{121} 1^{23} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 18 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 10 & 10 & 12 & 42 & 76 & 14 \\ 18 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[475] \ 1^{122} 3^{121} 1^{44} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 15 & 3 \\ 2 & 1 & 3 & 14 & 14 & 3 \\ 2 & 3 & 1 & 20 & 20 & 3 \\ 15 & 14 & 20 & 106 & 108 & 22 \\ 15 & 14 & 20 & 108 & 106 & 22 \\ 3 & 3 & 3 & 22 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[477] \ 1^{123} 3^{121} 1^{23} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -786, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 15 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 10 & 10 & 12 & 42 & 66 & 14 \\ 15 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[478] \ 1^{123} 3^{121} 1^{24} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -790, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 18 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 10 & 10 & 12 & 42 & 78 & 14 \\ 18 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[479] \ 1^{123} 3^{121} 1^{23} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -803, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 16 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 11 & 10 & 12 & 42 & 66 & 14 \\ 16 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[480] \ 1^{124} 3^{121} 1^{23} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -804, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 17 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 10 & 10 & 12 & 42 & 76 & 14 \\ 17 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[481] \ 21^{114} 3^{121} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -808, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 20 & 3 \\ 3 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 5 & 5 & 6 & 9 & 44 & 7 \\ 20 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[482] \ 21^{114} 3^{121} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -824, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 18 & 3 \\ 3 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 38 & 7 \\ 18 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[483] \quad 1^{122} 3^{121} 1^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -880, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 16 & 3 \\ 2 & 1 & 3 & 14 & 14 & 3 \\ 2 & 3 & 1 & 20 & 20 & 3 \\ 16 & 14 & 20 & 106 & 108 & 22 \\ 16 & 14 & 20 & 108 & 106 & 22 \\ 3 & 3 & 3 & 22 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[484] \quad 1^{123} 3^{121} 1^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -882, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 17 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 12 & 12 & 16 & 70 & 100 & 18 \\ 17 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[485] \quad 1^{114} 3^{121} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -907, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 19 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 5 & 5 & 6 & 9 & 44 & 7 \\ 19 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[486] \quad 21^{124} 3^{121} 1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -940, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 20 & 3 \\ 3 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 10 & 10 & 12 & 42 & 88 & 14 \\ 20 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[487] \quad 1^{124} 3^{121} 1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -940, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 19 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 9 & 10 & 12 & 42 & 88 & 14 \\ 19 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[488] \quad 1^{124} 3^{121} 1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -988, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 19 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 10 & 10 & 12 & 42 & 88 & 14 \\ 19 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[489] \quad 1^{123} 3^{121} 1^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -1011, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 18 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 13 & 12 & 16 & 70 & 100 & 18 \\ 18 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[490] \quad 1^{114} 3^{121} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -1018, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 20 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 5 & 5 & 6 & 9 & 44 & 7 \\ 20 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[491] \quad 1^{1^{24}} 1^{124} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -1192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 20 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 10 & 10 & 12 & 42 & 88 & 14 \\ 20 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [492] \quad 1^{111} 3^1 141^1 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 2 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 3 & 4 & 0 & 4 & 3 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [493] \quad 1^{111} 3^1 14141^1 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[496] \quad 1^{111} 3151^1 23^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 13 & 4 \\ 3 & 0 & 5 & 3 & 13 & 5 \\ 9 & 0 & 13 & 13 & 43 & 15 \\ 3 & 0 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[497] \quad 1^{111} 3^1 1^1 5 131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 2 & 3 & 1 & 0 & 2 & 1 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[498] \quad 1^{1^{12}} 31^1 5 13^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 1 & 7 & 3 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 5 & 7 & 5 & 0 & 4 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[499] \quad 1^{1^{13}} 31^1 5 13^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 1 & 14 & 3 \\ 1 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 9 & 14 & 10 & 0 & 8 & 74 & 18 \\ 2 & 3 & 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[501] \quad 1^{111} 31^1 5 13^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 5 & 2 \\ 2 & 0 & 4 & 6 & 3 & 12 & 6 \\ 1 & 0 & 1 & 3 & 0 & 5 & 2 \\ 4 & 0 & 5 & 12 & 5 & 17 & 9 \\ 2 & 0 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[500] \quad 1^{1^{12}} 31^1 5 13^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 9 & 2 \\ 2 & 0 & 4 & 6 & 3 & 20 & 6 \\ 1 & 0 & 1 & 3 & 0 & 8 & 2 \\ 6 & 0 & 9 & 20 & 8 & 49 & 15 \\ 2 & 0 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[502] \quad & 1^{11}3^11^15131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 10, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 2 & 4 & 3 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 3 & 4 & 4 & 0 & 3 & 6 & 6 \\ 2 & 3 & 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[503] \quad & 1^{14}431^1513^11^{44} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 128 & 24 \\ 1 & 1 & 0 & 12 & 0 & 12 & 2 \\ 8 & 16 & 0 & 128 & 12 & 126 & 24 \\ 2 & 3 & -1 & 24 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[504] \quad & 1^{14}331^1513^11^{43} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 6 & 2 \\ 1 & 1 & 1 & 16 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 96 & 24 \\ 1 & 1 & 0 & 12 & 0 & 9 & 2 \\ 6 & 12 & 0 & 96 & 9 & 70 & 18 \\ 2 & 3 & -1 & 24 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[505] \quad & 1^{11}231^1513^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 1 & 4 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 16 & 6 \\ 1 & 1 & 0 & 3 & 0 & 6 & 2 \\ 4 & 8 & 0 & 16 & 6 & 30 & 12 \\ 2 & 3 & -1 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[506] \quad & 1^{13}231^1513^11^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 4 & 2 \\ 1 & 1 & 1 & 12 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 48 & 18 \\ 1 & 1 & 0 & 9 & 0 & 6 & 2 \\ 4 & 8 & 0 & 48 & 6 & 30 & 12 \\ 2 & 3 & -1 & 18 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[507] \quad & 1^{11}431^1513^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 1 & 4 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 32 & 6 \\ 1 & 1 & 0 & 3 & 0 & 12 & 2 \\ 8 & 16 & 0 & 32 & 12 & 126 & 24 \\ 2 & 3 & -1 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[508] \quad & 1^{13}331^1513^11^{33} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 72 & 18 \\ 1 & 1 & 0 & 9 & 0 & 9 & 2 \\ 6 & 12 & 0 & 72 & 9 & 70 & 18 \\ 2 & 3 & -1 & 18 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[509] \quad & 1^{14}231^1513^11^{42} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 4 & 2 \\ 1 & 1 & 1 & 16 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 64 & 24 \\ 1 & 1 & 0 & 12 & 0 & 6 & 2 \\ 4 & 8 & 0 & 64 & 6 & 30 & 12 \\ 2 & 3 & -1 & 24 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[510] \quad & 1^{12}231^1513^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 1 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 & 5 & 2 \\ 4 & 7 & 5 & 17 & 4 & 19 & 9 \\ 1 & 1 & 2 & 4 & 0 & 4 & 2 \\ 4 & 7 & 5 & 19 & 4 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[511] \quad 1^{113} 31^1 513^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 24 & 6 \\ 1 & 1 & 0 & 3 & 0 & 9 & 2 \\ 6 & 12 & 0 & 24 & 9 & 70 & 18 \\ 2 & 3 & -1 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[512] \quad 1^{123} 31^1 513^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 10 & 2 \\ 2 & 0 & 4 & 6 & 3 & 24 & 6 \\ 1 & 0 & 1 & 3 & 0 & 10 & 2 \\ 8 & 0 & 10 & 24 & 10 & 74 & 18 \\ 2 & 0 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[513] \quad 1^{111} 31^1 513^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 8 & 6 \\ 1 & 1 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 8 & 3 & 6 & 6 \\ 2 & 3 & -1 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[514] \quad 1^{122} 31^1 513^1 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 6 & 32 & 12 \\ 1 & 1 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 0 & 32 & 6 & 30 & 12 \\ 2 & 3 & -1 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[515] \quad 2^1 3^1 1513^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 1 & 5 & 2 \\ 4 & 7 & 5 & 17 & 5 & 19 & 9 \\ 1 & 2 & 1 & 5 & 0 & 5 & 2 \\ 4 & 7 & 5 & 19 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[516] \quad 1^{122} 31^1 513^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 1 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 & 5 & 2 \\ 4 & 7 & 5 & 17 & 4 & 19 & 9 \\ 1 & 1 & 2 & 4 & 0 & 4 & 2 \\ 5 & 7 & 5 & 19 & 4 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[517] \quad 1^{111} 31^1 513^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 4 & 2 \\ 4 & 4 & 4 & 6 & 3 & 8 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 8 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[518] \quad 1^{113} 31^1 513^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 2 & 0 & 8 & 2 \\ 7 & 11 & 9 & 11 & 8 & 49 & 15 \\ 2 & 3 & 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[520] \quad 1^{12} 3^{11} 5^{12} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 2 & 7 & 3 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 5 & 7 & 5 & 0 & 5 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[519] \quad 1^{12} 3^{11} 5^{12} 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 2 & 8 & 3 \\ 1 & 2 & 0 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 6 & 8 & 8 & 0 & 6 & 30 & 12 \\ 2 & 3 & 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[521] \quad 1^{12} 3^{11} 5^{13} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 3 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[522] \quad 1^{11} 3^{11} 5^{13} 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 2 & 4 & 2 & 11 & 3 \\ 1 & 2 & 0 & 4 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 3 & 20 & 6 \\ 1 & 2 & 1 & 3 & 0 & 8 & 2 \\ 8 & 11 & 9 & 20 & 8 & 49 & 15 \\ 2 & 3 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[523] \quad 1^{11} 3^{11} 5^{13} 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 2 & 0 & 8 & 2 \\ 8 & 11 & 9 & 11 & 8 & 49 & 15 \\ 2 & 3 & 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[524] \quad 1^{11} 3^{11} 5^{13} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 34, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 3 & 8 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 8 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[525] \quad 1^{11} 3^{11} 5^{13} 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 2 & 4 & 2 & 14 & 3 \\ 1 & 2 & 0 & 4 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 3 & 24 & 6 \\ 1 & 2 & 1 & 3 & 0 & 10 & 2 \\ 9 & 14 & 10 & 24 & 10 & 74 & 18 \\ 2 & 3 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[526] \quad 1^{13} 3^{11} 5^{13} 13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 2 & 12 & 3 \\ 1 & 2 & 0 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 12 & 12 & 0 & 9 & 70 & 18 \\ 2 & 3 & 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[527] \quad 1^{112}3^{11}5^{12}131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 46, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[528] \quad 1^{13}3^{11}5^{13}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 2 & 11 & 3 \\ 1 & 2 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 8 & 11 & 9 & 0 & 8 & 49 & 15 \\ 2 & 3 & 2 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[529] \quad 1^{112}3^{11}5^{11}131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 3 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[530] \quad 1^{112}3^{11}5^{11}131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[531] \quad 1^{111}3^{11}5^{11}131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 52, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 4 & 2 \\ 3 & 4 & 4 & 6 & 3 & 8 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 8 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[532] \quad 1^{112}3^{11}5^{11}131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 55, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 2 & 1 & 4 & 3 & 13 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 13 & 4 \\ 1 & 3 & 0 & 3 & 1 & 11 & 3 \\ 6 & 13 & 0 & 13 & 11 & 49 & 15 \\ 2 & 4 & -1 & 4 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[533] \quad 1^{112}3^{11}5^{11}131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 58, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 7 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 7 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[534] \quad 1^{112}3^{11}5^{11}131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 63, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 3 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[535] \quad 1^{113}3^{113}15^{13}131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 66, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 2 & 4 & 2 & 12 & 3 \\ 1 & 2 & 0 & 4 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 3 & 24 & 6 \\ 1 & 2 & 1 & 3 & 0 & 9 & 2 \\ 9 & 12 & 12 & 24 & 9 & 70 & 18 \\ 2 & 3 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[536] \quad 1^{114}3^{114}15^{13}131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 69, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 11 & 2 \\ 1 & 1 & 2 & 4 & 2 & 15 & 3 \\ 1 & 2 & 0 & 4 & 1 & 13 & 2 \\ 2 & 4 & 4 & 6 & 3 & 28 & 6 \\ 1 & 2 & 1 & 3 & 0 & 11 & 2 \\ 11 & 15 & 13 & 28 & 11 & 97 & 21 \\ 2 & 3 & 2 & 6 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[537] \quad 1^{112}3^{111}15^{11}131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 90, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 5 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[538] \quad 1^{123}3^{112}15^{12}131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 90, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 2 & 11 & 3 \\ 1 & 2 & 0 & 8 & 1 & 9 & 2 \\ 5 & 8 & 8 & 30 & 6 & 40 & 12 \\ 1 & 2 & 1 & 6 & 0 & 8 & 2 \\ 7 & 11 & 9 & 40 & 8 & 49 & 15 \\ 2 & 3 & 2 & 12 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[539] \quad 1^{114}3^{114}15^{14}131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 94, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 12 & 2 \\ 1 & 1 & 2 & 4 & 2 & 16 & 3 \\ 1 & 2 & 0 & 4 & 1 & 16 & 2 \\ 2 & 4 & 4 & 6 & 3 & 32 & 6 \\ 1 & 2 & 1 & 3 & 0 & 12 & 2 \\ 12 & 16 & 16 & 32 & 12 & 126 & 24 \\ 2 & 3 & 2 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[540] \quad 1^{123}3^{113}15^{13}131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 95, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 8 & 2 \\ 1 & 1 & 2 & 7 & 2 & 11 & 3 \\ 1 & 2 & 0 & 5 & 1 & 9 & 2 \\ 4 & 7 & 5 & 17 & 5 & 31 & 9 \\ 1 & 2 & 1 & 5 & 0 & 8 & 2 \\ 8 & 11 & 9 & 31 & 8 & 49 & 15 \\ 2 & 3 & 2 & 9 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[541] \quad 1^{112}3^{112}15^{12}131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[542] \quad 1^{14}3^{113}15^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 10 & 2 \\ 1 & 1 & 2 & 4 & 2 & 15 & 3 \\ 1 & 2 & 0 & 4 & 1 & 13 & 2 \\ 2 & 4 & 4 & 6 & 3 & 28 & 6 \\ 1 & 2 & 1 & 3 & 0 & 11 & 2 \\ 10 & 15 & 13 & 28 & 11 & 97 & 21 \\ 2 & 3 & 2 & 6 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[551] $1^{124}3^1 1^1 5 131^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 188, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 10 & 2 \\ 1 & 1 & 2 & 7 & 2 & 15 & 3 \\ 1 & 2 & 0 & 5 & 1 & 13 & 2 \\ 4 & 7 & 5 & 17 & 5 & 43 & 9 \\ 1 & 2 & 1 & 5 & 0 & 11 & 2 \\ 10 & 15 & 13 & 43 & 11 & 97 & 21 \\ 2 & 3 & 2 & 9 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[552] $1^{123}3^1 1^1 5 131^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 207, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 7 & 2 \\ 1 & 1 & 2 & 7 & 2 & 11 & 3 \\ 1 & 2 & 0 & 5 & 1 & 9 & 2 \\ 4 & 7 & 5 & 17 & 5 & 31 & 9 \\ 1 & 2 & 1 & 5 & 0 & 8 & 2 \\ 7 & 11 & 9 & 31 & 8 & 49 & 15 \\ 2 & 3 & 2 & 9 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[553] $1^{114}3^1 1^1 5 131^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 297, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 10 & 2 \\ 1 & 1 & 2 & 4 & 2 & 15 & 3 \\ 1 & 2 & 0 & 4 & 1 & 13 & 2 \\ 3 & 4 & 4 & 6 & 3 & 28 & 6 \\ 1 & 2 & 1 & 3 & 0 & 11 & 2 \\ 10 & 15 & 13 & 28 & 11 & 97 & 21 \\ 2 & 3 & 2 & 6 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[554] $1^{114}3^1 1^1 5 131^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 298, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 11 & 2 \\ 1 & 1 & 2 & 4 & 2 & 16 & 3 \\ 1 & 2 & 0 & 4 & 1 & 16 & 2 \\ 3 & 4 & 4 & 6 & 3 & 32 & 6 \\ 1 & 2 & 1 & 3 & 0 & 12 & 2 \\ 11 & 16 & 16 & 32 & 12 & 126 & 24 \\ 2 & 3 & 2 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

4.928 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$T = 10 \quad [0] 1^{111}1^1 31^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 & 2 & 3 \\ 2 & 2 & 1 & 9 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 22 & 9 & 9 & 10 \\ 3 & 4 & 0 & 9 & 2 & 4 & 4 \\ 2 & 4 & 0 & 9 & 4 & 2 & 4 \\ 3 & 4 & -1 & 10 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

4.929 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + su2n3mix

$T = 12 \quad [0] 1^{122}3^1 1^1 5 13^1 1^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0$

[17] $1^{1122}3^1 1^1 5 13^1 1^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$

[16] $1^{133}3^1 1^1 5 13^1 1^{331} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.6$

[15] $1^{142}3^1 1^1 5 13^1 1^{421} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.5$

[14] $1^{1121}3^1 1^1 5 13^1 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$

[13] $1^{1123}1^1 1^1 5 13^1 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$

$$\begin{aligned}
[12] \quad & 1^{132}3^1 1^1 5 13^1 1^{321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.3 \\
[11] \quad & 1^{143}3^1 1^1 5 13^1 1^{431} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.8 \\
[10] \quad & 1^{144}3^1 1^1 5 13^1 1^{441} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 9.0 \\
[9] \quad & 1^{1113}3^1 1^1 5 13^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[8] \quad & 1^{114}3^1 1^1 5 13^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[7] \quad & 1^{1221}3^1 1^1 5 13^1 1^{222} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[6] \quad & 1^{1112}3^1 1^1 5 13^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[5] \quad & 1^{1141}3^1 1^1 5 13^1 1^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[4] \quad & 1^{113}3^1 1^1 5 13^1 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8 \\
[3] \quad & 1^{1131}3^1 1^1 5 13^1 1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8 \\
[2] \quad & 1^{111}3^1 1^1 5 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[1] \quad & 1^{1111}3^1 1^1 5 13^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0
\end{aligned}$$

4.930 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$\begin{aligned}
T = 8 \quad [0] \quad & 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1] \quad & 1^{12}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[2] \quad & 1^{13}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[3] \quad & 1^{12}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$(1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 21^{1^3 3^{11} 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 3 & 4 \\ 5 & 4 & 5 & 3 & 3 & 4 \\ 6 & 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1^3 3^{11} 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1^4 3^{11} 1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 12 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1^2 3^{11} 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 21^{1^3 3^{11} 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 3 \\ 4 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1^2 3^{11} 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 7 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1^1 3^{11} 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 6 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[20] \ 21^{1^3 3^{11} 1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 3 & 4 \\ 5 & 4 & 17 & 3 & 3 & 4 \\ 15 & 17 & 49 & 9 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 & 2 \\ 3 & 3 & 9 & 2 & 0 & 2 \\ 4 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{14}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 2 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 12 & 16 & 46 & 8 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[22] $21^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 3 \\ 4 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{14}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 2 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 11 & 16 & 46 & 8 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[24] $21^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 2 & 3 \\ 4 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 11 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[26] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 3 & 4 \\ 5 & 4 & 9 & 3 & 3 & 4 \\ 9 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 4 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[28] $21^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 3 & 4 \\ 5 & 4 & 13 & 3 & 3 & 4 \\ 12 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 2 & 0 & 2 \\ 4 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[29] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 3 \\ 4 & 4 & 9 & 3 & 3 & 4 \\ 6 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[30] $21^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) =$

$(1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 2 & 3 \\ 4 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix} b_0$$

$$[31] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 168, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 8 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix} b_0$$

[33] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 10 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 9 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix} b_0$$

[35] $21^{14}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 2 & 3 \\ 4 & 4 & 16 & 3 & 3 & 4 \\ 12 & 16 & 46 & 8 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[36] $21^{12\text{g}}3^{11\text{g}}1^{11\text{g}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 3 & 4 \\ 5 & 4 & 14 & 3 & 3 & 4 \\ 15 & 14 & 37 & 8 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 4 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 268, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 2 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 11 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{13}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 2 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 13 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[49] $21^{16}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 18 & 2 & 2 & 3 \\ 4 & 4 & 24 & 3 & 3 & 4 \\ 18 & 24 & 106 & 12 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[50] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[51] $21^{12}3^{11}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 21 & 3 & 3 & 4 \\ 5 & 4 & 22 & 3 & 3 & 4 \\ 21 & 22 & 85 & 12 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 & 2 \\ 3 & 3 & 12 & 2 & 0 & 2 \\ 4 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[52] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 14 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[53] $21^{12}3^{11}1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 24 & 3 & 3 & 4 \\ 5 & 4 & 26 & 3 & 3 & 4 \\ 24 & 26 & 118 & 14 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 & 2 \\ 3 & 3 & 14 & 2 & 0 & 2 \\ 4 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[54] $21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 2 & 2 & 3 \\ 4 & 4 & 17 & 3 & 3 & 4 \\ 12 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[55] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 2 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 15 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[56] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 2 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 16 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

$$[57] \quad 21^{17}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 2 & 3 \\ 4 & 4 & 28 & 3 & 3 & 4 \\ 21 & 28 & 145 & 14 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 17 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 21^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 2 & 3 \\ 4 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 21^{13}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 2 & 3 \\ 4 & 4 & 22 & 3 & 3 & 4 \\ 15 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 18 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 21^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 2 & 3 \\ 4 & 4 & 26 & 3 & 3 & 4 \\ 18 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [63] \quad 1^{11}3^1 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [64] \quad 1^{11}5^{13} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 7 & 0 & 0 & 4 \\ 6 & 7 & 9 & 0 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{13}5^{13} 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 0 & 4 \\ 18 & 21 & 97 & 15 & 0 & 21 \\ 3 & 3 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 21 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{12} 5^{13} 1^{12} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 12 & 14 & 42 & 10 & 0 & 14 \\ 3 & 3 & 10 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 14 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{11} 5^{13} 1^{13} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 6 & 7 & 9 & 5 & 0 & 7 \\ 3 & 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[69] $3^{11} 5^{13} 1^{12} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 2 & 4 & 8 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 6 & 8 & 6 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 8 & 0 & 4 \end{pmatrix}_{b_0}$$

[70] $3^{11} 5^{13} 1^{11} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 2 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 & 3 \\ 3 & 4 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

[68] $3^{11} 5^{13} 1^{13} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 2 & 4 & 12 & 0 & 4 \\ 3 & 4 & 1 & 9 & 0 & 3 \\ 9 & 12 & 9 & 34 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [75] $1^{12} 3^{11} 5^{13} 1^{11} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 5 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{14} 3^{11} 5^{13} 1^{12} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 2 & 3 \\ 1 & 2 & 0 & 10 & 2 & 1 & 2 \\ 10 & 14 & 10 & 74 & 8 & 10 & 18 \\ 1 & 1 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 1 & 10 & 1 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

4.931 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{so}_8$

$T = 9$ [0] $1^{13} 1^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

4.932 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{144}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 11 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 10 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 11 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{123}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 7 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 4 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 5 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 4 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{121}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 & 4 \\ 7 & 8 & 10 & 4 & 4 & 6 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 & 2 \\ 2 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{123}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 16 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 10 & 12 & 0 & 6 & 6 & 25 & 9 \\ 3 & 4 & 0 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{121} 3^{111} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 & 0 & 4 \\ 4 & 4 & 1 & 2 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{132} 3^{111} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 5 & 3 \\ 3 & 4 & 12 & 3 & 3 & 8 & 4 \\ 10 & 12 & 25 & 6 & 6 & 18 & 9 \\ 2 & 3 & 6 & 0 & 2 & 4 & 2 \\ 2 & 3 & 6 & 2 & 0 & 4 & 2 \\ 5 & 8 & 18 & 4 & 4 & 10 & 6 \\ 3 & 4 & 9 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{132} 3^{111} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 4 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 2 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{122} 3^{111} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 7 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 21^1 3^{111} 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 4 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 21^1 3^{122} 3^{111} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 2 & 9 & 3 \\ 4 & 4 & 12 & 3 & 3 & 13 & 4 \\ 9 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 9 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{132} 3^{111} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 8 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 8 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{132} 3^{111} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 8 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 9 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & 0 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 7 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 10 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 7 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 8 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 12 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 4 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 2 & 10 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 9 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 10 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 5 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{124} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 7 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 11 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{111} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 10 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[45] \ 21^{123} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 15 & 3 \\ 4 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 15 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[35] \ 21^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 6 & 3 \\ 4 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{112} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{113} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 8 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{12}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 6 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 10 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{11}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 15 & 4 \\ 5 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 15 & 17 & 12 & 9 & 9 & 49 & 13 \\ 4 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{113}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{131}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 & 4 \\ 10 & 12 & 25 & 6 & 6 & 9 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 & 2 \\ 3 & 4 & 9 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 7 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 9 & 12 & 0 & 6 & 6 & 25 & 9 \\ 3 & 4 & 0 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 8 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{142}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 3 & 8 & 4 \\ 10 & 12 & 25 & 6 & 6 & 18 & 9 \\ 2 & 3 & 6 & 0 & 2 & 4 & 2 \\ 2 & 3 & 6 & 2 & 0 & 4 & 2 \\ 6 & 8 & 18 & 4 & 4 & 10 & 6 \\ 3 & 4 & 9 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{14}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 12 & 16 & 0 & 8 & 8 & 46 & 12 \\ 3 & 4 & 0 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{124}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 7 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 7 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 7 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 5 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 11 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{124}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{122}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{122}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 11 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{112}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[82] \ 21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 9 & 4 \\ 5 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 9 & 9 & 6 & 5 & 5 & 13 & 7 \\ 4 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{122}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 5 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{143}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 11 & 16 & 0 & 8 & 8 & 46 & 12 \\ 3 & 4 & 0 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{113}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 8 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{114}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 11 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{123}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 15 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{1123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[96] $21^{113}3^{11}1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 12 & 4 \\ 5 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 12 & 13 & 9 & 7 & 7 & 28 & 10 \\ 4 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 8 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[100] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[104] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 8 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -192, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -192, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 16 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 14 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 8 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{13} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 13 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[135] & 21^{122}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 6 & 3 \\ 4 & 4 & 8 & 3 & 3 & 8 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[136] & 1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -312, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 14 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[137] & 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[140] & 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 8.3 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[143] & 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 12 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[141] & 21^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 9 & 3 \\ 4 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[138] & 21^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 18 & 4 \\ 5 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 18 & 18 & 12 & 10 & 10 & 58 & 14 \\ 4 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[139] & 21^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 9 & 3 \\ 4 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[142] \overset{1^2}{1^{112}3^{11}1^{12}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -368$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 13 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -384$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.3$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix} b_0$$

[149] $21^{123}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -384$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.3$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 9 & 3 \\ 4 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix} b_0$$

[147] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{112}3^{111}1^{12} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_2 + \textcolor{brown}{s}u_2 \rightarrow (-3) + \textcolor{red}{s}u_2 \rightarrow (-3) + \textcolor{blue}{s}u_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -392, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[150] $1^{123}3^{11}1^{11} + [\text{su}_2, \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -392, (n_+, n_-, n_0) =$

$(1, 9, 0), \quad c_{\text{ext}} = 8.3$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix} b_0$$

$$[152] \overset{1^2}{1^{11}3^3} \overset{1^1}{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix} b_0$$

$$[153] \ 1^{1123} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -408, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{13} 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 14 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & 0 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{1123} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -444, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[158] $21^{112}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 12 & 3 \\ 4 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[159] $21^{114}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 12 & 3 \\ 4 & 4 & 4 & 3 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 15 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & 0 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[161] \ 21^{12} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 24 & 4 \\ 5 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 24 & 26 & 18 & 14 & 14 & 118 & 20 \\ 4 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[162] \ 21^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 21 & 4 \\ 5 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 21 & 22 & 15 & 12 & 12 & 85 & 17 \\ 4 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{23} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 12 & 3 \\ 4 & 4 & 8 & 3 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[172] $21^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 9 & 3 \\ 4 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{113}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 15 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{113}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 16 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -556, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{114}3^{13}1^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3)+\textcolor{red}{su}_2 \rightarrow (-3)+\textcolor{blue}{su}_2] \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-560, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}}=8.0$$

$$T_{\min}=9.97 \rightarrow 10, \ T_{\max}=10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{113}3^{13}1^{12}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3)+\textcolor{red}{su}_2 \rightarrow (-3)+\textcolor{blue}{su}_2] \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-564, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}}=7.8$$

$$T_{\min}=9.96 \rightarrow 10, \ T_{\max}=10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{113}3^{12}1^{12}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3)+\textcolor{red}{su}_2 \rightarrow (-3)+\textcolor{blue}{su}_2] \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-576, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}}=7.8$$

$$T_{\min}=10, \ T_{\max}=10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{124}3^{11}1^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3)+\textcolor{red}{su}_2 \rightarrow (-3)+\textcolor{blue}{su}_2] \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-584, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}}=8.5$$

$$T_{\min}=9.92 \rightarrow 10, \ T_{\max}=10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[180] \ 21^{115}3^{11}1^1+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3)+\textcolor{red}{su}_2 \rightarrow (-3)+\textcolor{blue}{su}_2] \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-592, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}}=8.1$$

$$T_{\min}=9.95 \rightarrow 10, \ T_{\max}=10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 15 & 3 \\ 4 & 4 & 4 & 3 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 15 & 20 & 15 & 10 & 10 & 73 & 15 \\ 3 & 4 & 3 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[181] \ 21^{124}3^{11}1^{12}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3)+\textcolor{red}{su}_2 \rightarrow (-3)+\textcolor{blue}{su}_2] \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-624, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}}=8.5$$

$$T_{\min}=9.90 \rightarrow 10, \ T_{\max}=10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 18 & 3 \\ 4 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[182] \ 21^{123}3^{11}1^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3)+\textcolor{red}{su}_2 \rightarrow (-3)+\textcolor{blue}{su}_2] \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-640, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}}=8.3$$

$$T_{\min}=9.89 \rightarrow 10, \ T_{\max}=10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 12 & 3 \\ 4 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{114}3^{11}1^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{green}{su}_2+\textcolor{orange}{su}_2 \rightarrow (-3)+\textcolor{red}{su}_2 \rightarrow (-3)+\textcolor{blue}{su}_2] \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-684, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}}=8.0$$

$$T_{\min}=9.95 \rightarrow 10, \ T_{\max}=10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 3 \\ 2 & 4 & 1 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 \\ 17 & 26 & 18 & 14 & 14 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{114}3^{11}1^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2\rightarrow(-3)+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -956, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 \\ 3 & 4 & 4 & 3 & 3 \\ 3 & 4 & 1 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 \\ 17 & 26 & 18 & 14 & 14 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[195] $21^{117}3^{11}1^1+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2\rightarrow(-3)+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 \\ 4 & 4 & 4 & 3 & 3 \\ 3 & 4 & 1 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 \\ 21 & 28 & 21 & 14 & 14 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[194] $21^{124}3^{11}1^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2\rightarrow(-3)+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 \\ 4 & 4 & 8 & 3 & 3 \\ 6 & 8 & 10 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 4 & 2 & 0 \\ 15 & 21 & 30 & 11 & 11 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[193] $21^{113}3^{11}1^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2\rightarrow(-3)+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 \\ 4 & 4 & 4 & 3 & 3 \\ 3 & 4 & 1 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 \\ 15 & 22 & 15 & 12 & 12 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{114}3^{11}1^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2\rightarrow(-3)+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 \\ 3 & 4 & 4 & 3 & 3 \\ 3 & 4 & 1 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 \\ 18 & 26 & 18 & 14 & 14 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[197] $21^{114}3^{11}1^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2\rightarrow(-3)+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -1168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 \\ 4 & 4 & 4 & 3 & 3 \\ 3 & 4 & 1 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 \\ 15 & 21 & 15 & 11 & 11 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[198] $21^{114}3^{11}1^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2\rightarrow(-3)+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -1536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 \\ 4 & 4 & 4 & 3 & 3 \\ 3 & 4 & 1 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 \\ 18 & 26 & 18 & 14 & 14 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

4.933 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} + \text{su2n3mix}$

$$T = 8 \quad [1] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[0] $1^{122} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 5 & 5 & 4 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 5 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 9 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{132} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 10 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{131} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 11 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 3 & 5 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{131} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 3 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

- [10] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 3 & 5 & 4 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [11] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 12 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [12] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 11 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 6 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [13] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [14] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [15] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 8 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 6 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [16] $1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 6 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 12 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 6 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [17] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 4 & 5 & 4 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [18] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 13 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 3 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[19] $1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 4 & 5 & 4 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[21] $21^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 6 & 4 \\ 5 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 \\ 4 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 4 & 5 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{14}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 11 & 16 & 46 & 8 & 3 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 11 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 6 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{14}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 11 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 6 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 8 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{12}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 6 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & 0 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 11 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{12}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 7 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 5 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{12}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 8 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{1^{21}} 1^{11} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 10 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{1^{31}} 1^{12} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{1^{11}} 1^{11} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{1^{23}} 1^{12} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 8 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{1^{33}} 1^{12} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 9 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 11 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 9 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[41] $1^{1^{13}} 1^{12} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 5 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{1^{41}} 1^{13} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 10 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{1^{11}} 1^{11} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & 0 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

- [44] $1^{1^{22}} 1^{11} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [45] $1^{1^{13}} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 25 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 14 & 2 \\ 17 & 25 & 34 & 14 & 111 & 19 \\ 3 & 4 & 6 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$
- [46] $1^{1^{22}} 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$
- [47] $1^{1^{31}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 3 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [48] $1^{1^{11}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 6 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [49] $1^{1^{41}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 10 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 3 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [50] $1^{1^{14}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [51] $1^{1^3} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [55] $1^{1^{31}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 8 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1^{23}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 9 & 3 \\ 3 & 4 & 9 & 3 & 11 & 4 \\ 8 & 9 & 13 & 5 & 17 & 7 \\ 2 & 3 & 5 & 0 & 6 & 2 \\ 9 & 11 & 17 & 6 & 20 & 8 \\ 3 & 4 & 7 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1^23^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1^{11}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1^{23}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 7 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1^{22}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 7 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 13 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1^{21}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1^{32}3^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 6 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 6 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1^{13}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 9 & 11 & 0 & 6 & 20 & 8 \\ 3 & 4 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{141}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 12 & 16 & 46 & 8 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{143}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 16 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 3 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{122}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 7 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[64] $21^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 6 & 4 \\ 5 & 4 & 9 & 3 & 7 & 4 \\ 9 & 9 & 13 & 5 & 11 & 7 \\ 3 & 3 & 5 & 0 & 4 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 \\ 4 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 7 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{132}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 10 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 10 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 10 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{141}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 6 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 14 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 6 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 6 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 8 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 11 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 3 \\ 3 & 4 & 9 & 3 & 11 & 4 \\ 6 & 9 & 13 & 5 & 17 & 7 \\ 2 & 3 & 5 & 0 & 6 & 2 \\ 8 & 11 & 17 & 6 & 20 & 8 \\ 3 & 4 & 7 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{121}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 11 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 3 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{122}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 8 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{122}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 \\ 3 & 4 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{111}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{122}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{123}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{111}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[85] $21^{113}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 12 & 4 \\ 5 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 12 & 12 & 8 & 7 & 25 & 9 \\ 4 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{132}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 9 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 6 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{121}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{122}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 6 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 13 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 6 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{142}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 114, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 10 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 6 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{122}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 115, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 6 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 7 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{122}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 116, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{121}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 116, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{124}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 116, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 13 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{14}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 118, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 9 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 9 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 6 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 6 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{112}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[100] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 10 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 7 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[102] $1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 11 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{131}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 131, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 8 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{131}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 133, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{141}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 134, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 6 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 12 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 6 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{123}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 135, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[108] \quad 21^{113}3^{111}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 136, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 6 & 4 \\ 5 & 4 & 17 & 3 & 7 & 4 \\ 15 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 6 & 7 & 21 & 4 & 7 & 5 \\ 4 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{131}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 136, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 9 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{122}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 138, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 7 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{123}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 139, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 7 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{133}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 12 & 15 & 0 & 8 & 39 & 11 \\ 3 & 4 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 21^{113}3^{11}1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 6 & 4 \\ 5 & 4 & 13 & 3 & 7 & 4 \\ 12 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 6 & 7 & 16 & 4 & 7 & 5 \\ 4 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{13}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 13 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{13}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 14 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 3 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 11 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 12 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[120] \quad & 1^{12} 1^{12} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 152, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 5 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[123] \quad & 21^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 18 & 3 & 6 & 4 \\ 5 & 4 & 18 & 3 & 7 & 4 \\ 18 & 18 & 58 & 10 & 22 & 14 \\ 3 & 3 & 10 & 0 & 4 & 2 \\ 6 & 7 & 22 & 4 & 7 & 5 \\ 4 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[124] \quad & 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 3 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 10 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[122] \quad & 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 3 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 12 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 156, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 156, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[127] \quad & 21^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 18 & 4 \\ 5 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 18 & 20 & 14 & 11 & 69 & 15 \\ 4 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[128] \quad & 1^{12} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 161, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[129] \quad & 1^{14} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 162, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 3 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 163, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 11 & 15 & 0 & 8 & 39 & 11 \\ 3 & 4 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{41}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 164, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 5 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 12 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 21^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 15 & 4 \\ 5 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 15 & 16 & 11 & 9 & 44 & 12 \\ 4 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 21^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 11 & 3 \\ 4 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 168, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 8 & 3 \\ 4 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 169, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 10 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 170, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{23}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 170, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{\overset{131}{141}3^{\overset{11}{11}}1^{\overset{1}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 171, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 11 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{\overset{143}{132}3^{\overset{11}{11}}1^{\overset{1}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 173, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 10 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{\overset{111}{121}3^{\overset{11}{11}}1^{\overset{1}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 174, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 8 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{\overset{113}{11}3^{\overset{11}{11}}1^{\overset{12}{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 175, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 13 & 17 & 0 & 10 & 55 & 13 \\ 3 & 4 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{\overset{114}{12}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 176, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 13 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 1^{\overset{112}{112}3^{\overset{11}{11}}1^{\overset{1}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 180, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 21^{\overset{111}{11}3^{\overset{11}{11}}1^{\overset{1}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 180, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 3 \\ 4 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 1^{\overset{124}{112}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[147] \quad 1^{\overset{131}{13}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 183, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 12 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 3 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1^{132}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 1^{113}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1^{132}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 8 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{113}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 4 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 13 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 21^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 5 & 3 \\ 4 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 11 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 6 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[156] \quad & 1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 200, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 11 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[157] \quad & 1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 6 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 10 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 6 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[158] \quad & 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 2 & 6 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 14 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 6 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[159] \quad & 1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 6 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 6 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[160] \quad & 1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 204, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 7 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[162] \quad & 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 204, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 2 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 14 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[161] \quad & 21^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 204, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 2 & 5 & 3 \\ 4 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[163] \quad & 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 205, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 7 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[164] \quad & 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 206, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 9 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 11 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 9 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{143}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 9 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 9 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{123}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 12 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{132}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 9 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 10 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 9 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{114}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 12 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 15 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[173] \quad & 1^{13} 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 219, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[174] \quad & 1^{12} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 222, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 3 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 13 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[175] \quad & 1^{41} 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 223, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 3 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 13 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 3 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[176] \quad & 1^{31} 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 226, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 10 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 3 & 7 & 16 & 4 & 7 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[177] \quad & 1^{32} 1^{14} 2^3 1^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 17 & 2 & 8 & 3 \\ 3 & 4 & 26 & 3 & 11 & 4 \\ 17 & 26 & 118 & 14 & 50 & 20 \\ 2 & 3 & 14 & 0 & 6 & 2 \\ 8 & 11 & 50 & 6 & 20 & 8 \\ 3 & 4 & 20 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[178] \quad & 1^{12} 3^{11} 1^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 10 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[179] \quad & 1^{32} 1^{14} 2^3 1^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 2 & 8 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 11 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[180] \quad & 1^{42} 1^{14} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 234, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 18 & 2 & 6 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 6 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[181] \quad & 1^{14} 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 236, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 14 & 19 & 0 & 10 & 64 & 14 \\ 3 & 4 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[182] \quad 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 240, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 9 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{142}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 240, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 8 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 243, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 21^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 5 & 3 \\ 4 & 4 & 12 & 3 & 7 & 4 \\ 9 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 249, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[190] \quad 1^{143}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 250, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 3 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[189] \quad 1^{143}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 250, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 17 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 3 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 250, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 14 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[191] \quad 1^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 13 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[192] \quad 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 20 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 6 & 8 & 10 & 4 & 40 & 6 \\ 2 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[193] \quad 1^{113}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[194] \quad 21^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 5 & 3 \\ 4 & 4 & 18 & 3 & 7 & 4 \\ 12 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[196] \quad 1^{114}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 4 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{133}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 15 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 3 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[199] \quad 21^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 21 & 4 \\ 5 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 21 & 21 & 14 & 12 & 80 & 16 \\ 4 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[198] \quad 1^{114}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{123}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 14 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[200] $1^{123}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[201] $1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 11 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[202] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[203] $1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 2 & 6 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 16 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 6 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[204] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[205] $1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 20 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 2 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[206] $1^{142}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 2 & 9 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 15 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 9 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[207] $1^{113}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[208] \quad & 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 284, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[209] \quad & 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[210] \quad & 1^{141}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 2 & 6 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 6 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[211] \quad & 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 13 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[212] \quad & 1^{123}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 292, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[213] \quad & 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 295, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 6 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 13 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 6 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[214] \quad & 21^{123}3^{11}1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 298, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 21 & 3 & 6 & 4 \\ 5 & 4 & 22 & 3 & 7 & 4 \\ 21 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 6 & 7 & 27 & 4 & 7 & 5 \\ 4 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[215] \quad & 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[216] \quad & 21^{141}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0
\end{aligned}$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 5 & 3 \\ 4 & 4 & 16 & 3 & 7 & 4 \\ 12 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[217] \quad 1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[218] \quad 21^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 8 & 3 \\ 4 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[219] \quad 1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 16 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[220] \quad 1^{133}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 305, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 14 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[221] \quad 1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 11 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[222] \quad 1^{133}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 15 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[223] \quad 1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 15 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[224] \quad 21^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 14 & 3 \\ 4 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 14 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[225] \quad 1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 321, (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 13 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[226] \quad 1^{124} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[227] \quad 21^{122} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 8 & 3 \\ 4 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[228] \quad 1^{131} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 14 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[229] \quad 1^{143} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 9 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 13 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 9 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[230] \quad 1^{131} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 13 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[231] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 19 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 2 & 3 & 4 & 0 & 16 & 2 \\ 19 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[232] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[233] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 16 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 3 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[234] $21^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 20 & 3 \\ 4 & 4 & 8 & 3 & 29 & 4 \\ 6 & 8 & 10 & 4 & 40 & 6 \\ 2 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[235] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 15 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 355, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 13 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 13 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[240] $21^{132}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 8 & 3 \\ 4 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[241] $21^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 5 & 3 \\ 4 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[242] $21^{151}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 5 & 3 \\ 4 & 4 & 20 & 3 & 7 & 4 \\ 15 & 20 & 73 & 10 & 25 & 15 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 25 & 4 & 7 & 5 \\ 3 & 4 & 15 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[244] \quad & 1^{14}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 374, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 18 & 2 & 3 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 18 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 3 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[243] \quad & 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 374, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 16 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[247] \quad & 21^{123}3^{11}1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 378, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 24 & 4 \\ 5 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 24 & 25 & 17 & 14 & 111 & 19 \\ 4 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[246] \quad & 21^{123}3^{11}1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 378, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 24 & 3 & 6 & 4 \\ 5 & 4 & 26 & 3 & 7 & 4 \\ 24 & 26 & 118 & 14 & 32 & 20 \\ 3 & 3 & 14 & 0 & 4 & 2 \\ 6 & 7 & 32 & 4 & 7 & 5 \\ 4 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[245] \quad & 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 378, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[248] \quad & 21^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 384, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 2 & 8 & 3 \\ 4 & 4 & 13 & 3 & 11 & 4 \\ 9 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[249] \quad & 1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 388, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 12 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[250] \quad & 1^{143}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 390, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 16 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[252] \quad & 1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 396, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[251] \quad & 1^{141}3^{111}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 396, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 2 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 16 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[254] \quad & 1^{113}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 404, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 13 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[253] \quad & 1^{114}3^{111}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 404, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[255] \quad & 21^{112}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 412, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 11 & 3 \\ 4 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[256] \quad & 1^{124}3^{111}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 416, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[258] \quad & 21^{142}3^{111}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 424, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 2 & 8 & 3 \\ 4 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[257] \quad & 1^{143}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 424, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 17 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[259] \quad & 1^{133}3^{111}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 431, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 17 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[261] \quad & 1^{142}3^{111}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 432, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 8 & 3 \\ 3 & 4 & 26 & 3 & 11 & 4 \\ 18 & 26 & 118 & 14 & 50 & 20 \\ 2 & 3 & 14 & 0 & 6 & 2 \\ 8 & 11 & 50 & 6 & 20 & 8 \\ 3 & 4 & 20 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[260] $21^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 432, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$

[260] $21^{123}11^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 27 & 4 \\ 5 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 27 & 29 & 20 & 16 & 148 & 22 \\ 4 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[262] \quad 1^{131} 3^{111} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 436, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

[262] $1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[263] \quad 1^{113} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 439, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

[263] $1^{1133}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 439, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[264] \quad 1^{142}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 440, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

[264] $1^{142\text{g}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 14 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[265] \quad 1^{141} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 446, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

[265] $1^{1413}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 14 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[266] \quad 21^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 450, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$(1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 14 & 3 \\ 4 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix} b_0$$

[267] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

[267] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 16 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[268] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 457, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

[268] $1^{13}\textcolor{brown}{3}^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 457, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 18 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 \end{pmatrix} b,$$

[269] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$[269] \quad 1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 458, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$(1, 8, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 16 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 16 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[271] \quad 21^{16} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 460, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 5 & 3 \\ 4 & 4 & 24 & 3 & 7 & 4 \\ 18 & 24 & 106 & 12 & 30 & 18 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 30 & 4 & 7 & 5 \\ 3 & 4 & 18 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[272] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[273] \quad 21^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 11 & 3 \\ 4 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[274] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[275] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[276] \quad 1^{24} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[277] \quad 1^{41} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[278] \quad 21^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 11 & 3 \\ 4 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[279] \quad 1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 502, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 16 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[280] \quad 1^{142}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 15 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[281] \quad 1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 556, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[282] \quad 21^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 5 & 3 \\ 4 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[283] \quad 21^{171}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 5 & 3 \\ 4 & 4 & 28 & 3 & 7 & 4 \\ 21 & 28 & 145 & 14 & 35 & 21 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 35 & 4 & 7 & 5 \\ 3 & 4 & 21 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 17 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{143}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 20 & 29 & 0 & 16 & 148 & 22 \\ 3 & 4 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 601, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[287] \quad 21^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$\begin{aligned}
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 2 & 5 & 3 \\ 4 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[288] & 1^{13} 1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 16 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[289] & 21^{13} 2^3 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 2 & 8 & 3 \\ 4 & 4 & 17 & 3 & 11 & 4 \\ 12 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[290] & 1^{14} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 20 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 2 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[291] & 21^{14} 2^3 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 2 & 8 & 3 \\ 4 & 4 & 26 & 3 & 11 & 4 \\ 18 & 26 & 118 & 14 & 50 & 20 \\ 2 & 3 & 14 & 0 & 6 & 2 \\ 8 & 11 & 50 & 6 & 20 & 8 \\ 3 & 4 & 20 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[292] & 1^{13} 1^{14} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 657, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 19 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 2 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 19 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[293] & 1^{14} 1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[294] & 1^{41} 1^{14} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 18 & 2 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[295] & 21^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 14 & 3 \\ 4 & 4 & 4 & 3 & 19 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[296] & 21^{12} 4^3 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 2 & 14 & 3 \\ 4 & 4 & 8 & 3 & 19 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[297] \quad & 21^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 696, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 14 & 3 \\ 4 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[298] \quad & 21^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 724, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 17 & 3 \\ 4 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[299] \quad & 1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 745, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 19 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[300] \quad & 21^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 770, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 2 & 5 & 3 \\ 4 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[301] \quad & 21^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 774, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 2 & 17 & 3 \\ 4 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[302] \quad & 1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 832, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[303] \quad & 21^{142}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 864, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 2 & 8 & 3 \\ 4 & 4 & 21 & 3 & 11 & 4 \\ 15 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[304] \quad & 21^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 964, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 2 & 5 & 3 \\ 4 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[305] \quad & 21^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 1020, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 17 & 3 \\ 4 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[306] $21^{1143}1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 20 & 3 \\ 4 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [312] $1^{113}1^{11}513^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 1 & 3 & 2 \\ 4 & 0 & 5 & 17 & 5 & 11 & 9 \\ 1 & 0 & 1 & 5 & 0 & 3 & 2 \\ 2 & 0 & 3 & 11 & 3 & 5 & 5 \\ 2 & 0 & 2 & 9 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[311] $1^{123}1^{12}513^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 1 & 3 & 2 \\ 8 & 0 & 10 & 74 & 10 & 22 & 18 \\ 1 & 0 & 1 & 10 & 0 & 3 & 2 \\ 2 & 0 & 3 & 22 & 3 & 5 & 5 \\ 2 & 0 & 2 & 18 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[313] $1^{113}1^{11}513^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 2 & 4 & 3 \\ 1 & 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 3 & 5 & 5 \\ 2 & 3 & 2 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[314] $1^{131}1^{121}513^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 3 & 2 \\ 1 & 1 & 2 & 12 & 2 & 4 & 3 \\ 1 & 2 & 0 & 12 & 1 & 3 & 2 \\ 8 & 12 & 12 & 70 & 9 & 21 & 18 \\ 1 & 2 & 1 & 9 & 0 & 3 & 2 \\ 3 & 4 & 3 & 21 & 3 & 5 & 5 \\ 2 & 3 & 2 & 18 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[315] $1^{121}1^{11}513^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 3 & 2 \\ 1 & 1 & 2 & 8 & 2 & 4 & 3 \\ 1 & 2 & 0 & 8 & 1 & 3 & 2 \\ 5 & 8 & 8 & 30 & 6 & 14 & 12 \\ 1 & 2 & 1 & 6 & 0 & 3 & 2 \\ 3 & 4 & 3 & 14 & 3 & 5 & 5 \\ 2 & 3 & 2 & 12 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[316] $1^{111}1^{11}513^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 3 & 7 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 7 & 3 & 5 & 5 \\ 2 & 3 & 2 & 6 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[317] $1^{121}1^{131}513^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 3 & 2 \\ 1 & 1 & 2 & 8 & 2 & 4 & 3 \\ 1 & 2 & 0 & 8 & 1 & 3 & 2 \\ 7 & 8 & 8 & 30 & 6 & 14 & 12 \\ 1 & 2 & 1 & 6 & 0 & 3 & 2 \\ 3 & 4 & 3 & 14 & 3 & 5 & 5 \\ 2 & 3 & 2 & 12 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[318] \quad 1^{111}3^11^1513^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 3 & 2 \\ 4 & 4 & 4 & 6 & 3 & 7 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 7 & 3 & 5 & 5 \\ 2 & 3 & 2 & 6 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{141}3^11^1513^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 3 & 2 \\ 1 & 1 & 2 & 16 & 2 & 4 & 3 \\ 1 & 2 & 0 & 16 & 1 & 3 & 2 \\ 11 & 16 & 16 & 126 & 12 & 28 & 24 \\ 1 & 2 & 1 & 12 & 0 & 3 & 2 \\ 3 & 4 & 3 & 28 & 3 & 5 & 5 \\ 2 & 3 & 2 & 24 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[322] \quad 1^{122}3^11^1513^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 2 & 8 & 3 \\ 1 & 2 & 0 & 5 & 1 & 7 & 2 \\ 4 & 7 & 5 & 17 & 5 & 23 & 9 \\ 1 & 2 & 1 & 5 & 0 & 6 & 2 \\ 6 & 8 & 7 & 23 & 6 & 27 & 11 \\ 2 & 3 & 2 & 9 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[320] \quad 1^{123}3^11^1513^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 2 & 8 & 3 \\ 1 & 2 & 0 & 0 & 1 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 6 & 8 & 7 & 0 & 6 & 27 & 11 \\ 2 & 3 & 2 & 0 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[321] \quad 1^{112}3^11^1513^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 3 & 15 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 7 & 15 & 6 & 27 & 11 \\ 2 & 3 & 2 & 6 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{131}3^11^1513^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 3 & 2 \\ 1 & 1 & 2 & 12 & 2 & 4 & 3 \\ 1 & 2 & 0 & 12 & 1 & 3 & 2 \\ 9 & 12 & 12 & 70 & 9 & 21 & 18 \\ 1 & 2 & 1 & 9 & 0 & 3 & 2 \\ 3 & 4 & 3 & 21 & 3 & 5 & 5 \\ 2 & 3 & 2 & 18 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{121}3^11^1513^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 3 & 2 \\ 1 & 1 & 2 & 8 & 2 & 4 & 3 \\ 1 & 2 & 0 & 8 & 1 & 3 & 2 \\ 6 & 8 & 8 & 30 & 6 & 14 & 12 \\ 1 & 2 & 1 & 6 & 0 & 3 & 2 \\ 3 & 4 & 3 & 14 & 3 & 5 & 5 \\ 2 & 3 & 2 & 12 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{111}3^11^1513^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 3 & 2 \\ 3 & 4 & 4 & 6 & 3 & 7 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 7 & 3 & 5 & 5 \\ 2 & 3 & 2 & 6 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{13}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 2 & 12 & 3 \\ 1 & 2 & 0 & 0 & 1 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 12 & 11 & 0 & 9 & 65 & 17 \\ 2 & 3 & 2 & 0 & 2 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[327] \quad 1^{12}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 7 & 2 \\ 3 & 4 & 4 & 6 & 3 & 15 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 7 & 15 & 6 & 27 & 11 \\ 2 & 3 & 2 & 6 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

4.934 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{12}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}5^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}3^{11}5^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}5^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 4 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 5 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 6 & 7 & 6 & 4 & 7 & 5 & 5 \\ 4 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12}3^{11}5^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 5 & 3 \\ 3 & 4 & 9 & 3 & 3 & 8 & 4 \\ 5 & 9 & 13 & 5 & 5 & 12 & 7 \\ 2 & 3 & 5 & 0 & 2 & 4 & 2 \\ 3 & 3 & 5 & 2 & 0 & 4 & 2 \\ 5 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 7 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{13}3^{11}5^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 8 & 1 & 3 \\ 3 & 4 & 12 & 3 & 11 & 0 & 4 \\ 10 & 12 & 25 & 6 & 24 & 0 & 9 \\ 2 & 3 & 6 & 0 & 6 & 0 & 2 \\ 8 & 11 & 24 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 1 & 3 \\ 3 & 4 & 8 & 3 & 7 & 0 & 4 \\ 7 & 8 & 10 & 4 & 10 & 0 & 6 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 5 & 7 & 10 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 4 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 4 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{141} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 15 & 1 & 3 \\ 3 & 4 & 8 & 3 & 20 & 0 & 4 \\ 5 & 8 & 10 & 4 & 28 & 0 & 6 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 15 & 20 & 28 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 2 \\ 3 & 3 & 2 & 2 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{211} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 8 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 8 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 3 & 3 & 6 & 2 & 0 & 7 & 2 \\ 8 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 8 & 1 & 3 \\ 3 & 4 & 8 & 3 & 11 & 0 & 4 \\ 7 & 8 & 10 & 4 & 16 & 0 & 6 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 8 & 11 & 16 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1^{21}1^1 3^{11}1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 2 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 & 5 \\ 2 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1^{22}1^1 3^{11}1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 4 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 4 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1^{11}1^1 3^{11}1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 3 & 7 & 0 & 4 \\ 4 & 4 & 1 & 2 & 5 & 0 & 3 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 5 & 7 & 5 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1^{11}1^1 3^{11}1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 7 & 0 & 4 \\ 3 & 5 & 4 & 3 & 6 & 0 & 4 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 5 & 7 & 6 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1^{21}1^1 3^{11}1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1^{11}1^1 3^{11}1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 6 & 1 & 3 \\ 3 & 4 & 14 & 3 & 7 & 0 & 4 \\ 12 & 14 & 37 & 8 & 17 & 0 & 11 \\ 2 & 3 & 8 & 0 & 4 & 0 & 2 \\ 6 & 7 & 17 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1^{12}1^1 3^{11}1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 11 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1^{21}1^1 3^{11}1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 4 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 2 \\ 3 & 3 & 2 & 2 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$1^{312}$$

$$[24] \ 1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 9 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 9 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 3 & 3 & 6 & 2 & 0 & 7 & 2 \\ 9 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$1^{312}$$

$$[22] \ 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 7 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 7 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 7 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$1^{312}$$

$$[21] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 4 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 4 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 4 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{211}$$

$$[25] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$1^{211}$$

$$[27] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 2 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{211}$$

$$[26] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 3 & 12 & 0 & 4 \\ 2 & 4 & 1 & 2 & 8 & 0 & 3 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 9 & 12 & 8 & 7 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}$$

$$[28] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 8 & 4 \\ 5 & 9 & 13 & 5 & 11 & 12 & 7 \\ 2 & 3 & 5 & 0 & 4 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 10 & 5 \\ 5 & 8 & 12 & 4 & 10 & 10 & 6 \\ 3 & 4 & 7 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$1^{212}$$

$$[30] \ 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1^{12}1^1}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 5 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 6 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1^{11}1^1}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^{11}1^1}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 & 2 \\ 3 & 3 & 4 & 2 & 0 & 2 & 2 \\ 3 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1^{11}1^1}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 2 & 0 & 8 & 2 \\ 9 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1^{11}1^1}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 5 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1^{11}1^1}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 1 & 3 \\ 3 & 4 & 14 & 3 & 7 & 0 & 4 \\ 11 & 14 & 37 & 8 & 17 & 0 & 11 \\ 2 & 3 & 8 & 0 & 4 & 0 & 2 \\ 6 & 7 & 17 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1^{11}1^1}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 2 & 0 & 8 & 2 \\ 11 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1^{11}1^1}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 3 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 3 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 4 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 21^{11} 3^{11} 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 6 & 9 & 4 \\ 5 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 9 & 9 & 6 & 5 & 11 & 13 & 7 \\ 4 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 4 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 8 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 & 5 \\ 4 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 3 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{141} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 & 4 \\ 13 & 21 & 76 & 11 & 11 & 15 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 & 2 \\ 3 & 3 & 11 & 2 & 0 & 2 & 2 \\ 3 & 4 & 15 & 2 & 2 & 1 & 3 \\ 3 & 4 & 16 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 3 & 11 & 0 & 4 \\ 4 & 4 & 1 & 2 & 8 & 0 & 3 \\ 2 & 3 & 2 & 0 & 6 & 0 & 2 \\ 8 & 11 & 8 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1^{12}2^3}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 3 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1^{11}1^3}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 2 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1^{12}3}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 17 & 1 & 3 \\ 3 & 4 & 8 & 3 & 25 & 0 & 4 \\ 5 & 8 & 10 & 4 & 34 & 0 & 6 \\ 2 & 3 & 4 & 0 & 14 & 0 & 2 \\ 17 & 25 & 34 & 14 & 111 & 0 & 19 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1^{11}1^3}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & 0 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1^{11}1^3}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 3 & 3 & 0 & 2 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1^{12}3}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 3 & 3 & 0 & 2 & 0 & 4 & 2 \\ 6 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1^{21}3}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1^{13}3}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 3 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{112} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 3 & 3 & 0 & 2 & 0 & 10 & 2 \\ 12 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{131} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 & 2 \\ 3 & 3 & 6 & 2 & 0 & 2 & 2 \\ 3 & 4 & 9 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 3 & 16 & 0 & 4 \\ 4 & 4 & 1 & 2 & 11 & 0 & 3 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 12 & 16 & 11 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{123} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 3 & 3 & 4 & 2 & 0 & 12 & 2 \\ 14 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{123} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 3 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{213} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 3 & 3 & 4 & 2 & 0 & 12 & 2 \\ 15 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1212} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 6 & 3 \\ 3 & 4 & 12 & 3 & 3 & 8 & 4 \\ 8 & 12 & 25 & 6 & 6 & 18 & 9 \\ 2 & 3 & 6 & 0 & 2 & 4 & 2 \\ 3 & 3 & 6 & 2 & 0 & 4 & 2 \\ 6 & 8 & 18 & 4 & 4 & 10 & 6 \\ 3 & 4 & 9 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{1212} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 8 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 16 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 3 & 3 & 0 & 2 & 0 & 6 & 2 \\ 9 & 12 & 0 & 6 & 6 & 25 & 9 \\ 3 & 4 & 0 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 11 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{114} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{231} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 3 & 12 & 0 & 4 \\ 4 & 4 & 1 & 2 & 8 & 0 & 3 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 10 & 12 & 8 & 7 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 1 & 3 \\ 3 & 4 & 14 & 3 & 7 & 0 & 4 \\ 10 & 14 & 37 & 8 & 17 & 0 & 11 \\ 2 & 3 & 8 & 0 & 4 & 0 & 2 \\ 5 & 7 & 17 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 4 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{124} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 16 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{123} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 3 & 15 & 0 & 4 \\ 4 & 4 & 1 & 2 & 11 & 0 & 3 \\ 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 11 & 15 & 11 & 8 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 4 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 & 2 \\ 16 & 21 & 14 & 12 & 80 & 14 & 16 \\ 4 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 8 & 12 & 8 & 7 & 25 & 8 & 9 \\ 2 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 2 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 5 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 3 & 11 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 0 & 2 \\ 8 & 11 & 0 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 17 & 3 \\ 3 & 4 & 0 & 3 & 11 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 14 & 2 \\ 8 & 11 & 0 & 6 & 20 & 50 & 8 \\ 17 & 26 & 0 & 14 & 50 & 118 & 20 \\ 3 & 4 & 0 & 2 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{11} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 2 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 & 2 \\ 14 & 21 & 14 & 12 & 80 & 14 & 16 \\ 2 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 & 5 \\ 2 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{141} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 1 & 3 \\ 3 & 4 & 0 & 3 & 16 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 0 & 2 \\ 13 & 16 & 0 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 5 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 6 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 3 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 3 & 3 & 0 & 2 & 0 & 8 & 2 \\ 12 & 16 & 0 & 8 & 8 & 46 & 12 \\ 3 & 4 & 0 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{141} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 & 4 \\ 11 & 16 & 46 & 8 & 8 & 12 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 & 2 \\ 3 & 3 & 8 & 0 & 2 & 2 & 2 \\ 3 & 4 & 12 & 2 & 0 & 1 & 3 \\ 3 & 4 & 12 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{121} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 1 & 3 \\ 3 & 4 & 0 & 3 & 12 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 0 & 2 \\ 9 & 12 & 0 & 7 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{113} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{112} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{111} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 2 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 5 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 & 5 \\ 2 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{112} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 2 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{121} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 3 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 6 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{312} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 6 & 3 \\ 3 & 4 & 16 & 3 & 3 & 8 & 4 \\ 11 & 16 & 46 & 8 & 8 & 24 & 12 \\ 2 & 3 & 8 & 0 & 2 & 4 & 2 \\ 3 & 3 & 8 & 0 & 2 & 4 & 2 \\ 6 & 8 & 24 & 4 & 4 & 10 & 6 \\ 3 & 4 & 12 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{122} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{1^{13}3^{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 3 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{1^{12}3^{11}1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 14 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 6 & 7 & 5 & 4 & 7 & 22 & 5 \\ 14 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{1^{13}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{1^{12}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 3 & 3 & 0 & 2 & 0 & 7 & 2 \\ 9 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{1^{14}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{1^{11}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 8 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 6 & 7 & 0 & 4 & 7 & 11 & 5 \\ 8 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{1^{12}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{1^{12}3^{11}1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{1123} 1^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 3 & 16 & 0 & 4 \\ 2 & 4 & 1 & 2 & 11 & 0 & 3 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 12 & 16 & 11 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{1112} 1^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{1112} 1^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{1112} 1^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 3 & 3 & 0 & 2 & 0 & 5 & 2 \\ 7 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{1112} 1^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 6 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{1112} 1^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 3 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[126] \ 21^{113} 1^{11} 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 15 & 3 & 4 \\ 5 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 & 2 \\ 15 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 4 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{1121} 1^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 2 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 12 & 16 & 11 & 9 & 44 & 11 & 12 \\ 2 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{114} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 0 & 8 & 2 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 9 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 6 & 7 & 0 & 4 & 7 & 5 & 5 \\ 3 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{113} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 3 & 3 & 2 & 2 & 0 & 4 & 2 \\ 7 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 3 & 3 & 4 & 2 & 0 & 7 & 2 \\ 11 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 6 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{112} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 2 & 0 & 8 & 2 \\ 10 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 5 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{113} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 3 & 20 & 0 & 4 \\ 4 & 4 & 1 & 2 & 14 & 0 & 3 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 15 & 20 & 14 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{122} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 6 & 7 & 0 & 4 & 7 & 10 & 5 \\ 6 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & 0 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{113} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{122} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 7 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 6 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 6 & 7 & 10 & 4 & 7 & 11 & 5 \\ 7 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 1^{131} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 1 & 3 \\ 3 & 4 & 0 & 3 & 17 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 0 & 2 \\ 13 & 17 & 0 & 10 & 55 & 0 & 13 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{131} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 3 & 15 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 0 & 2 \\ 11 & 15 & 0 & 8 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 1 & 3 \\ 3 & 4 & 9 & 3 & 11 & 0 & 4 \\ 6 & 9 & 13 & 5 & 17 & 0 & 7 \\ 2 & 3 & 5 & 0 & 6 & 0 & 2 \\ 8 & 11 & 17 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{114} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 3 & 19 & 0 & 4 \\ 4 & 4 & 1 & 2 & 14 & 0 & 3 \\ 2 & 3 & 2 & 0 & 10 & 0 & 2 \\ 14 & 19 & 14 & 10 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 1^{112} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 10 & 5 \\ 5 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & 0 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1^{122} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 3 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[147] \quad 1^{121} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 & 8 \\ 2 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 1^{111} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 2 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 1^{112} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{1211} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 3 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 10 & 6 & 3 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 6 & 7 & 0 & 4 & 7 & 21 & 5 \\ 12 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[158] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 5 & 7 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 6 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 7 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 1^{122} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 13 & 1 & 3 \\ 3 & 4 & 8 & 3 & 16 & 0 & 4 \\ 6 & 8 & 10 & 4 & 22 & 0 & 6 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 13 & 16 & 22 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 11 & 1 & 3 \\ 3 & 4 & 8 & 3 & 16 & 0 & 4 \\ 5 & 8 & 10 & 4 & 22 & 0 & 6 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 11 & 16 & 22 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{1^{123}} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 6 & 2 \\ 6 & 7 & 0 & 4 & 7 & 15 & 5 \\ 9 & 12 & 0 & 6 & 15 & 25 & 9 \\ 3 & 4 & 0 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[159] \quad 1^{1^{111}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[161] \quad 1^{1^{12}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 4 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 13 & 16 & 0 & 9 & 44 & 11 & 12 \\ 4 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[160] \quad 1^{1^{112}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[164] \quad 1^{1^{123}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 3 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[162] \quad 1^{1^{114}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 3 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[163] \quad 1^{1^{214}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 0 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{1^{113}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{1^{122}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 7 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 6 & 7 & 0 & 4 & 7 & 11 & 5 \\ 7 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{1^{13}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 0 & 9 & 2 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{1^{112}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -118, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{1^{114}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -121, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 11 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 4 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 11 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{1^{111}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -122, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 & 5 \\ 3 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{1^{124}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -122, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 8 & 2 \\ 6 & 7 & 0 & 4 & 7 & 20 & 5 \\ 12 & 16 & 0 & 8 & 20 & 46 & 12 \\ 3 & 4 & 0 & 2 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{1^{231}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -123, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 10 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 10 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{1^{31}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 1 & 3 \\ 3 & 4 & 0 & 3 & 16 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 0 & 2 \\ 12 & 16 & 0 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{1^{212}} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 3 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{1^{112}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 11 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 11 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{1^{13}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{1^{221}} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[178] \quad 1^{1^{13}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 3 & 3 & 0 & 2 & 0 & 12 & 2 \\ 15 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{1^{13}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 3 & 3 & 0 & 2 & 0 & 9 & 2 \\ 12 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^{1^{214}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[181] \quad 21^{1^{112}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 5 & 6 & 3 \\ 4 & 4 & 9 & 3 & 7 & 8 & 4 \\ 6 & 9 & 13 & 5 & 11 & 12 & 7 \\ 2 & 3 & 5 & 0 & 4 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 10 & 5 \\ 6 & 8 & 12 & 4 & 10 & 10 & 6 \\ 3 & 4 & 7 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[182] \quad 1^{\overline{112}3^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{\overline{112}3^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 10 & 5 \\ 6 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & 0 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{\overline{113}3^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 3 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{\overline{112}3^{\overline{11}1^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 3 & 21 & 0 & 4 \\ 2 & 4 & 1 & 2 & 14 & 0 & 3 \\ 2 & 3 & 2 & 0 & 12 & 0 & 2 \\ 15 & 21 & 14 & 12 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 1^{\overline{112}3^{\overline{11}1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 5 & 7 & 0 & 4 & 7 & 11 & 5 \\ 7 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 1^{\overline{113}3^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 3 & 3 & 2 & 2 & 0 & 6 & 2 \\ 10 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 1^{\overline{112}3^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[194] \quad 1^{\overline{13}3^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 11 & 15 & 0 & 8 & 39 & 11 & 11 \\ 2 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{122}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 5 & 3 \\ 3 & 4 & 0 & 3 & 11 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 4 & 2 \\ 8 & 11 & 0 & 6 & 20 & 16 & 8 \\ 5 & 8 & 0 & 4 & 16 & 10 & 6 \\ 3 & 4 & 0 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 3 & 3 & 0 & 2 & 0 & 7 & 2 \\ 10 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 3 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{113}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 5 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{141}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 3 & 19 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 0 & 2 \\ 14 & 19 & 0 & 10 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{113}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 8 & 3 \\ 3 & 4 & 0 & 3 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 6 & 2 \\ 5 & 7 & 0 & 4 & 7 & 15 & 5 \\ 8 & 12 & 0 & 6 & 15 & 25 & 9 \\ 3 & 4 & 0 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[191] $21^{113}3^{11}1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 6 & 12 & 4 \\ 5 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 12 & 13 & 9 & 7 & 16 & 28 & 10 \\ 4 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{1121}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 2 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{1111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -145, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 5 & 7 & 0 & 4 & 7 & 11 & 5 \\ 6 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -145, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 4 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{1231}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 3 & 3 & 4 & 2 & 0 & 6 & 2 \\ 10 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{1111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -147, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 8 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 8 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[202] \ 1^{1123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -147, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 8 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 8 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -149, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[205] \ 1^{1131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 3 & 20 & 0 & 4 \\ 2 & 4 & 1 & 2 & 14 & 0 & 3 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 15 & 20 & 14 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{1111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 2 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 17 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 3 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 14 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 4 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 14 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 3 & 3 & 4 & 2 & 0 & 12 & 2 \\ 16 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{113}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 6 & 2 \\ 5 & 7 & 0 & 4 & 7 & 15 & 5 \\ 9 & 12 & 0 & 6 & 15 & 25 & 9 \\ 3 & 4 & 0 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 0 & 7 & 2 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[214] \quad 1^{1123} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 16 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 6 & 7 & 0 & 4 & 7 & 27 & 5 \\ 16 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & 0 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[215] \quad 1^{1123} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[216] \quad 1^{1123} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[217] \quad 1^{1111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -169, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[218] \quad 1^{1143} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[219] \quad 1^{1211} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[220] \quad 1^{1143} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 15 & 3 \\ 3 & 4 & 0 & 3 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 6 & 7 & 0 & 4 & 7 & 26 & 5 \\ 15 & 21 & 0 & 11 & 26 & 76 & 16 \\ 3 & 4 & 0 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[221] \quad 1^{1211} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 9 & 11 & 0 & 6 & 20 & 8 & 8 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[223] \quad 1^{122} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 3 & 3 & 4 & 2 & 0 & 7 & 2 \\ 10 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[224] \quad 1^{141} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 1 & 3 \\ 3 & 4 & 0 & 3 & 21 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 0 & 2 \\ 16 & 21 & 0 & 12 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[225] \quad 1^{113} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[226] \quad 1^{122} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 1 & 3 \\ 3 & 4 & 8 & 3 & 16 & 0 & 4 \\ 6 & 8 & 10 & 4 & 22 & 0 & 6 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 12 & 16 & 22 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[227] \quad 1^{112} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[222] \quad 1^{114} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 3 & 3 & 0 & 2 & 0 & 11 & 2 \\ 15 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & 0 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[228] \quad 1^{114} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[229] \quad 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 9 & 12 & 0 & 7 & 25 & 8 & 9 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{1121}3^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 2 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 2 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 3 & 3 & 0 & 2 & 0 & 10 & 2 \\ 13 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 6 & 3 \\ 3 & 4 & 0 & 3 & 11 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 4 & 2 \\ 9 & 11 & 0 & 6 & 20 & 16 & 8 \\ 6 & 8 & 0 & 4 & 16 & 10 & 6 \\ 3 & 4 & 0 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -181, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -182, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 8 & 2 \\ 5 & 7 & 0 & 4 & 7 & 20 & 5 \\ 12 & 16 & 0 & 8 & 20 & 46 & 12 \\ 3 & 4 & 0 & 2 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{133}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 1 & 3 \\ 3 & 4 & 0 & 3 & 20 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 0 & 2 \\ 15 & 20 & 0 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 11 & 16 & 0 & 9 & 44 & 11 & 12 \\ 2 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{122}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 3 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 3 & 11 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 6 & 2 \\ 8 & 11 & 0 & 6 & 20 & 24 & 8 \\ 8 & 12 & 0 & 6 & 24 & 25 & 9 \\ 3 & 4 & 0 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 2 & 2 \\ 14 & 19 & 0 & 10 & 64 & 14 & 14 \\ 2 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 8 & 2 \\ 5 & 7 & 0 & 4 & 7 & 20 & 5 \\ 11 & 16 & 0 & 8 & 20 & 46 & 12 \\ 3 & 4 & 0 & 2 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{132}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 3 & 15 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 4 & 2 \\ 11 & 15 & 0 & 8 & 39 & 22 & 11 \\ 5 & 8 & 0 & 4 & 22 & 10 & 6 \\ 3 & 4 & 0 & 2 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 16 & 5 \\ 11 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{124}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 6 & 7 & 0 & 4 & 7 & 22 & 5 \\ 14 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & 0 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{114}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 5 & 7 & 0 & 4 & 7 & 22 & 5 \\ 14 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & 0 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 10 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 16 & 5 \\ 10 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[254] \ 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 9 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[255] \ 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 3 & 3 & 0 & 2 & 0 & 9 & 2 \\ 13 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[256] \ 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -210, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[257] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -211, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 & 8 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 11 & 3 \\ 3 & 4 & 0 & 3 & 11 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 9 & 2 \\ 8 & 11 & 0 & 6 & 20 & 33 & 8 \\ 11 & 17 & 0 & 9 & 33 & 49 & 13 \\ 3 & 4 & 0 & 2 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[258] \ 1^{114} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 18 & 3 \\ 3 & 4 & 0 & 3 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 3 & 3 & 0 & 2 & 0 & 14 & 2 \\ 18 & 26 & 0 & 14 & 14 & 118 & 20 \\ 3 & 4 & 0 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -216, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 12 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 12 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{1111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[261] $21^{1112}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 12 & 3 \\ 4 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 12 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 4 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{1121}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[268] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 3 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{1^{122}} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 6 & 3 \\ 3 & 4 & 0 & 3 & 11 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 4 & 2 \\ 8 & 11 & 0 & 6 & 20 & 16 & 8 \\ 6 & 8 & 0 & 4 & 16 & 10 & 6 \\ 3 & 4 & 0 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[271] \quad 1^{1^{121}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 & 2 \\ 15 & 21 & 14 & 12 & 80 & 14 & 16 \\ 3 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[274] \quad 1^{1^{112}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 5 & 7 & 0 & 4 & 7 & 22 & 5 \\ 12 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & 0 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[273] \quad 1^{1^{114}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 11 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 11 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[272] \quad 1^{1^{112}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 12 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 12 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[275] \quad 1^{1^{214}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[276] \quad 1^{1^{131}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 14 & 1 & 3 \\ 3 & 4 & 8 & 3 & 20 & 0 & 4 \\ 5 & 8 & 10 & 4 & 28 & 0 & 6 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 14 & 20 & 28 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[277] \quad 1^{1^{121}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -233, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 12 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[278] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 14 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 3 & 3 & 0 & 2 & 0 & 10 & 2 \\ 14 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[281] \quad 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 14 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 14 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[280] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 20 & 1 & 3 \\ 3 & 4 & 8 & 3 & 29 & 0 & 4 \\ 5 & 8 & 10 & 4 & 40 & 0 & 6 \\ 2 & 3 & 4 & 0 & 16 & 0 & 2 \\ 20 & 29 & 40 & 16 & 148 & 0 & 22 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 22 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[279] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[282] \quad 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[283] \quad 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 9 & 3 \\ 3 & 4 & 0 & 3 & 11 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 6 & 2 \\ 8 & 11 & 0 & 6 & 20 & 24 & 8 \\ 9 & 12 & 0 & 6 & 24 & 25 & 9 \\ 3 & 4 & 0 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 10 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 10 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{11123} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 14 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 6 & 7 & 5 & 4 & 7 & 22 & 5 \\ 14 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[287] \quad 1^{1143} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 2 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{141} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 3 & 24 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 0 & 2 \\ 17 & 24 & 0 & 13 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[288] \quad 1^{123} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 10 & 3 \\ 3 & 4 & 0 & 3 & 11 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 7 & 2 \\ 8 & 11 & 0 & 6 & 20 & 25 & 8 \\ 10 & 13 & 0 & 7 & 25 & 28 & 10 \\ 3 & 4 & 0 & 2 & 8 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[289] \quad 1^{123} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 11 & 3 \\ 3 & 4 & 0 & 3 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 8 & 2 \\ 8 & 11 & 0 & 6 & 20 & 32 & 8 \\ 11 & 16 & 0 & 8 & 32 & 46 & 12 \\ 3 & 4 & 0 & 2 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{141} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 5 & 3 \\ 3 & 4 & 0 & 3 & 19 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 4 & 2 \\ 14 & 19 & 0 & 10 & 64 & 28 & 14 \\ 5 & 8 & 0 & 4 & 28 & 10 & 6 \\ 3 & 4 & 0 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{113} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 3 & 25 & 0 & 4 \\ 2 & 4 & 1 & 2 & 17 & 0 & 3 \\ 2 & 3 & 2 & 0 & 14 & 0 & 2 \\ 18 & 25 & 17 & 14 & 111 & 0 & 19 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[293] \quad 1^{11123} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 8 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 8 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 9 & 3 \\ 3 & 4 & 0 & 3 & 11 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 7 & 2 \\ 8 & 11 & 0 & 6 & 20 & 25 & 8 \\ 9 & 13 & 0 & 7 & 25 & 28 & 10 \\ 3 & 4 & 0 & 2 & 8 & 10 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 12 & 3 \\ 3 & 4 & 0 & 3 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 8 & 2 \\ 8 & 11 & 0 & 6 & 20 & 32 & 8 \\ 12 & 16 & 0 & 8 & 32 & 46 & 12 \\ 3 & 4 & 0 & 2 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

[297] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 12 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[298] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{124}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[300] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 16 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 3 & 3 & 0 & 2 & 0 & 12 & 2 \\ 16 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 1 & 3 \\ 3 & 4 & 0 & 3 & 25 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 0 & 2 \\ 18 & 25 & 0 & 14 & 111 & 0 & 19 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 8 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 8 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[303] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 9 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[304] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 14 & 20 & 0 & 11 & 69 & 14 & 15 \\ 2 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 15 & 3 & 3 \end{pmatrix}_{b_0}$$

[306] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 12 & 15 & 0 & 8 & 39 & 11 & 11 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

[305] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 5 & 7 & 0 & 4 & 7 & 27 & 5 \\ 14 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & 0 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[307] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 6 & 3 \\ 3 & 4 & 0 & 3 & 15 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 4 & 2 \\ 12 & 15 & 0 & 8 & 39 & 22 & 11 \\ 6 & 8 & 0 & 4 & 22 & 10 & 6 \\ 3 & 4 & 0 & 2 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 13 & 16 & 0 & 9 & 44 & 11 & 12 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[309] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 13 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 6 & 7 & 0 & 4 & 7 & 21 & 5 \\ 13 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[310] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 5 & 7 & 0 & 4 & 7 & 22 & 5 \\ 13 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & 0 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[311] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[312] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 13 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 13 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[313] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 14 & 1 & 3 \\ 3 & 4 & 8 & 3 & 20 & 0 & 4 \\ 6 & 8 & 10 & 4 & 28 & 0 & 6 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 14 & 20 & 28 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

[314] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[315] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[316] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[317] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 5 & 7 & 0 & 4 & 7 & 21 & 5 \\ 11 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[318] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[319] \ 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -323, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 11 & 16 & 0 & 9 & 44 & 11 & 12 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[320] \ 21^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -324, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 6 & 3 \\ 4 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -328, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -330, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 11 & 15 & 0 & 8 & 39 & 11 & 11 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[324] \ 1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -342, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 6 & 3 \\ 3 & 4 & 0 & 3 & 15 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 4 & 2 \\ 11 & 15 & 0 & 8 & 39 & 22 & 11 \\ 6 & 8 & 0 & 4 & 22 & 10 & 6 \\ 3 & 4 & 0 & 2 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -349, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 11 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 11 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[326] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[327] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 12 & 16 & 0 & 9 & 44 & 11 & 12 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[328] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[329] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 17 & 1 & 3 \\ 3 & 4 & 8 & 3 & 24 & 0 & 4 \\ 5 & 8 & 10 & 4 & 34 & 0 & 6 \\ 2 & 3 & 4 & 0 & 13 & 0 & 2 \\ 17 & 24 & 34 & 13 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

[330] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[331] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[332] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 5 & 7 & 0 & 4 & 7 & 21 & 5 \\ 13 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[333] $21^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 3 & 3 \\ 4 & 4 & 4 & 3 & 7 & 4 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[334] $1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 6 & 3 \\ 3 & 4 & 0 & 3 & 20 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 4 & 2 \\ 15 & 20 & 0 & 11 & 69 & 28 & 15 \\ 6 & 8 & 0 & 4 & 28 & 10 & 6 \\ 3 & 4 & 0 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[335] $1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 17 & 25 & 0 & 14 & 111 & 17 & 19 \\ 2 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

[336] $1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 16 & 3 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 & 2 \\ 16 & 21 & 14 & 12 & 80 & 14 & 16 \\ 3 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 17 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[338] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 2 & 2 \\ 17 & 24 & 0 & 13 & 100 & 17 & 18 \\ 2 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 18 & 3 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 3 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[341] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 13 & 3 \\ 3 & 4 & 0 & 3 & 11 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 9 & 2 \\ 8 & 11 & 0 & 6 & 20 & 33 & 8 \\ 13 & 17 & 0 & 9 & 33 & 49 & 13 \\ 3 & 4 & 0 & 2 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[340] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -410, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 15 & 21 & 0 & 12 & 80 & 14 & 16 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[343] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 0 & 9 & 2 & 2 \\ 13 & 17 & 12 & 9 & 49 & 13 & 13 \\ 3 & 4 & 3 & 2 & 13 & 3 & 3 \end{pmatrix}_{b_0}$$

[345] $1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 16 & 21 & 0 & 12 & 80 & 14 & 16 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[344] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[346] $21^{112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 6 & 3 \\ 4 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[347] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 5 & 7 & 0 & 4 & 7 & 21 & 5 \\ 12 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[348] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 14 & 3 \\ 3 & 4 & 0 & 3 & 11 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 11 & 2 \\ 8 & 11 & 0 & 6 & 20 & 41 & 8 \\ 14 & 21 & 0 & 11 & 41 & 76 & 16 \\ 3 & 4 & 0 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[350] \quad 1^{1121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 13 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[352] \quad 1^{1121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[351] \quad 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 12 & 3 \\ 3 & 4 & 0 & 3 & 11 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 9 & 2 \\ 8 & 11 & 0 & 6 & 20 & 33 & 8 \\ 12 & 17 & 0 & 9 & 33 & 49 & 13 \\ 3 & 4 & 0 & 2 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[353] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 18 & 3 \\ 3 & 4 & 0 & 3 & 11 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 14 & 2 \\ 8 & 11 & 0 & 6 & 20 & 50 & 8 \\ 18 & 26 & 0 & 14 & 50 & 118 & 20 \\ 3 & 4 & 0 & 2 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[354] \quad 1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -475, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[355] \quad 1^{141} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 2 & 2 \\ 14 & 19 & 0 & 10 & 64 & 14 & 14 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[356] \quad 1^{142} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 6 & 3 \\ 3 & 4 & 0 & 3 & 19 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 4 & 2 \\ 14 & 19 & 0 & 10 & 64 & 28 & 14 \\ 6 & 8 & 0 & 4 & 28 & 10 & 6 \\ 3 & 4 & 0 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[357] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -491, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 5 & 7 & 0 & 4 & 7 & 26 & 5 \\ 14 & 21 & 0 & 11 & 26 & 76 & 16 \\ 3 & 4 & 0 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[358] $1^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 5 & 7 & 0 & 4 & 7 & 27 & 5 \\ 15 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & 0 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{1121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 12 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{131}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 14 & 20 & 0 & 11 & 69 & 14 & 15 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 15 & 3 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{141}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 16 & 2 & 2 \\ 20 & 29 & 0 & 16 & 148 & 20 & 22 \\ 2 & 4 & 0 & 2 & 20 & 1 & 3 \\ 3 & 4 & 0 & 2 & 22 & 3 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 16 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 5 & 7 & 0 & 4 & 7 & 27 & 5 \\ 16 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & 0 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{142}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 6 & 3 \\ 3 & 4 & 0 & 3 & 24 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 4 & 2 \\ 17 & 24 & 0 & 13 & 100 & 34 & 18 \\ 6 & 8 & 0 & 4 & 34 & 10 & 6 \\ 3 & 4 & 0 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 0 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{131}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 15 & 20 & 0 & 11 & 69 & 14 & 15 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 15 & 3 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{114}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 3 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 5 & 7 & 0 & 4 & 7 & 26 & 5 \\ 15 & 21 & 0 & 11 & 26 & 76 & 16 \\ 3 & 4 & 0 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[367] $1^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 17 & 3 \\ 3 & 4 & 0 & 3 & 7 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 14 & 2 \\ 5 & 7 & 0 & 4 & 7 & 32 & 5 \\ 17 & 26 & 0 & 14 & 32 & 118 & 20 \\ 3 & 4 & 0 & 2 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{131}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -614, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 17 & 25 & 0 & 14 & 111 & 17 & 19 \\ 3 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

[369] $21^{1121}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 8 & 3 & 3 \\ 4 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{114}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -630, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[371] $1^{124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 15 & 3 \\ 3 & 4 & 0 & 3 & 11 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 11 & 2 \\ 8 & 11 & 0 & 6 & 20 & 41 & 8 \\ 15 & 21 & 0 & 11 & 41 & 76 & 16 \\ 3 & 4 & 0 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[372] $21^{1121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -660, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 11 & 3 & 3 \\ 4 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[373] $21^{1112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -676, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 9 & 3 \\ 4 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[374] \quad 1^{131} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 18 & 25 & 0 & 14 & 111 & 17 & 19 \\ 3 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[375] \quad 1^{114} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -770, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 18 & 3 \\ 3 & 4 & 0 & 3 & 7 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 14 & 2 \\ 5 & 7 & 0 & 4 & 7 & 32 & 5 \\ 18 & 26 & 0 & 14 & 32 & 118 & 20 \\ 3 & 4 & 0 & 2 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[376] \quad 1^{141} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -771, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 2 & 2 \\ 17 & 24 & 0 & 13 & 100 & 17 & 18 \\ 3 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 18 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[377] \quad 1^{141} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 16 & 2 & 2 \\ 20 & 29 & 0 & 16 & 148 & 20 & 22 \\ 3 & 4 & 0 & 2 & 20 & 1 & 3 \\ 3 & 4 & 0 & 2 & 22 & 3 & 3 \end{pmatrix}_{b_0}$$

4.935 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 3 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

4.936 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

4.937 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [1] \quad 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

4.938 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 7 \quad [1] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{12} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 6 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 17 & 23 & 100 & 28 & 18 \\ 6 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 5 & 5 & 4 & 6 & 4 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 21^{13} 3^{11} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 6 & 4 \\ 5 & 4 & 23 & 7 & 4 \\ 24 & 23 & 100 & 28 & 18 \\ 6 & 7 & 28 & 7 & 5 \\ 4 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 9 & 14 & 37 & 17 & 11 \\ 5 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 10 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 10 & 16 & 6 \\ 8 & 12 & 16 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 10 & 16 & 6 \\ 10 & 12 & 16 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 5 & 5 & 4 & 12 & 4 \\ 9 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 9 & 12 & 16 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 10 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 13 & 14 & 7 \\ 6 & 8 & 14 & 12 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 5 & 5 & 4 & 15 & 4 \\ 13 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 13 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 4 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^3 3^1 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{u}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[19] $1^{111}3^{11}1^{12} + [\textcolor{red}{5}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{green}{u}_2 + \textcolor{orange}{5}\textcolor{orange}{u}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 11 & 14 & 37 & 17 & 11 \\ 5 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$

[29] $1^{12} 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

$$[28] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{su}}_2 \quad T = 7, \, T_{\max} = 10, \, \Delta = -86, \, \det = -19, \, (n_+, n_-, n_0) = (1, 7, 0), \, c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 11 & 17 & 49 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{1^2} 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

$$[26] \quad 1^{\overline{1^2}} \overline{1^1} \overline{1^1} \overline{1^1} \overline{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 5 & 9 & 13 & 20 & 7 \\ 9 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{1\overline{1}}3^{1\overline{1}}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 4 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[24] $21^1 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

$$[23] \quad 1^{12} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \, T_{\max} = 10, \, \Delta = -86, \, \det = -19, \, (n_+, n_-, n_0) = (1, 7, 0), \, c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{122} 3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \, T_{\max} = 10, \, \Delta = -86, \, \det = -19, \, (n_+, n_-, n_0) = (1, 7, 0), \, c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 11 & 17 & 49 & 36 & 13 \\ 9 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

- [31] $1^{13} 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 12 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$
- [30] $1^{33} 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 7 & 8 & 10 & 22 & 6 \\ 12 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$
- [32] $1^{11} 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.64 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [34] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 9 & 14 & 37 & 17 & 11 \\ 4 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$
- [35] $1^{32} 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 11 & 14 & 37 & 17 & 11 \\ 6 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$
- [36] $1^{14} 1^{123} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 8 & 13 & 28 & 46 & 10 \\ 15 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$
- [33] $1^{11} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$
- [37] $1^{33} 1^{144} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 16 & 24 & 4 \\ 11 & 16 & 46 & 68 & 12 \\ 16 & 24 & 68 & 100 & 18 \\ 3 & 4 & 12 & 18 & 3 \end{pmatrix}_{b_0}$
- [42] $1^{12} 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 7 & 0 & 7 & 5 \\ 3 & 4 & 0 & 5 & 3 \end{pmatrix}_{b_0}$
- [44] $1^{42} 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 12 & 14 & 37 & 17 & 11 \\ 6 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$
- [43] $1^{22} 1^{131} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 25 & 15 & 9 \\ 6 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{13} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 10 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{21} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 25 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{141} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 21 & 12 & 4 \\ 13 & 21 & 76 & 44 & 16 \\ 8 & 12 & 44 & 25 & 9 \\ 3 & 4 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{142} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 6 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 19 & 27 & 133 & 33 & 21 \\ 6 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{124} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 7 & 8 & 10 & 34 & 6 \\ 17 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{131} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 8 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{121} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 10 & 14 & 37 & 17 & 11 \\ 5 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{14} 3^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 15 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 37 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 8 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{144} 3^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 17 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 13 & 21 & 76 & 89 & 16 \\ 17 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 5 & 9 & 13 & 17 & 7 \\ 7 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{14} 3^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 5 & 24 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 17 & 24 & 21 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{122} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 5 & 8 & 10 & 16 & 6 \\ 9 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 6 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

- [62] $1^{12}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -37$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 7 & 8 & 10 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [63] $1^{121}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -37$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 15 & 23 & 100 & 28 & 18 \\ 5 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$
- [64] $1^{134}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 9 & 12 & 25 & 51 & 9 \\ 16 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$
- [65] $1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 8 & 12 & 25 & 24 & 9 \\ 9 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$
- [70] $1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 10 & 10 & 6 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$
- [69] $1^{111}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 13 & 14 & 7 \\ 6 & 8 & 14 & 12 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [68] $1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$
- [66] $1^{141}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 12 & 16 & 46 & 20 & 12 \\ 4 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$
- [67] $1^{131}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 9 & 12 & 25 & 15 & 9 \\ 4 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$
- [74] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -41$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 8 & 3 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$
- [73] $1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -41$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 8 & 3 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{133} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 12 & 20 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 14 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{112} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 1 & 8 & 3 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{141} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 11 & 16 & 46 & 20 & 12 \\ 6 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{122} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{121} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 8 & 13 & 28 & 28 & 10 \\ 9 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{123} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 8 & 25 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 17 & 25 & 34 & 111 & 19 \\ 3 & 4 & 6 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{121} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 7 & 8 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{142} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 11 & 16 & 46 & 32 & 12 \\ 9 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 13 & 21 & 76 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 8 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 7 & 4 \\ 9 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 4 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 11 & 16 & 46 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 8 & 9 & 13 & 17 & 7 \\ 9 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 5 & 9 & 13 & 32 & 7 \\ 14 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 7 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{131} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 15 & 22 & 85 & 27 & 17 \\ 6 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{141} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 9 & 3 \\ 3 & 4 & 21 & 12 & 4 \\ 15 & 21 & 76 & 44 & 16 \\ 9 & 12 & 44 & 25 & 9 \\ 3 & 4 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{11} 1^{132} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 4 & 22 & 11 & 4 \\ 13 & 22 & 85 & 42 & 17 \\ 7 & 11 & 42 & 20 & 8 \\ 3 & 4 & 17 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{12} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 9 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{133} 1^{142} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 12 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 14 & 21 & 76 & 59 & 16 \\ 12 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{13} 1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 8 & 13 & 28 & 37 & 10 \\ 12 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{13} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 12 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{11} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 14 & 7 \\ 6 & 8 & 14 & 12 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{11} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{13} 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 11 & 0 & 20 & 8 \\ 3 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{12} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 13 & 17 & 7 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{142} 1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 11 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

[106] $1^{11} 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 10 & 17 & 49 & 36 & 13 \\ 8 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 13 & 20 & 7 \\ 10 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[110] $21^{14} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 33 & 6 & 4 \\ 5 & 4 & 32 & 7 & 4 \\ 33 & 32 & 193 & 39 & 25 \\ 6 & 7 & 39 & 7 & 5 \\ 4 & 4 & 25 & 5 & 3 \end{pmatrix}_{b_0}$$

[119] $21^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 12 & 12 & 8 & 25 & 9 \\ 4 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[118] $1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 1 & 8 & 3 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[111] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 8 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[112] $1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 11 & 0 & 20 & 8 \\ 3 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[117] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 8 & 3 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[116] \quad & 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 8 & 3 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[113] \quad & 21^{141} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 8 & 3 \\ 4 & 4 & 21 & 12 & 4 \\ 15 & 21 & 76 & 44 & 16 \\ 8 & 12 & 44 & 25 & 9 \\ 3 & 4 & 16 & 9 & 3 \end{pmatrix}_{b_0} \\
[114] \quad & 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 8 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0} \\
[115] \quad & 1^{133} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 12 & 20 & 4 \\ 8 & 12 & 25 & 42 & 9 \\ 14 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 8 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 13 & 17 & 7 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0} \\
[120] \quad & 1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 11 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0} \\
[121] \quad & 1^{133} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 8 & 12 & 25 & 33 & 9 \\ 12 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0} \\
[122] \quad & 1^{121} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[123] \quad & 1^{133} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 10 & 12 & 25 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[124] \quad & 1^{143} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 5 & 24 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 16 & 24 & 21 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[131] $1^{1^2} 1^{1^2} 3^1 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 15 & 23 & 100 & 28 & 18 \\ 4 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$

[129] $1^{1^2} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 13 & 17 & 7 \\ 9 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$

[128] $1^{1^2} 3^1 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 18 & 3 \\ 3 & 4 & 8 & 25 & 4 \\ 6 & 8 & 10 & 34 & 6 \\ 18 & 25 & 34 & 111 & 19 \\ 3 & 4 & 6 & 19 & 3 \end{pmatrix}_{b_0}$

[127] $1^{1^2} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 8 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[132] $1^{1^2} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[133] $1^{1^2} 3^1 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 73 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[130] $1^{1^2} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 13 & 14 & 7 \\ 6 & 8 & 14 & 12 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$

[134] $1^{1^2} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$

[135] $1^{1^2} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 12 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$

[136] $1^{1^2} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[137] $1^{1^2} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 7 & 8 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{134} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 11 & 16 & 46 & 44 & 12 \\ 12 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{13} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 14 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 37 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8 \\ T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 10 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5 \\ T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 25 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 16 & 25 & 34 & 111 & 19 \\ 3 & 4 & 6 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8 \\ T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 12 & 17 & 49 & 48 & 13 \\ 13 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 14 & 21 & 76 & 26 & 16 \\ 6 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5 \\ T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 13 & 18 & 58 & 22 & 14 \\ 6 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[148] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8 \\ T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 11 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^{22}

$$[147] \ 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^{32}

$$[159] \ 1^{122} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 7 & 8 & 10 & 16 & 6 \\ 8 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

 1^{11}

$$[160] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

 1^4

$$[150] \ 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

 1^{31}

$$[151] \ 1^{141} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 8 & 3 \\ 3 & 4 & 21 & 12 & 4 \\ 14 & 21 & 76 & 44 & 16 \\ 8 & 12 & 44 & 25 & 9 \\ 3 & 4 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

 1^{13}

$$[152] \ 1^{111} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 10 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

 1^{13}

$$[153] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

 1^{21}

$$[154] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 9 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

 1^{22}

$$[155] \ 1^{133} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 12 & 20 & 4 \\ 8 & 12 & 25 & 42 & 9 \\ 13 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

 1^{41}

$$[156] \ 1^{131} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 10 & 12 & 25 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[157] $1^{1^2} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 1 & 8 & 3 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{1^3} 1^{13} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 14 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{1^{21}} 1^{21} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{1^{34}} 1^{34} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 7 & 8 & 10 & 22 & 6 \\ 13 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{1^{12}} 1^{12} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[163] $21^{1^{11}} 1^{11} 3^1 1^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 18 & 4 \\ 5 & 4 & 21 & 20 & 4 \\ 18 & 21 & 76 & 74 & 16 \\ 18 & 20 & 74 & 69 & 15 \\ 4 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{1^2} 1^{12} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 11 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{1^{12}} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 11 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{1^{22}} 1^{22} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 11 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{1^{12}} 1^{12} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 11 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{1^{24}} 1^{24} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 13 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[170] $21^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 5 & 12 & 4 \\ 6 & 5 & 4 & 12 & 4 \\ 12 & 12 & 12 & 25 & 9 \\ 4 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 10 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 1 & 8 & 3 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 11 & 17 & 49 & 48 & 13 \\ 12 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 37 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{14}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 13 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[178] $1^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 16 & 26 & 118 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{131}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 9 & 12 & 25 & 15 & 9 \\ 6 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 10 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[181] $21^{1\bar{1}3^1 1^{1\bar{2}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 13 & 7 & 4 \\ 12 & 13 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$

[182] $1^{1\bar{1}3^1 1^{1\bar{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 8 & 9 & 13 & 32 & 7 \\ 15 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$

[183] $1^{1\bar{1}3^1 1^{1\bar{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 8 & 9 & 13 & 20 & 7 \\ 9 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$

[184] $1^{1\bar{1}3^1 1^{1\bar{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 12 & 20 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 15 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0}$

[185] $1^{1\bar{1}3^1 1^{1\bar{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$

[187] $1^{1\bar{2}3^1 1^{1\bar{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 7 & 8 & 10 & 28 & 6 \\ 15 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$

[188] $1^{1\bar{2}1^1 3^1 1^{1\bar{3}}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 16 & 23 & 100 & 28 & 18 \\ 5 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$

[186] $1^{1\bar{2}3^1 1^{1\bar{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 5 & 8 & 10 & 28 & 6 \\ 15 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$

[190] $1^{1\bar{3}2^1 3^1 1^{1\bar{2}}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 4 & 22 & 11 & 4 \\ 15 & 22 & 85 & 42 & 17 \\ 8 & 11 & 42 & 20 & 8 \\ 3 & 4 & 17 & 8 & 3 \end{pmatrix}_{b_0}$

[189] $1^{1\bar{3}1^1 3^1 1^{1\bar{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 13 & 17 & 49 & 36 & 13 \\ 10 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$

[191] $1^{1\bar{1}1^1 3^1 1^{1\bar{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 5 & 9 & 13 & 20 & 7 \\ 8 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$

[192] $21^{1\bar{1}3^11^{1\bar{3}1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 6 & 4 \\ 5 & 4 & 17 & 7 & 4 \\ 15 & 17 & 49 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 4 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{1\bar{2}2}3^11^{1\bar{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 7 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{1\bar{3}2}3^11^{1\bar{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 9 & 12 & 25 & 24 & 9 \\ 7 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{1\bar{1}2}3^11^{1\bar{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 7 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{1\bar{4}2}3^11^{1\bar{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 12 & 16 & 46 & 32 & 12 \\ 7 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{1\bar{1}2}3^11^{1\bar{1}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 13 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[197] $21^{1\bar{1}1}3^11^{1\bar{1}3} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 18 & 4 \\ 5 & 4 & 9 & 20 & 4 \\ 9 & 9 & 13 & 32 & 7 \\ 18 & 20 & 32 & 69 & 15 \\ 4 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{1\bar{1}2}3^11^{1\bar{1}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 5 & 9 & 13 & 26 & 7 \\ 11 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{1\bar{2}1}3^11^{1\bar{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 11 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{1\bar{2}1}3^11^{1\bar{3}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 17 & 23 & 100 & 28 & 18 \\ 5 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

[202] $21^{1\bar{2}3}1^{1\bar{2}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 6 & 4 \\ 5 & 4 & 18 & 7 & 4 \\ 18 & 18 & 58 & 22 & 14 \\ 6 & 7 & 22 & 7 & 5 \\ 4 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{123}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 10 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{121}3^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[205] $21^{113}3^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 15 & 16 & 11 & 44 & 12 \\ 4 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{123}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 12 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{123}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{123}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{133}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 17 & 15 & 4 \\ 11 & 17 & 49 & 45 & 13 \\ 11 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{121}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 28 & 16 & 10 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[210] $21^{113}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{123}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 13 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{133}3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 17 & 15 & 4 \\ 10 & 17 & 49 & 45 & 13 \\ 10 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[213] $21^{133}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 14 & 3 \\ 4 & 4 & 12 & 20 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 14 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{14}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 15 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 5 & 8 & 10 & 16 & 6 \\ 8 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{121}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 28 & 28 & 10 \\ 8 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{123}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 100 & 13 & 18 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{131}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 25 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{132}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 10 & 12 & 25 & 24 & 9 \\ 8 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 11 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 73 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{143}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 19 & 31 & 172 & 17 & 24 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{131}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 13 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{14} 1^{13} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 15 & 0 & 39 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$

[226] $1^{24} 1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 11 & 17 & 49 & 60 & 13 \\ 15 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$

[227] $1^{42} 1^{13} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 10 & 12 & 25 & 15 & 9 \\ 6 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[228] $1^{21} 1^{13} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 85 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$

[225] $21^{112} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 9 & 11 & 4 \\ 6 & 9 & 13 & 17 & 7 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$

[230] $1^{12} 1^{11} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 55 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$

[231] $1^{23} 1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 12 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$

[232] $1^{33} 1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 8 & 9 & 13 & 26 & 7 \\ 12 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$

[233] $21^{113} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 18 & 20 & 14 & 69 & 15 \\ 4 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$

[234] $1^{24} 1^{12} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 1 & 14 & 3 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$

[235] $1^{23} 1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 1 & 14 & 3 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$

[236] $1^{1^{23}}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{1^{13}}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{1^{11}}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 8 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{1^{13}}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 55 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{1^{13}}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 15 & 0 & 39 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[242] $21^{1^{11}}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{1^{21}}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 12 & 18 & 58 & 22 & 14 \\ 4 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{1^{31}}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 21 & 5 & 3 \\ 3 & 4 & 32 & 7 & 4 \\ 21 & 32 & 193 & 39 & 25 \\ 5 & 7 & 39 & 7 & 5 \\ 3 & 4 & 25 & 5 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{1^{31}}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 9 & 12 & 25 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{1^{12}}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{1^{14}}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 5 & 9 & 13 & 38 & 7 \\ 16 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{1^{24}} 1^{1^{22}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 13 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{1^{42}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 11 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 13 & 21 & 76 & 59 & 16 \\ 11 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{1^{21}} 3^1 1^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 133 & 15 & 21 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{1^{11}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 13 & 20 & 7 \\ 8 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

[251] $21^{1^{11}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{1^{22}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 7 & 8 & 10 & 16 & 6 \\ 9 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{1^{32}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 9 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 12 & 17 & 49 & 33 & 13 \\ 9 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{1^{21}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 11 & 13 & 28 & 28 & 10 \\ 9 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{1^{43}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 13 & 21 & 76 & 74 & 16 \\ 14 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[256] $1^{1^{23}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 7 & 8 & 10 & 22 & 6 \\ 11 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{1^{31}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 10 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 16 & 0 & 44 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 44 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 12 & 16 & 46 & 20 & 12 \\ 6 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 9 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 13 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 4 & 4 & 1 & 20 & 3 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 12 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{143}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 14 & 21 & 76 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[267] $21^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 11 & 3 \\ 4 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 11 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[268] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 44 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{1^{32}} 1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 10 & 13 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{1^{24}} 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{1^{24}} 1^{113} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 4 & 4 & 1 & 17 & 3 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{1^{12}} 1^{1123} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{1^3} 1^{1123} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 15 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{1^{34}} 1^{143} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 14 & 21 & 76 & 74 & 16 \\ 15 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[276] $1^{1^{42}} 1^{121} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 14 & 18 & 58 & 22 & 14 \\ 6 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{1^{32}} 1^{131} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 12 & 17 & 49 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{1^{31}} 1^{131} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 12 & 17 & 49 & 36 & 13 \\ 8 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{1^{22}} 1^{122} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 8 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{1^{33}} 1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 8 & 9 & 13 & 20 & 7 \\ 10 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{1^{22}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -126, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 13 & 21 & 76 & 41 & 16 \\ 8 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{1^{11}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -126, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 8 & 13 & 28 & 25 & 10 \\ 7 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{1^3} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -127, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 18 & 26 & 17 & 128 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{1^{12}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -127, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{1^{24}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -127, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{1^{11}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -127, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{1^{21}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -128, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 12 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{1^{12}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -129, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 13 & 20 & 7 \\ 9 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{1^{32}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -129, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 9 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 12 & 17 & 49 & 36 & 13 \\ 9 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{1^{42}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -132, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 11 & 13 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[289] \ 1^{1^{41}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -132, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 14 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

1^{22}

$$[294] \ 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 8 & 12 & 25 & 24 & 9 \\ 8 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

1^{43}

$$[291] \ 1^{133} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 10 & 12 & 25 & 33 & 9 \\ 11 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

1^{42}

$$[292] \ 1^{132} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 8 & 3 \\ 3 & 4 & 22 & 11 & 4 \\ 16 & 22 & 85 & 42 & 17 \\ 8 & 11 & 42 & 20 & 8 \\ 3 & 4 & 17 & 8 & 3 \end{pmatrix}_{b_0}$$

1^{31}

$$[293] \ 1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 13 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

1^{33}

$$[295] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 10 & 13 & 28 & 25 & 10 \\ 9 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

1^{33}

$$[296] \ 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 9 & 12 & 25 & 24 & 9 \\ 9 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[297] \ 21^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 8 & 7 & 4 \\ 6 & 8 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[298] \ 21^{111} 3^1 1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 15 & 4 \\ 5 & 4 & 17 & 16 & 4 \\ 15 & 17 & 49 & 48 & 13 \\ 15 & 16 & 48 & 44 & 12 \\ 4 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

1^{23}

$$[299] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 13 & 20 & 7 \\ 10 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

1^{21}

$$[300] \ 1^{131} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 17 & 27 & 133 & 33 & 21 \\ 5 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

1^{21}

$$[301] \ 1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 11 & 17 & 49 & 36 & 13 \\ 8 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{1^3} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 12 & 17 & 49 & 21 & 13 \\ 4 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[303] \quad 21^{1^1} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 12 & 4 \\ 5 & 4 & 13 & 12 & 4 \\ 12 & 13 & 28 & 28 & 10 \\ 12 & 12 & 28 & 25 & 9 \\ 4 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[304] \quad 1^{1^3} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 21 & 4 & 3 \\ 3 & 4 & 32 & 7 & 4 \\ 21 & 32 & 193 & 39 & 25 \\ 4 & 7 & 39 & 7 & 5 \\ 3 & 4 & 25 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[305] \quad 1^{1^{43}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -137, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 11 & 13 & 28 & 25 & 10 \\ 9 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[306] \quad 1^{1^{31}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 10 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[310] \quad 1^{1^{33}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 28 & 28 & 10 \\ 10 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[311] \quad 1^{1^{32}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 10 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[308] \quad 1^{1^{21}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[307] \quad 1^{1^{11}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 8 & 13 & 28 & 28 & 10 \\ 8 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[309] \quad 1^{1^{32}} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 8 & 3 \\ 3 & 4 & 26 & 11 & 4 \\ 17 & 26 & 118 & 50 & 20 \\ 8 & 11 & 50 & 20 & 8 \\ 3 & 4 & 20 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^{1^{22}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 9 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[313] \ 1^{1^{11}} 3^{1^1} 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 13 & 20 & 7 \\ 8 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[315] \ 1^{1^{41}} 3^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 11 & 16 & 46 & 20 & 12 \\ 5 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[314] \ 1^{1^{23}} 3^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 11 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[318] \ 1^{1^{12}} 3^{1^1} 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 5 & 9 & 13 & 26 & 7 \\ 10 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[316] \ 1^{1^{41}} 3^{1^1} 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[319] \ 21^{1^{11}} 3^{1^1} 1^{1^{44}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 18 & 21 & 4 \\ 5 & 4 & 21 & 24 & 4 \\ 18 & 21 & 76 & 89 & 16 \\ 21 & 24 & 89 & 100 & 18 \\ 4 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[317] \ 1^{1^{41}} 3^{1^1} 1^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 6 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 18 & 26 & 118 & 32 & 20 \\ 6 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{1^{13}} 3^{1^1} 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 5 & 9 & 13 & 32 & 7 \\ 13 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{1^{34}} 3^{1^1} 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 8 & 12 & 25 & 51 & 9 \\ 15 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[320] \ 1^{1^{32}} 3^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 10 & 12 & 25 & 24 & 9 \\ 9 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{1^{41}} 3^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 12 & 16 & 46 & 20 & 12 \\ 5 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[324] \ 21^{12}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 21 & 21 & 14 & 80 & 16 \\ 4 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{12}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 9 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 13 & 17 & 49 & 36 & 13 \\ 9 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 13 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 13 & 20 & 7 \\ 9 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[329] \ 1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 1 & 14 & 3 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[330] \ 1^{121}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 16 & 23 & 100 & 13 & 18 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[331] \ 1^{121}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 9 & 13 & 28 & 28 & 10 \\ 9 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[332] \ 1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 20 & 0 & 69 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[333] \ 1^{12}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[334] \ 21^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 3 \\ 4 & 4 & 18 & 7 & 4 \\ 12 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^{32}

$$[335] \ 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 9 & 12 & 25 & 24 & 9 \\ 8 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

 1^{31}

$$[338] \ 1^{123} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -157, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 19 & 28 & 154 & 16 & 22 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

 1^{12}

$$[337] \ 1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -157, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 21 & 0 & 80 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

 1^{14}

$$[336] \ 1^{143} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -157, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 19 & 0 & 64 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

 1^{13}

$$[339] \ 1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 5 & 8 & 10 & 28 & 6 \\ 14 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

 1^{42}

$$[340] \ 1^{143} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 12 & 16 & 46 & 44 & 12 \\ 10 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

 1^{32}

$$[341] \ 1^{133} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 9 & 12 & 25 & 33 & 9 \\ 10 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

 1^{12}

$$[342] \ 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 10 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

 1^{22}

$$[343] \ 1^{123} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 10 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

 1^{24}

$$[344] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 1 & 14 & 3 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[345] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[346] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 8 & 19 & 4 \\ 7 & 8 & 10 & 28 & 6 \\ 14 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

[347] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 9 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 12 & 16 & 46 & 32 & 12 \\ 9 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

[348] $1^{1^4} 1^{1^3} 3^1 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 15 & 22 & 85 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

[352] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 9 & 13 & 28 & 28 & 10 \\ 8 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 11 & 13 & 28 & 28 & 10 \\ 10 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 11 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 8 & 12 & 25 & 51 & 9 \\ 17 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 7 & 8 & 10 & 22 & 6 \\ 12 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{1^4} 1^{1^3} 3^1 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 6 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 16 & 22 & 85 & 27 & 17 \\ 6 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[358] $21^{133}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 27 & 6 & 4 \\ 5 & 4 & 27 & 7 & 4 \\ 27 & 27 & 133 & 33 & 21 \\ 6 & 7 & 33 & 7 & 5 \\ 4 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{143}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 17 & 26 & 118 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 76 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 7 & 9 & 13 & 26 & 7 \\ 11 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{134}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 9 & 12 & 25 & 51 & 9 \\ 17 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

[362] $21^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 12 & 4 \\ 5 & 4 & 9 & 12 & 4 \\ 9 & 9 & 13 & 20 & 7 \\ 12 & 12 & 20 & 25 & 9 \\ 4 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 4 & 4 & 1 & 17 & 3 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 8 & 9 & 13 & 26 & 7 \\ 13 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[367] $1^{1^4} 3^{1^1} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{1^{24}} 3^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 20 & 3 \\ 3 & 4 & 8 & 29 & 4 \\ 6 & 8 & 10 & 40 & 6 \\ 20 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

[369] $1^{1^{43}} 3^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 11 & 13 & 28 & 37 & 10 \\ 12 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{1^{13}} 3^{1^{11}} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[371] $21^{1^{11}} 3^{1^{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 16 & 4 \\ 9 & 9 & 13 & 26 & 7 \\ 15 & 16 & 26 & 44 & 12 \\ 4 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[373] $1^{1^{32}} 3^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 28 & 28 & 10 \\ 9 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[372] $1^{1^4} 3^{1^{11}} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 15 & 21 & 76 & 26 & 16 \\ 4 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[374] $1^{1^{14}} 3^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 8 & 29 & 4 \\ 5 & 8 & 10 & 40 & 6 \\ 20 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

[375] $1^{1^{41}} 3^{1^{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 22 & 5 & 3 \\ 3 & 4 & 32 & 7 & 4 \\ 22 & 32 & 193 & 39 & 25 \\ 5 & 7 & 39 & 7 & 5 \\ 3 & 4 & 25 & 5 & 3 \end{pmatrix}_{b_0}$$

[376] $1^{1^{43}} 3^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 9 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 15 & 21 & 76 & 41 & 16 \\ 9 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

[377] $1^{1^{13}} 3^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 21 & 0 & 80 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[378] \ 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -178, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 20 & 0 & 69 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[379] \ 1^{13} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -178, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[380] \ 21^{12} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -179, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5 \\ T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 21 & 6 & 4 \\ 5 & 4 & 22 & 7 & 4 \\ 21 & 22 & 85 & 27 & 17 \\ 6 & 7 & 27 & 7 & 5 \\ 4 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[381] \ 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -180, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 12 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[382] \ 1^{14} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -181, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 11 & 16 & 46 & 32 & 12 \\ 8 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[383] \ 21^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -181, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 9 & 12 & 4 \\ 6 & 9 & 13 & 20 & 7 \\ 8 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[384] \ 21^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -181, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8 \\ T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 8 & 3 \\ 4 & 4 & 17 & 12 & 4 \\ 12 & 17 & 49 & 36 & 13 \\ 8 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[385] \ 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -181, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 21 & 0 & 80 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{13} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -181, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8 \\ T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 12 & 19 & 4 \\ 10 & 12 & 25 & 42 & 9 \\ 14 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{23} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -181, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8 \\ T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 8 & 12 & 25 & 33 & 9 \\ 11 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -182, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5 \\ T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 5 & 8 & 10 & 28 & 6 \\ 13 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[392] \quad 1^{1^3} 1^3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 15 & 22 & 85 & 27 & 17 \\ 4 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[390] \quad 21^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[391] \quad 21^{112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 4 & 11 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[393] \quad 1^{11} 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 10 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[395] \quad 1^{142} 1^{41} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 15 & 21 & 76 & 26 & 16 \\ 6 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[396] \quad 21^{131} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 3 \\ 4 & 4 & 12 & 7 & 4 \\ 9 & 12 & 25 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[389] \quad 1^{124} 1^{23} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 12 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[394] \quad 1^{13} 1^{31} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 18 & 27 & 133 & 33 & 21 \\ 4 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[397] \quad 1^{14} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -186, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 20 & 0 & 69 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[398] \quad 1^{123} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -186, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 14 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[399] \quad 1^{142} 1^{32} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -187, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 11 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 13 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[400] \ 1^{114}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -187, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 5 & 9 & 13 & 38 & 7 \\ 15 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[401] \ 1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -187, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 13 & 17 & 49 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[402] \ 1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -188, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 11 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[403] \ 1^{131}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -189, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 18 & 27 & 133 & 15 & 21 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[404] \ 1^{123}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -193, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 17 & 23 & 100 & 13 & 18 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[405] \ 1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -196, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 11 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[406] \ 1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -197, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[407] \ 1^{131}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -198, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 32 & 3 & 4 \\ 21 & 32 & 193 & 18 & 25 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[408] \ 1^{142}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -198, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 12 & 16 & 46 & 32 & 12 \\ 8 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[409] \ 1^{134}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -199, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 8 & 12 & 25 & 51 & 9 \\ 16 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[410] \ 21^{111}3^11^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -199, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 21 & 4 \\ 5 & 4 & 17 & 24 & 4 \\ 15 & 17 & 49 & 72 & 13 \\ 21 & 24 & 72 & 100 & 18 \\ 4 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{1^{14}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -201, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 19 & 4 \\ 5 & 8 & 10 & 28 & 6 \\ 14 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{1^{12}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -201, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 8 & 29 & 4 \\ 5 & 8 & 10 & 40 & 6 \\ 18 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{1^{21}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -201, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 14 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{1^{13}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -202, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 15 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{1^{31}} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -202, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 20 & 31 & 172 & 17 & 24 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[416] \ 1^{1^{34}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -204, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 9 & 12 & 25 & 33 & 9 \\ 12 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[420] \ 1^{1^{41}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -207, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 85 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[419] \ 1^{1^{14}} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -207, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 19 & 26 & 17 & 128 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[421] \ 1^{1^{31}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -207, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 10 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 14 & 21 & 76 & 59 & 16 \\ 10 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[418] \ 1^{1^{44}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -207, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 10 & 12 & 25 & 33 & 9 \\ 12 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[417] \ 1^{1^{21}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -207, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 10 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[422] \ 1^{1^{12}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 8 & 13 & 28 & 37 & 10 \\ 11 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{1^{14}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[424] \ 1^{1^{32}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 10 & 13 & 28 & 37 & 10 \\ 11 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[425] \ 1^{1^{14}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 6 & 9 & 13 & 32 & 7 \\ 15 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[426] \ 1^{1^{13}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[427] \ 1^{1^{24}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 15 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[428] \ 1^{1^{44}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 11 & 13 & 28 & 46 & 10 \\ 15 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[429] \ 1^{1^{24}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 7 & 9 & 13 & 26 & 7 \\ 13 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[430] \ 21^{1^{11}} 3^{1^{133}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 18 & 4 \\ 5 & 4 & 17 & 20 & 4 \\ 15 & 17 & 49 & 60 & 13 \\ 18 & 20 & 60 & 69 & 15 \\ 4 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[431] \ 1^{1^{12}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 11 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[432] \ 1^{1^{43}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 9 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 13 & 17 & 49 & 33 & 13 \\ 9 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

 1^{13}

$$[433] \ 1^{124} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 3 \\ 3 & 4 & 8 & 29 & 4 \\ 5 & 8 & 10 & 40 & 6 \\ 19 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

 1^{21}

$$[434] \ 1^{141} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 5 & 3 \\ 3 & 4 & 31 & 7 & 4 \\ 19 & 31 & 172 & 38 & 24 \\ 5 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^{21}

$$[435] \ 1^{141} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 13 & 21 & 76 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^{41}

$$[436] \ 1^{123} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 154 & 16 & 22 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[437] \ 21^{122} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 8 & 11 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 8 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

 1^{41}

$$[439] \ 1^{131} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 13 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[440] \ 21^{111} 3^1 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 18 & 4 \\ 5 & 4 & 13 & 20 & 4 \\ 12 & 13 & 28 & 46 & 10 \\ 18 & 20 & 46 & 69 & 15 \\ 4 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

 1^{21}

$$[441] \ 1^{132} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 11 & 17 & 49 & 33 & 13 \\ 7 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

 1^{23}

$$[442] \ 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 7 & 9 & 13 & 32 & 7 \\ 14 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

 1^{14}

$$[438] \ 1^{124} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 8 & 13 & 28 & 55 & 10 \\ 17 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

 1^{31}

$$[443] \ 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$\begin{aligned}
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 23 & 3 & 3 \\ 3 & 4 & 36 & 3 & 4 \\ 23 & 36 & 238 & 20 & 28 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0} \\
[444] & 21^{12}3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 24 & 25 & 17 & 111 & 19 \\ 4 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0} \\
[445] & 21^{12}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 5 & 3 \\ 4 & 4 & 13 & 7 & 4 \\ 9 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^4 \\ 1^{13}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\ \begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[447] & 21^{11}3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 12 & 15 & 4 \\ 5 & 4 & 13 & 16 & 4 \\ 12 & 13 & 28 & 37 & 10 \\ 15 & 16 & 37 & 44 & 12 \\ 4 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{23} \\ 1^{12}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\ \begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 7 & 9 & 13 & 26 & 7 \\ 12 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[449] & 21^{12}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 8 & 3 \\ 4 & 4 & 13 & 12 & 4 \\ 9 & 13 & 28 & 28 & 10 \\ 8 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{31} \\ 1^{13}3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\ T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\ \begin{pmatrix} 1 & 3 & 18 & 5 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 18 & 27 & 133 & 33 & 21 \\ 5 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[451] & 1^{13}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 12 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{33} \\ 1^{133}3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\ T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\ \begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 9 & 12 & 25 & 33 & 9 \\ 11 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[453] & 1^{143}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 11 & 3 \\ 3 & 4 & 21 & 15 & 4 \\ 14 & 21 & 76 & 56 & 16 \\ 11 & 15 & 56 & 39 & 11 \\ 3 & 4 & 16 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{11} \\ 1^{132}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\ T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \end{matrix}
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 10 & 17 & 49 & 48 & 13 \\ 10 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

[454] $1^{13}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 26 & 0 & 128 & 20 \\ 3 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[455] $1^{133}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 11 & 3 \\ 3 & 4 & 17 & 15 & 4 \\ 13 & 17 & 49 & 45 & 13 \\ 11 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[458] $21^{131}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 5 & 3 \\ 4 & 4 & 27 & 7 & 4 \\ 18 & 27 & 133 & 33 & 21 \\ 5 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[457] $21^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 8 & 3 \\ 4 & 4 & 13 & 11 & 4 \\ 9 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

[461] $1^{133}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 17 & 15 & 4 \\ 12 & 17 & 49 & 45 & 13 \\ 11 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[459] $1^{132}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 13 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

[460] $1^{143}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 11 & 16 & 46 & 44 & 12 \\ 11 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[463] $1^{134}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 16 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 10 & 17 & 49 & 72 & 13 \\ 16 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

[464] $1^{143}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 12 & 16 & 46 & 44 & 12 \\ 12 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[462] $1^{143}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 24 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[465] $1^{123}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -233, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 8 & 13 & 28 & 46 & 10 \\ 14 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

[467] $1^{1^{21}} 1^{1^{42}} 3^1 1^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 7 & 3 \\ 3 & 4 & 26 & 11 & 4 \\ 16 & 26 & 118 & 50 & 20 \\ 7 & 11 & 50 & 20 & 8 \\ 3 & 4 & 20 & 8 & 3 \end{pmatrix}_{b_0}$$

[466] $21^{1^{23}} 3^1 1^{1^{41}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 24 & 6 & 4 \\ 5 & 4 & 26 & 7 & 4 \\ 24 & 26 & 118 & 32 & 20 \\ 6 & 7 & 32 & 7 & 5 \\ 4 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

[468] $1^{1^{12}} 1^{1^{13}} 3^1 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[470] $1^{1^{32}} 3^1 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 12 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

[469] $1^{1^{41}} 1^{1^{14}} 3^1 1^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 118 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

[471] $21^{1^{12}} 3^1 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[472] $1^{1^{24}} 1^{1^{14}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 12 & 19 & 4 \\ 8 & 12 & 25 & 42 & 9 \\ 14 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

[473] $1^{1^{24}} 1^{1^{14}} 3^1 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -243, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 7 & 9 & 13 & 38 & 7 \\ 17 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[475] $1^{1^{23}} 1^{1^{24}} 3^1 1^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 6 & 8 & 10 & 34 & 6 \\ 16 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[474] $1^{1^{14}} 1^{1^{23}} 3^1 1^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 26 & 0 & 128 & 20 \\ 3 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[476] $21^{1^{24}} 3^1 1^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 20 & 3 \\ 4 & 4 & 8 & 29 & 4 \\ 6 & 8 & 10 & 40 & 6 \\ 20 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[479] \ 1^{13} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -247, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.87 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 25 & 0 & 111 & 19 \\ 3 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[478] \ 1^{142} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -247, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 13 & 21 & 76 & 59 & 16 \\ 10 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[477] \ 1^{141} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -247, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.87 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 23 & 5 & 3 \\ 3 & 4 & 36 & 7 & 4 \\ 23 & 36 & 238 & 44 & 28 \\ 5 & 7 & 44 & 7 & 5 \\ 3 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[480] \ 1^{122} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -247, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 8 & 13 & 28 & 37 & 10 \\ 10 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[481] \ 1^{113} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -249, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 6 & 9 & 13 & 32 & 7 \\ 13 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[485] \ 1^{144} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -250, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 16 & 19 & 4 \\ 12 & 16 & 46 & 56 & 12 \\ 13 & 19 & 56 & 64 & 14 \\ 3 & 4 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[482] \ 1^{134} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -250, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 12 & 19 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 13 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[483] \ 1^{114} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -250, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 13 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[484] \ 1^{124} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -250, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 19 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 13 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[487] \ 1^{13} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -252, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 19 & 27 & 133 & 15 & 21 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[488] \ 1^{124} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -252, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 17 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[486] \quad 1^{1^{21}} 1^{141} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 16 & 26 & 118 & 32 & 20 \\ 5 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[489] \quad 1^{1^4} 1^{113} 3^{11} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[490] \quad 1^{1^3} 1^{113} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[491] \quad 21^{141} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -254, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 3 \\ 4 & 4 & 16 & 7 & 4 \\ 12 & 16 & 46 & 20 & 12 \\ 5 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[493] \quad 1^{1^{23}} 1^{122} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 12 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[492] \quad 1^{1^{43}} 1^{142} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 12 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 15 & 21 & 76 & 59 & 16 \\ 12 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[495] \quad 1^{1^{14}} 1^{143} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 24 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[494] \quad 1^{1^{22}} 1^{143} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 4 & 21 & 15 & 4 \\ 13 & 21 & 76 & 56 & 16 \\ 10 & 15 & 56 & 39 & 11 \\ 3 & 4 & 16 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[496] \quad 1^{1^{42}} 1^{143} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -258, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 13 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 15 & 21 & 76 & 74 & 16 \\ 13 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[497] \quad 1^{1^{13}} 1^{113} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -258, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 6 & 9 & 13 & 32 & 7 \\ 14 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[498] \quad 21^{1^{23}} 3^{11} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 27 & 29 & 20 & 148 & 22 \\ 4 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[500] \ 1^{1^{31}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 15 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[499] \ 1^{1^{42}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 8 & 3 \\ 3 & 4 & 26 & 11 & 4 \\ 18 & 26 & 118 & 50 & 20 \\ 8 & 11 & 50 & 20 & 8 \\ 3 & 4 & 20 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[501] \ 1^{1^{13}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -263, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[502] \ 1^{1^{13}} 3^{1^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -263, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 24 & 35 & 23 & 231 & 27 \\ 3 & 4 & 3 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[503] \ 21^{1^{32}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 8 & 3 \\ 4 & 4 & 12 & 11 & 4 \\ 9 & 12 & 25 & 24 & 9 \\ 8 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[504] \ 1^{1^{41}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 18 & 26 & 118 & 32 & 20 \\ 4 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[507] \ 1^{1^{32}} 3^{1^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 11 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[505] \ 1^{1^{14}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 25 & 0 & 111 & 19 \\ 3 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[506] \ 1^{1^{32}} 3^{1^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 13 & 17 & 49 & 48 & 13 \\ 13 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[516] \ 1^{1^{42}} 3^{1^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 11 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 15 & 21 & 76 & 59 & 16 \\ 11 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[508] \ 1^{1^{22}} 3^{1^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 10 & 13 & 28 & 37 & 10 \\ 13 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[509] \quad 1^{1^{41}} 1^{13} 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 5 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 19 & 27 & 133 & 33 & 21 \\ 5 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[510] \quad 1^{1^{43}} 1^{13} 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 12 & 16 & 46 & 44 & 12 \\ 11 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[511] \quad 1^{1^{44}} 1^{12} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 11 & 13 & 28 & 37 & 10 \\ 13 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[512] \quad 1^{1^{21}} 1^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 11 & 17 & 49 & 48 & 13 \\ 10 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[513] \quad 1^{1^{14}} 1^{13} 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[514] \quad 1^{1^{32}} 1^{14} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 11 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 14 & 21 & 76 & 59 & 16 \\ 11 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[515] \quad 1^{1^{12}} 1^{13} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[517] \quad 1^{1^{12}} 1^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 10 & 17 & 49 & 60 & 13 \\ 13 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[518] \quad 1^{1^{24}} 1^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 7 & 9 & 13 & 32 & 7 \\ 15 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[519] \quad 1^{1^{13}} 1^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[520] \quad 21^{1^{43}} 1^{14} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 36 & 6 & 4 \\ 5 & 4 & 36 & 7 & 4 \\ 36 & 36 & 238 & 44 & 28 \\ 6 & 7 & 44 & 7 & 5 \\ 4 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[521] \quad 1^{1^{32}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -277, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 8 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 14 & 21 & 76 & 41 & 16 \\ 8 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[522] \quad 1^{1^{41}} 3^{1^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 & 3 \\ 3 & 4 & 32 & 3 & 4 \\ 22 & 32 & 193 & 18 & 25 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[523] \quad 1^{1^{33}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 12 & 17 & 49 & 48 & 13 \\ 12 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[524] \quad 1^{1^{41}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -281, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 16 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[525] \quad 1^{1^{14}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[526] \quad 1^{1^{14}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 6 & 9 & 13 & 38 & 7 \\ 17 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[527] \quad 1^{1^{12}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 15 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[528] \quad 1^{1^{14}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[529] \quad 1^{1^{13}} 3^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[530] \quad 21^{1^{13}} 3^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 30 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 30 & 30 & 20 & 165 & 23 \\ 4 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[531] \quad 1^{1^{24}} 3^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 19 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 14 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[532] \ 1^{1^{31}} 1^{141} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 14 & 21 & 76 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[533] \ 1^{1^{22}} 1^{122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 11 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[534] \ 1^{1^{41}} 1^{143} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 21 & 31 & 172 & 17 & 24 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[535] \ 1^{1^{13}} 1^{114} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 6 & 9 & 13 & 38 & 7 \\ 16 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[536] \ 1^{1^{13}} 1^{124} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 16 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[539] \ 1^{1^2} 1^{113} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 19 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[538] \ 1^{1^{34}} 1^{144} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 16 & 19 & 4 \\ 11 & 16 & 46 & 56 & 12 \\ 14 & 19 & 56 & 64 & 14 \\ 3 & 4 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[537] \ 1^{1^3} 1^{113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 23 & 35 & 23 & 231 & 27 \\ 3 & 4 & 3 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[541] \ 1^{1^{33}} 1^{122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 10 & 13 & 28 & 37 & 10 \\ 12 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[540] \ 21^{1^{12}} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 11 & 3 \\ 4 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 11 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[544] \ 1^{1^4} 1^{141} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 24 & 4 & 3 \\ 3 & 4 & 36 & 7 & 4 \\ 24 & 36 & 238 & 44 & 28 \\ 4 & 7 & 44 & 7 & 5 \\ 3 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[543] \ 21^{113}3^11^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 15 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[542] \ 21^{123}3^11^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 3 \\ 4 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 14 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[546] \ 1^{133}3^11^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -301, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 10 & 13 & 28 & 46 & 10 \\ 14 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[545] \ 1^{114}3^11^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -301, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 22 & 31 & 0 & 188 & 24 \\ 3 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[547] \ 1^{114}3^11^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -303, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[550] \ 1^{141}3^11^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 21 & 4 & 3 \\ 3 & 4 & 31 & 7 & 4 \\ 21 & 31 & 172 & 38 & 24 \\ 4 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[549] \ 21^{134}3^11^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 17 & 3 \\ 4 & 4 & 12 & 24 & 4 \\ 9 & 12 & 25 & 51 & 9 \\ 17 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[548] \ 21^{133}3^11^{141} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 30 & 6 & 4 \\ 5 & 4 & 31 & 7 & 4 \\ 30 & 31 & 172 & 38 & 24 \\ 6 & 7 & 38 & 7 & 5 \\ 4 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[551] \ 1^{142}3^11^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 7 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 14 & 21 & 76 & 41 & 16 \\ 7 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[553] \ 1^{114}3^11^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 15 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[552] \ 1^{132}3^11^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 12 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

[554] $1^{13}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 165 & 23 \\ 3 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[555] $1^{13}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -313, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 29 & 0 & 148 & 22 \\ 3 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[557] $1^{134}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 12 & 19 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 14 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

[556] $1^{124}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 6 & 8 & 10 & 34 & 6 \\ 17 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[558] $1^{141}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -317, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 15 & 21 & 76 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[559] $1^{132}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 12 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 13 & 17 & 49 & 48 & 13 \\ 12 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

[560] $21^{113}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 20 & 3 \\ 4 & 4 & 4 & 30 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[562] $1^{113}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -322, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

[561] $1^{141}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -322, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 & 3 \\ 3 & 4 & 36 & 3 & 4 \\ 24 & 36 & 238 & 20 & 28 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[563] $1^{113}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[564] $21^{123}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 11 & 3 \\ 4 & 4 & 8 & 15 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 11 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[565] \quad 1^{1^{43}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -328, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 13 & 17 & 49 & 60 & 13 \\ 14 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[566] \quad 1^{1^{12}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -328, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 8 & 13 & 28 & 46 & 10 \\ 13 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[567] \quad 21^{1^{41}} 3^1 1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 24 & 5 & 3 \\ 4 & 4 & 36 & 7 & 4 \\ 24 & 36 & 238 & 44 & 28 \\ 5 & 7 & 44 & 7 & 5 \\ 3 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[568] \quad 21^{1^{42}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 8 & 3 \\ 4 & 4 & 16 & 11 & 4 \\ 12 & 16 & 46 & 32 & 12 \\ 8 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[569] \quad 1^{1^{24}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 9 & 13 & 28 & 46 & 10 \\ 15 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[570] \quad 1^{1^{22}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 9 & 13 & 28 & 46 & 10 \\ 13 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[571] \quad 1^{1^{32}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -337, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 12 & 17 & 49 & 60 & 13 \\ 13 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[572] \quad 1^{1^{31}} 3^1 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 20 & 5 & 3 \\ 3 & 4 & 31 & 7 & 4 \\ 20 & 31 & 172 & 38 & 24 \\ 5 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[573] \quad 1^{1^{14}} 3^1 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -342, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 21 & 30 & 0 & 165 & 23 \\ 3 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[574] \quad 21^{1^{51}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.1$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 3 \\ 4 & 4 & 20 & 7 & 4 \\ 15 & 20 & 73 & 25 & 15 \\ 5 & 7 & 25 & 7 & 5 \\ 3 & 4 & 15 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[575] \quad 1^{1^{42}} 3^1 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -346, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 15 & 21 & 76 & 41 & 16 \\ 8 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0} \\
[576] \quad & 21^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 5 & 3 \\ 4 & 4 & 17 & 7 & 4 \\ 12 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0} \\
[577] \quad & 1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 29 & 0 & 148 & 22 \\ 3 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0} \\
[578] \quad & 1^{141}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 17 & 26 & 118 & 32 & 20 \\ 5 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0} \\
[579] \quad & 1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0} \\
[580] \quad & 1^{143}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -353, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 11 & 3 \\ 3 & 4 & 21 & 15 & 4 \\ 15 & 21 & 76 & 56 & 16 \\ 11 & 15 & 56 & 39 & 11 \\ 3 & 4 & 16 & 11 & 3 \end{pmatrix}_{b_0} \\
[581] \quad & 1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 11 & 17 & 49 & 60 & 13 \\ 14 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0} \\
[582] \quad & 1^{141}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 24 & 5 & 3 \\ 3 & 4 & 36 & 7 & 4 \\ 24 & 36 & 238 & 44 & 28 \\ 5 & 7 & 44 & 7 & 5 \\ 3 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0} \\
[583] \quad & 1^{144}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 16 & 19 & 4 \\ 12 & 16 & 46 & 56 & 12 \\ 14 & 19 & 56 & 64 & 14 \\ 3 & 4 & 12 & 14 & 3 \end{pmatrix}_{b_0} \\
[584] \quad & 1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -360, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0} \\
[585] \quad & 21^{131}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 5 & 3 \\ 4 & 4 & 22 & 7 & 4 \\ 15 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0} \\
[587] \quad & 1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[586] \quad 1^{1^{13}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -367, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 8 & 13 & 28 & 55 & 10 \\ 16 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[588] \quad 1^{1^{14}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -369, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[589] \quad 1^{1^{13}} 3^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -373, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.87 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[591] \quad 21^{1^{13}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -376, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 11 & 3 \\ 4 & 4 & 12 & 15 & 4 \\ 9 & 12 & 25 & 33 & 9 \\ 11 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[590] \quad 1^{1^{24}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -376, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 17 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 10 & 13 & 28 & 55 & 10 \\ 17 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[592] \quad 21^{1^{32}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -379, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 8 & 3 \\ 4 & 4 & 17 & 11 & 4 \\ 12 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[595] \quad 1^{1^{14}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -381, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[593] \quad 21^{1^{33}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -381, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 11 & 3 \\ 4 & 4 & 17 & 15 & 4 \\ 12 & 17 & 49 & 45 & 13 \\ 11 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[594] \quad 21^{1^{13}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -381, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 3 \\ 4 & 4 & 9 & 20 & 4 \\ 6 & 9 & 13 & 32 & 7 \\ 14 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[596] \quad 1^{1^{22}} 3^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -382, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 13 & 21 & 76 & 74 & 16 \\ 13 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[597] \quad 1^{1^{14}} 3^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -387, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[598] $1^{14}3^{11}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 22 & 34 & 0 & 208 & 26 \\ 3 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

[600] $21^{13}3^{11}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -391, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 34 & 4 \\ 3 & 4 & 1 & 23 & 3 \\ 33 & 34 & 23 & 208 & 26 \\ 4 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[599] $1^{123}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -391, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 9 & 13 & 28 & 46 & 10 \\ 14 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

[601] $21^{142}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -394, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 8 & 3 \\ 4 & 4 & 26 & 11 & 4 \\ 18 & 26 & 118 & 50 & 20 \\ 8 & 11 & 50 & 20 & 8 \\ 3 & 4 & 20 & 8 & 3 \end{pmatrix}_{b_0}$$

[602] $1^{123}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 10 & 13 & 28 & 46 & 10 \\ 15 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

[604] $1^{114}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[603] $1^{134}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 17 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 11 & 17 & 49 & 72 & 13 \\ 17 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

[605] $1^{124}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 8 & 13 & 28 & 55 & 10 \\ 15 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[606] $1^{141}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -408, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 5 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 18 & 26 & 118 & 32 & 20 \\ 5 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

[607] $1^{133}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -409, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 12 & 17 & 49 & 60 & 13 \\ 15 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

[608] $1^{13}3^{11}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -411, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 35 & 0 & 231 & 27 \\ 3 & 4 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

[610] $21^{142}3^11^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 11 & 3 \\ 4 & 4 & 21 & 16 & 4 \\ 15 & 21 & 76 & 59 & 16 \\ 11 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

[609] $21^{122}3^11^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 11 & 3 \\ 4 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 11 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

[611] $21^{113}3^11^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[613] $1^{141}3^11^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 21 & 5 & 3 \\ 3 & 4 & 31 & 7 & 4 \\ 21 & 31 & 172 & 38 & 24 \\ 5 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

[612] $1^{143}3^11^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 14 & 21 & 76 & 74 & 16 \\ 14 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[615] $21^{152}3^11^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 8 & 3 \\ 4 & 4 & 20 & 11 & 4 \\ 15 & 20 & 73 & 40 & 15 \\ 8 & 11 & 40 & 20 & 8 \\ 3 & 4 & 15 & 8 & 3 \end{pmatrix}_{b_0}$$

[614] $21^{113}3^11^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 3 \\ 4 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

[617] $1^{133}3^11^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 11 & 17 & 49 & 60 & 13 \\ 13 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

[616] $1^{134}3^11^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 10 & 17 & 49 & 72 & 13 \\ 15 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

[618] $1^{114}3^11^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -431, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[619] $1^{143}3^11^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -434, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 15 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 15 & 21 & 76 & 74 & 16 \\ 15 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[620] $1^{14}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 23 & 34 & 0 & 208 & 26 \\ 3 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

[621] $1^{14}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -438, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[622] $21^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 17 & 3 \\ 4 & 4 & 9 & 24 & 4 \\ 6 & 9 & 13 & 38 & 7 \\ 17 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[625] $1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 13 & 21 & 76 & 89 & 16 \\ 16 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

[624] $1^{14}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 21 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[623] $1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 9 & 13 & 28 & 55 & 10 \\ 16 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[626] $21^{14}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -440, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 19 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[627] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 13 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 14 & 21 & 76 & 74 & 16 \\ 13 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[628] $21^{14}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -446, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 11 & 3 \\ 4 & 4 & 16 & 15 & 4 \\ 12 & 16 & 46 & 44 & 12 \\ 11 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[629] $1^{14}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -447, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[630] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 17 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 13 & 17 & 49 & 72 & 13 \\ 17 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

[631] $21^{16}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -454, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 5 & 3 \\ 4 & 4 & 24 & 7 & 4 \\ 18 & 24 & 106 & 30 & 18 \\ 5 & 7 & 30 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[632] \ 21^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -455, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 11 & 3 \\ 4 & 4 & 17 & 16 & 4 \\ 12 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[633] \ 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -463, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 1 & 23 & 3 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[634] \ 1^{133} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -466, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 12 & 17 & 49 & 60 & 13 \\ 14 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[635] \ 21^{143} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -469, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 39 & 4 \\ 5 & 4 & 4 & 39 & 4 \\ 3 & 4 & 1 & 26 & 3 \\ 39 & 39 & 26 & 280 & 30 \\ 4 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[636] \ 1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -469, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 26 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 1 & 26 & 3 \\ 26 & 39 & 26 & 280 & 30 \\ 3 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[637] \ 21^{124} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -470, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 3 \\ 4 & 4 & 8 & 19 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 14 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[638] \ 1^{144} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 13 & 17 & 49 & 60 & 13 \\ 15 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[639] \ 1^{124} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 9 & 13 & 28 & 55 & 10 \\ 17 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[640] \ 1^{143} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 14 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 15 & 21 & 76 & 74 & 16 \\ 14 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[641] \ 1^{143} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 25 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 1 & 26 & 3 \\ 25 & 39 & 26 & 280 & 30 \\ 3 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[642] \ 1^{144} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -487, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 16 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 15 & 21 & 76 & 89 & 16 \\ 16 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[643] \ 1^{14} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -492, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[645] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -499, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 11 & 17 & 49 & 72 & 13 \\ 15 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[644] \ 21^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -499, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 3 \\ 4 & 4 & 21 & 7 & 4 \\ 15 & 21 & 76 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[646] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -504, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 16 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 12 & 17 & 49 & 72 & 13 \\ 16 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[648] \ 21^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -504, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 17 & 3 \\ 4 & 4 & 8 & 24 & 4 \\ 6 & 8 & 10 & 34 & 6 \\ 17 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[647] \ 21^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -504, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 26 & 3 \\ 4 & 4 & 4 & 39 & 4 \\ 3 & 4 & 1 & 26 & 3 \\ 26 & 39 & 26 & 280 & 30 \\ 3 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[649] \ 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -517, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 26 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 26 & 39 & 0 & 280 & 30 \\ 3 & 4 & 0 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[650] \ 21^{14} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -520, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 14 & 3 \\ 4 & 4 & 12 & 19 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 14 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[652] \ 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -532, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 26 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 3 & 4 & 1 & 26 & 3 \\ 26 & 39 & 26 & 280 & 30 \\ 3 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[651] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -532, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 17 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 14 & 21 & 76 & 89 & 16 \\ 17 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[653] \ 21^{16} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -534, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 8 & 3 \\ 4 & 4 & 24 & 11 & 4 \\ 18 & 24 & 106 & 48 & 18 \\ 8 & 11 & 48 & 20 & 8 \\ 3 & 4 & 18 & 8 & 3 \end{pmatrix}_{b_0}$$

- [654] $1^{14}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -535$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 1 & 23 & 3 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$
- [655] $21^{15}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -536$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.1$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 11 & 3 \\ 4 & 4 & 20 & 15 & 4 \\ 15 & 20 & 73 & 55 & 15 \\ 11 & 15 & 55 & 39 & 11 \\ 3 & 4 & 15 & 11 & 3 \end{pmatrix}_{b_0}$
- [656] $21^{14}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -539$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 21 & 5 & 3 \\ 4 & 4 & 31 & 7 & 4 \\ 21 & 31 & 172 & 38 & 24 \\ 5 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$
- [657] $1^{13}4^31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -544$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 11 & 17 & 49 & 72 & 13 \\ 16 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$
- [658] $21^{14}2^31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -549$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 8 & 3 \\ 4 & 4 & 21 & 11 & 4 \\ 15 & 21 & 76 & 41 & 16 \\ 8 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$
- [659] $1^{14}4^31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -568$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 13 & 21 & 76 & 89 & 16 \\ 15 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$
- [660] $21^{14}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -571$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 11 & 3 \\ 4 & 4 & 21 & 15 & 4 \\ 15 & 21 & 76 & 56 & 16 \\ 11 & 15 & 56 & 39 & 11 \\ 3 & 4 & 16 & 11 & 3 \end{pmatrix}_{b_0}$
- [661] $21^{12}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -581$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 14 & 3 \\ 4 & 4 & 13 & 20 & 4 \\ 9 & 13 & 28 & 46 & 10 \\ 14 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$
- [662] $21^{17}1^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -584$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 21 & 5 & 3 \\ 4 & 4 & 28 & 7 & 4 \\ 21 & 28 & 145 & 35 & 21 \\ 5 & 7 & 35 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$
- [663] $21^{14}4^31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -590$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 14 & 3 \\ 4 & 4 & 16 & 19 & 4 \\ 12 & 16 & 46 & 56 & 12 \\ 14 & 19 & 56 & 64 & 14 \\ 3 & 4 & 12 & 14 & 3 \end{pmatrix}_{b_0}$
- [664] $21^{14}1^31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -594$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 18 & 5 & 3 \\ 4 & 4 & 26 & 7 & 4 \\ 18 & 26 & 118 & 32 & 20 \\ 5 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$
- [665] $1^{13}4^31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -603$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 12 & 17 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 12 & 17 & 49 & 72 & 13 \\ 17 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0} \\
[666] & 21^{1^{14}3^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -624, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 17 & 3 \\ 4 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0} \\
[667] & 1^{1^{44}3^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -639, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 14 & 21 & 76 & 89 & 16 \\ 16 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0} \\
[668] & 21^{1^{63}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 11 & 3 \\ 4 & 4 & 24 & 15 & 4 \\ 18 & 24 & 106 & 66 & 18 \\ 11 & 15 & 66 & 39 & 11 \\ 3 & 4 & 18 & 11 & 3 \end{pmatrix}_{b_0} \\
[669] & 21^{1^{14}3^1 1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 23 & 3 \\ 4 & 4 & 4 & 34 & 4 \\ 3 & 4 & 1 & 23 & 3 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0} \\
[670] & 21^{1^{72}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 21 & 8 & 3 \\ 4 & 4 & 28 & 11 & 4 \\ 21 & 28 & 145 & 56 & 21 \\ 8 & 11 & 56 & 20 & 8 \\ 3 & 4 & 21 & 8 & 3 \end{pmatrix}_{b_0} \\
[671] & 1^{1^{44}3^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -676, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 17 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 15 & 21 & 76 & 89 & 16 \\ 17 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0} \\
[672] & 21^{1^{54}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -680, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 14 & 3 \\ 4 & 4 & 20 & 19 & 4 \\ 15 & 20 & 73 & 70 & 15 \\ 14 & 19 & 70 & 64 & 14 \\ 3 & 4 & 15 & 14 & 3 \end{pmatrix}_{b_0} \\
[673] & 21^{1^{14}3^1 1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -696, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 20 & 3 \\ 4 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0} \\
[674] & 21^{1^{33}3^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -701, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 14 & 3 \\ 4 & 4 & 17 & 20 & 4 \\ 12 & 17 & 49 & 60 & 13 \\ 14 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0} \\
[675] & 21^{1^{24}3^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -719, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 17 & 3 \\ 4 & 4 & 13 & 24 & 4 \\ 9 & 13 & 28 & 55 & 10 \\ 17 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0} \\
[676] & 21^{1^{43}3^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -741, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 14 & 3 \\ 4 & 4 & 21 & 20 & 4 \\ 15 & 21 & 76 & 74 & 16 \\ 14 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0} \\
[677] & 21^{1^{73}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -776, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 21 & 11 & 3 \\ 4 & 4 & 28 & 15 & 4 \\ 21 & 28 & 145 & 77 & 21 \\ 11 & 15 & 77 & 39 & 11 \\ 3 & 4 & 21 & 11 & 3 \end{pmatrix}_{b_0} \\
[678] & 21^{134}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -919, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 17 & 3 \\ 4 & 4 & 17 & 24 & 4 \\ 12 & 17 & 49 & 72 & 13 \\ 17 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0} \\
[679] & 21^{144}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -1039, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 17 & 3 \\ 4 & 4 & 21 & 24 & 4 \\ 15 & 21 & 76 & 89 & 16 \\ 17 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0} \\
\\
T = 8 & [693] 1^{131}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 3 & 3 & 15 & 4 \\ 3 & 1 & 3 & 9 & 3 \\ 3 & 3 & 1 & 14 & 3 \\ 15 & 9 & 14 & 49 & 15 \\ 4 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0} \\
[680] & 1^{113}3^{121}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 2 \\ 1 & 1 & 3 & 1 & 5 & 3 \\ 4 & 4 & 5 & 5 & 10 & 6 \\ 2 & 2 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[681] & 1^{123}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 5 & 3 & 4 & 5 & 16 & 5 \\ 8 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0} \\
[682] & 1^{131}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 7 & 7 & 6 & 25 & 17 & 9 \\ 4 & 4 & 5 & 17 & 10 & 6 \\ 2 & 2 & 2 & 9 & 6 & 3 \end{pmatrix}_{b_0} \\
[683] & 1^{113}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 3 & 8 & 2 \\ 1 & 1 & 3 & 1 & 8 & 3 \\ 7 & 5 & 8 & 8 & 23 & 9 \\ 2 & 2 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[684] & 1^{122}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 6 & 7 & 6 & 25 & 17 & 9 \\ 5 & 4 & 5 & 17 & 10 & 6 \\ 2 & 2 & 2 & 9 & 6 & 3 \end{pmatrix}_{b_0} \\
[685] & 1^{122}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 3 & 8 & 2 \\ 3 & 1 & 3 & 1 & 8 & 3 \\ 6 & 5 & 8 & 8 & 23 & 9 \\ 2 & 2 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[686] & 1^{122}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 2 \\ 3 & 1 & 3 & 1 & 5 & 3 \\ 5 & 4 & 5 & 5 & 10 & 6 \\ 2 & 2 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[687] $1^{122} 1^{11} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 7 & 10 & 14 & 19 & 8 \\ 8 & 13 & 19 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$

[688] $1^{13} 1^{12} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 3 & 4 & 5 & 16 & 5 \\ 8 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$

[689] $1^{11} 1^{11} 3^{121} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$

[690] $1^{11} 1^{11} 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[691] $1^{111} 1^{11} 3^{121} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 8 & 3 \\ 4 & 5 & 0 & 8 & 23 & 9 \\ 2 & 2 & -1 & 3 & 9 & 3 \end{pmatrix}_{b_0}$

[692] $1^{122} 1^{11} 3^{121} 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 13 & 25 & 92 & 49 & 20 \\ 8 & 13 & 49 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$

[696] $1^{112} 1^{11} 3^{121} 21^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 5 & 0 & 33 & 29 & 11 \\ 4 & 5 & 0 & 29 & 23 & 9 \\ 2 & 2 & -1 & 11 & 9 & 3 \end{pmatrix}_{b_0}$

[694] $1^{111} 1^{11} 3^{121} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 4 & 2 & 4 & 2 & 4 \\ 4 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[695] $1^{111} 1^{11} 3^{121} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 4 & 3 \\ 2 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$

[697] $1^{111} 1^{11} 3^{121} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 17 & 5 \\ 6 & 6 & 0 & 17 & 42 & 12 \\ 2 & 2 & -1 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[704] $1^{11}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 10 & 4 \\ 3 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 8 & 3 \\ 10 & 8 & 8 & 22 & 10 \\ 4 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[718] $1^{12}3^{12}21^{24} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 1 & 6 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 54 & 10 \\ 8 & 11 & 0 & 54 & 103 & 19 \\ 2 & 2 & -1 & 10 & 19 & 3 \end{pmatrix}_{b_0}$

[705] $1^{12}3^{12}21^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 10 \\ 2 & 2 & 0 & 12 & 4 & 4 \\ 2 & 2 & -1 & 10 & 4 & 3 \end{pmatrix}_{b_0}$

[703] $1^{13}23^{12}21^{33} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & 9 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 60 & 15 \\ 6 & 8 & 0 & 60 & 56 & 14 \\ 2 & 2 & -1 & 15 & 14 & 3 \end{pmatrix}_{b_0}$

[702] $1^{11}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 5 \\ 4 & 5 & 0 & 13 & 23 & 9 \\ 2 & 2 & -1 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[701] $1^{13}3^{12}21^{32} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 39 & 15 \\ 4 & 5 & 0 & 39 & 23 & 9 \\ 2 & 2 & -1 & 15 & 9 & 3 \end{pmatrix}_{b_0}$

[700] $1^{122}3^{12}21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 10 & 9 & 2 \\ 8 & 10 & 10 & 58 & 46 & 14 \\ 7 & 7 & 9 & 46 & 35 & 11 \\ 2 & 2 & 2 & 14 & 11 & 3 \end{pmatrix}_{b_0}$

[699] $1^{13}3^{12}21^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 18 & 15 \\ 2 & 2 & 0 & 18 & 4 & 4 \\ 2 & 2 & -1 & 15 & 4 & 3 \end{pmatrix}_{b_0}$

[698] $1^{121}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 20 & 4 \\ 3 & 1 & 3 & 16 & 3 \\ 3 & 3 & 1 & 16 & 3 \\ 20 & 16 & 16 & 94 & 20 \\ 4 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$

[706] $1^{141}3^{12}21^{42} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 12 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 12 & 0 & 110 & 52 & 20 \\ 4 & 5 & 0 & 52 & 23 & 9 \\ 2 & 2 & -1 & 20 & 9 & 3 \end{pmatrix}_{b_0}$$

[707] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 5 \\ 2 & 2 & 0 & 6 & 4 & 4 \\ 2 & 2 & -1 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[708] $1^{112}3^{11}21^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 5 & 0 & 33 & 14 & 11 \\ 2 & 2 & 0 & 14 & 4 & 4 \\ 2 & 2 & -1 & 11 & 4 & 3 \end{pmatrix}_{b_0}$$

[709] $1^{114}3^{11}21^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 12 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 12 & 0 & 110 & 24 & 20 \\ 2 & 2 & 0 & 24 & 4 & 4 \\ 2 & 2 & -1 & 20 & 4 & 3 \end{pmatrix}_{b_0}$$

[710] $1^{12}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 1 & 1 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 35 & 11 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[711] $1^{1123}3^{11}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 20 & 5 \\ 6 & 8 & 0 & 20 & 56 & 14 \\ 2 & 2 & -1 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[712] $1^{11231}3^{11}21^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 26 & 10 \\ 4 & 5 & 0 & 26 & 23 & 9 \\ 2 & 2 & -1 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[713] $1^{124}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 3 & 1 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 35 & 11 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[714] $1^{111}3^{11}21^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & 7 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 45 & 52 & 13 \\ 6 & 8 & 0 & 52 & 56 & 14 \\ 2 & 2 & -1 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[715] $1^{124}3^{11}21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 1 & 9 & 2 \\ 1 & 2 & 0 & 3 & 15 & 2 \\ 3 & 1 & 3 & 1 & 15 & 3 \\ 11 & 9 & 15 & 15 & 83 & 17 \\ 2 & 2 & 2 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[719] $1^{11423}3^{11}21^{43} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 1 & 12 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 12 & 0 & 110 & 80 & 20 \\ 6 & 8 & 0 & 80 & 56 & 14 \\ 2 & 2 & -1 & 20 & 14 & 3 \end{pmatrix}_{b_0}$$

[716] $1^{113}3^{121}21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 27 & 5 \\ 8 & 11 & 0 & 27 & 103 & 19 \\ 2 & 2 & -1 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[717] $1^{122}3^{121}21^{23} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 40 & 10 \\ 6 & 8 & 0 & 40 & 56 & 14 \\ 2 & 2 & -1 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[721] $1^{113}3^{121}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 4 \\ 3 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 11 & 7 & 10 & 25 & 11 \\ 4 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[722] $1^{111}3^{121}21^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 4 & 0 & 16 & 21 & 8 \\ 4 & 5 & 0 & 21 & 23 & 9 \\ 2 & 2 & -1 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[720] $1^{113}3^{121}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 4 \\ 3 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 7 & 5 & 6 & 9 & 7 \\ 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[725] $1^{113}3^{121}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[723] $1^{131}3^{121}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 5 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[724] $1^{133}3^{121}21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 5 & 8 & 2 \\ 1 & 2 & 0 & 10 & 12 & 2 \\ 7 & 5 & 10 & 33 & 44 & 11 \\ 9 & 8 & 12 & 44 & 56 & 14 \\ 2 & 2 & 2 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[726] $1^{122}3^{121}21^{24} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 48 & 10 \\ 8 & 9 & 0 & 48 & 83 & 17 \\ 2 & 2 & -1 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[727] $1^{113}3^{121}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 5 & 4 & 4 \\ 2 & 3 & -1 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[728] \quad 1^{11} 3^{11} 21^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 4 & 0 & 16 & 10 & 8 \\ 2 & 2 & 0 & 10 & 4 & 4 \\ 2 & 2 & -1 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[729] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 2 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[730] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 8 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 8 & 6 \\ 3 & 5 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[731] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[732] \quad 1^{14} 3^{12} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 1 & 1 & 3 & 1 & 12 & 3 \\ 10 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[733] \quad 1^{124} 3^{12} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 3 & 1 & 3 & 1 & 12 & 3 \\ 10 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[734] \quad 21^{113} 3^{12} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 5 & 15 & 8 & 5 \\ 12 & 15 & 32 & 18 & 12 \\ 6 & 8 & 18 & 8 & 6 \\ 4 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[735] \quad 21^{111} 3^{12} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 4 & 2 \\ 2 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 2 \\ 4 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[736] \quad 21^{131} 3^{12} 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 4 & 2 \\ 2 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 10 & 10 & 15 & 88 & 32 & 18 \\ 4 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[737] \quad 1^{111} 3^{12} 21^{43} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 1 & 10 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 88 & 72 & 18 \\ 6 & 8 & 0 & 72 & 56 & 14 \\ 2 & 2 & -1 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

[740] $1^{134} 3^{121} 1^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 7 & 10 & 14 & 42 & 8 \\ 17 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

[738] $1^{111} 3^{121} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 5 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[739] $1^{11} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 5 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[743] $1^{13} 3^{121} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 17 & 4 \\ 3 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 14 & 3 \\ 17 & 13 & 14 & 65 & 17 \\ 4 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[742] $1^{121} 3^{121} 1^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 7 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 45 & 34 & 13 \\ 4 & 5 & 0 & 34 & 23 & 9 \\ 2 & 2 & -1 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

[741] $1^{13} 3^{121} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 10 & 2 \\ 1 & 2 & 0 & 3 & 13 & 2 \\ 1 & 1 & 3 & 1 & 13 & 3 \\ 10 & 10 & 13 & 13 & 74 & 16 \\ 2 & 2 & 2 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[746] $1^{112} 3^{121} 1^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 8 & 0 & 70 & 42 & 16 \\ 4 & 5 & 0 & 42 & 23 & 9 \\ 2 & 2 & -1 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

[747] $1^{123} 3^{121} 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 45 & 16 & 13 \\ 2 & 2 & 0 & 16 & 4 & 4 \\ 2 & 2 & -1 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

[744] $1^{141} 3^{121} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 22 & 4 \\ 3 & 1 & 3 & 14 & 3 \\ 3 & 3 & 1 & 20 & 3 \\ 22 & 14 & 20 & 106 & 22 \\ 4 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[745] $1^{121} 3^{121} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 4 \\ 3 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 12 & 3 \\ 14 & 10 & 12 & 42 & 14 \\ 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[748] $1^{121} 3^{121} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 4 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
[749] \quad & 1^{123} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 10 & 2 \\ 2 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 5 & 5 & 13 & 33 & 7 \\ 10 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[751] \quad & 1^{12} 3^1 21^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 8 & 0 & 70 & 20 & 16 \\ 2 & 2 & 0 & 20 & 4 & 4 \\ 2 & 2 & -1 & 16 & 4 & 3 \end{pmatrix}_{b_0} \\
& 1^{31} \\
[750] \quad & 1^{12} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 3 & 3 & 18 & 4 \\ 3 & 1 & 3 & 12 & 3 \\ 3 & 3 & 1 & 16 & 3 \\ 18 & 12 & 16 & 70 & 18 \\ 4 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0} \\
& 1^2 \\
[752] \quad & 1^1 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 2 & 10 & 4 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0} \\
& 1^{13} \\
[753] \quad & 1^{113} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 4 & 4 & 7 & 16 & 36 & 8 \\ 10 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0} \\
& 1^2 \\
[754] \quad & 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 2 \\ 5 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[755] \quad & 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 2 \\ 4 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
& 1^{22} \\
[761] \quad & 1^{131} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 10 & 5 & 2 \\ 1 & 2 & 0 & 15 & 8 & 2 \\ 9 & 10 & 15 & 88 & 47 & 18 \\ 6 & 5 & 8 & 47 & 23 & 9 \\ 2 & 2 & 2 & 18 & 9 & 3 \end{pmatrix}_{b_0} \\
& 1^{22} \\
[756] \quad & 1^{113} 3^1 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 5 & 15 & 28 & 5 \\ 8 & 15 & 34 & 63 & 12 \\ 15 & 28 & 63 & 116 & 22 \\ 3 & 5 & 12 & 22 & 4 \end{pmatrix}_{b_0} \\
& 1^{22} \\
[760] \quad & 1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 4 & 5 & 2 & 13 & 4 \\ 11 & 18 & 13 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[759] $1^{11}3^{12}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 5 & 10 & 14 & 19 & 8 \\ 7 & 13 & 19 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

[758] $1^{11}3^{12}21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 24 & 5 \\ 8 & 9 & 0 & 24 & 83 & 17 \\ 2 & 2 & -1 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[757] $1^{11}3^{12}21^{42} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 10 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 88 & 47 & 18 \\ 4 & 5 & 0 & 47 & 23 & 9 \\ 2 & 2 & -1 & 18 & 9 & 3 \end{pmatrix}_{b_0}$$

[762] $1^{11}3^{12}21^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 88 & 22 & 18 \\ 2 & 2 & 0 & 22 & 4 & 4 \\ 2 & 2 & -1 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

[769] $1^{11}3^{12}21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 5 & 15 & 28 & 5 \\ 9 & 15 & 34 & 63 & 12 \\ 17 & 28 & 63 & 116 & 22 \\ 3 & 5 & 12 & 22 & 4 \end{pmatrix}_{b_0}$$

[774] $1^{11}3^{12}21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 5 & 10 & 14 & 20 & 8 \\ 8 & 13 & 20 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

[768] $1^{11}3^{12}21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 4 & 13 & 2 \\ 1 & 2 & 0 & 7 & 17 & 2 \\ 4 & 4 & 7 & 16 & 47 & 8 \\ 13 & 13 & 17 & 47 & 127 & 21 \\ 2 & 2 & 2 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

[763] $1^{11}3^{12}21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 5 & 10 & 14 & 34 & 8 \\ 13 & 23 & 34 & 78 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[764] $1^{11}3^{12}21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 8 & 15 & 34 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

[765] $1^{11}3^{12}21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 7 & 4 & 7 & 16 & 25 & 8 \\ 9 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[766] $1^{11}3^{12}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 13 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 24 & 10 \\ 3 & 5 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[767] \ 1^{1^{31}} 3^{1^{21}} 21^{1^{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 6 & 5 & 5 & 13 & 13 & 7 \\ 4 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[771] \ 1^{1^{11}} 3^{1^{21}} 21^{1^{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 2 \\ 2 & 1 & 3 & 1 & 5 & 3 \\ 4 & 4 & 5 & 5 & 10 & 6 \\ 2 & 2 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[772] \ 1^{1^{12}} 3^{1^{21}} 21^{1^{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 5 & 5 & 13 & 13 & 7 \\ 5 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[770] \ 1^{1^{12}} 3^{1^{21}} 21^{1^{22}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 4 & 4 & 7 & 16 & 25 & 8 \\ 7 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[773] \ 1^{1^{12}} 3^{1^{21}} 21^{1^{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 2 \\ 2 & 1 & 3 & 1 & 5 & 3 \\ 5 & 4 & 5 & 5 & 10 & 6 \\ 2 & 2 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[782] \ 1^{1^2} 3^{1^{21}} 21^{1^{21}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 4 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[779] \ 1^1 3^{1^{21}} 21^{1^{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 3 & 5 & 2 & 6 & 4 \\ 4 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[778] \ 1^{1^{21}} 3^{1^{21}} 21^{1^{41}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 12 & 2 & 2 \\ 1 & 2 & 0 & 16 & 4 & 2 \\ 10 & 12 & 16 & 110 & 24 & 20 \\ 3 & 2 & 4 & 24 & 4 & 4 \\ 2 & 2 & 2 & 20 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[777] \ 1^{1^2} 3^{1^{21}} 21^{1^{21}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 4 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[776] \ 1^{1^4} 3^{1^{21}} 21^{1^{41}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 4 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[775] \ 1^{\overline{1}^3} 3^{\overline{1}^{11}} 21^{\overline{1}^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 9 & 15 & 34 & 18 & 12 \\ 4 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[780] \ 1^{\overline{1}^1} 3^{\overline{1}^{11}} 21^{\overline{1}^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 4 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[781] \ 1^{\overline{1}^{11}} 3^{\overline{1}^{11}} 21^{\overline{1}^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 2 \\ 1 & 2 & 0 & 8 & 4 & 2 \\ 5 & 6 & 8 & 26 & 12 & 10 \\ 3 & 2 & 4 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[783] \ 1^{\overline{1}^{21}} 3^{\overline{1}^{11}} 21^{\overline{1}^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 7 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[784] \ 1^{\overline{1}^3} 3^{\overline{1}^{11}} 21^{\overline{1}^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 3 & 4 & 5 & 19 & 5 \\ 9 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[785] \ 1^{\overline{1}^{34}} 3^{\overline{1}^{11}} 21^{\overline{1}^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 5 & 3 & 4 & 5 & 19 & 5 \\ 10 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[787] \ 1^{\overline{1}^{31}} 3^{\overline{1}^{11}} 21^{\overline{1}^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 7 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[786] \ 1^{\overline{1}^{11}} 3^{\overline{1}^{11}} 21^{\overline{1}^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 10 & 20 & 58 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[788] \ 1^{\overline{1}^{31}} 3^{\overline{1}^{11}} 21^{\overline{1}^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 35, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 21 & 4 \\ 3 & 1 & 3 & 15 & 3 \\ 3 & 3 & 1 & 18 & 3 \\ 21 & 15 & 18 & 97 & 21 \\ 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[789] \ 1^{\overline{1}^{11}} 3^{\overline{1}^{11}} 21^{\overline{1}^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 36, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 13 & 25 & 94 & 48 & 20 \\ 7 & 13 & 48 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

[790] $1^{14}3^{121}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 1 & 13 & 2 \\ 1 & 2 & 0 & 3 & 17 & 2 \\ 1 & 1 & 3 & 1 & 17 & 3 \\ 13 & 13 & 17 & 17 & 127 & 21 \\ 2 & 2 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[791] $1^{131}3^{121}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 24 & 4 \\ 3 & 1 & 3 & 18 & 3 \\ 3 & 3 & 1 & 20 & 3 \\ 24 & 18 & 20 & 130 & 24 \\ 4 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[792] $1^{122}3^{121}1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 8 & 15 & 34 & 30 & 12 \\ 8 & 13 & 30 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

[794] $1^{112}3^{121}1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 8 & 15 & 32 & 29 & 12 \\ 8 & 13 & 29 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

[795] $1^{111}3^{121}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[796] $1^{111}3^{121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 6 & 4 & 7 & 16 & 14 & 8 \\ 4 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[797] $1^{13}3^{121}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 2 & 17 & 4 \\ 14 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[798] $1^{121}3^{121}1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 14 & 5 & 2 \\ 9 & 8 & 14 & 70 & 28 & 16 \\ 4 & 4 & 5 & 28 & 10 & 6 \\ 2 & 2 & 2 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[799] $1^{122}3^{121}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 4 & 5 & 2 & 10 & 4 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[800] $1^{112}3^{121}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 2 \\ 5 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[801] $1^{111}3^{121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 4 & 4 & 7 & 16 & 14 & 8 \\ 4 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[802] \ 1^{1^{22}} 1^{1^{31}} 3^{1^{21}} 21^{1^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 7 & 8 & 9 & 40 & 22 & 12 \\ 5 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[803] \ 1^{1^{11}} 3^{1^{21}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 2 \\ 4 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[804] \ 1^{1^{42}} 1^{1^{11}} 3^{1^{21}} 21^{1^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 7 & 4 & 7 & 16 & 14 & 8 \\ 5 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[805] \ 1^{1^{22}} 1^{1^{11}} 3^{1^{21}} 21^{1^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 5 & 4 & 7 & 16 & 14 & 8 \\ 5 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[806] \ 1^{1^{12}} 1^{1^{11}} 3^{1^{21}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[807] \ 1^{1^{42}} 1^{1^{12}} 3^{1^{21}} 21^{1^{32}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 16 & 25 & 94 & 48 & 20 \\ 8 & 13 & 48 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[793] \ 1^{1^{12}} 1^{1^{11}} 3^{1^{21}} 21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 6 & 10 & 14 & 19 & 8 \\ 8 & 13 & 19 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[808] \ 21^{1^{23}} 3^{1^{21}} 21^{1^{41}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 6 & 4 \\ 5 & 5 & 30 & 8 & 5 \\ 24 & 30 & 134 & 36 & 24 \\ 6 & 8 & 36 & 8 & 6 \\ 4 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[809] \ 1^{1^{11}} 1^{1^{11}} 3^{1^{21}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 3 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[810] \ 1^{1^{41}} 1^{1^{31}} 3^{1^{21}} 21^{1^{41}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 25 & 4 \\ 3 & 1 & 3 & 17 & 3 \\ 3 & 3 & 1 & 22 & 3 \\ 25 & 17 & 22 & 137 & 25 \\ 4 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[811] \ 1^{121} 3^{121} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 4 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[812] \ 1^{111} 3^{121} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[813] \ 1^{14} 3^{121} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 1 & 0 & 7 & 2 \\ 9 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[814] \ 1^3 3^{121} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 2 & 14 & 4 \\ 11 & 18 & 14 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[815] \ 1^{112} 3^{121} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 2 & 1 & 0 & 7 & 2 \\ 7 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[816] \ 1^{31} 3^{121} 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 11 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[817] \ 1^{133} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 5 & 3 & 4 & 5 & 13 & 5 \\ 7 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[818] \ 1^{13} 3^{121} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 5 & 10 & 18 & 5 \\ 5 & 10 & 14 & 28 & 8 \\ 11 & 18 & 28 & 48 & 14 \\ 3 & 5 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[819] \ 1^{12} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 3 & 4 & 5 & 13 & 5 \\ 6 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[820] \ 1^{11} 3^{121} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 11 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[821] \quad 1^{11} 3^{12} 21^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 2 & 13 & 4 \\ 10 & 18 & 13 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[822] \quad 1^{112} 3^{13} 21^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 2 & 1 & 0 & 8 & 2 \\ 9 & 8 & 12 & 8 & 56 & 14 \\ 2 & 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[823] \quad 1^{121} 3^{121} 21^{121} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[824] \quad 1^{122} 3^{121} 21^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 5 & 6 & 8 & 26 & 26 & 10 \\ 6 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[825] \quad 1^{111} 3^{121} 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[826] \quad 1^{13} 3^{121} 21^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 48 & 14 \\ 3 & 5 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[827] \quad 1^{122} 3^{121} 21^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 48 & 14 \\ 3 & 5 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[838] \quad 1^{132} 3^{121} 21^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 10 & 11 & 13 & 81 & 31 & 17 \\ 5 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[830] \quad 1^{141} 3^{121} 21^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 10 & 15 & 34 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[840] \quad 1^{132} 3^{121} 21^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 8 & 7 & 11 & 45 & 23 & 13 \\ 5 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

[841] $1^{123} 1^1 3^1 21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 5 & 15 & 18 & 5 \\ 8 & 15 & 34 & 42 & 12 \\ 11 & 18 & 42 & 48 & 14 \\ 3 & 5 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[831] $1^{132} 1^1 3^1 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 11 & 20 & 62 & 40 & 16 \\ 8 & 13 & 40 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

[837] $1^{111} 1^1 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 13 & 25 & 94 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[836] $1^{111} 1^1 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 3 & 4 & 5 & 9 & 5 \\ 4 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[835] $1^{144} 1^{12} 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 10 & 15 & 2 \\ 8 & 5 & 10 & 33 & 54 & 11 \\ 11 & 9 & 15 & 54 & 83 & 17 \\ 2 & 2 & 2 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[834] $1^{121} 1^{131} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 6 & 7 & 6 & 25 & 17 & 9 \\ 4 & 4 & 5 & 17 & 10 & 6 \\ 2 & 2 & 2 & 9 & 6 & 3 \end{pmatrix}_{b_0}$$

[832] $1^{134} 1^{11} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 7 & 10 & 14 & 34 & 8 \\ 15 & 23 & 34 & 78 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[839] $1^{132} 1^{131} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 7 & 7 & 6 & 25 & 17 & 9 \\ 5 & 4 & 5 & 17 & 10 & 6 \\ 2 & 2 & 2 & 9 & 6 & 3 \end{pmatrix}_{b_0}$$

[833] $1^{132} 1^1 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 7 & 10 & 14 & 20 & 8 \\ 8 & 13 & 20 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

[828] $1^{132} 1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 5 & 3 & 4 & 5 & 9 & 5 \\ 5 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[829] $1^{132} 1^{11} 3^1 21^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 15 & 25 & 94 & 48 & 20 \\ 8 & 13 & 48 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[842] \ 1^{14}3^{121}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 28 & 4 \\ 3 & 1 & 3 & 20 & 3 \\ 3 & 3 & 1 & 24 & 3 \\ 28 & 20 & 24 & 174 & 28 \\ 4 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[843] \ 1^{123}3^{121}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 4 & 5 & 2 & 14 & 4 \\ 11 & 18 & 14 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[844] \ 1^{123}3^{121}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 4 & 5 & 2 & 17 & 4 \\ 14 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[845] \ 1^{112}3^{121}1^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 10 & 20 & 58 & 54 & 16 \\ 10 & 18 & 54 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[846] \ 1^{12}3^{121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 9 & 15 & 32 & 18 & 12 \\ 4 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[847] \ 1^{12}3^{121}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[848] \ 1^{131}3^{121}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 9 & 15 & 34 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[853] \ 1^{112}3^{121}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 9 \\ 2 & 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[852] \ 1^{133}3^{121}1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 5 & 5 & 2 \\ 1 & 2 & 0 & 10 & 8 & 2 \\ 7 & 5 & 10 & 33 & 29 & 11 \\ 7 & 5 & 8 & 29 & 23 & 9 \\ 2 & 2 & 2 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[851] \ 1^{142}3^{121}1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 5 & 5 & 2 \\ 1 & 2 & 0 & 10 & 8 & 2 \\ 8 & 5 & 10 & 33 & 29 & 11 \\ 6 & 5 & 8 & 29 & 23 & 9 \\ 2 & 2 & 2 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

[850] $1^{133} 3^{121} 1^{143} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 11 & 20 & 62 & 56 & 16 \\ 11 & 18 & 56 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

[849] $1^{133} 3^{121} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 9 \\ 2 & 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[855] $1^{133} 3^{121} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 3 & 4 & 5 & 23 & 5 \\ 11 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[854] $1^{111} 3^{121} 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 8 & 15 & 32 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

[857] $1^{144} 3^{121} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 15 & 2 \\ 2 & 3 & 4 & 5 & 24 & 5 \\ 11 & 9 & 15 & 24 & 83 & 17 \\ 2 & 2 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[856] $1^{144} 3^{121} 1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 5 & 10 & 14 & 36 & 8 \\ 14 & 23 & 36 & 80 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[861] $1^{133} 3^{121} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 2 & 1 & 0 & 10 & 2 \\ 10 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[860] $1^{244} 3^{121} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 4 & 5 & 2 & 24 & 4 \\ 20 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

[859] $1^{344} 3^{121} 1^{23} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 6 & 4 & 7 & 16 & 36 & 8 \\ 11 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[858] $1^{144} 3^{121} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 2 & 18 & 4 \\ 14 & 23 & 18 & 80 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[870] \quad 1^{1^{41}} 3^{1^{142}} 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 12 & 20 & 62 & 40 & 16 \\ 7 & 13 & 40 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[868] \quad 1^{1^{22}} 3^{1^{142}} 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 10 & 12 & 16 & 110 & 64 & 20 \\ 7 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[867] \quad 1^{1^{21}} 3^{1^{122}} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 6 & 10 & 14 & 20 & 8 \\ 7 & 13 & 20 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[866] \quad 1^{1^{22}} 3^{1^{123}} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 6 & 10 & 14 & 34 & 8 \\ 13 & 23 & 34 & 78 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[865] \quad 1^{1^{14}} 3^{1^{112}} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 2 & 1 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 9 & 35 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[864] \quad 1^{1^{12}} 3^{1^{112}} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 2 & 1 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 9 & 35 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[863] \quad 1^{1^{32}} 3^{1^{122}} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 10 & 9 & 2 \\ 9 & 10 & 10 & 58 & 46 & 14 \\ 7 & 7 & 9 & 46 & 35 & 11 \\ 2 & 2 & 2 & 14 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[862] \quad 1^{1^{11}} 3^{1^{112}} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 3 & 5 & 2 & 10 & 4 \\ 7 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[869] \quad 1^{1^{31}} 3^{1^{132}} 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 9 & 15 & 34 & 30 & 12 \\ 7 & 13 & 30 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[871] \quad 1^{1^{42}} 3^{1^{141}} 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 11 & 10 & 15 & 88 & 32 & 18 \\ 5 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[873] $1^{1^{24}}3^1 21^{134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 5 & 15 & 23 & 5 \\ 8 & 15 & 34 & 54 & 12 \\ 14 & 23 & 54 & 80 & 18 \\ 3 & 5 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[872] $1^{1^{42}}3^1 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 10 & 15 & 34 & 30 & 12 \\ 8 & 13 & 30 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

[874] $1^{1^{13}}3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 2 & 1 & 0 & 7 & 2 \\ 8 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[875] $1^{1^4}3^{12} 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 2 & 21 & 4 \\ 17 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[876] $1^{1^3}3^{12} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 1 & 0 & 7 & 2 \\ 8 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[877] $1^{1^{21}}3^{12} 21^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 9 & 2 & 2 \\ 1 & 2 & 0 & 12 & 4 & 2 \\ 8 & 9 & 12 & 61 & 18 & 15 \\ 3 & 2 & 4 & 18 & 4 & 4 \\ 2 & 2 & 2 & 15 & 4 & 3 \end{pmatrix}_{b_0}$$

[878] $1^{1^{33}}3^{12} 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 5 & 5 & 13 & 23 & 7 \\ 8 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[879] $1^{1^{13}}3^{12} 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 5 & 5 & 13 & 23 & 7 \\ 8 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[882] $1^{1^{12}}3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 23 & 0 & 78 & 18 \\ 3 & 5 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[880] $1^{1^{12}}3^{12} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 6 & 8 & 9 & 40 & 39 & 12 \\ 7 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[881] $1^{1^{14}}3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 23 & 0 & 80 & 18 \\ 3 & 5 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[888] \ 1^{1^{32}} 3^{1^{21}} 21^{1^{21}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 6 & 4 & 7 & 16 & 14 & 8 \\ 5 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[890] \ 1^{1^{31}} 3^{1^{21}} 21^{1^{21}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 13 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[883] \ 1^{1^{21}} 3^{1^{21}} 21^{1^{21}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 5 & 4 & 7 & 16 & 14 & 8 \\ 4 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[884] \ 1^{1^{32}} 3^{1^{21}} 21^{1^{22}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 10 & 15 & 32 & 29 & 12 \\ 8 & 13 & 29 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[885] \ 1^{1^{41}} 3^{1^{21}} 21^{1^{41}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 14 & 5 & 2 \\ 10 & 8 & 14 & 70 & 28 & 16 \\ 4 & 4 & 5 & 28 & 10 & 6 \\ 2 & 2 & 2 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[886] \ 1^{1^{41}} 3^{1^{21}} 21^{1^{21}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 9 & 10 & 10 & 58 & 26 & 14 \\ 5 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[887] \ 1^{1^{33}} 3^{1^{21}} 21^{1^{23}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 5 & 10 & 18 & 5 \\ 7 & 10 & 14 & 28 & 8 \\ 11 & 18 & 28 & 48 & 14 \\ 3 & 5 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[889] \ 1^{1^{32}} 3^{1^{21}} 21^{1^{32}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 9 & 5 & 2 \\ 1 & 2 & 0 & 12 & 8 & 2 \\ 8 & 9 & 12 & 61 & 39 & 15 \\ 6 & 5 & 8 & 39 & 23 & 9 \\ 2 & 2 & 2 & 15 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[893] \ 1^{1^{13}} 3^{1^{21}} 21^{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 2 & 1 & 3 & 1 & 9 & 3 \\ 8 & 7 & 9 & 9 & 35 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[892] \ 1^{1^{24}} 3^{1^{21}} 21^{1^{34}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 13 & 2 \\ 1 & 0 & 2 & 7 & 13 & 2 \\ 1 & 2 & 0 & 11 & 17 & 2 \\ 7 & 7 & 11 & 45 & 77 & 13 \\ 13 & 13 & 17 & 77 & 127 & 21 \\ 2 & 2 & 2 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$

[891] $1^{121} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 2 \\ 1 & 2 & 0 & 8 & 4 & 2 \\ 6 & 6 & 8 & 26 & 12 & 10 \\ 3 & 2 & 4 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[895] $1^{141} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 2 \\ 1 & 2 & 0 & 8 & 4 & 2 \\ 8 & 6 & 8 & 26 & 12 & 10 \\ 3 & 2 & 4 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[894] $1^{143} 3^{121} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 8 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 10 & 9 & 2 \\ 10 & 10 & 10 & 58 & 46 & 14 \\ 8 & 7 & 9 & 46 & 35 & 11 \\ 2 & 2 & 2 & 14 & 11 & 3 \end{pmatrix}_{b_0}$$

[898] $1^{111} 3^{121} 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 5 \\ 4 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[897] $1^{112} 3^{121} 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 3 & 5 & 2 & 10 & 4 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[896] $1^{112} 3^{121} 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 5 \\ 5 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[900] $1^{113} 3^{121} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 23 & 0 & 78 & 18 \\ 3 & 5 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[899] $1^{123} 3^{121} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 8 & 9 & 40 & 39 & 12 \\ 8 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[901] $1^{113} 3^{121} 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 5 & 10 & 14 & 42 & 8 \\ 16 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

[904] $1^{134} 3^{121} 21^{144} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 5 & 20 & 23 & 5 \\ 11 & 20 & 62 & 72 & 16 \\ 14 & 23 & 72 & 80 & 18 \\ 3 & 5 & 16 & 18 & 4 \end{pmatrix}_{b_0}$$

[903] $1^{13}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 3 & 8 & 2 \\ 2 & 1 & 3 & 1 & 8 & 3 \\ 7 & 5 & 8 & 8 & 23 & 9 \\ 2 & 2 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[902] $1^{12}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 3 & 8 & 2 \\ 2 & 1 & 3 & 1 & 8 & 3 \\ 6 & 5 & 8 & 8 & 23 & 9 \\ 2 & 2 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[906] $1^{21}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[905] $1^{41}3^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[908] $1^{12}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 2 & 17 & 4 \\ 13 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[907] $1^{12}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 1 & 0 & 7 & 2 \\ 7 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[909] $1^{14}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 2 & 1 & 0 & 7 & 2 \\ 9 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[910] $1^{41}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 8 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[913] $1^{22}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 6 & 8 & 26 & 18 & 10 \\ 5 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[914] $1^{23}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 6 & 10 & 14 & 34 & 8 \\ 14 & 23 & 34 & 78 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[911] \ 1^{124} 3^{121} 11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 90, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 4 & 5 & 2 & 18 & 4 \\ 14 & 23 & 18 & 80 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[912] \ 1^{122} 3^{121} 12^{22} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 90, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5 \\ T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 6 & 10 & 14 & 20 & 8 \\ 8 & 13 & 20 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[917] \ 1^{123} 3^{121} 12^{22} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 5 & 4 & 7 & 16 & 25 & 8 \\ 8 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[918] \ 1^{121} 3^{121} 14^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 13 & 25 & 92 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[919] \ 1^{131} 3^{121} 12^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 10 & 15 & 32 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[915] \ 1^{123} 3^{121} 13^{33} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5 \\ T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 7 & 7 & 11 & 45 & 59 & 13 \\ 10 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[916] \ 1^{133} 3^{121} 12^{22} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0 \\ T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 6 & 4 & 7 & 16 & 25 & 8 \\ 8 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[920] \ 21^{122} 3^{121} 12^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5 \\ T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 7 & 2 \\ 2 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 5 & 5 & 13 & 23 & 7 \\ 7 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[922] \ 1^{13} 3^{121} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5 \\ T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 15 & 25 & 94 & 30 & 20 \\ 4 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[923] \ 1^{133} 3^{121} 13^{32} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 8 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 10 & 11 & 13 & 81 & 55 & 17 \\ 8 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[924] $1^{131}3^{121}21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 2 \\ 1 & 2 & 0 & 8 & 4 & 2 \\ 7 & 6 & 8 & 26 & 12 & 10 \\ 3 & 2 & 4 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[921] $1^{13}3^{121}21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 58 & 24 & 16 \\ 4 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[925] $1^{111}3^{121}21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 8 & 15 & 32 & 29 & 12 \\ 7 & 13 & 29 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

[926] $1^{143}3^{121}21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 7 & 4 & 7 & 16 & 21 & 8 \\ 7 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[927] $1^{143}3^{121}21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 5 & 15 & 18 & 5 \\ 10 & 15 & 34 & 42 & 12 \\ 11 & 18 & 42 & 48 & 14 \\ 3 & 5 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[928] $1^{112}3^{121}21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 4 & 4 & 7 & 16 & 21 & 8 \\ 6 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[929] $1^{112}3^{121}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 2 & 13 & 4 \\ 11 & 18 & 13 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[930] $1^{124}3^{121}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 4 & 5 & 2 & 17 & 4 \\ 15 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[931] $1^{114}3^{121}21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 17 & 2 \\ 2 & 2 & 1 & 0 & 13 & 2 \\ 13 & 13 & 17 & 13 & 127 & 21 \\ 2 & 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[932] $1^{112}3^{121}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 2 & 1 & 0 & 9 & 2 \\ 8 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[933] $1^{121}3^{121}21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 9 & 15 & 32 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[934] \quad 1^{124} 3^{121} 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 4 & 5 & 2 & 21 & 4 \\ 17 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[935] \quad 1^{123} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 5 & 6 & 8 & 26 & 40 & 10 \\ 9 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[936] \quad 1^{123} 3^{121} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[937] \quad 1^{123} 3^{121} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[944] \quad 1^{132} 3^{121} 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 9 & 12 & 61 & 27 & 15 \\ 5 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[943] \quad 1^{132} 3^{121} 21^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 8 & 3 \\ 3 & 5 & 30 & 13 & 5 \\ 17 & 30 & 134 & 58 & 24 \\ 8 & 13 & 58 & 24 & 10 \\ 3 & 5 & 24 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[942] \quad 1^{121} 3^{121} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 5 & 5 & 13 & 13 & 7 \\ 4 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[941] \quad 1^{123} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 5 & 5 & 13 & 28 & 7 \\ 10 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[940] \quad 1^{132} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 9 & 15 & 34 & 30 & 12 \\ 8 & 13 & 30 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[939] \quad 1^{123} 3^{121} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 6 & 5 & 5 & 13 & 28 & 7 \\ 9 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[938] $1^{1^{32}} 1^{121} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 6 & 5 & 5 & 13 & 13 & 7 \\ 5 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[948] $1^{1^{22}} 1^{11} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 9 & 15 & 32 & 29 & 12 \\ 8 & 13 & 29 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

[946] $1^{1^{22}} 1^{11} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 11 & 20 & 58 & 39 & 16 \\ 8 & 13 & 39 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

[949] $1^{1^{22}} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 5 \\ 5 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[950] $1^{1^{22}} 1^{121} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 5 & 5 & 13 & 13 & 7 \\ 5 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[947] $1^{1^{21}} 1^{121} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 5 & 5 & 13 & 13 & 7 \\ 4 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[945] $1^{1^{11}} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 5 \\ 4 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[952] $1^{1^{33}} 1^{122} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 0 & 2 & 7 & 7 & 2 \\ 1 & 2 & 0 & 11 & 9 & 2 \\ 8 & 7 & 11 & 45 & 41 & 13 \\ 8 & 7 & 9 & 41 & 35 & 11 \\ 2 & 2 & 2 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[951] $1^{1^{22}} 1^{122} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 7 & 7 & 2 \\ 1 & 2 & 0 & 11 & 9 & 2 \\ 7 & 7 & 11 & 45 & 41 & 13 \\ 7 & 7 & 9 & 41 & 35 & 11 \\ 2 & 2 & 2 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[953] $1^{1^{34}} 1^1 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 7 & 10 & 14 & 36 & 8 \\ 14 & 23 & 36 & 80 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[957] $1^{1^{13}}3^{1^{12}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 13 & 5 \\ 7 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[956] $1^{1^{22}}3^{1^{12}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 13 & 5 \\ 6 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[955] $21^{1^{11}}3^{1^{11}}21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 4 & 2 \\ 2 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 5 \\ 4 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[954] $1^{1^{13}}3^{1^{12}}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 28 & 0 & 120 & 22 \\ 3 & 5 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[958] $1^{1^{13}}3^{1^{12}}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[959] $1^{1^{34}}3^{1^{12}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 5 & 3 & 4 & 5 & 16 & 5 \\ 9 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[961] $1^{1^{34}}3^{1^{12}}21^{1^{22}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 6 & 4 & 7 & 16 & 25 & 8 \\ 9 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[962] $1^{1^{22}}3^{1^{12}}21^{1^{22}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 5 & 4 & 7 & 16 & 25 & 8 \\ 7 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[960] $1^{1^2}3^{1^{12}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 3 & 4 & 5 & 16 & 5 \\ 7 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[963] $1^{1^3}3^{1^{12}}21^{1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 2 & 24 & 4 \\ 19 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

[964] $1^{113}3^{14}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 11 & 2 \\ 1 & 2 & 0 & 1 & 16 & 2 \\ 1 & 2 & 1 & 0 & 11 & 2 \\ 12 & 11 & 16 & 11 & 103 & 19 \\ 2 & 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[965] $1^{123}3^{121}1^{133} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 0 & 2 & 9 & 8 & 2 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 8 & 9 & 12 & 61 & 60 & 15 \\ 9 & 8 & 12 & 60 & 56 & 14 \\ 2 & 2 & 2 & 15 & 14 & 3 \end{pmatrix}_{b_0}$

[966] $1^{112}3^{121}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 2 & 3 & 4 & 5 & 20 & 5 \\ 9 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$

[967] $1^{113}3^{121}1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 28 & 0 & 116 & 22 \\ 3 & 5 & 0 & 22 & 4 \end{pmatrix}_{b_0}$

[968] $1^{113}3^{121}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 2 & 1 & 0 & 10 & 2 \\ 11 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[969] $1^{114}3^{121}1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 28 & 0 & 116 & 22 \\ 3 & 5 & 0 & 22 & 4 \end{pmatrix}_{b_0}$

[970] $1^{121}3^{121}1^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 11 & 20 & 58 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$

[971] $1^{114}3^{121}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 3 & 4 & 5 & 26 & 5 \\ 12 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$

[972] $1^{113}3^{121}1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 2 & 11 & 2 \\ 1 & 2 & 0 & 1 & 16 & 2 \\ 1 & 2 & 1 & 0 & 11 & 2 \\ 11 & 11 & 16 & 11 & 103 & 19 \\ 2 & 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[975] $1^{142}3^{121}1^{142} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 12 & 20 & 62 & 40 & 16 \\ 8 & 13 & 40 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

[973] $1^{142}3^{121}21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 12 & 12 & 16 & 110 & 36 & 20 \\ 5 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[974] $1^{111}3^{121}21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 5 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[976] $21^{121}3^{121}21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 4 & 2 \\ 2 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 5 & 5 & 13 & 13 & 7 \\ 4 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[977] $1^{122}3^{121}21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 5 & 10 & 18 & 5 \\ 6 & 10 & 14 & 28 & 8 \\ 10 & 18 & 28 & 48 & 14 \\ 3 & 5 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

[978] $1^{142}3^{121}21^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 12 & 20 & 62 & 56 & 16 \\ 10 & 18 & 56 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

[979] $1^{122}3^{121}21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 10 & 6 & 2 \\ 1 & 2 & 0 & 10 & 6 & 2 \\ 8 & 10 & 10 & 58 & 36 & 14 \\ 6 & 6 & 6 & 36 & 20 & 8 \\ 2 & 2 & 2 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

[980] $1^{122}3^{121}21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 5 & 6 & 2 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 6 & 5 & 5 & 13 & 18 & 7 \\ 6 & 6 & 6 & 18 & 20 & 8 \\ 2 & 2 & 2 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[985] $21^{114}3^{121}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 12 & 2 \\ 2 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 2 & 1 & 0 & 12 & 2 \\ 12 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[981] $1^{112}3^{121}21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 2 & 14 & 4 \\ 10 & 18 & 14 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[982] $1^{112}3^{121}21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 5 & 6 & 2 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 5 & 5 & 13 & 18 & 7 \\ 6 & 6 & 6 & 18 & 20 & 8 \\ 2 & 2 & 2 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[983] $1^{1^{32}}3^121^{133} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 5 & 15 & 18 & 5 \\ 9 & 15 & 34 & 42 & 12 \\ 10 & 18 & 42 & 48 & 14 \\ 3 & 5 & 12 & 14 & 4 \end{pmatrix}_{b_0}$

[984] $1^{1^4}3^121^{141} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 15 & 25 & 92 & 30 & 20 \\ 4 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$

[986] $1^{1^{32}}3^121^{142} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 6 & 2 \\ 1 & 0 & 2 & 12 & 5 & 2 \\ 1 & 2 & 0 & 16 & 8 & 2 \\ 11 & 12 & 16 & 110 & 52 & 20 \\ 6 & 5 & 8 & 52 & 23 & 9 \\ 2 & 2 & 2 & 20 & 9 & 3 \end{pmatrix}_{b_0}$

[987] $1^{1^{43}}3^121^{133} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 11 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 13 & 20 & 58 & 54 & 16 \\ 11 & 18 & 54 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$

[988] $1^{1^{43}}3^121^{132} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 5 & 5 & 2 \\ 1 & 2 & 0 & 10 & 8 & 2 \\ 8 & 5 & 10 & 33 & 29 & 11 \\ 7 & 5 & 8 & 29 & 23 & 9 \\ 2 & 2 & 2 & 11 & 9 & 3 \end{pmatrix}_{b_0}$

[989] $1^{1^{32}}3^121^{132} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 5 & 5 & 2 \\ 1 & 2 & 0 & 10 & 8 & 2 \\ 7 & 5 & 10 & 33 & 29 & 11 \\ 6 & 5 & 8 & 29 & 23 & 9 \\ 2 & 2 & 2 & 11 & 9 & 3 \end{pmatrix}_{b_0}$

[990] $1^{1^{21}}3^121^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 7 & 7 & 11 & 45 & 23 & 13 \\ 4 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$

[991] $1^{1^{43}}3^121^{142} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 8 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 15 & 9 & 2 \\ 11 & 10 & 15 & 88 & 57 & 18 \\ 8 & 7 & 9 & 57 & 35 & 11 \\ 2 & 2 & 2 & 18 & 11 & 3 \end{pmatrix}_{b_0}$

[992] $1^{1^{34}}3^121^{124} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 6 & 4 & 7 & 16 & 43 & 8 \\ 12 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$

[995] $1^{1^{11}}3^121^{121} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 6 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$

[994] $1^{1^{21}} 1^{13} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 9 & 10 & 15 & 88 & 32 & 18 \\ 4 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[993] $1^{1^{12}} 1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 4 & 4 & 7 & 16 & 36 & 8 \\ 9 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[996] $1^{1^{44}} 1^3 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 5 & 15 & 23 & 5 \\ 10 & 15 & 34 & 54 & 12 \\ 14 & 23 & 54 & 80 & 18 \\ 3 & 5 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[997] $1^{1^3} 1^{13} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 2 & 21 & 4 \\ 16 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[1000] $1^{1^{13}} 1^{13} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 2 & 14 & 4 \\ 11 & 18 & 14 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[999] $1^{1^{23}} 1^{111} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 5 & 4 & 7 & 16 & 21 & 8 \\ 7 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[998] $1^{1^{32}} 1^{111} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 6 & 4 & 7 & 16 & 21 & 8 \\ 6 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[1001] $1^{1^{13}} 1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 2 & 1 & 0 & 9 & 2 \\ 9 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1002] $1^{1^4} 1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 1 & 0 & 9 & 2 \\ 10 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1003] $1^{1^{22}} 1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 5 & 5 & 13 & 23 & 7 \\ 7 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[1004] $1^{1^{24}}3^{1^{12}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 5 & 5 & 13 & 23 & 7 \\ 9 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[1005] $1^{1^{31}}3^{1^{13}}21^{1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 9 & 2 & 2 \\ 1 & 2 & 0 & 12 & 4 & 2 \\ 9 & 9 & 12 & 61 & 18 & 15 \\ 3 & 2 & 4 & 18 & 4 & 4 \\ 2 & 2 & 2 & 15 & 4 & 3 \end{pmatrix}_{b_0}$$

[1006] $1^{1^{13}}3^{1^{13}}21^{1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 1 & 10 & 2 \\ 1 & 2 & 0 & 3 & 13 & 3 \\ 2 & 1 & 3 & 1 & 13 & 3 \\ 10 & 10 & 13 & 13 & 74 & 16 \\ 2 & 2 & 2 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[1007] $1^{1^{33}}3^{1^{13}}21^{1^{32}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 8 & 7 & 11 & 45 & 34 & 13 \\ 7 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

[1008] $1^{1^{31}}3^{1^{14}}21^{1^{41}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 3 & 2 \\ 1 & 0 & 2 & 12 & 2 & 2 \\ 1 & 2 & 0 & 16 & 4 & 2 \\ 11 & 12 & 16 & 110 & 24 & 20 \\ 3 & 2 & 2 & 24 & 4 & 4 \\ 2 & 2 & 2 & 20 & 4 & 3 \end{pmatrix}_{b_0}$$

[1009] $1^{1^{12}}3^{1^{13}}21^{1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 \\ 8 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[1010] $1^{1^{21}}3^{1^{12}}21^{1^{31}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 145$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 14 & 25 & 94 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[1012] $1^{1^{12}}3^{1^{11}}21^{1^{24}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 5 & 10 & 14 & 42 & 8 \\ 15 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

[1013] $1^{1^{12}}3^{1^{11}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 13 & 5 \\ 6 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[1011] $1^{1^{23}}3^{1^{11}}21^{1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 13 & 5 \\ 7 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1014] \quad & 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 5 & 10 & 18 & 5 \\ 6 & 10 & 14 & 28 & 8 \\ 11 & 18 & 28 & 48 & 14 \\ 3 & 5 & 8 & 14 & 4 \end{pmatrix}_{b_0} \\
[1015] \quad & 1^{121} 3^1 21^{142} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 8 & 5 & 2 \\ 1 & 2 & 0 & 14 & 8 & 2 \\ 10 & 8 & 14 & 70 & 42 & 16 \\ 7 & 5 & 8 & 42 & 23 & 9 \\ 2 & 2 & 2 & 16 & 9 & 3 \end{pmatrix}_{b_0} \\
[1016] \quad & 1^{133} 3^1 21^{144} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 0 & 2 & 8 & 11 & 2 \\ 1 & 2 & 0 & 14 & 16 & 2 \\ 9 & 8 & 14 & 70 & 86 & 16 \\ 11 & 11 & 16 & 86 & 103 & 19 \\ 2 & 2 & 2 & 16 & 19 & 3 \end{pmatrix}_{b_0} \\
[1017] \quad & 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 5 & 6 & 8 & 26 & 38 & 10 \\ 8 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0} \\
[1018] \quad & 1^{12} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 3 & 4 & 5 & 19 & 5 \\ 8 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0} \\
[1019] \quad & 1^{133} 3^1 21^{143} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 11 & 3 \\ 3 & 5 & 25 & 18 & 5 \\ 14 & 25 & 92 & 68 & 20 \\ 11 & 18 & 68 & 48 & 14 \\ 3 & 5 & 20 & 14 & 4 \end{pmatrix}_{b_0} \\
[1020] \quad & 21^{123} 3^1 21^{134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 11 & 2 \\ 2 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 7 & 7 & 11 & 45 & 70 & 13 \\ 11 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0} \\
[1021] \quad & 1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 11 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0} \\
[1022] \quad & 1^{114} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 1 & 0 & 10 & 2 \\ 11 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1023] \quad & 1^{114} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 2 & 1 & 0 & 12 & 2 \\ 11 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1024] \quad & 1^{122} 3^1 21^{143} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5
\end{aligned}$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 0 & 2 & 8 & 8 & 2 \\ 1 & 2 & 0 & 14 & 12 & 2 \\ 9 & 8 & 14 & 70 & 64 & 16 \\ 9 & 8 & 12 & 64 & 56 & 14 \\ 2 & 2 & 2 & 16 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1025] \ 1^{1^{33}} 3^{1^{12}} 21^{1^{23}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 6 & 4 & 7 & 16 & 32 & 8 \\ 9 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1026] \ 1^{1^{13}} 3^{1^{12}} 21^{1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 14 \\ 2 & 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1027] \ 1^{1^{33}} 3^{1^{42}} 21^{1^{43}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 9 & 2 \\ 1 & 0 & 2 & 12 & 8 & 2 \\ 1 & 2 & 0 & 16 & 12 & 2 \\ 11 & 12 & 16 & 110 & 80 & 20 \\ 9 & 8 & 12 & 80 & 56 & 14 \\ 2 & 2 & 2 & 20 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1028] \ 1^{1^{13}} 3^{1^{12}} 21^{1^{23}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 4 & 4 & 7 & 16 & 32 & 8 \\ 9 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1030] \ 1^{1^{14}} 3^{1^{12}} 21^{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 3 & 4 & 5 & 16 & 5 \\ 9 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1029] \ 1^{1^{22}} 3^{1^{12}} 21^{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 3 & 4 & 5 & 16 & 5 \\ 7 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1034] \ 1^{1^{14}} 3^{1^{13}} 21^{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1035] \ 1^{1^{44}} 3^{1^{23}} 21^{1^{44}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 12 & 2 \\ 1 & 0 & 2 & 8 & 11 & 2 \\ 1 & 2 & 0 & 14 & 16 & 2 \\ 10 & 8 & 14 & 70 & 86 & 16 \\ 12 & 11 & 16 & 86 & 103 & 19 \\ 2 & 2 & 2 & 16 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1031] \ 1^{1^{21}} 3^{1^{12}} 21^{1^{41}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 5 & 30 & 8 & 5 \\ 16 & 30 & 134 & 36 & 24 \\ 5 & 8 & 36 & 8 & 6 \\ 3 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1032] \ 1^{1^{41}} 3^{1^{11}} 21^{1^{31}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 13 & 20 & 58 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1033] \ 1^{13} 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1036] \ 1^{13} 3^{12} 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 5 & 0 & 33 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 33 & 0 & 164 & 26 \\ 3 & 5 & 0 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1037] \ 1^{13} 3^{12} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 1 & 10 & 2 \\ 1 & 2 & 0 & 3 & 13 & 2 \\ 2 & 1 & 3 & 1 & 13 & 3 \\ 11 & 10 & 13 & 13 & 74 & 16 \\ 2 & 2 & 2 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1038] \ 1^{121} 3^{12} 21^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 9 & 7 & 11 & 45 & 23 & 13 \\ 4 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1039] \ 1^{11} 3^{12} 21^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 13 & 20 & 58 & 39 & 16 \\ 8 & 13 & 39 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1040] \ 1^{121} 3^{12} 21^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 9 & 7 & 11 & 45 & 23 & 13 \\ 5 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1043] \ 1^{122} 3^{12} 21^{143} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 0 & 2 & 8 & 8 & 2 \\ 1 & 2 & 0 & 14 & 12 & 2 \\ 10 & 8 & 14 & 70 & 64 & 16 \\ 10 & 8 & 12 & 64 & 56 & 14 \\ 2 & 2 & 2 & 16 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1041] \ 1^{123} 3^{12} 21^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 5 & 30 & 8 & 5 \\ 18 & 30 & 134 & 36 & 24 \\ 4 & 8 & 36 & 8 & 6 \\ 3 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1042] \ 1^{123} 3^{12} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 14 \\ 2 & 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1046] \ 1^{123} 3^{12} 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 33 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 33 & 0 & 164 & 26 \\ 3 & 5 & 0 & 26 & 4 \end{pmatrix}_{b_0}$$

[1044] $1^{133}3^121^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 5 & 15 & 18 & 5 \\ 9 & 15 & 34 & 42 & 12 \\ 11 & 18 & 42 & 48 & 14 \\ 3 & 5 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[1045] $1^{113}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 1 & 0 & 9 & 2 \\ 9 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1047] $1^{114}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 2 & 1 & 0 & 9 & 2 \\ 10 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1048] $1^{131}3^121^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 58 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[1049] $1^{122}3^121^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 5 & 5 & 13 & 23 & 7 \\ 9 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[1050] $1^{122}3^121^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 5 & 5 & 13 & 23 & 7 \\ 7 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[1053] $1^{113}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 1 & 0 & 10 & 2 \\ 10 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1051] $1^{12}3^121^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 2 & 24 & 4 \\ 18 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

[1052] $1^{113}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 1 & 0 & 10 & 2 \\ 9 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1055] $1^{12}3^121^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 \\ 9 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1054] \quad 1^{11} 3^{14} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 17 & 4 \\ 15 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1056] \quad 1^{124} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 5 & 4 & 7 & 16 & 36 & 8 \\ 11 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1057] \quad 1^{121} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 11 & 20 & 58 & 39 & 16 \\ 7 & 13 & 39 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1058] \quad 1^{141} 3^{121} 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 9 & 2 & 2 \\ 1 & 2 & 0 & 12 & 4 & 2 \\ 10 & 9 & 12 & 61 & 18 & 15 \\ 3 & 2 & 4 & 18 & 4 & 4 \\ 2 & 2 & 2 & 15 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1059] \quad 1^{123} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 5 & 4 & 7 & 16 & 36 & 8 \\ 10 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1062] \quad 21^{121} 3^{121} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 4 & 2 \\ 2 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1061] \quad 1^{124} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 \\ 10 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1060] \quad 1^{123} 3^{121} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 6 & 10 & 14 & 42 & 8 \\ 16 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1065] \quad 1^{133} 3^{121} 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 12 & 20 & 58 & 54 & 16 \\ 11 & 18 & 54 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1063] \quad 1^{132} 3^{121} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 7 & 6 & 8 & 26 & 26 & 10 \\ 6 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[1064] $1^{123}3^{121}21^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 7 & 6 & 8 & 26 & 26 & 10 \\ 7 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[1066] $1^{123}3^{121}21^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 5 & 5 & 13 & 33 & 7 \\ 10 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[1067] $1^{113}4^{121}21^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 4 & 13 & 2 \\ 1 & 2 & 0 & 7 & 17 & 2 \\ 4 & 4 & 7 & 16 & 47 & 8 \\ 12 & 13 & 17 & 47 & 127 & 21 \\ 2 & 2 & 2 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

[1070] $1^{122}3^{121}21^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 6 & 8 & 26 & 32 & 10 \\ 7 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1068] $1^{113}3^{121}21^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 17 & 4 \\ 13 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[1069] $1^{123}3^{121}21^{142} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 7 & 3 \\ 3 & 5 & 30 & 13 & 5 \\ 16 & 30 & 134 & 58 & 24 \\ 7 & 13 & 58 & 24 & 10 \\ 3 & 5 & 24 & 10 & 4 \end{pmatrix}_{b_0}$$

[1077] $1^{113}3^{121}21^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 12 & 20 & 58 & 39 & 16 \\ 8 & 13 & 39 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

[1071] $1^{113}3^{121}21^{142} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 14 & 25 & 92 & 49 & 20 \\ 8 & 13 & 49 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

[1075] $1^{121}3^{121}21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 7 & 6 & 8 & 26 & 18 & 10 \\ 5 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[1074] $1^{121}3^{121}21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 7 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[1073] $1^{122} 3^{121} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 6 & 6 & 8 & 26 & 26 & 10 \\ 6 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[1072] $1^{143} 3^{121} 21^{143} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 12 & 20 & 62 & 56 & 16 \\ 11 & 18 & 56 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

[1078] $1^{143} 3^{121} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 8 & 6 & 8 & 26 & 26 & 10 \\ 7 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[1079] $1^{141} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 9 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[1076] $1^{132} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 8 & 9 & 40 & 22 & 12 \\ 5 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[1080] $1^{131} 3^{121} 21^{141} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 14 & 25 & 92 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[1082] $1^{141} 3^{121} 21^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 16 & 25 & 94 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[1081] $1^{131} 3^{121} 21^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 8 & 7 & 11 & 45 & 23 & 13 \\ 4 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

[1083] $1^{121} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 8 & 10 & 10 & 58 & 26 & 14 \\ 4 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[1085] $1^{122} 3^{121} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 5 & 4 & 7 & 16 & 21 & 8 \\ 6 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[1084] $1^{113}3^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 6 & 4 & 7 & 16 & 21 & 8 \\ 7 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[1086] $1^{123}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 5 & 5 & 13 & 23 & 7 \\ 8 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[1088] $1^{123}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 15 & 25 & 94 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[1087] $1^{123}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 8 & 7 & 11 & 45 & 59 & 13 \\ 11 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[1089] $1^{113}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 17 & 4 \\ 14 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[1090] $1^{122}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 5 & 6 & 2 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 5 & 5 & 5 & 13 & 18 & 7 \\ 6 & 6 & 6 & 18 & 20 & 8 \\ 2 & 2 & 2 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[1091] $1^{113}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 5 & 25 & 18 & 5 \\ 13 & 25 & 92 & 68 & 20 \\ 10 & 18 & 68 & 48 & 14 \\ 3 & 5 & 20 & 14 & 4 \end{pmatrix}_{b_0}$$

[1098] $1^{123}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 5 & 6 & 8 & 26 & 54 & 10 \\ 12 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1099] $1^{123}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 8 & 3 \\ 3 & 5 & 30 & 13 & 5 \\ 18 & 30 & 134 & 58 & 24 \\ 8 & 13 & 58 & 24 & 10 \\ 3 & 5 & 24 & 10 & 4 \end{pmatrix}_{b_0}$$

[1097] $1^{141}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 10 & 10 & 10 & 58 & 26 & 14 \\ 5 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[1096] $1^{141} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 10 & 10 & 10 & 58 & 26 & 14 \\ 4 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[1095] $1^{123} 3^{121} 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 8 & 6 & 8 & 26 & 46 & 10 \\ 11 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1093] $1^{124} 3^{121} 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 6 & 10 & 14 & 42 & 8 \\ 17 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

[1092] $1^{121} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[1094] $1^{122} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 8 & 6 & 8 & 26 & 18 & 10 \\ 5 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[1100] $1^{122} 3^{121} 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 7 & 7 & 11 & 45 & 34 & 13 \\ 6 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

[1101] $1^{113} 3^{121} 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 1 & 13 & 2 \\ 1 & 2 & 0 & 3 & 17 & 2 \\ 2 & 1 & 3 & 1 & 17 & 3 \\ 12 & 13 & 17 & 17 & 127 & 21 \\ 2 & 2 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[1102] $1^{113} 3^{121} 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 18 & 4 \\ 13 & 23 & 18 & 80 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[1103] $1^{114} 3^{121} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1104] $1^{133} 3^{121} 21^{134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 5 & 15 & 23 & 5 \\ 9 & 15 & 34 & 54 & 12 \\ 13 & 23 & 54 & 80 & 18 \\ 3 & 5 & 12 & 18 & 4 \end{pmatrix}_{b_0} \\
[1105] & 1^{123} 3^{124} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 6 & 10 & 14 & 36 & 8 \\ 13 & 23 & 36 & 80 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0} \\
[1106] & 1^{124} 3^{124} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 17 \\ 2 & 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0} \\
[1107] & 1^{143} 3^{121} 21^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 5 & 20 & 23 & 5 \\ 12 & 20 & 62 & 72 & 16 \\ 13 & 23 & 72 & 80 & 18 \\ 3 & 5 & 16 & 18 & 4 \end{pmatrix}_{b_0} \\
[1108] & 1^{13} 3^{121} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0} \\
[1109] & 1^{144} 3^{121} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 2 & 1 & 0 & 12 & 2 \\ 12 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1110] & 1^{123} 3^{121} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 4 & 4 & 7 & 16 & 43 & 8 \\ 12 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0} \\
[1112] & 1^{13} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 3 & 4 & 5 & 23 & 5 \\ 10 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0} \\
[1111] & 1^{14} 3^{121} 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 18 & 4 \\ 14 & 23 & 18 & 80 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0} \\
[1113] & 1^{132} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 7 & 7 & 2 \\ 1 & 2 & 0 & 11 & 9 & 2 \\ 8 & 7 & 11 & 45 & 41 & 13 \\ 7 & 7 & 9 & 41 & 35 & 11 \\ 2 & 2 & 2 & 13 & 11 & 3 \end{pmatrix}_{b_0} \\
[1114] & 1^{123} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 4 & 3 & 4 & 5 & 20 & 5 \\ 9 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1115] \quad 1^{124} 3^{121} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 6 & 10 & 14 & 36 & 8 \\ 14 & 23 & 36 & 80 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1116] \quad 1^{143} 3^{121} 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 0 & 2 & 7 & 7 & 2 \\ 1 & 2 & 0 & 11 & 9 & 2 \\ 9 & 7 & 11 & 45 & 41 & 13 \\ 8 & 7 & 9 & 41 & 35 & 11 \\ 2 & 2 & 2 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1117] \quad 1^{14} 3^{121} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1118] \quad 1^{14} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 2 & 3 & 4 & 5 & 27 & 5 \\ 12 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1120] \quad 1^{41} 3^{121} 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 3 & 2 \\ 1 & 0 & 2 & 12 & 2 & 2 \\ 1 & 2 & 0 & 16 & 4 & 2 \\ 12 & 12 & 16 & 110 & 24 & 20 \\ 3 & 2 & 4 & 24 & 4 & 4 \\ 2 & 2 & 2 & 20 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1122] \quad 1^{124} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 6 & 8 & 26 & 40 & 10 \\ 10 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1119] \quad 1^{144} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 8 & 6 & 8 & 26 & 40 & 10 \\ 10 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1121] \quad 1^{124} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 5 & 13 & 2 \\ 1 & 2 & 0 & 5 & 17 & 2 \\ 5 & 5 & 5 & 13 & 43 & 7 \\ 13 & 13 & 17 & 43 & 127 & 21 \\ 2 & 2 & 2 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1123] \quad 1^{14} 3^{121} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 225, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 4 & 9 & 2 \\ 1 & 2 & 0 & 7 & 15 & 2 \\ 4 & 4 & 7 & 16 & 39 & 8 \\ 11 & 9 & 15 & 39 & 83 & 17 \\ 2 & 2 & 2 & 8 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1124] \quad 1^{41} 3^{121} 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 15 & 25 & 92 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1125] \quad 1^{1^{13}} 3^{1^{12}} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 3 & 4 & 5 & 16 & 5 \\ 8 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1126] \quad 1^{1^{23}} 3^{1^{21}} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 3 & 4 & 5 & 16 & 5 \\ 8 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1127] \quad 21^{1^{12}} 3^{1^{21}} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 7 & 2 \\ 2 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 3 & 4 & 5 & 16 & 5 \\ 7 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1128] \quad 1^{1^{21}} 3^{1^{21}} 21^{1^{21}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 7 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1129] \quad 1^{1^{31}} 3^{1^{21}} 21^{1^{21}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 9 & 10 & 10 & 58 & 26 & 14 \\ 4 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1130] \quad 1^{1^{44}} 3^{1^{21}} 21^{1^{22}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 8 & 6 & 8 & 26 & 38 & 10 \\ 10 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1131] \quad 1^{1^{23}} 3^{1^{21}} 21^{1^{22}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 6 & 6 & 8 & 26 & 38 & 10 \\ 9 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1132] \quad 1^{1^{34}} 3^{1^{21}} 21^{1^{34}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 5 & 15 & 23 & 5 \\ 9 & 15 & 34 & 54 & 12 \\ 14 & 23 & 54 & 80 & 18 \\ 3 & 5 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1133] \quad 1^{1^{23}} 3^{1^{21}} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 4 & 3 & 4 & 5 & 19 & 5 \\ 9 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1134] \quad 1^{1^{14}} 3^{1^{21}} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 3 & 4 & 5 & 19 & 5 \\ 10 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1135] $1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 1 & 13 & 2 \\ 1 & 2 & 0 & 3 & 17 & 2 \\ 2 & 1 & 3 & 1 & 17 & 3 \\ 13 & 13 & 17 & 17 & 127 & 21 \\ 2 & 2 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[1136] $1^{24} 3^1 21^{134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 7 & 9 & 2 \\ 1 & 2 & 0 & 11 & 15 & 2 \\ 7 & 7 & 11 & 45 & 63 & 13 \\ 11 & 9 & 15 & 63 & 83 & 17 \\ 2 & 2 & 2 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[1137] $1^{43} 3^1 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 8 & 6 & 8 & 26 & 32 & 10 \\ 8 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1138] $1^{23} 3^1 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 6 & 8 & 26 & 32 & 10 \\ 8 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1139] $21^{141} 3^1 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 4 & 2 \\ 2 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 10 & 10 & 10 & 58 & 26 & 14 \\ 4 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[1140] $1^{32} 3^1 21^{142} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 8 & 5 & 2 \\ 1 & 2 & 0 & 14 & 8 & 2 \\ 9 & 8 & 14 & 70 & 42 & 16 \\ 6 & 5 & 8 & 42 & 23 & 9 \\ 2 & 2 & 2 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

[1141] $1^{32} 3^1 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 9 & 5 & 2 \\ 1 & 2 & 0 & 12 & 8 & 2 \\ 9 & 9 & 12 & 61 & 39 & 15 \\ 6 & 5 & 8 & 39 & 23 & 9 \\ 2 & 2 & 2 & 15 & 9 & 3 \end{pmatrix}_{b_0}$$

[1143] $1^{22} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 5 & 5 & 13 & 28 & 7 \\ 8 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1144] $1^{22} 3^1 21^{133} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 7 & 7 & 11 & 45 & 59 & 13 \\ 9 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[1142] $1^{31} 3^1 21^{142} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 7 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 14 & 25 & 92 & 49 & 20 \\ 7 & 13 & 49 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1149] \quad 1^{1^{12}} 3^{1^{12}} 21^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 3 & 4 & 5 & 16 & 5 \\ 7 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1145] \quad 1^{1^{32}} 3^{1^{12}} 21^{1^{42}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 15 & 9 & 2 \\ 10 & 10 & 15 & 88 & 57 & 18 \\ 7 & 7 & 9 & 57 & 35 & 11 \\ 2 & 2 & 2 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1146] \quad 1^{1^{12}} 3^{1^{12}} 21^{1^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 2 & 21 & 4 \\ 15 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1147] \quad 1^{1^{24}} 3^{1^{12}} 21^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 3 & 4 & 5 & 16 & 5 \\ 9 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1148] \quad 1^{1^{13}} 3^{1^{12}} 21^{1^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 1 & 11 & 2 \\ 1 & 2 & 0 & 3 & 16 & 2 \\ 2 & 1 & 3 & 1 & 16 & 3 \\ 11 & 11 & 16 & 16 & 103 & 19 \\ 2 & 2 & 2 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1150] \quad 1^{1^{42}} 3^{1^{12}} 21^{1^{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 9 & 8 & 9 & 40 & 22 & 12 \\ 5 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1151] \quad 1^{1^{31}} 3^{1^{12}} 21^{1^{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1152] \quad 1^{1^{44}} 3^{1^{12}} 21^{1^{33}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 9 & 7 & 11 & 45 & 59 & 13 \\ 11 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1153] \quad 1^{1^{22}} 3^{1^{12}} 21^{1^{32}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 9 & 11 & 13 & 81 & 55 & 17 \\ 7 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1154] \quad 1^{1^{13}} 3^{1^{12}} 21^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 249, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 10 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[1155] $1^{131} 3^{12} 21^{141} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 5 & 30 & 8 & 5 \\ 17 & 30 & 134 & 36 & 24 \\ 5 & 8 & 36 & 8 & 6 \\ 3 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$

[1156] $1^{133} 3^{12} 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 9 & 9 & 12 & 61 & 48 & 15 \\ 8 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$

[1157] $1^{123} 3^{12} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 9 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$

[1158] $1^{144} 3^{12} 21^{144} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 5 & 20 & 23 & 5 \\ 12 & 20 & 62 & 72 & 16 \\ 14 & 23 & 72 & 80 & 18 \\ 3 & 5 & 16 & 18 & 4 \end{pmatrix}_{b_0}$

[1159] $21^{123} 3^{12} 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 11 & 2 \\ 2 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 6 & 8 & 26 & 54 & 10 \\ 11 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$

[1160] $1^{143} 3^{12} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 12 & 14 & 0 & 96 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1161] $21^{132} 3^{12} 21^{143} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 8 & 2 \\ 2 & 0 & 2 & 10 & 8 & 2 \\ 1 & 2 & 0 & 15 & 12 & 2 \\ 10 & 10 & 15 & 88 & 72 & 18 \\ 8 & 8 & 12 & 72 & 56 & 14 \\ 2 & 2 & 2 & 18 & 14 & 3 \end{pmatrix}_{b_0}$

[1162] $1^{131} 3^{12} 21^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 8 & 9 & 12 & 61 & 27 & 15 \\ 4 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$

[1163] $1^{122} 3^{12} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 6 & 8 & 26 & 32 & 10 \\ 9 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$

[1164] $1^{132} 3^{12} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 6 & 8 & 26 & 32 & 10 \\ 7 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1167] \ 1^{123} 3^{121} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 6 & 5 & 5 & 13 & 28 & 7 \\ 10 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1168] \ 1^{123} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 5 & 5 & 13 & 28 & 7 \\ 9 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1165] \ 1^{123} 3^{121} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 3 & 4 & 5 & 23 & 5 \\ 10 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1166] \ 1^{124} 3^{121} 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 4 & 13 & 2 \\ 1 & 2 & 0 & 7 & 17 & 2 \\ 5 & 4 & 7 & 16 & 47 & 8 \\ 13 & 13 & 17 & 47 & 127 & 21 \\ 2 & 2 & 2 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1170] \ 1^{141} 3^{121} 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 263, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 11 & 11 & 13 & 81 & 31 & 17 \\ 5 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1171] \ 1^{142} 3^{121} 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 263, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 15 & 25 & 92 & 49 & 20 \\ 8 & 13 & 49 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1169] \ 1^{131} 3^{121} 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 263, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 10 & 9 & 12 & 61 & 27 & 15 \\ 5 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1174] \ 1^{124} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 264, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 5 & 11 & 2 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 6 & 5 & 5 & 13 & 33 & 7 \\ 11 & 11 & 11 & 33 & 71 & 15 \\ 2 & 2 & 2 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1172] \ 1^{124} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 264, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 5 & 11 & 2 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 11 & 11 & 11 & 33 & 71 & 15 \\ 2 & 2 & 2 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1173] \ 1^{123} 3^{121} 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 264, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 1 & 9 & 2 \\ 1 & 2 & 0 & 3 & 15 & 2 \\ 2 & 1 & 3 & 1 & 15 & 3 \\ 11 & 9 & 15 & 15 & 83 & 17 \\ 2 & 2 & 2 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[1175] $1^{1^{33}}3^{1^{21}}1^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 8 & 7 & 11 & 45 & 59 & 13 \\ 10 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[1177] $21^{1^{32}}3^{1^{21}}1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 7 & 2 \\ 2 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 8 & 9 & 40 & 39 & 12 \\ 7 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[1176] $1^{1^{14}}3^{1^{21}}1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 3 & 4 & 5 & 26 & 5 \\ 11 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1178] $1^{1^{14}}3^{1^{21}}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 17 & 2 \\ 1 & 2 & 1 & 0 & 13 & 2 \\ 13 & 13 & 17 & 13 & 127 & 21 \\ 2 & 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1179] $1^{1^{42}}3^{1^{21}}1^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 9 & 5 & 2 \\ 1 & 2 & 0 & 12 & 8 & 2 \\ 10 & 9 & 12 & 61 & 39 & 15 \\ 6 & 5 & 8 & 39 & 23 & 9 \\ 2 & 2 & 2 & 15 & 9 & 3 \end{pmatrix}_{b_0}$$

[1181] $1^{1^{41}}3^{1^{21}}1^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 10 & 12 & 16 & 110 & 36 & 20 \\ 4 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1180] $1^{1^{13}}3^{1^{21}}1^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 6 & 8 & 26 & 46 & 10 \\ 10 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1182] $1^{1^{13}}3^{1^{21}}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 17 & 2 \\ 1 & 2 & 1 & 0 & 13 & 2 \\ 12 & 13 & 17 & 13 & 127 & 21 \\ 2 & 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1184] $1^{1^{14}}3^{1^{21}}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 2 & 1 & 0 & 12 & 2 \\ 12 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1185] $1^{1^{13}}3^{1^{21}}1^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 5 & 6 & 8 & 26 & 52 & 10 \\ 11 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[1183] $1^{143}3^{121}21^{142} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 8 & 2 \\ 1 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 12 & 12 & 16 & 110 & 64 & 20 \\ 8 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

[1186] $1^{123}3^{121}21^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 7 & 7 & 11 & 45 & 52 & 13 \\ 9 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[1187] $1^{133}3^{121}21^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 7 & 6 & 8 & 26 & 38 & 10 \\ 9 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[1188] $1^{134}3^{121}21^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 7 & 6 & 8 & 26 & 38 & 10 \\ 10 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[1189] $1^{134}3^{121}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 19 \\ 2 & 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1190] $1^{143}3^{121}21^{143} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 11 & 3 \\ 3 & 5 & 25 & 18 & 5 \\ 15 & 25 & 92 & 68 & 20 \\ 11 & 18 & 68 & 48 & 14 \\ 3 & 5 & 20 & 14 & 4 \end{pmatrix}_{b_0}$$

[1191] $1^{124}3^{121}21^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 5 & 5 & 13 & 33 & 7 \\ 11 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[1192] $1^{134}3^{121}21^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 1 & 11 & 2 \\ 1 & 2 & 0 & 3 & 16 & 2 \\ 2 & 1 & 3 & 1 & 16 & 3 \\ 12 & 11 & 16 & 16 & 103 & 19 \\ 2 & 2 & 2 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

[1193] $21^{113}3^{121}21^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 11 & 2 \\ 2 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 27 & 5 \\ 11 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1194] $1^{144}3^{121}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1195] $1^{124}3^{121}21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 5 & 4 & 7 & 16 & 32 & 8 \\ 10 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[1196] $1^{112}3^{121}21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 6 & 4 & 7 & 16 & 32 & 8 \\ 10 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[1197] $21^{131}3^{121}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 4 & 2 \\ 2 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[1198] $1^{143}3^{121}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 2 & 1 & 0 & 12 & 2 \\ 11 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1199] $1^{112}3^{121}21^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 10 & 15 & 2 \\ 7 & 5 & 10 & 33 & 54 & 11 \\ 11 & 9 & 15 & 54 & 83 & 17 \\ 2 & 2 & 2 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[1200] $1^{112}3^{121}21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 4 & 9 & 2 \\ 1 & 2 & 0 & 7 & 15 & 2 \\ 6 & 4 & 7 & 16 & 39 & 8 \\ 11 & 9 & 15 & 39 & 83 & 17 \\ 2 & 2 & 2 & 8 & 17 & 3 \end{pmatrix}_{b_0}$$

[1201] $1^{131}3^{121}21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 10 & 10 & 15 & 88 & 32 & 18 \\ 4 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[1202] $1^{113}3^{121}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 2 & 21 & 4 \\ 16 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[1203] $21^{131}3^{121}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 4 & 2 \\ 2 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 9 & 12 & 61 & 27 & 15 \\ 4 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[1205] $1^{142}3^{121}21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 8 & 2 \\ 1 & 0 & 2 & 12 & 8 & 2 \\ 1 & 2 & 0 & 16 & 12 & 2 \\ 10 & 12 & 16 & 110 & 80 & 20 \\ 8 & 8 & 12 & 80 & 56 & 14 \\ 2 & 2 & 2 & 20 & 14 & 3 \end{pmatrix}_{b_0}$$

[1206] $1^{12} 3^{12} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 5 & 6 & 8 & 26 & 40 & 10 \\ 8 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[1204] $1^{12} 3^{12} 21^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 0 & 2 & 8 & 8 & 2 \\ 1 & 2 & 0 & 14 & 12 & 2 \\ 9 & 8 & 14 & 70 & 64 & 16 \\ 8 & 8 & 12 & 64 & 56 & 14 \\ 2 & 2 & 2 & 16 & 14 & 3 \end{pmatrix}_{b_0}$$

[1208] $1^{12} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 3 & 4 & 5 & 19 & 5 \\ 8 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1209] $1^{12} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 4 & 3 & 4 & 5 & 19 & 5 \\ 10 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1207] $1^{13} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 3 & 4 & 5 & 19 & 5 \\ 9 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1211] $1^{14} 3^{12} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 8 & 9 & 40 & 39 & 12 \\ 9 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[1212] $1^{12} 3^{12} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 8 & 9 & 40 & 39 & 12 \\ 7 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[1210] $1^{14} 3^{12} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 2 & 21 & 4 \\ 17 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[1213] $21^{122} 3^{12} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 7 & 2 \\ 2 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 6 & 8 & 26 & 32 & 10 \\ 7 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1215] $1^{12} 3^{12} 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 9 & 7 & 11 & 45 & 34 & 13 \\ 6 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1214] \quad 1^{1^{43}} 3^{1^{21}} 1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 9 & 7 & 11 & 45 & 34 & 13 \\ 7 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1218] \quad 1^{1^{33}} 3^{1^{21}} 1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 7 & 6 & 8 & 26 & 40 & 10 \\ 9 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1216] \quad 1^{1^{24}} 3^{1^{21}} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 4 & 3 & 4 & 5 & 20 & 5 \\ 10 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1217] \quad 1^{1^{14}} 3^{1^{21}} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 20 & 5 \\ 10 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1219] \quad 1^{1^{22}} 3^{1^{21}} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 297, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 8 & 9 & 40 & 39 & 12 \\ 7 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1220] \quad 1^{1^{41}} 3^{1^{21}} 1^{411} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 5 & 3 \\ 3 & 5 & 30 & 8 & 5 \\ 18 & 30 & 134 & 36 & 24 \\ 5 & 8 & 36 & 8 & 6 \\ 3 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1221] \quad 1^{1^{21}} 3^{1^{21}} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 9 & 11 & 13 & 81 & 31 & 17 \\ 4 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1222] \quad 1^{1^{23}} 3^{1^{21}} 1^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 7 & 13 & 2 \\ 1 & 2 & 0 & 11 & 17 & 2 \\ 7 & 7 & 11 & 45 & 77 & 13 \\ 12 & 13 & 17 & 77 & 127 & 21 \\ 2 & 2 & 2 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1223] \quad 1^{1^{22}} 3^{1^{21}} 1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 305, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 8 & 9 & 12 & 61 & 48 & 15 \\ 7 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1225] \quad 1^{1^{43}} 3^{1^{21}} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 8 & 9 & 40 & 39 & 12 \\ 8 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1224] \quad 1^{132} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 8 & 9 & 40 & 39 & 12 \\ 8 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1227] \quad 1^{143} 3^1 21^{143} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 0 & 2 & 8 & 8 & 2 \\ 1 & 2 & 0 & 14 & 12 & 2 \\ 10 & 8 & 14 & 70 & 64 & 16 \\ 9 & 8 & 12 & 64 & 56 & 14 \\ 2 & 2 & 2 & 16 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1228] \quad 1^{113} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 3 & 5 & 2 & 24 & 4 \\ 19 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1226] \quad 1^{123} 3^1 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 5 & 4 & 7 & 16 & 32 & 8 \\ 9 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1230] \quad 1^{131} 3^1 21^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 10 & 5 & 2 \\ 1 & 2 & 0 & 15 & 8 & 2 \\ 10 & 10 & 15 & 88 & 47 & 18 \\ 6 & 5 & 8 & 47 & 23 & 9 \\ 2 & 2 & 2 & 18 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1229] \quad 1^{121} 3^1 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 8 & 7 & 11 & 45 & 34 & 13 \\ 6 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1231] \quad 1^{132} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 10 & 6 & 2 \\ 1 & 2 & 0 & 10 & 6 & 2 \\ 9 & 10 & 10 & 58 & 36 & 14 \\ 6 & 6 & 6 & 36 & 20 & 8 \\ 2 & 2 & 2 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1233] \quad 1^{112} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 5 & 5 & 13 & 28 & 7 \\ 8 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1232] \quad 1^{114} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 3 & 5 & 2 & 24 & 4 \\ 20 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1234] \quad 1^{113} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 20 & 5 \\ 9 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[1235] $1^{142} 3^{121} 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 6 & 2 \\ 1 & 0 & 2 & 12 & 5 & 2 \\ 1 & 2 & 0 & 16 & 8 & 2 \\ 12 & 12 & 16 & 110 & 52 & 20 \\ 6 & 5 & 8 & 52 & 23 & 9 \\ 2 & 2 & 2 & 20 & 9 & 3 \end{pmatrix}_{b_0}$$

[1237] $1^{122} 3^{121} 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 6 & 8 & 26 & 32 & 10 \\ 7 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1236] $1^{144} 3^{121} 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 8 & 6 & 8 & 26 & 32 & 10 \\ 9 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1238] $21^{113} 3^{121} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 9 & 2 \\ 2 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 3 & 4 & 5 & 19 & 5 \\ 9 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1239] $1^{134} 3^{121} 21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 8 & 7 & 11 & 45 & 52 & 13 \\ 10 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[1240] $1^{113} 3^{121} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 2 & 3 & 4 & 5 & 27 & 5 \\ 11 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1243] $1^{123} 3^{121} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 5 & 5 & 13 & 28 & 7 \\ 10 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1241] $1^{131} 3^{121} 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 9 & 12 & 61 & 27 & 15 \\ 4 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[1242] $1^{123} 3^{121} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 5 & 5 & 13 & 28 & 7 \\ 9 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1245] $1^{114} 3^{121} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 2 & 3 & 4 & 5 & 30 & 5 \\ 13 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[1244] $1^{141} 3^{121} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 10 & 9 & 12 & 61 & 27 & 15 \\ 4 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[1248] $1^{144} 3^{121} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 21 \\ 2 & 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1247] $1^{144} 3^{121} 1^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 10 & 2 \\ 1 & 0 & 2 & 10 & 8 & 2 \\ 1 & 2 & 0 & 15 & 12 & 2 \\ 11 & 10 & 15 & 88 & 72 & 18 \\ 10 & 8 & 12 & 72 & 56 & 14 \\ 2 & 2 & 2 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

[1246] $1^{124} 3^{121} 1^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 13 & 2 \\ 1 & 0 & 2 & 7 & 13 & 2 \\ 1 & 2 & 0 & 11 & 17 & 2 \\ 8 & 7 & 11 & 45 & 77 & 13 \\ 13 & 13 & 17 & 77 & 127 & 21 \\ 2 & 2 & 2 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$

[1249] $1^{121} 3^{121} 1^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 346, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 8 & 5 & 2 \\ 1 & 2 & 0 & 14 & 8 & 2 \\ 10 & 8 & 14 & 70 & 42 & 16 \\ 6 & 5 & 8 & 42 & 23 & 9 \\ 2 & 2 & 2 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

[1250] $1^{113} 3^{121} 1^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 4 & 4 & 7 & 16 & 43 & 8 \\ 11 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[1251] $1^{124} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 5 & 5 & 5 & 13 & 38 & 7 \\ 11 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1252] $1^{122} 3^{121} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 6 & 8 & 26 & 32 & 10 \\ 8 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1253] $1^{113} 3^{121} 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 4 & 3 & 4 & 5 & 27 & 5 \\ 12 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1254] $1^{131} 3^{121} 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 11 & 10 & 15 & 88 & 32 & 18 \\ 4 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[1255] $1^{14} 1^{12} 3^1 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 15 & 2 \\ 3 & 3 & 4 & 5 & 24 & 5 \\ 11 & 9 & 15 & 24 & 83 & 17 \\ 2 & 2 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[1256] $1^{24} 1^{12} 3^1 21^{134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 355, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 7 & 7 & 11 & 45 & 70 & 13 \\ 12 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

[1257] $1^{34} 1^{12} 3^1 21^{144} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 0 & 2 & 8 & 9 & 2 \\ 1 & 2 & 0 & 14 & 15 & 2 \\ 9 & 8 & 14 & 70 & 78 & 16 \\ 11 & 9 & 15 & 78 & 83 & 17 \\ 2 & 2 & 2 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[1259] $1^{23} 1^{12} 3^1 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 6 & 8 & 26 & 40 & 10 \\ 9 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[1258] $1^{14} 1^{24} 3^1 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 5 & 13 & 2 \\ 1 & 2 & 0 & 5 & 17 & 2 \\ 4 & 5 & 5 & 13 & 43 & 7 \\ 13 & 13 & 17 & 43 & 127 & 21 \\ 2 & 2 & 2 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[1260] $1^{43} 1^{12} 3^1 21^{133} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 9 & 7 & 11 & 45 & 52 & 13 \\ 9 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[1261] $1^{42} 1^{12} 3^1 21^{142} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 15 & 9 & 2 \\ 11 & 10 & 15 & 88 & 57 & 18 \\ 7 & 7 & 9 & 57 & 35 & 11 \\ 2 & 2 & 2 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[1262] $1^{33} 1^{12} 3^1 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 374, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 7 & 6 & 8 & 26 & 46 & 10 \\ 10 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1263] $1^{13} 1^{24} 3^1 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 5 & 13 & 2 \\ 1 & 2 & 0 & 5 & 17 & 2 \\ 4 & 5 & 5 & 13 & 43 & 7 \\ 12 & 13 & 17 & 43 & 127 & 21 \\ 2 & 2 & 2 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[1264] $1^{23} 1^{13} 3^1 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 5 & 4 & 7 & 16 & 43 & 8 \\ 11 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[1265] $1^{124} 3^{121} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 383, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 3 & 4 & 5 & 23 & 5 \\ 11 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1267] $1^{124} 3^{121} 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 7 & 6 & 8 & 26 & 40 & 10 \\ 10 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[1266] $1^{142} 3^{121} 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 10 & 6 & 2 \\ 1 & 2 & 0 & 10 & 6 & 2 \\ 10 & 10 & 10 & 58 & 36 & 14 \\ 6 & 6 & 6 & 36 & 20 & 8 \\ 2 & 2 & 2 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

[1268] $1^{124} 3^{121} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 5 & 11 & 2 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 10 & 11 & 11 & 33 & 71 & 15 \\ 2 & 2 & 2 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[1269] $1^{124} 3^{121} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 4 & 3 & 4 & 5 & 26 & 5 \\ 12 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1270] $1^{124} 3^{121} 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 5 & 6 & 8 & 26 & 60 & 10 \\ 13 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[1272] $1^{132} 3^{121} 21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 0 & 2 & 9 & 8 & 2 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 10 & 9 & 12 & 61 & 60 & 15 \\ 9 & 8 & 12 & 60 & 56 & 14 \\ 2 & 2 & 2 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[1271] $21^{132} 3^{121} 21^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 7 & 2 \\ 2 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 9 & 9 & 12 & 61 & 48 & 15 \\ 7 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[1273] $1^{124} 3^{121} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 405, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 4 & 3 & 4 & 5 & 30 & 5 \\ 13 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[1274] $21^{113} 3^{121} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 10 & 2 \\ 2 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 3 & 4 & 5 & 23 & 5 \\ 10 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1275] $1^{1^{44}} 3^{1^{23}} 21^{1^{34}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 407, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 12 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 9 & 7 & 11 & 45 & 70 & 13 \\ 12 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

[1277] $1^{1^{12}} 3^{1^{21}} 21^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 3 & 4 & 5 & 23 & 5 \\ 9 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1276] $1^{1^{31}} 3^{1^{41}} 21^{1^{31}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 10 & 11 & 13 & 81 & 31 & 17 \\ 4 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

[1278] $1^{1^{33}} 3^{1^{32}} 21^{1^{43}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 0 & 2 & 10 & 8 & 2 \\ 1 & 2 & 0 & 15 & 12 & 2 \\ 10 & 10 & 15 & 88 & 72 & 18 \\ 9 & 8 & 12 & 72 & 56 & 14 \\ 2 & 2 & 2 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

[1281] $1^{1^{33}} 3^{1^{32}} 21^{1^{33}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 0 & 2 & 9 & 8 & 2 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 9 & 9 & 12 & 61 & 60 & 15 \\ 9 & 8 & 12 & 60 & 56 & 14 \\ 2 & 2 & 2 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[1279] $21^{1^{23}} 3^{1^{21}} 21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 9 & 2 \\ 2 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 5 & 5 & 13 & 28 & 7 \\ 9 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1280] $1^{1^{44}} 3^{1^{24}} 21^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 4 & 5 & 5 & 13 & 38 & 7 \\ 12 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1282] $1^{1^{44}} 3^{1^{32}} 21^{1^{33}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 0 & 2 & 9 & 8 & 2 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 10 & 9 & 12 & 61 & 60 & 15 \\ 10 & 8 & 12 & 60 & 56 & 14 \\ 2 & 2 & 2 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[1283] $1^{1^{42}} 3^{1^{32}} 21^{1^{32}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 425, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 10 & 9 & 12 & 61 & 48 & 15 \\ 7 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[1285] $1^{1^{22}} 3^{1^{22}} 21^{1^{33}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 7 & 7 & 11 & 45 & 52 & 13 \\ 8 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[1284] $1^{143} 3^{121} 1^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 8 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 11 & 11 & 13 & 81 & 55 & 17 \\ 8 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[1286] $21^{141} 3^{121} 1^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 4 & 2 \\ 2 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 12 & 12 & 16 & 110 & 36 & 20 \\ 4 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1287] $1^{141} 3^{121} 1^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 11 & 11 & 13 & 81 & 31 & 17 \\ 4 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

[1289] $1^{123} 3^{121} 21^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 6 & 8 & 26 & 46 & 10 \\ 9 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1288] $1^{141} 3^{121} 1^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 11 & 12 & 16 & 110 & 36 & 20 \\ 4 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1290] $21^{124} 3^{121} 21^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 12 & 2 \\ 2 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 6 & 6 & 8 & 26 & 52 & 10 \\ 12 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[1291] $1^{113} 3^{121} 21^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 449, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 3 & 4 & 5 & 23 & 5 \\ 11 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1293] $1^{123} 3^{121} 21^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 7 & 6 & 8 & 26 & 54 & 10 \\ 12 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1292] $1^{123} 3^{121} 21^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 5 & 6 & 8 & 26 & 54 & 10 \\ 11 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1294] $1^{123} 3^{121} 21^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 6 & 8 & 26 & 46 & 10 \\ 11 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1296] $1^{1^{13}3^1 21^{113}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 3 & 4 & 5 & 23 & 5 \\ 10 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1295] $1^{1^{14}3^1 21^{114}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 2 & 3 & 4 & 5 & 30 & 5 \\ 12 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[1297] $1^{1^{24}3^1 21^{124}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 4 & 9 & 2 \\ 1 & 2 & 0 & 7 & 15 & 2 \\ 5 & 4 & 7 & 16 & 39 & 8 \\ 11 & 9 & 15 & 39 & 83 & 17 \\ 2 & 2 & 2 & 8 & 17 & 3 \end{pmatrix}_{b_0}$$

[1299] $1^{1^{42}3^1 21^{142}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 6 & 2 \\ 1 & 0 & 2 & 10 & 5 & 2 \\ 1 & 2 & 0 & 15 & 8 & 2 \\ 11 & 10 & 15 & 88 & 47 & 18 \\ 6 & 5 & 8 & 47 & 23 & 9 \\ 2 & 2 & 2 & 18 & 9 & 3 \end{pmatrix}_{b_0}$$

[1298] $1^{1^{24}3^1 21^{124}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 5 & 4 & 7 & 16 & 43 & 8 \\ 12 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[1300] $1^{1^{24}3^1 21^{112}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 5 & 11 & 2 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 5 & 5 & 5 & 13 & 33 & 7 \\ 11 & 11 & 11 & 33 & 71 & 15 \\ 2 & 2 & 2 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[1302] $21^{1^{23}3^1 21^{123}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 10 & 2 \\ 2 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 6 & 8 & 26 & 46 & 10 \\ 10 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1301] $1^{1^{34}3^1 21^{123}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 7 & 6 & 8 & 26 & 52 & 10 \\ 12 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[1303] $1^{1^{43}3^1 21^{132}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 475, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 8 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 10 & 9 & 12 & 61 & 48 & 15 \\ 8 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[1304] $1^{1^{32}3^1 21^{132}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 477, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 9 & 9 & 12 & 61 & 48 & 15 \\ 7 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0} \\
[1305] \quad & 21^{141} 3^{121} 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 4 & 2 \\ 2 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 11 & 11 & 13 & 81 & 31 & 17 \\ 4 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0} \\
[1306] \quad & 1^{122} 3^{121} 21^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 8 & 7 & 11 & 45 & 52 & 13 \\ 9 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0} \\
[1307] \quad & 21^{142} 3^{121} 1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 7 & 2 \\ 2 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 11 & 11 & 13 & 81 & 55 & 17 \\ 7 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0} \\
[1309] \quad & 1^{142} 3^{121} 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 11 & 12 & 16 & 110 & 64 & 20 \\ 7 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0} \\
[1308] \quad & 1^{124} 3^{121} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 6 & 6 & 8 & 26 & 52 & 10 \\ 12 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0} \\
[1310] \quad & 1^{142} 3^{121} 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 10 & 11 & 13 & 81 & 55 & 17 \\ 7 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0} \\
[1311] \quad & 1^{124} 3^{121} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 5 & 5 & 5 & 13 & 38 & 7 \\ 12 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0} \\
[1312] \quad & 1^{124} 3^{121} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 4 & 5 & 5 & 13 & 38 & 7 \\ 11 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0} \\
[1313] \quad & 1^{122} 3^{121} 21^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 9 & 7 & 11 & 45 & 52 & 13 \\ 10 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0} \\
[1315] \quad & 1^{142} 3^{121} 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 504, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 11 & 11 & 13 & 81 & 55 & 17 \\ 7 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[1314] $1^{143} 3^{121} 1^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 504, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 9 & 2 \\ 1 & 0 & 2 & 12 & 8 & 2 \\ 1 & 2 & 0 & 16 & 12 & 2 \\ 12 & 12 & 16 & 110 & 80 & 20 \\ 9 & 8 & 12 & 80 & 56 & 14 \\ 2 & 2 & 2 & 20 & 14 & 3 \end{pmatrix}_{b_0}$$

[1316] $1^{124} 3^{121} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 13 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 7 & 6 & 8 & 26 & 60 & 10 \\ 13 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[1317] $1^{123} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 6 & 6 & 8 & 26 & 52 & 10 \\ 11 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[1318] $21^{124} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 12 & 2 \\ 2 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 5 & 5 & 5 & 13 & 38 & 7 \\ 12 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1319] $1^{133} 3^{121} 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 533, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 8 & 7 & 11 & 45 & 70 & 13 \\ 11 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

[1320] $1^{114} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 3 & 3 & 4 & 5 & 26 & 5 \\ 11 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1321] $1^{141} 3^{121} 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 542, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 4 & 2 \\ 1 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 12 & 12 & 16 & 110 & 36 & 20 \\ 4 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1322] $1^{113} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 542, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 27 & 5 \\ 11 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1323] $1^{123} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 6 & 8 & 26 & 46 & 10 \\ 10 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1324] $1^{114} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 3 & 3 & 4 & 5 & 26 & 5 \\ 12 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1325] $1^{123}3^{121}1^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 7 & 9 & 2 \\ 1 & 2 & 0 & 11 & 15 & 2 \\ 8 & 7 & 11 & 45 & 63 & 13 \\ 11 & 9 & 15 & 63 & 83 & 17 \\ 2 & 2 & 2 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[1326] $1^{123}3^{121}21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 550, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 7 & 6 & 8 & 26 & 46 & 10 \\ 11 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1327] $21^{142}3^{121}21^{142} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 558, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 7 & 2 \\ 2 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 12 & 12 & 16 & 110 & 64 & 20 \\ 7 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

[1328] $1^{123}3^{121}21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 6 & 8 & 26 & 54 & 10 \\ 11 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1329] $1^{113}3^{121}21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 578, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 27 & 5 \\ 12 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1330] $1^{123}3^{121}21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 7 & 7 & 11 & 45 & 70 & 13 \\ 11 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

[1331] $21^{114}3^{121}21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 12 & 2 \\ 2 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 3 & 3 & 4 & 5 & 26 & 5 \\ 12 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1332] $1^{123}3^{121}21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 614, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 6 & 8 & 26 & 54 & 10 \\ 12 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1333] $1^{122}3^{121}21^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 11 & 2 \\ 1 & 0 & 2 & 8 & 9 & 2 \\ 1 & 2 & 0 & 14 & 15 & 2 \\ 10 & 8 & 14 & 70 & 78 & 16 \\ 11 & 9 & 15 & 78 & 83 & 17 \\ 2 & 2 & 2 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[1334] $1^{124}3^{121}21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 644, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 5 & 6 & 8 & 26 & 60 & 10 \\ 12 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0} \\
[1335] \quad & 21^{143} 3^{121} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 644, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 13 & 2 \\ 2 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 3 & 3 & 4 & 5 & 30 & 5 \\ 13 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0} \\
[1336] \quad & 1^{143} 3^{121} 1^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 9 & 2 \\ 1 & 0 & 2 & 10 & 8 & 2 \\ 1 & 2 & 0 & 15 & 12 & 2 \\ 11 & 10 & 15 & 88 & 72 & 18 \\ 9 & 8 & 12 & 72 & 56 & 14 \\ 2 & 2 & 2 & 18 & 14 & 3 \end{pmatrix}_{b_0} \\
[1337] \quad & 1^{134} 3^{121} 1^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 673, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 12 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 8 & 7 & 11 & 45 & 70 & 13 \\ 12 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0} \\
[1338] \quad & 1^{142} 3^{121} 1^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 7 & 2 \\ 1 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 12 & 12 & 16 & 110 & 64 & 20 \\ 7 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0} \\
[1339] \quad & 1^{13} 3^{121} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 695, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 3 & 3 & 4 & 5 & 30 & 5 \\ 12 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0} \\
[1340] \quad & 21^{124} 3^{121} 1^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 710, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 13 & 2 \\ 2 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 6 & 6 & 8 & 26 & 60 & 10 \\ 13 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0} \\
[1341] \quad & 1^{123} 3^{121} 1^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 740, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 6 & 6 & 8 & 26 & 60 & 10 \\ 12 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0} \\
[1342] \quad & 1^{14} 3^{121} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 3 & 3 & 4 & 5 & 30 & 5 \\ 13 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0} \\
[1343] \quad & 1^{124} 3^{121} 1^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 836, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 13 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 6 & 6 & 8 & 26 & 60 & 10 \\ 13 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0} \\
T = 9 \quad [1344] \quad & 1^{11} 41^{11} 23^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 0 & 2 & 5 & 2 \\ 3 & 2 & 4 & 8 & 4 \\ 7 & 5 & 8 & 12 & 8 \\ 3 & 2 & 4 & 8 & 4 \end{pmatrix}_{b_0} \\
[1345] \quad 1^1 41^{11} 23^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 0 & 2 & 3 & 2 \\ 3 & 2 & 4 & 6 & 4 \\ 6 & 3 & 6 & 6 & 6 \\ 3 & 2 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[1346] \quad 23^1 141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 & 3 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 2 & 2 & 0 & 3 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0} \\
[1347] \quad 1^1 41^{11} 23^{11} 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 9 & 6 & 3 \\ 2 & 0 & 5 & 3 & 2 \\ 9 & 5 & 22 & 12 & 10 \\ 6 & 3 & 12 & 6 & 6 \\ 3 & 2 & 10 & 6 & 4 \end{pmatrix}_{b_0} \\
[1348] \quad 1^1 41^{12} 23^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 11 & 3 \\ 2 & 0 & 2 & 6 & 2 \\ 3 & 2 & 4 & 12 & 4 \\ 11 & 6 & 12 & 30 & 12 \\ 3 & 2 & 4 & 12 & 4 \end{pmatrix}_{b_0} \\
[1349] \quad 1^1 41^{12} 23^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 0 & 2 & 8 & 2 \\ 3 & 2 & 4 & 14 & 4 \\ 12 & 8 & 14 & 40 & 14 \\ 3 & 2 & 4 & 14 & 4 \end{pmatrix}_{b_0} \\
[1350] \quad 1^1 41^{13} 23^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 16 & 3 \\ 2 & 0 & 2 & 9 & 2 \\ 3 & 2 & 4 & 18 & 4 \\ 16 & 9 & 18 & 70 & 18 \\ 3 & 2 & 4 & 18 & 4 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[1351] \quad 23^1 141^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 3 & 2 \\ 1 & 0 & 2 & 4 & 3 & 2 \\ 2 & 2 & 0 & 7 & 3 & 2 \\ 5 & 4 & 7 & 16 & 9 & 8 \\ 3 & 3 & 3 & 9 & 4 & 4 \\ 2 & 2 & 2 & 8 & 4 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \\
[1352] \quad 23^1 141^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 2 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[1353] \quad 1^1 41^{12} 23^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 2 & 1 & 0 & 2 & 2 \\ 6 & 3 & 2 & 6 & 6 \\ 3 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[1354] \quad 1^{11} 41^{12} 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 5 & 4 & 0 & 4 & 10 & 6 \\ 2 & 2 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[1355] \quad 23^1 141^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 5 & 3 & 2 \\ 1 & 0 & 2 & 7 & 3 & 2 \\ 2 & 2 & 0 & 6 & 3 & 2 \\ 5 & 7 & 6 & 25 & 12 & 9 \\ 3 & 3 & 3 & 12 & 4 & 4 \\ 2 & 2 & 2 & 9 & 4 & 3 \end{pmatrix}_{b_0} \\
[1356] \quad & 1^{14} 41^{22} 23^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 8 & 11 & 3 \\ 2 & 0 & 5 & 6 & 2 \\ 8 & 5 & 24 & 28 & 10 \\ 11 & 6 & 28 & 30 & 12 \\ 3 & 2 & 10 & 12 & 4 \end{pmatrix}_{b_0} \\
[1362] \quad & 1^{41} 41^{14} 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 16 & 2 & 2 \\ 3 & 4 & 1 & 32 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 16 & 32 & 0 & 190 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 2 & 4 & 0 & 24 & 2 & 3 \end{pmatrix}_{b_0} \\
[1358] \quad & 1^{21} 41^{12} 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 8 & 2 & 2 \\ 3 & 4 & 1 & 16 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 16 & 0 & 46 & 8 & 12 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 0 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1359] \quad & 1^{12} 41^{11} 23^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 4 & 7 & 2 \\ 3 & 4 & 1 & 8 & 10 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 7 & 10 & 1 & 16 & 21 & 9 \\ 2 & 4 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0} \\
[1360] \quad & 1^{12} 41^{11} 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 4 & 6 & 2 \\ 3 & 4 & 1 & 8 & 10 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 6 & 10 & 0 & 16 & 20 & 8 \\ 2 & 4 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
[1361] \quad & 1^{11} 41^{12} 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 4 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 4 & 6 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 2 & 2 & 0 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[1357] \quad & 1^{31} 41^{13} 23^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 12 & 2 & 2 \\ 3 & 4 & 1 & 24 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 12 & 24 & 0 & 106 & 12 & 18 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 0 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[1363] \quad & 1^{11} 41^{11} 23^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 6 & 2 & 2 \\ 3 & 4 & 1 & 8 & 2 & 4 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 6 & 8 & 2 & 14 & 4 & 8 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 0 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[1364] \quad & 1^{11} 41^{13} 23^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 17 & 3 \\ 2 & 0 & 2 & 11 & 2 \\ 3 & 2 & 4 & 20 & 4 \\ 17 & 11 & 20 & 84 & 20 \\ 3 & 2 & 4 & 20 & 4 \end{pmatrix}_{b_0} \\
[1365] \quad & 1^{14} 41^{14} 23^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 21 & 3 \\ 2 & 0 & 2 & 12 & 2 \\ 3 & 2 & 4 & 24 & 4 \\ 21 & 12 & 24 & 126 & 24 \\ 3 & 2 & 4 & 24 & 4 \end{pmatrix}_{b_0} \\
[1366] \quad & 1^{11} 41^{11} 23^{11} 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 10 & 6 & 3 \\ 2 & 0 & 7 & 3 & 2 \\ 10 & 7 & 28 & 14 & 12 \\ 6 & 3 & 14 & 6 & 6 \\ 3 & 2 & 12 & 6 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[1367] $1^{111}41^{41}23^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -38$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & 10 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 15 & 10 & 2 & 116 & 36 & 20 \\ 5 & 4 & 0 & 36 & 10 & 6 \\ 2 & 2 & 0 & 20 & 6 & 3 \end{pmatrix}_{b_0}$

[1368] $1^{111}41^{11}23^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -38$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 6 & 6 & 3 \\ 2 & 0 & 5 & 3 & 2 \\ 6 & 5 & 12 & 10 & 8 \\ 6 & 3 & 10 & 6 & 6 \\ 3 & 2 & 8 & 6 & 4 \end{pmatrix}_{b_0}$

[1369] $1^{141}2^{12}23^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -38$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 8 & 6 & 3 \\ 2 & 0 & 5 & 3 & 2 \\ 8 & 5 & 24 & 14 & 10 \\ 6 & 3 & 14 & 6 & 6 \\ 3 & 2 & 10 & 6 & 4 \end{pmatrix}_{b_0}$

[1370] $1^{111}41^{21}23^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -38$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & 6 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 9 & 6 & 2 & 36 & 20 & 12 \\ 5 & 4 & 0 & 20 & 10 & 6 \\ 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0}$

1^{111}
[1371] $2^3 1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 5 & 10 & 9 & 5 \\ 6 & 10 & 14 & 13 & 8 \\ 6 & 9 & 13 & 11 & 7 \\ 3 & 5 & 8 & 7 & 4 \end{pmatrix}_{b_0}$

[1372] $1^{141}2^{23}1^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 2 & 1 & 0 & 4 & 2 \\ 11 & 6 & 4 & 30 & 12 \\ 3 & 2 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1373] $1^{141}1^{14}23^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 3 & 22 & 3 \\ 2 & 0 & 2 & 14 & 2 \\ 3 & 2 & 4 & 26 & 4 \\ 22 & 14 & 26 & 144 & 26 \\ 3 & 2 & 4 & 26 & 4 \end{pmatrix}_{b_0}$

[1374] $1^{111}41^{12}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 4 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 4 & 1 & 11 & 4 & 7 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 2 & 2 & 0 & 7 & 2 & 3 \end{pmatrix}_{b_0}$

[1375] $1^{141}3^{23}1^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 0 & 1 & 9 & 2 \\ 2 & 1 & 0 & 6 & 2 \\ 16 & 9 & 6 & 70 & 18 \\ 3 & 2 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1376] $1^{112}41^{22}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 9 & 8 & 0 & 8 & 46 & 12 \\ 2 & 2 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1377] $1^{112}41^{11}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 1 & 4 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 16 & 6 \\ 7 & 6 & 1 & 16 & 21 & 9 \\ 2 & 2 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0}$

[1378] $1^{111}41^{21}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & 6 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 0 & 32 & 20 & 10 \\ 5 & 4 & 0 & 20 & 10 & 6 \\ 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[1380] $23^{11}141^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 1 & 0 & 6 & 2 \\ 6 & 6 & 7 & 6 & 23 & 9 \\ 2 & 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1379] $1^{112}41^{223}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 10 & 8 & 1 & 8 & 47 & 13 \\ 2 & 2 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1382] $1^{111}41^{31}23^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 12 & 8 & 2 & 70 & 28 & 16 \\ 5 & 4 & 0 & 28 & 10 & 6 \\ 2 & 2 & 0 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[1381] $1^{11}41^{31}23^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 6 & 3 \\ 2 & 0 & 10 & 3 & 2 \\ 14 & 10 & 74 & 24 & 18 \\ 6 & 3 & 24 & 6 & 6 \\ 3 & 2 & 18 & 6 & 4 \end{pmatrix}_{b_0}$$

[1383] $23^{11}141^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 3 & 2 \\ 2 & 2 & 0 & 4 & 3 & 2 \\ 3 & 3 & 4 & 5 & 6 & 5 \\ 3 & 3 & 3 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[1385] $1^{11}41^{123}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 2 & 1 & 0 & 4 & 2 \\ 9 & 6 & 4 & 18 & 10 \\ 3 & 2 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1384] $1^{112}41^{123}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 7 & 6 & 1 & 6 & 21 & 9 \\ 2 & 2 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1386] $1^{111}41^{11}23^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 4 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 12 & 6 \\ 5 & 4 & 0 & 12 & 10 & 6 \\ 2 & 2 & 0 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[1387] $1^{11}41^{11}23^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 8 & 6 & 3 \\ 2 & 0 & 6 & 3 & 2 \\ 8 & 6 & 18 & 12 & 10 \\ 6 & 3 & 12 & 6 & 6 \\ 3 & 2 & 10 & 6 & 4 \end{pmatrix}_{b_0}$$

[1388] $23^{11}141^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 3 & 2 \\ 1 & 0 & 2 & 5 & 3 & 2 \\ 2 & 2 & 0 & 5 & 3 & 2 \\ 4 & 5 & 5 & 13 & 9 & 7 \\ 3 & 3 & 3 & 9 & 4 & 4 \\ 2 & 2 & 2 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[1389] $1^{113}41^{423}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 2 & 14 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 17 & 14 & 1 & 14 & 161 & 23 \\ 2 & 2 & 0 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1390] $1^{141}4^{23}1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 21 & 3 \\ 2 & 0 & 1 & 12 & 2 \\ 2 & 1 & 0 & 8 & 2 \\ 21 & 12 & 8 & 126 & 24 \\ 3 & 2 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1391] $23^{11}141^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 8 & 2 \\ 2 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 1 & 0 & 8 & 2 \\ 7 & 8 & 8 & 8 & 37 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1392] $23^{11}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 5 & 5 & 9 & 5 \\ 3 & 5 & 2 & 7 & 4 \\ 6 & 9 & 7 & 11 & 7 \\ 3 & 5 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[1393] $1^{111}41^{11}23^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 4 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 4 & 1 & 11 & 12 & 7 \\ 5 & 4 & 0 & 12 & 10 & 6 \\ 2 & 2 & 0 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[1394] $1^{141}1^{11}23^11^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 6 & 3 \\ 2 & 0 & 4 & 3 & 2 \\ 7 & 4 & 12 & 10 & 8 \\ 6 & 3 & 10 & 6 & 6 \\ 3 & 2 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[1395] $1^{141}1^{11}23^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 5 & 6 & 3 \\ 2 & 0 & 3 & 3 & 2 \\ 5 & 3 & 6 & 8 & 6 \\ 6 & 3 & 8 & 6 & 6 \\ 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[1396] $1^{1141}1^{11}23^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 5 & 7 & 3 \\ 2 & 0 & 3 & 5 & 2 \\ 5 & 3 & 6 & 10 & 6 \\ 7 & 5 & 10 & 12 & 8 \\ 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[1397] $23^{121}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 5 & 10 & 9 & 5 \\ 6 & 10 & 14 & 14 & 8 \\ 6 & 9 & 14 & 11 & 7 \\ 3 & 5 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[1398] $23^{122}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 5 & 10 & 19 & 5 \\ 6 & 10 & 14 & 28 & 8 \\ 12 & 19 & 28 & 53 & 15 \\ 3 & 5 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

[1399] $23^{111}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 5 & 5 & 14 & 5 \\ 3 & 5 & 2 & 10 & 4 \\ 9 & 14 & 10 & 29 & 11 \\ 3 & 5 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[1400] $1^{121}41^{21}23^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -126$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 0 & 46 & 24 & 12 \\ 5 & 4 & 0 & 24 & 10 & 6 \\ 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0} \\
[1401] & 1^{122}41^{21}23^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 1 & 8 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 0 & 46 & 32 & 12 \\ 6 & 6 & 0 & 32 & 20 & 8 \\ 2 & 2 & 0 & 12 & 8 & 3 \end{pmatrix}_{b_0} \\
[1402] & 1^{113}41^{12}23^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 1 & 4 & 10 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 28 & 6 \\ 12 & 10 & 2 & 28 & 68 & 16 \\ 2 & 2 & 0 & 6 & 16 & 3 \end{pmatrix}_{b_0} \\
[1403] & 1^{121}41^{22}23^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 1 & 8 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 9 & 8 & 1 & 47 & 8 & 13 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 2 & 2 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[1404] & 1^{112}41^{11}23^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 4 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 16 & 6 \\ 6 & 6 & 0 & 16 & 20 & 8 \\ 2 & 2 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
[1405] & 1^{11}41^{11}23^11^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 12 & 8 & 3 \\ 2 & 0 & 8 & 4 & 2 \\ 12 & 8 & 42 & 24 & 14 \\ 8 & 4 & 24 & 12 & 8 \\ 3 & 2 & 14 & 8 & 4 \end{pmatrix}_{b_0} \\
& 1^{121} \\
[1406] & 23^1141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 5 & 15 & 9 & 5 \\ 9 & 15 & 32 & 20 & 12 \\ 6 & 9 & 20 & 11 & 7 \\ 3 & 5 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[1407] & 1^{141}3^123^11^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 6 & 3 \\ 2 & 0 & 8 & 3 & 2 \\ 13 & 8 & 60 & 22 & 16 \\ 6 & 3 & 22 & 6 & 6 \\ 3 & 2 & 16 & 6 & 4 \end{pmatrix}_{b_0} \\
[1408] & 1^{111}41^{31}23^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 11 & 8 & 1 & 67 & 28 & 15 \\ 5 & 4 & 0 & 28 & 10 & 6 \\ 2 & 2 & 0 & 15 & 6 & 3 \end{pmatrix}_{b_0} \\
[1409] & 1^{112}41^{22}23^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 1 & 6 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 0 & 32 & 40 & 10 \\ 9 & 8 & 0 & 40 & 46 & 12 \\ 2 & 2 & 0 & 10 & 12 & 3 \end{pmatrix}_{b_0} \\
[1410] & 1^{121}41^{31}23^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & 10 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 0 & 84 & 32 & 16 \\ 5 & 4 & 0 & 32 & 10 & 6 \\ 2 & 2 & 0 & 16 & 6 & 3 \end{pmatrix}_{b_0} \\
& 1^{131} \\
[1411] & 23^1141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 5 & 15 & 9 & 5 \\ 9 & 15 & 34 & 21 & 12 \\ 6 & 9 & 21 & 11 & 7 \\ 3 & 5 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[1412] & 1^{13} 41^{23} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 13 & 12 & 0 & 12 & 106 & 18 \\ 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1413] & 1^{14} 41^{23} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 2 & 1 & 0 & 6 & 2 \\ 13 & 7 & 6 & 40 & 14 \\ 3 & 2 & 2 & 14 & 4 \end{pmatrix}_{b_0} \\
[1414] & 1^{11} 41^{11} 23^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 9 & 3 \\ 2 & 0 & 4 & 6 & 2 \\ 7 & 4 & 12 & 16 & 8 \\ 9 & 6 & 16 & 18 & 10 \\ 3 & 2 & 8 & 10 & 4 \end{pmatrix}_{b_0} \\
[1415] & 1^{14} 41^{12} 23^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 11 & 3 \\ 2 & 0 & 4 & 6 & 2 \\ 7 & 4 & 12 & 20 & 8 \\ 11 & 6 & 20 & 30 & 12 \\ 3 & 2 & 8 & 12 & 4 \end{pmatrix}_{b_0} \\
[1416] & 1^{12} 23^1 141^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 6 & 3 & 2 \\ 1 & 0 & 2 & 6 & 3 & 2 \\ 2 & 2 & 0 & 8 & 3 & 2 \\ 6 & 6 & 8 & 26 & 12 & 10 \\ 3 & 3 & 3 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0} \\
[1417] & 1^{12} 23^1 141^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 2 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 1 & 0 & 9 & 2 \\ 9 & 9 & 11 & 9 & 56 & 14 \\ 2 & 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1418] & 1^{11} 41^{23} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 0 & 1 & 9 & 2 \\ 2 & 1 & 0 & 6 & 2 \\ 14 & 9 & 6 & 50 & 16 \\ 3 & 2 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[1419] & 1^{13} 41^{32} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 15 & 12 & 2 & 12 & 110 & 20 \\ 2 & 2 & 0 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1420] & 1^{11} 41^{11} 23^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 9 & 3 \\ 2 & 0 & 3 & 6 & 2 \\ 5 & 3 & 6 & 12 & 6 \\ 9 & 6 & 12 & 18 & 10 \\ 3 & 2 & 6 & 10 & 4 \end{pmatrix}_{b_0} \\
[1422] & 1^{111} 41^{21} 23^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & 6 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 8 & 6 & 1 & 33 & 20 & 11 \\ 5 & 4 & 0 & 20 & 10 & 6 \\ 2 & 2 & 0 & 11 & 6 & 3 \end{pmatrix}_{b_0} \\
[1421] & 1^{11} 41^{21} 23^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =
\end{aligned}$$

5.0

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 11 & 6 & 3 \\ 2 & 0 & 8 & 3 & 2 \\ 11 & 8 & 40 & 18 & 14 \\ 6 & 3 & 18 & 6 & 6 \\ 3 & 2 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1423] \ 1^{121} 41^{223} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 1 & 8 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 10 & 8 & 2 & 50 & 8 & 14 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 2 & 2 & 0 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1424] \ 2 \ 3^{141} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 5 & 20 & 9 & 5 \\ 12 & 20 & 62 & 28 & 16 \\ 6 & 9 & 28 & 11 & 7 \\ 3 & 5 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1425] \ 1^{13} 41^{123} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 9 & 8 & 2 & 8 & 38 & 12 \\ 2 & 2 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1426] \ 1^{13} 41^{323} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 14 & 12 & 1 & 12 & 107 & 19 \\ 2 & 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1427] \ 23^{12} 141^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 3 & 2 \\ 1 & 0 & 2 & 10 & 3 & 2 \\ 2 & 2 & 0 & 10 & 3 & 2 \\ 8 & 10 & 10 & 58 & 18 & 14 \\ 3 & 3 & 3 & 18 & 4 & 4 \\ 2 & 2 & 2 & 14 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1428] \ 1^1 41^{13} 23^{1123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 16 & 3 \\ 2 & 0 & 4 & 9 & 2 \\ 7 & 4 & 12 & 30 & 8 \\ 16 & 9 & 30 & 70 & 18 \\ 3 & 2 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1430] \ 1^{13} 41^{223} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 12 & 10 & 2 & 10 & 68 & 16 \\ 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1429] \ 2 \ 3^{112} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 5 & 5 & 14 & 5 \\ 3 & 5 & 2 & 11 & 4 \\ 9 & 14 & 11 & 29 & 11 \\ 3 & 5 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1431] \ 1^{131} 41^{31} 23^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 0 & 106 & 36 & 18 \\ 5 & 4 & 0 & 36 & 10 & 6 \\ 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1432] \ 1^{13} 41^{223} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 11 & 10 & 1 & 10 & 65 & 15 \\ 2 & 2 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1435] \ 2 \ 3^{122} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$\begin{aligned}
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 5 & 10 & 14 & 5 \\ 6 & 10 & 14 & 22 & 8 \\ 9 & 14 & 22 & 29 & 11 \\ 3 & 5 & 8 & 11 & 4 \end{pmatrix}_{b_0} \\
[1434] & 1^{141} 1^{123} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 8 & 3 \\ 2 & 0 & 4 & 4 & 2 \\ 7 & 4 & 12 & 14 & 8 \\ 8 & 4 & 14 & 12 & 8 \\ 3 & 2 & 8 & 8 & 4 \end{pmatrix}_{b_0} \\
[1433] & 1^{341} 1^{323} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 13 & 2 & 2 \\ 3 & 4 & 1 & 24 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 13 & 24 & 1 & 107 & 12 & 19 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 0 & 19 & 2 & 3 \end{pmatrix}_{b_0} \\
[1436] & 1^{141} 1^{21} 23^{11} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 10 & 6 & 3 \\ 2 & 0 & 6 & 3 & 2 \\ 10 & 6 & 30 & 16 & 12 \\ 6 & 3 & 16 & 6 & 6 \\ 3 & 2 & 12 & 6 & 4 \end{pmatrix}_{b_0} \\
[1440] & 1^{121} 41^{21} 23^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 10 & 8 & 2 & 50 & 24 & 14 \\ 5 & 4 & 0 & 24 & 10 & 6 \\ 2 & 2 & 0 & 14 & 6 & 3 \end{pmatrix}_{b_0} \\
[1439] & 1^{141} 1^{21} 23^{11} 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 14 & 6 & 3 \\ 2 & 0 & 8 & 3 & 2 \\ 14 & 8 & 54 & 20 & 16 \\ 6 & 3 & 20 & 6 & 6 \\ 3 & 2 & 16 & 6 & 4 \end{pmatrix}_{b_0} \\
[1438] & 1^{1241} 1^{23} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & 4 & 8 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 24 & 6 \\ 10 & 8 & 1 & 24 & 47 & 13 \\ 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0} \\
[1437] & 1^{1141} 1^{22} 23^{11} 1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 10 & 14 & 3 \\ 2 & 0 & 6 & 9 & 2 \\ 10 & 6 & 30 & 40 & 12 \\ 14 & 9 & 40 & 50 & 16 \\ 3 & 2 & 12 & 16 & 4 \end{pmatrix}_{b_0} \\
[1441] & 1^{132} 23^{11} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 12 & 9 & 3 \\ 4 & 5 & 20 & 14 & 5 \\ 12 & 20 & 58 & 42 & 16 \\ 9 & 14 & 42 & 29 & 11 \\ 3 & 5 & 16 & 11 & 4 \end{pmatrix}_{b_0} \\
[1442] & 1^{12} 23^{11} 141^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 3 & 2 \\ 1 & 0 & 2 & 8 & 3 & 2 \\ 2 & 2 & 0 & 9 & 3 & 2 \\ 7 & 8 & 9 & 40 & 15 & 12 \\ 3 & 3 & 3 & 15 & 4 & 4 \\ 2 & 2 & 2 & 12 & 4 & 3 \end{pmatrix}_{b_0} \\
[1443] & 1^{151} 23^{11} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.1 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 15 & 6 & 3 \\ 4 & 5 & 25 & 9 & 5 \\ 15 & 25 & 98 & 35 & 20 \\ 6 & 9 & 35 & 11 & 7 \\ 3 & 5 & 20 & 7 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& [1444] \ 23^{112} 141^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -208, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 0 & 1 & 2 & 1 & 10 & 2 \\ 1 & 0 & 2 & 2 & 11 & 2 \\ 2 & 2 & 0 & 1 & 12 & 2 \\ 1 & 2 & 1 & 0 & 11 & 2 \\ 10 & 11 & 12 & 11 & 76 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& [1445] \ 1^{141} 23^{111} 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -208, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 0 & 1 & 10 & 2 \\ 2 & 1 & 0 & 8 & 2 \\ 18 & 10 & 8 & 84 & 20 \\ 3 & 2 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
& [1446] \ 23^{132} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8 \\
& \quad T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 5 & 15 & 14 & 5 \\ 9 & 15 & 34 & 33 & 12 \\ 9 & 14 & 33 & 29 & 11 \\ 3 & 5 & 12 & 11 & 4 \end{pmatrix}_{b_0} \\
& [1447] \ 1^{1141} 23^{111} 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -217, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 19 & 3 \\ 2 & 0 & 1 & 12 & 2 \\ 2 & 1 & 0 & 8 & 2 \\ 19 & 12 & 8 & 98 & 22 \\ 3 & 2 & 2 & 22 & 4 \end{pmatrix}_{b_0} \\
& [1450] \ 1^{112} 41^{12} 23^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & 4 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 24 & 6 \\ 9 & 8 & 0 & 24 & 46 & 12 \\ 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0} \\
& [1449] \ 1^{131} 41^{41} 23^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8 \\
& \quad T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & 14 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 15 & 14 & 0 & 160 & 44 & 22 \\ 5 & 4 & 0 & 44 & 10 & 6 \\ 2 & 2 & 0 & 22 & 6 & 3 \end{pmatrix}_{b_0} \\
& [1448] \ 1^{114} 41^{12} 23^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 32 & 6 \\ 13 & 12 & 2 & 32 & 90 & 18 \\ 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0} \\
& [1452] \ 1^{1141} 223^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5 \\
& \quad T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 1 & 2 & 10 & 12 & 3 \\ 2 & 0 & 6 & 8 & 2 \\ 10 & 6 & 30 & 36 & 12 \\ 12 & 8 & 36 & 40 & 14 \\ 3 & 2 & 12 & 14 & 4 \end{pmatrix}_{b_0} \\
& [1451] \ 1^{141} 14^{23} 1^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5 \\
& \quad T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 1 & 2 & 7 & 21 & 3 \\ 2 & 0 & 4 & 12 & 2 \\ 7 & 4 & 12 & 40 & 8 \\ 21 & 12 & 40 & 126 & 24 \\ 3 & 2 & 8 & 24 & 4 \end{pmatrix}_{b_0} \\
& [1453] \ 23^{131} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -225, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 5 & 20 & 9 & 5 \\ 12 & 20 & 58 & 27 & 16 \\ 6 & 9 & 27 & 11 & 7 \\ 3 & 5 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
& [1454] \ 1^{141} 223^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -230, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5 \\
& \quad T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11] \\
& \quad \begin{pmatrix} 1 & 2 & 10 & 13 & 3 \\ 2 & 0 & 6 & 7 & 2 \\ 10 & 6 & 30 & 36 & 12 \\ 13 & 7 & 36 & 40 & 14 \\ 3 & 2 & 12 & 14 & 4 \end{pmatrix}_{b_0} \\
& [1455] \ 1^{114} 41^{42} 23^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -230, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0
\end{aligned}$$

5.5

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 9 & 8 & 1 & 47 & 24 & 13 \\ 5 & 4 & 0 & 24 & 10 & 6 \\ 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

1^{112}

$$[1467] \ 2 \ 3^1 \ 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 19 & 5 \\ 3 & 5 & 2 & 14 & 4 \\ 12 & 19 & 14 & 53 & 15 \\ 3 & 5 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1468] \ 1^{141} 41^{41} 23^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & 16 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 16 & 16 & 0 & 190 & 48 & 24 \\ 5 & 4 & 0 & 48 & 10 & 6 \\ 2 & 2 & 0 & 24 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1470] \ 1^{112} 41^{13} 23^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & 4 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 32 & 6 \\ 13 & 10 & 1 & 32 & 85 & 17 \\ 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1469] \ 1^{11} 41^{23} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 0 & 1 & 10 & 2 \\ 2 & 1 & 0 & 8 & 2 \\ 16 & 10 & 8 & 64 & 18 \\ 3 & 2 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1471] \ 1^{11} 41^{12} 23^1 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 14 & 3 \\ 2 & 0 & 4 & 9 & 2 \\ 7 & 4 & 12 & 26 & 8 \\ 14 & 9 & 26 & 50 & 16 \\ 3 & 2 & 8 & 16 & 4 \end{pmatrix}_{b_0}$$

1^{113}

$$[1472] \ 2 \ 3^1 \ 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 19 & 5 \\ 3 & 5 & 2 & 15 & 4 \\ 12 & 19 & 15 & 55 & 15 \\ 3 & 5 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1473] \ 1^{141} 41^{41} 23^1 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 18 & 6 & 3 \\ 2 & 0 & 11 & 3 & 2 \\ 18 & 11 & 112 & 30 & 22 \\ 6 & 3 & 30 & 6 & 6 \\ 3 & 2 & 22 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1474] \ 1^{114} 41^{423} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 2 & 16 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 18 & 16 & 1 & 16 & 191 & 25 \\ 2 & 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

1^{123}

$$[1475] \ 2 \ 3^1 \ 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 5 & 10 & 24 & 5 \\ 6 & 10 & 14 & 36 & 8 \\ 15 & 24 & 36 & 85 & 19 \\ 3 & 5 & 8 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[1477] \ 1^{131} 41^{323} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 & 2 \\ 1 & 0 & 1 & 12 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 14 & 12 & 2 & 110 & 12 & 20 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 2 & 2 & 0 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

[1476] $1^{113}41^{12}23^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & 4 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 28 & 6 \\ 10 & 10 & 0 & 28 & 64 & 14 \\ 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$

[1479] $1^{114}41^{423}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 2 & 16 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 19 & 16 & 2 & 16 & 194 & 26 \\ 2 & 2 & 0 & 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1478] $1^{141}1^{23}1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 5 & 13 & 3 \\ 2 & 0 & 3 & 7 & 2 \\ 5 & 3 & 6 & 18 & 6 \\ 13 & 7 & 18 & 40 & 14 \\ 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$

1^{123}
[1480] $2^31^{141}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 5 & 10 & 19 & 5 \\ 6 & 10 & 14 & 30 & 8 \\ 12 & 19 & 30 & 55 & 15 \\ 3 & 5 & 8 & 15 & 4 \end{pmatrix}_{b_0}$

[1481] $1^{141}1^{23}1^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 11 & 3 \\ 2 & 0 & 7 & 6 & 2 \\ 12 & 7 & 40 & 36 & 14 \\ 11 & 6 & 36 & 30 & 12 \\ 3 & 2 & 14 & 12 & 4 \end{pmatrix}_{b_0}$

[1484] $1^{113}41^{12}23^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 1 & 4 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 28 & 6 \\ 11 & 10 & 1 & 28 & 65 & 15 \\ 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$

1^{171}
[1482] $2^31^{141}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 21 & 6 & 3 \\ 4 & 5 & 35 & 9 & 5 \\ 21 & 35 & 194 & 49 & 28 \\ 6 & 9 & 49 & 11 & 7 \\ 3 & 5 & 28 & 7 & 4 \end{pmatrix}_{b_0}$

[1483] $1^{1141}1^{23}1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 5 & 12 & 3 \\ 2 & 0 & 3 & 8 & 2 \\ 5 & 3 & 6 & 18 & 6 \\ 12 & 8 & 18 & 40 & 14 \\ 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$

[1485] $1^{114}41^{223}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -299$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 12 & 12 & 1 & 12 & 87 & 17 \\ 2 & 2 & 0 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1486] $1^{112}41^{14}23^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 40 & 6 \\ 16 & 12 & 1 & 40 & 135 & 21 \\ 2 & 2 & 0 & 6 & 21 & 3 \end{pmatrix}_{b_0}$

[1487] $1^{114}41^{323}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -313$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 2 & 14 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 15 & 14 & 1 & 14 & 133 & 21 \\ 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$

$$[1488] \ 1^{113}41^{13}23^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 15 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 36 & 6 \\ 15 & 12 & 2 & 36 & 110 & 20 \\ 2 & 2 & 0 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1489] \ 1^{142}23^{11}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 9 & 3 \\ 4 & 5 & 25 & 14 & 5 \\ 15 & 25 & 92 & 53 & 20 \\ 9 & 14 & 53 & 29 & 11 \\ 3 & 5 & 20 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1490] \ 1^{114}41^{223}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 13 & 12 & 2 & 12 & 90 & 18 \\ 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1491] \ 1^{114}41^{323}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 2 & 14 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 16 & 14 & 2 & 14 & 136 & 22 \\ 2 & 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1492] \ 23^{141}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 6 & 3 \\ 4 & 5 & 25 & 9 & 5 \\ 15 & 25 & 92 & 34 & 20 \\ 6 & 9 & 34 & 11 & 7 \\ 3 & 5 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1493] \ 1^{141}223^{11}1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 11 & 3 \\ 2 & 0 & 6 & 6 & 2 \\ 10 & 6 & 30 & 32 & 12 \\ 11 & 6 & 32 & 30 & 12 \\ 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1494] \ 1^{121}41^{41}23^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 15 & 12 & 1 & 135 & 40 & 21 \\ 5 & 4 & 0 & 40 & 10 & 6 \\ 2 & 2 & 0 & 21 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1495] \ 1^{121}41^{31}23^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 5 & 2 \\ 1 & 0 & 1 & 10 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 13 & 10 & 2 & 88 & 32 & 18 \\ 5 & 4 & 0 & 32 & 10 & 6 \\ 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1496] \ 1^{11}41^{31}23^{11}1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 16 & 6 & 3 \\ 2 & 0 & 11 & 3 & 2 \\ 16 & 11 & 84 & 26 & 20 \\ 6 & 3 & 26 & 6 & 6 \\ 3 & 2 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1497] \ 1^{11}41^{12}23^{11}1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 13 & 3 \\ 2 & 0 & 4 & 7 & 2 \\ 7 & 4 & 12 & 24 & 8 \\ 13 & 7 & 24 & 40 & 14 \\ 3 & 2 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1498] \ 1^{121}41^{31}23^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & 10 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 12 & 10 & 1 & 85 & 32 & 17 \\ 5 & 4 & 0 & 32 & 10 & 6 \\ 2 & 2 & 0 & 17 & 6 & 3 \end{pmatrix}_{b_0} \\
[1499] & 1^{14} 41^{31} 23^{11} 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 15 & 6 & 3 \\ 2 & 0 & 9 & 3 & 2 \\ 15 & 9 & 70 & 24 & 18 \\ 6 & 3 & 24 & 6 & 6 \\ 3 & 2 & 18 & 6 & 4 \end{pmatrix}_{b_0} \\
[1500] & 1^{11} 41^{13} 23^{11} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 15 & 3 \\ 2 & 0 & 3 & 10 & 2 \\ 5 & 3 & 6 & 24 & 6 \\ 15 & 10 & 24 & 74 & 18 \\ 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0} \\
[1501] & 1^{114} 41^{12} 23^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 32 & 6 \\ 12 & 12 & 1 & 32 & 87 & 17 \\ 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0} \\
[1502] & 1^{11} 41^{12} 23^{11} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 14 & 3 \\ 2 & 0 & 3 & 9 & 2 \\ 5 & 3 & 6 & 20 & 6 \\ 14 & 9 & 20 & 50 & 16 \\ 3 & 2 & 6 & 16 & 4 \end{pmatrix}_{b_0} \\
[1503] & 1^{14} 41^{21} 23^{11} 1^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 14 & 8 & 3 \\ 2 & 0 & 8 & 4 & 2 \\ 14 & 8 & 54 & 28 & 16 \\ 8 & 4 & 28 & 12 & 8 \\ 3 & 2 & 16 & 8 & 4 \end{pmatrix}_{b_0} \\
[1504] & 1^{11} 41^{12} 23^{11} 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 16 & 3 \\ 2 & 0 & 3 & 10 & 2 \\ 5 & 3 & 6 & 22 & 6 \\ 16 & 10 & 22 & 64 & 18 \\ 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0} \\
[1505] & 1^{121} 41^{41} 23^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 16 & 12 & 2 & 138 & 40 & 22 \\ 5 & 4 & 0 & 40 & 10 & 6 \\ 2 & 2 & 0 & 22 & 6 & 3 \end{pmatrix}_{b_0} \\
[1506] & 1^{114} 2^3 1^{141} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 24 & 5 \\ 3 & 5 & 2 & 19 & 4 \\ 15 & 24 & 19 & 89 & 19 \\ 3 & 5 & 4 & 19 & 4 \end{pmatrix}_{b_0} \\
[1507] & 1^{113} 41^{13} 23^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -410, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 36 & 6 \\ 13 & 12 & 0 & 36 & 106 & 18 \\ 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0} \\
[1509] & 1^{11} 41^{13} 23^{11} 1^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 19 & 3 \\ 2 & 0 & 4 & 12 & 2 \\ 7 & 4 & 12 & 36 & 8 \\ 19 & 12 & 36 & 98 & 22 \\ 3 & 2 & 8 & 22 & 4 \end{pmatrix}_{b_0} \\
[1510] & 1^{141} 41^{42} 23^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 & 2 \\ 1 & 0 & 1 & 16 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 18 & 16 & 2 & 194 & 16 & 26 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 2 & 2 & 0 & 26 & 2 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{124} \\ [1508] \end{matrix} 2 \, 3^1 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 5 & 10 & 24 & 5 \\ 6 & 10 & 14 & 38 & 8 \\ 15 & 24 & 38 & 89 & 19 \\ 3 & 5 & 8 & 19 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{113} \\ [1511] \end{matrix} 2 \, 3^1 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -437, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 24 & 5 \\ 3 & 5 & 2 & 18 & 4 \\ 15 & 24 & 18 & 85 & 19 \\ 3 & 5 & 4 & 19 & 4 \end{pmatrix}_{b_0} \\
& [1512] \, 1^{141} 3^1 23^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -443, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 17 & 6 & 3 \\ 2 & 0 & 10 & 3 & 2 \\ 17 & 10 & 84 & 26 & 20 \\ 6 & 3 & 26 & 6 & 6 \\ 3 & 2 & 20 & 6 & 4 \end{pmatrix}_{b_0} \\
& [1513] \, 1^{131} 41^{31} 23^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -443, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 13 & 12 & 1 & 107 & 36 & 19 \\ 5 & 4 & 0 & 36 & 10 & 6 \\ 2 & 2 & 0 & 19 & 6 & 3 \end{pmatrix}_{b_0} \\
& [1514] \, 1^{113} 41^{13} 23^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 36 & 6 \\ 14 & 12 & 1 & 36 & 107 & 19 \\ 2 & 2 & 0 & 6 & 19 & 3 \end{pmatrix}_{b_0} \\
& [1515] \, 1^{113} 41^{14} 23^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 18 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 44 & 6 \\ 18 & 14 & 2 & 44 & 164 & 24 \\ 2 & 2 & 0 & 6 & 24 & 3 \end{pmatrix}_{b_0} \\
& [1516] \, 1^{141} 3^1 23^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -459, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 16 & 3 \\ 2 & 0 & 3 & 9 & 2 \\ 5 & 3 & 6 & 24 & 6 \\ 16 & 9 & 24 & 70 & 18 \\ 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0} \\
& [1517] \, 1^{141} 12^1 23^1 1^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 15 & 3 \\ 2 & 0 & 4 & 8 & 2 \\ 7 & 4 & 12 & 28 & 8 \\ 15 & 8 & 28 & 54 & 16 \\ 3 & 2 & 8 & 16 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{124} \\ [1518] \end{matrix} 2 \, 3^1 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 5 & 10 & 29 & 5 \\ 6 & 10 & 14 & 44 & 8 \\ 18 & 29 & 44 & 125 & 23 \\ 3 & 5 & 8 & 23 & 4 \end{pmatrix}_{b_0} \\
& [1519] \, 1^{114} 41^{13} 23^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 40 & 6 \\ 16 & 14 & 2 & 40 & 136 & 22 \\ 2 & 2 & 0 & 6 & 22 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1520] $1^{14}41^{13}23^{11}1^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 18 & 3 \\ 2 & 0 & 4 & 10 & 2 \\ 7 & 4 & 12 & 34 & 8 \\ 18 & 10 & 34 & 84 & 20 \\ 3 & 2 & 8 & 20 & 4 \end{pmatrix}_{b_0}$$

[1521] $1^{11}41^{12}23^{11}1^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 16 & 3 \\ 2 & 0 & 4 & 10 & 2 \\ 7 & 4 & 12 & 30 & 8 \\ 16 & 10 & 30 & 64 & 18 \\ 3 & 2 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[1522] $1^{131}41^{31}23^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 14 & 12 & 2 & 110 & 36 & 20 \\ 5 & 4 & 0 & 36 & 10 & 6 \\ 2 & 2 & 0 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1523] $1^{11}41^{31}23^{11}1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 6 & 3 \\ 2 & 0 & 12 & 3 & 2 \\ 18 & 12 & 98 & 28 & 22 \\ 6 & 3 & 28 & 6 & 6 \\ 3 & 2 & 22 & 6 & 4 \end{pmatrix}_{b_0}$$

[1524] $1^{114}41^{13}23^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 40 & 6 \\ 14 & 14 & 0 & 40 & 132 & 20 \\ 2 & 2 & 0 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[1525] $1^{141}41^{14}23^{11}1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 5 & 19 & 3 \\ 2 & 0 & 3 & 11 & 2 \\ 5 & 3 & 6 & 30 & 6 \\ 19 & 11 & 30 & 112 & 22 \\ 3 & 2 & 6 & 22 & 4 \end{pmatrix}_{b_0}$$

[1526] $1^{141}41^{14}23^{11}1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 20 & 6 & 3 \\ 2 & 0 & 12 & 3 & 2 \\ 20 & 12 & 126 & 32 & 24 \\ 6 & 3 & 32 & 6 & 6 \\ 3 & 2 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

[1527] $1^{131}41^{41}23^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -543, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & 14 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 16 & 14 & 1 & 161 & 44 & 23 \\ 5 & 4 & 0 & 44 & 10 & 6 \\ 2 & 2 & 0 & 23 & 6 & 3 \end{pmatrix}_{b_0}$$

[1528] $1^{141}41^{13}23^{11}1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 5 & 18 & 3 \\ 2 & 0 & 3 & 10 & 2 \\ 5 & 3 & 6 & 26 & 6 \\ 18 & 10 & 26 & 84 & 20 \\ 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[1529] $1^{11}41^{13}23^{11}1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -558, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 5 & 17 & 3 \\ 2 & 0 & 3 & 11 & 2 \\ 5 & 3 & 6 & 26 & 6 \\ 17 & 11 & 26 & 84 & 20 \\ 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[1530] $1^{113}41^{14}23^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 44 & 6 \\ 17 & 14 & 1 & 44 & 161 & 23 \\ 2 & 2 & 0 & 6 & 23 & 3 \end{pmatrix}_{b_0} \\
[1531] & 1^{11}41^{41}23^{11}1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -575, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 21 & 6 & 3 \\ 2 & 0 & 14 & 3 & 2 \\ 21 & 14 & 144 & 34 & 26 \\ 6 & 3 & 34 & 6 & 6 \\ 3 & 2 & 26 & 6 & 4 \end{pmatrix}_{b_0} \\
[1532] & 1^{114}41^{13}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 15 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 40 & 6 \\ 15 & 14 & 1 & 40 & 133 & 21 \\ 2 & 2 & 0 & 6 & 21 & 3 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[1533] & 23^{11}141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -629, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 5 & 5 & 29 & 5 \\ 3 & 5 & 2 & 22 & 4 \\ 18 & 29 & 22 & 125 & 23 \\ 3 & 5 & 4 & 23 & 4 \end{pmatrix}_{b_0} \\
[1535] & 1^{11}41^{14}23^{11}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -647, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 20 & 3 \\ 2 & 0 & 3 & 13 & 2 \\ 5 & 3 & 6 & 32 & 6 \\ 20 & 13 & 32 & 130 & 24 \\ 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0} \\
[1534] & 1^{141}41^{41}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -647, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 5 & 2 \\ 1 & 0 & 1 & 16 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 17 & 16 & 1 & 191 & 48 & 25 \\ 5 & 4 & 0 & 48 & 10 & 6 \\ 2 & 2 & 0 & 25 & 6 & 3 \end{pmatrix}_{b_0} \\
[1536] & 1^{131}41^{41}23^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 5 & 2 \\ 1 & 0 & 1 & 14 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 17 & 14 & 2 & 164 & 44 & 24 \\ 5 & 4 & 0 & 44 & 10 & 6 \\ 2 & 2 & 0 & 24 & 6 & 3 \end{pmatrix}_{b_0} \\
[1537] & 1^{114}41^{14}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & 4 & 16 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 48 & 6 \\ 17 & 16 & 0 & 48 & 190 & 24 \\ 2 & 2 & 0 & 6 & 24 & 3 \end{pmatrix}_{b_0} \\
[1538] & 1^{11}41^{13}23^{11}1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -689, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 19 & 3 \\ 2 & 0 & 3 & 12 & 2 \\ 5 & 3 & 6 & 28 & 6 \\ 19 & 12 & 28 & 98 & 22 \\ 3 & 2 & 6 & 22 & 4 \end{pmatrix}_{b_0} \\
[1539] & 1^{114}41^{14}23^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -698, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 19 & 2 \\ 1 & 0 & 1 & 4 & 16 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 48 & 6 \\ 19 & 16 & 2 & 48 & 194 & 26 \\ 2 & 2 & 0 & 6 & 26 & 3 \end{pmatrix}_{b_0} \\
[1540] & 1^{141}41^{14}23^{11}1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 5 & 21 & 3 \\ 2 & 0 & 3 & 12 & 2 \\ 5 & 3 & 6 & 32 & 6 \\ 21 & 12 & 32 & 126 & 24 \\ 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1541] $1^{11}4^{14}41^{14}23^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -770, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 18 & 2 \\ 1 & 0 & 1 & 4 & 16 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 48 & 6 \\ 18 & 16 & 1 & 48 & 191 & 25 \\ 2 & 2 & 0 & 6 & 25 & 3 \end{pmatrix}_{b_0}$$

[1542] $1^{11}4^{14}41^{14}23^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 5 & 2 \\ 1 & 0 & 1 & 16 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 18 & 16 & 2 & 194 & 48 & 26 \\ 5 & 4 & 0 & 48 & 10 & 6 \\ 2 & 2 & 0 & 26 & 6 & 3 \end{pmatrix}_{b_0}$$

[1543] $1^{11}4^{14}41^{14}23^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 22 & 3 \\ 2 & 0 & 3 & 14 & 2 \\ 5 & 3 & 6 & 34 & 6 \\ 22 & 14 & 34 & 144 & 26 \\ 3 & 2 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ [1544] $1^{11}4^{13}13^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 7 & 6 & 5 & 18 & 10 \\ 3 & 3 & 2 & 10 & 5 \end{pmatrix}_{b_0}$$

[1545] $1^{11}4^{13}13^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 6 & 5 & 3 & 10 & 8 \\ 3 & 3 & 2 & 8 & 5 \end{pmatrix}_{b_0}$$

[1546] $1^{11}4^{13}13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 11 & 9 & 7 & 48 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

[1547] $31^{11}3^{11}5^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 5 & 3 \\ 5 & 5 & 8 & 7 & 5 \\ 6 & 8 & 9 & 10 & 7 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 5 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

[1548] $1^{11}5^{11}13^{11}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 3 & 3 \\ 5 & 5 & 8 & 3 & 5 \\ 6 & 8 & 9 & 5 & 7 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 5 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[1549] $1^{12}4^{13}13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 11 & 10 & 6 & 46 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

[1550] $1^{12}4^{13}13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 12 & 11 & 8 & 60 & 18 \\ 3 & 3 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[1551] $1^{1^{12}} 5^{1^{11}} 13^{1^{11}} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 9 & 3 \\ 5 & 5 & 8 & 11 & 5 \\ 6 & 8 & 9 & 16 & 7 \\ 9 & 11 & 16 & 22 & 10 \\ 3 & 5 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[1552] $1^{1^{13}} 4^{1^{13}} 13^{1^{13}} 21^{1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 15 & 13 & 8 & 90 & 22 \\ 3 & 3 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[1553] $1^{1^{13}} 4^{1^{13}} 13^{1^{13}} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 13 & 13 & 9 & 76 & 20 \\ 3 & 3 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

[1556] $31^{1^{23}} 3^1 5^1 1^{1^{21}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 12 & 5 & 3 \\ 5 & 5 & 16 & 7 & 5 \\ 12 & 16 & 42 & 20 & 14 \\ 5 & 7 & 20 & 8 & 6 \\ 3 & 5 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[1555] $1^{1^{21}} 5^{1^{13}} 13^{1^{13}} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 12 & 3 & 3 \\ 5 & 5 & 16 & 3 & 5 \\ 12 & 16 & 42 & 10 & 14 \\ 3 & 3 & 10 & 1 & 3 \\ 3 & 5 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1554] $1^{1^{14}} 4^{1^{14}} 13^{1^{14}} 21^{1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & 0 & 10 & 2 \\ 16 & 14 & 10 & 108 & 24 \\ 3 & 3 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

[1557] $31^{1^{11}} 3^1 1^1 5^1 1^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 11 & 3 \\ 5 & 5 & 8 & 15 & 5 \\ 6 & 8 & 9 & 21 & 7 \\ 11 & 15 & 21 & 39 & 13 \\ 3 & 5 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

[1558] $1^{1^{13}} 4^{1^{13}} 13^{1^{13}} 21^{1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 16 & 15 & 9 & 106 & 24 \\ 3 & 3 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

[1559] $1^{1^{14}} 4^{1^{14}} 13^{1^{14}} 21^{1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & 0 & 11 & 2 \\ 17 & 16 & 11 & 126 & 26 \\ 3 & 3 & 2 & 26 & 5 \end{pmatrix}_{b_0}$$

[1560] $1^{1^{14}} 4^{1^{14}} 13^{1^{14}} 21^{1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & 0 & 10 & 2 \\ 17 & 17 & 10 & 126 & 26 \\ 3 & 3 & 2 & 26 & 5 \end{pmatrix}_{b_0}$$

[1561] $1^{1^{14}} 4^{1^{14}} 13^{1^{14}} 21^{1^{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & 0 & 11 & 2 \\ 20 & 18 & 11 & 168 & 30 \\ 3 & 3 & 2 & 30 & 5 \end{pmatrix}_{b_0}$$

[1562] $1^{14} 4^{13} 1^{13} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 136$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 18 & 18 & 12 & 148 & 28 \\ 3 & 3 & 2 & 28 & 5 \end{pmatrix}_{b_0}$$

[1564] $1^{31} 5^{13} 1^{13} 3^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 18 & 3 & 3 \\ 5 & 5 & 24 & 3 & 5 \\ 18 & 24 & 97 & 15 & 21 \\ 3 & 3 & 15 & 1 & 3 \\ 3 & 5 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[1563] $31^{13} 3^{13} 1^{31} 5^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 18 & 5 & 3 \\ 5 & 5 & 24 & 7 & 5 \\ 18 & 24 & 97 & 30 & 21 \\ 5 & 7 & 30 & 8 & 6 \\ 5 & 5 & 21 & 6 & 4 \end{pmatrix}_{b_0}$$

[1565] $1^{14} 4^{13} 1^{13} 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 171$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 21 & 20 & 12 & 190 & 32 \\ 3 & 3 & 2 & 32 & 5 \end{pmatrix}_{b_0}$$

$T = 11$ [1570] $1^{13} 1^{31} 3^{13} 1^{11} 5^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 10 & 6 & 3 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 7 & 10 & 6 & 34 & 23 & 12 \\ 4 & 6 & 5 & 23 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1573] $1^{11} 3^{11} 1^{11} 5^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 10 & 2 \\ 2 & 0 & 4 & 6 & 23 & 6 \\ 7 & 0 & 10 & 23 & 67 & 17 \\ 2 & 0 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[1572] $1^{13} 1^{11} 5^{11} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 4 & 3 \\ 2 & 0 & 3 & 4 & 5 & 5 \\ 2 & 0 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1571] $1^{11} 1^{11} 3^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1566] $1^{11} 1^{11} 3^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 2 \\ 2 & 0 & 4 & 6 & 7 & 6 \\ 2 & 0 & 3 & 7 & 5 & 5 \\ 2 & 0 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1567] $1^{11} 1^{11} 3^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 3 & 1 & 1 & 6 & 3 \\ 4 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[1568] $1^{11}1^13^11^15^113^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 6 & 2 \\ 2 & 0 & 4 & 6 & 15 & 6 \\ 5 & 0 & 6 & 15 & 29 & 11 \\ 2 & 0 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

[1569] $1^{13}3^11^35^113^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1574] $1^{11}1^13^11^15^113^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 8 & 2 \\ 2 & 0 & 4 & 6 & 19 & 6 \\ 6 & 0 & 8 & 19 & 46 & 14 \\ 2 & 0 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[1575] $1^{12}3^11^25^113^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 11 & 2 \\ 2 & 0 & 4 & 6 & 27 & 6 \\ 9 & 0 & 11 & 27 & 96 & 20 \\ 2 & 0 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[1601] $1^{14}1^13^11^15^113^11^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 1 & 16 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 60 & 24 \\ 4 & 7 & 0 & 60 & 27 & 11 \\ 2 & 3 & -1 & 24 & 11 & 4 \end{pmatrix}_{b_0}$$

[1607] $1^{14}2^13^11^15^113^11^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & 1 & 16 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 76 & 24 \\ 5 & 9 & 0 & 76 & 44 & 14 \\ 2 & 3 & -1 & 24 & 14 & 4 \end{pmatrix}_{b_0}$$

[1583] $1^{11}3^11^15^113^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 3 & 4 & 4 & 6 & 3 & 6 \\ 2 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

[1584] $1^{12}1^13^11^15^113^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 1 & 8 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 12 \\ 3 & 5 & 0 & 22 & 14 & 8 \\ 2 & 3 & -1 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1585] $1^{12}3^11^15^113^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 6 & 8 & 8 & 30 & 6 & 12 \\ 2 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1586] \ 1^{131} 3^1 15 131^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 1 & 12 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 45 & 18 \\ 4 & 7 & 0 & 45 & 27 & 11 \\ 2 & 3 & -1 & 18 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1587] \ 1^{123} 1^{11} 5 131^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 12 \\ 2 & 3 & 0 & 14 & 5 & 5 \\ 2 & 3 & -1 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1588] \ 1^{113} 1^{12} 5 131^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 3 & 2 \\ 6 & 0 & 6 & 34 & 15 & 12 \\ 2 & 0 & 3 & 15 & 5 & 5 \\ 2 & 0 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1589] \ 1^{123} 1^{11} 5 131^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 3 & 2 \\ 6 & 0 & 9 & 49 & 18 & 15 \\ 2 & 0 & 3 & 18 & 5 & 5 \\ 2 & 0 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1590] \ 1^{132} 3^1 15 131^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 1 & 12 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 69 & 18 \\ 6 & 11 & 0 & 69 & 65 & 17 \\ 2 & 3 & -1 & 18 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1591] \ 1^{133} 3^1 15 131^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 1 & 12 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 93 & 18 \\ 8 & 15 & 0 & 93 & 119 & 23 \\ 2 & 3 & -1 & 18 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1592] \ 1^{131} 31 5 13^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 1 & 12 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 18 \\ 3 & 5 & 0 & 33 & 14 & 8 \\ 2 & 3 & -1 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1593] \ 1^{122} 31 5 13^1 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 8 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 12 \\ 5 & 9 & 0 & 38 & 44 & 14 \\ 2 & 3 & -1 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1594] \ 1^{132} 31 5 13^1 1^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

5.8

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & 1 & 12 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 57 & 18 \\ 5 & 9 & 0 & 57 & 44 & 14 \\ 2 & 3 & -1 & 18 & 14 & 4 \end{pmatrix}_{b_0}$$

[1595] $1^{114}31513^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 1 & 4 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 6 \\ 9 & 17 & 0 & 35 & 152 & 26 \\ 2 & 3 & -1 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[1596] $1^{113}31513^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 6 \\ 7 & 13 & 0 & 27 & 90 & 20 \\ 2 & 3 & -1 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[1581] $1^{124}31513^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 1 & 8 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 70 & 12 \\ 9 & 17 & 0 & 70 & 152 & 26 \\ 2 & 3 & -1 & 12 & 26 & 4 \end{pmatrix}_{b_0}$$

[1580] $1^{133}1^15131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 9 & 12 & 12 & 70 & 9 & 18 \\ 2 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

[1597] $1^{133}31513^{1133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 81 & 18 \\ 7 & 13 & 0 & 81 & 90 & 20 \\ 2 & 3 & -1 & 18 & 20 & 4 \end{pmatrix}_{b_0}$$

[1579] $1^{113}3^115131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 3 & 1 & 1 & 13 & 3 \\ 9 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1578] $1^{123}3^115131^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 1 & 8 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 62 & 12 \\ 8 & 15 & 0 & 62 & 119 & 23 \\ 2 & 3 & -1 & 12 & 23 & 4 \end{pmatrix}_{b_0}$$

[1577] $1^{112}3^115131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 23 & 6 \\ 6 & 11 & 0 & 23 & 65 & 17 \\ 2 & 3 & -1 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1576] \ 1^{123} 31^1 513^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 1 & 8 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 12 \\ 7 & 13 & 0 & 54 & 90 & 20 \\ 2 & 3 & -1 & 12 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1612] \ 1^{113} 31^1 5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 6 \\ 2 & 3 & 0 & 7 & 5 & 5 \\ 2 & 3 & -1 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1613] \ 1^{111} 31^1 5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 6 \\ 4 & 7 & 0 & 15 & 27 & 11 \\ 2 & 3 & -1 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1614] \ 1^{122} 31^1 5131^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 1 & 8 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 46 & 12 \\ 6 & 11 & 0 & 46 & 65 & 17 \\ 2 & 3 & -1 & 12 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1615] \ 1^{121} 31^1 5131^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 30 & 12 \\ 4 & 7 & 0 & 30 & 27 & 11 \\ 2 & 3 & -1 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1600] \ 1^{143} 31^1 5131^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 16 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 24 \\ 2 & 3 & 0 & 28 & 5 & 5 \\ 2 & 3 & -1 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1599] \ 1^{134} 31^1 5131^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 1 & 12 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 105 & 18 \\ 9 & 17 & 0 & 105 & 152 & 26 \\ 2 & 3 & -1 & 18 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1602] \ 1^{142} 31^1 5131^{43} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 1 & 1 & 16 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 92 & 24 \\ 6 & 11 & 0 & 92 & 65 & 17 \\ 2 & 3 & -1 & 24 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1603] \ 1^{143} 31^1 5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 8 & 14 & 10 & 74 & 10 & 18 \\ 2 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1604] \ 1^{143}3^1 1^1 5 131^{44} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 1 & 16 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 124 & 24 \\ 8 & 15 & 0 & 124 & 119 & 23 \\ 2 & 3 & -1 & 24 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1598] \ 1^{112}3^1 1^1 5 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 19 & 6 \\ 5 & 9 & 0 & 19 & 44 & 14 \\ 2 & 3 & -1 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1605] \ 1^{111}3^1 1^1 5 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 11 & 6 \\ 3 & 5 & 0 & 11 & 14 & 8 \\ 2 & 3 & -1 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1606] \ 1^{141}3^1 1^1 5 13^1 1^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 1 & 16 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 24 \\ 3 & 5 & 0 & 44 & 14 & 8 \\ 2 & 3 & -1 & 24 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1611] \ 1^{144}3^1 1^1 5 13^1 1^{44} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 1 & 16 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 140 & 24 \\ 9 & 17 & 0 & 140 & 152 & 26 \\ 2 & 3 & -1 & 24 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1610] \ 1^{113}3^1 1^1 5 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 31 & 6 \\ 8 & 15 & 0 & 31 & 119 & 23 \\ 2 & 3 & -1 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1609] \ 1^{123}3^1 1^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 2 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1608] \ 1^{143}3^1 1^1 5 13^1 1^{43} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 1 & 1 & 16 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 108 & 24 \\ 7 & 13 & 0 & 108 & 90 & 20 \\ 2 & 3 & -1 & 24 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1582] \ 1^{133}3^1 1^1 5 131^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 18 \\ 2 & 3 & 0 & 21 & 5 & 5 \\ 2 & 3 & -1 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1622] $1^{114}31513^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 12 & 15 & 92 & 20 \\ 2 & 3 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[1621] $1^{11}3^{11}15131^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 3 & 2 \\ 4 & 0 & 5 & 17 & 11 & 9 \\ 2 & 0 & 3 & 11 & 5 & 5 \\ 2 & 0 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

[1620] $1^{111}3151^{123}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 2 & 1 & 9 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 9 & 0 & 27 & 13 & 11 \\ 2 & 5 & 0 & 13 & 5 & 5 \\ 2 & 4 & -1 & 11 & 5 & 4 \end{pmatrix}_{b_0}$$

[1619] $1^{114}3^{11}15131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 3 & 17 & 3 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 3 & 1 & 1 & 17 & 3 \\ 11 & 17 & 14 & 17 & 121 & 23 \\ 2 & 3 & 2 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[1618] $1^{131}31513^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 7 & 10 & 6 & 34 & 15 & 12 \\ 2 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1617] $1^{123}3^{11}15131^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 8 & 13 & 3 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 5 & 8 & 8 & 30 & 46 & 12 \\ 8 & 13 & 10 & 46 & 67 & 17 \\ 2 & 3 & 2 & 12 & 17 & 4 \end{pmatrix}_{b_0}$$

[1616] $1^{111}31513^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 2 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1623] $1^{111}31513^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 3 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[1624] $1^{111}31513^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 2 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1625] $1^{13}3^1 1^1 5 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 7 & 11 & 9 & 49 & 8 & 15 \\ 2 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

[1626] $1^{12}3^1 1^1 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 3 & 2 \\ 8 & 0 & 10 & 74 & 22 & 18 \\ 2 & 0 & 3 & 22 & 5 & 5 \\ 2 & 0 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1629] $1^{11}3^1 1^1 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[1628] $1^{11}3^1 1^1 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 4 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[1627] $1^{11}3^1 1^1 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1632] $1^{11}3^1 1^1 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 5 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[1631] $1^{14}3^1 1^1 5 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 10 & 15 & 13 & 97 & 11 & 21 \\ 2 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

[1630] $1^{12}3^1 1^1 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 3 & 8 & 3 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 2 & 3 & 1 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 27 & 11 \\ 2 & 3 & 2 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1633] $1^{11}3^1 1^1 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 6 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 5 & 0 & 14 & 8 \\ 2 & 3 & 2 & 0 & 8 & 4 \end{pmatrix}_{b_0} \\
[1640] & 1^{133}1^{151}3^{231}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 & 3 \\ 15 & 0 & 27 & 259 & 24 & 33 \\ 2 & 0 & 3 & 24 & 1 & 3 \\ 2 & 0 & 3 & 33 & 3 & 4 \end{pmatrix}_{b_0} \\
[1639] & 1^{121}3^{151}5^{13}1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 2 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0} \\
[1638] & 1^{123}1^{151}2^{231}1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 & 3 \\ 10 & 0 & 18 & 114 & 16 & 22 \\ 2 & 0 & 3 & 16 & 1 & 3 \\ 2 & 0 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0} \\
[1637] & 1^{113}1^{151}1^{231}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 5 & 0 & 9 & 27 & 8 & 11 \\ 2 & 0 & 3 & 8 & 1 & 3 \\ 2 & 0 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
[1636] & 1^{111}3^{151}5^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 2 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0} \\
[1635] & 1^{141}3^{151}5^{13}1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 & 2 \\ 1 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 12 & 16 & 16 & 126 & 28 & 24 \\ 2 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0} \\
[1634] & 1^{131}3^{151}5^{13}1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 2 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0} \\
[1644] & 1^{112}3^{151}5^{13}1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -25, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 2 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0} \\
[1643] & 1^{113}3^{151}5^{131}1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -25, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 23 & 6 \\ 9 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[1642] $1^{11}31^1513^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -25, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 4 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1641] $1^{12}31^1513^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -25, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 5 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1646] $1^{12}3^{11}1^1513^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 7 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1645] $1^{13}3^{11}1^1513^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 10 & 6 & 3 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 6 & 10 & 6 & 34 & 23 & 12 \\ 4 & 6 & 5 & 23 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1648] $1^{12}3^{11}1^1513^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 9 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 6 & 0 & 29 & 11 \\ 2 & 3 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[1647] $1^{12}3^{11}31^1513^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 7 & 5 & 17 & 11 & 9 \\ 2 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

[1649] $1^{12}3^{11}31^1513^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 7 & 0 & 27 & 11 \\ 2 & 3 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[1650] $1^{11}3^{11}31^1513^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1653] $1^{13}3^{11}31^1513^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 18 & 2 \\ 2 & 4 & 4 & 6 & 19 & 6 \\ 7 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1651] \ 1^{131} 31 \ 5 \ 13^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 6 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1652] \ 1^{111} 31 \ 5 \ 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1654] \ 1^{122} 31 \ 5 \ 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 4 & 7 & 5 & 17 & 23 & 9 \\ 6 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1655] \ 1^{121} 31 \ 5 \ 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -46, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 4 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1657] \ 1^{121} 31 \ 5 \ 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 6 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1658] \ 1^{111} 31^{151} 1231^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 2 & 0 & 4 & 2 & 12 & 4 \\ 7 & 0 & 12 & 12 & 46 & 14 \\ 2 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1659] \ 1^{13} 31 \ 5 \ 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 8 & 0 & 46 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1660] \ 1^{13} 31 \ 5 \ 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 8 & 0 & 46 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1656] \ 1^{112} 31^{151} 1231^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 16 & 4 \\ 7 & 16 & 0 & 16 & 74 & 18 \\ 2 & 4 & -1 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1661] \ 1^{1231} 1^{12} 5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 9 & 0 & 44 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1663] \ 2 \ 3 \ 1513^{141} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 3 & 2 \\ 2 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 12 & 16 & 16 & 126 & 28 & 24 \\ 3 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1665] \ 2 \ 3 \ 1513^{131} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 3 & 2 \\ 2 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1662] \ 2 \ 3 \ 1513^{111} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1664] \ 2 \ 3 \ 1513^{121} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 3 & 2 \\ 2 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1666] \ 1^{111} 3^{11} 5 \ 131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 11 & 6 \\ 5 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1668] \ 1^{131} 31^{13} 5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 8 & 11 & 9 & 49 & 18 & 15 \\ 2 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1667] \ 1^{141} 31^{14} 5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 10 & 14 & 10 & 74 & 22 & 18 \\ 2 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1671] $1^{121}31\ 5\ 13^11^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -57, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 7 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1670] $1^{131}31\ 5\ 13^11^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -57, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 8 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1669] $1^{131}31\ 5\ 13^11^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -57, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 6 & 10 & 6 & 34 & 15 & 12 \\ 3 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1672] $1^{121}31\ 5\ 13^11^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1673] $1^{113}31\ 5\ 13^11^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1674] $1^{113}31\ 5\ 13^11^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 13 & 3 \\ 8 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1675] $1^{111}31\ 5\ 13^11^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -66, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[1676] $1^{121}31\ 5\ 13^11^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

[1679] $1^{131}31\ 5\ 13^11^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 7 & 10 & 6 & 34 & 15 & 12 \\ 3 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1678] $1^{\overline{1143}} 1^{\overline{13}} 1^{\overline{13}} 5 131^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 3 & 17 & 3 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 3 & 1 & 1 & 17 & 3 \\ 11 & 17 & 14 & 17 & 121 & 23 \\ 2 & 3 & 2 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[1677] $1^{\overline{113}} 31^{\overline{13}} 5 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1680] $1^{\overline{13}} 31^{\overline{13}} 5 131^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 11 & 0 & 65 & 17 \\ 2 & 3 & 2 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[1681] $1^{\overline{121}} 31^{\overline{11}} 1^{\overline{11}} 5 131^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 5 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1682] $1^{\overline{121}} 31^{\overline{11}} 1^{\overline{11}} 5 131^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 6 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 17 & 9 \\ 4 & 6 & 5 & 17 & 14 & 8 \\ 2 & 3 & 2 & 9 & 8 & 4 \end{pmatrix}_{b_0}$$

[1683] $1^{\overline{13}} 31^{\overline{13}} 1^{\overline{13}} 5 131^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 13 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 13 & 10 & 0 & 67 & 17 \\ 2 & 3 & 2 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[1684] $1^{\overline{121}} 31^{\overline{11}} 1^{\overline{11}} 5 131^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -77, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 7 & 6 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 17 & 9 \\ 4 & 6 & 5 & 17 & 14 & 8 \\ 2 & 3 & 2 & 9 & 8 & 4 \end{pmatrix}_{b_0}$$

[1686] $1^{\overline{123}} 31^{\overline{13}} 5 131^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 30 & 9 \\ 8 & 11 & 8 & 30 & 46 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0}$$

[1685] $1^{\overline{122}} 31^{\overline{12}} 1^{\overline{12}} 5 131^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 29 & 9 \\ 7 & 10 & 9 & 29 & 44 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1687] \ 1^{112} 3^{11} 5^{12} 13^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 19 & 6 \\ 7 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1688] \ 1^{141} 3^{11} 5^{13} 11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 11 & 15 & 13 & 97 & 25 & 21 \\ 2 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1690] \ 1^{113} 3^{11} 5^{13} 11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 13 & 3 \\ 9 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1689] \ 1^{114} 3^{11} 5^{13} 11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 12 & 15 & 92 & 20 \\ 2 & 3 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1691] \ 1^{114} 3^{11} 5^{13} 11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 17 & 3 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 4 & 4 & 6 & 31 & 6 \\ 12 & 17 & 14 & 31 & 121 & 23 \\ 2 & 3 & 2 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1692] \ 1^{121} 3^{11} 5^{13} 11^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -90, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1695] \ 1^{112} 3^{11} 5^{13} 11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 5 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1694] \ 1^{122} 3^{11} 5^{13} 11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 9 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 6 & 9 & 6 & 24 & 29 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

[1693] $1^{122}31\ 5\ 13\ 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 4 & 7 & 5 & 17 & 23 & 9 \\ 5 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

[1699] $1^{133}1\ 5\ 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 13 & 0 & 90 & 20 \\ 2 & 3 & 2 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[1698] $1^{113}31\ 5\ 13\ 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 7 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[1697] $1^{113}31\ 5\ 13\ 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 8 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[1696] $1^{131}3\ 1\ 5\ 131^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 6 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

[1701] $1^{141}31\ 5\ 13\ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 8 & 14 & 10 & 74 & 22 & 18 \\ 3 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1700] $1^{131}31\ 5\ 13\ 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1702] $1^{143}31\ 5\ 13\ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 12 & 0 & 92 & 20 \\ 2 & 3 & 2 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[1703] $1^{141}31\ 5\ 13\ 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 3 & 2 \\ 1 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 11 & 16 & 16 & 126 & 28 & 24 \\ 3 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

[1704] $1^{113}3^{11}5^{12}13^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 23 & 6 \\ 8 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[1705] $1^{131}3^{11}5^{12}13^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -106, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 7 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

[1706] $1^{141}3^{11}5^{12}13^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 9 & 15 & 13 & 97 & 25 & 21 \\ 3 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

[1707] $1^{112}3^{11}5^{12}13^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

[1708] $1^{122}3^{11}5^{12}13^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 10 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 38 & 12 \\ 7 & 10 & 9 & 38 & 44 & 14 \\ 2 & 3 & 2 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[1709] $1^{131}3^{11}5^{12}13^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -127, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 8 & 12 & 12 & 70 & 33 & 18 \\ 4 & 6 & 5 & 33 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[1710] $1^{143}3^{11}5^{12}13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 0 & 17 & 3 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 17 & 14 & 0 & 121 & 23 \\ 2 & 3 & 2 & 0 & 23 & 4 \end{pmatrix}_{b_0}$$

[1712] $1^{141}3^{11}5^{12}13^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -130, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 14 & 6 & 3 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 8 & 14 & 10 & 74 & 34 & 18 \\ 4 & 6 & 5 & 34 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[1711] $1^{131}3^{11}5^{12}13^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -130, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 12 & 12 & 70 & 33 & 18 \\ 4 & 6 & 5 & 33 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1714] \quad 1^{113}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -135, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 27 & 6 \\ 10 & 14 & 13 & 27 & 90 & 20 \\ 2 & 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1713] \quad 1^{131}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -135, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 8 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1717] \quad 2^13^{122}15^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 6 & 2 \\ 2 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 5 & 7 & 5 & 17 & 23 & 9 \\ 6 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1716] \quad 1^{112}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 8 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1715] \quad 2^13^{112}15^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1718] \quad 1^{113}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -153, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 23 & 6 \\ 8 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1719] \quad 1^{131}3^{11}5^{13}1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -155, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1720] \quad 1^{141}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -161, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 9 & 14 & 10 & 74 & 22 & 18 \\ 3 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1721] \quad & 1^{112}3^11^5131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -162, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 7 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0} \\
[1722] \quad & 1^{131}3^11^5131^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -166, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 7 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0} \\
[1723] \quad & 1^{141}3^11^5131^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -171, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 14 & 6 & 3 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 9 & 14 & 10 & 74 & 34 & 18 \\ 4 & 6 & 5 & 34 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0} \\
[1724] \quad & 1^{123}3^11^5131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -171, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 13 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 36 & 9 \\ 9 & 13 & 10 & 36 & 67 & 17 \\ 2 & 3 & 2 & 9 & 17 & 4 \end{pmatrix}_{b_0} \\
[1725] \quad & 1^{123}3^11^513^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -172, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 30 & 9 \\ 7 & 11 & 8 & 30 & 46 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0} \\
[1726] \quad & 1^{113}3^11^5131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -177, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 23 & 6 \\ 9 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0} \\
[1727] \quad & 1^{141}3^11^513^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -182, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 10 & 15 & 13 & 97 & 25 & 21 \\ 3 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0} \\
[1728] \quad & 1^{133}3^11^5131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -183, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 13 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 11 & 9 & 49 & 59 & 15 \\ 8 & 13 & 10 & 59 & 67 & 17 \\ 2 & 3 & 2 & 15 & 17 & 4 \end{pmatrix}_{b_0} \\
[1729] \quad & 1^{141}3^11^5131^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -195, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 1 & 2 & 16 & 6 & 3 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 11 & 16 & 16 & 126 & 44 & 24 \\ 4 & 6 & 5 & 44 & 14 & 8 \\ 2 & 3 & 2 & 24 & 8 & 4 \end{pmatrix}_{b_0}$$

[1730] $1^{114}3^{11}5^{13}13^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -197, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 4 & 17 & 3 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 4 & 4 & 6 & 31 & 6 \\ 11 & 17 & 14 & 31 & 121 & 23 \\ 2 & 3 & 2 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

[1731] $1^{114}3^{11}5^{13}13^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -207, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 1 & 2 & 4 & 18 & 3 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 2 & 4 & 4 & 6 & 35 & 6 \\ 13 & 18 & 17 & 35 & 152 & 26 \\ 2 & 3 & 2 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[1732] $1^{141}3^{11}5^{13}13^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -217, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 15 & 6 & 3 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 9 & 15 & 13 & 97 & 39 & 21 \\ 4 & 6 & 5 & 39 & 14 & 8 \\ 2 & 3 & 2 & 21 & 8 & 4 \end{pmatrix}_{b_0}$$

[1733] $1^{123}3^{11}5^{13}13^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -238, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 13 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 36 & 9 \\ 8 & 13 & 10 & 36 & 67 & 17 \\ 2 & 3 & 2 & 9 & 17 & 4 \end{pmatrix}_{b_0}$$

[1734] $1^{141}3^{11}5^{13}13^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -272, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & 2 & 15 & 6 & 3 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 10 & 15 & 13 & 97 & 39 & 21 \\ 4 & 6 & 5 & 39 & 14 & 8 \\ 2 & 3 & 2 & 21 & 8 & 4 \end{pmatrix}_{b_0}$$

[1735] $1^{124}3^{11}5^{13}13^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -294, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 16 & 3 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 10 & 16 & 11 & 43 & 96 & 20 \\ 2 & 3 & 2 & 9 & 20 & 4 \end{pmatrix}_{b_0}$$

[1736] $1^{113}3^{11}5^{13}13^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -306, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 27 & 6 \\ 10 & 14 & 13 & 27 & 90 & 20 \\ 2 & 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[1737] $1^{124}3^{11}5^{13}13^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -325, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 7 & 17 & 3 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 4 & 7 & 5 & 17 & 48 & 9 \\ 11 & 17 & 14 & 48 & 121 & 23 \\ 2 & 3 & 2 & 9 & 23 & 4 \end{pmatrix}_{b_0}$$

[1738] $1^{114}3^{11}5^{13}13^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -333, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

[1754] $1^{111}3^11^151321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

[1751] $1^{11}6^{1^13^12^1}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[1752] $1^{131}3^11^151321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 1 & 15 & 6 & 3 \\ 8 & 12 & 15 & 79 & 36 & 21 \\ 4 & 6 & 6 & 36 & 15 & 9 \\ 2 & 3 & 3 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

[1753] $1^{141}3^11^151321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 1 & 2 & 16 & 6 & 3 \\ 1 & 2 & 1 & 20 & 6 & 3 \\ 11 & 16 & 20 & 142 & 48 & 28 \\ 4 & 6 & 6 & 48 & 15 & 9 \\ 2 & 3 & 3 & 28 & 9 & 5 \end{pmatrix}_{b_0}$$

[1758] $1^{113}3^11^151321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 6 & 3 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 6 & 0 & 15 & 9 \\ 2 & 3 & 3 & 0 & 9 & 5 \end{pmatrix}_{b_0}$$

[1757] $1^{111}3^11^151321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 4 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

[1756] $1^{121}3^11^151321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 5 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

[1755] $1^{121}3^11^151321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 7 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

[1761] $1^{123}3^11^151321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 2 & 2 & 18 & 5 & 4 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 8 & 18 & 10 & 90 & 26 & 22 \\ 3 & 5 & 3 & 26 & 6 & 6 \\ 2 & 4 & 2 & 22 & 6 & 5 \end{pmatrix}_{b_0}$$

[1759] $1^{11}3^11^151321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 2 & 2 & 9 & 5 & 4 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 9 & 5 & 21 & 13 & 11 \\ 2 & 5 & 3 & 13 & 6 & 6 \\ 2 & 4 & 2 & 11 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1760] \ 1^{11} 3^1 1^5 1321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 2 & 2 & 9 & 5 & 4 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 4 & 9 & 5 & 21 & 13 & 11 \\ 3 & 5 & 3 & 13 & 6 & 6 \\ 2 & 4 & 2 & 11 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1762] \ 1^{12} 3^1 1^5 1321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 2 & 2 & 18 & 5 & 4 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 10 & 18 & 10 & 90 & 26 & 22 \\ 2 & 5 & 3 & 26 & 6 & 6 \\ 2 & 4 & 2 & 22 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1763] \ 1^{11} 3^{121} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 5 & 5 & 3 & 7 & 7 \\ 3 & 3 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1769] \ 1^{112} 3^1 1^5 1321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 2 & 4 & 5 & 7 & 21 & 7 \\ 7 & 10 & 11 & 21 & 48 & 16 \\ 2 & 3 & 3 & 7 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[1768] \ 1^{122} 3^1 1^5 1321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 31 & 10 \\ 7 & 10 & 11 & 31 & 48 & 16 \\ 2 & 3 & 3 & 10 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[1767] \ 1^{111} 3^1 1^5 1321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 5 & 4 & 7 & 22 & 7 \\ 7 & 14 & 8 & 22 & 55 & 17 \\ 2 & 4 & 2 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[1766] \ 1^{122} 3^1 1^5 1321^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 10 & 3 \\ 1 & 2 & 1 & 10 & 11 & 3 \\ 5 & 8 & 10 & 34 & 42 & 14 \\ 7 & 10 & 11 & 42 & 48 & 16 \\ 2 & 3 & 3 & 14 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[1765] \ 1^{123} 1^1 5 1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 11 & 0 & 48 & 16 \\ 2 & 3 & 3 & 0 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[1764] \ 1^{111} 3^1 1^5 1321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 5 & 4 & 7 & 22 & 7 \\ 7 & 14 & 8 & 22 & 55 & 17 \\ 2 & 4 & 2 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[1770] $1^{11}3^{11}5^{12}1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 4 & 14 & 4 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 4 & 1 & 2 & 14 & 4 \\ 7 & 14 & 8 & 14 & 55 & 17 \\ 2 & 4 & 2 & 4 & 17 & 5 \end{pmatrix}_{b_0}$$

[1771] $1^{11}6^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 8 & 7 & 5 & 21 & 11 \\ 3 & 3 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[1772] $1^{11}6^{11}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 6 & 2 \\ 7 & 9 & 0 & 6 & 34 & 12 \\ 2 & 3 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1773] $1^{11}3^{11}5^{11}1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

[1774] $1^{131}3^{11}5^{131}1321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 1 & 15 & 6 & 3 \\ 9 & 12 & 15 & 79 & 36 & 21 \\ 4 & 6 & 6 & 36 & 15 & 9 \\ 2 & 3 & 3 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

[1775] $1^{121}3^{11}5^{121}1321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 6 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

[1776] $1^{133}3^{11}5^{133}1321^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 16 & 0 & 99 & 23 \\ 2 & 3 & 3 & 0 & 23 & 5 \end{pmatrix}_{b_0}$$

[1777] $1^{133}3^{11}5^{133}1321^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 30 & 7 \\ 10 & 14 & 16 & 30 & 99 & 23 \\ 2 & 3 & 3 & 7 & 23 & 5 \end{pmatrix}_{b_0}$$

[1778] $1^{123}3^{11}5^{123}1321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 14 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 60 & 14 \\ 10 & 14 & 16 & 60 & 99 & 23 \\ 2 & 3 & 3 & 14 & 23 & 5 \end{pmatrix}_{b_0}$$

[1779] $1^{11} 6^{1^{12} 3^{12} 1^1} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 11 & 11 & 6 & 42 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

[1780] $1^{11} 6^{1^{11} 3^{12} 21^1} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 7 & 2 \\ 9 & 12 & 1 & 7 & 56 & 16 \\ 2 & 3 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1781] $1^{1143} 1^{14} 15 1321^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 1 & 2 & 4 & 18 & 3 \\ 1 & 2 & 1 & 5 & 21 & 3 \\ 2 & 4 & 5 & 7 & 39 & 7 \\ 13 & 18 & 21 & 39 & 168 & 30 \\ 2 & 3 & 3 & 7 & 30 & 5 \end{pmatrix}_{b_0}$$

[1782] $1^{111} 3^{112} 1^1 5 1321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 9 & 5 & 21 & 36 & 11 \\ 7 & 14 & 8 & 36 & 55 & 17 \\ 2 & 4 & 2 & 11 & 17 & 5 \end{pmatrix}_{b_0}$$

[1783] $1^{112} 3^{112} 1^1 5 1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 21 & 7 \\ 7 & 10 & 11 & 21 & 48 & 16 \\ 2 & 3 & 3 & 7 & 16 & 5 \end{pmatrix}_{b_0}$$

[1784] $1^{113} 3^{113} 1^1 5 1321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 196, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 30 & 7 \\ 10 & 14 & 16 & 30 & 99 & 23 \\ 2 & 3 & 3 & 7 & 23 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [1785] $1^{141} 3^{115} 151321^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 8 & 4 & 3 \\ 2 & 2 & 11 & 4 & 4 \\ 8 & 11 & 35 & 14 & 15 \\ 4 & 4 & 14 & 4 & 6 \\ 3 & 4 & 15 & 6 & 6 \end{pmatrix}_{b_0}$$

[1786] $1^{11413} 1^{111} 151321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 3 \\ 2 & 2 & 5 & 4 & 4 \\ 4 & 5 & 7 & 6 & 7 \\ 4 & 4 & 6 & 4 & 6 \\ 3 & 4 & 7 & 6 & 6 \end{pmatrix}_{b_0}$$

[1789] $1^{121} 41^{131} 151321^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 1 & 0 & 9 & 3 & 2 \\ 8 & 6 & 9 & 47 & 20 & 17 \\ 4 & 3 & 3 & 20 & 7 & 7 \\ 2 & 2 & 2 & 17 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1788] \ 1^{11}41^{13}151321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 4 & 2 & 3 & 7 & 8 & 7 \\ 4 & 3 & 3 & 8 & 7 & 7 \\ 2 & 2 & 2 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1787] \ 1^{11}413^{11}151321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 2 & 2 & 3 & 3 & 6 & 5 \\ 4 & 3 & 3 & 6 & 7 & 5 \\ 2 & 2 & 2 & 5 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1790] \ 1^{11}41^{13}151321^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 12 & 4 & 3 \\ 2 & 2 & 16 & 4 & 4 \\ 12 & 16 & 74 & 20 & 22 \\ 4 & 4 & 20 & 4 & 6 \\ 3 & 4 & 22 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[1791] \ 1^{11}41^{13}151321^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 4 & 3 & 2 \\ 1 & 1 & 0 & 6 & 3 & 2 \\ 6 & 4 & 6 & 22 & 14 & 12 \\ 4 & 3 & 3 & 14 & 7 & 7 \\ 2 & 2 & 2 & 12 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1792] \ 1^{112}41^{13}151321^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 0 & 3 & 9 & 2 \\ 2 & 2 & 3 & 3 & 16 & 5 \\ 10 & 7 & 9 & 16 & 59 & 19 \\ 2 & 2 & 2 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [1793] \ 1^{121}3161^{123}1513^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 & 3 \\ 2 & 2 & 13 & 4 & 4 \\ 12 & 13 & 58 & 15 & 20 \\ 3 & 4 & 15 & 3 & 5 \\ 3 & 4 & 20 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[1796] \ 1^{11}6^{113}15131^{11}131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 1 & 5 & 4 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 5 & 3 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[1795] \ 1^{111}321^{16}131513^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 & 3 \\ 2 & 2 & 9 & 4 & 4 \\ 7 & 9 & 27 & 10 & 13 \\ 3 & 4 & 10 & 3 & 5 \\ 3 & 4 & 13 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[1794] \ 1^{121}3161^{123}1513^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 9 & 3 & 3 \\ 2 & 2 & 11 & 4 & 4 \\ 9 & 11 & 40 & 12 & 16 \\ 3 & 4 & 12 & 3 & 5 \\ 3 & 4 & 16 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[1797] \ 1^{112}3161^{123}1513^{1113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 0 & 3 & 9 & 2 \\ 2 & 2 & 3 & 3 & 16 & 5 \\ 10 & 7 & 9 & 16 & 59 & 19 \\ 2 & 2 & 2 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [1799] \quad 1^{1^{11}3^{11}151321} 8^{111} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

$$[1798] \quad 1^{1^{11}3^{11}151321} 8^{111} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

$$[1800] \quad 1^{1^{11}3^{11}151321} 8^{111} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

4.939 NHC(-2)(-3) + su₂ + su2n3mix + f₄

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 4 & 7 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

4.940 NHC(-2)(-3) + su₂ + su2n3mix + f₄ + su₂

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 & 4 \\ 3 & 5 & 4 & 6 & 0 & 3 & 4 \\ 4 & 7 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 5 & 7 & 0 & 17 & 4 \\ 3 & 5 & 4 & 6 & 0 & 15 & 4 \\ 4 & 7 & 6 & 7 & 0 & 21 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 21 & 0 & 49 & 13 \\ 3 & 4 & 4 & 5 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 0 & 9 & 4 \\ 3 & 5 & 4 & 6 & 0 & 9 & 4 \\ 4 & 7 & 6 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 11 & 0 & 13 & 7 \\ 3 & 4 & 4 & 5 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 7 & 0 & 13 & 4 \\ 3 & 5 & 4 & 6 & 0 & 12 & 4 \\ 4 & 7 & 6 & 7 & 0 & 16 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 16 & 0 & 28 & 10 \\ 3 & 4 & 4 & 5 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

4.941 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 & 4 \\ 3 & 5 & 4 & 6 & 0 & 3 & 4 \\ 4 & 7 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.942 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 4 & 7 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 1 & 3 \\ 3 & 4 & 14 & 7 & 0 & 4 \\ 9 & 14 & 37 & 17 & 0 & 11 \\ 4 & 7 & 17 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 3 & 4 & 1 & 5 & 0 & 3 \\ 4 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 2 & 4 & 1 & 5 & 0 & 3 \\ 5 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 1 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 \\ 6 & 9 & 13 & 11 & 0 & 7 \\ 4 & 7 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 1 & 8 & 0 & 3 \\ 8 & 12 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 8 & 12 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 12 & 0 & 4 \\ 5 & 9 & 13 & 20 & 0 & 7 \\ 8 & 12 & 20 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 16 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 10 & 16 & 15 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 1 & 11 & 0 & 3 \\ 10 & 16 & 11 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 16 & 0 & 4 \\ 5 & 9 & 13 & 26 & 0 & 7 \\ 10 & 16 & 26 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

4.943 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 3 & 4 \\ 3 & 5 & 4 & 6 & 0 & 3 & 4 \\ 4 & 7 & 6 & 7 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & 5 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.944 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2$

$T = 9$ [12] $1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 10 & 4 \\ 3 & 1 & 3 & 3 & 8 & 3 \\ 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 1 & 8 & 3 \\ 10 & 8 & 8 & 8 & 22 & 10 \\ 4 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{11} 3^{11} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 18 & 4 \\ 3 & 1 & 3 & 3 & 12 & 3 \\ 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 3 & 2 & 1 & 12 & 3 \\ 18 & 12 & 16 & 12 & 70 & 18 \\ 4 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 5 & 5 & 8 & 0 & 5 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 5 & 8 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 4 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

[0] $1^{11} 3^{11} 21^{1223} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 8 & 3 \\ 3 & 5 & 25 & 13 & 15 & 5 \\ 13 & 25 & 92 & 49 & 57 & 20 \\ 8 & 13 & 49 & 24 & 30 & 10 \\ 8 & 15 & 57 & 30 & 34 & 12 \\ 3 & 5 & 20 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[illegible]

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 8 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 8 & 4 & 3 \\ 4 & 5 & 0 & 8 & 23 & 13 & 9 \\ 2 & 3 & 0 & 4 & 13 & 5 & 5 \\ 2 & 2 & -1 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{124} 3^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 16 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 48 & 10 \\ 16 & 25 & 0 & 48 & 94 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{21} 3^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 1 & 7 & 8 & 2 \\ 1 & 2 & 0 & 3 & 9 & 9 & 2 \\ 1 & 1 & 3 & 1 & 9 & 9 & 3 \\ 7 & 7 & 9 & 9 & 35 & 39 & 11 \\ 6 & 8 & 9 & 9 & 39 & 40 & 12 \\ 2 & 2 & 2 & 3 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 4 & 4 & 3 \\ 2 & 2 & 0 & 4 & 4 & 6 & 4 \\ 2 & 3 & 0 & 4 & 6 & 5 & 5 \\ 2 & 2 & -1 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{132} 3^{133} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 6 & 2 \\ 1 & 0 & 1 & 9 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 18 & 54 & 15 \\ 2 & 2 & 0 & 18 & 4 & 16 & 4 \\ 6 & 7 & 0 & 54 & 16 & 45 & 13 \\ 2 & 2 & -1 & 15 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 0 & 1 & 3 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 17 & 7 & 5 \\ 6 & 6 & 0 & 17 & 42 & 17 & 12 \\ 2 & 3 & 0 & 7 & 17 & 5 & 5 \\ 2 & 2 & -1 & 5 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1211} 3^{1222} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 4 & 2 \\ 1 & 0 & 1 & 6 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 26 & 22 & 10 \\ 4 & 5 & 0 & 26 & 23 & 21 & 9 \\ 4 & 4 & 0 & 22 & 21 & 16 & 8 \\ 2 & 2 & -1 & 10 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1211} 3^{1411} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 3 & 2 \\ 1 & 2 & 0 & 14 & 5 & 4 & 2 \\ 10 & 8 & 14 & 70 & 28 & 22 & 16 \\ 4 & 4 & 5 & 28 & 10 & 9 & 6 \\ 4 & 3 & 4 & 22 & 9 & 5 & 5 \\ 2 & 2 & 2 & 16 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{211} 3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 20 & 7 & 4 \\ 3 & 1 & 3 & 16 & 5 & 3 \\ 3 & 3 & 1 & 16 & 6 & 3 \\ 20 & 16 & 16 & 94 & 32 & 20 \\ 7 & 5 & 6 & 32 & 9 & 7 \\ 4 & 3 & 3 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3^{121} 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 15 & 11 & 4 \\ 3 & 1 & 3 & 9 & 7 & 3 \\ 3 & 3 & 1 & 14 & 10 & 3 \\ 15 & 9 & 14 & 49 & 37 & 15 \\ 11 & 7 & 10 & 37 & 25 & 11 \\ 4 & 3 & 3 & 15 & 11 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{121} 3^{112} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 8 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 2 & 5 & 2 & 10 & 11 & 4 \\ 8 & 13 & 10 & 24 & 29 & 10 \\ 8 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3^{112} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 2 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 5 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{113} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 2 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 7 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 7 & 4 \\ 3 & 1 & 3 & 3 & 5 & 3 \\ 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 4 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 1 & 2 \\ 1 & 0 & 2 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & 1 & 0 & 4 & 0 & 2 \\ 4 & 4 & 5 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{122} 3^{121} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 5 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 3 & 1 & 3 & 1 & 8 & 7 & 3 \\ 6 & 5 & 8 & 8 & 23 & 21 & 9 \\ 5 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{111} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 1 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 2 & 2 & 1 & 0 & 4 & 2 & 2 \\ 4 & 4 & 5 & 4 & 10 & 4 & 6 \\ 1 & 2 & 1 & 2 & 4 & 0 & 2 \\ 2 & 2 & 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{111} 3^{121} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 1 & 1 & 3 & 1 & 5 & 4 & 3 \\ 4 & 4 & 5 & 5 & 10 & 9 & 6 \\ 3 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{11} 1^{12} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 22 & 10 \\ 2 & 2 & 0 & 12 & 4 & 10 & 4 \\ 4 & 4 & 0 & 22 & 10 & 16 & 8 \\ 2 & 2 & -1 & 10 & 4 & 8 & 3 \end{pmatrix}_{b_0}$

[37] $1^{121} 1^{11} 3^1 21^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 2 & 5 & 2 & 10 & 7 & 4 \\ 8 & 13 & 10 & 24 & 19 & 10 \\ 6 & 10 & 7 & 19 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$

[25] $1^{1111} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 2 & 1 & 0 & 4 & 2 & 2 \\ 5 & 4 & 5 & 4 & 10 & 4 & 6 \\ 2 & 2 & 1 & 2 & 4 & 0 & 2 \\ 2 & 2 & 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[36] $1^{111} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 5 & 4 & 5 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[35] $1^{111} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 1 & 1 & 3 & 1 & 8 & 7 & 3 \\ 7 & 5 & 8 & 8 & 23 & 21 & 9 \\ 6 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$

[24] $1^{111} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 3 & 1 & 3 & 1 & 5 & 4 & 3 \\ 5 & 4 & 5 & 5 & 10 & 9 & 6 \\ 4 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[39] $1^{11} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 & 11 & 4 \\ 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 3 & 1 & 4 & 10 & 3 \\ 4 & 2 & 4 & 2 & 10 & 4 \\ 11 & 7 & 10 & 10 & 25 & 11 \\ 4 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$

[38] $1^{111} 1^{11} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 & 7 & 4 \\ 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 3 & 1 & 4 & 6 & 3 \\ 4 & 2 & 4 & 2 & 6 & 4 \\ 7 & 5 & 6 & 6 & 9 & 7 \\ 4 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$

[42] $1^{123} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 1 & 2 \\ 1 & 0 & 2 & 5 & 10 & 0 & 2 \\ 1 & 2 & 0 & 5 & 13 & 0 & 7 \\ 4 & 5 & 5 & 13 & 33 & 0 & 16 \\ 11 & 10 & 13 & 33 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 7 & 16 & 0 & 3 \end{pmatrix}_{b_0}$

[44] $1^{11} 1^{11} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 5 & 10 & 14 & 12 & 7 & 8 \\ 5 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 7 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{11} 3^{11} 21^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 2 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11} 3^{11} 21^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 2 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 5 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11} 3^{11} 21^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 4 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{114} 3^{114} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 25 & 4 \\ 3 & 1 & 3 & 3 & 17 & 3 \\ 3 & 3 & 1 & 2 & 22 & 3 \\ 3 & 3 & 2 & 1 & 17 & 3 \\ 25 & 17 & 22 & 17 & 137 & 25 \\ 4 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{112} 3^{112} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 17 & 4 \\ 3 & 1 & 3 & 3 & 13 & 3 \\ 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 1 & 13 & 3 \\ 17 & 13 & 14 & 13 & 65 & 17 \\ 4 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{114} 3^{114} 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 1 & 1 & 3 & 1 & 5 & 11 & 3 \\ 4 & 4 & 5 & 5 & 10 & 23 & 6 \\ 9 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{114} 3^{114} 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 11 & 2 \\ 1 & 2 & 0 & 3 & 5 & 13 & 2 \\ 1 & 1 & 3 & 1 & 5 & 13 & 3 \\ 4 & 4 & 5 & 5 & 10 & 31 & 6 \\ 9 & 11 & 13 & 13 & 31 & 81 & 17 \\ 2 & 2 & 2 & 3 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{112} 3^{112} 21^{1112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 1 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 6 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{112} 3^{112} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 14 & 4 \\ 3 & 1 & 3 & 3 & 10 & 3 \\ 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 3 & 2 & 1 & 10 & 3 \\ 14 & 10 & 12 & 10 & 42 & 14 \\ 4 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{121}3^121^{211}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 0 & 1 & 6 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 14 & 10 \\ 2 & 2 & 0 & 12 & 4 & 6 & 4 \\ 2 & 3 & 0 & 14 & 6 & 5 & 5 \\ 2 & 2 & -1 & 10 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{122}3^121^{212}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 28 & 10 \\ 2 & 2 & 0 & 12 & 4 & 12 & 4 \\ 4 & 6 & 0 & 28 & 12 & 26 & 10 \\ 2 & 2 & -1 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{12}3^{114}21^{114}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 22 & 4 \\ 3 & 1 & 3 & 2 & 14 & 3 \\ 3 & 3 & 1 & 4 & 20 & 3 \\ 4 & 2 & 4 & 2 & 20 & 4 \\ 22 & 14 & 20 & 20 & 106 & 22 \\ 4 & 3 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{112}3^121^{112}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 14 & 4 \\ 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 3 & 1 & 4 & 12 & 3 \\ 4 & 2 & 4 & 2 & 12 & 4 \\ 14 & 10 & 12 & 12 & 42 & 14 \\ 4 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{1121}3^121^{131}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 0 & 1 & 3 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 20 & 7 & 5 \\ 6 & 8 & 0 & 20 & 56 & 20 & 14 \\ 2 & 3 & 0 & 7 & 20 & 5 & 5 \\ 2 & 2 & -1 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{124}3^121^{214}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 56 & 10 \\ 2 & 2 & 0 & 12 & 4 & 24 & 4 \\ 8 & 12 & 0 & 56 & 24 & 110 & 20 \\ 2 & 2 & -1 & 10 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{111}3^121^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 7 & 5 \\ 2 & 2 & 0 & 6 & 4 & 6 & 4 \\ 2 & 3 & 0 & 7 & 6 & 5 & 5 \\ 2 & 2 & -1 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{122}3^121^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 10 & 4 \\ 3 & 1 & 3 & 8 & 8 & 3 \\ 3 & 3 & 1 & 8 & 8 & 3 \\ 10 & 8 & 8 & 22 & 24 & 10 \\ 10 & 8 & 8 & 24 & 22 & 10 \\ 4 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

[55] $1^{131}3^121^{311}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 & 2 \\ 1 & 0 & 1 & 9 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 18 & 21 & 15 \\ 2 & 2 & 0 & 18 & 4 & 6 & 4 \\ 2 & 3 & 0 & 21 & 6 & 5 & 5 \\ 2 & 2 & -1 & 15 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{132} 3^1 21^{312} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 4 & 2 \\ 1 & 0 & 1 & 9 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 18 & 42 & 15 \\ 2 & 2 & 0 & 18 & 4 & 12 & 4 \\ 4 & 6 & 0 & 42 & 12 & 26 & 10 \\ 2 & 2 & -1 & 15 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 0 & 1 & 3 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 7 & 5 \\ 4 & 5 & 0 & 13 & 23 & 13 & 9 \\ 2 & 3 & 0 & 7 & 13 & 5 & 5 \\ 2 & 2 & -1 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 1 & 3 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 14 & 5 \\ 4 & 5 & 0 & 13 & 23 & 26 & 9 \\ 4 & 6 & 0 & 14 & 26 & 26 & 10 \\ 2 & 2 & -1 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{142} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 20 & 10 & 4 \\ 3 & 1 & 3 & 16 & 8 & 3 \\ 3 & 3 & 1 & 16 & 8 & 3 \\ 20 & 16 & 16 & 94 & 48 & 20 \\ 10 & 8 & 8 & 48 & 22 & 10 \\ 4 & 3 & 3 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{114} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 28 & 5 \\ 2 & 2 & 0 & 6 & 4 & 24 & 4 \\ 8 & 12 & 0 & 28 & 24 & 110 & 20 \\ 2 & 2 & -1 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 5 & 10 & 14 & 19 & 7 & 8 \\ 7 & 13 & 19 & 24 & 10 & 10 \\ 3 & 5 & 7 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{1312} 3^1 21^{1323} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 11 & 3 \\ 3 & 5 & 15 & 13 & 20 & 5 \\ 9 & 15 & 34 & 30 & 45 & 12 \\ 7 & 13 & 30 & 24 & 39 & 10 \\ 11 & 20 & 45 & 39 & 58 & 16 \\ 3 & 5 & 12 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{121} 3^1 21^{1121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 1 & 1 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{1312} 3^1 21^{1414} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 13 & 3 \\ 3 & 5 & 20 & 8 & 25 & 5 \\ 11 & 20 & 62 & 24 & 76 & 16 \\ 5 & 8 & 24 & 8 & 30 & 6 \\ 13 & 25 & 76 & 30 & 92 & 20 \\ 3 & 5 & 16 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1^{21}} 1^{111} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 0 & 2 & 3 & 5 & 1 & 2 \\ 1 & 2 & 0 & 4 & 8 & 3 & 2 \\ 2 & 3 & 4 & 5 & 13 & 4 & 5 \\ 6 & 5 & 8 & 13 & 23 & 8 & 9 \\ 2 & 1 & 3 & 4 & 8 & 1 & 3 \\ 2 & 2 & 2 & 5 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1^{213}} 1^{12} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 4 & 5 & 2 & 6 & 14 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 13 & 20 & 14 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{1^{11}} 1^{111} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 1 & 3 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 11 & 5 \\ 4 & 5 & 0 & 13 & 23 & 21 & 9 \\ 4 & 4 & 0 & 11 & 21 & 16 & 8 \\ 2 & 2 & -1 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1^{11}} 1^{12} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 2 & 5 & 2 & 6 & 14 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 11 & 20 & 14 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1^{111}} 1^{121} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 7 & 4 \\ 3 & 1 & 3 & 8 & 5 & 3 \\ 3 & 3 & 1 & 8 & 6 & 3 \\ 10 & 8 & 8 & 22 & 16 & 10 \\ 7 & 5 & 6 & 16 & 9 & 7 \\ 4 & 3 & 3 & 10 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1^{123}} 1^{111} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 7 & 2 \\ 2 & 1 & 3 & 1 & 5 & 7 & 3 \\ 5 & 4 & 5 & 5 & 10 & 14 & 6 \\ 6 & 4 & 7 & 7 & 14 & 16 & 8 \\ 2 & 2 & 2 & 3 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1^{111}} 1^{11} 3^1 21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 10 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 8 & 15 & 34 & 18 & 45 & 12 \\ 5 & 8 & 18 & 8 & 24 & 6 \\ 10 & 20 & 45 & 24 & 58 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1^{112}} 1^{111} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 7 & 2 \\ 2 & 1 & 3 & 1 & 5 & 7 & 3 \\ 4 & 4 & 5 & 5 & 10 & 14 & 6 \\ 5 & 4 & 7 & 7 & 14 & 16 & 8 \\ 2 & 2 & 2 & 3 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{11} 3^{12} 21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 11 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 9 & 15 & 34 & 18 & 45 & 12 \\ 5 & 8 & 18 & 8 & 24 & 6 \\ 11 & 20 & 45 & 24 & 58 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{123} 3^{12} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 29 & 10 \\ 10 & 15 & 0 & 29 & 32 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{123} 3^{12} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 8 & 2 \\ 1 & 0 & 2 & 1 & 7 & 8 & 2 \\ 1 & 2 & 0 & 3 & 9 & 9 & 2 \\ 2 & 1 & 3 & 1 & 9 & 9 & 3 \\ 7 & 7 & 9 & 9 & 35 & 39 & 11 \\ 8 & 8 & 9 & 9 & 39 & 40 & 12 \\ 2 & 2 & 2 & 3 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{111} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 11 & 5 \\ 2 & 2 & 0 & 6 & 4 & 10 & 4 \\ 4 & 4 & 0 & 11 & 10 & 16 & 8 \\ 2 & 2 & -1 & 5 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{112} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 14 & 4 \\ 3 & 1 & 3 & 8 & 10 & 3 \\ 3 & 3 & 1 & 8 & 12 & 3 \\ 10 & 8 & 8 & 22 & 32 & 10 \\ 14 & 10 & 12 & 32 & 42 & 14 \\ 4 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

[85] $1^{122} 3^{12} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 1 & 1 & 3 & 1 & 9 & 8 & 3 \\ 7 & 7 & 9 & 9 & 35 & 32 & 11 \\ 6 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{1112} 3^{12} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 8 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 8 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{141} 3^{12} 21^{1411} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 3 & 2 \\ 1 & 0 & 2 & 12 & 2 & 3 & 2 \\ 1 & 2 & 0 & 16 & 4 & 4 & 2 \\ 10 & 12 & 16 & 110 & 24 & 28 & 20 \\ 3 & 2 & 4 & 24 & 4 & 6 & 4 \\ 3 & 3 & 4 & 28 & 6 & 5 & 5 \\ 2 & 2 & 2 & 20 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

- [81] $1^{112} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 1 & 3 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 22 & 5 \\ 4 & 5 & 0 & 13 & 23 & 42 & 9 \\ 8 & 8 & 0 & 22 & 42 & 70 & 16 \\ 2 & 2 & -1 & 5 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$
- [87] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 11 & 4 \\ 3 & 1 & 3 & 5 & 7 & 3 \\ 3 & 3 & 1 & 6 & 10 & 3 \\ 7 & 5 & 6 & 9 & 17 & 7 \\ 11 & 7 & 10 & 17 & 25 & 11 \\ 4 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$
- [86] $1^{113} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 21 & 4 \\ 3 & 1 & 3 & 3 & 15 & 3 \\ 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 3 & 2 & 1 & 15 & 3 \\ 21 & 15 & 18 & 15 & 97 & 21 \\ 4 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$
- [83] $1^{122} 3^1 21^{213} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 36 & 10 \\ 2 & 2 & 0 & 12 & 4 & 16 & 4 \\ 6 & 7 & 0 & 36 & 16 & 45 & 13 \\ 2 & 2 & -1 & 10 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$
- [84] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 18 & 4 \\ 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 3 & 1 & 4 & 16 & 3 \\ 4 & 2 & 4 & 2 & 16 & 4 \\ 18 & 12 & 16 & 16 & 70 & 18 \\ 4 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$
- [88] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 1 & 1 & 3 & 1 & 5 & 12 & 3 \\ 4 & 4 & 5 & 5 & 10 & 27 & 6 \\ 9 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$
- [92] $1^{114} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 24 & 4 \\ 3 & 1 & 3 & 3 & 18 & 3 \\ 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 1 & 18 & 3 \\ 24 & 18 & 20 & 18 & 130 & 24 \\ 4 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$
- [91] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 21 & 4 \\ 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 3 & 1 & 4 & 18 & 3 \\ 4 & 2 & 4 & 2 & 18 & 4 \\ 21 & 15 & 18 & 18 & 97 & 21 \\ 4 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$
- [96] $1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 49 & 10 \\ 14 & 25 & 0 & 49 & 92 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$
- [97] $1^{1211} 3^1 21^{1411} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 & 2 \\ 1 & 0 & 2 & 8 & 4 & 3 & 2 \\ 1 & 2 & 0 & 14 & 5 & 4 & 2 \\ 9 & 8 & 14 & 70 & 28 & 22 & 16 \\ 4 & 4 & 5 & 28 & 10 & 9 & 6 \\ 2 & 3 & 4 & 22 & 9 & 5 & 5 \\ 2 & 2 & 2 & 16 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{121} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 & 2 \\ 1 & 2 & 0 & 5 & 5 & 1 & 2 \\ 6 & 5 & 5 & 13 & 13 & 5 & 7 \\ 4 & 4 & 5 & 13 & 10 & 4 & 6 \\ 1 & 2 & 1 & 5 & 4 & 0 & 2 \\ 2 & 2 & 2 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{123} 3^1 21^{1223} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 12 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 5 & 10 & 14 & 20 & 30 & 8 \\ 8 & 13 & 20 & 24 & 39 & 10 \\ 12 & 20 & 30 & 39 & 58 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[104] $1^{121} 3^1 21^{1222} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 8 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 5 & 10 & 14 & 20 & 22 & 8 \\ 8 & 13 & 20 & 24 & 29 & 10 \\ 8 & 15 & 22 & 29 & 32 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[103] $1^{123} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 8 & 2 \\ 1 & 2 & 0 & 3 & 5 & 9 & 2 \\ 2 & 1 & 3 & 1 & 5 & 9 & 3 \\ 5 & 4 & 5 & 5 & 10 & 22 & 6 \\ 8 & 8 & 9 & 9 & 22 & 40 & 12 \\ 2 & 2 & 2 & 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[102] $1^{121} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 1 & 2 \\ 1 & 0 & 2 & 1 & 4 & 0 & 2 \\ 1 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 1 & 3 & 1 & 5 & 0 & 3 \\ 5 & 4 & 5 & 5 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{121} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 & 2 \\ 1 & 2 & 0 & 5 & 5 & 1 & 2 \\ 4 & 5 & 5 & 13 & 13 & 5 & 7 \\ 5 & 4 & 5 & 13 & 10 & 4 & 6 \\ 2 & 2 & 1 & 5 & 4 & 0 & 2 \\ 2 & 2 & 2 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[100] $1^{114} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 8 & 2 \\ 1 & 2 & 0 & 3 & 5 & 9 & 2 \\ 2 & 1 & 3 & 1 & 5 & 9 & 3 \\ 4 & 4 & 5 & 5 & 10 & 22 & 6 \\ 9 & 8 & 9 & 9 & 22 & 40 & 12 \\ 2 & 2 & 2 & 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 5 & 0 & 13 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 19 & 10 \\ 6 & 10 & 0 & 19 & 14 & 8 \\ 3 & 5 & 0 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^{1212} 21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 11 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 8 & 15 & 34 & 18 & 45 & 12 \\ 5 & 8 & 18 & 8 & 24 & 6 \\ 11 & 20 & 45 & 24 & 58 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[94] $1^{121} 3^{1231} 21^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 & 3 \\ 3 & 5 & 10 & 23 & 5 & 5 \\ 5 & 10 & 14 & 34 & 8 & 8 \\ 13 & 23 & 34 & 78 & 17 & 18 \\ 3 & 5 & 8 & 17 & 2 & 4 \\ 3 & 5 & 8 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[99] $1^{111} 3^{111} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 1 & 2 \\ 1 & 0 & 2 & 1 & 4 & 0 & 2 \\ 1 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 1 & 3 & 1 & 5 & 0 & 3 \\ 4 & 4 & 5 & 5 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3^{111} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 7 & 4 \\ 3 & 1 & 3 & 5 & 5 & 3 \\ 3 & 3 & 1 & 6 & 6 & 3 \\ 7 & 5 & 6 & 9 & 11 & 7 \\ 7 & 5 & 6 & 11 & 9 & 7 \\ 4 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[107] $1^{11} 3^{111} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 & 3 \\ 3 & 5 & 5 & 8 & 5 & 5 \\ 2 & 5 & 2 & 6 & 4 & 4 \\ 5 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[109] $1^{1212} 3^{122} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 11 & 4 \\ 3 & 1 & 3 & 7 & 7 & 3 \\ 3 & 3 & 1 & 10 & 10 & 3 \\ 11 & 7 & 10 & 25 & 27 & 11 \\ 11 & 7 & 10 & 27 & 25 & 11 \\ 4 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

[106] $1^{41} 3^{121} 3^{121} 21^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 1 & 1 & 3 & 1 & 12 & 4 & 3 \\ 10 & 8 & 12 & 12 & 56 & 20 & 14 \\ 3 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[111] $1^{111} 3^{111} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 5 & 0 & 8 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 6 & 6 \\ 3 & 5 & 0 & 6 & 2 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[110] $1^{1212} 3^{122} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 14 & 4 \\ 3 & 1 & 3 & 7 & 10 & 3 \\ 3 & 3 & 1 & 10 & 12 & 3 \\ 11 & 7 & 10 & 25 & 34 & 11 \\ 14 & 10 & 12 & 34 & 42 & 14 \\ 4 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{112} 3^{114} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 22 & 4 \\ 3 & 1 & 3 & 5 & 14 & 3 \\ 3 & 3 & 1 & 6 & 20 & 3 \\ 7 & 5 & 6 & 9 & 34 & 7 \\ 22 & 14 & 20 & 34 & 106 & 22 \\ 4 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{111} 3^{111} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 3 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{112} 3^{114} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 22 & 5 \\ 2 & 2 & 0 & 6 & 4 & 20 & 4 \\ 8 & 8 & 0 & 22 & 20 & 70 & 16 \\ 2 & 2 & -1 & 5 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{113} 3^{113} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{111} 3^{113} 21^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 10 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 5 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 10 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{112} 3^{113} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 13 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{123} 3^{111} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 5 & 4 & 5 & 5 & 10 & 9 & 6 \\ 5 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{111} 3^{111} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[117] $1^{\overline{111}} 3^{\overline{11}} 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 4 & 4 & 5 & 5 & 10 & 9 & 6 \\ 2 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[116] $1^{\overline{112}} 3^{\overline{11}} 21^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 1 & 12 & 0 & 2 \\ 1 & 2 & 1 & 0 & 8 & 0 & 2 \\ 9 & 8 & 12 & 8 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{\overline{111}} 3^{\overline{11}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 1 & 2 \\ 1 & 0 & 2 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & 1 & 0 & 4 & 0 & 2 \\ 5 & 4 & 5 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[114] $1^{\overline{123}} 3^{\overline{11}} 21^{\overline{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 48 & 10 \\ 15 & 25 & 0 & 48 & 94 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[115] $1^{\overline{1312}} 3^{\overline{11}} 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 5 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 4 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[125] $1^{\overline{1211}} 3^{\overline{11}} 21^{\overline{1212}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 8 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 22 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 8 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[126] $1^{\overline{111}} 3^{\overline{112}} 3^{\overline{11}} 21^{\overline{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 0 & 1 & 3 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 18 & 5 \\ 4 & 5 & 0 & 13 & 23 & 34 & 9 \\ 6 & 7 & 0 & 18 & 34 & 45 & 13 \\ 2 & 2 & -1 & 5 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[127] $1^{\overline{112}} 3^{\overline{11}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.3$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 17 & 4 \\ 3 & 1 & 3 & 8 & 13 & 3 \\ 3 & 3 & 1 & 8 & 14 & 3 \\ 10 & 8 & 8 & 22 & 40 & 10 \\ 17 & 13 & 14 & 40 & 65 & 17 \\ 4 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[128] $1^{12} 3^{112} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 2 & 5 & 2 & 6 & 8 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 8 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[129] $1^{113} 3^{113} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 21 & 4 \\ 3 & 1 & 3 & 8 & 15 & 3 \\ 3 & 3 & 1 & 8 & 18 & 3 \\ 10 & 8 & 8 & 22 & 48 & 10 \\ 21 & 15 & 18 & 48 & 97 & 21 \\ 4 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

[130] $1^{212} 3^{1111} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 7 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 4 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 7 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[132] $1^{111} 3^{111} 21^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 & 3 \\ 3 & 5 & 20 & 8 & 5 & 5 \\ 10 & 20 & 58 & 24 & 15 & 16 \\ 5 & 8 & 24 & 8 & 6 & 6 \\ 3 & 5 & 15 & 6 & 2 & 4 \\ 3 & 5 & 16 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[131] $1^{211} 3^{111} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 7 & 10 & 14 & 12 & 7 & 8 \\ 5 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 7 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[135] $1^{111} 3^{113} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 0 & 1 & 3 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 18 & 5 \\ 2 & 2 & 0 & 6 & 4 & 16 & 4 \\ 6 & 7 & 0 & 18 & 16 & 45 & 13 \\ 2 & 2 & -1 & 5 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[134] $1^{111} 3^{111} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 12 & 6 \\ 6 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[133] $1^{112} 3^{113} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 11 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[136] $1^{112} 3^{112} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 12 & 6 \\ 6 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[137] $1^{114} 3^{11} 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 15 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 5 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 15 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[138] $1^{111} 3^{11} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 8 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 5 & 10 & 14 & 12 & 22 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 8 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[139] $1^{114} 3^{11} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 28 & 4 \\ 3 & 1 & 3 & 3 & 20 & 3 \\ 3 & 3 & 1 & 2 & 24 & 3 \\ 3 & 3 & 2 & 1 & 20 & 3 \\ 28 & 20 & 24 & 20 & 174 & 28 \\ 4 & 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[140] $1^{114} 3^{11} 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 1 & 4 & 12 & 2 \\ 1 & 2 & 0 & 3 & 5 & 16 & 2 \\ 1 & 1 & 3 & 1 & 5 & 16 & 3 \\ 4 & 4 & 5 & 5 & 10 & 36 & 6 \\ 12 & 12 & 16 & 16 & 36 & 110 & 20 \\ 2 & 2 & 2 & 3 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[141] $1^{111} 3^{11} 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 5 & 10 & 14 & 12 & 8 & 8 \\ 5 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 8 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[142] $1^{114} 3^{11} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 28 & 4 \\ 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 3 & 1 & 4 & 24 & 3 \\ 4 & 2 & 4 & 2 & 24 & 4 \\ 28 & 20 & 24 & 24 & 174 & 28 \\ 4 & 3 & 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[143] $1^{112} 3^{11} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 6 & 4 \\ 4 & 3 & 4 & 0 & 6 & 5 & 5 \\ 2 & 2 & 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{111} 3^{11} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 6 & 4 \\ 3 & 3 & 4 & 0 & 6 & 5 & 5 \\ 2 & 2 & 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{11} 1^{123} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 50 & 10 \\ 2 & 2 & 0 & 12 & 4 & 22 & 4 \\ 8 & 10 & 0 & 50 & 22 & 88 & 18 \\ 2 & 2 & -1 & 10 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{31} 1^{131} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 1 & 1 & 3 & 1 & 13 & 4 & 3 \\ 10 & 10 & 13 & 13 & 74 & 23 & 16 \\ 3 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{112} 1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 17 & 4 \\ 3 & 1 & 3 & 5 & 13 & 3 \\ 3 & 3 & 1 & 6 & 14 & 3 \\ 7 & 5 & 6 & 9 & 27 & 7 \\ 17 & 13 & 14 & 27 & 65 & 17 \\ 4 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[148] $1^{213} 1^{113} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 21 & 4 \\ 3 & 1 & 3 & 7 & 15 & 3 \\ 3 & 3 & 1 & 10 & 18 & 3 \\ 11 & 7 & 10 & 25 & 51 & 11 \\ 21 & 15 & 18 & 51 & 97 & 21 \\ 4 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[149] $1^{114} 1^{13} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 25 & 4 \\ 3 & 1 & 3 & 2 & 17 & 3 \\ 3 & 3 & 1 & 4 & 22 & 3 \\ 4 & 2 & 4 & 2 & 22 & 4 \\ 25 & 17 & 22 & 22 & 137 & 25 \\ 4 & 3 & 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[152] $1^{213} 1^{12} 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 17 & 3 \\ 3 & 5 & 10 & 8 & 30 & 5 \\ 6 & 10 & 14 & 12 & 44 & 8 \\ 5 & 8 & 12 & 8 & 36 & 6 \\ 17 & 30 & 44 & 36 & 134 & 24 \\ 3 & 5 & 8 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[150] $1^{112} 1^1 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 6 & 3 \\ 3 & 5 & 10 & 8 & 10 & 5 \\ 5 & 10 & 14 & 12 & 16 & 8 \\ 5 & 8 & 12 & 8 & 12 & 6 \\ 6 & 10 & 16 & 12 & 14 & 8 \\ 3 & 5 & 8 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[153] $1^{141} 1^{113} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 1 & 2 \\ 1 & 0 & 2 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 1 & 13 & 0 & 2 \\ 2 & 2 & 1 & 0 & 10 & 0 & 2 \\ 11 & 10 & 13 & 10 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{112} 1^{12} 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$\begin{aligned}
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 5 & 16 & 3 \\ 3 & 5 & 10 & 8 & 30 & 5 \\ 5 & 10 & 14 & 12 & 44 & 8 \\ 5 & 8 & 12 & 8 & 36 & 6 \\ 16 & 30 & 44 & 36 & 134 & 24 \\ 3 & 5 & 8 & 6 & 24 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{132} \\ [155] 1^{11} 3^{1242} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 16 & 6 & 3 \\ 3 & 5 & 10 & 28 & 10 & 5 \\ 5 & 10 & 14 & 42 & 16 & 8 \\ 16 & 28 & 42 & 116 & 42 & 22 \\ 6 & 10 & 16 & 42 & 14 & 8 \\ 3 & 5 & 8 & 22 & 8 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{11} \\ [157] 1^{121} 3^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 & 2 \\ 1 & 0 & 1 & 3 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 24 & 7 & 5 \\ 8 & 9 & 0 & 24 & 83 & 24 & 17 \\ 2 & 3 & 0 & 7 & 24 & 5 & 5 \\ 2 & 2 & -1 & 5 & 17 & 5 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{13} \\ [151] 1^{13} 3^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 2 & 5 & 2 & 6 & 12 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 12 & 18 & 34 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{211} \\ [156] 1^{13} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 3 \\ 3 & 5 & 5 & 8 & 5 & 5 \\ 4 & 5 & 2 & 6 & 4 & 4 \\ 5 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{11} \\ [158] 1^{11} 3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 2 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 8 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{41} \\ [159] 1^{113} 3^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 & 2 \\ 1 & 0 & 2 & 3 & 10 & 1 & 2 \\ 1 & 2 & 0 & 4 & 13 & 3 & 2 \\ 2 & 3 & 4 & 5 & 23 & 4 & 5 \\ 11 & 10 & 13 & 23 & 74 & 13 & 16 \\ 2 & 1 & 3 & 4 & 13 & 1 & 3 \\ 2 & 2 & 2 & 5 & 16 & 3 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{12} \\ [160] 1^{11} 3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 2 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{112} \\ [161] 1^{12} 3^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 14 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[165] $1^{1^{214}} 3^{1^{124}} 21^{1^{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 22 & 4 \\ 3 & 1 & 3 & 7 & 14 & 3 \\ 3 & 3 & 1 & 10 & 20 & 3 \\ 11 & 7 & 10 & 25 & 54 & 11 \\ 22 & 14 & 20 & 54 & 106 & 22 \\ 4 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}$$

[164] $1^{1^{212}} 3^{1^{122}} 21^{1^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 17 & 14 & 4 \\ 3 & 1 & 3 & 13 & 10 & 3 \\ 3 & 3 & 1 & 14 & 12 & 3 \\ 17 & 13 & 14 & 65 & 54 & 17 \\ 14 & 10 & 12 & 54 & 42 & 14 \\ 4 & 3 & 3 & 17 & 14 & 4 \end{pmatrix}_{b_0}$$

[163] $1^{1^{112}} 3^{1^{112}} 21^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 14 & 4 \\ 3 & 1 & 3 & 5 & 10 & 3 \\ 3 & 3 & 1 & 6 & 12 & 3 \\ 7 & 5 & 6 & 9 & 22 & 7 \\ 14 & 10 & 12 & 22 & 42 & 14 \\ 4 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[162] $1^{1^{113}} 3^{1^{113}} 21^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 9 & 15 & 0 & 18 & 34 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[167] $1^{1^{131}} 3^{1^{12}} 21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 1 & 2 \\ 1 & 0 & 2 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & 1 & 0 & 7 & 0 & 2 \\ 8 & 7 & 9 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{1^{31}} 3^{1^{12}} 21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 8 & 7 & 9 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{1^{14}} 3^{1^{124}} 21^{1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 28 & 4 \\ 3 & 1 & 3 & 8 & 20 & 3 \\ 3 & 3 & 1 & 8 & 24 & 3 \\ 10 & 8 & 8 & 22 & 64 & 10 \\ 28 & 20 & 24 & 64 & 174 & 28 \\ 4 & 3 & 3 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

[168] $1^{1^{31}} 3^{1^{1131}} 21^{1^{1131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 & 2 \\ 1 & 0 & 2 & 1 & 8 & 1 & 2 \\ 1 & 2 & 0 & 3 & 12 & 3 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 & 3 \\ 9 & 8 & 12 & 12 & 56 & 12 & 14 \\ 2 & 1 & 3 & 3 & 12 & 1 & 3 \\ 2 & 2 & 2 & 3 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{1^{14}} 3^{1^{113}} 21^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 2 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 10 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{11}3^{13}21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 12 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 9 & 15 & 34 & 18 & 45 & 12 \\ 4 & 8 & 18 & 8 & 24 & 6 \\ 12 & 20 & 45 & 24 & 58 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[169] $1^{11}3^{11}21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 3 & 5 & 2 & 6 & 7 & 4 \\ 4 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[172] $1^{12}3^{11}21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 2 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 8 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{311}3^{11}21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 6 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{121}3^{11}21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 4 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 2 & 3 & 4 & 5 & 13 & 11 & 5 \\ 6 & 5 & 8 & 13 & 23 & 21 & 9 \\ 4 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{212}3^{11}21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 4 & 5 & 2 & 6 & 8 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 8 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[178] $1^{222}3^{11}21^{1323} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 11 & 3 \\ 3 & 5 & 15 & 13 & 20 & 5 \\ 8 & 15 & 34 & 30 & 45 & 12 \\ 8 & 13 & 30 & 24 & 39 & 10 \\ 11 & 20 & 45 & 39 & 58 & 16 \\ 3 & 5 & 12 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[179] $1^{113}3^{11}21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 8 & 2 \\ 1 & 2 & 0 & 3 & 5 & 9 & 2 \\ 2 & 1 & 3 & 1 & 5 & 9 & 3 \\ 4 & 4 & 5 & 5 & 10 & 22 & 6 \\ 8 & 8 & 9 & 9 & 22 & 40 & 12 \\ 2 & 2 & 2 & 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{11} 3^{1213} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 10 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 22 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 10 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{11} 3^{1212} 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 13 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 6 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 13 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[182] $1^{113} 3^{1124} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 8 & 2 \\ 1 & 2 & 0 & 3 & 5 & 9 & 2 \\ 2 & 1 & 3 & 1 & 5 & 9 & 3 \\ 5 & 4 & 5 & 5 & 10 & 22 & 6 \\ 9 & 8 & 9 & 9 & 22 & 40 & 12 \\ 2 & 2 & 2 & 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{112} 3^{1213} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 18 & 4 \\ 3 & 1 & 3 & 7 & 12 & 3 \\ 3 & 3 & 1 & 10 & 16 & 3 \\ 11 & 7 & 10 & 25 & 44 & 11 \\ 18 & 12 & 16 & 44 & 70 & 18 \\ 4 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

[187] $1^{112} 3^{1142} 21^{1142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 6 & 3 \\ 3 & 5 & 5 & 28 & 10 & 5 \\ 2 & 5 & 2 & 21 & 8 & 4 \\ 17 & 28 & 21 & 116 & 42 & 22 \\ 6 & 10 & 8 & 42 & 14 & 8 \\ 3 & 5 & 4 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[186] $1^{112} 3^{1113} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 18 & 4 \\ 3 & 1 & 3 & 5 & 12 & 3 \\ 3 & 3 & 1 & 6 & 16 & 3 \\ 7 & 5 & 6 & 9 & 28 & 7 \\ 18 & 12 & 16 & 28 & 70 & 18 \\ 4 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[185] $1^{122} 3^{1112} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 4 & 5 & 2 & 10 & 11 & 4 \\ 8 & 13 & 10 & 24 & 29 & 10 \\ 9 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[184] $1^{111} 3^{1111} 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 4 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 2 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{1^{222}} 1^{111} 3^1 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 5 & 4 & 7 & 16 & 14 & 11 & 8 \\ 5 & 4 & 5 & 14 & 10 & 9 & 6 \\ 4 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[176] $1^{1^{13}} 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 10 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$

[191] $1^{1^{21}} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[198] $1^{1^{12}} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[190] $1^{1^{21}} 1^{11} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 8 & 3 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 6 & 10 & 14 & 19 & 7 & 8 \\ 8 & 13 & 19 & 24 & 10 & 10 \\ 3 & 5 & 7 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$

[199] $1^{1^{12}} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[194] $1^{1^{31}} 1^{11} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 2 & 5 & 2 & 17 & 4 & 4 \\ 14 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$

[195] $1^{1^{12}} 1^{113} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 17 & 6 \\ 6 & 7 & 6 & 0 & 17 & 25 & 9 \\ 2 & 2 & 2 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0}$

[196] $1^{1^{23}} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 5 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{1111} 3^{11} 21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$
6.0

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 3 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 3 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{1111} 3^{11} 21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$
6.0

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 2 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{1123} 3^{11} 21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$
6.8

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 17 & 6 \\ 7 & 7 & 6 & 0 & 17 & 25 & 9 \\ 2 & 2 & 2 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{1121} 3^{11} 21^{1221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$
6.0

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 & 3 \\ 3 & 5 & 15 & 13 & 5 & 5 \\ 8 & 15 & 32 & 29 & 11 & 12 \\ 8 & 13 & 29 & 24 & 10 & 10 \\ 3 & 5 & 11 & 10 & 2 & 4 \\ 3 & 5 & 12 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[200] $1^{111} 3^{11} 21^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$
6.8

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 25 & 5 \\ 2 & 2 & 0 & 6 & 4 & 22 & 4 \\ 8 & 10 & 0 & 25 & 22 & 88 & 18 \\ 2 & 2 & -1 & 5 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{1212} 3^{11} 21^{1112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$
6.0

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 4 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[202] $1^{113} 3^{11} 21^{1213} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$
7.3

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 9 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 5 & 10 & 14 & 12 & 24 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 9 & 15 & 24 & 18 & 34 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[203] $1^{114} 3^{11} 21^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$
7.0

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 28 & 4 \\ 3 & 1 & 3 & 7 & 20 & 3 \\ 3 & 3 & 1 & 10 & 24 & 3 \\ 11 & 7 & 10 & 25 & 68 & 11 \\ 28 & 20 & 24 & 68 & 174 & 28 \\ 4 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[204] $1^{1112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{1111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 3 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 2 & 5 & 2 & 10 & 4 & 4 \\ 8 & 13 & 10 & 24 & 10 & 10 \\ 3 & 5 & 4 & 10 & 2 & 4 \\ 3 & 5 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[207] $1^{121} 3^1 21^{1222} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 8 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 6 & 10 & 14 & 20 & 22 & 8 \\ 7 & 13 & 20 & 24 & 29 & 10 \\ 8 & 15 & 22 & 29 & 32 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[208] $1^{112} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 14 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 14 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[209] $1^{122} 3^1 21^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 2 & 3 \\ 3 & 5 & 10 & 23 & 5 & 5 \\ 6 & 10 & 14 & 34 & 8 & 8 \\ 13 & 23 & 34 & 78 & 17 & 18 \\ 2 & 5 & 8 & 17 & 2 & 4 \\ 3 & 5 & 8 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[210] $1^{112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 4 & 10 & 13 & 6 \\ 5 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 2 \\ 5 & 4 & 5 & 4 & 10 & 13 & 6 \\ 5 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{122} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{121} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 3 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{122} 1^{111} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 2 & 5 & 2 & 10 & 11 & 4 \\ 8 & 13 & 10 & 24 & 29 & 10 \\ 9 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[215] $1^{122} 1^{111} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 11 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 2 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 11 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[216] $1^{112} 1^{123} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 2 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 6 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{113} 1^{124} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 24 & 4 \\ 3 & 1 & 3 & 8 & 18 & 3 \\ 3 & 3 & 1 & 8 & 20 & 3 \\ 10 & 8 & 8 & 22 & 56 & 10 \\ 24 & 18 & 20 & 56 & 130 & 24 \\ 4 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

[218] $1^{213} 1^{122} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 18 & 4 \\ 3 & 1 & 3 & 10 & 12 & 3 \\ 3 & 3 & 1 & 12 & 16 & 3 \\ 14 & 10 & 12 & 42 & 56 & 14 \\ 18 & 12 & 16 & 56 & 70 & 18 \\ 4 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[220] $1^{411} 1^{1211} 3^1 21^{1411} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 3 & 2 \\ 1 & 0 & 2 & 8 & 4 & 3 & 2 \\ 1 & 2 & 0 & 14 & 5 & 4 & 2 \\ 10 & 8 & 14 & 70 & 28 & 22 & 16 \\ 4 & 4 & 5 & 28 & 10 & 9 & 6 \\ 3 & 3 & 4 & 22 & 9 & 5 & 5 \\ 2 & 2 & 2 & 16 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{14} 1^{121} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 2 & 5 & 2 & 6 & 16 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 16 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{\overline{1}^{211}} \ 3^{\overline{1}} \ 21^{\overline{1311}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 & 3 \\ 3 & 5 & 15 & 8 & 5 & 5 \\ 8 & 15 & 34 & 18 & 12 & 12 \\ 5 & 8 & 18 & 8 & 6 & 6 \\ 3 & 5 & 12 & 6 & 2 & 4 \\ 3 & 5 & 12 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{\overline{1}^{24}} \ 3^{\overline{1}} \ 21^{\overline{1124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 18 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 2 & 5 & 2 & 10 & 22 & 4 \\ 8 & 13 & 10 & 24 & 58 & 10 \\ 18 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{\overline{1}^{21}} \ 3^{\overline{1}} \ 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 5 & 4 & 5 & 5 & 10 & 9 & 6 \\ 3 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{\overline{1}^{21}} \ 3^{\overline{1}} \ 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 3 & 2 \\ 1 & 0 & 2 & 5 & 4 & 3 & 2 \\ 1 & 2 & 0 & 5 & 5 & 4 & 2 \\ 4 & 5 & 5 & 13 & 13 & 10 & 7 \\ 5 & 4 & 5 & 13 & 10 & 9 & 6 \\ 3 & 3 & 4 & 10 & 9 & 5 & 5 \\ 2 & 2 & 2 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[224] \ 1^{\overline{1}^{14}} \ 3^{\overline{1}} \ 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 62 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{\overline{1}^{21}} \ 3^{\overline{1}} \ 21^{\overline{1221}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 3 & 2 \\ 1 & 0 & 2 & 4 & 7 & 3 & 2 \\ 1 & 2 & 0 & 7 & 9 & 4 & 2 \\ 4 & 4 & 7 & 16 & 25 & 11 & 8 \\ 7 & 7 & 9 & 25 & 35 & 16 & 11 \\ 3 & 3 & 4 & 11 & 16 & 5 & 5 \\ 2 & 2 & 2 & 8 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{\overline{1}^{312}} \ 3^{\overline{1}} \ 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 4 & 2 \\ 1 & 0 & 2 & 5 & 4 & 3 & 2 \\ 1 & 2 & 0 & 5 & 5 & 4 & 2 \\ 6 & 5 & 5 & 13 & 13 & 10 & 7 \\ 4 & 4 & 5 & 13 & 10 & 9 & 6 \\ 4 & 3 & 4 & 10 & 9 & 5 & 5 \\ 2 & 2 & 2 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{\overline{1}^{12}} \ 3^{\overline{1}} \ 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 4 & 4 & 5 & 5 & 10 & 9 & 6 \\ 4 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{\overline{122}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 6 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{\overline{112}} 3^{\overline{124}} 1^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 8 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{\overline{122}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 2 & 5 & 2 & 10 & 8 & 4 \\ 8 & 13 & 10 & 24 & 20 & 10 \\ 6 & 10 & 8 & 20 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[231] $1^{\overline{112}} 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 62 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[232] $1^{\overline{114}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 8 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{\overline{1213}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 4 & 5 & 2 & 6 & 12 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 12 & 18 & 34 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[234] $1^{\overline{121}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 5 & 0 & 13 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 10 & 10 \\ 3 & 5 & 0 & 10 & 2 & 4 \\ 3 & 5 & 0 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[235] $1^{\overline{112}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 6 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{\overline{121}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1221}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 5 & 10 & 14 & 20 & 8 & 8 \\ 8 & 13 & 20 & 24 & 10 & 10 \\ 3 & 5 & 8 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[238] $1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 1 & 2 \\ 1 & 0 & 2 & 4 & 7 & 0 & 2 \\ 1 & 2 & 0 & 7 & 9 & 0 & 2 \\ 5 & 4 & 7 & 16 & 25 & 0 & 8 \\ 8 & 7 & 9 & 25 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 8 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{124} 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 18 & 3 \\ 3 & 5 & 10 & 8 & 30 & 5 \\ 6 & 10 & 14 & 12 & 44 & 8 \\ 4 & 8 & 12 & 8 & 36 & 6 \\ 18 & 30 & 44 & 36 & 134 & 24 \\ 3 & 5 & 8 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[262] $1^{122} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 4 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{123} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 6 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{11} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 & 3 \\ 3 & 5 & 5 & 8 & 5 & 5 \\ 3 & 5 & 2 & 6 & 4 & 4 \\ 4 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[242] $1^{141} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 & 2 \\ 1 & 0 & 2 & 1 & 8 & 1 & 2 \\ 1 & 2 & 0 & 3 & 12 & 3 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 & 3 \\ 10 & 8 & 12 & 12 & 56 & 12 & 14 \\ 2 & 1 & 3 & 3 & 12 & 1 & 3 \\ 2 & 2 & 2 & 3 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{131} 3^1 21^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 & 3 \\ 3 & 5 & 15 & 8 & 5 & 5 \\ 9 & 15 & 34 & 18 & 12 & 12 \\ 4 & 8 & 18 & 8 & 6 & 6 \\ 3 & 5 & 12 & 6 & 2 & 4 \\ 3 & 5 & 12 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[244] $1^{132} 3^1 21^{1312} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 6 & 3 \\ 3 & 5 & 15 & 8 & 10 & 5 \\ 9 & 15 & 34 & 18 & 24 & 12 \\ 4 & 8 & 18 & 8 & 12 & 6 \\ 6 & 10 & 24 & 12 & 14 & 8 \\ 3 & 5 & 12 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[245] $1^{142} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 6 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 40 & 14 \\ 6 & 6 & 8 & 0 & 40 & 26 & 10 \\ 2 & 2 & 2 & 0 & 14 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{133} 3^1 21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 9 & 3 \\ 3 & 5 & 15 & 8 & 15 & 5 \\ 9 & 15 & 34 & 18 & 36 & 12 \\ 4 & 8 & 18 & 8 & 18 & 6 \\ 9 & 15 & 36 & 18 & 34 & 12 \\ 3 & 5 & 12 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{122} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 9 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 22 & 8 \\ 4 & 8 & 12 & 8 & 18 & 6 \\ 9 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{141} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 11 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 33 & 15 \\ 4 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 15 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{122} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 4 & 8 & 2 \\ 5 & 6 & 8 & 26 & 12 & 28 & 10 \\ 3 & 2 & 4 & 12 & 4 & 12 & 4 \\ 5 & 6 & 8 & 28 & 12 & 26 & 10 \\ 2 & 2 & 2 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{134} 3^1 21^{1314} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 12 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 9 & 15 & 34 & 18 & 48 & 12 \\ 4 & 8 & 18 & 8 & 24 & 6 \\ 12 & 20 & 48 & 24 & 62 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[251] \ 1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 14 & 4 \\ 3 & 1 & 3 & 10 & 10 & 3 \\ 3 & 3 & 1 & 12 & 12 & 3 \\ 14 & 10 & 12 & 42 & 44 & 14 \\ 14 & 10 & 12 & 44 & 42 & 14 \\ 4 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[252] \ 1^{141} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 1 & 2 \\ 1 & 0 & 2 & 1 & 7 & 0 & 2 \\ 1 & 2 & 0 & 3 & 9 & 0 & 2 \\ 2 & 1 & 3 & 1 & 9 & 0 & 3 \\ 9 & 7 & 9 & 9 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 2 & 3 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[253] \ 1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 6 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[256] \ 1^{\overline{1}^{22}} \overline{1}^{\overline{1}^{22}} 21^{\overline{1212}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 6 & 3 \\ 3 & 5 & 10 & 8 & 10 & 5 \\ 6 & 10 & 14 & 12 & 16 & 8 \\ 4 & 8 & 12 & 8 & 12 & 6 \\ 6 & 10 & 16 & 12 & 14 & 8 \\ 3 & 5 & 8 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[255] \ 1^{\overline{1}^{321}} \overline{1}^{\overline{1}^{321}} 21^{\overline{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 1 & 2 \\ 1 & 0 & 2 & 10 & 7 & 0 & 2 \\ 1 & 2 & 0 & 10 & 9 & 0 & 2 \\ 9 & 10 & 10 & 58 & 46 & 0 & 14 \\ 7 & 7 & 9 & 46 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 14 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[263] \ 1^{\overline{1}^{41}} \overline{1}^{\overline{1}^{41}} 21^{\overline{1411}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 & 3 \\ 3 & 5 & 20 & 8 & 5 & 5 \\ 12 & 20 & 62 & 24 & 16 & 16 \\ 4 & 8 & 24 & 8 & 6 & 6 \\ 3 & 5 & 16 & 6 & 2 & 4 \\ 3 & 5 & 16 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[257] \ 1^{\overline{1}^{21}} \overline{1}^{\overline{1}^{21}} 21^{\overline{1211}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 6 & 10 & 14 & 12 & 8 & 8 \\ 4 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 8 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[258] \ 1^{\overline{1}^{44}} \overline{1}^{\overline{1}^{44}} 21^{\overline{1414}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 12 & 3 \\ 3 & 5 & 20 & 8 & 20 & 5 \\ 12 & 20 & 62 & 24 & 64 & 16 \\ 4 & 8 & 24 & 8 & 24 & 6 \\ 12 & 20 & 64 & 24 & 62 & 16 \\ 3 & 5 & 16 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{\overline{1}^{114}} \overline{1}^{\overline{1}^{114}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 8 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{\overline{1}^{42}} \overline{1}^{\overline{1}^{42}} 21^{\overline{1412}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 6 & 3 \\ 3 & 5 & 20 & 8 & 10 & 5 \\ 12 & 20 & 62 & 24 & 32 & 16 \\ 4 & 8 & 24 & 8 & 12 & 6 \\ 6 & 10 & 32 & 12 & 14 & 8 \\ 3 & 5 & 16 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[261] \ 1^{\overline{1}^{121}} \overline{1}^{\overline{1}^{121}} 21^{\overline{1331}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 & 3 \\ 3 & 5 & 20 & 18 & 5 & 5 \\ 10 & 20 & 58 & 54 & 15 & 16 \\ 10 & 18 & 54 & 48 & 14 & 14 \\ 3 & 5 & 15 & 14 & 2 & 4 \\ 3 & 5 & 16 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[254] \ 1^{\overline{1}^{121}} \overline{1}^{\overline{1}^{121}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 1 & 2 \\ 1 & 0 & 2 & 1 & 7 & 0 & 2 \\ 1 & 2 & 0 & 3 & 9 & 0 & 2 \\ 2 & 1 & 3 & 1 & 9 & 0 & 3 \\ 7 & 7 & 9 & 9 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{\overline{1}122} 3^{\overline{1}} 21^{\overline{1}22} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 5 & 0 & 13 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 20 & 10 \\ 6 & 10 & 0 & 20 & 14 & 8 \\ 3 & 5 & 0 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[265] $1^{\overline{1}122} 3^{\overline{1}} 21^{\overline{1}112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 5 & 4 & 5 & 5 & 10 & 18 & 6 \\ 6 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{\overline{1}132} 3^{\overline{1}} 21^{\overline{1}32} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 5 & 0 & 23 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 78 & 34 & 18 \\ 6 & 10 & 0 & 34 & 14 & 8 \\ 3 & 5 & 0 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[267] $1^{\overline{1}212} 3^{\overline{1}} 21^{\overline{1}312} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.3$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 6 & 3 \\ 3 & 5 & 15 & 8 & 10 & 5 \\ 8 & 15 & 34 & 18 & 24 & 12 \\ 5 & 8 & 18 & 8 & 12 & 6 \\ 6 & 10 & 24 & 12 & 14 & 8 \\ 3 & 5 & 12 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[268] $1^{\overline{1}122} 3^{\overline{1}} 21^{\overline{1}222} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 6 & 3 \\ 3 & 5 & 10 & 13 & 10 & 5 \\ 5 & 10 & 14 & 20 & 16 & 8 \\ 8 & 13 & 20 & 24 & 20 & 10 \\ 6 & 10 & 16 & 20 & 14 & 8 \\ 3 & 5 & 8 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[269] $1^{\overline{1}114} 3^{\overline{1}} 21^{\overline{1}112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 8 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{\overline{1}12} 3^{\overline{1}} 21^{\overline{1}112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 7 & 2 \\ 2 & 3 & 4 & 5 & 9 & 11 & 5 \\ 4 & 4 & 5 & 9 & 10 & 14 & 6 \\ 5 & 4 & 7 & 11 & 14 & 16 & 8 \\ 2 & 2 & 2 & 5 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{\overline{1}12} 3^{\overline{1}} 21^{\overline{1}121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 9 & 35 & 16 & 11 \\ 2 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{114} 1^1 3^1 21^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 12 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 5 & 10 & 14 & 12 & 32 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 12 & 20 & 32 & 24 & 62 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[273] $1^{123} 1^{12} 3^1 21^{1124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 17 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 2 & 5 & 2 & 10 & 22 & 4 \\ 8 & 13 & 10 & 24 & 58 & 10 \\ 17 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$

[274] $1^{1123} 1^{112} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 2 \\ 5 & 4 & 5 & 4 & 10 & 13 & 6 \\ 6 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$

[275] $1^{1311} 1^1 3^1 21^{1211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 5 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 7 & 10 & 14 & 12 & 8 & 8 \\ 5 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 8 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$

[276] $1^{1111} 1^{112} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 4 & 10 & 13 & 6 \\ 4 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$

[277] $1^{123} 1^1 3^1 21^{1123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 2 & 5 & 2 & 10 & 12 & 4 \\ 8 & 13 & 10 & 24 & 30 & 10 \\ 9 & 15 & 12 & 30 & 34 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$

[278] $1^{1111} 1^{121} 3^1 21^{1211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 3 & 2 \\ 1 & 2 & 0 & 8 & 4 & 4 & 2 \\ 5 & 6 & 8 & 26 & 12 & 14 & 10 \\ 3 & 2 & 4 & 12 & 4 & 6 & 4 \\ 3 & 3 & 4 & 14 & 6 & 5 & 5 \\ 2 & 2 & 2 & 10 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[279] $1^{131} 1^{112} 3^1 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 & 2 \\ 1 & 0 & 2 & 3 & 8 & 1 & 2 \\ 1 & 2 & 0 & 4 & 12 & 3 & 2 \\ 2 & 3 & 4 & 5 & 20 & 4 & 5 \\ 9 & 8 & 12 & 20 & 56 & 12 & 14 \\ 2 & 1 & 3 & 4 & 12 & 1 & 3 \\ 2 & 2 & 2 & 5 & 14 & 3 & 3 \end{pmatrix}_{b_0}$

[281] $1^{1214} 1^1 3^1 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 4 & 5 & 2 & 6 & 16 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 16 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[280] $1^{141} 3^{11} 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 3 & 2 \\ 1 & 0 & 2 & 1 & 13 & 3 & 2 \\ 1 & 2 & 0 & 3 & 17 & 4 & 2 \\ 1 & 1 & 3 & 1 & 17 & 4 & 3 \\ 13 & 13 & 17 & 17 & 127 & 30 & 21 \\ 3 & 3 & 4 & 4 & 30 & 5 & 5 \\ 2 & 2 & 2 & 3 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[283] $1^{113} 3^{11} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 14 & 25 & 0 & 30 & 92 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[284] $1^{113} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 24 & 4 \\ 3 & 1 & 3 & 5 & 18 & 3 \\ 3 & 3 & 1 & 6 & 20 & 3 \\ 7 & 5 & 6 & 9 & 38 & 7 \\ 24 & 18 & 20 & 38 & 130 & 24 \\ 4 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[282] $1^{112} 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 9 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[285] $1^{141} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 1 & 2 \\ 1 & 0 & 2 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & 1 & 0 & 7 & 0 & 2 \\ 9 & 7 & 9 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{111} 3^{11} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 3 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[287] $1^{111} 3^{11} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 5 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{121} 3^{11} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 5 & 0 & 18 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 13 & 14 \\ 3 & 5 & 0 & 13 & 2 & 4 \\ 3 & 5 & 0 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[289] $1^{1^{21}} 1^{1^{12}} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 7 & 7 & 9 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[300] $1^{1^{312}} 1^1 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 6 & 3 \\ 3 & 5 & 10 & 8 & 10 & 5 \\ 7 & 10 & 14 & 12 & 16 & 8 \\ 5 & 8 & 12 & 8 & 12 & 6 \\ 6 & 10 & 16 & 12 & 14 & 8 \\ 3 & 5 & 8 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[292] $1^{1^{23}} 1^1 3^1 21^{1223} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 9 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 5 & 10 & 14 & 20 & 24 & 8 \\ 8 & 13 & 20 & 24 & 30 & 10 \\ 9 & 15 & 24 & 30 & 34 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[293] $1^{1^{213}} 1^1 3^1 21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 9 & 3 \\ 3 & 5 & 15 & 8 & 15 & 5 \\ 8 & 15 & 34 & 18 & 36 & 12 \\ 5 & 8 & 18 & 8 & 18 & 6 \\ 9 & 15 & 36 & 18 & 34 & 12 \\ 3 & 5 & 12 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[291] $1^{1^{214}} 1^{1^{13}} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 25 & 4 \\ 3 & 1 & 3 & 7 & 17 & 3 \\ 3 & 3 & 1 & 10 & 22 & 3 \\ 11 & 7 & 10 & 25 & 61 & 11 \\ 25 & 17 & 22 & 61 & 137 & 25 \\ 4 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

[301] $1^{1^{112}} 1^{1111} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 7 & 2 \\ 4 & 4 & 7 & 16 & 14 & 18 & 8 \\ 4 & 4 & 5 & 14 & 10 & 14 & 6 \\ 5 & 4 & 7 & 18 & 14 & 16 & 8 \\ 2 & 2 & 2 & 8 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{1^{111}} 1^{1111} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 7 & 2 \\ 6 & 4 & 7 & 16 & 14 & 18 & 8 \\ 4 & 4 & 5 & 14 & 10 & 14 & 6 \\ 5 & 4 & 7 & 18 & 14 & 16 & 8 \\ 2 & 2 & 2 & 8 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{1^{113}} 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 21 & 4 \\ 3 & 1 & 3 & 5 & 15 & 3 \\ 3 & 3 & 1 & 6 & 18 & 3 \\ 7 & 5 & 6 & 9 & 33 & 7 \\ 21 & 15 & 18 & 33 & 97 & 21 \\ 4 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[298] $1^{1^{123}} 1^1 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 30 & 10 \\ 9 & 15 & 0 & 30 & 34 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[297] $1^{112} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 28 & 6 \\ 10 & 8 & 14 & 0 & 28 & 70 & 16 \\ 2 & 2 & 2 & 0 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{112} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 14 & 6 \\ 5 & 4 & 7 & 0 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{123} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 5 & 4 & 5 & 5 & 10 & 27 & 6 \\ 9 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{123} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 14 & 6 \\ 6 & 4 & 7 & 0 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 4 & 4 & 5 & 5 & 10 & 9 & 6 \\ 3 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[307] $1^{122} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 29 & 10 \\ 9 & 15 & 0 & 29 & 32 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[306] $1^{122} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 5 & 4 & 5 & 5 & 10 & 9 & 6 \\ 4 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[305] $1^{213} 3^1 21^{1314} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 14 & 3 \\ 3 & 5 & 15 & 8 & 25 & 5 \\ 8 & 15 & 34 & 18 & 57 & 12 \\ 5 & 8 & 18 & 8 & 30 & 6 \\ 14 & 25 & 57 & 30 & 92 & 20 \\ 3 & 5 & 12 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[304] \quad 1^{114} 3^1 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 11 & 2 \\ 1 & 0 & 2 & 1 & 4 & 11 & 2 \\ 1 & 2 & 0 & 3 & 5 & 13 & 2 \\ 2 & 1 & 3 & 1 & 5 & 13 & 3 \\ 5 & 4 & 5 & 5 & 10 & 31 & 6 \\ 11 & 11 & 13 & 13 & 31 & 81 & 17 \\ 2 & 2 & 2 & 3 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[303] \quad 1^{124} 3^1 21^{1224} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 15 & 3 \\ 3 & 5 & 10 & 13 & 25 & 5 \\ 5 & 10 & 14 & 20 & 38 & 8 \\ 8 & 13 & 20 & 24 & 49 & 10 \\ 15 & 25 & 38 & 49 & 92 & 20 \\ 3 & 5 & 8 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{122} 3^1 21^{1223} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 11 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 5 & 10 & 14 & 20 & 30 & 8 \\ 8 & 13 & 20 & 24 & 39 & 10 \\ 11 & 20 & 30 & 39 & 58 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[311] \quad 1^{121} 3^1 21^{1321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 & 3 \\ 3 & 5 & 15 & 13 & 5 & 5 \\ 8 & 15 & 34 & 30 & 12 & 12 \\ 8 & 13 & 30 & 24 & 10 & 10 \\ 3 & 5 & 12 & 10 & 2 & 4 \\ 3 & 5 & 12 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[313] \quad 1^{1111} 3^1 21^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 6 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 4 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^{121} 3^1 21^{1121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 4 & 5 & 2 & 10 & 4 & 4 \\ 8 & 13 & 10 & 24 & 10 & 10 \\ 3 & 5 & 4 & 10 & 2 & 4 \\ 3 & 5 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[309] \quad 1^{1111} 3^1 21^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 5 & 4 & 7 & 16 & 14 & 11 & 8 \\ 5 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[310] \quad 1^{1111} 3^1 21^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 4 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[314] $1^{131} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 1 & 2 \\ 1 & 0 & 2 & 1 & 7 & 0 & 2 \\ 1 & 2 & 0 & 3 & 9 & 0 & 2 \\ 2 & 1 & 3 & 1 & 9 & 0 & 3 \\ 8 & 7 & 9 & 9 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[315] $1^{124} 3^{1213} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 24 & 4 \\ 3 & 1 & 3 & 10 & 18 & 3 \\ 3 & 3 & 1 & 12 & 20 & 3 \\ 14 & 10 & 12 & 42 & 76 & 14 \\ 24 & 18 & 20 & 76 & 130 & 24 \\ 4 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[316] $1^{112} 1^{11} 3^1 21^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 11 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 5 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 11 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[322] $1^{131} 1^{11} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 & 2 \\ 1 & 2 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 8 & 9 \\ 2 & 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 2 & 2 & 0 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[317] $1^{13} 1^{11} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 2 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[318] $1^{12} 1^{11} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 2 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 11 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[319] $1^{123} 1^{111} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 21 & 9 \\ 6 & 4 & 7 & 0 & 21 & 16 & 8 \\ 2 & 2 & 2 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[320] $1^{121} 1^{11} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 & 2 \\ 1 & 2 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 8 & 9 \\ 2 & 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 2 & 2 & 0 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[321] $1^{132} 1^{111} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 21 & 9 \\ 5 & 4 & 7 & 0 & 21 & 16 & 8 \\ 2 & 2 & 2 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

¹²⁴

$$[323] \ 1^{\underline{1}} 3^{\underline{1}} 21^{\underline{1124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 2 & 5 & 2 & 10 & 16 & 4 \\ 8 & 13 & 10 & 24 & 40 & 10 \\ 12 & 20 & 16 & 40 & 62 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

¹¹²

$$[325] \ 1^{\underline{12}} 3^{\underline{1}} 21^{\underline{1113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 14 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 14 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

¹¹²

$$[326] \ 1^{\underline{1112}} 3^{\underline{1}} 21^{\underline{1212}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 6 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 4 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 6 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

¹²²²

$$[327] \ 1^{\underline{1222}} 3^{\underline{1}} 21^{\underline{1322}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 6 & 3 \\ 3 & 5 & 15 & 13 & 10 & 5 \\ 8 & 15 & 34 & 30 & 24 & 12 \\ 8 & 13 & 30 & 24 & 20 & 10 \\ 6 & 10 & 24 & 20 & 14 & 8 \\ 3 & 5 & 12 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

¹²²²

$$[328] \ 1^{\underline{1112}} 3^{\underline{1}} 21^{\underline{1212}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 6 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 5 & 4 & 5 & 14 & 10 & 18 & 6 \\ 6 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

¹²²²

$$[324] \ 1^{\underline{1222}} 3^{\underline{1}} 21^{\underline{1122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 4 & 5 & 2 & 10 & 8 & 4 \\ 8 & 13 & 10 & 24 & 20 & 10 \\ 6 & 10 & 8 & 20 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

¹¹⁴

$$[333] \ 1^{\underline{112}} 3^{\underline{1}} 21^{\underline{1113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 2 & 1 & 3 & 1 & 5 & 11 & 3 \\ 4 & 4 & 5 & 5 & 10 & 23 & 6 \\ 9 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

¹¹²

$$[332] \ 1^{\underline{112}} 3^{\underline{1}} 21^{\underline{1113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 7 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 2 & 1 & 3 & 1 & 5 & 11 & 3 \\ 4 & 4 & 5 & 5 & 10 & 23 & 6 \\ 7 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[331] $1^{1124} 3^{123} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 39 & 10 \\ 13 & 20 & 0 & 39 & 58 & 16 \\ 3 & 5 & 0 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[330] $1^{124} 3^{1113} 21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 2 & 1 & 3 & 1 & 5 & 11 & 3 \\ 5 & 4 & 5 & 5 & 10 & 23 & 6 \\ 9 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[329] $1^{212} 3^{11314} 21^{1314} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 13 & 3 \\ 3 & 5 & 15 & 8 & 25 & 5 \\ 8 & 15 & 34 & 18 & 57 & 12 \\ 5 & 8 & 18 & 8 & 30 & 6 \\ 13 & 25 & 57 & 30 & 92 & 20 \\ 3 & 5 & 12 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[335] $1^{112} 3^{121} 21^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 13 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 5 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 13 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[334] $1^{214} 3^{1114} 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 18 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 4 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 18 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[336] $1^{41} 3^{113} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 10 & 9 & 10 & 9 & 51 & 0 & 13 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$$

[338] $1^{131} 3^{113} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 1 & 2 \\ 1 & 0 & 2 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & 1 & 0 & 9 & 0 & 2 \\ 9 & 9 & 10 & 9 & 51 & 0 & 13 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{41} 3^{113} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 1 & 16 & 0 & 2 \\ 1 & 2 & 1 & 0 & 11 & 0 & 2 \\ 12 & 11 & 16 & 11 & 103 & 0 & 19 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

[340] $1^{\overline{114}} 1^{\overline{13}} 3^{\overline{1}} 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 7 & 25 & 4 \\ 3 & 1 & 3 & 5 & 17 & 3 \\ 3 & 3 & 1 & 6 & 22 & 3 \\ 7 & 5 & 6 & 9 & 39 & 7 \\ 25 & 17 & 22 & 39 & 137 & 25 \\ 4 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$

[339] $1^{\overline{111}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 & 3 \\ 3 & 5 & 5 & 8 & 5 & 5 \\ 3 & 5 & 2 & 6 & 4 & 4 \\ 5 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$

[341] $1^{\overline{124}} 1^{\overline{114}} 3^{\overline{1}} 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 12 & 2 \\ 1 & 0 & 2 & 1 & 4 & 12 & 2 \\ 1 & 2 & 0 & 3 & 5 & 16 & 2 \\ 2 & 1 & 3 & 1 & 5 & 16 & 3 \\ 5 & 4 & 5 & 5 & 10 & 36 & 6 \\ 12 & 12 & 16 & 16 & 36 & 110 & 20 \\ 2 & 2 & 2 & 3 & 6 & 20 & 3 \end{pmatrix}_{b_0}$

[343] $1^{\overline{124}} 1^{\overline{114}} 3^{\overline{1}} 21^{\overline{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 12 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 5 & 10 & 14 & 20 & 32 & 8 \\ 8 & 13 & 20 & 24 & 40 & 10 \\ 12 & 20 & 32 & 40 & 62 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$

[344] $1^{\overline{124}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 40 & 10 \\ 12 & 20 & 0 & 40 & 62 & 16 \\ 3 & 5 & 0 & 10 & 16 & 4 \end{pmatrix}_{b_0}$

[342] $1^{\overline{214}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1314}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 12 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 8 & 15 & 34 & 18 & 48 & 12 \\ 5 & 8 & 18 & 8 & 24 & 6 \\ 12 & 20 & 48 & 24 & 62 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[345] $1^{\overline{111}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 13 & 6 \\ 4 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$

[346] $1^{\overline{123}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 13 & 6 \\ 6 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$

[347] $1^{\overline{123}} 1^{\overline{1112}} 3^{\overline{1}} 21^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 5 & 2 \\ 5 & 4 & 5 & 4 & 10 & 13 & 6 \\ 6 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[348] $1^{1112} 3^{11} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 4 & 10 & 13 & 6 \\ 4 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{1121} 3^{11} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 3 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{1121} 3^{12} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 7 & 35 & 16 & 11 \\ 2 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[352] $1^{131} 3^{11} 21^{1131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 & 3 \\ 3 & 5 & 5 & 18 & 5 & 5 \\ 2 & 5 & 2 & 14 & 4 & 4 \\ 11 & 18 & 14 & 48 & 14 & 14 \\ 3 & 5 & 4 & 14 & 2 & 4 \\ 3 & 5 & 4 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[349] $1^{1121} 3^{142} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 9 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{111} 3^{1213} 21^{1112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -93$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 10 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 4 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 10 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[354] $1^{113} 3^{11} 21^{1114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 11 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 16 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 11 & 20 & 16 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[355] $1^{131} 3^{11} 21^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 & 3 \\ 3 & 5 & 10 & 18 & 5 & 5 \\ 5 & 10 & 14 & 28 & 8 & 8 \\ 11 & 18 & 28 & 48 & 14 & 14 \\ 3 & 5 & 8 & 14 & 2 & 4 \\ 3 & 5 & 8 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[356] $1^{122} 1^{111} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 5 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 2 & 3 & 4 & 5 & 13 & 11 & 5 \\ 6 & 5 & 8 & 13 & 23 & 21 & 9 \\ 5 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{214} 1^{123} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 25 & 4 \\ 3 & 1 & 3 & 10 & 17 & 3 \\ 3 & 3 & 1 & 12 & 22 & 3 \\ 14 & 10 & 12 & 42 & 78 & 14 \\ 25 & 17 & 22 & 78 & 137 & 25 \\ 4 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

[358] $1^{213} 1^{123} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 21 & 4 \\ 3 & 1 & 3 & 10 & 15 & 3 \\ 3 & 3 & 1 & 12 & 18 & 3 \\ 14 & 10 & 12 & 42 & 66 & 14 \\ 21 & 15 & 18 & 66 & 97 & 21 \\ 4 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[359] $1^{111} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 3 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{122} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{112} 1^{112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 13 & 6 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{122} 1^{112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 13 & 6 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{13} 1^{113} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 2 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 9 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{123} 1^{112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 10 & 5 \\ 5 & 4 & 5 & 9 & 10 & 13 & 6 \\ 6 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[374] \quad 1^{1^{141}} 3^1 21^{1^{21}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 3 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 9 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[373] \quad 1^{1^{22}} 3^1 21^{1^{21}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[372] \quad 1^{1^{11}} 3^1 21^{1^{212}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 5 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[371] \quad 1^{1^{211}} 3^1 21^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 4 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 3 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[370] \quad 1^{1^{22}} 3^1 21^{1^{122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 5 & 2 \\ 1 & 0 & 2 & 3 & 7 & 4 & 2 \\ 1 & 2 & 0 & 4 & 9 & 7 & 2 \\ 2 & 3 & 4 & 5 & 16 & 11 & 5 \\ 7 & 7 & 9 & 16 & 35 & 25 & 11 \\ 5 & 4 & 7 & 11 & 25 & 16 & 8 \\ 2 & 2 & 2 & 5 & 11 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[369] \quad 1^{1^{13}} 3^1 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 7 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[368] \quad 1^{1^{41}} 3^1 21^{1^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 1 & 2 \\ 1 & 0 & 2 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 1 & 14 & 0 & 2 \\ 2 & 2 & 1 & 0 & 12 & 0 & 2 \\ 12 & 12 & 14 & 12 & 96 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[367] \quad 1^{1^{12}} 3^1 21^{1^{212}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 4 & 8 & 2 \\ 5 & 6 & 8 & 26 & 12 & 28 & 10 \\ 3 & 2 & 4 & 12 & 4 & 12 & 4 \\ 6 & 6 & 8 & 28 & 12 & 26 & 10 \\ 2 & 2 & 2 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[366] \quad 1^{1^{111}} 3^{1^{21}} 21^{1^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 4 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[365] \quad 1^{1^{123}} 3^{1^{21}} 21^{1^{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 17 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 3 & 5 & 2 & 10 & 22 & 4 \\ 8 & 13 & 10 & 24 & 58 & 10 \\ 17 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[375] \quad 1^{1^{32}} 3^{1^{21}} 21^{1^{132}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 6 & 3 \\ 3 & 5 & 5 & 18 & 10 & 5 \\ 2 & 5 & 2 & 14 & 8 & 4 \\ 11 & 18 & 14 & 48 & 28 & 14 \\ 6 & 10 & 8 & 28 & 14 & 8 \\ 3 & 5 & 4 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[377] \quad 1^{1^{231}} 3^{1^{21}} 21^{1^{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 4 & 5 & 2 & 17 & 4 & 4 \\ 14 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[379] \quad 1^{1^{122}} 3^{1^{21}} 21^{1^{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 4 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 9 & 35 & 16 & 11 \\ 4 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[380] \quad 1^{1^{141}} 3^{1^{21}} 21^{1^{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 3 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 9 & 7 & 9 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[378] \quad 1^{1^{213}} 3^{1^{21}} 21^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 4 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[376] \quad 1^{1^{112}} 3^{1^{21}} 21^{1^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 10 & 2 \\ 1 & 2 & 0 & 3 & 5 & 15 & 2 \\ 2 & 1 & 3 & 1 & 5 & 15 & 3 \\ 4 & 4 & 5 & 5 & 10 & 32 & 6 \\ 9 & 10 & 15 & 15 & 32 & 88 & 18 \\ 2 & 2 & 2 & 3 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[381] \ 1^{\overline{1}^{132}} \ 3^{\overline{1}} \ 21^{\overline{1232}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 6 & 3 \\ 3 & 5 & 10 & 18 & 10 & 5 \\ 5 & 10 & 14 & 28 & 16 & 8 \\ 11 & 18 & 28 & 48 & 28 & 14 \\ 6 & 10 & 16 & 28 & 14 & 8 \\ 3 & 5 & 8 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[382] \ 1^{\overline{1}^{112}} \ 3^{\overline{1}} \ 21^{\overline{1412}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 6 & 3 \\ 3 & 5 & 20 & 8 & 10 & 5 \\ 11 & 20 & 62 & 24 & 32 & 16 \\ 5 & 8 & 24 & 8 & 12 & 6 \\ 6 & 10 & 32 & 12 & 14 & 8 \\ 3 & 5 & 16 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[383] \ 1^{\overline{1}^{1111}} \ 3^{\overline{1}} \ 21^{\overline{1121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 3 & 2 \\ 1 & 0 & 2 & 3 & 5 & 3 & 2 \\ 1 & 2 & 0 & 4 & 8 & 4 & 2 \\ 2 & 3 & 4 & 5 & 13 & 7 & 5 \\ 6 & 5 & 8 & 13 & 23 & 13 & 9 \\ 3 & 3 & 4 & 7 & 13 & 5 & 5 \\ 2 & 2 & 2 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[384] \ 1^{\overline{1}^{114}} \ 3^{\overline{1}} \ 21^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 13 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[385] \ 1^{\overline{1}^{14}} \ 2^{\overline{12}} \ 3^{\overline{1}} \ 21^{\overline{1114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -102, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 18 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 2 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 18 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{\overline{1}^{12}} \ 2^{\overline{12}} \ 3^{\overline{1}} \ 21^{\overline{1114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -102, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 16 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 2 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 16 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{\overline{1}^{131}} \ 3^{\overline{1}} \ 21^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -102, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 5 & 0 & 18 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 14 & 14 \\ 3 & 5 & 0 & 14 & 2 & 4 \\ 3 & 5 & 0 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{\overline{1}^{122}} \ 2^{\overline{112}} \ 3^{\overline{1}} \ 21^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -103, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 5 & 2 \\ 5 & 4 & 5 & 4 & 10 & 13 & 6 \\ 5 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[389] $1^{\overline{111}} 3^{\overline{1}} 21^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[390] $1^{\overline{122}} 3^{\overline{1}} 21^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[391] $1^{\overline{12}} 3^{\overline{112}} 3^{\overline{1}} 21^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 4 & 10 & 13 & 6 \\ 5 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[397] $1^{\overline{13}} 3^{\overline{114}} 3^{\overline{1}} 21^{\overline{1114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 11 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 16 & 2 \\ 2 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 11 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[396] $1^{\overline{13}} 3^{\overline{1}} 21^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 18 & 4 \\ 9 & 9 & 12 & 0 & 18 & 61 & 15 \\ 2 & 2 & 2 & 0 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[395] $1^{\overline{113}} 3^{\overline{1112}} 3^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 7 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 7 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[394] $1^{\overline{123}} 3^{\overline{1112}} 3^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 7 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[393] $1^{\overline{123}} 3^{\overline{11}} 3^{\overline{1}} 21^{\overline{1124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 14 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 2 & 5 & 2 & 10 & 19 & 4 \\ 8 & 13 & 10 & 24 & 49 & 10 \\ 14 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[392] $1^{123} 1^{11} 3^{121} 21^{1123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 2 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 12 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[399] $1^{212} 1^{11} 3^{121} 21^{1212} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 9 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 22 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 9 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[398] $1^{112} 1^{11} 3^{121} 21^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 3 & 5 & 2 & 6 & 8 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 8 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[400] $1^{123} 1^{123} 3^{121} 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 57 & 11 \\ 10 & 10 & 15 & 0 & 57 & 88 & 18 \\ 2 & 2 & 2 & 0 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[401] $1^{122} 1^{121} 3^{121} 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 9 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 25 & 11 \\ 5 & 4 & 7 & 0 & 25 & 16 & 8 \\ 2 & 2 & 2 & 0 & 11 & 8 & 3 \end{pmatrix}_{b_0}$$

[402] $1^{33} 1^{11} 3^{121} 21^{1133} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 12 & 3 \\ 3 & 5 & 5 & 18 & 20 & 5 \\ 2 & 5 & 2 & 14 & 15 & 4 \\ 11 & 18 & 14 & 48 & 54 & 14 \\ 12 & 20 & 15 & 54 & 58 & 16 \\ 3 & 5 & 4 & 14 & 16 & 4 \end{pmatrix}_{b_0}$$

[414] $1^{321} 1^{1211} 3^{121} 21^{1311} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 3 & 2 \\ 1 & 0 & 2 & 7 & 4 & 3 & 2 \\ 1 & 2 & 0 & 11 & 5 & 4 & 2 \\ 8 & 7 & 11 & 45 & 23 & 18 & 13 \\ 5 & 4 & 5 & 23 & 10 & 9 & 6 \\ 3 & 3 & 4 & 18 & 9 & 5 & 5 \\ 2 & 2 & 2 & 13 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[416] $1^{123} 1^{113} 3^{121} 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 22 & 6 \\ 8 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[412] $1^{13} 1^{112} 3^{121} 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 7 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[413] \quad 1^{1211} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 3 & 2 \\ 1 & 0 & 2 & 6 & 5 & 3 & 2 \\ 1 & 2 & 0 & 8 & 8 & 4 & 2 \\ 5 & 6 & 8 & 26 & 26 & 14 & 10 \\ 6 & 5 & 8 & 26 & 23 & 13 & 9 \\ 3 & 3 & 4 & 14 & 13 & 5 & 5 \\ 2 & 2 & 2 & 10 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[418] \quad 1^{133} 3^1 21^{1133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 9 & 3 \\ 3 & 5 & 5 & 18 & 15 & 5 \\ 2 & 5 & 2 & 14 & 12 & 4 \\ 11 & 18 & 14 & 48 & 42 & 14 \\ 9 & 15 & 12 & 42 & 34 & 12 \\ 3 & 5 & 4 & 14 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[411] \quad 1^{132} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 5 & 0 & 18 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 28 & 14 \\ 6 & 10 & 0 & 28 & 14 & 8 \\ 3 & 5 & 0 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[410] \quad 1^{1321} 3^1 21^{1421} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 & 3 \\ 3 & 5 & 20 & 13 & 5 & 5 \\ 11 & 20 & 62 & 40 & 16 & 16 \\ 8 & 13 & 40 & 24 & 10 & 10 \\ 3 & 5 & 16 & 10 & 2 & 4 \\ 3 & 5 & 16 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[409] \quad 1^{1321} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 7 & 10 & 14 & 20 & 8 & 8 \\ 8 & 13 & 20 & 24 & 10 & 10 \\ 3 & 5 & 8 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[408] \quad 1^{114} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 10 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 12 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 10 & 15 & 12 & 18 & 34 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[407] \quad 1^{1112} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 4 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 5 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[406] \quad 1^{1112} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 5 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[405] $1^{11} 3^{132} 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 6 & 3 \\ 3 & 5 & 0 & 28 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 28 & 0 & 116 & 42 & 22 \\ 6 & 10 & 0 & 42 & 14 & 8 \\ 3 & 5 & 0 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[404] $1^{11} 3^{123} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 7 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 7 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[403] $1^{11} 3^{141} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 1 & 13 & 0 & 2 \\ 1 & 2 & 1 & 0 & 10 & 0 & 2 \\ 11 & 10 & 13 & 10 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[417] $1^{121} 3^{122} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 5 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 2 & 1 & 3 & 1 & 9 & 8 & 3 \\ 7 & 7 & 9 & 9 & 35 & 32 & 11 \\ 5 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[415] $1^{231} 3^{1331} 21^{1331} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 & 3 \\ 3 & 5 & 15 & 18 & 5 & 5 \\ 8 & 15 & 34 & 42 & 12 & 12 \\ 11 & 18 & 42 & 48 & 14 & 14 \\ 3 & 5 & 12 & 14 & 2 & 4 \\ 3 & 5 & 12 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[419] $1^{114} 3^{113} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 9 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[420] $1^{111} 3^{111} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -109$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 8 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 8 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[423] $1^{111} 3^{111} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 3 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 3 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[424] $1^{11}3^{121}1^{1314} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 15 & 3 \\ 3 & 5 & 15 & 8 & 25 & 5 \\ 9 & 15 & 34 & 18 & 57 & 12 \\ 4 & 8 & 18 & 8 & 30 & 6 \\ 15 & 25 & 57 & 30 & 92 & 20 \\ 3 & 5 & 12 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[422] $1^{12}1^{11}3^{121}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 11 & 4 \\ 4 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[421] $1^{111}3^{121}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 4 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 4 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[425] $1^{1231}1^{11}3^{121}1^{1231} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 & 3 \\ 3 & 5 & 10 & 23 & 5 & 5 \\ 6 & 10 & 14 & 34 & 8 & 8 \\ 14 & 23 & 34 & 78 & 17 & 18 \\ 3 & 5 & 8 & 17 & 2 & 4 \\ 3 & 5 & 8 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[426] $1^{1222}1^{1211}3^{121}1^{1211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 3 & 2 \\ 1 & 2 & 0 & 8 & 5 & 4 & 2 \\ 6 & 6 & 8 & 26 & 18 & 14 & 10 \\ 5 & 4 & 5 & 18 & 10 & 9 & 6 \\ 4 & 3 & 4 & 14 & 9 & 5 & 5 \\ 2 & 2 & 2 & 10 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[427] $1^{1211}1^{1211}3^{121}1^{1211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 3 & 2 \\ 1 & 0 & 2 & 6 & 4 & 3 & 2 \\ 1 & 2 & 0 & 8 & 5 & 4 & 2 \\ 8 & 6 & 8 & 26 & 18 & 14 & 10 \\ 4 & 4 & 5 & 18 & 10 & 9 & 6 \\ 3 & 3 & 4 & 14 & 9 & 5 & 5 \\ 2 & 2 & 2 & 10 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[428] $1^{1222}1^{121}3^{121}1^{1223} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 11 & 3 \\ 3 & 5 & 10 & 13 & 20 & 8 \\ 6 & 10 & 14 & 20 & 30 & 8 \\ 8 & 13 & 20 & 24 & 39 & 10 \\ 11 & 20 & 30 & 39 & 58 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[429] $1^{1222}1^{11}3^{121}1^{1222} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 9 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 6 & 10 & 14 & 20 & 22 & 8 \\ 8 & 13 & 20 & 24 & 29 & 10 \\ 9 & 15 & 22 & 29 & 32 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[430] \ 1^{\overline{1214}} 3^{\overline{1}} 21^{\overline{1213}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 10 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 24 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 10 & 15 & 24 & 18 & 34 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[433] \ 1^{\overline{1214}} 3^{\overline{1}} 21^{\overline{1113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 16 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 4 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 16 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[432] \ 1^{\overline{1113}} 3^{\overline{1}} 21^{\overline{1114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 5 & 0 & 8 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 36 & 6 \\ 17 & 30 & 0 & 36 & 134 & 24 \\ 3 & 5 & 0 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[431] \ 1^{\overline{1113}} 3^{\overline{1}} 21^{\overline{1113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[434] \ 1^{\overline{1113}} 3^{\overline{1}} 21^{\overline{1113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 2 & 1 & 3 & 1 & 5 & 11 & 3 \\ 4 & 4 & 5 & 5 & 10 & 23 & 6 \\ 8 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[435] \ 1^{\overline{1114}} 3^{\overline{1}} 21^{\overline{1114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -118, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 28 & 4 \\ 3 & 1 & 3 & 5 & 20 & 3 \\ 3 & 3 & 1 & 6 & 24 & 3 \\ 7 & 5 & 6 & 9 & 44 & 7 \\ 28 & 20 & 24 & 44 & 174 & 28 \\ 4 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[436] \ 1^{\overline{1212}} 3^{\overline{1}} 21^{\overline{1112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 6 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[437] \ 1^{\overline{1114}} 3^{\overline{1}} 21^{\overline{1112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 8 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[438] \ 1^{\overline{134}} 3^{\overline{1}} 21^{\overline{1134}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -122, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 12 & 3 \\ 3 & 5 & 5 & 18 & 20 & 5 \\ 2 & 5 & 2 & 14 & 16 & 4 \\ 11 & 18 & 14 & 48 & 56 & 14 \\ 12 & 20 & 16 & 56 & 62 & 16 \\ 3 & 5 & 4 & 14 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[443] \quad 1^{13} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 1 & 2 \\ 1 & 0 & 2 & 1 & 10 & 0 & 2 \\ 1 & 2 & 0 & 3 & 13 & 0 & 2 \\ 2 & 1 & 3 & 1 & 13 & 0 & 3 \\ 10 & 10 & 13 & 13 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[439] \quad 1^{13} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 & 3 \\ 3 & 5 & 15 & 13 & 5 & 5 \\ 10 & 15 & 32 & 29 & 11 & 12 \\ 8 & 13 & 29 & 24 & 10 & 10 \\ 3 & 5 & 11 & 10 & 2 & 4 \\ 3 & 5 & 12 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[440] \quad 1^{1111} 3^1 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 5 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 4 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[441] \quad 1^{1111} 3^1 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 6 & 4 & 7 & 16 & 14 & 11 & 8 \\ 5 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[442] \quad 1^{121} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 13 & 23 & 0 & 78 & 17 & 18 \\ 3 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[445] \quad 1^{12} 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 18 & 3 \\ 3 & 5 & 10 & 8 & 30 & 5 \\ 6 & 10 & 14 & 12 & 44 & 8 \\ 5 & 8 & 12 & 8 & 36 & 6 \\ 18 & 30 & 44 & 36 & 134 & 24 \\ 3 & 5 & 8 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[444] \quad 1^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 2 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 4 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[446] \quad 1^1 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 & 3 \\ 3 & 5 & 5 & 18 & 5 & 5 \\ 4 & 5 & 2 & 14 & 4 & 4 \\ 11 & 18 & 14 & 48 & 14 & 14 \\ 3 & 5 & 4 & 14 & 2 & 4 \\ 3 & 5 & 4 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[447] $1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 32 & 11 \\ 6 & 6 & 8 & 0 & 32 & 26 & 10 \\ 2 & 2 & 2 & 0 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[448] $1^{212} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 6 & 3 \\ 3 & 5 & 10 & 8 & 10 & 5 \\ 6 & 10 & 14 & 12 & 16 & 8 \\ 5 & 8 & 12 & 8 & 12 & 6 \\ 6 & 10 & 16 & 12 & 14 & 8 \\ 3 & 5 & 8 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[449] $1^{131} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 10 & 10 & 13 & 13 & 74 & 23 & 16 \\ 2 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[450] $1^{1122} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 6 & 2 \\ 1 & 2 & 0 & 4 & 8 & 8 & 2 \\ 2 & 3 & 4 & 5 & 13 & 14 & 5 \\ 6 & 5 & 8 & 13 & 23 & 26 & 9 \\ 6 & 6 & 8 & 14 & 26 & 26 & 10 \\ 2 & 2 & 2 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[451] $1^{114} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[452] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 8 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[453] $1^{141} 3^1 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 2 & 5 & 2 & 21 & 4 & 4 \\ 17 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[454] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 3 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 8 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[455] $1^{11} 3^{113} 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 14 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 14 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[456] $1^{11} 3^{113} 21^{1214} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 14 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 5 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 14 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[457] $1^{12} 3^{114} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 16 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[459] $1^{112} 3^{114} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 23 & 6 \\ 9 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[458] $1^{112} 3^{114} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 23 & 6 \\ 9 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[464] $1^{11} 3^{122} 21^{1234} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 13 & 3 \\ 3 & 5 & 10 & 18 & 25 & 5 \\ 6 & 10 & 14 & 28 & 38 & 8 \\ 10 & 18 & 28 & 48 & 68 & 14 \\ 13 & 25 & 38 & 68 & 92 & 20 \\ 3 & 5 & 8 & 14 & 20 & 4 \end{pmatrix}_{b_0}$$

[460] $1^{111} 3^{123} 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 29 & 9 \\ 7 & 5 & 10 & 0 & 29 & 33 & 11 \\ 2 & 2 & 2 & 0 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[461] $1^{111} 3^{121} 21^{1122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 4 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 2 & 1 & 3 & 1 & 8 & 7 & 3 \\ 6 & 5 & 8 & 8 & 23 & 21 & 9 \\ 4 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[462] $1^{123} 1^{11} 3^1 21^{1213} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 12 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 30 & 8 \\ 4 & 8 & 12 & 8 & 24 & 6 \\ 12 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[463] $1^{13} 1^{12} 3^1 21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 18 & 4 \\ 4 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[467] $1^{141} 1^{131} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 10 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 9 & 13 \\ 2 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[466] $1^{123} 1^{11} 3^1 21^{1224} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 14 & 3 \\ 3 & 5 & 10 & 13 & 25 & 5 \\ 5 & 10 & 14 & 20 & 38 & 8 \\ 8 & 13 & 20 & 24 & 49 & 10 \\ 14 & 25 & 38 & 49 & 92 & 20 \\ 3 & 5 & 8 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[465] $1^{123} 1^{11} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 39 & 10 \\ 12 & 20 & 0 & 39 & 58 & 16 \\ 3 & 5 & 0 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[468] $1^{113} 1^{112} 3^1 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 7 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 7 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[469] $1^{13} 1^{131} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 10 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 9 & 13 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[470] $1^{123} 1^{112} 3^1 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 7 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 5 & 4 & 5 & 5 & 10 & 18 & 6 \\ 7 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[471] $1^{113} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 47 & 9 \\ 10 & 10 & 15 & 0 & 47 & 88 & 18 \\ 2 & 2 & 2 & 0 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

[472] $1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -133$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 2 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[473] $1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 18 & 3 \\ 3 & 5 & 0 & 13 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 58 & 10 \\ 18 & 30 & 0 & 58 & 134 & 24 \\ 3 & 5 & 0 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

[474] $1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 5 & 4 & 5 & 5 & 10 & 18 & 6 \\ 8 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[475] $1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 6 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[476] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 15 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[477] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 12 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 12 & 18 & 34 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[478] $1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 28 & 4 \\ 3 & 1 & 3 & 10 & 20 & 3 \\ 3 & 3 & 1 & 12 & 24 & 3 \\ 14 & 10 & 12 & 42 & 88 & 14 \\ 28 & 20 & 24 & 88 & 174 & 28 \\ 4 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

[479] $1^{131} 3^{1241} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 & 3 \\ 3 & 5 & 10 & 28 & 5 & 5 \\ 5 & 10 & 14 & 42 & 8 & 8 \\ 16 & 28 & 42 & 116 & 21 & 22 \\ 3 & 5 & 8 & 21 & 2 & 4 \\ 3 & 5 & 8 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[484] $1^{141} 3^{1241} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 & 3 \\ 3 & 5 & 10 & 23 & 5 & 5 \\ 5 & 10 & 14 & 36 & 8 & 8 \\ 14 & 23 & 36 & 80 & 18 & 18 \\ 3 & 5 & 8 & 18 & 2 & 4 \\ 3 & 5 & 8 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[480] $1^{111} 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 \\ 1 & 2 & 0 & 0 & 8 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 13 \\ 3 & 3 & 4 & 0 & 13 & 5 \\ 2 & 2 & 2 & 0 & 9 & 5 \end{pmatrix}_{b_0}$$

[481] $1^{131} 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 1 & 13 & 0 & 2 \\ 1 & 2 & 1 & 0 & 10 & 0 & 2 \\ 10 & 10 & 13 & 10 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[482] $1^{122} 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 23 & 11 \\ 4 & 5 & 5 & 0 & 23 & 13 & 7 \\ 2 & 2 & 2 & 0 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

[483] $1^{132} 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 13 & 9 \\ 4 & 3 & 4 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & 0 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[485] $1^{131} 3^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 78 & 17 & 18 \\ 3 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[486] $1^{141} 3^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 2 & 5 & 2 & 18 & 4 & 4 \\ 14 & 23 & 18 & 80 & 18 & 18 \\ 3 & 5 & 4 & 18 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[487] $1^{1^{31}} 1^{12} 3^{121} 21^{1141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 & 3 \\ 3 & 5 & 5 & 33 & 5 & 5 \\ 2 & 5 & 2 & 24 & 4 & 4 \\ 19 & 33 & 24 & 164 & 24 & 26 \\ 3 & 5 & 4 & 24 & 2 & 4 \\ 3 & 5 & 4 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[488] $1^{1^{214}} 1^{13} 1^{21} 1^{1224} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 12 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 6 & 10 & 14 & 20 & 32 & 8 \\ 7 & 13 & 20 & 24 & 40 & 10 \\ 12 & 20 & 32 & 40 & 62 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[489] $1^{1^{212}} 1^{11} 3^{121} 21^{1213} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 11 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 11 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[490] $1^{1^{111}} 1^{13} 1^{21} 1^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 3 & 5 & 2 & 10 & 4 & 4 \\ 7 & 13 & 10 & 24 & 10 & 10 \\ 3 & 5 & 4 & 10 & 2 & 4 \\ 3 & 5 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[491] $1^{1^{112}} 1^{11} 3^{121} 21^{1122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 3 & 5 & 2 & 10 & 8 & 4 \\ 7 & 13 & 10 & 24 & 20 & 10 \\ 6 & 10 & 8 & 20 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[492] $1^{1^{113}} 1^{13} 1^{21} 1^{1123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 3 & 5 & 2 & 10 & 12 & 4 \\ 7 & 13 & 10 & 24 & 30 & 10 \\ 9 & 15 & 12 & 30 & 34 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[493] $1^{1^{114}} 1^{13} 1^{21} 1^{1124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 16 & 4 \\ 7 & 13 & 10 & 24 & 40 & 10 \\ 12 & 20 & 16 & 40 & 62 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[494] $1^{1^{212}} 1^{13} 1^{21} 1^{1222} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 6 & 3 \\ 3 & 5 & 10 & 13 & 10 & 5 \\ 6 & 10 & 14 & 20 & 16 & 8 \\ 7 & 13 & 20 & 24 & 20 & 10 \\ 6 & 10 & 16 & 20 & 14 & 8 \\ 3 & 5 & 8 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[495] $1^{1^{213}} 1^{13} 1^{21} 1^{1223} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 9 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 6 & 10 & 14 & 20 & 24 & 8 \\ 7 & 13 & 20 & 24 & 30 & 10 \\ 9 & 15 & 24 & 30 & 34 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[496] $1^{11} 3^{1142} 21^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 6 & 3 \\ 3 & 5 & 0 & 28 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 28 & 0 & 116 & 42 & 22 \\ 6 & 10 & 0 & 42 & 14 & 8 \\ 3 & 5 & 0 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[497] $1^{11} 3^{1142} 21^{1242} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 6 & 3 \\ 3 & 5 & 10 & 23 & 10 & 5 \\ 5 & 10 & 14 & 36 & 16 & 8 \\ 14 & 23 & 36 & 80 & 36 & 18 \\ 6 & 10 & 16 & 36 & 14 & 8 \\ 3 & 5 & 8 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[498] $1^{11} 3^{1142} 21^{1131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 2 & 5 & 2 & 17 & 4 & 4 \\ 13 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[499] $1^{11} 3^{1142} 21^{1142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 6 & 3 \\ 3 & 5 & 5 & 23 & 10 & 5 \\ 2 & 5 & 2 & 18 & 8 & 4 \\ 14 & 23 & 18 & 80 & 36 & 18 \\ 6 & 10 & 8 & 36 & 14 & 8 \\ 3 & 5 & 4 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[500] $1^{11} 3^{1142} 21^{1142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 6 & 3 \\ 3 & 5 & 5 & 28 & 10 & 5 \\ 2 & 5 & 2 & 21 & 8 & 4 \\ 16 & 28 & 21 & 116 & 42 & 22 \\ 6 & 10 & 8 & 42 & 14 & 8 \\ 3 & 5 & 4 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[503] $1^{11} 3^{1142} 21^{1112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 10 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 10 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[501] $1^{11} 3^{1142} 21^{1114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 4 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[502] $1^{11} 3^{1142} 21^{1114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 13 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[506] \ 1^{\overline{1421}} \ 3^{\overline{1}} \ 21^{\overline{1321}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 & 3 \\ 3 & 5 & 15 & 13 & 5 & 5 \\ 10 & 15 & 34 & 30 & 12 & 12 \\ 8 & 13 & 30 & 24 & 10 & 10 \\ 3 & 5 & 12 & 10 & 2 & 4 \\ 3 & 5 & 12 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[504] \ 1^{\overline{141}} \ 3^{\overline{1}} \ 21^{\overline{1141}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 & 2 \\ 1 & 0 & 2 & 3 & 11 & 1 & 2 \\ 1 & 2 & 0 & 4 & 16 & 3 & 2 \\ 2 & 3 & 4 & 5 & 27 & 4 & 5 \\ 12 & 11 & 16 & 27 & 103 & 16 & 19 \\ 2 & 1 & 3 & 4 & 16 & 1 & 3 \\ 2 & 2 & 2 & 5 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[505] \ 1^{\overline{1241}} \ 3^{\overline{1}} \ 21^{\overline{1341}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 & 3 \\ 3 & 5 & 15 & 23 & 5 & 5 \\ 8 & 15 & 34 & 54 & 12 & 12 \\ 14 & 23 & 54 & 80 & 18 & 18 \\ 3 & 5 & 12 & 18 & 2 & 4 \\ 3 & 5 & 12 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[512] \ 1^{\overline{114}} \ 3^{\overline{1}} \ 21^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 26 & 6 \\ 10 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[514] \ 1^{\overline{114}} \ 3^{\overline{1}} \ 21^{\overline{1114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 2 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 12 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[513] \ 1^{\overline{1241}} \ 3^{\overline{1}} \ 21^{\overline{123}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 1 & 2 \\ 1 & 0 & 2 & 4 & 10 & 0 & 2 \\ 1 & 2 & 0 & 7 & 13 & 0 & 2 \\ 5 & 4 & 7 & 16 & 36 & 0 & 8 \\ 11 & 10 & 13 & 36 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 8 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[507] \ 1^{\overline{121}} \ 3^{\overline{1}} \ 21^{\overline{121}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 8 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[508] \ 1^{\overline{124}} \ 3^{\overline{1}} \ 21^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 8 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 8 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[509] \ 1^{\overline{132}} \ 3^{\overline{1}} \ 21^{\overline{121}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -152, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 8 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[510] \quad 1^{1^{124}} 3^1 21^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 26 & 6 \\ 10 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[511] \quad 1^{1^{12}} 3^{1^{12}} 3^1 21^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 6 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[516] \quad 1^{1^{13}} 3^{1^{13}} 3^1 21^{1^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 17 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 2 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 17 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[515] \quad 1^{1^{213}} 3^1 21^{1^{213}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 9 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 24 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 9 & 15 & 24 & 18 & 34 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[520] \quad 1^{1^{214}} 3^1 21^{1^{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 4 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 13 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[519] \quad 1^{1^{121}} 3^1 21^{1^{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 3 & 2 \\ 1 & 0 & 2 & 1 & 5 & 3 & 2 \\ 1 & 2 & 0 & 3 & 8 & 4 & 2 \\ 2 & 1 & 3 & 1 & 8 & 4 & 3 \\ 6 & 5 & 8 & 8 & 23 & 13 & 9 \\ 3 & 3 & 4 & 4 & 13 & 5 & 5 \\ 2 & 2 & 2 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[518] \quad 1^{1^{141}} 3^1 21^{1^{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 1 & 2 \\ 1 & 0 & 2 & 1 & 10 & 0 & 2 \\ 1 & 2 & 0 & 3 & 13 & 0 & 2 \\ 2 & 1 & 3 & 1 & 13 & 0 & 3 \\ 11 & 10 & 13 & 13 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[517] \quad 1^{1^{132}} 3^1 21^{1^{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 4 & 2 \\ 1 & 0 & 2 & 1 & 5 & 3 & 2 \\ 1 & 2 & 0 & 3 & 8 & 4 & 2 \\ 2 & 1 & 3 & 1 & 8 & 4 & 3 \\ 7 & 5 & 8 & 8 & 23 & 13 & 9 \\ 4 & 3 & 4 & 4 & 13 & 5 & 5 \\ 2 & 2 & 2 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[523] \quad 1^{\overline{141}} 3^{\overline{1}} 21^{\overline{141}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 80 & 18 & 18 \\ 3 & 5 & 0 & 18 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[522] \quad 1^{\overline{222}} 1^{\overline{121}} 3^{\overline{1}} 21^{\overline{1221}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 4 & 2 \\ 1 & 0 & 2 & 4 & 7 & 3 & 2 \\ 1 & 2 & 0 & 7 & 9 & 4 & 2 \\ 5 & 4 & 7 & 16 & 25 & 11 & 8 \\ 7 & 7 & 9 & 25 & 35 & 16 & 11 \\ 4 & 3 & 4 & 11 & 16 & 5 & 5 \\ 2 & 2 & 2 & 8 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[521] \quad 1^{\overline{112}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{1113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 4 & 4 & 5 & 5 & 10 & 27 & 6 \\ 8 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[524] \quad 1^{\overline{112}} 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1212}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 7 & 2 \\ 5 & 4 & 7 & 16 & 14 & 18 & 8 \\ 4 & 4 & 5 & 14 & 10 & 14 & 6 \\ 5 & 4 & 7 & 18 & 14 & 16 & 8 \\ 2 & 2 & 2 & 8 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[525] \quad 1^{\overline{113}} 1^{\overline{1112}} 3^{\overline{1}} 21^{\overline{1113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 2 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 8 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[526] \quad 1^{\overline{114}} 1^{\overline{1112}} 3^{\overline{1}} 21^{\overline{1113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -157, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 11 & 2 \\ 1 & 0 & 2 & 1 & 4 & 11 & 2 \\ 1 & 2 & 0 & 3 & 5 & 13 & 2 \\ 2 & 1 & 3 & 1 & 5 & 13 & 3 \\ 4 & 4 & 5 & 5 & 10 & 31 & 6 \\ 11 & 11 & 13 & 13 & 31 & 81 & 17 \\ 2 & 2 & 2 & 3 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[527] \quad 1^{\overline{124}} 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -158, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 15 & 3 \\ 3 & 5 & 2 & 13 & 25 & 5 \\ 2 & 5 & 2 & 10 & 19 & 4 \\ 8 & 13 & 10 & 24 & 49 & 10 \\ 15 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[528] \quad 1^{\overline{143}} 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1143}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -158, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 9 & 3 \\ 3 & 5 & 5 & 23 & 15 & 5 \\ 2 & 5 & 2 & 18 & 12 & 4 \\ 14 & 23 & 18 & 80 & 54 & 18 \\ 9 & 15 & 12 & 54 & 34 & 12 \\ 3 & 5 & 4 & 18 & 12 & 4 \end{pmatrix}_{b_0}$$

[535] $1^{131} 1^{12} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 12 & 14 \\ 2 & 1 & 3 & 0 & 12 & 1 & 3 \\ 2 & 2 & 2 & 0 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[538] $1^{213} 1^{113} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 4 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 9 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[537] $1^{113} 1^{112} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 18 & 6 \\ 7 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[536] $1^{123} 1^{112} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 18 & 6 \\ 7 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[529] $1^{132} 1^{131} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 10 & 4 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 10 & 10 & 13 & 13 & 74 & 23 & 16 \\ 4 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[530] $1^{141} 1^{131} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 19 & 13 \\ 3 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[531] $1^{112} 1^{113} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 8 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[532] $1^{114} 1^{14} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 24 & 4 \\ 12 & 12 & 16 & 0 & 24 & 110 & 20 \\ 2 & 2 & 2 & 0 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[533] $1^{131} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 19 & 13 \\ 4 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[534] $1^{131} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 1 & 2 \\ 1 & 0 & 2 & 4 & 10 & 0 & 2 \\ 1 & 2 & 0 & 7 & 13 & 0 & 2 \\ 5 & 4 & 7 & 16 & 36 & 0 & 8 \\ 10 & 10 & 13 & 36 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 8 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[543] $1^{111} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 8 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 3 & 5 & 2 & 10 & 11 & 4 \\ 7 & 13 & 10 & 24 & 29 & 10 \\ 8 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[542] $1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 3 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[541] $1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 5 & 0 & 23 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 80 & 36 & 18 \\ 6 & 10 & 0 & 36 & 14 & 8 \\ 3 & 5 & 0 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[540] $1^{114} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 15 & 25 & 0 & 30 & 92 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[539] $1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 4 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 9 & 35 & 16 & 11 \\ 4 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[546] $1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 3 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[545] $1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -163$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 4 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 9 & 7 & 9 & 9 & 35 & 16 & 11 \\ 4 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[544] $1^{121} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -163$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 4 \\ 4 & 5 & 2 & 17 & 4 & 4 \\ 15 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[547] $1^{132} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 6 & 2 \\ 1 & 2 & 0 & 0 & 10 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 38 & 13 \\ 6 & 6 & 8 & 0 & 38 & 26 & 10 \\ 2 & 2 & 2 & 0 & 13 & 10 & 3 \end{pmatrix}_{b_0}$$

[548] $1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 26 & 9 \\ 6 & 6 & 8 & 0 & 26 & 26 & 10 \\ 2 & 2 & 2 & 0 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[549] $1^{124} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 27 & 6 \\ 10 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[550] $1^{111} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 4 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 4 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[551] $1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 18 & 4 \\ 10 & 9 & 12 & 0 & 18 & 61 & 15 \\ 2 & 2 & 2 & 0 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[552] $1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 3 & 5 & 2 & 10 & 4 & 4 \\ 8 & 13 & 10 & 24 & 10 & 10 \\ 3 & 5 & 4 & 10 & 2 & 4 \\ 3 & 5 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[554] \quad 1^{132} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -168, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 6 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 2 & 1 & 3 & 1 & 9 & 8 & 3 \\ 8 & 7 & 9 & 9 & 35 & 32 & 11 \\ 6 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[553] \quad 1^{121} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -168, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 3 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 3 & 3 & 4 & 5 & 9 & 7 & 5 \\ 5 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[556] \quad 1^{131} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -171, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 28 & 13 \\ 4 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[555] \quad 1^{111} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -171, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 2 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 5 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[560] \quad 1^{132} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 5 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 2 & 1 & 3 & 1 & 8 & 7 & 3 \\ 7 & 5 & 8 & 8 & 23 & 21 & 9 \\ 5 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[559] \quad 1^{123} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 2 & 1 & 3 & 1 & 8 & 7 & 3 \\ 6 & 5 & 8 & 8 & 23 & 21 & 9 \\ 6 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[558] \quad 1^{144} 3^1 21^{1144} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 12 & 3 \\ 3 & 5 & 5 & 23 & 20 & 5 \\ 2 & 5 & 2 & 18 & 16 & 4 \\ 14 & 23 & 18 & 80 & 72 & 18 \\ 12 & 20 & 16 & 72 & 62 & 16 \\ 3 & 5 & 4 & 18 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[557] \quad 1^{123} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 7 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 2 & 1 & 3 & 1 & 9 & 8 & 3 \\ 7 & 7 & 9 & 9 & 35 & 32 & 11 \\ 7 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[562] \quad 1^{1112} 3^{12} 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 2 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 7 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[561] \quad 1^{134} 1^{11} 3^{12} 21^{1134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 15 & 3 \\ 3 & 5 & 5 & 18 & 25 & 5 \\ 2 & 5 & 2 & 14 & 19 & 4 \\ 11 & 18 & 14 & 48 & 68 & 14 \\ 15 & 25 & 19 & 68 & 92 & 20 \\ 3 & 5 & 4 & 14 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[563] \quad 1^{1111} 3^{12} 21^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -175, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 5 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[567] \quad 1^{11} 3^{12} 21^{1213} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 13 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 13 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[565] \quad 1^{112} 1^{12} 3^{12} 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 14 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 14 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[566] \quad 1^{113} 1^{11} 3^{12} 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 15 & 4 \\ 4 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[564] \quad 1^{141} 1^{12} 3^{12} 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 12 & 14 \\ 2 & 1 & 3 & 0 & 12 & 1 & 3 \\ 2 & 2 & 2 & 0 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[568] \quad 1^{114} 1^{11} 3^{12} 21^{1114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 16 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 16 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[569] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 23 & 6 \\ 8 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[570] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[571] $1^{241} 3^1 21^{1141} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 4 & 5 & 2 & 21 & 4 & 4 \\ 17 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[572] $1^{241} 3^1 21^{1141} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 4 & 5 & 2 & 18 & 4 & 4 \\ 14 & 23 & 18 & 80 & 18 & 18 \\ 3 & 5 & 4 & 18 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[573] $1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 21 & 9 \\ 5 & 4 & 7 & 0 & 21 & 16 & 8 \\ 2 & 2 & 2 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[578] $1^{121} 3^1 21^{1221} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 3 & 2 \\ 1 & 0 & 2 & 4 & 7 & 3 & 2 \\ 1 & 2 & 0 & 7 & 9 & 4 & 2 \\ 5 & 4 & 7 & 16 & 25 & 11 & 8 \\ 8 & 7 & 9 & 25 & 35 & 16 & 11 \\ 3 & 3 & 4 & 11 & 16 & 5 & 5 \\ 2 & 2 & 2 & 8 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[574] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 18 & 6 \\ 6 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[575] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 8 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[576] $1^{124} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 18 & 6 \\ 8 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[577] \quad 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 6 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[580] \quad 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 5 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[579] \quad 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 4 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 8 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[581] \quad 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 3 & 5 & 2 & 10 & 8 & 4 \\ 8 & 13 & 10 & 24 & 20 & 10 \\ 6 & 10 & 8 & 20 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[582] \quad 1^{132} 3^1 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 23 & 11 \\ 5 & 5 & 5 & 0 & 23 & 13 & 7 \\ 2 & 2 & 2 & 0 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[590] \quad 1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 3 & 3 & 4 & 5 & 9 & 7 & 5 \\ 5 & 4 & 5 & 9 & 10 & 9 & 6 \\ 4 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[589] \quad 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 49 & 10 \\ 15 & 25 & 0 & 49 & 92 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[587] \quad 1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 3 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 4 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

^{1¹²²}

$$[586] \ 1^{11} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 3 & 5 & 2 & 10 & 11 & 4 \\ 8 & 13 & 10 & 24 & 29 & 10 \\ 9 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

^{1¹²²}

$$[585] \ 1^{11} 3^1 21^{1123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 11 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 11 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

^{1¹²⁴}

$$[584] \ 1^{113} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 10 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 5 & 4 & 5 & 5 & 10 & 27 & 6 \\ 10 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

^{1²¹²}

$$[588] \ 1^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

^{1¹²²}

$$[583] \ 1^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 10 & 5 \\ 5 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

^{1¹¹²}

$$[591] \ 1^{12} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 16 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 3 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 16 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

^{1²⁴}

$$[592] \ 1^{11} 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 15 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 6 & 10 & 14 & 12 & 38 & 8 \\ 4 & 8 & 12 & 8 & 30 & 6 \\ 15 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

^{1¹²²}

$$[593] \ 1^{122} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 5 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[594] $1^{1111} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 3 & 2 \\ 1 & 0 & 2 & 4 & 5 & 3 & 2 \\ 1 & 2 & 0 & 7 & 8 & 4 & 2 \\ 4 & 4 & 7 & 16 & 21 & 11 & 8 \\ 6 & 5 & 8 & 21 & 23 & 13 & 9 \\ 3 & 3 & 4 & 11 & 13 & 5 & 5 \\ 2 & 2 & 2 & 8 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[595] $1^{1214} 1^1 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 12 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 32 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 12 & 20 & 32 & 24 & 62 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[597] $1^{1121} 1^{142} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 9 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[596] $1^{121} 1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[598] $1^{1212} 1^{11} 3^1 21^{1223} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 11 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 6 & 10 & 14 & 20 & 30 & 8 \\ 7 & 13 & 20 & 24 & 39 & 10 \\ 11 & 20 & 30 & 39 & 58 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[600] $1^{131} 1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[599] $1^{132} 1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[603] $1^{113} 1^{112} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 7 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[602] \quad 1^{112} 3^{1213} 21^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 7 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[601] \quad 1^{113} 3^{113} 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 32 & 6 \\ 10 & 10 & 15 & 0 & 32 & 88 & 18 \\ 2 & 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[604] \quad 1^{111} 3^{1131} 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 28 & 0 & 116 & 21 & 22 \\ 3 & 5 & 0 & 21 & 2 & 4 \\ 3 & 5 & 0 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[605] \quad 1^{141} 3^{1141} 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -203, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 & 2 \\ 1 & 0 & 2 & 1 & 13 & 3 & 2 \\ 1 & 2 & 0 & 3 & 17 & 4 & 2 \\ 2 & 1 & 3 & 1 & 17 & 4 & 3 \\ 13 & 13 & 17 & 17 & 127 & 30 & 21 \\ 2 & 3 & 4 & 4 & 30 & 5 & 5 \\ 2 & 2 & 2 & 3 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[607] \quad 1^{112} 3^{112} 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 7 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[608] \quad 1^{122} 3^{1122} 21^{1222} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 6 & 3 \\ 3 & 5 & 10 & 13 & 10 & 5 \\ 6 & 10 & 14 & 20 & 16 & 8 \\ 8 & 13 & 20 & 24 & 20 & 10 \\ 6 & 10 & 16 & 20 & 14 & 8 \\ 3 & 5 & 8 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[606] \quad 1^{123} 3^{1123} 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 41 & 11 \\ 8 & 7 & 11 & 0 & 41 & 45 & 13 \\ 2 & 2 & 2 & 0 & 11 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[609] \quad 1^{141} 3^{1141} 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 1 & 2 \\ 1 & 0 & 2 & 1 & 13 & 0 & 2 \\ 1 & 2 & 0 & 3 & 17 & 0 & 2 \\ 2 & 1 & 3 & 1 & 17 & 0 & 3 \\ 13 & 13 & 17 & 17 & 127 & 0 & 21 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 21 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[610] \quad 1^{113} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 39 & 9 \\ 9 & 9 & 12 & 0 & 39 & 61 & 15 \\ 2 & 2 & 2 & 0 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[611] \quad 1^{113} 3^1 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 10 & 2 \\ 1 & 0 & 2 & 1 & 4 & 10 & 2 \\ 1 & 2 & 0 & 3 & 5 & 15 & 2 \\ 2 & 1 & 3 & 1 & 5 & 15 & 3 \\ 4 & 4 & 5 & 5 & 10 & 32 & 6 \\ 10 & 10 & 15 & 15 & 32 & 88 & 18 \\ 2 & 2 & 2 & 3 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[612] \quad 1^{1112} 3^1 21^{1212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -208, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 7 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 7 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[613] \quad 1^{131} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -211, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 33 & 16 \\ 4 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 16 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[614] \quad 1^{114} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -212, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 5 & 0 & 8 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 36 & 6 \\ 18 & 30 & 0 & 36 & 134 & 24 \\ 3 & 5 & 0 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[615] \quad 1^{1112} 3^1 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -214, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 2 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[616] \quad 1^{114} 3^1 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 4 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 12 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[617] \quad 1^{1112} 3^1 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[618] \quad 1^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 10 & 5 \\ 5 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[619] \quad 21^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 4 & 5 & 2 \\ 2 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[620] \quad 1^{112} 3^1 21^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 16 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 3 & 5 & 2 & 10 & 22 & 4 \\ 7 & 13 & 10 & 24 & 58 & 10 \\ 16 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[621] \quad 1^{123} 3^1 21^{1123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 3 & 5 & 2 & 10 & 12 & 4 \\ 8 & 13 & 10 & 24 & 30 & 10 \\ 9 & 15 & 12 & 30 & 34 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[622] \quad 1^{121} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -217, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 4 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 2 & 1 & 3 & 1 & 12 & 4 & 3 \\ 9 & 8 & 12 & 12 & 56 & 20 & 14 \\ 4 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[624] \quad 1^{1111} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 3 & 2 \\ 1 & 0 & 2 & 4 & 5 & 3 & 2 \\ 1 & 2 & 0 & 7 & 8 & 4 & 2 \\ 6 & 4 & 7 & 16 & 21 & 11 & 8 \\ 6 & 5 & 8 & 21 & 23 & 13 & 9 \\ 3 & 3 & 4 & 11 & 13 & 5 & 5 \\ 2 & 2 & 2 & 8 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[623] \quad 1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 28 & 0 & 116 & 21 & 22 \\ 3 & 5 & 0 & 21 & 2 & 4 \\ 3 & 5 & 0 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[625] \quad 1^{112} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 11 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 11 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[626] \quad 1^{122} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -220, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 6 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 2 & 1 & 3 & 1 & 9 & 8 & 3 \\ 7 & 7 & 9 & 9 & 35 & 32 & 11 \\ 6 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[628] \quad 1^{131} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -222, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 & 3 \\ 3 & 5 & 0 & 33 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 19 & 33 & 0 & 164 & 24 & 26 \\ 3 & 5 & 0 & 24 & 2 & 4 \\ 3 & 5 & 0 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[627] \quad 1^{132} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -222, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 5 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 3 & 3 & 4 & 5 & 13 & 11 & 5 \\ 7 & 5 & 8 & 13 & 23 & 21 & 9 \\ 5 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[632] \quad 1^{141} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 1 & 14 & 0 & 2 \\ 1 & 2 & 1 & 0 & 12 & 0 & 2 \\ 12 & 12 & 14 & 12 & 96 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[631] \quad 1^{213} 3^1 21^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 12 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 12 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[630] \quad 1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 26 & 6 \\ 9 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[629] \quad 1^{213} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 4 & 8 & 2 \\ 6 & 6 & 8 & 26 & 12 & 28 & 10 \\ 3 & 2 & 4 & 12 & 4 & 12 & 4 \\ 7 & 6 & 8 & 28 & 12 & 26 & 10 \\ 2 & 2 & 2 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[633] \quad 1^{13} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 & 2 \\ 1 & 0 & 2 & 1 & 11 & 3 & 2 \\ 1 & 2 & 0 & 3 & 16 & 4 & 2 \\ 2 & 1 & 3 & 1 & 16 & 4 & 3 \\ 11 & 11 & 16 & 16 & 103 & 27 & 19 \\ 2 & 3 & 4 & 4 & 27 & 5 & 5 \\ 2 & 2 & 2 & 3 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[634] \quad 1^{41} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

6.0

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 1 & 17 & 0 & 2 \\ 1 & 2 & 1 & 0 & 13 & 0 & 2 \\ 13 & 13 & 17 & 13 & 127 & 0 & 21 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 21 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[635] \quad 1^{1^{132}} 3^1 21^{1^{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 4 & 2 \\ 1 & 0 & 2 & 3 & 5 & 3 & 2 \\ 1 & 2 & 0 & 4 & 8 & 4 & 2 \\ 3 & 3 & 4 & 5 & 13 & 7 & 5 \\ 7 & 5 & 8 & 13 & 23 & 13 & 9 \\ 4 & 3 & 4 & 7 & 13 & 5 & 5 \\ 2 & 2 & 2 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[636] \quad 1^{1^{211}} 3^1 21^{1^{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 3 & 2 \\ 1 & 0 & 2 & 3 & 5 & 3 & 2 \\ 1 & 2 & 0 & 4 & 8 & 4 & 2 \\ 4 & 3 & 4 & 5 & 13 & 7 & 5 \\ 6 & 5 & 8 & 13 & 23 & 13 & 9 \\ 3 & 3 & 4 & 7 & 13 & 5 & 5 \\ 2 & 2 & 2 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[638] \quad 1^{1^{111}} 3^1 21^{1^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 3 & 2 \\ 1 & 0 & 2 & 5 & 4 & 3 & 2 \\ 1 & 2 & 0 & 5 & 5 & 4 & 2 \\ 4 & 5 & 5 & 13 & 13 & 10 & 7 \\ 4 & 4 & 5 & 13 & 10 & 9 & 6 \\ 3 & 3 & 4 & 10 & 9 & 5 & 5 \\ 2 & 2 & 2 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[637] \quad 1^{1^{13}} 3^1 21^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 4 & 4 & 5 & 5 & 10 & 27 & 6 \\ 9 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[639] \quad 1^{1^{12}} 3^1 21^{1^{212}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 6 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 6 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[643] \quad 1^{1^{21}} 3^1 21^{1^{241}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 & 3 \\ 3 & 5 & 10 & 28 & 5 & 5 \\ 5 & 10 & 14 & 42 & 8 & 8 \\ 15 & 28 & 42 & 116 & 21 & 22 \\ 3 & 5 & 8 & 21 & 2 & 4 \\ 3 & 5 & 8 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[642] \quad 1^{1^{42}} 3^1 21^{1^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 7 & 9 & 0 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[641] \quad 1^{1^{41}} 3^1 21^{1^{24}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 13 & 1 & 2 \\ 1 & 0 & 2 & 4 & 13 & 0 & 2 \\ 1 & 2 & 0 & 7 & 17 & 0 & 2 \\ 5 & 4 & 7 & 16 & 47 & 0 & 8 \\ 13 & 13 & 17 & 47 & 127 & 0 & 21 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 8 & 21 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[640] \ 1^{121} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[645] \ 1^{124} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 22 & 6 \\ 9 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[644] \ 1^{113} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 8 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[646] \ 1^{111} 3^1 21^{1122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 5 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 2 & 1 & 3 & 1 & 8 & 7 & 3 \\ 6 & 5 & 8 & 8 & 23 & 21 & 9 \\ 5 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[647] \ 1^{1121} 3^1 21^{1131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 2 & 3 & 4 & 5 & 20 & 7 & 5 \\ 9 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[648] \ 1^{114} 3^1 21^{1114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 19 & 4 \\ 4 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[649] \ 1^{113} 3^1 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 10 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 4 & 4 & 5 & 5 & 10 & 27 & 6 \\ 10 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[650] \ 1^{12} 3^1 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 & 3 \\ 3 & 5 & 0 & 33 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 20 & 33 & 0 & 164 & 24 & 26 \\ 3 & 5 & 0 & 24 & 2 & 4 \\ 3 & 5 & 0 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[651] $1^{142} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 32 & 14 \\ 5 & 4 & 7 & 0 & 32 & 16 & 8 \\ 2 & 2 & 2 & 0 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

[652] $1^{213} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 4 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 8 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[653] $1^{144} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 16 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 16 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[654] $1^{31} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 2 & 5 & 2 & 21 & 4 & 4 \\ 16 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[655] $1^{222} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 5 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 4 & 3 & 4 & 5 & 13 & 11 & 5 \\ 6 & 5 & 8 & 13 & 23 & 21 & 9 \\ 5 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[656] $1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 4 & 2 \\ 1 & 0 & 2 & 1 & 13 & 3 & 2 \\ 1 & 2 & 0 & 3 & 17 & 4 & 2 \\ 2 & 1 & 3 & 1 & 17 & 4 & 3 \\ 13 & 13 & 17 & 17 & 127 & 30 & 21 \\ 4 & 3 & 4 & 4 & 30 & 5 & 5 \\ 2 & 2 & 2 & 3 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[657] $1^{221} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 3 & 2 \\ 1 & 0 & 2 & 4 & 7 & 3 & 2 \\ 1 & 2 & 0 & 7 & 9 & 4 & 2 \\ 5 & 4 & 7 & 16 & 25 & 11 & 8 \\ 7 & 7 & 9 & 25 & 35 & 16 & 11 \\ 3 & 3 & 4 & 11 & 16 & 5 & 5 \\ 2 & 2 & 2 & 8 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[658] $1^{123} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 39 & 11 \\ 8 & 8 & 9 & 0 & 39 & 40 & 12 \\ 2 & 2 & 2 & 0 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[659] \quad 1^{114} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -251, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 11 & 2 \\ 1 & 0 & 2 & 1 & 4 & 10 & 2 \\ 1 & 2 & 0 & 3 & 5 & 15 & 2 \\ 2 & 1 & 3 & 1 & 5 & 15 & 3 \\ 4 & 4 & 5 & 5 & 10 & 32 & 6 \\ 11 & 10 & 15 & 15 & 32 & 88 & 18 \\ 2 & 2 & 2 & 3 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[660] \quad 1^{1112} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 6 & 2 \\ 1 & 2 & 0 & 4 & 8 & 8 & 2 \\ 4 & 3 & 4 & 5 & 13 & 14 & 5 \\ 6 & 5 & 8 & 13 & 23 & 26 & 9 \\ 6 & 6 & 8 & 14 & 26 & 26 & 10 \\ 2 & 2 & 2 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[661] \quad 1^{1121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -253, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 3 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 2 & 3 & 4 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[664] \quad 1^{141} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 17 & 4 & 4 \\ 15 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[662] \quad 1^{141} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 20 & 14 \\ 3 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[663] \quad 1^{1212} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 4 & 8 & 2 \\ 6 & 6 & 8 & 26 & 12 & 28 & 10 \\ 3 & 2 & 4 & 12 & 4 & 12 & 4 \\ 6 & 6 & 8 & 28 & 12 & 26 & 10 \\ 2 & 2 & 2 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[665] \quad 1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 1 & 2 \\ 1 & 2 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 15 & 17 \\ 2 & 1 & 3 & 0 & 15 & 1 & 3 \\ 2 & 2 & 2 & 0 & 17 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[666] \quad 1^{121} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 20 & 14 \\ 4 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[667] $1^{124} 3^{1123} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 64 & 11 \\ 11 & 12 & 16 & 0 & 64 & 110 & 20 \\ 2 & 2 & 2 & 0 & 11 & 20 & 3 \end{pmatrix}_{b_0}$$

[668] $1^{123} 3^{1133} 21^{1133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 9 & 3 \\ 3 & 5 & 5 & 18 & 15 & 4 \\ 3 & 5 & 2 & 14 & 12 & 4 \\ 10 & 18 & 14 & 48 & 42 & 14 \\ 9 & 15 & 12 & 42 & 34 & 12 \\ 3 & 5 & 4 & 14 & 12 & 4 \end{pmatrix}_{b_0}$$

[669] $1^{121} 3^{1131} 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 & 3 \\ 3 & 5 & 5 & 18 & 5 & 5 \\ 3 & 5 & 2 & 14 & 4 & 4 \\ 10 & 18 & 14 & 48 & 14 & 14 \\ 3 & 5 & 4 & 14 & 2 & 4 \\ 3 & 5 & 4 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[670] $1^{1421} 3^{11431} 21^{1431} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 & 3 \\ 3 & 5 & 20 & 18 & 5 & 5 \\ 12 & 20 & 62 & 56 & 16 & 16 \\ 10 & 18 & 56 & 48 & 14 & 14 \\ 3 & 5 & 16 & 14 & 2 & 4 \\ 3 & 5 & 16 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[671] $1^{1222} 3^{11232} 21^{1232} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 6 & 3 \\ 3 & 5 & 10 & 18 & 10 & 5 \\ 6 & 10 & 14 & 28 & 16 & 8 \\ 10 & 18 & 28 & 48 & 28 & 14 \\ 6 & 10 & 16 & 28 & 14 & 8 \\ 3 & 5 & 8 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

[672] $1^{1221} 3^{11231} 21^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 & 3 \\ 3 & 5 & 10 & 18 & 5 & 5 \\ 6 & 10 & 14 & 28 & 8 & 8 \\ 10 & 18 & 28 & 48 & 14 & 14 \\ 3 & 5 & 8 & 14 & 2 & 4 \\ 3 & 5 & 8 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[673] $1^{114} 3^{11114} 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 18 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 3 & 5 & 2 & 6 & 22 & 4 \\ 4 & 8 & 6 & 8 & 36 & 6 \\ 18 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[674] $1^{124} 3^{1112} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 36 & 8 \\ 9 & 10 & 10 & 0 & 36 & 58 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[675] $1^{124} 3^1 21^{1124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 16 & 4 \\ 8 & 13 & 10 & 24 & 40 & 10 \\ 12 & 20 & 16 & 40 & 62 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[676] $1^{122} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 39 & 11 \\ 7 & 8 & 9 & 0 & 39 & 40 & 12 \\ 2 & 2 & 2 & 0 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

[677] $1^{141} 3^1 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 16 & 19 \\ 2 & 1 & 3 & 0 & 16 & 1 & 3 \\ 2 & 2 & 2 & 0 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

[678] $1^{113} 3^1 21^{1113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[679] $1^{124} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 52 & 9 \\ 12 & 12 & 16 & 0 & 52 & 110 & 20 \\ 2 & 2 & 2 & 0 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[680] $1^{142} 3^1 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 33 & 16 \\ 5 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 16 & 7 & 3 \end{pmatrix}_{b_0}$$

[681] $1^{124} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 34 & 9 \\ 9 & 7 & 11 & 0 & 34 & 45 & 13 \\ 2 & 2 & 2 & 0 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[683] $21^{1111} 3^1 21^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 4 & 3 & 2 \\ 2 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 3 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[682] $1^{112} 3^1 21^{1113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 3 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 7 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[684] \quad 1^{111} 3^{1213} 21^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 14 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 6 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 14 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[685] \quad 1^{131} 3^{1142} 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 4 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 11 & 10 & 13 & 13 & 74 & 23 & 16 \\ 4 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[687] \quad 1^{124} 1^{12} 3^{11} 21^{1124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 18 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 3 & 5 & 2 & 10 & 22 & 4 \\ 8 & 13 & 10 & 24 & 58 & 10 \\ 18 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[686] \quad 1^{1112} 3^{11} 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 5 & 4 & 5 & 9 & 10 & 18 & 6 \\ 8 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[688] \quad 1^{1112} 3^{11} 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 4 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[691] \quad 1^{1111} 3^{11} 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 3 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 3 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[692] \quad 1^{131} 3^{11} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 19 & 13 \\ 3 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[690] \quad 1^{131} 3^{11} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 19 & 13 \\ 4 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[689] $1^{1111} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 4 & 3 & 4 & 5 & 9 & 7 & 5 \\ 5 & 4 & 5 & 9 & 10 & 9 & 6 \\ 4 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[693] $1^{1113} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 11 & 2 \\ 1 & 0 & 2 & 1 & 4 & 12 & 2 \\ 1 & 2 & 0 & 3 & 5 & 16 & 2 \\ 2 & 1 & 3 & 1 & 5 & 16 & 3 \\ 4 & 4 & 5 & 5 & 10 & 36 & 6 \\ 11 & 12 & 16 & 16 & 36 & 110 & 20 \\ 2 & 2 & 2 & 3 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[694] $1^{114} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 13 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[695] $1^{1111} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -279$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 5 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[696] $1^{1231} 3^1 21^{1241} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 & 3 \\ 3 & 5 & 10 & 28 & 5 & 5 \\ 6 & 10 & 14 & 42 & 8 & 8 \\ 16 & 28 & 42 & 116 & 21 & 22 \\ 3 & 5 & 8 & 21 & 2 & 4 \\ 3 & 5 & 8 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[697] $1^{131} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 & 3 \\ 3 & 5 & 5 & 18 & 5 & 5 \\ 3 & 5 & 2 & 14 & 4 & 4 \\ 11 & 18 & 14 & 48 & 14 & 14 \\ 3 & 5 & 4 & 14 & 2 & 4 \\ 3 & 5 & 4 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[698] $1^{1112} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -285$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 4 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 9 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[699] $1^{1112} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -285$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 9 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 3 & 3 & 4 & 5 & 9 & 18 & 5 \\ 5 & 4 & 5 & 9 & 10 & 23 & 6 \\ 9 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[700] $1^{11} 3^1 21^{1123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -285, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 13 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 13 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[701] $1^{142} 1^{131} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 23 & 16 \\ 4 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[702] $1^{121} 1^{11} 3^1 21^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 17 & 4 & 4 \\ 13 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[703] $1^{114} 1^{112} 3^1 21^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 3 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 9 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[704] $1^{113} 1^{114} 3^1 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 11 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[705] $1^{132} 1^1 3^1 21^{1132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 6 & 3 \\ 3 & 5 & 5 & 18 & 10 & 5 \\ 3 & 5 & 2 & 14 & 8 & 4 \\ 11 & 18 & 14 & 48 & 28 & 14 \\ 6 & 10 & 8 & 28 & 14 & 8 \\ 3 & 5 & 4 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

[706] $1^{113} 1^{11} 3^1 21^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[707] $1^{123} 1^{112} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 34 & 9 \\ 8 & 7 & 11 & 0 & 34 & 45 & 13 \\ 2 & 2 & 2 & 0 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[708] \quad 1^{121} 3^1 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 3 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 2 & 1 & 3 & 1 & 12 & 4 & 3 \\ 10 & 8 & 12 & 12 & 56 & 20 & 14 \\ 3 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[709] \quad 1^{121} 3^1 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 4 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 2 & 1 & 3 & 1 & 12 & 4 & 3 \\ 10 & 8 & 12 & 12 & 56 & 20 & 14 \\ 4 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[710] \quad 1^{142} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 14 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 38 & 18 \\ 4 & 5 & 5 & 0 & 38 & 13 & 7 \\ 2 & 2 & 2 & 0 & 18 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[711] \quad 1^{131} 3^1 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -301, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 3 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 10 & 10 & 13 & 13 & 74 & 23 & 16 \\ 3 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[712] \quad 1^{113} 3^1 21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 9 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[713] \quad 1^{132} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 32 & 14 \\ 5 & 4 & 7 & 0 & 32 & 16 & 8 \\ 2 & 2 & 2 & 0 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[714] \quad 1^{141} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 4 & 2 \\ 1 & 0 & 2 & 0 & 13 & 5 & 2 \\ 1 & 2 & 0 & 0 & 17 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 43 & 21 \\ 4 & 5 & 5 & 0 & 43 & 13 & 7 \\ 2 & 2 & 2 & 0 & 21 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[715] \quad 1^{214} 3^1 21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 4 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 10 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[716] $1^{11} 3^{112} 21^{1123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 11 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 15 & 4 \\ 7 & 13 & 10 & 24 & 39 & 10 \\ 11 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[717] $1^{121} 3^{1131} 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -312$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 20 & 14 \\ 3 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[718] $1^{131} 3^{1131} 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -312$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 17 & 4 & 4 \\ 14 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[719] $1^{121} 3^{1111} 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 3 & 2 \\ 1 & 0 & 2 & 3 & 5 & 3 & 2 \\ 1 & 2 & 0 & 4 & 8 & 4 & 2 \\ 3 & 3 & 4 & 5 & 13 & 7 & 5 \\ 6 & 5 & 8 & 13 & 23 & 13 & 9 \\ 3 & 3 & 4 & 7 & 13 & 5 & 5 \\ 2 & 2 & 2 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[720] $1^{122} 3^{1122} 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 32 & 11 \\ 6 & 6 & 8 & 0 & 32 & 26 & 10 \\ 2 & 2 & 2 & 0 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[721] $1^{141} 3^{131} 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 3 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 11 & 10 & 13 & 13 & 74 & 23 & 16 \\ 3 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[722] $1^{113} 3^{113} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -319$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 27 & 6 \\ 9 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[723] $1^{122} 3^{1122} 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 13 & 3 \\ 3 & 5 & 5 & 18 & 25 & 5 \\ 3 & 5 & 2 & 14 & 19 & 4 \\ 10 & 18 & 14 & 48 & 68 & 14 \\ 13 & 25 & 19 & 68 & 92 & 20 \\ 3 & 5 & 4 & 14 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[724] \quad 1^{131} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -321, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 3 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 2 & 1 & 3 & 1 & 12 & 4 & 3 \\ 9 & 8 & 12 & 12 & 56 & 20 & 14 \\ 3 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[725] \quad 1^{241} 3^1 21^{1241} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 & 3 \\ 3 & 5 & 10 & 28 & 5 & 5 \\ 6 & 10 & 14 & 42 & 8 & 8 \\ 17 & 28 & 42 & 116 & 21 & 22 \\ 3 & 5 & 8 & 21 & 2 & 4 \\ 3 & 5 & 8 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[726] \quad 1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 27 & 6 \\ 10 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[727] \quad 1^{142} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -327, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 28 & 13 \\ 5 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[728] \quad 1^{132} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -327, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 28 & 13 \\ 5 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[729] \quad 1^{142} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 4 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 9 & 7 & 9 & 16 & 35 & 16 & 11 \\ 4 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[730] \quad 1^{221} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 3 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 4 & 3 & 4 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[731] \quad 1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 8 & 2 \\ 1 & 2 & 0 & 0 & 8 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 42 & 9 \\ 10 & 8 & 14 & 0 & 42 & 70 & 16 \\ 2 & 2 & 2 & 0 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

[732] $1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 32 & 6 \\ 11 & 10 & 15 & 0 & 32 & 88 & 18 \\ 2 & 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[733] $1^{122} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 6 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 40 & 14 \\ 6 & 6 & 8 & 0 & 40 & 26 & 10 \\ 2 & 2 & 2 & 0 & 14 & 10 & 3 \end{pmatrix}_{b_0}$$

[734] $1^{131} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 4 & 2 \\ 1 & 0 & 2 & 1 & 11 & 3 & 2 \\ 1 & 2 & 0 & 3 & 16 & 4 & 2 \\ 2 & 1 & 3 & 1 & 16 & 4 & 3 \\ 12 & 11 & 16 & 16 & 103 & 27 & 19 \\ 4 & 3 & 4 & 4 & 27 & 5 & 5 \\ 2 & 2 & 2 & 3 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

[735] $1^{121} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 24 & 17 \\ 3 & 3 & 4 & 0 & 24 & 5 & 5 \\ 2 & 2 & 2 & 0 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

[736] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 5 & 4 & 5 & 9 & 10 & 18 & 6 \\ 7 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[737] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 4 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 7 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[738] $1^{123} 3^1 21^{1124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 14 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 3 & 5 & 2 & 10 & 19 & 4 \\ 8 & 13 & 10 & 24 & 49 & 10 \\ 14 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[739] $1^{11} 3^1 21^{1123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 12 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[740] $1^{113} 3^{113} 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 14 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 14 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[741] $1^{213} 3^{113} 21^{1224} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 14 & 3 \\ 3 & 5 & 10 & 13 & 25 & 8 \\ 6 & 10 & 14 & 20 & 38 & 8 \\ 7 & 13 & 20 & 24 & 49 & 10 \\ 14 & 25 & 38 & 49 & 92 & 20 \\ 3 & 5 & 8 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[742] $1^{142} 3^{113} 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 13 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 46 & 16 \\ 6 & 6 & 8 & 0 & 46 & 26 & 10 \\ 2 & 2 & 2 & 0 & 16 & 10 & 3 \end{pmatrix}_{b_0}$$

[743] $1^{114} 3^{113} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 10 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[744] $1^{214} 3^{113} 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 15 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 6 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 15 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[745] $1^{124} 3^{113} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 57 & 11 \\ 11 & 10 & 15 & 0 & 57 & 88 & 18 \\ 2 & 2 & 2 & 0 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[746] $1^{122} 3^{1111} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 5 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 3 & 3 & 4 & 5 & 13 & 11 & 5 \\ 6 & 5 & 8 & 13 & 23 & 21 & 9 \\ 5 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[747] $1^{114} 3^{113} 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -362, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 12 & 2 \\ 1 & 0 & 2 & 1 & 4 & 12 & 2 \\ 1 & 2 & 0 & 3 & 5 & 16 & 2 \\ 2 & 1 & 3 & 1 & 5 & 16 & 3 \\ 4 & 4 & 5 & 5 & 10 & 36 & 6 \\ 12 & 12 & 16 & 16 & 36 & 110 & 20 \\ 2 & 2 & 2 & 3 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[748] \quad 1^{1112} 3^1 21^{1122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 7 & 2 \\ 1 & 0 & 2 & 3 & 5 & 6 & 2 \\ 1 & 2 & 0 & 4 & 8 & 8 & 2 \\ 3 & 3 & 4 & 5 & 13 & 14 & 5 \\ 6 & 5 & 8 & 13 & 23 & 26 & 9 \\ 7 & 6 & 8 & 14 & 26 & 26 & 10 \\ 2 & 2 & 2 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[749] \quad 1^{114} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -371, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 31 & 6 \\ 10 & 11 & 13 & 0 & 31 & 81 & 17 \\ 2 & 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[750] \quad 1^{124} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -384, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 36 & 8 \\ 10 & 10 & 10 & 0 & 36 & 58 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[751] \quad 1^{12} 3^1 21^{1114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 17 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 3 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 17 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[752] \quad 1^{1112} 3^1 21^{1122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 6 & 2 \\ 1 & 2 & 0 & 4 & 8 & 8 & 2 \\ 3 & 3 & 4 & 5 & 13 & 14 & 5 \\ 6 & 5 & 8 & 13 & 23 & 26 & 9 \\ 6 & 6 & 8 & 14 & 26 & 26 & 10 \\ 2 & 2 & 2 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[753] \quad 1^{1112} 3^1 21^{1113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 3 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 8 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[754] \quad 1^{1112} 3^1 21^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -391, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 7 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[755] \quad 21^{1112} 3^1 21^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -396, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 4 & 6 & 2 \\ 2 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[756] $1^{12} 3^{11} 21^{1141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 & 3 \\ 3 & 5 & 5 & 33 & 5 & 5 \\ 3 & 5 & 2 & 24 & 4 & 4 \\ 19 & 33 & 24 & 164 & 24 & 26 \\ 3 & 5 & 4 & 24 & 2 & 4 \\ 3 & 5 & 4 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[757] $1^{132} 3^{11} 21^{1142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 6 & 3 \\ 3 & 5 & 5 & 23 & 10 & 5 \\ 3 & 5 & 2 & 18 & 8 & 4 \\ 13 & 23 & 18 & 80 & 36 & 18 \\ 6 & 10 & 8 & 36 & 14 & 8 \\ 3 & 5 & 4 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[758] $1^{131} 3^{11} 21^{1141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 18 & 4 & 4 \\ 13 & 23 & 18 & 80 & 18 & 18 \\ 3 & 5 & 4 & 18 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[759] $1^{142} 3^{11} 21^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 6 & 2 \\ 1 & 0 & 2 & 0 & 12 & 6 & 2 \\ 1 & 2 & 0 & 0 & 14 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 52 & 18 \\ 6 & 6 & 8 & 0 & 52 & 26 & 10 \\ 2 & 2 & 2 & 0 & 18 & 10 & 3 \end{pmatrix}_{b_0}$$

[760] $1^{141} 3^{11} 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 23 & 16 \\ 3 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[762] $1^{141} 3^{11} 21^{1141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 & 3 \\ 3 & 5 & 5 & 33 & 5 & 5 \\ 3 & 5 & 2 & 24 & 4 & 4 \\ 20 & 33 & 24 & 164 & 24 & 26 \\ 3 & 5 & 4 & 24 & 2 & 4 \\ 3 & 5 & 4 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[761] $1^{141} 3^{11} 21^{1141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 18 & 4 & 4 \\ 14 & 23 & 18 & 80 & 18 & 18 \\ 3 & 5 & 4 & 18 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[764] $1^{114} 3^{11} 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[763] \ 1^{1121} 3^1 21^{1121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -429, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 4 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 8 & 7 & 9 & 16 & 35 & 16 & 11 \\ 4 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[766] \ 1^{113} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -431, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 47 & 9 \\ 11 & 10 & 15 & 0 & 47 & 88 & 18 \\ 2 & 2 & 2 & 0 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[765] \ 1^{114} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -431, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 31 & 6 \\ 11 & 11 & 13 & 0 & 31 & 81 & 17 \\ 2 & 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[767] \ 1^{124} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -432, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 55 & 11 \\ 10 & 11 & 13 & 0 & 55 & 81 & 17 \\ 2 & 2 & 2 & 0 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[768] \ 1^{1121} 3^1 21^{1131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -432, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 4 & 3 & 4 & 5 & 20 & 7 & 5 \\ 9 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[769] \ 1^{131} 3^1 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -439, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 23 & 16 \\ 3 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[770] \ 1^{142} 3^1 21^{142} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -442, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 6 & 3 \\ 3 & 5 & 5 & 23 & 10 & 5 \\ 3 & 5 & 2 & 18 & 8 & 4 \\ 14 & 23 & 18 & 80 & 36 & 18 \\ 6 & 10 & 8 & 36 & 14 & 8 \\ 3 & 5 & 4 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[771] \ 1^{142} 3^1 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -444, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 14 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 38 & 18 \\ 5 & 5 & 5 & 0 & 38 & 13 & 7 \\ 2 & 2 & 2 & 0 & 18 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[772] \ 1^{\overline{112}} \ 3^{\overline{1}} \ 21^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -446, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[773] \ 1^{\overline{131}} \ 3^{\overline{1}} \ 21^{\overline{142}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -447, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 0 & 16 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 43 & 19 \\ 5 & 4 & 7 & 0 & 43 & 16 & 8 \\ 2 & 2 & 2 & 0 & 19 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[774] \ 1^{\overline{113}} \ 3^{\overline{1}} \ 21^{\overline{1124}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -450, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 14 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 3 & 5 & 2 & 10 & 19 & 4 \\ 7 & 13 & 10 & 24 & 49 & 10 \\ 14 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[775] \ 1^{\overline{142}} \ 3^{\overline{1}} \ 21^{\overline{142}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -451, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 0 & 15 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 39 & 17 \\ 5 & 4 & 7 & 0 & 39 & 16 & 8 \\ 2 & 2 & 2 & 0 & 17 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[776] \ 1^{\overline{123}} \ 3^{\overline{1}} \ 21^{\overline{123}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -451, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 48 & 11 \\ 9 & 9 & 12 & 0 & 48 & 61 & 15 \\ 2 & 2 & 2 & 0 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[777] \ 1^{\overline{1121}} \ 3^{\overline{1}} \ 21^{\overline{1121}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -460, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 3 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 8 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[778] \ 1^{\overline{124}} \ 3^{\overline{1}} \ 21^{\overline{123}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 55 & 11 \\ 11 & 11 & 13 & 0 & 55 & 81 & 17 \\ 2 & 2 & 2 & 0 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[779] \ 1^{\overline{142}} \ 3^{\overline{1}} \ 21^{\overline{121}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 33 & 15 \\ 5 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 15 & 7 & 3 \end{pmatrix}_{b_0}$$

[780] $1^{114} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -472$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 12 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[781] $1^{12} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -478$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 18 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 3 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 18 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[782] $1^{131} 3^1 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -482$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 3 & 2 \\ 1 & 0 & 2 & 1 & 11 & 3 & 2 \\ 1 & 2 & 0 & 3 & 16 & 4 & 2 \\ 2 & 1 & 3 & 1 & 16 & 4 & 3 \\ 11 & 11 & 16 & 16 & 103 & 27 & 19 \\ 3 & 3 & 4 & 4 & 27 & 5 & 5 \\ 2 & 2 & 2 & 3 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

[783] $1^{141} 3^1 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -491$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 3 & 2 \\ 1 & 0 & 2 & 1 & 13 & 3 & 2 \\ 1 & 2 & 0 & 3 & 17 & 4 & 2 \\ 2 & 1 & 3 & 1 & 17 & 4 & 3 \\ 13 & 13 & 17 & 17 & 127 & 30 & 21 \\ 3 & 3 & 4 & 4 & 30 & 5 & 5 \\ 2 & 2 & 2 & 3 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[784] $1^{131} 3^1 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 3 & 5 & 2 & 21 & 4 & 4 \\ 16 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[785] $21^{1121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 7 & 3 & 2 \\ 2 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[786] $1^{11} 3^1 21^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 15 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 3 & 5 & 2 & 10 & 19 & 4 \\ 8 & 13 & 10 & 24 & 49 & 10 \\ 15 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[787] $1^{114} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -508$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 36 & 6 \\ 12 & 12 & 16 & 0 & 36 & 110 & 20 \\ 2 & 2 & 2 & 0 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[788] $1^{132} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -520$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 13 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 46 & 16 \\ 6 & 6 & 8 & 0 & 46 & 26 & 10 \\ 2 & 2 & 2 & 0 & 16 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[789] \ 1^{11} 3^{11} 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 3 & 5 & 2 & 21 & 4 & 4 \\ 17 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[790] \ 1^{141} 3^{11} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 3 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 0 & 0 & 14 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 26 & 18 \\ 3 & 3 & 4 & 0 & 26 & 5 & 5 \\ 2 & 2 & 2 & 0 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[791] \ 1^{121} 3^{11} 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 4 & 3 & 4 & 5 & 20 & 7 & 5 \\ 10 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[792] \ 1^{131} 3^{11} 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 3 & 2 \\ 1 & 0 & 2 & 1 & 11 & 3 & 2 \\ 1 & 2 & 0 & 3 & 16 & 4 & 2 \\ 2 & 1 & 3 & 1 & 16 & 4 & 3 \\ 12 & 11 & 16 & 16 & 103 & 27 & 19 \\ 3 & 3 & 4 & 4 & 27 & 5 & 5 \\ 2 & 2 & 2 & 3 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[793] \ 1^{121} 3^{11} 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 3 & 3 & 4 & 5 & 20 & 7 & 5 \\ 10 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[794] \ 1^{121} 3^{11} 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 3 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[795] \ 1^{131} 3^{11} 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -556, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 3 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 27 & 19 \\ 3 & 3 & 4 & 0 & 27 & 5 & 5 \\ 2 & 2 & 2 & 0 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[796] \ 1^{132} 3^{11} 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -580, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 6 & 2 \\ 1 & 0 & 2 & 0 & 11 & 6 & 2 \\ 1 & 2 & 0 & 0 & 16 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 54 & 19 \\ 6 & 6 & 8 & 0 & 54 & 26 & 10 \\ 2 & 2 & 2 & 0 & 19 & 10 & 3 \end{pmatrix}_{b_0}$$

[797] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -640$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 64 & 11 \\ 12 & 12 & 16 & 0 & 64 & 110 & 20 \\ 2 & 2 & 2 & 0 & 11 & 20 & 3 \end{pmatrix}_{b_0}$$

[798] $1^{1121} 3^1 21^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -672$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 3 & 3 & 4 & 5 & 20 & 7 & 5 \\ 9 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[799] $1^{141} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -715$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 3 & 2 \\ 1 & 0 & 2 & 0 & 13 & 3 & 2 \\ 1 & 2 & 0 & 0 & 17 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 30 & 21 \\ 3 & 3 & 4 & 0 & 30 & 5 & 5 \\ 2 & 2 & 2 & 0 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[800] $1^{142} 3^1 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -796$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 6 & 2 \\ 1 & 0 & 2 & 0 & 13 & 6 & 2 \\ 1 & 2 & 0 & 0 & 17 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 60 & 21 \\ 6 & 6 & 8 & 0 & 60 & 26 & 10 \\ 2 & 2 & 2 & 0 & 21 & 10 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [801] $1^{1221} 3^1 1^5 1321^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 12, 1)$, $c_{\text{ext}} = 6.5$

[805] $1^{1111} 3^1 1^5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 4$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 7 & 3 \\ 2 & 4 & 3 & 4 & 5 & 7 & 5 \\ 3 & 4 & 4 & 4 & 7 & 6 & 6 \\ 2 & 3 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[806] $1^{111} 3^1 1^5 131^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 5 & 2 \\ 2 & 0 & 4 & 6 & 7 & 12 & 6 \\ 2 & 0 & 3 & 7 & 5 & 11 & 5 \\ 4 & 0 & 5 & 12 & 11 & 17 & 9 \\ 2 & 0 & 2 & 6 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[808] $1^{1114} 3^1 1^5 131^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 3 & 6 & 15 & 3 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 3 & 1 & 1 & 6 & 15 & 3 \\ 4 & 6 & 5 & 6 & 14 & 39 & 8 \\ 9 & 15 & 13 & 15 & 39 & 97 & 21 \\ 2 & 3 & 2 & 3 & 8 & 21 & 4 \end{pmatrix}_{b_0}$$

[809] $1^{1112} 3^1 1^5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} =$

6.5

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 3 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 3 & 5 & 2 \\ 1 & 3 & 1 & 1 & 4 & 7 & 3 \\ 3 & 4 & 3 & 4 & 5 & 11 & 5 \\ 5 & 7 & 5 & 7 & 11 & 17 & 9 \\ 2 & 3 & 2 & 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[810] \ 1^{\overline{114}1} 3^{\overline{1}12} 5^{\overline{1}13} 1^{\overline{1}13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 3 & 4 & 15 & 3 \\ 1 & 2 & 0 & 1 & 3 & 13 & 2 \\ 1 & 3 & 1 & 1 & 4 & 15 & 3 \\ 3 & 4 & 3 & 4 & 5 & 25 & 5 \\ 9 & 15 & 13 & 15 & 25 & 97 & 21 \\ 2 & 3 & 2 & 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[811] \ 1^{\overline{141}1} 3^{\overline{1}13} 1^{\overline{1}13} 1^{\overline{1}13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & 2 & 3 & 17 & 4 & 3 \\ 1 & 2 & 0 & 1 & 14 & 4 & 2 \\ 1 & 3 & 1 & 1 & 17 & 4 & 3 \\ 11 & 17 & 14 & 17 & 121 & 31 & 23 \\ 2 & 4 & 4 & 4 & 31 & 6 & 6 \\ 2 & 3 & 2 & 3 & 23 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[807] \ 1^{\overline{112}3} 1^{\overline{1}12} 5^{\overline{1}13} 1^{\overline{1}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 3 & 6 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 3 & 1 & 1 & 6 & 7 & 3 \\ 4 & 6 & 5 & 6 & 14 & 17 & 8 \\ 5 & 7 & 5 & 7 & 17 & 17 & 9 \\ 2 & 3 & 2 & 3 & 8 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[812] \ 1^{\overline{113}3} 1^{\overline{1}12} 5^{\overline{1}13} 1^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 3 & 6 & 11 & 3 \\ 1 & 2 & 0 & 1 & 5 & 9 & 2 \\ 1 & 3 & 1 & 1 & 6 & 11 & 3 \\ 4 & 6 & 5 & 6 & 14 & 28 & 8 \\ 7 & 11 & 9 & 11 & 28 & 49 & 15 \\ 2 & 3 & 2 & 3 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[813] \ 1^{\overline{113}3} 1^{\overline{1}12} 5^{\overline{1}13} 1^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 3 & 4 & 11 & 3 \\ 1 & 2 & 0 & 1 & 3 & 9 & 2 \\ 1 & 3 & 1 & 1 & 4 & 11 & 3 \\ 3 & 4 & 3 & 4 & 5 & 18 & 5 \\ 7 & 11 & 9 & 11 & 18 & 49 & 15 \\ 2 & 3 & 2 & 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[850] \ 1^{\overline{132}1} 3^{\overline{1}13} 1^{\overline{1}32} 1^{\overline{1}32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 & \\ 6 & 12 & 0 & 70 & 57 & 24 & 18 & \\ 5 & 9 & 0 & 57 & 44 & 19 & 14 & \\ 2 & 4 & 0 & 24 & 19 & 6 & 6 & \\ 2 & 3 & -1 & 18 & 14 & 6 & 4 & \end{pmatrix}_{b_0}$$

$$[842] \ 1^{\overline{112}1} 3^{\overline{1}13} 1^{\overline{1}13} 1^{\overline{1}13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 11 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 & \\ 2 & 4 & 0 & 6 & 23 & 8 & 6 & \\ 6 & 11 & 0 & 23 & 63 & 23 & 17 & \\ 2 & 4 & 0 & 8 & 23 & 6 & 6 & \\ 2 & 3 & -1 & 6 & 17 & 6 & 4 & \end{pmatrix}_{b_0}$$

$$[856] \ 1^{\overline{113}1} 3^{\overline{1}13} 1^{\overline{1}13} 1^{\overline{1}13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 & 2 \\ 1 & 1 & 1 & 4 & 13 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 8 & 6 \\ 7 & 13 & 0 & 27 & 90 & 27 & 20 \\ 2 & 4 & 0 & 8 & 27 & 6 & 6 \\ 2 & 3 & -1 & 6 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[822] \ 1^{1122} 3^1 1^1 5 13 1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 4 & 2 \\ 1 & 1 & 1 & 4 & 11 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 23 & 16 & 6 \\ 6 & 11 & 0 & 23 & 65 & 46 & 17 \\ 4 & 8 & 0 & 16 & 46 & 30 & 12 \\ 2 & 3 & -1 & 6 & 17 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[823] \ 1^{1133} 3^1 1^1 5 13 1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 6 & 2 \\ 1 & 1 & 1 & 4 & 13 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 24 & 6 \\ 7 & 13 & 0 & 27 & 90 & 81 & 20 \\ 6 & 12 & 0 & 24 & 81 & 70 & 18 \\ 2 & 3 & -1 & 6 & 20 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[815] \ 1^{1143} 3^1 1^1 5 13 1^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 6 & 2 \\ 1 & 1 & 1 & 4 & 17 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 24 & 6 \\ 9 & 17 & 0 & 35 & 152 & 105 & 26 \\ 6 & 12 & 0 & 24 & 105 & 70 & 18 \\ 2 & 3 & -1 & 6 & 26 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[816] \ 1^{1343} 1^1 5 13 1^{314} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 96 & 18 \\ 2 & 3 & 0 & 21 & 5 & 28 & 5 \\ 8 & 16 & 0 & 96 & 28 & 126 & 24 \\ 2 & 3 & -1 & 18 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[817] \ 1^{1111} 3^1 1^1 5 13 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 1 & 1 & 4 & 7 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 8 & 6 \\ 4 & 7 & 0 & 15 & 27 & 15 & 11 \\ 2 & 4 & 0 & 8 & 15 & 6 & 6 \\ 2 & 3 & -1 & 6 & 11 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[818] \ 1^{1134} 3^1 1^1 5 13 1^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 8 & 2 \\ 1 & 1 & 1 & 4 & 13 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 32 & 6 \\ 7 & 13 & 0 & 27 & 90 & 108 & 20 \\ 8 & 16 & 0 & 32 & 108 & 126 & 24 \\ 2 & 3 & -1 & 6 & 20 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[819] \ 1^{1211} 3^1 1^1 5 13 1^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & 1 & 8 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 16 & 12 \\ 3 & 5 & 0 & 22 & 14 & 11 & 8 \\ 2 & 4 & 0 & 16 & 11 & 6 & 6 \\ 2 & 3 & -1 & 12 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[820] \ 1^{1311} 3^1 1^1 5 13 1^{321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 & 2 \\ 1 & 1 & 1 & 12 & 7 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 45 & 24 & 18 \\ 4 & 7 & 0 & 45 & 27 & 15 & 11 \\ 2 & 4 & 0 & 24 & 15 & 6 & 6 \\ 2 & 3 & -1 & 18 & 11 & 6 & 4 \end{pmatrix}_{b_0}$$

[821] $1^{133}3^11^15131^{313}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.6$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 72 & 18 \\ 2 & 3 & 0 & 21 & 5 & 21 & 5 \\ 6 & 12 & 0 & 72 & 21 & 70 & 18 \\ 2 & 3 & -1 & 18 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[814] $1^{114}3^11^1513^11^{114}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 8 & 2 \\ 1 & 1 & 1 & 4 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 11 & 32 & 6 \\ 3 & 5 & 0 & 11 & 14 & 44 & 8 \\ 8 & 16 & 0 & 32 & 44 & 126 & 24 \\ 2 & 3 & -1 & 6 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[853] $1^{1231}3^11^1513^11^{231}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 & 2 \\ 1 & 1 & 1 & 8 & 13 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 16 & 12 \\ 7 & 13 & 0 & 54 & 90 & 27 & 20 \\ 2 & 4 & 0 & 16 & 27 & 6 & 6 \\ 2 & 3 & -1 & 12 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[852] $1^{1141}3^11^1513^11^{141}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 & 2 \\ 1 & 1 & 1 & 4 & 17 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 8 & 6 \\ 9 & 17 & 0 & 35 & 152 & 35 & 26 \\ 2 & 4 & 0 & 8 & 35 & 6 & 6 \\ 2 & 3 & -1 & 6 & 26 & 6 & 4 \end{pmatrix}_{b_0}$$

[851] $1^{1121}3^11^1513^11^{121}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & 1 & 4 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 19 & 8 & 6 \\ 5 & 9 & 0 & 19 & 44 & 19 & 14 \\ 2 & 4 & 0 & 8 & 19 & 6 & 6 \\ 2 & 3 & -1 & 6 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[855] $1^{1214}3^11^1513^11^{214}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 8 & 2 \\ 1 & 1 & 1 & 8 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 64 & 12 \\ 3 & 5 & 0 & 22 & 14 & 44 & 8 \\ 8 & 16 & 0 & 64 & 44 & 126 & 24 \\ 2 & 3 & -1 & 12 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[854] $1^{141}3^11^15131^{411}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 & 2 \\ 1 & 1 & 1 & 16 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 32 & 24 \\ 2 & 3 & 0 & 28 & 5 & 7 & 5 \\ 2 & 4 & 0 & 32 & 7 & 6 & 6 \\ 2 & 3 & -1 & 24 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[857] $1^{1212}3^11^15131^{222}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 30 & 12 \\ 4 & 7 & 0 & 30 & 27 & 11 \\ 4 & 8 & 0 & 32 & 30 & 12 \\ 2 & 3 & -1 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

[847] $1^{1224}31^1513^1224 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 8 & 2 \\ 1 & 1 & 1 & 8 & 9 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 64 & 12 \\ 5 & 9 & 0 & 38 & 44 & 76 & 14 \\ 8 & 16 & 0 & 64 & 76 & 126 & 24 \\ 2 & 3 & -1 & 12 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[848] $1^{1144}31^1513^1144 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 8 & 2 \\ 1 & 1 & 1 & 4 & 17 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 32 & 6 \\ 9 & 17 & 0 & 35 & 152 & 140 & 26 \\ 8 & 16 & 0 & 32 & 140 & 126 & 24 \\ 2 & 3 & -1 & 6 & 26 & 24 & 4 \end{pmatrix}_{b_0}$$

[847] $1^{1232}31^1513^11232 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 4 & 2 \\ 1 & 1 & 1 & 8 & 13 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 32 & 12 \\ 7 & 13 & 0 & 54 & 90 & 54 & 20 \\ 4 & 8 & 0 & 32 & 54 & 30 & 12 \\ 2 & 3 & -1 & 12 & 20 & 12 & 4 \end{pmatrix}_{b_0}$$

[824] $1^{1124}31^1513^1124 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 8 & 2 \\ 1 & 1 & 1 & 4 & 9 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 19 & 32 & 6 \\ 5 & 9 & 0 & 19 & 44 & 76 & 14 \\ 8 & 16 & 0 & 32 & 76 & 126 & 24 \\ 2 & 3 & -1 & 6 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[825] $1^{1111}31^1513^1111 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & 1 & 4 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 11 & 8 & 6 \\ 3 & 5 & 0 & 11 & 14 & 11 & 8 \\ 2 & 4 & 0 & 8 & 11 & 6 & 6 \\ 2 & 3 & -1 & 6 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[826] $1^{1213}1^15131^{211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & 1 & 8 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 16 & 12 \\ 2 & 3 & 0 & 14 & 5 & 7 & 5 \\ 2 & 4 & 0 & 16 & 7 & 6 & 6 \\ 2 & 3 & -1 & 12 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[827] $1^{1323}1^15131^{312} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 4 & 2 \\ 1 & 1 & 1 & 12 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 48 & 18 \\ 2 & 3 & 0 & 21 & 5 & 14 & 5 \\ 4 & 8 & 0 & 48 & 14 & 30 & 12 \\ 2 & 3 & -1 & 18 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[828] $1^{1313}1^15131^{311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 24 & 18 \\ 2 & 3 & 0 & 21 & 5 & 7 & 5 \\ 2 & 4 & 0 & 24 & 7 & 6 & 6 \\ 2 & 3 & -1 & 18 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[829] $1^{111}3^1 1^1 5 13 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 8 & 6 \\ 2 & 3 & 0 & 7 & 5 & 7 & 5 \\ 2 & 4 & 0 & 8 & 7 & 6 & 6 \\ 2 & 3 & -1 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[830] $1^{1242}3^1 1^1 5 13 1^{242} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 4 & 2 \\ 1 & 1 & 1 & 8 & 17 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 70 & 32 & 12 \\ 9 & 17 & 0 & 70 & 152 & 70 & 26 \\ 4 & 8 & 0 & 32 & 70 & 30 & 12 \\ 2 & 3 & -1 & 12 & 26 & 12 & 4 \end{pmatrix}_{b_0}$$

[831] $1^{1311}3^1 1^1 5 13 1^{311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 & 2 \\ 1 & 1 & 1 & 12 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 24 & 18 \\ 3 & 5 & 0 & 33 & 14 & 11 & 8 \\ 2 & 4 & 0 & 24 & 11 & 6 & 6 \\ 2 & 3 & -1 & 18 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[840] $1^{1221}3^1 1^1 5 13 1^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 & 2 \\ 1 & 1 & 1 & 8 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 16 & 12 \\ 5 & 9 & 0 & 38 & 44 & 19 & 14 \\ 2 & 4 & 0 & 16 & 19 & 6 & 6 \\ 2 & 3 & -1 & 12 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[839] $1^{1411}3^1 1^1 5 13 1^{421} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 & 2 \\ 1 & 1 & 1 & 16 & 7 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 60 & 32 & 24 \\ 4 & 7 & 0 & 60 & 27 & 15 & 11 \\ 2 & 4 & 0 & 32 & 15 & 6 & 6 \\ 2 & 3 & -1 & 24 & 11 & 6 & 4 \end{pmatrix}_{b_0}$$

[838] $1^{1443}3^1 1^1 5 13 1^{414} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 128 & 24 \\ 2 & 3 & 0 & 28 & 5 & 28 & 5 \\ 8 & 16 & 0 & 128 & 28 & 126 & 24 \\ 2 & 3 & -1 & 24 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[837] $1^{1212}3^1 1^1 5 13 1^{212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 4 & 2 \\ 1 & 1 & 1 & 8 & 5 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 32 & 12 \\ 3 & 5 & 0 & 22 & 14 & 22 & 8 \\ 4 & 8 & 0 & 32 & 22 & 30 & 12 \\ 2 & 3 & -1 & 12 & 8 & 12 & 4 \end{pmatrix}_{b_0}$$

[836] $1^{1414}3^1 1^1 5 13 1^{414} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 8 & 2 \\ 1 & 1 & 1 & 16 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 128 & 24 \\ 3 & 5 & 0 & 44 & 14 & 44 & 8 \\ 8 & 16 & 0 & 128 & 44 & 126 & 24 \\ 2 & 3 & -1 & 24 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[835] $1^{1314}31^1513^13^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 8 & 2 \\ 1 & 1 & 1 & 12 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 96 & 18 \\ 3 & 5 & 0 & 33 & 14 & 44 & 8 \\ 8 & 16 & 0 & 96 & 44 & 126 & 24 \\ 2 & 3 & -1 & 18 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[834] $1^{124}3^11^15131^{214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 8 & 2 \\ 1 & 1 & 1 & 8 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 64 & 12 \\ 2 & 3 & 0 & 14 & 5 & 28 & 5 \\ 8 & 16 & 0 & 64 & 28 & 126 & 24 \\ 2 & 3 & -1 & 12 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[833] $1^{1313}31^1513^11^{313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.6$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 6 & 2 \\ 1 & 1 & 1 & 12 & 5 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 72 & 18 \\ 3 & 5 & 0 & 33 & 14 & 33 & 8 \\ 6 & 12 & 0 & 72 & 33 & 70 & 18 \\ 2 & 3 & -1 & 18 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[832] $1^{1131}3^11^15131^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 & 2 \\ 1 & 1 & 1 & 4 & 15 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 31 & 8 & 6 \\ 8 & 15 & 0 & 31 & 119 & 31 & 23 \\ 2 & 4 & 0 & 8 & 31 & 6 & 6 \\ 2 & 3 & -1 & 6 & 23 & 6 & 4 \end{pmatrix}_{b_0}$$

[858] $1^{1241}31^1513^11^{241} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 & 2 \\ 1 & 1 & 1 & 8 & 17 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 70 & 16 & 12 \\ 9 & 17 & 0 & 70 & 152 & 35 & 26 \\ 2 & 4 & 0 & 16 & 35 & 6 & 6 \\ 2 & 3 & -1 & 12 & 26 & 6 & 4 \end{pmatrix}_{b_0}$$

[846] $1^{1223}31^1513^11^{223} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 6 & 2 \\ 1 & 1 & 1 & 8 & 9 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 48 & 12 \\ 5 & 9 & 0 & 38 & 44 & 57 & 14 \\ 6 & 12 & 0 & 48 & 57 & 70 & 18 \\ 2 & 3 & -1 & 12 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[845] $1^{1213}31^1513^11^{213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 6 & 2 \\ 1 & 1 & 1 & 8 & 5 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 48 & 12 \\ 3 & 5 & 0 & 22 & 14 & 33 & 8 \\ 6 & 12 & 0 & 48 & 33 & 70 & 18 \\ 2 & 3 & -1 & 12 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[844] $1^{1222}31^1513^11^{222} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 4 & 2 \\ 1 & 1 & 1 & 8 & 9 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 32 & 12 \\ 5 & 9 & 0 & 38 & 44 & 38 & 14 \\ 4 & 8 & 0 & 32 & 38 & 30 & 12 \\ 2 & 3 & -1 & 12 & 14 & 12 & 4 \end{pmatrix}_{b_0}$$

[843] $1^{1122}3^11^15131^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & 1 & 4 & 7 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 16 & 6 \\ 4 & 7 & 0 & 15 & 27 & 30 & 11 \\ 4 & 8 & 0 & 16 & 30 & 30 & 12 \\ 2 & 3 & -1 & 6 & 11 & 12 & 4 \end{pmatrix}_{b_0}$$

[841] $1^{122}3^11^15131^{212} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 32 & 12 \\ 2 & 3 & 0 & 14 & 5 & 14 & 5 \\ 4 & 8 & 0 & 32 & 14 & 30 & 12 \\ 2 & 3 & -1 & 12 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[859] $1^{1122}3^11^1513^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 3 & 5 & 2 \\ 2 & 3 & 1 & 1 & 4 & 7 & 3 \\ 2 & 4 & 3 & 4 & 5 & 11 & 5 \\ 2 & 7 & 5 & 7 & 11 & 17 & 9 \\ 2 & 3 & 2 & 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[860] $1^{1111}3^11^1513^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 4 & 2 \\ 2 & 4 & 4 & 6 & 7 & 8 & 6 \\ 3 & 4 & 3 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 8 & 7 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[861] $1^{1122}3^11^1513^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 3 & 5 & 2 \\ 2 & 4 & 4 & 6 & 7 & 12 & 6 \\ 3 & 4 & 3 & 7 & 5 & 11 & 5 \\ 4 & 7 & 5 & 12 & 11 & 17 & 9 \\ 2 & 3 & 2 & 6 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[862] $1^{1122}3^11^1513^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 4 & 11 & 3 \\ 1 & 2 & 0 & 1 & 3 & 9 & 2 \\ 2 & 3 & 1 & 1 & 4 & 11 & 3 \\ 2 & 4 & 3 & 4 & 5 & 18 & 5 \\ 8 & 11 & 9 & 11 & 18 & 49 & 15 \\ 2 & 3 & 2 & 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[863] $1^{1111}3^11^1513^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 4 & 2 \\ 3 & 4 & 4 & 6 & 7 & 8 & 6 \\ 2 & 4 & 3 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 8 & 7 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[864] $1^{1122}3^11^1513^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 52, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 5 & 2 \\ 1 & 1 & 2 & 4 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 3 & 5 & 2 \\ 3 & 4 & 4 & 6 & 7 & 12 & 6 \\ 2 & 4 & 3 & 7 & 5 & 11 & 5 \\ 5 & 7 & 5 & 12 & 11 & 17 & 9 \\ 2 & 3 & 2 & 6 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[865] \quad 1^{1111}3^1 1^1 5 131^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 6 & 4 & 3 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 2 & 4 & 4 & 6 & 11 & 8 & 6 \\ 4 & 6 & 5 & 11 & 14 & 11 & 8 \\ 3 & 4 & 4 & 8 & 11 & 6 & 6 \\ 2 & 3 & 2 & 6 & 8 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[866] \quad 1^{1112}3^1 1^1 5 131^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 87, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 2 & 4 & 4 & 6 & 11 & 12 & 6 \\ 4 & 6 & 5 & 11 & 14 & 17 & 8 \\ 4 & 7 & 5 & 12 & 17 & 17 & 9 \\ 2 & 3 & 2 & 6 & 8 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[867] \quad 2^1 3^{1111} 15 131^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 108, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 4 & 2 \\ 3 & 4 & 4 & 6 & 7 & 8 & 6 \\ 3 & 4 & 3 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 8 & 7 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

4.945 NHC(-2)(-3) + su₂ + su_{2n3mix} + su₂ + su₂→(-3)

$$T = 12 \quad [9] \quad 1^{141}3^1 1^1 5 13^{11} 1^{411} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 8.0$$

$$[17] \quad 1^{1112}3^1 1^1 5 13^{11} 1^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 7.5$$

$$[8] \quad 1^{1131}3^1 1^1 5 13^{11} 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 7.0$$

$$[7] \quad 1^{121}3^1 1^1 5 13^{11} 1^{211} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 7.5$$

$$[6] \quad 1^{1111}3^1 1^1 5 13^{11} 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 7.0$$

$$[5] \quad 1^{144}3^1 1^1 5 13^{11} 1^{414} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 9.0$$

$$[4] \quad 1^{124}3^1 1^1 5 13^{11} 1^{214} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 8.5$$

$$[3] \quad 1^{122}3^1 1^1 5 13^{11} 1^{212} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 8.0$$

$$[2] \quad 1^{1121}3^1 1^1 5 13^{11} 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 7.0$$

$$[11] \quad 1^{1411}3^1 1^1 5 13^{11} 1^{421} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 8.0$$

$$[1] \quad 1^{1311}3^1 1^1 5 13^{11} 1^{321} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 7.8$$

$$[0] \quad 1^{134}3^1 1^1 5 13^{11} 1^{314} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 8.8$$

$$\begin{aligned}
[12] \quad & 1^{111}3^11^1513^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[10] \quad & 1^{1212}3^11^1513^11^{222} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[13] \quad & 1^{1122}3^11^1513^11^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5 \\
[14] \quad & 1^{133}3^11^1513^11^{1313} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.6 \\
[15] \quad & 1^{131}3^11^1513^11^{1311} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8 \\
[16] \quad & 1^{132}3^11^1513^11^{1312} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.3
\end{aligned}$$

4.946 NHC(-2)(-3) + su₂ + su₂n₃mix + su₂ + su₂n₃mix

$$\begin{aligned}
T = 12 \quad [4] \quad & 1^{1111}3^11^1513^11^{1211} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[10] \quad & 1^{1114}3^11^1513^11^{1141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[3] \quad & 1^{1112}3^11^1513^11^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[5] \quad & 1^{1113}3^11^1513^11^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[0] \quad & 1^{131}3^11^1513^11^{1311} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8 \\
[1] \quad & 1^{1211}3^11^1513^11^{1211} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5 \\
[2] \quad & 1^{1212}3^11^1513^11^{1212} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5 \\
[6] \quad & 1^{1111}3^11^1513^11^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[7] \quad & 1^{1114}3^11^1513^11^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[8] \quad & 1^{1111}3^11^1513^11^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[9] \quad & 1^{1212}3^11^1513^11^{1212} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0
\end{aligned}$$

4.947 NHC(-2)(-3) + su₂ + su₂n₃mix + su₂ → (-3)

$$\begin{aligned}
T = 8 \quad [1] \quad & 1^{113}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 5 & 5 & 4 & 6 & 3 & 4 \\ 6 & 7 & 6 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[0] \quad & 1^{113}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 10 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 9 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 11 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 3 & 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{21} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 25, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 3 & 5 & 4 & 6 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 5 & 8 & 10 & 22 & 4 & 6 \\ 12 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 11 & 17 & 49 & 21 & 9 & 13 \\ 6 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 5 & 8 & 10 & 10 & 4 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 12 & 14 & 37 & 17 & 8 & 11 \\ 6 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 8 & 12 & 25 & 15 & 6 & 9 \\ 6 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{143}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 11 & 16 & 46 & 8 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 4 & 5 & 4 & 6 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

- [19] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 4 & 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [18] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [17] $21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 3 & 4 \\ 5 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 \\ 3 & 3 & 2 & 4 & 0 & 2 \\ 4 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [22] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 4 & 5 & 4 & 6 & 3 & 4 \\ 6 & 7 & 6 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [23] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 13 & 21 & 76 & 11 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [24] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 11 & 16 & 46 & 20 & 8 & 12 \\ 6 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [25] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 11 & 14 & 37 & 17 & 8 & 11 \\ 6 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [27] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [26] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 8 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{12}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 0 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 11 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 7 & 8 & 10 & 4 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{12}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 10 & 4 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 5 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 6 & 9 & 13 & 11 & 5 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 8 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{11}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 10 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

- [36] $1^{1^{31}} 1^{1213} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [35] $1^{1^{21}} 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [39] $1^{1^{11}} 1^{113} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 37 & 8 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [38] $1^{1^{23}} 1^{1323} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 8 & 12 & 25 & 24 & 6 & 9 \\ 9 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$
- [40] $1^{1^{33}} 1^{1423} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 2 & 3 \\ 3 & 4 & 16 & 11 & 3 & 4 \\ 11 & 16 & 46 & 32 & 8 & 12 \\ 9 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$
- [41] $1^{1^{13}} 1^{1223} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 5 & 8 & 10 & 16 & 4 & 6 \\ 9 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$
- [43] $1^{1^{41}} 1^{1313} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 10 & 12 & 25 & 15 & 6 & 9 \\ 5 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [42] $1^{1^{11}} 1^{113} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 0 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [44] $1^{1^{22}} 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 \\ 9 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 2 & 3 \\ 3 & 4 & 8 & 25 & 3 & 4 \\ 5 & 8 & 10 & 34 & 4 & 6 \\ 17 & 25 & 34 & 111 & 14 & 19 \\ 2 & 3 & 4 & 14 & 0 & 2 \\ 3 & 4 & 6 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{131}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 25 & 6 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 6 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 13 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{112}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{141}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 10 & 12 & 25 & 6 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

- [53] $1^{12}1^{11}3^{11}1^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 \\ 9 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$
- [54] $1^{12}1^{11}3^{11}1^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 9 & 2 & 3 \\ 3 & 4 & 9 & 11 & 3 & 4 \\ 8 & 9 & 13 & 17 & 5 & 7 \\ 9 & 11 & 17 & 20 & 6 & 8 \\ 2 & 3 & 5 & 6 & 0 & 2 \\ 3 & 4 & 7 & 8 & 2 & 3 \end{pmatrix}_{b_0}$
- [55] $1^{13}1^{11}3^{11}1^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 8 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [56] $1^{12}1^{11}3^{11}1^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 15 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 7 & 8 & 10 & 28 & 4 & 6 \\ 15 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$
- [57] $1^{12}1^{11}3^{11}1^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 13 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 7 & 8 & 10 & 22 & 4 & 6 \\ 13 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$
- [58] $1^{13}1^{11}3^{11}1^{12}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 6 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 15 & 22 & 85 & 27 & 12 & 17 \\ 6 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [59] $1^{12}1^{11}3^{11}1^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 6 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [60] $1^{12}1^{11}3^{11}1^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 0 & 8 & 2 & 3 \end{pmatrix}_{b_0}$
- [61] $1^{14}1^{11}3^{11}1^{12}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 16 & 26 & 118 & 14 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{141}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 2 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 12 & 16 & 46 & 8 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{132}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 7 & 8 & 10 & 16 & 4 & 6 \\ 8 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[64] $21^{113}3^{111}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 3 & 4 \\ 5 & 4 & 9 & 7 & 3 & 4 \\ 9 & 9 & 13 & 11 & 5 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 \\ 3 & 3 & 5 & 4 & 0 & 2 \\ 4 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{111}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{111}3^{111}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 7 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{132}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 10 & 12 & 25 & 24 & 6 & 9 \\ 8 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{111}3^{111}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 \\ 10 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{122}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{113}3^{111}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 10 & 12 & 0 & 25 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 0 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 8 & 9 & 13 & 11 & 5 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 6 & 8 & 10 & 10 & 4 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{141}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 14 & 21 & 76 & 26 & 11 & 16 \\ 6 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{121}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 11 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 5 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 2 & 3 \\ 3 & 4 & 9 & 11 & 3 & 4 \\ 6 & 9 & 13 & 17 & 5 & 7 \\ 8 & 11 & 17 & 20 & 6 & 8 \\ 2 & 3 & 5 & 6 & 0 & 2 \\ 3 & 4 & 7 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 5 & 8 & 10 & 22 & 4 & 6 \\ 12 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 11 & 17 & 49 & 9 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 104, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 104, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 0 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 104, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 8 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 108, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 11 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 109, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 112, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 \\ 8 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 112, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 3 & 4 \\ 5 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 12 & 12 & 8 & 25 & 7 & 9 \\ 3 & 3 & 2 & 7 & 0 & 2 \\ 4 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 13 & 18 & 58 & 22 & 10 & 14 \\ 6 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 9 & 12 & 25 & 15 & 6 & 9 \\ 6 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{121} 3^{111} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 37 & 8 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{142} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 10 & 12 & 25 & 15 & 6 & 9 \\ 6 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{122} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 7 & 9 & 13 & 11 & 5 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{121} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 6 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{122} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 5 & 8 & 10 & 22 & 4 & 6 \\ 11 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{124} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 13 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{14} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 12 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{112} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 12 & 0 & 25 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 0 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{12\bar{1}}3^{1\bar{1}}1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 122, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 9 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{12\bar{2}}3^{1\bar{1}}1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 123, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 9 & 13 & 28 & 16 & 7 & 10 \\ 6 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{1\bar{2}}3^{1\bar{1}}1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 124, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{1\bar{1}}3^{1\bar{1}}1^{1\bar{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 124, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 16 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{1\bar{1}}3^{1\bar{1}}1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 125, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 10 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{12\bar{1}}3^{1\bar{1}}1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 126, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 7 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{12\bar{1}}3^{1\bar{1}}1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 129, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 11 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{1\bar{1}}3^{1\bar{1}}1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 129, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 6 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[104] \quad & 1^{131}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 131, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 8 & 12 & 25 & 15 & 6 & 9 \\ 5 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[105] \quad & 1^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 133, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 37 & 8 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[106] \quad & 1^{141}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 134, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 6 & 2 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 12 & 16 & 46 & 20 & 8 & 12 \\ 6 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[107] \quad & 1^{123}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 135, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 15 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 5 & 8 & 10 & 28 & 4 & 6 \\ 15 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[108] \quad & 1^{131}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 136, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 5 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 9 & 12 & 25 & 15 & 6 & 9 \\ 5 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[109] \quad & 21^{113}3^{11}1^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 136, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 15 & 6 & 3 & 4 \\ 5 & 4 & 17 & 7 & 3 & 4 \\ 15 & 17 & 49 & 21 & 9 & 13 \\ 6 & 7 & 21 & 7 & 4 & 5 \\ 3 & 3 & 9 & 4 & 0 & 2 \\ 4 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[110] \quad & 1^{122}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 138, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 9 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 7 & 8 & 10 & 16 & 4 & 6 \\ 9 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[111] \quad & 1^{123}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 139, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 11 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 7 & 8 & 10 & 22 & 4 & 6 \\ 11 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[112] \quad & 1^{13}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 12 & 15 & 0 & 39 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 0 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 144, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 13 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[114] \quad 21^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 144, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 3 & 4 \\ 5 & 4 & 13 & 7 & 3 & 4 \\ 12 & 13 & 28 & 16 & 7 & 10 \\ 6 & 7 & 16 & 7 & 4 & 5 \\ 3 & 3 & 7 & 4 & 0 & 2 \\ 4 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 145, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 14 & 22 & 85 & 12 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 146, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 \\ 12 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 146, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 11 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 147, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 \\ 11 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 150, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 12 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{1112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 152, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 9 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 152, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 5 & 8 & 10 & 16 & 4 & 6 \\ 8 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 21^{123}3^{11}1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 18 & 6 & 3 & 4 \\ 5 & 4 & 18 & 7 & 3 & 4 \\ 18 & 18 & 58 & 22 & 10 & 14 \\ 6 & 7 & 22 & 7 & 4 & 5 \\ 3 & 3 & 10 & 4 & 0 & 2 \\ 4 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 10 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{121}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 2 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 12 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{124}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 4 & 4 & 1 & 14 & 2 & 3 \\ 16 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 4 & 4 & 1 & 14 & 2 & 3 \\ 14 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 21^{113}3^{11}1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 3 & 4 \\ 5 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 18 & 20 & 14 & 69 & 11 & 15 \\ 3 & 3 & 2 & 11 & 0 & 2 \\ 4 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{112}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 76 & 11 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[130] \quad & 1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 163, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 0 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[131] \quad & 1^{14}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 164, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 5 & 2 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 12 & 16 & 46 & 20 & 8 & 12 \\ 5 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[132] \quad & 1^{11}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 12 & 16 & 11 & 44 & 9 & 12 \\ 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[134] \quad & 21^{12}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 11 & 2 & 3 \\ 4 & 4 & 8 & 16 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 11 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[133] \quad & 21^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 15 & 3 & 4 \\ 5 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 15 & 16 & 11 & 44 & 9 & 12 \\ 3 & 3 & 2 & 9 & 0 & 2 \\ 4 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[135] \quad & 21^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 168, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 8 & 2 & 3 \\ 4 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 8 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[136] \quad & 1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 169, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 10 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[137] \quad & 1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 170, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 \\ 12 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[138] \quad & 1^{12}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 170, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 9 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 6 & 8 & 10 & 16 & 4 & 6 \\ 9 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[139] \quad & 1^{131} 1^{141} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 171, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 11 & 16 & 46 & 20 & 8 & 12 \\ 5 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[140] \quad & 1^{143} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 173, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 9 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 10 & 12 & 25 & 24 & 6 & 9 \\ 9 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[141] \quad & 1^{111} 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 174, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 8 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[142] \quad & 1^{13} 1^{11} 3^{11} 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 175, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 13 & 17 & 0 & 55 & 10 & 13 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[143] \quad & 1^{14} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 176, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 13 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 0 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[144] \quad & 1^{12} 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 180, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 8 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[145] \quad & 21^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 180, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 3 \\ 4 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[146] \quad & 1^{24} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 13 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 \\ 13 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[147] \quad & 1^{31} 1^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 183, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 2 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 12 & 17 & 49 & 9 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1^{133}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 185, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 9 & 12 & 25 & 24 & 6 & 9 \\ 9 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 1^{113}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 187, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 11 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1^{132}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 188, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 8 & 12 & 25 & 24 & 6 & 9 \\ 8 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{124}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 190, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 4 & 4 & 1 & 17 & 2 & 3 \\ 18 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 6 & 8 & 10 & 16 & 4 & 6 \\ 8 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 2 & 3 \\ 3 & 4 & 21 & 11 & 3 & 4 \\ 13 & 21 & 76 & 41 & 11 & 16 \\ 8 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 21^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 2 & 3 \\ 4 & 4 & 9 & 7 & 3 & 4 \\ 6 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 199, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 11 & 13 & 28 & 16 & 7 & 10 \\ 6 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 200, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 0 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 1^{132}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 12 & 17 & 49 & 21 & 9 & 13 \\ 6 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[158] \quad 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 10 & 13 & 28 & 16 & 7 & 10 \\ 6 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[159] \quad 1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 14 & 18 & 58 & 22 & 10 & 14 \\ 6 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[160] \quad 1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 14 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[162] \quad 1^{134}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 2 & 3 \\ 3 & 4 & 8 & 19 & 3 & 4 \\ 7 & 8 & 10 & 28 & 4 & 6 \\ 14 & 19 & 28 & 64 & 10 & 14 \\ 2 & 3 & 4 & 10 & 0 & 2 \\ 3 & 4 & 6 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[161] \quad 21^{123}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 2 & 3 \\ 4 & 4 & 8 & 7 & 3 & 4 \\ 6 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[163] \quad 1^{134}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 7 & 8 & 10 & 22 & 4 & 6 \\ 12 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 12 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 0 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[164] \quad & 1^{143} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 206, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 9 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 11 & 13 & 28 & 25 & 7 & 10 \\ 9 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[166] \quad & 1^{143} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 206, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 9 & 2 & 3 \\ 3 & 4 & 16 & 11 & 3 & 4 \\ 12 & 16 & 46 & 32 & 8 & 12 \\ 9 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[167] \quad & 1^{132} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 212, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 8 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 9 & 12 & 25 & 24 & 6 & 9 \\ 8 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[168] \quad & 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 214, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 9 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 10 & 13 & 28 & 25 & 7 & 10 \\ 9 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[169] \quad & 1^{114} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 216, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 14 & 2 & 3 \\ 3 & 4 & 4 & 19 & 3 & 4 \\ 4 & 4 & 1 & 14 & 2 & 3 \\ 14 & 19 & 14 & 64 & 10 & 14 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 4 & 3 & 14 & 2 & 3 \end{pmatrix}_{b_0} \\
[170] \quad & 1^{121} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 216, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 5 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 12 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[171] \quad & 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 217, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 11 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[172] \quad & 1^{113} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 217, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 15 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[173] \quad & 1^{123} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 219, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 5 & 8 & 10 & 22 & 4 & 6 \\ 11 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{12}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 222, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 2 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 13 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 223, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 2 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 13 & 17 & 49 & 9 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 226, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 10 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 10 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[178] \quad 1^{142}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 8 & 2 & 3 \\ 3 & 4 & 26 & 11 & 3 & 4 \\ 17 & 26 & 118 & 50 & 14 & 20 \\ 8 & 11 & 50 & 20 & 6 & 8 \\ 2 & 3 & 14 & 6 & 0 & 2 \\ 3 & 4 & 20 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{142}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 2 & 3 \\ 3 & 4 & 16 & 11 & 3 & 4 \\ 11 & 16 & 46 & 32 & 8 & 12 \\ 8 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^{141}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 234, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 6 & 2 & 3 \\ 3 & 4 & 26 & 7 & 3 & 4 \\ 18 & 26 & 118 & 32 & 14 & 20 \\ 6 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[181] \quad 1^{14}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 236, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & 0 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 9 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{142}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 2 & 3 \\ 3 & 4 & 16 & 11 & 3 & 4 \\ 12 & 16 & 46 & 32 & 8 & 12 \\ 8 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 9 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 4 & 4 & 1 & 14 & 2 & 3 \\ 15 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[186] $21^{131}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 2 & 3 \\ 4 & 4 & 12 & 7 & 3 & 4 \\ 9 & 12 & 25 & 15 & 6 & 9 \\ 5 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 249, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 5 & 8 & 10 & 28 & 4 & 6 \\ 14 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 76 & 11 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{143}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 17 & 26 & 118 & 14 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{123}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 2 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 14 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[191] \quad 1^{121}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 251, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 13 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[192] \quad 1^{124}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 256, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 20 & 2 & 3 \\ 3 & 4 & 8 & 29 & 3 & 4 \\ 6 & 8 & 10 & 40 & 4 & 6 \\ 20 & 29 & 40 & 148 & 16 & 22 \\ 2 & 3 & 4 & 16 & 0 & 2 \\ 3 & 4 & 6 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[193] \quad 1^{113}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 257, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 12 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[194] \quad 21^{121}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 260, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 2 & 3 \\ 4 & 4 & 18 & 7 & 3 & 4 \\ 12 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[196] \quad 1^{131}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 261, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 15 & 22 & 85 & 12 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{114}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 261, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 4 & 4 & 1 & 17 & 2 & 3 \\ 17 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[199] \quad 1^{114}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 264, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 4 & 4 & 19 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 14 & 19 & 14 & 64 & 10 & 14 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 4 & 3 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[198] \quad 1^{112}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 264, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 14 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[197] \quad 21^{123}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 264, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 3 & 4 \\ 5 & 4 & 4 & 21 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 21 & 21 & 14 & 80 & 12 & 16 \\ 3 & 3 & 2 & 12 & 0 & 2 \\ 4 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[200] \quad 1^{123}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 12 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[201] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 11 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[202] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 13 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[203] \quad 1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 6 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 16 & 22 & 85 & 27 & 12 & 17 \\ 6 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[204] \quad 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 15 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[205] \quad 1^{124}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 2 & 3 \\ 3 & 4 & 8 & 29 & 3 & 4 \\ 5 & 8 & 10 & 40 & 4 & 6 \\ 20 & 29 & 40 & 148 & 16 & 22 \\ 2 & 3 & 4 & 16 & 0 & 2 \\ 3 & 4 & 6 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[206] \quad 1^{142}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 9 & 2 & 3 \\ 3 & 4 & 21 & 11 & 3 & 4 \\ 15 & 21 & 76 & 41 & 11 & 16 \\ 9 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[207] \quad 1^{113}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 11 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 284, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 14 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 11 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 5 & 8 & 10 & 28 & 4 & 6 \\ 13 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 15 & 21 & 76 & 26 & 11 & 16 \\ 6 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 292, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 11 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 295, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 13 & 17 & 49 & 21 & 9 & 13 \\ 6 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[214] $21^{12}3^{11}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 298, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 21 & 6 & 3 & 4 \\ 5 & 4 & 22 & 7 & 3 & 4 \\ 21 & 22 & 85 & 27 & 12 & 17 \\ 6 & 7 & 27 & 7 & 4 & 5 \\ 3 & 3 & 12 & 4 & 0 & 2 \\ 4 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 2 & 3 \\ 3 & 4 & 8 & 19 & 3 & 4 \\ 5 & 8 & 10 & 28 & 4 & 6 \\ 14 & 19 & 28 & 64 & 10 & 14 \\ 2 & 3 & 4 & 10 & 0 & 2 \\ 3 & 4 & 6 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 12 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[215] \quad & 21^{1413^{11}1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 5 & 2 & 3 \\ 4 & 4 & 16 & 7 & 3 & 4 \\ 12 & 16 & 46 & 20 & 8 & 12 \\ 5 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[218] \quad & 1^{123^{11}1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 304, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 21 & 0 & 80 & 12 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 0 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[219] \quad & 21^{1123^{11}1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 304, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 8 & 2 & 3 \\ 4 & 4 & 4 & 11 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[220] \quad & 1^{133^{11}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 305, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 20 & 0 & 69 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 0 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[221] \quad & 1^{1213^{11}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 306, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 11 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[222] \quad & 1^{133^{11}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 307, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 15 & 20 & 0 & 69 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 0 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[223] \quad & 1^{123^{11}1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 310, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 15 & 21 & 0 & 80 & 12 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 0 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[224] \quad & 21^{1123^{11}1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 320, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 14 & 2 & 3 \\ 4 & 4 & 4 & 21 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[225] \quad & 1^{1413^{11}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 321, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 13 & 21 & 76 & 26 & 11 & 16 \\ 5 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[226] \quad 1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 322, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 15 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[227] \quad 21^{1223}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 328, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 2 & 3 \\ 4 & 4 & 8 & 11 & 3 & 4 \\ 6 & 8 & 10 & 16 & 4 & 6 \\ 8 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[228] \quad 1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 331, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 14 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[229] \quad 1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 331, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 9 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 13 & 17 & 49 & 33 & 9 & 13 \\ 9 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[230] \quad 1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 334, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 13 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[231] \quad 1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 334, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 2 & 3 \\ 3 & 4 & 8 & 29 & 3 & 4 \\ 5 & 8 & 10 & 40 & 4 & 6 \\ 19 & 29 & 40 & 148 & 16 & 22 \\ 2 & 3 & 4 & 16 & 0 & 2 \\ 3 & 4 & 6 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[232] \quad 1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 336, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 14 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[233] \quad 1^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 337, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 16 & 22 & 85 & 12 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[234] \quad 21^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 352, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 20 & 2 & 3 \\ 4 & 4 & 8 & 29 & 3 & 4 \\ 6 & 8 & 10 & 40 & 4 & 6 \\ 20 & 29 & 40 & 148 & 16 & 22 \\ 2 & 3 & 4 & 16 & 0 & 2 \\ 3 & 4 & 6 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[235] \quad 1^{13} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 15 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[236] \quad 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 355, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 13 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[237] \quad 1^{14} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 16 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{31} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 12 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{42} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 13 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[240] \quad 21^{132} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 8 & 2 & 3 \\ 4 & 4 & 12 & 11 & 3 & 4 \\ 9 & 12 & 25 & 24 & 6 & 9 \\ 8 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[241] \quad 21^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 2 & 3 \\ 4 & 4 & 13 & 7 & 3 & 4 \\ 9 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[242] \quad 21^{151} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 2 & 3 \\ 4 & 4 & 20 & 7 & 3 & 4 \\ 15 & 20 & 73 & 25 & 10 & 15 \\ 5 & 7 & 25 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 15 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[243] \quad 1^{41} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 374, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 18 & 26 & 118 & 14 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[244] \quad 1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 374, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 2 & 3 \\ 3 & 4 & 8 & 24 & 3 & 4 \\ 6 & 8 & 10 & 34 & 4 & 6 \\ 16 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[245] \quad 21^{123}3^{11}1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 6 & 3 & 4 \\ 5 & 4 & 26 & 7 & 3 & 4 \\ 24 & 26 & 118 & 32 & 14 & 20 \\ 6 & 7 & 32 & 7 & 4 & 5 \\ 3 & 3 & 14 & 4 & 0 & 2 \\ 4 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[246] \quad 1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 18 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[247] \quad 21^{123}3^{11}1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 3 & 4 \\ 5 & 4 & 4 & 25 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 24 & 25 & 17 & 111 & 14 & 19 \\ 3 & 3 & 2 & 14 & 0 & 2 \\ 4 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[248] \quad 21^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 8 & 2 & 3 \\ 4 & 4 & 13 & 11 & 3 & 4 \\ 9 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[249] \quad 1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 12 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 24 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & 0 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 2 & 3 \\ 3 & 4 & 8 & 24 & 3 & 4 \\ 5 & 8 & 10 & 34 & 4 & 6 \\ 17 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 2 & 3 \\ 3 & 4 & 26 & 7 & 3 & 4 \\ 16 & 26 & 118 & 32 & 14 & 20 \\ 5 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 13 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{14} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 2 & 3 \\ 3 & 4 & 4 & 19 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 19 & 14 & 64 & 10 & 14 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 4 & 3 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[255] \quad 21^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 412, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 2 & 3 \\ 4 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 11 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[256] \quad 1^{24} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 2 & 3 \\ 3 & 4 & 8 & 19 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 14 & 19 & 28 & 64 & 10 & 14 \\ 2 & 3 & 4 & 10 & 0 & 2 \\ 3 & 4 & 6 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[257] \quad 1^{14} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & 0 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[258] \quad 21^{14} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 8 & 2 & 3 \\ 4 & 4 & 16 & 11 & 3 & 4 \\ 12 & 16 & 46 & 32 & 8 & 12 \\ 8 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[259] \quad 1^{13} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 431, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 25 & 0 & 111 & 14 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & 0 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[260] \quad 21^{12} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 3 & 4 \\ 5 & 4 & 4 & 29 & 3 & 4 \\ 3 & 4 & 1 & 20 & 2 & 3 \\ 27 & 29 & 20 & 148 & 16 & 22 \\ 3 & 3 & 2 & 16 & 0 & 2 \\ 4 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[261] \quad 1^{42} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 8 & 2 & 3 \\ 3 & 4 & 26 & 11 & 3 & 4 \\ 18 & 26 & 118 & 50 & 14 & 20 \\ 8 & 11 & 50 & 20 & 6 & 8 \\ 2 & 3 & 14 & 6 & 0 & 2 \\ 3 & 4 & 20 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[262] \quad 1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 15 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[263] \quad 1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 439, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 17 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[264] \quad 1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 8 & 2 & 3 \\ 3 & 4 & 21 & 11 & 3 & 4 \\ 14 & 21 & 76 & 41 & 11 & 16 \\ 8 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[265] \quad 1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 14 & 21 & 76 & 26 & 11 & 16 \\ 5 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[266] \quad 21^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 2 & 3 \\ 4 & 4 & 8 & 20 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 14 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[267] \quad 1^{12}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 16 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[268] \quad 1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 457, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 18 & 25 & 0 & 111 & 14 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & 0 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[269] \quad 1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 2 & 3 \\ 3 & 4 & 8 & 24 & 3 & 4 \\ 5 & 8 & 10 & 34 & 4 & 6 \\ 16 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[270] \quad & 1^{131}3^{111}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 459, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 5 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 16 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[271] \quad & 21^{161}3^{111}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 460, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.2 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 5 & 2 & 3 \\ 4 & 4 & 24 & 7 & 3 & 4 \\ 18 & 24 & 106 & 30 & 12 & 18 \\ 5 & 7 & 30 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 18 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[272] \quad & 1^{113}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 461, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 14 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[273] \quad & 21^{113}3^{111}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 468, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 11 & 2 & 3 \\ 4 & 4 & 4 & 15 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 11 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[274] \quad & 1^{114}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 469, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 17 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 17 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[275] \quad & 1^{113}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 472, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 15 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 15 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[276] \quad & 1^{124}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 484, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 17 & 2 & 3 \\ 3 & 4 & 8 & 24 & 3 & 4 \\ 6 & 8 & 10 & 34 & 4 & 6 \\ 17 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[277] \quad & 1^{141}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 489, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 5 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 15 & 21 & 76 & 26 & 11 & 16 \\ 5 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[278] \quad & 21^{123}3^{111}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 492, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 11 & 2 & 3 \\ 4 & 4 & 8 & 15 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 11 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[279] $1^{143}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 502, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 16 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{142}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 544, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 2 & 3 \\ 3 & 4 & 21 & 11 & 3 & 4 \\ 15 & 21 & 76 & 41 & 11 & 16 \\ 8 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 556, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 17 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[282] $21^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 562, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 2 & 3 \\ 4 & 4 & 17 & 7 & 3 & 4 \\ 12 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[283] $21^{171}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 564, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.3$
 $T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 21 & 5 & 2 & 3 \\ 4 & 4 & 28 & 7 & 3 & 4 \\ 21 & 28 & 145 & 35 & 14 & 21 \\ 5 & 7 & 35 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 21 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 571, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 2 & 3 \\ 3 & 4 & 26 & 7 & 3 & 4 \\ 17 & 26 & 118 & 32 & 14 & 20 \\ 5 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[285] $1^{143}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 596, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 20 & 29 & 0 & 148 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & 0 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 601, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 18 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[287] $21^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 604, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 2 & 3 \\ 4 & 4 & 22 & 7 & 3 & 4 \\ 15 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 608, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 16 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[289] $21^{132}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 616, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 8 & 2 & 3 \\ 4 & 4 & 17 & 11 & 3 & 4 \\ 12 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 628, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 2 & 3 \\ 3 & 4 & 4 & 29 & 3 & 4 \\ 2 & 4 & 1 & 20 & 2 & 3 \\ 20 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[291] $21^{142}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 648, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 8 & 2 & 3 \\ 4 & 4 & 26 & 11 & 3 & 4 \\ 18 & 26 & 118 & 50 & 14 & 20 \\ 8 & 11 & 50 & 20 & 6 & 8 \\ 2 & 3 & 14 & 6 & 0 & 2 \\ 3 & 4 & 20 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[292] $1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 657, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 2 & 3 \\ 3 & 4 & 4 & 29 & 3 & 4 \\ 2 & 4 & 1 & 20 & 2 & 3 \\ 19 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 661, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 17 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 664, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 5 & 2 & 3 \\ 3 & 4 & 26 & 7 & 3 & 4 \\ 18 & 26 & 118 & 32 & 14 & 20 \\ 5 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[295] $21^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 2 & 3 \\ 4 & 4 & 4 & 19 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 19 & 14 & 64 & 10 & 14 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 4 & 3 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[297] $21^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 696, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 2 & 3 \\ 4 & 4 & 8 & 19 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 14 & 19 & 28 & 64 & 10 & 14 \\ 2 & 3 & 4 & 10 & 0 & 2 \\ 3 & 4 & 6 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[296] $21^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 2 & 3 \\ 4 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[298] $21^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 2 & 3 \\ 4 & 4 & 4 & 25 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 17 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 2 & 3 \\ 3 & 4 & 4 & 29 & 3 & 4 \\ 3 & 4 & 1 & 20 & 2 & 3 \\ 19 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[300] $21^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 2 & 3 \\ 4 & 4 & 21 & 7 & 3 & 4 \\ 15 & 21 & 76 & 26 & 11 & 16 \\ 5 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[301] $21^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 774, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 17 & 2 & 3 \\ 4 & 4 & 8 & 24 & 3 & 4 \\ 6 & 8 & 10 & 34 & 4 & 6 \\ 17 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 2 & 3 \\ 3 & 4 & 4 & 29 & 3 & 4 \\ 3 & 4 & 1 & 20 & 2 & 3 \\ 20 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[303] $21^{142}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 864, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 8 & 2 & 3 \\ 4 & 4 & 21 & 11 & 3 & 4 \\ 15 & 21 & 76 & 41 & 11 & 16 \\ 8 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[304] $21^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 964, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 5 & 2 & 3 \\ 4 & 4 & 26 & 7 & 3 & 4 \\ 18 & 26 & 118 & 32 & 14 & 20 \\ 5 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[305] $21^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 1020, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 2 & 3 \\ 4 & 4 & 4 & 24 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 17 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[306] $21^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 20 & 2 & 3 \\ 4 & 4 & 4 & 29 & 3 & 4 \\ 3 & 4 & 1 & 20 & 2 & 3 \\ 20 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [312] \quad 1^{12} 3^1 1^1 5 \quad 13^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 3 & 1 & 2 \\ 8 & 0 & 10 & 74 & 22 & 10 & 18 \\ 2 & 0 & 3 & 22 & 5 & 3 & 5 \\ 1 & 0 & 1 & 10 & 3 & 0 & 2 \\ 2 & 0 & 2 & 18 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[311] \quad 1^{11} 3^1 1^1 5 \quad 13^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 3 & 1 & 2 \\ 4 & 0 & 5 & 17 & 11 & 5 & 9 \\ 2 & 0 & 3 & 11 & 5 & 3 & 5 \\ 1 & 0 & 1 & 5 & 3 & 0 & 2 \\ 2 & 0 & 2 & 9 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[315] \quad 1^{13} 1^1 3^1 1^1 5 \quad 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 1 & 2 \\ 1 & 1 & 2 & 12 & 4 & 2 & 3 \\ 1 & 2 & 0 & 12 & 3 & 1 & 2 \\ 8 & 12 & 12 & 70 & 21 & 9 & 18 \\ 3 & 4 & 3 & 21 & 5 & 3 & 5 \\ 1 & 2 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & 2 & 18 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[313] \quad 1^{14} 1^1 3^1 1^1 5 \quad 13^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 3 & 1 & 2 \\ 1 & 1 & 2 & 16 & 4 & 2 & 3 \\ 1 & 2 & 0 & 16 & 3 & 1 & 2 \\ 11 & 16 & 16 & 126 & 28 & 12 & 24 \\ 3 & 4 & 3 & 28 & 5 & 3 & 5 \\ 1 & 2 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & 2 & 24 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[317] \quad 1^{12} 1^1 3^1 1^1 5 \quad 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 1 & 2 \\ 1 & 1 & 2 & 8 & 4 & 2 & 3 \\ 1 & 2 & 0 & 8 & 3 & 1 & 2 \\ 5 & 8 & 8 & 30 & 14 & 6 & 12 \\ 3 & 4 & 3 & 14 & 5 & 3 & 5 \\ 1 & 2 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[318] \quad 1^{11} 1^1 3^1 1^1 5 \quad 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 7 & 3 & 6 \\ 3 & 4 & 3 & 7 & 5 & 3 & 5 \\ 1 & 2 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{12} 1^1 3^1 1^1 5 \quad 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 1 & 2 \\ 1 & 1 & 2 & 8 & 4 & 2 & 3 \\ 1 & 2 & 0 & 8 & 3 & 1 & 2 \\ 7 & 8 & 8 & 30 & 14 & 6 & 12 \\ 3 & 4 & 3 & 14 & 5 & 3 & 5 \\ 1 & 2 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[316] \quad 1^{11} 1^1 3^1 1^1 5 \quad 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 0 & 4 & 2 & 3 \\ 1 & 2 & 0 & 0 & 3 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 3 & 0 & 5 & 3 & 5 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 2 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[314] \quad 1^{111}3^11^1513^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 3 & 1 & 2 \\ 4 & 4 & 4 & 6 & 7 & 3 & 6 \\ 3 & 4 & 3 & 7 & 5 & 3 & 5 \\ 1 & 2 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[320] \quad 1^{112}3^11^1513^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 2 & 4 & 8 & 2 & 3 \\ 1 & 2 & 0 & 4 & 7 & 1 & 2 \\ 2 & 4 & 4 & 6 & 15 & 3 & 6 \\ 6 & 8 & 7 & 15 & 27 & 6 & 11 \\ 1 & 2 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & 2 & 6 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[321] \quad 1^{122}3^11^1513^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 1 & 2 \\ 1 & 1 & 2 & 7 & 8 & 2 & 3 \\ 1 & 2 & 0 & 5 & 7 & 1 & 2 \\ 4 & 2 & 5 & 17 & 23 & 5 & 9 \\ 4 & 7 & 5 & 17 & 23 & 5 & 9 \\ 6 & 8 & 7 & 23 & 27 & 6 & 11 \\ 1 & 2 & 1 & 5 & 6 & 0 & 2 \\ 2 & 3 & 2 & 9 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[322] \quad 1^{123}3^11^1513^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 0 & 8 & 2 & 3 \\ 1 & 2 & 0 & 0 & 7 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 8 & 7 & 0 & 27 & 6 & 11 \\ 1 & 2 & 1 & 0 & 6 & 0 & 2 \\ 2 & 3 & 2 & 0 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{131}3^11^1513^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 1 & 2 \\ 1 & 1 & 2 & 12 & 4 & 2 & 3 \\ 1 & 2 & 0 & 12 & 3 & 1 & 2 \\ 9 & 12 & 12 & 70 & 21 & 9 & 18 \\ 3 & 4 & 3 & 21 & 5 & 3 & 5 \\ 1 & 2 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & 2 & 18 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{111}3^11^1513^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 7 & 3 & 6 \\ 3 & 4 & 3 & 7 & 5 & 3 & 5 \\ 1 & 2 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{121}3^11^1513^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 1 & 2 \\ 1 & 1 & 2 & 8 & 4 & 2 & 3 \\ 1 & 2 & 0 & 8 & 3 & 1 & 2 \\ 6 & 8 & 8 & 30 & 14 & 6 & 12 \\ 3 & 4 & 3 & 14 & 5 & 3 & 5 \\ 1 & 2 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{13}3^11^1513^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 0 & 12 & 2 & 3 \\ 1 & 2 & 0 & 0 & 11 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 12 & 11 & 0 & 65 & 9 & 17 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 3 & 2 & 0 & 17 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{112}3^11^{12}5^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 100, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 1 & 2 \\ 1 & 1 & 2 & 4 & 8 & 2 & 3 \\ 1 & 2 & 0 & 4 & 7 & 1 & 2 \\ 3 & 4 & 4 & 6 & 15 & 3 & 6 \\ 6 & 8 & 7 & 15 & 27 & 6 & 11 \\ 1 & 2 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & 2 & 6 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

4.948 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 4 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 5 & 5 & 4 & 6 & 3 & 3 & 4 \\ 6 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 4 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 3 & 8 & 4 \\ 5 & 9 & 13 & 5 & 5 & 12 & 7 \\ 3 & 3 & 5 & 0 & 2 & 4 & 2 \\ 2 & 3 & 5 & 2 & 0 & 4 & 2 \\ 5 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 7 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 3 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 1 & 3 \\ 3 & 4 & 8 & 7 & 3 & 0 & 4 \\ 7 & 8 & 10 & 10 & 4 & 0 & 6 \\ 5 & 7 & 10 & 7 & 4 & 0 & 5 \\ 2 & 3 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 2 & 1 & 3 \\ 3 & 4 & 12 & 11 & 3 & 0 & 4 \\ 10 & 12 & 25 & 24 & 6 & 0 & 9 \\ 8 & 11 & 24 & 20 & 6 & 0 & 8 \\ 2 & 3 & 6 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1^{212}} 1^{1111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 4 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 4 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1^{41}} 1^{123} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 2 & 1 & 3 \\ 3 & 4 & 8 & 20 & 3 & 0 & 4 \\ 5 & 8 & 10 & 28 & 4 & 0 & 6 \\ 15 & 20 & 28 & 69 & 11 & 0 & 15 \\ 2 & 3 & 4 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1^{211}} 1^{132} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 8 & 12 & 25 & 6 & 6 & 27 & 9 \\ 3 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 8 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1^{11}} 1^{113} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 2 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1^{11}} 1^{112} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1^{11}} 1^{113} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 4 & 3 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1^{321}} 1^{122} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 2 & 1 & 3 \\ 3 & 4 & 8 & 11 & 3 & 0 & 4 \\ 7 & 8 & 10 & 16 & 4 & 0 & 6 \\ 8 & 11 & 16 & 20 & 6 & 0 & 8 \\ 2 & 3 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1^{31}} 1^{121} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 2 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 7 & 3 & 0 & 4 \\ 4 & 4 & 1 & 5 & 2 & 0 & 3 \\ 5 & 7 & 5 & 7 & 4 & 0 & 5 \\ 2 & 3 & 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 4 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 4 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 0 & 4 \\ 3 & 5 & 4 & 6 & 3 & 0 & 4 \\ 5 & 7 & 6 & 7 & 4 & 0 & 5 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 2 & 1 & 3 \\ 3 & 4 & 14 & 7 & 3 & 0 & 4 \\ 12 & 14 & 37 & 17 & 8 & 0 & 11 \\ 6 & 7 & 17 & 7 & 4 & 0 & 5 \\ 2 & 3 & 8 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 2 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 11 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 7 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 7 & 8 & 10 & 10 & 4 & 12 & 6 \\ 7 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 7 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 4 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 4 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{312}$$

$$[23] \ 1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 9 & 12 & 25 & 6 & 6 & 27 & 9 \\ 3 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 9 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$1^{211}$$

$$[21] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 2 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$1^{21}$$

$$[26] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 1 & 3 \\ 3 & 4 & 4 & 12 & 3 & 0 & 4 \\ 2 & 4 & 1 & 8 & 2 & 0 & 3 \\ 9 & 12 & 8 & 25 & 7 & 0 & 9 \\ 2 & 3 & 2 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 9 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}$$

$$[25] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 3 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{211}$$

$$[27] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$1^{121}$$

$$[28] \ 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 6 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 5 & 8 & 10 & 10 & 4 & 12 & 6 \\ 6 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 6 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$1^{212}$$

$$[29] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}$$

$$[30] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 5 & 3 \\ 3 & 4 & 9 & 7 & 3 & 8 & 4 \\ 5 & 9 & 13 & 11 & 5 & 12 & 7 \\ 5 & 7 & 11 & 7 & 4 & 10 & 5 \\ 2 & 3 & 5 & 4 & 0 & 4 & 2 \\ 5 & 8 & 12 & 10 & 4 & 10 & 6 \\ 3 & 4 & 7 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 & 6 \\ 3 & 3 & 4 & 0 & 2 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 & 2 \\ 3 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 11 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 3 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 5 & 9 & 13 & 11 & 5 & 6 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 1 & 3 \\ 3 & 4 & 14 & 7 & 3 & 0 & 4 \\ 11 & 14 & 37 & 17 & 8 & 0 & 11 \\ 6 & 7 & 17 & 7 & 4 & 0 & 5 \\ 2 & 3 & 8 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 9 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 21^{113} 1^{1111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 3 & 9 & 4 \\ 5 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 3 & 3 & 2 & 4 & 0 & 5 & 2 \\ 9 & 9 & 6 & 11 & 5 & 13 & 7 \\ 4 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 4 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 8 & 9 & 13 & 11 & 5 & 6 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 4 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 0 & 4 \\ 4 & 5 & 4 & 6 & 3 & 0 & 4 \\ 5 & 7 & 6 & 7 & 4 & 0 & 5 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{141} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 & 4 \\ 13 & 21 & 76 & 11 & 11 & 15 & 16 \\ 3 & 3 & 11 & 0 & 2 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 & 2 \\ 3 & 4 & 15 & 2 & 2 & 1 & 3 \\ 3 & 4 & 16 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 6 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 4 & 5 & 4 & 3 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 0 & 4 \\ 4 & 5 & 4 & 6 & 3 & 0 & 4 \\ 6 & 7 & 6 & 7 & 4 & 0 & 5 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{123} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 2 & 1 & 3 \\ 3 & 4 & 8 & 25 & 3 & 0 & 4 \\ 5 & 8 & 10 & 34 & 4 & 0 & 6 \\ 17 & 25 & 34 & 111 & 14 & 0 & 19 \\ 2 & 3 & 4 & 14 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 19 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 2 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 2 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{122} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{122} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 8 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

 1^{211}

$$[52] \ 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 1 & 3 \\ 3 & 4 & 4 & 11 & 3 & 0 & 4 \\ 4 & 4 & 1 & 8 & 2 & 0 & 3 \\ 8 & 11 & 8 & 20 & 6 & 0 & 8 \\ 2 & 3 & 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

 1^{111}

$$[51] \ 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 7 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 0 & 5 \\ 2 & 3 & 0 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

 1^{111}

$$[58] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

 1^{111}

$$[57] \ 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & 0 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

 1^{112}

$$[63] \ 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 12 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

 1^{13}

$$[59] \ 1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

 1^{213}

$$[60] \ 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

 1^{211}

$$[61] \ 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 & 9 \\ 3 & 3 & 6 & 0 & 2 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 & 2 \\ 3 & 4 & 9 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 6 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 1 & 3 \\ 3 & 4 & 4 & 16 & 3 & 0 & 4 \\ 4 & 4 & 1 & 11 & 2 & 0 & 3 \\ 12 & 16 & 11 & 44 & 9 & 0 & 12 \\ 2 & 3 & 2 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{112} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 14 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{213} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 15 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 3 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 8 & 9 & 13 & 11 & 5 & 6 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{1212} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 3 & 8 & 4 \\ 8 & 12 & 25 & 6 & 6 & 18 & 9 \\ 3 & 3 & 6 & 0 & 2 & 4 & 2 \\ 2 & 3 & 6 & 2 & 0 & 4 & 2 \\ 6 & 8 & 18 & 4 & 4 & 10 & 6 \\ 3 & 4 & 9 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1212} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{113} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 9 & 12 & 0 & 6 & 6 & 25 & 9 \\ 3 & 4 & 0 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{114} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 16 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 11 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{114} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 2 & 1 & 3 \\ 3 & 4 & 4 & 12 & 3 & 0 & 4 \\ 4 & 4 & 1 & 8 & 2 & 0 & 3 \\ 10 & 12 & 8 & 25 & 7 & 0 & 9 \\ 2 & 3 & 2 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 9 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1211} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 1 & 3 \\ 3 & 4 & 14 & 7 & 3 & 0 & 4 \\ 10 & 14 & 37 & 17 & 8 & 0 & 11 \\ 5 & 7 & 17 & 7 & 4 & 0 & 5 \\ 2 & 3 & 8 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$1^{211}$$

$$[84] \ 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{213}$$

$$[82] \ 1^{111}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 4 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$1^{121}$$

$$[81] \ 1^{121}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 3 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 5 & 8 & 10 & 10 & 4 & 6 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{114}$$

$$[80] \ 1^{124}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$1^{112}$$

$$[79] \ 1^{124}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 16 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$1^{114}$$

$$[78] \ 1^{124}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}$$

$$[88] \ 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{12}$$

$$[87] \ 1^{111}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 & 3 \\ 14 & 21 & 14 & 80 & 12 & 14 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 & 2 \\ 2 & 4 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 3 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 2 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{1121}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 4 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 & 3 \\ 16 & 21 & 14 & 80 & 12 & 14 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 & 2 \\ 4 & 4 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 3 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{1411}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 1 & 3 \\ 3 & 4 & 0 & 16 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 13 & 16 & 0 & 44 & 9 & 0 & 12 \\ 2 & 3 & 0 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{1121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 2 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{1211}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 11 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 0 & 8 \\ 2 & 3 & 0 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1231}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 17 & 3 \\ 3 & 4 & 0 & 11 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 50 & 8 \\ 2 & 3 & 0 & 6 & 0 & 14 & 2 \\ 17 & 26 & 0 & 50 & 14 & 118 & 20 \\ 3 & 4 & 0 & 8 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{1131}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 2 & 1 & 3 \\ 3 & 4 & 4 & 15 & 3 & 0 & 4 \\ 4 & 4 & 1 & 11 & 2 & 0 & 3 \\ 11 & 15 & 11 & 39 & 8 & 0 & 11 \\ 2 & 3 & 2 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 11 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{1112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{1211} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 2 & 6 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 6 & 8 & 10 & 10 & 4 & 12 & 6 \\ 5 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 6 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{114} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 12 & 16 & 0 & 8 & 8 & 46 & 12 \\ 3 & 4 & 0 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{1111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{1411} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 & 4 \\ 11 & 16 & 46 & 8 & 8 & 12 & 12 \\ 3 & 3 & 8 & 0 & 2 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 & 2 \\ 3 & 4 & 12 & 2 & 2 & 1 & 3 \\ 3 & 4 & 12 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{1121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 9 & 12 & 0 & 25 & 7 & 0 & 9 \\ 2 & 3 & 0 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 9 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 2 & 3 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 6 & 9 & 13 & 11 & 5 & 6 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 5 & 9 & 13 & 11 & 5 & 6 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 2 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 2 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{113} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{112} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{142} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 6 & 3 \\ 3 & 4 & 16 & 3 & 3 & 8 & 4 \\ 11 & 16 & 46 & 8 & 8 & 24 & 12 \\ 3 & 3 & 8 & 0 & 2 & 4 & 2 \\ 2 & 3 & 8 & 2 & 0 & 4 & 2 \\ 6 & 8 & 24 & 4 & 4 & 10 & 6 \\ 3 & 4 & 12 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{\overline{112}3} 1^{\overline{11}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 14 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 6 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 14 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\overline{113}3} 1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overline{112}3} 1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 9 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overline{113}3} 1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 8 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 8 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & 0 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overline{112}3} 1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overline{114}3} 1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overline{113}3} 1^{\overline{11}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{112}3} 1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 1 & 3 \\ 3 & 4 & 4 & 16 & 3 & 0 & 4 \\ 2 & 4 & 1 & 11 & 2 & 0 & 3 \\ 12 & 16 & 11 & 44 & 9 & 0 & 12 \\ 2 & 3 & 2 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{1112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 7 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{1123} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 6 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{112} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[126] \ 21^{113} 3^{11} 1^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 3 & 3 & 4 \\ 5 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 15 & 16 & 11 & 44 & 9 & 11 & 12 \\ 3 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 4 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{1121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 & 3 \\ 12 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 2 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{111} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{1^{14}3^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{1^{12}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{1^{11}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{1^{11}3^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 3 & 4 & 0 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{1^{13}3^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 7 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^{1^{214}3^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 11 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{1^{12}3^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 10 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{1^{123}3^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 6 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 2 & 1 & 3 \\ 3 & 4 & 4 & 20 & 3 & 0 & 4 \\ 4 & 4 & 1 & 14 & 2 & 0 & 3 \\ 15 & 20 & 14 & 69 & 11 & 0 & 15 \\ 2 & 3 & 2 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{1211} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 6 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 5 & 8 & 10 & 10 & 4 & 12 & 6 \\ 5 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 6 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 6 & 3 \\ 3 & 4 & 0 & 7 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 6 & 8 & 0 & 10 & 4 & 10 & 6 \\ 3 & 4 & 0 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{113} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 2 & 7 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 6 & 8 & 10 & 10 & 4 & 12 & 6 \\ 6 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 7 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[147] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 8 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 & 2 \\ 2 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1^{144} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 2 & 1 & 3 \\ 3 & 4 & 4 & 19 & 3 & 0 & 4 \\ 4 & 4 & 1 & 14 & 2 & 0 & 3 \\ 14 & 19 & 14 & 64 & 10 & 0 & 14 \\ 2 & 3 & 2 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 14 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{1111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{131}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 1 & 3 \\ 3 & 4 & 0 & 17 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 13 & 17 & 0 & 55 & 10 & 0 & 13 \\ 2 & 3 & 0 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 13 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 2 & 1 & 3 \\ 3 & 4 & 9 & 11 & 3 & 0 & 4 \\ 6 & 9 & 13 & 17 & 5 & 0 & 7 \\ 8 & 11 & 17 & 20 & 6 & 0 & 8 \\ 2 & 3 & 5 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{112}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 5 & 8 & 0 & 10 & 4 & 10 & 6 \\ 3 & 4 & 0 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{131}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 15 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 0 & 11 \\ 2 & 3 & 0 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 11 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{112}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -102, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{1111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{1211} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 3 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 1^{123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 12 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{1211} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 2 & 7 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 6 & 8 & 10 & 10 & 4 & 12 & 6 \\ 5 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 7 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{1223} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 2 & 1 & 3 \\ 3 & 4 & 8 & 16 & 3 & 0 & 4 \\ 6 & 8 & 10 & 22 & 4 & 0 & 6 \\ 13 & 16 & 22 & 44 & 9 & 0 & 12 \\ 2 & 3 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 2 & 1 & 3 \\ 3 & 4 & 8 & 16 & 3 & 0 & 4 \\ 5 & 8 & 10 & 22 & 4 & 0 & 6 \\ 11 & 16 & 22 & 44 & 9 & 0 & 12 \\ 2 & 3 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[158] \quad 1^{123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 0 & 6 & 2 \\ 9 & 12 & 0 & 15 & 6 & 25 & 9 \\ 3 & 4 & 0 & 5 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 4 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 13 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 4 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[163] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{124}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{11123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 7 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & 0 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{11123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -118, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{11123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -121, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 11 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 4 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 11 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{11123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -122, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 3 & 4 & 0 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{1143} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -122, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 20 & 5 \\ 2 & 3 & 0 & 4 & 0 & 8 & 2 \\ 12 & 16 & 0 & 20 & 8 & 46 & 12 \\ 3 & 4 & 0 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{11113} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -123, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 2 & 3 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 10 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{1231} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 1 & 3 \\ 3 & 4 & 0 & 16 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 12 & 16 & 0 & 44 & 9 & 0 & 12 \\ 2 & 3 & 0 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{1223} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{1112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 11 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 11 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -125, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{113} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 15 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 12 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -131, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[181] \ 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[182] \ 21^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 2 & 6 & 3 \\ 4 & 4 & 9 & 7 & 3 & 8 & 4 \\ 6 & 9 & 13 & 11 & 5 & 12 & 7 \\ 5 & 7 & 11 & 7 & 4 & 10 & 5 \\ 2 & 3 & 5 & 4 & 0 & 4 & 2 \\ 6 & 8 & 12 & 10 & 4 & 10 & 6 \\ 3 & 4 & 7 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 6 & 3 \\ 3 & 4 & 0 & 7 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 6 & 8 & 0 & 10 & 4 & 10 & 6 \\ 3 & 4 & 0 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 1 & 3 \\ 3 & 4 & 4 & 21 & 3 & 0 & 4 \\ 2 & 4 & 1 & 14 & 2 & 0 & 3 \\ 15 & 21 & 14 & 80 & 12 & 0 & 16 \\ 2 & 3 & 2 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 16 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 7 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & 0 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 1^{114} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 10 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[197] \quad 1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 8 & 3 \\ 3 & 4 & 0 & 7 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 0 & 6 & 2 \\ 8 & 12 & 0 & 15 & 6 & 25 & 9 \\ 3 & 4 & 0 & 5 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[192] \quad 1^{141} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 19 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 0 & 14 \\ 2 & 3 & 0 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 14 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{13}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 11 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 2 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 11 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{122}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{113}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 10 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{112}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 5 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{12}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 2 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{122}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 5 & 3 \\ 3 & 4 & 0 & 11 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 16 & 8 \\ 2 & 3 & 0 & 6 & 0 & 4 & 2 \\ 5 & 8 & 0 & 16 & 4 & 10 & 6 \\ 3 & 4 & 0 & 8 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[190] $21^{113}3^{11}1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 3 & 12 & 4 \\ 5 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 3 & 3 & 2 & 4 & 0 & 7 & 2 \\ 12 & 13 & 9 & 16 & 7 & 28 & 10 \\ 4 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 6 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 6 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & 0 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 4 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 10 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 8 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 8 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 8 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 5 & 2 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 1 & 3 \\ 3 & 4 & 4 & 20 & 3 & 0 & 4 \\ 2 & 4 & 1 & 14 & 2 & 0 & 3 \\ 15 & 20 & 14 & 69 & 11 & 0 & 15 \\ 2 & 3 & 2 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 14 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 4 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 14 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 16 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{113}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 0 & 6 & 2 \\ 9 & 12 & 0 & 15 & 6 & 25 & 9 \\ 3 & 4 & 0 & 5 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 2 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 17 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 16 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 16 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & 0 & 5 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{1^{214}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -166, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{1^{12}} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -166, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{1^{111}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -169, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{1^{14}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -171, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{1^{211}} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 2 & 3 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[220] \ 1^{1^{124}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 15 & 3 \\ 3 & 4 & 0 & 7 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 26 & 5 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 15 & 21 & 0 & 26 & 11 & 76 & 16 \\ 3 & 4 & 0 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{1^{131}} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 6 & 8 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 & 2 \\ 3 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[224] \ 1^{1^{141}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 1 & 3 \\ 3 & 4 & 0 & 21 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 16 & 21 & 0 & 80 & 12 & 0 & 16 \\ 2 & 3 & 0 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 16 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 15 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & 0 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{113} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 10 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 2 & 1 & 3 \\ 3 & 4 & 8 & 16 & 3 & 0 & 4 \\ 6 & 8 & 10 & 22 & 4 & 0 & 6 \\ 12 & 16 & 22 & 44 & 9 & 0 & 12 \\ 2 & 3 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 6 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{113} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{114} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -178, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -179, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 9 & 12 & 0 & 25 & 7 & 8 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 3 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{123} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 6 & 3 \\ 3 & 4 & 0 & 11 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 6 & 16 & 8 \\ 2 & 3 & 0 & 6 & 0 & 4 & 2 \\ 6 & 8 & 0 & 16 & 4 & 10 & 6 \\ 3 & 4 & 0 & 8 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{1121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 & 3 \\ 11 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 2 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{113} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{113} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 13 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{113} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -181, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{114} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -182, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 20 & 5 \\ 2 & 3 & 0 & 4 & 0 & 8 & 2 \\ 12 & 16 & 0 & 20 & 8 & 46 & 12 \\ 3 & 4 & 0 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 1 & 3 \\ 3 & 4 & 0 & 20 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 15 & 20 & 0 & 69 & 11 & 0 & 15 \\ 2 & 3 & 0 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{111} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -187, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{1121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{124}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 8 & 3 \\ 3 & 4 & 0 & 11 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 24 & 8 \\ 2 & 3 & 0 & 6 & 0 & 6 & 2 \\ 8 & 12 & 0 & 24 & 6 & 25 & 9 \\ 3 & 4 & 0 & 8 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 5 & 3 \\ 3 & 4 & 0 & 15 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 22 & 11 \\ 2 & 3 & 0 & 8 & 0 & 4 & 2 \\ 5 & 8 & 0 & 22 & 4 & 10 & 6 \\ 3 & 4 & 0 & 11 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{14}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 14 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 & 2 \\ 2 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 14 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{113}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 20 & 5 \\ 2 & 3 & 0 & 4 & 0 & 8 & 2 \\ 11 & 16 & 0 & 20 & 8 & 46 & 12 \\ 3 & 4 & 0 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{1^{124}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 11 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{1^{122}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{1^{124}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 14 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & 0 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{1^{144}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 14 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & 0 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{1^{111}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 3 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 5 & 8 & 10 & 10 & 4 & 6 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{1^{122}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{1^{123}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 10 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 10 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{1^{121}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[255] \quad 1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 13 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[256] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[257] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[259] \quad 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 11 & 3 \\ 3 & 4 & 0 & 11 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 0 & 9 & 2 \\ 11 & 17 & 0 & 33 & 9 & 49 & 13 \\ 3 & 4 & 0 & 8 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[260] \quad 1^{114} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 18 & 26 & 0 & 14 & 14 & 118 & 20 \\ 3 & 4 & 0 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[258] \quad 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 8 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 & 2 \\ 3 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[264] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[263] \quad 1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[262] \quad & 21^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 12 & 3 \\ 4 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 12 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[261] \quad & 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 12 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 12 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[266] \quad & 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[265] \quad & 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 4 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[267] \quad & 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[268] \quad & 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -221, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[271] \quad & 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 6 & 3 \\ 3 & 4 & 0 & 11 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 16 & 8 \\ 2 & 3 & 0 & 6 & 0 & 4 & 2 \\ 6 & 8 & 0 & 16 & 4 & 10 & 6 \\ 3 & 4 & 0 & 8 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[270] \quad & 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 & 3 \\ 15 & 21 & 14 & 80 & 12 & 14 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 & 2 \\ 3 & 4 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 3 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{112}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 12 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 12 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 11 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 11 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{112}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 12 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & 0 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[276] $1^{123}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 2 & 1 & 3 \\ 3 & 4 & 8 & 20 & 3 & 0 & 4 \\ 5 & 8 & 10 & 28 & 4 & 0 & 6 \\ 14 & 20 & 28 & 69 & 11 & 0 & 15 \\ 2 & 3 & 4 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{1231}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -233, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 & 3 \\ 12 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{123}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 14 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[280] \quad 1^{1^{141}} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 2 & 1 & 3 \\ 3 & 4 & 8 & 29 & 3 & 0 & 4 \\ 5 & 8 & 10 & 40 & 4 & 0 & 6 \\ 20 & 29 & 40 & 148 & 16 & 0 & 22 \\ 2 & 3 & 4 & 16 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 22 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[279] \quad 1^{1^{114}} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 14 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 14 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[281] \quad 1^{1^{112}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[282] \quad 1^{1^{114}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[283] \quad 1^{1^{123}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 9 & 3 \\ 3 & 4 & 0 & 11 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 24 & 8 \\ 2 & 3 & 0 & 6 & 0 & 6 & 2 \\ 9 & 12 & 0 & 24 & 6 & 25 & 9 \\ 3 & 4 & 0 & 8 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{1^{113}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 10 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 10 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{1^{21}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{1^{14}} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 2 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[287] \quad 1^{1123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 14 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 6 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 14 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[288] \quad 1^{142} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 5 & 3 \\ 3 & 4 & 0 & 19 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 28 & 14 \\ 2 & 3 & 0 & 10 & 0 & 4 & 2 \\ 5 & 8 & 0 & 28 & 4 & 10 & 6 \\ 3 & 4 & 0 & 14 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[289] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 10 & 3 \\ 3 & 4 & 0 & 11 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 25 & 8 \\ 2 & 3 & 0 & 6 & 0 & 7 & 2 \\ 10 & 13 & 0 & 25 & 7 & 28 & 10 \\ 3 & 4 & 0 & 8 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{123} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 11 & 3 \\ 3 & 4 & 0 & 11 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 32 & 8 \\ 2 & 3 & 0 & 6 & 0 & 8 & 2 \\ 11 & 16 & 0 & 32 & 8 & 46 & 12 \\ 3 & 4 & 0 & 8 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{141} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 24 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 0 & 18 \\ 2 & 3 & 0 & 13 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 18 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{113} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 2 & 1 & 3 \\ 3 & 4 & 4 & 25 & 3 & 0 & 4 \\ 2 & 4 & 1 & 17 & 2 & 0 & 3 \\ 18 & 25 & 17 & 111 & 14 & 0 & 19 \\ 2 & 3 & 2 & 14 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 19 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[293] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 8 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 8 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[294] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[295] \quad 1^{1^{122}} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 9 & 3 \\ 3 & 4 & 0 & 11 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 25 & 8 \\ 2 & 3 & 0 & 6 & 0 & 7 & 2 \\ 9 & 13 & 0 & 25 & 7 & 28 & 10 \\ 3 & 4 & 0 & 8 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[296] \quad 1^{1^{124}} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 12 & 3 \\ 3 & 4 & 0 & 11 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 32 & 8 \\ 2 & 3 & 0 & 6 & 0 & 8 & 2 \\ 12 & 16 & 0 & 32 & 8 & 46 & 12 \\ 3 & 4 & 0 & 8 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[297] \quad 1^{1^{121}} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 & 3 \\ 12 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{1^{113}} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[299] \quad 1^{1^{214}} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[300] \quad 1^{1^{114}} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 16 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[301] \quad 1^{1^{141}} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 1 & 3 \\ 3 & 4 & 0 & 25 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 18 & 25 & 0 & 111 & 14 & 0 & 19 \\ 2 & 3 & 0 & 14 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 19 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{1^{111}} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 8 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 8 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[303] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 9 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[304] $1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 20 & 0 & 69 & 11 & 14 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 15 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[305] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 14 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & 0 & 5 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[306] $1^{141}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 12 & 15 & 0 & 39 & 8 & 11 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 11 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[307] $1^{142}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 6 & 3 \\ 3 & 4 & 0 & 15 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 12 & 15 & 0 & 39 & 8 & 22 & 11 \\ 2 & 3 & 0 & 8 & 0 & 4 & 2 \\ 6 & 8 & 0 & 22 & 4 & 10 & 6 \\ 3 & 4 & 0 & 11 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 13 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[309] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 13 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 21 & 5 \\ 13 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[310] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 13 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 13 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & 0 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[311] \ 1^{114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -299, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[312] \ 1^{1121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -301, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 & 3 \\ 13 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[313] \ 1^{123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 2 & 1 & 3 \\ 3 & 4 & 8 & 20 & 3 & 0 & 4 \\ 6 & 8 & 10 & 28 & 4 & 0 & 6 \\ 14 & 20 & 28 & 69 & 11 & 0 & 15 \\ 2 & 3 & 4 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[314] \ 1^{123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[316] \ 1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[315] \ 1^{1112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[317] \ 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -315, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 11 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[318] \ 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{1^{121}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[320] \quad 21^{1^{111}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 6 & 3 \\ 4 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[321] \quad 1^{1^{121}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[322] \quad 1^{1^{131}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 11 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 11 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{1^{113}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{1^{132}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 6 & 3 \\ 3 & 4 & 0 & 15 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 22 & 11 \\ 2 & 3 & 0 & 8 & 0 & 4 & 2 \\ 6 & 8 & 0 & 22 & 4 & 10 & 6 \\ 3 & 4 & 0 & 11 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{1^{124}} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 11 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 11 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{1^{114}} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[327] $1^{1^{131}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 12 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[328] $1^{1^{121}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 & 3 \\ 11 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[329] $1^{1^{141}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 2 & 1 & 3 \\ 3 & 4 & 8 & 24 & 3 & 0 & 4 \\ 5 & 8 & 10 & 34 & 4 & 0 & 6 \\ 17 & 24 & 34 & 100 & 13 & 0 & 18 \\ 2 & 3 & 4 & 13 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 18 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[330] $1^{1^{121}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[331] $1^{1^{123}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[332] $1^{1^{114}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 13 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 13 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[333] $21^{1^{111}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 3 & 3 \\ 4 & 4 & 4 & 7 & 3 & 4 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[334] $1^{1^{142}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 6 & 3 \\ 3 & 4 & 0 & 20 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 15 & 20 & 0 & 69 & 11 & 28 & 15 \\ 2 & 3 & 0 & 11 & 0 & 4 & 2 \\ 6 & 8 & 0 & 28 & 4 & 10 & 6 \\ 3 & 4 & 0 & 15 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[335] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 25 & 0 & 111 & 14 & 17 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 2 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 19 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[336] $1^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 2 & 3 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 & 3 \\ 16 & 21 & 14 & 80 & 12 & 14 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 & 2 \\ 3 & 4 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 3 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{143}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 17 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[338] $1^{143}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 17 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 & 2 \\ 2 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 18 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[340] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 13 & 3 \\ 3 & 4 & 0 & 11 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 0 & 9 & 2 \\ 13 & 17 & 0 & 33 & 9 & 49 & 13 \\ 3 & 4 & 0 & 8 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[341] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -410, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 15 & 21 & 0 & 80 & 12 & 14 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[343] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[344] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[345] $1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 16 & 21 & 0 & 80 & 12 & 14 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[346] $21^{1112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 6 & 3 \\ 4 & 4 & 4 & 7 & 3 & 8 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[347] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 12 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[348] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 14 & 3 \\ 3 & 4 & 0 & 11 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 41 & 8 \\ 2 & 3 & 0 & 6 & 0 & 11 & 2 \\ 14 & 21 & 0 & 41 & 11 & 76 & 16 \\ 3 & 4 & 0 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 13 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[352] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 12 & 3 \\ 3 & 4 & 0 & 11 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 0 & 9 & 2 \\ 12 & 17 & 0 & 33 & 9 & 49 & 13 \\ 3 & 4 & 0 & 8 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 11 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{124}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 18 & 3 \\ 3 & 4 & 0 & 11 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 50 & 8 \\ 2 & 3 & 0 & 6 & 0 & 14 & 2 \\ 18 & 26 & 0 & 50 & 14 & 118 & 20 \\ 3 & 4 & 0 & 8 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{144}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -475, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 14 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 14 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{142}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 6 & 3 \\ 3 & 4 & 0 & 19 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 28 & 14 \\ 2 & 3 & 0 & 10 & 0 & 4 & 2 \\ 6 & 8 & 0 & 28 & 4 & 10 & 6 \\ 3 & 4 & 0 & 14 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -491, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 26 & 5 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 14 & 21 & 0 & 26 & 11 & 76 & 16 \\ 3 & 4 & 0 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[358] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 15 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 15 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & 0 & 5 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{1121}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 12 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 20 & 0 & 69 & 11 & 14 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 15 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{141}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 2 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 20 & 29 & 0 & 148 & 16 & 20 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 & 2 \\ 2 & 4 & 0 & 20 & 2 & 1 & 3 \\ 3 & 4 & 0 & 22 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 16 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 16 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & 0 & 5 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{142}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 6 & 3 \\ 3 & 4 & 0 & 24 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 34 & 18 \\ 2 & 3 & 0 & 13 & 0 & 4 & 2 \\ 6 & 8 & 0 & 34 & 4 & 10 & 6 \\ 3 & 4 & 0 & 18 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 15 & 20 & 0 & 69 & 11 & 14 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 15 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 15 & 3 \\ 3 & 4 & 0 & 7 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 26 & 5 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 15 & 21 & 0 & 26 & 11 & 76 & 16 \\ 3 & 4 & 0 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[367] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 17 & 3 \\ 3 & 4 & 0 & 7 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 32 & 5 \\ 2 & 3 & 0 & 4 & 0 & 14 & 2 \\ 17 & 26 & 0 & 32 & 14 & 118 & 20 \\ 3 & 4 & 0 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -614, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 25 & 0 & 111 & 14 & 17 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 3 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 19 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[369] $21^{121}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 2 & 3 & 3 \\ 4 & 4 & 4 & 11 & 3 & 4 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -630, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[371] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 15 & 3 \\ 3 & 4 & 0 & 11 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 41 & 8 \\ 2 & 3 & 0 & 6 & 0 & 11 & 2 \\ 15 & 21 & 0 & 41 & 11 & 76 & 16 \\ 3 & 4 & 0 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[372] $21^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -660, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 2 & 3 & 3 \\ 4 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 11 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[373] $21^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -676, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 9 & 3 \\ 4 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[374] $1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 18 & 25 & 0 & 111 & 14 & 17 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 3 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 19 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[375] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -770, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 18 & 3 \\ 3 & 4 & 0 & 7 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 32 & 5 \\ 2 & 3 & 0 & 4 & 0 & 14 & 2 \\ 18 & 26 & 0 & 32 & 14 & 118 & 20 \\ 3 & 4 & 0 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[376] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -771, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 17 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 & 2 \\ 3 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 18 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[377] $1^{141}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 20 & 29 & 0 & 148 & 16 & 20 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 & 2 \\ 3 & 4 & 0 & 20 & 2 & 1 & 3 \\ 3 & 4 & 0 & 22 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.949 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + su2n3mix

$T = 8$ [2] $1^{1212}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[0] $1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 26 & 5 \\ 15 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{1212}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 6 & 3 \\ 3 & 4 & 8 & 12 & 7 & 4 \\ 5 & 8 & 10 & 16 & 10 & 6 \\ 8 & 12 & 16 & 25 & 15 & 9 \\ 6 & 7 & 10 & 15 & 7 & 5 \\ 3 & 4 & 6 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{1211}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 5 & 3 \\ 3 & 4 & 8 & 12 & 7 & 4 \\ 5 & 8 & 10 & 16 & 10 & 6 \\ 8 & 12 & 16 & 25 & 15 & 9 \\ 5 & 7 & 10 & 15 & 7 & 5 \\ 3 & 4 & 6 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{1122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 2 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{1112}3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$

[6] $1^{1221}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 9 & 6 & 3 \\ 3 & 4 & 8 & 12 & 7 & 4 \\ 6 & 8 & 10 & 16 & 10 & 6 \\ 9 & 12 & 16 & 25 & 15 & 9 \\ 6 & 7 & 10 & 15 & 7 & 5 \\ 3 & 4 & 6 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[10] $1^{1211}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 9 & 5 & 3 \\ 3 & 4 & 8 & 12 & 7 & 4 \\ 6 & 8 & 10 & 16 & 10 & 6 \\ 9 & 12 & 16 & 25 & 15 & 9 \\ 5 & 7 & 10 & 15 & 7 & 5 \\ 3 & 4 & 6 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[7] $1^{1122}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[8] $1^{1112}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 8 & 0 & 12 & 10 & 6 \\ 6 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[9] $1^{111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 3 & 5 & 4 & 3 & 6 & 4 \\ 3 & 3 & 3 & 0 & 4 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[11] $1^{111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 3 & 5 & 4 & 3 & 6 & 4 \\ 3 & 3 & 3 & 0 & 4 & 2 \\ 4 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[12] $1^{111}3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 4 & 3 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 9 & 14 & 37 & 17 & 8 & 11 \\ 4 & 7 & 17 & 7 & 4 & 5 \\ 3 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$

[14] $1^{111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 6 & 9 & 13 & 11 & 11 & 7 \\ 6 & 7 & 11 & 7 & 9 & 5 \\ 4 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[13] $1^{111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 5 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

- [16] $21^{113}3^{111}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 12 & 4 \\ 5 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 12 & 12 & 8 & 15 & 25 & 9 \\ 4 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$
- [17] $1^{122}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 20 & 5 \\ 11 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$
- [15] $1^{111}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 8 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{111}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 8 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$
- [19] $1^{112}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 8 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$
- [20] $1^{121}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{111}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 5 & 3 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 11 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 3 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [22] $1^{111}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 4 & 5 & 4 & 6 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 \\ 3 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [23] $1^{111}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [24] $1^{121}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 23 & 8 \\ 9 & 12 & 0 & 23 & 25 & 9 \\ 3 & 4 & 0 & 8 & 9 & 3 \end{pmatrix}_{b_0}$

[25] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 25 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 14 & 2 \\ 16 & 25 & 34 & 14 & 111 & 19 \\ 3 & 4 & 6 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 8 & 0 & 12 & 10 & 6 \\ 5 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 9 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 2 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 9 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 12 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 12 & 0 & 25 & 23 & 9 \\ 8 & 11 & 0 & 23 & 20 & 8 \\ 3 & 4 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 13 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 13 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 6 & 3 \\ 3 & 4 & 4 & 12 & 7 & 4 \\ 4 & 4 & 1 & 8 & 5 & 3 \\ 9 & 12 & 8 & 25 & 15 & 9 \\ 6 & 7 & 5 & 15 & 7 & 5 \\ 3 & 4 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 10 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 3 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{131}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 8 & 12 & 25 & 6 & 15 & 9 \\ 3 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 20 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 3 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 21 & 5 \\ 13 & 17 & 0 & 21 & 55 & 13 \\ 3 & 4 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

- [35] $1^{131}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 4 & 3 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 9 & 12 & 25 & 15 & 6 & 9 \\ 4 & 7 & 15 & 7 & 4 & 5 \\ 3 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [36] $1^{111}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 4 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [37] $1^{122}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 4 & 6 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 6 & 8 & 10 & 10 & 10 & 6 \\ 4 & 7 & 10 & 7 & 9 & 5 \\ 6 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$
- [38] $1^{112}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 4 & 7 & 5 & 7 & 14 & 5 \\ 9 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$
- [39] $1^{141}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 4 & 3 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 12 & 16 & 46 & 20 & 8 & 12 \\ 4 & 7 & 20 & 7 & 4 & 5 \\ 3 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [40] $1^{111}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 3 & 4 & 1 & 5 & 5 & 3 \\ 4 & 7 & 5 & 7 & 9 & 5 \\ 6 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$
- [41] $1^{121}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 4 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 \\ 4 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [42] $1^{111}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 7 & 2 \\ 8 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [43] $1^{121}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 8 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [44] $1^{111}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & 0 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

- [45] $1^{1^{222}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 4 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$
- [46] $1^{1^{11}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [47] $1^{1^{23}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 9 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 9 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$
- [48] $21^{1^{123}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 8 & 8 & 3 \\ 4 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 8 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$
- [49] $1^{3^{11}} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 & 5 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 11 & 16 & 46 & 8 & 20 & 12 \\ 3 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [50] $1^{3^{11}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 3 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 8 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 3 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [51] $1^{1^{22}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 10 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 9 & 2 \\ 10 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [52] $1^{1^{12}} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 5 & 8 & 10 & 4 & 16 & 6 \\ 3 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [54] $1^{1^{11}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 4 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 3 & 3 & 5 & 0 & 4 & 2 \\ 4 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [53] $1^{1^{12}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 2 & 4 & 1 & 8 & 8 & 3 \\ 8 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$

[55] $1^{1^{212}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 8 & 12 & 25 & 6 & 24 & 9 \\ 3 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{1^{12}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{1^{211}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{1^{31}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 14 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{1^{21}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 5 & 8 & 10 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{1^{12}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 10 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{1^{22}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 4 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 9 & 13 & 28 & 16 & 16 & 10 \\ 6 & 7 & 16 & 7 & 9 & 5 \\ 4 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{1^{32}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 15 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 42 & 11 \\ 11 & 16 & 0 & 42 & 44 & 12 \\ 3 & 4 & 0 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{1^{213}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{1^{22}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{1^{13}}3^{1^{13}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{1^{11}}3^{1^{11}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 2 & 4 & 1 & 5 & 5 & 3 \\ 6 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{1^{12}}3^{1^{12}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{1^{21}}3^{1^{21}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 11 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 3 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{1^{12}}3^{1^{12}}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 12 & 2 \\ 14 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{1^{14}}3^{1^{14}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 13 & 2 \\ 16 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{1^{312}}3^{1^{312}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 8 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 11 & 16 & 46 & 8 & 32 & 12 \\ 3 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{1^{112}}3^{1^{112}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 6 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 \\ 3 & 4 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{1^{123}}3^{1^{123}}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 19 & 3 \\ 3 & 4 & 0 & 11 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 56 & 8 \\ 19 & 29 & 0 & 56 & 148 & 22 \\ 3 & 4 & 0 & 8 & 22 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{1^{113}}3^{1^{113}}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 \\ 10 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 0 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{1^{12}} 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 4 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{1^{123}} 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 6 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{1^{111}} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 7 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 8 & 12 & 8 & 25 & 23 & 9 \\ 7 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{1^{113}} 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 21 & 5 \\ 13 & 17 & 0 & 21 & 55 & 13 \\ 3 & 4 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{1^{311}} 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{1^{123}} 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{1^{121}} 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 9 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 \\ 3 & 4 & 0 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{1^{312}} 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 6 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 8 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 6 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{1^{22}} 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{1^{23}} 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 6 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

- [85] $1^{1^2 1^4} 1^{113} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 4 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$
- [86] $1^{1^2 1^2} 1^{1123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [87] $1^{1^3} 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [88] $1^{1^2} 1^{1113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$
- [89] $1^{1^2 2} 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 23 & 8 \\ 9 & 12 & 0 & 23 & 25 & 9 \\ 3 & 4 & 0 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$
- [90] $1^{1^2 3} 1^{123} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 17 & 3 \\ 3 & 4 & 0 & 11 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 48 & 8 \\ 17 & 25 & 0 & 48 & 111 & 19 \\ 3 & 4 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$
- [100] $1^{1^2 1^2} 1^{1123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 9 & 2 \\ 11 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [92] $1^{311} 1^{132} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 7 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 3 & 3 & 6 & 0 & 6 & 2 \\ 7 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [93] $1^{1^2 1^1} 1^{1123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 \\ 7 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [94] $1^{211} 1^{142} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 7 & 3 \\ 3 & 4 & 26 & 3 & 11 & 4 \\ 16 & 26 & 118 & 14 & 50 & 20 \\ 3 & 3 & 14 & 0 & 6 & 2 \\ 7 & 11 & 50 & 6 & 20 & 8 \\ 3 & 4 & 20 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{1^{124}} 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 31 & 8 \\ 13 & 16 & 0 & 31 & 44 & 12 \\ 3 & 4 & 0 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{1^{111}} 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 7 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 8 & 13 & 28 & 7 & 25 & 10 \\ 3 & 3 & 7 & 0 & 6 & 2 \\ 7 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{2^{14}} 1^{123} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 25 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 14 & 2 \\ 18 & 25 & 34 & 14 & 111 & 19 \\ 3 & 4 & 6 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{2^{11}} 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 7 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 3 & 3 & 4 & 0 & 6 & 2 \\ 7 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{1^{121}} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 7 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 14 & 5 \\ 7 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{4^{11}} 1^{142} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 7 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 3 & 3 & 8 & 0 & 6 & 2 \\ 7 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{1^{122}} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{2^{12}} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 8 & 3 \\ 3 & 4 & 9 & 3 & 11 & 4 \\ 7 & 9 & 13 & 5 & 17 & 7 \\ 3 & 3 & 5 & 0 & 6 & 2 \\ 8 & 11 & 17 & 6 & 20 & 8 \\ 3 & 4 & 7 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[102] $1^{1^{123}} 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 25 & 5 \\ 14 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & 0 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[104] $1^{1^{12}} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{112} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 7 & 2 \\ 9 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[106] $1^{112} 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 10 & 2 \\ 12 & 17 & 0 & 10 & 55 & 13 \\ 3 & 4 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{121} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 4 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 4 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{111} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{111} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 3 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[110] $1^{124} 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 6 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[111] $1^{122} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[112] $1^{112} 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 21 & 5 \\ 12 & 17 & 0 & 21 & 55 & 13 \\ 3 & 4 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{112} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[114] $1^{121} 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 4 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 12 & 18 & 58 & 10 & 22 & 14 \\ 3 & 3 & 10 & 0 & 4 & 2 \\ 4 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[115] $1^{1^{122}} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 15 & 5 \\ 9 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 0 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[116] $1^{1^{113}} 1^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 8 & 2 \\ 11 & 15 & 0 & 8 & 39 & 11 \\ 3 & 4 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[117] $1^{1^{111}} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[118] $1^{1^{211}} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 7 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[119] $1^{1^{212}} 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[120] $1^{1^{213}} 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 10 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 10 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[121] $1^{1^{212}} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 4 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$

[122] $1^{1^{212}} 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 6 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 4 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 6 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[125] $1^{1^{211}} 1^{121} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 12 & 18 & 58 & 10 & 22 & 14 \\ 3 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[123] $1^{1^{212}} 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 13 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 13 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[124] $1^{\overline{1^{211}}}\overline{1^{121}}3^{\overline{11}}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[126] $1^{\overline{1^{211}}}\overline{1^{111}}3^{\overline{11}}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 4 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[128] $1^{\overline{1^{12}}}\overline{1^{123}}3^{\overline{11}}1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 14 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[127] $1^{\overline{1^{12}}}\overline{1^{133}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 13 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[131] $1^{\overline{1^{411}}}\overline{1^{131}}3^{\overline{11}}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 10 & 12 & 25 & 6 & 15 & 9 \\ 3 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[130] $1^{\overline{1^{312}}}\overline{1^{123}}3^{\overline{11}}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 7 & 8 & 10 & 4 & 16 & 6 \\ 3 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[129] $1^{\overline{1^{111}}}\overline{1^{123}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 10 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[132] $1^{\overline{1^{112}}}\overline{1^{123}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 13 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 13 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[133] $1^{\overline{1^{122}}}\overline{1^{112}}3^{\overline{11}}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 14 & 5 \\ 8 & 11 & 0 & 14 & 20 & 8 \\ 3 & 4 & 0 & 5 & 8 & 3 \end{pmatrix}_{b_0}$

[136] $1^{\overline{1^{112}}}\overline{1^{124}}3^{\overline{11}}1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 3 & 3 & 4 & 0 & 16 & 2 \\ 18 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[134] $1^{1^{14}} 1^{124} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[135] $1^{1^{13}} 1^{112} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 9 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[137] $1^{1^{11}} 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 10 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 10 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[138] $1^{1^{13}} 1^{112} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[139] $1^{1^{11}} 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 7 & 9 & 13 & 5 & 11 & 7 \\ 3 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[140] $1^{1^{14}} 1^{124} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 19 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 3 & 3 & 4 & 0 & 16 & 2 \\ 19 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[141] $1^{1^{11}} 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 \\ 3 & 4 & 0 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[142] $1^{1^{14}} 1^{124} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 17 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[143] $1^{1^{11}} 1^{131} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 11 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{1^{11}} 1^{131} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 9 & 12 & 25 & 6 & 15 & 9 \\ 3 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{\overline{114}3^{\overline{11}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{\overline{114}3^{\overline{11}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{\overline{114}3^{\overline{11}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 4 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{\overline{111}3^{\overline{11}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 \\ 8 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 0 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[150] $1^{\overline{131}3^{\overline{11}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 4 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 4 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{\overline{124}3^{\overline{11}1}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 18 & 3 \\ 3 & 4 & 0 & 7 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 31 & 5 \\ 18 & 25 & 0 & 31 & 111 & 19 \\ 3 & 4 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{\overline{123}3^{\overline{11}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 9 & 2 \\ 12 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{\overline{121}3^{\overline{11}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 6 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 7 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 6 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{\overline{113}3^{\overline{11}1}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 4 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{\overline{112}3^{\overline{11}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 5 \\ 9 & 11 & 0 & 14 & 20 & 8 \\ 3 & 4 & 0 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{\overline{1}^{214}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[157] $1^{\overline{1}^{222}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{\overline{1}^{114}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 10 & 2 \\ 14 & 19 & 0 & 10 & 64 & 14 \\ 3 & 4 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{\overline{1}^{12}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 16 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{\overline{1}^{11}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 8 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{\overline{1}^{213}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{\overline{1}^{23}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[162] $21^{\overline{1}^{13}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 15 & 4 \\ 5 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 15 & 16 & 11 & 20 & 44 & 12 \\ 4 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[163] $1^{\overline{1}^{412}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 10 & 12 & 25 & 6 & 24 & 9 \\ 3 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{\overline{1}^{113}} 1^{\overline{1}^{12}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 11 & 2 \\ 14 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 14 & 2 \\ 17 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[167] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 118$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[166] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 118$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[168] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 2 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[169] $1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 122$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 5 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 12 & 16 & 46 & 8 & 20 & 12 \\ 3 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[170] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 125$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 \\ 9 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 0 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[172] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 19 & 5 \\ 11 & 15 & 0 & 19 & 39 & 11 \\ 3 & 4 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$

[171] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 9 & 2 \\ 12 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[173] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 4 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$

[176] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 8 & 2 \\ 10 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[175] $1^{1^{212}3^{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 10 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 8 & 2 \\ 10 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{1^{122}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{1^{122}3^{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[178] $1^{1^{221}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 5 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 7 & 9 & 13 & 11 & 11 & 7 \\ 6 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{1^{411}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 11 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 3 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{1^{214}3^{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{1^{111}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 8 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{1^{13}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 16 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{1^{121}3^{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 3 & 4 & 1 & 5 & 5 & 3 \\ 6 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{1^{112}3^{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 9 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{1^{124}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 17 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 30 & 5 \\ 17 & 24 & 0 & 30 & 100 & 18 \\ 3 & 4 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{1^{212}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 3 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{1^{14}}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{1^{212}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{1^{13}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 4 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{1^{41}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 4 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 3 & 3 & 11 & 0 & 4 & 2 \\ 4 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{1^{123}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 10 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 10 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{1^{112}}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{1^{31}}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 4 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 4 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{1^{13}}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 12 & 2 \\ 15 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{1^{211}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 7 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[196] $1^{1^{211}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 & 7 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 11 & 17 & 49 & 9 & 33 & 13 \\ 3 & 3 & 9 & 0 & 6 & 2 \\ 7 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[199] $1^{1^{211}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 6 & 8 & 10 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[198] $1^{1^{13}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 13 & 2 \\ 16 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[197] $1^{1^{32}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 11 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 22 & 8 \\ 8 & 11 & 0 & 22 & 20 & 8 \\ 3 & 4 & 0 & 8 & 8 & 3 \end{pmatrix}_{b_0}$

[200] $1^{1^{221}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 6 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 9 & 13 & 28 & 16 & 16 & 10 \\ 6 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[201] $21^{1^{112}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 5 & 14 & 3 \\ 4 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 14 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$

[202] $1^{1^{211}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[203] $1^{1^{111}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 5 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[204] $1^{1^{111}}3^{1^11^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 6 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[205] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}1}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=149$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.0$
 $T_{\min}=9.88\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}1}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=150$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=7.0$
 $T_{\min}=9.99\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 14 & 21 & 76 & 11 & 41 & 16 \\ 3 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}2}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=150$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.0$
 $T_{\min}=9.88\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}1}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=150$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.5$
 $T_{\min}=9.99\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 9 & 13 & 28 & 7 & 25 & 10 \\ 3 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}1}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=153$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.8$
 $T_{\min}=9.96\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 3 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}2}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=153$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.0$
 $T_{\min}=9.91\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 16 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 26 & 5 \\ 16 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}1}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=155$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.0$
 $T_{\min}=9.97\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 4 & 0 & 11 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 39 & 8 \\ 13 & 20 & 0 & 39 & 69 & 15 \\ 3 & 4 & 0 & 8 & 15 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}1}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=155$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.0$
 $T_{\min}=9.97\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 5 \\ 8 & 11 & 0 & 14 & 20 & 8 \\ 3 & 4 & 0 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{\overline{1}123}3^{\overline{1}1}1^{\overline{1}1}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=156$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.0$
 $T_{\min}=9.92\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 20 & 5 \\ 12 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{\overline{1}122}3^{\overline{1}1}1^{\overline{1}2}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T=8$, $T_{\max}=10$, $\Delta=-59$, $\det=161$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=6.0$
 $T_{\min}=9.96\rightarrow 10$, $T_{\max}=10$, $T\in[10,10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 14 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{\overline{114}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 \\ 13 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{\overline{124}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 165$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 7 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 24 & 5 \\ 14 & 19 & 0 & 24 & 64 & 14 \\ 3 & 4 & 0 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{\overline{14}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 167$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{\overline{14}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 168$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 13 & 2 \\ 17 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 14 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 14 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{\overline{1211}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 8 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 3 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{\overline{311}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 173$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 14 & 21 & 76 & 11 & 26 & 16 \\ 3 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{\overline{121}} 3^{\overline{11}} 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 173$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 13 & 18 & 58 & 10 & 22 & 14 \\ 3 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{1^{312}2^3}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 10 & 13 & 28 & 7 & 25 & 10 \\ 3 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[227] $1^{1^{412}2^3}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 8 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 3 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[225] $1^{1^{114}2^3}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 19 & 5 \\ 12 & 15 & 0 & 19 & 39 & 11 \\ 3 & 4 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$

[228] $1^{1^{122}2^3}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 31 & 8 \\ 11 & 16 & 0 & 31 & 44 & 12 \\ 3 & 4 & 0 & 8 & 12 & 3 \end{pmatrix}_{b_0}$

[230] $1^{1^{214}2^3}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 178$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 20 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 6 & 8 & 10 & 4 & 40 & 6 \\ 3 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[229] $1^{1^{411}2^3}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 178$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 5 & 3 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 14 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 3 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$

[231] $1^{1^{112}2^3}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 179$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$

[232] $1^{1^{113}2^3}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 19 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 2 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 19 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[234] $1^{1^{311}2^3}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 181$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 17 & 26 & 118 & 14 & 32 & 20 \\ 3 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[233] $1^{1^{311}2^3}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 181$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 10 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[237] $1^{1^{12}} 1^{1^{12}} 3^{1^{12}} 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 183$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 14 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 14 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$

[236] $1^{1^{12}} 1^{1^{12}} 3^{1^{12}} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 183$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[235] $1^{1^{14}} 1^{1^{14}} 3^{1^{14}} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 183$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 11 & 2 \\ 15 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[240] $21^{1^{11}} 1^{1^{11}} 3^{1^{11}} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 8 & 5 & 3 \\ 4 & 4 & 4 & 12 & 7 & 4 \\ 3 & 4 & 1 & 8 & 5 & 3 \\ 8 & 12 & 8 & 25 & 15 & 9 \\ 5 & 7 & 5 & 15 & 7 & 5 \\ 3 & 4 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[239] $1^{1^{11}} 1^{1^{11}} 3^{1^{11}} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 6 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[238] $1^{1^{12}} 1^{1^{12}} 3^{1^{12}} 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 16 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[241] $1^{1^{13}} 1^{1^{13}} 3^{1^{13}} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 185$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[244] $1^{1^{12}} 1^{1^{12}} 3^{1^{12}} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 4 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$

[243] $1^{1^{14}} 1^{1^{14}} 3^{1^{14}} 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 16 & 2 \\ 20 & 29 & 0 & 16 & 148 & 22 \\ 3 & 4 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[242] $1^{1^{12}} 1^{1^{12}} 3^{1^{12}} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$

[245] $1^{\overline{1^{412}}} 1^{\overline{1^{1211}}} 3^{\overline{1^{11}}} 1^{\overline{1^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 187$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 6 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 11 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 6 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{\overline{1^{112}}} 3^{\overline{1^{11}}} 1^{\overline{1^{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 191$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 26 & 5 \\ 14 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{\overline{1^{124}}} 3^{\overline{1^{11}}} 1^{\overline{1^{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 193$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 6 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{\overline{1^{124}}} 3^{\overline{1^{12}}} 3^{\overline{1^{11}}} 1^{\overline{1^{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 193$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 18 & 3 \\ 3 & 4 & 0 & 11 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 48 & 8 \\ 18 & 25 & 0 & 48 & 111 & 19 \\ 3 & 4 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{\overline{1^{214}}} 3^{\overline{1^{11}}} 3^{\overline{1^{11}}} 1^{\overline{1^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 193$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 13 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 4 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 13 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{\overline{1^{112}}} 3^{\overline{1^{11}}} 3^{\overline{1^{11}}} 1^{\overline{1^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 194$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 13 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{\overline{1^{113}}} 3^{\overline{1^{11}}} 3^{\overline{1^{11}}} 1^{\overline{1^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 197$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{\overline{1^{114}}} 3^{\overline{1^{11}}} 3^{\overline{1^{11}}} 1^{\overline{1^{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 198$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 12 & 2 \\ 16 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{\overline{1^{213}}} 3^{\overline{1^{12}}} 3^{\overline{1^{11}}} 1^{\overline{1^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{\overline{1^{113}}} 3^{\overline{1^{14}}} 3^{\overline{1^{11}}} 1^{\overline{1^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 13 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{1^{213}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 13 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 10 & 2 \\ 13 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[256] $1^{1^{122}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 201$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{1^{213}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 13 & 2 \\ 16 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{1^{213}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{1^{223}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 31 & 8 \\ 12 & 16 & 0 & 31 & 44 & 12 \\ 3 & 4 & 0 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{1^{41}}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 4 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 3 & 3 & 14 & 0 & 4 & 2 \\ 4 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{1^{124}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 210$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 20 & 5 \\ 13 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{1^{124}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 213$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 11 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 30 & 8 \\ 12 & 15 & 0 & 30 & 39 & 11 \\ 3 & 4 & 0 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{1^{311}}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 216$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 7 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 14 & 21 & 76 & 11 & 41 & 16 \\ 3 & 3 & 11 & 0 & 6 & 2 \\ 7 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{1^{14}}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 217$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 20 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 2 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 218$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 10 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 10 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 218$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 10 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 10 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 219$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 14 & 2 \\ 18 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[268] $1^{122}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 220$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 11 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 22 & 8 \\ 8 & 11 & 0 & 22 & 20 & 8 \\ 3 & 4 & 0 & 8 & 8 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{112}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 221$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 9 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{113}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 19 & 5 \\ 11 & 15 & 0 & 19 & 39 & 11 \\ 3 & 4 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{311}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 225$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{311}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 225$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 226$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 \\ 11 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 230$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{114} 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 231, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 26 & 5 \\ 16 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[276] $21^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 5 & 3 \\ 4 & 4 & 9 & 7 & 7 & 4 \\ 6 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{1321} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 10 & 13 & 28 & 16 & 16 & 10 \\ 6 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{113} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 \\ 12 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{124} 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 25 & 5 \\ 15 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & 0 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{214} 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 15 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 15 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{114} 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[282] $1^{312} 1^{132} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 12 & 17 & 49 & 9 & 33 & 13 \\ 3 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[283] $1^{113} 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 15 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{112} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 249, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[285] $1^{\overline{113}} 3^{\overline{11}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{\overline{113}} 3^{\overline{11}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 26 & 5 \\ 15 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{\overline{112}} 3^{\overline{11}} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{\overline{121}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 10 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[289] $1^{\overline{131}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 13 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{\overline{112}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 273$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 25 & 5 \\ 13 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & 0 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[291] $1^{\overline{114}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 275$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 13 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 13 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[292] $1^{\overline{132}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 279$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 13 & 17 & 49 & 9 & 33 & 13 \\ 3 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{\overline{123}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 280$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{\overline{114}} 3^{\overline{11}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 281$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{\overline{114}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 281$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[296] $1^{\overline{1211}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 288$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 5 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 9 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[297] $21^{\overline{11113}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 288$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 5 & 5 & 3 \\ 4 & 4 & 4 & 7 & 7 & 4 \\ 3 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[298] $1^{\overline{124}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[299] $1^{\overline{113}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 299$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[300] $1^{\overline{123}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 299$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 11 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 30 & 8 \\ 11 & 15 & 0 & 30 & 39 & 11 \\ 3 & 4 & 0 & 8 & 11 & 3 \end{pmatrix}_{b_0}$

[301] $1^{\overline{114}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 7 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 24 & 5 \\ 14 & 19 & 0 & 24 & 64 & 14 \\ 3 & 4 & 0 & 5 & 14 & 3 \end{pmatrix}_{b_0}$

[302] $1^{\overline{1411}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 308$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 3 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[303] $21^{\overline{12113}} 3^{\overline{11}} 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 312$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 5 & 5 & 3 \\ 4 & 4 & 8 & 7 & 7 & 4 \\ 6 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[304] $1^{\overline{113}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 314$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 16 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[305] $1^{\overline{1124}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 317$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 13 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 13 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[306] \ 1^{\overline{1}^{411}} 3^{\overline{1}^{131}} 3^{\overline{1}^{11}} 1^{\overline{1}^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 16 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[307] \ 1^{\overline{1}^{113}} 3^{\overline{1}^{11}} 1^{\overline{1}^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[308] \ 1^{\overline{1}^{124}} 3^{\overline{1}^{11}} 1^{\overline{1}^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 20 & 3 \\ 3 & 4 & 0 & 11 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 56 & 8 \\ 20 & 29 & 0 & 56 & 148 & 22 \\ 3 & 4 & 0 & 8 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[309] \ 1^{\overline{1}^{112}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[310] \ 1^{\overline{1}^{214}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 17 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[311] \ 1^{\overline{1}^{123}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 11 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 39 & 8 \\ 14 & 20 & 0 & 39 & 69 & 15 \\ 3 & 4 & 0 & 8 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[312] \ 1^{\overline{1}^{123}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 342, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 16 & 3 \\ 3 & 4 & 0 & 11 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 47 & 8 \\ 16 & 24 & 0 & 47 & 100 & 18 \\ 3 & 4 & 0 & 8 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[313] \ 1^{\overline{1}^{412}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 342, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 15 & 21 & 76 & 11 & 41 & 16 \\ 3 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[314] \ 1^{\overline{1}^{112}} 3^{\overline{1}^{11}} 1^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[315] \ 1^{\overline{1}^{113}} 3^{\overline{1}^{11}} 1^{\overline{1}^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$\begin{aligned}
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[326] & 21^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 5 & 8 & 3 \\ 4 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0} \\
[327] & 1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 442, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 7 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 31 & 5 \\ 18 & 25 & 0 & 31 & 111 & 19 \\ 3 & 4 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0} \\
[328] & 21^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 5 & 5 & 3 \\ 4 & 4 & 13 & 7 & 7 & 4 \\ 9 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0} \\
[329] & 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 19 & 3 \\ 3 & 4 & 0 & 7 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 36 & 5 \\ 19 & 29 & 0 & 36 & 148 & 22 \\ 3 & 4 & 0 & 5 & 22 & 3 \end{pmatrix}_{b_0} \\
[330] & 21^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 5 & 11 & 3 \\ 4 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0} \\
[331] & 1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 505, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[332] & 1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 30 & 5 \\ 17 & 24 & 0 & 30 & 100 & 18 \\ 3 & 4 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0} \\
[333] & 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 512, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 17 & 3 \\ 3 & 4 & 0 & 11 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 47 & 8 \\ 17 & 24 & 0 & 47 & 100 & 18 \\ 3 & 4 & 0 & 8 & 18 & 3 \end{pmatrix}_{b_0} \\
[334] & 1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 599, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 7 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 36 & 5 \\ 20 & 29 & 0 & 36 & 148 & 22 \\ 3 & 4 & 0 & 5 & 22 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$T = 12 \quad [337] \quad 1^{\overline{1114}3} 1^{\overline{113}} 5 13^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 3 & 4 & 17 & 3 \\ 1 & 2 & 0 & 1 & 3 & 14 & 2 \\ 1 & 3 & 1 & 1 & 4 & 17 & 3 \\ 3 & 4 & 3 & 4 & 5 & 28 & 5 \\ 11 & 17 & 14 & 17 & 28 & 121 & 23 \\ 2 & 3 & 2 & 3 & 5 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[338] \quad 1^{\overline{1111}3} 1^{\overline{111}} 5 13^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 3 & 5 & 2 \\ 2 & 4 & 4 & 6 & 7 & 11 & 6 \\ 3 & 4 & 3 & 7 & 5 & 10 & 5 \\ 4 & 6 & 5 & 11 & 10 & 14 & 8 \\ 2 & 3 & 2 & 6 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

4.950 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + su2n3mix + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{1211}3} 1^{\overline{111}} 5 13^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 5 & 1 & 3 \\ 3 & 4 & 8 & 12 & 7 & 0 & 4 \\ 5 & 8 & 10 & 16 & 10 & 0 & 6 \\ 8 & 12 & 16 & 25 & 15 & 0 & 9 \\ 5 & 7 & 10 & 15 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 9 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}3} 1^{\overline{111}} 5 13^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 3 & 4 \\ 3 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 2 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}3} 1^{\overline{111}} 5 13^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 3 & 4 \\ 3 & 3 & 3 & 0 & 4 & 2 & 2 \\ 4 & 7 & 6 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1112}3} 1^{\overline{111}} 5 13^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1121}3} 1^{\overline{111}} 5 13^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 1 & 3 \\ 3 & 4 & 4 & 7 & 12 & 0 & 4 \\ 2 & 4 & 1 & 5 & 8 & 0 & 3 \\ 5 & 7 & 5 & 7 & 15 & 0 & 5 \\ 9 & 12 & 8 & 15 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1111}3} 1^{\overline{111}} 5 13^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 1 & 3 \\ 3 & 4 & 0 & 8 & 7 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 8 & 0 & 12 & 10 & 0 & 6 \\ 5 & 7 & 0 & 10 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1211} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -17, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 5 & 8 & 10 & 4 & 10 & 12 & 6 \\ 3 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -17, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 0 & 4 \\ 4 & 5 & 4 & 6 & 3 & 0 & 4 \\ 5 & 7 & 6 & 7 & 4 & 0 & 5 \\ 3 & 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -17, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 3 & 3 & 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 4 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 3 & 3 & 3 & 4 & 0 & 2 & 2 \\ 4 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 & 1 & 3 \\ 3 & 4 & 14 & 7 & 3 & 0 & 4 \\ 10 & 14 & 37 & 17 & 8 & 0 & 11 \\ 5 & 7 & 17 & 7 & 4 & 0 & 5 \\ 3 & 3 & 8 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{211} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 4 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 6 & 8 & 10 & 4 & 10 & 12 & 6 \\ 3 & 3 & 4 & 0 & 4 & 5 & 2 \\ 4 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 12 & 0 & 25 & 23 & 8 & 9 \\ 8 & 11 & 0 & 23 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 8 & 1 & 3 \\ 3 & 4 & 0 & 9 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 5 & 1 & 3 \\ 3 & 4 & 4 & 7 & 7 & 0 & 4 \\ 4 & 4 & 1 & 5 & 5 & 0 & 3 \\ 5 & 7 & 5 & 7 & 9 & 0 & 5 \\ 5 & 7 & 5 & 9 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -41, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 7 & 2 & 2 \\ 8 & 12 & 0 & 7 & 25 & 8 & 9 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1211} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -41, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 5 & 3 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 & 4 \\ 8 & 13 & 28 & 7 & 16 & 9 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 & 5 \\ 3 & 4 & 9 & 2 & 5 & 1 & 3 \\ 3 & 4 & 10 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -41, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 5 & 2 \\ 5 & 7 & 0 & 4 & 7 & 11 & 5 \\ 6 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -51, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 & 2 \\ 8 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1211} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -52, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 2 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 & 5 \\ 3 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -70, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 4 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -74, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 1 & 3 \\ 3 & 4 & 4 & 7 & 21 & 0 & 4 \\ 2 & 4 & 1 & 5 & 14 & 0 & 3 \\ 5 & 7 & 5 & 7 & 26 & 0 & 5 \\ 15 & 21 & 14 & 26 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -74, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 10 & 5 \\ 6 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & 0 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 1 & 3 \\ 3 & 4 & 0 & 7 & 12 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 0 & 5 \\ 9 & 12 & 0 & 15 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 5 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -85, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 12 & 1 & 3 \\ 3 & 4 & 4 & 7 & 16 & 0 & 4 \\ 2 & 4 & 1 & 5 & 11 & 0 & 3 \\ 5 & 7 & 5 & 7 & 20 & 0 & 5 \\ 12 & 16 & 11 & 20 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -87, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 5 & 2 \\ 5 & 7 & 0 & 4 & 7 & 11 & 5 \\ 7 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -92, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{1111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 9 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{1112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 3 & 3 & 2 & 0 & 4 & 10 & 2 \\ 4 & 7 & 5 & 4 & 7 & 22 & 5 \\ 12 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[41] $1^{1121}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 & 2 \\ 14 & 21 & 14 & 12 & 80 & 14 & 16 \\ 3 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{1111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 2 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 & 5 \\ 8 & 12 & 0 & 15 & 25 & 8 & 9 \\ 2 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{1111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 & 2 \\ 8 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{1112}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 7 & 11 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 0 & 5 \\ 8 & 11 & 0 & 14 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 5 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{1111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 5 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 5 & 1 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 16 & 5 \\ 5 & 7 & 0 & 9 & 7 & 16 & 5 \\ 8 & 13 & 0 & 16 & 16 & 28 & 10 \\ 3 & 4 & 0 & 5 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -110, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 6 & 2 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 & 8 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -119, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 9 & 2 & 2 \\ 11 & 16 & 0 & 9 & 44 & 11 & 12 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -125, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -129, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 9 & 5 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 & 5 \\ 3 & 4 & 0 & 5 & 5 & 1 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 4 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -137, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 & 5 \\ 8 & 12 & 0 & 15 & 25 & 8 & 9 \\ 3 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 2 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 5 & 5 \\ 8 & 11 & 0 & 14 & 20 & 8 & 8 \\ 2 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 10 & 5 \\ 5 & 7 & 0 & 9 & 7 & 10 & 5 \\ 5 & 8 & 0 & 10 & 10 & 10 & 6 \\ 3 & 4 & 0 & 5 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 7 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 & 2 \\ 7 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{121} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 14 & 1 & 3 \\ 3 & 4 & 4 & 7 & 21 & 0 & 4 \\ 2 & 4 & 1 & 5 & 14 & 0 & 3 \\ 5 & 7 & 5 & 7 & 26 & 0 & 5 \\ 14 & 21 & 14 & 26 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -160, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 7 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 11 & 5 \\ 5 & 7 & 0 & 9 & 7 & 11 & 5 \\ 6 & 9 & 0 & 11 & 11 & 13 & 7 \\ 3 & 4 & 0 & 5 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -167, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[62] \quad & 1^{1112} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0} \\
[63] \quad & 1^{1121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 & 5 \\ 9 & 12 & 0 & 15 & 25 & 8 & 9 \\ 3 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 3 \end{pmatrix}_{b_0} \\
[64] \quad & 1^{1111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 5 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 & 5 \\ 3 & 4 & 0 & 5 & 5 & 1 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[65] \quad & 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -211, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 & 5 \\ 11 & 16 & 0 & 20 & 44 & 11 & 12 \\ 2 & 4 & 0 & 5 & 11 & 1 & 3 \\ 3 & 4 & 0 & 5 & 12 & 3 & 3 \end{pmatrix}_{b_0} \\
[66] \quad & 1^{1112} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 7 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 10 & 5 \\ 5 & 7 & 0 & 9 & 7 & 10 & 5 \\ 6 & 8 & 0 & 10 & 10 & 10 & 6 \\ 3 & 4 & 0 & 5 & 5 & 6 & 3 \end{pmatrix}_{b_0} \\
[67] \quad & 1^{1121} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -264, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0} \\
[68] \quad & 1^{1112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -271, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0} \\
[69] \quad & 1^{1121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 11 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{1121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -314$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 5 & 5 \\ 8 & 11 & 0 & 14 & 20 & 8 & 8 \\ 3 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 16 & 5 \\ 5 & 7 & 0 & 9 & 7 & 16 & 5 \\ 9 & 13 & 0 & 16 & 16 & 28 & 10 \\ 3 & 4 & 0 & 5 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{1121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -375$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 & 5 \\ 11 & 16 & 0 & 20 & 44 & 11 & 12 \\ 3 & 4 & 0 & 5 & 11 & 1 & 3 \\ 3 & 4 & 0 & 5 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [73] $1^{1111} 3^1 1^{11} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 12, 1)$, $c_{\text{ext}} = 7.0$

4.951 NHC(-2)(-3) + su₂ + su₃

$T = 7$ [0] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 8 & 5 & 4 \\ 8 & 8 & 10 & 6 & 6 \\ 5 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -9$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 4 & 6 & 4 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{141} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -9$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 16 & 9 & 4 \\ 9 & 16 & 46 & 24 & 12 \\ 5 & 9 & 24 & 12 & 6 \\ 3 & 4 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -9$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 49 & 27 & 13 \\ 7 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -10$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.60 \rightarrow 10$, $T_{\max} = 12$, $T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 3 \end{pmatrix}_{b_0}$$

- [5] $1^{11} 1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -10$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 1 \end{pmatrix}_{b_0}$
- [6] $1^{11} 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{12} 1^{31} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 8 & 5 & 4 \\ 7 & 8 & 10 & 6 & 6 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{11} 1^{22} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 4 & 3 & 4 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [9] $1^{11} 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{112} 1^{43} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 9 & 9 & 13 & 12 & 7 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{13} 1^{41} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 13 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{12} 1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{11} 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{12} 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.60 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 5 & 4 \\ 5 & 8 & 10 & 6 & 6 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{122} 1^{32} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.7$
 $T_{\min} = 10.60 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$

- [14] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 8 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$
- [28] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$
- [17] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 9 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$
- [19] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$
- [20] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 5 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$
- [21] $21^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 5 & 13 & 4 \\ 6 & 5 & 4 & 12 & 4 \\ 12 & 13 & 12 & 27 & 9 \\ 4 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [22] $21^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 5 & 4 & 4 \\ 6 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$
- [23] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$
- [24] $1^1 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$
- [25] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 14 & 19 & 73 & 12 & 15 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$
- [26] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{12} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 7 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{122} 3^1 1^{142} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 8 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{121} 3^1 1^{143} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 8 & 8 & 10 & 12 & 6 \\ 8 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{122} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 4 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{141} 3^1 1^{142} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 16 & 9 & 4 \\ 12 & 16 & 46 & 24 & 12 \\ 7 & 9 & 24 & 12 & 6 \\ 3 & 4 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 21^1 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{111} 3^1 1^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 9 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[40] $21^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 27 & 3 & 4 \\ 5 & 4 & 24 & 4 & 4 \\ 27 & 24 & 121 & 15 & 19 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 17 & 24 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 25 & 9 & 9 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{11} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 12 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 17 & -1 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{141} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 16 & 9 & 4 \\ 11 & 16 & 46 & 24 & 12 \\ 6 & 9 & 24 & 12 & 6 \\ 3 & 4 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{11} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 4 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{11} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{121} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{13} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 11 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{11} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 10 & 0 & 21 & 7 \\ 3 & 4 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 5 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{11} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 10 & 0 & 21 & 7 \\ 3 & 4 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[61] $21^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 8 & 5 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 13 & 9 & 7 \\ 5 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 6 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 9 & 12 & 4 & 5 \\ 6 & 12 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 5 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{11} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 10 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[73] $21^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 14 & 4 & 4 \\ 9 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[71] $1^1 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[78] $1^1 31^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 6 & 12 & 25 & 27 & 9 \\ 7 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{112} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{112} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[76] $1^1 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{112} 31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{112} 31^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[80] $21^{141} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 4 & 16 & 9 & 4 \\ 12 & 16 & 46 & 24 & 12 \\ 6 & 9 & 24 & 12 & 6 \\ 3 & 4 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[81] $1^1 1^1 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$

[84] $1^{14} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 15 & 21 & 18 & 75 & 15 \\ 3 & 4 & 4 & 15 & 1 \end{pmatrix}_{b_0}$

[85] $1^{21} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$

[86] $1^{23} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[87] $1^{23} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 8 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$

[88] $1^{11} 3^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 4 & 8 & 10 & 12 & 6 \\ 5 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$

[89] $1^{14} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 9 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 1 \end{pmatrix}_{b_0}$

[90] $1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 5 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[91] $1^{131} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 6 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 1 \end{pmatrix}_{b_0}$

[92] $1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 7 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$

[93] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 13 & 12 & 7 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{132} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 10 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{11} 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 3 \\ 4 & 9 & 12 & 4 & 5 \\ 5 & 12 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 7 & 12 & 25 & 18 & 9 \\ 5 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 8 & 0 & 9 & 5 \\ 3 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 9 & 5 \\ 3 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 11 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{12} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 8 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{111} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 15 & 24 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 6 & 6 & 4 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[106] $1^1 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 9 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 14 & 21 & 18 & 75 & 15 \\ 3 & 4 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{121} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 13 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 4 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 8 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 12 & 21 & 18 & 75 & 15 \\ 3 & 4 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 9 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{1^{11}} 1^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 9 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{1^{13}} 1^{43} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 8 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{1^{12}} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 4 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{1^{12}} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 8 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{1^3} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{1^{11}} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{1^{12}} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{1^{11}} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 17 & 24 & 121 & 15 & 19 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{1^{11}} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{1^{11}} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{1^{11}} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

- [126] $21^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [127] $1^1 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 13 & 12 & 7 \\ 4 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 1 \end{pmatrix}_{b_0}$
- [128] $1^{13} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 24 & 8 \\ 3 & 4 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [129] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 10 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [131] $1^1 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 4 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [133] $1^{12} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 9 & 5 \\ 3 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$
- [134] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 5 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$
- [132] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 28 & 21 & 10 \\ 5 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 1 \end{pmatrix}_{b_0}$
- [140] $1^{11} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [146] $1^{11} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [135] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 12 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 5 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{112} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 13 & 12 & 7 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 37 & 9 & 11 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{112} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[143] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{111} 3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 16 & 24 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[141] $21^{12} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 15 & 14 & 9 & 36 & 10 \\ 4 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{121} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 8 & 8 & 10 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[157] $1^1 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 4 & 8 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{121} 3^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[155] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 4 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[154] $1^{13} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 11 & 19 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 10 & 17 & 49 & 15 & 13 \\ 4 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{121} 3^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 10 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 9 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 14 & 19 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{12} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 13 & 21 & 18 & 75 & 15 \\ 3 & 4 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{12} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[160] $1 \overset{1^{11}}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 5 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[161] $1 \overset{1^2}{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[162] $1 \overset{1^1}{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 5 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[163] $1 \overset{1^{11}}{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[164] $1 \overset{1^{21}}{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 6 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[165] $1 \overset{1^{32}}{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[171] $1 \overset{1^{13}}{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[166] $1 \overset{1^{21}}{1} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[170] $1 \overset{1^{12}}{1} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 10 & 18 & 58 & 24 & 14 \\ 4 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[169] $1 \overset{1^3}{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 12 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[168] $1 \overset{1^{31}}{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 7 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 9 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 7 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{122} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{123} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 5 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{113} 3 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 5 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{13} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{13} 3^{142} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 49 & 15 & 13 \\ 5 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{111} 3^{12} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 5 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[183] $21^{111} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 13 & 5 & 4 \\ 12 & 13 & 28 & 12 & 10 \\ 6 & 5 & 12 & 3 & 3 \\ 4 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{11} 3^{122} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 13 & 9 & 7 \\ 5 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{113} 3^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 17 & 40 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{132} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 6 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{12} 3^{111} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{121} 3^{121} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{111} 3^{131} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{12} 3^{131} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 7 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 18 & 24 & 121 & 15 & 19 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 13 & 12 & 7 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{132} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 13 & 9 & 7 \\ 5 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{21} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 9 & 24 & 4 & 5 \\ 10 & 24 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 10 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 8 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{11}3^{12}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{12}3^{12}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{12}3^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 11 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[208] $21^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 9 & 9 & 6 & 12 & 6 \\ 4 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{121}3^{133}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 8 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{13}3^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{13}3^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 12 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{112}3^{122}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{121}3^{111}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{141}3^{143}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 8 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{122} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 11 & 18 & 58 & 24 & 14 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 4 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 28 & 18 & 10 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 7 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 8 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{13} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 10 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{114} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 14 & 0 & 36 & 10 \\ 3 & 4 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{11} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 14 & 0 & 36 & 10 \\ 3 & 4 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{132} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 6 & 3 \\ 3 & 4 & 27 & 8 & 4 \\ 19 & 27 & 133 & 36 & 21 \\ 6 & 8 & 36 & 9 & 5 \\ 3 & 4 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 8 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 5 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 24 & 0 & 5 \\ 10 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{112} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 13 & 12 & 7 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 8 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 9 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 11 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{112} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 2 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{141} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 13 & 21 & 76 & 33 & 16 \\ 7 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{11}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 28 & 12 & 10 \\ 5 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{131}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 9 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 13 & 21 & 76 & 18 & 16 \\ 4 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 15 & 19 & 73 & 12 & 15 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 8 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{11}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 4 & 8 & 10 & 18 & 6 \\ 7 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 24 & 8 \\ 3 & 4 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 12 & 0 & 24 & 8 \\ 3 & 4 & 0 & 8 & 2 \end{pmatrix}_{b_0} \\
[245] & 21^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 21 & 4 & 4 \\ 18 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 4 & 16 & 2 & 1 \end{pmatrix}_{b_0} \\
[246] & 1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 5 & 9 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[247] & 1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0} \\
[248] & 21^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 6 & 6 & 4 \\ 5 & 4 & 5 & 5 & 4 \\ 6 & 5 & 4 & 6 & 4 \\ 6 & 5 & 6 & 3 & 3 \\ 4 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0} \\
[252] & 1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0} \\
[253] & 1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[254] & 1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0} \\
[255] & 1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0} \\
[256] & 1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0} \\
[257] & 1^{121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[258] & 1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 9 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{111} 3^{1^{13}} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 11 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{12} 3^{1^{11}} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 8 & 14 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{12} 3^{1^{11}} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 27 & 9 \\ 3 & 4 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{111} 3^{1^{12}} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{112} 3^{1^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 8 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{113} 3^{1^4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 15 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3^{1^2} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -69, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 9 & 20 & 4 & 5 \\ 10 & 20 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 5 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{122} 3^{1^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 4 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{132} 3^{1^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 7 & 12 & 25 & 27 & 9 \\ 7 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{113} 3^{1^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 9 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{112} 3^{1^{14}} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 11 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 7 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[274] \ 21^{131} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.17 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 12 & 9 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 6 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 24 & 8 \\ 3 & 4 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[277] \ 21^{113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.23 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 15 & 17 & 12 & 48 & 12 \\ 4 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.23 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[281] \ 1^{132} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 12 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[276] $21^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 9 & 3 & 4 \\ 8 & 9 & 12 & 4 & 5 \\ 9 & 12 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 4 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 13 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[284] $1^{122} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 12 & 18 & 58 & 24 & 14 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 11 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[287] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 13 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[299] $1^{121} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[301] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 13 & 19 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{121} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 10 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{132} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 7 & 12 & 25 & 27 & 9 \\ 8 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 14 & 0 & 36 & 10 \\ 3 & 4 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 14 & 0 & 36 & 10 \\ 3 & 4 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{121} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 4 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 20 & 28 & 154 & 18 & 22 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{141} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 4 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 16 & 26 & 118 & 18 & 20 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{121} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[290] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{121} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 12 & 19 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[302] $1^1 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 3 \\ 4 & 9 & 20 & 4 & 5 \\ 8 & 20 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 5 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[307] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{131} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 8 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{121} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 8 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{122} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 14 & 18 & 58 & 24 & 14 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{122} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 13 & 18 & 58 & 24 & 14 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 9 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[316] $21^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 12 & 13 & 9 & 27 & 9 \\ 4 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 13 & 0 & 27 & 9 \\ 3 & 4 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 7 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[308] $1^{132} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 10 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{11}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{131}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 18 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{132}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 27 & 8 & 4 \\ 17 & 27 & 133 & 36 & 21 \\ 5 & 8 & 36 & 9 & 5 \\ 3 & 4 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[312] $1^{111}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 12 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{12}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 10 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{113}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 16 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{141}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 10 & 16 & 46 & 12 & 12 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{111}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{121}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 58 & 12 & 14 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[324] $1^{111}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 11 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[325] $1^{111} 1^3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 7 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[326] $1^{14} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 16 & 0 & 45 & 11 \\ 3 & 4 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[330] $1^{121} 1^4 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[328] $1^{111} 1^{32} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 37 & 9 & 11 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[327] $1^{12} 1^{11} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 8 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[329] $1^{111} 1^2 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[336] $21^{111} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 5 & 9 & 4 \\ 6 & 5 & 4 & 9 & 4 \\ 9 & 9 & 9 & 12 & 6 \\ 4 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[335] $1^{121} 1^{11} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[334] $1^{121} 1^1 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 10 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[333] $1^{121} 1^{31} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[332] $1^{121} 1^1 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 7 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[331] $1^{121} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 17 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[337] $1^{132} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[342] $1^{112} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[343] $1^{132} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[338] $21^{111} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 17 & 4 \\ 9 & 9 & 13 & 27 & 7 \\ 15 & 17 & 27 & 48 & 12 \\ 4 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[339] $1^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 11 & 13 & 28 & 12 & 10 \\ 5 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[340] $1 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 7 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[341] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 48 & 12 \\ 3 & 4 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[344] $1^{122} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 7 & 13 & 28 & 18 & 10 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[345] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 13 & 0 & 27 & 9 \\ 3 & 4 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{113}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[348] $1^{113}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 7 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[349] $1^{113}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 7 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[350] $1^{13}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[351] $1^{111}3^11^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 9 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{132}3^11^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 5 & 3 \\ 3 & 4 & 27 & 8 & 4 \\ 18 & 27 & 133 & 36 & 21 \\ 5 & 8 & 36 & 9 & 5 \\ 3 & 4 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[352] $1^{122}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 28 & 18 & 10 \\ 4 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[353] $1^{112}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 9 & 20 & 4 & 5 \\ 9 & 20 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 5 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[355] $1^{14}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[360] $1^{12}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 13 & 0 & 27 & 9 \\ 3 & 4 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{111}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[357] & 21^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 19 & 4 & 4 \\ 21 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0} \\
[358] & 1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0} \\
[359] & 1^{122}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0} \\
[356] & 1^{131}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 25 & 9 & 9 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0} \\
[361] & 21^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 9 & 4 & 4 \\ 9 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 4 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[362] & 1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 12 & 14 & 37 & 9 & 11 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0} \\
[363] & 1^{11}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0} \\
[364] & 1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 9 & 14 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0} \\
[365] & 1^{111}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 15 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0} \\
[366] & 1^{132}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 8 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0} \\
[367] & 1^{11}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 7 & 13 & 28 & 18 & 10 \\ 4 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[368] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 85 & 15 & 17 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[369] $1^{131} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 25 & 9 & 9 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[370] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 4 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 16 & 23 & 100 & 15 & 18 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[373] $1^{132} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 7 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[372] $21^{111} 31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 9 & 4 \\ 5 & 4 & 17 & 9 & 4 \\ 15 & 17 & 49 & 27 & 13 \\ 9 & 9 & 27 & 12 & 6 \\ 4 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[374] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[376] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[375] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 57 & 13 \\ 3 & 4 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[378] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[377] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 7 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[380] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[379] $1^{\overline{31}} 3^{\overline{113}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 15 & 36 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[382] $21^{\overline{112}} 31^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[381] $21^{\overline{113}} 31^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 9 & 5 & 4 \\ 6 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[383] $1^{\overline{132}} 31^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 4 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[388] $1^{\overline{111}} 3^{\overline{113}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 14 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[387] $1^{\overline{111}} 3^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[386] $1^{\overline{111}} 3^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[385] $1^{\overline{113}} 31^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[384] $1^{\overline{14}} 3^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 75 & 15 \\ 3 & 4 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[389] $1^{\overline{111}} 3^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 5 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{\overline{121}} 3^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{122} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 11 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[397] $21^{121} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 7 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{122} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{132} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 10 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[398] $1^{122} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[399] $1^{122} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 28 & 18 & 10 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{113} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 14 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 1 \end{pmatrix}_{b_0}$$

[401] $1^{113} 3^1 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 16 & 32 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[403] $1^{11} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

[406] $1^{12} 3^{13} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[407] $1^{12} 3^{12} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{12} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 10 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[409] $1^{12} 3^{12} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[410] $1^{12} 3^{12} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 14 & 22 & 85 & 30 & 17 \\ 6 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[411] $1^{12} 3^{12} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[402] $1^{13} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[404] $1^{11} 3^{12} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

[405] $1^{13} 3^{13} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{14} 3^{11} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 45 & 11 \\ 3 & 4 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

[412] $1^{122}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[413] $1^{122}3^11^{23} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[415] $1^{12}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{132}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 13 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[418] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[423] $1^{13}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[422] $21^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 5 & 4 \\ 9 & 9 & 13 & 9 & 7 \\ 6 & 5 & 9 & 3 & 3 \\ 4 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[424] $1^{141}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[421] $1^{13}3^11^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 10 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[420] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 7 & 12 & 25 & 18 & 9 \\ 4 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[419] $1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 5 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[425] $1^{122} 3^1 1^{33} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[426] $1^{122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[427] $1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[429] $1^{131} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[430] $1^{122} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[428] $1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[431] $21^{111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[432] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 8 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[433] $1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[434] $1^{113} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 11 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 1 \end{pmatrix}_{b_0}$$

[435] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 8 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[436] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 28 & 21 & 10 \\ 7 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[437] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 45 & 11 \\ 3 & 4 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[440] $21^{11} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 4 & 4 \\ 15 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[442] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[443] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 11 & 17 & 49 & 15 & 13 \\ 4 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[444] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 25 & 9 & 9 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[445] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 13 & 21 & 76 & 15 & 16 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[446] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 9 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[447] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[439] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[441] \ 1^{\overline{11}} 3^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -115, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 7 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[438] \ 21^{\overline{1231}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -115, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.04 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 14 & 4 & 4 \\ 15 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 4 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[448] \ 1^{\overline{132}} 3^{\overline{133}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -116, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.07 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 10 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[449] \ 1^{\overline{122}} 3^{\overline{142}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -117, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.25 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[451] \ 1^{\overline{12}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -120, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[452] \ 1^{\overline{13}} 3^{\overline{14}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -120, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.47 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[450] \ 21^{\overline{111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -120, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[453] \ 1^{\overline{131}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -121, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.27 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 9 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[454] \ 1^{\overline{111}} 3^{\overline{132}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -121, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[455] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -121, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.24 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 17 & 0 & 48 & 12 \\ 3 & 4 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[456] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -121, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 9 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[458] $1^{113} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[459] $1^{11} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 9 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[457] $1^{113} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[461] $21^{111} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[462] $1^{13} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[460] $21^{11} 31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 4 & 4 \\ 12 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 4 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[464] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 9 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[465] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 7 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[463] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[466] $1^{113} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 13 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 1 \end{pmatrix}_{b_0}$$

[467] $1^{12} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 18 & 0 & 57 & 13 \\ 3 & 4 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[468] $1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[469] $1^{132} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 9 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[470] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 11 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[471] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[477] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[476] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[473] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 11 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

[475] $1^{111} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 12 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[472] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

[474] $21^{123} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 18 & 18 & 12 & 57 & 13 \\ 4 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[480] $1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[479] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[481] $1^{131} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 25 & 9 & 9 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[482] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[483] $1^{131} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 25 & 9 & 9 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[484] $1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 9 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[486] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 14 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[485] $1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 11 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[487] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[488] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 10 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[489] $1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[490] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[491] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 11 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[497] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[496] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[495] $1^{141} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 9 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[494] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 28 & 18 & 10 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[493] $1^{132} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 11 & 17 & 49 & 24 & 13 \\ 4 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[492] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[498] $1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 11 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

[499] $1^{12} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

[500] $1^{121} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[501] $1^{112}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[502] $1^{131}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 15 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[503] $1^{14}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 14 & 21 & 76 & 18 & 16 \\ 4 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[504] $1^{131}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 25 & 9 & 9 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[505] $1^{12}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[506] $21^{111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[507] $21^{121}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 23 & 4 & 4 \\ 15 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[508] $1^{131}3^11^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 25 & 9 & 9 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{131}3^11^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 18 & 27 & 133 & 18 & 21 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[514] $1^{121}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 5 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[513] $1^{21}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 13 & 32 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[515] $1^{114}3^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 22 & 2 & 3 \\ 4 & 9 & 44 & 4 & 5 \\ 22 & 44 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 5 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[510] $1^{13}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 48 & 12 \\ 3 & 4 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[512] $1^{111}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 7 & 13 & 28 & 21 & 10 \\ 5 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[511] $1^{131}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 5 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[517] $1^{14}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 18 & 0 & 57 & 13 \\ 3 & 4 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[516] $1^{14}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 12 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{13}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 18 & 0 & 57 & 13 \\ 3 & 4 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[518] $1^{112}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[520] $1^{141}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 11 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[521] $1^{132}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 12 & 17 & 49 & 24 & 13 \\ 7 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[522] $1^{12}3^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 12 & 22 & 85 & 30 & 17 \\ 4 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[523] $1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[524] $1^{132} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 13 & 22 & 85 & 30 & 17 \\ 4 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[525] $1^{132} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 7 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[526] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 48 & 12 \\ 3 & 4 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[527] $1^{11} 3^1 1^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 9 & 17 & 49 & 24 & 13 \\ 4 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[528] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 10 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[529] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 28 & 18 & 10 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[530] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[531] $1^{113} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 15 & 32 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[533] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[535] $1^{14} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[534] $1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[532] $1^{14} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

[537] $1^{13} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 12 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[538] $1^{131} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 12 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[536] $1^{14} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 21 & 0 & 75 & 15 \\ 3 & 4 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[539] $1^{114} 3 1^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 20 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 1 \end{pmatrix}_{b_0}$$

[540] $1 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 3 \\ 4 & 9 & 48 & 4 & 5 \\ 20 & 48 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 5 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[541] $1^{13} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[542] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[543] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 28 & 18 & 10 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[544] $1^{121} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[545] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[546] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[547] $1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 11 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[549] $1^{121} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 14 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[548] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

[550] $1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 10 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[551] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 11 & 13 & 28 & 18 & 10 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[552] $1^{131} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[553] $1^{131} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[554] $1^{141} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 9 & 16 & 46 & 12 & 12 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[555] $1^1 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 10 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[556] $1^{121}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[557] $21^{112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[558] $1^{121}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 58 & 12 & 14 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[559] $1^{141}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 11 & 16 & 46 & 12 & 12 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[560] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[561] $1^{131}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 12 & 17 & 49 & 15 & 13 \\ 4 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[562] $1^{132}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[563] $1^{132}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[564] $1^{113}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 13 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[565] $1^{113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 10 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[566] $1^{12}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[568] $1^{131} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 4 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 22 & 32 & 193 & 21 & 25 \\ 4 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[567] $1^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 10 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[569] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[570] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[571] $1^{132} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 9 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[572] $1^{141} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 4 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 20 & 31 & 172 & 21 & 24 \\ 4 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[573] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[574] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 13 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[575] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[576] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 13 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[577] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 12 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[578] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[579] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[582] $1^{111}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 9 & 17 & 49 & 27 & 13 \\ 5 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[583] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 14 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[580] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 18 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[581] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 48 & 0 & 5 \\ 20 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[584] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 15 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[585] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 10 & 17 & 49 & 24 & 13 \\ 4 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[587] $1^{12}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 12 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[586] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[588] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[589] $1^{\overline{11}} 1^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 9 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[590] $1^{\overline{12}} 1^{\overline{13}} 3 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[591] $1^{\overline{14}} 1^{\overline{13}} 3 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 16 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[592] $1^{\overline{14}} 1^{\overline{13}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 13 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[593] $1^{\overline{121}} 1^{\overline{13}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 28 & 21 & 10 \\ 5 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[594] $1^{\overline{13}} 1^{\overline{12}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[595] $21^{\overline{1231}} 1^{\overline{13}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 13 & 5 & 4 \\ 9 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[596] $1^{\overline{131}} 1^{\overline{13}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[597] $1^{\overline{112}} 1^{\overline{13}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[601] $1^{\overline{12}} 1^{\overline{14}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 13 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[600] $1^{\overline{132}} 1^{\overline{122}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[599] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[602] $1^{121} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 28 & 21 & 10 \\ 8 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[603] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[604] $1^{141} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 12 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[605] $1^{132} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[598] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[606] $1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[607] $1^{13} 1 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 13 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[608] $1^{13} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 49 & 15 & 13 \\ 4 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[609] $1^{121} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 4 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 17 & 23 & 100 & 15 & 18 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[610] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[611] $1^{121} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 11 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[612] $1^{131} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 49 & 27 & 13 \\ 7 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[613] $1^{121} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[614] $1^{132} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[615] $1^{132} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[616] $21^{111} 31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 12 & 4 \\ 5 & 4 & 9 & 13 & 4 \\ 9 & 9 & 13 & 21 & 7 \\ 12 & 13 & 21 & 27 & 9 \\ 4 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[617] $1^{111} 3 1^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 11 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[618] $1^{111} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[619] $21^{121} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 8 & 4 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[620] $21^{132} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[621] $1^{111} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 28 & 30 & 10 \\ 7 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[622] $1^{132} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 13 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[623] $1^{132} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 16 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[625] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 14 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[624] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[627] $21^{12} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 21 & 22 & 15 & 84 & 16 \\ 4 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[628] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 21 & 0 & 75 & 15 \\ 3 & 4 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[626] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[629] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 15 & 21 & 76 & 18 & 16 \\ 4 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[630] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 11 & 17 & 49 & 39 & 13 \\ 7 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[631] $21^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 9 & 4 \\ 5 & 4 & 9 & 9 & 4 \\ 9 & 9 & 13 & 15 & 7 \\ 9 & 9 & 15 & 12 & 6 \\ 4 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[632] $1^{11} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[633] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$

$$\begin{aligned}
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 7 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0} \\
[635] & 21^{111} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 12 & 9 & 4 \\ 5 & 4 & 13 & 9 & 4 \\ 12 & 13 & 28 & 21 & 10 \\ 9 & 9 & 21 & 12 & 6 \\ 4 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0} \\
[634] & 21^{112} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0} \\
[636] & 1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0} \\
[637] & 1^{121} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 18 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0} \\
[638] & 1^1 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 14 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 1 \end{pmatrix}_{b_0} \\
[639] & 1^{141} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0} \\
[640] & 1^{14} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0} \\
[641] & 1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 13 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0} \\
[642] & 1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0} \\
[643] & 1^{114} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 16 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 1 \end{pmatrix}_{b_0} \\
[644] & 1^{114} 3 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 18 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -1 & 1 \end{pmatrix}_{b_0}$$

[645] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[646] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[647] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[648] $1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 11 & 17 & 49 & 39 & 13 \\ 10 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[651] $1^{114} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 20 & 40 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[649] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[650] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[652] $21^{111} 31^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 12 & 4 \\ 5 & 4 & 17 & 13 & 4 \\ 15 & 17 & 49 & 39 & 13 \\ 12 & 13 & 39 & 27 & 9 \\ 4 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[654] $1^{132} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[653] $1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[655] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 13 & 17 & 49 & 27 & 13 \\ 8 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[657] $1^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 21 & 0 & 75 & 15 \\ 3 & 4 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[658] $21^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[659] $1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[656] $1^{132} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 10 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[660] $1^{121} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 58 & 12 & 14 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[661] $1^{131} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[662] $1^{121} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[663] $1^{141} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 10 & 16 & 46 & 12 & 12 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[664] $1^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

[665] $1^{141} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 12 & 16 & 46 & 12 & 12 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[666] $1^{141} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 12 & 16 & 46 & 12 & 12 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[667] $1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 11 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[668] $1^{114} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 9 & 44 & 4 & 5 \\ 21 & 44 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 5 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[669] $1^{121} 3^1 1^{43} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 11 & 13 & 28 & 21 & 10 \\ 8 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[670] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[671] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 13 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[672] $1^{131} 3^1 1^{32} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 85 & 15 & 17 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[673] $1^{112} 3^1 1^{23} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[674] $1^{121} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 17 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[675] $1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[678] $1^{121} 3^1 1^{42} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 11 & 13 & 28 & 21 & 10 \\ 7 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[677] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 18 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[676] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 7 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[679] $21^{122} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[681] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

[680] $21^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 9 & 16 & 4 & 5 \\ 12 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & 0 & 2 \\ 4 & 5 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

[682] $1^{141} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 20 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[683] $1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 49 & 27 & 13 \\ 5 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[684] $1^{141} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 10 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[685] $1^{141} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 14 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[686] $1^{132} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 15 & 22 & 85 & 30 & 17 \\ 6 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[687] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[688] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[689] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[690] $1^{13}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[691] $1^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[692] $1^{31}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 9 & 44 & 4 & 5 \\ 18 & 44 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 5 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[693] $21^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 15 & 3 & 4 \\ 8 & 9 & 20 & 4 & 5 \\ 15 & 20 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 4 & 5 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[694] $1^{141}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 24 & 4 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 24 & 36 & 238 & 24 & 28 \\ 4 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[695] $1^{121}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[696] $1^{141}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 11 & 16 & 46 & 12 & 12 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[697] $1^{131}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -225, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 49 & 27 & 13 \\ 5 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[698] $1^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[699] $1^{141}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 14 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[700] $1^{12}3^{12}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[701] $1^{13}3^{12}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 14 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[702] $1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 11 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[703] $1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 14 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

[704] $1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 10 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[705] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[707] $21^{122}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[708] $1^{11}3^{132}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 9 & 17 & 49 & 39 & 13 \\ 7 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[706] $21^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[709] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[710] $1^{121}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 14 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[711] $1^{141} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 14 & 21 & 76 & 15 & 16 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[712] $1^{11} 3 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 11 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[713] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 14 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[714] $1^{141} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 4 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 17 & 26 & 118 & 18 & 20 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[715] $1^{121} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 20 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[716] $1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[718] $1^{121} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 17 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[717] $1^{21} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 16 & 40 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[719] $1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[721] $1^{141} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 7 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 14 & 21 & 76 & 33 & 16 \\ 7 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[720] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[722] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 7 & 3 \\ 3 & 4 & 4 & 17 & 8 & 4 \\ 13 & 17 & 12 & 48 & 24 & 13 \\ 7 & 8 & 24 & 9 & 5 & 3 \\ 3 & 4 & 13 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[723] $1^{141} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 17 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[724] $21^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[725] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[726] $1^{132} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 15 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[727] $21^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 17 & 5 & 4 \\ 12 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[728] $1^{132} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 14 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[732] $21^{111} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 12 & 4 \\ 5 & 4 & 13 & 13 & 4 \\ 12 & 13 & 28 & 30 & 10 \\ 12 & 13 & 30 & 27 & 9 \\ 4 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[730] $21^{123} 3 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 18 & 4 & 4 \\ 18 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[729] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 15 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[731] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[733] $1^{114} 3 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -245, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 19 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 1 \end{pmatrix}_{b_0}$$

[734] $1^{122} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[735] $1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 28 & 21 & 10 \\ 7 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[736] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[737] $1^{13} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[738] $1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 15 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[739] $1^{114} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 17 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[740] $1^{12} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[741] $1^{12} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 13 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[742] $1^{121} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 16 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[743] $1^{11} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 18 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[744] $1^{131} 3 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 13 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[745] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[746] $1^{131} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 19 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[747] $1^{131} 3 1^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[748] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[749] $1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -257, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 17 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[751] $1^{12} 3 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 12 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[750] $1^{121} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 15 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[752] $1^{114} 3 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 19 & 40 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[753] $1^{13} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 18 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$

[755] $1^{141} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 12 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[754] $1^{14} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

[756] $1^{141} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 11 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[757] $1^{132} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 12 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[758] $21^{122} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 13 & 8 & 4 \\ 9 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[764] $1^{121} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 18 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[763] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[762] $1^{121} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 19 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[761] $1^{122} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[760] $1^{131} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 4 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 19 & 27 & 133 & 18 & 21 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[759] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[765] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -269, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 16 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[766] $1^{1^{11}} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -270, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 17 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[767] $1^{1^{32}} 3 1^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 10 & 17 & 49 & 39 & 13 \\ 7 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[768] $1^{1^{32}} 3 1^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 6 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 16 & 22 & 85 & 30 & 17 \\ 6 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[770] $1^{1^{22}} 3 1^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[769] $1^{1^{32}} 3 1^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 11 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[771] $1^{1^{14}} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 16 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[774] $1^{1^{21}} 3 1^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[772] $1^{1^{21}} 3 1^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 19 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[773] $1^{1^{21}} 3 1^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[775] $1^{1^{41}} 3 1^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -277, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 13 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[776] $1^{1^{31}} 3 1^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 13 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[778] $21^{113}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 9 & 32 & 4 & 5 \\ 24 & 32 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 5 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[777] $1^{141}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 12 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[779] $1^{141}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 12 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[780] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[781] $1^{143}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -282, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 17 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$

[782] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -282, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[783] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[784] $1^{114}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 18 & 40 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[785] $1^{131}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 7 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[786] $1^{143}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

[787] $21^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[788] $1^{131}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 16 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[789] \ 1^{131} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[790] \ 1^{131} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[791] \ 21^{121} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 18 & 4 & 4 \\ 12 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[792] \ 21^{132} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 6 & 3 \\ 4 & 4 & 22 & 8 & 4 \\ 15 & 22 & 85 & 30 & 17 \\ 6 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[793] \ 1^{12} 3 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[794] \ 1^{14} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[796] \ 21^{14} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 21 & 5 & 4 \\ 15 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[795] \ 21^{114} 31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.53 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 33 & 3 & 4 \\ 8 & 9 & 44 & 4 & 5 \\ 33 & 44 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 4 & 5 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[797] \ 1^{132} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -298, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 12 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[798] \ 1^{131} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -299, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 4 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 85 & 15 & 17 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[799] \ 1^{121} 3 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 16 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[800] $1^{132} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 12 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[801] $21^{131} 31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 12 & 4 & 4 \\ 9 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[802] $1^{131} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[803] $1^{141} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 13 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[804] $21^{121} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 13 & 4 & 4 \\ 9 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[805] $1^{113} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[806] $1^{132} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -312, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 13 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[807] $21^{13} 31^{121} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 23 & 4 & 4 \\ 24 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[808] $1^{14} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[809] $1^3 3 1^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 15 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[810] $1^{13} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[811] $1^{141} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 15 & 21 & 76 & 15 & 16 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[812] $1^{141} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 12 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[815] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 13 & 17 & 49 & 27 & 13 \\ 7 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[814] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 17 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[813] $1^{121} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 20 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[817] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[816] $1^{131} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 16 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[821] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[820] $1^{141} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 4 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 21 & 31 & 172 & 21 & 24 \\ 4 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[819] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 13 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[818] $21^{1231} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 22 & 4 & 4 \\ 21 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

- [822] $1^{132} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -332$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.25 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 10 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$
- [823] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -335$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$
- [824] $1^{12} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -336$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$
- [825] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -339$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$
- [827] $1^{14} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -340$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$
- [826] $21^{14} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -340$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 28 & 4 & 4 \\ 30 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 4 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$
- [828] $21^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -343$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$
- [829] $21^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -345$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$
- [830] $1^{141} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -349$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 7 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 7 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$
- [832] $21^{131} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -352$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 & 3 \\ 4 & 4 & 32 & 4 & 4 \\ 21 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$
- [831] $21^{132} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -352$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 12 & 8 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$
- [834] $1^{12} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -354$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

[833] $1^{12} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 18 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[835] $1^{14} 3^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 18 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[836] $1^{13} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[837] $1^{14} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 20 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[838] $1^{14} 3^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 18 & 26 & 118 & 18 & 20 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[839] $1^{12} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 14 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[840] $21^{13} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 9 & 24 & 4 & 5 \\ 18 & 24 & 52 & 9 & 12 \\ 3 & 4 & 9 & 0 & 2 \\ 4 & 5 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[841] $1^{14} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -366, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[842] $1^{14} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 14 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[843] $1^{13} 3^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 15 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[844] $1^{14} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

[845] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[846] $1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -372, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 9 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 13 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[847] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[848] $21^{113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[849] $21^{123} 31^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 26 & 4 & 4 \\ 24 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 4 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[850] $1^{141} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 15 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[851] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[852] $1^{141} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 13 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[853] $1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 11 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[854] $1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 11 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[855] $1^{141} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 13 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[856] $1^{132} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 12 & 17 & 49 & 39 & 13 \\ 10 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[857] $1^{13} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -393, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[858] $1^{13} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -393, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 19 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$

[859] $1^{141} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -395, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 15 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[860] $1^{131} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -399, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 20 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[862] $21^{113} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[861] $1^{131} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 16 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[863] $1^{141} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 14 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[864] $1^{131} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -411, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 17 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[865] $1^{12} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 19 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[866] $1^{14} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[867] $1^{13} 3^{121} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[868] $1^{14} 3^{111} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -420, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 18 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$

[869] $1^{141} 3^{111} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 18 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[870] $1^{141} 3^{111} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 21 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[871] $21^{113} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -433, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 21 & 3 & 4 \\ 8 & 9 & 28 & 4 & 5 \\ 21 & 28 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 5 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[872] $1^{131} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[873] $1^{132} 3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 9 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 12 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[874] $1^{131} 3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -443, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 19 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[875] $1^{131} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 17 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[876] $1^{131} 3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 19 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[877] $1^{12} 3^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 20 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[878] $1^{132} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 13 & 17 & 49 & 39 & 13 \\ 10 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[879] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 14 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[880] $1^{141} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -460, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 15 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[881] $1^{131} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -464, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 18 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[882] $1^{13} 1 3 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 17 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[883] $21^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[884] $1^{131} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[885] $21^{141} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 16 & 4 & 4 \\ 12 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[886] $1^{13} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -481, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[887] $21^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -481, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 4 & 17 & 8 & 4 \\ 12 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[888] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -486, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 17 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

- [889] $21^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -487, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$
- [890] $1^{141}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 15 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$
- [892] $1^{131}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 20 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$
- [891] $1^{131}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 18 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$
- [893] $21^{13}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 27 & 3 & 4 \\ 5 & 4 & 27 & 4 & 4 \\ 27 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 4 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$
- [894] $1^{121}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -501, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 20 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$
- [895] $1^{141}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 16 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$
- [896] $1^{131}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -511, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 21 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$
- [897] $1^{14}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -514, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 21 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$
- [898] $1^{131}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -515, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 22 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$
- [899] $1^{131}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 21 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$
- [900] $1^{13}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$$\begin{aligned}
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[901] & 21^{131} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 17 & 4 & 4 \\ 12 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0} \\
[902] & 1^{141} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -524, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 16 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0} \\
[903] & 1^{141} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -534, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0} \\
[904] & 1^{141} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -534, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 19 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0} \\
[905] & 131^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 20 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0} \\
[906] & 21^{131} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -547, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 4 & 27 & 4 & 4 \\ 18 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0} \\
[907] & 1^{131} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -555, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 19 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0} \\
[909] & 1^{141} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -556, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 18 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0} \\
[908] & 1^{141} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -556, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 17 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0} \\
[910] & 1^{141} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 18 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0} \\
[911] & 21^{141} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -574, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 9 & 32 & 4 & 5 \\ 24 & 32 & 94 & 12 & 16 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 5 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[912] $1^{141} 3^{121} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -576, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 19 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[914] $21^{141} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 6 & 3 \\ 4 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[913] $1^{131} 3^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 21 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$

[915] $1^{131} 3^{141} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -591, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 22 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[917] $1^{141} 3^{121} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 17 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[916] $1^{141} 3^{121} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 22 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[918] $21^{131} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 22 & 4 & 4 \\ 15 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[919] $21^{114} 31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -610, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 9 & 40 & 4 & 5 \\ 30 & 40 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 4 & 5 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[920] $21^{143} 31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 33 & 3 & 4 \\ 5 & 4 & 32 & 4 & 4 \\ 33 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 4 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[921] $1^{14} 3^{131} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -624, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 20 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$

[922] $21^{13} 31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 31 & 4 & 4 \\ 30 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 4 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

- [923] $1^{141}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -643$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 19 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$
- [924] $1^{141}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -645$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 22 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$
- [925] $1^{141}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -646$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 22 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$
- [926] $1^{141}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -656$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 21 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$
- [927] $1^{141}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -664$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 18 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$
- [928] $21^{132}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -673$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 12 & 9 & 3 \\ 4 & 4 & 17 & 13 & 4 \\ 12 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$
- [929] $1^{141}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -683$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 20 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$
- [930] $21^{151}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -688$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.1$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 20 & 4 & 4 \\ 15 & 20 & 73 & 15 & 15 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$
- [931] $1^{141}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -690$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 21 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$
- [932] $1^{141}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -691$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 21 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$
- [933] $1^{141}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -696$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 20 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$
- [934] $21^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -703$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 27 & 3 & 4 \\ 8 & 9 & 36 & 4 & 5 \\ 27 & 36 & 121 & 15 & 19 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 5 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[935] \ 1^{141} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -736, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 22 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 22 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[936] \ 1^{141} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -744, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 23 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[937] \ 1^{141} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -754, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 23 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[939] \ 1^{141} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 23 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[938] \ 1^{141} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 21 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[940] \ 21^{141} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -787, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 21 & 4 & 4 \\ 15 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[941] \ 1^{141} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -800, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 24 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[942] \ 1^{141} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -838, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 24 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[943] \ 21^{115} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -844, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 9 & 40 & 4 & 5 \\ 30 & 40 & 148 & 15 & 20 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 5 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[945] \ 1^{141} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 24 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[944] \ 21^{14} 31^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 5 & 36 & 3 & 4 \\ 5 & 4 & 36 & 4 & 4 \\ 36 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 4 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0} \\
[946] \quad & 21^{141}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -904, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 24 & 3 & 3 \\ 4 & 4 & 36 & 4 & 4 \\ 24 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0} \\
[947] \quad & 21^{161}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -952, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 4 & 24 & 4 & 4 \\ 18 & 24 & 106 & 18 & 18 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0} \\
[948] \quad & 21^{141}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -964, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 4 & 26 & 4 & 4 \\ 18 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0} \\
[949] \quad & 21^{141}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -1003, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 & 3 \\ 4 & 4 & 31 & 4 & 4 \\ 21 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0} \\
[950] \quad & 21^{171}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -1264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 & 3 \\ 4 & 4 & 28 & 4 & 4 \\ 21 & 28 & 145 & 21 & 21 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0} \\
\\
T = 8 \quad [955] \quad & 21^1 4^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 5 & 6 & 6 & 4 \\ 5 & 4 & 7 & 7 & 4 \\ 6 & 7 & 8 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 4 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[956] \quad & 1^{11} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[954] \quad & 1^1 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[951] \quad & 1^1 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 0 & -1 & 3 \end{pmatrix}_{b_0} \\
[952] \quad & 21^{11} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 11 & 7 & 4 \\ 12 & 11 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0} \\
[953] \quad & 21^1 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[957] $1^{\overline{1}} \overline{1}^1 3 2 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

[958] $1^{\overline{1}} \overline{1}^1 4 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[959] $1^{\overline{1}} \overline{1}^1 4 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 7 & 8 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[961] $2 1^{\overline{1}} \overline{1}^1 4 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[963] $1^{\overline{1}} \overline{1}^1 4 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[962] $1^{\overline{1}} \overline{1}^1 3 2 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[960] $1^{\overline{1}} \overline{1}^1 4 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[966] $1^{\overline{1}} \overline{1}^1 4 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[965] $1^{\overline{1}} \overline{1}^1 4 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[964] $2 1^{\overline{1}} \overline{1}^1 4 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[967] \ 1 \overset{11}{4} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[968] \ 1 \overset{11}{3} 21^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 1 & 3 \\ 3 & 5 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[969] \ 21 \overset{11}{4} 1^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 3 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[970] \ 1 \overset{12}{4} 1^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 6 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[971] \ 1 \overset{11}{4} 1^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[972] \ 21 \overset{11}{4} 1^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[973] \ 1 \overset{11}{3} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[974] \ 1 \overset{11}{4} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[975] \ 1 \overset{131}{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 7 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[976] \ 21 \overset{11}{4} 1^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 8 & 7 & 4 \\ 9 & 8 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 4 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[982] \ 21 \overset{11}{4} 1^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 4 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 4 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[979] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 7 & 7 & 8 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[977] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[980] $21^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 7 & 3 \\ 4 & 4 & 4 & 10 & 4 \\ 3 & 4 & 1 & 7 & 3 \\ 7 & 10 & 7 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[978] $1^1 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 4 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[981] $1^1 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 5 & 4 & 4 \\ 6 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[983] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[988] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 7 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[987] $1^1 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[986] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 10 & 7 & 4 \\ 9 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[985] $1^1 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 13 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[984] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[992] $221^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 6 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 6 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[991] $1^{11} 4 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[990] $1^{11} 4 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 6 & 10 & 21 & 8 & 9 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[989] $21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[993] $21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[994] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[998] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$

[997] $1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$

[996] $141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[995] $21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[999] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1000] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 9 & 10 & 21 & 8 & 9 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1001] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 5 & 3 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[1002] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 13 & 6 & 7 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[1003] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 13 & 11 & 7 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[1004] $1^{12} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 7 & 8 & 13 & 6 & 7 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[1005] $1^{11} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 8 & -1 & 4 \end{pmatrix}_{b_0}$$

[1007] $1^{13} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 10 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1008] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 13 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[1009] $21^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 13 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1010] \quad & 1^{111} 4^{12} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 4 & 7 & 8 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[1011] \quad & 221^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 3 & 5 & 3 \\ 6 & 13 & 8 & 12 & 7 \\ 3 & 8 & 4 & 6 & 4 \\ 5 & 12 & 6 & 7 & 5 \\ 3 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[1006] \quad & 1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 8 & -1 & 4 \end{pmatrix}_{b_0} \\
[1012] \quad & 1^{11} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[1013] \quad & 1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 17 & 21 & 101 & 31 & 19 \\ 5 & 7 & 31 & 7 & 5 \\ 3 & 4 & 19 & 5 & 3 \end{pmatrix}_{b_0} \\
[1014] \quad & 21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 17 & 21 & 101 & 31 & 19 \\ 5 & 7 & 31 & 7 & 5 \\ 3 & 4 & 19 & 5 & 3 \end{pmatrix}_{b_0} \\
[1015] \quad & 1^{11} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0} \\
[1016] \quad & 1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[1017] \quad & 21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[1018] \quad & 1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[1021] \quad & 1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 11 & 18 & 70 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1020] \ 1^{111} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 11 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1019] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1022] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1023] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1024] \ 21^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1025] \ 1^{111} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 10 & 7 & 4 \\ 6 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1026] \ 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1027] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 13 & 6 & 7 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1030] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1031] \ 21^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 18 & 6 & 4 \\ 5 & 4 & 18 & 7 & 4 \\ 18 & 18 & 70 & 26 & 16 \\ 6 & 7 & 26 & 7 & 5 \\ 4 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1044] \quad 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1043] \quad 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1042] \quad 21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1045] \quad 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 8 & 10 & 21 & 8 & 9 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1041] \quad 1^{12} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 6 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1046] \quad 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1047] \quad 21^{12} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 15 & 3 & 4 \\ 10 & 15 & 45 & 11 & 13 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1048] \quad 21^{12} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 7 & 5 & 3 \\ 4 & 4 & 10 & 7 & 4 \\ 7 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1049] \quad 21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 8 & 3 & 4 \\ 9 & 8 & 13 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1050] \quad 1^{13} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1051] $21^{11}41^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1052] $1^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[1053] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1054] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1055] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 15 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1056] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 70 & 14 & 16 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1057] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 14 & 25 & 121 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1058] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 7 & 4 \\ 9 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 4 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[1059] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1061] $1^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 7 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1060] $21^{11}41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 7 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1062] $1^{11} 1^{11} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[1063] $1^{11} 1^4 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 6 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1064] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 12 & 21 & 88 & 15 & 18 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1065] $1^{11} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1066] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[1067] $1^{11} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 11 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1068] $1^{11} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1069] $1^{11} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 4 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[1070] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[1071] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 10 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1072] $1^{11} \overset{1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[1074] $1^{13} \overset{1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 121 & 18 & 21 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$

[1073] $1^{11} \overset{1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$

[1076] $21^{11} \overset{1}{4} 1^{11} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 12 & 6 & 4 \\ 8 & 13 & 20 & 12 & 7 \\ 12 & 20 & 26 & 16 & 10 \\ 6 & 12 & 16 & 7 & 5 \\ 4 & 7 & 10 & 5 & 3 \end{pmatrix}_{b_0}$

[1075] $1^{111} \overset{1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 8 & 10 & 6 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[1077] $21^{21} \overset{1}{4} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 10 & 7 & 4 \\ 8 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[1078] $1^{111} \overset{1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 10 & 7 & 4 \\ 8 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[1079] $21^{11} \overset{1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 5 & 2 & 3 \\ 4 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1080] $1^{11} \overset{14}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 11 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1081] $1^{14} \overset{1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1082] $21^{11} \overset{1}{4} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1083] $21^{11} 4^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 22 & 3 & 4 \\ 21 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 4 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1084] $1^{11} 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1085] $21^{11} 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1086] $1^{11} 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 6 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[1088] $1^{11} 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 7 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1087] $1^{11} 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 15 & 18 & 70 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[1090] $1^{13} 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 13 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1089] $1^{11} 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 9 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1091] $1^{11} 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 13 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1095] $1^4 4^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 7 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1093] $1^{12}_{1^{11}}41^{1^{21}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 8 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1092] $1^{12}_{1^{21}}41^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 11 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1094] $1^{13}_{1^2}41^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[1097] $1^{12}_{1^{11}}41^{1^{31}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 9 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1096] $1^{13}_{1^3}41^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1101] $21^{11}_{1^{21}}41^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 15 & 3 & 4 \\ 15 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1100] $21^{11}_{1^{11}}41^{1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 3 & 4 \\ 15 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1099] $21^{11}_{1^{21}}41^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 15 & 6 & 4 \\ 5 & 4 & 15 & 7 & 4 \\ 15 & 15 & 45 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 4 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1098] $21^{11}_{1^{11}}41^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 9 & 3 & 4 \\ 9 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[1103] $1^{14}_{1^{41}}41^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 16 & 28 & 158 & 20 & 24 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[1102] $1^{11}_{1^{11}}41^{1^{31}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 10 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1104] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 7 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1108] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1107] $21^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[1106] $1^{14} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 12 & 19 & 65 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1105] $1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[1109] $21^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1110] $21^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 12 & 7 & 4 \\ 12 & 12 & 26 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1111] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 9 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1112] $1^{13} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 10 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1113] $1^{13} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 12 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 13 & 15 & 45 & 11 & 13 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1125] $21^{11} 41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 3 & 4 \\ 12 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1126] $1^{12} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 38 & 10 & 12 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1127] $1^{12} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 15 & 25 & 132 & 19 & 22 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1128] $1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 12 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1130] $1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1129] $1^{12} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 15 & 25 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1131] $1^{12} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 12 & 14 & 38 & 10 & 12 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1135] $1^{12} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 13 & 15 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1134] $1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1133] $21^{21} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1132] $21^{31}41^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 13 & 15 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1136] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 15 & 18 & 70 & 14 & 16 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1137] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[1138] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 8 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[1139] $1^{111}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 26 & 16 & 10 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1140] $1^{111}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1141] $21^{111}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[1142] $1^{111}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 10 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1143] $21^{11}41^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 12 & 4 & 4 \\ 12 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1144] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1145] $21^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1146] $21^{11} 4 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 26 & 3 & 4 \\ 24 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 4 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1147] $21^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 12 & 3 & 4 \\ 12 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1148] $21^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[1150] $1^{12} 4 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 14 & 17 & 61 & 13 & 15 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1149] $1^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 9 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[1153] $1^{14} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 14 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[1152] $21^{11} 4 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 24 & 3 & 4 \\ 21 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 4 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1154] $1^{13} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1155] $1^{12} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 38 & 10 & 12 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1156] $1^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 12 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[1157] $1^{13}_{1^{31}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 15 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1158] $1^{12}_{1^3} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1151] $21^{1^{31}} 4 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 27 & 3 & 4 \\ 5 & 4 & 30 & 3 & 4 \\ 27 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 4 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1159] $1^{12}_{1^2} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1160] $1^{14}_{1^{31}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 17 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1161] $1^{11}_{1^{21}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 12 & 15 & 45 & 11 & 13 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1162] $1^{12}_{1^2} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 11 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[1163] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1164] $1^{14}_{1^4} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1165] $1^{12}_{1^{21}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 10 & 12 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1166] $1^{11}_{1^{31}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 70 & 14 & 16 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1167] $1 \overset{1^{11}}{4} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 9 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1168] $1 \overset{1^{21}}{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1169] $21 \overset{1^{21}}{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 10 & 12 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1170] $1 \overset{1^3}{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 13 & 15 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1171] $1 \overset{1^2}{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 45 & 11 & 13 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1172] $21 \overset{1^{11}}{3^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 14 & 18 & 70 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[1173] $221 \overset{1^{11}}{1^{11}} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1174] $21 \overset{1^{11}}{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1175] $1 \overset{1^{31}}{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 14 & 18 & 70 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[1176] $1 \overset{1^{11}}{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1177] $1 \overset{1^{41}}{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 12 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1178] \ 21^{1^{11}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 4 & 7 & 7 & 4 \\ 5 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1179] \ 1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 11 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1180] \ 21^{1^{21}} 4 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 19 & 3 & 4 \\ 18 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1181] \ 1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1182] \ 21^{21} 4 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1183] \ 1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 9 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1184] \ 1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 12 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1185] \ 1^{14} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 65 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1186] \ 21^{1^{31}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 22 & 3 & 4 \\ 15 & 22 & 97 & 16 & 19 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1187] \ 21^{1^{31}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1188] $1^{13}_{1^1} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 11 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1189] $21^{11}_{1^1} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 16 & 3 & 4 \\ 15 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1190] $21^{12}_{1^1} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 3 \\ 4 & 4 & 11 & 3 & 4 \\ 8 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1193] $21^{11}_{1^1} 4 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 20 & 3 & 4 \\ 18 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1191] $1^{13}_{1^4} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 29 & 3 & 4 \\ 17 & 29 & 169 & 21 & 25 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1192] $21^{12}_{1^1} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 27 & 3 & 4 \\ 24 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 4 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1195] $1^{13}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 12 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1194] $1^{13}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 10 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1196] $1^{21}_{1^{11}} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 11 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1197] $1^{12}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1198] $1^{13}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 13 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1199] $1^{11} 4^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 23 & 3 & 4 \\ 21 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 4 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1200] $1^{14} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 15 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1201] $21^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 8 & 5 & 3 \\ 4 & 4 & 12 & 7 & 4 \\ 8 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1202] $1^{13} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1203] $1^{12} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 13 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1204] $1^{14} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 90 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1205] $21^{11} 4^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 90 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1206] $21^{21} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 11 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1207] $1^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 11 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1208] $1^{13} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 10 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[1209] $1^{12}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 10 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1210] $1^{13}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 13 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1211] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 65 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1212] $1^{14}_{1^{41}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 33 & 3 & 4 \\ 19 & 33 & 212 & 23 & 28 \\ 3 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1213] $1^{12}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1214] $21^{21}_{1^{11}} 4 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1215] $31^{11}_{1^1} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 6 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 6 & 3 & 4 & 6 & 4 \\ 3 & 6 & 3 & 6 & 3 & 3 \\ 2 & 4 & 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[1216] $1^{13}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1217] $1^{13}_{1^2} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1218] $1^{12}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 13 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1219] $1^{13}_{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

[1230] $1^{14}_{1^{11}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 12 & 19 & 65 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1231] $1^{12}_{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[1233] $1^{13}_{1^{31}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 16 & 21 & 88 & 15 & 18 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1232] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 14 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1235] $1^{13}_{1^{21}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 61 & 12 & 15 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1234] $1^{14}_{1^{21}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 15 & 22 & 90 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1236] $21^{14}_{1^1} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1237] $1^{13}_{1^{41}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 18 & 24 & 121 & 18 & 21 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1238] $21^{41}_{1^1} 4 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 18 & 24 & 121 & 18 & 21 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1239] $1^{13}_{1^{21}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 13 & 18 & 61 & 12 & 15 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1240] $1^{12}_{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1241] \quad 1^{12}_{1^{31}} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1242] \quad 21^{31}_{1^{11}} 4^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1243] \quad 1^{12}_{1^{31}} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 17 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1245] \quad 1^{13}_{1^{21}} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1244] \quad 21^{21}_{1^{11}} 4^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1250] \quad 1^{13}_{1^2} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1246] \quad 21^{13}_{1^{31}} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 15 & 18 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1247] \quad 1^{14}_{1^1} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 12 & 19 & 65 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1248] \quad 21^{41}_{1^{11}} 4^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 19 & 25 & 132 & 19 & 22 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1249] \quad 1^{12}_{1^{41}} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 19 & 25 & 132 & 19 & 22 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1251] $1^{13}_{1^2} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 13 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[1252] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 65 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1253] $1^{14}_{1^2} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$

[1254] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 12 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1255] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 16 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1256] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 225, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 15 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1258] $1^{12}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1257] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 65 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1259] $21^{12}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 19 & 3 & 4 \\ 13 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1260] $1^{12}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 15 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1261] $1^{12}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1264] $1^{13}_{1^{31}} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 15 & 18 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1263] $1^{14}_{1^{21}} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 90 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1262] $1^{12}_{1^2} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1265] $21^{13}_{1^{11}} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 3 \\ 4 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1266] $1^{13}_{1^{31}} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 16 & 21 & 88 & 15 & 18 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1267] $1^{14}_{1^{21}} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1268] $21^{21}_{1^{11}} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1269] $221^{11}_{1^1} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 13 & 12 & 12 & 7 \\ 5 & 12 & 8 & 10 & 6 \\ 5 & 12 & 10 & 7 & 5 \\ 3 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[1270] $21^{12}_{1^{11}} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 11 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1271] $1^{13}_{1^3} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 16 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1272] $1^{12}_{1^2} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$\begin{aligned}
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[1273] & 221^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 8 & 5 & 3 \\ 6 & 13 & 20 & 12 & 7 \\ 8 & 20 & 26 & 16 & 10 \\ 5 & 12 & 16 & 7 & 5 \\ 3 & 7 & 10 & 5 & 3 \end{pmatrix}_{b_0} \\
[1275] & 1^{14} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 17 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0} \\
[1274] & 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 18 & 24 & 121 & 18 & 21 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[1277] & 21^{31} 41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 17 & 25 & 121 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[1276] & 1^{14} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 17 & 25 & 121 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[1278] & 1^{14} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 65 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[1279] & 1^{12} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 17 & 22 & 97 & 16 & 19 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0} \\
[1280] & 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 13 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[1281] & 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 16 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0} \\
[1282] & 1^{12} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1283] $1^{12}_{1^{31}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 97 & 16 & 19 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1284] $1^{14}_{1^4} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 19 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0}$

[1285] $1^{14}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 90 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$

[1286] $1^{12}_{1^{41}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 19 & 25 & 132 & 19 & 22 \\ 2 & 3 & 19 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$

[1287] $21^{13}_{1^{41}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 20 & 2 & 3 \\ 4 & 4 & 29 & 3 & 4 \\ 20 & 29 & 169 & 21 & 25 \\ 2 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$

[1288] $21^{13}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 13 & 18 & 61 & 12 & 15 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1289] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 17 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1290] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1291] $1^{13}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 13 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

[1292] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 14 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1293] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 15 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1294] \ 1^{13}_{1^{21}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 288, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 17 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1295] \ 1^{14}_{1^{31}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 289, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 18 & 25 & 121 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1296] \ 21^{13}_{1^{21}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 291, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 17 & 3 & 4 \\ 12 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1297] \ 21^{15}_{1^{21}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 296, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.1$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 15 & 20 & 73 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1298] \ 1^{14}_{1^{21}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 297, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 15 & 22 & 90 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1299] \ 1^{13}_{1^{31}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 306, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1300] \ 21^{31}_{1^{21}} 4^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 306, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1301] \ 1^{14}_{1^{21}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 310, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 18 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1302] \ 1^{14}_{1^{21}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 310, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 90 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1303] \ 1^{14}_{1^{12}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 314, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.87 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 15 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

- [1304] $1^{13}_{1^1} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [1305] $21^{41}_{1^1} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 158 & 20 & 24 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$
- [1306] $1^{14}_{1^{41}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 158 & 20 & 24 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$
- [1307] $1^{13}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 97 & 15 & 19 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$
- [1308] $1^{13}_{1^1} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 15 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [1309] $1^{13}_{1^{31}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 19 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$
- [1310] $1^{14}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 15 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$
- [1311] $1^{14}_{1^2} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$
- [1312] $21^{14}_{1^{41}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 20 & 2 & 3 \\ 4 & 4 & 28 & 3 & 4 \\ 20 & 28 & 158 & 20 & 24 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$
- [1313] $1^{14}_{1^{31}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 17 & 25 & 121 & 17 & 21 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$
- [1314] $1^{13}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 354, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 16 & 23 & 97 & 15 & 19 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1315] $21^{14}_{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 4 & 19 & 3 & 4 \\ 14 & 19 & 65 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1316] $31^{11}_{11} 31^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 3 & 2 \\ 3 & 4 & 4 & 9 & 6 & 4 \\ 3 & 4 & 0 & 6 & 3 & 2 \\ 6 & 9 & 6 & 13 & 9 & 7 \\ 3 & 6 & 3 & 9 & 3 & 3 \\ 2 & 4 & 2 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[1317] $1^{13}_{14} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 29 & 3 & 4 \\ 21 & 29 & 169 & 21 & 25 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1318] $21^{14}_{11} 4 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 29 & 3 & 4 \\ 21 & 29 & 169 & 21 & 25 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1319] $1^{14}_{12} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 19 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1320] $1^{13}_{12} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 17 & 23 & 97 & 15 & 19 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1321] $1^{14}_{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1322] $1^{13}_{12} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1323] $1^{14}_{13} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 18 & 25 & 121 & 17 & 21 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1324] $1^{14}_{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 16 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1326] \ 21 \overset{11}{\overset{31}{3}} 4 \ 1 \overset{14}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 379, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 20 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1325] \ 1 \overset{131}{4} \ 1 \overset{11}{4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 379, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 20 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1327] \ 1 \overset{141}{\underset{13}{4}} 1 + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 381, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 18 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1328] \ 1 \overset{141}{\underset{12}{3}} 4 \ 1 \overset{11}{2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}\textcolor{teal}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 387, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1329] \ 21^{1\overline{3}1} 4 \ 1^{1\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 390, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 26 & 3 & 4 \\ 18 & 26 & 130 & 18 & 22 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1330] \ 1^{14} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 392, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 17 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1332] \ 21^{13} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 402, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.72 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 14 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1331] \ 1 \overset{1^3}{\underset{1^3}{3}} 4 \ 1 \overset{1^1}{\underset{1^1}{2}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 402, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 130 & 18 & 22 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1334] \ 21^{16}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 406, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 18 & 24 & 106 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1333] \begin{smallmatrix} 1^4 \\ 1^4 1 \end{smallmatrix} 4 \begin{smallmatrix} 1^4 \\ 1^4 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \, T_{\max} = 11, \, \Delta = -84, \, \det = 406, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.81 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$$

$$\left(\begin{array}{ccccc} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 158 & 20 & 24 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{array} \right)_{b_0}$$

$$[1335] \begin{smallmatrix} 1^4 \\ 1^3 1 \\ 1^3 1 \end{smallmatrix} 4 \begin{smallmatrix} 1^4 \\ 1^3 1 \\ 1^3 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\ T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 19 & 26 & 130 & 18 & 22 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1336] $21^{14}_{1^1} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 21 & 3 & 4 \\ 15 & 21 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1337] $1^{14}_{1^1} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 425, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1338] $1^{13}_{1^3} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1339] $1^{14}_{1^{21}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 132 & 17 & 22 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1340] $1^{14}_{1^{31}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 21 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1341] $21^{14}_{1^{31}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 431, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 25 & 3 & 4 \\ 18 & 25 & 121 & 17 & 21 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1342] $21^{14}_{1^{21}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 4 & 22 & 3 & 4 \\ 16 & 22 & 90 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1343] $21^{13}_{1^{21}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 435, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 4 & 23 & 3 & 4 \\ 16 & 23 & 97 & 15 & 19 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1344] $1^{13}_{1^{41}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 447, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 29 & 3 & 4 \\ 21 & 29 & 169 & 21 & 25 \\ 2 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1345] $1^{14}_{1^1} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1346] $1^{14}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 18 & 27 & 132 & 17 & 22 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1347] $1^{14}_{1^{41}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 & 3 \\ 3 & 4 & 33 & 3 & 4 \\ 23 & 33 & 212 & 23 & 28 \\ 3 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1348] $21^{41}_{1^{11}} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 & 3 \\ 3 & 4 & 33 & 3 & 4 \\ 23 & 33 & 212 & 23 & 28 \\ 3 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1349] $1^{14}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1350] $1^{14}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 494, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 19 & 27 & 132 & 17 & 22 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1351] $1^{14}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1352] $21^{17}_{1^1} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 28 & 3 & 4 \\ 21 & 28 & 145 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1353] $1^{14}_{1^{31}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 20 & 30 & 169 & 20 & 25 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1354] $1^{14}_{1^{31}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 21 & 30 & 169 & 20 & 25 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1355] $21^{14}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1356] $1^{14}_{1^3} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$$

[1357] $21^{14} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 23 & 2 & 3 \\ 4 & 4 & 33 & 3 & 4 \\ 23 & 33 & 212 & 23 & 28 \\ 2 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1358] $1^{14} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 23 & 2 & 3 \\ 3 & 4 & 33 & 3 & 4 \\ 23 & 33 & 212 & 23 & 28 \\ 2 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1359] $21^{14} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 19 & 2 & 3 \\ 4 & 4 & 27 & 3 & 4 \\ 19 & 27 & 132 & 17 & 22 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1360] $21^{14} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 30 & 3 & 4 \\ 21 & 30 & 169 & 20 & 25 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [1361] $1^{11} 2 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

[1362] $1^{11} 2 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[1363] $1^{11} 3 1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[1364] $1^{11} 1^{11} 3 1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 9 & 1 & 7 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 4 & 7 & 0 & 4 \end{pmatrix}_{b_0}$$

[1365] $1^{11} 4 1 13 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[1366] $1^{11} 4 1 13 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 6 & 4 & 6 \\ 4 & 2 & 4 \\ 6 & 4 & 6 \end{pmatrix}_{b_0}$$

[1367] $1^{\overline{1}} 4^{11} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 4 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1372] $21^{\overline{1}} 5^{\overline{11}} 1^{\overline{1}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 6 & 6 & 4 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[1369] $1^{\overline{1}} 4^{11} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 4 & 4 & 0 & 4 \end{pmatrix}_{b_0}$

[1370] $1^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[1371] $1^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1368] $1^{\overline{1}} 5^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 4 & 6 & 4 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[1373] $21^{\overline{1}} 5^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 6 & 6 & 4 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[1374] $21^{\overline{1}} 5^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1375] $1^{\overline{1}} 5^{\overline{1}} 1^{\overline{11}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[1376] $1^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 12 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$

[1377] $21^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 6 & 10 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1390] \ 21^{\overline{1}1} 51^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1389] \ 1^{\overline{1}1} 2321^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 6 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1392] \ 21^{\overline{1}1} 51^{\overline{1}1} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 7 & 4 & 5 \\ 3 & 3 & 4 & 1 & 3 \\ 3 & 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1391] \ 21^{\overline{1}1} 51^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 7 & 2 & 3 \\ 4 & 4 & 10 & 3 & 4 \\ 7 & 10 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1397] \ 1^{\overline{1}1} 4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 6 & 6 \\ 4 & 3 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1396] \ 221^{\overline{1}1} 51^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 5 & 5 & 7 & 4 & 5 \\ 2 & 3 & 4 & 1 & 3 \\ 3 & 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1395] \ 1^{\overline{1}1} 51^{\overline{1}1} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1394] \ 21^{\overline{1}1} 51^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1393] \ 1^{\overline{1}1} 51^{\overline{1}1} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1398] \ 1^{\overline{1}1} 41^{\overline{1}1} 31^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 3 & 3 \\ 5 & 3 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$$

[1400] $1^{\overline{1}11} 4^{12} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 7 & 4 \\ 3 & 1 & 5 & 3 \\ 7 & 5 & 12 & 8 \\ 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$$

[1399] $1^{\overline{1}11} 4^{12} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 4 & 4 \\ 3 & 1 & 4 & 3 \\ 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$$

[1401] $1^{\overline{1}} 5^{11} 1^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1403] $1^{\overline{1}11} 5^{12,1^2} 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[1402] $1^{\overline{1}11} 4^{12} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[1404] $21^{\overline{2}11} 5^{11} 1^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 8 & 10 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1405] $1^{\overline{1}11} 5^{12,1^1} 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[1406] $1^{\overline{1}} 5^{13,1^1} 1^{\overline{1}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 10 & 17 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1407] $1^{\overline{1}11} 4^{131^1} 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 8 & 6 \\ 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[1409] $1^{\overline{1}11} 5^{11} 1^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[1408] $21^{\overline{1}11} 5^{11} 1^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[1411] & 1^{12} \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 14 & 23 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0} \\
[1410] & 1^{12} \overset{1}{5} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 6 & 11 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[1412] & 1^{12} \overset{1}{5} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 8 & 14 & 38 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1413] & 21^4 \overset{1}{5} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 16 & 20 & 90 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[1414] & 1^{12} \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 20 & 32 & 234 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0} \\
[1415] & 1^{11} \overset{1}{4} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 6 & 1 & 14 & 8 \\ 3 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0} \\
[1416] & 21^{12} \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 17 & 3 & 4 \\ 12 & 17 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[1417] & 15^1 \overset{1}{1} 1^{11} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1418] & 1^{12} \overset{1}{5} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 121 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[1419] & 1^{12} \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 150 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1420] $21^{11} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 2 & 3 \\ 4 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1421] $1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 9 & 7 \\ 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

[1422] $1^{12} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 18 & 29 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$

[1423] $1^{12} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$

[1424] $21^{11} 5 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 7 & 11 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1425] $1^{12} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[1426] $1^{12} 5 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 10 & 18 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1427] $1^{12} 3 1^1 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 10 & 4 \\ 3 & 1 & 7 & 3 \\ 10 & 7 & 21 & 11 \\ 4 & 3 & 11 & 5 \end{pmatrix}_{b_0}$

[1428] $1^{13} 5 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 12 & 21 & 88 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$

[1429] $1^{12} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[1430] $21^{13} 5 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1431] \ 1^{12} \overset{1^1}{5} \overset{1^1}{1^3, 1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.52 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1432] \ 1^{12} \overset{1^1}{5} \overset{1^1}{1^3, 1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -85, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 14 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1433] \ 1^1 \overset{1^1}{5} 1^{12} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -86, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1435] \ 21^2 \overset{1^1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1434] \ 1^{12} \overset{1^1}{5} \overset{1^1}{1^2, 1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1437] \ 21^1 \overset{1^1}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -95, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 40 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1436] \ 1^{13} \overset{1^1}{5} \overset{1^1}{1^1, 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -95, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1438] \ 1^{11} \overset{1^1}{1^1} \overset{1^1}{31^1} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 27 & 11 \\ 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1439] \ 21^{12} \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1440] \ 1^{13} \overset{1^1}{5} \overset{1^1}{1^4, 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 16 & 27 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1441] $221 \overset{1}{\overset{1}{1}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 5 & 2 & 3 \\ 6 & 13 & 12 & 5 & 7 \\ 5 & 12 & 8 & 4 & 6 \\ 2 & 5 & 4 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1443] $1 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$

[1442] $21 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 121 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$

[1444] $21 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 8 & 2 & 3 \\ 4 & 4 & 11 & 3 & 4 \\ 8 & 11 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1445] $1 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$

[1446] $1 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1447] $21 \overset{1}{\overset{1}{1^2}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1448] $1 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$

[1449] $1 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 18 & 30 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$

[1450] $1 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 16 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$

[1451] $1 \overset{1}{\overset{1}{1^3}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 20 & 33 & 232 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1452] \quad 1^{13} \begin{smallmatrix} 1^1 \\ 1^4, 1^4 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^4 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 22 & 36 & 281 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 33 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1453] \quad 21^{13} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 18 & 24 & 121 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1454] \quad 21^{13} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1455] \quad 1^{13} \begin{smallmatrix} 1^1 \\ 1^3, 1^3 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^3 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 18 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1456] \quad 21^{13} \begin{smallmatrix} 1^1 \\ 1^3 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^3 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1457] \quad 21^{13} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 3 \\ 4 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1458] \quad 221^{12} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 2 & 3 \\ 6 & 13 & 24 & 5 & 7 \\ 10 & 24 & 38 & 8 & 12 \\ 2 & 5 & 8 & 0 & 2 \\ 3 & 7 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1459] \quad 21^{13} \begin{smallmatrix} 1^1 \\ 1^2 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^2 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 13 & 18 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1460] \quad 221^{13} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 2 & 3 \\ 6 & 13 & 36 & 5 & 7 \\ 15 & 36 & 88 & 12 & 18 \\ 2 & 5 & 12 & 0 & 2 \\ 3 & 7 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1461] \quad 1^{11} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[1462] \quad & 21 \overset{1}{1} \overset{11}{5} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[1468] \quad & 1^1 \overset{1}{5} 1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[1463] \quad & 1^1 \overset{1}{5} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[1464] \quad & 1^1 \overset{1}{5} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & 5 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 7 & -1 & 3 \end{pmatrix}_{b_0} \\
[1465] \quad & 1^1 \overset{13}{5} 131^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 13 & 21 & 97 & 18 & 21 \\ 2 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0} \\
[1466] \quad & 1^1 \overset{11}{5} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 7 & 7 & 9 & 6 & 7 \\ 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[1467] \quad & 1^1 \overset{1}{5} 1^{12} 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & 10 & 4 & 6 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[1469] \quad & 1^1 \overset{1}{5} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[1471] \quad & 1^1 \overset{1}{5} 1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0} \\
[1470] \quad & 1^1 \overset{1}{5} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & 2 & 2 & 4 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[1472] $1 \overset{1}{\underset{11}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 9 & 6 & 7 \\ 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$

[1474] $1 \overset{1}{\underset{11}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[1473] $1 \overset{1}{\underset{11}{5}} 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 5 & 2 & 3 \end{pmatrix}_{b_0}$

[1475] $1 \overset{1}{\underset{11}{12}} 5 131^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 8 & 14 & 42 & 12 & 14 \\ 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$

[1476] $1 \overset{1}{\underset{11}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[1477] $1 \overset{1}{\underset{11}{11}} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 9 & 6 & 7 \\ 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$

[1478] $1 \overset{1}{\underset{11}{13}} 5 131^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 12 & 21 & 97 & 18 & 21 \\ 3 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[1479] $1 \overset{1}{\underset{12}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 5 & 11 & 0 & 5 \\ 6 & 11 & 22 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 10 & -1 & 4 \end{pmatrix}_{b_0}$

[1480] $2 \overset{1}{\underset{11}{3}} 1 \overset{1}{\underset{11}{5}} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 9 & 6 & 7 \\ 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$

[1481] $1 \overset{1}{\underset{11}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 2 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[1482] $21 \overset{1}{\underset{11}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 9 & 6 & 7 \\ 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[1493] $1^{11} 5^{11} 1^{21} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 15$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 9 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 81 & 12 & 17 \\ 1 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & -1 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

[1494] $1^{11} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 16$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 9 & 6 & 7 \\ 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[1495] $1^{11} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 17$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 5 & 8 & 0 & 5 \\ 4 & 8 & 9 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

[1496] $1^{11} 5^{11} 1^{21} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 19$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 6 & 0 & 31 & 8 & 11 \\ 1 & 1 & 0 & 8 & 0 & 2 \\ 2 & 2 & -1 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[1497] $1^{13} 5^{11} 1^{21} 231^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 19$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 12 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 12 & 0 & 115 & 16 & 21 \\ 1 & 1 & 0 & 16 & 0 & 2 \\ 2 & 2 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1498] $2^3 3^1 5^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 20$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 42 & 12 & 14 \\ 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1499] $1^{13} 5^{11} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 21$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 5 & 19 & 0 & 5 \\ 10 & 19 & 65 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 17 & -1 & 4 \end{pmatrix}_{b_0}$$

[1500] $21^{12} 5^{11} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 22$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1501] $2^3 3^1 5^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 22$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1502] $1^{12} 5^{14} 131^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 24$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 12 & 14 & 42 & 12 & 14 \\ 2 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1503] $1^{12} 5^1 1^{23} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 9 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 9 & 0 & 66 & 12 & 16 \\ 1 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & -1 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1504] $1^{12} 5^1 1^{31} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 5 & 21 & 0 & 5 \\ 10 & 21 & 70 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 4 \end{pmatrix}_{b_0}$$

[1505] $1^{12} 5^1 1^{31} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 5 & 16 & 0 & 5 \\ 8 & 16 & 42 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 14 & -1 & 4 \end{pmatrix}_{b_0}$$

[1506] $1^{12} 5^1 1^{31} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 42 & 12 & 14 \\ 2 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1507] $1^{12} 5^1 1^{31} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1508] $1^{11} 5^1 1^{31} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 5 & 6 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[1509] $1^{12} 5^1 1^{23} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[1510] $1^{12} 5^1 1^{31} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 42 & 12 & 14 \\ 2 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1511] $1^{13} 5^1 1^{31} 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 14 & 21 & 97 & 18 & 21 \\ 2 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[1514] $1^{12} 5^1 1^{31} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 5 & 16 & 5 & 5 \\ 8 & 16 & 42 & 12 & 14 \\ 3 & 5 & 12 & 1 & 3 \\ 3 & 5 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1515] $1^{12} 5^1 1^{31} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 4 & 8 & 9 & 6 & 7 \\ 3 & 5 & 6 & 1 & 3 \\ 3 & 5 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1513] \quad 1^1 \bar{5} 131^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 5 & 24 & 5 & 5 \\ 12 & 24 & 97 & 18 & 21 \\ 3 & 5 & 18 & 1 & 3 \\ 3 & 5 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1512] \quad 1^1 \bar{5} 131^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 5 & 32 & 5 & 5 \\ 16 & 32 & 174 & 24 & 28 \\ 3 & 5 & 24 & 1 & 3 \\ 3 & 5 & 28 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1516] \quad 2 \bar{3} 1 \bar{5} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 97 & 18 & 21 \\ 3 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1517] \quad 1^1 \bar{5} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 5 & 27 & 0 & 5 \\ 14 & 27 & 130 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 24 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1519] \quad 1^{12} \bar{5} 1^3 231^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 1 & 15 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 15 & 0 & 218 & 20 & 28 \\ 1 & 1 & 0 & 20 & 0 & 2 \\ 2 & 2 & -1 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1518] \quad 1^{12} \bar{5} 1^2 231^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 1 & 12 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 130 & 16 & 22 \\ 1 & 1 & 0 & 16 & 0 & 2 \\ 2 & 2 & -1 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1520] \quad 1^1 \bar{5} 1^1 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 5 & 15 & 0 & 5 \\ 8 & 15 & 37 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 13 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1521] \quad 2 \bar{3} 1 \bar{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 5 & 5 & 3 & 5 \\ 3 & 5 & 2 & 2 & 4 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1522] \quad 1^1 \bar{5} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 5 & 24 & 0 & 5 \\ 12 & 24 & 97 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 21 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1523] \quad 1^1 \bar{5} 1^1 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 5 & 15 & 5 & 5 \\ 8 & 15 & 37 & 11 & 13 \\ 3 & 5 & 11 & 1 & 3 \\ 3 & 5 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

[1524] $1^{11} 5^1 1^1 231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 4 & 5 & 5 & 18 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 8 & -1 & 3 \end{pmatrix}_{b_0}$$

[1525] $1^{11} 5^1 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 5 & 29 & 0 & 5 \\ 14 & 29 & 137 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -1 & 4 \end{pmatrix}_{b_0}$$

[1526] $1^{11} 5^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 4 & 5 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[1527] $1^{11} 5^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 5 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[1528] $1^{13} 5^1 131^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 97 & 18 & 21 \\ 2 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[1529] $2^1 3^1 15^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 5 & 10 & 3 & 5 \\ 6 & 10 & 14 & 4 & 8 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$$

[1530] $21^{11} 5^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1532] $1^{11} 5^1 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 7 & 8 & 11 & 72 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1531] $1^{11} 5^1 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 4 & 5 & 8 & 24 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[1533] $1^{13} 5^1 131^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 16 & 21 & 97 & 18 & 21 \\ 2 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[1534] $1^{13} 5 1^{23} 1^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 1 & 15 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 15 & 0 & 193 & 20 & 27 \\ 1 & 1 & 0 & 20 & 0 & 2 \\ 2 & 2 & -1 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[1535] $1^{11} 5 1^{23} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 5 & 20 & 0 & 5 \\ 10 & 20 & 63 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 17 & -1 & 4 \end{pmatrix}_{b_0}$$

[1536] $1^{112} 3 2 2 1^{15} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 3 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 11 & 3 \\ 6 & 11 & 0 & 11 & 49 & 13 \\ 2 & 3 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[1537] $1^{11} 5 1^{23} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 5 & 25 & 0 & 5 \\ 12 & 25 & 97 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 21 & -1 & 4 \end{pmatrix}_{b_0}$$

[1538] $1^{11} 5 1^{14} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 5 & 32 & 0 & 5 \\ 16 & 32 & 174 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 28 & -1 & 4 \end{pmatrix}_{b_0}$$

[1539] $2^{13} 3 1 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 5 & 15 & 3 & 5 \\ 9 & 15 & 34 & 6 & 12 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

[1540] $2^{11} 3 1 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1541] $1^{11} 5 1^{12} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 5 & 6 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[1542] $1^{11} 5 1^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 6 & 7 & 10 & 52 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[1543] $1^{12} 5 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 4 & 6 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[1544] $1^{12} 5 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 6 & 8 & 8 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[1545] $1^{\overline{13}} 5^{\overline{1}} 1^{\overline{3}} 231^{\overline{31}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 75$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 1 & 18 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 18 & 0 & 295 & 24 & 33 \\ 1 & 1 & 0 & 24 & 0 & 2 \\ 2 & 2 & -1 & 33 & 2 & 3 \end{pmatrix}_{b_0}$$

[1546] $2^{\overline{11}} 3^{\overline{11}} 15^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 78$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1548] $1^{\overline{11}} 5^{\overline{1}} 1^{\overline{2}} 231^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 80$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 5 & 30 & 5 & 5 \\ 16 & 30 & 154 & 22 & 26 \\ 3 & 5 & 22 & 1 & 3 \\ 3 & 5 & 26 & 3 & 4 \end{pmatrix}_{b_0}$$

[1547] $1^{\overline{12}} 5^{\overline{1}} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 80$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 5 & 7 & 7 & 33 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[1549] $1^{\overline{11}} 5^{\overline{1}} 131^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 85$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 7 & 8 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[1550] $1^{\overline{11}} 5^{\overline{1}} 131^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 85$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 8 & 9 & 14 & 97 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1551] $2^{\overline{14}} 3^{\overline{11}} 15^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 86$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 5 & 20 & 3 & 5 \\ 12 & 20 & 62 & 8 & 16 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0}$$

[1552] $1^{\overline{11}} 5^{\overline{1}} 1^{\overline{2}} 231^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 88$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 5 & 6 & 8 & 31 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[1553] $2^{\overline{11}} 3^{\overline{11}} 15^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 89$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 5 & 5 & 8 & 5 \\ 3 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[1554] $2^{\overline{12}} 3^{\overline{11}} 15^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 98$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1555] \ 2 \overset{1}{3} \overset{12}{15} \overset{1}{1} \overset{11}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1556] \ 1 \overset{1}{1} \overset{1}{5} \overset{1}{1} \overset{12}{231} \overset{12}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 6 & 7 & 11 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1557] \ 1 \overset{12}{1} \overset{1}{5} \overset{1}{131} \overset{12}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 5 & 7 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1558] \ 21 \overset{13}{1} \overset{1}{5} \overset{1}{131} \overset{14}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[1559] \ 2 \overset{15}{3} \overset{1}{1} \overset{1}{5} \overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 5 & 25 & 3 & 5 \\ 15 & 25 & 98 & 10 & 20 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 5 & 20 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1560] \ 2 \overset{131}{3} \overset{1}{1} \overset{1}{5} \overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 3 \\ 4 & 5 & 15 & 8 & 5 \\ 9 & 15 & 34 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1561] \ 21 \overset{12}{1} \overset{1}{5} \overset{1}{131} \overset{12}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1562] \ 21 \overset{12}{1} \overset{1}{5} \overset{1}{131} \overset{11}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1563] \ 1 \overset{1}{1} \overset{1}{5} \overset{1}{1} \overset{2}{231} \overset{12}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 5 & 30 & 0 & 5 \\ 16 & 30 & 154 & 0 & 26 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 26 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1565] \ 1 \overset{12}{1} \overset{1}{5} \overset{1}{131} \overset{12}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 6 & 8 & 10 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[1564] $1^{12} 5^{14} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 10 & 12 & 90 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1566] $2^1 3^{141} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 3 \\ 4 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[1568] $1^{15} 5^{13} 231^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 & 3 \\ 3 & 5 & 45 & 5 & 5 \\ 24 & 45 & 349 & 33 & 39 \\ 3 & 5 & 33 & 1 & 3 \\ 3 & 5 & 39 & 3 & 4 \end{pmatrix}_{b_0}$$

[1567] $1^{12} 5^{13} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 7 & 9 & 11 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1569] $1^{11} 5^{12} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 9 & 12 & 81 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[1570] $2^1 3^{16} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 5 & 30 & 3 & 5 \\ 18 & 30 & 142 & 12 & 24 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 5 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

[1571] $1^{12} 5^{12} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 7 & 9 & 13 & 73 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[1572] $2^1 3^{151} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.1$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 3 \\ 4 & 5 & 25 & 8 & 5 \\ 15 & 25 & 98 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[1573] $2^1 3^{112} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 5 & 5 & 13 & 5 \\ 3 & 5 & 2 & 10 & 4 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[1574] $1^{15} 5^{12} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 5 & 35 & 0 & 5 \\ 18 & 35 & 202 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 30 & -1 & 4 \end{pmatrix}_{b_0}$$

[1575] $1^{11} 3221^{11} 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 1 & 2 \\ 4 & 18 & 6 & 25 & 0 & 8 \\ 1 & 6 & 0 & 8 & 0 & 2 \\ 5 & 25 & 8 & 31 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[1576] $2 \overset{12}{3} 1 \overset{1}{5} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1577] $1 \overset{12}{5} 1 \overset{1}{3} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 8 & 10 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1578] $2 \overset{17}{3} 1 \overset{1}{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 5 & 35 & 3 & 5 \\ 21 & 35 & 194 & 14 & 28 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 5 & 28 & 2 & 4 \end{pmatrix}_{b_0}$$

[1579] $1 \overset{13}{5} 1 \overset{1}{2} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 7 & 10 & 11 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1580] $1 \overset{11}{5} 1 \overset{1}{2} 2 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 9 & 10 & 15 & 108 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

[1581] $1 \overset{12}{5} 1 \overset{1}{4} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 9 & 11 & 15 & 117 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[1582] $1 \overset{11}{5} 1 \overset{1}{2} 2 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 5 & 40 & 0 & 5 \\ 20 & 40 & 258 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 34 & -1 & 4 \end{pmatrix}_{b_0}$$

[1583] $1 \overset{12}{5} 1 \overset{1}{3} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 9 & 11 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1584] $1 \overset{13}{5} 1 \overset{1}{3} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 11 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1585] $1 \overset{13}{5} 1 \overset{1}{4} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 9 & 12 & 13 & 112 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

[1586] $2 \overset{12}{3} 1 \overset{11}{5} 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 217, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1587] $22 \overset{111}{3} 151^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 8 \\ 5 & 13 & 9 & 11 & 7 \\ 5 & 13 & 11 & 8 & 6 \\ 3 & 8 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

[1589] $1 \overset{12}{1} 5 \overset{11}{1} 2 \overset{11}{2} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 9 & 11 & 13 & 101 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1588] $21 \overset{13}{1} 5 \overset{11}{1} 131^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1590] $1 \overset{12}{1} 5 \overset{11}{1} 131^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 10 & 12 & 18 & 150 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1591] $1 \overset{13}{1} 5 \overset{11}{1} 131^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 8 & 11 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1592] $21 \overset{13}{1} 5 \overset{11}{1} 131^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1593] $1 \overset{11}{1} 5 \overset{11}{1} 3 \overset{11}{2} 31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 5 & 45 & 0 & 5 \\ 24 & 45 & 349 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 39 & -1 & 4 \end{pmatrix}_{b_0}$$

[1594] $2 \overset{13}{3} 1 \overset{11}{5} 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 3 & 3 & 1 & 16 & 3 \\ 13 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1595] $1 \overset{11}{1} 5 \overset{11}{1} 3 \overset{11}{2} 31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 19 & 0 & 2 \\ 12 & 13 & 19 & 190 & 0 & 26 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 26 & -1 & 3 \end{pmatrix}_{b_0}$$

[1596] $22 \ 3 \ 151^{\overset{1}{1}\overset{1}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 5 & 5 & 3 \\ 5 & 9 & 12 & 11 & 7 \\ 5 & 12 & 8 & 11 & 6 \\ 5 & 11 & 11 & 9 & 7 \\ 3 & 7 & 6 & 7 & 4 \end{pmatrix}_{b_0}$

[1597] $1^{\overset{1}{1}\overset{1}{3}} 5 131^{\overset{1}{1}\overset{1}{3}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 277, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 9 & 12 & 15 & 115 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 21 & -1 & 3 \end{pmatrix}_{b_0}$

[1598] $1^{\overset{1}{1}\overset{1}{4}} 5 131^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 10 & 13 & 16 & 141 & 0 & 23 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 23 & -1 & 3 \end{pmatrix}_{b_0}$

[1599] $1^{\overset{1}{1}\overset{1}{2}} 5 1^{\overset{1}{1}\overset{1}{2}} 231^{\overset{1}{1}\overset{1}{2}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 10 & 12 & 16 & 130 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$

[1600] $1^{\overset{1}{1}\overset{1}{2}} 5 1^{\overset{1}{1}\overset{1}{3}} 231^{\overset{1}{1}\overset{1}{3}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 14 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 12 & 14 & 17 & 181 & 0 & 25 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 25 & -1 & 3 \end{pmatrix}_{b_0}$

[1601] $1^{\overset{1}{1}\overset{1}{3}} 5 131^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 10 & 13 & 18 & 148 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$

[1602] $1^{\overset{1}{1}\overset{1}{5}} 1^{\overset{1}{1}\overset{1}{3}} 231^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 26 & 1 & 3 \\ 3 & 5 & 50 & 0 & 5 \\ 26 & 50 & 419 & 0 & 43 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 43 & -1 & 4 \end{pmatrix}_{b_0}$

[1603] $22 \ 3 \ 151^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 330, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 3 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$

[1604] $22 \ 3 \ 151^{\overset{1}{1}\overset{1}{2}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 5 & 10 & 3 \\ 6 & 14 & 13 & 26 & 8 \\ 5 & 13 & 9 & 22 & 7 \\ 10 & 26 & 22 & 41 & 13 \\ 3 & 8 & 7 & 13 & 4 \end{pmatrix}_{b_0}$

[1605] $1^{\overset{1}{1}\overset{1}{2}} 5 1^{\overset{1}{1}\overset{1}{2}} 231^{\overset{1}{1}\overset{1}{3}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 19 & 0 & 2 \\ 11 & 13 & 19 & 165 & 0 & 25 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 25 & -1 & 3 \end{pmatrix}_{b_0}$

[1606] $1^{\overset{1}{1}\overset{1}{3}} 5 131^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$\begin{aligned}
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 14 & 0 & 2 \\ 1 & 2 & 0 & 19 & 0 & 2 \\ 11 & 14 & 19 & 176 & 0 & 26 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 26 & -1 & 3 \end{pmatrix}_{b_0} \\
[1607] & 1^{12} 3 2 2 1^{12} 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 10 & 1 & 2 \\ 4 & 18 & 6 & 50 & 0 & 8 \\ 1 & 6 & 0 & 16 & 0 & 2 \\ 10 & 50 & 16 & 130 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[1608] & 1^{13} 2 3 1 5 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 363, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 3 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0} \\
[1609] & 1^{121} 2 2 3 1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 398, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 10 & 5 & 3 \\ 6 & 14 & 26 & 13 & 8 \\ 10 & 26 & 42 & 22 & 14 \\ 5 & 13 & 22 & 8 & 6 \\ 3 & 8 & 14 & 6 & 4 \end{pmatrix}_{b_0} \\
[1610] & 1^{12} 5 1^1 3 2 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 15 & 0 & 2 \\ 1 & 2 & 0 & 20 & 0 & 2 \\ 13 & 15 & 20 & 218 & 0 & 28 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 28 & -1 & 3 \end{pmatrix}_{b_0} \\
[1611] & 1^{13} 5 1^1 3 2 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 16 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 13 & 16 & 18 & 211 & 0 & 27 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 27 & -1 & 3 \end{pmatrix}_{b_0} \\
[1612] & 1^{14} 2 3 1 5 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 3 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0} \\
[1613] & 1^{12} 5 1^1 3 2 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 1 & 2 \\ 1 & 0 & 2 & 16 & 0 & 2 \\ 1 & 2 & 0 & 23 & 0 & 2 \\ 14 & 16 & 23 & 261 & 0 & 31 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 31 & -1 & 3 \end{pmatrix}_{b_0} \\
[1614] & 1^{12} 5 1^1 3 2 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 549, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 2 & 17 & 0 & 2 \\ 1 & 2 & 0 & 26 & 0 & 2 \\ 15 & 17 & 26 & 310 & 0 & 34 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 34 & -1 & 3 \end{pmatrix}_{b_0} \\
[1615] & 1^{13} 5 1^1 3 2 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 1 & 2 \\ 1 & 0 & 2 & 17 & 0 & 2 \\ 1 & 2 & 0 & 21 & 0 & 2 \\ 14 & 17 & 21 & 250 & 0 & 30 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 30 & -1 & 3 \end{pmatrix}_{b_0} \\
[1616] & 1^{13} 5 1^1 3 2 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 664, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 2 & 18 & 0 & 2 \\ 1 & 2 & 0 & 24 & 0 & 2 \\ 15 & 18 & 24 & 295 & 0 & 33 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 33 & -1 & 3 \end{pmatrix}_{b_0} \\
[1617] & 1^{13}3221^{13}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 15 & 1 & 2 \\ 4 & 18 & 6 & 75 & 0 & 8 \\ 1 & 6 & 0 & 24 & 0 & 2 \\ 15 & 75 & 24 & 295 & 0 & 33 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & 33 & -1 & 3 \end{pmatrix}_{b_0} \\
[1618] & 22 \ 3 \ 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 683, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 15 & 5 & 3 \\ 6 & 14 & 39 & 13 & 8 \\ 15 & 39 & 97 & 33 & 21 \\ 5 & 13 & 33 & 8 & 6 \\ 3 & 8 & 21 & 6 & 4 \end{pmatrix}_{b_0} \\
[1619] & 1^{13}51^13231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 748, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 1 & 2 \\ 1 & 0 & 2 & 19 & 0 & 2 \\ 1 & 2 & 0 & 27 & 0 & 2 \\ 16 & 19 & 27 & 346 & 0 & 36 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 36 & -1 & 3 \end{pmatrix}_{b_0} \\
[1620] & 22 \ 3 \ 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 1082, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 20 & 5 & 3 \\ 6 & 14 & 52 & 13 & 8 \\ 20 & 52 & 174 & 44 & 28 \\ 5 & 13 & 44 & 8 & 6 \\ 3 & 8 & 28 & 6 & 4 \end{pmatrix}_{b_0} \\
[1621] & 22 \ 3 \ 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 1595, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 25 & 5 & 3 \\ 6 & 14 & 65 & 13 & 8 \\ 25 & 65 & 273 & 55 & 35 \\ 5 & 13 & 55 & 8 & 6 \\ 3 & 8 & 35 & 6 & 4 \end{pmatrix}_{b_0} \\
T = 11 & [1622] 1^{13}31^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[1623] & 1^{11}31^15131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0} \\
[1624] & 1^{11}51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0} \\
[1625] & 2321^161^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[1626] & 1^{11}31^15131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -37, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 6 & -1 & 4 \end{pmatrix}_{b_0}$$

[1628] $2 \overset{11}{3} 21 \overset{11}{6} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1627] $2 \overset{11}{3} 21 \overset{11}{6} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 1 & 7 & 3 \\ 7 & 6 & 7 & 19 & 9 \\ 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[1629] $2 \overset{11}{3} 21 \overset{11}{6} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -57, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 3 & 3 & 1 & 9 & 3 \\ 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[1630] $2 \overset{11}{3} 21 \overset{11}{6} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1631] $2 \overset{11}{3} 21 \overset{11}{6} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -78, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 11 & 3 \\ 10 & 8 & 11 & 37 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[1632] $41 \overset{11}{2} 31 \overset{11}{5} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 8 & 1 & 3 \\ 9 & 51 & 48 & 0 & 15 \\ 8 & 48 & 42 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 15 & 14 & -1 & 4 \end{pmatrix}_{b_0}$$

[1633] $2 \overset{112}{3} 21 \overset{11}{6} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1634] $2 \overset{112}{3} 21 \overset{11}{6} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 15 & 3 \\ 13 & 10 & 15 & 63 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1635] $41 \overset{112}{2} 31 \overset{11}{5} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 16 & 1 & 3 \\ 9 & 51 & 96 & 0 & 15 \\ 16 & 96 & 174 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 15 & 28 & -1 & 4 \end{pmatrix}_{b_0}$$

[1636] $2 \overset{11}{3} 21 \overset{11}{6} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -175, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 11 & 3 \\ 3 & 3 & 1 & 13 & 3 \\ 12 & 11 & 13 & 56 & 16 \\ 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[1637] $41 \overset{11}{2} 31 \overset{11}{5} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -186, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 24 & 1 & 3 \\ 9 & 51 & 144 & 0 & 15 \\ 24 & 144 & 394 & 0 & 42 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 15 & 42 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1638] \ 41^{1^4} 231 51^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -270, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 32 & 1 & 3 \\ 9 & 51 & 192 & 0 & 15 \\ 32 & 192 & 702 & 0 & 56 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 15 & 56 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1639] \ 2^1 3^{1^2} 21^{1^1} 61^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -277, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 17 & 3 \\ 15 & 13 & 17 & 86 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1640] \ 1^{1^1} 31^1 5^{1^1} 13221^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 12, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[1642] \ 1^{1^1} 6^{1^1} 131^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1641] \ 1^{1^1} 6^{1^1} 12321^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1643] \ 1^{1^1} 31^1 5^{1^1} 1321^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 12, \Delta = -9, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 1 & 5 & 0 & 3 \\ 2 & 4 & 5 & 7 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & 7 & -1 & 5 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [1644] \ 1^{1^1} 6^{1^1} 131^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 13, T_{\max} = 13, \Delta = -8, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [1645] \ 1^{1^1} 6^{1^1} 131^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 6.0$$

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$$T = 8 \quad [0] \ 1^{1^3} 31^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 12 & 13 & 0 & 4 \\ 6 & 12 & 25 & 27 & 0 & 9 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1^1} 31^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 3 \\ 4 & 9 & 12 & 4 & 0 & 5 \\ 5 & 12 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 7 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 9 & 17 & 15 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 4 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 11 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 15 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{121}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{122}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 1 & 3 \\ 3 & 4 & 18 & 8 & 0 & 4 \\ 10 & 18 & 58 & 24 & 0 & 14 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 14 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 10 & 24 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 14 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{122}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 10 & 18 & 0 & 6 \\ 7 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 11 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{14}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 11 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{112}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 8 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 11 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{13}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{112}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 7 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{122}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 28 & 18 & 0 & 10 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{113}3^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 15 & 36 & 133 & 18 & 0 & 21 \\ 2 & 4 & 13 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 21 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{112}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 18 & 12 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 9 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 11 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 15 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 7 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 9 & 17 & 27 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{113}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 13 & 32 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 18 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 9 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{132}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 1 & 3 \\ 3 & 4 & 22 & 8 & 0 & 4 \\ 12 & 22 & 85 & 30 & 0 & 17 \\ 4 & 8 & 30 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 17 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 28 & 21 & 0 & 10 \\ 5 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{114}31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 1 & 3 \\ 4 & 9 & 48 & 4 & 0 & 5 \\ 20 & 48 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 28 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 14 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{121} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 16 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 22 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{141} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 11 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 12 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{121} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 13 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 18 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 14 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 19 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 28 & 30 & 0 & 10 \\ 7 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 18 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 25 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{141} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 11 & 21 & 76 & 15 & 0 & 16 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{132} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 9 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{114} 31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 16 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 22 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{131} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 85 & 15 & 0 & 17 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 17 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{131} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 27 & 4 & 0 & 4 \\ 15 & 27 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 21 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{131} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 1 & 3 \\ 3 & 4 & 32 & 4 & 0 & 4 \\ 18 & 32 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 25 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{141} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 26 & 4 & 0 & 4 \\ 14 & 26 & 118 & 18 & 0 & 20 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 20 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{141} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 31 & 4 & 0 & 4 \\ 17 & 31 & 172 & 21 & 0 & 24 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 24 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{141} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 1 & 3 \\ 3 & 4 & 36 & 4 & 0 & 4 \\ 20 & 36 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 28 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

4.953 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.954 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.955 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & 3 & -4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 & 3 \\ 8 & 14 & 9 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 3 & 10 & -4 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1211} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 5 & 3 \\ 3 & 4 & 8 & 9 & 0 & 9 & 4 \\ 4 & 8 & 10 & 12 & 0 & 12 & 6 \\ 5 & 9 & 12 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 19 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 21 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{131} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 & 4 \\ 3 & 5 & 4 & 15 & 0 & 3 & 4 \\ 9 & 17 & 15 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 12 & 0 & 1 & 3 \\ 3 & 4 & 4 & 12 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1211} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 5 & 3 \\ 3 & 4 & 8 & 4 & 0 & 9 & 4 \\ 4 & 8 & 10 & 6 & 0 & 12 & 6 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 6 & 0 & 13 & 7 \\ 3 & 4 & 6 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^1 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 5 & 3 \\ 3 & 4 & 19 & 4 & 0 & 9 & 4 \\ 11 & 19 & 73 & 12 & 0 & 33 & 15 \\ 2 & 4 & 12 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 33 & 6 & 0 & 13 & 7 \\ 3 & 4 & 15 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1112} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -137, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 7 & 3 \\ 3 & 4 & 14 & 4 & 0 & 13 & 4 \\ 8 & 14 & 37 & 9 & 0 & 33 & 11 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 33 & 9 & 0 & 28 & 10 \\ 3 & 4 & 11 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 4 & 2 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1211} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 5 & 3 \\ 4 & 9 & 8 & 4 & 0 & 12 & 5 \\ 3 & 8 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 9 & 6 & 0 & 13 & 7 \\ 3 & 5 & 4 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1212} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 13 & 3 \\ 3 & 4 & 13 & 4 & 0 & 23 & 4 \\ 7 & 13 & 28 & 9 & 0 & 54 & 10 \\ 2 & 4 & 9 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 54 & 15 & 0 & 100 & 18 \\ 3 & 4 & 10 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 3 & 5 & 4 & 12 & 0 & 3 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 9 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{111}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{111}} 31^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -353, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 14 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 15 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -359, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 5 & 3 \\ 4 & 9 & 12 & 4 & 0 & 12 & 5 \\ 5 & 12 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 6 & 0 & 13 & 7 \\ 3 & 5 & 7 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{\overline{112}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overline{112}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{\overline{111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 3 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{114} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 21 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1212} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 4 & 3 \\ 3 & 4 & 8 & 4 & 0 & 8 & 4 \\ 4 & 8 & 10 & 6 & 0 & 12 & 6 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 6 & 0 & 10 & 6 \\ 3 & 4 & 6 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -425, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1212} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 9 & 0 & 13 & 4 \\ 4 & 8 & 10 & 12 & 0 & 18 & 6 \\ 5 & 9 & 12 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 12 & 3 \\ 3 & 4 & 14 & 4 & 0 & 22 & 4 \\ 8 & 14 & 37 & 9 & 0 & 57 & 11 \\ 2 & 4 & 9 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 57 & 15 & 0 & 85 & 17 \\ 3 & 4 & 11 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 5 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1212} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 7 & 3 \\ 3 & 4 & 8 & 4 & 0 & 13 & 4 \\ 4 & 8 & 10 & 6 & 0 & 18 & 6 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 9 & 0 & 28 & 10 \\ 3 & 4 & 6 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 8 & 3 \\ 3 & 4 & 9 & 4 & 0 & 14 & 4 \\ 5 & 9 & 13 & 6 & 0 & 24 & 7 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 24 & 9 & 0 & 37 & 11 \\ 3 & 4 & 7 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 5 & 4 & 0 & 28 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 30 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 6 & 0 & 13 & 7 \\ 3 & 4 & 7 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 8 & 3 \\ 3 & 4 & 14 & 4 & 0 & 14 & 4 \\ 8 & 14 & 37 & 9 & 0 & 39 & 11 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 39 & 9 & 0 & 37 & 11 \\ 3 & 4 & 11 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1112} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 5 & 4 & 0 & 26 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 24 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 5 & 4 & 0 & 23 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 24 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 21 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 11 & 3 \\ 3 & 4 & 9 & 4 & 0 & 21 & 4 \\ 5 & 9 & 13 & 6 & 0 & 33 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 15 & 0 & 76 & 16 \\ 3 & 4 & 7 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 13 & 4 \\ 5 & 9 & 13 & 6 & 0 & 21 & 7 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 9 & 0 & 28 & 10 \\ 3 & 4 & 7 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 10 & 3 \\ 3 & 4 & 14 & 4 & 0 & 18 & 4 \\ 8 & 14 & 37 & 9 & 0 & 48 & 11 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 48 & 12 & 0 & 58 & 14 \\ 3 & 4 & 11 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 16 & 3 \\ 3 & 4 & 9 & 4 & 0 & 28 & 4 \\ 5 & 9 & 13 & 6 & 0 & 48 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 48 & 18 & 0 & 154 & 22 \\ 3 & 4 & 7 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 4 & 0 & 17 & 4 \\ 5 & 9 & 13 & 6 & 0 & 27 & 7 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 12 & 0 & 49 & 13 \\ 3 & 4 & 7 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 1 & 6 & 0 & 12 & 3 \\ 5 & 9 & 6 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 3 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 21 & 4 \\ 2 & 4 & 1 & 6 & 0 & 15 & 3 \\ 5 & 9 & 6 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 33 & 0 & 76 & 16 \\ 3 & 4 & 3 & 6 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1565, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{1212} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1664, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 10 & 3 \\ 3 & 4 & 13 & 4 & 0 & 18 & 4 \\ 7 & 13 & 28 & 9 & 0 & 42 & 10 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 42 & 12 & 0 & 58 & 14 \\ 3 & 4 & 10 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{1113} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1709, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{1112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1760, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 10 & 3 \\ 3 & 4 & 9 & 4 & 0 & 18 & 4 \\ 5 & 9 & 13 & 6 & 0 & 30 & 7 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 30 & 12 & 0 & 58 & 14 \\ 3 & 4 & 7 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{1112} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1781, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 13 & 3 \\ 3 & 4 & 9 & 4 & 0 & 23 & 4 \\ 5 & 9 & 13 & 6 & 0 & 39 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 39 & 15 & 0 & 100 & 18 \\ 3 & 4 & 7 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{1212} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1784, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 7 & 3 \\ 3 & 4 & 13 & 4 & 0 & 13 & 4 \\ 7 & 13 & 28 & 9 & 0 & 30 & 10 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 30 & 9 & 0 & 28 & 10 \\ 3 & 4 & 10 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{1113} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1820, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{113} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1865, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 3 & 0 & 33 & 4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{1114} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 30 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1113} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 12 & 3 \\ 3 & 4 & 9 & 4 & 0 & 22 & 4 \\ 5 & 9 & 13 & 6 & 0 & 36 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 36 & 15 & 0 & 85 & 17 \\ 3 & 4 & 7 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1212} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 10 & 3 \\ 3 & 4 & 18 & 4 & 0 & 18 & 4 \\ 10 & 18 & 58 & 12 & 0 & 60 & 14 \\ 2 & 4 & 12 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 60 & 12 & 0 & 58 & 14 \\ 3 & 4 & 14 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{1114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 36 & 4 \\ 3 & 5 & 4 & 3 & 0 & 36 & 4 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1114} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 3 & 0 & 24 & 3 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1114} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 14 & 3 \\ 3 & 4 & 9 & 4 & 0 & 26 & 4 \\ 5 & 9 & 13 & 6 & 0 & 42 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 42 & 18 & 0 & 118 & 20 \\ 3 & 4 & 7 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{1114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3017$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1113}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 15 & 3 \\ 3 & 4 & 9 & 4 & 0 & 27 & 4 \\ 5 & 9 & 13 & 6 & 0 & 45 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 45 & 18 & 0 & 133 & 21 \\ 3 & 4 & 7 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{1113}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3365$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 18 & 3 \\ 3 & 4 & 9 & 4 & 0 & 32 & 4 \\ 5 & 9 & 13 & 6 & 0 & 54 & 7 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 54 & 21 & 0 & 193 & 25 \\ 3 & 4 & 7 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -4373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 17 & 3 \\ 3 & 4 & 9 & 4 & 0 & 31 & 4 \\ 5 & 9 & 13 & 6 & 0 & 51 & 7 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 51 & 21 & 0 & 172 & 24 \\ 3 & 4 & 7 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{1114}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -5216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 20 & 3 \\ 3 & 4 & 9 & 4 & 0 & 36 & 4 \\ 5 & 9 & 13 & 6 & 0 & 60 & 7 \\ 2 & 4 & 6 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 60 & 24 & 0 & 238 & 28 \\ 3 & 4 & 7 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

4.956 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10$ [0] $1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 220$, $(n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 24 & 11 \\ 5 & 9 & 6 & 6 & 0 & 24 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1112} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 18 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1113} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 27 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 21 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 36 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 796, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 21 & 7 \\ 7 & 13 & 12 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 988, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1032, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 6 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 6 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1122} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 30 & 10 \\ 7 & 13 & 12 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1180, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 30 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2368, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 15 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 7 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.957 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] 1^{\overline{1111}} 31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 12 & 4 & 0 & 12 & 4 & 5 \\ 5 & 12 & 13 & 6 & 0 & 15 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 15 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 3 & 0 & 6 & 0 & 2 \\ 3 & 5 & 7 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 651, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] 1^{\overline{1141}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 813, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 21 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 18 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] 1^{\overline{1121}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] 1^{\overline{1131}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 17 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 15 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.958 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{e}_6$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 & 3 \\ 3 & 4 & 9 & 8 & 0 & 5 & 4 \\ 5 & 9 & 13 & 12 & 0 & 9 & 7 \\ 4 & 8 & 12 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 9 & 6 & 0 & 3 & 3 \\ 3 & 4 & 7 & 5 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 2 & 3 \\ 4 & 9 & 36 & 4 & 0 & 4 & 5 \\ 14 & 36 & 121 & 15 & 0 & 15 & 19 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 5 & 19 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 2 & 3 \\ 4 & 9 & 12 & 4 & 0 & 4 & 5 \\ 5 & 12 & 13 & 6 & 0 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 5 & 7 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 2 & 3 \\ 4 & 9 & 20 & 4 & 0 & 4 & 5 \\ 8 & 20 & 37 & 9 & 0 & 9 & 11 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 5 & 11 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 2 & 3 \\ 4 & 9 & 28 & 4 & 0 & 4 & 5 \\ 11 & 28 & 73 & 12 & 0 & 12 & 15 \\ 2 & 4 & 12 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 3 & 0 & 0 & 2 \\ 3 & 5 & 15 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{122}} 3^1 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 4 & 3 \\ 3 & 4 & 13 & 5 & 0 & 8 & 4 \\ 7 & 13 & 28 & 12 & 0 & 18 & 10 \\ 3 & 5 & 12 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 18 & 6 & 0 & 9 & 5 \\ 3 & 4 & 10 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{132}} 3^1 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 4 & 3 \\ 3 & 4 & 17 & 5 & 0 & 8 & 4 \\ 9 & 17 & 49 & 15 & 0 & 24 & 13 \\ 3 & 5 & 15 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 24 & 6 & 0 & 9 & 5 \\ 3 & 4 & 13 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 2 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 & 4 \\ 8 & 14 & 37 & 9 & 0 & 9 & 11 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 4 & 11 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{113}} 3^1 1^{\overline{211}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 2 & 3 \\ 4 & 9 & 32 & 4 & 0 & 4 & 5 \\ 13 & 32 & 100 & 15 & 0 & 15 & 18 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 5 & 18 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{111}} 3^1 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{1111}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{114}} 3^1 1^{\overline{311}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 2 & 3 \\ 4 & 9 & 44 & 4 & 0 & 4 & 5 \\ 18 & 44 & 193 & 21 & 0 & 21 & 25 \\ 2 & 4 & 21 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 3 & 0 & 0 & 2 \\ 3 & 5 & 25 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -645, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 \\ 5 & 9 & 13 & 9 & 0 & 6 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 4 & 7 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -651, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 31^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -690, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 2 & 3 \\ 4 & 9 & 40 & 4 & 0 & 4 & 5 \\ 16 & 40 & 154 & 18 & 0 & 18 & 22 \\ 2 & 4 & 18 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 3 & 0 & 0 & 2 \\ 3 & 5 & 22 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{141} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 2 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 & 4 \\ 11 & 21 & 76 & 18 & 0 & 15 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 4 & 16 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -789, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 \\ 5 & 9 & 13 & 6 & 0 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 2 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 & 4 \\ 7 & 13 & 28 & 12 & 0 & 9 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 4 & 10 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 2 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 & 4 \\ 9 & 17 & 49 & 15 & 0 & 12 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 3 & 0 & 0 & 2 \\ 3 & 4 & 13 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 6 & 0 & 12 & 6 \\ 3 & 4 & 7 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.959 NHC(-2)(-3) + su₂ + su₃ + e₆ + su₃ + su₂

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 31^{\overline{1111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 12 & 4 & 0 & 4 & 12 & 5 \\ 5 & 12 & 13 & 6 & 0 & 6 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 15 & 6 & 0 & 6 & 13 & 7 \\ 3 & 5 & 7 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 31^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}} 31^{\overline{1111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}} 31^{\overline{1111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}} 31^{\overline{1111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 651, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1114}} 31^{\overline{1111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 813, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 21 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 15 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 15 & 18 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & 3 & -4 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1112}} 31^{\overline{1111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 9 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 9 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{1113}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 12 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.960 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_3$

$T = 10$ [0] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 22, \Delta = -100, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 & 4 \\ 5 & 9 & 13 & 9 & 0 & 6 & 6 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.961 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_3 + \text{f}_4$

$T = 8$ [0] $1^{132}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 12 & 13 & 0 & 4 \\ 6 & 12 & 25 & 27 & 0 & 9 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{121}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 3 \\ 4 & 9 & 12 & 4 & 0 & 5 \\ 5 & 12 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 7 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 9 & 17 & 15 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 12 & -3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 11 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 15 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 4 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{122}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 1 & 3 \\ 3 & 4 & 18 & 8 & 0 & 4 \\ 10 & 18 & 58 & 24 & 0 & 14 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 14 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 10 & 24 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 14 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 10 & 18 & 0 & 6 \\ 7 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 11 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 11 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 8 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 11 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 7 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 28 & 18 & 0 & 10 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{113}3^11^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 15 & 36 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 21 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 18 & 12 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 9 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 11 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 15 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 7 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 9 & 17 & 27 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 9 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{113}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 13 & 32 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 18 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 1 & 3 \\ 3 & 4 & 22 & 8 & 0 & 4 \\ 12 & 22 & 85 & 30 & 0 & 17 \\ 4 & 8 & 30 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 17 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{121}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 28 & 21 & 0 & 10 \\ 5 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{114}3^11^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 1 & 3 \\ 4 & 9 & 48 & 4 & 0 & 5 \\ 20 & 48 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 28 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 14 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{131}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{141}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 11 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{121}3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 16 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 22 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{131}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{113}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 12 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{114}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 14 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 19 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{121}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 13 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 18 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{122}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 28 & 30 & 0 & 10 \\ 7 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{114}3^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 18 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 25 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{141}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 11 & 21 & 76 & 15 & 0 & 16 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{132}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 9 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{114}3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 16 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 22 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{131}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 85 & 15 & 0 & 17 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 17 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{131}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 27 & 4 & 0 & 4 \\ 15 & 27 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 21 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{131}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 1 & 3 \\ 3 & 4 & 32 & 4 & 0 & 4 \\ 18 & 32 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 25 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{141}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 26 & 4 & 0 & 4 \\ 14 & 26 & 118 & 18 & 0 & 20 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 20 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{141}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1643, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 31 & 4 & 0 & 4 \\ 17 & 31 & 172 & 21 & 0 & 24 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 24 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{141}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 1 & 3 \\ 3 & 4 & 36 & 4 & 0 & 4 \\ 20 & 36 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 28 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

4.962 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$

$T = 9 \quad [0] 1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.963 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$

$T = 9$ [0] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 19 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 21 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 11 & 19 & 73 & 12 & 0 & 33 & 15 \\ 2 & 4 & 12 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 33 & 6 & 0 & 13 & 7 \\ 3 & 4 & 15 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{121}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -173$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{121}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -191$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.4$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 3 & 5 & 4 & 12 & 0 & 3 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 14 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 15 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 3 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{112}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1212}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 9 & 0 & 13 & 4 \\ 4 & 8 & 10 & 12 & 0 & 18 & 6 \\ 5 & 9 & 12 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 18 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -573, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11111} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -605, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 8 & 3 \\ 3 & 4 & 9 & 4 & 0 & 14 & 4 \\ 5 & 9 & 13 & 6 & 0 & 24 & 7 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 24 & 9 & 0 & 37 & 11 \\ 3 & 4 & 7 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -626, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -650, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 5 & 4 & 0 & 28 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 30 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 6 & 0 & 13 & 7 \\ 3 & 4 & 7 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11111} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 8 & 3 \\ 3 & 4 & 14 & 4 & 0 & 14 & 4 \\ 8 & 14 & 37 & 9 & 0 & 39 & 11 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 39 & 9 & 0 & 37 & 11 \\ 3 & 4 & 11 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -723, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1112} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -738, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -744, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 5 & 4 & 0 & 23 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 24 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -779, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 21 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -828, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -845, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -941, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 13 & 4 \\ 5 & 9 & 13 & 6 & 0 & 21 & 7 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 9 & 0 & 28 & 10 \\ 3 & 4 & 7 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{112} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -962, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 10 & 3 \\ 3 & 4 & 14 & 4 & 0 & 18 & 4 \\ 8 & 14 & 37 & 9 & 0 & 48 & 11 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 48 & 12 & 0 & 58 & 14 \\ 3 & 4 & 11 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -980, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 4 & 0 & 17 & 4 \\ 5 & 9 & 13 & 6 & 0 & 27 & 7 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 12 & 0 & 49 & 13 \\ 3 & 4 & 7 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{112} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -980, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 16 & 3 \\ 3 & 4 & 9 & 4 & 0 & 28 & 4 \\ 5 & 9 & 13 & 6 & 0 & 48 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 48 & 18 & 0 & 154 & 22 \\ 3 & 4 & 7 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 1 & 6 & 0 & 12 & 3 \\ 5 & 9 & 6 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 3 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 21 & 4 \\ 2 & 4 & 1 & 6 & 0 & 15 & 3 \\ 5 & 9 & 6 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 33 & 0 & 76 & 16 \\ 3 & 4 & 3 & 6 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1212} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 10 & 3 \\ 3 & 4 & 13 & 4 & 0 & 18 & 4 \\ 7 & 13 & 28 & 9 & 0 & 42 & 10 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 42 & 12 & 0 & 58 & 14 \\ 3 & 4 & 10 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 10 & 3 \\ 3 & 4 & 9 & 4 & 0 & 18 & 4 \\ 5 & 9 & 13 & 6 & 0 & 30 & 7 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 30 & 12 & 0 & 58 & 14 \\ 3 & 4 & 7 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 7 & 3 \\ 3 & 4 & 13 & 4 & 0 & 13 & 4 \\ 7 & 13 & 28 & 9 & 0 & 30 & 10 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 30 & 9 & 0 & 28 & 10 \\ 3 & 4 & 10 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 13 & 3 \\ 3 & 4 & 9 & 4 & 0 & 23 & 4 \\ 5 & 9 & 13 & 6 & 0 & 39 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 39 & 15 & 0 & 100 & 18 \\ 3 & 4 & 7 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 3 & 0 & 33 & 4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 30 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 12 & 3 \\ 3 & 4 & 9 & 4 & 0 & 22 & 4 \\ 5 & 9 & 13 & 6 & 0 & 36 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 36 & 15 & 0 & 85 & 17 \\ 3 & 4 & 7 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{11212} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 10 & 3 \\ 3 & 4 & 18 & 4 & 0 & 18 & 4 \\ 10 & 18 & 58 & 12 & 0 & 60 & 14 \\ 2 & 4 & 12 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 60 & 12 & 0 & 58 & 14 \\ 3 & 4 & 14 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 36 & 4 \\ 3 & 5 & 4 & 3 & 0 & 36 & 4 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1114} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 3 & 0 & 24 & 3 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 14 & 3 \\ 3 & 4 & 9 & 4 & 0 & 26 & 4 \\ 5 & 9 & 13 & 6 & 0 & 42 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 42 & 18 & 0 & 118 & 20 \\ 3 & 4 & 7 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1114} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 15 & 3 \\ 3 & 4 & 9 & 4 & 0 & 27 & 4 \\ 5 & 9 & 13 & 6 & 0 & 45 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 45 & 18 & 0 & 133 & 21 \\ 3 & 4 & 7 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 18 & 3 \\ 3 & 4 & 9 & 4 & 0 & 32 & 4 \\ 5 & 9 & 13 & 6 & 0 & 54 & 7 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 54 & 21 & 0 & 193 & 25 \\ 3 & 4 & 7 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1114} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 17 & 3 \\ 3 & 4 & 9 & 4 & 0 & 31 & 4 \\ 5 & 9 & 13 & 6 & 0 & 51 & 7 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 51 & 21 & 0 & 172 & 24 \\ 3 & 4 & 7 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 20 & 3 \\ 3 & 4 & 9 & 4 & 0 & 36 & 4 \\ 5 & 9 & 13 & 6 & 0 & 60 & 7 \\ 2 & 4 & 6 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 60 & 24 & 0 & 238 & 28 \\ 3 & 4 & 7 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$4.964 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \quad 1^{113} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{magenta}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{magenta}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{magenta}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 2 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 6 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{magenta}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 2 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{magenta}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 3 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 5 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 5 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.965} \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{\overline{121}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 9 & 0 & 3 & 4 \\ 4 & 8 & 10 & 12 & 0 & 4 & 6 \\ 5 & 9 & 12 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 8 & 4 & 0 & 4 & 5 \\ 3 & 8 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 2 & 3 \\ 3 & 4 & 19 & 4 & 0 & 3 & 4 \\ 11 & 19 & 73 & 12 & 0 & 11 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 11 & 2 & 0 & 0 & 2 \\ 3 & 4 & 15 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 2 & 3 \\ 3 & 4 & 8 & 4 & 0 & 3 & 4 \\ 4 & 8 & 10 & 6 & 0 & 4 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 6 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 2 & 3 \\ 3 & 4 & 4 & 8 & 0 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 \\ 4 & 8 & 6 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 2 & 3 \\ 4 & 9 & 12 & 4 & 0 & 4 & 5 \\ 5 & 12 & 13 & 6 & 0 & 5 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 2 & 0 & 0 & 2 \\ 3 & 5 & 7 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 8 & 0 & 3 & 4 \\ 5 & 9 & 13 & 12 & 0 & 5 & 7 \\ 4 & 8 & 12 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 4 & 0 & 0 & 2 \\ 3 & 4 & 7 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 \\ 5 & 9 & 6 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 \\ 3 & 4 & 3 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 2 & 3 \\ 3 & 4 & 14 & 4 & 0 & 3 & 4 \\ 8 & 14 & 37 & 9 & 0 & 8 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 8 & 2 & 0 & 0 & 2 \\ 3 & 4 & 11 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 3 & 4 \\ 5 & 9 & 13 & 6 & 0 & 5 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 2 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 2 & 3 \\ 3 & 4 & 4 & 13 & 0 & 3 & 4 \\ 2 & 4 & 1 & 9 & 0 & 2 & 3 \\ 7 & 13 & 9 & 27 & 0 & 7 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 7 & 0 & 0 & 2 \\ 3 & 4 & 3 & 9 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 2 & 3 \\ 3 & 4 & 13 & 4 & 0 & 3 & 4 \\ 7 & 13 & 28 & 9 & 0 & 7 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 7 & 2 & 0 & 0 & 2 \\ 3 & 4 & 10 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{121}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 2 & 3 \\ 3 & 4 & 28 & 4 & 0 & 3 & 4 \\ 16 & 28 & 154 & 18 & 0 & 16 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 16 & 2 & 0 & 0 & 2 \\ 3 & 4 & 22 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 2 & 3 \\ 3 & 4 & 17 & 4 & 0 & 3 & 4 \\ 9 & 17 & 49 & 12 & 0 & 9 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 9 & 2 & 0 & 0 & 2 \\ 3 & 4 & 13 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 1 & 2 & 3 \\ 3 & 4 & 36 & 4 & 0 & 3 & 4 \\ 20 & 36 & 238 & 24 & 0 & 20 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 20 & 2 & 0 & 0 & 2 \\ 3 & 4 & 28 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.966 NHC(-2)(-3) + su₂ + su₃ + f₄ + su₂→(-3) + su₂

$$T = 10 \quad [0] \quad 1^{11}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 & 3 \\ 5 & 9 & 6 & 12 & 0 & 5 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 6 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & 6 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 3 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 5 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 15 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 7 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.967 NHC(-2)(-3) + su₂ + su₃ + f₄ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{11}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 8 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 5 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}1}3^{\overline{112}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 11 & 3 \\ 2 & 4 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 11 & 11 & 0 & 55 & 13 \\ 3 & 4 & 3 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{112}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 6 & 3 \\ 3 & 4 & 4 & 5 & 0 & 11 & 4 \\ 2 & 4 & 1 & 3 & 0 & 8 & 3 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 9 & 0 & 20 & 8 \\ 3 & 4 & 3 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 5 & 0 & 7 & 4 \\ 2 & 4 & 1 & 3 & 0 & 5 & 3 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 6 & 0 & 7 & 5 \\ 3 & 4 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 8 & 4 \\ 5 & 9 & 13 & 6 & 0 & 14 & 7 \\ 2 & 4 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 14 & 5 & 0 & 12 & 6 \\ 3 & 4 & 7 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 12 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{113}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 20 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 18 & 14 & 0 & 69 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{111}1}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 4 & 3 \\ 3 & 4 & 4 & 9 & 0 & 7 & 4 \\ 2 & 4 & 1 & 6 & 0 & 5 & 3 \\ 5 & 9 & 6 & 12 & 0 & 11 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 11 & 0 & 7 & 5 \\ 3 & 4 & 3 & 6 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}1}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 3 & 0 & 8 & 3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{112}3}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 16 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 15 & 11 & 0 & 44 & 12 \\ 3 & 4 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{111}1}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 4 & 0 & 7 & 4 \\ 5 & 9 & 13 & 6 & 0 & 11 & 7 \\ 2 & 4 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 5 & 0 & 7 & 5 \\ 3 & 4 & 7 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{111}1}3^{\overline{11}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 12 & 4 \\ 5 & 9 & 13 & 6 & 0 & 20 & 7 \\ 2 & 4 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 20 & 8 & 0 & 25 & 9 \\ 3 & 4 & 7 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

4.968 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$

$T = 9$ [0] $1^{\overline{111}1}3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 2 & 3 \\ 4 & 9 & 12 & 4 & 0 & 4 & 5 \\ 5 & 12 & 13 & 6 & 0 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 5 & 7 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}3}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}1}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

13.2

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 2 & 3 \\ 4 & 9 & 28 & 4 & 0 & 4 & 5 \\ 11 & 28 & 73 & 12 & 0 & 12 & 15 \\ 2 & 4 & 12 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 3 & 0 & 0 & 2 \\ 3 & 5 & 15 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 2 & 3 \\ 4 & 9 & 20 & 4 & 0 & 4 & 5 \\ 8 & 20 & 37 & 9 & 0 & 9 & 11 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 5 & 11 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{122} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 4 & 3 \\ 3 & 4 & 13 & 5 & 0 & 8 & 4 \\ 7 & 13 & 28 & 12 & 0 & 18 & 10 \\ 3 & 5 & 12 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 18 & 6 & 0 & 9 & 5 \\ 3 & 4 & 10 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{132} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 4 & 3 \\ 3 & 4 & 17 & 5 & 0 & 8 & 4 \\ 9 & 17 & 49 & 15 & 0 & 24 & 13 \\ 3 & 5 & 15 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 24 & 6 & 0 & 9 & 5 \\ 3 & 4 & 13 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 2 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 & 4 \\ 8 & 14 & 37 & 9 & 0 & 9 & 11 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 4 & 11 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 1^{211} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 2 & 3 \\ 4 & 9 & 32 & 4 & 0 & 4 & 5 \\ 13 & 32 & 100 & 15 & 0 & 15 & 18 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 5 & 18 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^1 1^{311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 2 & 3 \\ 4 & 9 & 44 & 4 & 0 & 4 & 5 \\ 18 & 44 & 193 & 21 & 0 & 21 & 25 \\ 2 & 4 & 21 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 3 & 0 & 0 & 2 \\ 3 & 5 & 25 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 \\ 5 & 9 & 13 & 9 & 0 & 6 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 4 & 7 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^1 1^{211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 2 & 3 \\ 4 & 9 & 40 & 4 & 0 & 4 & 5 \\ 16 & 40 & 154 & 18 & 0 & 18 & 22 \\ 2 & 4 & 18 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 3 & 0 & 0 & 2 \\ 3 & 5 & 22 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 2 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 & 4 \\ 11 & 21 & 76 & 18 & 0 & 15 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 4 & 16 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 \\ 5 & 9 & 13 & 6 & 0 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 2 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 & 4 \\ 7 & 13 & 28 & 12 & 0 & 9 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 4 & 10 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{131} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 2 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 & 4 \\ 9 & 17 & 49 & 15 & 0 & 12 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 3 & 0 & 0 & 2 \\ 3 & 4 & 13 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 6 & 0 & 12 & 6 \\ 3 & 4 & 7 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

4.969 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + NHC(-2)(-3)

$$T = 11 \quad [0] \quad 1^{11} 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 2 & 3 & 0 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 10 & 3 & 3 & 0 & 4 \\ 6 & 10 & 21 & 6 & 8 & 0 & 9 \\ 2 & 3 & 6 & 0 & 2 & 0 & 2 \\ 2 & 3 & 8 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 9 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -23, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 3 & 0 & 4 \\ 10 & 17 & 61 & 10 & 13 & 0 & 15 \\ 2 & 3 & 10 & 0 & 2 & 0 & 2 \\ 2 & 3 & 13 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 15 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 0 & 4 \\ 4 & 7 & 8 & 4 & 5 & 0 & 6 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 2 & 3 & 5 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 6 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} \overset{1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -50, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 0 & 4 \\ 8 & 14 & 38 & 8 & 10 & 0 & 12 \\ 2 & 3 & 8 & 0 & 2 & 0 & 2 \\ 2 & 3 & 10 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 12 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} \overset{1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 2 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 0 & 4 \\ 12 & 21 & 88 & 12 & 15 & 0 & 18 \\ 2 & 3 & 12 & 0 & 2 & 0 & 2 \\ 2 & 3 & 15 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 18 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

4.970 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$

$$T = 8 \quad [0] \ 1^{11} \overset{121}{3} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{11}{3} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{141} \overset{11}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 16 & 9 & 0 & 4 \\ 9 & 16 & 46 & 24 & 0 & 12 \\ 5 & 9 & 24 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{132} \overset{11}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 12 & 13 & 0 & 4 \\ 6 & 12 & 25 & 27 & 0 & 9 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} \overset{11}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 4 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} \overset{11}{3} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 25, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} \overset{11}{3} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} \overset{11}{3} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 24 & 4 & 0 & 4 \\ 17 & 24 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 5 \\ 12 & 25 & 20 & 16 & 9 \\ 9 & 20 & 13 & 12 & 7 \\ 8 & 16 & 12 & 9 & 5 \\ 5 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 4 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 9 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 39, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 11 & 13 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 13 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 \\ 3 & 5 & 4 & 3 & 0 \\ 3 & 4 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{113}} 3 1^{\overline{41}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 45 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 5 & 0 & 0 & 4 \\ 3 & 5 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{11}} 3 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 1 & 3 \\ 3 & 4 & 19 & 0 & 0 & 4 \\ 11 & 19 & 73 & 0 & 0 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{12}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 8 & 5 & 0 & 4 \\ 5 & 8 & 10 & 6 & 0 & 6 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.52 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 10 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{\overline{11}} 3 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.42 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{\overline{132}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.42 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 1 & 3 \\ 3 & 4 & 12 & 8 & 0 & 4 \\ 6 & 12 & 25 & 18 & 0 & 9 \\ 5 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

- [27] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 15 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [29] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$
- [32] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [31] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 9 & 12 & 0 & 0 & 5 \\ 5 & 12 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [30] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 1 & 3 \\ 4 & 9 & 36 & 0 & 0 & 5 \\ 14 & 36 & 121 & 0 & 0 & 19 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [33] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 12 & 4 & 0 & 5 \\ 6 & 12 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.28 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 21^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.06 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 13 & 15 & 7 \\ 15 & 17 & 15 & 12 & 6 \\ 7 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 10 & 6 & 0 & 6 \\ 3 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 65, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 10 & 18 & 0 & 6 \\ 8 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 1 & 3 \\ 3 & 4 & 17 & 0 & 0 & 4 \\ 9 & 17 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

- [49] $1^{11} 1^3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [52] $1^{112} 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 8 & 1 & 1 & 3 \\ 4 & 9 & 20 & 0 & 0 & 5 \\ 8 & 20 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [51] $1^1 1^3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [50] $1^{113} 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 11 & 1 & 1 & 3 \\ 4 & 9 & 28 & 0 & 0 & 5 \\ 11 & 28 & 73 & 0 & 0 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [53] $1^{111} 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [54] $1^{111} 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [55] $1^{111} 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [56] $1^{111} 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [57] $1^{112} 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 10 & 1 & 1 & 3 \\ 4 & 9 & 24 & 0 & 0 & 5 \\ 10 & 24 & 58 & 0 & 0 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 14 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [58] $1^{111} 1^{21} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 7 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [59] $1^{111} 1^{31} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 11 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 8 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 4 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 3 \\ 4 & 9 & 12 & 4 & 0 & 5 \\ 5 & 12 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 1 & 3 \\ 3 & 4 & 14 & 0 & 0 & 4 \\ 8 & 14 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 9 & 0 & 0 & 4 \\ 5 & 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 9 & 17 & 15 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 3 & 4 & 1 & 12 & 0 & 3 \\ 9 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{\overset{11}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 4 & 5 & 4 & 6 & 0 & 4 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{\overset{11}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 1 & 3 \\ 3 & 4 & 13 & 0 & 0 & 4 \\ 7 & 13 & 28 & 0 & 0 & 10 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{\overset{11}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 11 & 1 & 6 \\ 15 & 20 & 16 & 0 & 8 \\ 11 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{\overset{131}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 20 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 20 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{\overset{121}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{\overset{111}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{\overset{121}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 7 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{\overset{112}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{\overset{111}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 11 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{\overset{111}{1}} 3^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 14 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 24 & 4 & 0 & 4 \\ 15 & 24 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 12 & 9 & 0 & 4 \\ 7 & 12 & 25 & 18 & 0 & 9 \\ 5 & 9 & 18 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 48 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 48 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 13 & 9 & 0 & 7 \\ 4 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 12 & 21 & 18 & 75 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 4 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 58 & 12 & 0 & 14 \\ 3 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 3 & 4 & 1 & 9 & 0 & 3 \\ 7 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 5 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 11 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 1 & 6 \\ 15 & 20 & 33 & 0 & 8 \\ 23 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 9 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 15 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 1 & 3 \\ 4 & 9 & 36 & 0 & 0 & 5 \\ 15 & 36 & 133 & 0 & 0 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 21 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 13 & 6 & 0 & 7 \\ 3 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

- [100] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 5 & 8 & 10 & 12 & 0 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 24 & 4 & 0 & 4 \\ 16 & 24 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{113} 3 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 1 & 3 \\ 4 & 9 & 32 & 0 & 0 & 5 \\ 13 & 32 & 100 & 0 & 0 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [102] $1^{122} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 \\ 5 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 26 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 26 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [107] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 13 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 13 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [108] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [109] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [110] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 1 & 3 \\ 3 & 4 & 28 & 0 & 0 & 4 \\ 16 & 28 & 154 & 0 & 0 & 22 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[111] 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 28 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[112] 1^{112} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 10 & 24 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[113] 1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 10 & 18 & 0 & 6 \\ 7 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[115] 21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[116] 1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 1 & 3 \\ 3 & 4 & 18 & 0 & 0 & 4 \\ 10 & 18 & 58 & 0 & 0 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[117] 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 9 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[114] 1^{112} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[118] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 23 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[119] 1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 11 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[120] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

- [121] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 10 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [122] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 1 & 3 \\ 4 & 9 & 40 & 0 & 0 & 5 \\ 16 & 40 & 154 & 0 & 0 & 22 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [123] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 6 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [124] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 6 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [125] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 1 & 3 \\ 3 & 4 & 23 & 0 & 0 & 4 \\ 13 & 23 & 100 & 0 & 0 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [126] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [127] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 29 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 29 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [128] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [129] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 11 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [130] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 10 & 17 & 15 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$
- [131] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 28 & 9 & 0 & 10 \\ 3 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{141} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 11 & 21 & 76 & 15 & 0 & 16 \\ 3 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{122} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 5 & 8 & 10 & 18 & 0 & 6 \\ 7 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{122} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 5 & 8 & 10 & 12 & 0 & 6 \\ 4 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{112} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 7 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{13} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 1 & 3 \\ 3 & 4 & 22 & 0 & 0 & 4 \\ 12 & 22 & 85 & 0 & 0 & 17 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{14} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 1 & 3 \\ 3 & 4 & 26 & 0 & 0 & 4 \\ 14 & 26 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{113} 3 \ 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 46 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 180, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.51 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 184, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.28 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 15 & 12 & 7 \\ 20 & 25 & 20 & 16 & 9 \\ 15 & 20 & 13 & 12 & 7 \\ 12 & 16 & 12 & 9 & 5 \\ 7 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 185, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.18 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 12 & 13 & 0 & 4 \\ 7 & 12 & 25 & 27 & 0 & 9 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{114} 3 1^4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 186, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 1 & 3 \\ 4 & 9 & 48 & 0 & 0 & 5 \\ 20 & 48 & 238 & 0 & 0 & 28 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 28 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 188, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.30 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 49 & 12 & 0 & 13 \\ 3 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 8 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[148] \ 1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 191, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 8 & 13 & 12 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 191, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 196, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 196, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 199, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 28 & 18 & 0 & 10 \\ 5 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 1 & 3 \\ 4 & 9 & 44 & 0 & 0 & 5 \\ 18 & 44 & 193 & 0 & 0 & 25 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{113} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 16 & 32 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{13} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 45 & 1 & 5 & 8 \\ 11 & 20 & 84 & 0 & 8 & 34 \\ 45 & 84 & 346 & 0 & 34 & -1 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 8 & 34 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 18 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 6 & 9 & 13 & 21 & 0 & 7 \\ 7 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 28 & 21 & 0 & 10 \\ 6 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{123} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 8 & 13 & 28 & 18 & 0 & 10 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 7 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 28 & 18 & 0 & 10 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 7 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 5 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 1 & 3 \\ 3 & 4 & 12 & 8 & 0 & 4 \\ 8 & 12 & 25 & 18 & 0 & 9 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 10 & 12 & 25 & 9 & 0 & 9 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 85 & 15 & 0 & 17 \\ 3 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{11} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 12 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 6 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{11} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 13 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 22 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 22 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 18 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 25 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 25 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 9 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 10 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 7 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 6 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 24 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 38 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 38 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 6 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

- [183] $21^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 13 & 17 & 27 & 15 & 7 \\ 17 & 20 & 33 & 17 & 8 \\ 27 & 33 & 49 & 27 & 13 \\ 15 & 17 & 27 & 12 & 6 \\ 7 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$
- [184] $1^{113}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 15 & 36 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 21 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [185] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 18 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$
- [186] $1^{113}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$
- [187] $1^{13}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 1 & 3 \\ 3 & 4 & 27 & 0 & 0 & 4 \\ 15 & 27 & 133 & 0 & 0 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [188] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 47 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 47 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$
- [189] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 9 & 17 & 49 & 24 & 0 & 13 \\ 5 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [190] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [191] $1^{141}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 & 1 & 3 \\ 3 & 4 & 31 & 4 & 0 & 4 \\ 17 & 31 & 172 & 21 & 0 & 24 \\ 3 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [192] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [193] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 9 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 12 & 8 & 0 & 4 \\ 7 & 12 & 25 & 18 & 0 & 9 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[195] $21^{112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 30 & 12 & 7 \\ 20 & 25 & 40 & 16 & 9 \\ 30 & 40 & 58 & 24 & 14 \\ 12 & 16 & 24 & 9 & 5 \\ 7 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 38 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 18 & 12 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[198] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 36 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 36 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 28 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 28 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 27 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 27 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[202] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 24 & 12 & 6 \\ 16 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 1 & 3 \\ 3 & 4 & 32 & 0 & 0 & 4 \\ 18 & 32 & 193 & 0 & 0 & 25 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 9 & 17 & 49 & 39 & 0 & 13 \\ 8 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[204] 1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 46 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 46 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[206] 1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 11 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[207] 1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 10 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[208] 1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[209] 1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 301, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 9 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[210] 1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 12 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[213] 1^{141} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 1 & 3 \\ 3 & 4 & 26 & 4 & 0 & 4 \\ 14 & 26 & 118 & 18 & 0 & 20 \\ 3 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[212] 1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 14 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[211] 1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 11 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[214] 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 6 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 1 & 3 \\ 3 & 4 & 31 & 0 & 0 & 4 \\ 17 & 31 & 172 & 0 & 0 & 24 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 37 & 1 & 5 & 5 \\ 11 & 20 & 67 & 0 & 8 & 8 \\ 37 & 67 & 217 & 0 & 27 & 27 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 8 & 27 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{131} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 7 & 12 & 25 & 9 & 0 & 9 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 7 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 9 & 17 & 27 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 & 5 \\ 12 & 25 & 92 & 0 & 9 & 9 \\ 43 & 92 & 325 & 0 & 33 & 33 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 33 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 13 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{132} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 9 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 56 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 56 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 9 & 12 & 25 & 9 & 0 & 9 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 386$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 8 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{132} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 390$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 11 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 392$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 10 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 394$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 10 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 395$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 12 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 1 & 3 \\ 3 & 4 & 36 & 0 & 0 & 4 \\ 20 & 36 & 238 & 0 & 0 & 28 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{114} 3 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -1 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{141} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 406$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 11 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{141} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 16 & 4 & 0 & 4 \\ 9 & 16 & 46 & 12 & 0 & 12 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{121} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 16 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{131} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 8 & 12 & 25 & 9 & 0 & 9 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 14 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 28 & 30 & 0 & 10 \\ 8 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 45 & 12 & 7 \\ 20 & 25 & 60 & 16 & 9 \\ 45 & 60 & 133 & 36 & 21 \\ 12 & 16 & 36 & 9 & 5 \\ 7 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 14 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 429$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 10 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[254] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 429$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 37 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 13 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 12 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{114} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 20 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 12 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 448$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 48 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 48 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 453$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 455$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 10 & 17 & 27 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 456$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 8 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{121} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 456$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 13 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 458$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 12 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 471$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 28 & 30 & 0 & 10 \\ 7 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 478$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 11 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 488$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 9 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 491$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 8 & 13 & 28 & 30 & 0 & 10 \\ 7 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 493$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 18 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[272] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 498$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 8 & 13 & 28 & 21 & 0 & 10 \\ 5 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{114} 3^1 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 21 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[274] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 503$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 15 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[275] $1^{114}3^{11}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{114}3^{11}3^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{14}3^{111}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 13 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{141}3^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 11 & 21 & 76 & 15 & 0 & 16 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{114}3^{11}3^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 18 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{114}3^{11}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 16 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{132}3^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 9 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{14}3^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 14 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{121}3^{11}3^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 17 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{141}3^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 14 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 13 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 11 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 11 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{141} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 16 & 4 & 0 & 4 \\ 12 & 16 & 46 & 12 & 0 & 12 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{141} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 16 & 4 & 0 & 4 \\ 10 & 16 & 46 & 12 & 0 & 12 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 13 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[291] $21^{114} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 60 & 12 & 7 \\ 20 & 25 & 80 & 16 & 9 \\ 60 & 80 & 238 & 48 & 28 \\ 12 & 16 & 48 & 9 & 5 \\ 7 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 15 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[293] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 10 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 645, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 55 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 55 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

- [295] $1^{131} 3 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 85 & 15 & 0 & 17 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [296] $1^{114} 3 1^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 19 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [297] $1^{114} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 47 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 47 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [298] $1^{114} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 669, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 17 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [300] $1^{121} 3 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 12 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [299] $1^{121} 3 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 20 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [302] $1^{121} 3 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 17 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [303] $1^{131} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 13 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [304] $1^{141} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 16 & 4 & 0 & 4 \\ 11 & 16 & 46 & 12 & 0 & 12 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [301] $1^{131} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 10 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [305] $1^{121} 3 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 14 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[306] \ 1^{114} 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 16 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[307] \ 1^{132} 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 10 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[308] \ 1^{114} 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[309] \ 1^{141} 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 730, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 12 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[310] \ 1^{121} 3 \ 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 16 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[311] \ 1^{114} 3 \ 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 18 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[312] \ 1^{121} 3 \ 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 15 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[313] \ 1^{141} 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 790, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 13 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[314] \ 1^{131} 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 11 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[317] \ 1^{121} 3 \ 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

[365] $1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 19 & 3 & 0 & 4 \\ 11 & 19 & 68 & 13 & 0 & 16 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

4.971 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{e}_6$

$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.972 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{f}_4$

$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{violet}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.973 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -51, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -87, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 1 & 5 \\ 12 & 25 & 76 & 0 & 0 & 9 \\ 35 & 76 & 217 & 0 & 0 & 27 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 27 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -127, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 1 & 5 \\ 12 & 25 & 36 & 0 & 0 & 9 \\ 17 & 36 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 13 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -131, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -133, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 1 & 5 \\ 11 & 20 & 16 & 0 & 0 & 8 \\ 8 & 16 & 10 & 0 & 0 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[9] 1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 1 & 5 \\ 12 & 25 & 56 & 0 & 0 & 9 \\ 26 & 56 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 20 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[10] 1^{112} 3 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 1 & 5 \\ 12 & 25 & 72 & 0 & 0 & 9 \\ 34 & 72 & 202 & 0 & 0 & 26 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[11] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[12] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 10 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 10 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[13] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 9 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[14] 1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 1 & 5 \\ 11 & 20 & 50 & 0 & 0 & 8 \\ 26 & 50 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 20 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[15] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 1 & 5 \\ 11 & 20 & 33 & 0 & 0 & 8 \\ 17 & 33 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 1 & 5 \\ 12 & 25 & 108 & 0 & 0 & 9 \\ 51 & 108 & 457 & 0 & 0 & 39 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 1 & 5 \\ 12 & 25 & 92 & 0 & 0 & 9 \\ 43 & 92 & 325 & 0 & 0 & 33 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 1 & 5 \\ 12 & 25 & 112 & 0 & 0 & 9 \\ 52 & 112 & 478 & 0 & 0 & 40 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 1 & 5 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 68 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 52 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 1 & 5 \\ 12 & 25 & 128 & 0 & 0 & 9 \\ 60 & 128 & 634 & 0 & 0 & 46 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 19 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 18 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 18 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

4.974 NHC(-2)(-3) + su₂ + su₃ + so₈ + su₂

$T = 9$ [0] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 4 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{\textcolor{brown}{1}}1^{\textcolor{blue}{1}}31^{\textcolor{teal}{1}\textcolor{teal}{1}\textcolor{teal}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}\textcolor{teal}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{u}_3 + \textcolor{red}{s}\textcolor{red}{o}_8 + \textcolor{blue}{s}\textcolor{blue}{u}_2$ $T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 4 & 3 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{\textcolor{brown}{11}} 3^{\textcolor{blue}{11}} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{blue}{u}_2 + \textcolor{brown}{5}\textcolor{blue}{u}_3 + \textcolor{red}{5}\textcolor{blue}{o}_8 + \textcolor{blue}{5}\textcolor{blue}{u}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -62$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 2 & -2 & 4 & 2 \end{pmatrix}$$

4.975 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 8 \quad [0] \, 1^{\overline{11}} \, 3 \, 1^{\overline{1}} + [\overline{\text{su}}_2 \cdot \overline{\text{g}}_2] + \overline{\text{su}}_2 + \overline{\text{su}}_3 + \overline{\text{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}1} \overline{1} \ 3 \ 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & 0 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 \end{pmatrix} b_0$$

[3] $1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 & 5 & 4 \\ 3 & 5 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 4 & 5 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & 4 & 3 \end{pmatrix} b_0$$

$$[4] \ 1^{\text{red}} 1^{\text{blue}} 3 \ 1^{\text{red}} 1^{\text{blue}} + [\text{red} 2 \cdot \text{blue} 2] + \text{red} 2 + \text{red} 3 + \text{red} 2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -1 & 7 & 3 \end{pmatrix} b_0$$

[5] $1^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 2 & 0 & 3 \end{pmatrix} b_0$$

[6] $1^{\text{11}} 1^{\text{111}} 3^{\text{11}} + [\text{5u}_2 \cdot \text{g}_2] + \text{5u}_2 + \text{5u}_3 + \text{5u}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 & 5 & 4 \\ 4 & 5 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 4 & 5 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{12} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 4 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{12} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & 0 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -1 & 5 & 3 \end{pmatrix} b_m$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 7 & 5 \\ 1 & 1 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 0 & 7 & 4 & 5 & 5 \\ 2 & 2 & -1 & 5 & 2 & 5 & 3 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -1 & 5 & 3 \end{pmatrix} b_0$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 4 & 0 & 2 & 4 \\ 3 & 5 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 4 & -1 & 8 & 4 \end{pmatrix} b_0$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 3 & 5 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 4 & -1 & 8 & 4 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix} b_0$$

$$45] 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{blue}{1}\textcolor{red}{1}} 21^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{green}{1}} + [\textcolor{red}{5}u_2. \\ T_{\min} = 10, T_{\max} = 10$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 7 & 0 & 5 & 5 \\ 2 & 2 & 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [48] \quad 1^{11} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 6 & 4 & 11 & 6 \\ 4 & 2 & 8 & 4 \\ 11 & 8 & 19 & 11 \\ 6 & 4 & 11 & 6 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 3 & 4 & 2 & 1 & 2 & 4 \\ 3 & 4 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 2 & 3 \\ 3 & 3 & 1 & 2 & 2 & 4 \\ 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 6 & 4 & 3 & 6 \\ 4 & 2 & 2 & 4 \\ 3 & 2 & 1 & 3 \\ 6 & 4 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{11} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 4 \\ 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 2 & 1 & 4 & 4 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 4 & 4 & 1 & 2 & 4 \\ 3 & 4 & 4 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 6 & 7 & 9 & 1 & 6 & 7 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 4 & 6 & 1 & 2 & 4 \\ 3 & 4 & 7 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 3 & 5 & 4 \\ 3 & 1 & 2 & 4 & 3 \\ 3 & 2 & 1 & 3 & 3 \\ 5 & 4 & 3 & 6 & 6 \\ 4 & 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 & 5 & 5 \\ 3 & 3 & 0 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[56] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 5 & 3 \\ 6 & 6 & 1 & 5 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 4 & 3 & 1 & 5 & 4 & 5 \\ 3 & 3 & 1 & 4 & 2 & 4 \\ 3 & 3 & 0 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 1 & 0 & 2 \\ 3 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[59] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 4 & 2 & 6 & 6 \\ 4 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

[60] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 5 & 5 & 4 \\ 3 & 1 & 4 & 3 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 5 & 3 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[66] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 4 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 4 & 4 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[61] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 3 & 4 & 3 \\ 5 & 3 & 3 & 4 & 5 \\ 4 & 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

[62] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 1 & 1 & 5 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 5 & 5 & 0 & 8 & 11 & 6 \\ 6 & 6 & 1 & 11 & 14 & 8 \\ 3 & 3 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 5 & 7 & 4 \\ 3 & 1 & 3 & 5 & 3 \\ 5 & 3 & 3 & 7 & 5 \\ 7 & 5 & 7 & 12 & 8 \\ 4 & 3 & 5 & 8 & 5 \end{pmatrix}_{b_0}$$

[64] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 4 & 4 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 4 & 4 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

- [65] $1^{111} 1^{11} 31^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 5 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 2 & 5 & 4 \\ 5 & 5 & 0 & 5 & 8 & 6 \\ 3 & 3 & 0 & 4 & 6 & 4 \end{pmatrix}_{b_0}$
- [67] $1^{111} 1^{11} 31^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 3 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0}$
- [69] $1^{111} 1^{11} 31^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 6 & 3 \\ 2 & 1 & 1 & 6 & 5 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 6 & 6 & 1 & 14 & 12 & 8 \\ 6 & 5 & 1 & 12 & 9 & 7 \\ 3 & 3 & 0 & 8 & 7 & 4 \end{pmatrix}_{b_0}$
- [70] $1^{111} 1^{11} 31^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 5 & 1 & 5 & 9 & 7 \\ 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [68] $1^{111} 1^{11} 4131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 4 \\ 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [74] $1^{112} 1^{11} 31^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 2 & 10 & 4 \\ 3 & 1 & 1 & 7 & 3 \\ 2 & 1 & 0 & 4 & 2 \\ 10 & 7 & 4 & 21 & 11 \\ 4 & 3 & 2 & 11 & 5 \end{pmatrix}_{b_0}$
- [71] $1^{111} 1^{11} 31^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 9 & 4 \\ 9 & 8 & 1 & 9 & 27 & 11 \\ 3 & 3 & 0 & 4 & 11 & 4 \end{pmatrix}_{b_0}$
- [72] $1^{111} 1^{11} 31^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 8 & 3 \\ 9 & 8 & 1 & 8 & 27 & 11 \\ 3 & 3 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$
- [73] $1^{111} 1^{11} 4131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 6 & 6 & 1 & 6 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [75] $1^{111} 1^{11} 4131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 7 & 4 \\ 6 & 6 & 1 & 7 & 14 & 8 \\ 3 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [76] \quad 1^1 5^1 1^1 231^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 2 & 0 & 11 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 15 & 11 & 0 & 37 & 13 \\ 3 & 5 & 4 & -1 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{11} 5^1 1^1 231^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 \\ 1 & 2 & 0 & 3 & 0 & 8 \\ 1 & 1 & 3 & 1 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 5 & 6 & 8 & 8 & 0 & 31 \\ 2 & 2 & 2 & 3 & -1 & 11 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^1 5^1 1^1 231^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 2 & 0 & 15 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 15 & 0 & 63 & 17 \\ 3 & 5 & 4 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[79] \quad 2^1 3^1 15^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[80] \quad 2^1 3^1 15^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^1 5^1 1^2 231^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 2 & 0 & 26 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 18 & 35 & 26 & 0 & 202 & 30 \\ 3 & 5 & 4 & -1 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[82] \quad 2^1 3^1 15^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 6 & 3 \\ 4 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[83] \quad 2^1 3^1 15^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 6 & 3 \\ 4 & 5 & 10 & 3 & 10 & 5 \\ 6 & 10 & 14 & 4 & 16 & 8 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 10 & 16 & 4 & 14 & 8 \\ 3 & 5 & 8 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[84] \quad 2^1 3^1 15^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.8

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 9 & 3 \\ 4 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{\overline{1}1^1} 5^{\overline{1}3} 231^{\overline{1}14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 26 & 3 \\ 3 & 5 & 5 & 0 & 50 & 5 \\ 2 & 5 & 2 & 0 & 37 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 26 & 50 & 37 & 0 & 419 & 43 \\ 3 & 5 & 4 & -1 & 43 & 4 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{\overline{1}1^1} 5^{\overline{1}2} 231^{\overline{1}14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 5 & 5 & 0 & 40 & 5 \\ 2 & 5 & 2 & 0 & 30 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 20 & 40 & 30 & 0 & 258 & 34 \\ 3 & 5 & 4 & -1 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[87] \ 2^{\overline{1}123} 3^{\overline{1}1} 5^{\overline{1}1} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 9 & 3 \\ 4 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[88] \ 2^{\overline{1}114} 3^{\overline{1}1} 5^{\overline{1}1} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 12 & 3 \\ 4 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[89] \ 2^{\overline{1}111} 3^{\overline{1}1} 5^{\overline{1}1} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 & 3 \\ 4 & 5 & 5 & 8 & 5 & 5 \\ 3 & 5 & 2 & 6 & 4 & 4 \\ 5 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[90] \ 2^{\overline{1}133} 3^{\overline{1}1} 5^{\overline{1}1} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 9 & 3 \\ 4 & 5 & 15 & 3 & 15 & 5 \\ 9 & 15 & 34 & 6 & 36 & 12 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 15 & 36 & 6 & 34 & 12 \\ 3 & 5 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[91] \ 2^{\overline{1}124} 3^{\overline{1}1} 5^{\overline{1}1} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 12 & 3 \\ 4 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[92] \ 2^{\overline{1}134} 3^{\overline{1}1} 5^{\overline{1}1} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 12 & 3 \\ 4 & 5 & 15 & 3 & 20 & 5 \\ 9 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 12 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

- [93] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{15}{5} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -232, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 2 & 15 & 3 \\ 4 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 2 & 20 & 4 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 15 & 25 & 20 & 10 & 98 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [94] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{15}{5} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -250, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 2 & 15 & 3 \\ 4 & 5 & 10 & 3 & 25 & 5 \\ 6 & 10 & 14 & 4 & 40 & 8 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 15 & 25 & 40 & 10 & 98 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [95] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{14}{4} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 12 & 2 & 12 & 3 \\ 4 & 5 & 20 & 3 & 20 & 5 \\ 12 & 20 & 62 & 8 & 64 & 16 \\ 2 & 3 & 8 & 0 & 8 & 2 \\ 12 & 20 & 64 & 8 & 62 & 16 \\ 3 & 5 & 16 & 2 & 16 & 4 \end{pmatrix}_{b_0}$
- [96] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{13}{5} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.9$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 2 & 15 & 3 \\ 4 & 5 & 15 & 3 & 25 & 5 \\ 9 & 15 & 34 & 6 & 60 & 12 \\ 2 & 3 & 6 & 0 & 10 & 2 \\ 15 & 25 & 60 & 10 & 98 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [97] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{11}{6} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -298, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 2 & 18 & 3 \\ 4 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 2 & 24 & 4 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 18 & 30 & 24 & 12 & 142 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$
- [98] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{12}{6} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 2 & 18 & 3 \\ 4 & 5 & 10 & 3 & 30 & 5 \\ 6 & 10 & 14 & 4 & 48 & 8 \\ 2 & 3 & 4 & 0 & 12 & 2 \\ 18 & 30 & 48 & 12 & 142 & 24 \\ 3 & 5 & 8 & 2 & 24 & 4 \end{pmatrix}_{b_0}$
- [99] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{14}{5} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 12 & 2 & 15 & 3 \\ 4 & 5 & 20 & 3 & 25 & 5 \\ 12 & 20 & 62 & 8 & 80 & 16 \\ 2 & 3 & 8 & 0 & 10 & 2 \\ 15 & 25 & 80 & 10 & 98 & 20 \\ 3 & 5 & 16 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [100] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{13}{6} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -346, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 2 & 18 & 3 \\ 4 & 5 & 15 & 3 & 30 & 5 \\ 9 & 15 & 34 & 6 & 72 & 12 \\ 2 & 3 & 6 & 0 & 12 & 2 \\ 18 & 30 & 72 & 12 & 142 & 24 \\ 3 & 5 & 12 & 2 & 24 & 4 \end{pmatrix}_{b_0}$
- [101] $2 \ 3 \ 15 1^{\overset{1}{1} \overset{11}{7} \overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 21 & 3 \\ 4 & 5 & 5 & 3 & 35 & 5 \\ 3 & 5 & 2 & 2 & 28 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 21 & 35 & 28 & 14 & 194 & 28 \\ 3 & 5 & 4 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[102] $2 \overset{1^{127}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -394, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 21 & 3 \\ 4 & 5 & 10 & 3 & 35 & 5 \\ 6 & 10 & 14 & 4 & 56 & 8 \\ 2 & 3 & 4 & 0 & 14 & 2 \\ 21 & 35 & 56 & 14 & 194 & 28 \\ 3 & 5 & 8 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[103] $2 \overset{1^{137}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 21 & 3 \\ 4 & 5 & 15 & 3 & 35 & 5 \\ 9 & 15 & 34 & 6 & 84 & 12 \\ 2 & 3 & 6 & 0 & 14 & 2 \\ 21 & 35 & 84 & 14 & 194 & 28 \\ 3 & 5 & 12 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[104] $22 \overset{1^{1111}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -568, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 13 & 8 \\ 5 & 13 & 9 & 11 & 11 & 7 \\ 5 & 13 & 11 & 8 & 11 & 6 \\ 5 & 13 & 11 & 11 & 9 & 7 \\ 3 & 8 & 7 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [105] $1 \overset{1^1}{11} \overset{1^1}{5} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 1 & 0 & 6 & 3 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 6 & 4 & 0 & 12 & 8 \\ 2 & 2 & 3 & 2 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

[106] $1 \overset{1^1}{11} \overset{1^1}{5} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 1 & 1 & 0 & 11 & 3 \\ 1 & 1 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 11 & 7 & 0 & 40 & 14 \\ 2 & 2 & 3 & 2 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

[107] $1 \overset{1^1}{11} \overset{1^1}{5} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 1 & 1 & 0 & 12 & 3 \\ 1 & 1 & 1 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 12 & 9 & 0 & 56 & 16 \\ 2 & 2 & 3 & 2 & -1 & 16 & 4 \end{pmatrix}_{b_0}$$

[108] $1 \overset{1^1}{12} \overset{1^1}{5} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 1 & 1 & 0 & 12 & 3 \\ 1 & 1 & 1 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 12 & 8 & 0 & 54 & 16 \\ 2 & 2 & 3 & 2 & -1 & 16 & 4 \end{pmatrix}_{b_0}$$

[109] $1 \overset{1^1}{12} \overset{1^1}{5} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 1 & 1 & 0 & 17 & 3 \\ 1 & 1 & 1 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 17 & 11 & 0 & 102 & 22 \\ 2 & 2 & 3 & 2 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{13} \overset{1^1}{5} \ 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 87, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 1 & 1 & 0 & 18 & 3 \\ 1 & 1 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 18 & 12 & 0 & 124 & 24 \\ 2 & 2 & 3 & 2 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [111] \ 1^{11} \overset{1^1}{6} \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -4, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 1 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 5 & 5 \\ 2 & 3 & 2 & 2 & -1 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11} \overset{1^1}{6} \ 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -16, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 1 & 0 & 8 & 3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 6 & 5 & 0 & 25 & 11 \\ 2 & 3 & 2 & 2 & -1 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{11} \overset{1^1}{6} \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -16, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 5 & 4 & 0 & 12 & 8 \\ 2 & 3 & 2 & 2 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [114] \ 1^{11} \overset{1^{12}31}{6} \ 131 \overset{1^1}{6} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -3, \ \det = 0, \ (n_+, n_-, n_0) = (1, 16, 1), \ c_{\text{ext}} = 7.0$$

4.976 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \ 1^{111} \overset{1^1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -33, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1^1}{3} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -33, \ \det = -5, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} \overset{1^1}{3} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -33, \ \det = -7, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} \overset{1^1}{3} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -33, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 3 & 6 & 0 & 4 \\ 3 & 4 & 3 & 0 & 3 & 0 & 2 \\ 4 & 5 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 4 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 3 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -1 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 3 & 6 & 0 & 4 \\ 3 & 4 & 3 & 0 & 3 & 0 & 2 \\ 4 & 5 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 4 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 8 & 13 & 12 & 0 & 28 & 9 & 10 \\ 3 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -1 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{1121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 6 & 3 \\ 3 & 4 & 9 & 0 & 8 & 9 & 4 \\ 5 & 9 & 13 & 0 & 12 & 15 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 12 & 0 & 10 & 12 & 6 \\ 6 & 9 & 15 & 0 & 12 & 13 & 7 \\ 3 & 4 & 7 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 3 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1111} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -264, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -271, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1112} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 8 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 21 & 7 \\ 8 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -348, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1112} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 3 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 21 & 7 \\ 8 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [41] \quad 1^{111} 3^{21111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 5 & 5 & 0 & 5 & 10 & 5 \\ 2 & 5 & 2 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 4 & 0 & 2 & 7 & 4 \\ 5 & 10 & 7 & 0 & 7 & 14 & 8 \\ 3 & 5 & 4 & -1 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [45] \quad 1^{111} 4^{131111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 & 4 \\ 3 & 1 & 1 & 4 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 4 & 4 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1111} 4^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 5 & 4 & 4 \\ 3 & 1 & 4 & 3 & 4 & 3 \\ 5 & 4 & 6 & 5 & 5 & 6 \\ 5 & 3 & 5 & 3 & 4 & 5 \\ 4 & 4 & 5 & 4 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{111} 1^{111} 3^{111} 4^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -3, \ \det = -9, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 & 2 & 4 \\ 3 & 1 & 3 & 4 & 1 & 3 \\ 5 & 3 & 3 & 4 & 2 & 5 \\ 4 & 4 & 4 & 3 & 2 & 5 \\ 2 & 1 & 2 & 2 & 0 & 2 \\ 4 & 3 & 5 & 5 & 2 & 5 \end{pmatrix}_{b_0}$$

4.977 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 16, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 6 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 1^{111} 3^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 36, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 9 & 3 \\ 8 & 13 & 12 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 4 & -1 & 3 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 1^{111} 3^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 64, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 6 & 6 & 3 \\ 6 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 1^{111} 3^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 92, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 6 & 9 & 6 & 0 & 6 & 13 & 6 & 7 \\ 3 & 4 & 3 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 1^{111} 3^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 6 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 3 & 0 & 6 & 1 & 6 & 3 \\ 6 & 9 & 6 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 1^{111} 3^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 9 & 6 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 21 & 10 \\ 6 & 9 & 0 & 0 & 6 & 21 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11111} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 144, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 & 3 \end{pmatrix} b_0$$

$$[7] \quad 1^{\overline{11111}} 3 \quad 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 3 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 6 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -1 & 3 & 3 & 7 & 3 \end{pmatrix} b_0$$

[8] $1^{11111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 6 & 6 & 3 \\ 6 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 3 & 7 & 7 & 3 \end{pmatrix}_b$$

[9] $1^{11111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10, T_{\max} = 10, \Delta = -3, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 6 & 6 & 3 \\ 6 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 3 & 7 & 7 & 3 \end{pmatrix} b$$

$$T = 11 \quad [10] \, 1^{\textcolor{blue}{11}} 4^{\textcolor{blue}{11111} 3^{\textcolor{blue}{1}}} 1^{\textcolor{red}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 9.0$$

4.978 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 4 & -1 & 3 & 3 & 2 & 3 \end{pmatrix} b_0$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 0 & -1 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_1}$$

4.979 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9, \quad [0] \, 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \, T_{\max} = 10, \, \Delta = -35, \, \det = -11, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 2 \\ 3 & 4 & 0 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & 0 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^1 4^{11} 13^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 1 & 4 \\ 3 & 1 & 1 & 4 & 0 & 3 \\ 2 & 1 & 0 & 2 & 0 & 2 \\ 5 & 4 & 2 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^1 5^1 1^1 23^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 5 & 5 & 0 & 15 & 4 & 5 \\ 2 & 5 & 2 & 0 & 11 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 8 & 15 & 11 & 0 & 37 & 10 & 13 \\ 2 & 4 & 3 & 0 & 10 & 1 & 3 \\ 3 & 5 & 4 & -1 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

4.980 NHC(-2)(-3) + su₂ + su₃ + su₂ + su₂→(-3) + su₂

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

4.981 NHC(-2)(-3) + su₂ + su₃ + su₂→(-3)

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 61, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 3 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 3 & 4 & 2 & 1 & 0 & 4 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[6] \quad & 1^{\overline{11}} 4 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{\overline{11}} 4 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 2 & 1 & 4 \\ 3 & 1 & 1 & 0 & 3 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{\overline{11}} 4 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 5 & 1 & 4 \\ 3 & 1 & 4 & 0 & 3 \\ 5 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 6 & 0 & 5 \end{pmatrix}_{b_0} \\
T = 11 \quad [10] \quad & 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 7 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 & 2 \\ 5 & 6 & 7 & 35 & 0 & 6 & 11 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 6 & 0 & 0 & 2 \\ 2 & 2 & 2 & 11 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{\overline{11}} 5 123^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 3 & 5 \\ 1 & 1 & 0 & 4 & 0 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 & 0 & 2 \\ 2 & 2 & -1 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{\overline{11}} 5 123^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 6 & 0 & 31 & 8 & 7 & 11 \\ 1 & 1 & 0 & 8 & 0 & 2 & 2 \\ 1 & 1 & 0 & 7 & 2 & 0 & 2 \\ 2 & 2 & -1 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 13 & 0 & 4 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 7 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[12] \quad & 21^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[11] & 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[16] & 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 4 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[17] & 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[18] & 1^{12} \overset{1}{5} 13^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[19] & 1^{12} \overset{1}{5} 13^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 8 & 14 & 42 & 12 & 10 & 14 \\ 3 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[20] & 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 4 & 7 & 9 & 6 & 5 & 7 \\ 3 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[21] & 1^{13} \overset{1}{5} 13^1 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 12 & 21 & 97 & 18 & 15 & 21 \\ 3 & 3 & 18 & 1 & 3 & 3 \\ 2 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[22] & 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 5 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 5 & 11 & 0 & 4 & 5 \\ 6 & 11 & 22 & 0 & 7 & 10 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 7 & 0 & 1 & 3 \\ 3 & 5 & 10 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{13} \overset{12}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -10, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 2 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -10, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 5 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} \overset{12}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -10, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 6 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11} \overset{13}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 5 & 19 & 0 & 4 & 5 \\ 10 & 19 & 65 & 0 & 12 & 17 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 5 & 17 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11} \overset{11}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} \overset{11}{5} 123^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 1 & 2 & 1 & 0 & 0 \\ 2 & 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} \overset{11}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 4 & 5 \\ 2 & 5 & 2 & 2 & 3 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 2 & 1 & 3 \\ 3 & 5 & 4 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^1 5 123^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 4 & 5 \\ 2 & 5 & 2 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 5 & 4 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} 5 13^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 12 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{12} 5 13^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 2 \\ 2 & 2 & 4 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 5 & 8 & 4 & 5 \\ 2 & 5 & 2 & 6 & 3 & 4 \\ 5 & 8 & 6 & 8 & 5 & 6 \\ 2 & 4 & 3 & 5 & 1 & 3 \\ 3 & 5 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{12} 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 & 2 \\ 6 & 8 & 8 & 49 & 0 & 7 & 13 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 7 & 0 & 0 & 2 \\ 2 & 2 & 2 & 13 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^1 5 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 5 & 27 & 0 & 4 & 5 \\ 14 & 27 & 130 & 0 & 17 & 24 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 17 & 0 & 1 & 3 \\ 3 & 5 & 24 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 21 \overset{1}{5} 13 \overset{1}{1} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 2 & 5 & 2 & 6 & 3 & 4 \\ 5 & 8 & 6 & 8 & 5 & 6 \\ 2 & 4 & 3 & 5 & 1 & 3 \\ 3 & 5 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1 \overset{1}{5} 13 \overset{1}{1} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 5 & 13 & 0 & 4 & 5 \\ 6 & 13 & 25 & 0 & 8 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 3 \\ 3 & 5 & 11 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1 \overset{1}{12} 5 13 \overset{1}{1} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[42] \quad 1 \overset{1}{13} 5 13 \overset{1}{1} \overset{13}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 2 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[43] \quad 1 \overset{1}{5} 13 \overset{1}{1} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 5 & 8 & 0 & 4 & 5 \\ 4 & 8 & 9 & 0 & 5 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 5 & 7 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1 \overset{1}{5} 13 \overset{1}{1} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 7 & 8 & 11 & 72 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1 \overset{1}{13} 5 13 \overset{1}{1} \overset{13}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 5 & 21 & 0 & 4 & 5 \\ 10 & 21 & 70 & 0 & 13 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 13 & 0 & 1 & 3 \\ 3 & 5 & 18 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1 \overset{1}{13} 5 13 \overset{1}{1} \overset{13}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 16 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 2 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{11}_2 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 5 & 16 & 0 & 4 & 5 \\ 8 & 16 & 42 & 0 & 10 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 10 & 0 & 1 & 3 \\ 3 & 5 & 14 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{11}_3 5 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 5 & 24 & 0 & 4 & 5 \\ 12 & 24 & 97 & 0 & 15 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 5 & 21 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[50] $21^{11}_1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{11}_2 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 1 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 & 2 \\ 4 & 5 & 6 & 21 & 0 & 5 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11}_3 5 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 5 & 29 & 0 & 4 & 5 \\ 14 & 29 & 137 & 0 & 18 & 25 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 18 & 0 & 1 & 3 \\ 3 & 5 & 25 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{11}_1 5^1 1^1 23^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 4 & 5 & 5 & 18 & 0 & 5 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 8 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{12}_2 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 1 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 & 2 \\ 5 & 7 & 7 & 33 & 0 & 6 & 11 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 6 & 0 & 0 & 2 \\ 2 & 2 & 2 & 11 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{11}_1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 3 & 4 & 5 & 11 & 0 & 4 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 7 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{11}_4 5 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) =$

$$(1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 5 & 32 & 0 & 4 & 5 \\ 16 & 32 & 174 & 0 & 20 & 28 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 20 & 0 & 1 & 3 \\ 3 & 5 & 28 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^1 5^1 1^1 23^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 5 & 15 & 0 & 4 & 5 \\ 8 & 15 & 37 & 0 & 10 & 13 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 10 & 0 & 1 & 3 \\ 3 & 5 & 13 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^1 1^2 5^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 & 2 \\ 4 & 6 & 6 & 21 & 0 & 5 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 21^1 1^2 5^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^1 1^1 5^1 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 1 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 & 2 \\ 6 & 7 & 10 & 52 & 0 & 8 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 8 & 0 & 0 & 2 \\ 2 & 2 & 2 & 14 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^1 1^1 5^1 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 & 2 \\ 5 & 6 & 9 & 36 & 0 & 7 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 7 & 0 & 0 & 2 \\ 2 & 2 & 2 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^1 5^1 1^1 23^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 5 & 20 & 0 & 4 & 5 \\ 10 & 20 & 63 & 0 & 13 & 17 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 13 & 0 & 1 & 3 \\ 3 & 5 & 17 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[63] \quad 21^1 1^2 5^1 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 10 & 12 & 42 & 10 & 14 \\ 2 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^1 1^2 5^1 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 & 2 \\ 8 & 10 & 12 & 90 & 0 & 10 & 18 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 10 & 0 & 0 & 2 \\ 2 & 2 & 2 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{13}_{1^4} 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 1 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 & 2 \\ 9 & 12 & 13 & 112 & 0 & 11 & 20 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 11 & 0 & 0 & 2 \\ 2 & 2 & 2 & 20 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{12}_{1^3} 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 7 & 9 & 11 & 68 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{12}_{1^2} 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -154, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 & 2 \\ 6 & 8 & 10 & 50 & 0 & 8 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 8 & 0 & 0 & 2 \\ 2 & 2 & 2 & 14 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[68] $21^{13}_5 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -180, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 2 & 3 \\ 3 & 1 & 3 & 13 & 2 & 3 \\ 2 & 3 & 1 & 14 & 3 & 3 \\ 13 & 13 & 14 & 65 & 12 & 17 \\ 2 & 2 & 3 & 12 & 1 & 3 \\ 3 & 3 & 3 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{13}_{1^3} 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -180, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 1 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 & 2 \\ 8 & 11 & 12 & 88 & 0 & 10 & 18 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 10 & 0 & 0 & 2 \\ 2 & 2 & 2 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{13}_{1^2} 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -187, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 7 & 10 & 11 & 68 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[70] $21^{13}_5 5^{11} 13^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -197, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{12}_{1^4} 5^{11} 13^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -214, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 1 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 & 2 \\ 9 & 11 & 15 & 117 & 0 & 12 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 12 & 0 & 0 & 2 \\ 2 & 2 & 2 & 21 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{12}5^{11}13^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 & 2 \\ 8 & 10 & 14 & 93 & 0 & 11 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 11 & 0 & 0 & 2 \\ 2 & 2 & 2 & 19 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{13}5^{11}13^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -305, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 1 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 & 2 \\ 10 & 13 & 16 & 141 & 0 & 13 & 23 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 13 & 0 & 0 & 2 \\ 2 & 2 & 2 & 23 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{13}5^{11}13^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 1 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 & 2 \\ 9 & 12 & 15 & 115 & 0 & 12 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 12 & 0 & 0 & 2 \\ 2 & 2 & 2 & 21 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{13}5^{11}13^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -427, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 1 & 2 \\ 1 & 0 & 2 & 14 & 0 & 1 & 2 \\ 1 & 2 & 0 & 19 & 0 & 2 & 2 \\ 11 & 14 & 19 & 176 & 0 & 15 & 26 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 15 & 0 & 0 & 2 \\ 2 & 2 & 2 & 26 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.982 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$

$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 3 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{\overline{11}} 4^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -5, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 5 & 4 \\ 3 & 1 & 1 & 0 & 4 & 3 \\ 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 2 & 0 & 6 & 6 \\ 4 & 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 1 & 5 & 3 \\ 5 & 7 & 0 & 6 & 5 & 9 & 7 \\ 3 & 4 & 0 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{11}} 5^{\overline{12}} 13^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 1 & 5 & 3 \\ 6 & 7 & 0 & 6 & 5 & 9 & 7 \\ 3 & 4 & 0 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{12}} 5^{\overline{12}} 13^{\overline{11}} 1^{\overline{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 12, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 3 & 1 & 10 & 3 \\ 10 & 14 & 0 & 12 & 10 & 42 & 14 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^1 \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 8 & 5 \\ 2 & 3 & 1 & 0 & 2 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 5 & 3 \\ 4 & 8 & 5 & 0 & 5 & 9 & 7 \\ 3 & 5 & 3 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^1 \overset{1}{5} 13^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 4 & 3 \\ 3 & 5 & 11 & 0 & 4 & 8 & 5 \\ 6 & 11 & 22 & 0 & 7 & 16 & 10 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 7 & 0 & 1 & 5 & 3 \\ 4 & 8 & 16 & 0 & 5 & 9 & 7 \\ 3 & 5 & 10 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 \overset{1}{5} 1^4 23^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 5 & 5 & 0 & 4 & 15 & 5 \\ 2 & 5 & 2 & 0 & 3 & 11 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 10 & 3 \\ 8 & 15 & 11 & 0 & 10 & 37 & 13 \\ 3 & 5 & 4 & -1 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^1 \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 2 & 4 & 3 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13} \overset{1}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 17, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 18 & 3 \\ 2 & 3 & 0 & 3 & 1 & 15 & 3 \\ 15 & 21 & 0 & 18 & 15 & 97 & 21 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^1 \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 18, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 5 & 3 & 0 & 4 & 16 & 5 \\ 2 & 3 & 1 & 0 & 2 & 10 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 10 & 3 \\ 8 & 16 & 10 & 0 & 10 & 42 & 14 \\ 3 & 5 & 3 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} \overset{13}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 20, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 3 & 1 & 10 & 3 \\ 11 & 14 & 0 & 12 & 10 & 42 & 14 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^1 \overset{1}{5} 13^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 23, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 4 & 5 & 5 \\ 2 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 4 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[19] \begin{matrix} 1^1 \\ 1^1 \\ 3^1 \end{matrix} 5 13^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 4 & 3 \\ 3 & 5 & 19 & 0 & 4 & 8 & 5 \\ 10 & 19 & 65 & 0 & 12 & 27 & 17 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 12 & 0 & 1 & 5 & 3 \\ 4 & 8 & 27 & 0 & 5 & 9 & 7 \\ 3 & 5 & 17 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[20] \begin{matrix} 1^1 \\ 1^1 \\ 3^1 \end{matrix} 5 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 5 & 3 & 0 & 4 & 24 & 5 \\ 2 & 3 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 15 & 3 \\ 12 & 24 & 15 & 0 & 15 & 97 & 21 \\ 3 & 5 & 3 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[22] \begin{matrix} 1^1 \\ 1^1 \\ 12^1 \end{matrix} 5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 \\ 1 & 1 & 1 & 0 & 0 & 1 & 6 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 \\ 1 & 1 & 2 & 1 & 0 & 0 & 5 \\ 4 & 5 & 6 & 6 & 0 & 5 & 21 \\ 2 & 2 & 2 & 2 & -1 & 2 & 9 \end{pmatrix}_{b_0}$$

$$[23] \begin{matrix} 1^2 \\ 1^1 \\ 1^1 \end{matrix} 5 13^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 26, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 4 & 10 & 5 \\ 2 & 5 & 2 & 2 & 3 & 8 & 4 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 4 & 3 & 2 & 1 & 6 & 3 \\ 6 & 10 & 8 & 4 & 6 & 14 & 8 \\ 3 & 5 & 4 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[24] \begin{matrix} 1^3 \\ 1^1 \\ 1^1 \end{matrix} 5 13^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 4 & 15 & 5 \\ 2 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 4 & 3 & 2 & 1 & 9 & 3 \\ 9 & 15 & 12 & 6 & 9 & 34 & 12 \\ 3 & 5 & 4 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[25] \begin{matrix} 1^4 \\ 1^1 \\ 1^1 \end{matrix} 5 13^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 4 & 20 & 5 \\ 2 & 5 & 2 & 2 & 3 & 16 & 4 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 4 & 3 & 2 & 1 & 12 & 3 \\ 12 & 20 & 16 & 8 & 12 & 62 & 16 \\ 3 & 5 & 4 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[26] \begin{matrix} 1^1 \\ 1^{12} \\ 1^1 \end{matrix} 5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 5 & 7 & 7 & 7 & 0 & 6 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{11}_{1^{12}} 5^{11} 13^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 6 & 2 \\ 2 & 2 & 4 & 5 & 0 & 3 & 12 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 5 & 2 \\ 4 & 5 & 6 & 12 & 0 & 5 & 21 & 9 \\ 2 & 2 & 2 & 5 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{11}_{1^{11}} 5^{11} 13^{11} 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 4 & 3 \\ 3 & 5 & 13 & 0 & 4 & 8 & 5 \\ 6 & 13 & 25 & 0 & 8 & 17 & 11 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 5 & 3 \\ 4 & 8 & 17 & 0 & 5 & 9 & 7 \\ 3 & 5 & 11 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{11}_{1^{12}} 5^{11} 13^{11} 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 8 & 3 \\ 3 & 5 & 13 & 0 & 4 & 16 & 5 \\ 6 & 13 & 25 & 0 & 8 & 34 & 11 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 10 & 3 \\ 8 & 16 & 34 & 0 & 10 & 42 & 14 \\ 3 & 5 & 11 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{11}_{1^{13}} 5^{11} 13^{11} 1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 12 & 3 \\ 3 & 5 & 13 & 0 & 4 & 24 & 5 \\ 6 & 13 & 25 & 0 & 8 & 51 & 11 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 15 & 3 \\ 12 & 24 & 51 & 0 & 15 & 97 & 21 \\ 3 & 5 & 11 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{11}_{1^{13}} 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 6 & 7 & 10 & 9 & 0 & 8 & 52 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{11}_{1^{11}} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 4 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11}_{1^{11}} 5^{11} 13^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 5 & 2 \\ 2 & 2 & 4 & 5 & 0 & 3 & 9 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 9 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 5 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{11}_{1^{11}} 5^{11} 13^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 4 & 3 \\ 3 & 5 & 8 & 0 & 4 & 8 & 5 \\ 4 & 8 & 9 & 0 & 5 & 11 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 5 & 3 \\ 4 & 8 & 11 & 0 & 5 & 9 & 7 \\ 3 & 5 & 7 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^1 \begin{smallmatrix} 1 \\ 12 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 10 & 3 \\ 3 & 5 & 8 & 0 & 4 & 21 & 5 \\ 4 & 8 & 9 & 0 & 5 & 28 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 13 & 3 \\ 10 & 21 & 28 & 0 & 13 & 70 & 18 \\ 3 & 5 & 7 & -1 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^1 \begin{smallmatrix} 1 \\ 12 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 11 \\ 12 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 5 & 8 & 0 & 4 & 16 & 5 \\ 4 & 8 & 9 & 0 & 5 & 22 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 10 & 3 \\ 8 & 16 & 22 & 0 & 10 & 42 & 14 \\ 3 & 5 & 7 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^1 \begin{smallmatrix} 1 \\ 13 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 12 & 3 \\ 3 & 5 & 8 & 0 & 4 & 24 & 5 \\ 4 & 8 & 9 & 0 & 5 & 33 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 15 & 3 \\ 12 & 24 & 33 & 0 & 15 & 97 & 21 \\ 3 & 5 & 7 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^1 \begin{smallmatrix} 1 \\ 13 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 11 \\ 14 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 5 & 8 & 0 & 4 & 29 & 5 \\ 4 & 8 & 9 & 0 & 5 & 39 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 18 & 3 \\ 14 & 29 & 39 & 0 & 18 & 137 & 25 \\ 3 & 5 & 7 & -1 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^1 \begin{smallmatrix} 12 \\ 13 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 11 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 7 & 9 & 11 & 10 & 0 & 9 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^1 \begin{smallmatrix} 13 \\ 13 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 12 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 8 & 11 & 12 & 11 & 0 & 10 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$4.983 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$$

$$T = 10 \quad [0] \quad 1^1 \begin{smallmatrix} 11 \\ 13 \end{smallmatrix} 3^1 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & -1 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

4.984 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 11 \quad [4] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 4 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 3 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 2 & 2 & 4 & 2 & 3 & 4 \\ 3 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 10 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 7 & 10 & 0 & 9 & 22 & 10 \\ 3 & 4 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 2 & 4 & 2 & 9 & 4 \\ 8 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 6 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 3 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

- [6] $1^{12} \overset{11}{5} 13^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 3 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [8] $1^{11} \overset{12}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 10 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 8 & 10 & 0 & 9 & 22 & 10 \\ 3 & 4 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$
- [9] $1^{11} \overset{11}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 5 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 3 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [10] $1^{12} \overset{12}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 15 & 3 \\ 12 & 17 & 0 & 15 & 65 & 17 \\ 3 & 4 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}$
- [11] $1^{11} \overset{11}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 2 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 5 & 0 & 4 & 4 \\ 2 & 2 & 2 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$
- [12] $1^{11} \overset{11}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$
- [13] $1^{11} \overset{121}{5} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 5 & 11 & 0 & 12 & 5 \\ 6 & 11 & 22 & 0 & 23 & 10 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 23 & 0 & 22 & 10 \\ 3 & 5 & 10 & -1 & 10 & 4 \end{pmatrix}_{b_0}$
- [16] $1^{12} \overset{131}{5} 13^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 3 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [15] $1^{12} \overset{121}{5} 13^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 3 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{13}_{1^1} 5^{121} 13^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 3 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{13}_{1^{31}} 5^{131} 13^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 6 & 3 \\ 3 & 5 & 19 & 0 & 12 & 5 \\ 10 & 19 & 65 & 0 & 39 & 17 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 39 & 0 & 22 & 10 \\ 3 & 5 & 17 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{13}_{1^{11}} 5^{131} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 5 & 3 & 0 & 12 & 5 \\ 2 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 7 & 0 & 22 & 10 \\ 3 & 5 & 3 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{12}_{1^{11}} 5^{131} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 & 2 \\ 1 & 2 & 0 & 6 & 0 & 3 & 2 \\ 4 & 6 & 6 & 21 & 0 & 10 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 10 & 0 & 4 & 4 \\ 2 & 2 & 2 & 9 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[19] $1^{13}_{1^{11}} 5^{131} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 4 & 0 & 3 & 2 \\ 2 & 2 & 4 & 5 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 6 & 0 & 4 & 4 \\ 2 & 2 & 2 & 5 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{13}_{1^1} 5^{131} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 5 & 3 & 9 & 5 & 5 \\ 2 & 5 & 2 & 2 & 7 & 4 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 9 & 7 & 4 & 11 & 7 & 7 \\ 3 & 5 & 4 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{12}_{1^1} 5^{131} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 15 & 3 & 3 \\ 13 & 17 & 0 & 15 & 65 & 17 & 17 \\ 3 & 4 & 0 & 3 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{13}_{1^1} 5^{131} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 21 & 3 \\ 17 & 24 & 0 & 21 & 130 & 24 \\ 3 & 4 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{13} \overset{131}{5} 13^{11} \overset{31}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 3 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{13} \overset{1}{1^{21}} 5 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 5 & 16 & 0 & 7 & 5 \\ 8 & 16 & 42 & 0 & 20 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 20 & 0 & 8 & 6 \\ 3 & 5 & 14 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{13} \overset{1}{1^{21}} 5 13^{11} 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 4 & 3 \\ 3 & 5 & 16 & 5 & 7 & 5 \\ 8 & 16 & 42 & 12 & 20 & 14 \\ 3 & 5 & 12 & 1 & 5 & 3 \\ 4 & 7 & 20 & 5 & 8 & 6 \\ 3 & 5 & 14 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{13} \overset{1}{1^{12}} 5 13^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 8 & 3 \\ 3 & 5 & 8 & 5 & 15 & 5 \\ 4 & 8 & 9 & 6 & 21 & 7 \\ 3 & 5 & 6 & 1 & 11 & 3 \\ 8 & 15 & 21 & 11 & 39 & 13 \\ 3 & 5 & 7 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{13} \overset{1}{1^{31}} 5 13^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 5 & 24 & 0 & 7 & 5 \\ 12 & 24 & 97 & 0 & 30 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 30 & 0 & 8 & 6 \\ 3 & 5 & 21 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{13} \overset{1}{1^{11}} 5 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 5 & 8 & 0 & 7 & 5 \\ 4 & 8 & 9 & 0 & 10 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 10 & 0 & 8 & 6 \\ 3 & 5 & 7 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{13} \overset{1}{1^{31}} 5 13^{11} 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 4 & 3 \\ 3 & 5 & 24 & 5 & 7 & 5 \\ 12 & 24 & 97 & 18 & 30 & 21 \\ 3 & 5 & 18 & 1 & 5 & 3 \\ 4 & 7 & 30 & 5 & 8 & 6 \\ 3 & 5 & 21 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{13} \overset{1}{1^{11}} 5 13^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 4 & 3 \\ 3 & 5 & 8 & 5 & 7 & 5 \\ 4 & 8 & 9 & 6 & 10 & 7 \\ 3 & 5 & 6 & 1 & 5 & 3 \\ 4 & 7 & 10 & 5 & 8 & 6 \\ 3 & 5 & 7 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

- [32] $1^1 \begin{smallmatrix} 1 \\ 12 \end{smallmatrix} 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 3 & 0 & 20 & 5 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 65 & 17 \\ 3 & 5 & 3 & -1 & 17 & 4 \end{pmatrix}_{b_0}$
- [34] $1^1 \begin{smallmatrix} 1 \\ 11 \end{smallmatrix} 5 13^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 5 & 13 & 0 & 12 & 5 \\ 6 & 13 & 25 & 0 & 25 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 25 & 0 & 22 & 10 \\ 3 & 5 & 11 & -1 & 10 & 4 \end{pmatrix}_{b_0}$
- [33] $1^{12} \begin{smallmatrix} 1 \\ 11 \end{smallmatrix} 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 & 2 \\ 1 & 2 & 0 & 9 & 0 & 3 & 2 \\ 5 & 7 & 9 & 36 & 0 & 13 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 13 & 0 & 4 & 4 \\ 2 & 2 & 2 & 12 & -1 & 4 & 3 \end{pmatrix}_{b_0}$
- [35] $1^{11} \begin{smallmatrix} 1 \\ 11 \end{smallmatrix} 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 8 & 0 & 3 & 2 \\ 4 & 5 & 8 & 24 & 0 & 11 & 10 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 11 & 0 & 4 & 4 \\ 2 & 2 & 2 & 10 & -1 & 4 & 3 \end{pmatrix}_{b_0}$
- [36] $1^1 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 13^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 14 & 5 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 14 & 11 & 6 & 29 & 11 \\ 3 & 5 & 4 & 2 & 11 & 4 \end{pmatrix}_{b_0}$
- [37] $1^{11} \begin{smallmatrix} 1 \\ 11 \end{smallmatrix} 5 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 2 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$
- [38] $1^1 \begin{smallmatrix} 1 \\ 13 \end{smallmatrix} 5 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 3 & 0 & 28 & 5 \\ 2 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 28 & 17 & 0 & 130 & 24 \\ 3 & 5 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$
- [39] $1^1 \begin{smallmatrix} 1 \\ 121 \end{smallmatrix} 5 13^1 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 6 & 3 \\ 3 & 5 & 21 & 0 & 12 & 5 \\ 10 & 21 & 70 & 0 & 41 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 41 & 0 & 22 & 10 \\ 3 & 5 & 18 & -1 & 10 & 4 \end{pmatrix}_{b_0}$
- [40] $1^{11} \begin{smallmatrix} 1 \\ 11 \end{smallmatrix} 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

7.0

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 & 2 \\ 1 & 2 & 0 & 5 & 0 & 3 & 2 \\ 3 & 4 & 5 & 11 & 0 & 8 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 8 & 0 & 4 & 4 \\ 2 & 2 & 2 & 7 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[41] \ 1 \overset{1}{\underset{12}{5}} 13 \overset{1}{1} \overset{122}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 5 & 13 & 0 & 20 & 5 \\ 6 & 13 & 25 & 0 & 42 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 42 & 0 & 65 & 17 \\ 3 & 5 & 11 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[42] \ 1 \overset{1}{\underset{1}{5}} 13 \overset{1}{1} \overset{113}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 5 & 3 & 19 & 5 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 19 & 15 & 8 & 55 & 15 \\ 3 & 5 & 4 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[43] \ 1 \overset{1}{\underset{12}{5}} 13 \overset{1}{1} \overset{111}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 5 & 8 & 0 & 15 & 5 \\ 4 & 8 & 9 & 0 & 21 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 15 & 21 & 0 & 39 & 13 \\ 3 & 5 & 7 & -1 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[44] \ 1 \overset{1}{\underset{11}{5}} 13 \overset{1}{1} \overset{111}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 5 & 8 & 0 & 12 & 5 \\ 4 & 8 & 9 & 0 & 16 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 16 & 0 & 22 & 10 \\ 3 & 5 & 7 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[46] \ 1 \overset{1}{\underset{121}{5}} 13 \overset{1}{1} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 6 & 0 & 3 & 2 \\ 4 & 5 & 6 & 21 & 0 & 11 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 11 & 0 & 4 & 4 \\ 2 & 2 & 2 & 9 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[45] \ 1 \overset{1}{\underset{121}{5}} 13 \overset{1}{1} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.5

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 & 2 \\ 1 & 2 & 0 & 7 & 0 & 3 & 2 \\ 5 & 7 & 7 & 33 & 0 & 13 & 11 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 13 & 0 & 4 & 4 \\ 2 & 2 & 2 & 11 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[47] \ 1 \overset{1}{\underset{121}{5}} 13 \overset{1}{1} \overset{121}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.5

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 6 & 3 \\ 3 & 5 & 16 & 0 & 12 & 5 \\ 8 & 16 & 42 & 0 & 32 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 32 & 0 & 22 & 10 \\ 3 & 5 & 14 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[48] \ 1 \overset{1}{\underset{131}{5}} 13 \overset{1}{1} \overset{141}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.8

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 6 & 3 \\ 3 & 5 & 29 & 0 & 12 & 5 \\ 14 & 29 & 137 & 0 & 57 & 25 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 57 & 0 & 22 & 10 \\ 3 & 5 & 25 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{11} \begin{smallmatrix} 1^1 \\ 5 \\ 1^{11} \end{smallmatrix} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 2 & 4 & 5 & 0 & 12 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 7 & 12 & 0 & 21 & 9 \\ 2 & 2 & 2 & 5 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{11} \begin{smallmatrix} 1^1 \\ 5 \\ 1^{13} \end{smallmatrix} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 9 & 2 \\ 2 & 2 & 4 & 5 & 0 & 18 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 9 & 18 & 0 & 49 & 13 \\ 2 & 2 & 2 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11} \begin{smallmatrix} 1^1 \\ 5 \\ 1^{13} \end{smallmatrix} 13^1 1^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 14 & 3 \\ 3 & 5 & 13 & 0 & 28 & 5 \\ 6 & 13 & 25 & 0 & 59 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 28 & 59 & 0 & 130 & 24 \\ 3 & 5 & 11 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{12} \begin{smallmatrix} 1^1 \\ 5 \\ 1^{31} \end{smallmatrix} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 & 2 \\ 1 & 2 & 0 & 8 & 0 & 3 & 2 \\ 6 & 8 & 8 & 49 & 0 & 16 & 13 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 16 & 0 & 4 & 4 \\ 2 & 2 & 2 & 13 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{11} \begin{smallmatrix} 1^1 \\ 5 \\ 1^{31} \end{smallmatrix} 13^1 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 6 & 3 \\ 3 & 5 & 24 & 0 & 12 & 5 \\ 12 & 24 & 97 & 0 & 48 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 48 & 0 & 22 & 10 \\ 3 & 5 & 21 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{11} \begin{smallmatrix} 1^1 \\ 5 \\ 1^{12} \end{smallmatrix} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 1 & 1 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 7 & 0 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{11} \begin{smallmatrix} 1^4 \\ 5 \\ 1^1 \end{smallmatrix} 13^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 24 & 5 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 24 & 19 & 10 & 89 & 19 \\ 3 & 5 & 4 & 2 & 19 & 4 \end{pmatrix}_{b_0}$$

[56] $1^{11} \begin{smallmatrix} 1^1 \\ 5 \\ 1^{12} \end{smallmatrix} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 2 & 4 & 5 & 0 & 15 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 15 & 0 & 33 & 11 \\ 2 & 2 & 2 & 5 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{13} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 & 2 \\ 1 & 2 & 0 & 11 & 0 & 3 & 2 \\ 7 & 10 & 11 & 68 & 0 & 18 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 18 & 0 & 4 & 4 \\ 2 & 2 & 2 & 16 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{13} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 & 2 \\ 1 & 2 & 0 & 9 & 0 & 3 & 2 \\ 5 & 6 & 9 & 36 & 0 & 14 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 14 & 0 & 4 & 4 \\ 2 & 2 & 2 & 12 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{13} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 & 2 \\ 1 & 2 & 0 & 10 & 0 & 3 & 2 \\ 6 & 7 & 10 & 52 & 0 & 17 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 17 & 0 & 4 & 4 \\ 2 & 2 & 2 & 14 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{13} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 5 & 8 & 0 & 20 & 5 \\ 4 & 8 & 9 & 0 & 27 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 27 & 0 & 65 & 17 \\ 3 & 5 & 7 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{13} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 5 & 8 & 0 & 23 & 5 \\ 4 & 8 & 9 & 0 & 32 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 23 & 32 & 0 & 92 & 20 \\ 3 & 5 & 7 & -1 & 20 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{12} \overset{1}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 & 2 \\ 1 & 2 & 0 & 13 & 0 & 3 & 2 \\ 7 & 9 & 13 & 73 & 0 & 19 & 17 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 19 & 0 & 4 & 4 \\ 2 & 2 & 2 & 17 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{13} \overset{1}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 & 2 \\ 1 & 2 & 0 & 14 & 0 & 3 & 2 \\ 8 & 11 & 14 & 93 & 0 & 21 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 21 & 0 & 4 & 4 \\ 2 & 2 & 2 & 19 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{13} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 8 & 12 & 10 & 0 & 70 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{12} \overset{1}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 & 2 \\ 1 & 2 & 0 & 10 & 0 & 3 & 2 \\ 6 & 8 & 10 & 50 & 0 & 16 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 16 & 0 & 4 & 4 \\ 2 & 2 & 2 & 14 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{12} \overset{1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 1 & 1 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 10 & 10 & 0 & 65 & 15 \\ 2 & 2 & 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{12} \overset{1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 9 & 8 & 0 & 47 & 13 \\ 2 & 2 & 2 & 2 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{12} \overset{1}{5} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -142, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 5 & 8 & 0 & 28 & 5 \\ 4 & 8 & 9 & 0 & 38 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 28 & 38 & 0 & 130 & 24 \\ 3 & 5 & 7 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{12} \overset{1}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 & 2 \\ 1 & 2 & 0 & 11 & 0 & 3 & 2 \\ 7 & 9 & 11 & 68 & 0 & 19 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 19 & 0 & 4 & 4 \\ 2 & 2 & 2 & 16 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{13} \overset{1}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 & 2 \\ 1 & 2 & 0 & 12 & 0 & 3 & 2 \\ 8 & 11 & 12 & 88 & 0 & 21 & 18 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 21 & 0 & 4 & 4 \\ 2 & 2 & 2 & 18 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{12} \overset{1}{5} 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 & 2 \\ 1 & 2 & 0 & 17 & 0 & 3 & 2 \\ 9 & 11 & 17 & 124 & 0 & 25 & 22 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 25 & 0 & 4 & 4 \\ 2 & 2 & 2 & 22 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{12} \overset{1}{5} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -171, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 & 2 \\ 1 & 2 & 0 & 14 & 0 & 3 & 2 \\ 8 & 10 & 14 & 93 & 0 & 22 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 22 & 0 & 4 & 4 \\ 2 & 2 & 2 & 19 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{12} \overset{11}{5} 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -187, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 13 & 11 & 0 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{13} \overset{11}{5} 13^1 1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -197, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 & 2 \\ 1 & 2 & 0 & 15 & 0 & 3 & 2 \\ 9 & 12 & 15 & 115 & 0 & 24 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 24 & 0 & 4 & 4 \\ 2 & 2 & 2 & 21 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{13} \overset{11}{5} 13^1 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -203, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 & 2 \\ 1 & 2 & 0 & 18 & 0 & 3 & 2 \\ 10 & 13 & 18 & 148 & 0 & 27 & 24 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 27 & 0 & 4 & 4 \\ 2 & 2 & 2 & 24 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{13} \overset{11}{5} 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -264, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 1 & 1 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 14 & 12 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

4.985 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{su}_2$

$T = 12$ [0] $1^{11} \overset{11}{5} 13^1 1^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2}$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 2 & 2 & 4 & 2 & 3 & 6 & 4 \\ 3 & 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1] $1^1 \overset{11}{5} 13^1 1^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2}$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 5 & 3 & 9 & 5 & 5 \\ 2 & 5 & 2 & 2 & 7 & 4 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 9 & 7 & 4 & 11 & 7 & 7 \\ 3 & 5 & 4 & 2 & 7 & 2 & 4 \\ 3 & 5 & 4 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

[2] $1^1 \overset{12}{5} 13^1 1^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2}$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 5 & 5 & 3 & 9 & 10 & 5 \\ 2 & 5 & 2 & 2 & 7 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 9 & 7 & 4 & 11 & 14 & 7 \\ 6 & 10 & 8 & 4 & 14 & 14 & 8 \\ 3 & 5 & 4 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overset{13}{\underset{1}{1}}} 5 \ 13^{\overset{1113}{\underset{1}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 51, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 5 & 5 & 3 & 9 & 15 & 5 \\ 2 & 5 & 2 & 2 & 7 & 12 & 4 \\ 2 & 3 & 2 & 0 & 4 & 6 & 2 \\ 5 & 9 & 7 & 4 & 11 & 21 & 7 \\ 9 & 15 & 12 & 6 & 21 & 34 & 12 \\ 3 & 5 & 4 & 2 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overset{21}{\underset{1}{1}}} 5 \ 13^{\overset{1121}{\underset{1}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 71, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 & 3 \\ 3 & 5 & 5 & 3 & 14 & 5 & 5 \\ 2 & 5 & 2 & 2 & 11 & 4 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 14 & 11 & 6 & 29 & 11 & 11 \\ 3 & 5 & 4 & 2 & 11 & 2 & 4 \\ 3 & 5 & 4 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overset{31}{\underset{1}{1}}} 5 \ 13^{\overset{1131}{\underset{1}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 107, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 & 3 \\ 3 & 5 & 5 & 3 & 19 & 5 & 5 \\ 2 & 5 & 2 & 2 & 15 & 4 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 & 2 \\ 11 & 19 & 15 & 8 & 55 & 15 & 15 \\ 3 & 5 & 4 & 2 & 15 & 2 & 4 \\ 3 & 5 & 4 & 2 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

4.986 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overset{1}{\underset{11}{1}}} 4 \ 1^{\overset{1}{\underset{11}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -21, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overset{1}{\underset{1}{1}}} 4 \ 1^{\overset{1}{\underset{11}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overset{1}{\underset{1}{1}}} 4 \ 1^{\overset{1}{\underset{11}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -40, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overset{1}{\underset{11}{1}}} 4 \ 1^{\overset{1}{\underset{11}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -61, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 4 & 7 & 8 & 0 & 5 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overset{11}{\underset{11}{1}}} 4 \ 1^{\overset{1}{\underset{11}{1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 5 & 7 & 8 & 5 & 0 & 6 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 11 & 3 & 0 & 4 \\ 6 & 11 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12} \ 5^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 8 & 14 & 38 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13} \ 5^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 21 & 3 & 0 & 4 \\ 12 & 21 & 88 & 12 & 0 & 18 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{13} \ 5^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 3 & 0 & 4 \\ 10 & 18 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 21^{11} \ 5^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 3 \\ 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[19] \ 21^{12} \ 5^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 3 \\ 4 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[20] \ 21^{13} \ 5^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 2 & 3 \\ 4 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.987 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11} \ 4^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \ 4^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -33$, $\det = 15$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & -1 & 0 & 3 \end{pmatrix}_{b_0}$

$$[4] \begin{matrix} 1^1 \\ 1^1 \end{matrix} 4 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{su}}_3 + \textcolor{red}{\mathbf{su}}_3 + \textcolor{blue}{\mathbf{su}}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -33, \quad \det = 18, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 5 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \begin{array}{c} 1^1 \\ 1^{11} 4^1 1^{11} \end{array} + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}u_2 + \textcolor{orange}{5}u_3 + \textcolor{red}{5}u_3 + \textcolor{blue}{5}u_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 3 & 0 & 7 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 6 & 5 & 0 & 8 & 6 \\ 3 & 4 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[5] \begin{array}{c} 1^1 \\ 1^{11} 4 1^{11} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{blue}{u}_2 + \textcolor{orange}{s} \textcolor{blue}{u}_3 + \textcolor{red}{s} \textcolor{blue}{u}_3 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \\ 1^1 \end{array} \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -33, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1 \overset{11}{\overset{111}{\underset{111}{4}}} 1^1 + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{u}_2 + \textcolor{orange}{s} \textcolor{orange}{u}_3 + \textcolor{red}{s} \textcolor{red}{u}_3 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 & 4 \\ 4 & 7 & 8 & 0 & 5 & 5 & 6 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 3 & 4 & 5 & 0 & 2 & 1 & 3 \\ 3 & 4 & 6 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \begin{array}{c} 1^1 \\ 1^{111} 4^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_2 + \textcolor{orange}{s}u_3 + \textcolor{red}{s}u_3 + \textcolor{blue}{s}u_2 \\ 1^1 \end{array} \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -33, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1 \overset{1}{\overset{11}{\underset{11}{4}}} + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}u_2 + \textcolor{orange}{5}u_3 + \textcolor{red}{5}u_3 + \textcolor{blue}{5}u_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 91, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 4 & 1 & 2 & 0 & 5 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 3 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1 \overset{11}{\overset{11}{\underset{11}{1}}} 4 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -33, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 4 & 1 & 2 & 0 & 5 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 3 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.988 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [1] \quad 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 5 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & -1 & 3 & 4 & 3 \end{pmatrix} b_0$$

$$[2] \quad 1^{\overline{11}} 4^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 0 & 5 & 4 \\ 2 & 3 & 1 & 3 & 0 & 0 & 3 & 3 \\ 2 & 3 & 3 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 3 & 2 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad \begin{array}{c} 1^1 \\ 1^{111} \\ 1^{11} \end{array} 4 \quad \begin{array}{c} 1^{11} \\ 1^{11} \end{array} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{u}_3 + \textcolor{red}{s}\textcolor{red}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{u}_2 + \textcolor{violet}{s}\textcolor{violet}{u}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 7 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 6 & 5 & 0 & 8 & 5 & 6 \\ 2 & 4 & 3 & 2 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad \begin{array}{c} 1^{11} \\ 1^{111} 4 1^{11} \\ 1^{11} \end{array} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 5 & 3 & 0 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 6 & 5 & 0 & 8 & 5 & 6 \\ 3 & 4 & 3 & 2 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{1111} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -81, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 & 4 & 4 \\ 4 & 7 & 8 & 0 & 5 & 5 & 5 & 6 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 & 2 \\ 3 & 4 & 5 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 5 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & 6 & -1 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

4.989 $\text{NHC}(-2)(-3) + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_3 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_2$

$$T = 11 \quad [0] \, 1^{\overset{11}{\underset{11}{1}}} 4 \, 1^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2 + \textcolor{brown}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 11.0$$

$$[1] \begin{matrix} 1^{11} \\ 1^{1111} \\ 1^{11} \end{matrix} 4 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 11.0$$

4.990 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$

$$T = 10 \quad [0] \quad 1 \overset{1}{\underset{1}{5}} 1 \overset{1}{2} 1 + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{u}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 3 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 10 & 3 & 0 & 4 \\ 8 & 10 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 85, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 5 & 7 & 8 & 4 & 0 & 6 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 11 & 3 & 0 & 4 \\ 7 & 11 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 13 & 17 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 15 & 3 & 0 & 4 \\ 9 & 15 & 40 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 196, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 10 & 14 & 38 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{2^1 21} \overset{11}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 2 & 1 & 3 \\ 6 & 13 & 12 & 5 & 0 & 7 \\ 5 & 12 & 8 & 4 & 0 & 6 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 316, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 18 & 3 & 0 & 4 \\ 12 & 18 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 381, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 21 & 3 & 0 & 4 \\ 15 & 21 & 88 & 12 & 0 & 18 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{13} \overset{2}{5} 1^2 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 2 & 1 & 3 \\ 6 & 13 & 24 & 5 & 0 & 7 \\ 10 & 24 & 38 & 8 & 0 & 12 \\ 2 & 5 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{13} \overset{2}{5} 1^2 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 901, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 2 & 1 & 3 \\ 6 & 13 & 36 & 5 & 0 & 7 \\ 15 & 36 & 88 & 12 & 0 & 18 \\ 2 & 5 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

4.991 NHC(-2)(-3) + $\textcolor{red}{su}_2 + (-3) \rightarrow (-2)$

$$T = 7 \quad [0] \ 1^{11} 31^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -21, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -24, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 6 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 31^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -29, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 3 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 31^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -51, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 5 & 0 & 4 \\ 6 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 31^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -66, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.21 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 10 & 0 & 4 \\ 12 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} 31^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -66, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

- [7] $1^{12}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 17 & 0 & 4 \\ 15 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{13}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 9 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{112}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 9 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 18 & 1 & 4 \\ 5 & 4 & 15 & 0 & 4 \\ 18 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 1 & 4 \\ 5 & 4 & 9 & 0 & 4 \\ 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{12}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 13 & 0 & 4 \\ 12 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 9 & 0 & 4 \\ 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{14}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 18 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{13}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 8 & 0 & 4 \\ 6 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{113}31^{21}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 15 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{13}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 4 & 8 & 9 & 1 & 4 \\ 8 & 9 & 12 & 0 & 5 \\ 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 7 & -1 & 1 \end{pmatrix}_{b_0} \\
[17] \quad 1^{11}3^{12}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 1 \end{pmatrix}_{b_0} \\
[18] \quad 131^{11}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 8 & 6 & 1 & 4 \\ 8 & 9 & 8 & 0 & 5 \\ 6 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0} \\
[19] \quad 1^{11}31^{12}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 14 & 0 & 4 \\ 15 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0} \\
[20] \quad 1^{11}4^{13}1^{12}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 21 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -1 & 1 \end{pmatrix}_{b_0} \\
[21] \quad 1^{11}31^{13}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 9 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0} \\
[22] \quad 1^{11}4^{13}1^{12}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0} \\
[24] \quad 1^{11}31^{12}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 13 & 0 & 4 \\ 9 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0} \\
[23] \quad 1^{11}3^{12}1^{12}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 23 & 0 & 4 \\ 15 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0} \\
[25] \quad 1^{11}4^{13}1^{12}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 18 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 2 \end{pmatrix}_{b_0} \\
[26] \quad 1^{12}31^{12}2^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 18 & 1 & 4 \\ 5 & 4 & 18 & 0 & 4 \\ 18 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[27] \ 1^{12}31^{12}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 18 & 0 & 4 \\ 12 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{14}31^{12}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 24 & 1 & 4 \\ 5 & 4 & 26 & 0 & 4 \\ 24 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}31^{13}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 21 & 1 & 4 \\ 5 & 4 & 19 & 0 & 4 \\ 21 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{13}31^{12}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 21 & 1 & 4 \\ 5 & 4 & 22 & 0 & 4 \\ 21 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{13}31^{14}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 16 & 0 & 4 \\ 12 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 31^{11}1^{12}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -409, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 \\ 3 & 4 & 0 & 3 & 1 \\ 3 & 3 & 3 & 1 & 1 \\ 1 & 1 & 1 & 1 & -2 \\ 2 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11}31^{14}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 27 & 1 & 4 \\ 5 & 4 & 24 & 0 & 4 \\ 27 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[34] \ 22^1 1^{11}3 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 7 & 7 & 7 & 4 \\ 7 & 5 & 8 & 7 & 5 \\ 7 & 8 & 4 & 7 & 4 \\ 7 & 7 & 7 & 5 & 5 \\ 4 & 5 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11}31^{13}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 17 & 0 & 4 \\ 12 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{12}31^{13}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 24 & 1 & 4 \\ 5 & 4 & 23 & 0 & 4 \\ 24 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{12}31^{13}2^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$\begin{aligned}
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 22 & 0 & 4 \\ 15 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0} \\
[37] \quad & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 32 & 0 & 4 \\ 21 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0} \\
[38] \quad & 1^{13}31^{15}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 20 & 0 & 4 \\ 15 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0} \\
[40] \quad & 1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -586, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 27 & 1 & 4 \\ 5 & 4 & 27 & 0 & 4 \\ 27 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[41] \quad & 1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 27 & 0 & 4 \\ 18 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[42] \quad & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -651, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 30 & 1 & 4 \\ 5 & 4 & 31 & 0 & 4 \\ 30 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0} \\
[43] \quad & 1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 21 & 0 & 4 \\ 15 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0} \\
[44] \quad & 1^{12}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -666, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 30 & 1 & 4 \\ 5 & 4 & 28 & 0 & 4 \\ 30 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0} \\
[45] \quad & 31^{112}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -724, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 3 & 6 & 1 & 2 \\ 3 & 1 & 4 & 6 & 1 & 3 \\ 3 & 4 & 0 & 6 & 1 & 2 \\ 6 & 6 & 6 & 10 & 2 & 6 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 6 & 0 & 1 \end{pmatrix}_{b_0} \\
[46] \quad & 1^{13}31^{16}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 24 & 0 & 4 \\ 18 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0} \\
[47] \quad & 1^{13}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -861, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 4 & 5 & 33 & 1 & 4 \\ 5 & 4 & 32 & 0 & 4 \\ 33 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{12}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -864, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 26 & 0 & 4 \\ 18 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{13}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -984, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 31 & 0 & 4 \\ 21 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{14}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1006, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 36 & 1 & 4 \\ 5 & 4 & 36 & 0 & 4 \\ 36 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{14}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1024, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 24 & 1 & 3 \\ 4 & 4 & 36 & 0 & 4 \\ 24 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{13}31^{17}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1044, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 21 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[53] $31^{113}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 9 & 1 & 2 \\ 3 & 1 & 4 & 9 & 1 & 3 \\ 3 & 4 & 0 & 9 & 1 & 2 \\ 9 & 9 & 9 & 25 & 3 & 9 \\ 1 & 1 & 1 & 3 & -2 & 0 \\ 2 & 3 & 2 & 9 & 0 & 1 \end{pmatrix}_{b_0}$$

[54] $31^{114}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1984, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 12 & 1 & 2 \\ 3 & 1 & 4 & 12 & 1 & 3 \\ 3 & 4 & 0 & 12 & 1 & 2 \\ 12 & 12 & 12 & 46 & 4 & 12 \\ 1 & 1 & 1 & 4 & -2 & 0 \\ 2 & 3 & 2 & 12 & 0 & 1 \end{pmatrix}_{b_0}$$

[55] $31^{115}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -2929, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 15 & 1 & 2 \\ 3 & 1 & 4 & 15 & 1 & 3 \\ 3 & 4 & 0 & 15 & 1 & 2 \\ 15 & 15 & 15 & 73 & 5 & 15 \\ 1 & 1 & 1 & 5 & -2 & 0 \\ 2 & 3 & 2 & 15 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [56] $1^{211}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{11}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{11}4^{11}12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[59] $21^141^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 13 & 12 & 6 & 7 \\ 6 & 12 & 8 & 5 & 6 \\ 3 & 6 & 5 & 0 & 2 \\ 4 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[60] $22^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 13 & 8 & 6 & 7 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 6 & 3 & 0 & 2 \\ 3 & 7 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{11}4^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{11}41^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 3 & 1 & 3 \\ 6 & 13 & 8 & 0 & 7 \\ 3 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{11}4^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11}4^{11}12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{12}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{11}4^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

- [69] $21^{11}41^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 13 & 20 & 6 & 7 \\ 12 & 20 & 26 & 8 & 10 \\ 3 & 6 & 8 & 0 & 2 \\ 4 & 7 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [70] $1^{11}2^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [72] $1^{13}2^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [71] $1^{12}2^{12}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [73] $1^{12}2^{13}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$
- [74] $1^{11}2^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$
- [75] $1^{11}2^{111}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 8 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [76] $1^{211}2^{12}41^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [77] $1^{12}2^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$
- [78] $1^{141}1^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [79] $1^{11}41^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$\begin{aligned}
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 3 & 4 & 1 & 2 \\ 1 & 0 & 3 & 4 & 0 & 2 \\ 3 & 3 & 3 & 8 & 0 & 3 \\ 4 & 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[80] \quad & 22^1 1^{11} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 8 & 5 & 6 \\ 3 & 6 & 5 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[81] \quad & 1^{13} 4^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[82] \quad & 22^1 1^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 8 & 3 & 3 \\ 6 & 13 & 20 & 6 & 7 \\ 8 & 20 & 26 & 8 & 10 \\ 3 & 6 & 8 & 0 & 2 \\ 3 & 7 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[83] \quad & 21^{11} 41^{14} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 13 & 56 & 6 & 7 \\ 30 & 56 & 212 & 23 & 28 \\ 3 & 6 & 23 & 0 & 2 \\ 4 & 7 & 28 & 2 & 3 \end{pmatrix}_{b_0} \\
[84] \quad & 21^{11} 41^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 13 & 32 & 6 & 7 \\ 18 & 32 & 68 & 13 & 16 \\ 3 & 6 & 13 & 0 & 2 \\ 4 & 7 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[85] \quad & 1^{11} 41^{11} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 8 & 1 & 3 \\ 6 & 13 & 20 & 0 & 7 \\ 8 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[86] \quad & 1^{13} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[87] \quad & 1^{12} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[88] \quad & 21^{11} 41^{13} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 13 & 44 & 6 & 7 \\ 24 & 44 & 130 & 18 & 22 \\ 3 & 6 & 18 & 0 & 2 \\ 4 & 7 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[89] \quad & 1^{13} 4^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{13} 2^{14} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 18 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{12} 2^{13} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{12} 2^{14} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 19 & 25 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{14} 2^{12} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{12} 2^{13} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{11} 2^{112} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 4 & 19 & 0 & 4 \\ 13 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{14} 2^{11} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{14} 1^{12} 2^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 13 & 24 & 0 & 7 \\ 10 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[98] $22^1 1^{12} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 10 & 3 & 3 \\ 6 & 13 & 24 & 6 & 7 \\ 10 & 24 & 38 & 10 & 12 \\ 3 & 6 & 10 & 0 & 2 \\ 3 & 7 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{13} 2^{12} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{14} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 17 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{13} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{14} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 20 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{14} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{13} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 447, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 21 & 29 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{14} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 20 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[106] \ 22^1 1^{12} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 475, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 13 & 3 & 3 \\ 6 & 13 & 32 & 6 & 7 \\ 13 & 32 & 68 & 13 & 16 \\ 3 & 6 & 13 & 0 & 2 \\ 3 & 7 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{11} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 26 & 0 & 4 \\ 18 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{11} 41^{12} 22^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 13 & 1 & 3 \\ 6 & 13 & 32 & 0 & 7 \\ 13 & 32 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{14} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 20 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{\textcolor{blue}{1}}41^{\textcolor{red}{13}}22^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 553, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 3 \\ 6 & 13 & 36 & 0 & 7 \\ 15 & 36 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 22^{\textcolor{blue}{1}}1^{\textcolor{red}{13}}41^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 15 & 3 & 3 \\ 6 & 13 & 36 & 6 & 7 \\ 15 & 36 & 88 & 15 & 18 \\ 3 & 6 & 15 & 0 & 2 \\ 3 & 7 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\textcolor{red}{14}}4^{\textcolor{blue}{2^1 1^4}}1^{\textcolor{blue}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 23 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\textcolor{blue}{11}}4^{\textcolor{red}{2^1 1^4}}1^{\textcolor{blue}{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 23 & 1 & 3 \\ 4 & 4 & 33 & 0 & 4 \\ 23 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 22^{\textcolor{blue}{1}}1^{\textcolor{red}{13}}41^{\textcolor{blue}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 846, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 18 & 3 & 3 \\ 6 & 13 & 44 & 6 & 7 \\ 18 & 44 & 130 & 18 & 22 \\ 3 & 6 & 18 & 0 & 2 \\ 3 & 7 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\textcolor{blue}{11}}41^{\textcolor{red}{13}}22^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 18 & 1 & 3 \\ 6 & 13 & 44 & 0 & 7 \\ 18 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\textcolor{blue}{1}}41^{\textcolor{red}{14}}22^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 20 & 1 & 3 \\ 6 & 13 & 48 & 0 & 7 \\ 20 & 48 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 22^{\textcolor{blue}{1}}1^{\textcolor{red}{14}}41^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 910, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 20 & 3 & 3 \\ 6 & 13 & 48 & 6 & 7 \\ 20 & 48 & 158 & 20 & 24 \\ 3 & 6 & 20 & 0 & 2 \\ 3 & 7 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 22^{\textcolor{blue}{1}}1^{\textcolor{red}{14}}41^{\textcolor{blue}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 23 & 3 & 3 \\ 6 & 13 & 56 & 6 & 7 \\ 23 & 56 & 212 & 23 & 28 \\ 3 & 6 & 23 & 0 & 2 \\ 3 & 7 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\textcolor{blue}{11}}41^{\textcolor{red}{14}}22^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$\begin{aligned}
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 23 & 1 & 3 \\ 6 & 13 & 56 & 0 & 7 \\ 23 & 56 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 28 & -1 & 3 \end{pmatrix}_{b_0} \\
[120] & 1^1 41^{15} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 25 & 1 & 3 \\ 6 & 13 & 60 & 0 & 7 \\ 25 & 60 & 248 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 30 & -1 & 3 \end{pmatrix}_{b_0} \\
[121] & 1^1 41^{16} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1876, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 30 & 1 & 3 \\ 6 & 13 & 72 & 0 & 7 \\ 30 & 72 & 358 & 0 & 36 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 36 & -1 & 3 \end{pmatrix}_{b_0} \\
[122] & 1^1 41^{17} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 2513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 35 & 1 & 3 \\ 6 & 13 & 84 & 0 & 7 \\ 35 & 84 & 488 & 0 & 42 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 42 & -1 & 3 \end{pmatrix}_{b_0} \\
& T = 9 \quad [123] \quad 22^{11} 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[125] & 1^{11} 51^{14} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[124] & 21^{12} 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 10 & 3 & 4 \\ 9 & 10 & 19 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[127] & 1^1 51^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[126] & 22^{11} 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[129] & 1^{12} 51^{21} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[128] $1^{11} 1^{11} 5 12^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$

[130] $1^{11} 5 12^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$

[131] $1^{11} 5 12^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$

[132] $1^{11} 5 122^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$

[133] $1^{11} 5 12^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$

[134] $21^{11} 5 12^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[135] $1^{11} 5 12^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$

[136] $1^{12} 5 12^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[140] $1^{11} 5 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$

[139] $21^{11} 5 1^{11} 22^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 1 & 2 \\ 3 & 4 & 5 & 6 & 0 & 4 \\ 2 & 5 & 3 & 4 & 0 & 3 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{11}5^{1^4}1^{2^1}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[137] $1^{12}5^{1^1}1^{2^1}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 6 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[142] $1^{11}5^{1^3}1^{2^1}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[141] $1^{11}5^{1^1}1^{3^1}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 13 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{11}5^{1^4}1^{3^1}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 19 & 25 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{11}5^{1^1}1^{1^1}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[143] $1^{12}5^{1^2}1^{2^1}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{1^2}5^{1^1}2^{1^1}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 13 & 5 & 0 & 7 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[147] $22^{1^1}1^{1^1}5^{1^1}1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -49$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 7 & 10 & 19 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[148] $22^{11}1^35^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -49$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 17 & 22 & 115 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[149] $21^{11}5^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[150] $1^{13}5^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{11}5^{12}1^12^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{11}5^{13}1^12^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{11}5^{13}1^12^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{11}5^{21}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{11}5^{11}1^12^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{11}5^{21}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[157] $22^{11}1^{11}5^{11}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 57 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[158] $21^{11}1^{11}5^{11}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 7 & 2 & 3 \\ 4 & 4 & 10 & 3 & 4 \\ 7 & 10 & 19 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{11}5^{11}1^{11}2^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{11}5^{11}1^{11}2^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[161] $22^{11}1^{11}5^{11}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 36 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{13}5^{12}1^{11}12^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[163] $22^{11}1^{11}5^{11}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 59 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{13}5^{12}1^{11}12^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{11}5^{11}1^{11}2^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{11} 5^{12} 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{11} 5^{12} 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[168] $21^{12} 5^{12} 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{12} 5^{14} 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{12} 5^{21} 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{12} 5^{11} 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{11} 5^{13} 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{11} 5^{12} 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{11} 5^{12} 1^{32} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[175] $21^{12}5^{21}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 14 & 20 & 82 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{12}5^{13}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 13 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{12}5^{13}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[178] $21^{13}5^{11}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[179] $22^{11}5^{12}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 14 & 20 & 82 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{21}5^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 7 & 1 & 3 \\ 6 & 13 & 17 & 0 & 7 \\ 7 & 17 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{12}5^{13}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{21}5^{11}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{13}5^{11}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{13} 5^{2^1 21^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{12} 5^{1^1} 1^{32^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 15 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[185] $21^{12} 5^{2^1 1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 10 & 14 & 36 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{12} 5^{1^4} 1^{2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 150 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[188] $22^{1^1} 3^{1^1} 5^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 19 & 26 & 148 & 19 & 24 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{12} 5^{1^2} 1^{2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{12} 5^{2^1 21^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{12} 5^{1^3} 1^{2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{1^1} 5^{1^2} 1^{2^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{13}5^{12}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[194] $22^11^{13}5^{13}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 16 & 24 & 113 & 16 & 21 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{13}5^{14}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{13}5^{13}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{12}5^{12}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[198] $21^{13}5^{21}1^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 30 & 3 & 4 \\ 21 & 30 & 187 & 21 & 27 \\ 2 & 3 & 21 & 0 & 2 \\ 3 & 4 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{12}5^{14}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 21 & 29 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{12}5^{21}1^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{12}5^{13}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[202] $1 \overset{1}{\underset{13}{5}} 1^3 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[203] $1 \overset{1}{\underset{13}{5}} 1^2 2^1 4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 24 & 32 & 234 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[204] $1 \overset{1}{\underset{13}{5}} 1^2 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[205] $1 \overset{1}{\underset{13}{5}} 1^2 2^1 12 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 14 & 1 & 3 \\ 6 & 13 & 34 & 0 & 7 \\ 14 & 34 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[206] $1 \overset{1}{\underset{13}{5}} 1^2 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[207] $1 \overset{1}{\underset{13}{5}} 1^2 2^1 2^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[208] $22 \overset{1}{\underset{13}{5}} 1^3 5 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 21 & 30 & 187 & 21 & 27 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[209] $21 \overset{1}{\underset{13}{5}} 1^2 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 13 & 18 & 59 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[210] $1 \overset{1}{\underset{13}{5}} 1^3 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{2^1 21^{12}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 13 & 24 & 0 & 7 \\ 10 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{1^3 5^1 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{1^4 5^1 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 20 & 30 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[214] $21^{1^3 5^1 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 113 & 16 & 21 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{2^1 21^{12}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 12 & 1 & 3 \\ 6 & 13 & 29 & 0 & 7 \\ 12 & 29 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{1^2 5^1 3^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{1^3 5^1 3^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{2^1 21^3} 5^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{1^4 5^1 3^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 23 & 33 & 232 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{13} 5^{2^1 21^4} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -461, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 26 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 26 & 36 & 281 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 33 & -1 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{13} 5^{2^1 21^{13}} 1^3 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 21 & 1 & 3 \\ 6 & 13 & 51 & 0 & 7 \\ 21 & 51 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{2^1 21^{13}} 5^{1^1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 3 \\ 6 & 13 & 36 & 0 & 7 \\ 15 & 36 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{2^1 21^{13}} 5^{1^2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -633, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 19 & 1 & 3 \\ 6 & 13 & 46 & 0 & 7 \\ 19 & 46 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{2^1 21^{13}} 5^{1^1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -657, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 17 & 1 & 3 \\ 6 & 13 & 41 & 0 & 7 \\ 17 & 41 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [225] $1^{11} 5^{1^1 31} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{11} 5^{1^1 31} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

[227] $1^{11} 5^{1^1 31} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{2^1 1^1} 5^{131} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 2 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{\overline{1}^1} 5^{2^1 1^1} 131^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 8 & 0 & 5 \\ 5 & 8 & 9 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

[230] $1^{\overline{1}^1} 5^{1^1 1^1 3^1} 12^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{\overline{1}^1} 5^{2^1 1^1} 131^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 5 & 13 & 0 & 5 \\ 7 & 13 & 25 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 4 \end{pmatrix}_{b_0}$$

[232] $1^{\overline{1}^1} 5^{2^1 1^1} 131^{\overline{1}^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 5 & 18 & 0 & 5 \\ 9 & 18 & 49 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

[233] $1^{\overline{1}^1} 5^{2^1 1^1} 131^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 4 & 4 & 5 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{\overline{1}^1} 5^{2^1 1^1} 131^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 5 & 5 & 8 & 24 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[235] $1^{\overline{1}^2} 5^{2^1 1^1} 131^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 5 & 6 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{\overline{1}^2} 5^{2^1 1^2} 131^{\overline{1}^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 5 & 21 & 0 & 5 \\ 12 & 21 & 70 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 4 \end{pmatrix}_{b_0}$$

[237] $1^{\overline{1}^1} 5^{2^1 1^2} 131^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 7 & 6 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{12} 5^{211^2} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 5 & 26 & 0 & 5 \\ 14 & 26 & 106 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 4 \end{pmatrix}_{b_0}$$

[239] $1^{12} 5^{211^1} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 6 & 7 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{11} 5^{211^2} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 7 & 12 & 57 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{12} 5^{211^1} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 7 & 8 & 12 & 57 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{12} 5^{211^2} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 8 & 8 & 10 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{13} 5^{211^1} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 7 & 9 & 10 & 52 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{11} 5^{211^3} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 10 & 8 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{13} 5^{211^1} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 8 & 10 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{11} 5^{211^3} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 11 & 9 & 16 & 104 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[247] \quad 1^{12} 5^{2^1 1^2} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 9 & 9 & 13 & 73 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[248] \quad 1^{13} 5^{2^1 1^2} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 9 & 10 & 11 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[249] \quad 1^{12} 5^{2^1 1^2} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 10 & 10 & 16 & 102 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{12} 5^{2^1 1^3} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 11 & 10 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^{13} 5^{2^1 1^2} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 10 & 11 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{12} 5^{2^1 1^3} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 456, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 12 & 11 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{13} 5^{2^1 1^3} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 12 & 12 & 15 & 115 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{13} 5^{2^1 1^2} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 11 & 12 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[255] \ 1^{13} 5^{2^{13}} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 613, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 13 & 13 & 18 & 148 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

4.992 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{e}_6$

$$T = 8 \quad [0] \ 1^{13} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -143, \ \det = 925, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 3 & 4 \\ 8 & 9 & 8 & 0 & 4 & 5 \\ 6 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

4.993 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{13} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 725, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 31, \ T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 3 & 4 \\ 8 & 9 & 8 & 0 & 4 & 5 \\ 6 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & -2 & 0 \\ 4 & 5 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

4.994 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 1^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 1^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 299, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 1^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 325, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 1^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 434, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.17 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 1^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 454, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111}31^{132^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 499, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.15 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{131}1^{112^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 525, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

4.995 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{2^11^1}5 \quad 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 2 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12^2}5 \quad 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 9 & 9 & 0 & 36 & 12 \\ 2 & 2 & 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12^2}5 \quad 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 9 & 13 & 13 & 0 & 73 & 17 \\ 2 & 2 & 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12^2}5 \quad 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 17 & 2 \\ 1 & 1 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 12 & 11 & 17 & 17 & 0 & 124 & 22 \\ 2 & 2 & 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{2^11^1}5 \quad 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -33, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 13 & 5 \\ 2 & 5 & 2 & 0 & 10 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 10 & 0 & 25 & 11 \\ 3 & 5 & 4 & -1 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{2^11^1}5 \quad 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -53, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 18 & 5 \\ 2 & 5 & 2 & 0 & 14 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 18 & 14 & 0 & 49 & 15 \\ 3 & 5 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\mathbf{1}^2\mathbf{1}^2} \ 5 \ 13^{\mathbf{1}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -96, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 26 & 5 \\ 2 & 5 & 2 & 0 & 20 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 26 & 20 & 0 & 106 & 22 \\ 3 & 5 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

$$\mathbf{4.996} \quad \text{NHC}(-2)(-3) + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 12 \quad [0] \ 1^{\mathbf{2}^1\mathbf{1}^1} \ 5 \ 13^{\mathbf{1}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 47, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 5 & 5 & 0 & 13 & 4 & 5 \\ 2 & 5 & 2 & 0 & 10 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 7 & 13 & 10 & 0 & 25 & 8 & 11 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 5 & 4 & -1 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\mathbf{2}^1\mathbf{1}^2} \ 5 \ 13^{\mathbf{1}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 122, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 5 & 5 & 0 & 26 & 4 & 5 \\ 2 & 5 & 2 & 0 & 20 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 26 & 20 & 0 & 106 & 16 & 22 \\ 2 & 4 & 3 & 0 & 16 & 1 & 3 \\ 3 & 5 & 4 & -1 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$\mathbf{4.997} \quad \text{NHC}(-2)(-3) + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \ 1^{\mathbf{1}^1\mathbf{3}^1\mathbf{1}^1} \ 5 \ 12^{\mathbf{1}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 4 & 5 \\ 2 & 5 & 2 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 5 & 4 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\mathbf{1}^1\mathbf{3}^1\mathbf{1}^1} \ 5 \ 12^{\mathbf{1}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\mathbf{2}^1\mathbf{1}^1} \ 5 \ 13^{\mathbf{1}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -31, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 8 & 0 & 4 & 5 \\ 5 & 8 & 9 & 0 & 5 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 5 & 7 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\mathbf{2}^1\mathbf{1}^1} \ 5 \ 13^{\mathbf{1}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -31, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 2 \\ 3 & 2 & 4 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\mathbf{2}^1\mathbf{1}^1} \ 5 \ 13^{\mathbf{1}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -76, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 4 & 4 & 5 & 11 & 0 & 4 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 7 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}^1} 5 \ 13^{\overline{1}^1} 1^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -76, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 5 & 13 & 0 & 4 & 5 \\ 7 & 13 & 25 & 0 & 8 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 3 \\ 3 & 5 & 11 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1}^1} 5 \ 13^{\overline{1}^1} 1^{\overline{1}^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -91, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 5 & 18 & 0 & 4 & 5 \\ 9 & 18 & 49 & 0 & 11 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 11 & 0 & 1 & 3 \\ 3 & 5 & 15 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1}^2} 5 \ 13^{\overline{1}^1} 1^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -91, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 & 2 \\ 5 & 6 & 6 & 21 & 0 & 5 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1}^2} 5 \ 13^{\overline{1}^1} 1^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -151, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 & 2 \\ 7 & 6 & 9 & 36 & 0 & 7 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 7 & 0 & 0 & 2 \\ 2 & 2 & 2 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1}^2} 5 \ 13^{\overline{1}^1} 1^{\overline{1}^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -151, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 5 & 21 & 0 & 4 & 5 \\ 12 & 21 & 70 & 0 & 13 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 13 & 0 & 1 & 3 \\ 3 & 5 & 18 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1}^2} 5 \ 13^{\overline{1}^1} 1^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -226, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 & 2 \\ 8 & 8 & 10 & 50 & 0 & 8 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 8 & 0 & 0 & 2 \\ 2 & 2 & 2 & 14 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1}^2} 5 \ 13^{\overline{1}^1} 1^{\overline{1}^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -226, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 5 & 26 & 0 & 4 & 5 \\ 14 & 26 & 106 & 0 & 16 & 22 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 16 & 0 & 1 & 3 \\ 3 & 5 & 22 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1}^2} 5 \ 13^{\overline{1}^1} 1^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -271, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.00 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 9 & 10 & 11 & 68 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{12} \overset{2^1 1^3}{5} \ 13^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \rightarrow \textcolor{red}{(-3)} \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -371, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 12.00 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 & 2 \\ 11 & 10 & 14 & 93 & 0 & 11 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 11 & 0 & 0 & 2 \\ 2 & 2 & 2 & 19 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{13} \overset{2^1 1^3}{5} \ 13^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \rightarrow \textcolor{red}{(-3)} \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -476, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 1 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 & 2 \\ 12 & 12 & 15 & 115 & 0 & 12 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 12 & 0 & 0 & 2 \\ 2 & 2 & 2 & 21 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.998 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [0] \ 1^{\overset{2^1 1^1}{1}} \ 5 \ 13^1 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \rightarrow \textcolor{red}{(-3)} + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 47, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 5 & 5 & 0 & 4 & 13 & 5 \\ 2 & 5 & 2 & 0 & 3 & 10 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 13 & 10 & 0 & 8 & 25 & 11 \\ 3 & 5 & 4 & -1 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overset{2^1 1^2}{1}} \ 5 \ 13^1 1^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \rightarrow \textcolor{red}{(-3)} + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 122, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 5 & 5 & 0 & 4 & 26 & 5 \\ 2 & 5 & 2 & 0 & 3 & 20 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 16 & 3 \\ 14 & 26 & 20 & 0 & 16 & 106 & 22 \\ 3 & 5 & 4 & -1 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

4.999 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \text{su2n3mix}$

$$T = 11 \quad [0] \ 1^{\overset{2^1 1^1}{1}} \ 5 \ 13^1 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -50, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 17 & 5 \\ 2 & 5 & 2 & 0 & 13 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 13 & 0 & 44 & 14 \\ 3 & 5 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overset{2^1 1^2}{1}} \ 5 \ 13^1 1^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -131, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 0 & 23 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 30 & 23 & 0 & 141 & 25 \\ 3 & 5 & 4 & -1 & 25 & 4 \end{pmatrix}_{b_0}$$

4.1000 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overset{1^1}{1}} \ 5 \ 1^{12} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 13, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overset{1^1}{1}} \ 5 \ 12^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{(-3)} \rightarrow \textcolor{brown}{(-2)} + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 40, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1}{5} 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 3 & 4 \\ 8 & 10 & 21 & 0 & 6 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{5} 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 85, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 5 & 7 & 8 & 0 & 4 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 11 & 0 & 3 & 4 \\ 7 & 11 & 21 & 0 & 6 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \overset{1}{5} 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 15 & 0 & 3 & 4 \\ 9 & 15 & 40 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} \overset{1}{5} 1^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 13 & 17 & 61 & 0 & 10 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 15 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 10 & 14 & 38 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} \overset{21}{5} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 285, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 2 & 3 \\ 6 & 13 & 12 & 0 & 5 & 7 \\ 5 & 12 & 8 & 0 & 4 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 7 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13} \overset{1}{5} 1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 316, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 12 & 18 & 61 & 0 & 10 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 15 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{13}5^1 1^3 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 15 & 21 & 88 & 0 & 12 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{12} 1^{21} 1^{12} 5^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 2 & 3 \\ 6 & 13 & 24 & 0 & 5 & 7 \\ 10 & 24 & 38 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 8 & 0 & 0 & 2 \\ 3 & 7 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{12} 1^{21} 1^{13} 5^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 901, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 2 & 3 \\ 6 & 13 & 36 & 0 & 5 & 7 \\ 15 & 36 & 88 & 0 & 12 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 12 & 0 & 0 & 2 \\ 3 & 7 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.1001 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{su}_3$ 2mix

$T = 7 \quad [0] 1^{11} 31^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 4 & 3 \\ 4 & 4 & 8 & 5 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 1^{11} 31^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 4 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 31^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{14} 1^{11} 31^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 18 & 4 & 4 \\ 5 & 4 & 21 & 4 & 4 \\ 18 & 21 & 76 & 15 & 16 \\ 4 & 4 & 15 & 0 & 2 \\ 4 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 31^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 4 & 3 \\ 4 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 31^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 1^{11} 31^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 9 & 4 & 4 \\ 5 & 4 & 9 & 4 & 4 \\ 9 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 4 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

- [7] $1^{131}31^{112^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 15 & 4 & 4 \\ 5 & 4 & 17 & 4 & 4 \\ 15 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 4 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{11}31^{1112^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 7 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$
- [9] $1^{121}31^{112^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 & 4 \\ 5 & 4 & 13 & 4 & 4 \\ 12 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 4 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{121}31^{122^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 18 & 4 & 4 \\ 5 & 4 & 18 & 4 & 4 \\ 18 & 18 & 58 & 12 & 14 \\ 4 & 4 & 12 & 0 & 2 \\ 4 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{131}31^{122^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -399, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 21 & 4 & 4 \\ 5 & 4 & 22 & 4 & 4 \\ 21 & 22 & 85 & 15 & 17 \\ 4 & 4 & 15 & 0 & 2 \\ 4 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{141}31^{122^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 24 & 4 & 4 \\ 5 & 4 & 26 & 4 & 4 \\ 24 & 26 & 118 & 18 & 20 \\ 4 & 4 & 18 & 0 & 2 \\ 4 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{131}31^{132^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 27 & 4 & 4 \\ 5 & 4 & 27 & 4 & 4 \\ 27 & 27 & 133 & 18 & 21 \\ 4 & 4 & 18 & 0 & 2 \\ 4 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{141}31^{132^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -759, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 30 & 4 & 4 \\ 5 & 4 & 31 & 4 & 4 \\ 30 & 31 & 172 & 21 & 24 \\ 4 & 4 & 21 & 0 & 2 \\ 4 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{141}31^{142^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1024, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 36 & 4 & 4 \\ 5 & 4 & 36 & 4 & 4 \\ 36 & 36 & 238 & 24 & 28 \\ 4 & 4 & 24 & 0 & 2 \\ 4 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$

- $T = 8$ [16] $1^{11}4^{11}12^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [17] $1^{11}41^{11}22^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 7 & 7 & 4 \\ 5 & 4 & 7 & 6 & 4 \\ 7 & 7 & 7 & 10 & 5 \\ 7 & 6 & 10 & 8 & 6 \\ 4 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{11}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

- [19] $1^{\overline{2}1^{11}} 4^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [20] $1^{\overline{2}1^{12}} 4^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{\overline{2}1^{13}} 4^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$
- [22] $1^{\overline{2}1^{11}} 4^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 8 & 3 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [24] $1^{\overline{2}1^{12}} 4^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 8 & 3 & 3 \\ 4 & 4 & 11 & 3 & 4 \\ 8 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$
- [23] $1^{\overline{2}1^{14}} 4^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$
- [25] $1^{\overline{2}1^{13}} 4^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 11 & 3 & 3 \\ 4 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$
- [26] $1^{\overline{2}1^{12}} 4^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 11 & 3 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 11 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$
- [27] $1^{\overline{2}1^{14}} 4^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 14 & 3 & 3 \\ 4 & 4 & 19 & 3 & 4 \\ 14 & 19 & 65 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$
- [28] $1^{\overline{2}1^{13}} 4^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 14 & 3 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 14 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$
- [29] $1^{\overline{2}1^{14}} 4^{\overline{1}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 17 & 3 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

$$\begin{aligned}
T = 9 \quad [30] \quad 1^{11} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 11 & 13 & 40 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[31] \quad 1^{11} \overset{1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[32] \quad 1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 8 & 10 & 21 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[33] \quad 1^{11} \overset{1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[34] \quad 1^{12} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[35] \quad 1^{12} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 10 & 15 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[36] \quad 1^{21} \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 5 & 7 \\ 5 & 12 & 8 & 4 & 6 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[37] \quad 1^{13} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 12 & 18 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[38] \quad 1^{21} \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 6 & 10 & 3 & 3 \\ 6 & 13 & 24 & 5 & 7 \\ 10 & 24 & 38 & 8 & 12 \\ 3 & 5 & 8 & 0 & 2 \\ 3 & 7 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[39] \quad 1^{21} \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 1 & 6 & 15 & 3 & 3 \\ 6 & 13 & 36 & 5 & 7 \\ 15 & 36 & 88 & 12 & 18 \\ 3 & 5 & 12 & 0 & 2 \\ 3 & 7 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
T = 10 \quad [40] \quad 1^{21} \overset{1}{5} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 5 & 8 & 9 & 6 & 7 \\ 3 & 5 & 6 & 1 & 3 \\ 3 & 5 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[41] \ 1^1 5^{21^2} 131^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 5 & 16 & 5 & 5 \\ 10 & 16 & 42 & 12 & 14 \\ 3 & 5 & 12 & 1 & 3 \\ 3 & 5 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[42] \ 1^1 5^{21^3} 131^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 5 & 24 & 5 & 5 \\ 15 & 24 & 97 & 18 & 21 \\ 3 & 5 & 18 & 1 & 3 \\ 3 & 5 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1002 NHC(-2)(-3) + $\mathfrak{su}_3$

$$T = 6 \quad [0] \ 1^{111} 31^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 76, T \in [14, 76]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 9 & 12 & 5 \\ 7 & 12 & 12 & 6 \\ 3 & 5 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 76, T \in [14, 76]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 4 \\ 5 & 4 & 0 & 2 \\ 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 10, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 9 & 16 & 5 \\ 7 & 16 & 27 & 9 \\ 3 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^1 31^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 10, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 5 & 4 \\ 5 & 5 & 3 & 3 \\ 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^1 31^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 76, T \in [14, 76]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 21 & 7 \\ 3 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^1 31^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 76, T \in [14, 76]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 21 & 7 \\ 3 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^1 31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.45 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 3 & 3 \\ 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 3 & 3 \\ 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 21 & 7 \\ 3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 21 & 7 \\ 3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^1 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 9 & 5 \\ 3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 8 & 4 \\ 8 & 8 & 9 & 5 \\ 3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 76, T \in [10, 76]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 21^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 76, T \in [10, 76]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 21, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 9 & 4 \\ 9 & 9 & 12 & 6 \\ 3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 21^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 23, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 5 & 4 \\ 3 & 5 & 3 & 3 \\ 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 21^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 12 & 5 \\ 6 & 12 & 12 & 6 \\ 3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

- [21] $1^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 12 & 5 \\ 6 & 12 & 12 & 6 \\ 3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{11}13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 76$, $T \in [13, 76]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 24 & 8 \\ 3 & 4 & 8 & 0 \end{pmatrix}_{b_0}$
- [23] $2131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 76$, $T \in [13, 76]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 8 & 5 \\ 3 & 8 & 3 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$
- [24] $1^131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 76$, $T \in [13, 76]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 8 & 5 \\ 3 & 8 & 3 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$
- [25] $1^{11}13^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 27$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 76$, $T \in [12, 76]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 9 & 12 & 5 \\ 5 & 12 & 12 & 6 \\ 3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$
- [26] $21^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 29$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 5 & 4 \\ 6 & 5 & 3 & 3 \\ 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [27] $1^131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 29$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 9 & 4 \\ 5 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{11}13^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 31$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 76$, $T \in [14, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 16 & 4 \\ 8 & 16 & 45 & 11 \\ 3 & 4 & 11 & -1 \end{pmatrix}_{b_0}$
- [29] $1^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 31$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 8 & 4 \\ 7 & 8 & 9 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 31$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 8 & 4 \\ 5 & 8 & 9 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [31] $21^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 35$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 76$, $T \in [12, 76]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 10 & 4 \\ 12 & 10 & 21 & 7 \\ 4 & 4 & 7 & 1 \end{pmatrix}_{b_0}$
- [32] $1^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 35$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 9 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [33] $1^{11}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 35$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 39, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[35] $1^{12} 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 40, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$

[36] $1^1 1^1 3^{15} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 40, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 76, T \in [14, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 20 & 4 \\ 10 & 20 & 72 & 14 \\ 3 & 4 & 14 & -1 \end{pmatrix}_{b_0}$

[37] $1^{12} 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 9 & 4 \\ 7 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[38] $21^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$

[39] $1^{13} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 12 & 4 \\ 7 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$

[40] $21^{11} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 45, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.53 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 21 & 4 \\ 18 & 21 & 75 & 15 \\ 4 & 4 & 15 & 2 \end{pmatrix}_{b_0}$

[41] $1^{11} 1^1 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 46, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 13 & 4 \\ 7 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$

[42] $21^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 47, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$

[43] $21^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 54, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 9 & 4 \\ 9 & 9 & 12 & 6 \\ 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[44] $1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 56, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$

[45] $1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 56, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$

- [46] $1^{13}31^{12}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [47] $1^{13}31^{14}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [48] $21^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 14 & 4 \\ 9 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [49] $1^{14}31^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 59$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 76$, $T \in [12, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 16 & 4 \\ 9 & 16 & 45 & 11 \\ 3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$
- [50] $1^{13}31^{13}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 60$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 12 & 4 \\ 9 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [51] $21^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 62$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 17 & 4 \\ 15 & 17 & 48 & 12 \\ 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [52] $21^{11}31^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 63$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 9 & 4 \\ 6 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [55] $1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.25 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [53] $1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.25 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [54] $1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.25 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 13 & 4 \\ 8 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [56] $21^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 65$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 13 & 4 \\ 12 & 13 & 27 & 9 \\ 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [57] $1^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 65$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 17 & 4 \\ 9 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [58] $21^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 9 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 70, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$

[60] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 71, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$

[61] $1^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 74, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 13 & 4 \\ 9 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$

[62] $1^{12}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 76, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$

[63] $1^{14}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 79, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 45 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$

[64] $1^{11}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 86, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 21 & 4 \\ 11 & 21 & 75 & 15 \\ 3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$

[65] $21^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 87, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 8 & 9 & 4 \\ 8 & 9 & 12 & 5 \\ 9 & 12 & 12 & 6 \\ 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$

[66] $1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 91, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$

[67] $1^{14}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 91, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 16 & 4 \\ 11 & 16 & 45 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$

[68] $21^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 92, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 14 & 4 \\ 15 & 14 & 36 & 10 \\ 4 & 4 & 10 & 2 \end{pmatrix}_{b_0}$

[69] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 95, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$

[70] $1^{14}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 95, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 45 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$

- [71] $1^{12} 1^3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 22 & 4 \\ 12 & 22 & 84 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$
- [72] $131^{11} 2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 9 & 8 & 5 \\ 6 & 8 & 3 & 3 \\ 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$
- [73] $21^{13} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 108, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [74] $1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 109, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 17 & 4 \\ 11 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [75] $1^{12} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 111, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [76] $21^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 13 & 4 \\ 9 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [77] $1^{12} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 18 & 4 \\ 13 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [79] $1^{13} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 17 & 4 \\ 12 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [78] $1^{12} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [80] $1^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 120, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [81] $1^{13} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 121, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [82] $21^{12} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 135, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 18 & 4 \\ 18 & 18 & 57 & 13 \\ 4 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [83] $1^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 136, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

- $$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 22 & 4 \\ 13 & 22 & 84 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$
- [84] $1^1 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 139, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 26 & 4 \\ 14 & 26 & 117 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$
- [85] $21^{12} 31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 144, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 18 & 4 \\ 12 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$
- [86] $1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 146, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$
- [87] $1^{13} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 160, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 84 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$
- [88] $21^{14} 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 164, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 16 & 4 \\ 12 & 16 & 45 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$
- [89] $21^{12} 31^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 164, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 22 & 4 \\ 21 & 22 & 84 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$
- [90] $1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 164, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$
- [91] $1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 174, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$
- [92] $1^{13} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 176, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 22 & 4 \\ 15 & 22 & 84 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$
- [93] $21^{12} 31^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 179, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 26 & 4 \\ 24 & 26 & 117 & 19 \\ 4 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$
- [94] $1^{14} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 179, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 26 & 4 \\ 15 & 26 & 117 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$
- [95] $1^{13} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 184, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 84 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$
- [96] $21^{13} 31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 191, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.03 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 211, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 26 & 4 \\ 16 & 26 & 117 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[98] \ 221^{11}3 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 234, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.03 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 3 \\ 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 235, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 10.04 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 26 & 4 \\ 17 & 26 & 117 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[100] \ 21^{15}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 236, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 20 & 4 \\ 15 & 20 & 72 & 14 \\ 3 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[101] \ 21^{13}31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 248, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 22 & 4 \\ 15 & 22 & 84 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 251, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 117 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[103] \ 21^{14}31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 279, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 10.02 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[104] \ 21^{14}31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 368, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 26 & 4 \\ 18 & 26 & 117 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [105] \ 141^{13}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 13 & 4 \\ 8 & 13 & 39 & 11 \\ 3 & 4 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{11}3131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 \\ 1 & 4 & 0 & 0 \\ 1 & 0 & 0 & -3 \\ 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{11}321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{11}321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$$

- [109] $1^1 31^2 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$
- [110] $1^1 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 21 & 7 \\ 2 & 3 & -1 & 7 & -1 \end{pmatrix}_{b_0}$
- [111] $1^1 321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [112] $1^1 331^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 9 & 3 \\ 5 & 9 & 9 & 24 & 8 \\ 2 & 3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$
- [113] $1^1 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [114] $1^1 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$
- [115] $1^1 321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 9 & 5 \\ 2 & 3 & -1 & 5 & -1 \end{pmatrix}_{b_0}$
- [116] $1^1 3131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & 2 \end{pmatrix}_{b_0}$
- [117] $1^1 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 23$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 23$, $T \in [11, 23]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [118] $21^1 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 23$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 23$, $T \in [11, 23]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [119] $1^1 41^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 23$, $\Delta = -114$, $\det = -10$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 39 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$
- [120] $21^1 341^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 23$, $\Delta = -114$, $\det = -10$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$

$$\begin{aligned}
& T_{\min} = 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 39 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0} \\
[125] & 31^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[122] & 1^1 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 23, T \in [11, 23] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[123] & 1^1 13131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0} \\
[124] & 1^1 1^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0} \\
[121] & 1^1 1^2 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[126] & 1^1 1^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[127] & 1^1 1^1 31^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0} \\
[128] & 1^1 1^1 31^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14.82 \rightarrow 15, T_{\max} = 19, T \in [15, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 3 & 0 & 3 & 3 \\ 2 & 3 & -1 & 3 & -2 \end{pmatrix}_{b_0} \\
[129] & 1^1 1^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[130] & 1^1 1^2 31^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 4 & 6 & 6 & 9 & 5 \\ 2 & 3 & 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[131] & 1^1 1^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[137] & 1^1 1^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[136] $1^1 2141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[135] $1^1 1^3 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[132] $1^1 1^2 321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 2 & 6 & 9 & 5 \\ 2 & 2 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[134] $1^1 1^2 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 4 & 2 & 5 & 3 \\ 2 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[133] $1^1 1^4 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[138] $12^1 3^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

[139] $1^1 31^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[140] $1^1 1^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 7 & 4 \\ 7 & 7 & 7 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

[145] $1^1 1^1 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[144] $1^1 1^3 321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 60 & 14 \\ 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[141] $2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{12}321^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 25 & 9 \\ 2 & 2 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 3 & 3 \\ 2 & 3 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{13}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 20 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[148] \ 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[151] \ 2141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[147] \ 141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 20 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[146] \ 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{13}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[153] \ 21^141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -17, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 23, \ T \in [11, 23]$$

$$\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 7 & 4 \\ 6 & 7 & 7 & 5 \\ 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[152] \ 2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -17, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[155] \ 21^141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 23, \ T \in [11, 23]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{12}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

- [157] $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 3 & 6 & 6 & 9 & 5 \\ 2 & 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$ $1^1 31^1 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$
- [159] $131^1 31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [160] $31^1 231^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 5 & 3 & 3 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$
- [161] $1^1 31^{231} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 19$, $T \in [13, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 21 & 7 \\ 2 & 3 & -1 & 7 & 0 \end{pmatrix}_{b_0}$
- [162] $2^1 2^1 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 4 & 2 & 6 & 9 & 5 \\ 2 & 2 & 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [163] $1^1 41^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 23$, $\Delta = -114$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 23$, $T \in [11, 23]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [164] $1^1 321^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 4 & 3 \\ 1 & 4 & 0 & 4 & 2 \\ 2 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$
- [156] $1^{111} 3131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [158] $1^1 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 23$, $\Delta = -114$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 23$, $T \in [11, 23]$
 $\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [165] $1^{111} 3^1 21^2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.24 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 4 & 0 & 15 & 7 \\ 2 & 2 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [175] $1^1 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$
- [170] $1^1 31^3 31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3}$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 51 & 11 \\ 2 & 3 & -1 & 11 & -1 \end{pmatrix}_{b_0}$$

[171] $1^{11}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[172] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 5 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[174] $2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[169] $1^{11}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[176] $131^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{11}3131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[166] $1^3321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{13}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[183] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 20 & 8 \\ 3 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

[181] $1^131^{23}12^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 6 & 0 & 21 & 7 \\ 2 & 3 & -1 & 7 & -1 \end{pmatrix}_{b_0}$$

[182] $21^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 20 & 8 \\ 3 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[180] \quad 2 \overset{1}{3} 1^{\overset{11}{1}} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[184] \quad 1 \overset{12}{1} 4 \overset{1}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 1 \overset{12}{1} 4 \overset{1}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 20 & 8 \\ 3 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 21 \overset{11}{1} 4 \overset{1}{1} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$$

$$\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 13 & 12 & 7 \\ 6 & 12 & 7 & 5 \\ 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[179] \quad 131 \overset{11}{1} 31 \overset{11}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 1 & 4 & 0 & 6 & 2 \\ 3 & 6 & 6 & 12 & 6 \\ 2 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[178] \quad 1 \overset{11}{1} 31 \overset{1}{1} 31 \overset{1}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 12 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[177] \quad 21 \overset{11}{1} 321 \overset{1}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 6 & 3 & 4 \\ 6 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 21 \overset{11}{1} 321 \overset{12}{1} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 31 & 11 \\ 4 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 1 \overset{12}{1} 321 \overset{1}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 12 & 6 \\ 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[189] \quad 221 \overset{1}{1} 41 \overset{11}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[190] \quad 13131 \overset{11}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[191] \quad 21 \overset{1}{1} 41 \overset{11}{1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{11} 3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{11} 3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 3 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{11} 3 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{12} 3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{11} 3 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 15 & 5 \\ 7 & 15 & 33 & 11 \\ 3 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

[198] $12 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 1 & 4 \\ 5 & 4 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[199] $21^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 5 & 10 & 5 \\ 9 & 10 & 13 & 7 \\ 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{12} 3 21^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 44 & 12 \\ 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{12} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{12} 3 1^{11} 3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 27 & 9 \\ 2 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{11}3^{11}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{13}31^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 51 & 11 \\ 2 & 3 & -1 & 11 & -1 \end{pmatrix}_{b_0}$$

[206] $1^{12}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 27 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{11}31^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 36 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{13}3^{11}21^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 39 & 11 \\ 2 & 2 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{11}3^{11}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 15 & 5 \\ 7 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{12}3^{14}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 7 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{12}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{13}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[213] $31^{11}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{12}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 9 & 5 \\ 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{11}3^{14}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 15 & 5 \\ 11 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{13}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 7 & 0 & 24 & 8 \\ 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 2 \ 2 \ 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 21^{12}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[217] \ 21^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[220] \ 2 \ 2 \ 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{13}3 \ 21^{11}4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 87 & 17 \\ 2 & 2 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[222] \ 2 \ 3 \ 1^{12}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 3 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 9 & 5 \\ 2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{11}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[224] \ 1^{12}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{11}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{14}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

- [227] $1^{11}31^331^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 72 & 14 \\ 2 & 3 & -1 & 14 & 1 \end{pmatrix}_{b_0}$
- [228] $1^21^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 23$, $\Delta = -114$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [229] $2^22^131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.64 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [230] $131^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [231] $1^{13}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 48 & 12 \\ 2 & 3 & -1 & 12 & 1 \end{pmatrix}_{b_0}$
- [232] $1^{11}31^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 12 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [233] $31^{11}31^13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -110$, $\det = -45$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 3 & 4 \\ 3 & 7 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [234] $12^331^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 5 & 3 \\ 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$
- [240] $1^{11}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 5 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$
- [239] $1^{11}31^{11}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 6 \\ 2 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [235] $1^3321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 20 & 5 \\ 9 & 20 & 57 & 15 \\ 3 & 5 & 15 & 2 \end{pmatrix}_{b_0}$
- [236] $131^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$

$$[238] \ 1^{\overline{13}} 3^{\overline{11}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -47, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.13 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[242] \ 21^{\overline{11}} 1^{\overline{1}} 41^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -47, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 23, \ T \in [11, 23]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{\overline{13}} 3131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -47, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.13 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 24 & 8 \\ 2 & 3 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{\overline{21}} 4^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -47, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.30 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[247] \ 2^{\overline{1}} 3^{\overline{1}} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 13, \ \Delta = -97, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 13, \ T \in [11, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{\overline{13}} 321^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[245] \ 2^{\overline{11}} 2^{\overline{11}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.17 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{\overline{12}} 321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{\overline{12}} 321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{\overline{13}} 321^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -51, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{\overline{11}} 321^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -51, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{\overline{11}} 321^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -51, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0} \\
[252] \quad & 1^{14}31^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 19, T \in [12, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 75 & 15 \\ 2 & 3 & -1 & 15 & 1 \end{pmatrix}_{b_0} \\
[251] \quad & 1^{13}1^{11}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[253] \quad & 1^{12}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[254] \quad & 1^{12}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 25 & 9 \\ 2 & 2 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[255] \quad & 2231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[259] \quad & 1^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 12 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0} \\
[256] \quad & 131^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 19, T \in [12, 19] \\
& \begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 4 & 12 & 9 & 27 & 9 \\ 2 & 4 & 2 & 9 & 1 \end{pmatrix}_{b_0} \\
[258] \quad & 1^{12}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 20 & 5 \\ 10 & 20 & 61 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0} \\
[261] \quad & 31^{11}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[260] \quad & 1^{12}3131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 9 & 5 \\ 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0} \\
[263] \quad & 1^{14}3131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 19, T \in [12, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 45 & 11 \\ 2 & 3 & -1 & 11 & 1 \end{pmatrix}_{b_0} \\
[264] \quad & 1^{11}1321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 25 & 5 \\ 11 & 25 & 91 & 19 \\ 3 & 5 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[257] \ 1^{1^2} 3^{1^2} 21^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[262] \ 1^{1^2} 3^{1^2} 21^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 4 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[266] \ 2^{1^1} 2^{1^1} 31^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[265] \ 2^{1^2} 2^{1^2} 31^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{1^2} 3^{1^2} 21^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 12 & 6 \\ 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{1^2} 31^{1^2} 31^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 57 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[269] \ 131^{1^1} 31^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 3 & 9 & 6 & 12 & 6 \\ 2 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{1^4} 3^{1^4} 21^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 8 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{1^1} 31^{1^2} 31^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 36 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{1^2} 3^{1^2} 21^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 6 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{1^1} 3^{1^1} 21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[274] \quad 2 \overset{1}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[275] \quad 12 \overset{1}{3} 1 \overset{13}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 15 & 5 \\ 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[276] \quad 1 \overset{12}{3} 21 \overset{13}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 5 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[277] \quad 2 \overset{1}{2} 31 \overset{12}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 4 & 8 & 3 \\ 2 & 4 & 0 & 8 & 2 \\ 6 & 8 & 8 & 25 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[279] \quad 31 \overset{11}{2} 231 \overset{11}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 3 & 3 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[278] \quad 1 \overset{12}{2} 31 \overset{11}{3} 1 \overset{11}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 27 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[281] \quad 1 \overset{1}{3} 21 \overset{14}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 61 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[282] \quad 1 \overset{13}{1} 321 \overset{13}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[283] \quad 1 \overset{14}{1} 321 \overset{13}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[284] \quad 1 \overset{11}{1} 31 \overset{11}{3} 1 \overset{11}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 4 & 9 & 6 & 12 & 6 \\ 2 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[285] \quad 2 \overset{1}{3} 1 \overset{11}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[286] \quad 1 \overset{13}{1} 321 \overset{11}{+} [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{13}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[287] $1^{12}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 6 & 12 & 9 & 27 & 9 \\ 2 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{11}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 20 & 5 \\ 10 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[289] $1^{13}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 60 & 14 \\ 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[290] $31^{11}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[293] $2^13^12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 9 & 5 \\ 2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{12}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{14}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 61 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[295] $21^{12}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[294] $2^12^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{12}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 5 & 12 & 9 & 27 & 9 \\ 2 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{11}31^331^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 72 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{11} 2^2 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 4 & 3 \\ 2 & 4 & 0 & 4 & 2 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[303] \quad 1^{12} 31^3 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 99 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{13} 3131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 24 & 8 \\ 2 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[301] \quad 1^{13} 31^2 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 84 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[299] \quad 1^{11} 31^2 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 36 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[300] \quad 1^{13} 1^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 5 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[304] \quad 21^{12} 321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[305] \quad 1^{13} 1^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 7 & 0 & 24 & 8 \\ 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[306] \quad 2^2 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[307] \quad 1^{11} 1^4 321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 20 & 5 \\ 13 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[308] \quad 1^{11} 1^2 321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[309] $1^{14}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$

[310] $1^{12}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$

[311] $1^{11}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 39 & 11 \\ 2 & 2 & -1 & 11 & 2 \end{pmatrix}_{b_0}$

[312] $31^{11}231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[313] $221^{11}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 7 & 5 \\ 3 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[314] $1^{11}321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 25 & 5 \\ 12 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$

[315] $1^{13}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$

[316] $1^{12}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$

[317] $31^{11}231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 5 & 7 & 3 \\ 5 & 5 & 10 & 5 \\ 7 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$

[318] $1^{12}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 6 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$

[319] $1^{13}31^131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 48 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$

[321] $31^{11}31^13 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -110, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

- [320] $1^{12}31^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 27 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$
- [323] $1^{14}31^{23}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 18 & 0 & 117 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$
- [324] $1^{111}13^22^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [322] $1^{12}31^{23}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 57 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$
- [325] $1^{13}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [328] $1^{14}3131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 45 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$
- [327] $1^{11}31^331^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 72 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$
- [326] $2^22^31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [329] $2^22^31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$
- [330] $221^{11}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 23$, $\Delta = -114$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 23$, $T \in [11, 23]$
 $\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 8 & 11 & 6 \\ 5 & 11 & 7 & 5 \\ 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$
- [331] $231^{11}3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.62 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [332] $1^{13}1^321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 6 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[333] $1^{12}3^{12}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[334] $1^{11}3^{12}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 25 & 5 \\ 13 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[336] $1^{14}3^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 75 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[335] $21^{13}3^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[337] $2^2 2^2 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[338] $31^{12}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 5 & 8 & 3 \\ 5 & 5 & 10 & 5 \\ 8 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{13}3^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 132 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[340] $1^{13}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[341] $1^{12}3^{13}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[342] $2^2 2^2 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 61 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[343] $1^{13}3^{13}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[344] $1^{13}3^{14}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[345] \ 1^{11}3^{21}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[346] \ 1^{12}3^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[347] \ 1^{12}3^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 99 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[348] \ 1^{11}3^{21}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[350] \ 1^{12}3^{12}3^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 57 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[349] \ 1^{13}3^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 48 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[351] \ 1^{11}3^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 7 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[352] \ 1^{13}3^{12}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 84 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[353] \ 2^2 2^3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[354] \ 1^{14}3^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 21 & 0 & 171 & 23 \\ 2 & 3 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[355] \ 21^{13}3^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[356] $2^1 2^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 5 & 2 \\ 2 & 1 & 4 & 7 & 3 \\ 2 & 4 & 0 & 7 & 2 \\ 5 & 7 & 7 & 15 & 7 \\ 2 & 3 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[357] $13^1 2^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 4 & 6 & 3 \\ 2 & 4 & 0 & 6 & 2 \\ 4 & 6 & 6 & 9 & 5 \\ 2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[358] $21^{13} 321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[359] $1^{12} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -167, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[360] $2^1 2^1 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[361] $2^1 3^1 1^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 2 & 5 & 2 \\ 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 5 & 5 & 7 & 5 \\ 2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[362] $1^{14} 31^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 75 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[363] $1^{12} 31^3 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 99 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{14} 31^2 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 117 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[367] $1^{13} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[366] $2^1 2^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[365] $2^1 2^1 31^{15} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

- [368] $1^{13}31^331^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 132 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$
- [369] $1^{13}3^{14}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -191$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.03 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [370] $1^{13}31^231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -195$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.31 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 84 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$
- [371] $31^{13}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -195$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.64 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$
- [372] $31^{12}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -200$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$
- [373] $1^{14}31^331^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -239$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 171 & 23 \\ 2 & 3 & -1 & 23 & 2 \end{pmatrix}_{b_0}$
- [374] $2^12^131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -240$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.03 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [375] $1^{14}31^231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -248$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.31 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 117 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$
- [376] $1^{13}31^331^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -248$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.31 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 132 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$
- [377] $2^12^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -249$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.3$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 141 & 23 \\ 3 & 5 & 23 & 3 \end{pmatrix}_{b_0}$
- [378] $2^12^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -249$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.07 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$
- [379] $1^{14}31^331^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -315$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.27 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 171 & 23 \\ 2 & 3 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[380] \quad 2 \overset{1}{2} 31^{\overset{17}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -327, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6 \\ T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 193 & 27 \\ 3 & 5 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[381] \quad 31^{\overset{13}{1}} 231^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -335, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 5 & 15 & 3 \\ 5 & 5 & 20 & 5 \\ 15 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [382] \quad 1^{\overset{1}{1}} 3 221^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[383] \quad 1^{\overset{1}{1}} 2321^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[384] \quad 1^{\overset{1}{1}} 2321^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[385] \quad 1^{\overset{1}{1}} 3 221^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[386] \quad 1^{\overset{1}{1}} 2321^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[387] \quad 1^{\overset{1}{1}} 4131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[388] \quad 1^{\overset{1}{1}} 4131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 10.57 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[393] \quad 1^{\overset{1}{1}} 4131^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[395] \quad 1^{\overset{1}{1}} 41^{\overset{13}{1}} 231^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 3 & 3 \\ 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[392] $1^{\overline{1}1231^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 2 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$

[391] $1^{\overline{1}2}4131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 6 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

[394] $21^{\overline{1}41^{\overline{1}31}12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 2 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$

[396] $1^{\overline{1}4}4131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[397] $21^{\overline{1}51^{\overline{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[398] $21^{\overline{1}51^{\overline{1}2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[399] $1^{\overline{1}131^{\overline{1}}}41^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 3 & 3 \\ 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[390] $1^{\overline{1}41231^{\overline{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$

[389] $21^{\overline{1}41^{\overline{1}31}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 4 & 6 & 1 & 4 \\ 6 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 4 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[402] $221^{\overline{1}51^{\overline{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[403] $21^{\overline{1}41^{\overline{1}31}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 7 & 5 \\ 7 & 8 & 6 \\ 5 & 6 & 4 \end{pmatrix}_{b_0}$

[404] $21^{\overline{1}51^{\overline{1}1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 6 & 4 \\ 6 & 6 & 7 & 5 \\ 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[405] $1 \overset{1}{\underset{1,1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[406] $131 \overset{1}{\underset{1,1}{4}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[407] $21 \overset{1}{\underset{1}{5}} 1^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 6 & 4 \\ 6 & 6 & 7 & 5 \\ 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[408] $1 \overset{1}{\underset{1,1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[401] $21 \overset{1}{\underset{1}{5}} 1^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[400] $1 \overset{1}{\underset{1}{5}} 1^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[409] $1 \overset{1}{\underset{1}{4}} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[410] $31 \overset{1}{\underset{1}{4}} 1314^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[411] $1 \overset{1}{\underset{1,1}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[412] $41 \overset{1}{\underset{1}{2}} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 26 & 1 & 0 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & -1 & -1 & 1 \end{pmatrix}_{b_0}$

[413] $14 \overset{1}{\underset{1}{4}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 4 & 5 \\ 4 & 1 & 3 \\ 5 & 3 & 4 \end{pmatrix}_{b_0}$

[414] $1 \overset{1}{\underset{1}{4}} \overset{1}{\underset{1}{3}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

- [415] $141^{1231^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 6 & 5 \\ 6 & 5 & 5 \\ 5 & 5 & 4 \end{pmatrix}_{b_0}$
- [417] $1^{11}4^{131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [418] $1^{131^1}4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 4 & 3 & 1 & 4 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [419] $1^{131^1}4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 4 & 5 \\ 4 & 1 & 3 \\ 5 & 3 & 4 \end{pmatrix}_{b_0}$
- [420] $1^{12}4^{131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 3 & 2 \\ 4 & 3 & 7 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [421] $1^{14}1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 4 & 5 \\ 4 & 1 & 3 \\ 5 & 3 & 4 \end{pmatrix}_{b_0}$
- [422] $1^{11}4^{131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [416] $21^{11}5^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [423] $1^{11}4^{131^1}1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 6 & 0 & 33 & 9 \\ 2 & 2 & 0 & 9 & 1 \end{pmatrix}_{b_0}$
- [424] $1314^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [425] $1^{13}3221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$
- [426] $1^{12}4^{131^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 2 & 2 & 9 & 5 \\ 2 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$
- [427] $1^{11}4^{131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 2 \end{pmatrix} b_0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix} b_0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 5 & 2 \\ 7 & 5 & 25 & 9 \\ 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix} b_0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 4 & 1 & 24 & 8 \\ 2 & 2 & 0 & 8 & 1 \end{pmatrix}_b$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 4 & 1 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix} b_0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix} b_0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix} b_0$$

$$9] 1^{\text{blue}} 4^{\text{green}} 131^{\text{red}} +$$

$$T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 1 & 3 \\ 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[440] \quad 1 \overset{1}{1} 1^{\overset{1}{1}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[441] \quad 1 \overset{1}{1} 1^{\overset{1}{1}} 3 1^{\overset{1}{1}} 4 \quad 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[442] \quad 1 4 1 3 1^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 3 & 2 \\ 3 & 3 & 1 & 3 \\ 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[444] \quad 1 \overset{1}{1} 3 1^{\overset{1}{1}} 4 \quad 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[445] \quad 1 \overset{1}{1} 3 1^{\overset{1}{1}} 4 \quad 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[446] \quad 1 \overset{1}{1} 1^{\overset{1}{1}} 4 1^{\overset{1}{1}} 2 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 2 & 3 \\ 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[447] \quad 1 \overset{1}{1} 1^{\overset{1}{1}} 4 1^{\overset{1}{1}} 2 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 2 & 6 & 4 \\ 6 & 8 & 6 \\ 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[448] \quad 3 1^{\overset{1}{1}} 1^{\overset{1}{1}} 3 1^{\overset{1}{1}} 4 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 9 & 5 \\ 2 & 2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[443] \quad 1 4 1^{\overset{1}{1}} 2 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 0 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[449] \quad 1 \overset{1}{1} 1^{\overset{1}{1}} 4 1^{\overset{1}{1}} 2 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[450] \quad 3 1^{\overset{1}{1}} 1^{\overset{1}{1}} 3 1^{\overset{1}{1}} 4 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 6 & 4 & 3 \\ 6 & 13 & 6 & 7 \\ 4 & 6 & 1 & 3 \\ 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[460] \quad 1 \overset{1}{1} 1^{\overset{1}{1}} 4 1 3 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 5 & 1 & 24 & 8 \\ 2 & 2 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[453] $1^{\overline{1}1} 4^{\overline{1}1} 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[455] $1^{\overline{1}3} 4^{\overline{1}1} 131^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 5 & 2 \\ 7 & 5 & 23 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[456] $1^{\overline{1}1} 4^{\overline{1}1} 131^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 7 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[457] $1^{\overline{1}1} 4^{\overline{1}1} 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[458] $21^{\overline{1}1} 5^{\overline{1}1} 1^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 7 & 5 \\ 3 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[459] $1^{\overline{1}1} 4^{\overline{1}1} 1231^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[452] $1^{\overline{1}1} 5^{\overline{1}1} 1^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[461] $2141^{\overline{1}1} 31^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[462] $1^{\overline{1}1} 4^{\overline{1}1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[463] $131^{\overline{1}1} 4^{\overline{1}1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[451] $1^{\overline{1}1} 4^{\overline{1}1} 31^{\overline{1}1} 1^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 3 & 1 & 12 & 6 \\ 2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[454] $2^1 3^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

[465] $1^1 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[467] $1^1 1^{11} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 19 & 7 \\ 2 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$

[468] $21^1 1^1 4131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[469] $1^1 1^{11} 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 2 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[470] $31^1 1^1 4^1 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[464] $1^1 1^{11} 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 2 \\ 4 & 3 & 1 & 3 \\ 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[466] $2141^1 131^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 2 \\ 4 & 3 & 1 & 3 \\ 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[471] $1321^1 41^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[472] $131^1 1^1 41^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[473] $1^1 1^{12} 31^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 2 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$

[474] $31^1 1^1 31^1 41^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[475] $1^{\overline{11}} 4 131^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 3 & 2 & 2 & 3 & 3 \\ 2 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[477] $1^{\overline{11}} 41^{\overline{1}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 9 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{\overline{12}} 4 131^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 4 & 2 \\ 6 & 4 & 13 & 7 \\ 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[480] $1^{\overline{1}} 4 131^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 17 & 7 \\ 3 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[479] $1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 9 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[476] $1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 4 & 2 \\ 6 & 4 & 13 & 7 \\ 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[481] $221^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 3 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[482] $1^{\overline{1}} 41^{\overline{12}} 231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 5 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[483] $1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 5 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[484] $1^{\overline{1}} 4 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[485] $1^{\overline{12}} 4 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 4 & 1 & 12 & 6 \\ 2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[486] $1^{\overline{1}} 41^{\overline{12}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 3 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[489] $1^{\overline{1}1231^1} 4^{\overline{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 3 & 2 & 15 & 7 \\ 2 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$

[487] $1^{\overline{1}1231^1} 4^{\overline{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 4 & 2 & 27 & 9 \\ 2 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$

[488] $1^{\overline{1}1231^1} 4^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 3 & 2 \\ 6 & 3 & 7 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[490] $1^{\overline{1}41^1} 1^{\overline{1}231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[491] $31^{\overline{1}1} 31^{\overline{1}1} 41^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 7 & 5 \\ 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[493] $21^{\overline{1}1} 4131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 6 & 1 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

[494] $2^{\overline{1}11} 3^{\overline{1}1} 141^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[492] $21^{\overline{1}1} 41^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 8 & 1 & 10 & 6 \\ 2 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$

[495] $2^{\overline{1}11} 3^{\overline{1}1} 141^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[496] $2^{\overline{1}1} 3^{\overline{1}1} 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 13 & 6 & 7 \\ 3 & 6 & 1 & 3 \\ 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$

[497] $21^{\overline{1}1} 41^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 8 & 0 & 9 & 5 \\ 2 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$

[498] $41^{\overline{1}1} 2^{\overline{1}1} 3^{\overline{1}1} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$

- [499] $1^{\overline{1}^2}4131^{\overline{1}^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 23 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [500] $221^{\overline{1}^1}51^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 2 & 3 \\ 6 & 13 & 5 & 7 \\ 2 & 5 & 0 & 2 \\ 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
- [501] $1^{\overline{1}^1}4131^{\overline{1}^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 3 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$
- [502] $1^{\overline{1}^{\overline{1}^1}31^{\overline{1}^1}}41^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$
- [503] $1^{\overline{1}^{\overline{1}^3}31^{\overline{1}^1}}41^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 4 & 2 \\ 8 & 4 & 17 & 7 \\ 3 & 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [504] $141^{\overline{1}^1}31^{\overline{1}^1}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [505] $31^{\overline{1}^1}41^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [506] $1^{\overline{1}^3}4131^{\overline{1}^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 33 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [507] $2^{\overline{1}^{\overline{1}^1}}3141^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [508] $41^{\overline{1}^1}231^{\overline{1}^1}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 4 & 3 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 3 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 9 & 5 \\ 2 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [510] $1^{\overline{1}^3}4131^{\overline{1}^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 5 & 2 & 27 & 9 \\ 2 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [511] $1^{\overline{1}^2}4131^{\overline{1}^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 4 & 2 & 15 & 7 \\ 2 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{\overline{1^4 31^1}} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 5 & 2 \\ 10 & 5 & 31 & 9 \\ 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[513] $1^{\overline{1^{12} 31^1}} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 11 & 7 \\ 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[512] $1^{\overline{1^4 1^{12} 31^1}} 41^{12} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 11 & 7 \\ 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[514] $1^{\overline{1^4 1^{12} 31^1}} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 6 & 2 \\ 3 & 4 & 1 & 10 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 10 & 1 & 20 & 8 \\ 2 & 4 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[515] $2^{\overline{1^4 1^1}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 3 & 3 \\ 5 & 8 & 6 & 6 \\ 3 & 6 & 1 & 3 \\ 3 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[516] $1^{\overline{1^{11} 31^1}} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 6 & 1 & 34 & 10 \\ 2 & 2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[517] $1^{\overline{1^{12} 31^1}} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 31 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[518] $1^{\overline{1^4 1^{12} 31^1}} 41^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 37 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[519] $1^{\overline{1^{11} 31^1}} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[520] $1^{\overline{1^{12} 31^1}} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 3 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[521] $1^{\overline{1^4 1^{12} 31^1}} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[522] $2^{\overline{1^{12}}} 3^1 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[523] $1^{\overline{1}1^{12}31^1} 4^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 19 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[524] $31^1 1^{\overline{1}31^1} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 8 & 6 & 6 \\ 4 & 6 & 1 & 3 \\ 3 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[525] $1^{\overline{1}1^{11}31^1} 4^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 20 & 8 \\ 2 & 2 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[526] $2^1 3^1 1^{\overline{1}41^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 4 & 2 \\ 3 & 5 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[527] $1^{\overline{1}41^1} 1^{\overline{1}231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 8 & 0 & 9 & 5 \\ 2 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[528] $1^{\overline{1}41^1} 1^{\overline{1}231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 13 & 12 & 7 \\ 6 & 12 & 8 & 6 \\ 3 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[532] $1^{\overline{1}4} 1^{\overline{1}3} 4131^{\overline{1}4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 47 & 13 \\ 3 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[531] $1^{\overline{1}41^1} 1^{\overline{1}231^1} 1^{\overline{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 21 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[529] $141^{\overline{1}1} 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[530] $1^{\overline{1}1^{13}31^1} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 21 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[533] $31^1 1^{\overline{1}31^1} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[534] $141^{\overline{1}1} 1^{\overline{1}231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

- [535] $1^{11} 4^{1231^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 4 & 2 & 13 & 7 \\ 2 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$
- [536] $1^{13} 2^3 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$
- [538] $1^{11} 4^{1231^1} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 5 & 2 & 23 & 9 \\ 2 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [539] $1^{13} 4^{1231^1} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 29 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [537] $1^{14} 4^{131^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 6 & 2 \\ 11 & 6 & 35 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [540] $31^{11} 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$
- [541] $1321^{11} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 7 & 5 \\ 3 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$
- [542] $1^{11} 4^{1231^1} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 6 & 2 & 37 & 11 \\ 2 & 2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$
- [543] $141^{11} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 6 & 1 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$
- [544] $1^{13} 4^{131^1} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 41 & 13 \\ 3 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$
- [545] $1^{14} 41^{11} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 8 & 1 & 10 & 6 \\ 2 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [546] $31^{11} 31^{141^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 5 & 9 & 3 \\ 5 & 5 & 12 & 5 \\ 9 & 12 & 21 & 9 \\ 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$
- [547] $41^{11} 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 25 & 9 \\ 2 & 10 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[548] \ 1^{\overline{1}1431^{\overline{1}}} 4^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 43 & 13 \\ 3 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[549] \ 41^{\overline{1}1} 2231^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 28 & 0 & 25 & 9 \\ 2 & 10 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[550] \ 1^{\overline{1}1431^{\overline{1}}} 4^{\overline{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 55 & 15 \\ 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[551] \ 41^{\overline{1}1} 232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 9 & 5 \\ 2 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[552] \ 41^{\overline{1}1} 2231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[553] \ 41^{\overline{1}1} 322 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 4 & 7 & 4 \\ 3 & 8 & 7 & 4 & 4 \\ 2 & 6 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [554] \ 1^{\overline{1}1} 41321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[559] \ 21^{\overline{1}1} 5131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 5 & 8 & 5 \\ 6 & 8 & 8 & 6 \\ 4 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[558] \ 231^{\overline{1}1} 51^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[557] \ 21^{\overline{1}1} 4141^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 15 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[560] \ 21^{\overline{1}1} 5131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

- [561] $1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [556] $21^{\overline{1}} \overline{5} 1^{\overline{1}} 3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 3 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [555] $1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [562] $1^{\overline{1}} \overline{4} 1321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$
- [563] $1^{\overline{1}} \overline{4} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 3 & 4 & 3 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [566] $1^{\overline{1}} \overline{5} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [567] $21^{\overline{1}} \overline{4} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 4 & 5 & 3 & 4 \end{pmatrix}_{b_0}$
- [568] $1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [565] $21^{\overline{1}} \overline{4} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 3 & 4 & 3 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [564] $1^{\overline{1}} \overline{4} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [569] $1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$
- [570] $1^{\overline{1}} \overline{4} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
[571] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0} \\
[573] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0} \\
[572] & 21^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[574] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 9 & 5 \\ 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0} \\
[575] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[576] & 21^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[577] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[578] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
[579] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[580] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[581] & 1^1 1^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[582] \ 21^1 \bar{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 3 & 5 \\ 3 & 3 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[583] \ 21^1 \bar{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[585] \ 21^1 \bar{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[586] \ 21 \bar{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[587] \ 1^1 \bar{1} 322151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[588] \ 1^1 \bar{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[589] \ 1^1 \bar{3} 221^1 \bar{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 17 & 7 \\ 2 & 0 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[590] \ 131^1 \bar{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[591] \ 1^1 \bar{4} 1^1 \bar{3} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 12 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[592] \ 1^1 \bar{4} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[584] \ 1^1 \bar{5} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[593] \ 1^1 \bar{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

- [594] $31^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 3 & 0 & 9 & 12 & 6 \\ 2 & 0 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [595] $1^1 41321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$
- [596] $1^1 51231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [597] $1^{11} 321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 10 & 5 & 7 & 6 \\ 1 & 5 & 0 & 4 & 2 \\ 2 & 7 & 4 & 4 & 4 \\ 2 & 6 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [598] $1^1 51^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 3 & 3 \\ 7 & 26 & 8 & 10 \\ 3 & 8 & 1 & 3 \\ 3 & 10 & 3 & 3 \end{pmatrix}_{b_0}$
- [599] $231^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [600] $31^1 4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 0 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [601] $21^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [602] $1^{11} 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$
- [604] $21^1 5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [603] $1^1 51231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$
- [605] $1^{12} 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 29 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0} \\
[606] & 1 \overset{1}{5} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 8 & 5 \\ 4 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[607] & 1 \overset{1}{5} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[608] & 1 \overset{1}{4} 1 \overset{1}{4} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 13 & 5 \\ 8 & 13 & 29 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0} \\
[609] & 1 \overset{1}{4} 1 \overset{1}{4} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 9 & 5 \\ 6 & 9 & 11 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0} \\
[610] & 231 \overset{1}{5} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[611] & 1 \overset{1}{5} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[612] & 21 \overset{1}{5} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[613] & 21 \overset{1}{5} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 3 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[614] & 31 \overset{1}{3} 1 \overset{1}{5} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.40 \rightarrow 13, T_{\max} = 15, T \in [13, 15] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
[615] & 1 \overset{1}{5} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 3 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[616] & 2151 \overset{1}{2} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 5 & 4 & 6 \\ 1 & 5 & 0 & 3 & 2 \\ 1 & 4 & 3 & 0 & 2 \\ 2 & 6 & 2 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[617] $1^{11}4^{11}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 9 & 5 \\ 5 & 9 & 11 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[618] $1^{11}5^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 5 & 4 & 6 \\ 1 & 5 & 0 & 3 & 2 \\ 1 & 4 & 3 & 0 & 2 \\ 2 & 6 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[620] $141^{11}31^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 13 & 6 & 12 & 7 \\ 1 & 6 & 0 & 4 & 2 \\ 3 & 12 & 4 & 9 & 5 \\ 2 & 7 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[621] $21^{11}51^{11}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 11 & 3 & 4 \\ 11 & 26 & 8 & 10 \\ 3 & 8 & 1 & 3 \\ 4 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

[619] $31^{11}4^{11}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 9 & 0 & 12 & 6 \\ 2 & 5 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[624] $15^{11}1^13^12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[622] $1^{11}5^{11}1231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[623] $15^{11}131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 13 & 5 \\ 6 & 13 & 24 & 10 \\ 3 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[625] $31^{11}4^{11}141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 5 & 8 & 9 & 5 \\ 2 & 8 & 4 & 9 & 4 \\ 3 & 9 & 9 & 12 & 6 \\ 2 & 5 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[626] $1^{11}3221^{11}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & 6 & 0 & 3 \\ 4 & 6 & 17 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[628] $1^{11}4^{11}3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[627] $2^13^1151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

- [630] $1^1 3221^1 51^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [629] $41^1 231^1 41^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 4 & 1 & 2 \\ 3 & 10 & 17 & 0 & 6 \\ 4 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$
- [634] $21^1 1^1 41^1 41^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$
- [631] $1^1 1^1 4131^1 41^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 7 & 0 & 20 & 8 \\ 2 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [632] $31^1 1^1 31^1 51^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 5 & 2 & 3 \\ 5 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [633] $1^1 1^1 3221^1 51^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 5 & 0 & 8 & 30 & 10 \\ 2 & 0 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [636] $1^1 41^1 3141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 7 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$
- [635] $1^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 2 & 3 \\ 7 & 26 & 8 & 10 \\ 2 & 8 & 1 & 3 \\ 3 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$
- [638] $1^1 41^1 1^1 3141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & 0 & 2 \\ 2 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [637] $1^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 6 & 2 & 3 \\ 6 & 19 & 8 & 9 \\ 2 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$
- [639] $141^1 31^1 41^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 13 & 1 & 12 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 12 & 0 & 9 & 5 \\ 2 & 7 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$
- [640] $21^1 5^1 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

- [641] $1^{\overline{1}1}3221^{\overline{1}5}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 24 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$
- [642] $1^{\overline{1}1}5131^{\overline{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 24 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$
- [643] $2^{\overline{1}1}32151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$
- [644] $1^{\overline{1}1}51^{\overline{1}1}32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$
- [645] $31^{\overline{1}1}4141^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 4 & 0 & 9 & 12 & 6 \\ 2 & 0 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [646] $31^{\overline{1}1}4141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 7 & 0 & 4 & 4 \\ 2 & 5 & -1 & 4 & 2 \end{pmatrix}_{b_0}$
- [647] $2^{\overline{1}1}32151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [648] $1^{\overline{1}1}51^{\overline{1}1}32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [649] $31^{\overline{1}1}4141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 5 & 8 & 0 & 5 \\ 2 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [650] $2^{\overline{1}1}3151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 5 & 8 & 5 \\ 5 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$
- [651] $1321^{\overline{1}1}51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 26 & 0 & 10 \\ 1 & 0 & -3 & -1 \\ 3 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [652] $22^{\overline{1}1}3151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 7 & 4 \\ 3 & 2 & 7 & 4 & 4 \\ 2 & 2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[653] \quad 1^1 2321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 3 & 3 \\ 8 & 26 & 8 & 10 \\ 3 & 8 & 1 & 3 \\ 3 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[654] \quad 31^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 13 & 10 & 0 & 7 \\ 2 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[655] \quad 1^1 3221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 1 & 5 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & 0 & 2 \\ 2 & 8 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[656] \quad 31^1 321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 7 & 4 & 3 \\ 7 & 19 & 8 & 9 \\ 4 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[657] \quad 21^1 5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 18 & 5 \\ 9 & 18 & 48 & 14 \\ 3 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[658] \quad 1^1 5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 18 & 5 \\ 9 & 18 & 48 & 14 \\ 3 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[659] \quad 1^1 3221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[660] \quad 1^{11} 41^{13} 131^{14} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 69 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[662] \quad 1^1 1^1 32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[663] \quad 232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[661] \quad 31^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 5 & 1 & 16 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 16 & 0 & 37 & 11 \\ 2 & 5 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[665] \ 1^1 \bar{5} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 23 & 5 \\ 11 & 23 & 80 & 18 \\ 3 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[664] \ 21^1 \bar{5} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 23 & 5 \\ 11 & 23 & 80 & 18 \\ 3 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[666] \ 1^{12} 3221^1 \bar{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 11 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[667] \ 2321^1 \bar{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 3 & 3 \\ 7 & 19 & 8 & 9 \\ 3 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[668] \ 31^{11} 32151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[669] \ 1^1 \bar{5} 131^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[670] \ 31^{11} 3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 5 & 8 & 5 \\ 6 & 8 & 8 & 6 \\ 3 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[671] \ 2^1 \bar{3} 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 5 & 13 & 5 \\ 8 & 13 & 24 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[672] \ 22^1 \bar{3} 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 14 & 13 & 8 \\ 5 & 13 & 8 & 6 \\ 3 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[673] \ 1^1 \bar{5} 1^1 32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[674] \ 1^1 \bar{5} 131^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[675] \ 1^{13} 3221^1 \bar{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 14 & 0 & 74 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[676] \quad 22 \overset{11}{3} 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 9 & 12 & 7 \\ 5 & 12 & 8 & 6 \\ 3 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[677] \quad 41 \overset{11}{1} 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 25 & 9 \\ 2 & 10 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[678] \quad 2 \overset{13}{3} 1 \overset{11}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 5 & 18 & 5 \\ 11 & 18 & 48 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[679] \quad 22 \overset{112}{3} 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 10 & 3 \\ 6 & 14 & 26 & 8 \\ 10 & 26 & 41 & 13 \\ 3 & 8 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[680] \quad 321 \overset{11}{1} 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 8 & 7 & 3 \\ 8 & 17 & 16 & 9 \\ 7 & 16 & 11 & 7 \\ 3 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[681] \quad 231 \overset{11}{1} 3151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 35 & 0 & 32 & 10 \\ 2 & 11 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[682] \quad 22 \overset{13}{3} 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 15 & 3 \\ 6 & 14 & 39 & 8 \\ 15 & 39 & 96 & 20 \\ 3 & 8 & 20 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [683] \quad 1 \overset{11}{4} 13221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 11, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[684] \quad 1 \overset{11}{5} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[685] \quad 1 \overset{11}{5} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 3 & 6 \\ 2 & 3 & 0 & 2 \\ 3 & 6 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[686] \quad 1 \overset{11}{4} 14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[687] \quad 1 \overset{11}{1} 315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

- [688] $21^1 5131^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 7 & 5 \\ 7 & 9 & 7 \\ 5 & 7 & 5 \end{pmatrix}_{b_0}$
- [689] $1^1 6131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [690] $1^1 61231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [691] $2321^1 61^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 6 & 6 \\ 2 & 5 & 0 & 2 & 2 \\ 2 & 6 & 2 & 3 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [692] $21^1 61^1 232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 6 & 6 \\ 2 & 5 & 0 & 2 & 2 \\ 2 & 6 & 2 & 3 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [693] $21^1 6131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 \end{pmatrix}_{b_0}$
- [694] $1^1 315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -67, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [695] $21^1 6131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 2 & 3 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [696] $21^1 61231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$
- [699] $1^1 51321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$
- [700] $1^1 414131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -63, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [697] $1^1 512321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -62, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[698] \ 1^1 315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[703] \ 21^1 61^1 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 1 & 6 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 3 \\ 2 & 6 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[701] \ 1^1 1^1 3221^1 6^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 1 & 6 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 3 \\ 2 & 6 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[702] \ 1^1 413151^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix}_{b_0}$$

$$[705] \ 1^1 413151^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[704] \ 1^1 414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[710] \ 1^1 413221^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 11, \Delta = -36, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 4 & 4 \\ 3 & 4 & 3 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[706] \ 21^1 61^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[707] \ 1^1 413151^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[708] \ 1^1 612231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[709] \ 1^1 61231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[711] $1^{11}4132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[718] $1^{11}51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0}$

[719] $1^{11}414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$

[712] $1^{11}51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$

[713] $21^{11}51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0}$

[714] $1^{11}512321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$

[715] $21^{11}51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$

[716] $1^{11}51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 6 & 9 & 6 \\ 4 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0}$

[717] $1^{11}414131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$

[722] $21^{11}61231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$

[723] $1^{11}414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[720] $232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[721] $1 \overset{21}{\underset{1}{6}} 131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[726] $1 \overset{21}{\underset{1}{6}} 131^{\overset{11}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[727] $1 \overset{1}{\underset{1}{6}} 1231^{\overset{11}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[724] $1^{\overset{1}{1}} 413151^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[725] $21^{\overset{1}{1}} 61231^{\overset{11}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[728] $1^{\overset{1}{1}} 414141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[730] $31^{\overset{1}{1}} 41^{\overset{1}{1}} 3151^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 6 & 0 & 7 & 5 \\ 2 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[729] $1^{\overset{1}{1}} 414141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 4 & 0 & 15 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[736] $221^{\overset{1}{1}} 6131^{\overset{11}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[731] $31^{\overset{1}{1}} 3161^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 7 & 3 \\ 6 & 7 & 21 & 9 \\ 3 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[732] $31^{\overset{1}{1}} 32161^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[733] $21^1 \overset{1}{6} 131^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[734] $41^1 231^{\overset{1}{1}} 51^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 5 & 0 & 2 \\ 4 & 5 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[735] $1^{\overset{1}{1}} 414141^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[737] $221^1 \overset{1}{6} 131^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

[739] $1^{\overset{1}{1}} \overset{1}{5} 1321^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 6 & 15 & 6 \\ 7 & 15 & 28 & 12 \\ 3 & 6 & 12 & 5 \end{pmatrix}_{b_0}$$

[738] $21^1 \overset{1}{5} 1321^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 6 & 15 & 6 \\ 7 & 15 & 28 & 12 \\ 3 & 6 & 12 & 5 \end{pmatrix}_{b_0}$$

[740] $31^1 3221^{\overset{1}{1}} 61^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 9 & 4 & 3 \\ 9 & 34 & 10 & 12 \\ 4 & 10 & 1 & 3 \\ 3 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[742] $1^1 41^1 32151^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 3 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[741] $1^1 41^1 4141^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[743] $1^1 41^1 32151^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[746] $1^1 \overset{1}{6} 1^{\overset{1}{1}} 3221^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 4 & 1 & 18 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[744] $1^{\overset{1}{1}} 41^1 3151^{\overset{1}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[745] $41^1 231^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 51 & 0 & 15 \\ 1 & 0 & -3 & -1 \\ 3 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

[747] $161232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[748] $231^1 3161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 7 & 3 \\ 8 & 7 & 21 & 9 \\ 3 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[750] $1^1 41321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 3 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[749] $1^{31} 6^{321} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 7 & 4 & 3 \\ 7 & 19 & 8 & 9 \\ 4 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[751] $41^1 31^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 32 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 32 & 0 & 37 & 11 \\ 2 & 10 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[752] $31^1 41^1 3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 5 & 1 & 3 & 2 \\ 5 & 8 & 1 & 6 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 \\ 2 & 6 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[754] $1^{13} 6^{3221} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

[753] $1^1 41321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 6 & 0 & 6 \\ 1 & 0 & -2 & 0 \\ 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

[755] $41^1 31^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 5 & 1 & 4 & 2 \\ 5 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 32 & 0 & 25 & 9 \\ 2 & 12 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[756] $1^{23} 6^{321} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 71, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 7 & 3 & 3 \\ 7 & 19 & 8 & 9 \\ 3 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[757] $216131^{11} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[758] \quad 16 \overset{1}{1} 131 \overset{1}{1} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 22 & 25 & 10 \\ 6 & 25 & 21 & 9 \\ 3 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[759] \quad 41 \overset{1}{1} 231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 9 & 9 & 3 \\ 9 & 51 & 48 & 15 \\ 9 & 48 & 41 & 13 \\ 3 & 15 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[760] \quad 41 \overset{1}{1} 232151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -90, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 25 & 9 \\ 2 & 10 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[761] \quad 41 \overset{1}{1} 231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 9 & 8 & 3 \\ 9 & 51 & 48 & 15 \\ 8 & 48 & 41 & 13 \\ 3 & 15 & 13 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [762] \quad 1^1 31 \overset{1}{5} 1321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[763] \quad 1^1 4141321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[764] \quad 1^1 513221 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[765] \quad 1^1 31 \overset{1}{5} 1321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[766] \quad 1^1 31 \overset{1}{5} 1321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[768] \quad 1^1 6131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[769] \quad 21^1 613141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 7 & 5 \\ 2 & 3 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[767] \ 1^{\overline{1}1321} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 13, \ \Delta = -65, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[770] \ 1^{\overline{1}131} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 13, \ \Delta = -64, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[772] \ 1^{\overline{1}131} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 13, \ \Delta = -64, \ \det = -5, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[773] \ 1^{\overline{1}1321} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 13, \ \Delta = -65, \ \det = -5, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[774] \ 1^{\overline{1}} \ 6 \ 12321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -5, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[771] \ 1^{\overline{1}1321} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 13, \ \Delta = -65, \ \det = -5, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[775] \ 21^{\overline{1}} \ 5131^{\overline{1}} \ 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 4 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[776] \ 41^{\overline{1}} \ 414141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 2 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[777] \ 1^{\overline{1}} \ 5132151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[778] \ 21^{\overline{1}} \ 61^{\overline{1}} \ 23141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 14, \ \Delta = -85, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 4 & 12 & 3 & 4 \\ 12 & 34 & 10 & 12 \\ 3 & 10 & 1 & 3 \\ 4 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[779] \ 1^{\overline{1}} \ 61^{\overline{1}} \ 23141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, \ T_{\max} = 14, \ \Delta = -85, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 26 & 1 & 14 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 14 & 0 & 7 & 5 \\ 2 & 10 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

- [780] $1\overset{1}{5}131\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [784] $1\overset{1}{5}132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$
- [781] $21\overset{1}{5}131\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [783] $1\overset{1}{5}131\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$
- [782] $1\overset{1}{5}131\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$
- [785] $1\overset{1}{5}132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [787] $1\overset{1}{5}13221^{\overset{1}{1}\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$
- [788] $1\overset{1}{5}13221^{\overset{1}{1}\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 7 & 10 & 7 \\ 4 & 10 & 10 & 8 \\ 3 & 7 & 8 & 6 \end{pmatrix}_{b_0}$
- [789] $1\overset{1}{4}141321^{\overset{1}{1}\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 2 & 4 \\ 2 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0}$
- [790] $21\overset{1}{5}13221^{\overset{1}{1}\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$
- [791] $1\overset{1}{5}131\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$
- [786] $1\overset{1}{6}123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[792] $21^1 6^{11} 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 4 & 3 \\ 5 & 4 & 9 & 7 \\ 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

[793] $1^1 6^{11} 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[794] $1^1 5^{11} 1321^{11} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 9 & 3 & 3 \\ 9 & 51 & 12 & 15 \\ 3 & 12 & 1 & 3 \\ 3 & 15 & 3 & 4 \end{pmatrix}_{b_0}$$

[795] $21^1 6^{11} 13141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 8 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 0 & 2 \\ 2 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[796] $21^1 6^{11} 131^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 35 & 11 \\ 2 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[797] $1^1 5^{11} 13221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 7 & 17 & 7 \\ 7 & 17 & 32 & 14 \\ 3 & 7 & 14 & 6 \end{pmatrix}_{b_0}$$

[798] $21^1 5^{11} 13221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 7 & 17 & 7 \\ 7 & 17 & 32 & 14 \\ 3 & 7 & 14 & 6 \end{pmatrix}_{b_0}$$

[799] $151^{11} 231^{11} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 51 & 0 & 15 \\ 1 & 0 & -3 & -1 \\ 3 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

[800] $1^{11} 4141^{11} 3151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 4 & 0 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[801] $51^{11} 2231^{11} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 7 & 2 & 3 \\ 7 & 7 & 3 & 7 \\ 2 & 3 & 0 & 2 \\ 3 & 7 & 2 & 4 \end{pmatrix}_{b_0}$$

[802] $161^{11} 23141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 26 & 1 & 7 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 7 & 0 & 0 & 2 \\ 2 & 10 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[803] $1^1 1^1 5 1321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 9 & 2 & 3 \\ 9 & 51 & 12 & 15 \\ 2 & 12 & 1 & 3 \\ 3 & 15 & 3 & 4 \end{pmatrix}_{b_0}$

[804] $1^1 1^1 5 12321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[805] $1^{11} 41 4 14141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 6 & 6 & 6 \\ 4 & 6 & 2 & 4 \\ 3 & 6 & 4 & 4 \end{pmatrix}_{b_0}$

[806] $1^1 4141^1 3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 12 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 3 & 8 & 0 & 4 \end{pmatrix}_{b_0}$

[807] $1^1 413221^1 61^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -77, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[808] $1^1 4141^1 3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -77, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 17 & 0 & 9 \\ 1 & 0 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$

[809] $1^{11} 41 4 14141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 3 & 6 & 5 \\ 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$

$T = 12$ [810] $1^1 231 5 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 12, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[815] $1^1 1^{131} 6 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 13, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[813] $1^1 4 1414131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 13, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[816] $1^1 1^{12321} 6 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 13, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[814] $1^1 1^{12321} 6 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 13, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 3 & 6 \end{pmatrix}_{b_0}$

[818] $1^1 1^{12321} 6 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 13, \Delta = -38, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 3 & 5 \\ 4 & 5 & 7 \end{pmatrix}_{b_0}$$

$$[817] \ 1^{\overline{1}} \overline{5}^{131} 1315131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 14, \ \Delta = -61, \ \det = 3, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[819] \ 1^{\overline{1}} \overline{5}^{131} 13161^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[820] \ 1^{\overline{1}} \overline{6}^{131} 13141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 14, \ \Delta = -63, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[821] \ 1^{\overline{1}} \overline{7}^{1321} 1231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 14, \ \Delta = -64, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[822] \ 21^{\overline{1}} \overline{6}^{131} 13151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[823] \ 1^{\overline{1}} \overline{4}^{141} 13221^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 12, \ \Delta = -6, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & 3 & 2 & 3 & 5 \\ 2 & 3 & 2 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[824] \ 1^{\overline{1}} \overline{6}^{131} 13141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 14, \ \Delta = -63, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[825] \ 1^{\overline{1}} \overline{5}^{131} 1315131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 14, \ \Delta = -61, \ \det = 5, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.60 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[827] \ 1^{\overline{1}} \overline{6}^{131} 13141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 14, \ \Delta = -63, \ \det = 5, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.60 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[826] \ 1^{\overline{1}} \overline{6}^{131} 13141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 14, \ \Delta = -63, \ \det = 5, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.60 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[832] \ 1^{\overline{1}} \overline{6}^{12231} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, \ T_{\max} = 15, \ \Delta = -80, \ \det = 6, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix}_{b_0}$$

- [853] $1^1 4 14 14 13 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 13, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 4 & 4 \\ 3 & 4 & 3 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}_{b_0}$
- [849] $1^1 5 13 21 6 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$
- [854] $1^1 6 1^1 22 31 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 18 & 1 & 4 & 8 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 8 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [855] $1^1 5 13 15 13 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 14, \Delta = -61, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$
- [856] $1^1 5 13 21 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 1 & 3 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 \\ 2 & 6 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [857] $3 1^1 5 13 16 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [858] $2 1^1 6 13 15 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$
- [859] $1^1 6 1^1 22 31 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 4 & 0 & 18 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$
- [860] $3 1^1 6 13 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [862] $2 16 1^1 23 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 10 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 3 \end{pmatrix}_{b_0}$
- [861] $1^1 5 13 21 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 10 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 3 \end{pmatrix}_{b_0}$
- [863] $16 1^1 22 31 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 18 & 1 & 0 & 8 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 8 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[864] \ 1^{1^1 512321^1} 6^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[865] \ 1^{1^1 512321^1} 6^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[867] \ 1^{1^1 4141^1 32161^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -84, \det = 77, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 33 & 8 & 0 & 11 \\ 1 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 11 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[869] \ 1^{1^1 4141321^1 61^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -84, \det = 77, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[868] \ 16^{1^1} 131^{1^1} 4141^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -83, \det = 77, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 38 & 1 & 8 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 8 & 0 & 0 & 2 \\ 2 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[866] \ 15^{1^1} 13221^{1^1} 61^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 77, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[871] \ 31^{1^1} 51321^{1^1} 61^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 9 & 3 & 3 \\ 9 & 34 & 10 & 12 \\ 3 & 10 & 1 & 3 \\ 3 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[870] \ 321^{1^1} 613151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 13 & 1 & 5 & 7 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 0 & 2 \\ 2 & 7 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[872] \ 31^{1^1 51321^1} 6^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 168, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 & 2 \\ 7 & 19 & 1 & 6 & 9 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 0 & 2 \\ 2 & 9 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [873] \ 1^{1^1} 4141413221^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 13, \Delta = -5, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.0$$

$$[874] \ 1^{1^1 12321^1} 7^{1^1} 1231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 14, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[875] \ 1^{1^1} 513151321^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[876] \ 1^1 1^1 3161231 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[877] \ 1^1 1^1 1^1 5^{1^1} 131 \overset{1^1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[878] \ 1^1 1^1 1^1 5^{1^1} 131 \overset{1^1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[879] \ 1^1 1^1 3216131 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[880] \ 21^1 51316131 \overset{1^1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[881] \ 1^1 1^1 1^1 5^{1^1} 131 \overset{1^1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[884] \ 1^1 414141321 \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 13, \Delta = -5, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[885] \ 1^1 51316131 \overset{1^1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[883] \ 1^1 1^1 3221 \overset{1^1}{8} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[882] \ 1^1 51316131 \overset{1^1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[886] \ 1^1 1^1 321 \overset{1^1}{8} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[887] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 8^{\overline{1}} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[894] \ 1^{\overline{1}} 3161231 \overline{1}^{\overline{1}} 5 \overline{1}^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[893] \ 1^{\overline{1}} 3216131 \overline{1}^{\overline{1}} 5 \overline{1}^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[892] \ 1^{\overline{1}} 5 \overline{1}^{\overline{1}} 131 \overline{1}^{\overline{1}} 6 \overline{1}^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[888] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 \overline{1}^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[889] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 6 \overline{1}^{\overline{1}} 12315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[890] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 6 \overline{1}^{\overline{1}} 1315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[891] \ 1^{\overline{1}} 5 \overline{1}^{\overline{1}} 131 \overline{1}^{\overline{1}} 6 \overline{1}^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[895] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 \overline{1}^{\overline{1}} 13151321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 7 & 7 & 7 \\ 3 & 7 & 3 & 5 \\ 3 & 7 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[900] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 3216131 \overline{1}^{\overline{1}} 5 \overline{1}^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[899] \ 21^{\overline{1}} \overline{1}^{\overline{1}} 6 \overline{1}^{\overline{1}} 1315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 4 & 3 \\ 5 & 4 & 9 & 7 \\ 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

[898] $1^{\overline{1}1}6^{11}1315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$

[897] $1^{\overline{1}1}6^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[896] $1^{\overline{1}1}3161231^{11}51^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[901] $1^{\overline{1}1}6^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$

[902] $41^{\overline{1}1}316^{11}13151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.91 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 6 & 0 & 2 & 3 \end{pmatrix}_{b_0}$

[903] $1^{\overline{1}1}414141^{\overline{1}1}3161^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 16, \Delta = -82, \det = -137, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.77 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 7 & 1 & 1 & 2 \\ 7 & 61 & 10 & 0 & 15 \\ 1 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

$T = 14$ [904] $1^{\overline{1}1}41414141321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 14, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 5.0$

[905] $1^{\overline{1}1}6^{11}123151321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[911] $1^{\overline{1}1}2321613151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[906] $1^{\overline{1}1}6^{11}13151321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0}$

[908] $1^{\overline{1}1}6^{11}1315131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[907] $1^{\overline{1}1}8^{11}12321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[910] $1^{\overline{1}1315131} 6^{11} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 3 & 6 \end{pmatrix}_{b_0}$$

[909] $1^{\overline{1}12321} 8^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[912] $1^{\overline{1}32171231} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

[913] $1^{\overline{1}1315131} 6^{11} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 3 & 5 \\ 4 & 5 & 7 \end{pmatrix}_{b_0}$$

[917] $1^{\overline{1}1321} 8^{11} 123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -62, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[916] $1^{\overline{1}1321} 8^{11} 123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -62, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[915] $1^{\overline{1}4141414131} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 3 & 5 & 5 \\ 3 & 5 & 4 & 6 \\ 3 & 5 & 6 & 7 \end{pmatrix}_{b_0}$$

[914] $1^{\overline{1}5131513221} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 14, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 8 & 8 & 8 \\ 3 & 8 & 4 & 6 \\ 3 & 8 & 6 & 7 \end{pmatrix}_{b_0}$$

[919] $1^{\overline{1}32171231} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[918] $1^{\overline{1}6123216131} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[920] $1^{\overline{1}5131} 6^{11} 13151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[921] $21^{\overline{1}} 6 13151321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 15, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 6 & 6 \\ 2 & 2 & 2 & 6 & 5 \end{pmatrix}_{b_0}$

[922] $1^{\overline{11}} 23216131^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 15, \Delta = -32, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[923] $1^{\overline{1}} 6 13151321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 15, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$

[924] $1^{\overline{1}} 6 131513151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 17, \Delta = -81, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[925] $1^{\overline{11}} 32171231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 16, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[926] $31^{\overline{1}31} 6 13161^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 16, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 5 & -1 & 4 \end{pmatrix}_{b_0}$

[927] $1^{\overline{1}} 6^{\overline{15131}} 13151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 17, \Delta = -81, \det = 17, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.59 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0}$

[928] $1^{\overline{1}} 61^{\overline{1231513151}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 17, \Delta = -82, \det = 77, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.91 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$

[929] $1^{\overline{1}} 51321^{\overline{1}} 613151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 17, \Delta = -82, \det = 135, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.80 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 12 & 1 & 3 \\ 12 & 51 & 0 & 15 \\ 1 & 0 & -2 & 0 \\ 3 & 15 & 0 & 4 \end{pmatrix}_{b_0}$

[930] $1^{\overline{1}} 51321^{\overline{1}} 613151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 14, T_{\max} = 17, \Delta = -82, \det = 137, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.77 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 11 & 1 & 3 \\ 11 & 42 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 3 & 14 & 0 & 4 \end{pmatrix}_{b_0}$

$T = 15$ [931] $1^{\overline{1}} 414141414131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 15, T_{\max} = 15, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$

[935] $1^{\overline{1}} 6^{\overline{1}31} 13161231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 15, T_{\max} = 16, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[934] $1^{11}23217123151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[933] $1^{11}311^{11}6131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -34, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 \\ 1 & 2 & 0 & 2 \\ 1 & 0 & 0 & -3 \\ 2 & 4 & 2 & -1 \\ 5 & & & \end{pmatrix}_{b_0}$

[937] $1^{11}311^{11}6131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -34, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 4 & 2 & 2 & 5 \end{pmatrix}_{b_0}$

[936] $23218123151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 5 & 1 & 0 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 5 & 0 & 0 & 4 \end{pmatrix}_{b_0}$

[939] $23218123151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$

[938] $1^{11}613161231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -35, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 4 & 0 & 2 & 5 \end{pmatrix}_{b_0}$

[940] $1^{11}512321^{11}8^{11}1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[941] $1^{11}6131513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 3 & 7 & 5 \\ 3 & 2 & 7 & 7 & 7 \\ 2 & 2 & 5 & 7 & 6 \end{pmatrix}_{b_0}$

[942] $21^{11}6131513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 3 & 2 & 4 & 7 & 7 \\ 2 & 2 & 3 & 7 & 6 \end{pmatrix}_{b_0}$

[943] $1^{11}512321^{11}8^{11}1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[944] $1^{11}23217123151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[945] $21^1 6^{11} 131612321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 15, T_{\max} = 16, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 2 & 6 & 4 \\ 5 & 6 & 13 & 9 \\ 3 & 4 & 9 & 6 \end{pmatrix}_{b_0}$

[946] $1^1 6^{11} 1316123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 15, T_{\max} = 18, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[947] $2^1 3^{11} 2181231^1 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 15, T_{\max} = 17, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 7 & 0 & 5 & 5 \\ 2 & 5 & 0 & 5 & 4 \end{pmatrix}_{b_0}$

[948] $1^1 6^{11} 2321613151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 15, T_{\max} = 18, \Delta = -81, \det = -77, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17.91 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$

[949] $1^1 6^{11} 13161^1 23151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 15, T_{\max} = 18, \Delta = -79, \det = -135, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17.80 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 10 & 1 & 3 \\ 10 & 28 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$

[950] $1^1 6^{11} 2321^1 613151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 15, T_{\max} = 18, \Delta = -81, \det = -137, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17.77 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 11 & 1 & 3 \\ 11 & 42 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 3 & 14 & 0 & 4 \end{pmatrix}_{b_0}$

$T = 16$ [954] $1^1 6^{11} 131613141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 17, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 3 & 4 \\ 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0}$

[952] $1^1 8^{13151321} 1231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 17, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[957] $1^1 6^{11} 12321712321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[955] $1^1 2321812231^1 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[956] $1^1 8^{11} 1321^1 12321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[953] $1^1 3151316131^1 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 17, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[958] $1^1 2^1 3^1 21812321^1 61^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 18, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[959] $1^1 1^1 6^1 131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -33, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 5 & -1 & 3 & 5 \end{pmatrix}_{b_0}$$

[960] $1^1 1^1 3151316131^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[961] $1^1 1^1 5^1 1321^1 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[962] $1^1 1^1 2321812231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[963] $1^1 1^1 8^1 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 19, \Delta = -80, \det = 17, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18.59 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[964] $1^1 61^1 23217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 19, \Delta = -80, \det = 77, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18.91 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

[965] $1^1 513221^1 8123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 19, \Delta = -80, \det = 213, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18.70 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 1 & 15 & 1 & 3 \\ 15 & 79 & 0 & 19 \\ 1 & 0 & -2 & 0 \\ 3 & 19 & 0 & 4 \end{pmatrix}_{b_0}$$

[966] $1^1 513221^1 8123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 19, \Delta = -80, \det = 215, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18.69 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 1 & 14 & 1 & 3 \\ 14 & 68 & 1 & 18 \\ 1 & 1 & -2 & 0 \\ 3 & 18 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 17$ [967] $1^1 1^1 223151316131^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 17, T_{\max} = 17, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$

[968] $1^1 32161316131^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[969] $1^1 1^1 2321^1 12321^1 61^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[970] $1^{11}2315131613151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

[971] $1^{11}3216131613151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[972] $1^{11}513218^{1232161^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 17, T_{\max} = 20, \Delta = -79, \det = -17, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 19.59 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 5 & 4 & 0 & 5 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 5 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[973] $1^{161^1}232181223151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 17, T_{\max} = 20, \Delta = -79, \det = -77, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 19.91 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

[974] $1^{11}513218^{12321^1}61^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 17, T_{\max} = 20, \Delta = -79, \det = -77, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 19.91 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[975] $1^{161}232181^{1223151^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 17, T_{\max} = 20, \Delta = -79, \det = -213, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 19.70 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 1 & 13 & 1 & 3 \\ 13 & 50 & 1 & 16 \\ 1 & 1 & -2 & 0 \\ 3 & 16 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 18$ [976] $21^{161}1316131513221^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 18, T_{\max} = 18, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 5.0$

[979] $1^{131513218123151^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 18, T_{\max} = 19, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[978] $1^{13221(12)}12231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 18, T_{\max} = 19, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

[980] $21^{161}131613151321^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 18, T_{\max} = 18, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 4 & 3 & 10 & 8 \\ 2 & 3 & 2 & 8 & 6 \end{pmatrix}_{b_0}$$

[981] $1^{11}23216131613151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 18, T_{\max} = 18, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[982] \ 1^{11}31513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 18, \ T_{\max} = 19, \ \Delta = -30, \ \det = 8, \ (n_+, n_-, n_0) = (1, 18, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, \ T_{\max} = 19, \ T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [983] \ 1^{11}2231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 19, \ T_{\max} = 19, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 19, 1), \ c_{\text{ext}} = 5.0$$

$$[986] \ 1^1316123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 19, \ T_{\max} = 20, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[987] \ 1^{11}612321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 19, \ T_{\max} = 20, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[988] \ 1^{11}613151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 19, \ T_{\max} = 20, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[989] \ 1^{11}231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 19, \ T_{\max} = 19, \ \Delta = -2, \ \det = -4, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, \ T_{\max} = 19, \ T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[990] \ 1^{11}316123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 19, \ T_{\max} = 20, \ \Delta = -29, \ \det = -8, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [991] \ 1^{11}23151316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 20, \ T_{\max} = 20, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 20, 1), \ c_{\text{ext}} = 5.0$$

$$[993] \ 1^{11}3151316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 20, \ T_{\max} = 20, \ \Delta = -1, \ \det = 4, \ (n_+, n_-, n_0) = (1, 20, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [994] \ 1^{11}232161316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 21, \ T_{\max} = 21, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 21, 1), \ c_{\text{ext}} = 5.0$$

$$[995] \ 231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 21, \ T_{\max} = 23, \ \Delta = -59, \ \det = -3, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, \ T_{\max} = 23, \ T \in [23, 23]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 5 & 3 & 2 & 5 \\ 1 & 3 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 5 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[996] $1^{1^1 31^1 5^{1^1} 13221} (12) 1223151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 21, T_{\max} = 23, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[997] $1^{1^1 1^1 232171232181231} 51^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 21, T_{\max} = 21, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$T = 22$ [998] $21^{1^1 1^1 1^1 13221} (12) 122315131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 23, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[999] $1^{1^1} (12) 122315131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 23, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[1000] $1^{1^1 1^1 2321} 8^{1^1 1^1 2321} 123218123151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 22, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

[1001] $21^{1^1 1^1 1^1 13221} (12) 122315131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 23, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[1002] $1^{1^1} (12) 122315131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 23, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 5 & 3 & 6 & 6 \\ 2 & 4 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$T = 23$ [1004] $1^{1^1 1^1 1^1 1^1 315131613161316131} 51^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$

[1005] $21^{1^1 1^1 1^1 13221} (12) 12231513221^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$

[1003] $1^{1^1} (12) 12231513221^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$

[1006] $1^{1^1} (12) 1223151321^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 2 & 6 & 5 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 6 & 3 & 7 & 7 \\ 2 & 5 & 2 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[1007] \quad 21 \quad \begin{pmatrix} 1^1 & 1^1 & 1^1 & 1^1 & 1^1 \\ 5 & 13 & 22 & 1 & 1 \end{pmatrix} \quad 1223151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 23, \quad T_{\max} = 23, \quad \Delta = -1, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 23, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, \quad T_{\max} = 23, \quad T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [1008] \quad 1^{11} 2315132181232181231 \quad 5 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 24, \quad T_{\max} = 24, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 24, 1), \quad c_{\text{ext}} = 5.0$$

$$[1009] \quad 1^{11} 315132181232181231 \quad 5 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 24, \quad T_{\max} = 24, \quad \Delta = 0, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 24, \quad T_{\max} = 24, \quad T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 26 \quad [1010] \quad 1^{11} 232171232181232181231 \quad 5 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 26, \quad T_{\max} = 26, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 26, 1), \quad c_{\text{ext}} = 5.0$$

$$T = 28 \quad [1011] \quad 1 \quad \begin{pmatrix} 1^1 & 1^1 & 1^1 & 1^1 & 1^1 \\ 5 & 13 & 22 & 1 & 1 \end{pmatrix} \quad 1223151321812321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 28, \quad T_{\max} = 28, \quad \Delta = 0, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 34 \quad [1012] \quad 1 \quad \begin{pmatrix} 1^{11} & 1^{11} & 1^{11} & 1^{11} & 1^{11} \\ 23 & 15 & 13 & 22 & 1 \end{pmatrix} \quad (12) \quad 12231513221(12)12231 \quad 5 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 34, \quad T_{\max} = 34, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 34, 1), \quad c_{\text{ext}} = 5.0$$

4.1003 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$

$$T = 7 \quad [0] \quad 1^{111} 3 \quad 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -172, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \quad 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -172, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.32 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & -4 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 \quad 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -172, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 21^{11} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -172, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -3 & -1 \\ 3 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \quad 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, \quad T_{\max} = 100, \quad \Delta = -172, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -3 & -1 \\ 3 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 8 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 11 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 2 & 4 & 3 & -3 & -1 \\ 3 & 5 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{15} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 72 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 14 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -215, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{12}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -330, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{13}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 21^{11}13^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -3 & -1 \\ 4 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{14}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -626, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 19 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [22] \ 1^{11}13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 5 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11}13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 5 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11}13^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 5 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11}13^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

- [26] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$
- [28] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 6 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & 6 & -4 & -1 \end{pmatrix}_{b_0}$
- [27] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$
- [29] $2231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [30] $1^{11}3131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 93$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & 2 & -4 & -1 \end{pmatrix}_{b_0}$
- [31] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 102$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 2 & -4 & -1 \end{pmatrix}_{b_0}$
- [32] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -2 & 0 \\ 4 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$
- [33] $1^{12}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 169$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 3 \\ 1 & 1 & 0 & 2 & -3 & -1 \\ 2 & 2 & -1 & 3 & -1 & 0 \end{pmatrix}_{b_0}$
- [34] $1^131^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 204$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 3 & -4 & -1 \end{pmatrix}_{b_0}$
- [35] $1^131^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 219$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$
- [36] $1^{12}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 228$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 9 & -4 & -1 \end{pmatrix}_{b_0}$

- [37] $1^{13} 3^{21} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 228$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 15 & 3 & 5 \\ 1 & 1 & 0 & 3 & -3 & -1 \\ 2 & 2 & -1 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 282$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$
- $T = 9$ [39] $1^1 3^{221} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -3$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 19$, $T_{\max} = 26$, $T \in [19, 26]$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$
- [40] $1^{11} 2^3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -11$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 16.82 \rightarrow 17$, $T_{\max} = 32$, $T \in [17, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & 2 & -1 & -3 \end{pmatrix}_{b_0}$$
- [41] $1^{11} 2^3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -22$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$
- [42] $1^1 4^{131} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$
- [43] $1^{11} 4^{131} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^{11} 4^{131} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^1 4^{131} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -106$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{11} 4^{131} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -109$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$
- $T = 10$ [47] $1^1 4^{1321} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 3 & 3 & 2 & -2 & -4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -137, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [50] \quad 1^{\overline{1}} \overline{5} 12321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 24, T \in [21, 24]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{\overline{1}} \overline{5} 1321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 5 & 15 & 6 & 5 \\ 15 & 42 & 15 & 12 \\ 6 & 15 & 4 & 4 \\ 5 & 12 & 4 & -3 \end{pmatrix}_{b_0}$$

4.1004 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_3 + \textcolor{brown}{e}_6 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{\overline{111}} \overline{3} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 5 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{121}} \overline{3} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 5 & 4 \\ 4 & 8 & 9 & 0 & 6 & 6 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 4 & 3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 3 \\ 3 & 4 & 5 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} \overline{3} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} \overline{3} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} \overline{3} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{\overline{111}} 3^{\overline{11}} 2^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 & 0 \\ 3 & 8 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 705, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1005 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1006 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1007 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 8 \quad [1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{violet}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{violet}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 302$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.86 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -171$, $\det = 330$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.68 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -196$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -354$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.07 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

4.1008 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

16.0

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

4.1009 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 8 \quad [0] 1^{14} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 24, T_{\max} = 98, T \in [24, 98]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 11 & 21 & 75 & 0 & 18 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 18 & 0 & 4 & 4 \\ 3 & 4 & 15 & -4 & 4 & -11 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 98, T \in [16, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 5 & 6 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] 1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 48 & 0 & 15 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 15 & 0 & 4 & 4 \\ 3 & 4 & 12 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 172, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 2 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 178, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 181, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 184, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.91 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 184, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 196, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 5 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{131} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 198, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.36 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 6 & 12 & 24 & 0 & 9 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 8 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{111}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 10 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 18 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 24 & 0 & 58 & 14 \\ 3 & 4 & 5 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{113}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{123}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{12}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 9 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{123}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 15 & 3 \\ 3 & 4 & 8 & 0 & 27 & 4 \\ 4 & 8 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 36 & 0 & 133 & 21 \\ 3 & 4 & 5 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{114}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{115}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 15 & 0 & 73 & 15 \\ 3 & 4 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{116}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 18 & 0 & 106 & 18 \\ 3 & 4 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{117}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 21 & 0 & 145 & 21 \\ 3 & 4 & 2 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{121}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 57 & 0 & 12 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 13 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{115}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 487, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 2 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{131}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 48 & 0 & 27 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 27 & 0 & 13 & 7 \\ 3 & 4 & 12 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{123}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 9 & 0 & 30 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 5 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{112}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{114}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{131}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 84 & 0 & 15 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 4 & 16 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{123} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [77] $1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & 4 & -1 \\ 2 & 3 & 0 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & -1 & 5 & -4 \end{pmatrix}_{b_0}$$

[78] $1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[79] 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 2 & -4 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[80] 1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[81] 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 13 & 4 \\ 5 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 10 & 0 \\ 13 & 15 & 0 & 10 & 38 & 12 \\ 4 & 4 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[82] 1^{12} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 & 2 \\ 2 & 4 & 4 & 12 & 0 & 6 & 4 \\ 1 & 4 & 0 & 9 & 0 & 3 & 2 \\ 4 & 12 & 9 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 12 & 0 & 4 & 4 \\ 2 & 4 & 2 & 9 & -4 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[83] 1^{12} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 68, T \in [19, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 12 & 0 & 28 & 10 \\ 2 & 4 & 2 & 3 & -4 & 10 & -6 \end{pmatrix}_{b_0}$$

$$[84] 1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 7 & 0 \\ 5 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[85] 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[86] 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 3 & 3 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[87] 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 68, T \in [15, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 3 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

- [89] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 9 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [88] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$
- [90] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$
- [91] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [92] $1^{112} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [93] $1^{11} 31^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & 2 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$
- [94] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [95] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [97] $1^{112} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

- [98] $1^{111} 3^{13} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [99] $1^{113} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [100] $1^{114} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [101] $1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -177$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$
- [102] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{115} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 1 & 3 \\ 2 & 4 & 2 & 6 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$
- [105] $1^{111} 3^{13} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 8 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{112} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [108] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.0

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{111} 3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{111} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 7 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{113} 31^{11} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 3 & 0 & 15 & 3 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 12 & 0 & 49 & 13 \\ 2 & 3 & 3 & 2 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

[112] $1^{121} 3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 4 & 3 \\ 1 & 1 & 0 & 2 & -3 & 1 & -1 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 \\ 2 & 2 & -1 & 3 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{111} 31^{11} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 3 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[114] $1^{113} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 11 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{112} 3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[116] & 1^{11} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0} \\
[117] & 1^{12} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 13.16 \rightarrow 14, T_{\max} = 68, T \in [14, 68] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 12 & 3 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 3 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0} \\
[118] & 1^1 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 13.29 \rightarrow 14, T_{\max} = 68, T \in [14, 68] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -4 & 4 & -1 \end{pmatrix}_{b_0} \\
[119] & 1^{11} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -537, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 13.04 \rightarrow 14, T_{\max} = 68, T \in [14, 68] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 9 & 0 & 4 & 4 \\ 2 & 4 & 2 & 6 & -4 & 4 & -1 \end{pmatrix}_{b_0} \\
[120] & 1^{12} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 9 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0} \\
[121] & 1^{14} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0 \\
& T_{\min} = 13.03 \rightarrow 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 14 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0} \\
[122] & 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5 \\
& T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[123] & 1^{12} 3^{14} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5 \\
& T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[124] & 1^{13} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8
\end{aligned}$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -615, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{112} 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{11} 31^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -762, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & 2 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{112} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 10 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{114} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -771, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{112} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -875, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 11 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{113} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -876, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 12 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{112} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -876, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 12 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{111} 31^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

12.0

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 13 & 7 \\ 2 & 4 & 2 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[134] $1^{113} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -915, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{111} 31^{13} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1059, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{113} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{114} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 15 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{113} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{113} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 13 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{111} 31^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & 2 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

[142] $1^{113} 3^{13} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1520$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 14 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1643$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{113} 3^{14} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1683$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 15 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1740$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 16 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1920$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{114} 3^{13} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -2171$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 17 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{114} 3^{14} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -2496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 18 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [149] $1^{131} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 17.33 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & -3 \end{pmatrix}_{b_0}$$

[150] $1^{11} 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 17.33 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & -3 \end{pmatrix}_{b_0}$$

[151] $1^{11} 1^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{11} 1^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 \\ 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{11} 1^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 7 & 0 & 5 & 4 \\ 5 & 7 & 7 & 0 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 5 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{11} 1^{11} 4 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{11} 1^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 7 & 0 & 9 & 4 \\ 5 & 7 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 11 & 0 & 13 & 7 \\ 3 & 4 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[156] $1^{11} 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{11} 1^{11} 4 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.05 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 15 & 11 & 0 & 45 & 13 \\ 3 & 4 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[158] $1^{11} 1^{11} 4 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [168] \quad 1^{13} \overset{1}{5} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 18, T_{\max} = 23, T \in [18, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & -1 & 0 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[169] \quad 1^{12} \overset{1}{5} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[170] \quad 1^{11} \overset{1}{5} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.1010 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{111} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1011 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$

$$T = 9 \quad [0] \quad 1^{111} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \overset{1}{3} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

4.1012 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] 1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 \\ 3 & 5 & 3 & 0 & 3 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 5 & 6 & 0 & 3 & 4 \\ 3 & 4 & 3 & -4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] 1^{1111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 & 4 \\ 8 & 14 & 36 & 0 & 9 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 3 & 3 \\ 2 & 4 & 9 & 0 & 3 & 1 & 3 \\ 3 & 4 & 10 & -4 & 3 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] 1^{1121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 9 & 4 \\ 5 & 9 & 12 & 0 & 12 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 12 & 6 \\ 5 & 9 & 15 & 0 & 12 & 13 & 7 \\ 3 & 4 & 6 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] 1^{1111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 2 & -4 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] 1^{111} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 19 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 21 & 4 \\ 11 & 19 & 12 & 0 & 21 & 73 & 15 \\ 3 & 4 & 2 & -4 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1\overline{3}1} 3 1^{1\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -107, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 2 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 & 4 \\ 9 & 17 & 48 & 0 & 15 & 12 & 12 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 15 & 0 & 4 & 3 & 4 \\ 2 & 4 & 12 & 0 & 3 & 1 & 3 \\ 3 & 4 & 12 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1\overline{1}1\overline{1}} 3 1^{1\overline{3}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 19 & 9 & 4 \\ 2 & 4 & 0 & 0 & 12 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 19 & 12 & 0 & 73 & 33 & 15 \\ 5 & 9 & 6 & 0 & 33 & 13 & 7 \\ 3 & 4 & 2 & -4 & 15 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1\overline{1}2\overline{1}} 3 1^{1\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 5 & 9 & 6 & 0 & 12 & 13 & 7 \\ 3 & 4 & 2 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1\overline{1}1} 3 1^{1\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1\overline{1}1\overline{2}} 3 1^{1\overline{2}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -137, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 7 & 3 \\ 3 & 4 & 4 & 0 & 14 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 33 & 11 \\ 7 & 13 & 9 & 0 & 33 & 28 & 10 \\ 3 & 4 & 2 & -4 & 11 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1\overline{1}3} 3 1^{1\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 9 & 3 \\ 3 & 4 & 4 & 0 & 5 & 17 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 15 & 4 \\ 9 & 17 & 12 & 0 & 15 & 49 & 13 \\ 3 & 4 & 2 & -4 & 4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1\overline{1}2\overline{1}} 3 1^{1\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1\overline{1}1\overline{1}} 3 1^{1\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 5 & 3 \\ 4 & 9 & 4 & 0 & 8 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 9 & 4 \\ 5 & 12 & 6 & 0 & 9 & 13 & 7 \\ 3 & 5 & 2 & -4 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1\overline{1}2\overline{2}} 3 1^{1\overline{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -200, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 13 & 3 \\ 3 & 4 & 4 & 0 & 13 & 23 & 4 \\ 2 & 4 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 54 & 10 \\ 13 & 23 & 15 & 0 & 54 & 100 & 18 \\ 3 & 4 & 2 & -4 & 10 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 2 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 12 & 0 & 4 & 3 & 4 \\ 2 & 4 & 9 & 0 & 3 & 1 & 3 \\ 3 & 4 & 9 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 2 & -4 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 8 & 3 \\ 3 & 4 & 4 & 0 & 5 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 15 & 4 \\ 8 & 14 & 9 & 0 & 15 & 37 & 11 \\ 3 & 4 & 2 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 15 & 7 \\ 5 & 12 & 6 & 0 & 15 & 13 & 7 \\ 3 & 5 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 12 & 4 \\ 7 & 13 & 9 & 0 & 12 & 28 & 10 \\ 3 & 4 & 2 & -4 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 10 & 6 \\ 3 & 4 & 2 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{114}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 11 & 21 & 18 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 4 & 8 & 6 & 0 & 12 & 10 & 6 \\ 3 & 4 & 2 & -4 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{112}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 2 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 6 & 12 & 9 & 0 & 9 & 25 & 9 \\ 3 & 4 & 2 & -4 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 7 & 3 \\ 3 & 4 & 9 & 0 & 8 & 13 & 4 \\ 5 & 9 & 12 & 0 & 12 & 21 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 18 & 6 \\ 7 & 13 & 21 & 0 & 18 & 28 & 10 \\ 3 & 4 & 6 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 12 & 3 \\ 3 & 4 & 4 & 0 & 14 & 22 & 4 \\ 2 & 4 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 57 & 11 \\ 12 & 22 & 15 & 0 & 57 & 85 & 17 \\ 3 & 4 & 2 & -4 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 16 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 8 & 16 & 12 & 0 & 12 & 46 & 12 \\ 3 & 4 & 2 & -4 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 7 & 3 \\ 3 & 4 & 4 & 0 & 8 & 13 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 18 & 6 \\ 7 & 13 & 9 & 0 & 18 & 28 & 10 \\ 3 & 4 & 2 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 12 & 0 & 6 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 6 & 3 \\ 5 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 6 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 8 & 3 \\ 3 & 4 & 4 & 0 & 9 & 14 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 24 & 7 \\ 8 & 14 & 9 & 0 & 24 & 37 & 11 \\ 3 & 4 & 2 & -4 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 10 & 3 \\ 3 & 4 & 4 & 0 & 5 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 18 & 4 \\ 10 & 18 & 12 & 0 & 18 & 58 & 14 \\ 3 & 4 & 2 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 16 & 3 \\ 3 & 4 & 4 & 0 & 5 & 28 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 30 & 4 \\ 16 & 28 & 18 & 0 & 30 & 154 & 22 \\ 3 & 4 & 2 & -4 & 4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{111} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 39 & 11 \\ 8 & 14 & 9 & 0 & 39 & 37 & 11 \\ 3 & 4 & 2 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 2 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{112} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 2 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 14 & 3 \\ 3 & 4 & 4 & 0 & 5 & 26 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 24 & 4 \\ 14 & 26 & 18 & 0 & 24 & 118 & 20 \\ 3 & 4 & 2 & -4 & 4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 13 & 3 \\ 3 & 4 & 4 & 0 & 5 & 23 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 24 & 4 \\ 13 & 23 & 15 & 0 & 24 & 100 & 18 \\ 3 & 4 & 2 & -4 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1113} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 12 & 3 \\ 3 & 4 & 4 & 0 & 5 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 21 & 4 \\ 12 & 22 & 15 & 0 & 21 & 85 & 17 \\ 3 & 4 & 2 & -4 & 4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 2 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 13 & 4 \\ 5 & 9 & 12 & 0 & 6 & 21 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 9 & 3 \\ 7 & 13 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 6 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 21 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 33 & 7 \\ 11 & 21 & 15 & 0 & 33 & 76 & 16 \\ 3 & 4 & 2 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 21 & 7 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 2 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 10 & 3 \\ 3 & 4 & 4 & 0 & 14 & 18 & 4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 48 & 11 \\ 10 & 18 & 12 & 0 & 48 & 58 & 14 \\ 3 & 4 & 2 & -4 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 27 & 7 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 2 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 4 & 0 & 9 & 28 & 4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 48 & 7 \\ 16 & 28 & 18 & 0 & 48 & 154 & 22 \\ 3 & 4 & 2 & -4 & 7 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 4 & 17 & 4 \\ 5 & 9 & 12 & 0 & 6 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 12 & 3 \\ 9 & 17 & 27 & 0 & 12 & 49 & 13 \\ 3 & 4 & 6 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 11 & 3 \\ 3 & 4 & 9 & 0 & 4 & 21 & 4 \\ 5 & 9 & 12 & 0 & 6 & 33 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 15 & 3 \\ 11 & 21 & 33 & 0 & 15 & 76 & 16 \\ 3 & 4 & 6 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 2 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 15 & 3 \\ 3 & 4 & 4 & 0 & 5 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 27 & 4 \\ 15 & 27 & 18 & 0 & 27 & 133 & 21 \\ 3 & 4 & 2 & -4 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1122} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 10 & 3 \\ 3 & 4 & 4 & 0 & 13 & 18 & 4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 42 & 10 \\ 10 & 18 & 12 & 0 & 42 & 58 & 14 \\ 3 & 4 & 2 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 2 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 4 & 0 & 9 & 18 & 4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 30 & 7 \\ 10 & 18 & 12 & 0 & 30 & 58 & 14 \\ 3 & 4 & 2 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 13 & 3 \\ 3 & 4 & 4 & 0 & 9 & 23 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 39 & 7 \\ 13 & 23 & 15 & 0 & 39 & 100 & 18 \\ 3 & 4 & 2 & -4 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1122} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 30 & 10 \\ 7 & 13 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 2 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1113} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 2 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{113} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 18 & 3 \\ 3 & 4 & 4 & 0 & 5 & 32 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 33 & 4 \\ 18 & 32 & 21 & 0 & 33 & 193 & 25 \\ 3 & 4 & 2 & -4 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 4 & 21 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 11 & 21 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 2 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{114} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 17 & 3 \\ 3 & 4 & 4 & 0 & 5 & 31 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 30 & 4 \\ 17 & 31 & 21 & 0 & 30 & 172 & 24 \\ 3 & 4 & 2 & -4 & 4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1113} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 12 & 3 \\ 3 & 4 & 4 & 0 & 9 & 22 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 36 & 7 \\ 12 & 22 & 15 & 0 & 36 & 85 & 17 \\ 3 & 4 & 2 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1122} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 60 & 14 \\ 10 & 18 & 12 & 0 & 60 & 58 & 14 \\ 3 & 4 & 2 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{114} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 4 & 0 & 9 & 36 & 4 \\ 2 & 4 & 0 & 0 & 6 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 60 & 7 \\ 20 & 36 & 24 & 0 & 60 & 238 & 28 \\ 3 & 4 & 2 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [80] \quad 1^{\overline{111}} 3 \, 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \, T_{\max} = 13, \, \Delta = -84, \, \det = 44, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \, T_{\max} = 13, \, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{\overline{111}} 3 \, 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \, T_{\max} = 13, \, \Delta = -84, \, \det = 72, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \, T_{\max} = 13, \, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{\overline{111}} 3 \, 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \, T_{\max} = 13, \, \Delta = -84, \, \det = 82, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \, T_{\max} = 13, \, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{\overline{111}} 31^{\overline{1}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \, T_{\max} = 14, \, \Delta = -91, \, \det = 111, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.62 \rightarrow 14, \, T_{\max} = 14, \, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 9 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{\overline{11}} 31^{\overline{121}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \, T_{\max} = 14, \, \Delta = -91, \, \det = 138, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.35 \rightarrow 14, \, T_{\max} = 14, \, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 6 & 0 & 22 & 12 & 8 \\ 2 & 6 & 3 & 3 & 0 & 12 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 8 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{\overline{111}} 3 \, 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \, T_{\max} = 13, \, \Delta = -84, \, \det = 224, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.89 \rightarrow 13, \, T_{\max} = 13, \, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 7 & 0 \\ 4 & 6 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 7 & 9 & 14 & 8 \\ 3 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{\overline{111}} 3 \, 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \, T_{\max} = 13, \, \Delta = -84, \, \det = 239, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \, T_{\max} = 13, \, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{\overline{111}} 3 \, 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \, T_{\max} = 13, \, \Delta = -84, \, \det = 268, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.97 \rightarrow 13, \, T_{\max} = 13, \, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 269, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{111231} 1^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 273, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 9 \\ 2 & 4 & 3 & 3 & 0 & 3 & 12 \\ 1 & 3 & 3 & 0 & 0 & 3 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 & 9 \\ 4 & 9 & 12 & 9 & 0 & 9 & 28 \\ 2 & 3 & 3 & 2 & -4 & 3 & 10 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 296, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{1131} 1^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 6 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 2 & 6 & 3 & 3 & 0 & 4 & 6 \\ 2 & 6 & 3 & 3 & 0 & 6 & 4 \\ 2 & 4 & 2 & 2 & -4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{1111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 320, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{1111} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 363, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 \\ 2 & 4 & 4 & 3 & 0 & 6 & 9 \\ 1 & 4 & 0 & 3 & 0 & 3 & 6 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 2 & 6 & 3 & 3 & 0 & 4 & 9 \\ 3 & 9 & 6 & 6 & 0 & 9 & 13 \\ 2 & 4 & 2 & 2 & -4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 374, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 3 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{1111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 381, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{1111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 704$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111} 31^{112} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1065$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 9 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 15 & 4 \\ 5 & 15 & 9 & 9 & 0 & 15 & 37 & 11 \\ 2 & 4 & 2 & 2 & -4 & 4 & 11 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{111} 31^{113} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1659$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 21 & 4 \\ 7 & 21 & 12 & 12 & 0 & 21 & 73 & 15 \\ 2 & 4 & 2 & 2 & -4 & 4 & 15 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111} 31^{114} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 2145$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 15 & 15 & 0 & 27 & 121 & 19 \\ 2 & 4 & 2 & 2 & -4 & 4 & 19 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [100] $1^{111} 4^{1311} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -24$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 7 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 5 & 7 & 5 & 0 & 6 & 8 & 6 \\ 3 & 4 & 2 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{11} 31^{111} 4^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -48$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 5 & 3 \\ 4 & 4 & 3 & 0 & 5 & 7 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 5 & 7 & 5 & 0 & 6 & 8 & 6 \\ 3 & 4 & 2 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.1013 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10$ [0] $1^{111} 31^{1111} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 6 & 9 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 9 & 13 & 7 \\ 3 & 4 & 2 & -4 & 4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 166$, $(n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & -4 & 4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11111} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 6 & 3 \\ 8 & 14 & 9 & 0 & 9 & 37 & 24 & 11 \\ 5 & 9 & 6 & 0 & 6 & 24 & 13 & 7 \\ 3 & 4 & 2 & -4 & 3 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1112} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 12 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & -4 & 4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1113} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 27 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 18 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & -4 & 4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & -4 & 3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 15 & 0 & 21 & 15 & 85 & 17 \\ 3 & 4 & 2 & -4 & 4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 36 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 36 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 24 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & -4 & 4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$(1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 15 & 13 & 7 \\ 3 & 4 & 2 & -4 & 7 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1014 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$T = 10$ [0] $1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 6 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 15 & 6 & 7 \\ 5 & 12 & 6 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 6 & 0 & 2 \\ 3 & 5 & 2 & -4 & 7 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & -4 & 4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 507$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 2 & -4 & 4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 651$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{1141}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 813$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 & 3 \\ 11 & 21 & 18 & 0 & 15 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{1121}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1173$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 & 3 \\ 7 & 13 & 12 & 0 & 9 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{1131^1}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1227$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.8$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 & 3 \\ 9 & 17 & 15 & 0 & 12 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1015 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{111}3^11^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 86$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1131^1}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 126$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 3 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 130$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 5 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 146$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 226$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1131^1}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 296$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 328$, $(n_+, n_-, n_0) =$

$(1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 8 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^{12} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 416, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 6 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^{11} 5^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 416, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \overset{12}{1} \overset{11}{1} \overset{21}{3} \overset{21}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 7 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{112} 3^{121} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 626, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{112} 3^{112} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 9 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^{121}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 10 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1016 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + su2n3mix

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 5 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 5 & 5 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & 2 & -4 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.1017 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{121}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 & 3 \\ 3 & 4 & 8 & 0 & 9 & 5 & 4 \\ 4 & 8 & 9 & 0 & 12 & 6 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 9 & 7 \\ 3 & 5 & 6 & 0 & 9 & 3 & 3 \\ 3 & 4 & 5 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{114}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 2 & 3 \\ 4 & 9 & 4 & 0 & 36 & 4 & 5 \\ 2 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 36 & 15 & 0 & 121 & 15 & 19 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 5 & 2 & -4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 5 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 3 & 4 & 2 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 2 & 3 \\ 4 & 9 & 4 & 0 & 20 & 4 & 5 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 20 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 5 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.8$

$T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 2 & 3 \\ 4 & 9 & 4 & 0 & 28 & 4 & 5 \\ 2 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 28 & 12 & 0 & 73 & 12 & 15 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 5 & 2 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 2 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -267$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.7$

$T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 5 & 0 & 13 & 8 & 4 \\ 3 & 5 & 3 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 18 & 10 \\ 4 & 8 & 6 & 0 & 18 & 9 & 5 \\ 3 & 4 & 3 & -4 & 10 & 5 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -294$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 5 & 0 & 17 & 8 & 4 \\ 3 & 5 & 3 & 0 & 15 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 24 & 13 \\ 4 & 8 & 6 & 0 & 24 & 9 & 5 \\ 3 & 4 & 3 & -4 & 13 & 5 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -402$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.79 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -420$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.8$

$T_{\min} = 13.09 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 2 & 3 \\ 4 & 9 & 4 & 0 & 32 & 4 & 5 \\ 2 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 32 & 15 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 5 & 2 & -4 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 12 & 6 \\ 3 & 4 & 2 & -4 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114}31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -555, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 2 & 3 \\ 4 & 9 & 4 & 0 & 44 & 4 & 5 \\ 2 & 4 & 0 & 0 & 21 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 44 & 21 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 5 & 2 & -4 & 25 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -645, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -651, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -690, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 2 & 3 \\ 4 & 9 & 4 & 0 & 40 & 4 & 5 \\ 2 & 4 & 0 & 0 & 18 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 40 & 18 & 0 & 154 & 18 & 22 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 5 & 2 & -4 & 22 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{141}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 18 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -789, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{131} 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 2 & -4 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 1 & 3 \\ 4 & 9 & 0 & 4 & 12 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 7 & 0 & 0 \\ 5 & 12 & 0 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 39, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 & 0 \\ 4 & 8 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 126, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 182, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 31^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 477, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & 2 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 1^{111} 31^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 486, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{11}1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 22$, $\Delta = -100$, $\det = 774$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 9 & 7 \\ 2 & 6 & 3 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 2 & 2 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [31] $1^{11}1^{11}4^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -36$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 & 4 \\ 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}1^{11}4^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -108$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.22 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 4 & 3 & 0 & 5 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11}1^{11}4^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -210$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 4 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 2 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{11}1^{11}4^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -378$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.25 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 2 & 3 \\ 4 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 3 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 5 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1018 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 10$ [0] $1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 15 & 6 & 13 & 7 \\ 3 & 5 & 2 & -4 & 7 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 6 & 0 & 9 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 2 & -4 & 4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & -4 & 4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 18 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 15 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.1019 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1020 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 3 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -363, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 3 & 9 & 6 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 \\ 2 & 4 & 2 & 6 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1021 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$T = 10$ [0] $1^{111}3^11^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 1 & 3 \\ 2 & 3 & 3 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 1 & 3 \\ 2 & 4 & 2 & 3 & -4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 7 & 5 & 14 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 3 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^11^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^11^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 328, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 15 & 0 & 10 & 8 & 38 & 12 \\ 3 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

4.1022 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & 2 & -4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 6 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 2 & 2 & 0 & 0 & 3 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 2 & -4 & 2 & 3 \end{pmatrix}_{b_0}$$

4.1023 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 3 & 2 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 7 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 14 & 2 \\ 1 & 3 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 14 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

12.0

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}3^{11}1^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 11 & 2 \\ 1 & 3 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 11 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & 2 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}3^{11}2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}3^{11}1^{12}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -337, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 1 & 3 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 8 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 8 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}3^{11}2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}3^{11}2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111}3^{11}2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111}3^{11}2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 11 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

4.1024 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{11} 3^1 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 5 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1025 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [8] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 1 & 3 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 1 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11} 31^{11} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [17] $1^{11} 4^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.20 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{11} 31^{11} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{11} 31^{11} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.18 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1026 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{111} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 \\ 3 & 4 & 3 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & 2 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1027 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 5 & 0 & 8 & 9 & 4 \\ 3 & 5 & 3 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 12 & 5 \\ 5 & 9 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & 3 & -4 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 14 & 3 \\ 4 & 9 & 4 & 0 & 4 & 36 & 5 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 14 & 36 & 15 & 0 & 15 & 121 & 19 \\ 3 & 5 & 2 & -4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 13 & 7 \\ 3 & 5 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 3 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 8 & 3 \\ 4 & 9 & 4 & 0 & 4 & 20 & 5 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 20 & 9 & 0 & 9 & 37 & 11 \\ 3 & 5 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 11 & 3 \\ 4 & 9 & 4 & 0 & 4 & 28 & 5 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 11 & 28 & 12 & 0 & 12 & 73 & 15 \\ 3 & 5 & 2 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 36 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 2 & -4 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{122} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 5 & 0 & 8 & 13 & 4 \\ 3 & 5 & 3 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 18 & 5 \\ 7 & 13 & 12 & 0 & 18 & 28 & 10 \\ 3 & 4 & 3 & -4 & 5 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 5 & 0 & 8 & 17 & 4 \\ 3 & 5 & 3 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 24 & 5 \\ 9 & 17 & 15 & 0 & 24 & 49 & 13 \\ 3 & 4 & 3 & -4 & 5 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 13 & 3 \\ 4 & 9 & 4 & 0 & 4 & 32 & 5 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 32 & 15 & 0 & 15 & 100 & 18 \\ 3 & 5 & 2 & -4 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 9 & 6 \\ 3 & 5 & 3 & 0 & 9 & 4 & 4 \\ 3 & 4 & 2 & -4 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 18 & 3 \\ 4 & 9 & 4 & 0 & 4 & 44 & 5 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 44 & 21 & 0 & 21 & 193 & 25 \\ 3 & 5 & 2 & -4 & 2 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -645, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -651, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -690, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 16 & 3 \\ 4 & 9 & 4 & 0 & 4 & 40 & 5 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 16 & 40 & 18 & 0 & 18 & 154 & 22 \\ 3 & 5 & 2 & -4 & 2 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 18 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -789, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 9 & 0 & 4 & 0 & 12 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 7 & 0 & 14 & 8 \\ 3 & 5 & -1 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 9 & 0 & 4 & 0 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 477, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 6 & 3 \\ 2 & 6 & 3 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 9 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 774, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 9 & 3 \\ 3 & 9 & 6 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \quad 1^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 3 & 0 & 4 & 5 & 4 \\ 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{\overline{11}} 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{\overline{11}} 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{\overline{11}} 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -378, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 6 & 3 \\ 4 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 6 & 9 & 5 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1028 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 13 & 15 & 7 \\ 5 & 12 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 5 & 2 & -4 & 2 & 7 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 3 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 6 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 2 & -4 & 2 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 15 & 3 \\ 11 & 21 & 18 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 2 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 2 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 15 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 2 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

4.1029 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1030 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 16, \ \Delta = -96, \ \det = 36, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 16, \ T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

(1, 10, 0), $c_{\text{ext}} = 16.0$

$T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

4.1033 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7$

$T = 8$ [0] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 68$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & -4 & -2 \\ 3 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & -1 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -4 & -2 \\ 3 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 170$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -4 & -2 \\ 4 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{12} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 261$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 3 \\ 1 & 1 & 0 & 2 & -5 & -3 \\ 2 & 2 & -1 & 3 & -3 & -2 \end{pmatrix}_{b_0}$$

[5] $1^{13} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 358$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.61 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 15 & 3 & 5 \\ 1 & 1 & 0 & 3 & -5 & -3 \\ 2 & 2 & -1 & 5 & -3 & -2 \end{pmatrix}_{b_0}$$

$T = 9$ [6] $1^{11} 3 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -9$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 102$, $T \in [19, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & -4 \end{pmatrix}_{b_0}$$

4.1034 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{111} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 103$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 3 & 0 & -2 \\ 3 & 8 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

4.1035 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 103, \quad \Delta = -198, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, \quad T_{\max} = 103, \quad T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 1 & -3 \\ 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

4.1036 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -169, \quad \det = -113, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -169, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 1 & -3 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -169, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 6 & -2 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -169, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.90 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 3 & -2 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -169, \quad \det = -180, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.84 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 3 & -2 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -169, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.67 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -169, \quad \det = -228, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.79 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -169, \quad \det = -228, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{113} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{114} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.07 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{115} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 15.19 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 15 & -2 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -2 & 20 & -2 \end{pmatrix}_{b_0}$$

[14] $1^{121} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 14.62 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 4 & 3 \\ 1 & 1 & 0 & 2 & -5 & 1 & -3 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 \\ 2 & 2 & -1 & 3 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{112} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -641, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{112} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

14.0

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{114} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -3008, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

4.1037 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & -3 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -2 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}} 3 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 420, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 464, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 589, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 1088, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1038 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -109, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & -3 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -109, \ \det = -202, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1039 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\overset{11}{11}} 3^{\overset{11}{1}} 21^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1040 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^{\overset{11}{11}} 3^{\overset{11}{1}} 21^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1041 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{\overset{11}{11}} 3^{\overset{11}{1}} 21^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overset{11}{11}} 3^{\overset{11}{1}} 21^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overset{11}{11}} 3^{\overset{11}{1}} 21^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overset{11}{11}} 3^{\overset{11}{1}} 21^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overset{11}{11}} 3^{\overset{11}{1}} 21^{\overset{11}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1042 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1043 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

4.1044 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [1] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 3 & 0 & -2 \\ 4 & 8 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1045 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1046 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1047 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

4.1048 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 11$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 0 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1049 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$T = 10$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 10, T_{\max} = 101, \Delta = -151, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 0 & 4 & 4 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 2 & 0 & 0 & -2 \\ 2 & 4 & 0 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 & 1 \\ 3 & 5 & -1 & -2 & 2 & -1 & -2 & 2 \end{pmatrix}_{b_0}$$

4.1050 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \text{su2n3mix}$

$T = 9$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2$ $T = 9, T_{\max} = 102, \Delta = -172, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 2 & 2 \\ 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & -1 & -2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2$ $T = 9, T_{\max} = 102, \Delta = -172, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{112}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{112}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{112}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -977, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -1280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

4.1051 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2\text{n3mix} + \textcolor{blue}{\mathfrak{su}}_2$

$T = 10 \quad [0] \quad 1^{111}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 6 & 0 & -2 \end{pmatrix}_{b_0}$$

4.1052 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_3$

$T = 9 \quad [0] \quad 1^{111}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{12}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 14.94 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 2 & -3 \\ 2 & 4 & 0 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.03 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & -2 \\ 3 & 4 & -1 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.18 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & -2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

4.1053 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$T = 10 \quad [1] 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 9 & 0 & 4 & 0 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.75 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 3 & 0 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1054 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 11 \quad [0] 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1055 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

4.1056 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 102, \Delta = -157, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.28 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

4.1057 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & -1 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -3 & -1 \\ 4 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 3 \\ 1 & 1 & 0 & 2 & -4 & -2 \\ 2 & 2 & -1 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{13}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 15 & 3 & 5 \\ 1 & 1 & 0 & 3 & -4 & -2 \\ 2 & 2 & -1 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [6] \ 1^{\overline{1}} 3^{\overline{1}} 221^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18, T_{\max} = 99, T \in [18, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

4.1058 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}'_7$ + NHC(-2)(-3)

$$T = 10 \quad [0] \quad 1^{111} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \, T_{\max} = 69, \, \Delta = -121, \, \det = 82, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, \, T_{\max} = 69, \, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 3 & 0 & -1 \\ 3 & 8 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.1059 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}'_7$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{so}_8 \quad T = 9, \, T_{\max} = 102, \, \Delta = -170, \, \det = -245, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, \, T_{\max} = 102, \, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

4.1060 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \, T_{\max} = 99, \, \Delta = -141, \, \det = -94, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.70 \rightarrow 14, \, T_{\max} = 99, \, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \, T_{\max} = 99, \, \Delta = -141, \, \det = -103, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, \, T_{\max} = 99, \, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \, T_{\max} = 99, \, \Delta = -141, \, \det = -126, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.81 \rightarrow 14, \, T_{\max} = 99, \, T \in [14, 99]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \, T_{\max} = 99, \, \Delta = -141, \, \det = -127, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.02 \rightarrow 15, \, T_{\max} = 99, \, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 3 & -1 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \, T_{\max} = 99, \, \Delta = -141, \, \det = -148, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, \, T_{\max} = 99, \, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 3 & -1 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \, T_{\max} = 99, \, \Delta = -141, \, \det = -166, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.76 \rightarrow 14, \, T_{\max} = 99, \, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{114} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -2110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -2464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

4.1061 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10 \quad [0] 1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & -2 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{113} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 485, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.1062 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.1063 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1064 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 11 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1065 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 528, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 807, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1066 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$T = 11 \quad [0] \quad 1^{111}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.1067 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \text{su2n3mix}$

$T = 10 \quad [0] \quad 1^{111}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

4.1068 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10 \quad [0] \quad 1^{111}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 3 & 0 & -1 \\ 4 & 8 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1069 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1070 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1071 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 528, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11} 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 807, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{12} 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

4.1072 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 11$ [0] $1^{11} 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 0 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

4.1073 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$T = 10$ [0] $1^{11} 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 2 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

4.1074 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \text{su2n3mix}$

$T = 9$ [0] $1^{11} 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -144, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.97 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -144, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -144, \ \det = -510, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{112}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -144, \ \det = -806, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{112}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -144, \ \det = -1056, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

4.1075 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_{2n3\text{mix}} + \textcolor{blue}{\mathfrak{su}}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 150, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

4.1076 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 100, \ \Delta = -150, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.08 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 100, \ \Delta = -150, \ \det = -57, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.05 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}3^1 21^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 100, \ \Delta = -150, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.11 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 1 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.29 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.05 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1077 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 10 \quad [0] 1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 9 & 0 & 4 & 0 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1078 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 11 \quad [0] 1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1079 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] 1^{111} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \rightarrow (-3) \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1080 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{21} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1081 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$

$$T = 7 \quad [1] \quad 21^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

- [8] $1^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 8 & -3 & 0 \end{pmatrix}_{b_0}$
- [9] $1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -79$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 97$, $T \in [14, 97]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 11 & -3 & -1 \end{pmatrix}_{b_0}$
- [10] $1^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -88$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 97$, $T \in [14, 97]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 72 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 14 & -3 & -1 \end{pmatrix}_{b_0}$
- [11] $1^{15}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -93$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 2 & 4 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$
- [12] $1^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 6 & -3 & 0 \end{pmatrix}_{b_0}$
- [13] $1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -112$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -148$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & -3 & 0 \end{pmatrix}_{b_0}$
- [15] $1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -169$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -228$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & -3 & 1 \end{pmatrix}_{b_0}$
- [17] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -259$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & -3 & 1 \end{pmatrix}_{b_0}$
- [18] $1^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -289$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 15 & -3 & 1 \end{pmatrix}_{b_0}$

- [19] $1^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -372$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & -3 & 1 \end{pmatrix}_{b_0}$
- [20] $21^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -379$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -2 & 0 \\ 4 & 5 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{14}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -487$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 19 & -3 & 1 \end{pmatrix}_{b_0}$
- $T = 8$ [22] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 5 & 13 & 9 & 5 \\ 13 & 30 & 20 & 10 \\ 9 & 20 & 12 & 6 \\ 5 & 10 & 6 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -3 & -1 \\ 5 & 10 & -1 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$
- [25] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 35$, $T \in [14, 35]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & -3 & 0 \end{pmatrix}_{b_0}$
- [26] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 10 & 18 & 12 & 6 \\ 18 & 30 & 20 & 10 \\ 12 & 20 & 12 & 6 \\ 6 & 10 & 6 & 3 \end{pmatrix}_{b_0}$
- [27] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & -3 & 2 \end{pmatrix}_{b_0}$
- [29] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$
- [30] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 4 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 6 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11} 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11} 3 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 83, T \in [14, 83]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[36] \ 2^2 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 10 & 19 & 12 & 6 \\ 19 & 30 & 20 & 10 \\ 12 & 20 & 12 & 6 \\ 6 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11} 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11} 3 2 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 7 & 15 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11} 3^2 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 7 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 5 & 15 & 0 & 5 \\ 7 & 15 & 31 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 11 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{11} 3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 5 & 25 & 0 & 5 \\ 11 & 25 & 91 & 0 & 19 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 19 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 5 & 20 & 0 & 5 \\ 9 & 20 & 57 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 15 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{12} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 3 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{12} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 3 & 5 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{13} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{13} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 15 & 3 & 5 \\ 1 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & -1 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{13} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 4 & 7 & 6 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 8 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{13} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{12} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

- [52] $1^{12}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 188$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 4 & 6 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$
- [53] $1^{12}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 5 & 7 & 11 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 219$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$
- [55] $1^{13}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 283$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 5 & 8 & 9 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$
- [56] $2^2 2^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 353$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 \\ 2 & 3 & 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [57] $1^{13}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 363$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 6 & 9 & 12 & 60 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$
- [58] $2^2 2^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 548$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 1 & 2 \\ 2 & 1 & 4 & 6 & 1 & 3 \\ 2 & 4 & 0 & 6 & 1 & 2 \\ 4 & 6 & 6 & 9 & 2 & 5 \\ 1 & 1 & 1 & 2 & -3 & -1 \\ 2 & 3 & 2 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$
- $T = 9$ [59] $1^{11}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -17$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{11}4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{11}4131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{11}4131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$$T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{11} 41^{1231^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{11} 4^{131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11} 4^{131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{141} 1^{1231^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 7 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [67] \ 1^{11} 5^{131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 & 7 \\ 9 & 13 & 0 & 7 & 1 \\ 1 & 0 & -3 & -1 & 3 \\ 5 & 7 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} 5^{1231^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 23 & 1 & 5 & 18 \\ 23 & 90 & 0 & 18 & 1 \\ 1 & 0 & -3 & -1 & 3 \\ 5 & 18 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

4.1082 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4$

$$T = 8 \quad [0] \ 1^{111} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 101, \Delta = -169, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \ 1^{11} 3^{21} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 & 10 \\ 13 & 30 & 20 & 0 & 10 & 6 \\ 9 & 20 & 12 & 0 & 6 & 1 \\ 1 & 0 & 0 & -5 & -3 & 1 \\ 5 & 10 & 6 & -3 & 1 & 1 \end{pmatrix}_{b_0}$$

4.1083 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{111} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 5 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 5 & 4 \\ 4 & 8 & 9 & 0 & 6 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 4 & 3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 3 \\ 3 & 4 & 5 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 2 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1084 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$

$T = 8 \quad [0] 1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 3 & 0 & 0 & 2 \\ 3 & 5 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 3 & 0 & 1 & 2 \\ 1 & 1 & 3 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 5 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 31^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 82, \Delta = -124, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 82, T \in [14, 82]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

4.1085 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.1086 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1087 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 8$ [0] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 15.51 \rightarrow 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{123}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 5 & 6 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 75$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

- [4] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 82$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 48 & 0 & 15 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 15 & 0 & 4 & 4 \\ 3 & 4 & 12 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 137$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

- [14] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 5 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 6 & 12 & 24 & 0 & 9 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 8 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 159$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 10 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{122}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 18 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 24 & 0 & 58 & 14 \\ 3 & 4 & 5 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$
- [22] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 171$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

- [24] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 178$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 192$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [27] $1^{123}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 197$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 15 & 3 \\ 3 & 4 & 8 & 0 & 27 & 4 \\ 4 & 8 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 36 & 0 & 133 & 21 \\ 3 & 4 & 5 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$
- [28] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$
- [29] $1^{115}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 203$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 15 & 0 & 73 & 15 \\ 3 & 4 & 2 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$
- [30] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 210$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$
- [31] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 226$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 232$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{116}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 18 & 0 & 106 & 18 \\ 3 & 4 & 2 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

- [34] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 243$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$
- [37] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 266$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{117}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 275$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.69 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 21 & 0 & 145 & 21 \\ 3 & 4 & 2 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$
- [39] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 292$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [40] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 294$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 306$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [42] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 312$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [43] $1^{115}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 338$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 2 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

- [44] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 340$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 371$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{121}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 381$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 57 & 0 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 13 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [47] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 411$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [48] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [49] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$
- [50] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 450$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [51] $1^{131}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 463$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 48 & 0 & 27 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 27 & 0 & 13 & 7 \\ 3 & 4 & 12 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [52] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 489$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$
- [53] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 490$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{123}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 9 & 0 & 30 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 5 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 507$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 576$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 579$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 587$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{112}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 588$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 588$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{131}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 607$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 84 & 0 & 15 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 4 & 16 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 626$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

- [64] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 635$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 656$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 770$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$
- [67] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 771$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$
- [68] $1^{123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 798$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$
- [69] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$
- [70] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 915$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$
- [71] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1136$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$
- [72] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1251$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$
- [73] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114}3^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1643, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{114}3^11^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [76] $1^{12}3^{11}21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 5 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111}3^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{111}3^11^131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{11}3^121^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 5 & 3 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11}3^11^131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 3 & 3 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{111}3^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{111}3^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 23 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{13}3^121^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 9 & 0 & 3 & 2 \\ 5 & 8 & 9 & 39 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 9 & 0 & 1 & 3 \\ 2 & 2 & 2 & 11 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

- [85] $1^{11}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 0 & 7 & 0 & 3 & 2 \\ 3 & 4 & 7 & 15 & 0 & 7 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 7 & 0 & 1 & 3 \\ 2 & 2 & 2 & 7 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [86] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 4 & 0 & 1 & 3 \\ 2 & 2 & 2 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [87] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 3 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [83] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$
- [88] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$
- [89] $1^{112}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 13 & 14 & 1 & 28 & 7 \\ 14 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 58 & 14 \\ 7 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [90] $1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [91] $1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -71$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [92] $1^{11}31^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$
- [93] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 7 & 0 & 2 & 4 \\ 3 & 5 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

- [95] $1^{11} 3 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 4 & 0 & 1 & 3 \\ 2 & 3 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [94] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [96] $1^{11} 3 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 4 & 0 & 4 & 3 \\ 1 & 4 & 0 & 4 & 0 & 4 & 2 \\ 2 & 4 & 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 7 & 0 & 5 & 5 \\ 2 & 3 & 2 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$
- [100] $1^{111} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [97] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 18 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [102] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [101] $1^{12} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 3 & 0 & 5 & 2 \\ 1 & 1 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{114} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 66 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 66 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [99] $1^{112} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$
- [103] $1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 2 \\ 1 & 2 & 0 & 4 & 0 & 1 & 2 \\ 2 & 3 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{11} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 3 & 0 & 7 & 2 \\ 1 & 1 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{11} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{12} 3^{11} 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 8 & 0 & 3 & 2 \\ 4 & 6 & 8 & 25 & 0 & 8 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 8 & 0 & 1 & 3 \\ 2 & 2 & 2 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{112} 3^{11} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 18 & 0 & 94 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{11} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 3 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{114} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 9 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 3 & 4 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 16 & 0 & 41 & 11 \\ 2 & 2 & 2 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11} 3^{11} 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 5 & 25 & 0 & 5 & 5 \\ 11 & 25 & 91 & 0 & 19 & 19 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 19 & 0 & 2 & 4 \\ 3 & 5 & 19 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{11} 3^{11} 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 5 & 15 & 0 & 5 & 5 \\ 7 & 15 & 31 & 0 & 11 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 11 & 0 & 2 & 4 \\ 3 & 5 & 11 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 14 & 8 \\ 3 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{111} 3^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 14 & 1 & 14 & 7 \\ 14 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 15 & 0 & 13 & 7 \\ 7 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{11} 3^{21} 1^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 5 & 20 & 0 & 5 & 5 \\ 9 & 20 & 57 & 0 & 15 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 15 & 0 & 2 & 4 \\ 3 & 5 & 15 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{121} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 & 2 \\ 1 & 2 & 0 & 8 & 0 & 1 & 2 \\ 4 & 6 & 8 & 25 & 0 & 6 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 6 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{111} 3^{11} 1^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 6 & 0 & 1 & 3 \\ 2 & 4 & 2 & 6 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{13} 3^{21} 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 25 & 17 & 0 & 104 & 20 \\ 3 & 5 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{113} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{111} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[126] $1^{111} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{112} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{111} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 11 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{113} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[124] $1^{111} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[123] $1^{112} 3^{21} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 52 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{113} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 3 & 0 & 15 & 3 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 12 & 0 & 49 & 13 \\ 2 & 3 & 3 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[128] $1^{121} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 5 & 0 & 1 & 2 \\ 3 & 5 & 5 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{11} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 12 \\ 3 & 5 & 3 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{11} 3^{21} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 19 & 0 & 92 & 20 \\ 3 & 5 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{111} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[132] $1^{14} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 3 & 0 & 10 & 2 \\ 1 & 1 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{112} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{111} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 3 & 4 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 11 & 0 & 16 & 8 \\ 2 & 2 & 2 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{111} 3^{21} 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 4 & 0 & 3 & 2 \\ 1 & 2 & 0 & 7 & 0 & 4 & 2 \\ 3 & 4 & 7 & 15 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 11 & 0 & 5 & 5 \\ 2 & 2 & 2 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{121} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 4 & 3 \\ 1 & 1 & 0 & 2 & -2 & 1 & 0 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 \\ 2 & 2 & -1 & 3 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{111} 3^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 58 & 16 \\ 3 & 5 & 3 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{111} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 5 & 5 \\ 2 & 2 & 2 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{112} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{12} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 3 & 0 & 8 & 2 \\ 1 & 1 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[142] $1^{12}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 14 & 0 & 62 & 16 \\ 3 & 5 & 3 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{113}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{112}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{112}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 14 & 0 & 62 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{13}3^{11}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[148] $1^{121}3^{11}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 5 & 0 & 3 & 2 \\ 1 & 2 & 0 & 5 & 0 & 4 & 2 \\ 3 & 5 & 5 & 12 & 0 & 10 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 10 & 0 & 5 & 5 \\ 2 & 2 & 2 & 6 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[146] $1^{113}3^{11}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 6 & 2 \\ 2 & 3 & 4 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 13 & 0 & 25 & 9 \\ 2 & 2 & 2 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{111}3^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -314$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.31 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[150] $1^{112}3^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -332$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 12 & 3 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{12}3^{11}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 11 & 2 \\ 1 & 1 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{113}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 25 & 0 & 188 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{112}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -356$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 5 & 2 \\ 2 & 3 & 4 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 10 & 0 & 13 & 7 \\ 2 & 2 & 2 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{113}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -381$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 13 & 0 & 74 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{112}321^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -386$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 14 & 2 \\ 2 & 3 & 4 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 14 & 22 & 0 & 70 & 16 \\ 2 & 2 & 2 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[156] $1^1 31^{111} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[157] $1^{112}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{112}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{114}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{114}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{113}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{112} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{112} 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{1131} 1^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 9 & 0 & 4 & 4 \\ 2 & 4 & 2 & 6 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{14} 3 \ 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 35 & 24 & 0 & 202 & 28 \\ 3 & 5 & 3 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{112} 3 \ 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{114} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 27 & 16 & 0 & 122 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{1131} 1^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{121} 3 \ 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & 0 & 2 & 6 & 0 & 3 & 2 \\ 1 & 2 & 0 & 8 & 0 & 4 & 2 \\ 4 & 6 & 8 & 25 & 0 & 14 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 14 & 0 & 5 & 5 \\ 2 & 2 & 2 & 9 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{12} 3 \ 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 25 & 18 & 0 & 94 & 20 \\ 3 & 5 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{114} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{112} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{113} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{13} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 \\ 1 & 2 & 0 & 3 & 0 & 12 \\ 1 & 1 & 3 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 6 & 9 & 12 & 12 & 0 & 61 \\ 2 & 2 & 2 & 2 & -3 & 15 \end{pmatrix}_{b_0}$$

[175] $1^{13} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -565, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 30 & 21 & 0 & 142 & 24 \\ 3 & 5 & 3 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 42 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 42 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{112} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -581, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 21 & 0 & 142 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{112} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -586, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 6 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 3 & 4 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 14 & 0 & 26 & 10 \\ 2 & 2 & 2 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[181] \ 1^{\overline{11}13} 31^{\overline{12}31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -587, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{\overline{11}3} 321^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -591, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 104 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{\overline{11}2} 321^{\overline{11}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 11 & 2 \\ 2 & 3 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 18 & 0 & 45 & 13 \\ 2 & 2 & 2 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{\overline{14}3} 21^{\overline{11}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{\overline{12}3} 21^{\overline{11}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 30 & 22 & 0 & 134 & 24 \\ 3 & 5 & 3 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{\overline{11}2} 321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{\overline{11}3} 321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{\overline{11}2} 321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{\overline{11}3} 321^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

- [187] $1^{114} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 67 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 67 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [191] $1^{13} 3 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -671, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 \\ 1 & 2 & 0 & 3 & 0 & 15 \\ 1 & 1 & 3 & 0 & 0 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 7 & 10 & 15 & 15 & 0 & 88 \\ 2 & 2 & 2 & 2 & -3 & 18 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$
- [192] $1^{111} 31^{111} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -699, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 15 & 0 & 13 & 7 \\ 2 & 4 & 2 & 6 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [193] $1^{113} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$
- [194] $1^{113} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$
- [195] $1^{111} 31^{13} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$
- [196] $1^{113} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 9 & 2 \\ 2 & 3 & 4 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 17 & 0 & 40 & 12 \\ 2 & 2 & 2 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{114} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 20 & 0 & 158 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{13} 3 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -895, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 35 & 25 & 0 & 188 & 28 \\ 3 & 5 & 3 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$
- [201] $1^{114} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[202] \ 1^{114} 3 \ 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{113} 3 \ 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{113} 3 \ 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

$$[203] \ 2 \ 2 \ 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -901, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 2 \\ 2 & 3 & 3 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{114} 3 \ 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -926, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 10 & 2 \\ 2 & 3 & 4 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 20 & 0 & 58 & 14 \\ 2 & 2 & 2 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[205] \ 1^{114} 3 \ 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 57 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{113} 3 \ 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{14} 3 \ 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 3 & 0 & 16 & 2 \\ 1 & 1 & 3 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[208] \ 1^{14} 3 \ 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -950, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 5 & 5 & 0 & 40 & 5 \\ 2 & 5 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 40 & 28 & 0 & 254 & 32 \\ 3 & 5 & 3 & -3 & 32 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{114}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -966$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 28 & 0 & 254 & 32 \\ 3 & 4 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{111}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -1007$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 15 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & 2 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[212] $1^{114}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1016$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 24 & 0 & 202 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{113}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1016$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{113}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 9 & 2 \\ 1 & 2 & 0 & 4 & 0 & 12 & 2 \\ 2 & 3 & 4 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 21 & 0 & 61 & 15 \\ 2 & 2 & 2 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{114}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 68 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{113}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.04 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{113}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{114}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{113}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1181$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.0

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 15 & 2 \\ 2 & 3 & 4 & 4 & 0 & 25 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 25 & 0 & 88 & 18 \\ 2 & 2 & 2 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{113} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{114} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{114} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{114} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 11 & 2 \\ 1 & 2 & 0 & 4 & 0 & 13 & 2 \\ 2 & 3 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 24 & 0 & 81 & 17 \\ 2 & 2 & 2 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{114} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 69 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{114} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{114} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[228] 1^{114} 3 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 0 & 2 & 3 & 0 & 12 & 2 \\ 1 & 2 & 0 & 4 & 0 & 16 & 2 \\ 2 & 3 & 4 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 28 & 0 & 110 & 20 \\ 2 & 2 & 2 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[229] 1^{114} 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 70 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[230] 1^{114} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[231] 1^{114} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [232] 1^{11} 4 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[233] 1^{11} 4 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[234] 1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 117, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 5 & 7 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[235] 1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[236] 1^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{\overset{1}{\color{blue}{1}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}}\overset{1}{\color{blue}{4}}\overset{1}{\color{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 222, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}}\overset{1}{\color{red}{4}}\overset{1}{\color{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}}\overset{1}{\color{red}{4}}\overset{1}{\color{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}}\overset{1}{\color{red}{4}}\overset{1}{\color{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 338, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [241] $1^{\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}}\overset{1}{\color{red}{5}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}}\overset{1}{\color{red}{5}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

4.1088 NHC(-2)(-3) + $\textcolor{red}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$

$T = 9$ [0] $1^{\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}}\overset{1}{\color{red}{3}}\overset{1}{\color{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}\overset{1}{\color{red}{1}}}\overset{1}{\color{red}{3}}\overset{1}{\color{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.1089 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 9$ [0] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 19 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 21 & 4 \\ 11 & 19 & 12 & 0 & 21 & 73 & 15 \\ 3 & 4 & 2 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 2 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 19 & 9 & 4 \\ 2 & 4 & 0 & 0 & 12 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 12 & 0 & 73 & 33 & 15 \\ 5 & 9 & 6 & 0 & 33 & 13 & 7 \\ 3 & 4 & 2 & -3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -173$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 2 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 12 & 0 & 4 & 3 & 4 \\ 2 & 4 & 9 & 0 & 3 & 1 & 3 \\ 3 & 4 & 9 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 8 & 3 \\ 3 & 4 & 4 & 0 & 5 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 15 & 4 \\ 8 & 14 & 9 & 0 & 15 & 37 & 11 \\ 3 & 4 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 12 & 4 \\ 7 & 13 & 9 & 0 & 12 & 28 & 10 \\ 3 & 4 & 2 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -331, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 2 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -338, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 7 & 3 \\ 3 & 4 & 9 & 0 & 8 & 13 & 4 \\ 5 & 9 & 12 & 0 & 12 & 21 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 18 & 6 \\ 7 & 13 & 21 & 0 & 18 & 28 & 10 \\ 3 & 4 & 6 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 12 & 0 & 6 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 6 & 3 \\ 5 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 6 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 8 & 3 \\ 3 & 4 & 4 & 0 & 9 & 14 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 24 & 7 \\ 8 & 14 & 9 & 0 & 24 & 37 & 11 \\ 3 & 4 & 2 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 10 & 3 \\ 3 & 4 & 4 & 0 & 5 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 18 & 4 \\ 10 & 18 & 12 & 0 & 18 & 58 & 14 \\ 3 & 4 & 2 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 16 & 3 \\ 3 & 4 & 4 & 0 & 5 & 28 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 30 & 4 \\ 16 & 28 & 18 & 0 & 30 & 154 & 22 \\ 3 & 4 & 2 & -3 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 39 & 11 \\ 8 & 14 & 9 & 0 & 39 & 37 & 11 \\ 3 & 4 & 2 & -3 & 11 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 2 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1112} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 2 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1112} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 13 & 3 \\ 3 & 4 & 4 & 0 & 5 & 23 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 24 & 4 \\ 13 & 23 & 15 & 0 & 24 & 100 & 18 \\ 3 & 4 & 2 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 12 & 3 \\ 3 & 4 & 4 & 0 & 5 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 21 & 4 \\ 12 & 22 & 15 & 0 & 21 & 85 & 17 \\ 3 & 4 & 2 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 2 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 13 & 4 \\ 5 & 9 & 12 & 0 & 6 & 21 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 9 & 3 \\ 7 & 13 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 6 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 21 & 7 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 2 & -3 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1112} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -962, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 10 & 3 \\ 3 & 4 & 4 & 0 & 14 & 18 & 4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 48 & 11 \\ 10 & 18 & 12 & 0 & 48 & 58 & 14 \\ 3 & 4 & 2 & -3 & 11 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1112} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 4 & 0 & 9 & 28 & 4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 48 & 7 \\ 16 & 28 & 18 & 0 & 48 & 154 & 22 \\ 3 & 4 & 2 & -3 & 7 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 27 & 7 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 2 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 4 & 17 & 4 \\ 5 & 9 & 12 & 0 & 6 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 12 & 3 \\ 9 & 17 & 27 & 0 & 12 & 49 & 13 \\ 3 & 4 & 6 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 11 & 3 \\ 3 & 4 & 9 & 0 & 4 & 21 & 4 \\ 5 & 9 & 12 & 0 & 6 & 33 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 15 & 3 \\ 11 & 21 & 33 & 0 & 15 & 76 & 16 \\ 3 & 4 & 6 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 2 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 15 & 3 \\ 3 & 4 & 4 & 0 & 5 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 27 & 4 \\ 15 & 27 & 18 & 0 & 27 & 133 & 21 \\ 3 & 4 & 2 & -3 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1122} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 10 & 3 \\ 3 & 4 & 4 & 0 & 13 & 18 & 4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 42 & 10 \\ 10 & 18 & 12 & 0 & 42 & 58 & 14 \\ 3 & 4 & 2 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 2 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 4 & 0 & 9 & 18 & 4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 30 & 7 \\ 10 & 18 & 12 & 0 & 30 & 58 & 14 \\ 3 & 4 & 2 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1122} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 30 & 10 \\ 7 & 13 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 2 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 13 & 3 \\ 3 & 4 & 4 & 0 & 9 & 23 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 39 & 7 \\ 13 & 23 & 15 & 0 & 39 & 100 & 18 \\ 3 & 4 & 2 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 2 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 18 & 3 \\ 3 & 4 & 4 & 0 & 5 & 32 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 33 & 4 \\ 18 & 32 & 21 & 0 & 33 & 193 & 25 \\ 3 & 4 & 2 & -3 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 17 & 3 \\ 3 & 4 & 4 & 0 & 5 & 31 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 30 & 4 \\ 17 & 31 & 21 & 0 & 30 & 172 & 24 \\ 3 & 4 & 2 & -3 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 12 & 3 \\ 3 & 4 & 4 & 0 & 9 & 22 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 36 & 7 \\ 12 & 22 & 15 & 0 & 36 & 85 & 17 \\ 3 & 4 & 2 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1122} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 60 & 14 \\ 10 & 18 & 12 & 0 & 60 & 58 & 14 \\ 3 & 4 & 2 & -3 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 4 & 0 & 9 & 36 & 4 \\ 2 & 4 & 0 & 0 & 6 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 60 & 7 \\ 20 & 36 & 24 & 0 & 60 & 238 & 28 \\ 3 & 4 & 2 & -3 & 7 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 3 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 5 & 5 & 0 & 5 & 10 & 5 \\ 2 & 5 & 1 & 0 & 4 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 7 & 4 \\ 5 & 10 & 7 & 0 & 7 & 14 & 8 \\ 3 & 5 & 3 & -3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 0 & 1 & 1 & 2 \\ 2 & 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 3 & 0 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 3 21^{1311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 5 & 20 & 0 & 5 & 5 & 5 \\ 9 & 20 & 57 & 0 & 15 & 15 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 15 & 0 & 2 & 4 & 4 \\ 2 & 5 & 15 & 0 & 4 & 2 & 4 \\ 3 & 5 & 15 & -3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111} 3 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 3 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 4 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 7 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111} 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 20 & 5 \\ 9 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 0 & 63 & 15 \\ 5 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 3 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 5 & 5 & 0 & 5 & 15 & 5 \\ 2 & 5 & 1 & 0 & 4 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 11 & 4 \\ 7 & 15 & 11 & 0 & 11 & 32 & 12 \\ 3 & 5 & 3 & -3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1111} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 1 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 7 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & 4 & -3 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{12}321^{1113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 5 & 5 & 0 & 5 & 25 & 5 \\ 2 & 5 & 1 & 0 & 4 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 18 & 4 \\ 12 & 25 & 18 & 0 & 18 & 94 & 20 \\ 3 & 5 & 3 & -3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111}321^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 10 & 5 \\ 2 & 5 & 1 & 0 & 7 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 16 & 8 \\ 5 & 10 & 7 & 0 & 16 & 14 & 8 \\ 3 & 5 & 3 & -3 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{13}321^{1114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 5 & 5 & 0 & 5 & 35 & 5 \\ 2 & 5 & 1 & 0 & 4 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 25 & 4 \\ 17 & 35 & 25 & 0 & 25 & 188 & 28 \\ 3 & 5 & 3 & -3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{111}321^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 390, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 5 & 5 & 0 & 10 & 15 & 5 \\ 2 & 5 & 1 & 0 & 7 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 23 & 8 \\ 7 & 15 & 11 & 0 & 23 & 32 & 12 \\ 3 & 5 & 3 & -3 & 8 & 12 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111}321^{1123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 630, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 9 & 3 \\ 3 & 5 & 5 & 0 & 15 & 20 & 5 \\ 2 & 5 & 1 & 0 & 11 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 45 & 12 \\ 9 & 20 & 15 & 0 & 45 & 58 & 16 \\ 3 & 5 & 3 & -3 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{12}321^{1114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 5 & 5 & 0 & 5 & 30 & 5 \\ 2 & 5 & 1 & 0 & 4 & 22 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 22 & 4 \\ 14 & 30 & 22 & 0 & 22 & 134 & 24 \\ 3 & 5 & 3 & -3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{111}321^{1122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 720, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 15 & 5 \\ 2 & 5 & 1 & 0 & 11 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 34 & 12 \\ 7 & 15 & 11 & 0 & 34 & 32 & 12 \\ 3 & 5 & 3 & -3 & 12 & 12 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{1111}321^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & 4 & -3 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{111}3^{11}2^{1133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 840$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.00 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 20 & 5 \\ 2 & 5 & 1 & 0 & 15 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 15 & 0 & 58 & 60 & 16 \\ 9 & 20 & 15 & 0 & 60 & 58 & 16 \\ 3 & 5 & 3 & -3 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

4.1090 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{113}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 3 & 4 \\ 2 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 392$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 462$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 512$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 3 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 5 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 6 & -3 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 622$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 2 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 1152$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 15 & 5 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 5 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1091 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 3 & 4 \\ 5 & 9 & 12 & 0 & 12 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 4 & 6 \\ 2 & 3 & 5 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 3 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 4 & 6 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 19 & 3 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 12 & 0 & 73 & 11 & 15 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 3 & 4 & 2 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 3 & 4 \\ 4 & 8 & 9 & 0 & 6 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 2 & 3 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & 5 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[7] \quad & 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 6 & 0 & 13 & 5 & 7 \\ 2 & 4 & 2 & 0 & 5 & 0 & 2 \\ 3 & 5 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 3 & 4 \\ 4 & 8 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 5 & 7 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 5 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 3 & 4 \\ 5 & 9 & 12 & 0 & 6 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 2 & 3 \\ 2 & 3 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 6 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 3 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 8 & 11 \\ 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 3 & 4 & 2 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 3 & 4 \\ 7 & 13 & 27 & 0 & 9 & 7 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 1 & 2 & 3 \\ 2 & 3 & 7 & 0 & 2 & 0 & 2 \\ 3 & 4 & 9 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.7 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 3 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 7 & 10 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 2 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[15] $1^{112}3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 2 & 3 \\ 3 & 4 & 4 & 0 & 28 & 3 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 18 & 0 & 154 & 16 & 22 \\ 2 & 3 & 2 & 0 & 16 & 0 & 2 \\ 3 & 4 & 2 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 3 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 9 & 13 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 3 & 4 & 2 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{114}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 3 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 11 & 16 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 3 & 4 & 2 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 18 & 3 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 10 & 14 \\ 2 & 3 & 2 & 0 & 10 & 0 & 2 \\ 3 & 4 & 2 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 23 & 3 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 13 & 18 \\ 2 & 3 & 2 & 0 & 13 & 0 & 2 \\ 3 & 4 & 2 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{113}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 22 & 3 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 12 & 17 \\ 2 & 3 & 2 & 0 & 12 & 0 & 2 \\ 3 & 4 & 2 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{113}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 4 & 0 & 27 & 3 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 15 & 21 \\ 2 & 3 & 2 & 0 & 15 & 0 & 2 \\ 3 & 4 & 2 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{114}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 26 & 3 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 14 & 20 \\ 2 & 3 & 2 & 0 & 14 & 0 & 2 \\ 3 & 4 & 2 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 2 & 0 & 3 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 3 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 5 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 5 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & -3 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 5 & 15 & 7 \\ 2 & 3 & 2 & 0 & 5 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 15 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1093 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 8 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 8 & 5 & 0 & 9 & 12 & 6 \\ 3 & 4 & 2 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 11 & 3 \\ 10 & 17 & 11 & 0 & 11 & 55 & 13 \\ 3 & 4 & 2 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 7 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 4 & 7 & 5 & 0 & 6 & 7 & 5 \\ 3 & 4 & 2 & -3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1123}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 5 & 0 & 4 & 11 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 6 & 11 & 9 & 0 & 8 & 20 & 8 \\ 3 & 4 & 3 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 6 & 0 & 5 & 7 & 5 \\ 3 & 4 & 3 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 14 & 7 \\ 5 & 8 & 5 & 0 & 14 & 12 & 6 \\ 3 & 4 & 2 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 20 & 4 \\ 2 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 18 & 4 \\ 11 & 20 & 14 & 0 & 18 & 69 & 15 \\ 3 & 4 & 2 & -3 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 12 & 4 \\ 2 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 12 & 4 \\ 7 & 12 & 8 & 0 & 12 & 25 & 9 \\ 3 & 4 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 5 & 0 & 5 & 7 & 5 \\ 3 & 4 & 2 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 4 & 7 & 4 \\ 5 & 9 & 12 & 0 & 6 & 11 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 5 & 3 \\ 4 & 7 & 11 & 0 & 5 & 7 & 5 \\ 3 & 4 & 6 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 12 & 8 & 0 & 8 & 25 & 9 \\ 3 & 4 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{112}3^{\overline{1}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 9 & 3 \\ 3 & 4 & 4 & 0 & 5 & 16 & 4 \\ 2 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 15 & 4 \\ 9 & 16 & 11 & 0 & 15 & 44 & 12 \\ 3 & 4 & 2 & -3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 4 & 3 \\ 3 & 4 & 4 & 0 & 9 & 7 & 4 \\ 2 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 11 & 7 \\ 4 & 7 & 5 & 0 & 11 & 7 & 5 \\ 3 & 4 & 2 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{111}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 12 & 4 \\ 2 & 4 & 0 & 0 & 6 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 20 & 7 \\ 7 & 12 & 8 & 0 & 20 & 25 & 9 \\ 3 & 4 & 2 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

4.1094 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 5 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 3 & 4 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^{\overline{1}}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 2 & 3 \\ 4 & 9 & 4 & 0 & 28 & 4 & 5 \\ 2 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 28 & 12 & 0 & 73 & 12 & 15 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 5 & 2 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 2 & 3 \\ 4 & 9 & 4 & 0 & 20 & 4 & 5 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 20 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 5 & 2 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -182, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.9$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 5 & 0 & 13 & 8 & 4 \\ 3 & 5 & 3 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 18 & 10 \\ 4 & 8 & 6 & 0 & 18 & 9 & 5 \\ 3 & 4 & 3 & -3 & 10 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 2 & -3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 5 & 0 & 17 & 8 & 4 \\ 3 & 5 & 3 & 0 & 15 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 24 & 13 \\ 4 & 8 & 6 & 0 & 24 & 9 & 5 \\ 3 & 4 & 3 & -3 & 13 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -315, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 2 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 2 & 3 \\ 4 & 9 & 4 & 0 & 32 & 4 & 5 \\ 2 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 32 & 15 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 5 & 2 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 12 & 6 \\ 3 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -399, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.53 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^1 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -426, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 2 & 3 \\ 4 & 9 & 4 & 0 & 44 & 4 & 5 \\ 2 & 4 & 0 & 0 & 21 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 44 & 21 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 5 & 2 & -3 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -512, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 2 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 2 & 3 \\ 4 & 9 & 4 & 0 & 40 & 4 & 5 \\ 2 & 4 & 0 & 0 & 18 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 40 & 18 & 0 & 154 & 18 & 22 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 5 & 2 & -3 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -527, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 18 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -621, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{121} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -707, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{131} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -713, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 14, \quad \Delta = -95, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 2 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

4.1095 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{133} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 35, \quad T \in [14, 35]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 3 & 4 \\ 6 & 12 & 24 & 0 & 6 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 8 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{113} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{123} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 98, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.22 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 118, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{113} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 128, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 35, \quad T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 208, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 35, \quad T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{123} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 278, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 35, \quad T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 7 & 13 & 27 & 0 & 7 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 9 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 328, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 9 & 17 & 48 & 0 & 9 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 12 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{13}3^{121}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3^{121}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -52, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 2 & 6 & 4 & 0 & 4 \\ 4 & 6 & 7 & 5 & 0 & 5 \\ 2 & 4 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}3^{121}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 8 & 6 & 5 & 0 & 6 \\ 2 & 6 & 1 & 4 & 0 & 3 \\ 2 & 5 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}3^{11}1^{11}3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 \\ 2 & 3 & 3 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{12}3^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 8 & 6 & 20 & 0 & 6 \\ 2 & 6 & 1 & 13 & 0 & 3 \\ 8 & 20 & 13 & 43 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11}3^{11}1^{11}3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -149, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 2 & 2 & 0 & 2 \\ 2 & 4 & 2 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^1 3 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 2 & 6 & 1 & 5 & 0 & 3 \\ 4 & 10 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^1 3^1 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^1 1^1 3 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 8 & 6 & 15 & 0 & 6 \\ 2 & 6 & 1 & 9 & 0 & 3 \\ 6 & 15 & 9 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^1 1^1 3^1 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 3 & 9 & 6 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 \\ 2 & 4 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [20] \quad 1^1 1^1 4 1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^1 1^1 3^1 1^1 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 4 & 3 & 0 & 0 & 4 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [22] \quad 1^1 1^1 5 1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 8 & 5 \\ 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 10 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.1096 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1097 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 3 & 8 & 4 \\ 5 & 9 & 12 & 0 & 5 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 4 & 8 & 12 & 0 & 4 & 10 & 6 \\ 3 & 4 & 6 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 3 & 8 & 4 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 4 & 8 & 6 & 0 & 4 & 10 & 6 \\ 3 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 3 & 19 & 4 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 11 & 19 & 12 & 0 & 11 & 73 & 15 \\ 3 & 4 & 2 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 5 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 & 4 \\ 4 & 8 & 9 & 0 & 4 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 1 & 3 \\ 3 & 4 & 5 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 5 & 2 \\ 5 & 12 & 6 & 0 & 5 & 13 & 7 \\ 3 & 5 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 8 & 0 & 3 & 9 & 4 \\ 4 & 8 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 9 & 12 & 0 & 5 & 13 & 7 \\ 3 & 4 & 5 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 & 4 \\ 5 & 9 & 12 & 0 & 5 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 1 & 3 \\ 3 & 4 & 6 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -386, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 3 & 14 & 4 \\ 2 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 8 & 14 & 9 & 0 & 8 & 37 & 11 \\ 3 & 4 & 2 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -416, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 & 4 \\ 7 & 13 & 27 & 0 & 7 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 2 & 2 \\ 2 & 4 & 9 & 0 & 2 & 1 & 3 \\ 3 & 4 & 9 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -626, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 3 & 13 & 4 \\ 2 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 13 & 9 & 0 & 7 & 28 & 10 \\ 3 & 4 & 2 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -776, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 3 & 17 & 4 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 9 & 17 & 12 & 0 & 9 & 49 & 13 \\ 3 & 4 & 2 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 16 & 3 \\ 3 & 4 & 4 & 0 & 3 & 28 & 4 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 16 & 2 \\ 16 & 28 & 18 & 0 & 16 & 154 & 22 \\ 3 & 4 & 2 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 3 & 21 & 4 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 11 & 21 & 15 & 0 & 11 & 76 & 16 \\ 3 & 4 & 2 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 3 & 18 & 4 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 10 & 2 \\ 10 & 18 & 12 & 0 & 10 & 58 & 14 \\ 3 & 4 & 2 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 3 & 23 & 4 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 13 & 23 & 15 & 0 & 13 & 100 & 18 \\ 3 & 4 & 2 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 3 & 22 & 4 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 12 & 22 & 15 & 0 & 12 & 85 & 17 \\ 3 & 4 & 2 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{113}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 3 & 27 & 4 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 15 & 2 \\ 15 & 27 & 18 & 0 & 15 & 133 & 21 \\ 3 & 4 & 2 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 3 & 26 & 4 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 14 & 26 & 18 & 0 & 14 & 118 & 20 \\ 3 & 4 & 2 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{113}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 3 & 32 & 4 \\ 2 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 18 & 2 \\ 18 & 32 & 21 & 0 & 18 & 193 & 25 \\ 3 & 4 & 2 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 3 & 31 & 4 \\ 2 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 17 & 2 \\ 17 & 31 & 21 & 0 & 17 & 172 & 24 \\ 3 & 4 & 2 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 3 & 36 & 4 \\ 2 & 4 & 0 & 0 & 2 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 20 & 2 \\ 20 & 36 & 24 & 0 & 20 & 238 & 28 \\ 3 & 4 & 2 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11}4^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111}1^{13}1^{11}4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 5 & 0 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111}1^{13}1^{11}4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 8 & 0 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

4.1098 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{113}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 5 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 6 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 2 & 4 & 3 & 0 & 2 & 3 & 1 \\ 3 & 4 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 5 & 9 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 & 4 & 4 \\ 5 & 9 & 12 & 0 & 5 & 6 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 6 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & 6 & -3 & 2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 5 & 15 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1099 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{12}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 8 & 0 & 3 & 5 & 4 \\ 4 & 8 & 9 & 0 & 4 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & 5 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 4 & 5 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -544$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 12 & 6 \\ 3 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.1100 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + su2n3mix

$T = 8$ [0] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 12 & 0 & 29 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 29 & 0 & 64 & 14 \\ 3 & 4 & 6 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 102$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 20 & 8 \\ 3 & 4 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 107$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 113$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 16 & 0 & 25 & 9 \\ 3 & 4 & 5 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 132$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 16 & 8 & 0 & 25 & 9 \\ 3 & 5 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 142$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 \\ 3 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 27 & 17 & 0 & 151 & 21 \\ 3 & 4 & 2 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{121}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 9 & 0 & 10 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 5 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 15 & 4 \\ 5 & 9 & 12 & 0 & 23 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 23 & 0 & 39 & 11 \\ 3 & 4 & 6 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 \\ 5 & 9 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 8 \\ 3 & 4 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 5 & 9 & 12 & 0 & 11 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 6 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 22 & 14 & 0 & 97 & 17 \\ 3 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 27 & 0 & 16 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 9 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{114}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 11 & 0 & 39 & 11 \\ 3 & 4 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{114}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 14 & 0 & 64 & 14 \\ 3 & 4 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{1123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1123}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 14 & 0 & 80 & 16 \\ 3 & 4 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{1123}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 31 & 20 & 0 & 188 & 24 \\ 3 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{1123}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 923, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 26 & 17 & 0 & 128 & 20 \\ 3 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 100 & 18 \\ 3 & 4 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 17 & 0 & 111 & 19 \\ 3 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 20 & 0 & 165 & 23 \\ 3 & 4 & 2 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{113}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1667, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 2 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1708, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 19 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 2 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 2 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [32] $1^{111}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 54 & 0 & 217 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11}31^{113}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 8 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 4 & 0 & 9 & 5 \\ 2 & 4 & 2 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{112}31^{13}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 7 & 3 \\ 2 & 4 & 3 & 3 & 0 & 8 & 3 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 8 & 7 & 0 & 15 & 7 \\ 2 & 3 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^{13}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{11}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{112}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 44 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[38] 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[39] 1^{112} 3^{11} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 84 & 0 & 520 & 42 \\ 5 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[40] 1^{111} 31^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[41] 1^{111} 31^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 3 \\ 2 & 4 & 3 & 3 & 0 & 5 & 3 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 \\ 2 & 3 & 3 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[42] 1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[43] 1^{11} 31^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 2 & 6 & 3 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[44] 1^{11} 3^{11} 1^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 14 & 2 \\ 1 & 3 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 14 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[45] 1^{11} 3^{11} 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[46] 1^{112} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 24 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^{11}1^{13}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 11 & 2 \\ 1 & 3 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 11 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{11}3^{11}1^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 1 & 3 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 8 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{11}3^{11}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 16 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11}3^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{11}3^{11}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 17 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{11}3^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{11}3^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{11}3^{11}3^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 5 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 2 \\ 3 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 2 & 6 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{11}3^{11}3^{11}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 45 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{11}3^{11}3^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -430, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 11 & 4 \\ 1 & 4 & 0 & 3 & 0 & 7 & 2 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{\overline{112}3^{\overline{1}}21^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{\overline{112}3^{\overline{1}}21^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -584, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 35 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{\overline{112}3^{\overline{1}}21^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -608, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{\overline{112}3^{\overline{1}}21^{\overline{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -608, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.02 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{\overline{111}31^{\overline{12}3^{\overline{1}}1^{\overline{1}}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -679, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 17 & 4 \\ 1 & 4 & 0 & 3 & 0 & 10 & 2 \\ 1 & 3 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 10 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{\overline{112}3^{\overline{1}}21^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -768, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 36 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{\overline{111}31^{\overline{13}3^{\overline{1}}1^{\overline{1}}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -886, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 23 & 4 \\ 1 & 4 & 0 & 3 & 0 & 13 & 2 \\ 1 & 3 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 23 & 13 & 13 & 0 & 88 & 16 \\ 2 & 4 & 2 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{\overline{111}31^{\overline{14}3^{\overline{1}}1^{\overline{1}}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -1051, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 29 & 4 \\ 1 & 4 & 0 & 3 & 0 & 16 & 2 \\ 1 & 3 & 3 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 29 & 16 & 16 & 0 & 140 & 20 \\ 2 & 4 & 2 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [65] \ 1^{\overline{111}4^{\overline{13}1^{\overline{11}}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 24, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{\overline{111}4^{\overline{13}1^{\overline{1}}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 56, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{\overline{1^1 3^1 1^1}} 4^{\overline{1^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{\overline{1^1 3^1 1^1}} 4^{\overline{1^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{\overline{1^{11} 1^1}} 4^{\overline{1^{11} 1^1}} 1^{\overline{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{\overline{1^1 3^1 1^1}} 4^{\overline{1^{11}}} 1^{\overline{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{\overline{1^{11} 1^1 3^1 1^1}} 4^{\overline{1^{11}}} 1^{\overline{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{\overline{1^{11} 1^1}} 4^{\overline{1^{11} 1^1}} 1^{\overline{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [73] $1^{\overline{1^{11} 5^1 1^1 1^1}} 1^{\overline{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 9 & 1 & 14 & 5 \\ 9 & 13 & 0 & 22 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 33 & 11 \\ 5 & 7 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{\overline{1^{12} 5^1 1^1 1^1}} 1^{\overline{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 32 & 0 & 70 & 16 \\ 5 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.1101 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + **su2n3mix** + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 4 & 4 & 0 & 8 & 5 & 4 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 9 & 6 \\ 3 & 5 & 3 & 0 & 9 & 4 & 4 \\ 3 & 4 & 2 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 & 4 \\ 2 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 11 & 0 & 55 & 11 & 13 \\ 2 & 4 & 3 & 0 & 11 & 1 & 3 \\ 3 & 4 & 2 & -3 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 0 & 7 & 5 & 4 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 6 & 5 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -3 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{121}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 9 & 0 & 20 & 8 & 8 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 3 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 3 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 9 & 4 \\ 2 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 14 & 6 \\ 5 & 9 & 6 & 0 & 14 & 13 & 7 \\ 3 & 4 & 2 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 & 3 \\ 3 & 4 & 4 & 0 & 12 & 5 & 4 \\ 2 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 12 & 9 \\ 3 & 5 & 3 & 0 & 12 & 4 & 4 \\ 3 & 4 & 2 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 & 3 \\ 3 & 4 & 4 & 0 & 20 & 5 & 4 \\ 2 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 18 & 15 \\ 3 & 5 & 3 & 0 & 18 & 4 & 4 \\ 3 & 4 & 2 & -3 & 15 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{1111}1}3^{\overline{1}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{1111}1}3^{\overline{1}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 & 4 \\ 5 & 9 & 12 & 0 & 11 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 5 & 5 \\ 2 & 4 & 6 & 0 & 5 & 1 & 3 \\ 3 & 4 & 6 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1111}1}3^{\overline{1}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 2 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{1111}1}3^{\overline{1}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 & 3 \\ 3 & 4 & 4 & 0 & 16 & 5 & 4 \\ 2 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 15 & 12 \\ 3 & 5 & 3 & 0 & 15 & 4 & 4 \\ 3 & 4 & 2 & -3 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{1111}1}3^{\overline{1}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 9 & 4 \\ 2 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 11 & 5 \\ 5 & 9 & 6 & 0 & 11 & 13 & 7 \\ 3 & 4 & 2 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{1111}1}3^{\overline{1}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 4 & 0 & 12 & 9 & 4 \\ 2 & 4 & 0 & 0 & 8 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 20 & 9 \\ 5 & 9 & 6 & 0 & 20 & 13 & 7 \\ 3 & 4 & 2 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1102 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_3$

$T = 9$ [0] $1^{\overline{1111}1}3^{\overline{1}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 & 4 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{1111}1}3^{\overline{1}1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

13.2

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 5 & 0 & 12 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 5 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 10 & 5 \\ 4 & 8 & 6 & 0 & 10 & 9 & 5 \\ 3 & 4 & 3 & -3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 9 & 0 & 20 & 8 & 8 \\ 2 & 4 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 3 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1103 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [8] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 10 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 11 & 0 & 31 & 11 \\ 3 & 4 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 5 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 3 & 0 & 0 \\ 2 & 3 & 3 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 3 & 0 & 0 \\ 2 & 4 & 3 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{12} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 3 & 0 & 5 & 2 \\ 1 & 1 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{11} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{11} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 13 & 7 \\ 3 & 5 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[25] 1^{11} 3^{11} 2^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 31 & 11 \\ 3 & 5 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[26] 1^{11} 3^{11} 2^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 57 & 15 \\ 3 & 5 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[27] 1^{11} 3^{11} 2^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 4 & 4 \\ 2 & 2 & 2 & 4 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [28] 1^{11} 4^{11} 131^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] 1^{11} 3^{11} 4^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[30] 1^{11} 3^{11} 4^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1104 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] 1^{11} 3^{11} 3^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 13 & 7 \\ 3 & 5 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3^{11} 3^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 3 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 11 & 3 \\ 4 & 9 & 4 & 0 & 4 & 28 & 5 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 11 & 28 & 12 & 0 & 12 & 73 & 15 \\ 3 & 5 & 2 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 8 & 3 \\ 4 & 9 & 4 & 0 & 4 & 20 & 5 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 20 & 9 & 0 & 9 & 37 & 11 \\ 3 & 5 & 2 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -182, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.9$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 5 & 0 & 8 & 13 & 4 \\ 3 & 5 & 3 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 18 & 5 \\ 7 & 13 & 12 & 0 & 18 & 28 & 10 \\ 3 & 4 & 3 & -3 & 5 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 36 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 2 & -3 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 5 & 0 & 8 & 17 & 4 \\ 3 & 5 & 3 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 24 & 5 \\ 9 & 17 & 15 & 0 & 24 & 49 & 13 \\ 3 & 4 & 3 & -3 & 5 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -315, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 2 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

14.0

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 13 & 3 \\ 4 & 9 & 4 & 0 & 4 & 32 & 5 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 32 & 15 & 0 & 15 & 100 & 18 \\ 3 & 5 & 2 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 9 & 6 \\ 3 & 5 & 3 & 0 & 9 & 4 & 4 \\ 3 & 4 & 2 & -3 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 18 & 3 \\ 4 & 9 & 4 & 0 & 4 & 44 & 5 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 44 & 21 & 0 & 21 & 193 & 25 \\ 3 & 5 & 2 & -3 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 2 & -3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 16 & 3 \\ 4 & 9 & 4 & 0 & 4 & 40 & 5 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 16 & 40 & 18 & 0 & 18 & 154 & 22 \\ 3 & 5 & 2 & -3 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{114} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 18 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{113}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & -3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1105 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{123}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 8 & 3 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 4 & 5 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 5 & 6 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1106 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$

$T = 9 \quad [0] \quad 1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 8 & 5 & 0 & 5 & 12 & 6 \\ 3 & 4 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 12 & 5 & 0 & 5 & 12 & 6 \\ 3 & 5 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{121}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 5 & 0 & 8 & 7 & 4 \\ 3 & 5 & 3 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 10 & 5 \\ 4 & 7 & 6 & 0 & 10 & 7 & 5 \\ 3 & 4 & 3 & -3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{112}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 5 & 0 & 4 & 11 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 6 & 11 & 9 & 0 & 8 & 20 & 8 \\ 3 & 4 & 3 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 4 & 7 & 6 & 0 & 5 & 7 & 5 \\ 3 & 4 & 3 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

4.1107 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$

$T = 9 \quad [0] \quad 1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 2 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 12 & 6 \\ 3 & 5 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1108 NHC(-2)(-3) + $\mathfrak{su}_3$ + NHC(-2)(-3)

$$T = 10 \quad [1] 21^{11} 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[0] 1^{11} 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 4 & 6 & 7 & 4 & 0 & 5 \\ 2 & 4 & 4 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 5 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] 21^{11} 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 3 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] 21^{11} 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 3 \\ 4 & 4 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [5] 1^{11} 6 13 15 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 2 & 3 & 1 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1109 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} \quad 5 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 1 & 2 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \quad 5 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 3 & 0 & 10 & 4 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 2 & 3 & 2 & 1 & 1 & 8 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 6 & 10 & 6 & 8 & 0 & 21 & 9 \\ 3 & 4 & 2 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} \quad 5 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -23, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 3 & 0 & 17 & 4 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 2 & 3 & 2 & 1 & 1 & 13 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 10 & 17 & 10 & 13 & 0 & 61 & 15 \\ 3 & 4 & 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \quad 5 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 2 & 3 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 4 & 7 & 4 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \quad 5 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 2 & 3 & 2 & 1 & 1 & 10 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 8 & 14 & 8 & 10 & 0 & 38 & 12 \\ 3 & 4 & 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \quad 5 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 2 & 3 & 2 & 1 & 1 & 15 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 12 & 21 & 12 & 15 & 0 & 88 & 18 \\ 3 & 4 & 2 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

4.1110 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 7 \quad [0] \quad 1^3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 71, T \in [15, 71]$$

$$\begin{pmatrix} 5 & 11 & 18 & 5 \\ 11 & 20 & 34 & 8 \\ 18 & 34 & 57 & 13 \\ 5 & 8 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 71, T \in [14, 71]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 12 & 6 \\ 6 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

- [2] $1^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14, T_{\max} = 71, T \in [14, 71]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 12 & 6 \\ 6 & 9 & 6 & 0 \end{pmatrix}_{b_0}$
- [3] $1^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 5 & 11 & 19 & 5 \\ 11 & 20 & 34 & 8 \\ 19 & 34 & 57 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [4] $1^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 10 & 15 & 25 & 6 \\ 15 & 20 & 34 & 8 \\ 25 & 34 & 57 & 13 \\ 6 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [5] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 8 & 5 \\ 12 & 25 & 16 & 9 \\ 8 & 16 & 9 & 5 \\ 5 & 9 & 5 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 10 & 16 & 23 & 6 \\ 16 & 25 & 36 & 9 \\ 23 & 36 & 48 & 12 \\ 6 & 9 & 12 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [8] $1^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 71, T \in [14, 71]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & -1 \end{pmatrix}_{b_0}$
- [9] $1^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
 $\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 17 & 8 \\ 10 & 17 & 12 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 5 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{13}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 71, T \in [13, 71]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 71, T \in [11, 71]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{15}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 71, T \in [15, 71]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 72 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{14}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 71, T \in [14, 71]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 17 & 8 \\ 9 & 17 & 12 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 17 & 8 \\ 12 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 10 & 15 & 1 & 6 \\ 15 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 36 & 9 \\ 18 & 36 & 48 & 12 \\ 5 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 17 & 8 \\ 13 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

- [25] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 71$, $T \in [13, 71]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 0 \end{pmatrix}_{b_0}$
- [26] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [27] $21^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -65$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 13 & 17 & 15 & 7 \\ 17 & 20 & 17 & 8 \\ 15 & 17 & 12 & 6 \\ 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [28] $1^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -67$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 71$, $T \in [13, 71]$
 $\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 12 & 6 \\ 5 & 9 & 6 & 1 \end{pmatrix}_{b_0}$
- [29] $1^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -71$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 36 & 9 \\ 17 & 36 & 48 & 12 \\ 5 & 9 & 12 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{11} 1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [31] $21^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -77$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 13 & 20 & 12 & 7 \\ 20 & 25 & 16 & 9 \\ 12 & 16 & 9 & 5 \\ 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{11} 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -83$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [33] $21^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -86$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 20 & 17 & 8 \\ 12 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{12} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [35] $1^{11} 1^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 1 \end{pmatrix}_{b_0}$
- [36] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[48] $21^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -275$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$

$$\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 12 & 6 \\ 7 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{12}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -284$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.04 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -339$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.6$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{14}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -348$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.6$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{13}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -396$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 71$, $T \in [11, 71]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{14}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -508$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.6$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 71$, $T \in [11, 71]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [54] $1^{11}41^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 3$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 18$, $T_{\max} = 23$, $T \in [18, 23]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 39 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & -4 \end{pmatrix}_{b_0}$$

[55] $131^131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 15$, $T_{\max} = 20$, $T \in [15, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[56] $21^{11}41^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 23$, $T \in [12, 23]$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 4 & 4 \\ 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{11}41^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 23$, $T \in [12, 23]$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 4 & 4 \\ 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{11}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 7 & 10 & 13 & 7 & 7 \\ 2 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 21^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 1 & 1 & 3 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 131^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 21^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 5 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 1^{\mathbf{1}} 3 1^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 17, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 21^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 17, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 1^{\mathbf{1}} 3 1^{\mathbf{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 17, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 & 3 \\ 1 & 4 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 21^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 17, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 4 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 21^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 3 & 3 \\ 2 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 21^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 21^{\mathbf{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{\mathbf{11}} 1^{\mathbf{1}} 3 21^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 2 & 1 & 3 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[70] $1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[71] $1^1 3 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 0 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 0 & 11 & 5 \\ 2 & 3 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[72] $1^1 3 3 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 9 & 0 & 24 & 9 & 8 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[75] $1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 4 & 3 \\ 2 & 5 & 4 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[76] $1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[77] $1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[78] $1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[79] $2^1 2^1 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[80] $1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[82] \quad 21^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 5 & 10 & 5 & 5 \\ 9 & 10 & 13 & 7 & 7 \\ 3 & 5 & 7 & 0 & 2 \\ 4 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 12 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[89] \quad 2 2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 0 & 3 & 1 & 2 \\ 2 & 1 & 3 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{11} 141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 4 & 4 \\ 5 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 6 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{11}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[92] \ 21^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 6 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{11}321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 4 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{11}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 3 \\ 2 & 2 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{11}1131^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 6 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{11}321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 3 & 3 \\ 2 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{11}113131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

- [105] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 2 & -2 & -1 \end{pmatrix}_{b_0}$$
- [104] $1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 8 & 7 & 4 \\ 8 & 13 & 16 & 12 & 7 \\ 8 & 16 & 12 & 12 & 6 \\ 7 & 12 & 12 & 9 & 5 \\ 4 & 7 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$
- [106] $1^{12}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 3 & 6 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [108] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [107] $1^{11}31^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [109] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 7 & 8 & 1 & 4 \\ 7 & 10 & 13 & 0 & 6 \\ 8 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 6 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [110] $1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$
- [111] $1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [112] $31^{11}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 5 & 3 & 4 & 3 \\ 5 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 4 & 5 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$
- [113] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 1 & 4 & 3 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{11} 2^2 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 4 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{11} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 5 & 25 & 0 & 5 \\ 11 & 25 & 91 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{12} 31^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 9 & 6 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{11} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 1 & 4 & 3 \\ 2 & 5 & 4 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{11} 31^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{11} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 7 & 15 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{11} 31^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{12} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 57$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 37 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 57$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{11} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

[125] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 4 & 4 \\ 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[126] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{11}321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 6 & 10 & 13 & 7 & 7 \\ 2 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{11}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 5 & 20 & 0 & 5 \\ 9 & 20 & 57 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{11}321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 5 & 10 & 13 & 7 & 7 \\ 2 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{11}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 5 & 15 & 0 & 5 \\ 7 & 15 & 31 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{12}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[133] $131^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 3 \\ 5 & 13 & 16 & 12 & 7 \\ 5 & 16 & 12 & 12 & 6 \\ 4 & 12 & 12 & 9 & 5 \\ 3 & 7 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{11}321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 5 & 10 & 13 & 7 & 7 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{14}31^{13}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 45 & 12 & 11 \\ 2 & 3 & 0 & 12 & 2 & 2 \\ 2 & 3 & -1 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{13}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 13 & 0 & 60 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{13}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 3 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{12}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 9 & 6 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 3 & 5 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{13}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 24 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{13}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 24 & 9 & 8 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{12}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{13}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{13}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 4 & 7 & 6 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[146] $21^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[145] $21^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

- [147] $1^{13}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 125$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [148] $1^{12}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 10 & 0 & 37 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$
- [149] $1^{12}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 132$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 4 & 6 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [150] $1^{12}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 5 & 7 & 11 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$
- [151] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [152] $2^2 2^2 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 1 & 4 & 3 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$
- [153] $1^{13}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 182$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 13 & 0 & 60 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [154] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 188$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [155] $1^{13}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 5 & 8 & 9 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [156] $2^2 2^2 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 211$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 5 & 5 \\ 6 & 10 & 13 & 7 & 7 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$
- [157] $1^{12}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 212$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 12 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[158] $2 \overset{1}{2} 31^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 249, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 \\ 2 & 3 & 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[159] $1 \overset{1}{1} 321^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 6 & 9 & 12 & 60 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[160] $2 \overset{1}{2} 2 31^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 1 & 2 \\ 2 & 1 & 4 & 6 & 1 & 3 \\ 2 & 4 & 0 & 6 & 1 & 2 \\ 4 & 6 & 6 & 9 & 2 & 5 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [161] $21^{\overset{1}{1}} 41^{\overset{1}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 7 & 2 & 5 \\ 7 & 8 & 1 & 6 \\ 2 & 1 & -2 & 0 \\ 5 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[162] $1 \overset{1}{1} 4 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[163] $1 \overset{1}{1} 4 131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[164] $1 \overset{1}{1} 4 131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[165] $1 \overset{1}{1} 4 131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & -2 & 0 \\ 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[166] $1 \overset{1}{1} 41^{\overset{1}{1}} 231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 5 & 2 & 5 \\ 1 & 2 & -2 & 0 \\ 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[167] $1 \overset{1}{1} 4 131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[168] $1 \overset{1}{1} 4 131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{\overline{11}} 4^{\overline{11}31^{\overline{1}}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 8 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 4 & 2 & 5 \\ 4 & 1 & 1 & 3 \\ 2 & 1 & -2 & 0 \\ 5 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{\overline{1}41^{\overline{1}231^{\overline{11}}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 2 & 6 & 2 & 4 \\ 6 & 8 & 1 & 6 \\ 2 & 1 & -2 & 0 \\ 4 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{\overline{1}31^{\overline{1}}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 14 & 10 & 8 \\ 10 & 5 & 5 \\ 8 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{\overline{1}4131^{\overline{11}}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{\overline{1}31^{\overline{1}}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 13 & 9 & 1 & 7 \\ 9 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 7 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{\overline{11}4131^{\overline{1}}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{\overline{11}4131^{\overline{1}}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 3 & 3 & 1 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{\overline{11}131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 2 & 3 & 0 & 2 \\ 4 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{\overline{11}131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 4 & 5 & 5 \\ 4 & 1 & 3 & 3 \\ 5 & 3 & 1 & 3 \\ 5 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{\overline{1}41^1231^{\overline{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 4 & 1 & 0 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{\overline{11}131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 7 & 1 & 5 \\ 7 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{\overline{11}131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{\overline{11}14131^{\overline{12}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 3 & 2 & 2 & 3 & 1 & 3 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{\overline{11}131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 4 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{\overline{1}41^1231^{\overline{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{\overline{11}131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[190] $1^{\overline{11}131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 4 & 1 & 2 \\ 8 & 4 & 17 & 1 & 7 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{\overline{11}131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{\overline{11}} 4^{\overline{11}} 131^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 6 & 1 & 3 \\ 2 & 0 & 3 & 1 & 2 \\ 6 & 3 & 7 & 1 & 5 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 3 & 2 & 3 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[195] \ 31^{\overline{11}} 31^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -100, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 3 & 2 & 3 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 4 & 2 & 4 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.15 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 5 & 3 & 2 & 7 & 2 & 5 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[198] \ 2^{\overline{14}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -186, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 1 & 2 \\ 3 & 4 & 4 & 3 & 2 & 4 \\ 2 & 4 & 0 & 3 & 1 & 2 \\ 2 & 3 & 3 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 4 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{\overline{14}} 41^{\overline{11}} 231^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -205, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 2 \\ 3 & 4 & 1 & 6 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 2 & 3 \\ 1 & 2 & 0 & 2 & -2 & 0 \\ 2 & 4 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{\overline{14}} 41^{\overline{11}} 231^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -217, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.19 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{\overline{14}} 41^{\overline{11}} 231^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -250, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 1 & 2 \\ 3 & 4 & 1 & 6 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 6 & 1 & 4 & 2 & 4 \\ 1 & 2 & 0 & 2 & -2 & 0 \\ 2 & 4 & 0 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [202] \quad 21^{\underline{1}^1} 5^{\underline{1}^1} 131^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 3 & 5 & 5 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[203] \quad 1^{\underline{1}^1} 1^{\underline{1}^1} 3221^{\underline{1}^1} 51^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 10 & 21 & 0 & 6 \\ 9 & 21 & 42 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[204] \quad 1^{\underline{1}^1} 4^{\underline{1}^1} 141^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[205] \quad 1^{\underline{1}^1} 5^{\underline{1}^1} 1231^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[206] \quad 1^{\underline{1}^1} 4^{\underline{1}^1} 141^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[207] \quad 1^{\underline{1}^1} 4^{\underline{1}^1} 141^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 4 & 0 & 5 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[208] \quad 1^{\underline{1}^1} 5^{\underline{1}^1} 131^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[209] \quad 1^{\underline{1}^1} 4^{\underline{1}^1} 1^{\underline{1}^1} 31^{\underline{1}^1} 41^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 2 \\ 3 & 13 & 6 & 12 & 0 & 7 \\ 1 & 6 & 0 & 4 & 0 & 2 \\ 3 & 12 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 7 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[210] \quad 1^{\underline{1}^1} 1^{\underline{1}^1} 4^{\underline{1}^1} 141^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 4 & 5 \\ 3 & 5 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[211] \quad 31^{\underline{1}^1} 4^{\underline{1}^1} 141^{\underline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 1 & 2 \\ 2 & 5 & 8 & 9 & 0 & 5 \\ 2 & 8 & 4 & 9 & 0 & 4 \\ 3 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 5 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[212] $2 \overset{11}{3} 1 \overset{1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[213] $1^1 41^1 31^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 25 & 9 \\ 3 & 12 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[214] $1^1 51^1 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 7 & 1 & 2 & 3 \\ 7 & 26 & 0 & 8 & 10 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 8 & 0 & 0 & 2 \\ 3 & 10 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.1111 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$

$T = 8 \quad [1] 1^1 11^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^1 11^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^1 11^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 302, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^1 11^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9 \quad [5] 1^1 11^1 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 3 & 0 & -2 & 0 \\ 3 & 5 & 3 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[6] $1^1 11^1 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[7] $1^1 11^1 31^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.07 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

4.1112 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

4.1113 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_7$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_7 \quad T = 9, T_{\max} = 103, \Delta = -198, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -5 & -3 \\ 2 & 2 & -1 & -1 & 0 & -3 & -3 \end{pmatrix}_{b_0}$$

4.1114 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}'_7$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}'_7 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -4 & -2 \\ 2 & 2 & -1 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

4.1115 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 1 & 1 & 3 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 2 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 5 & 10 & 5 & 0 & 5 \\ 5 & 10 & 13 & 7 & 0 & 7 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 7 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 82, \Delta = -124, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 82, T \in [14, 82]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

4.1116 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.1117 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9 \quad [0] \quad 1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1118 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 8 \quad [0] \quad 1^{11}31^{12}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 67, T \in [15, 67]$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{12}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 67, T \in [12, 67]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \ 1^{11} 3^1 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 10 & 13 & 0 & 0 & 6 \\ 5 & 13 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 3^1 2^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^1 3^1 2^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 1 & 1 & 3 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 2 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 10 & 1 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 0 & 11 & 5 \\ 3 & 6 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 5 & 3 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 3 & 5 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 7 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [21] \ 1^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 14 & 10 & 2 & 8 \\ 10 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 8 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[22] 1^{1^{11}31^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -2 & 1 & 0 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[23] 1^{1^{11}31^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

4.1119 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] 1^{1^{11}31^1} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{1^{11}31^1} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 28 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 28 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{1^{11}31^1} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{1^{11}31^1} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{1^{11}31^1} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 10 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{1^{11}31^1} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -1 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

- [6] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 19 & 5 \\ 12 & 25 & 16 & 0 & 40 & 9 \\ 8 & 16 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 40 & 24 & 0 & 58 & 14 \\ 5 & 9 & 5 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -190$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 20 & 5 \\ 12 & 25 & 16 & 0 & 40 & 9 \\ 8 & 16 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 40 & 24 & 0 & 58 & 14 \\ 5 & 9 & 5 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 9 & 5 \\ 9 & 13 & 10 & 0 & 14 & 7 \\ 6 & 10 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 13 & 7 \\ 5 & 7 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 6 & 5 \\ 9 & 13 & 10 & 0 & 10 & 7 \\ 6 & 10 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 7 & 0 & 5 & 5 \\ 5 & 7 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 15 & 5 \\ 9 & 13 & 10 & 0 & 24 & 7 \\ 6 & 10 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 16 & 0 & 38 & 12 \\ 5 & 7 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

4.1120 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1121 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \text{mix}$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 7 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1122 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 6 & 5 \\ 12 & 25 & 12 & 0 & 12 & 9 \\ 6 & 12 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 7 & 0 & 4 & 4 \\ 5 & 9 & 4 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & 5 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

4.1123 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 14 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 14 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 5 & 9 & 12 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

- [4] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 20 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 24 & 15 & 0 & 121 & 19 \\ 3 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 2 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [11] $1 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 9 & 0 & 0 & 4 \\ 5 & 9 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 5 & 0 & 0 & 4 \\ 3 & 5 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 13 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 0 & 0 & 73 & 15 \\ 3 & 4 & -1 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

- [25] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 10 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 5 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$
- [27] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [29] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 49$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$
- [31] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 37 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 37 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$
- [33] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [35] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[38] $1^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[41] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 12 & 15 & 6 \\ 15 & 17 & 15 & 13 & 7 \\ 7 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 8 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

- [46] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [47] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [48] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 49 & 13 \\ 3 & 4 & -1 & -2 & 13 & 1 \end{pmatrix}_{b_0}$
- [49] $1^{11} 1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 8 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 8 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [51] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [50] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 28 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 28 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$
- [52] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 19 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 19 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$
- [53] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [54] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [55] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [56] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 78, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 20 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.49 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.49 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 83, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 4 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 0 & 37 & 11 \\ 3 & 4 & -1 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 85, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -1 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 48 & 0 & 15 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 4 & 4 \\ 3 & 4 & 12 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -1 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 20 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 20 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 15 & 0 & 121 & 19 \\ 3 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{12} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 97$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 12 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 48 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 48 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 99$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 12 & 21 & 75 & 0 & 18 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 4 & 4 \\ 3 & 4 & 15 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 5 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 30 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 29 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{111} 3 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 24 & 15 & 0 & 121 & 19 \\ 3 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{122} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 5 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 26 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 26 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 9 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 16 & 3 \\ 3 & 4 & 0 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 0 & 0 & 154 & 22 \\ 3 & 4 & -1 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[114] $21^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 0 & 58 & 14 \\ 3 & 4 & -1 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 9 & 14 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 150, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{14}} 3^{\overline{111}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 151, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 155, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.34 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{\overline{112}} 3^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 157, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{\overline{114}} 3^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 158, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 38 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{\overline{11}} 3^{\overline{111}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 6 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{\overline{12}} 3^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 165, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 0 & 0 & 100 & 18 \\ 3 & 4 & -1 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 165, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 6 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{\overline{111}} 3^{\overline{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.31 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{\overline{111}} 3^{\overline{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.31 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 29 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 10 & 17 & 48 & 0 & 15 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 4 & 4 \\ 3 & 4 & 12 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{121}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 5 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

- [135] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [142] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 17 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$
- [141] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$
- [140] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 0 & 0 & 85 & 17 \\ 3 & 4 & -1 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [139] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 0 & 0 & 118 & 20 \\ 3 & 4 & -1 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [143] $21^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 13 & 7 \\ 7 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$
- [144] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [145] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 40 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [146] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [147] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [148] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 191$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 8 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 9 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 196$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 196$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 199$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 201$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 5 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 203$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 39 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[155] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 205$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[158] $1^{11} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 45 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 210$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{112} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 6 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{12} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{122} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{11} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{121} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{122} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{123} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{113} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{113} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{112} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{112} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 22 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{131} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 7 & 12 & 24 & 0 & 9 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 8 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 38 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[184] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 27 & 7 \\ 17 & 20 & 17 & 33 & 8 \\ 15 & 17 & 12 & 27 & 6 \\ 27 & 33 & 27 & 49 & 13 \\ 7 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 0 & 0 & 133 & 21 \\ 3 & 4 & -1 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 47 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 47 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 5 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

- [190] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 6 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [191] $1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$
- [192] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [193] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$
- [194] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$
- [195] $21^{112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 13 & 20 & 12 & 30 & 7 \\ 20 & 25 & 16 & 40 & 9 \\ 12 & 16 & 9 & 24 & 5 \\ 30 & 40 & 24 & 58 & 14 \\ 7 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$
- [196] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 57 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [197] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$
- [198] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 36 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$
- [199] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$
- [200] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 28 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[201] $21^{111}31^{111}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=286, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=10.0$
 $T_{\min}=11.07 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 24 & 6 \\ 16 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{111}31^{12}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=286, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=10.0$
 $T_{\min}=11.24 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 27 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13}31^{14}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=288, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=10.8$
 $T_{\min}=11.40 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 18 & 3 \\ 3 & 4 & 0 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 0 & 0 & 193 & 25 \\ 3 & 4 & -1 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{123}31^{111}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=290, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=12.0$
 $T_{\min}=11.14 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 8 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111}31^{14}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=290, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=10.0$
 $T_{\min}=11.25 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 46 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{123}31^{12}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=293, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=12.0$
 $T_{\min}=11.82 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{12}31^{111}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=295, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=10.5$
 $T_{\min}=11.26 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{112}31^{11}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=296, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=10.5$
 $T_{\min}=11.16 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{131}31^{11}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=301, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=12.0$
 $T_{\min}=11.22 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{113}31^{111}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]+[\textcolor{blue}{su}_3+\textcolor{orange}{so}_8+\textcolor{red}{su}_2] \quad T=8, T_{\max}=12, \Delta=-92, \det=303, (n_+, n_-, n_0)=(1, 8, 0), c_{\text{ext}}=10.8$
 $T_{\min}=11.17 \rightarrow 12, T_{\max}=12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 8 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 17 & 3 \\ 3 & 4 & 0 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 0 & 0 & 172 & 24 \\ 3 & 4 & -1 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 37 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 48 & 0 & 27 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 13 & 7 \\ 3 & 4 & 12 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 11 & 18 & 57 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 381$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[235] $1^{114} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 22 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 386$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 8 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 390$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 392$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 394$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 395$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 20 & 3 \\ 3 & 4 & 0 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 0 & 0 & 238 & 28 \\ 3 & 4 & -1 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 8 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 45 & 7 \\ 20 & 25 & 16 & 60 & 9 \\ 12 & 16 & 9 & 36 & 5 \\ 45 & 60 & 36 & 133 & 21 \\ 7 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{114} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{131} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 84 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{114} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 21 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[262] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 10 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 10 & 17 & 48 & 0 & 27 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 13 & 7 \\ 3 & 4 & 12 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{112} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{122} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[272] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 498$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 21 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[274] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 503$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[275] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 528$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[276] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 528$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[277] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 535$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 549$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 564$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 566$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 13 & 22 & 84 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 4 & 16 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{14} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{14} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[291] $21^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 60 & 7 \\ 20 & 25 & 16 & 80 & 9 \\ 12 & 16 & 9 & 48 & 5 \\ 60 & 80 & 48 & 238 & 28 \\ 7 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[293] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 645, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[295] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 19 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[297] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 19 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[298] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 669, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 17 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{112} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[300] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[301] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[302] $1^{113} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{112} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[304] $1^{114} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{112} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 680$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{114} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 717$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[307] $1^{123} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 722$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{114} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 724$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 17 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[309] $1^{114} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 730$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{112} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 744$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 756$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 24 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [352] \quad 1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 7 & 0 & 21 & 4 \\ 4 & 7 & 7 & 0 & 31 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 31 & 0 & 101 & 19 \\ 3 & 4 & 5 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[353] \quad 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[358] \quad 1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[356] \quad 21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 6 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[357] \quad 21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 12 & 6 \\ 12 & 13 & 10 & 14 & 7 \\ 8 & 10 & 4 & 9 & 4 \\ 12 & 14 & 9 & 13 & 7 \\ 6 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[355] \quad 1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 6 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[354] \quad 1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 12 & 6 \\ 12 & 13 & 10 & 14 & 7 \\ 8 & 10 & 4 & 9 & 4 \\ 12 & 14 & 9 & 13 & 7 \\ 6 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[359] \quad 21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 15 & 7 \\ 14 & 13 & 10 & 14 & 7 \\ 9 & 10 & 4 & 9 & 4 \\ 15 & 14 & 9 & 13 & 7 \\ 7 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[360] \quad 1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 7 & 0 & 5 & 4 \\ 4 & 7 & 7 & 0 & 6 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 5 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[361] \quad 1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 8 & 0 & 21 & 9 \\ 3 & 4 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[362] \ 21^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 9 & 7 \\ 14 & 13 & 10 & 10 & 7 \\ 9 & 10 & 4 & 7 & 4 \\ 9 & 10 & 7 & 5 & 5 \\ 7 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 4 & 4 \\ 6 & 10 & 20 & 0 & 7 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 7 & 0 & 1 & 3 \\ 3 & 4 & 8 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[365] \ 1^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 9 & 5 \\ 9 & 13 & 10 & 14 & 7 \\ 6 & 10 & 4 & 9 & 4 \\ 9 & 14 & 9 & 13 & 7 \\ 5 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[367] \ 1^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 20 & 6 \\ 12 & 13 & 10 & 24 & 7 \\ 8 & 10 & 4 & 16 & 4 \\ 20 & 24 & 16 & 38 & 12 \\ 6 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[368] \ 21^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 20 & 6 \\ 12 & 13 & 10 & 24 & 7 \\ 8 & 10 & 4 & 16 & 4 \\ 20 & 24 & 16 & 38 & 12 \\ 6 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 7 & 0 & 8 & 4 \\ 4 & 7 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 11 & 0 & 13 & 7 \\ 3 & 4 & 5 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[370] \ 1^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[371] \ 1^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 4 & 7 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[372] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[373] \ 1^{\frac{1}{12}} 4^{\frac{1}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 7 & 0 & 10 & 4 \\ 4 & 7 & 7 & 0 & 15 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 15 & 0 & 21 & 9 \\ 3 & 4 & 5 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[374] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 4 & 7 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[375] \ 1^{\frac{1}{13}} 4^{\frac{1}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 13 & 0 & 61 & 15 \\ 3 & 4 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[376] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 9 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[377] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 7 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 4 & 7 & 4 \\ 7 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[378] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 11 & 4 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 7 & 0 & 21 & 9 \\ 3 & 4 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[379] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 7 & 0 & 9 & 4 \\ 4 & 7 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 11 & 0 & 13 & 7 \\ 3 & 4 & 5 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[380] \ 1^{\frac{1}{13}} 4^{\frac{1}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 14 & 0 & 70 & 16 \\ 3 & 4 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 11 & 0 & 45 & 13 \\ 3 & 4 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[391] \ 1^{\overline{111}}_1 4^{\overline{11}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 16 & 5 \\ 9 & 13 & 10 & 24 & 7 \\ 6 & 10 & 4 & 16 & 4 \\ 16 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[392] \ 1^{\overline{13}}_1 4^{\overline{11}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -229, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 3 & 0 & 24 & 4 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 18 & 0 & 121 & 21 \\ 3 & 4 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[393] \ 1^{\overline{111}}_1 4^{\overline{11}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -254, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 7 & 0 & 12 & 4 \\ 4 & 7 & 7 & 0 & 16 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 16 & 0 & 26 & 10 \\ 3 & 4 & 5 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[394] \ 21^{\overline{11}}_1 4^{\overline{111}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -254, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 24 & 7 \\ 14 & 13 & 10 & 24 & 7 \\ 9 & 10 & 4 & 16 & 4 \\ 24 & 24 & 16 & 38 & 12 \\ 7 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[395] \ 1^{\overline{13}}_1 4^{\overline{11}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -265, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 61 & 15 \\ 3 & 4 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[396] \ 1^{\overline{111}}_1 4^{\overline{11}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -270, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 7 & 0 & 15 & 4 \\ 4 & 7 & 7 & 0 & 21 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 21 & 0 & 45 & 13 \\ 3 & 4 & 5 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[397] \ 1^{\overline{13}}_1 4^{\overline{11}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -286, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 15 & 0 & 88 & 18 \\ 3 & 4 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[398] \ 21^{\overline{111}}_1 4^{\overline{11}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -300, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 7 & 5 \\ 10 & 13 & 10 & 10 & 7 \\ 7 & 10 & 4 & 7 & 4 \\ 7 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[399] \ 1^{\overline{12}}_1 4^{\overline{11}}_1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -325, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 16 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 10 & 0 & 45 & 13 \\ 3 & 4 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[400] \ 1^{12} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 25 & 4 \\ 2 & 3 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 25 & 19 & 0 & 132 & 22 \\ 3 & 4 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[401] \ 1^{14} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -346, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 22 & 4 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 14 & 0 & 90 & 18 \\ 3 & 4 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[402] \ 21^{111} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -382, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 16 & 5 \\ 10 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 16 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[403] \ 1^{12} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -406, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 3 & 0 & 19 & 4 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 13 & 0 & 68 & 16 \\ 3 & 4 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[404] \ 1^{12} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -409, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 22 & 4 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 16 & 0 & 97 & 19 \\ 3 & 4 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[405] \ 1^{13} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -426, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 3 & 0 & 20 & 4 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 12 & 0 & 70 & 16 \\ 3 & 4 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[406] \ 1^{14} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -439, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 3 & 0 & 25 & 4 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 25 & 17 & 0 & 121 & 21 \\ 3 & 4 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[407] \ 1^{14} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -454, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 3 & 0 & 28 & 4 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 20 & 0 & 158 & 24 \\ 3 & 4 & 2 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[408] \ 1^{14} \overset{1}{4} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 24 & 4 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 14 & 0 & 101 & 19 \\ 3 & 4 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[409] \ 1^{13} \overset{1^1}{4} \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -579, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 23 & 4 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 15 & 0 & 97 & 19 \\ 3 & 4 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{13} \overset{1^1}{4} \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -651, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 3 & 0 & 29 & 4 \\ 2 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 21 & 0 & 169 & 25 \\ 3 & 4 & 2 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{13} \overset{1^1}{4} \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -654, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 26 & 4 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 18 & 0 & 130 & 22 \\ 3 & 4 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{14} \overset{1^1}{4} \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -734, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 27 & 4 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 17 & 0 & 132 & 22 \\ 3 & 4 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{14} \overset{1^1}{4} \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -881, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 3 & 0 & 30 & 4 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 30 & 20 & 0 & 169 & 25 \\ 3 & 4 & 2 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{14} \overset{1^1}{4} \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -950, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 3 & 0 & 33 & 4 \\ 2 & 3 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 33 & 23 & 0 & 212 & 28 \\ 3 & 4 & 2 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [415] \ 1^{11} \overset{1^{11}31^1}{4} \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 1 & 2 & 5 \\ 4 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 5 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.1124 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6$

$$T = 9 \quad [0] \ 1^{111} \overset{1^1}{3} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1^1}{3} 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{c}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{c}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.1125 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4$

$T = 9$ [0] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1126 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$

$T = 9$ [0] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{113}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -87$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 28 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -127$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 10 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -133$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -1 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 19 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -190$, $(n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 20 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -1 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -1 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 30 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 29 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 38 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -442, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 40 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 39 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -539, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -539, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 19 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -735, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 165, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 9 & 1 & 5 \\ 9 & 13 & 10 & 14 & 0 & 7 \\ 6 & 10 & 4 & 9 & 0 & 4 \\ 9 & 14 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

4.1127 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$

$$T = 9 \quad [0] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -62, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -62, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

4.1128 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 85, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 88, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{13}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 89, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 98, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 99, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[8] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 15 & 9 & 7 \\ 17 & 20 & 17 & 11 & 8 \\ 15 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 7 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[10] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 16 & 12 & 8 & 6 \\ 16 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 7 & 13 & 27 & 0 & 7 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 9 & 17 & 48 & 0 & 9 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [13] $1^{11}5^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.1129 $\text{NHC}(-2)(-3) + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1130 $\text{NHC}(-2)(-3) + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{so}_8$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1131 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 1 & 2 \\ 2 & 4 & 0 & 2 & 2 & 1 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 \\ 3 & 4 & 0 & 3 & 3 & 7 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 6 \\ 2 & 3 & 0 & 3 & 1 & 5 \\ 4 & 7 & 0 & 6 & 5 & 9 \\ 3 & 4 & -1 & 2 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 \\ 3 & 4 & 0 & 3 & 3 & 15 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 10 \\ 2 & 3 & 0 & 3 & 1 & 9 \\ 8 & 15 & 0 & 10 & 9 & 41 \\ 3 & 4 & -1 & 2 & 3 & 13 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 \\ 3 & 4 & 0 & 3 & 3 & 11 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 8 \\ 2 & 3 & 0 & 3 & 1 & 7 \\ 6 & 11 & 0 & 8 & 7 & 22 \\ 3 & 4 & -1 & 2 & 3 & 10 \end{pmatrix}_{b_0}$$

4.1132 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 8 \quad [0] \quad 1^{11} \overset{1}{3} 1^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1}{3} 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \overset{1}{3} 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 8 & 0 & 25 & 9 \\ 3 & 5 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{3} 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 15 & 0 & 64 & 14 \\ 3 & 4 & 3 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 12 & 0 & 29 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 19 & 29 & 0 & 64 & 14 \\ 3 & 4 & 6 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 25 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 27 & 0 & 16 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 9 & 0 & 10 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 5 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 9 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 16 & 0 & 25 & 9 \\ 3 & 4 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 8 & 0 & 25 & 9 \\ 3 & 5 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 17 & 0 & 151 & 21 \\ 3 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 15 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 12 & 0 & 39 & 11 \\ 3 & 4 & 3 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 6 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 5 & 9 & 12 & 0 & 11 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 9 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 9 & 0 & 10 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 5 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 15 & 4 \\ 5 & 9 & 12 & 0 & 23 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 23 & 0 & 39 & 11 \\ 3 & 4 & 6 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 \\ 5 & 9 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 17 & 0 & 20 & 8 \\ 3 & 4 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 5 & 9 & 12 & 0 & 11 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 \\ 5 & 9 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 17 & 0 & 20 & 8 \\ 3 & 4 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 14 & 0 & 97 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 12 & 0 & 29 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 29 & 0 & 64 & 14 \\ 3 & 4 & 6 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 17 & 0 & 151 & 21 \\ 3 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 15 & 4 \\ 5 & 9 & 12 & 0 & 23 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 23 & 0 & 39 & 11 \\ 3 & 4 & 6 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 27 & 0 & 16 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 2 & -2 & 26 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{11} 1^{14} 3^{11} 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 2208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 23 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 2 & -2 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [76] $1^{12} 5^{13} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 8 & 14 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{11} 5^{13} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{13} 5^{13} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -133, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 10 & 18 & 0 & 13 & 62 & 16 \\ 3 & 4 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

4.1133 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_3$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 7 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1134 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3$

$T = 8$ [0] $1^{11} 3^{11} 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{13} 3^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 10 & 0 & 0 & 4 \\ 6 & 10 & 21 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 11 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 11 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 0 & 9 & 5 \\ 3 & 4 & -1 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{14}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 0 & 0 & 75 & 15 \\ 3 & 4 & -1 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

- [13] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [17] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [21] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [22] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [23] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 0 & 57 & 13 \\ 3 & 4 & -1 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 0 & 0 & 117 & 19 \\ 3 & 4 & -1 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 0 & 0 & 84 & 16 \\ 3 & 4 & -1 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[39] $21^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 12 & 6 \\ 7 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [42] $1^1 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 5 \\ 12 & 25 & 12 & 12 & 9 \\ 6 & 12 & 4 & 7 & 4 \\ 6 & 12 & 7 & 4 & 4 \\ 5 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 0 & 2 \\ 2 & 4 & 2 & -1 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 8 & 8 & 7 & 1 & 4 \\ 8 & 13 & 16 & 12 & 0 & 7 \\ 8 & 16 & 12 & 12 & 0 & 6 \\ 7 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 7 & 6 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 10 & 1 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 0 & 12 & 6 \\ 3 & 6 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{11}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 3 & 2 \\ 3 & 3 & 0 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[48] $21^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 13 & 20 & 9 & 9 & 7 \\ 20 & 25 & 12 & 12 & 9 \\ 9 & 12 & 4 & 7 & 4 \\ 9 & 12 & 7 & 4 & 4 \\ 7 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 3 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 6 & 0 & 0 & 12 & 6 \\ 2 & 4 & 2 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 0 & 3 & 3 & 3 & 2 \\ 2 & 3 & 0 & 3 & 3 & 3 & 2 \\ 2 & 3 & -1 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{11} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{11} 31 \ 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

4.1135 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31 \ 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 31 \ 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 31 \ 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{11} 41 \ 1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 1 & 5 \\ 12 & 25 & 12 & 12 & 0 & 9 \\ 6 & 12 & 4 & 7 & 0 & 4 \\ 6 & 12 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1136 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 12 & 6 \\ 3 & 5 & 2 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 4 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 4 & 8 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 2 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 0 & 6 & 12 & 6 \\ 3 & 4 & -1 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.1137 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + (-3)→(-2)

$T = 8 \quad [0] 1^{11} 3 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 9 \quad [3] 1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{21} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

4.1138 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$T = 7 \quad [0] 1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 8 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{113} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 12 & 27 & 6 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1^2} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 5 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 5 & 9 & 12 & 24 & 6 \\ 9 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.60 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 9 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 5 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -13, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -13, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{122} 3^1 1^{23} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 27 & 18 & 9 \\ 7 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 9 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^1 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^1 3^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 5 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 7 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[26] $21^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 13 & 5 & 4 \\ 12 & 13 & 27 & 12 & 9 \\ 6 & 5 & 12 & 4 & 4 \\ 4 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[27] $21^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 5 & 3 & 4 & 4 \\ 4 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 14 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3^1 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{122} 3^1 1^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 8 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 12 & 9 & 6 \\ 3 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 4 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{14} 3^1 1^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 7 & 9 & 12 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 12 & 12 & 6 \\ 8 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 17 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{111} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[44] $21^{14} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 27 & 24 & 15 & 121 & 19 \\ 4 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{111} 3^1 1^2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{113} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{\mathbf{11}} 1^{\mathbf{11}} 3^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[39] $21^{\mathbf{13}} 31^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 4 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{\mathbf{111}} 1^{\mathbf{12}} 31^{\mathbf{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{\mathbf{112}} 1^{\mathbf{3}} 31^{\mathbf{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 9 & 16 & 6 & 22 & 8 \\ 3 & 5 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{\mathbf{11}} 1^{\mathbf{22}} 31^{\mathbf{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 12 & 9 & 6 \\ 5 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{\mathbf{11}} 1^{\mathbf{12}} 31^{\mathbf{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{\mathbf{111}} 1^{\mathbf{12}} 31^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 12 & 6 & 6 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{\mathbf{11}} 1^{\mathbf{1}} 31^{\mathbf{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{\mathbf{112}} 1^{\mathbf{21}} 31^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 12 & 12 & 6 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{\mathbf{114}} 1^{\mathbf{13}} 31^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 12 & 24 & 6 \\ 11 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{\mathbf{11}} 1^{\mathbf{1}} 31^{\mathbf{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{\overline{1}} 1^{\overline{21}} 3^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{\overline{1}} 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.55 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{\overline{1}} 1^{\overline{21}} 3^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 11 & 17 & 48 & 15 & 12 \\ 4 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{\overline{1}} 1^{\overline{21}} 3^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{\overline{1}} 1^{\overline{11}} 3^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[60] \ 21^{\overline{123}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.55 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{\overline{1}} 1^{\overline{21}} 3^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -23, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{\overline{113}} 1^{\overline{13}} 3^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -24, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 12 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{\overline{1}} 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -24, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -24, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{\overline{111}} 1^{\overline{1}} 3^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -24, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 12 & 6 & 13 & 7 \\ 3 & 5 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{11} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{11} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{11} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{11} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[73] $21^{11} 3^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{11} 3^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 15 & 0 & 52 & 12 \\ 3 & 4 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{12} 3^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 9 & 6 & 5 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{11} 3^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 7 & 13 & 27 & 27 & 9 \\ 6 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{12} 3^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[78] $13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{121} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 9 & 6 & 5 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 12 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{14} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 15 & 21 & 75 & 18 & 15 \\ 4 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{112} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 10 & 0 & 22 & 8 \\ 3 & 4 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 8 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 5 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{11} 3^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 12 & 12 & 6 \\ 4 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[89] $21^{114} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 16 & 4 \\ 6 & 9 & 12 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 22 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 12 & 9 & 6 \\ 5 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 9 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 12 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 27 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 6 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 12 & 6 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[98] $21^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 8 & 9 & 3 & 4 \\ 8 & 9 & 12 & 4 & 5 \\ 9 & 12 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 4 & 5 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 5 & 9 & 12 & 18 & 6 \\ 7 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 7 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 36 & 9 & 10 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 9 & 9 & 13 & 7 & 2 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[103] $1^1 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{131} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 48 & 12 & 12 \\ 4 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 36 & 9 & 10 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{12} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 8 & 13 & 27 & 12 & 9 \\ 4 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{112} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 8 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{14}3^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.6$
 $T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 14 & 21 & 75 & 18 & 15 \\ 3 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{112}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 13 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.6$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 12 & 21 & 75 & 18 & 15 \\ 3 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 4 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{113}3^{134}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -37$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.46 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 8 & 9 & 12 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{111}3^{113}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -37$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 9 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -37$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 4 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{121}3^{141}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -37$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 8 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -37$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 8 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{111}3^{113}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 17 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{111}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -39$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 12 & 6 & 6 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{111} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{111} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 8 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[126] $21^{14} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 21 & 5 & 4 \\ 15 & 21 & 75 & 18 & 15 \\ 3 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[131] $1^1 3^1 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 4 & 8 & 9 & 12 & 5 \\ 5 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 5 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[128] $21^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{13} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{12} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 27 & 12 & 9 \\ 3 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[133] $1^1 3^1 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 4 & 8 & 9 & 6 & 5 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{112}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 12 & 21 & 6 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 5 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{111}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 9 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[145] $21^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 14 & 4 & 4 \\ 15 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 4 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{112}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{12}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 8 & 14 & 36 & 9 & 10 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{111}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{121}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 5 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{111} 3^1 1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{111} 3^1 1^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 12 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 5 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{111} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 14 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 10 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[155] $1^1 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 4 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{112}3^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{111}3^{124}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 9 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 27 & 18 & 9 \\ 4 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{112}3^{124}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 8 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{14}3^{12}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 13 & 21 & 75 & 18 & 15 \\ 3 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{112}3^{123}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 12 & 12 & 6 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{11}3^{12}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 12 & 9 & 6 \\ 3 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[163] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 5 & 9 & 12 & 9 & 6 \\ 3 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{113}3^{112}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 12 & 18 & 6 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[164] $1^{122}3^{12}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 7 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 12 & 12 & 6 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[168] $1^{122} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[170] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 12 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[171] $1^{122} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{12} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 4 & 8 & 9 & 24 & 5 \\ 10 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 15 & 0 & 52 & 12 \\ 3 & 4 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{13} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 12 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 7 & 13 & 27 & 12 & 9 \\ 3 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 7 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 9 & 14 & 36 & 9 & 10 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 48 & 27 & 12 \\ 5 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{131} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 4 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 12 & 6 & 6 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 5 & 9 & 12 & 6 & 6 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{131} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 24 & 9 & 8 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{131} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 24 & 9 & 8 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 12 & 15 & 6 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{13} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 5 & 5 & 3 & 15 & 3 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[187] $21^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 5 & 13 & 4 \\ 6 & 5 & 3 & 12 & 3 \\ 12 & 13 & 12 & 28 & 10 \\ 4 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 6 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 13 & 24 & 9 & 52 & 12 \\ 3 & 5 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 12 & 9 & 6 \\ 3 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 8 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{12} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 12 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 18 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[200] $1^1 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 10 & 24 & 12 & 58 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 24 & 12 & 58 & 14 \\ 3 & 5 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 27 & 9 & 9 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{121}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 12 & 6 & 6 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{12}31^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 12 & 12 & 6 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 12 & 12 & 6 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{121}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[209] $21^{131}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 9 & 4 & 4 \\ 9 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 4 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 12 & 17 & 48 & 15 & 12 \\ 4 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 12 & 6 & 6 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 57 & 12 & 13 \\ 4 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 8 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 7 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 10 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 11 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 8 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 4 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 27 & 18 & 9 \\ 5 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{112} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 8 & 16 & 6 & 22 & 8 \\ 3 & 5 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{123} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 19 & 3 \\ 3 & 4 & 8 & 27 & 4 \\ 6 & 8 & 9 & 36 & 5 \\ 19 & 27 & 36 & 133 & 21 \\ 3 & 4 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 5 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{114} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 12 & 33 & 6 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{11} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 8 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 9 & 6 & 5 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 9 & 13 & 27 & 12 & 9 \\ 3 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 11 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 9 & 6 & 5 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 5 & 5 & 3 & 12 & 3 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{11} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{113} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 9 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[250] $1 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 7 & 13 & 27 & 18 & 9 \\ 4 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{111} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[246] $21^{11} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 18 & 21 & 15 & 76 & 16 \\ 4 & 4 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{11} 1^{\textcolor{red}{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{12} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 4 & 5 & 3 & 18 & 3 \\ 13 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{121} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{111} 3^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 15 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{122} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 27 & 18 & 9 \\ 6 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{13} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 10 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{12} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 8 & 13 & 27 & 12 & 9 \\ 3 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[247] $21^{11} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 6 & 4 \\ 5 & 4 & 5 & 5 & 4 \\ 6 & 5 & 3 & 6 & 3 \\ 6 & 5 & 6 & 4 & 4 \\ 4 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{121} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{12} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 12 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{12} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 9 & 13 & 27 & 12 & 9 \\ 4 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[261] $1^1 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 12 & 15 & 6 \\ 9 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 7 & 16 & 6 & 22 & 8 \\ 3 & 5 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 15 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -69, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 20 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 20 & 9 & 37 & 11 \\ 3 & 5 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{131}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 9 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{123}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 7 & 13 & 27 & 27 & 9 \\ 7 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{122}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 4 & 8 & 9 & 12 & 5 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{122}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 7 & 13 & 27 & 18 & 9 \\ 5 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{121}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 11 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{131}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 9 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[276] $21^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 9 & 12 & 4 \\ 6 & 9 & 12 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[278] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 12 & 6 & 13 & 7 \\ 4 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{13} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[281] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 4 & 4 \\ 15 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 4 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[283] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 12 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{122} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 12 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 48 & 15 & 12 \\ 4 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 11 & 13 & 27 & 12 & 9 \\ 4 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{111} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 13 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[290] $1^{111} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{123} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 8 & 13 & 27 & 27 & 9 \\ 7 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{131} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 11 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{112} 3^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{112} 3^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{111} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{111} 3^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[299] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 20 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

[302] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[301] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 20 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 20 & 9 & 37 & 11 \\ 3 & 5 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{122} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 13 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[308] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 8 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{122} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 14 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 2 \end{pmatrix}_{b_0} \\
[317] \quad & 21^{11} 31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.06 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 4 & 4 \\ 12 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 4 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0} \\
[320] \quad & 1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.06 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 2 \end{pmatrix}_{b_0} \\
[313] \quad & 1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.06 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[312] \quad & 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.52 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0} \\
[318] \quad & 1^{113} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.57 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 18 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[311] \quad & 1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.06 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 12 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[310] \quad & 1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.22 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 11 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 0 & 2 \end{pmatrix}_{b_0} \\
[315] \quad & 1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.06 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 27 & 27 & 9 \\ 10 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0} \\
[314] \quad & 1^{12} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.06 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 27 & 12 & 9 \\ 4 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0} \\
[316] \quad & 1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.06 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0} \\
[319] \quad & 1^{123} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.28 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 8 & 27 & 4 \\ 5 & 8 & 9 & 36 & 5 \\ 17 & 27 & 36 & 133 & 21 \\ 3 & 4 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[322] $1^{114} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 10 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 36 & 18 & 133 & 21 \\ 3 & 5 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[324] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[325] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[326] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 11 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[327] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 7 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[328] $1^{14} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

[329] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[330] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 8 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[331] $1^{111} 3 1^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[332] $1^{112} 3^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[338] $1^{112} 3^4 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[337] $21^{111} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 5 & 4 \\ 9 & 9 & 12 & 9 & 6 \\ 6 & 5 & 9 & 4 & 4 \\ 4 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[336] $1^{112} 3^4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 17 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[335] $1^{112} 3^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 7 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[334] $1^{112} 3^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[333] $1^{112} 3^4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[341] $1^{124} 3^4 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 5 & 5 & 3 & 12 & 3 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[339] $1^{13} 3^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

[342] $1^{121} 3^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[343] $1^{123} 3^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[344] $1^{123} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[345] $21^{111} 31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 9 & 4 \\ 5 & 4 & 17 & 9 & 4 \\ 15 & 17 & 48 & 27 & 12 \\ 9 & 9 & 27 & 13 & 7 \\ 4 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[340] $1^{11} 3 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 7 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 7 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[348] $1^{131} 3 1^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[349] $1^{131} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[350] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 24 & 9 & 8 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[351] $1^{12} 3 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

[352] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[353] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 9 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{123} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 8 & 27 & 4 \\ 5 & 8 & 9 & 36 & 5 \\ 18 & 27 & 36 & 133 & 21 \\ 3 & 4 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[356] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 20 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 20 & 9 & 37 & 11 \\ 3 & 5 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[355] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 4 & 8 & 9 & 18 & 5 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[357] $1^{14} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 46 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[359] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{113} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[360] $1^{113} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[361] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[362] $1^{121} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 27 & 21 & 9 \\ 7 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[363] $1^{111} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 15 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{112} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[365] & 21^{1331^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 21 & 19 & 12 & 73 & 15 \\ 4 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[366] & 1^{11} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 4 & 8 & 9 & 18 & 5 \\ 7 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 1 \end{pmatrix}_{b_0} \\
[367] & 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[368] & 1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 0 & 2 \end{pmatrix}_{b_0} \\
[369] & 1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 27 & 18 & 9 \\ 7 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0} \\
[370] & 1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 8 & 13 & 27 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0} \\
[373] & 21^{1131^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 9 & 6 & 13 & 7 \\ 4 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[372] & 1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[358] & 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 12 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[374] & 21^{111} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 17 & 4 \\ 9 & 9 & 12 & 27 & 6 \\ 15 & 17 & 27 & 49 & 13 \\ 4 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0} \\
[375] & 1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [376] $1^{123} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -93$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 7 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [377] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -94$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 13 & 0 & 2 \end{pmatrix}_{b_0}$
- [378] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -94$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 9 & 13 & 27 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$
- [379] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -95$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 7 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 1 \end{pmatrix}_{b_0}$
- [380] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -95$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 27 & 9 & 9 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$
- [381] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 5 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$
- [382] $1^1 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 15 & 36 & 18 & 133 & 21 \\ 3 & 5 & 2 & 21 & 2 \end{pmatrix}_{b_0}$
- [383] $21^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -97$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$
- [385] $1^{123} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -97$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 4 & 8 & 9 & 18 & 5 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$
- [384] $21^{121} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -97$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 9 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$
- [390] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [389] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.6$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 15 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{111} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 14 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 5 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{131} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 24 & 9 & 8 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[391] \ 1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 12 & 15 & 6 \\ 5 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[392] \ 1^{112} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 \\ 6 & 8 & 6 & 10 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[399] \ 1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[398] \ 21^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 8 & 4 \\ 6 & 9 & 12 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[397] \ 1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[396] \ 1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 7 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[395] \ 1^{123} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[393] \ 1^{112} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 10 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 27 & 30 & 9 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 36 & 18 & 133 & 21 \\ 3 & 5 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[402] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[405] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 12 & 24 & 9 & 52 & 12 \\ 3 & 5 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[413] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[412] $1^{121} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 11 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[411] $1 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 10 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[410] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 9 & 13 & 27 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[409] $1^{121} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[407] $1^{123} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 9 & 30 & 5 \\ 14 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[406] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 48 & 12 & 12 \\ 4 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[403] $1^{113} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 32 & 15 & 100 & 18 \\ 3 & 5 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[404] $1^{114} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{143} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[415] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{143} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[418] $1^{123} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[419] $1^{121} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[420] $1^{123} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[421] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[426] $1^{123} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 4 & 8 & 9 & 18 & 5 \\ 7 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[425] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[424] $1^{114} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[423] $21^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 5 & 9 & 4 \\ 6 & 5 & 3 & 9 & 3 \\ 9 & 9 & 9 & 13 & 7 \\ 4 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[422] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[427] $1^{122} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[428] $1^{122} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[429] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[430] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[431] $1^{113} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[432] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 27 & 18 & 9 \\ 5 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[433] $21^{111} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[434] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[435] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 48 & 27 & 12 \\ 7 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[436] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 24 & 9 & 52 & 12 \\ 3 & 5 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[437] $1^{14} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

[438] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 8 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[439] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 12 & 21 & 6 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[442] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[440] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 27 & 18 & 9 \\ 5 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[449] $21^{11} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 15 & 17 & 12 & 49 & 13 \\ 4 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[448] & 21^{1231^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 15 & 14 & 9 & 37 & 11 \\ 4 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[447] & 1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 11 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0} \\
[446] & 1^{11} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 7 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[445] & 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 9 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[444] & 1^{122} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 27 & 30 & 9 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0} \\
[441] & 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 13 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[443] & 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[450] & 1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 27 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0} \\
[451] & 1^{122} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 11 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0} \\
[452] & 1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[453] & 1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0} \\
[454] \quad & 21^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -120, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.10 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 9 & 4 & 4 \\ 6 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0} \\
[455] \quad & 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.11 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 8 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0} \\
[456] \quad & 1^{121} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.24 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 12 & 0 & 2 \end{pmatrix}_{b_0} \\
[457] \quad & 1^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.44 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0} \\
[458] \quad & 1^{13} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.27 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[460] \quad & 1^{131} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0} \\
[461] \quad & 1^{11} 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 9 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0} \\
[459] \quad & 1^{131} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0} \\
[467] \quad & 21^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.32 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 8 & 3 & 4 & 4 \\ 4 & 5 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[462] \quad & 1^{111} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.74 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 7 & 13 & 27 & 21 & 9 \\ 5 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0} \\
[463] \quad & 1^{12} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.26 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[464] $1^{13} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 2 \end{pmatrix}_{b_0}$$

[465] $1^{122} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 27 & 18 & 9 \\ 6 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[466] $21^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 12 & 13 & 9 & 28 & 10 \\ 4 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[468] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 11 & 24 & 9 & 52 & 12 \\ 3 & 5 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[469] $1^{121} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

[471] $1^{123} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 9 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[470] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 12 & 15 & 6 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[473] $1^{12} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[472] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[475] $1^{121} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{11} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[477] & 21^{1231^{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 18 & 4 & 4 \\ 18 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 4 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0} \\
[476] & 1^{11} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0} \\
[479] & 1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[474] & 1^{121} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 11 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0} \\
[480] & 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 12 & 15 & 6 \\ 8 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0} \\
[481] & 1^{122} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0} \\
[482] & 1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0} \\
[483] & 1^{113} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[484] & 1^{113} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[485] & 1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0} \\
[486] & 1^{111} 3 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 9 & 17 & 48 & 27 & 12 \\ 5 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[488] $1^{11} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[487] $1^{113} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 14 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[489] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[493] $1^{13} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 10 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[492] $1^{131} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[491] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 11 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[490] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[498] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[497] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 27 & 18 & 9 \\ 6 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[496] $1^{123} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 4 & 8 & 9 & 24 & 5 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[495] $1^{121} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 27 & 21 & 9 \\ 8 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[494] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 9 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[499] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 48 & 27 & 12 \\ 5 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[501] $1^{12} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[500] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[510] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 15 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[508] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[507] $21^{112} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[506] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[505] $21^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[502] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[504] $1^{14}3^{113}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 4 & 5 & 3 & 18 & 3 \\ 14 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[503] $1^{121}3^{111}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 27 & 21 & 9 \\ 5 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[511] $1^{113}3^{123}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[512] $1^{113}3^{13}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[513] $1^{112}3^{12}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[514] $1^{111}3^{112}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 12 & 21 & 6 \\ 7 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[515] $1^{132}3^{131}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

[516] $1^{12}3^{113}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 13 & 32 & 15 & 100 & 18 \\ 3 & 5 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[517] $1^{114}3^{14}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 22 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[518] $1^{14}3^{111}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 12 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{12}3^{131}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 13 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

[520] $1^{12} 3^{141} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

[521] $1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[522] $1^{114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 11 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[527] $1^{123} 3^{112} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 4 & 8 & 9 & 30 & 5 \\ 13 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[524] $1^{123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 7 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[525] $1^{12} 3^{123} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 4 & 8 & 9 & 30 & 5 \\ 12 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[526] $1^{13} 3^{141} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

[528] $1^{123} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 7 & 8 & 9 & 24 & 5 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[523] $1^{11} 3^{114} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[529] $1^{11} 3^{123} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 4 & 8 & 9 & 24 & 5 \\ 9 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[530] $1^{122} 3^{111} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[531] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[532] $1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[533] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 32 & 15 & 100 & 18 \\ 3 & 5 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[534] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[535] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[536] $1^{14} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 46 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[537] $1^{114} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 2 \end{pmatrix}_{b_0}$$

[538] $1^{14} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 15 & 0 & 2 \end{pmatrix}_{b_0}$$

[539] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 12 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[540] $1^{13} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[542] $1^1 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 9 & 4 & 48 & 5 \\ 2 & 4 & 0 & 24 & 2 \\ 20 & 48 & 24 & 238 & 28 \\ 3 & 5 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[541] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 3 \\ 4 & 9 & 4 & 48 & 5 \\ 2 & 4 & 0 & 24 & 2 \\ 22 & 48 & 24 & 238 & 28 \\ 3 & 5 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

[543] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[544] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[545] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[546] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[547] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[548] $1^{112} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 14 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[549] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[550] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[551] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 11 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[552] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[553] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 11 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[554] $1^{113} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[555] $1^{113} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[563] $21^{121} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 8 & 4 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[556] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[557] $1^{13} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[558] $1^{114} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 9 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[559] $1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[560] $1^{114} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[561] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[562] $1^1 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[565] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[564] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 32 & 15 & 100 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[567] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[566] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[568] $1^{12} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[575] $1^{123} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 9 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[576] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[569] $1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 4 & 4 & 0 & 21 & 2 \\ 20 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[570] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[571] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[572] $1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 4 & 4 & 0 & 21 & 2 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[573] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 48 & 27 & 12 \\ 5 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[574] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 13 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[577] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[578] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[579] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[580] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[581] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[583] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 16 & 32 & 12 & 94 & 16 \\ 3 & 5 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[584] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 12 & 27 & 6 \\ 9 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[585] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 9 & 4 & 48 & 5 \\ 2 & 4 & 0 & 24 & 2 \\ 21 & 48 & 24 & 238 & 28 \\ 3 & 5 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

[582] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 32 & 12 & 94 & 16 \\ 3 & 5 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[586] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 15 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[587] $1^{123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 4 & 8 & 9 & 24 & 5 \\ 10 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[588] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[589] $1^{12}3^11^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 12 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[593] $1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 2 \end{pmatrix}_{b_0}$$

[590] $1^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[591] $1^{11}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 9 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[592] $1^{122}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 27 & 30 & 9 \\ 7 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[594] $1^{131}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 13 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[595] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 12 & 21 & 6 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[596] $1^{131}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[597] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[598] $21^{12} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[599] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[602] $1^{14} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 13 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[603] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 12 & 21 & 6 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[604] $1^{114} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 12 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[605] $1^{123} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[606] $1^{122} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[607] $1^{123} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[600] $1^{121} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[601] $1^{121} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[608] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[611] $1^{13} 1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 13 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[610] $1^{112} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[609] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[612] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 12 & 27 & 6 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[613] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 11 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[614] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 7 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[615] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[617] $1^{111} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 11 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[616] $21^{111} 31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 9 & 4 \\ 5 & 4 & 13 & 9 & 4 \\ 12 & 13 & 27 & 21 & 9 \\ 9 & 9 & 21 & 13 & 7 \\ 4 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[619] $1^{123} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[618] $1^{123} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[620] $1^{11} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[622] $21^{123} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[621] $21^{123} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 13 & 12 & 4 \\ 9 & 13 & 27 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[624] $1^{111} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 27 & 30 & 9 \\ 7 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[625] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 13 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[623] $1^{123} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 16 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[626] $1^{14} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 14 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[627] $1^{14} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 46 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[628] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[629] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 15 & 0 & 2 \end{pmatrix}_{b_0} \\
[630] \quad & 21^{123} 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -187, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 12, \quad T \in [10, 12] \\
& \begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 22 & 4 & 4 \\ 21 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 4 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0} \\
[631] \quad & 1^{123} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -188, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.21 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 7 & 13 & 27 & 39 & 9 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 1 \end{pmatrix}_{b_0} \\
[632] \quad & 1^{14} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -188, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.26 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 4 & 5 & 3 & 18 & 3 \\ 15 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0} \\
[633] \quad & 1^{113} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -189, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.34 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0} \\
[634] \quad & 21^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -189, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.11 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 9 & 9 & 4 \\ 5 & 4 & 9 & 9 & 4 \\ 9 & 9 & 12 & 15 & 6 \\ 9 & 9 & 15 & 13 & 7 \\ 4 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0} \\
[635] \quad & 1^{122} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -190, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.84 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 27 & 30 & 9 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0} \\
[636] \quad & 21^{111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -192, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.12 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 9 & 12 & 4 \\ 5 & 4 & 9 & 13 & 4 \\ 9 & 9 & 12 & 21 & 6 \\ 12 & 13 & 21 & 28 & 10 \\ 4 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0} \\
[637] \quad & 21^{121} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -192, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.17 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 18 & 4 & 4 \\ 12 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0} \\
[638] \quad & 1^{112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.10 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[639] \quad & 1^{112} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.10 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[640] \quad & 1^3 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.55 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[641] $1^{14} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 46 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[642] $1^{14} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[643] $1^{123} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[644] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[645] $1^{114} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 15 & 32 & 12 & 94 & 16 \\ 3 & 5 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[646] $1^{114} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 14 & 32 & 12 & 94 & 16 \\ 3 & 5 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[647] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[648] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[649] $1^{131} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[650] $1^{123} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 27 & 39 & 9 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[651] $1^{13} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 2 \end{pmatrix}_{b_0}$$

- [652] $1^{114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -204$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 20 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 1 \end{pmatrix}_{b_0}$
- [653] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -204$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [654] $1^{123}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -205$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [655] $1^{11}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -205$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$
- [656] $21^{111}31^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -205$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 12 & 15 & 4 \\ 5 & 4 & 13 & 17 & 4 \\ 12 & 13 & 27 & 39 & 9 \\ 15 & 17 & 39 & 49 & 13 \\ 4 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$
- [660] $1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -207$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.6$
 $T_{\min} = 10.04 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 15 & 0 & 2 \end{pmatrix}_{b_0}$
- [659] $1^{131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -207$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 48 & 27 & 12 \\ 7 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$
- [658] $21^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -207$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [657] $1^{123}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -207$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 10 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$
- [661] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -207$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 8 & 9 & 12 & 27 & 6 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [662] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -208$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [663] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -208$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[664] \ 1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[665] \ 1^{114} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 10 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[666] \ 1^{114} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[667] \ 1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[668] \ 1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[669] \ 1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[670] \ 1^{114} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 21 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[672] \ 1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 12 & 21 & 6 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[671] \ 1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[677] \ 1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[674] \ 1^{112} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[675] \ 1^{131} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 13 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[676] \ 1^{131} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[673] \ 1^{121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 27 & 21 & 9 \\ 7 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[678] \ 1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 18 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[679] \ 1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 12 & 21 & 6 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[680] \ 1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[683] \ 1^{143} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[681] \ 21^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 13 & 8 & 4 \\ 9 & 13 & 27 & 18 & 9 \\ 6 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[682] \ 21^{112} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 3 & 12 & 4 \\ 8 & 9 & 4 & 16 & 5 \\ 3 & 4 & 0 & 6 & 2 \\ 12 & 16 & 6 & 22 & 8 \\ 4 & 5 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[685] \ 1^{114} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 20 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[684] \ 1^{114} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 14 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[686] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 10 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[687] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 12 & 27 & 6 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[688] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 9 & 30 & 5 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[689] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[690] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[691] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[692] $1^{141} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[693] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[694] $1^{13} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 18 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[695] $21^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 15 & 4 \\ 8 & 9 & 4 & 20 & 5 \\ 3 & 4 & 0 & 9 & 2 \\ 15 & 20 & 9 & 37 & 11 \\ 4 & 5 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[698] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[696] $1^{114} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 4 & 4 & 0 & 24 & 2 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[697] $1^{114} 3^1 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[699] $1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -225, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 12 & 27 & 6 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[700] $1^{114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[701] $1^{131} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[702] $1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[703] $1^{131} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[705] $1^{123} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[704] $1^{12} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[706] $1^{123} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 10 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[707] $1^{131} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[708] $1^{111} 3^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 7 & 13 & 27 & 39 & 9 \\ 9 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[709] $21^{122} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[710] $21^{121} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 13 & 4 & 4 \\ 9 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[711] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[715] $1^{131} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[712] $1^{112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[713] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[714] $1^{11} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 11 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[719] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[716] $1^{12} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[717] $1^{112} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[718] $1^{112}3^11^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 20 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[720] $1^{113}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[721] $1^{114}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[722] $1^{131}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[723] $1^{114}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 12 & 33 & 6 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[724] $1^{123}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 7 & 8 & 9 & 24 & 5 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[725] $1^{114}3^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 17 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[726] $21^{131}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 12 & 4 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[727] $1^{123}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[728] $21^{131}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[729] $1^{113}3^11^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[730] $1^{123} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 14 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[731] $21^{123} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 18 & 18 & 12 & 58 & 14 \\ 4 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[732] $1^{112} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[733] $21^{111} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 12 & 4 \\ 5 & 4 & 13 & 13 & 4 \\ 12 & 13 & 27 & 30 & 9 \\ 12 & 13 & 30 & 28 & 10 \\ 4 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[734] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[735] $1^{114} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -245, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 20 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[737] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 12 & 21 & 6 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[736] $1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[738] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[739] $1^{114} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 19 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[740] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[741] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[742] $1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[743] $1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[744] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[745] $1^{11}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[746] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[747] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[748] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 19 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[749] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[750] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 27 & 30 & 9 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[751] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -257, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 17 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[752] $1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 12 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[753] $1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[754] $1^{14} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 19 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[755] $1^{13} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

[756] $1^{14} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 17 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 2 \end{pmatrix}_{b_0}$$

[757] $1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 12 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[758] $1^{14} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[759] $21^{122} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[760] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[766] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[765] $1^{113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[761] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[762] $1^{122}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 27 & 30 & 9 \\ 11 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[763] $1^{112}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 18 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[764] $1^{112}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 19 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[767] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -269, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[768] $1^{12}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -270, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[769] $1^{123}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 7 & 13 & 27 & 39 & 9 \\ 10 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[770] $1^{123}31^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 9 & 30 & 5 \\ 16 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[771] $1^{123}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[772] $1^{122}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 27 & 30 & 9 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[776] $1^{112}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 19 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[773] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[774] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[775] $1^{114} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[777] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -277, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[778] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 13 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[779] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 27 & 30 & 9 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[780] $21^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 24 & 4 \\ 8 & 9 & 4 & 32 & 5 \\ 3 & 4 & 0 & 15 & 2 \\ 24 & 32 & 15 & 100 & 18 \\ 4 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[781] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 12 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[782] $1^{114} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[783] $1^{14} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -282, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[784] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -282, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[785] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[786] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 18 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[787] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[788] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[789] $21^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[792] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[791] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[790] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 16 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[793] $21^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[794] $21^{123} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 8 & 22 & 4 \\ 6 & 8 & 9 & 30 & 5 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[795] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

- [796] $1^{12} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$
- [797] $21^{14} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 15 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$
- [798] $21^{14} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.53 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 33 & 4 \\ 8 & 9 & 4 & 44 & 5 \\ 3 & 4 & 0 & 21 & 2 \\ 33 & 44 & 21 & 193 & 25 \\ 4 & 5 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$
- [799] $1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -298, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$
- [800] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -299, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [801] $1^{112} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [802] $21^{113} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [803] $1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$
- [804] $1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [805] $1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$
- [806] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$
- [807] $21^{112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[808] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -312, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[809] $21^{13} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 24 & 23 & 15 & 100 & 18 \\ 4 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[810] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[811] $1^{13} 1^3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 15 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[812] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[814] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 12 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[813] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[817] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 20 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[816] $1^{12} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[815] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 12 & 27 & 6 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[819] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[818] $1^{122} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[823] $21^{12} 31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 21 & 22 & 15 & 85 & 17 \\ 4 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[820] $1^{113} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 13 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[821] $1^{113} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[822] $1^{114} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 4 & 4 & 0 & 21 & 2 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[824] $1^{123} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 10 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[825] $1^{122} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -335, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 27 & 30 & 9 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[826] $1^{12} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[827] $1^{113} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[828] $1^{14} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[829] $21^{14} 31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 30 & 28 & 18 & 154 & 22 \\ 4 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[830] $21^{131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -343, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 4 & 17 & 9 & 4 \\ 12 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[831] $21^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -345, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[832] $1^{114}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[833] $21^{123}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[834] $21^{113}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[835] $1^{12}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[836] $1^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[837] $1^{14}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 18 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[838] $1^{113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[839] $1^{14}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[842] $21^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 18 & 4 \\ 8 & 9 & 4 & 24 & 5 \\ 3 & 4 & 0 & 9 & 2 \\ 18 & 24 & 9 & 52 & 12 \\ 4 & 5 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

- [841] $1^{12} 1^3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 14 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$
- [840] $1^{14} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$
- [843] $1^{14} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -366, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$
- [844] $1^{13} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$
- [845] $1^{14} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$
- [846] $1^{14} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$
- [847] $1^{13} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [848] $1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -372, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$
- [849] $1^{13} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [850] $21^{12} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 24 & 26 & 18 & 118 & 20 \\ 4 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$
- [851] $21^{13} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 17 & 4 & 4 \\ 12 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$
- [852] $1^{14} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$\begin{aligned}
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0} \\
[863] & 21^{13} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0} \\
[865] & 1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[866] & 1^{13} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -411, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 17 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0} \\
[868] & 1^{14} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[867] & 1^{12} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0} \\
[869] & 1^{13} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0} \\
[870] & 1^{14} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -420, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0} \\
[871] & 1^{14} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 18 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0} \\
[872] & 1^{14} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 21 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0} \\
[873] & 21^{13} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -433, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 8 & 3 & 21 & 4 \\ 8 & 9 & 4 & 28 & 5 \\ 3 & 4 & 0 & 12 & 2 \\ 21 & 28 & 12 & 73 & 15 \\ 4 & 5 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[875] & 1^{13} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[874] $1^{123} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[876] $1^{113} 3 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -443, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 19 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[878] $1^{113} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[877] $1^{113} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[879] $1^{12} 3 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[880] $1^{123} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 27 & 39 & 9 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[881] $1^{114} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[882] $1^{114} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -460, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 15 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[883] $1^{113} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -464, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[884] $1 3 1^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 17 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[885] $1^{113} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[886] & 21^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0} \\
[887] & 21^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[889] & 21^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -481, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0} \\
[888] & 1^{13}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -481, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0} \\
[890] & 1^{14}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -486, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[891] & 21^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -487, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0} \\
[892] & 1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[894] & 21^{133}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 27 & 27 & 18 & 133 & 21 \\ 4 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0} \\
[895] & 1^{113}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0} \\
[893] & 1^{113}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 20 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[896] & 1^{112}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -501, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [897] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [898] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -511, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$
- [899] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -514, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [900] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -515, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$
- [901] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 21 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$
- [902] $21^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [903] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$
- [904] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -524, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [905] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -534, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$
- [906] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -534, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [907] $1 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 24 & 2 \\ 20 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [908] $21^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -547, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[909] \ 1^{113} 3 \ 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -555, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[910] \ 1^{114} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -556, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[911] \ 1^{114} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -556, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[912] \ 1^{114} 3 \ 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 18 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[913] \ 21^{114} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -574, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 3 & 24 & 4 \\ 8 & 9 & 4 & 32 & 5 \\ 3 & 4 & 0 & 12 & 2 \\ 24 & 32 & 12 & 94 & 16 \\ 4 & 5 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[914] \ 1^{114} 3 \ 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -576, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 19 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[916] \ 21^{114} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[915] \ 1^{13} 3 \ 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[917] \ 1^{113} 3 \ 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -591, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[918] \ 1^{114} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[919] \ 1^{114} 3 \ 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 22 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0} \\
[920] \quad & 21^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[921] \quad & 21^{14}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -610, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 8 & 3 & 30 & 4 \\ 8 & 9 & 4 & 40 & 5 \\ 3 & 4 & 0 & 18 & 2 \\ 30 & 40 & 18 & 154 & 22 \\ 4 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[922] \quad & 21^{14}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 33 & 32 & 21 & 193 & 25 \\ 4 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[923] \quad & 1^{14}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -624, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0} \\
[924] \quad & 21^{13}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 30 & 31 & 21 & 172 & 24 \\ 4 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0} \\
[925] \quad & 1^{14}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -643, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 19 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0} \\
[926] \quad & 1^{13}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -645, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0} \\
[927] \quad & 1^{14}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0} \\
[928] \quad & 1^{14}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 24 & 2 \\ 21 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0} \\
[929] \quad & 1^{14}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0} \\
[930] \quad & 21^{123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -673, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[943] \ 1^{14} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -800, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 24 & 2 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[944] \ 1^{14} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -838, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[945] \ 21^{15} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -844, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 3 & 30 & 4 \\ 8 & 9 & 4 & 40 & 5 \\ 3 & 4 & 0 & 15 & 2 \\ 30 & 40 & 15 & 148 & 20 \\ 4 & 5 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[946] \ 21^{14} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 36 & 4 \\ 5 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 36 & 36 & 24 & 238 & 28 \\ 4 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[947] \ 1^{14} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[948] \ 21^{14} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -904, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 24 & 3 \\ 4 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[949] \ 21^{16} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -952, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 24 & 18 & 106 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[950] \ 21^{14} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -964, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[951] \ 21^{14} 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -1003, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[952] \ 21^{17} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -1264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 28 & 21 & 145 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [954] \ 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[955] $21^{1^{11}} 4 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 6 & 4 \\ 5 & 4 & 7 & 7 & 4 \\ 6 & 7 & 7 & 10 & 5 \\ 6 & 7 & 10 & 8 & 6 \\ 4 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[958] $21^{1^{11}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[953] $1^{1^{11}} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[956] $1^{1^{11}} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[957] $21^{1^{12}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 7 & 11 & 4 \\ 6 & 7 & 7 & 16 & 5 \\ 12 & 11 & 16 & 28 & 10 \\ 4 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[960] $1^{1^{12}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 7 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[961] $1^{1^{12}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 5 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[959] $1^{1^{11}} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[964] $21^{1^{11}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[962] $1^{1^{11}} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[963] $1^1 4^1 1^1 + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{5}\textcolor{blue}{u}_3 + \textcolor{red}{5}\textcolor{blue}{u}_2 \quad T = 8, \, T_{\max} = 11, \, \Delta = -84, \, \det = 7, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$
 $\left(\begin{array}{cccc} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{array} \right)_{b_0}$

[965] $1 \overset{1^1}{1^1} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\left(\begin{smallmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{smallmatrix} \right)_{b_0}$

[967] $1^1 4^1 1^1 + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{5}\textcolor{blue}{u}_3 + \textcolor{red}{5}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[966] $21^{\overline{1}1} \overline{1} 4 \overline{1}^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[968] $1^1 1^1 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[969] $1 \overset{11}{4} 1^1 + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{blue}{u}_3 + \textcolor{red}{5}\textcolor{brown}{u}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 10, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

$$[970] \ 1^{\overline{11}} \overline{3} \ 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[971] $21^{\overline{1}^{11}}41^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \, T_{\max} = 11, \, \Delta = -84, \, \det = 12, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 4 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[972] $1 \overset{1^1}{\underset{1^1}{2}} 4 \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 11, \, \Delta = -84, \, \det = 13, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 6 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[973] $1\overset{1}{4}1\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$

[974] $21^1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 3 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 1 & 3 \\ 4 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[975] \ 1^{11} 1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 7 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[976] \ 1^{11} 1^1 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 5 & 4 \\ 5 & 7 & 7 & 6 & 5 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[977] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & -3 & 0 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[978] \ 2 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 15, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 7 & 8 & 4 \\ 6 & 7 & 7 & 11 & 5 \\ 9 & 8 & 11 & 13 & 7 \\ 4 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[983] \ 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 \\ 1 & 2 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 \\ 2 & 1 & 3 & 0 & 1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[984] \ 2 1^{11} 1^2 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 & 3 \\ 4 & 4 & 10 & 4 & 4 \\ 7 & 10 & 20 & 7 & 8 \\ 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[980] \ 1^1 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 5 & 4 \\ 4 & 7 & 7 & 6 & 5 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[979] \ 2 1^1 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 4 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[981] \ 1^{11} 1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 7 & 13 & 4 \\ 4 & 7 & 7 & 20 & 5 \\ 9 & 13 & 20 & 40 & 12 \\ 3 & 4 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[982] \ 2 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 5 & 3 & 4 & 4 \\ 4 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[985] $1^{11} 4^{12} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 4 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[987] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 5 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[986] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 5 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$

[988] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[989] $1^{111} 4^{112} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 9 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[990] $1^{111} 4^{111} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 7 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[991] $21^1 4^{111} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 4 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 3 & 4 & 5 & 1 & 3 \\ 4 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[992] $1^{111} 4^{112} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 6 & 10 & 8 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[993] $221^{11} 4^{111} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 6 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 6 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[994] $1^{111} 4^{111} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[995] $21^{11} 4^{111} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[996] & 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[997] & 21^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 4 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[998] & 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[999] & 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0} \\
[1000] & 14^{11} 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[1001] & 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[1004] & 14^{11} 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 7 & 5 & 5 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[1005] & 14^{11} 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 4 & 7 & 7 & 11 & 5 \\ 5 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[1006] & 14^{11} 1^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 5 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[1003] & 1^{12} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 7 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1002] $1^{11} 1^{13} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 9 & 10 & 8 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1007] $1^{11} 1^{12} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 6 & 7 & 7 & 10 & 5 \\ 4 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[1008] $1^{11} 1^1 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -1 & 8 & 4 \end{pmatrix}_{b_0}$

[1009] $1^{11} 1^1 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -1 & 8 & 4 \end{pmatrix}_{b_0}$

[1010] $1^{13} 1^{12} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 10 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1011] $1^{11} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 6 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[1012] $221^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 8 & 7 \\ 5 & 12 & 7 & 6 & 5 \\ 3 & 8 & 6 & 4 & 4 \\ 3 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

[1013] $21^{11} 1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 6 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[1014] $1^{11} 1^{12} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1015] $21^{11} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 9 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 5 & 9 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$

[1016] $21^{14} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 7 & 21 & 4 \\ 5 & 7 & 7 & 31 & 5 \\ 17 & 21 & 31 & 101 & 19 \\ 3 & 4 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1017] \ 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 9 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 5 & 9 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1018] \ 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 7 & 21 & 4 \\ 5 & 7 & 7 & 31 & 5 \\ 17 & 21 & 31 & 101 & 19 \\ 3 & 4 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1019] \ 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1020] \ 1^{\overline{111}} 3^{\overline{21}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1021] \ 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1022] \ 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 5 & 7 & 7 & 26 & 5 \\ 11 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1023] \ 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 11 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1028] \ 1^{\overline{12}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 7 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1029] \ 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1027] \ 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 6 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[1026] $1^{111} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 6 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[1025] $21^{111} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 6 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[1024] $1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1033] $1^{112} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1032] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 9 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1031] $21^{113} 4^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 18 & 4 \\ 5 & 4 & 7 & 18 & 4 \\ 6 & 7 & 7 & 26 & 5 \\ 18 & 18 & 26 & 70 & 16 \\ 4 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1030] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 8 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1034] $21^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 6 & 5 & 3 & 4 & 4 \\ 4 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[1035] $21^{114} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 3 & 33 & 4 \\ 3 & 3 & 0 & 23 & 2 \\ 30 & 33 & 23 & 212 & 28 \\ 4 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1036] $21^{11} 4^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 3 & 21 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 18 & 21 & 11 & 76 & 16 \\ 4 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1037] $21^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[1038] & 1^{\frac{1}{13}} 4^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 10 & 17 & 13 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1039] & 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 7 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[1040] & 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[1041] & 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 4 & 7 & 7 & 5 & 5 \\ 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[1042] & 1^{\frac{1}{12}} 4^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 7 & 10 & 8 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1046] & 21^{\frac{1}{1}} 41^{\frac{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[1047] & 1^{\frac{1}{11}} 4^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.63 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[1045] & 1^{\frac{1}{12}} 4^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 10 & 8 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1043] & 1^{\frac{1}{12}} 41^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[1044] & 1^{\frac{1}{11}} 41^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1048] $21^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1049] $1 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1050] $21^{11} 4 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 7 & 3 \\ 4 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 7 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[1051] $21^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 3 & 8 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 9 & 8 & 6 & 13 & 7 \\ 4 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1052] $21^{11} 4 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 9 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1053] $1^{13} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 9 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1054] $1^{12} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1055] $1^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 8 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1056] $1^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1057] $1^{14} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 15 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1058] $1^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 11 & 18 & 14 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1059] $21^{11} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 7 & 9 & 4 \\ 6 & 7 & 7 & 11 & 5 \\ 9 & 9 & 11 & 13 & 7 \\ 4 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[1060] $1^{14} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 14 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1061] $1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1062] $21^{11} 41^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1063] $1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 9 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1064] $1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[1065] $1^4 41^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1066] $1^{13} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 12 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1067] $1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 7 & 21 & 4 \\ 4 & 7 & 7 & 31 & 5 \\ 14 & 21 & 31 & 101 & 19 \\ 3 & 4 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1069] $21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1070] $1^{111} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 4 & 7 & 7 & 15 & 5 \\ 7 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[1068] $1^{111} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 7 & 9 & 4 \\ 4 & 7 & 7 & 11 & 5 \\ 6 & 9 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$

[1071] $1^{111} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 4 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[1072] $1^{13} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 10 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1073] $1^{111} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[1074] $1^{12} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 10 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1075] $1^{13} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 14 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1076] $1^{112} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 7 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[1078] $1^{111} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 4 & 7 & 7 & 10 & 5 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[1077] $21^{111} 4^{11} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 6 & 12 & 4 \\ 8 & 13 & 12 & 20 & 7 \\ 6 & 12 & 7 & 16 & 5 \\ 12 & 20 & 16 & 26 & 10 \\ 4 & 7 & 5 & 10 & 3 \end{pmatrix}_{b_0}$

[1079] $21^{12} 4^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 8 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[1080] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1081] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1082] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 8 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[1084] $21^{11} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 14 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1083] $1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 14 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1085] $21^{11} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 16 & 2 \\ 21 & 22 & 16 & 97 & 19 \\ 4 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1088] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 6 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[1086] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1087] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1090] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 7 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1089] $1^{111} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 5 & 7 & 7 & 26 & 5 \\ 15 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1093] $1^{111} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 9 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1092] $1^{13} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 13 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1091] $1^{111} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 13 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1097] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 11 & 19 & 13 & 68 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1096] $1^{13} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1095] $1^{12} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1094] $1^4 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 7 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1099] $1^{13} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1098] $1^{12} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 9 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1103] $21^1 4^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 3 & 17 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 15 & 17 & 9 & 49 & 13 \\ 4 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1100] $21^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 9 & 9 & 5 & 13 & 7 \\ 4 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1101] $21^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 15 & 4 \\ 5 & 4 & 7 & 15 & 4 \\ 6 & 7 & 7 & 21 & 5 \\ 15 & 15 & 21 & 45 & 13 \\ 4 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1102] $21^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 15 & 15 & 11 & 45 & 13 \\ 4 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1105] $1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 10 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1104] $1^{14} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 16 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1106] $1^{12} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1109] $21^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[1107] $1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[1110] $1^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 9 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1108] $1^{14} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 12 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1112] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 7 & 12 & 4 \\ 6 & 7 & 7 & 16 & 5 \\ 12 & 12 & 16 & 26 & 10 \\ 4 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$

[1111] $21^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1113] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 9 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1114] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 10 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1115] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 12 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1116] $21^{11} 4 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 12 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1117] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[1118] $21^{11} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 10 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1119] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 10 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1120] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 5 & 10 & 3 \\ 4 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 10 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$

[1121] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 16 & 2 \\ 13 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1122] $1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 10 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1123] $1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1124] $21^{12} 4^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 10 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1125] $1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1126] $1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 13 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1127] $21^{11} 4^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 3 & 13 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 12 & 13 & 7 & 28 & 10 \\ 4 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1128] $1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 11 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1129] $1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 19 & 2 \\ 15 & 25 & 19 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1130] $1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 12 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1131] $1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1132] $1^{12} 4^{14} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 12 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1133] $1^{12} 4^{14} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 13 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 11 & 13 & 7 & 28 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1136] $1^{11} 4^{12} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 11 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1137] $21^{13} 4^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 13 & 17 & 13 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1135] $21^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 11 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1134] $1^{12} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 13 & 17 & 13 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1138] $1^{12} 4^{14} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 9 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1140] $1^{11} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 15 & 18 & 14 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1139] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 13 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 13 & 7 & 28 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1145] $1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 4 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1141] $1^{11} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 4 & 7 & 7 & 26 & 5 \\ 12 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1142] $21^{11} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 12 & 12 & 8 & 26 & 10 \\ 4 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1143] $1^{113} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 10 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1144] $21^{111} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 4 & 7 & 4 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[1146] $1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 9 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1147] $21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1149] $21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 12 & 12 & 8 & 26 & 10 \\ 4 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1148] $21^{11} 41^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 3 & 26 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 24 & 26 & 18 & 130 & 22 \\ 4 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1150] $21^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 3 \\ 4 & 4 & 7 & 9 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 6 & 9 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[1151] $1^{12} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 14 & 17 & 13 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1152] $1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1153] $1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 13 & 18 & 14 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1154] \ 1^{14} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 16 & 8 & 46 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1155] \ 1^{12} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 13 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 10 & 13 & 7 & 28 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1156] \ 21^{11} 4^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 27 & 30 & 20 & 169 & 25 \\ 4 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1157] \ 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 15 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1158] \ 1^{12} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 11 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1159] \ 1^{13} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1160] \ 21^{11} 4^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 21 & 24 & 14 & 101 & 19 \\ 4 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1162] \ 1^{14} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 17 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1165] \ 1^{14} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 16 & 8 & 46 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1167] \ 1^{12} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

[1174] $221^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1175] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1176] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 5 & 7 & 7 & 26 & 5 \\ 14 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1177] $21^{13}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 5 & 7 & 7 & 26 & 5 \\ 14 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1178] $21^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1179] $1^{13}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 12 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1180] $21^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[1181] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 11 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1182] $21^{11}41^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 18 & 19 & 13 & 68 & 16 \\ 4 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1186] $21^{12}41^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 12 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1185] $1^{13}41^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 12 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1184] $1^{111} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 12 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1183] $1^{111} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 9 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1188] $1^{14} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 13 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1189] $21^{13} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1187] $21^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 16 & 2 \\ 15 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1190] $1^{13} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 10 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1191] $21^{11} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 15 & 16 & 10 & 45 & 13 \\ 4 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1192] $21^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1193] $21^{11} 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 24 & 27 & 17 & 132 & 22 \\ 4 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1194] $1^{13} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 3 & 3 & 0 & 21 & 2 \\ 17 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1195] $21^{11} 4 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 18 & 20 & 12 & 70 & 16 \\ 4 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1196] \ 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 12 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1197] \ 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 10 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1198] \ 1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 11 & 19 & 13 & 68 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1199] \ 1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 12 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1201] \ 21^{11} 4 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 3 & 23 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 21 & 23 & 15 & 97 & 19 \\ 4 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1200] \ 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 13 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1206] \ 21^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 5 & 8 & 3 \\ 4 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1202] \ 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 13 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1205] \ 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 14 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1203] \ 21^{12} 4 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 14 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 3 & 3 & 0 & 23 & 2 \\ 19 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[1219] \ 1^{13}_{1^1} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 171, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1218] \ 31^{11} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -80, \ \det = 171, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 6 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 6 & 3 & 6 & 3 & 3 \\ 3 & 6 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1217] \ 1^{13}_{1^{12}} 4^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 171, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 14 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1220] \ 1^{12}_{1^{11}} 4^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 176, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.70 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1221] \ 1^{13}_{1^{13}} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 178, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 13 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1222] \ 21^{12}_{1^1} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 181, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 4 & 3 & 13 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 9 & 13 & 7 & 28 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1223] \ 1^{13}_{1^1} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 183, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 13 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1224] \ 1^{12}_{1^{11}} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 184, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 11 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1225] \ 1^{13}_{1^1} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 186, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 12 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1228] \ 1^{13}_{1^{13}} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.81 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 15 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[illegible]

[1248] $1^{13}_{1^{14}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 16 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1249] $21^{13}_{1^{13}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 15 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1250] $1^{13}_{1^{13}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 14 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1251] $1^{14}_{1^{12}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 13 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1252] $1^{12}_{1^{14}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 19 & 2 \\ 19 & 25 & 19 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1253] $21^{14}_{1^{11}} 4 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 19 & 2 \\ 19 & 25 & 19 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1254] $1^{14}_{1^{11}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 13 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1255] $1^{14}_{1^1} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 13 & 21 & 11 & 76 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1256] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 12 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1257] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 16 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1258] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 225, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

[1269] $1^{14}_{1^{12}} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 17 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[1270] $21^{13}_{1^{11}} 4 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1271] $221^{11}_{1^{11}} 41^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 13 & 12 & 12 & 7 \\ 5 & 12 & 7 & 10 & 5 \\ 5 & 12 & 10 & 8 & 6 \\ 3 & 7 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[1273] $21^{12}_{1^{11}} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 11 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1272] $1^{12}_{1^{13}} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 16 & 2 \\ 15 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[1274] $1^{13}_{1^{14}} 4 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 17 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1275] $221^{11}_{1^{11}} 41^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 5 & 8 & 3 \\ 6 & 13 & 12 & 20 & 7 \\ 5 & 12 & 7 & 16 & 5 \\ 8 & 20 & 16 & 26 & 10 \\ 3 & 7 & 5 & 10 & 3 \end{pmatrix}_{b_0}$

[1276] $1^{13}_{1^{14}} 4 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 18 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1277] $1^{14}_{1^1} 41^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 14 & 21 & 11 & 76 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1278] $21^{13}_{1^1} 41^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 17 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1279] $1^{14}_{1^{13}} 4 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 17 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$

[illegible]

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 13 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1291] \ 1^{14} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 274, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 17 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1292] \ 1^{14} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 281, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 14 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1293] \ 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 282, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.87 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 14 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1294] \ 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 282, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 14 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1295] \ 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 285, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 15 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1296] \ 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 288, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 17 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1297] \ 1^{14} 4^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 289, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 18 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1298] \ 21^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 291, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 12 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1299] \ 21^{15} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 296, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.1$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 15 & 20 & 10 & 73 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1300] \ 1^{14} 4^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 297, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 15 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[illegible]

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 19 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1312] \quad 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 15 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1313] \quad 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 18 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1314] \quad 21^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 20 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1315] \quad 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 17 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1316] \quad 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 354, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 16 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1317] \quad 21^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 4 & 3 & 19 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 14 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1319] \quad 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 3 & 3 & 0 & 21 & 2 \\ 21 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1318] \quad 21^{14} 4 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 3 & 3 & 0 & 21 & 2 \\ 21 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1320] \quad 31^{11} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 3 & 6 \\ 3 & 6 & 3 & 9 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[1321] \quad 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 19 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1322] $1^{13}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 17 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1323] $1^{14}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 16 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1324] $1^{13}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 17 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1325] $1^{14}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 18 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1326] $1^{14}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 16 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1327] $21^{13}_{1^{11}} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 20 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1328] $1^{14}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 20 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1329] $1^{14}_{1^{14}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 19 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1330] $1^{13}_{1^{14}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 2 & 3 & 0 & 21 & 2 \\ 19 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1331] $21^{13}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 18 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1332] $1^{14}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 17 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1334] $1^{13}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 18 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1333] $21^{13}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 14 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1336] $1^{14}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 20 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1335] $21^{16}_{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 18 & 24 & 12 & 106 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1337] $1^{13}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 19 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1338] $21^{14}_{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 15 & 21 & 11 & 76 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1339] $1^{14}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 425, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 18 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1340] $1^{13}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 2 & 3 & 0 & 21 & 2 \\ 20 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1341] $1^{14}_{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 17 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1342] $1^{14}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 21 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1343] \ 21 \overset{13}{\overset{14}{4}} 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 431, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 4 & 3 & 25 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 18 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1344] $21^{\overline{12}} 4^1 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 434, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\left(\begin{array}{ccccc} 1 & 4 & 2 & 16 & 3 \\ 4 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 16 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{array} \right)_{b_0}$

$$[1345] \ 21^{13}4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 435, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.72 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 16 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1346] \begin{matrix} 1^4 \\ 1^{13} 4 \end{matrix} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 11, \, \Delta = -84, \, \det = 447, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.80 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 2 & 3 & 0 & 21 & 2 \\ 21 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1347] \quad \begin{array}{c} 1^1 \\ 1^{14} \end{array} 4 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{brown}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 461, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.97 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\left(\begin{array}{ccccc} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 20 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{array} \right)_{b_0}$$

$$[1348] \ 1 \overset{1^3}{\underset{1^2}{4}} \ 1 \overset{11}{2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}\textcolor{teal}{u}_3 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 469, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\left(\begin{array}{ccccc} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 18 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{array} \right)_{b_0}$$

$$[1350] \ 1 \overset{14}{4} \ 1 \overset{11}{4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 481, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 3 & 3 & 0 & 23 & 2 \\ 23 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1349] $21^{14} 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 11, \, \Delta = -84, \, \det = 481, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$
 $\left(\begin{array}{ccccc} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 3 & 3 & 0 & 23 & 2 \\ 23 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{array} \right)_{b_0}$

$$[1351] \ 1 \overset{1^2}{\underset{13}{4}} 1 \overset{11}{} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 492, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 19 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1352] \ 1 \overset{1^4}{\overset{14}{\underset{12}{4}}} 1 \overset{11}{\overset{1}{\underset{12}{4}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 494, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 19 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1353] $1^{14} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 21 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1354] $21^{17}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 21 & 28 & 14 & 145 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1355] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 20 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1356] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 21 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1357] $21^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 17 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1358] $21^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 23 & 3 \\ 4 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 23 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1359] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 22 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1360] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 23 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1361] $21^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 3 \\ 4 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 19 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1362] $21^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 21 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [1364] $1^{11}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[1363] $1^{11} 2^1 3^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

[1365] $1^{11} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[1366] $1^{11} 4^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$

[1367] $1^{11} 4^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 6 & 7 & 1 & 9 & 7 \\ 3 & 4 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

[1368] $1^{11} 4^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 6 & 4 & 6 \\ 4 & 2 & 4 \\ 6 & 4 & 6 \end{pmatrix}_{b_0}$

[1369] $1^{11} 4^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 4 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1375] $1^{11} 4^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 3 & 4 & 1 & 2 & 4 \\ 3 & 4 & 0 & 4 & 4 \end{pmatrix}_{b_0}$

[1377] $1^{11} 5^1 1^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1376] $1^{11} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 7 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1370] $1^{11} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 6 & 4 & 4 \\ 4 & 6 & 7 & 4 & 5 \\ 2 & 4 & 4 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1371] $1^{11} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 4 & 4 \\ 4 & 6 & 7 & 4 & 5 \\ 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1372] $21^{\overline{1}^{11}} 5 1^{\overline{1}^1} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 6 & 4 & 4 \\ 6 & 6 & 7 & 4 & 5 \\ 3 & 4 & 4 & 1 & 3 \\ 4 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[1373] $21^{\overline{1}^{11}} 5 1^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 6 & 4 & 4 \\ 6 & 6 & 7 & 4 & 5 \\ 3 & 4 & 4 & 1 & 3 \\ 4 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[1374] $21^{\overline{1}^{11}} 5 1^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 4 & 2 \\ 4 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1381] $1^{\overline{1}^{11}} 3 221^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 1 & 4 & 0 & 2 & 4 \\ 2 & 2 & 3 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[1380] $1^{\overline{1}^{11}} 5 1^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1379] $1^{\overline{1}^{11}} 5 1^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1378] $21^{\overline{1}^{11}} 5 1^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1383] $21^{\overline{1}^{11}} 5 1^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1382] $1^{\overline{1}^{11}} 5 1^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[1385] $1^{\overline{1}^{11}} 4 131^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 3 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 0 & 2 \\ 3 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[1384] $1^{\overline{1}^{131^1}} 4 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 3 & 4 \\ 3 & 1 & 2 & 3 \\ 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[1387] $1^{11} 4^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[1388] $1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[1386] $1^{11} 5^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1390] $21^{11} 5^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 4 & 2 \\ 6 & 7 & 4 & 8 & 6 \\ 4 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1389] $1^{11} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1391] $1^{11} 4^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 5 & 5 \\ 3 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[1392] $1^{12} 5^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1394] $1^{11} 2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1393] $21^{11} 5^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1395] $21^{11} 5^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 7 & 4 & 5 \\ 3 & 3 & 4 & 1 & 3 \\ 3 & 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[1396] $21^{11}_{1^1} 5^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 7 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1397] $1^1 5^1 1^1 2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 4 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1398] $221^{11}_{5^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 5 & 5 & 7 & 4 & 5 \\ 2 & 3 & 4 & 1 & 3 \\ 3 & 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1399] $1^1 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 6 & 6 \\ 4 & 3 & 6 & 5 \end{pmatrix}_{b_0}$

[1400] $1^1 5^1 1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1401] $21^{11}_{5^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1402] $1^1 4^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 3 & 3 \\ 5 & 3 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$

[1403] $1^{11} 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 4 & 4 \\ 3 & 1 & 4 & 3 \\ 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$

[1404] $1^1 4^{11} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 7 & 4 \\ 3 & 1 & 5 & 3 \\ 7 & 5 & 12 & 8 \\ 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$

[1405] $1^1 5^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1407] $1^{11}_{1^2, 1^1} 5^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1408] $1^{11} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$

[1406] $21^{21} 5^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1409] $1^{11} 5^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1410] $1^{13} 5^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1411] $1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 8 & 6 \\ 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$

[1413] $21^{11} 5^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1412] $1^{11} 5^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1415] $1^{11} 5^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1414] $1^{12} 5^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 10 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1416] $1^{12} 5^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1418] $21^{14} 5^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 16 & 20 & 12 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1417] \ 1^{12} \overset{1^4}{5} \overset{1^4, 1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 16 & 20 & 12 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1420] \ 1^{11} \overset{1^{11}}{4} \overset{1^{11}}{131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 6 & 1 & 14 & 8 \\ 3 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1419] \ 21^{12} \overset{1^3}{5} \overset{1^3}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 12 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1421] \ 1^1 \overset{1^1}{5} \overset{1^1}{1^1} 1^{11} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1422] \ 1^{14} \overset{1^4}{5} \overset{1^4, 1^1}{1^1} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 14 & 24 & 14 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1423] \ 1^{12} \overset{1^4}{5} \overset{1^4, 1^1}{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1424] \ 21^{11} \overset{1^1}{5} \overset{1^1}{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1425] \ 1^{11} \overset{1^{11} 31^1}{4} \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 9 & 7 \\ 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1426] \ 1^{12} \overset{1^4}{5} \overset{1^4, 1^1}{1^3, 1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 14 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1429] \ 1^{12} \overset{1^3}{5} \overset{1^3, 1^1}{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1428] & 1^{12} 5^{1^1} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 7 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1427] & 21^{1^1} 5^{1^1} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 7 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1430] & 1^{1^2} 5^{1^1} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1431] & 1^{1^1} 4^{1^{12} 31^1} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 10 & 4 \\ 3 & 1 & 7 & 3 \\ 10 & 7 & 21 & 11 \\ 4 & 3 & 11 & 5 \end{pmatrix}_{b_0} \\
[1432] & 1^{1^3} 5^{1^1} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 12 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1433] & 1^{1^2} 5^{1^2} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1434] & 1^{1^2} 5^{1^3} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1435] & 21^{1^3} 5^{1^1} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1436] & 1^{1^2} 5^{1^3} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1437] & 15^{1^1} 1^{1^2} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1438] $1^{12} \overset{1^2}{5} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1439] $21^{\overset{1^1}{2} \overset{1^1}{5} 1^{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1440] $1^{13} \overset{1^1}{5} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 9 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1441] $21^{\overset{1^1}{1} \overset{1^1}{5} 1^{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 9 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1442] $1^{11} \overset{1^{11} 31^1}{4} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 27 & 11 \\ 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[1443] $21^{\overset{1^2}{12} \overset{1^1}{5} 1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1444] $1^{13} \overset{1^4}{5} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1445] $221^{\overset{1^1}{11} \overset{1^1}{5} 1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 2 & 5 & 3 \\ 6 & 13 & 5 & 12 & 7 \\ 2 & 5 & 0 & 4 & 2 \\ 5 & 12 & 4 & 8 & 6 \\ 3 & 7 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1446] $1^{13} \overset{1^2}{5} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1447] $21^{\overset{1^4}{13} \overset{1^1}{5} 1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 17 & 24 & 14 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1448] $21^{\overset{1^1}{12} \overset{1^1}{5} 1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1449] \ 1^{13} \overset{1^3}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -127, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1450] \ 1^{13} \overset{1^1}{5} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1451] \ 21^2 \overset{1^1}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -149, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 12 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1452] \ 1^{13} \overset{1^2}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -149, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 12 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1453] \ 1^{13} \overset{1^4}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 14 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1454] \ 1^{13} \overset{1^3}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1455] \ 1^{13} \overset{1^4}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 16 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1456] \ 1^{13} \overset{1^4}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 18 & 24 & 14 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1457] \ 21^4 \overset{1^1}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 18 & 24 & 14 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1458] \ 1^{13} \overset{1^3}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -175, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 15 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1459] $21^{3\frac{1}{1}51^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 15 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1460] $21^{13\frac{1}{1}51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 15 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1461] $21^{13\frac{1}{1}51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1462] $221^{12\frac{1}{1}51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 2 & 10 & 3 \\ 6 & 13 & 5 & 24 & 7 \\ 2 & 5 & 0 & 8 & 2 \\ 10 & 24 & 8 & 38 & 12 \\ 3 & 7 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1463] $21^{13\frac{1}{1}51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1464] $221^{13\frac{1}{1}51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 2 & 15 & 3 \\ 6 & 13 & 5 & 36 & 7 \\ 2 & 5 & 0 & 12 & 2 \\ 15 & 36 & 12 & 88 & 18 \\ 3 & 7 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [1466] $21^{1\frac{11}{1}5131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1465] $1^{11\frac{1}{1}5131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1471] $1^{1\frac{12}{1}5131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 4 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1472] $1^1 51^1 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 3 & 3 & 0 & 4 & 10 & 6 \\ 2 & 2 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1469] $1^{11} \bar{5} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[1468] $1^{11} \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 7 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1467] $1^{13} \bar{5} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 1 & 18 & 3 \\ 13 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1470] $1^{11} \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1473] $1^{11} \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[1474] $1^{11} \bar{5} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[1475] $1^{11} \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[1477] $1^{11} \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[1476] $1^{11} \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1478] $1^{11} \bar{5} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[1479] $1^{11} 4^{11} 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 6 & 4 \\ 3 & 2 & 5 & 4 \\ 6 & 5 & 10 & 8 \\ 4 & 4 & 8 & 6 \end{pmatrix}_{b_0}$$

[1483] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[1482] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 1 & 6 & 3 \\ 4 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1481] $1^{12} 5^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 1 & 12 & 3 \\ 8 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1480] $1^{13} 5^{11} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 3 & 3 & 1 & 18 & 3 \\ 12 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1484] $1^{12} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 4 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[1485] $2^{11} 3^{11} 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1488] $21^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 6 & 5 & 5 \\ 2 & 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[1487] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1486] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 6 & 5 & 5 \\ 2 & 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[1493] $1^{11} 5^{12} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[1494] $1^{11} 5^{12} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 \\ 2 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[1495] $1^{11} 5^{12} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1491] $21^{11} 5^{12} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 \\ 2 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[1490] $1^{11} 5^{12} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1489] $1^{11} 5^{12} 13^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 3 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[1492] $231^{11} 5^{12} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1496] $1^{12} 5^{12} 13^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 1 & 12 & 3 \\ 9 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1497] $1^{12} 5^{12} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 6 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1498] $1^{11} 4^{12} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 2 & 5 & 4 \\ 5 & 5 & 7 & 7 \\ 4 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

[1499] $1^{11} 5^{12} 23^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 12 & 2 \\ 8 & 9 & 0 & 12 & 81 & 17 \\ 2 & 2 & -1 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1500] \ 1^{11} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1501] \ 1^{11} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1503] \ 1^{11} 5^{11} 1^{11} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 8 & 2 \\ 5 & 6 & 0 & 8 & 31 & 11 \\ 2 & 2 & -1 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1502] \ 1^{13} 5^{11} 1^{11} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 16 & 2 \\ 9 & 12 & 0 & 16 & 115 & 21 \\ 2 & 2 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1504] \ 2^3 3^1 5^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 1 & 12 & 3 \\ 10 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1505] \ 1^{13} 5^{11} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 7 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1506] \ 2^3 3^1 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1507] \ 21^{12} 5^{11} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1508] \ 1^{12} 5^{11} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 1 & 12 & 3 \\ 12 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1509] \ 1^{12} 5^{11} 1^{11} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 12 & 2 \\ 7 & 9 & 0 & 12 & 66 & 16 \\ 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1510] $1^{1^2}_1 5^{13} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1511] $1^{1^2}_1 5^{13} 13^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$

[1512] $1^{1^2}_1 5^{12} 13^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[1514] $1^{1^1}_{1^1} 5^{13} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1513] $1^{1^3}_{1^1} 5^{13} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 7 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1515] $1^{1^1}_{1^1} 5^{11} 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$

[1516] $1^{1^3}_1 5^{13} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 1 & 18 & 3 \\ 14 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[1517] $1^{1^3}_1 5^{12} 13^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 1 & 12 & 3 \\ 11 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[1518] $1^{1^1}_{1^1} 5^{13} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 4 & 8 & 8 & 6 & 6 \\ 3 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$

[1519] $1^{1^2}_1 5^{13} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 5 & 8 & 10 & 5 \\ 4 & 8 & 8 & 12 & 6 \\ 6 & 10 & 12 & 14 & 8 \\ 3 & 5 & 6 & 8 & 4 \end{pmatrix}_{b_0}$

[1520] $1^{1^3}_{1^1} 5^{13} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 5 & 8 & 15 & 5 \\ 4 & 8 & 8 & 18 & 6 \\ 9 & 15 & 18 & 34 & 12 \\ 3 & 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1521] \ 1^{\overline{1}^4} 5 131^{\overline{1}^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 5 & 8 & 20 & 5 \\ 4 & 8 & 8 & 24 & 6 \\ 12 & 20 & 24 & 62 & 16 \\ 3 & 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1522] \ 2^{\overline{1}^{13}} 3^{\overline{1}^1} 151^{\overline{1}^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 3 & 3 & 1 & 18 & 3 \\ 15 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1523] \ 1^{\overline{1}^4} 5 131^{\overline{1}^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 10 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1524] \ 1^{\overline{1}^{12}} 5^{\overline{1}^1} 1^{\overline{2}^{23}} 1^{\overline{1}^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 16 & 2 \\ 10 & 12 & 0 & 16 & 130 & 22 \\ 2 & 2 & -1 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1525] \ 1^{\overline{1}^{12}} 5^{\overline{1}^1} 1^{\overline{3}^{23}} 1^{\overline{1}^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 20 & 2 \\ 13 & 15 & 0 & 20 & 218 & 28 \\ 2 & 2 & -1 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[1526] \ 1^{\overline{1}^{11}} 5^{\overline{1}^1} 1^{\overline{1}^{23}} 1^{\overline{1}^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 15 & 0 & 37 & 13 \\ 3 & 5 & -1 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[1527] \ 1^{\overline{1}^{13}} 5 131^{\overline{1}^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1528] \ 2^{\overline{1}^{11}} 3^{\overline{1}^1} 151^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1529] \ 1^{\overline{1}^{11}} 5^{\overline{1}^1} 1^{\overline{1}^{23}} 1^{\overline{1}^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 1 & 11 & 3 \\ 8 & 15 & 11 & 37 & 13 \\ 3 & 5 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[1530] \ 1^{\overline{1}^{11}} 5^{\overline{1}^1} 1^{\overline{1}^{23}} 1^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 5 & 0 & 18 & 8 \\ 2 & 2 & 2 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[1531] $1^{1^3} 5^{1^4} 13^{1^4} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 20 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 20 & 8 & 62 & 16 \\ 3 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[1533] $1^{1^2} 5^{1^2} 13^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1532] $1^{1^1} 5^{1^1} 13^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1534] $1^{1^3} 5^{1^3} 13^{1^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[1535] $2^1 3^1 15^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$

[1536] $21^{1^1} 5^{1^1} 13^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1537] $1^{1^1} 5^{1^1} 13^{1^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 7 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$

[1538] $1^{1^4} 5^{1^4} 13^{1^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 10 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$

[1539] $1^{1^4} 5^{1^4} 13^{1^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 1 & 18 & 3 \\ 16 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[1540] $1^{1^1} 5^{1^2} 23^{1^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 20 & 2 \\ 12 & 15 & 0 & 20 & 193 & 27 \\ 2 & 2 & -1 & 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1541] $1^{1^1} 5^{1^1} 23^{1^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 63 & 17 \\ 3 & 5 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[1543] $1^{1^1} 5^1 1^{231} 13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 61$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 25 & 0 & 97 & 21 \\ 3 & 5 & -1 & 21 & 4 \end{pmatrix}_{b_0}$$

[1542] $1^{121} 3221^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 61$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 11 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 11 & 0 & 49 & 11 & 13 \\ 1 & 3 & 0 & 11 & 1 & 3 \\ 2 & 3 & -1 & 13 & 3 & 3 \end{pmatrix}_{b_0}$$

[1544] $1^{1^4} 5^1 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 62$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 3 & 20 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 20 & 8 & 62 & 16 \\ 3 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1545] $2^1 3^1 1^1 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 65$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1546] $2^1 3^1 1^1 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 67$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1547] $1^{1^3} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 69$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 9 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1548] $1^{1^2} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 69$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1550] $1^{1^2} 5^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1549] $1^{1^3} 5^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 9 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1551] $1^{1^3} 5^1 3^1 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 75$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 24 & 2 \\ 15 & 18 & 0 & 24 & 295 & 33 \\ 2 & 2 & -1 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1552] \quad & 2 \overset{1^{11}}{3} \overset{1^1}{15} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[1553] \quad & 1 \overset{1^2}{5} 1^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[1554] \quad & 1 \overset{1^1}{5} 1^{231} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 1 & 22 & 3 \\ 16 & 30 & 22 & 154 & 26 \\ 3 & 5 & 3 & 26 & 4 \end{pmatrix}_{b_0} \\
[1555] \quad & 1 \overset{1^3}{5} 1^{13} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 11 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0} \\
[1556] \quad & 1 \overset{1^4}{5} 1^{13} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0} \\
[1557] \quad & 2 \overset{1^{14}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 5 & 3 & 20 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 20 & 8 & 62 & 16 \\ 3 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[1558] \quad & 1 \overset{1^{11}}{5} 1^{1231} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 0 & 31 & 11 \\ 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0} \\
[1559] \quad & 2 \overset{1^{11}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 5 & 8 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 \\ 3 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0} \\
[1560] \quad & 2 \overset{1^{12}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 6 & 3 \\ 4 & 5 & 8 & 10 & 5 \\ 5 & 8 & 8 & 12 & 6 \\ 6 & 10 & 12 & 14 & 8 \\ 3 & 5 & 6 & 8 & 4 \end{pmatrix}_{b_0} \\
[1561] \quad & 2 \overset{1^{12}}{3} \overset{1^1}{15} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[1562] \quad & 21 \overset{1^1}{13} 5 1^{14} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[1563] $1^{11} 5^1 1^{12} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 11 & 0 & 50 & 14 \\ 2 & 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[1564] $1^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 9 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1565] $2^1 3^{11} 15^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 113$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 3 \\ 4 & 5 & 8 & 15 & 5 \\ 5 & 8 & 8 & 18 & 6 \\ 9 & 15 & 18 & 34 & 12 \\ 3 & 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1566] $2^1 3^{15} 15^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 113$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.1$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 5 & 3 & 25 & 5 \\ 2 & 3 & 0 & 10 & 2 \\ 15 & 25 & 10 & 98 & 20 \\ 3 & 5 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1567] $21^{12} 5^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 118$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1568] $21^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 118$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1569] $1^{11} 5^1 2^{12} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 122$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 30 & 0 & 154 & 26 \\ 3 & 5 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

[1570] $1^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 132$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1571] $1^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 132$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 12 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1572] $2^1 3^{11} 15^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 134$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 12 & 3 \\ 4 & 5 & 8 & 20 & 5 \\ 5 & 8 & 8 & 24 & 6 \\ 12 & 20 & 24 & 62 & 16 \\ 3 & 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[1573] $1^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 11 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1574] \ 1^{11} 5^1 1^3 231^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 5 & 5 & 45 & 5 \\ 3 & 5 & 1 & 33 & 3 \\ 24 & 45 & 33 & 349 & 39 \\ 3 & 5 & 3 & 39 & 4 \end{pmatrix}_{b_0}$$

$$[1575] \ 1^{11} 5^1 1^2 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 9 & 12 & 0 & 81 & 17 \\ 2 & 2 & 2 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1576] \ 2^1 3^1 15^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 5 & 3 & 30 & 5 \\ 2 & 3 & 0 & 12 & 2 \\ 18 & 30 & 12 & 142 & 24 \\ 3 & 5 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[1577] \ 1^{12} 5^1 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1578] \ 2^1 3^1 15^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 5 & 15 & 3 \\ 4 & 5 & 8 & 25 & 5 \\ 5 & 8 & 8 & 30 & 6 \\ 15 & 25 & 30 & 98 & 20 \\ 3 & 5 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1579] \ 2^1 3^1 15^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 8 & 3 & 3 \\ 4 & 5 & 13 & 5 & 5 \\ 8 & 13 & 24 & 10 & 10 \\ 3 & 5 & 10 & 2 & 4 \\ 3 & 5 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1580] \ 1^{11} 5^1 1^2 231^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 35 & 0 & 202 & 30 \\ 3 & 5 & -1 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[1581] \ 1^{11} 3221^{11} 51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 5 & 2 \\ 4 & 18 & 6 & 0 & 25 & 8 \\ 1 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 25 & 8 & 0 & 31 & 11 \\ 2 & 8 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1582] \ 2^1 3^1 15^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1583] \ 1^{12} 5^1 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 13 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1584] $2 \overset{1^{17}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 5 & 3 & 35 & 5 \\ 2 & 3 & 0 & 14 & 2 \\ 21 & 35 & 14 & 194 & 28 \\ 3 & 5 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[1586] $1 \overset{1^3}{5} \overset{1^2}{13} 1^2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1585] $1 \overset{1^1}{11} 5 \overset{1^2}{1} 2 2 3 1^2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 10 & 15 & 0 & 108 & 20 \\ 2 & 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[1587] $1 \overset{1^2}{12} 5 \overset{1^4}{13} 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 14 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1588] $1 \overset{1^1}{5} \overset{1^2}{1} 2 2 3 1^4 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 40 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 40 & 0 & 258 & 34 \\ 3 & 5 & -1 & 34 & 4 \end{pmatrix}_{b_0}$$

[1589] $1 \overset{1^2}{12} 5 \overset{1^3}{13} 1^4 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 15 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[1591] $1 \overset{1^3}{13} 5 \overset{1^4}{13} 1^2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 14 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1590] $1 \overset{1^3}{13} 5 \overset{1^3}{13} 1^2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1592] $2 \overset{1^{12}}{3} \overset{1^1}{15} 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 217, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1593] $2 2 \overset{1^{11}}{3} 1 5 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 8 \\ 5 & 13 & 8 & 11 & 6 \\ 5 & 13 & 11 & 9 & 7 \\ 3 & 8 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

[1594] $2 1 \overset{1^1}{13} 5 \overset{1^3}{13} 1^3 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1596] $1^{12} \overset{1^4}{5} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[1595] $1^{12} \overset{1^1}{5} 1^{23} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 11 & 13 & 0 & 101 & 19 \\ 2 & 2 & 2 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[1597] $1^{13} \overset{1^2}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 14 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1598] $21^{13} \overset{1^1}{5} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1599] $1^{11} \overset{1^1}{5} 1^3 231^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 5 & 0 & 45 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 349 & 39 \\ 3 & 5 & -1 & 39 & 4 \end{pmatrix}_{b_0}$$

[1600] $2^3 \overset{1^3}{3} \overset{1^1}{15} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 3 & 3 & 1 & 16 & 3 \\ 13 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1601] $1^{11} \overset{1^1}{5} 1^3 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 13 & 19 & 0 & 190 & 26 \\ 2 & 2 & 2 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[1602] $22^3 \overset{1^{11}}{3} 151^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 5 & 5 & 3 \\ 5 & 9 & 12 & 11 & 7 \\ 5 & 12 & 8 & 11 & 6 \\ 5 & 11 & 11 & 9 & 7 \\ 3 & 7 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

[1603] $1^{13} \overset{1^3}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 277, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1604] $1^{13} \overset{1^4}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 16 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1605] $1^{12} 5^1 1^2 231^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 12 & 16 & 0 & 130 & 22 \\ 2 & 2 & 2 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

[1606] $1^{12} 5^1 1^3 231^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 14 & 2 \\ 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 14 & 17 & 0 & 181 & 25 \\ 2 & 2 & 2 & -1 & 25 & 3 \end{pmatrix}_{b_0}$

[1607] $1^{13} 5^1 131^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$

[1608] $1^{11} 5^1 1^3 231^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 26 & 3 \\ 3 & 5 & 0 & 50 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 419 & 43 \\ 3 & 5 & -1 & 43 & 4 \end{pmatrix}_{b_0}$

[1609] $2^3 3^1 15^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 330, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 3 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$

[1610] $22^1 3^1 151^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 10 & 5 & 3 \\ 6 & 14 & 26 & 13 & 8 \\ 10 & 26 & 41 & 22 & 13 \\ 5 & 13 & 22 & 9 & 7 \\ 3 & 8 & 13 & 7 & 4 \end{pmatrix}_{b_0}$

[1611] $1^{12} 5^1 1^2 231^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 13 & 19 & 0 & 165 & 25 \\ 2 & 2 & 2 & -1 & 25 & 3 \end{pmatrix}_{b_0}$

[1612] $1^{13} 5^1 131^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$

[1613] $1^{12} 3221^{12} 51^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 10 & 2 \\ 4 & 18 & 6 & 0 & 50 & 8 \\ 1 & 6 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 50 & 16 & 0 & 130 & 22 \\ 2 & 8 & 2 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

[1614] $2^3 3^1 15^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 363, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 3 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[1615] $22^1 3^1 151^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 398, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 10 & 3 \\ 6 & 14 & 13 & 26 & 8 \\ 5 & 13 & 8 & 22 & 6 \\ 10 & 26 & 22 & 42 & 14 \\ 3 & 8 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[1616] $1^{1^1} 1^2 5^1 1^3 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 15 & 2 \\ 1 & 2 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 15 & 20 & 0 & 218 & 28 \\ 2 & 2 & 2 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[1617] $1^{1^1} 1^3 5^1 1^3 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 16 & 18 & 0 & 211 & 27 \\ 2 & 2 & 2 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[1618] $1^{1^1} 1^2 5^1 1^3 231^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 16 & 23 & 0 & 261 & 31 \\ 2 & 2 & 2 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[1619] $2^{1^4} 3^{1^1} 15^1 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 3 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[1620] $1^{1^1} 1^2 5^1 1^3 231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 549, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 17 & 2 \\ 1 & 2 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 17 & 26 & 0 & 310 & 34 \\ 2 & 2 & 2 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

[1621] $1^{1^1} 1^3 5^1 1^3 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 0 & 17 & 2 \\ 1 & 2 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 17 & 21 & 0 & 250 & 30 \\ 2 & 2 & 2 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[1622] $1^{1^1} 1^3 5^1 1^3 231^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 664, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 18 & 2 \\ 1 & 2 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 18 & 24 & 0 & 295 & 33 \\ 2 & 2 & 2 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[1623] $1^{1^1} 1^3 3221^{13} 51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 15 & 2 \\ 4 & 18 & 6 & 0 & 75 & 8 \\ 1 & 6 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 75 & 24 & 0 & 295 & 33 \\ 2 & 8 & 2 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[1624] $22^1 3^{1^1} 151^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 683, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 15 & 3 \\ 6 & 14 & 13 & 39 & 8 \\ 5 & 13 & 8 & 33 & 6 \\ 15 & 39 & 33 & 97 & 21 \\ 3 & 8 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1625] $1^{1^1} 1^3 5^1 1^3 231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 748, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 0 & 19 & 2 \\ 1 & 2 & 0 & 0 & 27 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 19 & 27 & 0 & 346 & 36 \\ 2 & 2 & 2 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& [1626] \quad 22 \quad 3 \quad 151 \overset{114}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 1082, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0 \\
& \quad T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 6 & 5 & 20 & 3 \\ 6 & 14 & 13 & 52 & 8 \\ 5 & 13 & 8 & 44 & 6 \\ 20 & 52 & 44 & 174 & 28 \\ 3 & 8 & 6 & 28 & 4 \end{pmatrix}_{b_0} \\
& [1627] \quad 22 \quad 3 \quad 151 \overset{115}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 1595, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.1 \\
& \quad T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 6 & 5 & 25 & 3 \\ 6 & 14 & 13 & 65 & 8 \\ 5 & 13 & 8 & 55 & 6 \\ 25 & 65 & 55 & 273 & 35 \\ 3 & 8 & 6 & 35 & 4 \end{pmatrix}_{b_0} \\
& T = 11 \quad [1632] \quad 1 \overset{11}{1} 5 1 \overset{13}{2} 3 2 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 6 & 0 & 28 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 28 & 0 & 106 & 24 \\ 3 & 6 & -1 & 24 & 5 \end{pmatrix}_{b_0} \\
& [1631] \quad 1 \overset{11}{1} 3 1 \overset{11}{5} 1 3 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -37, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
& [1630] \quad 1 \overset{12}{1} 5 1 3 2 1 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 4 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
& [1629] \quad 1 \overset{11}{1} 3 1 \overset{11}{5} 1 3 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -37, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
& [1634] \quad 1 \overset{11}{1} 5 1 3 2 1 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0} \\
& [1633] \quad 1 \overset{11}{1} 5 1 3 2 1 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 3 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
& [1636] \quad 1 \overset{13}{1} 5 1 3 2 1 \overset{13}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.8 \\
& \quad T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 6 & 3 & 18 & 6 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 18 & 6 & 43 & 15 \\ 3 & 6 & 2 & 15 & 5 \end{pmatrix}_{b_0} \\
& [1637] \quad 1 \overset{12}{1} 5 1 3 2 1 \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.5 \\
& \quad T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 6 & 3 & 12 & 6 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 12 & 4 & 18 & 10 \\ 3 & 6 & 2 & 10 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[1635] $1^{1^4} 5^{1^1} 1321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 6 & 3 & 24 & 6 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 24 & 8 & 78 & 20 \\ 3 & 6 & 2 & 20 & 5 \end{pmatrix}_{b_0}$

[1638] $1^{1^1} 5^{1^1} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 6 & 3 & 6 & 6 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 6 & 2 & 3 & 5 \\ 3 & 6 & 2 & 5 & 5 \end{pmatrix}_{b_0}$

[1639] $1^{1^{11}} 4^{1^{11}} 14131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 2 & 1 & 3 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[1640] $1^{1^1} 5^{1^1} 1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$

[1641] $1^{1^1} 5^{1^1} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 6 & 6 & 16 & 6 \\ 3 & 6 & 2 & 12 & 4 \\ 8 & 16 & 12 & 38 & 14 \\ 3 & 6 & 4 & 14 & 5 \end{pmatrix}_{b_0}$

[1642] $21^{1^1} 5^{1^1} 1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$

[1643] $1^{1^1} 2321^{1^1} 61^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[1644] $1^{1^1} 5^{1^1} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 4 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$

[1645] $1^{1^1} 4^{1^1} 14131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 2 & 3 & 1 & 2 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[1646] $1^{1^1} 5^{1^1} 12321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$

[1647] $1^{1^1} 31^{1^1} 5^{1^1} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -37, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1648] \ 1^{\overline{11}} \overline{5}^{\overline{1}} 1321^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 7 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[1649] \ 1^{\overline{11}} \overline{4}^{\overline{1}} 141^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1650] \ 1^{\overline{11}} \overline{5}^{\overline{1}} 1^{\overline{1}} 2321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 4 & 8 & 0 & 22 & 10 \\ 2 & 2 & 3 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1651] \ 2^{\overline{11}} \overline{3}^{\overline{1}} 2161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1652] \ 2^{\overline{11}} \overline{3}^{\overline{1}} 21^{\overline{1}} 61^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 1 & 7 & 3 \\ 7 & 6 & 7 & 19 & 9 \\ 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1653] \ 1^{\overline{14}} \overline{5}^{\overline{1}} 1321^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 6 & 9 & 24 & 6 \\ 4 & 9 & 9 & 28 & 7 \\ 12 & 24 & 28 & 78 & 20 \\ 3 & 6 & 7 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[1654] \ 1^{\overline{13}} \overline{5}^{\overline{1}} 1321^{\overline{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 6 & 9 & 18 & 6 \\ 4 & 9 & 9 & 21 & 7 \\ 9 & 18 & 21 & 43 & 15 \\ 3 & 6 & 7 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[1655] \ 1^{\overline{12}} \overline{5}^{\overline{1}} 1321^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 6 & 9 & 12 & 6 \\ 4 & 9 & 9 & 14 & 7 \\ 6 & 12 & 14 & 18 & 10 \\ 3 & 6 & 7 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[1656] \ 1^{\overline{11}} \overline{5}^{\overline{1}} 1321^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 6 & 9 & 6 & 6 \\ 4 & 9 & 9 & 7 & 7 \\ 3 & 6 & 7 & 3 & 5 \\ 3 & 6 & 7 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1657] \ 1^{\overline{12}} \overline{5}^{\overline{1}} 1321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 6 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

[1658] $1^{11} \overset{1}{5} 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -28$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 5 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

[1659] $1^{11} \overset{1}{5} 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -32$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$$

[1660] $1^{11} \overset{2}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -37$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

[1661] $1^{11} \overset{3}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -42$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

[1662] $1^{11} \overset{4}{5} 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -43$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$$

[1663] $1^{11} \overset{11}{4} 141^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -33$, $\det = -50$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 5 & 6 & 1 & 6 & 15 & 9 \\ 2 & 3 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[1664] $2321^{11} \overset{1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -57$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 3 & 3 & 1 & 9 & 3 \\ 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[1665] $1^{11} \overset{1}{5} 1^{11} 2321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -32$, $\det = -63$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 13 & 0 & 54 & 16 \\ 2 & 2 & 3 & -1 & 16 & 4 \end{pmatrix}_{b_0}$$

[1666] $1^{12} \overset{1}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -63$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 9 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[1667] $2^3 3 216 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -67$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1668] $2 \overset{1^{11}}{3} 21 \overset{1^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -78, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 11 & 3 \\ 10 & 8 & 1 & 37 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[1669] $1 \overset{1^2}{5} 1321 \overset{1^3}{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 12 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1670] $1 \overset{1^2}{5} 1321 \overset{1^2}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 10 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[1671] $1 \overset{1^3}{5} 1321 \overset{1^2}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -87, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 11 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[1672] $1 \overset{1^3}{5} 1321 \overset{1^4}{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -87, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 15 & 14 & 24 & 122 & 26 \\ 3 & 3 & 4 & 26 & 5 \end{pmatrix}_{b_0}$$

[1673] $41 \overset{1^1}{231} 51 \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 8 & 3 \\ 9 & 51 & 0 & 48 & 15 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 48 & 0 & 42 & 14 \\ 3 & 15 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

[1674] $2 \overset{1^{12}}{3} 21 \overset{1^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1675] $1 \overset{1^3}{5} 1321 \overset{1^3}{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 13 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1676] $1 \overset{1^4}{5} 1321 \overset{1^3}{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -118, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 14 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1677] $2 \overset{1^{12}}{3} 21 \overset{1^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 15 & 3 \\ 13 & 10 & 15 & 63 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1678] $1^{12}_{1^1} 5^{14} 1321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -118$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 16 & 14 & 24 & 122 & 26 \\ 3 & 3 & 4 & 26 & 5 \end{pmatrix}_{b_0}$$

[1679] $41^{12}_{1^1} 231 5^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -126$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 16 & 3 \\ 9 & 51 & 0 & 96 & 15 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 96 & 0 & 174 & 28 \\ 3 & 15 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$

[1680] $1^{13}_{1^1} 5^{13} 1321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -147$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 14 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

[1681] $1^{13}_{1^1} 5^{13} 1321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -172$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 15 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

[1682] $2^3 3 21^{11} 6 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -175$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 11 & 3 \\ 3 & 3 & 1 & 13 & 3 \\ 12 & 11 & 13 & 56 & 16 \\ 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[1683] $1^{13}_{1^1} 5^{13} 1321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -182$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 26 & 4 \\ 17 & 17 & 26 & 153 & 29 \\ 3 & 3 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

[1684] $1^{13}_{1^1} 5^{13} 1321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -183$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 16 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

[1685] $41^{13}_{1^1} 231 5^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -186$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 24 & 3 \\ 9 & 51 & 0 & 144 & 15 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 144 & 0 & 394 & 42 \\ 3 & 15 & -1 & 42 & 4 \end{pmatrix}_{b_0}$$

[1686] $1^{13}_{1^1} 5^{13} 1321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -213$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 26 & 4 \\ 18 & 17 & 26 & 153 & 29 \\ 3 & 3 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

[1687] $41^{14}_{1^1} 231 5^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -270$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 32 & 3 \\ 9 & 51 & 0 & 192 & 15 \\ 1 & 0 & -3 & 0 & -1 \\ 32 & 192 & 0 & 702 & 56 \\ 3 & 15 & -1 & 56 & 4 \end{pmatrix}_{b_0}$$

[1688] $2^3 3 21^{12} 6 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -277$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 17 & 3 \\ 15 & 13 & 17 & 86 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1689] \quad 1^{11} 31^1 513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[1693] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 4 & 2 & 2 & 4 \\ 3 & 4 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1694] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 0 & 2 \\ 4 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1695] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1696] \quad 1^{12} 6^{11} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 4 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1699] \quad 1^{11} 6^{11} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 3 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1698] \quad 1^{11} 6^{11} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1697] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1700] \quad 1^{11} 5^{11} 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1701] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 6 \\ 2 & 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[1702] $1^{11}_{131} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1703] $21^{111}_{111} 5^{111} 13221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0}$

[1704] $1^{12}_{11} 5^{12} 13221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 22 & 5 \\ 12 & 12 & 22 & 88 & 24 \\ 3 & 3 & 5 & 24 & 6 \end{pmatrix}_{b_0}$

[1705] $1^{11}_{11} 5^{12} 13221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 3 & 14 & 5 \\ 8 & 7 & 14 & 33 & 15 \\ 3 & 3 & 5 & 15 & 6 \end{pmatrix}_{b_0}$

[1706] $1^{11}_{131} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 2 & 2 & 4 \\ 3 & 4 & 3 & 4 & 5 \end{pmatrix}_{b_0}$

[1707] $1^{11}_{111} 6^{11} 1^{11} 2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 8 & 3 \\ 3 & 2 & 5 & 5 & 12 & 5 \\ 4 & 4 & 8 & 12 & 22 & 10 \\ 2 & 2 & 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$

[1708] $21^{111}_{111} 6^{111} 1^{11} 2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 5 & 6 & 3 \\ 2 & 1 & 4 & 4 & 3 \\ 5 & 4 & 9 & 10 & 7 \\ 6 & 4 & 10 & 10 & 8 \\ 3 & 3 & 7 & 8 & 5 \end{pmatrix}_{b_0}$

[1709] $1^{12}_{11} 5^{14} 13221^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 3 & 28 & 5 \\ 16 & 14 & 28 & 138 & 30 \\ 3 & 3 & 5 & 30 & 6 \end{pmatrix}_{b_0}$

[1710] $1^{11}_{11} 6^{11} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$

[1711] $21^{11}_{11} 6^{11} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 5 & 4 & 3 \\ 2 & 1 & 4 & 4 & 3 \\ 5 & 4 & 9 & 8 & 7 \\ 4 & 4 & 8 & 6 & 6 \\ 3 & 3 & 7 & 6 & 5 \end{pmatrix}_{b_0}$

[1712] $1^{11}_{11} 31^{11} 5^{11} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -9, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & -1 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1713] \quad 1 \overset{1^1}{\underset{1^1}{6}} 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 4 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[1717] \quad 1 \overset{1^3}{\underset{1^1}{5}} 13221^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 7 & 10 & 21 & 7 \\ 4 & 10 & 10 & 24 & 8 \\ 9 & 21 & 24 & 52 & 18 \\ 3 & 7 & 8 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[1715] \quad 1 \overset{1^1}{\underset{1^1}{5}} 13221^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 7 & 10 & 7 & 7 \\ 4 & 10 & 10 & 8 & 8 \\ 3 & 7 & 8 & 4 & 6 \\ 3 & 7 & 8 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[1716] \quad 1 \overset{1^4}{\underset{1^1}{5}} 13221^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 7 & 10 & 28 & 7 \\ 4 & 10 & 10 & 32 & 8 \\ 12 & 28 & 32 & 94 & 24 \\ 3 & 7 & 8 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[1714] \quad 1 \overset{1^2}{\underset{1^1}{5}} 13221^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 7 & 10 & 14 & 7 \\ 4 & 10 & 10 & 16 & 8 \\ 6 & 14 & 16 & 22 & 12 \\ 3 & 7 & 8 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[1718] \quad 1 \overset{1^{11}321}{\underset{1^1}{6}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 4 & 4 \\ 3 & 2 & 5 & 4 \\ 4 & 5 & 4 & 6 \\ 4 & 4 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[1719] \quad 1 \overset{1^{11}321}{\underset{1^1}{6}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 4 & 1 & 3 & 5 \\ 3 & 4 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1720] \quad 1 \overset{1^1}{\underset{1^1}{11}} 6 12321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 7 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[1721] \quad 1 \overset{1^{11}31}{\underset{1^1}{6}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 2 & 3 & 6 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[1722] \quad 1 \overset{1^{11}2321^1}{\underset{1^1}{6}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 8 & 12 & 38 & 14 \\ 3 & 3 & 4 & 14 & 5 \end{pmatrix}_{b_0}$$

[1723] $1^{11} \overset{1}{6} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 1 & 5 & 13 & 3 \\ 3 & 2 & 5 & 5 & 19 & 5 \\ 6 & 7 & 13 & 19 & 54 & 16 \\ 2 & 2 & 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[1724] $21^{11} \overset{1}{6} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 5 & 10 & 3 \\ 2 & 1 & 4 & 8 & 3 \\ 5 & 4 & 9 & 18 & 7 \\ 10 & 8 & 18 & 34 & 14 \\ 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[1725] $1^1 \overset{1}{6} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 8 & 12 & 38 & 14 \\ 3 & 3 & 4 & 14 & 5 \end{pmatrix}_{b_0}$$

[1726] $1^1 \overset{1}{6} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$$

[1727] $1^{12} \overset{1}{6} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 6 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

[1728] $1^1 \overset{1}{6} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 2 & 3 & 6 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$$

[1729] $1^{11} \overset{1}{6} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 5 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

[1730] $1^{11} \overset{1}{6} 12321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$$

[1731] $1^{11} \overset{1}{6} 12321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

[1733] $1^1 \overset{1}{6} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$$

[1732] $1^{11} \overset{1^3}{6} 12321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$

[1734] $1^1 \overset{1^1}{6} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 2 & 3 & 10 & 4 \\ 2 & 3 & 1 & 8 & 3 \\ 7 & 10 & 8 & 26 & 12 \\ 3 & 4 & 3 & 12 & 5 \end{pmatrix}_{b_0}$

[1735] $1^1 \overset{1^3}{6} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 22 & 4 \\ 15 & 12 & 22 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$

[1736] $1^1 \overset{1^3}{6} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 13 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$

[1737] $1^{11} \overset{1^4}{6} 12321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$

[1738] $1^1 \overset{1^2}{6} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 2 & 3 & 10 & 4 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 10 & 8 & 26 & 12 \\ 3 & 4 & 3 & 12 & 5 \end{pmatrix}_{b_0}$

[1739] $1^1 \overset{1^4}{6} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 22 & 4 \\ 16 & 12 & 22 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$

[1740] $1^{11} \overset{1^1}{6} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 12 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$

[1741] $1^{11} \overset{1^1}{6} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 15 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$

[1742] $1^{11} \overset{1^2}{6} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 18 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$

[1743] $1^{11} \overset{1^1}{6} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 13 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[1744] \ 1^{\overline{11}} \overset{1^2}{6} \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 14 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[1745] \ 1^{\overline{11}} \overset{1^3}{6} \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 15 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[1746] \ 1^{\overline{11}} \overset{1^2}{6} \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 117, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 16 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

$$[1747] \ 1^{\overline{11}} \overset{1^3}{6} \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 19 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 19 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

$$[1748] \ 1^{\overline{11}} \overset{1^3}{6} \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 138, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 17 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

$$[1749] \ 1^{\overline{11}} \overset{1^4}{6} \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 145, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 18 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

$$[1750] \ 1^{\overline{11}} \overset{1^4}{6} \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 155, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 20 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [1752] \ 1^{\overline{11}31} \overset{1^4}{6} \ 13141^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 0 \\ 1 & 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 \\ 2 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1753] \ 1^{\overline{11}31} \overset{1^4}{6} \ 13141^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 4 & 0 & 2 & 2 \\ 2 & 4 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1754] \ 1^{\overline{11}2321} \overset{1^4}{6} \ 131^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -8, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$$[1755] \ 1^{11} 5^{11} 1315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 \\ 2 & 3 & 5 & 0 & 6 \\ 2 & 2 & 3 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[1757] \ 1^{11} 6^{11} 13141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1756] \ 1^{11} 6^{11} 13141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 2 & 0 & 4 & 2 & 2 & 4 \\ 2 & 0 & 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1758] \ 1^{11} 6^{11} 131^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 2 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 5 & 12 & 0 & 8 & 42 & 14 \\ 2 & 4 & -1 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1759] \ 1^{11} 6^{11} 131^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 5 & 0 & 8 & 6 & 22 & 10 \\ 2 & 0 & 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [1760] \ 1^{11} 4^{11} 1414141^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 6.0$$

$$[1762] \ 1^{11} 8^{11} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 6 \\ 2 & 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1763] \ 1^{11} 8^{11} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1764] \ 1^{12} 3161^{11} 23151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 0 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 34 & 12 \\ 2 & 3 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[1765] $1^{11}321613151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 3 & 5 \\ 3 & 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$

[1766] $1^{11}321^{11}613151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 5 & 0 & 12 & 8 \\ 3 & 3 & 0 & 8 & 5 \end{pmatrix}_{b_0}$

[1767] $1^{11}612315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 2 & 5 & 4 \\ 5 & 5 & 7 & 7 \\ 4 & 4 & 7 & 6 \end{pmatrix}_{b_0}$

[1768] $1^{11}316123151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 3 & 5 \\ 3 & 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$

[1769] $1^{112}3161^{11}23151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 6 & 9 & 0 & 9 & 34 & 12 \\ 2 & 3 & 0 & 3 & 12 & 4 \end{pmatrix}_{b_0}$

[1770] $1^{111}3151321^{11}61^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 2 & 3 & 10 & 4 \\ 3 & 2 & 8 & 4 \\ 10 & 8 & 25 & 13 \\ 4 & 4 & 13 & 6 \end{pmatrix}_{b_0}$

[1771] $1^{112}3151321^{11}61^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 2 & 3 & 13 & 4 \\ 3 & 2 & 10 & 4 \\ 13 & 10 & 43 & 17 \\ 4 & 4 & 17 & 6 \end{pmatrix}_{b_0}$

[1772] $1^{12}3161^{11}23151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 10 & 0 & 36 & 14 \\ 3 & 3 & 0 & 14 & 5 \end{pmatrix}_{b_0}$

[1773] $1^{11}3161^{11}23151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 42, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 0 & 7 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 7 & 0 & 21 & 11 \\ 3 & 3 & 0 & 11 & 5 \end{pmatrix}_{b_0}$

[1774] $1^{11}321^{11}613151^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 60, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 9 & 0 & 29 & 13 \\ 3 & 3 & 0 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[1775] \ 1^{11} 6^{1^{13}3151321^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 82, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 14.95 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 18 & 4 \\ 3 & 2 & 15 & 4 \\ 18 & 15 & 88 & 24 \\ 4 & 4 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[1776] \ 1^{11} 3161^{123151} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 87, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 14.99 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 0 & 8 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 8 & 0 & 27 & 13 \\ 3 & 3 & 0 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[1777] \ 1^{13} 3161^{123151} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 87, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 14.99 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 0 & 14 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 14 & 0 & 63 & 19 \\ 3 & 3 & 0 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[1778] \ 1^{11} 6^{1^{12}3151321^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 92, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 14.91 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 15 & 4 \\ 3 & 2 & 13 & 4 \\ 15 & 13 & 58 & 20 \\ 4 & 4 & 20 & 6 \end{pmatrix}_{b_0}$$

$$[1779] \ 1^{12} 3161^{123151} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 107, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 14.92 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 11 & 0 & 42 & 16 \\ 3 & 3 & 0 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[1780] \ 1^{12} 321^{1613151} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 115, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 14.99 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 0 & 13 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 13 & 0 & 56 & 18 \\ 3 & 3 & 0 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[1781] \ 1^{12} 3161^{123151} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 172, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 14.98 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 0 & 12 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 12 & 0 & 52 & 18 \\ 3 & 3 & 0 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[1782] \ 1^{13} 3161^{123151} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 182, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 14.96 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 15 & 0 & 73 & 21 \\ 3 & 3 & 0 & 21 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [1784] \ 1^{11} 4 \ 141414141^{131} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 2), \ c_{\text{ext}} = 6.0$$

$$[1783] \ 21^{11} 6 \ 131513221^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 6.0$$

$$[1785] \ 1^{11} 6 \ 131513221^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 6.0$$

[1786] $1^{1^{11}2315131}6^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 6.0$

[1794] $1^{1^{11}23151321^1}6^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 9 & 9 & 61 & 21 \\ 3 & 3 & 2 & 21 & 6 \end{pmatrix}_{b_0}$$

[1795] $1^{1^{11}23151321^1}6^{1^1}13151321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0}$$

[1796] $21^{1^{11}23151321^{11}}6^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 6 & 5 & 6 \\ 2 & 2 & 3 & 5 & 3 & 5 \\ 2 & 2 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[1797] $1^{1^{12}23151321^{12}}6^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -5, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 3 & 14 & 5 \\ 8 & 7 & 14 & 33 & 15 \\ 3 & 3 & 5 & 15 & 6 \end{pmatrix}_{b_0}$$

[1798] $31^{1^{11}31}6^{1^1}13161^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 8 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 8 & 0 & 6 & 6 \\ 2 & 2 & 5 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1799] $1^{1^{11}315131}6^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -7, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$T = 16$ [1800] $1^{1^{11}31}6^{131612321^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 16, T_{\max} = 16, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 6.0$

[1804] $1^{1^{151^1}2321}8^{1231^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 16, T_{\max} = 17, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 5 & 2 & 0 & 4 & 7 & 7 \\ 2 & 2 & 0 & 2 & 7 & 4 \end{pmatrix}_{b_0}$$

[1806] $1^{1^{151^1}2321}8^{1231^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 16, T_{\max} = 17, \Delta = -29, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 3 & 7 & 0 & 18 & 10 \\ 2 & 2 & 2 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[1805] $1^{1^{11}31}6^{1^1}1316131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 16, T_{\max} = 16, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 & 2 \\ 2 & 4 & 1 & 2 & 2 & 4 \\ 2 & 4 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

[1807] $1^{11} 2^3 2181231^1 5^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 5 & 0 & 9 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 13 & 9 \\ 3 & 5 & 0 & 9 & 5 \end{pmatrix}_{b_0}$$

[1809] $1^{11} 6^{31} 1316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = -4, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 \\ 2 & 5 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[1810] $1^{11} 6^{31} 1316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = -4, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 \\ 3 & 5 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[1808] $1^{11} 6^{31} 13161231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 1 & 2 & 3 & 5 \\ 2 & 4 & 0 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[1811] $1^{11} 6^{31} 1316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = -4, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 5 & 2 \\ 5 & 9 & 5 & 5 & 21 & 11 \\ 2 & 4 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[1812] $1^{11} 6^{31} 1316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = -4, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 5 & 2 \\ 4 & 9 & 5 & 5 & 21 & 11 \\ 2 & 4 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

$T = 17$ [1823] $1^{11} 8^{11} 5^{13} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 0 & 6 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 8 & 8 \\ 3 & 4 & 0 & 8 & 6 \end{pmatrix}_{b_0}$$

[1822] $1^{11} 6^{11} 12321712321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 2 & 16 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 14 & 16 & 9 & 97 & 25 \\ 3 & 4 & 2 & 25 & 6 \end{pmatrix}_{b_0}$$

[1821] $1^{11} 8^{11} 5^{13} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 4 & 5 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 5 & 4 \\ 2 & 5 & 0 & 5 & 6 & 6 \\ 2 & 4 & 0 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1824] \quad 1^{13} 232181^{11} 223151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 2 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 18 & 0 & 86 & 24 \\ 3 & 4 & 0 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[1825] \quad 1^{11} 6^{11} 131613141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 3 & 5 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 3 & 5 \\ 2 & 5 & -1 & 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1826] \quad 1^{14} 232181^{11} 223151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 23 & 0 & 132 & 30 \\ 3 & 4 & 0 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[1827] \quad 1^{11} 13232181^{11} 223151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 4 & 16 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 16 & 4 \\ 8 & 16 & 0 & 16 & 78 & 20 \\ 2 & 4 & 0 & 4 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[1828] \quad 1^{11} 6^{11} 1316131^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 3 & 1 & 3 & 14 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 10 & 3 \\ 5 & 14 & 0 & 10 & 46 & 16 \\ 2 & 5 & -1 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [1829] \quad 1^{11} 6^{11} 13161315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -6, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 6.0$$

$$T = 19 \quad [1830] \quad 1^{11} 6^{11} 131613161231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 6.0$$

$$[1831] \quad 1^{11} 6^{11} 13161316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 6.0$$

$$[1832] \quad 1^{11} 6^{11} 13161316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 6.5$$

$$T = 20 \quad [1834] \quad 1^{11} 6^{11} 1316131613141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 6.0$$

$$[1835] \quad 1^{11} 6^{11} 13161316131^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 6.0$$

4.1139 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6$

$$T = 8 \quad [0] \ 1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 12 & 0 & 4 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 6 & 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 4 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 4 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 48 & 15 & 0 & 12 \\ 3 & 5 & 15 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 9 & 6 & 0 & 5 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 9 & 12 & 0 & 5 \\ 5 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 36 & 9 & 0 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{122}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 1 & 3 \\ 3 & 4 & 8 & 18 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 10 & 18 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 14 & -4 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 \\ 3 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 \\ 3 & 5 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 24 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 27 & 18 & 0 & 9 \\ 4 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 11 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 16 & -4 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 7 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 9 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 27 & 9 & 0 & 9 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 7 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 5 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 57 & 12 & 0 & 13 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 48 & 12 & 0 & 12 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 27 & 21 & 0 & 9 \\ 5 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 48 & 27 & 0 & 12 \\ 5 & 9 & 27 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 18 & -4 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 9 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 13 & -4 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 4 & 8 & 9 & 30 & 0 & 5 \\ 12 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 17 & -4 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 7 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 9 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{112}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 22 & -4 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 9 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 11 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{131}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 18 & -4 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 19 & -4 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 27 & 30 & 0 & 9 \\ 7 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{114}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 25 & -4 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{123}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 9 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 22 & -4 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 12 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 17 & -4 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{113}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 32 & 0 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 32 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 25 & -4 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 14 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{114}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 17 & 31 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 24 & -4 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{114} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2464$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 36 & 0 & 4 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

4.1140 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 14$, $\Delta = -92$, $\det = 470$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 2 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1141 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$

$T = 9$ [0] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.1142 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$

$T = 9$ [0] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 16.86 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & 3 & -4 & 4 & -4 \end{pmatrix}_{b_0}$$

- [1] $1^{1111}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 16.71 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 2 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 & 4 \\ 8 & 14 & 36 & 9 & 0 & 9 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 1 & 3 \\ 3 & 4 & 10 & 3 & -4 & 3 & -4 \end{pmatrix}_{b_0}$$
- [2] $1^{11}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [3] $1^{1111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{1121}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 5 & 3 \\ 3 & 4 & 9 & 8 & 0 & 9 & 4 \\ 5 & 9 & 12 & 12 & 0 & 15 & 6 \\ 4 & 8 & 12 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$
- [5] $1^{1111}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & 3 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{1111}31^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 19 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 12 & 21 & 0 & 73 & 15 \\ 3 & 4 & 2 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{131}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.32 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 2 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 & 4 \\ 9 & 17 & 48 & 15 & 0 & 12 & 12 \\ 3 & 5 & 15 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 3 & 0 & 1 & 3 \\ 3 & 4 & 12 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{1121}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 5 & 3 \\ 3 & 4 & 4 & 8 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 6 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 12 & 0 & 13 & 7 \\ 3 & 4 & 2 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$
- [9] $1^{1111}31^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -116$, $(n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 5 & 3 \\ 3 & 4 & 4 & 19 & 0 & 9 & 4 \\ 2 & 4 & 0 & 12 & 0 & 6 & 2 \\ 11 & 19 & 12 & 73 & 0 & 33 & 15 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 33 & 0 & 13 & 7 \\ 3 & 4 & 2 & 15 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] 1^{111} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 7 & 3 \\ 3 & 4 & 4 & 14 & 0 & 13 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 33 & 0 & 28 & 10 \\ 3 & 4 & 2 & 11 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 5 & 0 & 17 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 15 & 0 & 49 & 13 \\ 3 & 4 & 2 & 4 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 5 & 3 \\ 4 & 9 & 4 & 8 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 8 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 9 & 0 & 13 & 7 \\ 3 & 5 & 2 & 4 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[15] 1^{112} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 13 & 3 \\ 3 & 4 & 4 & 13 & 0 & 23 & 4 \\ 2 & 4 & 0 & 9 & 0 & 15 & 2 \\ 7 & 13 & 9 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 54 & 0 & 100 & 18 \\ 3 & 4 & 2 & 10 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 & 6 \\ 3 & 5 & 9 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 \\ 3 & 4 & 6 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] 1^{121} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 2 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 & 9 \\ 3 & 5 & 12 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 1 & 3 \\ 3 & 4 & 9 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 2 & 4 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1111} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 5 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 15 & 0 & 37 & 11 \\ 3 & 4 & 2 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 5 & 3 \\ 4 & 9 & 4 & 12 & 0 & 12 & 5 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 12 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 15 & 0 & 13 & 7 \\ 3 & 5 & 2 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 12 & 0 & 28 & 10 \\ 3 & 4 & 2 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 \\ 3 & 4 & 6 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 18 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1122} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 4 & 3 \\ 3 & 4 & 4 & 8 & 0 & 8 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 6 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 12 & 0 & 10 & 6 \\ 3 & 4 & 2 & 6 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -425, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & 3 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 7 & 3 \\ 3 & 4 & 9 & 8 & 0 & 13 & 4 \\ 5 & 9 & 12 & 12 & 0 & 21 & 6 \\ 4 & 8 & 12 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 12 & 3 \\ 3 & 4 & 4 & 14 & 0 & 22 & 4 \\ 2 & 4 & 0 & 9 & 0 & 15 & 2 \\ 8 & 14 & 9 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 57 & 0 & 85 & 17 \\ 3 & 4 & 2 & 11 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & 3 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1122}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 7 & 3 \\ 3 & 4 & 4 & 8 & 0 & 13 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 4 & 8 & 6 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 18 & 0 & 28 & 10 \\ 3 & 4 & 2 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 12 & 6 & 0 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 6 & 0 & 13 & 7 \\ 3 & 4 & 6 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 14 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 5 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 24 & 0 & 37 & 11 \\ 3 & 4 & 2 & 7 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 10 & 3 \\ 3 & 4 & 4 & 5 & 0 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & 2 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 16 & 3 \\ 3 & 4 & 4 & 5 & 0 & 28 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 30 & 0 & 154 & 22 \\ 3 & 4 & 2 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 2 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 8 & 3 \\ 3 & 4 & 4 & 14 & 0 & 14 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 39 & 0 & 37 & 11 \\ 3 & 4 & 2 & 11 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{112} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 13 & 3 \\ 3 & 4 & 4 & 5 & 0 & 23 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 24 & 0 & 100 & 18 \\ 3 & 4 & 2 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{114} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 14 & 3 \\ 3 & 4 & 4 & 5 & 0 & 26 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 24 & 0 & 118 & 20 \\ 3 & 4 & 2 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 12 & 3 \\ 3 & 4 & 4 & 5 & 0 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 21 & 0 & 85 & 17 \\ 3 & 4 & 2 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & 3 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 13 & 4 \\ 5 & 9 & 12 & 6 & 0 & 21 & 6 \\ 2 & 4 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 9 & 0 & 28 & 10 \\ 3 & 4 & 6 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 21 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 33 & 0 & 76 & 16 \\ 3 & 4 & 2 & 7 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 5 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 2 & 7 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 10 & 3 \\ 3 & 4 & 4 & 14 & 0 & 18 & 4 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 8 & 14 & 9 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 48 & 0 & 58 & 14 \\ 3 & 4 & 2 & 11 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 16 & 3 \\ 3 & 4 & 4 & 9 & 0 & 28 & 4 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 5 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 48 & 0 & 154 & 22 \\ 3 & 4 & 2 & 7 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 5 & 9 & 6 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 2 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 4 & 0 & 17 & 4 \\ 5 & 9 & 12 & 6 & 0 & 27 & 6 \\ 2 & 4 & 6 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 12 & 0 & 49 & 13 \\ 3 & 4 & 6 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 11 & 3 \\ 3 & 4 & 9 & 4 & 0 & 21 & 4 \\ 5 & 9 & 12 & 6 & 0 & 33 & 6 \\ 2 & 4 & 6 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 15 & 0 & 76 & 16 \\ 3 & 4 & 6 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 15 & 3 \\ 3 & 4 & 4 & 5 & 0 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & 2 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 10 & 3 \\ 3 & 4 & 4 & 13 & 0 & 18 & 4 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 7 & 13 & 9 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 42 & 0 & 58 & 14 \\ 3 & 4 & 2 & 10 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 10 & 3 \\ 3 & 4 & 4 & 9 & 0 & 18 & 4 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 5 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 30 & 0 & 58 & 14 \\ 3 & 4 & 2 & 7 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 13 & 3 \\ 3 & 4 & 4 & 9 & 0 & 23 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 39 & 0 & 100 & 18 \\ 3 & 4 & 2 & 7 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 7 & 3 \\ 3 & 4 & 4 & 13 & 0 & 13 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 9 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 30 & 0 & 28 & 10 \\ 3 & 4 & 2 & 10 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1113} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & 3 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{113} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 18 & 3 \\ 3 & 4 & 4 & 5 & 0 & 32 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 33 & 0 & 193 & 25 \\ 3 & 4 & 2 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{114} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 17 & 3 \\ 3 & 4 & 4 & 5 & 0 & 31 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 30 & 0 & 172 & 24 \\ 3 & 4 & 2 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 12 & 3 \\ 3 & 4 & 4 & 9 & 0 & 22 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 36 & 0 & 85 & 17 \\ 3 & 4 & 2 & 7 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1122} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 10 & 3 \\ 3 & 4 & 4 & 18 & 0 & 18 & 4 \\ 2 & 4 & 0 & 12 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 60 & 0 & 58 & 14 \\ 3 & 4 & 2 & 14 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 20 & 3 \\ 3 & 4 & 4 & 5 & 0 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 3 & 5 & 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 36 & 0 & 238 & 28 \\ 3 & 4 & 2 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 2 & 4 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{1114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2996$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 14 & 3 \\ 3 & 4 & 4 & 9 & 0 & 26 & 4 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 5 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 42 & 0 & 118 & 20 \\ 3 & 4 & 2 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{1114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3017$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 17 & 3 \\ 3 & 4 & 4 & 9 & 0 & 31 & 4 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 21 & 0 & 172 & 24 \\ 3 & 4 & 2 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 15 & 3 \\ 3 & 4 & 4 & 9 & 0 & 27 & 4 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 5 & 9 & 6 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 45 & 0 & 133 & 21 \\ 3 & 4 & 2 & 7 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{1113}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3365$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 18 & 3 \\ 3 & 4 & 4 & 9 & 0 & 32 & 4 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 5 & 9 & 6 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 54 & 0 & 193 & 25 \\ 3 & 4 & 2 & 7 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{1114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -4373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 17 & 3 \\ 3 & 4 & 4 & 9 & 0 & 31 & 4 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 5 & 9 & 6 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 51 & 0 & 172 & 24 \\ 3 & 4 & 2 & 7 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{1114}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -5216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 20 & 3 \\ 3 & 4 & 4 & 9 & 0 & 36 & 4 \\ 2 & 4 & 0 & 6 & 0 & 24 & 2 \\ 5 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 60 & 0 & 238 & 28 \\ 3 & 4 & 2 & 7 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

4.1143 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{111}31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{magenta}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 6 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11111}} 3 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 24 & 11 \\ 5 & 9 & 6 & 6 & 0 & 24 & 13 & 7 \\ 3 & 4 & 2 & 3 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1112}} 3 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 12 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 2 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1113}} 3 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 18 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 2 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 6 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & 3 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}} 3 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 15 & 21 & 0 & 15 & 85 & 17 \\ 3 & 4 & 2 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1114}} 3 1^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 24 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 24 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 2 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

- [9] $1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & 3 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1112}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 9 & 12 & 0 & 21 & 28 & 10 \\ 3 & 4 & 2 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 9 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 15 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 6 & 3 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{\overline{1122}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 12 & 0 & 28 & 30 & 10 \\ 7 & 13 & 9 & 12 & 0 & 30 & 28 & 10 \\ 3 & 4 & 2 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{\overline{1114}} 3 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 31 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 21 & 30 & 0 & 21 & 172 & 24 \\ 3 & 4 & 2 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & 3 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 15 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & 7 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1144 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 6 & 0 & 6 & 3 & 2 \\ 5 & 12 & 6 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 6 & 0 & 6 & 0 & 2 \\ 3 & 5 & 2 & 7 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 2 & 4 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 2 & 4 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 651, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1141}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 813, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 21 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 18 & 15 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1121}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1145 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -6, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 & 3 \\ 3 & 4 & 8 & 9 & 0 & 5 & 4 \\ 4 & 8 & 9 & 12 & 0 & 6 & 5 \\ 5 & 9 & 12 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 9 & 0 & 3 & 3 \\ 3 & 4 & 5 & 7 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -51, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 5 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 12 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 5 & 2 & 7 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -51, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 2 & 3 \\ 4 & 9 & 4 & 36 & 0 & 4 & 5 \\ 2 & 4 & 0 & 15 & 0 & 3 & 2 \\ 14 & 36 & 15 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 5 & 2 & 19 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 3 & 4 & 2 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 4 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

14.8

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 2 & 3 \\ 4 & 9 & 4 & 28 & 0 & 4 & 5 \\ 2 & 4 & 0 & 12 & 0 & 3 & 2 \\ 11 & 28 & 12 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 12 & 0 & 0 & 2 \\ 3 & 5 & 2 & 15 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 2 & 3 \\ 4 & 9 & 4 & 20 & 0 & 4 & 5 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 20 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 5 & 2 & 11 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] 1^{122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 5 & 13 & 0 & 8 & 4 \\ 3 & 5 & 3 & 12 & 0 & 6 & 3 \\ 7 & 13 & 12 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 18 & 0 & 9 & 5 \\ 3 & 4 & 3 & 10 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] 1^{132} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 5 & 17 & 0 & 8 & 4 \\ 3 & 5 & 3 & 15 & 0 & 6 & 3 \\ 9 & 17 & 15 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 24 & 0 & 9 & 5 \\ 3 & 4 & 3 & 13 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[11] 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 14 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 2 & 11 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] 1^{113} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 2 & 3 \\ 4 & 9 & 4 & 32 & 0 & 4 & 5 \\ 2 & 4 & 0 & 15 & 0 & 3 & 2 \\ 13 & 32 & 15 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 5 & 2 & 18 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & 2 & 4 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114}31^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 2 & 3 \\ 4 & 9 & 4 & 44 & 0 & 4 & 5 \\ 2 & 4 & 0 & 21 & 0 & 3 & 2 \\ 18 & 44 & 21 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 21 & 0 & 0 & 2 \\ 3 & 5 & 2 & 25 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114}31^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 2 & 3 \\ 4 & 9 & 4 & 40 & 0 & 4 & 5 \\ 2 & 4 & 0 & 18 & 0 & 3 & 2 \\ 16 & 40 & 18 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 18 & 0 & 0 & 2 \\ 3 & 5 & 2 & 22 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{141}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 & 3 \\ 11 & 21 & 18 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 3 & 16 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 2 & 7 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 & 3 \\ 7 & 13 & 12 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 3 & 10 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{131}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -933, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 & 3 \\ 9 & 17 & 15 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 3 & 13 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -1200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 12 & 6 \\ 3 & 4 & 2 & 7 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.1146 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 10$ [0] $1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 0 & 3 & 6 & 2 \\ 5 & 12 & 6 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 15 & 0 & 6 & 13 & 7 \\ 3 & 5 & 2 & 7 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 2 & 4 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 507$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 651$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{1114}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 813$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 18 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 18 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & 3 & -4 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 15 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 15 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.1147 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 & 3 \\ 5 & 9 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1148 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{123}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 12 & 0 & 4 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 6 & 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 48 & 15 & 0 & 12 \\ 3 & 5 & 15 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 9 & 6 & 0 & 5 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 15 & -3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 9 & 12 & 0 & 5 \\ 5 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 36 & 9 & 0 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{122}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 1 & 3 \\ 3 & 4 & 8 & 18 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 10 & 18 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 14 & -3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 \\ 3 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 \\ 3 & 5 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 24 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 14 & -3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 27 & 18 & 0 & 9 \\ 4 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 11 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 16 & -3 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 11 & -3 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 7 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 9 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 13 & -3 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 27 & 9 & 0 & 9 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 7 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 21 & -3 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 5 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{121}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 57 & 12 & 0 & 13 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 48 & 12 & 0 & 12 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 15 & -3 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 27 & 21 & 0 & 9 \\ 5 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{131}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 48 & 27 & 0 & 12 \\ 5 & 9 & 27 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 9 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 13 & -3 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 18 & -3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{123}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 4 & 8 & 9 & 30 & 0 & 5 \\ 12 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 17 & -3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 7 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 28 & -3 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 9 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{114}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 11 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 9 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{112}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 22 & -3 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{131}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 19 & -3 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{112}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 18 & -3 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{122}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 27 & 30 & 0 & 9 \\ 7 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 25 & -3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 16 & -3 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{123}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 9 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 22 & -3 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 12 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 32 & 0 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 32 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 25 & -3 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 14 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1643, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 17 & 31 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 24 & -3 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 36 & 0 & 4 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1149 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{so}_8$

$T = 9 \quad [0] \quad 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1150 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9$ [0] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 19 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 12 & 21 & 0 & 73 & 15 \\ 3 & 4 & 2 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 5 & 3 \\ 3 & 4 & 4 & 19 & 0 & 9 & 4 \\ 2 & 4 & 0 & 12 & 0 & 6 & 2 \\ 11 & 19 & 12 & 73 & 0 & 33 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 33 & 0 & 13 & 7 \\ 3 & 4 & 2 & 15 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -173$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 & 6 \\ 3 & 5 & 9 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 \\ 3 & 4 & 6 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{121}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 2 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 & 9 \\ 3 & 5 & 12 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 3 & 0 & 1 & 3 \\ 3 & 4 & 9 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & 2 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 5 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 15 & 0 & 37 & 11 \\ 3 & 4 & 2 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 12 & 0 & 28 & 10 \\ 3 & 4 & 2 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 \\ 3 & 4 & 6 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{112}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1122}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 7 & 3 \\ 3 & 4 & 9 & 8 & 0 & 13 & 4 \\ 5 & 9 & 12 & 12 & 0 & 21 & 6 \\ 4 & 8 & 12 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 12 & 6 & 0 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 6 & 0 & 13 & 7 \\ 3 & 4 & 6 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 14 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 5 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 24 & 0 & 37 & 11 \\ 3 & 4 & 2 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 10 & 3 \\ 3 & 4 & 4 & 5 & 0 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & 2 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 16 & 3 \\ 3 & 4 & 4 & 5 & 0 & 28 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 30 & 0 & 154 & 22 \\ 3 & 4 & 2 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 2 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 8 & 3 \\ 3 & 4 & 4 & 14 & 0 & 14 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 39 & 0 & 37 & 11 \\ 3 & 4 & 2 & 11 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -738, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -744, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 13 & 3 \\ 3 & 4 & 4 & 5 & 0 & 23 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 24 & 0 & 100 & 18 \\ 3 & 4 & 2 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -779, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 12 & 3 \\ 3 & 4 & 4 & 5 & 0 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 21 & 0 & 85 & 17 \\ 3 & 4 & 2 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -828, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & 3 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -845, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 13 & 4 \\ 5 & 9 & 12 & 6 & 0 & 21 & 6 \\ 2 & 4 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 9 & 0 & 28 & 10 \\ 3 & 4 & 6 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -941, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 5 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 2 & 7 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -962, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 10 & 3 \\ 3 & 4 & 4 & 14 & 0 & 18 & 4 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 8 & 14 & 9 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 48 & 0 & 58 & 14 \\ 3 & 4 & 2 & 11 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1112} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -980, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 16 & 3 \\ 3 & 4 & 4 & 9 & 0 & 28 & 4 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 5 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 48 & 0 & 154 & 22 \\ 3 & 4 & 2 & 7 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{1113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 5 & 9 & 6 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 2 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\overline{1113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 4 & 0 & 17 & 4 \\ 5 & 9 & 12 & 6 & 0 & 27 & 6 \\ 2 & 4 & 6 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 12 & 0 & 49 & 13 \\ 3 & 4 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{\overline{1114}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 11 & 3 \\ 3 & 4 & 9 & 4 & 0 & 21 & 4 \\ 5 & 9 & 12 & 6 & 0 & 33 & 6 \\ 2 & 4 & 6 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 15 & 0 & 76 & 16 \\ 3 & 4 & 6 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{\overline{1113}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{\overline{113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 15 & 3 \\ 3 & 4 & 4 & 5 & 0 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & 2 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{\overline{1122}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 10 & 3 \\ 3 & 4 & 4 & 13 & 0 & 18 & 4 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 7 & 13 & 9 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 42 & 0 & 58 & 14 \\ 3 & 4 & 2 & 10 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{\overline{1113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & 3 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{\overline{1112}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 10 & 3 \\ 3 & 4 & 4 & 9 & 0 & 18 & 4 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 5 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 30 & 0 & 58 & 14 \\ 3 & 4 & 2 & 7 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 7 & 3 \\ 3 & 4 & 4 & 13 & 0 & 13 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 9 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 30 & 0 & 28 & 10 \\ 3 & 4 & 2 & 10 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1122} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 13 & 3 \\ 3 & 4 & 4 & 9 & 0 & 23 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 39 & 0 & 100 & 18 \\ 3 & 4 & 2 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & 3 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 18 & 3 \\ 3 & 4 & 4 & 5 & 0 & 32 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 33 & 0 & 193 & 25 \\ 3 & 4 & 2 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 17 & 3 \\ 3 & 4 & 4 & 5 & 0 & 31 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 30 & 0 & 172 & 24 \\ 3 & 4 & 2 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 12 & 3 \\ 3 & 4 & 4 & 9 & 0 & 22 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 36 & 0 & 85 & 17 \\ 3 & 4 & 2 & 7 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1122} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 10 & 3 \\ 3 & 4 & 4 & 18 & 0 & 18 & 4 \\ 2 & 4 & 0 & 12 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 60 & 0 & 58 & 14 \\ 3 & 4 & 2 & 14 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 20 & 3 \\ 3 & 4 & 4 & 9 & 0 & 36 & 4 \\ 2 & 4 & 0 & 6 & 0 & 24 & 2 \\ 5 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 60 & 0 & 238 & 28 \\ 3 & 4 & 2 & 7 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

4.1151 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 6 & 0 & 4 & 3 & 4 \\ 2 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 3 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 5 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 6 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & 3 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 5 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & 7 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1152 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 8 & 0 & 3 & 4 \\ 5 & 9 & 12 & 12 & 0 & 5 & 6 \\ 4 & 8 & 12 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 3 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 2 & 3 \\ 3 & 4 & 4 & 19 & 0 & 3 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 11 & 19 & 12 & 73 & 0 & 11 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 11 & 0 & 0 & 2 \\ 3 & 4 & 2 & 15 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 2 & 3 \\ 3 & 4 & 4 & 8 & 0 & 3 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 4 & 8 & 6 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 2 & 3 \\ 3 & 4 & 8 & 4 & 0 & 3 & 4 \\ 4 & 8 & 9 & 6 & 0 & 4 & 5 \\ 2 & 4 & 6 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 5 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 5 & 12 & 6 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & 2 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 2 & 3 \\ 3 & 4 & 4 & 17 & 0 & 3 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 9 & 17 & 12 & 49 & 0 & 9 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 9 & 0 & 0 & 2 \\ 3 & 4 & 2 & 13 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 2 & 3 \\ 3 & 4 & 4 & 21 & 0 & 3 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 11 & 21 & 15 & 76 & 0 & 11 & 16 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 11 & 0 & 0 & 2 \\ 3 & 4 & 2 & 16 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 2 & 3 \\ 3 & 4 & 4 & 18 & 0 & 3 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 10 & 18 & 12 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 10 & 0 & 0 & 2 \\ 3 & 4 & 2 & 14 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 2 & 3 \\ 3 & 4 & 4 & 23 & 0 & 3 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 13 & 23 & 15 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 13 & 0 & 0 & 2 \\ 3 & 4 & 2 & 18 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 2 & 3 \\ 3 & 4 & 4 & 22 & 0 & 3 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 12 & 22 & 15 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 12 & 0 & 0 & 2 \\ 3 & 4 & 2 & 17 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 2 & 3 \\ 3 & 4 & 4 & 27 & 0 & 3 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 15 & 27 & 18 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 15 & 0 & 0 & 2 \\ 3 & 4 & 2 & 21 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 2 & 3 \\ 3 & 4 & 4 & 26 & 0 & 3 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 14 & 26 & 18 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 14 & 0 & 0 & 2 \\ 3 & 4 & 2 & 20 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{113}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 2 & 3 \\ 3 & 4 & 4 & 32 & 0 & 3 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 & 2 \\ 18 & 32 & 21 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 18 & 0 & 0 & 2 \\ 3 & 4 & 2 & 25 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 2 & 3 \\ 3 & 4 & 4 & 31 & 0 & 3 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 & 2 \\ 17 & 31 & 21 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 17 & 0 & 0 & 2 \\ 3 & 4 & 2 & 24 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 2 & 3 \\ 3 & 4 & 4 & 36 & 0 & 3 & 4 \\ 2 & 4 & 0 & 24 & 0 & 2 & 2 \\ 20 & 36 & 24 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 20 & 0 & 0 & 2 \\ 3 & 4 & 2 & 28 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1153 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 3 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 3 & 4 & 4 \\ 5 & 9 & 12 & 6 & 0 & 5 & 6 & 6 \\ 2 & 4 & 6 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & 3 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 5 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 15 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1154 NHC(-2)(-3) + su₃ + su₂ + f₄ + su2n3mix

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 8 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & 2 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 3 & 0 & 11 & 2 \\ 2 & 4 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 11 & 11 & 0 & 55 & 13 \\ 3 & 4 & 2 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 5 & 0 & 7 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 6 & 0 & 7 & 5 \\ 3 & 4 & 2 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 5 & 4 & 0 & 11 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 8 & 4 \\ 2 & 4 & 0 & 6 & 0 & 5 & 2 \\ 5 & 9 & 6 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 14 & 0 & 12 & 6 \\ 3 & 4 & 2 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 3 & 5 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & 2 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}3^1}1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 20 & 4 \\ 2 & 4 & 0 & 3 & 0 & 14 & 2 \\ 3 & 5 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 14 & 18 & 0 & 69 & 15 \\ 3 & 4 & 2 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 4 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 5 & 0 & 7 & 5 \\ 3 & 4 & 2 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 4 & 0 & 7 & 4 \\ 5 & 9 & 12 & 6 & 0 & 11 & 6 \\ 2 & 4 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 5 & 0 & 7 & 5 \\ 3 & 4 & 6 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 2 & 4 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 8 & 0 & 25 & 9 \\ 3 & 4 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}3^1}1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 5 & 0 & 16 & 4 \\ 2 & 4 & 0 & 3 & 0 & 11 & 2 \\ 3 & 5 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 11 & 15 & 0 & 44 & 12 \\ 3 & 4 & 2 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 4 & 3 \\ 3 & 4 & 4 & 9 & 0 & 7 & 4 \\ 2 & 4 & 0 & 6 & 0 & 5 & 2 \\ 5 & 9 & 6 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 11 & 0 & 7 & 5 \\ 3 & 4 & 2 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}3^1}1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 12 & 4 \\ 2 & 4 & 0 & 6 & 0 & 8 & 2 \\ 5 & 9 & 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 20 & 0 & 25 & 9 \\ 3 & 4 & 2 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

4.1155 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}3^1}1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 5 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 12 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 5 & 2 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -119$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 3 & 4 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -126$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 4 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 2 & 3 \\ 4 & 9 & 4 & 28 & 0 & 4 & 5 \\ 2 & 4 & 0 & 12 & 0 & 3 & 2 \\ 11 & 28 & 12 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 12 & 0 & 0 & 2 \\ 3 & 5 & 2 & 15 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 2 & 3 \\ 4 & 9 & 4 & 20 & 0 & 4 & 5 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 20 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 5 & 2 & 11 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -182$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.9$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 5 & 13 & 0 & 8 & 4 \\ 3 & 5 & 3 & 12 & 0 & 6 & 3 \\ 7 & 13 & 12 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 18 & 0 & 9 & 5 \\ 3 & 4 & 3 & 10 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & 3 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{132}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 5 & 17 & 0 & 8 & 4 \\ 3 & 5 & 3 & 15 & 0 & 6 & 3 \\ 9 & 17 & 15 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 24 & 0 & 9 & 5 \\ 3 & 4 & 3 & 13 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 14 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 2 & 11 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 2 & 3 \\ 4 & 9 & 4 & 32 & 0 & 4 & 5 \\ 2 & 4 & 0 & 15 & 0 & 3 & 2 \\ 13 & 32 & 15 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 5 & 2 & 18 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 2 & 3 \\ 4 & 9 & 4 & 44 & 0 & 4 & 5 \\ 2 & 4 & 0 & 21 & 0 & 3 & 2 \\ 18 & 44 & 21 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 21 & 0 & 0 & 2 \\ 3 & 5 & 2 & 25 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 2 & 3 \\ 4 & 9 & 4 & 40 & 0 & 4 & 5 \\ 2 & 4 & 0 & 18 & 0 & 3 & 2 \\ 16 & 40 & 18 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 18 & 0 & 0 & 2 \\ 3 & 5 & 2 & 22 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -527, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 & 3 \\ 11 & 21 & 18 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 3 & 16 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -621, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 2 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -707, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 & 3 \\ 7 & 13 & 12 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 3 & 10 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -713, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 & 3 \\ 9 & 17 & 15 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 3 & 13 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -944, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 12 & 6 \\ 3 & 4 & 2 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

4.1156 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 11 \quad [0] \ 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 1 & 3 & 0 \\ 2 & 3 & 2 & 3 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -15, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 1 & 3 \\ 3 & 4 & 3 & 10 & 3 & 0 & 4 \\ 2 & 3 & 0 & 6 & 2 & 0 & 2 \\ 6 & 10 & 6 & 21 & 8 & 0 & 9 \\ 2 & 3 & 2 & 8 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 9 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -23, \ (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 1 & 3 \\ 3 & 4 & 3 & 17 & 3 & 0 & 4 \\ 2 & 3 & 0 & 10 & 2 & 0 & 2 \\ 10 & 17 & 10 & 61 & 13 & 0 & 15 \\ 2 & 3 & 2 & 13 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 15 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} \ 5 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -44, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 0 & 4 \\ 2 & 3 & 0 & 4 & 2 & 0 & 2 \\ 4 & 7 & 4 & 8 & 5 & 0 & 6 \\ 2 & 3 & 2 & 5 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 6 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} \ 5 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -50, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 0 & 4 \\ 2 & 3 & 0 & 8 & 2 & 0 & 2 \\ 8 & 14 & 8 & 38 & 10 & 0 & 12 \\ 2 & 3 & 2 & 10 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 12 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} \ 5 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -60, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 0 & 4 \\ 2 & 3 & 0 & 12 & 2 & 0 & 2 \\ 12 & 21 & 12 & 88 & 15 & 0 & 18 \\ 2 & 3 & 2 & 15 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 18 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

4.1157 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} \ 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 14 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 14 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \ 3 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{14} \ 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 16 & 0 & 4 \\ 5 & 9 & 12 & 24 & 0 & 6 \\ 9 & 16 & 24 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{123} \ 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 12 & 0 & 4 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 6 & 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} \ 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 20 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 0 & -2 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 24 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 17 & 24 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 9 & 0 & 0 & 4 \\ 5 & 9 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 5 & 0 & 0 & 4 \\ 3 & 5 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

- [13] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 13 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 13 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 1 & 3 \\ 3 & 4 & 0 & 19 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 11 & 19 & 0 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 4 & -2 & 1 \end{pmatrix}_{b_0}$
- [22] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 13 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$
- [23] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 10 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 5 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [25] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 15 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 5 & 8 & 9 & 18 & 0 & 5 \\ 6 & 12 & 18 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 37 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 37 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 12 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[41] $21^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 12 & 15 & 6 \\ 15 & 17 & 15 & 13 & 7 \\ 7 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11} 1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{11} 1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{122} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 8 & 13 & 27 & 18 & 0 & 9 \\ 4 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{112} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{11} 1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 \\ 4 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 17 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 9 & 17 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 8 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 28 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{13} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 19 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

- [57] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 20 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 7 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{111} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 11 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{121} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 9 & 6 & 0 & 5 \\ 3 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 8 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{111} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [63] $1^1 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 4 & 5 & 3 & 6 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [64] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{11} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 14 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 14 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{111} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 48 & 15 & 0 & 12 \\ 3 & 5 & 15 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 48 & 12 & 0 & 12 \\ 3 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 7 & 13 & 0 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 20 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 20 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 14 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 24 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 15 & 24 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{1121} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 7 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{1211} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 9 & 6 & 0 & 5 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 97$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 12 & 0 & 4 \\ 5 & 9 & 12 & 18 & 0 & 6 \\ 7 & 12 & 18 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 48 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 48 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 99$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 12 & 21 & 75 & 18 & 0 & 15 \\ 3 & 5 & 18 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 9 & 12 & 0 & 5 \\ 5 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 \\ 3 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 3 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 27 & 9 & 0 & 9 \\ 3 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 5 & 8 & 9 & 12 & 0 & 5 \\ 5 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 11 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 36 & 9 & 0 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 15 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

- [98] $1^{113}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 30 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$
- [99] $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{112}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 5 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 \\ 3 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [102] $1^{122}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 5 & 8 & 9 & 12 & 0 & 5 \\ 4 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{113}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 29 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$
- [104] $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{111}3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 24 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 16 & 24 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 26 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 26 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [107] $1^{123}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 \\ 3 & 5 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [108] $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 24 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 24 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 1 & 3 \\ 3 & 4 & 0 & 28 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 16 & 28 & 0 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 27 & 18 & 0 & 9 \\ 4 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 9 & 14 & 36 & 9 & 0 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{112} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 1 & 3 \\ 3 & 4 & 0 & 18 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 10 & 18 & 0 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[117] $21^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 150$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

- [119] $1^{14}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 11 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [120] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [121] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [122] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 38 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [123] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 6 & 9 & 12 & 9 & 0 & 6 \\ 3 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [125] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 6 & 9 & 12 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [124] $1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 1 & 3 \\ 3 & 4 & 0 & 23 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 13 & 23 & 0 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$
- [128] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [127] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 29 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [126] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [129] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 11 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 10 & 17 & 48 & 15 & 0 & 12 \\ 3 & 5 & 15 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 3 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 27 & 18 & 0 & 9 \\ 5 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 5 & 8 & 9 & 6 & 0 & 5 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 4 & 8 & 9 & 12 & 0 & 5 \\ 5 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 7 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 17 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 1 & 3 \\ 3 & 4 & 0 & 22 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 12 & 22 & 0 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 1 & 3 \\ 3 & 4 & 0 & 26 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 14 & 26 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

[143] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 13 & 7 \\ 7 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 12 & 0 & 4 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 7 & 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 40 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 3 & 4 & 0 & 12 & 0 & 2 \\ 9 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 8 & 13 & 27 & 12 & 0 & 9 \\ 3 & 5 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

- [151] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [150] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [152] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [153] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 5 & 8 & 9 & 18 & 0 & 5 \\ 7 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 10 & -2 & 1 \end{pmatrix}_{b_0}$
- [154] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 39 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 39 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$
- [155] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 16 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$
- [156] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [157] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 45 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$
- [158] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 9 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$
- [159] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 6 & 9 & 12 & 21 & 0 & 6 \\ 7 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 1 \end{pmatrix}_{b_0}$
- [160] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 27 & 21 & 0 & 9 \\ 6 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{12}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 7 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{12}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 8 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 7 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{12}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 7 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{12}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 27 & 9 & 0 & 9 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{12}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 8 & 12 & 18 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{12}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{11}3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{11}3^1 1^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{11}3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 3 & 4 & 0 & 15 & 0 & 2 \\ 12 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{112} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 22 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{131} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 7 & 12 & 24 & 9 & 0 & 8 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 38 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 6 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[183] $21^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 27 & 7 \\ 17 & 20 & 17 & 33 & 8 \\ 15 & 17 & 12 & 27 & 6 \\ 27 & 33 & 27 & 49 & 13 \\ 7 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{13} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 1 & 3 \\ 3 & 4 & 0 & 27 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 15 & 27 & 0 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 254$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 47 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 47 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 257$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 5 & 8 & 9 & 24 & 0 & 5 \\ 9 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 267$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 6 & 9 & 12 & 15 & 0 & 6 \\ 5 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

- [191] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 3 & 4 & 0 & 21 & 0 & 2 \\ 17 & 31 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$
- [192] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 5 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [193] $1^{123} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 7 & 12 & 18 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [194] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [195] $21^{112} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 30 & 7 \\ 20 & 25 & 16 & 40 & 9 \\ 12 & 16 & 9 & 24 & 5 \\ 30 & 40 & 24 & 58 & 14 \\ 7 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$
- [196] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 57 & 12 & 0 & 13 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 36 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$
- [199] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$
- [202] $21^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 24 & 6 \\ 16 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [200] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 28 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 27 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 288$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 1 & 3 \\ 3 & 4 & 0 & 32 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 18 & 32 & 0 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 8 & 13 & 27 & 39 & 0 & 9 \\ 9 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 46 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 293$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 11 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 295$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 10 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 296$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 301$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 48 & 12 & 0 & 12 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 303$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 12 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 3 & 4 & 0 & 18 & 0 & 2 \\ 14 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 14 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 6 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 309$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 8 & 13 & 27 & 9 & 0 & 9 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 309$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 1 & 3 \\ 3 & 4 & 0 & 31 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 17 & 31 & 0 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 319$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 37 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 320$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 7 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 326$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 27 & 21 & 0 & 9 \\ 5 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 331$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 48 & 27 & 0 & 12 \\ 5 & 9 & 27 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 333$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 13 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 9 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 8 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[226] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 4 & 8 & 9 & 30 & 0 & 5 \\ 13 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 4 & 8 & 9 & 30 & 0 & 5 \\ 12 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 7 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 9 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 9 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 11 & 18 & 57 & 12 & 0 & 13 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 15 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 22 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[235] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 8 & 13 & 27 & 21 & 0 & 9 \\ 5 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 11 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 10 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 12 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 1 & 3 \\ 3 & 4 & 0 & 36 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 20 & 36 & 0 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{14} 3^1 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 11 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 9 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 14 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 9 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{12} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 9 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{13} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 8 & 13 & 27 & 30 & 0 & 9 \\ 7 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 45 & 7 \\ 20 & 25 & 16 & 60 & 9 \\ 12 & 16 & 9 & 36 & 5 \\ 45 & 60 & 36 & 133 & 21 \\ 7 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 10 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[254] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 20 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 21 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[262] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 10 & 17 & 48 & 12 & 0 & 12 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 10 & 17 & 48 & 27 & 0 & 12 \\ 5 & 9 & 27 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 12 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 27 & 30 & 0 & 9 \\ 7 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 11 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 27 & 30 & 0 & 9 \\ 8 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 18 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[272] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 498, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 8 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 21 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[274] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 15 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[275] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[276] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[277] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 13 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 9 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 14 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 17 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 14 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 13 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 11 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[291] $21^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 60 & 7 \\ 20 & 25 & 16 & 80 & 9 \\ 12 & 16 & 9 & 48 & 5 \\ 60 & 80 & 48 & 238 & 28 \\ 7 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 15 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[293] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 10 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 645, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 20 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[295] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 12 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 19 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[297] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 19 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[298] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 669, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 17 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{112} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 17 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[300] $1^{112} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 20 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[301] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[302] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[304] $1^{112} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{112} 3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 16 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[307] $1^{123} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 10 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{114} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 17 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[309] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 730, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 12 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{112} 3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 16 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 18 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[312] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 15 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 790, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 13 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[314] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[315] $1^{112} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 19 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[316] $1^{112} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 18 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[317] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[318] $1^{113} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 816, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 888, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 12 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 14 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{113} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 1 & 3 \\ 3 & 4 & 4 & 36 & 0 & 4 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 24 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [352] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[353] \quad 21^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 15 & 7 \\ 14 & 13 & 10 & 14 & 7 \\ 9 & 10 & 4 & 9 & 4 \\ 15 & 14 & 9 & 13 & 7 \\ 7 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[354] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 7 & 5 & 0 & 4 \\ 4 & 7 & 7 & 6 & 0 & 5 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[355] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 5 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[356] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[358] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 8 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 5 & 8 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[357] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 7 & 8 & 0 & 4 \\ 4 & 7 & 7 & 11 & 0 & 5 \\ 5 & 8 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[359] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 4 & 7 & 7 & 5 & 0 & 5 \\ 2 & 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[360] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 9 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 5 & 9 & 5 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[362] \ 1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -139, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 3 & 11 & 0 & 4 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 6 & 11 & 7 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 3 & 12 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 7 & 12 & 8 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -325, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 3 & 16 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 9 & 16 & 10 & 45 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[365] \ 1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -406, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 19 & 0 & 4 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 11 & 19 & 13 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

4.1158 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6$

$$T = 9 \quad [0] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -931, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.1159 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1160 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{113}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 28 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 10 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 19 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 20 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 10 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 9 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 26 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 17 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 30 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 29 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 38 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 40 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 39 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 19 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 18 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1161 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & 0 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.1162 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.67 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

- [2] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [4] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [3] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 5 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 5 & 3 & 4 & 6 & 4 \\ 4 & 5 & 3 & 6 & 4 & 4 \\ 3 & 4 & 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [7] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 3 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [6] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 5 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 4 & 6 & 4 \\ 4 & 5 & 3 & 6 & 4 & 4 \\ 3 & 4 & 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [10] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -1 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [11] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

$$[12] \quad 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

$$[14] \quad 1^{111} 3 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 30, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 36, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

[17] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

[18] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -1 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

$$[19] \quad 1^{121} 1^{111} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 63, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 69, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{12} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.5$$

- [22] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$
- [23] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$
- [24] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$
- [25] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [26] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [27] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [28] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$
- [29] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [30] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 16 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$
- [31] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \, T_{\max} = 10, \, \Delta = -63, \, \det = 181, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, \, T_{\max} = 10, \, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$

[33] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_3 + \textcolor{orange}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$

[34] $1^{111} \ 3 \ 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 248, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$

[35] $1^{112} 1^{111} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 285, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$
 $\left(\begin{array}{cccccc} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -1 & 7 & 10 & 3 \end{array} \right)_{b_0}$

$$T = 9 \quad [37] \quad 1^{\textcolor{red}{11}} 3^{21} \textcolor{red}{1}^{\textcolor{red}{11}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{u}_3 + \textcolor{orange}{s} \textcolor{orange}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{11} 1^{11} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -35$, $\det = -2$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$

$$[38] \quad 1^{111} 3 2 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 4 & 2 \\ 2 & 3 & 0 & 4 & 5 & 7 & 5 \\ 2 & 3 & 0 & 4 & 7 & 5 & 5 \\ 2 & 2 & -1 & 2 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{11} 1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 1 & 3 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[40] $1^1 \overset{1^1}{3} 21^{\overset{1^1 1^1}{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\left(\begin{array}{ccccc} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & 0 & 4 & 2 \\ 3 & 5 & -1 & 4 & 4 \end{array} \right)_{b_0}$

[41] $1^{11} 1^1 3 2 1^{111} + [\textcolor{red}{s} \textcolor{red}{u} 2 \cdot \textcolor{blue}{g} 2] + \textcolor{green}{s} \textcolor{green}{u} 3 + \textcolor{orange}{s} \textcolor{orange}{u} 2 + \textcolor{red}{s} \textcolor{red}{u} 2$ $T = 9$, $T_{\max} = 10$, $\Delta = -35$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 6 & 4 & 11 & 6 \\ 4 & 2 & 8 & 4 \\ 11 & 8 & 19 & 11 \\ 6 & 4 & 11 & 6 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{111} 4^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -2 & 1 & 1 & 0 \\ 6 & 7 & 1 & 9 & 6 & 7 \\ 3 & 4 & 1 & 6 & 2 & 4 \\ 3 & 4 & 0 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{11} 3^{11} 221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -6, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 1 & 4 & 0 & 2 & 5 & 4 \\ 2 & 3 & 5 & 0 & 5 & 6 & 6 \\ 2 & 2 & 3 & -1 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{111} 4^{111} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -2 & 1 & 1 & 0 \\ 3 & 4 & 1 & 2 & 4 & 4 \\ 3 & 4 & 1 & 4 & 2 & 4 \\ 3 & 4 & 0 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11} 4^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 3 & 2 & 1 & 2 & 1 & 3 \\ 1 & 1 & 2 & 0 & 1 & 2 \\ 2 & 1 & 1 & 1 & 0 & 2 \\ 3 & 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{11} 4^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 5 & 3 \\ 6 & 6 & 1 & 5 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{11} 4^{11} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 4 & 3 & 1 & 5 & 4 & 5 \\ 3 & 3 & 1 & 4 & 2 & 4 \\ 3 & 3 & 0 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11} 4^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 & 5 & 5 \\ 3 & 3 & 0 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{11} 4^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 3 & 5 & 4 \\ 3 & 1 & 2 & 4 & 3 \\ 3 & 2 & 1 & 3 & 3 \\ 5 & 4 & 3 & 6 & 6 \\ 4 & 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11} 4^{11} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 4 & 2 & 6 & 6 \\ 4 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{11} 4^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 2 & 3 & 5 & 5 & 4 \\ 3 & 1 & 4 & 3 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 5 & 3 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0} \\
[67] & 1^{11} 4^{111} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 4 & 4 & 3 \\ 5 & 3 & 3 & 4 & 5 \\ 4 & 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 & 5 \end{pmatrix}_{b_0} \\
[66] & 1^{11} 4^{111} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 4 & 4 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 4 & 4 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0} \\
[65] & 1^{111} 4^{111} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 5 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 2 & 5 & 4 \\ 5 & 5 & 0 & 5 & 8 & 6 \\ 3 & 3 & 0 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
[64] & 1^{111} 4^{111} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 1 & 1 & 5 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 5 & 5 & 0 & 8 & 11 & 6 \\ 6 & 6 & 1 & 11 & 14 & 8 \\ 3 & 3 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0} \\
[63] & 1^{11} 4^{111} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 2 & 4 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 4 & 4 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[62] & 1^{112} 4^{112} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 5 & 7 & 4 \\ 3 & 1 & 3 & 5 & 3 \\ 5 & 3 & 3 & 7 & 5 \\ 7 & 5 & 7 & 12 & 8 \\ 4 & 3 & 5 & 8 & 5 \end{pmatrix}_{b_0} \\
[68] & 1^{111} 4^{111} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 3 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[71] & 1^{111} 4^{111} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 4 \\ 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[70] & 1^{111} 4^{111} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 6 & 3 \\ 2 & 1 & 1 & 6 & 5 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 6 & 6 & 1 & 14 & 12 & 8 \\ 6 & 5 & 1 & 12 & 9 & 7 \\ 3 & 3 & 0 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[69] & 1^{111} 4^{111} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 5 & 1 & 5 & 9 & 7 \\ 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{111} 4^{111} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 6 & 6 & 1 & 6 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{111} 4^{111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 9 & 4 \\ 9 & 8 & 1 & 9 & 27 & 11 \\ 3 & 3 & 0 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{112} 31^{11} 4^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 10 & 4 \\ 3 & 1 & 1 & 7 & 3 \\ 2 & 1 & 0 & 4 & 2 \\ 10 & 7 & 4 & 21 & 11 \\ 4 & 3 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[73] $1^{111} 4^{111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 8 & 3 \\ 9 & 8 & 1 & 8 & 27 & 11 \\ 3 & 3 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{111} 4^{111} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 7 & 4 \\ 6 & 6 & 1 & 7 & 14 & 8 \\ 3 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [77] $1^{11} 4^{111} 13221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 7.0$

[78] $1^{11} 4^{111} 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 6 & 5 & 4 \\ 3 & 2 & 5 & 5 & 4 \\ 6 & 5 & 10 & 9 & 8 \\ 5 & 5 & 9 & 7 & 7 \\ 4 & 4 & 8 & 7 & 6 \end{pmatrix}_{b_0}$$

[79] $1^{11} 5^{11} 1^{11} 231^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 11 & 4 \\ 8 & 15 & 0 & 11 & 37 & 13 \\ 3 & 5 & -1 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

[80] $1^{11} 5^{11} 1^{11} 231^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 8 & 3 \\ 5 & 6 & 8 & 0 & 8 & 31 & 11 \\ 2 & 2 & 2 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{\overline{1}^1} 5^1 1^{\overline{2}^1} 231^{\overline{1}^1 \overline{1}^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -47, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 15 & 4 \\ 10 & 20 & 0 & 15 & 63 & 17 \\ 3 & 5 & -1 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[82] \ 2^{\overline{1}^{\overline{1}^1} \overline{1}^1} 3^{\overline{1}^1} 151^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -63, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[83] \ 2^{\overline{1}^{\overline{1}^1} \overline{1}^1} 3^{\overline{1}^1} 151^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -88, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 3 \\ 4 & 5 & 3 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 4 & 4 \\ 3 & 5 & 2 & 4 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{\overline{1}^1} 5^1 1^{\overline{2}^1} 231^{\overline{1}^1 \overline{1}^2 \overline{1}^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -96, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 26 & 4 \\ 18 & 35 & 0 & 26 & 202 & 30 \\ 3 & 5 & -1 & 4 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[85] \ 2^{\overline{1}^{\overline{1}^2} \overline{1}^1} 3^{\overline{1}^1} 151^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -106, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 6 & 3 \\ 4 & 5 & 3 & 5 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 8 & 4 \\ 6 & 10 & 4 & 8 & 14 & 8 \\ 3 & 5 & 2 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[86] \ 2^{\overline{1}^{\overline{1}^2} \overline{1}^1} 3^{\overline{1}^1} 151^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -124, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 6 & 3 \\ 4 & 5 & 3 & 10 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 6 & 10 & 4 & 14 & 16 & 8 \\ 6 & 10 & 4 & 16 & 14 & 8 \\ 3 & 5 & 2 & 8 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[87] \ 2^{\overline{1}^{\overline{1}^3} \overline{1}^1} 3^{\overline{1}^1} 151^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -136, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 9 & 3 \\ 4 & 5 & 3 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 9 & 15 & 6 & 12 & 34 & 12 \\ 3 & 5 & 2 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{\overline{1}^1} 5^1 1^{\overline{3}^1} 231^{\overline{1}^1 \overline{1}^2 \overline{1}^4} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -143, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 26 & 3 \\ 3 & 5 & 0 & 5 & 50 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 37 & 4 \\ 26 & 50 & 0 & 37 & 419 & 43 \\ 3 & 5 & -1 & 4 & 43 & 4 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{\overline{1}^1} 5^1 1^{\overline{2}^1} 231^{\overline{1}^1 \overline{1}^2 \overline{1}^4} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -152, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 30 & 4 \\ 20 & 40 & 0 & 30 & 258 & 34 \\ 3 & 5 & -1 & 4 & 34 & 4 \end{pmatrix}_{b_0}$$

- [90] $2 \begin{smallmatrix} 1^{123} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 9 & 3 \\ 4 & 5 & 3 & 10 & 15 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 6 & 10 & 4 & 14 & 24 & 8 \\ 9 & 15 & 6 & 24 & 34 & 12 \\ 3 & 5 & 2 & 8 & 12 & 4 \end{pmatrix}_{b_0}$
- [91] $2 \begin{smallmatrix} 1^{114} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 & 12 & 3 \\ 4 & 5 & 3 & 5 & 20 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 3 & 5 & 2 & 2 & 16 & 4 \\ 12 & 20 & 8 & 16 & 62 & 16 \\ 3 & 5 & 2 & 4 & 16 & 4 \end{pmatrix}_{b_0}$
- [93] $2 \begin{smallmatrix} 1^{133} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 9 & 9 & 3 \\ 4 & 5 & 3 & 15 & 15 & 5 \\ 2 & 3 & 0 & 6 & 6 & 2 \\ 9 & 15 & 6 & 34 & 36 & 12 \\ 9 & 15 & 6 & 36 & 34 & 12 \\ 3 & 5 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$
- [92] $2 \begin{smallmatrix} 1^{1111} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 & 3 & 3 \\ 4 & 5 & 8 & 5 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 & 6 \\ 3 & 5 & 6 & 2 & 4 & 4 \\ 3 & 5 & 6 & 4 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 & 4 \end{pmatrix}_{b_0}$
- [94] $2 \begin{smallmatrix} 1^{124} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 12 & 3 \\ 4 & 5 & 3 & 10 & 20 & 5 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 6 & 10 & 4 & 14 & 32 & 8 \\ 12 & 20 & 8 & 32 & 62 & 16 \\ 3 & 5 & 2 & 8 & 16 & 4 \end{pmatrix}_{b_0}$
- [95] $2 \begin{smallmatrix} 1^{134} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 9 & 12 & 3 \\ 4 & 5 & 3 & 15 & 20 & 5 \\ 2 & 3 & 0 & 6 & 8 & 2 \\ 9 & 15 & 6 & 34 & 48 & 12 \\ 12 & 20 & 8 & 48 & 62 & 16 \\ 3 & 5 & 2 & 12 & 16 & 4 \end{pmatrix}_{b_0}$
- [96] $2 \begin{smallmatrix} 1^{115} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -232, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 & 15 & 3 \\ 4 & 5 & 3 & 5 & 25 & 5 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 3 & 5 & 2 & 2 & 20 & 4 \\ 15 & 25 & 10 & 20 & 98 & 20 \\ 3 & 5 & 2 & 4 & 20 & 4 \end{pmatrix}_{b_0}$
- [97] $2 \begin{smallmatrix} 1^{125} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -250, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 15 & 3 \\ 4 & 5 & 3 & 10 & 25 & 5 \\ 2 & 3 & 0 & 4 & 10 & 2 \\ 6 & 10 & 4 & 14 & 40 & 8 \\ 15 & 25 & 10 & 40 & 98 & 20 \\ 3 & 5 & 2 & 8 & 20 & 4 \end{pmatrix}_{b_0}$
- [98] $2 \begin{smallmatrix} 1^{144} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 12 & 3 \\ 4 & 5 & 3 & 20 & 20 & 5 \\ 2 & 3 & 0 & 8 & 8 & 2 \\ 12 & 20 & 8 & 62 & 64 & 16 \\ 12 & 20 & 8 & 64 & 62 & 16 \\ 3 & 5 & 2 & 16 & 16 & 4 \end{pmatrix}_{b_0}$$

[99] $2 \overset{1^{135}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.9$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 15 & 3 \\ 4 & 5 & 3 & 15 & 25 & 5 \\ 2 & 3 & 0 & 6 & 10 & 2 \\ 9 & 15 & 6 & 34 & 60 & 12 \\ 15 & 25 & 10 & 60 & 98 & 20 \\ 3 & 5 & 2 & 12 & 20 & 4 \end{pmatrix}_{b_0}$$

[100] $2 \overset{1^{116}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -298, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 18 & 3 \\ 4 & 5 & 3 & 5 & 30 & 5 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 3 & 5 & 2 & 2 & 24 & 4 \\ 18 & 30 & 12 & 24 & 142 & 24 \\ 3 & 5 & 2 & 4 & 24 & 4 \end{pmatrix}_{b_0}$$

[101] $2 \overset{1^{126}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 18 & 3 \\ 4 & 5 & 3 & 10 & 30 & 5 \\ 2 & 3 & 0 & 4 & 12 & 2 \\ 6 & 10 & 4 & 14 & 48 & 8 \\ 18 & 30 & 12 & 48 & 142 & 24 \\ 3 & 5 & 2 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[102] $2 \overset{1^{145}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 15 & 3 \\ 4 & 5 & 3 & 20 & 25 & 5 \\ 2 & 3 & 0 & 8 & 10 & 2 \\ 12 & 20 & 8 & 62 & 80 & 16 \\ 15 & 25 & 10 & 80 & 98 & 20 \\ 3 & 5 & 2 & 16 & 20 & 4 \end{pmatrix}_{b_0}$$

[103] $2 \overset{1^{136}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -346, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 18 & 3 \\ 4 & 5 & 3 & 15 & 30 & 5 \\ 2 & 3 & 0 & 6 & 12 & 2 \\ 9 & 15 & 6 & 34 & 72 & 12 \\ 18 & 30 & 12 & 72 & 142 & 24 \\ 3 & 5 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[104] $2 \overset{1^{117}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 21 & 3 \\ 4 & 5 & 3 & 5 & 35 & 5 \\ 2 & 3 & 0 & 2 & 14 & 2 \\ 3 & 5 & 2 & 2 & 28 & 4 \\ 21 & 35 & 14 & 28 & 194 & 28 \\ 3 & 5 & 2 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[105] $2 \overset{1^{127}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -394, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 21 & 3 \\ 4 & 5 & 3 & 10 & 35 & 5 \\ 2 & 3 & 0 & 4 & 14 & 2 \\ 6 & 10 & 4 & 14 & 56 & 8 \\ 21 & 35 & 14 & 56 & 194 & 28 \\ 3 & 5 & 2 & 8 & 28 & 4 \end{pmatrix}_{b_0}$$

[106] $2 \overset{1^{137}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 21 & 3 \\ 4 & 5 & 3 & 15 & 35 & 5 \\ 2 & 3 & 0 & 6 & 14 & 2 \\ 9 & 15 & 6 & 34 & 84 & 12 \\ 21 & 35 & 14 & 84 & 194 & 28 \\ 3 & 5 & 2 & 12 & 28 & 4 \end{pmatrix}_{b_0}$$

[107] $22 \overset{1^{111}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -568, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 13 & 8 \\ 5 & 13 & 8 & 11 & 11 & 6 \\ 5 & 13 & 11 & 9 & 11 & 7 \\ 5 & 13 & 11 & 11 & 9 & 7 \\ 3 & 8 & 6 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [108] \quad 1^{11} 1^{11} 4 \quad 141321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[109] \quad 1^{11} 1^{11} 5 \quad 1321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 5 & 7 & 4 \\ 3 & 2 & 5 & 3 & 7 & 5 \\ 5 & 5 & 7 & 7 & 10 & 8 \\ 3 & 3 & 4 & 5 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{11} 1^{11} 4 \quad 14131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 & 2 \\ 2 & 2 & 1 & 3 & 2 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{11} 1^{11} 5 \quad 1321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 5 & 7 & 4 \\ 3 & 2 & 5 & 3 & 7 & 5 \\ 4 & 5 & 7 & 7 & 10 & 8 \\ 3 & 3 & 4 & 5 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{11} 1^{11} 4 \quad 141^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 1 & 2 & 2 & 4 & 4 \\ 3 & 4 & 0 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{11} 1^{11} 5 \quad 1321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 6 & 3 & 6 & 6 & 6 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 6 & 2 & 3 & 5 & 5 \\ 3 & 6 & 2 & 5 & 3 & 5 \\ 3 & 6 & 2 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{11} 1^{11} 5 \quad 1321^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 6 & 3 & 18 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 18 & 6 & 43 & 15 & 15 \\ 3 & 6 & 2 & 15 & 3 & 5 \\ 3 & 6 & 2 & 15 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{11} 1^{11} 5 \quad 1321^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 6 & 3 & 12 & 6 & 6 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 12 & 4 & 18 & 10 & 10 \\ 3 & 6 & 2 & 10 & 3 & 5 \\ 3 & 6 & 2 & 10 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{11} 1^{11} 5 \quad 1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 2 & 0 & 12 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 12 & 0 & 29 & 13 \\ 3 & 3 & 4 & 0 & 13 & 5 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 6 & 9 & 12 & 6 & 6 \\ 4 & 9 & 9 & 14 & 7 & 7 \\ 6 & 12 & 14 & 18 & 10 & 10 \\ 3 & 6 & 7 & 10 & 3 & 5 \\ 3 & 6 & 7 & 10 & 5 & 5 \end{pmatrix}_{b_0}$$

[127] $1^{1^{11}} 5^{131} 1321^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 & 3 \\ 3 & 6 & 9 & 18 & 6 & 6 \\ 4 & 9 & 9 & 21 & 7 & 7 \\ 9 & 18 & 21 & 43 & 15 & 15 \\ 3 & 6 & 7 & 15 & 3 & 5 \\ 3 & 6 & 7 & 15 & 5 & 5 \end{pmatrix}_{b_0}$$

[128] $1^{12} 5^{1^{13}} 1321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 2 & 5 & 19 & 4 \\ 3 & 2 & 5 & 3 & 19 & 5 \\ 13 & 12 & 19 & 19 & 79 & 21 \\ 3 & 3 & 4 & 5 & 21 & 5 \end{pmatrix}_{b_0}$$

[129] $1^{13} 5^{1^{13}} 1321^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 2 & 5 & 26 & 4 \\ 3 & 2 & 5 & 3 & 26 & 5 \\ 17 & 17 & 26 & 26 & 153 & 29 \\ 3 & 3 & 4 & 5 & 29 & 5 \end{pmatrix}_{b_0}$$

[130] $1^{13} 5^{1^{13}} 1321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 21 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 21 & 0 & 106 & 24 \\ 3 & 3 & 4 & 0 & 24 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [132] $1^{111} 4^{111} 14141^{131^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 7.0$

[133] $1^{11} 4^{111} 1414131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 7.0$

[131] $1^{111} 4^{111} 14141^{131^{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 7.0$

[135] $1^{1^{11}} 6^{1^{11}} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 13, T_{\max} = 13, \Delta = -4, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & 1 & 0 & 4 & 0 & 2 \\ 3 & 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

[134] $1^{1^{11}} 6^{1^{11}} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 13, T_{\max} = 13, \Delta = -4, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 2 & 4 \\ 3 & 4 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[136] $1^{1^{11}} 6^{1^{11}} 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 13, T_{\max} = 13, \Delta = -4, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 2 & 3 & 4 & 6 & 4 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 2 & 2 & 6 & 4 \\ 5 & 6 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

[138] $1^{1^{11}} 6^{1^{11}} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 13, T_{\max} = 13, \Delta = -4, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 2 & 3 & 0 & 6 & 4 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 6 & 6 & 0 & 10 & 8 \\ 3 & 4 & 3 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{\overset{11}{\underset{1131}{1}}} \ 6 \ 131^{\overset{12}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 3 \\ 2 & 2 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 8 & 0 & 26 & 12 \\ 3 & 4 & 3 & 0 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{\overset{11}{\underset{1131}{1}}} \ 6 \ 131^{\overset{12}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 2 & 3 & 4 & 10 & 4 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 2 & 2 & 10 & 4 \\ 7 & 10 & 8 & 10 & 26 & 12 \\ 3 & 4 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{\overset{11}{\underset{11131}{1}}} \ 6 \ 131^{\overset{12}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 7 & 3 \\ 2 & 2 & 3 & 6 & 10 & 4 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 6 & 6 & 10 & 18 & 8 \\ 7 & 10 & 8 & 18 & 26 & 12 \\ 3 & 4 & 3 & 8 & 12 & 5 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [143] \ 1^{\overset{11231}{\underset{11}{1}}} \ 6 \ 131^{\overset{11}{\underset{11}{1}}} \ 6 \ 131^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.5$$

$$[144] \ 1^{\overset{11231}{\underset{11}{1}}} \ 6 \ 131^{\overset{11}{\underset{11}{1}}} \ 6 \ 131^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.0$$

$$[142] \ 1^{\overset{11231}{\underset{11}{1}}} \ 6 \ 131^{\overset{12}{\underset{11}{1}}} \ 6 \ 131^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.5$$

$$[141] \ 1^{\overset{11131}{\underset{11}{1}}} \ 6 \ 131^{\overset{11}{\underset{11}{1}}} \ 6 \ 131^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.0$$

$$T = 17 \quad [145] \ 1^{\overset{11231}{\underset{11}{1}}} \ 6 \ 1316131^{\overset{11}{\underset{11}{1}}} \ 41^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 7.0$$

$$T = 19 \quad [146] \ 1^{\overset{11231}{\underset{11}{1}}} \ 6 \ 1316131^{\overset{12}{\underset{11}{1}}} \ 6 \ 131^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 19, T_{\max} = 19, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 19, 2), c_{\text{ext}} = 7.5$$

$$T = 20 \quad [147] \ 1^{\overset{11231}{\underset{11}{1}}} \ 6 \ 13161316131^{\overset{11}{\underset{11}{1}}} \ 41^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 20, T_{\max} = 20, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 20, 2), c_{\text{ext}} = 7.0$$

4.1163 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overset{111}{\underset{11}{1}}} \ 31^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overset{11}{\underset{11}{1}}} \ 3 \ 1^{\overset{11}{\underset{11}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 6 & 0 & 4 \\ 4 & 5 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 3 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 4 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 4 & 5 & 3 & 4 & 6 & 0 & 4 \\ 4 & 5 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 3 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & 0 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{121} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 12 & 3 & 4 \\ 8 & 13 & 0 & 12 & 28 & 9 & 10 \\ 3 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -1 & 4 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1121} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 12 & 15 & 7 \\ 5 & 8 & 0 & 12 & 10 & 12 & 6 \\ 6 & 9 & 0 & 15 & 12 & 13 & 7 \\ 3 & 4 & -1 & 7 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1211} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 12 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 15 & 7 \\ 6 & 9 & 0 & 12 & 15 & 13 & 7 \\ 3 & 4 & -1 & 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 0 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 0 & 12 & 13 & 7 \\ 3 & 4 & -1 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{112} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 0 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 9 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -1 & 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -55, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 3 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -1 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -77, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -87, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -87, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 5 & 0 & 5 & 5 & 10 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 & 7 & 4 \\ 3 & 5 & 0 & 4 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 7 & 14 & 8 \\ 3 & 5 & -1 & 4 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [44] \quad 1^{11} 1^{1111} 31^1 4^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 5 & 4 & 4 \\ 3 & 1 & 4 & 3 & 4 & 3 \\ 5 & 4 & 6 & 5 & 5 & 6 \\ 5 & 3 & 5 & 3 & 4 & 5 \\ 4 & 4 & 5 & 4 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11} 1^{1111} 31^1 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 4 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 4 & 4 & 2 & 3 & 4 & 5 \\ 5 & 3 & 2 & 4 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{11} 1^{11} 4^1 131^{1111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 & 4 \\ 3 & 1 & 1 & 4 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 4 & 4 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

4.1164 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{1111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 6 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{1111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 9 & 3 \\ 8 & 13 & 0 & 12 & 9 & 9 & 28 & 10 \\ 3 & 4 & -1 & 4 & 3 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11111} 3^1 1^{1111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11111} 3^1 1^{1111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 6 & 7 \\ 3 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1121}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 9 & 6 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 21 & 10 \\ 6 & 9 & 0 & 0 & 6 & 21 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 3 & 6 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 15 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 224, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 7 & 7 \\ 3 & 4 & -1 & 3 & 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 364, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 400, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 11, 1), \ c_{\text{ext}} = 9.0$$

4.1165 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$T = 11 \quad [7] \quad 1^{11} 4^{11} 13^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 1 & 4 \\ 3 & 1 & 1 & 4 & 0 & 3 \\ 2 & 1 & 0 & 2 & 0 & 2 \\ 5 & 4 & 2 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^{11} 5^{11} 1^{11} 23^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 5 & 0 & 5 & 15 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 11 & 3 & 4 \\ 8 & 15 & 0 & 11 & 37 & 10 & 13 \\ 2 & 4 & 0 & 3 & 10 & 1 & 3 \\ 3 & 5 & -1 & 4 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1167 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$

$$T = 10 \quad [1] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 3 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 \\ 3 & 4 & -1 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.1168 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 3 & 4 & -1 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 61, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 1 & 0 & 0 \\ 3 & 4 & 1 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 4 \\ 3 & 1 & 1 & 0 & 3 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 1 & 4 \\ 3 & 1 & 4 & 0 & 3 \\ 5 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [9] \quad 1^{11} 5^{12} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 & 2 \\ 2 & 3 & 0 & 4 & 5 & 3 & 5 \\ 1 & 1 & 0 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & 2 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 7 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[12] $1^{11} \overset{1}{5} 1^{11} 23^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 8 & 2 & 2 \\ 5 & 6 & 0 & 8 & 31 & 7 & 11 \\ 1 & 1 & 0 & 2 & 7 & 0 & 2 \\ 2 & 2 & -1 & 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{11} \overset{1}{5} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[14] $21^{11} \overset{1}{5} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[15] $1^{11} \overset{1}{5} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 7 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{11} \overset{1}{5} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 4 & 2 & 4 & 2 & 3 & 4 \\ 2 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{11} \overset{1}{5} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{12} \overset{1}{5} 13^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 9 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{11} \overset{1}{5} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 3 & 3 & 1 & 6 & 3 & 3 \\ 4 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{12} \overset{1}{5} 13^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 3 & 3 & 1 & 12 & 3 & 3 \\ 8 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13} \underset{1^1}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 3 & 3 & 1 & 18 & 3 & 3 \\ 12 & 21 & 18 & 97 & 15 & 21 \\ 2 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} \underset{1^1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 5 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12} \underset{1^1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 4 & 5 & 2 & 2 & 3 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} \underset{1^1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 6 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{13} \underset{1^1}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 14 & 21 & 18 & 97 & 15 & 21 \\ 2 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} \underset{1^1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} \underset{1^1}{5} 123^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} \underset{1^{11}}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{\frac{1}{1^3}} 5 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 7 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{\frac{1}{1^1}} 5 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\frac{1}{1^1}} 5 123^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 5 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\frac{1}{1^1}} 5 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 5 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{\frac{1}{1^4}} 5 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 12 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{\frac{1}{1^2}} 5 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{\frac{1}{1^4}} 5 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 10 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{\frac{1}{1^1}} 5 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 5 & 8 & 8 & 6 & 5 & 6 \\ 2 & 5 & 6 & 2 & 3 & 4 \\ 2 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{12} \overset{1^3}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 9 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$

[37] $21^1 \overset{1^1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 5 & 8 & 8 & 6 & 5 & 6 \\ 2 & 5 & 6 & 2 & 3 & 4 \\ 2 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[36] $1^1 \overset{1^1}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$

[35] $1^1 \overset{1^1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 2 & 4 & 2 & 3 & 4 \\ 2 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[41] $1^{12} \overset{1^3}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 11 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$

[42] $1^{13} \overset{1^3}{5} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 21 & 18 & 97 & 15 & 21 \\ 2 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[43] $1^1 \overset{1^1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[45] $1^{11} \overset{1^4}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 10 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 10 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 2 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$

[44] $1^1 \overset{1^2}{5} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$\begin{aligned}
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0} \\
[46] & 1^{1^2}_{1^1} 5^{1^2} 13^{1^1} 1^{1^2} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0} \\
[47] & 1^{1^4}_{1^1} 5^{1^4} 13^{1^1} 1^{1^3} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 16 & 21 & 18 & 97 & 15 & 21 \\ 2 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0} \\
[48] & 1^{1^3}_{1^1} 5^{1^3} 13^{1^1} 1^{1^3} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -44, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0} \\
[49] & 1^{1^2}_{1^1} 5^{1^2} 13^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 2 \\ 6 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[50] & 21^{1^1}_{1^1} 5^{1^1} 13^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[51] & 1^{1^1}_{1^1} 5^{1^1} 13^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -47, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 5 & 0 & 18 & 5 & 8 \\ 1 & 1 & 2 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & -1 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[52] & 1^{1^3}_{1^1} 5^{1^3} 13^{1^1} 1^{1^4} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 5 & 3 & 20 & 4 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 20 & 8 & 62 & 12 & 16 \\ 2 & 4 & 2 & 12 & 1 & 3 \\ 3 & 5 & 2 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[53] & 1^{1^1}_{1^1} 5^{1^1} 13^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{12} 5^{12} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{14} 5^{14} 13^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 5 & 3 & 20 & 4 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 12 & 20 & 8 & 62 & 12 & 16 \\ 2 & 4 & 2 & 12 & 1 & 3 \\ 3 & 5 & 2 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[56] \quad 21^{12} 5^{12} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{12} 5^{12} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 7 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 5^{11} 13^1 23^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 15 & 0 & 37 & 10 & 13 \\ 2 & 4 & 0 & 10 & 1 & 3 \\ 3 & 5 & -1 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{13} 5^{13} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 9 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 5^{12} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 8 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 5^{11} 13^1 23^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 10 & 20 & 0 & 63 & 13 & 17 \\ 2 & 4 & 0 & 13 & 1 & 3 \\ 3 & 5 & -1 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$

[illegible]

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{12} \overset{1^4}{5} 1^3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -214, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 2 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 14 & 12 & 16 & 70 & 13 & 18 \\ 2 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12} \overset{1^3}{5} 1^3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 2 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 13 & 12 & 16 & 70 & 13 & 18 \\ 2 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13} \overset{1^4}{5} 1^3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -305, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 2 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 16 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{13} \overset{1^3}{5} 1^3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -314, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 2 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13} \overset{1^4}{5} 1^3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -427, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 2 & 3 \\ 2 & 1 & 3 & 17 & 2 & 3 \\ 2 & 3 & 1 & 22 & 3 & 3 \\ 18 & 17 & 22 & 137 & 18 & 25 \\ 2 & 2 & 3 & 18 & 1 & 3 \\ 3 & 3 & 3 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1169 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} \overset{1^{11}}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \overset{1^{11}}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1^{11}}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 5 & 4 \\ 3 & 1 & 1 & 0 & 4 & 3 \\ 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 2 & 0 & 6 & 6 \\ 4 & 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [10] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 5 & 3 \\ 6 & 7 & 6 & 0 & 5 & 9 & 7 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 2 & 3 & 0 & 3 \\ 2 & 3 & 2 & 1 & 2 & 0 & 3 \\ 2 & 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 5 & 3 \\ 5 & 7 & 6 & 0 & 5 & 9 & 7 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 5^{12} 13^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 3 & 0 & 3 & 14 & 4 \\ 2 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 10 & 3 \\ 10 & 14 & 12 & 0 & 10 & 42 & 14 \\ 3 & 4 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[12] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 13^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 4 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[13] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 & 5 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 2 & 0 & 3 & 2 & 4 \\ 3 & 5 & 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[14] \begin{matrix} 1^1 \\ 1^1 \end{matrix} 5 1^1 23^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 4 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 3 & 11 & 4 \\ 2 & 4 & 0 & 3 & 1 & 10 & 3 \\ 8 & 15 & 0 & 11 & 10 & 37 & 13 \\ 3 & 5 & -1 & 4 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[15] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 2 & 4 & 0 & 3 & 2 & 4 \\ 3 & 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[16] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 17, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 3 & 0 & 3 & 21 & 4 \\ 2 & 3 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 15 & 3 \\ 15 & 21 & 18 & 0 & 15 & 97 & 21 \\ 3 & 4 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[17] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 5 & 3 & 0 & 4 & 10 & 5 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 6 & 3 \\ 6 & 10 & 4 & 0 & 6 & 14 & 8 \\ 3 & 5 & 2 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[18] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 3 & 0 & 3 & 14 & 4 \\ 2 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 10 & 3 \\ 11 & 14 & 12 & 0 & 10 & 42 & 14 \\ 3 & 4 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[19] $1^1_1 5^{13} 13^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 5 & 3 & 0 & 4 & 15 & 5 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 9 & 3 \\ 9 & 15 & 6 & 0 & 9 & 34 & 12 \\ 3 & 5 & 2 & 0 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{11}_{1^1} 5^{12} 13^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[22] $1^1_1 5^{13} 13^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{13}_1 5^{13} 13^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 2 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 7 & 10 & 4 & 14 & 6 & 8 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 & 3 \\ 3 & 5 & 2 & 8 & 3 & 2 & 4 \\ 3 & 5 & 2 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{12}_1 5^{13} 13^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 26, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 5 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 5 & 2 & 2 & 3 & 8 & 4 \\ 2 & 4 & 2 & 3 & 1 & 6 & 3 \\ 6 & 10 & 4 & 8 & 6 & 14 & 8 \\ 3 & 5 & 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{13}_1 5^{13} 13^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 9 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{14}_1 5^{13} 13^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 5 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 5 & 2 & 2 & 3 & 16 & 4 \\ 2 & 4 & 2 & 3 & 1 & 12 & 3 \\ 12 & 20 & 8 & 16 & 12 & 62 & 16 \\ 3 & 5 & 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{11}_{1^1} 5^{12} 13^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 3 & 2 & 4 & 2 & 3 & 6 & 4 \\ 2 & 2 & 3 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12} \overset{11}{5} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 3 & 0 & 2 & 8 & 3 \\ 2 & 3 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 7 & 3 \\ 8 & 8 & 8 & 0 & 7 & 22 & 10 \\ 3 & 3 & 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} \overset{12}{5} 13^{11} \overset{121}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 5 & 10 & 4 & 14 & 6 & 8 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 & 3 \\ 3 & 5 & 2 & 8 & 3 & 2 & 4 \\ 3 & 5 & 2 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{12} \overset{11}{5} 13^{11} \overset{122}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 5 & 3 & 10 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 & 4 & 2 \\ 5 & 10 & 4 & 14 & 6 & 16 & 8 \\ 2 & 4 & 2 & 6 & 1 & 6 & 3 \\ 6 & 10 & 4 & 16 & 6 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{13} \overset{11}{5} 13^{11} \overset{123}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.3$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 9 & 3 \\ 3 & 5 & 3 & 10 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 2 & 6 & 2 \\ 5 & 10 & 4 & 14 & 6 & 24 & 8 \\ 2 & 4 & 2 & 6 & 1 & 9 & 3 \\ 9 & 15 & 6 & 24 & 9 & 34 & 12 \\ 3 & 5 & 2 & 8 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} \overset{13}{5} 13^{11} \overset{12}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 3 & 0 & 2 & 7 & 3 \\ 2 & 3 & 1 & 0 & 3 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 8 & 3 \\ 9 & 7 & 10 & 0 & 8 & 25 & 11 \\ 3 & 3 & 3 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11} \overset{11}{5} 13^{11} \overset{11}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 5 & 3 \\ 5 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11} \overset{11}{5} 13^{11} \overset{111}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 3 & 2 & 4 & 2 & 3 & 6 & 4 \\ 2 & 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} \overset{11}{5} 13^{11} \overset{111}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[35] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 8 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[36] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 5 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 3 & 8 & 4 \\ 2 & 4 & 2 & 3 & 1 & 6 & 3 \\ 6 & 10 & 4 & 8 & 6 & 14 & 8 \\ 3 & 5 & 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[37] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 9 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[38] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 5 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 3 & 5 & 2 & 2 & 3 & 16 & 4 \\ 2 & 4 & 2 & 3 & 1 & 12 & 3 \\ 11 & 20 & 8 & 16 & 12 & 62 & 16 \\ 3 & 5 & 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[39] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 3 & 0 & 2 & 10 & 3 \\ 2 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 10 & 3 \\ 11 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[40] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 3 & 0 & 2 & 13 & 3 \\ 2 & 3 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 12 & 3 \\ 13 & 13 & 14 & 0 & 12 & 65 & 17 \\ 3 & 3 & 3 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$4.1170 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$$

$$T = 10 \quad [1] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

4.1171 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_{2n3\text{mix}}$

$$T = 11 \quad [4] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 2 & 2 & 4 & 2 & 3 & 4 \\ 3 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 9 & 0 & 22 & 10 \\ 3 & 4 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 7 & 6 & 9 & 5 & 7 \\ 3 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 2 & 4 & 2 & 9 & 4 \\ 8 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 6 & 7 & 6 & 9 & 5 & 7 \\ 3 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

- [6] $1^{12}_{1^1} 5^{11} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 9 & 14 & 12 & 42 & 10 & 14 \\ 3 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$
- [8] $1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 7 & 6 & 9 & 5 & 7 \\ 3 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$
- [9] $1^{11}_{1^1} 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 10 & 9 & 0 & 22 & 10 \\ 3 & 4 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$
- [10] $1^{12}_{1^1} 5^{12} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 17 & 15 & 0 & 65 & 17 \\ 3 & 4 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$
- [11] $1^{11}_{1^1} 5^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$
- [12] $1^{11}_{1^1} 5^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$
- [13] $1^{121}_{1^1} 5^{11} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 4 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$
- [15] $1^{13}_{1^1} 5^{13} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 14 & 21 & 18 & 97 & 15 & 21 \\ 3 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$
- [16] $1^{12}_{1^1} 5^{131} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 11 & 14 & 12 & 42 & 10 & 14 \\ 3 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{12}_{1^1} 5^{1^{21}} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 14 & 12 & 42 & 10 & 14 \\ 3 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{11}_{1^1} 5^{1^{11}} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 3 & 0 & 9 & 5 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 4 & 0 & 11 & 7 \\ 3 & 5 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{13}_{1^1} 5^{1^{31}} 13^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 7 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{12}_{1^1} 5^{1^{11}} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 7 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{11}_{1^1} 5^{1^{11}} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 2 & 4 & 2 & 3 & 4 \\ 3 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{11}_{1^1} 5^{1^{11}} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{13}_{1^1} 5^{1^{13}} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 3 & 0 & 24 & 4 \\ 2 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 17 & 24 & 21 & 0 & 130 & 24 \\ 3 & 4 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{12}_{1^1} 5^{1^{13}} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 17 & 15 & 0 & 65 & 17 \\ 3 & 4 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{131}_{1^1} 5^{13} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 15 & 6 & 34 & 9 & 12 \\ 3 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$

[28] $1^{121}_{1^1} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 10 & 4 & 14 & 6 & 8 \\ 3 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$

[30] $1^{121}_{1^1} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 5 & 8 & 10 & 4 & 5 \\ 4 & 8 & 8 & 12 & 5 & 6 \\ 6 & 10 & 12 & 14 & 6 & 8 \\ 3 & 4 & 5 & 6 & 1 & 3 \\ 3 & 5 & 6 & 8 & 3 & 4 \end{pmatrix}_{b_0}$

[29] $1^{111}_{1^1} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 \\ 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[26] $1^{131}_{1^1} 5^{13} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 3 & 3 \\ 3 & 5 & 8 & 15 & 4 & 5 \\ 4 & 8 & 8 & 18 & 5 & 6 \\ 9 & 15 & 18 & 34 & 9 & 12 \\ 3 & 4 & 5 & 9 & 1 & 3 \\ 3 & 5 & 6 & 12 & 3 & 4 \end{pmatrix}_{b_0}$

[25] $1^{112}_{1^1} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 6 & 3 \\ 3 & 5 & 8 & 5 & 9 & 5 \\ 4 & 8 & 8 & 6 & 11 & 6 \\ 3 & 5 & 6 & 2 & 7 & 4 \\ 6 & 9 & 11 & 7 & 11 & 7 \\ 3 & 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0}$

[24] $1^{131}_{1^1} 5^{13} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 15 & 3 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 21 & 18 & 97 & 15 & 21 \\ 3 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[27] $1^{111}_{1^1} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 4 & 8 & 8 & 6 & 5 & 6 \\ 3 & 5 & 6 & 2 & 3 & 4 \\ 3 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[32] $1^{112}_{1^1} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 3 & 0 & 14 & 5 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 14 & 6 & 0 & 29 & 11 \\ 3 & 5 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11}_{1^1} 5^{11} 13^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{12}_{1^1} 5^{11} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 9 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11}_{1^1} 5^{11} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 7 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{12}_{1^1} 5^{11} 13^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 5 & 0 & 8 & 6 \\ 3 & 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{113}_{1^1} 5^{11} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 3 & 0 & 19 & 5 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 19 & 8 & 0 & 55 & 15 \\ 3 & 5 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{121}_{1^1} 5^{11} 13^1 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 5 & 3 \\ 3 & 5 & 3 & 15 & 9 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 8 & 15 & 6 & 34 & 21 & 12 \\ 5 & 9 & 4 & 21 & 11 & 7 \\ 3 & 5 & 2 & 12 & 7 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{1^{12}}_5 13^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 5 & 3 & 10 & 14 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 10 & 4 & 14 & 22 & 8 \\ 8 & 14 & 6 & 22 & 29 & 11 \\ 3 & 5 & 2 & 8 & 11 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{1^3}_5 13^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 \\ 3 & 5 & 2 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{1^{11}}_5 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{1^{12}}_5 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 7 & 4 \\ 6 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{1^{21}}_5 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{1^{21}}_5 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 6 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{1^{21}}_5 13^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 6 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{1^{31}}_5 13^1 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 5 & 3 \\ 3 & 5 & 3 & 20 & 9 & 5 \\ 2 & 3 & 0 & 8 & 4 & 2 \\ 11 & 20 & 8 & 62 & 28 & 16 \\ 5 & 9 & 4 & 28 & 11 & 7 \\ 3 & 5 & 2 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

- [50] $1^{11}_{1^1} 5^{13} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 9 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$
- [49] $1^{11}_{1^1} 5^{11} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 7 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$
- [52] $1^{11}_{1^1} 5^{13} 13^1 1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 3 & 10 & 19 & 5 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 5 & 10 & 4 & 14 & 30 & 8 \\ 11 & 19 & 8 & 30 & 55 & 15 \\ 3 & 5 & 2 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$
- [51] $1^{12}_{1^1} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 9 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$
- [53] $1^{131}_{1^1} 5^{13} 13^1 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 5 & 3 \\ 3 & 5 & 3 & 15 & 9 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 9 & 15 & 6 & 34 & 21 & 12 \\ 5 & 9 & 4 & 21 & 11 & 7 \\ 3 & 5 & 2 & 12 & 7 & 4 \end{pmatrix}_{b_0}$$
- [54] $1^{11}_{1^1} 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 9 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$
- [55] $1^{14}_{1^1} 5^{13} 13^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 5 & 3 & 5 & 24 & 5 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 14 & 24 & 10 & 19 & 89 & 19 \\ 3 & 5 & 2 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$
- [57] $1^{13}_{1^1} 5^{121} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 & 3 \\ 2 & 1 & 3 & 13 & 2 & 3 \\ 2 & 3 & 1 & 14 & 3 & 3 \\ 12 & 13 & 14 & 65 & 12 & 17 \\ 3 & 2 & 3 & 12 & 1 & 3 \\ 3 & 3 & 3 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$
- [58] $1^{11}_{1^1} 5^{12} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 8 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11}_{1^1} 5^{121} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 8 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11}_{1^1} 5^{131} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 9 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11}_{1^1} 5^{12} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{11}_{1^1} 5^{13} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 9 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{13}_{1^1} 5^{121} 13^1 1^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 14 & 15 & 18 & 97 & 15 & 21 \\ 3 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{12}_{1^1} 5^{121} 13^1 1^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 12 & 12 & 16 & 70 & 13 & 18 \\ 3 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{11}_{1^1} 5^{13} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 9 & 13 & 0 & 44 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{12}_{1^1} 5^{121} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{12}_{1^1} 5^{13} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -123$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 11 & 0 & 39 & 13 \\ 3 & 3 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{12}_{1^1} 5^{13} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -123$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 11 & 0 & 39 & 13 \\ 3 & 3 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[68] $1^{13}_{1^1} 5^{13} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -142$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 \\ 3 & 5 & 2 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{13}_{1^1} 5^{31} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -143$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 & 3 \\ 2 & 1 & 3 & 13 & 2 & 3 \\ 2 & 3 & 1 & 14 & 3 & 3 \\ 13 & 13 & 14 & 65 & 12 & 17 \\ 3 & 2 & 3 & 12 & 1 & 3 \\ 3 & 3 & 3 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{12}_{1^1} 5^{31} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -143$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 11 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{12}_{1^1} 5^{31} 13^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -151$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 & 3 \\ 2 & 1 & 3 & 14 & 2 & 3 \\ 2 & 3 & 1 & 20 & 3 & 3 \\ 15 & 14 & 20 & 106 & 16 & 22 \\ 3 & 2 & 3 & 16 & 1 & 3 \\ 3 & 3 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{12}_{1^1} 5^{31} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -171$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 13 & 12 & 16 & 70 & 13 & 18 \\ 3 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{12}_{1^1} 5^{13} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -187$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 12 & 15 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{13}_{1^1} 5^{131} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -197, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 3 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{13}_{1^1} 5^{131} 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -203, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 & 3 \\ 2 & 1 & 3 & 17 & 2 & 3 \\ 2 & 3 & 1 & 22 & 3 & 3 \\ 17 & 17 & 22 & 137 & 18 & 25 \\ 3 & 2 & 3 & 18 & 1 & 3 \\ 3 & 3 & 3 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{13}_{1^1} 5^{131} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -264, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 17 & 0 & 92 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

4.1172 NHC(-2)(-3) + su₃ + su₂ + su₂n3mix + su₂

$T = 12$ [0] $1^{11}_{1^1} 5^{111} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 2 & 4 & 2 & 6 & 4 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{11}_{1^1} 5^{111} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 5 & 2 & 2 & 7 & 4 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 & 7 \\ 3 & 5 & 2 & 4 & 7 & 2 & 4 \\ 3 & 5 & 2 & 4 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{12}_{1^1} 5^{112} 13^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 5 & 9 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 4 & 2 \\ 2 & 5 & 2 & 2 & 7 & 8 & 4 \\ 5 & 9 & 4 & 7 & 11 & 14 & 7 \\ 6 & 10 & 4 & 8 & 14 & 14 & 8 \\ 3 & 5 & 2 & 4 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{13}_{1^1} 5^{113} 13^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 5 & 9 & 15 & 5 \\ 2 & 3 & 0 & 2 & 4 & 6 & 2 \\ 2 & 5 & 2 & 2 & 7 & 12 & 4 \\ 5 & 9 & 4 & 7 & 11 & 21 & 7 \\ 9 & 15 & 6 & 12 & 21 & 34 & 12 \\ 3 & 5 & 2 & 4 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

[4] $1^{21}_{1^1} 5^{121} 13^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 71, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 & 2 \\ 2 & 5 & 2 & 2 & 11 & 4 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 & 11 \\ 3 & 5 & 2 & 4 & 11 & 2 & 4 \\ 3 & 5 & 2 & 4 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1^{31}} 5^{1^{11}} 13^{1^{11}} 1^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 107, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 & 2 \\ 2 & 5 & 2 & 2 & 15 & 4 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 & 15 \\ 3 & 5 & 2 & 4 & 15 & 2 & 4 \\ 3 & 5 & 2 & 4 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

4.1173 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{1^{11}} 4^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1^{11}} 4^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 5 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1^{11}} 4^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1^{11}} 4^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -61, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.97 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1^{11}} 4^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 5 & 7 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 21^{1^{11}} 5^{1^{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1^{11}} 5^{1^{11}} 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 4 & 6 & 7 & 4 & 0 & 5 \\ 2 & 4 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 3 & 21 & 0 & 4 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 12 & 21 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13} \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 3 & 18 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 10 & 18 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 21^{11} \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 3 \\ 4 & 4 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 2 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 21^{12} \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 2 & 3 \\ 4 & 4 & 3 & 8 & 3 & 4 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 8 & 4 & 10 & 4 & 6 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 21^{13} \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 2 & 3 \\ 4 & 4 & 3 & 12 & 3 & 4 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 12 & 6 & 25 & 6 & 9 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 2 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

4.1174 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{11} \quad 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \quad 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \quad 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \quad 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 3 & 5 & 0 & 7 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 6 & 0 & 8 & 6 \\ 3 & 4 & 2 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 5 & 0 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 0 & 7 & 4 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 3 & 4 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 91, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 0 & 7 & 4 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 2 & 4 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 4 & 0 & 7 & 4 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 2 & 4 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.1175 NHC(-2)(-3) + su₃ + su₂ + su₃ + su₂ + su₂

$T = 11$ [4] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 3 & 5 & 0 & 7 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 5 & 6 & 0 & 8 & 5 & 6 \\ 2 & 4 & 2 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & 4 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 3 & 5 & 0 & 7 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 5 & 6 & 0 & 8 & 5 & 6 \\ 3 & 4 & 2 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & 4 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 5 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & 3 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 0 & 5 & 4 \\ 2 & 3 & 0 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 0 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & 3 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 3 & 4 & 2 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 5 & 5 & 0 & 8 & 5 & 6 \\ 2 & 4 & 2 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

4.1176 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 11, 1), \quad c_{\text{ext}} = 11.0$$

$$[1] \quad 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 11, 1), \quad c_{\text{ext}} = 11.0$$

4.1177 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$

$$T = 10 \quad [0] \quad 1^{11} 5^{11} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5^{11} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 5^{11} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 10 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 8 & 10 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{5} 1^{12} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 85, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 5 & 7 & 4 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 1^{12} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 3 & 11 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 7 & 11 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} \overset{1}{5} 1^{32} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 3 & 17 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 13 & 17 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \overset{1}{5} 1^{12} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 3 & 15 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 9 & 15 & 8 & 40 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 1^{22} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 196, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 10 & 14 & 8 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{2^1 21^{11}} \overset{1}{5} \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 5 & 1 & 3 \\ 6 & 13 & 5 & 12 & 0 & 7 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 5 & 12 & 4 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13} \overset{1}{5} 1^{22} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 316, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 3 & 18 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 12 & 18 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{13} \overset{1}{5} 1^{32} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 381, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 0 & 4 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 15 & 21 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 2^{121} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 10 & 1 & 3 \\ 6 & 13 & 5 & 24 & 0 & 7 \\ 2 & 5 & 0 & 8 & 0 & 2 \\ 10 & 24 & 8 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 2^{121} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 901, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 15 & 1 & 3 \\ 6 & 13 & 5 & 36 & 0 & 7 \\ 2 & 5 & 0 & 12 & 0 & 2 \\ 15 & 36 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

4.1178 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 7 \quad [0] \ 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -17, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -20, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 9 & 9 & 12 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -24, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -28, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 4 & 4 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -29, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 4 & 8 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -29, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.59 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 7 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{13}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 10 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.23 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 6 & 9 & 12 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -74, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 7 & 9 & 12 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 21^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 13^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -77, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 7 & 13 & 27 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 21^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -79, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{12}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -84, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{13}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -87, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 13^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -98, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 9 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 21^{11}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 3 & 4 \\ 15 & 17 & 48 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 4 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 21^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 12 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $21^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 13 & 3 & 4 \\ 9 & 13 & 27 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $21^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -314, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 17 & 3 & 4 \\ 12 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [44] $1^{13}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[45] $31^{11}31^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 3 & 7 & 3 & 3 & 0 & 3 \\ 3 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[46] $31^{11}31^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[47] $31^{11}31^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 4 & 6 & 1 & 5 & 3 \\ 5 & 10 & 5 & 7 & 5 \\ 3 & 6 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[48] $31^{11}31^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 & 2 \\ 3 & 4 & 4 & 6 & 3 & 4 \\ 3 & 4 & 0 & 3 & 2 & 2 \\ 3 & 6 & 3 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [49] $1^{14}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{14}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{\overline{1}} 4 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -65, \ \det = -13, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 4 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [53] \ 1^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[55] \ 21^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 5 & 8 & 4 & 5 \\ 6 & 8 & 8 & 5 & 6 \\ 3 & 4 & 5 & 1 & 3 \\ 4 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[54] \ 21^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{\overline{1}} 5 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 4, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 4, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{\overline{1}} 4 13^{\overline{1}} 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 11, \ \Delta = -38, \ \det = 8, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{\overline{1}} 5 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 13, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 13, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 3 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 5 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{\overline{1}} \overline{5} 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 21^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 5 & 8 & 8 & 5 & 6 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[64] \quad 21^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 5 & 8 & 8 & 5 & 6 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [67] \quad 1^{\overline{1}} \overline{4} 1413^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{\overline{1}} \overline{4} 1413^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{\overline{1}} \overline{4} 13^{\overline{1}} 1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 6 & 3 & 6 \\ 3 & 1 & 3 \\ 6 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{\overline{1}} \overline{6} 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[68] \ 21^1 6123^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -3, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[71] \ 1^1 3^1 1^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -39, \ \det = -8, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[73] \ 1^1 413^1 1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -12, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[72] \ 1^1 1^1 413^1 1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -12, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[74] \ 1^1 413^1 1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -13, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 21^1 6123^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -13, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 8 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 6 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[76] \ 1^1 6123^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -13, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [77] \ 21^1 61^1 23^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 4 & 12 & 3 & 6 & 4 \\ 12 & 34 & 10 & 18 & 12 \\ 3 & 10 & 1 & 5 & 3 \\ 6 & 18 & 5 & 8 & 6 \\ 4 & 12 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[78] \ 1^1 61^1 23^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 & 2 \\ 4 & 26 & 1 & 14 & 12 & 10 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 14 & 0 & 7 & 6 & 5 \\ 2 & 12 & 0 & 6 & 4 & 4 \\ 2 & 10 & -1 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{\overline{1}^1 3^1 2^1} 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -38, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[80] $1^{\overline{1}^1} 6^1 123^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[81] $1^1 14141^{\overline{1}^1} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 134, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 8 & 1 & 4 & 3 \\ 8 & 17 & 0 & 8 & 9 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 8 & 0 & 2 & 4 \\ 3 & 9 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$T = 13$ [83] $1^{\overline{1}^1} 5^1 131513^{\overline{1}^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 5 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 6 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

[82] $1^{\overline{1}^1} 5^1 131513^{\overline{1}^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[84] $1^{\overline{1}^1} 6^1 131^1 513^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -22, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[85] $1^{\overline{1}^1} 6^1 1^1 231^1 513^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -108, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 9 & 3 & 1 & 3 \\ 9 & 34 & 10 & 0 & 12 \\ 3 & 10 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 12 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 14$ [89] $1^{\overline{1}^1} 6^1 1231513^{\overline{1}^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 0 & 2 \\ 4 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[88] $1^{\overline{1}^1} 3^1 216131^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

[86] $1^{11}3^{11}2161315^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 2 & 3 & 0 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[87] $1^{11}6^{11}131513^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 4 & 4 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

$T = 15$ [91] $1^{11}3^{11}21712315^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

[90] $1^{11}3^{11}21712315^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 2 & 3 & 0 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.1179 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$T = 8$ [0] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 3 & 0 & 4 \\ 4 & 8 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 208, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 5 & 9 & 12 & 5 & 0 & 6 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 278, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 3 & 0 & 4 \\ 7 & 13 & 27 & 7 & 0 & 9 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 328, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 9 & 17 & 48 & 9 & 0 & 12 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1180 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{so}}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -392, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1181 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 3 & 0 & 8 & 4 \\ 5 & 9 & 12 & 5 & 0 & 12 & 6 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 4 & 0 & 10 & 6 \\ 3 & 4 & 6 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 8 & 4 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 4 & 0 & 10 & 6 \\ 3 & 4 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 3 & 0 & 14 & 4 \\ 2 & 4 & 0 & 2 & 0 & 9 & 2 \\ 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 8 & 0 & 37 & 11 \\ 3 & 4 & 2 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 2 & 3 \\ 3 & 4 & 13 & 3 & 0 & 4 & 4 \\ 7 & 13 & 27 & 7 & 0 & 9 & 9 \\ 2 & 3 & 7 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 2 & 0 & 1 & 3 \\ 3 & 4 & 9 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 3 & 0 & 13 & 4 \\ 2 & 4 & 0 & 2 & 0 & 9 & 2 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 7 & 0 & 28 & 10 \\ 3 & 4 & 2 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{112}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 3 & 0 & 28 & 4 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 2 & 3 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 16 & 0 & 154 & 22 \\ 3 & 4 & 2 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 3 & 0 & 17 & 4 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 9 & 0 & 49 & 13 \\ 3 & 4 & 2 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 11 & 0 & 76 & 16 \\ 3 & 4 & 2 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 2 & 3 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 10 & 0 & 58 & 14 \\ 3 & 4 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 3 & 0 & 23 & 4 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 13 & 0 & 100 & 18 \\ 3 & 4 & 2 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 3 & 0 & 22 & 4 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 12 & 0 & 85 & 17 \\ 3 & 4 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 2 & 3 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 15 & 0 & 133 & 21 \\ 3 & 4 & 2 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 3 & 0 & 26 & 4 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 14 & 0 & 118 & 20 \\ 3 & 4 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 3 & 0 & 32 & 4 \\ 2 & 4 & 0 & 2 & 0 & 21 & 2 \\ 2 & 3 & 2 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 18 & 0 & 193 & 25 \\ 3 & 4 & 2 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 3 & 0 & 31 & 4 \\ 2 & 4 & 0 & 2 & 0 & 21 & 2 \\ 2 & 3 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 17 & 0 & 172 & 24 \\ 3 & 4 & 2 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 3 & 0 & 36 & 4 \\ 2 & 4 & 0 & 2 & 0 & 24 & 2 \\ 2 & 3 & 2 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 20 & 0 & 238 & 28 \\ 3 & 4 & 2 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

4.1182 $\text{NHC}(-2)(-3) + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{113}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 3 & 2 \\ 2 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 3 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & 2 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 5 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 5 & 0 & 6 & 6 & 6 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & 2 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 5 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 6 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 5 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1183 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{123}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -34, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 8 & 3 & 0 & 5 & 4 & 4 \\ 4 & 8 & 9 & 4 & 0 & 6 & 5 & 5 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 & -3 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 5 & 2 & -3 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 4 & 5 & 5 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 & 2 \\ 3 & 5 & 2 & 2 & -3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -274, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

4.1184 NHC(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{so}_8$

$$T = 8 \quad [0] \ 1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 3 & 0 & 4 \\ 4 & 8 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[8] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 9 & 7 \\ 17 & 20 & 17 & 11 & 8 \\ 15 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 7 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 5 & 9 & 12 & 5 & 0 & 6 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[10] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 12 & 8 & 6 \\ 16 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 3 & 0 & 4 \\ 7 & 13 & 27 & 7 & 0 & 9 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 9 & 17 & 48 & 9 & 0 & 12 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1185 $\text{NHC}(-2)(-3) + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \mathfrak{f}_4$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8 + \mathfrak{f}_4$ $T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1186 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{so}_8$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1187 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 1 & 0 & 2 & 4 \\ 3 & 4 & 0 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 2 & 4 \\ 3 & 1 & 0 & 1 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 4 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overset{1}{\underset{11}{1}}} 4 \overset{1}{1} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 5 & 4 \\ 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 4 & 0 & 6 & 6 \\ 4 & 3 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [15] \quad 1^{\overset{1}{\underset{11}{1}}} 5 \overset{1}{1} 123^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 & 8 \\ 1 & 1 & 0 & 2 & 0 & 7 \\ 5 & 6 & 0 & 8 & 7 & 31 \\ 2 & 2 & -1 & 2 & 2 & 11 \end{pmatrix}_{b_0}$$

$$[12] \quad 21^{\overset{1}{\underset{11}{1}}} 5 \overset{1}{1} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overset{1}{\underset{11}{1}}} 5 \overset{1}{1} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 4 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overset{1}{\underset{11}{1}}} 5 \overset{1}{1} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 7 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overset{1}{\underset{11}{1}}} 5 \overset{1}{1} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{\overset{1}{\underset{11}{1}}} 5 \overset{1}{1} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 4 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{\overset{1}{\underset{11}{1}}} 5 \overset{1}{1} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} \overset{1}{5} 123 \overset{1}{1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 & 4 & 2 \\ 1 & 1 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} \overset{1}{5} 13 \overset{1}{1} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 9 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 3 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 4 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12} \overset{1}{5} 13 \overset{1}{1} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 3 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 8 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{13} \overset{1}{5} 13 \overset{1}{1} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 3 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 12 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12} \overset{1}{5} 13 \overset{1}{1} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 4 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 3 & 4 & 0 & 5 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{11} \overset{1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[25] $1^{11} \overset{12}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 6 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[26] $1^{13} \overset{12}{5} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 14 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[29] $1^{11} \overset{1}{5} 123^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 1 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \\ 2 & 2 & 1 & 0 & 1 & 0 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$

[32] $1^{11} \overset{1}{5} 123^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[28] $1^{11} \overset{1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[30] $1^{13} \overset{1}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 7 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$

[27] $1^{11} \overset{1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[31] $1^{11} \overset{1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{12} \overset{14}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 12 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} \overset{12}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 10 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^1 \overset{1}{5} 1^1 23^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 3 & 3 & 4 & 0 & 4 & 10 & 6 \\ 2 & 2 & 2 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^1 \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 5 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^1 \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 5 & 8 & 8 & 5 & 6 & 6 \\ 2 & 4 & 5 & 1 & 3 & 3 \\ 2 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^1 \overset{4}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 10 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^1 \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{12} \overset{3}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 9 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[36] $1^{1^1} 5^{1^1} 13^{1^1} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 5 & 8 & 8 & 5 & 6 & 6 \\ 2 & 4 & 5 & 1 & 3 & 3 \\ 2 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{1^3} 5^{1^3} 13^{1^1} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 11 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{1^3} 5^{1^3} 13^{1^1} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 15 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{1^1} 5^{1^1} 13^{1^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{1^4} 5^{1^4} 13^{1^1} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 10 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{1^2} 5^{1^2} 13^{1^1} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 8 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{1^4} 5^{1^4} 13^{1^1} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 16 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{1^2} 5^{1^2} 13^{1^1} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 6 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{1^3}_1 5^1 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 9 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[50] $21^{11}_1 5^1 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{1^2}_1 5^1 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{1^3}_1 5^1 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 3 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 12 & 3 \\ 11 & 20 & 8 & 12 & 62 & 16 \\ 3 & 5 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{11}_1 5^1 1^1 23^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 5 & 5 & 0 & 5 & 18 & 8 \\ 2 & 2 & 2 & -1 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{1^4}_1 5^1 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 12 & 3 \\ 12 & 20 & 8 & 12 & 62 & 16 \\ 3 & 5 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[56] $1^{11}_1 5^1 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[55] $1^{1^2}_1 5^1 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 8 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[59] $1^{1^1}_1 5^1 1^1 23^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 4 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 10 & 3 \\ 8 & 15 & 0 & 10 & 37 & 13 \\ 3 & 5 & -1 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{12} \overset{1}{5} 13^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 7 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 21^{12} \overset{1}{5} 13^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 8 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} \overset{3}{5} 13^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 9 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} \overset{12}{5} 13^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 8 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} \overset{1}{5} 1^{12} 23^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 4 & 20 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 13 & 3 \\ 10 & 20 & 0 & 13 & 63 & 17 \\ 3 & 5 & -1 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{12} \overset{14}{5} 13^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 12 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[64] \quad 21^{12} \overset{1}{5} 13^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{12} \overset{13}{5} 13^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 2 & 3 & 1 & 15 & 3 \\ 16 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{13} \overset{1}{5} 13^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 2 & 3 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{13} \overset{1}{5} 13^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -427, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 3 & 22 & 3 \\ 2 & 2 & 3 & 1 & 18 & 3 \\ 18 & 17 & 22 & 18 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [83] \ 1^{11} \overset{1}{6} 123^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[78] \ 21^1 \overset{1}{6} 1^{13} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 0 & 4 & 0 & 4 & 4 \\ 2 & 2 & 0 & 3 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11} 413^1 15^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 3 & 3 & 0 & 1 & 2 & 4 \\ 3 & 2 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{12} \overset{1}{6} 123^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 4 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{11} 413^1 15^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 0 & 0 & 1 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 3 & 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{11} 413^1 15^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 6 & 3 & 7 & 6 \\ 3 & 1 & 3 & 3 \\ 7 & 3 & 7 & 7 \\ 6 & 3 & 7 & 6 \end{pmatrix}_{b_0} \\
[80] & 1^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 9 & 4 \\ 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 3 & 1 & 3 & 7 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 9 & 6 & 7 & 7 & 19 & 9 \\ 4 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0} \\
[79] & 1^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 6 123^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[77] & 21^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 6 123^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 3 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[88] & 1^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 41^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 51^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 6 & 3 & 9 & 6 \\ 3 & 1 & 5 & 3 \\ 9 & 5 & 11 & 9 \\ 6 & 3 & 9 & 6 \end{pmatrix}_{b_0} \\
[87] & 1^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 12 & 4 \\ 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 3 & 1 & 3 & 9 & 3 \\ 3 & 2 & 3 & 1 & 9 & 3 \\ 12 & 9 & 9 & 9 & 34 & 12 \\ 4 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0} \\
[86] & 21^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 0 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 4 & 0 & 6 & 0 & 10 & 6 \\ 2 & 2 & 0 & 3 & 0 & 6 & 3 \end{pmatrix}_{b_0} \\
[92] & 1^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 6 & 3 \\ 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 7 & 6 & 7 & 7 & 19 & 9 \\ 3 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0} \\
[91] & 1^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 6 123^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[90] & 1^{\overline{11}} \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 6 123^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{\overline{11}} \overset{1^2}{6} \overset{1^1}{1^{\overline{13^1 21^1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 1 & 9 & 3 \\ 10 & 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{\overline{11}} \overset{1^1}{41^{\overline{13^1}}} \overset{1^1}{15^{\overline{1}}} \overset{1^2}{1^{\overline{12}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 14, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 6 & 3 & 16 & 6 & 3 \\ 3 & 1 & 8 & 3 & 3 \\ 16 & 8 & 38 & 16 & 6 \\ 6 & 3 & 16 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{\overline{12}} \overset{1^1}{41^{\overline{13^1}}} \overset{1^1}{15^{\overline{1}}} \overset{1^3}{1^{\overline{13}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 14, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 6 & 3 & 23 & 6 & 3 \\ 3 & 1 & 11 & 3 & 3 \\ 23 & 11 & 83 & 23 & 3 \\ 6 & 3 & 23 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{\overline{1^3 1^2 1^1}} \overset{1^1}{6} \overset{1^1}{1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 6 & 3 \\ 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 6 & 6 & 7 & 7 & 19 & 9 \\ 3 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{\overline{1^1}} \overset{1^1}{6} \overset{1^1}{123^{\overline{1}}} \overset{1^1}{1^{\overline{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{\overline{1^3}} \overset{1^3}{6} \overset{1^1}{1^{\overline{113^1 21^1}}} \overset{1^1}{1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 11 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{\overline{11}} \overset{1^3}{41^{\overline{13^1}}} \overset{1^1}{15^{\overline{1}}} \overset{1^1}{1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 & 4 & 2 \\ 4 & 4 & 0 & 0 & 4 & 10 & 6 \\ 2 & 2 & 0 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{\overline{111}} \overset{1^1}{41^{\overline{13^1}}} \overset{1^1}{15^{\overline{1}}} \overset{1^1}{1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 4 & 2 \\ 2 & 1 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 4 & 4 & 10 & 6 \\ 2 & 2 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[100] \quad & 1^{11} \overset{1^4}{6} 123^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 31, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 10 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[102] \quad & 1^{11} \overset{1^1}{6} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 44, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 1 & 9 & 3 \\ 9 & 9 & 9 & 3 & 12 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0} \\
[101] \quad & 21^{11} \overset{1^1 3^1 21^1}{6} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 44, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 1 & 9 & 3 \\ 4 & 8 & 1 & 0 & 18 & 6 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 0 & 34 & 12 \\ 3 & 6 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0} \\
[105] \quad & 21^{11} \overset{1^1 3^1 21^1}{6} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 1 & 10 & 3 \\ 4 & 8 & 1 & 0 & 18 & 6 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 1 & 0 & 35 & 13 \\ 3 & 6 & 0 & 0 & 13 & 4 \end{pmatrix}_{b_0} \\
[106] \quad & 1^{11} \overset{1^4}{6} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 2 & 2 & 3 & 1 & 12 & 3 \\ 14 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0} \\
[103] \quad & 1^{11} \overset{1^2}{6} 123^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[104] \quad & 21^{11} \overset{1^1}{6} 123^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 8 & 1 & 0 & 10 & 6 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 1 & 0 & 9 & 7 \\ 3 & 6 & 0 & 0 & 7 & 4 \end{pmatrix}_{b_0} \\
[107] \quad & 1^{11} \overset{1^1 3^1 21^1}{6} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 47, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 1 & 9 & 3 \\ 8 & 9 & 9 & 3 & 34 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0} \\
[108] \quad & 1^{11} \overset{1^4}{6} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 15 & 3 \\ 2 & 2 & 3 & 1 & 13 & 3 \\ 14 & 10 & 15 & 13 & 63 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{\overline{11}} \overset{1^2}{6} \overset{1^1}{1^{\overline{11}} 3^{\overline{121}} 2^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 58, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 10 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\overline{11}} 4^{\overline{13}} 1^{\overline{15}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 58, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 3 & 2 \\ 5 & 3 & 0 & 3 & 6 & 6 \\ 3 & 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{\overline{11}} \overset{1^1}{6} 1^{\overline{123}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 58, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{\overline{11}} 4^{\overline{13}} 1^{\overline{15}} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 58, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 3 & 2 \\ 7 & 4 & 0 & 3 & 12 & 8 \\ 3 & 2 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overline{11}} \overset{1^1}{6} \overset{1^1}{1^{\overline{11}} 3^{\overline{121}} 2^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 67, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 9 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overline{11}} 4^{\overline{13}} 1^{\overline{15}} 1^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 76, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 0 & 0 & 1 & 7 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 10 & 7 & 0 & 4 & 28 & 12 \\ 3 & 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overline{11}} \overset{1^3}{6} 1^{\overline{123}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 76, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 9 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overline{11}} 4^{\overline{13}} 1^{\overline{15}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 76, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 6 & 5 & 0 & 4 & 12 & 8 \\ 3 & 2 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[118] \quad & 1^{\overset{1^3}{123^1 21^1}} 6^{\overset{1^3}{1}} 1^{\overset{1^3}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 91, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 15 & 3 \\ 2 & 2 & 3 & 1 & 13 & 3 \\ 13 & 10 & 15 & 13 & 63 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[117] \quad & 1^{\overset{1^2}{11^1 6^{12} 123^1 1^{12}}}} 6^{\overset{1^2}{1}} 1^{\overset{1^2}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 91, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 8 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[119] \quad & 1^{\overset{1^1}{11^1 41^1 31^1 15^1 1^{12}}}} 6^{\overset{1^1}{1}} 1^{\overset{1^1}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 0 & 0 & 1 & 6 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 8 & 6 & 0 & 4 & 18 & 10 \\ 3 & 2 & 0 & 2 & 10 & 4 \end{pmatrix}_{b_0} \\
[120] \quad & 1^{\overset{1^3}{11^1 31^1 21^1}} 6^{\overset{1^3}{1}} 1^{\overset{1^3}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 110, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 2 & 2 & 3 & 1 & 12 & 3 \\ 13 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0} \\
[121] \quad & 1^{\overset{1^2}{11^1 31^1 21^1}} 6^{\overset{1^2}{1}} 1^{\overset{1^2}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 145, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 2 & 2 & 3 & 1 & 12 & 3 \\ 12 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0} \\
[122] \quad & 1^{\overset{1^1}{11^1 31^1 21^1}} 6^{\overset{1^1}{1}} 1^{\overset{1^1}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 150, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 2 & 2 & 3 & 1 & 12 & 3 \\ 11 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0} \\
[123] \quad & 1^{\overset{1^4}{11^1 31^1 21^1}} 6^{\overset{1^4}{1}} 1^{\overset{1^4}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 195, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 17 & 3 \\ 2 & 2 & 3 & 1 & 15 & 3 \\ 16 & 13 & 17 & 15 & 86 & 20 \\ 3 & 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0} \\
[124] \quad & 1^{\overset{1^3}{11^1 31^1 21^1}} 6^{\overset{1^3}{1}} 1^{\overset{1^3}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 232, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 17 & 3 \\ 2 & 2 & 3 & 1 & 15 & 3 \\ 15 & 13 & 17 & 15 & 86 & 20 \\ 3 & 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{\overset{1^2}{11^1 31^1 21^1}} 6^{\overset{1^2}{1}} 1^{\overset{1^2}{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 239, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0
\end{aligned}$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 17 & 3 \\ 2 & 2 & 3 & 1 & 15 & 3 \\ 14 & 13 & 17 & 15 & 86 & 20 \\ 3 & 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [126] \quad 1^{11} 6^{11} 13^1 5^1 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [129] \quad 1^{11} 6^{13} 13^1 5^1 13^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 4 & 12 & 4 \\ 2 & 2 & 4 & 2 & 10 & 4 \\ 9 & 7 & 12 & 10 & 29 & 13 \\ 3 & 3 & 4 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{11} 6^{13} 13^1 5^1 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 \\ 2 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 7 & 6 & 10 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

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$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 \\ 3 & 4 & -1 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 1^{11} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 1^{11} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & 2 & 3 & 1 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 2 & 5 & 4 \\ 3 & 1 & 0 & 1 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 & 2 \\ 5 & 4 & 0 & 2 & 6 & 6 \\ 4 & 3 & 0 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 6 & 5 & 0 & 9 & 7 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 5^{11} 13^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 & 3 \\ 2 & 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 5^{12} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 7 & 6 & 5 & 0 & 9 & 7 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 5^{12} 13^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \\ 2 & 3 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 12 & 10 & 0 & 42 & 14 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{21} 5^{11} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 5 & 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 2 & 3 & 0 & 2 & 4 \\ 3 & 5 & 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} 23^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 5 & 0 & 4 & 5 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 10 & 3 \\ 2 & 5 & 0 & 3 & 2 & 11 & 4 \\ 8 & 15 & 0 & 10 & 11 & 37 & 13 \\ 3 & 5 & -1 & 3 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 2 & 0 & 2 & 3 \\ 2 & 3 & 1 & 3 & 0 & 4 & 3 \\ 2 & 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 3 & 0 & 2 & 4 \\ 3 & 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 17, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 3 & 0 & 21 & 4 \\ 2 & 3 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 15 & 21 & 18 & 15 & 0 & 97 & 21 \\ 3 & 4 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 5 & 3 & 4 & 0 & 10 & 5 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 2 & 4 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 10 & 4 & 6 & 0 & 14 & 8 \\ 3 & 5 & 2 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \\ 2 & 3 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 14 & 12 & 10 & 0 & 42 & 14 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 & 3 \\ 7 & 10 & 4 & 6 & 14 & 8 & 8 \\ 3 & 5 & 2 & 3 & 8 & 2 & 4 \\ 3 & 5 & 2 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 2 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}_{1^1} 5^{12} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 2 & 0 & 5 & 3 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 6 & 5 & 0 & 9 & 7 \\ 3 & 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13}_{1^1} 5^{13} 13^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 5 & 3 & 4 & 0 & 15 & 5 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 2 & 4 & 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 6 & 9 & 0 & 34 & 12 \\ 3 & 5 & 2 & 3 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{12}_{1^1} 5^{13} 13^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 26, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 4 & 5 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 3 & 6 & 3 \\ 2 & 5 & 2 & 3 & 2 & 8 & 4 \\ 6 & 10 & 4 & 6 & 8 & 14 & 8 \\ 3 & 5 & 2 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{13}_{1^1} 5^{13} 13^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 2 & 5 & 2 & 3 & 2 & 12 & 4 \\ 9 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12}_{1^1} 5^{13} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 2 & 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 2 & 4 & 3 & 2 & 6 & 4 \\ 6 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{14}_{1^1} 5^{13} 13^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 4 & 5 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 3 & 12 & 3 \\ 2 & 5 & 2 & 3 & 2 & 16 & 4 \\ 12 & 20 & 8 & 12 & 16 & 62 & 16 \\ 3 & 5 & 2 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{12}_{1^1} 5^{13} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 2 & 0 & 8 & 3 \\ 2 & 3 & 1 & 3 & 0 & 8 & 3 \\ 2 & 2 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 8 & 8 & 7 & 0 & 22 & 10 \\ 3 & 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11}_{1^1} 5^{13} 13^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 & 3 \\ 5 & 10 & 4 & 6 & 14 & 8 & 8 \\ 3 & 5 & 2 & 3 & 8 & 2 & 4 \\ 3 & 5 & 2 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[29] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 6 & 3 \\ 5 & 10 & 4 & 6 & 14 & 16 & 8 \\ 6 & 10 & 4 & 6 & 16 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[31] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 2 & 0 & 7 & 3 \\ 2 & 3 & 1 & 3 & 0 & 10 & 3 \\ 2 & 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 10 & 8 & 0 & 25 & 11 \\ 3 & 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[30] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 9.3$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 4 & 10 & 15 & 5 \\ 2 & 3 & 0 & 2 & 4 & 6 & 2 \\ 2 & 4 & 2 & 1 & 6 & 9 & 3 \\ 5 & 10 & 4 & 6 & 14 & 24 & 8 \\ 9 & 15 & 6 & 9 & 24 & 34 & 12 \\ 3 & 5 & 2 & 3 & 8 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[32] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 2 & 0 & 5 & 3 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 6 & 5 & 0 & 9 & 7 \\ 3 & 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[33] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 2 & 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 2 & 4 & 3 & 2 & 6 & 4 \\ 5 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[34] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 60, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[35] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 8 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 3 & 5 & 2 & 3 & 2 & 12 & 4 \\ 8 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{\frac{1^{12}}{1^1}} 5 13^{\frac{1^{112}}{1^1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 4 & 5 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 3 & 6 & 3 \\ 3 & 5 & 2 & 3 & 2 & 8 & 4 \\ 6 & 10 & 4 & 6 & 8 & 14 & 8 \\ 3 & 5 & 2 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{\frac{1^{13}}{1^1}} 5 13^{\frac{1^{113}}{1^1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 3 & 5 & 2 & 3 & 2 & 12 & 4 \\ 9 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{\frac{1^{13}}{1^1}} 5 13^{\frac{1^{114}}{1^1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 11 & 3 \\ 3 & 5 & 3 & 4 & 5 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 3 & 12 & 3 \\ 3 & 5 & 2 & 3 & 2 & 16 & 4 \\ 11 & 20 & 8 & 12 & 16 & 62 & 16 \\ 3 & 5 & 2 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{\frac{1^{13}}{1^1}} 5 13^{\frac{1^{12}}{1^1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 2 & 0 & 10 & 3 \\ 2 & 3 & 1 & 3 & 0 & 12 & 3 \\ 2 & 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 12 & 10 & 0 & 42 & 14 \\ 3 & 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{\frac{1^{13}}{1^1}} 5 13^{\frac{1^{12}}{1^1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 2 & 0 & 13 & 3 \\ 2 & 3 & 1 & 3 & 0 & 14 & 3 \\ 2 & 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 14 & 12 & 0 & 65 & 17 \\ 3 & 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

4.1189 $\text{NHC}(-2)(-3) + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$

$$T = 10 \quad [1] \quad 1^{\frac{1^{11}}{1^1}} 3^{\frac{1^{11}}{1^1}} 1^{\frac{1^{11}}{1^1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{\frac{1^{11}}{1^1}} 3^{\frac{1^{11}}{1^1}} 1^{\frac{1^{11}}{1^1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 2 & 0 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

4.1190 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 7 \quad [0] \ 1^{\overline{11}} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 8 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 14 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 10 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{13}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 7 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{12}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{21}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{23}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 12 & 11 & 6 \\ 7 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{23}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 9 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 7 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 10 & 17 & 48 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 11 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 7 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 4 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 6 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 6 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 20 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 7 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 5 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 8 & 2 \\ 8 & 16 & 8 & 25 & 9 \\ 3 & 5 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 16 & 5 \\ 9 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 5 & 9 & 12 & 17 & 6 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 11 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 7 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 7 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

- [40] $1^{14} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 10 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{14} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 25 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 4 & 4 & 0 & 26 & 2 \\ 25 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 1 \end{pmatrix}_{b_0}$$
- [38] $1^{14} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 12 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{14} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 12 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^{14} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 16 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{14} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 17 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^{14} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [33] $1^{14} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 4 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [32] $1^{14} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 10 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$
- [41] $1^{14} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 6 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^{14} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 13 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[43] $21^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 9 & 8 & 5 & 12 & 6 \\ 4 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{14}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 14 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 8 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[49] $13^{11}1^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 4 & 8 & 9 & 16 & 5 \\ 7 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{24}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 10 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 4 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 6 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 7 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{13} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.56 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 24 & 6 & 8 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 8 & 2 \\ 7 & 16 & 8 & 25 & 9 \\ 3 & 5 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 21^1 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 4 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 4 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{12} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -37, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.46 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 16 & 5 \\ 7 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{12} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -37, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 8 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{14} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -37, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 12 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{12} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -37, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.46 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 16 & 5 \\ 8 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -37, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.46 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 16 & 5 \\ 10 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -37, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 6 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{11} 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -38, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.26 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 5 & 9 & 12 & 29 & 6 \\ 10 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 7 & 0 & 7 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 12 & 11 & 6 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 9 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[68] $21^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[71] $21^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 12 & 4 \\ 8 & 9 & 4 & 16 & 5 \\ 3 & 4 & 0 & 8 & 2 \\ 12 & 16 & 8 & 25 & 9 \\ 4 & 5 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 5 & 2 \\ 6 & 12 & 5 & 12 & 6 \\ 3 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 8 & 13 & 27 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{121}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 9 & 10 & 5 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{12}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 9 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 7 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{111} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 18 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 10 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 10 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[82] \quad 21^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 16 & 5 \\ 8 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 11 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 7 & 9 & 12 & 29 & 6 \\ 13 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[86] \quad 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 12 & 5 & 12 & 6 \\ 3 & 5 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 9 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 7 & 8 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 7 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 6 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 12 & 17 & 6 \\ 10 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 7 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 8 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 12 & 17 & 6 \\ 7 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

 1^{22}

$$[101] \ 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 7 & 9 & 12 & 23 & 6 \\ 10 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

 1^{14}

$$[99] \ 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 16 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

 1^{13}

$$[100] \ 1^{123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 9 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

 1^{12}

$$[98] \ 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 7 & 0 & 7 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

 1^{21}

$$[102] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

 1^{14}

$$[103] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 12 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

 1^{22}

$$[104] \ 1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 12 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

 1^1

$$[105] \ 1^{123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

 1^{12}

$$[106] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 12 & 11 & 6 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

 1^1

$$[107] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[108] $13^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 4 & 8 & 9 & 10 & 5 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 1 \end{pmatrix}_{b_0}$

[109] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$

[110] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 16 & 5 \\ 8 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$

[111] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 8 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$

[112] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 8 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$

[114] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$

[115] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 5 & 9 & 12 & 23 & 6 \\ 10 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$

[113] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 10 & 13 & 27 & 16 & 9 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$

[116] $13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$

[117] $21^{143} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 5 & 19 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 14 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$

[118] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$

[119] $1^{11}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{11}3^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{121}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 7 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{11}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 13 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{11}3^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 5 & 9 & 12 & 23 & 6 \\ 8 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{131}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 24 & 6 & 8 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{131}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{121}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 7 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{122}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 16 & 5 \\ 9 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[128] $21^{113}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 5 & 7 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[129] $21^{113}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 7 & 4 \\ 9 & 9 & 12 & 11 & 6 \\ 6 & 7 & 11 & 7 & 5 \\ 4 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{1^{22}} 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{1^{11}} 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{1^{12}} 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 13 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{1^{11}} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 12 & 11 & 6 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{1^{11}} 1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 18 & 0 & 72 & 14 \\ 3 & 4 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{1^{14}} 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{1^{11}} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{1^{11}} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 9 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{1^{31}} 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 7 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{1^{14}} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[140] $21^{1^{13}} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 11 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

- [141] $13^{\overline{11}}1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 5 & 9 & 12 & 17 & 6 \\ 6 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$
- [142] $1^{\overline{11}}3^{\overline{11}}1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 19 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [143] $1^{\overline{11}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 11 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [144] $1^{\overline{11}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [145] $1^{\overline{11}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 12 & 11 & 6 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$
- [146] $21^{\overline{12}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 5 & 11 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 8 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$
- [147] $1^{\overline{12}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$
- [148] $1^{\overline{11}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 7 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [149] $1^{\overline{12}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 9 & 10 & 5 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$
- [150] $1^{\overline{12}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 6 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$
- [151] $1^{\overline{12}}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 5 & 9 & 12 & 17 & 6 \\ 7 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{121}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 11 & 17 & 48 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[154] $1^{124}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 5 & 9 & 12 & 29 & 6 \\ 12 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{121}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 9 & 13 & 27 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[155] $1^{111}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 13 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{121}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{121}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{112}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 15 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{111}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 17 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{122}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 12 & 11 & 6 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{124}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 17 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[163] $1^{11}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 16 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[164] $1^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[165] $13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 13 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[166] $13^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 6 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{12}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[168] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 5 & 9 & 12 & 29 & 6 \\ 11 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

[169] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 5 & 9 & 12 & 23 & 6 \\ 9 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[170] $1^{12}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{13}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 18 & 0 & 72 & 14 \\ 3 & 4 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 27 & 16 & 9 \\ 6 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 11 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[171] \quad 1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 18 & 0 & 72 & 14 \\ 3 & 4 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{13} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 8 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{122} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{124} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 22 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 22 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[178] \quad 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[180] \quad 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 7 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[181] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[182] \quad 1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 6 & 9 & 12 & 29 & 6 \\ 14 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 6 & 9 & 12 & 29 & 6 \\ 12 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{1^{21}} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{1^{21}} 1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 18 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{1^{21}} 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[188] $21^{1^{43}} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 30 & 27 & 17 & 151 & 21 \\ 4 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{1^{13}} 1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 19 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[189] $21^{1^{13}} 1^{11} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 13 & 7 & 4 \\ 12 & 13 & 27 & 16 & 9 \\ 6 & 7 & 16 & 7 & 5 \\ 4 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[190] $1^{1^{21}} 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{1^{31}} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 10 & 13 & 27 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{1^{11}} 13^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 7 & 13 & 27 & 16 & 9 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{1^{12}} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 7 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{1^{13}} 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{1^{23}} 1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 18 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{1^{12}} 1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 10 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{1^{23}} 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 13 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{1^{23}} 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{1^{13}} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{1^{11}} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{1^{12}} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{1^{13}} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 11 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{1^{14}} 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 10 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{1^{14}} 1^{14} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 10 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{1^{22}} 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 10 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 12 & 17 & 6 \\ 9 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 12 & 17 & 6 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 27 & 16 & 9 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{12}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 23 & 0 & 120 & 18 \\ 3 & 4 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{11}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

^{1¹²}

$$[217] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

^{1²⁴}

$$[218] \ 1^{11} 4^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 7 & 9 & 12 & 29 & 6 \\ 14 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

^{1²³}

$$[219] \ 1^{11} 3^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 7 & 9 & 12 & 23 & 6 \\ 11 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

^{1²}

$$[220] \ 1^{11} 3^3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 12 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

^{1¹³}

$$[223] \ 1^{11} 3^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

^{1²}

$$[221] \ 1^{11} 3^3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

^{1¹²}

$$[224] \ 1^{11} 3^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 15 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

^{1¹⁴}

$$[225] \ 1^{11} 3^3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 20 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[226] \ 21^{11} 3^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 12 & 12 & 8 & 25 & 9 \\ 4 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

^{1¹}

$$[227] \ 1^{11} 3^3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 7 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

^{1¹¹}

$$[228] \ 1^{11} 3^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[229] \quad 1^{1^2} 1^{11} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 15 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[230] \quad 1^{1^1} 1^{11} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 14 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[222] \quad 1^{1^{22}} 1^{14} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 12 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[231] \quad 1^{1^{12}} 1^{13} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 10 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[232] \quad 1^{1^{31}} 1^{13} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 12 & 17 & 48 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[233] \quad 1^{1^1} 1^{13} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[234] \quad 1^{1^{24}} 1^{13} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 7 & 9 & 12 & 23 & 6 \\ 12 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[235] \quad 1^{1^{11}} 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 8 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[236] \quad 1^{1^{23}} 1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.53 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[237] \quad 1^{1^{14}} 1^1 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 23 & 0 & 120 & 18 \\ 3 & 4 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{13}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{114}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 6 & 9 & 12 & 29 & 6 \\ 13 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{121}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 8 & 13 & 27 & 16 & 9 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[242] $13^11^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 8 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{111}3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 11 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{111}3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 14 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{112}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 9 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{113}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 10 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{113}3^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 23 & 0 & 120 & 18 \\ 3 & 4 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{112}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 7 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{112}3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 12 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{1^{12}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{1^{12}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{1^{11}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{1^{11}3^11^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 14 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[253] $21^{1^{11}3^11^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -146, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & 0 & 5 & 2 \\ 9 & 12 & 5 & 12 & 6 \\ 4 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{1^{14}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 15 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{1^{13}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[256] $21^{1^{11}3^11^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[257] $21^{1^{11}3^11^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 21 & 24 & 17 & 100 & 18 \\ 4 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{1^{12}3^11^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{1^{12}3^11^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 19 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{11}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -153, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[261] \ 1^{11}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -156, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 12 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[262] \ 21^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -156, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 24 & 22 & 14 & 97 & 17 \\ 4 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[263] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -156, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{11}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -156, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 16 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[265] \ 21^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -156, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 9 & 7 & 4 \\ 6 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{11}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -156, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 17 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -159, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.28 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{11}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -160, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.28 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[269] \ 1^{11}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -160, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 20 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -162, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{113}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -163, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.37 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[273] \ 21^{123}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.12 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 18 & 17 & 11 & 55 & 13 \\ 4 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[275] \ 21^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{123}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.34 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[274] \ 1^{113}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.12 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 13 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{121}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.07 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 10 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{121}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -167, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.10 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{13}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -171, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[279] \ 13^{11}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -171, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 9 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{11}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -171, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[281] \ 1^{12}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -172, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[282] $21^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 15 & 16 & 11 & 44 & 12 \\ 4 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[283] $13^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 10 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[284] $21^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{14}1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{12}1^{11}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[287] $21^{11}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 18 & 20 & 14 & 69 & 15 \\ 4 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[288] $21^{12}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 3 \\ 4 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{11}1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 9 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[290] $1^{14}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 11 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[292] $21^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[304] \quad & 21^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 11 & 3 \\ 4 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 11 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0} \\
[307] \quad & 1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0} \\
[306] \quad & 1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0} \\
[305] \quad & 1^{12}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[308] \quad & 21^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 14 & 3 \\ 4 & 4 & 9 & 19 & 4 \\ 6 & 9 & 12 & 29 & 6 \\ 14 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0} \\
[313] \quad & 1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[312] \quad & 1^{112}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 19 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0} \\
[311] \quad & 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[310] \quad & 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[309] \quad & 1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 10 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[314] \quad & 1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 17 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[315] \quad 1^{11} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[318] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[317] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[316] \quad 1^{12} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 16 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[320] \quad 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 13 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{12} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[322] \quad 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 12 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[321] \quad 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 14 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{11} 1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 11 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -233, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{11} 1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.53 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 13 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[329] \ 1^{12} 1^{13} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[330] \ 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 13 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[331] \ 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 10 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[332] \ 1^{11} 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -242, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[333] \ 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -243, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[334] \ 1^{12} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[335] \ 1^{13} 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 15 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[336] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 26 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 4 & 4 & 0 & 26 & 2 \\ 26 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

[339] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[340] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[337] $13^11^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 11 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[338] $13^11^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 18 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[341] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 11 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[342] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[343] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[344] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[345] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{113}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[348] $21^{121}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -266, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 3 \\ 4 & 4 & 13 & 7 & 4 \\ 9 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[349] $1^{143}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -270, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[350] $21^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[351] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[352] $1^{114}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[353] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 13 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 15 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[355] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[356] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 19 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[357] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[358] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[359] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[361] $1^{11} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[360] $1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[362] $1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[363] $1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[365] $1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[366] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 12 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[367] $1^{13} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -297, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[368] $1^{13} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[378] \ 1^{1^4} 1^{1^3} 3^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[379] \ 1^{1^1} 1^{1^2} 3^1 1^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 19 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[383] \ 1^{1^1} 1^{1^4} 3^1 1^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -319, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 3 & 4 & 0 & 26 & 2 \\ 22 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[382] \ 1^{1^{1^2}} 1^{1^4} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -319, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 12 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[384] \ 1^{1^{1^3}} 1^{1^4} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 13 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[385] \ 1^{1^{1^3}} 1^{1^2} 3^1 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 18 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{1^{1^3}} 1^{1^3} 3^1 1^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{1^{1^3}} 1^{1^2} 3^1 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[389] \ 1^{1^{1^3}} 1^{1^3} 3^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -329, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{1^{1^1}} 1^{1^3} 3^1 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -329, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{1^2}1^{1^2}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 17 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{1^4}3^11^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{1^4}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 19 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[392] $1^11^11^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 13 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[394] $1^{1^4}3^11^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -341, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{1^4}3^11^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -343, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{1^3}3^11^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 21 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[398] $1^{1^2}3^11^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[397] $1^13^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 14 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{1^2}3^11^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[402] $1^{1^3}3^11^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[399] $1^{112}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 21 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{112}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 20 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[404] $21^{112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[403] $1^{112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 15 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[405] $1^{11}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[407] $1^{13}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -359, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{112}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -359, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 21 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[406] $1^{112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -359, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[409] $1^{11}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -360, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 15 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[410] $1^{114}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 23 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 4 & 4 & 0 & 23 & 2 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[411] $1^{114}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 13 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[415] \quad 1^{12} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[412] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[413] \quad 1^{11} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 18 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[414] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[417] \quad 1^{12} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 18 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[416] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 14 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[418] \quad 21^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[419] \quad 1^{14} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[420] \quad 1^{24} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -387, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[422] \quad 21^{13} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 27 & 26 & 17 & 128 & 20 \\ 4 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[421] \quad 1^{14} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[423] $1^{13}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -407, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[424] $1^{14}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -411, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 19 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[425] $1^{14}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[426] $1^{12}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 22 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[429] $13^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 17 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[427] $21^{12}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 24 & 25 & 17 & 111 & 19 \\ 4 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[428] $1^{14}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[430] $1^{13}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[431] $1^{11}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -419, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 15 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[433] $1^{12}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[432] $1^{13}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[434] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 14 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[435] $21^{113}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -431, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 23 & 3 \\ 4 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 23 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[436] $1^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[438] $21^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[437] $21^{14}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 33 & 31 & 20 & 188 & 24 \\ 4 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[439] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -440, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[440] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -443, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[441] $1^{12}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[442] $1^{14}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[443] $13^11^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -459, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 20 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[444] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -459, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[445] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -461, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[446] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -464, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[447] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[448] $1^{14}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[449] $1^{14}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 39 & 0 & 280 & 30 \\ 3 & 4 & -1 & 30 & 2 \end{pmatrix}_{b_0}$$

[450] $13^11^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 16 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[452] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -479, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 17 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[451] $21^{123}1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -479, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 27 & 29 & 20 & 148 & 22 \\ 4 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[453] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 15 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[454] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -491, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[455] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -491, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 15 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[456] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 18 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[457] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 22 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[458] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -497, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[459] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[460] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 19 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[461] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -507, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[462] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -513, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[463] $1^{123}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -514, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[464] $1^{143}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[465] $1^{143}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -528, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[466] $1^{114}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -531, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 20 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[467] \ 1^{1^{11}} 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -535, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 3 & 4 & 0 & 26 & 2 \\ 23 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[468] \ 1^{1^{13}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[469] \ 1^{1^2} 1^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 19 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[470] \ 1^{1^4} 1^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[471] \ 1^{1^{13}} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -547, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 21 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[472] \ 1^{1^{14}} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -556, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[473] \ 1^{1^3} 1^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -560, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[474] \ 1^{1^{14}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[475] \ 1^{1^{14}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -572, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 17 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[476] \ 1^{1^{14}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -576, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[477] $1^{13}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{13}1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -587, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 19 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[479] $1^{13}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -594, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[480] $1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[481] $21^{13}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 30 & 30 & 20 & 165 & 23 \\ 4 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[482] $1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -603, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 22 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[483] $1^{12}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -611, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[484] $1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -615, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[485] $1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -616, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 23 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[486] $1^{114}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[487] $1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 24 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[488] $1^{1^3}3^11^{1^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -627, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 23 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[489] $1^{1^4}3^11^{1^3} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -633, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[490] $1^{1^1}3^11^{1^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -634, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 39 & 0 & 280 & 30 \\ 3 & 4 & -1 & 30 & 2 \end{pmatrix}_{b_0}$$

[491] $1^{1^4}3^11^{1^2} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -636, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[492] $21^{1^3}3^11^{1^3} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 20 & 3 \\ 4 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[493] $21^{1^3}3^11^{1^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -644, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[494] $1^{1^2}3^11^{1^3} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -648, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[495] $1^{1^4}3^11^{1^2} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -648, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[496] $21^{1^4}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -655, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[497] $1^{1^4}3^11^{1^3} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[498] $1^{1^3}3^11^{1^2} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -668, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[499] $1^{1^4} 1^{1^4} 3^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -668, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[500] $1^{1^4} 1^{1^4} 3^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -672, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 0 & 26 & 2 \\ 22 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

[502] $1^{1^4} 3^{1^3} 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -691, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 20 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[501] $1^{1^4} 3^{1^3} 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -691, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 21 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[503] $1^{1^3} 1^{1^3} 3^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -701, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[504] $1^{1^3} 3^{1^3} 1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -704, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 24 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[505] $21^{1^3} 3^{1^3} 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -711, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 3 \\ 4 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[506] $1^{1^4} 3^{1^3} 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -719, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[507] $1^{1^4} 3^{1^3} 1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -719, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 3 & 4 & 0 & 26 & 2 \\ 24 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

[508] $21^{1^4} 3^{1^3} 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -736, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 36 & 4 \\ 5 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 36 & 35 & 23 & 231 & 27 \\ 4 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{1^4} 3^{1^3} 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -744, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[510] $21^{13}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -759, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 33 & 34 & 23 & 208 & 26 \\ 4 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[511] $1^{12}1^{14}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -763, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 21 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[512] $1^{14}1^{13}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -767, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[513] $1^{12}1^{14}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -772, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 39 & 0 & 280 & 30 \\ 3 & 4 & -1 & 30 & 2 \end{pmatrix}_{b_0}$$

[514] $1^{11}1^{14}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 0 & 26 & 2 \\ 23 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

[515] $1^{14}1^{13}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -791, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[516] $1^{13}1^{14}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -803, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[517] $1^{14}1^{13}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -811, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[518] $1^{14}1^{13}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -816, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{13}1^{14}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -819, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[520] $1^{12}1^{14}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -864, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 0 & 26 & 2 \\ 24 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 4 & 3 & 23 & 3 \\ 4 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [533] \quad 31^{11}31^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 4 & 2 \\ 3 & 4 & 4 & 6 & 5 & 4 \\ 3 & 4 & 0 & 3 & 4 & 2 \\ 3 & 6 & 3 & 3 & 5 & 3 \\ 4 & 5 & 4 & 5 & 4 & 4 \\ 2 & 4 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[534] \quad 31^{11}31^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 6 & 3 & 3 & 6 & 3 \\ 4 & 6 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [535] \quad 1^{11}5^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[536] \quad 1^{13}5^{11}13^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 1 & 21 & 3 \\ 14 & 24 & 21 & 130 & 24 \\ 3 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[537] \quad 1^{12}5^{11}13^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 3 & 3 & 1 & 15 & 3 \\ 10 & 17 & 15 & 65 & 17 \\ 3 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[538] \quad 1^{11}5^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 3 & 3 & 1 & 9 & 3 \\ 6 & 10 & 9 & 22 & 10 \\ 3 & 4 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[539] \quad 1^{11}5^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 10 & 9 & 22 & 10 \\ 3 & 4 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[540] \quad 1^{11}5^{11}2^{23}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 22 & 2 \\ 13 & 16 & 0 & 22 & 229 & 29 \\ 2 & 2 & -1 & 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[541] \quad 1^{13}5^{11}13^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 1 & 21 & 3 \\ 16 & 24 & 21 & 130 & 24 \\ 3 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[542] \quad 1^{11}5^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 1 & 9 & 3 \\ 8 & 10 & 9 & 22 & 10 \\ 3 & 4 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[543] \ 1^{12} 5^{12} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 1 & 15 & 3 \\ 12 & 17 & 15 & 65 & 17 \\ 3 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[544] \ 1^{11} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 5 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[545] \ 1^{11} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 4 & 8 & 8 & 5 & 6 \\ 3 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[546] \ 1^{12} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 5 & 8 & 9 & 5 \\ 4 & 8 & 8 & 11 & 6 \\ 6 & 9 & 11 & 11 & 7 \\ 3 & 5 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[547] \ 1^{13} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 5 & 8 & 14 & 5 \\ 4 & 8 & 8 & 17 & 6 \\ 9 & 14 & 17 & 29 & 11 \\ 3 & 5 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[548] \ 1^{14} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 5 & 8 & 19 & 5 \\ 4 & 8 & 8 & 23 & 6 \\ 12 & 19 & 23 & 55 & 15 \\ 3 & 5 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[549] \ 1^{11} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[550] \ 1^{12} 5^{13} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 1 & 15 & 3 \\ 14 & 17 & 15 & 65 & 17 \\ 3 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[551] \ 1^{11} 5^{13} 23^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 26 & 2 \\ 16 & 19 & 0 & 26 & 339 & 35 \\ 2 & 2 & -1 & 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[552] \ 1^{12} 5^{13} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 1 & 15 & 3 \\ 13 & 17 & 15 & 65 & 17 \\ 3 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[553] $1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 3 & 9 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 9 & 4 & 11 & 7 \\ 3 & 5 & 2 & 7 & 4 \end{pmatrix}_{b_0}$$

[554] $1^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 9 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 9 & 4 & 11 & 7 \\ 3 & 5 & 2 & 7 & 4 \end{pmatrix}_{b_0}$$

[555] $1^{13} \overset{1}{5} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 1 & 21 & 3 \\ 17 & 24 & 21 & 130 & 24 \\ 3 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[556] $1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 5 & 8 & 6 \\ 3 & 3 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

[557] $21^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 37, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[558] $1^{13} \overset{1}{5} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 1 & 21 & 3 \\ 18 & 24 & 21 & 130 & 24 \\ 3 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[559] $1^{12} \overset{1}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 14 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 14 & 6 & 29 & 11 \\ 3 & 5 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

[560] $1^{13} \overset{1}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 14 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 14 & 6 & 29 & 11 \\ 3 & 5 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

[562] $1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[561] $1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

- [563] $1^{1^3} 5^{13} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 19 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 19 & 8 & 55 & 15 \\ 3 & 5 & 2 & 15 & 4 \end{pmatrix}_{b_0}$
- [564] $1^{1^2} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 8 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$
- [565] $1^{1^4} 5^{13} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 3 & 19 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 19 & 8 & 55 & 15 \\ 3 & 5 & 2 & 15 & 4 \end{pmatrix}_{b_0}$
- [566] $1^{1^3} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 13 & 3 \\ 11 & 9 & 13 & 44 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$
- [567] $1^{1^4} 5^{13} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 3 & 24 & 5 \\ 2 & 3 & 0 & 10 & 2 \\ 14 & 24 & 10 & 89 & 19 \\ 3 & 5 & 2 & 19 & 4 \end{pmatrix}_{b_0}$
- [568] $21^{1^2} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 15 & 3 \\ 12 & 12 & 15 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$
- [570] $1^{1^3} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 125, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 11 & 3 \\ 11 & 10 & 11 & 39 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$
- [569] $1^{1^2} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 125, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 11 & 3 \\ 10 & 10 & 11 & 39 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$
- [571] $21^{1^2} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 11 & 3 \\ 10 & 10 & 11 & 39 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$
- [572] $1^{1^2} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 15 & 3 \\ 12 & 12 & 15 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$
- [573] $1^{1^3} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

[584] $1^{13}51^323^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -62, \det = 629, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 18 & 2 \\ 1 & 2 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 18 & 23 & 0 & 290 & 32 \\ 2 & 2 & 2 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [585] $21^{1^13^121^1}61^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 6 & 3 \\ 5 & 4 & 0 & 6 & 10 & 6 \\ 2 & 2 & 0 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[586] $1^{11}6^{1^13^121^1}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 3 & 3 & 15 & 4 \\ 3 & 1 & 3 & 11 & 3 \\ 3 & 3 & 1 & 12 & 3 \\ 15 & 11 & 12 & 55 & 15 \\ 4 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[587] $1^{11}413^1151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 6 & 10 & 6 \\ 10 & 16 & 10 \\ 6 & 10 & 6 \end{pmatrix}_{b_0}$$

[588] $1^{111}41^13^1151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 5 & 2 \\ 6 & 5 & 0 & 5 & 20 & 8 \\ 2 & 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[589] $1^{1^1}6^{1^1}123^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[590] $1^{11}41^13^1151^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 6 & 19 & 6 \\ 19 & 57 & 19 \\ 6 & 19 & 6 \end{pmatrix}_{b_0}$$

[591] $21^{11}6^{1^13^121^1}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 11 & 3 \\ 4 & 8 & 1 & 18 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 18 & 1 & 35 & 13 \\ 3 & 6 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[592] $1^{11}413^1151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[593] $1^{12}41^13^1151^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 6 & 26 & 6 \\ 26 & 108 & 26 \\ 6 & 26 & 6 \end{pmatrix}_{b_0}$$

[594] $1^{11} 6^{11} 123^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 5 & 8 & 6 \\ 3 & 3 & 3 & 6 & 4 \end{pmatrix}_{b_0}$

[595] $1^1 413^1 15^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 2 & 0 & 2 & 4 \\ 3 & 2 & 0 & 4 & 4 \end{pmatrix}_{b_0}$

[596] $21^{11} 6^{11} 123^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -37, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 8 & 1 & 10 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 10 & 1 & 9 & 7 \\ 3 & 6 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

[597] $1^{11} 41^1 3^1 15^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 0 & 0 & 20 & 8 \\ 2 & 2 & 0 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[598] $21^{11} 6^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 8 & 1 & 18 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 18 & 0 & 34 & 12 \\ 3 & 6 & 0 & 12 & 4 \end{pmatrix}_{b_0}$

[602] $1^1 41^1 3^1 15^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 4 & 0 & 14 & 8 \\ 3 & 2 & 0 & 8 & 4 \end{pmatrix}_{b_0}$

[603] $1^{11} 6^{11} 123^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[599] $1^{11} 6^{11} 123^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[600] $1^1 6^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 10 & 14 & 60 & 16 \\ 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$

[601] $1^{11} 413^1 15^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 4 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$

[604] $1^{12} 41^1 3^1 15^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 7 & 0 & 0 & 32 & 10 \\ 2 & 2 & 0 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[605] $1^{11} \overset{1^2}{6} 123^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 8 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[607] $1^{11} \overset{1^1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 11 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 11 & 11 & 12 & 55 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[606] $1^{11} \overset{1^2}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 11 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 12 & 11 & 12 & 55 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[609] $1^{11} \overset{1^3}{6} 123^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 13 & 3 \\ 11 & 9 & 13 & 44 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[608] $1^{11} 41^1 3^1 15^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 6 & 0 & 22 & 10 \\ 3 & 2 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[610] $1^4 41^1 3^1 15^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -125, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 5 & 0 & 20 & 10 \\ 3 & 2 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[612] $1^{11} \overset{1^4}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -141, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 16 & 13 & 16 & 83 & 19 \\ 3 & 3 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[611] $1^{11} \overset{1^1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -141, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 13 & 13 & 16 & 83 & 19 \\ 3 & 3 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[613] $1^{11} \overset{1^2}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -157, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 14 & 13 & 16 & 83 & 19 \\ 3 & 3 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[614] $1^{11} \overset{1^3}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -157, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 15 & 13 & 16 & 83 & 19 \\ 3 & 3 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[615] $1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -173$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 15 & 15 & 20 & 119 & 23 \\ 3 & 3 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[616] $1^{11} 41^{13} 1^{15} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -189$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 7 & 0 & 28 & 12 \\ 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[617] $1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -205$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 15 & 20 & 119 & 23 \\ 3 & 3 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[618] $1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -221$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 18 & 15 & 20 & 119 & 23 \\ 3 & 3 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[619] $1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -221$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 17 & 15 & 20 & 119 & 23 \\ 3 & 3 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[620] $1^{11} 41^{13} 1^{15} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -272$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 8 & 0 & 38 & 14 \\ 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [621] $1^{11} 6^{12} 1^{14} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12$, $T_{\max} = 14$, $\Delta = -58$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[622] $1^{11} 6^{12} 1^{14} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12$, $T_{\max} = 14$, $\Delta = -58$, $\det = 64$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 2 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 7 & 10 & 0 & 10 & 56 & 14 \\ 2 & 3 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[623] $1^{11} 6^{12} 1^{14} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12$, $T_{\max} = 14$, $\Delta = -58$, $\det = 64$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[624] $1^{11} 41^{13} 1^{15} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 12$, $T_{\max} = 14$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 13.94 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 17 & 0 & 16 & 9 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 16 & 0 & 14 & 8 \\ 3 & 9 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

[625] $1^{11}6^{11}23^{11}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 7 & 13 & 0 & 12 & 79 & 17 \\ 2 & 3 & -1 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[626] $1^{11}6^{11}23^{11}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 157, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 8 & 13 & 0 & 12 & 79 & 17 \\ 2 & 3 & -1 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$T = 13$ [627] $1^{11}6^{11}1315^{11}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 15, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$T = 14$ [628] $1^{11}6^{11}1315^{11}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -30, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 4 & 4 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

[629] $1^{11}6^{11}1315^{11}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 11 & 4 \\ 7 & 7 & 11 & 26 & 12 \\ 3 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0}$$

[630] $1^{11}6^{11}1315^{11}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -30, \det = 43, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 11 & 4 \\ 8 & 7 & 11 & 26 & 12 \\ 3 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0}$$

[631] $1^{11}6^{11}1315^{11}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -30, \det = 50, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 16 & 4 \\ 11 & 9 & 16 & 53 & 17 \\ 3 & 3 & 4 & 17 & 5 \end{pmatrix}_{b_0}$$

[632] $1^{11}6^{11}23^{11}151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -31, \det = 98, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 2 & 3 & 17 & 4 \\ 3 & 2 & 14 & 4 \\ 17 & 14 & 72 & 22 \\ 4 & 4 & 22 & 6 \end{pmatrix}_{b_0}$$

4.1191 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{114}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 19 & 0 & 4 \\ 5 & 9 & 12 & 29 & 0 & 6 \\ 10 & 19 & 29 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 6 & 11 & 9 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 8 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 8 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 4 & 8 & 9 & 16 & 0 & 5 \\ 7 & 12 & 16 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 7 & 16 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 7 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 16 & 27 & 17 & 151 & 0 & 21 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 21 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 8 & 7 & 0 & 4 \\ 4 & 8 & 9 & 10 & 0 & 5 \\ 4 & 7 & 10 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{113}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 15 & 0 & 4 \\ 5 & 9 & 12 & 23 & 0 & 6 \\ 8 & 15 & 23 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11131^11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11131^11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 \\ 5 & 9 & 12 & 11 & 0 & 6 \\ 4 & 7 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11231^11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 11 & 0 & 4 \\ 5 & 9 & 12 & 17 & 0 & 6 \\ 6 & 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11131^113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 13 & 22 & 14 & 97 & 0 & 17 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11231^11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 11 & 0 & 4 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 6 & 11 & 8 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 8 & -3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11131^11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 7 & 12 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{12131^11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 7 & 0 & 4 \\ 7 & 13 & 27 & 16 & 0 & 9 \\ 4 & 7 & 16 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{11131^112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 10 & 17 & 11 & 55 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{11131^11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 15 & 0 & 4 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 8 & 15 & 11 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 11 & -3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{111431^11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 10 & 19 & 14 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 14 & -3 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{1123}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 9 & 16 & 11 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1123}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 12 & 21 & 14 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{1133}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 20 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 11 & 20 & 14 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 15 & -3 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{1123}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 18 & 31 & 20 & 188 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 24 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{1123}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 923, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 15 & 26 & 17 & 128 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1143}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 24 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 13 & 24 & 17 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 18 & -3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{1133}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 25 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 14 & 25 & 17 & 111 & 0 & 19 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 19 & -3 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1133}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 30 & 0 & 4 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 17 & 30 & 20 & 165 & 0 & 23 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 23 & -3 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{1133}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1667, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 35 & 0 & 4 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 20 & 35 & 23 & 231 & 0 & 27 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{1143}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1708, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 29 & 0 & 4 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 16 & 29 & 20 & 148 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 22 & -3 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{114}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 1 & 3 \\ 3 & 4 & 4 & 34 & 0 & 4 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 19 & 34 & 23 & 208 & 0 & 26 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 26 & -3 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{114}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 1 & 3 \\ 3 & 4 & 4 & 39 & 0 & 4 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 22 & 39 & 26 & 280 & 0 & 30 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 30 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1192 NHC(-2)(-3) + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 8 & 0 & 5 & 4 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 5 & 8 & 5 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 9 & 0 & 4 & 4 \\ 3 & 4 & 2 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 2 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 & 4 \\ 2 & 4 & 0 & 11 & 0 & 3 & 2 \\ 10 & 17 & 11 & 55 & 0 & 11 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 11 & 0 & 1 & 3 \\ 3 & 4 & 2 & 13 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 & 3 \\ 3 & 4 & 4 & 7 & 0 & 5 & 4 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 4 & 7 & 5 & 7 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 5 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 6 & 11 & 9 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 8 & 0 & 1 & 3 \\ 3 & 4 & 3 & 8 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 4 & 7 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 3 & 5 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 8 & 0 & 9 & 4 \\ 2 & 4 & 0 & 5 & 0 & 6 & 2 \\ 5 & 8 & 5 & 12 & 0 & 14 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 14 & 0 & 13 & 7 \\ 3 & 4 & 2 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{1111}1}3^{\overline{111}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 & 3 \\ 3 & 4 & 4 & 20 & 0 & 5 & 4 \\ 2 & 4 & 0 & 14 & 0 & 3 & 2 \\ 11 & 20 & 14 & 69 & 0 & 18 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 18 & 0 & 4 & 4 \\ 3 & 4 & 2 & 15 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{1111}1}3^{\overline{111}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 & 3 \\ 3 & 4 & 4 & 12 & 0 & 5 & 4 \\ 2 & 4 & 0 & 8 & 0 & 3 & 2 \\ 7 & 12 & 8 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 12 & 0 & 4 & 4 \\ 3 & 4 & 2 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{1111}1}3^{\overline{111}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 2 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 & 4 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 4 & 7 & 5 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 2 & 5 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{1111}1}3^{\overline{111}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 & 4 \\ 5 & 9 & 12 & 11 & 0 & 6 & 6 \\ 4 & 7 & 11 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 5 & 0 & 1 & 3 \\ 3 & 4 & 6 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1111}1}3^{\overline{111}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 2 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 & 4 \\ 2 & 4 & 0 & 8 & 0 & 3 & 2 \\ 7 & 12 & 8 & 25 & 0 & 8 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 8 & 0 & 1 & 3 \\ 3 & 4 & 2 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{1111}1}3^{\overline{111}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 & 3 \\ 3 & 4 & 4 & 16 & 0 & 5 & 4 \\ 2 & 4 & 0 & 11 & 0 & 3 & 2 \\ 9 & 16 & 11 & 44 & 0 & 15 & 12 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 15 & 0 & 4 & 4 \\ 3 & 4 & 2 & 12 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{1111}1}3^{\overline{111}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 5 & 3 \\ 3 & 4 & 4 & 7 & 0 & 9 & 4 \\ 2 & 4 & 0 & 5 & 0 & 6 & 2 \\ 4 & 7 & 5 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 11 & 0 & 13 & 7 \\ 3 & 4 & 2 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{1111}1}3^{\overline{111}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 5 & 3 \\ 3 & 4 & 4 & 12 & 0 & 9 & 4 \\ 2 & 4 & 0 & 8 & 0 & 6 & 2 \\ 7 & 12 & 8 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 20 & 0 & 13 & 7 \\ 3 & 4 & 2 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1193 NHC(-2)(-3) + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 & 4 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 5 & 8 & 5 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 2 & 6 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 5 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 5 & 12 & 5 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 0 & 2 \\ 3 & 5 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 5 & 7 & 0 & 8 & 4 \\ 3 & 5 & 3 & 6 & 0 & 6 & 3 \\ 4 & 7 & 6 & 7 & 0 & 10 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 10 & 0 & 9 & 5 \\ 3 & 4 & 3 & 5 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 6 & 11 & 9 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 3 & 8 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 4 & 7 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1194 NHC(-2)(-3) + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

- $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
- $$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 8 & 16 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [3] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
- $$T_{\min} = 11.24 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 5 & 11 & 1 & 7 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$
- [4] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
- $$T_{\min} = 11.46 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 19 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 11 & 19 & 15 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 42$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
- $$T_{\min} = 11.29 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 19 & 0 & 4 \\ 5 & 9 & 12 & 29 & 0 & 6 \\ 10 & 19 & 29 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
- $$T_{\min} = 11.56 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
- $$T_{\min} = 11.25 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 6 & 8 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 49$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
- $$T_{\min} = 11.76 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 5 & 11 & 9 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
- $$T_{\min} = 11.24 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 5 & 11 & 10 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
- $$T_{\min} = 11.24 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 7 & 0 & 4 \\ 7 & 13 & 27 & 16 & 0 & 9 \\ 5 & 7 & 16 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
- $$T_{\min} = 11.24 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$
- $$\begin{pmatrix} 5 & 11 & 1 & 25 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
- $$T_{\min} = 11.77 \rightarrow 12$$
- ,
- $T_{\max} = 13$
- ,
- $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 7 & 0 & 4 \\ 4 & 8 & 9 & 10 & 0 & 5 \\ 5 & 7 & 10 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 9 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 6 & 11 & 9 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 4 & 8 & 9 & 16 & 0 & 5 \\ 7 & 12 & 16 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 8 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 7 & 16 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 6 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 16 & 27 & 17 & 151 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{\overline{11}13}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 \\ 5 & 9 & 12 & 11 & 0 & 6 \\ 4 & 7 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{\overline{11}23}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 11 & 0 & 4 \\ 5 & 9 & 12 & 17 & 0 & 6 \\ 7 & 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{\overline{11}23}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 11 & 0 & 4 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 6 & 11 & 8 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{\overline{11}13}1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 13 & 22 & 14 & 97 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{\overline{11}43}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 19 & 0 & 4 \\ 5 & 9 & 12 & 29 & 0 & 6 \\ 11 & 19 & 29 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{\overline{11}13}1^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 17 & 27 & 17 & 151 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{\overline{11}13}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 15 & 0 & 4 \\ 5 & 9 & 12 & 23 & 0 & 6 \\ 9 & 15 & 23 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{\overline{11}13}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 7 & 12 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{\overline{12}13}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 7 & 0 & 4 \\ 7 & 13 & 27 & 16 & 0 & 9 \\ 4 & 7 & 16 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{\overline{11}13}1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 1 & 3 \\ 3 & 4 & 4 & 39 & 0 & 4 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 22 & 39 & 26 & 280 & 0 & 30 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 30 & -2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{11} 1^{14} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 1952$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 34 & 0 & 4 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 20 & 34 & 23 & 208 & 0 & 26 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 26 & -2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{11} 1^{14} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 2208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 1 & 3 \\ 3 & 4 & 4 & 39 & 0 & 4 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 23 & 39 & 26 & 280 & 0 & 30 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 30 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1195 NHC(-2)(-3) + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{so}_8$ + $\mathfrak{so}_8$

$T = 9$ [0] $1^{11} 1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 9 & 7 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 7 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1196 NHC(-2)(-3) + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{su}_2$

$T = 11$ [1] $1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 4 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 3 & 3 & 1 & 5 & 3 \\ 4 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[0] $1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 3 & 10 & 0 & 4 \\ 2 & 3 & 1 & 9 & 0 & 3 \\ 7 & 10 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 10 & 0 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 4 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[4] $1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 3 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

- [5] $1^{11}_{1^1} 5^{12} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 22 & 9 & 10 \\ 2 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$
- [6] $1^{11}_{1^1} 5^{12} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 6 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 3 & 3 & 1 & 5 & 3 \\ 6 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [7] $1^{12}_{1^1} 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 9 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 3 & 3 & 1 & 10 & 3 \\ 9 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$
- [8] $1^{11}_{1^1} 5^{121} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 10 & 0 & 4 \\ 2 & 3 & 1 & 9 & 0 & 3 \\ 8 & 10 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 10 & 0 & 4 \end{pmatrix}_{b_0}$
- [9] $1^{11}_{1^1} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 3 & 3 & 1 & 5 & 3 \\ 5 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [10] $1^{12}_{1^1} 5^{121} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 3 & 17 & 0 & 4 \\ 2 & 3 & 1 & 15 & 0 & 3 \\ 12 & 17 & 15 & 65 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 17 & 0 & 4 \end{pmatrix}_{b_0}$
- [12] $1^{11}_{1^1} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [11] $1^{11}_{1^1} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 3 & 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$
- [13] $1^{11}_{1^1} 5^{12} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 4 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 4 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$

- [14] $1^{12}_{1^1} 5^{12} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 3 & 3 & 1 & 10 & 3 \\ 10 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$
- [15] $1^{12}_{1^1} 5^{13} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 3 & 3 & 1 & 10 & 3 \\ 11 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$
- [16] $1^{13}_{1^1} 5^{12} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 3 & 3 & 1 & 15 & 3 \\ 14 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$
- [17] $1^{11}_{1^1} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 9 & 0 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 5 & 9 & 4 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 7 & 0 & 4 \end{pmatrix}_{b_0}$$
- [18] $1^{13}_{1^1} 5^{13} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 7 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 7 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$
- [19] $1^{12}_{1^1} 5^{11} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 7 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$
- [20] $1^{11}_{1^1} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$
- [21] $1^{11}_{1^1} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 2 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$
- [22] $1^{13}_{1^1} 5^{31} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 3 & 24 & 0 & 4 \\ 2 & 3 & 1 & 21 & 0 & 3 \\ 17 & 24 & 21 & 130 & 0 & 24 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 24 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12}_{1^1} 5^{131} 13^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 3 & 17 & 0 & 4 \\ 2 & 3 & 1 & 15 & 0 & 3 \\ 13 & 17 & 15 & 65 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 17 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{113}_{1^1} 5^{131} 13^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 9 & 3 \\ 3 & 5 & 8 & 4 & 15 & 5 \\ 4 & 8 & 8 & 5 & 18 & 6 \\ 3 & 4 & 5 & 1 & 9 & 3 \\ 9 & 15 & 18 & 9 & 34 & 12 \\ 3 & 5 & 6 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{121}_{1^1} 5^{131} 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 5 & 8 & 9 & 5 & 5 \\ 4 & 8 & 8 & 11 & 6 & 6 \\ 6 & 9 & 11 & 11 & 7 & 7 \\ 3 & 5 & 6 & 7 & 2 & 4 \\ 3 & 5 & 6 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{112}_{1^1} 5^{131} 13^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 6 & 3 \\ 3 & 5 & 8 & 4 & 10 & 5 \\ 4 & 8 & 8 & 5 & 12 & 6 \\ 3 & 4 & 5 & 1 & 6 & 3 \\ 6 & 10 & 12 & 6 & 14 & 8 \\ 3 & 5 & 6 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111}_{1^1} 5^{131} 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 4 & 8 & 8 & 5 & 6 & 6 \\ 3 & 4 & 5 & 1 & 3 & 3 \\ 3 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{113}_{1^1} 5^{131} 13^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 4 & 2 & 1 & 9 & 3 \\ 9 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{113}_{1^1} 5^{131} 13^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 15 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 3 & 3 & 1 & 15 & 3 \\ 15 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{112}_{1^1} 5^{131} 13^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 4 & 2 & 1 & 6 & 3 \\ 6 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[25] $1 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 13 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[32] $1 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 13 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 12 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 5 & 3 & 14 & 0 & 5 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 8 & 14 & 6 & 29 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 11 & 0 & 4 \end{pmatrix}_{b_0}$$

[33] $1 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 13 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 5 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

[34] $1 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 13 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 9 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[35] $1 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 13 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 7 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[36] $1 \begin{smallmatrix} 12 \\ 11 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 13 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 \\ 2 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

[37] $1 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 13 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 5 & 0 & 3 \\ 5 & 5 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

[38] $1 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 5 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 13 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 5 & 3 & 19 & 0 & 5 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 11 & 19 & 8 & 55 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 15 & 0 & 4 \end{pmatrix}_{b_0}$$

- [39] $1^{1^{12}}_1 5^{1^{13}} 13^{1^{113}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 5 & 3 & 9 & 15 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 9 & 4 & 11 & 21 & 7 \\ 8 & 15 & 6 & 21 & 34 & 12 \\ 3 & 5 & 2 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$
- [40] $1^{1^{11}}_1 5^{1^{13}} 13^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$
- [41] $1^{1^{21}}_1 5^{1^{13}} 13^{1^{122}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 5 & 3 \\ 3 & 5 & 3 & 14 & 10 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 8 & 14 & 6 & 29 & 22 & 11 \\ 5 & 10 & 4 & 22 & 14 & 8 \\ 3 & 5 & 2 & 11 & 8 & 4 \end{pmatrix}_{b_0}$$
- [42] $1^{1^3}_1 5^{1^{13}} 13^{1^{131}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 \\ 2 & 5 & 2 & 15 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$
- [44] $1^{1^{11}}_1 5^{1^{13}} 13^{1^{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 3 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$
- [43] $1^{1^{21}}_1 5^{1^{13}} 13^{1^{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 9 & 4 & 11 & 7 & 7 \\ 3 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$
- [46] $1^{1^{12}}_1 5^{1^{13}} 13^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$
- [45] $1^{1^{12}}_1 5^{1^{13}} 13^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 8 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$
- [47] $1^{1^{12}}_1 5^{1^{13}} 13^{1^{112}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 6 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{13}_{1^1} 5^{13} 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 3 & 9 & 20 & 5 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 5 & 9 & 4 & 11 & 28 & 7 \\ 11 & 20 & 8 & 28 & 62 & 16 \\ 3 & 5 & 2 & 7 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{11}_{1^1} 5^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{31}_{1^1} 5^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 9 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{13}_{1^1} 5^{12} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 9 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{31}_{1^1} 5^{13} 1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 5 & 3 \\ 3 & 5 & 3 & 19 & 10 & 5 \\ 2 & 3 & 0 & 8 & 4 & 2 \\ 11 & 19 & 8 & 55 & 30 & 15 \\ 5 & 10 & 4 & 30 & 14 & 8 \\ 3 & 5 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{13}_{1^1} 5^{13} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 9 & 15 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 9 & 4 & 11 & 21 & 7 \\ 9 & 15 & 6 & 21 & 34 & 12 \\ 3 & 5 & 2 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{21}_{1^1} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 1 & 3 \\ 2 & 1 & 3 & 7 & 0 & 3 \\ 2 & 3 & 1 & 9 & 0 & 3 \\ 8 & 7 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 10 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{14}_{1^1} 5^{13} 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 5 & 3 & 24 & 5 & 5 \\ 2 & 3 & 0 & 10 & 2 & 2 \\ 14 & 24 & 10 & 89 & 19 & 19 \\ 2 & 5 & 2 & 19 & 2 & 4 \\ 3 & 5 & 2 & 19 & 4 & 4 \end{pmatrix}_{b_0}$$

[56] $1^{13}_{1^1} 5^{12}_{1^1} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 14 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 12 & 13 & 14 & 12 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{11}_{1^1} 5^{12}_{1^1} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 8 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{11}_{1^1} 5^{12}_{1^1} 13^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[59] $1^{11}_{1^1} 5^{13}_{1^1} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 9 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{11}_{1^1} 5^{13}_{1^1} 13^1 1^{121} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 \\ 3 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{11}_{1^1} 5^{13}_{1^1} 13^1 1^{121} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 14 & 6 & 29 & 11 & 11 \\ 3 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{12}_{1^1} 5^{12}_{1^1} 13^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 3 & 16 & 3 \\ 3 & 2 & 3 & 1 & 13 & 3 \\ 12 & 12 & 16 & 13 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{13}_{1^1} 5^{12}_{1^1} 13^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 14 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

- [64] $1^{11}_{1^1} 5^{131} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 1 & 3 \\ 2 & 1 & 3 & 9 & 0 & 3 \\ 2 & 3 & 1 & 13 & 0 & 3 \\ 11 & 9 & 13 & 44 & 0 & 14 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$
- [65] $1^{12}_{1^1} 5^{112} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$
- [67] $1^{12}_{1^1} 5^{121} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 1 & 3 \\ 2 & 1 & 3 & 10 & 0 & 3 \\ 2 & 3 & 1 & 11 & 0 & 3 \\ 10 & 10 & 11 & 39 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 13 & 0 & 4 \end{pmatrix}_{b_0}$$
- [66] $1^{12}_{1^1} 5^{131} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 1 & 3 \\ 2 & 1 & 3 & 10 & 0 & 3 \\ 2 & 3 & 1 & 11 & 0 & 3 \\ 11 & 10 & 11 & 39 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 13 & 0 & 4 \end{pmatrix}_{b_0}$$
- [68] $1^{11}_{1^1} 5^{131} 13^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -142, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 \\ 3 & 5 & 2 & 15 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$
- [70] $1^{13}_{1^1} 5^{113} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 14 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 13 & 13 & 14 & 12 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$
- [69] $1^{12}_{1^1} 5^{113} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$
- [71] $1^{12}_{1^1} 5^{113} 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 3 & 20 & 3 \\ 3 & 2 & 3 & 1 & 16 & 3 \\ 15 & 14 & 20 & 16 & 106 & 22 \\ 3 & 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$
- [72] $1^{12}_{1^1} 5^{113} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -171, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 3 & 16 & 3 \\ 3 & 2 & 3 & 1 & 13 & 3 \\ 13 & 12 & 16 & 13 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[73] 1^{12} 5^{13} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -187, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 1 & 3 \\ 2 & 1 & 3 & 12 & 0 & 3 \\ 2 & 3 & 1 & 15 & 0 & 3 \\ 13 & 12 & 15 & 65 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 17 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[74] 1^{13} 5^{13} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -197, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[75] 1^{13} 5^{13} 13^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -203, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 3 & 22 & 3 \\ 3 & 2 & 3 & 1 & 18 & 3 \\ 17 & 17 & 22 & 18 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[76] 1^{13} 5^{13} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -264, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 1 & 3 \\ 2 & 1 & 3 & 15 & 0 & 3 \\ 2 & 3 & 1 & 17 & 0 & 3 \\ 15 & 15 & 17 & 92 & 0 & 20 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 20 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [77] 1^{16} 6^{11} 123^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[78] 1^{11} 6^{11} 123^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 4 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[79] 1^{11} 6^{11} 13^1 21^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 6 & 3 \\ 2 & 3 & 1 & 3 & 7 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 6 & 6 & 7 & 7 & 19 & 9 \\ 3 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[80] 1^{11} 6^{11} 13^1 21^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 3 & 2 & 3 & 1 & 9 & 3 \\ 8 & 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

- [81] $1^{11}41^{13}1^{11}51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 5 & 4 \\ 3 & 2 & 0 & 0 & 5 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$
- [82] $1^{11}6^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 6 & 3 \\ 2 & 3 & 1 & 3 & 7 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 7 & 6 & 7 & 7 & 19 & 9 \\ 3 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$
- [83] $1^{11}6^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 8 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$
- [84] $1^{11}413^{11}51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 21, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \\ 1 & 2 & 0 & 0 & 3 & 0 & 2 \\ 2 & 2 & 0 & 0 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$
- [85] $1^{11}6^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 17 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 12 & 13 & 17 & 15 & 86 & 20 \\ 3 & 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$
- [86] $1^{11}6^{11}123^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 5 & 0 & 3 \\ 5 & 5 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$
- [87] $1^{11}6^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 3 & 2 & 3 & 1 & 9 & 3 \\ 9 & 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$
- [89] $1^{11}41^{13}1^{11}51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 11 & 4 \\ 5 & 6 & 0 & 0 & 11 & 20 & 8 \\ 2 & 2 & 0 & 0 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$
- [88] $1^{11}6^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 10 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[90] $1^{\overline{11}} \begin{smallmatrix} 11 \\ 1^2 3^1 21^1 \end{smallmatrix} 6 \cdot 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 15 & 3 \\ 3 & 2 & 3 & 1 & 13 & 3 \\ 11 & 10 & 15 & 13 & 63 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[91] $1^{\overline{121}} 41^{\overline{131}} 1^{\overline{151}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 53, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 0 & 7 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 0 & 0 & 32 & 7 & 10 \\ 1 & 2 & 0 & 0 & 7 & 0 & 2 \\ 2 & 2 & 0 & 0 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{\overline{11}} \begin{smallmatrix} 11 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \cdot 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 9 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[93] $1^{\overline{111}} 41^{\overline{131}} 1^{\overline{151}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 0 & 3 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 8 & 4 \\ 4 & 4 & 0 & 0 & 8 & 10 & 6 \\ 2 & 2 & 0 & 0 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{\overline{11}} \begin{smallmatrix} 11 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \cdot 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 83, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 1 & 3 \\ 2 & 1 & 3 & 11 & 0 & 3 \\ 2 & 3 & 1 & 12 & 0 & 3 \\ 11 & 11 & 12 & 55 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 15 & 0 & 4 \end{pmatrix}_{b_0}$$

[95] $1^{\overline{11}} \begin{smallmatrix} 11 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \cdot 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 110, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 11 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[96] $1^{\overline{11}} \begin{smallmatrix} 11 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \cdot 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 123, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 1 & 3 \\ 2 & 1 & 3 & 13 & 0 & 3 \\ 2 & 3 & 1 & 16 & 0 & 3 \\ 13 & 13 & 16 & 83 & 0 & 19 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 19 & 0 & 4 \end{pmatrix}_{b_0}$$

[97] $1^{\overline{11}} \begin{smallmatrix} 11 \\ 1^1 3^1 21^1 \end{smallmatrix} 6 \cdot 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 17 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 13 & 13 & 17 & 15 & 86 & 20 \\ 3 & 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [100] \quad 1^{11}_1 6 \quad 131513^{11}_1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 2 & 4 & 5 & 4 \\ 3 & 2 & 4 & 2 & 4 & 4 \\ 3 & 2 & 4 & 2 & 4 & 4 \\ 3 & 3 & 4 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{11}_1 6 \quad 131513^{11}_1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 2 & 4 & 0 & 4 \\ 3 & 2 & 4 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{11}_1 6 \quad 131513^{11}_1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 4 & 12 & 4 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 7 & 7 & 12 & 10 & 29 & 13 \\ 3 & 3 & 4 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{11}_1 6 \quad 131513^{11}_1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 2 & 11 & 5 & 4 \\ 7 & 7 & 11 & 26 & 11 & 12 \\ 3 & 2 & 5 & 11 & 3 & 5 \\ 3 & 3 & 4 & 12 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{11}_1 6 \quad 131513^{11}_1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -23, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 7 & 6 & 10 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

4.1197 NHC(-2)(-3) + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11}_1 5 \quad 13^{11}_1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 3 & 2 & 3 & 1 & 3 & 5 & 3 \\ 2 & 2 & 4 & 3 & 2 & 6 & 4 \\ 5 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}_1 5 \quad 13^{11}_1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 & 7 \\ 2 & 5 & 2 & 7 & 2 & 4 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}_1 5 \quad 13^{11}_1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 5 & 3 & 9 & 5 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 & 4 & 2 \\ 5 & 9 & 4 & 11 & 7 & 14 & 7 \\ 2 & 5 & 2 & 7 & 2 & 8 & 4 \\ 6 & 10 & 4 & 14 & 8 & 14 & 8 \\ 3 & 5 & 2 & 7 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 9 & 3 \\ 3 & 5 & 3 & 9 & 5 & 15 & 5 \\ 2 & 3 & 0 & 4 & 2 & 6 & 2 \\ 5 & 9 & 4 & 11 & 7 & 21 & 7 \\ 2 & 5 & 2 & 7 & 2 & 12 & 4 \\ 9 & 15 & 6 & 21 & 12 & 34 & 12 \\ 3 & 5 & 2 & 7 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

[4] $1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} 12^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 71, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 & 11 \\ 2 & 5 & 2 & 11 & 2 & 4 & 4 \\ 3 & 5 & 2 & 11 & 4 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[5] $1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 & 15 \\ 2 & 5 & 2 & 15 & 2 & 4 & 4 \\ 3 & 5 & 2 & 15 & 4 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

4.1198 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3$

$T = 7$ [0] $1^{\frac{1}{1^1}} 3^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 7, T_{\max} = 19, \Delta = -102, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 7 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{\frac{1}{1^1}} 3^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 7, T_{\max} = 19, \Delta = -102, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} 3^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 19, T \in [15, 19]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 5 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{\frac{1}{1^1}} 13^{\frac{1}{1^1}} 3^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 8 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} 3^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{\frac{1}{1^1}} 12^{\frac{1}{1^1}} 3^{\frac{1}{1^1}} 11^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 5 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 4 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[8] $21^{13} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 5 & 12 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 17.59 \rightarrow 18, T_{\max} = 19, T \in [18, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 45 & 11 \\ 3 & 4 & -1 & 11 & -5 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 5 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 17.59 \rightarrow 18, T_{\max} = 19, T \in [18, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 10 & 0 & 21 & 7 \\ 3 & 4 & -1 & 7 & -5 \end{pmatrix}_{b_0}$$

[14] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 8 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 5 & 0 & 3 & 3 \\ 3 & 4 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[34] $21^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 5 & 3 & 3 & 3 \\ 4 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 12 & 6 & 12 & 6 \\ 3 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[37] $21^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 10 & 0 & 21 & 7 \\ 3 & 4 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 21 & 7 \\ 3 & 4 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[40] $1^1 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 12 & 6 & 12 & 6 \\ 3 & 5 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[45] $21^{123} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 8 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^1 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{111} 3^1 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[52] $21^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 9 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 8 & 0 & 9 & 5 \\ 3 & 4 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[57] $21^{11} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 9 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 9 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 27 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 1^3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{12} 1^3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{13} 1^3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 12 & 0 & 24 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{12} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 9 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{111} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{111} 1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{13} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 48 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 45 & 11 \\ 3 & 4 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[75] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 9 & 6 & 12 & 6 \\ 4 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 21 & 0 & 75 & 15 \\ 3 & 4 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 57 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[81] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 12 & 6 & 12 & 6 \\ 4 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 24 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 27 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{13} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 24 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[87] $21^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 15 & 14 & 9 & 36 & 10 \\ 4 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 14 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[90] $21^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{14} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 45 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 13 & 0 & 27 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 84 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -167, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 13 & 0 & 27 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[95] $21^{11} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 8 & 3 & 3 & 3 \\ 4 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 48 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[97] $21^{11} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 18 & 0 & 57 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{14} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 45 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{14} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 26 & 0 & 117 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{14} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.53 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 21 & 0 & 75 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -230, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 17 & 0 & 48 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{12} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 18 & 0 & 57 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 48 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 22 & 0 & 84 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 18 & 0 & 57 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -297, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 21 & 0 & 75 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -329, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 26 & 0 & 117 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 84 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 21 & 0 & 75 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 22 & 0 & 84 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -425, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 26 & 0 & 117 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -497, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 26 & 0 & 117 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [114] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 \\ 1 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

- [116] $1^{11}31^{231^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 6$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 4 & 6 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [117] $21^1 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [118] $1^1 4^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [119] $1^{11}31^{131^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 0 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 0 & 12 & 6 \\ 2 & 3 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [120] $1^1 31^{131^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & -1 & 2 \end{pmatrix}_{b_0}$
- [121] $1^{11}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 12 & 15 & 6 \\ 5 & 6 & 0 & 15 & 12 & 6 \\ 2 & 3 & -1 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [122] $1^{13}31^{131^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [123] $21^1 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [124] $1^{11} 4^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [127] $1^{11}31^{131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & -1 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [126] $1^{11}31^{231^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

- [125] $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 9 & 2 \\ 7 & 9 & 0 & 9 & 36 & 10 \\ 2 & 3 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$
 $1^{12}31^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 6 & 3 \\ 3 & 6 & 0 & 6 & 9 & 5 \\ 2 & 3 & -1 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [128] $1^{13}31^131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 24 & 8 \\ 2 & 3 & -1 & 3 & 8 & 2 \end{pmatrix}_{b_0}$
- [129] $1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [130] $1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [131] $1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [132] $1^{111}31^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 12 & 2 \\ 8 & 12 & 0 & 12 & 72 & 14 \\ 2 & 3 & -1 & 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [133] $1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [134] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$
- [135] $21^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [136] $1^{12}31^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 6 & 0 & 9 & 5 \\ 2 & 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$
- [137] $1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 4 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1 \overset{1^3}{1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 7 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[142] \ 21 \overset{1^{11}}{1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[139] \ 1 \ 4 \ 1^{\overset{11}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 31 \overset{1^{11}}{1^1} 231 \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 5 & 3 & 4 & 3 \\ 5 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[141] \ 21 \overset{1^{11}}{1^1} 4 \ 1^{\overset{11}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 3 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 1 \overset{1^1}{1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[144] \ 14 \overset{1^1}{1^1} 1^{\overset{11}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[145] \ 1 \overset{1^{11}}{1^1} 31 \overset{1^{11}}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[146] \ 1 \overset{1^{11}}{1^1} 31 \overset{1^{12}}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 6 & 2 \\ 5 & 6 & 0 & 6 & 12 & 6 \\ 2 & 3 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[147] \ 1 \overset{1^{12}}{1^1} 31 \overset{1^{12}}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 6 & 3 \\ 4 & 6 & 0 & 6 & 9 & 5 \\ 2 & 3 & -1 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[148] \ 1 \overset{1^1}{1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

- $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 0 & 6 & 12 & 6 \\ 2 & 3 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [160] $1^{11}31^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 63$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 9 & 2 \\ 6 & 9 & 0 & 9 & 36 & 10 \\ 2 & 3 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [161] $31^{11}231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -82$, $\det = 63$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 7 & 3 \\ 7 & 9 & 7 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [162] $1^{11}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 72$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 6 & 2 \\ 4 & 6 & 0 & 6 & 12 & 6 \\ 2 & 3 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [163] $21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 76$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [164] $21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [165] $1^{14}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 87$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 18 & 3 \\ 7 & 15 & 0 & 18 & 75 & 15 \\ 2 & 3 & -1 & 3 & 15 & 2 \end{pmatrix}_{b_0}$
- [166] $21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [167] $221^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 93$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 5 & 3 & 3 \\ 5 & 8 & 11 & 6 & 6 \\ 5 & 11 & 7 & 6 & 5 \\ 3 & 6 & 6 & 3 & 3 \\ 3 & 6 & 5 & 3 & 2 \end{pmatrix}_{b_0}$
- [168] $1^{11}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 9 & 3 \\ 4 & 6 & 0 & 9 & 12 & 6 \\ 2 & 3 & -1 & 3 & 6 & 2 \end{pmatrix}_{b_0}$
- [169] $1^{11}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 12 & 15 & 6 \\ 3 & 6 & 0 & 15 & 12 & 6 \\ 2 & 3 & -1 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [170] $1^{12}31^{13}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[171] \quad & 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0} \\
[172] \quad & 1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 12 & 15 & 6 \\ 4 & 6 & 0 & 15 & 12 & 6 \\ 2 & 3 & -1 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[173] \quad & 1^{13}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 15 & 3 \\ 6 & 12 & 0 & 15 & 48 & 12 \\ 2 & 3 & -1 & 3 & 12 & 2 \end{pmatrix}_{b_0} \\
[174] \quad & 1^{12}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 12 & 3 \\ 5 & 9 & 0 & 12 & 27 & 9 \\ 2 & 3 & -1 & 3 & 9 & 2 \end{pmatrix}_{b_0} \\
[175] \quad & 1^{12}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[176] \quad & 1^{12}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 12 & 3 \\ 6 & 9 & 0 & 12 & 27 & 9 \\ 2 & 3 & -1 & 3 & 9 & 2 \end{pmatrix}_{b_0} \\
[177] \quad & 1^{111}2^31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 3 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 0 & 3 & 2 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[178] \quad & 1^{14}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.61 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 3 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 18 & 3 \\ 8 & 15 & 0 & 18 & 75 & 15 \\ 2 & 3 & -1 & 3 & 15 & 2 \end{pmatrix}_{b_0} \\
[179] \quad & 1^{13}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 15 & 3 \\ 7 & 12 & 0 & 15 & 48 & 12 \\ 2 & 3 & -1 & 3 & 12 & 2 \end{pmatrix}_{b_0} \\
[180] \quad & 31^{111}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 5 & 4 & 7 & 3 \\ 5 & 5 & 5 & 10 & 5 \\ 4 & 5 & 1 & 7 & 3 \\ 7 & 10 & 7 & 13 & 7 \\ 3 & 5 & 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[181] \quad & 31^{111}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 5 & 4 & 4 & 3 \\ 5 & 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 4 & 3 \\ 4 & 5 & 4 & 1 & 3 \\ 3 & 5 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [182] \quad 31^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 6 & 4 \\ 4 & 4 & 1 & 5 & 3 \\ 5 & 6 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[183] \quad 21^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 31^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 5 & 3 \\ 4 & 2 & 5 & 5 & 4 \\ 4 & 5 & 1 & 5 & 3 \\ 5 & 5 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 31^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 & 3 \\ 5 & 8 & 6 & 6 & 6 \\ 4 & 6 & 1 & 3 & 3 \\ 3 & 6 & 3 & 3 & 3 \\ 3 & 6 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 21^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[189] \quad 21^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[190] \quad 31^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 3 \\ 5 & 5 & 7 & 5 & 5 \\ 5 & 7 & 7 & 5 & 5 \\ 4 & 5 & 5 & 1 & 3 \\ 3 & 5 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [191] \quad 1^{11}41^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[192] $1^{\overline{1}}1^{\overline{1}}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

$$[193] \quad 1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -67, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [194] \quad 1^{\overline{1}1} 6 1231^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_3 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -65, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [195] \quad 1^{\overline{1}} \overline{5} 131 \overline{5} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 12, \quad T_{\max} = 14, \quad \Delta = -61, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[196] $1^{\overline{1}} 1^{\overline{1}} 5 131 5 1^{\overline{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 12, T_{\max} = 14, \Delta = -61, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

$$T = 13 \quad [200] \quad 1^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 5^{1321^{\textcolor{teal}{1}}} 6^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_3 + \textcolor{brown}{su}_3 \quad T = 13, \quad T_{\max} = 15, \quad \Delta = -59, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 1 & 2 \\ 4 & 10 & 1 & 6 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[199] $21^{\overline{1}} 1^{\overline{1}} 6 13 15 1^{\overline{1}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{brown}{s}u_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

$$[198] \quad 1 \overset{1}{\underset{1}{6}} 131 \overset{1}{5} 1 \overset{1}{} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[197] $21^1 61^1 231 51^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}u_3 + \textcolor{brown}{s}u_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 & 1 & 2 \\ 4 & 10 & 1 & 6 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[201] $1^{\overline{1}} 6^{1\overline{1}} 123151^{\overline{1}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{brown}{s}u_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 6 & 2 & 6 \\ 2 & 0 & 2 \\ 6 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[202] \ 21^{1^1} 513161^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 11 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{1^1} 613151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{1^1} 613151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[206] \ 21^{1^1} 613151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 8 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 6 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{1^1} 613151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[205] \ 1^{1^1} 6123151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[208] \ 1513161^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[209] \ 1^{1^1} 5^{1^1} 61321^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 7 & 1 & 3 & 3 \\ 7 & 19 & 1 & 6 & 9 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 1 & 3 \\ 3 & 9 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[210] \ 1^{1^1} 61^{1^1} 223151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -85, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 9 & 1 & 1 & 3 \\ 9 & 34 & 1 & 0 & 12 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 12 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [211] \ 1^{1^1} 6^{1^1} 131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [212] \quad 1^{\overline{1}} 6 13161^{\overline{1}} 23151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 16, T_{\max} = 18, \Delta = -58, \det = 168, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 17.93 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 10 & 1 & 2 & 3 \\ 10 & 28 & 1 & 6 & 12 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 0 & 2 \\ 3 & 12 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.1199 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$

4.1200 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 2 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1201 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{121}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 8 & 0 & 9 & 4 \\ 3 & 5 & 3 & 6 & 0 & 9 & 3 \\ 4 & 8 & 6 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 3 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{114}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 36 & 15 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & 2 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 28 & 12 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 9 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & 2 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 14 & 9 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 2 & 10 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -267$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.7$

$T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 8 & 0 & 13 & 4 \\ 3 & 5 & 3 & 6 & 0 & 12 & 3 \\ 4 & 8 & 6 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 18 & 0 & 28 & 10 \\ 3 & 4 & 3 & 5 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -294$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 5 & 8 & 0 & 17 & 4 \\ 3 & 5 & 3 & 6 & 0 & 15 & 3 \\ 4 & 8 & 6 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 24 & 0 & 49 & 13 \\ 3 & 4 & 3 & 5 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -402$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.79 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & 2 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -420$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.8$

$T_{\min} = 13.09 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 32 & 15 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 9 & 0 & 4 & 4 \\ 3 & 4 & 2 & 6 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -555, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 44 & 21 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & 2 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -645, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 & 7 \\ 3 & 4 & 3 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -651, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 2 & 6 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -690, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 40 & 18 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & 2 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 18 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & 2 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -789, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 & 7 \\ 3 & 4 & 2 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 2 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

4.1202 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 6 & 0 & 13 & 15 & 7 \\ 5 & 12 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 5 & 2 & 2 & -4 & 7 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 228, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 231, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & 6 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 507, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & 2 & -4 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 651, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 15 & 3 \\ 11 & 21 & 18 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & 2 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & 2 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 15 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & 2 & -4 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

4.1203 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1204 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 92, \ \Delta = -130, \ \det = -27, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.11 \rightarrow 15, \ T_{\max} = 92, \ T \in [15, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & 2 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 92, \ \Delta = -130, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 92, \ \Delta = -130, \ \det = -405, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

15.0

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1205 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

4.1206 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}} 3^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

4.1207 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9 \quad [0] 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & 2 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 28 & 12 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{112}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 9 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.9$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 8 & 0 & 13 & 4 \\ 3 & 5 & 3 & 6 & 0 & 12 & 3 \\ 4 & 8 & 6 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 18 & 0 & 28 & 10 \\ 3 & 4 & 3 & 5 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 14 & 9 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 2 & 10 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{123}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 5 & 8 & 0 & 17 & 4 \\ 3 & 5 & 3 & 6 & 0 & 15 & 3 \\ 4 & 8 & 6 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 24 & 0 & 49 & 13 \\ 3 & 4 & 3 & 5 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -315$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -321$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 32 & 15 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 9 & 0 & 4 & 4 \\ 3 & 4 & 2 & 6 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -399$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -426$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 44 & 21 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & 2 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 7 & 0 \\ 3 & 4 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 2 & 6 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1114} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 18 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & 2 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{114} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 18 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 2 & 6 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1208 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 8 & 0 & 3 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 4 & 8 & 6 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 2 & 0 & 0 & 2 \\ 3 & 5 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 5 & 9 & 6 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 \\ 3 & 4 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1209 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 5 & 5 & 0 & 12 & 6 \\ 3 & 5 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 5 & 8 & 0 & 7 & 4 \\ 3 & 5 & 3 & 6 & 0 & 6 & 3 \\ 4 & 8 & 6 & 9 & 0 & 10 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 10 & 0 & 7 & 5 \\ 3 & 4 & 3 & 5 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 5 & 4 & 0 & 11 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -346, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

4.1210 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 21, \ \Delta = -104, \ \det = -15, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.20 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & 2 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 21, \ \Delta = -104, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 21, \ \Delta = -104, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.55 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1211 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 5, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 17.60 \rightarrow 18, \ T_{\max} = 20, \ T \in [18, 20]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & -4 \end{pmatrix}_{b_0}$$

- [2] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 10 & 0 & 0 & 4 \\ 6 & 10 & 21 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$
- [1] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 7 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$
- [3] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [5] $1^{111} 3 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 1 & 3 \\ 4 & 9 & 16 & 0 & 0 & 5 \\ 7 & 16 & 27 & 0 & 0 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 9 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{12} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 1 & 3 \\ 3 & 4 & 0 & 21 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 11 & 21 & 0 & 75 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 12 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{12} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 5 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 101, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 101, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 17 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 9 & 17 & 0 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 104, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.48 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 14 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 14 & 0 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 117, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 117, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 125, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 7 & 13 & 0 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 1 & 3 \\ 3 & 4 & 0 & 18 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 10 & 18 & 0 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 1 & 3 \\ 3 & 4 & 0 & 22 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 12 & 22 & 0 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 1 & 3 \\ 3 & 4 & 0 & 26 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 14 & 26 & 0 & 117 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[39] $21^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 12 & 6 \\ 7 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [42] \quad 1 \overset{1}{4} 1 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 5 \\ 12 & 25 & 12 & 12 & 9 \\ 6 & 12 & 4 & 7 & 4 \\ 6 & 12 & 7 & 4 & 4 \\ 5 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 21 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 13 & 20 & 9 & 9 & 7 \\ 20 & 25 & 12 & 12 & 9 \\ 9 & 12 & 4 & 7 & 4 \\ 9 & 12 & 7 & 4 & 4 \\ 7 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[45] \quad 1 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1212 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$

$$T = 9 \quad [0] \quad 1 \overset{1}{11} 3 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1 \overset{1}{11} 3 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1 \overset{1}{11} 3 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{so}_8} \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{so}_8} \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 1 & 5 \\ 12 & 25 & 12 & 12 & 0 & 9 \\ 6 & 12 & 4 & 7 & 0 & 4 \\ 6 & 12 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1213 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] 1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 6 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 8 & 3 \\ 3 & 4 & 0 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 9 & 0 & 36 & 10 \\ 3 & 4 & -1 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 0 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 6 & 0 & 12 & 6 \\ 3 & 4 & -1 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.1214 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 2 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 6 & 0 & 14 & 8 \\ 3 & 4 & 3 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overset{1}{1}\overset{11}{4}} 1^{\overset{1}{1}\overset{1}{4}} 1^{\overset{1}{1}\overset{1}{4}} 1^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 0 & 4 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overset{1}{1}\overset{11}{4}} 1^{\overset{1}{1}\overset{1}{4}} 1^{\overset{1}{1}\overset{1}{4}} 1^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overset{1}{1}\overset{11}{4}} 1^{\overset{1}{1}\overset{11}{4}} 1^{\overset{1}{1}\overset{1}{4}} 1^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{\overset{1}{1}\overset{11}{4}} 1^{\overset{1}{1}\overset{11}{4}} 1^{\overset{1}{1}\overset{1}{4}} 1^{\overset{1}{1}\overset{1}{4}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 4 & 9 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 7 & 3 \\ 5 & 9 & 0 & 7 & 12 & 8 \\ 3 & 5 & -1 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [26] \ 1^{\overset{1}{1}\overset{11}{5}} 1^{\overset{1}{1}\overset{1}{13}} 1^{\overset{1}{1}\overset{1}{15}} 1^{\overset{1}{1}\overset{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -31, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [27] \ 1^{\overset{1}{1}\overset{11}{6}} 1^{\overset{1}{1}\overset{1}{12}} 1^{\overset{1}{1}\overset{1}{13}} 1^{\overset{1}{1}\overset{1}{15}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{\overset{1}{1}\overset{12}{5}} 1^{\overset{1}{1}\overset{1}{6}} 1^{\overset{1}{1}\overset{1}{13}} 1^{\overset{1}{1}\overset{1}{21}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 15 & 6 \\ 2 & 0 & 6 & 2 \\ 15 & 6 & 31 & 15 \\ 6 & 2 & 15 & 6 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{\overset{1}{1}\overset{1}{5}} 1^{\overset{1}{1}\overset{1}{13}} 1^{\overset{1}{1}\overset{1}{21}} 1^{\overset{1}{1}\overset{1}{6}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 44, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 9 & 3 \\ 2 & 1 & 1 & 0 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 9 & 0 & 0 & 34 & 12 \\ 3 & 3 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{\overset{1}{1}\overset{1}{6}} 1^{\overset{1}{1}\overset{1}{11}} 1^{\overset{1}{1}\overset{1}{22}} 1^{\overset{1}{1}\overset{1}{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 188, (n_+, n_-, n_0) =$$

$$(1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 9 & 1 & 1 & 9 & 3 \\ 9 & 34 & 1 & 0 & 36 & 12 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 36 & 0 & 0 & 34 & 12 \\ 3 & 12 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

4.1215 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 2 \\ 3 & 4 & 2 & 0 & 2 & 1 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 4 & 2 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 3 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -1 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 3 & 0 & 5 & 7 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 4 & 7 & 5 & 0 & 6 & 8 & 6 \\ 3 & 4 & 2 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 5 & 0 & 8 & 5 & 6 \\ 3 & 4 & 2 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 91, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 3 & 0 & 4 & 7 & 4 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 5 & 3 \\ 4 & 7 & 5 & 0 & 5 & 8 & 6 \\ 3 & 4 & 2 & -1 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 3 & 0 & 4 & 7 & 4 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 5 & 3 \\ 5 & 7 & 5 & 0 & 5 & 8 & 6 \\ 3 & 4 & 2 & -1 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [12] \ 1^{111} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 & 4 \\ 3 & 5 & 0 & 3 & 4 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 5 & 0 & 4 & 5 & 9 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 7 & 3 \\ 2 & 5 & 0 & 3 & 2 & 7 & 4 \\ 5 & 9 & 0 & 7 & 7 & 12 & 8 \\ 3 & 5 & -1 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

4.1216 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [1] \ 1^{111} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 3 & 0 & 5 & 7 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 6 & 3 \\ 4 & 7 & 5 & 0 & 6 & 8 & 5 \\ 3 & 4 & 2 & 0 & 3 & 5 & 1 \\ 3 & 4 & 2 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 & 5 \\ 2 & 3 & 0 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 4 & 2 & 0 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 4 \\ 3 & 4 & 2 & -1 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -1 & 3 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 3 & 0 & 5 & 7 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 6 & 3 & 4 \\ 4 & 7 & 5 & 0 & 6 & 8 & 5 & 6 \\ 2 & 4 & 2 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & 2 & -1 & 4 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -81, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 & 4 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 7 & 5 & 0 & 8 & 5 & 5 & 6 \\ 3 & 4 & 2 & 0 & 5 & 1 & 3 & 3 \\ 2 & 4 & 2 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & 2 & -1 & 6 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

4.1217 NHC(-2)(-3) + su₃ + su₃ + su₂ + su₂ + su₂ + su₂

$T = 11$ [1] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 11.0$

[0] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 11.0$

4.1218 NHC(-2)(-3) + su₃ + su₃ + su₃

$T = 8$ [0] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 4 & 5 & 3 & 6 & 0 & 3 \\ 5 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 5 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 \\ 5 & 8 & 3 & 3 & 3 & 3 \\ 3 & 5 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 7 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

- [5] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [7] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 6 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 0 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 0 & 27 & 9 \\ 3 & 4 & -1 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{13} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 0 & 84 & 16 \\ 3 & 4 & -1 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 5 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 0 & 57 & 13 \\ 3 & 4 & -1 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 11 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[23] $21^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 10 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 0 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 0 & 27 & 9 \\ 3 & 4 & -1 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{14} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 21 & 0 & 0 & 75 & 15 \\ 3 & 4 & -1 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 0 & 48 & 12 \\ 3 & 4 & -1 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{12} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 18 & 0 & 0 & 57 & 13 \\ 3 & 4 & -1 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{13} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 22 & 0 & 0 & 84 & 16 \\ 3 & 4 & -1 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{14} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 26 & 0 & 0 & 117 & 19 \\ 3 & 4 & -1 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

4.1219 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 4 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 3 & 0 & 3 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1220 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 1 & 3 \\ 4 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 9 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 1 & 4 \\ 8 & 9 & 4 & 4 & 0 & 5 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{12}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 1 & 3 \\ 4 & 4 & 4 & 14 & 0 & 4 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 9 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{12}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 1 & 3 \\ 4 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 6 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 1 & 3 \\ 4 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 1 & 4 \\ 5 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 9 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{111}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 1 & 3 \\ 4 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 6 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

4.1221 NHC(-2)(-3) + $\textcolor{red}{su}_3 + \textcolor{red}{su}_3 + \textcolor{red}{su}_3$ + su3n2mix

$T = 8 \quad [0] \quad 1^{111}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 & 4 \\ 5 & 4 & 4 & 5 & 4 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 6 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{111}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 4 & 3 \\ 4 & 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.1222 NHC(-2)(-3) + $\textcolor{red}{su}_3 + (-3) \rightarrow (-2)$

$T = 7 \quad [0] \quad 1^{111}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111}312^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 3 & 8 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 5 & 0 & 4 \\ 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{13}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

- $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 17 & 0 & 4 \\ 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{13}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{12}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{13}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$
- [9] $1^{11}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 9 & 1 & 4 \\ 5 & 4 & 9 & 0 & 4 \\ 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{12}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 13 & 0 & 4 \\ 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{11}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 9 & 0 & 4 \\ 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{13}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 8 & 0 & 4 \\ 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{13}31^{13}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 9 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{11}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 13 & 0 & 4 \\ 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{12}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 18 & 0 & 4 \\ 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0} \\
[16] & 1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 17 & 0 & 4 \\ 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0} \\
[17] & 31^{11}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -561, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0} \\
[18] & 1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -596, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 22 & 0 & 4 \\ 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0} \\
T = 8 & [19] 1^{12}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[20] & 21^{11}41^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 8 & 6 & 1 & 4 \\ 8 & 13 & 12 & 0 & 7 \\ 6 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[21] & 1^{11}4^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[22] & 1^{11}2^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 7 & 0 & 4 \\ 6 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[23] & 1^{11}4^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[24] & 1^{21}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[25] & 1^{12}2^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [26] $22^1 1^1 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$
- [27] $21^1 41^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 13 & 12 & 6 & 7 \\ 6 & 12 & 7 & 5 & 5 \\ 3 & 6 & 5 & 0 & 2 \\ 4 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$
- [28] $1^1 2^1 1^1 4 \quad 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 4 & 10 & 0 & 4 \\ 7 & 10 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$
- [30] $1^1 2^1 1^1 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [29] $1^1 41^{11} 22^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [31] $1^1 2^1 1^1 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$
- [32] $1^1 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$
- [33] $1^1 2^1 1^1 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^1 41^{11} 22^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [35] $22^1 1^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 7 & 5 & 5 \\ 3 & 6 & 5 & 0 & 2 \\ 3 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^1 41^{11} 22^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
T = 9 \quad [37] \quad 21^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[38] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[39] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[40] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[41] \quad 21^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 1 & 5 & 0 & 3 \\ 4 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[42] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[43] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 5 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[44] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 13 & 5 & 0 & 7 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[45] \quad 22^1 1^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 6 & 2 & 3 & 3 \\ 6 & 13 & 5 & 6 & 7 \\ 2 & 5 & 0 & 2 & 2 \\ 3 & 6 & 2 & 0 & 2 \\ 3 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
T = 11 \quad [46] \quad 1^1 6^1 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [47] \quad 1^{1^1 1^1 5^{131}} 6^{12^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 6 \\ 2 & 0 & 2 \\ 6 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1^1 1^1 5^{131}} 6^{12^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

4.1223 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{11} 4^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{2^1 1^{11}} 4^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

4.1224 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1^1} 51^{1^1} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

- [1] $1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$
- [2] $1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 6 & 0 & 21 & 9 \\ 3 & 4 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$
- [3] $1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 4 & 0 & 8 & 6 \\ 3 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$
- [4] $1^{12} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 11 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 11 & 6 & 0 & 21 & 9 \\ 3 & 4 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$
- [6] $1^{12} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 17 & 10 & 0 & 61 & 15 \\ 3 & 4 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$
- [5] $1^{13} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 15 & 8 & 0 & 40 & 12 \\ 3 & 4 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$
- [7] $1^{12} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 14 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 14 & 8 & 0 & 38 & 12 \\ 3 & 4 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{2^1 21^{11}} \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 2 & 1 & 5 & 3 \\ 6 & 13 & 5 & 0 & 12 & 7 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 4 & 0 & 8 & 6 \\ 3 & 7 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{13} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 18 & 10 & 0 & 61 & 15 \\ 3 & 4 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} \overset{1}{5} 1^3 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 21 & 12 & 0 & 88 & 18 \\ 3 & 4 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{2^{12} 21^{12}} \overset{1}{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 10 & 3 \\ 6 & 13 & 5 & 0 & 24 & 7 \\ 2 & 5 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 8 & 0 & 38 & 12 \\ 3 & 7 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{2^{12} 21^{13}} \overset{1}{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 901, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 15 & 3 \\ 6 & 13 & 5 & 0 & 36 & 7 \\ 2 & 5 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 36 & 12 & 0 & 88 & 18 \\ 3 & 7 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [13] \ 1^{2^1 1^1} \overset{1}{6} 13^1 \overset{1}{5} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 10, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 12 & 6 \\ 2 & 0 & 5 & 2 \\ 12 & 5 & 20 & 12 \\ 6 & 2 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} \overset{1}{1} \overset{1}{5} 13^1 \overset{1}{6} 12^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1225 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11} 31^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 9 & 3 \\ 4 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 3 & 4 \\ 8 & 9 & 4 & 0 & 4 & 5 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 4 & 5 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 9 & 3 \\ 4 & 4 & 4 & 0 & 14 & 4 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}31^{12}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 153, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 3 \\ 4 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 237, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 9 & 4 \\ 5 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 9 & 6 & 0 & 12 & 6 \\ 4 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}31^{111}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 453, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

4.1226 NHC(-2)(-3) + $\mathfrak{su}_3$ + su3n2mix

$$T = 7 \quad [0] \ 1^{11}31^{13}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 4 & 3 \\ 4 & 4 & 12 & 5 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}31^{12}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -65, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.2$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 4 & 3 \\ 4 & 4 & 8 & 5 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}31^{12}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.63 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 0 & 2 \\ 4 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.47 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -156, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 9 & 4 & 4 \\ 5 & 4 & 9 & 4 & 4 \\ 9 & 9 & 12 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{11} 4^{11} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4^{11} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 4 & 4 & 2 & 0 & 2 \\ 4 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 4^{11} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \quad 1^{11} 5^{11} 1^{11} 2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [10] \quad 1^{11} 6^{11} 13^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 5 & 0 & 1 & 3 \\ 3 & 6 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1227 NHC(-2)(-3) + $\mathfrak{su}_3$ + su3n2mix + $\mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{111} 31^{11} 2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 6 & 4 \\ 5 & 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 4 & 4 & 3 & 0 & 3 & 2 \\ 6 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 & 3 \\ 4 & 4 & 4 & 5 & 4 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

5 NHC $(-2)(-3)(-2)$ External (163 combos, 2345 bases)

5.1 NHC $(-2)(-3)(-2)$

$$T = 12 \quad [167] \quad 1^{\overline{1}1} 4 14 13 22 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[166] \quad 1^{\overline{1}1} 3 1 5 13 22 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 4 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [178] \quad 1^{\overline{1}1} 4 14 14 13 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 5 \\ 3 & 5 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [190] \quad 1^{\overline{1}1} 23 15 13 1^{\overline{1}} 6 13 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 15, \Delta = -4, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 16 \quad [194] \quad 1^{\overline{1}1} 6 13 16 12 32 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 17 \quad [197] \quad 1^{\overline{1}1} 23 15 13 21^{\overline{1}} 8 12 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [200] \quad 1^{\overline{1}1} 6 13 16 13 15 13 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 18, T_{\max} = 18, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 19 \quad [204] \quad 1^{\overline{1}1} 8 12 32 17 12 32 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 19, T_{\max} = 19, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$[203] \quad 1^{\overline{1}1} 6 13 16 13 16 12 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 20 \quad [205] \quad 1^{1^{2321}} 8^{1^{12321}} 8^{1^{1231}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] \quad T = 20, \quad T_{\max} = 20, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 2 & 2 & 2 & 4 \\ 2 & 3 & 3 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$T = 22 \quad [208] \quad 1^{1^{1321}} 8^{1^{12321}} 8^{1^{12315131}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] \quad T = 22, \quad T_{\max} = 22, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 22, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$T = 23 \quad [209] \quad 1^{1^{1321}} 8^{1^{12321}} 8^{1^{123216131}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] \quad T = 23, \quad T_{\max} = 23, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 23, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, \quad T_{\max} = 23, \quad T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

5.2 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{1^{11}} 3^{1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 2 & 3 \\ 4 & 9 & 7 & 0 & 8 & 4 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 4 & -4 & 4 & 2 & -12 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

5.3 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{1^{11}} 3^{1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 & 3 \\ 4 & 9 & 7 & 0 & 4 & 8 & 5 \\ 3 & 7 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 3 & 8 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & -4 & 2 & 4 & -12 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

5.4 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$

$$T = 12 \quad [188] \quad 1^{1^{12}} 3^{1^{11}} 5^{1^{12}} 13^{1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 7 & 6 & 0 & 18 & 10 \\ 2 & 3 & 3 & 0 & 10 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[187] \quad 1^{1^{11}} 3^{1^{11}} 5^{1^{12}} 13^{1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 13 \quad [195] \quad 1^{1^{12}2321^1} 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 2 & 9 & 3 \\ 2 & 2 & 2 & 15 & 4 \\ 12 & 9 & 15 & 67 & 21 \\ 3 & 3 & 4 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[196] \quad 1^{11} 6^{1^{12}2321^1} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 2 & 12 & 3 \\ 2 & 2 & 2 & 19 & 4 \\ 14 & 12 & 19 & 106 & 26 \\ 3 & 3 & 4 & 26 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[194] \quad 1^{11} 6^{1^{11}2321} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 2 & 9 & 4 \\ 6 & 6 & 9 & 22 & 12 \\ 3 & 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[193] \quad 1^{11} 6^{1^{11}2321} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 14 \quad [197] \quad 1^{1^{11}2321} 7^{11} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [199] \quad 1^{11} 6^{1^{11}315131} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 2 & 9 & 4 \\ 6 & 6 & 9 & 22 & 12 \\ 3 & 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[200] \quad 1^{11} 6^{1^{11}315131} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

5.5 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [72] \quad 1^{1231^1} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 0 & 7 & 2 & 3 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 5 & 9 \\ 1 & 2 & 1 & 0 & 5 & 0 & 2 \\ 2 & 3 & 2 & 0 & 9 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[71] \quad 1^{11} 3^{11} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 4 & 2 & 3 \\ 1 & 2 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 2 & 0 & 6 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

5.6 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 7 & 4 & 0 & 8 & 5 \\ 3 & 7 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 2 & -4 & 4 & -12 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

Non-unimodular ($|\det| > 1$) — 70725 bases

5.7 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{red}{\mathfrak{sp}_1} \cdot \textcolor{green}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 6 & 5 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

5.8 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 \\ 3 & 4 & -4 & 3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 1 & -3 & 1 & -1 \\ 3 & 8 & 3 & 6 & 1 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

5.9 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 6 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 3 & 2 \\ 3 & 5 & 0 & 6 & 4 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 6 & 4 & 4 \\ 3 & 4 & -4 & 4 & 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 6 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 6 & 2 \\ 3 & 5 & 0 & 6 & 4 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

5.10 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 1 & 6 & 0 & 4 \\ 1 & 0 & 0 & 1 & -3 & 1 & 0 & -1 \\ 3 & 8 & 3 & 6 & 1 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

5.11 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \quad 1^{11} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 6 & 6 & 4 & 4 & 4 \\ 3 & 4 & -4 & 4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 9 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 9 & 6 & 4 & 4 & 4 \\ 3 & 4 & -4 & 7 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

5.12 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 1 & -1 \\ 3 & 8 & 3 & 0 & 6 & 1 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

5.13 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \quad 1^{11} 3^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 2 & -1 \\ 4 & 4 & 0 & 2 & 0 & 2 \\ 5 & 6 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{11} 2^1 3^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 11, T_{\max} = 28, \Delta = -115, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 28, T \in [15, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 1 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} \ 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 11, \ T_{\max} = 26, \ \Delta = -113, \ \det = -224, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.93 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 0 & 2 \\ 5 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

5.14 $\text{NHC}(-2)(-3) + \textcolor{red}{e}'_7 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \ 1^{11} \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 224, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.93 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 0 & 2 \\ 5 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \ 1^{11} \ 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 11, \ T_{\max} = 14, \ \Delta = -87, \ \det = -36, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 1 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

5.15 $\text{NHC}(-2)(-3) + \textcolor{red}{e}_8 + \text{NHC}(-2)(-3)(-2)$

$$T = 11 \quad [0] \ 1^{11} \ 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_8 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 11, \ T_{\max} = 106, \ \Delta = -228, \ \det = -416, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.65 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 2 & -4 \\ 4 & 4 & 0 & 2 & 0 & 2 \\ 5 & 6 & -1 & -4 & 2 & -3 \end{pmatrix}_{b_0}$$

5.16 $\text{NHC}(-2)(-3) + \textcolor{red}{f}_4 + \text{NHC}(-2)(-3)(-2)$

$$T = 9 \quad [0] \ 1^{11} \ 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 \\ 3 & 4 & -3 & 3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \ 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & -3 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} \ 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 3 & 8 & 3 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 1 & -3 & 1 & -1 \\ 3 & 8 & 3 & 6 & 1 & 4 & 4 \\ 3 & 5 & 0 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

5.17 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 6 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 9 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 9 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & 7 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

5.18 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 4 & 2 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 & 4 \\ 3 & 5 & 2 & -4 & 4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

5.19 $\text{NHC}(-2)(-3)(-2)$

$$T = 6 \quad [0] \ 21^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 33, T \in [14, 33]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 7 & 4 \\ 4 & 7 & 9 & 5 \\ 3 & 4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.56 \rightarrow 10, T_{\max} = 33, T \in [10, 33]$$

$$\begin{pmatrix} 5 & 7 & 5 \\ 7 & 8 & 6 \\ 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 13^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 10, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 33, T \in [11, 33]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 7 & 5 \\ 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 13^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 33, T \in [11, 33]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 33, T \in [10, 33]$$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

- $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 33, T \in [13, 33]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 9 & 11 & 5 \\ 4 & 11 & 9 & 5 \\ 3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$
- [5] $2131^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 33, T \in [13, 33]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 9 & 11 & 5 \\ 4 & 11 & 9 & 5 \\ 3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$
- [7] $21^{11}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 27, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 33, T \in [11, 33]$
 $\begin{pmatrix} 4 & 8 & 5 & 4 \\ 8 & 9 & 7 & 5 \\ 5 & 7 & 4 & 4 \\ 4 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [8] $21^{11}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 33, T \in [10, 33]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [9] $131^{11}2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 96, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 33, T \in [11, 33]$
 $\begin{pmatrix} 4 & 8 & 8 & 4 \\ 8 & 9 & 11 & 5 \\ 8 & 11 & 9 & 5 \\ 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$
- [10] $221^{11}3 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 216, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 33, T \in [10, 33]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 2 \\ 3 & 3 & 4 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $T = 7 \quad [11] \quad 1^1 1^1 3 2 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$
- [12] $21^1 3 2 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 4 & 5 & 5 & 4 \end{pmatrix}_{b_0}$
- [13] $1^1 1^1 3 2 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [14] $1^1 3 1^1 3 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 2 \end{pmatrix}_{b_0}$
- [15] $1^1 1^1 4 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 7, T_{\max} = 19, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [16] $21^1 1^1 4 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 7, T_{\max} = 19, \Delta = -109, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [17] $1^1 3 1^1 3 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[18] $1^1 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 7 & 5 \\ 3 & 7 & 5 & 5 \\ 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[19] $31^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 14, \Delta = -98, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11} 3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 13, \Delta = -97, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[22] $21^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 4 & 5 & 4 \\ 5 & 5 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[23] $2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[25] $1^1 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[24] $2141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[28] $131^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 13, \Delta = -97, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 4 & 3 \\ 1 & 3 & 4 & 1 & 3 \\ 2 & 3 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[27] $1^1 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[26] $2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[29] $221^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

- $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 3 & 3 \\ 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- $[31] \quad 2 \overset{1}{3} \overset{11}{1} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 14$, $\Delta = -98$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 14$, $T \in [11, 14]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 2 & 2 & 4 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $[30] \quad 21 \overset{1}{1} 321 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- $[32] \quad 21 \overset{1}{1} 41 \overset{1}{1} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 19$, $\Delta = -109$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 8 & 4 \\ 8 & 13 & 15 & 7 \\ 8 & 15 & 13 & 7 \\ 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- $[33] \quad 31 \overset{1}{1} 231 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 14$, $\Delta = -98$, $\det = -35$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 14$, $T \in [12, 14]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 2 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $[34] \quad 2 \overset{1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $[35] \quad 2 \overset{1}{2} 31 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 5 & 7 & 5 \\ 4 & 7 & 5 & 5 \\ 3 & 5 & 5 & 4 \end{pmatrix}_{b_0}$
- $[36] \quad 131 \overset{1}{1} 31 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 13$, $\Delta = -97$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 4 & 2 \\ 2 & 6 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- $[37] \quad 1 \overset{21}{4} 1 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 19$, $\Delta = -109$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- $[38] \quad 2 \overset{1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$
- $[39] \quad 31 \overset{1}{1} 231 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 14$, $\Delta = -98$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 14$, $T \in [11, 14]$
 $\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$
- $[40] \quad 221 \overset{1}{1} 41 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 19$, $\Delta = -109$, $\det = -88$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 13 & 15 & 7 \\ 6 & 15 & 13 & 7 \\ 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- $[41] \quad 13 \overset{1}{2} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[42] & 31^{11}31^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.48 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 4 & 2 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[43] & 231^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 14, T \in [11, 14] \\
& \begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
T = 8 & [44] 1^1 2^1 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[45] & 1^1 4 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[46] & 21^1 4 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[47] & 1^1 3 2 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 10, \Delta = -61, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[48] & 1^1 4 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[49] & 21^1 4 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[50] & 1 3 1^1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[51] & 1^1 4 1^1 2 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 3 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[55] & 3 1^1 3 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 14, T \in [12, 14] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[54] $1^{\overline{11}} 1^{\overline{1}} 4 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$

[58] $1^{\overline{1}} 14131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[57] $1^{\overline{1}} 41231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[56] $1^{\overline{11}} 41231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$

[52] $1^{\overline{11}} 31^{\overline{1}} 1^{\overline{1}} 4 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$

[53] $1^{\overline{1}} 31^{\overline{1}} 1^{\overline{1}} 4 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[60] $1^{\overline{1}} 41^{\overline{1}} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

[59] $1^{\overline{1}} 23141^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[61] $31^{\overline{1}} 31^{\overline{1}} 41^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 5 & 9 & 5 \\ 6 & 9 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$

[62] $1^{\overline{11}} 23141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[65] $1^{\overline{1}} 41^{\overline{1}} 2^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$

[66] $21^{\overline{11}} 4131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 6 & 1 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

- [63] $141^{11}32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [64] $2^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 9 & 5 \\ 6 & 9 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$
- [67] $31^{31}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [68] $21^{11}41^{11}31 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 & 2 \\ 2 & 3 & 3 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$
- [69] $31^{11}31^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [70] $23141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 5 & 7 & 5 \\ 4 & 7 & 5 & 5 \\ 3 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$
- [71] $1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$
- [72] $1^{11}31^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$
- [73] $2141^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$
- [74] $41^{11}322 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 4 & 8 & 4 \\ 3 & 8 & 8 & 5 & 5 \\ 2 & 6 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$
- [76] $141^{11}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 6 & 1 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$
- [75] $31^{11}3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[78] $3131^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 13 & 15 & 7 \\ 6 & 15 & 13 & 7 \\ 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$

[77] $1321^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 13 & 15 & 7 \\ 6 & 15 & 13 & 7 \\ 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$

[79] $141^{11}31^13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 4 & 2 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$

[80] $31^{11}413 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 4 & 2 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$

[81] $41^{11}2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 28 & 1 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[82] $41^{11}232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -91, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 5 & 2 \\ 4 & 12 & 5 & 10 & 6 \\ 2 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

$T = 9$ [84] $1^14141^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 4 & 5 & 5 & 4 \end{pmatrix}_{b_0}$

[83] $1^14141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[85] $1^141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[86] $21^14141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 5 & 6 & 5 \\ 5 & 6 & 5 & 5 \\ 4 & 5 & 5 & 4 \end{pmatrix}_{b_0}$

[87] $1^14141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 6 & 5 \\ 3 & 6 & 5 & 5 \\ 3 & 5 & 5 & 4 \end{pmatrix}_{b_0}$

[88] $21^15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

- [89] $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
 $1^1 1^1 5 13 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [90] $1^1 1^1 5 123 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [91] $1^1 1^1 23 15 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [92] $1^1 1^1 5 13 1^1 3 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [93] $3 1^1 3 15 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [95] $1^1 1^1 3 2 1^1 5 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$
- [94] $2 1^1 1^1 5 13 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 4 \\ 2 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [96] $3 14 14 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [97] $1^1 1^1 5 13 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [98] $23 2 15 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [99] $1^1 4 1^1 3 14 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 5 & 1 & 1 & 3 \\ 2 & 7 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 223151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 3 & 4 \\ 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 11^1 3221^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[102] \quad 31^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 1 & -1 \\ 3 & 9 & 1 & 13 & 7 \\ 2 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[103] \quad 2321^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 2^1 3^{141} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 6 & 8 & 6 \\ 4 & 8 & 6 & 6 \\ 3 & 6 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[105] \quad 31^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 1 & -1 \\ 2 & 7 & 1 & 5 & 5 \\ 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[106] \quad 22^1 3^{11} 151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 8 & 4 \\ 3 & 2 & 8 & 5 & 5 \\ 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 221^{11} 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[108] \quad 2^1 3^{11} 2^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 4 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[110] \quad 41^{11} 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 28 & 1 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[109] \quad 151^{11} 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 6 & 20 & 8 \\ 1 & 6 & 0 & 6 & 2 \\ 4 & 20 & 6 & 18 & 8 \\ 2 & 8 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 321^{\overline{1}} 4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 8 & 8 & 3 \\ 8 & 17 & 19 & 9 \\ 8 & 19 & 17 & 9 \\ 3 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [113] \ 1^{\overline{1}} 31\overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\overline{1}} 413221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 11, \Delta = -31, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overline{1}} 414131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[120] \ 21\overline{6} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 21^{\overline{1}} 61231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 221^{\overline{1}} 6131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overline{1}} 4131\overline{5} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{1}} 61231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{1}} 612231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{\overline{21}} 6131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[124] $1 \overset{1}{\underset{1}{6}} 131^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 2 & 2 \\ 3 & 4 & 12 & 4 & 6 \\ 1 & 2 & 4 & 0 & 2 \\ 2 & 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[122] $21 \overset{1}{\underset{1}{6}} 131^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[123] $2321 \overset{1}{\underset{1}{6}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[125] $31 \overset{1}{\underset{1}{6}} 131^1 61 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 3 & 4 & 12 & 4 & 6 \\ 1 & 1 & 4 & 0 & 2 \\ 2 & 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[128] $21 \overset{1}{\underset{1}{6}} 161232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[129] $1 \overset{1}{\underset{1}{6}} 1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[127] $1 \overset{1}{\underset{1}{6}} 1^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[126] $21 \overset{1}{\underset{1}{6}} 1^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[130] $1 \overset{1}{\underset{1}{6}} 1^1 3221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[131] $1^1 41321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 2 & 2 \\ 4 & 2 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[132] $1 \overset{1}{\underset{1}{6}} 1^1 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[133] $41^1 31^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 4 & 2 \\ 5 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 32 & 1 & 26 & 10 \\ 2 & 12 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[134] $221^{11} 61231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 12 & 1 & 10 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[135] $1^{221^{11}} 6 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 12 & 1 & 10 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[136] $1^1 414141^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 14 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 3 & 4 & 1 & 3 \\ 2 & 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[137] $2221^{11} 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 1 & 20 & 8 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 20 & 1 & 18 & 8 \\ 2 & 8 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[140] $1^{11} 3221^{11} 6 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 12 & 1 & 10 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[139] $2161^{11} 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 12 & 1 & 10 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[138] $41^{11} 314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 28 & 1 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[141] $51^{11} 223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 7 & 3 \\ 7 & 7 & 9 & 7 \\ 7 & 9 & 7 & 7 \\ 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[142] $4131^{11} 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 1 & -1 \\ 5 & 40 & 1 & 38 & 12 \\ 2 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [143] $1^{11} 3151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 12, \Delta = -34, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[145] $1^{11} 6 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[146] $1^{\overline{1}1321} 1^{\overline{1}6} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[147] $1^{\overline{1}131} 1^{\overline{1}6} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[144] $1^{\overline{1}14141321} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 2 & 4 \\ 3 & 4 & 4 & 6 \end{pmatrix}_{b_0}$$

[148] $21^{\overline{1}71231} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{\overline{1}712231} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[150] $21^{\overline{1}613141} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 2 & 4 & 1 & 8 & 6 \\ 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[151] $221^{\overline{1}71231} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[152] $21^{\overline{1}712231} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{\overline{1}6123141} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[154] $21^{\overline{1}71232} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[155] $21^{\overline{1}71231} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[156] & 2321\overset{1}{7}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 2 & 3 & 4 & 3 & 4 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[157] & 1\overset{1}{4}141\overset{1}{3}151\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 8 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[159] & 1\overset{1}{6}1\overset{1}{2}3141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 26 & 1 & 7 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 7 & 1 & 1 & 3 \\ 2 & 10 & -1 & 3 & 3 \end{pmatrix}_{b_0} \\
[158] & 1\overset{1}{7}221\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 4 & 6 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[160] & 21\overset{1}{7}3221\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0} \\
[161] & 1\overset{1}{4}13221\overset{1}{1}61\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[162] & 221\overset{1}{6}13141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[163] & 2\overset{1}{3}1\overset{1}{4}141\overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 5 & 4 \\ 2 & 5 & 5 & 3 & 5 \\ 2 & 5 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[164] & 161\overset{1}{2}223141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 28 & 7 & 30 & 10 \\ 1 & 7 & 0 & 7 & 2 \\ 5 & 30 & 7 & 28 & 10 \\ 2 & 10 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[165] & 15131\overset{1}{4}141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 8 & 8 & 3 \\ 8 & 17 & 19 & 9 \\ 8 & 19 & 17 & 9 \\ 3 & 9 & 9 & 4 \end{pmatrix}_{b_0} \\
T = 12 & [168] 1\overset{1}{6}2321\overset{1}{1}31\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 13, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[171] \ 1^{\overline{1}} \overline{6} \ 13141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{\overline{1}} \overline{6} \ 13141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{\overline{1}} \overline{7} \ 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{\overline{21}} \overline{8} \ 1232^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[173] \ 2321^{\overline{1}} \overline{8} \ 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 2 & 3 & 4 & 3 & 4 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{\overline{4}} 14141^{\overline{1}} 32161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 33 & 8 & 0 & 11 \\ 1 & 8 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 11 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{\overline{4}} 14141321^{\overline{1}} 61^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 61^{\overline{1}} 22314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 220, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 6 & 1 & 6 & 2 \\ 6 & 64 & 1 & 66 & 16 \\ 1 & 1 & -3 & 1 & -1 \\ 6 & 66 & 1 & 64 & 16 \\ 2 & 16 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[177] \ 5131^{\overline{1}} 414141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 312, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 10 & 10 & 3 \\ 10 & 21 & 23 & 11 \\ 10 & 23 & 21 & 11 \\ 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [179] \ 1^{\overline{1}} \overline{7} \ 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 14, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{\overline{1}} \overline{8} \ 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{\overline{1}} \overline{8} \ 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$\begin{aligned}
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[181] & 1^{1^1} 2^{1^1} 3^{1^1} 8^{1^1} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[182] & 1^{1^1} 8^{1^1} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[184] & 2^{1^1} 3^{1^1} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 13, \Delta = -2, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 & 6 \\ 2 & 6 & 3 & 6 & 5 \\ 2 & 6 & 6 & 4 & 6 \\ 2 & 6 & 5 & 6 & 6 \end{pmatrix}_{b_0} \\
T = 14 & [187] 1^{1^1} 8^{1^1} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[186] & 1^{1^1} 6^{1^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0} \\
[185] & 1^{1^1} 8^{1^1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[188] & 31^{1^1} 6^{1^1} 13161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.80 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 5 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
T = 15 & [189] 1^{1^1} 6^{1^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 15, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0 \\
[191] & 1^{1^1} 6^{1^1} 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 16, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0} \\
[192] & 1^{1^1} 6^{1^1} 13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 16, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
[193] & 1^{1^1} 8^{1^1} 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 16, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [195] \quad 1^{\mathbf{1}^1 3151321} 8^{1231^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [196] \quad 1^{\mathbf{1}^1 223151321} 8^{1231^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 17, T_{\max} = 17, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$$

$$[198] \quad 1^{\mathbf{1}^1 321} 8^{123216131^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [199] \quad 1^{\mathbf{1}^1 31} 6^{131613151321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 18, T_{\max} = 18, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0$$

$$[201] \quad 1^{\mathbf{1}^1 321} 8^{1232171231^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [202] \quad 1^{\mathbf{1}^1 31} 6^{1316131612321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 19, T_{\max} = 19, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$$

$$T = 21 \quad [206] \quad 1^{\mathbf{1}^1 31} 6^{131613161315131^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 21, T_{\max} = 21, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$$

$$T = 22 \quad [207] \quad 1^{\mathbf{1}^1 321} 8^{123218123151321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 22, T_{\max} = 22, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 4.0$$

$$T = 24 \quad [210] \quad 1^{\mathbf{1}^1 321} 8^{12321812321712321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 24, T_{\max} = 24, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0$$

$$T = 25 \quad [211] \quad 1^{\mathbf{1}^1 2321} 8^{1232181232181231^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 25, T_{\max} = 25, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 4.0$$

$$T = 27 \quad [212] \quad 1^{\mathbf{1}^1 321} 8^{12321812321812315131^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 27, T_{\max} = 27, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 4.0$$

5.20 NHC(-2)(-3)(-2) + e₆

$$T = 7 \quad [1] \quad 21^{\mathbf{1}^1 31^{111}} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -3 & -1 \\ 3 & 5 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{\mathbf{1}^1 11^1 31^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -3 & -1 \\ 3 & 5 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 7 & 0 & 5 \\ 3 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 2 & 4 & 4 & -3 & -1 \\ 3 & 5 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

[5] $21^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -440, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 8 & 3 & 4 \\ 8 & 9 & 11 & 4 & 5 \\ 8 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -3 & -1 \\ 4 & 5 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$T = 8$ [6] $1^1 1^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -137, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.14 \rightarrow 16, T_{\max} = 98, T \in [16, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & -3 & -1 \\ 2 & 2 & -1 & 3 & -1 & -3 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

[8] $1^1 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [9] $1^{11} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & -5 \end{pmatrix}_{b_0}$$

[10] $1^1 1^1 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $21^1 41^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & 0 & 3 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1}31^{\overline{1}1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

5.21 NHC(-2)(-3)(-2) + e₆ + e₆

$$T = 8 \quad [0] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 103, \Delta = -214, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 5 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

5.22 NHC(-2)(-3)(-2) + e₆ + f₄

$$T = 8 \quad [0] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 5 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

5.23 NHC(-2)(-3)(-2) + e₆ + NHC(-2)(-2)(-3)

$$T = 10 \quad [0] \ 1^{\overline{1}111^{\overline{1}1}} 31^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 4 & 3 \\ 3 & 5 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 3 & 4 & 0 \\ 3 & 4 & 4 & -4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

5.24 NHC(-2)(-3)(-2) + e₆ + NHC(-2)(-3)

$$T = 9 \quad [0] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 2 & 3 \\ 4 & 9 & 7 & 0 & 8 & 4 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 5 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.25 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

5.26 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

5.27 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{\overline{1}} 4^{\overline{1}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 544, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 5 & 10 & 10 & 1 & 5 \\ 10 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

5.28 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

5.29 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

5.30 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{116}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 21, T_{\max} = 94, T \in [21, 94]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 24 & 0 & 106 & 18 \\ 3 & 4 & 4 & -4 & 18 & -8 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19.56 \rightarrow 20, T_{\max} = 94, T \in [20, 94]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 7 & 3 \\ 4 & 9 & 7 & 0 & 16 & 5 \\ 3 & 7 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 16 & 13 & 0 & 28 & 10 \\ 3 & 5 & 4 & -4 & 10 & -7 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.89 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 \\ 3 & 5 & 4 & -4 & 3 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 20.83 \rightarrow 21, T_{\max} = 94, T \in [21, 94]$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 5 & -1 & 0 & -8 \end{pmatrix}_{b_0}$$

[4] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.62 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 9 & 7 & 0 & 16 & 5 \\ 3 & 7 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 16 & 10 & 0 & 22 & 8 \\ 3 & 5 & 4 & -4 & 8 & -2 \end{pmatrix}_{b_0}$$

[5] $1^{115}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 20 & 0 & 73 & 15 \\ 3 & 4 & 4 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.02 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 12 & 3 \\ 4 & 9 & 7 & 0 & 28 & 5 \\ 3 & 7 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 28 & 22 & 0 & 85 & 17 \\ 3 & 5 & 4 & -4 & 17 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 3 \\ 4 & 9 & 7 & 0 & 8 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 12 \\ 3 & 4 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 5 & 3 \\ 4 & 9 & 7 & 0 & 12 & 5 \\ 3 & 7 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 9 & 0 & 13 & 7 \\ 3 & 5 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{112}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 28 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 14 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 225, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 8 & 3 \\ 4 & 9 & 7 & 0 & 20 & 5 \\ 3 & 7 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 14 & 0 & 37 & 11 \\ 3 & 5 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 10 & 3 \\ 4 & 9 & 7 & 0 & 24 & 5 \\ 3 & 7 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 24 & 18 & 0 & 58 & 14 \\ 3 & 5 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 16 \\ 3 & 4 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 1000, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 1228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 1304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [30] $1^{111}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 \\ 2 & 3 & 3 & 3 & -4 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 3 \\ 2 & 6 & 4 & 4 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 \\ 2 & 4 & 2 & 4 & -4 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{14}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.8

$$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 4 & 0 & 21 & 2 \\ 2 & 6 & 4 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 30 & 0 & 172 & 24 \\ 2 & 4 & 2 & 4 & -4 & 24 & -2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^1 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^1 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^1 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 12 & 0 & 22 & 8 \\ 2 & 4 & 2 & 4 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[36] \ 1^1 31^{13} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -724, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 4 & 0 & 9 & 2 \\ 2 & 6 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 9 & 18 & 0 & 52 & 12 \\ 2 & 4 & 2 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{12} 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 18 & 0 & 58 & 14 \\ 2 & 4 & 2 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11} 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 4 \\ 1 & 4 & 0 & 4 & 0 & 9 & 2 \\ 2 & 6 & 4 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 15 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^1 31^{14} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 12 & 24 & 0 & 94 & 16 \\ 2 & 4 & 2 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{13} 31^{13} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 4 & 0 & 18 & 2 \\ 2 & 6 & 4 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 27 & 0 & 133 & 21 \\ 2 & 4 & 2 & 4 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 21 & 0 & 73 & 15 \\ 2 & 4 & 2 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 4 & 0 & 15 & 2 \\ 2 & 6 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 24 & 0 & 100 & 18 \\ 2 & 4 & 2 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{14}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 0 & 36 & 4 \\ 1 & 4 & 0 & 4 & 0 & 24 & 2 \\ 2 & 6 & 4 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 36 & 0 & 238 & 28 \\ 2 & 4 & 2 & 4 & -4 & 28 & -1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 4 & 0 & 15 & 2 \\ 2 & 6 & 4 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 27 & 0 & 121 & 19 \\ 2 & 4 & 2 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{13}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 6 & 0 & 33 & 4 \\ 1 & 4 & 0 & 4 & 0 & 21 & 2 \\ 2 & 6 & 4 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 21 & 33 & 0 & 193 & 25 \\ 2 & 4 & 2 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{12}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 4 & 0 & 18 & 2 \\ 2 & 6 & 4 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 18 & 30 & 0 & 154 & 22 \\ 2 & 4 & 2 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [47] \ 1^{12}4 \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 \\ 4 & 5 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 10 & 0 & 21 & 9 \\ 3 & 4 & 4 & -3 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{11}4 \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 4 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{11}4 \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

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$$T = 9 \quad [0] \quad 1^{1132} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 13 & 4 \\ 3 & 6 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 13 & 0 & 27 & 28 & 10 \\ 3 & 4 & 4 & -4 & 9 & 10 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.89 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 5 & 3 \\ 4 & 9 & 7 & 0 & 4 & 12 & 5 \\ 3 & 7 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 12 & 9 & 0 & 6 & 13 & 7 \\ 3 & 5 & 4 & -4 & 3 & 7 & -6 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1153} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.9$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 6 & 3 \\ 3 & 4 & 6 & 0 & 20 & 12 & 4 \\ 3 & 6 & 4 & 0 & 20 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 20 & 0 & 73 & 45 & 15 \\ 6 & 12 & 12 & 0 & 45 & 25 & 9 \\ 3 & 4 & 4 & -4 & 15 & 9 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1144} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 6 & 0 & 16 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 48 & 12 \\ 8 & 16 & 16 & 0 & 48 & 46 & 12 \\ 3 & 4 & 4 & -4 & 12 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1112} 3^1 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 23 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1112} 3^1 1^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 6 & 0 & 14 & 13 & 4 \\ 3 & 6 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & 4 & -4 & 11 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1113} 3^1 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 6 & 0 & 5 & 14 & 4 \\ 3 & 6 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & 4 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1152} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 6 & 0 & 20 & 8 & 4 \\ 3 & 6 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 20 & 0 & 73 & 30 & 15 \\ 4 & 8 & 8 & 0 & 30 & 10 & 6 \\ 3 & 4 & 4 & -4 & 15 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1151} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.1$$

$$T_{\min} = 13.50 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 20 & 0 & 73 & 15 & 15 \\ 2 & 4 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & 4 & -4 & 15 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1143} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 6 & 0 & 16 & 12 & 4 \\ 3 & 6 & 4 & 0 & 16 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 36 & 12 \\ 6 & 12 & 12 & 0 & 36 & 25 & 9 \\ 3 & 4 & 4 & -4 & 12 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1121} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.56 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & 4 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1142} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 6 & 0 & 16 & 8 & 4 \\ 3 & 6 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 24 & 12 \\ 4 & 8 & 8 & 0 & 24 & 10 & 6 \\ 3 & 4 & 4 & -4 & 12 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1133} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 27 & 9 \\ 6 & 12 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1141} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -168, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & 4 & -4 & 12 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -192, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.56 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 6 & 0 & 5 & 17 & 4 \\ 3 & 6 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & 4 & -4 & 4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1132} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

13.3

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 6 & 0 & 12 & 8 & 4 \\ 3 & 6 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & 4 & -4 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 0 & 5 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{121} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1123} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 6 & 0 & 8 & 22 & 4 \\ 3 & 6 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & 4 & -4 & 6 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1133} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 6 & 0 & 12 & 17 & 4 \\ 3 & 6 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & 4 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 6 & 0 & 5 & 13 & 4 \\ 3 & 6 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & 4 & -4 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1113}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 4 & 32 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1111}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 6 & 0 & 14 & 9 & 4 \\ 3 & 6 & 4 & 0 & 14 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 14 & 0 & 37 & 24 & 11 \\ 5 & 9 & 9 & 0 & 24 & 13 & 7 \\ 3 & 4 & 4 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1122}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 8 & 13 & 4 \\ 3 & 6 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1123}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 9 & 3 \\ 3 & 4 & 6 & 0 & 18 & 17 & 4 \\ 3 & 6 & 4 & 0 & 18 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 54 & 14 \\ 9 & 17 & 17 & 0 & 54 & 49 & 13 \\ 3 & 4 & 4 & -4 & 14 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1123}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 6 & 0 & 5 & 18 & 4 \\ 3 & 6 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & 4 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1134}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 6 & 0 & 12 & 21 & 4 \\ 3 & 6 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & 4 & -4 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1143}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 22 & 4 \\ 3 & 6 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & 4 & -4 & 16 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1112} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -584, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -648, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 6 & 0 & 5 & 22 & 4 \\ 3 & 6 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & 4 & -4 & 4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -648, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1123} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -672, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 8 & 17 & 4 \\ 3 & 6 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 6 & 0 & 5 & 26 & 4 \\ 3 & 6 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & 4 & -4 & 4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 6 & 0 & 9 & 21 & 4 \\ 3 & 6 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & 4 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 6 & 0 & 9 & 13 & 4 \\ 3 & 6 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1124} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.53 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 6 & 0 & 8 & 26 & 4 \\ 3 & 6 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & 4 & -4 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 6 & 0 & 9 & 17 & 4 \\ 3 & 6 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & 4 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -912, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1124} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 6 & 0 & 8 & 21 & 4 \\ 3 & 6 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 4 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1122} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1016, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 6 & 0 & 13 & 18 & 4 \\ 3 & 6 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & 4 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1112} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1024, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 6 & 0 & 9 & 18 & 4 \\ 3 & 6 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & 4 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1152, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{1113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -1152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 22 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{1114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -1160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & 4 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{1123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -1344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 6 & 0 & 13 & 17 & 4 \\ 3 & 6 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & 4 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{1124}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -1376$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.52 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 6 & 0 & 13 & 21 & 4 \\ 3 & 6 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & 4 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{1114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -1448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 4 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{1133}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -1488$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 6 & 0 & 17 & 22 & 4 \\ 3 & 6 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & 4 & -4 & 13 & 17 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{1113}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -1520$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 6 & 0 & 9 & 22 & 4 \\ 3 & 6 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & 4 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{1133}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -1680$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 26 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1144} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 6 & 0 & 21 & 26 & 4 \\ 3 & 6 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & 4 & -4 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1114} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 4 & 31 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1123} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 6 & 0 & 13 & 22 & 4 \\ 3 & 6 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & 4 & -4 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 6 & 0 & 9 & 26 & 4 \\ 3 & 6 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & 4 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1134} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 6 & 0 & 17 & 21 & 4 \\ 3 & 6 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & 4 & -4 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1144} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -2176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.53 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & 4 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1134} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -2400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 6 & 0 & 17 & 26 & 4 \\ 3 & 6 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & 4 & -4 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1124}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -2424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 6 & 0 & 13 & 26 & 4 \\ 3 & 6 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & 4 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [66] $1^{1131}1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 9 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{131}1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 6 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 9 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

5.34 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{11111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 1472, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

5.35 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{13}} 1^{\overline{111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -86, \ \det = 724, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{13}} 3 1^{\overline{111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -86, \ \det = 848, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

5.36 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [2] \ 1^{\overline{1111}} 3 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.49 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 14 & 4 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 5 & 2 & 3 \\ 4 & 9 & 7 & 0 & 12 & 4 & 5 \\ 3 & 7 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 5 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -217, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 8 & 2 & 3 \\ 4 & 9 & 7 & 0 & 20 & 4 & 5 \\ 3 & 7 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 20 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 5 & 4 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -220, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 10 & 2 & 3 \\ 4 & 9 & 7 & 0 & 24 & 4 & 5 \\ 3 & 7 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 24 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 5 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -400, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -436, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.28 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -436, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11111^1} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11311^1} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1012, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 25, \quad T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 25 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11211^1} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1012, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11211^1} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1024, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11311^1} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11311^1} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1744, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11411^1} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11311^1} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1876, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1141} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -2704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1141} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -2752, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1141} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -3028, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [25] \quad 1^{1111} 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 31^{112} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 168, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.57 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 12 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 31^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 492, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [28] \quad 1^{111} 4^{131^{11}} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -94, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 4 & 5 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

5.37 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

5.38 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 492, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.39 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{131}} 2^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 2 & 2 & 1 & 2 & 4 & 4 \\ 2 & 2 & 0 & 4 & -2 & 0 \\ 3 & 3 & -1 & 4 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{131}} 2^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

5.40 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$T = 10$ [0] $1^1 3^1 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 3^1 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 9 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

5.41 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^1 3^1 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 14, \Delta = -89, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

5.42 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$T = 9$ [0] $1^1 3^1 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 8 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 12 & 0 & 25 & 9 \\ 2 & 4 & 2 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 3^1 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 5 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

5.43 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3$

$T = 8$ [0] $1^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 5 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 12 & 0 & 24 & 8 \\ 3 & 4 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 0 & 9 & 5 \\ 3 & 4 & 4 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [9] $1^{111}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 3 \\ 2 & 6 & 4 & 4 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 4 & 0 & 4 & 0 & 3 \\ 2 & 6 & 4 & 4 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 2 & 6 & 3 & 6 & 0 & 3 \\ 2 & 4 & 2 & 4 & -4 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -696$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [13] $1^{11}4^1 1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 35$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}4^1 1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 344$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 2 & 3 \\ 4 & 4 & 6 & 0 & 3 & 4 \\ 4 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

5.44 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10$, $T_{\max} = 15$, $\Delta = -94$, $\det = 172$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.45 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

5.46 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9$ [1] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 14 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 5 & 3 \\ 4 & 9 & 7 & 0 & 4 & 12 & 5 \\ 3 & 7 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 12 & 9 & 0 & 6 & 13 & 7 \\ 3 & 5 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 8 & 3 \\ 4 & 9 & 7 & 0 & 4 & 20 & 5 \\ 3 & 7 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 8 & 20 & 14 & 0 & 9 & 37 & 11 \\ 3 & 5 & 4 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 16 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 8 & 16 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & 4 & -4 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 10 & 3 \\ 4 & 9 & 7 & 0 & 4 & 24 & 5 \\ 3 & 7 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 10 & 24 & 18 & 0 & 12 & 58 & 14 \\ 3 & 5 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 9 & 8 & 4 \\ 3 & 6 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 6 & 0 & 4 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 6 & 12 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & 4 & -4 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 4 & 8 & 4 \\ 3 & 6 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & 4 & -4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 4 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & 4 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1113} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1876, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 22 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1114} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -2704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 26 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1114} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -2752, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 4 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1114} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -3028, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 4 & 31 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [25] \quad 1^{1111} 31^{13} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & 3 & 0 & 3 & 1 \\ 2 & 3 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 168, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.57 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 4 & 12 & 6 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & 2 & 4 & -4 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 492, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [28] \quad 1^{1111} 31^{11} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 4 & 5 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 7 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

5.47 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

5.48 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 492, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.49 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 0 & 5 & 8 & 4 \\ 3 & 6 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & 4 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -660, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 \\ 2 & 3 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & 4 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [6] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -264, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 4 & 6 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

5.50 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 492, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

5.51 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

5.52 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$

$$T = 7 \quad [0] \quad 21^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -2 & 0 \\ 3 & 5 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -2 & 0 \\ 3 & 5 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 7 & 0 & 5 \\ 3 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 2 & 4 & 4 & -2 & 0 \\ 3 & 5 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 21^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 8 & 3 & 4 \\ 8 & 9 & 11 & 4 & 5 \\ 8 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -2 & 0 \\ 4 & 5 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^1 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 7 & 0 & 5 \\ 3 & 7 & 5 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 5 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 2^2 2^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [12] \ 1^{11}31^{11}4^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 21^1 41^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & 0 & 3 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}31^{11}4^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11}41^{11}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 3 \\ 6 & 13 & 15 & 0 & 7 \\ 6 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

5.53 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{e}_6$

$$T = 8 \quad [0] \ 1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 5 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

5.54 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

5.55 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 2 & 3 \\ 4 & 9 & 7 & 0 & 8 & 4 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 5 & 4 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

5.56 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{1}} 3^{\overline{1}} 2^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 0 & 2 \\ 3 & 5 & 5 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [5] $1^{11}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -108$, $\det = 424$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 10 & 1 & 5 \\ 10 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

5.57 NHC(-2)(-3)(-2) + $\textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$

$T = 9$ [0] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -342$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^1321^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -80$, $\det = 144$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 5 & 0 & 0 & 4 & 2 \\ 2 & 5 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & 5 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11}321^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -80$, $\det = 256$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 7 & 0 & 5 & 10 & 5 \\ 3 & 7 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & 5 & -3 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

5.58 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_{2 \rightarrow (-3)}$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 9, \ T_{\max} = 21, \ \Delta = -110, \ \det = -134, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 6 & 5 \\ 12 & 25 & 23 & 0 & 12 & 9 \\ 11 & 23 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 11 & 0 & 4 & 4 \\ 5 & 9 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

5.59 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -412, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

5.60 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 16.29 \rightarrow 17, \ T_{\max} = 20, \ T \in [17, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 \\ 3 & 5 & 4 & -3 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 18.57 \rightarrow 19, \ T_{\max} = 20, \ T \in [19, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 7 & 3 \\ 4 & 9 & 7 & 0 & 16 & 5 \\ 3 & 7 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 16 & 13 & 0 & 28 & 10 \\ 3 & 5 & 4 & -3 & 10 & -6 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{15}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.3$$

$$T_{\min} = 14.60 \rightarrow 15, \ T_{\max} = 20, \ T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 20 & 0 & 73 & 15 \\ 3 & 4 & 4 & -3 & 15 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 30, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 20, \ T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 9 & 7 & 0 & 16 & 5 \\ 3 & 7 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 10 & 0 & 22 & 8 \\ 3 & 5 & 4 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{14}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 16 & 0 & 46 & 12 \\ 3 & 4 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 60, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 63, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.11 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 12 & 3 \\ 4 & 9 & 7 & 0 & 28 & 5 \\ 3 & 7 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 28 & 22 & 0 & 85 & 17 \\ 3 & 5 & 4 & -3 & 17 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 3 \\ 4 & 9 & 7 & 0 & 8 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 5 & 3 \\ 4 & 9 & 7 & 0 & 12 & 5 \\ 3 & 7 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 9 & 0 & 13 & 7 \\ 3 & 5 & 4 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{112}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 28 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 14 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 8 & 3 \\ 4 & 9 & 7 & 0 & 20 & 5 \\ 3 & 7 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 14 & 0 & 37 & 11 \\ 3 & 5 & 4 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{114}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1016, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [28] $1^{11}3^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 2 & 4 \\ 3 & 5 & 5 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{111}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 5 & 7 & 0 & 10 & 5 \\ 3 & 7 & 5 & 0 & 10 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 10 & 0 & 14 & 8 \\ 3 & 5 & 5 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^{21}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 5 & 7 & 0 & 20 & 5 \\ 3 & 7 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 20 & 0 & 62 & 16 \\ 3 & 5 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11}3^{21}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 5 & 7 & 0 & 25 & 5 \\ 3 & 7 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 25 & 0 & 92 & 20 \\ 3 & 5 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

- [34] $1^{11}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 5 & 7 & 0 & 15 & 5 \\ 3 & 7 & 5 & 0 & 15 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 15 & 0 & 32 & 12 \\ 3 & 5 & 5 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{112}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 6 \\ 1 & 2 & 0 & 2 & 0 & 8 \\ 1 & 2 & 2 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 4 & 6 & 8 & 6 & 0 & 26 \\ 2 & 2 & 2 & 2 & -3 & 10 \end{pmatrix}_{b_0}$$
- [38] $1^{11}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 5 & 7 & 0 & 20 & 5 \\ 3 & 7 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 20 & 0 & 58 & 16 \\ 3 & 5 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{112}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 \\ 1 & 2 & 0 & 2 & 0 & 5 \\ 1 & 2 & 2 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 3 & 5 & 5 & 5 & 0 & 13 \\ 2 & 2 & 2 & 2 & -3 & 7 \end{pmatrix}_{b_0}$$
- [40] $1^{113}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 7 \\ 1 & 2 & 0 & 2 & 0 & 6 \\ 1 & 2 & 2 & 0 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 4 & 7 & 6 & 7 & 0 & 25 \\ 2 & 2 & 2 & 2 & -3 & 9 \end{pmatrix}_{b_0}$$
- [41] $1^{113}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 9 \\ 1 & 2 & 0 & 2 & 0 & 12 \\ 1 & 2 & 2 & 0 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 6 & 9 & 12 & 9 & 0 & 61 \\ 2 & 2 & 2 & 2 & -3 & 15 \end{pmatrix}_{b_0}$$
- [42] $1^{12}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 5 & 7 & 0 & 25 & 5 \\ 3 & 7 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 25 & 25 & 0 & 94 & 20 \\ 3 & 5 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$
- [43] $1^{131}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -340$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^{11}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -384$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 4 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{113}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 8 & 0 & 40 & 12 \\ 2 & 2 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[46] $1^1 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 6 & 12 & 0 & 22 & 8 \\ 2 & 4 & 2 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{12} 321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 5 & 7 & 0 & 30 & 5 \\ 3 & 7 & 5 & 0 & 30 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 30 & 30 & 0 & 134 & 24 \\ 3 & 5 & 5 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{114} 321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -472$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 12 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[49] $1^1 31^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -564$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 4 & 0 & 9 & 2 \\ 2 & 6 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 9 & 18 & 0 & 52 & 12 \\ 2 & 4 & 2 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{114} 321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -568$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{12} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -600$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 18 & 0 & 58 & 14 \\ 2 & 4 & 2 & 4 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114} 321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -624$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 11 & 0 & 81 & 17 \\ 2 & 2 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[53] $2^1 2^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 & 1 & 3 & 3 \\ 1 & 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{11} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -748$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 4 \\ 1 & 4 & 0 & 4 & 0 & 9 & 2 \\ 2 & 6 & 4 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 15 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[55] $1^1 31^{114} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -760$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 12 & 24 & 0 & 94 & 16 \\ 2 & 4 & 2 & 4 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{13}31^{113}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -960, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 4 & 0 & 18 & 2 \\ 2 & 6 & 4 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 27 & 0 & 133 & 21 \\ 2 & 4 & 2 & 4 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

[57] $2^1 2^1 31^{1112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1056, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 3 & 1 & 6 & 3 \\ 2 & 4 & 0 & 4 & 1 & 6 & 2 \\ 2 & 3 & 4 & 1 & 1 & 6 & 3 \\ 1 & 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & 3 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{11}31^{113}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 21 & 0 & 73 & 15 \\ 2 & 4 & 2 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{12}31^{113}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 4 & 0 & 15 & 2 \\ 2 & 6 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 24 & 0 & 100 & 18 \\ 2 & 4 & 2 & 4 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{14}31^{114}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 0 & 36 & 4 \\ 1 & 4 & 0 & 4 & 0 & 24 & 2 \\ 2 & 6 & 4 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 36 & 0 & 238 & 28 \\ 2 & 4 & 2 & 4 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{11}31^{114}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 4 & 0 & 15 & 2 \\ 2 & 6 & 4 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 15 & 27 & 0 & 121 & 19 \\ 2 & 4 & 2 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{13}31^{114}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1996, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 6 & 0 & 33 & 4 \\ 1 & 4 & 0 & 4 & 0 & 21 & 2 \\ 2 & 6 & 4 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 33 & 0 & 193 & 25 \\ 2 & 4 & 2 & 4 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{12}31^{114}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2056, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 4 & 0 & 18 & 2 \\ 2 & 6 & 4 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 18 & 30 & 0 & 154 & 22 \\ 2 & 4 & 2 & 4 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [64] $1^{11} 4^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 4 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 7 & 0 & 8 & 6 \\ 3 & 4 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[65] 1^{111} 41^{111} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 672, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 5 & 3 \\ 6 & 13 & 15 & 0 & 12 & 7 \\ 6 & 15 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 12 & 0 & 8 & 6 \\ 3 & 7 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

5.61 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] 1^{11} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & 5 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 7 & 0 & 10 & 5 & 5 \\ 3 & 7 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 10 & 0 & 14 & 7 & 8 \\ 2 & 5 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & 5 & -3 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

5.62 NHC(-2)(-3)(-2) + f₄ + su₂ + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 6 & 0 & 5 & 14 & 4 \\ 3 & 6 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & 4 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 23 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 6 & 0 & 14 & 13 & 4 \\ 3 & 6 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & 4 & -3 & 11 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 6 & 0 & 5 & 17 & 4 \\ 3 & 6 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & 4 & -3 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 0 & 5 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1122} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1133} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -192, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 6 & 0 & 12 & 17 & 4 \\ 3 & 6 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1121} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1123} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 6 & 0 & 8 & 22 & 4 \\ 3 & 6 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & 4 & -3 & 6 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 4 & 32 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 6 & 0 & 5 & 13 & 4 \\ 3 & 6 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & 4 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 6 & 0 & 14 & 9 & 4 \\ 3 & 6 & 4 & 0 & 14 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 24 & 11 \\ 5 & 9 & 9 & 0 & 24 & 13 & 7 \\ 3 & 4 & 4 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

11.2

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1123} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 8 & 13 & 4 \\ 3 & 6 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1123} 31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 9 & 3 \\ 3 & 4 & 6 & 0 & 18 & 17 & 4 \\ 3 & 6 & 4 & 0 & 18 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 54 & 14 \\ 9 & 17 & 17 & 0 & 54 & 49 & 13 \\ 3 & 4 & 4 & -3 & 14 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 6 & 0 & 5 & 18 & 4 \\ 3 & 6 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & 4 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1134} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 6 & 0 & 12 & 21 & 4 \\ 3 & 6 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & 4 & -3 & 9 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1143} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 22 & 4 \\ 3 & 6 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & 4 & -3 & 16 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{112} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{113} 3^1 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 6 & 0 & 5 & 22 & 4 \\ 3 & 6 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & 4 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1123} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 8 & 17 & 4 \\ 3 & 6 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1114} 3^1 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 6 & 0 & 5 & 26 & 4 \\ 3 & 6 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & 4 & -3 & 4 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1114} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 6 & 0 & 9 & 21 & 4 \\ 3 & 6 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & 4 & -3 & 7 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1112} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 6 & 0 & 9 & 13 & 4 \\ 3 & 6 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & 4 & -3 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1124} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 6 & 0 & 8 & 26 & 4 \\ 3 & 6 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & 4 & -3 & 6 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1113} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 6 & 0 & 9 & 17 & 4 \\ 3 & 6 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & 4 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1124} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 6 & 0 & 8 & 21 & 4 \\ 3 & 6 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{1113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -704, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{1113} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -736, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 4 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{1122} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -792, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 6 & 0 & 13 & 18 & 4 \\ 3 & 6 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & 4 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{1112} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -800, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 6 & 0 & 9 & 18 & 4 \\ 3 & 6 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & 4 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -888, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{1122} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{1113} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 22 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{1123} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -1040, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 6 & 0 & 13 & 17 & 4 \\ 3 & 6 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & 4 & -3 & 10 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1124} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1056, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 6 & 0 & 13 & 21 & 4 \\ 3 & 6 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & 4 & -3 & 10 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 4 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1133} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1152, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 6 & 0 & 17 & 22 & 4 \\ 3 & 6 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & 4 & -3 & 13 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 6 & 0 & 9 & 22 & 4 \\ 3 & 6 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & 4 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1133} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & 4 & -3 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1320, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 26 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1144} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 6 & 0 & 21 & 26 & 4 \\ 3 & 6 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & 4 & -3 & 16 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 4 & 31 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1123} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 6 & 0 & 13 & 22 & 4 \\ 3 & 6 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & 4 & -3 & 10 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1134} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1424, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 6 & 0 & 17 & 21 & 4 \\ 3 & 6 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & 4 & -3 & 13 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 6 & 0 & 9 & 26 & 4 \\ 3 & 6 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & 4 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1144} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & 4 & -3 & 16 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1134} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 6 & 0 & 17 & 26 & 4 \\ 3 & 6 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & 4 & -3 & 13 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1124} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1880, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 6 & 0 & 13 & 26 & 4 \\ 3 & 6 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & 4 & -3 & 10 & 20 & 1 \end{pmatrix}_{b_0}$$

5.63 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

5.64 NHC(-2)(-3)(-2) + f₄ + su₂ + su₂→(-3)

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 3 & 4 \\ 3 & 6 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 4 & 6 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 14 & 3 & 4 \\ 3 & 6 & 4 & 0 & 14 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 8 & 11 \\ 2 & 3 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 13 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 17 & 3 & 4 \\ 3 & 6 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 9 & 13 \\ 2 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 10 & 14 \\ 2 & 3 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -552, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 11 & 16 \\ 2 & 3 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -824, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 22 & 3 & 4 \\ 3 & 6 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 22 & 0 & 85 & 12 & 17 \\ 2 & 3 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1072, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 26 & 3 & 4 \\ 3 & 6 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 26 & 0 & 118 & 14 & 20 \\ 2 & 3 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

5.65 NHC(-2)(-3)(-2) + f₄ + su₂ + su₂n3mix

$$T = 9 \quad [0] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 0 & 5 & 7 & 4 \\ 3 & 6 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 6 & 4 \\ 4 & 7 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & 4 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 6 & 0 & 12 & 7 & 4 \\ 3 & 6 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 15 & 9 \\ 4 & 7 & 7 & 0 & 15 & 7 & 5 \\ 3 & 4 & 4 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -216, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 8 & 16 & 4 \\ 3 & 6 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 22 & 6 \\ 9 & 16 & 16 & 0 & 22 & 44 & 12 \\ 3 & 4 & 4 & -3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 7 & 4 \\ 3 & 6 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 10 & 6 \\ 4 & 7 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

11.2

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 11 & 4 \\ 3 & 6 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 17 & 7 \\ 6 & 11 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & 4 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}1} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 4 & 7 & 4 \\ 3 & 6 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 5 & 3 \\ 4 & 7 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}1} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1122}1} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 6 & 0 & 8 & 11 & 4 \\ 3 & 6 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 16 & 6 \\ 6 & 11 & 11 & 0 & 16 & 20 & 8 \\ 3 & 4 & 4 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}1} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 9 & 7 & 4 \\ 3 & 6 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1122}1} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 6 & 0 & 4 & 11 & 4 \\ 3 & 6 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 8 & 3 \\ 6 & 11 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & 4 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1123}1} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 6 & 0 & 18 & 7 & 4 \\ 3 & 6 & 4 & 0 & 18 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 22 & 14 \\ 4 & 7 & 7 & 0 & 22 & 7 & 5 \\ 3 & 4 & 4 & -3 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1113}1} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 11 & 3 \\ 8 & 15 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & 4 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1112}1} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

11.2

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 14 & 3 \\ 12 & 21 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1121}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 6 & 0 & 13 & 7 & 4 \\ 3 & 6 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 16 & 10 \\ 4 & 7 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1122}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 6 & 0 & 13 & 11 & 4 \\ 3 & 6 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 25 & 10 \\ 6 & 11 & 11 & 0 & 25 & 20 & 8 \\ 3 & 4 & 4 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1114}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 19 & 4 \\ 3 & 6 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 14 & 3 \\ 10 & 19 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1112}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 16 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 11 & 3 \\ 9 & 16 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1131}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 6 & 0 & 17 & 7 & 4 \\ 3 & 6 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 21 & 13 \\ 4 & 7 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{1141}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -808, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 6 & 0 & 21 & 7 & 4 \\ 3 & 6 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 26 & 16 \\ 4 & 7 & 7 & 0 & 26 & 7 & 5 \\ 3 & 4 & 4 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{1113}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 6 & 0 & 22 & 7 & 4 \\ 3 & 6 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 22 & 0 & 85 & 27 & 17 \\ 4 & 7 & 7 & 0 & 27 & 7 & 5 \\ 3 & 4 & 4 & -3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{1113}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.2

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 20 & 4 \\ 3 & 6 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 14 & 3 \\ 11 & 20 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11113311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 25 & 4 \\ 3 & 6 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 17 & 3 \\ 14 & 25 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11141311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 6 & 0 & 26 & 7 & 4 \\ 3 & 6 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 26 & 0 & 118 & 32 & 20 \\ 4 & 7 & 7 & 0 & 32 & 7 & 5 \\ 3 & 4 & 4 & -3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11114311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 17 & 3 \\ 13 & 24 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11114311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 6 & 0 & 4 & 29 & 4 \\ 3 & 6 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 20 & 3 \\ 16 & 29 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

5.66 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [1] \ 1^{1111311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 14 & 4 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{1111311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 5 & 2 & 3 \\ 4 & 9 & 7 & 0 & 12 & 4 & 5 \\ 3 & 7 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 5 & 4 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

- [3] $1^{1141}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 8 & 2 & 3 \\ 4 & 9 & 7 & 0 & 20 & 4 & 5 \\ 3 & 7 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 20 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 5 & 4 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{1121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{112}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 10 & 2 & 3 \\ 4 & 9 & 7 & 0 & 24 & 4 & 5 \\ 3 & 7 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 24 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 5 & 4 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{1131}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{1121}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{1111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{1121}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

12.7

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{1121}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -788, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{1131}31^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -788, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{1121}31^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{1131}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1088, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{1131}31^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{1141}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11\bar{1}1^1} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11\bar{1}1^1} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2096, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11\bar{1}1^1} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11\bar{1}1^1} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [24] \quad 1^{11\bar{1}1^1} 4^1 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 4 & 5 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

5.67 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11\bar{1}1^1} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 672, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

5.68 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{11\bar{1}1^1} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 21, \ \Delta = -111, \ \det = 104, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 21, \ T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{13}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 2 & 6 & 1 & 6 & 0 & 3 \\ 4 & 10 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

5.69 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -110, \ \det = -134, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 6 & 5 \\ 12 & 25 & 23 & 0 & 12 & 9 \\ 11 & 23 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 11 & 0 & 4 & 4 \\ 5 & 9 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

5.70 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 5 & 4 \\ 3 & 6 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 8 & 4 \\ 3 & 6 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 14 & 4 \\ 3 & 6 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 8 & 2 \\ 8 & 14 & 14 & 0 & 8 & 37 & 11 \\ 3 & 4 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 13 & 4 \\ 3 & 6 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 3 & 17 & 4 \\ 3 & 6 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 17 & 0 & 9 & 49 & 13 \\ 3 & 4 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 10 & 2 \\ 10 & 18 & 18 & 0 & 10 & 58 & 14 \\ 3 & 4 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 11 & 2 \\ 11 & 21 & 21 & 0 & 11 & 76 & 16 \\ 3 & 4 & 4 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 22 & 4 \\ 3 & 6 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 12 & 2 \\ 12 & 22 & 22 & 0 & 12 & 85 & 17 \\ 3 & 4 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -1072, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 26 & 4 \\ 3 & 6 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 14 & 2 \\ 14 & 26 & 26 & 0 & 14 & 118 & 20 \\ 3 & 4 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

5.71 NHC(-2)(-3)(-2) + f₄ + su₂→(-3) + su₃

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 4 & 5 \\ 3 & 7 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 8 & 4 \\ 3 & 6 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

5.72 NHC(-2)(-3)(-2) + f₄ + su₂n3mix

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 8 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 17 & 0 & 55 & 13 \\ 3 & 4 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 11 & 4 \\ 3 & 6 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 8 \\ 3 & 4 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 15 & 0 & 39 & 11 \\ 3 & 4 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 19 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 19 & 0 & 64 & 14 \\ 3 & 4 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 16 & 0 & 44 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 21 & 0 & 80 & 16 \\ 3 & 4 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 20 & 0 & 69 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 716, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 740, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 25 & 4 \\ 3 & 6 & 4 & 0 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 25 & 0 & 111 & 19 \\ 3 & 4 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 1016, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 29 & 4 \\ 3 & 6 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 29 & 0 & 148 & 22 \\ 3 & 4 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [13] \ 1^{11}31^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 4 \\ 1 & 4 & 0 & 4 & 0 & 4 & 2 \\ 2 & 6 & 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{12}31^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 14 & 4 \\ 1 & 4 & 0 & 4 & 0 & 10 & 2 \\ 2 & 6 & 4 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 14 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[15] $1^1 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.18 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 8 & 4 \\ 1 & 4 & 0 & 4 & 0 & 4 & 2 \\ 2 & 6 & 4 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 8 & 0 & 9 & 5 \\ 2 & 4 & 2 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[16] $1^1 3^1 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 8 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 12 & 0 & 25 & 9 \\ 2 & 4 & 2 & 4 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[17] $1^1 3^1 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 5 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[18] $1^1 3^1 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 11 & 4 \\ 1 & 4 & 0 & 4 & 0 & 7 & 2 \\ 2 & 6 & 4 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 11 & 0 & 20 & 8 \\ 2 & 4 & 2 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [19] $1^1 3^1 1^1 4^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 41$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

5.73 NHC(-2)(-3)(-2) + f₄ + su₂n3mix + su₂

$T = 9$ [0] $1^1 1^1 3^1 1^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 6 & 0 & 7 & 5 & 4 \\ 3 & 6 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 15 & 5 \\ 3 & 5 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & 4 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^1 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 6 & 0 & 7 & 12 & 4 \\ 3 & 6 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 15 & 5 \\ 6 & 12 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 4 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[2] $1^1 1^1 1^1 2^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 6 & 0 & 16 & 8 & 4 \\ 3 & 6 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 16 & 0 & 44 & 22 & 12 \\ 4 & 8 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & 4 & -3 & 12 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 7 & 8 & 4 \\ 3 & 6 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1121}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 6 & 0 & 11 & 9 & 4 \\ 3 & 6 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 9 & 0 & 17 & 13 & 7 \\ 3 & 4 & 4 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 6 & 0 & 11 & 8 & 4 \\ 3 & 6 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 16 & 8 \\ 4 & 8 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & 4 & -3 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 7 & 9 & 4 \\ 3 & 6 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{1121}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 11 & 4 & 4 \\ 3 & 6 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 6 & 0 & 7 & 18 & 4 \\ 3 & 6 & 4 & 0 & 7 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 22 & 5 \\ 10 & 18 & 18 & 0 & 22 & 58 & 14 \\ 3 & 4 & 4 & -3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1113} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 & 4 \\ 3 & 6 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & 4 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1121} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 7 & 13 & 4 \\ 3 & 6 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1122} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 6 & 0 & 11 & 13 & 4 \\ 3 & 6 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 25 & 8 \\ 7 & 13 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & 4 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1141} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 19 & 4 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1121} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 7 & 17 & 4 \\ 3 & 6 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 21 & 5 \\ 9 & 17 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1114} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -808, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 6 & 0 & 7 & 21 & 4 \\ 3 & 6 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 26 & 5 \\ 11 & 21 & 21 & 0 & 26 & 76 & 16 \\ 3 & 4 & 4 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11131} 3^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -848, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 6 & 0 & 7 & 22 & 4 \\ 3 & 6 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 27 & 5 \\ 12 & 22 & 22 & 0 & 27 & 85 & 17 \\ 3 & 4 & 4 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1131} 3^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -976, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1131} 3^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.68 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 25 & 4 & 4 \\ 3 & 6 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & 4 & -3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1141} 3^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 6 & 0 & 7 & 26 & 4 \\ 3 & 6 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 32 & 5 \\ 14 & 26 & 26 & 0 & 32 & 118 & 20 \\ 3 & 4 & 4 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1141} 3^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & 4 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1141} 3^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1656, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 6 & 0 & 29 & 4 & 4 \\ 3 & 6 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & 4 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

5.74 NHC(-2)(-3)(-2) + f₄ + su₂n₃mix + su₃

$$T = 9 \quad [0] \ 1^{1111} 3^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 17 & 0 & 55 & 11 & 13 \\ 2 & 4 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -468, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

12.2

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

5.75 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 0 & 24 & 8 \\ 3 & 4 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 9 & 5 \\ 3 & 4 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \ 1^{113}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 \\ 1 & 2 & 0 & 2 & 0 & 3 \\ 1 & 2 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 1 & 3 \\ 3 & 5 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 6 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 \\ 1 & 2 & 0 & 2 & 0 & 4 \\ 1 & 2 & 2 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 2 & 3 & 4 & 3 & 0 & 4 \\ 2 & 2 & 2 & 2 & -3 & 4 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 \\ 1 & 0 & 2 & 2 & 0 & 6 \\ 1 & 2 & 0 & 2 & 0 & 8 \\ 1 & 2 & 2 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 4 & 6 & 8 & 6 & 0 & 25 \\ 2 & 2 & 2 & 2 & -3 & 9 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 \\ 2 & 4 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 3 \\ 2 & 6 & 4 & 4 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 \\ 2 & 4 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -540$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [19] $1^{11}4^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 24$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{11}31^{11}4^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 268$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.09 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 2 & 3 \\ 4 & 4 & 6 & 0 & 3 & 4 \\ 4 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

5.76 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -412$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

5.77 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9$ [1] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 14 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 5 & 3 \\ 4 & 9 & 7 & 0 & 4 & 12 & 5 \\ 3 & 7 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 12 & 9 & 0 & 6 & 13 & 7 \\ 3 & 5 & 4 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 16 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 8 & 16 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & 4 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 8 & 3 \\ 4 & 9 & 7 & 0 & 4 & 20 & 5 \\ 3 & 7 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 8 & 20 & 14 & 0 & 9 & 37 & 11 \\ 3 & 5 & 4 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 10 & 3 \\ 4 & 9 & 7 & 0 & 4 & 24 & 5 \\ 3 & 7 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 10 & 24 & 18 & 0 & 12 & 58 & 14 \\ 3 & 5 & 4 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 9 & 8 & 4 \\ 3 & 6 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -244, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 6 & 0 & 4 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 6 & 12 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 4 & 8 & 4 \\ 3 & 6 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 23 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -788, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -788, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 4 & 32 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 18 & 32 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -3 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1112} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -800, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1088, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1113} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1360, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 4 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & 4 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 22 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 26 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 4 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1114} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.10 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 4 & 31 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [24] \quad 1^{1111} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 4 & 5 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 7 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

5.78 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 6 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

5.79 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 2^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 3 & 4 \\ 3 & 6 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 9 & 4 & 5 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

5.80 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \text{n3mix}$

$$T = 9 \quad [0] \quad 1^{11} 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 11 & 2 \\ 10 & 17 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & 4 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 4 & 7 & 4 \\ 3 & 6 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 5 & 2 \\ 4 & 7 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 8 & 2 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

5.81 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 2^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 0 & 5 & 8 & 4 \\ 3 & 6 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.30 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -640, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

5.82 $\text{NHC}(-2)(-3)(-2) + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \ 21^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 10, \ T_{\max} = 11, \ \Delta = -29, \ \det = 288, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 3 & 3 \\ 4 & 4 & 6 & 4 & 3 & 4 & 4 \\ 4 & 6 & 4 & 4 & 3 & 4 & 4 \\ 3 & 4 & 4 & 1 & 3 & 3 & 3 \\ 2 & 3 & 3 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

5.83 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{so}_8$

$$T = 7 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -5, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 15 & 6 \\ 16 & 25 & 23 & 9 \\ 15 & 23 & 20 & 8 \\ 6 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 11 & 5 \\ 12 & 25 & 23 & 9 \\ 11 & 23 & 20 & 8 \\ 5 & 9 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -27, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 7 & 0 & 5 \\ 3 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -28, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.14 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 7 & 4 & 5 \\ 7 & 8 & 6 & 6 \\ 4 & 6 & 0 & 2 \\ 5 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -38, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 13 & 20 & 17 & 7 \\ 20 & 25 & 23 & 9 \\ 17 & 23 & 20 & 8 \\ 7 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 5 & 7 & 5 & 5 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 0 & 2 \\ 5 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -56, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 5 & 12 & 12 & 5 \\ 12 & 25 & 27 & 9 \\ 12 & 27 & 25 & 9 \\ 5 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -60, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 10 & 15 & 15 & 6 \\ 15 & 20 & 22 & 8 \\ 15 & 22 & 20 & 8 \\ 6 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -64, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 11 & 5 \\ 11 & 20 & 22 & 8 \\ 11 & 22 & 20 & 8 \\ 5 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 21^1 31^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -64, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.31 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 13 & 17 & 17 & 7 \\ 17 & 20 & 22 & 8 \\ 17 & 22 & 20 & 8 \\ 7 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 21^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -184, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 16 & 6 \\ 16 & 20 & 22 & 8 \\ 16 & 22 & 20 & 8 \\ 6 & 8 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 21^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -248, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 13 & 20 & 20 & 7 \\ 20 & 25 & 27 & 9 \\ 20 & 27 & 25 & 9 \\ 7 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [13] \ 1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, \ T_{\max} = 11, \ \Delta = -87, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 2 & 3 & 0 & 4 & 3 & 3 \\ 2 & 2 & -1 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 21^1 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -106, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 15 & 7 \\ 12 & 15 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -106, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 15 & 7 \\ 12 & 15 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 31^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

- [16] $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$ $1^1 31^1 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 0 & -2 & 1 \end{pmatrix}_{b_0}$
- [18] $1^1 321^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -87$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 7 & 0 & 5 \\ 3 & 7 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 5 & -2 & 3 \end{pmatrix}_{b_0}$
- [19] $1^1 41^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -106$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [20] $1^1 31^1 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 4 & 7 & 8 & 1 & 4 \\ 7 & 10 & 13 & 0 & 6 \\ 8 & 13 & 12 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 4 & 6 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [21] $1^1 321^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -87$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$
- [22] $1^1 31^1 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 40$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [23] $1^1 3131^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 4 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 12 & 1 & 10 & 6 \\ 3 & 6 & -1 & 6 & 1 \end{pmatrix}_{b_0}$
- [24] $1^1 41^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -106$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 15 & 7 \\ 9 & 15 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [25] $1^1 31^1 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 11 & 5 \\ 2 & 3 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [26] $21^1 41^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -106$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 13 & 14 & 14 & 7 \\ 14 & 13 & 15 & 7 \\ 14 & 15 & 13 & 7 \\ 7 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [27] $1^1 31^1 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 7 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 13 & 6 \\ 7 & 12 & 13 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 131^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 13 & 6 \\ 4 & 12 & 13 & 10 & 6 \\ 3 & 6 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^1 31^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 21^{11} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 10 & 5 \\ 10 & 13 & 15 & 7 \\ 10 & 15 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 \\ 3 & 5 & 5 & 0 & 2 \\ 3 & 5 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{111} 2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [33] \quad 21^1 41^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 1 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 2 & 1 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 21^1 41^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^1 31^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^1 41 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 1 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^1 31^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^1 31^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 3 & 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{\overline{11}} 4 131^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{\overline{11}} 3 \ 221^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 5 & 0 & 4 & 4 \\ 2 & 2 & 3 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[43] \ 2 \ 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 4 & 1 & 2 \\ 3 & 3 & 4 & 4 & 2 & 4 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 2 \ 3 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 9 & 5 & 5 \\ 6 & 9 & 13 & 6 & 7 \\ 3 & 5 & 6 & 0 & 2 \\ 3 & 5 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[44] \ 21^{\overline{11}} 41^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & 0 & 3 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[46] \ 31^{\overline{11}} 31^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 6 & 6 & 4 & 3 \\ 6 & 13 & 15 & 6 & 7 \\ 6 & 15 & 13 & 6 & 7 \\ 4 & 6 & 6 & 0 & 2 \\ 3 & 7 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{\overline{11}} 41^{\overline{11}} 231^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 2 \\ 3 & 4 & 1 & 6 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 6 & 1 & 4 & 2 & 4 \\ 1 & 2 & 0 & 2 & -2 & 0 \\ 2 & 4 & 0 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{\overline{11}} 41^{\overline{11}} 231^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 3 \\ 6 & 13 & 15 & 0 & 7 \\ 6 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[49] \ 31^{\overline{11}} 31^{\overline{11}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 2 & 4 \\ 3 & 4 & 0 & 4 & 1 & 2 \\ 3 & 6 & 4 & 4 & 2 & 4 \\ 1 & 2 & 1 & 2 & -2 & 0 \\ 2 & 4 & 2 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [50] \quad 2 \quad 3 \quad 21 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 1 & 3 & 3 \\ 3 & 6 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

5.84 NHC(-2)(-3)(-2) + so₈ + e₆

$$T = 8 \quad [0] \quad 1 \quad 1^{11} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 & 5 \\ 12 & 25 & 23 & 0 & 9 & 9 \\ 11 & 23 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & -6 & -4 & -4 \\ 5 & 9 & 8 & -4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 & 5 \\ 12 & 25 & 27 & 0 & 9 & 9 \\ 12 & 27 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -6 & -4 & -4 \\ 5 & 9 & 9 & -4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 & 5 \\ 11 & 20 & 22 & 0 & 8 & 8 \\ 11 & 22 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & -6 & -4 & -4 \\ 5 & 8 & 8 & -4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{111} \quad 31^{11} \quad 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

5.85 NHC(-2)(-3)(-2) + so₈ + e₆ + su₂

$$T = 9 \quad [0] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -442$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

5.86 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{e}_6$ + $\mathfrak{su}_3$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

5.87 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$

$T = 8$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 81$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 \\ 12 & 25 & 27 & 0 & 9 \\ 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [3] $1^{111}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

5.88 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{112}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -342$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

5.89 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 6 & 5 \\ 12 & 25 & 23 & 0 & 12 & 9 \\ 11 & 23 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 11 & 0 & 4 & 4 \\ 5 & 9 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

5.90 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$

$T = 9$ [0] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -412$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

5.91 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 8$ [0] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 8 & -2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 76$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 5 \\ 7 & 8 & 6 & 0 & 6 \\ 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 \\ 12 & 25 & 27 & 0 & 9 \\ 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 4 & 12 & 1 & 10 & 12 & 6 \\ 5 & 12 & 0 & 12 & 11 & 5 \\ 3 & 6 & -1 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

5.92 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 4 & 5 \\ 7 & 8 & 6 & 0 & 6 & 6 \\ 4 & 6 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 4 & 0 & 2 & 4 \\ 5 & 6 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 10 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 11 & 5 \\ 7 & 8 & 6 & 0 & 15 & 6 \\ 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 15 & 9 & 0 & 22 & 10 \\ 5 & 6 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 7 & 5 \\ 7 & 8 & 6 & 0 & 9 & 6 \\ 4 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 5 & 0 & 8 & 6 \\ 5 & 6 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 18 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 5 & 5 \\ 7 & 8 & 6 & 0 & 6 & 6 \\ 4 & 6 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 4 & 0 & 2 & 4 \\ 5 & 6 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{112}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 26 & 5 \\ 7 & 8 & 6 & 0 & 33 & 6 \\ 4 & 6 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 33 & 19 & 0 & 122 & 22 \\ 5 & 6 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{112}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 27 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 12 & 5 \\ 7 & 8 & 6 & 0 & 15 & 6 \\ 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 9 & 0 & 22 & 10 \\ 5 & 6 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}} 3^{\overline{112}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 19 & 5 \\ 7 & 8 & 6 & 0 & 24 & 6 \\ 4 & 6 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 24 & 14 & 0 & 62 & 16 \\ 5 & 6 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

5.93 NHC(-2)(-3)(-2) + so₈ + so₈ + su₂ → (-3)

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 6 & 5 \\ 12 & 25 & 23 & 0 & 12 & 9 \\ 11 & 23 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 11 & 0 & 4 & 4 \\ 5 & 9 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 4 & 5 \\ 7 & 8 & 6 & 0 & 5 & 6 \\ 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 1 & 3 \\ 5 & 6 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

5.94 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_{2n3\text{mix}}$

$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 11 & 5 \\ 7 & 8 & 6 & 0 & 14 & 6 \\ 4 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 14 & 8 & 0 & 21 & 9 \\ 5 & 6 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

5.95 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$

$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 1 & 5 \\ 12 & 25 & 23 & 0 & 0 & 9 \\ 11 & 23 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 7 & 5 \\ 7 & 8 & 6 & 0 & 9 & 6 \\ 4 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 5 & 0 & 7 & 5 \\ 5 & 6 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 1 & 5 \\ 11 & 20 & 22 & 0 & 0 & 8 \\ 11 & 22 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$T = 10 \quad [4] \quad 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 6 & 5 \\ 9 & 13 & 15 & 0 & 10 & 7 \\ 9 & 15 & 13 & 0 & 10 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 10 & 0 & 4 & 4 \\ 5 & 7 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

5.96 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 15 & 13 & 6 \\ 16 & 25 & 23 & 20 & 9 \\ 15 & 23 & 20 & 17 & 8 \\ 13 & 20 & 17 & 13 & 7 \\ 6 & 9 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112} 3 \ 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 28 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 28 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 15 & 23 & 6 \\ 16 & 25 & 23 & 36 & 9 \\ 15 & 23 & 20 & 33 & 8 \\ 23 & 36 & 33 & 49 & 13 \\ 6 & 9 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 15 & 36 & 6 \\ 16 & 25 & 23 & 56 & 9 \\ 15 & 23 & 20 & 50 & 8 \\ 36 & 56 & 50 & 118 & 20 \\ 6 & 9 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 21^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 17 & 27 & 7 \\ 20 & 25 & 23 & 36 & 9 \\ 17 & 23 & 20 & 33 & 8 \\ 27 & 36 & 33 & 49 & 13 \\ 7 & 9 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 17 & 15 & 7 \\ 20 & 25 & 23 & 20 & 9 \\ 17 & 23 & 20 & 17 & 8 \\ 15 & 20 & 17 & 13 & 7 \\ 7 & 9 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 7 & 5 \\ 7 & 8 & 6 & 9 & 6 \\ 5 & 6 & 0 & 5 & 2 \\ 7 & 9 & 5 & 8 & 6 \\ 5 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 5 & 5 \\ 7 & 8 & 6 & 6 & 6 \\ 5 & 6 & 0 & 4 & 2 \\ 5 & 6 & 4 & 2 & 4 \\ 5 & 6 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{113} 3 \ 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 120, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 28 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 3 \ 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 120, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{112} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 140, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 23 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 140, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 19 & 5 \\ 7 & 8 & 6 & 24 & 6 \\ 5 & 6 & 0 & 14 & 2 \\ 19 & 24 & 14 & 62 & 16 \\ 5 & 6 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 164, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 12 & 5 \\ 7 & 8 & 6 & 15 & 6 \\ 5 & 6 & 0 & 9 & 2 \\ 12 & 15 & 9 & 22 & 10 \\ 5 & 6 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 21^{112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 17 & 42 & 7 \\ 20 & 25 & 23 & 56 & 9 \\ 17 & 23 & 20 & 50 & 8 \\ 42 & 56 & 50 & 118 & 20 \\ 7 & 9 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 16 & 0 & 46 & 12 \\ 3 & 4 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 16 & 0 & 46 & 12 \\ 3 & 4 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 32 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$

[42] $1^{113} 1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{507} \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{508} + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$

[43] $1^{11}3^{11}3^{11} + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{5o}_7 \cdot \textcolor{green}{5u}_2] + \textcolor{brown}{5o}_8 + \textcolor{red}{5u}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 272, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{112} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 11, \, \Delta = -85, \, \det = 296, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$
 $\left(\begin{array}{cccccc} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{0} & \frac{11}{18} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{4}{4} & \frac{0}{0} & \frac{18}{18} & \frac{4}{4} \\ \frac{3}{1} & \frac{6}{0} & \frac{4}{0} & \frac{0}{-4} & \frac{18}{0} & \frac{4}{-2} \\ \frac{11}{3} & \frac{18}{4} & \frac{18}{4} & \frac{0}{-2} & \frac{58}{14} & \frac{14}{2} \end{array} \right)_{b_0}$

[44] $1^{14} 3^1 1^{12} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{s}o_7 \cdot \textcolor{green}{s}u_2] + \textcolor{orange}{s}o_8 + \textcolor{red}{s}u_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 296, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{113} 3^1 1^{13} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{so}}_7 \cdot \textcolor{teal}{\mathbf{su}}_2] + \textcolor{brown}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{4} & \frac{18}{27} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{3}{4} & 0 & \frac{27}{27} & \frac{4}{4} \\ \frac{3}{1} & \frac{6}{0} & \frac{4}{0} & 0 & \frac{27}{-4} & \frac{4}{0} \\ \frac{1}{18} & \frac{0}{27} & \frac{0}{27} & -\frac{4}{0} & 0 & -\frac{2}{21} \\ \frac{18}{3} & \frac{27}{4} & \frac{27}{4} & 0 & \frac{133}{21} & \frac{21}{2} \end{pmatrix}_{b_0}$

[47] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 332, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{0} & \frac{11}{21} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{6}{4} & \frac{0}{0} & \frac{21}{21} & \frac{4}{4} \\ \frac{3}{1} & \frac{6}{0} & \frac{4}{0} & \frac{0}{-4} & \frac{21}{0} & \frac{4}{-2} \\ \frac{11}{3} & \frac{21}{4} & \frac{21}{4} & \frac{0}{-2} & \frac{76}{16} & \frac{16}{2} \end{pmatrix}_{b_0}$$

[48] $1^{113} 1^{12} 3 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{s}o_7 \cdot \textcolor{teal}{s}u_2] + \textcolor{brown}{s}o_8 + \textcolor{red}{s}u_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{0} & \frac{11}{17} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{6}{4} & \frac{0}{0} & \frac{17}{17} & \frac{4}{4} \\ \frac{3}{3} & \frac{6}{0} & \frac{4}{0} & \frac{0}{-4} & \frac{0}{0} & \frac{4}{-2} \\ \frac{11}{3} & \frac{17}{4} & \frac{17}{4} & \frac{0}{-2} & \frac{49}{13} & \frac{13}{2} \end{pmatrix}_{b_0}$

[49] $1^{14} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 21 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$

[50] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 384, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{113}3^{11}3^{112} + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{5}o_7 \cdot \textcolor{green}{5}u_2] + \textcolor{brown}{5}o_8 + \textcolor{red}{5}u_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$

[52] $1^{113} 3^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 404, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{0} & \frac{18}{32} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{3}{4} & \frac{0}{0} & \frac{32}{32} & \frac{4}{-2} \\ \frac{1}{1} & \frac{0}{0} & \frac{0}{-4} & \frac{0}{0} & \frac{0}{25} & \frac{-2}{25} \\ \frac{18}{3} & \frac{32}{4} & \frac{32}{4} & \frac{0}{-2} & \frac{193}{25} & \frac{25}{2} \end{pmatrix}_{b_0}$$

[53] $1^{14} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 428, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$

[54] $1^{113}3^{11}3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$

[55] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\left(\begin{array}{cccccc} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{array} \right)_{b_0}$

[56] $1^{11}3^{12}3^11^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 32 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$

[57] $1^{11}1^{14}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 472, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 24 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$

[58] $1^{11} 1^4 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 508, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 21 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$

[59] $1^{11}3^{12}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 512, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$

[60] $1^{11} 1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$

$$[71] \ 1^{114} 3^{113} 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 23 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{114} 3^{113} 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{114} 3^{113} 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 876, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{114} 3^{113} 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 904, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{114} 3^{113} 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [76] \ 2^1 3^{141} 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -288, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 1 & 5 & 3 \\ 3 & 6 & 6 & 5 & 3 & 5 \\ 3 & 6 & 6 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

5.97 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 15 & 8 & 6 \\ 16 & 25 & 23 & 12 & 9 \\ 15 & 23 & 20 & 11 & 8 \\ 8 & 12 & 11 & 4 & 4 \\ 6 & 9 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 21^{11} 3^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 17 & 9 & 7 \\ 20 & 25 & 23 & 12 & 9 \\ 17 & 23 & 20 & 11 & 8 \\ 9 & 12 & 11 & 4 & 4 \\ 7 & 9 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 5 \\ 7 & 8 & 6 & 5 & 6 \\ 5 & 6 & 0 & 3 & 2 \\ 4 & 5 & 3 & 1 & 3 \\ 5 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^1 41^{11} 23^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 1 & 3 \\ 6 & 13 & 15 & 0 & 0 & 7 \\ 6 & 15 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 7 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 41^{11} 23^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 6 & 2 & 0 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 3 & 6 & 1 & 4 & 2 & 0 & 4 \\ 1 & 2 & 0 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

5.98 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^1 41^{11} 23^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -560, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 1 & 5 & 3 \\ 6 & 13 & 15 & 0 & 0 & 12 & 7 \\ 6 & 15 & 13 & 0 & 0 & 12 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 12 & 0 & 0 & 8 & 6 \\ 3 & 7 & 7 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 41^{11} 23^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -576, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 1 & 3 & 2 \\ 3 & 4 & 1 & 6 & 2 & 0 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 3 & 6 & 1 & 4 & 2 & 0 & 6 & 4 \\ 1 & 2 & 0 & 2 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 6 & 2 & 0 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

5.99 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^1 1^1 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

5.100 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [1] \quad 1^1 1^1 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 21 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & 4 & -2 & 2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^1 1^1 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 4 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 & 5 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 2 & 2 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 5 & 5 & 6 & 13 & 7 \\ 3 & 4 & 4 & -2 & 2 & 2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = 0, \ \det = -576, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 4 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 & 7 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 2 & 2 & 1 & 9 & 3 \\ 7 & 13 & 13 & 0 & 7 & 7 & 9 & 28 & 10 \\ 3 & 4 & 4 & -2 & 2 & 2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

5.102 NHC(-2)(-3)(-2) + so₈ + su₂→(-3) + su₂n3mix

$$T = 9 \quad [0] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -60, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 7 & 12 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 4 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -60, \ \det = -468, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 3 & 16 & 4 \\ 3 & 6 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 \\ 9 & 16 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 4 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -60, \ \det = -616, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 20 & 4 \\ 3 & 6 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 11 & 2 \\ 11 & 20 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 4 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

5.103 NHC(-2)(-3)(-2) + so₈ + su₂→(-3) + su₂n3mix + su₂

$$T = 10 \quad [0] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -30, \ \det = 344, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 12 & 12 & 0 & 7 & 25 & 8 & 9 \\ 2 & 4 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & -2 & 2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -30, \ \det = 720, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 & 2 \\ 9 & 16 & 16 & 0 & 9 & 44 & 11 & 12 \\ 2 & 4 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & -2 & 2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

5.104 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 8 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 11 & 5 \\ 7 & 8 & 6 & 14 & 6 \\ 5 & 6 & 0 & 8 & 2 \\ 11 & 14 & 8 & 21 & 9 \\ 5 & 6 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 17 & 0 & 55 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 17 & 0 & 55 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 11 & 4 \\ 3 & 6 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 11 & 0 & 20 & 8 \\ 3 & 4 & 4 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -87$, $\det = 732$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 29 & 4 \\ 3 & 6 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 29 & 29 & 0 & 148 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -87$, $\det = 876$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 29 & 4 \\ 3 & 6 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 29 & 0 & 148 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [22] $1^{111}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 8 & 5 & 0 & 23 & 9 \\ 2 & 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{112}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 8 & 0 & 56 & 14 \\ 2 & 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{113}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 \\ 3 & 5 & 5 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{111}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 5 & 7 & 0 & 18 & 5 \\ 3 & 7 & 5 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 18 & 0 & 48 & 14 \\ 3 & 5 & 5 & -2 & 14 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{113}3^121^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 16 & 11 & 0 & 103 & 19 \\ 2 & 2 & 2 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{111}3^121^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -220$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 5 & 7 & 0 & 23 & 5 \\ 3 & 7 & 5 & 0 & 23 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 23 & 23 & 0 & 78 & 18 \\ 3 & 5 & 5 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 5 & 7 & 0 & 28 & 5 \\ 3 & 7 & 5 & 0 & 28 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 28 & 28 & 0 & 116 & 22 \\ 3 & 5 & 5 & -2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [29] \quad 1^{141}1^{11}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 5 & 2 \\ 3 & 4 & 1 & 6 & 2 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 & 1 & 0 \\ 3 & 6 & 1 & 4 & 2 & 6 & 4 \\ 1 & 2 & 0 & 2 & -2 & 2 & 0 \\ 5 & 6 & 1 & 6 & 2 & 5 & 5 \\ 2 & 4 & 0 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{141}1^{11}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 6 & 3 \\ 6 & 13 & 15 & 0 & 12 & 7 \\ 6 & 15 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 12 & 0 & 8 & 6 \\ 3 & 7 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{141}1^{11}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 4 & 2 \\ 3 & 4 & 1 & 6 & 2 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 3 & 6 & 1 & 4 & 2 & 6 & 4 \\ 1 & 2 & 0 & 2 & -2 & 2 & 0 \\ 4 & 6 & 0 & 6 & 2 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

5.105 NHC(-2)(-3)(-2) + so₈ + su₂n3mix + su₂

$$T = 10 \quad [1] \quad 1^{1112}3^{11}21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 8 & 2 \\ 1 & 2 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 5 & 0 & 23 & 26 & 9 \\ 4 & 6 & 8 & 6 & 0 & 26 & 26 & 10 \\ 2 & 2 & 2 & 2 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{1111}3^{11}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 2 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 5 & 0 & 23 & 13 & 9 \\ 2 & 3 & 4 & 3 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & 2 & -2 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 & 6 \\ 2 & 5 & 5 & 0 & 6 & 2 & 4 \\ 3 & 5 & 5 & -2 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 5 & 7 & 0 & 8 & 20 & 5 \\ 3 & 7 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 24 & 6 \\ 9 & 20 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 5 & -2 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 5 & 7 & 0 & 8 & 10 & 5 \\ 3 & 7 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 12 & 6 \\ 5 & 10 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & 5 & -2 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{121}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 2 & 0 & 12 & 4 \\ 1 & 2 & 2 & 0 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & 0 \\ 6 & 8 & 12 & 8 & 0 & 56 & 20 \\ 2 & 3 & 4 & 3 & 0 & 20 & 5 \\ 2 & 2 & 2 & 2 & -2 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 5 & 7 & 0 & 8 & 15 & 5 \\ 3 & 7 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 18 & 6 \\ 7 & 15 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 5 & -2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{121}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 7 & 0 & 23 & 5 & 5 \\ 3 & 7 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 23 & 23 & 0 & 78 & 17 & 18 \\ 2 & 5 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 5 & -2 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

5.106 NHC(-2)(-3)(-2) + so₈ + su₂n₃mix + su₂ + su₂

$$T = 11 \quad [0] \ 1^{11}3^{121}21^{1311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 7 & 0 & 23 & 5 & 5 \\ 3 & 7 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 23 & 23 & 0 & 78 & 17 & 18 \\ 2 & 5 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 5 & -2 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{121}21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 & 10 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 & 10 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 & 12 & 6 \\ 2 & 5 & 5 & 0 & 6 & 2 & 7 & 4 \\ 5 & 10 & 10 & 0 & 12 & 7 & 14 & 8 \\ 3 & 5 & 5 & -2 & 6 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{121}21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 & 20 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 & 20 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 & 24 & 6 \\ 2 & 5 & 5 & 0 & 6 & 2 & 15 & 4 \\ 9 & 20 & 20 & 0 & 24 & 15 & 58 & 16 \\ 3 & 5 & 5 & -2 & 6 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{121}21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 & 15 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 & 15 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 & 18 & 6 \\ 2 & 5 & 5 & 0 & 6 & 2 & 11 & 4 \\ 7 & 15 & 15 & 0 & 18 & 11 & 32 & 12 \\ 3 & 5 & 5 & -2 & 6 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

5.107 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 7 & 9 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 3 & 4 \\ 3 & 6 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 20 & 3 & 4 \\ 3 & 6 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 20 & 0 & 69 & 11 & 15 \\ 2 & 3 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

5.108 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 3 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 7 & 8 & 9 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 4 & -2 & 9 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 720, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 3 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 4 & -2 & 12 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

5.109 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix + su2n3mix

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 7 & 4 \\ 3 & 6 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 8 & 0 & 12 & 10 & 6 \\ 4 & 7 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & 4 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 7 & 12 & 4 \\ 3 & 6 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 7 & 0 & 7 & 15 & 5 \\ 7 & 12 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 4 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}123\overline{11}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 7 & 16 & 4 \\ 3 & 6 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 7 & 0 & 7 & 20 & 5 \\ 9 & 16 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 4 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

5.110 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 1 & 3 \\ 4 & 9 & 7 & 0 & 0 & 5 \\ 3 & 7 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 15 & 13 & 6 \\ 15 & 20 & 22 & 17 & 8 \\ 15 & 22 & 20 & 17 & 8 \\ 13 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 5 \\ 7 & 8 & 6 & 0 & 6 \\ 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 4 & 8 & 5 \\ 7 & 8 & 6 & 9 & 6 \\ 4 & 6 & 0 & 5 & 2 \\ 8 & 9 & 5 & 7 & 5 \\ 5 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 4 & 7 & 5 \\ 7 & 8 & 6 & 9 & 6 \\ 4 & 6 & 0 & 5 & 2 \\ 7 & 9 & 5 & 7 & 5 \\ 5 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 \\ 12 & 25 & 27 & 0 & 9 \\ 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 9 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 5 & 7 & 5 \\ 7 & 8 & 6 & 9 & 6 \\ 5 & 6 & 0 & 5 & 2 \\ 7 & 9 & 5 & 7 & 5 \\ 5 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 15 & 1 & 6 \\ 15 & 20 & 22 & 0 & 8 \\ 15 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 12 & 0 & 24 & 8 \\ 3 & 4 & 4 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 15 & 12 & 6 \\ 15 & 20 & 22 & 17 & 8 \\ 15 & 22 & 20 & 17 & 8 \\ 12 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^1 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 5 \\ 11 & 20 & 22 & 17 & 8 \\ 11 & 22 & 20 & 17 & 8 \\ 9 & 17 & 17 & 12 & 6 \\ 5 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 8 & 0 & 9 & 5 \\ 3 & 4 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{112} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 8 & 0 & 9 & 5 \\ 3 & 4 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 21^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 20 & 12 & 7 \\ 20 & 25 & 27 & 16 & 9 \\ 20 & 27 & 25 & 16 & 9 \\ 12 & 16 & 16 & 9 & 5 \\ 7 & 9 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{113} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 24 & 8 \\ 3 & 4 & 4 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 21^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 16 & 12 & 6 \\ 16 & 20 & 22 & 17 & 8 \\ 16 & 22 & 20 & 17 & 8 \\ 12 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 428, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [29] \quad 1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 15 & 10 & 7 \\ 12 & 15 & 13 & 10 & 7 \\ 8 & 10 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 21^{11} 4 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 15 & 10 & 7 \\ 12 & 15 & 13 & 10 & 7 \\ 8 & 10 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[34] \quad 21^{11} 4 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 14 & 9 & 7 \\ 14 & 13 & 15 & 10 & 7 \\ 14 & 15 & 13 & 10 & 7 \\ 9 & 10 & 10 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 7 & 5 \\ 9 & 13 & 15 & 10 & 7 \\ 9 & 15 & 13 & 10 & 7 \\ 7 & 10 & 10 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 21^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 10 & 7 & 5 \\ 10 & 13 & 15 & 10 & 7 \\ 10 & 15 & 13 & 10 & 7 \\ 7 & 10 & 10 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

5.111 NHC(-2)(-3)(-2) + so₈ + su₃ + e₆

$$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

5.112 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

5.113 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 1 & 5 \\ 12 & 25 & 23 & 0 & 0 & 9 \\ 11 & 23 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 7 & 4 & 7 & 1 & 5 \\ 7 & 8 & 6 & 9 & 0 & 6 \\ 4 & 6 & 0 & 5 & 0 & 2 \\ 7 & 9 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 1 & 5 \\ 11 & 20 & 22 & 0 & 0 & 8 \\ 11 & 22 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

5.114 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 17 & 1 & 7 \\ 20 & 25 & 23 & 0 & 9 \\ 17 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 17 & 1 & 7 \\ 17 & 20 & 22 & 0 & 8 \\ 17 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{\overline{211}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 10 & 1 & 5 \\ 10 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

5.115 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$

- $T = 7$ [0] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -10$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 5 \\ 7 & 8 & 6 & 6 \\ 4 & 6 & 2 & 4 \\ 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$
- [1] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 7 & 5 \\ 7 & 8 & 9 & 6 \\ 7 & 9 & 8 & 6 \\ 5 & 6 & 6 & 4 \end{pmatrix}_{b_0}$
- [2] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 5 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 2 & 4 \\ 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$
- [3] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 11 & 5 \\ 7 & 8 & 15 & 6 \\ 11 & 15 & 22 & 10 \\ 5 & 6 & 10 & 4 \end{pmatrix}_{b_0}$
- [4] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.64 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 5 & 5 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 2 & 4 \\ 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$
- [7] $21^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 15 & 4 \\ 5 & 4 & 6 & 14 & 4 \\ 5 & 6 & 4 & 14 & 4 \\ 15 & 14 & 14 & 37 & 11 \\ 4 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$
- [6] $21^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 3 & 4 \\ 5 & 4 & 6 & 4 & 4 \\ 5 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 1 & 3 \\ 4 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [5] $21^{12} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 4 & 6 & 23 & 4 \\ 4 & 6 & 4 & 23 & 4 \\ 15 & 23 & 23 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -34$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 10 & 5 \\ 7 & 8 & 12 & 6 \\ 10 & 12 & 14 & 8 \\ 5 & 6 & 8 & 4 \end{pmatrix}_{b_0}$
- [8] $1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -34$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 11 & 5 \\ 7 & 8 & 12 & 6 \\ 11 & 12 & 14 & 8 \\ 5 & 6 & 8 & 4 \end{pmatrix}_{b_0}$
- [10] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -45$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 12 & 5 \\ 7 & 8 & 15 & 6 \\ 12 & 15 & 22 & 10 \\ 5 & 6 & 10 & 4 \end{pmatrix}_{b_0}$
- [11] $1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -49$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 14 & 5 \\ 7 & 8 & 15 & 6 \\ 14 & 15 & 22 & 10 \\ 5 & 6 & 10 & 4 \end{pmatrix}_{b_0}$$

[12] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 15 & 5 \\ 7 & 8 & 18 & 6 \\ 15 & 18 & 34 & 12 \\ 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$

[13] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 13 & 5 \\ 7 & 8 & 15 & 6 \\ 13 & 15 & 22 & 10 \\ 5 & 6 & 10 & 4 \end{pmatrix}_{b_0}$

[15] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 16 & 5 \\ 7 & 8 & 18 & 6 \\ 16 & 18 & 34 & 12 \\ 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$

[14] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 19 & 5 \\ 7 & 8 & 24 & 6 \\ 19 & 24 & 62 & 16 \\ 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[16] $21^{11} 31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 9 & 4 \\ 5 & 4 & 6 & 9 & 4 \\ 5 & 6 & 4 & 9 & 4 \\ 9 & 9 & 9 & 13 & 7 \\ 4 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[17] $21^{11} 31^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 18 & 4 \\ 5 & 4 & 6 & 21 & 4 \\ 5 & 6 & 4 & 21 & 4 \\ 18 & 21 & 21 & 76 & 16 \\ 4 & 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[18] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 20 & 5 \\ 7 & 8 & 24 & 6 \\ 20 & 24 & 62 & 16 \\ 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[19] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 20 & 5 \\ 7 & 8 & 24 & 6 \\ 20 & 24 & 62 & 16 \\ 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[20] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 17 & 5 \\ 7 & 8 & 21 & 6 \\ 17 & 21 & 44 & 14 \\ 5 & 6 & 14 & 4 \end{pmatrix}_{b_0}$

[21] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 21 & 5 \\ 7 & 8 & 24 & 6 \\ 21 & 24 & 62 & 16 \\ 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[25] $21^{13} 31^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 4 & 6 & 32 & 4 \\ 4 & 6 & 4 & 32 & 4 \\ 21 & 32 & 32 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$

[26] $21^{11} 31^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

- [22] $21^{111}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 4 & 6 & 4 & 4 \\ 4 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$
- [23] $21^{11}31^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 4 \\ 5 & 4 & 6 & 13 & 4 \\ 5 & 6 & 4 & 13 & 4 \\ 12 & 13 & 13 & 28 & 10 \\ 4 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [24] $21^{13}31^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 24 & 4 \\ 5 & 4 & 6 & 23 & 4 \\ 5 & 6 & 4 & 23 & 4 \\ 24 & 23 & 23 & 100 & 18 \\ 4 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$
- [27] $1^{12}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 19 & 5 \\ 7 & 8 & 21 & 6 \\ 19 & 21 & 44 & 14 \\ 5 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$
- [28] $1^{12}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 18 & 5 \\ 7 & 8 & 21 & 6 \\ 18 & 21 & 44 & 14 \\ 5 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$
- [29] $21^{111}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 4 & 6 & 9 & 4 \\ 4 & 6 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [30] $21^{112}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 4 & 6 & 8 & 4 \\ 4 & 6 & 4 & 8 & 4 \\ 6 & 8 & 8 & 10 & 6 \\ 3 & 4 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$
- [31] $1^{12}31^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 24 & 5 \\ 7 & 8 & 30 & 6 \\ 24 & 30 & 94 & 20 \\ 5 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$
- [32] $1^{13}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 22 & 5 \\ 7 & 8 & 27 & 6 \\ 22 & 27 & 74 & 18 \\ 5 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$
- [33] $21^{123}1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 18 & 4 \\ 5 & 4 & 6 & 18 & 4 \\ 5 & 6 & 4 & 18 & 4 \\ 18 & 18 & 18 & 58 & 14 \\ 4 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$
- [34] $1^{13}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 23 & 5 \\ 7 & 8 & 27 & 6 \\ 23 & 27 & 74 & 18 \\ 5 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$
- [35] $1^{12}31^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 25 & 5 \\ 7 & 8 & 30 & 6 \\ 25 & 30 & 94 & 20 \\ 5 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[36] $1^{12} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -157, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 31 & 5 \\ 7 & 8 & 39 & 6 \\ 31 & 39 & 164 & 26 \\ 5 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[37] $21^{113} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 4 & 6 & 12 & 4 \\ 4 & 6 & 4 & 12 & 4 \\ 9 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{12} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 26 & 5 \\ 7 & 8 & 30 & 6 \\ 26 & 30 & 94 & 20 \\ 5 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[39] $21^{112} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 4 & 6 & 18 & 4 \\ 4 & 6 & 4 & 18 & 4 \\ 12 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{14} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 27 & 5 \\ 7 & 8 & 33 & 6 \\ 27 & 33 & 112 & 22 \\ 5 & 6 & 22 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{12} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 32 & 5 \\ 7 & 8 & 39 & 6 \\ 32 & 39 & 164 & 26 \\ 5 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{13} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 29 & 5 \\ 7 & 8 & 36 & 6 \\ 29 & 36 & 134 & 24 \\ 5 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[46] $21^{112} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 4 & 6 & 13 & 4 \\ 4 & 6 & 4 & 13 & 4 \\ 9 & 13 & 13 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[45] $21^{123} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 21 & 4 \\ 5 & 4 & 6 & 22 & 4 \\ 5 & 6 & 4 & 22 & 4 \\ 21 & 22 & 22 & 85 & 17 \\ 4 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[44] $21^{143} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 33 & 4 \\ 5 & 4 & 6 & 32 & 4 \\ 5 & 6 & 4 & 32 & 4 \\ 33 & 32 & 32 & 193 & 25 \\ 4 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[43] $21^{114} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 4 & 6 & 16 & 4 \\ 4 & 6 & 4 & 16 & 4 \\ 12 & 16 & 16 & 46 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{12} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -225, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 33 & 5 \\ 7 & 8 & 39 & 6 \\ 33 & 39 & 164 & 26 \\ 5 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 39 & 5 \\ 7 & 8 & 48 & 6 \\ 39 & 48 & 254 & 32 \\ 5 & 6 & 32 & 4 \end{pmatrix}_{b_0}$$

[49] $21^{13} 31^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 27 & 4 \\ 5 & 4 & 6 & 27 & 4 \\ 5 & 6 & 4 & 27 & 4 \\ 27 & 27 & 27 & 133 & 21 \\ 4 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 36 & 5 \\ 7 & 8 & 45 & 6 \\ 36 & 45 & 214 & 30 \\ 5 & 6 & 30 & 4 \end{pmatrix}_{b_0}$$

[51] $21^{12} 31^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 24 & 4 \\ 5 & 4 & 6 & 26 & 4 \\ 5 & 6 & 4 & 26 & 4 \\ 24 & 26 & 26 & 118 & 20 \\ 4 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 40 & 5 \\ 7 & 8 & 48 & 6 \\ 40 & 48 & 254 & 32 \\ 5 & 6 & 32 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 34 & 5 \\ 7 & 8 & 42 & 6 \\ 34 & 42 & 182 & 28 \\ 5 & 6 & 28 & 4 \end{pmatrix}_{b_0}$$

[54] $21^{15} 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 4 & 6 & 20 & 4 \\ 4 & 6 & 4 & 20 & 4 \\ 15 & 20 & 20 & 73 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[55] $21^{13} 31^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 4 & 6 & 27 & 4 \\ 4 & 6 & 4 & 27 & 4 \\ 18 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 37 & 5 \\ 7 & 8 & 45 & 6 \\ 37 & 45 & 214 & 30 \\ 5 & 6 & 30 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 43 & 5 \\ 7 & 8 & 54 & 6 \\ 43 & 54 & 314 & 36 \\ 5 & 6 & 36 & 4 \end{pmatrix}_{b_0}$$

[58] $21^{13} 31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 4 & 6 & 17 & 4 \\ 4 & 6 & 4 & 17 & 4 \\ 12 & 17 & 17 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 44 & 5 \\ 7 & 8 & 54 & 6 \\ 44 & 54 & 314 & 36 \\ 5 & 6 & 36 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -373, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 41 & 5 \\ 7 & 8 & 51 & 6 \\ 41 & 51 & 272 & 34 \\ 5 & 6 & 34 & 4 \end{pmatrix}_{b_0}$$

[61] $21^{116} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 4 & 6 & 24 & 4 \\ 4 & 6 & 4 & 24 & 4 \\ 18 & 24 & 24 & 106 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[63] $21^{113} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 4 & 6 & 22 & 4 \\ 4 & 6 & 4 & 22 & 4 \\ 15 & 22 & 22 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[62] $21^{131} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 30 & 4 \\ 5 & 4 & 6 & 31 & 4 \\ 5 & 6 & 4 & 31 & 4 \\ 30 & 31 & 31 & 172 & 24 \\ 4 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[64] $21^{143} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 36 & 4 \\ 5 & 4 & 6 & 36 & 4 \\ 5 & 6 & 4 & 36 & 4 \\ 36 & 36 & 36 & 238 & 28 \\ 4 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 48 & 5 \\ 7 & 8 & 60 & 6 \\ 48 & 60 & 382 & 40 \\ 5 & 6 & 40 & 4 \end{pmatrix}_{b_0}$$

[67] $21^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 4 & 6 & 21 & 4 \\ 4 & 6 & 4 & 21 & 4 \\ 15 & 21 & 21 & 76 & 16 \\ 3 & 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[66] $21^{114} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 24 & 3 \\ 4 & 4 & 6 & 36 & 4 \\ 4 & 6 & 4 & 36 & 4 \\ 24 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[68] $21^{117} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 4 & 6 & 28 & 4 \\ 4 & 6 & 4 & 28 & 4 \\ 21 & 28 & 28 & 145 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[70] $21^{114} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 4 & 6 & 26 & 4 \\ 4 & 6 & 4 & 26 & 4 \\ 18 & 26 & 26 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[69] $21^{114} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 4 & 6 & 31 & 4 \\ 4 & 6 & 4 & 31 & 4 \\ 21 & 31 & 31 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$T = 8$ [72] $1^1 3^1 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 2 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 2 2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 & 5 \\ 3 & 5 & 5 & 4 & 4 & 4 \\ 3 & 5 & 5 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{11} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 6 & 5 & 0 & 10 & 33 & 11 \\ 2 & 2 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{12} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 4 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{13} 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 6 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{14} 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 16 & 3 \\ 8 & 12 & 0 & 16 & 110 & 20 \\ 2 & 2 & -1 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{11} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 4 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[83] $21^{11} 3 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 4 \\ 5 & 5 & 7 & 15 & 5 \\ 5 & 7 & 5 & 15 & 5 \\ 12 & 15 & 15 & 32 & 12 \\ 4 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[84] $1^{11} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 6 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{12} 3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 8 & 8 & 0 & 14 & 70 & 16 \\ 2 & 2 & -1 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{13} 3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 15 & 3 \\ 8 & 10 & 0 & 15 & 88 & 18 \\ 2 & 2 & -1 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[89] $21^{12} 3 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 24 & 4 \\ 5 & 5 & 7 & 30 & 5 \\ 5 & 7 & 5 & 30 & 5 \\ 24 & 30 & 30 & 134 & 24 \\ 4 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[90] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[91] $2 2 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 2 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 2 & 0 & 3 & 2 \\ 3 & 3 & 4 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

- [92] $1^{12} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$
- [93] $1^{12} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$
- [94] $1^{11} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 62$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$
- [95] $2 \overset{11}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -60$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [96] $1^{11} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 70$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$
- [97] $1^{13} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 75$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$
- [98] $2 \overset{111}{2} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 \\ 3 & 5 & 5 & 2 & 4 \\ 3 & 5 & 5 & 4 & 4 \end{pmatrix}_{b_0}$
- [100] $2 \overset{111}{2} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 5 & 7 & 10 & 5 \\ 4 & 7 & 5 & 10 & 5 \\ 6 & 10 & 10 & 14 & 8 \\ 3 & 5 & 5 & 8 & 4 \end{pmatrix}_{b_0}$
- [101] $2 \overset{112}{2} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 2 & 0 & 6 & 2 \\ 6 & 6 & 8 & 6 & 26 & 10 \\ 2 & 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$
- [99] $2 \overset{112}{2} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 5 & 7 & 10 & 5 \\ 4 & 7 & 5 & 10 & 5 \\ 6 & 10 & 10 & 14 & 8 \\ 3 & 5 & 5 & 8 & 4 \end{pmatrix}_{b_0}$
- [102] $1^{12} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{12} 3^{12} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[104] \ 2 \ 2 \ 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 5 & 7 & 15 & 5 \\ 4 & 7 & 5 & 15 & 5 \\ 9 & 15 & 15 & 34 & 12 \\ 3 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{12} 3^{14} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 4 & 2 \\ 2 & 3 & 3 & 1 & 6 & 3 \\ 4 & 6 & 4 & 6 & 10 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 0 & 3 & 10 & 2 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 10 & 15 & 10 & 15 & 73 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[109] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 0 & 3 & 12 & 2 \\ 2 & 3 & 3 & 1 & 18 & 3 \\ 12 & 18 & 12 & 18 & 106 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[110] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 3 & 21 & 3 \\ 2 & 3 & 0 & 3 & 14 & 2 \\ 2 & 3 & 3 & 1 & 21 & 3 \\ 14 & 21 & 14 & 21 & 145 & 21 \\ 2 & 3 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 0 & 3 & 8 & 2 \\ 2 & 3 & 3 & 1 & 12 & 3 \\ 8 & 12 & 8 & 12 & 46 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 9 & 3 \\ 2 & 3 & 0 & 3 & 6 & 2 \\ 2 & 3 & 3 & 1 & 9 & 3 \\ 6 & 9 & 6 & 9 & 25 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 2 \overset{1}{\underset{111}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[114] \quad 2 \overset{1}{\underset{114}{2}} 31^{\textcolor{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 62 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[115] \quad 2 \overset{1}{\underset{112}{2}} 31^{\textcolor{red}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 62 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[116] \quad 1 \overset{1}{\underset{13}{13}} 3 21^{\textcolor{red}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[117] \quad 1 \overset{1}{\underset{13}{12}} 3 21^{\textcolor{red}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[118] \quad 1 \overset{1}{\underset{14}{13}} 3 21^{\textcolor{red}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[119] \quad 1 \overset{1}{\underset{13}{12}} 3 21^{\textcolor{red}{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[120] \quad 1 \overset{1}{\underset{14}{12}} 3 21^{\textcolor{red}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[121] \quad 2 \overset{1}{\underset{13}{2}} 2 31^{\textcolor{red}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 2 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 2 & 0 & 9 & 2 \\ 9 & 9 & 12 & 9 & 61 & 15 \\ 2 & 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[122] \quad 2 \overset{1}{\underset{115}{2}} 2 31^{\textcolor{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 98 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[123] \ 2 \overset{1^{11}}{3} \ 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -60, \ \det = 192, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 3 & 5 & 2 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 4 & 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1 \overset{1^{14}}{4} \ 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 196, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 5 & 7 & 15 & 5 \\ 4 & 7 & 5 & 15 & 5 \\ 9 & 15 & 15 & 32 & 12 \\ 3 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[126] \ 1 \overset{1^{13}}{3} \ 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 203, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 1 \overset{1^{14}}{2} \ 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 206, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[128] \ 2 \ 2 \ 31^{16} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 5 & 7 & 30 & 5 \\ 4 & 7 & 5 & 30 & 5 \\ 18 & 30 & 30 & 142 & 24 \\ 3 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[129] \ 1 \overset{1^{13}}{3} \ 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 234, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 1 \overset{1^{14}}{3} \ 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 237, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[131] \ 2 \ 2 \ 31^{17} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 276, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 5 & 7 & 35 & 5 \\ 4 & 7 & 5 & 35 & 5 \\ 21 & 35 & 35 & 194 & 28 \\ 3 & 5 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[132] \ 1 \overset{1^{14}}{4} \ 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 283, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[133] $1^{13} 3^{14} 21^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[134] $2^2 2 31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 2 & 2 & 0 & 2 & 16 & 2 \\ 1 & 2 & 2 & 0 & 12 & 2 \\ 12 & 12 & 16 & 12 & 110 & 20 \\ 2 & 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[135] $2^2 2 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 58 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[136] $2^2 3^{11} 2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 0 & 3 & 7 & 2 \\ 2 & 3 & 3 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[137] $2^2 2 31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 94 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[138] $1^{14} 3^{14} 21^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[139] $2^2 2 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 92 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[140] $2^2 3^{11} 2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 11 & 3 \\ 2 & 3 & 0 & 3 & 9 & 2 \\ 2 & 3 & 3 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 40 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[141] $2^2 2 31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 5 & 7 & 30 & 5 \\ 4 & 7 & 5 & 30 & 5 \\ 18 & 30 & 30 & 134 & 24 \\ 3 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[142] $2^2 3^{11} 2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 14 & 3 \\ 2 & 3 & 0 & 3 & 11 & 2 \\ 2 & 3 & 3 & 1 & 14 & 3 \\ 10 & 14 & 11 & 14 & 65 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [143] \ 1^{11} 2^1 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[144] $1^{13}2^13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[145] $1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}}3^{\textcolor{blue}{1}}221^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 3 & 0 & 2 & 4 \end{pmatrix} b_0$$

[146] $1^{12}2^13^{11}21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_b$$

[147] $1^{14}2^13^{12}21^{12} + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_b$$

[148] $1^{11}1^13221^{11} + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{5}07 \cdot \textcolor{blue}{5}u_2] + \textcolor{brown}{5}u_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[149] $1^{11} 3^{221} + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{50}_7 \cdot \textcolor{green}{5u}_2] + \textcolor{blue}{5u}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 1 & 9 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 4 & 9 & 0 & 20 & 10 \\ 2 & 2 & 3 & 0 & 10 & 4 \end{pmatrix}$$

[150] $1^{\textcolor{brown}{1}\textcolor{blue}{1}} 3^{221\textcolor{blue}{1}\textcolor{brown}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[151] $1^{11} 1^{12} 3^{22} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[152] $1^{11} 1^{12} 3^{221} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 1 & 9 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 4 & 9 & 0 & 20 & 10 \\ 2 & 2 & 3 & 0 & 10 & 4 \end{pmatrix}$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 1 & 19 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 19 & 0 & 104 & 22 \\ 2 & 2 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{13} 3 \ 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 1 & 1 & 15 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 9 & 15 & 0 & 70 & 18 \\ 2 & 2 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{14} 3 \ 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 11 & 16 & 0 & 90 & 20 \\ 2 & 2 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{13} 3 \ 221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 1 & 19 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 10 & 19 & 0 & 104 & 22 \\ 2 & 2 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{14} 3 \ 221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 1 & 1 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 12 & 12 & 20 & 0 & 126 & 24 \\ 2 & 2 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[168] \ 41^{11} 1^{14} 232 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -59, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 4 & 2 \\ 4 & 10 & 5 & 12 & 12 & 6 \\ 2 & 5 & 0 & 5 & 4 & 2 \\ 4 & 12 & 5 & 10 & 12 & 6 \\ 4 & 12 & 4 & 12 & 10 & 6 \\ 2 & 6 & 2 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [175] \ 2 \ 3 \ 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 \\ 3 & 6 & 6 & 3 & 5 \\ 3 & 6 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[169] \ 2 \ 3 \ 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 6 & 8 & 12 & 6 \\ 4 & 8 & 6 & 12 & 6 \\ 6 & 12 & 12 & 18 & 10 \\ 3 & 6 & 6 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[170] \ 2 \ 3 \ 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 6 & 8 & 18 & 6 \\ 4 & 8 & 6 & 18 & 6 \\ 9 & 18 & 18 & 43 & 15 \\ 3 & 6 & 6 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[171] \ 2 \ 3 \ 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 6 & 8 & 24 & 6 \\ 4 & 8 & 6 & 24 & 6 \\ 12 & 24 & 24 & 78 & 20 \\ 3 & 6 & 6 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[172] \ 2 \ 3 \ 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 6 & 8 & 30 & 6 \\ 4 & 8 & 6 & 30 & 6 \\ 15 & 30 & 30 & 123 & 25 \\ 3 & 6 & 6 & 25 & 5 \end{pmatrix}_{b_0}$$

[173] $2 \ 3 \ 21^{141} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 6 & 8 & 36 & 6 \\ 4 & 8 & 6 & 36 & 6 \\ 18 & 36 & 36 & 178 & 30 \\ 3 & 6 & 6 & 30 & 5 \end{pmatrix}_{b_0}$$

[174] $2 \ 3 \ 21^{141} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 6 & 8 & 42 & 6 \\ 4 & 8 & 6 & 42 & 6 \\ 21 & 42 & 42 & 243 & 35 \\ 3 & 6 & 6 & 35 & 5 \end{pmatrix}_{b_0}$$

[176] $2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 4 & 12 & 3 \\ 2 & 4 & 4 & 2 & 16 & 4 \\ 8 & 16 & 12 & 16 & 62 & 16 \\ 2 & 4 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

[177] $2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 2 & 4 & 4 \\ 2 & 4 & 3 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[178] $2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 4 & 9 & 3 \\ 2 & 4 & 4 & 2 & 12 & 4 \\ 6 & 12 & 9 & 12 & 34 & 12 \\ 2 & 4 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

[179] $2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 4 & 6 & 3 \\ 2 & 4 & 4 & 2 & 8 & 4 \\ 4 & 8 & 6 & 8 & 14 & 8 \\ 2 & 4 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [180] $1^{11} 31 \ 5 \ 131^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{11} 31 \ 5 \ 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[182] $1^{13} 31 \ 5 \ 131^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[183] $1^{13} 31 \ 5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{brown}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{14} 31^{12} 5^{13} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{12} 31^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [186] \ 1^{11} 31^{11} 5^{11} 13221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[192] \ 2^{1^{11} 1^{11} 4^{14} 141} 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 2 & 5 & 4 & 4 \\ 2 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 4 & 5 & 3 & 5 \\ 2 & 5 & 4 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[191] \ 2^{1^{12} 1^{11} 4^{14} 141} 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 2 & 5 & 8 & 4 \\ 2 & 5 & 5 & 3 & 10 & 5 \\ 4 & 10 & 8 & 10 & 18 & 10 \\ 2 & 5 & 4 & 5 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[190] \ 2^{1^{13} 1^{11} 4^{14} 141} 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 3 & 5 & 5 & 15 & 5 \\ 2 & 5 & 2 & 5 & 12 & 4 \\ 2 & 5 & 5 & 3 & 15 & 5 \\ 6 & 15 & 12 & 15 & 43 & 15 \\ 2 & 5 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[189] \ 2^{1^{14} 1^{11} 4^{14} 141} 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 3 & 5 & 5 & 20 & 5 \\ 2 & 5 & 2 & 5 & 16 & 4 \\ 2 & 5 & 5 & 3 & 20 & 5 \\ 8 & 20 & 16 & 20 & 78 & 20 \\ 2 & 5 & 4 & 5 & 20 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [198] \ 1^{11} 6^{2315131} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$$

$$T = 16 \quad [202] \ 1^{11} 6^{14} 13161^{12} 2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 5.0$$

$$[201] \ 1^{11} 6^{14} 131612321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 5.0$$

$$T = 17 \quad [203] \ 1^{11} 8^{23151321} 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

5.116 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 7 & 5 & 5 \\ 7 & 8 & 9 & 6 & 6 \\ 7 & 9 & 8 & 5 & 6 \\ 5 & 6 & 5 & 0 & 2 \\ 5 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 4 & 5 & 5 \\ 7 & 8 & 6 & 6 & 6 \\ 4 & 6 & 2 & 4 & 4 \\ 5 & 6 & 4 & 0 & 2 \\ 5 & 6 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 11 & 5 & 5 \\ 7 & 8 & 15 & 6 & 6 \\ 11 & 15 & 22 & 9 & 10 \\ 5 & 6 & 9 & 0 & 2 \\ 5 & 6 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 5 & 5 \\ 7 & 8 & 6 & 6 & 6 \\ 5 & 6 & 2 & 4 & 4 \\ 5 & 6 & 4 & 0 & 2 \\ 5 & 6 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 19 & 5 & 5 \\ 7 & 8 & 24 & 6 & 6 \\ 19 & 24 & 62 & 14 & 16 \\ 5 & 6 & 14 & 0 & 2 \\ 5 & 6 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 12 & 5 & 5 \\ 7 & 8 & 15 & 6 & 6 \\ 12 & 15 & 22 & 9 & 10 \\ 5 & 6 & 9 & 0 & 2 \\ 5 & 6 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [6] \ 2 \ 3 \ 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -288, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 3 & 5 & 5 \\ 3 & 6 & 6 & 5 & 1 & 3 \\ 3 & 6 & 6 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

5.117 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 2 \ 3 \ 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 2 & 3 & 3 \\ 2 & 3 & 0 & 3 & 3 & 2 & 2 \\ 2 & 3 & 3 & 1 & 2 & 3 & 3 \\ 2 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 3 & 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 2 \ 3 \ 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 5 & 3 \\ 2 & 3 & 0 & 3 & 4 & 5 & 2 \\ 2 & 3 & 3 & 1 & 6 & 5 & 3 \\ 4 & 6 & 4 & 6 & 10 & 10 & 6 \\ 4 & 5 & 5 & 5 & 10 & 8 & 6 \\ 2 & 3 & 2 & 3 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[3] \quad & 2 \overset{1}{\underset{1^{1112}}{3}} 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 2 & 4 & 2 \\ 2 & 3 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 2 & 3 & 1 & 6 & 3 \\ 4 & 6 & 4 & 6 & 6 & 10 & 6 \\ 2 & 3 & 2 & 3 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[2] \quad & 2 \overset{1}{\underset{1^{1111}}{3}} 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 & 2 \\ 2 & 3 & 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
[4] \quad & 2 \overset{11}{\underset{1^{1111}}{3}} 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 3 & 2 & 5 & 2 \\ 2 & 3 & 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 3 & 1 & 5 & 3 \\ 4 & 5 & 5 & 5 & 8 & 6 & 6 \\ 2 & 3 & 2 & 3 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[5] \quad & 2 \overset{11}{\underset{1^{1112}}{3}} 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 3 & 8 & 3 \\ 2 & 3 & 0 & 3 & 2 & 7 & 2 \\ 2 & 3 & 3 & 1 & 3 & 8 & 3 \\ 2 & 3 & 2 & 3 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 8 & 21 & 9 \\ 2 & 3 & 2 & 3 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
T = 10 \quad & [9] \quad 1 \overset{122}{\underset{1^{134}}{2}} 3 \overset{112}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 0 & 10 & 14 & 3 \\ 1 & 2 & 0 & 1 & 6 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 52 & 12 \\ 8 & 14 & 10 & 0 & 52 & 74 & 18 \\ 2 & 3 & 2 & 0 & 12 & 18 & 4 \end{pmatrix}_{b_0} \\
[8] \quad & 1 \overset{121}{\underset{1^{132}}{2}} 3 \overset{111}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 7 & 3 \\ 1 & 2 & 0 & 1 & 6 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 26 & 12 \\ 4 & 7 & 5 & 0 & 26 & 17 & 9 \\ 2 & 3 & 2 & 0 & 12 & 9 & 4 \end{pmatrix}_{b_0} \\
[7] \quad & 1 \overset{112}{\underset{1^{14}}{2}} 3 \overset{112}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 0 & 4 & 14 & 3 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 24 & 6 \\ 8 & 14 & 10 & 0 & 24 & 74 & 18 \\ 2 & 3 & 2 & 0 & 6 & 18 & 4 \end{pmatrix}_{b_0} \\
[6] \quad & 1 \overset{111}{\underset{1^{112}}{2}} 3 \overset{111}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 0 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 12 & 6 \\ 4 & 7 & 5 & 0 & 12 & 17 & 9 \\ 2 & 3 & 2 & 0 & 6 & 9 & 4 \end{pmatrix}_{b_0} \\
[10] \quad & 1 \overset{122}{\underset{1^{144}}{2}} 3 \overset{122}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 2 & 0 & 14 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 14 & 10 & 0 & 74 & 76 & 18 \\ 8 & 14 & 10 & 0 & 76 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{124} 2^1 3^{112} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 0 & 7 & 14 & 3 \\ 1 & 2 & 0 & 1 & 5 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 38 & 9 \\ 8 & 14 & 10 & 0 & 38 & 74 & 18 \\ 2 & 3 & 2 & 0 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{122} 2^1 3^{111} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 19 & 9 \\ 4 & 7 & 5 & 0 & 19 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [38] \ 2^1 3^{141} 21^{1167} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 21 & 3 \\ 4 & 6 & 8 & 36 & 42 & 6 \\ 4 & 8 & 6 & 36 & 42 & 6 \\ 18 & 36 & 36 & 178 & 210 & 30 \\ 21 & 42 & 42 & 210 & 243 & 35 \\ 3 & 6 & 6 & 30 & 35 & 5 \end{pmatrix}_{b_0}$$

$$[40] \ 2^1 3^{1136} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 18 & 3 \\ 4 & 6 & 8 & 18 & 36 & 6 \\ 4 & 8 & 6 & 18 & 36 & 6 \\ 9 & 18 & 18 & 43 & 90 & 15 \\ 18 & 36 & 36 & 90 & 178 & 30 \\ 3 & 6 & 6 & 15 & 30 & 5 \end{pmatrix}_{b_0}$$

$$[13] \ 2^1 3^{1133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 9 & 3 \\ 4 & 6 & 8 & 18 & 18 & 6 \\ 4 & 8 & 6 & 18 & 18 & 6 \\ 9 & 18 & 18 & 43 & 45 & 15 \\ 9 & 18 & 18 & 45 & 43 & 15 \\ 3 & 6 & 6 & 15 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[27] \ 2^1 3^{1141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 & 3 \\ 4 & 6 & 8 & 24 & 6 & 6 \\ 4 & 8 & 6 & 24 & 6 & 6 \\ 12 & 24 & 24 & 78 & 20 & 20 \\ 3 & 6 & 6 & 20 & 3 & 5 \\ 3 & 6 & 6 & 20 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[32] \ 2^1 3^{1151} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 & 3 \\ 4 & 6 & 8 & 30 & 6 & 6 \\ 4 & 8 & 6 & 30 & 6 & 6 \\ 15 & 30 & 30 & 123 & 25 & 25 \\ 3 & 6 & 6 & 25 & 3 & 5 \\ 3 & 6 & 6 & 25 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[31] \ 2^1 3^{1147} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 21 & 3 \\ 4 & 6 & 8 & 24 & 42 & 6 \\ 4 & 8 & 6 & 24 & 42 & 6 \\ 12 & 24 & 24 & 78 & 140 & 20 \\ 21 & 42 & 42 & 140 & 243 & 35 \\ 3 & 6 & 6 & 20 & 35 & 5 \end{pmatrix}_{b_0}$$

$$[34] \ 2^1 3^{1156} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.4$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 & 3 \\ 4 & 6 & 8 & 42 & 6 & 6 \\ 4 & 8 & 6 & 42 & 6 & 6 \\ 21 & 42 & 42 & 243 & 35 & 35 \\ 3 & 6 & 6 & 35 & 3 & 5 \\ 3 & 6 & 6 & 35 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[28] \ 2 \overset{1^{141}}{3} \overset{1^{141}}{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = 0, \ \det = -144, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 12 & 3 \\ 4 & 6 & 8 & 24 & 24 & 6 \\ 4 & 8 & 6 & 24 & 24 & 6 \\ 12 & 24 & 24 & 78 & 80 & 20 \\ 12 & 24 & 24 & 80 & 78 & 20 \\ 3 & 6 & 6 & 20 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[41] \ 2 \overset{1^{1111}}{3} \overset{1^{1111}}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = 0, \ \det = -256, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 2 & 4 & 4 \\ 2 & 4 & 3 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 & 2 \\ 2 & 4 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [42] \ 1 \overset{1^{11}}{1^{12}} 31 \overset{1^{11}}{5} 1321 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 6.5$$

$$[43] \ 1 \overset{1^{12}}{1^{14}} 31 \overset{1^{12}}{5} 131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 0 & 4 & 14 & 3 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 24 & 6 \\ 8 & 14 & 10 & 0 & 24 & 74 & 18 \\ 2 & 3 & 2 & 0 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[44] \ 1 \overset{1^{11}}{1^{12}} 31 \overset{1^{11}}{5} 131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 0 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 12 & 6 \\ 4 & 7 & 5 & 0 & 12 & 17 & 9 \\ 2 & 3 & 2 & 0 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [46] \ 1 \overset{1^{112}}{1^{112}} 3 \overset{1^{112}}{6} 131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 13, 1), \ c_{\text{ext}} = 6.8$$

$$[45] \ 1 \overset{1^{111}}{1^{11}} 3 \overset{1^{111}}{6} 131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 13, 1), \ c_{\text{ext}} = 6.0$$

5.118 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \ 1 \overset{1^{11}}{1^{12}} 3 \overset{1^{11}}{1} 5 131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 7.5$$

5.119 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1 \overset{1^1}{1^{11}} 3 \overset{1^1}{1} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 5 \\ 7 & 8 & 6 & 5 & 6 \\ 4 & 6 & 2 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1 \overset{1^1}{1^{11}} 3 \overset{1^1}{1} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 25, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 7 & 4 & 5 \\ 7 & 8 & 9 & 5 & 6 \\ 7 & 9 & 8 & 5 & 6 \\ 4 & 5 & 5 & 1 & 3 \\ 5 & 6 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1 \overset{1^{12}}{1^{11}} 3 \overset{1^{12}}{1} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 27, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

- $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 4 & 5 \\ 7 & 8 & 6 & 5 & 6 \\ 6 & 6 & 2 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [3] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 11 & 4 & 5 \\ 7 & 8 & 15 & 5 & 6 \\ 11 & 15 & 22 & 8 & 10 \\ 4 & 5 & 8 & 1 & 3 \\ 5 & 6 & 10 & 3 & 4 \end{pmatrix}_{b_0}$
- [4] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 5 & 4 & 5 \\ 7 & 8 & 6 & 5 & 6 \\ 5 & 6 & 2 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [5] $21^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 15 & 3 & 4 \\ 5 & 4 & 6 & 14 & 3 & 4 \\ 5 & 6 & 4 & 14 & 3 & 4 \\ 15 & 14 & 14 & 37 & 8 & 11 \\ 3 & 3 & 3 & 8 & 0 & 2 \\ 4 & 4 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [6] $21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 4 \\ 5 & 4 & 6 & 4 & 3 & 4 \\ 5 & 6 & 4 & 4 & 3 & 4 \\ 3 & 4 & 4 & 1 & 2 & 3 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 4 & 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [7] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 11 & 4 & 5 \\ 7 & 8 & 12 & 5 & 6 \\ 11 & 12 & 14 & 6 & 8 \\ 4 & 5 & 6 & 1 & 3 \\ 5 & 6 & 8 & 3 & 4 \end{pmatrix}_{b_0}$
- [8] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 10 & 4 & 5 \\ 7 & 8 & 12 & 5 & 6 \\ 10 & 12 & 14 & 6 & 8 \\ 4 & 5 & 6 & 1 & 3 \\ 5 & 6 & 8 & 3 & 4 \end{pmatrix}_{b_0}$
- [9] $21^{11}3^{11}1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 18 & 3 & 4 \\ 5 & 4 & 6 & 21 & 3 & 4 \\ 5 & 6 & 4 & 21 & 3 & 4 \\ 18 & 21 & 21 & 76 & 11 & 16 \\ 3 & 3 & 3 & 11 & 0 & 2 \\ 4 & 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$
- [10] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 16 & 4 & 5 \\ 7 & 8 & 18 & 5 & 6 \\ 16 & 18 & 34 & 9 & 12 \\ 4 & 5 & 9 & 1 & 3 \\ 5 & 6 & 12 & 3 & 4 \end{pmatrix}_{b_0}$
- [11] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 15 & 4 & 5 \\ 7 & 8 & 18 & 5 & 6 \\ 15 & 18 & 34 & 9 & 12 \\ 4 & 5 & 9 & 1 & 3 \\ 5 & 6 & 12 & 3 & 4 \end{pmatrix}_{b_0}$
- [12] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$\begin{aligned}
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 14 & 4 & 5 \\ 7 & 8 & 15 & 5 & 6 \\ 14 & 15 & 22 & 8 & 10 \\ 4 & 5 & 8 & 1 & 3 \\ 5 & 6 & 10 & 3 & 4 \end{pmatrix}_{b_0} \\
[13] & 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 12 & 4 & 5 \\ 7 & 8 & 15 & 5 & 6 \\ 12 & 15 & 22 & 8 & 10 \\ 4 & 5 & 8 & 1 & 3 \\ 5 & 6 & 10 & 3 & 4 \end{pmatrix}_{b_0} \\
[14] & 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 13 & 4 & 5 \\ 7 & 8 & 15 & 5 & 6 \\ 13 & 15 & 22 & 8 & 10 \\ 4 & 5 & 8 & 1 & 3 \\ 5 & 6 & 10 & 3 & 4 \end{pmatrix}_{b_0} \\
[15] & 1^{14} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 4 & 5 \\ 7 & 8 & 24 & 5 & 6 \\ 20 & 24 & 62 & 12 & 16 \\ 4 & 5 & 12 & 1 & 3 \\ 5 & 6 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[16] & 1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 19 & 4 & 5 \\ 7 & 8 & 24 & 5 & 6 \\ 19 & 24 & 62 & 13 & 16 \\ 4 & 5 & 13 & 1 & 3 \\ 5 & 6 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[17] & 21^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 9 & 3 & 4 \\ 5 & 4 & 6 & 9 & 3 & 4 \\ 5 & 6 & 4 & 9 & 3 & 4 \\ 9 & 9 & 9 & 13 & 5 & 7 \\ 3 & 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[18] & 1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 4 & 5 \\ 7 & 8 & 24 & 5 & 6 \\ 20 & 24 & 62 & 13 & 16 \\ 4 & 5 & 13 & 1 & 3 \\ 5 & 6 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[20] & 1^{13} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 21 & 4 & 5 \\ 7 & 8 & 24 & 5 & 6 \\ 21 & 24 & 62 & 13 & 16 \\ 4 & 5 & 13 & 1 & 3 \\ 5 & 6 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[19] & 21^{13} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 15 & 3 & 4 \\ 5 & 4 & 6 & 17 & 3 & 4 \\ 5 & 6 & 4 & 17 & 3 & 4 \\ 15 & 17 & 17 & 49 & 9 & 13 \\ 3 & 3 & 3 & 9 & 0 & 2 \\ 4 & 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[21] & 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 17 & 4 & 5 \\ 7 & 8 & 21 & 5 & 6 \\ 17 & 21 & 44 & 11 & 14 \\ 4 & 5 & 11 & 1 & 3 \\ 5 & 6 & 14 & 3 & 4 \end{pmatrix}_{b_0} \\
[22] & 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 19 & 4 & 5 \\ 7 & 8 & 21 & 5 & 6 \\ 19 & 21 & 44 & 11 & 14 \\ 4 & 5 & 11 & 1 & 3 \\ 5 & 6 & 14 & 3 & 4 \end{pmatrix}_{b_0} \\
[23] & 21^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 12 & 3 & 4 \\ 5 & 4 & 6 & 13 & 3 & 4 \\ 5 & 6 & 4 & 13 & 3 & 4 \\ 12 & 13 & 13 & 28 & 7 & 10 \\ 3 & 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[24] & 21^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 3 \\ 4 & 4 & 6 & 4 & 3 & 4 \\ 4 & 6 & 4 & 4 & 3 & 4 \\ 3 & 4 & 4 & 1 & 2 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[25] & 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 18 & 4 & 5 \\ 7 & 8 & 21 & 5 & 6 \\ 18 & 21 & 44 & 11 & 14 \\ 4 & 5 & 11 & 1 & 3 \\ 5 & 6 & 14 & 3 & 4 \end{pmatrix}_{b_0} \\
[26] & 21^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 3 \\ 4 & 4 & 6 & 8 & 3 & 4 \\ 4 & 6 & 4 & 8 & 3 & 4 \\ 6 & 8 & 8 & 10 & 4 & 6 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[27] & 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 3 \\ 4 & 4 & 6 & 9 & 3 & 4 \\ 4 & 6 & 4 & 9 & 3 & 4 \\ 6 & 9 & 9 & 13 & 5 & 7 \\ 2 & 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[28] & 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 24 & 4 & 5 \\ 7 & 8 & 30 & 5 & 6 \\ 24 & 30 & 94 & 16 & 20 \\ 4 & 5 & 16 & 1 & 3 \\ 5 & 6 & 20 & 3 & 4 \end{pmatrix}_{b_0} \\
[29] & 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 22 & 4 & 5 \\ 7 & 8 & 27 & 5 & 6 \\ 22 & 27 & 74 & 14 & 18 \\ 4 & 5 & 14 & 1 & 3 \\ 5 & 6 & 18 & 3 & 4 \end{pmatrix}_{b_0} \\
[30] & 21^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 9 & 2 & 3 \\ 4 & 4 & 6 & 12 & 3 & 4 \\ 4 & 6 & 4 & 12 & 3 & 4 \\ 9 & 12 & 12 & 25 & 6 & 9 \\ 2 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[31] \quad 21^{12}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 232, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 18 & 3 & 4 \\ 5 & 4 & 6 & 18 & 3 & 4 \\ 5 & 6 & 4 & 18 & 3 & 4 \\ 18 & 18 & 18 & 58 & 10 & 14 \\ 3 & 3 & 3 & 10 & 0 & 2 \\ 4 & 4 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 238, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 4 & 5 \\ 7 & 8 & 27 & 5 & 6 \\ 23 & 27 & 74 & 14 & 18 \\ 4 & 5 & 14 & 1 & 3 \\ 5 & 6 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 259, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 25 & 4 & 5 \\ 7 & 8 & 30 & 5 & 6 \\ 25 & 30 & 94 & 16 & 20 \\ 4 & 5 & 16 & 1 & 3 \\ 5 & 6 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{12}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 268, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 26 & 4 & 5 \\ 7 & 8 & 30 & 5 & 6 \\ 26 & 30 & 94 & 16 & 20 \\ 4 & 5 & 16 & 1 & 3 \\ 5 & 6 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 21^{114}3^11^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 280, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 3 \\ 4 & 4 & 6 & 16 & 3 & 4 \\ 4 & 6 & 4 & 16 & 3 & 4 \\ 12 & 16 & 16 & 46 & 8 & 12 \\ 2 & 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{14}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 297, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 27 & 4 & 5 \\ 7 & 8 & 33 & 5 & 6 \\ 27 & 33 & 112 & 17 & 22 \\ 4 & 5 & 17 & 1 & 3 \\ 5 & 6 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 21^{112}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 312, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 3 \\ 4 & 4 & 6 & 18 & 3 & 4 \\ 4 & 6 & 4 & 18 & 3 & 4 \\ 12 & 18 & 18 & 58 & 10 & 14 \\ 2 & 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{13}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 340, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 29 & 4 & 5 \\ 7 & 8 & 36 & 5 & 6 \\ 29 & 36 & 134 & 19 & 24 \\ 4 & 5 & 19 & 1 & 3 \\ 5 & 6 & 24 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 21^{115}3^11^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 352, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.1$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 3 \\ 4 & 4 & 6 & 20 & 3 & 4 \\ 4 & 6 & 4 & 20 & 3 & 4 \\ 15 & 20 & 20 & 73 & 10 & 15 \\ 2 & 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 21^{112}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 360, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$\begin{aligned}
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 9 & 2 & 3 \\ 4 & 4 & 6 & 13 & 3 & 4 \\ 4 & 6 & 4 & 13 & 3 & 4 \\ 9 & 13 & 13 & 28 & 7 & 10 \\ 2 & 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[41] \quad & 21^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 360, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 21 & 3 & 4 \\ 5 & 4 & 6 & 22 & 3 & 4 \\ 5 & 6 & 4 & 22 & 3 & 4 \\ 21 & 22 & 22 & 85 & 12 & 17 \\ 3 & 3 & 3 & 12 & 0 & 2 \\ 4 & 4 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0} \\
[42] \quad & 21^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 424, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 24 & 3 & 4 \\ 5 & 4 & 6 & 26 & 3 & 4 \\ 5 & 6 & 4 & 26 & 3 & 4 \\ 24 & 26 & 26 & 118 & 14 & 20 \\ 3 & 3 & 3 & 14 & 0 & 2 \\ 4 & 4 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0} \\
[43] \quad & 21^{116}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 440, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 18 & 2 & 3 \\ 4 & 4 & 6 & 24 & 3 & 4 \\ 4 & 6 & 4 & 24 & 3 & 4 \\ 18 & 24 & 24 & 106 & 12 & 18 \\ 2 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[44] \quad & 1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = \\
& 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 34 & 4 & 5 \\ 7 & 8 & 42 & 5 & 6 \\ 34 & 42 & 182 & 22 & 28 \\ 4 & 5 & 22 & 1 & 3 \\ 5 & 6 & 28 & 3 & 4 \end{pmatrix}_{b_0} \\
[45] \quad & 21^{117}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 544, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 21 & 2 & 3 \\ 4 & 4 & 6 & 28 & 3 & 4 \\ 4 & 6 & 4 & 28 & 3 & 4 \\ 21 & 28 & 28 & 145 & 14 & 21 \\ 2 & 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[46] \quad & 21^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 544, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 3 \\ 4 & 4 & 6 & 17 & 3 & 4 \\ 4 & 6 & 4 & 17 & 3 & 4 \\ 12 & 17 & 17 & 49 & 9 & 13 \\ 2 & 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[47] \quad & 21^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 640, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 3 \\ 4 & 4 & 6 & 22 & 3 & 4 \\ 4 & 6 & 4 & 22 & 3 & 4 \\ 15 & 22 & 22 & 85 & 12 & 17 \\ 2 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0} \\
[48] \quad & 21^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 744, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 3 \\ 4 & 4 & 6 & 21 & 3 & 4 \\ 4 & 6 & 4 & 21 & 3 & 4 \\ 15 & 21 & 21 & 76 & 11 & 16 \\ 2 & 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[49] \quad & 21^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 984, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 18 & 2 & 3 \\ 4 & 4 & 6 & 26 & 3 & 4 \\ 4 & 6 & 4 & 26 & 3 & 4 \\ 18 & 26 & 26 & 118 & 14 & 20 \\ 2 & 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$T = 9 \quad [50] \quad 1^1 \overset{1}{1} 3^1 21^{11} + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{11}3^{1^1}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{teal}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -11$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{11}3^{11}2^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix} b_0$$

[53] $1^{11}3^{12}21^{11} + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{11}3^121^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 1 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[55] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix} b_0$$

[56] $1^{11}3^{12}21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix} b_0$$

[57] $1^{12} 1^{11} 3^1 2^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 1 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{12}3^{13}21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$[67] \quad 1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -427, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 1 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 18 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & 0 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{14} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -450, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 1 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 18 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -538, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 20 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 1 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 20 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & 0 & 28 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [70] \quad 1^{11} 3^1 1^1 5^{13} 1321^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -3, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 6.0$$

$$T = 14 \quad [73] \quad 1^{11} 7^{2321} 123^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 14, 1), \quad c_{\text{ext}} = 6.0$$

5.120 NHC(-2)(-3)(-2) + su₂ + su₂ → (-3) + su₂

$$T = 10 \quad [0] \quad 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 3 & 0 & 2 & 15 & 3 \\ 2 & 3 & 1 & 1 & 4 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 18 & 4 \\ 15 & 15 & 18 & 0 & 18 & 97 & 21 \\ 3 & 3 & 3 & 0 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 1 & 4 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 6 & 4 \\ 5 & 5 & 6 & 0 & 6 & 9 & 7 \\ 3 & 3 & 3 & 0 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 3 & 0 & 2 & 10 & 3 \\ 2 & 3 & 1 & 1 & 4 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 12 & 4 \\ 10 & 10 & 12 & 0 & 12 & 42 & 14 \\ 3 & 3 & 3 & 0 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 3 & 0 & 2 & 20 & 3 \\ 2 & 3 & 1 & 1 & 4 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 24 & 4 \\ 20 & 20 & 24 & 0 & 24 & 174 & 28 \\ 3 & 3 & 3 & 0 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[6] \quad & 1^{112} 1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 5 & 3 \\ 6 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{112} 1^{111} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 7 & 3 \\ 2 & 3 & 1 & 1 & 6 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 17 & 7 \\ 8 & 7 & 10 & 0 & 17 & 25 & 11 \\ 3 & 3 & 3 & 0 & 7 & 11 & 4 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{122} 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 12 & 3 \\ 2 & 1 & 3 & 0 & 5 & 13 & 3 \\ 2 & 3 & 1 & 1 & 6 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 27 & 7 \\ 12 & 13 & 14 & 0 & 27 & 65 & 17 \\ 3 & 3 & 3 & 0 & 7 & 17 & 4 \end{pmatrix}_{b_0} \\
[5] \quad & 1^1 1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 5 & 3 \\ 4 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[4] \quad & 1^{122} 1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 14 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 14 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{114} 1^{114} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 20 & 3 \\ 2 & 1 & 3 & 0 & 3 & 20 & 3 \\ 2 & 3 & 1 & 1 & 2 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 20 & 3 \\ 20 & 20 & 24 & 0 & 20 & 174 & 28 \\ 3 & 3 & 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{113} 1^{113} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 15 & 3 \\ 2 & 1 & 3 & 0 & 3 & 15 & 3 \\ 2 & 3 & 1 & 1 & 2 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 15 & 3 \\ 15 & 15 & 18 & 0 & 15 & 97 & 21 \\ 3 & 3 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{112} 1^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 10 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 10 & 3 \\ 10 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[12] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 5 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 5 & 3 \\ 5 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 5 & 3 \\ 5 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 11 & 7 \\ 5 & 5 & 6 & 0 & 11 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[15] $1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 10 & 3 \\ 11 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 7 & 3 \\ 2 & 1 & 3 & 0 & 5 & 8 & 3 \\ 2 & 3 & 1 & 1 & 6 & 8 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 16 & 7 \\ 7 & 8 & 8 & 0 & 16 & 22 & 10 \\ 3 & 3 & 3 & 0 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 19 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 68 & 11 \\ 19 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 28 & 7 \\ 13 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 17 & 3 \\ 2 & 1 & 3 & 0 & 5 & 18 & 3 \\ 2 & 3 & 1 & 1 & 6 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 38 & 7 \\ 17 & 18 & 20 & 0 & 38 & 130 & 24 \\ 3 & 3 & 3 & 0 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{111} 3^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 10 & 3 \\ 9 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{112} 3^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 10 & 3 \\ 10 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{112} 3^{112} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 10 & 3 \\ 2 & 1 & 3 & 0 & 7 & 10 & 3 \\ 2 & 3 & 1 & 1 & 10 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 34 & 11 \\ 10 & 10 & 12 & 0 & 34 & 42 & 14 \\ 3 & 3 & 3 & 0 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{114} 3^{114} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 20 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 68 & 11 \\ 20 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{113} 3^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 15 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 51 & 11 \\ 15 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{111} 3^{111} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 5 & 3 \\ 2 & 1 & 3 & 0 & 7 & 5 & 3 \\ 2 & 3 & 1 & 1 & 10 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 17 & 11 \\ 5 & 5 & 6 & 0 & 17 & 9 & 7 \\ 3 & 3 & 3 & 0 & 11 & 7 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{112} 3^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 10 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 22 & 7 \\ 10 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{114} 3^{114} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 3 & 0 & 3 & 15 & 3 \\ 2 & 3 & 1 & 1 & 2 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 15 & 3 \\ 16 & 15 & 18 & 0 & 15 & 97 & 21 \\ 3 & 3 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{1^{12}2} 1^{111} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 7 & 3 \\ 2 & 3 & 1 & 1 & 6 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 17 & 7 \\ 8 & 7 & 10 & 0 & 17 & 25 & 11 \\ 3 & 3 & 3 & 0 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{1^{12}2} 1^{113} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 14 & 3 \\ 2 & 1 & 3 & 0 & 3 & 15 & 3 \\ 2 & 3 & 1 & 1 & 2 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 15 & 3 \\ 14 & 15 & 18 & 0 & 15 & 97 & 21 \\ 3 & 3 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{1^{14}2} 1^{113} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 39 & 7 \\ 18 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{1^{12}2} 1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 12 & 3 \\ 2 & 1 & 3 & 0 & 5 & 13 & 3 \\ 2 & 3 & 1 & 1 & 6 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 27 & 7 \\ 12 & 13 & 14 & 0 & 27 & 65 & 17 \\ 3 & 3 & 3 & 0 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{1^{12}2} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 6 & 3 \\ 2 & 1 & 3 & 0 & 5 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 11 & 7 \\ 6 & 5 & 6 & 0 & 11 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[33] $1^{1^{13}2} 1^{113} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 3 & 0 & 3 & 15 & 3 \\ 2 & 3 & 1 & 1 & 2 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 15 & 3 \\ 15 & 15 & 18 & 0 & 15 & 97 & 21 \\ 3 & 3 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[34] $1^{1^{11}2} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 11 & 7 \\ 5 & 5 & 6 & 0 & 11 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[35] $1^{1^{13}2} 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 15 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 33 & 7 \\ 15 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[36] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 3 & 0 & 8 & 15 & 3 \\ 2 & 3 & 1 & 1 & 8 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 48 & 10 \\ 15 & 15 & 18 & 0 & 48 & 97 & 21 \\ 3 & 3 & 3 & 0 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 8 & 3 \\ 2 & 3 & 1 & 1 & 6 & 8 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 16 & 7 \\ 8 & 8 & 8 & 0 & 16 & 22 & 10 \\ 3 & 3 & 3 & 0 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 3 & 0 & 8 & 20 & 3 \\ 2 & 3 & 1 & 1 & 8 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 64 & 10 \\ 20 & 20 & 24 & 0 & 64 & 174 & 28 \\ 3 & 3 & 3 & 0 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 3 & 0 & 8 & 10 & 3 \\ 2 & 3 & 1 & 1 & 8 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 32 & 10 \\ 10 & 10 & 12 & 0 & 32 & 42 & 14 \\ 3 & 3 & 3 & 0 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 7 & 3 \\ 2 & 3 & 1 & 1 & 6 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 17 & 7 \\ 8 & 7 & 10 & 0 & 17 & 25 & 11 \\ 3 & 3 & 3 & 0 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 10 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 22 & 7 \\ 10 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 3 & 0 & 7 & 10 & 3 \\ 2 & 3 & 1 & 1 & 10 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 34 & 11 \\ 10 & 10 & 12 & 0 & 34 & 42 & 14 \\ 3 & 3 & 3 & 0 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{13} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 19 & 3 \\ 2 & 1 & 3 & 0 & 3 & 20 & 3 \\ 2 & 3 & 1 & 1 & 2 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 20 & 3 \\ 19 & 20 & 24 & 0 & 20 & 174 & 28 \\ 3 & 3 & 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 28 & 7 \\ 13 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 17 & 3 \\ 2 & 1 & 3 & 0 & 5 & 18 & 3 \\ 2 & 3 & 1 & 1 & 6 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 38 & 7 \\ 17 & 18 & 20 & 0 & 38 & 130 & 24 \\ 3 & 3 & 3 & 0 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 11 & 3 \\ 2 & 1 & 3 & 0 & 7 & 10 & 3 \\ 2 & 3 & 1 & 1 & 10 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 34 & 11 \\ 11 & 10 & 12 & 0 & 34 & 42 & 14 \\ 3 & 3 & 3 & 0 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{114} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 278, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 3 & 0 & 3 & 20 & 3 \\ 2 & 3 & 1 & 1 & 2 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 20 & 3 \\ 20 & 20 & 24 & 0 & 20 & 174 & 28 \\ 3 & 3 & 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 278, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 15 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 33 & 7 \\ 15 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 20 & 3 \\ 2 & 1 & 3 & 0 & 5 & 20 & 3 \\ 2 & 3 & 1 & 1 & 6 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 44 & 7 \\ 20 & 20 & 24 & 0 & 44 & 174 & 28 \\ 3 & 3 & 3 & 0 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{122} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 324, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 12 & 3 \\ 2 & 1 & 3 & 0 & 10 & 12 & 3 \\ 2 & 3 & 1 & 1 & 12 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 56 & 14 \\ 12 & 12 & 16 & 0 & 56 & 70 & 18 \\ 3 & 3 & 3 & 0 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 324, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 12 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 28 & 7 \\ 12 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[52] \quad & 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 326, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 15 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0} \\
[53] \quad & 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 372, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 10 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 22 & 7 \\ 10 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0} \\
[55] \quad & 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 390, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 20 & 3 \\ 2 & 1 & 3 & 0 & 5 & 20 & 3 \\ 2 & 3 & 1 & 1 & 6 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 44 & 7 \\ 20 & 20 & 24 & 0 & 44 & 174 & 28 \\ 3 & 3 & 3 & 0 & 7 & 28 & 4 \end{pmatrix}_{b_0} \\
[54] \quad & 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 390, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 11 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 22 & 7 \\ 11 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0} \\
[56] \quad & 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 400, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 39 & 7 \\ 18 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0} \\
[57] \quad & 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 400, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 16 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0} \\
[58] \quad & 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 438, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 10 & 3 \\ 2 & 3 & 1 & 1 & 12 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 44 & 14 \\ 11 & 10 & 12 & 0 & 44 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 14 & 4 \end{pmatrix}_{b_0} \\
[59] \quad & 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 438, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 68 & 11 \\ 20 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[60] $1^{113} 3^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 28 & 7 \\ 13 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{113} 3^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 13 & 3 \\ 2 & 3 & 1 & 1 & 6 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 27 & 7 \\ 13 & 13 & 14 & 0 & 27 & 65 & 17 \\ 3 & 3 & 3 & 0 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{123} 3^{112} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 13 & 3 \\ 2 & 1 & 3 & 0 & 10 & 13 & 3 \\ 2 & 3 & 1 & 1 & 12 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 54 & 14 \\ 13 & 13 & 14 & 0 & 54 & 65 & 17 \\ 3 & 3 & 3 & 0 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{123} 3^{112} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 518, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 13 & 3 \\ 2 & 1 & 3 & 0 & 10 & 12 & 3 \\ 2 & 3 & 1 & 1 & 12 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 56 & 14 \\ 13 & 12 & 16 & 0 & 56 & 70 & 18 \\ 3 & 3 & 3 & 0 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{122} 3^{112} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 564, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 10 & 3 \\ 2 & 3 & 1 & 1 & 12 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 44 & 14 \\ 10 & 10 & 12 & 0 & 44 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{123} 3^{112} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 17 & 3 \\ 2 & 1 & 3 & 0 & 10 & 17 & 3 \\ 2 & 3 & 1 & 1 & 12 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 78 & 14 \\ 17 & 17 & 22 & 0 & 78 & 137 & 25 \\ 3 & 3 & 3 & 0 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{113} 3^{112} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 17 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 39 & 7 \\ 17 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{113} 3^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 692, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 15 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 33 & 7 \\ 15 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[68] $1^{114} 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 758, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 16 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 33 & 7 \\ 16 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{124} 1^{123} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 806, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 16 & 3 \\ 2 & 1 & 3 & 0 & 10 & 15 & 3 \\ 2 & 3 & 1 & 1 & 12 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 66 & 14 \\ 16 & 15 & 18 & 0 & 66 & 97 & 21 \\ 3 & 3 & 3 & 0 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{114} 1^{113} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 870, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 39 & 7 \\ 18 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{123} 1^{123} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 884, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 15 & 3 \\ 2 & 1 & 3 & 0 & 10 & 15 & 3 \\ 2 & 3 & 1 & 1 & 12 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 66 & 14 \\ 15 & 15 & 18 & 0 & 66 & 97 & 21 \\ 3 & 3 & 3 & 0 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{114} 1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 900, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 18 & 3 \\ 2 & 3 & 1 & 1 & 6 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 38 & 7 \\ 18 & 18 & 20 & 0 & 38 & 130 & 24 \\ 3 & 3 & 3 & 0 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{124} 1^{124} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 900, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 18 & 3 \\ 2 & 1 & 3 & 0 & 10 & 18 & 3 \\ 2 & 3 & 1 & 1 & 12 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 76 & 14 \\ 18 & 18 & 20 & 0 & 76 & 130 & 24 \\ 3 & 3 & 3 & 0 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{124} 1^{123} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 918, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 18 & 3 \\ 2 & 1 & 3 & 0 & 10 & 17 & 3 \\ 2 & 3 & 1 & 1 & 12 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 78 & 14 \\ 18 & 17 & 22 & 0 & 78 & 137 & 25 \\ 3 & 3 & 3 & 0 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{114} 1^{114} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 1140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 20 & 3 \\ 2 & 1 & 3 & 0 & 5 & 20 & 3 \\ 2 & 3 & 1 & 1 & 6 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 44 & 7 \\ 20 & 20 & 24 & 0 & 44 & 174 & 28 \\ 3 & 3 & 3 & 0 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 1332, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 20 & 3 \\ 2 & 1 & 3 & 0 & 10 & 20 & 3 \\ 2 & 3 & 1 & 1 & 12 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 88 & 14 \\ 20 & 20 & 24 & 0 & 88 & 174 & 28 \\ 3 & 3 & 3 & 0 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [77] $1^{112} 3^1 1^5 131^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$

5.121 NHC(-2)(-3)(-2) + su₂ + su₂ → (-3) + su₂ + su₂

$T = 10$ [0] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.0$
[1] $1^{111} 3^1 21^{1111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.0$
[2] $1^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.5$
[4] $1^{1121} 3^1 21^{1111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.5$
[5] $1^{131} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.8$
[6] $1^{141} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$
[3] $1^{121} 3^1 21^{1121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.5$

5.122 NHC(-2)(-3)(-2) + su₂ + su₂ → (-3) + su₂ → (-3)

$T = 9$ [3] $1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 5 \\ 7 & 8 & 6 & 5 & 5 & 6 \\ 6 & 6 & 2 & 3 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[0] $1^{123} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 4 & 5 \\ 7 & 8 & 12 & 5 & 5 & 6 \\ 11 & 12 & 14 & 6 & 6 & 8 \\ 4 & 5 & 6 & 1 & 3 & 3 \\ 4 & 5 & 6 & 3 & 1 & 3 \\ 5 & 6 & 8 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 5 \\ 7 & 8 & 6 & 5 & 5 & 6 \\ 4 & 6 & 2 & 3 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{14} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 16 & 4 & 4 & 5 \\ 7 & 8 & 18 & 5 & 5 & 6 \\ 16 & 18 & 34 & 9 & 9 & 12 \\ 4 & 5 & 9 & 1 & 3 & 3 \\ 4 & 5 & 9 & 3 & 1 & 3 \\ 5 & 6 & 12 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[4] \quad & 21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 3 & 4 \\ 5 & 4 & 6 & 4 & 3 & 3 & 4 \\ 5 & 6 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 & 2 \\ 4 & 4 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{14}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 4 & 4 & 5 \\ 7 & 8 & 24 & 5 & 5 & 6 \\ 20 & 24 & 62 & 12 & 12 & 16 \\ 4 & 5 & 12 & 1 & 3 & 3 \\ 4 & 5 & 12 & 3 & 1 & 3 \\ 5 & 6 & 16 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[5] \quad & 1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 5 \\ 7 & 8 & 12 & 5 & 5 & 6 \\ 10 & 12 & 14 & 6 & 6 & 8 \\ 4 & 5 & 6 & 1 & 3 & 3 \\ 4 & 5 & 6 & 3 & 1 & 3 \\ 5 & 6 & 8 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[8] \quad & 21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 9 & 3 & 3 & 4 \\ 5 & 4 & 6 & 9 & 3 & 3 & 4 \\ 5 & 6 & 4 & 9 & 3 & 3 & 4 \\ 9 & 9 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 3 & 5 & 2 & 0 & 2 \\ 4 & 4 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[7] \quad & 21^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 15 & 3 & 3 & 4 \\ 5 & 4 & 6 & 17 & 3 & 3 & 4 \\ 5 & 6 & 4 & 17 & 3 & 3 & 4 \\ 15 & 17 & 17 & 49 & 9 & 9 & 13 \\ 3 & 3 & 3 & 9 & 0 & 2 & 2 \\ 3 & 3 & 3 & 9 & 2 & 0 & 2 \\ 4 & 4 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[13] \quad & 21^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 2 & 3 \\ 4 & 4 & 6 & 8 & 3 & 3 & 4 \\ 4 & 6 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[16] \quad & 21^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 21 & 2 & 2 & 3 \\ 4 & 4 & 6 & 28 & 3 & 3 & 4 \\ 4 & 6 & 4 & 28 & 3 & 3 & 4 \\ 21 & 28 & 28 & 145 & 14 & 14 & 21 \\ 2 & 3 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 4 & 21 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[9] \quad & 21^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 18 & 2 & 2 & 3 \\ 4 & 4 & 6 & 24 & 3 & 3 & 4 \\ 4 & 6 & 4 & 24 & 3 & 3 & 4 \\ 18 & 24 & 24 & 106 & 12 & 12 & 18 \\ 2 & 3 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 4 & 18 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[10] \quad & 21^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.1 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 2 & 3 \\ 4 & 4 & 6 & 20 & 3 & 3 & 4 \\ 4 & 6 & 4 & 20 & 3 & 3 & 4 \\ 15 & 20 & 20 & 73 & 10 & 10 & 15 \\ 2 & 3 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 4 & 15 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[11] $21^{114}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

[11] $21^{114}3^{11}1^{1} + [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{blue}{\mathbf{so}_7} \cdot \textcolor{green}{\mathbf{su}_2}] + \textcolor{blue}{\mathbf{su}_2} + \textcolor{red}{\mathbf{su}_2} \rightarrow (-3) + \textcolor{blue}{\mathbf{su}_2} \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 2 & 3 \\ 4 & 4 & 6 & 16 & 3 & 3 & 4 \\ 4 & 6 & 4 & 16 & 3 & 3 & 4 \\ 12 & 16 & 16 & 46 & 8 & 8 & 12 \\ 2 & 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[12] $21^{1133}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 2 & 2 & 3 \\ 4 & 4 & 6 & 12 & 3 & 3 & 4 \\ 4 & 6 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 3 & 6 & 0 & 2 & 2 \\ 3 & 4 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[15] $21^{113}3^{111}1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 3 & 3 & 4 \\ 5 & 4 & 6 & 13 & 3 & 3 & 4 \\ 5 & 6 & 4 & 13 & 3 & 3 & 4 \\ 12 & 13 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 3 & 7 & 2 & 0 & 2 \\ 4 & 4 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[14] $21^{1113}3^{111}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 2 & 3 \\ 4 & 4 & 6 & 4 & 3 & 3 & 4 \\ 4 & 6 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 4 & 3 & 2 & 2 & 3 \end{pmatrix} b_0$$

[17] $21^{1113}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 2 & 3 \\ 4 & 4 & 6 & 9 & 3 & 3 & 4 \\ 4 & 6 & 4 & 9 & 3 & 3 & 4 \\ 6 & 9 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 4 & 7 & 2 & 2 & 3 \end{pmatrix} b_0$$

[18] $21^{1123}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 2 & 2 & 3 \\ 4 & 4 & 6 & 13 & 3 & 3 & 4 \\ 4 & 6 & 4 & 13 & 3 & 3 & 4 \\ 9 & 13 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 4 & 10 & 2 & 2 & 3 \end{pmatrix} b_0$$

[19] 21¹¹³³1¹¹1¹¹ + [su₂·s₀₇·su₂] + su₂ + su₂ → (-3) + su₂ → (-3) $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -832$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 2 & 3 \\ 4 & 4 & 6 & 17 & 3 & 3 & 4 \\ 4 & 6 & 4 & 17 & 3 & 3 & 4 \\ 12 & 17 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 4 & 13 & 2 & 2 & 3 \end{pmatrix} b_0$$

[20] 21¹¹⁴3¹¹1¹¹+[su₂so₇su₂]+su₂+su₂→(−3)+su₂→(−3) $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -1088$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 2 & 3 \\ 4 & 4 & 6 & 21 & 3 & 3 & 4 \\ 4 & 6 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}$$

5.123 NHC(-2)(-3)(-2) + su₂ + su₂→(-3) + su₂→(-3) + su₂

$$T = 10 \quad [3] \quad 1^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 10 & 5 \\ 7 & 8 & 6 & 5 & 5 & 12 & 6 \\ 4 & 6 & 2 & 3 & 3 & 8 & 4 \\ 4 & 5 & 3 & 1 & 3 & 6 & 3 \\ 4 & 5 & 3 & 3 & 1 & 6 & 3 \\ 10 & 12 & 8 & 6 & 6 & 14 & 8 \\ 5 & 6 & 4 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{113}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 15 & 5 \\ 7 & 8 & 6 & 5 & 5 & 18 & 6 \\ 4 & 6 & 2 & 3 & 3 & 12 & 4 \\ 4 & 5 & 3 & 1 & 3 & 9 & 3 \\ 4 & 5 & 3 & 3 & 1 & 9 & 3 \\ 15 & 18 & 12 & 9 & 9 & 34 & 12 \\ 5 & 6 & 4 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{114}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 20 & 5 \\ 7 & 8 & 6 & 5 & 5 & 24 & 6 \\ 4 & 6 & 2 & 3 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 3 & 12 & 3 \\ 4 & 5 & 3 & 3 & 1 & 12 & 3 \\ 20 & 24 & 16 & 12 & 12 & 62 & 16 \\ 5 & 6 & 4 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{131}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 16 & 4 & 4 & 5 & 5 \\ 7 & 8 & 18 & 5 & 5 & 6 & 6 \\ 16 & 18 & 34 & 9 & 9 & 12 & 12 \\ 4 & 5 & 9 & 1 & 3 & 3 & 3 \\ 4 & 5 & 9 & 3 & 1 & 3 & 3 \\ 5 & 6 & 12 & 3 & 3 & 2 & 4 \\ 5 & 6 & 12 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{132}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 16 & 4 & 4 & 10 & 5 \\ 7 & 8 & 18 & 5 & 5 & 12 & 6 \\ 16 & 18 & 34 & 9 & 9 & 24 & 12 \\ 4 & 5 & 9 & 1 & 3 & 6 & 3 \\ 4 & 5 & 9 & 3 & 1 & 6 & 3 \\ 10 & 12 & 24 & 6 & 6 & 14 & 8 \\ 5 & 6 & 12 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 5 & 5 \\ 7 & 8 & 6 & 5 & 5 & 6 & 6 \\ 6 & 6 & 2 & 3 & 3 & 4 & 4 \\ 4 & 5 & 3 & 1 & 3 & 3 & 3 \\ 4 & 5 & 3 & 3 & 1 & 3 & 3 \\ 5 & 6 & 4 & 3 & 3 & 2 & 4 \\ 5 & 6 & 4 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 10 & 5 \\ 7 & 8 & 6 & 5 & 5 & 12 & 6 \\ 6 & 6 & 2 & 3 & 3 & 8 & 4 \\ 4 & 5 & 3 & 1 & 3 & 6 & 3 \\ 4 & 5 & 3 & 3 & 1 & 6 & 3 \\ 10 & 12 & 8 & 6 & 6 & 14 & 8 \\ 5 & 6 & 4 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{113}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 15 & 5 \\ 7 & 8 & 6 & 5 & 5 & 18 & 6 \\ 6 & 6 & 2 & 3 & 3 & 12 & 4 \\ 4 & 5 & 3 & 1 & 3 & 9 & 3 \\ 4 & 5 & 3 & 3 & 1 & 9 & 3 \\ 15 & 18 & 12 & 9 & 9 & 34 & 12 \\ 5 & 6 & 4 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 5 & 5 \\ 7 & 8 & 6 & 5 & 5 & 6 & 6 \\ 4 & 6 & 2 & 3 & 3 & 4 & 4 \\ 4 & 5 & 3 & 1 & 3 & 3 & 3 \\ 4 & 5 & 3 & 3 & 1 & 3 & 3 \\ 5 & 6 & 4 & 3 & 3 & 2 & 4 \\ 5 & 6 & 4 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 4 & 10 & 5 \\ 7 & 8 & 12 & 5 & 5 & 12 & 6 \\ 11 & 12 & 14 & 6 & 6 & 16 & 8 \\ 4 & 5 & 6 & 1 & 3 & 6 & 3 \\ 4 & 5 & 6 & 3 & 1 & 6 & 3 \\ 10 & 12 & 16 & 6 & 6 & 14 & 8 \\ 5 & 6 & 8 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 4 & 5 & 5 \\ 7 & 8 & 12 & 5 & 5 & 6 & 6 \\ 11 & 12 & 14 & 6 & 6 & 8 & 8 \\ 4 & 5 & 6 & 1 & 3 & 3 & 3 \\ 4 & 5 & 6 & 3 & 1 & 3 & 3 \\ 5 & 6 & 8 & 3 & 3 & 2 & 4 \\ 5 & 6 & 8 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{114} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 20 & 5 \\ 7 & 8 & 6 & 5 & 5 & 24 & 6 \\ 6 & 6 & 2 & 3 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 3 & 12 & 3 \\ 4 & 5 & 3 & 3 & 1 & 12 & 3 \\ 20 & 24 & 16 & 12 & 12 & 62 & 16 \\ 5 & 6 & 4 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 21^{1121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 128, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 2 & 6 & 3 \\ 4 & 4 & 6 & 8 & 3 & 3 & 9 & 4 \\ 4 & 6 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 21^{113} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 128, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 3 & 9 & 4 \\ 5 & 4 & 6 & 4 & 3 & 3 & 9 & 4 \\ 5 & 6 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 9 & 9 & 9 & 6 & 5 & 5 & 13 & 7 \\ 4 & 4 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{124} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 20 & 5 \\ 7 & 8 & 12 & 5 & 5 & 24 & 6 \\ 10 & 12 & 14 & 6 & 6 & 32 & 8 \\ 4 & 5 & 6 & 1 & 3 & 12 & 3 \\ 4 & 5 & 6 & 3 & 1 & 12 & 3 \\ 20 & 24 & 32 & 12 & 12 & 62 & 16 \\ 5 & 6 & 8 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{141} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 20 & 4 & 4 & 5 & 5 \\ 7 & 8 & 24 & 5 & 5 & 6 & 6 \\ 20 & 24 & 62 & 12 & 12 & 16 & 16 \\ 4 & 5 & 12 & 1 & 3 & 3 & 3 \\ 4 & 5 & 12 & 3 & 1 & 3 & 3 \\ 5 & 6 & 16 & 3 & 3 & 2 & 4 \\ 5 & 6 & 16 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[15] $1^{123}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 5 & 5 \\ 7 & 8 & 12 & 5 & 5 & 6 & 6 \\ 10 & 12 & 14 & 6 & 6 & 8 & 8 \\ 4 & 5 & 6 & 1 & 3 & 3 & 3 \\ 4 & 5 & 6 & 3 & 1 & 3 & 3 \\ 5 & 6 & 8 & 3 & 3 & 2 & 4 \\ 5 & 6 & 8 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{122}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 10 & 5 \\ 7 & 8 & 12 & 5 & 5 & 12 & 6 \\ 10 & 12 & 14 & 6 & 6 & 16 & 8 \\ 4 & 5 & 6 & 1 & 3 & 6 & 3 \\ 4 & 5 & 6 & 3 & 1 & 6 & 3 \\ 10 & 12 & 16 & 6 & 6 & 14 & 8 \\ 5 & 6 & 8 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{123}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 15 & 5 \\ 7 & 8 & 12 & 5 & 5 & 18 & 6 \\ 10 & 12 & 14 & 6 & 6 & 24 & 8 \\ 4 & 5 & 6 & 1 & 3 & 9 & 3 \\ 4 & 5 & 6 & 3 & 1 & 9 & 3 \\ 15 & 18 & 24 & 9 & 9 & 34 & 12 \\ 5 & 6 & 8 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[19] $21^{11}3^{11}1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 12 & 4 \\ 5 & 4 & 6 & 4 & 3 & 13 & 4 \\ 5 & 6 & 4 & 4 & 3 & 13 & 4 \\ 3 & 4 & 4 & 1 & 2 & 9 & 3 \\ 3 & 3 & 3 & 2 & 0 & 7 & 2 \\ 3 & 3 & 3 & 2 & 0 & 7 & 2 \\ 12 & 13 & 13 & 9 & 7 & 28 & 10 \\ 4 & 4 & 4 & 3 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[22] $21^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 6 & 3 \\ 4 & 4 & 6 & 4 & 3 & 9 & 4 \\ 4 & 6 & 4 & 4 & 3 & 9 & 4 \\ 3 & 4 & 4 & 1 & 2 & 6 & 3 \\ 2 & 3 & 3 & 2 & 0 & 5 & 2 \\ 2 & 3 & 3 & 2 & 0 & 5 & 2 \\ 6 & 9 & 9 & 6 & 5 & 13 & 7 \\ 3 & 4 & 4 & 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[21] $21^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 2 & 3 & 3 \\ 4 & 4 & 6 & 4 & 3 & 3 & 4 & 4 \\ 4 & 6 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[20] $21^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 2 & 3 & 3 \\ 4 & 4 & 6 & 8 & 3 & 3 & 4 & 4 \\ 4 & 6 & 4 & 8 & 3 & 3 & 4 & 4 \\ 6 & 8 & 8 & 10 & 4 & 4 & 6 & 6 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 4 & 2 & 0 & 2 & 2 \\ 3 & 4 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[23] $21^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 1024, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 2 & 9 & 3 \\ 4 & 4 & 6 & 4 & 3 & 3 & 13 & 4 \\ 4 & 6 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

5.124 NHC(-2)(-3)(-2) + su₂ + su₂→(-3) + su₂→(-3) + su₂ + su₂

$$T = 10 \quad [7] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$$

$$[2] \quad 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.8$$

$$[4] \quad 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[10] \quad 1^{141} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 10.0$$

$$[9] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[6] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$$

$$[5] \quad 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[0] \quad 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[1] \quad 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[8] \quad 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$$

$$[3] \quad 1^{122} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

5.125 NHC(-2)(-3)(-2) + su₂ + su₂→(-3) + su₂n3mix

$$T = 9 \quad [1] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 6 \\ 11 & 15 & 22 & 8 & 17 & 10 \\ 4 & 5 & 8 & 1 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 7 \\ 5 & 6 & 10 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 14 & 4 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 6 \\ 14 & 15 & 22 & 8 & 17 & 10 \\ 4 & 5 & 8 & 1 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 7 \\ 5 & 6 & 10 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 7 & 4 & 9 & 5 \\ 7 & 8 & 9 & 5 & 11 & 6 \\ 7 & 9 & 8 & 5 & 10 & 6 \\ 4 & 5 & 5 & 1 & 6 & 3 \\ 9 & 11 & 10 & 6 & 11 & 7 \\ 5 & 6 & 6 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[3] \quad & 1^{131} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 4 & 9 & 5 \\ 7 & 8 & 18 & 5 & 11 & 6 \\ 16 & 18 & 34 & 9 & 21 & 12 \\ 4 & 5 & 9 & 1 & 6 & 3 \\ 9 & 11 & 21 & 6 & 11 & 7 \\ 5 & 6 & 12 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[4] \quad & 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 4 & 9 & 5 \\ 7 & 8 & 6 & 5 & 11 & 6 \\ 6 & 6 & 2 & 3 & 7 & 4 \\ 4 & 5 & 3 & 1 & 6 & 3 \\ 9 & 11 & 7 & 6 & 11 & 7 \\ 5 & 6 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[5] \quad & 21^{11} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 12 & 4 \\ 5 & 4 & 6 & 4 & 3 & 12 & 4 \\ 5 & 6 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 3 & 2 & 0 & 7 & 2 \\ 12 & 12 & 12 & 8 & 7 & 25 & 9 \\ 4 & 4 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{121} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 4 & 9 & 5 \\ 7 & 8 & 12 & 5 & 11 & 6 \\ 11 & 12 & 14 & 6 & 14 & 8 \\ 4 & 5 & 6 & 1 & 6 & 3 \\ 9 & 11 & 14 & 6 & 11 & 7 \\ 5 & 6 & 8 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 4 & 9 & 5 \\ 7 & 8 & 6 & 5 & 11 & 6 \\ 4 & 6 & 2 & 3 & 7 & 4 \\ 4 & 5 & 3 & 1 & 6 & 3 \\ 9 & 11 & 7 & 6 & 11 & 7 \\ 5 & 6 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{122} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 4 & 14 & 5 \\ 7 & 8 & 12 & 5 & 17 & 6 \\ 11 & 12 & 14 & 6 & 22 & 8 \\ 4 & 5 & 6 & 1 & 9 & 3 \\ 14 & 17 & 22 & 9 & 29 & 11 \\ 5 & 6 & 8 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{122} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 4 & 14 & 5 \\ 7 & 8 & 6 & 5 & 17 & 6 \\ 6 & 6 & 2 & 3 & 11 & 4 \\ 4 & 5 & 3 & 1 & 9 & 3 \\ 14 & 17 & 11 & 9 & 29 & 11 \\ 5 & 6 & 4 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{132} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 4 & 14 & 5 \\ 7 & 8 & 18 & 5 & 17 & 6 \\ 16 & 18 & 34 & 9 & 33 & 12 \\ 4 & 5 & 9 & 1 & 9 & 3 \\ 14 & 17 & 33 & 9 & 29 & 11 \\ 5 & 6 & 12 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{112} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 14 & 5 \\ 7 & 8 & 6 & 5 & 17 & 6 \\ 4 & 6 & 2 & 3 & 11 & 4 \\ 4 & 5 & 3 & 1 & 9 & 3 \\ 14 & 17 & 11 & 9 & 29 & 11 \\ 5 & 6 & 4 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{141} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 20 & 4 & 9 & 5 \\ 7 & 8 & 24 & 5 & 11 & 6 \\ 20 & 24 & 62 & 12 & 28 & 16 \\ 4 & 5 & 12 & 1 & 6 & 3 \\ 9 & 11 & 28 & 6 & 11 & 7 \\ 5 & 6 & 16 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 9 & 5 \\ 7 & 8 & 6 & 5 & 11 & 6 \\ 5 & 6 & 2 & 3 & 7 & 4 \\ 4 & 5 & 3 & 1 & 6 & 3 \\ 9 & 11 & 7 & 6 & 11 & 7 \\ 5 & 6 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 15 & 4 & 9 & 5 \\ 7 & 8 & 18 & 5 & 11 & 6 \\ 15 & 18 & 34 & 9 & 21 & 12 \\ 4 & 5 & 9 & 1 & 6 & 3 \\ 9 & 11 & 21 & 6 & 11 & 7 \\ 5 & 6 & 12 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 26 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 6 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 26 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 4 & 9 & 5 \\ 7 & 8 & 21 & 5 & 11 & 6 \\ 19 & 21 & 44 & 11 & 24 & 14 \\ 4 & 5 & 11 & 1 & 6 & 3 \\ 9 & 11 & 24 & 6 & 11 & 7 \\ 5 & 6 & 14 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 13 & 4 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 6 \\ 13 & 15 & 22 & 8 & 17 & 10 \\ 4 & 5 & 8 & 1 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 7 \\ 5 & 6 & 10 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 12 & 4 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 6 \\ 12 & 15 & 22 & 8 & 17 & 10 \\ 4 & 5 & 8 & 1 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 7 \\ 5 & 6 & 10 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 21 & 5 \\ 7 & 8 & 6 & 5 & 26 & 6 \\ 4 & 6 & 2 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 14 & 3 \\ 21 & 26 & 16 & 14 & 69 & 17 \\ 5 & 6 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{1^{123}}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 22 & 5 \\ 7 & 8 & 6 & 5 & 26 & 6 \\ 6 & 6 & 2 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 14 & 3 \\ 22 & 26 & 16 & 14 & 69 & 17 \\ 5 & 6 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{1^{14}}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 24 & 5 \\ 7 & 8 & 6 & 5 & 29 & 6 \\ 4 & 6 & 2 & 3 & 19 & 4 \\ 4 & 5 & 3 & 1 & 15 & 3 \\ 24 & 29 & 19 & 15 & 89 & 19 \\ 5 & 6 & 4 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{1^{42}}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 20 & 4 & 14 & 5 \\ 7 & 8 & 24 & 5 & 17 & 6 \\ 20 & 24 & 62 & 12 & 44 & 16 \\ 4 & 5 & 12 & 1 & 9 & 3 \\ 14 & 17 & 44 & 9 & 29 & 11 \\ 5 & 6 & 16 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{1^{22}}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 14 & 5 \\ 7 & 8 & 12 & 5 & 17 & 6 \\ 10 & 12 & 14 & 6 & 22 & 8 \\ 4 & 5 & 6 & 1 & 9 & 3 \\ 14 & 17 & 22 & 9 & 29 & 11 \\ 5 & 6 & 8 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[24] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 15 & 4 \\ 5 & 4 & 6 & 4 & 3 & 16 & 4 \\ 5 & 6 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 3 & 2 & 0 & 9 & 2 \\ 15 & 16 & 16 & 11 & 9 & 44 & 12 \\ 4 & 4 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{1^{14}}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 31 & 5 \\ 7 & 8 & 6 & 5 & 38 & 6 \\ 6 & 6 & 2 & 3 & 24 & 4 \\ 4 & 5 & 3 & 1 & 20 & 3 \\ 31 & 38 & 24 & 20 & 149 & 25 \\ 5 & 6 & 4 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{1^{12}}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 25 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 4 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 25 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{1^{14}}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 27 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 4 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 27 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[28] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 8 & 3 \\ 4 & 4 & 6 & 4 & 3 & 12 & 4 \\ 4 & 6 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 26 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 4 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 26 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 4 & 9 & 5 \\ 7 & 8 & 21 & 5 & 11 & 6 \\ 18 & 21 & 44 & 11 & 24 & 14 \\ 4 & 5 & 11 & 1 & 6 & 3 \\ 9 & 11 & 24 & 6 & 11 & 7 \\ 5 & 6 & 14 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 17 & 4 & 9 & 5 \\ 7 & 8 & 21 & 5 & 11 & 6 \\ 17 & 21 & 44 & 11 & 24 & 14 \\ 4 & 5 & 11 & 1 & 6 & 3 \\ 9 & 11 & 24 & 6 & 11 & 7 \\ 5 & 6 & 14 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 27 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 6 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 27 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 26 & 5 \\ 7 & 8 & 12 & 5 & 32 & 6 \\ 10 & 12 & 14 & 6 & 40 & 8 \\ 4 & 5 & 6 & 1 & 17 & 3 \\ 26 & 32 & 40 & 17 & 105 & 21 \\ 5 & 6 & 8 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 21 & 5 \\ 7 & 8 & 6 & 5 & 26 & 6 \\ 5 & 6 & 2 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 14 & 3 \\ 21 & 26 & 16 & 14 & 69 & 17 \\ 5 & 6 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 30 & 5 \\ 7 & 8 & 6 & 5 & 38 & 6 \\ 4 & 6 & 2 & 3 & 24 & 4 \\ 4 & 5 & 3 & 1 & 20 & 3 \\ 30 & 38 & 24 & 20 & 149 & 25 \\ 5 & 6 & 4 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 27 & 5 \\ 7 & 8 & 12 & 5 & 32 & 6 \\ 10 & 12 & 14 & 6 & 40 & 8 \\ 4 & 5 & 6 & 1 & 17 & 3 \\ 27 & 32 & 40 & 17 & 105 & 21 \\ 5 & 6 & 8 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 31 & 5 \\ 7 & 8 & 6 & 5 & 38 & 6 \\ 4 & 6 & 2 & 3 & 24 & 4 \\ 4 & 5 & 3 & 1 & 20 & 3 \\ 31 & 38 & 24 & 20 & 149 & 25 \\ 5 & 6 & 4 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 24 & 5 \\ 7 & 8 & 6 & 5 & 29 & 6 \\ 5 & 6 & 2 & 3 & 19 & 4 \\ 4 & 5 & 3 & 1 & 15 & 3 \\ 24 & 29 & 19 & 15 & 89 & 19 \\ 5 & 6 & 4 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 22 & 5 \\ 7 & 8 & 6 & 5 & 26 & 6 \\ 5 & 6 & 2 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 14 & 3 \\ 22 & 26 & 16 & 14 & 69 & 17 \\ 5 & 6 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 24 & 5 \\ 7 & 8 & 12 & 5 & 29 & 6 \\ 10 & 12 & 14 & 6 & 38 & 8 \\ 4 & 5 & 6 & 1 & 15 & 3 \\ 24 & 29 & 38 & 15 & 89 & 19 \\ 5 & 6 & 8 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[42] $21^{1121}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 5 & 3 \\ 4 & 4 & 6 & 8 & 3 & 7 & 4 \\ 4 & 6 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[41] $21^{1111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 5 & 3 \\ 4 & 4 & 6 & 4 & 3 & 7 & 4 \\ 4 & 6 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 22 & 4 & 9 & 5 \\ 7 & 8 & 27 & 5 & 11 & 6 \\ 22 & 27 & 74 & 14 & 31 & 18 \\ 4 & 5 & 14 & 1 & 6 & 3 \\ 9 & 11 & 31 & 6 & 11 & 7 \\ 5 & 6 & 18 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 23 & 4 & 9 & 5 \\ 7 & 8 & 27 & 5 & 11 & 6 \\ 23 & 27 & 74 & 14 & 31 & 18 \\ 4 & 5 & 14 & 1 & 6 & 3 \\ 9 & 11 & 31 & 6 & 11 & 7 \\ 5 & 6 & 18 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 31 & 5 \\ 7 & 8 & 12 & 5 & 38 & 6 \\ 10 & 12 & 14 & 6 & 48 & 8 \\ 4 & 5 & 6 & 1 & 20 & 3 \\ 31 & 38 & 48 & 20 & 149 & 25 \\ 5 & 6 & 8 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -530, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 26 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 5 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 26 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{141} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 27 & 4 & 9 & 5 \\ 7 & 8 & 33 & 5 & 11 & 6 \\ 27 & 33 & 112 & 17 & 38 & 22 \\ 4 & 5 & 17 & 1 & 6 & 3 \\ 9 & 11 & 38 & 6 & 11 & 7 \\ 5 & 6 & 22 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 21^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 8 & 3 \\ 4 & 4 & 6 & 4 & 3 & 11 & 4 \\ 4 & 6 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 27 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 5 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 27 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 11 & 3 \\ 4 & 4 & 6 & 4 & 3 & 16 & 4 \\ 4 & 6 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 31 & 5 \\ 7 & 8 & 6 & 5 & 38 & 6 \\ 5 & 6 & 2 & 3 & 24 & 4 \\ 4 & 5 & 3 & 1 & 20 & 3 \\ 31 & 38 & 24 & 20 & 149 & 25 \\ 5 & 6 & 4 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

5.126 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 9 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 12 & 6 \\ 11 & 15 & 22 & 8 & 17 & 18 & 10 \\ 4 & 5 & 8 & 1 & 6 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 14 & 7 \\ 9 & 12 & 18 & 6 & 14 & 14 & 8 \\ 5 & 6 & 10 & 3 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 9 & 17 & 5 \\ 7 & 8 & 6 & 5 & 11 & 21 & 6 \\ 4 & 6 & 2 & 3 & 7 & 13 & 4 \\ 4 & 5 & 3 & 1 & 6 & 11 & 3 \\ 9 & 11 & 7 & 6 & 11 & 24 & 7 \\ 17 & 21 & 13 & 11 & 24 & 44 & 14 \\ 5 & 6 & 4 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 9 & 17 & 5 \\ 7 & 8 & 6 & 5 & 11 & 21 & 6 \\ 5 & 6 & 2 & 3 & 7 & 13 & 4 \\ 4 & 5 & 3 & 1 & 6 & 11 & 3 \\ 9 & 11 & 7 & 6 & 11 & 24 & 7 \\ 17 & 21 & 13 & 11 & 24 & 44 & 14 \\ 5 & 6 & 4 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

5.127 NHC(-2)(-3)(-2) + su₂ + su₂ → (-3) + su_{2n3mix} + su₂ + su₂

$$T = 10 \quad [1] \ 1^{1121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.0$$

$$[2] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.0$$

$$[3] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.5$$

$$[0] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.5$$

$$[4] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.0$$

$$[5] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.0$$

5.128 NHC(-2)(-3)(-2) + su₂ + su_{2n3mix}

$$T = 8 \quad [0] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 9 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 19 & 24 & 62 & 27 & 16 \\ 9 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 7 & 8 & 5 \\ 7 & 8 & 9 & 11 & 6 \\ 7 & 9 & 8 & 10 & 6 \\ 8 & 11 & 10 & 11 & 7 \\ 5 & 6 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 21 & 9 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 21 & 24 & 62 & 27 & 16 \\ 9 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 8 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 19 & 24 & 62 & 27 & 16 \\ 8 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 17, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 7 & 9 & 5 \\ 7 & 8 & 9 & 11 & 6 \\ 7 & 9 & 8 & 10 & 6 \\ 9 & 11 & 10 & 11 & 7 \\ 5 & 6 & 6 & 7 & 4 \end{pmatrix}_{b_0} \\
[5] & 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 9 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 11 & 15 & 22 & 17 & 10 \\ 9 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[6] & 1^{124} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 31 & 5 \\ 7 & 8 & 12 & 38 & 6 \\ 11 & 12 & 14 & 48 & 8 \\ 31 & 38 & 48 & 149 & 25 \\ 5 & 6 & 8 & 25 & 4 \end{pmatrix}_{b_0} \\
[7] & 1^{133} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 26 & 5 \\ 7 & 8 & 18 & 32 & 6 \\ 15 & 18 & 34 & 60 & 12 \\ 26 & 32 & 60 & 105 & 21 \\ 5 & 6 & 12 & 21 & 4 \end{pmatrix}_{b_0} \\
[8] & 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 22 & 5 \\ 7 & 8 & 12 & 26 & 6 \\ 11 & 12 & 14 & 32 & 8 \\ 22 & 26 & 32 & 69 & 17 \\ 5 & 6 & 8 & 17 & 4 \end{pmatrix}_{b_0} \\
[9] & 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 13 & 5 \\ 7 & 8 & 15 & 17 & 6 \\ 11 & 15 & 22 & 26 & 10 \\ 13 & 17 & 26 & 29 & 11 \\ 5 & 6 & 10 & 11 & 4 \end{pmatrix}_{b_0} \\
[13] & 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 8 & 5 \\ 7 & 8 & 12 & 11 & 6 \\ 10 & 12 & 14 & 14 & 8 \\ 8 & 11 & 14 & 11 & 7 \\ 5 & 6 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[12] & 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 8 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 5 & 6 & 2 & 7 & 4 \\ 8 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[11] & 1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 8 & 5 \\ 7 & 8 & 18 & 11 & 6 \\ 15 & 18 & 34 & 21 & 12 \\ 8 & 11 & 21 & 11 & 7 \\ 5 & 6 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[10] & 1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 8 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 20 & 24 & 62 & 28 & 16 \\ 8 & 11 & 28 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
[15] & 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 14 & 9 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 14 & 15 & 22 & 17 & 10 \\ 9 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[14] & 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 19 & 14 & 5 \\ 7 & 8 & 21 & 17 & 6 \\ 19 & 21 & 44 & 37 & 14 \\ 14 & 17 & 37 & 29 & 11 \\ 5 & 6 & 14 & 11 & 4 \end{pmatrix}_{b_0} \\
[16] & 1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 21 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 6 & 6 & 2 & 16 & 4 \\ 21 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[17] & 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 9 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 4 & 6 & 2 & 7 & 4 \\ 9 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[18] & 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 21 & 10 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 21 & 24 & 62 & 27 & 16 \\ 10 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
[19] & 1^{133} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 27 & 5 \\ 7 & 8 & 18 & 32 & 6 \\ 15 & 18 & 34 & 60 & 12 \\ 27 & 32 & 60 & 105 & 21 \\ 5 & 6 & 12 & 21 & 4 \end{pmatrix}_{b_0} \\
[21] & 21^{123} 3^1 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 21 & 4 \\ 5 & 4 & 6 & 4 & 21 & 4 \\ 5 & 6 & 4 & 4 & 21 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 21 & 21 & 21 & 14 & 80 & 16 \\ 4 & 4 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0} \\
[20] & 21^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 12 & 4 \\ 5 & 4 & 6 & 4 & 12 & 4 \\ 5 & 6 & 4 & 4 & 12 & 4 \\ 3 & 4 & 4 & 1 & 8 & 3 \\ 12 & 12 & 12 & 8 & 25 & 9 \\ 4 & 4 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[22] & 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 9 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 6 & 6 & 2 & 7 & 4 \\ 9 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[23] & 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 9 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 20 & 24 & 62 & 27 & 16 \\ 9 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
[24] & 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 12 & 8 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 12 & 15 & 22 & 17 & 10 \\ 8 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[25] & 1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 14 & 15 & 5 \\ 7 & 8 & 15 & 17 & 6 \\ 14 & 15 & 22 & 26 & 10 \\ 15 & 17 & 26 & 29 & 11 \\ 5 & 6 & 10 & 11 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [26] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 16 & 5 \\ 7 & 8 & 6 & 20 & 6 \\ 4 & 6 & 2 & 12 & 4 \\ 16 & 20 & 12 & 41 & 13 \\ 5 & 6 & 4 & 13 & 4 \end{pmatrix}_{b_0}$
- [27] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 13 & 14 & 5 \\ 7 & 8 & 15 & 17 & 6 \\ 13 & 15 & 22 & 26 & 10 \\ 14 & 17 & 26 & 29 & 11 \\ 5 & 6 & 10 & 11 & 4 \end{pmatrix}_{b_0}$
- [28] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 17 & 5 \\ 7 & 8 & 6 & 20 & 6 \\ 6 & 6 & 2 & 12 & 4 \\ 17 & 20 & 12 & 41 & 13 \\ 5 & 6 & 4 & 13 & 4 \end{pmatrix}_{b_0}$
- [30] $1^{121} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 11 & 9 & 5 \\ 7 & 8 & 12 & 11 & 6 \\ 11 & 12 & 14 & 14 & 8 \\ 9 & 11 & 14 & 11 & 7 \\ 5 & 6 & 8 & 7 & 4 \end{pmatrix}_{b_0}$
- [29] $1^{112} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 14 & 5 \\ 7 & 8 & 6 & 17 & 6 \\ 4 & 6 & 2 & 11 & 4 \\ 14 & 17 & 11 & 29 & 11 \\ 5 & 6 & 4 & 11 & 4 \end{pmatrix}_{b_0}$
- [31] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 12 & 14 & 5 \\ 7 & 8 & 15 & 17 & 6 \\ 12 & 15 & 22 & 26 & 10 \\ 14 & 17 & 26 & 29 & 11 \\ 5 & 6 & 10 & 11 & 4 \end{pmatrix}_{b_0}$
- [33] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 38 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 6 & 6 & 2 & 29 & 4 \\ 38 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0}$
- [32] $1^{112} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 14 & 5 \\ 7 & 8 & 6 & 17 & 6 \\ 6 & 6 & 2 & 11 & 4 \\ 14 & 17 & 11 & 29 & 11 \\ 5 & 6 & 4 & 11 & 4 \end{pmatrix}_{b_0}$
- [37] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 5 & 9 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 5 & 6 & 2 & 7 & 4 \\ 9 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0}$
- [36] $1^{141} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 20 & 10 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 20 & 24 & 62 & 28 & 16 \\ 10 & 11 & 28 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$
- [35] $1^{121} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 10 & 10 & 5 \\ 7 & 8 & 12 & 11 & 6 \\ 10 & 12 & 14 & 14 & 8 \\ 10 & 11 & 14 & 11 & 7 \\ 5 & 6 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[34] & 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 10 & 5 \\ 7 & 8 & 18 & 11 & 6 \\ 15 & 18 & 34 & 21 & 12 \\ 10 & 11 & 21 & 11 & 7 \\ 5 & 6 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[38] & 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 10 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 5 & 6 & 2 & 7 & 4 \\ 10 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[39] & 1^{12} 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 17 & 8 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 17 & 21 & 44 & 24 & 14 \\ 8 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0} \\
[40] & 1^{12} 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 24 & 8 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 24 & 30 & 94 & 34 & 20 \\ 8 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0} \\
[41] & 1^{141} 1^{131} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 9 & 5 \\ 7 & 8 & 18 & 11 & 6 \\ 16 & 18 & 34 & 21 & 12 \\ 9 & 11 & 21 & 11 & 7 \\ 5 & 6 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[42] & 1^{132} 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 14 & 5 \\ 7 & 8 & 12 & 17 & 6 \\ 11 & 12 & 14 & 22 & 8 \\ 14 & 17 & 22 & 29 & 11 \\ 5 & 6 & 8 & 11 & 4 \end{pmatrix}_{b_0} \\
[43] & 1^{121} 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 9 & 5 \\ 7 & 8 & 12 & 11 & 6 \\ 10 & 12 & 14 & 14 & 8 \\ 9 & 11 & 14 & 11 & 7 \\ 5 & 6 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[44] & 1^{122} 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 21 & 5 \\ 7 & 8 & 12 & 26 & 6 \\ 10 & 12 & 14 & 32 & 8 \\ 21 & 26 & 32 & 69 & 17 \\ 5 & 6 & 8 & 17 & 4 \end{pmatrix}_{b_0} \\
[45] & 1^{11} 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 20 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 4 & 6 & 2 & 16 & 4 \\ 20 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[46] & 1^{13} 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 22 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 4 & 6 & 2 & 16 & 4 \\ 22 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[47] & 1^{111} 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 12 & 9 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 12 & 15 & 22 & 17 & 10 \\ 9 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[48] & 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 16 & 5 \\ 7 & 8 & 6 & 20 & 6 \\ 5 & 6 & 2 & 12 & 4 \\ 16 & 20 & 12 & 41 & 13 \\ 5 & 6 & 4 & 13 & 4 \end{pmatrix}_{b_0} \\
[49] & 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 19 & 5 \\ 7 & 8 & 6 & 23 & 6 \\ 4 & 6 & 2 & 15 & 4 \\ 19 & 23 & 15 & 55 & 15 \\ 5 & 6 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[50] & 1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 26 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 6 & 6 & 2 & 20 & 4 \\ 26 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0} \\
[51] & 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 13 & 9 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 13 & 15 & 22 & 17 & 10 \\ 9 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[52] & 1^{134} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 23 & 5 \\ 7 & 8 & 12 & 26 & 6 \\ 11 & 12 & 14 & 32 & 8 \\ 23 & 26 & 32 & 69 & 17 \\ 5 & 6 & 8 & 17 & 4 \end{pmatrix}_{b_0} \\
[53] & 1^{123} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 19 & 5 \\ 7 & 8 & 6 & 23 & 6 \\ 6 & 6 & 2 & 15 & 4 \\ 19 & 23 & 15 & 55 & 15 \\ 5 & 6 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[54] & 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 13 & 10 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 13 & 15 & 22 & 17 & 10 \\ 10 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[55] & 1^{132} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 29 & 14 & 5 \\ 7 & 8 & 36 & 17 & 6 \\ 29 & 36 & 134 & 63 & 24 \\ 14 & 17 & 63 & 29 & 11 \\ 5 & 6 & 24 & 11 & 4 \end{pmatrix}_{b_0} \\
[56] & 1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 9 & 5 \\ 7 & 8 & 18 & 11 & 6 \\ 15 & 18 & 34 & 21 & 12 \\ 9 & 11 & 21 & 11 & 7 \\ 5 & 6 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[57] & 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 21 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 4 & 6 & 2 & 16 & 4 \\ 21 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[58] \quad & 1^{1^{142}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 14 & 5 \\ 7 & 8 & 18 & 17 & 6 \\ 16 & 18 & 34 & 33 & 12 \\ 14 & 17 & 33 & 29 & 11 \\ 5 & 6 & 12 & 11 & 4 \end{pmatrix}_{b_0} \\
[59] \quad & 1^{1^{13}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 22 & 8 & 5 \\ 7 & 8 & 27 & 11 & 6 \\ 22 & 27 & 74 & 31 & 18 \\ 8 & 11 & 31 & 11 & 7 \\ 5 & 6 & 18 & 7 & 4 \end{pmatrix}_{b_0} \\
[61] \quad & 1^{1^{134}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 27 & 5 \\ 7 & 8 & 12 & 32 & 6 \\ 11 & 12 & 14 & 40 & 8 \\ 27 & 32 & 40 & 105 & 21 \\ 5 & 6 & 8 & 21 & 4 \end{pmatrix}_{b_0} \\
[60] \quad & 1^{1^{123}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 22 & 5 \\ 7 & 8 & 12 & 26 & 6 \\ 10 & 12 & 14 & 32 & 8 \\ 22 & 26 & 32 & 69 & 17 \\ 5 & 6 & 8 & 17 & 4 \end{pmatrix}_{b_0} \\
[62] \quad & 1^{1^{133}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 19 & 5 \\ 7 & 8 & 12 & 23 & 6 \\ 11 & 12 & 14 & 30 & 8 \\ 19 & 23 & 30 & 55 & 15 \\ 5 & 6 & 8 & 15 & 4 \end{pmatrix}_{b_0} \\
[63] \quad & 1^{1^{112}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 17 & 5 \\ 7 & 8 & 6 & 20 & 6 \\ 5 & 6 & 2 & 12 & 4 \\ 17 & 20 & 12 & 41 & 13 \\ 5 & 6 & 4 & 13 & 4 \end{pmatrix}_{b_0} \\
[64] \quad & 1^{1^{112}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 14 & 5 \\ 7 & 8 & 6 & 17 & 6 \\ 5 & 6 & 2 & 11 & 4 \\ 14 & 17 & 11 & 29 & 11 \\ 5 & 6 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
[65] \quad & 1^{1^{141}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 19 & 9 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 19 & 21 & 44 & 24 & 14 \\ 9 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0} \\
[66] \quad & 1^{1^{123}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 22 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 6 & 6 & 2 & 16 & 4 \\ 22 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[67] \quad & 1^{1^{113}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 15 & 5 \\ 7 & 8 & 6 & 17 & 6 \\ 5 & 6 & 2 & 11 & 4 \\ 15 & 17 & 11 & 29 & 11 \\ 5 & 6 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
[68] \quad & 1^{1^{133}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 15 & 15 & 5 \\ 7 & 8 & 18 & 17 & 6 \\ 15 & 18 & 34 & 33 & 12 \\ 15 & 17 & 33 & 29 & 11 \\ 5 & 6 & 12 & 11 & 4 \end{pmatrix}_{b_0} \\
[69] & 1^{142} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 15 & 5 \\ 7 & 8 & 24 & 17 & 6 \\ 20 & 24 & 62 & 44 & 16 \\ 15 & 17 & 44 & 29 & 11 \\ 5 & 6 & 16 & 11 & 4 \end{pmatrix}_{b_0} \\
[70] & 1^{123} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 15 & 5 \\ 7 & 8 & 12 & 17 & 6 \\ 10 & 12 & 14 & 22 & 8 \\ 15 & 17 & 22 & 29 & 11 \\ 5 & 6 & 8 & 11 & 4 \end{pmatrix}_{b_0} \\
[72] & 1^{14} 1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 24 & 5 \\ 7 & 8 & 6 & 29 & 6 \\ 4 & 6 & 2 & 19 & 4 \\ 24 & 29 & 19 & 89 & 19 \\ 5 & 6 & 4 & 19 & 4 \end{pmatrix}_{b_0} \\
[71] & 1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 28 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 4 & 6 & 2 & 21 & 4 \\ 28 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0} \\
[73] & 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 14 & 5 \\ 7 & 8 & 12 & 17 & 6 \\ 10 & 12 & 14 & 22 & 8 \\ 14 & 17 & 22 & 29 & 11 \\ 5 & 6 & 8 & 11 & 4 \end{pmatrix}_{b_0} \\
[74] & 1^{141} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 9 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 20 & 24 & 62 & 28 & 16 \\ 9 & 11 & 28 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
[75] & 1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 34 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 6 & 6 & 2 & 25 & 4 \\ 34 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0} \\
[76] & 21^{113} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 15 & 4 \\ 5 & 4 & 6 & 4 & 16 & 4 \\ 5 & 6 & 4 & 4 & 16 & 4 \\ 3 & 4 & 4 & 1 & 11 & 3 \\ 15 & 16 & 16 & 11 & 44 & 12 \\ 4 & 4 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0} \\
[79] & 1^{133} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 19 & 5 \\ 7 & 8 & 18 & 23 & 6 \\ 16 & 18 & 34 & 45 & 12 \\ 19 & 23 & 45 & 55 & 15 \\ 5 & 6 & 12 & 15 & 4 \end{pmatrix}_{b_0} \\
[77] & 1^{124} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 24 & 5 \\ 7 & 8 & 6 & 29 & 6 \\ 6 & 6 & 2 & 19 & 4 \\ 24 & 29 & 19 & 89 & 19 \\ 5 & 6 & 4 & 19 & 4 \end{pmatrix}_{b_0} \\
[78] & 1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 30 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 6 & 6 & 2 & 21 & 4 \\ 30 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

 1^{14}

$$[80] 1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 27 & 8 & 5 \\ 7 & 8 & 33 & 11 & 6 \\ 27 & 33 & 112 & 38 & 22 \\ 8 & 11 & 38 & 11 & 7 \\ 5 & 6 & 22 & 7 & 4 \end{pmatrix}_{b_0}$$

 1^{13}

$$[81] 1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 29 & 8 & 5 \\ 7 & 8 & 36 & 11 & 6 \\ 29 & 36 & 134 & 41 & 24 \\ 8 & 11 & 41 & 11 & 7 \\ 5 & 6 & 24 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[82] 21^{11} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 6 & 4 \\ 5 & 4 & 6 & 13 & 7 & 4 \\ 5 & 6 & 4 & 13 & 7 & 4 \\ 12 & 13 & 13 & 28 & 16 & 10 \\ 6 & 7 & 7 & 16 & 7 & 5 \\ 4 & 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^{14}

$$[83] 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 27 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 4 & 6 & 2 & 20 & 4 \\ 27 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

 1^{12}

$$[84] 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 25 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 4 & 6 & 2 & 20 & 4 \\ 25 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

 1^{132}

$$[85] 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 15 & 14 & 5 \\ 7 & 8 & 18 & 17 & 6 \\ 15 & 18 & 34 & 33 & 12 \\ 14 & 17 & 33 & 29 & 11 \\ 5 & 6 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

 1^{121}

$$[86] 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 24 & 9 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 24 & 30 & 94 & 34 & 20 \\ 9 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

 1^{132}

$$[87] 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 14 & 5 \\ 7 & 8 & 21 & 17 & 6 \\ 18 & 21 & 44 & 37 & 14 \\ 14 & 17 & 37 & 29 & 11 \\ 5 & 6 & 14 & 11 & 4 \end{pmatrix}_{b_0}$$

 1^{134}

$$[88] 1^{124} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 24 & 5 \\ 7 & 8 & 12 & 29 & 6 \\ 11 & 12 & 14 & 38 & 8 \\ 24 & 29 & 38 & 89 & 19 \\ 5 & 6 & 8 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[89] 21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 8 & 3 \\ 4 & 4 & 6 & 4 & 12 & 4 \\ 4 & 6 & 4 & 4 & 12 & 4 \\ 3 & 4 & 4 & 1 & 8 & 3 \\ 8 & 12 & 12 & 8 & 25 & 9 \\ 3 & 4 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[90] 21^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 14 & 3 \\ 4 & 4 & 6 & 4 & 21 & 4 \\ 4 & 6 & 4 & 4 & 21 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 14 & 21 & 21 & 14 & 80 & 16 \\ 3 & 4 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 17 & 14 & 5 \\ 7 & 8 & 21 & 17 & 6 \\ 17 & 21 & 44 & 37 & 14 \\ 14 & 17 & 37 & 29 & 11 \\ 5 & 6 & 14 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 15 & 31 & 5 \\ 7 & 8 & 18 & 38 & 6 \\ 15 & 18 & 34 & 72 & 12 \\ 31 & 38 & 72 & 149 & 25 \\ 5 & 6 & 12 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 31 & 5 \\ 7 & 8 & 6 & 38 & 6 \\ 6 & 6 & 2 & 24 & 4 \\ 31 & 38 & 24 & 149 & 25 \\ 5 & 6 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 26 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 4 & 6 & 2 & 20 & 4 \\ 26 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[95] \quad 21^{113} 3^1 1^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 18 & 4 \\ 5 & 4 & 6 & 4 & 20 & 4 \\ 5 & 6 & 4 & 4 & 20 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 18 & 20 & 20 & 14 & 69 & 15 \\ 4 & 4 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 17 & 9 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 17 & 21 & 44 & 24 & 14 \\ 9 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 10 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 18 & 21 & 44 & 24 & 14 \\ 10 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 26 & 9 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 26 & 30 & 94 & 34 & 20 \\ 9 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 15 & 5 \\ 7 & 8 & 21 & 17 & 6 \\ 19 & 21 & 44 & 37 & 14 \\ 15 & 17 & 37 & 29 & 11 \\ 5 & 6 & 14 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 23 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 6 & 6 & 2 & 16 & 4 \\ 23 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

[100] $1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 18 & 9 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 18 & 21 & 44 & 24 & 14 \\ 9 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0}$$

[102] $1^{134} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 16 & 24 & 5 \\ 7 & 8 & 18 & 29 & 6 \\ 16 & 18 & 34 & 57 & 12 \\ 24 & 29 & 57 & 89 & 19 \\ 5 & 6 & 12 & 19 & 4 \end{pmatrix}_{b_0}$$

[104] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 26 & 10 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 26 & 30 & 94 & 34 & 20 \\ 10 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

[103] $1^{142} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 20 & 14 & 5 \\ 7 & 8 & 24 & 17 & 6 \\ 20 & 24 & 62 & 44 & 16 \\ 14 & 17 & 44 & 29 & 11 \\ 5 & 6 & 16 & 11 & 4 \end{pmatrix}_{b_0}$$

[105] $1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 19 & 5 \\ 7 & 8 & 6 & 23 & 6 \\ 5 & 6 & 2 & 15 & 4 \\ 19 & 23 & 15 & 55 & 15 \\ 5 & 6 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[106] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 21 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 5 & 6 & 2 & 16 & 4 \\ 21 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

[107] $1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 25 & 9 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 25 & 30 & 94 & 34 & 20 \\ 9 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

[108] $1^{124} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 38 & 5 \\ 7 & 8 & 12 & 47 & 6 \\ 10 & 12 & 14 & 58 & 8 \\ 38 & 47 & 58 & 229 & 31 \\ 5 & 6 & 8 & 31 & 4 \end{pmatrix}_{b_0}$$

[109] $1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 19 & 5 \\ 7 & 8 & 12 & 23 & 6 \\ 10 & 12 & 14 & 30 & 8 \\ 19 & 23 & 30 & 55 & 15 \\ 5 & 6 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 32 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 4 & 6 & 2 & 25 & 4 \\ 32 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0}$$

[111] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 34 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 4 & 6 & 2 & 25 & 4 \\ 34 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0}$$

[114] $1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 20 & 5 \\ 7 & 8 & 6 & 23 & 6 \\ 5 & 6 & 2 & 15 & 4 \\ 20 & 23 & 15 & 55 & 15 \\ 5 & 6 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[115] $1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 20 & 5 \\ 7 & 8 & 12 & 23 & 6 \\ 10 & 12 & 14 & 30 & 8 \\ 20 & 23 & 30 & 55 & 15 \\ 5 & 6 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

[116] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 28 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 5 & 6 & 2 & 21 & 4 \\ 28 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

[117] $1^{133} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 15 & 20 & 5 \\ 7 & 8 & 18 & 23 & 6 \\ 15 & 18 & 34 & 45 & 12 \\ 20 & 23 & 45 & 55 & 15 \\ 5 & 6 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[112] $21^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 5 & 3 \\ 4 & 4 & 6 & 18 & 7 & 4 \\ 4 & 6 & 4 & 18 & 7 & 4 \\ 12 & 18 & 18 & 58 & 22 & 14 \\ 5 & 7 & 7 & 22 & 7 & 5 \\ 3 & 4 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{143} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 20 & 20 & 5 \\ 7 & 8 & 24 & 23 & 6 \\ 20 & 24 & 62 & 60 & 16 \\ 20 & 23 & 60 & 55 & 15 \\ 5 & 6 & 16 & 15 & 4 \end{pmatrix}_{b_0}$$

[118] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 6 & 27 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 6 & 6 & 2 & 20 & 4 \\ 27 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[119] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 26 & 5 \\ 7 & 8 & 12 & 32 & 6 \\ 10 & 12 & 14 & 40 & 8 \\ 26 & 32 & 40 & 105 & 21 \\ 5 & 6 & 8 & 21 & 4 \end{pmatrix}_{b_0}$$

[120] $1^{141} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 34 & 8 & 5 \\ 7 & 8 & 42 & 11 & 6 \\ 34 & 42 & 182 & 48 & 28 \\ 8 & 11 & 48 & 11 & 7 \\ 5 & 6 & 28 & 7 & 4 \end{pmatrix}_{b_0}$$

[121] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 30 & 5 \\ 7 & 8 & 6 & 38 & 6 \\ 4 & 6 & 2 & 24 & 4 \\ 30 & 38 & 24 & 149 & 25 \\ 5 & 6 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[122] $1^{143} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$\begin{aligned}
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 23 & 19 & 5 \\ 7 & 8 & 27 & 23 & 6 \\ 23 & 27 & 74 & 65 & 18 \\ 19 & 23 & 65 & 55 & 15 \\ 5 & 6 & 18 & 15 & 4 \end{pmatrix}_{b_0} \\
[123] & 21^{1122} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 11 & 3 \\ 4 & 4 & 6 & 8 & 16 & 4 \\ 4 & 6 & 4 & 8 & 16 & 4 \\ 6 & 8 & 4 & 10 & 22 & 6 \\ 11 & 16 & 16 & 22 & 44 & 12 \\ 3 & 4 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0} \\
[124] & 1^{133} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 19 & 5 \\ 7 & 8 & 18 & 23 & 6 \\ 15 & 18 & 34 & 45 & 12 \\ 19 & 23 & 45 & 55 & 15 \\ 5 & 6 & 12 & 15 & 4 \end{pmatrix}_{b_0} \\
[125] & 1^{113} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 33 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 4 & 6 & 2 & 25 & 4 \\ 33 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0} \\
[126] & 21^{1111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 5 & 3 \\ 4 & 4 & 6 & 9 & 7 & 4 \\ 4 & 6 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 5 & 7 & 7 & 11 & 7 & 5 \\ 3 & 4 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[127] & 1^{112} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 23 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 5 & 6 & 2 & 16 & 4 \\ 23 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[128] & 1^{113} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 22 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 5 & 6 & 2 & 16 & 4 \\ 22 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[129] & 21^{123} 3^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 24 & 4 \\ 5 & 4 & 6 & 4 & 25 & 4 \\ 5 & 6 & 4 & 4 & 25 & 4 \\ 3 & 4 & 4 & 1 & 17 & 3 \\ 24 & 25 & 25 & 17 & 111 & 19 \\ 4 & 4 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0} \\
[130] & 1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 31 & 5 \\ 7 & 8 & 6 & 38 & 6 \\ 4 & 6 & 2 & 24 & 4 \\ 31 & 38 & 24 & 149 & 25 \\ 5 & 6 & 4 & 25 & 4 \end{pmatrix}_{b_0} \\
[131] & 1^{112} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 36 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 4 & 6 & 2 & 29 & 4 \\ 36 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0} \\
[132] & 1^{123} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 27 & 5 \\ 7 & 8 & 12 & 32 & 6 \\ 10 & 12 & 14 & 40 & 8 \\ 27 & 32 & 40 & 105 & 21 \\ 5 & 6 & 8 & 21 & 4 \end{pmatrix}_{b_0} \\
[133] & 21^{1111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 5 & 3 \\ 4 & 4 & 6 & 4 & 7 & 4 \\ 4 & 6 & 4 & 4 & 7 & 4 \\ 3 & 4 & 4 & 1 & 5 & 3 \\ 5 & 7 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{133} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 19 & 5 \\ 7 & 8 & 27 & 23 & 6 \\ 22 & 27 & 74 & 65 & 18 \\ 19 & 23 & 65 & 55 & 15 \\ 5 & 6 & 18 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{142} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 10 & 5 \\ 7 & 8 & 27 & 11 & 6 \\ 23 & 27 & 74 & 31 & 18 \\ 10 & 11 & 31 & 11 & 7 \\ 5 & 6 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{143} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 20 & 19 & 5 \\ 7 & 8 & 24 & 23 & 6 \\ 20 & 24 & 62 & 60 & 16 \\ 19 & 23 & 60 & 55 & 15 \\ 5 & 6 & 16 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{131} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 9 & 5 \\ 7 & 8 & 27 & 11 & 6 \\ 22 & 27 & 74 & 31 & 18 \\ 9 & 11 & 31 & 11 & 7 \\ 5 & 6 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{142} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 14 & 5 \\ 7 & 8 & 27 & 17 & 6 \\ 23 & 27 & 74 & 48 & 18 \\ 14 & 17 & 48 & 29 & 11 \\ 5 & 6 & 18 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{114} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 24 & 5 \\ 7 & 8 & 6 & 29 & 6 \\ 5 & 6 & 2 & 19 & 4 \\ 24 & 29 & 19 & 89 & 19 \\ 5 & 6 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[141] \ 21^{1121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 5 & 3 \\ 4 & 4 & 6 & 8 & 7 & 4 \\ 4 & 6 & 4 & 8 & 7 & 4 \\ 6 & 8 & 8 & 10 & 10 & 6 \\ 5 & 7 & 7 & 10 & 7 & 5 \\ 3 & 4 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{113} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 29 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 5 & 6 & 2 & 21 & 4 \\ 29 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{132} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 14 & 5 \\ 7 & 8 & 27 & 17 & 6 \\ 22 & 27 & 74 & 48 & 18 \\ 14 & 17 & 48 & 29 & 11 \\ 5 & 6 & 18 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{131} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 9 & 5 \\ 7 & 8 & 27 & 11 & 6 \\ 23 & 27 & 74 & 31 & 18 \\ 9 & 11 & 31 & 11 & 7 \\ 5 & 6 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

[144] $1^{124} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 24 & 5 \\ 7 & 8 & 12 & 29 & 6 \\ 10 & 12 & 14 & 38 & 8 \\ 24 & 29 & 38 & 89 & 19 \\ 5 & 6 & 8 & 19 & 4 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 30 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 5 & 6 & 2 & 21 & 4 \\ 30 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

[146] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 37 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 4 & 6 & 2 & 29 & 4 \\ 37 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0}$$

[147] $1^{134} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 15 & 24 & 5 \\ 7 & 8 & 18 & 29 & 6 \\ 15 & 18 & 34 & 57 & 12 \\ 24 & 29 & 57 & 89 & 19 \\ 5 & 6 & 12 & 19 & 4 \end{pmatrix}_{b_0}$$

[148] $1^{142} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 34 & 14 & 5 \\ 7 & 8 & 42 & 17 & 6 \\ 34 & 42 & 182 & 74 & 28 \\ 14 & 17 & 74 & 29 & 11 \\ 5 & 6 & 28 & 11 & 4 \end{pmatrix}_{b_0}$$

[149] $21^{131} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 5 & 3 \\ 4 & 4 & 6 & 12 & 7 & 4 \\ 4 & 6 & 4 & 12 & 7 & 4 \\ 9 & 12 & 12 & 25 & 15 & 9 \\ 5 & 7 & 7 & 15 & 7 & 5 \\ 3 & 4 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[150] $1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 29 & 9 & 5 \\ 7 & 8 & 36 & 11 & 6 \\ 29 & 36 & 134 & 41 & 24 \\ 9 & 11 & 41 & 11 & 7 \\ 5 & 6 & 24 & 7 & 4 \end{pmatrix}_{b_0}$$

[151] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 314, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 26 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 5 & 6 & 2 & 20 & 4 \\ 26 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[152] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 38 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 4 & 6 & 2 & 29 & 4 \\ 38 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0}$$

[153] $1^{144} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 20 & 24 & 5 \\ 7 & 8 & 24 & 29 & 6 \\ 20 & 24 & 62 & 76 & 16 \\ 24 & 29 & 76 & 89 & 19 \\ 5 & 6 & 16 & 19 & 4 \end{pmatrix}_{b_0}$$

[154] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 342, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 10 & 31 & 5 \\ 7 & 8 & 12 & 38 & 6 \\ 10 & 12 & 14 & 48 & 8 \\ 31 & 38 & 48 & 149 & 25 \\ 5 & 6 & 8 & 25 & 4 \end{pmatrix}_{b_0} \\
[155] & 21^{1^{112}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 8 & 3 \\ 4 & 4 & 6 & 4 & 11 & 4 \\ 4 & 6 & 4 & 4 & 11 & 4 \\ 3 & 4 & 4 & 1 & 8 & 3 \\ 8 & 11 & 11 & 8 & 20 & 8 \\ 3 & 4 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0} \\
& 1^{141} \\
[156] & 1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 27 & 9 & 5 \\ 7 & 8 & 33 & 11 & 6 \\ 27 & 33 & 112 & 38 & 22 \\ 9 & 11 & 38 & 11 & 7 \\ 5 & 6 & 22 & 7 & 4 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[157] & 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 358, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 27 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 5 & 6 & 2 & 20 & 4 \\ 27 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0} \\
[160] & 21^{1^{141}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 12 & 5 & 3 \\ 4 & 4 & 6 & 16 & 7 & 4 \\ 4 & 6 & 4 & 16 & 7 & 4 \\ 12 & 16 & 16 & 46 & 20 & 12 \\ 5 & 7 & 7 & 20 & 7 & 5 \\ 3 & 4 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0} \\
[159] & 21^{1^{12}3^1 1^{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 27 & 4 \\ 5 & 4 & 6 & 4 & 29 & 4 \\ 5 & 6 & 4 & 4 & 29 & 4 \\ 3 & 4 & 4 & 1 & 20 & 3 \\ 27 & 29 & 29 & 20 & 148 & 22 \\ 4 & 4 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0} \\
[158] & 21^{1^{12}3^1 1^{141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 24 & 6 & 4 \\ 5 & 4 & 6 & 26 & 7 & 4 \\ 5 & 6 & 4 & 26 & 7 & 4 \\ 24 & 26 & 26 & 118 & 32 & 20 \\ 6 & 7 & 7 & 32 & 7 & 5 \\ 4 & 4 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0} \\
& 1^{143} \\
[161] & 1^{143} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 27 & 19 & 5 \\ 7 & 8 & 33 & 23 & 6 \\ 27 & 33 & 112 & 80 & 22 \\ 19 & 23 & 80 & 55 & 15 \\ 5 & 6 & 22 & 15 & 4 \end{pmatrix}_{b_0} \\
[162] & 21^{1^{122}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 8 & 3 \\ 4 & 4 & 6 & 8 & 11 & 4 \\ 4 & 6 & 4 & 8 & 11 & 4 \\ 6 & 8 & 8 & 10 & 16 & 6 \\ 8 & 11 & 11 & 16 & 20 & 8 \\ 3 & 4 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
& 1^{113} \\
[163] & 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 33 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 5 & 6 & 2 & 25 & 4 \\ 33 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0} \\
& 1^{142} \\
[164] & 1^{142} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 27 & 14 & 5 \\ 7 & 8 & 33 & 17 & 6 \\ 27 & 33 & 112 & 59 & 22 \\ 14 & 17 & 59 & 29 & 11 \\ 5 & 6 & 22 & 11 & 4 \end{pmatrix}_{b_0} \\
[165] & 21^{1^{122}3^1 1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 4 & 9 & 8 & 3 \\ 4 & 4 & 6 & 13 & 11 & 4 \\ 4 & 6 & 4 & 13 & 11 & 4 \\ 9 & 13 & 13 & 28 & 25 & 10 \\ 8 & 11 & 11 & 25 & 20 & 8 \\ 3 & 4 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0} \\
[166] \quad & 21^{11213^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 9 & 5 & 3 \\ 4 & 4 & 6 & 13 & 7 & 4 \\ 4 & 6 & 4 & 13 & 7 & 4 \\ 9 & 13 & 13 & 28 & 16 & 10 \\ 5 & 7 & 7 & 16 & 7 & 5 \\ 3 & 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0} \\
[168] \quad & 21^{11123^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 11 & 3 \\ 4 & 4 & 6 & 4 & 16 & 4 \\ 4 & 6 & 4 & 4 & 16 & 4 \\ 3 & 4 & 4 & 1 & 11 & 3 \\ 11 & 16 & 16 & 11 & 44 & 12 \\ 3 & 4 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0} \\
[167] \quad & 21^{11513^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 5 & 3 \\ 4 & 4 & 6 & 20 & 7 & 4 \\ 4 & 6 & 4 & 20 & 7 & 4 \\ 15 & 20 & 20 & 73 & 25 & 15 \\ 5 & 7 & 7 & 25 & 7 & 5 \\ 3 & 4 & 4 & 15 & 5 & 3 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[169] \quad & 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 34 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 5 & 6 & 2 & 25 & 4 \\ 34 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[170] \quad & 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 31 & 5 \\ 7 & 8 & 6 & 38 & 6 \\ 5 & 6 & 2 & 24 & 4 \\ 31 & 38 & 24 & 149 & 25 \\ 5 & 6 & 4 & 25 & 4 \end{pmatrix}_{b_0} \\
& 1^{141} \\
[171] \quad & 1^{141} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 34 & 9 & 5 \\ 7 & 8 & 42 & 11 & 6 \\ 34 & 42 & 182 & 48 & 28 \\ 9 & 11 & 48 & 11 & 7 \\ 5 & 6 & 28 & 7 & 4 \end{pmatrix}_{b_0} \\
[172] \quad & 21^{11133^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 11 & 3 \\ 4 & 4 & 6 & 4 & 15 & 4 \\ 4 & 6 & 4 & 4 & 15 & 4 \\ 3 & 4 & 4 & 1 & 11 & 3 \\ 11 & 15 & 15 & 11 & 39 & 11 \\ 3 & 4 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[173] \quad & 1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 38 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 5 & 6 & 2 & 29 & 4 \\ 38 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0} \\
[174] \quad & 21^{11313^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 5 & 3 \\ 4 & 4 & 6 & 22 & 7 & 4 \\ 4 & 6 & 4 & 22 & 7 & 4 \\ 15 & 22 & 22 & 85 & 27 & 17 \\ 5 & 7 & 7 & 27 & 7 & 5 \\ 3 & 4 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0} \\
[175] \quad & 21^{11133^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 17 & 3 \\ 4 & 4 & 6 & 4 & 25 & 4 \\ 4 & 6 & 4 & 4 & 25 & 4 \\ 3 & 4 & 4 & 1 & 17 & 3 \\ 17 & 25 & 25 & 17 & 111 & 19 \\ 3 & 4 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0} \\
[176] \quad & 21^{11313^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 4 & 12 & 5 & 3 \\ 4 & 4 & 6 & 17 & 7 & 4 \\ 4 & 6 & 4 & 17 & 7 & 4 \\ 12 & 17 & 17 & 49 & 21 & 13 \\ 5 & 7 & 7 & 21 & 7 & 5 \\ 3 & 4 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0} \\
[177] \quad & 21^{114}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 14 & 3 \\ 4 & 4 & 6 & 4 & 19 & 4 \\ 4 & 6 & 4 & 4 & 19 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 14 & 19 & 19 & 14 & 64 & 14 \\ 3 & 4 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0} \\
[178] \quad & 21^{113}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 14 & 3 \\ 4 & 4 & 6 & 4 & 20 & 4 \\ 4 & 6 & 4 & 4 & 20 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 14 & 20 & 20 & 14 & 69 & 15 \\ 3 & 4 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0} \\
[179] \quad & 21^{141}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 872, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 5 & 3 \\ 4 & 4 & 6 & 21 & 7 & 4 \\ 4 & 6 & 4 & 21 & 7 & 4 \\ 15 & 21 & 21 & 76 & 26 & 16 \\ 5 & 7 & 7 & 26 & 7 & 5 \\ 3 & 4 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0} \\
[180] \quad & 21^{141}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 1000, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 18 & 5 & 3 \\ 4 & 4 & 6 & 26 & 7 & 4 \\ 4 & 6 & 4 & 26 & 7 & 4 \\ 18 & 26 & 26 & 118 & 32 & 20 \\ 5 & 7 & 7 & 32 & 7 & 5 \\ 3 & 4 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0} \\
[181] \quad & 21^{114}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 1072, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 17 & 3 \\ 4 & 4 & 6 & 4 & 24 & 4 \\ 4 & 6 & 4 & 4 & 24 & 4 \\ 3 & 4 & 4 & 1 & 17 & 3 \\ 17 & 24 & 24 & 17 & 100 & 18 \\ 3 & 4 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0} \\
[182] \quad & 21^{114}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 1120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 20 & 3 \\ 4 & 4 & 6 & 4 & 29 & 4 \\ 4 & 6 & 4 & 4 & 29 & 4 \\ 3 & 4 & 4 & 1 & 20 & 3 \\ 20 & 29 & 29 & 20 & 148 & 22 \\ 3 & 4 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0} \\
T = 9 \quad [183] \quad & 1^{112}3^{11}21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 1 & 0 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 15 & 3 \\ 2 & 3 & 0 & 4 & 5 & 24 & 5 \\ 8 & 9 & 0 & 15 & 24 & 83 & 17 \\ 2 & 2 & -1 & 3 & 5 & 17 & 3 \end{pmatrix}_{b_0} \\
[184] \quad & 1^{12}3^{11}21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 7 & 7 & 9 & 0 & 9 & 35 & 11 \\ 2 & 2 & 2 & 0 & 3 & 11 & 3 \end{pmatrix}_{b_0} \\
[185] \quad & 1^{114}3^{11}21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 2 \\ 1 & 2 & 0 & 1 & 7 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 47 & 8 \\ 13 & 13 & 17 & 0 & 47 & 127 & 21 \\ 2 & 2 & 2 & 0 & 8 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[186] \begin{matrix} 1^{13} \\ 1^3 3^1 2^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \end{matrix} \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$[187] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{1}\textcolor{blue}{2}\textcolor{red}{3}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{s}\textcolor{red}{o}\textcolor{blue}{7} \cdot \textcolor{blue}{s}\textcolor{red}{u}_2] + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 36 & 8 \\ 10 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & 0 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[189] \text{ } 1^{14} 3^{12} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \, T_{\max} = 10, \, \Delta = -31, \, \det = -18, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 0 & 1 & 3 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 17 & 3 \\ 13 & 13 & 17 & 0 & 17 & 127 & 21 \\ 2 & 2 & 2 & 0 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{112} 3^1 21^{122} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{5}\textcolor{violet}{0}_7 \cdot \textcolor{teal}{5}\textcolor{teal}{u}_2] + \textcolor{orange}{5}\textcolor{orange}{u}_2 + \textcolor{red}{5}\textcolor{red}{u}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\left(\begin{array}{cccccc} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 1 & 7 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 25 & 8 \\ 7 & 7 & 9 & 0 & 25 & 35 & 11 \\ 2 & 2 & 2 & 0 & 8 & 11 & 3 \end{array} \right)_{b_0}$$

$$[196] \ 1^{111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & 0 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{111} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{507} \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 4 & 7 & 0 & 16 & 14 & 8 \\ 4 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & 0 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{\overline{111}} \ 3^{\overline{11}} \ 21^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 14 & 8 \\ 4 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & 0 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{\overline{111}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{s}07 \cdot \textcolor{blue}{s}u_2] + \textcolor{blue}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 6 \\ 2 & 2 & 2 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{111} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 4 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 6 \\ 2 & 2 & 2 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{121} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 14 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 8 & 14 & 0 & 70 & 28 & 16 \\ 4 & 4 & 5 & 0 & 28 & 10 & 6 \\ 2 & 2 & 2 & 0 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{121} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 5 & 0 & 13 & 13 & 7 \\ 4 & 4 & 5 & 0 & 13 & 10 & 6 \\ 2 & 2 & 2 & 0 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{112} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 7 & 2 \\ 7 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 16 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{113} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 10 & 2 \\ 10 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{112} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 7 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{121} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & 0 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{121} 3^1 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 14 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 8 & 14 & 0 & 70 & 28 & 16 \\ 4 & 4 & 5 & 0 & 28 & 10 & 6 \\ 2 & 2 & 2 & 0 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{111} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 14 & 8 \\ 4 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & 0 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 7 & 2 \\ 8 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{114} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 13 & 2 \\ 13 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 19 & 5 \\ 8 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{122} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 41 & 13 \\ 7 & 7 & 9 & 0 & 41 & 35 & 11 \\ 2 & 2 & 2 & 0 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 1 & 7 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 25 & 8 \\ 7 & 7 & 9 & 0 & 25 & 35 & 11 \\ 2 & 2 & 2 & 0 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 9 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & 0 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{121} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 4 & 4 & 5 & 0 & 13 & 10 & 6 \\ 2 & 2 & 2 & 0 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 11 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{121} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 32 & 8 \\ 9 & 8 & 12 & 0 & 32 & 56 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 9 & 10 & 0 & 9 & 51 & 13 \\ 2 & 2 & 2 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{113} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 10 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{122} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 16 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{114} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 10 & 2 \\ 11 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{113} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 9 & 10 & 0 & 9 & 51 & 13 \\ 2 & 2 & 2 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{122} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 23 & 7 \\ 7 & 7 & 9 & 0 & 23 & 35 & 11 \\ 2 & 2 & 2 & 0 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{123} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 36 & 8 \\ 10 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & 0 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{114} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 43 & 8 \\ 12 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -177, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 26 & 5 \\ 11 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{121} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 33 & 7 \\ 10 & 10 & 13 & 0 & 33 & 74 & 16 \\ 2 & 2 & 2 & 0 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 36 & 8 \\ 11 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & 0 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 11 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 12 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{141} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 8 & 9 & 0 & 40 & 22 & 12 \\ 4 & 4 & 5 & 0 & 22 & 10 & 6 \\ 2 & 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{121} 3^{121} 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{121} 3^{121} 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 10 & 10 & 10 & 0 & 58 & 26 & 14 \\ 4 & 4 & 5 & 0 & 26 & 10 & 6 \\ 2 & 2 & 2 & 0 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{114} 3^{121} 14 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -227, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 13 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{113}3^121^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 27 & 5 \\ 11 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

1^{123}

6.0
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 19 & 5 \\ 9 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{114} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 12 & 2 \\ 12 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{124} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 0 & 5 & 13 & 2 \\ 1 & 2 & 0 & 1 & 5 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 43 & 7 \\ 13 & 13 & 17 & 0 & 43 & 127 & 21 \\ 2 & 2 & 2 & 0 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{132} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 1 & 15 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 57 & 18 \\ 7 & 7 & 9 & 0 & 57 & 35 & 11 \\ 2 & 2 & 2 & 0 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{124} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 2 \\ 1 & 2 & 0 & 1 & 7 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 47 & 8 \\ 13 & 13 & 17 & 0 & 47 & 127 & 21 \\ 2 & 2 & 2 & 0 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{132} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 41 & 13 \\ 7 & 7 & 9 & 0 & 41 & 35 & 11 \\ 2 & 2 & 2 & 0 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 16 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{132} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 32 & 10 \\ 7 & 7 & 9 & 0 & 32 & 35 & 11 \\ 2 & 2 & 2 & 0 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 22 & 12 \\ 4 & 4 & 5 & 0 & 22 & 10 & 6 \\ 2 & 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{131} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 32 & 18 \\ 4 & 4 & 5 & 0 & 32 & 10 & 6 \\ 2 & 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{123} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 16 & 5 \\ 8 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 30 & 5 \\ 12 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{124} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{124} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -337, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 33 & 7 \\ 11 & 10 & 13 & 0 & 33 & 74 & 16 \\ 2 & 2 & 2 & 0 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[256] $1^{132} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 39 & 12 \\ 7 & 7 & 9 & 0 & 39 & 35 & 11 \\ 2 & 2 & 2 & 0 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 43 & 8 \\ 11 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 19 & 5 \\ 9 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{123} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -363, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 43 & 8 \\ 11 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{124} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 26 & 5 \\ 12 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{122} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 32 & 10 \\ 7 & 7 & 9 & 0 & 32 & 35 & 11 \\ 2 & 2 & 2 & 0 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{123} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 32 & 8 \\ 9 & 8 & 12 & 0 & 32 & 56 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{131} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 27 & 15 \\ 4 & 4 & 5 & 0 & 27 & 10 & 6 \\ 2 & 2 & 2 & 0 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{123} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 1 & 5 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 28 & 7 \\ 9 & 9 & 10 & 0 & 28 & 51 & 13 \\ 2 & 2 & 2 & 0 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{141} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 32 & 18 \\ 4 & 4 & 5 & 0 & 32 & 10 & 6 \\ 2 & 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{131} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 27 & 15 \\ 4 & 4 & 5 & 0 & 27 & 10 & 6 \\ 2 & 2 & 2 & 0 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[268] $1^{142} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -427, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 1 & 15 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 57 & 18 \\ 7 & 7 & 9 & 0 & 57 & 35 & 11 \\ 2 & 2 & 2 & 0 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{124} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -457, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 23 & 5 \\ 11 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{142} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 7 & 2 \\ 1 & 2 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 48 & 15 \\ 7 & 7 & 9 & 0 & 48 & 35 & 11 \\ 2 & 2 & 2 & 0 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{141} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 1 & 13 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 31 & 17 \\ 4 & 4 & 5 & 0 & 31 & 10 & 6 \\ 2 & 2 & 2 & 0 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{113} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{124} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 46 & 10 \\ 11 & 10 & 13 & 0 & 46 & 74 & 16 \\ 2 & 2 & 2 & 0 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{132} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 7 & 2 \\ 1 & 2 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 48 & 15 \\ 7 & 7 & 9 & 0 & 48 & 35 & 11 \\ 2 & 2 & 2 & 0 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 43 & 8 \\ 12 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[276] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 23 & 5 \\ 11 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 6 & 12 & 2 \\ 1 & 2 & 0 & 1 & 8 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 52 & 10 \\ 12 & 12 & 14 & 0 & 52 & 96 & 18 \\ 2 & 2 & 2 & 0 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{124} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 1 & 5 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 38 & 7 \\ 12 & 12 & 14 & 0 & 38 & 96 & 18 \\ 2 & 2 & 2 & 0 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -578, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 27 & 5 \\ 11 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{142} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -578, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 7 & 2 \\ 1 & 2 & 0 & 1 & 13 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 55 & 17 \\ 7 & 7 & 9 & 0 & 55 & 35 & 11 \\ 2 & 2 & 2 & 0 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{141} 3^1 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 4 & 2 \\ 1 & 2 & 0 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 36 & 20 \\ 4 & 4 & 5 & 0 & 36 & 10 & 6 \\ 2 & 2 & 2 & 0 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[282] $1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 46 & 10 \\ 10 & 10 & 13 & 0 & 46 & 74 & 16 \\ 2 & 2 & 2 & 0 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[283] $1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -632, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 26 & 5 \\ 12 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -687, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[285] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 6 & 11 & 2 \\ 1 & 2 & 0 & 1 & 8 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 54 & 10 \\ 12 & 11 & 16 & 0 & 54 & 103 & 19 \\ 2 & 2 & 2 & 0 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{142} 3^1 21^{142} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -762, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 7 & 2 \\ 1 & 0 & 2 & 0 & 12 & 7 & 2 \\ 1 & 2 & 0 & 1 & 16 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 64 & 20 \\ 7 & 7 & 9 & 0 & 64 & 35 & 11 \\ 2 & 2 & 2 & 0 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -842, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -938, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 13 & 2 \\ 1 & 0 & 2 & 0 & 6 & 13 & 2 \\ 1 & 2 & 0 & 1 & 8 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 60 & 10 \\ 13 & 13 & 17 & 0 & 60 & 127 & 21 \\ 2 & 2 & 2 & 0 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

5.129 NHC(-2)(-3)(-2) + su₂ + su_{2n3mix} + su₂

$T = 10$ [0] $1^{121} 3^1 21^{1121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 9 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 0 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) =$

$(1, 10, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 13 & 6 \\ 4 & 5 & 5 & 0 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{122} 3^1 21^{121} + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{5}0_7 \cdot \textcolor{green}{5}u_2] + \textcolor{brown}{5}u_2 + \textcolor{red}{5}u_2 + \textcolor{blue}{5}u_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 0 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{12} 3^{121} 21^{112} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{5}\textcolor{red}{0}_7 \cdot \textcolor{green}{5}\textcolor{blue}{u}_2] + \textcolor{orange}{5}\textcolor{blue}{u}_2 + \textcolor{red}{5}\textcolor{blue}{u}_2 + \textcolor{blue}{5}\textcolor{blue}{u}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 16 & 3 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 7 & 11 \\ 1 & 2 & 1 & 0 & 3 & 7 & 0 & 2 \\ 2 & 2 & 2 & 0 & 5 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^1 21^{112} + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{5}07 \cdot \textcolor{green}{5}u_2] + \textcolor{orange}{5}u_2 + \textcolor{red}{5}u_2 + \textcolor{blue}{5}u_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 18 & 6 \\ 5 & 6 & 8 & 0 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1111} 3^{121} 21^{1112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 9 & 11 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 14 & 6 \\ 5 & 4 & 7 & 0 & 11 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 5 & 6 & 8 & 3 \end{pmatrix} b_0$$

$$[5] \quad 1^{1111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 0 & 0 & 10 & 9 & 6 \\ 2 & 3 & 4 & 0 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{1112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

5.130 NHC(-2)(-3)(-2) + su₂ + su₂n3mix + su₂ + su₂

$$\begin{aligned}
T=10 \quad [0] \quad & 1^{1111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T=10, T_{\max}=10, \Delta=0, \det=0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}}=8.0 \\
[1] \quad & 1^{1112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T=10, T_{\max}=10, \Delta=0, \det=0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}}=8.5 \\
[2] \quad & 1^{1121} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T=10, T_{\max}=10, \Delta=0, \det=0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}}=8.0 \\
[3] \quad & 1^{1121} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T=10, T_{\max}=10, \Delta=0, \det=0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}}=8.5
\end{aligned}$$

5.131 NHC(-2)(-3)(-2) + su₂ + su₂n3mix + su₂ → (-3)

$$\begin{aligned}
T=9 \quad [1] \quad & 1^{1131} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-32, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 14 & 9 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 \\ 14 & 15 & 22 & 17 & 8 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 \\ 5 & 6 & 10 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[2] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-32, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 9 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 \\ 11 & 15 & 22 & 17 & 8 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 \\ 5 & 6 & 10 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[0] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-32, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 7 & 9 & 4 & 5 \\ 7 & 8 & 9 & 11 & 5 & 6 \\ 7 & 9 & 8 & 10 & 5 & 6 \\ 9 & 11 & 10 & 11 & 6 & 7 \\ 4 & 5 & 5 & 6 & 1 & 3 \\ 5 & 6 & 6 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[3] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 \\ 4 & 6 & 2 & 7 & 3 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 \\ 5 & 6 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[4] \quad & 21^{113} 3^{11} 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 12 & 3 & 4 \\ 5 & 4 & 6 & 4 & 12 & 3 & 4 \\ 5 & 6 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 4 & 1 & 8 & 2 & 3 \\ 12 & 12 & 12 & 8 & 25 & 7 & 9 \\ 3 & 3 & 3 & 2 & 7 & 0 & 2 \\ 4 & 4 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[5] \quad & 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 9 & 4 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 \\ 6 & 6 & 2 & 7 & 3 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 \\ 5 & 6 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.5
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 9 & 4 & 5 \\ 7 & 8 & 12 & 11 & 5 & 6 \\ 11 & 12 & 14 & 14 & 6 & 8 \\ 9 & 11 & 14 & 11 & 6 & 7 \\ 4 & 5 & 6 & 6 & 1 & 3 \\ 5 & 6 & 8 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
& [9] \ 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 14 & 4 & 5 \\ 7 & 8 & 6 & 17 & 5 & 6 \\ 6 & 6 & 2 & 11 & 3 & 4 \\ 14 & 17 & 11 & 29 & 9 & 11 \\ 4 & 5 & 3 & 9 & 1 & 3 \\ 5 & 6 & 4 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
& [10] \ 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 14 & 4 & 5 \\ 7 & 8 & 12 & 17 & 5 & 6 \\ 11 & 12 & 14 & 22 & 6 & 8 \\ 14 & 17 & 22 & 29 & 9 & 11 \\ 4 & 5 & 6 & 9 & 1 & 3 \\ 5 & 6 & 8 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
& [8] \ 1^{12} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 14 & 4 & 5 \\ 7 & 8 & 6 & 17 & 5 & 6 \\ 4 & 6 & 2 & 11 & 3 & 4 \\ 14 & 17 & 11 & 29 & 9 & 11 \\ 4 & 5 & 3 & 9 & 1 & 3 \\ 5 & 6 & 4 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
& [7] \ 1^{132} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 14 & 4 & 5 \\ 7 & 8 & 18 & 17 & 5 & 6 \\ 16 & 18 & 34 & 33 & 9 & 12 \\ 14 & 17 & 33 & 29 & 9 & 11 \\ 4 & 5 & 9 & 9 & 1 & 3 \\ 5 & 6 & 12 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
& [12] \ 1^{111} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 \\ 5 & 6 & 2 & 7 & 3 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 \\ 5 & 6 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
& [11] \ 1^{131} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 9 & 4 & 5 \\ 7 & 8 & 18 & 11 & 5 & 6 \\ 15 & 18 & 34 & 21 & 9 & 12 \\ 9 & 11 & 21 & 11 & 6 & 7 \\ 4 & 5 & 9 & 6 & 1 & 3 \\ 5 & 6 & 12 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
& [13] \ 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 9 & 4 & 5 \\ 7 & 8 & 12 & 11 & 5 & 6 \\ 10 & 12 & 14 & 14 & 6 & 8 \\ 9 & 11 & 14 & 11 & 6 & 7 \\ 4 & 5 & 6 & 6 & 1 & 3 \\ 5 & 6 & 8 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
& [14] \ 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -139, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 26 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 6 & 6 & 2 & 20 & 3 & 4 \\ 26 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[15] \quad & 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -144, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 19 & 4 & 5 \\ 7 & 8 & 6 & 23 & 5 & 6 \\ 4 & 6 & 2 & 15 & 3 & 4 \\ 19 & 23 & 15 & 55 & 12 & 15 \\ 4 & 5 & 3 & 12 & 1 & 3 \\ 5 & 6 & 4 & 15 & 3 & 4 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 13 & 9 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 \\ 13 & 15 & 22 & 17 & 8 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 \\ 5 & 6 & 10 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{141} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 19 & 9 & 4 & 5 \\ 7 & 8 & 21 & 11 & 5 & 6 \\ 19 & 21 & 44 & 24 & 11 & 14 \\ 9 & 11 & 24 & 11 & 6 & 7 \\ 4 & 5 & 11 & 6 & 1 & 3 \\ 5 & 6 & 14 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 12 & 9 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 \\ 12 & 15 & 22 & 17 & 8 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 \\ 5 & 6 & 10 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -187, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 21 & 4 & 5 \\ 7 & 8 & 6 & 26 & 5 & 6 \\ 4 & 6 & 2 & 16 & 3 & 4 \\ 21 & 26 & 16 & 69 & 14 & 17 \\ 4 & 5 & 3 & 14 & 1 & 3 \\ 5 & 6 & 4 & 17 & 3 & 4 \end{pmatrix}_{b_0} \\
[20] \quad & 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -189, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 22 & 4 & 5 \\ 7 & 8 & 6 & 26 & 5 & 6 \\ 6 & 6 & 2 & 16 & 3 & 4 \\ 22 & 26 & 16 & 69 & 14 & 17 \\ 4 & 5 & 3 & 14 & 1 & 3 \\ 5 & 6 & 4 & 17 & 3 & 4 \end{pmatrix}_{b_0} \\
[22] \quad & 1^{142} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 14 & 4 & 5 \\ 7 & 8 & 24 & 17 & 5 & 6 \\ 20 & 24 & 62 & 44 & 12 & 16 \\ 14 & 17 & 44 & 29 & 9 & 11 \\ 4 & 5 & 12 & 9 & 1 & 3 \\ 5 & 6 & 16 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
[21] \quad & 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 14 & 4 & 5 \\ 7 & 8 & 12 & 17 & 5 & 6 \\ 10 & 12 & 14 & 22 & 6 & 8 \\ 14 & 17 & 22 & 29 & 9 & 11 \\ 4 & 5 & 6 & 9 & 1 & 3 \\ 5 & 6 & 8 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
[23] \quad & 21^{113} 3^{11} 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 15 & 3 & 4 \\ 5 & 4 & 6 & 4 & 16 & 3 & 4 \\ 5 & 6 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 4 & 1 & 11 & 2 & 3 \\ 15 & 16 & 16 & 11 & 44 & 9 & 12 \\ 3 & 3 & 3 & 2 & 9 & 0 & 2 \\ 4 & 4 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 31 & 4 & 5 \\ 7 & 8 & 6 & 38 & 5 & 6 \\ 6 & 6 & 2 & 24 & 3 & 4 \\ 31 & 38 & 24 & 149 & 20 & 25 \\ 4 & 5 & 3 & 20 & 1 & 3 \\ 5 & 6 & 4 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{112} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 25 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 4 & 6 & 2 & 20 & 3 & 4 \\ 25 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{114} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 27 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 4 & 6 & 2 & 20 & 3 & 4 \\ 27 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 21^{1111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 8 & 2 & 3 \\ 4 & 4 & 6 & 4 & 12 & 3 & 4 \\ 4 & 6 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 4 & 1 & 8 & 2 & 3 \\ 8 & 12 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 26 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 4 & 6 & 2 & 20 & 3 & 4 \\ 26 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 17 & 9 & 4 & 5 \\ 7 & 8 & 21 & 11 & 5 & 6 \\ 17 & 21 & 44 & 24 & 11 & 14 \\ 9 & 11 & 24 & 11 & 6 & 7 \\ 4 & 5 & 11 & 6 & 1 & 3 \\ 5 & 6 & 14 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{131} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 9 & 4 & 5 \\ 7 & 8 & 21 & 11 & 5 & 6 \\ 18 & 21 & 44 & 24 & 11 & 14 \\ 9 & 11 & 24 & 11 & 6 & 7 \\ 4 & 5 & 11 & 6 & 1 & 3 \\ 5 & 6 & 14 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{113} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 19 & 4 & 5 \\ 7 & 8 & 6 & 23 & 5 & 6 \\ 5 & 6 & 2 & 15 & 3 & 4 \\ 19 & 23 & 15 & 55 & 12 & 15 \\ 4 & 5 & 3 & 12 & 1 & 3 \\ 5 & 6 & 4 & 15 & 3 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[32] \quad & 1^{124} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -301, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 27 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 6 & 6 & 2 & 20 & 3 & 4 \\ 27 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0} \\
[33] \quad & 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -314, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 26 & 4 & 5 \\ 7 & 8 & 12 & 32 & 5 & 6 \\ 10 & 12 & 14 & 40 & 6 & 8 \\ 26 & 32 & 40 & 105 & 17 & 21 \\ 4 & 5 & 6 & 17 & 1 & 3 \\ 5 & 6 & 8 & 21 & 3 & 4 \end{pmatrix}_{b_0} \\
[34] \quad & 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 21 & 4 & 5 \\ 7 & 8 & 6 & 26 & 5 & 6 \\ 5 & 6 & 2 & 16 & 3 & 4 \\ 21 & 26 & 16 & 69 & 14 & 17 \\ 4 & 5 & 3 & 14 & 1 & 3 \\ 5 & 6 & 4 & 17 & 3 & 4 \end{pmatrix}_{b_0} \\
[35] \quad & 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -349, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 30 & 4 & 5 \\ 7 & 8 & 6 & 38 & 5 & 6 \\ 4 & 6 & 2 & 24 & 3 & 4 \\ 30 & 38 & 24 & 149 & 20 & 25 \\ 4 & 5 & 3 & 20 & 1 & 3 \\ 5 & 6 & 4 & 25 & 3 & 4 \end{pmatrix}_{b_0} \\
[36] \quad & 1^{124} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -364, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 27 & 4 & 5 \\ 7 & 8 & 12 & 32 & 5 & 6 \\ 10 & 12 & 14 & 40 & 6 & 8 \\ 27 & 32 & 40 & 105 & 17 & 21 \\ 4 & 5 & 6 & 17 & 1 & 3 \\ 5 & 6 & 8 & 21 & 3 & 4 \end{pmatrix}_{b_0} \\
[37] \quad & 1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -387, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 31 & 4 & 5 \\ 7 & 8 & 6 & 38 & 5 & 6 \\ 4 & 6 & 2 & 24 & 3 & 4 \\ 31 & 38 & 24 & 149 & 20 & 25 \\ 4 & 5 & 3 & 20 & 1 & 3 \\ 5 & 6 & 4 & 25 & 3 & 4 \end{pmatrix}_{b_0} \\
[38] \quad & 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -392, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 22 & 4 & 5 \\ 7 & 8 & 6 & 26 & 5 & 6 \\ 5 & 6 & 2 & 16 & 3 & 4 \\ 22 & 26 & 16 & 69 & 14 & 17 \\ 4 & 5 & 3 & 14 & 1 & 3 \\ 5 & 6 & 4 & 17 & 3 & 4 \end{pmatrix}_{b_0} \\
[40] \quad & 21^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -400, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 5 & 2 & 3 \\ 4 & 4 & 6 & 8 & 7 & 3 & 4 \\ 4 & 6 & 4 & 8 & 7 & 3 & 4 \\ 6 & 8 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[39] \quad & 21^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -400, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 5 & 2 & 3 \\ 4 & 4 & 6 & 4 & 7 & 3 & 4 \\ 4 & 6 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 9 & 4 & 5 \\ 7 & 8 & 27 & 11 & 5 & 6 \\ 23 & 27 & 74 & 31 & 14 & 18 \\ 9 & 11 & 31 & 11 & 6 & 7 \\ 4 & 5 & 14 & 6 & 1 & 3 \\ 5 & 6 & 18 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 9 & 4 & 5 \\ 7 & 8 & 27 & 11 & 5 & 6 \\ 22 & 27 & 74 & 31 & 14 & 18 \\ 9 & 11 & 31 & 11 & 6 & 7 \\ 4 & 5 & 14 & 6 & 1 & 3 \\ 5 & 6 & 18 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 31 & 4 & 5 \\ 7 & 8 & 12 & 38 & 5 & 6 \\ 10 & 12 & 14 & 48 & 6 & 8 \\ 31 & 38 & 48 & 149 & 20 & 25 \\ 4 & 5 & 6 & 20 & 1 & 3 \\ 5 & 6 & 8 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -530, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 26 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 5 & 6 & 2 & 20 & 3 & 4 \\ 26 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{141}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 27 & 9 & 4 & 5 \\ 7 & 8 & 33 & 11 & 5 & 6 \\ 27 & 33 & 112 & 38 & 17 & 22 \\ 9 & 11 & 38 & 11 & 6 & 7 \\ 4 & 5 & 17 & 6 & 1 & 3 \\ 5 & 6 & 22 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 21^{112}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 8 & 2 & 3 \\ 4 & 4 & 6 & 4 & 11 & 3 & 4 \\ 4 & 6 & 4 & 4 & 11 & 3 & 4 \\ 3 & 4 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 27 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 5 & 6 & 2 & 20 & 3 & 4 \\ 27 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 21^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 11 & 2 & 3 \\ 4 & 4 & 6 & 4 & 16 & 3 & 4 \\ 4 & 6 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 4 & 1 & 11 & 2 & 3 \\ 11 & 16 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -738, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 31 & 4 & 5 \\ 7 & 8 & 6 & 38 & 5 & 6 \\ 5 & 6 & 2 & 24 & 3 & 4 \\ 31 & 38 & 24 & 149 & 20 & 25 \\ 4 & 5 & 3 & 20 & 1 & 3 \\ 5 & 6 & 4 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

5.132 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 9 & 4 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 & 6 \\ 11 & 15 & 22 & 17 & 8 & 9 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 & 3 \\ 4 & 6 & 9 & 7 & 3 & 2 & 4 \\ 5 & 6 & 10 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 9 & 4 & 12 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 6 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 12 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 7 & 9 & 4 & 4 & 5 \\ 7 & 8 & 9 & 11 & 5 & 6 & 6 \\ 7 & 9 & 8 & 10 & 5 & 5 & 6 \\ 9 & 11 & 10 & 11 & 6 & 7 & 7 \\ 4 & 5 & 5 & 6 & 1 & 3 & 3 \\ 4 & 6 & 5 & 7 & 3 & 2 & 4 \\ 5 & 6 & 6 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 9 & 4 & 9 & 5 \\ 7 & 8 & 15 & 11 & 5 & 12 & 6 \\ 11 & 15 & 22 & 17 & 8 & 18 & 10 \\ 9 & 11 & 17 & 11 & 6 & 14 & 7 \\ 4 & 5 & 8 & 6 & 1 & 6 & 3 \\ 9 & 12 & 18 & 14 & 6 & 14 & 8 \\ 5 & 6 & 10 & 7 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 13 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 4 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 13 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1211} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 9 & 4 & 12 & 5 \\ 7 & 8 & 12 & 11 & 5 & 15 & 6 \\ 10 & 12 & 14 & 14 & 6 & 18 & 8 \\ 9 & 11 & 14 & 11 & 6 & 17 & 7 \\ 4 & 5 & 6 & 6 & 1 & 8 & 3 \\ 12 & 15 & 18 & 17 & 8 & 22 & 10 \\ 5 & 6 & 8 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 9 & 4 & 5 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 & 6 \\ 6 & 6 & 2 & 7 & 3 & 4 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 & 3 \\ 5 & 6 & 4 & 7 & 3 & 2 & 4 \\ 5 & 6 & 4 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{111}} 3^{\overline{11}1} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$[8] \quad 1^{112} 3^{11} 1^1 + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{red}{s} \textcolor{blue}{o}_7 \cdot \textcolor{red}{s} \textcolor{blue}{u}_2] + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{red}{s} \textcolor{blue}{u}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$[9] \quad 1^{\overline{113}} 2^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 18 & 5 \\ 7 & 8 & 6 & 11 & 5 & 21 & 6 \\ 4 & 6 & 2 & 7 & 3 & 13 & 4 \\ 9 & 11 & 7 & 11 & 6 & 24 & 7 \\ 4 & 5 & 3 & 6 & 1 & 11 & 3 \\ 18 & 21 & 13 & 24 & 11 & 44 & 14 \\ 5 & 6 & 4 & 7 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[10] \overset{111}{1} \overset{1111}{1} \overset{3}{3} \overset{1}{1} \overset{11}{1} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 12 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 4 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 12 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[111] \overset{111}{1} \overset{1112}{1} \overset{311}{3} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 16 & 5 \\ 7 & 8 & 6 & 11 & 5 & 21 & 6 \\ 4 & 6 & 2 & 7 & 3 & 13 & 4 \\ 9 & 11 & 7 & 11 & 6 & 24 & 7 \\ 4 & 5 & 3 & 6 & 1 & 11 & 3 \\ 16 & 21 & 13 & 24 & 11 & 44 & 14 \\ 5 & 6 & 4 & 7 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[12] \begin{array}{l} 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \\ (1, 10, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\ \left(\begin{array}{cccccc} 5 & 7 & 4 & 14 & 4 & 5 & 5 \\ 7 & 8 & 6 & 17 & 5 & 6 & 6 \\ 4 & 6 & 2 & 11 & 3 & 4 & 4 \\ 14 & 17 & 11 & 29 & 9 & 11 & 11 \\ 4 & 5 & 3 & 9 & 1 & 3 & 3 \\ 5 & 6 & 4 & 11 & 3 & 2 & 4 \\ 5 & 6 & 4 & 11 & 3 & 4 & 4 \end{array} \right)_{b_0} \end{array} \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 200, (n_+, n_-, n_0) =$$

$$[16] \begin{array}{l} 1^{1111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s} \textcolor{red}{o}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \\ (1, 10, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\ \begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 5 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 & 6 \\ 5 & 6 & 2 & 7 & 3 & 4 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 & 3 \\ 5 & 6 & 4 & 7 & 3 & 2 & 4 \\ 5 & 6 & 4 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0} \end{array} \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) =$$

$$[13] \begin{array}{l} 1^{11112} \\ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \\ (1, 10, 0), c_{\text{ext}} = 8.0 \end{array} \quad T = 10, T_{\text{max}} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) =$$

$$T_{\text{min}} = 10, T_{\text{max}} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 13 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 5 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 13 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 12 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 5 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 12 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[15] $1^{1112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 10 & 5 \\ 7 & 8 & 6 & 11 & 5 & 12 & 6 \\ 5 & 6 & 2 & 7 & 3 & 8 & 4 \\ 9 & 11 & 7 & 11 & 6 & 14 & 7 \\ 4 & 5 & 3 & 6 & 1 & 6 & 3 \\ 10 & 12 & 8 & 14 & 6 & 14 & 8 \\ 5 & 6 & 4 & 7 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 17 & 5 \\ 7 & 8 & 6 & 11 & 5 & 21 & 6 \\ 4 & 6 & 2 & 7 & 3 & 13 & 4 \\ 9 & 11 & 7 & 11 & 6 & 24 & 7 \\ 4 & 5 & 3 & 6 & 1 & 11 & 3 \\ 17 & 21 & 13 & 24 & 11 & 44 & 14 \\ 5 & 6 & 4 & 7 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 17 & 5 \\ 7 & 8 & 6 & 11 & 5 & 21 & 6 \\ 5 & 6 & 2 & 7 & 3 & 13 & 4 \\ 9 & 11 & 7 & 11 & 6 & 24 & 7 \\ 4 & 5 & 3 & 6 & 1 & 11 & 3 \\ 17 & 21 & 13 & 24 & 11 & 44 & 14 \\ 5 & 6 & 4 & 7 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

5.133 NHC(-2)(-3)(-2) + su₂ + su₂n3mix + su₂ → (-3) + su₂ + su₂

$T = 10$ [4] $1^{1211} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$

[5] $1^{1121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$

[3] $1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$

[2] $1^{1121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$

[1] $1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$

[0] $1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$

5.134 NHC(-2)(-3)(-2) + su₂ + su₂n3mix + su₂n3mix

$T = 9$ [0] $1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 8 & 17 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 5 & 6 & 2 & 7 & 12 & 4 \\ 8 & 11 & 7 & 11 & 23 & 7 \\ 17 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -18, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 16 & 14 & 5 \\ 7 & 8 & 6 & 20 & 17 & 6 \\ 5 & 6 & 2 & 12 & 11 & 4 \\ 16 & 20 & 12 & 41 & 35 & 13 \\ 14 & 17 & 11 & 35 & 29 & 11 \\ 5 & 6 & 4 & 13 & 11 & 4 \end{pmatrix}_{b_0}$$

$$1^{1121}$$

$$[2] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -18, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 10 & 16 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 5 & 6 & 2 & 7 & 12 & 4 \\ 10 & 11 & 7 & 11 & 23 & 7 \\ 16 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

$$1^{112}$$

$$[3] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -27, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 16 & 14 & 5 \\ 7 & 8 & 6 & 20 & 17 & 6 \\ 4 & 6 & 2 & 12 & 11 & 4 \\ 16 & 20 & 12 & 41 & 35 & 13 \\ 14 & 17 & 11 & 35 & 29 & 11 \\ 5 & 6 & 4 & 13 & 11 & 4 \end{pmatrix}_{b_0}$$

$$1^{113}$$

$$[4] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -72, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 8 & 22 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 5 & 6 & 2 & 7 & 16 & 4 \\ 8 & 11 & 7 & 11 & 30 & 7 \\ 22 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$1^{111}$$

$$[5] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -99, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 16 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 4 & 6 & 2 & 7 & 12 & 4 \\ 9 & 11 & 7 & 11 & 23 & 7 \\ 16 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

$$1^{111}$$

$$[6] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 9 & 5 \\ 7 & 8 & 6 & 11 & 11 & 6 \\ 4 & 6 & 2 & 7 & 7 & 4 \\ 9 & 11 & 7 & 11 & 13 & 7 \\ 9 & 11 & 7 & 13 & 11 & 7 \\ 5 & 6 & 4 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$1^{111}$$

$$[8] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -141, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 20 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 4 & 6 & 2 & 7 & 16 & 4 \\ 9 & 11 & 7 & 11 & 30 & 7 \\ 20 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$1^{113}$$

$$[7] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -141, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 22 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 4 & 6 & 2 & 7 & 16 & 4 \\ 9 & 11 & 7 & 11 & 30 & 7 \\ 22 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$1^{1111}$$

$$[9] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -162, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 16 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 5 & 6 & 2 & 7 & 12 & 4 \\ 9 & 11 & 7 & 11 & 23 & 7 \\ 16 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 17 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 5 & 6 & 2 & 7 & 12 & 4 \\ 9 & 11 & 7 & 11 & 23 & 7 \\ 17 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 9 & 5 \\ 7 & 8 & 6 & 11 & 11 & 6 \\ 5 & 6 & 2 & 7 & 7 & 4 \\ 9 & 11 & 7 & 11 & 13 & 7 \\ 9 & 11 & 7 & 13 & 11 & 7 \\ 5 & 6 & 4 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1211} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 9 & 9 & 5 \\ 7 & 8 & 12 & 11 & 11 & 6 \\ 10 & 12 & 14 & 14 & 14 & 8 \\ 9 & 11 & 14 & 11 & 13 & 7 \\ 9 & 11 & 14 & 13 & 11 & 7 \\ 5 & 6 & 8 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 21 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 4 & 6 & 2 & 7 & 16 & 4 \\ 9 & 11 & 7 & 11 & 30 & 7 \\ 21 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 14 & 9 & 5 \\ 7 & 8 & 6 & 17 & 11 & 6 \\ 5 & 6 & 2 & 11 & 7 & 4 \\ 14 & 17 & 11 & 29 & 20 & 11 \\ 9 & 11 & 7 & 20 & 11 & 7 \\ 5 & 6 & 4 & 11 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 21 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 5 & 6 & 2 & 7 & 16 & 4 \\ 9 & 11 & 7 & 11 & 30 & 7 \\ 21 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

5.135 NHC(-2)(-3)(-2) + su₃

$$T = 7 \quad [0] \quad 21^{11} 31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 5 & 3 & 4 \\ 8 & 9 & 7 & 4 & 5 \\ 5 & 7 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 4 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 6 & 5 & 4 \\ 3 & 6 & 4 & 5 & 4 \\ 4 & 5 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 12 & 4 \\ 9 & 12 & 12 & 24 & 8 \\ 3 & 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 6 & 13 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 10 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 7 & 4 & 5 \\ 3 & 7 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 2 \\ 3 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 7 & 0 & 5 \\ 3 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 7 & 15 & 5 \\ 7 & 8 & 21 & 6 \\ 15 & 21 & 43 & 13 \\ 5 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 4 & 5 \\ 7 & 8 & 6 & 6 \\ 4 & 6 & 1 & 3 \\ 5 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[9] $21^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 5 & 6 & 4 \\ 5 & 4 & 6 & 5 & 4 \\ 5 & 6 & 4 & 5 & 4 \\ 6 & 5 & 5 & 3 & 3 \\ 4 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 7 & 5 \\ 7 & 8 & 9 & 6 \\ 7 & 9 & 7 & 5 \\ 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^1 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 6 & 5 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 1 & 3 \\ 5 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{12} 3^1 1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 7 & 12 & 5 \\ 7 & 8 & 12 & 6 \\ 12 & 12 & 13 & 7 \\ 5 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 11 & 5 \\ 7 & 8 & 15 & 6 \\ 11 & 15 & 21 & 9 \\ 5 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

- [14] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -31$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 5 & 7 & 5 & 5 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 1 & 3 \\ 5 & 6 & 3 & 3 \end{pmatrix}_{b_0}$
- [16] $1^{11} 3^1 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 6 & 8 & 4 \\ 3 & 6 & 4 & 8 & 4 \\ 6 & 8 & 8 & 9 & 5 \\ 3 & 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 5 & 4 \\ 3 & 6 & 4 & 5 & 4 \\ 3 & 5 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 7 & 15 & 5 \\ 7 & 8 & 15 & 6 \\ 15 & 15 & 21 & 9 \\ 5 & 6 & 9 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 5 & 7 & 10 & 5 \\ 7 & 8 & 12 & 6 \\ 10 & 12 & 13 & 7 \\ 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [19] $1^{12} 3^1 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.49 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 5 & 7 & 11 & 5 \\ 7 & 8 & 12 & 6 \\ 11 & 12 & 13 & 7 \\ 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [22] $21^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 5 & 3 & 4 \\ 5 & 4 & 6 & 4 & 4 \\ 5 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [21] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [24] $1^1 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 6 & 8 & 4 \\ 3 & 6 & 4 & 8 & 4 \\ 4 & 8 & 8 & 9 & 5 \\ 3 & 4 & 4 & 5 & 1 \end{pmatrix}_{b_0}$
- [23] $1^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -54$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 12 & 5 \\ 7 & 8 & 15 & 6 \\ 12 & 15 & 21 & 9 \\ 5 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 14 & 5 \\ 7 & 8 & 15 & 6 \\ 14 & 15 & 21 & 9 \\ 5 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 6 & 8 & 4 \\ 3 & 6 & 4 & 8 & 4 \\ 5 & 8 & 8 & 9 & 5 \\ 3 & 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 15 & 5 \\ 7 & 8 & 18 & 6 \\ 15 & 18 & 33 & 11 \\ 5 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 13 & 5 \\ 7 & 8 & 15 & 6 \\ 13 & 15 & 21 & 9 \\ 5 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11} 1^3 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 5 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 12 & 4 \\ 8 & 12 & 12 & 24 & 8 \\ 3 & 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 16 & 5 \\ 7 & 8 & 18 & 6 \\ 16 & 18 & 33 & 11 \\ 5 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[34] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 4 & 6 & 5 & 4 \\ 4 & 6 & 4 & 5 & 4 \\ 3 & 5 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 12 & 4 \\ 7 & 12 & 12 & 24 & 8 \\ 3 & 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 13 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 9 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[37] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 5 & 9 & 4 \\ 5 & 4 & 6 & 9 & 4 \\ 5 & 6 & 4 & 9 & 4 \\ 9 & 9 & 9 & 12 & 6 \\ 4 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{14} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 20 & 5 \\ 7 & 8 & 24 & 6 \\ 20 & 24 & 61 & 15 \\ 5 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 6 & 17 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 12 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{12} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 17 & 5 \\ 7 & 8 & 21 & 6 \\ 17 & 21 & 43 & 13 \\ 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 13 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 7 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 19 & 5 \\ 7 & 8 & 21 & 6 \\ 19 & 21 & 43 & 13 \\ 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 6 & 13 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 8 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{12} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 18 & 5 \\ 7 & 8 & 21 & 6 \\ 18 & 21 & 43 & 13 \\ 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[45] \ 21^{11} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 4 \\ 5 & 4 & 6 & 13 & 4 \\ 5 & 6 & 4 & 13 & 4 \\ 12 & 13 & 13 & 27 & 9 \\ 4 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[46] \ 21^{11} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 5 & 15 & 4 \\ 5 & 4 & 6 & 17 & 4 \\ 5 & 6 & 4 & 17 & 4 \\ 15 & 17 & 17 & 48 & 12 \\ 4 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{11} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.03 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 17 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 9 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 4 & 6 & 4 & 4 \\ 4 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 22 & 5 \\ 7 & 8 & 27 & 6 \\ 22 & 27 & 73 & 17 \\ 5 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 6 & 17 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 11 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 8 & 3 & 4 \\ 8 & 9 & 11 & 4 & 5 \\ 8 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & 0 & 2 \\ 4 & 5 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[52] \quad 21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 4 & 6 & 9 & 4 \\ 4 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 6 & 17 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 10 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 23 & 5 \\ 7 & 8 & 27 & 6 \\ 23 & 27 & 73 & 17 \\ 5 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 21^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 4 & 6 & 8 & 4 \\ 4 & 6 & 4 & 8 & 4 \\ 6 & 8 & 8 & 9 & 5 \\ 3 & 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 27 & 5 \\ 7 & 8 & 33 & 6 \\ 27 & 33 & 111 & 21 \\ 5 & 6 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 21^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 4 & 6 & 12 & 4 \\ 4 & 6 & 4 & 12 & 4 \\ 9 & 12 & 12 & 24 & 8 \\ 3 & 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 21^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 4 & 6 & 13 & 4 \\ 4 & 6 & 4 & 13 & 4 \\ 9 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 21^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 4 & 6 & 17 & 4 \\ 4 & 6 & 4 & 17 & 4 \\ 12 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [60] \quad 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 7 & 4 \\ 6 & 7 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 4 \\ 5 & 4 & 5 & 4 & 4 \\ 5 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 5 & 6 & 4 \\ 5 & 4 & 5 & 7 & 4 \\ 5 & 5 & 4 & 7 & 4 \\ 6 & 7 & 7 & 7 & 5 \\ 4 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 7 & 4 \\ 5 & 7 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 5 & 6 & 4 \\ 5 & 4 & 6 & 7 & 4 \\ 5 & 6 & 4 & 7 & 4 \\ 6 & 7 & 7 & 7 & 5 \\ 4 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 4 \\ 5 & 4 & 6 & 3 & 4 \\ 5 & 6 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 4 & 7 & 4 \\ 5 & 7 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 21^{11} 4 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 8 & 6 & 4 \\ 8 & 13 & 15 & 12 & 7 \\ 8 & 15 & 13 & 12 & 7 \\ 6 & 12 & 12 & 7 & 5 \\ 4 & 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 21^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 3 \\ 4 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 21^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 5 & 3 \\ 4 & 4 & 6 & 7 & 4 \\ 4 & 6 & 4 & 7 & 4 \\ 5 & 7 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 221^{111}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 6 & 5 & 3 \\ 6 & 13 & 15 & 12 & 7 \\ 6 & 15 & 13 & 12 & 7 \\ 5 & 12 & 12 & 7 & 5 \\ 3 & 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 31^{111}31^13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 6 & 4 \\ 3 & 4 & 0 & 4 & 3 & 2 \\ 3 & 6 & 4 & 4 & 6 & 4 \\ 3 & 6 & 3 & 6 & 3 & 3 \\ 2 & 4 & 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [74] \quad 1^{11}5^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [76] \quad 21^{11}61^{11}2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 & 2 \\ 4 & 10 & 1 & 12 & 6 & 6 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 12 & 1 & 10 & 6 & 6 \\ 2 & 6 & 0 & 6 & 3 & 3 \\ 2 & 6 & 0 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{11}3221^{11}6^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 & 2 \\ 4 & 10 & 1 & 12 & 6 & 6 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 12 & 1 & 10 & 6 & 6 \\ 2 & 6 & 0 & 6 & 3 & 3 \\ 2 & 6 & 0 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{11}6^{11}1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{11}3221^{11}6^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 6 & 1 & 19 & 9 \\ 3 & 3 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [79] \quad 1^14141^{11}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 8 & 8 & 1 & 3 \\ 8 & 17 & 19 & 0 & 9 \\ 8 & 19 & 17 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 9 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

5.136 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 7 & 4 & 0 & 5 \\ 3 & 7 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{112}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 8 & 0 & 4 \\ 3 & 6 & 4 & 8 & 0 & 4 \\ 4 & 8 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{112}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 13 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 13 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{113}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 17 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 17 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

5.137 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] \quad 1^{\overline{1111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.138 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$

$T = 9 \quad [0] \quad 1^{\overline{111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

5.139 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [1] \quad 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{teal}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.49 \rightarrow 14, \quad T_{\max} = 25, \quad T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 4 & 0 & 14 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\text{5u}_2 \cdot \text{5o}_7 \cdot \text{5u}_2] + \text{5u}_3 + \text{c}_6 + \text{5u}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 5 & 3 \\ 4 & 9 & 7 & 4 & 0 & 12 & 5 \\ 3 & 7 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 9 & 6 & 0 & 13 & 7 \\ 3 & 5 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}}31^{\overline{11}} + [\overline{\text{su}}_2 \cdot \overline{\text{so}}_7 \cdot \overline{\text{su}}_2] + \overline{\text{su}}_3 + \overline{\text{e}}_6 + \overline{\text{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 4 & 0 & 5 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{112}31^{111} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{5}\textcolor{red}{0}\textcolor{blue}{7} \cdot \textcolor{blue}{5}\textcolor{blue}{u}_2] + \textcolor{brown}{5}\textcolor{brown}{u}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{5}\textcolor{blue}{u}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 8 & 3 \\ 4 & 9 & 7 & 4 & 0 & 20 & 5 \\ 3 & 7 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 14 & 9 & 0 & 37 & 11 \\ 3 & 5 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -220$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{112} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{5}\textcolor{red}{0}\textcolor{blue}{7} \cdot \textcolor{blue}{5}\textcolor{red}{u}_2] + \textcolor{red}{5}\textcolor{blue}{u}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{5}\textcolor{red}{u}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 10 & 3 \\ 4 & 9 & 7 & 4 & 0 & 24 & 5 \\ 3 & 7 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 24 & 18 & 12 & 0 & 58 & 14 \\ 3 & 5 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{1114}}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{teal}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 4 & 0 & 16 & 4 \\ 3 & 6 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 4 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{1112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_3 + \textcolor{violet}{c}_6 + \textcolor{teal}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 6 & 9 & 0 & 8 & 4 \\ 3 & 6 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 4 & 6 & -4 & 6 & 0 \end{pmatrix} b_0$$

$$[9] \ 1^{\overline{1113}} 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 6 & 4 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1112}} 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -400, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 6 & 4 & 0 & 8 & 4 \\ 3 & 6 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1112}} 3 1^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -436, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.28 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 6 & 4 & 0 & 23 & 4 \\ 3 & 6 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}} 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -436, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -592, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{1113}} 3 1^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -1012, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 6 & 4 & 0 & 32 & 4 \\ 3 & 6 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{\overline{1112}} 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -1012, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 4 & 0 & 13 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{1112}} 3 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -1024, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 4 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 6 & 4 & 0 & 17 & 4 \\ 3 & 6 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1113}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 6 & 4 & 0 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1780, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 4 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1113}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1876, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 6 & 4 & 0 & 22 & 4 \\ 3 & 6 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1114}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 6 & 4 & 0 & 26 & 4 \\ 3 & 6 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1114}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 6 & 4 & 0 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1114}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3028, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 6 & 4 & 0 & 31 & 4 \\ 3 & 6 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

5.140 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

5.141 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 492, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.142 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{112}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 5 & 0 & 8 & 4 \\ 3 & 6 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 4 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -660$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -816$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

5.143 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 492$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

5.144 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3$

$T = 10$ [0] $1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 864$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

5.145 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$

$T = 8$ [0] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 5 \\ 4 & 9 & 7 & 4 & 0 & 5 & 4 \\ 3 & 7 & 4 & 4 & 0 & 4 & 2 \\ 2 & 4 & 4 & 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 & 0 \\ 3 & 5 & 4 & 2 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 & 4 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 & 0 \\ 3 & 4 & 4 & 3 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 164$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$

$T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 & 4 \\ 3 & 4 & 6 & 8 & 0 & 4 & 4 \\ 3 & 6 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 8 & 9 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 & 0 \\ 3 & 4 & 4 & 5 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{112}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 13 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 13 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{113}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 17 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 17 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

5.146 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$

$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

5.147 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 5 & 3 \\ 4 & 9 & 7 & 4 & 0 & 12 & 5 \\ 3 & 7 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 9 & 6 & 0 & 13 & 7 \\ 3 & 5 & 4 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 4 & 0 & 14 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 4 & 0 & 5 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 4 & 0 & 16 & 4 \\ 3 & 6 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 8 & 3 \\ 4 & 9 & 7 & 4 & 0 & 20 & 5 \\ 3 & 7 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 14 & 9 & 0 & 37 & 11 \\ 3 & 5 & 4 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 6 & 9 & 0 & 8 & 4 \\ 3 & 6 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 10 & 3 \\ 4 & 9 & 7 & 4 & 0 & 24 & 5 \\ 3 & 7 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 24 & 18 & 12 & 0 & 58 & 14 \\ 3 & 5 & 4 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 4 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 6 & 4 & 0 & 8 & 4 \\ 3 & 6 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 6 & 4 & 0 & 23 & 4 \\ 3 & 6 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -788, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 4 & 0 & 13 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -788, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 6 & 4 & 0 & 32 & 4 \\ 3 & 6 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1112} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -800, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 4 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1088, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 6 & 4 & 0 & 17 & 4 \\ 3 & 6 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 4 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1113} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1360, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 6 & 4 & 0 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1364, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 4 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 4 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 6 & 4 & 0 & 22 & 4 \\ 3 & 6 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 6 & 4 & 0 & 26 & 4 \\ 3 & 6 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 6 & 4 & 0 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1114} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.10 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 6 & 4 & 0 & 31 & 4 \\ 3 & 6 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

5.148 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 6 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

5.149 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 7 & 4 & 0 & 4 & 5 \\ 3 & 7 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 6 & 8 & 0 & 3 & 4 \\ 3 & 6 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 8 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

5.150 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + su2n3mix

$$T = 9 \quad [0] \ 1^{\overline{1111}} 3^1 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 4 & 0 & 17 & 4 \\ 3 & 6 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 4 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -468, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.15 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 6 & 4 & 0 & 7 & 4 \\ 3 & 6 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -528, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 4 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

5.151 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{12}} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 5 & 0 & 8 & 4 \\ 3 & 6 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.30 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -640, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

5.152 NHC(-2)(-3)(-2) + $\textcolor{blue}{su}_3$ + $\textcolor{blue}{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 1 & 3 \\ 4 & 9 & 7 & 0 & 0 & 5 \\ 3 & 7 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 7 & 4 & 0 & 5 \\ 3 & 7 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 37, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.09 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 15 & 13 & 6 \\ 15 & 20 & 22 & 17 & 8 \\ 15 & 22 & 20 & 17 & 8 \\ 13 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 5 \\ 7 & 8 & 6 & 0 & 6 \\ 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 55, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 11 & 1 & 5 \\ 7 & 8 & 15 & 0 & 6 \\ 11 & 15 & 21 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 63, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 7 & 4 & 5 \\ 7 & 8 & 9 & 6 & 6 \\ 7 & 9 & 7 & 5 & 5 \\ 4 & 6 & 5 & 0 & 2 \\ 5 & 6 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$[19] \ 1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 21^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 20 & 12 & 7 \\ 20 & 25 & 27 & 16 & 9 \\ 20 & 27 & 25 & 16 & 9 \\ 12 & 16 & 16 & 9 & 5 \\ 7 & 9 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 12 & 0 & 4 \\ 7 & 12 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 21^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 16 & 12 & 6 \\ 16 & 20 & 22 & 17 & 8 \\ 16 & 22 & 20 & 17 & 8 \\ 12 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 13 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 13 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 6 & 13 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 8 & 13 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 17 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 17 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 428, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 6 & 17 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 10 & 17 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

5.153 $\text{NHC}(-2)(-3)(-2) + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6$

$$T = 9 \quad [0] \ 1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

5.154 $\text{NHC}(-2)(-3)(-2) + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{f}}_4$

$$T = 9 \quad [0] \ 1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

5.155 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 1 & 5 \\ 12 & 25 & 23 & 0 & 0 & 9 \\ 11 & 23 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 7 & 7 & 4 & 1 & 5 \\ 7 & 8 & 9 & 6 & 0 & 6 \\ 7 & 9 & 7 & 5 & 0 & 5 \\ 4 & 6 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 1 & 5 \\ 11 & 20 & 22 & 0 & 0 & 8 \\ 11 & 22 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

5.156 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 2 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 8 & 5 & 0 & 24 & 10 \\ 2 & 2 & 2 & 2 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 5 & 4 & 0 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 6 & 6 & 0 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 9 & 7 & 0 & 36 & 12 \\ 2 & 2 & 2 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 7 & 2 \\ 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 7 & 7 & 0 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 10 & 8 & 0 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 14 & 3 \end{pmatrix}_h$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 13 & 2 \\ 1 & 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 13 & 9 & 0 & 73 & 17 \\ 2 & 2 & 2 & 2 & -1 & 17 & 3 \end{pmatrix}_h$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 11 & 10 & 0 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{12} \overset{1}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 2 & 0 & 14 & 2 \\ 1 & 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 14 & 11 & 0 & 93 & 19 \\ 2 & 2 & 2 & 2 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{13} \overset{1}{5} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -242, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 11 & 12 & 11 & 0 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{13} \overset{1}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 15 & 2 \\ 1 & 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 15 & 12 & 0 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{13} \overset{1}{5} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -338, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 0 & 13 & 2 \\ 1 & 2 & 0 & 2 & 0 & 18 & 2 \\ 1 & 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 13 & 18 & 13 & 0 & 148 & 24 \\ 2 & 2 & 2 & 2 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

5.157 NHC(-2)(-3)(-2) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 12$ [0] $1^{11} \overset{1}{5} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 \\ 1 & 2 & 0 & 2 & 0 & 3 & 5 \\ 1 & 1 & 2 & 0 & 0 & 1 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 1 & 0 & 1 & 5 \\ 3 & 4 & 5 & 4 & 0 & 5 & 11 \\ 2 & 2 & 2 & 2 & -1 & 3 & 7 \end{pmatrix}_{b_0}$$

[1] $1^{12} \overset{1}{5} 131^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 1 & 0 & 1 & 10 & 3 \\ 6 & 8 & 10 & 8 & 0 & 10 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

5.158 NHC(-2)(-3)(-2) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$T = 12$ [0] $1^{11} \overset{1}{5} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s07} \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 \\ 1 & 2 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

5.159 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 9.0$$

5.160 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^1 \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \quad 1^{11} \overset{1}{6} 123^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

5.161 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 1 \\ 1 & 1 & 2 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 \\ 1 & 1 & 2 & 1 & 0 & 0 & 1 \\ 1 & 2 & 1 & 1 & 0 & 1 & 0 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [1] \quad 1^{11} \overset{1}{6} 1^{11} 6^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 4 & 5 & 5 & 4 & 0 & 5 & 18 & 8 \\ 2 & 2 & 2 & 2 & -1 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1}{6} 123^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{6} 1^{11} 6^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 5 & 6 & 6 & 6 & 0 & 6 & 28 & 10 \\ 2 & 2 & 2 & 2 & -1 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$5.162 \quad \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 12 \quad [0] \quad 1^{\overline{11}} \overset{1^1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 9.0$$

$$T = 13 \quad [1] \quad 1^{\overline{11}} \overset{1^1}{6} 123^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 9.0$$

$$5.163 \quad \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$$

$$T = 12 \quad [0] \quad 1^{\overline{11}} \overset{1^1}{6} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 8 & 7 & 0 & 42 & 12 \\ 2 & 2 & 2 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} \overset{1^1}{6} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 124, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 8 & 11 & 8 & 0 & 61 & 15 \\ 2 & 2 & 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

6 NHC $(-2)(-2)(-3)$ External (186 combos, 2947 bases)

6.1 NHC $(-2)(-2)(-3)$

$$T = 12 \quad [374] \quad 1^{\overline{1}} 231 \overline{5} 1321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 4 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[375] \quad 1^{\overline{1}} 31 \overline{5} 13221^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 2 & 5 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 5 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 14 \quad [456] \quad 1^{\overline{1}} \overline{7}^{12321} 12321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ 3 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 15 \quad [498] \quad 1^{\overline{1}} \overline{8}^{12321} 12321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 4 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[499] \quad 1^{\overline{1}} \overline{6}^{12315131} 131^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 16 \quad [500] \quad 1^{\overline{1}} \overline{6}^{131} 131612321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 16, T_{\max} = 16, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 17 \quad [504] \quad 1^{\overline{1}} \overline{8}^{123151321} 1231^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[505] \quad 1^{\overline{1}} \overline{8}^{13151321} 12321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ 3 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$[506] \quad 1^{\overline{1}} \overline{8}^{13151321} 12321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [509] \quad 1^{\overline{1}12321} 8^{\overline{1}} 123216131^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[510] \quad 1^{\overline{1}12321} 8^{\overline{1}} 123216131^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 5 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$[511] \quad 1^{\overline{1}131} 6^{\overline{1}} 13161315131^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$T = 19 \quad [514] \quad 1^{\overline{1}12321} 8^{\overline{1}} 1232171231^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$[515] \quad 1^{\overline{1}12321} 8^{\overline{1}} 1232171231^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 5 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[516] \quad 1^{\overline{1}1321} 8^{\overline{1}} 12321712321^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

6.2 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{1}11} 3^{\overline{1}1} 1^{\overline{1}1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 4 \\ 3 & 3 & 4 & -4 & 4 & -3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

6.3 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [3] \quad 1^{\overline{1}11} 5^{\overline{1}1} 13^{\overline{1}1} 1^{\overline{1}1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 4 & 0 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[2] \quad 1^{\overline{1}12} 5^{\overline{1}1} 13^{\overline{1}1} 1^{\overline{1}1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 4 & 0 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 13 & 3 \\ 10 & 12 & 21 & 0 & 13 & 70 & 18 \\ 3 & 3 & 5 & -1 & 3 & 18 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[1] \ 1^{12} \overset{1}{5} 13^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 3 & 5 & 3 & 5 & 0 & 4 & 13 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[0] \ 1^{11} \overset{1}{5} 13^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 2 & 4 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

6.4 NHC(-2)(-2)(-3) + su₃ + su₂n3mix + su₂

$$T = 12 \quad [0] \ 1^{12} \overset{1}{5} 13^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 2 & 3 \\ 2 & 1 & 4 & 0 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 12 & 5 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 7 & 12 & 0 & 22 & 9 & 10 \\ 2 & 2 & 5 & 0 & 9 & 2 & 4 \\ 3 & 3 & 5 & -1 & 10 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[1] \ 1^{12} \overset{1}{5} 13^1 1^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 4 & 0 & 12 & 2 & 3 \\ 3 & 4 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 10 & 12 & 20 & 0 & 65 & 15 & 17 \\ 2 & 2 & 5 & 0 & 15 & 2 & 4 \\ 3 & 3 & 5 & -1 & 17 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[2] \ 1^{13} \overset{1}{5} 13^1 1^{131} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 4 & 0 & 17 & 2 & 3 \\ 3 & 4 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 17 & 28 & 0 & 130 & 21 & 24 \\ 2 & 2 & 5 & 0 & 21 & 2 & 4 \\ 3 & 3 & 5 & -1 & 24 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

6.5 NHC(-2)(-2)(-3)

$$T = 6 \quad [0] \ 21^1 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 6, \ T_{\max} = 26, \ \Delta = -116, \ \det = 5, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \ T_{\max} = 26, \ T \in [10, 26]$$

$$\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 21^1 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 6, \ T_{\max} = 26, \ \Delta = -116, \ \det = 7, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, \ T_{\max} = 26, \ T \in [12, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} 31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 6, \ T_{\max} = 26, \ \Delta = -116, \ \det = 11, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.45 \rightarrow 10, \ T_{\max} = 26, \ T \in [10, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 10.43 \rightarrow 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 10.25 \rightarrow 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 21 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 9.83 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 12.08 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 21 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 12.08 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 21 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 31, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 10.45 \rightarrow 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 4 & 3 & 8 & 4 \\ 3 & 1 & 5 & 3 \\ 8 & 5 & 9 & 5 \\ 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 21 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 9.89 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 21 \overset{1}{3} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 47, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 9.79 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 131 \overset{1}{1} 2 + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 10.42 \rightarrow 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 4 & 6 & 8 & 4 \\ 6 & 4 & 9 & 4 \\ 8 & 9 & 9 & 5 \\ 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 221 \overset{1}{1} 3 + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 234, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 9.75 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [14] \quad 21 \overset{1}{3} 321 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

- [15] $1 \overset{21^1}{4} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -5$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [16] $131^1 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -5$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 3 & 3 \\ 2 & 3 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [17] $1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [18] $1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [19] $1 \overset{21^1}{4} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13$, $T_{\max} = 19$, $T \in [13, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [20] $1^1 1^1 321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 6 & 4 \\ 3 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{11} 1^1 3131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 3 & -1 & 2 \end{pmatrix}_{b_0}$
- [22] $21^{11} 1^1 41^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 6 & 8 & 4 \\ 6 & 8 & 13 & 6 \\ 8 & 13 & 13 & 7 \\ 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [23] $1^1 1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 1 & 3 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [24] $1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.50 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [25] $1^1 1^1 1^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [26] $1^1 1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

- $$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$
- [27] $1^1 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [28] $1^1 131^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$
- [29] $1^{11} 3131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 0 & 2 \\ 2 & 3 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$
- [30] $21^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$
- [32] $1^1 1^1 41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 19$, $T \in [12, 19]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [31] $21^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 19$, $T \in [12, 19]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^1 1^1 41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -14$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$
- [33] $21^1 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -14$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$
- [35] $21^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$
- [36] $1^1 1^1 31^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 4 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$
- [37] $21^1 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^1 1^1 41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$

- $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$
- [39] $1^1 31^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [40] $31^1 231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- 1^1
 [41] $2231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- 1^1
 [42] $21^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [43] $21^{11} 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 1 & 2 \\ 2 & 3 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [45] $21^{11} 41^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- 21^1
 [44] $1^1 4^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- 1^{11}
 [46] $2^1 2^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 4 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [47] $131^1 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 4 & 4 \\ 1 & 3 & 4 & 0 & 2 \\ 2 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- 1^{111}
 [48] $1^1 4^1 12 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -26$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$
- [49] $22141^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -26$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$
- 1^1
 [50] $1^1 4^1 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -26$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 2 \ 2 \ 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1 \ 4 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 4 & 3 & 8 & 4 \\ 3 & 1 & 5 & 3 \\ 8 & 5 & 9 & 5 \\ 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 131^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 7 & 4 \\ 2 & 3 & 7 & 3 & 3 \\ 2 & 3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 221^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 131^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 1 & 4 & 3 \\ 2 & 6 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1 \ 4 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 2 \ 3 \ 1^{11} 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 2 \ 3 \ 1^{11} 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 2 \ 2 \ 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 31^{11} 231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 1 & 4 & 3 \\ 3 & 1 & 4 & 0 & 2 \\ 2 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[61] \quad 31^{11} 31^{11} 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[63] \quad 1 \ 4 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

- [62] $131^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [64] $2\overset{11}{3}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 4 & 4 & 4 \\ 1 & 3 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [65] $31^{11}231^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -51$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [66] $2\overset{1}{3}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [67] $221^{11}41^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -62$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 7 & 4 \\ 3 & 2 & 7 & 3 & 3 \\ 2 & 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [68] $2\overset{1}{3}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [69] $221^{11}41^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -66$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [70] $31^{11}31^13 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.44 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 3 & 4 \\ 3 & 7 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [71] $2\overset{11}{3}1^13 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -79$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [72] $31^{11}231^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -84$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.24 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 2 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [73] $31^{11}31^13 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 6 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [74] $2\overset{1}{3}1^{11}3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$

- $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [75] $221^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 8 & 13 & 6 \\ 6 & 13 & 13 & 7 \\ 3 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [76] $21^{11}412 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 6 & 4 \\ 2 & 7 & 4 & 7 & 4 \\ 2 & 6 & 7 & 3 & 3 \\ 2 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
- [77] $1^{11}41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 4 & 6 & 8 & 4 \\ 6 & 4 & 9 & 4 \\ 8 & 9 & 9 & 5 \\ 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [78] $131^{11}31 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 6 & 4 \\ 2 & 7 & 4 & 7 & 4 \\ 2 & 6 & 7 & 3 & 3 \\ 2 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
- [79] $13^{11}22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [80] $2^{11}31 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [81] $2^{11}313 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [82] $231^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 & 2 \\ 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [83] $31^{11}313 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -234$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 6 & 4 \\ 3 & 7 & 4 & 7 & 4 \\ 3 & 6 & 7 & 3 & 3 \\ 2 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [84] $141^{11}22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -234$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 6 & 4 \\ 3 & 7 & 4 & 7 & 4 \\ 3 & 6 & 7 & 3 & 3 \\ 2 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [85] $1321^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -234$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [86] $41^{11}222 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -416$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 10 & 13 & 12 & 6 \\ 4 & 13 & 10 & 13 & 6 \\ 4 & 12 & 13 & 9 & 5 \\ 2 & 6 & 6 & 5 & 2 \end{pmatrix}_{b_0} \\
& [87] \quad 2^{21^{11}} 3^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 8 & 11 & 10 & 6 \\ 4 & 11 & 8 & 11 & 6 \\ 4 & 10 & 11 & 7 & 5 \\ 2 & 6 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
& [88] \quad 231^{11} 23 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -650, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.17 \rightarrow 11, T_{\max} = 13, T \in [11, 13] \\
& \begin{pmatrix} 0 & 5 & 5 & 5 & 2 \\ 5 & 8 & 11 & 10 & 6 \\ 5 & 11 & 8 & 11 & 6 \\ 5 & 10 & 11 & 7 & 5 \\ 2 & 6 & 6 & 5 & 2 \end{pmatrix}_{b_0} \\
& T = 8 \quad [90] \quad 1^1 2^1 3^1 21^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
& [89] \quad 1^1 3^1 221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
& [91] \quad 21^1 41^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 4 & 4 \\ 3 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
& [92] \quad 21^1 4131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 5 & 7 & 5 \\ 7 & 9 & 7 \\ 5 & 7 & 5 \end{pmatrix}_{b_0} \\
& [93] \quad 21^{11} 51^1 22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 20, T \in [15, 20] \\
& \begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 4 & 5 & 6 & 4 \\ 3 & 5 & 4 & 5 & 4 \\ 2 & 6 & 5 & 3 & 3 \\ 2 & 4 & 4 & 3 & -1 \end{pmatrix}_{b_0} \\
& [94] \quad 1^1 2^1 3^1 21^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -61, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
& [95] \quad 1^1 5^1 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& 21^1 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0} \\
& [96] \quad 21^1 5^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0} \\
& [97] \quad 1^1 3^1 221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 4 & -1 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [98] $1^1 2^1 3^1 2^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -61, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [99] $41^1 2231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [100] $1^1 4^1 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 4 & 4 \\ 4 & 5 & 5 \\ 4 & 5 & 5 \end{pmatrix}_{b_0}$
- [101] $1^1 4^1 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [103] $21^1 5^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [102] $1^1 5^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [104] $1^1 4^1 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 3 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [105] $221^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [106] $1^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [107] $21^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [108] $1^1 5^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [109] $41^1 231^1 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 13, \Delta = -95, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 4 & 3 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[110] \quad 221 \overset{1}{5} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[111] \quad 1 \overset{1}{5} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[112] \quad 1 \overset{1}{5} 12 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[113] \quad 21 \overset{1}{5} 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[114] \quad 21 \overset{1}{5} 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 1 & 6 & 3 \\ 5 & 6 & 13 & 7 \\ 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[115] \quad 21 \overset{1}{4} 1^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[116] \quad 1 \overset{1}{5} 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[120] \quad 1 \overset{1}{5} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[121] \quad 1 \overset{1}{5} 12 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[119] \quad 1 \overset{1}{4} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 3 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[118] \quad 1 \overset{1}{4} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

- [117] $221\bar{5}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$
- [122] $1^1 41^{11} 231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$
- [123] $31^1 3141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [124] $1^1 \bar{5}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [125] $31^1 31^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 3 & 1 & 4 & 0 & 2 \\ 2 & 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [126] $1^1 41^{11} 231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [127] $221^1 \bar{5}1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 5 & 3 & 3 \\ 5 & 8 & 7 & 6 \\ 3 & 7 & 4 & 4 \\ 3 & 6 & 4 & 3 \end{pmatrix}_{b_0}$
- [128] $21^1 \bar{5}1^{11} 22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 & 2 \\ 3 & 4 & 7 & 4 & 4 \\ 3 & 7 & 4 & 5 & 4 \\ 2 & 4 & 5 & 3 & 3 \\ 2 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [129] $21^1 \bar{5}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [130] $1^1 \bar{5}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [131] $1^1 \bar{5}1^{11} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 6 & 8 & 4 \\ 6 & 4 & 9 & 4 \\ 8 & 9 & 13 & 7 \\ 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [132] $21^1 \bar{5}1^{11} 22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 5 & 6 & 4 \\ 2 & 5 & 4 & 5 & 4 \\ 3 & 6 & 5 & 3 & 3 \\ 2 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [133] $1^1 41^{11} 231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[134] & 31^1 31^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 3 & 4 \\ 1 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[135] & 2^1 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 4 & 3 & 3 & 4 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[136] & 2^1 3^1 141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[137] & 221^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 6 & 3 \\ 6 & 6 & 13 & 7 \\ 3 & 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[138] & 31^1 31^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 2 & 7 & 4 \\ 6 & 7 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[139] & 41^1 1231^1 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 4 & 3 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 3 & 5 & 1 & 5 & 3 \\ 4 & 12 & 5 & 9 & 5 \\ 2 & 6 & 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[140] & 1^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[141] & 3131^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[142] & 21^1 51^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[143] & 1^1 5^1 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[144] & 31^1 4^1 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[146] & 2^1 3^1 141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[145] & 31^{11}3141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 1 & 4 & 3 \\ 3 & 1 & 4 & 0 & 2 \\ 2 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[147] & 2^1 3^{141} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 8 & 8 & 6 \\ 4 & 8 & 5 & 5 \\ 3 & 6 & 5 & 4 \end{pmatrix}_{b_0} \\
[148] & 141^{11}31^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.44 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[149] & 2221^{151}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 18 & 1 & 8 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 8 & -1 & 2 \end{pmatrix}_{b_0} \\
[150] & 21^1 51^{11}2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0} \\
[151] & 1^1 51^{11} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0} \\
[152] & 31^1 31^{141} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.10 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 8 & 8 & 6 \\ 5 & 8 & 5 & 5 \\ 3 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[153] & 31^{11}3141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[154] & 41^1 2^1 3^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[155] & 31^1 4^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 2 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[156] & 221^{11} 1^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[157] & 141^{11}31^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 3 & 4 \\ 3 & 7 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0} \\
[158] & 31^{11} 4^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0} \\
[159] & 31^{11} 3141^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 2 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[160] & 41^{11} 3^{22} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 5 & 8 & 5 \\ 3 & 8 & 8 & 4 & 4 \\ 2 & 6 & 5 & 4 & 3 \end{pmatrix}_{b_0} \\
[161] & 141^{13} 31^{11} 3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0} \\
[162] & 1^{221} 5^{11} 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 8 & 13 & 6 \\ 6 & 13 & 13 & 7 \\ 3 & 6 & 7 & 3 \end{pmatrix}_{b_0} \\
[163] & 3131^{11} 41^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 8 & 13 & 6 \\ 6 & 13 & 13 & 7 \\ 3 & 6 & 7 & 3 \end{pmatrix}_{b_0} \\
[164] & 412^{111} 3^2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[165] & 41^{11} 232 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 1 & 5 & 3 \\ 4 & 12 & 5 & 9 & 5 \\ 2 & 6 & 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[166] & 31^{111} 4^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.42 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[167] & 2221^{11} 51^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 1 & 20 & 8 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 20 & 1 & 17 & 7 \\ 2 & 8 & 0 & 7 & 2 \end{pmatrix}_{b_0} \\
[168] & 41^{11} 31^{13} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 5 & 2 \\ 4 & 10 & 13 & 12 & 6 \\ 4 & 13 & 10 & 13 & 6 \\ 5 & 12 & 13 & 12 & 6 \\ 2 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[169] \ 31^{11}413 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 & 2 \\ 3 & 4 & 7 & 3 & 4 \\ 3 & 7 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[170] \ 41^{11}2231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 41^{11}232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[172] \ 321^{11}41^13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 5 & 3 & 5 & 2 \\ 5 & 13 & 6 & 15 & 7 \\ 3 & 6 & 1 & 6 & 3 \\ 5 & 15 & 6 & 12 & 6 \\ 2 & 7 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[173] \ 41^{11}322 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 4 & 3 & 2 \\ 3 & 5 & 9 & 7 & 5 \\ 4 & 9 & 10 & 9 & 6 \\ 3 & 7 & 9 & 4 & 4 \\ 2 & 5 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 2221^{11}51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 18 & 21 & 0 & 8 \\ 4 & 21 & 18 & 1 & 8 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[175] \ 41^{11}322 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 4 & 3 & 2 \\ 4 & 10 & 13 & 8 & 6 \\ 4 & 13 & 10 & 9 & 6 \\ 3 & 8 & 9 & 4 & 4 \\ 2 & 6 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 14131^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[177] \ 41^{11}322 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[178] \ 41^{11}31^{11}32 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 5 & 4 & 5 & 2 \\ 5 & 13 & 13 & 15 & 7 \\ 4 & 13 & 10 & 13 & 6 \\ 5 & 15 & 13 & 12 & 6 \\ 2 & 7 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [179] \ 1^141321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[180] \ 21^{11}51^132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[181] & 1^{\overline{21}} 5^{11} 131^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[182] & 21^{\overline{1}} 51231^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[183] & 1^{\overline{1}} 41321^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[184] & 1^{\overline{1}} 5131^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 4 & 4 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0} \\
[185] & 41^{\overline{1}} 23141^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[186] & 21^{\overline{1}} 5131^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 5 & 5 \\ 4 & 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[187] & 151^{\overline{1}} 2231^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & 6 & 3 & 3 \\ 4 & 6 & 18 & 6 & 8 \\ 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[188] & 1^{\overline{1}} 51231^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[189] & 31^{\overline{1}} 3151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 2 & 4 & 4 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0} \\
[190] & 21^{\overline{1}} 51^{\overline{1}} 231^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 9 & 5 & 4 \\ 9 & 19 & 11 & 9 \\ 5 & 11 & 5 & 5 \\ 4 & 9 & 5 & 4 \end{pmatrix}_{b_0} \\
[195] & 21^{\overline{1}} 5131^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 5 \\ 3 & 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[191] & 21^{\overline{1}} 5131^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

- [192] $1^{\overline{1}}\overline{5}1231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [193] $1^{\overline{1}}\overline{5}131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [194] $2^{\overline{1}}3^{\overline{1}}1\overline{5}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [197] $31^{\overline{1}}321^{\overline{1}}\overline{5}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 3 & 2 \\ 4 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [196] $1^{\overline{1}}\overline{5}1231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [198] $1^{\overline{1}}\overline{5}131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 5 \\ 3 & 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [199] $321^{\overline{1}}4141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 17 & 9 \\ 3 & 4 & 9 & 4 \end{pmatrix}_{b_0}$
- [200] $2^{\overline{1}}3^{\overline{1}}1\overline{5}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$
- [201] $21^{\overline{1}}\overline{5}131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [202] $21^{\overline{1}}\overline{5}1^{\overline{1}}231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 10 & 4 & 4 & 6 \\ 2 & 4 & 0 & 2 & 2 \\ 1 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [203] $2^{\overline{1}}3^{\overline{1}}1\overline{5}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -17$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 5 & 5 \\ 3 & 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [204] $1^{\overline{1}}\overline{4}1^{\overline{1}}3141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 5 & 1 & 0 & 2 \\ 2 & 7 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[205] & 31^1 32151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[206] & 1^1 5 131^1 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[207] & 1^1 5 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 19 & 11 & 9 \\ 3 & 11 & 5 & 5 \\ 3 & 9 & 5 & 4 \end{pmatrix}_{b_0} \\
[208] & 1^1 41^1 3141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 7 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[209] & 1^1 3221^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 8 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[210] & 1^1 5 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 1 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[211] & 31^1 321^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 7 & 5 & 3 \\ 7 & 19 & 11 & 9 \\ 5 & 11 & 5 & 5 \\ 3 & 9 & 5 & 4 \end{pmatrix}_{b_0} \\
[212] & 1^1 3221^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 1 & 5 & 8 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 5 & 1 & 0 & 2 \\ 2 & 8 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[213] & 1^1 5 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 3 & 4 & 10 & 5 & 6 \\ 1 & 1 & 5 & 0 & 2 \\ 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[214] & 31^1 3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 4 & 3 \\ 5 & 4 & 5 & 5 \\ 3 & 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[215] & 1^1 41^1 3141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & 6 & 0 & 3 \\ 3 & 6 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 7 & -1 & 3 \end{pmatrix}_{b_0} \\
[216] & 31^1 4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 13 & 1 & 9 & 7 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 9 & 1 & 4 & 4 \\ 2 & 7 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[217] \quad 2 \quad 3 \quad 21^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 2 & 4 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[218] \quad 41^{11} 231^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 4 & 1 & 2 \\ 3 & 10 & 17 & 0 & 6 \\ 4 & 17 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 6 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[221] \quad 2 \quad 3 \quad 2151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[219] \quad 31 \quad 4 \quad 141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 7 & 1 & 4 & 4 \\ 2 & 5 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[220] \quad 1^{11} 51 \quad 3 \quad 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[222] \quad 2321^{11} 51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[223] \quad 31^{11} 4 \quad 141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 13 & 10 & 0 & 7 \\ 2 & 10 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[225] \quad 1^{11} 3221^{11} 51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 18 & 6 & 8 \\ 1 & 0 & 6 & 0 & 2 \\ 2 & 0 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[226] \quad 2 \quad 3 \quad 21^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 3 & 7 & 5 \\ 4 & 7 & 6 & 6 \\ 3 & 5 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[224] \quad 231^{11} 3151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 33 & 1 & 11 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[227] \quad 2321^{11} 51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 4 & 2 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 4 & 4 & 10 & 5 & 6 \\ 2 & 1 & 5 & 0 & 2 \\ 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

- [228] $31^{\overline{11}}4141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 3 & 1 & 2 \\ 2 & 5 & 10 & 0 & 5 \\ 3 & 10 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [229] $151^{\overline{11}}32 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$
- [230] $232151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$
- [231] $31^{\overline{11}}4141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 5 & 8 & 0 & 5 \\ 2 & 8 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 5 & 5 & -1 & 3 \end{pmatrix}_{b_0}$
- [232] $2321^{\overline{11}}51^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 7 & 4 & 3 \\ 7 & 19 & 11 & 9 \\ 4 & 11 & 5 & 5 \\ 3 & 9 & 5 & 4 \end{pmatrix}_{b_0}$
- [233] $31321^{\overline{11}}51^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [234] $15131^{\overline{11}}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [235] $31^{\overline{11}}32151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [236] $223151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -55$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 5 & 8 & 5 \\ 3 & 2 & 8 & 4 & 4 \\ 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$
- [237] $2^{\overline{11}}3^{\overline{11}}41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 2 & 4 & 4 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [238] $151^{\overline{11}}232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [239] $31^{\overline{11}}513 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[240] $22 \overset{1}{3} \overset{1}{151} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[241] $2 \overset{1}{\overset{1}{3}} \overset{1}{\overset{1}{41}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[242] $31 \overset{1}{5} \overset{1}{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[243] $41 \overset{231}{4} \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 5 & 3 & 3 & 2 \\ 5 & 13 & 10 & 9 & 7 \\ 3 & 10 & 5 & 8 & 5 \\ 3 & 9 & 8 & 4 & 4 \\ 2 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[244] $15 \overset{1}{131} \overset{1}{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 2 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[245] $31 \overset{11}{13} 32151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 2 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[246] $22 \overset{1}{3} \overset{1}{151} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 9 & 14 & 7 \\ 6 & 14 & 14 & 8 \\ 3 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

[247] $31 \overset{321}{5} \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 5 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 5 & 5 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[248] $41 \overset{11}{231} 141 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[249] $231 \overset{11}{13} 151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 33 & 1 & 35 & 11 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 35 & 1 & 32 & 10 \\ 2 & 11 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[250] $151 \overset{11}{2231} \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 18 & 21 & 5 & 8 \\ 4 & 21 & 18 & 6 & 8 \\ 1 & 5 & 6 & 0 & 2 \\ 2 & 8 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

- [251] $151^{11}232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [252] $31321^{11}51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 8 & 3 \\ 7 & 19 & 25 & 9 \\ 8 & 25 & 26 & 10 \\ 3 & 9 & 10 & 3 \end{pmatrix}_{b_0}$
- [253] $41^231^{11}4 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 5 & 3 & 2 \\ 3 & 5 & 10 & 7 & 5 \\ 5 & 10 & 13 & 10 & 7 \\ 3 & 7 & 10 & 4 & 4 \\ 2 & 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [254] $321^{11}4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 7 & 8 & 3 \\ 7 & 12 & 17 & 8 \\ 8 & 17 & 17 & 9 \\ 3 & 8 & 9 & 4 \end{pmatrix}_{b_0}$
- [255] $215131^{11}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -89, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [256] $1^{31}5^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -89, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [257] $41^14^{11}132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$
- [258] $151^{11}3^{11}22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$
- [259] $231^{11}3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 33 & 36 & 0 & 11 \\ 5 & 36 & 33 & 1 & 11 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 11 & 11 & -1 & 3 \end{pmatrix}_{b_0}$
- [260] $31^{321}5^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 5 & 5 & 2 & 2 \\ 5 & 8 & 11 & 4 & 6 \\ 5 & 11 & 8 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [261] $41^231^{11}4^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 5 & 5 & 3 & 2 \\ 5 & 13 & 16 & 9 & 7 \\ 5 & 16 & 13 & 10 & 7 \\ 3 & 9 & 10 & 4 & 4 \\ 2 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [262] $21^{11}4132^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 12 & 15 & 14 & 8 \\ 4 & 15 & 12 & 15 & 8 \\ 4 & 14 & 15 & 11 & 7 \\ 2 & 8 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [263] \quad 1^1 4^{11} 13221^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[264] \quad 1^1 31^1 5^{11} 131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[269] \quad 21^1 612^1 3^1 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[265] \quad 21^1 612231^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 7 & 5 \\ 7 & 9 & 7 \\ 5 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[266] \quad 21^1 513141^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[267] \quad 1^1 6^{21} 1231^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 7 & 5 \\ 7 & 9 & 7 \\ 5 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[268] \quad 1^1 31^1 5^{11} 131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^1 31^1 5^{11} 131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[277] \quad 2161^1 2231^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[278] \quad 1^1 6^{11} 3221^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[271] \quad 151^1 23141^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 6 & 4 \\ 6 & 6 & 6 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[272] \quad 41^1 231^1 5^{11} 1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 51 & 15 \\ 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned} [273] \quad & 141^1 131^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 5 & 6 & 5 \\ 6 & 6 & 6 \\ 5 & 6 & 5 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [274] \quad & 41^1 232151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [275] \quad & 1^1 413151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 5 & 4 & 5 \\ 4 & 2 & 4 \\ 5 & 4 & 5 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [276] \quad & 41^1 314141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [284] \quad & 1^1 413151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 2 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 5 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [279] \quad & 16^1 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [280] \quad & 21^1 161231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [281] \quad & 23^1 2161^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [282] \quad & 21^1 61^1 2231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 4 & 2 & 1 & 2 \\ 4 & 10 & 7 & 0 & 6 \\ 2 & 7 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 4 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [283] \quad & 1^1 612231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [285] \quad & 1^1 413151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [286] \quad & 4131^1 4141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{\overline{1}321^1} 1^{\overline{1}2} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 4 & 5 & 0 & 4 \\ 2 & 5 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[289] $21^{\overline{1}61231^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[287] $21^{\overline{1}61^1 232} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 7 & 4 & 6 \\ 2 & 7 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 2 & 6 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{\overline{1}41^1 32151^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[292] $1^{\overline{1}5131^1 41^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[291] $31^{\overline{1}413151^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 8 & 5 \\ 8 & 9 & 7 \\ 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[297] $1^{\overline{1}321^1} 1^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 1 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{\overline{1}5131^1 41^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{\overline{1}41321^1 51^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[294] $21^{\overline{1}321^1} 1^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{\overline{1}41^1 32151^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{\overline{1}5131^1 41^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[301] & 141^{1321}151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 3 & 1 & 2 \\ 4 & 2 & 4 & 0 & 4 \\ 3 & 4 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[300] & 141^{131}51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[302] & 1^{1321}1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 3 & 4 & 10 & 5 & 6 \\ 1 & 1 & 5 & 0 & 2 \\ 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[303] & 1^{141321}151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 2 & 3 \\ 5 & 3 & 2 & 5 \\ 2 & 2 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[304] & 21^{1321}1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[305] & 1^{13221}1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 10 & 4 & 0 & 6 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[298] & 1^{141}132151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[306] & 151^{1231}141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 8 & 8 & 6 \\ 6 & 8 & 6 & 6 \\ 3 & 6 & 6 & 4 \end{pmatrix}_{b_0} \\
[307] & 41^{1231}51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 5 & 0 & 2 \\ 4 & 5 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[308] & 1^1 61232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[309] & 1^{2321}1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [311] $1^{\overline{1}}41321^{\overline{1}}51^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 2 & 7 & 4 \\ 7 & 7 & 7 \\ 4 & 7 & 5 \end{pmatrix}_{b_0}$
- [312] $3141^{\overline{1}}3151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 5 & 10 & 5 \\ 10 & 13 & 9 \\ 5 & 9 & 4 \end{pmatrix}_{b_0}$
- [310] $1^{\overline{1}}3221^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 4 & 0 & 2 \\ 4 & 4 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [313] $1^{\overline{1}}2321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 4 & 4 & 10 & 5 & 6 \\ 2 & 1 & 5 & 0 & 2 \\ 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [314] $31^{\overline{1}}41^{\overline{1}}3151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 3 & 5 & 1 & 2 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [315] $41^{\overline{1}}4141^{\overline{1}}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 4 & 7 & 5 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 7 & 4 & 4 & 4 \\ 2 & 5 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [316] $31^{\overline{1}}41^{\overline{1}}3151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 5 & 3 & 1 & 2 \\ 5 & 8 & 7 & 0 & 6 \\ 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [317] $1^{\overline{1}}2^{\overline{1}}321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [318] $41^{\overline{1}}31^{\overline{1}}513 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [319] $2161^{\overline{1}}2^{\overline{1}}32 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [320] $41^{\overline{1}}31^{\overline{1}}4141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 5 & 1 & 2 \\ 4 & 26 & 33 & 0 & 10 \\ 5 & 33 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [321] $31^{\overline{1}}413151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 0 & 4 \\ 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 2 \end{pmatrix}_{b_0} \\
[322] & 41^1 31^1 4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 5 & 4 & 1 & 2 \\ 5 & 38 & 33 & 0 & 12 \\ 4 & 33 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 12 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[328] & 41^{11} 314141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[327] & 41^{11} 232151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[326] & 1^{11} 3221^{11} 6 \quad 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[325] & 2161^{11} 2231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[324] & 151^{11} 23141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 3 & 7 & 5 \\ 6 & 7 & 6 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[323] & 41^{11} 23151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 9 & 3 \\ 8 & 42 & 49 & 14 \\ 9 & 49 & 51 & 15 \\ 3 & 14 & 15 & 4 \end{pmatrix}_{b_0} \\
[329] & 2161^{11} 232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[330] & 1^{11} 2321^{11} 6 \quad 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[331] & 41^{11} 31513 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[332] & 51^{11} 223141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 6 & 7 & 3 \\ 6 & 4 & 8 & 6 \\ 7 & 8 & 7 & 7 \\ 3 & 6 & 7 & 4 \end{pmatrix}_{b_0} \\
[333] & 4131^{11} 4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 218, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 38 & 41 & 0 & 12 \\ 5 & 41 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 12 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[334] $41 \overset{1}{4} 1413 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[335] $3141 \overset{1}{3} 151 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[336] $2231 \overset{1}{3} 1316 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 251, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 7 & 1 & 7 & 2 \\ 7 & 33 & 6 & 35 & 11 \\ 1 & 6 & 0 & 6 & 2 \\ 7 & 35 & 6 & 32 & 10 \\ 2 & 11 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [337] $1 \overset{1}{3} 15 1321 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[338] $1 \overset{1}{3} 15 1321 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

[339] $1 \overset{1}{3} 6 131 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[341] $1 \overset{1}{3} 15 1321 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

[340] $1 \overset{1}{3} 6 131 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[342] $1 \overset{21}{7} 12 \overset{1}{3} 2 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[343] $1 \overset{1}{3} 6 131 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[344] $1 \overset{1}{3} 6 131 \overset{1}{1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[345] $21^1 1^1 7^{11} 12^1 3^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[346] $23^1 21^1 7^1 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[347] $15131^1 4141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 17 & 9 \\ 3 & 4 & 9 & 4 \end{pmatrix}_{b_0}$$

[349] $321^1 513151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[348] $51231^1 4141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 17 & 9 \\ 3 & 4 & 9 & 4 \end{pmatrix}_{b_0}$$

[350] $1^{23} 21^1 7^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 7 & 4 & 6 \\ 2 & 7 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 2 & 6 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[351] $31^1 5^1 1321^1 5 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 5 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 5 & 3 & 3 & 3 & 5 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[353] $3151^1 23151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 7 & 1 & 68 & 1 & 16 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[352] $1^1 4141^1 3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 8 & 3 & 0 & 6 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[354] $1^1 4141^1 3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 6 & 2 & 3 \\ 6 & 9 & 3 & 7 \\ 2 & 3 & 0 & 2 \\ 3 & 7 & 2 & 4 \end{pmatrix}_{b_0}$$

[357] $1^1 7^1 12^1 3^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 2 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[356] $31^1 5^1 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[355] \quad 21 \overset{1}{7} 12 \overset{1}{3} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[358] \quad 21 \overset{1}{7} \overset{1}{2} 1 \overset{1}{3} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[359] \quad 1 \overset{1}{2} 3 \overset{1}{2} 1 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 18 & 5 & 10 & 8 \\ 1 & 5 & 0 & 3 & 2 \\ 2 & 10 & 3 & 4 & 4 \\ 2 & 8 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[360] \quad 51 \overset{1}{1} 231 \overset{1}{4} 141 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 7 & 7 & 3 \\ 7 & 12 & 10 & 8 \\ 7 & 10 & 7 & 7 \\ 3 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[361] \quad 31 \overset{1}{5} 13151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -41, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[362] \quad 1 \overset{1}{1} 4141 \overset{1}{1} 3151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 2 & 9 & 4 \\ 9 & 18 & 10 \\ 4 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[363] \quad 2 \overset{1}{1} \overset{1}{3} \overset{1}{4} 141 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 6 & 4 & 5 \\ 2 & 6 & 3 & 5 & 5 \\ 2 & 4 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[364] \quad 2 \overset{1}{1} \overset{1}{3} \overset{1}{4} 141 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[365] \quad 31 \overset{1}{5} 1321 \overset{1}{1} 5 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -167, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 5 & 5 & 2 & 2 \\ 5 & 3 & 6 & 2 & 5 \\ 5 & 6 & 3 & 3 & 5 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 5 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[366] \quad 51 \overset{1}{1} 2314141 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -188, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 6 & 7 & 3 \\ 6 & 4 & 8 & 6 \\ 7 & 8 & 7 & 7 \\ 3 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[367] \quad 321 \overset{1}{1} 513151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -218, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 38 & 41 & 0 & 12 \\ 5 & 41 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 12 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[368] & 51231^{11}4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -218, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 7 & 8 & 3 \\ 7 & 12 & 17 & 8 \\ 8 & 17 & 17 & 9 \\ 3 & 8 & 9 & 4 \end{pmatrix}_{b_0} \\
[369] & 15131^{11}4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -218, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 7 & 8 & 3 \\ 7 & 12 & 17 & 8 \\ 8 & 17 & 17 & 9 \\ 3 & 8 & 9 & 4 \end{pmatrix}_{b_0} \\
[370] & 14141^{11}3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -248, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[371] & 21^{11}414132^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -416, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 16 & 19 & 18 & 10 \\ 4 & 19 & 16 & 19 & 10 \\ 4 & 18 & 19 & 15 & 9 \\ 2 & 10 & 10 & 9 & 5 \end{pmatrix}_{b_0} \\
[372] & 3151^{11}23151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -458, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 68 & 71 & 0 & 16 \\ 7 & 71 & 68 & 1 & 16 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 16 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
T = 12 & [373] 1^{11}231513221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.0 \\
[376] & 1^{11}6^{11}131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[377] & 1^{11}316^{11}131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[378] & 1^{11}6^{11}131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
[379] & 21^{11}61315^{11}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[380] & 41^{11}31513151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[381] & 31513161^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[382] & 1 \overset{321}{6} 13151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 10 & 5 \\ 10 & 18 & 10 \\ 5 & 10 & 5 \end{pmatrix}_{b_0} \\
[383] & 321^{11} 6123151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 10 & 5 \\ 10 & 18 & 10 \\ 5 & 10 & 5 \end{pmatrix}_{b_0} \\
[384] & 161^{11} 231 \overset{11}{5} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 5 & 2 & 2 \\ 2 & 1 & 6 & 3 & 3 \\ 5 & 6 & 18 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 2 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[385] & 21^{11} \overset{11}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 9 & 5 \\ 9 & 13 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[386] & 1^{11} \overset{11}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 12 & 5 \\ 12 & 24 & 12 \\ 5 & 12 & 5 \end{pmatrix}_{b_0} \\
[387] & 32161^{11} 23151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 12 & 5 \\ 12 & 24 & 12 \\ 5 & 12 & 5 \end{pmatrix}_{b_0} \\
[388] & 31 \overset{11}{6} 13151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[389] & 1 \overset{11}{6} 131 \overset{11}{5} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[390] & 31 \overset{11}{5} 132161^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[391] & 31 \overset{11}{5} 1321^{11} 61^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 18 & 5 & 5 & 8 \\ 1 & 5 & 0 & 2 & 2 \\ 2 & 5 & 2 & 0 & 2 \\ 2 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[392] & 61^{11} 231 \overset{11}{5} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 3 & 6 & 3 & 2 \\ 3 & 5 & 13 & 7 & 5 \\ 6 & 13 & 28 & 13 & 10 \\ 3 & 7 & 13 & 4 & 4 \\ 2 & 5 & 10 & 4 & 3 \end{pmatrix}_{b_0} \\
[394] & 31 \overset{11}{6} 13151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 41, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[393] $321 \overset{11}{6} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 41, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[395] $31 \overset{11}{6} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 41, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[396] $321 \overset{1}{6} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 13 & 6 & 0 & 7 \\ 2 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[397] $31 \overset{1}{5} 1321^1 61^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 5 & 2 & 2 \\ 1 & 0 & 5 & 1 & 2 \\ 5 & 5 & 18 & 6 & 8 \\ 2 & 1 & 6 & 0 & 2 \\ 2 & 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[398] $231 \overset{11}{5} 13161^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 61, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.69 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[399] $321 \overset{1}{6} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 5 & 1 & 2 \\ 2 & 1 & 6 & 0 & 3 \\ 5 & 6 & 13 & 1 & 7 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[400] $31 \overset{1}{6} \overset{151321}{6} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 85, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.64 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 19 & 7 & 0 & 9 \\ 2 & 7 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[401] $31 \overset{1}{6} \overset{151321}{6} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 122, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 7 & 1 & 2 \\ 2 & 1 & 7 & 0 & 3 \\ 7 & 7 & 19 & 1 & 9 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[402] $41 \overset{11}{3} 1513151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 131, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[403] $21 \overset{11}{6} \overset{15131}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 8 & 11 & 0 & 6 \\ 4 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[404] $161 \overset{11}{231} \overset{1}{5} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 167, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 5 & 2 & 2 \\ 5 & 18 & 21 & 5 & 8 \\ 5 & 21 & 18 & 6 & 8 \\ 2 & 5 & 6 & 0 & 2 \\ 2 & 8 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[405] $61^{11}22315^{11}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 6 & 6 & 2 & 2 \\ 6 & 28 & 31 & 6 & 10 \\ 6 & 31 & 28 & 7 & 10 \\ 2 & 6 & 7 & 0 & 2 \\ 2 & 10 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[406] $61^{11}2315^{11}132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -82, \det = 207, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.68 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 6 & 6 & 3 & 2 \\ 6 & 28 & 31 & 12 & 10 \\ 6 & 31 & 28 & 13 & 10 \\ 3 & 12 & 13 & 4 & 4 \\ 2 & 10 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[407] $321^{11}6123151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -82, \det = 218, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 13 & 1 & 7 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[408] $1^{32111}6^{11}13151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 218, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 13 & 1 & 7 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[409] $3221^{11}613151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 416, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[410] $32161^{11}23151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -82, \det = 458, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.76 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 19 & 22 & 0 & 9 \\ 7 & 22 & 19 & 1 & 9 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[411] $1^{115132111}6^{11}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 458, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.76 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 19 & 22 & 0 & 9 \\ 7 & 22 & 19 & 1 & 9 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[412] $3161^{11}223151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -81, \det = 732, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 9 & 9 & 1 & 2 \\ 9 & 34 & 37 & 0 & 12 \\ 9 & 37 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 13$ [413] $1^{112321}7^{11}1231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13, T_{\max} = 14, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

[414] $23218^{11}1231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[415] $23218^{11}1231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[416] & 1^{\overline{1}} \overline{7}^{12321} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
[417] & 21^{\overline{1}} 71231 \overline{5}^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[420] & 41^{\overline{1}} 31^{\overline{1}} \overline{6}^{115131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[418] & 41^{\overline{1}} 316123151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[419] & 41^{\overline{1}} 321613151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[422] & 21^{\overline{1}} 71^{\overline{1}} 231 \overline{5}^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 5 & 2 & 2 & 2 \\ 5 & 18 & 9 & 5 & 8 \\ 2 & 9 & 4 & 3 & 4 \\ 2 & 5 & 3 & 0 & 2 \\ 2 & 8 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[421] & 31^{\overline{1}} \overline{5}^{1321} 71^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[423] & 232181231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[424] & 41^{\overline{1}} 31613151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -17, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.59 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 4 & 2 & 1 & 2 \\ 4 & 10 & 5 & 0 & 6 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[425] & 413161^{\overline{1}} 23151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 5 & 12 & 5 \\ 12 & 24 & 12 \\ 5 & 12 & 5 \end{pmatrix}_{b_0} \\
[426] & 31^{\overline{1}} \overline{6}^{15131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -26, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 3 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[427] & 31^{\overline{1}} \overline{5}^{13221} 71^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16]
\end{aligned}$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 & 2 \\ 6 & 28 & 6 & 6 & 10 \\ 1 & 6 & 0 & 2 & 2 \\ 2 & 6 & 2 & 0 & 2 \\ 2 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[430] $1^1 7 1231^1 5 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -31$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.90 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[428] $31^1 5 1322171^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -31$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.90 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[429] $1^1 31^1 5 1321^1 7 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -31$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.65 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 18 & 5 & 5 & 8 \\ 1 & 5 & 0 & 2 & 2 \\ 2 & 5 & 2 & 0 & 2 \\ 2 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[431] $31^1 6 131^1 11 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 15$, $\Delta = -59$, $\det = -35$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 5 & 0 & 4 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

[432] $231^1 5 1321^1 71^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -40$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.60 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 6 & 1 & 3 & 2 \\ 6 & 28 & 6 & 12 & 10 \\ 1 & 6 & 0 & 3 & 2 \\ 3 & 12 & 3 & 4 & 4 \\ 2 & 10 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[435] $41^1 31^1 6 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -41$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.83 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 4 & 1 & 2 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[433] $321^1 7 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -41$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.83 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[434] $4131^1 6 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -41$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.83 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[436] $41^1 321^1 613151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -43$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.65 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 7 & 1 & 2 \\ 4 & 10 & 17 & 0 & 6 \\ 7 & 17 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[437] $41^1 321^1 613151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -44$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.91 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 4 & 1 & 2 \\ 7 & 26 & 17 & 0 & 10 \\ 4 & 17 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[438] $321^1 7 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -46$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.65 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 13 & 6 & 0 & 7 \\ 2 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[439] $31 \overset{1}{5} 13221 \overset{1}{7} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.65 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 6 & 2 & 2 \\ 1 & 0 & 6 & 1 & 2 \\ 6 & 6 & 28 & 7 & 10 \\ 2 & 1 & 7 & 0 & 2 \\ 2 & 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[440] $2 \overset{1}{1} \overset{1}{4} 14141 \overset{1}{3} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 5 & 6 \\ 2 & 7 & 4 & 6 & 6 \\ 2 & 5 & 6 & 3 & 5 \\ 2 & 6 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

[441] $31 \overset{1}{1} \overset{1}{5} 131 \overset{1}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 2 & 4 & 4 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[442] $231 \overset{1}{5} 132171 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -61, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.69 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[444] $31 \overset{1}{1} \overset{1}{5} 131 \overset{1}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[443] $2 \overset{1}{1} \overset{1}{4} 14141 \overset{1}{3} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 & 6 \\ 2 & 6 & 4 & 6 & 6 \\ 2 & 6 & 6 & 3 & 5 \\ 2 & 6 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

[445] $321 \overset{1}{5} 1321 \overset{1}{7} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -85, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 19 & 7 & 0 & 9 \\ 2 & 7 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[446] $41 \overset{1}{1} 321613151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -131, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[447] $41 \overset{1}{1} 31 \overset{1}{5} 131 \overset{1}{6} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -131, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[448] $41 \overset{1}{1} 316123151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -131, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[449] $71 \overset{1}{1} 2231 \overset{1}{5} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -188, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 7 & 3 & 2 \\ 7 & 40 & 43 & 14 & 12 \\ 7 & 43 & 40 & 15 & 12 \\ 3 & 14 & 15 & 4 & 4 \\ 2 & 12 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[450] $171^{11}2231^1 5^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -192$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.67 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 6 & 6 & 2 & 2 \\ 6 & 28 & 31 & 6 & 10 \\ 6 & 31 & 28 & 7 & 10 \\ 2 & 6 & 7 & 0 & 2 \\ 2 & 10 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[453] $41321^{11}613151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -416$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.67 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[452] $3221^{11}7123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -416$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.67 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[451] $21^{11}41414132^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 13$, $\Delta = 0$, $\det = -416$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 20 & 23 & 22 & 12 \\ 4 & 23 & 20 & 23 & 12 \\ 4 & 22 & 23 & 19 & 11 \\ 2 & 12 & 12 & 11 & 6 \end{pmatrix}_{b_0}$$

[454] $413161^{11}23151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -458$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.76 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 19 & 22 & 0 & 9 \\ 7 & 22 & 19 & 1 & 9 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[455] $32171^{11}223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -732$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.63 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 9 & 9 & 1 & 2 \\ 9 & 34 & 37 & 0 & 12 \\ 9 & 37 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 14$ [459] $1^{11}8^{13221} 12321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

[458] $1^{11}6^{1315131} 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -30$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

[457] $1^{12321} 8^{1231} 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[462] $1^{11}8^{13221} 12321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

[461] $1^{12321} 8^{1231} 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

[460] $1^{\overline{1}1315131} 6^{11} 131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -30, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[463] $21^{\overline{1}1} 81231^{\overline{1}1} 5^{11} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 5 & 5 & 7 & 5 \\ 2 & 5 & 4 & 5 & 4 \\ 3 & 7 & 5 & 4 & 4 \\ 2 & 5 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[464] $41^{\overline{1}1} 3217123151^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0}$$

[465] $231^{\overline{1}1} 5^{11} 1321^{\overline{1}1} 81^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[466] $21^{\overline{1}1} 81^{\overline{1}1} 231^{\overline{1}1} 5^{11} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 6 & 2 & 3 & 2 \\ 6 & 28 & 11 & 12 & 10 \\ 2 & 11 & 4 & 5 & 4 \\ 3 & 12 & 5 & 4 & 4 \\ 2 & 10 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[467] $21^{\overline{1}11} 8^{11} 51321^{\overline{1}1} 123 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 13 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0}$$

[471] $21^{\overline{1}1} 5^{11} 1321^{\overline{1}1} 8^{11} 123 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 7 & 4 & 1 & 2 \\ 7 & 19 & 13 & 0 & 9 \\ 4 & 13 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[468] $21^{\overline{1}1} 8^{321} 123151^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 5 & 4 & 1 & 2 \\ 5 & 13 & 11 & 0 & 7 \\ 4 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[469] $231^{\overline{1}1} 5^{11} 13221^{\overline{1}1} 81^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 7 & 1 & 3 & 2 \\ 7 & 40 & 7 & 14 & 12 \\ 1 & 7 & 0 & 3 & 2 \\ 3 & 14 & 3 & 4 & 4 \\ 2 & 12 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[470] $321^{\overline{1}1} 8^{11} 123151^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[472] $1^{\overline{1}1} 231^{\overline{1}1} 5^{11} 1321^{\overline{1}1} 8^{11} 1^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 40, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 6 & 1 & 3 & 2 \\ 6 & 28 & 6 & 12 & 10 \\ 1 & 6 & 0 & 3 & 2 \\ 3 & 12 & 3 & 4 & 4 \\ 2 & 10 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[473] $321 \overset{11}{8} 1223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.83 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[474] $3221 \overset{11}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.83 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[475] $321 \overset{11}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.83 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[476] $41^1 321^1 7123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.91 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 4 & 1 & 2 \\ 7 & 26 & 17 & 0 & 10 \\ 4 & 17 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[477] $321^1 \overset{11}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 13 & 6 & 0 & 7 \\ 2 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[478] $321^1 \overset{11}{8} 1223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 13 & 6 & 0 & 7 \\ 2 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[479] $2231 \overset{11}{5} 1321^1 81^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 51, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 1 & 4 & 2 \\ 7 & 40 & 7 & 21 & 12 \\ 1 & 7 & 0 & 4 & 2 \\ 4 & 21 & 4 & 10 & 6 \\ 2 & 12 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[480] $231 \overset{11}{5} 1322181^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 61, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.69 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[481] $181231 \overset{11}{5} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 61, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.69 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[482] $3221 \overset{11}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 26 & 8 & 0 & 10 \\ 2 & 8 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[483] $231 \overset{11}{5} 13221^1 81^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.62 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 7 & 3 & 2 \\ 1 & 0 & 7 & 2 & 2 \\ 7 & 7 & 40 & 15 & 12 \\ 3 & 2 & 15 & 4 & 4 \\ 2 & 2 & 12 & 4 & 3 \end{pmatrix}_{b_0}$

- [484] $321 \overset{1^1 51321^1}{8} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 85, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 19 & 7 & 0 & 9 \\ 2 & 7 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [485] $3221 \overset{1^1 51321^1}{8} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 85, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 19 & 7 & 0 & 9 \\ 2 & 7 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [486] $2231 \overset{1^{11}}{5} 132181^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 101, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 11 & 4 & 13 & 7 \\ 1 & 4 & 0 & 4 & 2 \\ 4 & 13 & 4 & 10 & 6 \\ 2 & 7 & 2 & 6 & 3 \end{pmatrix}_{b_0}$
- [487] $321 \overset{1^1 513221^1}{8} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 9 & 2 & 1 & 2 \\ 9 & 34 & 9 & 0 & 12 \\ 2 & 9 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [488] $3221 \overset{1^1}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 118, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.66 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 7 & 1 & 2 \\ 2 & 1 & 8 & 0 & 3 \\ 7 & 8 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 10 & 0 & 3 \end{pmatrix}_{b_0}$
- [489] $41 \overset{1^1}{3217123151^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 131, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.95 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [490] $21 \overset{1^1 51321}{8} 123 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 143, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.59 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 8 & 11 & 0 & 6 \\ 4 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [491] $81 \overset{1^1}{1^1 22315} 1322 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 8 & 8 & 4 & 2 \\ 8 & 54 & 57 & 24 & 14 \\ 8 & 57 & 54 & 25 & 14 \\ 4 & 24 & 25 & 10 & 6 \\ 2 & 14 & 14 & 6 & 3 \end{pmatrix}_{b_0}$
- [492] $321 \overset{1^1 513221^1}{8} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 181, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 9 & 1 & 2 \\ 2 & 1 & 9 & 0 & 3 \\ 9 & 9 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 12 & 0 & 3 \end{pmatrix}_{b_0}$
- [493] $181 \overset{1^1}{1^1 22315} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 188, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 7 & 3 & 2 \\ 7 & 40 & 43 & 14 & 12 \\ 7 & 43 & 40 & 15 & 12 \\ 3 & 14 & 15 & 4 & 4 \\ 2 & 12 & 12 & 4 & 3 \end{pmatrix}_{b_0}$
- [495] $1 \overset{3221^1}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 416, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0}$
- [494] $3221 \overset{1^1}{8} 1223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 416, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[496] & 322181^{11}223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 732, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16.63 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 9 & 9 & 1 & 2 \\ 9 & 34 & 37 & 0 & 12 \\ 9 & 37 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[497] & 1^{11}513221^{11}8^{11}123 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 732, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16.63 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 9 & 9 & 1 & 2 \\ 9 & 34 & 37 & 0 & 12 \\ 9 & 37 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 16 & [501] 1^{11}3151321^{11}8^{11}1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[502] & 1^{11}3151321^{11}8^{11}1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
T = 17 & [503] 1^{11}23151321^{11}8^{11}12321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0 \\
[507] & 31^{31}6^{11}1316131^{11}51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -56, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17.93 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 6 & 4 & 5 \\ 2 & 6 & 3 & 5 & 5 \\ 2 & 4 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[508] & 31^{31}6^{11}1316131^{11}51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
T = 18 & [512] 1^{11}3221^{11}(12)12231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
T = 19 & [513] 1^{11}2321^{11}8^{11}12321712321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0 \\
[517] & 3221^{11}(12)12231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 20.92 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 3 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[518] & 3221^{11}(12)12231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} =
\end{aligned}$$

4.0

$$T_{\min} = 20.89 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 5 & 0 & 4 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[519] \quad 3221 \quad (12) \quad 1223 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20.93 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 2 & 4 & 4 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[520] \quad 3221 \quad (12) \quad 1223 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [521] \quad 1^1 8^{1^1} 12321 8^{1^1} 12321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

$$T = 22 \quad [522] \quad 1^1 8^{1^1} 12321 8^{1^1} 12315131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 4.0$$

$$T = 23 \quad [523] \quad 1^1 8^{1^1} 12321 8^{1^1} 123216131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 23, T_{\max} = 23, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0$$

6.6 NHC(-2)(-2)(-3) + e₆

$$T = 7 \quad [0] \quad 21^1 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 4 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & -3 & -1 \\ 3 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 4 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & -3 & -1 \\ 3 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 3 & 5 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 4 & 9 & 9 & 4 & 5 \\ 2 & 3 & 4 & -3 & -1 \\ 3 & 4 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 21^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -475, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 4 \\ 6 & 4 & 9 & 3 & 4 \\ 8 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & -3 & -1 \\ 4 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [7] \ 1^{11}31^131^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, T_{\max} = 98, T \in [16, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 1 & 4 & 0 & 3 \\ 2 & 3 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 3 & -4 & -3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 3 & 6 & 5 & 5 & 5 \\ 2 & 3 & 5 & -2 & 0 \\ 3 & 4 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 2231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 5 & 5 & 5 \\ 3 & 3 & 5 & -2 & 0 \\ 3 & 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}31^131^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 4 & 4 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}31^131^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 4 & 7 & 0 & 4 \\ 2 & 3 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 4 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}31^131^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^131^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 4 & 4 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [16] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & 0 & 2 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11}3221^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 4 & 0 & -2 & 0 \\ 2 & 2 & 4 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & 3 \\ 1 & 2 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 0 & 3 & 1 & -3 & -1 \\ 2 & 0 & 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 2 & 2 \\ 2 & 3 & 3 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [22] \quad 1^{11}41321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 2 & 0 & 3 \\ 2 & 2 & 1 & 2 & 3 \\ 1 & 0 & 2 & -4 & -2 \\ 3 & 3 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

6.7 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 103, \Delta = -212, \det = 460, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

6.8 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{11} 3^{11} 2^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{f}_4 \quad T = 9, \quad T_{\max} = 101, \quad \Delta = -158, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 5 & 5 & 0 & 5 \\ 2 & 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

6.9 NHC(-2)(-2)(-3) + $\mathbf{e}_6$ + NHC(-2)(-2)(-3)

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -83, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 6 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.10 NHC(-2)(-2)(-3) + $\mathbf{e}_6$ + NHC(-2)(-3)

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.33 \rightarrow 17, \quad T_{\max} = 24, \quad T \in [17, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -4 & 4 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -129, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 4 & 3 \\ 3 & 3 & 5 & 0 & 4 & 3 & 3 \\ 3 & 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -130, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 3 & 5 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -3 & 3 & 0 & -1 \\ 3 & 6 & 8 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 5 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -275, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 24, \quad T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -322, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 428, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.11 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 51, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 150, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 12 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 294, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 721, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

6.12 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 266, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 3 & 5 & 0 & 4 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 326, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

6.13 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 135, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 6 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 238, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 242, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 253, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.68 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 6 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11} 3^1 2^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.60 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 1 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 1 & 0 & -3 & 1 & -1 \\ 1 & 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 2^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.68 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 3 \\ 1 & 2 & 1 & 1 & 3 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 2^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -105, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.69 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 5 & 5 & 0 & 5 \\ 2 & 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} 3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] 1^{11} 31^{11} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.14 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] 1^{111} 31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{114} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -4 & 34 & -1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] 1^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[6] 1^{111} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[7] 1^{1111} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -525, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 9 & 3 & 0 & 6 & 4 \\ 4 & 9 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 8 & 3 & 0 & 4 & 4 \\ 3 & 4 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[9] 1^{113} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -653, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[10] 1^{111} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[11] 1^{112} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [12] 1^{11} 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 3 & 1 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & -3 & 1 & 0 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 \\ 2 & 2 & 3 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

6.15 NHC(-2)(-2)(-3) + $\mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2$

$$T = 10 \quad [0] 1^{11} 3 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 15 & 7 \\ 9 & 12 & 20 & 0 & 15 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 17 & 5 \\ 8 & 10 & 17 & 0 & 12 & 24 & 6 \\ 12 & 17 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 27 & 7 \\ 17 & 24 & 36 & 0 & 27 & 49 & 13 \\ 5 & 6 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 300, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.64 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 15 & 7 \\ 9 & 15 & 17 & 0 & 15 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 832, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.52 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 17 & 5 \\ 9 & 13 & 18 & 0 & 15 & 27 & 7 \\ 11 & 18 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 27 & 7 \\ 17 & 27 & 33 & 0 & 27 & 49 & 13 \\ 5 & 7 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

6.16 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{113}} 3^{\overline{121}} 21^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -107, \ \det = 138, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.70 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 5 & 5 & 0 & 3 & 5 \\ 2 & 3 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

6.17 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{111}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -274, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.03 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -559, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 25, \ \Delta = -113, \ \det = 87, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.84 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 3 & 1 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & -3 & 1 & 0 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

6.18 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$

$$T = 8 \quad [1] \ 1^{111} 3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 25, T_{\max} = 92, T \in [25, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & -1 & 0 & -12 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.91 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 16 & 0 & 28 & 10 \\ 3 & 3 & 5 & -4 & 10 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.27 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 6 & 16 & 0 & 22 & 8 \\ 3 & 3 & 5 & -4 & 8 & -3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 10 & 0 & 22 & 8 \\ 3 & 4 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 3 & 3 & 4 & -3 & 3 & -1 \\ 3 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 21^1 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 3 & 3 & 4 & -3 & 3 & -1 \\ 3 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{13} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 28 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 15 & 28 & 0 & 85 & 17 \\ 3 & 3 & 5 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.20 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 4 \\ 3 & 3 & 5 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{1111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 4 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 2 & 3 & 4 & -3 & 3 & -1 \\ 4 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 18 & 27 & 0 & 133 & 21 \\ 3 & 3 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 14 \\ 3 & 3 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 7 \\ 3 & 3 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{114}31^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 20 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 24 & 36 & 0 & 238 & 28 \\ 3 & 3 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{112}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 6 \\ 3 & 3 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 142, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 0 & 46 & 12 \\ 3 & 3 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 7 \\ 3 & 3 & 5 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{115} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 151, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.1$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 15 & 20 & 0 & 73 & 15 \\ 3 & 3 & 4 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{116} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 18 & 24 & 0 & 106 & 18 \\ 3 & 3 & 4 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{117} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 175, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.3$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 21 & 28 & 0 & 145 & 21 \\ 3 & 3 & 4 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 198, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{14} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 239, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 4 & 5 & 9 & 0 & 40 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 21 & 40 & 0 & 172 & 24 \\ 3 & 3 & 5 & -4 & 24 & -1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 239, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.98 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 20 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 9 & 20 & 0 & 37 & 11 \\ 3 & 3 & 5 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^1 1^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 250, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 2 & 3 & 4 & -3 & 3 & -1 \\ 3 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 253, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 10 \\ 3 & 3 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{12} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 256, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 4 & 5 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 24 & 0 & 58 & 14 \\ 3 & 3 & 5 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 296, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 4 & 5 & 9 & 0 & 28 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 12 & 28 & 0 & 73 & 15 \\ 3 & 3 & 5 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 306, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.54 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{114} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 314, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 27 & 24 & 0 & 121 & 19 \\ 3 & 4 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 319, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 15 & 36 & 0 & 121 & 19 \\ 3 & 3 & 5 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 336, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 19 & 0 & 73 & 15 \\ 3 & 4 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{112} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 336, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 379, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 17 \\ 3 & 3 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 382, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{13} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 436, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.98 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 4 & 5 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 18 & 36 & 0 & 133 & 21 \\ 3 & 3 & 5 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{112} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 440, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{112} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 450, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{12} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 455, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 32 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 15 & 32 & 0 & 100 & 18 \\ 3 & 3 & 5 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{114} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 505, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{114} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 513, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{12} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 620, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.01 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 4 & 5 & 9 & 0 & 40 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 18 & 40 & 0 & 154 & 22 \\ 3 & 3 & 5 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 630, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.53 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{114} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 634, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.45 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{14} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 688, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 20 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 \\ 4 & 5 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 24 & 48 & 0 & 238 & 28 \\ 3 & 3 & 5 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{114}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 743, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 18 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 4 & 5 & 9 & 0 & 44 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 21 & 44 & 0 & 193 & 25 \\ 3 & 3 & 5 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 30 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{113}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 846, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[55] $21^{1111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1076, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 6 & 4 \\ 6 & 4 & 9 & 3 & 6 & 4 \\ 8 & 9 & 9 & 4 & 8 & 5 \\ 3 & 3 & 4 & -3 & 3 & -1 \\ 6 & 6 & 8 & 3 & 4 & 4 \\ 4 & 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{114}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1098, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [58] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 1 & 3 & 0 & -3 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 \\ 2 & 2 & 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{112}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 18.27 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 1 & 4 & 1 & 4 & 0 & 9 & 3 \\ 2 & 3 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & 3 & -4 & 10 & -6 \end{pmatrix}_{b_0}$$

- [60] $1^{11}321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 16.05 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 5 & 5 & 0 & 5 \\ 2 & 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$
- [61] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 3 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [62] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$
- [63] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -71$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 3 & 3 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$
- [64] $1^{11}31^{12}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 9 & 0 & 37 & 11 \\ 2 & 3 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$
- [65] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$
- [66] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [67] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -323$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 4 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$
- [68] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -473$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [69] $1^{11}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 30 & 18 & 0 & 154 & 22 \\ 2 & 3 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{14}31^{12}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 18 & 24 & 18 & 0 & 118 & 20 \\ 2 & 3 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{13}31^{112}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 12 & 4 \\ 2 & 7 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 4 & 2 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{112}31^{12}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 18 & 12 & 0 & 58 & 14 \\ 2 & 3 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{13}31^{114}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 7 & 3 & 0 & 33 & 4 \\ 2 & 7 & 4 & 4 & 0 & 33 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 4 & 2 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{112}31^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 15 & 24 & 15 & 0 & 100 & 18 \\ 2 & 3 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{113}31^{12}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -731, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 3 & 0 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 21 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 15 & 21 & 15 & 0 & 85 & 17 \\ 2 & 3 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{11}31^{112}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -746, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 7 & 3 & 0 & 15 & 4 \\ 2 & 7 & 4 & 4 & 0 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12}31^{113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -809, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 7 & 3 & 0 & 24 & 4 \\ 2 & 7 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 4 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{13}31^{113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -833, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 7 & 3 & 0 & 18 & 4 \\ 2 & 7 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 4 & 2 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{131}1^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 7 & 3 & 0 & 24 & 4 \\ 2 & 7 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{113}31^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 18 & 27 & 18 & 0 & 133 & 21 \\ 2 & 3 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{113}31^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 7 & 3 & 0 & 21 & 4 \\ 2 & 7 & 4 & 4 & 0 & 21 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{113}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1379, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 33 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 21 & 0 & 193 & 25 \\ 2 & 3 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{114}31^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 21 & 30 & 21 & 0 & 172 & 24 \\ 2 & 3 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{123}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1700, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 7 & 3 & 0 & 30 & 4 \\ 2 & 7 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{113}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1862, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 7 & 3 & 0 & 27 & 4 \\ 2 & 7 & 4 & 4 & 0 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 4 & 2 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{114}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 3 & 0 & 24 & 3 \\ 2 & 4 & 4 & 4 & 0 & 36 & 4 \\ 1 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 36 & 24 & 0 & 238 & 28 \\ 2 & 3 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

6.19 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{121} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 12 & 9 & 10 \\ 3 & 6 & 5 & 0 & 12 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 6 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 6 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.20 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{114} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -4 & 34 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{c}_6+\textcolor{red}{su}_2+\textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -317$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{c}_6+\textcolor{red}{su}_2+\textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{c}_6+\textcolor{red}{su}_2+\textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -439$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{c}_6+\textcolor{red}{su}_2+\textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -525$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -3 & 3 & 0 & -1 \\ 3 & 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{c}_6+\textcolor{red}{su}_2+\textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -526$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^11^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{c}_6+\textcolor{red}{su}_2+\textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -653$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.8$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{c}_6+\textcolor{red}{su}_2+\textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -709$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^11^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{c}_6+\textcolor{red}{su}_2+\textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -848$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.5$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [12] \quad 1^{11} 3^{11} 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 3 & 0 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

6.21 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 31^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 15 & 7 \\ 9 & 12 & 20 & 0 & 15 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 17 & 5 \\ 8 & 10 & 17 & 0 & 12 & 24 & 6 \\ 12 & 17 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 27 & 7 \\ 17 & 24 & 36 & 0 & 27 & 49 & 13 \\ 5 & 6 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 15 & 7 \\ 9 & 15 & 17 & 0 & 15 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 17 & 5 \\ 9 & 13 & 18 & 0 & 15 & 27 & 7 \\ 11 & 18 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 27 & 7 \\ 17 & 27 & 33 & 0 & 27 & 49 & 13 \\ 5 & 7 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

6.22 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{1121} 3^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 1 & 3 \\ 3 & 3 & 4 & -4 & 14 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1121} 3^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 3 & 4 & -4 & 4 & 10 & -3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1121} 3^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 12 & 7 \\ 4 & 6 & 8 & 0 & 12 & 10 & 6 \\ 3 & 3 & 4 & -4 & 7 & 6 & -3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{114} 3^1 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 3 & 5 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 18 & 26 & 0 & 24 & 118 & 20 \\ 3 & 3 & 4 & -4 & 4 & 20 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 17.92 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 5 \\ 4 & 5 & 9 & 0 & 4 & 20 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 9 & 20 & 0 & 9 & 37 & 11 \\ 3 & 3 & 5 & -4 & 3 & 11 & -5 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{114} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 15 & 3 \\ 3 & 5 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 12 & 16 & 0 & 46 & 60 & 12 \\ 11 & 15 & 21 & 0 & 60 & 76 & 16 \\ 3 & 3 & 4 & -4 & 12 & 16 & -3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 6 & 12 & 0 & 6 & 13 & 7 \\ 3 & 3 & 5 & -4 & 3 & 7 & -3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3^1 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 5 & 3 \\ 2 & 1 & 5 & 0 & 9 & 6 & 3 \\ 4 & 5 & 9 & 0 & 16 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 16 & 0 & 28 & 21 & 10 \\ 5 & 6 & 12 & 0 & 21 & 13 & 7 \\ 3 & 3 & 5 & -4 & 10 & 7 & -4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 4 & 5 & 9 & 0 & 4 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 \\ 10 & 12 & 24 & 0 & 12 & 58 & 14 \\ 3 & 3 & 5 & -4 & 3 & 14 & -3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1111} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -92, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 6 & 0 & 6 & 12 & 4 \\ 3 & 6 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 12 & 4 \\ 7 & 12 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & 4 & -4 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1111} 31^{132} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -125, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.58 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 6 & 0 & 15 & 12 & 4 \\ 3 & 6 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 49 & 39 & 13 \\ 7 & 12 & 13 & 0 & 39 & 28 & 10 \\ 3 & 4 & 4 & -4 & 13 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{113} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -127, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 16.23 \rightarrow 17, \quad T_{\max} = 21, \quad T \in [17, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 15 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 4 & 5 & 9 & 0 & 4 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 15 & 18 & 36 & 0 & 18 & 133 & 21 \\ 3 & 3 & 5 & -4 & 3 & 21 & -4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 31^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.59 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 6 & 4 \\ 3 & 3 & 8 & 0 & 6 & 4 & 4 \\ 3 & 3 & 5 & -4 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{112} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -165, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.23 \rightarrow 16, \quad T_{\max} = 21, \quad T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 13 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 4 & 5 & 9 & 0 & 4 & 32 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 13 & 15 & 32 & 0 & 15 & 100 & 18 \\ 3 & 3 & 5 & -4 & 3 & 18 & -3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{114} 31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -190, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.48 \rightarrow 17, \quad T_{\max} = 21, \quad T \in [17, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 20 & 3 \\ 2 & 1 & 5 & 0 & 3 & 24 & 3 \\ 4 & 5 & 9 & 0 & 4 & 48 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 24 & 3 \\ 20 & 24 & 48 & 0 & 24 & 238 & 28 \\ 3 & 3 & 5 & -4 & 3 & 28 & -4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 3 & 4 & -4 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 31^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.37 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 8 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 9 & 4 \\ 5 & 6 & 12 & 0 & 9 & 13 & 7 \\ 3 & 3 & 5 & -4 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.68 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 6 & 4 \\ 3 & 6 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -209, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -215, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 9 & 3 \\ 2 & 1 & 5 & 0 & 9 & 12 & 3 \\ 3 & 5 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 36 & 9 \\ 9 & 12 & 17 & 0 & 36 & 49 & 13 \\ 3 & 3 & 4 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112} 31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -236, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 15 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{113} 31^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 15 & 3 \\ 2 & 1 & 5 & 0 & 6 & 18 & 3 \\ 3 & 5 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 45 & 7 \\ 15 & 18 & 27 & 0 & 45 & 133 & 21 \\ 3 & 3 & 4 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114} 31^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 20 & 3 \\ 2 & 1 & 5 & 0 & 6 & 24 & 3 \\ 3 & 5 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 60 & 7 \\ 20 & 24 & 36 & 0 & 60 & 238 & 28 \\ 3 & 3 & 4 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{112} 31^{122} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 60 & 14 \\ 10 & 12 & 18 & 0 & 60 & 58 & 14 \\ 3 & 3 & 4 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 3 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1121} 3^1 1^{121} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 5 & 3 \\ 2 & 1 & 5 & 0 & 12 & 6 & 3 \\ 3 & 5 & 4 & 0 & 18 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 30 & 14 \\ 5 & 6 & 9 & 0 & 30 & 13 & 7 \\ 3 & 3 & 4 & -4 & 14 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1122} 3^1 1^{121} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -254, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 7 & 3 \\ 2 & 1 & 5 & 0 & 12 & 9 & 3 \\ 3 & 5 & 4 & 0 & 18 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 42 & 14 \\ 7 & 9 & 13 & 0 & 42 & 28 & 10 \\ 3 & 3 & 4 & -4 & 14 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1124} 3^1 1^{122} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -260, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 14 & 3 \\ 2 & 1 & 5 & 0 & 12 & 18 & 3 \\ 3 & 5 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 84 & 14 \\ 14 & 18 & 26 & 0 & 84 & 118 & 20 \\ 3 & 3 & 4 & -4 & 14 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -261, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1122} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -262, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.56 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 18 & 6 \\ 7 & 9 & 13 & 0 & 18 & 28 & 10 \\ 3 & 3 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1112} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -262, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.56 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 4 & 6 & 8 & 0 & 6 & 10 & 6 \\ 3 & 3 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1124} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.64 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 14 & 3 \\ 2 & 1 & 5 & 0 & 6 & 18 & 3 \\ 3 & 5 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 36 & 6 \\ 14 & 18 & 26 & 0 & 36 & 118 & 20 \\ 3 & 3 & 4 & -4 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1123}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.64 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 12 & 6 \\ 4 & 6 & 8 & 0 & 12 & 10 & 6 \\ 3 & 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -272, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.69 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 6 & 9 & 12 & 0 & 9 & 25 & 9 \\ 3 & 3 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1123}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -278, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 18 & 6 \\ 6 & 9 & 12 & 0 & 18 & 25 & 9 \\ 3 & 3 & 4 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1114}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 \\ 8 & 12 & 16 & 0 & 12 & 46 & 12 \\ 3 & 3 & 4 & -4 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1133}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 27 & 9 \\ 6 & 9 & 12 & 0 & 27 & 25 & 9 \\ 3 & 3 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1124}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -292, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 8 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 24 & 6 \\ 8 & 12 & 16 & 0 & 24 & 46 & 12 \\ 3 & 3 & 4 & -4 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1123}31^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 14 & 24 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{113}31^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.91 \rightarrow 15, \quad T_{\max} = 21, \quad T \in [15, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 18 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 4 & 5 & 9 & 0 & 4 & 44 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 18 & 21 & 44 & 0 & 21 & 193 & 25 \\ 3 & 3 & 5 & -4 & 3 & 25 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{1134}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -302$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 12 & 3 \\ 3 & 5 & 4 & 0 & 12 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 36 & 9 \\ 8 & 12 & 16 & 0 & 36 & 46 & 12 \\ 3 & 3 & 4 & -4 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{1115}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$

$T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 10 & 15 & 20 & 0 & 15 & 73 & 15 \\ 3 & 3 & 4 & -4 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{11125}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 10 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 30 & 6 \\ 10 & 15 & 20 & 0 & 30 & 73 & 15 \\ 3 & 3 & 4 & -4 & 6 & 15 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{1114}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 8 & 3 \\ 2 & 1 & 5 & 0 & 12 & 12 & 3 \\ 3 & 5 & 4 & 0 & 16 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 12 & 16 & 0 & 46 & 48 & 12 \\ 8 & 12 & 16 & 0 & 48 & 46 & 12 \\ 3 & 3 & 4 & -4 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{1135}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.9$

$T_{\min} = 13.19 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 10 & 3 \\ 2 & 1 & 5 & 0 & 9 & 15 & 3 \\ 3 & 5 & 4 & 0 & 12 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 45 & 9 \\ 10 & 15 & 20 & 0 & 45 & 73 & 15 \\ 3 & 3 & 4 & -4 & 9 & 15 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{11116}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -326$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 13.24 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 3 & 5 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 12 & 18 & 24 & 0 & 18 & 106 & 18 \\ 3 & 3 & 4 & -4 & 3 & 18 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{11126}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -332$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.29 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 12 & 3 \\ 2 & 1 & 5 & 0 & 6 & 18 & 3 \\ 3 & 5 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 36 & 6 \\ 12 & 18 & 24 & 0 & 36 & 106 & 18 \\ 3 & 3 & 4 & -4 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{11111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -333$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 0 & 6 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1117} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -352, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 13.44 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 3 & 5 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 14 & 21 & 28 & 0 & 21 & 145 & 21 \\ 3 & 3 & 4 & -4 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1127} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -358, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.49 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 14 & 3 \\ 2 & 1 & 5 & 0 & 6 & 21 & 3 \\ 3 & 5 & 4 & 0 & 8 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 42 & 6 \\ 14 & 21 & 28 & 0 & 42 & 145 & 21 \\ 3 & 3 & 4 & -4 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111} 31^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 15 & 7 \\ 5 & 6 & 12 & 0 & 15 & 13 & 7 \\ 3 & 3 & 5 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{114} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -404, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 16 & 30 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -4 & 3 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -443, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 21 & 7 \\ 7 & 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 3 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1112} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -445, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 3 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -445, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 17 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 3 & 5 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 17 & 21 & 31 & 0 & 21 & 172 & 24 \\ 3 & 3 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1114} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -459, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 33 & 7 \\ 11 & 15 & 21 & 0 & 33 & 76 & 16 \\ 3 & 3 & 4 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1111} 31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -477, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 6 & 0 & 12 & 9 & 4 \\ 3 & 6 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & 4 & -4 & 10 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1134} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -477, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.70 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 11 & 3 \\ 2 & 1 & 5 & 0 & 9 & 15 & 3 \\ 3 & 5 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 45 & 9 \\ 11 & 15 & 21 & 0 & 45 & 76 & 16 \\ 3 & 3 & 4 & -4 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1111} 31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 30 & 10 \\ 7 & 12 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1123} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1123} 31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -506, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 12 & 3 \\ 2 & 1 & 5 & 0 & 12 & 15 & 3 \\ 3 & 5 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 72 & 14 \\ 12 & 15 & 22 & 0 & 72 & 85 & 17 \\ 3 & 3 & 4 & -4 & 14 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -508, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 12 & 15 & 22 & 0 & 15 & 85 & 17 \\ 3 & 3 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 27 & 7 \\ 9 & 12 & 17 & 0 & 27 & 49 & 13 \\ 3 & 3 & 4 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{1123} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 24 & 6 \\ 9 & 12 & 17 & 0 & 24 & 49 & 13 \\ 3 & 3 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{112} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 12 & 21 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -569, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.66 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 13 & 24 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{111} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -695, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 12 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 36 & 7 \\ 12 & 15 & 22 & 0 & 36 & 85 & 17 \\ 3 & 3 & 4 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -703, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 12 & 17 & 0 & 12 & 49 & 13 \\ 3 & 3 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{113} 31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -703, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 9 & 3 \\ 2 & 1 & 5 & 0 & 15 & 12 & 3 \\ 3 & 5 & 4 & 0 & 22 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 66 & 17 \\ 9 & 12 & 17 & 0 & 66 & 49 & 13 \\ 3 & 3 & 4 & -4 & 17 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 30 & 10 \\ 7 & 9 & 13 & 0 & 30 & 28 & 10 \\ 3 & 3 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -782, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 30 & 6 \\ 11 & 15 & 21 & 0 & 30 & 76 & 16 \\ 3 & 3 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{113} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -909, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{114} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -947, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 17 & 3 \\ 2 & 1 & 5 & 0 & 6 & 21 & 3 \\ 3 & 5 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 51 & 7 \\ 17 & 21 & 31 & 0 & 51 & 172 & 24 \\ 3 & 3 & 4 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1123} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 12 & 3 \\ 2 & 1 & 5 & 0 & 9 & 15 & 3 \\ 3 & 5 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 51 & 10 \\ 12 & 15 & 22 & 0 & 51 & 85 & 17 \\ 3 & 3 & 4 & -4 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1123} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -955, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 9 & 3 \\ 2 & 1 & 5 & 0 & 9 & 12 & 3 \\ 3 & 5 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 39 & 10 \\ 9 & 12 & 17 & 0 & 39 & 49 & 13 \\ 3 & 3 & 4 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{114} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -965, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 11 & 15 & 21 & 0 & 15 & 76 & 16 \\ 3 & 3 & 4 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{113} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 17 & 30 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1124} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1012, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 17 & 3 \\ 2 & 1 & 5 & 0 & 9 & 21 & 3 \\ 3 & 5 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 72 & 10 \\ 17 & 21 & 31 & 0 & 72 & 172 & 24 \\ 3 & 3 & 4 & -4 & 10 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{114} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1016, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 14 & 3 \\ 2 & 1 & 5 & 0 & 6 & 18 & 3 \\ 3 & 5 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 42 & 7 \\ 14 & 18 & 26 & 0 & 42 & 118 & 20 \\ 3 & 3 & 4 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1114} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1018, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 3 & 5 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 14 & 18 & 26 & 0 & 18 & 118 & 20 \\ 3 & 3 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{1124} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1028, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 11 & 3 \\ 2 & 1 & 5 & 0 & 9 & 15 & 3 \\ 3 & 5 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 48 & 10 \\ 11 & 15 & 21 & 0 & 48 & 76 & 16 \\ 3 & 3 & 4 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1144} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1034, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 18 & 3 \\ 3 & 5 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 96 & 16 \\ 14 & 18 & 26 & 0 & 96 & 118 & 20 \\ 3 & 3 & 4 & -4 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{114} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 18 & 33 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1133} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1264, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 87 & 17 \\ 12 & 15 & 22 & 0 & 87 & 85 & 17 \\ 3 & 3 & 4 & -4 & 17 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{1133} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1276, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 51 & 13 \\ 9 & 12 & 17 & 0 & 51 & 49 & 13 \\ 3 & 3 & 4 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1134} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1475, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 15 & 3 \\ 3 & 5 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 63 & 13 \\ 11 & 15 & 21 & 0 & 63 & 76 & 16 \\ 3 & 3 & 4 & -4 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{114} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1476, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{1134} 31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1522, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 18 & 3 \\ 3 & 5 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 102 & 17 \\ 14 & 18 & 26 & 0 & 102 & 118 & 20 \\ 3 & 3 & 4 & -4 & 17 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{1124} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1522, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 14 & 3 \\ 2 & 1 & 5 & 0 & 9 & 18 & 3 \\ 3 & 5 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 60 & 10 \\ 14 & 18 & 26 & 0 & 60 & 118 & 20 \\ 3 & 3 & 4 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{1134} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1528, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 14 & 3 \\ 2 & 1 & 5 & 0 & 12 & 18 & 3 \\ 3 & 5 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 78 & 13 \\ 14 & 18 & 26 & 0 & 78 & 118 & 20 \\ 3 & 3 & 4 & -4 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{1144} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 11 & 3 \\ 2 & 1 & 5 & 0 & 15 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 78 & 16 \\ 11 & 15 & 21 & 0 & 78 & 76 & 16 \\ 3 & 3 & 4 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{1144} 31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 14 & 3 \\ 2 & 1 & 5 & 0 & 18 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 18 & 26 & 0 & 118 & 120 & 20 \\ 14 & 18 & 26 & 0 & 120 & 118 & 20 \\ 3 & 3 & 4 & -4 & 20 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [96] \quad 1^{111} 31^{112} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 73, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 15 & 4 \\ 5 & 9 & 15 & 9 & 0 & 15 & 37 & 11 \\ 2 & 3 & 4 & 2 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{111} 31^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 145, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 6 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{111} 31^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 151, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{112}31^{112}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 12 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 18 & 4 \\ 6 & 12 & 18 & 12 & 0 & 18 & 58 & 14 \\ 2 & 3 & 4 & 2 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{112}31^{113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 388, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 24 & 4 \\ 8 & 15 & 24 & 15 & 0 & 24 & 100 & 18 \\ 2 & 3 & 4 & 2 & -4 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{113}31^{113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 409, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 18 & 27 & 18 & 0 & 27 & 133 & 21 \\ 2 & 3 & 4 & 2 & -4 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{111}31^{121}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 583, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 9 & 0 & 37 & 24 & 11 \\ 3 & 6 & 9 & 6 & 0 & 24 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{114}31^{114}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 24 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 36 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 36 & 4 \\ 12 & 24 & 36 & 24 & 0 & 36 & 238 & 28 \\ 2 & 3 & 4 & 2 & -4 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{111}31^{122}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 700, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 9 & 0 & 37 & 39 & 11 \\ 5 & 9 & 15 & 9 & 0 & 39 & 37 & 11 \\ 2 & 3 & 4 & 2 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{113}31^{114}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 769, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 33 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 33 & 4 \\ 11 & 21 & 33 & 21 & 0 & 33 & 193 & 25 \\ 2 & 3 & 4 & 2 & -4 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{111}31^{111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 844, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 6 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1036, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 4 & 2 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

6.23 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 15 & 7 \\ 9 & 12 & 20 & 0 & 15 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 17 & 5 \\ 8 & 10 & 17 & 0 & 12 & 24 & 6 \\ 12 & 17 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 27 & 7 \\ 17 & 24 & 36 & 0 & 27 & 49 & 13 \\ 5 & 6 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 15 & 7 \\ 9 & 15 & 17 & 0 & 15 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 17 & 5 \\ 9 & 13 & 18 & 0 & 15 & 27 & 7 \\ 11 & 18 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 27 & 7 \\ 17 & 27 & 33 & 0 & 27 & 49 & 13 \\ 5 & 7 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

6.24 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 6 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 3 1^{\overline{114}} 1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 & 3 \\ 14 & 24 & 26 & 0 & 18 & 118 & 18 & 20 \\ 2 & 3 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1121}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 253, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 & 3 \\ 7 & 12 & 13 & 0 & 9 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}} 3^{\overline{1121}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 325, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 & 3 \\ 13 & 24 & 23 & 0 & 15 & 100 & 15 & 18 \\ 2 & 3 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 409, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 447, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 3 & 4 & 0 & 6 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 15 & 6 & 12 & 6 \\ 3 & 3 & 4 & -4 & 7 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 & 3 \\ 4 & 5 & 9 & 0 & 12 & 12 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 15 & 6 & 7 \\ 5 & 6 & 12 & 0 & 15 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 3 & 5 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 499, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 3 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{112}} 3^{\overline{1121}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 760, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11111} 3^1 1^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 772, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 1^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 820, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 & 3 \\ 12 & 21 & 22 & 0 & 15 & 85 & 15 & 17 \\ 2 & 3 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3^1 1^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 928, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 33 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 & 3 \\ 18 & 33 & 32 & 0 & 21 & 193 & 21 & 25 \\ 2 & 3 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 3^1 1^{1211} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 931, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 21 & 9 & 10 \\ 5 & 9 & 9 & 0 & 21 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 9 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^{1221} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 964, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 30 & 9 & 10 \\ 7 & 12 & 13 & 0 & 30 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1345, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 27 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 18 & 21 \\ 2 & 3 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{114}3^11^{141} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1621, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 & 3 \\ 17 & 30 & 31 & 0 & 21 & 172 & 21 & 24 \\ 2 & 3 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{114}3^11^{141} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 36 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 24 & 3 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 24 & 28 \\ 2 & 3 & 4 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

6.25 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] 1^{114}3^11^{141}3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{114}3^11^{141}3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{114}3^11^{123}3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{114}3^11^{141}3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 543, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{13}3^11^{114}3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 744, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.26 NHC(-2)(-2)(-3) + e₆ + su₂ + su2n3mix

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^{111} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 8 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 8 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 3 & 4 & 2 & -4 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

6.27 NHC(-2)(-2)(-3) + e₆ + su₂ + su₃

$$T = 9 \quad [0] \quad 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 16.20 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 9 & 13 & 0 & 12 & 27 & 9 \\ 3 & 3 & 4 & -4 & 4 & 9 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 5 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 9 & 7 \\ 3 & 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 3 & 4 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 6 & 12 & 0 & 6 & 12 & 6 \\ 3 & 3 & 5 & -4 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -4 & 14 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 12 & 6 \\ 5 & 6 & 9 & 0 & 12 & 12 & 6 \\ 3 & 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -143, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 4 & 5 & 9 & 0 & 28 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 12 & 28 & 0 & 73 & 12 & 15 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 5 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 4 & 5 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 3 & 5 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -257, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 12 & 10 \\ 3 & 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 3 & 4 & -4 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -3 & 3 & 0 & -1 \\ 3 & 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -281, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 \\ 3 & 5 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 18 & 16 \\ 3 & 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 3 & 4 & -4 & 16 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -287, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 8 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 4 & 5 & 9 & 0 & 20 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 9 & 20 & 0 & 37 & 9 & 11 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 3 & 5 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -299, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -317, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 3 & 4 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -317, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 19 & 0 & 73 & 12 & 15 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -318, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 3 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -329, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 15 & 13 \\ 3 & 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 3 & 4 & -4 & 13 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 4 & 5 & 9 & 0 & 24 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 24 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 5 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -351, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -378, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 6 & 9 & 0 & 15 & 12 & 6 \\ 3 & 3 & 4 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -383, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -392, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 3 & 4 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1131} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 9 & 9 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & -4 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1141} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -428, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 12 & 16 & 0 & 46 & 12 & 12 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -4 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -542, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 15 & 14 & 0 & 37 & 9 & 11 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{131} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -563, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 15 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 & 3 \\ 4 & 5 & 9 & 0 & 36 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 18 & 36 & 0 & 133 & 18 & 21 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 3 & 5 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{121} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 13 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 \\ 4 & 5 & 9 & 0 & 32 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 15 & 32 & 0 & 100 & 15 & 18 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 3 & 5 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{112} 31^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -632, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 24 & 26 & 0 & 118 & 18 & 20 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{121} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -674, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 16 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 & 3 \\ 4 & 5 & 9 & 0 & 40 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 18 & 40 & 0 & 154 & 18 & 22 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 3 & 5 & -4 & 22 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 5 & 3 \\ 2 & 1 & 5 & 0 & 9 & 6 & 3 \\ 3 & 5 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 21 & 10 \\ 5 & 6 & 9 & 0 & 21 & 12 & 6 \\ 3 & 3 & 4 & -4 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -698, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{141} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -878, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 20 & 2 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 & 3 \\ 4 & 5 & 9 & 0 & 48 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 24 & 48 & 0 & 238 & 24 & 28 \\ 2 & 3 & 4 & 0 & 24 & 0 & 2 \\ 3 & 3 & 5 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -887, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 15 & 17 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 3 & 4 & -4 & 17 & 2 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{1131^1}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -894, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 5 & 3 \\ 2 & 1 & 5 & 0 & 12 & 6 & 3 \\ 3 & 5 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 27 & 13 \\ 5 & 6 & 9 & 0 & 27 & 12 & 6 \\ 3 & 3 & 4 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{112}31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{1141^1}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -950, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 2 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 21 & 31 & 0 & 172 & 21 & 24 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 3 & 4 & -4 & 24 & 2 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{1141^1}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -972, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 5 & 3 \\ 2 & 1 & 5 & 0 & 15 & 6 & 3 \\ 3 & 5 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 33 & 16 \\ 5 & 6 & 9 & 0 & 33 & 12 & 6 \\ 3 & 3 & 4 & -4 & 16 & 6 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{112}31^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 21 & 22 & 0 & 85 & 15 & 17 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{131}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1043, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 18 & 2 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 & 3 \\ 4 & 5 & 9 & 0 & 44 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 21 & 44 & 0 & 193 & 21 & 25 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 3 & 5 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{1131^1}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1085, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 12 & 13 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{114}31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 6 & 0 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 30 & 28 & 0 & 154 & 18 & 22 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{113} 31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 24 & 23 & 0 & 100 & 15 & 18 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1141} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1478, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 15 & 16 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 3 & 4 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1141} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1652, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 18 & 26 & 0 & 118 & 18 & 20 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 3 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113} 31^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 27 & 3 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{113} 31^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1847, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 30 & 31 & 0 & 172 & 21 & 24 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114} 31^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2162, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 33 & 3 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 33 & 32 & 0 & 193 & 21 & 25 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{114} 31^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 36 & 3 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 3 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 31^{112} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 12 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 6 & 9 & 6 & 0 & 9 & 12 & 6 \\ 2 & 3 & 4 & 2 & -4 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 354, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 456, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 3 & 6 & 3 & 0 & 9 & 3 & 3 \\ 2 & 3 & 4 & 2 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111}31^{112}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 606, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 15 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 15 & 9 & 0 & 37 & 9 & 11 \\ 1 & 3 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111}31^{113}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 21 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 21 & 12 & 0 & 73 & 12 & 15 \\ 1 & 3 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{111}31^{114}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1686, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 27 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 27 & 15 & 0 & 121 & 15 & 19 \\ 1 & 3 & 3 & 3 & 0 & 15 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

6.28 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112}31^{1114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 18 & 2 \\ 14 & 24 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 31^{1112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 253, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 31^{1112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 24 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 31^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 409, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 447, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 6 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 3 & 4 & -4 & 3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 31^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 6 & 15 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 6 & 12 & 0 & 15 & 6 & 13 & 7 \\ 3 & 3 & 5 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 499, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

- [9] $1^{112}31^{1112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{1111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 772, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{112}31^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 820, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 21 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{114}31^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 928, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 33 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 3 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{1111}31^{1211} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 931, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 9 & 21 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 21 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 10 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{1111}31^{1212} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 964, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 9 & 30 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 30 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{1111}31^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{113}31^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1345, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1621, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 30 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2164, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 24 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

6.29 NHC(-2)(-2)(-3) + $\mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 84, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{121} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 12 & 9 & 10 \\ 3 & 3 & 5 & 0 & 12 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 10 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 498, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 15 & 14 & 0 & 37 & 9 & 9 & 11 \\ 2 & 3 & 4 & 0 & 9 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 11 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1141} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 534, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 21 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 18 & 15 & 16 \\ 3 & 3 & 5 & 0 & 18 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 15 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 16 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 555, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{11}1} 3^{\overline{11}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 678, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 17 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 15 & 12 & 13 \\ 3 & 3 & 5 & 0 & 15 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 12 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 13 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{11}1} 3^{\overline{11}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 894, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

6.30 NHC(-2)(-2)(-3) + $\mathbf{e}_6$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{1}1} 3^{\overline{1}1} 2^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 2 & 4 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}1} 3^{\overline{1}1} 2^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.41 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 2 & 1 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 2 & 4 & 0 & -2 & 0 \\ 3 & 3 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}1} 3^{\overline{1}1} 2^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -62, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 3 & 6 & 5 & 5 & 3 & 5 \\ 2 & 3 & 5 & -2 & 2 & 0 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}1} 3^{\overline{1}1} 1^{\overline{1}1} 3^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -179, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}1} 3^{\overline{1}1} 1^{\overline{1}1} 3^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -345, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.31 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 138, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 0 & 4 \\ 3 & 6 & 5 & 5 & 3 & 0 & 5 \\ 2 & 3 & 5 & -2 & 2 & 0 & 0 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

6.32 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 6 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 & 3 \\ 2 & 3 & 6 & 3 & 0 & 3 & 4 \\ 2 & 3 & 4 & 2 & -4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 6 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 & 3 \\ 1 & 3 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 4 & 2 & -4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{12}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 9 & 15 & 9 & 0 & 8 & 37 & 11 \\ 2 & 3 & 4 & 2 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 543, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 9 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 744, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 4 & 2 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

6.33 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 524, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 4 & 2 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.34 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11}3^{12}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 0 & 4 & 2 & 4 \\ 2 & 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & -2 & 2 & 0 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 284, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

6.35 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 9 & 5 & 0 & 12 & 6 \\ 2 & 3 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 14 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 24 & 14 & 0 & 97 & 17 \\ 2 & 3 & 4 & 2 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 4 & 2 & -4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 12 & 4 \\ 2 & 7 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 4 & 2 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 11 & 3 \\ 2 & 4 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 11 & 18 & 11 & 0 & 55 & 13 \\ 2 & 3 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^1 1^{11}31^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 8 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 8 & 0 & 25 & 9 \\ 2 & 3 & 4 & 2 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^1 3^1 1^{111}31^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 4 & 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

6.36 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + su2n3mix + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111}3^1 1^{111}31^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 8 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 8 & 0 & 25 & 12 & 9 \\ 2 & 3 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 3 & 4 & 2 & -4 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

6.37 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + su2n3mix + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{111}3^1 1^{111}31^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

6.38 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11}3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 \\ 3 & 3 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^1 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 16 & 0 & 27 & 9 \\ 3 & 3 & 5 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

- [3] $1^{111}31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [5] $1^{11}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 8 & 0 & 3 & 3 \\ 3 & 3 & 5 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{112}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 18 & 0 & 57 & 13 \\ 3 & 3 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 \\ 3 & 3 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 6 & 8 & 0 & 9 & 5 \\ 3 & 3 & 4 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$
- [11] $1^{113}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 9 & 12 & 0 & 24 & 8 \\ 3 & 3 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$
- [12] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

- [13] $1^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 204$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$
- [14] $1^{111}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 274$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 12 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 300$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 316$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 0 & 27 & 9 \\ 3 & 3 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{113}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 442$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 15 & 22 & 0 & 84 & 16 \\ 3 & 3 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{113}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 445$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 48 & 12 \\ 3 & 3 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$
- $T = 9$ [19] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 4 & 0 & 4 & 4 \\ 3 & 4 & 4 & 1 & 3 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & -2 & 0 \\ 3 & 4 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -39$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 2 & 1 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & -2 & 0 \\ 3 & 4 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 1 & 3 & 0 & -3 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [22] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

6.39 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + [\textcolor{blue}{su}_2 \cdot \textcolor{green}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 4 & 3 \\ 3 & 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 3 & 4 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + [\textcolor{blue}{su}_2 \cdot \textcolor{green}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.40 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -559, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 2^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 3 & 0 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

6.41 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 16.20 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 27 & 12 & 9 \\ 3 & 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 3 & 4 & -4 & 9 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 3 & 5 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -65, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 9 & 3 \\ 5 & 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 3 & 4 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -78, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.87 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 4 & 5 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 3 & 5 & -4 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1112} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 12 & 18 & 0 & 12 & 58 & 14 \\ 3 & 3 & 4 & -4 & 2 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1112} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -138, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 12 & 6 \\ 4 & 6 & 8 & 0 & 12 & 10 & 6 \\ 3 & 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -143, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 4 & 5 & 9 & 0 & 4 & 28 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 11 & 12 & 28 & 0 & 12 & 73 & 15 \\ 3 & 3 & 5 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

13.0

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 12 & 0 & 6 & 13 & 7 \\ 3 & 3 & 5 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -257, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 12 & 3 \\ 7 & 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 3 & 4 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 9 & 3 & 0 & 6 & 4 \\ 4 & 9 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 8 & 3 & 0 & 4 & 4 \\ 3 & 4 & 5 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 18 & 3 \\ 11 & 15 & 21 & 0 & 18 & 76 & 16 \\ 3 & 3 & 4 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111}31^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 4 & 5 & 9 & 0 & 4 & 20 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 8 & 9 & 20 & 0 & 9 & 37 & 11 \\ 3 & 3 & 5 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111}31^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 27 & 9 & 9 \\ 2 & 3 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 3 & 4 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 11 & 21 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & 4 & -4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 3 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 15 & 3 \\ 9 & 12 & 17 & 0 & 15 & 49 & 13 \\ 3 & 3 & 4 & -4 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}31^{112} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 4 & 5 & 9 & 0 & 4 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 12 & 24 & 0 & 12 & 58 & 14 \\ 3 & 3 & 5 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 15 & 6 \\ 5 & 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 3 & 4 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 4 & 6 & 8 & 0 & 6 & 10 & 6 \\ 3 & 3 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1113} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 6 & 9 & 12 & 0 & 9 & 25 & 9 \\ 3 & 3 & 4 & -4 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1114} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -428, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 8 & 12 & 16 & 0 & 12 & 46 & 12 \\ 3 & 3 & 4 & -4 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{112} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -542, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 8 & 15 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{113} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -563, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 15 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 4 & 5 & 9 & 0 & 4 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 15 & 18 & 36 & 0 & 18 & 133 & 21 \\ 3 & 3 & 5 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{112} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 13 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 4 & 5 & 9 & 0 & 4 & 32 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 13 & 15 & 32 & 0 & 15 & 100 & 18 \\ 3 & 3 & 5 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{112} 31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -632, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 14 & 24 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -674, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 16 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 4 & 5 & 9 & 0 & 4 & 40 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 16 & 18 & 40 & 0 & 18 & 154 & 22 \\ 3 & 3 & 5 & -4 & 2 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 21 & 6 \\ 7 & 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 3 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -698, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 7 & 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 3 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{114} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -878, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 20 & 3 \\ 2 & 1 & 5 & 0 & 3 & 24 & 3 \\ 4 & 5 & 9 & 0 & 4 & 48 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 20 & 24 & 48 & 0 & 24 & 238 & 28 \\ 3 & 3 & 5 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -887, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 12 & 15 & 22 & 0 & 15 & 85 & 17 \\ 3 & 3 & 4 & -4 & 2 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -894, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 27 & 6 \\ 9 & 12 & 17 & 0 & 27 & 49 & 13 \\ 3 & 3 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{114} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -950, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 17 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 3 & 5 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 17 & 21 & 31 & 0 & 21 & 172 & 24 \\ 3 & 3 & 4 & -4 & 2 & 24 & -1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1114} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -972, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 33 & 6 \\ 11 & 15 & 21 & 0 & 33 & 76 & 16 \\ 3 & 3 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{112} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -974, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 12 & 21 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{113} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1043, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 18 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 4 & 5 & 9 & 0 & 4 & 44 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 18 & 21 & 44 & 0 & 21 & 193 & 25 \\ 3 & 3 & 5 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 9 & 12 & 17 & 0 & 12 & 49 & 13 \\ 3 & 3 & 4 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 16 & 30 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 13 & 24 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{114} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1478, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 11 & 15 & 21 & 0 & 15 & 76 & 16 \\ 3 & 3 & 4 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{114} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1652, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 3 & 5 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 14 & 18 & 26 & 0 & 18 & 118 & 20 \\ 3 & 3 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{113} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1847, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 17 & 30 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2162, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 18 & 33 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{114} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 31^{112} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 12 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 4 & 12 & 12 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & 4 & 2 & -4 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111} 31^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 201, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 12 & 9 & 6 \\ 2 & 3 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 3 & 4 & 2 & -4 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11} 31^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 354, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 4 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 31^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 456, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.60 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 3 & 9 & 3 \\ 3 & 6 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111}31^{112}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 606, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 15 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 9 & 2 \\ 5 & 15 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 4 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111}31^{113}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1200, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 21 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 12 & 2 \\ 7 & 21 & 21 & 12 & 0 & 12 & 73 & 15 \\ 2 & 4 & 4 & 2 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{111}31^{114}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1686, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 27 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 15 & 2 \\ 9 & 27 & 27 & 15 & 0 & 15 & 121 & 19 \\ 2 & 4 & 4 & 2 & -4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

6.42 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & -4 & 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112}31^{1114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 136, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 18 & 3 \\ 14 & 24 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 2 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}31^{1112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 253, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 12 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113}31^{1112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 325, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 24 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 2 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

- [4] $1^{\overline{111}} 3 1^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 409, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{\overline{1111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 447, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 & 15 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 3 & 4 & -4 & 6 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{\overline{1111}} 3 1^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 6 & 12 & 0 & 6 & 13 & 15 & 7 \\ 5 & 6 & 12 & 0 & 6 & 15 & 13 & 7 \\ 3 & 3 & 5 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{\overline{1111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 499, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{\overline{1111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{112}} 3 1^{\overline{1112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 772, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{112}} 3 1^{\overline{1113}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 820, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 21 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 928, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 21 & 3 \\ 18 & 33 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 2 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 3^1 1^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 931, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 6 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 21 & 10 \\ 5 & 9 & 9 & 0 & 6 & 21 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 10 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 964, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 9 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 30 & 10 \\ 7 & 12 & 13 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1345, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1621, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 21 & 3 \\ 17 & 30 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2164, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 24 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

6.43 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{1121} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1121} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 12 & 3 & 3 \\ 7 & 9 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1121} 3^1 1^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 498, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 15 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 3 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1141} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 534, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 21 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 18 & 3 & 3 \\ 11 & 15 & 21 & 0 & 18 & 76 & 15 & 16 \\ 2 & 3 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111} 3^1 1^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 555, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1131} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 678, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 15 & 3 & 3 \\ 9 & 12 & 17 & 0 & 15 & 49 & 12 & 13 \\ 2 & 3 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3^1 1^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 894, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

6.44 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 2 & 4 & 0 & 4 & 2 & 4 \\ 3 & 4 & 4 & 1 & 3 & 3 & 4 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & -2 & 2 & 0 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 284, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

6.45 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 4 & 2 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

6.46 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 68, T \in [15, 68]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 3 & 8 & 0 & 3 & 3 & 3 \\ 3 & 3 & 5 & -4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 12 & 0 & 6 & 12 & 6 \\ 3 & 3 & 5 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.0

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -507, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 3 & 4 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \ 1^{11} 3^1 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 3 & 0 & 4 \\ 2 & 4 & 2 & 1 & 4 & 0 & 4 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 & 1 \\ 3 & 4 & 4 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} 3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 558, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

6.47 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1112}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 9 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 3 & 4 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 498, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 15 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 2 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1114}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 534, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 15 & 2 \\ 11 & 15 & 21 & 0 & 18 & 15 & 76 & 16 \\ 3 & 3 & 4 & -4 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 555, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1113}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 678, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 12 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 3 & 4 & -4 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 894, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

6.48 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 216, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 324, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

6.49 NHC(-2)(-2)(-3) + e₆ + (-3)→(-2)

$$T = 8 \quad [0] \ 1^{\overline{131}} 3^{\overline{111}} 2^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 98, \ \Delta = -143, \ \det = 1219, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 1 & 4 & 4 \\ 6 & 4 & 9 & 3 & 0 & 4 & 4 \\ 8 & 9 & 9 & 4 & 0 & 5 & 5 \\ 3 & 3 & 4 & -3 & 0 & -1 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 4 & 4 & 5 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

6.50 NHC(-2)(-2)(-3) + e₇

$$T = 9 \quad [0] \ 1^{\overline{13}} 3^{\overline{221}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_7 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -23, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.91 \rightarrow 17, \ T_{\max} = 101, \ T \in [17, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 & -1 \\ 1 & 1 & 4 & 0 & -4 & -2 & -2 \\ 2 & 2 & 4 & -1 & -2 & -3 & -3 \end{pmatrix}_{b_0}$$

6.51 NHC(-2)(-2)(-3) + e'₇

$$T = 9 \quad [0] \ 1^{\overline{13}} 3^{\overline{221}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}'_7 \quad T = 9, \ T_{\max} = 93, \ \Delta = -133, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 16, \ T_{\max} = 93, \ T \in [16, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 & -1 \\ 1 & 1 & 4 & 0 & -3 & -1 & -1 \\ 2 & 2 & 4 & -1 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

6.52 NHC(-2)(-2)(-3) + f₄

$$T = 7 \quad [1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -2, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14, \ T_{\max} = 97, \ T \in [14, 97]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 & 4 \\ 4 & 9 & 9 & 4 & 5 & 5 \\ 3 & 3 & 4 & -2 & 0 & 0 \\ 3 & 4 & 5 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \ 21^{\overline{131}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -2, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14, \ T_{\max} = 97, \ T \in [14, 97]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 & 4 \\ 4 & 9 & 9 & 4 & 5 & 5 \\ 3 & 3 & 4 & -2 & 0 & 0 \\ 3 & 4 & 5 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{13}} 3^{\overline{11}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 97, \ T \in [13, 97]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & -3 & -3 \\ 3 & 3 & 5 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -49, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 4 & 9 & 9 & 4 & 5 \\ 2 & 3 & 4 & -2 & 0 \\ 3 & 4 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 21^{111} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 4 \\ 6 & 4 & 9 & 3 & 4 \\ 8 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & -2 & 0 \\ 4 & 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [7] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 2 & 6 & 0 & 4 \\ 3 & 6 & 5 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 2 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 18 & 6 \\ 12 & 13 & 21 & 7 \\ 18 & 21 & 30 & 10 \\ 6 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 5 & 9 & 13 & 5 \\ 9 & 13 & 21 & 7 \\ 13 & 21 & 30 & 10 \\ 5 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11}321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 5 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 2^12^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 19 & 6 \\ 12 & 13 & 21 & 7 \\ 19 & 21 & 30 & 10 \\ 6 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 \\ 2 & 4 & 4 & 4 & 0 \\ 1 & 3 & 4 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 4 & 7 & 0 & 4 \\ 2 & 3 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 4 & 4 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 2^12^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 345, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 1 & 4 & 1 & 3 \\ 2 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 31^{111}231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 787, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 & 1 & 3 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [22] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 2 & 3 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

6.53 NHC(-2)(-2)(-3) + f₄ + e₆

$$T = 8 \quad [0] \quad 1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{11} 3^{21} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{c}_6 \quad T = 9, \quad T_{\max} = 101, \quad \Delta = -158, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & -2 & 0 \\ 3 & 4 & 5 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

6.54 NHC(-2)(-2)(-3) + f₄ + f₄

$$T = 8 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 \quad T = 8, \quad T_{\max} = 100, \quad \Delta = -160, \quad \det = 264, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.42 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{11} 3^{21} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -109, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 13 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 13 & 21 & 30 & 0 & 10 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

6.55 NHC(-2)(-2)(-3) + f₄ + f₄ + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 92, \quad \Delta = -130, \quad \det = -584, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.42 \rightarrow 14, \quad T_{\max} = 92, \quad T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 9 & 3 & 0 & 6 & 4 \\ 4 & 9 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 8 & 3 & 0 & 4 & 4 \\ 3 & 4 & 5 & 0 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

6.56 NHC(-2)(-2)(-3) + f₄ + f₄ + su₂ → (-3)

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 3 & 4 \\ 4 & 9 & 9 & 4 & 0 & 4 & 5 \\ 2 & 3 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 5 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

6.57 NHC(-2)(-2)(-3) + f₄ + NHC(-2)(-3)

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -87, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

6.58 NHC(-2)(-2)(-3) + f₄ + so₈

$$T = 8 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -136, \quad \det = 106, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 6 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 6 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 1 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 1 & 3 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -2 & 1 & 0 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 1 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 3 & 2 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 5 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 13 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 13 & 21 & 30 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

13.2

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 7 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}31^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.59 NHC(-2)(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{111}31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{114}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -3 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -249, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111}31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}1^1}3^11^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -393$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 9 & 3 & 0 & 6 & 4 \\ 4 & 9 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 8 & 3 & 0 & 4 & 4 \\ 3 & 4 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}1^1}3^11^{\overline{112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{113}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -557$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{112}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -662$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

6.60 NHC(-2)(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{\overline{111}1^1}3^11^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -130$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 3 & 4 \\ 4 & 9 & 9 & 4 & 0 & 4 & 5 \\ 2 & 3 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{113}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 6 & 5 \\ 8 & 10 & 17 & 0 & 8 & 6 \\ 12 & 17 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 12 & 0 & 4 & 4 \\ 5 & 6 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{113}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -330$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{11} 3^{11} 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 13 & 1 & 6 & 5 \\ 9 & 13 & 21 & 0 & 9 & 7 \\ 13 & 21 & 30 & 0 & 13 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 13 & 0 & 4 & 4 \\ 5 & 7 & 10 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

6.61 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 15 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 15 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

6.62 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{11} 3^{11} 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 0 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

6.63 NHC(-2)(-2)(-3) + f₄ + su₂

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 16 & 0 & 28 & 10 \\ 3 & 3 & 5 & -3 & 10 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 6 & 16 & 0 & 22 & 8 \\ 3 & 3 & 5 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 10 & 0 & 22 & 8 \\ 3 & 4 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 28 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 15 & 28 & 0 & 85 & 17 \\ 3 & 3 & 5 & -3 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 8 & 0 & 4 & 4 \\ 3 & 3 & 5 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 4 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 2 & 3 & 4 & -2 & 3 & 0 \\ 4 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 18 & 27 & 0 & 133 & 21 \\ 3 & 3 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{117}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 21 & 28 & 0 & 145 & 21 \\ 3 & 3 & 4 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 12 & 18 & 0 & 58 & 14 \\ 3 & 3 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{114}3^11^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 20 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 24 & 36 & 0 & 238 & 28 \\ 3 & 3 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{114}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 46 & 12 \\ 3 & 3 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 6 \\ 3 & 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{115}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 15 & 20 & 0 & 73 & 15 \\ 3 & 3 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{116}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 18 & 24 & 0 & 106 & 18 \\ 3 & 3 & 4 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 7 \\ 3 & 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 99, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 12 & 0 & 13 & 7 \\ 3 & 3 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 155, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 176, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{14} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 186, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 4 & 5 & 9 & 0 & 40 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 21 & 40 & 0 & 172 & 24 \\ 3 & 3 & 5 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 186, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 9 & 20 & 0 & 37 & 11 \\ 3 & 3 & 5 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^1 1^1 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 191, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 2 & 3 & 4 & -2 & 3 & 0 \\ 3 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 196, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 10 \\ 3 & 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{12} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 200, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 4 & 5 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 12 & 24 & 0 & 58 & 14 \\ 3 & 3 & 5 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 228, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 4 & 5 & 9 & 0 & 28 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 12 & 28 & 0 & 73 & 15 \\ 3 & 3 & 5 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{114} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 239, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 27 & 24 & 0 & 121 & 19 \\ 3 & 4 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 239, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 242, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 15 & 36 & 0 & 121 & 19 \\ 3 & 3 & 5 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 260, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 19 & 0 & 73 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 260, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 294, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 294, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 15 & 22 & 0 & 85 & 17 \\ 3 & 3 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{13} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 340, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 4 & 5 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 18 & 36 & 0 & 133 & 21 \\ 3 & 3 & 5 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{112} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 344, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{112} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 351, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{12}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 354, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 32 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 15 & 32 & 0 & 100 & 18 \\ 3 & 3 & 5 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{114}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{114}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{12}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 4 & 5 & 9 & 0 & 40 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 18 & 40 & 0 & 154 & 22 \\ 3 & 3 & 5 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{114}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{113}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{14}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 20 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 \\ 4 & 5 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 24 & 48 & 0 & 238 & 28 \\ 3 & 3 & 5 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{13}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 578, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 18 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 4 & 5 & 9 & 0 & 44 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 21 & 44 & 0 & 193 & 25 \\ 3 & 3 & 5 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{114}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{113}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{113}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 659, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[51] $21^{111}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 840, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 6 & 4 \\ 6 & 4 & 9 & 3 & 6 & 4 \\ 8 & 9 & 9 & 4 & 8 & 5 \\ 3 & 3 & 4 & -2 & 3 & 0 \\ 6 & 6 & 8 & 3 & 4 & 4 \\ 4 & 4 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{114}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 855, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{114}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1016, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [54] $1^1321^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 2 & 4 \\ 3 & 4 & 5 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{12}321^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 5 & 3 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 3 & 2 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[56] $1^1321^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 & 3 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & 2 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{111}31^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{11}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 2 & 6 & 0 & 11 & 4 \\ 3 & 6 & 5 & 0 & 15 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 15 & 0 & 32 & 12 \\ 3 & 4 & 5 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11}321^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 2 & 6 & 0 & 15 & 4 \\ 3 & 6 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 20 & 0 & 58 & 16 \\ 3 & 4 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{11}321^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 2 & 6 & 0 & 19 & 4 \\ 3 & 6 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 25 & 0 & 92 & 20 \\ 3 & 4 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{11}321^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 2 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{12}321^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 2 & 6 & 0 & 22 & 4 \\ 3 & 6 & 5 & 0 & 30 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 22 & 30 & 0 & 134 & 24 \\ 3 & 4 & 5 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{13}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 0 & 9 & 3 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 8 & 9 & 0 & 40 & 12 \\ 2 & 3 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 7 & 3 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 4 & 7 & 0 & 16 & 8 \\ 2 & 3 & 2 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{12}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 3 & 0 & 8 & 3 \\ 1 & 2 & 0 & 2 & 0 & 6 & 2 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 8 & 0 & 26 & 10 \\ 2 & 3 & 2 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{112}31^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{111}31^{1231} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 9 & 0 & 37 & 11 \\ 2 & 3 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{14}321^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 0 & 13 & 3 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 11 & 13 & 0 & 81 & 17 \\ 2 & 3 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{12}321^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 3 & 0 & 14 & 3 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 14 & 8 & 14 & 0 & 70 & 16 \\ 2 & 3 & 2 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{12}321^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 0 & 11 & 3 \\ 1 & 2 & 0 & 2 & 0 & 7 & 2 \\ 1 & 3 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 7 & 11 & 0 & 45 & 13 \\ 2 & 3 & 2 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{13}321^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 3 & 0 & 12 & 3 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 12 & 0 & 61 & 15 \\ 2 & 3 & 2 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114}31^{12}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 18 & 24 & 18 & 0 & 118 & 20 \\ 2 & 3 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112}31^{14}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 30 & 18 & 0 & 154 & 22 \\ 2 & 3 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{112}31^{12}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 18 & 12 & 0 & 58 & 14 \\ 2 & 3 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{13}321^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -495, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 0 & 15 & 3 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 10 & 15 & 0 & 88 & 18 \\ 2 & 3 & 2 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{11231^{13}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 24 & 15 & 0 & 100 & 18 \\ 2 & 3 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{11331^{12}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 3 & 0 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 21 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 21 & 15 & 0 & 85 & 17 \\ 2 & 3 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{14321^{114}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -566, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 0 & 16 & 3 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 3 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 16 & 0 & 110 & 20 \\ 2 & 3 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[81] $2^1 2^1 31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -781, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 & 3 \\ 2 & 4 & 1 & 4 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 3 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{11331^{13}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -908, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 18 & 27 & 18 & 0 & 133 & 21 \\ 2 & 3 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{11431^{13}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1048, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 21 & 30 & 21 & 0 & 172 & 24 \\ 2 & 3 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{11331^{14}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1048, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 33 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 21 & 0 & 193 & 25 \\ 2 & 3 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[85] $2^1 2^1 31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1054, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 3 & 1 & 6 & 3 \\ 2 & 4 & 1 & 4 & 1 & 6 & 3 \\ 2 & 3 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 3 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{11431^{14}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 3 & 0 & 24 & 3 \\ 2 & 4 & 4 & 4 & 0 & 36 & 4 \\ 1 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 36 & 24 & 0 & 238 & 28 \\ 2 & 3 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[87] $2 \ 2 \ 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 6 & 2 \\ 2 & 1 & 4 & 3 & 1 & 9 & 3 \\ 2 & 4 & 1 & 4 & 1 & 9 & 3 \\ 2 & 3 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & 1 & -3 & 3 & -1 \\ 6 & 9 & 9 & 3 & 25 & 9 & 9 \\ 2 & 3 & 3 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[88] $2 \ 2 \ 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -2146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 8 & 2 \\ 2 & 1 & 4 & 3 & 1 & 12 & 3 \\ 2 & 4 & 1 & 4 & 1 & 12 & 3 \\ 2 & 3 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & 1 & -3 & 4 & -1 \\ 8 & 12 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 3 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[89] $2 \ 2 \ 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -2965, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 10 & 2 \\ 2 & 1 & 4 & 3 & 1 & 15 & 3 \\ 2 & 4 & 1 & 4 & 1 & 15 & 3 \\ 2 & 3 & 4 & 0 & 1 & 15 & 2 \\ 1 & 1 & 1 & 1 & -3 & 5 & -1 \\ 10 & 15 & 15 & 15 & 5 & 73 & 15 \\ 2 & 3 & 3 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

6.64 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$

$T = 9 \quad [0] \ 1^{1111} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$

$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -2 & 3 & 0 & 0 \\ 3 & 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 0 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

6.65 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$T = 9 \quad [0] \ 1^{1111} 31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{114} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -3 & 34 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -249$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -331$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -393$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -2 & 3 & 0 & 0 \\ 3 & 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{113}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -557$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -662$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

6.66 NHC(-2)(-2)(-3) + f₄ + su₂ + su₂→(-3)

$$T = 9 \quad [0] \quad 1^{112}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 4 & 6 \\ 2 & 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 3 \\ 3 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{112}3^{11}1^2 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 2 & 3 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 12 & 18 & 0 & 58 & 10 & 14 \\ 2 & 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 3 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{114}3^{11}1^4 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 20 & 2 & 3 \\ 2 & 1 & 5 & 0 & 24 & 2 & 3 \\ 3 & 5 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 24 & 36 & 0 & 238 & 20 & 28 \\ 2 & 2 & 3 & 0 & 20 & 0 & 2 \\ 3 & 3 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113}3^{11}1^3 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 2 & 3 \\ 3 & 5 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 18 & 27 & 0 & 133 & 15 & 21 \\ 2 & 2 & 3 & 0 & 15 & 0 & 2 \\ 3 & 3 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}3^{11}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -242, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 2 & 3 \\ 3 & 5 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 7 & 10 \\ 2 & 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -356, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 2 & 3 \\ 3 & 5 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 12 & 17 & 0 & 49 & 9 & 13 \\ 2 & 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113}3^{11}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 2 & 3 \\ 3 & 5 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 15 & 22 & 0 & 85 & 12 & 17 \\ 2 & 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -416, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 2 & 3 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 15 & 21 & 0 & 76 & 11 & 16 \\ 2 & 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 3 & 4 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114}3^{11}1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -544, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 2 & 3 \\ 2 & 1 & 5 & 0 & 21 & 2 & 3 \\ 3 & 5 & 4 & 0 & 31 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 21 & 31 & 0 & 172 & 17 & 24 \\ 2 & 2 & 3 & 0 & 17 & 0 & 2 \\ 3 & 3 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114}3^{11}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -636, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 2 & 3 \\ 3 & 5 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 18 & 26 & 0 & 118 & 14 & 20 \\ 2 & 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 3 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [17] \quad 1^{11}3^{11}1^{13}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11}3^{11}1^{13}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11}1^13^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 6 & 3 \\ 2 & 2 & 6 & 4 & 0 & 9 & 4 \\ 4 & 6 & 8 & 6 & 0 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 15 & 9 & 0 & 22 & 10 \\ 3 & 4 & 6 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}1^13^{11}1^{12}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 349, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}1^13^{11}1^{13}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 412, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11}1^13^{11}1^{12}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 12 & 3 \\ 2 & 2 & 6 & 4 & 0 & 18 & 4 \\ 4 & 6 & 8 & 6 & 0 & 30 & 6 \\ 2 & 4 & 6 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 18 & 30 & 18 & 0 & 94 & 20 \\ 3 & 4 & 6 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11}1^13^{11}1^{13}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 18 & 3 \\ 2 & 2 & 6 & 4 & 0 & 27 & 4 \\ 4 & 6 & 8 & 6 & 0 & 45 & 6 \\ 2 & 4 & 6 & 1 & 0 & 27 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 27 & 45 & 27 & 0 & 214 & 30 \\ 3 & 4 & 6 & 3 & -3 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11}1^14^{11}1^{14}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 24 & 3 \\ 2 & 2 & 6 & 4 & 0 & 36 & 4 \\ 4 & 6 & 8 & 6 & 0 & 60 & 6 \\ 2 & 4 & 6 & 1 & 0 & 36 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 24 & 36 & 60 & 36 & 0 & 382 & 40 \\ 3 & 4 & 6 & 3 & -3 & 40 & 2 \end{pmatrix}_{b_0}$$

6.67 NHC(-2)(-2)(-3) + f₄ + su₂ + su₂→(-3) + su₃

$$T = 10 \quad [0] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 302, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 2 & 6 & 3 \\ 2 & 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 6 & 9 & 0 & 6 & 5 & 12 & 6 \\ 3 & 3 & 4 & -3 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 7 & 9 & 10 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

6.68 NHC(-2)(-2)(-3) + f₄ + su₂ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 3 & 5 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 8 & 3 \\ 7 & 8 & 12 & 0 & 8 & 25 & 9 \\ 3 & 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 6 & 0 & 15 & 6 & 4 \\ 3 & 6 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 15 & 17 & 0 & 49 & 21 & 13 \\ 4 & 6 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1113}1^{\overline{113}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -41, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 14 & 3 \\ 11 & 18 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}3^{\overline{111}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 5 & 3 \\ 4 & 6 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{113}3^{\overline{11}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 4 & 3 \\ 2 & 1 & 5 & 0 & 9 & 5 & 3 \\ 3 & 5 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 25 & 15 & 9 \\ 4 & 5 & 7 & 0 & 15 & 7 & 5 \\ 3 & 3 & 4 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1113}3^{\overline{111}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -140, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 8 & 3 \\ 3 & 5 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 20 & 7 \\ 7 & 8 & 12 & 0 & 20 & 25 & 9 \\ 3 & 3 & 4 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{112}3^{\overline{111}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 8 & 3 \\ 3 & 5 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 28 & 10 \\ 7 & 8 & 12 & 0 & 28 & 25 & 9 \\ 3 & 3 & 4 & -3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1112}3^{\overline{11}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -168, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 8 & 3 \\ 3 & 5 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 17 & 7 \\ 6 & 8 & 11 & 0 & 17 & 20 & 8 \\ 3 & 3 & 4 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1112}3^{\overline{111}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 9 & 6 & 4 \\ 3 & 6 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 11 & 7 \\ 4 & 6 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1113}1^{\overline{121}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -183, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 6 & 0 & 12 & 6 & 4 \\ 3 & 6 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 16 & 10 \\ 4 & 6 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1121}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -188, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 5 & 3 \\ 3 & 5 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 10 & 6 \\ 4 & 5 & 7 & 0 & 10 & 7 & 5 \\ 3 & 3 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1113}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 5 & 3 \\ 3 & 5 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 11 & 7 \\ 4 & 5 & 7 & 0 & 11 & 7 & 5 \\ 3 & 3 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1114}3^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 19 & 3 \\ 2 & 1 & 5 & 0 & 3 & 23 & 4 \\ 3 & 5 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 23 & 3 \\ 19 & 23 & 34 & 0 & 23 & 208 & 26 \\ 3 & 3 & 4 & -3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 3 & 5 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 5 & 3 \\ 4 & 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 3 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1113}3^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -247, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 11 & 3 \\ 9 & 15 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1113}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -257, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1122}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -300, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 8 & 3 \\ 3 & 5 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 16 & 6 \\ 6 & 8 & 11 & 0 & 16 & 20 & 8 \\ 3 & 3 & 4 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1113}3^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 17 & 3 \\ 3 & 5 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 17 & 3 \\ 14 & 17 & 25 & 0 & 17 & 111 & 19 \\ 3 & 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11123} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -327, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 11 & 3 \\ 3 & 5 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 11 & 3 \\ 9 & 11 & 16 & 0 & 11 & 44 & 12 \\ 3 & 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11123} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -348, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 3 & 5 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 8 & 3 \\ 6 & 8 & 11 & 0 & 8 & 20 & 8 \\ 3 & 3 & 4 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11143} 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -383, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 4 & 3 \\ 2 & 1 & 5 & 0 & 21 & 5 & 3 \\ 3 & 5 & 4 & 0 & 31 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 21 & 31 & 0 & 172 & 38 & 24 \\ 4 & 5 & 7 & 0 & 38 & 7 & 5 \\ 3 & 3 & 4 & -3 & 24 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11123} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -383, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 4 & 3 \\ 2 & 1 & 5 & 0 & 9 & 5 & 3 \\ 3 & 5 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 16 & 10 \\ 4 & 5 & 7 & 0 & 16 & 7 & 5 \\ 3 & 3 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11123} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -399, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 11 & 3 \\ 3 & 5 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 26 & 7 \\ 9 & 11 & 16 & 0 & 26 & 44 & 12 \\ 3 & 3 & 4 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11123} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -404, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 8 & 3 \\ 3 & 5 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 25 & 10 \\ 6 & 8 & 11 & 0 & 25 & 20 & 8 \\ 3 & 3 & 4 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113} 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -432, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 4 & 3 \\ 2 & 1 & 5 & 0 & 15 & 5 & 3 \\ 3 & 5 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 15 & 22 & 0 & 85 & 27 & 17 \\ 4 & 5 & 7 & 0 & 27 & 7 & 5 \\ 3 & 3 & 4 & -3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 11 & 3 \\ 3 & 5 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 11 & 3 \\ 8 & 11 & 15 & 0 & 11 & 39 & 11 \\ 3 & 3 & 4 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11313^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -531, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 4 & 3 \\ 2 & 1 & 5 & 0 & 12 & 5 & 3 \\ 3 & 5 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 12 & 17 & 0 & 49 & 21 & 13 \\ 4 & 5 & 7 & 0 & 21 & 7 & 5 \\ 3 & 3 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11133^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 14 & 3 \\ 3 & 5 & 4 & 0 & 9 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 32 & 7 \\ 11 & 14 & 20 & 0 & 32 & 69 & 15 \\ 3 & 3 & 4 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11233^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -572, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 9 & 3 \\ 2 & 1 & 5 & 0 & 9 & 11 & 3 \\ 3 & 5 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 37 & 10 \\ 9 & 11 & 16 & 0 & 37 & 44 & 12 \\ 3 & 3 & 4 & -3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11143^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -572, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 14 & 3 \\ 3 & 5 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 14 & 3 \\ 10 & 14 & 19 & 0 & 14 & 64 & 14 \\ 3 & 3 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11143^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -623, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 13 & 3 \\ 2 & 1 & 5 & 0 & 6 & 17 & 3 \\ 3 & 5 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 38 & 7 \\ 13 & 17 & 24 & 0 & 38 & 100 & 18 \\ 3 & 3 & 4 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11133^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 14 & 3 \\ 3 & 5 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 14 & 3 \\ 11 & 14 & 20 & 0 & 14 & 69 & 15 \\ 3 & 3 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11143^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -647, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 4 & 3 \\ 2 & 1 & 5 & 0 & 15 & 5 & 3 \\ 3 & 5 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 15 & 21 & 0 & 76 & 26 & 16 \\ 4 & 5 & 7 & 0 & 26 & 7 & 5 \\ 3 & 3 & 4 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{11143^1}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 4 & 3 \\ 2 & 1 & 5 & 0 & 18 & 5 & 3 \\ 3 & 5 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 18 & 26 & 0 & 118 & 32 & 20 \\ 4 & 5 & 7 & 0 & 32 & 7 & 5 \\ 3 & 3 & 4 & -3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{111431}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -824, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 16 & 3 \\ 2 & 1 & 5 & 0 & 3 & 20 & 3 \\ 3 & 5 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 20 & 3 \\ 16 & 20 & 29 & 0 & 20 & 148 & 22 \\ 3 & 3 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{111431}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -943, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 13 & 3 \\ 2 & 1 & 5 & 0 & 3 & 17 & 3 \\ 3 & 5 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 17 & 3 \\ 13 & 17 & 24 & 0 & 17 & 100 & 18 \\ 3 & 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [36] \quad 1^{1111}31^{13}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 4 & 4 & 0 & 3 & 5 \\ 1 & 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 4 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{111}31^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 101, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 7 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 11 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 11 & 4 \\ 4 & 7 & 11 & 7 & 0 & 11 & 20 & 8 \\ 2 & 3 & 4 & 2 & -3 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111}31^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 104, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 8 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 8 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 3 & 4 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{111}31^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 365, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 4 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 5 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{111}31^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 480, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 7 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 11 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 18 & 7 \\ 4 & 7 & 11 & 7 & 0 & 18 & 20 & 8 \\ 2 & 3 & 4 & 2 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

6.69 NHC(-2)(-2)(-3) + f₄ + su₂ + su_{2n3mix} + su₃

$$T = 10 \quad [0] \quad 1^{111}31^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 389, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 8 & 3 & 3 \\ 7 & 12 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 3 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

6.70 NHC(-2)(-2)(-3) + f₄ + su₂ + su₃

$$T = 9 \quad [0] \quad 1^{1121} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 12 & 6 \\ 5 & 6 & 9 & 0 & 12 & 12 & 6 \\ 3 & 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{113} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 19 & 0 & 73 & 12 & 15 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1111} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 6 & 9 & 0 & 15 & 12 & 6 \\ 3 & 3 & 4 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^{1121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 14 & 0 & 37 & 9 & 11 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 24 & 26 & 0 & 118 & 18 & 20 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 5 & 3 \\ 2 & 1 & 5 & 0 & 9 & 6 & 3 \\ 3 & 5 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 21 & 10 \\ 5 & 6 & 9 & 0 & 21 & 12 & 6 \\ 3 & 3 & 4 & -3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -686, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 5 & 3 \\ 2 & 1 & 5 & 0 & 12 & 6 & 3 \\ 3 & 5 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 12 & 17 & 0 & 49 & 27 & 13 \\ 5 & 6 & 9 & 0 & 27 & 12 & 6 \\ 3 & 3 & 4 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 5 & 3 \\ 2 & 1 & 5 & 0 & 15 & 6 & 3 \\ 3 & 5 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 15 & 21 & 0 & 76 & 33 & 16 \\ 5 & 6 & 9 & 0 & 33 & 12 & 6 \\ 3 & 3 & 4 & -3 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112} 3^1 1^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112} 3^1 1^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 22 & 0 & 85 & 15 & 17 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^11^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 6 & 0 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 30 & 28 & 0 & 154 & 18 & 22 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^11^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -934, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 23 & 0 & 100 & 15 & 18 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^{131} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 27 & 3 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{113}3^11^{141} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 30 & 31 & 0 & 172 & 21 & 24 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114}3^11^{131} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 33 & 3 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 33 & 32 & 0 & 193 & 21 & 25 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{114}3^11^{141} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 36 & 3 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 3 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [22] $1^{11}3^121^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 1 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 3 & 4 & 0 & 4 & 4 & 4 \\ 2 & 3 & 2 & 2 & -3 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11}3^121^{1113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 2 & 6 & 0 & 4 & 15 & 4 \\ 3 & 6 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 0 & 2 & 15 & 4 \\ 9 & 15 & 20 & 0 & 15 & 57 & 15 \\ 3 & 4 & 5 & -3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{111}3^{11}21^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 365, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 4 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 3 & 2 & 2 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

6.71 NHC(-2)(-2)(-3) + f₄ + su₂ + su₃ + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 2 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 302, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 6 & 2 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 5 & 6 \\ 2 & 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 3 & 4 & -3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 9 & 7 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 5 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

6.72 NHC(-2)(-2)(-3) + f₄ + su₂ + su₃ + su2n3mix

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 389, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 8 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 8 & 2 \\ 7 & 12 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

6.73 NHC(-2)(-2)(-3) + f₄ + su₂→(-3)

$$T = 8 \quad [0] \quad 1^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 13^{11}1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 4 & 9 & 9 & 4 & 4 & 5 \\ 2 & 3 & 4 & -2 & 2 & 0 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 21^{111}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 3 & 4 \\ 6 & 4 & 9 & 3 & 3 & 4 \\ 8 & 9 & 9 & 4 & 4 & 5 \\ 3 & 3 & 4 & -2 & 2 & 0 \\ 3 & 3 & 4 & 2 & 0 & 2 \\ 4 & 4 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11}3^{11}21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 6 & 1 & 5 \\ 10 & 18 & 11 & 0 & 8 \\ 6 & 11 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 8 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}31^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 2 & 6 & 4 & 0 & 4 \\ 4 & 6 & 8 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^1 3 1^{11} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -250, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 8 & 11 & 5 & 0 & 6 \\ 4 & 11 & 8 & 6 & 0 & 6 \\ 2 & 5 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 3 1^{11} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -269, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

6.74 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{f}_4$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 2 & 0 & 0 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

6.75 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -130, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 2 & 0 & 0 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{11} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 6 & 5 \\ 8 & 10 & 17 & 0 & 8 & 6 \\ 12 & 17 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 12 & 0 & 4 & 4 \\ 5 & 6 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 1^{11} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -330, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.68 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

6.76 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 \\ 3 & 5 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 6 & 8 & 0 & 4 & 10 & 6 \\ 3 & 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{11} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -122, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 15 & 3 \\ 2 & 1 & 5 & 0 & 2 & 18 & 3 \\ 3 & 5 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 15 & 2 \\ 15 & 18 & 27 & 0 & 15 & 133 & 21 \\ 3 & 3 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 2 & 12 & 3 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 12 & 18 & 0 & 10 & 58 & 14 \\ 3 & 3 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{114}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 20 & 3 \\ 2 & 1 & 5 & 0 & 2 & 24 & 3 \\ 3 & 5 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 20 & 2 \\ 20 & 24 & 36 & 0 & 20 & 238 & 28 \\ 3 & 3 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 7 & 12 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 2 & 9 & 3 \\ 3 & 5 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 9 & 13 & 0 & 7 & 28 & 10 \\ 3 & 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 2 & 12 & 3 \\ 3 & 5 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 12 & 17 & 0 & 9 & 49 & 13 \\ 3 & 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 1^{11} 3^{11} 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 2 & 15 & 3 \\ 3 & 5 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 15 & 22 & 0 & 12 & 85 & 17 \\ 3 & 3 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 2 & 15 & 3 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 15 & 21 & 0 & 11 & 76 & 16 \\ 3 & 3 & 4 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 1^{11} 3^{11} 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 17 & 3 \\ 2 & 1 & 5 & 0 & 2 & 21 & 3 \\ 3 & 5 & 4 & 0 & 3 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 17 & 2 \\ 17 & 21 & 31 & 0 & 17 & 172 & 24 \\ 3 & 3 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 1^{11} 3^{11} 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 2 & 18 & 3 \\ 3 & 5 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 18 & 26 & 0 & 14 & 118 & 20 \\ 3 & 3 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [17] \quad 1^{11} 1^{11} 3^{11} 1^{11} 3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 1^{11} 3^{11} 1^{11} 3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 4 & 2 & -3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 6 & 3 \\ 2 & 2 & 6 & 4 & 0 & 9 & 4 \\ 4 & 6 & 8 & 6 & 0 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 15 & 9 & 0 & 22 & 10 \\ 3 & 4 & 6 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 349, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 9 \\ 2 & 4 & 4 & 4 & 0 & 3 & 15 \\ 1 & 3 & 4 & 0 & 0 & 2 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 8 \\ 5 & 9 & 15 & 9 & 0 & 8 & 37 \\ 2 & 3 & 4 & 2 & -3 & 2 & 11 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 412, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 6 \\ 2 & 4 & 4 & 4 & 0 & 3 & 9 \\ 1 & 3 & 4 & 0 & 0 & 2 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 5 \\ 3 & 6 & 9 & 6 & 0 & 5 & 13 \\ 2 & 3 & 4 & 2 & -3 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 12 & 3 \\ 2 & 2 & 6 & 4 & 0 & 18 & 4 \\ 4 & 6 & 8 & 6 & 0 & 30 & 6 \\ 2 & 4 & 6 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 18 & 30 & 18 & 0 & 94 & 20 \\ 3 & 4 & 6 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 18 & 3 \\ 2 & 2 & 6 & 4 & 0 & 27 & 4 \\ 4 & 6 & 8 & 6 & 0 & 45 & 6 \\ 2 & 4 & 6 & 1 & 0 & 27 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 27 & 45 & 27 & 0 & 214 & 30 \\ 3 & 4 & 6 & 3 & -3 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{114}31^{14}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 24 & 3 \\ 2 & 2 & 6 & 4 & 0 & 36 & 4 \\ 4 & 6 & 8 & 6 & 0 & 60 & 6 \\ 2 & 4 & 6 & 1 & 0 & 36 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 24 & 36 & 60 & 36 & 0 & 382 & 40 \\ 3 & 4 & 6 & 3 & -3 & 40 & 2 \end{pmatrix}_{b_0}$$

6.77 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{111}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 1 & 3 \\ 2 & 3 & 4 & 0 & 2 & 3 & 0 \\ 3 & 4 & 4 & -3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 302, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 0 & 2 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 5 & 6 & 12 & 6 \\ 3 & 3 & 4 & -3 & 2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 13 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 12 & 13 & 0 & 7 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

6.78 NHC(-2)(-2)(-3) + $\mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$

$$T = 9 \quad [0] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 3 & 4 & -3 & 2 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 4 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 4 & 0 & 0 & 2 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 & 2 \\ 3 & 3 & 5 & -3 & 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 6 & 8 & 0 & 4 & 9 & 5 & 5 \\ 3 & 3 & 4 & -3 & 2 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -210, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 2 & 9 & 3 \\ 3 & 5 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 9 & 13 & 0 & 7 & 27 & 9 \\ 3 & 3 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

6.79 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_3$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{111}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 3 \\ 2 & 3 & 4 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 6 \\ 3 & 6 & 4 & 0 & 3 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 3 \\ 2 & 3 & 4 & 0 & 2 & 0 & 3 \\ 3 & 6 & 5 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 302, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 6 & 9 & 0 & 5 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & 4 & -3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^11^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 7 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 622$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

6.80 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_{2n3\text{mix}}$

$T = 8$ [0] $1^{123}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 20$, $T \in [15, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 \\ 4 & 5 & 9 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 11 & 20 & 0 & 44 & 12 \\ 3 & 3 & 5 & -3 & 12 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 70$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{112}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 70$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 14 & 21 & 0 & 80 & 16 \\ 3 & 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 70$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 20 & 30 & 0 & 165 & 23 \\ 3 & 3 & 4 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{114}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 70$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 22 & 3 \\ 2 & 1 & 5 & 0 & 26 & 3 \\ 3 & 5 & 4 & 0 & 39 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 26 & 39 & 0 & 280 & 30 \\ 3 & 3 & 4 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 71$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{113}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 106$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 5 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 18 & 20 & 0 & 69 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

- [8] $1^{111}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 5 \\ 3 & 3 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 8 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 132$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 8 & 16 & 0 & 25 & 9 \\ 3 & 3 & 5 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{112}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 182$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 3 & 5 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 8 \\ 3 & 3 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{111}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 219$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 16 & 0 & 44 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 11 & 16 & 0 & 44 & 12 \\ 3 & 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{113}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 11 & 15 & 0 & 39 & 11 \\ 3 & 3 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{114}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 294$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 14 & 19 & 0 & 64 & 14 \\ 3 & 3 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{113}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 322$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 17 & 25 & 0 & 111 & 19 \\ 3 & 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

- [18] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 378$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 14 & 20 & 0 & 69 & 15 \\ 3 & 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{114}3^11^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 420$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 19 & 3 \\ 2 & 1 & 5 & 0 & 23 & 3 \\ 3 & 5 & 4 & 0 & 34 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 19 & 23 & 34 & 0 & 208 & 26 \\ 3 & 3 & 4 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{114}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 532$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 17 & 24 & 0 & 100 & 18 \\ 3 & 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{114}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 574$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 20 & 29 & 0 & 148 & 22 \\ 3 & 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$
- $T = 9$ [22] $1^{11}31^{11}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -35$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 5 & 4 \\ 2 & 7 & 4 & 4 & 0 & 5 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [23] $1^{111}31^{12}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 10 & 3 \\ 2 & 4 & 4 & 4 & 0 & 17 & 4 \\ 1 & 3 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 17 & 10 & 0 & 48 & 12 \\ 2 & 3 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{111}3^11^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 14 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 24 & 14 & 0 & 97 & 17 \\ 2 & 3 & 4 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{11}3^11^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 9 & 5 & 0 & 12 & 6 \\ 2 & 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [26] $1^{111}3^11^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$
- [27] $1^{11}3^11^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -129$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 12 & 4 \\ 2 & 7 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{111}31^{13}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 3 \\ 2 & 4 & 4 & 4 & 0 & 5 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{112}31^{14}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 19 & 3 \\ 2 & 4 & 4 & 4 & 0 & 32 & 4 \\ 1 & 3 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 32 & 19 & 0 & 175 & 23 \\ 2 & 3 & 4 & 2 & -3 & 23 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}31^{113}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 7 & 3 \\ 2 & 4 & 4 & 4 & 0 & 11 & 4 \\ 1 & 3 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 7 & 0 & 20 & 8 \\ 2 & 3 & 4 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{111}31^{113}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 8 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 8 & 0 & 25 & 9 \\ 2 & 3 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{111}31^{123}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 11 & 3 \\ 2 & 4 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 18 & 11 & 0 & 55 & 13 \\ 2 & 3 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{113}31^{113}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 13 & 3 \\ 2 & 4 & 4 & 4 & 0 & 17 & 4 \\ 1 & 3 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 13 & 17 & 13 & 0 & 60 & 14 \\ 2 & 3 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[34] $1^131^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 8 & 4 \\ 2 & 7 & 4 & 4 & 0 & 8 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 8 & 4 & 0 & 9 & 5 \\ 2 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 11 & 4 \\ 2 & 7 & 4 & 4 & 0 & 11 & 4 \\ 1 & 3 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 11 & 7 & 0 & 20 & 8 \\ 2 & 4 & 4 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{112}31^{113}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 10 & 3 \\ 2 & 4 & 4 & 4 & 0 & 14 & 4 \\ 1 & 3 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 14 & 10 & 0 & 37 & 11 \\ 2 & 3 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^1 3^1 1^{11} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 6 & 0 \\ 2 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11} 31^{13} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 16 & 3 \\ 2 & 4 & 4 & 4 & 0 & 26 & 4 \\ 1 & 3 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 26 & 16 & 0 & 117 & 19 \\ 2 & 3 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11} 31^{12} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 3 & 0 & 13 & 3 \\ 2 & 4 & 4 & 4 & 0 & 20 & 4 \\ 1 & 3 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 20 & 13 & 0 & 71 & 15 \\ 2 & 3 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} 31^{12} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 16 & 3 \\ 2 & 4 & 4 & 4 & 0 & 23 & 4 \\ 1 & 3 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 23 & 16 & 0 & 100 & 18 \\ 2 & 3 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11} 431^{12} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 19 & 3 \\ 2 & 4 & 4 & 4 & 0 & 26 & 4 \\ 1 & 3 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 19 & 26 & 19 & 0 & 135 & 21 \\ 2 & 3 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{11} 31^{13} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -972, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 19 & 3 \\ 2 & 4 & 4 & 4 & 0 & 29 & 4 \\ 1 & 3 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 29 & 19 & 0 & 152 & 22 \\ 2 & 3 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11} 31^{14} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 3 & 0 & 22 & 3 \\ 2 & 4 & 4 & 4 & 0 & 35 & 4 \\ 1 & 3 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 35 & 22 & 0 & 216 & 26 \\ 2 & 3 & 4 & 2 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{11} 431^{13} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 22 & 3 \\ 2 & 4 & 4 & 4 & 0 & 32 & 4 \\ 1 & 3 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 22 & 32 & 22 & 0 & 193 & 25 \\ 2 & 3 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{11} 431^{14} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 13 & 2 \\ 1 & 1 & 4 & 3 & 0 & 25 & 3 \\ 2 & 4 & 4 & 4 & 0 & 38 & 4 \\ 1 & 3 & 4 & 0 & 0 & 25 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 25 & 38 & 25 & 0 & 263 & 29 \\ 2 & 3 & 4 & 2 & -3 & 29 & 1 \end{pmatrix}_{b_0}$$

6.81 NHC(-2)(-2)(-3) + f₄ + su2n3mix + so₈

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 15 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 15 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

6.82 NHC(-2)(-2)(-3) + f₄ + su2n3mix + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 3 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 6 & 15 & 4 \\ 3 & 6 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 21 & 5 \\ 9 & 15 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{131} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 18 & 20 & 0 & 69 & 14 & 15 \\ 2 & 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 1 & 5 & 0 & 5 & 9 & 3 \\ 3 & 5 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 15 & 5 \\ 6 & 9 & 12 & 0 & 15 & 25 & 9 \\ 3 & 3 & 4 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}3^11^{\overline{111}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -140, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 5 & 3 \\ 2 & 1 & 5 & 0 & 8 & 6 & 3 \\ 3 & 5 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 20 & 9 \\ 5 & 6 & 9 & 0 & 20 & 13 & 7 \\ 3 & 3 & 4 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}3^11^{\overline{111}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 28 & 9 \\ 7 & 9 & 13 & 0 & 28 & 28 & 10 \\ 3 & 3 & 4 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1112}3^11^{\overline{11}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -168, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 5 & 3 \\ 2 & 1 & 5 & 0 & 8 & 6 & 3 \\ 3 & 5 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 17 & 8 \\ 5 & 6 & 9 & 0 & 17 & 13 & 7 \\ 3 & 3 & 4 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}3^11^{\overline{111}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 6 & 9 & 4 \\ 3 & 6 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}3^11^{\overline{112}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -183, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 6 & 12 & 4 \\ 3 & 6 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 16 & 5 \\ 7 & 12 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1112}3^11^{\overline{1}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -188, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 4 & 3 \\ 2 & 1 & 5 & 0 & 5 & 6 & 3 \\ 3 & 5 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 10 & 5 \\ 4 & 6 & 8 & 0 & 10 & 10 & 6 \\ 3 & 3 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}3^11^{\overline{1}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 5 & 3 \\ 2 & 1 & 5 & 0 & 5 & 6 & 3 \\ 3 & 5 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 11 & 5 \\ 5 & 6 & 9 & 0 & 11 & 13 & 7 \\ 3 & 3 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1141}3^11^{\overline{13}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 19 & 2 & 3 \\ 2 & 1 & 5 & 0 & 23 & 3 & 3 \\ 3 & 5 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 19 & 23 & 34 & 0 & 208 & 23 & 26 \\ 2 & 3 & 4 & 0 & 23 & 1 & 3 \\ 3 & 3 & 4 & -3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111}3^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 5 & 1 & 3 \\ 3 & 3 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111}3^{11}1^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -247, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 15 & 16 & 0 & 44 & 11 & 12 \\ 2 & 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111}3^{11}1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -257, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1122}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -300, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 4 & 3 \\ 2 & 1 & 5 & 0 & 8 & 6 & 3 \\ 3 & 5 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 16 & 8 \\ 4 & 6 & 8 & 0 & 16 & 10 & 6 \\ 3 & 3 & 4 & -3 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1131}3^{11}1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 & 3 \\ 3 & 5 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 17 & 25 & 0 & 111 & 17 & 19 \\ 2 & 3 & 4 & 0 & 17 & 1 & 3 \\ 3 & 3 & 4 & -3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1121}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -327, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 11 & 16 & 0 & 44 & 11 & 12 \\ 2 & 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 3 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1121}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -348, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 3 & 5 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 8 & 8 \\ 2 & 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 3 & 4 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1114}3^{11}1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -383, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 17 & 3 \\ 2 & 1 & 5 & 0 & 5 & 21 & 3 \\ 3 & 5 & 4 & 0 & 7 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 38 & 5 \\ 17 & 21 & 31 & 0 & 38 & 172 & 24 \\ 3 & 3 & 4 & -3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11123}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -383, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 7 & 3 \\ 2 & 1 & 5 & 0 & 5 & 9 & 3 \\ 3 & 5 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 16 & 5 \\ 7 & 9 & 13 & 0 & 16 & 28 & 10 \\ 3 & 3 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11123}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -399, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 5 & 3 \\ 2 & 1 & 5 & 0 & 11 & 6 & 3 \\ 3 & 5 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 11 & 16 & 0 & 44 & 26 & 12 \\ 5 & 6 & 9 & 0 & 26 & 13 & 7 \\ 3 & 3 & 4 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11123}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -404, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 9 & 3 \\ 3 & 5 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 25 & 8 \\ 7 & 9 & 13 & 0 & 25 & 28 & 10 \\ 3 & 3 & 4 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -432, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 12 & 3 \\ 2 & 1 & 5 & 0 & 5 & 15 & 3 \\ 3 & 5 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 27 & 5 \\ 12 & 15 & 22 & 0 & 27 & 85 & 17 \\ 3 & 3 & 4 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{113}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 2 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 & 3 \\ 3 & 5 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 11 & 15 & 0 & 39 & 11 & 11 \\ 2 & 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 3 & 4 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1113}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -531, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 1 & 5 & 0 & 5 & 12 & 3 \\ 3 & 5 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 21 & 5 \\ 9 & 12 & 17 & 0 & 21 & 49 & 13 \\ 3 & 3 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{113}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 5 & 3 \\ 2 & 1 & 5 & 0 & 14 & 6 & 3 \\ 3 & 5 & 4 & 0 & 20 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 14 & 20 & 0 & 69 & 32 & 15 \\ 5 & 6 & 9 & 0 & 32 & 13 & 7 \\ 3 & 3 & 4 & -3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{114}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -572, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 & 3 \\ 3 & 5 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 14 & 19 & 0 & 64 & 14 & 14 \\ 2 & 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 3 & 4 & -3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11223^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -572, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 7 & 3 \\ 2 & 1 & 5 & 0 & 11 & 9 & 3 \\ 3 & 5 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 11 & 16 & 0 & 44 & 37 & 12 \\ 7 & 9 & 13 & 0 & 37 & 28 & 10 \\ 3 & 3 & 4 & -3 & 12 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11413^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -623, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 5 & 3 \\ 2 & 1 & 5 & 0 & 17 & 6 & 3 \\ 3 & 5 & 4 & 0 & 24 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 17 & 24 & 0 & 100 & 38 & 18 \\ 5 & 6 & 9 & 0 & 38 & 13 & 7 \\ 3 & 3 & 4 & -3 & 18 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11313^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -635, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 14 & 20 & 0 & 69 & 14 & 15 \\ 2 & 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 3 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11143^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -647, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 11 & 3 \\ 2 & 1 & 5 & 0 & 5 & 15 & 3 \\ 3 & 5 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 26 & 5 \\ 11 & 15 & 21 & 0 & 26 & 76 & 16 \\ 3 & 3 & 4 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11143^1}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -776, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 14 & 3 \\ 2 & 1 & 5 & 0 & 5 & 18 & 3 \\ 3 & 5 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 32 & 5 \\ 14 & 18 & 26 & 0 & 32 & 118 & 20 \\ 3 & 3 & 4 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11413^1}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -824, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 2 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 & 3 \\ 3 & 5 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 20 & 29 & 0 & 148 & 20 & 22 \\ 2 & 3 & 4 & 0 & 20 & 1 & 3 \\ 3 & 3 & 4 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11413^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -943, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 2 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 17 & 24 & 0 & 100 & 17 & 18 \\ 2 & 3 & 4 & 0 & 17 & 1 & 3 \\ 3 & 3 & 4 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [36] \quad 1^{1111}31^131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 3 & 4 & 2 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 7 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 11 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 7 & 0 & 20 & 11 & 8 \\ 2 & 3 & 6 & 3 & 0 & 11 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 8 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 8 & 0 & 25 & 12 & 9 \\ 2 & 3 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 365, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 5 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 6 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & 4 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 480, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 7 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 11 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 7 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 7 & 0 & 20 & 18 & 8 \\ 3 & 6 & 9 & 6 & 0 & 18 & 13 & 7 \\ 2 & 3 & 4 & 2 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

6.83 NHC(-2)(-2)(-3) + f₄ + su₂n₃mix + su₂ + su₃

$$T = 10 \quad [0] \ 1^{111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 389, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 8 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

6.84 NHC(-2)(-2)(-3) + f₄ + su₂n₃mix + su₃

$$T = 9 \quad [0] \ 1^{112}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 3 & 5 & 4 & 0 & 11 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 9 & 8 \\ 3 & 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 3 & 4 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -89, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 0 & 2 \\ 3 & 3 & 4 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -97, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 & 3 \\ 4 & 5 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 5 & 12 & 0 & 12 & 5 & 6 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 3 & 5 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 1^{\overset{1}{11}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 & 3 \\ 3 & 5 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 6 & 5 \\ 3 & 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 3 & 4 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overset{1}{11}} 3^{\overset{1}{1}} 1^{\overset{1}{11}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 1^{\overset{1}{131}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 18 & 20 & 0 & 69 & 14 & 15 \\ 2 & 3 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & 4 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 1^{\overset{1}{11}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 4 & 5 & 9 & 0 & 16 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 16 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 0 & 2 \\ 3 & 3 & 5 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 1^{\overset{1}{11}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 8 & 0 & 12 & 5 & 6 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 1^{\overset{1}{11}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 5 & 3 \\ 2 & 1 & 5 & 0 & 5 & 6 & 3 \\ 3 & 5 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 11 & 5 \\ 5 & 6 & 9 & 0 & 11 & 12 & 6 \\ 3 & 3 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -362, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 3 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overset{1}{111}} 3^{\overset{1}{1}} 1^{\overset{1}{121}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 15 & 16 & 0 & 44 & 11 & 12 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}3^{11}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

6.85 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_{2n3\text{mix}}$ + $\mathbf{su}_3$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{111}3^{11}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 389, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

6.86 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_3$

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 3 \\ 3 & 3 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 16 & 0 & 27 & 9 \\ 3 & 3 & 5 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 8 & 0 & 3 & 3 \\ 3 & 3 & 5 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 12 & 18 & 0 & 57 & 13 \\ 3 & 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 6 \\ 3 & 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 24 & 8 \\ 3 & 3 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 0 & 9 & 5 \\ 3 & 3 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 13 & 0 & 27 & 9 \\ 3 & 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 15 & 22 & 0 & 84 & 16 \\ 3 & 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 48 & 12 \\ 3 & 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [21] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 2 & 4 & 4 & 0 & 4 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 10 & 1 & 5 \\ 9 & 13 & 16 & 0 & 7 \\ 10 & 16 & 18 & 1 & 8 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 1 & 3 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 13 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 13 & 21 & 30 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{12}321^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 5 \\ 1 & 2 & 0 & 2 & 0 & 5 \\ 1 & 3 & 2 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 3 & 5 & 5 & 0 & 12 & 6 \\ 2 & 3 & 2 & 2 & -3 & 6 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11}321^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 2 & 0 & 1 \\ 1 & 3 & 2 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 1 & 3 & 0 & 0 \\ 2 & 3 & 2 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}321^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 2 & 6 & 0 & 11 & 4 \\ 3 & 6 & 5 & 0 & 15 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 15 & 0 & 31 & 11 \\ 3 & 4 & 5 & -3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 7 & 0 & 3 & 4 \\ 2 & 3 & 7 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 4 & 3 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11}321^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 2 & 6 & 0 & 15 & 4 \\ 3 & 6 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 20 & 0 & 57 & 15 \\ 3 & 4 & 5 & -3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11}321^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 2 & 6 & 0 & 19 & 4 \\ 3 & 6 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 25 & 0 & 91 & 19 \\ 3 & 4 & 5 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 3 & 3 \\ 2 & 3 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112}321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 3 & 1 & 0 & 5 & 2 \\ 1 & 3 & 0 & 2 & 0 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11}321^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{13}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 0 & 9 & 3 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 8 & 9 & 0 & 39 & 11 \\ 2 & 3 & 2 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 7 & 3 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 4 & 7 & 0 & 15 & 7 \\ 2 & 3 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{12}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 3 & 0 & 8 & 3 \\ 1 & 2 & 0 & 2 & 0 & 6 & 2 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 8 & 0 & 25 & 9 \\ 2 & 3 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 12 & 6 \\ 2 & 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{13}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[44] \ 2^1 2^1 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 & 3 \\ 2 & 4 & 1 & 4 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

6.87 NHC(-2)(-2)(-3) + f₄ + su₃ + so₈

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{11} 3^1 21^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 & 0 & 1 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -2 & 0 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 1 & 0 & -2 \\ 2 & 2 & 3 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

6.88 NHC(-2)(-2)(-3) + f₄ + su₃ + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 12 & 6 \\ 4 & 6 & 8 & 0 & 12 & 10 & 6 \\ 3 & 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

- [3] $1^{111}31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -234$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 27 & 9 & 9 \\ 2 & 3 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{113}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 11 & 21 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & 4 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 3 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -294$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 15 & 6 \\ 5 & 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -318$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -381$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{112}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -423$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 8 & 15 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -486$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 14 & 24 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -539$, $(n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 21 & 6 \\ 7 & 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1113} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -686, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 27 & 6 \\ 9 & 12 & 17 & 0 & 27 & 49 & 13 \\ 3 & 3 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1114} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 33 & 6 \\ 11 & 15 & 21 & 0 & 33 & 76 & 16 \\ 3 & 3 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112} 3^1 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112} 3^1 1^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 12 & 21 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^1 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 16 & 30 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -934, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 13 & 24 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 3^1 1^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113} 3^1 1^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 17 & 30 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114}31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 18 & 33 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114}31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [22] \ 1^{111}321^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 3 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 1 & 3 & 0 & 4 & 1 & 3 \\ 2 & 3 & 2 & 2 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}321^{1131} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 2 & 6 & 0 & 15 & 4 & 4 \\ 3 & 6 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 15 & 20 & 0 & 57 & 15 & 15 \\ 2 & 4 & 5 & 0 & 15 & 2 & 4 \\ 3 & 4 & 5 & -3 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111}321^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 365, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 4 & 7 & 4 \\ 2 & 4 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 3 & 2 & 2 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

6.89 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111}3^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 302, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 2 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 6 & 5 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 2 & 3 \\ 2 & 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 3 & 4 & -3 & 6 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}3^1}1^{\overline{112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 7 & 10 \\ 2 & 3 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}3^1}1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

6.90 NHC(-2)(-2)(-3) + f₄ + su₃ + su₂ + su₂n3mix

$$T = 10 \quad [0] \ 1^{\overline{111}3^1}1^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 389, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 8 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 8 & 3 \\ 7 & 12 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

6.91 NHC(-2)(-2)(-3) + f₄ + su₃ + su₂ → (-3)

$$T = 9 \quad [0] \ 1^{\overline{11}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 3 \\ 3 & 5 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 3 \\ 4 & 5 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}3^1}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 9 & 4 & 5 \\ 2 & 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -194, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 3 \\ 3 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -210, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 5 & 6 \\ 2 & 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 2 & 3 \\ 3 & 5 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 27 & 7 & 9 \\ 2 & 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 3 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

6.92 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_3$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 302, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 5 & 6 & 6 \\ 2 & 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 6 & 2 & 1 & 3 \\ 3 & 3 & 4 & -3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 484, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 12 & 13 & 0 & 9 & 7 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

6.93 NHC(-2)(-2)(-3) + f₄ + su₃ + su₂n3mix

$$T = 9 \quad [0] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 3 & 5 & 4 & 0 & 5 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 9 & 3 \\ 6 & 8 & 11 & 0 & 9 & 20 & 8 \\ 3 & 3 & 4 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -89, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 3 & 5 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 7 & 8 & 12 & 0 & 8 & 25 & 9 \\ 3 & 3 & 4 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -97, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.55 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 5 & 12 & 0 & 5 & 12 & 6 \\ 3 & 3 & 5 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 3 & 5 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 7 & 0 & 6 & 7 & 5 \\ 3 & 3 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -127, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 4 & 6 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -157, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 14 & 2 \\ 11 & 18 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -175, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 4 & 5 & 9 & 0 & 4 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 7 & 8 & 16 & 0 & 8 & 25 & 9 \\ 3 & 3 & 5 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -202, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 9 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -329, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 5 & 3 \\ 3 & 5 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 11 & 6 \\ 4 & 5 & 7 & 0 & 11 & 7 & 5 \\ 3 & 3 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -362, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 3 & 5 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 4 & 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 3 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -466, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 11 & 2 \\ 9 & 15 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

6.94 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 389, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 8 & 3 & 2 \\ 7 & 12 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

6.95 NHC(-2)(-2)(-3) + f₄ + su₃ + su₃

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 3 & 8 & 0 & 3 & 3 & 3 \\ 3 & 3 & 5 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 12 & 0 & 6 & 12 & 6 \\ 3 & 3 & 5 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 3 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -441, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

6.96 NHC(-2)(-2)(-3) + f₄ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 31^{11} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 1 & 4 \\ 6 & 4 & 9 & 3 & 0 & 4 \\ 8 & 9 & 9 & 4 & 0 & 5 \\ 3 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

6.97 NHC(-2)(-2)(-3) + NHC(-2)(-3)

$$T = 11 \quad [0] \quad 1^{11} 5^{11} 131^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 4 & 2 & 0 & 3 \\ 3 & 4 & 5 & 5 & 0 & 5 \\ 2 & 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

6.98 NHC(-2)(-2)(-3) + so₈

$$T = 7 \quad [1] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 11 & 16 & 6 \\ 11 & 10 & 17 & 6 \\ 16 & 17 & 25 & 9 \\ 6 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 13 & 12 & 17 & 7 \\ 12 & 10 & 17 & 6 \\ 17 & 17 & 20 & 8 \\ 7 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 5 \\ 8 & 10 & 17 & 6 \\ 12 & 17 & 25 & 9 \\ 5 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 12 & 15 & 6 \\ 12 & 13 & 18 & 7 \\ 15 & 18 & 20 & 8 \\ 6 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 11 & 15 & 6 \\ 11 & 10 & 17 & 6 \\ 15 & 17 & 20 & 8 \\ 6 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 9 & 11 & 5 \\ 9 & 13 & 18 & 7 \\ 11 & 18 & 20 & 8 \\ 5 & 7 & 8 & 2 \end{pmatrix}_{b_0}$$

[12] $21^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 12 & 20 & 7 \\ 12 & 10 & 17 & 6 \\ 20 & 17 & 25 & 9 \\ 7 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[13] $1^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 8 & 11 & 5 \\ 8 & 10 & 17 & 6 \\ 11 & 17 & 20 & 8 \\ 5 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[14] $1^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 9 & 12 & 5 \\ 9 & 13 & 21 & 7 \\ 12 & 21 & 25 & 9 \\ 5 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[15] $21^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 12 & 16 & 6 \\ 12 & 13 & 18 & 7 \\ 16 & 18 & 20 & 8 \\ 6 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[16] $21^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 15 & 17 & 7 \\ 15 & 13 & 18 & 7 \\ 17 & 18 & 20 & 8 \\ 7 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[17] $21^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 12 & 16 & 6 \\ 12 & 10 & 17 & 6 \\ 16 & 17 & 20 & 8 \\ 6 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[18] $21^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 15 & 20 & 7 \\ 15 & 13 & 21 & 7 \\ 20 & 21 & 25 & 9 \\ 7 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$T = 8$ [19] $1^1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 2 & 6 & 0 & 4 \\ 3 & 6 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 3 \end{pmatrix}_{b_0}$$

[20] $1^1 1^1 3 2 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 10 & 5 & 0 & 6 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 6 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[21] $2 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 8 & 16 & 6 \\ 8 & 5 & 13 & 5 \\ 16 & 13 & 25 & 9 \\ 6 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[22] $1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 8 & 16 & 6 \\ 8 & 5 & 13 & 5 \\ 16 & 13 & 25 & 9 \\ 6 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[23] $2 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 9 & 14 & 7 \\ 9 & 5 & 11 & 5 \\ 14 & 11 & 13 & 7 \\ 7 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[24] $2 2 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 1 & 4 & 1 & 3 \\ 2 & 1 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11} 1^1 3 1^1 3 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 7 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 7 & 13 & 10 & 13 & 6 \\ 8 & 12 & 13 & 12 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[27] $2 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 15 & 7 \\ 12 & 15 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[26] $1^1 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 15 & 7 \\ 12 & 15 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[28] $2 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 5 & 7 & 12 & 5 \\ 7 & 5 & 13 & 5 \\ 12 & 13 & 25 & 9 \\ 5 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11} 1^1 3 2 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 3 & 6 & 5 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{11} 1^1 3 1^1 3 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 6 & 6 & 0 & 11 & 5 \\ 2 & 3 & 3 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

- [31] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 12 & 1 & 12 & 6 \\ 3 & 6 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [33] $1^{11}3131^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 0 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [32] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [35] $1^{11}321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 2 & 3 & 2 & -1 & 3 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{11}321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [38] $1^1 4^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 5 & 6 & 12 & 5 \\ 6 & 5 & 13 & 5 \\ 12 & 13 & 25 & 9 \\ 5 & 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [37] $1^1 4^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [39] $2^1 2^1 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 5 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [40] $1^1 4^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 6 & 9 & 5 \\ 6 & 5 & 11 & 5 \\ 9 & 11 & 13 & 7 \\ 5 & 5 & 7 & 3 \end{pmatrix}_{b_0}$
- [41] $1^{11}321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$

- [42] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 1 & 8 & 4 \\ 7 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 12 & 1 & 12 & 6 \\ 4 & 6 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [43] $1^{11}3^{11}21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 41$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 3 & 0 & 4 & 3 & 2 \\ 2 & 2 & 0 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [46] $1^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 15 & 7 \\ 9 & 15 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [45] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 13 & 16 & 7 \\ 8 & 12 & 16 & 12 & 6 \\ 4 & 6 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [44] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 4 & 1 & 3 \\ 5 & 13 & 13 & 0 & 7 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [47] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [48] $131^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 13 & 16 & 7 \\ 5 & 12 & 16 & 12 & 6 \\ 3 & 6 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [49] $31^{11}231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 & 1 & 3 \\ 1 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [50] $21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 67$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 10 & 5 \\ 9 & 13 & 15 & 7 \\ 10 & 15 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [51] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 8 & 7 & 8 & 4 \\ 8 & 13 & 13 & 15 & 7 \\ 7 & 13 & 10 & 13 & 6 \\ 8 & 15 & 13 & 12 & 6 \\ 4 & 7 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [52] $21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 83$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 13 & 9 & 20 & 7 \\ 9 & 5 & 13 & 5 \\ 20 & 13 & 25 & 9 \\ 7 & 5 & 9 & 3 \end{pmatrix}_{b_0}$
- [53] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0} \\
[54] \quad & 131^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 4 & 5 & 3 \\ 5 & 13 & 13 & 15 & 7 \\ 4 & 13 & 10 & 13 & 6 \\ 5 & 15 & 13 & 12 & 6 \\ 3 & 7 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[55] \quad & 21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 13 & 15 & 14 & 7 \\ 15 & 13 & 15 & 7 \\ 14 & 15 & 13 & 7 \\ 7 & 7 & 7 & 3 \end{pmatrix}_{b_0} \\
[56] \quad & 1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 4 & 4 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0} \\
[57] \quad & 21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 7 & 10 & 5 \\ 7 & 5 & 11 & 5 \\ 10 & 11 & 13 & 7 \\ 5 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[58] \quad & 1^{111}2^231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 2 & 6 & 4 & 4 \\ 4 & 6 & 5 & 5 & 5 \\ 3 & 4 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[59] \quad & 1^{111}2^231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 1 & 4 & 1 & 3 \\ 2 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0} \\
[60] \quad & 31^{111}231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 553, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 & 1 & 3 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0} \\
T = 9 \quad [61] \quad & 1^{11}4^{11}131^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 2 & 4 & 2 & 4 \\ 4 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 4 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[62] \quad & 1^{11}2^321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0} \\
[63] \quad & 1^{11}31^{11}4^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 3 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[64] \quad & 1^{11}31^{11}4^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{1^{11}31^1} 4^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 1 & 2 & 2 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 3 & 2 & 2 & 0 & 2 \\ 3 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[66] $31^{11}31^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 3 & 2 & 4 \\ 3 & 4 & 1 & 4 & 1 & 3 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[67] $31^{11}31^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 6 & 4 & 3 \\ 5 & 8 & 13 & 5 & 6 \\ 6 & 13 & 13 & 6 & 7 \\ 4 & 5 & 6 & 0 & 2 \\ 3 & 6 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[68] $2^1 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 4 & 3 & 3 \\ 5 & 8 & 8 & 5 & 6 \\ 4 & 8 & 5 & 5 & 5 \\ 3 & 5 & 5 & 0 & 2 \\ 3 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[69] $31^{11}31^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 3 & 2 & 4 \\ 3 & 7 & 4 & 4 & 2 & 4 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & -2 & 0 \\ 2 & 4 & 4 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[70] $31^{11}31^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 2 & 4 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & -2 & 0 \\ 2 & 3 & 4 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [71] $21^{11}51^{11}231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 9 & 5 & 3 & 4 \\ 9 & 19 & 11 & 7 & 9 \\ 5 & 11 & 5 & 5 & 5 \\ 3 & 7 & 5 & 0 & 2 \\ 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{11}51231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 10 & 5 & 0 & 6 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 6 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[73] $2^1 3^1 1^{11} 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 & 4 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[74] $2^1 3^1 1^{11} 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 4 & 4 & 5 & 5 & 5 \\ 3 & 2 & 5 & 0 & 2 \\ 3 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [75] \quad 1^{11}31^151321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 2 & 1 & 5 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 5 & 5 \\ 2 & 3 & 4 & -1 & 5 & 4 \end{pmatrix}_{b_0}$$

6.99 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 135, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 6 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 238, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.55 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 12 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 7 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 242, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 253, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.68 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 6 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.96 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -391, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.100 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -90, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.29 \rightarrow 15, \quad T_{\max} = 93, \quad T \in [15, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -4 & 34 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -139$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -317$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{113}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -439$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^1+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -525$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 9 & 5 \\ 9 & 13 & 21 & 0 & 15 & 7 \\ 12 & 21 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 20 & 0 & 13 & 7 \\ 5 & 7 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{112}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -526$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{113}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -653$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.8$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -709$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

6.101 $\text{NHC}(-2)(-2)(-3) + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 15 & 7 \\ 9 & 12 & 20 & 0 & 15 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 17 & 5 \\ 8 & 10 & 17 & 0 & 12 & 24 & 6 \\ 12 & 17 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 27 & 7 \\ 17 & 24 & 36 & 0 & 27 & 49 & 13 \\ 5 & 6 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 15 & 7 \\ 9 & 15 & 17 & 0 & 15 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 17 & 5 \\ 9 & 13 & 18 & 0 & 15 & 27 & 7 \\ 11 & 18 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 27 & 7 \\ 17 & 27 & 33 & 0 & 27 & 49 & 13 \\ 5 & 7 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

6.102 $\text{NHC}(-2)(-2)(-3) + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -559, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

6.103 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 6 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 12 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 6 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] 1^{111} 3^1 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 7 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3^1 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.104 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] 1^{111} 3^1 1^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{114} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -3 & 34 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -186$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -249$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{113}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -331$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^1+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -393$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 9 & 5 \\ 9 & 13 & 21 & 0 & 15 & 7 \\ 12 & 21 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 20 & 0 & 13 & 7 \\ 5 & 7 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{112}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{113}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -557$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

6.105 NHC(-2)(-2)(-3) + $\mathbf{so}_8$ + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 6 & 5 \\ 9 & 13 & 21 & 0 & 9 & 7 \\ 12 & 21 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 4 & 4 \\ 5 & 7 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 6 & 5 \\ 8 & 10 & 17 & 0 & 8 & 6 \\ 12 & 17 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 12 & 0 & 4 & 4 \\ 5 & 6 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

6.106 NHC(-2)(-2)(-3) + $\mathbf{so}_8$ + $\mathbf{f}_4$ + $\mathbf{su}_2 \mathbf{n}_3 \mathbf{mix}$

$$T = 9 \quad [0] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 15 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 15 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

6.107 NHC(-2)(-2)(-3) + $\mathbf{so}_8$ + $\mathbf{f}_4$ + $\mathbf{su}_3$

$$T = 9 \quad [0] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -213$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 13.24 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

6.108 NHC(-2)(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8$

$T = 8$ [0] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 77$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 115$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 1 \\ 1 & 1 & 2 & 0 & 1 & -2 \\ 2 & 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 12 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 139$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 8 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 1 & 3 \\ 4 & 10 & 1 & 12 & 0 & 6 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 12 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 0 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 3 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 0 & 2 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 6 & 9 & 1 & 5 \\ 6 & 5 & 11 & 0 & 5 \\ 9 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 6 & 12 & 1 & 5 \\ 6 & 5 & 13 & 0 & 5 \\ 12 & 13 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -209, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 7 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [12] \ 1^{\overline{11}} 3^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 1 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

6.109 NHC(-2)(-2)(-3) + so₈ + so₈ + su₂

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -55, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 10 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 1 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & -2 & 1 \\ 1 & 2 & 1 & 0 & 2 & 1 & 0 \\ 2 & 2 & 2 & -1 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

13.0

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 2 & 0 \\ 2 & 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 18 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 2 & 0 \\ 1 & 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 \ 1^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -223, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 3 & 0 \\ 4 & 3 & 3 & 0 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 3 & 0 \\ 2 & 3 & 3 & 0 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -287, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 37 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 37 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -290, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 45 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 46 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 46 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -342, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 27 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -349, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 18 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -349, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 45 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -405, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -411, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 36 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -431, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 3 & 0 \\ 3 & 3 & 3 & 0 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -556, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 27 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{113} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -565, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 36 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

6.110 NHC(-2)(-2)(-3) + so₈ + so₈ + su₂ → (-3)

$$T = 9 \quad [0] \ 1^{113} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 6 & 5 \\ 8 & 10 & 17 & 0 & 8 & 6 \\ 12 & 17 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 0 & 4 & 4 \\ 5 & 6 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113} 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -214, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 6 & 1 & 6 & 5 \\ 8 & 10 & 9 & 0 & 8 & 6 \\ 6 & 9 & 5 & 1 & 7 & 5 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 7 & 0 & 3 & 3 \\ 5 & 6 & 5 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

6.111 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 16 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 16 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 12 & 9 & 1 & 9 & 7 \\ 12 & 10 & 9 & 0 & 8 & 6 \\ 9 & 9 & 5 & 1 & 7 & 5 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 8 & 7 & 0 & 3 & 3 \\ 7 & 6 & 5 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 15 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 15 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

6.112 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 11 & 17 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 6 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 12 & 17 & 25 & 0 & 0 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 10 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 1 & 5 \\ 9 & 13 & 18 & 0 & 0 & 7 \\ 11 & 18 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 7 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 & -1 \\ 2 & 2 & 2 & -1 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -201, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 10 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{11} 4 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 117, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 6 & 12 & 1 & 6 & 5 \\ 6 & 5 & 13 & 0 & 7 & 5 \\ 12 & 13 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 12 & 0 & 4 & 4 \\ 5 & 5 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 237, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 6 & 9 & 1 & 6 & 5 \\ 6 & 5 & 11 & 0 & 7 & 5 \\ 9 & 11 & 13 & 0 & 10 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 10 & 0 & 4 & 4 \\ 5 & 5 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 6 & 5 \\ 9 & 13 & 15 & 0 & 9 & 7 \\ 9 & 15 & 13 & 0 & 10 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 10 & 0 & 4 & 4 \\ 5 & 7 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

6.113 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{111} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 1 & 6 & 6 \\ 12 & 13 & 18 & 0 & 7 & 7 \\ 15 & 18 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 7 & 8 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 13 & 6 & 6 \\ 11 & 10 & 17 & 12 & 6 & 6 \\ 16 & 17 & 25 & 20 & 9 & 9 \\ 13 & 12 & 20 & 13 & 7 & 7 \\ 6 & 6 & 9 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{114} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 23 & 6 & 6 \\ 11 & 10 & 17 & 24 & 6 & 6 \\ 16 & 17 & 25 & 36 & 9 & 9 \\ 23 & 24 & 36 & 49 & 13 & 13 \\ 6 & 6 & 9 & 13 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 13 & 7 \\ 6 & 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 1 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 2 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 11 & 6 \\ 12 & 13 & 18 & 12 & 7 \\ 15 & 18 & 20 & 16 & 8 \\ 11 & 12 & 16 & 10 & 6 \\ 6 & 7 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 40, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 13 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 13 & 15 & 17 & 13 & 7 \\ 6 & 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 40, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 40, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 41, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 42, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 62 & 6 \\ 11 & 10 & 17 & 60 & 6 \\ 16 & 17 & 25 & 96 & 9 \\ 62 & 60 & 96 & 346 & 34 \\ 6 & 6 & 9 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 42, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 36 & 6 \\ 11 & 10 & 17 & 36 & 6 \\ 16 & 17 & 25 & 56 & 9 \\ 36 & 36 & 56 & 118 & 20 \\ 6 & 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{11}1} 3 1^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 49 & 6 \\ 11 & 10 & 17 & 48 & 6 \\ 16 & 17 & 25 & 76 & 9 \\ 49 & 48 & 76 & 217 & 27 \\ 6 & 6 & 9 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{11}1} 3 1^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 5 & 0 \\ 4 & 5 & 7 & 0 & 5 & 24 & 8 \\ 2 & 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 3 & 2 \\ 2 & 1 & 3 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 2 & 0 \\ 2 & 2 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 3 & 2 \\ 2 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overline{11}1} 3 1^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 55, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 19 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 21^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 12 & 20 & 15 & 7 \\ 12 & 10 & 17 & 12 & 6 \\ 20 & 17 & 25 & 20 & 9 \\ 15 & 12 & 20 & 13 & 7 \\ 7 & 6 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 3 & 2 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{113} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & -2 \\ 8 & 9 & 12 & 0 & 25 & 9 & 9 \\ 3 & 3 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & -2 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{114} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 20 & 3 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 & 3 \\ 3 & 5 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & -2 \\ 20 & 24 & 36 & 0 & 238 & 28 & 28 \\ 3 & 3 & 4 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & -2 \\ 5 & 6 & 9 & 0 & 13 & 7 & 7 \\ 3 & 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{113} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 3 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 & 3 \\ 3 & 5 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & -2 \\ 15 & 18 & 27 & 0 & 133 & 21 & 21 \\ 3 & 3 & 4 & -2 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 12 & 15 & 35 & 6 & 6 \\ 12 & 13 & 18 & 42 & 7 & 7 \\ 15 & 18 & 20 & 50 & 8 & 8 \\ 35 & 42 & 50 & 118 & 20 & 20 \\ 6 & 7 & 8 & 20 & 3 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 4 & 0 \\ 6 & 4 & 5 & 0 & 4 & 12 & 6 \\ 2 & 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{112} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & -2 \\ 10 & 12 & 18 & 0 & 58 & 14 & 14 \\ 3 & 3 & 4 & -2 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & -2 \\ 3 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 3 & 2 \\ 1 & 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[38] \quad 21^{111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 12 & 20 & 27 & 7 \\ 12 & 10 & 17 & 24 & 6 \\ 20 & 17 & 25 & 36 & 9 \\ 27 & 24 & 36 & 49 & 13 \\ 7 & 6 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 2 & 3 & 3 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{114} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 61 & 6 \\ 12 & 13 & 18 & 72 & 7 \\ 15 & 18 & 20 & 84 & 8 \\ 61 & 72 & 84 & 346 & 34 \\ 6 & 7 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 37 & 6 \\ 12 & 13 & 18 & 42 & 7 \\ 15 & 18 & 20 & 50 & 8 \\ 37 & 42 & 50 & 118 & 20 \\ 6 & 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 8 & 0 & 10 & 6 \\ 3 & 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 24 & 6 \\ 12 & 13 & 18 & 27 & 7 \\ 15 & 18 & 20 & 33 & 8 \\ 24 & 27 & 33 & 49 & 13 \\ 6 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 6 & 0 \\ 6 & 6 & 9 & 0 & 6 & 40 & 10 \\ 2 & 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{111} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 4 & 0 \\ 3 & 4 & 5 & 0 & 4 & 12 & 6 \\ 2 & 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 23 & 6 \\ 12 & 13 & 18 & 27 & 7 \\ 15 & 18 & 20 & 33 & 8 \\ 23 & 27 & 33 & 49 & 13 \\ 6 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{111} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 4 & 3 & 3 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 15 & 17 & 15 & 7 \\ 15 & 13 & 18 & 15 & 7 \\ 17 & 18 & 20 & 17 & 8 \\ 15 & 15 & 17 & 13 & 7 \\ 7 & 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 48 & 6 \\ 12 & 13 & 18 & 57 & 7 \\ 15 & 18 & 20 & 67 & 8 \\ 48 & 57 & 67 & 217 & 27 \\ 6 & 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{111} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 6 & 0 \\ 8 & 6 & 9 & 0 & 6 & 40 & 10 \\ 2 & 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 9 & 13 & 0 & 28 & 10 \\ 3 & 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 5 & 0 \\ 7 & 5 & 7 & 0 & 5 & 24 & 8 \\ 2 & 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 124, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 3 & 0 & 6 & 4 & 4 \\ 2 & 2 & 2 & -1 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 136, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 49 & 6 \\ 12 & 13 & 18 & 57 & 7 \\ 15 & 18 & 20 & 67 & 8 \\ 49 & 57 & 67 & 217 & 27 \\ 6 & 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 3 & 3 & 3 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 13 & 0 & 28 & 10 \\ 3 & 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 145, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{111} 3^1 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 146, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 6 & 0 \\ 7 & 6 & 9 & 0 & 6 & 40 & 10 \\ 2 & 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 146, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 36 & 6 \\ 12 & 13 & 18 & 42 & 7 \\ 15 & 18 & 20 & 50 & 8 \\ 36 & 42 & 50 & 118 & 20 \\ 6 & 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 159, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 5 & 0 \\ 5 & 5 & 7 & 0 & 5 & 24 & 8 \\ 2 & 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{111} 3 \ 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 4 \\ 5 & 4 & 5 & 0 & 4 & 12 \\ 2 & 2 & 2 & -1 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{114} 3 \ 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 16 & 0 & 46 & 12 \\ 3 & 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 2 & 3 & 3 & 0 & 2 & 6 \\ 3 & 3 & 3 & 0 & 6 & 4 \\ 2 & 2 & 2 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{112} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{111} 3 \ 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 4 \\ 4 & 4 & 5 & 0 & 4 & 12 \\ 2 & 2 & 2 & -1 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3 \ 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 \\ 1 & 2 & 0 & 1 & 1 & 7 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 5 \\ 6 & 5 & 7 & 0 & 5 & 24 \\ 2 & 2 & 2 & -1 & 0 & 8 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{113} 3 \ 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 15 & 22 & 0 & 85 & 17 \\ 3 & 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112} 3 \ 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{112} 3 \ 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

- [76] $1^{113} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 19 & 0 & 73 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [77] $1^{112} 3 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [78] $21^{111} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 194$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 12 & 16 & 24 & 6 \\ 12 & 13 & 18 & 27 & 7 \\ 16 & 18 & 20 & 33 & 8 \\ 24 & 27 & 33 & 49 & 13 \\ 6 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$
- [79] $1^{112} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 204$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [80] $1^{113} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [81] $1^{113} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 209$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 22 & 0 & 85 & 17 \\ 3 & 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [82] $1^{112} 3 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 220$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [83] $21^{123} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 230$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 13 & 12 & 20 & 42 & 7 \\ 12 & 10 & 17 & 36 & 6 \\ 20 & 17 & 25 & 56 & 9 \\ 42 & 36 & 56 & 118 & 20 \\ 7 & 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$
- [85] $1^{113} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 234$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [84] $1^{114} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 234$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [86] $21^{114} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 13 & 12 & 20 & 72 & 7 \\ 12 & 10 & 17 & 60 & 6 \\ 20 & 17 & 25 & 96 & 9 \\ 72 & 60 & 96 & 346 & 34 \\ 7 & 6 & 9 & 34 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{112}3^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[88] $21^{111}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 27 & 7 \\ 15 & 13 & 18 & 27 & 7 \\ 17 & 18 & 20 & 33 & 8 \\ 27 & 27 & 33 & 49 & 13 \\ 7 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{114}3^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 18 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{112}3^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{112}3^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{113}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 19 & 0 & 73 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{114}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{114}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{114}3^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[96] $21^{113}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 12 & 20 & 57 & 7 \\ 12 & 10 & 17 & 48 & 6 \\ 20 & 17 & 25 & 76 & 9 \\ 57 & 48 & 76 & 217 & 27 \\ 7 & 6 & 9 & 27 & 3 \end{pmatrix}_{b_0}$$

- [97] $1^{114} 3 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$
- [98] $1^{112} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [99] $1^{113} 3 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [100] $1^{112} 3 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [101] $1^{114} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 346, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [102] $1^{113} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [103] $1^{114} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 355, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [104] $21^{113} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 57 & 7 \\ 15 & 13 & 18 & 57 & 7 \\ 17 & 18 & 20 & 67 & 8 \\ 57 & 57 & 67 & 217 & 27 \\ 7 & 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$
- [105] $1^{114} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 374, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [107] $1^{114} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 30 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [106] $1^{113} 3 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{114} 3^{112} 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

6.114 NHC(-2)(-2)(-3) + $\mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$

$T = 9$ [0] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 1 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 1 & 2 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 1 & 2 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 1 & 0 \\ 4 & 3 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 1 & 2 & 0 & 1 & 3 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 1 & 0 \\ 2 & 3 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 1 & 2 & 0 & 1 & 3 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 18 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[7] 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 1 & 0 \\ 3 & 3 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 1 & 2 & 0 & 1 & 3 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[9] 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[10] 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -556, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 27 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[11] 1^{113} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -565, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 36 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

6.115 NHC(-2)(-2)(-3) + so₈ + su₂ → (-3)

$$T = 8 \quad [0] 1^{113} 1^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 8 & 6 \\ 11 & 10 & 17 & 8 & 6 \\ 16 & 17 & 25 & 12 & 9 \\ 8 & 8 & 12 & 4 & 4 \\ 6 & 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] 1^{113} 1^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] 1^{113} 1^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 8 & 6 \\ 12 & 13 & 18 & 9 & 7 \\ 15 & 18 & 20 & 11 & 8 \\ 8 & 9 & 11 & 4 & 4 \\ 6 & 7 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 5 & 8 & 6 & 1 & 5 \\ 8 & 10 & 9 & 0 & 6 \\ 6 & 9 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 6 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 21^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 85, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 12 & 20 & 9 & 7 \\ 12 & 10 & 17 & 8 & 6 \\ 20 & 17 & 25 & 12 & 9 \\ 9 & 8 & 12 & 4 & 4 \\ 7 & 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 5 & 5 & 0 & 6 & 4 \\ 3 & 3 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 21^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 114, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 16 & 8 & 6 \\ 12 & 13 & 18 & 9 & 7 \\ 16 & 18 & 20 & 11 & 8 \\ 8 & 9 & 11 & 4 & 4 \\ 6 & 7 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 159, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 15 & 17 & 9 & 7 \\ 15 & 13 & 18 & 9 & 7 \\ 17 & 18 & 20 & 11 & 8 \\ 9 & 9 & 11 & 4 & 4 \\ 7 & 7 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 21^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 399, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 15 & 20 & 9 & 7 \\ 15 & 13 & 21 & 9 & 7 \\ 20 & 21 & 25 & 12 & 9 \\ 9 & 9 & 12 & 4 & 4 \\ 7 & 7 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [10] \ 1^{11}3^{11}21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 10 & 5 & 0 & 4 & 6 \\ 2 & 5 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 6 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}3^{11}21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -21, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 & 3 \\ 3 & 4 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}3^{11}21^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -29, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 3 & 6 & 5 & 5 & 3 & 5 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

6.116 NHC(-2)(-2)(-3) + so₈ + su₂→(-3) + so₈

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -214, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 6 & 1 & 6 & 5 \\ 8 & 10 & 9 & 0 & 8 & 6 \\ 6 & 9 & 5 & 1 & 7 & 5 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 7 & 0 & 3 & 3 \\ 5 & 6 & 5 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -241, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

6.117 NHC(-2)(-2)(-3) + so₈ + su₂→(-3) + su₂→(-3)

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 2 & 3 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

6.118 NHC(-2)(-2)(-3) + so₈ + su₂→(-3) + su₂n3mix

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 2 & 8 & 3 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 8 & 12 & 0 & 7 & 25 & 9 \\ 3 & 3 & 4 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 2 & 14 & 3 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 14 & 21 & 0 & 12 & 80 & 16 \\ 3 & 3 & 4 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -89, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 4 & 6 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & 4 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 2 & 5 & 3 \\ 3 & 5 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 5 & 7 & 0 & 4 & 7 & 5 \\ 3 & 3 & 4 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 2 & 11 & 3 \\ 3 & 5 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 11 & 16 & 0 & 9 & 44 & 12 \\ 3 & 3 & 4 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -322, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 2 & 17 & 3 \\ 3 & 5 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 17 & 25 & 0 & 14 & 111 & 19 \\ 3 & 3 & 4 & -2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -364, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 2 & 14 & 3 \\ 3 & 5 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 14 & 20 & 0 & 11 & 69 & 15 \\ 3 & 3 & 4 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 13 & 3 \\ 2 & 1 & 5 & 0 & 2 & 17 & 3 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 13 & 2 \\ 13 & 17 & 24 & 0 & 13 & 100 & 18 \\ 3 & 3 & 4 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}3^{11}1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -560, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 16 & 3 \\ 2 & 1 & 5 & 0 & 2 & 20 & 3 \\ 3 & 5 & 4 & 0 & 3 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 16 & 2 \\ 16 & 20 & 29 & 0 & 16 & 148 & 22 \\ 3 & 3 & 4 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

6.119 NHC(-2)(-2)(-3) + so₈ + su₂n3mix

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 3 & 3 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 5 & 7 & 0 & 7 & 5 \\ 3 & 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 3 & 3 & 1 & 6 & 4 \\ 3 & 1 & 3 & 0 & 5 & 3 \\ 3 & 3 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 5 & 5 & 0 & 6 & 4 \\ 4 & 3 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 21 & 6 \\ 11 & 10 & 17 & 20 & 6 \\ 16 & 17 & 25 & 32 & 9 \\ 21 & 20 & 32 & 37 & 11 \\ 6 & 6 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 4 & 3 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 12 & 9 & 1 & 7 \\ 12 & 10 & 9 & 0 & 6 \\ 9 & 9 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 6 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 41, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 21 & 6 \\ 12 & 13 & 18 & 24 & 7 \\ 15 & 18 & 20 & 28 & 8 \\ 21 & 24 & 28 & 37 & 11 \\ 6 & 7 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 20 & 6 \\ 12 & 13 & 18 & 24 & 7 \\ 15 & 18 & 20 & 28 & 8 \\ 20 & 24 & 28 & 37 & 11 \\ 6 & 7 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 2 & 0 \\ 3 & 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 20 & 30 & 0 & 165 & 23 \\ 3 & 3 & 4 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 8 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{114}3^11^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 22 & 3 \\ 2 & 1 & 5 & 0 & 26 & 3 \\ 3 & 5 & 4 & 0 & 39 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 26 & 39 & 0 & 280 & 30 \\ 3 & 3 & 4 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 14 & 21 & 0 & 80 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 2 & 0 \\ 2 & 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 20 & 0 & 69 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 11 & 16 & 0 & 44 & 12 \\ 3 & 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^11^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 20 & 0 & 69 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{111}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 7 & 0 & 7 & 5 \\ 3 & 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 3 & 5 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 8 & 11 & 0 & 20 & 8 \\ 3 & 3 & 4 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[20] $21^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 25 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 17 & 25 & 0 & 111 & 19 \\ 3 & 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 14 & 20 & 0 & 69 & 15 \\ 3 & 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 14 & 20 & 0 & 69 & 15 \\ 3 & 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11} 1^{11} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 19 & 3 \\ 2 & 1 & 5 & 0 & 23 & 3 \\ 3 & 5 & 4 & 0 & 34 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 23 & 34 & 0 & 208 & 26 \\ 3 & 3 & 4 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 17 & 24 & 0 & 100 & 18 \\ 3 & 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 1^{11} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 20 & 29 & 0 & 148 & 22 \\ 3 & 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 17 & 24 & 0 & 100 & 18 \\ 3 & 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 1^{11} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 20 & 29 & 0 & 148 & 22 \\ 3 & 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [38] $1^1 3^1 21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 6 & 0 & 6 & 4 \\ 3 & 6 & 5 & 0 & 8 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 8 & 6 \\ 3 & 4 & 5 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{14} 3^1 21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 10 & 5 & 0 & 44 & 6 \\ 2 & 5 & 1 & 1 & 22 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 18 & 44 & 22 & 0 & 192 & 26 \\ 3 & 6 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

- [39] $1^{12}3^{11}21^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 10 & 5 & 0 & 24 & 6 \\ 2 & 5 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 24 & 12 & 0 & 56 & 14 \\ 3 & 6 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$
- [41] $1^{11}3^{11}21^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 10 & 5 & 0 & 20 & 6 \\ 2 & 5 & 1 & 1 & 9 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 20 & 9 & 0 & 35 & 11 \\ 3 & 6 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$
- [42] $1^{11}3^{11}21^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 2 & 6 & 0 & 17 & 4 \\ 3 & 6 & 5 & 0 & 23 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 23 & 0 & 78 & 18 \\ 3 & 4 & 5 & -2 & 18 & 3 \end{pmatrix}_{b_0}$
- [43] $1^{12}3^{11}21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 10 & 5 & 0 & 36 & 6 \\ 2 & 5 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 16 & 36 & 16 & 0 & 116 & 20 \\ 3 & 6 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$
- [44] $1^{11}3^{11}21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 17 & 3 \\ 4 & 10 & 5 & 0 & 40 & 6 \\ 2 & 5 & 1 & 1 & 19 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 17 & 40 & 19 & 0 & 151 & 23 \\ 3 & 6 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$
- [45] $1^{13}3^{11}21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -37$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 3 & 0 & 10 & 3 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 9 & 10 & 0 & 51 & 13 \\ 2 & 3 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [46] $1^{11}3^{11}21^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 5 & 3 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 4 & 5 & 0 & 10 & 6 \\ 2 & 3 & 2 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [47] $1^{12}3^{11}21^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 24 & 3 \\ 4 & 10 & 5 & 0 & 56 & 6 \\ 2 & 5 & 1 & 1 & 26 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 24 & 56 & 26 & 0 & 292 & 32 \\ 3 & 6 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$
- [48] $1^{12}3^{11}21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 0 & 9 & 3 \\ 1 & 2 & 0 & 2 & 0 & 7 & 2 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 7 & 9 & 0 & 35 & 11 \\ 2 & 3 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$
- [49] $1^{14}3^{11}21^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 0 & 14 & 3 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 12 & 14 & 0 & 96 & 18 \\ 2 & 3 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{11}1^{12}3^{12}1^{13}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 3 & 0 & 12 & 3 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 8 & 12 & 0 & 56 & 14 \\ 2 & 3 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{11}1^{13}3^{12}1^{13}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 0 & 13 & 3 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 10 & 13 & 0 & 74 & 16 \\ 2 & 3 & 2 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{11}1^{13}3^{12}1^{14}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -421$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 0 & 16 & 3 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 3 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 11 & 16 & 0 & 103 & 19 \\ 2 & 3 & 2 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{11}1^{14}3^{12}1^{14}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -429$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 3 & 0 & 17 & 3 \\ 1 & 2 & 0 & 2 & 0 & 13 & 2 \\ 1 & 3 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 13 & 17 & 0 & 127 & 21 \\ 2 & 3 & 2 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

6.120 NHC(-2)(-2)(-3) + $\mathbf{so}_8$ + $\mathbf{su}_{2n3\text{mix}}$ + $\mathbf{so}_8$

$T = 9$ [0] $1^{11}1^{11}3^{11}1^{11}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 12 & 9 & 1 & 9 & 7 \\ 12 & 10 & 9 & 0 & 8 & 6 \\ 9 & 9 & 5 & 1 & 7 & 5 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 8 & 7 & 0 & 3 & 3 \\ 7 & 6 & 5 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

6.121 NHC(-2)(-2)(-3) + $\mathbf{so}_8$ + $\mathbf{su}_{2n3\text{mix}}$ + $\mathbf{su}_2 \rightarrow (-3)$

$T = 9$ [1] $1^{11}1^{12}3^{11}1^{12}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 2 & 3 \\ 2 & 1 & 5 & 0 & 14 & 2 & 3 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 12 & 14 & 21 & 0 & 80 & 12 & 16 \\ 2 & 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[0] $1^{11}1^{11}3^{11}1^{11}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 2 & 3 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 8 & 12 & 0 & 25 & 7 & 9 \\ 2 & 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 3 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11}1^{11}3^{11}1^{11}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 7 & 0 & 7 & 4 & 5 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 5 & 2 & 3 \\ 3 & 5 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 7 & 0 & 7 & 4 & 5 \\ 2 & 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112}3^{11}1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 11 & 2 & 3 \\ 3 & 5 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 11 & 16 & 0 & 44 & 9 & 12 \\ 2 & 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113}3^{11}1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 5 & 0 & 17 & 2 & 3 \\ 3 & 5 & 4 & 0 & 25 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 14 & 17 & 25 & 0 & 111 & 14 & 19 \\ 2 & 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 3 & 4 & -2 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{113}3^{11}1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 14 & 2 & 3 \\ 3 & 5 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 14 & 20 & 0 & 69 & 11 & 15 \\ 2 & 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 3 & 4 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{114}3^{11}1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 2 & 3 \\ 2 & 1 & 5 & 0 & 17 & 2 & 3 \\ 3 & 5 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 13 & 17 & 24 & 0 & 100 & 13 & 18 \\ 2 & 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 3 & 4 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{114}3^{11}1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 2 & 3 \\ 2 & 1 & 5 & 0 & 20 & 2 & 3 \\ 3 & 5 & 4 & 0 & 29 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 16 & 20 & 29 & 0 & 148 & 16 & 22 \\ 2 & 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 3 & 4 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

6.122 NHC(-2)(-2)(-3) + so₈ + su₂n₃mix + su₂n₃mix

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 6 & 4 \\ 3 & 6 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 7 & 0 & 7 & 9 & 5 \\ 4 & 6 & 7 & 0 & 9 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}3^{\overline{11}1^{\overline{11}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 4 & 3 \\ 2 & 1 & 5 & 0 & 8 & 5 & 3 \\ 3 & 5 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 8 & 12 & 0 & 25 & 15 & 9 \\ 4 & 5 & 7 & 0 & 15 & 7 & 5 \\ 3 & 3 & 4 & -2 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1112}3^{\overline{11}1^{\overline{11}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -325, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 1 & 5 & 0 & 5 & 11 & 3 \\ 3 & 5 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 7 & 0 & 7 & 20 & 5 \\ 9 & 11 & 16 & 0 & 20 & 44 & 12 \\ 3 & 3 & 4 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

6.123 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 8 \quad [3] \ 1^{\overline{111}3^{\overline{11}1^{\overline{12}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 2 & 0 \\ 3 & 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 11 & 16 & 13 & 6 \\ 11 & 10 & 17 & 12 & 6 \\ 16 & 17 & 25 & 20 & 9 \\ 13 & 12 & 20 & 12 & 6 \\ 6 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{112}3^{\overline{11}1^{\overline{12}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 12 & 18 & 0 & 57 & 13 \\ 3 & 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 6 & 9 & 0 & 12 & 6 \\ 3 & 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 1 & 3 \\ 2 & 1 & 5 & 0 & 0 & 3 \\ 3 & 5 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 3 & 3 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 1 & 3 \\ 2 & 1 & 5 & 0 & 0 & 3 \\ 4 & 5 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 3 & 3 & 5 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

- [7] $21^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 12 & 16 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 16 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 5 & 0 & 3 & 3 \\ 3 & 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [9] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 41$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 12 & 15 & 1 & 6 \\ 12 & 13 & 18 & 0 & 7 \\ 15 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{12}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 16 & 0 & 27 & 9 \\ 3 & 3 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 11 & 15 & 12 & 6 \\ 11 & 10 & 17 & 12 & 6 \\ 15 & 17 & 20 & 17 & 8 \\ 12 & 12 & 17 & 12 & 6 \\ 6 & 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$
- [15] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 8 & 12 & 10 & 5 \\ 8 & 10 & 17 & 12 & 6 \\ 12 & 17 & 25 & 20 & 9 \\ 10 & 12 & 20 & 12 & 6 \\ 5 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$
- [18] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 11 & 15 & 1 & 6 \\ 11 & 10 & 17 & 0 & 6 \\ 15 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.28 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 15 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 12 & 14 & 1 & 7 \\ 12 & 10 & 13 & 0 & 6 \\ 14 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 68, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.35 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 70, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 1 & 2 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 12 & 1 & 7 \\ 14 & 13 & 13 & 0 & 7 \\ 12 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 0 & 2 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 15 & 13 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 13 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 3 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 8 & 1 & 5 \\ 9 & 13 & 13 & 0 & 7 \\ 8 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3 1^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0 \\ T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 9 & 1 & 5 \\ 8 & 10 & 13 & 0 & 6 \\ 9 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

- [30] $1^{111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [29] $1^{111}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 6 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [28] $1^{111}31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [32] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 89$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$
- [31] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 89$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 10 & 5 \\ 9 & 13 & 18 & 15 & 7 \\ 11 & 18 & 20 & 17 & 8 \\ 10 & 15 & 17 & 12 & 6 \\ 5 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [33] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{112}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 9 & 5 \\ 3 & 3 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [37] $1^{112}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 18 & 0 & 57 & 13 \\ 3 & 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 9 & 0 & 12 & 6 \\ 3 & 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [38] $21^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 109$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 12 & 20 & 15 & 7 \\ 12 & 10 & 17 & 12 & 6 \\ 20 & 17 & 25 & 20 & 9 \\ 15 & 12 & 20 & 12 & 6 \\ 7 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[41] $1^1 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 5 \\ 9 & 13 & 18 & 15 & 7 \\ 11 & 18 & 20 & 17 & 8 \\ 9 & 15 & 17 & 12 & 6 \\ 5 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 8 & 0 & 9 & 5 \\ 3 & 3 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 9 & 13 & 0 & 27 & 9 \\ 3 & 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[48] $21^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 16 & 12 & 6 \\ 12 & 10 & 17 & 12 & 6 \\ 16 & 17 & 20 & 17 & 8 \\ 12 & 12 & 17 & 12 & 6 \\ 6 & 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 13 & 0 & 27 & 9 \\ 3 & 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 12 & 0 & 24 & 8 \\ 3 & 3 & 4 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 21^{111} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 15 & 17 & 15 & 7 \\ 15 & 13 & 18 & 15 & 7 \\ 17 & 18 & 20 & 17 & 8 \\ 15 & 15 & 17 & 12 & 6 \\ 7 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[52] \ 21^{111} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 15 & 20 & 12 & 7 \\ 15 & 13 & 21 & 12 & 7 \\ 20 & 21 & 25 & 16 & 9 \\ 12 & 12 & 16 & 9 & 5 \\ 7 & 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 15 & 22 & 0 & 84 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 17 & 0 & 48 & 12 \\ 3 & 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 22 & 0 & 84 & 16 \\ 3 & 3 & 4 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 17 & 0 & 48 & 12 \\ 3 & 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [58] \ 1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 8 & 16 & 8 & 6 \\ 8 & 5 & 13 & 7 & 5 \\ 16 & 13 & 25 & 12 & 9 \\ 8 & 7 & 12 & 4 & 4 \\ 6 & 5 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[57] \ 21^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 8 & 16 & 8 & 6 \\ 8 & 5 & 13 & 7 & 5 \\ 16 & 13 & 25 & 12 & 9 \\ 8 & 7 & 12 & 4 & 4 \\ 6 & 5 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[60] \ 21^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 15 & 9 & 7 \\ 12 & 15 & 13 & 10 & 7 \\ 8 & 9 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 15 & 9 & 7 \\ 12 & 15 & 13 & 10 & 7 \\ 8 & 9 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 9 & 1 & 5 \\ 6 & 5 & 11 & 0 & 5 \\ 9 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 7 & 5 \\ 9 & 13 & 15 & 9 & 7 \\ 9 & 15 & 13 & 10 & 7 \\ 7 & 9 & 10 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[63] \ 21^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 9 & 20 & 9 & 7 \\ 9 & 5 & 13 & 7 & 5 \\ 20 & 13 & 25 & 12 & 9 \\ 9 & 7 & 12 & 4 & 4 \\ 7 & 5 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 4 & 0 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 4 & 0 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{111} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 9 & 7 & 5 \\ 6 & 5 & 11 & 7 & 5 \\ 9 & 11 & 13 & 10 & 7 \\ 7 & 7 & 10 & 4 & 4 \\ 5 & 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 4 & 0 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 7 & 0 & 7 & 5 \\ 3 & 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 21^{11} \overset{1}{4} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 10 & 7 & 5 \\ 9 & 13 & 15 & 9 & 7 \\ 10 & 15 & 13 & 10 & 7 \\ 7 & 9 & 10 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[70] $21^{111}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -151$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 15 & 14 & 9 & 7 \\ 15 & 13 & 15 & 9 & 7 \\ 14 & 15 & 13 & 10 & 7 \\ 9 & 7 & 10 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[71] $21^{111}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -391$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 7 & 10 & 7 & 5 \\ 7 & 5 & 11 & 7 & 5 \\ 10 & 11 & 13 & 10 & 7 \\ 7 & 7 & 10 & 4 & 4 \\ 5 & 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

6.124 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{e}_6$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -559$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.125 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{f}_4$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.126 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 11 & 17 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 6 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 12 & 17 & 25 & 0 & 0 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 8 & 12 & 10 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 12 & 17 & 25 & 20 & 0 & 9 \\ 10 & 12 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 1 & 5 \\ 9 & 13 & 18 & 0 & 0 & 7 \\ 11 & 18 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 7 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 10 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

6.127 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + (-3)→(-2)

$T = 8$ [0] $1^{11} 3 1^{11} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 12 & 20 & 1 & 7 \\ 12 & 10 & 17 & 0 & 6 \\ 20 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 12 & 16 & 1 & 6 \\ 12 & 13 & 18 & 0 & 7 \\ 16 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{11} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 15 & 17 & 1 & 7 \\ 15 & 13 & 18 & 0 & 7 \\ 17 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3 1^{11} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 15 & 20 & 1 & 7 \\ 15 & 13 & 21 & 0 & 7 \\ 20 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11} 4 \ 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 10 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 4 \ 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -313, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 10 & 1 & 5 \\ 7 & 5 & 11 & 0 & 5 \\ 10 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

6.128 NHC(-2)(-2)(-3) + su₂

$$T = 7 \quad [0] \ 1^{12} 3 \ 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 2 & 0 & 6 & 4 \\ 2 & 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 21^{11} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 18 & 4 \\ 12 & 12 & 18 & 58 & 14 \\ 3 & 3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.46 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 4 & 4 \\ 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

- [8] $21^{11}31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 5 & 12 & 4 \\ 3 & 1 & 5 & 9 & 3 \\ 5 & 5 & 4 & 13 & 4 \\ 12 & 9 & 13 & 28 & 10 \\ 4 & 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [9] $1^{12}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 4 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{11}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -33$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{13}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -41$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [16] $1^{13}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 7 & 5 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{12}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 6 & 6 & 0 & 22 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [17] $21^{12}31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 5 & 24 & 4 \\ 3 & 1 & 5 & 18 & 3 \\ 5 & 5 & 4 & 26 & 4 \\ 24 & 18 & 26 & 118 & 20 \\ 4 & 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$
- [18] $1^{12}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -55$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 5 & 4 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{12} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 7 & 8 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{14} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 6 & 0 & 40 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[21] $21^{11} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 6 & 4 \\ 6 & 4 & 6 & 6 & 4 \\ 5 & 6 & 4 & 5 & 4 \\ 6 & 6 & 5 & 4 & 4 \\ 4 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 4 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 5 & 4 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[24] $21^{12} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 4 & 6 & 4 & 13 & 4 \\ 9 & 12 & 13 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{13} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 7 & 5 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 4 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[26] $21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 4 & 9 & 4 \\ 6 & 6 & 9 & 13 & 7 \\ 3 & 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{14} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

- [30] $1^{13} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 7 & 5 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [29] $1^{11} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 5 & 7 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [31] $21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 4 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [32] $1^{13} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [33] $1^{14} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 9 & 6 & 0 & 40 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [34] $21^{12} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 21 & 4 \\ 6 & 4 & 6 & 21 & 4 \\ 5 & 6 & 4 & 22 & 4 \\ 21 & 21 & 22 & 85 & 17 \\ 4 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$
- [35] $1^{12} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 6 & 6 & 0 & 22 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^{11} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 5 & 7 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{14} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 8 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [38] $21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 9 & 4 \\ 6 & 4 & 6 & 9 & 4 \\ 5 & 6 & 4 & 9 & 4 \\ 9 & 9 & 9 & 13 & 7 \\ 4 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [39] $1^{13} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

- [40] $21^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 4 & 4 & 4 \\ 3 & 3 & 4 & 1 & 3 \\ 3 & 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$
- [41] $1^{13}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [43] $1^{13}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 6 & 6 & 0 & 22 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [44] $1^{14}31^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^{12}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 6 & 6 & 0 & 22 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [45] $1^{12}31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 7 & 8 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [46] $21^{14}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 6 & 24 & 4 \\ 4 & 6 & 4 & 26 & 4 \\ 18 & 24 & 26 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$
- [47] $21^{112}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 4 & 8 & 4 \\ 6 & 6 & 8 & 10 & 6 \\ 3 & 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$
- [48] $1^{12}31^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 10 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [49] $1^{13}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

- [50] $1^{13} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -153$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [52] $1^{14} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -154$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 8 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [51] $1^{13} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -154$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [53] $21^{12} 31^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -156$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 6 & 18 & 4 \\ 4 & 6 & 4 & 18 & 4 \\ 12 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$
- [55] $1^{13} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -161$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [54] $1^{12} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -161$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 7 & 8 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [56] $21^{112} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -163$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 15 & 4 \\ 6 & 4 & 6 & 15 & 4 \\ 5 & 6 & 4 & 14 & 4 \\ 15 & 15 & 14 & 37 & 11 \\ 4 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$
- [58] $21^{114} 31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -167$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 1 & 5 & 21 & 3 \\ 4 & 5 & 4 & 31 & 4 \\ 21 & 21 & 31 & 172 & 24 \\ 3 & 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$
- [59] $1^{14} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -167$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [57] $21^{112} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -167$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 1 & 5 & 9 & 3 \\ 4 & 5 & 4 & 13 & 4 \\ 9 & 9 & 13 & 28 & 10 \\ 3 & 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [60] $21^{13} 31^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -172$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 6 & 21 & 4 \\ 4 & 6 & 4 & 22 & 4 \\ 15 & 21 & 22 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

- [61] $1^{12} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 7 & 8 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$
- [62] $21^{12} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 6 & 5 & 18 & 4 \\ 6 & 4 & 6 & 18 & 4 \\ 5 & 6 & 4 & 18 & 4 \\ 18 & 18 & 18 & 58 & 14 \\ 4 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$
- [63] $21^{13} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 1 & 5 & 9 & 3 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 9 & 12 & 25 & 9 \\ 3 & 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [64] $1^{14} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 10 & 8 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$
- [65] $21^{12} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 6 & 24 & 4 \\ 4 & 6 & 4 & 23 & 4 \\ 15 & 24 & 23 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [66] $21^{13} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 22 & 4 \\ 15 & 15 & 22 & 85 & 17 \\ 3 & 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$
- [67] $1^{13} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -193, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$
- [68] $1^{14} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -194, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 10 & 8 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$
- [69] $1^{12} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -194, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 8 & 10 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$
- [70] $1^{13} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$
- [71] $21^{13} 31^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 6 & 5 & 30 & 4 \\ 6 & 4 & 6 & 30 & 4 \\ 5 & 6 & 4 & 31 & 4 \\ 30 & 30 & 31 & 172 & 24 \\ 4 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

- [72] $1^{12} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -222, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 8 & 10 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [73] $1^{14} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$
- [74] $1^{14} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [75] $1^{13} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [76] $1^{13} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [77] $21^{14} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -254, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 16 & 4 \\ 12 & 12 & 16 & 46 & 12 \\ 3 & 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$
- [78] $1^{13} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [79] $21^{13} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -281, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 17 & 4 \\ 12 & 12 & 17 & 49 & 13 \\ 3 & 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$
- [80] $1^{14} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [81] $21^{13} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 6 & 27 & 4 \\ 4 & 6 & 4 & 27 & 4 \\ 18 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$
- [82] $1^{13} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -305, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{14} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -305, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 11 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[85] $21^{13} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 27 & 4 \\ 6 & 4 & 6 & 27 & 4 \\ 5 & 6 & 4 & 27 & 4 \\ 27 & 27 & 27 & 133 & 21 \\ 4 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[86] $21^{13} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 24 & 4 \\ 6 & 4 & 6 & 24 & 4 \\ 5 & 6 & 4 & 23 & 4 \\ 24 & 24 & 23 & 100 & 18 \\ 4 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{13} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 11 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[88] $21^{14} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 6 & 30 & 4 \\ 4 & 6 & 4 & 31 & 4 \\ 21 & 30 & 31 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[89] $21^{15} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 20 & 4 \\ 15 & 15 & 20 & 73 & 15 \\ 3 & 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[90] $21^{13} 31^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 6 & 33 & 4 \\ 4 & 6 & 4 & 32 & 4 \\ 21 & 33 & 32 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{13} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 11 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[92] $21^{14} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -386, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 1 & 5 & 18 & 3 \\ 4 & 5 & 4 & 26 & 4 \\ 18 & 18 & 26 & 118 & 20 \\ 3 & 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{14} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 12 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0} \\
[94] \quad & 21^{114}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 21 & 4 \\ 15 & 15 & 21 & 76 & 16 \\ 3 & 3 & 4 & 16 & 3 \end{pmatrix}_{b_0} \\
[95] \quad & 21^{114}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 6 & 5 & 30 & 4 \\ 6 & 4 & 6 & 30 & 4 \\ 5 & 6 & 4 & 28 & 4 \\ 30 & 30 & 28 & 154 & 22 \\ 4 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0} \\
[96] \quad & 21^{116}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -454, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 1 & 5 & 18 & 3 \\ 4 & 5 & 4 & 24 & 4 \\ 18 & 18 & 24 & 106 & 18 \\ 3 & 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[97] \quad & 21^{143}1^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 24 & 3 \\ 3 & 4 & 6 & 36 & 4 \\ 4 & 6 & 4 & 36 & 4 \\ 24 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[98] \quad & 21^{114}31^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 6 & 5 & 36 & 4 \\ 6 & 4 & 6 & 36 & 4 \\ 5 & 6 & 4 & 36 & 4 \\ 36 & 36 & 36 & 238 & 28 \\ 4 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[99] \quad & 21^{114}31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 6 & 5 & 33 & 4 \\ 6 & 4 & 6 & 33 & 4 \\ 5 & 6 & 4 & 32 & 4 \\ 33 & 33 & 32 & 193 & 25 \\ 4 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[100] \quad & 221^{111}3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -531, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 & 3 \\ 3 & 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[101] \quad & 21^{117}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 1 & 5 & 21 & 3 \\ 4 & 5 & 4 & 28 & 4 \\ 21 & 21 & 28 & 145 & 21 \\ 3 & 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
T = 8 \quad & [102] \quad 2 \overset{11}{3} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 2 & 3 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[103] \quad & 1^{11}31^131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0} \\
[104] \quad & 1^{11}31^131^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[105] \quad & \begin{smallmatrix} 1^{11} \\ 2 \ 3 \ 2 \\ 1^{111} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 4 & 2 & 5 & 3 \\ 2 & 4 & 1 & 3 & 5 & 3 \\ 2 & 2 & 3 & 0 & 5 & 2 \\ 4 & 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[106] \quad & \begin{smallmatrix} 1^{12} \\ 2 \ 3 \ 2 \\ 1^{112} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 10 & 3 \\ 2 & 4 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 0 & 10 & 2 \\ 8 & 10 & 10 & 10 & 38 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[107] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{111} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 & 3 \\ 2 & 4 & 1 & 3 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 1 & 3 \\ 2 & 3 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[108] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{112} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 4 & 2 & 6 & 3 \\ 2 & 4 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 6 & 4 & 10 & 6 \\ 2 & 3 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[109] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{113} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 4 & 2 & 9 & 3 \\ 2 & 4 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 9 & 6 & 25 & 9 \\ 2 & 3 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[110] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{114} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 12 & 3 \\ 2 & 4 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 12 & 8 & 46 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[111] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{115} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 15 & 3 \\ 2 & 4 & 1 & 3 & 15 & 3 \\ 2 & 2 & 3 & 0 & 10 & 2 \\ 10 & 15 & 15 & 10 & 73 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[112] \quad & \begin{smallmatrix} 1^{11} \\ 2 \ 3 \ 2 \\ 1^{112} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 4 & 2 & 8 & 3 \\ 2 & 4 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 0 & 7 & 2 \\ 6 & 8 & 8 & 7 & 21 & 9 \\ 2 & 3 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[113] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{116} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 4 & 2 & 18 & 3 \\ 2 & 4 & 1 & 3 & 18 & 3 \\ 2 & 2 & 3 & 0 & 12 & 2 \\ 12 & 18 & 18 & 12 & 106 & 18 \\ 2 & 3 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[114] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{117} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 4 & 2 & 21 & 3 \\ 2 & 4 & 1 & 3 & 21 & 3 \\ 2 & 2 & 3 & 0 & 14 & 2 \\ 14 & 21 & 21 & 14 & 145 & 21 \\ 2 & 3 & 3 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[115] \quad 2 \overset{1^{12}}{\underset{1^{113}}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 13 & 3 \\ 2 & 4 & 1 & 3 & 13 & 3 \\ 2 & 2 & 3 & 0 & 12 & 2 \\ 10 & 13 & 13 & 12 & 61 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[116] \quad 2 \overset{1^{11}}{\underset{1^{113}}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 11 & 3 \\ 2 & 4 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 0 & 9 & 2 \\ 8 & 11 & 11 & 9 & 40 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[117] \quad 2 \overset{1^{11}}{\underset{1^{114}}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 14 & 3 \\ 2 & 4 & 1 & 3 & 14 & 3 \\ 2 & 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 14 & 11 & 65 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[118] \quad 2 \overset{1^{12}}{\underset{1^{114}}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 4 & 2 & 16 & 3 \\ 2 & 4 & 1 & 3 & 16 & 3 \\ 2 & 2 & 3 & 0 & 14 & 2 \\ 12 & 16 & 16 & 14 & 90 & 18 \\ 2 & 3 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [120] \quad 1 \overset{1^{11}}{\underset{1^{12}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 3 & -1 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[124] \quad 1 \overset{1^1}{\underset{1^{11}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[123] \quad 1 \overset{1^{12}}{\underset{1^{13}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 6 & 2 \\ 6 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[122] \quad 1 \overset{1^1}{\underset{1^{11}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[119] \quad 1 \overset{1^1}{\underset{1^{11}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 2 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[121] \quad & 1^{11} 2^1 3^1 21^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[127] \quad & 1^{12} 2^1 3^1 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0} \\
[128] \quad & 1^{12} 2^1 3^1 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[129] \quad & 1^{13} 2^1 3^1 21^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0} \\
[130] \quad & 1^{14} 2^1 3^1 21^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0} \\
[131] \quad & 1^{11} 2^1 3^1 21^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{13} 2^1 3^1 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 9 & 2 \\ 6 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{11} 2^1 3^1 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0} \\
[132] \quad & 1^{13} 2^1 3^1 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 1 & 9 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 3 & -1 & 15 & 4 \end{pmatrix}_{b_0} \\
[133] \quad & 1^{14} 2^1 3^1 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 10 & 2 \\ 8 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[134] \ 1^{\overline{11}} 2^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{11} 2^3 3^{21} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{14} 2^3 3^{21} 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 1 & 1 & 13 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 3 & -1 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{12} 2^3 3^{21} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 5 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[138] $1^{12}2^321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 4 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$

$$[139] \quad 1^{13} 2^3 3^{21} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 6 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{12} 2^3 3^{12} 21^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -42, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 6 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{12}2^33^{21}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -42, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 5 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{12}2^321^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 7 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{13} 2^3 3^{21} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 10 & 3 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 6 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{13} 2^3 3^{21} 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 8 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 7 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 9 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 9 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 14 & 3 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 8 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_6}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 8 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix} b_i$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 9 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix}_b$$

$$[1^{14} 2^3 3^{21} 21^{13} + (-2)^{\text{min}} = 10, T_{\max} = 10,$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 10 & 0 & 12 & 12 & 70 & 18 \end{pmatrix}$$

$$\left(\begin{array}{cccccc} 10 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{array} \right)_{b_0}$$

$$\left(\begin{array}{cccccc} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 16 & 3 \\ 1 & 0 & 2 & 0 & 16 & 2 \end{array} \right)$$

$$\begin{pmatrix} 1 & 0 & 2 & 0 & 16 & 2 \\ 11 & 0 & 16 & 16 & 126 & 24 \\ 2 & 0 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$T_{\text{min}} = 9.93 \rightarrow 10, \quad T_{\text{max}} = 10.07 \rightarrow 10$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 14 & 3 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 9 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10.09 \rightarrow 10,$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 10 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[155] $1^{14}2^3 3 21^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 14 & 3 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 10 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[156] $1^{14}2^3 3 21^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 16 & 3 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 12 & 0 & 16 & 16 & 126 & 24 \\ 2 & 0 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[157] $1^{14}2^3 3 21^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 11 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ [158] $231^{11}13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 566, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 5 & 2 \\ 5 & 33 & 36 & 0 & 35 & 11 \\ 5 & 36 & 33 & 1 & 35 & 11 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 35 & 35 & 0 & 33 & 11 \\ 2 & 11 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [163] $1^{11}231^5 5 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[162] $1^{13}231^5 5 1321^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$

[164] $1^{12}231^5 5 13221^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 2), c_{\text{ext}} = 5.5$

[161] $1^{14}231^5 5 1321^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$

[165] $1^{12}31^5 5 13221^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.5$

[166] $1^{13}31^5 5 13221^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[160] $1^{11}31^5 5 13221^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[159] $1^{11}31^5 5 13221^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[175] $1^{12}31^5 5 1321^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 1 & 10 & 3 \\ 6 & 0 & 8 & 10 & 34 & 14 \\ 2 & 0 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[170] $1^{13}31^5 5 1321^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 2 & 1 & 15 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 12 & 15 & 0 & 79 & 21 \\ 2 & 3 & 4 & -1 & 21 & 5 \end{pmatrix}_{b_0}$$

[171] $1^{14}31^5 5 1321^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 1 & 16 & 3 \\ 9 & 0 & 15 & 16 & 106 & 24 \\ 2 & 0 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0} \\
[172] & 1^{11} 31 5 1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 2 & 1 & 5 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0} \\
[173] & 1^{11} 31 5 1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 2 & 0 & 4 & 1 & 2 & 4 \\ 2 & 0 & 4 & 2 & 4 & 5 \end{pmatrix}_{b_0} \\
[174] & 1^{11} 31 5 1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 1 & 5 & 3 \\ 3 & 0 & 4 & 5 & 7 & 7 \\ 2 & 0 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0} \\
[169] & 1^{14} 31 5 1321^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 2 & 1 & 20 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 16 & 20 & 0 & 142 & 28 \\ 2 & 3 & 4 & -1 & 28 & 5 \end{pmatrix}_{b_0} \\
[168] & 1^{12} 31 5 1321^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 2 & 1 & 10 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 8 & 10 & 0 & 34 & 14 \\ 2 & 3 & 4 & -1 & 14 & 5 \end{pmatrix}_{b_0} \\
[167] & 1^{13} 31 5 1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 1 & 15 & 3 \\ 9 & 0 & 12 & 15 & 79 & 21 \\ 2 & 0 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0} \\
[176] & 1^{11} 31 5 1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 5 & 4 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 0 & 5 & 4 & 7 & 7 \\ 2 & 0 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0} \\
[177] & 1^{11} 31 5 1321^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 9 & 4 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 4 & 0 & 9 & 5 & 21 & 11 \\ 2 & 0 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0} \\
[178] & 1^{12} 31 5 1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 14 & 4 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 6 & 0 & 14 & 9 & 58 & 18 \\ 2 & 0 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0} \\
[180] & 1^{12} 31 5 1321^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 9 & 4 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 5 & 0 & 9 & 5 & 21 & 11 \\ 2 & 0 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[179] $1^{11}31^15^{12}1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 13 & 4 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 6 & 0 & 13 & 6 & 43 & 15 \\ 2 & 0 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[181] $1^{12}31^15^{12}1321^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 18 & 4 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 8 & 0 & 18 & 10 & 90 & 22 \\ 2 & 0 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[182] $1^{12}31^15^{12}1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 14 & 4 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 7 & 0 & 14 & 9 & 58 & 18 \\ 2 & 0 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[183] $1^{13}31^15^{12}1321^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 19 & 4 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 9 & 0 & 19 & 13 & 113 & 25 \\ 2 & 0 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

[184] $1^{13}31^15^{12}1321^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 18 & 4 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 9 & 0 & 18 & 10 & 90 & 22 \\ 2 & 0 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

6.129 NHC(-2)(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$T = 8$ [0] $1^{11}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 2 & 1 & 0 & 0 & -2 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 1 & 2 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 1 & -2 \\ 2 & 2 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 1 & -2 \\ 2 & 2 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

[13] $1^{111} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & 2 & -1 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 & 2 \\ 1 & 2 & 0 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 5 & 5 & 7 & 0 & 24 & 5 & 8 \\ 1 & 2 & 1 & 0 & 5 & -2 & 0 \\ 2 & 2 & 2 & -1 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 5 & 0 & 12 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & 2 & -1 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 4 & 6 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 & 2 \\ 2 & 2 & 2 & -1 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 4 & 4 & 5 & 0 & 12 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & 2 & -1 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 & 2 \\ 1 & 2 & 0 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 6 & 5 & 7 & 0 & 24 & 5 & 8 \\ 1 & 2 & 1 & 0 & 5 & -2 & 0 \\ 2 & 2 & 2 & -1 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

[19] $21^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 12 & 16 & 24 & 6 \\ 12 & 13 & 18 & 27 & 7 \\ 16 & 18 & 20 & 33 & 8 \\ 24 & 27 & 33 & 49 & 13 \\ 6 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[20] $21^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 27 & 7 \\ 15 & 13 & 18 & 27 & 7 \\ 17 & 18 & 20 & 33 & 8 \\ 27 & 27 & 33 & 49 & 13 \\ 7 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[21] $21^{112} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 42 & 7 \\ 15 & 13 & 18 & 42 & 7 \\ 17 & 18 & 20 & 50 & 8 \\ 42 & 42 & 50 & 118 & 20 \\ 7 & 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

6.130 NHC(-2)(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 9$ $[0] 1^{11} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 2 & 0 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 3 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 3 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 4 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 3 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 4 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 3 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 4 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

6.131 NHC(-2)(-2)(-3) + $\mathbf{su}_2$ + $\mathbf{su}_2$

$T = 9$ [0] $1^{11} 3^1 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

6.132 NHC(-2)(-2)(-3) + $\mathbf{su}_2$ + $\mathbf{su}_{2n3\text{mix}}$

$T = 12$ [0] $1^{11} 3^1 1^5 13221^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 2), c_{\text{ext}} = 6.0$

[1] $1^{11} 3^1 1^5 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$

[2] $1^{11} 3^1 1^5 1321^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$

[3] $1^{11} 3^1 1^5 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$

6.133 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{14}} 3^{\overline{12}} 1^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.6$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 10 & 2 \\ 9 & 11 & 10 & 0 & 10 & 71 & 13 \\ 2 & 2 & 2 & -1 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

6.134 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 3 & 0 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

6.135 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3$

$$T = 7 \quad [0] \ 21^{\overline{12}} 31^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 21 & 4 \\ 3 & 1 & 5 & 15 & 3 \\ 5 & 5 & 4 & 22 & 4 \\ 21 & 15 & 22 & 84 & 16 \\ 4 & 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

- [1] $1^{11} 1^3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 3 & 5 & 4 & 5 & 4 \\ 3 & 3 & 5 & 3 & 3 \\ 3 & 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [2] $1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 13 & 4 \\ 9 & 9 & 13 & 27 & 9 \\ 3 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [3] $1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [4] $1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 2 & 0 & 5 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$
- [5] $1^{11} 1^3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{111} 1^3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 7 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 4 & 5 & 9 & 16 & 5 \\ 7 & 9 & 16 & 27 & 9 \\ 3 & 3 & 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 8 & 4 \\ 6 & 6 & 8 & 9 & 5 \\ 3 & 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [8] $21^{13} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 6 & 15 & 4 \\ 4 & 6 & 4 & 17 & 4 \\ 12 & 15 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [9] $1^{111} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 9 & 12 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 3 & 6 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [11] $1^1 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$

- [14] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & -1 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{\overline{112}} 3^{\overline{1}} 1^{\overline{12}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 11 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 18 & 4 \\ 11 & 12 & 18 & 57 & 13 \\ 3 & 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 9 & 4 \\ 6 & 6 & 9 & 12 & 6 \\ 3 & 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [16] $21^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 3 & 8 & 3 & 4 \\ 3 & 1 & 5 & 3 & 3 \\ 8 & 5 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [17] $21^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 9 & 12 & 5 \\ 6 & 6 & 12 & 12 & 6 \\ 3 & 3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$
- [19] $21^{\overline{11}} 3^{\overline{1}} 1^{\overline{12}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -25$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 3 & 5 & 12 & 4 \\ 3 & 1 & 5 & 9 & 3 \\ 5 & 5 & 4 & 13 & 4 \\ 12 & 9 & 13 & 27 & 9 \\ 4 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [21] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 4 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{\overline{13}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 15 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 9 & 15 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$
- [23] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{14} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 10 & 8 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 2 & 4 & 0 & 5 & 3 \\ 2 & 2 & 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 9 & 8 & 5 \\ 4 & 3 & 8 & 3 & 3 \\ 3 & 3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 1 & 4 \\ 3 & 1 & 4 & 0 & 3 \\ 5 & 4 & 4 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 4 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{14} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 9 & 6 & 0 & 39 & 9 \\ 2 & 2 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 4 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 6 & 15 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 10 & 15 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 21^{11} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 15 & 4 \\ 3 & 1 & 5 & 12 & 3 \\ 5 & 5 & 4 & 17 & 4 \\ 15 & 12 & 17 & 48 & 12 \\ 4 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 6 & 6 & 4 \\ 3 & 6 & 4 & 5 & 4 \\ 4 & 6 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

- [34] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 3 & 5 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 4 & 5 & 9 & 12 & 5 \\ 6 & 6 & 12 & 12 & 6 \\ 3 & 3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{111}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 5 & 7 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$
- [36] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 3 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [38] $21^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$
- [40] $1^{12}131^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 18 & 4 \\ 10 & 12 & 18 & 57 & 13 \\ 3 & 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{11}131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 9 & 4 \\ 5 & 6 & 9 & 12 & 6 \\ 3 & 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [41] $21^131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 6 & 4 \\ 4 & 6 & 4 & 5 & 4 \\ 3 & 6 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$
- [44] $1^1131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 3 & 5 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{113}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 14 & 3 \\ 2 & 1 & 5 & 15 & 3 \\ 3 & 5 & 4 & 22 & 4 \\ 14 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^1131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 6 & 4 \\ 3 & 6 & 4 & 5 & 4 \\ 3 & 6 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

- [46] $1^1 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 5 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 4 & 5 & 9 & 12 & 5 \\ 5 & 6 & 12 & 12 & 6 \\ 3 & 3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$
- [45] $1^{11} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 5 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$
- [48] $1^1 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 4 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 8 & 4 \\ 4 & 6 & 8 & 9 & 5 \\ 3 & 3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$
- [47] $21^1 13^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 4 & 5 & 4 \\ 3 & 3 & 5 & 3 & 3 \\ 3 & 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [50] $1^{11} 13^1 3^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 12 & 4 \\ 8 & 9 & 12 & 24 & 8 \\ 3 & 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [49] $1^{11} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 3 & 3 & 0 & 3 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$
- [51] $1^{13} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 8 & 7 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$
- [52] $1^{11} 12^1 3^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 8 & 4 \\ 5 & 6 & 8 & 9 & 5 \\ 3 & 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [53] $21^1 13^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 6 & 5 & 3 & 4 \\ 6 & 4 & 6 & 3 & 4 \\ 5 & 6 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [57] $1^{13} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 7 & 5 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$
- [56] $21^{12} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 18 & 4 \\ 12 & 12 & 18 & 57 & 13 \\ 3 & 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$

- [55] $1^{12}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 6 & 6 & 0 & 21 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{11}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 3 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{111}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 8 & 12 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 7 & 12 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [60] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [61] $1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 13 & 4 \\ 8 & 9 & 13 & 27 & 9 \\ 3 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [62] $1^{11}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 5 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [63] $21^{111}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 5 & 12 & 4 \\ 6 & 4 & 6 & 12 & 4 \\ 5 & 6 & 4 & 13 & 4 \\ 12 & 12 & 13 & 27 & 9 \\ 4 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [64] $1^{12}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 5 & 4 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{11}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 13 & 4 \\ 7 & 9 & 13 & 27 & 9 \\ 3 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [68] $1^{12}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 7 & 8 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{14} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 6 & 0 & 39 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 12 & 4 \\ 7 & 9 & 12 & 24 & 8 \\ 3 & 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 11 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 17 & 4 \\ 11 & 12 & 17 & 48 & 12 \\ 3 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 5 & 4 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 4 & 5 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{12} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 10 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{13} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 7 & 5 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 4 & 5 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{14} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 11 & 10 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{11} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 6 & 9 & 0 & 39 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{11} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 5 & 7 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{13} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 9 & 9 & 0 & 51 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[80] $21^{12} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 4 & 6 & 4 & 13 & 4 \\ 9 & 12 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[79] $21^{11} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 8 & 6 & 4 \\ 3 & 1 & 5 & 3 & 3 \\ 8 & 5 & 9 & 8 & 5 \\ 6 & 3 & 8 & 3 & 3 \\ 4 & 3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 5 & 15 & 3 \\ 3 & 5 & 4 & 22 & 4 \\ 13 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

[82] $21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 3 & 8 & 9 & 4 \\ 3 & 1 & 5 & 6 & 3 \\ 8 & 5 & 9 & 12 & 5 \\ 9 & 6 & 12 & 12 & 6 \\ 4 & 3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[83] $21^{11} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 4 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{12} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 5 & 15 & 3 \\ 3 & 5 & 4 & 22 & 4 \\ 12 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

[85] $21^{11} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 5 & 6 & 4 \\ 6 & 4 & 6 & 6 & 4 \\ 5 & 6 & 4 & 5 & 4 \\ 6 & 6 & 5 & 3 & 3 \\ 4 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{11} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 17 & 4 \\ 9 & 12 & 17 & 48 & 12 \\ 3 & 3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{12} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 6 & 6 & 0 & 21 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 5 & 7 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 17 & 4 \\ 10 & 12 & 17 & 48 & 12 \\ 3 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{14} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 8 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{13} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 11 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{13} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 7 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[94] $21^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 4 & 9 & 4 \\ 6 & 6 & 9 & 12 & 6 \\ 3 & 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 6 & 9 & 0 & 39 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{12} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 6 & 6 & 0 & 21 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{14} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 12 & 12 & 0 & 93 & 15 \\ 2 & 2 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{12} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 7 & 8 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

- [98] $21^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 5 & 9 & 4 \\ 6 & 4 & 6 & 9 & 4 \\ 5 & 6 & 4 & 9 & 4 \\ 9 & 9 & 9 & 12 & 6 \\ 4 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [99] $21^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$
- [100] $1^{12}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 10 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{13}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 7 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [102] $21^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 4 \\ 6 & 4 & 9 & 3 & 4 \\ 8 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$
- [104] $1^{13}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 9 & 0 & 51 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{14}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 8 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{12}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 7 & 8 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{14}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 11 & 10 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$
- [107] $21^{112}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 4 & 8 & 4 \\ 6 & 6 & 8 & 9 & 5 \\ 3 & 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$
- [108] $1^{13}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 11 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{12} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 11 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[110] $21^{112} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 1 & 5 & 9 & 3 \\ 4 & 5 & 4 & 13 & 4 \\ 9 & 9 & 13 & 27 & 9 \\ 3 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{12} 3 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 9 & 9 & 0 & 51 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[112] $21^{113} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 1 & 5 & 9 & 3 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 9 & 12 & 24 & 8 \\ 3 & 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[113] $21^{113} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 22 & 4 \\ 15 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[114] $1^{14} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 12 & 12 & 0 & 93 & 15 \\ 2 & 2 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{14} 3 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 11 & 10 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{12} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 10 & 11 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{14} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 12 & 12 & 0 & 93 & 15 \\ 2 & 2 & 2 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[118] $21^{113} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 17 & 4 \\ 12 & 12 & 17 & 48 & 12 \\ 3 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 8 \quad [119] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 6 & 5 & 6 & 4 \\ 6 & 4 & 6 & 6 & 4 \\ 5 & 6 & 4 & 7 & 4 \\ 6 & 6 & 7 & 7 & 5 \\ 4 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
& [120] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 5 & 2 & 3 \\ 5 & 5 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [121] \, 1^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [122] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 4 & 3 & 3 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [125] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 6 & 6 & 4 \\ 4 & 6 & 4 & 7 & 4 \\ 5 & 6 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
& [123] \, 1^{12} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 6 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
& [124] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 3 & 5 & 6 & 4 \\ 3 & 1 & 4 & 5 & 3 \\ 5 & 4 & 4 & 7 & 4 \\ 6 & 5 & 7 & 7 & 5 \\ 4 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
& [126] \, 1^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -1 & 3 \end{pmatrix}_{b_0} \\
& [129] \, 1^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [128] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 4 & 2 & 3 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 2 & 4 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [127] \, 1^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 2 & 4 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[130] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[131] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[132] $221^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 6 & 4 \\ 5 & 7 & 8 & 11 & 6 \\ 5 & 6 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[133] $1^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 5 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[134] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[135] $221^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 6 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 5 & 6 & 12 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[136] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 5 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[137] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 6 & 5 & 3 & 4 \\ 6 & 4 & 6 & 3 & 4 \\ 5 & 6 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[138] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 3 \\ 2 & 1 & 4 & 5 & 3 \\ 4 & 4 & 4 & 7 & 4 \\ 5 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[139] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 1 & 5 & 2 & 3 \\ 4 & 5 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[140] $221^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 5 & 3 & 5 & 3 \\ 5 & 8 & 7 & 10 & 6 \\ 3 & 7 & 4 & 7 & 4 \\ 5 & 10 & 7 & 7 & 5 \\ 3 & 6 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[141] $31^{11}31^{11}3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 3 & 4 \\ 3 & 4 & 1 & 4 & 3 & 3 \\ 3 & 6 & 4 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[142] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 1 & 5 & 5 & 3 \\ 4 & 5 & 4 & 7 & 4 \\ 5 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[143] $31^{11}31^{11}3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 3 & 6 & 4 \\ 3 & 7 & 4 & 4 & 6 & 4 \\ 3 & 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 6 & 3 & 3 & 3 \\ 2 & 4 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[144] $221^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 5 & 6 & 5 & 3 \\ 5 & 8 & 13 & 10 & 6 \\ 6 & 13 & 13 & 12 & 7 \\ 5 & 10 & 12 & 7 & 5 \\ 3 & 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [145] $1^{11}51^{11}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 5 & 2 & 3 \\ 5 & 5 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[148] $21^{11}51^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[146] $21^{11}51^{11}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 1 & 5 & 3 \\ 4 & 4 & 5 & 7 & 5 \\ 3 & 3 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{11}51^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[150] $21^{11}51^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 4 & 2 & 3 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{11}51^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[151] $21^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[152] $21^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[153] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 1 & 5 & 2 & 3 \\ 4 & 5 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[154] $221^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 6 & 2 & 3 \\ 5 & 8 & 13 & 4 & 6 \\ 6 & 13 & 13 & 5 & 7 \\ 2 & 4 & 5 & 0 & 2 \\ 3 & 6 & 7 & 2 & 3 \end{pmatrix}_{b_0}$

$T = 10$ [155] $2^{\frac{1}{1}} 3^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 & 4 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[156] $21^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[157] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 2 & 4 & 4 & 4 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[158] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} 231^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 19 & 11 & 7 & 9 \\ 3 & 11 & 5 & 5 & 5 \\ 3 & 7 & 5 & 1 & 3 \\ 3 & 9 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[159] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 4 & 0 & 3 \\ 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$

[160] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 \\ 1 & 2 & 0 & 2 & 0 \\ 1 & 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$

[161] $1^{11} 5^{11} 131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 5 & 5 \\ 3 & 2 & 5 & 1 & 3 \\ 3 & 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[162] $1^{11} 5^{11} 1^{11} 231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 4 & 4 & 0 \\ 1 & 4 & 0 & 2 & 0 \\ 1 & 4 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[163] $2^{11} 3^{11} 151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[164] $2^{11} 3^{11} 151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 2 & 6 & 2 & 4 \\ 4 & 6 & 5 & 3 & 5 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 4 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[165] $2^{11} 3^{11} 151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 2 & 6 & 6 & 4 \\ 4 & 6 & 5 & 8 & 5 \\ 5 & 6 & 8 & 8 & 6 \\ 3 & 4 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[166] $2^{11} 3^{11} 151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 2 & 6 & 10 & 4 \\ 4 & 6 & 5 & 13 & 5 \\ 8 & 10 & 13 & 24 & 10 \\ 3 & 4 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

[167] $22^{11} 3^{11} 151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 5 & 3 \\ 5 & 9 & 14 & 11 & 7 \\ 6 & 14 & 14 & 13 & 8 \\ 5 & 11 & 13 & 8 & 6 \\ 3 & 7 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[168] $22^{11} 3^{11} 151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 371, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 5 & 5 & 3 \\ 5 & 9 & 12 & 11 & 7 \\ 5 & 12 & 9 & 12 & 7 \\ 5 & 11 & 12 & 8 & 6 \\ 3 & 7 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

[169] $22^{11} 3^{11} 151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 10 & 3 \\ 5 & 9 & 14 & 22 & 7 \\ 6 & 14 & 14 & 26 & 8 \\ 10 & 22 & 26 & 41 & 13 \\ 3 & 7 & 8 & 13 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [170] $1^{11} 41^{11} 321^{11} 51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 6 & 1 & 3 \\ 5 & 8 & 8 & 0 & 6 \\ 6 & 8 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 6 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

[171] $2^{11} 3^{11} 2161^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[172] $1^{11}2321^161^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -58$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 2 & 3 & 3 \\ 7 & 19 & 7 & 7 & 9 \\ 2 & 7 & 1 & 3 & 3 \\ 3 & 7 & 3 & 1 & 3 \\ 3 & 9 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[173] $41^{11}2315^11^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -134$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 8 & 9 & 1 & 3 \\ 8 & 42 & 49 & 0 & 14 \\ 9 & 49 & 51 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 14 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

6.136 NHC(-2)(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$

$T = 8$ [0] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 \\ 3 & 5 & 4 & 5 & 0 \\ 3 & 3 & 5 & 3 & 0 \\ 1 & 0 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 \\ 4 & 5 & 9 & 4 & 0 \\ 2 & 3 & 4 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

[1] $1^{12}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 4 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 4 & 5 & 9 & 16 & 0 & 5 \\ 7 & 9 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 5 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 15 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 166$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{112}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 10 & 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 190$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 8 & 0 & 4 \\ 4 & 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 6 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 3 & 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 4 & 5 & 9 & 12 & 0 & 5 \\ 5 & 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 5 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 13 & 0 & 4 \\ 7 & 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{113}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 442, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 1 & 3 \\ 2 & 1 & 5 & 15 & 0 & 3 \\ 3 & 5 & 4 & 22 & 0 & 4 \\ 12 & 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 445, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 17 & 0 & 4 \\ 9 & 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

6.137 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{e}_6 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 4 & 3 \\ 3 & 3 & 5 & 3 & 0 & 4 & 3 & 3 \\ 3 & 3 & 4 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 326, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.138 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 8 & 11 & 9 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 11 & 17 & 20 & 17 & 0 & 8 \\ 9 & 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 6 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -559, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.139 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -5, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 16.20 \rightarrow 17, \ T_{\max} = 23, \ T \in [17, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 & 3 \\ 3 & 5 & 4 & 13 & 0 & 5 & 4 \\ 7 & 9 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 3 & 4 & 9 & -4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 3 & 5 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -65, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 3 & 5 & 4 & 5 & 0 & 9 & 4 \\ 3 & 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 3 & 4 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 3 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -78, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.87 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 \\ 4 & 5 & 9 & 12 & 0 & 4 & 5 \\ 5 & 6 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 3 & 5 & 6 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1112}} 3 1^{\overline{12}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 10 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 3 & 5 & 4 & 4 & 0 & 18 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 18 & 12 & 0 & 58 & 14 \\ 3 & 3 & 4 & 2 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -138, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 4 & 3 \\ 2 & 1 & 5 & 6 & 0 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 8 & 4 \\ 5 & 6 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 6 & 8 & 12 & 0 & 10 & 6 \\ 3 & 3 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3 1^{\overline{113}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -143, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 11 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 4 & 5 & 9 & 4 & 0 & 28 & 5 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 12 & 28 & 12 & 0 & 73 & 15 \\ 3 & 3 & 5 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 6 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 5 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 12 & 6 & 0 & 13 & 7 \\ 3 & 3 & 5 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{112}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -257, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 \\ 3 & 5 & 4 & 5 & 0 & 13 & 4 \\ 3 & 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 12 & 0 & 28 & 10 \\ 3 & 3 & 4 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 \\ 3 & 5 & 4 & 5 & 0 & 21 & 4 \\ 3 & 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 15 & 21 & 18 & 0 & 76 & 16 \\ 3 & 3 & 4 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111}31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 8 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 \\ 4 & 5 & 9 & 4 & 0 & 20 & 5 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 9 & 20 & 9 & 0 & 37 & 11 \\ 3 & 3 & 5 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111}31^{121} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 6 & 12 & 0 & 3 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 9 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 3 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 19 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 3 & 4 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 3 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 3 & 5 & 4 & 5 & 0 & 17 & 4 \\ 3 & 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 15 & 0 & 49 & 13 \\ 3 & 3 & 4 & 3 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{112}31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 10 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 4 & 5 & 9 & 4 & 0 & 24 & 5 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 24 & 12 & 0 & 58 & 14 \\ 3 & 3 & 5 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 3 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 5 & 3 \\ 2 & 1 & 5 & 6 & 0 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 9 & 4 \\ 5 & 6 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 3 & 4 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 3 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1112} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 4 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 3 & 5 & 4 & 4 & 0 & 8 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 6 & 8 & 6 & 0 & 10 & 6 \\ 3 & 3 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1111} 31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 12 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 6 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 \\ 3 & 5 & 4 & 4 & 0 & 12 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 9 & 12 & 9 & 0 & 25 & 9 \\ 3 & 3 & 4 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1114} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 8 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 3 & 5 & 4 & 4 & 0 & 16 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 12 & 0 & 46 & 12 \\ 3 & 3 & 4 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -542, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 3 & 0 & 15 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 15 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{113} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -563, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 15 & 3 \\ 2 & 1 & 5 & 3 & 0 & 18 & 3 \\ 4 & 5 & 9 & 4 & 0 & 36 & 5 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 18 & 36 & 18 & 0 & 133 & 21 \\ 3 & 3 & 5 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{112} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 13 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 \\ 4 & 5 & 9 & 4 & 0 & 32 & 5 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 15 & 32 & 15 & 0 & 100 & 18 \\ 3 & 3 & 5 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{112} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -632, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 6 & 3 & 0 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 26 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 24 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{112} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -674, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 16 & 3 \\ 2 & 1 & 5 & 3 & 0 & 18 & 3 \\ 4 & 5 & 9 & 4 & 0 & 40 & 5 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 18 & 40 & 18 & 0 & 154 & 22 \\ 3 & 3 & 5 & 2 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 7 & 3 \\ 2 & 1 & 5 & 6 & 0 & 9 & 3 \\ 3 & 5 & 4 & 9 & 0 & 13 & 4 \\ 5 & 6 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 21 & 0 & 28 & 10 \\ 3 & 3 & 4 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -698, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 \\ 3 & 5 & 4 & 4 & 0 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 9 & 0 & 28 & 10 \\ 3 & 3 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{114} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -878, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 20 & 3 \\ 2 & 1 & 5 & 3 & 0 & 24 & 3 \\ 4 & 5 & 9 & 4 & 0 & 48 & 5 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 24 & 48 & 24 & 0 & 238 & 28 \\ 3 & 3 & 5 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1113} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -887, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 12 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 \\ 3 & 5 & 4 & 4 & 0 & 22 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 15 & 22 & 15 & 0 & 85 & 17 \\ 3 & 3 & 4 & 2 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -894, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 9 & 3 \\ 2 & 1 & 5 & 6 & 0 & 12 & 3 \\ 3 & 5 & 4 & 9 & 0 & 17 & 4 \\ 5 & 6 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 27 & 0 & 49 & 13 \\ 3 & 3 & 4 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{112} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 3 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 18 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1114} 31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -950, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 17 & 3 \\ 2 & 1 & 5 & 3 & 0 & 21 & 3 \\ 3 & 5 & 4 & 4 & 0 & 31 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 21 & 31 & 21 & 0 & 172 & 24 \\ 3 & 3 & 4 & 2 & -4 & 24 & -1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1114} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -972, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 11 & 3 \\ 2 & 1 & 5 & 6 & 0 & 15 & 3 \\ 3 & 5 & 4 & 9 & 0 & 21 & 4 \\ 5 & 6 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 15 & 21 & 33 & 0 & 76 & 16 \\ 3 & 3 & 4 & 6 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{112} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -974, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 6 & 3 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 22 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 21 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{113} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1043, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 18 & 3 \\ 2 & 1 & 5 & 3 & 0 & 21 & 3 \\ 4 & 5 & 9 & 4 & 0 & 44 & 5 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 21 & 44 & 21 & 0 & 193 & 25 \\ 3 & 3 & 5 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 9 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 3 & 5 & 4 & 4 & 0 & 17 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 12 & 0 & 49 & 13 \\ 3 & 3 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 6 & 3 & 0 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 28 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 30 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{113}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 6 & 3 & 0 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 23 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 24 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{114}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1478, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 11 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 \\ 3 & 5 & 4 & 4 & 0 & 21 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 15 & 21 & 15 & 0 & 76 & 16 \\ 3 & 3 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{114}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1652, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 14 & 3 \\ 2 & 1 & 5 & 3 & 0 & 18 & 3 \\ 3 & 5 & 4 & 4 & 0 & 26 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 18 & 26 & 18 & 0 & 118 & 20 \\ 3 & 3 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{113}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 6 & 3 & 0 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 27 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1847, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 6 & 3 & 0 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 31 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 30 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{114}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2162, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 6 & 3 & 0 & 33 & 4 \\ 3 & 6 & 4 & 4 & 0 & 32 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 33 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 6 & 3 & 0 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 36 & 4 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

6.140 NHC(-2)(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 31^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 31^{\overline{1114}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 18 & 3 \\ 14 & 24 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & 2 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 31^{\overline{1112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 253, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 12 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{113}} 31^{\overline{1112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 24 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 31^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 409, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1111}} 31^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 447, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 6 & 3 \\ 5 & 6 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 3 & 4 & 6 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1111}} 31^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 6 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 12 & 5 \\ 2 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 6 & 0 & 13 & 15 & 7 \\ 5 & 6 & 12 & 6 & 0 & 15 & 13 & 7 \\ 3 & 3 & 5 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1111}} 31^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 499, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{1111}} 31^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 516, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{\overline{112}} 31^{\overline{1112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 760, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{\overline{1111}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 772, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overline{112}} 31^{\overline{1113}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 820, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 21 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{114}} 31^{\overline{1113}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 928, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 33 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 33 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{1111}} 31^{\overline{1121}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 931, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 9 & 0 & 28 & 21 & 10 \\ 5 & 9 & 9 & 6 & 0 & 21 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 10 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overline{1111}} 31^{\overline{1122}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 964, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 9 & 0 & 28 & 30 & 10 \\ 7 & 12 & 13 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1345, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{1114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1621, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 21 & 3 \\ 17 & 30 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{1114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

6.141 NHC(-2)(-2)(-3) + $\mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1121} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 3 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^1 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 498, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 15 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 3 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 534, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 21 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 18 & 0 & 76 & 15 & 16 \\ 2 & 3 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 555, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1131}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 678, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 15 & 0 & 49 & 12 & 13 \\ 2 & 3 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 894, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

6.142 NHC(-2)(-2)(-3) + su₃ + e₆ + su₃

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.54 \rightarrow 15, \quad T_{\max} = 68, \quad T \in [15, 68]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 8 & 5 & 5 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 3 & 0 & 3 & 3 & 3 \\ 3 & 3 & 5 & 2 & -4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 & 2 \\ 3 & 3 & 4 & 3 & -4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -282, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 5 & 5 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 6 & 0 & 12 & 6 & 6 \\ 3 & 3 & 5 & 2 & -4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -312, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -357, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -507, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 0 & 2 \\ 3 & 3 & 4 & 6 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -573, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -651, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

6.143 NHC(-2)(-2)(-3) + $\mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 84, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 9 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 3 & 4 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 498, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 15 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 534, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 21 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 15 & 2 \\ 11 & 15 & 21 & 18 & 0 & 15 & 76 & 16 \\ 3 & 3 & 4 & 3 & -4 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 555, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 678, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 12 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 3 & 4 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 894, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

6.144 $\text{NHC}(-2)(-2)(-3) + \textcolor{brown}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 216, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 324, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

6.145 $\text{NHC}(-2)(-2)(-3) + \textcolor{brown}{su}_3 + \textcolor{red}{f}_4$

$$T = 8 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{12}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 4 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 4 & 5 & 9 & 16 & 0 & 5 \\ 7 & 9 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^3 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 15 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 8 & 0 & 4 \\ 4 & 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{112}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 10 & 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 6 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 3 & 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 4 & 5 & 9 & 12 & 0 & 5 \\ 5 & 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 215$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 13 & 0 & 4 \\ 7 & 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 343$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 17 & 0 & 4 \\ 9 & 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{113}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 343$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 1 & 3 \\ 2 & 1 & 5 & 15 & 0 & 3 \\ 3 & 5 & 4 & 22 & 0 & 4 \\ 12 & 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

6.146 NHC(-2)(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{f}_4$ + $\mathbf{so}_8$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 8 & 11 & 9 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 11 & 17 & 20 & 17 & 0 & 8 \\ 9 & 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 6 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.147 NHC(-2)(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{f}_4$ + $\mathbf{su}_2$

$T = 9$ [0] $1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 4 & 3 \\ 2 & 1 & 5 & 6 & 0 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 8 & 4 \\ 5 & 6 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 12 & 0 & 10 & 6 \\ 3 & 3 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -234, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 6 & 12 & 0 & 3 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -241, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 3 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 19 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -245, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 3 & 4 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -294, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 5 & 3 \\ 2 & 1 & 5 & 6 & 0 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 9 & 4 \\ 5 & 6 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 3 & 4 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -318, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -381, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -423, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

12.7

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 3 & 0 & 15 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}3^11^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 6 & 3 & 0 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 26 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 24 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 7 & 3 \\ 2 & 1 & 5 & 6 & 0 & 9 & 3 \\ 3 & 5 & 4 & 9 & 0 & 13 & 4 \\ 5 & 6 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 13 & 21 & 0 & 28 & 10 \\ 3 & 3 & 4 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -686, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 9 & 3 \\ 2 & 1 & 5 & 6 & 0 & 12 & 3 \\ 3 & 5 & 4 & 9 & 0 & 17 & 4 \\ 5 & 6 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 27 & 0 & 49 & 13 \\ 3 & 3 & 4 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{114}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 11 & 3 \\ 2 & 1 & 5 & 6 & 0 & 15 & 3 \\ 3 & 5 & 4 & 9 & 0 & 21 & 4 \\ 5 & 6 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 15 & 21 & 33 & 0 & 76 & 16 \\ 3 & 3 & 4 & 6 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 3 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 18 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^11^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 6 & 3 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 22 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 6 & 3 & 0 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 28 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -934, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 6 & 3 & 0 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 23 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 6 & 3 & 0 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 27 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 6 & 3 & 0 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 31 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{113}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 6 & 3 & 0 & 33 & 4 \\ 3 & 6 & 4 & 4 & 0 & 32 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 6 & 3 & 0 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 36 & 4 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.148} \quad \mathbf{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \ 1^{113}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 6 & 3 & 0 & 33 & 4 \\ 3 & 6 & 4 & 4 & 0 & 43 & 4 \\ 2 & 3 & 4 & 0 & 0 & 32 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 23 & 43 & 43 & 0 & 12 & 3 \\ 2 & 3 & 3 & 2 & 0 & 20 & 2 \\ 3 & 4 & 4 & 2 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 32 & 3 \\ 3 & 4 & 6 & 3 & 0 & 63 & 4 \\ 3 & 6 & 4 & 4 & 0 & 53 & 4 \\ 2 & 3 & 4 & 0 & 0 & 32 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 3 & 0 & 43 & 4 \\ 2 & 3 & 3 & 2 & 0 & 30 & 2 \\ 3 & 4 & 4 & 2 & -3 & 42 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 302, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 22 & 3 \\ 2 & 1 & 5 & 6 & 0 & 32 & 3 \\ 3 & 5 & 4 & 9 & 0 & 43 & 4 \\ 5 & 6 & 9 & 12 & 0 & 65 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 6 & 0 & 12 & 3 \\ 2 & 2 & 3 & 5 & 0 & 20 & 2 \\ 3 & 3 & 4 & 6 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 484, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 3 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 9 & 0 & 28 & 7 & 10 \\ 2 & 3 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

6.149 NHC(-2)(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{f}_4$ + $\mathbf{su}_2$ + $\mathbf{su}_2\mathbf{n}_3\mathbf{mix}$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 389, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 12 & 12 & 8 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

6.150 NHC(-2)(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -54, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.52 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 2 & 3 \\ 3 & 5 & 4 & 5 & 0 & 3 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 2 & 3 \\ 4 & 5 & 9 & 4 & 0 & 4 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 3 & 5 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}} 3^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 2 & 3 \\ 3 & 5 & 4 & 8 & 0 & 3 & 4 \\ 4 & 6 & 8 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 2 & 3 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -210, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 2 & 3 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 6 & 9 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 3 & 4 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 2 & 3 \\ 2 & 1 & 5 & 9 & 0 & 2 & 3 \\ 3 & 5 & 4 & 13 & 0 & 3 & 4 \\ 7 & 9 & 13 & 27 & 0 & 7 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 3 & 4 & 9 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

6.151 NHC(-2)(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 194, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 244, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 302, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 5 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 6 & 0 & 2 & 1 & 3 \\ 3 & 3 & 4 & 6 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 484, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 12 & 13 & 9 & 0 & 7 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

6.152 NHC(-2)(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 3 & 0 & 8 & 3 \\ 3 & 5 & 4 & 5 & 0 & 11 & 4 \\ 3 & 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 11 & 9 & 0 & 20 & 8 \\ 3 & 3 & 4 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 8 & 3 \\ 3 & 5 & 4 & 4 & 0 & 12 & 4 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 8 & 12 & 8 & 0 & 25 & 9 \\ 3 & 3 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 5 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 5 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 3 & 5 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 3 & 0 & 5 & 3 \\ 3 & 5 & 4 & 5 & 0 & 7 & 4 \\ 3 & 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 7 & 6 & 0 & 7 & 5 \\ 3 & 3 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 7 & 4 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 3 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 20 & 4 \\ 2 & 3 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 18 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 8 & 3 \\ 4 & 5 & 9 & 4 & 0 & 16 & 5 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 8 & 16 & 8 & 0 & 25 & 9 \\ 3 & 3 & 5 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -202, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 8 & 4 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -329, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 4 & 3 \\ 2 & 1 & 5 & 6 & 0 & 5 & 3 \\ 3 & 5 & 4 & 9 & 0 & 7 & 4 \\ 5 & 6 & 9 & 12 & 0 & 11 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 7 & 11 & 0 & 7 & 5 \\ 3 & 3 & 4 & 6 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -362, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 4 & 3 \\ 2 & 1 & 5 & 3 & 0 & 5 & 3 \\ 3 & 5 & 4 & 4 & 0 & 7 & 4 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 7 & 5 & 0 & 7 & 5 \\ 3 & 3 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}3^1}1^{\overline{112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -466, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 6 & 3 & 0 & 15 & 4 \\ 3 & 6 & 4 & 4 & 0 & 16 & 4 \\ 2 & 3 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}3^1}1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

6.153 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_{2n3\text{mix}} + \mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}3^1}1^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 389, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

6.154 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}3^1}1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -95, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 3 & 5 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 5 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 12 & 6 & 0 & 12 & 6 \\ 3 & 3 & 5 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -441, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

6.155 NHC(-2)(-2)(-3) + su₃ + NHC(-2)(-3)

$$T = 11 \quad [0] \ 1^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 2 & 1 & 3 \\ 2 & 1 & 4 & 2 & 3 & 0 & 3 \\ 3 & 4 & 4 & 3 & 3 & 0 & 4 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

6.156 NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 1 & 3 \\ 2 & 1 & 5 & 0 & 0 & 3 \\ 3 & 5 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 6 & 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 11 & 16 & 13 & 6 \\ 11 & 10 & 17 & 12 & 6 \\ 16 & 17 & 25 & 20 & 9 \\ 13 & 12 & 20 & 12 & 6 \\ 6 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 4 & 0 & 5 & 2 & 3 \\ 1 & 2 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 2 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} 3^{11} 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 11 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 11 & 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 1 & 3 \\ 2 & 1 & 5 & 0 & 0 & 3 \\ 4 & 5 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 16 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 16 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 15 & 1 & 6 \\ 12 & 13 & 18 & 0 & 7 \\ 15 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

- [10] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{12}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 4 & 5 & 9 & 16 & 0 & 5 \\ 7 & 9 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 11 & 15 & 12 & 6 \\ 11 & 10 & 17 & 12 & 6 \\ 15 & 17 & 20 & 17 & 8 \\ 12 & 12 & 17 & 12 & 6 \\ 6 & 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [15] $1^{11}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 8 & 12 & 10 & 5 \\ 8 & 10 & 17 & 12 & 6 \\ 12 & 17 & 25 & 20 & 9 \\ 10 & 12 & 20 & 12 & 6 \\ 5 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 1 & 0 & 0 & -2 \\ 2 & 2 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$
- [17] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 12 & 15 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 11 & 15 & 1 & 6 \\ 11 & 10 & 17 & 0 & 6 \\ 15 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 13 & 12 & 14 & 1 & 7 \\ 12 & 10 & 13 & 0 & 6 \\ 14 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{111}31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 68$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.35 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 15 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 70, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 1 & 2 & 0 & 0 & -2 \\ 2 & 2 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 12 & 1 & 7 \\ 14 & 13 & 13 & 0 & 7 \\ 12 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 15 & 13 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 13 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 8 & 1 & 5 \\ 9 & 13 & 13 & 0 & 7 \\ 8 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 9 & 1 & 5 \\ 8 & 10 & 13 & 0 & 6 \\ 9 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 4 & 5 & 9 & 12 & 0 & 5 \\ 6 & 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 6 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 4 & 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.10 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 6 & 15 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 10 & 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 10 & 5 \\ 9 & 13 & 18 & 15 & 7 \\ 11 & 18 & 20 & 17 & 8 \\ 10 & 15 & 17 & 12 & 6 \\ 5 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{112}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 8 & 0 & 4 \\ 4 & 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{112}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 10 & 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[38] $21^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 12 & 20 & 15 & 7 \\ 12 & 10 & 17 & 12 & 6 \\ 20 & 17 & 25 & 20 & 9 \\ 15 & 12 & 20 & 12 & 6 \\ 7 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 4 & 5 & 9 & 12 & 0 & 5 \\ 5 & 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 6 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 3 & 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

- [41] $1^{11} 1^3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 5 \\ 9 & 13 & 18 & 15 & 7 \\ 11 & 18 & 20 & 17 & 8 \\ 9 & 15 & 17 & 12 & 6 \\ 5 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^{112} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 8 & 0 & 4 \\ 5 & 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{112} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 13 & 0 & 4 \\ 8 & 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [44] $1^{111} 1^1 3^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^{111} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 8 & 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{111} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 6 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [47] $21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 16 & 12 & 6 \\ 12 & 10 & 17 & 12 & 6 \\ 16 & 17 & 20 & 17 & 8 \\ 12 & 12 & 17 & 12 & 6 \\ 6 & 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [48] $1^{111} 1^1 3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [49] $1^{112} 1^1 3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 13 & 0 & 4 \\ 7 & 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [50] $1^{113} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 7 & 9 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$
- [51] $21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 15 & 17 & 15 & 7 \\ 15 & 13 & 18 & 15 & 7 \\ 17 & 18 & 20 & 17 & 8 \\ 15 & 15 & 17 & 12 & 6 \\ 7 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[52] $21^{113}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 205$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 15 & 20 & 12 & 7 \\ 15 & 13 & 21 & 12 & 7 \\ 20 & 21 & 25 & 16 & 9 \\ 12 & 12 & 16 & 9 & 5 \\ 7 & 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 216$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 1 & 3 \\ 2 & 1 & 5 & 15 & 0 & 3 \\ 3 & 5 & 4 & 22 & 0 & 4 \\ 13 & 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{113}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 17 & 0 & 4 \\ 9 & 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{113}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 1 & 3 \\ 2 & 1 & 5 & 15 & 0 & 3 \\ 3 & 5 & 4 & 22 & 0 & 4 \\ 12 & 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{113}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 269$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 17 & 0 & 4 \\ 10 & 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [57] $1^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 4 & 3 & 0 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

6.157 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -559$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.158 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.159 NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 11 & 17 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 12 & 17 & 25 & 0 & 0 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 11 & 9 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 11 & 17 & 20 & 17 & 0 & 8 \\ 9 & 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 10 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 12 & 17 & 25 & 20 & 0 & 9 \\ 10 & 12 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 1 & 5 \\ 9 & 13 & 18 & 0 & 0 & 7 \\ 11 & 18 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 10 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 10 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

6.160 NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{14} 3 \ 1^{131} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.6$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 9 & 11 & 10 & 0 & 71 & 10 & 13 \\ 1 & 1 & 2 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & -1 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \ 2 \ 3 \ 1^{111} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 3 \\ 3 & 2 & 4 & 4 & 6 & 4 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 3 & 1 & 6 & 3 \\ 5 & 6 & 7 & 6 & 9 & 7 \\ 3 & 4 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 2 \ 3 \ 1^{112} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 10 & 3 \\ 3 & 2 & 4 & 4 & 12 & 4 \\ 3 & 4 & 4 & 3 & 14 & 4 \\ 3 & 4 & 3 & 1 & 12 & 3 \\ 10 & 12 & 14 & 12 & 42 & 14 \\ 3 & 4 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 5 \ 1^{1231} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 1 & 2 \\ 3 & 10 & 4 & 4 & 0 & 3 & 6 \\ 1 & 4 & 0 & 2 & 0 & 2 & 2 \\ 1 & 4 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 2 & 1 & 0 & 0 & 2 \\ 2 & 6 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 2 \ 3 \ 1^{113} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 15 & 3 \\ 3 & 2 & 4 & 4 & 18 & 4 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 3 & 1 & 18 & 3 \\ 15 & 18 & 21 & 18 & 97 & 21 \\ 3 & 4 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 2 \ 3 \ 1^{111} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 2 & 6 & 2 & 4 & 4 \\ 4 & 6 & 5 & 3 & 5 & 5 \\ 2 & 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 5 & 2 & 2 & 4 \\ 3 & 4 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[9] $2^{\overset{1}{1}\overset{112}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 2 & 6 & 2 & 8 & 4 \\ 4 & 6 & 5 & 3 & 10 & 5 \\ 2 & 2 & 3 & 0 & 4 & 2 \\ 6 & 8 & 10 & 4 & 14 & 8 \\ 3 & 4 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[10] $2^{\overset{1}{1}\overset{113}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 2 & 6 & 2 & 12 & 4 \\ 4 & 6 & 5 & 3 & 15 & 5 \\ 2 & 2 & 3 & 0 & 6 & 2 \\ 9 & 12 & 15 & 6 & 34 & 12 \\ 3 & 4 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[11] $2^{\overset{1}{1}\overset{111}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 2 & 3 & 4 & 6 & 4 \\ 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 3 & 1 & 6 & 3 \\ 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[12] $2^{\overset{1}{1}\overset{114}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 2 & 6 & 2 & 16 & 4 \\ 4 & 6 & 5 & 3 & 20 & 5 \\ 2 & 2 & 3 & 0 & 8 & 2 \\ 12 & 16 & 20 & 8 & 62 & 16 \\ 3 & 4 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[13] $2^{\overset{1}{1}\overset{115}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 15 & 3 \\ 3 & 2 & 6 & 2 & 20 & 4 \\ 4 & 6 & 5 & 3 & 25 & 5 \\ 2 & 2 & 3 & 0 & 10 & 2 \\ 15 & 20 & 25 & 10 & 98 & 20 \\ 3 & 4 & 5 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[14] $2^{\overset{1}{1}\overset{116}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 18 & 3 \\ 3 & 2 & 6 & 2 & 24 & 4 \\ 4 & 6 & 5 & 3 & 30 & 5 \\ 2 & 2 & 3 & 0 & 12 & 2 \\ 18 & 24 & 30 & 12 & 142 & 24 \\ 3 & 4 & 5 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[15] $2^{\overset{1}{1}\overset{1111}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -190, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 3 \\ 3 & 2 & 6 & 6 & 4 & 4 \\ 4 & 6 & 5 & 8 & 5 & 5 \\ 5 & 6 & 8 & 8 & 6 & 6 \\ 3 & 4 & 5 & 6 & 2 & 4 \\ 3 & 4 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[16] $2^{\overset{1}{1}\overset{117}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 21 & 3 \\ 3 & 2 & 6 & 2 & 28 & 4 \\ 4 & 6 & 5 & 3 & 35 & 5 \\ 2 & 2 & 3 & 0 & 14 & 2 \\ 21 & 28 & 35 & 14 & 194 & 28 \\ 3 & 4 & 5 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[17] $22^{\overset{1}{1}\overset{1111}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -696, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 5 & 5 & 3 \\ 5 & 9 & 14 & 11 & 11 & 7 \\ 6 & 14 & 14 & 13 & 13 & 8 \\ 5 & 11 & 13 & 8 & 11 & 6 \\ 5 & 11 & 13 & 11 & 9 & 7 \\ 3 & 7 & 8 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

6.161 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 3 & 0 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

6.162 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 2 & 3 \\ 2 & 2 & 4 & 4 & 3 & 4 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 4 & 0 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

6.163 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [4] \quad 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 2 & 4 & 3 \\ 2 & 2 & 4 & 4 & 3 & 6 & 4 \\ 3 & 4 & 4 & 3 & 3 & 7 & 4 \\ 3 & 4 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 3 & 1 & 5 & 3 \\ 4 & 6 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[5] \quad & 1^{12}_{1^1} 5^{11} 13^1 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 & 2 & 8 & 3 \\ 2 & 2 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 3 & 4 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 3 & 1 & 10 & 3 \\ 8 & 12 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{11}_{1^1} 5^{11} 13^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 4 & 0 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & 5 & -1 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{13}_{1^1} 5^{11} 13^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 4 & 0 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 29 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 18 & 3 \\ 14 & 17 & 29 & 0 & 18 & 137 & 25 \\ 3 & 3 & 5 & -1 & 3 & 25 & 4 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{13}_{1^1} 5^{11} 13^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 & 2 & 12 & 3 \\ 2 & 2 & 4 & 4 & 3 & 18 & 4 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 3 & 1 & 15 & 3 \\ 12 & 18 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{11}_{1^{11}} 5^{11} 13^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{12}_{1^1} 5^{11} 13^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 1 & 4 & 0 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 11 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 7 & 3 \\ 6 & 8 & 11 & 0 & 7 & 22 & 10 \\ 3 & 3 & 5 & -1 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{13}_{1^1} 5^{11} 13^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 4 & 0 & 2 & 13 & 3 \\ 3 & 4 & 5 & 0 & 4 & 19 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 13 & 19 & 0 & 12 & 65 & 17 \\ 3 & 3 & 5 & -1 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{12}_{1^{11}} 5^{11} 13^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 5 & 7 & 6 & 9 & 0 & 7 & 36 & 12 \\ 2 & 2 & 2 & 2 & -1 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 3 & 4 & 4 & 5 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 6 & 5 & 6 & 0 & 5 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 0 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 8 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 5 & 3 \\ 4 & 5 & 8 & 0 & 5 & 9 & 7 \\ 3 & 3 & 5 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 1 & 4 & 0 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 16 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 10 & 3 \\ 8 & 10 & 16 & 0 & 10 & 42 & 14 \\ 3 & 3 & 5 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{13} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 6 & 9 & 7 & 10 & 0 & 8 & 52 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 4 & 0 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 24 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 15 & 24 & 0 & 15 & 97 & 21 \\ 3 & 3 & 5 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 5 & 6 & 6 & 0 & 5 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[19] \begin{matrix} 1^1 \\ 1^{12} 5 13^1 1^{11} \end{matrix} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{teal}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 6 & 2 \\ 5 & 7 & 7 & 7 & 0 & 6 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[21] \begin{matrix} 1^1 \\ 1^3 5 \\ 1^{12} \end{matrix} 13^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 14 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 11 & 2 \\ 8 & 11 & 10 & 14 & 0 & 11 & 93 & 19 \\ 2 & 2 & 2 & 2 & -1 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12} \overset{1}{5} \overset{1}{13} 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{teal}{su}_2 \quad T = 12, \, T_{\max} = 12, \, \Delta = 0, \, \det = 86, \, (n_+, n_-, n_0) = (1, 12, 0), \, c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, \, T_{\max} = 12, \, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 6 & 8 & 8 & 10 & 0 & 8 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13} \overset{1}{5} \overset{1}{13} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 9 & 2 \\ 7 & 10 & 9 & 11 & 0 & 9 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1 \overset{1}{2} 5 \overset{1}{13} 1 \overset{12}{2} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 121, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 9 & 2 \\ 7 & 9 & 10 & 11 & 0 & 9 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{13} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 10 & 2 \\ 8 & 11 & 11 & 12 & 0 & 10 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad {}^{1\overline{1}3}_{{}_{1\overline{1}3}}{}^{\overline{1}1}\overline{1}{}^{1\overline{1}3} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{teal}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$
$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$
$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 12 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 12 & 2 \\ 9 & 12 & 12 & 15 & 0 & 12 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

6.164 NHC(-2)(-2)(-3) + su₃ + su₂n₃mix

$$T = 11 \quad [0] \quad 1^{12} \overset{1}{5} 13 \overset{1}{1} 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 10 & 3 \\ 2 & 2 & 4 & 4 & 15 & 4 \\ 3 & 4 & 4 & 3 & 17 & 4 \\ 3 & 4 & 3 & 1 & 15 & 3 \\ 10 & 15 & 17 & 15 & 65 & 17 \\ 3 & 4 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 6 & 3 \\ 2 & 2 & 4 & 4 & 9 & 4 \\ 3 & 4 & 4 & 3 & 10 & 4 \\ 3 & 4 & 3 & 1 & 9 & 3 \\ 6 & 9 & 10 & 9 & 22 & 10 \\ 3 & 4 & 4 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} \overset{1}{5} 13 \overset{1}{1} 1^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 14 & 3 \\ 2 & 2 & 4 & 4 & 21 & 4 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 3 & 1 & 21 & 3 \\ 14 & 21 & 24 & 21 & 130 & 24 \\ 3 & 4 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 4 & 3 \\ 2 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 5 & 5 & 7 & 5 \\ 3 & 2 & 5 & 1 & 5 & 3 \\ 4 & 5 & 7 & 5 & 8 & 6 \\ 3 & 3 & 5 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 13 \overset{1}{1} 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 8 & 3 \\ 2 & 1 & 4 & 2 & 10 & 3 \\ 3 & 4 & 5 & 5 & 15 & 5 \\ 3 & 2 & 5 & 1 & 11 & 3 \\ 8 & 10 & 15 & 11 & 39 & 13 \\ 3 & 3 & 5 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13} \overset{1}{5} 13 \overset{1}{1} 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 12 & 3 \\ 2 & 1 & 4 & 2 & 15 & 3 \\ 3 & 4 & 5 & 5 & 23 & 5 \\ 3 & 2 & 5 & 1 & 17 & 3 \\ 12 & 15 & 23 & 17 & 92 & 20 \\ 3 & 3 & 5 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 4 & 0 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 12 & 0 & 22 & 10 \\ 3 & 3 & 5 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 5 & 7 & 0 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{12} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 4 & 0 & 12 & 3 \\ 3 & 4 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 12 & 20 & 0 & 65 & 17 \\ 3 & 3 & 5 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[10] $1^{13} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 10 & 8 & 12 & 0 & 70 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{12} \overset{1}{5} 13^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 6 & 8 & 0 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{13} \overset{1}{5} 13^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 4 & 0 & 17 & 3 \\ 3 & 4 & 5 & 0 & 28 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 17 & 28 & 0 & 130 & 24 \\ 3 & 3 & 5 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[13] $1^{13} \overset{1}{5} 13^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 11 & 16 & 0 & 117 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{12} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 9 & 12 & 0 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{13} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -154, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 11 & 10 & 13 & 0 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{13} \overset{1}{5} 13^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -188, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 0 & 13 & 2 \\ 1 & 2 & 0 & 2 & 0 & 13 & 2 \\ 1 & 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 13 & 13 & 17 & 0 & 141 & 23 \\ 2 & 2 & 2 & 2 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{13} \overset{1}{5} 13^{12} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -242, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 12 & 14 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

6.165 NHC(-2)(-2)(-3) + su₃ + su₃

$T = 10 \quad [0] \quad 21^{11} \overset{1}{5} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 3 & 3 & 4 \\ 3 & 1 & 5 & 2 & 2 & 3 \\ 5 & 5 & 4 & 3 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 & 2 \\ 3 & 2 & 3 & 2 & 0 & 2 \\ 4 & 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} \overset{1}{5} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 4 & 2 & 0 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[2] $21^{11} \overset{1}{5} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 1 & 5 & 2 & 2 & 3 \\ 4 & 5 & 4 & 3 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 & 2 \\ 2 & 2 & 3 & 2 & 0 & 2 \\ 3 & 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

6.166 NHC(-2)(-2)(-3) + su₃ + (-3)→(-2)

$T = 10 \quad [0] \quad 1^{11} \overset{1}{5} 1^{12} 2^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 & 0 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{21} 21^{11} \overset{1}{5} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 2 & 1 & 3 \\ 5 & 8 & 13 & 4 & 0 & 6 \\ 6 & 13 & 13 & 5 & 0 & 7 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 7 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

6.167 NHC(-2)(-2)(-3) + su₃ + (-3)→(-2) + su₂

$T = 11 \quad [0] \quad 1^{11} \overset{1}{5} 1^{12} 2^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 1 & 4 & 2 & 0 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 1 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

6.168 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{red}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 7 & 3 & 4 \\ 3 & 5 & 0 & 7 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{red}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{red}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 1 & 4 & 3 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{red}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 1 & 4 & 3 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{red}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{red}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 7 & 0 & 4 \\ 3 & 8 & 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 2^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{red}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 3 & 3 & 9 \\ 27 & 33 & 1 & 4 & 2 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 4 & 2 & 0 & 13 \\ 9 & 11 & -1 & 1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{red}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 428, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 7 & 3 & 4 \\ 2 & 6 & 3 & 0 & 7 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [8] \quad 1^{11} 1^{11} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -41, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 3 & 3 & 0 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 1^{11} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 & 2 & 3 \\ 4 & 4 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 4 & 0 & 1 & 4 & 2 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 1^{11} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -215, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 & 2 & 3 \\ 4 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 4 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

6.169 $\text{NHC}(-2)(-3) + \textcolor{green}{e}_6 + \text{NHC}(-2)(-2)(-3) + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 4 \\ 3 & 4 & -4 & 3 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 1^{11} 3^{111} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 9 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 15 & 9 & 37 & 11 \\ 3 & 4 & -4 & 3 & 4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 1^{11} 3^{111} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & -4 & 3 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 6 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

6.170 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^1 1^{\overline{11}} + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + [(-2) \cdot \textcolor{red}{\mathfrak{s}}\mathfrak{p}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{s}}\mathfrak{u}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & -4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 4 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3 1^{111} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{red}{s} \textcolor{blue}{p}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s} \textcolor{blue}{u}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -92, \quad \det = 326, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 & 3 \\ 3 & 4 & -4 & 3 & 4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

6.171 $\mathbf{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathbf{NHC}(-2)(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{s}\textcolor{blue}{u}_2 + [(-2) \cdot \textcolor{blue}{s}\textcolor{blue}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{121} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{s}u_2 + [(-2) \cdot \textcolor{blue}{s}p_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -83, \quad \det = 150, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 15 & 9 & 11 \\ 2 & 4 & 0 & 9 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 15 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 9 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 11 & 3 & 4 & 2 & 0 \end{pmatrix} b_0$$

$$[2] \quad 1^{1111} 3 1^{11} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{s} \textcolor{blue}{u}_2 + [(-2) \cdot \textcolor{blue}{s} \textcolor{red}{p}_1 \cdot \textcolor{green}{g}_2] \quad T = 10, \, T_{\max} = 13, \, \Delta = -83, \, \det = 294, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 4 & 2 & 0 \end{pmatrix} b_0$$

$$[3] \quad 1^{1111} 3 1^{111} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{s}u_2 + [(-2) \cdot \textcolor{blue}{s}p_1 \cdot \textcolor{violet}{g}_2] \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -83, \quad \det = 721, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 9 & 6 & 7 \\ 2 & 4 & 0 & 6 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 7 & 3 & 4 & 2 & 0 \end{pmatrix} b_0$$

6.172 $\mathbf{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathbf{NHC}(-2)(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{teal}{1}\textcolor{brown}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{c}_6 + \textcolor{brown}{s}\textcolor{brown}{u}_3 + [(-2) \cdot \textcolor{red}{s}\textcolor{blue}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 266, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 3 & 4 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 326, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.173 NHC(-2)(-3) + $\textcolor{green}{e}_7$ + NHC(-2)(-2)(-3)

$$T = 10 \quad [0] \ 1^{11} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 6, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 16.33 \rightarrow 17, \ T_{\max} = 100, \ T \in [17, 100]$$

$$\begin{pmatrix} 14 & 13 & 1 & 7 & 8 \\ 13 & 12 & 1 & 7 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & -2 & 0 \\ 8 & 8 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 159, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 100, \ T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 4 & 8 & 0 & 4 & 1 & 9 & 5 \\ 3 & 4 & -1 & -2 & 0 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 334, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 5 \\ 7 & 8 & 1 & 5 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 13 & 7 \\ 5 & 6 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \ 1^{11} 3^{111} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -3, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.33 \rightarrow 19, \ T_{\max} = 24, \ T \in [19, 24]$$

$$\begin{pmatrix} 63 & 62 & 1 & 15 \\ 62 & 61 & 1 & 15 \\ 1 & 1 & -3 & -1 \\ 15 & 15 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -130, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 24, \ T \in [16, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 4 & 8 & 0 & 4 & 1 & 9 & 5 \\ 3 & 4 & -1 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -275, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.83 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 5 \\ 7 & 8 & 1 & 5 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 13 & 7 \\ 5 & 6 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1231} 1^{111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 48, \ \Delta = -119, \ \det = -525, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.05 \rightarrow 16, \ T_{\max} = 48, \ T \in [16, 48]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 7 & 0 & 4 \\ 2 & 6 & 0 & 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

6.174 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -62, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 1 & 5 \\ 7 & 8 & 1 & 5 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 6 & 0 & -1 \\ 9 & 12 & 0 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & -1 & 7 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -81, \ \det = 51, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 1 & 5 \\ 7 & 8 & 1 & 5 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 6 & 0 & 0 \\ 9 & 12 & 0 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

6.175 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$

$$T = 11 \quad [0] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -62, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 9 & 5 \\ 7 & 8 & 1 & 5 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 6 & 0 & 13 & 7 \\ 5 & 6 & -1 & -1 & 0 & 7 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 12, \ T_{\max} = 15, \ \Delta = -81, \ \det = 51, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 9 & 5 \\ 7 & 8 & 1 & 5 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 6 & 0 & 13 & 7 \\ 5 & 6 & -1 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

6.176 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{NHC}(-2)(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 17.33 \rightarrow 18, \ T_{\max} = 24, \ T \in [18, 24]$$

$$\begin{pmatrix} 63 & 62 & 1 & 15 \\ 62 & 61 & 1 & 15 \\ 1 & 1 & -3 & -1 \\ 15 & 15 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.20 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 4 & 8 & 0 & 4 & 1 & 9 & 5 \\ 3 & 4 & -1 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 275, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 5 \\ 7 & 8 & 1 & 5 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 13 & 7 \\ 5 & 6 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -216, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.98 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 21 & 27 & 1 & 33 & 9 \\ 27 & 33 & 1 & 42 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 42 & 0 & 49 & 13 \\ 9 & 11 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1231} 1^{111} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -428, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 7 & 0 & 4 \\ 2 & 6 & 0 & 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.177} \quad \text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \text{NHC}(-2)(-2)(-3) + \textcolor{red}{\mathfrak{su}}_2$$

$$T = 11 \quad [0] \quad 1^{111} 3 \quad 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 1 & 5 \\ 7 & 8 & 1 & 5 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 6 & 0 & 0 \\ 9 & 12 & 0 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.178} \quad \text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 + \text{NHC}(-2)(-2)(-3)$$

$$T = 11 \quad [0] \quad 1^{111} 3 \quad 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 9 & 5 \\ 7 & 8 & 1 & 5 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 6 & 0 & 13 & 7 \\ 5 & 6 & -1 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.179} \quad \text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{e}}_8 + \text{NHC}(-2)(-2)(-3)$$

$$T = 11 \quad [0] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_8 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.13 \rightarrow 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 14 & 13 & 1 & 7 & 8 \\ 13 & 12 & 1 & 7 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & -5 & -3 \\ 8 & 8 & -1 & -3 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_8 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -246, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.86 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 4 & -5 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 4 & 8 & 0 & 4 & 1 & 9 & 5 \\ 3 & 4 & -1 & -5 & 0 & 5 & -4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_8 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -511, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 5 \\ 7 & 8 & 1 & 5 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 6 & -4 \\ 9 & 12 & 0 & 6 & 13 & 7 \\ 5 & 6 & -1 & -4 & 7 & -3 \end{pmatrix}_{b_0}$$

$$\mathbf{6.180} \quad \mathbf{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathbf{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$$

$$T = 12 \quad [0] \ 1^{\overline{111}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 106, \Delta = -196, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.67 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 1 & 5 \\ 7 & 8 & 1 & 5 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 6 & 0 & -4 \\ 9 & 12 & 0 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & -4 & 7 & 0 & -3 \end{pmatrix}_{b_0}$$

$$\mathbf{6.181} \quad \mathbf{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathbf{NHC}(-2)(-2)(-3)$$

$$T = 12 \quad [0] \ 1^{\overline{111}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 106, \Delta = -196, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.67 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 9 & 5 \\ 7 & 8 & 1 & 5 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 0 & 6 & -4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 6 & 0 & 13 & 7 \\ 5 & 6 & -1 & -4 & 0 & 7 & -3 \end{pmatrix}_{b_0}$$

$$\mathbf{6.182} \quad \mathbf{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathbf{NHC}(-2)(-2)(-3)$$

$$T = 9 \quad [0] \ 1^{\overline{111}} \ 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.183} \quad \mathbf{NHC}(-2)(-3) + \mathbf{NHC}(-2)(-2)(-3)$$

$$T = 11 \quad [0] \ 1^{\overline{1}} \ 5 \ 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 3 & 5 & 0 & 5 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 5 & 3 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 3 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$\mathbf{6.184} \quad \mathbf{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathbf{NHC}(-2)(-2)(-3)$$

$$T = 10 \quad [0] \ 1^{\overline{111}} \ 3 \ 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} \ 3 \ 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 5 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 15 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 15 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 9 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 11 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 7 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.185 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 0 & 4 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.186 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{NHC}(-2)(-2)(-3)$

$$T = 11 \quad [0] \quad 1^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 2 & 3 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

7
NHC
 $(-2)(-4)$
External
(1 combos, 1 bases)

7.1
nhc₂₄

$$\begin{aligned}
 T = 9 \quad [0] \quad 1^{\textcolor{blue}{1}}2231321^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{red}{u}\textcolor{red}{8} \cdot \textcolor{blue}{s}\textcolor{blue}{o}\textcolor{blue}{16}] \quad T = 9, \, T_{\max} = 12, \, \Delta = -85, \, \det = -4, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 5.0 \\
 T_{\min} = 12, \, T_{\max} = 12, \, T \in [12, 12] \\
 \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 2 & 0 \\ 1 & 2 & 2 & 4 \\ 2 & 0 & 4 & 4 \end{pmatrix}_{b_0}
 \end{aligned}$$